

UDC

Universal Decimal Classification

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Universal Decimal Classification

Standard Edition

Volume 1 – Systematic tables

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FOREWORD

Overview

This edition of the UDC has been created from the entire content of the UDC Master Reference File database (over 66 700 entries). It takes account of official amendments up to and including those in *Extensions and Corrections to the UDC*, Number 26, November 2004. It supersedes *Universal Decimal Classification*, BS 1000M:1993.

UDC is an indexing and retrieval language for classifying information on all subjects and in all forms. It was originally derived from Dewey Decimal Classification (DDC), and managed by the organization known as the International Federation for Information and Documentation (FID). It was first published, in French, from 1904 to 1907.

The preparation and publication of the English versions of UDC as BS 1000 was at the original joint request of the British Society for International Bibliography (BSIB) and the Association of Special Libraries and Information Bureaux (ASLIB), now amalgamated as Aslib, the Association for Information Management. It was approved by Lake Placid Education Foundation, New York, then proprietors of the Dewey Decimal Classification. DDC is now published by Forest Press, a division of OCLC Online Computer Library Center, Inc., Albany, NY, USA, the current owner of the copyright in DDC.

On 1 January 1992, all rights and responsibilities for UDC were transferred to the UDC Consortium (UDCC), whose founder members were:

- AENOR (Asociación Española de Normalización y Certificación), Spain
- Bohn Stafleu Van Loghum, Netherlands
- BSI (British Standards Institution), UK
- CLPCF (Centre de Lecture publique de la Communauté française de Belgique), Belgium
- FID (International Federation for Information and Documentation)
- INFOSTA-NIPDOK (Information Science and Technology Association), Japan.

Since 1992 UDC has been maintained in English in a database called the UDC MRF. The UDC MRF was derived from the machine-readable files of the first edition of BS 1000M: 1985, and was then amplified and updated. The classification is regularly updated by UDCC (<http://www.udcc.org>) and a new database file is released and distributed to licence holders annually. During the period 1993–2005 UDC has been extensively revised and amended. The list of classes that have undergone major change is published at http://www.udcc.org/major_changes.htm.

This edition is published under general licence from the UDCC, whose Executive Committee authorizes editions in various languages and at various levels of inclusiveness. This edition corresponds to what was previously known as the 'medium' edition and appears in two volumes, of which this is *Volume 1 – Systematic tables*. The alphabetical subject index constitutes *Volume 2*.

Typographic presentation

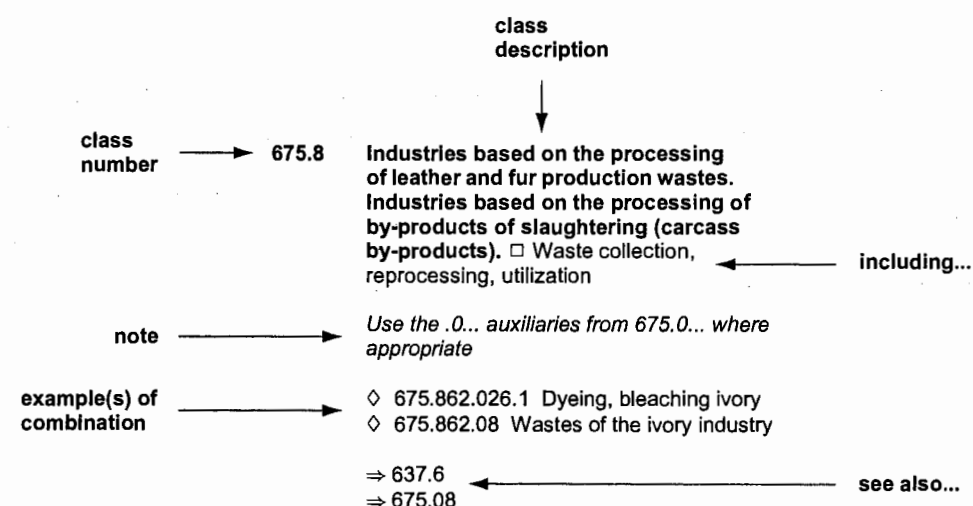
Layout

A UDC entry consists of three main elements:

- the class number (notation), which appears in the number column at the left
- the class description (caption), which appears in the text column at the right
- various amplifications, which appear indented within the text column.

The latter include notes, cross-references, instructions for synthesizing and examples of compound notation obtained. Class description can be extended with other concepts that are included in the class; in this edition the word 'Including' is replaced by the symbol □ (square). It should be noted that synthesis is available to the user wherever it results in a meaningful combination, and the compound numbers shown are only examples from a potentially vast number. In this edition, examples are indented and they are preceded by the symbol ◇ (diamond).

Sample entry layout



Special auxiliary tables, usually placed at the beginning of the main class in which they are valid, are marked with a vertical line to the left of the class number.

Summary of symbols used

In addition to the UDC notation, including the signs in Tables 1a to 1k, the following symbols are used in this edition:

sign		meaning
⇒	(double arrow)	see also
◇	(diamond)	example(s) of combination
□	(square)	including
≡	(approximately equal to)	subdivide as
	(sidelining)	special auxiliaries
*	(asterisk)	non-UDC notation (see Table 1h)

Acknowledgements

The UDC was originally derived from the Dewey Decimal Classification, and the schedules and tables in this edition incorporate some material published in later editions of DDC © 1911, 1913, 1915, 1919, 1922, 1927, 1932, 1942, 1951–52, 1958, 1965, 1971, 1979, 1989, variously by Melvil Dewey, Lake Placid [Club] Education Foundation, Forest Press and OCLC Online Computer Library Center, Inc., Albany, NY, USA.

It is possible that some proprietary names occur in this publication. No reflections on their legal status, either by the British Standards Institution or by any of its cooperating organizations, are intended, and none should be inferred from this publication.

Many place names are included. The names used, the selection and sequence of regions, descriptive expressions and relations implied by the numeric hierarchy do not constitute any endorsement of their national or international status either by the British Standards Institution or by any of its cooperating organizations. No opinions about any of these matters should be inferred from this publication.

BSI would like to express their gratitude to Aida Slavic-Overfield from the BSI UDC advisory board. Aida was not only able to provide invaluable advice during the development stages of this new edition but also committed a tremendous amount of time and effort in editing the proofs. Thanks are also due to Rich Cutler from Helius for all his technical expertise.

INTRODUCTION

1 History of UDC

1.1 The use of decimal notation as a code for expressing the concepts in a documentary classification was first proposed by the physicist André Made Ampère (1775–1836) and popularized by the American librarian Melvil Dewey in the later nineteenth century. Dewey's scheme was developed and used in the library of Amherst College, Massachusetts, and first published in 1876. The first edition, entitled 'Classification and subject index', was brief, with its ten pages of tables containing 919 headings, and lacked some of the features later associated with the scheme (it was not called decimal, and contained no decimal fractions); nor was the order of subjects particularly innovative, deriving from a tradition going back to the Paris booksellers of the seventeenth century; but it still contained the beginnings of a system that was to prove immensely serviceable and influential. Later, the three-digit minimum and the decimal point after the third digit were introduced, the scheme was expanded in successive editions, and its use spread rapidly throughout the USA and then in other English-speaking countries. It played an important part in establishing the norm of a systematic code denoting the subject as a primary means of arranging and retrieving literature in libraries grouping together works on similar subjects, irrespective of marks identifying individual documents.

1.2 In 1895, a further step in the development of decimal classification was taken by Paul Otlet (1869–1944), a young Belgian barrister already noted for his work in bibliography in the social sciences. Under the aegis of the newly founded Institut International de Bibliographie (IIB) in Brussels, Otlet and his collaborator Henry La Fontaine (1854–1943) were working on the projected Universal Bibliographic Repertory, which was intended to become a comprehensive classified index to all published information. A means of arranging the entries would be needed, and Otlet, having heard of Dewey's Decimal Classification, now in its fifth edition (1894), had obtained a copy and been deeply impressed by it. He wrote to Melvil Dewey in 1895 and obtained permission to translate it into French. Otlet and La Fontaine saw in Decimal Classification a taxonomy of human knowledge that could be expressed 'in an international language – that of numbers'; they saw too that, because of the extensibility of decimal numbers, it could readily accommodate the detail needed for bibliographic rather than strictly library use. The idea outgrew the plan of mere translation, and a number of radical innovations were made, adapting the purely enumerative classification (in which all the subjects envisaged are already listed and coded) into one which allows for synthesis (that is, the construction of compound numbers to denote interrelated subjects that could never be exhaustively foreseen); various possible relations between subjects were identified, and symbols assigned to represent them. They also realized that characteristics common to many subjects could be assembled as a separate list; repeated patterns of digits where the same characteristic of division was applied were already present to some extent in Dewey's first edition, but now the further step was taken of detaching the terminal digits and listing them as tables of auxiliary numbers, which could be added as required by the user. This synthetic principle meant that a level of detail could be obtained much greater than what was actually displayed in the published scheme: higher precision combined with economy of presentation. At the same time, Otlet and La Fontaine were expanding the purely enumerative content, to provide for the extensive requirements of the Repertory, and the result was a scheme both more copious and more sophisticated than its predecessor. It contained about 33 000 subdivisions. It was published by the IIB, in French, from 1904 to 1907, and was called the 'Handbook to the Universal Bibliographic Repertory' (*Manuel du Répertoire Bibliographique Universel*). It was in fact the first edition of UDC.

1.3 The scheme continued to expand, though interrupted by the First World War, and work progressed on a second edition, in which editorship was shared with Frits Donker Duyvis of the Dutch Patent Office, who was responsible for the extensive revision and expansion of the science and technology sections. It was published from 1927 to 1933, and by now had over 70 000 subdivisions. The Universal Bibliographic Repertory, the impetus for the scheme, was proving unmanageable, and this edition was now offered independently under the title 'Classification Décimale Universelle', the name by which (in various translations) it is still known. This edition became the master version of UDC, and remained so until 1993, when a new database became the authoritative source; from 1933 to 1993 the second French edition, as modified and augmented by approved amendments collected in various supplements and later in the serial *Extensions and Corrections to the UDC*, was the official source for all UDC editions. A third edition, the first in German, was next begun, under the editorship of Carl Walther, and published from 1934 to 1951; this was probably double the size of the second, about 140 000 subdivisions.

1.4 Interest in UDC in the United Kingdom was particularly promoted by the enthusiasm of Dr S.C. Bradford (1878–1948), who was keeper of the Science Museum Library from 1925 to 1937, and responsible for its adoption of UDC in 1928. The *Classification for Works on Pure and Applied Science in the Science Museum Library*, third edition (but the first to use UDC) was published in 1936, and was the first completed (though selective) edition of UDC in English. It was in fact an abridged edition (about 6000 subdivisions) with emphasis on science and technology. Work was also under way on a translation of the still authoritative second (French) edition, and this was to be the beginning of a fourth full edition of UDC, the first in English. Parts of this were published in 1936 to 1939, jointly by the British Society for International Bibliography and the Association of Special Libraries and Information Bureaux (BSIB and ASLIB later united to form Aslib, the Society for Information Management). At their request, the British Standards Institution (BSI) assumed responsibility for publishing UDC, and, after issuing slightly corrected versions of the existing parts in 1943, continued to produce English editions in the BS 1000 series.

1.5 The Institut International de Bibliographie (IIB) in 1931 became the Institut International de Documentation (IID), now with its headquarters at The Hague; in 1937 it changed to the Fédération Internationale de Documentation (FID), and in 1988 changed again to the International Federation for Information and Documentation (FID) – the name it held until the Federation's dissolution in 2000. FID remained for many years the centre for the management and maintenance of UDC. In 1991, acting on the recommendations of the specially formed Task Force for UDC System Development, it held talks with several UDC publishers about restructuring the finance, management and maintenance of the scheme to ensure its future into the next century. As a result of these talks, the UDC Consortium (UDCC) was formed, with FID, BSI and four other publishers as founder members (see Foreword). All rights were transferred to the UDCC on 1 January 1992. Its first priority was the creation of a database of 60 000 entries, known as the Master Reference File (MRF), which was completed in the spring of 1993 and is now the authoritative statement of the content of UDC. The former full editions (totalling about 220 000 entries) remain valid.

1.6 Since August 1949, authorized amendments have appeared in *Extensions and Corrections to the UDC*, issued at first six-monthly, later annually.

1.7 The working languages of FID were English, French and German, and these remained the medium for the maintenance and management of UDC (including all amendments appearing in *Extensions and Corrections to the UDC*) until 1992, when the UDCC took over. The official language of the Consortium is English. The Master Reference File is currently in English, but is planned to be extended to include French and German, and perhaps other languages. Since its early editions, UDC has been translated into many other languages and is used in over one

hundred countries worldwide. There are editions of various lengths in 39 different languages listed in the 'UDC Translations: A survey report and bibliography' (*Extensions and Corrections to the UDC*, 26 (2004), 58–69; an updated version of the same report is available at <http://dlist.sir.arizona.edu/649/>).

2 Characteristics of UDC

2.1 Nature of classification

Classification is a means of bringing order to a multiplicity of concepts or items of information, by arranging them into classes. A *class* is a group of concepts that have at least one thing in common. This shared property gives the class its identity. A class may be further divided into smaller classes (or subclasses), and so on, until no further subdivision is feasible. So classification is likely to be hierarchic, with each level of division (except the lowest) divided into its logical subsets.

Classes may consist of various kinds of concept, such as physical things (objects, persons, places, etc.) and their parts, activities, processes, abstract ideas; for example:

- buildings (schools, churches, houses, etc.) – things
- parts of buildings (doors, walls, stairways, etc.) – parts
- building services (joinery, glazing, plumbing, etc.) – activities
- architectural styles (classical, Georgian, etc.) – abstract ideas.

2.2 Kinds of classification

Classifications may be designed for various purposes. They include scientific and documentary classifications. *Scientific* classifications arrange the phenomena of the natural world as an aid to systematic study. They include the arrangements in systematic botany and zoology, and the table of chemical elements, and they often form the basis of field guides. The other kind of classification is designed for *information organization and retrieval* – in other words, for locating knowledge recorded in various information resources (objects, images, sounds, printed or digital documents), hence its name: *documentary* classification. The above categories of classification scheme are not mutually exclusive. Scientific classifications may be incorporated into documentary ones, as they have been to some extent in UDC, for example at 549 and (though not explicitly) parts of 58 and 59. Bibliographic and library classifications may each incorporate some of the other's features, and in practice are not limited to a single application.

2.2.1 In one respect, UDC differs markedly from *scientific* classifications, which tend to be taxonomies – that is, isolating phenomena and allocating to each a unique place in the scheme. UDC, like any documentary classification, is an *aspect classification*, in which a phenomenon is classed according to the context or discipline in which it is considered (so that its various aspects occur at many different places). For example, 'coal' has no single place in UDC. The petrological aspect is at 552.574, the economic geology aspect at 553.94, the mining aspect at 622.33, or as dust at 622.411.52, the soil mechanics aspect at 624.131.27, the agricultural aspect (fertilizers) at 631.878, the fuels aspect at 662.66 and 662.74, or as briquettes at 662.814, and coal for pig iron production is at 669.162.16. More peripheral aspects are at still other numbers. Theoretically, general documents covering all of those aspects should be classed at all of those numbers; but in a given collection, with a particular bias, it is often possible to select one number as the main place for a subject.

2.2.2 Documentary classifications tend to be either for library use – for the physical arrangement of an actual collection and as a guide to its coverage; or bibliographic – for the detailed indexing and description of documents, not confined to any one collection. They will often be enumerative – exhaustively listing the categories into which the universe of knowledge has been divided; or faceted – identifying characteristics common to many categories and arranging them in lists or tables each representing a facet (that is, the total of concepts obtained by applying a particular characteristic of division). Thus, the compiler has analysed subjects into

their elements for the user to recombine or synthesize, and faceted classification is also called analytico-synthetic. UDC has evolved from a project to develop an enumerative into a faceted classification.

2.2.3 Classifications may be special – concentrating on, or biased towards, a particular subject; or general – covering the universe of knowledge. UDC is a general classification scheme that is particularly detailed and sophisticated. It can therefore be used both for information organization in covering all subjects, or most of them, and in document collections which are more specialized but still cover a range of other subjects in less detail. UDC was designed for bibliographic use, but has proved eminently suitable for library use. Though general, it has been found useful in special collections and services, and a number of specialized selections (known as special-subject editions) have appeared.

2.3 UDC notation

Notation is a code that symbolizes the subject of each class and its place in the sequence. A simple list of named classes filed alphabetically would not fulfil the purpose of keeping related items together or separating unrelated ones. This can be done by using a notation which has an inherent order, such as numerals, alphabetic notation or a mixture (alpha-numeric). When such a code is assigned to each class, it expresses and fixes the order of classes (that is the *filing order*), and enables automatic sorting of entries. Notation with variable length can also express the position in the hierarchy, with each extra character representing a lower level; this is called *expressive notation*. Arabic numerals, arranged as decimal fractions, are ideal for this purpose and are the basis of the notation in UDC.

Notation is an artificial indexing language, from which many of the ambiguities of natural language have been eliminated. For example, the term 'paraffin' has both a technical sense (a series of saturated aliphatic hydrocarbons of the general formula C_nH_{2n+2}) and a popular one (= kerosine, a petroleum fraction with a particular boiling range), while 'kerosine' has at times been known as 'petroleum', a term now used as the general name for mineral oils. But in UDC there is no such ambiguity; in the context of mineral oil technology, 'mineral oil' generally (petroleum) is at 665.6, 'production of paraffins' is at 665.637.2, and 'production of kerosine' is at 665.634. Similarly, 'power' in mechanical engineering (the output of an engine) is at 621.1.018.7, and 'power' in electrical engineering is at 621.3.016.2, with 'power supply' at 621.311. Other aspects of these subjects may occur elsewhere, but in each case the class number represents a clearly defined concept, not a word or phrase whose meaning may vary according to context.

UDC notation is based on Arabic numerals (plus a few other common symbols). The numbers are arranged as decimal fractions, and this determines their filing order. One can think of them as following an imaginary nought and point, which for convenience are omitted (for example, 5 stands for 0.5). So they do not have the same order as integers, in which 6 would precede 59 by a long way; 5 is followed not immediately by 6, but by 50 to 59. Similarly, 59 is followed by all its subdivisions from 591 to 599 before you reach 6; and between 591 and 592 come all the subdivisions of 591 up to 591.9. After the third digit, there is a point, but it is **not** a decimal point – merely punctuation for ease of reading. A long code is easier to read if it is broken into small groups, so a point is added after every third digit; for example, in the UDC database the eleven-digit number 62138233332 becomes 621.382.333.32, which is far more manageable. As the whole class number is a decimal fraction, including the part before the first point, it is preferable to pronounce it not as an integer but as a series of digits ('six-two-one point three ...' rather than 'six hundred and twenty-one point three ...').

As an example, the following numbers would be in this order if they were integers (or integers plus decimals): 1, 02, 3, 6, 22, 37, 66, 82, 94, 210, 543, 655, 681.81, 728.1, 811, 929; but if they are UDC class numbers, they file as if they were all decimal fractions, thus:

class number	as if
02	0.02
1	0.1
210	0.210
22	0.22
3	0.3
37	0.37
543	0.543
6	0.6
655	0.655
66	0.66
681.81	0.68181
728.1	0.7281
811	0.811
82	0.82
929	0.929
94	0.94

Because decimal fractions are infinitely extensible, it is always possible to introduce further subdivisions without altering the ordinal value of the rest of the sequence. New developments, or increasingly detailed information, can therefore be accommodated in the scheme by the creation of new classes while preserving the stability of the rest. Such notation is said to be *hospitable*.

NOTE: For filing order in general, see 2.4.3.

2.4 Structure of UDC

It will be apparent from its history that UDC is a hybrid of two kinds of documentary classification scheme (enumerative and analytico-synthetic), and its structure reflects this. There are two kinds of table: the main and auxiliary tables (sometimes called the 'schedules' and 'tables' respectively), and, though the distinction is not hard and fast, they exemplify to some extent the enumerative and analytico-synthetic elements in UDC, and their associated notation (see Figure 1).

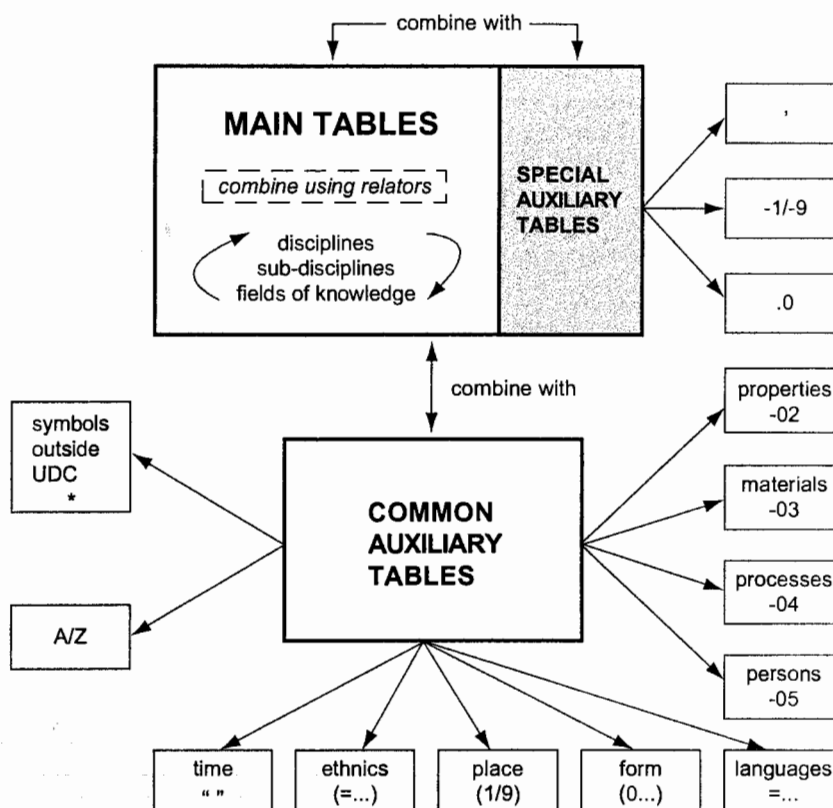


Figure 1. UDC macrostructure

2.4.1 Main tables

2.4.1.1 The main tables, and thus the primary notation, in UDC are the ones most closely related to DDC. In Dewey's scheme, the universe was divided into ten classes, each of which was then further divided. In UDC, one class is now vacant, since class 4 was cancelled in 1963 to make room for future developments, and its subject, linguistics, was merged into class 8 with literature. UDC therefore comprises the following top classes:

- 0 Generalities. Science and knowledge. Organization. Information. Documentation. Librarianship. Institutions. Publications
- 1 Philosophy. Psychology
- 2 Religion. Theology
- 3 Social sciences. Statistics. Politics. Economics. Trade. Law. Government. Military affairs. Welfare. Insurance. Education. Folklore
- [4] [Vacant]
- 5 Mathematics and natural sciences
- 6 Applied sciences. Medicine. Technology
- 7 The arts. Recreation. Entertainment. Sport
- 8 Language. Linguistics. Literature
- 9 Geography. Biography. History

Thus, each of these broadest classes is denoted by a single-digit Arabic number. Each of the ten theoretical classes, or nine occupied classes, may in turn be divided to form ten narrower classes (or *subclasses*). The *subclasses* accommodate more restricted concepts, as represented by longer numbers. For example, class 5 is divided into the following *subclasses*:

- 50 Generalities about the pure sciences
- 51 Mathematics
- 52 Astronomy. Astrophysics. Space research. Geodesy
- 53 Physics
- 54 Chemistry. Crystallography. Mineralogy
- 55 Earth sciences. Geology, meteorology etc.
- 56 Palaeontology
- 57 Biological sciences in general
- 58 Botany
- 59 Zoology

The next level of division gives three digits; for example, 57 is divided into:

- 572 Anthropology
- 573 General and theoretical biology
- 574 General ecology and biodiversity...
- 575 General genetics. General cytogenetics... Evolution etc.
- 576 Cellular and subcellular biology. Cytology
- 577 Material bases of life. Biochemistry... Biophysics
- 578 Virology
- 579 Microbiology

Each of these is a logical subdivision of the content of class 5 and 57 respectively, and each is denoted by a number beginning with 5; but each is also a more restrictive class than 5, and has a class number one digit longer. The main tables, in other words, are divided hierarchically, with the numeric hierarchy reflecting the conceptual hierarchy; the broadest classes are at the highest level, and the narrowest or most restrictive classes are at the lowest level, of the hierarchy. The length of the class number is indicative of the degree of detail. Class numbers of the same length, denoting a similar level of generality (or extension) are *coordinate*. Classes with shorter numbers, denoting greater extension, are *superordinate*. Classes with longer numbers, denoting greater specificity (or intension) are *subordinate*. 58 and 59 are coordinate. 5 is superordinate to 58 and 59. 591 is subordinate to 59. An

exhaustive set of coordinate classes (containing the full logical subdivisions of the superordinate class) is an *array*.

2.4.1.2 The parsing of an example will show the structure of a long class number from the main tables:

621.397.132.125 Phase Alternating Line (PAL) system

This string of twelve digits represents a chain in a hierarchy of concepts, in which each successive place implies a choice from the possible subclasses:

6	Applied sciences. Medicine. Technology
62	Engineering. Technology in general
621	Mechanical engineering ... Electrical engineering. Machinery
621.3	Electrical engineering
621.39	Telecommunication ... Telecontrol
621.397	Video technology. Television engineering
621.397.1	[vacant]
621.397.13	Television
621.397.132	Colour television
621.397.132.1	Simultaneous colour systems
621.397.132.12	With common transmission channel for the primary colour signals
621.397.132.125	Phase Alternating Line (PAL) system

The first digit, 6, denotes 'Applied sciences. Medicine. Technology'. A start has been made along a path down the hierarchy, which branches at each successive level: out of the universe of information, a class of concepts has been selected that belong to the applied sciences. The second digit, 2, shows that, of the ten possible subclasses 60 to 69, the one selected is 62, restricting us to engineering. Each succeeding digit further specifies the preceding concept, and symbolizes a more restrictive class, until the lowest level is reached. The meaning of each digit is determined by its place in the chain (2 means 'engineering' only if it follows 6). The full number 621.397.132.125 exhibits increasing specificity going towards the right.

2.4.1.3 Reality sometimes declines to fit into predetermined categories, and the universe cannot consistently be divided into decimal fractions. Not every number is necessarily occupied. Nought tends to be reserved for a special purpose, particularly when following a point, but most arrays occupy fewer than the nine remaining places. Occasionally they require more than nine places (e.g. ministries occupy the range 354.11/.86).

2.4.2 Auxiliary tables

2.4.2.1 UDC's most innovative and influential feature was its auxiliary notation, the signs and subdivisions provided to allow for the construction of compound numbers, or synthesis. The auxiliary tables list concepts which recur throughout several or all subjects, such as time and place, and are therefore applicable in a range of classes. They are added to a class number to express a more precise meaning.

Auxiliary notation is of two main kinds: the common and special auxiliaries. The *common auxiliaries*, as well as providing a means of expressing interrelations between subjects, denote *generally recurrent* characteristics (i.e. those that are applicable throughout the main tables). The *special auxiliaries* denote *locally recurrent characteristics* (i.e. those that are applicable in a limited range of the main tables).

There are also linking signs which enable combination of numbers from different parts of the tables. For convenience, these are also presented as tables. A number taken from a single place in the tables and cited on its own – whether a main number or an independent auxiliary – is a *simple number*, e.g. (410) and 622 are simple numbers. A number created by synthesizing, using elements from more

than one place in the tables, is a *compound number*, e.g. 622+669 and 622(410) are compound numbers.

2.4.2.2 Common auxiliaries. The common auxiliaries comprise two kinds of symbol: the signs and subdivisions. The common auxiliary *signs* are the plus, the stroke, the colon, the square brackets and the double colon, which serve as relators, linking UDC numbers (either main or auxiliary), but are not themselves numbers, do not represent classes, and cannot be subdivided. Not many kinds of relation are distinguished: the plus and the stroke represent kinds of *aggregation* (the sum of meanings of several UDC numbers), while the colon serves for most other relations. When UDC numbers are linked by colon, it merely shows that the subjects denoted by the numbers are related to each other in some way; it does not specify which influences the other(s), nor show the nature of the influence exerted – in short, it does not denote the *phase* of the relation. For example, the compound:

658.512.2:004 Industrial design in relation to computers and data processing

does not indicate whether computers are the instrument or the object of designing (computer-aided design or design of computers). Changing the order of the elements in such a compound does not in principle change the meaning: it is stated in Table 1b that the relation is reversible, and that A:B and B:A are of equal significance. In private use, however, the order might be made significant. Recent additions to the UDC vocabulary are the algebraic subgrouping and order-fixing signs (Table 1b, Sections 2 and 3). These to some extent afford the user a means of clarifying relations between subjects.

2.4.2.2.1 The lack of specificity in the common auxiliary signs, notably the colon, implies a correspondingly extensive usefulness: this device enables any concept in the whole classification to be related to any other, and so in a sense to qualify it. Such a copious range of qualifiers could hardly be achieved by other means.

2.4.2.2.2 The common auxiliary *subdivisions* consist of numeric tables, in which concepts are enumerated and arranged hierarchically; to this extent, they resemble the main tables, but they are distinguished by their own symbols either prefixed to, or enclosing, the number. Common auxiliary tables are as follows:

concept	symbol
the linking signs – Tables 1a and 1b	+, /, :
language of the document – Table 1c	= ...
form of the document – Table 1d	(0/09)
place – Table 1e	(1/9)
race, nationality etc. – Table 1f	(= ...)
time – Table 1g	" ... "
non-UDC codes etc. – Table 1h	#, A/Z
general characteristics – Table 1k, includes	-0...
properties	-02...
materials	-03...
processes	-04...
persons	-05...

It is important to note that two of the tables (language and form) denote characteristics of the document (or information carrier), e.g. the language in which it is written or spoken. The others denote aspects of the subject, e.g. the materials or persons involved. One must distinguish between:

811.112.2 German (as a subject – main number) and
 =112.2 German (language of document – auxiliary number), as in
 5=112.2 Scientific works in German.

Some features, such as time and space, are relevant to practically all phenomena, while others, such as language and documentary form, become relevant as soon as a phenomenon becomes the subject of a document. When a given characteristic of

division recurs throughout a classification, it is convenient and mnemonic if the resultant facet is expressed in the same notation wherever it occurs; but this also makes it possible for the digits expressing it to be detached and separately listed. By this means, they become available to be affixed, theoretically, to every class number in the main tables, so permitting more specific denotation of any concept in the scheme. Removed from their context, they require an extra symbol to identify the characteristic of division (a *facet indicator*); for example, the digits 1 to 9 have many different meanings, but when they are enclosed in parentheses, this announces that the place facet is to be enumerated. Similarly, quotation marks announce the time facet, and so on. Some of these symbols serve more than one purpose, and their meaning depends on what immediately follows; for example, an opening parenthesis may be followed by 0, denoting form, by 1 to 9, denoting place, or by the equals sign, denoting race and nationality. The numeric part of the auxiliary notation is punctuated after every third digit, as with the main tables.

2.4.2.2.3 The common auxiliary subdivisions fall into two groups: the independent and the dependent auxiliary tables. The *independent auxiliary tables*, though they may be affixed to any UDC number where appropriate, may also be used on their own, to form the whole class number for a document. These are Tables 1c to 1g, the auxiliaries of language, form, place, race and time. For example, if it were decided that the place facet was the only one that needed to be expressed (e.g. in classifying maps), a way of doing this would be to cite the place auxiliary alone, from Table 1e. The *dependent auxiliary tables* must always be affixed to a UDC number; these are, officially, Table 1k common auxiliaries of general characteristics: properties, processes, materials and persons. Additionally, Table 1h, which specifies means of adding non-UDC notation, is in practice dependent (though not described as such), since the asterisk and alphabetic extensions must be added to a UDC number.

2.4.2.2.4 Most of the independent auxiliary tables have symbols that enclose the number, thus demarcating it from adjacent numbers. These are *biterminal signs* – that is: signs with both an opening and a closing element. Tables 1d, 1e and 1f have parentheses, while Table 1g has quotation marks. The exception is Table 1c (Common auxiliaries of language), which has only an initial equals sign; this is compensated for by adding a colon at the end of the language notation in some positions, as in the example =133.1:641.5(083.1) given in the preamble to Table 1c. Because of their demarcation, auxiliaries with biterminal signs can be affixed to any part of a UDC number, at the beginning, middle or end (i.e. *prefixed, infixes or suffixed*, as well as being usable independently, e.g.

(410)	Great Britain. United Kingdom of Great Britain and Northern Ireland
(410)622.33	Britain – coal mining
622(410).33	Mining – Britain – coal
622.33(410)	Coal mining – Britain

This is in contrast with the dependent auxiliaries, which are used only as suffixes, e.g.

622.33-022.316	Coal mining – sustainable
----------------	---------------------------

though they may be followed by further auxiliaries, e.g.

622.33-022.316(410)	Coal mining – sustainable – Britain
---------------------	-------------------------------------

The asterisk, the point and the hyphen, therefore, can never occur at the beginning of a number.

2.4.2.3 Special auxiliaries. The special auxiliaries, unlike the common auxiliaries, are not listed in one place, and by definition do not have such extensive applicability. They occur at various places in the tables, and express concepts that are recurrent, but in a more limited subject range.

2.4.2.3.1 Of the notation used in special auxiliaries, the three main kinds are described and differentiated in auxiliary tables. Section II: the hyphen series -1/-9, the point-nought series .01/.09 and the apostrophe series '1/'9. (.0 is almost always used to introduce special auxiliary subdivisions, but there are a few exceptions, such as 626.0 and 669.05.) In this edition, special auxiliary numbers are recognizable by side-lining, for example under UDC 53:

53	Physics
53.02	General laws of phenomena
53.05	Observation and recording of phenomena. Visual indication of phenomena ...

The number **before** the side-lining is a main number, 53 'Physics'. The auxiliary notation .0... may be used either in the form in which it is shown, i.e.

53.05 Observation and recording of phenomena etc. (in general)

or detached and added to any direct decimal subdivision of the main number, e.g.

531	Mechanics
531.05	Observation and recording of mechanical phenomena
531.5	Gravity ...
531.5.05	Observation and recording of gravitational phenomena

2.4.2.3.2 Most special auxiliaries are enumerative, though there is one synthesizing sign, the apostrophe (which, however, is also used in some places to introduce enumerated subdivisions). Special auxiliaries are always listed as suffixes to other numbers, and cannot be used independently. A common form of presentation is a list immediately under the main number where they are valid. If no other indication is given, they are applicable also to all the direct divisions of that number. For example, at 621.3.01 to .095.4, the .0 element is applicable anywhere from 621.31 to 621.398, as well as 621.3 itself.

2.4.2.3.3 Special auxiliaries are applicable only where indicated, and the same notation may be used elsewhere with a different meaning. In a few cases, special auxiliaries have either an extended or a reduced range of applicability. This is always indicated by a note, e.g. at 52-1/-8 or 616. For instance, hyphen auxiliaries under 62 are applicable throughout the range 62 to 69 (as indicated in a running footnote), and the point-nought auxiliaries listed at 7, applicable throughout the class except under 77. Also, for example, the 666.1.03 and .05 auxiliaries are valid not only under 666.1, but throughout 666.11/.28, as is indicated in the note preceding them; and a note at 666.21/.28 draws attention to the availability of these auxiliaries in this section outside their expected range.

2.4.2.3.4 Special auxiliaries may also occur within other series of auxiliaries, as they do, for example, at the beginning of Table 1d: Common auxiliaries of form. The principle is the same: they may be used in the form in which they appear, e.g.

(0.035.22) Transparent, strip-form [documents]. Microfilm

or the special auxiliary element (beginning with .0) may be detached and affixed to any of the direct divisions of (0), e.g.

(05) Serial publications. Periodicals
(05.035.22) Periodicals on microfilm.

The compound thus formed is still a common auxiliary, which may qualify a main number, e.g.

5(05.035.22) Science periodicals on microfilm.

The -6 auxiliaries at 66.041 are an example of special auxiliaries within other special auxiliaries.

2.4.2.3.5 There is another device, though not an auxiliary table at all, that is similar to special auxiliaries in that it results in the same notation being used to denote a

given concept in more than one place. It is *parallel division*, signalled in the tables by the *subdivide-as* sign $\equiv$. This simply means that the number preceding $\equiv$ may be subdivided in a manner analogous to the number following it; this will result in an exactly analogous array, with the same concepts expressed by the same sequence of digits. A simple example is in 611 'Anatomy', parts of which are parallel to 616 'Pathology', where both are subdivided into particular organs. The compiler has opted to give more detail under 616 (the pathology of particular organs), but it would not matter which was chosen as the main place: the point is that they are parallel, and to enumerate organs fully in both places would be a waste of space and effort. Instead, there are instructions such as that under 611.2 'Respiratory system': 611.21/.26 $\equiv$ 616.21/.26, or under 611.3 'Digestive system': 611.3 $\equiv$ 616.3. In these cases, the 616 subdivisions are the *source numbers*, from which digits may be detached and added to the *target numbers* under 611; thus 616.21 gives us the analogous 611.21, while 616.31 gives us 611.31, and so on. The parallel arrays may be represented symmetrically:

611.21	Nose. Sinuses	616.21
.22	Larynx (voice-box)	.22
.23	Trachea (windpipe)	.23
.24	Lungs	.24
.25	Pleurae	.25
.26	Diaphragm	.26

The numbers in the left-hand column represent these items in the context of anatomy, while those in the right-hand column represent them in the context of pathology. The list of items is the same.

2.4.2.3.6 This is generally easy to follow when the source number and the target number are the same length. When they are of different lengths, then repunctuation may be necessary. An example of this is 611.81 'Central nervous system', divided like 616.831; thus 611.813 derives from 616.831.3. The parallelism here is not obvious to the eye, but the principle is the same: the digits that follow 611.81 are inferred from the digits that follow 616.831; thus, from 616.831.3 the final 3 is detached and added to 611.81 – hence: 611.813. It may be helpful in such cases to delete the points mentally; then 61181 $\equiv$ 616831 and from 6168313 we infer 611813. We then punctuate after every third digit – hence 611.813. Here too, the parallel arrays may be represented symmetrically:

611.813	Prosencephalon (forebrain) etc.	616.831.3
.814	Diencephalon (thalamencephalon)	.4
.815	Mesencephalon (midbrain) etc.	.5
.817	Metencephalon (hindbrain)	.7
.818	Myelencephalon etc.	.8
.819	Cerebral meninges	.9

and it becomes clear that the italicized digits are common to both number columns.

2.4.2.3.7 There is no limit to the kinds of notation where parallel division is used. They may be main numbers (611.81 $\equiv$ 616.831); they may be special auxiliaries (675.025 $\equiv$ 3675.055); they may be common auxiliaries ("511" $\equiv$ "411"); auxiliary numbers may derive from main numbers (-036.4 $\equiv$ 678.4) or vice versa (811 $\equiv$ =11/=9); and a number may be the source for the parallel division of one of its own subdivisions (546.289 $\equiv$ 546.28). This should cause no confusion, as in every case where parallel division is authorized, the fact is clearly indicated, often with examples.

2.4.3 Filing order

2.4.3.1 The filing order of UDC symbols is based on a progression from the general to the particular. Thus, the common auxiliaries (which are by definition general) come first, and an independent auxiliary used alone or cited first (see 2.4.4.1) files before a main number. Next, an aggregation of several numbers has a broader

meaning than a simple number, so compounds with the plus and the stroke – Table 1a – file before a single component number (622+669 comes before 622). Finally, a shorter number files before a longer number, because a number followed by an auxiliary (unless it is an aggregation) is more specific than the simple number, while in simple hierarchic division each successive digit further specifies the concept, increasing the particularity.

2.4.3.2 Arabic numerals arranged as decimal fractions have an inherent ordinal value; the other symbols have been given an arbitrary one. Table 1 shows the filing order of both simple and compound numbers.

Table 1. Filing order

symbol	example	
=	=112.2	German language
(0...)	(0.035.2)	Microform documents
(1/9)	(430)	Germany
(=...)	(=1.410)	British nationals
"..."	"18"	Nineteenth century
+	622+669	Mining and metallurgy
/	622/623	Mining and military engineering
simple number	622	Mining
:	622:338.3	Productivity in mining
=	622=112.2	Documents in German about mining
(0..)	622(0.035.2)	Microform documents about mining
(1/9)	622(430)	Mining in Germany
(=..)	622(=1.366)	Mining among the ancient British
"..."	622"18"	Mining in the nineteenth century
*	622*Fe ₂ O ₃	Mining of red haematite (Fe ₂ O ₃)
A/Z	622GOE	Mining of named ores: goethite
-0	622-057.2	Manual workers in mining
-1/-9	622-78	Protective devices and measures in mining
.0	622.03	Geological character of ore deposits
'	622'17	Tailings, waste, residues from mining
next simple number	622.3	Mining (extraction) of specific minerals...

2.4.3.3 The algebraic subgrouping sign (square brackets) does not affect the filing order, and may be ignored for this purpose, except where class numbers would be identical but for the square brackets. Then one may apply the rule of nothing-before-something, so that the one without brackets files first:

658.512.2:004-051	Industrial design in relation to: practitioners of data processing (e.g. designing for convenience of computer operators)
[658.512.2:004]-051	Practitioners of: industrial design in relation to data processing (e.g. practitioners of computer-aided design)

2.4.3.4 *Intercalation* of auxiliaries (see 2.4.5) may lead to class numbers with infixes at any point in the hierarchy. Such a compound, where the auxiliary separates the opening element from the closing element of the simple number, files after the opening element alone and the opening element plus suffix. For example, the compound 622.341.1(430) in Figure 2 could be rearranged as:

622(430).341.1 Mining – Germany – iron ores

so creating an order such as:

622 Mining
622(430) Mining – Germany
622(430).341.1 Mining – Germany – iron ores

2.4.3.5 In general, it will be seen that the filing order of UDC symbols is that in which they are displayed in the tables. The exceptions are (i) that the simple number comes between compounds with the stroke and compounds with the colon, and (ii) that square brackets have no inherent filing order.

2.4.4 Citation order

2.4.4.1 When an element of notation is selected as representing an aspect of a document's subject, and is incorporated into a class number for that subject, it is said to be cited. The order in which the elements are combined to make a compound number is the *citation order* (as each element symbolizes a facet of the subject, the citation order is also known as the *facet formula*).

2.4.4.2 In order to ensure a sequence from general to special, it is necessary that the filing order should be the reverse of the citation order (this is called the *principle of inversion*). In theory, either might be taken as the starting point, but here we have a specified filing order with both tables as a whole, and examples of synthesis given within the tables displayed in this order. So the simplest rule for citing elements in a compound is:

the citation order is the reverse of the filing order.

This means that the elements in a compound class number proceed from special to general. If one needed to cite, say, =112.2 and (430) and "18" and '17, all qualifying 622.341.1, the standard order would be: '17; "18"; (430); =112.2, thus:

622.341.1'17."18"(430)=112.2 Iron-ore mining – wastes – nineteenth century – Germany – in German

2.4.4.3 The standard citation order may not be satisfactory for all purposes. The reason would be the need, in particular collections, to bring together (or *collocate*) all references to a particular aspect of a subject, which would be separated if the standard order were followed. For example, in the standard citation order, time "... " would precede place (1/9) (the reverse of the filing order), and so, following the main number, a sequence of entries would be arranged primarily according to time, with any given time aspect divided secondarily according to place:

622"-..."	Mining at various times BC
622"-..."(1/9)	Mining, BC, in various places
622"-..."(410.197)	Mining, BC, in Cornwall
622"+..."	Mining at various times AD
622"+..."(1/9)	Mining, AD, in various places
622"+..."(410.197)	Mining, AD, in Cornwall
622"17"	Mining in the eighteenth century
622"17"(1/9)	Mining, eighteenth century, in various places
622"17"(410.197)	Mining, eighteenth century, in Cornwall

In this example, references to Cornwall are scattered through the sequence, separated by, first, the time element, and, second, most of the place sequence, since (1/9) represents many individual entries. If, for a particular purpose, it were necessary to collocate references to mining in Cornwall, it would be necessary to exchange the positions of the time and place auxiliaries in the citation order. Then the division would be primarily according to place and secondarily according to time:

622(1/9)	Mining in various places
622(1/9)"-..."	Mining in various places, BC
622(1/9)" + ..."	Mining in various places, AD
622(1/9)"17"	Mining in various places, eighteenth century
622(410.197)	Mining in Cornwall
622(410.197)"-..."	Mining in Cornwall, BC
622(410.197)" + ..."	Mining in Cornwall, AD
622(410.197)"17"	Mining in Cornwall, eighteenth century

In this example, references to Cornwall are brought together, but references to the eighteenth century are scattered.

2.4.4.4 One may summarize the matter of citation order as follows. If there is no other preference, follow the standard citation order (reverse of filing order). But to bring together aspects that would otherwise be scattered, vary the citation order so as to place the relevant notation nearer the beginning of the compound (so affecting the filing order of the class number). The filing order is prescribed, but the citation order is optional.

2.4.5 Intercalation

2.4.5.1 *Intercalation* is the use of an auxiliary as an infix rather than a prefix or suffix. In other words, certain auxiliaries may interrupt a main number, creating a compound such as 622(430).341.1 (mentioned in 2.4.3.4). The reason for this would be to create a more helpful arrangement of documents, or references to them, when classed. It is merely an extreme case of varying the citation order to produce a different sequence. In the example given in 2.4.4.3, the place auxiliary (410.197) was brought forward to precede the time auxiliary "17", giving 622(410.197)"17". Theoretically, it might be brought still further forward, and intercalated into the main number, giving 62(410.197)2"17". This would be useful only if one wanted to group all information on the subject 62 'Engineering', then subdivide it according to place (1/9) before proceeding to ...2 'Mining'. In practice, this is not likely, but there might be a need to group information about a more specific activity, such as mining, in a given area:

622(410)	Mining – Britain
622(410).333	Mining – Britain – coal
22(410).34	Mining – Britain – metal ores
622(430)	Mining – Germany
622(430).333	Mining – Germany – coal
622(430).34	Mining – Germany – metal ores

or about institutions, such as laws, that vary from country to country:

347.78(410)	Copyright laws – Britain
347.78(410)1	Literary copyright – Britain
347.78(410)5	Copyright of music – Britain
347.78(44)	Copyright laws – France
347.78(44)1	Literary copyright – France
347.78(44)5	Copyright of music – France

2.4.5.2 This can only be done with auxiliaries that have biterminal signs i.e. parentheses or quotation marks: the common auxiliaries of form, place, race and time.

3 Management of UDC

3.1 For many years, UDC was managed by the organization created by Otlet and La Fontaine, known since 1988 as the International Federation for Information and Documentation (FID). On 1 January 1992, all rights in UDC were transferred to the UDC Consortium (UDCC), of which FID was a founder member along with five UDC publishers, including BSI. Hence, since 1992 UDC has been owned by the UDC Consortium (UDCC), an international group of UDC publishers. The

Consortium has an Executive Committee and an Advisory Board, both with international membership, and an Editor in Chief. Its headquarters is at the Royal Library in The Hague. Each member publisher has the right to issue UDC editions in its own language.

3.2 One of the UDCC's first actions after its inauguration in 1992 was to create an internationally owned database which would represent the authoritative text of UDC. This is known as the Master Reference File (MRF). It is updated once a year according to the authorized amendments, as issued in the annual *Extensions and Corrections to the UDC*. The MRF in 2005 contains 66 736 entries, currently in English, with versions in French and German planned for the future. It is available for use under licence and users can choose the type of database exports. The database is in CDS/ISIS and users buying a UDC MRF licence are also provided with database definition files if they wish to use UDC in the database.

The UDCC maintains the Master Reference File (MRF), which is the authoritative statement of the content of UDC. This has superseded the Master Version of UDC which was formerly maintained by FID, and which consisted of the text of the French edition of 1927–1933, as modified by all approved amendments since that date. The MRF is the authority for, and source of, published UDC editions.

3.3 Published editions of UDC are authorized by the Executive Committee of the UDCC in various languages and at various levels of inclusiveness, and each Executive Member organization has the right to publish them in its own language. Other publishers can apply for licences to publish UDC editions. Most published editions are in a single language, and are known as the *language editions*, there have occasionally also been multilingual editions (printed and digital). BSI is the publisher of the English-language editions. The UDC MRF is also available in an online version: UDC Online (<http://www.udconline.net/>), with a powerful and user-friendly searching interface.

3.4 The scheme is under continuous revision, a state of affairs that has been maintained since the first edition began to appear in 1904. This means that later states of knowledge in a subject, and missing concepts previously overlooked, can be considered, and if necessary acted on, without waiting until a new edition is imminent. An Editor-in-Chief is appointed, who, supported by an editorial board, is charged to maintain the quality of the UDC and enhance its international credibility. Major revision proposals are published in *Extensions and Corrections* (see 3.5), along with the authorized amendments. Once amendments are authorized, they are entered in the database, and are taken into account whenever new editions are undertaken by the publishers.

3.5 The published amendments became known from early on as *Extensions et Corrections*. Starting in 1933 with a duplicated loose-leaf updating service called *Extensions et Corrections Décidées*, this developed into a formal periodical, *Extensions and Corrections to the UDC*, from 1949. This was at first semi-annual, and later annual, cumulating in three-year series; the earlier series have been collected into two larger volumes:

Cumulated UDC Supplement (~1964)
Cumulated UDC Supplement 1965–1975

The (so far) uncollected series start with Series 9, No. 3, dated 1977. The text of the entries in *Extensions and Corrections* used to be in any one of the three FID languages, or in bilingual or trilingual versions. From Series 15 (September 1993), the entries are in English. This series has also introduced a new section entitled 'Comments and Communications', including revision proposals.

3.6 To take full advantage of UDC's continuity of revision, users therefore have the obligation of annually updating their editions, and their classifying practice, by reference to *Extensions and Corrections*. The UDC publishers, such as BSI, from time to time replace published versions with new editions incorporating recent amendments, but inevitably there is some delay. It is necessary to weigh the

advantages of up-to-dateness against the onus of the updating procedure. The full editions appear in many sections at different stages of up-to-dateness, but with self-contained versions such as the present one, it is possible to make a decision to use UDC in a given, date,d edition.

3.7 Though the revision structure is still subject to change, the UDCC continues to welcome proposals for improving UDC from any of its users. Suggestions and inquiries about UDC can be addressed to the following:

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4 Further reading

UDC Online – the complete content of the UDC MRF, updated annually – <http://www.udconline.net/>

PD 1000: 2003, *Universal Decimal Classification, International Abridged edition*. London: British Standards Institution.

Extensions and corrections to the UDC [annual]. The Hague: UDC Consortium. Contains authorised amendments to the scheme. Since 1993, it also includes articles and proposals. Content is available at <http://www.udcc.org/ec.htm>

McIlwaine, I.C. (2000) *The Universal Decimal Classification: A guide to its use* (UDC P035). The Hague: UDC Consortium.

4.1 About UDC

Books

Broughton, V. (2004) *Essential Classification*. London: Facet Publishing, pp206–254.

Batley, S. (2005) *Classification in Theory and Practice*. Oxford: Chandos Publishing, pp81–109.

Foskett, A.C. (1996) *The Subject Approach to Information*, 5th edition. London: Library Association, pp281–294.

Marcella, R. and Newton, R. (1994) *A New Manual of Classification*. Aldershot: Gower.

Articles

Boyd Rayward, W. (1975) *The Universe of Information: The work of Paul Otlet for documentation and international organization* (FID 520). Moscow: VINITI.

Boyd Rayward, W. (1991) 'The case of Paul Otlet, pioneer of information science, internationalist, visionary: reflections on biography'. *Journal of Librarianship and Information Science*, 23, 135–145 (also at: http://people.lis.uiuc.edu/~wrayward/otlet/PAUL_OTLET_REFLECTIONS_ON_BIOG.HTM).

Gilchrist, A. (1992) 'UDC: The 1990s and Beyond'. In: Williamson, N.J. and Hudon, M. (eds). *Classification Research for Knowledge Representation and Organization*. Amsterdam and London: Elsevier (FID 698), pp69–78.

Gilchrist, A. and Strachan, D. (eds) (1990) *The UDC: Essays for a new decade*. London: Aslib.

McIlwaine, I.C. (1994) 'UDC: The present state and future developments'. In: *International Cataloguing and Bibliographic Control*, 23(2), 29–33.

McIlwaine, I.C. (1995) 'UDC Centenary: The present state and future prospects'. In: *Knowledge Organization*, 22 (2), 64–69.

McIlwaine, I.C. (1997) 'The Universal Decimal Classification: Some factors concerning its origins, development and influence'. In: *Journal of the American Society of Information Scientists*, 48(4), 331–339.

McIlwaine, I.C. (2000) 'UDC in the twenty-first century'. In: Marcella, R. and Maltby, A. (eds) *The Future of Classification*. Aldershot: Gower, pp93–104.

Slavic, A. (2004) 'UDC Implementation: From library shelves to a structured indexing language'. *International Cataloguing and Bibliographic Control*, 33(3), 60–65 (also at: <http://www.ifla.org/IV/ifla69/papers/032e-Slavic.pdf>).

Slavic, A. (2004) 'UDC Translations: A 2004 survey report and bibliography', *Extension and Corrections to the UDC*, 26, 58–80 (also at: <http://dlist.sir.arizona.edu/649/>).

Websites

UDCC homepage – <http://www.udcc.org>

- UDC publications – <http://www.udcc.org/pub.htm>
- Bibliography on UDC related publications – <http://www.udcc.org/bibliography.htm>
- UDC users – <http://www.udcc.org/use.htm>
- UDC MRF licence – <http://www.udcc.org/mrf.htm#licence>

UDC Online (BSI) – <http://www.udconline.net/>

Czech UDC on the Web – <http://aip.nkp.cz/mdt/>

Italian UDC on the Web – <http://mail.biocfarm.unibo.it/%7Espinelli/cdu/>

Swedish UDC on the Web – <http://www.hb.se/bhs/udk/>

SUMMARY OF THE CLASSIFICATION

The summary extends to second-level classes (two-digit numbers) except for class 62 and a few cases where notational anomalies prevent consistency. Class 62 is summarized to the third level.

Numbers enclosed in brackets [thus] denote classes that are currently vacant.

Common auxiliary tables

- 1a Coordination. Extension
- 1b Relation. Subgrouping. Order-fixing
- 1c Language
- 1d Form
- 1e Place
- 1f Ethnic grouping and nationality
- 1g Time
- 1h Subject specification by notations from non-UDC sources
- 1k General characteristics
Properties, Materials, Processes, Persons

Main tables (schedules)

- 0 **Generalities. Science and knowledge. Organization. Information. Documentation. Librarianship. Institutions. Publications**
- 00 Prolegomena. Fundamentals of knowledge and culture
Computer science and technology, Management, Metrology
- 01 Bibliography and bibliographies. Catalogues
- 02 Librarianship
- [03]
- 030 Encyclopaedias. General reference works
- [04]
- [05]
- 050 Serial publications. Periodicals (journals, magazines etc.)
- 06 Organizations and other types of cooperation. Associations. Congresses. Exhibitions. Museums
- [07]
- 070 Newspapers. Journalism. The press
- 08 Polygraphies. Collective works
- 09 Manuscripts. Rare and remarkable works
- 1 **Philosophy. Psychology**
- 11 Metaphysics
- [12]

- 13 Philosophy of mind and spirit. Metaphysics of spiritual life
- 14 Philosophical systems and points of view. *Weltanschauung*
- [15]
- 159.9 Psychology
- 16 Logic. Epistemology. Theory of knowledge. Methodology of logic
- 17 Moral philosophy. Ethics. Practical philosophy
- [18]
- [19]

- 2 Religion. Theology**
- 21 Prehistoric and primitive religions
- 22 Religions of the Far East
- 23 Religions of the Indian subcontinent
- 24 Buddhism
- 25 Religions of antiquity. Minor cults and religions
- 26 Judaism
- 27 Christianity
- 28 Islam
- 29 Modern spiritual movements

- 3 Social sciences. Statistics. Politics. Economics. Trade. Law. Government. Military affairs. Welfare. Insurance. Education. Folklore**
- 30 Theories, methodology and methods in the social sciences. Sociography
- 31 Demography. Sociology. Statistics
- 32 Politics
- 33 Economics. Economic science
- 34 Law. Jurisprudence
- 35 Public administration. Government. Military affairs
- 36 Safeguarding the mental and material necessities of life
Social work. Social aid. Housing. Insurance
- 37 Education. Teaching. Training. Leisure
- [38]
- 39 Ethnography. Customs. Manners. Traditions. Way of life. Folklore

- [4]

- 5 Mathematics and natural science**
- [50]
- 501 Generalities about the pure sciences
- 502 The environment and its protection
- 504 Threats to the environment
- 51 Mathematics

- 52 Astronomy. Astrophysics. Space research. Geodesy
- 53 Physics
- 54 Chemistry. Crystallography. Mineralogical sciences
- 55 Earth sciences. Geological sciences.
Meteorology etc.
- 56 Palaeontology
- 57 Biological sciences in general
Molecular biology. Virology. Microbiology
- 58 Botany
- 59 Zoology
- 6 Applied sciences. Medicine. Technology**
- 60 Biotechnology
- 61 Medical sciences
Anatomy. Physiology. Pharmacology. Pathology. Surgery. Orthopaedics. Ophthalmology. Gynaecology
- 62 Engineering. Technology in general
620 Materials testing. Commercial materials. Power stations. Economics of energy
621 Mechanical engineering in general. Nuclear technology. Electrical engineering. Machinery
623 Military engineering
624 Civil and structural engineering in general
625 Civil engineering of land transport. Railway engineering. Highway engineering
626 Hydraulic engineering in general
627 Natural waterway, port, harbour and shore engineering. Navigational, dredging, salvage and rescue facilities. Dams and hydraulic power plant
628 Public health engineering. Water. Sanitation. Illuminating engineering
629 Transport vehicle engineering
- 63 Agriculture and related sciences and techniques. Forestry. Farming. Wildlife exploitation. Veterinary science
- 64 Home economics. Domestic science. Housekeeping
Cooking. Catering. Laundry. Cleaning
- [65]
- 654 Telecommunication and telecontrol (organization, services)
- 655 Graphic industries. Printing. Publishing. Book trade
- 656 Transport and postal services. Traffic organization and control
- 657 Accountancy
- 658 Business management, administration. Commercial organization
See also: 005 Management (in general)
- 659 Publicity. Information work. Public relations
- 66 Chemical technology. Chemical and related industries
Industrial microbiology and mycology. Beverages. Stimulants. Food
- 67 Various industries, trades and crafts
Timber and woodworking. Leather and fur. Pulp, paper and board. Textiles. Macromoleculars, rubbers and plastics

- 68 Industries, crafts and trades for finished or assembled articles
Precision mechanisms. Instruments. Horology. Printing technology. Optical apparatus. Sound recording and reproduction. Metal work. Furniture industry. Saddlery. Shoemaking etc. Clothing industry. Toys
- 69 Building (construction) trade. Building materials. Building practice and procedure

- 7 The arts. Recreation. Entertainment. Sport**
- 71 Physical planning. Regional, town and country planning. Landscapes, parks, gardens
- 72 Architecture
- 73 Plastic arts
Sculpture. Numismatics. Metal arts
- 74 Drawing. Design. Applied arts and crafts
- 75 Painting
- 76 Graphic arts. Graphics
- 77 Photography and similar processes
- 78 Music
- 79 Recreation. Entertainment. Games. Sport

- 8 Language. Linguistics. Literature**
- 80 General questions relating to both linguistics and literature. Philology
- 81 Linguistics and languages
- 82 Literature
- [83/89]

- 9 Geography. Biography. History**
- [90]
- 902 Archaeology
- 903 Prehistory. Prehistoric and later remains
- 904 Cultural remains of historical times
- 908 Area studies. Study of a locality
- 91 Geography. Exploration of the Earth and of individual countries. Travel. Regional geography
- [92]
- 929 Biographical and related studies
- [93]
- 930 Science of history. Historiography
- 94 General history

Part 1

Auxiliary tables

SECTION I: COMMON AUXILIARY SIGNS AND SUBDIVISIONS

+ , / TABLE 1a. COORDINATION. EXTENSION

Filing order. The symbols in Table 1a extend rather than restrict the meaning of a number, and compounds containing them file before the simple number itself. First in the filing order comes the number followed by +...; secondly, the number followed by /...; thirdly, the simple number.

Use with biterminal auxiliaries. When the plus + and stroke or slash / are used to link auxiliary numbers with biterminal signs (i.e. parentheses or quotation marks), the resultant compound number is enclosed within a single set of signs, e.g.

(470+571) Russia. Russian Federation - not (470)+(571).

"19/20" The 20th and 21st centuries - not "19"/"20".

(410)(051) Britain - periodicals (place - form).

(410)(051)"20" Britain - periodicals - 21st century (place - form - time).

SECTION 1. COORDINATION. ADDITION

The coordination sign + (plus) connects two or more separated (non-consecutive) UDC numbers, to denote a compound subject for which no single number exists, e.g.

(44+460) France and Spain.

622+669 Mining and metallurgy.

SECTION 2. CONSECUTIVE EXTENSION

The extension sign / (stroke or slash) connects the first and last of a series of consecutive UDC numbers to denote a broad subject, or range of concepts, e.g.

(7/8) North and Central America, and South America. The Americas.

592/599 Systematic zoology (equivalent to 592+593.1/.99 +599).

629.734/.735 Heavier-than-air craft. Aeroplanes.

643/645 The home and household equipment (equivalent to 643+644+645).

If the number following the stroke is more than three digits long and begins with a group of digits in common with the preceding number, it may be abbreviated by omitting the digits common to both, so long as the first item after the stroke is a decimal point. Thus, in the third example above, compounded from 629.734 and 629.735, the group 629 is common to both and is not repeated; the group .735 following the stroke is coordinate with .734 preceding it. The use of the stroke in this way to form a range-number conceals all the component numbers except the first, and the information so classed files in only one place in a classified list. It is therefore inappropriate in cases where there is a need to retrieve information from all the component numbers, e.g.

546.32/.35.

The more important alkali metals might require separate class entries as follows.

546.32 Potassium.

546.33 Sodium.

546.34 Lithium.

546.35 Rubidium

;, [, :: TABLE 1b. RELATION. SUBGROUPING. ORDER-FIXING

SECTION 1. SIMPLE RELATION.

The relation sign : (colon) indicates relationship between two or more subjects by connecting their UDC numbers. Unlike the plus and stroke (Table 1a), the colon restricts rather than extends the subjects it connects. E.g.

17:7 Ethics in relation to art.

341.63(44:450) Arbitration of disputes between France and Italy.

628.463:692.758 Refuse collection systems using refuse chutes.

When the colon is used to link auxiliary numbers with biterminal signs (i.e. parentheses or quotation marks), the resultant compound number is enclosed within a single set of signs, as in the example (44:450) above.

SECTION 2. SUBGROUPING.

Square brackets [] may be used as a subgrouping device within a complex combination of UDC numbers, in order to clarify the relationship of the components. Subgrouping may be needed when a subject denoted by two or more UDC numbers linked by plus sign, stroke or colon is as a whole related to another number by colon, or is modified by a common or special auxiliary, e.g.

061.2(100):[54+66]IUPAC International Union of Pure and Applied Chemistry.

[622+669](485) Mining and metallurgy in Sweden.

004.384:[621.771.016.3:669.14] Computers in cold-rolling mills for steel.

783:[274/278] Protestant church music.

SECTION 3. ORDER-FIXING.

The double colon :: may be used to fix the order of the component numbers in a compound number, especially when the UDC is used in a computer-based information system, e.g.

061.2(100)::[54+66]IUPAC International Union of Pure and Applied Chemistry.

575::576.3 Cytogenetics.

77.044::355.4 War photography

=... TABLE 1c. COMMON AUXILIARIES OF LANGUAGE

The common auxiliaries of language denote the language or linguistic form of a document whose subject is denoted by a main UDC number.

Table 1c is the main place in the UDC tables for enumeration of languages, and serves as the source for the subdivision of Class 811 Languages (as subjects of study), Class 821 Literatures of individual languages, and (=...) Table 1f - Common auxiliaries of ethnic grouping

Although in theory the language of any document or item of information may be denoted, in practice it is useful only when there is a need to distinguish between those in different languages, e.g. to enable retrieval according to language or to provide a satisfactory filing order.

CITATION ORDER. In citation order, the language auxiliary normally comes last; it may, however, be cited medially or even first in a compound number if there is a need to file documents in order of language rather than subject. If necessary, it is separated from the following number by a colon (as in the second example below), e.g.

663.4(493)(075)=112.5 Brewing industry in Belgium - textbook - in Flemish.

(02.053.2)=111:... Children's books - in English - subject arrangement.

=111(02.053.2)... English language works - children's books - subject arrangement.

MULTILINGUAL DOCUMENTS. Multilingual documents may be denoted by =00 or by the auxiliaries of the individual languages in ascending numeric order, e.g.

53(035)=00 Multilingual handbook of physics.

53(035)=111=112.2=133.1 Handbook of physics in English, French and German

Systematic table

=... '0 ORIGINS AND PERIODS OF LANGUAGE. PHASES OF DEVELOPMENT

'03, '05, '07 and '09 are not generally applicable. They are reserved for specific applications (e.g. at Latin, Greek, Hebrew, Egyptian)

=... '01	Old period. Archaic period	=025	Adapted, edited, amended versions
=... '02	Classical period		⇒ (0.046)
=... '04	Middle period		⇒ (0.074)
=... '06	Modern period	=030	Translated documents. Translations
=... '08	Revived language		=030.1/.9 ⇒ =1/=9
=... '282	Dialects. Local and regional language. Variants and vernaculars		Denote the source language by =030.1/.9 and the target language directly by =1/=9
	=... '282 ⇒ 81'282		◇ 61=030.161.1 Medical documents translated from Russian
=00	MULTILINGUAL. POLYGLOT		◇ 61=030.161.1=133.1 Medical documents translated from Russian into French (filed near other medical translations from Russian)
	⇒ 81'246.3		◇ 61=133.1=030.161.1 Medical documents translated from Russian into French (filed near other medical documents in French)
	⇒ 81'374.8-022.218		⇒ 81'25
=02	ORIGINALS OR THEIR ADAPTATIONS (UNTRANSLATED)		⇒ 81'322
=021	Original versions (unadapted, unedited)		

=1/=9 LANGUAGES (NATURAL AND ARTIFICIAL)

=1/=8	NATURAL LANGUAGES	=112.22	Upper German
=1/=2	INDO-EUROPEAN LANGUAGES	=112.24	Middle German
=11	Germanic languages		(Not to be confused with =122.2'04 Middle High German)
=111	English	=112.28	Yiddish (Judeo-German)
	◇ =111'01 Anglo-Saxon (Old English)	=112.3/.4	Low German
	◇ =111'04 Middle English	=112.3	Plattdeutsch ('Low German' in the narrower sense)
=111.8	English based pidgins and creoles		Frisian
	Specify by (4/9) for geographical region and/or by name (alphabetic extension - Table 1h)	=112.4	Dutch (Flemish in Belgium and north-eastern France)
		=112.5	Dutch-based pidgins and creoles
=112	Other West Germanic languages		Specify by (4/9) for geographical region and/or by name (alphabetic extension - Table 1h)
=112.2	German (High German, Standard written German)	=112.58	
	◇ =112.2'04 Middle High German (c. 10th-14th century)		

=... Table 1c. Common auxiliaries of language

=112.6	Afrikaans	=14	Greek (Hellenic)
=113	North Germanic (Nordic) languages	◇ =14'02 Classical Greek	
=113.1	Old Norse	◇ =14'03 Koine (New Testament Greek)	
	<i>If preferred, Old Norse may be classed at =113'01</i>	◇ =14'04 Byzantine Greek	
		◇ =14'06 Modern Greek	
=113.2	Faeroese	Katharevousa	
=113.3	Icelandic	Dhimotiki (Demotic)	
=113.4	Danish	Yevanic (Judeo-Greek)	
=113.5	Norwegian	Celtic languages	
=113.52	Bokmål. Riksmål	Gaulish	
=113.53	Landsmål. Nynorsk	Goidelic group	
=113.55	Samnorsk	Irish	
=113.6	Swedish	Scots Gaelic	
=114	East Germanic languages	Manx	
=114.1	Burgundian	Other Goidelic languages	
=114.2	Ostrogothic	Brythonic group	
=114.3	Visigothic	Welsh	
=12	Italic languages	Breton	
=122	Umbro-Sabellian (Osco-Umbrian) languages	Cornish	
=122.1	Oscan	Other Brythonic languages	
=122.2	Sabellic	Slavic languages	
=122.3	Umbrian	East Slavic languages	
=122.4	Volscian	Russian	
=122.9	Other Umbro-Sabellian languages	Ukrainian	
=123	Faliscan	Byelorussian (White Russian)	
=124	Latin	West Slavic languages	
	◇ =124'01 Old Latin	Polish	
	◇ =124'02 Classical Latin	Polabian	
	◇ =124'03 Vulgar Latin	Czech	
	◇ =124'04 Mediaeval Latin	Slovak	
	◇ =124'06 Modern Latin (e.g. for scientific nomenclature)	Sorbian (Lusatian, Wendish)	
		Upper Sorbian	
		Lower Sorbian	
=13	Romance languages	South Slavic languages	
=131	Italo-Romance languages	Old Church Slavic	
=131.1	Italian	Bulgarian	
=131.6	Sardinian	Macedonian	
=131.8	Italkian (Judeo-Italian)	Serbo-Croatian	
=132	Rhaeto-Romance languages	Serbian	
=132.1	Friulian	Croatian	
=132.2	Ladin	Slovenian	
=132.3	Romansch	Baltic languages	
=133	Gallo-Romance languages	Old Prussian	
=133.1	French	Lithuanian	
	◇ =133.1'01 Old French	Kursh (Curonian)	
	◇ =133.1'03 Anglo-Norman	Latvian (Lettish)	
=133.18	French-based pidgins and creoles	Albanian	
	<i>Specify by (4/9) for geographical region and/or by name (alphabetic extension - Table 1h)</i>	Armenian	
=133.2	Provençal. Occitan	INDO-IRANIAN LANGUAGES	
=133.3	Corsican	Indic languages	
=134	Ibero-Romance languages	Sanskrit	
=134.1	Catalan	◇ =211'01 Vedic	
=134.2	Spanish	Pali	
=134.28	Judezmo (Judeo-Spanish. Ladino)	Prakrit	
=134.3	Portuguese	Ardhamagadhi	
=134.38	Portuguese-based pidgins and creoles	Avanti	
	<i>Specify by (4/9) for geographical region and/or by name (alphabetic extension - Table 1h)</i>	Maharashtri	
=134.4	Gallegan (Galician)	Magadhi	
		Sauraseni	
=135	Balkan Romance languages	Modern Indic languages	
=135.1	Romanian	Dardic group	
=135.2	Moldavian	Chitral	
=135.8	Dalmatian (Vegliote)	Kashmiri	
=138	Lingua Franca (Sabir)	Kohistani	
		Kunar	

=214.15	Pashai	=222.7	Mazanderani
=214.16	Sina	=222.8	Tajik (Tadzhik)
=214.2	Central group	=223.1	Talysh
=214.21/.22	Hindustani (Hindi-Urdu complex)	=223.2	Tat
=214.21	Hindi	=223.9	Other West Iranian languages
=214.22	Urdu	=23	Nuristani (Kafiri)
=214.23	Banjuri	=29	Dead Indo-European languages (not listed elsewhere)
=214.24	Bhili		Pre-Italic languages
=214.25	Gujarati	=291	Lepontic
=214.26	Khandesi	=291.1	Ligurian
=214.27	Punjabi (Panjabi)	=291.2	Raetic
=214.28	Rajasthani	=291.3	Sicel
=214.281	Marwari (Merwari)	=291.4	Venetic
=214.3	Eastern group	=291.5	Illyrian
=214.31	Assamese	=291.6	Messapian
=214.32	Bengali (Bangla)	=291.7	Other pre-Italic languages
=214.33	Bihari	=291.9	
=214.331	Bhojpuri	=292	Anatolian languages
=214.332	Magahi	=292.1	Hittite
=214.333	Maithili	=292.2	Luwian
=214.34	Khosali (East Hindi)	=292.3	Palaic
=214.341	Awadhi	=292.4	Lydian
=214.342	Chhattisgarhi	=293	Macedonian (ancient)
=214.35	Marathi	=294	Dacian
=214.351	Konkani (Gomantaki)	=295	Thracian-Phrygian
=214.36	Oriya	=296	Tocharian
=214.4	Northern group	=34	DEAD LANGUAGES OF UNKNOWN AFFILIATION, SPOKEN IN THE MEDITERRANEAN AND NEAR EAST (EXCEPT SEMITIC)
=214.41	Garhwali	=341.1	Sumerian
=214.42	Kumauni	=341.2	Elamite
=214.43	Nepali	=341.3	Kassite (Cossaeon)
=214.44	Pahari (Himachali)	=341.4	Hattic
=214.5	North-western group	=341.5	Hurrian
=214.51	Lahnda	=341.6	Urartean (Chaldic, Kaldic)
=214.52	Sindhi	=341.7	Carian
=214.58	Romany (Gipsy)	=341.8	Lyconian
=214.6	Singhalese-Maldivian group	=342.1	Etruscan
=214.61	Singhalese (Sinhala)	=342.2	Iberian
=214.62	Maldivian	=342.9	Other dead languages of the Mediterranean and Near East
=214.63	Vedda	=35	CAUCASIAN LANGUAGES
=22	Iranian languages	=351	North-eastern group
=221	Eastern Iranian languages	=351.1	Avar-Andi-Dido subgroup
=221.1	Northern group	=351.11	Andi
=221.11	Alanic	=351.12	Avar
=221.12	Khotanese (Saka)	=351.13	Dido
=221.13	Khwarizmian	=351.19	Other languages of the Avar-Andi-Dido subgroup
=221.14	Parthian	=351.2	Lak-Dargwa subgroup
=221.15	Sarmatian	=351.21	Lak
=221.16	Scythian	=351.22	Dargwa
=221.17	Sogdian	=351.3	Lezghian (Samur) subgroup
=221.18	Ossetic	=351.31	Agul
=221.19	Yaghnobi	=351.32	Lezghian
=221.3	Pamir (Ghalchah) group	=351.33	Rutul
=221.31	Pashto (Pushto)	=351.34	Tsakhur
=221.32	Shugni (Khugni)	=351.39	Other Lezghian languages
=221.323	Wakhi	=351.4	Vejnakh subgroup
=221.4	Central group	=351.41	Bats
=221.41	Ormuri	=351.42	Chechen
=221.42	Parachi	=351.43	Ingush
=222/=223	Western Iranian languages	=352	North-western group
=222.1	Persian	=352.1	Abasa
	◇ =222.1'01 Avestan		
	◇ =222.1'02 Old Persian		
	◇ =222.1'04 Pahlavi		
	◇ =222.1'06 Farsi		
=222.18	Judeo-Persian		
=222.2	Bakhtiari		
=222.3	Baluchi		
=222.4	Gilaki		
=222.5	Kurdish		
=222.6	Luri		

=... Table 1c. Common auxiliaries of language

=352.2	Abkhazian	=411.45	Tigrinya
=352.3	Circassian (Adyghe)	=411.49	Other Ethio-Semitic languages
=353	Southern (Kartvelian) group	=412	Egyptian-Coptic
=353.1	Georgian	◇ =412'01 Old Egyptian	◇ =412'04 Middle Egyptian
=353.2	Svan	◇ =412'05 New Egyptian	◇ =412'07 Late Egyptian (Demotic)
=353.3	Zan	◇ =412'09 Coptic	
=353.31	Laz (Chan)		
=353.32	Mingrelian	=413	Berber languages
=359	Other Caucasian languages	=413.1	Tamazight-Riff-Kabyle group
=361	Basque (Euskera, Euskara)	=413.11	Kabyle
=371	Burushaski	=413.21	Tuareg
=41	AFRO-ASIATIC (HAMITO-SEMITIC) LANGUAGES	=413.31	Zenaga
=411	Semitic languages	=413.4	Zenati group
=411.1	North Semitic languages	=413.9	Other Berber languages
=411.11	Akkadian	=414	Chadic languages
=411.111	Babylonian	=414.1	East Chadic languages
=411.112	Assyrian	=414.11	Bata-Tera group
=411.12	Amorite	=414.111	Bata subgroup
=411.13	Eblaite	=414.112	Bura subgroup
=411.14	Ugaritic	=414.112.1	Margi
=411.15	Canaanite	=414.113	Higi subgroup
=411.151	Moabite	=414.114	Tera subgroup
=411.152	Phoenician	=414.12	Daba-Gisiga-Matakam group
=411.153	Punic	=414.121	Daba subgroup
=411.16	Hebrew	=414.122	Gisiga-Matakam subgroup
◇ =411.16'02 Biblical Hebrew		=414.13	Mandara group
◇ =411.16'03 Mishnaic		=414.19	Other East Chadic languages
◇ =411.16'04 Mediaeval Hebrew. Rabbinic		=414.2	West Chadic (Plateau-Sahel) languages
◇ =411.16'08 Modern Hebrew		=414.21	Bana group
=411.17	Aramaic	=414.22	Bolewa-Plateau group
=411.171	West Aramaic	=414.221	Bolewa subgroup
=411.171.1	Palestinian Aramaic	=414.222	Plateau subgroup
◇ =411.171.1'02 Biblical Aramaic		=414.223	Ron subgroup
◇ =411.171.1'03 Galilean Aramaic (language of Jerusalem Talmud and Targumim)		=414.23	Hausa group
=411.171.2	Nabataean	=414.231	Hausa
=411.171.3	Palmyrene	=414.24	Kotoko group
=411.171.4	Samaritan	=414.251	Musgu
=411.172	East Aramaic	=414.26	Ngizim group
=411.172.1	Babylonian Aramaic (language of the Babylonian Talmud and Targumim)	=414.27	Sahel group
=411.172.2	Mandaean	=414.271	Sokoro subgroup
=411.172.3	Syriac	=414.272	Somrai subgroup
◇ =411.172.3'02 Classical Syriac		=414.273	Tuburi
◇ =411.172.3'06 Modern Syriac ('Assyrian')		=414.28	Warjawa-Gesawa group
=411.2	South-west Semitic languages	=414.29	Other West Chadic languages
=411.21	Arabic	=415	Cushitic languages
◇ =411.21'02 Classical Arabic		=415.1	Northern Cushitic group
◇ =411.21'06 Modern ('Standard') Arabic		=415.11	Beja
=411.216	Maltese	=415.2	Central Cushitic group (Agau)
=411.218	Arabic-based pidgins	=415.3	Eastern Cushitic group
<i>Specify by (4/9) for geographical region and/or by name (alphabetic extension - Table 1h)</i>		=415.31	Highland group
=411.3	South-east Semitic languages	=415.311	Derasa
=411.31	Hadhrami	=415.312	Haddiya
=411.32	Himyarite	=415.313	Sidamo
=411.33	Minaean	=415.32	Lowland group
=411.34	Sabaeen	=415.321	Afar-Saho group
=411.35	Mahri-Sokotri (Modern South Arabic)	=415.322	Galla (Afan. Oromo)
=411.4	Ethio-Semitic languages	=415.323	Somali
=411.41	Amharic	=415.4	Southern Cushitic group
=411.42	Ge'ez	=415.411	Iraku
=411.43	Gurage	=416	Omotc languages
=411.44	Tigre	=416.1	Eastern Omotic languages
		=416.11	Ari-Banna branch
		=416.2	Western Omotic languages
		=416.21	Gimojan group

=416.211	Welamo	=429	Other Nilo-Saharan languages
=416.22	Gonga group	=43	CONGO-KORDOFANIAN
=416.23	Majoid group		(NIGER-KORDOFANIAN) LANGUAGES
=416.9	Other Omotic languages	=431	Kordofanian languages
=419	Other Afro-Asiatic languages	=431.1	Katla group
=42	NILO-SAHARAN LANGUAGES	=431.2	Koalib group
=421.1	Songhai	=431.3	Talodi group
=422	Saharan branch	=431.4	Tegali group
=422.1	Kanuri	=431.5	Tumtum (Kadugli-Krongo) group
=422.2	Tebu (Tubu)	=432	Niger-Congo languages
=422.3	Teda	=432.1	Senegalo-Guinean (West Atlantic)
=422.4	Zaghawa		languages
=423	Maban branch	=432.11	Northern group
=423.1	Masalit	=432.111	Balanta
=424.1	Fur	=432.112	Diola (Jola)
=425	Koman	=432.113	Fula(Fulani)
=426	Chari-Nile branch	=432.114	Manjaku
=426.11	Berta group	=432.115	Serer
=426.2	Eastern Sudanic languages	=432.116	Wolof
=426.21	Meroitic	=432.12	Southern group
=426.22	Didinga-Murle (Surma) group	=432.121	Bullom
=426.23	Daju group	=432.122	Kissi
=426.24	Nubian group	=432.123	Limba
=426.25	Nyimang group	=432.124	Temne (Timne)
=426.26	Tama group	=432.19	Other Senegalo-Guinean languages
=426.27	Temein group	=432.2	Mande (Mali) languages
=426.28	Teuso	=432.21	North-western group
=426.29	Other Eastern Sudanic languages	=432.211	Mandekan
=426.3	Central Sudanic languages	=432.211.1	Bambara
=426.31	Bagirmi group	=432.211.2	Dyula
=426.32	Bongo group	=432.211.3	Malinka (Mandingo)
=426.33	Kara group	=432.212	Soninke
=426.34	Lendu	=432.213	Susu
=426.35	Mangbetu group	=432.214	Vai
=426.36	Mangbutu-Efe group	=432.22	South-western group
=426.37	Moru-Ma'di group	=432.221	Kpelle
=426.371	Lugbara (Logbara, Ma'adi)	=432.222	Loma
=426.372	Moru	=432.223	Mende
=426.38	Sara group	=432.23	Southern group
=426.381	Gambai	=432.231	Dan
=426.39	Other Central Sudanic languages	=432.232	Kweni
=426.4	Western Nilotic languages	=432.233	Mano
=426.41	Lwo group	=432.24	South-eastern group
=426.411	Acholi	=432.241	Bisa
=426.412	Adhola	=432.242	Busa
=426.413	Alur	=432.243	Samo
=426.414	Anuak	=432.29	Other Mande languages
=426.415	Lango	=432.3	Gur (Voltaic) languages
=426.416	Luo	=432.311	Bargu
=426.417	Shilluk	=432.321	Bobo (Boomu) group
=426.42	Dinka-Nuer group	=432.33	Grusi group
=426.421	Dinka	=432.331	Kabré
=426.422	Nuer	=432.332	Kasem
=426.423	Mabaan group	=432.34	Moré-Gurma group
=426.49	Other Western Nilotic languages	=432.341	Dagbani
=426.5	Eastern Nilotic (Nilo-Hamitic) languages	=432.342	Gurma
=426.511	Bari	=432.343	Moré
=426.521	Masai	=432.344	Nankani (Gurenne)
=426.53	Teso subgroup	=432.35	Kirma-Tyurama
=426.532	Lotuko	=432.36	Lobi group
=426.533	Teso	=432.361	Lobi(ri)
=426.534	Turkana	=432.37	Senufo group
=426.59	Other Eastern Nilotic languages	=432.371	Suppire-Mianka subgroup
=426.6	Southern Nilotic languages	=432.372	Tagbana-Dyimini subgroup
=426.61	Nandi (Kipsikis)	=432.39	Other Gur languages
=426.62	Pokot (Suk)	=432.4/.5	Kwa languages
		=432.41	Akan (Twi-Fanté; Volta-Comoe) group
		=432.411	Guang subgroup
		=432.412	Ona subgroup

=... Table 1c. Common auxiliaries of language

=432.413	Tano subgroup	=432.721	Mambile-Wute group
=432.413.1	Anyi (Baule)	=432.722	Tiv-Batu group
=432.413.2	Fanté	=432.722.1	Tiv
=432.413.3	Nzima	=432.73	Cross-River languages
=432.413.4	Twi	=432.731	Bendi (Boki-Ebekwara) group
=432.42	Akoko group	=432.732	Delta-Cross group
=432.43	Edo group	=432.732.1	Efik
=432.431	Bini	=432.732.2	Ogoni
=432.432	Isoko (Sobo, Urhobo)	=432.74	Jukunoid group
=432.44	Ewe group	=432.75	Plateau Benue-Congo languages
=432.441	Fon	=432.79	Other Bantoid languages (except Bantu)
=432.442	Gu		
=432.443	Mina	=432.8/.9	Bantu languages
=432.45	Ga-Adangme group		<i>The arrangement of Bantu languages is based on the distribution into fifteen zones given in 'A consolidated classification of the Bantu languages' by A.T. Cope, African Studies 30, 1971 pp. 213-236</i>
=432.46	Idoma group		
=432.47	Igbo (Ibo) group		
=432.48	Ijo group		
=432.481	Kalabari	=432.811	Bafia group
=432.482	Nembe	=432.812	Bassa group
=432.51	Kru group	=432.813	Bube-Benga group
=432.511	Bassa	=432.814	Duala group
=432.512	Bete	=432.815	Kaka group
=432.513	Grebo	=432.816	Lundu-Balong group
=432.514	Kran	=432.817	Maka-Njem group
=432.515	Kru	=432.818	Sanaga group
=432.52	Lagoon group	=432.819	Yaunde-Fang group
=432.53	Nupe-Gbari group	=432.819.1	Bulu
=432.531	Gbari	=432.819.2	Ewondo (Yaunde)
=432.532	Nupe	=432.819.3	Fang
=432.55	Central Togo languages	=432.821	Kele (Kota) group
=432.56	Yoruba group	=432.822	Mbete group
=432.561	Yoruba	=432.823	Myene group
=432.562	Igala	=432.824	Njabi group
=432.59	Other Kwa languages	=432.825	Shira-Punu group
=432.6	Adamawa-Eastern languages	=432.826	Teke group
=432.61	Adamawa branch	=432.827	Tende-Yanzi group
=432.611	Boa group	=432.828	Tsogo group
=432.612	Chamba group	=432.831	Bangi-Ntumba group
=432.613	Daka group	=432.831.1	Ngala
=432.614	Duru group	=432.832	Kuba group
=432.615	Jen group	=432.833	Losengo group
=432.616	Mbum group	=432.833.1	Lingala (Mangala)
=432.616.1	Mundang	=432.834	Mboshi group
=432.617	Mumuye group	=432.835	Mongo-Nkundu group
=432.618	Yungur group	=432.835.1	Mongo
=432.619	Other Adamawa languages	=432.835.2	Ngandu
=432.62	Eastern branch	=432.835.3	Nkundu
=432.621	Banda	=432.836	Ngombe group
=432.622	Gbaya-Ngbaka-Manza group	=432.837	Ngundi (Pande) group
=432.623	Mayogo group	=432.838	Soko-Kele group
=432.624	Mba group	=432.839	Tetela group
=432.625	Ndogo-Feroge group	=432.841	Bembe-Kabwari group
=432.626	Ngbandi group	=432.841.1	Bembe
=432.626.1	Sangho (Sango)	=432.841.2	Hunde
=432.627	Zande group	=432.842	Bira-Huku group
=432.629	Other Eastern languages	=432.843	Konjo (Nande) group
=432.7/.9	Benue-Congo languages	=432.844	Lega-Kalanga group
=432.7	Bantoid languages	=432.845	Mbolo-Ena group
=432.71	Bane languages	=432.846	Ruanda-Rundi group
=432.711	Ekoid group	=432.846.1	Ha
=432.711.1	Ekajuk	=432.846.2	Ruanda (Rwandese)
=432.712	Grasslands groups	=432.846.3	Rundi (Barundi)
=432.712.1	Bamun	=432.851	Chaga (Shaka) group
=432.712.2	Bamileke subgroup	=432.852	Haya-Jita group
=432.713	Jarawan group	=432.852.1	Haya
=432.714	Mamfe group	=432.852.2	Jita
=432.715	Misaje group	=432.853	Kikuyu-Kamba group
=432.716	Tikar group	=432.853.1	Embo
=432.719	Other Bane Bantoid languages	=432.853.2	Kamba
=432.72	Non-Bantu Bantoid languages	=432.853.3	Kikuyu
		=432.853.4	Meru

=432.854	Masaba-Luhya group	=432.934	Tumbuka group
=432.854.1	Luhya	=432.941	Makua group
=432.854.2	Nyore	=432.942	Matumbi (Ngindo) group
=432.855	Nyika-Taita group	=432.943	Yao group
=432.855.1	Pokomo	=432.951	Herero
=432.855.2	Taita (Dabida)	=432.952	Ndonga group
=432.856	Nyoro-Ganda group	=432.952.1	Kuanyama
=432.856.1	Chiga	=432.952.2	Ndonga
=432.856.2	Ganda	=432.953	Umbundu group
=432.856.3	Lugwere	=432.953.1	Mbundu
=432.856.4	Nyankole	=432.953.2	Nyaneka
=432.856.5	Nyoro	=432.954	Yeye
=432.856.6	Soga	=432.961	Chopi group
=432.856.7	Toro	=432.962	Nguni group
=432.857	Ragoli-Kuria group	=432.962.1	Ndebele (Tabele)
=432.857.1	Gusii	=432.962.2	Swazi (Siswati)
=432.857.2	Kuria	=432.962.3	Xhosa
=432.857.3	Ragoli (Logooli)	=432.962.4	Zulu
=432.861	Ilamba-Irangi (Nilamba) group	=432.963	Shona group
=432.862	Sukuma-Nyamwezi group	=432.963.1	Shanga
=432.862.1	Nyamwezi	=432.963.2	Shona
=432.862.2	Sukuma	=432.964	Sotho-Tswana group
=432.863	Tongwe group	=432.964.1	Northern Sotho
=432.871	Bena-Kinga (Hehe) group	=432.964.2	Southern Sotho (Sesotho)
=432.871.1	Kisi	=432.964.3	Tswana
=432.872	Gogo group	=432.965	Tswa-Ronga (Tsonga) group
=432.873	Pogolo group	=432.965.1	Ronga
=432.874	Shambala group	=432.965.2	Tsonga
=432.874.1	Asu	=432.965.3	Tswa
=432.874.2	Shambala	=432.966	Venda group
=432.875	Swahili group	=432.99	Other Bantu languages
=432.875.1	Ngwana	=45	KHOISAN LANGUAGES
=432.875.2	Swahili	=451	Northern Bushman (Bush-A) languages
=432.876	Zigula-Zaramo group	=451.1	Auen (Aukwe)
=432.881	Kongo group	=451.2	!Xu (!Kung)
=432.882	Mbala group	=452	Southern Bushman languages
=432.883	Mbundu group	=452.1	Taa group (Bush-B)
=432.884	Yaka group	=452.2	!Wi group (Bush-C)
=432.891	Chokwe-Luchazi group	=453	Hottentot (Khwe-Kovab, Central Khoisan) languages
=432.891.1	Luchazi	=453.1	Nama group
=432.891.2	Lwena (Luvale)	=453.2	Tshu-Khwe group
=432.892	Lozi group	=454	Sandawe
=432.893	Luyana group	=459	Other Khoisan languages
=432.893.1	Mashi	=51	URAL-ALTAIC LANGUAGES
=432.894	Subiya group	=511	Uralic languages
=432.911	Kaonde group	=511.1	Finno-Ugric languages
=432.912	Luba group	=511.11	Finnic languages
=432.912.1	Luba-Lulua	=511.111	Finnish
=432.913	Lunda group	=511.112	Karelian
=432.913.1	Lunda	=511.113	Estonian
=432.913.2	Luwunda (Ruund)	=511.114	Livonian
=432.914	Nkoya group	=511.115	Vepsian
=432.915	Pende group	=511.116	Votian
=432.916	Songe group	=511.117	Ingrian (Izhorian)
=432.916.1	Mbala	=511.12	Lappic
=432.916.2	Songe	=511.13	Permian languages
=432.921	Bemba group	=511.131	Votyak (Udmurt)
=432.922	Bisa-Lamba group	=511.132	Zyrian (Komi)
=432.922.1	Lamba	=511.14	Ugric languages
=432.923	Fipa-Mambwe group	=511.141	Hungarian
=432.923.1	Mambwe	=511.142	Ostyak
=432.924	Konde (Nyakyusa)	=511.143	Vogul (Mansi)
=432.925	Lenje-Tonga (Ila) group	=511.15	Volgaic languages
=432.925.1	Ila	=511.151	Cheremis
=432.925.2	Tonga	=511.152	Mordvin
=432.926	Nyiha-Safwa group	=511.152.1	Erzya
=432.926.1	Mwanga	=511.152.2	Moksha
=432.926.2	Nyiha		
=432.931	Manda group		
=432.932	Nyanja (Chewa) group		
=432.933	Senga-Sena (Nsenga) group		

=... Table 1c. Common auxiliaries of language

=511.2	Samoyedic languages	=512.39	Pao-an
=511.21	Ostyak Samoyed (Selkup)	=521	Japanese
=511.22	Sayan Samoyed	=526.531	Karamojong
=511.23	Tavgy Samoyed	=531	Korean
=511.24	Yenissei Samoyed	=541	Ainu
=511.25	Yurak Samoyed	=55	PALAEO-SIBERIAN LANGUAGES
=512	Altaic languages	=551	Chukchi-Kamtchatkan languages
=512.1	Turkic languages	=551.1	Chukchi
=512.111	Chuvash	=551.2	Kamchadal
=512.12	Central group	=551.3	Koryak
=512.121	Karakalpak	=552	Gilyak
=512.122	Kazakh	=553	Yenisei-Ostyak
=512.123	Nogai	=554	Yukaghir
=512.13	Eastern (Karluk) group	=56	ESKIMO-ALEUT LANGUAGES
=512.131	Khoton	=561	Aleut
=512.132	Uighur	=562	Inuit (Inupiak)
=512.133	Uzbek	=562.1	Greenlandic (Kalaallisut)
=512.14	Western group	=563	Yupik
=512.141	Bashkir	=569	Other Eskimo languages
=512.142	Karachay	=57	DRAVIDIAN LANGUAGES
=512.143	Karaite	=571	Northern branch
=512.144	Kumyk	=571.11	Brahui
=512.145	Tatar	=571.12	Kurukh (Oraon)
=512.15	Northern group	=571.13	Malto (Sauria)
=512.151	Altai	=572	Central branch
=512.152	Chulym	=572.11	Gadba (Ollari)
=512.153	Khakas (Abakan)	=572.12	Gondi
=512.154	Kirghiz	=572.13	Kolami
=512.155	Shor	=572.14	Konda
=512.156	Tuva (Soyon)	=572.15	Koya
=512.157	Yakut (Sakha)	=572.16	Kui (Khond)
=512.16	Southern group	=572.17	Kuwi
=512.161	Turkish (Osmanli)	=572.18	Manda
=512.162	Azerbaijani	=572.21	Naiki
=512.163	Khalaj	=572.22	Parji (Dhurva, Tagara)
=512.164	Turkmen	=572.23	Pengo
=512.165	Gagauz	=572.24	Telugu
=512.19	Other Turkic languages	=572.25	Tulu
=512.2	Tungus languages	=573	Southern branch
=512.21	Northern group	=573.11	Kannada (Kanarese)
=512.211	Even (Lamut)	=573.12	Kodagu
=512.212	Evenki	=573.13	Kota
=512.213	Manegir	=573.14	Malayalam
=512.214	Negidal	=573.15	Tamil
=512.215	Orochon	=573.16	Toda
=512.216	Solon	=58	SINO-TIBETAN LANGUAGES
=512.22	Southern group	=581	Chinese languages
=512.221	Nanaj subgroup	=581.11	Mandarin (Kuang-hua; Northern Chinese)
=512.221.1	Akani	=581.12	Cantonese (Yüeh)
=512.221.2	Birar	=581.13	Hakka
=512.221.3	Gold	=581.14	Hsiang
=512.221.4	Kile (Kire)	=581.15	Kan (Nan Ch'ang)
=512.221.5	Olcha	=581.16	Min Pei (Northern Min)
=512.221.6	Orok	=581.17	Min Man (Southern Min)
=512.221.7	Samagir	=581.171	Taiwanese
=512.222	Udihe subgroup	=581.18	Wu
=512.222.1	Udihe	=581.19	Other Chinese languages
=512.222.2	Oroch	=582	Kam-Tai languages
=512.223	Manchu subgroup	=582.1	Kadai group
=512.223.1	Manchu	=582.11	Li
=512.223.2	Ju-chen	=582.2	Kam-Sui group
=512.3	Mongolian languages		
=512.31	Buryat		
=512.32	Dagur		
=512.33	Khalkha		
=512.34	Khorchin		
=512.35	Mongour		
=512.36	Mongul		
=512.37	Oirat (Kalmyk)		
=512.38	Ordos		

=582.3	Tai languages	=612.4	Cambodian (Khmer)
=582.31	Central Tai (Nung-Tho) languages	=612.5	Khmuic
=582.32	Northern Tai (Dioi; Chung-Chia) languages	=612.6	Mon
=582.33	South-western Tai (Tai-Shan) languages	=612.7	Palaung-Wa (Salween)
=582.331	Lao (Laotian)	=612.71	Wa
=582.332	Lü	=612.8	Pearic
=582.333	Shan	=612.9	Viet-Muong languages
=582.334	Thai (Siamese)	=612.91	Vietnamese
=582.335	Tho	=612.92	Muong
=582.336	Yuan	=612.99	Other Mon-Khmer languages
=583	Miao-Yao languages	=613	Munda languages
=583.11	Laka	=613.1	Gorum (Pareng)
=583.12	Miao	=613.2	Gutob (Gadaba)
=583.13	Mien (Yao)	=613.3	Juang
=583.14	Punu	=613.4	Kharia
=584	Tibeto-Burman languages	=613.5	Kurku
=584.1	Bodo-Naga-Chin languages	=613.6	Mundari-Ho
=584.11	Bodo (Boro)	=613.61	Ho
=584.12	Garö	=613.7	Santali
=584.13	Naga (Tangsa) group	=613.8	Sora
=584.131	Lepcha	=613.9	Other Munda languages
=584.132	Tangsa	=614	Nicobarese group
=584.14	Kachin (Ching Pa'o)	=62	AUSTRONESIAN LANGUAGES
=584.15	Nung	=621	Malayo-Polynesian languages
=584.16	Rawang	=621.1	Formosan group
=584.2	Burmese-Lolo languages	=621.11	Atayalic group
=584.21	Burmese	=621.12	Paiwanic group
=584.22	Lolo-Moso group	=621.2	Hesperonesian group
=584.221	Akha	=621.21	Philippine group
=584.222	Lahu	=621.211	Bilaan
=584.223	Lisu	=621.212	Cordilleran group
=584.224	Lolo	=621.212.1	Banagic group
=584.225	Nosu	=621.212.11	Gaddang
=584.3	Gyarung-Mishmi (Himalayan) languages	=621.212.12	Ibanag
=584.31	Adi (Abor-Miri)	=621.212.2	Ifugao group
=584.32	Newari	=621.212.3	Igorot group
=584.4	Karen languages	=621.212.31	Bontoc
=584.41	Pho (Pwo)	=621.212.32	Ilocano
=584.42	Sgaw	=621.212.33	Kankana
=584.43	Taungthu (Pa'o)	=621.212.4	Kalinga group
=584.5	Naga-Kuki-Chin languages	=621.212.41	Maguindanao
=584.51	Chin languages	=621.212.42	Tausug (Sulu)
=584.511	Lushai	=621.212.5	Pangasinan
=584.512	Paite (Vuite)	=621.212.9	Other Cordilleran languages
=584.513	Thado	=621.213	Maranao group
=584.52	Kuki	=621.214	Murutic group
=584.53	Lakher	=621.215	Sulic group
=584.54	Manipuri (Meithei)	=621.215.1	Bukidnic
=584.55	Naga	=621.215.2	Dibabai (Manobo)
=584.551	Mikir	=621.215.3	Mesophilippine languages
=584.6	Tibetan (Bhotia)	=621.215.31	Bikol
◇ =584.6'02	Classical Tibetan	=621.215.32	Hanunocic group
=584.61	Central Tibetan	=621.215.33	Irayic group
=584.62	West Tibetan	=621.215.34	Mansakic group
=584.7	Dzorgaic (Ch'iang)	=621.215.35	Tagalic group
=61	AUSTRO-ASIATIC LANGUAGES	=621.215.351	Tagalog (Filipino)
=611	Malacca (Aslian) group	=621.215.352	Bisayan (Visayan) group
=611.1	Jakun	=621.215.352.1	Cebuano
=611.2	Sakai	=621.215.352.2	Ilonggo (Hiligaynon)
=611.3	Semang	=621.215.352.3	Waray (Samaron)
=612	Mon-Khmer languages	=621.215.36	Palawan
=612.1	Bahnaric	=621.215.37	Pampango
=612.11	Koho	=621.216	Chamorro
=612.2	Katuic		
=612.3	Khasi		

=... Table 1c. Common auxiliaries of language

=621.217	Palauan	=622.5	New Hebrides languages
=621.218	Gorontalo group	=622.6	New Caledonian languages
=621.219	Other Philippine languages	=622.8	Eastern Oceanic languages
=621.22	Northern Sulawesi (Celebes) group	=622.81	Melanesian languages
=621.221	Minhasa (Ton) group	=622.811	Fijian
=621.222	Tomini group	=622.82	Polynesian languages
=621.23	Central Sulawesi group	=622.821	Nuclear group
=621.231	Bungku-Laki-Mori group	=622.821.1	Easter Island
=621.231.1	Mori	=622.821.2	Hawaiian
=621.232	Loinang-Banggai group	=622.821.3	Maori
=621.233	Toradja group	=622.821.4	Marquesan
=621.233.1	Bare'e	=622.821.5	Rarotongan
=621.24	Southern Sulawesi group	=622.821.6	Tahitian
=621.241	Buginese	=622.821.7	Tuamotuan
=621.242	Makasarese	=622.821.9	Other Nuclear group languages
=621.243	Mandarese	=622.822	Samoic-Outlier group
=621.244	Muna-Butung group	=622.822.1	Samoan
=621.244.1	Butung	=622.822.9	Other languages belonging to the
=621.244.2	Muna		Samoic-Outlier group
=621.25	West Indonesian languages	=622.823	Tongic group
=621.251	Malay (Bahasa Indonesia; Bahasa Malaysia)	=622.823.1	Tongan (Uvean)
=621.252	Sumatran languages	=622.823.2	Niuean
=621.252.1	Achinese	=622.9	Other Oceanic languages
=621.252.2	Batak	=71	INDO-PACIFIC (NON-AUSTRONESIAN)
=621.252.3	Gayo		LANGUAGES
=621.252.4	Lampung	=711	Andamanese
=621.252.5	Lubu	=712	Timorese
=621.252.6	Minangkabau	=713	New Guinean (Papuan) languages
=621.252.7	Redjang	=713.1	Central New Guinean languages
=621.252.8	Toba	=713.18	Hiri Motu (Police Motu)
=621.252.9	Other Sumatran languages	=713.2	Highlands Guinean languages
=621.253	Javanese languages	=713.3	Southern New Guinean languages
=621.253.1	Javanese	=713.4	South-eastern New Guinean languages
	◊ =621.253.1'01 Kawi	=713.5	Northern New Guinean languages
=621.253.2	Madurese	=713.6	Huon-Finisterre languages
=621.253.3	Sundanese	=714	North-eastern New Guinean (Madang)
=621.254	Balinese languages	=715	Western New Guinean languages
=621.254.1	Balinese	=716	Bougainvillean
=621.254.2	Basak	=717	Other languages spoken in Central
=621.255	Borneo languages		Melanesia
=621.255.1	Land Dayak	=718	Tasmanian
=621.255.2	Sea Dayak (Iban)	=719	Other Indo-Pacific languages
=621.255.3	Maanyan (Dusun)	=72	AUSTRALIAN LANGUAGES
=621.255.9	Other Borneo languages	=721	Pama-Maric group
=621.256	Malayo-Polynesian languages of Indochina	=722	Pama-Nyungan group
=621.256.1	Cham	=729	Other Australian languages
=621.256.2	Chru (Kru)	=8	AMERICAN INDIAN (AMERINDIAN)
=621.256.3	Jarai		LANGUAGES
=621.256.4	Rade	=81/=82	NORTH-AMERICAN INDIAN
=621.258	Malagasy		LANGUAGES
=621.259	Other West Indonesian languages	=811	Athapaskan-Eyak languages
=621.3	East Indonesian languages	=811.11	Apache
=621.31	Ambon-Timor group	=811.12	Carrier
=621.32	Bima-Sumba group	=811.13	Chilcotin
=621.33	Sula-Batjan group	=811.14	Chipewyan
=621.4	Moluccan languages	=811.15	Dogrib (Hare)
=621.41	Biakic (Numfor)	=811.16	Hupa
=622	Oceanic languages	=811.17	Navajo
=622.1	Micronesian languages	=811.18	Sekani
=622.11	Gilbertese	=811.21	Tanana
=622.12	Kusaie	=811.22	Haida
=622.13	Marshallese		
=622.14	Nauru		
=622.15	Ponape		
=622.16	Truk		
=622.17	Yap		
=622.2	North-east New Guinean languages		
=622.3	Papua Austronesian		
=622.4	Solomons languages		

=811.23	Tlingit	=821.174	Quichéan languages
=811.29	Other Athapaskan languages	=821.174.1	Cakchiquel
=812	Algonquian languages	=821.174.2	Pokomam
=812.11	Arapaho	=821.174.3	Pokonchi
=812.12	Blackfoot (Siksika)	=821.174.4	Quiché
=812.13	Cheyenne	=821.174.5	Tzutuhil
=812.14	Cree	=821.174.6	Uspantec
=812.15	Delaware	=821.175	Tzeltal
=812.16	Fox	=821.176	Tzotzil
=812.17	Menominee	=821.18	Miwok
=812.18	Micmac	=821.21	Sahaptin-Nez-Perce
=812.21	Mohegan	=821.22	Totonacan
=812.22	Naskapi	=821.23	Tsimshian
=812.23	Ojibwa	=821.24	Zoque languages
=812.24	Passamaquoddy	=821.241	Mixe
=812.25	Penobscot	=821.242	Zoque
=812.26	Potawatomi	=821.25	Zufi
=812.27	Shawnee	=821.29	Other Penutian languages
=812.5	Kutenai	=822	Aztec-Tanoan languages
=813	Muskogean languages	=822.1	Tanoan languages
=813.1	Alabama	=822.11	Kiowa
=813.2	Choctaw (Chickasaw)	=822.12	Tewa
=813.3	Hitchiti	=822.13	Tiwa
=813.4	Koasati	=822.14	Towa
=813.5	Muskogee (Creek, Seminole)	=822.2	Uto-Aztec languages
=814	Caddoan languages	=822.21	Aztec (Nahuatl)
=814.1	Caddo	=822.22	Hopi
=814.2	Pawnee	=822.23	Numic (Plateau Shoshonian) languages
=814.3	Wichita	=822.231	Paiute
=815	Iroquois languages	=822.232	Shoshoni
=815.1	Cherokee	=822.24	Sonoran languages
=815.2	Mohawk	=822.241	Cora
=815.3	Oneida	=822.242	Huichola
=815.4	Seneca	=822.243	Pima
=815.5	Tuscarora	=822.244	Tarahumara
=816	Siouan languages	=822.245	Yaqui
=816.1	Chiwere	=822.25	Takic (South Californian Shoshonian) languages
=816.2	Crow	=822.251	Luiseno
=816.3	Dakota (Assiniboine)	=822.29	Other Uto-Aztec languages
=816.4	Dhegiha	=823	Oto-Manguean languages
=816.5	Hidatsa	=823.1	Amuzgo
=816.6	Osage	=823.2	Chinantec
=816.7	Winnebago	=823.3	Manguean
=817	Hokan languages	=823.4	Mixtec
=817.1	Tequistlatecan (Chontal)	=823.5	Otomian languages
=817.2	Tlapanecan	=823.51	Matlatzinca
=817.3	Washo	=823.52	Mazahua
=817.4	Yuman	=823.53	Otomi
=817.9	Other Hokan languages	=823.54	Pame
=818.1	Keresan	=823.6	Popolocan (Olmecan) languages
=821	Penutian languages	=823.61	Mazatec
=821.11	Araucanian	=823.62	Popoloc
=821.12	Chinook	=823.7	Zapotecan
=821.13	Chipayan	=824	Salish languages
=821.14	Huave	=824.1	Kalispel
=821.15	Maidu	=824.2	Lillooet
=821.16	Mayan languages	=824.3	Nikutamuk
=821.161	Chol	=824.4	Okanogan
=821.162	Chontal	=824.5	Shuswap
=821.163	Chortí	=824.9	Other Salish languages
=821.164	Chujean	=825.1	Tarascan
=821.165	Huastecan		
=821.166	Jacalteco		
=821.167	Kanjobalan		
=821.168	Kekchi		
=821.171	Ixil		
=821.172	Mamean		
=821.173	Maya		

=... Table 1c. Common auxiliaries of language

=826	Wakashan languages	=873.22	Guahiban
=826.1	Kwakiutlan	=873.23	Salivan
=826.2	Nootkan	=873.24	Tupi-Guarani languages
=829	Other North-American Indian languages	=873.241	Guarani
		=873.242	Tupi
=87	CENTRAL AND SOUTH AMERICAN INDIAN LANGUAGES	=873.3	Jivaroan
=871	Ge-Pano-Carib languages	=873.4	Macro-Tucanoan languages
=871.1	Macro-Carib languages	=873.41	Catuquina
=871.11	Carib	=873.42	Tucanoan
=871.12	Witotoan (Huitotoan)	=873.49	Other Macro-Tucanoan languages
=871.19	Other Macro-Carib languages	=873.9	Other Andean-Equatorial languages
=871.2	Macro-Ge-Bororo languages	=879	Other Central and South American Indian languages
=871.21	Bororo	=9	ARTIFICIAL LANGUAGES
=871.22	Macro-Ge languages	=92	ARTIFICIAL LANGUAGES FOR USE AMONG HUMAN BEINGS. INTERNATIONAL AUXILIARY LANGUAGES (INTERLANGUAGES)
=871.221	Carajá		Volapük (designed by Schleyer)
=871.222	Ge		Esperanto (designed by Zamenhof)
=871.29	Other Macro-Ge-Bororo languages		Systems derived from Volapük or Esperanto. □ Ido
=871.3	Macro-Panoan languages	=921	Langue bleue (designed by Bollack)
=871.31	Mataco	=922	Interlingue (Occidental)
=871.32	Pano-Tacana languages	=923	Interlingua (designed by Gode)
=871.321	Pano		Various a priori, eclectic and a posteriori systems
=871.322	Tacana		<i>Specify by alphabetic extension</i>
=871.33	Moseten	=924	A priori philosophical systems based on logical classification of ideas (using numbers, letters and other signs). □ Pasigraphie. Ao. Ro
=871.39	Other Macro-Panoan languages	=925	Eclectic, mixed systems (with rationalized grammar). □ Pasilingua. Dilpok. Langue universelle
=871.41	Nambicwara	=926	A posteriori systems (chiefly based on natural languages). □ Afrihili. Basic English.
=871.9	Other Ge-Pano-Carib languages	=929	Interlingua (simplified Latin, designed by Peano). Mundolingua. Neo. Novial
=872	Macro-Chibchan languages		ARTIFICIAL LANGUAGES USED TO INSTRUCT MACHINES. PROGRAMMING LANGUAGES. COMPUTER LANGUAGES
=872.1	Chibchan		<i>Use only when it is desired to denote the language in which the document being classed (e.g. a computer program) is written</i>
=872.2	Misumalpan		<i>Specify by alphabetic extension</i>
=872.3	Paezan languages	=929.1	◇ 004.912=93C++ Word-processing program written in C++
=872.31	Barbacoan	=929.2	⇒ 004.43
=872.32	Choco	=929.3	⇒ 004.655
=872.33	Jiraran		
=872.34	Paez-Coconuco (Inter-Andine)		
=872.4	Waican		
=872.9	Other Macro-Chibchan languages		
=873	Andean-Equatorial languages		
=873.1	Andean languages	=93	
=873.11	Cahuapanan		
=873.12	Quechumaran		
=873.121	Aymara		
=873.122	Quechua		
=873.129	Other Quechumaran languages		
=873.19	Other Andean languages		
=873.2	Equatorial languages		
=873.21	Arawakan languages		
=873.211	Chapacura-Wanhaman		
=873.212	Maipuran		
=873.219	Other Arawakan languages		

(0...) TABLE 1d. COMMON AUXILIARIES OF FORM

SCOPE. The common auxiliaries of form denote the form or presentation of documents. They are not used to denote the subject matter of documents.

Literary forms (poetry, plays, fiction etc.) are classed under 82-1/-9.

Kinds of form not listed in Table 1d may be denoted by (0:...), e.g.

929(0:82-31) Biography presented in novel form

CITATION ORDER. These auxiliaries are normally used following a subject notation, but, if desired, all documents of the same form of presentation (or their surrogates) may be grouped together under the appropriate form auxiliary, e.g.

54(035) Handbooks of chemistry.

54(038) Dictionaries of chemistry (filed beside other works on chemistry).

(038)54 Dictionaries of chemistry (filed beside other dictionaries).

(054) Newspapers (of all kinds).

(054)(44) French newspapers.

(085.6)685.53 Price lists of camping equipment (filed beside other price lists).

INNER AND OUTER FORM. Where more than one form aspect is to be expressed, distinction should be made between inner form, where the form influences the subject (e.g. historical presentation) and outer form, which expresses only the physical characteristics of the information carrier (e.g. a sound recording). Regardless of numerical order, inner form should occur next to the subject before outer form is expressed, e.g.

[792+82-2](091)(086.7) A spoken-word history of the theatre (subject - historical form - sound recording)

Systematic table

(0.0...)	PHYSICAL FEATURES, PRODUCTION AND USE CHARACTERISTICS, SUPPLEMENTARY MATTER, ETC. <i>Used as special auxiliaries with the form auxiliaries (01)/(09), or as (0.0...) when no particular form of document is concerned</i> ◇ (038.021.6) Miniature dictionaries ◇ (0.021.6) Miniature documents (general)	(0.025.4) Unillustrated documents (0.026) Documents according to their binding (0.026.2) Documents with fixed covers. Bound. Cased. Hardback (0.026.4) Sewn, stitched, stapled documents. Brochures (0.026.6) Loose-leaf documents (0.027) Kind of edition (0.027.2) Standard editions (0.027.3) Popular editions ⇒ (0.062) (0.027.5) Paperback editions (0.027.6) De luxe editions. Fine editions. Collectors', book-lovers' editions ⇒ 094.3 (0.027.8) Unauthorized editions. Pirated editions (0.028) Defective documents (0.028.1) Incomplete documents. Copies with missing pages (0.028.3) Damaged documents (0.028.5) Documents with printing defects, misprints, errata etc. (0.028.7) Forged documents. Fake documents (0.029) Documents with other physical features (0.029.1) Blank books. Albums ⇒ 686.82 (0.029.2) Gummed documents. □ Labels (0.029.4) Mounted, pasted-in documents
(0.02)	Documents according to physical, external form	(0.03) Documents according to method of production
(0.021)	Documents according to scope and size	(0.032) Handwritten documents (autograph, holograph copies). Manuscripts. Pictorial documents (drawings, paintings)
(0.021.2)	Large documents. Comprehensive documents	(0.033) Typescripts. Printouts. □ Output from typewriter, teleprinter, text processor, line printer, computer terminal etc.
(0.021.3)	Medium-sized documents	(0.034) Machine-readable documents
(0.021.4)	Small documents. Concise documents	(0.034.1) Perforated media. □ Punched cards. Punched tape
(0.021.6)	Miniature editions	
(0.021.8)	Documents with special format	
(0.022)	Documents according to colour and transparency	
(0.022.2)	One-colour documents	
(0.022.3)	Black-and-white documents	
(0.022.4)	Two-colour documents (other than black-and-white)	
(0.022.5)	Multicolour documents	
(0.022.7)	Transparent documents	
(0.022.8)	Opaque documents	
(0.023)	Documents according to their material (substance) (0.023) ≡ -03	
(0.023.552)	Documents on processed leather, parchment	
(0.024)	Documents according to their spatial impression ⇒ (086.4)	
(0.024.3)	Documents with stereo impression. □ Anaglyphs	
(0.024.5)	Documents with perspective impression	
(0.024.7)	Documents with panoramic impression	
(0.025)	Documents according to illustration	
(0.025.2)	Illustrated documents	

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	Optionally, monolingual general dictionaries may be denoted solely by common auxiliaries	(048.34)	Informative abstracts (summarizing arguments, conclusions)
		(048.4)	Summaries
	◇ (038)=111 English dictionary	(048.8)	Surveys. Compilations of information from several primary documents
	<i>In linguistics, priority may be given to either the language or the linguistic feature</i>	(048.82)	Uncritical surveys. Pure compilation
	◇ 811.135.1'373.421(038) Romanian - synonyms - dictionary	(048.83)	Critical surveys. Evaluative surveys. Analytico-synthetic surveys
	◇ 81'373.421=135.1(038) Synonyms - Romanian - dictionary	(048.84)	Factual surveys. Fact documentation
	⇒ 81'373	(048.85)	Predictive, prospective surveys
	⇒ 81'374	(049)	Other non-serial separates
(04)	NON-SERIAL SEPARATES. SEPARATA	(049.1)	Notifications. Announcements. Proclamations. Manifestos
	⇒ 82-4	(049.2)	Polemics. Disputations. Controversies. Diatribes. Justifications. Vindications. Rejoinders. Apologias
(041)	Pamphlets. Brochures		Declarations of opinion. □ Advice. Hints. Critiques. Appreciations
(042)	Addresses. Lectures. Speeches	(049.3)	Critical reviews
	⇒ (07)	(049.32)	Petitions. Requests. Applications
(042.3)	Lectures	(049.4)	Questionnaires. Inquiries. Opinion polls. Referenda
(042.4)	Lecture courses	(049.5)	
(042.5)	Speeches. Orations		
	⇒ 82-5		
(043)	Theses. Dissertations	(05)	SERIAL PUBLICATIONS. PERIODICALS
(043.2)	Theses. Academic theses		<i>Individual articles and contributions in periodicals and newspapers are classed using (045) and (046)</i>
(043.3)	Dissertations		<i>Denote periodicity of serial publications by "5..."</i>
(043.5)	Inaugural dissertations (academic)		◇ (05)"520*1" Published daily
(044)	Personal documents. Correspondence. Letters. Circulars		◇ (05)"530*1" Published weekly
	⇒ 82-6		◇ (05)"530*2" Published fortnightly
	⇒ 82-94		◇ (05)"540*1" Published monthly
	⇒ 929		◇ (05)"540*2" Published two-monthly
(044.2)	Personal letters. Private correspondence		◇ (05)"540*3" Published quarterly, three-monthly
(044.3)	Other personal documents. □ Expressions of wish (similar to wills but not legally binding)		◇ (05)"550*1" Published annually, yearly
	⇒ 347.69		◇ (05)-022.345.4 Irregular serials (published at irregular intervals)
(044.4)	Business letters. Official correspondence		
(044.6)	Circular letters		
(045)	Articles in serials, collections etc. Contributions	(051)	Periodicals (in the strict sense).
	⇒ (05)		□ Journals. Magazines
	⇒ (08)	(054)	Newspapers
(046)	Newspaper articles		<i>Newspapers in the strict sense, i.e. frequent publications presenting primarily a generality of news and current information</i>
(047)	Reports. Notices. Bulletins	(055)	News bulletins. Newsletters. Information bulletins
	⇒ (055)		⇒ (047)
(047.1)	Progress reports. Situation reports	(058)	Yearbooks. Directories
(047.2)	Travel reports. Reports of trips (business, study trips etc.)	(058.7)	Address books. Directories listing residents, streets etc. Trade directories
(047.3)	Special reports	(059)	Almanacs. Calendars (of all types)
(047.31)	Research reports. Reports of tests, experiments, projects etc.	(059.2)	Tear-off calendars
	⇒ (083.94)	(059.3)	Ephemerides. Calendars, diaries of coming events
(047.32)	Management and administrative reports		
(047.33)	Reports on accidents, damage and reconstruction works	(06)	DOCUMENTS RELATING TO SOCIETIES, ASSOCIATIONS, ORGANIZATIONS
(047.34)	Reports on production, technical processes		Documents concerning the organization and activities of societies etc.
(047.36)	Trade reports. Marketing reports	(062.13)	Statutes. Charter. Articles of association
(047.5)	Reports for the press. Press releases, hand-outs. Communiqués	(062.21)	Membership lists, rolls
(047.53)	Interviews	(062.532)	Calling notices. Invitations. Notices of meetings
(047.6)	Brief reports, notices. News items. Press cuttings	(062.533)	Agendas
		(062.534)	Discussion documents. Submissions. Working papers
(048)	Bibliographic descriptions. Abstracts. Summaries. Surveys	(062.536)	Motions. Proposals. Resolutions of assembly
	⇒ (01)	(062.537)	Records, minutes, reports of meetings
(048.1)	Bibliographic descriptions		
(048.2)	Annotations		
(048.3)	Abstracts		
(048.32)	Indicative abstracts (giving scope and content)		

(0...) Table 1d. Common auxiliaries of form

(062.551)	Reports on activities. Bulletins	(083)	Technical and normative documents. Tables. Lists etc.
(062.552)	Regular collections of (printed) work. □ 'Proceedings...'. 'Transactions...'	(083.1)	Recipes. Prescriptions. Directions
(07)	DOCUMENTS FOR INSTRUCTION, TEACHING, STUDY, TRAINING ⇒ 371.67	(083.12)	Recipes. Prescriptions
(072)	Documentary materials for teaching. Teaching aids ⇒ (084) ⇒ (086)	(083.13)	Directions for use. Advice. Instructions ⇒ (076)
(073)	Curricula. Syllabuses	(083.131)	Directions for use. User's guides. User's manuals (directions for the use of equipment, instruments, specialist documents, computer software etc.). Operating instructions ⇒ (035)
(075)	Educational texts. Schoolbooks. Texts for students	(083.132)	Practical advice, hints
(075.2)	Elementary school texts. Primers	(083.133)	Instructions <i>For operating instructions use (083.131)</i>
(075.3)	Secondary school texts	(083.15)	Rules. Regulations ⇒ (094.58)
(075.4)	Texts for self-instruction, private study, home study, correspondence courses, adult education	(083.17)	Timetables
(075.8)	Texts for university, higher education	(083.2)	Forms, blanks. Printed forms
(075.9)	Texts for short courses, crash courses, refresher courses	(083.4)	Tables (other than purely numeric)
(076)	Documents for practical instruction, training ⇒ (035) ⇒ (083.13)	(083.41)	Statistical tables
(076.1)	Documents with problems, questions	(083.44)	Classification tables
(076.2)	Documents with solutions, answers	(083.5)	Technical data tables. Calculation tables. Conversion tables ⇒ 519.66
(076.3)	Documents with questions and answers ⇒ 27-282.3	(083.53)	Numeric tables. Percentage tables. Ready reckoners
(076.5)	Practical exercises. □ Laboratory exercises. Field exercises	(083.57)	Nomograms
(076.6)	Programmed texts. Scrambled textbooks, with program frames ⇒ 37.018.43 ⇒ 37.091.315.7	(083.58)	Network diagrams. Critical path diagrams (arrow diagrams)
(078)	Educational documents and material not listed elsewhere	(083.7)	Terminological and normative documents
(078.7)	Case studies (as form) ⇒ 001.87	(083.71)	Definitions
(078.9)	Models (in the theoretical sense)	(083.72)	Nomenclatures
(079)	Documents connected with competitions, tests, examinations etc.	(083.73)	Codes. Cryptographies. Keys
(079.1)	Questions and exercises for tests, contests, competitions. Examination papers	(083.74)	Standards
(079.2)	Paper qualifications. Diplomas. Degrees. Certificates of proficiency etc.	(083.742)	Standards for objects or products. Specifications
(08)	COLLECTED AND POLYGRAPHIC WORKS. FORMS. LISTS. ILLUSTRATIONS. BUSINESS PUBLICATIONS ⇒ 08 ⇒ 82-82	(083.744)	Standards for methods or processes. Codes of practice. Guidelines
(081)	Individual polygraphies. Collected works of individual authors	(083.76)	Patterns. Samples
(081.1)	Collected works	(083.77)	Comparative presentation
(081.2)	Selected works (081.2) ≡ (082.2)	(083.78)	Estimates of cost. Tenders
(082)	Collective polygraphies. Collections of works by several authors	(083.8)	Enumerative records ⇒ (01) ⇒ 01
(082.1)	Series (works by various authors published under a general title). Publishers' series, sets, libraries	(083.81)	Lists in general. Schedules
(082.2)	Anthologies. Selections. Excerpts. Quotations	(083.82)	Inventories and catalogues
(082.21)	Anthologies. Selections	(083.822)	Inventories (records of permanent collections)
(082.22)	Excerpts. Quotations	(083.824)	Exhibition catalogues (records of temporary assemblages) <i>Class here catalogues of unique or individual items regardless of whether they are offered for sale. For catalogues of lines of articles or stock in trade see (085.2)</i>
(082.24)	Fragments	(083.84)	Stock records. Stores lists. Warehouse lists
		(083.86)	Tables of contents. Indexes of names, places, subjects
		(083.87)	Concordances ⇒ 001.86 ⇒ 014.5 ⇒ 27-272
		(083.89)	Registers of persons in a given category
		(083.9)	Plans. Projects. Programmes
		(083.92)	Plans
		(083.94)	Projects
		(083.97)	Programmes
		(084)	Images. Pictorial documents. Graphic documents
		(084.1)	Pictures. Illustrations
		(084.11)	Drawings. Sketches
		(084.12)	Photographic images
		(084.121)	Still photographs

(084.122)	Films. Motion pictures	(085.3)	House journals. Company magazines
(084.122.3)	Silent film	(085.5)	Prospectuses
(084.122.5)	Sound film	(085.6)	Price lists for goods
(084.127)	Survey photographs. Photograms (with known internal orientation)	(085.7)	Price lists for services. Tariffs
(084.14)	Magnetically recorded images. Video recordings ⇒ (086.8)	(085.8)	Advertisements
(084.17)	Printing blocks (half-tone blocks etc.). Electrotypes	(086)	Three-dimensional (solid) documents. Audio and audiovisual documents
(084.18)	Prints from blocks. Block-pulls	(086.3)	Specimens, objects as documents.
(084.2)	Schematic representations	(086.4)	□ Microscope slides. Museum pieces
(084.21)	Diagrams. Graphs	(086.42)	Solid, three-dimensional representations ⇒ (0.024)
(084.28)	Profiles (two-dimensional)	(086.43)	Modelling sheets. Flat designs for cut-out models. Networks plotted for solid geometric models
(084.29)	Block diagrams (two-dimensional). Flow charts (flow-sheets) ⇒ (086.45)	(086.44)	Globes ⇒ (084.3)
(084.3)	Cartographic images. □ Maps. Plans. Charts ⇒ (086.43) ⇒ 912	(086.45)	Reliefs (solid). □ Relief models. Relief maps
(084.3-1)	Maps etc. according to scale	(086.48)	Block diagrams (three-dimensional) ⇒ (084.29)
(084.3-12)	1:5 000 to > 1:20 000	(086.7)	Models. □ Maquettes
(084.3-14)	1:75 000 to > 1:250 000 (topographic survey maps)	(086.72)	Audio documents. Sound recordings
(084.3-16)	1:1 000 000 to > 1:20 000 000 (maps of countries, of continents)	(086.74)	Mechanical sound recordings. □ Gramophone records (discs etc.)
(084.3-18)	1:100 000 000 and smaller (world maps)	(086.76)	Magnetic sound recordings. □ Magnetic tape
(084.3-2)	According to form of representation	(086.8)	Optical sound recordings. □ CD (compact disc) <i>For sound on film see (084.122.5)</i>
(084.3-21)	Nonpictorial forms. Contour maps. Maps with hachures, hill-shading	(086.82)	Video recordings <i>Class here video recordings with or without sound</i>
(084.3-22)	Pictorial forms. Bird's-eye maps	(086.84)	Videodisc recordings
(084.3-223)	Airphoto mosaics	(086.86)	Video magnetic tape recordings
(084.3-3)	According to type of representation	(088)	Video optical tape recordings
(084.3-32)	Skeleton maps. Distribution maps	(088.2)	Attestations. Special-purpose documents
(084.3-34)	Statistical maps on the dot system. Dot maps	(088.22)	Certificates, licences etc. issued by authorities
(084.3-35)	Cartograms. Cartodiagrams	(088.24)	Identity and nationality certificates. Passports. Identity cards etc.
(084.3-36)	Structural maps. Maps of complex spatial division	(088.27)	Licences giving right to practise profession, conduct business, possess or drive vehicles etc. □ Driving licences. Firearms certificates (gun licences)
(084.3-37)	Survey maps	(088.4)	Certificates of property, ownership.
(084.3-4)	According to method of production, printing, reproduction	(088.7)	Identification, registration papers. Title deeds etc.
(084.3-41)	Originals. Map sketches, drawings	(088.72)	Contracts. Written agreements
(084.3-42)	Produced by photographic tracing processes. □ Blueprints (cyanotypes). Diazos	(088.77)	Trademarks, crests (letterhead, seal, etc.) and other commercial marks and devices
(084.3-43)	Multicoloured copies on plastics (proof sheets)	(088.8)	Trademarks. Commercial marks and devices ⇒ 347.772
(084.3-44)	Originals for printing (metals, stone etc.)	(088.82)	Marks and certifications of quantity and quality of goods ⇒ (083.74/.76)
(084.3-45)	Printed maps ⇒ 528.927 ⇒ 655.3.026	(088.83)	Patents and similar documents ⇒ 347.771
(084.3-5)	According to type of storage	(088.84)	Applications for patents and inventor's certificates
(084.3-52)	Not mounted. Flat. Rolled	(088.85)	Applications for patents
(084.3-54)	In book form	(088.86)	Patents
(084.3-56)	Mounted	(088.87)	Patent specifications
(084.31)	Sketch maps. Map drawings	(088.88)	Inventor's certificates
(084.32)	Map sheets. Individual sheets	(089.3)	Descriptions of inventions to inventor's certificates
(084.33)	Pocket maps		Proposals for innovations
(084.35)	Wall maps		Licences (patent, etc.) ⇒ (088.2)
(084.4)	Atlases		Curiosities. Miscellanies. Varia. Anecdotes. Facetiae. Aphorisms. Maxims etc.
(084.41)	General geographic atlases (nonthematic)		
(084.42)	Thematic atlases		
(084.5)	Pictorial posters		
(085)	Publications of business firms (industrial, commercial). Commercial publicity, information, advertising material		
(085.2)	Trade catalogues. Industrial catalogues <i>Class here catalogues of lines of items or stock-in-trade.</i> <i>For catalogues of unique or individual articles (assembled permanently or temporarily) see (083.822/.824)</i> ◇ 658.872(085.2) Mail order catalogues		

(0...) Table 1d. Common auxiliaries of form

(089.5)	Frauds. Deceptions. Literary, artistic, scientific hoaxes ⇒ 001.94/.98	(094.2)	International agreements and treaties. International laws. Enactments between independent states
(089.7)	Greetings and special-occasion messages (cards, leaflets etc.) Specify periods or special days by "..." especially "38..." ◇ (089.7)"383New Year" New Year greetings (cards etc.)	(094.4)	Law books and legal problems (textbooks)
		(094.5)	Individual laws. Simple laws and detailed regulations resulting from them. Bills (draft laws). Acts
(09)	PRESENTATION IN HISTORICAL FORM. LEGAL AND HISTORICAL SOURCES	(094.57)	Decrees, rescripts, edicts of central authority, arising from a law. Ordinances of the executive
(091)	Historical presentation in the strict sense ⇒ 93/94	(094.58)	Decisions of central authority in application of a law. Promulgations. Regulations
(092)	Biographical presentation ⇒ 929	(094.7)	Legal sources emanating from administrative authorities of various grades
(093)	Historical sources	(094.71)	Legal sources from delegated powers of the central government. □ Plebiscite. Referendum. Decrees of the senate
(093.2)	Official sources. Public records, documents, acts	(094.73)	Legal sources emanating from individual members of a federation (e.g. kingdoms, individual states, cantons, dominions)
(093.3)	Literary sources. Chronicles. Diaries	(094.74)	Legal sources emanating from provincial authorities
(093.4)	Wills. Testaments	(094.76)	Legal sources emanating from district, municipal and parochial authorities
(094)	Legal sources. Legal documents ⇒ 340.13 ⇒ 340.14	(094.77)	Legal sources emanating from executives of authorities
(094.1)	Enactments of absolute government in general. □ Royal decrees, edicts, ordinances	(094.8)	Legal sources emanating from courts of justice
		(094.9)	Collections of legal documents. Collections of rulings, legal decisions. Legal case-books

(1/9) TABLE 1e. COMMON AUXILIARIES OF PLACE

SCOPE. The place auxiliaries indicate the geographical range, locality or other spatial aspect of a subject denoted by a main UDC number, e.g.

- 331.2(44) Wages in France.
- 338.47(81) Economics of transport in Brazil.

They are the primary characteristic of division in 913 'Regional geography' and 94 'History'

CITATION ORDER. These auxiliaries are normally used following a subject notation. A sequence based on place can also be created by citing the place auxiliary first, e.g.

- 339.5.053(73) Trade balance - USA.
- (73)339.5.053 USA - trade balance.

The place auxiliary may also be intercalated in a UDC number to achieve a desired sequence, e.g.

- 354(44) Central administration in France.
- 354(44)51 French ministry of justice.

Exceptionally, the place auxiliary alone might be adequate for classifying some kinds of document where the place aspect is the only one likely to be sought (e.g. some maps).

FURTHER SUBDIVISION. In general, the political section of the following table, (4/9), is subdivided as far as counties, departments or equivalent administrative units.

Smaller units may be denoted in two ways by using the special auxiliaries under (1-2) and (1-3), either at the number for the high-level unit (country, state) or at that for the lower-level unit (county, department, etc.), e.g.

- (44) France.
- (44-2) Communes of France.
- (44-37) Arrondissements of France.
- (441.1) Département of Finistère.
- (441.1-2) Communes of Finistère.
- (441.1-37) Arrondissements of Finistère.

The names used, the selection and sequence of regions, descriptive expressions and relations implied by the numeric hierarchy do not constitute any endorsement of their national or international status either by the UDC Consortium or by any of its cooperating organizations. No opinions about any of these matters should be inferred from this publication

Systematic table

(1) PLACE AND SPACE IN GENERAL. LOCALIZATION. ORIENTATION

(1:....)	ONE PLACE IN RELATION TO ANOTHER ◇ (44:450) Relations between France and Italy ◇ 339.5(44:450) Trade relations between France and Italy	(1-021)	Biological zones
		(1-021.1)	Floral kingdoms ⇒ (207.3) ⇒ 58
		(1-021.11)	Holarctic kingdom (Europe, Northern Asia and North America)
		(1-021.12)	Paleotropical kingdom (South Asia and Africa)
(1=...)	PLACE WITH REFERENCE TO RACE. ETHNIC ZONES <i>Details by (=...) (Table 1f)</i> ◇ (494=112.2) German Switzerland ◇ (494=131.1) Italian Switzerland ◇ (494=133.1) French Switzerland	(1-021.13)	Neotropical kingdom (South America)
		(1-021.14)	Cape kingdom (S Africa)
		(1-021.15)	Australian kingdom
		(1-021.16)	Antarctic kingdom
		(1-021.2)	Faunal zones ⇒ (207.4) ⇒ 59
(1-0/-9)	BOUNDARIES AND SPATIAL FORMS OF VARIOUS KINDS <i>The (-0/-9) special auxiliaries may be combined with the (2/9) common auxiliaries to denote parts or aspects of specific areas. They may also be used with (1), with a general significance</i> ◇ (1-04) Boundaries in general ◇ (430-04) The boundaries of Germany	(1-021.21)	Palearctic (Europe, Asia and North Africa)
		(1-021.22)	Nearctic (North America)
		(1-021.23)	Neotropical (South America)
		(1-021.24)	Afro-Tropical. Ethiopian
		(1-021.25)	Oriental (Indo-Malayan)
		(1-021.26)	Australasian
		(1-04)	Limiting zones. Boundaries ⇒ (1-192)
		(1-05)	Areas or regions (of the Earth) according to shape
(1-0)	Zones	(1-051)	Elongated areas. Strips
(1-01)	Zones pertaining to particular places or disciplines	(1-052)	Areas with straight or slightly curved boundaries (except strips)
(1-02)	Zones defined for use in particular sciences		

(1/9) Table 1e. Common auxiliaries of place

(1-053)	Areas with mainly curved or non-straight boundaries (except strips)	(1-195.2)	In front. Front. Forward
(1-054)	Areas of special or irregular shape	(1-195.4)	Mid-length. Mid-part. Midships (amidships)
(1-056)	Areas according to spatial coherence, contiguity etc.	(1-195.6)	Behind. Back. Backward. Aft
(1-056.2)	Whole areas. Integral areas	(1-195.8)	In line. In series. End-to-end
(1-056.4)	Included areas (enclaves). Excluded areas (exclaves)	(1-196)	Sideways. Lateral relations
(1-056.42)	Enclaves	(1-196.2)	Right. Right side. Starboard. Toward the right
(1-056.44)	Exclaves	(1-196.4)	Centre-line. Mid-line. Median line
(1-07)	Military zones during war and postwar periods	(1-196.6)	Left. Left side. Port. Toward the left
(1-071)	Front	(1-196.8)	Side-by-side. Alongside. In parallel
(1-072)	Rear	(1-197)	Relative level (above/below)
(1-073)	Conquered territory. Surrendered territory	(1-197.2)	Below. Underneath. Downwards
(1-074)	Occupied zones	(1-197.4)	Mean or average level
(1-075)	Nonoccupied territory	(1-197.6)	Above. Upperside. Upwards
(1-076)	Disputed territory	(1-197.8)	One above the other. Superimposed. Layered. Stratified
(1-077)	Zones without military installations. Weapon-free zones. Demilitarized zones	(1-198)	Inclination. Gradient
(1-077.2)	Zones free of particular weapons. □ Non-nuclear zones (atomic-weapon-free zones) ◇ (430-077.2) Non-nuclear zones in Germany	(1-198.2)	Vertical
(1-077.4)	Zones free of particular military installations	(1-198.4)	Inclined. Sloping. Oblique. Skew
(1-077.6)	Disarmament zones. Demilitarized zones	(1-198.6)	Horizontal
(1-078)	Territories under control of a counter-government (rival government)	(1-199)	Position undefined. Free. Free-moving
(1-08)	Unknown, unexplored regions and zones	(1-2/-4)	Political units. Administrative units <i>The (-2/-4) auxiliaries are applicable to (3/9), in which case they refer to specific localities; used in the form in which they are listed here, they refer to the administrative unit in general and related affairs</i>
(1-1)	Orientation. Points of the compass. Relative position	(1-2)	Lowest administrative units. Localities. □ Districts (UK). Communes (France). Gemeinden (Germany)
(1-11)	East. Eastern <i>Use (...-11) only for relative position or orientation. For the East (Orient) conventionally defined see (5)</i>	(1-21)	Municipalities. Urban... Urban districts
(1-12)	South-east. South-eastern	(1-22)	Village communities, parishes. Rural... Rural districts
(1-13)	South. Southern	(1-24)	Joint local administrative units. □ Cantons (France). Ämter (Germany)
(1-14)	South-west. South-western	(1-25)	Capital cities. Metropolises
(1-15)	West. Western <i>Use (...-15) only for relative position or orientation. For The West (Occident) conventionally defined see (4+7)</i>	(1-28)	Intercommunal groupings. □ for vocational training, water supply
(1-16)	North-west. North-western	(1-29)	Independent functional authorities, jurisdictions
(1-17)	North. Northern	(1-291)	Dike-building associations. Improvement boards. Polder boards
(1-18)	North-east. North-eastern	(1-3)	Larger units within the state
(1-19)	Relative location, direction and orientation	(1-32)	Higher-level administrative units. Provinces
(1-191)	Internal spatial relationships. Inside. Within. Endo-...	(1-328)	Inter-provincial. Higher-order communal associations
(1-191.2)	Central. In the middle	(1-35)	Middle-level administrative units. □ Counties (UK). Départements (France). Regierungsbezirke (Germany). Amter (Denmark)
(1-191.4)	Movement in relation to the centre	(1-37)	Lower-level administrative units. □ Arrondissements (France). Kreise (Germany)
(1-191.42)	Centripetal	(1-4)	Units of highest (state) level. Nations. States. Confederations
(1-191.44)	Centrifugal	(1-41)	Confederations of states. Federations. Commonwealths
(1-191.6)	Eccentric. Not in the middle	(1-43)	◇ (494-41) Switzerland as a confederation ◇ (73-41) USA as a federation ◇ (94-41) Australia as a commonwealth Constituent states within a confederation. Federal states. Autonomous republics ◇ (494-43) Swiss cantons ◇ (73-43) States of the USA ◇ (94-43) Australian states ◇ (944-43) New South Wales as a state of the Commonwealth of Australia
(1-191.8)	Peripheral. At the edge		
(1-192)	Position in relation to a border or frontier		
(1-192.2)	Adjacent. Bordering. Touching		
(1-192.4)	Overlapping. Intruding into		
(1-192.6)	On this side. Cis-...		
(1-192.7)	On the other side. Trans-...		
(1-192.8)	Opposite. Facing		
(1-194)	External spatial relationships. Outside. Exo-...		
(1-194.2)	Alongside. Adjoining. Nearby. Close		
(1-194.4)	Half-way. Midway		
(1-194.6)	Far off. Far away. Distant. Wide apart		
(1-195)	Lengthwise. Longitudinal relations		

(1-44)	Empires. Combination of mother-country, dominions and colonies ◇ (41-44) British Commonwealth and Empire generally	(1-662)	Capitalist countries
(1-45)	Nonfederal sovereign states	(1-664)	Socialist countries
(1-46)	Territories geographically separated from the mother country but integrated in its administration	(1-67)	Groupings of states according to economic point of view
(1-5)	Dependent or semi-dependent territories	(1-672)	Countries with market economy
(1-51)	Dependent territory administered by a specific state ◇ (668.2-51:44) Dahomey under French administration	(1-674)	Countries with centrally planned economy
(1-52)	Colonized territory. Colonies ◇ (6-52) Colonized territory in Africa ◇ (676.2-52) Kenya as a colony	(1-69)	Groupings of states according to other points of view. □ From scientific or technical point of view <i>Specify by A/Z</i> ◇ (4-69ESA) Countries of the European Space Agency
(1-53)	Concessions (territorial) ◇ (512.317-53) The New Territories of Hong Kong as a leased territory	(1-7)	Places and areas according to privacy, publicness and other special features
(1-54)	Dependent or semi-dependent territories with special statute	(1-71)	Places of private, individual and familial activity. In private places. In private
(1-542)	Protectorates ◇ (676.1-542) Uganda as a protectorate	(1-72)	Places of public, collective, group activity. In public places. In public
(1-544)	Mandate (League of Nations). Trusteeship (United Nations) ◇ (569.1-544) Syria as a mandated territory	(1-75)	Areas designated for special treatment
(1-546)	Condominium ◇ (934-546:410+44) New Hebrides (now Vanuatu) as British-French condominium	(1-751)	Areas protected for specific scientific or recreational purposes
(1-55)	Territories in transition between dependence and independence ◇ (549.3-55) East Pakistan during its secession (becoming Bangladesh)	(1-751.2)	National parks (primarily large open spaces for aesthetic and recreational purposes, and including human cultural features) ⇒ 712.253
(1-6)	States or groupings of states from various points of view <i>The (1-6) auxiliaries apply when the territorial aspect is subsidiary</i> ◇ 669.18(4-672EU) Steel production in EU (European Union) countries	(1-751.3)	Nature reserves (primarily for the protection of plant and animal species and habitats, usually with limitation of human access). □ National Wildlife Refuges (USA). Special Protection Areas and Special Areas of Conservation (European Union). Ramsar sites (International Convention on Wetlands of International Importance, Ramsar, Iran, 1971)
(1-61)	Groupings from the point of view of sovereignty	(1-751.4)	Areas primarily for protection of landscapes with aesthetic value. □ Areas of Outstanding Natural Beauty (UK)
(1-612)	Groupings without surrender of sovereignty	(1-751.6)	Areas primarily for protection of inanimate features of scientific interest. □ Sites of Special Scientific Interest (UK)
(1-615)	Groupings with partial surrender of sovereignty	(1-77)	Areas according to stage of development
(1-618)	Groupings with total surrender of sovereignty	(1-773)	Developing countries. 'Third World' countries
(1-62)	Groupings of states according to military or defence view	(1-775)	Developed areas. Highly developed areas
(1-622)	Countries belonging to military or defence pacts <i>Specify by A/Z</i> ◇ (1-622) NATO Countries of the North Atlantic Treaty Organization ⇒ (1-65) ⇒ 327.51	(1-8)	Location. Source. Transit. Destination
(1-624)	Noncommitted countries. Nonaligned countries ⇒ (1-652) ⇒ 327.55	(1-81)	Location. Site. Place of manifestation
(1-65)	Wartime groupings of states	(1-82)	Source. Place of origin. Source of supply ◇ [663.551.5:663.43](410.5-82) Scotch whisky
(1-651)	Belligerent states	(1-83)	Place of transit. Transit area
(1-651.1)	Aggressor	(1-85)	Destination
(1-651.2)	Attacked	(1-86)	Place in a conventional sense, i.e. used as descriptive of some particular article ◇ 677.074.58(55-86) Persian carpets
(1-652)	Neutral states	(1-87)	Foreign. Abroad. Elsewhere than one's own country ⇒ -027.63
(1-66)	Groupings of states according to social and political system	(1-88)	Home. One's own country ◇ 27-46(1-88) Christian missions in the home country. Home missions
		(1-89)	Places in their earlier, historical status. Historical and traditional entities
		(1-9)	Regionalization according to specialized points of view

(1/9) Table 1e. Common auxiliaries of place

(1-92)	Regionalization of the earth according to physiogeographical features <i>If an arrangement according to physiographic feature as leading category is required, an alternative classification is provided at (29). If this alternative is selected, the auxiliary (1-92) should not be used</i>		<i>Further subdivision by three-figure and two-figure notation for degrees longitude and latitude respectively, delineating the western and southern boundaries of the area concerned</i>
	◇ (52-92) Physical regions of Japan	(161)	◇ (161.007.49) River Nahe and western Rhenipfalz area - 7°E, 49°N
	◇ (98-92) Arctic regions		
(100)	UNIVERSAL AS TO PLACE. INTERNATIONAL. ALL COUNTRIES IN GENERAL <i>Up to three countries may be denoted by the auxiliaries for the individual countries. Where more than three countries are concerned, one (or each, by multiple entry) may be specified, in relation to (100). The totality of all or many countries is denoted by (100) alone</i>	(162)	Places in the northern quadrant, 0° to 180° east of Greenwich ◇ (161.000.51) Kent, England - 0°E, 51°N ◇ (161.106.48) Ulan Bator, Mongolia - 106°E, 48°N
	◇ (44+460) France and Spain	(163)	Places in the southern quadrant, 0° to 180° east of Greenwich ◇ (163.026.18) Victoria Falls (R. Zambezi) - 26°E, 18°S ◇ (163.138.35) Gulf of St Vincent near Adelaide, South Australia - 138°E, 35°S
	◇ (44+100) France and various other countries		
	◇ 061.2(100) International organizations		
(15)	The universe, celestial or cosmic space generally. In Space. Cosmic. Extraterrestrial localities	(164)	Places in the southern quadrant, 0° to 180° west of Greenwich ◇ (164.070.16) Lake Titicaca - 70°W, 16°S ◇ (164.150.18) Tahiti - 150°W, 18°S
(151.2)	Specific parts of the cosmos		
(151.21)	In the Solar System ⇒ 523	(18)	SPATIAL MEASUREMENTS OR DIMENSIONS <i>Details by colon combination with (2...)</i> ◇ (23:181) Heights (above sea level) ◇ (24:181) Depths (below sea level) ◇ (24:183) Volume. Cubic capacity (of caves etc.) ◇ (25:182) Surface areas. Areas of land ◇ (26:181) Ocean or sea depths
(151.22)	Environs of the Solar System		
(151.23)	The local stellar system		
(151.24)	Region of the Galaxy (Milky Way) ⇒ 524.6		
(157)	Planets, their regions and localities		
(158)	The Moon, its regions and localities ⇒ 523.34		
(16)	PLACES AND AREAS OF THE EARTH ACCORDING TO GEOGRAPHICAL COORDINATES OR GRIDS ⇒ 528.28	(181)	One-dimensional measurements. Linear dimensions. Distances. Lengths etc. <i>For precise distances, add standard international abbreviation for unit, followed by actual figures</i> ◇ (181m100) 100 metres ◇ (181km427) 427 kilometres Short Medium length Long
(160)	Geographical meridians and parallels. Poles. Points on the Earth according to geographical coordinates		
(160.1)	Longitudes. Meridians ◇ (160.1-11) Eastern longitudes ◇ (160.1-15) Western longitudes	(181.3)	
(160.2)	Latitudes. Parallels ◇ (160.2-13) Southern latitudes ◇ (160.2-17) Northern latitudes	(181.5)	
(160.20)	Equator	(181.7)	
(160.23)	Tropics ◇ (160.23-13) Southern: Tropic of Capricorn ◇ (160.23-17) Northern: Tropic of Cancer	(182)	Two-dimensional measurements. Square dimensions. Areas <i>Details as (181)</i> ◇ (182m2000) 2000 square metres
(160.26)	Polar circles ◇ (160.26-13) South Polar Circle (Antarctic Circle) ◇ (160.26-17) North Polar Circle (Arctic Circle)	(183)	Three-dimensional measurements. Cubic dimensions. Volumes <i>Details as (181)</i> ◇ (183m150) 150 cubic metres
(160.27)	Poles ◇ (160.27-13) South Pole ◇ (160.27-17) North Pole		
(161/164)	Place according to quadrants <i>The Earth's surface is divided into four quadrants: two northern, 0°-180°E and 0°-180°W of Greenwich, and two southern, 0°-180°E and 0°-180°W of Greenwich, numbered (161) to (164)</i>	(188)	Relative size of places, areas, spaces (188.2) Very small. Part of a small area (188.3) Small. Part of a medium-sized area (188.5) Medium-sized. Moderate-sized. Part of a large area (188.7) Large. Part of a very large area (188.8) Very large. Huge areas

(2) PHYSIOGRAPHIC DESIGNATION

(20)	ECOSPHERE	(213.14)	Wet or humid subtropical regions. Cotton Belt type
(203)	In the atmosphere. In the air. Aerial ⇒ 551.510	(213.5)	Tropical regions. Torrid zone. Tropics
(203.1)	Thermosphere. Outermost atmospheric space	(213.52)	Hot dry or arid regions. Hot desert
(203.3)	Mesosphere (upper boundary: mesopause)	(213.521)	Deserts of Asia
(203.5)	Stratosphere (upper boundary: stratopause)	(213.521.1)	Syrian desert
(203.7)	Troposphere (upper boundary: tropopause)	(213.521.2)	Negev desert
(204)	Of the hydrosphere. In the water. Aquatic ⇒ (26) ⇒ (28) ⇒ 556	(213.521.4)	Rub'al Khali
(204.1)	Underwater	(213.521.6)	Thar desert (Great Indian desert)
(204.2)	Floating	(213.522)	Deserts of Africa
(205)	Lithosphere. Earth's mantle. Centrosphere	(213.522.2)	Sahara desert. □ Libyan desert. Arabian desert (Western desert). Nubian desert
(205.2)	Lithosphere. Outer crust. Surface layer	(213.522.4)	Kalahari desert
(205.4)	Centrosphere. Earth's core. Barysphere	(213.522.6)	Namib desert
(207)	Of, or in the spheres of Nature. Biosphere	(213.523)	Deserts of North America
(207.3)	In the plant kingdom	(213.523.4)	Mojave (Mohave) desert
(207.4)	In the animal kingdom	(213.523.6)	Colorado desert
(208)	In human surroundings. In human society. In social, artificial milieux	(213.524)	Deserts of South America
(21)	SURFACE OF THE EARTH IN GENERAL. LAND AREAS IN PARTICULAR. NATURAL ZONES AND REGIONS ⇒ 551.4 ⇒ 551.8	(213.524.2)	Atacama desert
(210)	Land formations	(213.525)	Deserts of Australia
(210.1)	Peninsulas	(213.525.2)	Great Sandy desert
(210.2)	Capes. Promontories	(213.525.4)	Gibson desert
(210.3)	Headlands. Tongues of land	(213.525.6)	Great Victoria desert
(210.4)	Isthmuses. Necks of land	(213.54)	Savanna. Tropical grassland regions with gallery forest
(210.5)	Coasts. Beaches. Banks. Shores. Littoral...	(213.55)	Monsoon regions
(210.7)	Islands	(213.56)	Hot humid regions. Equatorial forest regions. Tropical forest regions
(211/213)	Climatic zones	(215)	Hemispheres ◇ (215-11) Eastern hemisphere ◇ (215-13) Southern hemisphere ◇ (215-15) Western hemisphere ◇ (215-17) Northern hemisphere
(211)	Cold regions. Polar regions. Frigid regions ◇ (211-13) South polar regions. Antarctic regions ◇ (211-17) North polar regions. Arctic regions	(217)	Former continents. Pangaea. Laurentia. Fennosarmatia. Angaria (Angaraland). Gondwanaland. Serindia. Sinia. Philippina ⇒ (309)
(211.6)	Ice-cap climate	(218)	Individual crustal plates. Eurasian. African. Indian. Pacific. North American. South American. Nazca. Antarctic ⇒ 551.24
(211.7)	Tundra	(23)	ABOVE SEA LEVEL. SURFACE RELIEF. ABOVE GROUND GENERALLY. MOUNTAINS <i>Specify height by colon combination with (181)</i> ◇ (23:181) Heights above sea level ◇ (23:181m4500) Mountains with height of 4500 metres above sea level
(212)	Intermediate and subpolar regions. Temperate zones. Mid-latitude climatic regions ◇ (212-13) South temperate zone ◇ (212-17) North temperate zone	(23.0)	In mountainous country. In hill country
(212.3)	Cold intermediate regions. Subpolar regions. Coniferous Forest Belt	(23.01)	Lower mountains. Hills
(212.4)	Cool temperate regions	(23.02)	Medium mountains. Mid-mountain
(212.5)	Warm temperate regions	(23.03)	High mountains. Higher parts of mountains
(212.6)	Mid-latitude steppe regions	(23.04)	Orological formations. Mountain chains. Terraces. Valleys ⇒ 551.43
(212.7)	Mid-latitude desert regions	(23.042)	Mountain chains
(212.72)	Gobi (Shamo) desert	(23.044)	Terraces
(212.75)	Takla Makan desert	(23.045)	Plateaux
(212.77)	Kara Kum desert	(23.046)	Valleys ⇒ 551.43
(213)	Subtropical and tropical regions generally ◇ 69.03(213.5) Tropical building	(23.07)	Limits of altitude. Height zones
(213.1)	Subtropical regions generally	(23.071)	Tree zones. Limits of trees
(213.12)	Dry subtropical regions	(23.073)	Treeless vegetation zones. Limits of vegetation
		(23.075)	Snow zones. Limits of snow

(1/9) Table 1e. Common auxiliaries of place

(234)	Mountains of Europe	(234.321.5)	Cetic Alps
(234.1)	Mountains of the Iberian peninsula	(234.321.6)	Plessur Alps
(234.12)	Pyrenees. Monte Perdido. Maladetta	(234.321.7)	Salzburg Schist Alps
(234.13)	Catalonian Mountains. Montserrat	(234.321.8)	Eisenerzer Alpen
(234.17)	Sierra Nevada. Mulhacén	(234.322)	Northern limestone Alps
(234.2)	Mountains of France and Belgium	(234.322.1)	Allgäuer Alps
(234.23)	Armorican Massif	(234.322.11)	Rhätikon
(234.24)	Massif Central	(234.322.12)	Lechtaler Alpen
(234.241)	Western part of Massif Central. Monts de la	(234.322.2)	North Tyrolean limestone Alps
	Marche. Monts du Limousin	(234.322.21)	Wettersteingebirge
(234.242)	Monts d'Auvergne	(234.322.22)	Karwendelgebirge
(234.243)	Eastern part of Massif Central. Monts du	(234.322.24)	Kaisergebirge
	Vivaraïs. Monts du Lyonnais. Monts du	(234.322.3)	Salzburg limestone Alps
	Beaujolais. Monts du Charollais etc.	(234.322.4)	Austrian Alps
(234.243.1)	Cévennes	(234.322.45)	Moln Alps
(234.251)	Côte d'Or	(234.322.46)	Wienerwald
(234.252)	Morvan	(234.323)	Southern limestone Alps
(234.261)	Plateau de Langres	(234.323.1)	Lombardic Alps
(234.262)	Monts Faucilles	(234.323.2)	Etschbucht Alps (Alps of the Adige Bay)
(234.27)	Vosges	(234.323.3)	South Tyrolean highland
	⇒ (234.524)	(234.323.4)	Venezia Alps
(234.28)	Ardennes	(234.323.5)	Carnic Alps
	⇒ (234.512)	(234.323.6)	Julian Alps
(234.29)	Argonne	(234.323.61)	Karawanken
(234.3)	Alps	(234.351)	Jura Mountains
(234.31)	Western Alps		⇒ (234.547)
(234.311)	Inner Gneiss formation		⇒ (234.548)
(234.311.1)	Ligurian Alps	(234.351.1)	French Jura
(234.311.2)	Cottian Alps	(234.351.2)	Swiss Jura (Kettenjura)
(234.311.3)	Graian Alps	(234.351.3)	German Jura (Tafeljura)
(234.311.4)	Pennine Alps. □ Matterhorn	(234.371)	Mountains of Austria
(234.311.5)	Lepontine Alps		⇒ (234.322.4)
(234.311.51)	Ticino group	(234.371.1)	Leithagebirge
(234.311.52)	Simplon group	(234.372)	Mountains of Czechia and Slovakia
(234.311.53)	St Gotthard group		⇒ (234.58)
(234.311.54)	Adula group	(234.372.3)	Western Carpathians
(234.312)	Outer Gneiss formation		⇒ (234.421.1)
(234.312.1)	Maritime Alps	(234.372.4)	Beskids. Tatra. Slovak Ore Mountains
(234.312.2)	Alpes du Dauphiné	(234.373)	Mountains of Hungary
(234.312.3)	Savoy Alps	(234.373.1)	Bakony
(234.312.31)	Massif du Mont Blanc	(234.373.2)	Vértes
(234.312.32)	Dent du Midi group	(234.373.3)	Matra
(234.312.4)	Freiburg Alps	(234.373.4)	Bükk
(234.312.5)	Bernese Alps. □ Jungfrau	(234.373.5)	Hegyalja
(234.312.6)	Emmental Alps	(234.373.6)	Mecsek
(234.312.7)	Glarus Alps (Glarner Alpen)	(234.41)	Mountains of the Italian peninsula
(234.312.8)	Schwyz Alps	(234.411)	Apennines
(234.312.9)	Appenzeller Alps	(234.411.2)	Northern Apennines
(234.313)	French limestone Alps	(234.411.21)	Ligurian Apennines
(234.313.1)	Provençal Alps	(234.411.23)	Tuscan Apennines
(234.313.2)	Jura	(234.411.25)	Umbrian Apennines
(234.313.3)	Chablais	(234.411.3)	Central Apennines
(234.32)	Eastern Alps	(234.411.31)	Abruzzi. Gran Sasso
(234.321)	Central Gneiss Alps	(234.411.33)	Sabine Mountains
(234.321.1)	Rhaetic Alps	(234.411.35)	Neapolitan Apennines
(234.321.12)	Silvretta group	(234.411.4)	Southern Apennines
(234.321.13)	Bernina group	(234.413)	Mount Vesuvius
(234.321.15)	Oetztaier Alpen	(234.415)	Mount Etna
(234.321.16)	Adamello group	(234.42)	Mountains of the Balkan Peninsula
(234.321.17)	Ortles group	(234.421)	Mountains of Romania (Rumania)
(234.321.2)	Hohe Tauern	(234.421.1)	East Carpathians
(234.321.21)	Zillertal Alps	(234.421.2)	Transylvanian Alps
(234.321.22)	Venediger group	(234.422)	Mountains of Bosnia, Herzegovina and Albania
(234.321.23)	Glockner group	(234.422.1)	Dinaric Alps
(234.321.3)	Niedere Tauern	(234.422.2)	North Albanian Alps
(234.321.31)	Radstädter Tauern	(234.423)	Mountains of Bulgaria
(234.321.32)	Schladminger Tauern	(234.424)	Mountains of Turkey in Europe
(234.321.33)	Rottenmanner Tauern	(234.425)	Mountain ranges and mountains of Northern and Central Greece
(234.321.4)	Noric Alps		Pindus Mountains
(234.321.41)	Gurktal Alps	(234.425.1)	
(234.321.42)	Lavanttal Alps		<i>For Olympus (previously at this number) see (234.425.23)</i>

(234.425.2)	Mountains of Eastern Greece <i>For Ossa (previously at this number) see (234.425.25)</i>				□ Bodmin Moor. Dartmoor. Exmoor. Blackdown Hills. Quantock Hills. Mendip Hills. Cotswold Hills
(234.425.21)	Athos	(234.62)			Mountains of Wales. Brecon Beacons.
(234.425.23)	Olympus				Cambrian Mountains. Snowdonia
(234.425.25)	Ossa	(234.63)			Mountains of northern England
(234.425.27)	Pelion (Pilion)	(234.631)			Pennine Chain. The Peak. Ross Fell
(234.425.29)	Othris (Othrys) mountains	(234.632)			Cumbrian Mountains. Scafell Pike
(234.425.4)	Mountains of South East Greece <i>For Othris (Othrys) mountains (previously at this number) see (234.425.29)</i>	(234.64)			Mountains of the Scottish border. Cheviot Hills
		(234.65)			Scottish Highlands
(234.425.42)	Giona	(234.651)			Grampians. Ben Nevis. Cairngorms
(234.425.44)	Parnassus	(234.652)			Caledonian Highlands. Ben Wyvis
(234.425.46)	Helicon	(234.7)			Mountains of Scandinavia
(234.425.48)	Pentelicus	(234.72)			Dovre fjell. Snøhetta. Jostedalbreen
(234.425.49)	Hymettos	(234.73)			Langfjellet. Jotunheimen. Galdhøpiggen
(234.426)	Mountains of the Peloponnese. Taygetus	(234.74)			Baltic Shield
(234.5)	Mountains of Central Europe	(234.8)			Mountains of Russia in Europe
(234.51)	Rhine schist mountains	(234.81)			Valdai Hills
(234.511)	Hohes Venn	(234.84)			Volga Hills. Yergeni Heights (Ergeni Heights)
(234.512)	Eifel ⇒ (234.28)	(234.85)			Urals
		(234.851)			Northern Urals. Polar Urals
(234.513)	Hunsrück	(234.852)			Central Urals. Ural ore mountains
(234.515)	Rothaargebirge	(234.853)			Southern Urals. Ural forest areas
(234.516)	Siebengebirge	(234.86)			Crimean Mountains. Yaila Mountains.
(234.517)	Westerwald				Roman-Kosh
(234.518)	Taunus	(234.9)			Caucasus. Kavkaz range. □ Mount El'brus. Mount Kazbek
(234.52)	Mountains of the Upper Rhine Rift Valley				
(234.521)	Pfälzer Bergland	(235)			Mountains of Asia
(234.522)	Haardt	(235.1)			Mountains of western Asia. □ Anatolo-Iranian
(234.524)	Vosges (Vogesen) ⇒ (234.27)				Plateaux
		(235.11)			Black Sea coastal range. Pontine Mountains
(234.525)	Odenwald	(235.111)			Anatolian Plateau
(234.527)	Black Forest (Schwarzwald)	(235.12)			Taurus Mountains
(234.53)	Weserbergland and Harz	(235.123)			Mount Ararat
(234.531)	Teutoburger Wald	(235.125)			Armenian Plateau
(234.532)	Elge	(235.13)			Mountains of northern Iran
(234.534)	Wesergebirge	(235.131)			Elburz (Alborz) Mountains. Mount Damavand
(234.538)	Solling	(235.14)			Iranian Plateau
(234.539)	Harz	(235.15)			Mountains of southern Iran. Zagros Mountains
(234.54)	Hessisches Bergland. Swabian-Franconian basin. Alpenvorland	(235.16)			Lebanon (Jabal Lubnan). Anti-Lebanon. □ Mount Hermon (Jabal esh Sheikh)
					Mountains of the Sinai Peninsula
(234.541)	Meissner	(235.18)			Mountains of Central Asia (Asiatic Plateau)
(234.542)	Vogelsberg	(235.2)			Western area of Asiatic Plateau
(234.543)	Rhön	(235.21)			Pamirs
(234.544)	Spessart	(235.211)			Hindu Kush
(234.545)	Steigerwald	(235.212)			Trans-Alai Chain
(234.547)	Schwäbische Alb ⇒ (234.351)	(235.213)			Alai mountain range
		(235.214)			Tien Shan
(234.548)	Franconian Jura	(235.216)			Northern area of Asiatic Plateau
(234.549)	German Alpenvorland	(235.22)			Dzungarian Plateau
(234.55)	Mountains of Thuringia (Thüringen). Mountains of the western edge of the Bohemian basin	(235.221)			Altai (Altay) mountain range
		(235.222)			Sayan Mountains
(234.551)	Thüringerwald	(235.223)			Tannu-Ola
(234.553)	Fichtelgebirge	(235.224)			Khangai Mountains
(234.554)	Böhmerwald	(235.225)			Central area of Asiatic Plateau. Kunlun mountain range
(234.555)	Bayerischer Wald	(235.23)			Central Kunlun. Altun Shan (Altyn Tagh)
(234.56)	Mountains of Saxony (Sachsen)				Southern area of Asiatic Plateau
(234.562)	Vogtland	(235.232)			Karakoram Range
(234.563)	Erzgebirge	(235.24)			Tibetan Plateau
(234.564)	Elbsandsteingebirge	(235.241)			Himalayas
(234.565)	Lausitzer Bergland	(235.242)			Mount Everest (= Chomolungma [Tibetan] = Qomolangma [Chinese] = Sagarmatha [Nepalese])
(234.57)	Mountains of Silesia. Sudetes	(235.243)			K2
(234.571)	Isergebirge	(235.243.1)			Kanchen Junga
(234.572)	Riesengebirge				Nanga Parbat
(234.573)	Waldenburger Berge				Transhimalayas
(234.575)	Adlergebirge	(235.243.2)			Mountains of northern Asia
(234.58)	Mountains of the Bohemian-Moravian Basin	(235.243.3)			
(234.582)	Central Bohemian Range	(235.243.4)			
(234.6)	Mountains of Great Britain	(235.244)			
(234.61)	Highlands of south and south-west England.	(235.3)			

(1/9) Table 1e. Common auxiliaries of place

(235.31)	Byrranga Mountains. Mountains of Taimyr Peninsula	(236.81)	Drakensberg
(235.32)	Verkoyansk Mountains	(236.89)	Table Mountain
(235.33)	Kolyma Mountains	(237)	Mountains of North America
(235.34)	Stanovoi Mountains	(237.1)	Western Cordillera
(235.35)	Yablonoi Mountains	(237.11)	Alaskan Range. □ Mount McKinley
(235.4)	Mountains of eastern Asia	(237.12)	Rocky Mountains
(235.41)	Nan Ling	(237.13)	Cascade Range. □ Mount Rainier
(235.42)	Tsin Ling	(237.14)	Sierra Nevada. □ Mount Whitney
(235.43)	T'ai-hang Shan	(237.15)	Coast Range
(235.44)	'Beijing Grid' (Barrier Ranges)	(237.153)	Lower California Peninsula
(235.45)	Great Khingan Mountains	(237.16)	Sierra Madre
(235.46)	Little Khingan Mountains	(237.161)	Sierra Madre Oriental
(235.47)	Sikhote-Alin'	(237.162)	Sierra Madre Occidental
(235.48)	Mountains of south-eastern China	(237.162.3)	Colima Volcano complex. Paricutin
(235.5)	Mountains of southern Asia	(237.163)	Transverse Volcanic Axis (Sierra Volcanica Transversal). □ Citlaltépetl (Orizaba).
(235.51)	Mountains of the Indian Peninsula		Iztaccíhuatl - Popocatepetl. Nevado de Toluca
(235.511)	Deccan Plateau		Sierra Madre del Sul
(235.512)	Western Ghats. Eastern Ghats	(237.164)	Sierra Madre del Chiapas. □ Acatenengo.
(235.52)	Mountains of South East Asia	(237.165)	Agua. Atitlán. Pacaya. Tacaná. Tajumulco (volcanoes)
(235.521)	Mountains of Myanmar		Sierra de los Cuchumatanes
	<i>For mountains of the Malay Archipelago (previously at this number) see (235.522)</i>	(237.166)	Sierra de las Minas
(235.521.1)	Patkai, Naga, Chin Hills	(237.168)	Maya Mountains. □ Victoria Peak
	<i>For mountains of the Sunda Islands (previously at this number) see (235.522.5)</i>	(237.169)	Mountains of Honduras and El Salvador.
(235.521.2)	Arakan Yoma	(237.17)	□ Sierra del Espíritu Santo. Montañas de Comayagua. Cordillero Nombre de Dios.
	<i>For mountains of Sumatra, Barisan Chain/Pegunungan Barisan (previously at this number) see (235.522.3)</i>		Montañas del Patuca. Cordillera entre Ríos.
(235.522)	Mountains of Malaysia and Indonesia		Volcán Cerro Verde. Volcán Conchagua.
(235.522.3)	Mountains of Sumatra. Barisan Chain (Pegunungan Barisan)	(237.18)	Volcán Izalco. Volcán San Salvador. Volcán Santa Ana
(235.522.4)	Mountains of Borneo		Mountains of Nicaragua. □ Volcán Telica.
(235.522.45)	Mount Kinabalu		Volcán Rata. Volcán Hoyo. Volcán Momotombo
(235.522.5)	Mountains of the Sunda Islands	(237.181)	Cordillera Isabella
		(237.185)	Volcán Santiago. Volcán Masaya. Volcán Mombacho. Volcán Concepción. Volcán Maderas
(236)	Mountains of Africa		Cordillera Chontalina. Cordillera Yolaina
(236.1)	Atlas Mountains		Mountains of Costa Rica and Panama
(236.11)	Er Rif	(237.187)	Cordillera de Guanacaste
(236.12)	Middle Atlas (Moyen Atlas)	(237.19)	Cordillera Central. □ Llanura de Tortuguero.
(236.13)	High Atlas (Haut Atlas)	(237.191)	Volcán Irazú. Volcán Poás
(236.14)	Anti-Atlas	(237.193)	Cordillera de Talamanca. Volcán Barú
(236.15)	Tell Atlas (Atlas Tellien), Little Atlas		Serranía de San Blas
(236.16)	Saharan Atlas (Atlas Saharien)	(237.194)	Serranía de Majé
(236.2)	Mountains of the Sahara region	(237.195)	Serranía de Darién. Altos de Puna. Altas de Limón
(236.21)	Tibesti	(237.196)	Appalachian Mountains
(236.22)	Ahaggar (Hoggar) massif	(237.197)	Allegheny Mountains
(236.23)	Air		Mountains of the Greater Antilles
(236.3)	Mountains of the Sudan region, Guinea and the Cameroons	(237.2)	Mountains of Cuba
(236.31)	Fouta Djallon	(237.25)	Sierra Maestra
(236.32)	Adamawa (Adamaoua)	(237.4)	Sierra de los Organos
(236.33)	Darfur Plateau	(237.41)	Sierra del Trinidad
(236.5)	Mountains of north-eastern Africa	(237.411)	Mountains of Jamaica
(236.51)	Ethiopian Highlands	(237.413)	Blue Mountains
(236.52)	Somali Plateau	(237.415)	Cockpit country
(236.6)	Mountains of the Congo region	(237.42)	Mountains of the Dominican Republic
(236.61)	Lunda Plateau	(237.422)	Cordillera Septentrional
(236.62)	Katanga Plateau	(237.423)	Cordillera Central. □ Pico Duarte
(236.63)	Mitumba Mountains	(237.424)	Cordillera Oriental
(236.64)	Kibare Mountains	(237.424.1)	Sierra de Neiba
(236.65)	Virunga (Birunga) Mountains. □ Mount Karisimbi, Mount Miken, Mount Nyiragongo, Mount Nyamulagira	(237.424.2)	Sierra de Bahoruco
		(237.424.3)	Mountains of Puerto Rico
(236.7)	Mountains of eastern Africa	(237.424.4)	Cordillera Central. □ Cerro de Punta
(236.71)	Ruwenzori (Mountains of the Moon)	(237.424.5)	Mountains of the Lesser Antilles
(236.72)	Mount Kenya group	(237.425)	Soufrière Hills (Montserrat). □ Soufrière Hills volcano
(236.73)	Kilimanjaro group	(237.425.1)	Mountains of Dominica
(236.75)	Great Rift Valley	(237.5)	Pitons du Carbet (Martinique). □ La Pelée
(236.8)	Mountains of southern Africa. □ Stormberge. Kompasberg	(237.51)	
		(237.52)	
		(237.53)	

(237.54)	Mountains of Saint Lucia	(239.21)	Mountains of Sumatra. Barisan Chain
(237.56)	Mountains of Trinidad		(Pegunungan Barisan)
(238)	Mountains of South America	(239.3)	Mountains of New Zealand and Melanesia
(238.1)	South America, western area. South American	(239.4)	Mountains of Australia
	Cordillera. Andes	(239.41)	Mountains of eastern Australia. Great Dividing Range
(238.11)	Colombian Andes		Mountains of southern Australia. Flinders Ranges
(238.111)	Cordillera Occidental	(239.42)	Mountains of west Australia
(238.113)	Cordillera Central	(239.43)	West Australian tableland. Hamersley Range.
(238.113.2)	Sierra Nevada de Huila	(239.431)	Kenneth Range. Barlee Range. Waldberg Range. Robinson Ranges
(238.113.3)	Sierra Nevada de Tolima		Darling Range
(238.113.5)	Sierra Nevada de Ruiz	(239.432)	Mountains of central Australia. Macdonnell Ranges. Musgrave Ranges
(238.115)	Cordillera Oriental	(239.44)	Mountains of New Guinea. □ Bismark Range
(238.115.2)	Sierra Nevada de Cocuy		Mountains of Polynesia
(238.117)	Sierra Nevada de Santa Marta	(239.5)	Mountains of the Arctic
(238.117.1)	Pico Cristóbal Colón	(239.6)	Mountains of the Antarctic
(238.12)	Ecuador Andes	(239.8)	
(238.121)	Mount Cayambe	(239.9)	
(238.122)	Mount Pichincha	(24)	BELOW SEA LEVEL. UNDERGROUND. SUBTERRANEAN
(238.123)	Mount Antisana		<i>Specify depth by colon combination with (181)</i>
(238.124)	Mount Cotopaxi		◇ (24:181m200) 200 metres below sea level
(238.125)	Mount Chimborazo		⇒ 551.44
(238.13)	Peruvian-Bolivian Andes. Bolivian Plateau (Altiplano)	(25)	NATURAL FLAT GROUND (AT, ABOVE OR BELOW SEA LEVEL). THE GROUND IN ITS NATURAL CONDITION, CULTIVATED OR INHABITED
(238.131)	Cordillera Occidental		⇒ 551.43
(238.131.1)	Cordillera Blanca	(251)	Steppes. Pampas. Plains. Prairies. Savannas. Heathland
(238.131.12)	Mount Huascarán	(251.1)	Steppes proper
(238.131.2)	Mount Yerupaja	(251.2)	Pampas
(238.131.3)	Mount Coropuna	(251.3)	Prairies
(238.131.5)	Mount El Misti	(251.4)	Savannas
(238.131.7)	Mount Sajama	(251.5)	Landes (steppe-like regions in France)
(238.133)	Cordillera Central	(251.9)	Heathland
(238.135)	Cordillera Oriental	(252)	Unproductive land
(238.135.3)	Mount Auzangate	(252.32)	Stony land
(238.136)	Cordillera Real	(252.33)	Sandy land
(238.136.3)	Mount Illimarni	(252.331)	Dunes. Drifting sand
(238.1361)	Mount Ancohuma		⇒ (212)
(238.14)	Chilean-Argentinian Andes		⇒ (213)
(238.141)	Nevados de Cachi	(252.47)	Estuarine land
(238.142)	Mount Llullaillaco		⇒ (282.05)
(238.143)	Nevado Incahuasi		⇒ (285.3)
(238.144)	Nevado Ojos de Salado	(252.6)	Marshland. Peat bogs. Other freshwater formations
(238.145)	Cerro Bonete	(252.8)	Lava-covered areas
(238.146)	Cerro del Toro	(253)	Virgin (primeval) woods and forests. Jungle
(238.147)	Cerro Mercidario	(254)	Arable land, cleared or cultivated
(238.148)	Ceagua	(255)	Flooded land. Irrigated land. Drained land. □ Broads. Polders
(238.149)	Cerro Tupungato	(256)	Fenced, enclosed land
(238.15)	Mountains of Tierra del Fuego	(257)	Inhabited land. Built-up areas
(238.16)	Patagonian Cordillera	(258)	Fortified country
(238.2)	South America, northern area	(26)	OCEANS, SEAS AND INTERCONNECTIONS
(238.21)	Cordillera de la Costa (Venezuelan Coastal Range)		⇒ (285)
(238.212)	Cordillera la Merida		⇒ 551.46
(238.214)	Cordillera de la Costa (Venezuelan Coastal Range/Central Highlands)	(26.02)	Open sea. High seas
(238.22)	Guiana Highlands	(26.03)	Abyssal zone. Depths of the sea. Submarine...
(238.221)	Pacaraima Mountains	(26.04)	Gulfs. Bays. Bights. Creeks. Natural harbours.
(238.221.1)	Mount Roraima		Roadsteads
(238.221.3)	Pico da Neblina	(26.05)	Lagoons. Haffs. Salt ponds
(238.223)	Tumuc-Humac Mountains		
(238.3)	South America, Eastern area		
(238.31)	Brazilian Highlands		
(238.311)	Serra Geral de Goiás		
(238.312)	Chapada Diamantina		
(238.313)	Serra do Espinhaço		
(238.314)	Pico do Bandejas		
(238.315)	Serra de Mantiqueira		
(238.316)	Serra do Mar		
(239)	Mountains of Oceania, the Arctic, the Antarctic		
(239.1)	Mountains of the Malay Archipelago		
(239.2)	Mountains of the Sunda Islands		

(1/9) Table 1e. Common auxiliaries of place

(26.06)	Ocean and sea interconnections. Sea canals ⇒ (282.5) ⇒ 626.9	(261.68)	Seas and waters of the south-east coast of South America
(26.07)	Ocean and sea currents	(261.683)	Rio de la Plata
(261)	Atlantic Ocean	(261.684)	Bahia Blanca
(261.1)	North Atlantic Ocean ◇ (261.1.07) Gulf Stream	(261.685)	Golfo San Matías
(261.2)	North-east Atlantic ⇒ (268.4)	(261.686)	Golfo San Jorge
(261.24)	Baltic Sea. □ Gulf of Bothnia. Gulf of Finland. Gulf of Riga. Gulf of Gdansk	(261.687)	Bahia Grande ⇒ (269.528)
(261.247)	Belts and Sund (The Sound. Oresund)	(261.7)	East and south-east Atlantic
(261.248)	Kattegat	(261.74)	Seas and waters of the coast of west Africa ◇ (261.74.07) Guinea current
(261.249)	Kiel Canal (Baltic/North Sea)	(261.76)	Gulf of Guinea
(261.26)	North Sea	(261.77)	Seas and waters of the west coast of Equatorial Africa
(261.262)	German Bight. Friesian Island waters	(261.78)	Seas and waters of South-West Africa and the west coast of South Africa ◇ (261.78.07) Benguela current
(261.263)	Skagerrak	(262)	Mediterranean Sea
(261.265)	Dogger Bank	(262.1)	Western Mediterranean
(261.266)	Waters of the east coast of Britain	(262.11)	Seas and waters from Gibraltar to Corsica
(261.267)	Strait of Dover	(262.112)	Strait of Gibraltar
(261.268)	English Channel	(262.114)	Iberian Sea (Balearic Sea)
(261.27)	Irish Sea. Western waters of British Isles	(262.115)	Golfe du Lion (Gulf of Lions)
(261.273)	St George's Channel and adjacent Welsh and Irish coastal waters	(262.116)	Ligurian Sea
(261.274)	Irish Sea (central) and adjacent coastal waters	(262.13)	Tyrrhenian Sea
(261.275)	North Channel, with Firth of Clyde	(262.138)	Strait of Messina
(261.276)	Malin Sea	(262.16)	Seas and waters of the Algerian and north Tunisian coast
(261.28)	Bay of Biscay and adjacent French and Spanish coastal waters	(262.2)	Eastern Mediterranean in general
(261.29)	Waters of North west Africa ◇ (261.29.07) Canary Current	(262.22)	Ionian Sea. □ Gulf of Taranto
(261.4)	North-west Atlantic. Davis Strait	(262.26)	Seas and waters of east Tunisian and Libyan coasts
(261.43)	Sea of Labrador ◇ (261.43.07) Labrador current	(262.27)	Seas and waters of the Middle Eastern coast
(261.44)	Gulf of St Lawrence and adjacent coastal waters of Newfoundland and south-eastern Nova Scotia. □ Grand Banks	(262.3)	Adriatic Sea
(261.45)	Gulf of Maine	(262.32)	Gulf of Venice
(261.452)	Bay of Fundy	(262.4)	Aegean Sea
(261.46)	Waters of the east coast of the USA from Cape Cod to Chesapeake Bay. □ Long Island Sound. Nantucket Sound. Rhode Island Sound. Buzzards Bay. Delaware Bay. Narragansett Bay. New York Bay	(262.42)	Sea of Crete
(261.47)	Chesapeake Bay	(262.429)	Corinth Canal
(261.48)	Waters of the south-east coast of the USA from Chesapeake Bay to Florida Strait. □ Albermarle Sound. Pamlico Sound. Everglades	(262.5)	Black Sea
(261.5)	West and south-west North Atlantic ◇ (261.5.03) Puerto Rico Trench	(262.53)	Sea of Marmara
(261.52)	Sargasso Sea	(262.536)	Dardanelles
(261.53)	Bahama Sea	(262.538)	Bosporus
(261.538)	Straits of Florida	(262.54)	Sea of Azov
(261.54)	Gulf of Mexico	(262.8)	Inner seas of Eurasia
(261.545)	Gulf of Campeche	(262.81)	Caspian Sea
(261.548)	Yucatan Channel	(262.83)	Aral Sea
(261.55)	Caribbean Sea ◇ (261.55.03) Cayman Trench	(265)	Pacific Ocean
(261.552)	Gulf of Honduras	(265.1)	East and south-east Pacific ◇ (265.1.07) Peruvian (Humbolt) current
(261.554)	Gulf of Mosquitos	(265.13)	Gulf of Guayaquil
(261.555)	Gulf of Darien	(265.15)	Gulf of Panama
(261.556)	Gulf of Venezuela	(265.159)	Panama Canal
(261.56)	Seas and waters of the north-east coast of South America	(265.16)	Central American and south-west Mexican Pacific waters
(261.565)	Gulf of the Amazon	(265.17)	Gulf of California
(261.6)	South Atlantic Ocean ⇒ (269.4)	(265.2)	North and north-east Pacific
(261.67)	Seas and waters of the coast of Brazil ◇ (261.67.07) Brazil current	(265.22)	Seas and waters of coast of California and West United States of America ◇ (265.22.07) California current
		(265.222)	Puget Sound
		(265.225)	Juan de Fuca Strait
		(265.23)	West Canadian and south-east Alaskan coastal seas and waters
		(265.233)	Strait of Georgia
		(265.235)	Queen Charlotte Sound and Strait
		(265.236)	Inside Passage
		(265.238)	Hecate Strait
		(265.239)	Dixon Entrance
		(265.24)	Gulf of Alaska ◇ (265.24.07) Alaska current
		(265.242)	Cook Inlet

(265.244)	Bristol Bay	(267.335)	Gulf of Suez
(265.246)	Norton Sound	(267.337)	Gulf of Aqaba
(265.5)	West and north-west Pacific	(267.339)	Suez Canal
(265.51)	Bering Sea	(267.35)	The Gulf (Persian Gulf). Strait of Hormuz
(265.518)	Bering Strait	(267.36)	Gulf of Oman
(265.52)	Seas and waters off the south-east coast of Kamchatka	(267.37)	Arabian Sea
(265.53)	Sea of Okhotsk	(267.38)	Laccadive Sea
(265.538)	La Perouse (Soya) Strait	(267.6)	North-eastern Indian Ocean
(265.54)	Sea of Japan	(267.64)	Bay of Bengal
(265.546)	Tatar (Tartary) Strait	(267.65)	Andaman Sea
(265.547)	Tsugaru Strait	(267.654)	Gulf of Martaban
(265.548)	Korea Strait	(267.657)	Malacca Strait
(265.55)	East coast and inner seas of the Japanese islands	(267.67)	Seas and waters of south-west Sumatra
	◇ (265.55.03) Japan Trench	(267.7)	◇ (267.67.03) Java Trench
	◇ (265.55.07) Kuroshio (Kuro Siwo) current	(267.74)	East and south-east Indian Ocean
(265.555)	Inland Sea (Seto Naikai)	(267.748)	North-west Australian coastal seas and waters:
(265.56)	Yellow Sea	(267.75)	Melville Island to Northwest Cape
(265.57)	East China Sea		Timor Sea
(265.578)	Formosa Strait (Taiwan Strait)		West Australian coastal seas and waters:
(265.58)	Philippine Sea	(267.76)	Northwest Cape to Cape Leeuwin
	◇ (265.58.03) Mindanao Trench		◇ (267.75.07) West Australian current
(265.7)	South and south-west Pacific	(267.761)	South Australian coastal seas and waters:
	◇ (265.7.03) Mariana Trench		Cape Leeuwin to Cape Howe
(265.72)	South China Sea	(268)	Great Australian Bight. Bass Strait
(265.723)	Gulf of Thailand	(268.4)	Arctic Ocean
(265.724)	Malay Sea		European Arctic Ocean. □ Greenland Sea.
(265.727)	Singapore Strait	(268.45)	Denmark Strait. Norwegian Sea
(265.73)	Seas of the Malayan Archipelago in general	(268.46)	Barents Sea
(265.732)	Sulu Sea	(268.5)	White Sea (Beloye More)
(265.734)	Sulawesi Sea (Celebes Sea)	(268.52)	Asian Arctic Ocean
(265.735)	Molucca Sea	(268.53)	Kara Sea
(265.736)	Seram (Ceram) Sea	(268.55)	Laptev Sea
(265.738)	Makassar Strait	(268.56)	East Siberian Sea
(265.74)	Seas along the Sunda Islands	(268.7)	Chukchi Sea
(265.742)	Java Sea	(268.72)	American Arctic Ocean
(265.743)	Bali Sea. Bali Strait	(268.74)	Beaufort Sea
(265.744)	Flores Sea		Seas of the Canadian Archipelago: Viscount
(265.745)	Savu Sea	(268.75)	Melville Sound, Lancaster Sound, McClintock
(265.746)	Banda Sea	(268.77)	Channel, Gulf of Boothia
(265.747)	Arafura Sea. Gulf of Carpentaria	(268.78)	Hudson Bay. Hudson Strait
(265.76)	Coral Sea and adjoining seas of Melanesia.	(268.9)	Baffin Bay
	Solomon Sea. Torres Strait. Bismark Sea		Lincoln Sea
(265.762)	East of the East Queensland coast. □ Great Barrier Reef	(269)	Arctic Basin
(265.77)	Seas around Fiji	(269.4)	Southern Ocean (Antarctic Ocean)
	◇ (265.77.03) Tonga Trench. Kermadec Trench	(269.42)	Atlantic sector of Southern Ocean. Scotia Sea.
(265.78)	Tasman Sea	(269.46)	Lazarev Sea
	◇ (265.78.07) Australia current	(269.5)	Drake's Passage (Drake's Strait)
(265.782)	Seas, gulfs and bays of New South Wales coast	(269.52)	Weddell Sea
(265.788)	Cook Strait	(269.528)	Pacific sector of Southern Ocean.
(265.8)	Central Southern Pacific. Seas and waters around French Polynesia	(269.54)	Bellingshausen Sea
		(269.56)	Inner seas of Chilean archipelago
(267)	Indian Ocean	(269.7)	Strait of Magellan
(267.2)	West and south-west Indian Ocean		Amundsen Sea
(267.23)	East Madagascar coastal seas and waters		Ross Sea
(267.24)	South and south-east African seas and waters:		Indian Ocean sector of Southern Ocean
	Cape of Good Hope to Delagoa Bay	(28)	INLAND WATERS
	◇ (267.24.07) Mozambique current	(282)	Flowing waters. Watercourses.
(267.25)	East African seas and waters: Delagoa Bay to Cape Delgado. Mozambique Channel		Waterways. Rivers
(267.26)	Eastern African coastal waters, of Tanzania, Kenya, Somalia: Cape Delgado to Cape Guardafui (Ras Asir)	(282.03)	⇒ 556.5
	North-western Indian Ocean	(282.04)	Source or sources of river etc.
(267.3)	Gulf of Aden	(282.042)	Course of river
(267.32)	Strait of Bab el Mandeb	(282.043)	Upper course of river
(267.328)	Red Sea	(282.044)	Middle course of river
(267.33)		(282.05)	Lower course of river
			River mouths. Outflows. Course and arms of outflow. Deltas. Estuaries
		(282.2)	River basins and systems. Rivers with their sources, tributaries and outflows
		(282.24)	European rivers

(1/9) Table 1e. Common auxiliaries of place

(282.242)	Rivers of the British Isles	(282.244.31)	Rhône
(282.242.1)	Rivers of Scotland. □ Moray. Tay. Forth. Solway. Dee. Spey	(282.244.4)	Lake Geneva
(282.242.11)	Clyde	(282.245)	Rivers of Italy
(282.242.2)	Rivers of Ireland. □ Foyle. Bann. Boyne. Liffey. Slaney. Barrow. Blackwater. Lee	(282.245.1)	Rivers of northern Italy
(282.242.21)	Shannon	(282.245.111)	Lake Maggiore
(282.242.3)	Rivers of northern England and north Wales. □ Yorkshire Ouse. Dee. Ribble	(282.245.112)	Lake Como
(282.242.31)	Tyne	(282.245.113)	Lake Garda
(282.242.32)	Tees	(282.245.12)	Adige
(282.242.33)	Humber	(282.245.13)	Po
(282.242.34)	Trent	(282.245.2)	Rivers of central Italy
(282.242.35)	Mersey	(282.245.21)	Arno
(282.242.4)	Rivers of southern England and south Wales. □ Norfolk Ouse. Yare. Stour. Medway. Arun. Exe. Wye. Tawe. Taf	(282.245.22)	Tevere (Tiber)
(282.242.41)	Thames	(282.245.3)	Rivers of southern Italy
(282.242.42)	Severn	(282.245.4)	Rivers of Sicily
(282.242.43)	Avon	(282.245.5)	Rivers of Corsica and Sardinia
(282.243)	Rivers of central Europe	(282.246)	Rivers of the Iberian peninsula
(282.243.1)	Rhine	(282.246.1)	Iberian rivers flowing into the Mediterranean
◇ (282.243.1.03)	Sources of the Rhine	(282.246.11)	Ebro
◇ (282.243.1.042)	Upper Rhine (Lake Constance to Bingen). Including: Lake Constance (Bodensee). Lake Lucerne	(282.246.2)	Iberian rivers flowing into the Atlantic
◇ (282.243.1.043)	Middle Rhine (Bingen to Godesberg)	(282.246.21)	Guadalquivir
◇ (282.243.1.044)	Lower Rhine (Godesberg to the estuary)	(282.246.23)	Tagus (Tejo/Tajo)
(282.243.163)	Ruhr	(282.246.24)	Douro (Duro)
(282.243.17)	Rivers of the Rhine estuary	(282.247)	Rivers of eastern Europe and the Caucasus
(282.243.171)	IJssel. □ IJsselmeer (Zuider Zee)	(282.247.1)	Rivers of eastern Europe flowing into the Barents Sea and White Sea (Beloye More)
(282.243.2)	Rivers of north west Central Europe	(282.247.11)	Pechora
(282.243.21)	Ems	(282.247.12)	Mezen'
(282.243.22)	Weser	(282.247.13)	Severnaya Dvina (Northern Dvina)
(282.243.3)	Elbe	(282.247.14)	Onega
(282.243.31)	Vltava (Moldau)	(282.247.18)	Rivers of the Kola Peninsula
(282.243.35)	Saale	(282.247.2)	Rivers of eastern Europe flowing into the Baltic
(282.243.36)	Havel	(282.247.21)	Neva
(282.243.4/.6)	Rivers flowing into the Baltic	(282.247.211)	Onezhskoye Ozero (Lake Onega)
(282.243.4)	Rivers flowing into the Baltic between Lübeck and Gdańsk	(282.247.212)	Ladozhskoye Ozero (Lake Ladoga)
(282.243.41)	Trave	(282.247.216)	Ozero Il'men' (Lake Ilmen)
(282.243.5)	Oder	(282.247.224)	Narva
(282.243.55)	Lausitzer (Görlitzer) Neisse (Nysa)	(282.247.23)	Zapadnaya Dvina (Western Dvina)
(282.243.56)	Warta	(282.247.28)	Neman (Memel)
(282.243.6)	Rivers of north-east central Europe	(282.247.281)	Berezina
(282.243.61)	Vistula (Wisla/Weichsel)	(282.247.3)	Rivers of eastern Europe and the Caucasus flowing into the Black Sea and the Sea of Azov ⇒ (282.243.7)
(282.243.7)	Danube	(282.247.314)	Dnestr (Dniester)
◇ (282.243.7.03)	Sources of the Danube	(282.247.318)	Yuzhnyy Bug (Southern Bug)
◇ (282.243.7.042)	Upper Danube (Regensburg to Passau)	(282.247.32)	Dnepr (Dnieper)
◇ (282.243.7.043)	Middle Danube (Passau to the Iron Gate)	(282.247.322)	Pripyat' (Pripet)
◇ (282.243.7.044)	Lower Danube	(282.247.324)	Desna
◇ (282.243.7.05)	Danube Delta	(282.247.34)	Rivers of the Crimea
(282.244)	Rivers of western mainland Europe	(282.247.36)	Don
(282.244.1)	Rivers of western mainland Europe flowing into the North Sea or English Channel	(282.247.362)	Khoper
(282.244.11)	Meuse (Maas)	(282.247.364)	Severskiy Donets
(282.244.12)	Schelde	(282.247.366)	Manych
(282.244.13)	Somme	(282.247.38)	Kuban (Hypanis)
(282.244.14)	Seine	(282.247.4)	Rivers of Europe and the Caucasus flowing into the Caspian Sea ⇒ (282.255)
(282.244.2)	Rivers of western mainland Europe flowing into the Atlantic	(282.247.41)	Volga
(282.244.21)	Loire	(282.247.412)	Oka
(282.244.22)	Dordogne	(282.247.412.2)	Moskva
(282.244.23)	Garonne	(282.247.412.4)	Moksha
(282.244.3)	Rivers of western mainland Europe flowing into the Mediterranean	(282.247.413.5)	Rybinskoye Vodokhranilishche (Rybinsk Reservoir)
		(282.247.415)	Kama
		(282.247.42)	Ural
		(282.247.441)	Kura
		(282.247.444)	Terek
		(282.247.445)	Kuma
		(282.248)	Rivers of northern Europe
		(282.248.1)	Rivers of Norway
		(282.248.2)	Rivers of Sweden
		(282.248.3)	Rivers of Finland

(282.249)	Rivers of the Balkan Peninsula	(282.262.5)	Ogooué
(282.249.1)	Rivers of the north-west Balkan Peninsula. □ Neretva. Drin. Semeni	(282.262.6)	Niger
(282.249.3)	Rivers of Greece	(282.262.7)	Senegal
(282.25)	Rivers of Asia	(282.262.8)	Volta
(282.252)	Rivers of eastern Asia	(282.262.9)	Gambia
(282.252.2)	Huang Ho (Hwang Ho/Yellow River)	(282.263)	African rivers flowing into the Mediterranean
(282.252.3)	Ch'ang Chiang (Yangtze Kiang)	(282.263.1)	Nile
(282.252.4)	Rivers of Korea	(282.263.11)	Blue Nile
(282.252.41)	Yalou	(282.263.12)	White Nile
(282.253)	Rivers of southern Asia	(282.263.13)	Atbara
(282.253.1)	Rivers of Southeast Asia	(282.263.5)	Inland lakes of Africa. □ Lake Tana. Lake Turkana (Rudolf). Lake Victoria (Victoria Nyanza). Lake Malawi (Nyasa). Lake Tanganyika
(282.253.13)	Salween	(282.264)	Rivers of the African interior
(282.253.14)	Irrawaddy. □ Chindwin	(282.264.1)	Chari. Lake Chad
(282.253.15)	Chao Praya. □ Menam	(282.27)	Rivers of North America
(282.253.16)	Mekong	(282.271)	Rivers of North America flowing into the Pacific
(282.253.17)	Red River (Song Hong)	(282.271.1)	Fraser
(282.253.2)	Rivers of the Indian subcontinent	(282.271.2)	Columbia
(282.253.21)	Ganges system. □ Jumna. Hooghly	(282.271.3)	Sacramento
(282.253.22)	Brahmaputra system. □ Tsangpo, Dihang	(282.271.4)	Colorado
(282.253.24)	Indus system. □ Jhelum, Chenab, Ravi, Beas, Sutlej	(282.271.6)	Great Salt Lake
(282.253.27)	Rivers of Sri Lanka	(282.272)	Rivers of North America flowing into the Atlantic
(282.254)	Rivers of south-west Asia	(282.272.1)	St Lawrence
(282.254.1)	Rivers of Turkey	(282.272.11)	Ottawa
(282.254.11)	Menderes (Meander)	(282.272.115)	Great Lakes: Lake Superior, Lake Michigan, Lake Huron, Lake Erie, Lake Ontario
(282.254.2)	Rivers of the Levant	(282.272.117)	Niagara Falls
(282.254.21)	Orontes (Asi)	(282.272.2)	Connecticut
(282.254.23)	Jordan	(282.272.3)	Hudson
(282.254.231)	Sea of Galilee (Lake Tiberias)	(282.272.4)	Delaware
(282.254.233)	Dead Sea	(282.272.5)	Susquehanna
(282.254.3)	Rivers of Mesopotamia	(282.272.6)	Potomac
(282.254.31)	Euphrates (Furat)	(282.272.7)	Mississippi
(282.254.32)	Tigris (Dijlah)	(282.272.71)	Illinois
(282.254.5)	Rivers of the Arabian peninsula	(282.272.72)	Ohio
(282.254.6)	Rivers of the Iranian plateau	(282.272.73)	Missouri
(282.255)	Rivers of the inland drainage basins of middle and central Asia flowing into the Aral or Caspian Seas ⇒ (282.247.4)	(282.272.74)	Arkansas
(282.255.1)	Amu Darya (Oxus)	(282.272.75)	Red River
(282.255.2)	Syr Darya (Jaxartes)	(282.272.8)	Río Bravo del Norte (Río Grande). Río Conchos (Chihuahua)
(282.255.32)	Emba	(282.272.81/.82)	Rivers and inland waters of Mexico
(282.255.5)	Ozero Balkhash and its basin	(282.272.811)	Laguna de Chapala
(282.256)	Rivers of northern Asia flowing into the Arctic Ocean	(282.272.812)	Río de Conchos (Tamaulipas). □ Laguna Madre
(282.256.1)	Ob'	(282.272.813)	Río Pánuco
(282.256.16)	Irtys	(282.272.814)	Laguna de Tamiahua
(282.256.3)	Yenisei (Ulug-Chem)	(282.272.815)	Río de las Balsas
(282.256.341)	Ozero Baykal (Lake Baikal)	(282.272.816)	Río Papaloapan
(282.256.6)	Lena	(282.272.817)	Río Grijalva
(282.256.7)	Rivers flowing into the Laptev Sea east of the Lena	(282.272.818)	Río Usumacinta (Río Salinas)
(282.256.86)	Kolyma	(282.272.819)	Río Candelaria. □ Laguna de Terminos
(282.257.1)	Rivers flowing into the Bering Sea	(282.272.821)	Lake Bacalar
(282.257.2)	Rivers of Kamchatka flowing into the Pacific	(282.272.9)	Rivers of Guatemala
(282.257.3)	Rivers of the Kuriles	(282.272.91)	Río Polochic. Río Dulce. □ Lago de Izabál
(282.257.4)	Mainland rivers flowing into the Sea of Okhotsk	(282.272.92)	Río Motagua
(282.257.5)	Amur (Hilung-kiang)	(282.272.93)	Rivers of Belize
(282.257.58)	Ussuri (Wu-su-li Chiang)	(282.272.931)	Río Hondo
(282.257.6)	Rivers of Sakhalin	(282.272.933)	Belize River (Río Mopán)
(282.26)	Rivers of Africa	(282.272.94)	Rivers and inland waters of Honduras
(282.261)	Rivers flowing into the Indian Ocean	(282.272.941)	Río Ulúa. Lago de Yojoa
(282.261.3)	Zambezi	(282.272.943)	Río Aguán
(282.261.31)	Victoria Falls	(282.272.945)	Río Patuca
(282.261.32)	Lake Kariba	(282.272.946)	Laguna de Caratasca
(282.261.4)	Limpopo	(282.272.947)	Río Coco
(282.262)	African rivers flowing into the Atlantic	(282.272.95)	Rivers of El Salvador
(282.262.1)	Orange	(282.272.951)	Río Lempa (drains into Pacific). Lago de Guija. Embalse 15 de Septiembre 44. Embalse Cerrón Grande
(282.262.2)	Cunene (Kunene)		
(282.262.4)	Congo (Zaire)		

(1/9) Table 1e. Common auxiliaries of place

(282.272.953)	Río Jiboa. Lago de Ilopango	(282.283.1)	Lago Titicaca and associated waters
(282.272.96)	Rivers of Nicaragua, Costa Rica and Panamá	(282.283.11)	Lago Titicaca
(282.272.961)	Río Grande de Matagalpa. Río Tuma	(282.283.12)	Río Desaguadero
(282.272.962)	Río Escondido. Río Siquia. Río Mico. Río Rama	(282.283.13)	Lago de Poopó
(282.272.963)	Río San Juan. Lago de Nicaragua. Lago de Managua. Río San Carlos. Laguna de Arenal	(282.283.2)	Maritime lakes of Brazil
(282.272.964)	Río General	(282.283.21)	Lagoa dos Patos
(282.272.965)	Río Santa María	(282.283.22)	São Gonçales Canal
(282.272.966)	Río Chagres. Madden Lake ⇒ (265.159)	(282.283.23)	Lago Mirim
(282.272.967)	Río Chepo	(282.283.3)	Lago Mar Chiquita
(282.272.968)	Río Chucunaque. Río Tuirá	(282.283.4)	Lakes of the Chilean-Argentine Andes
(282.272.969)	Río Balsas	(282.283.41)	Lago Buenos Aires
(282.272.97)	Rivers of the Greater Antilles	(282.283.42)	Lago San Martín
(282.272.972)	Rivers of Jamaica. □ Black River, Río Grande	(282.283.44)	Lago Argentino
(282.273)	Rivers of North America flowing into the Arctic Ocean	(282.29)	Rivers of Oceania
(282.273.1)	Yukon	(282.293)	Rivers of Melanesia
(282.273.12)	Klondike River	(282.293.1)	Rivers of New Zealand
(282.273.2)	Mackenzie	(282.293.11)	Lakes of New Zealand. □ Lake Taupo. Lake Wakatipu
(282.273.21)	Peace River	(282.294)	Rivers of Australia
(282.273.22)	Great Slave Lake. Great Bear Lake	(282.294.1)	Murray
(282.273.222)	Athabasca River	(282.294.11)	Darling
(282.274)	Rivers of North America flowing into the Hudson Bay	(282.294.12)	Lachlan
(282.274.1)	Churchill	(282.294.13)	Murrumbidgee
(282.274.2)	Nelson-Saskatchewan system	(282.294.3)	Swan
(282.274.25)	Lake Winnipeg	(282.295)	Rivers of New Guinea
(282.274.3)	Albany	(282.296)	Rivers of Polynesia
(282.28)	Rivers and inland waters of South America	(282.3)	Waterfalls. Rapids. Cataracts. Cascades
(282.281)	South American rivers flowing into the Atlantic	(282.5)	Canals. Artificial waterways ⇒ (26.06) ⇒ 626.1 ⇒ 656.628
(282.281.1)	Rivers of northern South America	(285)	Still and stagnant waters. Lakes. Ponds. Swamps
(282.281.11)	Magdalena	(285.2)	Lakes other than those associated with river systems ⇒ 556.55
(282.281.112)	Cauca	(285.3)	Ponds. Swamps. Marshes. Fens ⇒ 556.56
(282.281.12)	Sinu	(285.32)	Saltwater swamps. Salt ponds. Mangrove swamps
(282.281.121)	Lago Ciénaga Grande	(285.33)	Freshwater swamps and ponds
(282.281.13)	Lago de Maracaibo	(289)	Brackish waters
(282.281.2)	Rivers of northeastern South America	(29)	THE WORLD ACCORDING TO PHYSIOGRAPHIC FEATURES
(282.281.21)	Orinoco		<i>This alternative arrangement makes provision for users wishing to organize the regions of the world by physiographic features in accordance with their position on the map, rather than either by a specific physiographic feature which is provided for by the numbers at (2/289) or by their continental or political designation, as provided for in (4/9).</i>
(282.281.22)	Angel Falls		<i>Just as (3) designates the areas of the world according to a Western view of ancient history, and provision is also made for collocating areas, irrespective of date, in (4/9), a similar provision is made here for those who view the world from the physical geographic point of view.</i>
(282.281.23)	Potaro River		<i>The same concepts may be expressed, if preferred, through the use of the remainder of the table together with the special auxiliary number -92 'from the physiographic point of view' (see (1-92))</i>
(282.281.232)	Kaieteur Falls		<i>If required, the areas listed here may be expanded in the usual manner through colon combination with the remaining numbers at (2) and those at (4/9).</i>
(282.281.24)	Demerara River		
(282.281.25)	Berbice River		
(282.281.26)	Essequibo River		
(282.281.27)	Corantijn River		
(282.281.28)	Suriname River		
(282.281.282)	Van Blommestein Meer		
(282.281.3)	Amazon (Amazonas)		
(282.281.31)	Río Negro (Brazil)		
(282.281.32)	Marañón-Ucayali. Solimoes		
(282.281.33)	Madeira		
(282.281.34)	Tapajos		
(282.281.35)	Xingu		
(282.281.4)	Tocantins		
(282.281.5)	São Francisco		
(282.281.6)	La Plata		
(282.281.61)	Uruguay		
(282.281.62)	Parana		
(282.281.621)	Paraguay		
(282.281.7)	Río Negro (Argentina)		
(282.281.8)	Río Chubut. Río Chico		
(282.281.81)	Lago Colhué Huape. Lago Musters		
(282.282)	South American rivers flowing into the Pacific		
(282.282.3)	Chilean lakes draining into the Pacific		
(282.282.31)	Lago Panguipulli		
(282.282.32)	Lago Ranco		
(282.282.33)	Lago de Puyehé		
(282.282.34)	Lago Llanquihue		
(282.283)	Lakes and inland waters of southern South America		

<i>If this alternative table is used the -92 auxiliary (see (1-92)) must not be used, and no other number in (2) may be used, except if coloned to a number listed under (29), and then only for the purpose of further specification</i>		(292.457)	Stara Planina (Balkan mountains)
		(292.46)	⇒ (234.42)
		(292.461)	Mediterranean Southern Europe
			Iberian Peninsula with Balearic Islands
			⇒ (46)
			⇒ (469)
		(292.462)	Central Mediterranean islands
		(292.463)	Appenine Peninsula
			⇒ (234.41)
			⇒ (450)
		(292.464)	Balkan Peninsula
			⇒ (234.42)
		(292.471)	Crimea
			⇒ (477.75)
		(292.472/.476)	Caucasus
			⇒ (234.9)
			⇒ (479)
		(292.472)	Precaucasian plains
		(292.473)	Greater Caucasus (Bol'shoy Kavkaz)
		(292.474)	Kolkhida (Rion) Lowlands
		(292.475)	Kura Depression
		(292.476)	Lesser Caucasus (Malyy Kavkaz)
		(292.48)	Russian Plain (East European Plain)
		(292.481)	Tundra and Forest Tundra region
		(292.482/.484)	Forest Zone (of the East European Plain)
		(292.482)	Northern and Central Tayga sub-region
		(292.483)	Central and Eastern Russian plain
		(292.484)	Western Russian Plain
		(292.485/.486)	Southern Russian Plain
		(292.485)	Forest-Steppe region
		(292.486)	Steppe region
		(292.49)	Ural region
			⇒ (234.85)
			⇒ (574.11)
		(292.491)	Tundra and forest tundra of Pay-Khoy Range and Polar Ural
			⇒ (234.851)
		(292.493)	Taiga of Subpolar and Northern Ural
		(292.495)	Forests of Middle Ural
		(292.497)	Forest steppe and steppe of Southern Ural
			⇒ (234.853)
		(292.498)	Semi-desert of Mugodzhary
		(292.5)	Physical regions of Asia. Asia as a physical region
			⇒ (5)
		(292.51)	Northern Asia
		(292.511/.516)	Siberia
			⇒ (571.1/.5)
		(292.511)	West Siberian Plain
		(292.511.1)	Tundra and forest tundra of West Siberia (Yamal-Gydan region)
		(292.511.2)	Taiga and swamp zone of West Siberia
		(292.511.6)	Forest steppe of West Siberia
		(292.511.7)	Steppe of West Siberia
		(292.512)	Middle Siberia
		(292.512.1)	Northern Middle Siberia (Taimyr Peninsula and North Siberian Lowland)
		(292.512.2/.3)	Middle Siberian Uplands
		(292.512.2)	Northern part of Middle Siberian Uplands
		(292.512.3)	Central Siberia (Southern part of Middle Siberian Uplands)
		(292.513)	Lena basin. Central Yakutia (Yakutia basin)
			⇒ (282.256.6)
		(292.514)	North-eastern Siberia
		(292.515)	Altai and Sayan Mountains
			⇒ (571.15)
		(292.516)	Baikal region and Transbaikalia
		(292.517)	Amur and Primor'ye Region
		(292.518)	Kamchatka. Kuriles
			⇒ (571.66)
		(292.52)	Mountainous Middle Asia
(292.2)	Arctic regions		
	⇒ (98)		
(292.21)	European Arctic islands		
(292.22)	Asian Arctic islands		
(292.23/.24)	Western Arctic islands (American Sector)		
	⇒ (987)		
(292.23)	Canadian Arctic islands. Baffin Island		
(292.24)	Greenland		
	⇒ (988)		
(292.3)	Antarctic regions. Antarctica		
	⇒ (99)		
(292.31)	Central Antarctic plateau		
(292.32)	Coastal regions of Eastern Antarctic		
(292.33)	West Antarctica. Lesser Antarctica		
(292.34)	Subantarctic islands		
(292.4/.5)	Physical regions of Eurasia		
(292.4)	Physical regions of Europe. Europe as a physical region		
	⇒ (4)		
(292.41)	Northern Europe		
(292.411)	Iceland		
	⇒ (491.1)		
(292.412)	Jan Mayen		
	⇒ (984)		
(292.413)	Faeroes		
	⇒ (491.2)		
(292.414)	Scandinavian Peninsula		
	⇒ (48)		
(292.415)	Finland and Karelia		
	⇒ (480)		
(292.416)	Kola Peninsula		
(292.42/.43)	Western and Northern Central Europe		
(292.421/.422)	British Isles		
	⇒ (410)		
(292.421)	Island of Ireland		
	⇒ (410.7)		
	⇒ (415)		
	⇒ (417)		
(292.422)	Great Britain and minor islands		
(292.423)	Atlantic France		
(292.43)	Northern Central Europe		
(292.431)	Northern Central European lowlands		
(292.432)	Hercynian Central Europe		
(292.44/.45)	Alps and Carpathian regions		
	⇒ (234.3)		
(292.441)	Jura		
	⇒ (234.313.2)		
(292.442)	Northern Alpine foreland		
(292.443)	Alps		
	⇒ (234.3)		
(292.444)	Lombardic Lowland (Venetian and Po plains)		
(292.45)	South-eastern Central Europe		
(292.451/.454)	Carpathians		
(292.451)	Western Carpathians		
	⇒ (234.372.3)		
(292.452)	Eastern Carpathians		
	⇒ (234.421.1)		
(292.453)	Southern Carpathians		
(292.454)	West Rumanian (Romanian) mountains.		
	Transylvania. Transylvanian Alps		
	⇒ (234.421)		
(292.455)	Pannonian Basin		
(292.456)	Lower Danubian plains. Walachia. Dobruja		

(1/9) Table 1e. Common auxiliaries of place

(292.524)	Kazakhstan-Dzungarian mountainous region ⇒ (574)	(292.63/.69)	Sub-Saharan Africa ⇒ (66/67)
(292.525)	Western Tien Shan ⇒ (235.216)	(292.63)	Region bordering the south of the Sahara. □ Sahel. Sudan (region)
(292.526)	Gissar-Alay mountainous region ⇒ (235.214)	(292.64/.65)	Equatorial Africa
(292.527)	Pamir high-mountainous region ⇒ (235.211) ⇒ (575.32)	(292.64)	Upper Guinea
(292.53)	Central Asia ⇒ (235.2)	(292.65)	Congo Basin and surrounding area. Lunda-Katanga. Gulf of Guinea islands. St Helena. Ascension ⇒ (282.262.4)
(292.535)	Tibet ⇒ (515)	(292.66)	Abyssinian highlands. Somalia
(292.536)	Himalayas ⇒ (235.243)	(292.67)	East African highlands. Rift valley ⇒ (676.1/.2)
(292.54/.55)	Western Asia	(292.68)	Southern Africa. Lowlands of Mozambique and Natal ⇒ (68...)
(292.54)	Anatolo-Iranian Plateaux	(292.69)	Madagascar. Islands in the western Indian Ocean ⇒ (267.23)
(292.541)	Anatolian Plateau		
(292.542)	Armenian Plateau		
(292.543)	Elburz Mountains and South Caspian coast ⇒ (235.131)	(292.7)	Physical regions of North and Central America. North and Central America as a physical region
(292.544)	Iranian Plateau ⇒ (235.15)		Alaskan Cordilleras ⇒ (237.11)
(292.545)	Hindu Kush ⇒ (235.212)	(292.71)	Canadian Cordilleras (Northern Rockies)
(292.55)	South-western Asia (Arabian-Mesopotamian regions)	(292.72)	Central Cordilleras (Southern Rockies)
(292.551)	Levant ⇒ (56)	(292.73)	Southern Cordilleras (Mexican Highlands)
(292.552)	Mesopotamia. Iraq ⇒ (567)	(292.74)	Mackenzie Basin. Canadian Shield/Laurentian Plateau. Labrador
(292.553)	Arabian Peninsula	(292.75)	⇒ (261.43) ⇒ (261.44) ⇒ (282.273.2)
(292.554)	Sinai Peninsula		
(292.56)	Southern Asia. Indian subcontinent (except Himalayas) ⇒ (54)	(292.76)	Interior plains. Great Plains. Great Lakes. St Lawrence region ⇒ (282.272.1)
(292.561)	Indus-Ganges Lowland. Indo-Gangetic Plains ⇒ (282.253.21) ⇒ (282.253.24)	(292.77)	Appalachians ⇒ (237.2)
(292.562)	Indian Peninsula. □ Deccan. Malabar Coast. Coromandel Coast ⇒ (235.511)	(292.78)	Coastal lowlands of North America
(292.563)	Sri Lanka (Ceylon) ⇒ (548.7)	(292.79)	Caribbean region. Central America. West Indies. Bermuda ⇒ (729)
(292.57)	East Asia	(292.792)	Hispaniola
(292.571)	Japanese archipelago ⇒ (52)	(292.797)	Grenadines as a whole
(292.573)	North-eastern China and Korea	(292.8)	Physical regions of South America. South America as a physical region ⇒ (238.1)
(292.575)	Eastern China	(292.81)	Northern Andes
(292.58)	Indo-China	(292.82)	Central Andes
(292.59)	South-east Asian islands	(292.83)	Subtropical Andes
(292.591)	Philippines ⇒ (599)	(292.84)	Patagonian Andes. Tierra del Fuego archipelago ⇒ (238.15) ⇒ (828.9)
(292.592)	Malay archipelago		
(292.593)	Lesser Sunda Islands ⇒ (594.71/.73)	(292.85)	Patagonia. Falkland Islands ⇒ (828) ⇒ (829.1)
(292.6)	Physical regions of Africa. Africa as a physical region ⇒ (6)	(292.86)	Interior plains of south-eastern South America. □ Pampas Sierras. La-Plata Plains. Marmoré. Mojos and Pantanal Plains
(292.61)	African western Mediterranean region. North West Africa	(292.88)	Amazon basin ⇒ (282.281.3)
(292.611)	Atlas Mountains ⇒ (236.1)	(292.89)	South America bordering on Caribbean Sea. Orinoco basin. Guiana Highlands
(292.612)	Azores, Madeira and Selvagen Islands ⇒ (649.8/.9)		
(292.613)	Canary Islands ⇒ (649)	(292.9)	Australasia. Physical regions of Australia and Oceania. Australia and Oceania as a physical region
(292.62)	Sahara. Nile valley ⇒ (282.263.1)	(292.91/.93)	Australia ⇒ (94)

(292.91)	West Australian Plateau ⇒ (941)	(292.95)	New Guinea ⇒ (954)
(292.92)	Interior Australian depression. Selwyn Heights	(292.96)	Melanesian archipelago. Islands of Melanesia
(292.93)	East Australian Highlands. Tasmania	(292.97)	Micronesia ⇒ (966/968)
(292.94/.99)	Oceania	(292.98)	Polynesia. Easter Island ⇒ (961/964)
(292.94)	New Zealand and adjacent islands ⇒ (931)	(292.99)	Hawaiian Islands ⇒ (739.9)
(292.95/.96)	Melanesia ⇒ (932/935)		

(3) PLACES OF THE ANCIENT AND MEDIAEVAL WORLD

Including countries known to Graeco-Roman antiquity, up to the fall of the western Roman Empire (ca. AD 476), and other places (e.g. pre-Columbian America) up to the 15th century. In the case of countries where this distinction is meaningless, all information may optionally be classed in (4/9), e.g. China at (510)

(309)	Legendary countries. conjectural countries. places of uncertain existence, e.g. Atlantis. Fabulous lands ⇒ (1-08) ⇒ (217) ⇒ 572.4	(367)	Regions of the Slavs
		(368)	Regions of the Nordic peoples. □ Scandinavians, Norse, Normans, Vikings
		(369.1)	Regions of the Huns
		(369.2)	Regions of the Avars
(31)	Ancient China and Japan	(37)	Italia. Ancient Rome and Italy
(315)	Cinarum regio. Ancient China	(375)	Etruria
		(376)	Roma. Latium and City of Rome
(32)	Aegyptus. Ancient Egypt	(38)	Graecia. Ancient Greece
(33)	Judaea. The Holy Land. Region of the Israelites. □ Jerusalem. Nazareth. Bethlehem. Sichem. Jericho	(391)	Graeciae insulae
		(392)	Asia Minor occidentalis. □ Phrygia. Troia (Troy). Lydia
(34)	Ancient India		
(35)	Medo-Persia	(393)	Asia Minor orientalis. □ Cappadocia.
(351)	Chaldaeae		Cilicia. Cyprus
(352)	Assyria. Assur. Nineveh	(393.7)	Cyprus
(353)	Media. Ecbatana		
(354)	Babylonia	(394)	Ancient Syria. Phoenicia. Palaestina. The Hittites. Arabia
(355)	Ancient Persia. Persepolis		Phoenicia. □ Tyre. Sidon. Tripolis. Byblos
(356)	Parthia. Hyrcania	(394.4)	Palaestina. Ancient Palestine
(357)	Sassania (Sasanid Persia)	(394.5)	Arabia. Arabia Deserta
(358)	Mesopotamia	(394.7)	
(359)	Susiana. Susa. Elam	(395)	Asia septentrionalis occidentalis. □ Scythia. Colchis. Armenia
(36)	Regions of the so-called barbarians		Asia meridionalis centralis. □ Hyrcania. Bactria
(363)	Regions of the Germanic tribes	(396)	
(363.1)	Regions of the Burgundians, Vandals, Alans, Gepids	(397)	Ancient Africa. □ Mauretania. Numidia.
(363.2)	Regions of the Angles, Saxons, Jutes		Karthago (Carthage). Libya. Aethiopia
(363.3)	Regions of the Alemanni, Heruli, Suevi	(397.3)	Africa propria. □ Karthago (Carthage)
(363.4)	Regions of the Franks	(398)	Europa meridionalis orientalis. □ Dacia.
(363.5)	Regions of the Langobardi (Lombards)		Vindelicia. Rhaetia. Noricum. Pannonia. Illyria.
(363.6)	Regions of the Goths		Dalmatia
(363.61)	Regions of the Ostrogoths (East Goths)	(399)	Other regions. Ancient geographical divisions other than those of classical antiquity
(363.62)	Regions of the Visigoths (West Goths)		Regions of ancient American cultures.
(364)	Regions of the Celts		Pre-Columbian America
(365)	Hispania. Iberia	(399.7)	
(366)	Britannia. Roman Britain		

(4/9) COUNTRIES AND PLACES OF THE MODERN WORLD

The following subdivisions represent the currently valid divisions of the countries of the modern world usually down to primary administrative level.

Some historical concepts are also included, and are described as such

Secondary and further administrative units may be denoted by the (1-2) and (1-3) special auxiliaries, or using alphabetic extension (Table 1h)

(4+7)	THE WEST (OCCIDENT) CONVENTIONALLY DEFINED ⇒ (1-15)	(4)	EUROPE
		(4-01)	Regions of Europe

(1/9) Table 1e. Common auxiliaries of place

(4-11)	Eastern Europe. □ (470) Russian Federation in Europe. (474.2) Estonia. (474.3) Latvia. (474.5) Lithuania. (476) Belarus. (477) Ukraine. (478.9) Moldova. (479.22) Georgia. (479.24) Azerbaijan. (479.25) Armenia ⇒ (474)	(410.152)	Humberside area <i>Class here also County of Humberside 1974-1996</i>
(4-13)	Southern Europe. □ (449.49) Monaco. (450) Italy. (454.4) San Marino. (456.31) Vatican City. (458.2) Malta. (460) Spain. (467.2) Andorra. (468.2) Gibraltar. (469) Portugal. (495) Greece	(410.152.21) (410.154) (410.156)	East Riding of Yorkshire North Yorkshire West Yorkshire area <i>Class here also Metropolitan County of West Yorkshire 1974-1986</i>
(4-15)	Western Europe. □ (410) United Kingdom. (417) Ireland. (430) Germany. (435.9) Luxembourg. (44) France. (492) Netherlands. (493) Belgium	(410.158) (410.16) (410.161)	South Yorkshire area <i>Class here also Metropolitan County of South Yorkshire 1974-1986</i> Northern region Cleveland area <i>Class here also County of Cleveland 1974-1996</i>
(4-17)	Northern Europe. □ (480) Finland. (481) Norway. (485) Sweden. (489) Denmark. (491.1) Iceland. (491.2) Faeroe Islands ⇒ (48)	(410.162) (410.164)	Durham (county) Tyne and Wear area <i>Class here also Metropolitan County of Tyne and Wear 1974-1986</i>
(4-191.2)	Central Europe. □ (436) Austria. (437.1/.2) Czech Republic. (437.6) Slovak Republic. (438) Poland. (439) Hungary. (494) Switzerland. (494.9) Liechtenstein. (496.5) Albania. (497.11 + 497.16) Serbia and Montenegro. Yugoslavia. (497.2) Bulgaria. (497.4) Slovenia. (497.5) Croatia. (497.6) Bosnia-Herzegovina. (497.7) Macedonia. (498) Romania	(410.166) (410.168) (410.17) (410.172) (410.174)	Northumberland Cumbria North west Lancashire Merseyside area <i>Class here also Metropolitan County of Merseyside 1974-1986</i>
(41)	BRITISH ISLES (GREAT BRITAIN, IRELAND AND MINOR ISLANDS) <i>Class here collective works on the British Isles and the Channel Islands</i> ◇ (41-44) British Commonwealth and Empire generally. NOTE: For specific dependencies see individual (4/9) numbers ⇒ (423.4) ⇒ (428.9)	(410.176) (410.178) (410.18) (410.181) (410.182) (410.184)	Greater Manchester area <i>Class here also Metropolitan County of Greater Manchester 1974-1986</i> Cheshire West Midlands Staffordshire Shropshire Hereford and Worcester area <i>Class here also County of Hereford and Worcester 1974-1998</i>
(410)	United Kingdom of Great Britain and Northern Ireland <i>Class here also Great Britain (England, Scotland and Wales)</i>	(410.184.2) (410.185) (410.186) (410.188)	Herefordshire Worcestershire Warwickshire West Midlands urban area <i>Class here also Metropolitan County of West Midlands 1974-1986</i>
(410.1)	England	(410.19)	South West
(410.11/.12)	South East	(410.191)	Gloucestershire
(410.111)	Greater London area	(410.192)	Avon area <i>Class here also County of Avon 1974-1996</i>
(410.111.11)	City of London		Wiltshire
(410.111.24)	City of Westminster	(410.193)	Somerset
(410.112)	Kent	(410.194)	Dorset
(410.113)	Essex	(410.195)	Devon
(410.114)	Hertfordshire	(410.196)	Cornwall
(410.115)	Bedfordshire	(410.197)	Isles of Scilly
(410.116)	Buckinghamshire	(410.198)	
(410.117)	Oxfordshire		
(410.118)	Berkshire area <i>Class here also County of Berkshire 1974-1998</i>	(410.3) (410.311)	Wales South Glamorgan area <i>Class here also County of South Glamorgan 1974-1996</i>
(410.121)	Hampshire		West Glamorgan area <i>Class here also County of West Glamorgan 1974-1996</i>
(410.122)	Isle of Wight	(410.312)	Mid Glamorgan area <i>Class here also County of Mid Glamorgan 1974-1996</i>
(410.124)	West Sussex		Gwent area <i>Class here also County of Gwent 1974-1996</i>
(410.126)	East Sussex	(410.313)	Powys
(410.128)	Surrey		Clwyd area <i>Class here also County of Clwyd 1974-1996</i>
(410.13)	East Anglia	(410.314)	Gwynedd area. □ Isle of Anglesey <i>Class here also County of Gwynedd 1974-1996</i>
(410.132)	Suffolk		
(410.134)	Norfolk	(410.315)	
(410.138)	Cambridgeshire	(410.316)	
(410.14)	East Midlands		
(410.141)	Northamptonshire		
(410.142)	Leicestershire		
(410.144)	Lincolnshire		
(410.146)	Nottinghamshire	(410.317)	
(410.148)	Derbyshire		
(410.15)	Yorkshire and Humberside		

(410.318)	Dyfed area <i>Class here also County of Dyfed 1974-1996</i>	(411.7)	Hebrides (Outer and Inner Hebrides)
(410.5)	Scotland	(412)	North central Scotland <i>Also used for the Highlands in general</i>
(410.511)	Lothian area <i>Class here also Lothian Region 1975-1996</i>	(412.1)	Inverness-shire. □ Rockall. St Kilda
(410.512)	Central area <i>Class here also Central Region 1975-1996</i>	(412.2)	Nairnshire
(410.513)	Fife	(412.3)	Moray (Elginshire)
(410.514)	Tayside area <i>Class here also Tayside Region 1975-1996</i>	(412.4)	Banffshire
(410.516)	Grampian area <i>Class here also Grampian Region 1975-1996</i>	(412.5)	Aberdeenshire
(410.518)	Highland	(412.6)	Kincardineshire
(410.521)	Orkney Islands	(413)	South central Scotland
(410.522)	Shetland Islands. □ Fair Isle. Foula	(413.1)	Angus (Forfarshire)
(410.523)	Western Isles. □ Outer Hebrides. Rockall. Rona. St Kilda	(413.2)	Perthshire
(410.524)	Strathclyde area <i>Class here also Strathclyde Region 1975-1996</i>	(413.3)	Fife
(410.526)	Dumfries and Galloway	(413.4)	Kinross-shire
(410.528)	The Scottish Borders <i>Class here also Borders Region 1975-1996</i>	(413.5)	Clackmannanshire
(410.7)	Northern Ireland	(413.6)	Stirlingshire
(410.712/.718)	Districts of Northern Ireland	(413.7)	Dunbartonshire
(410.712)	Districts in the north-eastern area of Northern Ireland	(413.8)	Argyllshire
(410.712.12)	Belfast	(413.9)	Bute, with Arran
(410.712.14)	Newtownabbey	(414)	Southern Scotland <i>Also used for the Lowlands in general</i>
(410.712.16)	Carrickfergus	(414.1)	Renfrewshire
(410.712.18)	Larne	(414.2)	Ayrshire
(410.712.22)	Moyle	(414.3)	Lanarkshire
(410.712.24)	Ballymoney	(414.42)	West Lothian (Linlithgowshire)
(410.712.26)	Ballymena	(414.45)	Midlothian
(410.712.28)	Antrim	(414.52)	East Lothian (Haddingtonshire)
(410.714)	Districts in the northern area of Northern Ireland	(414.55)	Berwickshire
(410.714.12)	Magherafelt	(414.62)	Peeblesshire (Tweeddale)
(410.714.14)	Coleraine	(414.65)	Selkirkshire
(410.714.16)	Limavady	(414.7)	Roxburghshire
(410.714.18)	Derry <i>Formerly: Londonderry</i>	(414.8)	Dumfriesshire
(410.716)	Districts in the western area of Northern Ireland	(414.92)	Kirkcudbrightshire
(410.716.12)	Dungannon	(414.95)	Wigtownshire
(410.716.14)	Cookstown	(415)	Ireland (geographical whole) ⇒ (292.421) ⇒ (410.7) ⇒ (417)
(410.716.16)	Omagh	(416)	Northern Ireland: traditional county divisions <i>The name 'Ulster' is sometimes used loosely as a synonym for Northern Ireland; class such information at (416). But historically, Ulster was a larger area; if in doubt, prefer (417-89ULS)</i>
(410.716.18)	Strabane	(416.1)	Antrim
(410.716.22)	Fermanagh	(416.2)	Londonderry (Derry)
(410.718)	Districts in the south-eastern area of Northern Ireland	(416.4)	Tyrone
(410.718.11)	Armagh	(416.5)	Down
(410.718.12)	Newry and Mourne	(416.6)	Armagh
(410.718.14)	Banbridge	(416.8)	Fermanagh
(410.718.16)	Down	(417)	Ireland. Republic of Ireland. Eire
(410.718.18)	Ards	◇ (417-89)	Historic and traditional divisions of Ireland. Including: Provinces. Kingdoms. Overlordships
(410.718.22)	North Down	◇ (417-89CON)	Connacht
(410.718.24)	Castlereagh	◇ (417-89LEI)	Leinster
(410.718.26)	Lisburn	◇ (417-89MUN)	Munster
(410.718.28)	Craigavon	◇ (417-89ULS)	Ulster ⇒ (415)
(411/429)	Traditional divisions of the United Kingdom. Republic of Ireland	⇒ (415)	Southern and Eastern Regional Authority
(411)	Scotland: traditional divisions	(417.7)	Dublin region <i>Class here the former Dublin county</i>
(411.1)	County of Zetland. Shetland Islands	(417.71)	Dublin (city)
(411.2)	County of Orkney. Orkney Islands	(417.711)	Fingal (county)
(411.3)	Caithness	(417.713)	South Dublin (county)
(411.4)	Sutherland	(417.715)	Dun Laoghaire/Rathdown (county)
(411.5/.6)	Ross and Cromarty (single county, 1889-1975)	(417.717)	Mid-east region
(411.5)	Cromarty (to 1889)	(417.72)	Meath (county)
(411.6)	Ross (to 1889)	(417.722)	

(1/9) Table 1e. Common auxiliaries of place

(417.724)	Kildare (county)	(423.47)	Alderney (Aurigny). □ Islets near Alderney:
(417.726)	Wicklow (county)		Burhou, les Casquets
(417.73)	South-east region	(423.5)	Devon
(417.731)	Carlow (county)	(423.7)	Cornwall. Isles of Scilly
(417.733)	Wexford (county)	(423.8)	Somerset
(417.735)	Kilkenny (county)	(424.1)	Gloucestershire
(417.737)	Waterford (county). □ Waterford (city)	(424.3)	Monmouthshire
(417.739)	Tipperary South Riding (county)	(424.4)	Herefordshire
(417.74)	Mid-west region	(424.5)	Shropshire
(417.742)	Tipperary North Riding (county)	(424.6)	Staffordshire
(417.744)	Limerick (county). □ Limerick (city)	(424.7)	Worcestershire
(417.746)	Clare (county)	(424.8)	Warwickshire
(417.75)	South-west region	(425.1)	Derbyshire
(417.752)	Cork (county). □ Cork city	(425.2)	Nottinghamshire
(417.755)	Kerry (county)	(425.3)	Lincolnshire
(417.8)	Border, Midlands Western Regional Authority	(425.42)	Leicestershire
(417.82)	Midlands region	(425.45)	Rutland
(417.821)	Laois (county)	(425.5)	Northamptonshire
	<i>Formerly known as Queen's County</i>	(425.62)	Huntingdon and Peterborough (single county, 1965-1974). Huntingdonshire (to 1965). Soke of Peterborough (county status, 1888-1965)
(417.823)	Offaly (county)		Bedfordshire
	<i>Formerly known as King's County</i>	(425.65)	Oxfordshire
(417.825)	Westmeath (county)	(425.72)	Buckinghamshire
(417.827)	Longford (county)	(425.75)	Hertfordshire
(417.84)	West region	(425.8)	Cambridgeshire and Isle of Ely
(417.842)	Roscommon (county)	(425.9)	Norfolk
(417.844)	Galway (county). □ Galway (city)	(426.1)	Suffolk
(417.846)	Mayo (county)	(426.4)	Essex
(417.86)	Border region	(426.7)	Cheshire
(417.861)	Sligo (county)	(427.1)	Lancashire
(417.863)	Leitrim (county)	(427.2)	Yorkshire
(417.864)	Donegal (county)	(427.4)	◇ (427.4-11) East Riding
(417.866)	Cavan (county)		◇ (427.4-15) West Riding
(417.867)	Monaghan (county)		◇ (427.4-17) North Riding
(417.869)	Louth (county)		Durham, county
(420)	England: traditional divisions	(428.1)	Northumberland
(421)	County of London (1899 to 1965)	(428.2)	Cumberland
(421.1)	London north of the Thames	(428.5)	Westmorland
(421.2)	Central London. □ City of London. Boroughs of Finsbury, Holborn	(428.8)	Isle of Man
(421.3)	West London. □ City of Westminster. Boroughs of Chelsea, Fulham, Hammersmith, Kensington, Paddington	(428.9)	
(421.4)	North London boroughs. □ Hackney. Hampstead. Islington. St Marylebone. St Pancras. Stoke Newington	(429)	Wales: traditional divisions
(421.5)	East London boroughs. □ Bethnal Green. Poplar. Shoreditch. Stepney	(429.1)	Anglesey
(421.6)	London south of the Thames. □ Battersea. Bermondsey. Camberwell. Deptford. Greenwich. Lambeth. Lewisham. Southwark. Wandsworth. Woolwich	(429.22)	Caernarvonshire
(421.9)	Middlesex	(429.25)	Merionethshire
(422.1)	Surrey	(429.32)	Denbighshire
(422.3)	Kent	(429.35)	Flintshire
(422.5)	Sussex	(429.4)	Montgomeryshire
	◇ (422.5-11) East Sussex	(429.5)	Cardiganshire
	◇ (422.5-15) West Sussex	(429.62)	Radnorshire
(422.7)	Hampshire	(429.65)	Brecknockshire
(422.8)	Isle of Wight	(429.7)	Glamorgan
(422.9)	Berkshire	(429.8)	Cararthenshire
(423.1)	Wiltshire	(429.9)	Pembrokeshire
(423.3)	Dorset	(430)	GERMANY. FEDERAL REPUBLIC OF GERMANY (BUNDESREPUBLIK DEUTSCHLAND)
(423.4)	Channel Islands		<i>Class here information about the Federal Republic of Germany from 1990, and about Germany as a whole at any period</i>
(423.41)	Jersey. □ Islets near Jersey: Ecréhous, les Minquiers etc.		◇ (430-25) Berlin as capital of Germany (option). NOTE: Class here if position following (430) is required. If in doubt, prefer (430.131)
(423.43)	Guernsey (Guernesey)	(430.1)	Western and southern Länder
(423.432)	Lihou		<i>Class here also information about Western Germany from 1945 and the Federal Republic of Germany, 1949-1990</i>
(423.44)	Herm		Land Schleswig-Holstein
(423.442)	Jethou	(430.111)	Land Hamburg
(423.45)	Sark (Sercq)	(430.113)	Land Bremen
(423.452)	Bréchou	(430.115)	

(430.117)	Land Niedersachsen (Lower Saxony) ◇ (430.117:210.7FRI) The East Frisian Islands		<i>required. If in doubt, prefer (430.2) for information about Eastern Germany as a whole.</i>
(430.117.1)	Regierungsbezirk Weser-Ems		<i>Use the subdivisions of (430.3) only for information specifically about the DDR and its parts</i>
(430.117.3)	Regierungsbezirk Lüneburg		
(430.117.5)	Regierungsbezirk Braunschweig		
(430.117.7)	Regierungsbezirk Hannover		◇ (430.3-25) Berlin (Ost). East Berlin as capital of the German Democratic Republic.
(430.119)	Land Nordrhein-Westfalen (North Rhine - Westphalia)		NOTE: If in doubt, class East Berlin at (430.131.2)
(430.119.1)	Regierungsbezirk Düsseldorf	(430.313)	Bezirk Rostock
(430.119.3)	Regierungsbezirk Münster	(430.315)	Bezirk Neubrandenburg
(430.119.5)	Regierungsbezirk Detmold	(430.317)	Bezirk Potsdam
(430.119.7)	Regierungsbezirk Arnsberg	(430.319)	Bezirk Frankfurt/Oder
(430.119.9)	Regierungsbezirk Köln	(430.321)	Bezirk Cottbus
(430.121)	Land Hessen	(430.323)	Bezirk Dresden
(430.121.1)	Regierungsbezirk Kassel	(430.325)	Bezirk Leipzig
(430.121.3)	Regierungsbezirk Giessen	(430.327)	Bezirk Karl-Marx-Stadt
(430.121.5)	Regierungsbezirk Darmstadt	(430.329)	Bezirk Gera
(430.123)	Land Rheinland-Pfalz (Rhineland-Palatinate)	(430.331)	Bezirk Suhle
(430.123.1)	Regierungsbezirk Trier	(430.333)	Bezirk Erfurt
(430.123.3)	Regierungsbezirk Koblenz	(430.335)	Bezirk Halle
(430.123.5)	Regierungsbezirk Rheinhessen-Pfalz	(430.337)	Bezirk Magdeburg
(430.125)	Saarland	(430.339)	Bezirk Schwerin
(430.127)	Land Baden-Württemberg		
(430.127.1)	Regierungsbezirk Karlsruhe	(431/435.8)	Individual states of Germany, -1934 <i>Only for classifying historical information</i>
(430.127.3)	Regierungsbezirk Stuttgart		
(430.127.5)	Regierungsbezirk Tübingen		
(430.127.7)	Regierungsbezirk Freiburg	(431)	PREUSSEN (PRUSSIA)
(430.129)	Land Bayern (Bavaria)	(431.1)	Ostpreussen (East Prussia) ⇒ (470.26)
(430.129.1)	Regierungsbezirk Unterfranken	(431.2)	Westpreussen (West Prussia)
(430.129.2)	Regierungsbezirk Oberfranken	(431.25)	Freie Stadt Danzig (Free City of Danzig)
(430.129.3)	Regierungsbezirk Mittelfranken	(431.3)	Posen
(430.129.4)	Regierungsbezirk Oberpfalz		<i>From 1722 partition of Poland to 1919</i>
(430.129.5)	Regierungsbezirk Niederbayern	(431.35)	Grenzmark Posen-Westpreussen
(430.129.7)	Regierungsbezirk Oberbayern	(431.4)	Schlesien (Silesia)
(430.129.9)	Regierungsbezirk Schwaben	(431.5)	Brandenburg
(430.131)	Land Berlin <i>Class here Berlin as a whole, at any period.</i> <i>Option: see (430-25)</i>	(431.6)	Pommern (Pomerania)
(430.131.1)	Western districts <i>Class here West Berlin 1945-1990</i>	(431.7)	Mecklenburg
(430.131.2)	Eastern districts <i>Class here East Berlin 1945-1990.</i> <i>Option: see (430.3-25)</i>	(431.8)	Sachsen (Provinz) (Saxony)
		(432.1)	Sachsen (Land) (Saxony)
		(432.2)	Thüringen (Thuringia)
		(433)	Bayern (Bavaria)
		(434.1)	Hessen (Hesse)
		(434.2)	Rheinprovinz (Rhine Province)
		(434.4)	Elsass-Lothringen <i>From 1870 to 1919</i>
(430.2)	Eastern Länder <i>Class here also Eastern Germany from 1945 and the German Democratic Republic (Deutsche Demokratische Republik) 1949-1990.</i> <i>Option: see (430.3)</i>	(434.45)	Elsass (Alsace)
(430.245)	Land Brandenburg	(434.5)	Lothringen (Lorraine)
(430.246)	Land Sachsen (Saxony)	(434.6)	Baden
(430.246.1)	Regierungsbezirk Dresden	(434.7)	Württemberg
(430.246.3)	Regierungsbezirk Chemnitz	(434.9)	Hohenzollern
(430.246.5)	Regierungsbezirk Leipzig	(435.1)	Schleswig-Holstein
(430.247)	Land Thüringen (Thuringia)	(435.15)	Hamburg
(430.248)	Land Sachsen-Anhalt (Saxony-Anhalt)	(435.2)	Oldenburg
(430.248.1)	Regierungsbezirk Magdeburg	(435.25)	Bremen
(430.248.3)	Regierungsbezirk Dessau	(435.3)	Hannover (Hanover)
(430.248.5)	Regierungsbezirk Halle	(435.4)	Braunschweig (Brunswick)
(430.249)	Land Mecklenburg-Vorpommern (Mecklenburg - West Pomerania)	(435.5)	Lippe
		(435.6)	Westfalen (Westphalia)
		(435.7)	Waldeck-Pyrmont
		(435.8)	Hessen-Nassau
		(435.9)	Luxembourg. Grand Duchy of Luxembourg
		(435.91)	Luxembourg (city)
		(435.92)	Diekirch
		(435.93)	Grevenmacher
(430.3)	Eastern Germany: historical entities <i>This section is an option that may be used if a separate historical sequence for information about Eastern Germany 1945-1990, particularly the German Democratic Republic (Deutsche Demokratische Republik) 1949-1990, is</i>	(436)	AUSTRIA. REPUBLIC OF AUSTRIA. REPUBLIK ÖSTERREICH <i>Primary administrative divisions: Bundesländer</i>
		(436.1)	Wien

(1/9) Table 1e. Common auxiliaries of place

(436.2)	Niederösterreich	(437.614)	Nitriansky (Nitra) (region). □ Komárno. Levice.
(436.3)	Burgenland		Nitra. Nové Zámky. (c)al'a. Topol'čany. Zlaté
(436.4)	Steiermark		Moravce (districts)
(436.5)	Kärnten	(437.64)	Regions of central Slovakia
(436.6)	Tirol		<i>Class here also former Kraj</i>
(436.7)	Vorarlberg		<i>Stredoslovenský</i>
(436.8)	Salzburg	(437.641)	(r)ilinský (Žilina) (region). □ Bytca. Cadca.
(436.9)	Oberösterreich		Dolný Kubín. Kysucké Nové Mesto. Liptovský
(437)	CZECHOSLOVAKIA, 1918-1992		Mikuláš. Martin. Námestovo. Ružomberok.
(437.3)	CZECH REPUBLIC. CZECHIA. ČESKÁ REPUBLIKA. CESCO, 1993-	(437.642)	Turčianske Teplice. Tvrdošín. (r)ilina (districts)
	<i>In 1992 the Federal Parliament of the</i>		Banskobystrický (Banská Bystrica) (region).
	<i>Czech and Slovak Federal Republic</i>		□ Banská Bystrica (district). Banská (c)tiavnica.
	<i>(formerly Czechoslovakia) decided to divide</i>		Brezno. Detva. Krupina. Lučenec. Poltár.
	<i>the country into two states; thus the Czech</i>	(437.67)	Revúca. Rimavská Sobota. Veľký Krtíš.
	<i>Republic and the Slovak Republic came into</i>		(r)arnovica. (r)jar nad Hronom. Zvolen (districts)
	<i>being on 1 January 1993.</i>		Regions of eastern Slovakia
	<i>For Slezsko (Silesia), previously at this</i>		<i>Class here also former Kraj</i>
	<i>number, see (437.33)</i>	(437.671)	<i>Východoslovenský</i>
⇒ (437)			Prešovský (Presov) (region). □ Bardejov.
(437.31)	Čechy (Bohemia)		Humenné. Kežmarok. Levoča. Medzilaborce.
(437.311)	Praha (kraj) (Prague region)		Poprad. Prešov. Sabinov. Snina. Stará
(437.312)	Středočeský kraj (Central Bohemia region)	(437.672)	L'ubovňa. Stropkov. Svidník. Vranov nad
(437.313)	Královéhradecký kraj (Hradec Králové region)		Topľou (districts)
(437.314)	Pardubický kraj (Pardubice region)		Košický (Košice) (region). □ Gelnica. Kosice I
(437.315)	Ústecký kraj (Ústí nad Labem region)		to IV (districts). Kosice-okolie. Michalovce.
(437.316)	Liberecký kraj (Liberec region)	(437.7)	Rožňava. Sobrance. Spišská Nová Ves.
(437.317)	Plzeňský kraj (Plzeň [Pilsen] region)		Trebišov (districts)
(437.318)	Karlovarský kraj (Karlovy Vary [Karlsbad] region)		ZAKARPATSKA UKRAJINA
(437.319)	Jihočeský kraj (south Bohemia region)		(TRANSCARPATHIAN UKRAINE/
(437.32)	Morava (Moravia)	(438)	CARPATHO-RUSSIA/RUTHENIA),
(437.322)	Jihomoravský kraj (South Moravia region)		1918-1938
(437.323)	Zlínský kraj (Zlín region)		POLAND. POLISH REPUBLIC
(437.324)	Kraj Vysočina (Vysocina region)		<i>Primary administrative divisions:</i>
(437.325)	Olomoucký kraj (Olomouc region)		<i>Wojewodztwa</i>
(437.326)	Moravskoslezský kraj (Moravian and Silesian region)	⇒ (475)	
(437.33)	Slezsko (Silesia)	(438.11)	Wojewodztwo warszawskie. Warszawa
(437.4)	Galicia (Galizien), -1919		(Warsaw)
(437.5)	Bukowina, -1919	(438.12)	Wojewodztwo łódzkie. Łódź
(437.6)	SLOVAKIA. SLOVAK REPUBLIC. SLOVENSKO. SLOVENSKÁ REPUBLIKA, -1993	(438.13)	Wojewodztwo kieleckie. Kielce
	<i>In 1992 the Federal Parliament of the</i>	(438.14)	Wojewodztwo lubelskie. Lublin
	<i>Czech and Slovak Federal Republic</i>	(438.15)	Wojewodztwo białostockie. Białystok
	<i>(formerly Czechoslovakia) decided to divide</i>	(438.16)	Wojewodztwo gdanskie. Gdańsk (Danzig)
	<i>the country into two states; thus the Czech</i>		⇒ (431.25)
	<i>Republic and the Slovak Republic came into</i>	(438.18)	Wojewodztwo olsztyńskie. Olsztyn (Allenstein)
	<i>being on 1 January 1993.</i>	(438.19)	Wojewodztwo opolskie. Opole (Oppeln)
	<i>For Slezsko (Silesia), previously at this</i>	(438.21)	Wojewodztwo bydgoskie. Bydgoszcz
	<i>number, see (437.33)</i>		(Bromberg)
⇒ (437)		(438.22)	Wojewodztwo poznańskie. Poznań (Posen)
(437.61)	Kraj Zapadoslovenský (west Slovakia)		⇒ (431.3)
	<i>Class here also former Kraj</i>	(438.23)	Wojewodztwo katowickie, formerly Śląsk
	<i>Západoslovenský</i>		(Silesia). Katowice
(437.611)	Bratislavský (Bratislava) (region). □ Bratislava	(438.24)	⇒ (431.4)
	I to Bratislava V. Malacky. Pezínok. Senec	(438.25)	Wojewodztwo rzeszowskie. Rzeszów
	(districts)	(438.26)	Wojewodztwo szczecińskie. Szczecin (Stettin)
(437.612)	Trnavský (Trnava) (region). □ Dunajská	(438.27)	Wojewodztwo wrocławskie. Wrocław (Breslau)
	Streda. Galanta. Hlohovec. Piešťany. Senica.	(438.31)	Wojewodztwo zielonogórskie. Zielona Góra
	Skalica. Trnava (districts)	(438.32/.44)	Wojewodztwo krakowskie. Kraków (Cracow)
(437.613)	Trenčiansky (Trenčín) (region). □ Bánovce nad	(438.32)	Historical divisions of Poland
	Bebravou. Ilava. Myjava. Nové Mesto nad	(438.33)	Lwów (Lemberg)
	Váhom. Partizánske. Považská Bystrica.	(438.34)	Stanisławów
	Prievidza. Púchov. Trenčín (districts)	(438.41)	Tarnopol
		(438.42)	Wołyń
		(438.43)	Polesie
		(438.44)	Nowogródek
			Wilno
		(439)	HUNGARY. HUNGARIAN REPUBLIC
			<i>Primary administrative divisions: megyék.</i>
			<i>Divide alphabetically where not otherwise</i>
			<i>provided for</i>

	◇ (439MIS) Miskolc megye	(442.4/5)	Haute-Normandie. Région
	◇ (439PEC) Pecs megye	(442.4)	Eure
	◇ (439SZE) Szeged megye	(442.5)	Seine-Maritime
(439.11)	Transdanubia. Dunantul (left bank of the Danube) area	(442.6)	Picardie. Région
(439.111)	Vas megye	(442.61)	Somme
(439.113)	Sopron vármegye (Oedenburg) (historical)	(442.62)	Aisne
(439.114)	Győr-Sopron megye	(442.63)	Oise
(439.115)	Komarom megye	(442.7/8)	Nord-Pas-de-Calais. Région
(439.117)	Veszprem megye	(442.7)	Pas de Calais
(439.118)	Fejer megye	(442.8)	Nord
(439.121)	Zala megye	(443.6)	Ile-de-France. Région
(439.123)	Somogy megye	(443.61)	Paris. Département
(439.125)	Tolna megye	(443.611)	Capitale de France (Paris)
(439.127)	Baranya megye	(443.62)	Yvelines
(439.13)	North Hungary	(443.63)	Seine-et-Marne
(439.131)	Nograd megye	(443.64)	Val-d'Oise
(439.133)	Heves megye	(443.65)	Hauts-de-Seine
(439.134)	Borsod-Abauj-Zemplen megye	(443.66)	Seine-Saint-Denis
(439.136)	Abauj-Torna varmegye (historical)	(443.67)	Val-de-Marne
(439.138)	Zemplen varmegye (historical)	(443.68)	Essonne
(439.14)	Alföld (Great Basin) area	(443.7)	Champagne-Ardenne. Région
(439.15)	Duna-Tisza area (Danube-Theiss basin)	(443.71)	Ardennes
(439.151)	Budapest	(443.72)	Marne
(439.153)	Pest megye	(443.73)	Aube
(439.155)	Bacs-Kiskun megye	(443.74)	Haute-Marne
(439.16)	Tisza-tul (left bank of the Theiss) area	(443.82)	Lorraine. Région
(439.161)	Szabolcs-Szatmar megye.	(443.821)	⇒ (434.4)
(439.163)	Szatmar-Ugocsa-Bereg-Ung (historical)	(443.822)	Meuse
(439.165)	Szabolcs varmegye (historical)	(443.823)	Meurthe-et-Moselle
(439.169)	Hajdu-Bihar megye	(443.824)	Moselle
(439.173)	Szolnok megye	(443.83)	Vosges
(439.175)	Bihar varmegye (historical)	(443.831)	Alsace. Région
(439.175)	Békés megye	(443.832)	⇒ (434.4)
(439.18)	Tisza-Maros (Theiss-Maros) basin		Bas-Rhin
(439.181)	Csongrad megye		Haut-Rhin
(439.183)	Csanad-Arad-Torontal varmegye (historical)		
(439.2)	Former provinces of Hungary, to 1919	(444.1/4)	Bourgogne (Burgundy). Région
(439.21)	Erdély (Transylvania)	(444.1)	Yonne
(439.22)	Szlovakia	(444.2)	Côte-d'Or
(439.23)	Karpatlja (Ruthenia)	(444.3)	Saône-et-Loire
(439.24)	Horvat-Szlavonország (Croatia, Slavonia)	(444.4)	Nièvre
(439.25)	Fiume	(444.5/8)	Franche-Comté. Région
(439.26)	Nyugat-Magyarország (Burgenland)	(444.5)	Haute-Saône
(439.5)	Former Austro-Hungarian provinces, to 1919	(444.6)	Doubs
(439.55)	Bosznia	(444.7)	Jura
(439.56)	Hercegovina	(444.8)	Territoire de Belfort
(44)	FRANCE. FRENCH REPUBLIC.	(445.1/5)	Centre. Région
	RÉPUBLIQUE FRANÇAISE	(445.1)	Eure-et-Loir
	<i>Primary administrative divisions:</i>	(445.2)	Loiret
	<i>départements grouped into conventional regions</i>	(445.3)	Loir-et-Cher
		(445.4)	Indre-et-Loire
		(445.51)	Indre
		(445.52)	Cher
(441.1/4)	Bretagne (Brittany)	(445.6)	Rhône-Alpes. Région
(441.1)	Finistère	(445.61)	Ain
(441.2)	Côtes d'Armor	(445.62)	Ardèche
(441.3)	Morbihan	(445.63)	Drôme
(441.4)	Ille-et-Vilaine	(445.64)	Isère
(441.5/9)	Pays de la Loire. Région	(445.65)	Savoie
(441.5)	Loire-Atlantique	(445.66)	Haute-Savoie
(441.6)	Mayenne	(445.67)	Rhône
(441.7)	Sarthe	(445.68)	Loire
(441.8)	Maine-et-Loire	(445.7/9)	Auvergne. Région
(441.9)	Vendée	(445.7)	Allier
(442.1/3)	Basse-Normandie. Région	(445.91)	Puy-de-Dôme
(442.1/5)	Normandie	(445.92)	Cantal
(442.1)	Manche	(445.93)	Haute-Loire
(442.2)	Calvados	(446.2/5)	Poitou-Charentes. Région
(442.3)	Orne		

(1/9) Table 1e. Common auxiliaries of place

(446.2)	Deux-Sèvres	(450.341)	Provincia di Venezia (Venice)
(446.3)	Vienne	(450.342)	Provincia di Verona
(446.4)	Charente-Maritime	(450.343)	Provincia di Vicenza
(446.5)	Charente	(450.344)	Provincia di Belluno
(446.6/.8)	Limousin. Région	(450.346)	Provincia di Treviso
(446.6)	Haute-Vienne	(450.347)	Provincia di Padova (Padua)
(446.7)	Corrèze	(450.348)	Provincia di Rovigo
(446.8)	Creuse	(450.36)	Regione del Friuli-Venezia Giulia
(447.3/.8)	Midi-Pyrénées. Région	(450.361)	Provincia di Trieste
(447.3)	Lot	(450.363)	Provincia di Pordenone
(447.4)	Aveyron	(450.365)	Provincia di Udine
(447.5)	Tarn-et-Garonne	(450.367)	Provincia di Gorizia
(447.71)	Gers	(450.42)	Regione della Liguria
(447.72)	Tarn	(450.421)	Provincia di Genova (Genoa)
(447.73)	Haute-Garonne	(450.423)	Provincia di Imperia
(447.74)	Ariège	(450.425)	Provincia di Savona
(447.8)	Hautes-Pyrénées	(450.427)	Provincia di La Spezia
(447.9)	Aquitaine. Région	(450.45)	Regione di Emilia e Romagna
(447.91)	Gironde	(450.451)	Provincia di Bologna
(447.92)	Dordogne	(450.452)	Provincia di Piacenza
(447.93)	Lot-et-Garonne	(450.453)	Provincia di Parma
(447.94)	Landes	(450.454)	Provincia di Reggio nell' Emilia
(447.95)	Pyrénées-Atlantiques	(450.455)	Provincia di Modena
(448.1/.9)	Languedoc-Rousillon. Région	(450.456)	Provincia di Ferrara
(448.12)	Lozère	(450.457)	Provincia di Ravenna
(448.3)	Gard	(450.458)	Provincia di Forlì
(448.4)	Hérault	(450.5/.6)	Central Italy
(448.7)	Aude	(450.52)	Regione della Toscana
(448.9)	Pyrénées-Orientales	(450.521)	Provincia di Firenze (Florence)
(449.1/.43)	Provence-Alpes-Côte d'Azur. Région	(450.522)	Provincia di Massa Carrara
(449.1)	Bouches-du-Rhône	(450.523)	Provincia di Lucca
(449.2)	Vaucluse	(450.524)	Provincia di Pistoia
(449.3)	Var	(450.525)	Provincia di Livorno (Leghorn)
(449.4)	Alpes-Maritimes	(450.526)	Provincia di Pisa
(449.42)	Alpes-de-Haute-Provence	(450.527)	Provincia di Arezzo
(449.43)	Hautes-Alpes	(450.528)	Provincia di Siena
(449.45)	Corse (Corsica). Région	(450.529)	Provincia di Grosseto
⇒ (459.9)		(450.55)	Regione dell' Umbria
(449.452)	Haute-Corse	(450.551)	Provincia di Perugia
(449.453)	Corse-du-Sud	(450.554)	Provincia di Terni
(449.49)	MONACO. PRINCIPALITY OF MONACO	(450.57)	Regione delle Marche (Marches)
(450)	ITALY. REPUBLIC OF ITALY.	(450.571)	Provincia di Ancona
	REPUBBLICA ITALIANA	(450.573)	Provincia di Macerata
	<i>Primary administrative divisions: province</i>	(450.575)	Provincia di Pesaro e Urbino
(450.2/.4)	Northern Italy	(450.577)	Provincia di Ascoli Piceno
(450.21)	Regione del Piemonte (Piedmont)	(450.62)	Regione del Lazio (Latium)
(450.211)	Provincia di Torino (Turin)	(450.621)	Provincia di Roma (Rome)
(450.213)	Provincia di Vercelli	(450.624)	Provincia di Viterbo
(450.214)	Provincia di Novara	(450.625)	Provincia di Rieti
(450.215)	Provincia di Cuneo	(450.627)	Provincia di Latina
(450.217)	Provincia di Asti	(450.628)	Provincia di Frosinone
(450.218)	Provincia di Alessandria	(450.65)	Regione degli Abruzzi
(450.23)	Regione della Valle D'Aosta	(450.651)	Provincia di L'Aquila
(450.231)	Provincia di Aosta	(450.653)	Provincia di Teramo
(450.25)	Regione della Lombardia	(450.655)	Provincia di Pescara
(450.251)	Provincia di Milano (Milan)	(450.657)	Provincia di Chieti
(450.252)	Provincia di Varese	(450.67)	Regione del Molise
(450.253)	Provincia di Como	(450.671)	Provincia di Campobasso
(450.254)	Provincia di Sondrio	(450.674)	Provincia di Isernia
(450.255)	Provincia di Bergamo	(450.7)	Southern Italy
(450.256)	Provincia di Brescia	(450.72)	Regione della Campania
(450.257)	Provincia di Pavia	(450.721)	Provincia di Napoli (Naples)
(450.258)	Provincia di Cremona	(450.723)	Provincia di Caserta
(450.259)	Provincia di Mantova (Mantua)	(450.725)	Provincia di Benevento
(450.32)	Regione del Trentino Alto Adige	(450.727)	Provincia di Avellino
(450.321)	Provincia di Trento (Trent)	(450.728)	Provincia di Salerno
(450.323)	Provincia di Bolzano	(450.75)	Regione della Puglia (Apulia)
(450.34)	Regione del Veneto	(450.751)	Provincia di Bari
		(450.753)	Provincia di Foggia
		(450.754)	Provincia di Taranto
		(450.756)	Provincia di Brindisi

(450.758)	Provincia di Lecce	(460.231)	Lérida
(450.77)	Regione della Basilicata	(460.233)	Gerona
(450.771)	Provincia di Potenza	(460.235)	Barcelona
(450.774)	Provincia di Matera	(460.237)	Tarragona
(450.78)	Regione della Calabria	(460.25)	Extremadura
(450.781)	Provincia di Reggio Calabria	(460.251)	Cáceres
(450.784)	Provincia di Cosenza	(460.253)	Badajoz
(450.786)	Provincia di Catanzaro	(460.27)	Madrid
(450.8)	Italian islands	(460.28)	Castilla-La Mancha
(450.82)	Regione della Sicilia (Sicily)	(460.281)	Guadalajara
(450.821)	Provincia di Palermo	(460.283)	Cuenca
(450.822)	Provincia di Trapani	(460.285)	Toledo
(450.823)	Provincia di Messina	(460.287)	Ciudad Real
(450.824)	Provincia di Agrigento	(460.288)	Albacete
(450.825)	Provincia di Caltanissetta	(460.31)	Comunidad Valenciana
(450.826)	Provincia di Enna	(460.311)	Castellón
(450.827)	Provincia di Catania	(460.313)	Valencia
(450.828)	Provincia di Ragusa	(460.315)	Alicante
(450.829)	Provincia di Siracusa (Syracuse)	(460.32)	Islas Baleares
(450.88)	Regione della Sardegna (Sardinia)	(460.33)	Región de Murcia
(450.881)	Provincia di Cagliari	(460.35)	Andalucía
(450.883)	Provincia di Oristano	(460.351)	Córdoba
(450.884)	Provincia di Sassari	(460.352)	Jaén
(450.886)	Provincia di Nuoro	(460.353)	Sevilla
(454.4)	SAN MARINO. REPUBLIC OF SAN MARINO	(460.354)	Huelva
(456.31)	VATICAN CITY STATE. STATO DELLA CITTA DEL VATICANO	(460.355)	Cádiz
(458.2)	MALTA. REPUBLIC OF MALTA	(460.356)	Málaga
(459.9)	CORSICA AS PART OF ITALY, -1768 ⇒ (449.45)	(460.357)	Granada
(46)	IBERIAN PENINSULA	(460.358)	Almería
(460)	SPAIN. KINGDOM OF SPAIN. REINO DE ESPAÑA <i>Administrative subdivision in comunidades autónomas and provincias, authorized by the Constitution of 1978</i>	(460.41)	Canary Islands. Canarias ⇒ (649.1/7)
(460.11)	Galicia	(460.411)	Santa Cruz de Tenerife
(460.111)	La Coruña	(460.412)	Gran Canaria
(460.113)	Lugo	(460.413)	Las Palmas
(460.115)	Pontevedra	(460.414)	Gomera
(460.117)	Orense	(460.415)	Hierro (Ferro)
(460.12)	Principado de Asturias	(460.416)	Fuerteventura
(460.13)	Cantabria	(460.417)	Lanzarote
(460.15)	País Vasco	(462.1)	León (historical)
(460.152)	Vizcaya	(463)	Castilla la Vieja (Old Castile) (historical)
(460.154)	Guipúzcoa	(463.01)	The two Castiles (historical)
(460.156)	Alava	(464)	Castilla la Nueva (New Castile) (historical)
(460.16)	Navarra	(465.2)	Aragón
(460.18)	Castilla y León	(465.21)	Zaragoza (Saragossa)
(460.181)	León	(465.22)	Huesca
(460.182)	Burgos	(467.2)	ANDORRA. THE PRINCIPALITY OF ANDORRA. PRINCIPAT D'ANDORRA
(460.183)	Palencia	(468.1)	Andalucía (Andalusia) (historical)
(460.184)	Zamora	(468.2)	Gibraltar (UK)
(460.185)	Valladolid	(469)	PORTUGAL. PORTUGUESE REPUBLIC
(460.186)	Soria	(469.11)	Minho (province)
(460.187)	Salamanca	(469.111)	Viana do Castelo
(460.188)	Segovia	(469.112)	Braga
(460.189)	Ávila	(469.12)	Douro Litoral (province)
(460.21)	Rioja	(469.121)	Porto (Oporto)
(460.22)	Aragón	(469.2)	Trás-os-Montes e Alto Douro
(460.222)	Huesca	(469.201)	Bragança (Braganza)
(460.224)	Zaragoza	(469.202)	Vila Real
(460.226)	Teruel	(469.3)	Beiras (as a whole)
(460.23)	Cataluña	(469.31)	Beira Alta (province)
		(469.311)	Guarda
		(469.312)	Viseu

(1/9) Table 1e. Common auxiliaries of place

(469.32)	Beira Litoral (province)	(470.318)	Kaluzhskaia oblast'
(469.321)	Aveiro		<i>Formerly Kaluga Province</i>
(469.322)	Coimbra	(470.319)	Orlovskaja oblast'
(469.325)	Leiria		<i>Formerly Orel Province</i>
(469.33)	Beira Baixa (province)	(470.32)	Central Black-Earth (Chernozem) Region
(469.331)	Castelo Branco	(470.322)	Lipetskaia oblast'
(469.41)	Estremadura (province)	(470.323)	Kurskaia oblast'
(469.411)	Lisboa (Lisbon)	(470.324)	Voronezhskaja oblast'
(469.412)	Setúbal	(470.325)	Belgorodskaja oblast'
(469.42)	Ribatejo (province)	(470.326)	Tambovskaja oblast'
(469.421)	Santarém	(470.33)	West Central Russia
(469.5)	Alentejo (as a whole)	(470.331)	Tverskaia oblast'
(469.51)	Alto Alentejo (province)		<i>Formerly Kalininskaja oblast'</i>
(469.511)	Portalegre	(470.332)	Smolenskaja oblast'
(469.512)	Évora	(470.333)	Brianskaia oblast'
(469.52)	Baixo Alentejo (province)	(470.34)	Volga-Vyatka Region
(469.521)	Beja	(470.341)	Nizhegorodskaja oblast'
(469.6)	Algarve (province)		◇ (470.341-25) Nizhnii Novgorod
(469.601)	Faro	(470.342)	Kirovskaja oblast'
(469.8)	Arquipélago da Madeira	(470.343)	Marii-El Republic
(469.81)	Funchal		<i>Formerly Mariiskaja A.S.S.R.</i>
(469.9)	Arquipélago dos Açores (Azores)	(470.344)	Chuvashia. Chuvash Republic
(469.91)	Horta		<i>Formerly Chuvashskaja A.S.S.R.</i>
(469.92)	Angra do Heroísmo	(470.345)	Mordvinia. Mordovian Republic
(469.93)	Ponta Delgada		<i>Formerly Mordovskaja A.S.S.R.</i>
(47+57)	FORMER UNION OF SOVIET SOCIALIST REPUBLICS (USSR)	(470.4)	Povolz'e. Region of Middle and Lower Volga
(47)	FORMER EUROPEAN USSR	(470.40/.43)	Middle Volga Region
(470+571)	RUSSIA. RUSSIAN FEDERATION <i>Formerly Russian Soviet Federal Socialist Republic RSFSR</i>	(470.40)	Penzenskaja oblast'
		(470.41)	Tatarstan. Tatarstan Republic
(470)	Russian Federation in Europe ◇ (470-25) Moscow. Moskva ⇒ (571)	(470.42)	Ul'ianovskaja oblast'
(470.0)	Non-chernozem (Non-black-earth) zone in Russia <i>Includes (470.1/.319); (470.33/.345); (470.51/.54)</i>	(470.43)	Samarskaja oblast'
			<i>Formerly Kuibyshevskaja oblast'</i>
(470.1/.6)	North European Russia	(470.44/.47)	Lower Volga Region
(470.1)	North-East European Russia	(470.44)	Saratovskaja oblast'
(470.11)	Arkhangel'skaja oblast'. Archangel	(470.45)	Volgogradskaja oblast'
(470.111)	Nenetskii avtonomnyi okrug	(470.46)	Astrakhanskaja oblast'
(470.12)	Vologodskaja oblast'	(470.47)	Kalmykia. Republic of Kalmykia
(470.13)	Komi		<i>Formerly Kalmytskaja A.S.S.R.</i>
	<i>Formerly Komi A.S.S.R.</i>	(470.5)	Ural Region. Central and North Ural
(470.21)	Murmanskaja oblast'	(470.51/.54)	Middle Ural Region
(470.22)	Karelia. Karelian Republic	(470.51)	Udmurtia. Udmurtskaja Respublika
	<i>Formerly Karel'skaja A.S.S.R.</i>	(470.53)	Permskaja oblast'
(470.23)	Leningradskaja oblast'	(470.531)	Komi-Permiatskii avtonomnyi okrug
	◇ (470.23-25) Sankt Peterburg. Saint Petersburg	(470.54/.56+470.58)	Zaural'e
(470.24)	Novgorodskaja oblast'		⇒ (571)
(470.25)	Pskovskaja oblast'	(470.54)	Sverdlovskaja oblast'
	<i>Capital Pskov</i>		◇ (470.54-25) Ekaterinenburg
(470.26)	Kaliningradskaja oblast'	(470.55/.58)	Southern Ural Region
	⇒ (474)	(470.55)	Cheliabinskaja oblast'
(470.3)	Central region of European Russia	(470.56)	Orenburgskaja oblast'
(470.31)	Central industrial region of European Russia	(470.57)	Bashkortostan
(470.311)	Moskovskaja oblast'		<i>Formerly Bashkirskaja A.S.S.R.</i>
(470.312)	Tul'skaja oblast'	(470.58)	Kurganskaja oblast'
	<i>Formerly Tula oblast'</i>	(470.6)	Northern Caucasus and Lower Don
(470.313)	Riazanskaja oblast'		⇒ (571)
(470.314)	Vladimirskaja oblast'	(470.61)	Rostovskaja oblast'. Lower Don
(470.315)	Ivanovskaja oblast'	(470.62/.67)	Northern Caucasus
(470.316)	Iaroslavskaja oblast'		⇒ (571)
(470.317)	Kostromskaja oblast'	(470.62)	Kubanskaja oblast'. Kuban' Region
		(470.620)	Krasnodarskii krai
		(470.621)	Adygea. Adyge republic
			<i>Formerly Adygeiskaiya avtonomnaia oblast'</i>
		(470.63)	Stavropol'skii krai and Karachay-Cherkessia
		(470.630)	Stavropol'skii krai
		(470.631)	Karachay-Cherkessia. Karachay-Cherkes Republic
			<i>Formerly Karachaevo-Cherkesskaja avtonomnaia oblast'</i>

(470.64)	Kabardino-Balkaria. Kabardin-Balkar Republic <i>Formerly Kabardino-Balkarskaia A.S.S.R.</i>	(477.62)	Donetskaia oblast'
(470.65)	North Ossetia. North Ossetian Republic (Alania) <i>Formerly Severo-Osetinskaia A.S.S.R.</i>	(477.63/.65)	Lower Dneper (Dnieper) region
(470.66)	Chechnia and Ingushetia <i>Formerly Checheno-Ingushskaia A.S.S.R.</i>	(477.63)	Dnepropetrovskaia oblast'
(470.661)	Chechnia. Chechen Republic	(477.64)	Zaporozhskaia oblast'
(470.662)	Ingushetia. Ingush Republic	(477.65)	Kirovogradskaia oblast'
(470.67)	Dagestan. Dagestan Republic <i>Formerly Dagestanskaia A.S.S.R.</i>	(477.7)	Southern Ukraine. Black Sea region
(474)	BALTIC STATES ⇒ (470.26)	(477.72)	Khersonskaia oblast'
(474.2)	ESTONIA. REPUBLIC OF ESTONIA ◇ (474.2-25) Tallinn	(477.73)	Nikolaievskaia oblast'
(474.211)	Harjumaa	(477.74)	Odesskaia oblast'
(474.212)	Lääne-Virumaa	(477.75)	Krymskaia oblast'. Crimea
(474.213)	Ida-Virumaa	(477.8)	Western Ukraine
(474.214)	Järvamaa	(477.81)	Rovenskaia oblast'
(474.215)	Raplamaa	(477.82)	Volynskaia oblast'
(474.216)	Läänemaa	(477.83)	L'vovskaia oblast'
(474.217)	Hiiumaa	(477.84)	Ternopol'skaia oblast'
(474.218)	Saaremaa	(477.85)	Chernovitskaia oblast'
(474.219)	Pärnumaa	(477.86)	Ivano-Frankovskaia oblast'
(474.220)	Viljandimaa	(477.87)	Zakarpatskaia oblast'
(474.221)	Valgamaa	(478)	MOLDOVA
(474.222)	Tartumaa	(479)	Caucasus as a whole <i>Including (470.62/.67) Northern Caucasus</i> ⇒ (57)
(474.223)	Jõgevamaa	(479.2)	Zakavkaz'e. Transcaucasia
(474.224)	Põlvamaa	(479.22)	Georgia. Republic of Georgia. Sakartvelo
(474.225)	Võrumaa	(479.223)	Adzharia <i>Formerly Adzharskaia A.S.S.R.</i>
(474.226)	Petserimaa (Setumaa) (historical)	(479.224)	Abkhazia <i>Formerly Abkhazskaia A.S.S.R.</i>
(474.3)	LATVIA. REPUBLIC OF LATVIA	(479.225)	South Ossetia <i>Formerly Osetinskaia avtonomnaia oblast'</i>
(474.5)	LITHUANIA. REPUBLIC OF LITHUANIA	(479.24)	Azerbaijan <i>Formerly Azerbaijan S.S.R.</i>
(475)	Poland: period of Russian domination, -1918 ⇒ (438)	(479.242)	Nakhichevan
(475.1)	Warszawa (Warsaw)	(479.243)	Nagorni Karabakh
(475.2)	Kalisz	(479.25)	Armenia. Republic of Armenia. Hayastan
(475.3)	Kielce	(48)	SCANDINAVIAN STATES ⇒ (4-17)
(475.4)	Lomza	(480)	FINLAND. REPUBLIC OF FINLAND. SUOMI
(475.5)	Lublin	(480*1/*9)	Traditional division: historical provinces 1 to 9 <i>See Suomi käsikirja (Handbook of Finland), Helsinki, 1968)</i>
(475.6)	Piotrkow (Petrokov)	(480*1)	Ahvenanmaa/Åland
(475.7)	Plock	(480*2)	Varsinais Suomi/Egentliga Finland
(475.8)	Radom	(480*3)	Uusimaa/Nyland
(475.91)	Suwalki	(480*4)	Satakunta/Satakunda (historical)
(475.92)	Siedlce	(480*41)	Satakunta/Satakunda (present)
(476)	BELARUS. REPUBLIC OF BELARUS	(480*42)	Pirkanmaa. Tammermaa
(476.1)	Minskaia oblast'	(480*5)	Häme/Tavastland (historical)
(476.2)	Gomel'skaia oblast'	(480*52)	Keski-Suomi/Mellersta Finland
(476.4)	Mogilevskaia oblast'	(480*6)	Savo/Savolax (historical)
(476.5)	Vitebskaia oblast'	(480*62)	Pohjois-Savo/Norra Savolax (present)
(476.6)	Grodnenskaia oblast'	(480*7)	Karjala/Karelen (Karelia, historical)
(476.7)	Brestskaia oblast'	(480*71)	Kymmenlaakso/Kymmenedalen
(477)	UKRAINE. UKRAINE	(480*72)	Etelä-Karjala/Södra Karelen (present)
(477.4)	Right Bank Ukraina. Central Ukraina	(480*73)	Pohjois-Karjala/Norra Karelen (present)
(477.41)	Kiyevskaia oblast'	(480*8)	Pohjanmaa/Osterbotten (historical)
(477.42)	Zhitomirskaia oblast'	(480*81)	Etelä-Pohjanmaa/Södra Osterbotten
(477.43)	Khmel'nitskaia oblast'	(480*82)	Keski-Pohjanmaa/Mellersta Osterbotten
(477.44)	Vinnitskaia oblast'	(480*83)	Pohjois-Pohjanmaa/Norra Osterbotten
(477.46)	Cherkasskaia oblast'	(480*84)	Kainuu/Kajanaland
(477.5)	Left Bank Ukraina	(480*9)	Lappi/Lappland (Finnish Lapland)
(477.51)	Chernigovskaia oblast'	(480.1/.9)	Administrative districts: läänit/läne
(477.52)	Sum'skaia oblast'	(480.1)	Uusimaa/Nyland. Helsinki/Helsingfors
(477.53)	Poltav'skaia oblast'	(480.2)	Turku ja Pori/Åbo och Björneborg
(477.54)	Khar'kov'skaia oblast'	(480.3)	Ahvenanmaa/Åland
(477.6)	Donets Basin		
(477.61)	Luganskaia oblast'		

(1/9) Table 1e. Common auxiliaries of place

(480.4)	Häme/Tavastehus	(485*25)	Norrbottn
(480.5)	Viipuri/Viborg, -1944	(485*51)	Häme (Etelä-Häme)/Tavastland (Södra Tavastland)(present)
(480.51)	Kymi/Kymmene, 1944-		
(480.6)	Mikkeli/St Michel	(485.6/.8)	Traditional division of Sweden: regions and provinces
(480.7)	Kuopio, -1960	(485.6)	Götaland
(480.71)	Kuopio, 1960-	(485.61)	Ostergötland
(480.75)	Pohjois-Karjala/Norra Karelen	(485.62)	Västergötland
(480.8)	Vaasa/Vasa, -1960	(485.63)	Dalsland
(480.81)	Vaasa/Vasa, 1960-	(485.64)	Småland
(480.87)	Keski Suomi/Mellersta Finland	(485.65)	Bohuslän
(480.9)	Oulu/Uleaborg, -1938	(485.66)	Halland
(480.91)	Oulu/Uleaborg, 1938-	(485.67)	Skåne
(480.99)	Lappi/Lappland (Finnish Lapland)	(485.68)	Blekinge
(481)	NORWAY. KINGDOM OF NORWAY <i>Primary administrative divisions: Fylker (five-figure subdivisions)</i>	(485.691)	Öland
		(485.692)	Gotland
(481.1)	Ostlandet	(485.7)	Svealand
(481.11)	Ostfold	(485.72)	Uppland
(481.12)	Akershus	(485.73)	Södermanland
(481.13)	Oslo	(485.74)	Västmanland
(481.14)	Hedmark	(485.75)	Närke
(481.15)	Oppland	(485.76)	Värmland
(481.16)	Buskerud	(485.77)	Dalarna
(481.17)	Vestfold	(485.8)	Norrland
(481.18)	Telemark	(485.81)	Gästrikland
(481.3)	Agder	(485.82)	Hälsingland
(481.31)	Aust-Agder	(485.83)	Medelpad
(481.32)	Vest-Agder	(485.84)	Ängermanland
(481.5)	Vestlandet	(485.85)	Jämtland
(481.51)	Rogaland	(485.86)	Härjedalen
(481.53)	Hordaland	(485.87)	Västerbotten
(481.54)	Bergen	(485.88)	Norrbottn
(481.55)	Sogn og Fjordane	(485.89)	Lappland (Swedish Lapland)
(481.56)	More og Romsdal	(489)	DENMARK. KINGDOM OF DENMARK ⇒ (491.2) ⇒ (988)
(481.6)	Trondelag	(489.1)	Sjælland og Lolland-Falster
(481.62)	Sor-Trondelag	(489.10)	Sjælland
(481.63)	Nord-Trondelag	(489.11)	Hovedstadsområdet
(481.7)	Nord-Norge	(489.111)	København
(481.71)	Nordland	(489.112)	Frederiksberg
(481.73)	Troms	(489.113)	Københavns amt
(481.74)	Finnmark	(489.115)	Frederiksborg amt
(485)	SWEDEN. KINGDOM OF SWEDEN	(489.117)	Roskilde amt
(485*01/*25)	Administrative division of Sweden: läne	(489.14)	Vestsjællands amt
(485*01)	Stockholm	(489.15)	Storströms amt
(485*03)	Uppsala	(489.151)	Sydsjælland
(485*05)	Ostergötland	(489.152)	Lolland
(485*06)	Jönköping	(489.153)	Falster
(485*07)	Kronoberg	(489.2)	Fyn. Fyns amt
(485*08)	Kalmar	(489.4/.6)	Jylland
(485*09)	Gotland	(489.4)	Nordjylland. Vestjylland
(485*10)	Blekinge	(489.43)	Nordjyllands amt
(485*11)	Kristianstad	(489.45)	Viborg amt
(485*12)	Malmöhus	(489.46)	Ringkøbing amt
(485*13)	Halland	(489.49)	Ribe amt
(485*14)	Göteborg och Bohus	(489.5)	Ostjylland
(485*15)	Älvsborg	(489.51)	Århus amt
(485*16)	Skaraborg	(489.53)	Vejle amt
(485*17)	Värmland	(489.6)	Sonderjylland. Sonderjyllands amt ◇ (489.6:210.7FRI) The North Frisian Islands
(485*18)	Örebro	(489.7)	Bornholm. Bornholms amt
(485*19)	Västmanland	(491.1)	ICELAND. REPUBLIC OF ICELAND <i>Denote Syslur (primary administrative divisions) by A/Z (Table 1h)</i> ◇ (491.1GUL) Gullbringusysla. Reykjavik
(485*20)	Kopparberg		
(485*21)	Gävleborg		
(485*22)	Västernorrland		
(485*23)	Jämtland		
(485*24)	Västerbotten		

(491.2)	FAROEES. THE FAROE ISLANDS.	(493.71)	Hasselt
	FAEROERNE/FOROYAR (DENMARK)	(493.72)	Maaseik
(492)	NETHERLANDS. KINGDOM OF THE	(493.73)	Tongeren/Tongres
	NETHERLANDS. KONINKRIJK DER	(493.781)	Enclave of Baarle-Hertog/Baarle-Duc, in Netherlands
	<i>Primary administrative divisions: Provinces</i>	(493.8)	Luxembourg/Luxemburg (province).
	<i>For further subdivision see Table 1h</i>		Ardennes/Ardennen. Belgian
	<i>example (492*...)</i>		Lorraine/Lotharingen
(492.61)	Zuid-Holland (South Holland)	(493.81)	Aarlen/Arlon
(492.62)	Noord-Holland (North Holland)	(493.82)	Bastogne/Bastenaken
(492.66)	Flevoland	(493.83)	Marche
(492.71)	Friesland	(493.84)	Neufchâteau
	◊ (492.71:210.7FRI) The West Frisian Islands	(493.85)	Virton
(492.72)	Groningen	(493.9)	Namur/Namen (province). Condroz.
(492.73)	Drenthe		Entre Sambre-et- Meuse/Land tussen
(492.74)	Overijssel		Samber en Maas. Marlagne
(492.82)	Gelderland	(493.91)	Dinant
(492.83)	Utrecht	(493.92)	Namur/Namen (arrondissement)
(492.91)	Zeeland	(493.93)	Philippeville
(492.93)	Noord-Brabant		
(492.94)	Limburg	(494)	SWITZERLAND. SWISS
(493)	BELGIUM. KINGDOM OF BELGIUM		CONFEDERATION
	<i>Provinces (four-figure subdivisions) and</i>		<i>Primary administrative divisions:</i>
	<i>arrondissements (five-figure subdivisions).</i>		<i>(Halb)Kanton</i>
	<i>Dual names: Dutch/French or</i>	(494.11/3)	German Switzerland
	<i>French/Dutch, according to predominant</i>	(494.11)	Schwyz
	<i>language of province</i>	(494.12)	Unterwalden
(493.1)	Antwerpen (province). Hageland	(494.121)	Unterwalden ob dem Wald (Obwalden)
(493.11)	Antwerpen/Anvers (arrondissement) (Antwerp)	(494.122)	Unterwalden nid dem Wald (Nidwalden)
(493.12)	Mechelen/Malines (Mechlin)	(494.13)	Uri
(493.13)	Turnhout	(494.21)	Appenzell
(493.2)	Brabant	(494.211)	Appenzell Inner-Rhoden
(493.21)	Brussel/Bruxelles (Brussels)	(494.212)	Appenzell Ausser-Rhoden
(493.22)	Leuven/Louvain	(494.22)	Aargau
(493.23)	Nijvel/Nivelles	(494.23)	Basel (canton)
(493.3)	West-Vlaanderen/Flandre Occidentale	(494.231)	Basel-Landschaft (half-canton)
	(West Flanders)	(494.232)	Basel-Stadt (half-canton)
(493.31)	Brugge/Bruges	(494.24)	Bern (canton)
(493.32)	Diksmuide/Dixmude	(494.25)	Glarus
(493.33)	Ieper/Ypres	(494.26)	Graubünden/Grisons/Grigioni/Grischun
(493.34)	Kortrijk/Courtrai	(494.27)	Luzern/Lucerne
(493.35)	Oostende/Ostende (Ostend)	(494.28)	Sankt Gallen
(493.36)	Roeselare/Roulers	(494.29)	Schaffhausen
(493.37)	Tielt	(494.31)	Thurgau
(493.38)	Veurne/Furnes	(494.32)	Solothurn
(493.4)	Oost-Vlaanderen/Flandre Orientale	(494.33)	Zug
	(East Flanders). Land van Waas/Pays	(494.34)	Zürich
	de Waes	(494.4)	French Switzerland
(493.41)	Aalst/Alost	(494.41)	Fribourg/Freiburg
(493.42)	Dendermonde/Termonde	(494.42)	Genève/Genf (Geneva)
(493.43)	Eeklo	(494.43)	Neuchâtel/Neuenburg (canton)
(493.44)	Gent/Gand (Ghent)	(494.44)	Valais/Wallis
(493.45)	Oudenaarde/Audenarde	(494.45)	Vaud/Waadt
(493.46)	Sint-Niklaas/Saint-Nicolas	(494.46)	Jura (canton)
(493.5)	Hainaut/Henegouwen. Borinage	(494.5)	Italian Switzerland. Ticino/Tessin
(493.51)	Ath/Aat	(494.9)	LIECHTENSTEIN. PRINCIPALITY OF
(493.52)	Charleroi		LIECHTENSTEIN
(493.53)	Mons/Bergen	(495)	GREECE. HELLENIC REPUBLIC
(493.54)	Soignies/Zinnik	(495.1)	Kentrikí Ellás kaí Évvoia (Central
(493.55)	Thuin		Greece and Euboea)
(493.56)	Tournai/Doornik	(495.11)	Attikí (Attica). □ Salamis. Aígina (Aegina)
(493.6)	Liège/Luik (province)	(495.12)	Fthiótis
(493.61)	Huy/Hoei	(495.13)	Aitolía kaí Akarnanía (Aetolia and Acarnania)
(493.62)	Liège/Luik (arrondissement)	(495.15)	Fokís
(493.63)	Verviers	(495.16)	Voiotía (Boeotia)
(493.64)	Waremmе/Borgworm	(495.17)	Evrytanía
(493.7)	Limburg/Limbourg.	(495.18)	Piraiéfs (Piraeus). □ Kythira
	Taxandrie/Taxandrie. Kempen/Campine	(495.19)	Évvoia (Euboea). □ Skyros

(1/9) Table 1e. Common auxiliaries of place

(495.2)	Peloponnísos (Peloponnese)	(496.526)	Lezhë
(495.21)	Argolis	(496.528)	Krujë
(495.22)	Achaía	(496.532)	Mat
(495.23)	Messinía	(496.534)	Tiranë
(495.24)	Arkadíá	(496.536)	Durrës
(495.25)	Lakonía	(496.538)	Lushnjë
(495.26)	Ilía	(496.542)	Elbasan
(495.27)	Korinthía	(496.544)	Librazhd
(495.3)	Thessalía	(496.546)	Pogradec
(495.31)	Lárisa	(496.548)	Gramsh
(495.32)	Tríkala	(496.552)	Berat
(495.34)	Kardítsa	(496.554)	Fier
(495.35)	Magnisía. □ Skiáthos. Skópelos. Halónnisos. Kyrá Panagía	(496.556)	Vlorë
(495.4)	Iónioi Nísoi (Ionian Islands)	(496.558)	Tepelenë
(495.41)	Kérkyra (Corfu). □ Páxos. Antípaxos	(496.562)	Skrapar
(495.43)	Lefkás (Santa Maura)	(496.564)	Korçë
(495.44)	Kefallinía (Cephalonia). □ Itháki	(496.566)	Kolonjë
(495.46)	Zákynthos	(496.568)	Përmet
(495.5)	Ipiros (Epirus)	(496.572)	Gjirokaštër
(495.51)	Ioánnina	(496.574)	Sarandë
(495.52)	Thesprotía	(497)	BALKAN STATES IN GENERAL.
(495.53)	Préveza		□ Dalmatia
(495.54)	Arta	(497.1)	YUGOSLAVIA (JUGOSLAVIJA).
(495.6)	Makedonía		FEDERAL REPUBLIC OF YUGOSLAVIA,
(495.611)	Flórina		-1991
(495.612)	Kozáni.		⇒ (497.11+497.16)
(495.613)	Kastoría	(497.11+497.16)	Serbia and Montenegro (Srbija i Crna Gora)
(495.614)	Grevená		<i>Known as Yugoslavia (Jugoslavija)</i>
(495.616)	Pélla		<i>1991-2003</i>
(495.617)	Imathía	(497.11)	Srbija (Serbia)
(495.618)	Piéria	(497.113)	Vojvodina (autonomous province)
(495.619)	Kilkís	(497.115)	Kosovo-Metohija (autonomous province)
(495.622)	Thessaloníki	(497.12)	Slovenia. Slovenija, -1991
(495.623)	Chalkidíki		⇒ (497.4)
(495.624)	Sérrai	(497.13)	Croatia. Hrvatska, -1991
(495.626)	Dráma		⇒ (497.5)
(495.627)	Kavála. □ Thásos	(497.15)	Bosnia and Herzegovina. Bosna i Hercegovina,
(495.631)	Hágion Oros		-1991
(495.7)	Thráki (Thrace)		⇒ (497.6)
(495.72)	Xánthi	(497.16)	Crna Gora (Montenegro)
(495.74)	Rodópi	(497.17)	Macedonia. Makedonija, -1991
(495.76)	Hévros. □ Samothráki		⇒ (497.7)
(495.8)	Nísoi Aigíou (Aegean Islands)	(497.2)	BULGARIA. REPUBLIC OF BULGARIA
(495.82)	Lésvos (Lesbos). □ Lésvos. Límnos. Hágios Efstrátios	(497.21)	Northern Bulgaria
(495.83)	Chíos. □ Chíos. Psará	(497.211)	Varna
(495.85)	Sámos. □ Sámos. Ikária	(497.212)	Vidin
(495.86)	Dodekánisos. □ Pátmos. Lipsói. Léros. Kálymnos. Kós. Astypalaía. Nisyros. Tilos. Chálki. Symi. Ródos. Kápathos. Kásos	(497.213)	Vratsa
(495.88)	Kykládes. □ Andros. Tínos. Kéa. Kythnos. Syros. Mykonos. Antíparos. Páros. Náxos. Mílos. Folégandros. Síkinos. Ios. Amorgós. Thíra (Santoríni). Anáfi	(497.214)	Pleven (Plevna)
(495.9)	Kríti (Crete)	(497.215)	Ruse (Ruschuk)
(495.91)	Chanía. □ Gávdos	(497.216)	Veliko Turnovo (Turnovo/Trnova)
(495.92)	Réthymnon	(497.217)	Shumen (Kolarovgrad/Sumla)
(495.93)	Hiráklion	(497.22)	Western Bulgaria
(495.94)	Lasíthion	(497.221)	Kyustendil (Keustendil)
(496.5)	ALBANIA. PEOPLE'S SOCIALIST REPUBLIC OF ALBANIA	(497.222)	Petrich
(496.512)	Tropojë	(497.223)	Sofiya (Sofia)
(496.514)	Shkodër	(497.23)	Southern Bulgaria
(496.516)	Pukë	(497.231)	Burgas
(496.518)	Kukës	(497.232)	Momchilgrad (Mastanli)
(496.522)	Dibër	(497.233)	Smolyan (Pashmakli)
(496.524)	Mirditë	(497.234)	Plovdiv (Philippopolis)
		(497.235)	Stara Zagora
		(497.236)	Khaskovo (Haskovo)
		(497.4)	SLOVENIA. REPUBLIC OF SLOVENIA.
			SLOVENIJA, 1991-
			<i>Slovenia is divided into 12 administrative</i>
			<i>(statistical) regions and 193 municipalities</i>
			<i>(source: 'Municipalities and statistical</i>
			<i>regions of Republic of Slovenia, 1 January</i>
			<i>2004' by Geodetska uprava Republike</i>

	<i>Slovenije in Statistični urad Republike Slovenije, http://www.stat.si/doc/pub/slo_figures_04.pdf, page 78)</i>	(497.524)	Međimurje. □ Čakovec (town)
	⇒ (497.12)	(497.525)	<i>Class here Medimurska županija</i>
(497.41)	Regions of north-eastern Slovenia	(497.525.1)	Podravina
(497.411)	Pomurska, Murska Sobota (town)	(497.525.2)	Koprivničko-križevačka županija. □ Koprivnica (town)
	<i>Class here: Prekmurje, Goričko</i>	(497.526)	Virovitičko-podravska županija. □ Virovitica (town)
(497.412)	Podravska, Maribor (town)	(497.527)	Bjelovarsko-bilogorska županija. □ Bjelovar (town)
	<i>Class here: Podravje, Prlekija</i>	(497.527.1)	Sisačko-moslavačka županija. □ Sisak (town)
(497.413)	Koroška, Slovenj Gradec (town)	(497.527.2)	Moslavina
(497.43)	Regions of eastern Slovenia	(497.528.3)	Pokuplje
(497.431)	Savinjska, Celje (town)	(497.529)	Banovina
(497.432)	Zasavska, Trbovlje (town)	(497.529.1)	Karlovačka županija. □ Karlovac (town)
	<i>Class here: Zasavje, Črni revir</i>	(497.54)	Kordun
(497.433)	Spodnjeposavska, Krško (town)		Istočna Hrvatska (Eastern Croatia).
(497.434)	Jugovzhodna Slovenija (Southeastern Slovenia), Novo Mesto (town)	(497.541)	Slavonija (Slavonia)
	<i>Class here: Dolenjska, Bela Krajina, Kočevska</i>	(497.542)	Zapadna Slavonija. □ Požega (town)
		(497.543)	<i>Class here Požeško-slavonska županija</i>
(497.45)	Regions of north-western and central Slovenia	(497.544)	Posavina. □ Slavonski-Brod (town)
(497.451)	Osrednjeslovenska (Central Slovenia)		<i>Class here Brodsko-posavska županija</i>
(497.451.1)	Ljubljana (state capital)	(497.56)	Baranja. □ Osijek (town)
(497.452)	Gorenjska, Kranj (town)	(497.561)	<i>Class here Osječko-baranjska županija</i>
		(497.562)	Zapadni Srijem. □ Vukovar (town)
(497.47)	Regions of western and south-western Slovenia		<i>Class here Vukovarsko-srijemska županija</i>
(497.471)	Notranjsko-kraška, Postojna (town)	(497.57)	Gorska Hrvatska (Mountain Croatia)
	<i>Class here: Notranjska</i>	(497.571)	Gorski Kotar. □ Rijeka (town)
(497.472)	Obalno-kraška, Koper/Capodistria (town)		<i>Class here Primorsko-goranska županija</i>
	<i>Class here: Primorska, Primorje, Kras</i>	(497.58)	Lika. □ Gospić (town)
(497.473)	Goriška, Nova Gorica (town)		<i>Class here Ličko-senjska županija</i>
	<i>Class here: Severna Primorska, Tolminska, Bovško, Idrijsko, Cerkljansko</i>	(497.581)	Istra (Istria). Hrvatsko primorje (Kvarner).
(497.5)	CROATIA. REPUBLIC OF CROATIA. HRVATSKA, 1991-	(497.581.1)	Kvarnerski otoci. □ Pazin (administrative centre). Pula (town)
	<i>Croatia is divided into 21 counties (županija) including Grad Zagreb; The following classification allows for the five major geographical areas: Central Croatia; Mountain Croatia; Istria and Kvarner; and Southern Croatian Coast, i.e. Dalmatia and Dalmatian Coast. Within these are the 20 traditional, functional regions, e.g. the subdivisions of (497.521); these are the focus of much literature. Finally, the 21 županija are allowed for, being the official administrative structure, though not of much documentary significance. Where these are coextensive with functional regions, they dealt with in 'class here' notes, as at (497.522); where several are contained within a region, they are enumerated, as in the subdivisions of (497.525); where a region is contained within a županija, the region is notationally subordinate, as at (497.529.1); finally, where there is no correspondence at all, the županija has its own class mark, as (497.526).</i>	(497.581.2)	Istra (Istria)
	⇒ (497.13)	(497.583)	<i>Class here Istarska županija</i>
(497.52)	Središnja Hrvatska (Central Croatia)	(497.584)	Hrvatsko primorje. Kvarner. Kvarnerski otoci (Kvarner islands)
(497.521)	Zagrebačka županija		⇒ (497.561)
(497.521.2)	Zagreb (capital). Grad Zagreb (City of Zagreb)	(497.585)	Južno Hrvatsko primorje (southern Croatian coast). Dalmacija (Dalmatia). Dalmatinski otoci (Dalmatian islands)
(497.521.3)	Žumberak	(497.581)	Sjeverna Dalmacija (Northern Dalmatia).
(497.521.5)	Prigorje	(497.581.1)	□ Sjevero-dalmatinski otoci (Northern Dalmatian islands)
(497.521.6)	Turopolje	(497.581.2)	Zadarsko-kninska županija. □ Zadar (town)
(497.522)	Hrvatsko zagorje. □ Krapina (town)	(497.583)	Šibenska županija. □ Šibenik (town)
	<i>Class here Krapinsko-zagorska županija</i>		Srednjsna Dalmacija (Central Dalmatia). □ Split (town). Srednje-dalmatinski otoci (Central Dalmatian islands)
(497.523)	Varaždinska županija. □ Varaždin (town)	(497.584)	<i>Class here Splitsko-dalmatinska županija</i>
			Južna Dalmacija (Southern Dalmatia).
		(497.6)	□ Dubrovnik (city). Južno-dalmatinski otoci (Southern Dalmatian islands)
			<i>Class here Dubrovačko-neretvanska županija</i>
		(497.7)	BOSNIA-HERZEGOVINA. REPUBLIC OF BOSNIA-HERZEGOVINA. BOSNA I HERCEGOVINA, 1991-
			⇒ (497.15)
		(497.7)	MACEDONIA. FORMER YUGOSLAV REPUBLIC OF MACEDONIA. MAKEDONIJA, 1991-
			⇒ (497.17)
		(498)	ROMANIA. REPUBLIC OF ROMANIA
		(498.1)	Muntenia (Great Wallachia)
		(498.11)	Ilfov. Bucuresti (Bucharest)
		(498.12)	Prahova. Ploesti
		(498.13)	Braila

(1/9) Table 1e. Common auxiliaries of place

(498.2)	Oltenia (Little Wallachia)	(511.31)	Jiangsu (Ciangsu, Kiangsu). □ Nanjing
(498.21)	Dolj (Doljiu). Craiova		(Nanching, Nanking). Shanghai
(498.3)	Moldava (Romanian Moldavia)	(511.32)	Anhui (Anhui). □ Anqing (Anching, Anking, Huaining)
(498.31)	Iasi	(511.33)	Jiangxi (Kiangsi). □ Nanchang
(498.32)	Covurlui. Galati	(512)	South-eastern China
(498.4)	Transylvania		⇒ (529)
	⇒ (439.21)	(512.1)	Min-Che
(498.41)	Maramures (Maramaros). Sighet	(512.11)	Fujian (Fuchien, Fukien). □ Fuzhou (Fuchow, Foochow)
(498.42)	Cojocna (Kolozs). Cluj	(512.12)	Zhejiang (Chechiang, Chekiang). □ Hangzhou (Hangchow)
	(Kolozsvár/Klausenburg)		
(498.43)	Bihor. Oradea (Nagyvarad/Grosswardein)	(512.2)	Hu region
(498.44)	Arad	(512.21)	Hubei (Hupei, Hupeh). □ Wuhan
(498.45)	Brasov (Brasso)	(512.22)	Hunan. □ Changsha
(498.46)	Sibiu (Szeben). Sibiu (Nagyszeben)	(512.3)	Guan (Kwang) region
(498.5)	Banat. Timis. Timisoara (Temesvar)	(512.31)	Guangdong (Kwangtung). □ Guangzhou (Kuangchow, Canton)
(498.6)	Bukovina, -1940. Southern Bukovina, 1940-	(512.311)	Hainan province
	⇒ (477.8)	(512.317)	Hong Kong, Special Administrative Region
(498.7)	Bessarabia, -1940	(512.318)	Macao (Macau), Special Administrative Region
	⇒ (478)	(512.32)	Guangxi Zhuang Zizhiqu (Kwangsi Chuang Autonomous Region)
(498.71)	Chisinau (now Kishinev)	(512.33)	Kwangchow Wang/Kouang Tchéou-wan (French, historical)
(498.72)	Akkerman (now Belgorod Dnestrovskiy)	(513)	South-western China
(498.8)	Dobrogea		
(498.81)	Danube Delta	(513.1)	Sichuan (Ssu-ch'uan, Szechwan)
(498.82)	Constanta	(513.2)	Yun-Gui (Yunkwei)
(5)	ASIA	(513.21)	Yunnan
	<i>Class here also Orient in general</i>	(513.22)	Guizhou (Kueichow)
(5-11)	Eastern Asia. Far East. □ (510) China.	(514)	North-western China
	(512.317) Hong Kong. (512.318) Macau.		⇒ (516)
	(517.3) Mongolia. (519.3) North Korea. (519.5) South Korea. (520) Japan. (529) Taiwan	(514.1)	Shaan-Gan (Shen-kan) region
(5-12)	South-east Asia. □ (591) Myanmar. (592.3) Singapore. (592.6) Brunei. (593) Thailand. (594) Indonesia. (595) Malaysia. (596) Cambodia. (597) Vietnam. (598) Laos. (599) Philippines	(514.11)	Shaanxi (Shensi)
		(514.12)	Gansu (Kansu)
(5-13)	South Asia. □ (540) India. (541.31) Bhutan. (541.35) Nepal. (548.82) Maldives. (548.7) Sri Lanka. (549.1) Pakistan. (549.3) Bangladesh	(515)	Xizang Zizhiqu. Tibet Autonomous Region
		(515.7)	Qinghai (Tsinghai, Koko Nor)
(5-15)	Western Asia. Near and/or Middle East. □ (532) Saudi Arabia. (536.2) United Arab Emirates. (533) Yemen. (535) Oman. (536.4) Qatar. (536.5) Bahrain. (536.8) Kuwait. (55) Iran. (564.3) Cyprus. (567) Iraq. (569.1) Syria. (569.3) Lebanon. (569.4) Israel. (569.5) Jordan	(516)	XinJiang Uygur Zizhiqu. Sinkiang Uighur Autonomous Region (Chinese Turkestan)
		(517)	Mongolia as a whole, -1924
(5-191.2)	Central Asia. □ (574) Kazakhstan. (575.1) Uzbekistan. (575.2) Kirghizia. (575.3) Tajikistan. (575.4) Turkmenistan. (581) Afghanistan	(517.3)	MONGOLIA. STATE OF MONGOLIA
		(517.3A/Z)	Provinces (aymags) of Mongolia. □ Arhangay. Bayanhongor. Bayan-Olgii. Bulgan. Darhan (autonomous municipality). Dornod. Dornogovi. Dundgovi. Dzavhan. Govi-Altay. Hentiy. Hovd. Hovsgol. Ovorangeay. Omnogovi. Selenge. Suhbaatar. Tov. Ulaanbaatar (Ulan-Bator; autonomous municipality). Uvs
(510)	CHINA. PEOPLE'S REPUBLIC OF CHINA	(517.4)	Nei Mongol Zizhiqu. Inner Mongolian Autonomous Region
	<i>Names of places are given primarily in Pinyin romanization (if available), followed in parentheses by other spellings with some currency</i>	(518)	Manchuria (historical). Manchukuo, 1931-1945
			<i>Now partly in Inner Mongolia (517.4), as well as provinces below</i>
(511)	North-eastern China		
	⇒ (518)	(518.1)	Liaoning
(511.1)	Hebei (Hopel)	(518.11)	Liaohsi (historical)
(511.12)	Beijing (Pei-ching/Peking)	(518.12)	Shenyang (Mukden)
(511.13)	Tianjin(T'ien-ching, Tientsin)	(518.14)	Liaodong area (Liaotung) (historical)
(511.2)	Former autonomous provinces	(518.15)	Lü-shun (Port Arthur) and Dalian (Talien, Luta, Dairen)
(511.21)	Shandong (Shantung). □ Jinan (Chinan)		
(511.218)	Weihai	(518.2)	Jilin (Chilin, Kirin)
(511.22)	Shanxi (Shansi). □ Taiyuan	(518.3)	Heilongjiang (Heilungkiang)
(511.23)	Henan (Honan). □ Kaifeng		
(511.3)	Klang-nan (historical)		

(519)	KOREA	(522.8)	Kagoshima-ken
(519.3)	NORTH KOREA. PEOPLE'S DEMOCRATIC REPUBLIC OF KOREA	(523)	Shikoku-chiho
(519.3A/Z)	Provinces of North Korea. □ Chagang-do. Ch'ongjin-si. Hamgyong-namdo. Hamgyong-pukdo. Hamhung-si. Hwanghae-namdo. Hwanghae-pukdo. Kaesong-chigu. Kangwon-do. P'yongan-namdo. P'yongan-pukdo. P'yongyang-si. Yanggang-do	(523.1)	Tokushima-ken
		(523.2)	Kagawa-ken
		(523.4)	Ehime-ken
		(523.5)	Kochi-ken (Koti-ken)
(519.5)	SOUTH KOREA. REPUBLIC OF KOREA	(524)	Hokkaido (Yezo)
(519.5A/Z)	Provinces (and Special Cities) of South Korea. □ Cheju-do. Cholla-namdo. Cholla-pukdo. Ch'ungch'ong-namdo. Ch'ungch'ong-pukdo. Kangwon-do. Kyonggi-do. Kyongsang-namdo. Kyongsang-pukdo. Inchon-si. Kwangju-si. Pusan-si. Soul-t'ukpyolsi (Seoul). Taegu-si	(524.11)	Hiyama
		(524.12)	Oshima (Osima)
		(524.21)	Shiribeshi-yama (Siribesi-yama)
		(524.22)	Iburi
		(524.31)	Ishikari (Isikari)
		(524.32)	Sorachi (Sorati)
		(524.41)	Kamikawa
		(524.42)	Rumoi (Rumoe)
		(524.51)	Soya
		(524.52)	Abashiri (Abasiri)
		(524.61)	Hidaka
		(524.62)	Tokachi (Tokati)
		(524.71)	Kushiro (Kusiro)
		(524.72)	Nemuro
		(524.728)	Habomai and Shikotan (Sikotan)
(52)	JAPAN AND ADJACENT ISLANDS	(527)	Island territories north of Japanese mainland, -1945
(520)	JAPAN. NIPPON (NIHON KOKU)	(527.1)	Kuriles (Chishima-retto), -1945 ⇒ (571.64)
(521)	Honshu (Honsyu/Hondo)	(527.5)	Karafuto ⇒ (571.64)
(521.1)	Tohoku (Ou-tiho/Ou-chiho)	(528)	Island territories south of Japanese mainland
(521.11)	Aomori-ken	(528.1)	Ogasawara-gunto (Bonin Islands). Kazan-retto (Volcano Islands)
(521.12)	Iwate-ken	(528.3)	Ryukyu Islands
(521.13)	Miyagi-ken	(529)	Taiwan. Republic of China
(521.14)	Akita-ken	(529.1)	Taiwan (Formosa)
(521.15)	Yamagata-ken	(529.8)	P'eng-hu Lieh-tao (Pescadores). Bokoto
(521.16)	Fukushima-ken (Hukusima-ken)	(53)	ARABIAN STATES AND TERRITORIES
(521.2)	Kanto-chiho	(532)	SAUDI ARABIA. KINGDOM OF SAUDI ARABIA
(521.22)	Ibaraki-ken		<i>Saudi Arabia is divided into thirteen manatiq (mintagah, sing.) (regions or emirates), each with a capital. Historically, it was divided into four provinces which roughly correspond with the four regional subdivisions below</i>
(521.23)	Tochigi-ken (Totigi-ken)	(532.2)	Northwest region. Al Hijaz (Hejaz) (historical)
(521.24)	Gumma-ken (Gunma-ken)	(532.21)	Makkah (emirate), Mecca (capital). □ Jiddah (Jedda), Taif. Qishran, Sirrayn, Abu Latt
(521.25)	Saitama-ken	(532.22)	Al Madina (emirate), Medina (capital), Medina Salah (archaeological site)
(521.27)	Tokyo-to	(532.23)	Tabuk (emirate and regional capital), Tiran, Sanafir, Al Hasani, Shaybara, Mashabih
(521.28)	Kanagawa-ken	(532.25)	Al Jawf (emirate), Sakaka (capital)
(521.29)	Chiba-ken (Tiba-ken)	(532.26)	Al Hudud ash Shamaliyah (emirate), Ar'ar (capital)
(521.3)	Chubu-chiho (Tyubu-tiho)	(532.27)	Ha'il (emirate and regional capital)
(521.4)	Hokuriko-chiho	(532.5)	Southern region (formerly Asir province)
(521.41)	Niigata-ken	(532.51)	'Asir (emirate), Abha (capital)
(521.42)	Toyama-ken	(532.52)	Al Bahah (emirate), Bagha (capital)
(521.43)	Ishikawa-ken (Isikawa-ken)	(532.53)	Jizan (emirate and regional capital), Farasan Islands
(521.44)	Fukui-ken (Hukui-ken)		
(521.5)	Tosan-chiho		
(521.51)	Yamanashi-ken (Yamanasi-ken)		
(521.52)	Nagano-ken		
(521.53)	Gifu-ken (Gihu-ken)		
(521.6)	Tokai-chiho		
(521.61)	Shizuoka-ken (Sizuoka-ken)		
(521.62)	Aichi-ken (Aiti-ken)		
(521.7)	Kinki-chiho		
(521.71)	Mie-ken (Miye-ken)		
(521.72)	Shiga-ken (Siga-ken)		
(521.73)	Kyoto-fu (Kyoto-hu)		
(521.74)	Osaka-fu (Osaka-hu)		
(521.75)	Hyogo-ken		
(521.76)	Nara-ken		
(521.77)	Wakayama-ken		
(521.8)	Chugoku-chiho (Tyugoku-tiho)		
(521.81)	Tottori-ken		
(521.82)	Shimane-ken (Simane-ken)		
(521.83)	Okayama-ken		
(521.84)	Hiroshima-ken (Hiroshima-ken)		
(521.85)	Yamaguchi-ken (Yamaguti-ken)		
(522)	Kyushu (Kyusu)		
(522.1)	Fukuoka-ken (Hukuoka-ken)		
(522.2)	Nagasaki-ken		
(522.3)	Saga-ken		
(522.5)	Kumamoto-ken		
(522.6)	Oita-ken		
(522.7)	Miyazaki-ken		

(532.6)	Central region. Najd (Nejd) (historical). North and central Arabian deserts. Badiyah ash Sham (Syrian Desert). An Nafud. Jabal Shammar. Desert regions of As Summan and Ad Dahna	(535.3)	Adh Dhahirah (Adh Dhahirah, A'Dhahirah, Az Zahirah), Ibri (capital)
(532.61)	Ar Riyad (emirate), Riyadh (state and regional capital)	(535.4)	Ad Dakhliyah (A'Dakhiya, Ad Dakhiliyah, Al Jauf, Al Joof, Dakhliya, Interior), Nizwa (capital)
(532.62)	Al Qasim (emirate), Buraidah (capital)	(535.5)	Ash Sharqiyah (Al Hajar, A'Shariqiyah, Eastern), Sur (capital), Masirah, Mahawt
(532.63)	Najran (emirate and regional capital)	(535.6)	Al Batinah, Sohar, Rustaq (capital)
(532.7)	Eastern region	(535.7)	Muscat (Masqat, Muscat and Matrah), Muscat (state and regional capital). Barka. Mutrah. Ruwi
(532.72)	Ash Sharqiyah, Dammam (capital), Al 'Arabiyyah, Abu 'Ali, Al Batinah, Tarut, Harqus, Karan, Kurayn, Al Jurayd. Al Ahsa (El Hasa), Dhahran. Al Hufuf (Hofuf). Al Bayad (Biyadh). □ Jabal Tuwayq	(535.8)	Musandam (Mussandam, Ru'us al-Jibal), Khasab (capital)
(533)	YEMEN. REPUBLIC OF YEMEN. AL-JAMHURIYA AL-YAMANIYA, 1990- <i>Yemen is divided into muhafazah (governates), except for San'a (city). The governates are subdivided into districts. Class here also Yemen (North); Yemen Arab Republic (to 1990); Yemen, Democratic; People's Democratic Republic of Yemen (to 1990)</i>	(536)	EAST ARABIAN STATES (GULF STATES)
(533.11/23)	Northern governates	(536.2)	UNITED ARAB EMIRATES. ITTIHAD AL-IMARAT AL-ARABIYAH. □ Abu Dhabi. Ajman. Dubai. Fujaira. Sharja. Umm al Qaiwain. Ras al Khaimah (member from 1972), 1971-
(533.11)	Sa'd ah (Sa'sah, Sa'ada, Saidah), Sa'dah (capital)		<i>Divided into emirates. Each emirate capital has the same name as the emirate. Formerly Trucial Coast/Trucial Oman/Trucial States</i>
(533.12)	Hajjah (Wayyah), Hajjah (capital)	(536.21)	Abu Dhabi (Abu Zabi, Abu Zaby). Abu al Abyad, Sir Bani Yas, Dalma, Sir Abu Nu'ayr
(533.13)	Al Mahwit, Al Mawhit (capital)	(536.22)	Dubay (Dubai, Dubayy). Deira <i>Dubay has one large section, plus a small enclave next to Oman</i>
(533.14)	Amran ('Amran)	(536.23)	Sharjah (Ash Shariqah, Ash Sharjah, Sharjah and Dependencies, Sharjah and Kalba) <i>Sharjah consists of a large section on the Persian Gulf, plus two enclaves on the Gulf of Oman (one of which is partly shared with Fujayrah). It shares the administration of Abu Musa Island with Iran</i>
(533.15)	San'a' (Sana'a), San'a' (capital)		
(533.16)	San'a' (city and state capital)	(536.24)	Ajmān ('Ajman) <i>Ajmān consists of three enclaves. The one containing the capital lies on the Persian Gulf; one is on the border between Fujayrah and Sharjah, and the last is partly shared with Oman</i>
(533.17)	Al Jawf, R'aydah (capital)		
(533.18)	Ma'rib, Ma'rib (capital)	(536.25)	Umm al Qaywayn (Qaiwan, Umm al Qaiwan, Umm al Qayqayn, Umm al Qaywayn, Umm el Quwain)
(533.19)	Al Hudaydah (Al Hodeida), Al Hudaydah (capital)	(536.26)	Ras al Khaymah (Ra's al Khayma, Ra's al Khaymah) <i>Ras al Khaymah consists of two sections separated by Fujayrah. It claims the Tunb Islands in the Persian Gulf</i>
(533.21)	Dhamar, Dhamar (capital)	(536.27)	Fujayrah (Al Fujayrah, Fujaira, Fujairah, Fujeira). □ Kalba, Khor Fakkan <i>Fujayrah has two sections. The one containing the capital is partly shared with Sharjah</i>
(533.22)	Al Bayda' (A Beida), Al Bayda (capital)		
(533.23)	Ta'izz (Ta'iz), Ta'izz (capital)		
(533.24)	Ibb, Ibb (capital)		
(533.25)	al-Dali (Ad Dali', al-Dhalea, Dala'a, Dhali), al-Dali (capital)		
(533.4)	Southern governates	(536.4)	QATAR. STATE OF QATAR. DAWLAT QUATAR <i>Qatar is divided into baladiyah (baladiyah, sing.) (municipalities). Qatar occupies a peninsula in the Persian Gulf, and neighbouring reefs and islets. It disputes the Hawar Islands with Bahrain. If Qatar's claim is valid, the Hawar Islands are in Al Jumaliyah municipality</i>
(533.41)	'Aden, Aden (capital). □ Crater. Socotra (Suqutra), 'Abd al Kuri, The Brothers (Samha and Darsa). Perim. Kamaram <i>From April 1st 1937 until 30th November 1967 Aden was a Crown Colony of the UK. Prior to 1937 it was administered by the Presidency of Bombay, India</i>		
(533.42)	Lahij (Lahej). □ Lahej (capital)		
(533.43)	Abyan, Zinzibar (capital)		
(533.44)	Shabwah (Shabwa), 'Ataq (capital)		
(533.45)	Hadramawt (Hadhramaut, Hadramawt, Hadramaut), Al Mukalla (capital)		
(533.46)	Al Mahrah (Mahrah), Al Ghaydah (capital)		
(535)	OMAN. SULTANATE OF OMAN. SALTANAT 'UMAN <i>Oman is divided into mintaqat (mintaqaq, sing.) (regions), or muhafazat (muhafazah, sing.) (provinces). Below the regions or provinces it is subdivided into wilayat (governorates), which have steadily increased in number since 1977, numbering 59 in 1996</i>		
(535.1)	Dhofar (Dhufar, Al Janubiyah, Southern Region, Zufar) Salalah (capital). Kuria Muria Islands, Halaniyat	(536.41)	Madinet ach Shamal (Ash Shamal)
(535.2)	Al Wusta (Central Oman, Oman Proper, Rub al Khali, 'Uman al-Wusta), Haima (capital)	(536.42)	Al Ghuwariyah
		(536.43)	Al Khawr (Al Khor)
		(536.44)	Al Jumaliyah (Al Gummaylah, Al Jumayliyah). Dukhan. □ Hawar Islands (disputed) ⇒ (536.552)
		(536.45)	Umm Salal (Um Salal, Umm Salal)

(536.46)	Ar Rayyan	(540.37)	Arrunachal Pradesh
(536.47)	Ad Dawhah (Doha) (state capital)	(540.39)	Nagaland
(536.48)	Jariyan al Batnah (Jarayan al Batna, Jerian al Batna)	(540.41)	Manipur
(536.49)	Al Wakrah. Umm Said	(540.43)	Mizoram
(536.5)	BAHRAIN. STATE OF BAHRAIN. DAWLAT AL-BAHRAIN <i>Bahrain is divided into manatiq (mintaqah, sing.) (regions). It consists entirely of islands, the largest being Bahrain itself. Near it are Al Murarra, Sitrah und Umm Nasan, all connected by causeways. Just off the coast of Qatar is Hawar Island and some neighbouring islets. Qatar also claims this group. All municipalities are administered from Manama</i>	(540.45)	Tripura
(536.51)	Northern governorate	(540.47)	Orissa
(536.511)	Al Mintaqah ash Shamaliyah	(540.49)	Madhya Pradesh
(536.512)	Jidd Hafs (Judd Hafs), Jidd Hafs (capital)	(540.51)	Gujarat
(536.52)	Capital governorate. Al Manamah (Manamah), Manama (state and regional capital)	(540.53)	Maharashtra
(536.53)	Muharraq governorate	(540.55)	Goa
(536.531)	Al Muharraq, Al Muharraq (capital)	(540.57)	Daman and Diu (territory)
(536.532)	Al Hadd (Hedd, Hidd), Al Hadd (capital)	(540.59)	Dadra and Nagar Haveli (territory)
(536.54)	Central governorate	(540.61)	Karnataka
(536.541)	Al Mintaqah al Wusta	(540.63)	Andhra Pradesh
(536.542)	Madinat 'Isa	(540.65)	Tamil Nadu
(536.543)	Sitrah (Sitra), Sitrah (capital)	(540.67)	Pondicherry (territory)
(536.55)	Southern governorate	(540.69)	Kerala. □ Malabar Coast
(536.551)	Ar Rifa' wa Mintaqah al Janubiyah (Rifaa, Rufaa, Rifa'a), Ar Rifa al Gharbi (capital)	(540.71)	Lakshadweep
(536.552)	Mintaqat Juzur Hawar (Huwar Islands, Howar Islands) (disputed) ⇒ (536.44)	(540.73)	Andaman and Nicobar Islands (territory)
(536.553)	Al Mintaqah al Gharbiyah	(541)	EASTERN INDIA AND NEIGHBOURING STATES
(536.554)	Madinat Hamat	(541.2)	Bengal (historical) ⇒ (549.3)
(536.8)	KUWAIT. STATE OF KUWAIT. DAWLAT AL-KUWAIT <i>Kuwait is divided into muhafazah (muhafazat, sing.) (provinces). The capitals have the same names as the provinces</i>	(541.31)	Bhutan. Kingdom of Bhutan. Druk-yul Administrative subdivisions of Bhutan. □ Bumthang. Chirang. Chukha. Dagana. Gasa/Punakha. Gayleghug. Ha. Lhuntsi. Mongar. Paro. Pema Gatsel. Samchi. Samdrup. Jonghkar. Shemgang. Tashigang. Thimphu. Tongsa. Wangdiphodrang
(536.81)	Al Jahrah. Bubiyan, Warbah Islands	(541.31A/Z)	
(536.82)	Al Farwaniyah	(541.35)	Nepal. Kingdom of Nepal Provinces of Nepal. □ Bagmati. Bheri. Dhaulagiri. Gandaki. Janakpur. Karnali. Kosi. Lumbini. Mahakali. Mechi. Narayani. Rapti. Sagarmatha. Seti
(536.83)	Al Kuwayt, Kuwait (state and province capital). Faylakah, Maskan, Awhah Islands	(541.35A/Z)	
(536.84)	Hawalli. As Salimya	(543)	CENTRAL INDIA
(536.85)	Al Ahmadi	(544)	Madhya Bharat. Rajasthan ⇒ (540.25)
(54)	STATES OF THE INDIAN SUBCONTINENT ◇ (54-076) Disputed territory in the Indian subcontinent. Including: Kashmir region	(544.63)	Jaipur
(540)	INDIA. REPUBLIC OF INDIA	(544.64)	Jodhpur
(540.11/.73)	States and Union Territories of India	(544.65)	Bikaner
(540.11)	Jammu and Kashmir	(544.8)	Ajmer-Merwara
(540.13)	Himachal Pradesh	(545)	NORTHERN INDIA
(540.15)	Punjab	(545.2)	East Punjab (historical) <i>Now divided into Punjab and Haryana. West Punjab is part of Pakistan (549.1)</i> ⇒ (540.15) ⇒ (540.21)
(540.17)	Meghalaya	(545.3)	Patiala and East Punjab States Union (PEPSU) (historical)
(540.19)	Chandigarh	(547)	WESTERN INDIA
(540.21)	Haryana	(547.1)	Mumbai State (Bombay State). □ Presidency of Bombay (historical) ⇒ (540.51) ⇒ (540.53)
(540.23)	Delhi (territory)	(548)	SOUTHERN INDIA AND ADJACENT ISLANDS Andhra Desha
(540.25)	Rajasthan	(548.7)	SRI LANKA. DEMOCRATIC SOCIALIST REPUBLIC OF SRI LANKA <i>Sri Lanka is divided into nine provinces, subdivided into 25 districts which are the primary administrative divisions. Subordinate to the districts there are pradeshiya mandalaya (divisional council) and pradeshiya mandalaya (village council).</i>
(540.27)	Uttar Pradesh		
(540.29)	Bihar		
(540.31)	Sikkim		
(540.33)	West Bengal		
(540.35)	Assam		

(1/9) Table 1e. Common auxiliaries of place

	<i>Since the late 19th century both palata (provinces) and districts have co-existed, with each province being a group of districts. The provinces had no administrative function until 1988. The Northern and Eastern provinces were temporarily merged into a North Eastern province, intended to placate Tamil demands for greater self-rule. The latest ISO 3166-2, December 15 1998, shows the Northern and Eastern as separate provinces</i>	(549.16)	Islāmābād Capital Territory. □ Islāmābād (national capital). Rawalpindi
		(549.17)	Punjab. □ Lahore (provincial capital).
		(549.18)	Bahāwalpur. Faisalābād. Multan. Sargodha
		(549.19)	Balochistān (Baluchistan). □ Quetta (provincial capital). Gwādar. Zhob
		(549.3)	Sindh (Sind). □ Karāchi (provincial capital). Sukkur. Hyderābād
(548.71)	Northern province	(549.32)	BANGLADESH. PEOPLE'S REPUBLIC OF BANGLADESH. GANA
(548.711)	Jaffna (provincial capital)	(549.33)	PRAJA-TANTRI BANGLADESH
(548.712)	Kilinochchi		Rājshāhi Division. □ Rājshāhi. Bogra. Rangpur
(548.713)	Mullaitivu	(549.34)	Dhaka Division. □ Dhaka (formerly Dacca), national capital. Mymensingh. Nārāyanganj
(548.714)	Mannar	(549.35)	Sylhet Division. □ Sylhet. Habiganj
(548.715)	Vavuniya		Chittagong Division. □ Chittagong. Comilla. Cox's Bāzār
(548.72)	North Central province	(549.37)	Barisāl Division. □ Barisāl. Patuākhāli
(548.721)	Anuradhapura (provincial capital)	(549.39)	Khulna Division. □ Khulna. Jessore
(548.722)	Polonnaruwa	(55)	IRAN. ISLAMIC REPUBLIC OF IRAN. PERSIA (HISTORICAL), 1979-
(548.73)	North Western province	(551)	Irak-Ajmi. Tehran. Esfahan (Isfahan)
(548.731)	Puttalam (provincial capital)	(552)	Taberistan. Amol
(548.732)	Kurunegala	(553)	Mazandaran. Sari. Gorgan (Gurgan/Asterabad)
(548.74)	Central province	(554)	Gilan. Rasht (Resht)
(548.741)	Matale	(555)	Azarbaijan (Azerbaijan). Tabriz (Tauris)
(548.742)	Kandy (provincial capital). □ Peradeniya	(556)	Kordestan (Kurdistan). Kermanshah
(548.743)	Nuwara Eliya	(558)	Fars (Farsistan). Shiraz. Takht-e Jamshid (Persepolis)
(548.75)	Eastern province		Kerman
(548.751)	Trincomalee (provincial capital)	(559.1)	Khuzestan. Abadan
(548.752)	Batticaloa	(559.2)	Khorasan. Mashhad (Meshed). Persian
(548.753)	Ampara	(559.3)	Turkestan (historical)
(548.76)	Western province	(56)	LEVANT. ASIA MINOR
(548.761)	Gampaha. □ Negombo	(560)	TURKEY. REPUBLIC OF TURKEY
(548.762)	Colombo (state and provincial capital). □ Dehilawa, Mount Lavinia	(560.1)	Marmara Region
(548.763)	Kalutara. □ Beruwela	(560.11)	European Turkey
(548.77)	Sabaragamuwa province	(560.112)	Edirne
(548.771)	Kegalle	(560.114)	Kirklareli
(548.772)	Ratnapura (provincial capital)	(560.116)	Tekirdag
(548.78)	Uva province	(560.118)	Istanbul
(548.781)	Badulla (provincial capital)	(560.122)	Kocaeli (Izmit)
(548.782)	Moneragala. □ Katagarama, Tissamaharama	(560.124)	Sakarya
(548.79)	Southern province	(560.126)	Bilecik
(548.791)	Galle (provincial capital). □ Unawatuna	(560.128)	Bursa
(548.792)	Hambantota	(560.132)	Balikesir
(548.793)	Matara. □ Tangalla	(560.134)	Canakkale
(548.82)	MALDIVES. REPUBLIC OF THE MALDIVES. □ Aliff. Baqa. Daalu. Faafu. Gaafu Aliff. Gaafu Daalu. Haa Aliff. Haa Daalu. Kaafu. Laamu. Laviyani. Male. Meemu. Naviyani. Noonu. Raa. Seenu. Shaviyani. Thaa. Waavu	(560.2)	Aegean Region
	<i>The Maldive Islands are a series of atolls, apart from Male, which is a city. The atolls have no administrative centres. If necessary, each atoll could be considered its capital</i>	(560.212)	Izmir (Smyrna)
(549)	PAKISTAN, 1947-1971	(560.214)	Aydin
(549.1)	PAKISTAN. ISLAMIC REPUBLIC OF PAKISTAN. ISLAM-I JAMHURIYA-E PAKISTAN, 1971-	(560.216)	Mugla
	◇ (549.1-076) Disputed territory under Pakistan's control. Including: Azad Kashmir.	(560.221)	Denizli
	Northern Areas	(560.223)	Manisa
(549.14)	North-West Frontier Province. □ Peshāwar (provincial capital)	(560.225)	Usak
(549.15)	Federally Administered Tribal Areas	(560.227)	Kütahya
		(560.229)	Afyonkarahisar
		(560.3)	Southern Anatolia Region
		(560.312)	Burdur
		(560.314)	Isparta
		(560.316)	Antalya
		(560.322)	Icel
		(560.324)	Adana
		(560.326)	Kahramanmaras
		(560.328)	Hatay
		(560.4)	Central Anatolia Region
		(560.412)	Cankiri
		(560.414)	Ankara
		(560.416)	Eskisehir

(560.418)	Konya	(567)	IRAQ. REPUBLIC OF IRAQ. AL JUMHURIYAH AL 'IRAQIYAH
(560.422)	Nigde		<i>Iraq is divided into 18 muhāfazāt (muhafazah, sing.) (provinces), which are enumerated here with their provincial capitals and some other major towns. Some alternative spellings are given, but variant names that refer to the ancient world, e.g. Babylon (now Babil), Nineveh (now Ninawa) are no longer included, as they are provided for under (3)</i>
(560.424)	Nevsehir		
(560.426)	Kirsehir		
(560.428)	Yozgat		
(560.432)	Kayseri		
(560.434)	Sivas		
(560.6)	Black Sea Region		
(560.612)	Bolu	(567.11)	Arbil. □ Ar Ramadi (provincial capital). Arbil (Irbil). Rayat
(560.614)	Zonguldak	(567.13)	As Sulaymaniyah. □ As Sulaymaniyah (provincial capital). Qal'at Dizah. Halabjah
(560.616)	Kastamonu	(567.15)	At Ta'mim. □ Kirkuk (Karkuk) (provincial capital)
(560.618)	Corum	(567.17)	Salah ad Din. □ Tikrit (provincial capital). Baiji (Baiji). Samarra
(560.622)	Sinop	(567.19)	Diyala. □ Ba'quba (provincial capital). Khanaqin. Mandali
(560.624)	Samsun	(567.21)	Baghdad. □ Baghdad (national and provincial capital)
(560.626)	Amasya	(567.23)	Babil. □ Al Hillah (provincial capital)
(560.628)	Tokat	(567.25)	Wasit. □ Al Kut (provincial capital). Al Hayy
(560.632)	Ordu	(567.27)	Al Qadisiyah. □ Ad Diwaniyah (provincial capital)
(560.634)	Giresun	(567.29)	Dhi Qar. □ An Nasiriyah (Nasiriya) (provincial capital)
(560.636)	Gümüşhane	(567.31)	Maysan. □ Al 'Amarah (provincial capital)
(560.638)	Trabzon (Trebizonde)	(567.33)	Al Basrah. □ Al Basrah (Basra) (provincial capital). As Zubayr. Umm Qasr
(560.642)	Rize	(567.35)	Al Muthanna. □ As Samawah (provincial capital). As Salman. Al Busayyah
(560.644)	Artvin	(567.37)	An Najaf. □ An Najaf (Najaf) (provincial capital)
(560.7)	Eastern Anatolia Region	(567.39)	Karbala'. □ Karbalā' (provincial capital). Al Hindiya
(560.712)	Malatya	(567.41)	Al Anbar. □ Ar Rāmadi (provincial capital). Al Qa'im. Haditha. Akashat. Ar Rutbah. Al Habbaniyah. An Nukhayb
(560.714)	Elazig	(567.43)	Ninawa. □ Al Mawsil (Mosul) (provincial capital). 'Aqrah. Sinjar
(560.716)	Tunceli	(567.45)	Dahuk. □ Dahuk (Dihok) (provincial capital). Zakho (Zakhu)
(560.718)	Erzincan		
(560.722)	Bingöl		
(560.724)	Erzurum		
(560.726)	Mus		
(560.728)	Bitlis		
(560.732)	Hakkari		
(560.734)	Van		
(560.736)	Agri		
(560.738)	Kars		
(560.8)	South Eastern Anatolia Region		
(560.812)	Gaziantep		
(560.814)	Adiyaman		
(560.816)	Sanliurfa		
(560.822)	Diyarbakir		
(560.824)	Mardin		
(560.826)	Siirt		
(564.3)	CYPRUS. REPUBLIC OF CYPRUS. KYPRIAKI DIMOKRATIA/KIBRIS ÇUMHURİYETİ		
	<i>Since 1974 the island of Cyprus has been partitioned. The larger part, in the south, is controlled by the Republic of Cyprus. The north constitutes the self-proclaimed Turkish Republic of Northern Cyprus (TRNC) which is recognized only by Turkey. The district of Kyrenia is entirely within the TRNC. Famagusta, Larnarca and Nicosia all have sections which lie within the TRNC. Two small coastal enclaves of Cyprus are British military bases, sovereign soil of the United Kingdom. One of them is surrounded by Limassol, the other borders on Famagusta and Larnaca. The border between the Greek and Turkish sectors of Cyprus is a buffer strip, administered by the United Nations. The administrative divisions are districts and the capitals have the same name as their districts</i>	(569.1)	SYRIA. SYRIAN ARAB REPUBLIC. AL-JUMHURIYA AL-ARABIYA AS-SURIYA
(564.31)	Paphos. □ Latsi. Polis		<i>Syria is divided into muhāfazāt (muhafazah, sing.) (provinces). The capitals have the same names as their provinces, except that Damascus is the capital of both Damascus and Rif Dimashq. The provinces are divided into mantika (districts) which are further subdivided into nahia (subdistricts)</i>
(564.32)	Nicosia (state capital). □ Kato Lakatamia	(569.11)	Northern provinces
(564.33)	Limassol. □ Akrotiri. Episkopi. Vasilikos	(569.111)	Idlib
(564.34)	Larnaca. □ Dhekelia. Protaras	(569.112)	Aleppo (Haleb, Halab). □ Ebla (archaeological site)
(564.35)	Famagusta. □ Agia Napa. Paralimni	(569.113)	Ar Raqqah (Al Rashid, Al Rakka, Raqqa)
(564.36)	Kyrenia. □ Morfou. Rizokarpaso	(569.114)	Al Hasakah (El Haseke, Hassakeh, Hazakieh, Al Jezira, Jezireh)
		(569.12)	Mediterranean (coastal) region
		(569.121)	Latakia (Al Lādhiqiyyah, El Ladhqiya, Lattakia, Lattakieh). □ Ras Shamra (Ugarit) (archaeological site)
		(569.122)	Tartus (Tartous)
		(569.13)	Central provinces
		(569.131)	Hamah (Hama). □ Apamea (archaeological site)
		(569.132)	Hims (Homs). □ Qala'at al-Hosn (Krak/Crac des Chevaliers) (archaeological site)
		(569.133)	Dayr az Zawr (Dayr az Zaur, Deir ez Zor). □ Palmyra (archaeological site)

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(571.64)	Sakhalinskaia oblast'	(575.32)	Gorno-Badakhskanskaya avtonomnaya oblast'.
(571.65)	Magadanskaya oblast'	(575.34)	Pamir
(571.651)	Chukotskii avtonomnyi okrug	(575.35)	Kulyabskaya oblast'
(571.651.8)	Ostrov Vrangelya (Wrangel Island)		Kurgan-T'yubinskaya oblast'
(571.66)	Kamchatskaya oblast'	(575.4)	TURKMENISTAN
	◊ (571.66-25) Petropavlovsk-Kamchatskii		<i>Formerly Turkmenistanskaya S.S.R.</i>
(571.661)	Koriakskii natsional'nyi okrug	(575.41)	Mariyskaya Oblast'
(574)	KAZAKHSTAN. REPUBLIC OF KAZAKHSTAN	(575.42)	Tashauzskaya oblast'
(574.1)	Western Kazakhstan	(575.43)	Chardzhouskaya oblast'
(574.11)	Ural'skaya oblast'	(575.44)	Ashkhabadskaya oblast'
(574.12)	Gur'yevskaya oblast'	(575.45)	Krasnovodskaya oblast'
(574.13)	Aktyubinskaya oblast'	(581)	AFGHANISTAN. ISLAMIC STATE OF AFGHANISTAN (AFGHANISTAN. DOWLAT-E ESLAMI-YE AFGHANISTAN)
(574.14)	Mangyshlakskaia oblast'		<i>Also referred to as Islamic Republic of Afghanistan and Islamic Emirate of Afghanistan</i>
(574.2)	Northern Kazakhstan	(581.1/.2)	Northern provinces (velayat)
(574.21)	Kustanayskaya oblast'	(581.11)	Bādghis. □ Qal'eh-ye Now
(574.22)	Severo-Kazakhstanskaya oblast'	(581.13)	Fāryāb. □ Meymaneh
(574.23)	Kokchetavskaya oblast'	(581.15)	Sar-e Pol. □ Sar-e Pol (town). Tokzar
(574.24)	Tselinogradskaya oblast'	(581.17)	Jowzjān. □ Sheberghān. Aqchah. Keleft
(574.25)	Pavlodarskaya oblast'	(581.19)	Balkh. □ Mazār-e Sharif. Jeyretān. Kholm
(574.26)	Turgayskaya oblast'	(581.21)	Samangan. □ Samangān (town)
(574.3)	Central Kazakhstan	(581.23)	Kondoz (Kundūz). □ Kondūz
(574.31)	Karagandinskaya oblast'	(581.25)	Baghlan. □ Baghlān (town). Pol-e Khomri
(574.32)	Dzhezkazganskaya oblast'	(581.27)	Takhār. □ Tāloqān
(574.4)	Eastern Kazakhstan	(581.28)	Badakhshān. □ Feyzābād. Eshkāshem
(574.41)	Semipalatinskaya oblast'	(581.3/.4)	Provinces (velayat) of the central mountainous region
(574.42)	Vostochno-Kazakhstanskaya oblast'	(581.31)	Herat. □ Herāt (town). Shindand. Towraghondī
(574.5)	Southern Kazakhstan	(581.32)	Ghowr (Ghor). □ Chagcharān
(574.51)	Alma-Atinskaya oblast'	(581.33)	Bāmīān. □ Bāmīān (town). Sayghan
(574.52)	Dzhambul'skaya oblast'	(581.34)	Parvān. □ Chārīkār
(574.53)	Chimkentskaya oblast'	(581.35)	Kapisa. □ Mahmūd-e Rāqī (Raqi)
(574.54)	Kzyl-Ordinskaya oblast'	(581.36)	Laghman. □ Mehtarlām
(574.57)	Taldy-Kurganskaya oblast'	(581.37)	Konar (Konarha). □ Asadābād
(575)	FORMER SOVIET CENTRAL ASIA	(581.38)	Vardak (Wardak). □ Kowt-e 'Ashrow
(575.1)	UZBEKISTAN	(581.41)	Lowgar. □ Barakī
(575.11)	Syr'-Dar'inskaya oblast'. Syr-Darya	(581.42)	Kābol. □ Kābul (provincial and national capital)
(575.111)	Tashkentskaya oblast'	(581.43)	Nangarhar. □ Jalālābād
(575.112)	Syrdar'inskaya oblast'	(581.44)	Orūzgān. □ Tarīn Kowt. Oruzgan (town)
(575.114)	Dzhizakskaya oblast'	(581.45)	Ghaznī. □ Ghaznī (town)
(575.12)	Fergana	(581.46)	Paktiā. □ Gardeyz. Khowst (Khost).
(575.121)	Ferganskaya oblast'		Jaji-Maydan
(575.122)	Andizhanskaya Oblast'	(581.48)	Paktīkā. □ Zareh Sharan. Orgūn. Tani
(575.123)	Namanganskaya oblast'	(581.5)	Southern provinces (velayat)
(575.14)	Zeravshan River. Zeravshan River Delta	(581.51)	Farāh. □ Farāh (town). Anar Darreh
(575.141)	Samarkandskaya oblast'	(581.53)	Nīmruz. □ Zaranj. Delārām
(575.144)	Navinskaya oblast'	(581.55)	Helmand. □ Lashkar Gāh
(575.146)	Bukharskaya oblast'	(581.57)	Kandahār (Qandahār). □ Kandahār
(575.15)	Middle Amu Darya region		(Qandahār) (town)
(575.151)	Surkhandar'inskaya oblast'	(581.59)	Zābol (Zābul). □ Qalāt
(575.152)	Kashkadar'inskaya oblast'	(591)	MYANMAR (BURMA). UNION OF MYANMAR. MYANMAR NAINGGANDAW
(575.17)	Lower Amu Darya region		<i>Myanmar is divided into taing (divisions or states). Generally speaking, states are semi-autonomous areas allocated to particular ethnic groups. The divisions and states are subdivided into layaing (districts) and substates. These are further subdivided into townships and villages; The districts have been listed together with other places of interest within a specific state. The problems of romanization and changed names are especially intense for Myanmar,</i>
(575.171)	Khorezmskaya Oblast'		
(575.172)	Karakalpakstan		
	<i>Formerly Karakalpakskaya A.S.S.R.</i>		
(575.2)	KYRGYZSTAN. REPUBLIC OF KYRGYZSTAN		
(575.21)	Narynskaya oblast'		
	<i>Tyan'-Shan'skaya/Tien-Shan Oblast', 1939-62</i>		
(575.22)	Oshskaya oblast'		
(575.23)	Issyk-Kul'skaya oblast'		
(575.24)	Talasskaya oblast'		
(575.3)	TAJIKISTAN. REPUBLIC OF TAJIKISTAN		
	<i>Formerly Tadzhikskaya S.S.R.</i>		
(575.31)	Leninabadskaya oblast'		

(1/9) Table 1e. Common auxiliaries of place

together with a lack of accurate detailed information, rendering it difficult to assign all places of interest to their correct district, so no attempt has been made to go beyond state of division level. There are countless changes of name and orthography, and only the principal differences have been noted			
(591.1)	North (formerly Upper Burma)	(593)	THAILAND. KINGDOM OF THAILAND
(591.11)	Kachin (state), Myitkyina (capital). □ Bhamo, Putao (districts)		Thailand (Siam) is divided into five unofficial but standardized regions. Each region is made up of changwats (provinces) except for Bangkok Metropolis, which is a municipality (maha nakhon). The capitals have the same name as their provinces. Sometimes the capital is given the generic prefix 'Muang'. The provinces are divided into amphoes (amphurs: districts) and king amphoes (subdistricts). These are in turn subdivided into tambons (communes) which are further subdivided into moobans (villages). In the early part of the 20th century there were groupings of provinces called monthons (administrative circles)
(591.12)	Sagaing (division), Sagaing (capital). □ Katha, Lower Chindwin, Shwebo, Upper Chindwin (districts)		
(591.13)	Chin (state), Haka (capital). □ Northern Chin Hills, Pakokku Hill Tracts, Southern Chin Hills (districts)		
(591.14)	Magwe (division), Magwe (capital). □ Minbu, Pakokku, Thayetmyo (districts). Bagan	(593.1)	Northern region
(591.15)	Mandalay (division), Mandalay (capital). □ Amarapura, Kyaukse, Meiktila, Myingyan, Yamethin (districts)	(593.111)	Mae Hong Son. □ Mae Sariang, Pai
(591.16)	Shan (state), Taunggyi (capital). □ Eastern, Northern, Palaung, Southern (districts). Hsipaw. Kalaw. Kengtung (Kyaingtong). Lahio. Mamgton. Namsang. Nyaungshwe (Yaungshwe). Pyin U Lwin (Maymyo)	(593.112)	Chiang Rai. □ Mae Sai, Mae Salong
(591.17)	Rakhine (Rakhaing) (state), Akyab (Sittwe) (capital). □ Kyaukpyu. Sandoway (Thandwe) (districts). Boronga (Baronga) Islands, Cheduba Islands, Ramree, Ye Kyun	(593.113)	Phayao
(591.18)	Kayah (Karenni) (state), Loi-kaw (capital). □ Bawlake, Kantarawaddy, Kyebogyi, Mong Pai (districts)	(593.114)	Chiang Mai. □ Chiang Khong, Chiang Saen, Chom Thong, Fang, Tha Tong
(591.3)	South (formerly Lower Burma)	(593.115)	Nan. □ Doi Phu National Park, Nong Bua
(591.31)	Ayeyarwaddy (Irrawaddy) (division), Bassein (capital). □ Bassein, Henzada, Ma-ubin, Myaungmya, Pyapon (districts). Great Coco. Little Coco. Prepara. Table Island	(593.116)	Lamphun (Hariphunchai, Lampoon)
(591.32)	Yangon (Rangoon) (division), Yangon (state and division capital)	(593.117)	Lampang
(591.33)	Bago (Pegu) (division), Pegu (capital). □ Hanthawaddy, Insein, Prome, Tharrawaddy (districts)	(593.118)	Phrae
(591.34)	Kayin (Karen, Kawthule) (state), Pa-an (capital). □ Salween (district)	(593.119)	Uttaradit
(591.35)	Mon (state), Moulmein (Mawlamyaing) (capital). Kyaikmaraw. Bilugyun Island	(593.121)	Tak. □ Mae Sot, Um Phang
(591.36)	Tanintharyi (Tenasserim) (division), Tavoy (Dawei) (capital). □ Amherst (Kyaikkami), Mergui (Myeik), Thaton, Toungoo (districts)	(593.122)	Sukhotai (Sawankhalokt, Sukhotai, Sukothai, Szargalok). □ Si Satchanalai-Chaliang
(592.3)	SINGAPORE. REPUBLIC OF SINGAPORE	(593.123)	Historical park (archaeological site)
(592.31)	Jurong		Phitsanulok (Bisnulok, Phisanulauk, Phitsnulok, Pitsanuloke). □ Phu Hin Rong Kla National Park
(592.33)	Bukit Panjang	(593.124)	Kamphaeng Phet (Gampheang Phet, Kambhengbhej)
(592.34)	Serangoon	(593.125)	Phichit
(592.36)	City	(593.126)	Phetchabun
(592.38)	Katong	(593.127)	Uthai Thani
(592.4)	Christmas Island (Australia)	(593.128)	Nakhon Sawan
(592.5)	⇒ (948)		
(592.6)	Cocos Islands (Keeling Islands) (Australia)		
	⇒ (948)		
(592.6+594.1+595.41/.43)	BORNEO	(593.3)	Northeastern region
(592.6)	BRUNEI. STATE OF BRUNEI DARUSSALAM	(593.311)	Nong Khai. □ Beung Kan, Sangkhom
(592.61)	Belait	(593.312)	Loei. □ Chiang Khan, Nam Nao National Park, Phu Luang National Park, Phu Reua National Park
(592.63)	Tutong		
(592.65)	Brunei Muara	(593.313)	Nong Bua Lam Phu
(592.67)	Temburong	(593.314)	Udon Thani. □ Ban Chiang
		(593.315)	Sakon Nakhon
		(593.316)	Nakhon Phanom. □ That Phanom
		(593.317)	Mukdahan
		(593.318)	Chaiyaphum
		(593.319)	Khon Kaen. □ Chonabot
		(593.321)	Kalasin
		(593.322)	Maha Sarakham
		(593.323)	Roi Et
		(593.324)	Yasothon
		(593.325)	Amnat Charoen
		(593.326)	Nakhon Ratchasima (Khorat, Nagara Sridharmaraj, Nakhon Sri Thammarat, Nakhonsrithamrat, Nakornsrihamaraj). □ Khao Yai National Park
			Buri Ram
			Surin
			Si Sa Ket (Khukhan, Khukhandh, Srisaket)
			Ubon Ratchathani
		(593.5)	Central region
		(593.511)	Chai Nat
		(593.512)	Lop Buri

(593.513)	Kanchanaburi (Kan Buri, Park Prag). □ Sai Yok National Park, Sangkhlaburi, Si Sawat, Thong Pha Phum	(593.821)	Satun. □ Ko Tarutao, Butong Group of islands, Ko Khao Yai, Ko Bulon Lae, Ko Laoling
(593.514)	Suphan Buri	(593.822)	Songkhla. □ Hat Yai, Ko Yo, Ko Nu, Ko Maeo Islands
(593.515)	Sing Buri	(593.823)	Pattani
(593.516)	Saraburi	(593.824)	Yala. □ Betong
(593.517)	Ang Thong	(593.825)	Narathiwat. □ Sungai Kolok
(593.518)	Nakhon Pathom	(594)	INDONESIA. REPUBLIC OF INDONESIA
(593.519)	Phra Nakhon Si Ayutthaya (Ayudhya, Ayutthaya, Phranakhomsri-ayuthaya)	(594.1)	Provinces of Kalimantan (the Indonesian part of Borneo)
(593.521)	Pathum Thani	(594.11)	Kalimantan Timur (East Kalimantan, East Borneo)
(593.522)	Nakhon Nayok	(594.13)	Kalimantan Selatan (South Kalimantan, South Borneo). □ Sebuksu
(593.523)	Prachin Buri	(594.15)	Kalimantan Tengah (Central Kalimantan, Central Borneo)
(593.524)	Sa Kaeo (Sa Kaew). □ Aranya Prathet	(594.17)	Kalimantan Barat (West Kalimantan, West Borneo). □ Padangtikar. Maya. Karimata
(593.525)	Ratchaburi	(594.2)	Provinces of Sulawesi (Celebes)
(593.526)	Samut Songkhram (Mae Klong, Samudrasonggram)	(594.21)	Sulawesi Utara (North Sulawesi, North Celebes). □ Kep. Talaud. Kep. Sangihe
(593.527)	Samut Sakhon (Krokrak Maha Chai, Mahachai, Samudrasagara)	(594.23)	Sulawesi Tengah (Central Sulawesi, Central Celebes). □ Kep. Togian. Peleng. Banggai
(593.528)	Nonthaburi	(594.25)	Sulawesi Tenggara (South-east Sulawesi, South-east Celebes). □ Wowoni. Butung.
(593.529)	Bangkok Metropolis (Krung Threp, Krung Thep Maha Nakhon, Phra Nakhon-Thonburi) (state capital)	(594.27)	Kabaena. Kep. Tukanbesi
(593.531)	Samut Prakan (Pak Nam, Samudh Prakarn, Samudraprakar, Samut Phrakhan)	(594.31)	Sulawesi Selatan (South Sulawesi, South Celebes). □ Salayar. Kep. Sabalana
(593.532)	Chachoengsao (Chaxerngsao, Pad Rew, Paed Piu, Petrieu, Shajeun Dhras)		Maluku (Moluccas). □ Morotai. Halmahera.
(593.533)	Phetchaburi (Bejraburi, Petchburi, Phet Buri, Tetchburit). □ Cha-am, Kaeng Krachan National Park		Kep. Bacan. Obi. Kep. Sula. Mangole. Sanana.
(593.534)	Prachuap Kiri Khan (Korh Luxk, Prachuab Girikhand, Prachuapkirikhan, Prachut Khirikhan). □ Bang Saphan, Hua Hin, Khao Sam Roi Yot National Park, Thap Sakae		Buru. Seram. Seram. Ambon. Kep. Kai. Kep. Aru. Kep. Tanimbar. Kep. Leti. Romang. Wetar
(593.6)	Eastern region	(594.4)	Provinces of Sumatera (Sumatra), and Special District
(593.61)	Chon Buri. □ Naklua, Pattaya, Si Racha. Ko Si Chang, Ko Khram, Ko Pai Islands	(594.41)	Aceh (Atjeh). □ Simeulue
(593.62)	Rayong. □ Ko Samet Island	(594.42)	Sumatera Utara (North Sumatra). □ Nias
(593.63)	Chanthaburi (Chantaburi, Muang Chan)	(594.43)	Sumatera Barat (West Sumatra). □ Siberut.
(593.64)	Trat. □ Ko Chang, Ko Kut, Ko Mak Islands <i>Ko Chang should not be confused with the island of the same name in Ranong province (593.812)</i>	(594.44)	Sipora. Pagai Selatan
(593.8)	Southern region		Riau. □ Kep. Bunguran. Kep. Anambas. Kep. Tambelan. Kep. Badas. Kep. Riau. Lingga.
(593.811)	Chumphon	(594.46)	Sinkep. Rangsang. Padang. Rupat
(593.812)	Ranong. □ Kawthaung, Saphaan Plaa. Laem Son National Park. Ko Phayam, Ko Chang Islands <i>Ko Chang should not be confused with the town of the same name in Trat province (593.64)</i>	(594.47)	Jambi
(593.813)	Surat Thani. □ Chaiya, Khao Sok National Park. Ko Samui (island), with Hat Chaweng, Hat Lamai, Na Thon (towns). Ko Phangan (island), with Thong Sala (town). Ko Tao. Ko Nangyuan. Ko Ang. Thong Islands	(594.48)	Sumatera Selatan (South Sumatra).
(593.814)	Phangnga. □ Khuraburi, Takua, Thai Muang. Ko Phra Thong, Ko Kho Kao, Similan (Muu Ko) Islands	(594.49)	□ Bangka. Belitung
(593.815)	Phuket. □ Karon, Kata, Laem Kha, Nai han, Nai Thon, Patong, Rawai, Surin. Ko Phuket, Ko Yao Yai, Ko Yao Noi, Ko Sire, Ko Kaeo	(594.5)	Benkulu
(593.816)	Pitsadan Islands	(594.51)	Lampung
(593.817)	Krabi. □ Khao Phanom Bencha National Park, Than Bokkharani National Park. Ko Lanta, Phi Phi Don, Phi Phi Le Islands	(594.53)	Provinces of Jawa (Java)
(593.817)	Nakhon Si Thammarat. □ Khao Luang National Park	(594.55)	Jakarta Raya
(593.818)	Trang (Tab Tiang). □ Ko Libong, Ko Muk Islands	(594.57)	Jawa Barat (West Jawa)
(593.819)	Phatthalung	(594.59)	Jawa Tengah (Central Jawa)
			Yogyakarta
			Jawa Timur (East Jawa). □ Bawean. Madura.
			Kep. Kangean
			Bali
		(594.61)	Provinces of the Lesser Sunda Islands
		(594.7)	Nusa Tenggara Barat (Western Lesser Sunda Islands). □ Lombok. Sumbawa
		(594.71)	Nusa Tenggara Timur (Eastern Lesser Sunda Islands). □ Flores. Kepulauan Solor.
		(594.73)	Kepulauan Alor. Sumba. Suwu. Roti. Tengora Timur
		(594.81)	Irian Jaya
		(594.9)	TIMOR-LESTE. DEMOCRATIC REPUBLIC OF TIMOR-LESTE (FORMERLY EAST TIMOR/ TIMOR LOROSA'E). □ Dili (capital city)

(1/9) Table 1e. Common auxiliaries of place

(595)	MALAYSIA. FEDERATION OF MALAYSIA	(596.37)	Mondol Kiri (Môndôl Kiri, Mondolkiri, Môndul Kiri) (province), Senmonorom (provincial capital)
(595.1)	Federal territories (wilayah persekutuan) and states (negeri) of Malaysia	(596.4)	Southern Cambodia
(595.11)	Melaka (Malacca)	(596.41)	Kaoh Kong (Kaôh Kong, Koh Kong) (province), Krong Koh Kong (provincial capital)
(595.12)	Pulau Pinang dan Seberang Perai	(596.42)	Kampot (province and provincial capital). □ Bokor (Preah Monivong) National Park, Thmei Island, Tonsay Island, Set (Sês) Island
(595.21)	Perak	(596.43)	Krong Preah Sihanouk (Kompong Som) (province), Sihanoukville (provincial capital). □ Ream National Park
(595.22)	Pahang	(596.44)	Krong Keb (autonomous municipality). □ Koh Tonsay (Rabbit Island)
(595.23)	Selangor	(596.45)	Kandal (province), Ta Khmau (provincial capital)
(595.231)	Wilayah Persekutuan Putrajaya (state capital)	(596.46)	Takev (Takêv, Takeo) (province and provincial capital). □ Ankor Borei (archaeological site)
(595.232)	Wilayah Persekutuan Kuala Lumpur	(597)	VIETNAM. SOCIALIST REPUBLIC OF VIETNAM
(595.24)	Negeri Sembilan		<i>Vietnam is divided into tỉnh (provinces), except for Da Nang, Haiphong, Hanoi en Ho Chi Minh which are thu do (formerly called thành phố; municipalities). Each is further subdivided into divisions. There has been considerable change in the structure of these administrative divisions in recent years, and full details of the present position, with maps, can be found at http://www.angelfire.com/co/hongnam/vnmap.html; The country has been divided into seven Economic Regions by the UNDP and these are used for the primary grouping. See: http://www.undp.org.vn/dmu/reference/en/provinces.htm</i>
(595.31)	Johor		
(595.32)	Terengganu (Trengganu)		
(595.33)	Kelantan		
(595.34)	Kedah		
(595.35)	Perlis		
(595.41)	Sabah		
(595.42)	Wilayah Persekutuan Labuan		
(595.43)	Sarawak		
(596/598)	INDO-CHINA		
(596)	CAMBODIA. KINGDOM OF CAMBODIA <i>Formerly Kampuchea, Khmer Republic Cambodia is divided into khét (provinces) and two autonomous municipalities, Krong Pailin and Phnom Penh</i>		
(596.1)	Northwestern Cambodia		
(596.11)	Banteay Meanchey (Bântéay Méanchey, Banteay Mean Cheay) (province), Sisophon (provincial capital). □ Poipet	(597.11)	North Mountain and Midland Region
(596.12)	Otdar Meanchey (Otdâr Méanchey, Oddâr Méanchey) (province), Samraong (provincial capital)	(597.111)	Provinces of North Mountain and Midlands Region
(596.13)	Siemreab (Siemréab, Siem Reap, Siem Réab, Siemreap) (province and provincial capital). □ Angkor (archaeological site)	(597.111.1)	Lai Chau (province and provincial capital). □ Dien Bien Phu
(596.14)	Prear Vihear (province), Phnum Tbêng Méanchey (provincial capital). □ Choam Ksant, Koh Ker (archaeological site)	(597.111.2)	Lao Cai (province and provincial capital). □ Bac Ha. Sa Pa
(596.15)	Bătdămbang (Bătdămbăng, Bat Dâm Băng) (province and provincial capital). □ Sneng	(597.111.3)	Ha Giang (province and provincial capital)
(596.16)	Krong Pailin (autonomous municipality)	(597.111.4)	Cao Bang (province and provincial capital)
(596.2)	Central Cambodia	(597.111.5)	Tuyen Quang (province and provincial capital)
(596.21)	Pouthisât (Pouthisat, Pursat) (province and provincial capital). □ Kompong Luong	(597.111.6)	Bac Can (province and provincial capital)
(596.22)	Kampong Chhnang (province and provincial capital)	(597.111.7)	Lang Son (province and provincial capital)
(596.23)	Kampong Thum (Kâmpóng Thum, Kompong Thom) (province and provincial capital)	(597.111.8)	Thai Nguyen (province), Song Cong (provincial capital)
(596.24)	Kampong Spœ (Kâmpóng Spœ, Kompong Speu) (province and provincial capital)	(597.111.9)	Vinh Phuc (province), Vinh Yen (provincial capital)
(596.25)	Phnom Penh (Phnum Pénh) (autonomous municipality and state capital)	(597.112.1)	Bac Giang (province and provincial capital)
(596.3)	Eastern Cambodia	(597.112.2)	Son La (province and provincial capital)
(596.31)	Kampong Cham (Kâmpóng Chhnăng, Kompong Chhang) (province and provincial capital). □ Memot	(597.112.3)	Yen Bai (province and provincial capital)
(596.32)	Prey Veng (province and provincial capital)	(597.112.4)	Phu Tho (province and provincial capital)
(596.33)	Svay Rieng (province and provincial capital)	(597.112.5)	Hoa Binh (province and provincial capital). □ Mai Chau
(596.34)	Kracheh (Krâchéh, Kratié, Kratiê, Krâ Chéh, Krachen) (province and provincial capital). □ Snuol	(597.112.6)	Bac Ninh (province and provincial capital)
(596.35)	Stoeng Treng (Stœng Trêng, Stung Treng) (province and provincial capital)	(597.112.7)	Quang Ninh (province), Hong Gai (Halong) (provincial capital). □ Bai Chay (Ba Che), Mong Cai
(596.36)	Rotanokiri (Rôtânôkiri, Râtânâkiri, Rotanah Kiri) (province), Bân Lung (provincial capital). □ Lumphat (former provincial capital)	(597.12)	Red River Delta Region
		(597.121)	Municipalities and provinces of the Red River Delta Region
		(597.121.1)	Hanoi (state capital and municipality)
		(597.121.2)	Ha Tay (province), Ha Dong (provincial capital). □ Son Tay
		(597.121.3)	Hung Yen (province and provincial capital)
		(597.121.4)	Hai Duong (province and provincial capital)
		(597.121.5)	Haiphong (municipality)

(597.121.6)	Ha Nam (province and provincial capital)	(597.172.3)	Tra Vinh (province and provincial capital)
(597.121.7)	Thai Binh (province and provincial capital)	(597.3)	South Vietnam. Republic of Vietnam (to 1975). Cochinchina (historical)
(597.121.8)	Ninh Binh (province), Ninh Binh Phan (provincial capital)	(597.5)	Annam (historical) <i>From 1954 to 1975, divided between North Vietnam and South Vietnam (597.3)/(597.7). Reunited from 1976 (597)</i>
(597.121.9)	Nam Dinh (province and provincial capital)	(597.7)	North Vietnam. Democratic Republic of Viet-Nam (to 1975). Tonkin (historical)
(597.13)	North Central Coast Region	(598)	LAOS. LAO PEOPLE'S DEMOCRATIC REPUBLIC <i>Laos is divided into khueng (provinces), except for Vientiane municipality, which is a kamping nakhon (municipality or prefecture). The secondary administrative divisions are districts. Several districts, in different provinces, have the same name, e.g. Viengkham, Viengthong, Xatsetha</i>
(597.131)	Thanh Hoa (province and provincial capital)		
(597.132)	Nghe An (province), Vinh (provincial capital)		
(597.133)	Ha Tinh (province and provincial capital)		
(597.134)	Quang Binh (province), Dong Hoi (provincial capital)		
(597.135)	Quang Tri (province), Dong Ha (provincial capital)		
(597.136)	Thua Thien-Hue (province), Hué (provincial capital). □ Nam Dong		
(597.14)	South Central Coast Region		
(597.141)	Da Nang (municipality)		
(597.142)	Quang Nam (province), Tam Ky (provincial capital). □ Hoi An, My Son		
(597.143)	Quang Ngai (province and provincial capital). □ Sa Huynh. Son My (Mai Lai)	(598.1)	Northern and Northwestern region
(597.144)	Binh Dinh (province), Qui Nhon (provincial capital)	(598.11)	Phongsali (Fong Sali, Phongsaly) (province), Phongsaly (district and provincial capital). □ Boon Neua (Booneua), Boontai, Khua, May, Nhot Ou, Samphanh (districts)
(597.145)	Phu Yen (province), Tuy Hoa (provincial capital)	(598.12)	Luangnamtha (Luang Namtha, Muong Luang Namtha, Namtha) (obsolete names: Hiua khong, Houa Khong, Upper Mekong) (province), Luang Namtha (district and provincial capital). □ Long, Nalae, Sing, Viengphoukha (districts)
(597.146)	Khanh Hoa (province), Nha Trang (provincial capital)		
(597.147)	Ninh Thuan (province), Phan Rang-Thap Cham (provincial capital)	(598.13)	Bokeo (province), Ban Houayxay (Huaixai, Huoixai) (district and provincial capital). □ Meung, Nam You, Paktha, Pha Oudom, Tonpheung (districts)
(597.148)	Bin Thuan (province), Phan Thiet (provincial capital). □ Ca Na, Mui Ne		
(597.15)	Central Highlands Region	(598.14)	Oudomxai (Oudomsai, Oudomsay, Oudomxay, Udom Xay) (province), Muang Xay (district and provincial capital). □ Beng, Hoon (Houn), La, Namor, Nga, Pakbeng, Xai (districts)
(597.151)	Kon Tum (province and provincial capital)		
(597.152)	Gia Lai (province), Pleiku (Play Ku) (provincial capital). □ Ban Don	(598.15)	Xaignabouri (Sayabouri, Xainyabuli, Xayabouri, Xayabury) (province), Muang Xayabury (Sayaboury) (district and provincial capital). □ Botene, Hongsa, Kenethao, Khop (Khorb), Ngeun, Parklai (Paklai), Phiang, Thongmyxay, Xienghone (districts)
(597.153)	Dac Lac (province), Buon Ma Thuot (provincial capital)		
(597.154)	Lam Dong (province), Da Lat (provincial capital)		
(597.16)	North East South Region	(598.2)	Central region
(597.161)	Tay Ninh (province and provincial capital)	(598.21)	Louangphrabang (Luang Phabang, Louangphrabang Luang Prabang) (province), Louangphrabang (district and provincial capital). □ Chomphet, Nambak, Nan, Ngoi, Pak Xeng, Park Ou (Pak Ou), Phonxay, Phoukhoun, Viengkham, Xieng Ngeun (districts)
(597.162)	Bin Phuoc (province), Dong Phu (provincial capital)		
(597.163)	Binh Duong (province), Thu Dau Mot (provincial capital)	(598.22)	Houaphan (Hua Phan, Huaphanh) (obsolete names: Sam Neua, Xam Nua) (province), Sam Neua (provincial capital). □ Huameuang, Muang Et, Sop Bao, Viengthong, Viengxai, Xamneua, Xamtay, Xiengkhor (districts)
(597.164)	Dong Nai (province), Bien Hoa (provincial capital)	(598.23)	Xiangkhoang (Xiang Khouang, Xieng Khouang) (province), Xiangkhoang (provincial capital). □ Kham, Khoun, Morkmay (Morkmai), Nonghed, Pek, Phaxay, Phookood, (districts). Phonsavan
(597.165)	Ho Chi Minh City (municipality) (formerly Saigon)		
(597.166)	Ba Ria-Vung Tau (province), Vung Tau (provincial capital). □ Con Dao Islands	(598.24)	Vientiane (Viangchan) (province), Muang Phôn-Hông (provincial capital). □ Feuang, Hinhurp, Kasy, Keo Oudom, Mad, Thoulakhom, Vangvieng, Viengkham, Xanakharm (Xanakhom) (districts)
(597.17)	Mekong River Delta Region	(598.25)	Vientiane (Kamphaeng Nakhon Viang Chan) (prefecture and state capital). □ Chanthabuly (Chanthabouly), Hadxaifong (Hadxaifong),
(597.171)	Provinces of the Mekong River Delta Region		
(597.171.1)	An Giang (province), Long Xuyen (provincial capital). □ Chau Doc		
(597.171.2)	Dong Thap (province), Sa Dec (provincial capital). □ Cao Lanh		
(597.171.3)	Long An (province), Tan An (provincial capital)		
(597.171.4)	Kien Giang (province), Rach Gia (provincial capital). □ Ha Tien. Phu Quoc, Rai, Tre Islands		
(597.171.5)	Can Tho (province and provincial capital)		
(597.171.6)	Vinh Long (province and provincial capital)		
(597.171.7)	Tien Giang (province), My Tho (provincial capital)		
(597.171.8)	Ben Tre (province and provincial capital)		
(597.171.9)	Ca Mau (province and provincial capital)		
(597.172.1)	Bac Lieu (province and provincial capital)		
(597.172.2)	Soc Trang (province and provincial capital)		

(1/9) Table 1e. Common auxiliaries of place

(598.26)	Mayparkngum (Maypakngum), Naxaithong, Sangthong, Sikhottabong, Sissattanak, Xaysetha (Xaysettha), Xaythany (Xaithany) (districts)	(6-15)	West Africa. □ (661.1) Mauritania. (662.2) Mali. (662.5) Burkina Faso. (662.6) Niger. (663) Senegal. (664) Sierra Leone. (665.1) The Gambia. (665.2) Guinea. (665.7) Guinea Bissau. (665.8) Cape Verde. (666.2) Liberia. (666.8) Côte d'Ivoire. (667) Ghana. (668.1) Togo. (668.2) Benin. (669.1) Nigeria. (669.95) São Tomé and Príncipe. (671.1) Cameroon. (671.8) Equatorial Guinea. (699.2) St Helena and Dependencies
(598.27)	Xaisômboun (Saysomboun, Xaysomboun) (province), Ban Mouang Cha (provincial capital). □ Hom, Longsane (Longsan, Longxan), Phun (Phoun), Thathom, (districts) Bolikhamxai (Bolikhamxai, Bolikhamxay, Borikhamzay) (province), Muang Pakxan (Paksane, Pakxanh) (provincial capital). □ Bolikhanh (Bolikanh, Bolikhan), Khamkheuth (Khamkeuth), Pakkading, Thaphabath, Viengthong (districts)	(6-17)	North Africa. □ (611) Tunisia. (612) Libya. (620) Egypt. (64) Morocco. (65) Algeria
(598.35)	Champasack, (Bassac, Champassack, Champassak) (province), Pakxé (Paksé) (provincial capital). □ Bachiangchaleunsook, Khong, Moonlapamok, Paksong (Pakxong), Pathoomphone, Phonthong, Sanasomboon, Sukhuma (districts)	(6-18)	North-east Africa. □ (642) Sudan. (63) Ethiopië. (635) Eritrea. (677) Somalia. (677.1) Djibouti
(598.36)	Attapu (Atpu, Attapeu, Attopeu, Muang Mai) (province), Attapu (Muang Samakhsai) (provincial capital). □ Phouvong, Samakhsay (Samakhsai), Sanamxay (Sanamxai), Sanxay, Xaysetha (Xaysettha) (districts)	(6-191.2)	Central Africa. □ (672.1) Gabon. (672.4) Congo. (673.13) Zaire. (674.1) Central African Republic. (674.3) Chad. (675.97) Burundi. (675.98) Rwanda. (689.1) Zimbabwe. (689.4) Zambia. (689.7) Malawi
(599)	PHILIPPINES. REPUBLIC OF THE PHILIPPINES	(61)	NORTH AFRICAN STATES IN GENERAL. MAGHREB. BARBARY STATES
(599.11)	Region I. Ilocos	(611)	TUNISIA. REPUBLIC OF TUNISIA. TUNIS. AL JUMHURIYAH AT-TUNISIYAH
(599.13)	Region II. Cagayan Valley. □ Babuyan Islands. Batan Islands	<i>Tunisia is divided into 24 governorates (wilayats), all of whose capitals have the same name as the governorate. They are further subdivided into districts</i>	
(599.15)	Region III. Central Luzon	(611A/Z)	Governorates of Tunisia. □ Aryanah. Bajah (Béja). Banzart (Bizerta). Bin Arus. Jundubah (Jendouba). Kaf (Le Kef). Madaniyin. Mahdiah. Munastir. Nabul (Nabeul). Qabis (Gabès) Qafsa (Gafsa). Qasrayn (Kassérine). Qayrawan (Kairouan). Qibili (Kébili). Safaqia (Sfax). Sidi Bu Zayd (Sidi Bouzid). Silyana. Susah (Sousse). Tattawin (tataouine). Tawzar (Tozeur). Tunis. Zaghwan (Zaghouan)
(599.17)	Region IVa. Metropolitan Manila Area	(611.1)	Governorates of Cap Bon Peninsula. □ Nabeul (Nabil, Nabol, Nabul), Zembra
(599.18)	Region IVb. Southern Tagalog. □ Lubang Islands. Mindoro. Marinduque Island. Tablas Island. Rombon Island. Sibuyan Island. Calamian Group. Linapacan Island. Palawan. Semirara Islands. Dumaran Island. Bugsuk Island. Balabac Island	(611.2)	Governorates of Northern Tunisia. □ Tunis (state capital and governorate), Carthage (archaeological site). Ariana (Al Arianah, Tunis Ariana, Al Aryanah). Béja (Bajah, Béjah). Ben Arous (Bin Arous, Tunis Ben Arous). Bizerte (Banzart, Bensert, Binzart, Biserta, Bizerta), La Galite. Jendouba (Jendoubah, Jenduba, Jondouba, Jundubah). Kairouan (Al Qayrawan, Al Qirwan, Qairouân). Le Kef (Al Kaf, Kaf, Kef). Manouba. Sidi Bou Zid (Qamudah, Sidi Bou Said, Sidi Buzid, Sidi Bu Zayd). Siliana (Silianah, Silyanah). Zaghwan
(599.21)	Region V. Bicol. □ Catanduanes Island. Batan Island. Rapu-Rapu Island. Burias Island. Ticao Island. Masbate Island	(611.3)	Governorates of Central Tunisia. □ Kairouan. Kassérine. Mahdia. Monastir. Sfax, Kerkenna Islands. Sousse (Sousa, Sussa, Susse, Susa, Susah)
(599.23)	Region VI. Western Visayas. □ Panay. Guimaras Island. West Negros	(611.4)	Governorates of Southern Tunisia. □ Gabes (Gabis, Qabis). Gafsa (Gafsah, Qafsah). Kebili (Kebilli, Qbili, Qibili). Médenine (Madanin, Medinin, Madaniyin), Djerba (Jerba). Tataouine (Foum Tataouine, Tatahouine, Tatuine, Tattawin). Tozeur (Tawzar, Touzar, Tuzar)
(599.25)	Region VII. Central Visayas. □ East Negros. Cebu. Bantayan Island. Bohol. Siquijor Island	(612)	LIBYA. LIBYAN ARAB JAMAHIRIYA
(599.27)	Region VIII. Eastern Visayas. □ Samar. Camotes Islands. Biliran Island. Leyte. Homonhon Island. Panaon Island	(612A/Z)	Governorates of Libya. □ Al-Bayda'. Al-Khums. Awbari (al-Khalij). Az-Zawiyah. Benghazi (Benghazi). Darnah. Gharyan. Misratah. Sabha. Tarabulus (Tripoli)
(599.31)	Region IX. Western Mindanao		
(599.311)	Subregion A. Sulu Archipelago. □ Samales Group. Jolo Group. Tapul Group. Pangutaran Group. Tawitawi Group		
(599.313)	Subregion B. □ Basilan Island. Pilas Group. Taptantana Group. Olutanga Island		
(599.33)	Region X. Northern Mindanao. □ Camiguin Island. Dinagat Island. Siargao Island. Bucas Grande Island		
(599.35)	Region XI. Southern Mindanao. □ Samal Island. Talikud Island. Sarangani Island		
(599.37)	Region XII. Central Mindanao		
(6)	AFRICA ◇ (6=111) Anglophone Africa ◇ (6=133.1) Francophone Africa		
(6-11)	East Africa. □ (676.1) Uganda. (676.2) Kenya. (678) Tanzania		
(6-13)	Southern Africa. □ (673) Angola. (679) Mozambique. (680) South Africa. (688.1) Namibia. (688.3) Botswana. (688.5) Lesotho. (688.7) Swaziland		

(62)	EGYPT AND SUDAN	(620.223)	Al Bahr al Ahmar (Red Sea) (governorate) Al Ghurdaqah (Hurghada) (governorate capital). □ Būr Safāga
(620)	EGYPT. ARAB REPUBLIC OF EGYPT. MISR. JUMHURIYAT MISR AL-ARABIYA <i>Egypt is divided into 27 muhāfazāt (muhafazah, sing.) governorates, grouped into Upper, Lower, Urban and Desert governorates. Here, the former UDC arrangement of a division into the 3 historical areas has been maintained, with the Upper and Lower governorates being placed under their respective headings. Alexandria and Cairo are Urban Governorates, and have been located in Lower Egypt; and Luxor and Suez, also Urban Governorates, have been located in Upper Egypt. There is a variety of transliterations from the Arabic, the most common of which are listed together with the anglicized form placed last in the list, where relevant</i>	(620.3) (620.311/.321) (620.311)	Upper Egypt Governorates of Upper Egypt Al Fayyūm (El Faiyum, el Fayoum, Faium, Faiyūm, Fayoum, Fayum, Fayyum) (governorate and governorate capital) Al Jīzah (El Giza, El Gīzah, Gizeh, Giza) (governorate and governorate capital) As Suways (El Suweiz, Es Suweis, Suez) (governorate and governorate capital) Banī Suwayf (Beni Suef) (governorate and governorate capital) Al Minyā (Menia, Minia, Minieh, Minya) (governorate and governorate capital). □ Beni Hasan (archaeological site). Mallawi. Tell al-Amarna (archaeological site) Asyūt (governorate and governorate capital) Sūhāj (Sawhāj, Sohag, Suhag) (governorate and governorate capital). □ Abydos (archaeological site) Qinā (Qena, Quena, Shamal Sīnā', Kena) (governorate and governorate capital). □ Dendara (archaeological site) Al Uqsar (Al Qusur, Al-Uqsar, Luxor) (governorate and governorate capital). □ Karnak (archaeological site). Valley of the Kings (archaeological site) Aswān (Assiut, Assyut, Asyout) (governorate and governorate capital). □ Abu Simbel (archaeological site). Philae (archaeological site). Wadi as-Subua
(620.1)	Lower Egypt. Masr el Bahri. Nile Delta	(620.318)	SUDAN. REPUBLIC OF THE SUDAN. JAMHURIYAT AS-SUDAN <i>The Sudan is divided into 26 states (wilayat), which are further subdivided into districts</i>
(620.111/.123)	Governorates of Lower Egypt	(624.1)	States of the Northern Sudan
(620.111)	Al Iskandariyah (Alexandria) (governorate and governorate capital). □ Rashid (Rosetta)	(624.11)	North Darfur (Darfur ash Shamaliyah, Shamal Darfur). □ Al-Fashir (capital)
(620.112)	Al Buhayrah (Behera, El Buhayra, Beheira) (governorate), Damanhūr (governorate capital)	(624.12)	Northern state (Ash Shamali, Ash Shamaliyah). □ Dongula (capital). Karima
(620.113)	Kafr ash Shaykh (Kafr ash Shaikh, Kafr ash-Shayk, Kafr el Sheik, Kafr el Sheikh) (governorate and governorate capital)	(624.13)	River Nile State (an Nil, Nahr an Nil, Nile, Nile River). □ Al-Damar (capital). Atbarah. Meroe (archaeological site). Shendi. Wadi Halfa
(620.114)	Ad Daqahliyah (Dacahlia, Dagahlia, Dakahlieh, Dakahliya, Daqahlīya, Dakahlia (governorate), Al Mansūrah (governorate capital)	(624.14)	Red Sea State (Al Bahr al Ahmar). □ Port Sudan (capital). Suakin
(620.115)	Dumyat (Dumiāt, Damietta) (governorate and governorate capital)	(624.2)	States of West and Central Sudan
(620.116)	Būr Saīd (Port Said) (governorate and governorate capital)	(624.21)	West Darfur (Darfur al Gharbiyah, Gharb Darfur). □ Geneina (capital)
(620.117)	Al Ismā'īliyah (Ismailia) (governorate and governorate capital)	(624.22)	South Darfur (Darfur al Janubiyah, Janub Darfur). □ Nyala (capital)
(620.118)	Ash Sharqīyah (Charkieh, Sharqia, Sharqīya, Sharquia, Sharkia) (governorate), Az Zaḡāziq (Zagaziz) (governorate capital)	(624.23)	North Kurdufan (Kurdufan ash Shamaliyah, Shamal Kurdufan, North Kordofan). □ Al-Obeid (capital)
(620.119)	Al Gharbiyah (al-Garbiya, El Gharbiya, Garbia, Gharbieh, Gharbiya, Gharbia) (governorate), Tantā (governorate capital). □ el-Mahalla el-Kubra	(624.24)	West Kurdufan (Gharb Kurdufan, Gharb Kurdufan, Kurdufan al Gharbiyah, West Kordofan). □ Al-Fula (capital)
(620.121)	Al Minvīyah (Menoufia, Menoufieh, Menufia, Menūfiya, Munufia, Minufia) (governorate), Shibīn al Kawm (governorate capital)	(624.25)	South Kurdufan (Janub Kurdufan, Janub Kurdufan, Kurdufan al Janubiyah, South Kordofan). □ Kadugli (capital)
(620.122)	Al Qalyūbiyah (Caliubia, Kaloubieh, Kalioubiya, Kalyoubia, Qaliyubia, Qalyubiya, Kalyubia) (governorate), Banha (governorate capital)	(624.3)	States of East Sudan
(620.123)	Al Qahirah (Cairo) (state capital, governorate and governorate capital). Giza, Memphis (archaeological site). Saqqara (archaeological site)	(624.31)	Khartoum (Al Khartum, Al Khurtum) (state capital). □ Omdurman
(620.2)	Middle Egypt	(624.32)	Gezira (Al Jazirah, El Gezira, Gezeira). □ Wad Medani (capital)
(620.21/.22)	Desert governorates	(624.33)	White Nile (An Bahr al Abyad, An Nil al-Abyad). □ Rabak (capital). Kosti
(620.21)	Western Desert governorates	(624.34)	Kassala (Kessala, Ash Sharqi, Ash Sharqiyah, Eastern) (state and capital)
(620.211)	Matrūh (Marsā Matrū, Mersa Matruh, Western Desert) (governorate), Marsa Matrūh (governorate capital). □ El-Alamein. Siwa		
(620.212)	Al Wādī al Jaded (El-Wadi El-Gidid, New Valley) (governorate), Al Khārijah (governorate capital). □ Al-Kharga. Mut		
(620.22)	Eastern Desert governorates		
(620.221)	Shamal Sīnā' (governorate), Al 'Arish (governorate capital)		
(620.222)	Janūb Sīnā' (Sina al-Janubiyah, Sina' al-Janūbiyah, Sinai al Janūbiya, South Sinai) (governorate), At Tur (governorate capital). □ Dahab. Nuweiba. Sharm el Sheik		

(1/9) Table 1e. Common auxiliaries of place

(624.35)	Gedarif (Al Qadarif, Gadarif, Gedaref, Gederaf) (state and capital)	(635.2)	Semenawi Keyih Bahri (Semenawi Keyh Bahri, Semien-Keih-Bahri, Northern Red Sea).
(624.36)	Sennar (Sinnar). □ Sinja (capital)		□ Massawa (provincial capital). Dahlak
(624.37)	Blue Nile (Al Wustá, Al Awsat, An Nil al Azraq, Central). □ Al-Damazin (capital)	(635.3)	Islands. Taulud Island
(624.38)	Upper Nile (A'Ali an Nil). □ Malakal (capital)		Gash Barka (Debubawi Gash Barka, Maekel, Maakel, Maekelay, Makelay, Central).
(624.4)	States of Southern Sudan	(635.4)	□ Barentu (provincial capital)
(624.41)	West Bahr-al-Ghazal (Bahr al-Ghazal al Gharbiyah, Gharb Bahr al-Ghazal). □ Wau (capital)	(635.5)	Maekel. □ Asmara (state and provincial capital)
(624.42)	North Bahr-al-Ghazal (North Bahr-al-Gazal, North Bahr el Ghazal, Shamal Bahr al Ghazal). □ Awil (capital)		Debub (Southern). □ Mendeferas (provincial capital). Adi Keyh. Dekemhare. Qohaito (archaeological site). Senafe
(624.43)	Warap (Warab) (state and capital)	(635.6)	Deubawi Keyih Bahri (Debubawi Keyh Bahri, Debub-Keih-Bahri, Southern Red Sea). □ Assab
(624.44)	Unity (Al-Wahdah, Wahda). □ Bantio (capital)	(64)	MOROCCO. KINGDOM OF MOROCCO. AL MAGHRIB. AL MAMLAKAH AL-MAGHRIBIYAH
(624.45)	Jungoli (Jonglei, Jonglie, Jonqley, Junglei, Junqali). □ Bor (capital)		<i>Morocco is divided into 16 regions or states (wilayat) and each wilaya is subdivided into prefectures and provinces. The prefectures tend to be the more urbanized divisions, centred on larger cities; the provinces are more agricultural and centred on smaller cities. Historically, Spain had possessions on the Mediterranean coast of Morocco. She still retains 2 enclaves, Ceuta and Melilla, in addition to a few small islands which are governed as overseas territories. The United Nations intends to hold a referendum to determine the future of the territory, known as Western Sahara. Pending the decision, other governments have withheld recognition of Morocco's sovereignty over Western Sahara. The region is known by the locals as the Sahrawi Arab Democratic Republic (SADR), supported by the Polisario political movement. In 2002 the situation is that Morocco claims it, but that the SADR continues to struggle to become an independent territory. The area consists of 4 provinces in Laâoune-Boujdour-Sakia El Hamra, Oued Ed-Dahab-Lagouira and Guelmim-Es-Semara: Boujdour, Es-Semara, Laâoune, Oued Ed-Dahab</i>
(624.46)	Lakes (Al Buhairat, Al Buhayrah, Buheirat, El Boheirat, El Buhayrat). □ Rumbek (capital)		
(624.47)	East Equatoria (Al Istiwa'iyah ash Sharqiya, Sharq al Istiwa'iyah). □ Kapoita (capital)		
(624.48)	Bahr-al-Jabel (Bahr el Gabel). □ Juba (capital)		
(624.49)	West Equatoria (Al Istiwa'iyah al Gharbiyah, Gharb al Istiwa'iyah). □ Yambio (capital)		
(63)	ETHIOPIA AND ERITREA		
(630)	ETHIOPIA. FEDERAL DEMOCRATIC REPUBLIC OF ETHIOPIA. ITYOP'YA. ITYOP'YA FEDERALAWI DEMOKRASİYAWI RIPEBLIK		
	<i>Ethiopia is divided into 11 Divisions, consisting of 9 states (kilil) and 2 administrations (astedader). These are further subdivided into districts (wereda). Addis Ababa, the national capital city, is situated within the state of Oromia, and acts as the capital. Somali is populated by the Somali peoples and has no administrative capital</i>		
(630.1)	Northern states		
(630.11)	Tigray (Tegré, Tigrá, Tigre). □ Mekele (capital). Adwa. Adigrat. Aksum. Alamata. Endaselasie. Lalibela. Maychew. Yeha		
(630.12)	Amhara (Amara). □ Bahir Dar (capital)	(641)	Atlantic Coast Regions
(630.13)	Afar. □ Asayita (capital). Dessie. Kombolcha	(641.1)	Gharb-Chrarda-Béni Hassen (region), Kénitra (Kenitra) (province and regional capital).
(630.14)	Benshangul-Gumuz (Benishangul-Gumuz, Binshangul Gumuz). □ Asosa (capital)		□ Sidi-Kacem (Sidi Qassem) (province)
(630.2)	Administrations	(641.2)	Rabat-Salé-Zemmour-Zaer (region), Rabat (Er Ribat) (state capital, province and regional capital). □ Salé-Al Jadida. SalT-Médina. Skhirate-Témara (provinces)
(630.21)	Addis Ababa (Addis Abeba, Adis-Abeba) (state capital)		Grand Casablanca (region), Casablanca (Dar el Beida, Ed Dar el Baida) (regional capital).
(630.22)	Dire Dawa	(641.3)	□ Aïn Chock-Hay Hassani. Aïn Sebaâ-Hay Mohammadi. Al Fida-Derb Sultan. Ben M'Sick-Médiouna. Casablanca-Anfa. Mechouar de Casablanca. Mohammedia. Moulay Rachid-Sidi Othmane. Sidi Bernoussi-Zenata (provinces)
(630.3)	Southern states		Chaouia-Ouadigha (region), Settât (province and regional capital). □ Ben Slimane. Khouribga (provinces)
(630.31)	Gambela Peoples (Gambela Hizboch). □ Gambela (capital)		
(630.32)	Southern Nations, Nationalities and Peoples (YeDebub Biheroch Bihereseboch na Hizboch). □ Awasa (capital) Arba Minch. Dila. Yirga Alem		
(630.33)	Oromia (Oromiya). □ Asela. Bahar Dar. Debri Zeyit. Goba. Gonder. Jimma. Nazret. Nekemte. Shashemene	(641.4)	
(630.34)	Harari People (Hareri Hizb). □ Harar (capital)		
(630.35)	Somali (Sumale)	(642)	Mediterranean Coast and The Rif Regions
(635)	ERITREA. REPUBLIC OF ERITREA. HAGERE ERTRA. DEWLET ERITREA	(642.1)	Tanger-Tétouan (region), Tanger (Tangier, Tanja) (regional capital). □ Chefchaouen (Chaouén, Châouén, Chechaouène, Chefchaouane, Xexauen). Fahs-Béni Makada. Larache (El-Aranch, Laarach). Tanger-Assilah. Tétouan (Tetouane, Tetuan, Tetuán) (provinces)
	<i>Eritrea is divided into six zoba (provinces, regions)</i>		
(635.1)	Anseba (Ansaba, A'Nseba). □ Keren (provincial capital). Nakfa		

(642.2)	Taza-Al Hoceima-Taounate (region), Al Hoceima (El Hoceïma, El Hosseïma, Alhucemas) (province and regional capital). □ Taza (province)	(651)	Northwestern provinces. □ Aïn Defla. Aïn Témouchent. Béchar. Blida (El Boulaïda, al-Boulaida). Chlef (El Asnam, Chelif, Ech Cheliff). Médéa (El Mediyya, Lemdiyya). Mostaganem (Mestghanem). Oran (Ouahrane, Ouahrane, Wahrane). Relizane (Ghelizane, Ghilizane). Tindouf. Tipaza. Tissemsilt. Tlemcen (Tilimsene, Tlemsane, Tilimsen)
(643)	Middle Atlas and the East Regions		
(643.1)	Oriental (region), Oujda (Oudjda) (regional capital). □ Berkane. Figuig. Jerada (Irada). Nador. Oujda-Angad (provinces)		Northeastern provinces. □ Alger (Dzayer, Djezaïr. Al-Jazair, al-Djazair) (state capital). Annaba. Batna. Béjaïa (Bougie). Bordj Bou Arréridj (Bordj Bou Arirj). Bouira. Boumerdès (Boumerdas) Constantine (Qoussantina, Qacentina). El Tarf (Et Tarf). Guelma. Jijel (Djijel). Khenchela (Khenchla). Mila. Msila. Oum el Bouaghi (Oum al Bouagui). Sétif (Stif). Skikda (Skidda). Souk Ahras (Souq Ahras). Tébessa (Tbessa). Tizi Ouzou
(643.2)	Meknès-Tafilalet (region), Meknès (Mequinez) (regional capital). □ Al Ismailia. El Hajeb. Errachidia (Rachidia, Rachidiya) Meknès-El Menzeh (provinces)	(652)	
(643.3)	Fès-Boulemane (Boulmane) (region), Fès (Fâs, Fez) (regional capital). □ Fès Jdid-Dar Dbibagh. Fès-Médina. Sefrou. Zouagha-My Yacoub (provinces). Volubilis (archaeological site)		
(644)	Marrakesh and Central Regions	(653)	Central provinces. □ Biskra (Beskra). Djelfa (El Djelfa). El Bayadh (El Bayad, El Beyyadh). El Oued (El Ouadi, El Wad). Illizi (Ilizi). Laghouat. Mascara (Mouaskar). Naama. Saïda. Sidi Bel Abbès. Tiaret (Tihert)
(644.1)	Marrakech-Tensift-Al Haouz (region), Marrakesh (regional capital). □ Al Haouz. Chichaoua. El Kelaâ des Sraghna (El Kelaâ des Srarhna, El Kelaâ Sraghna, Qel'at es Sraghna, Qel'at Sraghna) Errachidia. Essaouira (Eç Çauouira, Saouira). Ifrane. Khénifa. Marrakech-Médina. Marrakech-Ménara. Sidi Youssef Ben Ali (provinces)	(654)	Southern provinces. □ Adrar. Ghardaïa (Ghardaya). Ouargla (Wargla). Tamanghasset (Tamanrasset, Tamenghest)
(644.2)	Tadla-Azilal (region), Béni Mellal (province and regional capital). □ Azilal. El Jadida (Ej Jdida) (provinces)	(66/69)	STATES AND TERRITORIES SOUTH OF THE SAHARA. SUB-SAHARAN AFRICA
(645)	South Atlantic and Western Sahara Regions	(66)	WEST AFRICAN STATES AND TERRITORIES
(645.1)	Souss-Massa-Daraâ (region), Agadir (regional capital). □ Agadir-Ida ou Tanane. Chtouka-Aït Baha. Inezgane-Aït Melloul. Ouarzazate (Ouarzazat, Warzazat). Taroudannt (Taroudant). Tiznit. Zagora (provinces)	(661.2)	MAURITANIA. ISLAMIC AND AFRICAN REPUBLIC OF MAURITANIA. AL JUMHURIYAH AL ISLAMIYAH AL MURITANIYAH <i>Mauritania is divided into 12 Regions (wilayat) and one district (Nouakchott, the capital of the state). These are further subdivided into 53 moughataas (departments). Between 1976 and 1979 Mauritania annexed the Southern part of Spanish Sahara as Tiris el Gharbia Region. In August 1979 Mauritania withdrew from there, and it was annexed to Western Sahara by Morocco</i>
(645.2)	Doukala-Abda (region), Safi (Asfi, Saffi) (province and regional capital). □ El Jadida (province)		
(645.3)	Guelmim-Es-Semara (region), Guelmim (Goulimim, Goulimime, Goulimine) (province and regional capital). □ Assa-Zag. Es-Semara (Western Sahara). Tan-Tan (Tantane) (provinces)		
(645.4)	Laâyoune-Boujdour-Sakia El Hamra (region), Laâoune (La?youne) (province and regional capital) (Western Sahara). □ Boujdour (Western Sahara) (province)	(661.21)	Northern Regions
(645.5)	Oued Ed-Dahab-Lagouira (region), Dakhla (regional capital). □ Aousserd. Oued Ed-Dahab (Western Sahara) (provinces)	(661.211)	Tiris Zemmour, Zouérate (department and regional capital)
(647)	Spanish possessions on north African coast. □ Alhucemas. Ceuta. Chafarinas. Mellila. Penon de Velez de la Gomera ⇒ (460.37) ⇒ (460.38)	(661.212)	Dakhlet Nouadhibou, Nouadhibou (department and regional capital)
(648)	Spanish possessions in West Africa (historical) <i>See notes at Morocco (64) relating to Western Sahara (formerly Spanish Sahara) and Sidi Ifni</i>	(661.213)	Inchiri, Akjoujt (department and regional capital)
(65)	ALGERIA. DEMOCRATIC AND POPULAR REPUBLIC OF ALGERIA. AL JAZA'IR. AL JUMHURIYAH AL JAZA'IRIYAH AD DIMUQRATIYAH ASH SHA'BIYAH <i>Algeria is divided into 48 provinces (wilayat), all of whose capitals have the same name as the provinces. They are subdivided into subprefectures</i>	(661.214)	Adrar, Atar (department and regional capital). □ Aoujeft. Chinguetti (Chingueti, Chinguitti). Ouadane (departments)
		(661.22)	Nouakchott (state capital). □ Arafat. Dar Nanm. El Mina. Ksar. Riad (Riadh). Sebkh. Teyragh-Zeïn. Teyarett. Toujounine (departments)
		(661.23)	Southern Regions
		(661.231)	Trarza, Rosso (department and regional capital). □ Bir Moghreïn. Boutilimit. F'Dérik (F'Férick, F'Deirik. Keur-Macène. Méderdra. Ouad-naga. R'Kiz (departments)
		(661.232)	Brakna, Aleg (department and regional capital). □ Bababé. Boghé. Magta-Lahjar. M'Bagne (departments)
		(661.233)	Tagant, Tidjikdja (department and regional capital). □ Moudjéria. Tichitt (departments)

(1/9) Table 1e. Common auxiliaries of place

(661.234)	Assaba (Açaba, el-'Açaba), Kiffa (regional capital). □ Barkéol. Boumdeïd. Guérou (Guerrou). Kankossa. Kiffa (departments)	(662.526)	Sanguié, Réo (provincial capital)
(661.235)	Gorgol, Kaédi (department and regional capital). □ Maghama. M'Bout. Monguel (Mounguel) (departments)	(662.527)	Sissili, Léo (provincial capital)
(661.236)	Guidimaka (Guidimqagha, Guidimakha), Sélibaby (department and regional capital). □ Ould Yengé (departments)	(662.53)	Southern provinces
(661.237)	Hodh el Gharbi (Western Hodh), Aioun el Atrouss (department and regional capital). □ Kobenni (Koubeni, Koubenni, Kobeni). Tamchakett (Tamchekett). Tintane (departments)	(662.531)	Naouri (Nahouri), Pô (provincial capital)
(661.238)	Hodh ech Chargui (Hodh Charghi, Eastern Hodh), Néma (department and regional capital). □ Amourj. Bassikounou (Bassiknou). Djiguenni (Djigueni). Oualata. Timbédra (departments)	(662.532)	Ziro, Sapouy (provincial capital)
(662.1)	MALI. REPUBLIC OF MALI. RÉPUBLIQUE DE MALI <i>Mali is divided into eight regions and one district, Bamako, the state capital. The regions are also called First Region, Second Region, etc., in the order in which they appear in the schedule below. The capitals, which are also circles, have the same name as the Regions. The regions are subdivided into 49 cercles (circles). The circles and the district are subdivided into communes</i>	(662.533)	Zoundwéogo, Gourcy (provincial capital)
(662.11)	Kayes. □ Bafoulabé. Diéma. Kéniéba. Kita. Nioro. Yélimané (circles)	(662.534)	Bazéga (Bazèga), Kombissiri (provincial capital)
(662.12)	Koulikoro. □ Banamba. Dioïla. Kankaba. Kati. Kolokani. Nara (circles)	(662.535)	Kadiogo, Ouagadougou (State and provincial capital)
(662.13)	Sikasso. □ Bougouni. Kadiolo. Kolondiéba. Koutiala. Yanfolila. Yorosso (circles)	(662.536)	Boulkiemdé, Koudougou (provincial capital)
(662.14)	Segou. □ Barrouéli. Bla. Macina. Niono. San. Tominian (circles)	(662.537)	Kourwéogo, Bousé (provincial capital)
(662.15)	Mopti. □ Bandiagara. Bankass. Djenné. Douentza. Koro. Ténenkou. Youwarou (circles)	(662.538)	Passoré, Yako (provincial capital)
(662.16)	Timbuktu (Timbouctou, Tombouctou). □ Diré. Goundam. Gourma-Rharous. Niafinké (circles)	(662.54)	Northern provinces
(662.17)	Gao. □ Ansongo. Bourem. Ménaka (circles)	(662.541)	Zondoma, Gourcy (provincial capital)
(662.18)	Kidal. □ Abeïbara. Tessalit. Tin-Essako (circles)	(662.542)	Sourou, Tougan (provincial capital)
(662.19)	Bamako (Capital District and circle)	(662.543)	Yatenga, Ouahigouya (provincial capital)
(662.5)	BURKINA FASO. PEOPLE'S DEMOCRATIC REPUBLIC OF BURKINA FASO <i>Burkina Faso is divided into 45 provinces. These are further subdivided into districts. Historically, until 1960, it was Upper Volta</i>	(662.544)	Loroum, Titao (provincial capital)
(662.51)	Western provinces	(662.545)	Bam, Kongoussi (provincial capital)
(662.511)	Kossi, Nouna (provincial capital)	(662.546)	Sanmatenga, Kaya (provincial capital)
(662.512)	Banwa (Les Banwa), Solenzo (provincial capital)	(662.547)	Namentenga, Boulsa (provincial capital)
(662.513)	KénéDougou, Orodara (provincial capital)	(662.548)	Soum, Djibo (provincial capital)
(662.514)	Houet (Houé), Bobo-Dioulasso (provincial capital)	(662.549)	Oudalan, Gorom-Gorom (provincial capital)
(662.515)	Léraba, Sindou (provincial capital)	(662.55)	Eastern provinces
(662.516)	Komoé (Comoé), Banfora (provincial capital)	(662.551)	Séno, Dori (provincial capital)
(662.517)	Bougouriba, Diébougou (provincial capital)	(662.552)	Yagha, Sebba (provincial capital)
(662.518)	Poni, Gaoua (provincial capital)	(662.553)	Gnagna, Bogandé (provincial capital)
(662.519)	Noumbiel, Batié (provincial capital)	(662.554)	Komondjari, Gayéri (provincial capital)
(662.52)	Central provinces	(662.555)	Gourma, Fada-N'Gourma (provincial capital)
(662.521)	Ioba, Dano (provincial capital)	(662.556)	Tapoa, Doapaga (provincial capital)
(662.522)	Tui (Tuy), Houndé (provincial capital)	(662.557)	Kompienga, Pama (provincial capital)
(662.523)	Balé (Les Balés), Boromo (provincial capital)	(662.558)	Koulpélogo, Ouargaye (provincial capital)
(662.524)	Mouhoun, Dédougou (provincial capital)	(662.56)	Southeastern provinces
(662.525)	Nayala, Toma (provincial capital)	(662.561)	Boulgou, Tenkodogo (provincial capital)
		(662.562)	Kouritenga (Kourittenga), Koupéla (provincial capital)
		(662.563)	Ganzourgou, Zorgo (provincial capital)
		(662.564)	Oubritenga, Ziniaré (provincial capital)
		(662.6)	NIGER. REPUBLIC OF NIGER
		(662.6A/Z)	Départements of Niger. □ Agadez. Diffa. Dosso. Maradi. Niamey. Tahoua. Zinder
		(663)	SENEGAL. REPUBLIC OF SENEGAL
		(663.1)	Saint-Louis
		(663.2)	Dakar
		(663.3)	Thiès
		(663.4)	Diourbel
		(663.5)	Kaolack
		(663.6)	Fatick
		(663.7)	Tambacounda
		(663.8)	Ziguinchor
		(663.9)	Kolda
		(664)	SIERRA LEONE. REPUBLIC OF SIERRA LEONE
			<i>Sierra Leone is divided into three Provinces and one 'Area'. These are subdivided into districts. The tertiary divisions are chiefdoms</i>
		(664.1)	Western Area
		(664.11)	Freetown (state and area capital)
		(664.2)	Northern Province
		(664.21)	Kambia (district and district capital)
		(664.22)	Koinadugu (district). Kabala (district capital)
		(664.23)	Port Loko (district and district capital)
		(664.24)	Bombali (district). Makeni (provincial and district capital)
		(664.25)	Tonkolili (district). Magburaka (district capital)

(664.3)	Southern Province	(665.813)	Ilha de Santa Luzia
(664.31)	Bonthe (district and district capital)	(665.814)	Ilha de São Nicolau
(664.32)	Sherbro Urban (district), Bonthe (district capital)	(665.815)	Ilha do Sal
(664.33)	Bo (provincial capital, district and district capital)	(665.816)	Ilha da Boavista
(664.34)	Moyamba (district and district capital)	(665.82)	Grupo de Sotavento
(664.35)	Pujehun (district and district capital)	(665.821)	Ilha do Maio
(664.4)	Eastern Province	(665.822)	Ilha de Santiago
(664.41)	Kenema (provincial capital, district and district capital)	(665.823)	Ilha do Fogo
(664.42)	Kono (district), Sefadu (district capital)	(665.824)	Ilha Brava
(664.43)	Kailahun (district and district capital)		
(665)	UPPER GUINEA STATES AND TERRITORIES	(666.2)	LIBERIA. REPUBLIC OF LIBERIA
			<i>Liberia is divided into fifteen counties which are subdivided into districts</i>
(665.1)	GAMBIA. REPUBLIC OF THE GAMBIA	(666.21)	Western counties
(665.1A/Z)	Divisions of Gambia. □ Banjul (city) (Bathurst). North Bank. Lower River. MacCarthy Island. Upper River. Western	(666.211)	Grand Cape Mount, Robertsport (county capital)
			Bomi, Tubmanburg (county capital)
(665.2)	GUINEA. REPUBLIC OF GUINEA. RÉPUBLIQUE DE GUINÉE	(666.212)	Montserrado, Bensonville (county capital).
	<i>Formerly known as French Guinea, this was one of the territorial units of French West Africa. It attained independence in 1958 as the Republic of Guinea. Guinea is divided into régions administratives (administrative regions). ISO calls all except Conakry prefectures. This it calls a gouvernorat (governorate). The administrative regions are further subdivided into 175 districts. The regions are now grouped into seven governorates and one city - Conakry</i>	(666.213)	□ Monrovia (state capital)
		(666.214)	Margibi, Kakata (county capital)
(665.21)	Labé (Governorate). □ Koubia, Lélouma, Mali, Tougué (districts)	(666.215)	Gbarpolu, Bopolu (county capital)
(665.22)	Boké (Governorate). □ Boffa, Fria, Gaoual, Koundara (Youkounkoun) (districts)	(666.216)	Lofa, Voinjama (county capital)
(665.23)	Mamou (Governorate). □ Dalaba, Pita (districts)	(666.217)	Bong, Gbarnga (county capital)
(665.24)	Kindia (Governorate). □ Coyah, Dubréka, Forécariah, Téliimélé (districts)	(666.218)	Grand Bassa, Buchanan (county capital)
(665.25)	Conakry (City and state capital)	(666.23)	Eastern counties
(665.26)	Kankan (Governorate). □ Kérouané, Kouroussa, Mandiana, Pita, Siguiri (districts)	(666.231)	Nimba, Sanniquellie (county capital)
(665.27)	Faranah (Governorate). □ Dabola (district). Dinguiraye (district). Kissidougou (districts)	(666.232)	Rivercess (county and county capital)
(665.28)	Nzérékoré (Governorate). □ Beyla, Guéckédou (Guékédou), Lola, Macenta, Yomou, (districts)	(666.233)	Sinoe, Greenville (county capital)
		(666.234)	Grand Kru, Barclayville (county capital)
(665.7)	GUINEA-BISSAU. REPUBLIC OF GUINEA-BISSAU	(666.235)	Maryland, Harper (county capital)
	<i>Portuguese Guinea to 1974</i>	(666.236)	Grand Gedeh, Zwedru (county capital)
(665.711)	Concelho de Bafata	(666.237)	River Gee, Fish Town (county capital)
(665.712)	Concelho de Bissau	(666.8)	CÔTE D'IVOIRE (IVORY COAST). REPUBLIC OF CÔTE D'IVOIRE. RÉPUBLIQUE DE CÔTE D'IVOIRE
(665.713)	Concelho de Bolama		<i>Formerly part of French West Africa, the Ivory Coast gained independence in 1960. In 1986 its name was officially changed to Côte d'Ivoire. The Côte d'Ivoire is divided into 19 regions which are subdivided into departments. Although many sources list the departments as the primary regions, this is not the case since reorganization in 1997, with further adjustments in 2000. In 1983 President Félix Houphouët-Boigny announced plans to move the national capital from Abidjan to Yamoussoukrou. In 2004 Abidjan is still the administrative capital while Yamoussoukrou is the capital of state</i>
(665.714)	Circunscrição de Cacheu		
(665.715)	Circunscrição de Catio	(666.81)	Northern regions
(665.716)	Circunscrição de Farim	(666.811)	Denguete, Odienné (department and regional capital)
(665.717)	Circunscrição de Fulacunda		Savannes, Korhogo (department and regional capital). □ Boundiali, Ferkessédougou, Korhogo, Tengréla (departments)
(665.718)	Concelho de Gabu	(666.812)	Bafing (region), Toubia (department and regional capital)
(665.719)	Concelho de Mansoa		Worodougou (region), Séguéla (department and regional capital). □ Mankono (department)
(665.721)	Circunscrição de São Domingos	(666.813)	
(665.722)	Concelho de Bissora	(666.814)	
(665.79)	Arquipélago dos Bijagos (Bissagos Islands)	(666.82)	Western regions
(665.791)	Circunscrição de Bijagos	(666.821)	Dix-Huit Montagnes (region), Man (department and regional capital). □ Bangolo, Biankouma, Danané (departments)
(665.8)	CAPE VERDE. REPUBLIC OF CAPE VERDE		Moyen-Cavally (region), Guiglo (department and regional capital). □ Duékoué, Toulépleu (departments)
(665.81)	Grupo de Barlavento	(666.822)	Haut-Sassandra (Sassandra supérieur) (region), Daloa (department and regional capital). □ Issia, Vavoua (departments)
(665.811)	Ilha de Santo Antão		
(665.812)	Ilha de São Vicente	(666.823)	

(1/9) Table 1e. Common auxiliaries of place

(666.824)	Fromager (region), Gagnoa (department and regional capital). □ Oumé (department)		
(666.825)	Bas-Sassandra (Cavally inférieur) (region),. □ San Pédro (department and regional capital). Sassandra, Soubré (departments). Tabou		
(666.84)	Central regions		
(666.841)	Marahoué(region), Bouaflé (department and regional capital). □ Sinfra, Zuénoula (departments)		
(666.842)	Lacs (region), Yamoussoukro (department and state and regional capital). □ Tiébissou, Toumodi (departments)	(668.11)	broader eastern strip. The French mandate of Togo was administrated under Dahomey (Benin) and thus part of French west Africa. After World War II the mandate was extended as a UN Trusteeship Territory. Togo became independent in 1960. Togo is divided into five regions which are further subdivided into 30 prefectures and one commune. Lomé, the state capital, is the commune and is also the capital of the Golfe prefecture
(666.843)	Vallee du Bandama (region), Bouaké (department and regional capital). □ Béoumi, Dabakala, Katiola, Sakassou (departments)		Savanes Region. □ Kpendjal (prefecture), Mandouri (prefecture capital). Oti (prefecture), Mango (prefecture capital). Tandjouaré (prefecture and prefecture capital). Tône (prefecture), Dapaong (prefecture capital)
(666.844)	N'zi-Comoé (region), Dimbokro (department and regional capital). □ Bocanda, Bongouanou, Daoukro, Dimbokro, Mbahiakro (departments)	(668.12)	Kara Region. □ Assoli (prefecture), Bafilo (prefecture capital). Bassar (prefecture and prefecture capital). Binah (prefecture), Pagouda (prefecture capital). Dankpen (prefecture), Guérin-Kouka (prefecture capital). Doufelgou (prefecture), Niamtougou (prefecture capital). Kéran (prefecture), Kandé (prefecture capital). Kozah (prefecture), Kara (prefecture capital)
(666.86)	Eastern region		
(666.862)	Zanzan (region), Bondoukou (department and regional capital). □ Bouna, Tanda (departments)		Centre Region. □ Biitta. Sotouboua. Tchamba (prefectures and prefecture capitals). Tchaoudjo (prefecture), Sokodé (prefecture capital)
(666.88)	Southern regions	(668.13)	Plateaux Region. □ Agou (prefecture), Agou-Gadzépé (prefecture capital). Amou (prefecture), Amlamé (prefecture capital). Danyi (prefecture), Danyi-Apéyéme (prefecture capital). Est-Mono (prefecture), Elavagnon (prefecture capital). Haho (prefecture), Notsé (prefecture capital). Kloti (prefecture), Kpalimé (prefecture capital). Moyen-Mono (prefecture), Tohoun (prefecture capital). Ogou (prefecture), Atakpamé (prefecture capital). Wawa (prefecture), Badou (prefecture capital)
(666.881)	Sud-Bandama (region), Divo (department and regional capital). □ Lakota (department)		Maritime Region. □ Lomé (commune and state capital). Avé (prefecture), Kévé (prefecture capital). Golfe (prefecture), Lomé (state and prefecture capital and commune). Lacs (prefecture), Aného (prefecture capital). Vo (prefecture), Vogan (prefecture capital). Yoto (prefecture), Tabligbo (prefecture capital). Zio (prefecture), Tsévié (prefecture capital)
(666.882)	Agnéby (Agnébi) (region), Agboville (department and regional capital). □ Adzopé (department)	(668.14)	
(666.883)	Lagunes (region), Abidjan (department and administrative and regional capital). □ Alépé, Dabou, Grand-Lahou, Jacqueville, Tiassalé (departments)		
(666.884)	Moyen-Comoé(Comoé central) (region), Abengourou (department and regional capital). □ Adiaké, Agnibilékrou (departments)		
(666.885)	Sud-Comoé (region), Aboisso (department and regional capital). □ Grand Bassam (department)	(668.15)	
(667)	GHANA. REPUBLIC OF GHANA <i>The independent nation of Ghana was founded in 1957 through the merger of the British dependency of Gold Coast and the trust territory of British Togoland. Ghana is divided into ten regions. The regions are further subdivided into 110 districts</i>		
(667.1)	Northern Ghana		
(667.11)	Upper West Region. □ Bolgatanga (regional capital)	(668.2)	BENIN. REPUBLIC OF BENIN. RÉPUBLIQUE DU BÉNIN <i>Benin (formerly Dahomey) gained its independence from France in 1960. Benin is divided into 12 departments. Six of these departments have not had capitals officially designated, but the cities noted with a question mark have begun to provide department-level services. The departments are subdivided into communes</i>
(667.12)	Upper East Region. □ Wa (regional capital)		
(667.13)	Northern Region. □ Tamale (regional capital)		
(667.14)	Brong Ahafo Region. □ Sunyani (regional capital)		
(667.3)	Southern Ghana		
(667.31)	Ashanti Region. □ Kumasi (regional capital)		
(667.32)	Eastern Region. □ Koforidua (regional capital)		
(667.33)	Volta. □ Ho (regional capital)	(668.21)	Northern departments
(667.34)	Greater Accra. □ Accra (state and regional capital). Tema	(668.211)	Alibori (department), Kandi (department capital?). □ Banikoara. Gogounou. Karimama. Malanville. Segbana (communes)
(667.35)	Central Region. □ Cape Coast (regional capital)		Atacora (department), Natitingou (department capital). □ Boukoubé (Boukoubé). Coby (Cobli). Kérou. Kouandé. Matéri. Natitingou. Péhunco. Tanguéta. Toucountouna (communes)
(667.36)	Western Region.	(668.213)	Donga (department), Djougou (department capital?). □ Bassila. Copargo. Ouaké (communes)
(668.1)	TOGO. REPUBLIC OF TOGO. RÉPUBLIQUE TOGOLAISE <i>After the First World War, Togoland was a League of Nations territory mandated to Great Britain and France. They split it longitudinally and France received the</i>	(668.215)	

(668.217)	Borgou (department), Parakou (department capital). □ Bembéréké. Kalalé. N'Dali. Nikki. Parakou. Pèrèrè. Sinendé. Tchaourou (communes)	(669.148)	Adamawa. □ Yola (state capital)
(668.22)	Central departments	(669.151.19)	States mainly south of the Niger-Benue river system
(668.221)	Collines (department), Savalou (department capital?). □ Bantè. Dassa-ZoumF. Glazoué. Ouèssè. Savè (communes)	(669.152)	Taraba. □ Jalingo (state capital)
(668.223)	Zou (department), Abomey (Abomé) (department capital). □ Agbangnizoun. Bohicon. Covè. Djidja. Ouinhi. Zagnanado. Za-Kpota. Zogbodomey (communes)	(669.154)	Benue. □ Makurdi (state capital)
(668.225)	Plateau (department), Sakété (department capital?). □ Adja-Ouèrè. Ifangni. Kétou. Pobè (communes)	(669.156)	Cross River. □ Calabar (state capital)
(668.227)	Couffo (department), Dogbo (department capital?). □ Aplahoué. Djakotomey. Klouékanmè (Klouékanmey). Lalo. Toviklin (communes)	(669.158)	Akwa Ibom. □ Uyo (state capital)
(668.23)	Southern departments	(669.162)	Abia. □ Umuahia (state capital)
(668.231)	Mono (department), Lokossa (department capital). □ Athiémié. Bopa. Comè. Grand-Popo. Houéyogbé. Lokossa (communes)	(669.164)	Ebonyi. □ Abakaliki (state capital)
(668.233)	Atlantique (department), Ouidah (department capital?). □ Abomey-Calavi. Allada. Kpomassè. Sô-Ava. Toffo. Torri-Bossito. Zè (communes)	(669.166)	Enugu. □ Enugu (state capital)
(668.235)	Littoral (department), Cotonou (department capital).	(669.168)	Kogi. □ Lokoja (state capital)
(668.237)	Ouémè (department), Porto-Novo (state and department capital). □ Adjarra (Adjara). Adjohoun. Aguégués (Aguégué). Akpro-Missérété. Avrankou. Bonou. Dangbo. Sèmè-Kpodji (Sèmè-Podji) (communes)	(669.172)	Edo. □ Benin City (state capital)
(669.1)	NIGERIA. FEDERAL REPUBLIC OF NIGERIA <i>Nigeria is divided into 36 states and one territory. As there are marked cultural and religious differences between the north and south of the country, it seemed useful to divide the states in two groups: those mainly north of a line formed by the Niger and Benue rivers up to the confluence, and those mainly south of the line; but there is one state, Adamawa, that lies to either side in roughly equal proportions (so assignment to the north is somewhat arbitrary). The opportunity is taken to relocate Nigeria from (669) to (669.1), so that Sao Tomé e Príncipe at (669.95) no longer appears to be a subdivision of Nigeria</i>	(669.174)	Anambra. □ Awka (state capital)
(669.111.14)	States mainly north of the Niger-Benue river system	(669.176)	Imo. □ Owerri (state capital)
(669.112)	Kebbi. □ Birnin Kebbi (state capital)	(669.178)	Rivers. □ Port Harcourt (state capital)
(669.114)	Sokoto. □ Sokoto (state capital)	(669.182)	Bayelsa. □ Yenogoa (state capital)
(669.116)	Zamfara. □ Gusau (state capital)	(669.184)	Delta. □ Asaba (state capital)
(669.118)	Niger. □ Minna (state capital)	(669.186)	Ondo. □ Akure (state capital)
(669.122)	Abuja Federal Capital Territory. □ Abuja (national capital)	(669.188)	Osun. □ Oshogbo (state capital)
(669.124)	Nassarawa. □ Lafia (state capital)	(669.192)	Ekiti. □ Ado-Ekiti (state capital)
(669.126)	Kaduna. □ Kaduna (state capital). Zaria	(669.194)	Kwara. □ Ilorin (state capital)
(669.128)	Katsina. □ Katsina (state capital)	(669.196)	Oyo. □ Ibadan (state capital)
(669.132)	Jigawa. □ Dutse (state capital)	(669.198)	Ogun. □ Abeokuta (state capital)
(669.134)	Kano. □ Kano (state capital)	(669.199)	Lagos (state). □ Ikeja (state capital). Lagos (city)
(669.136)	Plateau. □ Jos (state capital)	(669.95)	SÃO TOMÉ E PRÍNCIPE. DEMOCRATIC REPUBLIC OF SÃO TOMÉ AND PRÍNCIPE
(669.138)	Bauchi. □ Bauchi (state capital)	(669.951)	Ilha de São Tomé
(669.142)	Gombe. □ Gombe (state capital)	(669.952)	Ilha do Príncipe
(669.144)	Yobe. □ Damaturu (state capital)	(669.953)	São João Baptista de Ajudá
(669.146)	Borno. □ Maiduguri (state capital)	(67)	EQUATORIAL, CENTRAL AND EASTERN AFRICAN STATES AND TERRITORIES
		(671.1)	CAMEROON. REPUBLIC OF CAMEROON. RÉPUBLIQUE DU CAMEROUN <i>Cameroon gained independence from France in 1960 and the following year Southern Cameroons was united with it. Northern Cameroons later joined Nigeria. Cameroon is divided into 10 provinces which are subdivided into departments. These are further subdivided into arrondissements and then into districts. The departments in Nord-Ouest and Sud-Ouest provinces, which used to be part of the British mandate, are sometimes called 'divisions'</i>
		(671.11)	Francophone provinces
		(671.111)	Extrême Nord (province). □ Diamaré (department), Maroua (provincial and department capital). Logone-et-Chari (department), Kousséri (department capital). Mayo-Danay (Mayo-Danai) (department), Yagoua (department capital). Mayo-Kani (Kaele) (department), Kaele (department capital). Mayo-Sava (department), Mora (department capital). Mayo-Tsanaga (Maya Tsanaaga, Mayo Sanaga, Mayo-Tsanga) (department), Mokolo (department capital)
		(671.112)	Nord (province). □ Bénoué (Benue) (department), Garoua (provincial and department capital). Faro (department), Poli (department capital), Mayo-Louti (Mayo-Luti) (department), Guider (department capital). Mayo-Rey (department), Tcholliré (department capital)

(1/9) Table 1e. Common auxiliaries of place

(671.113)	Adamaoua (Adamaua, Adamawa) (province). □ Vina (department), Ngaoundéré (provincial and department capital). Djerem (department), Tibati (department capital). Faro-Deo (Faro et Déo) (department), Tignere (department capital). Mayo-Banyo (Maya Banyo) (department), Banyo (department capital). Mbéré (department), Meinganga (department capital)	(671.133)	capital). Ngo-Ketunjia (Ngocketunja, Ngogetunja, Ngok Etunja)(department), Ndop (department capital) Sud Ouest (province), BuTa (provincial capital). □ Fako (department), Limbe (department capital). Koupé-Manengouba (department), Bangem (department capital). Lebialem (department), Menji (Koupé et Manengouba, Kupe-Manengouba) (department capital). Manyu (department), Mamfe (department capital). Meme (department), Kumba (department capital). Ndian (Ndan, Ndiang) (department), Mundemba (department capital)
(671.114)	Ouest (province). □ Mifi (department), Bafoussam (provincial and department capital). Bamboutos (department), Mbouda (department capital). Haut-Nkam (Upper Nkam) (department), Bafang (department capital). Hauts Plateaux (Upper Plateaux) (department), Baham (department capital). Koung-Khi (Koung-Ki) (department), Badjoun (department capital). Menoua (Ménoua)(department), Dschang (department capital). Ndé (department), Bangangte (department capital). Noun (Bamoun, Nun) (department), Foumban (department capital)	(671.8)	EQUATORIAL GUINEA. REPUBLIC OF EQUATORIAL GUINEA. REPUBLICA DE GUINEA ECUATORIAL <i>Equatorial Guinea gained independence from Spain in 1968. The division into two regions, the mainland territory of Río Muni and the island of Fernando Po, reflects the history. The regions consist of seven provinces.</i>
(671.115)	Littoral (Litoral) (province). □ Wouri (Wuri) (department), Douala (provincial and department capital). Moundou (Mungo)(department), Nkongsamba (department capital). Nkam (department), Yabassi (department capital). Sanaga-Maritime (department), Edéa (department capital)	(671.81) (671.811) (671.813)	Región Continental. Río Muni Kié-Ntem (division), Ebebiyín (division capital) Centro Sur (division), Evinayong (division capital) Litoral (division), Bata (division capital). □ Corisco, Elobey Chico, Elobey Grande Islands Wele-Nzás (division), Mongomo (division capital)
(671.116)	Centre (province). □ Mfoundi (Mfundi) (department), Yaoundé (state, provincial and department capital). Haute-Sanaga (Upper Sanaga) (department), Nanga-Eboko (department capital). Lekié (department), Monatele (department capital). Mbam-et-Inoubou (Mbam et Inougou) (department), Bafia (department capital). Mbam-et-Kim (department), Ntui (department capital). Mefou-et-Afamba (Mefou Afamba) (department), Mfou (department capital). Mefou-et-Akono (department), Ngoumou (department capital). Nyong-et-Kélé (Nyong-et-Kele) (department), Eseka (department capital). Nyong-et-Mfoumou (Myong and Mfoumou, Nyong et Mfoumo, Ny Est (province). □ Lom-et-Djerem (Lom-et-Jerem) (department), Bertoua (provincial and department capital). Boumba-et-Ngoko (Boumba-Nkogo, Bumba-Nkogo) (department), Yokadouma (department capital). Haut-Nyong (Upper Nyong) (department), Abong-Mbang (department capital). Kadey (Kaden) (department), Batouri (department capital)	(671.815) (671.817) (671.83) (671.831) (671.831.1) (671.831.3) (671.833)	Región Insular Bioko <i>Formerly Fernando Po (Poo)</i> Bioko Norte (division), Malabo (state and division capital) Bioko Sur (division), Malabo (division capital). □ Luba Annobón (division), San Antonio de Palea (division capital)
(671.117)	Sud (province). □ Mvila (Ntem) (department), Ebolowa (provincial and department capital). Dja-et-Lobo (Ja-et-Lobo) (department), Sangmelima (department capital). Océan (Kribi) (department), Kribi (department capital). Vallée-du-Ntem (department), Ambam (department capital)	(672.1)	GABON. GABONESE REPUBLIC. RÉPUBLIQUE GABONAISE <i>Gabon gained independence from France in 1960. It is divided into nine provinces which are further subdivided into departments</i>
(671.118)		(672.11)	Woleu-Ntem (province). □ Woleu (department), Oyem (provincial and department capital). Haut-Komo (department), Médouneu (department capital). Haut-Ntem (department), Minvoul (department capital). Ntem (department), Bitam (department capital). Okano (department), Mitzié (department capital)
(671.13)	Anglophone provinces	(672.12)	Estuaire (province), Libreville (state and provincial capital). □ Komo (department), Kango (department capital). Komo-Mondah (department), Ntoum (department capital). Noy (department), Cocobeach (department capital)
(671.131)	Nord Ouest (province). □ Mezam (department), Bamenda (provincial and department capital). Boyo (department), Fundong (department capital). Bui (Mbui)(department), Koumbo (department capital). Donga-Mantung (Ndonga Mendum, Nkambe) (department), NkamT (department capital). Mentchum (Mentchum, Mendum) (department), Wum (department capital). Momo (department), Mbengwi (department	(672.13) (672.14)	Moyen-OgoouT (province). □ Ogooué et des Lacs (department), Lambaréné (provincial and department capital). Abanga-Bigné (department), Ndjolé (department capital) Ogooué-Ivindo (province). □ Ivindo (department), Makokou (provincial and department capital). Lopé (department), Booué (department capital). Mvoung (department), Ovan (department capital). Zadié (department), Mékambo (department capital)

(672.15)	Ogooué-Lolo (province). □ Lolo-Bouendguidi (department), Koulamoutou (provincial and department capital). Lombo-Bouendguidi (department), Pana (department capital). Mouloundou (department), Lastoursville (department capital)	(672.432)	Niari (province), Dolisie (provincial capital). □ Banda, Divénié, Kibangou, Kimongo, Londéla-Kayes, Louvakou (Dolisie, Loubomo), Makabana, Mayoko, Mbinda, Moundoundou Nord, Moundoundou Sud, Moutamba (Mossendjo), Nyanga (Nianga), Yaya (districts)
(672.16)	Ogooué-Maritime (province). □ Bendjé (department), Port-Gentil (provincial and department capital). Étimboué (department), Ombout (department capital). Ndougou (department), Gamba (department capital). Ogoulou (department), Mimongo (department capital)	(672.433)	Lekoumou (province), Sibiti (provincial capital). □ Bambama, Komono, Mayéyé (Makabana), Zanaga (districts)
(672.17)	Ngounié (province), Masuku (provincial capital). □ Douya-Onoy (department), Mouila (provincial and department capital). Boumi-Louetsi (department), Mbigou (department capital). Dola (department), Ndendé (department capital). Louetsi-Wano (department), Lébamba (department capital). Ndolou (department), Mandji (department capital). Tsamba-Magotsi (department), Fougamou (department capital)	(672.434)	Pool (province), Kinkala (provincial capital). □ Boko, Boko-Songho, Djambala, Goma Tsé-Tsé (Ngamaba, Ngoma Tsé-Tsé), Igné (Ignie, P.K.45, PK Rouge), Kimba, Kinkala (Kibouendé), Kindamba, Louingui, Loumo, Makotipoko (Makotimpoko, Mokotipoko), Mayama, Mbandza-Ndouna (Mbanza-Ndouna), Mbon, Mindouli, Ngabé, Ngo, Ollombo, Ongoni (Ongogni, Ongonyi), Vindza (districts)
(672.18)	Haut-Ogooué (province). □ Léboumbi-Leyou (department), Moanda (department capital). Léconi-Djoué (department), Akiéni (department capital). Lékoko (department), Bakoumba (department capital). Mpassa (department), Franceville (department capital). Plateaux (department), Léconi (department capital). Sébé-Brikolo (department), Okondja (department capital)	(672.435)	Brazzaville (state capital). □ Bacongo, Makélékélé, Mfilou, Moundali, Ouenzé, Poto-Poto, Talangaï (districts)
(672.19)	Nyanga (province). □ Mougoutsi (department), Tchibanga (provincial and department capital). Basse-Banio (department), Mayumba (department capital). Douigni (department), Moabi (department capital). Haute-Banio (department), Ndindi (department capital)	(672.436)	Bouenza (province), Madingou (provincial capital). □ Kayes (Nkayi), Kingoué, Loudima, Mabombo, Madingou, Mfouati, Mouyondzi, Tsiaki, Yamba (districts)
(672.4)	CONGO. REPUBLIC OF THE CONGO <i>Congo gained independence from France in 1960. In 1970 its name was changed to People's Republic of Congo but this change was later reversed. Congo is divided into 10 regions and 1 commune, capital district or federal district (Brazzaville). The regions are divided into sub-prefectures, administrative control posts and communes. The sub-prefectures are further subdivided into cantons and communes</i>	(672.437)	Kouilou (province), Pointe-Noire (Tchamba Nzassi, Nzassi, Tchiamba-Nzassi (provincial capital). □ Hinda, Kakamoéka (Kalamoéka), Madingo-Kayes (Madingou Kayes), Mvouti (Mvoun), Nzambi (districts)
(672.41)	Northern regions	(673)	ANGOLA. REPUBLIC OF ANGOLA. REPUBLICA DE ANGOLA
(672.411)	Sangha (province), Ouessou (provincial capital). □ Mokéko, Ngbala, Pikounda (Pikoundo), Sembé, Souanké (districts)	(673.211)	Cabinda
(672.413)	Likouala (province), Impfondo (provincial capital). □ Bétou, Bouaniela (Bouagnela, Bouanéla, Bouanila), Dongou, Enyellé (Enyelé), Epéna, Impfondo, Liranga (districts)	(673.213)	Zaire
(672.415)	Cuvette Ouest (province), Ewo (provincial capital). □ Etoumbi, Kélé (Kélé), Mbama, Mbomo, Okoyo (districts)	(673.214)	Uige
(672.417)	Cuvette (province), Owandao (provincial capital). □ Boundji, Loukoléla, Makoua, Mossaka, Ngoko, Ntokou (Niokou, Tokou), Owando, Oyo, Tchikapika (districts)	(673.215)	Luanda
(672.43)	Southern regions	(673.216)	Bengo
(672.431)	Plateaux (province), Djambala (provincial capital). □ Abala, Allembé (Alembé), Gamboma, Lékana, Mpouya (districts)	(673.217)	Cuanza Norte
		(673.218)	Cuanza Sul
		(673.219)	Malanje
		(673.221)	Lunda Norte
		(673.222)	Lunda Sul
		(673.226)	Benguela
		(673.227)	Huambo
		(673.228)	Bié
		(673.229)	Moxico
		(673.235)	Namibe
			Formerly Moçamedes
		(673.236)	Huila
		(673.237)	Cunene
		(673.238)	Quando Cubango
		(674)	INNER AFRICAN (CENTRAL AFRICAN) TERRITORIES
		(674.1)	CENTRAL AFRICAN REPUBLIC <i>The Central African Republic was formerly known as Ubangi-Shari when it was one of the overseas divisions of French Equatorial Africa. Officially proclaimed the Central African Republic in 1958, from 1975-1979 it was renamed Central African Empire. The name Central African Republic was restored in 1979. The Central African Republic is divided into prefectures, except for Bangui which is an autonomous commune and Nana-Grébizi and Sangha-Mbaéré which are economic prefectures. The prefectures are further subdivided into sub-prefectures, 2002 totalling 71</i>

(1/9) Table 1e. Common auxiliaries of place

(674.11)	Western prefectures				<i>annexed by Libya. The area generally known as the Aozou (Aouzou) Strip was awarded to Chad by the International Court of Justice in 1993 and Libya withdrew its troops in 1994. The administrative divisions of Chad were completely re-organized in 1999 into departments which are subdivided into sub-prefectures. Except where noted, the capitals of the subprefectures are the same as the subprefecture. There is one exception N'Djaména which is a commune, subdivided into arrondissements</i>
(674.111)	Ouham-Pendé (prefecture), Bozoum (subprefecture and prefecture capital). □ Bocaranga, Kouï, Ngaoundaye, Paoua (subprefectures)				
(674.112)	Nana-Mambéré (Bouar-Baboua, Bouar-Baboua, Nana Mambéré) (prefecture), Bouar (subprefecture and prefecture capital). □ Abba, Baboua, Baoro (subprefectures)				
(674.113)	Mambéré-Kadéï (Haute-Sangha, Mambéré Kadéï) (prefecture), Berbérati (subprefecture and prefecture capital). □ Amada-Gaza, Carnot, Dédé-Mokouba, Gadzi, Gamboula, Sosso-Nakombo (subprefectures)	(674.31)			Northern departments
(674.114)	Sangha-Mbaéré (Sangha, Sangha-Économique, Sangha-M'Baéré) (economic prefecture), Nola (subprefecture and prefecture capital). □ Bambio, Bayanga (subprefectures)	(674.311)			(Bourkou-Ennedi-Tibesti, B.E.T.)
(674.115)	Lobaye (prefecture), Mbaïki (subprefecture and prefecture capital). □ Boda, Boganangone, Boganda, Mongoumba (subprefectures)	(674.313)			Tibesti (department), Bardaï (subprefecture and department capital). □ Aouzou, Zouar (subprefectures)
(674.116)	Ombella-M'Poko (Ombella-Mpoko, Ombelle Mpoko) (prefecture), Bimbo (subprefecture and prefecture capital). □ Boali, Bogangolo, Bossembélé, Damara, Yaloké (subprefectures)	(674.315)			Borkou (department), Faya-Largeau (subprefecture and department capital). □ Bourkou Yala (subprefecture), Kirdimi (subprefecture capital). Kouba Olanga (subprefecture)
(674.117)	Bangui (state capital and autonomous commune)				Ennedi (department), Fada (subprefecture and department capital). □ Gouro, Kalait (subprefectures). Ouadi Hawar (subprefecture), Bahaï (subprefecture capital). Ounianga Kébir (subprefecture)
(674.118)	Ouham (prefecture), Bossangoa (subprefecture and prefecture capital). □ Batangafo, Bouca, Kabo, Nana-Bakassa, Nanga-Boguila (subprefectures)	(674.33)			Central departments
(674.13)	Central prefectures	(674.331)			Kanem (department), Mao (subprefecture and department capital). □ Mondo, Nokou, Rig-Rig (subprefectures)
(674.131)	Nana-Grébizi (Gribingui, Gribingui-Économique, Ibingui, Nana Gribizi) (economic prefecture), Kaga Bondoro (subprefecture and prefecture capital). □ Mbrès (subprefecture)	(674.332)			Bar el Gazal (department), Moussoro (subprefecture and department capital).
(674.133)	Kémo (Kémo-Gribingui, Kémo-Ibingui) (prefecture), Sibut (subprefecture and prefecture capital). □ Dekoa, Mala, Ndkoukou (subprefectures)	(674.333)			□ Michémire, Salal (subprefectures)
(674.135)	Ouaka (prefecture), Bambari (subprefecture and prefecture capital). □ Bakala, Grimari, Ippy, Kouango, Markounda (subprefectures)	(674.334)			Batha Ouest (department), Ati (subprefecture and department capital). □ Djédaa (subprefecture). Fitri (subprefecture), Yao (subprefecture capital). Haraze Djombo Kibit, Koundjourou (subprefectures)
(674.137)	Bamingui-Bangoran (N'Dele, Ndélé) (prefecture), Ndélé (subprefecture and prefecture capital). □ Bamingui (subprefecture)	(674.335)			Batha Est (department), Oum Hadjer (subprefecture and department capital).
(674.15)	Eastern prefectures	(674.336)			□ Assinet, Oum Hadjer (subprefectures)
(674.151)	Vakaga (prefecture), Birao (subprefecture and prefecture capital). □ Ouanda Djallé (subprefecture)	(674.337)			Biltine (department, subprefecture and department capital). □ Am Zoer, Arada, Guéréda, Iriba (subprefectures)
(674.153)	Haute-Kotto (prefecture), Bria (subprefecture and prefecture capital). □ Ouadda, Yalinga (subprefectures)	(674.338)			Ouaddaï (department), Abéché (subprefecture and department capital). □ Abdi, Abougoudam, Am Dam, Chokoyan (subprefectures)
(674.155)	Basse-Kotto (prefecture), Mobaye (subprefecture and prefecture capital). □ Alindao, Kembé, Mingala, Satema, Zangba (subprefectures)	(674.339)			Assongha (department), Adré (subprefecture and department capital). □ Hadjer Hadid, Mabrone (subprefectures)
(674.157)	Mbomou (M'Bomou) (prefecture), Bangassou (subprefecture and prefecture capital). □ Bakouma, Gambo, Ouango, Rafan (subprefectures)	(674.340)			Sila (department), Goz Beïda (subprefecture and department capital). □ Adé, Koukou Angarana, Tissi (subprefectures)
(674.158)	Haut-Mbomou (Haut-M'bomou, Obo-Zemio) (prefecture), Obo (subprefecture and prefecture capital). □ Bambouti, Djemah, Zemio (subprefectures)	(674.35)			Southern departments
(674.3)	CHAD. REPUBLIC OF CHAD. RÉPUBLIQUE DU TCHAD <i>Chad gained full independence from France in 1960. In the mid 1970s a strip of territory along the Norther border of Chad was</i>	(674.351)			Salamat (department), Am Timan (department capital). □ Abou Deïa, Am Timan. Harazé (subprefectures)
		(674.352)			Guéra (department), Mongo (subprefecture and department capital). □ Bitkine, Mangalmé, Melfi (subprefectures)
		(674.353)			Lac Iro (department), Kyabé (subprefecture and department capital). □ Bohobé, Boum Kebir, Kyabé (subprefectures)
		(674.354)			Barh Kôh (department), Sarh (subprefecture and department capital). □ Dembo, Korbol, Maro (subprefectures)
		(674.355)			Mandoul (department), Koumra (subprefecture and department capital). □ Béjondo, Bouna, Goundi, Moïssala (subprefectures)

(674.356)	Logone Oriental (department), Doba (subprefecture and department capital). □ Bébedjia. Béboto. Bodo, Goré (subprefectures)		<i>the Katanga province was called Shaba and Haut and Bas Congo were Haut- and Bas-Zaïre respectively. The names of major cities have been provided with their colonial forms in brackets to assist reference</i>
(674.357)	Monts de Lam (department), Baïbokoum (subprefecture and department capital). □ Baïkoro. Bessao, Larmanaye (subprefectures)	(675.1)	Northern provinces
(674.358)	Tandjilé Est (department), Laï (subprefecture and department capital). □ Déressia (subprefecture). Dono Manga (subprefecture), Guidari (subprefecture capital)	(675.11)	Equateur (province), Mbandaka (Coquilhatville) (administrative zone and provincial capital). □ Basankusu, Befale, Bikoro, Boende, Bokungu, Bolomba, Bomongo, Bongandan, Bosobolo, Budjala, Bumba, Businga, Djolu, Gemena, Ikela, Ingende, Kungu, Libenge, Lisala, Lukolelave, Makanza, Mobayi-Mbongo, Monkot, Nzulu, Wangata, Wango, Yakoma (administrative zones)
(674.359)	Tandjilé Ouest (department), Kélo (subprefecture and department capital). □ Béré, Dafra (subprefectures)		Haut-Congo (province), Kisangani (Stanleyville) (administrative zone and provincial capital). □ Aketi, Ango, Aru, Bafwasende, Bambesa, Banalia, Basoko, Bondo, Buta, Djugu, Dungu, Faradje, Irumu, Isangi, Kabondo, Kisangani, Lubunga, Mahagi, Makiso, Mambasa, Mangobo, Niangara, Opala, Poko, Rungu, Tshopo, Ubundu, Wamba, Watsa, Yahuma (administrative zones)
(674.37)	Western departments		<i>There are two places called Basoko in the Democratic Republic of Congo, at (677.13) and (677.31); and two places called Kabondo, at (677.13) and (677.33)</i>
(674.371)	Lac (department), Bol (subprefecture and department capital). □ Baga Sola, Doum-Doum (subprefectures). Liwa (subprefecture), Guim (subprefecture capital), Ngouri (subprefecture)	(675.13)	Nord-Kivu (province), Goma (administrative zone and provincial capital). □ Beni, Lubero, Masisi, Rutshuru, Walikale (administrative zones)
(674.372)	Hadjer Lamis (department), Massaguet (subprefecture and department capital). □ Karal, Linia, Mandéla, Mani, Massakory (subprefectures)		Southern provinces
(674.373)	N'Djaména (Nyamena, Yamena) (state capital and commune)		Bandundu (Banningville) (province and provincial capital). □ Bagata, Basoko, Bolobo, Bulungu, Disasa, Feshi, Gungu, Idiofa, Inongo, Kahemba, Kasongo-Lunda, Kazamba, Kenge, Kiri, Kutu, Lukeni, Lukolela, Masi-Manimba, Mayoyo, Mushie, Nzinda, Oshwe, Popokabak (administrative zones)
(674.374)	Dababa (department), Bokoro (subprefecture and department capital). □ Doba, Gama, Moïto (subprefectures)	(675.15)	<i>There are two places called Basoko in the Democratic Republic of Congo, at (677.13) and (677.31)</i>
(674.375)	Baguirmi (department), Massénya Adré (subprefecture and department capital). □ Bousso, Dourbali (subprefectures)	(675.3)	Kinshasa (Leopoldville) (state capital). □ Bandalung, Barumbu, Bumbu, Gombe, Kalamu, Kasa-Vubu, Kimbanseke, Kintambo, Kisenso, Lemba, Limete, Lingwala, Makala, Maluku, Masina, Matete, Mont-Ngafu, Ndjili, Ngaba, Ngaliema, Ngiri-Ngiri, Nsele, Selembao (administrative zones)
(674.376)	Mayo Boneye (department), Bongor (subprefecture and department capital). □ Guélandeng, Kim (subprefectures). Rigaza (subprefecture), Biliam Oursi (subprefecture capital)	(675.31)	<i>There are two places called Kalamu in the Democratic Republic of Congo, at (677.32) and (677.33)</i>
(674.377)	Kabia (department), Gounou Gaya (subprefecture and department capital). □ Binder, Fianga (subprefectures)		Bas-Congo (province), Matadi (administrative zone and provincial capital). □ Kabondo, Kalamu, Kasangulu, Kimvula, Lukula, Luozi, Madimba, Mbanza-Ngungu, Moanda, Mvuzi, Nzadi, Nzanza, Seke-Banza, Songololo, Tshela (administrative zones)
(674.378)	Mayo-Dallah (Mayo Dala)(department), Pala (subprefecture and department capital). □ Gagai, Lagon, Léré (subprefectures)	(675.32)	<i>There are two places called Kabondo in the Democratic Republic of Congo, at (677.13) and (677.33); and two places called Kalamu, at (677.32) and (677.33)</i>
(674.379)	Logone Occidental (department), Moundou (subprefecture and department capital). □ Beinamar (subprefecture). Krim-Krim (subprefecture), Bao (subprefecture capital)		Kasai-Occidental (province), Kananga (Luluabourg) (administrative zone and provincial capital). □ Dekese, Demba, Dibaya, Dimbelenge, Ilebo, Katoka, Kazumba, Luebo, Luiza, Lukonga, Mweka, Ndesha, Nganza, Tshikapa (administrative zones)
(675)	DEMOCRATIC REPUBLIC OF CONGO <i>Formerly the colony of Belgian Congo. From 1926 to 1960. The Belgian mandate (late Trust Territory) of Ruanda-Urundi (now Rwanda) was also affiliated to the colony. It became independent in 1960, becoming Republic of the Congo. In July 1960 the southern province of Katanga seceded, and in August the mining State of South Kasai also proclaimed its independence. In early 1963 both were again incorporated into the Congo state. In 1964 the country's name became Democratic Republic of the Congo and in 1971 Republic of Zaïre was adopted as the official name of the state. In 1997 it changed back to the Democratic Republic of Congo; Congo is divided into eleven regions, or provinces, including the city of Kinshasa. The regions are divided into sub-regions which are divided into administrative zones, which are further subdivided into collectivities. Each collectivity contains groupings which in turn contain numerous villages. From 1972-1997</i>	(675.33)	Kasai-Oriental (province), Mbuji-Mayi (Bakwanga) (provincial capital). □ Bipemba, Dibindi, Diulu, Kabeya-Kamwanga, Kabinda,

(1/9) Table 1e. Common auxiliaries of place

	Kamiji, Kanshi, Katoko-Kombe, Katanda, Kole, Lodja, Lomela, Lubao, Lubefu, Lupatapata, Lusambo, Miabi, Muya, Mwene-Ditu, Ngandajika, Tshilenge (administrative zones)	(675.973.6)	Bujumbura Rural. □ Isale, Kabezi, Kanyosha, Mugongomanga, Muhuta, Mukiki, Mutambu, Mutimbuzi (communes)
(675.36)	Maniema (province), Kindu (administrative zone and provincial capital). □ Kabambare, Kasongo, Kibombo, Lubutu, Oangi, Punia (administrative zones)	(675.973.7)	Bururi. □ Burambi, Buyengero, Matana, Mubimbi, Mugamba, Rutovu Rumonge, Songa, Vyanda (communes)
(675.37)	Sud-Kivu (province), Bukavu (Cotermansville) (provincial capital). □ Bagira, Fizi, Ibanda, Ile Idjwi, Kabare, Kadutu, Kalehe, Mwenga, Shabunda, Uvira, Walungu (administrative zones)	(675.973.8)	Rutana. □ Bukemba, Gihao, Gitanga, Mpinga, Musongati (communes)
(675.38)	Katanga (province), Lubumbashi (+Isabethville) (provincial capital). □ Annexe, Bukama, Dilala, Dilolo, Kabalo, Kabongo, Kalemie, Kamalondo, Kambove, Kamina, Kampemba, Kaniama, Kapanga, Kasenga, Katuba, Kenya, Kikula, Kipushi, Kongolo, Likasi, Lubudi, Lubumbashi, Malemba Nkulu, Manika, Manono, Mitwaba, Moba, Mutshatsh, Nyunzu, Panda, Pweto, Ruashi, Sakanina, Sandoa, Tshituru (administrative zones)	(675.973.9)	Makamba. □ Kayogoro, Kibago, Mabanda, Nyanza-Lac, Vugizo (communes)
		(675.98)	RWANDA. REPUBLIC OF RWANDA <i>Rwanda was formerly the Northern division (Ruanda) of the Belgian administered Trust territory of Ruanda-Urundi. It achieved independence in 1962 and became the Republic of Rwanda. Rwanda is divided into 12 provinces which are further divided into districts and municipalities. The provincial capitals have the same name as their provinces and are municipalities</i>
(675.97)	BURUNDI. REPUBLIC OF BURUNDI <i>Burundi was formerly the southern division (Urundi) of the Belgian administered Trust Territory of Ruanda-Urundi. It became independent as the Kingdom of Burundi in 1960, and following the deposition of the monarch in 1966 it became the Republic of Burundi. Burundi is divided into provinces, and the capitals have the same name as the province. The regions are subdivided into 114 districts and the districts are divided into communes</i>	(675.981)	Western provinces
		(675.981.1)	Ruhengeri. □ Bugarura, Buhoma, Bukamba, Bukonya, Kinigi, Mutobo, Nyamugali, Nyarutovu (districts)
		(675.981.2)	Gisenyi (Kisenyi). □ Bigogwe, Gaseke, Gasize, Gatumba, Kageyo, Kanama, Kayove, Nyamyumba (districts)
		(675.981.3)	Kibuye. □ Budaha, Gisunzu, Itabire, Rusenye (districts)
		(675.981.4)	Cyangugu (Shangugu). □ Bugarama, Bukunzi, Cyanzarwe, Gashonga, Gatara, Impala, Nyamasheke (districts)
		(675.981.5)	Gikongoro. □ Kaduha, Karaba, Mudasonwa, Mushubi, Nshili, Nyaruguru (districts)
		(675.981.6)	Butare (Astrida). □ Nyanza (municipality). Gikonko, Kibingo, Kirihura, Maraba, Mugobmwa, Nyakizu, Save (districts)
(675.971)	Northern provinces	(675.981.7)	Gitarama. □ Kabagari, Kamonyi, Kayumbu, Muhanga, Ndiza, Ntenyo, Ntongwe, Ruhango, Ruyumba (districts)
(675.971.1)	Cibitoke. □ Buganda, Bukinanyana, Mabayi, Mugina, Murwi, Rugombo (communes)		
(675.971.2)	Bubanza. □ Gihanga, Mpanda, Musigati, Rugazi (communes)	(675.983)	Eastern provinces
(675.971.3)	Kayanza. □ Butaganzwa, Gahombo, Gatara, Kabarore, Matongo, Muhanga, Muruta, Rango (communes)	(675.983.2)	Kigali-Ville (state capital). □ Butamwa, Gikondo, Gisozi, Kacyiru, Kanombe, Kicukiro, Nyamirambo, Nyarugenge (districts)
	<i>There are two places called Butaganzwa in Burundi, at (675.971.3) and (675.973.1)</i>	(675.983.3)	Kigali-Rural (Kigali Ngali, Kigali Ngari, Kinyarwanda). □ Bicumbi, Buliza, Butaro, Cyeru, Gasabo, Gashora, Kabugad, Ngenda, Nyamata, Rulindo, Rushashi, Shyorongi (districts)
(675.971.4)	Ngozi. □ Busiga, Gashikanwa, Kiremba, Marangara, Murenza, Mwumba, Ruhororo, Tangara (communes)	(675.983.4)	Kibungo. □ Rwamagana (municipality). Cyarubare, Kayanza, Kigarama, Mirenge, Muhazi, Nyarubuye, Rukira, Rusumo (districts)
(675.971.5)	Kirundo. □ Bugabira, Busoni, Bwambarangwe, Gitobe, Ntega, Vumbi (communes)	(675.983.5)	Umutara (Mutara). □ Bugaragara, Gabiro, Kahi, Murambi, Muvumba, Rukara (districts)
(675.971.6)	Muyinga (Muhinga). □ Buhinyuza, Gashoho, Gasorwe, Giteranyi, Mwakiro (communes)	(675.983.6)	Byumba. □ Bungwe, Gatsibo, Humure, Kinyihara, Kisaro, Rebero, Rushaki, Rwamiko (districts)
(675.971.7)	Karuzi (Karusi). □ Bugenyuzi, Buhiga, Gihogazi, Gitaramuka, Mutumba, Nyabikere, Shombo (communes)		
(675.971.8)	Cankuzo. □ Cendajuru, Gisagara, Kigamba, Mishiha (communes)	(676.1)	UGANDA. REPUBLIC OF UGANDA <i>Uganda was formerly a protectorate, a British dependency within the realm popularly known as British East Africa in the colonial period. It achieved independence in 1962 and in 1967 was proclaimed the Republic of Uganda. Historically, it was divided into four regions and although these no longer have any administrative status they are still referred to. The primary administrative division is into districts which are further subdivided into counties, municipalities and one city council. The capitals of the districts have the same name</i>
(675.973)	Southern provinces		
(675.973.1)	Ruyigi. □ Butaganzwa, Butezi, Bweru, Gisuru, Kanyinya, Nyabitsinda (communes)		
	<i>There are two places called Butaganzwa in Burundi, at (675.971.3) and (675.973.1)</i>		
(675.973.2)	Gitega (Kitega). □ Bugendana, Bukirasazi, Buraza, Giheta, Gishubi, Itaba, Makebuko, Mutaho, Ryansoro (communes)		
(675.973.3)	Muramvya. □ Bukeye, Butihinda, Kiganda, Mbuye, Rutegama (communes)		
(675.973.4)	Mwaro. □ Bisoro, Gisozi, Kayokwe, Ndava, Nyabihanga, Rusaka (communes)		
(675.973.5)	Bujumbura Mairie (Usumbura) (state capital)		

	<i>as their districts with the exception of Kabarole. This information is based on the 2002 census</i>	(676.151.6)	Kampala (state capital and city council)
		(676.151.7)	Kalangala (district). □ Bujumba, Kyamuswa (counties)
(676.11)	Northern region	(676.151.8)	Mukono (district). □ Buikwe, Buvuma Islands, Nakifuma (counties)
(676.111.1)	Yumbe (district). □ Aringa (county)	(676.151.9)	Mubende (district). □ Busujju, Buwekula, Kassanda, Mityana (counties)
(676.111.2)	Arua (district and municipality). □ Ayivu, Koboko, Madi-Okollo, Maracha, Terego, Vurra (counties)	(676.152.1)	Mpigi (district). □ Butambala, Gomba, Mawokota (counties)
(676.111.3)	Nebbi (district). □ Jonam, Okoro, Padyere (counties)	(676.152.2)	Sembabule (district). □ Lwemiyaga, Mawogola (counties)
(676.111.4)	Gulu (district and municipality). □ Aswa, Kilac, Nwoya, Omoro (counties)	(676.152.3)	Masaka (district and municipality). □ Bukomansimbi, Bukoto, Kalungu
(676.111.5)	Moyo (district). □ Obongi, West Moyo (counties)	(676.152.4)	Rakai (district). □ Kabula, Kakuuto, Kooki, Kyotera
(676.111.6)	Adjumani (district). □ East Moyo (county)	(676.17)	Eastern region
(676.111.7)	Kitgum (district). □ Chua, Lamwo (counties)	(676.171.1)	Katakwi (district). □ Amuria, Kapelebyong, Usuk (counties)
(676.111.8)	Pader (district). □ Agago, Aruu (counties)	(676.171.2)	Kabarama (district). □ Kalaki (county)
(676.111.9)	Kotido (district). □ Dodoth, Jie, Labwor (counties)	(676.171.3)	Soroti (district and municipality). □ Kasilo, Serere, Soroti (counties)
(676.112.1)	Moroto (district and municipality). □ Bokora, Matheniko (counties)		<i>Soroti district is distinct from Soroti county</i>
	<i>Moroto district (676.112.1) is distinct from Moroto county (676.112.3)</i>	(676.171.4)	Kumi (district). □ Bukedea, Ngora (counties)
(676.112.2)	Nakapiripirit (district). □ Chekwii, Pian, Pokot (counties)	(676.171.5)	Kamuli (district). □ Budioppe, Bugabula, Bulamogi, Buzaaya (counties)
(676.112.3)	Lira (district and municipality). □ Dokolo, Erute, Kioga, Moroto, Otupe (counties)	(676.171.6)	Pallisa (district). □ Budaka, Butebo, Kibuku (counties)
	<i>Moroto district (676.112.1) is distinct from Moroto county (676.112.3)</i>	(676.171.7)	Kapchorwa (district). □ Kongasis, Kween, Tingey (counties)
(676.112.5)	Apac (district). □ Kole, Kwana, Maruzu, Oyam (counties)	(676.171.8)	Sironko (district). □ Budadiri, Bulambuli (counties)
(676.13)	Western region	(676.171.9)	Mbale (district and municipality). □ Bubulo, Bungokho, Mamnjiya (counties)
(676.131.1)	Masindi (district). □ Bujenje, Kibanda (counties)	(676.172.1)	Tororo (district and municipality). □ Bunyloe, Kisoko (West Budama), Tororo (counties)
(676.131.2)	Hoima (district). □ Bugahya, Buhaguzi (counties)		<i>Tororo district is distinct from Tororo county</i>
(676.131.3)	Kibale (district). □ Bugangaizi, Buyaga (counties)	(676.172.2)	Busia (district). □ Samia-Bugwe (county)
(676.131.4)	Bundibugyo (district). □ Bwamba, Ntoroko (counties)	(676.172.3)	Inganga (district). □ Bugweri, Busiki, Luuka (counties)
(676.131.5)	Kyenjojo (district). □ Kyaka, Mwenge (counties)	(676.172.4)	Bugiri (district). □ Bukooli (county)
(676.131.6)	Kabarole (district), Fort Portal (municipality and district capital). □ Bunyangabu, Burahya (counties)	(676.172.5)	Mayuge (district). □ Bunya (county)
(676.131.7)	Kamwenge (district). □ Kibaale, Kitawenda (counties)	(676.172.8)	Jinja (district and municipality). □ Butembe, Kagoma (counties)
(676.131.8)	Kasese (district). □ Bukonjo, Busongora (counties)	(676.2)	KENYA. REPUBLIC OF KENYA
(676.131.9)	Bushenyi (district). □ Buhweju, Bunyaruguru, Igara, Ruhinda, Sheema (counties)	(676.2A/Z)	Provinces (and Area) of Kenya. □ Central. Coast. Eastern. Nairobi (Area). North- Eastern. Nyanza. Rift Valley. Western
(676.132.1)	Mbarara (district and municipality). □ Bukanga, Ibanda, Isingiro, Kashari, Kazo, Nyabushozi, Rwampara (counties)	(677)	SOMALIA. SOMALI DEMOCRATIC REPUBLIC
(676.132.2)	Ntungamo (district). □ Kajara, Ruhaama, Rushenyi (counties)	(677A/Z)	Regions of Somalia. □ Bakool. Bari. Banaadir. Bay. Galguduud. Gedo. Hiraa. Juddaba. Dhexe. Jubbada Hoose. Mudug. Nugaal. Sanaag. Shabeellaha Dhexe. Shabeellaha Hoose. Togdheer. Woqoyi Galbeed
(676.132.3)	Rukungiri (district). □ Rubabo, Rujumbura (counties)	(677.1)	DJIBOUTI. REPUBLIC OF DJIBOUTI
(676.132.4)	Kanungu (district). □ Kinkiizi (county)	(677.1A/Z)	Cercles of Djibouti. □ Ali-Sahih. Dikhil. Djibouti. Obock. Tadjoura
(676.132.5)	Kisoro (district). □ Bufumbira (county)	(678)	TANZANIA. UNITED REPUBLIC OF TANZANIA
(676.132.6)	Kabale (district and municipality)	(678A/Z)	Regions of Tanzania. □ Arusha. Bukoba. Dar es Salaam. Dodoma. Iringa. Kagera. Kigoma. Kilimanjaro. Lindi. Mara. Mbeya. Morogoro. Mtwara. Mwanza. Pemba North. Pemba South. Pwani. Rukwo. Ruvuma. Shinyanga. Singida. Tabora. Tanga. Zanzibar North. Zanzibar Urban/West. Zanzibar Central/South
(676.15)	Central region	(679)	MOZAMBIQUE. REPUBLIC OF MOZAMBIQUE
(676.151.1)	Kiboga (district)	(679.1)	Niassa
(676.151.2)	Nakasongola (district). □ Buruli (county)		
(676.151.3)	Luwero (district). □ Bamunanika, Katikamu, Nakaseke (counties)		
(676.151.4)	Kayunga (district). □ Bbaale, Ntenjeru (counties)		
(676.151.5)	Wakiso (district). □ Entebbe (municipality). Busiuro, Kyadondo, (counties)		

(1/9) Table 1e. Common auxiliaries of place

(679.2)	Cabo Delgado	(684.4)	Eshowe. Inanda. Lower Tugela. Maphumulo.
(679.3)	Nampula		Mtunzini. Ndwedwe. □ Maudini. Stanger.
	<i>Formerly Moçambique</i>		Tongaat
(679.5)	Gaza	(684.5)	Pinetown. Port Shepstone. Umzinto.
(679.6)	Tete		Umbumbulu. □ Margate. Scottburgh. Uvongo
(679.74)	Sofala	(684.55)	Durban. □ Chatsworth. Umlazi
	<i>Formerly Beira</i>	(684.6)	Alfred. Mount Currie. Umzimkulu. □ Harding.
(679.77)	Manica		Kokstad
	<i>Formerly Vila Pery</i>	(684.7)	Bergville. Camperdown. Eatcourt. Ixopo.
(679.8)	Inhambane		Kliprivier. Kranskop. Lions River. Mooirivier.
(679.9)	Maputo		Mpendle. New Hanover. Polela. Richmond.
	<i>Formerly Lourenço Marques</i>		Umvoti. Underberg. Weenen. □ Colenso.
(68)	STATES AND TERRITORIES OF SOUTHERN AFRICA		Greytown. Howick. Ladysmith
		(684.75)	Pietermaritzburg
(680)	SOUTH AFRICA. REPUBLIC OF SOUTH AFRICA	(685)	Free State
			<i>Formerly Orange Free State</i>
(680A/Z)	Homelands of South Africa. □ Bophutatswana.	(685.1)	Bethlehem. Ficksburg. Fouriesburg. Frankfort.
	Ciskei. Gazankulu. Ka Ngwane. Kwa Ndebele.		Harrismith. Lindley. Reitz. Senekal. Vrede.
	Kwa Zulu. Lebowa. Qwaqwa. Transkei. Venda		Witsieshoek
(682.1)	Gauteng	(685.2)	Bothaville. Heilbron. Koppies. Kroonstad.
	<i>Formerly P.W.V. (Pretoria, Witwatersrand, Vereeniging)</i>	(685.3)	Parys. Sasolburg. Viljoenskroon. Vredefort
(682.111)	Pretoria		Brandfort. Bultfontein. Hennenman. Hoopstad.
(682.12)	Bronkhorstspuit. Cullinan. Wonderboom		Odendaalsrus. Theunissen. Ventersburg.
(682.13)	Witwatersrand. □ Alberton. Benoni. Boksburg.	(685.4)	Virginia. Welkom. Wesselsbron. Winburg.
	Brakpan. Carletonville. Germiston. Kempton	(685.5)	□ Allanridge
	Park. Krugersdorp. Nigel. Oberholzer.	(685.6)	Bloemfontein
	Randburg. Randfontein. Roodepoort. Springs.		Clocolan. Excelsior. Ladybrand. Marquard
	Westonaria	(685.7)	Dewetsdorp. Reddersburg. Rouxville.
(682.131)	Johannesburg. □ Soweto		Smithfield. Wepener. Zastron
(682.14)	Vereeniging	(685.8)	Bethulie. Edenburg. Fauresmith. Jagersfontein.
(682.141)	Heidelberg. Meyerton. Vanderbijlpark		Petrusburg. Philippolis. Trompsburg
(682.4)	North West	(686)	Boshof. Jacobsdal. Koffiefontein
(682.41)	Bafokeng. Brits. Koster. Lehutshhe. Madikwe.		Basutoland. Kaffraria (historical)
	Mankwe. Marico. Rustenburg. Swartruggens	(687)	Kaap Provinsie/Cape Province (historical).
(682.42)	Coligny. Delarayville. Ditsobotla. Klerksdorp.		□ Kaap die Goeie Hoop. Cape of Good Hope
	Lichtenburg. Molopo. Potchefstroom.	(687.1)	Northern Cape
	Ventersdorp. □ Orkney. Stilfontein	(687.11)	Barkly West. Hartswater. Hay. Herbert.
(682.43)	Bloemhof. Christiana. Ganyesa. Kudumane.		Kuruman. Postmasburg. Warrenton
	Schweizer-Reneke. Taung. Vryburg.	(687.112)	Kimberley
	Wolmaransstad	(687.12)	Gordonia. Kenhardt. Prieska. □ Upington
		(687.13)	Britstown. Colesberg. De Aar. Hanover.
(682.5)	North Province		Hopetown. Noupoot. Philipstown. Richmond
	<i>Formerly Northern Transvaal</i>	(687.17)	Calvinia. Carnarvon. Fraserburg.
(682.51)	Dzanani. Messina. Mutale. Soutpansberg.		Namakwaland. Sutherland. Victoria-West.
	Thohoyandou		Willistown
(682.52)	Bochum. Bolebedu. Giyani. Mokerong.	(687.3)	Western Cape
	Pietersburg. Seshago. Sekgosese. Vuwani	(687.31)	Clanwilliam. Vanrhynsdorp. Vredendal
(682.53)	Ellisras. Potgietersrus. Thabamooa.	(687.33)	Bredasburg. Caledon. Ceres. Heidelberg.
	Thabazimbi. Warmbad. Waterberg		Hermanus. Hopefield. Malmesbury. Montagu.
(682.54)	Letaba. Mhala. Namakgale. Naphuno. Nebo.		Moorreesburg. Paarl. Piketberg. Riversdal.
	Phalaborwa. Ritavi. Sekhukhuneland		Robertson. Somerset-West. Strand.
(682.6)	Mpumalanga		Stellenbosch. Swellendam. Tuilbagh.
	<i>Formerly Eastern Transvaal</i>		Vredenburg. Wellington. Worcester
(682.61)	Barberton. Groblersdal. Kamhlushwa.	(687.34)	George. Knysna. Mosselbaai. Uniondale
	Lydenburg. Moutse. Nelspruit. Nxikzi. Pilgrim's	(687.35)	Cape Peninsula: Bellville. Cape (district).
	Rest. Witrivier		Goodwood. Kuilsriver. Simonstown. Wynberg
(682.62)	Amersfoort. Balfour. Belfast. Bethal. Carolina.	(687.355)	Cape Town
	Delmas. Eerstehoek. Ermelo. Highveld Ridge.	(687.36)	Calitzdorp. Ladismith. Oudtshoorn
	Middelburg. Piet Retief. Standerton. Volksrust.	(687.38)	Beaufort-West. Laingsburg. Murraysburg.
	Wakkerstroom. Waterval Boven. Witbank		Prince Albert
(684)	KwaZulu/Natal (KwaZulu-Natal)	(687.5)	Eastern Cape
	<i>Formerly Natal (KwaZulu/Natal)</i>	(687.51)	Aberdeen. Hankey. Humansdorp. Joubertina.
(684.1)	Dannhauser. Dundee. Glencoe. Newcastle.		Willowmore
	Utrecht	(687.53)	Cradock. Graaff-Reinet. Hofmeyr. Jansenville.
(684.2)	Babanango. Ngotshe. Paulpietersburg.		Middelburg. Pearston. Somerset-East.
	Vryheid. □ Louwsburg		Steytlerville. Tarkastad
(684.3)	Hlabisa. Ingwavuma. Lower Umfolozi.	(687.55)	Adelaide. Albany. Alexandria. Bathurst.
	Mahlabitini. Mtonjaneni. Nongoma. Ubombo.		Bedford. Cathcart. Fort Beaufort. Hewu.
	□ Empangeni. Melmoth. Mtubatuba		Keiskamahoe. King William's Town. Kirkwood.

	Komga. Mdantsane. Middledrift. Mpopu. Peddie. Queenstown. Sterkstroom. Stutterheim. Uitenhage. Victoria-East	(687.551)	Port Elizabeth	(689)	RHODESIA AND NYASALAND. FEDERATION OF RHODESIA AND NYASALAND, -1963
(687.552)	East London	(689.1)		(689.1A/Z)	ZIMBABWE. REPUBLIC OF ZIMBABWE Administrative subdivisions of Zimbabwe. □ Manicaland. Mashonaland Central. Mashonaland East. Mashonaland West. Masvingo (Victoria). Matabeleland North. Matabeleland South. Midlands
(687.56)	Albert. Aliwal-North. Barkly-East. Elliot. Indwe. Lady Grey. Maclear. Molteno. Steynsburg. Venterstad. Wodehouse	(687.57)	Bizana. Cacadu. Cala. Centane. Cofimvaba. Engcobo. Gatyana. Gouwa. Herschel. Idutywa. Kwa Bhaca. Libode. Lusikisiki. Mount Fletcher. Matatiele. Maxesibeni. Mqanduli. Mzimvusu. Ngqeleni. Nqamakwe. Qumbu. Sipaqeni. Tabankulu. Tsolo. Tsomo. Umtata. Xhora <i>For Mount Currie see (684.6)</i> ⇒ (684.4)	(689.4)	ZAMBIA. REPUBLIC OF ZAMBIA <i>Northern Rhodesia to 1964</i> Provinces of Zambia. □ Central. Copperbelt. Eastern. Luapula. Lusaka. Northern. North Western. Southern. Western
(687.85)	British Bechuanaland, -1895	(689.7)		(689.7A/Z)	MALAWI. REPUBLIC OF MALAWI Regions and Districts of Malawi. □ Central Region: Dedza. Dowa. Kasungu. Lilongwe. Mchinji. Ncheu. Nkhokota. Ntchisi. Salima. Northern Region: Chitipa. Karonga. Mzimba. Nkhata Bay. Rumphi Southern Region: Blantyre. Chickwawa. Chiradzulu. Kasupe. Mangochi. Mulanje. Nasanje. Thyolo. Zomba
(688.1)	NAMIBIA	(69)		(691)	AFRICAN ISLANDS OF THE INDIAN AND SOUTH ATLANTIC OCEANS
(688.11)	Caprivi East. Damaraland. Grootfontein. Bushmanland: Kavangoland, Kakaoland. Owambo. Tsemeb	(691A/Z)		(691A/Z)	MADAGASCAR. DEMOCRATIC REPUBLIC OF MADAGASCAR Provinces of Madagascar. □ Antanarivo. Antsiranana. Fianarantsoa. Mahajanga. Toamasina. Toliara
(688.12)	Gobabis: Hereroland East, Hereroland West. Karibib. Okahandja. Omaruru. Otjiwarongo. Outjo	(694)		(694.1)	ISLANDS NORTH OF MADAGASCAR Comoro Islands. Archipel des Comores Comoros. Union of the Comoros Mayotte. Collectivité territoriale de Mayotte (France)
(688.125)	Walvis Bay	(694.11)		(694.12)	SEYCHELLES. REPUBLIC OF SEYCHELLES. □ Mahé. Praslin. Ile Silhouette. Cosmoledo Islands. Providence Island. Amirante Islands
(688.13)	Bethanien. Gibeon. Karasburg. Keetmanshoop. Lüderitz. Maltahöhe. Mariental. Namaland. Rehoboth. Swakopmund. Windhoek	(696)		(697)	BRITISH INDIAN OCEAN TERRITORY (BIOT) (UK). □ Diego Garcia. Peros Banhos. Salomon. Danger Island. Six Islands (Egmont Islands). Three Brothers (Trois Frères). Aldabra Islands. Farquhar Islands. Desroches
(688.3)	BOTSWANA. REPUBLIC OF BOTSWANA <i>Botswana is divided into districts, city and town councils. The districts are further subdivided into sub-districts, city councils, town councils and townships. These change frequently. Some sources place them on the same level as districts but the standards class them as subordinate, as does Law</i>	(698)		(698.1)	MASCARENE ISLANDS RÉUNION (FRANCE). □ Iles Glorieuses. Tromelin <i>Formerly Ile de Bourbon</i>
(688.342)	North-West (district), Maun (capital). □ Chobe, Gomare, Kasane, Okavango (sub-districts)	(698.2)		(699)	MAURITIUS. REPUBLIC OF MAURITIUS. □ Agalega. Rodrigues. Cargados Carajos Shoals
(688.344)	North-East (district), Francistown (capital and city council). □ Masunga (sub-district)	(699.1)		(699.11)	OUTLYING ISLANDS IN THE SOUTH INDIAN AND SOUTH ATLANTIC OCEANS ⇒ (99)
(688.346)	Central (district), Serowe (capital). □ Bobonong, Letlhakane, Mahalapye, Palapye, Tutume (sub-districts). Selebi-Phikwe, Sowa (town councils)	(699.12)		(699.13)	ISLANDS IN THE SOUTH INDIAN OCEAN Ile Amsterdam (Nouvelle Amsterdam/New Amsterdam) (France) Ile Saint-Paul (France) Iles Kerguelen (France) Heard and McDonald Islands (Australia) Heard Island (Australia) McDonald Islands (Australia) Iles Crozet (France) Prince Edward Islands (South Africa). □ Marion Island
(688.348)	Kgatleng (district), Mochudi (capital)	(699.14/.15)		(699.15)	
(688.352)	Kweneng (district), Molepolole (capital). □ Letlhakeng (sub-district)	(699.16)		(699.17)	
(688.354)	Kgalagadi (district), Tshabong (capital). □ Hukuntsi (sub-district)				
(688.356)	South-East (district), Gaborone (state and district capital and city council). □ Tlokweng (sub-district), Lobatse (town council)				
(688.358)	Southern (Ngwaketse) (district), Kanye (capital). □ Mabutsane, Good Hope (sub-districts). Jwaneng (town council)				
(688.359)	Ghanzi (Gantsi) (district and capital). □ Charles Hill (sub-district)				
(688.5)	LESOTHO				
(688.51)	Berea. Butha Buthe. Leribe				
(688.52)	Mafeteng. Maseru. Mohale's Hoek. □ Roma				
(688.53)	Mokhotlong. Qacha's Nek. Quthing. Thaba Tseka. □ Marakabei				
(688.7)	SWAZILAND				
(688.71)	Hhohho. Manzini. □ Mbabane. Mtsapa. Piggs Peak				
(688.72)	Lubombo. Shiselweni. □ Big Bend				

(1/9) Table 1e. Common auxiliaries of place

(699.2)	ST HELENA AND DEPENDENCIES (UK). □ St Helena. Ascension. Tristan da Cunha	(712.445)	Regina
(7)	NORTH AND CENTRAL AMERICA	(712.7)	Manitoba
(71)	CANADA	(712.71)	Northern region. □ Churchill, Port Nelson, Thompson ⇒ (282.274.1) ⇒ (282.274.2)
(711)	British Columbia ⇒ (237.12)	(712.72)	Central region. □ Dauphin, Flin Flon, The Pas ⇒ (282.274.25)
(711.1)	Northern Coastal region. □ Bella Coola, Kitimat, Ocean Falls, Prince Rupert ⇒ (265.235) ⇒ (265.236) ⇒ (265.238) ⇒ (265.239)	(712.73)	Southwestern region. □ Brandon, Minnedosa, Portage la Prairie
(711.12)	Queen Charlotte Islands	(712.74)	Southeastern region. □ Selkirk
(711.2)	Vancouver Island ⇒ (265.233) ⇒ (265.235)	(712.743)	Winnipeg
(711.28)	Victoria ⇒ (265.225)	(713)	Ontario <i>Upper Canada 1791-1841</i>
(711.3)	Southwestern region ⇒ (282.271.1)	(713.1)	Northern Ontario and Georgian Bay regions
(711.31)	Southern coastal region	(713.11/.14)	Northern Ontario region
(711.33)	Greater Vancouver Regional District. □ City of Vancouver	(713.11)	Northwestern Ontario. □ Kenora District, Rainy River District
(711.37)	Lower Fraser Valley	(713.12)	Thunder Bay District ⇒ (282.272.115)
(711.5)	Okanagan-Similkameen region. □ Armstrong, Enderby, Kelowna, Osoyoos, Penticton, Princeton, Vernon ⇒ (237.13)	(713.13)	Northeastern Ontario. □ Sudbury
(711.6)	Southeastern region ⇒ (282.271.2)	(713.14)	Clay Belt. □ Cochrane District. Timiskaming District. Nipissing District
(711.62)	West Kootenay region	(713.15)	Georgian Bay region. □ Parry Sound District. District Municipality of Muskoka. Simcoe County. Grey County
(711.65)	East Kootenay Regional District. □ Cranbrook, Invermere, Kimberley	(713.2/.8)	Southern Ontario
(711.68)	Columbia-Shuswap Regional District. □ Golden, Revelstoke, Salmon Arm	(713.2)	Lake Huron region. □ Bruce County. Huron County. Perth County. Middlesex County. London. Lambton County
(711.7)	Central interior region	(713.3)	Lake Erie region. □ Essex County. Windsor. Kent County. Elgin County, Saint Thomas. Regional Municipality of Haldimand-Norfolk. Regional Municipality of Niagara Niagara Falls (city) ⇒ (282.272.117)
(711.72)	Thompson-Nicola Regional District	(713.4)	West Central region. □ Dufferin County. Wellington County. Guelph. Regional Municipality of Waterloo. Kitchener-Waterloo, Waterloo (city), Kitchener. Oxford County. Brant County, Brantford
(711.75)	Cariboo Regional District	(713.5)	Lake Ontario region. □ Regional Municipality of Durham. Lennox and Addington County ⇒ (282.272.115)
(711.8)	Northern region. □ Mackenzie, McBride, Prince George, Smithers, Valemount, Vanderhoof	(713.54)	Metropolitan Toronto and Regional Municipality of York
(711.85)	Northwestern region. □ Hazelton, Stewart, Terrace	(713.6)	East central region. □ Haliburton County. Victoria County. Peterborough County, Peterborough
(711.87)	Northeastern region. □ Dawson Creek, Fort Nelson, Fort St. John, Tumbler Ridge ⇒ (282.273.21)	(713.7)	Saint Lawrence River region. □ Frontenac County. Kingston. United Counties of Leeds and Grenville. United Counties of Stormont, Dundas and Glengarry ⇒ (282.272.1)
(712.3/.7)	Prairie Provinces	(713.8)	Ottawa River region. □ Renfrew County. Lanark County. Regional Municipality of Ottawa-Carleton. United Counties of Prescott and Russell ⇒ (282.272.11)
(712.3)	Alberta	(713.84)	Ottawa
(712.31)	Northwestern Region ⇒ (282.273.21)	(714)	Quebec <i>Lower Canada 1791-1841</i> ⇒ (282.272.1)
(712.32)	Northeastern region. □ Fort McMurray ⇒ (282.273.22)	(714.1)	Northern region. □ New Quebec (Nouveau-Québec). Hudson Bay and Ungava Bay regions. Kativik Regional Administration. Caniapiscau (Regional County Municipality). Nouveau-Québec Administrative Region.
(712.33)	Central region. □ Drumheller, Red Deer ⇒ (237.12)		
(712.332)	Rocky Mountain Parks region. □ Banff		
(712.334)	Edmonton		
(712.338)	Calgary		
(712.34)	Southern region. □ Crowsnest Pass, Fort Macleod, Medicine Hat		
(712.345)	Lethbridge		
(712.4)	Saskatchewan		
(712.41)	Northern region		
(712.42)	Central region. □ Battlefords, Lloydminster, Prince Albert, Yorkton		
(712.425)	Saskatoon		
(712.43)	Southwestern region. □ Swift Current		
(712.44)	Southeastern region. □ Fort Qu'Appelle, Melville, Moose Jaw		

	Abitibi-Témiscamingue Administrative Region. Lac-Saint-Jean region. Chicoutimi region. Côte-Nord region ⇒ (268.75)	(716) (716.1)	Nova Scotia Northern Counties. □ Cumberland County, Amherst. Colchester County, Truro. Pictou County, New Glasgow, Pictou. Antigonish County
(714.2)	Western counties (Ottawa Valley counties). □ Pontiac Regional County Municipality. Antoine-Labelle Regional County Municipality. Papineau Regional County Municipality. Terrebonne region. Deux-Montagnes County. Vaudreuil-Soulanges Regional County Municipality ⇒ (282.272.11)	(716.2) (716.22) (716.225) (716.3)	Southern Counties. □ Guysborough County. Lunenburg County. Queens County, Liverpool. Shelburne County Halifax County Halifax-Dartmouth metropolitan area. □ Halifax Bay of Fundy counties. □ Yarmouth County. Digby County. Annapolis County. Kings County, Wolfville. Hants County, Windsor ⇒ (261.452)
(714.22)	Outaouais Administrative region		Cape Breton Island and Sable Island
(714.221)	Gatineau-Hull region. □ Aylmer, Gatineau National Capital Region in the province of Quebec, Hull	(716.9) (716.91)	Cape Breton Island. □ Inverness County. Victoria County. Cape Breton County, Glace Bay, Sydney, Louisbourg. Richmond County Sable Island
(714.27)	Montreal Administrative Region. □ Laval (Jésus Island)		Prince Edward Island
(714.28)	Montreal Urban Community (Communauté urbaine de Montréal). □ Bizard, Dorval, Montreal islands	(716.99) (717)	Newfoundland Province. □ Newfoundland. Labrador
(714.3)	Southwestern region. □ Huntingdon region. Beauharnois-Salaberry Regional County Municipality. Châteauguay region. Laprairie region. Napierville region. Verchères region. Chambly region. Saint-Jean region	(718) (718.1) (718.11) (718.2) (718.9)	Newfoundland Island. □ Corner Brook. Gander. Grand Falls St Johns Labrador. □ Battle Harbour. Goose Bay St Pierre and Miquelon <i>Overseas territory of France</i>
(714.4)	North central region. □ Lanaudière region. Joliette Regional County Municipality. Berthier region. Maskinongé Regional County Municipality. Mauricie region. Champlain region. Pontneuf region. Montmorency region. Charlevoix region	(719) (719.1) (719.2) (719.21)	Yukon. Northwest Territories Yukon Territory. □ Dawson. Whitehorse Northwest Territories Fort Smith region. □ Fort Providence, Fort Simpson, Fort Smith, Hay River, Pine Point, Rae, Rae Lakes, Snowdrift, Wrigley, Yellowknife ⇒ (282.273.22)
(714.47)	Québec region and Jacques-Cartier River Valley. □ La Jacques-Cartier Regional County Municipality, Québec Administrative Region		Inuvik region. □ Aklavik, Fort Franklin, Fort Norman, Inuvik, Paulatuk, Sach Harbour, Tuktoyaktuk ⇒ (282.273.2)
(714.471)	Québec Urban Community (Communauté urbaine de Québec). □ Ancienne-Lorette, Beauport, Cap-Rouge, Charlesbourg, Lac-Saint-Charles, Loretteville, Saint-Augustin-de-Desmaures, Saint-Foy, Saint-Émile, Sillery, Val-Bélair, Vanier	(719.3) (719.31)	Nunavut Baffin region. □ Arctic Bay, Cape Dorset, Hall Beach, Igloolik, Iqaluit (Frobisher Bay), Pangnirtung, Pond Inlet, Resolute, Sanikiluaq ⇒ (268.74) ⇒ (268.77)
(714.5)	South central region. □ Richelieu region. Saint-Hyacinthe and Bagot regions. Rouville region. Yamaska region. Nicolet region. Drummond and Athabasca regions. Wolfe region. Mégantic region. Lotbinière region. Lévis region	(719.33)	Keewatin region. □ Baker Lake, Chesterfield Inlet, Coral Harbour, Rankin Inlet, Repulse Bay Kitikmeot region. □ Bathurst Inlet, Cambridge Bay, Coppermine, Gjoa Haven, Holman, Pelly Bay, Spence Bay, Umingmaktok
(714.6)	Southern region. □ Iberville region. Missisquoi region. Shefford region. Brome region. Richmond region. Sherbrooke region. Stanstead region. Compton region. Frontenac region	(719.35)	
(714.7)	Southeastern region. □ Eastern Quebec	(72)	MEXICO. CENTRAL AMERICAN STATES. CARIBBEAN TERRITORIES
(715/717)	Atlantic Provinces. Maritime Provinces	(721/727)	MEXICO. UNITED STATES OF MEXICO
(715)	New Brunswick	(721) (721.1)	Northern states Tamaulipas. □ Ciudad Victoria (state capital, municipality and municipal capital). Guerrero (municipality), Nueva Ciudad Guerrero (municipal capital). Matamoros (municipality and municipal capital). Nuevo Laredo (municipality and municipal capital). Ocampo (municipality), Ciudad Ocampo (municipal capital), Reynosa. Tampico (municipalities and municipal capitals)
(715.1)	Restigouche County. Gloucester County, Bathurst		Nuevo León. □ Monterrey (state capital, municipality and municipal capital).
(715.2)	Eastern counties. □ Northumberland County. Kent County. Westmorland County, Sackville, Moncton		Aguaileguas. Galeana. Garza Garcia. Linas. Santa Catarina (municipalities and municipal capitals)
(715.3)	Southern Counties. □ Albert County. Charlotte County ⇒ (261.452)		
(715.32)	Saint John County. □ Saint John		
(715.4)	Central Counties. □ Kings County. Queens County. Sunbury County	(721.2)	
(715.5)	Western Counties. □ Carleton County. Victoria County. Madawaska County		
(715.51)	York County. □ Fredericton		

(1/9) Table 1e. Common auxiliaries of place

(721.3)	Coahuila de Zaragoza. □ Saltillo (state capital, municipality and municipal capital). Acuña. Arteaga (municipalities and municipal capitals). Cuatrociénegas (municipality), Cuatrociénegas de Carranza (municipal capital). Monclova (municipality and municipal capital). Muzquiz (municipality), Ciudad Melchor Muzquiz (municipal capital). Parras (municipality), Parras de la Fuente (municipal capital). Piedras Negras. Ramos Arizpe. Sabinas. Torreón (municipalities and municipal capitals)	(723.4)	Michoacán. □ Morelia (state capital, municipality and municipal capital). Angangueo (municipality), Mineral de Angangueo (municipal capital). Hidalgo (municipality), Ciudad Hidalgo (municipal capital). Cuitzeo (municipality), Cuitzeo de Porvenir (municipal capital). Charo (municipality and municipal capital). Ixtlán (municipality), Ixtlán de los Hervores (municipal capital). Jiquilpan (municipality), Jiquilpan de Juárez (municipal capital). La Piedad (municipality), La Piedad de Cabadas (municipal capital). Paracho (municipality), Paracho de Verduzco (municipal capital). Pátzcuaro (municipality and municipal capital). Salvador Escalante (municipality), Santa Clara de Cobre (municipal capital). Tlalpujahua (municipality), Tlalpujahua de Rayón (municipal capital). Tuxpán. Tzintzuntzan. Uruapan (municipalities and municipal capitals). Zamora (municipality), Zamora de Hidalgo (municipal capital). Zinapécuaro (municipality), Zinapécuaro de Figueroa (municipal capital), Zitácuaro (municipality), Heroica Zitácuaro (municipal capital)
(721.4)	Chihuahua. □ Chihuahua (state capital, municipality and municipal capital). Batopilas. Ciudad Juárez (municipalities and municipal capitals). Coyame (municipality), Santiago de Coyame (municipal capital). Cuauhtémoc. Hidalgo del Parral (municipalities and municipal capitals). Jiménez (municipality), José Mariano Jiménez (municipal capital), Casas Grandes (Paquimé) (archaeological site)	(723.5)	Nayarit. □ Tepic (state capital, municipality and municipal capital). Compostela. Ixtlán de Río. San Blas (municipalities and municipal capitals). Islas Tres Marías (Isla María Madre, Isla María Magdalena, Isla María Cleofas), Islas San Juanito, Isabela, Marietas
(721.5)	Sonora. □ Hermosilla (state capital, municipality and municipal capital). Alamos. Cananea (municipalities and municipal capitals). Guaymas (municipality), Heroica Guaymas (municipal capital). Huatabampo. Navojoa. Puerto Peñasco (municipalities and municipal capitals). Trincheras (municipality and archaeological site), Trincheras (municipal capital). Islas Tiburón, San Estaban, Pelicano, Lobos	(724)	Central states
(721.6)	Durango. □ (Victoria de) Durango (state capital, municipality and municipal capital). El Salto. Gómez Palacio. Mapimí (municipalities and municipal capitals)	(724.1)	Zacatecas. □ Zacatecas (state capital, municipality and municipal capital). Chalchihuites. Fresnillo. Guadalupe (municipalities and municipal capitals), Chicomoztoc (La Quemada) (archaeological site). Jerez (municipality), Jerez de García Salinas (municipal capital). Sombrerete. Tepetongo. Tlaltenango de Sánchez Roman. Villa García (municipalities and municipal capitals)
(722)	Baja California (Lower California) and adjacent islands	(724.2)	San Luis Potosí. □ San Luis Potosí (state capital, municipality and municipal capital). Catorce (municipality), Real de Catorce (municipal capital). Ciudad Valles. Guadalcázar. Matehuala. Santa María del Río. Tamasopo (municipalities and municipal capitals)
(722.1)	Baja California Norte. □ Mexicali (state capital, municipality and municipal capital). Ensenada. Tecate. Tijuana (municipalities and municipal capitals). Islas Ángel de la Guarda, Montague, San Lorenzo, Salsipudes, Smith, Cedros, Guadalupe	(724.3)	Aguascalientes. □ Aguascalientes (state capital, municipality and municipal capital). Rincón de Romos. San José de García. Valladolid (municipalities and municipal capitals)
(722.2)	Baja California Sud. □ La Paz (state capital, municipality and municipal capital). Loreto (municipality and municipal capital). Los Cabos (municipality), San José del Cabo (municipal capital). Mulegé (municipality), Santa Rosalía (municipal capital). Islas El Carmen, San José, Cerralvo, Espíritu Santo, San Marcos, Santa Margarita, Magdalena	(724.4)	Guanajuato. □ Guanajuato (state capital, municipality and municipal capital). Acámbaro (municipality and municipal capital). Allende (municipality), San Miguel de Allende (municipal capital). Valle de Santiago (municipality and municipal capital)
(723)	Central Pacific states	(724.5)	Querétaro. □ Querétaro (state capital, municipality and municipal capital). Amealco de Bonfil (municipality), Amealco (municipal capital). Cadereyta de Montes (municipality), Cadereyta (municipal capital). Tequisquiapan. San Joaquín (municipalities and municipal capitals). San Juan del Río (municipality), San Juan del Río del Centauro (municipal capital)
(723.1)	Sinaloa. □ Culiacán (Rosales) (state capital, municipality and municipal capital). Ahomé (municipality), Los Mochis (municipal capital). Cosala. El Fuerte. Guasave. Mazatlán. Rosario (municipalities and municipal capitals). Islas Talchichilte, de Altamira, Santa María, San Ignacio	(724.6)	Hidalgo. □ Pachuca (de Soto) (state capital, municipality and municipal capital). Actopan. Ajacuba. Huasca de Ocampo. Huichapan.
(723.2)	Jalisco. □ Guadalajara (state capital, municipality and municipal capital). Ahululco (municipality), Ahululco de Mercado (municipal capital). Chapala (municipality and municipal capital). Puerto Vallarta. San Juan de los Lagos. Tequila. Tlaquepaque. Zapopan (municipalities and municipal capitals)		
(723.3)	Colima. □ Colima (state capital, municipality and municipal capital). Manzanillo. Tecmán. Villa de Álvarez (municipalities and municipal capitals). Islas Revillagigedo (Socorro, Clarío, San Benedicto, Roca Partida)		

(724.7)	Ixquimilpan. Mineral de Chico. Tasquillo. Tula de Allende (municipalities and municipal capitals). Tulacingo de Bravo (municipality), Tulacingo (municipal capital)	(726.5)	(archaeological sites). Lerma (municipality) Lerma de Villada (municipal capital). Isla del Carmen Quintana Roo. □ Othon P. Blanco (municipality), Chetumal (state and municipal capital), Kohunlich (archaeological site). Benito Juárez (municipality), Cancún (municipal capital), Cobá, Tulum (archaeological sites). Felipe Carrillo Puerto. Kantunil. Isla Cozumel. Isla Mujeres (municipalities and municipal capitals). Islas Holbox, Contoy, Tamales, Cayo Chelén, Banco Chinchorro (Cayos Lobos Norte) (reef)
(724.8)	Tlaxcala. □ Tlaxcala (de Xicohtécatl) (state capital, municipality and municipal capital), Cacaxtla (archaeological site). Apizaco. Huamantla (municipalities and municipal capitals). Ixtacuixtla de Marino Matamoros (municipality), Villa Mariano Matamoros (municipal capital). Ocotlán. Santa Cruz Tlaxcala (municipalities and municipal capitals)	(726.6)	Yucatán. □ Mérida (state capital, municipality and municipal capital), Dzibilchaltún (archaeological site). Dzilam de Bravo. Kopoma (municipalities and municipal capitals). Mayapán (municipality and municipal capital and archaeological site). Progreso. Río Lagartos. Santa Clara. Tecoh. Telchac Puerto. Ticul (municipalities and municipal capitals), Kabáh, Labná, Sayil, Uxmal, Xlapak (archaeological sites). Tizimin. Valladolid (municipalities and municipal capitals), Chichén-Itzá (archaeological site)
(724.9)	Morelos. □ Cuernavaca (state capital, municipality and municipal capital). Amacuzac. Coatlán del Río. Cuautla Huitzilac. Jojutla. Miacatlán. Puente de Ixtla. Temixco. Tepoztlán. Tetecala (municipalities and municipal capitals), Xochicalco (archaeological site). Tlaltizapán. Tlaquiltenango. Tlayacapan. Xochitepec (municipalities and municipal capitals)	(727) (727.1)	Southern Pacific Coast states Guerrero. □ Chilpancingo (de los Bravo) (state capital, municipality and municipal capital). Acapulco de Juárez. Arcelia. Coyuca de Benitez. Iguala de la Independencia. Ixcateopan de Cuauhtémoc. Ixtapa (municipalities and municipal capitals). José Azueta (municipality), Zihuatanejo (municipal capital). Olinale. Ometepe. Petatlan. Taxco de Alarcon (municipalities and municipal capitals) Oaxaca. □ Oaxaca (de Juárez) (state capital, municipality and municipal capital), Monte Albán, Yagul (archaeological sites). Juchitán de Zaragoza. San Bartolo Coyotepec. San Pedro Mixtepec, Puerto Escondido. San Pedro Pochutla. Santa María el Tule. Santo Tomas Jalieza (municipalities and municipal capitals) Chiapas. □ Tuxtla Gutiérrez (state capital, municipality and municipal capital). Chamula. Chiapa de Corzo. Comitán de Domínguez (municipalities and municipal capitals), Chincultik (archaeological site). Ocosingo (municipality and municipal capital), Toniná (archaeological site). Palenque (municipality and municipal capital and archaeological site), Bonampak (archaeological site). San Cristóbal de las Casas (municipality and municipal capital). Tapachula (municipality) Tapachula de Cordova y Ordoñez (municipal capital), Izapa (archaeological site)
(725) (725.1)	México and Federal District México (state). □ Toluca (de Lerdo) (state capital, municipality and municipal capital), Calixtlahuaca (archaeological site). Acolman (municipality), Acolman de Nezahualcoyotl (municipal capital). Atlacomulco (municipality), Atlacomulco de Fabela (municipal capital). Chalma. Ixtapan de la Sal. Metepec. San Mateo Atenco (municipalities and municipal capitals). Teotihuacan (municipality) Teotihuacan de Arista (municipal capital). Tepozotlán (municipality and municipal capital). Texcoco (municipality), Texcoco de Mora (municipal capital). Valle de Bravo (municipality and municipal capital) Distrito Federal (Federal District). México City (Ciudad de México) (national capital)	(727.2)	CENTRAL AMERICAN STATES GUATEMALA. REPUBLIC OF GUATEMALA
(726) (726.1)	Southern Gulf states (and territory) Veracruz. □ Jalapa (Enríquez), (state capital, municipality and municipal capital). Catemaco. Nautla. Orizaba Papantla (municipalities and municipal capitals), Tajín (archaeological site). San Andrés Tuxtla. Santiago Tuxtla. Tecolutla. Tlacotalpan. Veracruz. Zempoala (municipalities and municipal capitals)	(728)	Northern Guatemala Petén (department). □ Flores (department capital). Bethel. Carmelita. Cruce dos Aguadas. Dolores. El Remate. La Libertad. Melchor de Mencos. Naranjo. Poptún. San Andrés. San José. San Luis. Santa Cruz. Santa Elena. Sayaxché. Aguatemala. Dos Pílas. El Ceibal. El Mirador. Motúl. Nakúm. Río Azul. Tikal. Topoxté. Uaxactún. Yaxha (archaeological sites)
(726.2)	Tabasco. □ Villahermosa (state capital, municipality and municipal capital). Comalcalco (municipality and municipal capital archaeological site). Emiliano Zapata. Paraíso (municipalities and municipal capitals)	(728.1)	
(726.3)	Campeche. □ Campeche (state capital, municipality and municipal capital). Calkini (municipality and municipal capital). Carmen (municipality), Ciudad del Carmen (municipal capital). Champotón. Hopelchén (municipalities and municipal capitals), Bolonchén de Rejón. Calakmul (municipality), Xpujil (municipal capital), Becán, Chicanná, Edzná, Xpujil	(728.11) (728.111)	

(1/9) Table 1e. Common auxiliaries of place

(728.13)	Western Guatemala	(728.2)	BELIZE
(728.131)	Huehuetenango (department). □ Huehuetenango (department capital). Aguatán. La Mesilla. San Juan Atitán. Todos Santos Cuchumatán. Zaceleu (archaeological site)	(728.21)	<i>British Honduras to 1973</i> Corozal (district). □ Corozal (district capital)
(728.132)	San Marcos (department). □ San Marcos (department capital)	(728.22)	Orange Walk (district). □ Orange Walk (district capital)
(728.133)	Quiché (department). □ Santa Cruz de Quiché (department capital). Chajul. Chichicastenango. Nebaj. Sacapulas. San Juan Cotzal. Uspantán. K'umarcaaj (Gumarcaah) (Utatlán) (archaeological site)	(728.23)	Belize (district). □ Belize (district capital). Altun Ha (archaeological site)
(728.134)	Totonicapán (department). □ Totonicapán (department capital). Momostenango. San Francisco El Alto	(728.24)	Cayo (district). □ Belmopan (state and district capital). San Ignacio. Caracol. El Pilar. Xunantunich (archaeological sites)
(728.135)	Quetzaltenango (department). □ Quetzaltenango (department capital). Zunil	(728.25)	Stann Creek (district). □ Dangriga (district capital)
(728.136)	Retalhuleu (department). □ Retalhuleu (department capital). Champerico. Abaj Takalik (archaeological site)	(728.26)	Toledo (district). □ Punta Gorda (district capital)
(728.137)	Suchitepéquez (department). □ Mazatenango (department capital). Santa Lucía Cotzumalguapa. Bilbao. Finca El Baul. Finca Las Ilusiones (archaeological sites)	(728.3)	HONDURAS. REPUBLIC OF HONDURAS
(728.15)	Central Guatemala	(728.31)	North Coast and Mosquitia
(728.151)	Alta Verapaz (department). □ Cobán (department capital). Cahabón. Fray Bartolomé de Las Casas. La Tinta. Lanquín. Panzós. Playa Grande. Raxrujá. San Cristóbal Verapaz. San Juan Chamelco. San Pedro Carchá. Senahú. Tactic. Telemán	(728.311)	Cortés (department). □ San Pedro Sula (department capital). Choloma. Cofradía. Corinto. La Lima. Omoa. Puerto Cortés
(728.152)	Baja Verapaz (department). □ Salamá (department capital). Cubulco. Rabinal	(728.312)	Yoro (department). □ Yoro (department capital). El Progreso. El Triunfo de la Cruz. Lancetilla. Olanchito. Punta Sal
(728.153)	Sololá (department). □ Sololá (department capital). Jaibalito. Panajachel. Paxanax. San Antonio Palopó. San Juan la Laguna. San Marcos la Laguna. San Pedro la Laguna. Santa Cruz la Laguna. Santiago Atitlán. Tzununá	(728.313)	Atlántida (department). □ La Ceiba (department capital). Tela
(728.154)	Chimaltenango (department). □ Chimaltenango (department capital). Los Encuentros. Tecpán. Iximché (archaeological site)	(728.314)	Colón (department). □ Trujillo (department capital). Palacios. Sonaguera
(728.155)	Sacatepéquez (department). □ Antigua Guatemala (department capital). Jocotenango. San Andrés Itzapa. Santa María de Jesús. Santiago Sacatepéquez	(728.315)	Gracias a Dios (department). □ Puerto Lempira (department capital). Ahuas. Brus Laguna. Plaplaya
(728.156)	Guatemala (department). □ Guatemala City (state and department capital)	(728.32)	Islas de la Bahía. Bay Islands (department). □ Roatán (department capital). Barbareta. Guanaja. Utila. Islas Santanilla
(728.157)	El Progreso (department). □ El Progreso (Guastatoya) (department capital)	(728.33)	Central and Western Highlands
(728.158)	Jalapa (department). □ Jalapa (department capital)	(728.331)	Copán (department). □ Santa Rosa de Copán (department capital). Copán Ruinas. Florida. La Entrada. Santa Rita. El Puente (archaeological site)
(728.17)	Eastern Guatemala	(728.332)	Santa Bárbara (department). □ Santa Bárbara (department capital). Colinas. Pito Solo. Villanueva
(728.171)	Izabal (department). □ Puerto Barrios (department capital). El Estor. Livingston. Río Dulce. Quiriguá (archaeological site)	(728.333)	Comayagua (department). □ Comayagua (department capital). Siguatepeque
(728.172)	Zacapa (department). □ Zacapa (department capital). Quiriguá (archaeological site)	(728.334)	Francisco Morozán (department). □ Distrito Central. Tegucigalpa (state and department capital). Comayagüela. Cedros. Curarén. Sabana Grande
(728.173)	Chiquimula (department). □ Chiquimula (department capital). Esquiulas	(728.335)	Ocatepeque (department). □ Nueva Ocatepeque (department capital). El Poy
(728.18)	Southern Guatemala	(728.336)	Lempira (department). □ Gracias (department capital). La Esperanza. Santa Cruz
(728.181)	Escuintla (department). □ Escuintla (department capital). La Democracia. Puerto Quetzal. Puerto San José	(728.337)	Intibucá (department). □ La Esperanza (department capital). Colomocagua. Intibucá
(728.182)	Santa Rosa (department). □ Cuilapa (department capital). Monterrico	(728.34)	Eastern Honduras
(728.183)	Jutiapa (department). □ Jutiapa (department capital)	(728.341)	Olancho (department). □ Juticalpa (department capital). Catacamas. El Boquerón. La Avispa. La Unión
		(728.35)	Southern Honduras
		(728.351)	La Paz (department). □ La Paz (department capital)
		(728.352)	Valle (department). □ Nacaome (department capital). El Amatillo. Goascorán. Langue. San Lorenzo. Isla el Tigre. Amapala
		(728.353)	Choluteca (department). □ Choluteca (department capital). Concepción de María. El Corpus. El Triunfo. Pespire. San Juancito. Santa Lucía. San Marcos de Colón. Valle de Angeles

(728.354)	El Paraíso (department). □ Yuscarán (department capital). Danlí. El Paraíso. Las Manos	(728.53)	Northern central area
(728.4)	EL SALVADOR. REPUBLIC OF EL SALVADOR	(728.531)	Jinotega (department). □ Jinotega (department capital)
(728.41)	Western region	(728.532)	Nueva Segovia (department). □ Ocotal (department capital)
(728.411)	Ahuachapán (department). □ Ahuachapán (department capital). Apaneca. Cara Sucia. La Hachadura. Playa Barra de Santiago. San Francisco Menéndez. Santa Cruz	(728.533)	Madriz (department). □ Somoto (department capital). Ocotal
(728.412)	Santa Ana (department). □ Santa Ana (department capital). Chalchuapa. Metapán. Textistepatque. Tazumal (archaeological site)	(728.534)	Estelí (department). □ Estelí (department capital)
(728.413)	Sonsonate (department). □ Sonsonate (department capital). Acajutla. Armenia. Juayúa. Los Cóbano. Los Remedios. Nahuizalco. Playa de Metalio	(728.535)	Matagalpa (department). □ Matagalpa (department capital)
(728.414)	La Libertad (department). □ Nueva San Salvador (Santa Tecla) (department capital). La Libertad. Playa San Diego. Quezaltepeque. Joya de Cerén. San Andrés (archaeological sites)	(728.536)	León (department). □ León (department capital). Corinto. León Viejo. Poneoya
(728.42)	Central area	(728.55)	Southern central area
(728.421)	Chalatenango (department). □ Chalatenango (department capital). Citalá. Concepción. Quezaltepeque. El Poy. La Palma. San Ignacio. Tejutla	(728.551)	Boaco (department). □ Boaco (department capital)
(728.422)	Cuscatlán (department). □ Cojutepeque (department capital). Apulo. Suchitoto. San Salvador Ciudad Vieja (archaeological site)	(728.552)	Chontales (department). □ Juigalpa (department capital)
(728.423)	San Salvador (department). □ San Salvador (state and department capital). Aguilaes. Apopa. Panchimalco. Cihuatán (archaeological site)	(728.553)	Managua (department). □ Managua (state and department capital)
(728.424)	La Paz. □ Zacatecoluca (department capital). Comalapa. La Herradura. Playa Costa del Sol. Playa San Marcelino	(728.554)	Masaya (department). □ Masaya (department capital)
(728.43)	Eastern region	(728.555)	Granada (department). □ Granada (department capital). Isla Zapatera
(728.431)	Cabañas (department). □ Sensuntepeque (department capital). Ilobasco. Victoria	(728.57)	Pacific area
(728.432)	San Vicente (department). □ San Vicente (department capital)	(728.571)	Chinandega (department). □ Chinandega (department capital)
(728.433)	Usulután (department). □ Usulután (department capital). El Triunfo. La Canoa. Playa El Espino. Puerto El Triunfo. Santiago de María	(728.573)	Carazo (department). □ Jinotepe (department capital)
(728.434)	San Miguel (department). □ San Miguel (department capital). Playa El Cuco. Quelepa. Sesorí	(728.575)	Rivas (department). □ Rivas (department capital). San Juan del Sur. Sapoa. Isla de Ometepe. Altagracia. Moyogalpa
(728.435)	Morazán (department). □ San Francisco Gotera (department capital). Arambala. Cacaopera. Corinto. Perquín	(728.6)	COSTA RICA. REPUBLIC OF COSTA RICA
(728.436)	La Unión (department). □ La Unión (department capital). Conchagua. El Amatillo. El Carmen. Intipucá. Lislique. Puerto Cutuco. Santa Rosa de Lima. Isla Conchagüita. Isla Zacatillo	(728.61)	Guanacaste (province). □ Liberia (canton and provincial capital). Abangares. Monteverde. Santa Elena. Bagaces. Cañas. Carrillo. Hojancha. Nosara. Sámara. La Cruz. Nandayure. Nicoya. Santa Cruz. El Coco. Panama. Tamarindo. Tilarán (cantons)
(728.5)	NICARAGUA. REPUBLIC OF NICARAGUA	(728.62)	Alajuela (province). □ Alajuela (canton and provincial capital). Alfaro Ruiz. Atenas. Grecia. Sarchí. Guatuso. Los Chiles. Naranjo. Sarchí. Orotina. Palmares. Poás. San Carlos. Fortuna. Upala. Valverde Vega (cantons)
(728.51)	Atlantic coast area	(728.63)	Heredia (province). □ Heredia (canton and provincial capital). Barva. Belén. Flores. San Pablo. San Rafael. Santa Barbara. Santo Domingo (canton). Sarapiquí (cantons)
(728.511)	Atlántico Norte (autonomous region). □ Puerto Cabezas (regional capital). Cayos Miskitis	(728.64)	Puntarenas (province). □ Puntarenas (canton and provincial capital). Montezuma. Aguirre. Dominical. Quepos. Buenos Aires. Corredores. Coto Brus. Esparza. Garabito. Jacó. Golfito. Canoas. Montes de Oro. Osa. Bahía Drake. Carate. Puerto Jiménez. Parrita (cantons). Isla Chira. Isla del Caño
(728.513)	Atlántico Sur (autonomous region). □ Bluefields (regional capital). El Rama. Isla Grande del Mús. Isla Pequeña del Mús	(728.65)	San José (province). □ San José (state capital, canton and provincial capital). Acosta (canton). Alajuelita. Aserri. Curridabat. Desamparados. Dota. Escazú. Goicoechea. León Cortés. Mora. Moravia. Pérez Zeledón. Puriscal. Santa Ana. Tarrazú. Tibás. Turrubares. Vásquez de Coronado (cantons)
(728.515)	Río San Juan (department). □ San Carlos (department capital). San Juan del Norte. Archipiélago de Solentiname. La Venada. Mancarrón. Mancarroncito. San Fernando	(728.66)	Cartago (province). □ Cartago (canton and provincial capital). Alvarado (canton). El Guarco (canton). Jiménez (canton). Guápiles. La Unión (canton). Oreamuno (canton). Turrialba (canton)
		(728.67)	Limón (province). □ Puerto Limón (canton and provincial capital). Bribri. Cahuita. Manzanillo. Guácimo (canton). Matina (canton). Pocicí (canton). Siquirres (canton). Talamanca (canton)

(1/9) Table 1e. Common auxiliaries of place

(728.7)	PANAMA. REPUBLIC OF PANAMA	(728.789)	Canal Area (formerly Panama Canal Zone) <i>Since December 31st 1999 part of Panamá, spanning a five mile strip on either side of the Canal</i>
(728.71)	Bocas del Toro (province). □ Bocas del Toro (provincial capital, district and district capital). Changuinola. Almirante. Chiriquí Grande (districts and district capitals). Islas Bastimentos, Colón	(728.79)	Darién (province). □ Chepigana (district), La Palma (provincial and district capital), Meteti, Puerto Quimba. Pinogana (district), El Real de Santa María (district capital), Yaviza <i>For Canal Area/Panama Canal Zone (previously at this number) see (728.789)</i>
(728.72)	Chiriquí (province). □ David (provincial capital, district and district capital). Alanje. Barú, Cerro Punta. Boquerón (districts and district capitals). Boquete (district), Bajo Boquete (district capital). Bugaba (district), La Concepción (district capital). Dolega. Gualaca. Remedios (districts and district capitals). Renacimiento (district), Río Sereno (district capital). San Félix (district), Las Lajas (district capital). San Lorenzo (district), Horconcitos (district capital). Tolé (district and district capital)	(728.793)	Kuna de Wargandi (intendency)
(728.722)	Ngöbe Buglé (intendency). □ Chichica (intendency capital). Bsiko (district and district capital). Kankintú (district), Birira (district capital). Kusapin (district and district capital). Mirono (district), Hato Pilón (district capital). Múna (district) Chichica (district capital). Nole Duima (district), Cerro Iglesias (district capital). Nürüm (district), Buenos Aires (district capital)	(728.795)	Emberá (intendency). □ Cémaco (district), Cirilo Guainora (intendency and district capital). Sambú (district). Río Sábalo (district capital), Puerto Indio
(728.73)	Veraguas (province). □ Santiago (district), Santiago de Veraguas (provincial and district capital). Atalaya. Calobre. Cañazas. La Mesa. Las Palmas. Montijo. Río de Jesús. San Francisco. Santa Fé. Soná (districts and district capitals)	(729)	CARIBBEAN TERRITORIES. WEST INDIES (ANTILLES)
(728.74)	Herrera (province). □ Chitré (provincial capital, district and district capital). Las Minas. Los Pozos. Ocu. Parita. Pesé. Santa María (districts and district capitals)	(729.11/5)	GREATER ANTILLES
(728.75)	Los Santos (province). □ Las Tablas (provincial capital, district and district capital). Guararé (district and district capital). Los Santos (district), La Villa de los Santos (district capital). Macaras. Pedasí. Pocrí. Tonosí (districts and district capitals). Islas Cañas, Iguana	(729.1)	CUBA. REPUBLIC OF CUBA
(728.76)	Coclé (province). □ Penonomé (provincial capital, district and district capital), El Valle. Aguadulce. Antón. La Pintada. Natá. Olá. Santa Isabel (districts and district capitals)	(729.11)	Western provinces
(728.77)	Colón (province). □ Colón (provincial capital, district and district capital). Chagres (district), Nuevo Chagres (district capital). Donoso (district), Miguel de la Borda (district capital). Portobelo (district and district capital), Nombre de Dios	(729.111)	La Habana. □ Havana (state and provincial capital), Bauta, Ensenada de la Broa <i>Ciudad de la Habana has provincial status, separate from the state La Habana. Havana is capital of both provinces, as well as the state capital. The two are here combined for convenience</i>
(728.78)	Panamá (province). □ Panamá (state and provincial capital, district and district capital), Panama Veijo (archaeological site). Arraiján. Balboa. Capira. Chame. Chepo. El Llano. Chimán. La Chorrera. San Carlos (districts and district capitals), Nueva Gorgona, Playa Coronado. San Miguelito (district), Bellisario Porras (district capital), Gamboa. Taboga (district and district capital)	(729.112)	Pinar del Río (province and provincial capital). □ Soroa, Viñales, Cayos de San Felipe
(728.783)	Kuna de Magungandé (intendency)	(729.113)	Isla de la Juventud (special municipality). □ Nueva Gerona (municipal capital), Archipiélago de los Canarreos, Cayo Largo, Islas de Mangles
(728.785)	Kuna Yala (intendency) (formerly, Comarca de San Blas). □ Narganá (intendency capital), Cartí Suitupo, El Porvenir. Islas Ailigandi, Corazon de Jesús, Nalunega, Narasgandup, Narganá, Ukuptupu, Wichubula, Cayos Holandeses	(729.12)	Central provinces
		(729.121)	Matanzas. □ Matanzas (provincial capital), Jovellanos, Playa Girón (Bay of Pigs), Varadero
		(729.122)	Cienfuegos. □ Cienfuegos (provincial capital)
		(729.123)	Villa Clara. □ Santa Clara (provincial capital), Cayo Barién (Caibarién), Cayo Conuco, Cayo Santa María
		(729.124)	Sancti Spíritus. □ Sancti Spíritus (provincial capital), Trinidad
		(729.13)	Western provinces
		(729.131)	Ciego de Avila. □ Ciego de Avila (provincial capital), Morón, Cayo Coco, Cayo Guillermo, Cayo Caballones, Cayos de Ana María
		(729.132)	Camagüey. □ Camagüey (provincial capital), Cayo Romano, Minas, Nuevitas, Santa Lucía, Cayo Anaclitos, Cayo Medio Luna
		(729.133)	Las Tunas. □ Las Tunas (provincial capital)
		(729.134)	Holguín. □ Holguín (provincial capital), Guardalavaca
		(729.135)	Granma. □ Bayamo (provincial capital), Marea de Portillo, Manzanillo
		(729.136)	Santiago de Cuba. □ Santiago de Cuba, Baconao, Siboney
		(729.137)	Guantanamo. □ Guantanamo (provincial capital), Baracoa
		(729.2)	JAMAICA
		(729.21)	Parishes of South and Southwest Jamaica
		(729.211)	Kingston (state and parish capital)
		(729.212)	St Andrew. □ Half Way Tree (capital), Constant Spring, Cross Roads, Palisadoes, Port Royal, Stony Hill
		(729.213)	Saint Catherine. □ Spanish Town (capital), Linstead, Old Harbour
		(729.214)	Clarendon. □ May Pen (capital)

(729.215)	Manchester. □ Mandeville (capital)	(729.347)	Santiago Rodriguez. □ San Ignacio de Sabaneta (provincial capital)
(729.216)	Saint Elizabeth. □ Black River (capital)	(729.348)	Santiago. □ Santiago de los Caballeros (provincial capital)
(729.22)	Parishes of West and Northwest Jamaica	(729.349)	La Vega. □ Concepción de la Vega (provincial capital)
(729.221)	Westmoreland. □ Savannah-la-Mar (capital)	(729.35)	Northern area
(729.222)	Hanover. □ Lucea (capital), Negril	(729.351)	Esipaillat. □ Moca (provincial capital)
(729.223)	St James. □ Montego Bay (capital)	(729.352)	Salcedo. □ Salcedo (provincial capital)
(729.224)	Trelawney. □ Falmouth (capital)	(729.353)	María Trinidad Sánchez. □ Trinidad Sánchez (provincial capital)
(729.23)	Parishes of North and Northeast Jamaica	(729.354)	Duarte. □ San Francisco de Macorís (provincial capital)
(729.231)	St Ann. □ Saint Ann's Bay (capital), Ocho Rios, Runaway Bay	(729.355)	Hato Mayor. □ Hato Mayor del Rey (provincial capital)
(729.232)	St Mary. □ Port Maria (capital), Clarendon, Oracabessa	(729.356)	Samaná. □ Santa Barbara de Samaná (provincial capital)
(729.233)	Portland. □ Port Antonio (capital)	(729.4)	HAITI. REPUBLIC OF HAITI
(729.24)	Parishes of East and Southeast Jamaica	(729.41)	Départements of Haiti
(729.241)	St Thomas. □ Morant Bay (capital), Bath, Golden Grove, Port Morant	(729.411)	Ouest. □ Port-au-Prince (state and département capital), Pétionville. Île de la Gonâve
(729.29)	CAYMAN ISLANDS (UK)	(729.412)	Centre. □ Hinche (département capital)
(729.291)	Grand Cayman Island	(729.413)	L'Artibonite. □ Gonaïves (département capital)
(729.291.1)	Western. □ George Town (state capital)	(729.414)	Nord. □ Cap Haïtien (département capital)
(729.291.2)	West End. □ West Bay	(729.415)	Nord-Ouest. □ Port de Paix (département capital). Île de la Tortue (Tortuga)
(729.291.3)	Creek	(729.416)	Nord-Est. □ Fort-Liberté (département capital)
(729.291.4)	Midland. □ Bodden Town	(729.417)	Sud-Est. □ Jacmel (département capital)
(729.291.5)	Eastern. □ East End	(729.418)	Sud. □ Les Cayes (département capital). Île à Vache
(729.291.6)	South Town. □ Prospect	(729.419)	Grand'Anse. □ Jérémie (département capital). Île Grande Cayemite
(729.292)	Little Cayman Island. Spot Bay (district)	(729.5)	PUERTO RICO. COMMONWEALTH OF PUERTO RICO (USA)
(729.293)	Cayman Brac Island. Stake Bay (district). □ Brac Reef Beach, Tiara Beach		<i>USA 1899 to 1952; associate of USA from 1952</i>
(729.3/.4)	HISPANIOLA	(729.51)	San Juan (state capital). □ Aguas Buenas. Barceloneta. Canóvanas. Carolina. Ceiba. Ciales. Condado. Corozal. Dorado. Fajardo. Gurabo. Juncos. Loíza. Luquillo. Morovis. Naranjito. Orocovis. Palmer. Río Grande. San Lorenzo. Trujillo Alto. Vega Alta. Vega Baja. Culebra, Mona, Vieques (islands) Bayamón. □ Cataño. Guaynabo. Manatí. San Antonio. Toa Alta. Toa Baja
(729.3)	DOMINICAN REPUBLIC	(729.52)	Humacao. □ Aibonito. Barranquitas. Caguas. Cayey. Cidra. Comerío. Las Piedras. Naguabo. Palmas del Mar. Punto Santiago. Yabucoa. Culebra, Vieques (islands)
(729.31)	Southern area	(729.53)	Guayama. □ Aguirre. Arroyo. Cayey. Jobos. Maunabo. Patillas. Salinas
(729.311)	Distrito Nacional. □ Santo Domingo (state capital). Boca Chica	(729.54)	Ponce. □ Adjuntas. Coamo. Ensenada. Guánica. Guayanilla. Jayuya. Juana Díaz. Peñuelas. Santa Isabel. Villalba. Yauco
(729.312)	San Cristóbal. □ San Cristóbal (provincial capital)	(729.55)	Mayagüez. □ Aguada, Añasco. Boquerón. Cabo Rojo. Hormigueros. Lajas. Las Marías. Maricao. Puerto Real. Rosario. Sabana Grande. San Germán. San Sebastián. Mona Island
(729.313)	Peravia. □ Baní (provincial capital)	(729.56)	Aguadilla. □ Aguada. Isabela. Moca. Rincón
(729.314)	Monseñor Nouel. □ Bonao (provincial capital)	(729.57)	Arecibo. □ Angeles. Camuy. Hatillo. Lares.
(729.315)	Sánchez Ramírez. □ Cotuí (provincial capital)	(729.58)	Playa de Vega Baja. Quebradillas. San Sebastián. Utuado
(729.316)	Monte Plata. □ Monte Plata (provincial capital)	(729.61)	Bahamas. Commonwealth of the Bahamas
(729.317)	San Pedro de Macorís. □ San Pedro de Macorís (provincial capital), Juan Dolio	(729.611)	Districts of the Bahamas
(729.32)	Eastern area	(729.611.11)	Grand Bahama (district). □ Freeport (chief town)
(729.321)	El Saibo. □ Santa Cruz de El Saibo (provincial capital)		
(729.322)	La Romana. □ La Romana (provincial capital), Casa de Campo. Isla Catalina		
(729.323)	La Altagracia. □ Salvaleón de Higüey (provincial capital), Punta Cana		
(729.33)	Southwestern area		
(729.331)	Azua. □ Azua de Compostela (provincial capital)		
(729.332)	Barahona. □ Santa Cruz de Barahona (provincial capital)		
(729.333)	Pedernales. □ Pedernales (provincial capital)		
(729.334)	Independencia. □ Jimaní (provincial capital)		
(729.335)	Bahoruco (Baoruco). □ Neiba (provincial capital)		
(729.34)	Northwestern area		
(729.341)	San Juan. □ San Juan de la Maguana (provincial capital)		
(729.342)	Elias Piña. □ Comendador (provincial capital)		
(729.343)	Dajabón. □ Dajabón (provincial capital)		
(729.344)	Monte Cristi. □ San Fernando de Monte Cristi (provincial capital)		
(729.345)	Puerto Plata. □ San Felipe de Puerto Plata (provincial capital)		
(729.346)	Valverde. □ Mao (provincial capital)		

(1/9) Table 1e. Common auxiliaries of place

(729.611.13)	Abaco (district). □ Great Abaco, Marsh Harbour (chief town). Little Abaco, Moore's Island (Mores Island)	(729.724.1)	Saint Kitts
(729.611.15)	Berry Islands (district). □ Bullocks Harbour (chief town)	(729.724.11)	Parishes of Saint Kitts
(729.611.17)	Biminis (district). □ Alice Town (chief town), Cay Sal Bank	(729.724.111)	Saint George Basseterre. □ Basseterre (state capital)
(729.611.19)	Harbour Island (district). □ Dunmore Town (chief town)	(729.724.112)	Saint Peter Basseterre. □ Monkey Hill Village (principal town)
(729.611.21)	New Providence Island (district). □ Nassau (state capital and chief town)	(729.724.113)	Trinity Palmetto Point. □ Boyd's (principal town)
(729.611.23)	Andros (district). □ Nicholls' Town (chief town)	(729.724.114)	Saint Thomas Middle Island. □ Middle Island (principal town)
(729.611.25)	Spanish Wells (district). □ Spanish Wells (chief town)	(729.724.115)	Saint Anne Sandy Point. □ Sandy Point (principal town)
(729.611.27)	Eleuthera (district). □ Governor's Harbour (chief town)	(729.724.116)	Saint John Capisterre. □ Sadlers (principal town)
(729.611.29)	Cat Island (district). □ Arthur's Town (chief town)	(729.724.117)	Christ Church Nicola Town. □ Mansion (principal town)
(729.611.31)	San Salvador (district). □ Cockburn Town (chief town)	(729.724.3)	Nevis
(729.611.33)	Rum Cay (district). □ Port Nelson (chief town)	(729.724.31)	Parishes of Nevis
(729.611.35)	Exuma (district). □ Great Exuma, George Town (chief town). Little Exuma (Exuma and Cays)	(729.724.311)	Saint Paul Charlestown. □ Charlestown (chief settlement)
(729.611.37)	Long Island (district). □ Clarence Town (chief town)	(729.724.312)	Saint John Figtree. □ Figtree (principal town)
(729.611.39)	Ragged Island (district). □ Duncan Town (chief town)	(729.724.313)	Saint George Gingerland. □ Gingerland (principal town)
(729.611.41)	Crooked Island (district). □ Colonel Hill (chief town)	(729.724.314)	Saint James Windward. □ Newcastle (principal town)
(729.611.43)	Long Cay (district). □ Albert Town (chief town)	(729.724.315)	Saint Thomas Lowland. □ Cotton Ground (principal town)
(729.611.45)	Acklins Island (district). □ Masons Bay (chief town)	(729.724.8)	Anguilla (UK)
(729.611.47)	Mayaguana (district). □ Abraham's Bay (chief town)	(729.724.81)	The Valley (administrative centre)
(729.611.49)	Cay Lobos (district). □ Cay Lobos (chief town)	(729.724.812)	Sandy ground
(729.611.51)	Inagua (district). □ Great Inagua, Matthew Town (chief town). Little Inagua	(729.724.813)	Offshore islands of Anguilla. □ Dog Island. Prickly Pear Cays. Scrub Island. Seal Islands. Sombrero Island
(729.68)	Turks and Caicos Islands (UK). □ Caicos Islands, Grand Caicos, East Caicos, North Caicos, Providenciales, West Caicos, South Caicos, Ambergris Cays, Long Cay, Seal Cay, French Cay. Turks Islands, Grand Turk, Cockburn Town (capital), Salt Cay	(729.726)	ANTIGUA AND BARBUDA
(729.71.8)	LESSER ANTILLES (EASTERN CARIBBEAN)	(729.726.1)	<i>UK to 1967; associate of UK from 1967</i>
(729.7)	LEEWARD ISLANDS	(729.726.11)	Antigua
(729.71)	VIRGIN ISLANDS OF THE UNITED STATES (AMERICAN VIRGIN ISLANDS) (USA)	(729.726.111)	Parishes of Antigua
(729.711)	Saint Thomas. □ Charlotte Amalie (capital). East End, Great and Little Saint James Islands, Thatch Cay. North Side, Hans Lollok, Inner Brass, Outer Brass. South Side. Tutu. West End, Savanna Island. Hassel Island, Water Island, Hassel, Water	(729.726.112)	Saint George. □ Cedar Grove. Piggotts
(729.712)	Saint John. □ Central, Congo Cay, Grass Cay, Lovango Cay, Mingo Cay. Coral Bay. Cruz Bay. East End	(729.726.113)	Saint John. □ St. John's (state capital). All Saints. Five Islands. Golden Grove
(729.713)	Saint Croix. □ Anna's Hope Village. Christiansted. East End, Buck Island. Frederiksted. Northcentral. Northwest. Sion Farm. Southwest	(729.726.114)	Saint Peter. □ Glanvilles. Pares. Parham
(729.72)	BRITISH VIRGIN ISLANDS (UK). □ Tortola, Road Town (capital). Anegada. Jost van Dyke. Virgin Gorda	(729.726.115)	Saint Philip. □ Freetown. Willikies
(729.724)	ST KITTS-NEVIS. FEDERATION OF ST KITTS-NEVIS <i>UK to 1967; associate of UK from 1967</i>	(729.726.116)	Saint Paul. □ All Saints. English Harbour. Falmouth. Liberta. Swetes
		(729.726.2)	Saint Mary. □ Bolans. Jolly Harbour. Old Road
		(729.726.21)	Barbuda
		(729.726.28)	Codrington
		(729.727)	Redonda
		(729.727.1)	MONTSERRAT (UK)
		(729.727.2)	Safe Zone. □ Saint John's (island centre, i.e. temporary capital). Saint Peter
		(729.727.3)	Central Zone (Daytime Entry Zone). □ Part of Saint George, Weekes, Elberton
		(729.74)	Exclusion Zone. □ Part of Saint George, Plymouth. Saint Anthony
		(729.741)	GUADELOUPE (DÉPARTEMENT OF FRANCE)
		(729.741.1)	Arondissements of Guadeloupe
		(729.741.3)	Basse-Terre. □ Basseterre (state capital). Marie Galante Island. Îles des Saintes
		(729.741.5)	Point-à-Pitre. □ Grande Terre. La Désirade. Îles de la Petite Terre
		(729.8)	Saint-Martin et Saint-Barthélemy
		(729.81)	WINDWARD ISLANDS
		(729.811)	MARTINIQUE (DÉPARTEMENT OF FRANCE)
			Arondissements of Martinique

(729.811.1)	Saint-Pierre. □ Bellefontaine. Case-Pilote. Fonds-Saint-Denis. Le Carbet. Le Morne-Rouge. Le Morne-Ver. Le Prêcheur. Saint-Pierre	(729.861.15)	Saint Thomas
(729.811.3)	Trinité. □ Basse-Pointe. Grand'Rivière. Gros-Morne. L'Ajoupa-Bouillon. La Trinité. Le Lorrain. Le Marigot. Le Robert. Macouba. Sainte-Marie	(729.861.16)	Saint Joseph
(729.811.5)	Fort-de-France. □ Fort-de France (capital). Le Lamentin. Saint-Joseph. Schoelcher	(729.861.17)	Saint John
(729.811.7)	Marin. □ Ducos. Le Diamant. Le François. Le Marin. Les Anses-d'Arlets. Les Trois-Îlets. Le Vauclin. Rivière-Pilote. Rivière-Salée. Sainte-Anne. Sainte-Luce. Saint Esprit	(729.861.18)	Saint Michael. □ Bridgetown (state capital)
(729.821)	DOMINICA. COMMONWEALTH OF DOMINICA	(729.861.19)	Saint George
(729.821.1)	Parishes of Dominica	(729.861.21)	Saint Philip
(729.821.111)	Saint John. □ Portsmouth	(729.861.22)	Christ Church. □ Hastings. Oistins. Saint Lawrence. Worthing
(729.821.112)	Saint Andrew. □ Marigot	(729.87)	TRINIDAD AND TOBAGO. REPUBLIC OF TRINIDAD AND TOBAGO
(729.821.113)	Saint Peter	(729.871)	Trinidad
(729.821.114)	Saint Joseph	(729.871.11)	Port of Spain (city and state capital)
(729.821.115)	Saint David. □ Rosalie	(729.871.12)	Tunapuna/Piarco (region). □ Tunapuna (capital)
(729.821.116)	Saint Paul	(729.871.13)	Arima (borough)
(729.821.117)	Saint George. □ Roseau (state capital)	(729.871.14)	San Juan/Laventille (region). □ Laventille (capital)
(729.821.118)	Saint Patrick	(729.871.15)	Sangre Grande (region). □ Sangre Grande
(729.821.119)	Saint Luke	(729.871.16)	Chaguanas (borough)
(729.821.121)	Saint Mark	(729.871.17)	Couva/Tabaquite/Talparo (region). □ Couva (capital)
(729.822)	SAINT LUCIA. STATE OF SAINT LUCIA	(729.871.18)	San Fernando (city)
(729.822.1)	Districts of Saint Lucia	(729.871.19)	Princes Town (region). □ Princes Town (capital)
(729.822.111)	Gros Îlet	(729.871.21)	Mayaro/Rio Clara (region). □ Rio Clara (capital)
(729.822.113)	Castries. □ Castries (state capital)	(729.871.22)	Diego Martin (region). □ Petit Valley
(729.822.115)	Anse-la-Raye. □ Marigot	(729.871.23)	Point Fortin (borough)
(729.822.117)	Dennery	(729.871.24)	Penal/Debe (region). □ Siparia (capital)
(729.822.119)	Canaries	(729.872)	Tobago
(729.822.121)	Soufrière	(729.872.1)	Eastern Tobago (region). □ Charlotteville. Roxborough
(729.822.123)	Micoud	(729.872.2)	Western Tobago (region). □ Scarborough (chief settlement). Plymouth
(729.822.125)	Choisil	(729.88)	NETHERLANDS ANTILLES (DUTCH ANTILLES). □ Part of St Martin/St Maarten (Netherlands)
(729.822.127)	Laborie	⇒ (729.741.5)	
(729.822.129)	Vieux Fort. □ Maria Island	(729.881)	Bovenwindse Eilanden (Leeward Islands)
(729.824)	SAINT VINCENT. SAINT VINCENT AND THE GRENADINES	(729.881.1)	Saba. □ Bottom (capital)
(729.824.1)	Parishes of Saint Vincent and the Grenadines	(729.881.2)	Sint Eustatius. □ Oranjestad (capital)
(729.824.11)	Saint George. □ Kingstown (state capital). Milligan Cay, Young Island	(729.881.3)	Sint Maarten. □ Philipsburg (capital)
(729.824.12)	Saint Andrew. □ Layou (principal town)	⇒ (729.741.5)	
(729.824.13)	Saint Patrick. □ Barrouallie (principal town)	(729.882)	Benedenwindse Eilanden (Windward Islands)
(729.824.14)	Saint David. □ Chateaubelair (principal town)	(729.882.1)	Curaçao. □ Willemstad (capital). Klein Curaçao
(729.824.15)	Charlotte. □ Georgetown (principal town)	(729.882.2)	Bonaire. □ Kralendijk (capital). Klein Bonaire
(729.824.17)	Grenadines. □ Port Elizabeth (principal town). Bequia, Canouan, Mustique, Union	(729.89)	ARUBA (NETHERLANDS). □ Oranjestad (state capital)
(729.828)	GRENADA	(729.9)	BERMUDAS (SOMERS ISLANDS) (UK)
(729.828.1)	Parishes of Grenada	(729.91)	Sandys. □ Boaz Island, Ireland Island, Somerset Island, Watford Island and the western tip of Bermuda Island (including the US Naval Air Station Annex)
(729.828.11)	Saint Patrick. □ Sauteurs. Ronde Island, Diamond Island, Les Tantes	(729.92)	Southampton
(729.828.12)	Saint Mark. □ Victoria	(729.93)	Warwick
(729.828.13)	Saint John	(729.94)	Paget
(729.828.14)	Saint Andrew	(729.95)	Pembroke
(729.828.15)	Saint George (state capital). □ Glover Island	(729.96)	Devonshire
(729.828.16)	Saint David	(729.97)	Smith's
(729.828.2)	Dependencies of Grenada	(729.98)	Hamilton. □ Hamilton (state capital and municipality)
(729.828.21)	Carriacou. □ Hillsborough. Petit Martinique	(729.99)	Saint George's. □ Saint George (municipality). Coney Island, Saint David's Island (including the US Naval Air Station), Saint George's Island
(729.828.3)	Grenadines of Grenada (uninhabited)		
⇒ (292.797)			
(729.86)	BARBADOS	(73)	UNITED STATES OF AMERICA (USA)
(729.861)	Parishes of Barbados		<i>The District of Columbia and the federal capital are classed at (735.3), and the state capitals are listed with the counties where</i>
(729.861.11)	Saint Lucy		
(729.861.12)	Saint Peter. □ Speightstown		
(729.861.13)	Saint Andrew		
(729.861.14)	Saint James. □ Holetown		

(1/9) Table 1e. Common auxiliaries of place

		they are located, with the addition of the words '(state capital)'; but, if preferred, they can be classed at federal or state level, using the special auxiliary -25, from (1-25). Class protected areas such as National Parks and National Wildlife Refuges preferably with the state or area in which they are situated, adding the subdivisions of (1-751) (Table 1e). Class collective works here	(734.411.2)	Berkshire County
			(734.411.4)	Franklin County
			(734.411.6)	Hampshire County
			(734.411.8)	Hampden County. □ Springfield
			(734.413.1)	Worcester County
			(734.413.3)	Middlesex County. □ Cambridge, Lexington, Lowell
			(734.413.5)	Essex County
			(734.413.7)	Suffolk County. □ Boston (state capital)
			(734.415.2)	Norfolk County
			(734.415.4)	Bristol County
			(734.415.6)	Plymouth County
			(734.415.8)	Barnstable County
			(734.417.1)	Dukes County
			(734.417.3)	Nantucket County
			(734.5)	Rhode Island
			(734.512.2)	Providence County. □ Providence (state capital), Pawtucket
			(734.512.3)	Kent County
			(734.512.4)	Bristol County
			(734.512.6)	Newport County. □ Newport (city)
			(734.512.8)	Washington County. □ Block Island
			(734.6)	Connecticut
			(734.611.2)	Litchfield County
			(734.611.4)	Hartford County. □ Hartford (state capital)
			(734.611.6)	Tolland County
			(734.611.8)	Windham County
			(734.613.1)	New London County
			(734.613.3)	Middlesex County
			(734.613.5)	New Haven County. □ New Haven (city)
			(734.613.7)	Fairfield County. □ Stamford
			(734.7/.9)	Middle Atlantic states
			(734.7/.8+737.1/.6)	States bordering the Great Lakes
			(734.7)	New York (state)
			(734.711+734.912)	Boroughs and counties of New York metropolitan area
			(734.711)	New York metropolitan area within New York state
			(734.711.1)	Borough of Manhattan (= New York County = Manhattan Island)
			(734.711.2/.5)	Boroughs and counties of Long Island
			(734.711.2)	Borough of Brooklyn (= Kings County)
			(734.711.3)	Borough of Queens (= Queens County)
			(734.711.4)	Nassau County
			(734.711.5)	Suffolk County
			(734.711.6)	Borough of Richmond (= Richmond County = Staten Island)
			(734.711.7/.8)	Mainland counties east of River Hudson
			(734.711.7)	Borough of the Bronx (= Bronx County)
			(734.711.8)	Westchester County
			(734.711.9)	Rockland County
			(734.712.1)	Orange County
			(734.712.2)	Putnam County
			(734.712.3)	Dutchess County
			(734.712.4)	Ulster County
			(734.712.5)	Sullivan County
			(734.712.6)	Delaware County
			(734.712.7)	Greene County
			(734.712.8)	Columbia County
			(734.713.1)	Rensselaer County
			(734.713.2)	Albany County. □ Albany (state capital)
			(734.713.3)	Schenectady County. □ Schenectady (city)
			(734.713.4)	Schoharie County
			(734.713.5)	Montgomery County
			(734.713.6)	Fulton County
			(734.713.7)	Saratoga County
			(734.713.8)	Washington County
			(734.714.1)	Warren County
			(734.714.2)	Hamilton County
			(734.714.3)	Essex County
(734)	States of the north-eastern USA			
(734.1/.6)	New England			
(734.1)	Maine			
(734.112.2)	Aroostook County			
(734.112.4)	Washington County			
(734.112.6)	Hancock County			
(734.112.8)	Penobscot County. □ Bangor			
(734.114.1)	Piscataquis County			
(734.114.3)	Somerset County			
(734.114.5)	Franklin County			
(734.114.7)	Kennebec County. □ Augusta (state capital)			
(734.116.2)	Waldo County			
(734.116.4)	Knox County			
(734.116.6)	Lincoln County			
(734.116.8)	Sagadahoc County			
(734.118.1)	Androscoggin County			
(734.118.3)	Oxford County			
(734.118.5)	Cumberland County. □ Portland			
(734.118.7)	York County			
(734.2)	New Hampshire			
(734.211.2)	Coos County			
(734.211.4)	Grafton County			
(734.211.6)	Carroll County			
(734.211.8)	Strafford County			
(734.213.1)	Belknap County			
(734.213.3)	Merrimack County. □ Concord (state capital)			
(734.213.5)	Sullivan County			
(734.213.7)	Cheshire County			
(734.215.2)	Hillsboro County. □ Manchester			
(734.215.4)	Rockingham County. □ Portsmouth			
(734.3)	Vermont			
(734.312.1)	Grand Isle County			
(734.312.3)	Franklin County			
(734.312.5)	Orleans County			
(734.312.7)	Essex County			
(734.314.2)	Chittenden County. □ Burlington			
(734.314.4)	Lamoille County			
(734.314.6)	Caledonia County			
(734.314.8)	Washington County. □ Montpelier (state capital)			
(734.316.1)	Addison County			
(734.316.3)	Orange County			
(734.316.5)	Windsor County			
(734.316.7)	Rutland County			
(734.318.2)	Bennington County			
(734.318.4)	Windham County			
(734.4)	Massachusetts			

(734.714.4)	Clinton County	(734.822.1)	Clinton County
(734.714.5)	Franklin County	(734.822.3)	Potter County
(734.714.6)	St. Lawrence County	(734.822.5)	Tioga County
(734.714.7)	Jefferson County	(734.822.7)	Clearfield County
(734.714.8)	Lewis County	(734.823.2)	Jefferson County
(734.715.1)	Herkimer County	(734.823.4)	Elk County
(734.715.2)	Oneida County. □ Utica	(734.823.6)	McKean County
(734.715.3)	Oswego County	(734.823.8)	Cameron County
(734.715.4)	Onondaga County. □ Syracuse	(734.824.1)	Warren County
(734.715.5)	Madison County	(734.824.3)	Forest County
(734.715.6)	Otsego County	(734.824.5)	Clarion County
(734.715.7)	Chenango County	(734.824.7)	Fulton County
(734.716.1)	Cortland County	(734.825.2)	Bedford County
(734.716.2)	Broome County. □ Binghamton	(734.825.4)	Huntingdon County
(734.716.3)	Tioga County	(734.825.6)	Blair County. □ Altoona
(734.716.4)	Tompkins County. □ Ithaca	(734.825.8)	Cambria County
(734.716.5)	Cayuga County	(734.826.1)	Somerset County
(734.716.6)	Wayne County	(734.826.3)	Fayette County
(734.716.7)	Seneca County	(734.826.5)	Greene County
(734.717.1)	Schuyler County	(734.826.7)	Washington County
(734.717.2)	Chemung County. □ Elmira	(734.827.2)	Westmoreland County
(734.717.3)	Steuben County	(734.827.4)	Allegheny County. □ Pittsburgh
(734.717.4)	Yates County	(734.827.6)	Indiana County
(734.717.5)	Ontario County	(734.827.8)	Beaver County
(734.717.6)	Monroe County. □ Rochester	(734.828.1)	Butler County
(734.717.7)	Livingston County	(734.828.3)	Armstrong County
(734.717.8)	Allegany County	(734.828.5)	Lawrence County
(734.718.1)	Wyoming County	(734.828.7)	Mercer County
(734.718.2)	Genesee County	(734.829.2)	Venango County
(734.718.3)	Orleans County	(734.829.4)	Crawford County
(734.718.4)	Niagara County. □ Niagara Falls (city)	(734.829.6)	Erie County
(734.718.5)	Erie County. □ Buffalo	(734.9)	New Jersey
(734.718.6)	Cattaraugus County	(734.912)	Counties of the New York metropolitan area within New Jersey
(734.718.7)	Chautauqua County		⇒ (734.711)
(734.8)	Pennsylvania		Bergen County
(734.811.2)	Philadelphia County. □ Philadelphia (city)	(734.912.2)	Passaic County. □ Paterson
(734.811.4)	Bucks County	(734.912.3)	Hudson County. □ Hoboken, Jersey City
(734.811.6)	Montgomery County	(734.912.4)	Essex County. □ Newark, East Orange,
(734.811.8)	Delaware County	(734.912.5)	Maplewood, Orange, South Orange, West Orange (The Oranges)
(734.812.1)	Chester County		Union County. □ Elizabeth, Rahway
(734.812.3)	Lancaster County. □ Lancaster (city)	(734.912.6)	Middlesex County. □ New Brunswick
(734.812.5)	Berks County	(734.912.8)	Somerset County
(734.812.7)	Schuylkill County	(734.913.2)	Hunterdon County
(734.813.2)	Dauphin County. □ Harrisburg (state capital)	(734.913.4)	Warren County
(734.813.4)	Lebanon County	(734.913.6)	Morris County
(734.813.6)	Wayne County	(734.913.8)	Sussex County
(734.813.8)	Pike County	(734.915.2)	Mercer County. □ Princeton, Trenton (state capital)
(734.814.1)	Monroe County	(734.915.4)	Monmouth County
(734.814.3)	Northampton County. □ Bethlehem		Ocean County
(734.814.5)	Carbon County	(734.915.6)	Burlington County
(734.814.7)	Lehigh County. □ Allentown	(734.915.8)	Camden County. □ Camden (city)
(734.815.2)	Northumberland County	(734.917.2)	Atlantic County. □ Atlantic City
(734.815.4)	Montour County	(734.917.4)	Gloucester County
(734.815.6)	Columbia County	(734.917.6)	Salem County
(734.815.8)	Luzerne County. □ Wilkes-Barre	(734.917.8)	Cumberland County. □ Bridgeton
(734.816.1)	Susquehanna County	(734.918.2)	Cape May County
(734.816.3)	Bradford County	(734.918.4)	
(734.816.5)	Sullivan County	(734.918.6)	
(734.816.7)	Wyoming County		
(734.817.2)	Lackawanna County. □ Scranton	(735)	States of the south-eastern USA (South Atlantic states)
(734.817.4)	York County		<i>Class here the southern states</i>
(734.817.6)	Adams County		
(734.817.8)	Franklin County	(735.1)	Delaware
(734.818.1)	Cumberland County	(735.112.3)	New Castle County. □ Wilmington
(734.818.3)	Perry County	(735.112.5)	Kent County. □ Dover (state capital)
(734.818.5)	Juniata County	(735.112.7)	Sussex County
(734.818.7)	Mifflin County		
(734.821.2)	Snyder County	(735.2)	Maryland
(734.821.4)	Union County	(735.212/213)	Counties east of Chesapeake Bay
(734.821.6)	Lycoming County	(735.212.2)	Worcester County
(734.821.8)	Centre County	(735.212.4)	Somerset County

(1/9) Table 1e. Common auxiliaries of place

(735.212.6)	Wicomico County	(735.418.2)	Preston County
(735.212.8)	Dorchester County	(735.418.4)	Tucker County
(735.213.1)	Caroline County	(735.418.5)	Randolph County
(735.213.3)	Talbot County	(735.418.6)	Pocahontas County
(735.213.5)	Queen Anne's County	(735.418.7)	Greenbrier County
(735.213.6)	Kent County	(735.418.8)	Pendleton County
(735.213.7)	Cecil County	(735.419.2)	Grant County
(735.214/217)	Counties west of Chesapeake Bay	(735.419.4)	Hardy County
(735.214.2)	St. Mary's County	(735.419.5)	Mineral County
(735.214.4)	Charles County	(735.419.6)	Hampshire County
(735.214.6)	Calvert County	(735.419.7)	Morgan County
(735.214.8)	Anne Arundel County. □ Annapolis (state capital)	(735.419.8)	Berkeley County
(735.215.2/4)	Counties in the Washington metropolitan area ⇒ (735.3) ⇒ (735.525.2/8)	(735.419.9)	Jefferson County
(735.215.2)	Prince George's County	(735.5)	Virginia
(735.215.4)	Montgomery County	(735.521.2/4)	Counties east of Chesapeake Bay
(735.215.6)	Howard County	(735.521.2)	Northampton County
(735.215.8)	Baltimore County	(735.521.4)	Accomack County
(735.217.2)	Independent City of Baltimore	(735.522/544)	Counties west of Chesapeake Bay and the Potomac
(735.217.4)	Harford County	(735.522.1)	Northumberland County
(735.217.6)	Carroll County	(735.522.3)	Lancaster County
(735.217.8)	Frederick County. □ Frederick (city)	(735.522.5)	Richmond County
(735.218.2)	Washington County. □ Hagerstown	(735.522.7)	Westmoreland County
(735.218.4)	Allegany County. □ Cumberland	(735.523.2)	King George County
(735.218.6)	Garrett County	(735.523.4)	Stafford County
(735.3)	District of Columbia. Washington DC	(735.523.6)	Prince William County
(735.4)	West Virginia	(735.523.8)	Independent City of Manassas
(735.411.2)	Hancock County	(735.524.1)	Independent City of Manassas Park
(735.411.4)	Brooke County	(735.524.3)	Fauquier County
(735.411.5)	Ohio County. □ Wheeling	(735.524.5)	Loudoun County
(735.411.6)	Marshall County	(735.525.2/8)	Counties and cities in the Washington metropolitan area ⇒ (735.215.2/4) ⇒ (735.3)
(735.411.7)	Wetzel County	(735.525.2)	Fairfax County
(735.411.8)	Tyler County	(735.525.4)	Independent City of Fairfax
(735.412.2)	Pleasants County	(735.525.6)	Independent City of Falls Church
(735.412.4)	Wood County. □ Parkersburg	(735.525.7)	Independent City of Alexandria
(735.412.5)	Ritchie County	(735.525.8)	Arlington County
(735.412.6)	Wirt County	(735.526.1)	Mathews County
(735.412.7)	Gilmer County	(735.526.3)	Gloucester County
(735.412.8)	Calhoun County	(735.526.5)	Middlesex County
(735.413.2)	Roane County	(735.526.7)	Essex County
(735.413.4)	Jackson County	(735.527.2)	King and Queen County
(735.413.5)	Mason County	(735.527.4)	King William County
(735.413.6)	Cabell County	(735.527.6)	Caroline County
(735.413.7)	Putnam County	(735.527.8)	Spotsylvania County
(735.413.8)	Kanawha County. □ Charleston (state capital)	(735.528.1)	Independent City of Fredericksburg
(735.414.2)	Boone County	(735.528.3)	Orange County
(735.414.4)	Lincoln County	(735.528.5)	Greene County
(735.414.5)	Wayne County	(735.528.7)	Madison County
(735.414.6)	Mingo County	(735.531.2)	Culpeper County
(735.414.7)	Logan County	(735.531.4)	Rappahannock County
(735.414.8)	Wyoming County	(735.531.6)	York County
(735.415.2)	McDowell County	(735.531.8)	Independent City of Poquoson
(735.415.4)	Monongalia County. □ Morgantown	(735.541.2)	Independent City of Hampton
(735.415.5)	Marion County	(735.541.4)	Independent City of Newport News
(735.415.6)	Taylor County	(735.541.6)	James City County
(735.415.7)	Doddridge County	(735.541.8)	Independent City of Williamsburg
(735.415.8)	Harrison County. □ Clarksburg	(735.542.1)	New Kent County
(735.416.2)	Barbour County	(735.542.3)	Charles City County
(735.416.4)	Lewis County	(735.542.5)	Henrico County
(735.416.5)	Upshur County	(735.542.7)	Independent City of Richmond (state capital)
(735.416.6)	Webster County	(735.543.2)	Goochland County
(735.416.7)	Braxton County	(735.543.4)	Hanover County
(735.416.8)	Clay County	(735.543.6)	Louisa County
(735.417.2)	Nicholas County	(735.543.8)	Fluvanna County
(735.417.4)	Fayette County	(735.544.1)	Albemarle County
(735.417.5)	Raleigh County	(735.544.3)	Independent City of Charlottesville
(735.417.6)	Mercer County	(735.544.5)	Nelson County
(735.417.7)	Summers County	(735.544.7)	Amherst County
(735.417.8)	Monroe County		

(735.546.1/.7)	South-eastern group of Independent Cities	(735.587.6)	Independent City of Lexington
(735.546.1)	Independent City of Norfolk	(735.587.8)	Independent City of Buena Vista
(735.546.3)	Independent City of Virginia Beach	(735.588.1)	Bath County
(735.546.5)	Independent City of Chesapeake	(735.588.3)	Highland County
(735.546.6)	Independent City of Portsmouth	(735.588.5)	Augusta County
(735.546.7)	Independent City of Suffolk	(735.588.7)	Independent City of Staunton
(735.547.2)	Isle of Wight County	(735.591.2)	Independent City of Waynesboro
(735.547.4)	Southampton County	(735.591.4)	Rockingham County
(735.547.6)	Independent City of Franklin	(735.591.6)	Independent City of Harrisonburg
(735.547.8)	Surry County	(735.591.8)	Page County
(735.548.1)	Sussex County	(735.592.1)	Shenandoah County
(735.548.3)	Greensville County	(735.592.3)	Warren County
(735.548.5)	Independent City of Emporia	(735.592.5)	Clarke County
(735.548.7)	Brunswick County	(735.592.7)	Frederick County
(735.561.2)	Dinwiddie County	(735.593.2)	Independent City of Winchester
(735.561.4)	Independent City of Petersburg	(735.6)	North Carolina
(735.561.6)	Prince George County	(735.611.2)	Currituck County
(735.561.8)	Independent City of Hopewell	(735.611.4)	Camden County
(735.562.1)	Chesterfield County	(735.611.6)	Pasquotank County
(735.562.3)	Independent City of Colonial Heights	(735.611.8)	Perquimans County
(735.562.5)	Powhatan County	(735.612.1)	Chowan County
(735.562.7)	Cumberland County	(735.612.3)	Gates County
(735.563.2)	Buckingham County	(735.612.5)	Hertford County
(735.563.4)	Appomattox County	(735.612.7)	Northampton County
(735.563.6)	Prince Edward County	(735.613.2)	Bertie County
(735.563.8)	Amelia County	(735.613.4)	Martin County
(735.564.1)	Nottoway County	(735.613.6)	Washington County
(735.564.3)	Lunenburg County	(735.613.8)	Tyrrell County
(735.564.5)	Mecklenburg County	(735.614.1)	Dare County
(735.564.7)	Charlotte County	(735.614.3)	Hyde County
(735.565.2)	Halifax County. □ South Boston	(735.614.5)	Beaufort County
(735.565.6)	Pittsylvania County	(735.614.7)	Craven County
(735.565.8)	Independent City of Danville	(735.615.2)	Pamlico County
(735.566.1)	Campbell County	(735.615.4)	Carteret County
(735.566.3)	Independent City of Lynchburg	(735.615.6)	Jones County
(735.566.5)	Bedford County	(735.615.8)	Onslow County
(735.566.7)	Independent City of Bedford	(735.616.1)	Pender County
(735.567.2)	Franklin County	(735.616.3)	New Hanover County. □ Wilmington
(735.567.4)	Henry County	(735.616.5)	Brunswick County
(735.567.6)	Independent City of Martinsville	(735.616.7)	Columbus County
(735.567.8)	Patrick County	(735.618.2)	Bladen County
(735.568.1)	Floyd County	(735.618.4)	Robeson County
(735.568.3)	Carroll County	(735.618.6)	Scotland County
(735.568.5)	Independent City of Galax	(735.618.8)	Hoke County
(735.568.7)	Grayson County	(735.622.1)	Richmond County
(735.581.2)	Smyth County	(735.622.3)	Moore County
(735.581.4)	Washington County	(735.622.5)	Lee County
(735.581.6)	Independent City of Bristol	(735.622.7)	Harnett County
(735.581.8)	Scott County	(735.623.2)	Cumberland County
(735.582.1)	Lee County	(735.623.4)	Sampson County
(735.582.3)	Wise County	(735.623.6)	Duplin County
(735.582.5)	Independent City of Norton	(735.623.8)	Lenoir County
(735.582.7)	Dickenson County	(735.624.1)	Greene County
(735.583.2)	Buchanan County	(735.624.3)	Wayne County
(735.583.4)	Russell County	(735.624.5)	Johnston County
(735.583.6)	Tazewell County	(735.624.7)	Wilson County
(735.583.8)	Bland County	(735.625.2)	Pitt County
(735.584.1)	Wythe County	(735.625.4)	Edgecombe County
(735.584.3)	Pulaski County	(735.625.6)	Nash County
(735.584.5)	Giles County	(735.625.8)	Halifax County
(735.584.7)	Montgomery County	(735.626.1)	Warren County
(735.585.2)	Independent City of Radford	(735.626.3)	Vance County
(735.585.4)	Roanoke County	(735.626.5)	Granville County
(735.585.6)	Independent City of Roanoke	(735.626.7)	Durham County
(735.585.8)	Independent City of Salem	(735.627.2)	Wake County. □ Raleigh (state capital)
(735.586.1)	Craig County	(735.627.4)	Franklin County
(735.586.3)	Alleghany County	(735.627.6)	Orange County
(735.586.5)	Independent City of Clifton Forge	(735.627.8)	Person County
(735.586.7)	Independent City of Covington	(735.628.1)	Caswell County
(735.587.2)	Botetourt County	(735.628.3)	Alamance County
(735.587.4)	Rockbridge County	(735.628.5)	Chatham County

(1/9) Table 1e. Common auxiliaries of place

(735.628.7)	Randolph County	(735.736.3)	Williamsburg County
(735.631.2)	Guilford County	(735.736.5)	Florence County
(735.631.4)	Rockingham County	(735.736.7)	Darlington County
(735.631.6)	Stokes County	(735.738.1)	Lee County
(735.631.8)	Surry County	(735.738.3)	Sumter County
(735.632.1)	Yadkin County	(735.738.5)	Richland County. □ Columbia (state capital)
(735.632.3)	Forsyth County. □ Winston; Salem	(735.738.7)	Calhoun County
(735.632.5)	Davidson County	(735.741.2)	Lexington County
(735.632.7)	Davie County	(735.741.4)	Aiken County
(735.633.2)	Rowan County	(735.741.6)	Barnwell County
(735.633.4)	Cabarrus County	(735.741.8)	Allendale County
(735.633.6)	Stanly County	(735.743.2)	Bamberg County
(735.633.8)	Montgomery County	(735.743.4)	Orangeburg County
(735.634.1)	Anson County	(735.743.6)	Clarendon County
(735.634.3)	Union County	(735.743.8)	Berkeley County
(735.634.5)	Mecklenburg County. □ Charlotte	(735.745.2)	Charleston County. □ Charleston (city)
(735.634.7)	Gaston County	(735.745.4)	Dorchester County
(735.635.2)	Cleveland County	(735.745.6)	Colleton County
(735.635.4)	Lincoln County	(735.745.8)	Hampton County
(735.635.6)	Catawba County	(735.747.2)	Jasper County
(735.635.8)	Iredell County	(735.747.4)	Beaufort County
(735.636.1)	Alexander County	(735.8)	Georgia
(735.636.3)	Wilkes County	(735.812.1)	Rabun County
(735.636.5)	Alleghany County	(735.812.3)	Towns County
(735.636.7)	Ashe County	(735.812.5)	Union County
(735.637.2)	Watauga County	(735.812.7)	Lumpkin County
(735.637.4)	Caldwell County	(735.813.2)	White County
(735.637.6)	Burke County	(735.813.4)	Habersham County
(735.637.8)	Avery County	(735.813.6)	Stephens County
(735.641.1)	Mitchell County	(735.813.8)	Hall County
(735.641.3)	Yancey County	(735.814.1)	Banks County
(735.641.5)	Madison County	(735.814.3)	Franklin County
(735.641.7)	Buncombe County. □ Asheville	(735.814.5)	Hart County
(735.643.2)	McDowell County	(735.814.7)	Jackson County
(735.643.4)	Rutherford County	(735.815.2)	Madison County
(735.643.6)	Polk County	(735.815.4)	Elbert County
(735.643.8)	Henderson County	(735.815.6)	Lincoln County
(735.644.1)	Transylvania County	(735.815.8)	Wilkes County
(735.644.3)	Haywood County	(735.816.1)	Oglethorpe County
(735.644.5)	Jackson County	(735.816.3)	Clarke County. □ Athens
(735.644.7)	Swain County	(735.816.5)	Barrow County
(735.645.2)	Graham County	(735.816.7)	Walton County
(735.645.4)	Macon County	(735.817.2)	Newton County
(735.645.6)	Clay County	(735.817.4)	Rockdale County
(735.645.8)	Cherokee County	(735.817.6)	De Kalb County
(735.7)	South Carolina	(735.817.8)	Gwinnett County
(735.721.2)	Oconee County	(735.818.1)	Fulton County. □ Atlanta (state capital)
(735.721.4)	Pickens County	(735.818.3)	Cobb County
(735.721.6)	Anderson County	(735.818.5)	Cherokee County
(735.721.8)	Greenville County. □ Greenville (city)	(735.818.7)	Forsyth County
(735.723.2)	Spartanburg County	(735.821.2)	Dawson County
(735.723.4)	Laurens County	(735.821.4)	Pickens County
(735.723.6)	Greenwood County	(735.821.6)	Gilmer County
(735.723.8)	Abbeville County	(735.821.8)	Fannin County
(735.725.2)	McCormick County	(735.822.1)	Murray County
(735.725.4)	Edgefield County	(735.822.3)	Whitfield County
(735.725.6)	Saluda County	(735.822.5)	Catoosa County
(735.725.8)	Newberry County	(735.822.7)	Walker County
(735.727.2)	Union County	(735.823.2)	Dade County
(735.727.4)	Cherokee County	(735.823.4)	Chattooga County
(735.727.6)	York County	(735.823.6)	Floyd County
(735.727.8)	Chester County	(735.823.8)	Gordon County
(735.732.1)	Fairfield County	(735.824.1)	Bartow County
(735.732.3)	Lancaster County	(735.824.3)	Polk County
(735.732.5)	Kershaw County	(735.824.5)	Paulding County
(735.732.7)	Chesterfield County	(735.824.7)	Douglas County
(735.734.1)	Marlboro County	(735.825.2)	Haralson County
(735.734.3)	Dillon County	(735.825.4)	Carroll County
(735.734.5)	Marion County	(735.825.6)	Heard County
(735.734.7)	Horry County	(735.825.8)	Coweta County
(735.736.1)	Georgetown County	(735.826.1)	Fayette County

(735.826.3)	Clayton County	(735.856.1)	Bacon County
(735.826.5)	Henry County	(735.856.3)	Pierce County
(735.826.7)	Spalding County	(735.856.5)	Ware County
(735.827.2)	Butts County	(735.856.7)	Clinch County
(735.827.4)	Jasper County	(735.858.2)	Echols County
(735.827.6)	Monroe County	(735.858.4)	Lanier County
(735.827.8)	Crawford County	(735.858.6)	Atkinson County
(735.832.1)	Upson County	(735.858.8)	Coffee County
(735.832.3)	Lamar County	(735.861.2)	Ben Hill County
(735.832.5)	Pike County	(735.861.4)	Irwin County
(735.832.7)	Meriwether County	(735.861.6)	Berrien County
(735.833.2)	Troup County	(735.861.8)	Lowndes County
(735.833.4)	Harris County	(735.864.1)	Brooks County
(735.833.6)	Talbot County	(735.864.3)	Cook County
(735.833.8)	Taylor County	(735.864.5)	Tift County
(735.834.1)	Muscogee County	(735.864.7)	Turner County
(735.834.3)	Chattahoochee County	(735.867.2)	Wilcox County
(735.834.5)	Marion County	(735.867.4)	Crisp County
(735.834.7)	Schley County	(735.867.6)	Dooly County
(735.835.2)	Macon County	(735.867.8)	Sumter County
(735.835.4)	Peach County	(735.872.1)	Webster County
(735.835.6)	Houston County	(735.872.3)	Stewart County
(735.835.8)	Bibb County. □ Macon	(735.872.5)	Quitman County
(735.836.1)	Twiggs County	(735.872.7)	Clay County
(735.836.3)	Bleckley County	(735.875.2)	Randolph County
(735.836.5)	Pulaski County	(735.875.4)	Terrell County
(735.836.7)	Dodge County	(735.875.6)	Lee County
(735.837.2)	Telfair County	(735.875.8)	Worth County
(735.837.4)	Wheeler County	(735.882.1)	Dougherty County
(735.837.6)	Laurens County	(735.882.3)	Calhoun County
(735.837.8)	Wilkinson County	(735.882.5)	Early County
(735.838.1)	Jones County	(735.882.7)	Miller County
(735.838.3)	Baldwin County	(735.885.2)	Baker County
(735.838.5)	Putnam County	(735.885.4)	Mitchell County
(735.838.7)	Hancock County	(735.885.6)	Colquitt County
(735.841.2)	Taliaferro County	(735.885.8)	Thomas County
(735.841.4)	Greene County	(735.887.1)	Grady County
(735.841.6)	Oconee County	(735.887.3)	Decatur County
(735.841.8)	Morgan County	(735.887.5)	Seminole County
(735.842.1)	Warren County	(735.9+736.1/.4) States around the Gulf of Mexico	
(735.842.3)	McDuffie County	(735.9)	Florida
(735.842.5)	Columbia County	(735.921.2)	Escambia County. □ Pensacola
(735.842.7)	Richmond County. □ Augusta	(735.921.4)	Santa Rosa County
(735.843.2)	Burke County	(735.921.6)	Okaloosa County
(735.843.4)	Jefferson County	(735.921.8)	Walton County
(735.843.6)	Glascock County	(735.922.1)	Holmes County
(735.843.8)	Washington County	(735.922.3)	Washington County
(735.844.1)	Johnson County	(735.922.5)	Bay County
(735.844.3)	Emanuel County	(735.922.7)	Gulf County
(735.844.5)	Jenkins County	(735.925.2)	Calhoun County
(735.844.7)	Screven County	(735.925.4)	Jackson County
(735.845.2)	Effingham County	(735.925.6)	Gadsden County
(735.845.4)	Bulloch County	(735.925.8)	Liberty County
(735.845.6)	Candler County	(735.926.1)	Franklin County
(735.845.8)	Treutlen County	(735.926.3)	Wakulla County
(735.846.1)	Montgomery County	(735.926.5)	Leon County. □ Tallahassee (state capital)
(735.846.3)	Toombs County	(735.926.7)	Jefferson County
(735.846.5)	Tattnall County	(735.931.2)	Taylor County
(735.846.7)	Evans County	(735.931.4)	Madison County
(735.847.2)	Bryan County	(735.931.6)	Hamilton County
(735.847.4)	Chatham County. □ Savannah	(735.931.8)	Suwannee County
(735.847.6)	Liberty County	(735.932.1)	Lafayette County
(735.847.8)	Long County	(735.932.3)	Dixie County
(735.852.1)	McIntosh County	(735.932.5)	Gilchrist County
(735.852.3)	Glynn County	(735.932.7)	Columbia County
(735.852.5)	Camden County	(735.935.2)	Baker County
(735.852.7)	Charlton County	(735.935.4)	Nassau County
(735.853.2)	Brantley County	(735.935.6)	Duval County. □ Jacksonville
(735.853.4)	Wayne County	(735.935.8)	Union County
(735.853.6)	Appling County	(735.936.1)	Bradford County
(735.853.8)	Jeff Davis County	(735.936.3)	Clay County

(1/9) Table 1e. Common auxiliaries of place

(735.936.5)	St. Johns County	(736.136.3)	Russell County
(735.936.7)	Levy County	(736.136.5)	Macon County
(735.941.2)	Alachua County	(736.136.7)	Elmore County
(735.941.4)	Putnam County	(736.137.2)	Coosa County
(735.941.6)	Flagler County	(736.137.4)	Tallapoosa County
(735.941.8)	Volusia County. □ Daytona Beach	(736.137.6)	Lee County
(735.944.1)	Lake County	(736.137.8)	Chambers County
(735.944.3)	Seminole County	(736.142.1)	Randolph County
(735.944.5)	Orange County. □ Orlando	(736.142.3)	Clay County
(735.944.7)	Osceola County	(736.142.5)	Talladega County
(735.947.2)	Brevard County	(736.142.7)	St. Clair County
(735.947.4)	Indian River County	(736.145.2)	Calhoun County
(735.947.6)	St. Lucie County	(736.145.4)	Cleburne County
(735.947.8)	Martin County	(736.145.6)	Cherokee County
(735.952.1)	Palm Beach County. □ West Palm Beach	(736.145.8)	De Kalb County
(735.952.3)	Broward County. □ Fort Lauderdale	(736.148.1)	Etowah County
(735.952.5)	Miami-Dade County (Dade County). □ Miami.	(736.148.3)	Blount County
	Miami Beach	(736.148.5)	Cullman County
(735.952.7)	Monroe County. □ Florida Keys. Key West	(736.148.7)	Winston County
(735.955.2)	Collier County. □ Ten Thousand Islands	(736.151.2)	Walker County
(735.955.4)	Hendry County	(736.151.4)	Jefferson County. □ Birmingham
(735.955.6)	Lee County	(736.151.6)	Shelby County
(735.955.8)	Charlotte County	(736.151.8)	Chilton County
(735.958.1)	Glades County	(736.154.1)	Bibb County
(735.958.3)	Okeechobee County	(736.154.3)	Tuscaloosa County
(735.958.5)	Highlands County	(736.154.5)	Pickens County
(735.958.7)	Hardee County	(736.154.7)	Lamar County
(735.961.2)	De Soto County	(736.157.2)	Fayette County
(735.961.4)	Sarasota County. □ Sarasota (city)	(736.157.4)	Marion County
(735.961.6)	Manatee County	(736.157.6)	Franklin County
(735.961.8)	Pinellas County	(736.157.8)	Colbert County
(735.964.1)	Hillsborough County. □ Tampa	(736.162.1)	Lauderdale County
(735.964.3)	Polk County	(736.162.3)	Lawrence County
(735.964.5)	Pasco County	(736.162.5)	Limestone County
(735.964.7)	Hernando County	(736.162.7)	Morgan County
(735.967.2)	Citrus County	(736.165.2)	Madison County. □ Huntsville
(735.967.4)	Sumter County	(736.165.4)	Marshall County
(735.967.6)	Marion County	(736.165.6)	Jackson County
(736)	States of the south central USA	(736.2)	Mississippi
	<i>Class here the Old South-west</i>	(736.221.2)	Jackson County
(736.1)	Alabama	(736.221.4)	Harrison County
(736.121.2)	Baldwin County	(736.221.6)	Hancock County
(736.121.4)	Mobile County. □ Mobile (city)	(736.221.8)	Pearl River County
(736.121.6)	Washington County	(736.224.1)	Stone County
(736.121.8)	Clarke County	(736.224.3)	George County
(736.122.1)	Monroe County	(736.224.5)	Greene County
(736.122.3)	Conecuh County	(736.224.7)	Perry County
(736.122.5)	Escambia County	(736.227.4)	Forrest County
(736.122.7)	Covington County	(736.227.6)	Lamar County
(736.125.2)	Geneva County	(736.227.8)	Marion County
(736.125.4)	Houston County	(736.231/.232)	Counties between the Mississippi and Pearl rivers
(736.125.6)	Henry County	(736.231.2)	Walthall County
(736.125.8)	Barbour County	(736.231.3)	Pike County
(736.126.1)	Dale County	(736.231.4)	Amite County
(736.126.3)	Coffee County	(736.231.6)	Wilkinson County
(736.126.5)	Pike County	(736.231.7)	Adams County. □ Natchez
(736.126.7)	Crenshaw County	(736.231.8)	Franklin County
(736.127.2)	Butler County	(736.232.1)	Lincoln County
(736.127.4)	Wilcox County	(736.232.2)	Lawrence County
(736.127.6)	Marengo County	(736.232.3)	Copiah County
(736.127.8)	Choctaw County	(736.232.5)	Jefferson County
(736.132.1)	Sumter County	(736.232.6)	Claiborne County
(736.132.3)	Greene County	(736.232.7/.8+736.251.8)	Counties of the Jackson metropolitan area
(736.132.5)	Hale County	(736.232.7)	Hinds County. □ Jackson (state capital)
(736.132.7)	Perry County	(736.232.8)	Madison County
(736.133.2)	Dallas County	(736.241/.244)	Counties at the confluence of the Yazoo and Mississippi rivers, and north-west of Big Black River
(736.133.4)	Autauga County		
(736.133.6)	Lowndes County		
(736.133.8)	Montgomery County. □ Montgomery (state capital)		
(736.136.1)	Bullock County	(736.241.2)	Warren County

(736.241.3)	Issaquena County	(736.315.4)	St. James Parish
(736.241.4)	Sharkey County	(736.315.6)	St. John the Baptist Parish
(736.241.6)	Yazoo County	(736.315.8)	St. Charles Parish
(736.241.8)	Holmes County	(736.332.1)	Orleans Parish (New Orleans)
(736.242.1)	Humphreys County	(736.332.3)	St. Bernard Parish
(736.242.3)	Washington County	(736.332.5)	Plaquemines Parish
(736.242.5)	Bolivar County	(736.332.7)	Jefferson Parish
(736.242.7)	Sunflower County	(736.334.2)	Lafourche Parish
(736.242.8)	Leflore County	(736.334.4)	Terrebonne Parish
(736.243.2)	Tallahatchie County	(736.334.6)	St. Mary Parish
(736.243.4)	Quitman County	(736.334.8)	Assumption Parish
(736.243.6)	Coahoma County	(736.336.1)	Iberville Parish
(736.243.7)	Tunica County	(736.336.3)	West Baton Rouge Parish
(736.243.8)	De Soto County	(736.336.5)	Pointe Coupee Parish
(736.244.1)	Tate County	(736.341/.354)	Western parishes and those south-west of Red River
(736.244.2)	Panola County	(736.341.2)	St. Landry Parish
(736.244.3)	Yalobusha County	(736.341.4)	Lafayette Parish
(736.244.5)	Grenada County	(736.341.6)	St. Martin Parish
(736.244.6)	Carroll County	(736.341.8)	Iberia Parish
(736.244.7)	Montgomery County	(736.343.1)	Vermilion Parish
(736.251/.277)	Counties west and north of Pearl River	(736.343.3)	Cameron Parish
(736.251.2)	Attala County	(736.343.5)	Calcasieu Parish
(736.251.4)	Leake County	(736.343.7)	Jefferson Davis Parish
(736.251.6)	Scott County	(736.345.2)	Acadia Parish
(736.251.8)	Rankin County	(736.345.4)	Evangeline Parish
	⇒ (736.232.7/.8+736.251.8)	(736.345.6)	Allen Parish
(736.254.1)	Smith County	(736.345.8)	Beauregard Parish
(736.254.3)	Simpson County	(736.352.1)	Vernon Parish
(736.254.5)	Jefferson Davis County	(736.352.3)	Sabine Parish
(736.254.7)	Covington County	(736.352.5)	De Soto Parish
(736.257.2)	Jones County	(736.352.7)	Red River Parish
(736.257.4)	Wayne County	(736.354.2)	Natchitoches Parish
(736.257.6)	Jasper County	(736.354.4)	Rapides Parish
(736.257.8)	Clarke County	(736.354.6)	Avoyelles Parish
(736.262.1)	Lauderdale County. □ Meridian	(736.362/.376)	Parishes mainly between Red River and Mississippi
(736.262.3)	Newton County	(736.362.1)	Concordia Parish
(736.262.5)	Neshoba County	(736.362.3)	Catahoula Parish
(736.262.7)	Kemper County	(736.362.5)	La Salle Parish
(736.265.2)	Winston County	(736.362.7)	Grant Parish
(736.265.4)	Noxubee County	(736.364.2)	Winn Parish
(736.265.6)	Lowndes County	(736.364.4)	Caldwell Parish
(736.265.8)	Oktibbeha County	(736.364.6)	Franklin Parish
(736.268.1)	Choctaw County	(736.364.8)	Tensas Parish
(736.268.3)	Webster County	(736.366.1)	Madison Parish
(736.268.5)	Calhoun County	(736.366.3)	East Carroll Parish
(736.268.7)	Chickasaw County	(736.366.5)	West Carroll Parish
(736.271.2)	Clay County	(736.366.7)	Morehouse Parish
(736.271.4)	Monroe County	(736.372.3)	Richland Parish
(736.271.6)	Itawamba County	(736.372.4)	Ouachita Parish
(736.271.8)	Lee County	(736.372.6)	Union Parish
(736.274.1)	Pontotoc County	(736.372.8)	Lincoln Parish
(736.274.3)	Lafayette County	(736.374.1)	Jackson Parish
(736.274.5)	Marshall County	(736.374.3)	Bienville Parish
(736.274.7)	Benton County	(736.374.5)	Claiborne Parish
(736.277.2)	Union County	(736.374.7)	Webster Parish
(736.277.3)	Tippah County	(736.376.2)	Bossier Parish
(736.277.5)	Prentiss County	(736.376.4)	Caddo Parish. □ Shreveport
(736.277.6)	Alcorn County		
(736.277.8)	Tishomingo County		
(736.3)	Louisiana	(736.4)	Texas
(736.311/.336)	South-eastern parishes	(736.411/.413)	Counties around the Gulf of Mexico
(736.311.2)	Washington Parish	(736.411.2)	Cameron County
(736.311.4)	St. Tammany Parish	(736.411.3)	Willacy County
(736.311.6)	Tangipahoa Parish	(736.411.4)	Kenedy County
(736.311.8)	Livingston Parish	(736.411.5)	Kleberg County
(736.313.1)	St. Helena Parish	(736.411.6)	Nueces County. □ Corpus Christi
(736.313.3)	East Feliciana Parish	(736.411.7)	San Patricio County
(736.313.5)	West Feliciana Parish	(736.411.8)	Bee County
(736.313.7)	East Baton Rouge Parish. □ Baton Rouge (state capital)	(736.412.2)	Refugio County
(736.315.2)	Ascension Parish	(736.412.3)	Aransas County
		(736.412.4)	Calhoun County

(1/9) Table 1e. Common auxiliaries of place

(736.412.5)	Goliad County	(736.424.3)	Kaufman County
(736.412.6)	Victoria County	(736.424.5)	Rockwall County
(736.412.7)	Jackson County	(736.424.6)	Dallas County. □ Dallas
(736.412.8)	Matagorda County	(736.424.8)	Tarrant County. □ Fort Worth
(736.413.2)	Wharton County	(736.425.3)	Johnson County
(736.413.3)	Fort Bend County	(736.425.4)	Ellis County
(736.413.4)	Brazoria County	(736.425.5)	Navarro County
(736.413.5)	Galveston County	(736.425.7)	Hill County
(736.413.6)	Harris County. □ Houston	(736.425.8)	McLennan County
(736.413.7)	Chambers County	(736.426.2)	Limestone County
(736.413.8)	Jefferson County	(736.426.3)	Falls County
(736.414/.419)	Eastern counties	(736.426.4)	Bell County
(736.414.2)	Hardin County	(736.426.5)	Milam County
(736.414.3)	Orange County	(736.426.6)	Williamson County
(736.414.4)	Jasper County	(736.426.7)	Bastrop County
(736.414.5)	Newton County	(736.426.8)	Caldwell County
(736.414.6)	Tyler County	(736.428.2/.7)	Austin-San Antonio region
(736.414.7)	Polk County	(736.428.2)	Travis County. □ Austin (state capital)
(736.414.8)	Liberty County	(736.428.3)	Hays County
(736.415.2)	Montgomery County	(736.428.5)	Comal County
(736.415.3)	San Jacinto County	(736.428.6)	Guadalupe County
(736.415.4)	Walker County	(736.428.7)	Bexar County. □ San Antonio
(736.415.5)	Trinity County	(736.432/.436)	Counties of the southern spur ⇒ (736.411/.413)
(736.415.6)	Angelina County	(736.432.1)	Medina County
(736.415.7)	San Augustine County	(736.432.3)	Uvalde County
(736.415.8)	Sabine County	(736.432.5)	Kinney County
(736.416.2)	Shelby County	(736.432.7)	Maverick County
(736.416.3)	Nacogdoches County	(736.433.2)	Zavala County
(736.416.4)	Cherokee County	(736.433.4)	Frio County
(736.416.5)	Rusk County	(736.433.6)	Atascosa County
(736.416.6)	Panola County	(736.433.8)	Wilson County
(736.416.7)	Gregg County	(736.434.1)	Karnes County
(736.416.8)	Harrison County	(736.434.3)	Live Oak County
(736.417.2)	Marion County	(736.434.5)	McMullen County
(736.417.3)	Cass County	(736.434.7)	La Salle County
(736.417.4)	Morris County	(736.435.2)	Dimmit County
(736.417.5)	Bowie County	(736.435.4)	Webb County
(736.417.6)	Red River County	(736.435.6)	Duval County
(736.417.7)	Franklin County	(736.435.8)	Jim Wells County
(736.417.8)	Titus County	(736.436.2)	Brooks County
(736.418.2)	Camp County	(736.436.3)	Jim Hogg County
(736.418.3)	Upshur County	(736.436.5)	Zapata County
(736.418.4)	Wood County	(736.436.7)	Starr County
(736.418.5)	Smith County	(736.436.8)	Hidalgo County
(736.418.6)	Henderson County	(736.441/.454)	Central and northern counties
(736.418.7)	Anderson County	(736.441/.445)	Counties north-west of the Colorado river
(736.418.8)	Freestone County	(736.441.2)	Burnet County
(736.418.9)	Leon County	(736.441.3)	Lampasas County
(736.419.1)	Houston County	(736.441.4)	Mills County
(736.419.2)	Madison County	(736.441.6)	Hamilton County
(736.419.4)	Robertson County	(736.441.8)	Coryell County
(736.419.6)	Brazos County	(736.442.1)	Bosque County
(736.419.7)	Grimes County	(736.442.3)	Somervell County
(736.419.8)	Waller County	(736.442.5)	Hood County
(736.421/.426)	Counties south-west of River Brazos	(736.442.6)	Parker County
(736.421.3)	Austin County	(736.442.7)	Wise County
(736.421.4)	Washington County	(736.443.2)	Denton County
(736.421.5)	Burleson County	(736.443.4)	Collin County
(736.421.6)	Lee County	(736.443.5)	Grayson County
(736.421.8)	Fayette County	(736.443.6)	Cooke County
(736.422.3)	Colorado County	(736.443.8)	Montague County
(736.422.4)	Lavaca County	(736.444.1)	Clay County
(736.422.5)	Gonzales County	(736.444.2)	Archer County
(736.422.6)	De Witt County	(736.444.3)	Jack County
(736.422.8)	Lamar County	(736.444.5)	Young County
(736.423.2)	Fannin County	(736.444.7)	Stephens County
(736.423.4)	Hunt County	(736.445.2)	Palo Pinto County
(736.423.5)	Delta County	(736.445.4)	Erath County
(736.423.6)	Hopkins County	(736.445.6)	Eastland County
(736.423.8)	Rains County	(736.445.7)	Comanche County
(736.424.2)	Van Zandt County		

(736.445.8)	Brown County	(736.475.4)	Kent County
(736.446/.447)	Counties south of the Colorado river	(736.475.6)	Fisher County
(736.446.1)	San Saba County	(736.475.8)	Scurry County
(736.446.3)	Llano County	(736.476.3)	Borden County
(736.446.5)	Blanco County	(736.476.5)	Dawson County
(736.446.6)	Kendall County	(736.476.7)	Gaines County
(736.446.7)	Gillespie County	(736.482/.488)	Western counties. □ Counties on the Edwards Plateau
(736.447.2)	Mason County	(736.482.2)	Andrews County
(736.447.4)	McCulloch County	(736.482.4)	Martin County
(736.447.6)	Concho County	(736.482.6)	Howard County
(736.447.7)	Tom Green County	(736.482.8)	Mitchell County
(736.447.8)	Coke County	(736.484.1)	Nolan County
(736.452/.454)	Counties north of the Colorado river	(736.484.3)	Sterling County
(736.452.1)	Runnels County	(736.484.5)	Glasscock County
(736.452.3)	Coleman County	(736.484.7)	Midland County
(736.452.5)	Callahan County	(736.485.2)	Ector County
(736.452.6)	Taylor County	(736.485.4)	Winkler County
(736.452.7)	Jones County	(736.485.6)	Loving County
(736.453.2)	Shackelford County	(736.485.8)	Ward County
(736.453.4)	Throckmorton County	(736.486.1)	Crane County
(736.453.6)	Haskell County	(736.486.3)	Upton County
(736.453.7)	Stonewall County	(736.486.5)	Reagan County
(736.453.8)	King County	(736.486.7)	Irion County
(736.454.1)	Knox County	(736.487.2)	Crockett County
(736.454.2)	Baylor County	(736.487.4)	Schleicher County
(736.454.3)	Wichita County	(736.487.6)	Menard County
(736.454.5)	Wilbarger County	(736.487.8)	Sutton County
(736.454.7)	Hardeman County	(736.488.1)	Kimble County
(736.454.8)	Foard County	(736.488.3)	Kerr County
(736.456/.476)	Northernmost (Panhandle) counties	(736.488.4)	Bandera County
(736.456.2)	Cottle County	(736.488.5)	Real County
(736.456.4)	Childress County	(736.488.7)	Edwards County
(736.456.6)	Collingsworth County	(736.488.8)	Val Verde County
(736.456.8)	Wheeler County	(736.492/.495)	Counties west of the Pecos river
(736.458.1)	Hemphill County	(736.492.2)	Terrell County
(736.458.3)	Lipscomb County	(736.492.4)	Pecos County
(736.458.5)	Ochiltree County	(736.492.6)	Brewster County
(736.458.7)	Hansford County	(736.494.1)	Presidio County
(736.462.2)	Sherman County	(736.494.3)	Jeff Davis County
(736.462.4)	Dallam County	(736.494.5)	Reeves County
(736.462.6)	Hartley County	(736.495.2)	Culberson County
(736.462.8)	Moore County	(736.495.4)	Hudspeth County
(736.464.1)	Hutchinson County	(736.495.6)	El Paso County. □ El Paso (city)
(736.464.3)	Roberts County	(736.6)	Oklahoma
(736.464.5)	Gray County	(736.612.2/.8)	North-western counties. □ Panhandle counties
(736.464.7)	Carson County	(736.612.2)	Cimarron County
(736.466.2)	Potter County	(736.612.3)	Texas County
(736.466.4)	Oldham County	(736.612.4)	Beaver County
(736.466.6)	Deaf Smith County	(736.612.5)	Harper County
(736.466.8)	Randall County	(736.612.6)	Woodward County
(736.468.1)	Armstrong County	(736.612.7)	Ellis County
(736.468.3)	Donley County	(736.612.8)	Dewey County
(736.468.5)	Hall County	(736.621/.628)	Counties south of Canadian river
(736.468.7)	Briscoe County	(736.621.2)	Roger Mills County
(736.471.2)	Swisher County	(736.621.4)	Custer County
(736.471.4)	Castro County	(736.621.6)	Beckham County
(736.471.6)	Parmer County	(736.621.8)	Washita County
(736.471.8)	Bailey County	(736.622.3)	Caddo County
(736.472.1)	Lamb County	(736.622.5)	Grady County
(736.472.3)	Hale County	(736.622.7)	McClain County
(736.472.5)	Floyd County	(736.622.9)	Garvin County
(736.472.7)	Motley County	(736.623.2)	Pontotoc County
(736.473.2)	Dickens County	(736.623.4)	Murray County
(736.473.4)	Crosby County	(736.623.6)	Carter County
(736.473.6)	Lubbock County	(736.623.8)	Stephens County
(736.473.8)	Hockley County	(736.625.1)	Comanche County
(736.474.1)	Cochran County	(736.625.3)	Kiowa County
(736.474.3)	Yoakum County	(736.625.5)	Greer County
(736.474.5)	Terry County	(736.625.7)	Harmon County
(736.474.7)	Lynn County	(736.626.2)	Jackson County
(736.475.2)	Garza County		

(1/9) Table 1e. Common auxiliaries of place

(736.626.4)	Tillman County	(736.716.3)	Izard County
(736.626.6)	Cotton County	(736.716.5)	Fulton County
(736.626.8)	Jefferson County	(736.716.7)	Sharp County
(736.627.1)	Love County	(736.718.2)	Randolph County
(736.627.3)	Marshall County	(736.718.4)	Lawrence County
(736.627.5)	Johnston County	(736.718.6)	Independence County
(736.627.8)	Bryan County	(736.718.8)	Stone County
(736.628.2)	Atoka County	(736.721.2)	Cleburne County
(736.628.4)	Coal County	(736.721.4)	White County
(736.628.6)	Pittsburg County	(736.721.6)	Prairie County
(736.631.2/.7)	South-eastern counties, south of the Arkansas river	(736.721.8)	Lonoke County
(736.631.2)	Haskell County	(736.723.1)	Pulaski County. □ Little Rock (state capital)
(736.631.3)	Latimer County	(736.723.3)	Faulkner County
(736.631.4)	Pushmataha County	(736.723.5)	Van Buren County
(736.631.5)	Choctaw County	(736.723.7)	Conway County
(736.631.6)	McCurtain County	(736.725.2)	Pope County
(736.631.7)	Le Flore County	(736.725.4)	Johnson County
(736.631.8)	Sequoyah County	(736.725.6)	Franklin County
(736.633/.643)	Central and northern counties, between the Arkansas and Canadian rivers	(736.725.8)	Crawford County
(736.633.2)	Muskogee County	(736.732/.738)	South-western counties (south of the Arkansas river)
(736.633.4)	McIntosh County	(736.732.1)	Sebastian County
(736.633.6)	Okmulgee County	(736.732.3)	Logan County
(736.633.8)	Creek County	(736.732.5)	Yell County
(736.635.1)	Okfuskee County	(736.732.7)	Perry County
(736.635.3)	Hughes County	(736.733.2)	Saline County
(736.635.5)	Seminole County	(736.733.4)	Hot Spring County
(736.635.7)	Pottawatomie County	(736.733.6)	Garland County
(736.637.2)	Cleveland County	(736.733.8)	Montgomery County
(736.637.4)	Oklahoma County. □ Oklahoma City (state capital)	(736.735.1)	Scott County
(736.637.6)	Lincoln County	(736.735.3)	Polk County
(736.637.8)	Logan County	(736.735.5)	Sevier County
(736.638.1)	Payne County	(736.735.7)	Howard County
(736.638.3)	Pawnee County	(736.736.2)	Pike County
(736.638.5)	Noble County	(736.736.4)	Clark County
(736.638.7)	Garfield County	(736.736.6)	Nevada County
(736.642.1)	Kingfisher County	(736.736.8)	Hempstead County
(736.642.3)	Canadian County	(736.738.1)	Little River County
(736.642.5)	Blaine County	(736.738.3)	Miller County
(736.642.7)	Major County	(736.738.5)	Lafayette County
(736.643.2)	Woods County	(736.738.7)	Columbia County
(736.643.4)	Alfalfa County	(736.751/.755)	South-eastern counties (south of the Arkansas river)
(736.643.6)	Grant County	(736.751.2)	Union County
(736.643.8)	Kay County	(736.751.4)	Ouachita County
(736.652/.656)	Counties north-east of Arkansas river	(736.751.6)	Calhoun County
(736.652.1)	Osage County	(736.751.8)	Bradley County
(736.652.3)	Tulsa County. □ Tulsa (city)	(736.753.1)	Ashley County
(736.652.5)	Washington County	(736.753.3)	Chicot County
(736.652.7)	Nowata County	(736.753.5)	Desha County
(736.654.2)	Rogers County	(736.753.7)	Drew County
(736.654.4)	Wagoner County	(736.755.2)	Lincoln County
(736.654.6)	Cherokee County	(736.755.3)	Cleveland County
(736.654.8)	Adair County	(736.755.4)	Dallas County
(736.656.1)	Delaware County	(736.755.6)	Grant County
(736.656.3)	Mayes County	(736.755.8)	Jefferson County
(736.656.5)	Craig County	(736.772/.778)	North-eastern counties (east of the White and Black rivers)
(736.656.7)	Ottawa County	(736.772.1)	Arkansas County
(736.7)	Arkansas	(736.772.3)	Monroe County
(736.712/.725)	North-western, northern and central counties (mainly north of the Arkansas river)	(736.772.5)	Phillips County
(736.712.1)	Benton County	(736.772.7)	Lee County
(736.712.3)	Washington County	(736.774.2)	St. Francis County
(736.712.5)	Madison County	(736.774.4)	Woodruff County
(736.712.7)	Carroll County	(736.774.6)	Cross County
(736.714.2)	Boone County	(736.774.8)	Crittenden County
(736.714.4)	Newton County	(736.776.1)	Mississippi County
(736.714.6)	Searcy County	(736.776.3)	Poinsett County
(736.714.8)	Marion County	(736.776.5)	Jackson County
(736.716.1)	Baxter County	(736.778.2)	Craighead County
		(736.778.4)	Greene County

(736.778.6)	Clay County	(736.854.2)	Roane County
(736.8)	Tennessee	(736.854.3)	Morgan County
(736.812.1)	Lake County	(736.854.4)	Scott County
(736.812.3)	Obion County	(736.854.6)	Campbell County
(736.812.5)	Dyer County	(736.856.1)	Anderson County
(736.812.7)	Lauderdale County	(736.856.3)	Knox County. □ Knoxville
(736.814.2)	Tipton County	(736.856.5)	Loudon County
(736.814.4)	Shelby County. □ Memphis	(736.856.7)	McMinn County
(736.814.6)	Fayette County	(736.858.2)	Polk County
(736.814.8)	Haywood County	(736.858.4)	Monroe County
(736.816.1)	Crockett County	(736.858.6)	Blount County
(736.816.3)	Gibson County	(736.858.7)	Sevier County
(736.816.5)	Weakley County	(736.861.1)	Cocke County
(736.816.7)	Carroll County	(736.861.2)	Greene County
(736.818.2)	Madison County	(736.861.4)	Hamblen County
(736.818.4)	Hardeman County	(736.861.6)	Jefferson County
(736.818.6)	McNairy County	(736.861.8)	Grainger County
(736.818.8)	Chester County	(736.863.1)	Union County
(736.822.2)	Henderson County	(736.863.2)	Claiborne County
(736.822.3)	Hardin County	(736.863.3)	Hancock County
(736.822.4)	Decatur County	(736.863.5)	Hawkins County
(736.822.6)	Benton County	(736.865.2)	Sullivan County
(736.822.8)	Henry County	(736.865.4)	Washington County
(736.823.1)	Stewart County	(736.865.6)	Unicoi County
(736.823.3)	Houston County	(736.865.7)	Carter County
(736.823.5)	Humphreys County	(736.865.8)	Johnson County
(736.823.7)	Perry County	(736.9)	Kentucky
(736.825.2)	Wayne County	(736.912/.918)	Counties of the Eastern Kentucky Coalfield
(736.825.4)	Lawrence County	(736.912.1)	McCreary County
(736.825.6)	Lewis County	(736.912.3)	Whitley County
(736.825.8)	Hickman County	(736.912.5)	Bell County
(736.827.1)	Dickson County	(736.912.7)	Harlan County
(736.827.3)	Montgomery County	(736.913.2)	Leslie County
(736.827.5)	Robertson County	(736.913.4)	Clay County
(736.827.7)	Cheatham County	(736.913.6)	Knox County
(736.832.3)	Davidson County. □ Nashville (state capital)	(736.913.8)	Laurel County
(736.832.4)	Williamson County	(736.914.1)	Jackson County
(736.832.5)	Maury County	(736.914.3)	Owsley County
(736.832.6)	Giles County	(736.914.5)	Perry County
(736.834.2)	Lincoln County	(736.914.7)	Letcher County
(736.834.4)	Marshall County	(736.915.2)	Knott County
(736.834.6)	Moore County	(736.915.4)	Breathitt County
(736.834.8)	Bedford County	(736.915.6)	Lee County
(736.836.1)	Rutherford County	(736.915.8)	Estill County
(736.836.3)	Wilson County	(736.916.1)	Powell County
(736.836.5)	Sumner County	(736.916.3)	Wolfe County
(736.836.7)	Macon County	(736.916.5)	Magoffin County
(736.838.2)	Trousdale County	(736.916.7)	Floyd County
(736.838.4)	Smith County	(736.916.8)	Pike County
(736.838.6)	De Kalb County	(736.917.1)	Martin County
(736.838.7)	Cannon County	(736.917.2)	Johnson County
(736.841.2)	Coffee County	(736.917.4)	Morgan County
(736.841.3)	Franklin County	(736.917.6)	Menifee County
(736.841.4)	Marion County	(736.917.8)	Rowan County
(736.841.5)	Grundy County	(736.918.1)	Elliott County
(736.843.1)	Warren County	(736.918.3)	Lawrence County
(736.843.2)	Van Buren County	(736.918.5)	Boyd County
(736.843.4)	White County	(736.918.7)	Carter County
(736.843.5)	Putnam County	(736.918.8)	Greenup County
(736.845.2)	Jackson County	(736.922/.938)	Counties of the Blue Grass Region
(736.845.4)	Clay County	(736.922.1)	Lewis County
(736.845.6)	Pickett County	(736.922.3)	Mason County
(736.845.8)	Overton County	(736.922.5)	Bracken County
(736.847.1)	Fentress County	(736.922.7)	Pendleton County
(736.847.3)	Cumberland County	(736.924.1)	Campbell County
(736.847.5)	Bledsoe County	(736.924.3)	Kenton County
(736.847.7)	Sequatchie County	(736.924.5)	Boone County
(736.852.1)	Hamilton County. □ Chattanooga	(736.924.7)	Gallatin County
(736.852.3)	Bradley County	(736.926.1)	Carroll County
(736.852.5)	Meigs County	(736.926.3)	Trimble County
(736.852.6)	Rhea County	(736.926.5)	Oldham County

(1/9) Table 1e. Common auxiliaries of place

(736.926.7)	Henry County	(736.966.5)	Muhlenberg County
(736.928.1)	Owen County	(736.966.7)	Hopkins County
(736.928.3)	Grant County	(736.981/.983)	Counties of the Mississippi Embayment (Jackson Purchase)
(736.928.5)	Harrison County	(736.981.2)	Marshall County
(736.928.7)	Robertson County	(736.981.4)	Calloway County
(736.931.2)	Nicholas County	(736.981.6)	Graves County
(736.931.4)	Fleming County	(736.981.8)	McCracken County
(736.931.6)	Bath County	(736.983.2)	Ballard County
(736.931.7)	Montgomery County	(736.983.4)	Carlisle County
(736.931.8)	Bourbon County	(736.983.6)	Hickman County
(736.933.2)	Scott County	(736.983.8)	Fulton County
(736.933.4)	Franklin County. □ Frankfort (state capital)		
(736.933.6)	Shelby County	(737)	States of the north central USA <i>Class here the Middle West</i>
(736.933.7)	Jefferson County. □ Louisville		
(736.933.8)	Spencer County	(737.1/.6)	Mid-western states bordering the Great Lakes ⇒ (734.7/.8+737.1/.6)
(736.935.2)	Anderson County		
(736.935.4)	Woodford County	(737.1)	Ohio
(736.935.6)	Fayette County. □ Lexington	(737.121.2)	Williams County
(736.935.7)	Clark County	(737.121.4)	Defiance County
(736.935.8)	Madison County	(737.121.6)	Paulding County
(736.937.2)	Jessamine County	(737.121.8)	Putnam County
(736.937.4)	Garrard County	(737.123.2)	Henry County
(736.937.6)	Lincoln County	(737.123.4)	Fulton County
(736.937.8)	Boyle County	(737.123.6)	Lucas County. □ Toledo
(736.938.1)	Mercer County	(737.123.8)	Wood County
(736.938.3)	Washington County	(737.125.2)	Hancock County
(736.938.5)	Nelson County	(737.125.4)	Wyandot County
(736.938.7)	Bullitt County	(737.125.6)	Seneca County
(736.941/.948)	Counties of the Pennyrile Region (Mississippi Plateaus)	(737.125.8)	Sandusky County
(736.941.2)	Meade County	(737.127.2)	Ottawa County
(736.941.4)	Breckinridge County	(737.127.4)	Erie County
(736.941.6)	Hardin County	(737.127.6)	Huron County
(736.941.8)	Larue County	(737.127.8)	Crawford County
(736.943.2)	Hart County	(737.129.2)	Richland County
(736.943.4)	Green County	(737.129.4)	Ashland County
(736.943.6)	Taylor County	(737.129.6)	Lorain County
(736.943.8)	Marion County	(737.129.8)	Cuyahoga County. □ Cleveland
(736.944.1)	Casey County	(737.132.1)	Medina County
(736.944.3)	Pulaski County	(737.132.3)	Summit County. □ Akron
(736.944.5)	Rockcastle County	(737.132.5)	Portage County
(736.944.7)	Wayne County	(737.132.7)	Geauga County
(736.945.2)	Clinton County	(737.134.1)	Lake County
(736.945.4)	Russell County	(737.134.3)	Ashtabula County
(736.945.6)	Adair County	(737.134.5)	Trumbull County
(736.945.8)	Cumberland County	(737.134.7)	Mahoning County. □ Youngstown
(736.946.1)	Monroe County	(737.136.1)	Columbiana County
(736.946.3)	Metcalfe County	(737.136.3)	Jefferson County
(736.946.5)	Barren County	(737.136.5)	Harrison County
(736.946.7)	Allen County	(737.136.7)	Carroll County
(736.946.8)	Warren County	(737.138.1)	Stark County
(736.947.2)	Simpson County	(737.138.3)	Wayne County
(736.947.4)	Logan County	(737.138.5)	Holmes County
(736.947.5)	Todd County	(737.138.7)	Tuscarawas County
(736.947.6)	Christian County	(737.141.2)	Coshocton County
(736.947.8)	Trigg County	(737.141.4)	Knox County
(736.948.1)	Caldwell County	(737.141.6)	Morrow County
(736.948.3)	Lyon County	(737.141.8)	Delaware County
(736.948.5)	Livingston County	(737.143.2)	Marion County
(736.948.7)	Crittenden County	(737.143.4)	Union County
(736.962/.966)	Counties of the Western Kentucky Coalfield	(737.143.6)	Logan County
(736.962.1)	Union County	(737.143.8)	Hardin County
(736.962.3)	Webster County	(737.145.2)	Allen County
(736.962.5)	Henderson County	(737.145.4)	Auglaize County
(736.962.7)	McLean County	(737.145.6)	Van Wert County
(736.964.1)	Daviess County. □ Owensboro	(737.145.8)	Mercer County
(736.964.3)	Hancock County	(737.147.2)	Darke County
(736.964.5)	Ohio County	(737.147.4)	Shelby County
(736.964.7)	Grayson County	(737.147.6)	Miami County
(736.966.1)	Edmonson County	(737.147.8)	Clark County
(736.966.3)	Butler County		

(737.152.1)	Champaign County	(737.232.5)	Monroe County
(737.152.3)	Madison County	(737.232.7)	Brown County
(737.152.5)	Franklin County. □ Columbus (state capital)	(737.234.1)	Johnson County
(737.152.7)	Fairfield County	(737.234.3)	Bartholomew County
(737.154.1)	Licking County	(737.234.5)	Shelby County
(737.154.3)	Perry County	(737.234.7)	Rush County
(737.154.5)	Muskingum County	(737.236.1)	Fayette County
(737.154.7)	Morgan County	(737.236.3)	Union County
(737.156.1)	Noble County	(737.236.5)	Wayne County
(737.156.3)	Guernsey County	(737.236.7)	Henry County
(737.156.5)	Belmont County	(737.238.1)	Hancock County
(737.156.7)	Monroe County	(737.238.3)	Marion County. □ Indianapolis (state capital)
(737.158.1)	Washington County	(737.238.5)	Hendricks County
(737.158.3)	Athens County	(737.238.7)	Putnam County
(737.158.5)	Meigs County	(737.241.2)	Parke County
(737.158.7)	Vinton County	(737.241.4)	Vermillion County
(737.161.2)	Hocking County	(737.241.6)	Warren County
(737.161.4)	Pickaway County	(737.241.8)	Fountain County
(737.161.6)	Ross County	(737.243.2)	Tippecanoe County
(737.161.8)	Highland County	(737.243.4)	Montgomery County
(737.163.2)	Fayette County	(737.243.6)	Boone County
(737.163.4)	Greene County	(737.243.8)	Clinton County
(737.163.6)	Clinton County	(737.245.2)	Tipton County
(737.163.8)	Warren County	(737.245.4)	Hamilton County
(737.165.2)	Montgomery County. □ Dayton	(737.245.6)	Madison County
(737.165.4)	Preble County	(737.245.8)	Delaware County. □ Muncie
(737.165.6)	Butler County	(737.247.2)	Randolph County
(737.165.8)	Hamilton County. □ Cincinnati	(737.247.4)	Jay County
(737.167.2)	Clermont County	(737.247.6)	Blackford County
(737.167.4)	Brown County	(737.247.8)	Grant County
(737.167.6)	Adams County	(737.252.1)	Howard County
(737.167.8)	Pike County	(737.252.3)	Carroll County
(737.169.2)	Scioto County	(737.252.5)	White County
(737.169.4)	Jackson County	(737.252.7)	Benton County
(737.169.6)	Gallia County	(737.254.1)	Newton County
(737.169.8)	Lawrence County	(737.254.3)	Jasper County
(737.2)	Indiana	(737.254.5)	Pulaski County
(737.212.1)	Switzerland County	(737.254.7)	Fulton County
(737.212.3)	Ohio County	(737.256.1)	Cass County
(737.212.5)	Dearborn County	(737.256.3)	Miami County
(737.212.7)	Franklin County	(737.256.5)	Wabash County
(737.214.1)	Decatur County	(737.256.7)	Huntington County
(737.214.3)	Ripley County	(737.258.1)	Wells County
(737.214.5)	Jennings County	(737.258.3)	Adams County
(737.214.7)	Jefferson County	(737.258.5)	Allen County. □ Fort Wayne
(737.216.1)	Scott County	(737.258.7)	De Kalb County
(737.216.3)	Clark County	(737.261.2)	Steuben County
(737.216.5)	Floyd County	(737.261.4)	Lagrange County
(737.216.7)	Harrison County	(737.261.6)	Noble County
(737.218.1)	Washington County	(737.261.8)	Whitley County
(737.218.3)	Jackson County	(737.263.2)	Kosciusko County
(737.218.5)	Lawrence County	(737.263.4)	Elkhart County
(737.218.7)	Orange County	(737.263.6)	St. Joseph County. □ South Bend
(737.221.2)	Crawford County	(737.263.8)	Marshall County
(737.221.4)	Perry County	(737.265.2)	Starke County
(737.221.6)	Spencer County	(737.265.4)	La Porte County
(737.221.8)	Dubois County	(737.265.6)	Porter County
(737.223.2)	Martin County	(737.265.8)	Lake County. □ Gary
(737.223.4)	Daviess County	(737.3)	Illinois
(737.223.6)	Pike County	(737.311.2)	Cook County. □ Chicago
(737.223.8)	Warrick County	(737.311.4)	Lake County
(737.225.2)	Vanderburgh County	(737.311.6)	McHenry County
(737.225.4)	Posey County	(737.311.8)	Boone County
(737.225.6)	Gibson County	(737.321.2)	Winnebago County
(737.225.8)	Knox County	(737.321.4)	Stephenson County
(737.227.2)	Sullivan County	(737.321.6)	Jo Daviess County
(737.227.4)	Vigo County	(737.321.8)	Carroll County
(737.227.6)	Clay County	(737.323.2)	Whiteside County
(737.227.8)	Greene County	(737.323.4)	Lee County
(737.232.1)	Owen County	(737.323.6)	Ogle County
(737.232.3)	Morgan County	(737.323.8)	De Kalb County

(1/9) Table 1e. Common auxiliaries of place

(737.325.2)	Kane County	(737.365.8)	Washington County
(737.325.4)	Du Page County	(737.367.2)	St. Clair County
(737.325.6)	Will County	(737.367.4)	Monroe County
(737.325.8)	Kendall County	(737.367.6)	Randolph County
(737.327.2)	Grundy County	(737.367.8)	Perry County
(737.327.4)	La Salle County	(737.372.1)	Franklin County
(737.327.6)	Putnam County	(737.372.3)	Jefferson County
(737.327.8)	Bureau County	(737.372.5)	Hamilton County
(737.332.1)	Henry County	(737.372.7)	Wayne County
(737.332.3+737.763.2)	Tri-city area (Davenport-Moline-Rock Island)	(737.374.1)	Edwards County
(737.332.3)	Rock Island County. □ Rock Island (city). Moline	(737.374.3)	Wabash County
(737.332.5)	Mercer County	(737.374.5)	White County
(737.332.7)	Henderson County	(737.374.7)	Gallatin County
(737.334.1)	Warren County	(737.376.1)	Hardin County
(737.334.3)	Knox County	(737.376.3)	Saline County
(737.334.5)	Stark County	(737.376.5)	Williamson County
(737.334.7)	Peoria County. □ Peoria (city)	(737.376.7)	Jackson County
(737.336.1)	Marshall County	(737.378.1)	Union County
(737.336.3)	Woodford County	(737.378.3)	Johnson County
(737.336.5)	Livingston County	(737.378.5)	Pope County
(737.336.7)	Kankakee County	(737.378.7)	Massac County
(737.338.1)	Iroquois County	(737.379.2)	Pulaski County
(737.338.3)	Ford County	(737.379.4)	Alexander County
(737.338.5)	McLean County	(737.4)	Michigan
(737.338.7)	Tazewell County	(737.421/.461)	Counties of the Lower Peninsula
(737.341.2)	Fulton County	(737.421.2)	Berrien County
(737.341.4)	McDonough County	(737.421.4)	Cass County
(737.341.6)	Hancock County	(737.421.6)	St. Joseph County
(737.341.8)	Adams County	(737.421.8)	Branch County
(737.343.2)	Brown County	(737.423.2)	Hillsdale County
(737.343.4)	Schuyler County	(737.423.4)	Lenawee County
(737.343.6)	Cass County	(737.423.6)	Monroe County
(737.343.8)	Mason County	(737.423.8)	Wayne County. □ Dearborn. Detroit
(737.345.2)	Menard County	(737.425.2)	Washtenaw County. □ Ann Arbor
(737.345.4)	Logan County	(737.425.4)	Jackson County
(737.345.6)	De Witt County	(737.425.6)	Calhoun County
(737.345.8)	Macon County	(737.425.8)	Kalamazoo County
(737.347.2)	Piatt County	(737.427.2)	Van Buren County
(737.347.4)	Champaign County	(737.427.4)	Allegan County
(737.347.6)	Vermillion County	(737.427.6)	Barry County
(737.347.8)	Edgar County	(737.427.8)	Eaton County
(737.352.1)	Douglas County	(737.432.1)	Ingham County. □ Lansing (state capital). East Lansing
(737.352.3)	Coles County	(737.432.3)	Livingston County
(737.352.5)	Moultrie County	(737.432.5)	Oakland County
(737.352.7)	Shelby County	(737.432.7)	Macomb County
(737.354.1)	Christian County	(737.434.1)	St. Clair County
(737.354.3)	Sangamon County. □ Springfield (state capital)	(737.434.3)	Lapeer County
(737.354.5)	Morgan County	(737.434.5)	Genesee County. □ Flint
(737.354.7)	Scott County	(737.434.7)	Shiawassee County
(737.356.1)	Pike County	(737.436.1)	Clinton County
(737.356.3)	Calhoun County	(737.436.3)	Ionia County
(737.356.5)	Greene County	(737.436.5)	Kent County. □ Grand Rapids
(737.356.7)	Jersey County	(737.436.7)	Ottawa County
(737.358.1)	Madison County	(737.438.1)	Muskegon County
(737.358.3)	Macoupin County	(737.438.3)	Oceana County
(737.358.5)	Montgomery County	(737.438.5)	Newaygo County
(737.358.7)	Bond County	(737.438.7)	Mecosta County
(737.361.2)	Fayette County	(737.441.2)	Montcalm County
(737.361.4)	Effingham County	(737.441.4)	Isabella County
(737.361.6)	Jasper County	(737.441.6)	Gratiot County
(737.361.8)	Cumberland County	(737.441.8)	Midland County
(737.363.2)	Clark County	(737.443.2)	Saginaw County
(737.363.4)	Crawford County	(737.443.4)	Tuscola County
(737.363.6)	Lawrence County	(737.443.6)	Sanilac County
(737.363.8)	Richland County	(737.443.8)	Huron County
(737.365.2)	Clay County	(737.445.2)	Bay County. □ Bay City
(737.365.4)	Marion County	(737.445.4)	Arenac County
(737.365.6)	Clinton County	(737.445.6)	Gladwin County
		(737.445.8)	Clare County
		(737.447.2)	Osceola County

(737.447.4)	Lake County	(737.538.3)	Dunn County
(737.447.6)	Mason County	(737.538.5)	Saint Croix County
(737.447.8)	Manistee County	(737.538.7)	Pierce County
(737.452.1)	Wexford County	(737.541.2)	Pepin County
(737.452.3)	Missaukee County	(737.541.4)	Buffalo County
(737.452.5)	Roscommon County	(737.541.6)	Trempealeau County
(737.452.7)	Ogemaw County	(737.541.8)	Jackson County
(737.454.1)	Iosco County	(737.543.2)	Wood County
(737.454.3)	Alcona County	(737.543.4)	Portage County
(737.454.5)	Oscoda County	(737.543.6)	Waupaca County
(737.454.7)	Crawford County	(737.543.8)	Outagamie County
(737.456.1)	Kalkaska County	(737.545.2)	Calumet County
(737.456.3)	Grand Traverse County	(737.545.4)	Manitowoc County
(737.456.5)	Benzie County	(737.545.6)	Sheboygan County
(737.456.7)	Leelanau County	(737.545.8)	Fond du Lac County
(737.458.1)	Antrim County	(737.547.2)	Winnebago County
(737.458.3)	Charlevoix County. □ Beaver Island	(737.547.4)	Waushara County
(737.458.5)	Otsego County	(737.547.6)	Green Lake County
(737.458.7)	Montmorency County	(737.547.8)	Marquette County
(737.461.2)	Alpena County	(737.552.1)	Adams County
(737.461.4)	Presque Isle County	(737.552.3)	Juneau County
(737.461.6)	Cheboygan County	(737.552.5)	Monroe County
(737.461.8)	Emmet County	(737.552.7)	La Crosse County
(737.472/478)	Counties of the Upper Peninsula	(737.554.1)	Vernon County
(737.472.1)	Mackinac County. □ Bois Blanc Island	(737.554.3)	Crawford County
(737.472.3)	Chippewa County. □ Drummond Island	(737.554.5)	Richland County
(737.472.5)	Luce County	(737.554.7)	Sauk County
(737.472.7)	Alger County	(737.556.1)	Columbia County
(737.474.1)	Schoolcraft County	(737.556.3)	Dodge County
(737.474.3)	Delta County	(737.556.5)	Washington County
(737.474.5)	Menominee County	(737.556.7)	Ozaukee County
(737.474.7)	Dickinson County	(737.558.1)	Milwaukee County. □ Milwaukee (city)
(737.476.1)	Marquette County	(737.558.3)	Waukesha County
(737.476.3)	Baraga County	(737.558.5)	Jefferson County
(737.476.5)	Iron County	(737.558.7)	Dane County. □ Madison (state capital)
(737.476.7)	Gogebic County	(737.561.2)	Iowa County
(737.478.1)	Ontonagon County	(737.561.4)	Grant County
(737.478.3)	Houghton County	(737.561.6)	Lafayette County
(737.478.5)	Keweenaw County. □ Isle Royale	(737.561.8)	Green County
(737.5)	Wisconsin	(737.563.2)	Rock County
(737.521.2+737.669.4)	Area of Superior (Wisconsin) and Duluth (Minnesota)	(737.563.4)	Walworth County
(737.521.2)	Douglas County. □ Superior	(737.563.6)	Racine County
(737.521.4)	Bayfield County	(737.563.8)	Kenosha County
(737.521.6)	Ashland County	(737.6)	Minnesota
(737.521.8)	Iron County	(737.621.2)	Houston County
(737.523.2)	Vilas County	(737.621.4)	Fillmore County
(737.523.4)	Florence County	(737.621.6)	Mower County
(737.523.6)	Marinette County	(737.621.8)	Freeborn County
(737.523.8)	Forest County	(737.623.2)	Faribault County
(737.525.2)	Oneida County	(737.623.4)	Martin County
(737.525.4)	Price County	(737.623.6)	Jackson County
(737.525.6)	Sawyer County	(737.623.8)	Nobles County
(737.525.8)	Washburn County	(737.625.2)	Rock County
(737.527.2)	Burnett County	(737.625.4)	Pipestone County
(737.527.4)	Polk County	(737.625.6)	Murray County
(737.527.6)	Barron County	(737.625.8)	Cottonwood County
(737.527.8)	Rusk County	(737.627.2)	Watsonwan County
(737.532.1)	Taylor County	(737.627.4)	Blue Earth County
(737.532.3)	Lincoln County	(737.627.6)	Waseca County
(737.532.5)	Langlade County	(737.627.8)	Steele County
(737.532.7)	Menominee County	(737.632.1)	Dodge County
(737.534.1)	Oconto County	(737.632.3)	Olmsted County
(737.534.3)	Door County	(737.632.5)	Winona County
(737.534.5)	Kewaunee County	(737.632.7)	Wabasha County
(737.534.7)	Brown County	(737.634.1)	Goodhue County
(737.536.1)	Shawano County	(737.634.3)	Rice County
(737.536.3)	Marathon County	(737.634.5)	Le Sueur County
(737.536.5)	Clark County	(737.634.7)	Nicollet County
(737.536.7)	Eau Claire County	(737.636.1)	Brown County
(737.538.1)	Chippewa County	(737.636.3)	Redwood County
		(737.636.5)	Lyon County

(1/9) Table 1e. Common auxiliaries of place

(737.636.7)	Lincoln County	(737.723.4)	Cherokee County
(737.638.1)	Yellow Medicine County	(737.723.6)	Dickinson County
(737.638.3)	Renville County	(737.723.8)	Clay County
(737.638.5)	Sibley County	(737.725.2)	Buena Vista County
(737.638.7)	McLeod County	(737.725.4)	Emmet County
(737.641)	The seven counties of the Metropolitan Council	(737.725.6)	Palo Alto County
(737.641.1)	Carver County	(737.725.8)	Pocahontas County
(737.641.2)	Scott County	(737.727.2)	Kossuth County
(737.641.3)	Dakota County	(737.727.4)	Humboldt County
(737.641.4)	Washington County	(737.727.6)	Winnebago County
(737.641.5/.6)	Area of the Twin Cities (Minneapolis and St. Paul)	(737.727.8)	Hancock County
(737.641.5)	Ramsey County. □ St. Paul (state capital)	(737.732.1)	Wright County
(737.641.6)	Hennepin County. □ Minneapolis	(737.732.3)	Worth County
(737.641.7)	Anoka County	(737.732.5)	Cerro Gordo County
(737.643.2)	Wright County	(737.732.7)	Franklin County
(737.643.4)	Meeker County	(737.734.1)	Mitchell County
(737.643.6)	Kandiyohi County	(737.734.3)	Floyd County
(737.643.8)	Swift County	(737.734.5)	Butler County
(737.645.2)	Chippewa County	(737.734.7)	Howard County
(737.645.4)	Lac qui Parle County	(737.736.1)	Chickasaw County
(737.645.6)	Big Stone County	(737.736.3)	Bremer County
(737.645.8)	Stevens County	(737.736.5)	Winneshiek County
(737.647.2)	Pope County	(737.736.7)	Fayette County
(737.647.4)	Stearns County	(737.738.1)	Allamakee County
(737.647.6)	Benton County	(737.738.3)	Clayton County
(737.647.8)	Sherburne County	(737.741/.763)	Central strip
(737.652.1)	Isanti County	(737.741.2)	Woodbury County. □ Sioux City
(737.652.3)	Chisago County	(737.741.4)	Monona County
(737.652.5)	Pine County	(737.741.6)	Harrison County
(737.652.7)	Carlton County	(737.741.8)	Ida County
(737.654.1)	Aitkin County	(737.743.2)	Crawford County
(737.654.3)	Kanabec County	(737.743.4)	Shelby County
(737.654.5)	Mille Lacs County	(737.743.6)	Sac County
(737.654.7)	Crow Wing County	(737.743.8)	Carroll County
(737.656.1)	Morrison County	(737.745.2)	Audubon County
(737.656.3)	Todd County	(737.745.4)	Calhoun County
(737.656.5)	Wadena County	(737.745.6)	Greene County
(737.656.7)	Otter Tail County	(737.745.8)	Guthrie County
(737.658.1)	Douglas County	(737.747.2)	Webster County
(737.658.3)	Grant County	(737.747.4)	Boone County
(737.658.5)	Traverse County	(737.747.6)	Dallas County
(737.658.7)	Wilkin County	(737.747.8)	Hamilton County
(737.661.2)	Clay County	(737.752.1)	Hardin County
(737.661.4)	Becker County	(737.752.3)	Story County
(737.661.6)	Hubbard County	(737.752.5)	Polk County. □ Des Moines (state capital)
(737.661.8)	Cass County	(737.752.7)	Grundy County
(737.663.2)	Itasca County	(737.754.1)	Marshall County
(737.663.4)	Beltrami County	(737.754.3)	Jasper County
(737.663.6)	Clearwater County	(737.754.5)	Black Hawk County
(737.663.8)	Mahnomen County	(737.754.7)	Tama County
(737.665.2)	Norman County	(737.756.1)	Poweshiek County
(737.665.4)	Polk County	(737.756.3)	Buchanan County
(737.665.6)	Red Lake County	(737.756.5)	Benton County
(737.665.8)	Pennington County	(737.756.7)	Iowa County
(737.667.2)	Marshall County	(737.758.1)	Delaware County
(737.667.4)	Kittson County	(737.758.3)	Linn County
(737.667.6)	Roseau County	(737.758.5)	Johnson County
(737.667.8)	Lake of the Woods County	(737.758.7)	Dubuque County
(737.669.2)	Koochiching County	(737.761.2)	Jones County
(737.669.4)	Saint Louis County. □ Duluth ⇒ (737.521.2+737.669.4)	(737.761.4)	Cedar County
(737.669.6)	Lake County	(737.761.6)	Jackson County
(737.669.8)	Cook County	(737.761.8)	Clinton County
(737.7)	Iowa	(737.763.2)	Scott County. □ Davenport ⇒ (737.332.3+737.763.2)
(737.721/.738)	Northern strip	(737.763.4)	Muscatine County
(737.721.2)	Lyon County	(737.765/.783)	Southern strip
(737.721.4)	Sioux County	(737.765.2)	Pottawattamie County
(737.721.6)	Plymouth County	(737.765.4)	Mills County
(737.721.8)	Osceola County	(737.765.6)	Fremont County
(737.723.2)	O'Brien County	(737.765.8)	Cass County
		(737.767.2)	Montgomery County

(737.767.4)	Page County	(737.835.6)	Warren County
(737.767.6)	Adair County	(737.835.8)	St. Charles County
(737.767.8)	Adams County	(737.842/.857)	Central counties, south and west of Missouri River
(737.772.1)	Taylor County	(737.842.1+738.143.4)	Metropolitan area of Kansas City, Missouri/Kansas
(737.772.3)	Madison County	(737.842.1)	Jackson County. □ Kansas City.
(737.772.5)	Union County		Independence
(737.772.7)	Ringgold County	(737.842.3)	Cass County
(737.774.1)	Warren County	(737.842.5)	Bates County
(737.774.3)	Clarke County	(737.842.7)	Vernon County
(737.774.5)	Decatur County	(737.844.1)	Lafayette County
(737.774.7)	Marion County	(737.844.3)	Johnson County
(737.776.1)	Lucas County	(737.844.5)	Henry County
(737.776.3)	Wayne County	(737.844.7)	Saint Clair County
(737.776.5)	Mahaska County	(737.846.1)	Saline County
(737.776.7)	Monroe County	(737.846.3)	Pettis County
(737.778.1)	Appanoose County	(737.846.5)	Benton County
(737.778.3)	Keokuk County	(737.846.7)	Hickory County
(737.778.5)	Wapello County	(737.848.1)	Cooper County
(737.778.7)	Davis County	(737.848.3)	Moniteau County
(737.781.2)	Washington County	(737.848.5)	Morgan County
(737.781.4)	Jefferson County	(737.848.7)	Camden County
(737.781.6)	Van Buren County	(737.851.2)	Cole County. □ Jefferson City (state capital)
(737.781.8)	Louisa County	(737.851.4)	Miller County
(737.783.2)	Henry County	(737.851.6)	Pulaski County
(737.783.4)	Des Moines County	(737.851.8)	Osage County
(737.783.6)	Lee County	(737.853.2)	Maries County
(737.8)	Missouri	(737.853.4)	Phelps County
(737.811/.835)	Northern counties, north and east of Missouri River	(737.853.6)	Gasconade County
(737.811.2)	Atchison County	(737.853.8)	Crawford County
(737.811.4)	Holt County	(737.855.2)	Franklin County
(737.811.6)	Nodaway County	(737.855.4)	Washington County
(737.811.8)	Andrew County	(737.855.6)	St. Louis County
(737.813.2)	Buchanan County	(737.855.8)	Independent City of St. Louis
(737.813.4)	Platte County	(737.857.2)	Jefferson County
(737.813.6)	Worth County	(737.857.4)	St. Francois County
(737.813.8)	Gentry County	(737.857.6)	Sainte Genevieve County
(737.815.2)	De Kalb County	(737.857.8)	Perry County
(737.815.4)	Clinton County	(737.862/.884)	Southern counties
(737.815.6)	Clay County	(737.862.1)	Barton County
(737.815.8)	Harrison County	(737.862.3)	Jasper County
(737.817.2)	Daviess County	(737.862.5)	Newton County
(737.817.4)	Caldwell County	(737.862.7)	McDonald County
(737.817.6)	Ray County	(737.864.1)	Cedar County
(737.817.8)	Mercer County	(737.864.3)	Dade County
(737.822.1)	Grundy County	(737.864.5)	Lawrence County
(737.822.3)	Livingston County	(737.864.7)	Barry County
(737.822.5)	Carroll County	(737.866.1)	Polk County
(737.822.7)	Putnam County	(737.866.3)	Greene County
(737.824.1)	Sullivan County	(737.866.5)	Christian County
(737.824.3)	Linn County	(737.866.7)	Stone County
(737.824.5)	Chariton County	(737.868.1)	Taney County
(737.824.7)	Schuyler County	(737.868.3)	Dallas County
(737.826.1)	Adair County	(737.868.5)	Laclede County
(737.826.3)	Macon County	(737.868.7)	Webster County
(737.826.5)	Randolph County	(737.871.2)	Wright County
(737.826.7)	Howard County	(737.871.4)	Douglas County
(737.828.1)	Boone County	(737.871.6)	Ozark County
(737.828.3)	Scotland County	(737.871.8)	Texas County
(737.828.5)	Knox County	(737.873.2)	Howell County
(737.828.7)	Shelby County	(737.873.4)	Dent County
(737.831.2)	Monroe County	(737.873.6)	Shannon County
(737.831.4)	Audrain County	(737.873.8)	Oregon County
(737.831.6)	Callaway County	(737.875.2)	Iron County
(737.831.8)	Clark County	(737.875.4)	Reynolds County
(737.833.2)	Lewis County	(737.875.6)	Carter County
(737.833.4)	Marion County	(737.875.8)	Ripley County
(737.833.6)	Ralls County	(737.877.2)	Madison County
(737.833.8)	Pike County	(737.877.4)	Wayne County
(737.835.2)	Lincoln County	(737.877.6)	Butler County
(737.835.4)	Montgomery County		

(1/9) Table 1e. Common auxiliaries of place

(737.877.8)	Bollinger County	(738.156.7)	Chase County
(737.882.1)	Stoddard County	(738.158.1)	Wabaunsee County
(737.882.3)	Cape Girardeau County	(738.158.3)	Lyon County
(737.882.5)	Scott County	(738.158.5)	Shawnee County. □ Topeka (state capital)
(737.882.7)	Mississippi County	(738.158.7)	Osage County
(737.884.1)	New Madrid County	(738.161.2)	Coffey County
(737.884.3)	Dunklin County	(738.161.4)	Douglas County
(737.884.5)	Pemiscot County	(738.161.6)	Franklin County
(738)	States of the western USA	(738.161.8)	Anderson County
	<i>Class here the West</i>	(738.163.2)	Johnson County
(738.1)	Kansas	(738.163.4)	Miami County
(738.121/.143)	Northern counties	(738.163.6)	Linn County
(738.121.2)	Cheyenne County	(738.172/.192)	Southern counties
(738.121.4)	Sherman County	(738.172.1)	Stanton County
(738.121.6)	Wallace County	(738.172.3)	Morton County
(738.121.8)	Rawlins County	(738.172.5)	Grant County
(738.123.2)	Thomas County	(738.172.7)	Stevens County
(738.123.4)	Logan County	(738.174.1)	Haskell County
(738.123.6)	Decatur County	(738.174.3)	Seward County
(738.123.8)	Sheridan County	(738.174.5)	Gray County
(738.125.2)	Gove County	(738.174.7)	Meade County
(738.125.4)	Norton County	(738.176.1)	Ford County
(738.125.6)	Graham County	(738.176.3)	Clark County
(738.125.8)	Trego County	(738.176.5)	Edwards County
(738.127.2)	Phillips County	(738.176.7)	Kiowa County
(738.127.4)	Rooks County	(738.178.1)	Comanche County
(738.127.6)	Ellis County	(738.178.3)	Stafford County
(738.127.8)	Smith County	(738.178.5)	Pratt County
(738.132.1)	Osborne County	(738.178.7)	Barber County
(738.132.3)	Russell County	(738.181.2)	Reno County
(738.132.5)	Jewell County	(738.181.4)	Kingman County
(738.132.7)	Mitchell County	(738.181.6)	Harper County
(738.134.1)	Lincoln County	(738.181.8)	Harvey County
(738.134.3)	Republic County	(738.183.2)	Sedgwick County. □ Wichita
(738.134.5)	Cloud County	(738.183.4)	Sumner County
(738.134.7)	Ottawa County	(738.183.6)	Butler County
(738.136.1)	Washington County	(738.183.8)	Cowley County
(738.136.3)	Clay County	(738.185.2)	Greenwood County
(738.136.5)	Riley County	(738.185.4)	Elk County
(738.136.7)	Geary County	(738.185.6)	Chautauqua County
(738.138.1)	Marshall County	(738.185.8)	Woodson County
(738.138.3)	Pottawatomie County	(738.187.2)	Wilson County
(738.138.5)	Nemaha County	(738.187.4)	Montgomery County
(738.138.7)	Jackson County	(738.187.6)	Allen County
(738.141.2)	Brown County	(738.187.8)	Neosho County
(738.141.4)	Doniphan County	(738.192.1)	Labette County
(738.141.6)	Atchison County	(738.192.3)	Bourbon County
(738.141.8)	Jefferson County	(738.192.5)	Crawford County
(738.143.2)	Leavenworth County	(738.192.7)	Cherokee County
(738.143.4)	Wyandotte County. □ Kansas City ⇒ (737.842.1+738.143.4)	(738.2)	Nebraska
(738.145/.163)	Central counties	(738.222/.231)	Eastern counties, north of Platte River
(738.145.2)	Greeley County	(738.222.1)	Dakota County
(738.145.4)	Hamilton County	(738.222.3)	Dixon County
(738.145.6)	Wichita County	(738.222.5)	Cedar County
(738.145.8)	Kearny County	(738.222.7)	Knox County
(738.147.2)	Scott County	(738.224.1)	Antelope County
(738.147.4)	Finney County	(738.224.3)	Pierce County
(738.147.6)	Lane County	(738.224.5)	Wayne County
(738.147.8)	Ness County	(738.224.7)	Thurston County
(738.152.1)	Hodgeman County	(738.226.1)	Burt County
(738.152.3)	Rush County	(738.226.3)	Cuming County
(738.152.5)	Pawnee County	(738.226.5)	Stanton County
(738.152.7)	Barton County	(738.226.7)	Madison County
(738.154.1)	Ellsworth County	(738.228.1)	Boone County
(738.154.3)	Rice County	(738.228.3)	Nance County
(738.154.5)	Saline County	(738.228.5)	Merrick County
(738.154.7)	McPherson County	(738.228.7)	Platte County
(738.156.1)	Dickinson County	(738.231.2)	Colfax County
(738.156.3)	Marion County	(738.231.4)	Dodge County
(738.156.5)	Morris County	(738.231.6)	Washington County
		(738.231.7)	Douglas County. □ Omaha

(738.231.8)	Sarpy County	(738.273.6)	Cheyenne County
(738.233/.248)	Eastern counties, south of Platte River	(738.273.8)	Kimball County
(738.233.2)	Cass County	(738.275.2)	Banner County
(738.233.4)	Otoe County	(738.275.4)	Scotts Bluff County
(738.233.6)	Johnson County	(738.275.6)	Sioux County
(738.233.8)	Nemaha County	(738.3)	South Dakota
(738.235.2)	Richardson County	(738.322/.346)	Counties east and north of Missouri River
(738.235.4)	Pawnee County	(738.322.1)	Roberts County
(738.235.6)	Gage County	(738.322.3)	Marshall County
(738.235.8)	Lancaster County. □ Lincoln (state capital)	(738.322.5)	Day County
(738.237.2)	Saunders County	(738.322.7)	Brown County
(738.237.4)	Butler County	(738.324.1)	Edmunds County
(738.237.6)	Seward County	(738.324.3)	McPherson County
(738.237.8)	Saline County	(738.324.5)	Campbell County
(738.242.1)	Jefferson County	(738.324.7)	Walworth County
(738.242.3)	Thayer County	(738.326.1)	Potter County
(738.242.5)	Fillmore County	(738.326.3)	Faulk County
(738.242.7)	York County	(738.326.5)	Spink County
(738.244.1)	Polk County	(738.326.7)	Clark County
(738.244.3)	Hamilton County	(738.328.1)	Codington County
(738.244.5)	Clay County	(738.328.3)	Grant County
(738.244.7)	Nuckolls County	(738.328.5)	Deuel County
(738.246.1)	Adams County	(738.328.7)	Hamlin County
(738.246.3)	Webster County	(738.331.2)	Brookings County
(738.246.5)	Franklin County	(738.331.4)	Kingsbury County
(738.246.7)	Kearney County	(738.331.6)	Beadle County
(738.248.1)	Phelps County	(738.331.8)	Hand County
(738.248.3)	Harlan County	(738.333.2)	Hyde County
(738.248.5)	Furnas County	(738.333.4)	Sully County
(738.248.7)	Gosper County	(738.333.6)	Hughes County. □ Pierre (state capital)
(738.251/.268)	Central counties	(738.333.8)	Buffalo County
(738.251.2)	Dawson County	(738.335.2)	Jerauld County
(738.251.4)	Buffalo County	(738.335.4)	Sanborn County
(738.251.6)	Hall County	(738.335.6)	Miner County
(738.251.8)	Howard County	(738.335.8)	Lake County
(738.253.2)	Sherman County	(738.337.2)	Moody County
(738.253.4)	Custer County	(738.337.4)	Minnehaha County. □ Sioux Falls
(738.253.6)	Valley County	(738.337.6)	McCook County
(738.253.8)	Greeley County	(738.337.8)	Hanson County
(738.255.2)	Wheeler County	(738.342.1)	Davison County
(738.255.4)	Garfield County	(738.342.3)	Aurora County
(738.255.6)	Loup County	(738.342.5)	Brule County
(738.255.8)	Blaine County	(738.342.7)	Douglas County
(738.257.2)	Brown County	(738.344.1)	Hutchinson County
(738.257.4)	Rock County	(738.344.3)	Turner County
(738.257.6)	Holt County	(738.344.5)	Lincoln County
(738.257.8)	Boyd County	(738.344.7)	Union County
(738.262.1)	Keya Paha County	(738.346.1)	Clay County
(738.262.3)	Cherry County	(738.346.3)	Yankton County
(738.262.5)	Thomas County	(738.346.5)	Bon Homme County
(738.262.7)	Hooker County	(738.346.7)	Charles Mix County
(738.264.1)	Grant County	(738.351/.355)	Counties between the Missouri and Cheyenne Rivers
(738.264.3)	Arthur County	(738.351.2)	Gregory County
(738.264.5)	McPherson County	(738.351.4)	Tripp County
(738.264.7)	Logan County	(738.351.6)	Todd County
(738.266.1)	Lincoln County	(738.351.8)	Mellette County
(738.266.3)	Hayes County	(738.353.2)	Jones County
(738.266.5)	Frontier County	(738.353.4)	Lyman County
(738.266.7)	Red Willow County	(738.353.6)	Stanley County
(738.268.1)	Hitchcock County	(738.353.8)	Haakon County
(738.268.3)	Dundy County	(738.355.2)	Jackson County
(738.268.5)	Chase County	(738.355.4)	Bennett County
(738.268.7)	Perkins County	(738.355.6)	Shannon County
(738.268.8)	Keith County	(738.355.8)	Fall River County
(738.271/.275)	Western (Panhandle) counties	(738.362/.366)	Counties west and north of Cheyenne River
(738.271.2)	Deuel County	(738.362.1)	Custer County
(738.271.4)	Garden County	(738.362.3)	Pennington County
(738.271.6)	Sheridan County	(738.362.5)	Lawrence County
(738.271.8)	Dawes County	(738.362.7)	Butte County
(738.273.2)	Box Butte County	(738.364.1)	Meade County
(738.273.4)	Morrill County		

(1/9) Table 1e. Common auxiliaries of place

(738.364.3)	Ziebach County	(738.624.1)	Phillips County
(738.364.5)	Dewey County	(738.624.3)	Blaine County
(738.364.7)	Corson County	(738.624.5)	Hill County
(738.366.1)	Perkins County	(738.624.7)	Liberty County
(738.366.3)	Harding County	(738.625.2)	Chouteau County
(738.4)	North Dakota	(738.626/.628)	Counties south of Missouri River
(738.422/.428)	Eastern counties, north and east of Missouri River	(738.626.1)	Judith Basin County
(738.422.1)	Richland County	(738.626.3)	Fergus County
(738.422.3)	Sargent County	(738.626.5)	Petroleum County
(738.422.5)	Ransom County	(738.626.7)	Garfield County
(738.422.7)	Cass County. □ Fargo	(738.628.1)	McCone County
(738.424.1)	Barnes County	(738.628.3)	Prairie County
(738.424.3)	Griggs County	(738.628.5)	Dawson County
(738.424.5)	Steele County	(738.628.6)	Richland County
(738.424.7)	Traill County	(738.628.7)	Wibaux County
(738.426.1)	Grand Forks County	(738.631/.637)	South-eastern counties
(738.426.3)	Nelson County	(738.631.2)	Fallon County
(738.426.5)	Ramsey County	(738.631.4)	Carter County
(738.426.7)	Walsh County	(738.631.6)	Powder River County
(738.428.1)	Pembina County	(738.631.8)	Custer County
(738.428.3)	Cavalier County	(738.633.2)	Rosebud County
(738.431/.444)	Central and western counties, north and east of Missouri River	(738.633.4)	Treasure County
(738.431.2)	Towner County	(738.633.6)	Big Horn County
(738.431.4)	Rolette County	(738.633.8)	Yellowstone County
(738.431.6)	Pierce County	(738.635.2)	Musselshell County
(738.431.8)	Benson County	(738.635.4)	Golden Valley County
(738.433.2)	Eddy County	(738.635.6)	Wheatland County
(738.433.4)	Foster County	(738.635.8)	Sweet Grass County
(738.433.6)	Wells County	(738.637.2)	Stillwater County
(738.433.8)	Kidder County	(738.637.4)	Carbon County
(738.435.2)	Stutsman County	(738.642/.653)	Western counties
(738.435.4)	La Moure County	(738.642.1)	Park County
(738.435.6)	Dickey County	(738.642.3)	Gallatin County
(738.435.8)	McIntosh County	(738.642.5)	Madison County
(738.437.2)	Logan County	(738.642.7)	Beaverhead County
(738.437.4)	Emmons County	(738.644.1)	Deer Lodge County
(738.437.6)	Burleigh County. □ Bismarck (state capital)	(738.644.3)	Silver Bow County. □ Butte
(738.437.8)	Sheridan County	(738.644.5)	Jefferson County
(738.442.1)	McLean County	(738.644.7)	Broadwater County
(738.442.3)	Ward County	(738.646.1)	Meagher County
(738.442.5)	McHenry County	(738.646.3)	Cascade County
(738.442.7)	Bottineau County	(738.646.5)	Lewis and Clark County. □ Helena (state capital)
(738.444.1)	Renville County	(738.646.7)	Powell County
(738.444.3)	Burke County	(738.648.1)	Granite County
(738.444.5)	Mountrail County	(738.648.3)	Ravalli County
(738.444.7)	Divide County	(738.648.5)	Missoula County
(738.444.8)	Williams County	(738.648.7)	Mineral County
(738.461/.467)	Counties south and west of Missouri River	(738.651.2)	Sanders County
(738.461.2)	McKenzie County	(738.651.4)	Lake County
(738.461.4)	Dunn County	(738.651.6)	Teton County
(738.461.6)	Mercer County	(738.651.8)	Pondera County
(738.461.8)	Oliver County	(738.653.2)	Toole County
(738.463.2)	Morton County	(738.653.4)	Glacier County
(738.463.4)	Grant County	(738.653.6)	Flathead County
(738.463.6)	Sioux County	(738.653.8)	Lincoln County
(738.463.8)	Adams County	(738.7)	Wyoming
(738.465.2)	Bowman County	(738.721.2)	Crook County
(738.465.4)	Slope County	(738.721.4)	Campbell County
(738.465.6)	Hettinger County	(738.721.6)	Weston County
(738.465.8)	Stark County	(738.721.8)	Niobrara County
(738.467.2)	Billings County	(738.723.2)	Converse County
(738.467.4)	Golden Valley County	(738.723.4)	Platte County
(738.6/.9+739.6)	Rocky Mountains states	(738.723.6)	Goshen County
(738.6)	Montana	(738.723.8)	Laramie County. □ Cheyenne (state capital)
(738.622/.624)	Counties north of Missouri River	(738.725.2)	Albany County
(738.622.1)	Sheridan County	(738.725.4)	Carbon County
(738.622.3)	Roosevelt County	(738.725.6)	Natrona County
(738.622.5)	Daniels County	(738.725.8)	Johnson County
(738.622.7)	Valley County	(738.727.2)	Sheridan County
		(738.727.4)	Big Horn County

(738.727.6)	Washakie County	(738.862.3)	Crowley County
(738.727.8)	Hot Springs County	(738.862.5)	Otero County
(738.732.1)	Park County	(738.862.7)	Las Animas County
(738.732.3)	Teton County	(738.864.1)	Bent County
(738.732.5)	Fremont County	(738.864.3)	Prowers County
(738.732.7)	Sublette County	(738.864.5)	Baca County
(738.734.1)	Lincoln County		
(738.734.3)	Sweetwater County	(738.9)	New Mexico
(738.734.5)	Uinta County	(738.922/.924)	Northeast New Mexico
		(738.922.1)	Colfax County
(738.8)	Colorado	(738.922.3)	Union County
(738.822/.826)	Colorado Plateau counties	(738.922.5)	Harding County
(738.822.1)	Moffat County	(738.922.7)	Mora County
(738.822.3)	Routt County	(738.924.1)	San Miguel County
(738.822.5)	Rio Blanco County	(738.924.3)	Guadalupe County
(738.822.7)	Garfield County	(738.924.5)	Quay County
(738.824.1)	Mesa County	(738.931/.933)	Eastern, Pecos Valley and central counties
(738.824.3)	Delta County	(738.931.2)	Curry County
(738.824.5)	Montrose County	(738.931.4)	Roosevelt County
(738.824.7)	Ouray County	(738.931.6)	De Baca County
(738.826.1)	San Miguel County	(738.931.8)	Chaves County
(738.826.3)	San Juan County	(738.933.2)	Lea County
(738.826.5)	Dolores County	(738.933.4)	Eddy County
(738.826.7)	Montezuma County	(738.933.6)	Otero County
(738.826.8)	La Plata County	(738.933.7)	Lincoln County
(738.828/.844)	Rocky Mountains counties	(738.933.8)	Torrance County
(738.828.1)	Archuleta County	(738.942)	Rocky Mountains counties
(738.828.3)	Conejos County	(738.942.1)	Santa Fe County. □ Santa Fe (state capital)
(738.828.5)	Costilla County	(738.942.3)	Taos County
(738.828.7)	Huerfano County	(738.942.5)	Rio Arriba County
(738.831.2)	Alamosa County	(738.942.7)	Los Alamos County
(738.831.4)	Rio Grande County	(738.942.8)	Sandoval County
(738.831.6)	Mineral County	(738.951.2/.8)	Rio Grande counties
(738.831.8)	Hinsdale County	(738.951.2)	Bernalillo County. □ Albuquerque
(738.833.2)	Gunnison County	(738.951.4)	Valencia County
(738.833.4)	Chaffee County	(738.951.6)	Socorro County
(738.833.6)	Saguache County	(738.951.7)	Sierra County
(738.833.8)	Fremont County	(738.951.8)	Doña Ana County
(738.835.2)	Custer County	(738.962/.964)	Western counties
(738.835.4)	Pueblo County	(738.962.1)	Luna County
(738.835.6)	El Paso County. □ Colorado Springs	(738.962.3)	Hidalgo County
(738.835.8)	Teller County	(738.962.5)	Grant County
(738.837.2)	Park County	(738.962.7)	Catron County
(738.837.4)	Lake County	(738.964.1)	Cibola County
(738.837.6)	Pitkin County	(738.964.3)	McKinley County
(738.837.8)	Eagle County	(738.964.5)	San Juan County
(738.842.1)	Summit County		
(738.842.3)	Clear Creek County	(739)	States of the Great Basin and Pacific Slope region of the USA, and the Pacific Ocean
(738.842.5)	Gilpin County		<i>Class here the New South-west</i>
(738.842.7)	Boulder County		
(738.844.1)	Grand County		
(738.844.3)	Jackson County	(739.1)	Arizona
(738.844.5)	Larimer County	(739.122/.124)	Counties of the Colorado Plateau and mountain region
(738.851/.864)	Great Plains counties		
(738.851.2)	Weld County	(739.122.1)	Coconino County
(738.851.4)	Logan County	(739.122.3)	Navajo County
(738.851.6)	Sedgwick County	(739.122.5)	Apache County
(738.851.8)	Phillips County	(739.122.7)	Greenlee County
(738.853.2)	Yuma County	(739.124.1)	Cochise County
(738.853.4)	Washington County	(739.124.3)	Graham County
(738.853.6)	Morgan County	(739.124.5)	Gila County
(738.853.8)	Adams County	(739.124.7)	Yavapai County
(738.855.2)	Arapahoe County	(739.124.8)	Mohave County
(738.855.4)	City and County of Denver (state capital)	(739.126/.128)	Counties of the Plains region
(738.855.5)	City and County of Broomfield	(739.126.1)	La Paz County
(738.855.6)	Jefferson County	(739.126.3)	Yuma County
(738.855.8)	Douglas County	(739.126.5)	Maricopa County. □ Phoenix (state capital)
(738.857.2)	Elbert County	(739.126.7)	Pinal County
(738.857.4)	Lincoln County	(739.128.1)	Pima County. □ Tucson
(738.857.6)	Kit Carson County	(739.128.3)	Santa Cruz County
(738.857.8)	Cheyenne County		
(738.862.1)	Kiowa County	(739.2)	Utah

(1/9) Table 1e. Common auxiliaries of place

(739.221/.225)	Counties of the north-eastern mountainous region	(739.427.2)	Yuba County
(739.221.2)	Uintah County	(739.427.4)	Butte County
(739.221.4)	Daggett County	(739.427.6)	Glenn County
(739.221.6)	Duchesne County	(739.427.8)	Lake County
(739.221.8)	Summit County	(739.429.2)	Colusa County
(739.223.2)	Wasatch County	(739.429.4)	Sutter County
(739.223.4)	Utah County	(739.429.6)	Placer County
(739.223.6)	Salt Lake County. □ Salt Lake City (state capital)	(739.432/.436)	Central counties
(739.223.8)	Davis County	(739.432.1)	El Dorado County
(739.225.2)	Morgan County	(739.432.3)	Amador County
(739.225.4)	Weber County	(739.432.5)	Alpine County
(739.225.6)	Rich County	(739.432.7)	Mono County
(739.225.8)	Cache County	(739.434.1)	Mariposa County
(739.232/.234)	Counties of the Great Basin region	(739.434.3)	Madera County
(739.232.1)	Box Elder County	(739.434.5)	Merced County
(739.232.3)	Tooele County	(739.434.7)	Stanislaus County
(739.232.5)	Juab County	(739.436.1)	Tuolumne County
(739.232.7)	Millard County	(739.436.3)	Calaveras County
(739.234.1)	Beaver County	(739.436.5)	San Joaquin County
(739.234.3)	Iron County	(739.436.7)	Sacramento County. □ Sacramento (state capital)
(739.234.5)	Washington County	(739.436.8)	Yolo County
(739.241/.245)	Counties of the Colorado Plateau and River	(739.441/.443)	Counties in the San Francisco Bay Area
(739.241.2)	Kane County	(739.441.2)	Solano County
(739.241.4)	Garfield County	(739.441.4)	Napa County
(739.241.6)	Wayne County	(739.441.6)	Sonoma County
(739.241.8)	Piute County	(739.441.8)	Marin County
(739.243.2)	Sevier County	(739.443.2)	San Francisco County (San Francisco)
(739.243.4)	Sanpete County	(739.443.4)	Contra Costa County
(739.243.6)	Carbon County	(739.443.6)	Alameda County. □ Oakland; Berkeley
(739.243.8)	Emery County	(739.443.7)	San Mateo County
(739.245.2)	Grand County	(739.443.8)	Santa Clara County
(739.245.4)	San Juan County	(739.451+739.455.4)	Counties of the Southern Coast Range
(739.3)	Nevada	(739.451.2)	Santa Cruz County. □ San Jose
(739.321.2/.8)	Eastern counties	(739.451.4)	San Benito County
(739.321.2)	Clark County. □ Las Vegas	(739.451.6)	Monterey County
(739.321.4)	Lincoln County	(739.453/.455)	South central counties
(739.321.6)	White Pine County	(739.453.2)	Fresno County. □ Fresno (city)
(739.321.8)	Elko County	(739.453.4)	Inyo County
(739.323.2/.8)	Central counties	(739.453.6)	Kings County
(739.323.2)	Eureka County	(739.453.8)	Tulare County
(739.323.4)	Lander County	(739.455.2)	Kern County. □ Bakersfield
(739.323.6)	Nye County	(739.455.4)	San Luis Obispo County
(739.323.8)	Esmeralda County	(739.462/.464)	Counties of Southern California
(739.325/.327)	Western counties	(739.462.1)	Santa Barbara County
(739.325.2)	Mineral County	(739.462.3)	Ventura County
(739.325.4)	Churchill County	(739.462.5)	Los Angeles County. □ Los Angeles (city); Pasadena
(739.325.6)	Pershing County	(739.462.7)	San Bernardino County
(739.325.8)	Humboldt County	(739.464.1)	Orange County
(739.327.2)	Washoe County. □ Reno	(739.464.3)	Riverside County
(739.327.4)	Storey County. □ Virginia City	(739.464.5)	San Diego County. □ San Diego (city)
(739.327.5)	Carson City (county and city; state capital)	(739.464.7)	Imperial County
(739.327.6)	Lyon County	(739.5/.8)	States of the Pacific North-west
(739.327.8)	Douglas County	(739.5)	Oregon
(739.4/.5+739.7/.8)	States of the Pacific coast	(739.522/.528)	Western counties. □ Counties west of the Willamette River
(739.4)	California	(739.522.1)	Coos County
(739.421/.429)	Northern counties	(739.522.3)	Curry County
(739.421.2)	Del Norte County	(739.522.5)	Josephine County
(739.421.4)	Siskiyou County	(739.522.7)	Jackson County
(739.421.6)	Modoc County	(739.524.1)	Douglas County
(739.421.8)	Lassen County	(739.524.3)	Lane County
(739.423.2)	Shasta County	(739.524.5)	Lincoln County
(739.423.4)	Trinity County	(739.524.7)	Benton County
(739.423.6)	Humboldt County	(739.526.1)	Polk County
(739.423.8)	Mendocino County	(739.526.3)	Yamhill County
(739.425.2)	Tehama County	(739.526.5)	Washington County
(739.425.4)	Plumas County	(739.526.7)	Tillamook County
(739.425.6)	Sierra County	(739.528.1)	Clatsop County
(739.425.8)	Nevada County	(739.528.3)	Columbia County

(739.531/.535)	Northern counties	(739.7)	Washington
(739.531.2)	Multnomah County. □ Portland	(739.722.1/.5)	Tri-county area of north-east Washington
(739.531.4)	Hood River County	(739.722.1)	Pend Oreille County
(739.531.6)	Clackamas County	(739.722.3)	Stevens County
(739.531.8)	Marion County. □ Salem (state capital)	(739.722.5)	Ferry County
(739.533.2)	Linn County	(739.731/.733)	Other northern counties and eastern counties between the Spokane and Snake Rivers
(739.533.4)	Jefferson County	(739.731.2)	Okanogan County
(739.533.6)	Wasco County	(739.731.4)	Douglas County
(739.533.8)	Sherman County	(739.731.6)	Grant County
(739.535.2)	Gilliam County	(739.731.8)	Franklin County
(739.535.4)	Morrow County	(739.733.2)	Adams County
(739.535.6)	Umatilla County	(739.733.4)	Lincoln County
(739.535.7)	Union County	(739.733.6)	Spokane County. □ Spokane (city)
(739.535.8)	Wallowa County	(739.733.8)	Whitman County
(739.542/.544)	Western and southern counties	(739.742/.748)	Counties along the southern border and Columbia River, and south-western area
(739.542.1)	Baker County	(739.742.1)	Asotin County
(739.542.3)	Grant County	(739.742.3)	Garfield County
(739.542.5)	Wheeler County	(739.742.5)	Columbia County
(739.542.7)	Crook County	(739.742.7)	Walla Walla County
(739.544.1)	Deschutes County	(739.744.1)	Benton County
(739.544.3)	Klamath County	(739.744.3)	Klickitat County
(739.544.5)	Lake County	(739.744.5)	Skamania County
(739.544.6)	Harney County	(739.744.7)	Clark County
(739.544.7)	Malheur County	(739.746.1)	Cowlitz County
(739.6)	Idaho	(739.746.3)	Wahkiakum County
(739.621/.638)	Counties of southern Idaho	(739.746.5)	Pacific County
(739.621.2)	Owyhee County	(739.746.7)	Lewis County
(739.621.4)	Canyon County	(739.748.1)	Yakima County
(739.621.6)	Payette County	(739.748.3)	Kittitas County
(739.621.8)	Washington County	(739.748.5)	Chelan County
(739.623.2)	Adams County	(739.751/.753+739.762.3)	Counties around Puget Sound
(739.623.4)	Gem County	(739.751.2)	Snohomish County
(739.623.6)	Ada County. □ Boise (state capital)	(739.751.4)	Skagit County
(739.623.8)	Elmore County	(739.751.6)	Whatcom County
(739.625.2)	Camas County	(739.751.8)	San Juan County
(739.625.4)	Blaine County	(739.753.2)	Island County
(739.625.6)	Minidoka County	(739.753.4)	Kitsap County
(739.625.8)	Lincoln County	(739.753.6)	King County. □ Seattle
(739.627.2)	Jerome County	(739.753.7)	Pierce County. □ Tacoma
(739.627.4)	Gooding County	(739.753.8)	Thurston County. □ Olympia (state capital)
(739.627.6)	Twin Falls County	(739.762.1/.8)	Counties of the Olympic Peninsula
(739.627.8)	Cassia County	(739.762.1)	Grays Harbor County
(739.632.1)	Oneida County	(739.762.3)	Mason County
(739.632.3)	Franklin County	(739.762.5)	Jefferson County
(739.632.5)	Bear Lake County	(739.762.8)	Clallam County
(739.632.7)	Caribou County		
(739.634.1)	Bannock County	(739.8)	Alaska
(739.634.3)	Power County	(739.812/.814)	Boroughs and census areas of the south-eastern arm of Alaska
(739.634.5)	Bingham County	(739.812.1)	City and Borough of Juneau. □ Juneau (state capital)
(739.634.7)	Bonneville County	(739.812.3)	Haines Borough
(739.636.1)	Teton County	(739.812.5)	Skagway-Hoonah-Angoon Census Area.
(739.636.3)	Madison County		□ Yakutat Borough
(739.636.5)	Fremont County	(739.812.7)	Sitka Borough
(739.636.7)	Clark County	(739.814.1)	Wrangell-Petersburg Census Area
(739.636.8)	Jefferson County	(739.814.3)	Prince of Wales-Outer Ketchikan Census Area
(739.638.1)	Butte County	(739.814.5)	Ketchikan Gateway Borough
(739.638.3)	Custer County	(739.822/.828)	Boroughs and census areas of the main part of Alaska (with associated islands)
(739.638.5)	Boise County	(739.822.1)	Valdez-Cordova Census Area
(739.638.6)	Valley County	(739.822.3)	Matanuska-Susitna Borough
(739.638.7)	Lemhi County	(739.822.5)	Anchorage Borough
(739.641/.645)	Counties of northern Idaho	(739.822.7)	Kenai Peninsula Borough
(739.641.2)	Idaho County	(739.823.2)	Kodiak Island Borough
(739.641.4)	Lewis County	(739.823.4)	Bristol Bay Borough
(739.641.6)	Nez Perce County	(739.823.6)	Lake and Peninsula Borough
(739.641.8)	Latah County	(739.823.8)	Aleutians East Borough
(739.643.2)	Clearwater County	(739.826.1)	Aleutians West Census Area
(739.643.4)	Shoshone County	(739.826.3)	Dillingham Census Area
(739.643.6)	Benewah County		
(739.643.8)	Kootenai County		
(739.645.2)	Bonner County		
(739.645.4)	Boundary County		

(1/9) Table 1e. Common auxiliaries of place

(739.826.5)	Bethel Census Area. □ Nunivak Island	(812.1)	Maranhão State. □ São Luis (state capital and municipality). Alcântara. Barreirinhas. Caxias. Imperatriz. Ribamar Fiquene. Tutóia (municipalities)
(739.826.7)	Wade Hampton Census Area		
(739.827.2)	Nome Census Area. □ St. Lawrence Island	(812.2)	Piauí State. □ Teresina (state capital and municipality). Luís Correia. Parnaíba. Pedro II. Picos. Piracuruca. Piripiri. Porto. São Gonçalo do Piauí. São Raimundo Nonato (municipalities)
(739.827.4)	Northwest Arctic Borough		
(739.827.6)	North Slope Borough		
(739.827.8)	Yukon-Koyukuk Census Area		
(739.827.9)	Denali Borough		
(739.828.1)	Fairbanks North Star Borough	(813.1)	Ceará State. □ Fortaleza (state capital and municipality). Aracati. Baturité. Beberibe. Camocim. Canindé. Caucaia. Crateús. Crato. Jijoca de Jericoacoara. Juazeiro do Norte. Paracuru. Quixadá. São Benedito. São Gonçalo do Amarante. Sobral Solonópole. Tianguá. Trairi. Ubajara (municipalities)
(739.828.3)	Southeast Fairbanks Census area		
(739.9)	Hawaii		
(739.911.2)	Honolulu County. □ Oahu Island, with City of Honolulu (state capital). Minor Hawaiian islands <i>The official limits of Honolulu County include the 'Island of Oahu and all other islands in the state of Hawaii not included in the other counties'</i>	(813.2)	Rio Grande do Norte State. □ Natal (state capital and municipality). Acari. Açu. Apodi. Areia Branca. Baía Formosa. Caicó. Canguaretama. Goianinha. Maxaranguape. Mossoró. Pau dos Ferros. São Rafael. São Miguel de Touros. Senador Georgino Avelino. Tibau. Tibau do Sul. Touros (municipalities)
(739.911.4)	Kauai County. □ Kauai Island. Niihau Island		
(739.911.6)	Maui County. □ Maui Island. Kahoolawe Island. Lanai Island. Molokai Island		
(739.911.8)	Hawaii County (= Hawaii Island; also known as Big Island)	(813.3)	Paraíba State. □ João Pessoa (state capital and municipality). Baía da Traição. Cabedelo. Cajazeiras. Campina Grande. João Pessoa. Patos. Pitimbu. Poço Dantas. Poço de José de Moura. Riachão do Poço. Sousa (municipalities)
(8)	SOUTH AMERICA. STATES AND REGIONS OF SOUTH AMERICA. LATIN AMERICA ⇒ (292.8)		
(81)	BRAZIL. FEDERATIVE REPUBLIC OF BRAZIL. BRAZIL. REPÚBLICA FEDERATIVA DO BRASIL <i>Brazil is divided into 26 estados (states) and one distrito federal (federal district). The states are subdivided into municípios (municipalities). The municípios are subdivided into distritos (districts). The federal district is subdivided into 13 administrative regions</i>	(813.4)	Pernambuco State. □ Recife (state capital and municipality). Arcoverde. Bezerros. Caruaru. Fernando de Noronha. Garanhuns. Goiana. Gravatá. Igarassu. Ipojuca. Itamaracá. Olinda. Ouricuri. Petrolina. Rio Formoso. São José da Coroa Grande. Serra Talhada. Sirinhaém. Tamandaré. Tracunhaém. Triunfo. Vitória de Santo Antão (municipalities)
(811)	Norte Region	(813.5)	Alagoas State. □ Maceió (state capital and municipality). Barra de Santo Antônio. Japaratinga. Maragogi. Marechal Deodoro. Paripueira. Penedo. Porto de Pedras. São Luís do Quitunde. São Miguel dos Milagres (municipalities)
(811.1)	Roraima State. □ Pôrto Velho (state capital and municipality). Guajara Mirim. Ji-Paraná. Porto Velho. Vilhena (municipalities) <i>Formerly Federal Territory of Guaporé</i>		
(811.2)	Acre State. □ Rio Branco (state capital and municipality). Assis Brasil. Brasiléia. Cruzeiro do Sul (municipalities)	(813.6)	Fernando de Noronha Territory
(811.3)	Amazonas State. □ Manaus (state capital and municipality). Apuí. Barcelos. Benjamin Constant. Humaitá. Manacapuru. Manicoré. Novo Ayrão. Parintins. São Gabriel da Cachoeira. Tabatinga. Tefé (municipalities)	(813.7)	Sergipe State. □ Aracaju (state capital and municipality). Barra dos Coqueiros. Estância. Laranjeiras. Neópolis. Propriá. São Cristóvão (municipalities)
(811.4)	Roraima Territory. □ Boa Vista (territory capital and municipality). Bonfim. Caracará (municipalities)		
(811.5)	Pará State. □ Belém (state capital and municipality). Alenquer. Altamira. Belterra. Capanema. Conceição do Araguaia. Itaituba. Marabá. Monte Alegre. Obidos. Oriximiná. Parauapebas. Redenção. Salinópolis. Santarém (municipalities)	(813.8)	Bahia State. □ Salvador (state capital). Cachoeira. Cairu. Camamu. Candéias. Canudos. Caravelas. Cipó. Conde. Euclides da Cunha. Feira de Santana. Ilhéus. Itabuna. Itacaré. Itaparica. Jacobina. Jequié. Juazeiro. Lençóis. Maragogipe. Mucugê. Palmeiras. Paulo Afonso. Pôrto Seguro. Prado. Salvador. Santa Cruz Cabralia. Santo Amaro. Senhor do Bonfim. Ubaitaba. Valença. Vitória da Conquista (municipalities)
(811.6)	Amapá Territory. □ Macapá (territory capital and municipality). Serra do Navio (municipality)		
(811.7)	Tocantins State. □ Palmas (state capital and municipality). Araguaína. Fátima. Gurupi. Lageado. Lagoa da Confusão. Miracema do Tocantins. Paraíso do Tocantins. Porto Nacional. Santa Tereza do Tocantins. Santa Terezinha do Tocantins. Tocantínia. Xambioá (municipalities) ⇒ (282.281.4)	(815)	Sudeste Region
(812/813)	Nordeste Region	(815.1)	Minas Gerais State. □ Belo Horizonte (state capital and municipality). Alto Capará. Araxá. Caeté. Cambuquira. Caxambu. Congonhas. Diamantina. Dionísio. Governador Valadares. Juiz de Fora. Lambari. Lavras. Manhumirim. Mariana. Marliéria. Montes Claros. Nova Lima. Ouro Preto. Pirapora. Poços de Caldas. Sabará. São Francisco. São João del Rei. São Lourenço. São Tomé das Letras. Serro. Sete Lagoas. Teófilo Otoni. Timóteo. Tiradentes. Três Corações. Uberaba. Uberlândia (municipalities)

- (815.2) Espírito Santo State. □ Vitória (state capital and municipality). Anchieta. Aracruz. Cachoeiro de Itapemirim. Conceição da Barra. Domingos Martins. Guarapari. Linhares. Piúma. Santa Leopoldina. Santa Maria do Jetibá. Santa Teresa. Sooretama. Venda Nova do Imigrante (municipalities). Trindade, Martim Vaz (offshore islands)
- (815.3) Rio de Janeiro State. □ Rio de Janeiro (state capital and municipality). Angra dos Reis. Araruama. Armação de Búzios. Arraial do Cabo. Cabo Frio. Campos dos Goytacazes. Itatiaia. Macaé. Mangaratiba. Maricá. Miguel Pereira. Niterói. Nova Friburgo. Nova Iguaçu. Parati. Petrópolis. Resende. Santo Antônio de Pádua. Saquarema. Teresópolis. Valença. Vassouras. Volta Redonda (municipalities)
- (815.6) São Paulo State. □ São Paulo (state capital and municipality). Águas de Lindóia. Americana. Aparecida. Atibaia. Bauru. Bertioga. Campinas. Campos do Jordão. Cananéia. Caraguatatuba. Guarujá. Holambra. Iguape. Ilha Comprida. Ilabela. Itanhaém. Itatinga. Jundiaí. Lindóia. Marília. Ourinhos. Peruíbe. Praia Grande. Presidente Epitácio. Presidente Prudente. Ribeirão Preto. Santa Bárbara D'Oeste. Santos. São Sebastião. São Vicente. Serra Negra. Sorocaba. Taubaté. Ubatuba (municipalities)
- (816)
(816.2) **Sul Region**
Paraná State. □ Curitiba (state capital and municipality). Antonina. Araucária. Esperança Nova. Foz do Iguaçu. Guarapuava. Guaraqueçaba. Guaratuba. Iguaçu. Irati. Lapa. Laranjeiras do Sul (Iguazú). Mallet. Maringá. Morretes. Paranaguá. Ponta Grossa. Prudentópolis (municipalities)
- (816.4) Santa Catarina State. □ Florianópolis (state capital and municipality). Aberlardo Luz. Águas Mornas. Balneário de Camboriú. Barra Velha. Blumenau. Bombinhas. Caçador. Chapecó. Criciúma. Dionísio Cerqueira. Garopaba. Imbituba. Itajaí. Joaçaba. Joinville. Lages. Laguna. Orleans. Penha. Piçarras. Pomerode. Porto Belo. Praia Grande. Santa Catarina. Santo Amaro da Imperatriz. São Francisco do Sul. São Joaquim. Treze Tílias. Urussanga. Videira (municipalities)
- (816.5) Rio Grande do Sul State. □ Porto Alegre (state capital and municipality). Acegua. Alegrete. Bagé. Barra do Quaraí. Bento Gonçalves. Cambará do Sul. Canela. Capão da Canoa. Caxias do Sul. Cruz Alta. Flores da Cunha. Gramado. Lavras do Sul. Nova Petrópolis. Passo Fundo. Pelotas. Rio Grande. Chuí. Santa Maria. Santana do Livramento. Santo Ângelo. São Borja. São Francisco de Paula. São José do Norte. São Lourenço do Sul. São Miguel das Missões. Torres. Tramandaí. Uruguaiana. Vacaria (municipalities)
- (817)
(817.1) **Centro-Oeste Region**
Mato Grosso do Sul State. □ Campo Grande (state capital and municipality). Anastácio. Aquidauana. Bela Vista. Bonito. Camapuã. Campo Grande. Corumbá. Coxim. Dourados. Jardim. Maracaju. Miranda. Mundo Novo. Ponta Porã (municipalities). Porto Murtinho (municipality, until 1979 part of Mato Grosso State)

- (817.2) Mato Grosso State. □ Cuiabá (state capital and municipality). Alta Floresta. Barão de Melgaço. Barra do Garças. Cáceres. Chapada dos Guimarães. Colider. Poconé. Rondonópolis. Santo Antônio do Leverger. São Félix do Araguaia. Sinop (municipalities)
- (817.3) Goiás State. □ Goiânia (state capital and municipality). Anápolis. Aruanã. Britânia. Caldas Novas. Goiás. Morrinhos. Pirenópolis. Pires do Rio. Rio Quente. São Domingos (municipalities)
- (817.4) Federal District. □ Brasília

(82)

ARGENTINA. ARGENTINE REPUBLIC. REPÚBLIC OF ARGENTINA

Argentina is divided into 23 provincias (provinces) and a distrito federal (federal district). The provinces are further into departamentos (departments), except for Distrito Federal, which has no internal divisions on this level. and Buenos Aires, which is divided into partidos (parts)

(821/822)

Litoral

Northeastern Argentina

Capital Federal District
Buenos Aires
Buenos Aires Province. □ La Plata (provincial capital). Adolfo Alsina. Adolfo González Chaves. Alberti. Almirante Brown. Avellaneda. Ayacucho. Azul (parts). Bahía Blanca (part and town). Balcarce. Baradero. Bartolomé Mitre. Benito Juárez. Berazategui. Berisso. Bolívar. Bragado. Brandsen. Campana. Cañuelas. Capitán Sarmiento. Carlos Casares. Carlos Tejedor. Carmen de Areco. Castelli. Chacabuco. Chascomús. Chivilcoy. Colón. Coronel de Marina L. Rosales. Coronel Dorrego. Coronel Pringles. Coronel Suárez. Daireux. Dolores. Ensenada. Escobar. Esteban Echeverría. Exaltación de la Cruz. Florencio Varela. Florentino Ameghino. General Alvarado. General Alvear. General Arenales. General Belgrano. General Guido. General Juan Madariaga. General La Madrid. General Las Heras. General Lavalle. General Paz. General Pinto. General Pueyrredón. General Rodríguez. General San Martín. General Sarmiento. General Viamonte. General Villegas. Guaminí. Hipólito Yrigoyen. Junín. La Costa. La Matanza. Lanús. La Plata, Mar del Plata. Laprida. Las Flores. Leandro N. Alem. Lincoln. Lobería. Lobos. Lomas de Zamora. Luján. Magdalena. Maipú. Mar Chiquita. Marcos Paz. Mercedes. Merlo. Monte. Monte Hermoso. Moreno. Morón. Navarro. Necochea. Nueve de Julio. Olavarría. Patagones. Pehuajó. Pellegrini. Pergamino. Pila. Pilar. Pinamar. Puán (parts). Quilmes (part and town). Ramallo. Rauch. Rivadavia. Rojas. Roque Pérez. Saavedra. Saladillo. Saliqueló. Salto. San Andrés de Giles. San Antonio de Areco. San Cayetano. San Fernando. San Isidro. San Nicolás. San Pedro. San Vicente. Suipacha. Tandil. Tapalqué. Tigre. Tordillo. Tornquist. Trenque Lauquen. Tres Arroyos. Tres de Febrero. Tres Lomas. Veinticinco de Mayo. Vicente López. Villa Gesell. Villarino. Zárate (parts)

(1/9) Table 1e. Common auxiliaries of place

(821.6)	Santa Fe Province. □ Santa Fe (provincial capital and department). Belgrano. Caseros. Castellanos. Constitución. Garay. General López. General Obligado. Iriondo. Las Colonias. Nueve de Julio (departments). Rosario (department and town). San Cristóbal. San Javier. San Jerónimo. San Justo. San Lorenzo. San Martín. Vera (departments)	(823.3)	San Javier. San Justo. Santa María. Sobremonte. Tercero Arriba. Totoral. Tulumba. Unión (departments)
(821.8)	Chaco Province. □ Resistencia (provincial capital). Almirante Brown. Bermejo. Chacabuco. Comandante Fernández. Doce de Octubre. Fray Justo Santa María de Oro. General Belgrano. General Donovan. General Güemes. Independencia. Libertad. Libertador General San Martín. Maipú. Mayor Luis J. Fontana. Nueve de Julio. O'Higgins. Presidencia de la Plaza. Primero de Mayo. Quitilipi. San Fernando. San Lorenzo. Sargento Cabral. Tapenagá. Veinticinco de Mayo (departments)	(825) (825.1)	San Luis Province. □ San Luis (provincial capital and department). Ayacucho. Belgrano. Chacabuco. Coronel Pringles. General Pederna. Gobernador Dupuy. Junín. Libertador General San Martín (departments)
(821.9)	Formosa Province. □ Formosa (provincial capital and department). Bermejo. Laishi. Matacos. Patiño. Pilagás. Pilcomayo. Pirane. Ramón Lista (departments)	(825.2)	Andina Region Mendoza Province. □ Mendoza (provincial capital and department). General Alvear. Godoy Cruz. Guaymallén. Junín. La Paz (departments). Las Heras (department and town). Lavalle. Luján de Cuyo. Maipú. Malargüe. Rivadavia. San Carlos. San Martín (departments). San Rafael (department and town). Santa Rosa. Tunuyán. Tupungato (departments)
(822) (822.1)	Mesopotamia Argentina Entre Ríos Province. □ Paraná (provincial capital). Colón. Concordia (department and town). Diamante. Federación. Federal. Feliciano. Gualaguay. Gualaguaychú. Islas del Ibicuy. La Paz. Nogoya. Paraná. Tala (departments). Concepción de Uruguay (department and town). Victoria. Villaguay (departments)	(825.3)	San Juan Province. □ San Juan (provincial capital and department). Albardón. Angaco. Calingasta. Caucete. Chimbos. Iglesia. Jáchal. Nueve de Julio. Pocito. Rawson. Rivadavia. San Martín. Santa Lucía. Sarmiento. Ullum. Valle Fértil. Veinticinco de Mayo. Zonda (departments)
(822.2)	Corrientes Province. □ Corrientes (provincial capital and department). Bella Vista. Berón de Astrada. Concepción. Curuzú Cuyatí. Empedrado. Esquina. General Alvear. General Paz (departments). Goya (department and town). Itati. Ituzaingó. Lavalle. Mburucuyá. Mercedes. Monte Caseros (departments). Paso de los Libres (department and town). Saladas. San Cosme. San Luis del Palmar. San Martín. San Miguel. San Roque. Santo Tomé. Sauce (departments)	(825.4)	La Rioja Province. □ La Rioja (provincial capital). Auroco. Castro Barros. Chamental. Chilecito. Coronel Felipe Varela. Famatina. General Angel V. Peñalosa. General Belgrano. General Juan F. Quiroga. General Lamadrid. General Ocampo. General San Martín. Independencia. Rosario Vera Peñaloza. Sanagasta. San Blas de los Sauces. Vinchina (departments)
(822.3)	Misiones Province. □ Posadas (provincial capital and department). Apóstoles. Caingúas. Candelaria. Concepción (departments). El Dorado (department and town). General Manuel Belgrano. Guaraní. Iguazú. Leandro N. Alem. Libertador General San Martín. Limay Mahuida. Montecarlo. Oberá. San Ignacio. San Javier. San Pedro. Veinticinco de Mayo (departments)	(826) (826.2)	Catamarca Province. □ Catamarca (provincial capital and department). Ambarto. Ancastí. Andalgalá. Antofagasta de la Sierra. Belén. Capayán. El Alto. Fray Mamerto Esquiú. La Paz. Paclín. Poman. Santa María. Santa Rosa (departments). Tinogasta (department and town). Valle Viejo (department)
(823) (823.1)	Centro Region La Pampa Province. □ Santa Rosa (provincial capital and department). Atreuco. Caleu Caleu. Catriló. Chalileo. Chapaleufú. Chical Cò. Conhelo. Curacó. Guatraché. Hucal. Loventué. Maracó. Puelén. Quemú Quemú. Rancul. Realicó. Toay. Trenel. Uhuel Calel. Utracán (departments) ⇒ (251.2)	(826.5)	Norte Region Santiago del Estero Province. □ Santiago del Estero (provincial capital and department). Aguirre. Alberdi. Atamsqui. Avellaneda (departments). Banda (department and town). Belgrano. Choya. Copo. Figueroa. General Taboada. Guayacán. Jiménez. Juan F. Ibarra. Loreto. Mitre. Moreno. Ojo de Agua. Pellegrini. Quebrachos. Río Hondo. Rivadavia. Robles. Salavina. San Martín. Sarmiento. Silipica (departments)
(823.2)	Córdoba Province. □ Córdoba (provincial capital and department). Calamuchita. Colón. Cruz del Eje. General Roca. General San Martín. Ischilín. Juárez Celman. Marcos Juárez. Minas. Pocho. Punilla (departments). Río Cuarto (department and town). Río Primero. Río Seco. Río Segundo. San Alberto.	(826.7)	Tucumán Province. □ San Miguel de Tucumán (provincial capital and department). Burruyacú. Chicligasta. Cruz Alta. Famallá. Graneros. Juan B. Alberdi. La Cocha. Leales. Lules. Monteros. El Alto. Simoca. Taffí del Valle. Taffí Viejo. Trancas. Yerba Buena (departments)
		(826.9)	Salta Province. □ Salta (provincial capital and department). Anta. Cachi. Cafayate. Cerrillos. Chicoana. General Güemes. General José de San Martín. Guachipas. Iruya. La Caldera. La Candelaria. La Poma. La Viña. Los Andes. Metán. Molinos. Orán. Rivadavia. Rosario de la Frontera. Rosario de Lerma. San Carlos. Santa Victoria (departments)
			Jujuy Province. □ San Salvador de Jujuy (provincial capital). Cochinoa. Doctor Manuel Belgrano. El Carmen. Humahuaca. Ledesma. Palpalá. Rinconada. San Antonio. San Pedro.

	Santa Bárbara. Santa Catalina. Susques. Tilcara. Tumbaya. Valle Grande. Yavi (departments)	(835.22)	Valparaíso Region. □ Valparaíso (regional capital, province and provincial capital). Quilpue. Viña del Mar. Los Andes (province and provincial capital). Petorca (province), La Ligua (provincial capital). Quillota (province and provincial capital). San Antonio (province and provincial capital). San Felipe de Aconcagua (province), San Felipe (provincial capital). Isla de Pascua (Easter Island/Rapa Nui) (province), Hanga Roa (provincial capital). Islas Juan Fernández
(828)	Patagonia Region		Región Metropolitana (Santiago). □ Puente Alto San Bernardo
(828.1)	Neuquén Province. □ Neuquén (provincial capital and department). Aluminé. Añelo. Catán Lil. Chos Malal. Collón Curá. Confluencia. Huiliches. Lacar. Loncopué. Los Lagos. Minas. Pehuenches. Picunches. Picún Leufú (departments). Zapala (department and town)	(835.23)	Libertador General Bernardo O'Higgins Region (or VI Region). □ Rancagua (regional capital and capital of Cachapoal province). Cardenal Caro (province), Pichilemo (provincial capital). Cachapoal (province), Rancagua (provincial capital). Colchagua (province), San Fernando (provincial capital)
(828.3)	Río Negro Province. □ Viedma (provincial capital). Adolfo Alsina. Avellaneda. Bariloche, San Carlos de Bariloche. Conesa. El Cuy. General Roca. Norquínco. Nueve de Julio. Pichi Mahuida. Pilnaniyeu. San Antonio, San Antonio Oeste. Valcheta. Veinticinco de Mayo (departments)	(835.24)	Maule Region. □ Talca (regional capital, province and provincial capital). Cauquenes (province and provincial capital). Curicó (province and provincial capital). Linares (province and provincial capital)
(828.6)	Chubut Province. □ Rawson (provincial capital and department), Trelew. Biedma. Cushamen. Escalante. Comodoro Rivadavia. Florentino Ameghino. Futaleufú. Gaimán. Gastre. Languineo. Mártires. Paso de los Indios (departments). Presidente Roque Sáenz Peña (department and town). Punilla. Río Senguerr. Sarmiento. Telsen. Tehuelches (departments)	(835.26)	Bío-Bío Region. □ Concepción (regional capital, province and provincial capital). Arauco (province), Lebu (provincial capital). Bío-Bío (province), Los Angeles (provincial capital). Concepción (province and provincial capital), Talcahuano. Nuble (province), Chillán (provincial capital)
(828.8)	Santa Cruz Province. □ Río Gallegos (provincial capital). Corpen Aike. Deseado. Güer Aike. Lago Argentino. Lago Buenos Aires. Magallanes. Río Chico (departments)	(835.28)	La Araucanía Region. □ Temuco (regional capital and capital of Cautín province). Cautín (province), Villarrica. Malleco (province), Angol (provincial capital)
(828.9)	Tierra del Fuego Province, Antártida y Islas del Atlántico Sur Nacional. □ Ushuaia (provincial capital and department). Río Grande (department and town)	(835.32)	Los Lagos Region with Isla de Chiloé. □ Puerto Montt (regional capital and provincial capital of Llanquihue). Chiloé (province), Castro (provincial capital). Llanquihue (province), Osorno (province and provincial capital). Palena (province), Chaitén (provincial capital). Valdivia (province and provincial capital)
(829)	BRITISH POSSESSIONS IN THE SOUTH ATLANTIC	(835.34)	Aisén del General Carlos Ibáñez del Campo Region. □ Coihaique (regional capital, province and provincial capital). Aisén (province), Puerto Aisén (provincial capital). General Carrera (province), Chile Chico (provincial capital). Capitán Prat (province), Cochrane (provincial capital)
(829.1)	FALKLAND ISLANDS. FALKLAND ISLANDS AND DEPENDENCIES (UK).		Magallanes y Antártica Chilena Region. □ Punta Arenas (regional capital and capital of Magallanes province). Antártica Chilena (province), Puerto Williams (provincial capital). Magallanes (province). Tierra del Fuego (province), Porvenir (provincial capital). Última Esperanza (province), Puerto Natales (provincial capital)
(829.15)	□ Port Stanley (capital). Goose Green South Georgia and South Sandwich Islands (UK)		
(829.3)	SOUTH SHETLAND ISLANDS	(835.36)	
(829.5)	SOUTH ORKNEY ISLANDS		
(83)	CHILE. REPUBLIC OF CHILE. REPÚBLICA DE CHILE		BOLIVIA. REPUBLIC OF BOLIVIA. REPÚBLICA DE BOLIVIA
	<i>Chile is divided into 13 regiones (regions). Regions are divided into 51 provincias (provinces), which are further subdivided into comunas (municipalities)</i>	(835.38)	<i>Bolivia is divided into nine departamentos (departments). The departments are divided into about 100 provincias (provinces). The provinces are subdivided into secciones (sections), also known as municipios (municipalities)</i>
(835)	Administrative divisions of Chile		
(835.12)	Tarapacá Region. □ Iquique (regional capital, province and provincial capital). Arica (province and provincial capital)		
(835.14)	Antofagasta Region. □ Antofagasta (regional capital, province and provincial capital). El Loa (province), Calama (provincial capital). Tocopila (province and provincial capital)	(84)	Northern region
(835.16)	Atacama Region. □ Copiapó (regional capital, province and provincial capital). Chañaral (province and provincial capital). Huasco (province), Vallenar (provincial capital). Los Islas Desaventurados	(844.1)	Pando (department). □ Cobija (department capital)
(835.18)	Coquimbo Region. □ La Serena (regional capital). Choapa (province), Illapel (provincial capital). Elqui (province), Coquimbo (provincial capital). Limarí (province), Ovalle (provincial capital)		

(1/9) Table 1e. Common auxiliaries of place

(844.2)	El Beni (department). □ Trinidad (department capital). Guayaramerin. Riberalta	(852)	Northern Peru
(845)	Andes region	(852.12)	Tumbes (department). □ Tumbes (department capital and province). Contralmirante Villar. Zarumilla (provinces)
(845.1)	La Paz (department). □ La Paz (state and department capital)	(852.14)	Piura (department). □ Piura (department capital and province). Ayabaca. Huancabamba. Morropón. Paita. Sechura. Sullana. Talara (provinces)
(845.2)	Oruro (department). □ Oruro (department capital). Huanuni	(852.16)	Lambayeque (department). □ Chiclayo (department capital and province). Ferreñafe. Lambayeque (provinces)
(845.3)	Potosí (department). □ Potosí (department capital)	(852.18)	Cajamarca (department). □ Cajamarca (department capital and province). Cajabamba. Celendin. Chota. Contumazá. Cutervo. Hualgayoc. Jaé. San Ignacio. San Marcos. San Miguel. San Pablo. Santa Cruz (provinces)
(846)	Central and southeastern region	(852.19)	La Libertad (department). □ Trujillo (department capital and province). Chanchan (ancient city). Moche pyramids (ancient site). Ascope. Bolívar. Chepén. Gran Chimú. Julcán. Otuzco. Pacasmayo. Pataz. Sánchez Carrion. Santiago de Chuco. Viru (provinces)
(846.1)	Cochabamba (department). □ Cochabamba (department capital). Punata. Quillacollo		Southern Peru
(846.2)	Chuquisaca (department). □ Sucre (department capital)		Arequipa (department). □ Arequipa (department capital and province). Camaná. Caravelí. Castilla. Caylloma. Condesuyos. Islay. La Unión (provinces)
(846.3)	Tarija (department). □ Tarija (department capital). Camiri. Montero		Moquegua (department). □ Moquegua (department capital). General Sánchez Cerro. Ilo. Mariscal Nieto (provinces)
(847)	Eastern region		Tacna (department). □ Tacna (department capital and province). Candarave. Jorge Basadre. Tarata (provinces)
(847.1)	Santa Cruz (department). □ Santa Cruz de la Sierra (department capital)		Puno (department). □ Puno (department capital and province). Azángaro. Carabaya. Chucuito. El Collao. Huancané. Lampa. Melgar. Moho. San Antonio de Putina. San Román. Sandía. Yunguyo (provinces)
(85)	PERU. REPUBLIC OF PERU. REPÚBLICA DE PERÚ		Cuzco (department). □ Cuzco (department capital, province and ancient city). Acomayo. Anta. Calca. Canas. Canchis. Chumbivilcas. Espinar. La Convención. Paruro. Paucartambo. Quispicanchi. Urubamba (provinces), Machu Picchu (ancient city)
	<i>Since 2002, Peru has been divided into 25 regions and one province. The regions are divided into 51 provincias (provinces), which are further subdivided into partidos or distritos (districts). Lima [Province] and Callao, however, are divided directly into districts</i>		Eastern Peru
(851)	Centro. Central Peru		Madre de Dios (department). □ Puerto Maldonado (department capital). Manu. Tahuamanu. Tambopata (provinces)
(851.11)	Ancash (department). □ Huaraz (department capital and province). Aija. Antonio Raymondi. Asunción. Bolognesi. Carhuaz. Carlos Fermín Fitzcarraldo. Casma. Corongo. Huari (provinces), Chavín de Huantar (ancient city). Huarmey. Huaylas. Mariscal Luzuriaga. Ocros. Pallasca. Pomabamba. Recuay. Santa. Chimbote. Sihuas. Yungay (provinces)		Ucuyali (department). □ Pucallpa (department capital). Atalaya. Coronel Portillo. Padre Abad. Purús (provinces)
(851.13)	Huánuco (department). □ Huánuco (department capital and province). Ambo. Dos de Mayo. Huacaybamba. Huamalies. Lauricocha. Leoncio Prado. Marañón. Pachitea. Puerto Inca. Yarowilca (provinces)		Loreto (department). □ Iquitos (department capital). Alto Amazonas. Loreto. Mariscal Ramón Castilla. Maynas. Requena. Ucayali (provinces)
(851.15)	Pasco (department). □ Cerro de Pasco (department capital and province). Daniel Alcides Carrión. Oxapampa (provinces)		San Martín (department). □ Moyobamba (department capital and province). Bellavista. El Dorado. Huallaga. Lamas. Mariscal Cáceres. Picota. Rioja. San Martín. Tocache (provinces)
(851.17)	Junín (department). □ Huancayo (department capital and province). Chanchamayo. Chupaca. Concepción. Huaral. Jauja. Junín. Satipo. Tarma. Yauli (provinces)		Amazonas (department). □ Chachapoyas (department capital and province). Bagua. Bongará. Condorcanqui. Luya. Rodríguez de Mendoza. Utcubamba (provinces)
(851.19)	Lima (capital, department and province). □ Barranca. Cañete. Canta. Huarochiri. Huaura. Oyon. Yauyos (provinces)		
(851.21)	Callao (constitutional province and provincial capital)		
(851.23)	Ica (department). □ Ica (department capital and province). Chincha. Nazca (provinces), Nazca (ancient city). Palpa. Pisco (provinces)		
(851.25)	Huancavelica (department). □ Huancavelica (department capital and province). Acobamba. Angaraes. Castrovirreyna. Churcampá. Huaytara. Tayacaja (provinces)		
(851.27)	Ayacucho (department). □ Ayacucho (department capital). Cangallo. Huamanga. Huanca Sancos. Huanta. La Mar. Lucanas. Parinacochas. Paucar del Sara Sara. Sucre. Víctor Fajardo. Vilcas Huamán (provinces)		
(851.29)	Apurímac (department). □ Abancay (department capital and province). Andahuaylas. Antabamba. Aymaraes. Chincheros. Cotabambas. Grau (provinces)		
		(853)	COLOMBIA. REPUBLIC OF COLOMBIA. REPÚBLICA DE COLOMBIA
		(853.11)	<i>Colombia is divided into 32 departamentos (departments) and one distrito capital (capital district). Subdivisions of the departments are called municipios (municipalities)</i>
		(853.13)	
		(853.15)	
		(853.17)	
		(853.19)	
		(854)	
		(854.12)	
		(854.14)	
		(854.16)	
		(854.18)	
		(854.19)	
		(862)	

(862.1)	Northern and Caribbean coast departments	(866.13)	Pichincha. □ Quito (state capital and canton). Cayambe. Majía. Pedro Moncayo. Rumiñahui. Santo Domingo de los Colorados. San Miguel de los Bancos (cantons)
(862.11)	Córdoba. □ Montería (department capital)	(866.14)	Cotopaxi. □ Latacunga (provincial capital and canton). La Mana (canton and town). Pangua (canton). El Corazón. Pujili (canton and town). Salcedo. San Miguel. Sigchos. Saquisilí (cantons)
(862.12)	Antioquia. □ Medellín (department capital)	(866.15)	◇ (866.14-751.2) Cotopaxi National Park Tungurahua. □ Ambato (provincial capital and canton). Baños. Cevallos. Mocha. Patate. Quero (cantons and towns). San Pedro de Pelileo. Pelileo. Santiago de Pillaro. Pillaro (cantons). Tisaleo (canton and town)
(862.13)	Sucre. □ Sincelejo (department capital)	(866.16)	Chimborazo. □ Riobamba (provincial capital and canton). Alausí (canton and town). Chambo (canton). Chunchi (canton and town). Colta. Villa La Unión. Cumandá (cantons). Guamote. Guano. Pallatanga. Penipe (cantons and towns)
(862.14)	Bolívar. □ Cartagena (department capital)	(866.17)	Cañar. □ Azogues (provincial capital and canton). Biblián. Cañar. Déleg. El Tambo. La Troncal (cantons)
(862.15)	Atlántico. □ Barranquilla (department capital)	(866.18)	Azuay. □ Cuenca (provincial capital and canton). Chordelec. El Pan. Girón. Guachapala. Gualaceo. Nabón. Oña. Paute. Pucará. San Fernando. Santa Isabel. Sevilla de Oro. Sigsig (cantons)
(862.16)	Magdalena. □ Santa Marta (department capital)	(866.19)	Loja. □ Loja (provincial capital and canton). Alamor. Calvas. Catamayo. Celica. Chaguarpamba. Gonzanamá. Macará. Paltas. Pindal. Puyango. Quilanga. Saraguro. Sozoranga. Zapotillo (cantons)
(862.17)	Cesar. □ Valledupar (department capital)	(866.2)	Coastal provinces
(862.18)	La Guajira. □ Ríoacha (department capital)	(866.22)	El Oro. □ Machalá (provincial capital and canton). Arenillas. Atahualpa. Balsas. Chilla. El Guabo. Huaquillas. Las Lajas. Marcabellí. Pasaje. Piñas. Portoviejo. Santa Rosa. Zaruma (cantons)
(862.19)	San Andrés y Providencia. □ Isla de San Andrés. San Andrés (department capital). Isla de Providencia. Cayos del Este Sudeste. Cayos de Albuquerque	(866.23)	Guayas. □ Guayaquil (provincial capital and canton). Balao. Balzar. Baquerizo Moreno. Colimes. Coronel Marcelino Maridueña. Daule. Durán. El Empalme. El Triunfo. Eloy Alfaro. General Antonio Elizade. Isidro Ayora. Jado. La Libertad. Lomas de Sargentillo. Milagro. Naranjal. Naranjito. Nobol. Palestina. Pedro Carbo. Playas. Salinas. Salitre. Samborondón. Santa Elena. Santa Lucía. Simón Bolívar. Urbina. Yaguachi (cantons)
(862.2)	Western and Pacific Coast departments	(866.24)	Los Ríos. □ Babahoyo (provincial capital and canton). Baba. Catarama. Mocache. Montalvo. Pueblo Viejo. Quevedo. Urdaneta. Valencia. Ventanas. Vines Palenque (cantons)
(862.21)	Nariño. □ Pasto (department capital)	(866.25)	Manabí. □ Portoviejo (provincial capital and canton). Bolívar. Chone. El Carmen. Flavio Alfaro. Junín. Jipijapa. Manta. Montecristi. Paján. Pichincha. Rocafuerte. Santa Ana. Sucre. Tosagua y 24 de Mayo (cantons)
(862.22)	Cauca. □ Popayán (department capital)	(866.26)	Esmeraldas. □ Esmeraldas (provincial capital and canton). Atacames. Eloy Alfaro. Muisne. Quinindé. San Lorenzo (cantons)
(862.23)	Valle de Cauca. □ Cali (department capital)	(866.3)	Eastern provinces
(862.24)	Chocó. □ Quibdó (department capital)	(866.31)	Napo. □ Tena (provincial capital and canton). Archidona. Baeza. El Chaco. Quijos (cantons)
(862.3)	Central departments	(866.32)	Sucumbios. □ Nueva Loja (provincial capital). Cascales. Gonzalo Pizarro. Lago Agrio. Putumayo. Shushufindi. Sucumbios (cantons)
(862.31)	Huila. □ Neiva (department capital)	(866.33)	Orellana. □ Francisco de Orellana (provincial capital and canton). Aguarico. La Joya de los Sachas. Loreto (cantons)
(862.32)	Tolima. □ Ibagué (department capital)		
(862.33)	Quindío. □ Armenia (department capital)		
(862.34)	Risaralda. □ Pereira (department capital)		
(862.35)	Cundinamarca. □ Distrito Capital de Santa Fe de Bogotá. Bogotá (state and department capital)		
(862.36)	Caldas. □ Manizales (department capital)		
(862.37)	Boyacá. □ Tunja (department capital)		
(862.38)	Santander. □ Bucaramanga (department capital)		
(862.39)	Norte de Santander. □ Cúcuta (department capital)		
(862.4)	Eastern departments		
(862.41)	Meta. □ Villavicencio (department capital)		
(862.42)	Casanare. □ Yopal		
(862.43)	Arauca. □ Arauca (department capital)		
(862.44)	Vichada. □ Puerto Carreño (department capital)		
(862.45)	Guainía. □ Obando (department capital)		
(862.5)	Southern departments		
(862.51)	Putumayo. □ Mocoa (department capital)		
(862.52)	Caquetá. □ Florencia (department capital)		
(862.53)	Guaviare. □ San José (department capital)		
(862.54)	Vaupés. □ Mitú (department capital)		
(862.55)	Amazonas. □ Leticia (department capital)		
(866)	ECUADOR. REPUBLIC OF ECUADOR. REPÚBLICA DE ECUADOR		
	<i>Ecuador is divided into 22 provincias (provinces). The provinces are subdivided into cantones (cantons), which are further subdivided into parroquias (parishes)</i>		
(866.1)	Sierra provinces		
(866.10)	Bolívar. □ Guaranda (provincial capital and canton). Caluma. Chillanes. San José de Chimbo. Echeandía. Las Naves. San Miguel (cantons)		
(866.11)	Carchi. □ Tulcán (provincial capital and canton). Bolívar. Guaranda. Espejo (cantons). El Angel. Mira (canton and town). Montúfar. San Gabriel. San Pedro de Huaca (cantons)		
(866.12)	Imbabura. □ Ibarra (provincial capital and canton). Antonio Ante (canton). Atuntaqui. Cotacachi. Otavalo. Pimampiro (cantons and towns). San Miguel de Urququí. Urququí (cantons)		

(1/9) Table 1e. Common auxiliaries of place

(866.34)	Pastaza. □ Puyo (provincial capital). Mera. Pastaza. Santa Clara (cantons)	
(866.36)	Morona-Santiago. □ Macas (provincial capital, canton and cantonal capital). Gualaquiza (canton and cantonal capital). General Leónidas, Limón Indanza (cantonal capital). Huamboya. Palora. Santiago. Sucúa. San Juan Bosco (cantons)	
(866.38)	Zamora-Chinchipe. □ Zamora (provincial capital and canton). Nangaritza, Guayzimi. Yacuambi (cantons)	
(866.4)	Galápagos. □ Isla San Cristóbal, Puerto Baquerizo Moreno (provincial capital). Isla Santa Cruz, Puerto Ayora. Isla Isabela, Puerto Villamil. Isla Fernandina, Narborough. Isla Santa María, Floreana. Isla Marchena, Bindloe. Isla Pinta, Abingdon. Isla Genovesa, Tower. Isla Española, Hood. Isla San Salvador, Isla Rábida, Isla Pinzón	(872.23)
(87)	VENEZUELA. BOLIVARIAN REPUBLIC OF VENEZUELA. REPÚBLICA BOLIVARIANA DE VENEZUELA <i>Venezuela is divided into 23 estados (states), one dependencias federales (federal dependencies), and one distrito capital (capital district). The states are divided into municipios (municipalities), or, in the case of territories/capital district, into departamentos (departments). The municipios are further subdivided into parroquias (parishes)</i>	
(872)	States and districts of Venezuela	
(872.15)	Amazonas. □ Puerto Ayacucho (state capital). Alto Orinoco (municipality). La Esmeralda (municipal capital). Atabapo (municipality). San Fernando de Atabapo (municipal capital). Atures (municipality). Puerto Ayacucho (municipal capital). Autana (municipality). Isla Ratón (municipal capital). Manapiare (municipality). San Juan de Manapiare (municipal capital). Maroa (municipality and municipal capital). Río Negro (municipality). San Carlos de Río Negro (municipal capital). Bolívar. □ Ciudad Bolívar (state and Heres municipal capital). Caroní (municipality). Ciudad Guayana (municipal capital). Cedeño (municipality). Cairaca del Orinoco (municipal capital). El Callao (municipality and municipal capital). Gran Sabana (municipality). Santa Elena de Uairén (municipal capital). Heres (municipality). Ciudad Bolívar (municipal capital). Padre Pedro Chien (municipality). El Palmar (municipal capital). Piar (municipality). Upata (municipal capital). Raúl Leoni (municipality). Ciudad Piar (municipal capital). Roscio (municipality). Guasipati (municipal capital). Sifontes (municipality). Tumeremo (municipal capital). Sucre (municipality). Maripa (municipal capital)	(872.24)
(872.17)	Delta Amacuro. □ Tucupita (state capital). Antonio Díaz (municipality). Curiapo (municipal capital). Casacoima (municipality). Sierra Imatac (municipal capital). Pedernales. Tucupita (municipalities and municipal capitals)	
(872.19)	Monagas. □ Maturín (state capital). Acosta (municipality). San Antonio (municipal capital). Aguasay (municipality and municipal capital). Bolívar (municipality). Caripito (municipal capital). Caripe (municipality and municipal capital). Cedeño (municipality). Caicaro (municipal capital). Ezequiel Zamora (municipality). Punta de Mata (municipal capital). Libertador (municipality). Temblador (municipal capital). Maturín (municipality and municipal capital). Piar (municipality). Aragua (municipality). Punceres (municipality). Quirquire (municipal capital). Santa Bárbara (municipality and municipal capital). Sotillo (municipality). Barrancas (municipal capital). Uracoa (municipality and municipal capital). Sucre. □ Cumaná (state, municipality and municipal capital). Andrés Eloy Blanco (municipality). Casanay (municipal capital). Andrés Mata (municipality). San José de Aerocuar (municipal capital). Arismendi (municipality). Río Caribe (municipal capital). Benítez (municipality). El Pilar (municipal capital). Bermúdez (municipality). Carúpano (municipal capital). Bolívar (municipality). Mariguiter (municipal capital). Cajigal (municipality). Yaguaraparo (municipal capital). Cruz Salmerón Acosta (municipality). Araya (municipal capital). Libertador (municipality). Tunapuy (municipal capital). Mariño (municipality). Irapa (municipal capital). Mejía (municipality). San Antonio del Golfo (municipal capital). Montes (municipality). Cumanacoa (municipal capital). Ribero (municipality). Cariaco (municipal capital). Valdez (municipality). Guira (municipal capital). Anzoátegui. □ Barcelona (state capital). Anaco (municipality and municipal capital). Aragua (municipality). Aragua de Barcelona (municipal capital). Diego Bautista Urbaneja (municipality). Lecherías (municipal capital). Fernando de Peñalver (municipality). Puerto Píritu (municipal capital). Francisco de Carmen Carvajal (municipality). Valle de Guanape (municipal capital). Francisco de Miranda (municipality). Pariguán (municipal capital). Guanta (municipality and municipal capital). Independencia (municipality). Soledad (municipal capital). José Gregorio Monagas (municipality). Mapire (municipal capital). Juan Antonio Sotillo (municipality). Puerto La Cruz (municipal capital). Juan Manuel Cajigal (municipality). Onoto (municipal capital). Libertad (municipality). San Mateo (municipal capital). Lobatera (municipality and municipal capital). Manuel Ezequiel Bruzual (municipality). Clarines (municipal capital). Pedro María Freitas (municipality). Cantaura (municipal capital). Píritu. San Cristóbal (municipalities and municipal capitals). San José de Guanipa (municipality). San José de Guanipa (El Tigrito) (municipal capital). San Juan de Capistrano (municipality). Bocha de Uchire (municipal capital). Santa Ana (municipality and municipal capital). Simón Bolívar (municipality). Barcelona (municipal capital). Simón Rodríguez (municipality). El Tigre (municipal capital). Sir Arthur McGregor (municipality). El Chaparro (municipal capital). Nueva Esparta. □ La Asunción (state capital). Antolín del Campo (municipality). Plaza Paraguachi (municipality and municipal capital). Arismendi (municipality). La Asunción (municipal capital). Díaz (municipality). San Juan Bautista (municipal capital). García (municipality). El	(872.25)
(872.22)		

(872.26)	Valle del Espíritu Santo (municipal capital). Gómez (municipality), Santa Ana (municipal capital). Maneiro (municipality), Pampatar (municipal capital). Marcano (municipality), Juan Griego (municipal capital). Mariño (municipality), Porlamar (municipal capital). Península de Macanao (municipality), Boca del Río (municipal capital). Tubores (municipality), Punta de Piedras (municipal capital). Villalba (municipality), San Pedro de Coche (municipal capital). Margarita, Cubagua, Coche Islands	(872.33)	(municipal capital). San Carlos de Austria (municipality), San Carlos (municipal capital). Tinaco (municipality and municipal capital) Portuguesa. □ Guanare (state capital, municipality and municipal capital). Agua Blanca. Araure (municipalities and municipal capitals). Esteller (municipality), Píritu (municipal capital). Guanarito (municipality and municipal capital). Monseñor José Vicente de Unda (municipality), Paraíso de Chebasquén (municipal capital). Ospino (municipality and municipal capital). Páez (municipality), Acarigua (municipal capital). Papelón (municipality and municipal capital). San Genaro de Boconoito (municipality), Boconoito (municipal capital). San Rafael de Ooto (municipality and municipal capital). Santa Rosalia (municipality), El Playon (municipal capital). Sucre (municipality), Biscucuy (municipal capital). Turén (municipality), Villa Bruzual (municipal capital)
(872.28)	Guárico. □ San Juan de los Morros (state capital and capital of Juan Germán Roscio municipality). Camaguán. Chaguaramas. El Socorro (municipalities and municipal capitals). Francisco de Miranda (municipality), Calabozo (municipal capital). José Félix Ribas (municipality), Tucupido (municipal capital). José Tadeo Monagas (municipality), Altavracia de Orituco (municipal capital). Juan Germán Roscio (municipality), San Juan de los Morros (municipal capital). Julián Mellado (municipality), El Sombrero (municipal capital). Las Mercedes (municipality and municipal capital). Leonardo Infante (municipality), Valle de la Pascua (municipal capital). Las Mercedes. Ortiz (municipalities and municipal capitals). Pedro Zaraza (municipality), Zaraza (municipal capital). San Gerónimo de Guayabal (municipality), Guayabal (municipal capital). San José de Guaribe. Santa María de Ipire (municipalities and municipal capitals)	(872.35)	Miranda. □ Los Teques (state capital). Acevedo (municipality), Caucagua (municipal capital). Andrés Bello (municipality), San José de Barlovento (municipal capital). Baruta (municipality), Nuestra Señora del Rosario (municipal capital). Brión (municipality), Higuero (municipal capital). Buroz (municipality), Mamporal (municipal capital). Carrizal. Chacao (municipalities and municipal capitals). Cristóbal Rojas (municipality), Charallave (municipal capital). El Hatillo (municipality and municipal capital). Guaicaipuro (municipality), Los Teques (municipal capital). Independencia (municipality), Santa Teresa del Tuy (municipal capital). Lander (municipality), Ocumare del Tuy (municipal capital). Los Salias (municipality), San Antonio de los Altos (municipal capital). Páez (municipality), Ro Chico (municipal capital). Paz Castillo (municipality), Santa Lucía (municipal capital). Pedro Gual (municipality), Cúpira (municipal capital). Plaza (municipality), Guarenas (municipal capital). Simón Bolívar (municipality), San Francisco de Yare (municipal capital). Sucre (municipality), Petare (municipal capital). Urdaneta (municipality), Cúa (municipal capital). Zamora (municipality), Guatire (municipal capital)
(872.29)	Apure. □ San Fernando (state capital). Achaguas. Biruaca (municipalities and municipal capitals). Muñoz (municipality), Bruzual (municipal capital). Páez (municipality), Guasdalito (municipal capital). Pedro Camejo (municipality), San Juan de Payara (municipal capital). Rómulo Gallegos (municipality), Elorza (municipal capital). San Fernando (municipality), San Fernando de Apure (municipal capital)	(872.36)	Distrito Federal. □ Libertador (municipality), Caracas (national and municipal capital) Vargas. □ La Guaira (state capital)
(872.32)	Barinas. □ Barinas (state capital, municipality and municipal capital). Alberto Arvelo Torrealba (municipality), Sabaneta (municipal capital). Andrés Eloy Blanco (municipality), El Cantón (municipal capital). Antonio José de Sucre (municipality), Socopo (municipal capital). Arismendi (municipality and municipal capital). Bolívar (municipality), Barinitas (municipal capital). Cruz Paredes (municipality), Barrancas (municipal capital). Ezequiel Zamora (municipality), Santa Bárbara (municipal capital). Obispos (municipality and municipal capital). Pedraza (municipality), Ciudad Bolivia (municipal capital). Rojas (municipality), Libertad (municipal capital). Sosa (municipality), Ciudad de Nutrias (municipal capital)	(872.37)	Aragua. □ Maracay (state capital). Bolívar (municipality), San Mateo (municipal capital). Camatagua (municipality and municipal capital). Francisco Linares Alcántara (municipality), Santa Rita (municipal capital). Girardot (municipality), Maracay (municipal capital). José Angel Lamas (municipality), Santa Cruz (municipal capital). José Félix Ribas (municipality), La Victoria (municipal capital). José Rafael Revenga (municipality), El Consejo (municipal capital). Libertador (municipality), Palo Negro (municipal capital). Mario Briceño Iragorry (municipality), El Limón (municipal capital). Ocumare de la Costa de Oro (municipality), Ocumare de la Costa (municipal capital). San Casimiro (municipality and municipal capital). San Sebastián (municipality and municipal capital). Santiago

	Mariño (municipality), Turmero (municipal capital), Santos Michelena (municipality), Las Tejerías (municipal capital), Sucre (municipality), Cagua (municipal capital), Tovar (municipality), La Colonia Tovar (municipal capital), Urdaneta (municipality), Barnacoas (municipal capital), Zamora (municipality), Villa de Cura (municipal capital)		
(872.39)	Carabobo. □ Valencia (state capital, municipality and municipal capital), Bejuma (municipality and municipal capital), Carlos Arvelo (municipality), Guigue (municipal capital), Caroní (municipality), Ciudad Guyana (municipal capital), Cedeño (municipality), Calcara del Orinoco (municipal capital), Diego Ibarra (municipality), Mariara (municipal capital), El Callao (municipality and municipal capital), Gran Sabana (municipality), Santa Elena de Uairén (municipal capital), Guacara (municipality and municipal capital), Juan José Mora (municipality), Morón (municipal capital), Libertador (municipality), Tocuyito (municipal capital), Los Guayos, Miranda, Montalbán, Naguanagua (municipalities and municipal capitals), Piar (municipality), Upata (municipal capital), Puerto Cabello, San Diego, San Joaquín (municipalities and municipal capitals)	(872.52)	Cruz de Taratara (municipal capital), Tocopero (municipality and municipal capital), Unión (municipality), Santa Cruz de Bucaral (municipal capital), Urumaco (municipality and municipal capital), Zamora (municipality), Puerto Cumarebo (municipal capital)
(872.42)	Yaracuy. □ San Felipe (state capital), Aristides Bastidas (municipality), San Pablo (municipal capital), Bolívar (municipality), Aroa (municipal capital), Bruzual (municipality), Chivacoa (municipal capital), Corocote, Independencia (municipalities and municipal capitals), José Antonio Páez (municipality), Sabana de Parra (municipal capital), La Trinidad (municipality), Boraure (municipal capital), Manuel Monge (municipality), Yuare (municipal capital), Nirgua (municipality and municipal capital), Peña (municipality), Yaritagua (municipal capital), San Felipe (municipality and municipal capital), Sucre (municipality), Guama (municipal capital), Urachiche (municipality and municipal capital), Verroes (municipality), Farriar (municipal capital)	(872.54)	Lara. □ Barquisimeto (state and municipal capital), Andrés Eloy Blanco (municipality), Sanare (municipal capital), Crespo (municipality), Duaca (municipal capital), Iribarren (municipality), Barquisimeto (municipal capital), Jiménez (municipality), Quibor (municipal capital), Morán (municipality), El Tocuyo (municipal capital), Palavecino (municipality), Cabudare (municipal capital), Simón Planas (municipality), Sarare (municipal capital), Torres (municipality), Carora (municipal capital), Urdaneta (municipality), Siquisique (municipal capital)
(872.44)	Falcón. □ Coro (state capital), Acosta (municipality), San Juan de los Cayos (municipal capital), Bolívar (municipality), San Luis (municipal capital), Buchivacoa (municipality), Capatárida (municipal capital), Cacique Manaure (municipality), Yaracal (municipal capital), Carirubana (municipality), Punto Fijo (municipal capital), Colina (municipality), La Vela de Coro (municipal capital), Dabajuro (municipality and municipal capital), Democracia (municipality), Pedregal (municipal capital), Falcón (municipality), Pueblo Nuevo (municipal capital), Federación (municipality), Churuguara (municipal capital), Jacura (municipality and municipal capital), Los Taques (municipality), Santa Cruz de los Taques (municipal capital), Mauroa (municipality), Mene de Mauroa (municipal capital), Miranda (municipality), Santa Ana de Coro (municipal capital), Monseñor Iturriza (municipality), Chichiriviche (municipal capital), Palmasola (municipality and municipal capital), Petit (municipality), Cabure (municipal capital), Píritu (municipality and municipal capital), San Francisco (municipality), Mirimire (municipal capital), Silver (municipality), Tucacas (municipal capital), Sucre (municipality), La	(872.62)	Trujillo. □ Trujillo (state capital, municipality and municipal capital), Andrés Bello (municipality), Santa Isabel (municipal capital), Boconó (municipality and municipal capital), Bolívar (municipality), Sabana Grande (municipal capital), Candelaria (municipality), Chejendé (municipal capital), Carache, Esuque (municipalities and municipal capitals), José Felipe Márques Cañizales (municipality), El Paradero (municipal capital), José Vicente Campo Elías (municipality), Campo Elías (municipal capital), La Ceiba (municipality), Santa Apolonia (municipal capital), Miranda (municipality), El Dividive (municipal capital), Monte Carmelo, Motatán, Pampán, Pampanito (municipalities and municipal capitals), Rafael Rangel (municipality), Betijoque (municipal capital), San Rafael de Carvajal (municipality), Carvajal (municipal capital), Sucre (municipality), Sabana de Mendoza (municipal capital), Urdaneta (municipality), La Quebrada (municipal capital), Valera (municipality and municipal capital)
			Mérida. □ Mérida (state capital), Alberto Adriani (municipality), El Vigía (municipal capital), Andrés Bello (municipality), La Azulita (municipal capital), Antonio Pinto Salinas (municipality), Santa Cruz de Mora (municipal capital), Aricagua (municipality and municipal capital), Arzobispo Chacón (municipality), Canagua (municipal capital), Campo Elías (municipality), Ejido (municipal capital), Caracciolo Parra Olmedo (municipality), Tucani (municipal capital), Cardenal Quintero (municipality), Santo Domingo (municipal capital), Guaraque (municipality and municipal capital), Julio César Salas (municipality), Arapuey (municipal capital), Justo Briceño (municipality), Torondoy (municipal capital), Libertador (municipality), Mérida (municipal capital), Miranda (municipality), Timotes (municipal capital), Obispo Ramos de Lora (municipality), Santa Elena de Arenales (municipal capital), Padre Noguera (municipality), Santa María de Caparo (municipal capital), Pueblo Llano (municipality and municipal capital), Rangel (municipality), Mucuchíes (municipal capital), Rivas Dávila (municipality), Bailadores (municipal capital), Santos Marquina (municipality), Tabay (municipal capital), Sucre (municipality),

(872.64)	Lagunillas (municipal capital). Tovar (municipality and municipal capital). Tulio Febres Cordero (municipality), Nueva Bolivia (municipal capital). Zea (municipality and municipal capital)	(873.9)	Bobores (municipal capital). Valmore Rodríguez (municipality), Bachaquero (municipal capital)
	Táchira. □ San Cristóbal (state capital, municipality and municipal capital). Andrés Bello (municipality), Cordero (municipal capital). Antonio Rómulo Costa (municipality), Las Mesas (municipal capital). Ayacucho (municipality), Colón (municipal capital). Bolívar (municipality), San Antonio del Táchira (municipal capital). Cárdenas (municipality), Tariba (municipal capital). Córdoba (municipality), Santa Ana del Táchira (municipal capital). Fernández Feo (municipality), San Rafael del Piñal (municipal capital). Francisco de Miranda (municipality), San José de Bolívar (municipal capital). García de Hevia (municipality), La Fria (municipal capital). Guasimos (municipality), Palmira (municipal capital). Independencia (municipality), Capacho Nuevo (municipal capital). Jáuregui (municipality), La Grita (municipal capital). José María Vargas (municipality), El Cobre (municipal capital). Junín (municipality), Rubio (municipal capital). Libertad (municipality), Capacho Viejo (municipal capital). Libertador (municipality), Abejales (municipal capital). Lobatera. Michelena (municipalities and municipal capitals). Panamericano (municipality), Coloncita (municipal capital). Pedro María Ureña (municipality), Ureña (municipal capital). Rafael Urdaneta (municipality), Delicias (municipal capital). Samuel Darío Maldonado (municipality), La Tendida (municipal capital). San Judas Tadeo (municipality), Umuquena (municipal capital). Seboruco (municipality and municipal capital). Simón Rodríguez (municipality), San Simón (municipal capital). Sucre (municipality), Queniquea (municipal capital). Torbes (municipality), San Josecito (municipality)	(88) (881) (881.11/.29) (881.11) (881.13) (881.15) (881.17) (881.19) (881.21) (881.23) (881.25) (881.27) (881.29) (882) (882.1) (882.11)	Dependencias Federales. □ La Tortuga. La Blanquilla. Los Roques. Isla de las Aves
(872.66)	Zulia. □ Maracaibo (state capital, municipality and municipal capital). Almirante Padilla (municipality), El Toro (municipal capital). Baralt (municipality), San Timoteo (municipal capital). Cabimas (municipality and municipal capital). Catatumbo (municipality), Encontrados (municipal capital). Colón (municipality), San Carlos del Zulia (municipal capital). Francisco Javier Pulgar (municipality), Pueblo Nuevo (El Chivo) (municipal capital). Jesús Enrique Lossada (municipality), La Concepción (municipal capital). Jesús María Semprún (municipality), Casigua el Cubo (municipal capital). La Cañada de Urdaneta (municipality), Concepción (municipal capital). Lagunillas (municipality), Ciudad Ojeda (municipal capital). Machiques de Perijá (municipality), Machiques (municipal capital). Mara (municipality), San Rafael de Moján (municipal capital). Miranda (municipality), Los Puertos de Altigracia (municipal capital). Páez (municipality), Sinamaica (municipal capital). Rosario de Perijá (municipality), La Villa del Rosario (municipal capital). San Francisco. Santa Rita (municipalities and municipal capitals). Simón Bolívar (municipality), Tía Juana (municipal capital). Sucre (municipality),	(883) (883.1/.2) (883.11) (883.12) (883.13) (883.14) (883.15) (883.16)	THE GUIANAS GUYANA. COOPERATIVE REPUBLIC OF GUYANA <i>Guyana is divided into 10 administrative regions</i> Divisions of Guyana Barima-Waini. □ Mabaruma (division capital). Matthew's Ridge. Moranhua. Port Kaituma Cuyuni-Mazaruni. □ Bartica (division capital). Arimu Mine. Issano. Kartabo Point. Peter's Mine Demerara-Mahaica. □ Georgetown (state capital). Paradise (division capital) East Berbice-Corentyne. □ New Amsterdam (division capital). Epira. Mara. Rose Hall. Springlands Essequibo Islands-West Demerara. □ Vreed en Hoop (division capital). Parika Mahaica-Berbice. □ Fort Wellington (division capital). Rosignol Pomeroon-Supenaam. □ Anna Regina (division capital). Charity. Spring Garden Potaro-Siparuni. □ Mahdia (division capital). Orinduik. Potero Landing Upper Demerara-Berbice. □ Linden (division capital). Ituni. Kwakwadi. Takama Upper Takutu-Upper Essequibo. □ Lethem (division capital). Dadanawa. Good Hope. Isherton. Kalkuni. Kurupukari
			FRENCH GUIANA. DEPARTMENT OF GUIANA (FRANCE). GUYANE FRANÇAISE <i>French Guiana is divided into two administrative regions (arrondissements)</i> Arrondissements of French Guiana Cayenne. □ Cayenne (state and arrondissement capital). Camopi. Guisanbourg. Iracoubo-Sinnamary. Kaw. Kourou. Macouria. Matoury. Montsinery. Ovanary. Regina. Rémine. Roura. Saint-Georges. Tonate. Isles Connétable, Isles Rémine. Iles du Salut, Devil's Island (Isle du Diable) Saint-Laurent-du-Maroni. □ Saint-Laurent-du-Maroni (arrondissement capital). Apatou. Grand Santi. Les Hattes. Mana. Saint Jean. Saül
			SURINAME. REPUBLIC OF SURINAME. REPUBLIEK SURINAME <i>Suriname is divided into ten distrikten (sing. distrikt: districts)</i> Districts of Suriname Nickerie. □ Nieuw Nickerie (district capital). Wageningen Coronie. □ Totness (district capital). Friendship. Jenny Saramacca. □ Groningen (district capital). Beskamp Wanga. □ Lelydorp Paramaribo (state capital and urban district) <i>The district of Sipaliwini is administered by Paramaribo</i> Commewijne. □ Nieuw Amsterdam (district capital)

(1/9) Table 1e. Common auxiliaries of place

(883.17)	Marowijne. □ Albina (district capital). Moengo	(899.33)	Río Negro. □ Fray Bentos (department capital). Algorta. Nuevo Berlín. Young
(883.18)	Para. □ Onverwacht (district capital).	(899.35)	Durazno. □ Durazno (department capital).
(883.19)	Paranam. Zanderij	(899.37)	Carmen. Sarandí del Yí
(883.21)	Brokopondo. □ Brokopondo (district capital). Brownswey	(899.39)	Treinta y Tres. □ Treinta y Tres (department capital). General Enrique Martínez. Vergara
	Sipaliwini. □ Anapaike. Apetina. Apoera. Bakhuis. Benzdorp. Bitagron. Cottica. Kajana. Kamp 52. Kwamalasamutu. Pelelu Tepu. Pokigron	(899.4)	Rocha. □ Rocha (department capital). La Paloma
(893)	PARAGUAY. REPUBLIC OF PARAGUAY <i>Paraguay is divided into 17 departamentos (departments) and one distrito capital (capital district)</i>	(899.41)	Southern departments
(893.1)	Northern departments	(899.43)	Lavalleja. □ Minas (department capital). José Battle y Ordoñez Solís
(893.11)	Alto Paraguay. □ Fuerte Olimpo (department capital). Capitán Pablo Lagerenza. Puerto Bahía Negra	(899.45)	Florida. □ Florida (department capital).
(893.12)	Boquerón. □ Filadelfia (department capital). Doctor Pedro P. Peña. General Eugenio A. Garay. Mariscal Estigarribia	(899.47)	Sarandí Grande, Casupa, Fray Marcos
(893.13)	Presidente Hayes. □ Pozo Colorado (department capital). Villa Hayes	(899.49)	Flores. □ Trinidad (department capital).
(893.14)	Concepción. □ Concepción (department capital). Puerto La Victoria. Puerto Pinasco	(899.51)	Soriano. □ Mercedes (department capital).
(893.15)	Amambay. □ Pedro Juan Caballero (department capital)	(899.53)	Soriano, Santa Catalina
(893.16)	San Pedro. □ San Pedro (department capital). Rosario	(899.54)	Colonia. □ Nueva Palmira, Carmelo, Juan Lacaze, Puerto del Sauce
(893.17)	Canendiyú. □ Salto del Guairá (department capital)	(899.55)	San José. □ San José (de Mayo) (department capital)
(893.3)	Central district and departments	(9)	Canelones. □ Canelones (department capital).
(893.31)	Asunción (capital district)		Atlántida, La Floresta. La Paz. Santa Lucía
(893.32)	Central. □ Mayor Pablo Lagerenza (department capital). Areguá. Limpio. Villeta. Ypacarai		Montevideo. □ Montevideo (state and department capital). Ramírez, Pocitos, Carrasco
(893.33)	Cordillera. □ Caacupé (department capital)		Maldonado. □ Maldonado (department capital).
(893.34)	Caaguazú. □ Coronel Orviedo (department capital)		Piriapolis. Solís. Punta del Este. La Sierra
(893.35)	Caazapá. □ Caazapá (department capital). Abai		STATES AND REGIONS OF THE SOUTH PACIFIC AND AUSTRALIA. ARCTIC. ANTARCTIC
(893.36)	Guairá. □ Villarrica (department capital)		<i>For the South Pacific, Australia, the Arctic and the Antarctic as physiogeographic regions, and for the physiogeographic units of these regions, use (292.94/.99), (292.91/.93), (292.2) and (292.3) and their subdivisions</i>
(893.37)	Alto Paraná. □ Ciudad del Este (department capital)	(931)	NEW ZEALAND
(893.5)	Southern departments	(931.2)	North Island
(893.51)	Paraguarí. □ Paraguarí (department capital)	(931.211)	Northland Region
(893.52)	Neembucú. □ Pilar (department capital)	(931.212)	Auckland Regional Authority
(893.53)	Misiones. □ San Juan Bautista (de las Misiones) (department capital). San Ignacio	(931.214)	Thames Valley Region
(893.54)	Itapúa. □ Encarnación (department capital)	(931.215)	Waikato Region
(899)	URUGUAY. EASTERN REPUBLIC OF URUGUAY. REPÚBLICA ORIENTAL DEL URUGUAY <i>Uruguay is divided into 19 departamentos (departments)</i>	(931.217)	Bay of Plenty Region
(899.2)	Northern departments	(931.218)	East Cape Region
(899.21)	Artigas. □ Artigas (department capital). Bella Unión. Tomás Gomensoro	(931.219)	Tongariro Region
(899.23)	Salto. □ Salto (department capital). Belé. Constitución. Lavalleja	(931.231)	Taranaki Region
(899.25)	Rivera. □ Rivera (department capital). Tranqueras	(931.251)	Hawke's Bay Region
(899.27)	Cerro Largo. □ Melo (department capital). Fraile Muerto. Río Branco	(931.261)	Wanganui Region
(899.29)	Tacuarembó. □ Tacuarembó (department capital). Paso de los Toros	(931.263)	Manawatu Region
(899.3)	Central departments	(931.265)	Horowhenua Region
(899.31)	Paysandú. □ Paysandú (department capital). Merinos. Porvenir	(931.267)	Wairarapa Region
		(931.268)	Wellington Region
		(931.3)	South Island. Stewart Island
		(931.312)	Nelson Region
		(931.314)	Marlborough Region
		(931.316)	West Coast Region
		(931.322)	Canterbury Region
		(931.324)	Aorangi Region
		(931.332)	Coastal-North Otago Region
		(931.334)	Clutha-Central Otago Region
		(931.353)	Southland Region
		(931.8)	Minor Islands. □ Chatham Islands, Kermadec Islands, Bounty Islands, Antipodes Islands, Auckland Islands, Campbell Island
		(932/935)	MELANESIA. STATES AND REGIONS OF MELANESIA
		(932)	New Caledonia. Territory of New Caledonia (France)
		(934)	Vanuatu. Republic of Vanuatu
		(935)	Solomon Islands

(938.2)	LORD HOWE ISLAND (AUSTRALIA)	(942.9)	Northern Territory
(938.4)	NORFOLK ISLAND (AUSTRALIA)	(942.91)	Southern district. □ Alice Springs. Uluru (Ayers Rock)
(94)	AUSTRALIA. COMMONWEALTH OF AUSTRALIA <i>The subdivision of Australia has been compiled using the Statistical Divisions of the 1991 Census. Each state is divided into the headings corresponding to these, with the exception of the category 'Offshore islands and migratory' used in every state, but not considered helpful for the present purpose, since many areas in this category have a moving, aboriginal population and are not settlements or towns. Islands that are inhabited have been listed under the appropriate Division of each State and the remainder are at (948). Major towns in each state have been listed, and all cities with a population of over 10,000 are included. The decision of what constitutes a major town, however, varies from state to state, since many in the South East of the continent are considerably larger than those of Western Australia or Northern Territory, for example</i>	(942.95)	Class here <i>Central Australia</i> (a term still in general use but of specific administrative meaning only between 1926 and 1931; today the term 'Central Australia' usually refers to Alice Springs and an area of about 700 km radius around it - generally a major tourist destination)
		(942.96)	Northern district. □ Arnhem Land. Jabiru. Katherine. Litchfield. Tennant Creek. Groote Eylandt. Melville Island. Bathurst Island. Darwin. □ Palmerston
		(943)	Queensland
		(943.1)	Brisbane. □ Beaudesert. Beenleigh. Caboolture. Ipswich. Logan. Redcliffe
		(943.2)	Southeastern Queensland
		(943.21)	Moreton. □ Caloundra. Gatton. Gold Coast. Maroochydore. Nambour. Surfers Paradise. Tewantin. Fraser Island. Moreton Island. Stradbroke Island
			⇒ (943.1)
		(943.25)	Wide Bay - Burnet. □ Bundaberg. Gympie. Hervey Bay. Kingaroy. Maryborough. Monto. Mundubberra. Wondai
(941)	Western Australia	(943.3)	Darling Downs. □ Chinchilla. Dalby.
(941.1)	Perth. □ Armadale. Belmont. Canning. Cockburn. Fremantle. Gosnells. Kwinana. Melville. Nedlands. Rockingham. Stirling. Subiaco. Wanneroo		Goondiwindi. Inglewood. Miles. Millmerran. Oakey. Pittsworth. Stanthorpe. Texas. Toowoomba. Warwick
(941.2)	South West. Great Southern. Midlands. Central	(943.4)	South West. □ Charleville. Cunnamulla.
(941.21)	South West. □ Bunbury. Bridgetown. Busselton. Collie. Mandurah. Manjimup ⇒ (941.1)	(943.5)	Longreach. Mitchell. Roma. St George
(941.23)	Lower Great Southern. □ Albany. Broomehill. Denmark. Gnowangerup. Katanning. King. Kojonup. Pallinup	(943.51)	Central Queensland
(941.25)	Upper Great Southern. □ Boddington. Coobaling. Narrogin. Wickopin. Williams	(943.55)	Fitzroy. □ Biloela. Emerald. Gladstone. Mount Morgan. Rockhampton. Yeppoon.
(941.27)	Midlands. □ Merredin. Northam. York	(943.6)	Northumberland Islands
(941.29)	Central. □ Carnarvon. Geraldton. Wiluna	(943.7)	Central West. □ Barcaldine. Blackall. Winton
(941.3)	Pilbara. □ Dampier. Karratha. Mount Magnet. Newman. Pannawonica. Paraburdoo. Port Hedland. Roebourne. Tom Price. Wickham. Barrow Island. Bernier and Dorre Islands. Dick Hartog Island	(943.8)	Mackay. □ Clermont. Mackay. Whitsunday (Cumberland) Islands
(941.4)	Kimberley. □ Broome. Derby. Halls Creek. Kununurra. Wyndham	(943.9)	Northern. □ Ayr. Bowen. Charters Towers. Ingham. Townsville. Hinchinbrook Island
(941.7)	South Eastern. □ Boulder. Coolgardie. Esperance. Kalgoorlie. Kambalda. Norseman	(943.9)	Far North. □ Atherton. Cairns. Mareeba. Torres Strait Islands. Tully
(942)	South Australia and the Northern Territory	(944)	North West. □ Cloncurry. Flinders. Hughenden. Innisfail. Mount Isa. Normanton. Richmond
(942.3)	South Australia	(944.1)	New South Wales
(942.31)	Adelaide. □ Elizabeth. Gawler. Noalunga. Salisbury		Sydney. □ Camden. The Entrance. Gosford. Hornsby. Liverpool. Katoomba. Parramatta. Penrith. Woy Woy
(942.32)	Outer Adelaide. □ Angaston. Barossa. Kapunda. Kingscote. Light. Mount Barker. Onkaparinga. Port Elliot. Strathalbyn. Tanunda. Victor Harbour. Kangaroo Island	(944.2)	Hunter. □ Cessnock. Dungog. Forster. Gloucester. Maitland. Musswellbrook. Newcastle. Scone. Singleton
(942.33)	Murray Lands. □ Barmera. Berri. Loxton. Mannum. Murray Bridge. Renmark. Tailem Bend	(944.3)	North Coast
(942.34)	South East. □ Bordertown. Kingston South East. Millicent. Mount Gambier. Naracoorte	(944.31)	Mid North Coast. □ Coffs Harbour. Grafton. Kempsey. Lord Howe Island. Port Macquarie. Taree
(942.35)	Yorke and Lower North. □ Clare. Kadina. Maitland. Moonta. Wallaroo	(944.35)	Richmond-Tweed. □ Ballina. Byron Bay. Casino. Kyogle. Lismore. Murwillumbah
(942.36)	Eyre. □ Ceduna. Cleve. Port Lincoln. Tumby Bay	(944.4)	Northern. □ Armidale. Barraba. Boggabri. Coonabarabran. Glen Innes. Gunnedah. Inverell. Moree. Murrumbidgee. Narrabri. Quirindi. Tamworth. Tenterfield
(942.37)	Northern. □ Quorn. Leigh Creek. Peterborough. Port Augusta. Port Pirie. Whyalla	(944.5)	Central West. □ Bathurst. Condobolin. Cowra. Crookwell. Dubbo. Forbes. Gilgandra. Grenfell. Lachlan. Lithgow. Mudgee. Orange. Parkes. Wellington
		(944.6)	Illawarra. □ Kiama. Mittagong. Moss Vale. Port Kembla. Shellharbour. Shoalhaven. Woolongong

(1/9) Table 1e. Common auxiliaries of place

(944.7)	South Eastern. □ Batemansbay. Bega. Bombala. Braidwood. Captains Flat. Cooma. Eden. Goulburn. Kiama. Moruya. Murrumburrah. Narooma. Nowra. Queanbeyan. Ulladulla. Yass. Young ⇒ (947)	(948)	Outlying islands of Australia. □ Ashmore and Cartier Islands. Coral Sea Islands ⇒ (592.4) ⇒ (592.5) ⇒ (699.14/.15) Norfolk Island
(944.8)	Southwestern New South Wales. Riverina	(948.2)	
(944.81)	Murray. □ Albury-Wodonga. Balranald. Corowa. Deniliquin. Wentworth	(954)	PAPUA NEW GUINEA. INDEPENDENT STATE OF PAPUA NEW GUINEA
(944.85)	Murrumbidgee. □ Cootamundra. Griffith. Gundagai. Kay. Junee. Leeton. Temora. Tumut. Wagga Wagga	(961/964)	POLYNESIA. STATES AND REGIONS OF POLYNESIA
(944.9)	Western New South Wales	(961.1)	Fiji. Republic of Fiji
(944.91)	North Western New South Wales. □ Bourke. Brewarrina. Cobar. Coonamble. Nyngan. Walgett. Warren	(961.2)	Tonga. Kingdom of Tonga
(944.95)	Far West New South Wales. □ Broken Hill	(961.3) (961.31)	Countries of the Samoan Islands Samoa. Independent State of Samoa (Samoa. Malo Sa'oloto Tuto'atasi o Samoa) <i>Formerly: Independent State of Western Samoa (Malolotuto'atasi o Samoa i Sisifo)</i> American Samoa (USA)
(945)	Victoria	(961.33)	
(945.1)	Melbourne. □ Cranbourne. Emerald. Healesville. Melton. Mornington. Pakenham. Rosebud. Sorrento. Sunbury. Werribee	(961.4)	Tuvalu. □ Niulakita (Nurakita), Nukulailai Atoll, Funafuti Atoll, Nukufetau Atoll, Vaitupu, Nui Atoll, Nanumanga, Niutao, Nanumea Atoll <i>Formerly Ellice Islands</i>
(945.2)	Barwon. □ Colac. Geelong. Queenscliffe. Torquay. Winchelsea ⇒ (945.1)	(961.6)	Howland Island. Baker Island (USA)
(945.3)	Loddon-Campaspe. □ Bendigo. Castlemaine. Charlton. Echua. Heathcote. Inglewood. Kyneton. Maldon. Maryborough. Rochester. Woodend	(961.7)	Wallis and Futuna Islands. Territory of Wallis and Futuna. Territoire de Wallis-et-Futuna (France)
(945.4)	Goulburn. □ Alexandra. Benalla. Broadford. Euroa. Kilmore. Kyabram. Mansfield. Nathalia. Numurkah. Rushworth. Seymour. Shepparton. Tatura. Yea	(961.9)	Line Islands (USA)
(945.5)	Ovens-Murray. □ Beechworth. Corryong. Mount Beauty. Myrtleford. Rutherglen. Tallangatta. Wangaratta. Yarrawonga ⇒ (944.81)	(962.3)	Cook Islands (New Zealand)
(945.6)	Gippsland. □ Bairnsdale. Korumburra. Leongatha. Moe. Morwell. Traralgon. Warragul. Wonthaggi. Yarram. French. Phillip Islands	(962.4)	Tokelau Islands (New Zealand)
(945.65)	East Gippsland. □ Lakes Entrance. Maffra. Orbost. Sale	(962.6)	Niue (New Zealand)
(945.7)	Western District. □ Camperdown. Casterton. Coleraine. Hamilton. Mortlake. Port Fairy. Portland. Terang. Warracknabeal	(963)	French Polynesia. Territory of French Polynesia. Territoire de la Polynésie Française (France). □ Îles Marquises. Îles Touamotu. Îles Gambier. Îles de la Société (Society Islands, including Tahiti). Îles Tubuai
(945.75)	Central Highlands. □ Ararat. Bacchus Marsh. Ballarat. Creswick. Daylesford	(964.1)	Pitcairn Islands (UK)
(945.8)	Wimmera. □ Dimboola. Donald. Horsham. Kaniva. Murtoa. Nhill. Saint Arnaud. Stawell. Warrnambool	(966/968)	MICRONESIA. STATES AND REGIONS OF MICRONESIA
(945.9)	Mallee. □ Birchip. Hopetoun. Irymple. Kerang. Merbein. Mildura. Ouyen. Red Cliffs. Robinvale. Sea Lake. Swan Hill	(966)	Trust Territory of the Pacific Islands (USA)
(946)	Tasmania	(966.1)	Federated States of Micronesia
(946.1)	Hobart. □ Kingston	(966.2)	Palau. Republic of Palau (Belau) (USA)
(946.2)	Southern Tasmania. □ Cygnet. Huonville. New Norfolk. Oatlands. Bruny Island ⇒ (946.1)	(966.3)	Northern Mariana Islands. Commonwealth of the Northern Mariana Islands
(946.3)	Northern Tasmania. □ Beaconsfield. Campbell Town. Deloraine. George Town. Launceston. Longford. Scottsdale. Westbury	(966.4)	Marshall Islands. Republic of the Marshall Islands
(946.5)	Mersey-Lyell. □ Burnie. Devonport. Latrobe. Penguin. Queenstown. Railton. Roseberry. Smithton. Somerset. Ulverstone. Wynyard. Zeehan	(967.2)	Guam (USA)
(946.7)	Offshore Islands. □ Bass Strait Islands. Furneaux Group. Hunter Group. Kent Group. Macquarie Island	(968.2)	Kiribati. Republic of Kiribati. □ Banaba (Ocean Island), the Gilbert Islands (Butaritari, Marakei, Abaiang, Tarawa, Maiana, Aberama, Kuria, Aranuka, Nonouti, Tabiteuea, Beru, Nikunau, Onotoa, Tamana, Arorae), the Phoenix Islands (Birnie, Rawaki/Phoenix, Enderbury, Kanton/Canton/Abariringa), Manra (Sidney), Orona (Hull), McKean, Nikumaroro/Gardner, certain of the Line Islands (Teraina/Washington, Tabuaeran/Fanning), Kiritimati/Christmas, Malden, Starbuck, Caoline, Flint, Vostok) <i>Formerly Gilbert Islands</i>
(947)	Australian Capital Territory		
(947.1)	Canberra		

(968.5)	Nauru. Republic of Nauru	(99)	ANTARCTIC TERRITORIES. SOUTH POLAR REGIONS
(97)	ISOLATED ISLANDS OF THE PACIFIC OCEAN		<i>In accordance with the 1961 Antarctic treaty, the territorial claims in Antarctica are now in abeyance. However, the colon combinations formerly listed here continue to be available at the user's discretion</i>
	<i>Class here those Pacific Islands which are not be found elsewhere in (5/9)</i>		
	⇒ (739.9)		
(971)	Wake Island (USA)		⇒ (160.26)
(972)	Johnston Atoll (USA)		⇒ (160.27)
(973)	Midway Islands (USA)		⇒ (211)
(976)	Clipperton (France)		⇒ (269)
(98)	ARCTIC TERRITORIES		⇒ (292.3)
(984)	European Arctic islands. □ Svalbard (Spitsbergen and Bjornoya/Bear Island) (Norway). Jan Mayen (Norway)		⇒ (699)
			⇒ (828.9)
(985)	Russian Arctic, including islands		⇒ (829.15)
(987)	North American Arctic, including islands		⇒ (829.3)
(988)	Greenland (Gronland) (Denmark)		⇒ (829.5)
	⇒ (489)		⇒ (835.38)
			⇒ (946.7)

(=...) TABLE 1f. COMMON AUXILIARIES OF RACE, ETHNIC GROUPING AND NATIONALITY

SCOPE. The concept of race is difficult to define and a sensitive subject, but the distinctive physical types of human are a fact, and important in (e.g.) medicine. Where race is defined by genetically determined physical characteristics, three races are commonly recognized - in alphabetical order, Caucasian, Mongoloid and Negroid. A fourth, Australoid, is recognized in some sources. In UDC, the system of dividing the ethnic auxiliaries like the language auxiliaries is useful for cultural-linguistic groupings ('Anglo-Saxon', Hispanic etc.), but is a clumsy way of denoting physically defined races, involving cumbersome range numbers such as (=11/=19) for caucasoids and (=415/=5) for negroids - while (=512.3) for mongoloids is positively misleading, since notationally it appears to exclude (=521) Japanese-speaking and (=581) Chinese-speaking peoples. The four races recognized in MeSH (Medical Subject Headings) are now enumerate at (=011/=014) and the derived numbers in (=11/=8) are clearly limited in linguistic-cultural concepts.

The common auxiliaries of ethnic grouping and nationality denote the nationality or ethnic aspects of a subject represented by a main UDC number, e.g.

398(=81/=82) North Amerindian (American Indian) folklore.

They are derived mainly from the common auxiliaries of language =... (Table 1c) and so may also usefully distinguish linguistic-cultural groups, e.g.

(=111) English speaking peoples (as opposed to the English).

Political nationality (citizenship of nation-states) may also be denoted, mainly by (=1:4/9) which derive from the common auxiliaries of place (4/9) (Table 1e), but it may be found that for some subjects the place auxiliaries themselves will often serve as well.

For ethnology as a main subject see 572.9

CITATION ORDER. In citation order, a common auxiliary of ethnic grouping or nationality normally follows a main UDC number; it may, however, be cited medially or even first in a compound number if there is a need to group together documents or references on particular ethnic groups or nationalities

⇒ 572.9

Systematic table

(=01)	Physically defined races	(=11/=8)	Linguistic-cultural groups, ethnic groups, peoples
(=011)	Caucasoid		(=11/=8) = =11/=8
(=012)	Mongoloid	(=11/=2)	Indo-European ethnic groups and peoples
(=013)	Negroid	(=11)	Germanic races and peoples
(=014)	Australoid	(=111)	English-speaking peoples
(=088)	Mixed race	(=112.2)	German-speaking peoples
(=1:1/9)	Peoples associated with particular places	(=13)	Romanic, 'Latin' races and peoples
	Denote details by colon combination with	(=133.1)	French-speaking, Gallic races and peoples
	place auxiliaries (Table 1e) and other ethnic	(=16)	Slavic (Slavonic) races and peoples
	auxiliaries from this table	(=2/=8)	Oriental, African and other ethnic groups and peoples
◇	(=1:100) Cosmopolitans		peoples
◇	(=1:2) Peoples of particular physiogeographic areas	(=21)	Indic races and peoples
◇	(=1:210.7) Islanders	(=214.58)	Romany peoples. Gipsies
◇	(=1:23) Mountain-dwellers	(=41)	Afro-Asiatic races and peoples
◇	(=1:24) Cave-dwellers. Troglodytes	(=411)	Semitic races and peoples
◇	(=1:253) Forest-dwellers. Jungle-dwellers	(=411.16)	Hebraic. Jews. Ethnically Jewish
◇	(=1:3) Particular nationalities of the ancient world	(=411.21)	See 26-051 for Jews by religion
◇	(=1:37) Ancient Romans	(=414/=45)	Arabic. Arabs. Arab
◇	(=1:4/9) Particular nationalities of the modern world		Various sub-Saharan African ethnic groups and peoples
◇	(=1:493) Belgians	(=521)	Japanese-speaking peoples. Ethnically Japanese people
◇	(=1:6=013) Black Africans	(=581)	Chinese-speaking peoples. Ethnically Chinese people
◇	(=1:73=013) Black Americans	(=62)	Austronesians
		(=622.82)	Polynesians
(=1-81)	Autochthonous (aboriginal) races and peoples	(=72)	Australian races and peoples. Australian aboriginal peoples
(=1-82)	Indigenes. Native, local-born inhabitants	(=81/=82)	North Amerindian (American Indian) peoples
	◇ (=1:72-82) Native Mexicans (Mexicans born in Mexico)	(=87)	Central and South Amerindian (American Indian) peoples

"..." TABLE 1g. COMMON AUXILIARIES OF TIME

SCOPE. The common auxiliaries of time denote the date, point of time or range of time of a subject represented by a main UDC number. They do not indicate the date of publication of a document, which is a cataloguing matter. The basis of date indication is the Christian calendar, but non-Christian systems of time reckoning are also allowed for (in "68" and "69"), as well as other time concepts, e.g. seasons and geologic time.

NOTATION. In calendar notation (see 'DATES' below), the point separates elements of different magnitudes (year-month-day). Otherwise, a point occurs after every third digit, as usual in UDC. Non-hierarchical numbering is introduced by the hash or asterisk, e.g. under "327" for months. Only arabic numerals are used, e.g. MM should be converted to "2000".

CITATION ORDER. The time auxiliary is normally cited after the main number, but since quotation marks are biterminal, they readily permit reversal of order or intercalation. E.g. 'Nineteenth-century chamber music' would normally be denoted by 785.7"18", but if a filing order giving higher priority to date is desired then 785"18"7, or even "18"785.7, is possible. Within the auxiliary itself, the time elements are cited in order of decreasing magnitude.

DATES. Dates are denoted by citing the ordinary calendar notation in the order year-month-day, enclosed in quotation marks, e.g.

"1898.12.11" 11 December 1898 CE.

The order of magnitudes (beginning with the greatest and ending with the least) corresponds to the principle of progression from general to special. In the interests of consistency, the year is always expressed as a four-digit number, and the month and day as two-digit numbers. The non-significant places are occupied by zeros, e.g.

"0435.08.04" 4 August 435 CE.

CHRISTIAN OR COMMON ERA AND PRE-CHRISTIAN ERAS. Dates BCE (Before Common era; also known as Before Christ/BC) and CE (Common era; also known as Anno Domini/AD) may be distinguished by prefixing the minus sign to dates BCE, e.g. "-0054" for 54 BCE, (and optionally the plus sign to dates CE, e.g. "+0043" for 43 CE). This method of distinction need only be used when references to both kinds of date are likely to occur.

"-" and "+" may be used without dates to denote the Before Common and Common era.

CENTURIES, DECADES. Centuries and decades may be denoted by 2 and 3 digits respectively, e.g.

"03" The 300s (loosely, the 4th century).

"19" The 1900s (loosely, the 20th century).

"192" The 1920s (1920-1929).

"200" The 2000s (first decade, i.e. 2000-2009).

RANGES OF TIME. Ranges of several centuries, decades or years may be denoted by the initial and final figures, using the slash or oblique stroke, e.g.

"04/14" The 5th to 15th centuries (the Middle Ages).

"1815/1830" The period from 1815 to 1830.

"625/627" Glacial and post-glacial periods.

When one of the limiting dates is undetermined, it is represented by three dots, e.g.

94(100)".../18" World history up to the (end of the) 19th century.

94(100)"19/..." World history from the (beginning of the) 20th century onwards.

SMALLER TIME DIVISIONS. If required, the exact hour, minute or second when an event occurred may be denoted, using two-digit notations separated by points, e.g.

"1898.12.07.15.46.03" 7 December 1898 at 15 hours, 46 minutes, 3 seconds

Systematic table

"-"	BCE (Before Christian era) (BC)
⇒ "679"	
"+"	CE (Christian era) (AD)

"0/2" DATES AND RANGES OF TIME (CE OR AD) IN CONVENTIONAL CHRISTIAN (GREGORIAN) RECKONING

Denote a millennium by 1 digit, a century by 2 digits, a decade by 3 digits and a year by 4 digits

"0"	FIRST MILLENNIUM CE	"000"	First decade of first century CE
"00"	First century CE	"0001"	First year. 1 CE

"..." Table 1g. Common auxiliaries of time

"01"	Second century CE	"15/20"	Sixteenth to twenty-first centuries CE. Modern times. □ (European) post-Renaissance period ⇒ "654"
"04/14"	Fifth to fifteenth centuries CE. □ (European) Middle Ages ⇒ "653"	"15"	Sixteenth century CE
"04"	Fifth century CE	"16"	Seventeenth century CE
"09"	Tenth century CE	"17"	Eighteenth century CE
"1"	SECOND MILLENNIUM CE	"18"	Nineteenth century CE
"10"	Eleventh century CE	"19"	Twentieth century CE
"14"	Fifteenth century CE	"2"	THIRD MILLENNIUM CE
		"20"	Twenty-first century CE

"3/7" TIME DIVISIONS OTHER THAN DATES IN CHRISTIAN (GREGORIAN) RECKONING

"3"	CONVENTIONAL TIME DIVISIONS AND SUBDIVISIONS: NUMBERED, NAMED ETC. <i>To indicate sub-units of time used generically, use the 3-digit subdivisions of "3" followed by the notation for the sub-unit</i> ◇ "329*01/*31" First to thirty-first (last) day of the month (i.e. any month) ◇ "347*0915" 09 15 hours (9.15 am) on any day	"329.2"	Tuesday
		"329.3"	Wednesday
		"329.4"	Thursday
		"329.5"	Friday
		"329.6"	Saturday
		"329.7"	Sunday
		"34"	Day and night phenomena. Hours or times of day
		"344"	Daytime. Daylight hours
"31"	General time indications	"344.2"	Daybreak. Dawn. Sunrise period
"317"	Conventional time indications ◇ "317GMT" Greenwich Mean Time	"344.3"	Morning
		"344.5"	Midday. Noon period
"32"	The year. Seasons and other divisions of the year	"344.7"	Afternoon
		"345"	Night-time. Hours of darkness or semi-darkness
"321/324"	Seasons ⇒ 581.54 ⇒ 591.54	"345.2"	Dusk. Sunset period
		"345.3"	Evening
"321"	Spring	"345.5"	Middle of the night. Midnight
"322"	Summer	"345.7"	The small hours, until dawn
"323"	Autumn (fall)	"347"	Specified times of day or periods of a day, expressed in hours, minutes, seconds ◇ "347*08" 08 00 hours (8 am), or the period from 08 00 to 08 59 ◇ "347*0915" 09 15 hours (9.15 am) ◇ "347*0915/1730" The period from 09 15 hours (9.15 am) to 17 30 (5.30 pm) ◇ "347*1425.10" 14 25 hours and 10 seconds (2.25 and 10 seconds pm)
"324"	Winter		
"325"	Quarters (quarter years, trimesters) ◇ "325*1/*4" First to fourth (last) quarter		
"326"	Periods shorter than seasons but longer than months		
"327"	Months. □ Calendar months (numbered or named) ◇ "327*01" Month 01. January ◇ "327*02" Month 02. February ◇ "327*03" Month 03. March ◇ "327*04" Month 04. April ◇ "327*05" Month 05. May ◇ "327*06" Month 06. June ◇ "327*07" Month 07. July ◇ "327*08" Month 08. August ◇ "327*09" Month 09. September ◇ "327*10" Month 10. October ◇ "327*11" Month 11. November ◇ "327*12" Month 12. December	"348"	Decimal divisions of hours ◇ "348.001/.009" Thousandths of hours ◇ "348.01/.09" Hundredths of hours ◇ "348.1/.9" Tenths of hours
"327.4"	Lunar (synodic) months	"36"	Times of peace, war, danger, emergency, difficulties
"328"	Weeks. □ Numbered weeks (of the year or month) ◇ "328*01/*53" First to fifty-third (last) week of the year ◇ "328*1/*5" First to fifth (last) week of the month	"362"	Peacetime. Time of no danger <i>Use this number only if it is clear that a chronological aspect is intended. If in doubt, use :327.57</i> ⇒ 341.231.12
		"363"	Time of danger, threat <i>Use this number only if it is clear that a chronological aspect is intended. If in doubt, use :355.271</i>
"328.6"	Weekend	"363.3"	Time of tension, e.g. cold war <i>Use this number only if it is clear that a chronological aspect is intended. If in doubt, use :327.54</i>
"329"	Days. □ Named and numbered days ◇ "329*001/*366" First to three-hundred-and-sixty-sixth (last) day of the year ◇ "329*01/*31" First to thirty-first (last) day of the month	"363.5"	Time of emergency <i>Use this number only if it is clear that a chronological aspect is intended. If in doubt, use :355.271</i> ⇒ -022.326.5 ⇒ 338.246.8 ⇒ 342.76
"329.1"	Monday		

"364"	Wartime <i>Use this number only if it is clear that a chronological aspect is intended. If in doubt, use :355/359</i> ⇒ 341.3 ⇒ 623	"374.7"	Time lost through personal reasons. □ Through parental needs, attendance at family events, weddings, funerals. Compassionate leave
"366"	Time of recovery, relaxation, détente, diminishing danger <i>Use this number only if it is clear that a chronological aspect is intended. If in doubt, use :327.56</i>	"374.8"	Time lost through wilful absenteeism, malingering, labour conflicts etc.
"367"	Times according to volume of use, load, demand	"374.9"	Time lost through other causes
"367.2"	Time of peak demand, maximum use. Peak times (peak hours). □ Rush hour	"375"	Time for work or availability for work outside working hours
"367.3"	Times of average demand or load, middling use	"375.2"	Overtime ⇒ 331.313.6 ⇒ 331.314.2
"367.4"	Off-peak times. Hours of low demand or load, little use. Slack periods	"375.4"	Time of readiness for work or duty. Time on standby, on call, on reserve
"368"	Times according to permission, prescription and prohibition	"376"	Training, instruction, learning, study time. Release periods for study <i>Denote duration by "376:4..."</i> ◇ "376:430*2" Two-week study period ◇ "376:45" School year. Study year ⇒ 371.23, 378.143
"368.2"	Permitted times or periods	"377"	Rest and recreation time. Spare time. Free time. Time outside working hours ⇒ "372.4" ⇒ "374.2"
"368.3"	Excess time. Periods beyond the permitted limit	"377.2"	Rest time. Sleep time
"368.5"	Times of prohibition. Prohibited times or periods	"377.4"	Recreation and leisure time outside working hours ⇒ "372.4" ⇒ "374.2"
"368.7"	Recommended or prescribed times or periods	"378"	Time for daily routine, meals etc. (other than working time and leisure time)
"37"	Time of work activity, occupation, production, daily routine	"378.1"	Waking, rising, dressing time. Getting-up time ⇒ "344.2"
"372"	Working hours. Service hours. Time of occupation ⇒ 331.31	"378.3"	Time of morning meal. Breakfast time
"372.1"	Starting time. Opening time. Clocking-on (clocking-in) time	"378.5"	Time of midday meal. Lunchtime
"372.2"	Working hours. Business hours. Real working time. Productive hours	"378.7"	Time of evening meal. Dinner time. Supper time
"372.3"	Hours of opening to the public (of shops, administration etc.). Service hours. Trading hours	"378.9"	Bedtime
"372.4"	Time for leisure facilities, during working hours ⇒ "374.2" ⇒ "377.4"	"38"	Holidays. Festive and commemorative occasions
"372.5"	Finishing time. Closing time. End of working hours. Clocking-off (clocking-out) time	"382"	Religious holidays, festive and commemorative occasions <i>Prefer main numbers where available</i> ⇒ "69"
"372.6"	Full-time (e.g. working, occupation)	"383"	Public, national or regional holidays (other than religious) ◇ "383:321"(410) Spring public holiday ('Bank holiday') in Britain ◇ "383:327*07.04"(73) July 4th (Independence Day) in USA
"372.7"	Part-time (e.g. working, occupation)	"385"	Personal private holidays, vacation or leave time ◇ "385:322" Summer holiday (summer vacation) time
"372.8"	Divided time. Shared time	"387"	Festive and commemorative occasions (other than holidays) <i>When in doubt, prefer main numbers</i>
"373"	Time for maintenance, renovation, repair, household work	"388"	Personal anniversaries ◇ 392.5"388" Wedding anniversary
"373.2"	Renovation, alteration, inspection time	"4"	DURATION. TIME-SPAN. PERIOD. TERM. AGES AND AGE-GROUPS <i>The subdivisions of "4" denote all kinds of duration, including periods or ages in relation to activities, processes, events, things, persons, groups, institutions, nations etc. The elements in the number denote, following an asterisk, firstly, unit of time; secondly, duration in terms of that unit; and then, separated by points, successively smaller time-units (mostly two digits). The slash is used to denote time-spans or age ranges.</i>
"373.3"	Maintenance time		
"373.5"	Repair time		
"373.8"	Time for household work. Cleaning time		
"374"	Lost time. Wasted time. Idle periods ⇒ 331.316		
"374.2"	Break time. Rest periods during working hours. Meal breaks. Tea, coffee breaks etc. Time spent on personal needs ⇒ "372.4" ⇒ "377" ⇒ "378" ⇒ 331.313.4		
"374.3"	Time spent on travel, transport, to and from workplace. Travelling time ⇒ 331.315		
"374.4"	Time lost through internal factors at workplace		
"374.5"	Time lost through external factors. □ Through weather, natural disasters, acts of war		
"374.6"	Time lost through illness, injury, accidents. Sick leave ⇒ 331.316.2		

"..." Table 1g. Common auxiliaries of time

	For properties related to temporal values see -022.3.	"430"	Duration of 1 to less than 10 weeks
	For human age-groups, prefer -053		◇ "430*1" Duration of 1 week
	◇ "414*03.14.46.012" Duration of 3 hours, 14 minutes, 46 seconds, 12 milliseconds	"431"	◇ "430*2" Duration of 2 weeks. Fortnights
	◇ "440*8" Period of 8 months		Duration of 10 to less than 100 weeks
	◇ "451*12.06" Period of 12 years, 6 months	"432"	◇ "431*40" Duration of 40 weeks
	◇ "463*07" Seven weeks old		Duration of 100 to less than 1000 weeks
	◇ "465*13/19" Age range 13 to 19 years	"433"	◇ "432*105" Duration of 105 weeks
	⇒ -053.6	"437"	Duration of 1000 or more weeks
"40"	Relative duration, period, term		The working week. The week conventionally defined in terms of occupied hours
"401"	Life or lifetime with definite term or deadline.		◇ "437.5" 5-day week
	Period of durability	"44"	◇ "437.7" 7-day week
	⇒ -027.45	"440"	Duration expressed in months
"401.3"	Economic life. Durability from economic point of view		Duration of 1 to less than 10 months
"401.4"	Useful life. Operating life. Durability in relation to use		◇ "440*1" Duration of 1 month
	⇒ 620.193.95		◇ "440*3" Duration of 3 months. Quarters (quarter years). Trimesters
"401.7"	Lifetime duration. For life. During one's lifetime		◇ "440*4" Duration of 4 months
	⇒ "408"		◇ "440*6" Duration of 6 months. Half years. Semesters
"403"	Short duration. Short-term	"441"	Duration of 10 to less than 100 months
"404"	Medium duration. Medium-term		◇ "441*33" Duration of 33 months
"405"	Long duration. Long-term	"442"	Duration of 100 to less than 1000 months
"408"	Indefinite, unlimited duration. Without specific term or deadline		◇ "442*120" Duration of 120 months
"409"	Definite duration. With specific term or deadline	"45"	Duration expressed in years, centuries, millenniums, etc.
"41"	Duration up to one day		"45..." corresponds broadly to "42..."
"411"	Duration in fractions of a second (of periods less than a second)		⇒ "0/2"
			⇒ "67"
"411.001/.009"	Duration expressed in milliseconds (of periods less than 1/100 of a second)	"450"	Duration of 1 to less than 10 years
"411.01/.09"	Duration expressed in centiseconds (of periods less than 1/10 of a second)		◇ "450*1" Duration of 1 year
"411.1/.9"	Duration expressed in deciseconds (of periods less than 1 second)		◇ "450*3" Duration of 3 years. Trienniums
"412"	Duration expressed in seconds (of periods less than 1 minute)		◇ "450*4" Duration of 4 years. Quadrenniums, e.g. Olympiads
	◇ "412*46" Duration of 46 seconds	"451"	◇ "450*5" Duration of 5 years. Quinquenniums
	◇ "412*46.012" Duration of 46 seconds, 12 milliseconds		Duration of 10 to less than 100 years
"413"	Duration expressed in minutes (of periods less than 1 hour)		◇ "451*10" Duration of 10 years. Decenniums. Decades
	◇ "413*14" Duration of 14 minutes		◇ "451*12.06" Duration of 12 years, 6 months (12.5 years)
	◇ "413*14.46.012" Duration of 14 minutes, 46 seconds, 12 milliseconds	"452"	◇ "451*35" Duration of 35 years (3.5 decades)
"414"	Duration expressed in hours		◇ "451*99" Duration of 99 years
	◇ "414*03" Duration of 3 hours		Duration in centuries (or 3-figure periods of years)
	◇ "414*03.14.46.012" Duration of 3 hours, 14 minutes, 46 seconds, 12 milliseconds	"453"	◇ "452*2" Duration of 2 centuries
"42"	Duration expressed in days		◇ "452*2.50" Duration of 250 years (2.5 centuries)
"420"	Duration of 1 to less than 10 days		◇ "452*9" Duration of 9 centuries
	◇ "420*1" Duration of 1 day		Duration in millenniums
	◇ "420*3" Duration of 3 days		◇ "453*1" Duration of 1 millennium (1000 years)
	◇ "420*3.12" Duration of 3 days, 12 hours	"454"	◇ "453*2.25" Duration of 2.25 millenniums (2250 years)
	◇ "420*9" Duration of 9 days	"455"	Duration in tens of thousands of years
"421"	Duration of 10 to less than 100 days	"456"	Duration in hundreds of thousands of years
	◇ "421*12" Duration of 12 days		Duration in millions of years
	◇ "421*40" Duration of 40 days		⇒ "67"
"422"	Duration of 100 to less than 1000 days	"46"	Ages and age groups of persons, animate and inanimate things
	◇ "422*280" Duration of 280 days		⇒ "712"
"423"	Duration of 1000 or more days	"461/464"	Age expressed in hours and fractions of hours, days, weeks, months
	⇒ "44"		"461/464" ≡ "41/44"
	⇒ "45"		Age of up to one day
"427"	The working day. The day conventionally defined in terms of occupied hours	"461"	◇ "461.4*03.15" Age of 3 hours, 15 minutes
	◇ "427.08" 8-hour day	"462"	Age in days
	◇ "427.12" 12-hour day		◇ "462.1*11" Age of 11 days. 11 days old
	⇒ "37"	"463"	Age in weeks
"43"	Duration expressed in weeks		◇ "463.0*5" Age of 5 weeks. 5 weeks old

"464"	Age in months ◇ "464.1*10" Age of 10 months. 10 months old	"521"	At intervals of 10 to less than 100 days ◇ "521*12" Every 12 days
"465"	Age expressed in years and decades ◇ "465*02.06" Age of 2 years, 6 months (2.5 years old) ◇ "465*06/11" Age group 6 to 11 years ◇ "465*06" Age of 6 years. 6 years old ◇ "465*13/19" Age group 13 to 19 years. Teenagers ⇒ -053.6 ◇ "465*9" Age of 90 years or more. Nonagenarians	"522"	At intervals of 100 to less than 1000 days ◇ "522*280" Every 280 days
"467"	Age expressed in centuries (to less than 1000 years) ◇ "467*1" Age range 100 to 199 years ◇ "467*110" Age of 110 years. 110 years old ◇ "467*8" Age of 800 years. 8 centuries old	"523"	At intervals of 1000 or more days
"468"	Age expressed in millennia (to less than 10 000 years) ◇ "468*1" Age of 1 millennium (1000 years). 1000 years old ⇒ "679"	"53"	Recurring at intervals of 1 week or more (expressed in weeks) <i>Details by colon combination with "3..."</i> ◇ "53:329" Recurring on a particular day of the week
"469"	Age expressed in larger time units, e.g. millions of years ⇒ "62" ⇒ "67"	"530"	At intervals of 1 to less than 10 weeks ◇ "530*1" Once per week. Weekly ◇ "530*2" Every 2 weeks. Fortnightly
"5"	PERIODICITY. FREQUENCY. RECURRENCE AT SPECIFIED INTERVALS	"531"	At intervals of 10 to less than 100 weeks ◇ "531*40" Every 40 weeks
"51"	Recurring at intervals of less than a day. More frequent than daily	"532"	At intervals of 100 to less than 1000 weeks
"511"	More frequent than once per second ◇ "511*001" Once per millisecond ◇ "511*25" Every 0.25 of a second (25 centiseconds). 4 times per second ◇ "511*5" Every 0.5 of a second (5 deciseconds). Twice per second	"533"	At intervals of 1000 or more weeks
"512"	At intervals of 1 second or more (to less than 1 minute) ◇ "512*01" Once per second. Every second ◇ "512*05" Every 5 seconds. 12 times per minute ◇ "512*12" Every 12 seconds. 5 times per minute ◇ "512*30" Every 30 seconds (half-minute). Twice per minute	"54"	Recurring at intervals of 1 month or more (expressed in months) <i>Details by colon combination with "3..."</i> ◇ "54:328" Recurring in a particular week of the month ◇ "54:329" Recurring on a particular day of the month
"513"	At intervals of 1 minute or more (to less than 1 hour) ◇ "513*02" Every 2 minutes ◇ "513*15" Every 15 minutes. Quarter-hourly. 4 times per hour ◇ "513*30" Every 30 minutes. Half-hourly. Twice per hour	"540"	At intervals of 1 to less than 10 months ◇ "540*1" Once per month. Monthly ◇ "540*2" Every 2 months ◇ "540*3" Every 3 months. Quarterly. Trimestrial ◇ "540*6" Every 6 months. Half-yearly. Semiannual
"514"	At intervals of 1 hour or more (to less than 1 day) ◇ "514*01" Once per hour. Hourly ◇ "514*03" Every 3 hours. 8 times per day ◇ "514*12" Every 12 hours. Twice per day	"541"	At intervals of 10 to less than 100 months ◇ "541*18" Every 18 months
"52"	Recurring at intervals of 1 day or more (expressed in days) <i>Details by colon combination with "3..."</i> ◇ "52:344" Recurring during daylight hours ◇ "52:345" Recurring during hours of darkness ◇ "52:347" Recurring at specific times of day	"542"	At intervals of 100 to less than 1000 months
"520"	At intervals of 1 to less than 10 days ◇ "520*1" Once per day. Daily ◇ "520*2" Every 2 days. Every other day ◇ "520*3" Every 3 days. Twice per week ◇ "520*9" Every 9 days	"543"	At intervals of 1000 or more months
		"55"	Recurring at intervals of 1 year or more (expressed in years, centuries, millennia) <i>Details by colon combination with "3..."</i> ◇ "55:327" Recurring in a particular month of the year ◇ "55:329" Recurring on a particular day of the year
		"550"	At intervals of 1 to less than 10 years ◇ "550*1" Once per year. Yearly. Annual ◇ "550*2" Every 2 years. Biennial ◇ "550*3" Every 3 years. Triennial
		"551"	At intervals of 10 to less than 100 years ◇ "551*10" Every 10 years. Decennial
		"552"	At intervals of 100 to less than 1000 years ◇ "552*1" Once per century. Every 100 years. Centennial ◇ "552*2" Every 2 centuries. Bicentennial
		"553"	At intervals of 1000 or more years ◇ "553*1" Once per millennium. Every 1000 years. Millennial
		"554"	At intervals of tens of thousands of years
		"555"	At intervals of hundreds of thousands of years
		"556"	At intervals of millions of years
		"56"	Recurring every generation, lifetime etc. Life-cycle "56" = "46"
		"6"	GEOLOGICAL, ARCHAEOLOGICAL AND CULTURAL TIME DIVISIONS. □ Universal reckoning: YBP (Years Before Present), MYBP (Million Years Before Present). Other non-Christian (non-Gregorian) time reckonings

"..." Table 1g. Common auxiliaries of time

"61/62"	Geological (lithological/biological/palaeoecological) time division. □ Aeons (eons). Eras. Periods ⇒ 551.7	"654"	Modern times (Renaissance to contemporary 21st century times) ⇒ "15/20"
"611"	Precambrian (more than 600 MYBP)	"67/69"	Time reckonings: universal, secular, non-Christian religious
"611.1"	Archaean. Archaeozoic	"67"	Universal time reckoning. Before Present <i>IMPORTANT: The 3-digit subdivisions of "67" provide a forward chronological sequence, avoiding problems of filing dates in reverse order. However, they overlap other time auxiliaries in meaning, and must be used with discretion. If "672/678" are used for geological periods, then "61/63" may not be used. If "679" is used, then calendar notation may not be used. Fourth and subsequent digits may be added (after a point) to denote Years Before Present (YBP) or Million Years Before Present (MYBP), but be aware that this will enumerate dates in reverse chronological order</i>
"611.2"	Eozoic. Proterozoic. Algonkian		◇ "676.4" 4 MYBP
"613"	Palaeozoic (600-220 MYBP)		◇ "676.3" 3 MYBP
"613.1"	Cambrian (600-490 MYBP)	"672"	10 billion (10 thousand million) YBP or more. The Beginning. Time of initial explosion of the Universe
"613.2"	Ordovician (490-430 MYBP)	"673"	Less than 10 billion to 1 billion YBP (1000 MYBP). □ Time of origin of the Solar System to time of origin of Kiruna iron ore
"613.3"	Silurian (430-400 MYBP)	"674"	Less than 1 billion YBP (1000 MYBP) to 100 MYBP. □ Cambrian and Carboniferous periods
"613.4"	Devonian (400-350 MYBP)	"675"	Less than 100 MYBP to 10 MYBP. □ Beginning of Tertiary period
"613.5"	Carboniferous (350-270 MYBP)	"676"	Less than 10 MYBP. □ Approximate time of transition from animal to human. Time of Homo habilis
"613.6"	Permian (270-220 MYBP)	"677"	Less than 1 MYBP to 100 000 YBP. □ Time of Pithecanthropus. Time of Swanscombe Man
"615"	Mesozoic (220-70 MYBP)	"678"	Less than 100 000 to 10 000 YBP. □ Time of late Neanderthal Man, e.g. at Mount Carmel.
"615.1"	Triassic (220-180 MYBP)	"679"	Time of Cro-Magnon Man Less than 10 000 YBP <i>Alternatively, dates BC and CE may be denoted by "-9999/-0001" and "0/2" (calendar notation); use only one option or the other</i>
"615.2"	Jurassic (180-135 MYBP)		◇ "679.8" 8000 YBP
"615.3"	Cretaceous (135-70 MYBP)		◇ "679.1" 1000 YBP
"62"	Cenozoic (Cainozoic). Neozoic (70 MYBP - present)	"68"	Secular time reckonings other than universal and the Christian (Gregorian) calendar <i>Specify by (3/9) or (=...), with actual dates denoted by four-digit numbers</i>
"623"	Tertiary (70-1 MYBP)		◇ "68(37)" Dates in ancient Roman time reckoning 'Ab Urbe Condita'
"623.1"	Palaeogene (70-25 MYBP). □ Palaeocene. Eocene. Oligocene		◇ "68(37)0001" Supposed year of the foundation of Rome, 1 AUC, i.e. 753 BC
"623.5"	Neocene (25-1 MYBP). □ Miocene. Pliocene		◇ "68(37)0753" 753 AUC in the Roman reckoning, i.e. 1 CE
"624/627"	Pleistocene. Quaternary (1 MYBP - present)		◇ "68(=821.16)" Time reckoning of the Mayas
"624"	Preglacial. Calabrian. Villafranchian. Blancan		
"625"	Glacial/Interglacial/Interstadial cycle (G/I/IS). Ice ages	"69"	Dates and time units in non-Christian (non-Gregorian) religious time reckonings <i>Add other time auxiliaries to the subdivisions as required, and denote actual dates by four-digit numbers</i> "69" ≡ 21/26 "69" ≡ 28/29
"626"	Late Glacial. Late Pleistocene. Late Quaternary (14 000-10 300 YBP)		
"627"	Postglacial. Flandrian. Holocene (10 300 YBP - present)	"694"	Buddhist time reckoning "694" is derived from 24 ◇ "694:327.02" Second month of the Buddhist year
"63"	Archaeological, prehistoric, protohistoric periods and ages <i>Details by (292...) for regions, (3/9) for countries</i> ⇒ 7.031.1		
"631/634"	Stone Age		
"631"	Eolithic period		
"632"	Palaeolithic period		
"632.2"	Lower Palaeolithic		
"632.3"	Middle Palaeolithic		
"632.5"	Upper Palaeolithic		
"632.8"	Epi-Palaeolithic		
"633"	Mesolithic period		
"634"	Neolithic period		
"634.2"	Pre-pottery Neolithic		
"634.3"	Early Neolithic		
"634.5"	Middle Neolithic		
"634.7"	Late Neolithic		
"636/637"	Copper and Bronze Ages		
"636"	Chalcolithic. Copper Age		
"637"	Bronze Age		
"637.3"	Early Bronze Age		
"637.5"	Middle Bronze Age		
"637.7"	Late Bronze Age		
"638"	Iron Age		
"638.3"	Early Iron Age		
"638.5"	Middle Iron Age		
"638.7"	Late Iron Age		
"639"	Protohistoric period (earliest historic period). Period of first written records (of uncertain date) <i>For later periods and records with ascertainable dates, use "-..."</i>		
"65"	Periods of civilizations and cultures <i>For specified periods by date use "-...", "0/2"</i>		
"652"	Ancient times (historic). Antiquity. □ Oriental antiquity. Classical antiquity		
"653"	Mediaeval times. Middle ages ⇒ "04/14"		

"696"	Jewish religious time reckoning "696" is derived from 26 ◇ "696:550*1" Once every Jewish year	"712.4"	Early period. Time of immaturity, youth, growth ⇒ -027.14 ⇒ -044.88 ⇒ 7.03'01
"698"	Islamic time reckoning "698" is derived from 28 ◇ "698*0001" Year of the Hejira (flight of Mohammed from Mecca to Medina), i.e. 622 CE ◇ "698:327*09" Ramadan, ninth month of the Islamic year	"712.7"	Period of flowering, flourishing. Peak. Heyday. Prime. Zenith. Maturity ⇒ 005.414 ⇒ 159.922.62 ⇒ 7.03'02'04
"7"	PHENOMENA IN TIME. PHENOMENOLOGY OF TIME <i>The subdivisions of "7" overlap with some of the common auxiliaries of properties, -02... and processes, -04... Use "7..." only if there is clearly a chronological aspect</i>	"712.8"	Late period. Period of ripeness, fulfilment ⇒ 7.03'06
		"712.9"	Final period. Period of decline. Completion. End ⇒ -043.92/93 ⇒ -044.94 ⇒ -044.967 ⇒ 005.416
"71"	Evolution. Development. Progression ⇒ -027.1	"713"	Aftermath. Period of successors, epigones. Subsequent. Following. Post-... ⇒ -021.68
"711"	Time of precursors, forerunners. Time of preparation. Preceding. Before. Pre-... ⇒ -021.67 ⇒ -027.14	"714"	Transition period. Period of diffusion, dispersal, dissemination ⇒ -027.19
"712"	Period of existence ⇒ -021.121 ⇒ -021.142.5	"715"	Period of reactivation, revival, renaissance, renewal ⇒ -043.95 ⇒ -048.37/38 ⇒ 005.418
"712.3"	Beginnings. Earliest phase. Time of origin, generation, creation, infancy. Embryonic, incipient ⇒ -027.14 ⇒ -043.83	"78"	Indefiniteness of time. Indeterminate, uncertain in time. Random in time ⇒ -022.345.4

***, A/Z TABLE 1h. SUBJECT SPECIFICATION BY NOTATIONS FROM NON-UDC SOURCES**

0. GENERAL

Non-UDC notation and alphabetic extension may be used to increase the specificity of a UDC class mark. If they relate to a facet denoted by a biterminal auxiliary (see Tables 1a and 1b), they should be placed within the biterminal signs, but otherwise they should be placed outside the signs. In the example (492...) below, the asterisked component is part of the place facet - Netherlands place name codes - and is therefore inside the parentheses; in the example 625.711.1(4)*E4, it is not part of the place facet, but introduces a new characteristic - highway identifiers - and is therefore placed after the closing parenthesis. Similarly, in (492.83Utrecht), the alpha component is part of the place facet, whereas in 005.71-027.563(430)AEG it is not: it introduces a new characteristic - names of firms.*

1. THE ASTERISK *

A. *The asterisk * introduces a notation which is not an authorized UDC number.*

B. *The asterisk may follow a UDC number to introduce a word, symbol or number from a non-UDC source, which is added in order to specify a subject. For all such codings, the asterisk serves to mark their separation from the UDC numbers, e.g.*

(492*...) Netherlands: local subdivisions (4-figure code numbers from the Ned. PTT Plaatsnamencode).

523.44*433 Minor planet Eros (IAU authorized number).

546.42.027*90 Strontium 90 (atomic mass number of isotope).

625.711.1(4)*E4 Europa highway 4 (E4), in road engineering.

66-97*C150 Temperature of 150° Centigrade.

796.83*kg51 Flyweight (maximum 51 kg), in boxing.

Where the use of a non-UDC code following the asterisk is recommended, the source of the code is stated in a note.

C. *The asterisk may also be used to indicate an unauthorized, locally assigned notation, usually resembling a UDC number, for a concept which is missing in UDC.*

2. DIRECT ALPHABETICAL SPECIFICATION

For alphabetical specification by proper names (or their abbreviations) and acronyms, the letter notations may be added direct to the UDC base number, with no asterisk or blank space intervening. Such alphabetical specification of the auxiliaries of place (Table 1e) should be within the brackets. E.g.

(492.83UTR) City of Utrecht.

005.71-027.563(430)AEG Business firms - Germany - AEG (the former Allgemeine Elektrizitätsgesellschaft).

597.556.255HIP The sea-horse, in zoology.

821.133.1MOL Works of Molière, in literature.

929NAP1 Biography of Napoleon I (Bonaparte).

-0... TABLE 1k. COMMON AUXILIARIES OF GENERAL CHARACTERISTICS

Systematic table

-02 COMMON AUXILIARIES OF PROPERTIES

The -02 auxiliaries denote general properties or attributes of entities

The -02 auxiliaries are applicable throughout the main tables. The -02 auxiliaries are not to be used independently, or cited first in a compound notation. They are always suffixed to a main number or notation which expresses the subject to be qualified. Since there is some repetition of terms between the various auxiliary tables, care should be taken in selecting the appropriate application e.g. to distinguish between audio-visual as a property (audio-visual training methods classified as 377.44-028.26) and as a form (a training video classified as 377.44(086.8))

-021	Properties of existence	-021.331	Whole. Complete. Full. Entire. Comprehensive ⇒ -024.13
-021.111	True. False. Fallacious	-021.332	Partial. Incomplete ⇒ -024.14
-021.121	Real. Actual. Unreal	-021.333	Ideal. Perfect
-021.131	Virtual	-021.341	Open ⇒ -025.28
-021.141	Imaginary	-021.342	Closed ⇒ -025.31
-021.142	Capacity for existence	-021.351	Limited. Limits. Unlimited
-021.142.1	Non-existent	-021.352	Free ⇒ -027.5
-021.142.3	Potential	-021.353	Infinite
-021.142.5	Existent	-021.361	Integral. Intrinsic
-021.143	Incidence	-021.362	Included. Contained. Excluded <i>In the sense of membership of a set. For physical containment see -024.78</i>
-021.144	Occurrence	-021.363	Complementarity. Complementary
-021.146	Presence (or not)	-021.364	Reciprocal. Reciprocation
-021.146.2	Present. At hand. With	-021.37	Properties of regularity
-021.146.4	Absent. Lacking. Without	-021.371	Regular. Usual. Ordinary
-021.151	Possible. Impossible ⇒ -027.233	-021.372	Irregular. Unusual. Extraordinary
-021.161	Available. Unavailable. Extant <i>For defunct, obsolete see -027.17</i>	-021.381	Simple. Simplified. Unified
-021.163	Lost	-021.383	Complex
-021.167	Definite. Indefinite	-021.385	Compound. Composite ⇒ -025.26
-021.171	Fictional <i>'Fictional' is understood here in the broadest sense of imagined, non-existent. For literary fiction use colon combination at -029:</i> ◇ 343.12-05-029:82-3 Fictional detectives ⇒ -029:82-3	-021.387	Constant. Stable. Unchanging. Invariable ⇒ -022.322
-021.181	Fantasy	-021.388	Variability. Variety. □ Inconstant. Unstable.
-021.191	Misleading. Hoax. Spurious	-021.4	Variable. Changing. Fluctuating
-021.2	Properties of relation	-021.411	Properties of value, quality
-021.21	Absolute	-021.412	Nature. Purpose. Essence. Character
-021.22	Relative	-021.412.1	Significance. Meaning Significant. Important. Critical ⇒ -028.43
-021.23	Equality. Inequality	-021.412.2	Insignificant. Trivial
-021.24	Degrees of equivalence	-021.413	Relevant. Irrelevant
-021.242	Equivalence. Synonymy	-021.414	Value. Status
-021.244	Non-equivalence. Antonymy	-021.415	Correct. Incorrect
-021.25	Degrees of likeness	-021.416	Precise. Imprecise. Exact. Inexact. Accurate. Inaccurate
-021.252	Similarity. Resemblance. Partial equivalence. Alike	-021.42	Degree of normality
-021.254	Dissimilarity. Difference. Unlike	-021.421	Normal
-021.261	Congruity. Congruence	-021.423	Abnormal
-021.262	Coincidence. Coincident	-021.425	Paranormal
-021.263	Proportion. Proportional	-021.427	Subnormal
-021.271	Analogy	-021.431	Rational. Irrational
-021.272	Comparison. Comparable. Comparability	-021.432	Logical. Illogical
-021.273	Interchangeability	-021.433	Axiomatic. Non-axiomatic
-021.29	Compatible. Incompatible ⇒ -021.482	-021.434	Perceivable. Comprehensible. Incomprehensible
-021.3	Properties of range	-021.451	Obscure
-021.311	General. Universal ⇒ -027.511	-021.452	Hidden
-021.321	Special. Specific ⇒ -027.512	-021.453	Secret. Esoteric. Clandestine

-0... Table 1k. Common auxiliaries of general characteristics

-021.461	Conditional. Unconditional	-022.227	All. Comprehensive. Total
-021.462	Dependent. Independent	-022.231	Least. Minimum
-021.463	Interdependent	-022.233	Optimum
-021.464	Self-contained. Self-supporting. Self-sufficient	-022.235	Most. Maximum
-021.465	Quality	-022.241	Uniqueness
-021.466	Pure. Unadulterated	-022.251	Rare. Rarity
-021.467	Impure. Mixed	-022.252	Dearth. Shortage. Insufficient. Uncommon
-021.471	Good. Benign	-022.253	Adequate. Sufficient
-021.472	Neutral. Indifferent	-022.255	Abundant. Plenteous. Common
-021.473	Bad. Malignant. Malign	-022.257	Excess. Extreme. Excessive
-021.474	Pleasant. Unpleasant	-022.258	Acute. Chronic
-021.475	Degree of advantage	-022.3	Properties related to temporal values
-021.475.2	Advantageous. Advantages, benefits of ...		<i>Care should be taken in the use of this array; many of the terms are applicable in more than one context (spatial or causal, as well as temporal). If the purely temporal aspect is intended, Table 1g should be preferred</i>
-021.475.4	Disadvantageous. Disadvantages of		
-021.476	Difficulties with ...		
-021.476	Unwanted		Duration. Durable
-021.477	Superior. Better		⇒ "4"
	⇒ -024.71		⇒ -027.45
-021.478	Inferior. Worse	-022.312	Long-lasting
	⇒ -024.72		Sustained. Sustainable
-021.479	Best. Ideal. Luxury		Non-durable. Disposable
-021.481	Worst	-022.314	⇒ -027.34
-021.482	Friendly. Accommodating	-022.316	Degree of permanence
	⇒ -021.29	-022.318	Perpetual. Eternal
-021.483	Tolerant. Intolerant		Persistent. Constant
-021.484	Resistant		⇒ -021.387
-021.485	Hostile	-022.32	Permanent. Enduring
-021.5	Properties of origin	-022.321	Semi-permanent
-021.51	Source	-022.322	Temporary
-021.52	Provenance		Urgent. Rapid-response.... □ Emergency....
-021.53	Authority	-022.324	Crisis.... Disaster
-021.54	Natural. Inherent	-022.325	Transient. Impermanent. Ephemeral
-021.55	Accidental	-022.326	Instantaneous
-021.56	Acquired	-022.326.5	Degree of continuity
-021.57	Planned. Contrived. Deliberate. Imposed		Continuous. Uninterrupted. Unbroken
-021.58	Synthetic. Manufactured. Artificial	-022.327	⇒ -022.332
-021.59	Spontaneous. Of own accord. On own initiative	-022.328	⇒ -025.32
-021.6	Properties of order (sequence, priority)	-022.33	Discontinuous. Interrupted. Intermittent
-021.61	Basic. Fundamental	-022.332	⇒ -022.334
-021.62	Principal. Main. Dominant. Primary		⇒ -025.33
-021.631	Subordinate	-022.334	Repetitive. Iterative
-021.632	Supplementary. Additional		Degree of recurrence or repeatability
-021.633	Subsidiary. Ancillary. Accessory. Auxiliary.		Nonrecurrent. Unrepeatable. One-off. Once
-021.634	Secondary	-022.338	Repeated. Repeatable. More than once
-021.635	Peripheral. Marginal	-022.34	Recurrent. Periodic
-021.64	Substitute. Replacement. Ersatz	-022.342	Regularity. Predictability
-021.65	Initial. First. Leading	-022.343	⇒ -021.371
-021.66	Intermediate	-022.344	⇒ -023.722
-021.67	Final. Terminal	-022.345	⇒ -025.23
-021.68	Before. Preliminary. Ante. Pre-		Regular. Predictable. Foreseeable. Expected
	⇒ -024.75		Irregular. Unpredictable. Unforeseeable.
	After. Post-	-022.345.2	Randomly occurring
	⇒ -024.76	-022.345.4	⇒ -025.22
-022.1	Properties of magnitude		Frequent
-022.11	Density. Frequency		Occasional
-022.12	Light		Infrequent
-022.15	Moderate	-022.346	Sequence. Sequential. Successive.
-022.17	Heavy	-022.348	Consecutive
-022.18	Excessive. Severe	-022.352	Simultaneous
-022.2	Properties of degree, quantity, number	-022.354	Synchronous
-022.211	Positive		Asynchronous. Dyschronous
-022.212	Negative	-022.356	
-022.213	Zero. None. Nil	-022.358	
-022.214	One. Single. Unitary. Sole. Alone. Solitary.	-022.362	
-022.215	Unilateral	-022.4	Properties of dimension
-022.217	Two. Double. Dual. Binary. Paired. Bilateral	-022.41	Dimensionless. Singularity
-022.218	Three. Treble. Triple	-022.43	High. Long. Tall. Height
-022.221	Many. Multiple. Multilateral	-022.45	Short. Low
-022.222	Few. Negligible	-022.46	Thick. Deep
-022.225	Some	-022.47	Breadth. Broad. Width. Wide
-022.225	Much. Many. Very	-022.48	Narrow. Thin. Shallow. Fine

-022.5	Properties of size		
-022.51	Small. Little. Tiny. Dwarf. Toy	-023.62	Chamfered
-022.52	Miniature	-023.63	Serrated
-022.53	Micro... Microscopic. Micrographic	-023.65	Slotted
-022.532	Nano...	-023.67	Spined
-022.54	Moderate	-023.7	Surface conditions
-022.55	Average. Mean. Median	-023.711	Flat. Level
-022.56	Large. Big		⇒ -023.4
-022.57	Giant. Oversize. Mammoth	-023.721	Curved
-022.58	Macroscopic	-023.722	Regular. Irregular
-022.59	Bulk. Mass		⇒ -021.371
			⇒ -022.344
			⇒ -025.53
-023	Properties of shape		
-023.11	Profile. Outline	-023.723	Smooth
-023.12	Contour	-023.725	Rough
-023.2	One-dimensional. Line. Linear	-023.727	Grained
-023.211	Straight	-023.728	Patterned
-023.221	Curved. Arced		⇒ -025.13
-023.222	Epicyclic		⇒ -027.28
-023.225	Hypocyclic	-023.731	Textured
-023.231	Parabolic	-023.732	Ribbed
-023.235	Hyperbolic	-023.733	Knurled
-023.237	Ogee	-023.734	Grooved
-023.251	Bent. Angled	-023.735	Scored
-023.255	Orthogonal	-023.737	Corrugated
-023.3	In the form of symbols, e.g. alpha characters, numerals	-023.8	Form
	◇ -023.3*#8 Figure-of-eight shaped	-023.811	Sheet. Plate
	◇ -023.3A A-shaped	-023.812	Foil
	◇ -023.3Pi Pi-shaped, π -shaped	-023.813	Pyramidal
-023.4	Two-dimensional. Plane. Planar	-023.814	Cuboid
-023.41	Fiat	-023.815	Block. Brick. Slab. Billet
	⇒ -023.711	-023.816	Octahedral
-023.42	Rectilinear	-023.817	Wedge-shaped
-023.43	Triangular	-023.818	Prismatic
-023.431	Scalene	-023.821	Cylindrical
-023.435	Isosceles	-023.822	Spherical
-023.437	Equilateral	-023.823	Balls. Beads
-023.44	Quadrangular. Tetragonal. Quadrilateral	-023.824	Spheroidal. Globular
-023.441	Rectangular	-023.825	Hemispherical
-023.442	Square	-023.826	Oblate
-023.443	Parallelogrammatic	-023.827	Prolate
-023.444	Rhomboidal	-023.831	Conical
-023.445	Orthorhombic	-023.832	Biconical
-023.446	Trapeziform	-023.834	Helical. Helicoid
-023.447	Trapezoidal	-023.835	Hyperboloidal
-023.448	Diamond shaped. Lozenge	-023.836	Ellipsoid. Ovoid
-023.46	Polygonal	-023.837	Paraboloidal
-023.462	Pentagonal	-023.838	Toroidal. Torus shaped
-023.463	Hexagonal	-023.839	Dumb-bell shaped
-023.464	Heptagonal	-023.841	Barrel shaped
-023.465	Octagonal	-023.842	Annular. Ring shaped
-023.466	Forked. Bifurcated	-023.843	Hollow
-023.467	Arrow shaped	-023.844	Open hollow. Concave
-023.468	Cross-shaped. Cruciform		⇒ -023.481
-023.47	Curvilinear	-023.845	Closed hollow
-023.472	Circular. Round	-023.846	Pipes
-023.473	Semicircular	-023.847	Tubes. Tubular
-023.474	Quarter-circular	-023.848	Hoses
-023.475	Fluted	-023.851	Spherical hollow
-023.477	Annular. Ring shaped	-023.852	Globes. Globular
-023.481	Concave	-023.853	Balloons
	⇒ -023.844	-023.861	Perforated
-023.482	Convex	-023.862	Perforated annular
-023.483	Concavo-convex	-023.871	Wires. Filaments
-023.485	Lenticular	-023.872	Unifilar
-023.487	Drop-shaped	-023.873	Multifilar
-023.488	Oval. Elliptical	-023.874	Cords
-023.5	Three-dimensional. Solid	-023.875	Braids
-023.6	Edge conditions	-023.876	Ropes
-023.61	Bevelled	-023.877	Cables
		-023.881	Webs
		-023.885	Nets. Meshes

-0... Table 1k. Common auxiliaries of general characteristics

-024	Properties of structure	-025	Properties of arrangement
-024.11	Analytic	-025.11	Configuration
-024.12	Synthetic	-025.12	Layout. Arrangement
-024.13	Complete. Whole	-025.13	Pattern. Design
	⇒ -021.331		⇒ -023.728
-024.14	Incomplete. Part	-025.14	Combined
	⇒ -021.332	-025.15	Coordinated
-024.15	Half. Semi. Demi-	-025.16	Ordered. Disordered
-024.16	Quarter	-025.17	Balance. Equilibrium. Balanced. Unbalanced.
-024.17	Fraction. Portion. Percentage		Disturbed
-024.21	Monobloc	-025.18	Harmony. Harmonious
-024.22	Sectional	-025.21	Systematic
-024.23	Cellular	-025.22	Random. Arbitrary
-024.24	Modular	-025.23	Regular. Irregular
-024.25	Layered. Laminated		⇒ -021.371
-024.26	Duplex		⇒ -022.344
-024.27	Triplex		⇒ -023.722
-024.28	Multiplex	-025.24	Homogeneous. Uniform
-024.29	Sandwich	-025.25	Heterogeneous
-024.31	Clad. Wrapped. Covered	-025.26	Hybrid
-024.32	Plated		⇒ -021.385
-024.33	Lined	-025.27	Clustered
-024.41	Appendages	-025.28	Open
-024.42	Remainder. Residue. Residual		⇒ -021.342
-024.5	Properties of position	-025.31	Closed. Bounded
	<i>There is some overlap with concepts in the</i>		⇒ -021.342
	<i>common auxiliaries of place (Table 1e), e.g.</i>	-025.32	Continuous. Uninterrupted
	<i>(5-191.2) Central Asia, 35-024.548 Central</i>		⇒ -022.332
	<i>administration. Where it is clear that a</i>	-025.33	Discontinuous. Intermittent
	<i>concept is spatial prefer Table 1e</i>		⇒ -022.334
-024.512	Front. Fore. Frontal	-025.35	Recursive
-024.522	Back. Tail. Dorsal. End	-025.36	Curved
-024.524	Side. Lateral	-025.37	Folded
-024.526	Left	-025.38	Labyrinthine
-024.528	Right	-025.39	Coupled
-024.532	Mid-line. Medial		⇒ -022.215
-024.534	Tip	-025.41	Parallel. Non-parallel
-024.536	Edge. Limit. Boundary	-025.42	Series
-024.538	Surface. Face. Facet. Superficial	-025.43	Tandem
-024.542	Corner	-025.44	Opposed. Opposite
-024.544	Top. Head	-025.45	Crossed
-024.546	Bottom. Base. Root	-025.46	Spaced. Intervals
-024.548	Centre. Central. Middle	-025.47	Staggered
-024.61	Inside. Interior. Internal	-025.48	Tiered. Stepped
-024.62	Outside. Exterior. External	-025.49	Columnar. Rows. Ranked
-024.63	Between. Inter-	-025.51	Split. Divided
-024.64	Near. Proximate	-025.52	Segmented. Articulated
-024.65	Interface. Abutting. Juxtaposition	-025.53	Twisted. Crooked
-024.66	Next to. Adjacent. Alongside. Beside	-025.54	Symmetrical. Symmetry
-024.67	Junction. Intersection	-025.55	Axial symmetry. Axisymmetry
-024.68	Touching. Tangential		⇒ -026.32
-024.71	Above. Over. Upper. Superior	-025.56	Asymmetry. Asymmetrical
	⇒ -021.477	-025.57	Unsymmetry. Unsymmetrical
-024.72	Below. Under. Underneath. Lower. Inferior	-025.61	Centred
	⇒ -021.478	-025.62	Concentric
-024.74	Beside	-025.63	Heliocentric
-024.75	Before. Anterior. In front of	-025.65	Eccentric
	⇒ -021.67		⇒ -026.294
-024.76	Behind. Posterior	-025.71	Vertical. Perpendicular
	⇒ -021.68	-025.72	Horizontal
-024.77	Along. Lengthways. Longitudinal	-025.73	Inclined. Oblique. Sloping
-024.78	Across. Transverse	-025.74	Skew
-024.79	Surrounding. Enclosing. Outside	-025.75	Gradient
-024.81	Contained. Enclosed. Inside	-025.76	Lateral
	⇒ -021.362	-026	Properties of action and movement
-024.82	Overlapping	-026.11	Passive
-024.83	Apart. Separated. Isolated	-026.12	Active
-024.84	Widespread. Generalised. Systemic. Pervasive	-026.13	Reactive. Responsive. Sensitive
-024.85	Displaced. Misplaced. Dislocated	-026.14	Non-reactive. Inert. Insensitive
-024.86	Close. Neighbouring. Local	-026.145	Dormant. Latent. Quiescent
-024.87	Far. Distant. Remote	-026.15	Proactive. Creative

-026.16	Stable. Unstable	-026.563	Brittle. Fragile
-026.17	Controlled. Uncontrolled	-026.564	Rigidity. Rigid. Stiff
	⇒ -021.59	-026.564.2	Flexibility. Flexible. Supple
-026.18	Progressive	-026.564.3	Porous. Permeable. Impermeable
	⇒ -026.42		⇒ 539.217
	⇒ -026.451	-026.564.4	Spongy
-026.19	Degree of alterability	-026.565	Stress. Tension. Pressure
-026.191	Reversible		⇒ 531.22
-026.193	Irreversible		⇒ 539.4
-026.195	Alterable	-026.565.2	High pressure
-026.197	Unalterable	-026.565.4	Low pressure
-026.21	Expanding. Contracting	-026.566	Compressibility. Compressed
-026.22	Pulsating		⇒ 532.12
-026.23	Growing. Increasing	-026.567	Deformability. Elasticity
-026.24	Diminishing. Decreasing		⇒ 539.3
-026.25	Accumulating		⇒ 539.52
-026.26	In motion. Moving. Motive. Mobile	-026.568	Cohesion. Cohesive
-026.27	At rest. Immobile. Stationary		⇒ 539.61
-026.28	Reciprocating	-026.569	Adhesion. Adhesive
-026.29	Rotary. Rotating. Semi-rotary		⇒ 539.61
-026.292	Counterrotating	-026.57	Acoustic properties
-026.294	Eccentric rotation		⇒ 534
	⇒ -025.65	-026.571	Audible. Inaudible
-026.31	Oscillating		⇒ 534-7
-026.312	Pendulum motion	-026.572	Sonic. Supersonic. Ultrasonic
-026.32	Axial	-026.573	Subsonic
	⇒ -025.55	-026.61	Optical properties
-026.321	Uniaxial		⇒ 535
-026.322	Biaxial	-026.611	Light. Dark
-026.323	Multiaxial	-026.612	Visibility. Visible. Invisible
-026.325	Coaxial	-026.613	Properties related to visible light. Hues.
-026.33	Centrifugal. Radial		Colours
-026.34	Centripetal		<i>The 7-digit divisions -026.613.1/8 denote</i>
-026.35	Vibratory. Shaking		<i>the main spectral groups of visible light</i>
-026.36	Degree of portability		<i>clearly distinguishable by the eye, as well</i>
-026.362	Transportable		<i>as the corresponding hues of surfaces, e.g.</i>
-026.37	Degree of movability		<i>-026.613.1 'Red', -026.613.6 'Blue', with</i>
-026.38	Detachable. Separable		<i>further specification by an eighth digit, e.g.</i>
-026.39	Manoeuvrable. Manipulable. Adjustable		<i>-026.613.11 'Magenta', -026.613.61 'Cyan'.</i>
-026.4	Properties of direction		<i>To denote the saturation and brightness of</i>
-026.41	Direct. Indirect		<i>a colour, a ninth digit may be added:.</i>
-026.42	Directional. Non-directional		<i>-026.613...1 Pure, strongly saturated.</i>
-026.43	Unidirectional		<i>-026.613...2 Moderately white-saturated or</i>
-026.44	Omnidirectional		<i>light.</i>
-026.45	Orientation		<i>-026.613...3 Moderately black-saturated or</i>
-026.451	Forward. Progressive		<i>dark.</i>
-026.452	Backward. Reverse. Regressive. Retrogressive		<i>-026.613...4 Strongly white-saturated, light.</i>
-026.453	Upward. Rising		<i>-026.613...5 Grey-toned. Dull.</i>
-026.454	Downward. Falling. Dropping		<i>-026.613...6 Strongly black-saturated, dark.</i>
-026.455	Sideways		<i>An unequivocal classification of colours can</i>
-026.46	Approaching. Towards. Convergent		<i>be obtained only by application of a colour</i>
-026.47	Receding. Away from. Divergent		<i>system, e.g. DIN, Munsell. The notations</i>
-026.48	Incoming		<i>used by such systems may be added</i>
-026.49	Outgoing		<i>following a hash or asterisk, as in the</i>
-026.5	Physical properties		<i>following examples for a specific blue</i>
-026.51	Speed. Velocity		<i>colour:.</i>
-026.512	Relative speed		<i>-026.613.6*DIN 17:4:3.</i>
-026.512.1	Fast		<i>-026.613.6*Munsell 4PB 4/8</i>
-026.512.3	Slow	-026.613.01	Colourless (in clear transparent bodies)
-026.514	Variable speed	-026.613.02	White
-026.516	Stationary. Idling	-026.613.021	Pure white. Standard white
-026.52	Volume. Capacity	-026.613.023	Daylight white (light sources)
-026.53	Mass	-026.613.024	Cold white (light sources)
	⇒ 531.42	-026.613.025	Warm white (light sources)
-026.54	Weight. Light. Heavy	-026.613.03	Grey
	⇒ 531.421	-026.613.033	Neutral grey (aselective grey)
-026.55	Density. Thick. Thin	-026.613.034	Selective grey
	⇒ 531.424	-026.613.037	Colour-tinted grey
-026.56	Mechanical properties	-026.613.04	Black
-026.561	Strength. Strong. Weak	-026.613.041	Perfect black
-026.562	Hard. Soft	-026.613.1	Red colour group

-0... Table 1k. Common auxiliaries of general characteristics

-026.613.11	Bluish red. Magenta. Crimson. Carmine	-026.73	Liquid. Liquefied
-026.613.114	Rose. Pink	⇒ 54-145	
-026.613.13	Bright red. Primary red. Scarlet	-026.731	Mobile
-026.613.15	Vermilion. Cinnabar	-026.732	Viscous
-026.613.17	Yellowish red	-026.741	Solutions
-026.613.174	Yellowish pink	-026.742	Colloids. Colloidal
-026.613.2	Orange colour group	⇒ 54-148	
-026.613.21	Reddish orange	-026.743	Suspensions
-026.613.215	Reddish brown	-026.744	Emulsions
-026.613.24	Orange	-026.745	Sols
-026.613.245	Brown	-026.746	Gels
-026.613.27	Yellowish orange	-026.747	Foams
-026.613.3	Yellow colour group	-026.751	Semi-colloidal
-026.613.31	Reddish yellow. Golden yellow	-026.752	Pastes
-026.613.33	Pure yellow. Primary yellow	-026.753	Mashes
-026.613.334	Cream	-026.754	Doughs
-026.613.335	Yellowish brown	-026.76	Solid
-026.613.36	Greenish yellow	⇒ 54-16	
-026.613.366	Olive	-026.761	Coarseness. Coarse
-026.613.38	Yellow-green	-026.762	Fineness. Fine
-026.613.4	Green colour group	-026.763	Crystalline
-026.613.41	Yellowish green. Leaf green	⇒ 54-162	
-026.613.43	Pure green. Primary green	⇒ 544.22	
-026.613.47	Sea green. Bluish green	-026.764	Amorphous
-026.613.55	Blue-green	⇒ 544.23	
-026.613.6	Blue colour group	-026.765	Glassy. Vitreous
-026.613.61	Greenish blue. Cyan. Ice blue	⇒ 544.236.2	
-026.613.612	Turquoise	-026.766	Bulk matter
-026.613.63	Pure blue. Primary blue	-026.767	Loose solids
-026.613.634	Sky blue. Azure	⇒ 54-169	
-026.613.75	Indigo blue	-026.771	Dusts. Powders. Powdery
-026.613.8	Violet colour group	-026.772	Grains. Granules. Granular. Particles. Particulate
-026.613.81	Bluish violet	-026.773	Shot. Grit. Gritty
-026.613.814	Lilac	-026.774	Splinters. Shavings
-026.613.82	Violet	-026.775	Chips. Flakes
-026.613.83	Bluish purple. Red-violet	-026.78	Chemical properties
-026.613.85	Purple	-026.781	Acidity. Alkalinity. Acidic. Basic
-026.613.87	Reddish purple	⇒ 544.362	
-026.614	Coloured. Monotone	-026.782	Volatility
-026.615	Reflectivity. Shiny. Dull. Matt	⇒ 54-145.56	
⇒ 535.312		-026.783	Flammability
-026.616	Fluorescent. Luminescent	⇒ 544.452.1	
⇒ 535.37		-026.784	Taste
-026.63	Radioactive properties	-026.785	Smell. Odour. Aroma
⇒ 539.16		-026.8	Biological properties
-026.65	Thermal properties	-026.81	Living. Non-living
⇒ 536		-026.82	Dead
-026.651	Heat levels. □ Hot. Warm		<i>For dead persons see -053.18</i>
-026.652	Heated. Re-heated	-026.83	Extinct
-026.653	Super-heated	⇒ 57.07	
-026.654	Moderate. Temperate	-026.85	Biodegradable
-026.655	Cold	-026.86	Toxicity. Toxic. Non-toxic
-026.656	Subzero. Freezing. Frozen. Cryo-		⇒ 615.9
-026.657	Conductivity	-026.87	Dangerous. Lethal. Life-threatening
⇒ 536.2		-026.9	Environmental properties
-026.66	Electromagnetic properties	-026.91	Indoor. Outdoor
⇒ 537			⇒ (203)
-026.661	Electrical properties		⇒ (204.1)
⇒ 537			⇒ (24)
-026.662	Magnetic properties	-026.911	Indoor
⇒ 537.6		-026.912	Outdoor
-026.67	Hydraulic properties	-027	Operational properties
-026.68	Pneumatic properties	-027.1	Properties of development
-026.7	Properties of material state. State of matter	-027.11	Undeveloped. Developed
-026.71	Fluid	-027.12	Developing
-026.72	Gaseous. Vaporous	-027.13	Underdeveloped
⇒ 54-13		-027.14	Early. Proto-. Prototypical
-026.721	Rarefied	-027.15	Advanced. Precocious
⇒ 54-132		-027.16	Delayed. Retarded
-026.722	Compressed	-027.17	Obsolete. Defunct. Outmoded
⇒ 54-133		-027.18	Instantaneous

-027.19	Transitional	-027.543	International. World-wide
-027.2	Properties of function	-027.544	Central. Centralized
-027.21	Theoretical. Pure (non-applied)	-027.545	Devolved. Decentralized
-027.22	Practical. Applied	-027.551	Private
-027.231	Feasible. Possible. Achievable	-027.552	Personal
-027.232	Manageable. Workable	-027.553	Domestic. Familial. Home-
-027.233	Degree of success	-027.554	Semi-private
-027.233.2	Successful	-027.555	Non-private. Public
-027.233.4	Unsuccessful. Failed	-027.556	Voluntary, optional
-027.234	Impossible. Unachievable	-027.557	Mandatory. Obligatory. Compulsory
	⇒ -021.151	-027.561	Professional
-027.235	Useful. Profitable. Utility	-027.562	Amateur
-027.236	Efficient. Inefficient	-027.563	Commercial
-027.241	Dual purpose	-027.564	Non-commercial
-027.242	Multifunctional. Multi-purpose	-027.564.2	Charitable
-027.243	All-purpose	-027.565	Free
-027.252	Protective. Resistant		⇒ -021.351
-027.253	Waterproof	-027.566	Fee-paying. Chargeable
-027.254	Drip-proof	-027.567	Subsidized
-027.255	Splash-proof	-027.568	Official. Authorized. Authoritative
-027.256	Spray-proof	-027.571	Semi-official
-027.257	Watertight	-027.572	Unofficial. Unauthorized
-027.258	Immersible	-027.575	Independent
-027.259	Submersible	-027.577	Authentic
-027.262	Gastight. Airtight	-027.581	Legal. With legal force
-027.263	Dust-proof	-027.582	Without legal force
-027.264	Weatherproof	-027.583	Illegal
-027.265	Stormproof	-027.584	Criminal
-027.266	Soundproof. Lightproof	-027.6	Properties of membership
-027.267	Fireproof. Heatproof	-027.61	Belonging. Native. Indigenous
-027.268	Flameproof	-027.63	Alien. Foreign
-027.272	Shockproof		⇒ (1-87)
-027.274	Explosion-proof	-027.65	Majority. Minority
-027.28	Design properties		
	⇒ -023.728	-028	Properties of style and presentation
	⇒ -025.13		<i>Properties from this array should not be</i>
-027.282	Foldable		<i>confused with the common auxiliaries of</i>
-027.283	Collapsible		<i>form (Table 1d)</i>
-027.284	Dismantable		◇ 070.48(084.1) Illustrations of newspapers
-027.285	Stackable		(form)
-027.286	Extensible. Extending. Telescopic		◇ 070.48-028.22 Illustrated newspapers
-027.287	Inflatable		(property)
-027.3	Properties of production	-028.11	Full. Complete
-027.31	New. Original	-028.12	Annotated
-027.32	Used. Second-hand		⇒ (0.072)
-027.33	Recycled. Recyclable. Re-usable		⇒ (048.2)
-027.332	Waste (as product)	-028.13	Abridged. Shortened. Edited
-027.34	Disposable	-028.14	Abstracted. Summarised
-027.35	Replaceable		⇒ (048.3)
-027.36	Reproduction. Facsimile. Copy	-028.15	Concise. Skeleton
	⇒ -022.343	-028.16	Verbal. Oral. Spoken word
-027.37	Mass-produced	-028.17	Aural. Audio-
-027.38	Pre-fabricated		⇒ (086.7)
-027.39	Customised	-028.21	Textual. Text-based. Written word
-027.41	Manual	-028.22	Visual. Graphic. Illustrated. Pictorial
-027.42	Mechanical		⇒ (084)
-027.43	Automatic. Semi-automatic	-028.23	Video-
-027.44	Computerised. Computer-aided		⇒ (084.14)
-027.45	Safety. Reliability. Durability		⇒ (086.8)
	⇒ -022.312	-028.24	Manuscript. Handwritten
-027.5	Organizational properties		⇒ (0.032)
-027.511	General. Generalized. Global		⇒ 09
-027.512	Special. Specialized. Specialist	-028.25	Print-, paper-based
-027.521	Individual		⇒ (0.036)
-027.522	Collective. Group	-028.26	Audio-visual
-027.531	Organized. Unorganized	-028.27	Electronic
	<i>For structural organization see -024</i>	-028.31	Elementary
-027.541	Local. Regional. Municipal. Provincial		⇒ (0.062)
	⇒ (1-2)	-028.32	Intermediate
-027.542	National		⇒ (0.063)
	⇒ (1-4)	-028.33	Advanced
			⇒ (0.064)

-0... Table 1k. Common auxiliaries of general characteristics

-028.41	Popular ⇒ (0.027.3)	-028.82	Alternative
-028.42	Learned. Scholarly. Academic	-028.83	Suppressed. Withheld
-028.43	Critical. Non-critical. Uncritical ⇒ -021.412.1	-028.84	Forgotten
-028.44	Descriptive	-028.85	Prohibited. Banned. Condemned
-028.45	Analytical	-028.86	Censored
-028.46	Experiential. Pragmatic	-029	Properties derived from other main classes
-028.47	Anecdotal		<i>In general, notions of attributes directly derived from major disciplines are best dealt with by direct use of the colon and this should be the preferred treatment e.g. 618.39-089.888:17 Ethical abortion, rather than 618.39-089.888-029:17.</i>
-028.5	Properties of authorship		<i>This auxiliary should only be used when the subject can only be adequately expressed by use of an attribute descriptor.</i>
-028.51	Anonymous		<i>Denote details by :0/9...</i>
-028.52	Pseudonymous	◇ -029:088	Miscellaneous
-028.53	Eponymous	◇ -029:1	Philosophical
-028.6	Properties of order, arrangement	◇ -029:133	Occult
-028.61	Literal	◇ -029:17	Ethical. Moral
-028.62	Alphabetical	◇ -029:2	Religious. Spiritual
-028.63	Numerical	◇ -029:3	Social
-028.64	Ordinal	◇ -029:32	Political
-028.65	Cardinal	◇ -029:33	Economic
-028.66	Alpha-numeric	◇ -029:5	Scientific
-028.67	Systematic. Classified ⇒ 025.4	◇ -029:51	Mathematical
-028.68	Taxonomic ⇒ 57.06	◇ -029:6	Technological
-028.7	Properties of content and position	◇ -029:7	Artistic
-028.71	Biased. Slanted	◇ -029:8	Literary
-028.72	Sensational	◇ -029:82-3	Fictional
-028.73	Orthodox	⇒ -021.171	
-028.74	Unorthodox	◇ -029:82-7	Humorous. Comic
-028.75	Conventional	◇ -029:9	Historical
-028.76	Unconventional		
-028.77	Experimental		
-028.78	Traditional. Conservative		
-028.79	Liberal. Reform		
-028.81	Radical		

-03 COMMON AUXILIARIES OF MATERIALS

SCOPE. The -03 auxiliaries denote the materials or constituents of which objects or products are made.

The main places for classing materials as such are in most cases the sections of 66 and 67 dealing with their manufacture or processing. Documents about materials in general, or all or many materials from a particular point of view (e.g. dangerous goods) are classed under 620.2, materials testing under 620.1, and chemical analysis under 543

USE. The -03 auxiliaries are applicable throughout the main tables (unless an exception is noted, e.g. at 613.2, 616-03) if the material aspect is secondary to the subject.

CITATION ORDER. The -03 auxiliaries are not to be used independently or cited first in a compound notation. They are always suffixed to a main number or notation which expresses the subject to be qualified.

COMPOSITE MATERIALS. Composite materials may be denoted by apostrophe combination, in which the apostrophe replaces -03 in the second and subsequent component numbers, e.g.

-036.674 Polyester resins.

-037.52 Artificial mineral fibres. Glass fibres. Ceramic fibres.

-036.674'752 Polyester and glass fibre

-032	Naturally occurring mineral materials	-032.32	Mineral oil. Petroleum
	⇒ -033	-032.34	Peat
	⇒ -034	-032.35	Coal, including lignite and anthracite
	⇒ -039.6	-032.36	Graphite
-032.1	Air. □ Air at atmospheric pressure.	-032.37	Bitumen (asphalt, pitch etc.)
	Compressed air. Rarefied air	-032.38	Fossil resins and copals
-032.2	Water (in any of its forms)	-032.386	Ozokerite
-032.23	Water vapour. Steam	-032.387	Amber
-032.24	Ice. Snow	-032.4	Ores. Metalliferous minerals
-032.25	Fresh water (liquid)	⇒ -034	
-032.26	Salt water. Brine	-032.41	Iron ores. Manganese ores. □ Haematite.
-032.27	Mineral waters (spring waters)		Manganite
	-032.27 ≡ 553.7	-032.42	Gold ores. Silver ores. □ Argentite
-032.3	Carbonaceous and hydrocarbon minerals	-032.43/49	Other ores (as materials)
-032.31	Natural gas		-032.43/49 ≡ 553.43/499

-032.5	Rock. Stone -032.5 $\equiv$ 553.5 $\Rightarrow$ -033.2	-034.7	Light metals. Light-metal alloys
-032.6	Earths. Nonmetalliferous minerals -032.6 $\equiv$ 553.6	-034.822	Uranium
-032.61	Earth (soil) in general. Clay etc. $\diamond$ -032.61/22 Earth and water. Mud. Adobe	-034.88	Alkali metals
-032.8	Precious stones. Semiprecious stones -032.81/.87 $\equiv$ 553.81/.87	-034.89	Alkaline-earth metals
-032.89	Other precious and semiprecious stones -032.89 $\equiv$ 549	-035	Materials of mainly organic origin $\Rightarrow$ -036 $\Rightarrow$ -037 $\Rightarrow$ -039.7
-033	Manufactured mineral-based materials	-035.2	Materials of plant (vegetable) origin <i>For wood use -035.3 and subdivisions</i>
-033.2	Hard-setting plasters, cements, compositions	-035.22	Grass. Herbage (fresh or dried). Turf
-033.21	Gypsum and anhydrite plasters. $\square$ Plaster of Paris	-035.23/.25	Specific crop materials and plant products -035.23/.25 $\equiv$ 633/635 $\diamond$ -035.237.1 Tobacco products, including nicotine
-033.22	Lime. Quicklime. Limewash. Fine stuff	-035.26	Stems. Twigs. Cane etc. $\square$ Hay. Bamboo -035.26 $\equiv$ 677.54
-033.24	Cement. $\square$ Portland cement. High-alumina cement -033.24 $\equiv$ 666.94	-035.27	Plant parts other than stems. $\square$ Foliage. Flowers. Juice -035.27 $\equiv$ 631.57
-033.25	Pozzolana and materials with similar effect -033.25 $\equiv$ 666.95	-035.3	Wood. Timber. Logwood etc.
-033.26	Hard-setting compositions. Imitation stone -033.26 $\equiv$ 666.96	-035.31	Hardwood
-033.27	Putty and similar oil-bound compositions	-035.32	Softwood
-033.3	Concrete (cement concrete)	-035.35	Cork
-033.32	Plain concrete (mass concrete)	-035.38	Wood waste. Sawdust. Shavings. Wood-wool
-033.33	Lightweight concretes. $\square$ No-fines concrete. Aerated concrete	-035.4	Pulp. Paper. Board
-033.34	Special concretes. $\square$ Fire-resisting concrete. Expansive concrete	-035.41	Pulp
-033.37	Reinforced concrete	-035.42/.46	Papers -035.42/.46 $\equiv$ 676.22/.26
-033.38	Prestressed concrete	-035.47	Board. Cardboard. Boxboards
-033.5	Glass and similar vitreous materials	-035.48	Building boards. $\square$ Fibreboards. Hardboard. Wallboard
-033.51	Ordinary rolled glass. $\square$ Plate glass	-035.5	Materials of animal origin
-033.52	High-quality glass. $\square$ Crown glass. Flint glass. Crystal glass	-035.51	Leather (natural leather)
-033.53	Strass	-035.511	Raw skins. Hides
-033.57	Enamel	-035.513	Soft leathers. $\square$ Kid leather. Morocco leather
-033.58	Fused quartz. Fused basalt etc. $\Rightarrow$ -037.51	-035.514	Hard, rigid leathers
-033.6	Ceramics. Pottery. Clayware in general $\Rightarrow$ -032.61	-035.52	Specially processed leathers. $\square$ Suède. Parchment -035.52 $\equiv$ 675.2
-033.62	Porcelain. Semiporcelain. China	-035.53	Imitation leather. Simulated leather, of plastics etc.
-033.63	Stoneware. $\square$ Technical, chemical, sanitary stoneware	-035.54	Furs
-033.64	Earthenware (porous ware). $\square$ Terracotta. Fireclay -033.64 $\equiv$ 666.64	-035.55	Hair. Feathers
-033.65	Special ceramic objects for electrical purposes. Ceramic insulators	-035.552	Hair. Bristles etc. $\Rightarrow$ -037.3
-033.7	Heavy clayware. Refractories. Cermets $\Rightarrow$ 666.7	-035.554	Feathers. Quills
-033.71	Bricks	-035.56	Bones. Teeth. Horns. Shell. Ivory. Tortoiseshell etc. -035.56 $\equiv$ 675.86
-033.76	Refractories	-035.57	Live animal produce as materials. $\square$ Milk. Eggs -035.57 $\equiv$ 637
-033.77	Chemically resistant clayware	-035.58	Dung
-033.78	Cermets. Ceramic-metal composites <i>Denote constituent metal by '4...', derived from -034...</i>	-035.6	Chemically processed materials, mainly organic
-034	Metals -034 $\equiv$ 669 $\Rightarrow$ -032.4 $\Rightarrow$ -033.78	-035.63/.64	Materials of the food and drink industry -035.63/.64 $\equiv$ 663/664
-034.1	Ferrous metals. Iron and steel	-035.66	Organoleptics. Odorants. Flavourings $\Rightarrow$ -035.8
-034.13	Cast iron	-035.67	Colorants. Dyes. Inks. Paints etc. -035.67 $\equiv$ 667
-034.14	Steel	-035.68	Distillates and residues of wood, coal etc.
-034.15	Alloy steel	-035.682	Tars. $\square$ Coal tar. Wood tar
-034.2/.8	Nonferrous metals	-035.683	Charcoal. Wood distillation residues
-034.3	Copper. Copper alloys	-035.684	Coke. Coal distillation residues
-034.4	Lead. Lead alloys	-035.8	Oils. Fats. Waxes. Gums. Natural resins
-034.5	Zinc. Zinc alloys	-035.81	Fats. Natural waxes -035.81 $\equiv$ 665.1
-034.6	Tin. Tin alloys	-035.82	Animal oils, fats and waxes -035.82 $\equiv$ 665.2
		-035.83	Vegetable oils, fats and waxes -035.83 $\equiv$ 665.3

-0... Table 1k. Common auxiliaries of general characteristics

-035.84	Gums. Natural resins. Oleoresins	-037.4	Man-made fibres. Chemical fibres
-035.841	Gums. □ Gum arabic. Tragacanth	⇒ -036	
-035.842/.847	Natural resins. Oleoresins	-037.45	Semisynthetic fibres (cellulosic fibres etc.).
	-035.842/.847 ≡ 665.944.2/.7	□ Rayon. Artificial silk	
-035.848	Rosin (colophony)	-037.47	Synthetic fibres
-035.85	Essential oils. Cosmetics	-037.474	Polyester fibres and fabrics
	-035.85 ≡ 665.5	-037.475	Polyamide fibres and fabrics. Nylon fibres
-036	Macromolecular materials. Rubbers and plastics	-037.477	Polyethylene fibres
-036.4	Rubber. Natural macromoleculars	-037.5	Mineral, metal and other fibres
	-036.4 ≡ 678.4	-037.51	-037.5 ≡ 677.5
-036.5	Plastics in general, especially semisynthetics	-037.52	Natural mineral fibres. □ Quartz fibres.
	-036.5 ≡ 678.5	Asbestos fibres	
⇒ -037.4		-037.53	Artificial mineral fibres. Glass fibres. Ceramic fibres
-036.6/.8	Synthetic plastics, resins, rubbers etc.	-037.6	Metal fibres
-036.6	Polycondensates (condensation resins, fibres).	-037.63	Mixed fabrics. Special fabrics (woven, knitted etc.)
	□ Polyesters. Polyamides. Nylon	-037.65	Felts, batts etc.
	-036.6 ≡ 678.6	-037.7	Gauze. Lace. Other openwork fabrics
-036.7	Polymerizates (addition resins and copolymers). □ Polyethylene. Polyvinyls.	-037.71	Cordage. Trimmage. Fancy fabrics
	Polystyrene	-037.75	Cordage. □ Rope. Twine. String
	-036.7 ≡ 678.7	-037.8	Ribbon. Tape. Webbing
-036.8	Mixed polymerizate-polycondensates. Other macromoleculars. □ Silicones	-037.86	Specially treated and finished textiles
	-036.8 ≡ 678.8	-037.87	Proofed, coated or bonded fabrics
-037	Textiles. Fibres. Yarns. Fabrics. Cloth	-039	Other materials
	-037.1/.3 ≡ 677.1/.3	-039.6	Inorganic chemicals and compounds
-037.1	Vegetable fibres. □ Bast. Flax. Hemp. Jute.		<i>Except those at -032/-034</i>
	Sisal	-039.6 ≡ 546	
-037.2	Vegetable seed hairs. □ Cotton. Kapok	-039.7	Organic chemicals and compounds
-037.3	Animal fibres and hairs. □ Wool. Silk		<i>Except those at -035/-036</i>
⇒ -035.55		-039.7 ≡ 547	

-04 COMMON AUXILIARIES OF RELATIONS, PROCESSES AND OPERATIONS

SCOPE. The -04 auxiliaries denote relations between concepts (particularly under -042) and processes, activities and operations in any discipline. They are applicable throughout the main tables. Where such processes etc. are already adequately denoted in a main class, that notation may be preferred. Care should be taken to distinguish between a process as a subsidiary aspect of a subject, and cases where the process or activity is itself the subject, e.g. appraisal of research and methodology in the social sciences would be classed using -04..., at 303.1-047.4 - as opposed to appraisal as a management technique, denoted by a main number at 005.96.

CITATION ORDER. The -04 auxiliaries are not to be used independently, or cited first in a compound notation. They are always suffixed to a main number.

-042	Phase relations	-042.75	Interdependence. Mutual dependence.
-042.1	Bias phase	'Symbiosis'	
	<i>Use form auxiliary (0.05) Documents designed for particular types of reader/audience</i>	-042.82	Complementarity
		-043	General processes
-042.2	Comparison phase	-043.2	Action
-042.3	Influence phase	-043.3	Inaction. Inertia
-042.4	Tool phase. Exposition phase	-043.4	Reaction. Response
	<i>Intended to provide for relations of the 'study sample' or 'case study' variety; e.g. gene mutation as demonstrated in drosophila, where drosophila is the vehicle or tool and not the principal focus of the study</i>	-043.5	Interaction
		-043.61	Preparing. Preparation
		-043.62	Starting
		-043.65	Stopping
		-043.7	Change
		-043.72	Transition
		-043.74	Variation. Alteration (change in one direction)
-042.5/.8	Other terms descriptive of relationships between entities	-043.76	Fluctuation (alternating change, bipolar variation)
	⇒ -021.2	-043.77	Rate of change
-042.52	Isolation	-043.78	Acceleration
-042.55	Separation	-043.79	Deceleration
-042.62	Association	-043.8/.9	Processes of existence
-042.65	Independence	-043.82	Life cycle
-042.72	Dependence	-043.83	Formation. Origination. Inception. Beginning. 'Birth'

-043.84	Actuation. Actualization. Realization	-044.87	Narrowing. Thinning
-043.85	Resting. Interphase	-044.88/.89	Processes related to size
-043.86	Development. Evolution	-044.88	Growth. Enlargement. Expanding. Expansion.
	⇒ -044.88		Swelling. Dilation
-043.865	Struggle	-044.89	Contraction. Contracting. Diminishing
-043.87	Reversion	-044.9	Processes related to shape
-043.88	Maturation	-044.92/.93	Change of form. Change of shape
-043.89	Climax	-044.922	Metamorphosis. Mutation
-043.92	Ageing	-044.923	Deformation. Change in outline
-043.93	Decline. Degeneration	-044.924/.927	One-dimensional change
-043.94	Finishing. Ending. Completion	-044.924	Straightening
-043.95	Cessation. 'Death'. Termination	-044.925	Curving. Arcing
-043.96	Regeneration. Renewal	-044.927	Bending
-043.97	Disappearance. Extinction. Loss	-044.932/.937	Two-dimensional change
-043.98	Deception. Falsification	-044.932	Flattening
-044.22	Processes of range	-044.933	Forking. Bifurcation
-044.225	Generalization	-044.934	Projection
-044.227	Specialization	-044.935	Curving. Curling
-044.235	Opening	-044.935.2	Rounding
-044.237	Closing. Closure	-044.935.4	In-curving
-044.245	Limitation. Restriction	-044.935.6	Out-curving
-044.247	Integration	-044.937	Drooping. Sagging. Hanging
-044.252	Inclusion	-044.94	Change of surface conditions
-044.257	Exclusion	-044.942	Levelling. Flattening
-044.262	Simplification	-044.943	Curving. Curvature
-044.267	Complication	-044.944	Smoothing
-044.3	Processes of value	-044.945	Roughening. Coarsening. Abrasion
-044.325	Qualitative change	-044.946	Grooving
-044.332	Differentiation. Adaptation	-044.947	Scoring
-044.337	Improvement. Amelioration	-044.948	Corrugation
-044.342	Worsening. Deterioration. Degeneration	-044.95	Change of shape
-044.347	Accommodation. Acceptance	-044.952	Erosion. Wearing away
-044.352	Toleration. Tolerance	-044.953	Atrophy
-044.357	Resistance. Hostility	-044.954	Deposition. Building up
-044.362	Abnormality	-044.955	Excrescence. Outgrowth
-044.367	Accident. Mischance	-044.956	Hollowing. Cavitation
-044.372	Crisis	-044.957	Ballooning
-044.377	Failure	-044.958	Perforation
-044.382	Recovery. Repair	-044.96	Change of structure
-044.4	Processes of ordering and sequence	-044.962	Construction. Synthesis
-044.42	Ordering. Sequencing	-044.963	Destruction. Deconstruction. Analysis
-044.44	Leading. Precedence. Domination.	-044.964	Aggregation. Agglomeration
	Predomination. Primacy	-044.965	Stratification
-044.46	Following. Subordination. Marginalization	-044.966	Dissociation. Detachment. Separation
-044.5	Processes of magnitude	-044.967	Disintegration. Decay
-044.52	Quantitative change	-044.968	Rupture. Bursting
-044.53	Increase. Supplementation. Augmentation	-044.969	Collapse
-044.55	Proliferation. Excess	-045.2	Change of position
-044.57	Decrease. Diminution. Diminishment	-045.22	Dislocation. Displacement
-044.6	Processes of number and degree	-045.23	Advance. Advancing. Advancement.
-044.62	Increase in number. Addition		Progression
-044.63	Decrease in number. Subtraction	-045.24	Retreat. Retreating. Regression
-044.64	Unification. Integration	-045.25	Rising. Rise
-044.65	Doubling. Duplication. Twinning	-045.26	Falling. Fall. Drop
-044.66	Trebling. Tripling	-045.27	Centralization
-044.67	Multiplication. Replication	-045.28	Marginalization
-044.68	Minimization	-045.32	Internalization
-044.69	Maximization	-045.33	Interpolation
-044.7	Processes related to time and chronology	-045.34	Extrapolation
-044.73	Retardation	-045.35	Insertion. Invasion. Penetration
-044.74	Endurance	-045.36	Immersion. Submersion
-044.75	Persistence	-045.37	Embedding
-044.76	Interruption	-045.38	Extraction. Removal
-044.77	Recurrence. Relapse	-045.4	Processes of arrangement
-044.8	Processes related to dimension	-045.42	Configuration
-044.82/.87	Change in dimension	-045.43	Design
-044.82	Lengthening. Elongation. Extension	-045.44	Spacing
-044.83	Stretching	-045.45	Combination
-044.84	Shortening	-045.46	Co-ordination
-044.85	Thickening. Deepening	-045.47	Clustering
-044.86	Broadening. Widening	-045.48	Localization

-0... Table 1k. Common auxiliaries of general characteristics

-045.52	Distribution. Dissemination. Scattering. Spread.	-047.25	Finding. Location
	Dispersal	-047.26	Observation
-045.53	Randomization	-047.27	Identification. Recognition
-045.54	Pervasion	-047.272	Acknowledgement
-045.55	Folding. Pleating	-047.275	Denial
-045.56	Unfolding. Invagination	-047.28	Verification. Confirmation
-045.57	Protrusion. Evagination	-047.29	Description
-045.58	Inversion	-047.32	Recording
-045.59	Eversion	-047.33	Marking. Tagging
-045.62	Joining. Merger. Fusion	-047.34	Matching. Mapping
-045.623	Amalgamation	-047.35	Co-ordination
-045.625	Assimilation	-047.36	Monitoring
-045.63	Splitting. Division	-047.37	Investigation. Research
-045.64	Segmentation. Articulation	-047.38	Examination
-045.65	Twisting. Torsion	-047.42	Experimentation. Experiments
-045.66	Inclination. Skewing	-047.43	Review. Appraisal
-045.7	Processes involving position and movement	-047.44	Analysis. Assessment. Evaluation
-045.72	Movement. Motion	-047.46	Deduction. Inference
-045.73	Convergence	-047.48	Explanation. Hypothesizing. Theorizing
-045.74	Divergence	-047.52	Clarification. Disambiguation
-045.75	Rotation. Counter-rotation	-047.54	Justification
-045.76	Oscillation	-047.56	Exposition
-045.77	Centrifugal motion	-047.58	Modelling
-045.78	Centripetal motion	-047.62	Idealization
-045.79	Vibration	-047.64	Control
-045.82	Internal movement		⇒ 62-5
-045.83	Flow		⇒ 681.5
-045.84	Reflux. Reverse flow	-047.645	Limitation. Restriction
-045.85	Circulation. Transport	-047.647	Containment
-045.86	Tropisms	-047.72	Forecasting. Prediction. Prognosis
-045.87	Locomotion	-047.74	Planning
-046	Processes related to physical properties	-047.75	Targeting. Purposive efforts
-046.22	Lightening. Weight decrease. Weight loss	-047.76	Retrospection
-046.23	Weight increase. Weight gain	-047.82	Conception. Design
-046.24	Thickening (density)	-047.84	Construction. Creation. Synthesis
	⇒ -044.85	-047.86	Destruction. Deletion
-046.25	Thinning (density). Dilution	-048.22	Introduction. Implementation
-046.26	Strengthening	-048.23	Realization. Execution
-046.27	Weakening	-048.24	Testing. Piloting. Trialling
-046.28	Hardening. Sclerosis	-048.25	Modification. Amendment. Adjustment
-046.29	Softening	-048.26	Judgment. Adjudication
-046.32	Stiffening. Rigidifying	-048.27	Supplementation. Augmentation
-046.33	Relaxation. Easing	-048.28	Withdrawal. Removal. Cut-back. 'Pruning'.
-046.34	Compression		Diminution
-046.35	Constriction	-048.32	Enhancement. Improvement
-046.36	Cohesion	-048.34	Optimization
-046.37	Adhesion	-048.35	Updating. Modernization
-046.38	Lightening (colour)	-048.36	Revision
-046.39	Darkening	-048.37	Renewal
-046.42	Colouring. Pigmentation	-048.38	Revival
-046.43	Discolouration	-048.42	Arrangement. Organization
-046.44	Reflection	-048.43	Ordering. Prioritizing
-046.45	Luminescence. Fluorescence	-048.44	Systematization. Grouping
-046.46	Heating. Warming	-048.442	Categorization
-046.47	Cooling. Chilling	-048.445	Classification
-046.48	Freezing	-048.46	Alignment
-046.52	Conduction (heat)	-048.47	Conformity
-046.54	Convection	-048.52	Collection. Aggregation. Gathering
-046.55	Radiation	-048.53	Dispersal. Dissemination
-046.6	Processes related to states of matter	-048.54	Generalization
-046.62	Vaporization. Evaporation	-048.55	Specialization
-046.63	Liquefaction. Melting	-048.56	Differentiation. Distinction
-046.64	Solution. Dissolution. Dissolving	-048.57	Grading. Ranking. Weighting
-046.65	Solidification	-048.58	Selection. Choice
-046.66	Pulverisation	-048.582	Inclusion
-046.67	Granulation	-048.583	Elimination. Discarding. Rejection
-047	General operations and activities	-048.62	Representation. Substitution. Replacement
-047.22	Awareness. Knowledge. Cognizance.	-048.63	Simulation
	Unawareness. Ignorance	-048.64	Misrepresentation
-047.23	Familiarization. Learning	-048.65	Promotion. Encouragement. Stimulation
-047.24	Searching	-048.66	Suppression. Discouragement

-048.67	Popularization	-048.94	Delegation. Subordination
-048.72	Allocation. Distribution	-049.2	Manipulation. Handling
-048.73	Centralization	-049.3	Maintenance. Servicing
-048.74	Decentralization		⇒ 62-7
-048.75	Conversion	-049.34	Preservation
-048.76	Alteration	-049.35	Conservation
-048.77	Modification	-049.36	Replenishment
-048.78	Improvement. Upgrading	-049.4	Care
-048.79	Downgrading	-049.5	Safety
-048.82	Complication	-049.6	Security
-048.85	Simplification. 'Streamlining'	-049.65	Protection
-048.87	Co-operation		⇒ 62-7
-048.88	Consultation	-049.7	Use. Exploitation
-048.92	Dissimulation	-049.8	Application
-048.93	Conflict. Antipathy		

-05 COMMON AUXILIARIES OF PERSONS AND PERSONAL CHARACTERISTICS

SCOPE. The -05 auxiliaries denote the persons concerned or their characteristics

USE. The -05 auxiliaries are applicable throughout the main tables, if the personal aspect is secondary to the subject.

If the main table for a particular subject provides no direct subdivision or special auxiliary for the personal aspect, -05 itself may be used to denote this, e.g.

324 Elections.

324-05 Persons connected with elections.

The roles of agent and target should usually be distinguished by -051 or -052, e.g.

324 Elections.

324-051 Electors, voters.

324-052 Elected representatives.

with any further personal characteristics denoted by adding the appropriate subdivision of -053/-058, e.g.

324-052-055.2 Female elected representatives.

If the main tables already provide a place for the personal aspect, then the subdivisions of -053/-058 may be added directly to this, e.g.

070.42-057.13 Freelance journalists.

347.96-055.2 Female lawyers.

616-083-055.1 Male nurses.

As with all common auxiliaries, the subdivisions of -05 may be combined with each other, or other auxiliaries, e.g.

647-053.6-055.2 Female adolescent domestic staff.

78.071-056.45(=411.16) Jewish musical prodigies.

CITATION ORDER. The -05 auxiliaries are not to be used independently or cited first in a compound notation. They are always suffixed to a main number or notation which expresses the subject to be qualified

-051	Persons as agents, doers, practitioners (studying, making, serving etc.)	-053.4	Preschool-age children. Toddlers
	◇ 616-051 Medical staff: doctors, nurses etc.	-053.5	School-age children (in general)
	◇ 929-051 Biographers	-053.6	⇒ -057.87
-052	Persons as targets, clients, users (studied, served etc.)		Young persons. Adolescents. Teenagers
	◇ 616-052 Medical patients	-053.66	⇒ -057.87
	◇ 929-052 Biographees	-053.67	Persons in early adolescence
-053	Persons according to age or age-groups	-053.8	Persons in late adolescence
	<i>Details by "4..."</i>	-053.81	Adults. Grown-ups
-053.1	Human beings, born or unborn, living or dead		Young adults. Persons in prime of life
-053.12	Ovum	-053.85	<i>Details by "465..."</i>
-053.13	Embryo (foetus)	-053.87	◇ -053.81 "465*20/29" Persons in their
-053.15	Alive. Living persons	-053.88	twenties
-053.18	Dead. Deceased persons	-053.9	Persons in middle age (in general)
-053.2	Children and infants (in general)		Persons in climacteric (change of life)
	⇒ -055.15	-054	Persons in late middle age (troisième âge)
	⇒ -055.25		Old persons. Persons in old age. Senior citizens
	⇒ -055.62		
-053.3	Infants. Babies		Persons according to ethnic characteristics, nationality, citizenship etc.
-053.31	New-born babies	-054.4	◇ -054(23) Mountain dwellers. Hill folk
-053.32	Premature babies	-054.5	◇ -054(=2/=8) Coloured persons
			Residents (national or non-national)
			Nationals. Citizens of a country

-0... Table 1k. Common auxiliaries of general characteristics

-054.51	Nationals by birth	-056.1	Persons according to general constitution, disposition or character
-054.52	Naturalized citizens		
	◊ -054.52-058.833 Naturalized by marriage	-056.12	Passive persons. □ Unreactive. Inactive.
-054.57	Members of national minorities		Neutral. Indifferent. Placid. Lazy. Apathetic
-054.6	Non-nationals. Foreigners. Aliens	-056.13	Moderately, normally active persons.
-054.62	Resident aliens		□ Reactive. Concerned. Interested
-054.64	Enemy aliens	-056.14	Intensively, abnormally, excessively active persons. □ Overreactive. Energetic.
-054.65	Prisoners of war		Hyperergic. Enthusiasts. Fanatics. Zealots
-054.68	Nonresidents. Visiting foreigners		Personal bias
	⇒ -057.68	-056.15	⇒ 316.353
-054.7	Expatriates. Exiles. Repatriates. Migrants.		Positive bias. Pro-... For... In favour of...
	Stateless persons	-056.153	Supporters
-054.72	Voluntary migrants. □ Immigrants. Emigrants		Neutral. Uncommitted. Neither for nor against
-054.73	Refugees. Evacuées. Displaced persons	-056.155	Negative bias. Anti-... Against... Opposed to...
-054.74	Compulsory exiles. Banned persons	-056.157	Persons according to laterality
	(proscribed from entry). Expellees. Deportees	-056.17	Right-handed persons. Dextrals
-054.75	Repatriates	-056.172	Left-handed persons. Sinistrals
-054.78	Stateless persons. Apatriates	-056.173	Ambidextrous persons
-055	Persons according to gender and kinship	-056.175	Persons according to physical state and health
-055.1/3	Persons according to gender (sex)	-056.2	Physically healthy persons
-055.1	Male persons. Men	-056.22	Persons according to body size, height, stature
-055.15	Boys	-056.23	Very small, very short persons. Dwarfs
-055.2	Female persons. Women	-056.231	Small persons. Short persons
-055.25	Girls	-056.232	Persons of average size or height
-055.26	Women as child-bearers. Pregnant women.	-056.233	Big persons. Tall persons
	Expectant mothers. Nursing mothers	-056.234	Very big, very tall persons. Giants
-055.3	Persons according to sexual orientation	-056.235	Physically ill persons. The sick. Invalids.
	<i>Class other sexual and quasi-sexual characteristics with the subject, e.g.</i>	-056.24	Convalescents
	<i>391.2 Transvestism.</i>	-056.25	Persons according to nourishment (nutritional state) or body weight
	<i>612.6.058 Hermaphroditism (intersexuality).</i>		Underweight persons. Weakened, wasted, debilitated by hunger etc. Starved.
	<i>616.89-008.442 Sexual psychopathology</i>	-056.253	Undernourished. Suffering from malnutrition.
-055.32	Heterosexual		Hypotrophic
-055.34	Homosexual. Gay		Persons of average weight. Normally nourished. Eutrophic
	◊ -055.34-055.1 Gay male	-056.255	Overweight persons. Overnourished. Fat.
	◊ -055.34-055.2 Lesbian		Obese. Hypertrophic
-055.36	Bisexual	-056.257	Physically handicapped persons. Disabled persons
-055.5/7	Persons according to family or other kinship relation		⇒ -058.65
-055.5	Persons in parental or ascendant relation.	-056.26	Blind persons
	Ancestors. Forebears		Deaf persons. Dumb persons. Deaf-mutes
-055.52	Parents		Persons with speech impediments.
	◊ -055.52-058.832 Single parents	-056.262	Speech-defective persons. □ Stammerers. Stutterers
	◊ -055.52-058.832-055.1 Single fathers	-056.263	Temporarily disabled persons. Injured. Wounded (curably)
	◊ -055.52-058.832-055.2 Single mothers	-056.264	Permanently disabled persons. Cripples
-055.53	Grandparents. Great-grandparents. Further forebears in direct ascendant line		Totally disabled persons
-055.54	Persons in indirect ascendant relation.	-056.265	Motor-handicapped persons
	□ Uncles. Aunts. Great-uncles. Great-aunts		Persons according to intelligence and mental or psychological state
-055.6	Persons in filial or descendant relation	-056.266	Persons according to intelligence
-055.62	Offspring. Children of... Filial	-056.268	<i>Denote level of intelligence either by the direct subdivisions below or by adding intelligence quotient (IQ) after an asterisk, see Table 1h</i>
-055.621	Children born in wedlock. Legitimate children	-056.29	◊ -056.31*IQ105 Persons with IQ of 105
-055.623	Children born out of wedlock. Natural children.	-056.3	Persons of low intelligence
	Illegitimate children. Bastard children		Persons of normal, average intelligence
-055.63	Grandchildren. Great-grandchildren. Further descendants in direct line	-056.31	Persons of high intelligence
-055.64	Persons in indirect descendant relation.		Mentally healthy persons
	□ Nephews. Nieces. Great-nephews.	-056.32	Persons with mental health problems. Mentally ill persons
	Great-nieces	-056.34	Persons with learning difficulties. Mentally handicapped persons
-055.7	Persons in collateral relation		Persons according to temperament, predisposition
-055.71	Persons as siblings	-056.313	Accident-prone (injury-prone) persons
	◊ -055.71-055.1 Brothers	-056.315	
	◊ -055.71-055.2 Sisters	-056.317	
-055.72	Persons in collateral relation of the same generation, or in more distant collateral relation. Cousins	-056.32	
		-056.34	
-055.76	Persons as twins, triplets etc., or as one of twins, triplets etc..	-056.36	
-056	Persons according to constitution, health, disposition, hereditary or other traits	-056.4	
		-056.42	

-056.43	Disease-prone (disorder-prone) persons. Persons with diathesis. Hypersensitive persons. Allergic persons	-057.17 -057.173 -057.175 -057.177	Managers in general. The management Lower management Middle management Higher management. Top management. □ Directorate. Board of directors. Chief scientific staff. Chief technical staff. Chief administrative staff Chairmen. Presidents Acting Chairmen. Vice Presidents Chief executive officers. Managing directors Deputy chairmen, presidents Directors. Board members Executive directors Non-executive directors Deputy directors. Assistant directors First-line managers. Supervisors People related to the organization <i>For persons officially on half-pay (previously at this number) see -057.116</i>
-056.45	Gifted persons. Precocious persons. Geniuses. Prodigies		
-056.47	Maladjusted persons. Misfits. Eccentrics. Cranks		
-056.48	Introverts. Extraverts	-057.177.1	
-056.49	Persons with behavioural problems	-057.177.11	
-056.7	Persons according to their inherited characteristics, constitution	-057.177.2 -057.177.21 -057.177.3 -057.177.31 -057.177.32 -057.177.321 -057.177.4 -057.18	
-056.8	Persons according to their acquired habits, behaviour, attitude		
-056.82	Persons characterized by particular daily, routine habits. □ Early risers, late risers. Active in daytime, nighttime. Indoor, outdoor types		
-056.83	Persons with persistent cravings. Addicts: alcohol, tobacco, narcotics etc.		
-056.84	Vegetarians		
-056.85	Total abstainers from stimulants. Teetotallers	-057.182	Agents. Representatives
-056.87	Persons well-disposed to their fellows. □ Sociable, friendly, sympathetic types. Benefactors. Humanitarians. Philanthropists	-057.184 -057.185 -057.187 -057.188 -057.19	Governors Shareholders Customers Suppliers Unemployed persons. Jobless, out-of-work persons Persons seeking work Applicants. Candidates Technical staff. Operatives. Manual workers (blue-collar workers) ⇒ 331.102.342 Specialized workers. Skilled workers. Craftsmen. Artisans Supervisory technical staff. □ Works managers. Foremen Executive technical staff. □ Craftsmen. Operatives Unskilled workers. □ Labourers Workers in training. Apprentice workers Clerical, administrative workers. Clerks. Office staff (white-collar workers) Clerical and related workers in commerce and industry Sales and service workers Officials of the political authority and officials of administrative authorities Heads of state. Monarchs. Presidents Ministers of state Persons in military service. Persons in civilian services. □ Armed forces personnel (servicemen, servicewomen). Police personnel. Fire service personnel Inspectors Professional or academic workers <i>Except those at -057.3...</i>
-056.88	Persons ill-disposed to their fellows. □ Unsociable, unfriendly, cruel types. Misanthropes. Bullies ⇒ 177.84 ⇒ 316.624.3 ⇒ 364.636		
-057	Persons according to occupation, work, livelihood, education <i>For specific occupations use main UDC numbers and -051 or -052</i> ◇ 658.87-051 Shopkeepers. Retailers ◇ 658.87-052 Customers. Clients ◇ 730-051 Sculptors ◇ 77-051 Photographers ◇ 77-052 Persons photographed. Photographic models	-057.21 -057.212 -057.214 -057.22 -057.23 -057.3 -057.32 -057.33 -057.34 -057.341 -057.343 -057.36 -057.39 -057.4 -057.5 -057.51 -057.52 -057.54 -057.56	
-057.1	Persons according to work, employment, administrative status ⇒ 331.102		
-057.11	Persons according to employment conditions		
-057.112	Persons in full-time employment		
-057.113	Persons in part-time employment. □ Half-day workers ⇒ -057.54		
-057.114	Persons in multiple employment. Persons with several jobs. Moonlighters ⇒ 331.218.2		
-057.115	Persons in temporary employment ('temps'). Relief workers. Substitute staff ⇒ -057.54		
-057.116	Persons officially on half-pay		
-057.117	Persons in casual employment. Jobbing workers		
-057.118	Persons in illicit or undeclared employment. Persons in the black economy		
-057.13	Self-employed persons. Independent workers. Freelance workers		
-057.15	Employers in general		
-057.16	Employees in general		
-057.162	Executives		
-057.164	Auxiliaries. Assistants		
-057.165	Technical staff		
-057.166	Ancillary staff (employees not engaged in the primary work process). Service staff. □ Stewards. Commissionaires. Janitors. Concierges. Caretakers. Messengers. Office cleaners		
-057.168	Persons concerned in surveillance. Security staff. Night-watchmen		

-0... Table 1k. Common auxiliaries of general characteristics

-057.57	Members of production or service cooperatives ⇒ 334.73	-058.17	Persons in marginal classes. Borderline cases as to class
-057.6	Persons according to domicile and its relation to work or other activity	-058.18	Classless persons (not belonging to any social class)
-057.61	Persons living at or near their workplace	-058.3	Persons according to economic categories (means, security, other rights) ⇒ -057.7
-057.62	Persons travelling to and from workplace. Commuters		
-057.65	Persons as occupants and managers of their homes. Householders (owners or tenants) ◇ -057.65-055.2 Housewives	-058.31 -058.32	Persons according to specific income levels Persons with high income. Persons of great means. Wealthy persons. The rich
-057.66	Persons without fixed abode. Persons with mobile dwellings and way of life. Travellers (sometimes called 'gipsies') Itinerants. Nomads <i>For gipsies as an ethnic group use (=214.58)</i> ⇒ -058.51	-058.33 -058.34 -058.38	Persons with middling income, average means Persons with low income, little means. Poor Persons with no income, without means. Destitutes
-057.68	Tourists. Travellers in general. Passengers. Persons travelling for reasons other than work ⇒ -054.68	-058.5 -058.51	Persons belonging to marginal socioeconomic categories. Fringe groups Isolated, drifting persons. Mendicants. Beggars. Vagabonds. Vagrants. Tramps. Derelicts. Down-and-outs ⇒ -057.66
-057.7	Persons according to income sources or insurance ⇒ -058.3	-058.53	Persons shunning the established order. □ Underground. Hippies. Social drop-outs
-057.71/77	Persons according to income sources	-058.57	Reformed, reclassified persons. Rehabilitated prisoners etc. Ex... ◇ -058.57-054.65 Ex-prisoners-of-war ◇ 343.23-051-058.57 Ex-criminals ◇ 343.261-052:321.64-058.57 Ex-detainees. Ex-political-prisoners
-057.71	Unpaid persons. Persons in honorary positions		Victims of circumstances
-057.73	Paid persons. Salaried persons	-058.6 -058.64	Victims of social or political circumstances ◇ -058.64-054.74 Persons banned or deported for social or political reasons
-057.75	Pensioners. □ Old-age pensioners. Pensioners retired because of sickness, accident, injury	-058.65	War victims ◇ -058.65-056.26 War-disabled
-057.77	Persons with other income sources	-058.66	Victims of disasters ◇ 550.348.436-058.66 Earthquake victims ◇ 656.1.08-058.66 Victims of road traffic accidents
-057.772	Persons with independent or private means, e.g. from shareholdings, property, legacies	-058.68	Victims of loneliness, neglect, ill-treatment
-057.775	Persons with income from social assistance	-058.7 -058.8	Dependent persons. Dependants
-057.78	Persons according to type of insurance		Persons according to marriage, family or other civil status ⇒ -055.5/7
-057.782	Compulsorily, legally insured persons	-058.83	Persons according to marital status or related civil status
-057.786	Voluntarily insured persons	-058.832	Single, unmarried persons ◇ -058.832-055.1 Bachelors ◇ -058.832-055.2 Spinsters
-057.8	Persons according to education, training, experience	-058.833	Married persons. Spouses ◇ -058.833-055.1 Husbands ◇ -058.833-055.2 Wives
-057.81	Illiterates	-058.835	Widowed persons
-057.82	Uneducated, untrained persons	-058.836	Divorced or separated persons
-057.83	Self-educated, self-taught, self-trained persons. Autodidacts. Self-made men in general	-058.836.2	Divorced persons. Divorcees. Legally separated persons
-057.84	Persons with school education only	-058.836.6	Persons separated de facto. Married persons living apart
-057.85	Persons with higher education (university or equivalent)	-058.837	Unmarried persons living together. Cohabitants. Partners
-057.86	Vocationally or professionally trained or qualified persons. Technical experts	-058.839	Bigamists. Polygamists
-057.87	Students. Schoolchildren. Pupils. Trainees	-058.85	Persons with quasi-parental relationships ⇒ -055.5
-057.874	Schoolchildren	-058.854	Step-parents ◇ -058.854-055.1 Stepfathers ◇ -058.854-055.2 Stepmothers
-057.875	Students in higher education (university or equivalent)	-058.855	Adoptive parents
-057.876	Vocational or professional trainees	-058.856	Foster-parents
-057.88	Persons with incomplete, unfinished study or training, at any level. Drop-outs	-058.86	Persons with quasi-filial relationships ⇒ -055.6
-057.9	Persons according to executive functions	-058.862	Orphans
-057.91	Office holders. Incumbents ◇ -057.91"403" Temporary incumbents ◇ -057.91"408" Permanent incumbents		
-057.911	Secretaries. Registrars		
-057.912	Treasurers		
-057.95	Persons according to membership of organizations		
-057.951	Ordinary members		
-057.953	Subscribing members. Foundation members		
-057.955	Honorary members		
-058	Persons according to social class, civil status		
-058.1	Persons according to social class or status		
-058.12	Upper-class persons		
-058.13	Middle-class persons		
-058.14	Lower-class persons		

-058.864	Step-children	-058.866	Foster-children
	◇ -058.864-055.1 Stepsons	-059	Other aspects of persons
	◇ -058.864-055.2 Stepdaughters	-059.1	Persons singly. Individual persons
-058.865	Adopted children	-059.2	Persons plurally or collectively. Groups. Parties

-1/-9, .01/.09, '0/'9 SECTION II: SPECIAL AUXILIARY SUBDIVISIONS

The special auxiliary subdivisions -1/-9, .01/.09 and '0/'9 are limited in their scope, and each series is used to denote recurrent concepts in that part of the main tables for which it is designed and scheduled, or in certain other sections where specifically indicated. Thus, unlike the common auxiliary signs and subdivisions of Section I, in the special auxiliaries the same notation may be used with different meanings in different parts of the main tables, e.g. -3 at 54, 62 and 82; .02 at 06, 334, 53, 54, 57, 621.3, 629, 656, 66, 677, and 7. It always, however, denotes the same recurrent concept at the main number where it is listed or at the subdivisions of that number

A. Three kinds of notation are in use as special auxiliary subdivisions.

(a) The hyphen series -1/-9 (but not -0, see Table 1k) are mainly analytical or differentiative in function, serving to indicate elements, components, properties and other details of the subject denoted by the main number to which they apply.

Examples: the series listed at 62-1/-8 and 66-9, applicable (with minor exceptions) throughout 62/69, denoting engineering, mechanical and processing details; the series 82-1/-9, applicable throughout 82, denoting literary form (poetry, drama, novels etc.).

(b) The point-nought series .01/.09 are more diversely used and often developed in much more detail than -1/-9, providing sets and subsets of recurrent concepts such as aspect studies, activities, processes, operations, plant and equipment.

Examples: 3.07/.08, applicable throughout 30/39 and developed in fuller detail under 35; 7.01/.09, applicable (with minor exceptions) throughout 71/79; series at many two-digit and lower-level divisions of Class 5 (e.g. 528, 53, 54, 556, 57/59), Class 8 and Class 9; and numerous series in the divisions of Class 6.

(c) The apostrophe series '0/'9 are usually more specific than -1/-9 and .01/.09, and are often synthetic or integrative in function, denoting compound subjects by compound notation. They are sometimes fully listed, but elsewhere are derived from main numbers by parallel division.

Examples: Synthesis of subdivisions of 329 'Political parties', e.g.

329.17'23'12 Nationalist-republican-liberal parties (from 329.17, 329.23 and 329.12).

Specification of compounds at 547 'Organic chemistry' by '11/'19 (derived from 546.1/.9) and '2/'8 (derived from 547.2/.8), e.g.

547.426.24'117 Nitroglycerine (from 546.17).

Specification of alloys at 669 'Metallurgy', e.g.

669.35'5'6 Copper-zinc-tin alloys (from 669.35, 669.5 and 669.6).

Listed apostrophe auxiliaries, as at 628.1'1 'Characteristics of waterworks systems', 81'0 'Phases of development of languages', 81'2'44 'Subject fields of linguistics' and 903'1 'Prehistoric culture forms'.

B. Special auxiliaries may be used either singly or in combination with other auxiliaries.

(a) Citing a single auxiliary, e.g.

329.052 Opposition parties.

329.12'23 Liberal-republican parties.

547.29-41 Reagents for organic acids.

547.29.07 Preparation of organic acids.

547.29'26 Acid-alcohol esters.

821.161.1-31 Russian novels.

821.161.1.09 Criticism of Russian literature.

(b) Citing several auxiliaries with the same symbol (a homogeneous chain), e.g.

329.053.052 Opposition splinter parties.

329.12'13'23 Liberal-progressive-republican parties.

62-242-436 Engine pistons - spherical.

678.6.067.029.4 Polycondensate mouldings - joining.

(c) Citing several auxiliaries with different symbols (a heterogeneous chain), e.g.

329.12'23.052 Liberal-republican opposition parties.

547.29'26.07-41 Acid-alcohol ester preparation - reagents.

821.161.1-31.09 Russian novels - criticism

Part 2

Main tables

0 GENERALITIES. SCIENCE AND KNOWLEDGE. ORGANIZATION. INFORMATION. DOCUMENTATION. LIBRARIANSHIP. INSTITUTIONS. PUBLICATIONS

00 PROLEGOMENA. FUNDAMENTALS OF KNOWLEDGE AND CULTURE. PROPAEDEUTICS

⇒ 008
⇒ 130.2
⇒ 165

001	SCIENCE AND KNOWLEDGE IN GENERAL. ORGANIZATION OF INTELLECTUAL WORK ◇ 001:061.1(100) International governmental scientific organizations, e.g. Unesco	001.6	Scientific laws. □ Ideal cases. Exceptions ⇒ 113
001.1	Concepts of science and knowledge ⇒ 165	001.8	Methodology. □ General study of method. Technical and scientific procedures in study, research, discussion. Analysis and synthesis
001.101	Knowledge	001.81	Technique of intellectual work
001.102	Information	001.811	General. □ Rules for citations, quotations, bibliographic references. Format of stationery
001.103	Data ⇒ 004.22	001.812	Working materials ⇒ 003.5 ⇒ 005.936 ⇒ 686.8
001.103.2	Metadata		
001.11	Definition of concepts. Nature, purpose, essence of knowledge	001.813	Cuttings ⇒ (04) ⇒ 025.17
001.12	Character of the objects examined by knowledge. □ Realities. Phenomena. Facts. Circumstances	001.814	Assembling of references. Research. □ Contents lists. Extracts. Summaries. Preliminary drafts, rough copy for literary work ⇒ 019.9 ⇒ 025.32
001.123	Origin and causes of phenomena		
001.124	Properties and effects of phenomena	001.815	Production of indexes. □ Author indexes. Subject indexes ⇒ 014.5 ⇒ 019.96 ⇒ 025.347
001.124.2	Useful, desirable effects		
001.124.4	Non-useful, undesirable effects	001.816	Work concerned with printing and publication. □ Author's directions to compositor or printer. Manuscripts. Proofs ⇒ 655.253 ⇒ 655.52 ⇒ 808.2
001.126	Application and utilization of phenomena		
001.18	Future of knowledge. □ Prediction. Forecasting. Futurology	001.817	Lecture work. □ Compilation and arrangement of lectures, dissertations ⇒ 069.1 ⇒ 808.5 ⇒ 82-5
001.19	Knowledge and the unknowable. Limits of knowledge	001.818	Evaluation, writing up of results. □ Presentation of information. Technique of report writing ⇒ (047) ⇒ (062.551)
001.2	Relations between individual disciplines ⇒ 001.82 ⇒ 025.4 ⇒ 168.2	001.82	Study of organization. □ Methodology. Analysis. Synthesis. Classification. Systematization ⇒ 025.4 ⇒ 168.2
001.3	Significance of science and knowledge in general. □ Value, utilization, status, defence and appreciation	001.83	Intellectual cooperation. Exchange of ideas
001.31	Advancement, promotion of science and knowledge in general	001.85	Scientific criticism
001.32	Learned, scientific societies. Academies ◇ 001.32(450) In Italy, e.g. Accademia dei Lincei	001.86	Concordances ⇒ 014.5 ⇒ 27-272
001.35	Criticisms of science. Objections to science	001.87	Case studies (as a subject), their compilation and use
001.4	Specialist terminology. Scientific nomenclature ⇒ (083.72) ⇒ 003.62	001.89	Organization of science and scientific work ⇒ 001.82 ⇒ 007 ⇒ 351.854
001.5	Scientific theories. Hypotheses, systems. □ Tracing of relationships between scientific facts ⇒ 162		
001.51	Processes in the formation of hypotheses, systems		
001.53	Comparative presentation ⇒ (083.77)		

001.891	Scientific research. □ Research methods ⇒ 001.891 ⇒ 001.891.5 ⇒ 069.8	002	DOCUMENTATION. BOOKS. WRITINGS. AUTHORSHIP ⇒ 003 ⇒ 005.92 ⇒ 01 ⇒ 02 ⇒ 37.091 ⇒ 322.7 ⇒ 659.2
001.891.3	Theoretical research		
001.891.5	Practical, active research. □ Experiments. Tests		
001.891.53	Laboratory tests		
001.891.54	Testing on models. Model tests		
001.891.55	Field tests. Workfloor, in situ tests		
001.891.58	Testing after modification, conversion, improvement	002.1	Documents (embodiments of information), as subject ◇ 002.1-028.17 Audio documents ◇ 002.1-028.25 Print-based and paper-based documents
001.891.7	Observational research. □ Field work. Observation in the field		
001.892	Development	002.2	Total literary output. □ Of persons, on subjects, of countries ◇ 002.2(494)(083.41) Statistics of literary output in Switzerland
001.893	Control. Supervision		
001.894	Inventions and discoveries. □ Techniques of inventing. Work in preparation of patenting and realization of inventions <i>Class form of documents on inventions and discoveries using Table 1d, e.g. (088.8). Denote the invention or discovery by :...</i> ◇ 001.894:621.11 Invention of the steam engine ◇ 001.894:778.5 Invention of cinematography ⇒ 347.77	002.6	Documentation centres <i>002.6.01/.05 ≡ 06.01/.05</i> ⇒ 005.572 ⇒ 026 ⇒ 366.61.07
001.894.2	Useful and profitable, applied inventions and discoveries	002.61	International documentation centres
001.894.4	Failed, unsuccessful inventions and discoveries	002.63	National documentation centres
001.894.6	Suppressed, withheld inventions and discoveries	002.66	Documentation centres of firms, institutions, factories, public authorities
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001.895	Innovations		
001.9	Dissemination of ideas	003.01	Origins, precursors of scripts. Early forms of scripts
001.91	Generalization. Extension of applicability	003.02	Emergence of writing
001.92	Dissemination of factual knowledge. □ Popularization of science ⇒ 001.31	003.03	Graphic expression of language
001.94	Reported phenomena not yet fully explained ◇ 001.94:633 Unexplained phenomena affecting field crops. Including: Crop circles ◇ 001.94(410.518NES) Loch Ness Monster ◇ 001.94UFO Unidentified flying objects (UFOs). 'Flying saucers' ⇒ 133	003.031	Order of letters or characters. Enumeration of signs and symbols
001.95	Deliberate scientific deceptions and hoaxes ◇ 001.95:569.89(410.126PIL) Eoanthropus Dawsonii (Piltdown Man) ⇒ (089.5)	003.032	Names of letters. Acrophony
001.97	Pseudo-scientific prophecies, fancies. □ 'Death ray' ⇒ 133 ⇒ 159.961 ⇒ 398.4 ⇒ 82-311.9	003.033	Conversion of script by decree, for native speakers of the language <i>E.g. romanization of Turkish in 1928</i>
001.98	Falsification of science. □ False doctrines, fallacies (due to insufficient knowledge, bigotry, excessive zeal etc.)	003.034	Transliteration <i>Representation of graphemes of source script by graphemes of target script</i> ◇ 003.332.4.034:003.344 Transliteration of Hebrew script into Roman script
001.99	Obstruction of and obstacles to the spread of ideas. □ Non-dissemination. Secrecy ⇒ 098.1 ⇒ 27-428.4:002 ⇒ 343.534 ⇒ 347.775 ⇒ 35.083.8	003.035	Transcription <i>Representation of sounds of source language by graphemes of target language and script</i> ◇ 003.324.1.035:003.349.3 Transcription of Chinese into Russian Cyrillic
		003.038	Adaptation of a script to different languages and dialects. □ By adding special letters, diacritical marks
		003.05	Means of producing signs and scripts ⇒ 003.5
		003.052	Handwriting using hand and fingers only
		003.053	Handwriting with devices
		003.055	Printing means <i>Details by :681.615/.616 and :681.62</i>
		003.056	Keyboarding. Typewriting <i>Details by :004..., :681.612/.613</i>
		003.057	Stamping etc. <i>Details by :681.626</i>

003.07	Uses and styles of writing	003.29	Codes. Technographies. Nonconventional writing systems
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003.074	Diplomatics. □ Diplomatic writing. Writing of records, charters and acts. Official scripts ⇒ 930.2	003.292	Alphabetic codes. Letter codes
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003.076	Handwriting. Cursive writing ⇒ 159.925.6	003.292.3	Alphanumeric codes
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003.08	Characteristics of writing ⇒ 655.24	003.295	Morse code <i>Denote mode of transmission by :654.9...</i>
003.081	Formation. Basic signs. Characters, letters. □ Capital letters (majuscules). Small letters (minuscules). Alphabets. Combinations, digraphs, ligatures	003.295.1	Mechanically readable codes
003.082	Diacritics. Accents. □ Diaeresis. Umlaut. Tilde. Grave	003.295.4	Electromagnetically readable codes
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003.085	Sentence or clause formation. Word separation ⇒ 81'367.5	003.3	Scripts
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003.09	Techniques and methods of deciphering scripts	003.315	Prehistoric writings. Embryo-writing. Protolinear writing
003.2	Writing systems. Graphic representations of concepts	003.32	Ideographic and logographic scripts ⇒ 003.214
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003.212	Pictographic writing systems. □ Hieroglyphs	003.323	Cuneiform scripts. □ Sumerian, Hittite cuneiforms
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003.26	Secret writing systems. Cryptographies ⇒ (083.73) ⇒ 003.26 ⇒ 004.056.55	003.324.25	Kanji <i>Chinese characters as used to write Japanese</i>
003.27	Stenographies. Shorthand writing systems	003.324.26	Kanamajiri <i>Kanji together with Hiragana</i>
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		003.326.3	Cyprian syllabic script
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		003.328	Pasigraphic scripts. Scripts for artificial universal languages. □ Bishop Wilkins 'Real character'
		003.329	Other logographic or ideographic scripts <i>Specify by (4/9)</i>
		003.33/.34	Alphabetic scripts
		003.33	Non-European alphabetic scripts
		003.332	Semitic scripts
		003.332.1	Early alphabetic scripts of the Mediterranean basin. □ Sinaitic script
		003.332.2	Phoenician scripts. □ Punic scripts
		003.332.3	Aramaic scripts
		003.332.4	Hebrew and Canaanitic scripts
		003.332.5	Arabic scripts <i>Denote adaptation to non-Semitic languages by .038</i>
			◇ 003.332.5.038:811.432.875 To Swahili

003.332.6	South Semitic scripts. □ Sabaeen, Himyaritic scripts	003.6	Other kinds of graphic representation of thought ◇ 003.6.079 Graffiti ⇒ (084.2) ⇒ (084.3) ⇒ 084 ⇒ 7.045
003.332.7	Ethiopic scripts. □ Amharic scripts		
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003.336.2	Kharosthi (Indo-Bactrian)	003.626	Lexicographical symbols
003.336.3	Brahmi and its offshoots. □ Gupta. Devanagari	003.628	Emblems. Allegories
003.336.4	North East Indian scripts. □ Kaithi scripts. Gujarati. Nepali scripts	003.63	Diagrams. □ Cartograms. Schemata
003.336.5	Tibetan scripts	003.64	Monograms. □ Anagrams. Chronograms. Signature as mark of origin (in porcelain, carpets, tapestries etc.)
003.336.6	South Indian scripts. □ Tamil	003.65	Property marks. Brands
003.337	Scripts of South East Asia (Indo-China) and the East Indian Archipelago. □ Burmese script. Cambodian script (Thai). Indonesian scripts. Scripts of the Philippine Islands and the Malay Peninsula	003.69	Other graphic signs. □ Signs made on houses by thieves and vagrants
003.339	Other non-European alphabetic scripts. □ Scripts of North American Indians. Scripts of Eskimos	004	COMPUTER SCIENCE AND TECHNOLOGY. COMPUTING. DATA PROCESSING <i>Class here also information technology (computing and telecommunications). Throughout 004 digital computing is assumed unless concept is qualified by :004.386 or :004.387</i> ⇒ 621.39
003.34	European alphabetic scripts	004.01	Documentation ◇ 004.415.2.01 Documentation of the software design process ◇ 004.415.3.01 Documentation of source code. Program comments ⇒ (083.131)
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003.346	Oghams	004.031.2	Offline. □ Batch
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003.349.2	Church Slavonic	004.032	System characteristics
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003.356	Systems using specific signs as numerals	004.032.26	Neural network
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		004.032.5	Scaleability

004.032.6	Multimedia ⇒ 004.357 ⇒ 004.4'27	004.057.4	Protocols
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004.04	Processing orientation	004.057.6	Conversion
004.041	Procedure-oriented	004.057.7	Emulation
004.042	Data-flow-oriented	004.057.8	Open systems. Open architecture
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004.045	Object-oriented	004.072	Memory action
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004.047	Logic-oriented	004.072.3	Writing
004.048	AI-oriented ⇒ 004.8	004.072.4	Accessing
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		004.085.2	Compact discs
		004.085.22	CD-ROM
		004.085.23	CD-R (CD-recordable)
		004.085.3	WORMs (Write once, read many times)
		004.085.4	Videodiscs
		004.085.7	Digital paper
		004.085.72	Digital paper tape
		004.085.74	Digital paper disks
		004.087	Electronic media
		004.087.2	Semiconductor memory. Solid-state disks. Flash disks
		004.087.4	Memory cards. PC cards
		004.087.5	Smart cards. IC cards
		004.2	Computer architecture
		004.22	Data representation ⇒ 001.103 ⇒ 621.3.037.3
		004.222	Numerical data
		004.222.2	Fixed-point number representation
		004.222.3	Floating-point number representation
		004.222.5	Overflow and underflow

004.223	Characters and similar data. □ Character sets. Fonts ◇ 004.223.057.2ANSI ANSI character set ◇ 004.223.057.2Unicode Unicode character set ⇒ 655.24	004.275.2	Shared-disk architecture
004.223.2	Alphanumerics <i>Denote particular alphabets by :003.33...</i> ◇ 004.223.2:003.332.5 Arabic alphanumerics	004.3	Computer hardware <i>The -1/-9 auxiliaries from 62-1/-8 and 66-9 are applicable throughout 004.3</i>
004.223.3	Graphic characters. □ Two-byte and three-byte representation <i>Denote particular scripts by :003.32...</i>	004.3'1	Production of computers
004.223.5	Special characters	004.3'12	Design considerations
004.223.6	Control characters	004.3'122	Noise reduction design
004.223.7	Escape characters. Escape keys	004.3'124	Thermal design. Cooling technology
004.225	Date and time data	004.3'14	Assembly technology for computers
004.225.2	Date representation ◇ 004.225.2.052"2000" Year 2000 problem	004.3'142	Packages and packaging
004.225.4	Time representation	004.3'142.2	Device-level packaging
004.23	Instruction set architecture	004.3'142.22	Die bonding. □ Wire bonding. Tape automated bonding (TAB) ⇒ 621.792
004.231	Kinds of instruction set	004.3'142.23	Flip chips
004.231.2	Complex instruction set computers (CISC)	004.3'142.24	Single inline packages (SIPs)
004.231.3	Reduced instruction set computers (RISC)	004.3'142.25	Dual inline packages(DIPs)
004.232	Instruction format	004.3'142.26	Pin-grid arrays
004.233	Kinds of instruction	004.3'142.27	Flat packages
004.233.2	Branch instructions	004.3'142.4	Board-level packaging. □ Through-hole mount.
004.233.3	Data manipulation instructions		Surface mount. Hybrid packaging
004.233.5	Input/output (I/O) instructions	004.3'142.6	Frame-level packaging
004.234	Registers in architecture	004.3'144	Computer components <i>Specify components by :..., in particular :621.3...</i>
004.235	Addressing schemes		◇ 004.3'144:621.3.049.75 Printed circuits
004.236	Subroutines in instruction set		◇ 004.3'144:621.3.049.771.15 Very large-scale integrated circuits
004.237	Interrupts		◇ 004.3'144:621.314 Power packs for computers
004.238	Process status		◇ 004.3'144:621.316.54 Switches
004.239	Memory protection		◇ 004.3'144:621.318.5 Relays
004.25	Memory system	004.3'2	Computer installations ◇ 004.3'2:692.5 Floor structuring for computer installations ◇ 004.3'2:697.9 Ventilation and air-conditioning for computer installations
004.252	Memory hierarchies		
004.254	Cache memory		
004.255	Virtual memory		
004.258	Memory control system		
004.27	Advanced architectures. Non-Von Neumann architectures	004.31	Processing units. Processing circuits
004.272	Parallel processing architectures ⇒ 004.032.24	004.312	Logic circuits
004.272.2	Parallel processing techniques	004.312.2	Combinational circuits
004.272.22	Pipeline control	004.312.22	Logic gates
004.272.23	Using attached support processors (ASPs)	004.312.222	AND-gates. OR-gates. NOT-gates
004.272.25	Vector processing	004.312.225	NAND-gates. NOR-gates
004.272.26	Using multiple processors	004.312.24	Multiplexers
004.272.3	Architecture designs for parallel processing	004.312.26	Encoders. Decoders
004.272.32	SIMD architecture <i>Single instruction stream, multiple data stream</i>	004.312.4	Sequential circuits
004.272.33	MISD architecture <i>Multiple instruction stream, single data stream</i>	004.312.42	Flip-flop in computers
004.272.34	MIMD architecture <i>Multiple instruction stream, multiple data stream</i>	004.312.43	Counters in computers
004.272.4	Kinds of parallel processor	004.312.44	Arithmetic and logic circuits for arithmetic units ⇒ 004.315
004.272.42	Array processor systems	004.312.46	Controlling circuits for mathematical operations
004.272.43	Multiprocessor systems	004.312.462	Wired logic
004.272.44	Dataflow systems	004.312.463	Pin-board logic
004.272.45	Interaction network architecture	004.312.466	Microprogram controlled logic
004.273	Software-oriented architecture	004.314	Registers
004.274	Dynamic architecture	004.314.2	General purpose registers
004.275	Clustering ⇒ 004.052.3 ⇒ 004.056.3 ⇒ 004.056.4	004.314.3	Accumulators
004.275.1	Shared-nothing architecture	004.314.4	Shift registers
		004.314.5	Floating point registers
		004.314.6	Stack registers
		004.315	Arithmetic units ⇒ 004.312.44
		004.315.2	Adders. Half adders
		004.315.4	Complementers
		004.315.5	Multippliers. Dividers
		004.315.7	Vector arithmetic units
		004.315.8	Shifters and comparators
		004.318	Central processing unit (CPU)

004.32	Computer pathways	004.355	Drives
004.322	Channels		<i>Specify by .08...</i>
004.324	Buses		◇ 004.355.083.4 Tape drives
004.326	Connectors to peripheral devices. Input-output pathways		◇ 004.355.083.723 Disk array drives. RAID (redundant arrays of inexpensive disks) configurations
004.326.1	Ports		◇ 004.355.083.74 Floppy disk drives
	◇ 004.326.1.621.384.3 Infrared ports		◇ 004.355.085.22 CD-ROM drives
	◇ 004.326.1.032.22 Serial ports	004.356	Peripherals for hard-copy output
	◇ 004.326.1.032.24 Parallel ports	004.356.2	Printers as peripherals
004.326.2	Card slots		⇒ 681.625.9
	◇ 004.326.2.004.087.4 Slots for PC cards	004.356.4	Output on photosensitive materials.
004.326.7	Docking stations. □ Docking cradles		□ Computer output microfilming (COM)
004.33	Memory units. Storage units	004.357	Peripherals for sound and multimedia. □ Voice input devices
	<i>Denote memory characteristics by .07... and memory storage media by .08...</i>	004.358	Peripherals for simulation and virtual reality.
			□ Data gloves. Data suits. Data helmets
004.332	Memory types according to access possibility	004.38	Computers. Kinds of computer
004.332.2	Direct access memory		<i>Denote computer generations by .032.8...</i>
004.332.3	Random access memory (RAM)	004.382	General purpose computers
004.332.33	Dynamic RAM (DRAM)	004.382.2	Supercomputers
004.332.34	Static RAM (SRAM)	004.382.3	Mainframes
004.332.5	Sequential access memory	004.382.4	Minicomputers
004.333	Read-only memory (ROM)	004.382.6	Work stations
004.334	Pushup memory. Pushdown memory	004.382.7	Personal computers. Microcomputers
004.335	Associative memory. Content addressable memory	004.382.72	Desk-top computers
004.337	Internal memory	004.382.73	Lap-top computers. □ Notebook computers
004.338	External memory	004.382.75	Palm-top computers
004.35	Peripherals. Input-output units	004.382.76	Pocketable computers
004.351	Readers	004.383	Special purpose computers
	◇ 004.351.081.1 Punched card readers	004.383.1	Word processors
004.352	Scanners	004.383.2	Server computers
004.352.2	Optical scanners		◇ 004.383.2.004.63 File server computer
004.352.22	Graphics scanners		◇ 004.383.2.004.738.52 Web server computers
004.352.24	Optical recognition scanners. Optical wands		⇒ 004.455.2
004.352.242	Optical character recognition (OCR)	004.383.3	Digital signal processors (DSPs)
004.352.243	Handwriting recognition	004.383.4	Simulators
004.352.246	Bar code recognition	004.383.5	Image processors
004.352.4	Scanners for specific media	004.383.6	Database machines
	◇ 004.352.4.083.3 for magnetic ink character recognition (MICR)	004.383.8	Artificial intelligence machines
	◇ 004.352.4.087.5 for IC cards	004.384	◇ 004.383.8.032.26 Neurocomputers
004.353	Computer consoles. Video terminals. Video display units. Visual display units	004.386	Industrial computers. Process computers
004.353.2	Display screens	004.387	Hybrid computers
004.353.22	Display resolution	004.388	Analog computers
004.353.24	Types of display	004.388.2	Computer-related machines
004.353.242	Bitmapped display	004.388.4	Calculators. Programmable calculators
004.353.244	Raster display		Games machines
004.353.245	Vector display. Calligraphic display	004.4	Software
004.353.25	Types of screen	004.4'2	Software development tools
004.353.253	Cathode-ray display	004.4'22	Computer-aided software engineering (CASE) tools
004.353.254	Flat-panel display	004.4'23	Programming tools
004.353.254.2	Plasma display	004.4'232	Editors
004.353.254.3	Liquid crystal display (LCD)	004.4'233	Debuggers
004.353.254.4	Electroluminescent display	004.4'234	Program browsing tools
004.353.254.5	Light-emitting diode display (LED)		⇒ 004.455.1
004.353.4	Computer keyboards	004.4'236	Visual programming tools
	⇒ 681.61.065	004.4'24	Software automation tools
004.353.44	Special keys	004.4'242	Program generators. Application generators.
004.353.45	Function keys		Automatic programming
004.354	Locating devices. Pointing devices. Graphic input devices	004.4'244	System generators
004.354.2	Light pens	004.4'27	Multimedia development tools
004.354.3	Touch-screens	004.4'272	Authoring systems
004.354.4	Graphics tablets. Graphic pads. Digitizer tablets	004.4'273	Image editors
004.354.5	Mice	004.4'274	Video editors
	◇ 004.354.5.621.384.3 Infrared mice	004.4'275	Animation software tools
004.354.6	Trackerballs	004.4'277	Tools for sound
004.354.7	Joy sticks	004.4'277.2	Sound editors
		004.4'277.4	Music creation tools. □ MIDI sequencers
		004.4'4	Programming language translation

004.4'41	Programming language translation processes	004.416	System and software maintenance
004.4'412	Lexical analysis	004.416.2	Correction. Debugging
004.4'413	Syntax analysis. Parsing	004.416.3	Software adaptation
004.4'414	Semantic analysis	004.416.6	Software enhancement
004.4'415	Code generation	004.418	Software localisation
004.4'416	Code optimization		
004.4'417	Use of a symbol table	004.42	Computer programming. Computer programs
004.4'418	Use of an intermediate language		⇒ 519.85
004.4'42	Programming language translators	004.421	Algorithms for program construction
004.4'422	Compilers		<i>Class algorithms for particular processes at the process, using .021</i>
004.4'423	Interpreters		◇ 004.424.5.021 Sort algorithms
004.4'424	Assemblers	004.421.2	Basic mathematical algorithms
004.4'426	Macro processors		<i>For mathematical theory of algorithms in general use 510.5...</i>
004.4'427	Preprocessors		<i>Specify mathematical process by :51...</i>
004.4'6	Runtime environment. □ Runtime storage allocation. Address computation. Execution control		◇ 004.421.2:517.443 Fast Fourier transform
			◇ 004.421.2:517.535 Algorithms for rational expression
			◇ 004.421.2:519.17 Graph algorithms
			⇒ 512.54.05
			⇒ 519.16
			⇒ 519.178
			⇒ 519.712
004.41	Software engineering		Evaluation expression algorithms
	<i>Class here computer system engineering</i>	004.421.5	Random number (pseudorandom number) generation algorithms
004.412	Software metrics		⇒ 004.056.55
004.412.2	Metrics based on program size. Lines of code (LOC) metrics	004.421.6	Symbol processing algorithms
004.412.3	Metrics based on function points	004.422	Components of computer programs
004.413	Software development management. Software project management	004.422.1	Atoms
004.413.2	Scheduling management	004.422.2	Constants
004.413.4	Risk analysis	004.422.3	Variables
004.413.5	Software project estimation techniques	004.422.32	Kinds of variable
004.414	Definition phase of system and software engineering	004.422.322	Global or local variable
004.414.2	Computer system analysis and design	004.422.324	Static or dynamic variable
004.414.22	System requirements analysis	004.422.326	Free or shared variable
	◇ 004.414.22:004.2 Computer architecture requirements	004.422.33	Default rule for variables
004.414.23	Modelling and simulation of computer systems	004.422.35	Declaration of variables
004.414.28	System specification	004.422.352	Explicit declaration
004.414.3	Software requirements analysis	004.422.353	Implicit declaration
004.414.32	Software prototyping	004.422.4	Expressions
004.414.38	Software requirements specification	004.422.42	Parts of expression
004.415	Development phase of system and software engineering	004.422.422	Operator
004.415.2	Software design	004.422.423	Operand
	◇ 004.415.2.01 Documentation of the software design process	004.422.44	Kinds of expression
	◇ 004.415.2.031.43 Real-time system design	004.422.442	M-expression
	◇ 004.415.2.041 Procedure-oriented design techniques	004.422.444	Lambda-expression
	◇ 004.415.2.043 Data-structure-oriented design techniques	004.422.5	Statements
	◇ 004.415.2.052.3 Fault-tolerant design	004.422.52	Executable statements
004.415.22	Abstraction	004.422.53	Declarative statements
004.415.23	Modularization	004.422.55	Horn clause
004.415.24	Information hiding	004.422.56	Macro instructions. Macro calls
004.415.25	Programming in-the-small	004.422.6	Data types
004.415.26	Program design languages. Pseudocode	004.422.61	Basic data types
004.415.28	Software design specification	004.422.612	Logical data type. Boolean data type
004.415.3	Coding	004.422.613	Numeric data types
	◇ 004.415.3.01 Documentation of source code. Program comments	004.422.614	Character data types
	⇒ 004.42	004.422.615	Enumerated data types
004.415.5	Software quality assurance. □ Verification and validation (VV)	004.422.616	Pointers
	⇒ 006.015.5	004.422.63	Structured data types. Data structures
004.415.52	Formal technical reviews. □ Walk-through	004.422.632	Arrays
004.415.53	Software testing	004.422.632.2	Character strings
004.415.532	Unit testing. Testing in the small	004.422.632.4	Multidimensional arrays
004.415.532.2	Black box testing	004.422.633	Records
004.415.532.3	White box testing	004.422.634	Sets
004.415.533	Integration testing	004.422.635	Dynamic data structures
004.415.535	Validation testing	004.422.635.2	Linked-list (chained-list) structure
004.415.538	System testing. Testing in the large	004.422.635.3	Tree structure
		004.422.635.32	Balanced tree (B-tree). □ B+tree
		004.422.635.33	Binary tree

004.422.635.5	Stream type structure	004.431	Low level languages
004.422.636	Abstract data types	004.431.2	Machine languages
004.422.636.2	Lists	004.431.4	Assembly languages
004.422.636.3	Stacks	004.432	High level languages
004.422.636.4	Queues	004.432.2	Procedural languages. Imperative languages
004.422.636.5	Dequeues	004.432.4	Nonprocedural languages. Declarative languages
004.422.636.7	Tables		
004.422.639	Other data types	004.432.42	Functional languages
004.422.8	Program units	004.432.45	Fourth generation languages (4GLs).
004.422.81	Main program		Conversational languages
004.422.83	Subroutines		⇒ 004.655
004.422.832	Coroutines	004.434	Problem oriented languages
004.422.833	Modules. Procedures		◇ 004.434:004.422.635.2 List processing languages
004.422.834	Functions		◇ 004.434:004.8 Programming languages for AI
004.422.837	Program macros		◇ 004.434:004.82 Knowledge representation languages
004.422.86	Parameters (arguments)		◇ 004.434:004.94 Simulation languages
004.423	Syntax and semantics of programs		◇ 004.434:005 Business application languages
004.423.2	Program syntax		◇ 004.434:5 Scientific application languages
004.423.22	Concrete syntax	004.435	Metalanguages
004.423.23	Abstract syntax	004.436	Description languages
004.423.24	Regular expression	004.436.2	Hardware description language
004.423.25	Context-free grammar	004.436.4	System description language
004.423.26	Attribute grammar	004.436.6	Page description languages
004.423.4	Program semantics	004.438	Individual programming languages
004.423.42	Operational semantics		<i>Specify by alphabetic extension (Table 1h).</i>
004.423.43	Denotational semantics		<i>If preferred, individual languages may be listed under the kind of language using alphabetic extension at the appropriate subdivisions of 004.43</i>
004.423.45	Algebraic semantics		⇒ =93
004.423.46	Axiomatic semantics		Markup languages. Tagging codes
004.424	Programming techniques	004.439	<i>Conventions for encoding text and other data (e.g. pictures, sound) in electronic documents to enable them to be output, displayed or searched in a particular way</i>
004.424.2	Basic control structures		<i>Specify named markup languages by direct alphabetical extension (Table 1h) or by and alphabetical extension</i>
004.424.22	Repetition. Loops. Iteration		◇ 004.439:004.55 Markup languages for hypermedia
004.424.23	Selection		◇ 004.439:004.55HTML Hypertext Markup Language
004.424.25	Decision tables		◇ 004.439:004.55HTML'232 HTML editors
004.424.27	Jump. GOTO. Branch		◇ 004.439:004.94VRML Virtual Reality Modeling Language
004.424.3	Subroutine techniques		◇ 004.439SGML Standard Generalized Markup Language
004.424.32	Subroutine calls		
004.424.33	Arguments	004.45	System software
004.424.35	Re-entrant routines	004.451	Operating systems
004.424.36	Recursive procedures	004.451.1	Input and output control
004.424.4	Search techniques		◇ 004.451.1:004.237 Interrupt I/O
004.424.42	Search key	004.451.2	Process management
004.424.43	Table search. Table look-up	004.451.21	Mutual exclusion
004.424.44	Set operation	004.451.22	Fairness
004.424.45	Linear search	004.451.23	Synchronization
004.424.46	Binary search		⇒ 004.032.32
004.424.47	Hashing	004.451.24	Interprocess communication
004.424.5	Sort techniques	004.451.25	Process control
	◇ 004.424.5:004.337 Internal sorting	004.451.26	Process scheduling
	◇ 004.424.5:004.338 External sorting	004.451.27	Deadlock prevention. Deadlock recovery
	◇ 004.424.5.032.24 Parallel sorting	004.451.3	Memory management
004.424.52	Minimum-comparison sorting	004.451.31	Memory allocation
004.424.53	Sorting network	004.451.33	Garbage collection
004.424.56	Merging	004.451.34	Memory compaction. Defragmentation
004.424.57	Permutation	004.451.35	Virtual memory management
004.424.6	Pattern matching	004.451.352	Paging
004.424.62	String pattern matching. String matching	004.451.353	Segmentation
004.424.64	Tree pattern matching		
004.424.7	Memory (storage) techniques		
004.424.72	Blocking		
004.424.74	Dynamic memory allocation		
004.424.75	Heap manipulation		
004.424.8	Input/output instructions		
004.428	Libraries of subroutines		
004.428.2	Standard subroutines		
004.428.4	User-defined subroutines		
004.43	Computer languages. □ Programming languages		
	<i>The 004.4'4... auxiliaries are particularly applicable here</i>		
	⇒ 004.655		
	⇒ 811.93		

004.451.354	Paged segmentation	004.491.42	Trojan horses ⇒ 004.056.54
004.451.36	Memory protection	004.491.43	Logic bombs
004.451.37	Secondary storage (auxiliary memory) management	004.492	Software against computer infections
004.451.4	Job management	004.492.2	Software to prevent infections
004.451.43	Spooling	004.492.3	Software to detect infections
004.451.44	Job scheduling	004.492.4	Software to remove infections
004.451.45	Multithreading	004.493	Unauthorised software to monitor machine activity. Spyware. □ Adware, adbots ⇒ 004.056.54
004.451.46	Multitasking. Multiprogramming		
004.451.47	Job control language (JCL). Command language	004.5	Human-computer interaction. Man-machine interface. User interface. User environment <i>For peripheral hardware use 004.35 and subdivisions</i>
004.451.48	Operator command		
004.451.5	Data management	004.51	Display interface
004.451.51	Volume management	004.512	Textual interfaces. Character-based user interfaces
004.451.52	File management. □ File creation. File deletion. File manipulation ◇ 004.451.52.056.3 File dumping ◇ 004.451.52.056.5 File protection	004.512.2	Command prompts
004.451.53	File organization. □ Directory structures	004.512.3	Menu interfaces
004.451.54	Record manipulation ⇒ 005.92	004.512.4	Conversational interfaces
004.451.56	Access methods	004.512.5	Forms dialogue interfaces
004.451.57	Buffering	004.514	Graphical user interfaces
004.451.6	Communication management	004.514.2	Pointers
004.451.62	Inter-process communication	004.514.4	Pop-up menus. Pull-down menus
004.451.622	Message passing	004.514.6	Windows environment. Wimp interface
004.451.624	Shared memory	004.514.62	Desktop manager
004.451.64	Privileged instructions. Hardware protection	004.514.64	Drag-and-drop
004.451.642	Operating modes. □ Monitor mode (supervisor mode, system mode), User mode	004.52	Sound interface
004.451.644	System calls (supervisor calls)	004.522	Speaking interface
004.451.7	Time-sharing processing ◇ 004.451.7:004.7 Data communication processing ◇ 004.451.7.031.43 Real-time processing. Transaction processing	004.523	Use of sounds other than speech
004.451.8	Design features of operating systems	004.55	Hypermedia. Hypertext
004.451.82	Layering	004.58	User help ⇒ (083.131)
004.451.83	Client-server design	004.582	On-screen help
004.451.84	System interfaces	004.584	Wizards
004.451.86	Subsystems	004.588	Tutorial programs
004.451.87	Kernel		
004.451.88	Microkernel	004.6	Data
004.451.9	Specific operating systems <i>Specify by alphabetic extension (Table 1h)</i> ◇ 004.451.9:UNIX UNIX operating system	004.62	Data handling
004.453	Program control software	004.622	Data preparation
004.453.2	Linkers	004.623	Downloading. Uploading
004.453.3	Relocators	004.624	Data export. Data import
004.453.4	Loaders	004.627	Data compression
004.453.5	Initial program loader	004.63	Files
004.454	Device drivers	004.632	File access
004.455	Client-server software	004.632.2	Serial access
004.455.1	Client software. □ Browsers ◇ 004.455.1:004.738.52 Worldwide Web browsers	004.632.3	Sequential access
004.455.2	Server software ⇒ 004.383.2	004.632.4	Random access. Direct access
004.457	Utility programs. □ Software agents. Daemons	004.632.5	Indexed sequential access
004.49	Computer infections ⇒ 004.056.53 ⇒ 004.056.54 ⇒ 004.056.57 ⇒ 004.493	004.633	File manipulation
004.491	Software to attack computer systems. Malicious software (malware)	004.633.2	Sort operation
004.491.2	Replicating infections	004.633.3	Merge operation
004.491.22	Computer viruses	004.633.4	Update operation
004.491.23	Computer worms	004.65	Database management systems (DBMS) ◇ 004.65:004.451 Database operating systems
004.491.4	Nonreplicating infections	004.651	Database file organization
		004.651.2	Partitioned organization
		004.651.3	Multilist organization
		004.651.4	Tree structure organization
		004.651.5	Dynamic file organization
		004.651.52	B-tree
		004.651.53	Self-organization tree
		004.651.54	Dynamic hash organization
		004.652	Database models
		004.652.2	Hierarchical model
		004.652.3	Network model
		004.652.4	Relational model

004.652.42	Relational integrity	004.728.7	Presentation layer
004.652.43	Relational algebra	004.728.8	Application layer
004.652.44	Relational calculus		
004.652.5	Object-oriented data model	004.73	Networks according to area covered
004.652.6	Logical database model	004.732	Local area networks (LANs)
004.652.7	Inverted file model	004.733	Metropolitan area networks (MANs)
004.652.8	Entity-relationship model	004.735	Wide area networks (WANs)
004.654	Data dependency	004.738	Network interconnection. Internetworking.
004.655	Database languages		□ Networks of networks, e.g. Arpanet and Internet, and national, state or academic networks
004.655.2	Data definition languages		◇ 004.738:378(410) UK Joint Academic Network - JANET
004.655.3	Data manipulation languages. □ Query languages. Standard query language (SQL)	004.738.1	Site. Service node. Sites by type of service
004.656	Data dictionaries		◇ 004.738.1FTP FTP site
004.657	Database query processing		◇ 004.738.1WWW World Wide Web
004.658	Database management	004.738.4	Intranets
004.658.2	Operation of database	004.738.45	Extranets
004.658.3	Restructuring of database	004.738.5	The Internet. □ World Wide Web (WWW)
004.658.4	Standardization of database		◇ 004.738.5:338.46 Internet service providers
004.658.6	Interoperable database		◇ 004.738.5:339 Electronic commerce (e-commerce). Including: dotcoms (Internet-based companies)
004.659	Search engines		◇ 004.738.5:339.371.4 Electronic retailing (e-tailing)
004.67	Systems for numeric data. □ Spreadsheet systems		◇ 004.738.5:655.41 Electronic publishing (e-publishing)
004.7	Computer communication. Computer networks		◇ 004.738.5.057.4 Internet protocol. TCP/IP
004.71	Computer communication hardware <i>Telecommunication equipment is classed under 621.39</i>		◇ 004.738.5.057.2 Internet and WWW standards
004.712	Network adapters. Network cards	004.75	Distributed processing systems
004.713	Data switches		◇ 004.75:004.451 Distributed operating systems
004.714	Hubs. □ Active hubs. Passive hubs		◇ 004.75:004.455 Client-server systems
004.715	Routers		⇒ 004.383.2
004.716	Network interconnection devices. □ Bridges. Gateways. Relays	004.77	General networking applications and services. □ Internet applications
004.717	Front-end processors. Front-end computers	004.771	Remote access. □ Telnet, Secure Shell (ssh), remote login (rlogin), remote shell (rsh), remote command prompt (rcp), remote executable (rexec)
004.72	Network architecture <i>Denote protocols by .057.4</i>		<i>Programs for executing commands on remote machine</i>
	◇ 004.72:004.451 Network operating systems	004.772	File transfer
	◇ 004.72.056.52 Restricted access in networks. Firewalls		◇ 004.772FSP File Service Protocol
004.722	Network topologies. □ FTP (file transfer protocol)		◇ 004.772FTP File Transfer Protocol
004.722.2	Point-to-point networks		◇ 004.772FTP File Transfer Protocol
004.722.22	Star networks	004.773	Message interchange
004.722.23	Ring networks	004.773.2	Bulletin boards
004.722.25	Tree networks	004.773.3	Electronic mail (e-mail)
004.722.4	Broadcast networks		◇ 004.773.3ELM Easy Learning Mail - ELM program
004.722.42	Token ring networks		◇ 004.773.3EUDORA Eudora mail program
004.722.43	Token bus networks	004.773.5	Computer conferencing. □ Video conferencing
004.722.45	Satellite networks	004.773.6	Instant messaging. Real-time messaging. Chat rooms
004.724	Data switching methods		◇ 004.773.6IRC Internet Relay Chat program
004.724.2	Circuit switching		◇ 004.773.6ICQ 'I-see-you'
004.724.3	Message switching	004.773.7	Newsgroups. Forums. Online discussion groups
004.724.4	Packet switching	004.774	HTTP application. World Wide Web in the strict sense. Web pages. Web resources. Web Content
004.725	Network parts		◇ 004.774:001.103.2 Web resources metadata
004.725.2	Hosts		◇ 004.774-026.11 Static web pages. Passive web
004.725.4	Nodes		◇ 004.774-026.12 Dynamic web pages. Active web
004.725.5	Local access networks	004.774.2	Semantic Web
004.725.7	Backbone networks		
004.728	Open systems interconnection (OSI) reference model		
004.728.1	Physical layer		
004.728.3	Data-link layer		
	◇ 004.728.3.057.4 Data-link control protocols		
004.728.4	Network layer		
004.728.5	Transport layer		
004.728.6	Session layer		

004.774.6	Tools and programs for web page design, web content publishing, organization and management	004.896	Artificial intelligence in industrial systems. □ Intelligent CAD/CAM. Intelligent robotics
	◇ 004.774.6:311 Web page access statistics programs	004.9	Application-oriented computer-based techniques
	◇ 004.774.6CMS Content Management Systems	004.91	Document processing and production
	◇ 004.774.6RSS Syndication of Web content using Really Simple Syndication (RSS) format	004.912	Word processing. Text processing
	◇ 004.774.6BLOG Weblog pages. Blogs. Interactive web journals	004.915	Desktop publishing
004.775	Search and information services. □ Wide Area Information System (WAIS), Gopher, web crawlers, search engines, intelligent agents, bots, directories, gateways, portals, vortals (vertical industry portals), webring	004.92	Computer graphics
004.777	Programmatic network interfaces. Including: Web services	004.921	Elements or objects in computer graphics
004.78	Online computing systems for specific use	004.922	Coordinates in computer graphics. □ Viewing transformation
	◇ 004.78:025.4.036 For information retrieval	004.923	Stereoscopic view
	◇ 004.78:336.717 For banking transactions	004.924	Input methods for graphics ⇒ 004.352.22 ⇒ 004.354.4
004.8	Artificial intelligence	004.925	Computer graphics techniques
004.81	Models of cognitive processes	004.925.2	Hidden line removal. Hidden surface removal ◇ 004.925.2.021 Hidden line algorithm
	Specify by :159.9...	004.925.3	Shading. □ Ray tracing
	◇ 004.81:159.942.52 Model of feeling	004.925.4	Texture mapping
	◇ 004.81:159.953 Model of memory	004.925.5	Colour display
	◇ 004.81:159.953.32 Model of association	004.925.6	Painting
	◇ 004.81:159.953.5 Model of learning	004.925.8	Geometric modelling
	◇ 004.81:159.953.6 Model of forgetting	004.925.82	Wireframe modelling
	◇ 004.81:159.955 Model of thinking	004.925.83	Surface modelling
004.82	Knowledge representation	004.925.84	Solid modelling
004.822	Knowledge networks. □ Semantic networks	004.925.86	Curve modelling
004.823	Frame systems. Frames. □ Schemata. Scripts	004.928	Animation
004.824	Multiple word	004.93	Pattern information processing
004.825	Production systems. Rule-based systems	004.93'1	Pattern recognition. Pattern matching
004.826	Blackboard model	004.93'11	Pattern space. □ Feature space. Distance
004.827	Representation of ambiguity. □ Uncertainty. Incompleteness of knowledge	004.93'12	Pattern discrimination
004.83	Reasoning	004.93'14	Clustering in patterns
004.832	Problem solving	004.932	Image processing
004.832.2	Search	004.932.1	Image sampling
004.832.22	Representation of problem	004.932.2	Image analysis
004.832.23	Search in problem space	004.932.4	Image editing. □ Image filtering. Image enhancement. Anti-aliasing
004.832.24	Search in game tree	004.932.7	Image types
004.832.25	Constraint-based search	004.932.72	Objects in images ◇ 004.932.72'1 Object recognition
004.832.28	Control strategies	004.932.75	Characters as images ◇ 004.932.75'1 Character recognition
004.832.3	Inference	004.934	Speech information processing
004.832.32	Method of inference	004.934.1	Words in speech ◇ 004.934.1'1 Word recognition
004.832.34	Inference based on ambiguous knowledge	004.934.2	Speech analysis
004.832.38	Control of inference	004.934.5	Speech synthesis
004.838	Kinds of reasoning	004.934.8	Speakers ◇ 004.934.8'1 Speaker recognition
004.838.2	Qualitative reasoning	004.94	Simulation
004.838.3	Analogical reasoning		⇒ 004.358
004.838.5	Abduction	004.942	⇒ 004.383.4
004.85	Learning		Simulation using mathematical models Class here computing aspects ⇒ 519.876.5
004.852	Statistical learning and parameter learning	004.946	Virtual reality
004.853	Knowledge acquisition and maintenance	005	MANAGEMENT
004.855	Learning and inductive inference		⇒ 37.091
004.855.2	Grammatical inference	005.1	Management theory
004.855.3	Concept learning	005.11	Schools of thought. □ Agency theory
004.855.5	Learning from examples	005.12	Principles. Basis
004.855.6	Learning by analogy. Learning by discovery	005.13	Causes. Drivers
004.89	Artificial intelligence application systems. Intelligent knowledge-based systems	005.2	Management agents. Mechanisms. Measures
004.891	Expert systems		
004.891.2	Consultation expert systems		
004.891.3	Diagnostic expert systems		

005.21	Strategies	005.334.4	Losses. Wastage. □ Insolvency
	⇒ 005.52	005.336	Positive factors. Opportunities. Advantages.
005.212	Visions		Strengths. Progress
005.213	Missions	005.336.1	Efficiency
	◇ 005.213(047.3) Mission statements	005.336.2	Competence. Incompetence
005.214	Goals. Aims	005.336.3	Quality
005.216	Objectives. Targets	005.336.4	Intellectual capital. Expertise. Know-how
005.216.1	Performance indicators	005.336.5	Skills
	⇒ 005.332.1		◇ 005.336.5:004 Computer skills. IT skills
	⇒ 005.962.13		◇ 005.336.5:005.57 Communication skills.
005.216.3	SMART		Interpersonal skills
005.218	Deliverables		⇒ 316.77
005.22	Methods. Techniques. Approaches	005.336.6	Goodwill. Reputation
005.24	Rights	005.337	Income. Incomings. Earnings. Returns.
005.25	Responsibilities. Functions. Obligations.		Revenue
	Roles		⇒ 338.515
005.254	Duties. Tasks	005.337.1	Bonuses. Allowances. Credit
005.3	Management activities	005.34	Management styles. Management approaches
005.31	Operational research (OR)		Conservative. Safe. Low-risk
	⇒ 303.7	005.341	Innovative. Risk-taking. Entrepreneurial
	⇒ 519.8	005.342	Competitive
005.311	Data collection. □ Surveys	005.343	Cooperative. Relationship based
005.311.1	Types and techniques of operational research	005.344	Task-focused
005.311.11	Qualitative operational research	005.345	Customer-focused. Market-driven
	◇ 005.311.11:303.62 Use of questionnaires	005.346	
005.311.12	Quantitative operational research	005.35	Social aspects of management. Social responsibility in management. Corporate responsibility
	◇ 005.311.12:519.852 Use of linear programming techniques		
005.311.121	Statistical techniques	005.4	Processes in management
	⇒ 519.2	005.41	Developmental processes. Life-cycles
005.311.2	Flow charts. Function diagrams (as subjects)	005.411	Start up. Foundation. Establishment.
	⇒ (084.21)		Inauguration
	⇒ (084.29)	005.412	Growth. Rise. Development
005.311.6	Decision theory. □ Decision models (as subject)	005.414	Maturity
005.311.7	Game theory	005.415	Decline
005.32	Organizational behaviour. Management psychology	005.416	Dissolution. Closure. Liquidation
	◇ 005.32:331.101.3 Motivation. Attitudes to work	005.418	Renewal. Reconstruction
	◇ 005.32:331.101.32 Job satisfaction	005.42	Interaction. Relations
005.321	Aspirations		◇ 005.42-024.61 Internal relations in the organization
005.322	Personal qualities	005.44	Globalisation
	◇ 005.322:316.46 Leadership qualities	005.5	Management operations. Direction
005.33	Conditions. Factors	005.51	Planning
005.331	Human assets. Stakeholders	005.511	Business planning
005.332	Forces		◇ 005.511(083.92) Business plans
	◇ 005.332:338.124.2 Recession as a factor in management	005.52	Analysis. Definition
005.332.1	Performance		◇ 005.52:005.33 Analysis of the situation.
	⇒ 005.216.1		SWOT analysis
	⇒ 005.962.13		◇ 005.52:005.334 Risk analysis. Risk management
005.332.2	Trends		◇ 005.52:330.133.1 Value analysis. Utility value cost analysis
	◇ 005.332.2:005.591.47 Downsizing as a trend in management	005.521	⇒ 303.7
005.332.3	Change. □ Variance. Breakpoints		Forecasting. Foresight. Scenario planning
	◇ 005.332.3:005.13 Change drivers	005.523	◇ 005.521:334.7 Business forecasting
005.332.4	Competition	005.525	Critical success factors (CSFs)
005.332.5	Openness. Transparency. Confidentiality.	005.53	Force field analysis
	Secrecy	005.54	Decision making. Design. Specification
	⇒ 001.9		Implementation. Execution
005.332.6	Demand		◇ 005.54:005.31 Use of OR in implementation studies
005.332.7	Events. Incidents. Occurrences. Happenings	005.55	Administration
005.332.8	Results. Outcomes. Effects. □ Impact. Output		⇒ 005.91
005.334	Negative factors. Threats. Problems. □ Risks.	005.551	Organization. Arrangement. □ Co-ordination.
	Disasters. Crises		Formation. Teambuilding
005.334.1	Deficiencies. Faults	005.552	Allocation. Assignment
005.334.2	Disputes. Disagreements. Conflicts	005.552.1	Distribution. Supply
005.334.3	Criticisms. Complaints. Grievances. Claims		

005.552.2	Demarcation. Division	005.642.5	Work study and analysis. □ Work study.
005.552.3	Scheduling. Routing. Queueing ⇒ 355.41		Method study. Simplification of work.
005.56	Collaboration. Cooperation. Participation		Time-and-motion study. Quantification and rating of work ⇒ 331.101.3 ⇒ 331.103.3
005.562	Deployment. Use. Delivery. □ Processing. Progressing. Expediting	005.642.6	Activity sampling. Ratio delay study. □ Predetermined motion-time (PMT) systems. Methods-time measurement (MTM). Work factor ⇒ 331.103.3
005.57	Communication. □ Informing. Notification	005.642.8	Standard and allowed time and their determination. □ Waiting time. Delay. Lost, unproductive, idle time ⇒ 331.103.28
005.571	Interviewing		
005.572	Consultation. Advice		
005.573	Discussion. Debate. □ Brainstorming		
005.574	Mediation. Conflict resolution. □ Negotiation. Bargaining ⇒ 316.485.6		
005.575	Liaison. □ Networking. Lobbying	005.65	Just-in-time management
005.58	Authority. Power	005.66	Reward systems ⇒ 005.88
005.581	Compulsion	005.662	Recognition systems
005.582	Intervention. Regulation. Manipulation	005.664	Monetary reward systems
005.582.1	Approval. Authorization. Inclusion		
005.582.2	Disapproval. Refusal. Exclusion	005.7	Organizational management (OM) ◇ 005.7:334.722.24 Organizational management in family-owned companies ◇ 005.7(084.2) Organization diagrams, graphs, charts
005.583	Empowerment. Delegation		
005.583.1	Motivating		
005.583.3	Assistance. Encouragement	005.71	Organizational types. Organizations. Bodies ◇ 005.71-021.131 Virtual organizations ◇ 005.71-022.5 Size of a firm ◇ 005.71-022.57:005.21 Strategic planning in very large organizations
005.584	Supervision ⇒ 005.935		
005.584.1	Monitoring. Inspection. Checking	005.72	Organizational structures ◇ 005.72:005.591.46 Decentralized structures ◇ 005.72:005.74 Organizational systems ◇ 005.72:334.7 Federations, alliances of organizations Rigid organizational structures Flexible organizational structures Flat management Pyramidal management
005.584.2	Discovery. Detection		
005.585	Evaluation. Assessment	005.73	Organizational culture Bureaucracy Hierarchical organizational culture Power-based culture Role-based culture
005.59	Problem solving. Troubleshooting. Following-up	005.731	
005.591	Adjustment. Modification. Revision ◇ 005.591:005.21 Modification of strategies	005.732	
005.591.1	Optimization. Improvement. Enhancement	005.733	
005.591.3	Expansion. □ Maximization. Extension	005.734	
005.591.4	Reorganization. Restructuring. Redesign. □ Conversion. Exchange. Transferral. Substitution. Relocation ◇ 005.591.4:005.58 Transfer of power ◇ 005.591.4:005.743 Departmentalization	005.74	Systems. Arrangements Organs. Executive bodies. □ Boards. Committees ◇ 005.742:005.591.45 Committees for cooperation. Joint committees ◇ 005.742:005.72 Corporate control ◇ 005.742-057.162 Executive officers. Executives ◇ 005.742-057.162:005.25 Responsibilities of executives ◇ 005.742-057.177.1 Board chairmen Organizational divisions. □ Teams. Units. Departments ◇ 005.743:005.334.3 Customers' complaints department ◇ 005.743-027.512 Specialized units. Including: Working groups. Technical committees Meetings. Congresses. Symposiums. Conferences ◇ 005.745:005.55 Conduct of meetings ◇ 005.745:81'253 Interpreting (oral translation) at meetings. Including: Simultaneous, consecutive interpreting at meetings ◇ 005.745(060.533) Agendas for meetings ◇ 005.745-021.372 Extraordinary meetings
005.591.41	Buy-out. Takeover		
005.591.43	Contracting out. Subcontracting. Outsourcing. □ Licensing. Franchising		
005.591.45	Concentration. Division		
005.591.452	Concentration. □ Integration. Merger. Amalgamation. Syndication. Partnership		
005.591.454	Division. □ Demerger. Dissolution of cartel		
005.591.46	Centralization. Decentralization		
005.591.47	Contraction. Downsizing		
005.591.5	Specialization. Focusing		
005.591.6	Modernization. Automation. Innovation		
005.591.61	Diversification. Experimentation		
005.6	Quality management. Total quality management (TQM)	005.743	
005.61	Productivity		
005.62	Continuous improvement		
005.63	Best values	005.745	
005.64	Teamwork ⇒ 005.961		
005.642	Group decision making		
005.642.1	Benchmarking		
005.642.2	Statistical process control		
005.642.3	'Cost of quality' measurement		
005.642.4	Process analysis. □ Pareto charts. Fish bone cause-and-effect diagrams. Control charts		

005.8	Project management ◇ 005.8:005.574 Negotiation tactics	005.922.52	Substitution as a method of record preservation ◇ 005.922.52:004.932 Digitization of images as a method of record preservation ◇ 005.922.52:778.14 Microfilming as a method of record preservation
005.81	Sequences. Flow		
005.82	Network techniques. Network diagrams. Gantt charts	005.922.7	Conservation
005.821	Activity-on-arrow systems. □ ADM (arrow diagrams). CPM (critical path analysis). PERT (Programme Evaluation and Review Technique)	005.922.72 005.923 005.923.1	Renovation. Restoration. Repair Agencies responsible for records Records creation agencies ◇ 005.923.1:354.11 Ministry of Foreign Affairs as a record creating agency
005.823	Activity-on-node networks. □ Precedence diagrams (PDM). Roy method, or method of potentials (MOM or MPM)	005.923.2	Records storage agencies ⇒ 930.253
005.828	Milestones	005.923.21	Records centres
005.83	Procedures. Processes	005.93	Plant management. Physical resources management
005.84	Formalities	005.931	Physical conditions. Physical environment
005.85	Restrictions. Controls	005.931.1	Management of adverse conditions. Damage limitation
005.86	Directions	005.931.11	Emergencies. Disasters. Crises
005.87	Deliverables	005.931.12	Accidents
005.88	Rewards. Incentives ⇒ 005.66	005.931.13 005.931.2 005.931.21 005.932	Breakdowns. Damage Management of deficiencies, shortages, lacks Errors. Defects. Faults Inventory management. Stock management. Resource management
005.89	Penalties		Ordering. Commissioning
005.9	Fields of management	005.932.1	Acquisition. Purchase. Buying
005.91	Administrative management. Secretariat	005.932.2	Selection. Choice
005.911	Levels of management ◇ 005.911-057.175 Middle management	005.932.3 005.932.4 005.932.5 005.932.6 005.933 005.933.1	Receipt. Collection Payment. Spending Dispatch Setting up. Fixing Assembly. Construction. Putting together. Installation
005.912	Office management. Office services ◇ 005.912:656.8-024.61 Internal mail services ◇ 005.912:656.874 Use of courier services ◇ 005.912:81'25 Office translating service ◇ 005.912-051 Office staff in general ◇ 005.912-057.164 Assistants. Secretarial staff	005.933.2 005.934 005.934.1 005.934.2 005.934.3 005.934.4 005.934.5 005.935 005.935.1 005.935.2 005.935.3	Disposition. Sorting Security measures. Safety precautions Oversight Maintenance Protection. Avoidance Renovation. Restoration. Repair Servicing Supervision of plant and production Observation Investigation Examination ⇒ 006.015.5
005.912.2	Interface with the public. □ Commissionaire service. Reception. Enquiries. Visits	005.935.31 005.935.32 005.935.33 005.935.34 005.935.35 005.935.38 005.936 005.936.1 005.936.2	Checking Sampling Testing. Tests Trials Discovery. Detection Decommissioning. Dismantling Physical assets. Property. Facilities Land Buildings. Sites ◇ 005.936.2:69.05 Site layouts
005.912.3	Counter duty. Desk duty		Offices
005.913	Registries		Headquarters. Head office
005.915	Financial management		Subsidiary sites Plant Fixtures and fittings Office equipment. Office machines ◇ 005.936.33:772.93 Electrographic office machines. Office copiers, duplicators
005.92	Records management ◇ 005.92:004.63 Records management of digital documents, files. Management of electronic records ⇒ 004.451.54 ⇒ 02	005.936.21 005.936.22 005.936.23 005.936.3 005.936.31 005.936.33	Stock. Inventory Supplies. Raw materials Work in progress. Semi-finished products Products. Goods. Merchandise. Wares By-products Waste. Residue. Refuse. Scrap
005.921	Records management policy decisions		
005.921.1	Retention. Discarding ◇ 005.921.1(083.81) Retention schedules ◇ 005.921.1'4" Retention periods ◇ 005.921.1-022.324 Permanent retention of records. Archiving		
005.921.8	Destruction. □ Shredding ⇒ 686.865.24		
005.922	Records management procedures ◇ 005.922:005.311 Records surveys ◇ 005.922:025.25 Accessioning, serial numbering of records ◇ 005.922:025.4 Classification and indexing of records ◇ 005.922:025.6 Loan of records, documents		
005.922.1	Security. Protection. Avoidance		
005.922.3	Storage ◇ 005.922.3:684.455.6 Storage of records in plan cabinets. Shallow drawer filing ◇ 005.922.3-025.76 Lateral filing of records. Including: Roll-out filing	005.936.4 005.936.41 005.936.42 005.936.43 005.936.44 005.936.5	
005.922.5	Preservation		

005.94	Knowledge management	005.962.11	Time management
	◇ 005.94:004.9 Automated knowledge management systems	005.962.13	Performance management
005.941	Corporate know-how. Organizational know-how		◇ 005.962.13:005.332.1 Performance measures
005.942	Advisory services. Consultancy services		⇒ 005.216.1
	◇ 005.942:334.7 Business consultancy	005.962.131	Appraisal. Feedback
005.95/96	Personnel management. Human Resources management	005.962.15	Staff discipline
005.95	Personnel management	005.963	Staff development
	◇ 005.95:005.55 Personnel administration. Including: Personnel files, records	005.963.1	Training. Training courses
	◇ 005.95:005.92 Management of personnel records	005.963.2	Mentoring
	◇ 005.95-051 Personnel managers	005.963.3	Visits, study trips
	⇒ 331.5	005.963.5	Workplace training
005.951	Posts. Positions	005.966	Careers. Career development
	◇ 005.951:005.52 Job analysis	005.966.1	Job shadowing
	◇ 005.951:005.53 Job specification	005.966.2	Voluntary work
	◇ 005.951:005.585 Job evaluation	005.966.3	Job mobility
005.952	Workforce market	005.966.32	Secondments
	◇ 005.952:005.53 Person specification	005.966.4	Job flexibility
005.952.2	Staff ratios	005.966.42	Job swaps
005.953	Recruitment	005.966.44	Job sharing
	◇ 005.953:316.356.2 Nepotism in recruitment	005.966.5	Promotion. Advancement
	◇ 005.953:331.5 Influence of the labour market on recruitment	006	STANDARDIZATION OF PRODUCTS, OPERATIONS, WEIGHTS, MEASURES AND TIME. □ Standards. Specifications. Codes of practice. Recommendations
	◇ 005.953:364.614.8 Equal opportunities in recruitment		⇒ (083)
005.953.2	Selection. □ Selection criteria. References. Curricula vitae	006.01	⇒ 658.516
	◇ 005.953.2:316.45 Group selection	006.011	Theory, principles, fundamentals, aims
	◇ 005.953.2:005.571 Assessment by interview	006.012	General theory, principles of standardization
005.954	Appointment of staff. Employment of staff		Specific bases and fundamentals of standardization
	◇ 005.954:349.22 Employment contracts		<i>Specify by .02...</i>
	◇ 005.954:331.31 Hours of work. Including: Flexitime	006.013	◇ 006.83.012.024 Legal bases for quality standardization
005.954.2	Probationary periods		Investigation of the usefulness and expediency of standardization in particular subject fields
005.954.4	Leave. Holiday entitlement		◇ 006.78.013 Need for dimensional standardization
005.954.6	Compensation. Benefits. □ Benefits in kind. Fringe benefits		⇒ 006.052
	◇ 005.954.6:336.77:365.262.1 Low-interest loans for home purchase	006.015	Aims of standardization, and associated activities
	◇ 005.954.6:625.712.63 Provision of car parking facilities	006.015.2	Variety reduction. Simplification. Unification. Rationalization
	◇ 005.954.6:656.03:336.773 Interest-free loans for travel, season tickets	006.015.3	Comparability. Interchangeability
005.955	Remuneration. Wages. Salaries	006.015.4	Predictability. Repeatability. □ Process validation
	◇ 005.955"530*1" Weekly wages	006.015.5	Quality. □ Quality assurance
	◇ 005.955-021.61 Basic pay		⇒ 004.415.5
005.956	Job termination	006.015.7	⇒ 005.935.3
005.956.2	Redundancy	006.015.8	Reliability
	◇ 005.956.2:347.447.84 Compensation. Redundancy payment	006.02	Safety
	◇ 005.956.2-027.556 Voluntary redundancy	006.02	Special aspects of standardization and standards
005.956.4	Resignation	006.022	Technical aspects
005.956.6	Retirement	006.024	Legal aspects
005.958	Problems with personnel	006.025	Economic aspects
005.958.2	Negative attitudes to employment. Unsatisfactory attitudes to work	006.027	Organizational and management aspects
005.958.4	Absenteeism	006.03	Authority and competence of standardizing body
005.96	Human resources management (HRM)		<i>Denote the body by (4/9) and/or alphabetic extension</i>
005.961	Work	006.032	World-wide
	◇ 005.961:005.81 Work flow		◇ 006.032ISO International Organization for Standardization
	◇ 005.961:005.336.1 Ergonomics	006.033	Regional. Multinational
	⇒ 005.64	006.034	National. State
005.962	Staff		◇ 006.034(410) British Standards Institution
	◇ 005.962:005.56 Staff supervision	006.036	Special subject field
	◇ 005.962:364.634 Workplace harassment	006.037	Specific firm, concern
005.962.1	Staff procedures	006.039	Other authorities

006.05	Creation of standards. Stages of standardization work	006.35	Standards in the strict sense
006.052	Investigation of the need for a standard. □ Establishment of programme, plan ⇒ 006.013		◇ 006.35(100)ISO ISO standards ◇ 006.35(4)CEN European standards ◇ 006.35(410)BS British Standards ◇ 006.35.037(492)Philips Company standards for the Netherlands branch of Philips
006.053	Preparation, drafting of standards. □ Technical-economic analysis	006.352	Standards without inherent legal force ◇ 006.352(430)DIN German (DIN) standards
006.054	Evaluation of comments. Modification of draft standards	006.354	Standards with legal force ◇ 006.354(47)GOST Soviet (GOST) standards
006.055	Approval of modified standards for publication		
006.056	Publication as standards	006.39	Other kinds of standard according to their nature
006.057	Revision of published standards. □ Modification. Amendment. Conversion ⇒ 006.067	006.4	Standards according to their status and scope
006.058	Adoption of standards prepared by other bodies	006.42	Fundamental standards <i>Valid in the whole of the subject field covered by the standardizing body</i> ◇ 006.42:003.62 Standards for symbols in general use
006.06	Methods of introduction of standards	006.44	Standards in a special subject field <i>Valid only in one field of activity of the standardizing body</i> ◇ 006.44:621.3 Standards in electrical engineering ◇ 006.44:621.882.218 Standards for wing screws ◇ 006.44:625.112 Standards for determining the gauge of railway lines
006.062	Propagation, promulgation of standardization and standards. □ By instruction and publicity, use of mass media, legal, administrative, other means <i>For referral services, use 659.2:006</i> ◇ 006.062:659.131 Propagation of standards by advertising in newspapers, magazines etc.	006.5	Standards according to degree of standardization
006.063	Certification and marking of products for conformity to standards ⇒ 347.772 ⇒ 658.782	006.52	Comprehensive standards <i>Covering all aspects and possible formulations of an object or process</i>
006.065	Application of standardization. Use of standards	006.55	Partially comprehensive standards <i>Covering only certain aspects and possible formulations of an object or process</i>
006.065.2	Voluntary application	006.57	Skeleton, abridged standards <i>Giving only a framework, so that the relevant object or process may be used together with other objects and processes, e.g. standards for assembly dimensions in measuring apparatus</i>
006.065.3	Application without legal obligation or enforcement. □ For efficiency	006.71.8	Standards according to content
006.065.5	Obligatory, mandatory application within certain fields of activity. □ In official trade. In public authorities	006.72	Communication standards. □ For nomenclature, terminology, definitions, symbols, units, numerals, documentation, classification, drawing, bibliography, correspondence
006.065.6	Obligatory, mandatory application in all fields of activity	006.73	Grading standards <i>Criteria for evaluating various objects, commercial categories etc. by quality</i> ⇒ 006.83
006.067	Maintenance and revision. □ Supervising conformity with, and level of, standards. Control and review of validity of older standards. Revision proposals ⇒ 006.057	006.74	Type standards <i>Standards for classing objects according to type, size, similar characteristics</i>
006.07	Effects, influences of standards and standardization	006.76	Standards for planning. □ Codes of practice for design and calculation. Regulations for design and construction
006.072	Effects on the manufacturer	006.77	Standards for design and construction <i>Laying down details of construction, design of an object and of its parts</i>
006.073	Effects on trade	006.78	Dimensional standards <i>Standardization of dimensions of objects and of their parts</i>
006.074	Effects on the consumer		
006.077	Effects on the legislator		
006.078	Effects on national and local authorities		
006.1	Activity of standardization in general		
006.3/8	Standards ⇒ (083.74)		
006.3	Standards in general. Standards according to their nature and validity. □ National, international standards		
006.32	Standards issued by world-wide organizations for regional organizations and national institutions; by regional organizations for national institutions		
006.322	Standards issued by world-wide organizations for regional and national institutions ◇ 006.322(100)ISO ISO recommendations		
006.323	Standards issued by regional organizations for national institutions ◇ 006.323(4)CEN CEN unification documents		

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|---|---|
| <p>006.82 Standards for materials
 <i>Standardization of physical, chemical, technological properties and composition of materials</i></p> <p>006.83 Standards for quality. □ Performance standards. Standards for reliability, durability of products
 ⇒ 006.73</p> <p>006.85 Standards for processes, operations.
 □ Standardization of work, manufacturing processes</p> <p>006.86 Standards for testing. □ Standardization of techniques of inspection and measurement</p> <p>006.87 Standards for delivery and service.
 □ Standardization of conditions of delivery, packing. Conditions of contract</p> <p>006.88 Safety standards. □ Standards for accident prevention, e.g. standards for protection against ionizing radiation
 ⇒ 614.8</p> <p>006.89 Standards with other contents</p> <p>006.9 Standardization of weights, measures and time measurement</p> <p>006.91 Metrology. Weights and measures in general
 <i>General matters only</i>
 ◇ 006.91:663.2 Measures for wine and wine production
 ⇒ 343.532
 ⇒ 351.821
 ⇒ 531.7
 ⇒ 681.2</p> <p>006.915 Various systems of weights and measures</p> <p>006.915.1 Metric systems
 ◇ 006.915.1SI Système Internationale</p> <p>006.915.2 Systems of national origin
 ◇ 006.915.2(410) The imperial (British) system</p> <p>006.92 Horology. Determination and standardization of time. □ Apparent and mean time. Equation of time. Causes of inequality
 ⇒ 521.9
 ⇒ 551.7.03</p> <p>006.922 Intervals of time</p> <p>006.922.1 Eras. Vaguely defined expanses of time
 <i>For geological eras see 551.7 and "6"</i></p> <p>006.922.2 Years. □ Solar year. Lunar year. Planetary year</p> <p>006.922.3 Shorter intervals. □ Months. Weeks</p> <p>006.922.4 Days. □ Sidereal days. Solar days</p> <p>006.922.5 Subdivisions of days. □ Hours</p> <p>006.924 Standard time. Time zones. Time signals
 ◇ 006.924(4-15) West European time
 ◇ 006.924(410) UK standard time
 ◇ 006.924(410)"322" British Summer Time</p> | <p>006.924.2 Standard time. Greenwich Mean Time</p> <p>006.924.4 Time signals</p> <p>006.95 Chronology. The calendar
 <i>For calendars as documents see (059) and 050.9</i>
 ⇒ 26-564
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 <i>Other than Christian</i>
 ◇ 006.951(354) Babylonian calendar
 ◇ 006.951(=411.16) Jewish calendar
 ⇒ 26-564</p> <p>006.954 Western calendars. The Christian calendar</p> <p>006.954.1 Primitive Christian calendars
 ◇ 006.954.1(=412'09) Coptic calendar. NOTE: (=412'09)derived from =412'09(Table 1c)</p> <p>006.954.2 Julian calendar. Julian reform. Julian date</p> <p>006.954.3 Gregorian calendar. Gregorian reform. Gregorian date</p> <p>006.954.4 Ecclesiastical calendar. The Christian church calendar. □ Determination of movable feasts and feast days. Easter cycle (Metonic cycle). Date of Easter
 <i>Alternatively see 27-564</i></p> <p>006.955 Calendar reform. □ Modern proposals for reforming the calendar</p> <p>007 ACTIVITY AND ORGANIZING. COMMUNICATION AND CONTROL THEORY GENERALLY (CYBERNETICS). 'HUMAN ENGINEERING'
 ⇒ 004
 ⇒ 005.1
 ⇒ 316.77
 ⇒ 519.7
 ⇒ 62-5
 ⇒ 621.316.7
 ⇒ 621.391
 ⇒ 681.5</p> <p>007.2 Variability, physical configuration and state. Transformation of state</p> <p>007.3 Directives, programming and physical organization of variability</p> <p>007.5 Self-acting systems
 ⇒ 681.51</p> <p>007.51 With human operators as links</p> <p>007.52 Without human operators. □ Automata. Robots
 ⇒ 004.8
 ⇒ 519.713</p> <p>008 CIVILIZATION. CULTURE. PROGRESS
 ⇒ 17.023.36</p> |
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01 BIBLIOGRAPHY AND BIBLIOGRAPHIES. CATALOGUES

- ⇒ 002
- ⇒ 025.3
- ⇒ 025.4
- ⇒ 930.255

011/016

BIBLIOGRAPHIES

Ideal lists of documents, not confined to an actual collection

011

UNIVERSAL AND GENERAL BIBLIOGRAPHIES

Not limited to a specific author, country of origin, subject or format

- 012 AUTHOR BIBLIOGRAPHIES. INDIVIDUAL BIBLIOGRAPHIES**
Bibliographies of editions of, or writings about, the works of a particular author, or particular anonymous works. Bibliographies of works published by a particular organization
- 013 COLLECTIVE BIBLIOGRAPHIES**
Bibliographies of works of a particular class of writer
Denote bibliographies of authors belonging to a particular profession, society or subject field by ...; of works in a particular language by ...; of authors of a particular period by "..."; of authors of a particular nationality by (1/9)
- 014 BIBLIOGRAPHIES OF WORKS OF PARTICULAR CHARACTERISTICS**
- 014.1 Bibliographies of anonymous and pseudonymous works**
- 014.3 Contents bibliographies.** □ Abstracts. Lists of works appearing in serials or polygraphies. Contents lists and indexes of all kinds
- 014.5 Concordances.** □ Analytical indexes to individual authors or works
- 015 BIBLIOGRAPHIES OF PLACE.**
 □ National bibliographies. Local and regional bibliographies
 ◇ 015(410) Bibliographies of works published in the UK, e.g. British National Bibliography
- 016 SPECIAL SUBJECT BIBLIOGRAPHIES**
 ◇ 016:011 Bibliographies of general bibliographies
 ◇ 016:087.5 Bibliographies of publications for children
 ◇ 016:34(05) Bibliographies of law journals
 ◇ 016:929 Biobibliographies
- 017/019 CATALOGUES.** □ Inventories of actual collections. Library catalogues. Publishers', booksellers' lists etc.
 ⇒ 025.3
- 017 CATALOGUES IN GENERAL. SUBJECT CATALOGUES.** □ Classified or systematic catalogues. Alphabetic subject catalogues
- 017.09 Kinds of catalogue within an individual library**
 017.091 Location indexes. Shelf lists
 017.092 Accessions catalogues
 017.099 Other kinds of catalogue, arranged for the particular needs of individual libraries
- 017.1 Catalogues of non-private libraries.**
 □ Catalogues of public libraries (municipal libraries), national libraries, provincial libraries, libraries of public bodies, associations, corporations, universities, learned societies
- 017.11 Union, central, joint subject catalogues.**
 □ Repertories. Catalogues of the holdings of more than one library
- 017.2 Catalogues of private libraries**
- 017.3 Auction catalogues. Catalogues of book sales**
- 017.4 Commercial, trade catalogues.**
 □ Publishers', printers', distributors', booksellers' catalogues
- 017.41 Collective trade catalogues.** □ Catalogues of distributors, booksellers etc. (not confined to a single printer or publisher)
- 017.42 Trade catalogues of individual presses, printers, publishers**
- 017.43 Catalogues of book bargains.**
 □ Second-hand, antiquarian book catalogues
- 018 NAME CATALOGUES.** □ Author catalogues. Alphabetic catalogues of persons and places. Catchword catalogues
 018.09 ≡ 017.09
- 018.1 Name catalogues of non-private libraries**
For scope of 'non-private' see 017.1
- 018.11 Union name catalogues**
- 018.2 Name catalogues of private libraries**
- 018.3 Auction name catalogues. Name catalogues of book sales**
- 018.4 Commercial, trade alphabetic catalogues.** □ Name catalogues of publishers, printers, distributors, booksellers
- 019 DICTIONARY CATALOGUES**
Catalogues incorporating author, title, catchword, alphabetic subject entries etc. in a single sequence
 019.09 ≡ 017.09
- 019.1 Dictionary catalogues of non-private libraries**
For scope of 'non-private' see 017.1
- 019.11 Dictionary union catalogues**
- 019.2 Dictionary catalogues of private libraries**
- 019.3 Auction dictionary catalogues. Dictionary catalogues of book sales**
- 019.4 Commercial, trade dictionary catalogues.** □ Publishers', printers', distributors', booksellers' dictionary catalogues
- 019.6 Union catalogues in general. Repertories**
 ⇒ 017.11
 ⇒ 018.11
 ⇒ 019.11
- 019.9 Bibliographies and catalogues according to their own characteristics as publications**
Alternative classification
- 019.91 According to frequency of publication**
 Periodic bibliographies. □ Bibliographies in periodical form: daily, weekly, monthly, yearly etc., in which new works are listed as they appear. Current awareness lists
 019.911 Nonperiodic bibliographies. Once-and-for-all bibliographies
- 019.912 According to degree of coverage**
 Exhaustive, comprehensive bibliographies and catalogues
 019.921 Selective, partial bibliographies and catalogues
- 019.922 According to degree of bibliographic detail**
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 019.931 *With full bibliographic details*
 Concise bibliographies and catalogues
 019.932 *With abbreviated or summarized entries*

019.94	According to analytical or critical character	019.954	Card bibliographies and catalogues.
019.941	Critical bibliographies and catalogues.		□ Catalogue cards in books ('prenatal' cataloguing)
019.942	Noncritical bibliographies and catalogues.	019.96	According to arrangement of entries
	□ Title lists	019.962	In alphabetic order of author's name
019.95	According to format, style of publication	019.963	In subject order, alphabetic or systematic (classified)
019.951	Separately published bibliographies and catalogues in volume form	019.964	In alphabetic order of title
019.952	Bibliographies and catalogues in periodicals, serials	019.965	Combinations of the above arrangements.
			Mixed entries

02 LIBRARIANSHIP

⇒ 002
⇒ 01
⇒ 37.091
⇒ 322.7

021	FUNCTION, VALUE, UTILITY, CREATION, DEVELOPMENT OF LIBRARIES	022.42	Shelving for public access <i>In the reading room etc.</i>
021.1	Function. Purpose. Utility (in general)	022.43	Storage, closed access. Stacks
021.2	Educational function of libraries ⇒ 027.6/.8 ⇒ 331.348 ⇒ 371.64	022.44	Open cases
021.3	Relation of libraries to educational institutions. □ To schools, young people. Children's sections. Juvenile libraries ⇒ 027.6/.8	022.47	Closed cases
021.4	Accessory services. Specialized functions. □ Extension work. Lectures. Talks. Discussions. Exhibitions	022.48	Special-purpose cases, shelves. □ Hanging shelves. Map cases. Rolling presses
021.6	Library expansion. Grouping. Cooperation	022.5	Reading rooms. Study. Workrooms
021.61	Storage areas. Book stores, depositories, repositories	022.6	Administrative and other areas. □ Accommodation for various services. Special collection rooms. Lecture, conference rooms
021.62	Branch libraries. Regional branches	022.7	Lighting ⇒ 628.9
021.63	Centralization. Structure of library systems	022.8	Heating. Ventilation ⇒ 628.8 ⇒ 697
021.64	Interlibrary cooperation	022.9	Furniture. Fixtures. Fittings. □ Steps, trolleys. Lifts ⇒ 684.4
021.65	Mobile libraries. Travelling libraries	023	LIBRARY ADMINISTRATION. STAFF. PERSONNEL <i>Details by -05... (Table 1k)</i>
021.7	Library publicity, propaganda. □ In the press. By talks. Via schools, societies. Leaflets	023.1	Library statutes, rules, regulations, charters, byelaws
021.8	Relations between libraries and public authorities	023.2	Appointment, tenure of officers. Conditions of appointment
021.84	Free copies. □ Compulsory remittance of publications. Legal deposit. Copyright deposit	023.3	Administrative councils. □ Library committees. Boards of trustees, of governors, of directors. Appointment. Functions
021.85	Exchange of publications	023.4	Executives: duties, responsibility, vocation. Profession of librarian. □ Chief, head librarian, keeper
021.89	Library legislation	023.5	Staff: functions, qualifications, seniority. □ Sublibrarians, deputies. Specialist, departmental librarians, librarians-in-charge. Cataloguers, stock editors. Assistant librarians. Library assistants, clerks, counter assistants. Messengers <i>Denote further particulars by .08... from 35</i> ◇ 023.5.087.41 Salaries of library staff
021.9	Finance, funds. □ Subscriptions, gifts, grants, rates		
022	LIBRARY SITE, BUILDING, PREMISES. EQUIPMENT ⇒ 727	024	RELATIONS WITH THE PUBLIC. REGULATIONS FOR LIBRARY USE
022.1	Situation. Site. Location. Accessibility		
022.2	Building materials. Protection of building		
022.3	Planning, utilization of space. □ Layout, plan. Number of rooms		
022.4	Organization of book storage. □ Accommodation of stock. Shelving ⇒ 025.8		

024.01	Prescribed uses of the library. Rules and regulations in general	025.3/.6	Departments concerned with exploitation, utilization of the literature
024.1	Conditions of admission. □ Qualifications of users, readers, borrowers. Enrolment, registration	025.3	Cataloguing. Cataloguing technique ⇒ 017/019
024.2	Charges, fees. Rented use. Free use	025.31	Cataloguing rules in general. □ General instructions. Cataloguing codes
024.3	Hours, times of opening	025.32	Principles of cataloguing. Entries
024.4	Periods of closing. □ Sundays, festivals, holidays. Stock checking periods	025.321	Style of cataloguing. □ Compilation of entries and references
024.5	Reference use. □ Use, consulting within the library ⇒ 025.5 ⇒ 027.081	025.321.3	Arrangement of entry. □ Conventional layout of information
024.6	Home use. □ Loan for use outside the library. Loan period. Overdues. Fines. Reservations ⇒ 025.6 ⇒ 027.082	025.321.4	Full bibliographic entries
024.7	Special privileges. Exemptions from regulations	025.321.5	Short-title, abbreviated entries
024.8	Abuses. □ Damage to stock. Defacement, destruction, loss, theft (by users) ◇ 024.8:62-759.8 Theft-prevention, security devices. Including: Tattling (affixing sensitive strips to books, to activate alarm if they are unauthorisedly removed) ⇒ 025.85	025.321.7	Cataloguer's interpolations. □ Completion of doubtful or incomplete items. Indication of mistakes in title. Use of square brackets and other methods
025	ADMINISTRATIVE DEPARTMENTS OF LIBRARIES	025.321.8	Added entries. References
025.1	Executive. General supervision	025.322	Analytical entries (with bibliographic details)
025.11	Finances. □ Budget. Bookkeeping. Accounts <i>Details by :657</i>	025.323	Heading. Entry word
025.12	Printing and publications. □ Library reports	025.324	Title. Description
025.13	Document reproduction and selection	025.326	Edition. □ Editor. Compiler
025.133	Reproduction. Duplication	025.327	Imprint. □ Printer. Publisher. Place. Date
025.135	Selection. Sorting	025.328	Notes. Annotations
025.15	Correspondence	025.329	Filing. □ Filing rules. Incorporation, withdrawal of entries
025.17	Treatment of special materials	025.33	Collation. □ Cataloguing of parts of a work. Recording of format, pagination, chapters
025.171	Manuscripts. Archives. Rarities ⇒ 930.25	025.34	Method of arrangement of catalogues. Rules for compilation
025.172	Pamphlets	025.341	Catalogues in order of accession or location
025.173	Serials. Official publications	025.341.1	Accessions catalogues
025.174	Broadside, broadsheets. Bills, handbills	025.341.2	Shelf lists
025.175	Cuttings, clippings	025.342	Author catalogues. Author entry
025.176	Maps. Charts	025.343	Subject catalogues
025.177	Art material. Audiovisual materials and services. □ Engravings. Photographs. Drawings. Slides. Illustrations. Media resources	025.343.1	Alphabetical subject catalogues
025.178	Music. □ Scores. Rolls	025.343.2	Classified, systematic catalogues
025.2	Acquisition and accessioning	025.343.3	In combined alphabetical and systematic order. Alphabetic-classed catalogues
025.21	Selection of material	025.344	Title catalogues. □ Alphabetical catalogues of the significant terms from titles, catchword catalogues
025.22	Acquisition of stock. □ Purchase. Exchange	025.345	Combined catalogues. □ Combinations of the above forms. Dictionary catalogues
025.24	Receipt and treatment of material on arrival. □ Collation	025.346	Catalogues of additions. Additions lists
025.25	Processing of new stock. □ Accessioning. Accessions registers	025.347	Alphabetic indexes. □ Of persons, subjects, places
025.26	Duplicate and exchange departments. □ Sale or exchange of surplus copies	025.348	Indexes arranged in numerical order
025.267	Clearinghouses for duplicates	025.35	External (physical) forms of catalogues
		025.352	Printed catalogues
		025.354	Card catalogues
		025.357	Supplements to the catalogue. Accessions bulletins
		025.4	Classification and indexing. □ Indexing and retrieval languages. Classifications, thesauruses etc. and their construction ⇒ 001.82 ⇒ 168.2
		025.4.01	Classification principles and procedure. □ Characteristics of division. Facets. Citation order. Filing order. Notation
		025.4.02	Practical application, uses of classification. □ Shelf arrangement. Catalogue arrangement. Bibliographic use
		025.4.03	Searching and retrieval techniques
		025.4.032	Manual search and retrieval
		025.4.034	Mechanical sorting. □ With punched card equipment
		025.4.036	Computer-aided search and retrieval

025.4.05	Indexing and retrieval languages for the whole of knowledge. □ Universal classifications. Universal thesauruses	027.082	Loan service ⇒ 024.6 ⇒ 027.3
025.4.06	Indexing and retrieval languages for special subjects. □ Special classifications. Special thesauruses	027.1	Private libraries
025.43	Controlled term lists. □ Thesauruses. Descriptor lists	027.2	Proprietary, semiprivate libraries. □ Libraries of learned societies, academies, clubs, associations, corporations, firms
025.44/.47	Classification systems	027.3	Public paying libraries. □ Subscription libraries. Circulating libraries ⇒ 027.082
025.44	Classifications with mainly alphabetic notations ◇ 025.44BC Bibliographic Classification (Bliss)	027.4	Free public libraries <i>Established and supported by individuals or institutions</i>
025.45/.46	Classifications with mainly numeric notations	027.5	Public authority libraries
025.45	Decimal classifications ◇ 025.45DDC Dewey Decimal Classification ◇ 025.45UDC Universal Decimal Classification	027.52	Local authority libraries. □ Municipal library systems
025.46	Systems based on non-decimal or non-arabic numerals	027.53	Regional authority libraries. □ Provincial library systems
025.47	Systems with mixed, alphanumeric notations ◇ 025.47CC Colon Classification (Ranganathan)	027.54	Central authority libraries. □ State, national, government libraries (supported by central government funds). Libraries of departments of state, ministries
025.48	Thesaurus-classification combinations. □ Thesaurofacet	027.6	Libraries for special classes of user. □ Libraries for prisons, hospitals, factories, the disabled <i>Details by :... or -05...</i> ◇ 027.6-056.262 Libraries for the blind (Braille libraries)
025.49	Other indexing and retrieval languages	027.62	Libraries according to the age and sex of users
025.5	Reference work. □ Reference collections. Information services. Advice to readers ⇒ 024.5	027.625	For children, juniors, juveniles, young people, adolescents. Library work with children ⇒ 021.3
025.6	Loans. □ Loan records. Charging and issuing systems and equipment. Book cards (charging cards). Membership cards ⇒ 024.6	027.63	Libraries for minority groups. □ For foreigners, particular ethnic groups
025.7/.9	Preservation, protection of books	027.64	Libraries for industrial, labouring, artisan groups. Workmen's libraries
025.7	Binding, upkeep, repair of books	027.7	Libraries of institutions of higher education. □ University libraries. College libraries. Polytechnic libraries ⇒ 378.16
025.8	Handling, treatment, shelving of books. □ Fixed and relative locations. Labelling, class-marking, shelf-marking ⇒ 022.4	027.8	Libraries of institutions of primary and secondary education. □ School libraries ⇒ 371.64
025.85	Care, preservation of books. □ Damage. Restoration. Cleaning. Protection ⇒ 024.8	027.9	Free reading rooms, newsrooms
025.855	Destruction, dispersal of collections. □ Deliberate destruction (of volumes not needed)	028	READING
025.9	Administration of library buildings. □ Maintenance. Cleaning. Removals	028.01	Taste, passion for reading ⇒ 090.1
026	SPECIAL LIBRARIES ⇒ 002.6	028.02	Reading psychology. □ Relationships between reader, author, book
026.06	Libraries and collections of audiovisual documents	028.1	Reading methods, technique. Behaviour with books. □ Skimming. Browsing. Note-taking. Evaluation of reading matter
026.062	Sound-record libraries. □ Gramophone records. Compact discs. Music tapes. Cassettes	028.2	Choice of editions. □ Criteria for selection of best edition: annotations, indexes, paper, type
026.064	Film libraries	028.3	Courses of reading. Prescribed books. Reading lists
026.07	Kinds of special library 026.07 = 027	028.4	Reading of various classes of literature ⇒ 82-1/-9
026.08	Kind of service (reference, loan) 026.08 = 027.08	028.41	Fiction. Novels
027	GENERAL LIBRARIES	028.42	Works in foreign language
027.02	Sphere of activity of general libraries	028.5	Reading of young people. Children's reading habits ⇒ 087.5
027.021	Learned, scholarly libraries. Research libraries		
027.022	Public, popular general libraries		
027.08	Kind of service		
027.081	Reference service ⇒ 024.5		

- 028.6 Teaching the art of reading.** □ Lectures.
Stimulus and guidance in schools
⇒ 37.016:003-028.31
- 028.7 Use of reference works**
- 028.8 Advice to readers. Suggestions for reading**
- 030 GENERAL REFERENCE WORKS.**
□ Encyclopaedias. Dictionaries
Class here works about encyclopaedias and other general reference works. For encyclopaedias and general reference works themselves see (03)
- 050 SERIAL PUBLICATIONS. PERIODICALS**
Their function, business and editorial management
050.1/.4 ≡ 070.1/.4
- 050.8 Yearbooks. Directories, address books**
Class here works about yearbooks, directories and address books. For yearbooks etc. themselves see (058)
- 050.9 Almanacs. Calendars**
Class here works about almanacs and calendars. For almanacs and calendars themselves see (059)
⇒ 006.95

06 ORGANIZATIONS OF A GENERAL NATURE. □ Associations. Museums

Class organizations limited by subject with the subject, using colon combination.

The 06.0... auxiliaries are not applicable at 069 and subdivisions

⇒ 001.32

⇒ 005.71

⇒ 347.19

⇒ 347.41

- 06.01 Rights and obligations of members.**
□ Voting rights. Subscriptions, contributions, dues, fees
For general character (previously at this number) see 005.72
- 06.015 Professional enrolment. Chartering. Registration**
- 06.02 Financial commitments**
For members (previously at this number) see -057.95
⇒ 334
⇒ 336
- 06.022 Assets.** □ Movable. Immovables
For admission, resignation, exclusion (previously at this number) see -021.362
06.022 ≡ 347.2/.5
- 06.023 Distribution of communal assets among members**
For kinds of membership (previously at this number) see -057.95
- 06.024 Grants**
For grouping of members, including: specialist groups, local groups (previously at this number) see 316.35
- 06.03 Benefits of membership.** □ Special rates and remissions for members
For corporate assets, property (previously at this number) see 005.936
- 06.05 Awards. Honours. Prizes. Distribution of prizes, awards**
For activities of organizations (previously at this number) see 005.5
◇ 06.05NOBEL:172.4 Nobel Peace Prize
- 061 ORGANIZATIONS AND OTHER TYPES OF COOPERATION**
Denote persons in relation to organizations by -05
⇒ 001.32
⇒ 005.71
⇒ 347.19
⇒ 347.471
- 061.1 Governmental organizations and cooperation.** □ Government, state, national, international organizations
◇ 061.1EU European Union
⇒ (1-4/-6)
⇒ (100)
⇒ (4/9)
⇒ 341.217
- 061.2 Nongovernmental organizations and cooperation.** □ Private organizations, societies, associations, movements etc.
⇒ 341.215.2
- 061.23 Organizations with humanitarian, philanthropic aims.** □ Rotary
Organizations with specialized aims and activities (previously at this number) - class with activity and add :061
◇ 792:061.23 Philanthropic organizations associated with the theatre and entertainment business. Including: Grand Order of Water Rats
- 061.25 Secret or semi-secret organizations and movements.** □ Oddfellows. Elks
For federations, alliances of organizations (previously at this number) see 005.72:334.7
- 061.251 Freemasonry**
- 061.27 Foundations. Endowments. Institutions**
◇ 061.27FOR Ford Foundation
◇ 061.27GUL Gulbenkian Foundation
⇒ 001.32
⇒ 378.3
- 069 MUSEUMS. PERMANENT EXHIBITIONS**
The auxiliaries 06.0... are not applicable at 069 and subdivisions
◇ 069:5 Of the pure sciences
◇ 069:7 Of the arts. Art galleries
◇ 069:929 Biographical museums. NOTE:
Includes exhibitions about particular persons, homes of the famous etc
◇ 069(1-21) Municipal museums
◇ 069(1-4) National museums
◇ 069-027.551 Private museums
◇ 069-027.555 Public museums

069.01	Theory of museums. □ Museography. Museology	069.62	Regulations. □ Visiting, access arrangements. Relations with the public. Admission. Hours of opening
069.029	Collections of living specimens <i>See examples under 58 and 59</i>	069.63	Management. Office staff
069.1	Aims. Purpose. Utility. □ Educational, instructive function of museums ⇒ 371.335 ⇒ 379.822	069.64	Finances
069.12	Educational activities of museums in general. □ Classes. Courses. Film and slide shows. Study groups. Guided tours. Lectures to parties of visitors	069.67	Decentralization. □ Branch museums. Annexes
069.15	Extramural, outside activities. □ Loan of exhibits. Travelling exhibitions (from permanent collections). Organized outings, excursions	069.68	Relations with other establishments. Cooperation between museums
069.23	Plans. Utilization of site. □ Arrangement of buildings. Room for expansion. Layout of rooms	069.7	Museum publications <i>Publications of particular museums are classed primarily at the number for the museum, with the appropriate (0...) auxiliary from Table 1d</i>
069.27	Special services	069.8	Research in museums ⇒ 001.891
069.271	Technical services. □ Workshops. Laboratories. Design studios. Warehouse, storage areas	069.9	Temporary exhibitions <i>Class permanent exhibitions with museums at 069. Note that trade fairs are sometimes called 'exhibitions' or 'shows'; if the purpose is predominantly commercial, class with trade fairs at 339.174</i>
069.272	Installations and services for the public. □ Information points. Restaurants. Cafeterias. Sales counters		◇ 069.9(100) World fairs, expositions (expos) ◇ 069.9(100)"1851"(410.111) The Great Exhibition (London, 1851)
069.29	Interior decoration. □ Evocative decor	070	NEWSPAPERS. THE PRESS. □ Journalism <i>Significance, ownership and control. Business and editorial management</i> ⇒ (054)
069.3	Equipment of various kinds. □ Fittings. Furniture ⇒ 645.4	070.1	Scope, theory, significance of the press. Relation to other professions
069.4	Procedures of arranging, setting up exhibits	070.11	Standards. Ethics. Cultural, educational influence. Sense of responsibility
069.41	Acquisition. □ Purchases. Legacies	070.12	Endowed newspapers. Independence from public opinion, support
069.42	Treatment	070.13	Freedom of the press. □ Censorship. Invasion of privacy, its prevention. Protection of secrets, of national interest. Anarchic, seditious influence ⇒ 342.732 ⇒ 351.751.5
069.424	Collection of specimens. Fieldwork ⇒ 57.082.11	070.133	Legal restraints on journalists
069.426	Accessioning. □ Description. Registration. Inventories. Shelf and case lists	070.14	Censorship in wartime ⇒ 355.244.1
069.427	Numbering, labelling of exhibits	070.15	Subservience to sectional interests. □ Political, commercial interests. Press as mouthpiece for pressure groups
069.44	Conservation of exhibits ⇒ 57.082.5	070.16	Sensational, 'yellow' press. □ False, slanted, partial news. News with emphasis on the lurid, shocking and scandalous
069.441	Cleaning	070.18	Newspapers from unknown presses and publishers. □ Secret, subversive, 'underground' press. Anonymous pamphlets. Handbills ⇒ 098.1
069.442	Disinfection. Destruction of parasites	070.19	Other general questions. □ Comparative scope, significance, influence of newspapers, periodicals, books
069.444	Repair, reconstruction in general	070.2	Ownership and control
069.445	Mounting, display. □ Casts. Models. Dioramas	070.22	Newspaper names, titles
069.5	Collections. Exhibits	070.23	Area of operation, circulation ◇ 070.23(1-22) Village newspapers
069.51	Museum pieces <i>Specify by :...</i> ◇ 069.51:64 Household items. Furniture ◇ 069.51:75 Paintings ◇ 069.51:903 Prehistoric remains		
069.53	Technique of exhibiting		
069.532	Selective or exhaustive presentation		
069.533	Originals or reproductions		
069.535	Neutral, nondecorative setting		
069.536	Atmospheric setting or decor. □ Use of colour, lighting, sound		
069.538	Museum catalogues		
069.6	Organization and management of museums		
069.61	Administrative organization		
069.611	Administration in the strict sense <i>069.611 ≡ 06.01</i>		
069.614	Administrative organs		

070.26	Syndicates. Syndication of copy	070.432	Editorials. □ Leading articles, leaders
070.3	Business management	070.433	Local interests. Discussion of local affairs
	⇒ 005	070.44	Various other departments
	⇒ 331		<i>Denote subjects not enumerated below by :...</i>
070.32	Printing		◇ 070.447:78 Music criticism
	⇒ 655.3	070.443	Financial news. Business news. □ Reports on stocks, shares. Shipping news
	⇒ 681.62	070.445	Special interests. □ Household, home features. Court, palace. Gardening. Women's pages. Children's sections. Games, puzzles
070.325	Frequency of issue		Recreation. Sport
070.327	Time of issue	070.446	Critical, review section. □ Critiques.
070.328	Size of edition. Number printed	070.447	Discussions. Analyses
070.33	Distribution. □ Circulation. Sale. Subscription. House delivery. Newsagents. Newsvendors. Promotions. Circulation boosting	070.448	Literary and other artistic features. □ Poetry. Fiction. Serials (serialized novels etc.)
070.36	Advertising department. □ Business pertaining to paper's advertising columns, small-ads etc.	070.449.7	Readers' queries. Questions answered. Information. □ Advice column. Agony column
	⇒ 659.1	070.449.8	Illustrated, picture sections. □ Cartoons. Comic strips
070.39	Foreign offices and agencies	070.45	Voluntary contributions. □ Correspondence. Letters to the editor
070.4	Editorial management	070.46	Announcement of public events
	⇒ 808.2	070.47	Special editions. □ Sunday issues. Supplementary editions, extras
070.41	General editorial practice. Editing rules. □ Order of sections and articles. Style. Newspaper diction. Journalesse	070.48	Special kinds of newspaper
070.42	Staff. Other contributors. Their organization and duties	070.481	Official journals, gazettes
070.421	Editors. Subeditors	070.482	Religious newspapers
070.422	Staff writers. □ Reporters. Columnists. Feature-writers. Journalists on contract	070.484	Foreign-language newspapers
070.423	Occasional contributors. □ Freelancers. Special correspondents. Syndicated copy		<i>For foreign-language communities. Papers in a language foreign to the country of publication</i>
070.43/.44	Individual departments	070.485	Advertisement papers. □ 'Advertisers'. Free, give-away newspapers containing mainly advertising
070.43	Editorial departments	070.486	Professional, technical papers
070.431	News. □ News desk. Sources and channels of news	070.487	Humorous, comic papers
070.431.2	News agency services	070.488	Literary papers
070.431.4	Official news section. □ Official announcements, communiqués	070.489	Other special kinds
070.431.5	Foreign news desk		
070.431.6	Home news desk		

08 POLYGRAPHIES. COLLECTIVE WORKS

⇒ (08)
⇒ 82-82

081	INDIVIDUAL POLYGRAPHY <i>Collected works of individual authors on various subjects</i>	087	DOCUMENTS OF PARTICULAR ORIGIN OR DESTINATION
082	COLLECTIVE POLYGRAPHY <i>Collective works by several authors on various subjects</i>	087.2	Publications for particular classes of reader ⇒ (0.05)
082.1	Publishers' series, sets, libraries. □ Trade series. Reprint series	087.5	Publications for young people. □ Children's, juvenile literature. Books for infants. Picture books. Story books ⇒ (0.053.2) ⇒ 373.3 ⇒ 398.2 ⇒ 82-93
082.2	Miscellanies. □ Commemorative volumes (Festschriften). Collections in honour of eminent scholars	087.6	Popular writing. Colportage literature
084	PICTORIAL MATERIAL. □ Collections of illustrations, plans, maps 084 ≡ (084) ⇒ 096.1	087.7	Official and semiofficial publications on events of general interest
086	DOCUMENTS OF PARTICULAR FORM. □ Objects as documents 086 ≡ (086)	088	MISCELLANIES. VARIA

09 MANUSCRIPTS. RARE AND REMARKABLE WORKS

Class here works about manuscripts, rare books etc.

The documents themselves are normally denoted using the (0...) auxiliaries of Table 1d, e.g. (0.032) or (0.068).

Alternatively they may be grouped together at the subdivisions of 09 with further specification by (0...), (1/9), "... and alphabetic extension

090.1	Bibliophily. Bibliomania. □ Book collectors and collections ⇒ 028.01 ⇒ 655.425	094.4	First editions. Editiones principes ⇒ (0.044)
091	MANUSCRIPTS ⇒ (0.032) ⇒ 75.05	094.5	Less valuable old books
091.0	Technical questions relating to manuscripts	095	BOOKS REMARKABLE FOR THEIR BINDING
091.01	Script. Writing of manuscripts ⇒ 003 ⇒ 930.1:003.072	096	BOOKS REMARKABLE FOR THEIR ILLUSTRATIONS OR FOR MATERIALS USED
091.07	Facsimiles. □ Reproduction of manuscripts by chromolithography, photography and other processes ⇒ 655.2/.3 ⇒ 774/776	096.1	Rare illustrated books ⇒ 084
091.1	Form of manuscript	096.2	Books of precious materials. □ On vellum, silk, bark. With silver, gold. With woven, carved, engraved text
091.11	Tablets. Single leaves	097	MARKS OF OWNERSHIP OR ORIGIN. □ Ex libris. Bookplates. Monograms, seals
091.12	Double, multiple leaves. □ Diptychs. Triptychs. Polyptychs. Palm leaf books	098	VARIOUS KINDS OF WORK REMARKABLE FOR INTRINSIC CHARACTERISTICS
091.13	Rolls. Volumina	098.1	Prohibited books. □ Works condemned or banned by temporal or spiritual powers. Suppressed, censored works. Library 'infernos' (collections not considered suitable for open display). Clandestine, secret, underground literature ⇒ 070.18 ⇒ 82-993
091.14	Codices. Volumes. Books with folded leaves	098.11	Prohibited by religious authorities. □ Index librorum prohibitorum ⇒ 27-428.4:002
091.2	Manuscripts according to writing surface. □ On papyrus, parchment, paper ⇒ 003.51	098.12	Prohibited by civil authorities ⇒ 351.751.5
091.3	Various kinds of manuscript	098.3	Lost works. Imaginary, suppositious and projected works. Works announced but never published
091.31	Manuscript decoration. □ Manuscripts with miniatures. Illuminated manuscripts ⇒ 096 ⇒ 75.05	098.5	Works with key <i>With detailed correspondence to real persons, events. Romans à clef</i>
091.33	Palimpsests. Re-used manuscripts	099	OTHER KINDS OF WORK, REMARKABLE FOR EXTRINSIC CHARACTERISTICS. CURIOSA
091.5	Autographs. Autography. □ Handwritten texts. Holographs. Signatures ⇒ 099.3	099.1	Works of exceptional format. □ Outsize, giant, mammoth. Minute, dwarf, microscopic. Micrographic editions ⇒ (0.021)
091.6	Printed works with manuscript notes by celebrated authors (other than the authors of the works)	099.2	Works which have fetched high prices
092	XYLOGRAPHIC BOOKS. □ Block books (books printed from carved wooden blocks)	099.3	Works with autograph dedication ⇒ 091.5
093	INCUNABULA <i>Works printed in Europe before 1500: originals; facsimiles of them; studies of them</i>	099.4	Works which have become rare through adventitious circumstances. □ Books not on the market. Out-of-print editions ⇒ 094.1
094	OTHER PRECIOUS, REMARKABLE OR RARE PRINTED WORKS	099.5	Works of value because of their provenance. □ Books formerly belonging to historic, royal, literary personages, great collections
094.1	Works of great rarity but later than incunabula. □ Masterpieces of typography. Works from the celebrated great presses. Limited editions ⇒ 099.4		
094.2	Unique copies. Sole extant copies. Unica ⇒ (0.068)		
094.3	Fine editions. □ De luxe editions. Collectors', book-lovers' editions ⇒ (0.027.6)		

1 PHILOSOPHY. PSYCHOLOGY

101	NATURE AND ROLE OF PHILOSOPHY	101.3	Scope and limits of philosophy
101.1	Nature of philosophy. Philosophy as a science. Philosophy as an art	101.8	Methods of philosophizing
101.2	Possibility of philosophy. How is philosophy possible?	101.9	Person and vocation of the philosopher

11 METAPHYSICS

111	GENERAL METAPHYSICS. ONTOLOGY	114	SPACE. PLACE, INTERNAL AND EXTERNAL. DIMENSION. EMPTY, VOID; FULL, PLENUM. INFINITE SPACE. SINGULARITY AND PLURALITY. EXTENSION
111.1	Concept of being, of primary substance (tode ti). Ontology		
111.11	Existence, thereness. Difference in essence		
111.12	Essential being. Essence. Quiddity	115	TIME. DURATION. COMPONENT OF TEMPORAL DURATION. ETERNITY. WORLD CREATED AB AETERNO. RELATION OF TIME TO MOTION
111.125	Possibility. □ Inner possibility. Divine essence and possibility		
111.3	Substance (ousia). Monad		
111.31	Difference between substance and accident. Appearance	115.4	The space-time continuum. Unity of space and time ⇒ (15) ⇒ 531.111.4
111.32	Primary substance, individual substance (protè ousia). Subsistence (substant being). Hypostasis. Person	116	MOTION. DEVELOPMENT. MOBILITY. TRANSIENT AND IMMANENT EFFECT. REMOTE ACTION. BECOMING, CHANGE, TRANSITION
111.33	Secondary substance, abstract essence (deutera ousia)		
111.4	Accidence. Qualities	117	MATTER. TRANSMUTATION OF MATTER. PRIMARY MATTER (FIRST PRINCIPLE). SUBSTANTIAL FORM. MATERIAL CAUSE. FORMAL CAUSE
111.5	Relations. Relativistic theories		
111.6	Acts. Activities		
111.61	Being in potentiality. Being in act	118	FORCE. ENERGY. DETERMINATION OF ACCIDENCES OF INORGANIC SUBSTANCES ⇒ 531.6 ⇒ 536.7
111.62	Motion. Change. Development		
111.7	Created and uncreated being. Immateriality. Pure mind and spirit		
111.8	Transcendental properties of being	119	QUANTITY. NUMBER. CONSTANT QUANTITY OF COEXISTENCE AND SUCCESSION. DISTINCTION FROM SUBSTANCE
111.81	Distinction between being and its transcendental properties		
111.82	Unity. Non-being, nonexistence. Plurality. Unity and indivisibility of being	122/129	SPECIAL METAPHYSICS
111.821	Real and logical differentiation. □ Singularity. Identity. Unity of composition, simplicity. Metaphysical composition. Plurality, unlimited number, possibility of an infinite multitude	122	CAUSALITY. CAUSATION. PRINCIPLES. CAUSA EFFICIENS. CONDITION. EFFECT. OCCASION
111.83	Truth. Falsity	123	FREEDOM AND NECESSITY ⇒ 2-145.3 ⇒ 27-423.4
111.84	Goodness. Benignity. Evil		
111.85	Beauty. Ugliness	123.1	Freedom. Indeterminism
111.852	Aesthetics (objective and subjective)	123.11	Chance
113/119	COSMOLOGY. PHILOSOPHY OF NATURE ⇒ 524.8 ⇒ 53.01/.05	123.2	Necessity
		123.21	Fatalism
		124	TELEOLOGY
		124.1	Order. Chaos
		124.2	Meaning. Meaningfulness
		124.3	Purpose. Purposefulness. □ Expediency. Nonexpediency. Dysteleology. Final cause. Entelechy
113	GENERAL LAWS OF NATURE. TRANSFORMATION AND TRANSIENCE OF MATTER. ORIGIN OF THE UNIVERSE. CREATION. COSMOGONY ⇒ 2-172.2 ⇒ 2-172.25 ⇒ 2-175 ⇒ 27-143	124.4	Causa exemplaris. Creative idea. Type. Ideal

124.5	Value. Norm	128	THE SOUL. NATURE OF LIFE AND DEATH
124.6	Determination. Destiny	129	ORIGIN AND DESTINY OF THE INDIVIDUAL SOUL. □ Transmigration of souls. Metempsychosis. Incarnation. Reincarnation. Immortality
125	FINITENESS, FINITY. INFINITENESS, INFINITY. THE INFINITE AND BOUNDLESS. UNIVERSE		
13 PHILOSOPHY OF MIND AND SPIRIT. METAPHYSICS OF SPIRITUAL LIFE			
130.1	General concepts and laws	133.4	Occult influence. □ Magic. Sorcery. Enchantment. Spells. Thaumaturgy. Theurgy ⇒ 2-545 ⇒ 398.47
130.11	Nature of mind and spirit	133.5	Special occult sciences ◇ 133.5:54 Alchemy
130.12	Coherence, continuity and laws of spiritual life	133.52	Astrology ⇒ 52
130.121	Phenomenology and morphology of mind and spirit	133.521	Theoretical astrology. □ Critique. Methods. Schools
130.122	Experience and understanding of the spiritual	133.522	Elements of astrology
130.123	Laws of spiritual life	133.522.1	Principles and survey of astrological tradition. Abbreviations. Symbols. Signs
130.123.1	Intensification of contradictions. Laws of intensification by contrast. Dialectics, real dialectics	133.522.2	The zodiac
130.123.2	Heterogeneity of goals. Goal displacement	133.522.3	The planets
130.123.3	Creative synthesis	133.522.4	Fixed stars. Other astral factors
130.123.4	Growth of spiritual values	133.522.5	Sublunary factors in general. □ Geocentrism. Heliocentrism
130.2	Philosophy of culture. Cultural systems. Theory of cultural complexes	133.522.6	Houses of the firmament and their meaning
130.3	Metaphysics of spiritual life	133.522.7	The aspects
130.31	The individual mind or spirit	133.522.8	Sensitive points. Other equidistances
130.32	The objective spirit (Hegel, Simmel, Sombart)	133.522.9	Other representations. Position and domination
130.33	The collective spirit. The cosmic spirit. Der Weltgeist (spirit of the universe)	133.526	Astrological technique. Horoscopes
133	THE PARANORMAL. THE OCCULT. PSI PHENOMENA ⇒ 001.94 ⇒ 398.4	133.527	General system of interpretation. Individual astrology. Microcosm. □ Ptolemy's genethliac. Individual readings, predictions
133.1	Concept and nature of the occult	133.528	Interpretation in universal astrology. Macrocosm. □ Ptolemy's catholicon. Investigation, prediction of world events
133.2	Supersensory, supernatural experience. Initiation. Spiritual training. Path of knowledge	133.529	Technique and interpretation of prognostic astrology. Forecasts. Predictions. □ Horoscopes for major events of individual life (marriage, career etc.)
133.3	Occult perception. Precognition. □ Clairvoyance. Clairaudience. Soothsaying ⇒ 159.961.6 ⇒ 2-544.8	133.6	Chiromancy. Palmistry. Chirolgy ⇒ 159.925.7
		133.9	Spiritualism (spiritism). □ Manifestations. Telekinesis. Ectoplasm

14 PHILOSOPHICAL SYSTEMS AND POINTS OF VIEW

140	POSSIBLE PHILOSOPHICAL ATTITUDES. SYSTEM TYPOLOGY	141.133	Panlogism. Absolutism
140.8	Weltanschauung. Philosophical outlook, view of the world	141.134	Objective idealism (Bergmann)
141	KINDS OF VIEWPOINT	141.135	Spiritualism (philosophical)
141.1	According to number and quality of principles	141.136	Immaterialism (Berkeley)
141.111	Monism	141.141	Animism. Hylozoism. Panpsychism
141.112	Dualism. □ Causalism. Occasionalism	141.142	Voluntarism
141.113	Pluralism. □ Triadic systems. Polyrealism	141.143	Alogism (Schopenhauer, Ed. v. Hartmann, Bahnsen)
141.12	Ontological materialism	141.144	Monadism. Monadology. Personalism
141.13	Metaphysical idealism	141.145	Organicism
141.131	Platonism. Neoplatonism	141.151	Conditionalism (Verworn)
141.132	Rationalism. Intellectualism. Universalism ⇒ 165.63	141.152	Mechanicalism (Mechanicism)
		141.153	Dynamism
		141.154	Energism (Ostwald, Ranzoli)
		141.155	Evolutionism
		141.156	Mobilism (Chide)
		141.157	Finalism (teleological viewpoint)

141.2	According to judgement of the value of the world	141.41	Theism
141.201	Problem of any value judgement of the world	141.411	Polytheism. Henotheism
141.201.1	Problem of the scale of values. Objective and subjective scale	141.412	Monotheism
141.201.2	Basis of evaluation: by principle, theoretical, factual-empirical	141.42	Deism
141.21	Optimism	141.43	Pantheism. Acosmism
141.22	Pessimism	141.44	Panentheism
141.23	Meliorism	141.45	Atheism
141.24	Indifference	141.5	According to attitude to problem of soul and free will
141.3	Special and mixed forms	⇒ 159.947.2	⇒ 17.021.2
141.30	Thomism	⇒ 27-187.3	⇒ 27-423.4
141.31	Scholasticism	141.7	According to attitude to human society
141.311	Terminism. □ Nominalism. Conceptualism	⇒ 172	⇒ 316.2
141.312	Conceptual realism	141.72	Feminism
141.319	Neoscholasticism	141.74	Racism
141.319.8	Philosophical anthropology	141.76	Nudism. Naturism
141.32	Existentialism	141.78	Postmodernism
141.33	Philosophical mysticism	⇒ 7.038.6	
⇒ 27-587		141.8	Socialist schools and systems
141.331	Earlier esoteric schools	⇒ 329.14	⇒ 330.86
141.331.5	Cabbala	141.81	Pre-Marxian socialist schools. Utopian socialism
141.332	Theosophy	⇒ 330.84	
141.333	Anthroposophy	141.811	Socialist doctrines up to the end of the 18th century
⇒ 279.99A		141.812	Saint-Simonism
141.336	Sufism	141.813	Fourierism
141.338	Eclecticism. Syncretism	141.814	Pre-Marxian English socialist doctrines.
141.339	Various esoteric schools		□ Owenism. Chartism
	<i>Specify by alphabetic extension (Table 1h)</i>	141.82	Scientific socialism, communism, Marxism. Marxism-Leninism
141.4	According to attitude to problem of the Deity	⇒ 330.85	
⇒ 2-14			
⇒ 2-145.2			
⇒ 2-145.3			
⇒ 252+253.5			

159.9 PSYCHOLOGY

Denote persons and personal characteristics by -05 (Table 1k)

⇒ 316.6
 ⇒ 591.51
 ⇒ 612.821
 ⇒ 616.89

159.9.01	Theories. Laws. Metaphysical psychology.	159.9.018	Psychological methods in general
	Rational psychology	⇒ 159.9.072.5	
159.9.015	Theories based on the exact sciences	159.9.018.2	Subjective. □ Introspection
159.9.015.4	Chemical theories	159.9.018.3	Objective
159.9.015.7	Biological theories	159.9.018.4	Analytical
159.9.016	Other theories	159.9.018.5	Synthetic
159.9.016.1	Body and soul. Body and mind	159.9.018.6	Descriptive
159.9.016.12	Dualistic theories. □ Interactionism.	159.9.018.7	Explanatory
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159.9.016.13	Monistic theories	⇒ 159.964.2	
159.9.016.132	Materialistic, mechanistic theories.	159.9.019.2	Structural psychology. □ Gestalt psychology
	Psychological materialism. □ Automatism	159.9.019.3	Functional psychology. □ Dynamic psychology.
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159.9.016.133	Spiritualistic theories. □ Idealism. Animism	159.9.019.4	Reaction psychology. □ Psychology of behaviour. Behaviourism. Reflex psychology
159.9.016.2	Intellectualistic theories	159.9.07	Experimental psychology. Psychological research
159.9.016.3	Psychomechanistic theories	159.9.072	Psychological experiment, investigation, tests, measurements
159.9.016.4	Psychophysical theories		
159.9.016.5	Functional theories. Functionalism	159.9.072.2	Psychological studies in laboratories
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- 159.9.072.42 Individual investigation
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 159.9.072.5 Methods of experimental psychology
 159.9.072.52 Subjective methods. □ Self-observation
 159.9.072.53 Objective methods
 159.9.072.532 Stimulation or impression methods
 159.9.072.533 Reaction methods. □ Interrogation. Association
 159.9.072.59 Various psychometric methods
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 159.9.078 Apparatus
- 159.91 Psychophysiology (physiological psychology). Mental physiology**
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- 159.913 Mental hygiene
- 159.92 Mental development and capacity. Comparative psychology**
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- 159.922 Characteristics of mental life. Consciousness. Mental development in humans. Differential psychology. Genetic psychology. Developmental psychology
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- 159.922.1 Sex psychology
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 159.922.23 Sensory influences
 159.922.24 Topographic influences. Landscape
 159.922.25 Meteorological influences (influences of climate, weather)
 159.922.26 Residential influences (influences of dwelling, domicile)
 159.922.27 Social influences. □ Neighbours. Associates
 ⇒ 316.614
 159.922.28 Influence of dress, clothing
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 159.922.3 Influence of ancestry. Mental heredity. Inherited characteristics
 159.922.4 Race psychology. Ethnopsychology. National psychology
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 159.922.41 Palaeopsychology. Psychology of primitives
 159.922.5 Influence of body characteristics. □ Growth. Deportment. Deformities
 159.922.6 Influence of age
 159.922.62 Maturity
 159.922.63 Old age. Senescence
 159.922.7 Child psychology. Paedology. Paedopsychology
 159.922.72 Intellectual development of the child
 159.922.73 Mental characteristics of the child. □ Influence of child's age. Psychology of infancy, childhood, puberty
 159.922.74 Children's point of view, attitudes
 159.922.75 Children's descriptions and representations
 159.922.76 Abnormal children
 ◇ 159.922.76-056.26 Children with physical defects. NOTE: Details by :616...
 ◇ 159.922.76-056.313 Retarded children
 ◇ 159.922.76-056.34 Children with mental defects. NOTE: Details by :616.89...
 ◇ 159.922.76-056.45 Precocious, highly gifted children
 ◇ 159.922.76-056.49 Children with behavioural problems
- 159.922.8 Psychology of adolescence
 159.922.8 ≡ 159.922.7
 159.923 Type psychology. Individual psychology. Psychology of individualities. Individuality. Personality. Character psychology. Characterology. Idiosyncrasies. Personal equation. Personality types
 159.923.2 Development, emergence of the personality. Self-awareness. Identity of the personality.
 □ The ego. Self and not-self. Self and body. The inner self. Self as person. Self and society
 159.923.3 Composition of the personality. Character traits. Psychogram
 159.923.31 Corporeal elements
 159.923.32 Moral and religious elements
 159.923.33 Social elements
 159.923.35 Mental elements
 159.923.37 Aesthetic elements
 159.923.38 Special aptitudes and gifts
 159.923.4 Temperaments
 159.923.42 Popular classifications
 159.923.423 Sanguine
 159.923.424 Choleric
 159.923.425 Phlegmatic
 159.923.426 Melancholic
 159.923.43 Scientific classifications
 159.923.434 Kretschmer's classification
 159.923.434.2 Schizothymic
 159.923.434.4 Viscous
 159.923.434.6 Cyclothymic
 159.923.5 Cultivation of the personality
 159.924 Genius
 159.924.2 Characteristic traits of genius
 159.924.3 Genius as hereditary, inherited
 159.924.6 Genius as pathological, morbid state
 159.924.62 Genius and insanity
 159.924.63 Genius and illness
 159.924.64 Genius and vice
 159.924.65 Genius and degeneracy
 159.924.7 Environment and training of genius.
 □ Upbringing. Home. Family. Education
 159.924.8 Classes of genius
 159.924.9 Geographic distribution of the gift of genius
 159.925 Study of expression. Physical manifestation of mentality. Bodily expression of character
 159.925.2 Physiognomy
 159.925.5 Phrenology. Craniology. Cerebral localization of faculties or functions
 159.925.6 Graphology
 159.925.7 Chiromnomy
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 159.925.8 Mime and mimicry. □ Voice. Gesture. Gait
 159.928 Various kinds of talent. Abilities. Aptitudes
 159.928.22 General abilities, giftedness
 159.928.23 Special capacity, aptitude, talent
 ⇒ 159.95
 159.929 Plant and animal behaviour. Biopsychology. Comparative psychology
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- 159.93 Sensation. Sensory perception**
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 ⇒ 612.84/.88
 159.931/.936 The senses (faculties)
 159.931 Vision. Sight
 159.932 Audition. Hearing
 159.933 Olfaction. Smell
 159.934 Gustation. Taste
 159.935 Tactile sense. Touch, Feeling
 159.936 Muscular, articular and organic senses. Equilibrium

- 159.937 Perception. Interpretation of percepts
 159.937.2 Elements of perception
 159.937.22 Sensory performance
 159.937.23 Apperception
 159.937.24 Preperception
 ⇒ 159.954.3
 159.937.3 Perceptual errors. Normal illusions
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 159.937.5 Special classes of perception
 159.937.51 Perception of colours. Colour as visual sensation. Psychological theory of colour
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 159.937.511 Appearance of the colours. General properties of colour sensations
 159.937.511.1 Free colour. Aperture colours
 159.937.511.2 Surface (or area) colours
 159.937.511.3 Solid (spatial) colours
 159.937.511.4 Chromatic and achromatic colours
 159.937.511.5 Pure colours. Dull colours. Broken colours. Clouded colours
 159.937.512 Attributes of colours. □ Warm and cold colours. Emergent and receding colours. Brightness
 159.937.513 Colour nomenclature. Names of colours
 159.937.514 Problems of colour of lighting (illumination), and of tinted spectacles. Colour sensitivity in altered environmental conditions. □ Local colour. Shadow perception. Colour fidelity (in reproductions)
 159.937.515 Effects of individual colours. □ Warning colours, signals. Congenial, relaxing colours. Spatial effects
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 159.937.515.5 Colour symbolism. □ Colours for sorrow, joy
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 159.937.522 Elements of space perception. □ Distance. Size, dimensions. Shape. Direction
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 159.937.53 Perception of time (duration) and rhythm
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 159.938.2 Psychophysics
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 159.938.342 Physiological and reflex reaction times
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 159.938.36 Conditions influencing reaction time
 159.938.362 Objective conditions. □ Types of stimulation and reaction. Environmental influences
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 159.938.363.5 Practice
 159.938.363.6 Health
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 159.938.363.8 Toxicity
 159.938.37 Special fields of measurement
 159.94 **Executive functions**
 159.942 Emotions. Affections. Sensibility. Feelings
 159.942.2 Properties of the sensibility. □ Affective quality. Intensity. Rhythm. Duration. Content
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159.952.6	Acquired inattention. Lack of concentration	159.963	Sleep and dreams
159.953	Memory and learning	159.963.2	Sleep
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159.953.3	Memory processes	159.963.23	Normal phenomena accompanying sleep. □ Brain activity during sleep
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159.953.4	Memory aids. Mnemonics. Mnemonic techniques	159.963.273	Sleeplessness. Wakefulness. Insomnia
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159.953.6	Forgetting. Lapses of memory. Forgetfulness	159.963.32	Classification of dreams. Types of dream
159.953.7	Loss of memory. Amnesia ⇒ 616.89-008.46	159.963.33	Contents of dreams
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159.954.2	Fancy. Fantasy. Image-inventing faculty	159.963.36	Dreams of different persons ◇ 159.963.36-053.2 Children's dreams ◇ 159.963.36-056.262 Dreams of blind persons ◇ 159.963.36-056.83 Dreams of addicts
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159.956.2	Intellectual or speculative intuition. □ Internal sense. Common sense	159.964.227	Re-education
159.956.3	Sense intuition	159.964.26	Psychoanalytical schools, systems (Freudian, Jungian, Adlerian etc.) Specify by name (alphabetic extension - Table 1h)
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159.96	Special mental states and processes	159.964.3	Other kinds of depth psychology
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159.961.2	Hallucinations. Illusions. □ Apparitions. Ghosts	159.97	Abnormal psychology ⇒ 616.89
159.961.3	Prophecy. Prediction. Oracles	159.972	Mental disorders. Insanity. Alienation
159.961.6	Telepathy (mindreading), clairvoyance, clairaudience as psychic phenomena ⇒ 133.3	159.973	Mental deficiency
159.961.8	Out-of-body experiences	159.974	Hypochondria. Melancholia
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159.962	Hypnotism. Suggestion	159.976	Ecstasy
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16 LOGIC. EPISTEMOLOGY. THEORY OF KNOWLEDGE. METHODOLOGY OF LOGIC

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161.11	Criteria and types of concept	161.114.2	General concept. Category concept. Type concept. Collective concept. Veritable concept
161.111	Comparison. Differentiation. Abstraction	161.114.3	Possible and impossible concepts. Compatibility of predicates. Contradictio in adjecto
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161.13	Categoremata. Predicables (modi praedicandi)	162.13	Contraposition. Inversion
161.14	Scope. Range of concepts	162.14	Subalternation. Rearrangement. Subordination. Dictum de omni et nullo. Superposition
161.15	Content. Simple and compound concepts	162.15	Opposition. Contrasting
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161.161	Coordination	162.3	Induction. Inductive reasoning (from particular to general)
161.162	Subordination	162.4	Inference by analogy
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161.201	Theories of judgement	164.01	Nature and role of symbolic logic
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161.201.2	Theories of content	164.021	Variable
161.201.3	Theory of total identity (Münsterberg)	164.022	Constants
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161.201.5	Existential theory (Brentano)	164.031	Argument
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161.223.2	Negative judgements	164.21	Calculus of predicates. Determination of a predicate according to the content
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161.26	Modality of statements. Degree of validity		
161.261	Apodictic judgements. Statement of necessity		
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164.31	Relations	165.243	Knowing of ideal facts. Appreciation of the essence
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165.02	Possibilities of knowledge	165.324	Problem of the external or the internal world. Immanence problem
165.021	Naive and critical knowledge	165.325	Anthropomorphism. Egomorphism
165.022	Possibility of a theory of knowledge	165.4	Certainty and validity of knowledge
165.023	Possibility of truth of knowledge	165.41	Degrees of certainty
165.023.1	Truth. Correctness	165.411	Absolute certainty
165.023.2	Error. Falsehood	165.412	Uncertainty. General doubt. Cartesian doubt. $\square$ Apories. Paradoxes. Antinomies
165.1	Conditions, sources, kinds of knowledge	165.413	Relative certainty. Probability
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165.15	Logical and teleological premises of knowledge	165.421	General validity
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165.171	Pure cognition $\diamond$ 165.171:114 Space as a form of pure cognition $\diamond$ 165.171:115 Time as a form of pure cognition	165.433	Criteria of the individual sciences
165.172	Rational knowledge. $\square$ Categories. Axioms	165.5	Value of knowledge. $\square$ Truth value. Ethical value
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165.191	Mythical and scientific knowledge $\Rightarrow$ 159.922.41	165.611	Intuitionism
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165.193	Intuitive knowledge	165.613	Mysticism
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165.195	Emotional knowledge. $\square$ Feeling and will as means of knowing. Inner experience	165.62	Phenomenology (Husserl, Heidegger)
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165.211	Immediate representation. Theory of naive representation	165.641	Sensualism
165.212	Symbolic representation. Symbolization $\Rightarrow$ 003.6	165.642	Psychologism
165.22	Arrangement. Disposition. Determination of unit	165.65	Criticism. $\square$ Neocriticism
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		165.73	Positivism. Agnosticism
		165.731	Positivism
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165.74	Pragmatism. Humanism. Activism	167.3	Synthetic procedure
165.741	Pragmatism (Pierce, James)	167.32	Synthetic methods of the rational sciences
165.742	Humanism. Hominism (FCS Schiller)	167.33	Inductive methods of the empirical sciences
165.743	Instrumentalism (John Dewey)	167.4	Axioms. Postulates
165.744	Philosophy of action (M Blondel). Activism (Eucken)	167.5	Hypothesis. Fiction. Utopia. Invention
165.745	Philosophy of life. Biologism	167.6	Laws. Rules
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165.811	Conscientualism	168.2	Division. Analysis. Classification ⇒ 001.82 ⇒ 025.4
165.812	Illusionism (Berkeley)	168.3	Theory of evidence (proof)
165.813	Immanence point of view (Schuppe, Rehmke)	168.32	Direct evidence (proof)
165.814	Solipsism	168.33	Indirect evidence (proof). □ Reductio ad absurdum. Circumstantial evidence
165.82	Realism	168.35	Theory of errors in evidence (proof). Logical fallacies. □ Circular arguments. Petitio principii (begging the question). Equivocation. Ambiguity. Hysteron proteron. Ignoratio elenchi (irrelevant conclusion). Jumping to conclusion ⇒ 161.25 ⇒ 162.5
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165.822	Critical realism	168.5	System in the sciences. Problem of a systematic order of the sciences
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17 MORAL PHILOSOPHY. ETHICS. PRACTICAL PHILOSOPHY

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□ Drunkenness. Alcoholism. Dipsomania |
| 177.2 | Small talk. Art of conversation. Gossip. Scandal-mongering | 178.2 | Total abstinence (teetotalism). Temperance, moderation |
| 177.3 | Sincerity. Candour. Frankness. Truthfulness, veracity. Lying, mendacity. Deceitfulness. Flattery. Slander. Calumny. Libel. Anonymous letters | 178.3 | In company. Social drinking |
| 177.4 | Luxury. Opulence. Sybaritism. Finery. Ostentation | 178.7 | Use of tobacco. Smoking. Tobacco craving |
| 177.5 | Social inequality. □ Caste feeling. Class-consciousness. Snobbery. Social climbing. Upstarts | 178.8 | Use of various stimulants and narcotics (opium, hashish, chloral, ether, cocaine etc.) |
| 177.6 | Love. Friendship. Affection. Coquetry. Flirtation | 178.9 | Other kinds of intemperance |
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| 177.84 | Antisocial attitudes. □ Spitefulness
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| 177.9 | Duties of justice to others. Integrity, probity. Theft. Deceit | | |
| 178 | ETHICS AND TEMPERANCE. □ Morality of intoxication, addiction | | |

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- 2-174 Spiritual realms. Heaven. Abode of the gods. Spirit world**
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- 2-175 Eschatology. End of the world. Last things**
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 2-175.25 Signs of the end times
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- 2-18 MAN. MANKIND. HUMANITY. DOCTRINAL ANTHROPOLOGY**
- 2-181 Origins of man**
 2-181.5 Creation of man
 2-181.7 Primeval paradise. Earthly paradise
- 2-182 Purpose, destiny of man**
2-183 Nature of man
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 2-183.7 Freedom. Free will. Predestination
- 2-184 Man's relation to god(s)**
 2-184.2 Religious feelings. Religious experience
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 2-184.35 Formal acknowledgement of faith. Religious conversion
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 2-184.4 Doubt. Unbelief
 2-184.5 Denial. Apostasy
 2-184.6 Grace
 2-184.7 Covenantal relations
 2-184.8 God's response to man
 2-184.85 God's gifts to man. Gifts of the Holy Spirit
- 2-185 Human experience. Life**
 2-185.2 Suffering
 2-185.3 Evil. The problem of evil. Sin
Class here general ideas of the nature and religious understanding of sin. Conformity to the specific requirements of religious law and practice should file with 2-42 religious observance and moral theology
 2-185.32 Origin and nature of sin. Nature of evil
 2-185.325 Original sin. Inherent sinfulness of mankind.
 Fall of man
 2-185.33 Repentance. Contrition
 2-185.35 Atonement for sin
 2-185.36 Forgiveness. Remission. Absolution
 2-185.5 Salvation. Soteriology
 2-185.52 Justification
 2-185.53 Redemption
 2-185.57 Reconciliation
- 2-186 Death. The dead**
For funerary customs and rituals see 2-557
- 2-187 Life after death. Afterlife. Hereafter**
 2-187.2 States of being. Forms of continued existence
 2-187.3 Immortality. Persistence of life
 2-187.4 Restoration (apokatastasis) of souls
 2-187.5 Reincarnation. Metempsychosis.
 Metamorphosis
 2-187.55 Birth and re-birth
 2-187.6 Resurrection. Rejuvenation. Restoration of the physical body
 2-187.7 Spiritual, immaterial body. Heavenly body
 2-187.8 Liberation of the soul from the body
 2-187.85 Annihilation. Cessation of existence.
 Non-being. Assimilation. Nirvana
- 2-188 Spiritual realms. Non-material world. Spirit world**
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 Access to the spirit world
 Preparation
 Judgement
 Journey to the spirit world. Ascension. Descent
 Heaven. The heavens. Paradise. Abode of the gods. Astral plane
 Abode of ancestors
 2-188.55 Intermediate state. Limbo. Purgatory
 2-188.6 Hell. Underworld. Hades. Sheol
 2-188.7 ⇒ 2-173
 Abode of ghosts
- 2-2 EVIDENCES OF RELIGION**
- 2-21 NATURAL THEOLOGY. EVIDENCE OF THE NATURAL WORLD. RATIONAL ARGUMENTS FOR BELIEF**
- 2-22 REVEALED THEOLOGY. REVELATION OF GOD(S)**

- 2-225 Theophany. Hierophany. Kratophany**
For revelation of gods in human form see 2-166
- 2-23 SACRED BOOKS. SCRIPTURES. RELIGIOUS TEXTS**
In most faiths there are central sacred books (which have been largely enumerated for each religion). Secondary types of material (religious law, liturgical material, etc.) are provided for as well. These should not be confused with the scriptures themselves (e.g. Torah is regarded as a work of scripture, and not a work of religious law, although it is in the form of law). Examples here are mainly taken from the Hebrew or Christian Bible and religious texts; these are to be understood as a pattern for the works of other faiths, which are enumerated at the appropriate place in the main table
- 2-232 Original texts. Canonical works of scripture**
- 2-233 Variants**
- 2-234 Translations (i.e. from the original text)**
Use common auxiliary Tables 1c and 1g to specify language and period as required
◇ 2-234"05"=124 Vulgate
- 2-235 Versions**
Forms of the text not translated from original sources
Use common auxiliary Tables 1c and 1g to specify language and period as required
- 2-236 Rewritings. Paraphrases**
2-236.5 Narrative summaries. 'Bible stories'
2-236.7 Parodies
- 2-24 SPECIFIC TEXTS. NAMED TEXTS AND BOOKS**
Use to enumerate specific texts, books, elements, etc. of the sacred writings of a particular religion
- 2-25 SECONDARY LITERATURE. PSEUDO-CANONICAL WORKS**
- 2-252 Apocrypha. Pseudepigrapha (other than as named)**
- 2-254 Commentary on sacred works**
E.g. Mishnah, Qu'uranic commentary
- 2-256 Interpretative works. Works of guidance on practice**
E.g. Talmud, sutras
- 2-262 Oral forms. Oral tradition**
- 2-264 Myths and legends**
- 2-265 Other literary forms**
2-265.2 Poetic works
2-265.3 Epics and sagas
2-265.4 Dramatic works
2-265.6 Ritual drama. Ritual re-enactments
- 2-266 Law. Legal codes. Legal texts**
- 2-268 Texts on 'secular' subjects (e.g. medicine, science)**
- 2-27 CRITICAL WORKS**
- 2-271 Aids to study**
- 2-271.4 Parallel forms. Interlinear texts. Harmonies
2-271.5 Annotated versions
- 2-272 Concordances**
2-272.4 Verbal concordances
2-272.5 Subject concordances
- 2-273 Commentaries**
- 2-274 General criticism**
2-274.2 Authorship, origin of texts
2-274.4 Canon. Canonicity
2-274.6 Authority, inerrancy of scripture
- 2-275 Textual criticism**
2-275.2 Textual apparatus
- 2-276 Literary criticism**
2-276.2 Origin and formation of scriptures
2-276.3 Documentary hypotheses
2-276.4 Form criticism. Formgeschichte
2-276.5 Redaction criticism
2-276.6 Literary content
2-276.63 Symbolism
2-276.64 Typology
2-276.66 Allegory
2-276.7 Historical criticism. Salvation history. Heilsgeschichte
- 2-277 Interpretation. Understanding. 'Bible study'**
2-277.2 Exegesis. Hermeneutics
2-277.3 Eisegesis
2-277.4 Literalist, fundamentalist interpretation
2-277.6 Allegorical interpretation
2-277.7 Demythologizing
- 2-278 Theology of sacred texts. Theological content**
Denote further detail by combining hyphen auxiliaries
◇ 27-278-31 Christology of the New Testament
- 2-279 Religion as demonstrated in sacred texts. Religious practice**
Denote further detail by combining hyphen auxiliaries
◇ 26-279-538 Sacrifice in the Old Testament
- 2-28 OTHER RELIGIOUS TEXTS**
- 2-282 Liturgical works. Liturgical texts**
This class provides for the collocation of religious texts if required. Normally liturgical materials are subordinated to the ritual to which they apply
2-282.2 Comprehensive liturgical works
2-282.3 Creeds and confessions of faith
2-282.4 Catechisms
2-282.5 Prayer books. Books of prayers
2-282.7 Service books. Missals. Breviaries
- 2-283 Historical documents. Evidential documents. Historical records. Contemporaneous accounts. Letters**
- 2-284 Works of doctrine, dogma. Articles of faith. Systematic theology. Dogmatics**
Normally the doctrine of say, a practice, will be subordinated to the practice e.g. 27-549-284 the doctrine of the Holy Communion. 2-284 may be used to collect all works of doctrine at this point e.g. 27-284-549 the doctrine of the Holy Communion

- 2-285** **Defence and exposition of doctrine**
General works only. Normally subordinate to the subject discussed, using hyphen combination. See note above at 2-284
- 2-285.2 Apologetics
2-285.4 Polemics
- 2-286** **Other forms of religious texts**
- 2-287** **Sermons as a form**
⇒ 2-475
- 2-29** **RELIGIOUS LITERATURE**
Imaginative literature designed to support the faith, or with religious subjects as a theme
- 2-291** **Religious poetry**
- 2-292** **Religious drama**
Religious plays. See 2-265.6 for ritual drama
- 2-293.1 Religious novels
- 2-3** **PERSONS IN RELIGION**
Religious officials, persons in authority in the faith etc., are dealt with under the institutional structure of the religion at 2-725/-727. However, there are some general classes of agents that are not linked in this way with a particular religious activity, and these are given here
Generally speaking, persons should be subordinated to the function they perform using the common auxiliaries of persons Table 1k-05 e.g.
2-468.6-051 healers = healing + persons.
2-788-051 monks = monastic orders + persons
- 2-31** **ORIGINATOR, FOUNDER, CENTRAL FIGURE OF THE FAITH**
- 2-312** **Founder's life**
2-312.3 Birth, nativity
2-312.4 Family of the founder
2-312.45 Father
2-312.47 Mother
2-312.5 Other significant individuals
2-312.55 Original followers, supporters
2-312.6 Initiation. Conversion. Enlightenment
2-312.7 Ministry. Public life
2-312.8 Death
2-312.9 Events after death
- 2-313** **Relics of the founder**
- 2-315** **Accounts, anecdotal material**
- 2-317** **Memoirs. Sayings. Deeds**
- 2-318** **Founder as model, example**
- 2-32** **MESSIAH(S)**
- 2-33** **INSPIRED PERSONS. MAJOR FIGURES**
- 2-335** **Scholars. Thinkers. Philosophers**
2-337 Reformers
- 2-34** **MARTYRS**
- 2-35** **ASCETICS. HERMITS. FAKIRS**
- 2-36** **SAINTS. BODDHISATVAS. ENLIGHTENED ONES**
- 2-37** **MAHATMAS. GURUS. WISE MEN**
- 2-38** **CHARISMATICS. PERSONS WITH SUPERNATURAL POWERS**
- 2-382** **Shamans. Witch doctors. Medicine men. Sorcerers**
⇒ 256
- 2-384** **Cultic priests. High priests**
- 2-386** **Thaumaturgers. Wonder workers. Miracle workers**
- 2-39** **THEOCRATIC SOVEREIGNS. DIVINELY APPOINTED KINGS. SACRAL KINGSHIP**
- 2-4** **RELIGIOUS ACTIVITIES. RELIGIOUS PRACTICE**
- 2-41** **SOCIAL BEHAVIOUR AND ACTIVITIES. NON-FORMAL RELIGIOUS PRACTICE AND OBSERVANCE**
- 2-42** **MORAL BEHAVIOUR. MORAL THEOLOGY**
Moral theology is distinguished from the theological discussion of the nature of Sin which is located at 2-185.3. This class is concerned with human ethical actions, and with right and wrong, virtue and vice, rather than inherent sinfulness
- 2-423** **Right and wrong. Good and evil**
2-423.3 Knowledge of right and wrong
2-423.35 Conscience. Guilt
2-423.4 Freedom to act. Free will. Human ethical actions
2-423.42 Influences on the exercise of free will
2-423.44 Determinism
2-423.45 Predestination
2-423.47 Temptation
2-423.5 Wrongdoing. Sin
2-423.53 Origin, source of wrongdoing
2-423.55 Nature of wrongdoing. Nature of sin. □ Gravity of sin
2-423.57 Specific sins
◇ 2-423.57-184.3 Sins against the faith
- 2-423.58 Seven deadly sins. Mortal sins
2-423.59 Venial sins
2-423.7 Good. Virtue. The good
2-423.73 Origin of good
2-423.75 Nature of virtue
2-423.77 Specific virtues
2-423.78 Cardinal virtues, etc.
2-423.79 Theological virtues. □ Faith. Hope. Charity
- 2-425** **Ethical systems. Ethical theory**
These should be derived from Philosophy
◇ 2-425:17.023.6 Casuistry. Situation ethics
◇ 2-425:17.026.4 Categorical imperativism. Kantian ethics
◇ 2-425:17.036.2 Utilitarianism
- 2-426** **Religious ethics. Ethics directly derived from religious systems**
- 2-427** **Informal promotion of the faith. Acting as exemplar or ambassador for the faith**
- 2-428** **Religious laws, obligations, moral precepts, moral laws. Regulations. Religious duties**
Many of these are not particularly 'moral', and if required, will be subordinated to the religious practice they refer to; so, dietary laws to dietary practice
2-428.2 Nature of religious law

- 2-428.3 Requirements of law
2-428.4 Prohibitions
For specific prohibitions use colon combination e.g. 2-428.4:002 prohibited books
- 2-428.5 Obedience to the law
2-428.6 Interpretation of law
⇒ 256
- 2-428.7 Exemption from law
2-428.8 Transgression of law
2-428.85 Mitigation
2-428.87 Indulgences
2-428.9 Amendment, cancellation of law
- 2-43 SOCIAL CUSTOMS AND PRACTICE. SOCIAL THEOLOGY**
This class is regarded as embracing pastoral theology, but there is a clear distinction between social behaviour (such as food laws, sexual customs) and pastoral behaviour (almsgiving, counselling)
- 2-44 SOCIAL BEHAVIOUR**
- 2-441 Self-emptying. Kenosis**
- 2-442 Food and diet**
2-442.4 Food laws
2-442.45 Dietary requirements. Dietary limitations
2-442.47 Abstinence. Fasting. Prohibition
2-442.5 Rules concerning specific foods and drinks
◇ 2-442.5:637.5 Rules concerning meat
◇ 2-442.5:663.2/.5 Rules concerning alcoholic drinks
- 2-443 Personal hygiene and appearance. Personal conduct**
2-443.3 Cleanliness. Washing
2-443.5 Clothing. Headgear
2-443.7 Cutting, wearing, arrangement of the hair
- 2-444 Social relationships**
2-444.2 Friendships
2-444.4 Conversation
2-444.6 Silence
- 2-445 Domestic customs and routines**
- 2-446 Medical care and treatment**
◇ 2-446:615.38 Attitudes to blood transfusion
◇ 2-446:615 Attitudes to drugs
- 2-447 Sexual relations, activity**
2-447.2 Segregation of sexes
2-447.4 Chastity. Purity
2-447.6 Sexual faithfulness. Promiscuity
2-447.7 Abstinence. Celibacy
2-447.8 Prostitution
- 2-45 MARRIAGE AND THE FAMILY**
- 2-452 Courtship. Selection of marriage partners. Betrothal**
2-452.2 Marriage contracts
- 2-453 Marital partners**
2-453.2 Monogamy
2-453.5 Polygamy. Polyandry
- 2-454 Divorce**
- 2-455 Children**
- 2-456 Pregnancy**
2-456.2 Abortion
2-456.5 Childbirth
- 2-457 Parenthood**
2-457.5 Adoption
- 2-458 Surrogacy**
- 2-46 CHARITY. SUPPORT FOR OTHERS. PASTORAL ACTIVITIES. PASTORAL THEOLOGY IN THE BROAD SENSE**
- 2-462 Agencies, institutions for pastoral care and support**
- 2-463 Charitable giving**
2-463.2 Tithes. Stewardship
2-463.4 Almsgiving
2-463.6 Surrender of wealth, possessions. Voluntary poverty
- 2-464 Practical support. Voluntary work**
- 2-465 Counselling. Advising**
- 2-466 Visiting**
- 2-467 Hospitality**
- 2-468 Care. Physical care and attendance**
2-468.2 Temporary, respite care
2-468.4 Residential care
2-468.45 Shelter. Housing
2-468.47 Hostels. Hospices
2-468.5 Nursing care. Medical care
2-468.6 Healing
2-468.65 Laying on of hands
- 2-47 EDUCATION**
⇒ 2-725:37
⇒ 2-75
- 2-472 Religious education**
◇ 2-472-053.2 Religious education for children. Sunday school
2-472.3 Preparation for initiation. Religious instruction.
□ Catechism
2-472.5 Continuing religious education
- 2-474 Scriptural study and interpretation (for education)**
⇒ 2-277
Preaching (as education and instruction)
⇒ 2-766
2-475 Homiletics. Theory of preaching
2-475.2 Sermons
2-475.5 *See 2-287 for sermons as a form*
- 2-48 RECIPIENTS OF PASTORAL CARE. SOCIAL TYPES AND GROUPS**
- 2-482 Children**
- 2-483 Elderly. Aged**
- 2-484 Victims**
2-484.3 Victims of crime
2-484.5 Victims of war
2-484.7 Oppressed groups. Liberation theology
2-484.75 Religious and politically oppressed
- 2-485 Minority groups**
2-485.5 Ethnic minority
- 2-486 Sick**
2-486.3 Mentally ill
2-486.5 Physically ill
2-486.7 Disabled
- 2-487 Dying**
- 2-488 Other disadvantaged groups**
2-488.2 Needy. Poor
2-488.3 Homeless
2-488.4 Jobless. Unemployed
2-488.5 Destitute

- 2-488.6 Prisoners
2-488.65 Prisoners' families
2-488.7 Widows and orphans
2-488.8 Foreigners. Aliens. Strangers
2-488.85 Refugees
- 2-5 WORSHIP BROADLY. CULT. RITES AND CEREMONIES**
- 2-52 (AGENTS OF WORSHIP)**
- 2-522 Religious sites, locations. Holy places**
2-522.2 Situation, position, location, orientation
2-522.4 Places of particular spirituality, holiness
2-522.5 Places with spiritual associations
2-522.52 Sites of visions
2-522.54 Places of martyrdom
2-522.58 Birthplace of holy persons
- 2-523 Buildings for religious use**
General only. For details of buildings use 72 Architecture
2-523.2 Parts of buildings. Rooms. Areas
2-523.3 Monuments. Shrines
2-523.4 Centres of worship (religious significance)
For religious architecture see 726
2-523.41 Chief centres of worship. □ Principal temples, mosques. Cathedrals
2-523.42 Local and peripheral centres of worship.
□ Temples. Mosques. Parish churches
2-523.43 Small, simple or austere centres of worship.
□ Separate, free-standing shrines and chapels
For shrines that are not buildings see 2-523.3
2-523.44 Sanctuary. Holy area
2-523.45 Appellation. Dedication
2-523.46 Side rooms. Chapels
2-523.47 Screens
2-523.472 Doors
2-523.48 Functional rooms. Robing rooms. Vestries
2-523.5 Subsidiary, attached structures
2-523.6 Monasteries. Community buildings
2-523.8 Buildings for secular purposes
- 2-524 Vehicles of cultic acts**
◇ 2-524:536.46 Fire
◇ 2-524:612.1 Blood
◇ 2-524:662 Ashes
◇ 2-524-032.2 Water
◇ 2-524-035.8 Oil
2-525.2 Incense
2-525.4 Vestments. Clothing. Robes. Head coverings
2-525.5 Masks
2-525.6 Headgear. Head-dresses
2-525.7 Religion draperies. Hangings. Textiles
- 2-526 Objects in worship. Furnishings and decorations**
2-526.1 Furniture. Platforms. Lecterns
2-526.2 Altars. Tables
2-526.3 Containers. Sacred vessels. Chalices. Cups
2-526.4 Aids to prayer. Prayer wheels. Rosaries
2-526.5 Lights
2-526.6 Symbols. Images
2-526.62 Pictorial symbols. Icons
2-526.63 Decorative devices and objects
2-526.64 Statuary. Statues
⇒ 730
2-526.65 Funerary monuments. Tombs and tombstones. Memorials
2-526.7 Relics. Reliquaries. Containers for religious objects
2-526.8 Objects worn on the body. Charms. Talismans
- 2-526.9 Musical instruments in worship
As part of the ritual e.g. horns, bells, shophar. See 2-535 for religious music
- 2-528 Liturgy**
General treatments of liturgy only. Liturgical texts should be subordinated to the rites and ceremonies they refer to using 2-282 e.g. 272-555-282 texts of the Catholic marriage service. Alternatively 2-282 may be used alone to collocate all the liturgical texts of a faith
2-528.5 Form of rite. Office
- 2-53 ACTS OF WORSHIP (BY PARTICIPANTS)**
2-532.2 Public worship. Communal worship. Services. (Usually assumed)
◇ 2-532.2"345" Night services. Vigils
Ceremonial. Ceremonies on the grand scale
State ceremonies
Partially public
Domestic observances. Family worship.
Worship in the home
Personal worship. Private devotions
Secret worship (by degree of compulsion)
Compulsory. Obligatory
Recommended
Voluntary. Optional
- 2-533 (Object of worship)**
Use colon combination with numbers from the whole classification
◇ 2-533-312.47 Mariolatry
◇ 271-533-526.62 Veneration of icons in the Orthodox church
- 2-534 Form of rites, worship (oral)**
2-534.2 Calling to worship, prayer
2-534.3 Prayer
Communal prayer assumed. See 2-58 for private prayer
2-534.35 Specific prayers
2-534.4 Reading of texts
2-534.5 Chants. Incantations
2-534.6 Prophecy
2-534.7 Glossolalia. Ecstatic utterance. Speaking in tongues
- 2-535 Music in worship**
The preferred location for religious music is in 783.
Music by function and purpose: in general subordinate to function e.g. 2-555-535
Wedding music. This location may be used to collate works of music if required. Obtain by hyphen combination e.g. 2-535-555
Wedding music
2-535.2 Solo singing
2-535.3 Choral music
2-535.35 Musical chants. Plainsong
2-535.36 Antiphonal singing
2-535.4 Congregational music. Public singing.
Community singing
Instrumental music
Liturgical settings
Of texts
Of services
Hymns
This class may be used to collate collections of hymns and hymnology if so desired
◇ 2-535.7:631.5 Harvest hymns
◇ 2-535.7-23 Hymns based on text from sacred books
◇ 2-535.7-544.5 Hymns of thanksgiving

- 2-535.75 Anthems
 2-535.8 Psalms
 2-535.85 Metrical psalms
 2-535.9 Religious musical works. Oratorios. Cantatas
- 2-536 Physical rites and ceremonies**
 2-536.1 Washing. Ablution. Immersion in water.
 Sprinkling of water
 2-536.2 Use of other materials, substances
 2-536.25 Oil. Anointing. Unction
 2-536.27 Incense. Censing
 2-536.3 Postures
 2-536.32 Orientation
 2-536.34 Bowing. Bending
 2-536.35 Kneeling. Genuflection
 2-536.36 Prostration
 2-536.37 Laying on of hands
 ⇒ 2-468.65
 2-536.4 Symbolic gestures
 2-536.5 Procession. Circumambulation
 2-536.6 Dance
 2-536.65 Drama. Symbolic enactment
 2-536.7 Eating. Food rites and ceremonies
 2-536.72 Shared food. Ritual meal. Breaking of bread
 2-536.74 Ritual fasting. Abstinence from food
 ⇒ 2-442.47
 2-536.76 Drinking
 2-536.765 Potions
 2-536.8 Marking of the body
 2-536.82 Ritual marking and scarring
 2-536.84 Body piercing
 2-536.86 Circumcision
 2-536.9 Control of the body
 2-536.92 Endurance of pain
 2-536.94 Fire-walking
 2-536.96 Levitation
- 2-537 Offerings**
 ◇ 2-537:58 Vegetable offerings
 ◇ 2-537:59 Animal offerings
 ◇ 2-537:599.89 Humans
 ◇ 2-537:633.1 Meal offerings
 ◇ 2-537-032.2 Libations
- 2-538 Sacrifices**
 2-538.2 Burnt offerings
 2-538.4 Whole burnt offerings
 2-538.6 (By subject of sacrifice)
 ◇ 2-538.6:599.89 Human sacrifice
 ⇒ 392.89
- 2-54 CEREMONIES BY PURPOSE**
 2-542.3 Personal
 2-542.5 Altruistic
 2-542.7 Vicarious
 2-543.2 Homage. Adoration. Veneration. Worship
 narrowly
 2-543.3 Profession of belief. Credo. Confession of faith
 2-543.4 Intercession. Influence
 2-543.5 Propitiation
 2-543.6 Petition (By subject of request)
 ◇ 2-543.6:551.578.1 Rain
 ◇ 2-543.6:57.017.5 Fertility
 2-543.7 Confession of sins
 2-543.8 Penitence
 2-544.2 Expiation. Atonement
 2-544.25 Scapegoat
 2-544.3 Absolution. Forgiveness
 2-544.4 Purification. Catharsis
 2-544.5 Thanksgiving
 2-544.55 Commemoration
 2-544.6 Dedication. Vow. Pledge
 2-544.7 Knowledge. Enlightenment. Insight
- 2-544.8 Knowledge of events. Divination. Augury.
 Soothsaying. Oracles
- 2-545 Control of events. Magic**
 2-545.2 Sympathetic magic
 2-545.3 Contagious magic
 2-545.5 Homeopathic. Imitative rites
 2-545.6 Binding and losing
 2-545.7 Generating rites
 2-546.2 Blessing. Sanctification. Consecration
 2-546.4 Curse. Anathema. Desecration
 2-546.6 Deconsecration
- 2-547 Evocation, 'calling-up' of spirits**
 2-547.5 Evocation of the dead. Necromancy
- 2-548 Elimination. 'De-desecration'. Casting out**
 ⇒ 2-544.25
 2-548.5 Exorcism
- 2-549 Communication with the deity. Communion with the deity**
 2-549.2 Evocation of the deity
 2-549.3 Coming of the deity. Presence of the deity
 2-549.4 Assimilation of the deity
 2-549.6 Departure of the deity
- 2-55 SACRAMENTS. SACRAMENTALS**
While this will differ from one religion to another, the concept of sacraments is inherent in many faiths. Individual sacraments are specified but the list is more comprehensive in coverage than will be required by any one faith. Use this location for general and comprehensive treatments of sacraments
- 2-552 Rites of passage. Stages in life**
2-553 Birth
 2-553.2 Pre-natal. Antenatal
 2-553.3 Peri-natal
 2-553.4 Post-natal
- 2-554 Adolescence. Puberty. Transition to adulthood**
- 2-555 Betrothal. Marriage. Wedding ceremonies**
 2-555.5 Procreation. Fertility. Childbirth
- 2-556 Old age**
- 2-557 Death. Funeral ceremonies**
 2-557.5 Disposal ceremonies. Burial. Cremation
- 2-558 Stages in spiritual life**
 2-558.2 Initiation
 2-558.3 Baptism. Naming ceremonies
 2-558.4 Confirmation. Adult re-initiation
 2-558.5 Ordination. Entry to priesthood
 2-558.6 Beatification
 2-558.7 Canonization
 2-558.8 Expulsion. Excommunication
- 2-559 Coronation. Enthronement**
- 2-56 CELEBRATION**
- 2-562 Feasts and festivals. Holy days**
*By specific characteristics
 Synthesize from common auxiliaries, special auxiliaries, and colon combination*
 ◇ 2-562:631.5 Harvest festivals
 ◇ 2-562"321" Spring festivals
 ◇ 2-562-31 Birth of the founder

- 2-563** **Special weekday. Day of rest. Sabbath**
2-563.5 Special times of day. Hours for prayer. Hours for worship
- 2-564** **Religious calendar. Religious year**
- 2-565** **Named festivals**
Specify by alphabetic extension A/Z (Table 1h)
- 2-567** **Special years. Holy years. Sabbatical years. Jubilees**
- 2-57** **PILGRIMAGE**
Place of pilgrimage: use common auxiliaries of place Table 1e.
Purpose of pilgrimage: obtain by hyphen combination with the rest of the class
◇ 2-57-543.8 Penitential pilgrimage
◇ 28-57(532) Muslim pilgrimage to Mecca
⇒ 2-522
- 2-58** **CONTEMPLATION AND REFLECTIVE RELIGION**
◇ 2-58-2 Devotional literature
◇ 2-58-282.5 Personal prayer books
- 2-582** **Religious experience**
- 2-583** **Meditation. Contemplation. Prayer in the personal sense**
⇒ 2-534.3
- 2-584** **Spiritual life**
2-584.5 Spiritual growth and development
2-584.7 Spiritual exercises
- 2-585** **Asceticism**
2-585.5 Self-denial
2-585.7 Mortification of the flesh
- 2-586** **Withdrawal. Retreat**
2-586.5 Hermitage. Isolation
- 2-587** **Mysticism. Altered states of consciousness. Mystic experiences**
◇ 2-587-184.27 False mystical experiences
2-587.3 Trances
2-587.5 Visions
2-587.6 Mystic phenomena
2-587.65 Stigmata
2-587.7 Direct experience of the divine
- 2-6** **PROCESSES IN RELIGION**
Most activities in religion will come under the heading of operations (religious practice, observance, ritual; the administration of churches; pastoral care; missions and so on). Some much more general 'activity' terms deal with the internal dynamics of religions, and their external relations with each other. These rather less concrete and internalized actions are analyzed as 'processes'. Although some are developmental, they should not be regarded as historical (history of a subject is more than adequately catered for in UDC by the application of common auxiliaries and the judicious use of the colon)
- 2-63** **DEVELOPMENTAL PROCESSES. INTERNAL PROCESSES**
- 2-632** **Origin of the faith. Genesis**
- 2-633** **Evolution. Development**
- 2-634** **Growth. Expansion**
- 2-635** **Decline. Contraction**
- 2-636** **Revival. Rejuvenation. Renaissance**
- 2-637** **Death. Extinction**
- 2-64** **RELATIVE PROCESSES. EXTERNAL PROCESSES. INTERACTIONS BETWEEN FAITHS**
- 2-65** **COMPARISON**
A caveat on the use of the term 'Comparative religion'. This usually means comprehensive rather than comparative; works of comparative religion are still usually descriptive in nature, traditionally used to handle non-Christian religion, and most of the literature will reflect this. Therefore 'comparison' should be used conservatively
- 2-66** **INFLUENCE**
- 2-662** **Influence on. Effect on**
- 2-663** **Dependence**
- 2-664** **Influenced by**
2-664.3 Syncretism. Borrowing
2-664.5 Assimilation
- 2-67** **RELATIONS BETWEEN FAITHS OR WITH SOCIETY IN GENERAL. RELIGION AND SOCIETY**
⇒ 322
- 2-672** **Isolation. Separation**
- 2-673** **Co-existence**
2-673.5 Tolerance. Irenics
- 2-674** **Conflict. Antagonism. Hostility**
2-674.5 Persecution
- 2-675** **Dialogue. Ecumenism**
- 2-676** **Co-operation**
- 2-677** **Inter-communion**
- 2-678** **Integration. Merger. Amalgamation**
- 2-679** **Re-union**
- 2-7** **RELIGIOUS ORGANIZATION AND ADMINISTRATION**
All things to do with the structure and organization of a religion. Subordinated to structure are the activities associated with the management and administration of churches, faiths and movements
- 2-72** **NATURE AND STRUCTURE OF THE ORGANIZED RELIGION, CHURCH. ECCLESIOLOGY**
◇ 2-72-1 Doctrine of the religious body, church
◇ 2-72-133/-134 Origin and nature of the church
◇ 2-72-144 Characteristics of the church
◇ 2-72-4 Practice of the religion. Religious custom and tradition
- 2-722** **Persons in religion**
Normally persons in religion with specific functions will be subordinate to the function they perform, and expressed by the use of the common auxiliaries of persons Table 1k-05 e.g.
2-544.8-051 Soothsayers.
2-76-051 Missionaries.

This location is used to deal with material on the 'clergy' or officers of the group generally, and to provide a single class for aggregating such literature

2-725

Officers of the religion. Priests. Ministers. Leaders

Most of the detail in this class will be provided by hyphen combination or by colon combination with appropriate main classes

- ◇ 2-725:331 Conditions of service. Appointment. Preferment. Tenure. Support and maintenance of ministry
- ◇ 2-725:331.2 Financial support. Salaries. Stipends
- ◇ 2-725:332.8 Physical support. Housing
- ◇ 2-725:37 Education and training of the clergy
- ◇ 2-725:377 Training colleges. Seminaries
- ◇ 2-725-428 Obligations
- ◇ 2-725-447.7 Celibacy of priests
- ◇ 2-725-45 Marriage of priests
- ◇ 2-725-544.6 Vows
- ◇ 2-725-584 Spiritual life

2-726

Levels of leadership

⇒ 2-732.2

- 2-726.1 Archbishops. Metropolitans
- 2-726.2 Senior. Bishops. Abbots
- 2-726.3 Priests. Ministers
- 2-726.4 Deacons. Assistant ministers
- 2-726.5 Lay assistants
- 2-726.6 Laity in general. Lay persons. Adherents
- 2-726.7 Elders. Senior members (either lay or ministers)

2-727

Non-adherents of the religion. Non-believers

2-73

GOVERNMENT OF THE RELIGION

2-732

Authority of the church (Agents of authority)

- ◇ 2-732-31 Authority of founder
- 2-732.1 Source of authority
- 2-732.2 Head of religion
- 2-732.3 Ruling bodies. Synods. Convocations
- ◇ 2-732.3-284 Doctrine, judgements of synods and councils
- 2-732.4 Councils. Bodies convened for the purpose of establishing doctrine and practice
- 2-732.5 Authority of officers, priests
- 2-732.6 Lay persons. Adherents

2-733

Religious office. Vocation

2-735

Management and administration

Detail in this class will usually be provided by colon combination

- ◇ 2-735:336 Temporal goods, wealth of the church
- ◇ 2-735:347.23 Church property, lands, patrimony
- ◇ 2-735:657 Accountancy
- ◇ 2-735:658.14/17 Financial matters
- ◇ 2-735:658.3 Personnel management

2-736

Other institutions, agencies, organizations within the religion or church

2-737

Congregations. Worshippers

2-74

LEGAL ADMINISTRATION. RELIGIOUS LAW. CANON LAW

In many cases, religious law deals only with discipline within the faith; in this case most detail can be obtained by hyphen combination with other parts of Class 2. Religious law may however also be the Law of the Land; in such case provision is made to append the detail of Class 34 if so desired

- ◇ 2-74-67 Religious law and the law of the land
- ⇒ 348

2-741

Theory of canon law. Religious jurisprudence

2-742

Sources of law

- ◇ 2-742-14 Divine inspiration. God as source of law
- ◇ 2-742-23 Texts. Religious texts
- ◇ 2-742-266 Codes of law
- ◇ 2-742-267 Commentary and interpretation. Legal texts

2-744

Legal practice and procedure.

- Verdict. Sentence

2-745

Religious courts

2-746

Religious judgements

2-747

Subjects of law

- ◇ 2-747:342 Public law
- ◇ 2-747:343 Criminal law
- ◇ 2-747:347 Civil law. Private law
- ◇ 2-747:347.2 Property law
- ◇ 2-747:347.6 Family law

2-747-1/-9

Topics in religious law

In general the law of specific concepts like those below should be regarded as indistinguishable from custom and practice, and placed with the practice; exceptions to this would be cases when the subject of discussion was very clearly the law itself

- ◇ 2-747-1/-9-442 Dietary laws
- ◇ 2-747-1/-9-443 Laws on personal hygiene

2-75

EDUCATION

Education as a function of the central church, religion.

For education at the local level, including religious instruction and Sunday schools, see 2-47 and its subdivisions

Denote details by colon combination with 37

⇒ 2-725:37

2-752

Secular education. Religious schools, foundations

2-754

Tertiary education. Colleges with religious association

2-756

Higher education. Religiously based universities

2-76

RECRUITMENT. MISSIONARY ACTIVITY. MISSIONS BROADLY. MISSIOLOGY

- ◇ 2-76-051 Missionaries
- ◇ 2-76-736 Missionary societies

2-762

Missionary work. Forms of missionary activity

2-763

Campaigns. Crusades

- ◇ 2-763(1-88) Home missions

- 2-764** Overseas missions. Missions in the narrow sense
- 2-765** Propaganda. Religious propaganda. Publicity
- 2-766** Evangelical activity. Evangelism
⇒ 2-475
- 2-767** Proselytizing. Conversion
- 2-768** Forced conversion. Conversion by the sword. Imposition of faith. Holy war
- 2-769** Religious conquest
- 2-77** **ORGANIZATIONAL STRUCTURE OF THE FAITH, RELIGION**
- 2-771** Administrative levels
- 2-772** Top level. See
- 2-773** Intermediate. Regional organization. Diocese. Circuit
- 2-774** Local. Parishes. Local community, group
- 2-775** Neighbourhood. House groups. Informal groups
- 2-78** **RELIGIOUS ORGANIZATIONS. RELIGIOUS SOCIETIES AND ASSOCIATIONS**
 ◇ 2-78-028.85 Prohibited, banned religious organizations
 ⇒ 35.078.17
 ◇ 2-78-05 Religious organizations for particular persons
 ◇ 2-78-055.2 Religious women's groups
 ◇ 2-78-053.2 Religious children's organizations
 ◇ 2-78-46 Religious associations according to activities or purpose
 ◇ 2-784-463 Religious charities
 ◇ 2-784-468.5 Religious medical agencies and organizations
 ◇ 2-784-488.85 Religious organizations for refugees
- 2-784** Religious groups and lay orders not defined by purpose or membership
These are numerous. The preferred treatment should be faith - religious groups - A/Z
 ◇ 233-784VHP Visva Hindu Parisad (VHP)
 ◇ 26-784BNA Bnai Brith
 ◇ 27-784ROS Rosicrucians
- 2-784.5** Secret societies
- 2-788** Religious orders proper. Monastic orders. Monasticism
(Persons in religious orders)
 ◇ 2-788-055.1 Monks
 ◇ 2-788-055.2 Nuns
- 2-788-1/-9** **Monastic religion and practice**
- 2-788.2** Enclosed orders. Separated from the world
- 2-788.3** Mendicant orders
- 2-788.4** Working orders
- 2-788.435** Militant, military
- 2-788.437** Teaching
- 2-788.461** Medical, nursing, caring. Hospitallers
- 2-788.5** Celibate orders
- 2-788.6** Silent orders
- 2-789** Named monastic orders
- 2-79** **SECTS. SECTARIAN MOVEMENTS**
Generally, where significant these will be specified or enumerated sub-classes of a particular religion. Use this location for general works on sectarian movements, or where the given sect is not otherwise specified
- 2-8** **RELIGIONS CHARACTERISED BY VARIOUS PROPERTIES**
It is not recommended that this facet be used for the generation of compound classes which ought really to be enumerated sub-classes of a religion; e.g. Orthodox Christianity, Reformed Judaism should be regarded as significant divisions of the faith, and treated as movements in their own right. The subtlety and complexity of relations between religious movements do not allow for the simplistic results given by over rigorous facet analysis where it is clearly not merited
- 2-81** **CLASSIFICATION OF RELIGIONS**
(Properties derived from previous facets)
 ◇ 2-81-154 Monotheistic religions
 ◇ 2-81-22 Revealed religions
- 2-82** **PROPERTIES RELATED TO MINISTRY**
- 2-823** Religions with hierarchical structure, levels of ministry. 'Episcopalian'
- 2-824** Religions with ministers, priests 'Presbyterian'
- 2-825** Religions without ministers, priests 'Congregational'
- 2-83** **PROPERTIES RELATING TO ORTHODOXY**
- 2-832** Ultraorthodox
- 2-833** Orthodox
- 2-834** Heterodox
- 2-835** Moderate
- 2-836** Reformed
- 2-837** Liberal
- 2-838** Reconstructed
- 2-84** **PROPERTIES RELATING TO THE STATE**
- 2-842** National religions
- 2-844** Established churches. State religions
- 2-846** Religions of specific regions or countries
Use common auxiliaries of place Table 1e for detail
 ◇ 2-846(1-773) Religions of the Third World
 ◇ 2-846(519) Religions of Korea
- 2-85** **PROPERTIES RELATING TO DEGREE OF OFFICIALDOM**
- 2-852** **Official. Mainstream**
 Philosophies. Schools of thought
Use for recognised parties within a religion
- 2-852.5**
- 2-853** **Popular. Folk religion. Popular religiosity**

- 2-853.5 Superstition
 ⇒ 133
 ⇒ 398
- 2-854 Dissident
- 2-855 Separatist
- 2-856 Unofficial. Breakaway
- 2-86 **FACTIONAL GROUPS. FACTIONS**
Groups that are a recognised part of a mainstream religion, but which form a distinctive and identifiable separate entity. There is a clear distinction between dissenting factions within a religion, and those that have broken away, which are classed at 2-856. Class here those that differ in matters of practice or observance, as distinct from those that have major differences of theology and doctrine. The latter should be regarded as heresies or schisms, and classed at 2-87
- 2-87 **SCHISMATIC GROUPS. HERESIES**
Groups that are in dispute with the major part of the faith, or do not accept significant doctrines
- 2-9 **HISTORY OF THE FAITH, RELIGION, DENOMINATION OR CHURCH**
- 21/29 **RELIGIOUS SYSTEMS. RELIGIONS AND FAITHS**
*There are numerous criteria for deciding what constitutes a religion; in this arrangement it is taken that, at the very broadest level, a religion is a system which contains some recognition of a spiritual or supernatural dimension in its view of the world.
 The fundamental arrangement is historical, but within that religions are organized in familial groups i.e. Hinduism, Jainism and Sikhism.
 Problems arise in the placing of 'new' groups that have roots in an older religion, but regard themselves as independent. This is particularly evident in the case of twentieth century movements. As a general rule, self-definition is taken as the authority. In order to assist the indexer, and to accommodate the enormous number of specialised terms in religious literature, a large number of examples of combination are given for a range of religions and faiths. It is intended that the indexer use these as a pattern for the development of the classes for all other religions as required*
- 21 **PREHISTORIC AND PRIMITIVE RELIGIONS**
- 212 **PREHISTORIC RELIGIONS**
- 213 **PRIMITIVE RELIGIONS**
In general only the containing class should be used, and no specific movements should be located here
- 22 **RELIGIONS ORIGINATING IN THE FAR EAST**
- 221 **RELIGIONS OF CHINA**
 ◇ 221-853 Folk religion of China. Popular Chinese religion
- 221.3 Taoism
- 221.5 The old State religion
- 221.7 Confucianism
- 223 **RELIGIONS OF KOREA**
- 225 **RELIGIONS OF JAPAN**
- 225.2 Shinto
- 225.22 Seicho-No-Je
- 225.24 Sekai Kyuseikyō
- 225.26 Zenrinkai
- 225.27 Tensho Kotai Jingukyo
- 225.28 Ennokyo
- 23 **RELIGIONS ORIGINATING IN INDIAN SUB-CONTINENT. HINDU RELIGION IN THE BROAD SENSE**
Hinduism is extremely difficult to define. Vedism and Brahmanism may be regarded as the religion of the archaic period, and, as a general rule 233 should be used for works on Hinduism
- 231 **VEDISM**
- 232 **BRAHMANISM**
- 233 **HINDUISM NARROWLY**
The authority for vocabulary, spelling, and definitions is Klostermaier, K. Hinduism. Oxford; Oneworld 1998
- 233-13 **The Holy. Brahma. Absolute being**
- 233-14 **God(s) and goddess(es)**
 ◇ 233-158DEV Devi
 ◇ 233-158GAN Ganesh
 ◇ 233-158KAL Kali
 ◇ 233-158SHI Shiva
 ◇ 233-158SHI-145.4 Shiva the destroyer
 ◇ 233-158VIS Vishnu
 ◇ 233-158VIS-144.2 Thousand names of Vishnu
 ◇ 233-158VIS-144.81 Lordship of Vishnu
 ◇ 233-158VIS-145.3 Vishnu the preserver
- 233-166 Incarnations. Avatars
 ◇ 233-166KRI Krishna
 ◇ 233-166RAM Rama
- 233-167 Spiritual beings
- 233-167.62 Asuras. Demons
- 233-167.8 Semi-divine beings. Apsaras. Nymphs
- 233-17 **The Universe**
- 233-172.4 Asat. The state before creation
- 233-175 End of the world
- 233-175.4 Pralaya. Annihilation. Universal dissolution
- 233-18 **Man. Mankind**
- 233-183 Atman. The individual soul
- 233-184.6 Anugraha. Grace
- 233-185 Human life
- 233-185.2 Suffering
- 233-185.3 Adharma. Unrighteousness. Sin
- 233-186 Death
- 233-187 Life after death
- 233-187.5 Reincarnation
- 233-187.55 Samsara. The cycle of birth and re-birth
- 233-187.6 Mukti. Moksa. The liberation of the soul
- 233-187.79 Aprakṛta sarira. The immaterial body
- 233-187.8 Ananda. Cessation of being
- 233-188 Spiritual realms
- 233-188.5 Pitra loka. World of ancestors
- 233-188.7 Narakas. Hells
- 233-188.75 Preta loka. World of ghosts

233-23	Sacred texts of Hinduism <i>All the sacred texts of Hinduism are collected here, although strictly speaking, the Vedas could be subordinated to Vedism. Many ancient texts are themselves interpretations, commentary, and development of earlier traditions, and it is difficult to draw the line between original text and developed text</i>	233-33	Inspired persons. Major figures
		233-335	Reformers ◇ 233-335G Gandhi ◇ 233-335T Tagore
233-24	Sruti. Revealed writings. Vedas in the widest sense	233-34	Martyrs
233-242	Vedas in the narrower sense. Samhitas <i>Collections of textual material; the texts themselves</i>	233-35	Ascetics. Hermits. Fakirs. Sadhus. Sannyasi
233-242.2	Trayi Vidya 'Triple doctrine'	233-36	Sants. Saints
233-242.3	Rig Veda	233-36R	Ramakrishna
233-242.4	Yajur Veda	233-37	Mahatmas. Gurus. Scholars
233-242.45	Krsna. 'Black tradition'	233-37R	Radhakrishnan
233-242.452	Taittiriya	233-4	Hindu religious practice
233-242.454	Maitrayani	233-42	Moral behaviour. Moral theology
233-242.455	Kathaka samhitas	233-423	Right and wrong
233-242.46	Sukla. 'White tradition'	233-423.5	Adharma. Unrighteousness. Evil
233-242.6	Samsa Veda	233-423.57	Aparadha. Specific sins <i>Specify by alphabetic extension A/Z (Table 1h)</i>
233-242.7	Atharva Veda		◇ 233-423.57A Avidya. Ignorance
233-243	Brahmanas <i>Manuals for Brahmins; interpretations of earlier texts; beginnings of hermeneutic tradition</i>	233-423.58	◇ 233-423.57H Himsa. Violence
233-245	Aranyakas	233-423.7	Mahapatakas. Great sins. Brahmanahatya.
233-247	Upanishads. Vedanta	233-423.77	Surapana. Steya. Guruvanganagana.
233-247.2	Brhadaranyaka	233-425	Mahapatahasamsarga
233-247.3	Chandogya	233-426	Dharma. Good. Virtue. Right living
233-247.4	Taittiriya	233-428	Specific virtues. Ahimsa. Aparigraha. Asteya.
233-247.5	Isa		Vidya
233-247.6	Svetasvatara	233-425	Ethical systems. Acara. Rules for living
233-247.7	Mahanarayana	233-426	Religious ethics
233-247.8	Mandukya	233-428	Obligations, moral precepts. Religious duties
233-25	Smrti. Tradition	233-43	Social customs and practice
233-256	Sutras <i>Texts connected with ritual practice, domestic rites, rites of passage, law and social relationships</i>	233-44	Social behaviour
233-256.2	Svanta-sutras	233-442	Food and diet
233-256.4	Gryha-sutras	233-442.45	Dietary requirements
233-256.6	Dharma-sutras	233-442.47	Upavasa. Fasting
233-265.3	Itihasa. Epics and sagas	233-45	Vivaha. Marriage
233-265.32	Ramayana	233-46	Charity
233-265.33	Mahabharata	233-463	Dana. Charitable giving
233-265.34	Bhagavadgita	233-5	Worship in Hinduism
233-265.35	Puranas <i>Wide range of material, including mythologies of the Hindu deities; links with sectarianism, and popular Hinduism</i>	233-52	Religious sites, locations
233-266	Smrtis. Codes of Law		◇ 233-52(282) Sacred rivers. Seven rivers.
233-268	Upavedas		Saptasindhu
233-271	Vedangas	233-523	Religious buildings
233-282	Tantras and Agamas. Ritual texts <i>Sectarian texts, largely associated with Shivaism, tantras are texts also used by Jains and Buddhists</i>	233-523.6	Ashrams. Monasteries. Community buildings
233-286	Non-Sanskrit texts. Vernacular traditions	233-524	Vehicles of cultic acts
233-286.5	Tiravaimoli. 'Tamil-veda'		◇ 233-524:662 Bhasma. Ashes
233-286.7	Other <i>Locate sectarian scriptures with sectarian movement e.g. 233.2-242 Vaisnava Samhitas.</i> <i>Use common auxiliaries Table 1c for specific languages e.g. 233-291=214.32</i> <i>Bengali devotional poetry</i>	233-525.2	◇ 233-524:663.2 Soma. Wine, alcohol
		233-526	Gandha. Perfumes. Incense
		233-526.4	Religious objects
		233-526.5	Mala. Rosaries
		233-526.6	Dipa. Lamps, lights
		233-526.8	Murti. Images of the deity. Astamurti
		233-53	Janeu. Personal adornments
		233-532.6	Acts of worship
		233-532.8	Nitya. Compulsory
		233-533	Naimittika. Voluntary devotions
		233-534	Objects of worship
		233-534.3	◇ 233-533-526.6 Puja. Worship of images
		233-534.35	Rites and rituals
		233-534.4	Prayer
		233-534.45	Specific prayers. Gayatri
		233-534.5	Reading of texts
		233-534.55	Specific readings
		233-535	Chants
			Mantras. Japa (repetition of mantra)
			Music
233-3	Persons in Hinduism		

233-535.4	Kirtana. Public singing	24	BUDDHISM
233-535.7	Stotra. Hymna. Stotramalas. Collections of hymns	24-1	THEORY AND PHILOSOPHY OF BUDDHISM. NATURE OF BUDDHISM
233-536	Ritual acts	24-17	The Buddhist view of the universe. Cosmology
233-536.2	Abhisekha. Anointing	24-172	The material world. The physical environment
233-536.3	Postures	24-172.2	Origins. Cosmogony. Creation of the world
233-536.36	Namaskara. Pranama. Prostration	24-18	Man. Mankind. Humanity
233-537	Offerings	24-183	The nature of man
	◇ 233-537:641 Prasada. Food offerings	24-183.2	Material body. Rupa
233-538	Medha. Sacrifices	24-183.3	The non-material. Nama
233-538.2	Burnt offerings. Agnihotra, fire sacrifice	24-183.4	Consciousness
233-538.25	Agnicayana, the fire offering	24-183.5	Atma. The soul. The spirit. Human spirit
233-538.6	Sacrifices according to subject		⇒ 24-584.5
	◇ 233-538.6:599.89 Purusamedha. Human sacrifice	24-183.55	Anatta. Anatma. No-self. Denial of personality
		24-183.7	Freedom. Free will. Predestination
233-54	Purpose of ceremony	24-185	Human experience. Life
233-544.6	Vrata. Vow, pledge	24-185.2	Suffering. Dukkha
233-546.2	Asirveda. Blessing. Benediction	24-185.325	Karma. Desire. Craving
233-549.2	Avahana. Invocation of the deity	24-186	Death. The dead
233-549.5	Visarjana. Departure of the deity		⇒ 24-557
233-552	Samskara. Rites of passage	24-186.5	End of life. Parinirvana (for the enlightened)
233-553	Jata-karma. Birth ritual	24-187	Life after death. Afterlife. Hereafter
233-555	Vivaha. Marriage ceremonies	24-187.2	States of being. Forms of continued existence. Transmigration
233-557	Sraddha. Antyesti. Mrtyu-samskara. Last rites. Funeral ceremonies	24-187.5	Rebirth. Reincarnation. Metempsychosis. Metamorphosis
233-557.5	Sati. Disposal of the body	24-187.55	Samsara. Cycle of birth and re-birth
233-558.2	Upanayana. Initiation	24-187.56	Ananda. Dependent origination
233-558.5	Diksa. Entry to religious community	24-187.6	Liberation of the soul
		24-187.85	Annihilation. Cessation of existence. Non-being. Assimilation. Nirvana. Nibbana
233-56	Feasts and festivals. Holy days	24-188	Non-material world. Spiritual realms. Planes of life. Destinies
	◇ 233-565 Named feasts and festivals	24-188.5	Deva-loka. Deva-worlds
	◇ 233-565DIV Divali	24-188.75	Abode of ghosts. Preta-loka
	◇ 233-565DUR Durga puja. Dassehra	24-2	EVIDENCES OF RELIGION IN BUDDHISM
	◇ 233-565HOL Holi	24-23	Buddhist Sacred books. Scriptures. Religious texts
	◇ 233-565JAG Jagannath	24-232	Original texts. Canonical works of scripture
	◇ 233-565MAK Makara-sankranti	24-233	Variants
	◇ 233-565NAG Naga Panchami	24-234	Translations (i.e. from the original text)
	◇ 233-565PON Pongal		<i>Denote language and period by Common auxiliary Tables 1c and 1g</i>
	◇ 233-565RAM Ramlila		◇ 24-234=111 Buddhist scriptures in English
	◇ 233-565VAI Vaisakhi	24-235	Versions. □ Devanagari Pali Series (Pali Text Society)
233-852.5	Philosophies. Schools of thought. Darsanas. 'Theories'	24-24	The Pali Canon (canon of the Theravada school). Tipitaka. The Three Baskets
	◇ 233-852.5NYA Nyaya	24-244	Vinaya pitaka. Basket of discipline
	◇ 233-852.5PUR Purva Mimamsa		⇒ 24-788
	◇ 233-852.5SAM Samkhya	24-244.2	Mahavagga
	◇ 233-852.5UTT Uttara Mimamsa. Vedanta	24-244.3	Cullavagga
	◇ 233-852.5VAI Vaisesika	24-244.4	Parajika
	◇ 233-852.5YOG Yoga	24-244.5	Pacittiya
		24-244.7	Parivara
233.2	Vishnuism	24-246	Sutta pitaka. Basket of instructions
233.3	Shivaism	24-246.2	Digha Nikaya
233.4	Shaktism. Shaktas	24-246.3	Majjhima Nikaya
233.5	Neo-Hinduism. Reform Hinduism	24-246.4	Samyutta Nikaya
233.7	Virashaivas. Lingayats		
	<i>Consider themselves independent of Hinduism, but most government and other official organizations regard them as sectarian</i>		
234	JAINISM		
234.3	Svetambara		
234.5	Sthanakavasis		
234.7	Digambaras		
235	SIKHISM		
239	OTHER RELIGIONS OF EASTERN ORIGIN		
239.4	Theosophy		
239.6	Sri Chinmoy		

24-246.5	Anguttara Nikaya	24-285.2	On the existence of suffering
24-246.6	Khuddaka Nikaya	24-285.3	On the origin of suffering
24-246.622	Khuddakapatha	24-285.4	On the knowledge of the cessation of suffering
24-246.624	Dhammapada	24-285.5	Eightfold path. Magga
24-246.625	Udana	24-285.52	Wisdom. Panna
24-246.626	Itivuttaka	24-285.523	Sammadiitthi. Right views, right understanding.
24-246.627	Sutta Nipata		Knowledge. Nana
24-246.632	Vimnavatthu	24-285.524	Sammasamkappa. Right aspirations, right thought
24-246.634	Petavatthu	24-285.53	Morality. Sila
24-246.635	Theragatha	24-285.533	Sammavaca. Right speech
24-246.637	Therigatha	24-285.534	Sammakamanta. Right conduct, right action
24-246.642	Jataka	24-285.535	Sammaajiva. Right livelihood
24-246.643	Nidessa	24-285.54	Meditation. Samadhi
24-246.644	Patisambhidamagga	24-285.543	Sammavayama. Right effort
24-246.652	Apadana	24-285.544	Sammasati. Right mindfulness
24-246.653	Buddhavamsa	24-285.545	Sammasamadhi. Right contemplation
24-246.654	Carriyapitaka		
24-248	Abhidhamma Pitaka. Basket of the higher dhamma	24-286	Other forms of religious texts
24-248.2	Dhammasangani	24-287	Sermons as a form
24-248.3	Vibhanga	24-3	PERSONS IN BUDDHISM
24-248.4	Dhatukatha	24-31	Buddha. Gautama Buddha. Adi Buddha
24-248.5	Puggalapannatti	24-31-1	Theology of the Buddha
24-248.6	Kathavatthu		◇ 24-31-144.3 Trikaya. Doctrine of the three bodies
24-248.7	Yamaka		
24-248.8	Patthana	24-312	Life of Buddha
24-252	Post-canonical Pali literature	24-312.47	Mother of Buddha. Foster mother.
24-252.2	Dipavamsa		Mahaprajapati
24-252.3	Mahavamsa	24-312.55	Original followers. Arhants
24-252.4	Callavamsa	24-312.6	Enlightenment
24-252.5	Dhammapada attakatha	24-312.9	Return of the Buddha [as Maitreya]
24-252.6	Milindipanha	24-315	Accounts, anecdotal material
24-252.7	Nettipakarana	24-317	Memoirs. Sayings. Deeds
24-252.8	Visuddhimagga	24-318	Buddha as model, example
24-252.9	Vimuttimaggga	24-35	Ascetics. Hermits
24-253	Sanskrit Buddhist literature. □ Mahayana texts	24-36	Saints. Boddhisatvas. Enlightened ones
	<i>Class here only if there is a requirement to keep all sacred texts together. If in doubt, prefer 242-24/-254</i>	24-365	Arahant (discipline at the final stage of enlightenment)
24-254	Commentary on sacred works	24-37	Acariya. Wise men
24-256	Interpretative works. Works of guidance on practice	24-4	RELIGIOUS ACTIVITIES. RELIGIOUS PRACTICE IN BUDDHISM
24-262	Oral forms. Oral tradition	24-42	Moral behaviour. Moral theology
24-264	Myths and legends	24-423.5	Adhamma. Wrong-doing
24-265	Other literary forms		◇ 24-423.5DOS Dosa. Hatred
24-265.2	Poetic works		◇ 24-423.5HIM Himsa. Violence
24-265.3	Epics and sagas		◇ 24-423.5LOB Lobha. Greed
24-266	Law. Legal codes. Legal texts		◇ 24-423.5MOH Moha. Delusion
24-278	Theology of sacred texts. Theological content	24-423.58	Five hindrances. Nivaranani
	<i>Add to 24-278 detail from the whole auxiliary schedule</i>	24-423.582	Sense desires. Covetousness
24-279	Religion as demonstrated in sacred texts. Religious practice	24-423.583	Malevolence
	<i>Add to 24-279 detail from the whole auxiliary schedule</i>	24-423.584	Sloth
24-28	Other religious texts	24-423.585	Restlessness
24-282	Liturgical works. Liturgical texts	24-423.586	Doubts
24-283	Historical documents. Evidential documents. Historical records. Contemporaneous accounts. Letters	24-423.7	Dhamma. Good. Virtue. The good. Right living
24-284	Works of doctrine, dogma. Articles of faith. Systematic theology. Dogmatics	24-423.77	Specific virtues
24-285	Four noble truths. Ariya sacca. Arya satya		◇ 24-423.77AHI Ahimsa. Non-harm. Respect for life
		24-423.78	Ten transcendental virtues. Paramis
		24-423.780	Dana. Generosity
		24-423.781	Sila. Morality
		24-423.782	Nekkhamma. Renunciation
		24-423.783	Panna. Wisdom
		24-423.784	Viriya. Energy
		24-423.785	Khanti. Patience
		24-423.786	Sacca. Truthfulness

- 24-423.787 Adhithana. Resolution
 24-423.788 Metta. Loving-kindness
 24-423.789 Upekkha. Equanimity
- 24-426 Buddhist ethics**
- 24-428 Religious laws, obligations, moral precepts, moral laws. Regulations. Religious duties**
- 24-43 Social customs and practice. Social theology**
 24-442.45 Dietary requirements. Dietary limitations
 24-442.47 Abstinence. Fasting. Prohibition
 24-442.5 Rules concerning specific foods and drinks
 ◇ 24-442.5:637.5 Rules concerning meat
 ◇ 24-442.5:663.2/.5 Rules concerning alcoholic drinks
- 24-443 Personal hygiene and appearance. Personal conduct**
 24-443.5 Clothing. Apparel
- 24-447 Sexual relations, activity**
 24-447.7 Abstinence. Celibacy
- 24-46 Obligation to others. Support of the monastic community**
- 24-5 RITES, CEREMONIES ETC.**
- 24-557 Funerary customs and rituals**
- 24-58 Contemplation and reflection**
- 24-583 Meditation. Contemplation**
- 24-584 Spiritual life**
 24-584.5 Spiritual growth and development. Anna. Knowledge. Gnosis
 24-584.52/.55 Stages of development
 24-584.52 Sotapanna (entering)
 24-584.53 Sahadagamin
 24-584.54 Anagamin (never-returner)
 24-584.55 Arantship
 24-584.7 Spiritual exercises
- 24-6 PROCESSES IN BUDDHISM**
- 24-64 Relative processes. External processes. Interactions between Buddhism and other faiths**
 24-664.3 Syncretism. Borrowing
 24-664.5 Assimilation
- 24-67 Relations between Buddhism and society in general**
 24-673.5 Tolerance. Irenics
- 24-674 Conflict. Antagonism. Hostility**
 24-674.5 Persecution
- 24-7 BUDDHIST ORGANIZATION AND ADMINISTRATION**
- 24-722 Persons in Buddhism**
Normally persons in religion with specific functions will be subordinate to the function they perform, and denoted by the common auxiliaries of persons (Table 1k-05); e.g. denote soothsayers by 24-544.8-051
- 24-725 Samgha. Community of disciples. 'Priesthood'**
 24-732.2 Head of religion
 24-732.4 Buddhist councils
 ⇒ 24-9-732.4
 24-732.6 Lay persons. Adherents
- 24-78 Buddhist organizations. Buddhist societies and associations**
- 24-788 Buddhist orders proper. Monastic orders. Monasticism**
 ◇ 24-788-05 Persons in Buddhist religious orders
 ◇ 24-788-055.1 Bhikkhus. Buddhist monks
 ◇ 24-788-055.2 Buddhist nuns
 ◇ 24-788-428 Buddhist monastic rules
 ⇒ 24-244
- 24-79 Sects. Sectarian movements**
- 24-9 HISTORY OF BUDDHISM**
 24-9-732.4 Buddhist councils. □ First council at Rajagṛha 544 BCE; Second council at Vaisali 444 BCE; Third council at Pataliputra 247 BCE; Fourth council 100 CE; Fifth council in Rangoon 1871 CE; Sixth council in Rangoon 1954-6 CE
 24-9-852.5 Historical schools of thought. The eighteen schools
- 241 HINAYANA BUDDHISM. THE LESSER VEHICLE. THERAVADA BUDDHISM. PALI SCHOOL**
These are not synonymous, but are interchangeable in common usage. This is the 'orthodox' Buddhism of the southern Buddhist world (India, Ceylon, Burma, Thailand)
- 242 MAHAYANA. THE GREAT VEHICLE**
'Reformed' Buddhism of the northern area (Tibet, Nepal, Korea, China and Japan) has more emphasis on popular participation and more traditional religious features than Hinayana. Since most examples of conventional 'religious' practice are to be found in Mahayana Buddhism, those concepts are enumerated at this location
- 242-162.6 Prajnaparamita. Mother of all the Buddhas
 242-163 Pantheistic hierarchy
- 242-24 Mahayana texts. Sanskrit Buddhist literature**
 242-253 Sutras
 242-254 Sastras
 242-257 Tantras
 242-31-1 Buddhology
 242-31-169 Mythical Buddhas
- 242-36 Bodhisattvas. Bodhisattva ideal**
- 242-5 Puja. Buddhist worship broadly**
 242-522 Buddhist holy places
 242-522.4 Places of particular spirituality, holiness
 242-523 Buildings for religious use
 242-523.3 Buddhist shrines
 242-523.4 Temples
 242-523.6 Monasteries. Community buildings
 242-524 Vehicles of cultic acts
 ◇ 242-524:536.46 Fire
 ◇ 242-524:612.1 Blood
 ◇ 242-524:662 Ashes
 ◇ 242-524-032.2 Water
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		27-245.28	Tobit
		27-245.32	Judith
		27-245.34	Esther
		27-245.36	Wisdom of Solomon
		27-245.38	Ecclesiasticus
		27-245.42	Baruch

27-245.44	Letter of Jeremiah	27-273	Bible commentaries
27-245.46	Song of the Three Holy Children	27-274.2	Authorship. Origin of texts
27-245.48	Susannah	27-274.4	Canon. Canonicity of the Bible
27-245.52	Bel and the dragon	27-274.6	Authority, inerrancy of Scripture
27-245.54	Prayer of Manasseh	27-275	Textual criticism
27-245.56	Maccabees I	27-276	Literary criticism
27-245.58	Maccabees II	27-277	Bible study. Interpretation
27-245.61.7	Old Testament pseudepigrapha	27-277.2	Biblical exegesis. Biblical hermeneutics
27-245.62	Book of Jubilees	27-278	Biblical theology
27-245.63	I Esdras	27-28	Non-Biblical Christian texts
27-245.64	III Maccabees	27-282	Liturgical texts
27-245.65	IV Maccabees	27-282.3	Creeds and confessions of faith
27-245.66	Life of Adam and Eve		◇ 27-282.3NIC Nicene Creed
27-245.67	Epistle of Aristes		◇ 27-282.3-144.89 The Trinity in the Creed.
27-245.68	Martyrdom of Isaiah		NOTE: Class here Filioque question
27-245.69	Psalms of Solomon	27-282.4	Catechisms
27-245.72	Book of Enoch	27-282.5	Prayer books
27-245.73	Assumption of Moses	27-282.7	Service books
27-245.74	Apocalypses of Baruch	27-284	Systematic Christian theology. Dogmatic theology
27-245.75	IV Esdras	27-285.2	Apologetics
27-245.76	Sibylline oracles	27-285.4	Christian polemics
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27-245.8	Other Old Testament pseudepigrapha		◇ 27-285.4:27-87 Against heretics
27-246	The New Testament	27-29	Christian literature
27-247	The Gospels	27-31	Jesus Christ. Christology
27-247.2	The Synoptic Gospels		◇ 27-31-144.894 Divinity of Christ
27-247.4	Matthew		◇ 27-31-144.894.2 Christ as Logos, Word of God
27-247.6	Mark		◇ 27-31-144.894.25 Pre-existence of the Logos
27-247.7	Luke		◇ 27-31-166 Incarnation of Christ. Humanity of Christ
27-247.8	John		◇ 27-31-185.53 Christ and Redemption
27-248.2	Acts of the Apostles		◇ 27-31-384 Priesthood of Christ
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27-248.4	Pauline epistles		◇ 27-31-543.4 Intercession of Christ
27-248.42	Romans	27-312	Life of Jesus
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27-248.48	Galatians	27-312.45	Joseph
27-248.52	Ephesians	27-312.47	Mary. The Virgin Mary. Theotokos (Mother of God). Mariology
27-248.54	Philippians		John the Baptist
27-248.56	Colossians	27-312.5	The disciples
27-248.58	Thessalonians I	27-312.55	Ministry. Public life
27-248.62	Thessalonians II	27-312.7	Passion of Christ. Trial and Crucifixion.
27-248.64	Timothy I	27-312.8	Shrouding. Entombment
27-248.66	Timothy II		Resurrection. Ascension
27-248.68	Titus	27-312.9	
27-248.72	Philemon	27-317	Words and sayings of Jesus. Deeds of Jesus
27-248.74	Hebrews	27-317.3	Parables
27-248.76	James	27-317.5	Miracles
27-248.78	Peter I	27-318	Jesus as example for Christians. Imitation of Christ
27-248.82	Peter II	27-32	Jesus as Messiah
27-248.83	John I		◇ 27-32-12 Concept of Messiah. Nature, understanding of Messiah
27-248.84	John II		◇ 27-32-23 The Messiah according to the Bible
27-248.86	John III		◇ 27-32-24:929.5 Genealogies of Christ
27-248.88	Jude		◇ 27-32-244 Messianic prophecies in the Bible
27-249	The revelation of John		
27-252	New Testament apocrypha and pseudepigrapha		
27-252.2	Apocryphal and pseudepigraphal gospels		
27-252.4	Pseudepigraphal epistles		
27-252.6	Other New Testament Apocrypha and Pseudepigrapha		
27-27	Critical works. Introductions to the Bible		
27-271.4	Parallel forms. Bible harmonies		
27-271.5	Annotated versions		
27-272	Biblical concordances		
27-272.4	Verbal concordances		
27-272.5	Subject concordances		

- 27-36 Saints**
 ◇ 27-36ATH St Athanasius
 ◇ 27-36BAS St Basil
 ◇ 27-36CYR St Cyril (Constantine)
 ◇ 27-36IGN St Ignatius
 ◇ 27-36JOS St Joseph
 ◇ 27-36MAX St Maximus
 ◇ 27-36MET St Methodius
 ◇ 27-36NIL St Nilus
 ◇ 27-36SERA St Seraphim
 ◇ 27-36SERG St Sergius
- 27-4 PRACTICAL THEOLOGY. CHRISTIAN PRACTICE**
- 27-42 Moral theology**
- 27-423 Right and wrong**
 27-423.3 Knowledge of right and wrong
 27-423.35 Conscience
 27-423.4 Free will. Freedom to act
 27-423.42 Factors affecting freewill. Ignorance. Fear
 27-423.45 Predestination
 27-423.5 Sin. Wrongdoing
 ◇ 27-423.5-051 Sinners
 27-423.57 Specific sins
 27-423.58 Deadly sins. Pride. Avarice. Envy. Lust. Anger. Gluttony. Sloth
 27-423.7 Virtues
 27-423.75 Specific virtues
Specify by alphabetic extension AVZ (Table 1h)
 27-423.77 Moral virtues. Humility. Penitence. Modesty. Zeal. Chastity. Poverty. Obedience
 27-423.78 Cardinal virtues. Justice. Prudence. Fortitude. Temperance
 27-423.79 Theological virtues. Faith. Hope. Charity
- 27-428 Moral laws, obligations, regulations. The Church's commandments**
 27-428-23 Commandments derived from the Bible
 27-428-242.6 The Ten Commandments
The order of commandments given is that of the Roman Catholic Church in order to accommodate the detail enumerated in the cancelled schedule. It is open to users to interpret the numbering of the commandments as they find most useful
 27-428-242.6.1 The first commandment. Adherence to, and transgression of the first commandment.
 □ Adoration. Prayer. Observance of festivals. Superstition. Idolatry. Magic. Divination
 27-428-242.6.2 The second commandment. Adherence to, and transgression of the second commandment.
 □ Tempting God. Sacrilege. Simony. Blasphemy. Imprecation. Oath. Swearing
 27-428-242.6.3 The third commandment. Adherence to, and transgression of the third commandment
 27-428-242.6.4 The fourth commandment. Adherence to, and transgression of the fourth commandment
 27-428-242.6.5 The fifth commandment. Adherence to, and transgression of the fifth commandment.
 □ Suicide. Homicide. Murder. Abortion. Duel. War
 27-428-242.6.6 The sixth commandment. Adherence to, and transgression of the sixth commandment
 27-428-242.6.7 The seventh commandment. Adherence to, and transgression of the seventh commandment
 27-428-242.6.8 The eighth commandment. Adherence to, and transgression of the eighth commandment.
 □ Lying. False judgements. False witness. Defamation. Calumny. Insolence. Contumely. Obligation to secrecy. Discretion. Hypocrisy. Boasting
 27-428-242.6.9 The ninth commandment. Adherence to, and transgression of the ninth commandment
 27-428-242.6.99 The tenth commandment. Adherence to, and transgression of the tenth commandment
 27-428.2 Nature of law
 27-428.3 Requirements of law
 27-428.6 Interpretation of law. Epiky
 27-428.7 Exemption from law
 27-428.8 Transgression of law
 27-428.85 Mitigation
 27-428.87 Indulgences
 27-428.9 Amendment, cancellation of law
- 27-43 Christian social customs and practice. Social theology**
- 27-442 Food and diet**
 27-442.47 Fasting
 27-443.5 Costume, appearance
- 27-447 Sexual activity and practice**
 27-447.7 Celibacy
- 27-45 Christian marriage and family**
- 27-452 Christian marriage**
 ⇒ 27-555
 27-452.5 Marriage contracts
- 27-454 Divorce**
- 27-46 Pastoral theology. Pastoral work. Christian charity**
- 27-463 Charitable giving**
 27-463.2 Tithes. Stewardship
- 27-464 Voluntary work**
- 27-468 Care. Physical care and attendance**
 27-468.5 Nursing
 27-468.6 Healing. Ministry of healing
- 27-47 Christian education**
- 27-472 Religious instruction**
- 27-475 Preaching. Religious oratory**
 27-475.2 Homiletics
 27-475.5 Sermons
 ◇ 27-475.5-23 Sermons based on Holy Scripture
- 27-48 Recipients of care. Groups in pastoral theology**
 27-484.7 Oppressed groups. Liberation theology
- 27-5 CHRISTIAN WORSHIP AND RITUAL**
- 27-522 Christian sites. Holy places**
 27-522.2 Location, orientation
- 27-523 Church buildings and their content**
 ◇ 27-523.7 Buildings by period and style
 Christian church buildings (religious significance)
For religious architecture see 726
 ◇ 27-523.4:7.033.2 Byzantine churches
 ◇ 27-523.4:7.033.5 Gothic churches
 27-523.41 Cathedrals
 27-523.42 Churches. Parish churches
 27-523.43 Separate, free-standing chapels. □ Chapels of ease
- 27-524 Vehicles of cultic acts**
 ◇ 27-524-032.2 Holy water
 27-525.4 Vestments and personal objects. Chasuble. Surplice. Biretta. Mitre. Crozier
 27-525.7 Church linen. Draperies. Altar cloths
 27-526.1 Furniture. Fonts. Lecterns. Pulpits. Prayer desks

- 27-526.2 Altars. Tables. Communion tables
 27-526.22 Tabernacles. Hanging pyxes
 27-526.24 Aumbries
 27-526.3 Vessels. □ Chalices. Ciboria. Monstrances
 27-526.5 Lights. Lighting
 27-526.6 Christian symbols and images
 ◇ 27-526.6-137 Veneration of images
 27-526.62 Icons. Frescoes. Paintings. Iconography
 27-526.63 Decorative objects and devices. Stained glass
 27-526.64 Effigies. Statues
 27-526.65 Funerary monuments. Tombstones
 27-526.7 Relics
- 27-528 Christian liturgy**
 ◇ 27-528:001.4 Liturgical language
 ◇ 27-528-274.2 Sources, origin of Christian liturgy
 ◇ 27-528-274.6 Authority of liturgy
 ◇ 27-528-276.63 Symbolism of liturgy
 ◇ 27-528-276.63:535.6 Liturgical colours
 27-528.1/.8 Traditional Christian liturgies
Most of these date from before the division of the Church; however libraries may prefer to subordinate them to particular denominations and churches
 27-528.1 Old Roman rite
 27-528.2 Gallican rite
 27-528.3 Mozarabian rite
 27-528.4 Ambrosian
 27-528.5 African liturgies
 27-528.55 Ethiopian, Coptic liturgies
 27-528.57 Egyptian, Alexandrine rite
 27-528.6 Syrian, Palestinian, Antiochene rites
 27-528.7 British rites. □ Sarum rite. York. Hereford
 27-528.75 Celtic rite
 27-528.8 Byzantine rite. Orthodox rites
- 27-53 Christian acts of worship**
 27-532.2 Public worship. Services. Offices
 ◇ 27-532.2"345" Night services. Vigils
 ◇ 27-532.2-282 Office books. Service books
 27-532.3 Family worship. Domestic worship. Worship in the home
 27-532.4 Private worship. Personal devotions
 27-534.3 Prayers. □ Orisons, litanies, prefaces
 27-534.35 Specific prayers
 ◇ 27-534.35AVE Ave Maria. Hail Mary
 ◇ 27-534.35PAT Pater Noster. Our Father. Lord's prayer
 27-534.4 Reading of texts. Lessons
 27-534.5 Chants
 27-534.6 Prophecy
 27-534.7 Glossolalia. Speaking in tongues
- 27-535 Music**
 27-535.35 Musical chants
 27-535.7 Hymns
 ◇ 27-535.7-23 Hymnbooks
 27-535.8 Psalms
- 27-536 Physical rites**
 27-536.1 Use of water. Asperges
 27-536.27 Censing
 27-536.3 Bodily postures
 27-536.34 Bowing
 27-536.35 Kneeling. Genuflection
 27-536.36 Prostration
 27-536.37 Laying on of hands
 ⇒ 27-468.65
 27-536.4 Symbolic gestures
 27-536.42 Sign of the Cross
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 27-536.5 Processions
- 27-537 Offerings**
- 27-54 Ceremonies by purpose**
 27-543.7 Confession
 27-543.8 Penance
 27-544.3 Absolution. Forgiveness
 27-544.6 Dedication. Vow. Pledge
 27-546.2 Blessing. Benediction
 27-546.4 Curse. Anathema. Desecration
 ◇ 27-546.4-526.6 Iconoclasm
 27-546.4-528.82 Eucharist of St John Chrysostom
 27-546.4-528.83 Eucharist of St Basil
 27-546.4-528.84 Eucharist of St James, the brother of the Lord
 27-546.4-528.85 Eucharist of the Presanctified
 27-548.5 Exorcism
- 27-549 Eucharist. Communion. Holy Communion. Mass**
 ◇ 27-549-544.55 Communion as commemoration
 27-549.3 Presence of Christ in the communion
 ◇ 27-549.3-284 Doctrine of Christ's presence. Transubstantiation. Consubstantiation
- 27-55 Sacraments**
Most religions have some concept of sacraments, although they are not necessarily identical in what is regarded as sacramental. Consequently this general position is provided for comprehensive treatments of sacraments. Individual sacraments are not necessarily adjacent for any given religion
 ◇ 27-55-282 Liturgy of the sacraments
- 27-555 Weddings**
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- 27-557 Funerals. Funeral rites**
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- 27-558 Stages in religious life**
 27-558.3 Baptism
 27-558.4 Confirmation
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- 27-559 Coronations**
- 27-562 Feasts and festivals**
- 27-563 Sundays. Sabbath day. Lord's Day**
 ◇ 27-563:331.311 Working on Sundays
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- 27-564 Church year. Christian calendar**
- 27-565 Major Fasts and Festivals**
 27-565.2 Advent
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 27-565.37 Septuagesima. Sexagesima. Quinquagesima
 27-565.4 Lent
 27-565.42 Ash Wednesday
 27-565.44 Palm Sunday
 27-565.5 Holy Week. Passiontide
 27-565.52 Good Friday
 27-565.54 Eastertide
 27-565.6 Easter Day
 ◇ 27-565.6-549 Easter communion
 27-565.72 Ascension

- 27-565.73 Pentecost. Whitsun
27-565.74 Trinity
27-565.75 Corpus Christi
27-565.76 All Saints
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27-565.79 Saint's Days
27-565.8 Festivals of Mary. Annunciation. Assumption. Immaculate conception
27-565.9 Other feasts and festivals
- 27-57 Pilgrimages**
- 27-58 Contemplative and reflective religion**
◇ 27-58-472 Spiritual guidance
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- 27-583 Meditation. Contemplation. Personal prayer**
- 27-584 Spiritual life**
27-584.5 Spiritual growth and development
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- 27-585 Asceticism**
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- 27-587 Mysticism. Altered states of consciousness. Divine ecstasies**
27-587.6 Mystic phenomena. Charisms
27-587.65 Stigmata. Stigmatization
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- 27-636 Christian revivals**
- 27-64 Relations of Christianity**
- 27-662 Influence exerted by Christian churches**
◇ 27-662:3 Influence of the Church on society. Including: Christian social movements
- 27-67 Relations of Christianity with other faiths**
◇ 27-67:26 Relations with Judaism
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- 27-673.5 Irenics
- 27-675 Ecumenism**
- 27-676 Co-operation. Church unity**
- 27-72 The Church. The Christian Church. Ecclesiology**
◇ 27-72-144 Characteristics of the Church. Unity. Sanctity. Catholicity. Apostolicity. Visibility
◇ 27-72-4 Customs and tradition of the Church
- 27-725 Clergy. Christian ministers**
◇ 27-725:37 Education of priests
◇ 27-725-12 Concept of priesthood
◇ 27-725-13 Holiness of priesthood
◇ 27-725-428 Religious duties of ministers, clergy
◇ 27-725-443.5 Personal appearance of clergy
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◇ 27-725-558.5 Ordination of priests
◇ 27-725-735 Benefices. Livings
- 27-726.1 Archbishops
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27-726.25 Senior priests. Archdeacons. Canons. Deans. Archpriests
27-726.3 Priests. Ministers. Vicars
27-726.4 Assistant ministers. Deacons
- 27-726.6 Lay persons. Church members. Believers
◇ 27-726.6-428.5 Religious duties of lay persons
◇ 27-726.6-427 Informal promotion of the Roman Catholic faith by lay persons.
NOTE: Class here the universal priesthood of the faithful, personal prelature and movements such as Opus Dei
- 27-727 Non-Christians. Non-believers**
- 27-732 Authority of the Church**
27-732.1 Source of authority. □ Apostolic succession
27-732.3 Ruling bodies of churches. Synods. Councils
◇ 27-732.3"325" Council of Nicaea
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- 27-733 Religious vocation**
- 27-74 Religious law. Christian religious law. Canon law**
◇ 27-74-725 Persons in canon law
The Bible as a source of Law
Codes of Law. □ Corpus Iuris Canonici
Commentary and interpretation. Legal texts
- 27-742-24 Christian religious courts**
27-742-266
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- 27-745 Education**
- 27-75 Sunday schools**
- 27-752 Christian missions**
27-76-051 Missionaries
- 27-763 Missionary campaigns. Crusades**
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- 27-774 Parishes**
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- 27-78 Christian associations and organizations**
◇ 27-78-055.1 Christian associations for men
◇ 27-78-055.2 Christian associations for women
◇ 27-78-053.2 Christian association for children, young people
◇ 27-78-46 Organizations for pastoral work
- 27-784 Religious groups and lay orders**
- 27-788 Monastic orders. Monasticism. Religious orders proper**
◇ 27-788-056 Clerks. Persons in orders
◇ 27-788-055.1 Male orders. Monks
◇ 27-788-055.2 Female orders. Nuns
Monastic religion and practice
Ascetics in general. □ Hermits. Recluses. Anchorites. Stylites
27-788-428.5 Duties of persons in orders. Obligations, rules
27-788-544.6 Vows
◇ 27-788-544.6-428.5 Obedience
◇ 27-788-544.6-447.4 Chastity
◇ 27-788-544.6-463.6 Poverty
- 27-788-558.2 Novitiate**
27-788-733 Vocation to orders

- 27-788-9 History of monasticism
27-788.4 Working Christian orders
 ◇ 27-788.435 Military orders. Including:
 Knights Templars
 ◇ 27-788.461 Medical, nursing orders.
 Including: Knights Hospitallers
- 27-789 Specific Christian monastic orders**
A number of the orders here named are only represented in the Roman Catholic church; however, that cannot be said for all of them, and consequently this location is provided to maintain a single unitary class for Monastic orders. Individual orders may of course be subordinated to Roman Catholicism if required e.g. 272-789.5 Jesuits
- 27-789.1 Early orders. Orders on the Eastern model
27-789.15 Rule of St Pachomius
 ⇒ 27-789.2-428
27-789.17 Rule of St Basil
 ◇ 27-789.17-028.11 Longer Rule
 ◇ 27-789.17-028.13 Shorter Rule
- 27-789.2 Benedictines
 ◇ 27-789.2-428 Rule of St Benedict
- 27-789.23 Camaldolese
27-789.24 Carthusians
27-789.25 Cistercians. 'White monks'
27-789.255 Trappists
- 27-789.3 The mendicant orders generally
27-789.32 Franciscans. Friars minor. Grey friars
27-789.325 Capuchins
27-789.33 Dominicans. Black friars. Jacobins
27-789.34 Carmelites. White friars
27-789.35 Servites
27-789.4 Augustinians
27-789.43 Premonstratensians. Norbertines. White canons
27-789.45 Ursulines
27-789.5 Jesuits
27-789.6 Lesser orders
27-789.62 Redemptorists
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27-789.9 Other orders
27-852.5 Christian schools of thought, philosophies
- 27-87 Heresy. Heresies**
- 27-872 Early heresies. Heresies of the patristic period**
- 27-873 Antinomianism
- 27-874 Gnosticism and related heresies**
27-874-3 Simon Magus
27-874.3 Valentinus. Valentinians
27-874.5 Basilides
27-874.6 Marcion. Marcionites
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27-875.2 Heresies on the nature of God and the Trinity
27-875.22 Docetism
27-875.23 Montanism
27-875.24 Adoptianism
27-875.25 Sabellianism
27-875.26 Arianism
27-875.27 Macedonianism
27-875.5 Heresies of the incarnation
27-875.53 Ebionites
27-875.54 Elkesaites
27-875.55 Nestorianism
27-875.56 Monophysites
- 27-875.57 Eutychianism
27-875.6 Heresies on the nature of the Church and believers
27-875.64 Novatianism
27-875.66 Donatism
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- 27-876 Heresies of later Church history**
27-876.2 Neo-Manicheism. Gnostic heresies. Dualistic beliefs
27-876.3 Paulicians
27-876.4 Cathari. Cathars
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27-876.7 Iconoclasts
- 27-877 Heresies of the 13th-15th centuries**
27-877.2 Hesychasts
27-877.3 Joachimites
27-877.4 Flagellants
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27-877.6 Wycliffites
27-877.7 Lollards
- 27-878 Heresies of the 16th century onwards**
27-878.2 Jansenists
27-878.3 Utraquists
27-878.4 Hussites
27-878.5 Bohemian Brethren
- 27-879 Other heresies**
- 27-9 CHURCH HISTORY. GENERAL HISTORY OF THE CHRISTIAN CHURCH**
27-9"/1054" Church history to the Great Schism
27-9"00" Primitive apostolic church. Church during the first century
27-9"01/07" History of the second to eighth centuries AD. Period of the Church fathers. Patristic period
27-9"01/07"-284 Patristics narrowly. Writings of the patristic period
 These may be most conveniently arranged by century, and then alphabetically by author
 ◇ 27-9"01"CLE Clement
 ◇ 27-9"01"TER Tertullian
 ◇ 27-9"02"ORI Origen
 ◇ 27-9"03"JER Jerome
27-9"0100/0325" Church in the ante-Nicene period
27-9"0325/1054" Church in the post-Nicene period
27-9"08/09"-36PHO Activities of St Photius
27-9"0858/1204" Period of estrangement
- 271/279 CHRISTIAN CHURCHES AND DENOMINATIONS**
- 271 EASTERN CHURCH**
- 271.2 Orthodox Church. Eastern Orthodox. Russian and Greek Orthodox churches**
 ◇ 271.2-1 The Orthodox 'Tradition'. NOTE: This is distinct from traditions, and incorporates the whole body of faith, the Bible, Fathers, Councils, etc. Scripture is part of Tradition and not distinct from it. Tradition comprises Scripture, Councils, Fathers, Liturgy, Canons, Icons
 ◇ 271.2-23 The Orthodox Bible. NOTE: This comprises the Septuagint, i.e. 39 books of the Old Testament, 10 Deutero-Canonical Books and New Testament
 ◇ 271.2-282.7-247 The Gospel Book
 ◇ 271.2-282.7-247-536.36 Prostration before the Gospel Book

- ◇ 271.2-284 Doctrinal statements. Symbolical Books
 - ◇ 271.2-423.461 'Chapel' of the Prothesis, Preparation
 - ◇ 271.2-423.463 Diakonikon
 - ◇ 271.2-472-022.43 The Longer Catechism
 - ◇ 271.2-523.47 Ikonostasis
 - ◇ 271.2-523.472 Holy Door. Royal Door
 - ◇ 271.2-523.472-525.7 Curtain across the Holy Door
 - ◇ 271.2-525.4 Stoles. Epitrachilia
 - ◇ 271.2-526.2 Holy Table. Holy Throne
 - ◇ 271.2-526.3 Spoon (for distributing Communion)
 - ◇ 271.2-526.4 Chaplets. Prayer ropes
 - ◇ 271.2-526.5 Candles
 - ◇ 271.2-528-534.5 Litanies
 - ◇ 271.2-528-549 The Holy Liturgy
 - ◇ 271.2-528-549-536.4-022.51 Little Entrance. Synaxis
 - ◇ 271.2-528-549-536.4-022.56 Great Entrance
 - ◇ 271.2-528-549.3 Epiclesis
 - ◇ 271.2-528.8 Byzantine rite
 - ◇ 271.2-532.4 Anthologies of ascetic and mystical texts. Philokalia
 - ◇ 271.2-534.35 Invocation of the Name
 - ◇ 271.2-534.35-312 The Jesus prayer
 - ◇ 271.2-536.1-032.2 Blessing of water
 - ◇ 271.2-536.4 Sign of the Cross. NOTE: Class here use of two fingers (Russian practice) and of three fingers (Greek practice)
 - ◇ 271.2-536.72:644.642-024.42 Antidoron. NOTE: Unconsecrated remainder of the loaf used for Communion, distributed at the end of the Eucharist
 - ◇ 271.2-55 Sacraments
 - ◇ 271.2-55-543.7-544.3 Penance
 - ◇ 271.2-55-549 Holy Communion
 - ◇ 271.2-55-555 Holy Matrimony
 - ◇ 271.2-55-557-536.25 Anointing of the sick
 - ◇ 271.2-55-558.3 Baptism
 - ◇ 271.2-55-558.4 Chrismation
 - ◇ 271.2-55-558.5 Ordination
 - ◇ 271.2-55-022.51 Sacramentalia. Sacramentals
 - ◇ 271.2-55-022.51-557 Funeral rites
 - ◇ 271.2-55-022.51-565.35-536.1-032.2 Blessing of water at Epiphany
 - ◇ 271.2-55-022.51-788-558.2 Monastic profession. Clothing of religious
 - ◇ 271.2-55-022.51-532.36-536.2 Anointing of the monarch
 - ◇ 271.2-563.5 The Divine Office
 - ◇ 271.2-563.51 First Hour
 - ◇ 271.2-563.52 Third Hour
 - ◇ 271.2-563.53 Mattins
 - ◇ 271.2-563.54 Sixth Hour
 - ◇ 271.2-563.55 Vespers
 - ◇ 271.2-563.56 Compline
 - ◇ 271.2-563.57 Midnight Office
 - ◇ 271.2-564.11.43 The Orthodox year. Kalendar. NOTE: The Orthodox Church year begins on Sept. 1
 - ◇ 271.2-564.11-565.8 Nativity of Our Lady (Sept. 8)
 - ◇ 271.2-564.12-565.9-312.8 Exaltation of the Holy Cross (Sept. 14)
 - ◇ 271.2-564.15-565.2 The fast of Christmas (Nov. 15-Dec. 24)
 - ◇ 271.2-564.17 Entry of Our Lady into the Temple (Nov. 21)
 - ◇ 271.2-564.19-565.3 Christmas
 - ◇ 271.2-564.21-565.35 Epiphany (Baptism of the Lord)
 - ◇ 271.2-564.23-565.36 Presentation in the Temple (Feb. 2)
 - ◇ 271.2-564.25-565.4 The Great Fast (7 weeks before Easter)
 - ◇ 271.2-564.27-564.17 Annunciation (March 25)
 - ◇ 271.2-564.29-565.44 Palm Sunday
 - ◇ 271.2-564.31-565.72 Ascension
 - ◇ 271.2-564.33-565.73 Pentecost
 - ◇ 271.2-564.35-565.79PET St Peter and St Paul (June 29)
 - ◇ 271.2-564.37-565.79-442.47 Fast of the Apostles. NOTE: Begins on Monday following Pentecost and ends June 28
 - ◇ 271.2-564.39-565.8ASS-442.47 Fast of the Assumption (Aug. 1-14)
 - ◇ 271.2-564.41-565.9TRA Transfiguration (Aug. 6)
 - ◇ 271.2-564.43-565.8ASS Assumption (Aug. 15)
 - ◇ 271.2-726.1 Patriarchs
 - ◇ 271.2-726.2 Bishops
 - ◇ 271.2-726.3 Priests
 - ◇ 271.2-726.3-447.7 Celibate priests. 'Black' priests
 - ◇ 271.2-726.3-45 Married priests. 'White' priests
 - ◇ 271.2-726.4 Deacons
 - ◇ 271.2-726.52 Subdeacons
 - ◇ 271.2-726.54 Reader. NOTE: These are 2 distinct minor orders in the Orthodox Church
 - ◇ 271.2-74 Canons of the Orthodox Church
 - ◇ 271.2-788 Monastic orders
 - ◇ 271.2-788(234.425.21) Mount Athos. The Great Lavra
 - ◇ 271.2-788(477.41) Petchersky Lavra. The monastery of the Caves
 - ◇ 271.2-788-726.7 Elders. Startsy (Starets sing.)
- | | |
|---|---|
| <ul style="list-style-type: none"> 271.2-1 271.2-23 271.2-282.7-247 271.2-284 271.2-423.461 271.2-423.463 271.2-472-022.43 271.2-523.47 271.2-523.472 271.2-525.4 271.2-526.2 271.2-526.3 | <ul style="list-style-type: none"> The Orthodox 'Tradition'
<i>This is distinct from traditions, and incorporates the whole body of faith, the Bible, Fathers, Councils, etc. Scripture is part of Tradition and not distinct from it. Tradition comprises Scripture, Councils, Fathers, Liturgy, Canons, Icons</i> The Orthodox Bible
<i>This comprises the Septuagint, i.e. 39 books of the Old Testament, 10 Deutero-Canonical Books and New Testament</i> The Gospel Book
◇ 271.2-282.7-247-536.36 Prostration before the Gospel Book Doctrinal statements. Symbolical Books 'Chapel' of the Prothesis, Preparation Diakonikon The Longer Catechism Ikonostasis Holy Door. Royal Door ◇ 271.2-523.472-525.7 Curtain across the Holy Door Stoles. Epitrachilia Holy Table. Holy Throne Spoon (for distributing Communion) |
|---|---|

- 271.2-526.4 Chaplets. Prayer ropes
 271.2-526.5 Candles
 271.2-528-534.5 Litanies
 271.2-528-549 The Holy Liturgy
 ◇ 271.2-528-549-536.4-022.51 Little Entrance. Synaxis
 ◇ 271.2-528-549-536.4-022.56 Great Entrance
 ◇ 271.2-528-549.3 Epiclesis
 271.2-528.8 Byzantine rite
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 271.2-536.4 Sign of the Cross
 Class here use of two fingers (Russian practice) and of three fingers (Greek practice)
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 Unconsecrated remainder of the loaf used for Communion, distributed at the end of the Eucharist
 271.2-55 Sacraments
 ◇ 271.2-55-543.7/-544.3 Penance
 ◇ 271.2-55-549 Holy Communion
 ◇ 271.2-55-555 Holy Matrimony
 ◇ 271.2-55-557-536.25 Anointing of the sick
 ◇ 271.2-55-558.3 Baptism
 ◇ 271.2-55-558.4 Chrismation
 ◇ 271.2-55-558.5 Ordination
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 ◇ 271.2-55-022.51-557 Funeral rites
 ◇ 271.2-55-022.51-565.35-536.1-032.2 Blessing of water at Epiphany
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 ◇ 271.2-563.54 Sixth Hour
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 271.2-726.3-447.7 Celibate priests. 'Black' priests
 271.2-726.3-45 Married priests. 'White' priests
 271.2-726.4 Deacons
 271.2-726.52 Subdeacons
 Subdeacons and Reader are distinct minor orders in the Orthodox Church. See also 271.2-726.54
 271.2-726.54 Reader
 See also 271.2-726.52
 271.2-74 Canons of the Orthodox Church
 271.2-788 Monastic orders
 ◇ 271.2-788(234.425.21) Mount Athos. The Great Lavra
 ◇ 271.2-788(477.41) Pechersky Lavra. The monastery of the Caves
 ◇ 271.2-788-726.7 Elders. Startsy (Starets sing.)
 271.21 **Eastern Patriarchates**
 ◇ 271.21(560) Constantinople
 ◇ 271.21(560)-726.1 Ecumenical (Universal) Patriarch
 ◇ 271.21(569.1) Antioch
 ◇ 271.21(569.4) Jerusalem
 ◇ 271.21(620) Alexandria
 271.22 **National churches**
 Use common auxiliaries of place (Table 1e)
 271.222 Autocephalous churches
 Use common auxiliaries of place (Table 1e)
 271.222(437.3+437.6) Orthodox Church of the Czech Republic and Slovakia
 271.222(438) Polish Orthodox Church
 ◇ 271.222(438)-726.1 Archbishop, Metropolitan of Poland
 271.222(470+571) Russian Orthodox Church
 ◇ 271.222(470+571)-726.1 Patriarch of the Russian Church
 ◇ 271.222(470+571)-86 Churches in Exile. Russian Orthodox Diaspora. Including: 1. The Moscow Patriarchate - Parishes in emigration who have chosen to maintain direct links with Russia (c. 300,000/400,000 members); 2. The Russian Orthodox Church Outside Russia (ROCOR). Karlovtsky Synod - Synod of Bishops of the Russian Church in exile, originally in Yugoslavia, now New York. Synod in Exile. Synodal Church (c. 150,000 members); 3. The Russian Orthodox Diocese in Western Europe. 'Paris Jurisdiction' (c. 50,000 members); 4. The Russian Orthodox Greek Catholic Church of America. 'The Metropolia'. The Orthodox Church of America (OCA) (1,000,000 members)
 ◇ 271.222(470+571)"1700/1917"-732.3 Spiritual College. Holy Synod
 ◇ 271.222(470+571)-86"16" Sectarians. Old Believers. Raskolniki (dissenters over ritual, not doctrine)

- ◇ 271.222(470+571)-86POP Popovtsky.
 NOTE: This sect has priests and bishops
 ◇ 271.222(470+571)-86BEZ Bezpopovtsky.
 NOTE: This sect has no priests. [NB both
 still exist]
 ◇ 271.222(470+571)-9"1917/1925" Years of
 struggle to preserve the liberty of the church
 ◇ 271.222(470+571)-9"1918.2.1"-546.4
 Anathema of St Tikhon
 ◇ 271.222(470+571)-9-672 The 'Living
 Church'
 ◇ 271.222(470+571)-9"1925/1943" Years of
 Metropolitan Sergius attempting *modus
 vivendi*
 ◇ 271.222(470+571)-9"1943/1959" Years of
 revival of church life under Stalin.
 Restoration of Patriarchate
 ◇ 271.222(470+571)-9"1959/1964" Years of
 persecution under Krushchev
 ◇ 271.222(470+571)-9"1964/1988" Revival
 years when dissident movement arises and
 is crushed
 ◇ 271.222(470+571)-9"1976" Christian
 Committee for the Defence of Believers'
 Rights
 ◇ 271.222(470-25) Patriarch of Moscow
 Ukraine Autocephalous Orthodox Church
 [NB Split into 2 groups, neither recognized
 by any other Orthodox Church]
 ◇ 271.222(477.41)-726.1 Patriarch of Kiev.
 From 1453 subordinate to the Patriarch of
 Moscow, formerly superior
 271.222(477) Georgian Orthodox Church
 ◇ 271.222(479.22)-726.1 Catholikos Patriarch
 271.222(495) Greek Orthodox Church
 ◇ 271.222(495)-726.1 Archbishop,
 Metropolitan of Greece
 271.222(496.5) Albanian Orthodox Church
 ◇ 271.222(496.5)-726.1 Archbishop,
 Metropolitan of Albania
 271.222(497.11) Serbian Orthodox Church
 ◇ 271.222(497.11)-726.1 Patriarch of the
 Serbian Church
 271.222(497.2) Bulgarian Orthodox Church
 ◇ 271.222(497.2)-726.1 Patriarch of the
 Bulgarian Church
 271.222(498) Romanian Orthodox Church
 ◇ 271.222(498)-726.1 Patriarch of the
 Romanian Church
 271.222(564.3) Cyprus Orthodox Church
 ◇ 271.222(564.3)-726.1 Archbishop,
 Metropolitan of Cyprus
 271.224 Autonomous churches
 Use common auxiliaries of place (Table 1e)
 ◇ 271.224(480) Orthodox Church of Finland
 ◇ 271.224(510) Orthodox Church of China
 ◇ 271.224(520) Orthodox Church of Japan
 ◇ 271.224(620SINAI) Orthodox Church of
 Sinai
 271.3 Separated churches
 271.4 Uniate churches
 271.6 The Church of the East (Nestorian)
 271.7 Monophysite rite
 271.72 Armenian. Alexandrian
 271.722 Coptic church
 271.723 Ethiopian church. Abyssinian church
 271.74 Antiochene church
 271.742 Syrian church
 271.744 Maronite church

272/279

WESTERN CHURCH

272/273

CATHOLIC CHURCH

272

ROMAN CATHOLIC CHURCH

272-282
272-282.7
272-549

Roman Catholic liturgical works

Missals

The Mass

◇ 272-549-282 Liturgy of the Mass

Influence exerted by the Roman Catholic
Church

◇ 272-662:3 Influence of the Roman Catholic
Church on society. Including: social
movements. Catholic Action

272-726.6-427

Informal promotion of the Roman Catholic faith
by lay persons

Class here the universal priesthood of the
faithful, personal prelature and movements
such as *Opus Dei*

272-732.2

The Papacy

◇ 272-732.2:321 Political, temporal power of
the Papacy

◇ 272-732.2-284 Doctrine of the Papacy

◇ 272-732.2-72 Structure of the Papacy

◇ 272-732.2-725 Persons in the Papal
household, administration

◇ 272-732.2-725.2 Cardinals. Including:
College of Cardinals. Conclave

◇ 272-732.2-725.4 Legates. Nuncios.
Apostolic delegates

◇ 272-732.2-725.6 Pronotaries apostolic

272-74

Roman Catholic canon law

272-742-266

Codes of Law. □ Codex Iuris Canonici (CIC)

272-745

Religious courts. Papal courts. Curia

272-745.2

Roman Congregations

272-745.4

Tribunals

272-745.5

Curial offices. □ Apostolic chancellery.

272-745.6

Apostolic camera

Commissions

273

NON-ROMAN CATHOLIC EPISCOPAL CHURCHES

273.2

Old Catholics

273.4

Anglican Church

273.4(4/9)

Anglican church in various countries

◇ 273.4(94) Anglican church in Australia

273.6

Aglipayans. Philippine Independent Church

274/278

PROTESTANT CHURCHES

274

PROTESTANTISM GENERALLY. PROTESTANTS. DISSENTERS. PURITANS

274.3

Minor movements in early Protestantism

274.36

Pietism

274.37

Labadists

274.38

Arminians. Remonstrants

274.5

Lutheran churches. Lutherans

275

RE-FORMED CHURCHES

275.2

Moravians

275.4

Calvinists

275.5

Huguenots

275.6

Presbyterians

276

ANABAPTISTS

276.2	Mennonite churches. Mennonites	284.5	Ahmadiya
277	FREE CHURCHES.		<i>Regarded as heretical by mainstream Islam</i>
	NON-CONFORMISTS	284.55	Lahore Group (Ahmadiya
277.2	Congregationalists		Anjuman-i-Isha'at-i-Islam)
277.4	Baptists	284.7	Druze
277.6	Methodists. Methodist churches	285	BABI-BAHA'I
278	OTHER PROTESTANT CHURCHES	285.5	Non-Bahai Babis
278.15	Unitarians	285.52	Azali Babis
278.17	Quakers. Society of Friends	286	BAHA'I
278.19	Salvation Army	286.2	BUPC (Baha'is under the provisions of
279	OTHER CHRISTIAN MOVEMENTS AND		the Covenant)
	CHURCHES	286.4	Orthodox Baha'is
	⇒ 27-78	286.6	Faith of God
279.12	Modern Christian movements	286.7	Orthodox Baha'i faith under the
	<i>These are treated separately from Christian</i>		Regency
	<i>groups per se, since these movements</i>	286.8	Charles Mason Remey Society
	<i>cross boundaries, and some may include</i>	29	MODERN SPIRITUAL MOVEMENTS
	<i>traditional church members. Some terms</i>		<i>There are a large number of these,</i>
	<i>may also be applicable to other faith groups</i>		<i>predominantly 20th century (most of the</i>
	<i>(e.g. fundamentalist)</i>		<i>preceding array are 19th century</i>
279.122	Born-again		<i>phenomena). They often do not have any</i>
279.123	Evangelical		<i>particular faith associations, although some</i>
279.124	Fundamentalist		<i>have grown out of a particular cultural</i>
279.125	Pentecostal		<i>context and may have features of the</i>
279.127	Charismatic		<i>dominant religion of the culture (e.g. New</i>
279.13	Adventists		<i>Age and Paganism). Some have very large</i>
279.14	Seventh Day Adventists		<i>numbers of adherents, perhaps because</i>
279.15	Pentecostal churches		<i>they are 'official' state religions, but even so</i>
	⇒ 279		<i>are not widely recognised outside their own</i>
279.153	Assemblies of God		<i>territories</i>
279.154	Church of God in Christ	299	PHILOSOPHICAL SYSTEMS
279.156	Universal Church of the Kingdom of God		<i>In general discussions of atheism, etc.</i>
279.16	Latter-day Saints. Mormons		<i>should be placed in philosophy, but</i>
279.167	RLDS (Reorganized Latter Day Saints)		<i>provision is made here, since they usually</i>
279.17	Jehovah's Witnesses		<i>deal with the same subject matter (e.g.</i>
279.19	Apostolic churches		<i>existence of God, etc.)</i>
279.21	Christian Science	299.2	Atheism
279.22	New Thought (derived from Christian	299.252	Shinto derived faiths
	Science)	299.252.5	Tenrikyo
279.222	Unity Church	299.252.6	Omoto-kyo
279.224	Religious Science	299.3	Agnosticism
279.226	Divine Science	299.4	Humanism
279.228	Restoration Movement (Christian Churches	299.42	P(erfect) L(ife) Kyodan
	and Churches of Christ)		<i>Japanese movement of Buddhist origin</i>
279.23	Free Catholics	299.5	Absence of religion. Secularism
279.25	Christengemeinschaft	299.572	Neo-Paganism
279.27	Swedenborgians	299.6	Irreligion
279.28	African Indigenous sects	299.7	Movements with Christian elements
279.285	African Initiated/Indigenous/Independent	299.72	Anthroposophical movement
	Churches (AICs)	299.725	Christian Community
279.287	Kimbanguist Church	299.73	Sea of Faith
279.99	Others. Deutsche Christen. Doukhobors	299.92	Cao Dai
	<i>Specify by alphabetic extension A/Z (Table</i>		<i>Syncretic and nationalist movement in</i>
	<i>1h)</i>		<i>Vietnam</i>
28	ISLAM	299.93	Scientology
282	SUNNI. SUNNITE ISLAM	299.935	Dianetics
284	SHI'A. SHI'ITE ISLAM	299.94	New Age

3 SOCIAL SCIENCES. STATISTICS. POLITICS. ECONOMICS. TRADE. LAW. GOVERNMENT. MILITARY AFFAIRS. WELFARE. INSURANCE. EDUCATION. FOLKLORE

3.07	ADMINISTRATIVE ARRANGEMENTS. AUTHORITIES. DEPARTMENTS <i>3.07 ≡ 35.07</i>	3.08	PERSONNEL OF AUTHORITIES. OFFICIALS. STAFF <i>3.08 ≡ 35.08</i>
3.071	Foundation. Development	3.083	Duties, obligations of officials
3.072	Relations with other organizations	3.085	Competence, powers of officials. Responsibilities. Official actions
3.076	Competence, sphere of authority. Duties. Powers	3.086	Responsibility, answerability of officials

30 THEORIES, METHODOLOGY AND METHODS IN SOCIAL SCIENCES IN GENERAL. SOCIOGRAPHY

303	METHODS OF THE SOCIAL SCIENCES	303.043	Unstructured
303.01	Theoretical aspects	303.05	Degree of systematizing
303.02	General features of research	303.052	Systematic
303.022	Qualitative	303.053	Unsystematic
303.023	Quantitative. Degree of quantification	303.06	Degree of automation
303.023.2	Categorical. Nominal	303.062	Conventional
303.023.22	Dichotomous	303.063	Mechanized
303.023.23	Cardinal	303.064	Electronic. Computer-aided
303.023.3	Ordinal	303.09	Special aspects and problems of social science models
303.023.4	Interval	303.092	Assumptions in empirical models
303.023.5	Ratio	303.092.2	Linear
303.024	Relations of quantities	303.092.3	Nonlinear (exponential, quadratic, log-linear)
303.024.2	Absolute quantitative values	303.092.4	Deterministic
303.024.3	Relative quantitative values. Proportions. Standardized values. Indices	303.092.5	Stochastic. Probabilistic. □ Discrete (binominal, Poisson-distributed, hypergeometric). Continuous (normal-distributed, t-distributed, F-distributed, chi-square-distributed). Bayesian
303.025	Empirical	303.092.6	With ML-estimators
303.026	Nonempirical	303.092.7	With other estimators
303.028	Open. Overt. Undisguised	303.092.8	Parametric
303.029	Covert. Disguised	303.092.9	Nonparametric
303.03	Degree of complexity and direction of research of the method	303.093	Kind of system in empirical models
303.031	Dimensionality	303.093.2	Closed (without external effects)
303.031.1	Unidimensional	303.093.3	Open (with external effects)
303.031.12	Quasi-unidimensional	303.093.6	Static
303.031.2	Bidimensional. Two-dimensional	303.093.7	Dynamic
303.031.3	Three-dimensional	303.094	Building empirical models
303.031.4	Multidimensional	303.094.2	Specification
303.032	Variability	303.094.3	Aggregation
303.032.1	Univariate	303.094.4	Identification
303.032.2	Bivariate	303.094.5	Estimation
303.032.4	Multivariate	303.094.6	Forecasting
303.035	Orientation	303.094.7	Simulation
303.035.2	Direction of analysis	303.094.8	Linkage
303.035.22	R-analysis type (oriented on variables)	303.1	General methodology of social research ⇒ 303.211 ⇒ 303.832
303.035.23	Q-analysis type (oriented on units)	303.2	Measurement
303.035.24	Three-mode. Three-way (simultaneous)	303.21	Theory of measurement
303.035.3	Direction of testing	303.211	Operational definition. Indicators
303.035.31	One-sided. Unilateral. Asymmetric. One-tailed	303.212	Measurement level (nominal, ordinal, interval, ratio)
303.035.32	Two-sided. Bilateral. Symmetric. Two-tailed	303.213	Discreteness and continuity of variables
303.035.33	Multi-sided. Multilateral	303.214	Measuring criteria ⇒ 303.714
303.035.4	Direction of factors		
303.035.42	Orthogonal		
303.035.43	Oblique		
303.039	Kind of control of the variables under study		
303.039.2	Partial (without indirect influences)		
303.039.3	Multiple (with multiple influences)		
303.039.4	Crossed. Multiway (with multiple effects)		
303.039.5	Canonical		
303.04	Degree of structuring		
303.042	Structured		

303.214.2	Reliability (retest, split half procedure, parallel test, inter-item consistency)	303.447.2	Nonexperimental. □ Correlation studies
303.214.3	Validity (structural, predictive, observational validity)	303.447.3	Experimental studies. □ Natural experiment.
303.214.4	Selectivity	303.448	Artificial experiment. Projective experiment
303.214.5	Degree of complexity		Method testing investigations
303.214.6	Homogeneity		⇒ 303.714
303.22	Measuring methods	303.5	Selection. Sampling
303.222	Free estimation method		◇ 303.5.01 Sampling theory
303.223	Index formation	303.51	Fixing investigation population. Universe of investigation
	⇒ 303.71	303.52	Selection methods
303.224	Scaling. Scaling models. □ Cumulative estimation. Discrimination technique. Scalogram analysis. Polarity profile	303.522	Sample size
		303.522.2	Total survey
		303.522.3	Representative sample
		303.522.4	Nonrepresentative sample
		303.523	Sample collecting
		303.523.2	Probability samples. Random samples
		303.523.3	Conscious samples. Intentional samples
303.4	Kinds of investigation. Research strategies. Investigation design	303.53	Sampling technique
303.42	Investigation methods	303.532	Sampling framework. Sampling basis. □ List sampling. Area sampling. Random walk
303.422	Biographies. Biograms		Selection of the investigating unit. Survey unit
	⇒ 303.686		
	⇒ 929		
303.423	Small-group research. □ Sociometric, genealogical investigations	303.533	
303.424	Organizational research. Institutional research	303.54	Projection methods
303.425	Survey. □ Opinion polling. Market research	303.55	Estimation of sample error
303.43	Research strategies	303.6	Survey methods
303.432	With regard to database. □ Primary, secondary investigations. Replication study	303.62	Inquiry methods. Interview. Questionnaire
303.433	With regard to scope	303.621	Kinds of inquiry
303.433.2	Individual. Case study. Monographic study	303.621.2	Degree of standardization
303.433.3	Multi-case investigation	303.621.3	Form of inquiry
303.436	Origin of data	303.621.32	Oral interview. □ Personal interview.
303.436.2	Archives. Libraries. Authorities		Telephone interview
303.436.3	Field	303.621.33	Written interview. □ Postal inquiry.
303.436.4	Laboratory		Questionnaire schedule
303.44	Investigation design	303.621.34	Answer-sheet interview
303.442	Qualitative investigations	303.621.35	Group interview. Group discussion methods
	⇒ 303.022	303.621.36	Special forms of questioning. □ In-depth interview. Motivation interview
303.442.2	Exploratory investigations	303.622	Questionnaire design. □ Open questions.
303.442.22	Descriptive investigations. Exploration of facts		Options, multiple choice questions
	⇒ 303.721	303.623	Interview strategy. □ Hard, soft interview
	⇒ 303.722	303.624	Interviewer
303.442.23	Field research. Analysis of social environment, milieu. 'Natural sociology'	303.625	Interviewee
	⇒ 303.433.2	303.626	Interview context. □ Timing. Location
	⇒ 303.686	303.627	Recording, writing up of interview. Accessories at interview
	⇒ 316.277		Sociology of the interview
303.442.3	Analytical investigations	303.628	
	⇒ 164	303.63	Observation methods
	⇒ 303.725.2	303.632	Kind of observation
303.442.4	Interpretative methods. Hermeneutic approach. Reconstructive methods. Investigation of meaning	303.632.2	Self-observation
	⇒ 316.257	303.632.3	Independent observation
	⇒ 316.277	303.632.4	Environmental observation
	⇒ 930.2		⇒ 303.688
303.443	Attendant, concomitant research. Scientific advice	303.633	Observers. □ Selection. Training
303.444	Prognostic investigations	303.634	Observation scheme
303.445	Nomologic studies. Parameter estimation	303.635	Recording the observation. □ Observation report. Technical accessories
303.446	Comparative investigations	303.64	Content analysis
303.446.2	Cultural comparison. Cross-sectional comparison. □ Intercultural comparison. Intracultural comparison	303.642	Kinds of content analysis
		303.643	Medium of content analysis
		303.643.2	Kind of medium. □ Written material. Audio material. Visual material
		303.643.3	Form of medium
303.446.3	Time comparison. □ Panel study, analysis. Trend study. Time series study. Cohort study, analysis		◇ 303.643.3:070 Newspapers as a medium
			◇ 303.643.3(044.4) Business letters, office correspondence as a medium
303.446.4	Historical comparison. Historiography	303.644	Unit of analysis
303.447	Hypothesis-testing investigations	303.645	Content analysis methods
	⇒ 303.714		◇ 303.645.063 Computerized content analysis

- 303.68 **Special survey methods**
 303.682 Sociometry
 303.683 Genealogical, ethnogenealogical methods
 303.684 Elite-identifying techniques
 303.685 Time-budget techniques
 303.686 Biographical methods. Oral history methods
 ⇒ 303.422
 ⇒ 303.621.3
 ⇒ 929
 ⇒ 930
- 303.687 Projective methods and techniques. □ Word association. Rorschach
 303.688 Nonreactive methods. Unobtrusive measures
- 303.7 Analysis procedures**
 ◇ 303.7.032.1 Univariate
 ◇ 303.7.032.2 Bivariate
 ◇ 303.7.032.4 Multivariate
- 303.71 Statistical measures. Test statistics. Representation of analysis outcome**
 303.711 Matrix of the raw data. Frequency distribution. Tables
 303.712 Descriptive statistical measures. E.g. distribution measures, measures of association
 ⇒ 303.721
 303.714 Measures and procedures of statistical tests
 ⇒ 303.214
 ⇒ 303.447
 303.715 Description or picture of the degree of equality. Representation of equations
 303.716 Graphic pictures and solutions
 303.717 Algorithms and programs
- 303.72 Kinds of analysis**
 303.721 Descriptive analysis. □ Tabular analysis
 303.722 Dimensional analysis. Techniques of multivariate reduction
 303.722.2 Factor analysis
 303.722.3 Multidimensional scaling. Smallest space analysis
 303.722.4 Cluster analysis. Numerical taxonomy
 303.722.8 Pattern recognition
 303.723 Correlation analysis
 ◇ 303.723.032.2 Bivariate (simple)
 ◇ 303.723.039.2 Partial
 ◇ 303.723.039.3 Multiple
 ◇ 303.723.039.5 Canonical
 303.724 Causal analysis. □ Variance analysis. Regression methods, path analysis
 303.725 Model analysis
 303.725.2 Theoretical analysis of models
 303.725.3 Empirical analysis of models
 303.725.33 Application of methods of building and solving econometric models
 303.725.34 Application of input-output analysis
 ◇ 303.725.34(083.41) Input-output tables
 ⇒ 330.44
 303.725.35 Application of operations research methods
Details by :519.8
 303.725.36 Application of system theory, control theory and cybernetics
Details by :519.71
 303.725.37 Application of information theory
Details by :519.72
 303.725.38 Impact analysis and application of simulation
 ⇒ 303.733.4
- 303.73 Levels of analysis**
 303.732 System levels
 303.732.2 Individual data analysis. Aggregate data analysis
 303.732.3 Multilevel analysis. Context analysis. Relation analysis
 303.732.4 System analysis
- 303.733 Time level
 303.733.2 Cross-sectional analysis
 303.733.3 Longitudinal analysis. Time series analysis. Dynamic analysis. □ Trend analysis. Markov chain analysis. Stochastic processes. Autoregressive models. Distributed lag models. Spectral analysis
 ⇒ 303.446.3
 303.733.4 Heuristic prognostic analysis. Planning games. Delphi method
 ⇒ 303.725.38
- 303.8 Research organization. Research procedure**
 ⇒ 001.891
- 303.82 Research organization**
 ◇ 303.82-057.177 Directors
 ◇ 303.82-057.4 Researchers
 ◇ 303.82-057.54 Consultants
 303.823 Organization of investigation
 ◇ 303.823-057.17 Managers, administrators of investigation
 303.823.2 Research planning
 303.823.4 Division of work. □ Individual research. Teamwork. Cooperation with external bodies
 303.824 Research report
- 303.83 Research procedure**
 303.832 Hypotheses
 303.832.2 Assembling of thesis. □ Evaluation of literature. Questioning of experts
 303.832.3 Evaluation of thesis
 303.833 Carrying out of investigation
 303.833.2 Operationalizing
 ⇒ 303.211
 303.833.3 Survey
 303.833.4 Processing. Coding
 303.833.5 Analysis. Interpretation
 303.833.6 Scientific discussion and criticism
 303.833.7 Phases of the investigation
 303.833.8 Error control. Error checking
- 304 SOCIAL QUESTIONS. SOCIAL PRACTICE. CULTURAL PRACTICE. WAY OF LIFE (LEBENSWEISE)**
For special questions use the relevant classes, e.g. 001.18, 308, 316, 33, 36
- 304.2 Social and cultural questions**
- 304.3 Way of life**
Use this class when all societal, social and cultural aspects of human activity in all fields of social and personal life are referred to. For way of life in the meaning of (outward) style of life use 316.728
- 304.4 General aspects of social and cultural policy and practice**
- 304.42 Theories regarding social and cultural policy**
- 304.44 Application of social science knowledge. Social practice. Cultural practice**
 ⇒ 316.42
 ⇒ 36
 304.442 Social planning. Cultural planning
 304.444 Possibilities for the conscious changing of the social and cultural realities. Social and cultural feasibility (Machbarkeit). Social and cultural desirability
- 304.5 Social science aspects of development in societies and between societies**
For special questions use the relevant classes, e.g. 316.43, 339.96

- 304.9 Critique and projection of the situation in society. Plans for the improvement of society. Social utopias**
- 305 GENDER STUDIES.** □ Sex role. Gender. Persons from an interdisciplinary viewpoint
Class here comprehensive works. Works dealing with a specific subject, e.g. Education, should be classed with the subject
- ◇ 305-055.1 Men's studies
 - ◇ 305-055.2 Women's studies. Women and society
- ⇒ 141.72
 ⇒ 347.156

- 305.5 Gender reassignment**
 ⇒ 616-089.844

- 308 SOCIOGRAPHY. DESCRIPTIVE STUDIES OF SOCIETY (BOTH QUALITATIVE AND QUANTITATIVE)**
Details by auxiliaries of place (1/9) (Table 1e), time "..." (Table 1g) etc.
- ◇ 308(430)"17" The social situation of Germany in the 18th century
- ⇒ 338

31 DEMOGRAPHY. SOCIOLOGY. STATISTICS

- | | |
|--|---|
| 311 STATISTICS AS A SCIENCE. STATISTICAL THEORY
⇒ 519.22 | 311.219.1 Currency. Topicality. Up-to-dateness. Recency of research |
| 311.1 Fundamentals, bases of statistics | 311.219.2 Preservation, storage of statistical documents |
| 311.11 Statistical measures. Unit. Concept. Criterion | 311.3 General organization of statistics. Official statistics |
| 311.12 Statistical grouping | ◇ 311.3:331.56 Organization of unemployment statistics |
| 311.13 Statistical series. Classification.
□ Qualitative, quantitative classification | ◇ 311.3(492) Organization of official statistics in the Netherlands |
| 311.14 Statistical mean. Averages. Weights. Ratio | 311.31 Kinds of official statistics |
| 311.141 Index numbers | 311.311 International statistics. Figures from international agencies |
| 311.15 Distribution. □ Standard deviation. Marginal values, limits. Quartile, percentile | 311.312 National statistics. Figures from national government agencies |
| 311.16 Correlation. Coefficient of correlation. Coefficient of variation | 311.313 Regional statistics. Figures from regional, provincial authorities, counties (UK), states (federal) |
| 311.17 Statistical analysis of complex movement phenomena | 311.314 Local statistics. Figures from local authorities, municipalities |
| 311.171 Fluctuations: short, medium term | 311.4 Private statistics |
| 311.172 Trend: long term. Fundamental direction of a movement | 311.41 Statistics of associations, unions (trade, professional etc.) |
| 311.174 Seasonal fluctuations. Periodic fluctuations | 311.42 Statistics of (single) organizations. Management, internal statistics |
| 311.175 Structural changes | 311.43 Statistics of religious organizations, churches |
| 311.2 Research technique. Preparation. Tabulation | 311.43 Statistics of religious organizations, churches |
| 311.21 Statistical research | 314/316 SOCIETY |
| 311.211 Plan. programme, preparation of survey. Object, scope, location, duration. □ Draft plan of enumeration. Error prediction. Cost estimates | 314 DEMOGRAPHY. POPULATION STUDIES |
| 311.212 Execution of survey. □ Drafting, printing, distribution and collection of questionnaires. Individual and collective forms. Survey processes, organizations, agents (enumerators). Centralization. Decentralization | 314.01 Subject, scope and aims of demography |
| 311.213 Kinds of research | 314.015 Demographic programmes |
| 311.213.1 Collection of primary statistical material | 314.018 Demometrics (measurement techniques) |
| 311.213.2 Representative methods. □ Sampling. Spot-checking | 314.02 Sources of demographic data, including vital statistics |
| 311.213.3 Inquiries. Evaluation | 314.04 Types and features of populations. Demographic phenomena. Impact on demographic development |
| 311.213.4 Single (one-off) and continuous investigations | 314.044 Voluntary action or change |
| 311.213.5 Collection of secondary statistical material | 314.045 Compulsory, forced action or change |
| 311.213.6 Monographs. Other methods not strictly statistical | 314.07 Demographic institutions |
| 311.214 Verification of material. Reliability, accuracy. Errors: systematic, accidental. Sources of error | <i>Auxiliaries 3.07... (as at 35.07...)</i> |
| 311.215 Formulæ for accuracy; for correction | 314.08 Demographic personnel |
| 311.217 Tabulation. Compilation of tables | <i>Auxiliaries 3.08... (as at 35.08...)</i> |
| 311.218 Graphic presentation. □ Diagrams. Cartograms. Histograms. Stereograms | 314.1 Population |
| | 314.112 Population theories |

314.113	Population policy	314.117.3	Households
◇ 314.113:323.12-025.44	Anti-racist population policy	314.117.31	Types of household
◇ 314.113:323.12	Racist population policy. Including: Ethnic cleansing	◇ 314.117.31:316.812	Households comprising family units
◇ 314.113:323.14	Racial population policy	◇ 314.117.31-05-022.214	Households comprising one person
◇ 314.113.015	Programmes of population policy	314.117.312	Alternative forms of household. □ Communes
⇒ 323.1		314.118	Life expectancy
314.114	Factors of demographic phenomena and their effects on social and economic development	314.12	Births
◇ 314.114:33	Economic factors and their effects		<i>For population theories (previously at this number) see 314.112</i>
314.116	Population change, growth, decline	314.14	Death. Morbidity
◇ 314.116(083.41)	Statistics of population growth, decline		<i>For branches of demography (previously at this number) see 314 and use colon combination</i>
◇ 314.116-022.252	Depopulation	⇒ 393	
◇ 314.116-022.257	Overpopulation		
◇ 314.116-026.16	Stable population	314.15	Migration. Movement of populations
◇ 314.116-026.241	Increase in population	◇ 314.15:343.343.62	Illegal immigration
◇ 314.116-026.243	Decrease in population	◇ 314.15"32"	Seasonal migration
314.116.1	Population explosion	◇ 314.15-022.324	Permanent migration
314.116.3	Population density	◇ 314.15-022.326	Temporary migration
314.117	Composition and distribution of the population	◇ 314.15-026.48	In-migration
	<i>Details by colon combination, (1/9) Table 1e, (=...) Table 1f, -05... Table 1k-05</i>	◇ 314.15-026.49	Out-migration
◇ 314.117-054.51	Autochthonous inhabitants (peoples whose residence is also their place of birth)	◇ 314.15.044	Voluntary migration
◇ 314.117-054.52	Allochthonous inhabitants (people whose place of residence is not their place of birth)	◇ 314.15.045	Forced migration
◇ 314.117-054.72-027.583	Illegal immigrants	314.151	Population transfer
◇ 314.117-055.1	Preponderance of males	314.151.1	Expatriation
◇ 314.117-055.2	Preponderance of females	314.151.3	Types of migrant
◇ 314.117-058.51	Wanderers. Vagrants		<i>Details by Table 1k-05</i>
314.117.1	Composition and distribution according to birthplace or domicile	◇ 314.151.3-054.72	Voluntary migrants. Including: Immigrants. Emigrants
314.117.11	Composition and distribution according to birthplace	◇ 314.151.3-054.73	Refugees. Evacuees. Displaced persons
314.117.13	Composition and distribution according to domicile (de jure)	◇ 314.151.3-054.74	Compulsory exiles. Banned persons (proscribed from entry). Expellees. Deportees
314.117.15	Composition and distribution according to residence (de facto)	◇ 314.151.3-054.75	Repatriates
314.117.17	Composition and distribution according to birthplace and residence	◇ 314.151.3-054.78	Stateless persons. Apatriates
		◇ 314.151.3-054.7	Expatriates. Exiles. Repatriates. Migrants. Stateless persons
		314.151.5	Repatriation
		314.151.6	Transit migration
		314.151.7	Re-emigration

316 SOCIOLOGY

For special sociologies not mentioned in the subdivisions or examples use :...

◇ 316:61 Medical sociology

⇒ 355.01

⇒ 37.013.78

⇒ 591.55

316.012	Macrosociology
316.013	Mesosociology
316.014	Microsociology
316.022	Majorities. Minorities
316.022.2	Majorities
316.022.4	Minorities
316.023	Groups according to size
316.023.2	Dyad. Triad
316.023.4	Small
316.023.5	Medium
316.023.6	Large
316.032	Primary
316.034	Secondary

316.1 Object and scope of sociology

316.2

Sociological points of view and trends

The names of individual sociologists in the following subdivisions are intended only to characterise more closely the type of tendency mentioned

Denote individual sociologists and their schools of thought by 316.2A/Z (alphabetic extension - Table 1h)

316.22

Mechanistic trend

316.222

Sociological mechanists (S.C.Haret, A.J.Lotka)

316.223

Sociological physicists (Carey)

316.224

Sociological energeticists (Solvay, Bekhterev, Ostwald, Winiarski)

316.225

Mathematical sociologists (Pareto, Carli)

- 316.23 Synthetic and geographical trends**
 316.232 Synthetico-geographical trend (Le Play)
 316.233 Geographical trend (Jevons, Moore, Huntington)
- 316.24 Biological trend**
 316.242 Organicists (Lilienfeld, Schäffle, Worms, Novikov, Fouillée)
 316.243 Race theory trend (Gobineau, Lapouge, Chamberlain)
 316.244 Darwinian trend. Evolutionary theories of sociology. Historical sociology (Morgan, Bachofen, Spencer, Aron, Ginsburg)
 ⇒ 316.423.2
 316.245 Sociobiological trend. Demographic trend (Coste)
 316.246 Biopsychological trend. Instinctivists
- 316.25 Sociological trends**
 316.252 Neopositivists (Roberty)
 316.253 Pure sociology (Durkheim)
 316.254 Theory of group conflict (Gumpłowicz, Oppenheimer)
 316.255 Formal sociology. Morphological trend (Simmel, G. Richard, Von Wiese, Park, Burgess)
 316.256 Institutional trend in sociology (Gehlen, Schelsky)
 316.257 Frankfurt School. Critical theory. Hermeneutic trend (Horkheimer, Adorno, Marcuse)
 316.258 Functionalism. Structural-functional sociology. Functional-structural sociology (Merton, Parsons, Luhmann)
 316.259 Cybernetics trends, general systems theories, and game theories in sociology (K.W. Deutsch, Rappoport)
- 316.26 Marxist-Leninist sociology (Marx, Engels, Lenin)**
- 316.27 Psychological trend**
 316.272 Sociology according to certain psychological factors (Tarde, Le Bon, Sighele)
 316.273 Ethnic psychology (Lazarus, Steinthal, W. Wundt)
 316.274 Introspectivists (Müller-Freienfels)
 316.275 Behaviourist and experimental sociology. Learning theoretical trend (Skinner, Cooley, Dollard, Bandura, Opp)
 316.276 Cognitive theories (Heider, Newcomb, Festinger, H.H. Kelley, Jones, Davis).
 □ Balance theory. Consistency theory. Dissonance theory. Attribution theory. Inference theory
 316.277 Phenomenological trend. Gestalt-theoretical trend (Mauß, G.H. Mead)
- 316.28 Trends based on communication theory**
 316.282 Action theory (Max Weber)
 316.283 Role theory. Reference group theory. Exchange theory (Homans, Blau, J.W. Thibaut, H.H. Kelley)
 316.285 Conflict theory. Theories of social inequality (Coser, Dahrendorf, S.N. Eisenstadt)
 316.286 Reflexive sociology. Paradigms based on communication theory (Habermas, Eisermann, Bendix, I. Eisenstadt)
- 316.3 Social structure. Society as a social system**
 Denote persons and personal characteristics by -05... (Table 1k)
- 316.32 Global societies**
The following classification of global societies enumerates abstract, theoretical types of society. For classing concrete societies use 316.32 and the common auxiliaries of place (1/9) (Table 1e) and time "..." (Table 1g).
To class a particular society as a specimen of a special type of society, use the notation for that type and the common auxiliaries of place and time
 ◇ 316.32(493) Belgian society
 ◇ 316.323.72(439) Hungary as a socialist society
 ◇ 316.324.7(439) Hungary as an industrial society
 ◇ 316.323.82(675)"18" Nineteenth century Zaire as a colonial society
 316.322 General typologies of global societies
 316.323 Socioeconomic types of global society. Socioeconomic formations
 316.323.2 Kinship societies. Primitive society
 316.323.3 Societies of Antiquity. Slave-owning societies
 316.323.4 Asiatic despotisms
 316.323.5 Feudal societies
 316.323.6 Capitalist societies
 316.323.62 Early capitalist societies
 316.323.63 High-capitalist societies
 316.323.64 Late capitalist societies
 316.323.65 Welfare State
 316.323.7 Socialist and communist societies
 316.323.72 Socialist societies
 316.323.73 Communist societies. Classless societies
 316.323.8 Colonial and neocolonial societies
 316.323.82 Colonial societies
 316.323.83 Neocolonial societies. Postcolonial societies
 316.324 Global societies according to stages of economic development
 316.324.2 Preagricultural societies
 316.324.22 Hunting and gathering societies
 316.324.23 Horticultural societies
 316.324.3 Nomadic societies. Herding societies
 316.324.4 Fishing societies. Maritime societies
 316.324.5/7 Agrar-industrial societies
 316.324.5 Agricultural, agrarian societies
 316.324.6 Preindustrial societies
 316.324.7 Industrial societies
 316.324.8 Postindustrial societies
 316.325 Societies according to specific criteria. E.g. theocracies, mass societies
- 316.33 Basic elements and subsystems of global societies as sociological categories**
 Base and superstructure
 Subsystems and institutions
 Sociology of economic subsystems and institutions. Economic structure. Sociology of economics
 ◇ 316.334.2:338.439 Nutritional sociology
 ◇ 316.334.2:339.13 Sociology of the market
 Sociology of work
 Industrial sociology. Sociology of the enterprise
 Sociology of political subsystems and institutions. Political sociology
 ◇ 316.334.3:321 Sociology of the state
 ◇ 316.334.3:324 Sociology of elections
 ◇ 316.334.3:329 Sociology of political parties
 Sociology of legal subsystems and institutions. Sociology of law
 ⇒ 316.354
 ⇒ 316.75

- 316.334.5 Sociology of ecological subsystems and institutions
⇒ 502.22
⇒ 504.61
- 316.334.52 Regional and community studies
- 316.334.54 Housing. Neighbourhood. Sociology of housing and dwelling
- 316.334.55/.56 Rurban sociology (country/town sociology). Community sociology
- 316.334.55 Rural communities. Rural sociology
- 316.334.56 Urban communities. Urban sociology
- 316.34 Social stratification. Social differentiation**
⇒ 316.443
- 316.342 Types of social differentiation. Types of social stratification
- 316.342.2 Social classes
- 316.342.3 Castes
- 316.342.4 Standings. Estates
- 316.342.5 Strata. Social levels
- 316.342.6 Socioeconomic groups
- 316.342.7 Pillarization ('Verzuiling'). Vertical categorization according to religion or ideology
- 316.343 Social stratification and class structure of different socioeconomic formations
◇ 316.343:27-725-05:321.17 Clergy in feudal societies
◇ 316.343:321.17 Social differentiation of feudal society
◇ 316.343:35-051 Civil servants
◇ 316.343:63-051 Farmers
◇ 316.343:63-051-058.12 Gentleman farmers
◇ 316.343"652" Social differentiation in societies of antiquity and slave-owning societies
◇ 316.343"653" Social stratification in feudal societies
◇ 316.343-051"652" Slave owners
◇ 316.343-052"652" Slaves
◇ 316.343-054.51 Autochthonous population. Aborigines
◇ 316.343-054.62 Colonials
◇ 316.343-057.177 Managers. Executives
◇ 316.343-057.3 Employees. Office workers. White-collar workers
◇ 316.343-057.66 Unsettled persons. Vagrants
◇ 316.343-058.12 Nobility. Aristocracy
◇ 316.343-058.13 Middle-class. Bourgeoisie
◇ 316.343-058.14 Workers. Proletarians
- 316.343.7 Social differentiation of the socialist and communist societies
- 316.343.72 Social differentiation under socialism
- 316.343.75 Social differentiation under communism
- 316.344 Social differentiation according to specific criteria
- 316.344.2 According to economic factors
- 316.344.22 According to the ownership of production means
- 316.344.23 According to economic status and income and wealth
- 316.344.24 According to occupation or employment
Specific occupations may be denoted using a special classification, e.g. the ILO classification, following the asterisk (Table 1h)
- 316.344.25 According to economic sectors, e.g. mining, agriculture
- 316.344.26 According to position within the production process and production cycle. E.g. workers in primary or secondary processes of production
- 316.344.27 According to participation in employment
⇒ 331
- 316.344.3 According to education and knowledge
- 316.344.32 Intellectuals. Literate. Illiterate
- 316.344.34 According to education, training and schooling
◇ 316.344.34:373 According to general school
◇ 316.344.34:377 According to vocational, technical, professional training
◇ 316.344.34:378.2 According to university and college qualifications
- 316.344.4 According to possession of authority and power
⇒ 316.46
- 316.344.42 Elites
- 316.344.43 Ruling classes and strata
- 316.344.44 Suppressed classes and strata
- 316.344.5 According to legal status, e.g. civil rights, personal freedom, privileges
- 316.344.6 According to health and working ability
- 316.344.7 Marginal strata. Alienated and excluded strata and groups
- 316.344.8 According to self-identification
- 316.346 Demographic differentiation
⇒ 314
- 316.346.2 According to sex. Sexes
◇ 316.346.2-055.1 Men
◇ 316.346.2-055.2 Women
- 316.346.3 According to age
- 316.346.32 Age groups
◇ 316.346.32-053.6 Youth sociology
◇ 316.346.32-053.9 Sociology of the elderly
- 316.346.36 Generations
- 316.347 Social differentiation according to specific national, racial, ethnic and religious origins
◇ 316.347(=1-86) Racial groupings. Ethnic communities. Ethnicity
- 316.35 Social groups. Social organizations**
Specify according to size by .023...
◇ 316.35.023.2 Dyads and triads
◇ 316.35.023.4 Small groups
◇ 316.35.023.5 Mesogroups. Mesoorganizations
◇ 316.35.023.6 Large groups. Large organizations
⇒ 061
⇒ 316.342.6
⇒ 316.45
- 316.351 Social groups according to cohesiveness
- 316.353 Groups according to disposition or interest
- 316.354 Purposive organized groups and organizations. Sociology of organizations
◇ 316.354:351/354 Sociology of public administration
◇ 316.354:355.1 Sociology of the armed forces
⇒ 061
⇒ 316.334.4
- 316.354.2 Regulative groups
- 316.354.4 Voluntary groups
- 316.356 Ascribed, nonvoluntary and predetermined groups
- 316.356.2 Consanguinity groups
- 316.356.4 Nonconsanguinity ascribed groups. □ Races. Nations
- 316.37 Individuals (as a basic element of social structure)**
- 316.4 Social processes. Social dynamics**
- 316.4.05 Types of social process
- 316.4.051 Processes in time
- 316.4.051.2 Recurring processes. Cyclic processes
- 316.4.051.3 Nonrecurring processes. One-way processes

- 316.4.051.6 Processes between and within generations
 316.4.051.62 Intragenerational processes. Intra-cohort processes
 ⇒ 316.477
 316.4.051.63 Intergenerational processes. Superseding of cohorts
 316.4.057 Processes according to way of manifestation
 316.4.057.2 Manifest processes
 316.4.057.4 Latent processes
 316.4.06 Function, effect and determinants of social processes
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	⇒ 328.124	329.17	Nationalist attitude
328.185	Corruption	329.18	Fascist attitude
328.187	Relations between parliament and voters, electorate	329.21	Monarchist attitude
328.188	Relations among parliament, the economy and administration. □ Protection. Nepotism. Family influence	329.23	Republican attitude
328.3	Functions of parliaments	329.26	Revanchist outlook
328.32	Formation of political climate, will, purpose	329.271	Adherents of a unitary state (unitarists)
328.34	Legislation. Passing of laws	329.272	Adherents of a federal state (federalists)
	⇒ 342.52	329.273	Separatist outlook. Secessionist outlook
328.36	Control over government. Watchdog function	329.275	Aim of unification of territories (unionists, territorial revisionists)
329	POLITICAL PARTIES AND MOVEMENTS	329.281	Syndicalist outlook
	<i>Individual parties may be arranged alphabetically by name or recognized abbreviation (Table 1h) following the common auxiliaries of place (1/9) (Table 1e) (and, if needed, the common auxiliaries of time "... (Table 1g))</i>	329.285	Anarchist outlook
	◇ 329(410)Lab British Labour Party	329.286	Nihilist outlook
	◇ 329(430)SPD Sozialdemokratische Partei Deutschlands	329.29	Advocates of a corporate state (corporatists)
	◇ 329(438)PZPR Polska Zjednoczona Partia Robotnicza (Polish United Workers' Party)	329.3	Parties and movements with religious outlook
	◇ 329(73)Dem Democratic Party, USA		◇ 329.3:271/279 Christian outlook
			◇ 329.3:28 Islamic, Muslim, Mohammedan outlook
329.05	Position of parties and movements	329.36	Parties and movements with antireligious, atheist, anticlerical outlook
329.051	Government parties. Parties in power	329.361	Antireligious outlook
329.052	Opposition parties	329.362	Anticlerical outlook
329.053	Splinter parties	329.4	Parties and movements with predominantly ethnic, racial, linguistic aims
329.054	Satellite parties		Racial separation (apartheid) movements
329.055	Tendencies within parties. Wings	329.42	Illegal parties and movements
329.055.1	Extreme right	329.61	Underground movements
329.055.2	Right	329.614	
329.055.3	Centre. Centrist	329.63	Parties and movements with special interests
329.055.4	Left	329.7	Political and national movements and alliances in various areas
329.055.5	Extreme left	329.71	Political mass organizations, e.g. in 'peoples' democracies'. □ 'Popular Front'. 'National Front'
329.058	Attitudes of parties to particular problems	329.73	National movements in general
329.058.1	Positive	329.733	Movements for national strength and territorial defence. Premilitary and paramilitary organizations
329.058.2	Negative		⇒ 355.233.11
329.058.3	Indifferent		

329.734	Movements for the preservation of public order	329.78	Political and national youth movements
329.735	Formations of parties and movements for defensive and offensive purposes		◇ 329.78-055.1 For boys
			◇ 329.78-055.2 For girls
329.75	Soldiers' and veterans' movements	329.8	Party structure. Party strategy. Party tactics
33 ECONOMICS. ECONOMIC SCIENCE			
330	ECONOMICS IN GENERAL	330.123.5	Goods for use (durables)
	⇒ 005	330.123.6	Services
	⇒ 316.334.2		⇒ 338.46
	⇒ 338	330.123.7	Means of production. □ Capital goods. Fuel. Raw materials
	⇒ 339		
330.1	Science of economics. Basic economic concepts, theory. Value. Capital. Funds	330.13	Profitability. Economic principle. Utility. Value. Value principle
330.101	Science of economics in general. Economics and its problems, uncertainties. Object and methods of political economy		⇒ 338.5
330.101.2	Empirico-descriptive economics	330.131	Basis of profitability. Economic principle. Economic calculation
330.101.3	Deductive rational economics	330.131.3	Concept of the economic dimension
330.101.4	Normative-teleological economics	330.131.5	Economic utility effect. Social utility
330.101.5	Special methodological problems	330.131.7	Risk, uncertainty, in economic sense
330.101.52	Statics. Dynamics. Equilibrium	330.132	Utility. Use. Usefulness. Exploitability. Usability
330.101.54	Macroeconomics. Microeconomics	330.132.2	Marginal utility
330.101.541	Macroeconomics	330.132.6	Sentimental value
	⇒ 330.5	330.133	Wares. Goods. Utility value and exchange value. Economic value
330.101.542	Microeconomics	330.133.1	Utility value
	⇒ 658	330.133.2	Yield value
330.101.8	Classification and subdivisions of economics. Theoretical, practical. Pure, applied. General, special. Relation of economics to theory of political economy. Interrelations of political economics, private economics, business economics (commercial economics). Relation of general to special political economics. Sectorial economics	330.133.3	Exchange value
		330.133.4	Replacement value
		330.133.6	Individual value and social value
		330.133.7	Measurement of value
		330.133.8	Commercial value of labour capacity
			⇒ 331.214
330.11	Economic phenomena and laws	330.135	Value principle
330.111	Concept of the economy. Definitions	330.138	Theories of value
330.111.2	Manner of production	330.138.1	Objective value theories
330.111.4	Factors, determinants in production. Productive forces	330.138.11	Production cost theory
		330.138.12	Replacement cost theory
		330.138.13	Theory of labour saving
		330.138.14	Labour theory of value
330.111.42	Distribution of factors in production	330.138.15	Theory of working time value (K. Marx)
330.111.44	Exchangeability of factors in production	330.138.2	Subjective value theories. □ Marginal utility theory (marginal value)
330.111.6	Circumstances of production		Negative value theories
330.111.62	Property ratios as production ratios. Ownership	330.138.8	
330.111.64	System of distribution	330.14	Capital. Funds for material production
330.111.66	Socioeconomic sectors		⇒ 330.32
330.111.8	Relation of productive forces to production ratios	330.14.01	Theories of capital. Concept and definition of capital
330.112	Purpose, aims of economic management		
330.112.1	Satisfaction of demand	330.142	Forms and kinds of capital
330.112.2	Creation of profit	330.142.1	Free capital
330.113	Concept of laws, principles in economics. Predictability. Regularity. Economic laws	330.142.2	Invested capital
		330.142.21	Productive capital
330.113.2	Universal character of economic laws	330.142.211	Investment capital. Fixed capital
330.113.4	Historical relativity of economic laws	330.142.211.2	Depreciation (decrease in value through wear and tear etc.)
330.113.6	Mode of effect of economic laws		Amortization
330.117	Subjects of economic management: individuals, groups, state	330.142.211.4	Circulating capital. Working capital. Floating capital
		330.142.212	⇒ 658.14/17
330.12	Objects of economic management. Goods. Services	330.142.22	Loan capital. □ Usury capital. Bank capital
330.121	Scarcity of goods. Rarity	330.142.23	Finance capital. Financial oligarchy. Finance capitalism
330.122	Free goods		Trade capital
330.123	Economic goods	330.142.24	Turnover capital
330.123.1	Renewable and nonrenewable goods	330.142.26	Commodity capital
330.123.2	Reproducible and nonreproducible goods	330.142.27	Fictitious capital. Watering stock
330.123.3	Stocks		
330.123.4	Goods for consumption (consumables)	330.142.28	

330.143	Capital profit. Capital gain. Rate of profit. Profit from capital investment. Increase in value. Value added. Rate of appreciation ⇒ 332.68 ⇒ 336.78	330.322.54	Measurement of investment profitability. Investment accounting. Estimation of returns on investments
330.143.1	Capital gain. Return on capital	330.322.55	Payback period of investments
330.143.12	Rate of gain. Rate of profit	330.322.7	Multiplier effect
330.143.2	Value added	330.322.8	Acceleration effect
330.144	Distribution of capital	330.322.9	Disinvestment. Realization, consumption of capital
330.144.1	National distribution of capital		
330.144.2	International distribution of capital		
330.145	Circulation of capital. Capital transfer		
330.146	Accumulation of capital. Capital formation		
330.147	Concentration and centralization of capital		
330.15	Natural factors. Soil. Nature. Natural forces. Natural wealth ⇒ 332.2 ⇒ 504.062	330.33	Cyclic behaviour of trade. Business cycles <i>Class here theoretical aspects only. For information of a descriptive nature use 338.1</i> ⇒ 338.1
330.16	Physio-psychological factors. Economic behaviour, conduct. Demands. Wishes. Inhibition, restraint. Economic instincts. Intuition	330.33.01	Business trends and crisis theory
330.161	Egoism. Altruism. Activities for personal benefit and that of others. Interests	330.33.012	Cyclic movement. Oscillation of business trends. Trade cycle. Business cycle. Economic cycle. Cyclic phenomena
330.162	Community spirit. Solidarity. Economic morality. Care	330.33.015	Causes, origin of cyclic movements ◇ 330.33.015:330.322 Causes due to investment ◇ 330.33.015:336.77 Causes due to credit
330.163	Demands		
330.163.1	Kinds of demand		
330.163.11	Individual demands		
330.163.12	Collective demands		
330.163.13	Subsistence demands		
330.163.14	Cultural demands		
330.163.15	Demands for luxuries		
330.163.2	Elasticity of demand		
330.163.3	Scales of demand. Urgency of demand. Satisfaction of demand. □ Gossen's first and second laws		
330.3	Dynamics of the economy. Economic movement	330.34	Economic development <i>Class here theoretical aspects only. For information of a descriptive nature use 338.1</i>
330.31	Economic circulation. Replacement ◇ 330.31:330.14 Circulation and replacement of capital	330.34.01	Theory of economic development
330.311	Simple replacement	330.34.011	Definition of economic development
330.313	Production process	330.34.014	Kinds of development in the economy
330.314	Circulation process	330.34.014.1	Stationary economy
		330.34.014.2	Advancing, expanding economy
330.32	New formation of capital	330.341	Factors influencing economic development ◇ 330.341:32 Political factors
330.322	Investments. Capital formation	330.341.1	Development of productive power. Technical progress. Innovation
330.322.01	Investment theory	330.341.2	Development of production ratios. Institutional changes
330.322.011	Investment criteria ⇒ 658	330.341.4	Structural changes
330.322.014	Volume of investment	330.341.42	Changes in the proportions of branches of the economy to each other. Structure of the economy according to economic sectors, branches. Division of labour in society
330.322.014.2	Gross investment		
330.322.014.3	Net investment	330.341.424	Industrialization
330.322.014.6	Investment quota	330.341.426	De-industrialization
330.322.1	Investment according to source of finance	330.341.44	Concentration. Centralization ⇒ 332.156
330.322.12	Private investment	330.341.46	Deconcentration. Decentralization. Hiving off. Dispersal ⇒ 332.156
330.322.14	Public investment. State investment	330.342	Developmental level. Stages of development of the economy. Economic systems
330.322.16	Investment with private and public participation	330.342.1	Economic stages according to development of production
330.322.2	Investment in material production	330.342.11	Primitive economic systems. □ Individual economy. Family economy. Tribal economy. Household economy. Primitive community economies. Early communism
330.322.3	Investment other than in material production ◇ 330.322.3:37 Investment in education	330.342.12	Slavery as economic system
330.322.4	Investment in proportion to investment requirement ◇ 330.322.4.053.2 Overinvestment ◇ 330.322.4.053.3 Underinvestment	330.342.13	Feudal economy
330.322.5	Profitability, efficiency of investments. Return on investments	330.342.14	Capitalist economy
		330.342.141	Classical capitalism. Capitalism of fair competition
		330.342.142	Monopoly capitalism. Mature capitalism
		330.342.143	State monopoly capitalism. Late capitalism
		330.342.146	Social market economy. Economy of the welfare state
		330.342.15	Socialist economy. Communist economy
		330.342.151	Socialist economy
		330.342.152	Communist economy
		330.342.17	System of free economy and controlled economy

330.342.172	Free economy. Laissez-faire	330.46	System theory, control theory, and cybernetics in economics. Economic cybernetics ⇒ 519.71
330.342.173	Controlled economy. Interventionism		
330.342.2	Economic stages according to productivity, to degree of industrialization	330.47	Economic informatics ⇒ 519.72
330.342.21	Undeveloped (underdeveloped) economy. Preindustrial economy	330.5	National property. National assets. Social product. Balance of the economy. Estimation of national product
330.342.22	Developing economy. Industrializing economy	330.5.051	Calculation
330.342.23	Developed economy. Industrialized economy	330.5.055.2	Increase
330.342.24	Fully developed economy. Highly industrialized economy	330.5.055.3	Decrease
330.342.3	Economic stages from individual economy to world economy (Bücher)	330.5.055.4	Stagnation
330.342.4	Economic stages according to other structural principles. □ Theories of List, Hildebrand, Rostow	330.52	National wealth. National property. National assets
330.35	Economic growth <i>Class here theoretical aspects only. For information of a descriptive nature use 338.1</i>	330.522	Real, actual national wealth. Actual assets of nation
330.35.01	Theories of growth <i>Specify by :330.8...</i>	330.522.2	Basic funds and means
330.35.011	Concept and definition of economic growth	330.522.4	Stocks, reserves. Material in hand
330.352.2	Extensive growth	330.522.6	Property of the population. Personal property
330.352.3	Intensive growth	330.522.7	Supply of the population with consumer goods
330.354	Factors determining economic growth ◇ 330.354:330.322 Investment as growth factor	330.524	Potential national wealth. Potential assets of nation ◇ 330.524:553 Mineral resources. Mineral wealth
330.356	Relationship of individual factors of economic growth	330.526	Division, distribution of national wealth. Distribution of national assets
330.356.2	Capital coefficient. Capital-output ratio	330.526.2	Distribution according to economic sector ◇ 330.526.2:63 Agricultural wealth
330.356.3	Capital productivity	330.526.3	Distribution according to economic units
330.356.4	Capital intensity. Capital-labour ratio	330.526.32	Private households
330.356.7	Production function	330.526.33	Business enterprises
330.357	Rate of growth in the economy. Growth tempo ◇ 330.357:330.322 Investment rate	330.526.34	Public enterprises. □ State enterprises. Regional government enterprises. Local government enterprises
330.36	Economic equilibrium. Proportionality of the economy <i>Class here theoretical aspects only. For information of a descriptive nature use 338.1</i>	330.526.35	Semipublic enterprises
330.36.01	Theory of equilibrium	330.526.36	Cooperatives ⇒ 334.73
330.36.012	Static equilibrium. Dynamic equilibrium	330.526.39	Other bodies. □ Churches. Charities
330.36.012.2	Static equilibrium	330.526.5	Redistribution of wealth. Principles of policy towards property ⇒ 338.23
330.36.012.4	Dynamic equilibrium	330.53	Balance of the economy. Social accounting. National accounting
330.362	Macroeconomic equilibrium. Factors determining equilibrium in general	330.532	Balance of the economy
330.364	Microeconomic equilibrium. Partial equilibrium	330.534	Social accounting. National accounts
330.366	Balanced economic growth. Stable economic growth	330.534.4	National budget ◇ 330.534.4(410) National budget of the United Kingdom
330.368	Economic growth without equilibrium. Unbalanced, disproportionate growth. Distorted economy	330.54	Social product (global product)
330.4	Mathematical economics ⇒ 303 ⇒ 519.2 ⇒ 519.86	330.542	Generation of the social product according to economic sectors ◇ 330.542:338.43 Agriculture ◇ 330.542:338.45 Industry
330.42	Mathematical economic theory	330.544	Application of social product
330.43	Econometrics	330.544.2	Consumption of the product ⇒ 338.054.23
330.44	Input-output analysis in economics. Input-output models ⇒ 303.725.34	330.546	Constituents of the social product ◇ 330.546:330.123.4 Goods for consumption ◇ 330.546:330.123.7 Means of production
330.45	Operations research in economics <i>Denote methodological aspects by :519.8... and applications by :33...</i> ◇ 330.45:519.852:338.55 Linear programming to determine imputed prices	330.55	Gross national product. Gross domestic product
		330.552	Generation of the gross national product and gross domestic product according to economic sectors ◇ 330.552:338.43 Agriculture ◇ 330.552:338.45 Industry ◇ 330.552:338.46 Services ◇ 330.552:338.47 Transport ◇ 330.552:339.1 Trade

330.554	Generation of the gross national product according to economic units 330.554 = 330.526.3	330.813	Physiocracy and other economic doctrines of the eighteenth century (Quesnay, Mirabeau, Le Trosne, Du Pont de Nemours, de Gournay etc.)
330.556	Utilization of the gross national product ◇ 330.556:330.322 Investment ◇ 330.556:330.567.4 Government spending. Public spending	330.818	Economic thought outside Europe
330.56	National income. Net national product	330.82	Classical and postclassical economic theories. Liberalism. Ethical, romantic, historical schools
330.562	Formation of the national income, of the total public income according to economic sectors ◇ 330.562:338.43 Agriculture ◇ 330.562:338.45 Industry	330.821	Classical and postclassical economic theories
330.563	Generation of the national income according to economic units 330.563 = 330.526.3	330.821.1	Classical economic theory (Petty, A. Smith, Ricardo)
330.564	Division, distribution of national income	330.821.5	Postclassical economic theory (Malthus, MacCulloch, Senior, James Mill, John S. Mill, Say, von Thünen etc.)
330.564.2	Total income of populace	330.822	Liberal optimists. Manchester school (Cobden, Bastiat, Carey)
330.564.22	Private income from wealth, from enterprise	330.823	Moderate individualism
330.564.24	Private income from independent work	330.824	Doctrine of the free economy (Gesell)
330.564.26	Private income from employment	330.826	Ethical school (Sismondi, Cohn etc.)
330.564.4	State revenue ⇒ 336.2	330.827	Romantic, neoromantic school (Müller, Haller, Spann)
330.564.6	Income of public enterprises in planned, socialist economies	330.828	Historico-sociological school. National reaction. Protectionism (Carey, List, Roscher, Hildebrand, Schmoller, Bücher, Sombart, Comte, Weber etc.)
330.564.62	Income of state enterprises, of nationalized industries	330.83	Economic doctrines of the twentieth century and their predecessors in the nineteenth century
330.564.64	Income of cooperative enterprises	330.831	Psychological school. Marginalism.
330.565	Redistribution of national and private income	330.831.4	Neoclassical school (Böhm-Bawerk, Haberler, Perroux etc.)
330.565.012.23	Redistribution in a market economy	330.831.8	Cambridge school (Marshall, Hawtrey etc.). Welfare economics (Pigou, Pareto etc.)
330.565.012.24	Redistribution in a planned economy	330.831.84	Neoliberalism (Eucken, Röpke, Hayek, von Mises)
330.566	Forms of income	330.831.84	Chicago School. Monetarists (M. Friedman etc.)
330.566.2	Kinds of income	330.831.9	Neoclassical synthesis (Samuelson, Hicks, Tobin etc.)
330.566.22	Primary income	330.832	Mathematical school. Theory of equilibrium.
330.566.24	Derived income	330.834	Econometric school (Walras, Pareto, Wicksell, Samuelson, Tinbergen, Frisch)
330.566.6	Forms of income	330.835	Keynes and Keynesianism. Neo-Keynesianism. Post-Keynesianism
330.566.62	Income in kind. □ As goods. As services	330.835	Theories of economic growth (Harrod, Domar, Kaldor, Rostow, Solow, Meade etc.)
330.566.63	Monetary income	330.836	⇒ 330.35.01
330.567	Use of the national income and public income	330.837	Technical economic doctrines. Technocracy (Theobald, Fourastié etc.)
330.567.2	Expenditure. Consumption. Savings. Economy of private households	330.838	Institutionalism, institutional economics (Elly, Commons, Veblen, J M Clark etc.).
330.567.22	Individual consumption. Private consumption	330.839	Neoinstitutionalism (Galbraith etc.)
330.567.25	Savings by private households	330.84	French sociological school (Simiand, J A Marchal, Lhomme etc.)
330.567.28	Economy of private households	330.841	Theory of managerial revolution (Berle, Means, Burnham etc.)
330.567.4	National consumption. State expenditure ◇ 330.567.4:001 On science ◇ 330.567.4:61 On health ⇒ 336.5	330.842	Pre-Marxian socialist schools. Utopian socialism. Fourierism
330.567.6	Accumulation of income. Capital formation. Reserves of wealth	330.843	⇒ 141.81
330.59	Standard of living ⇒ 331.215.4	330.844	Socialist economic doctrines to the end of the 18th century
330.8	History of economic theories, doctrines, dogmas <i>History or accounts of comprehensive theories are classed under 330.8. Class theoretical aspects of particular economic problems with the problem</i> ◇ 330.138 Theories of value	330.845	Saint-Simon school
330.81	Preclassical economic theories	330.852	Fourier school
330.811	Economic theories of antiquity and the mediaeval period	330.853	Pre-Marxian English socialist schools.
330.811.1	Economic thought of antiquity (Aristotle, Plato, Cato etc.)	330.854	□ Owenism. Chartism
330.811.2	Economic thought of the mediaeval period (St Thomas Aquinas, Oresme etc.)		Marxist-Leninist economic doctrine
330.812	Mercantilism and other economic doctrines of the sixteenth and seventeenth centuries. □ Bullionism. Cameralism. Colbertism		⇒ 141.82
			Origin of Marxist economic doctrine
			Development of Marxist economic doctrine to Lenin
			Origin of Marxist-Leninist economic doctrine

330.856	Development of Marxist-Leninist economic doctrine up to the end of the 1920s	331.101.21	Right to work
330.857	Development of Marxist-Leninist economic doctrine from the 1930s to the mid-50s	331.101.22	Obligation to work
330.858	Development of Marxist-Leninist economic doctrine since the mid-50s	331.101.23	Division of labour
330.86	Social-democratic and non-Marxist socialist economic theories	331.101.232	Division of labour according to sex
330.861	Social democratic economic theories.	331.101.234	Division of labour according to age groups
	Revisionist economic theories. Reformist economic theories	331.101.24	Work restrictions. Professional restrictive practices. □ Work prohibition ⇒ 338.246.84
330.862	Anarchist economic theories.	331.101.25	Effects of work on private life
	Anarcho-syndicalist economic theories	331.101.26	Labour economics
330.863	Rational socialist economic theory. □ Collin's school	331.101.262	Human resources
330.864	Fabian economic theory	331.101.264	Working and nonworking sections of the population
330.865	Corporate-socialist, guild-socialist economic theories	331.101.264.2	Working population ⇒ 331.522.4
330.866	Religious socialist movements. Ethical socialist economic theories	331.101.264.22	Sectors of the population contributing directly to the gross national product. Primary and secondary sectors of the working population
330.867	Radical economists and their predecessors (P.A. Baran, P.M. Sweezy, E.K. Hunt, H. Sherman, H. Gintis etc.)	331.101.264.24	Sectors not directly contributing to the gross national product. Tertiary and quaternary sector. □ Civil servants. Public service sector
330.868	State socialist doctrines	331.101.264.4	Nonworking sector of the population
330.87	Social-legalistic, solidarist, corporative, religious, cooperative, anthroposophic, collectivist, planning-based economic doctrines	331.101.264.6	Cost of nonproductive sectors to the economy
330.872	Socio-legalistic school	331.101.3	Attitude to work. Work motivation. Work morale. Work incentives
330.873	Solidarist economic doctrines	331.101.31	Positive attitude to work
330.874	Corporative movement (Mgr. W. von Ketteler, A.A.M. de Mun, de la Tour du Pin, etc.). Liège School. Belgian Christian democracy	331.101.32	Job satisfaction
	Religious economic doctrines	331.101.33	Indifferent attitude to work
330.875	Roman Catholic economic doctrines	331.101.34	Negative attitude to work
330.875.2	Protestant economic doctrines	331.101.35	Dissatisfaction at work
330.875.3	Non-Christian religious economic doctrines.	331.101.36	Effects of personal outlook on the job
330.875.5	□ Doctrine of Mahatma Gandhi	331.101.37	Effects of work environment on the job
330.876	Anthroposophic economic doctrines. Society as threefold commonwealth	331.101.38	Work incentives
330.877	Collectivism	331.101.39	Valuation of work by company and society
330.878	Doctrines of planned economy.	331.101.4	Humanization of work
	Budget-oriented economic doctrine	331.101.5	Impact of technological development on work
330.88	Most recent movements	331.101.52	Technology as an alleviating factor on human work
	<i>Denote movements by alphabetic extension (Table 1h)</i>	331.101.54	Technology as a threat to human work
331	LABOUR. EMPLOYMENT. WORK. LABOUR ECONOMICS. ORGANIZATION OF LABOUR	331.101.6	Work performance ⇒ 159.944
	<i>For human relations within the enterprise (personnel) see 658.3 which is subdivided parallel to 331</i>	331.101.62	Individual work performance
	⇒ 316.334.2	331.101.64	Collective work performance
	⇒ 351.83	331.101.68	Degree, rate of work performance. Intensity of work
331.07	Administrative organizations, authorities and official bodies in the field of labour	331.102	Type of work
	<i>Auxiliaries 3.07... (as at 35.07)</i>	331.102.1	Mode of employment
	◇ 331.07(100)ILO International Labour Office	331.102.12	Self-employed. Freelance work. □ Self-employed professions. Casual work
331.08	Staff in the field of labour	331.102.14	Work rendered under contractual employment
	<i>Auxiliaries 3.08... (as at 35.08)</i>	331.102.16	Commissioned and uncommissioned work
	⇒ 331.108.2	331.102.2	Vocational work and hobbies
331.1	Theory and organization of work. Relations between business firms and employees	331.102.22	Hobbies
	◇ 331.1-055.2 Women and work	331.102.24	Vocational work
331.101	Labour questions of a general nature. Theory of labour. Science of labour	331.102.3	Scope and content of work
331.101.1	Ergonomics	331.102.31	Creative and noncreative work
331.101.2	The role of work in society	331.102.312	Creative work
		331.102.314	Noncreative work
		331.102.32	Work according to level of complexity
		331.102.322	Specialized work
		331.102.323	Difficult work. Skilled work
		331.102.324	Work requiring a certain degree of training. Semiskilled work
		331.102.325	Simple work. Unskilled labour
		331.102.34	Physical work and mental work
		331.102.342	Physical work
		331.102.344	Mental work
		331.102.346	Elimination of discrimination between physical and mental work
		331.103	Organization of work. Industrial engineering ⇒ 658

331.103.1	Jobs. Posts	331.105.444	Company unions. Staff associations. Trade unions in one firm
331.103.11	Creation of individual jobs, posts. Allocation of tasks. Assignment of duties. Specialization	331.105.445	Trade unions according to membership of a profession or trade. ☐ Craft unions <i>Denote profession or trade by ...</i>
331.103.112	Homogeneous groupings of work components	331.105.446	Trade unions for specific groups <i>Specify by -05... (Table 1k)</i> ◇ 331.105.446-057.175 Trade unions for middle management
331.103.114	Heterogeneous groupings of work components	331.105.45	Compulsory trade union membership. Closed shop agreements
331.103.116	Designing the job according to qualification. ☐ Job enlargement. Job enrichment. Suiting the job to the worker	331.105.46	Shop stewards. Workplace representatives
331.103.12	Job analysis ⇒ 331.103.32	331.105.5	Professional organizations and associations of the self-employed
331.103.13	Job description	331.105.6	Cooperation between employers, employees and government. Cooperation between state and worker
331.103.14	Job grading	331.106	Contracts of employment ⇒ 349.22
331.103.15	Job evaluation	331.106.2	Types of contract of employment
331.103.2	Work process	331.106.22	Skeleton agreements. Framework agreements
331.103.22	Planning the work process. Initiation of work	331.106.24	Collective agreements. Company agreements
331.103.24	Carrying out of the work process	331.106.25	Wage agreements
331.103.242	Single operation, each person completing one part of the product	331.106.26	Individual contracts
331.103.244	Cooperation in a small team manufacturing one product or part of a product. Autonomous working groups	331.106.27	Contracts for specific jobs for specified periods. Contract work
331.103.246	Teamwork, where the individual worker is alienated or distanced from the end-product	331.106.28	Service contracts
331.103.25	Work with and without mechanical aids	331.106.29	Temporary work contracts
331.103.252	Manual work	331.106.4	Commencement and termination of agreements, contracts
331.103.253	Mechanized work	331.106.42	Negotiation strategy, tactics. Bargaining techniques
331.103.254	Conveyor-belt work	331.106.43	Recognition of negotiating authority
331.103.255	Automated work	331.106.44	Termination of contract of employment by employer ⇒ 331.108.644.7
331.103.27	Work rhythms	331.106.46	Termination of contract of employment by employee
331.103.28	Work pace ⇒ 331.101.6	331.106.47	Termination of contract of employment through retirement
331.103.3	Work study. Time and motion study. Work measurement	331.106.5	Conditions attached to the contract of employment. ☐ Probationary period. Fixed period
331.103.32	Work analysis. Work content ⇒ 331.103.12	331.107	Worker participation. Codetermination. Industrial democracy at work. Collaboration of workers in planning and control
331.103.34	Various methods. ☐ Bedaux system. Multitimer system. Motion time systems (MTM, QSK etc.)	331.107.1	Joint consultation
331.103.4	Introduction of the employee to the work process ⇒ 331.108.38 ⇒ 331.361	331.107.2	Works committee. Staff council. Works council
331.103.5	Special forms of work organization in centrally planned people's economies. ☐ Labour brigades	331.107.22	Council chairman
331.103.6	Employee suggestion schemes	331.107.23	Employer's representatives on council
331.104	Relations at work in general. ☐ Industrial relations. Labour relations	331.107.24	Delegated representatives of employees on council
331.104.2	Human relations in the workplace <i>For human relations in the enterprise (personnel) see 658.310.42</i>	331.107.25	Elected representatives of employees on council
331.105	Parties in a labour relationship	331.107.26	Jurisdiction and rights of council. ☐ Social benefits of the firm
331.105.2	Parties in an individual contract of employment ◇ 331.105.2-057.15 Employers ◇ 331.105.2-057.16 Employees	331.107.262	Suggestions made by staff
331.105.4	Parties in a collective agreement	331.107.264	Participation in decisions connected with promotions and recruitment
331.105.42	Employers' associations ⇒ 334.786	331.107.266	Participation in questions of business policy. ☐ Investments. Humanization of work
331.105.422	Central employers' organizations. National employers' organizations. Federations of employers' associations ◇ 331.105.422:329.1/6 Political outlook of national employers' organizations	331.107.3	Worker directors. Employees' representation on boards of directors
331.105.423	Employers' associations in a particular sector of industry	331.107.5	Co-ownership and copartnership
331.105.44	Employees' organizations. Trade unions	331.107.8	Self-administration. Workers' control. ☐ Control according to Yugoslav models
331.105.441	Central trade union organizations, federations	331.108	Personnel administration. Provision of the workforce. Work regulations. Work discipline
331.105.442	General unions. Trade unions covering related sectors of industry	331.108.2	Personnel management. Staff structure
331.105.443	Industrial unions. Trade unions in one sector of industry	331.108.22	Number of personnel. Number of staff
		331.108.23	Personnel structure. Staff structure

331.108.24	Staff turnover and wastage	331.109.334	Work-to-rule
331.108.26	Personnel administration. □ Personnel files, records, passes	331.109.336	Sit-down strikes
331.108.27	Staff reduction	331.109.34	Boycotts of firms by employees. Picketing
331.108.3	Appointment of new staff. Hiring ⇒ 331.5	331.109.35	Black lists and fair lists of employers
331.108.32	Aspects of appointment. □ Legal requirements, e.g. regarding employment of the disabled	331.109.36	Disruption of the production process by force
331.108.34	Recruitment <i>Denote method by ...</i> ◇ 331.108.34:331.53 Recruitment through employment agencies ◇ 331.108.34:659.1 Recruitment by advertising	331.109.362	Occupation of workplaces. Sit-ins
331.108.36	Applications	331.109.363	Continuation of production by workers. Work-ins
331.108.37	Selection of staff. □ Aptitude tests. References	331.109.364	Sabotage
331.108.38	Work familiarization of new staff. Induction ⇒ 331.103.4 ⇒ 331.361	331.109.37	Strike funds. Organization of financial support
331.108.4	Career structure	331.109.38	Political action. □ Lobbying parliament. Influencing economic partners, associations, etc. ⇒ 323.23
331.108.41	Seniority. Rank	331.109.4	Action by employers ⇒ 368.025.63
331.108.42	Duration of employment	331.109.41	Injunctions. Legal action
331.108.43	Staff assessment	331.109.42	Measures against internal effects of strikes
331.108.44	Junior staff career development. □ Overlooking, neglect of junior staff	331.109.422	Strike breaking
331.108.45	Staff training	331.109.423	Employment of outside workers. Blacklegging ⇒ 331.109.47
331.108.47	Transfers. Placements	331.109.43	Strike assistance for the employers. Reciprocal financial support
331.108.5	Advancement of experienced employees	331.109.44	Lock-outs
331.108.52	Promotion	331.109.45	Black lists and fair lists of employees
331.108.53	Pay increases	331.109.46	Self-help for affected employers. □ Organization of transport
331.108.54	Nonfinancial rewards. Distinctions. Medals. Diplomas	331.109.47	Use of troops for work in labour disputes
331.108.6	Morale at work. Discipline at work ⇒ 349.22	331.109.48	Political action. □ Lobbying parliament. Influencing economic partners, associations, etc. ⇒ 323.23
331.108.62	Offences against work discipline. □ Breach of contract or regulations. Absenteeism. Malingering	331.109.6	Mediation in labour disputes
331.108.64	Disciplinary measures. Penalties ⇒ 343	331.109.62	Disputes resolved in the workplace
331.108.641	Supervision ◇ 331.108.641-057.212 Supervisors	331.109.63	Adjournment of a strike. Suspension of strike action. Cooling-off period
331.108.642	Disciplinary procedure	331.109.64	External solutions. □ Arbitration services. Arbitration courts. Public enquiries. Court rulings ⇒ 347.918
331.108.643	Caution. Warnings	331.109.7	Restriction of right to strike ⇒ 343.46
331.108.644	Penalties	331.109.72	Abolition of right to strike
331.108.644.1	Fines	331.109.74	Conscription of labour by incorporation into armed services
331.108.644.2	Docking of pay. Suspension of salary	331.109.76	Legal ban on strikes
331.108.644.3	Pay reductions	331.2	Salaries. Wages. Remuneration. Pay ⇒ 349.232 ⇒ 35.087.4
331.108.644.4	Abolition of privileges	331.21	Wage administration
331.108.644.5	Demotion. Transfer into a lower or lesser paid job. Downgrading	331.211	Method of payment <i>Specify frequency of payment by "5..."</i>
331.108.644.6	Expulsion from workplace. Suspension	331.211.4	Advance on wages
331.108.644.7	Dismissal ⇒ 331.106.44	331.211.5	Money payment
331.109	Labour disputes ◇ 331.109-057.15:331.109-057.16 Disputes between employers and employees	331.211.52	Cash payment
331.109.3	Action by employees	331.211.54	Payment by bank transfer or by cheque
331.109.31	Legal action. Court orders	331.211.6	Payment in kind
331.109.32	Strikes ⇒ 323.264	331.211.7	Obligation to leave part of wages within the company. □ Truck system. Invested wages. Employees' shares
331.109.321	General strikes	331.212	Withholding of wages or of part of wages. Wage deduction ⇒ 331.108.644
331.109.322	Sectional strikes	331.212.2	Legally required deductions. □ Deductions of tax. Social contributions. Payments of debt, alimony etc.
331.109.323	Official strikes	331.212.4	Deduction of part of the wages with the agreement of the recipient. □ Union dues. Staff fund
331.109.324	Unofficial strikes. □ Wildcat strikes. Disruptive action	331.213	Lodging of claims for wages due
331.109.325	Sympathy strikes	331.214	Economic theory of wages
331.109.33	Passive resistance. Noncooperation with management ⇒ 323.25		
331.109.332	Go-slow		

331.214.1	Iron law of wages	331.224.6	Special payments
331.214.2	Wage-fund theory	331.224.64	Holiday bonus
331.214.3	Theory of relationship between productivity and wages	331.224.66	Additional month's pay, i.e. 'thirteenth salary'
331.214.4	Theory of exploitation. Power-political wage theory	331.224.7	Additional allowances because of conditions of work. ☐ Night work allowance. 'Dirty money'
331.214.5	Theory of wage movement	331.224.8	Compensation, settlement for premature dismissal. ☐ Redundancy pay. 'Golden handshake'
331.214.51	Doctrine of falling wage ratio (proportion of wages to national income)	331.225	Bonus systems. Bonuses. Other additions to basic pay
331.214.52	Doctrine of rising wage ratio (proportion of wages to national income)	331.225.3	Premium bonus schemes
331.214.7	Economic principle of remuneration according to work	331.225.4	Productivity bonuses. Bonuses for increased efficiency
331.215	Pay levels. Wage levels	331.225.5	Bonuses for employees' inventions and suggestions for improvement. Suggestion schemes
331.215.1	Fair wage. Relation between wage and work assessment	331.225.6	Gratuities
331.215.2	Fluctuations in wage levels. Pay increases, rises. Pay reductions, cuts	331.225.7	Tips
331.215.22	Pay increases	331.226	Family allowances (as wage element). Supplements for dependants
331.215.23	Pay reductions	331.228	Compensation funds. Funds for making up wage loss
331.215.24	Fixing of wage rates	331.23	Forms of payment
331.215.3	Relation between pay, productivity and production costs	331.231	Piecework payment. Piece rate. Contract payment. Payment by result. Flat rates
331.215.4	Relation of pay to cost of living. Nominal wage. Real wage ⇒ 330.59 ⇒ 339.133.3	331.232	Time-based payment. Pay per unit of time. ☐ Hourly rate. Daily rate. Weekly wage. Monthly salary. Annual salary Specify by "... (Table 1g)
331.215.44	Nominal wage	331.236	◇ 331.232"375.2" Overtime pay
331.215.45	Real wage		Subcontract system. Group contract payment. Sweating system
331.215.5	Maximum wage. Minimum wage	331.24	Profit-sharing
331.215.52	Maximum wage	331.25	Provision by the firm for old age and retirement. Superannuation schemes.
331.215.53	Minimum wage	331.252	Company pensions
331.215.54	Average wage	331.255	Interruption of pension accumulation through change of employment
331.215.56	Gap between maximum wage and minimum wage. ☐ Levelling of wages	331.257	Transitional wage at early retirement
331.215.6	Gross pay. Net pay	331.27	Half-pay. Reservists' pay
331.215.62	Gross pay	331.28	Professional remuneration. Fees, e.g. of doctors, lawyers, architects, writers, artists
331.215.63	Net pay. Take-home pay	331.29	Organization of wages and payments. Wage planning. ☐ Wage councils. Wage boards
331.215.7	Wage components	331.3	Option to choose between wage level and other conditions of employment. ☐ Choice between wage level and reduction in working hours, increased leave entitlement, improvements in pension arrangements
331.215.72	Basic wage. Basic pay. Fixed wage	331.31	Working conditions other than pay
331.215.74	Wage supplements. Variable components of wages	331.311	Hours of work ⇒ 349.235 ⇒ 35.083.53
331.215.8	Pay differentials		Hours of definite work periods Specify by "... (Table 1g)
331.216	Continued payment of wages in case of sickness, accident etc. Sick pay		◇ 331.311"329.7" Sunday work ⇒ 27-563:331.331
331.217	Calculation of wages		◇ 331.311"345" Night work
331.218	Receipt of pay from more than one source. Double and multiple earnings		◇ 331.311"427" Working hours per day
331.218.2	Individuals in receipt of more than one wage		◇ 331.311"437" Working hours per week
331.218.3	Collective or family income. Total earnings of man and wife		◇ 331.311"437.5" Five-day week
331.219	Employers' declarations to financial authorities or to the social insurance about wages paid	331.312	Hours of work according to fixed schedule or without timetable
331.22	Types of wages. Additions to pay	331.312.2	Work without fixed hours and schedule
331.221	Payment systems	331.312.4	Flexible working time. Flexitime
331.221.1	Wage systems. ☐ Wage scales. Wage rates	331.312.6	Fixed working hours
331.221.2	Wages related to length of service or to age	331.312.62	Work with fixed timetable
331.221.4	Wages related to performance		
331.221.5	Wage related to qualifications		
331.221.6	Wages with reference to the wage levels of comparable groups. Wage comparison		
331.221.7	Wages with reference to the regional level. ☐ Regional supplements. Local weighting ⇒ 331.215		
331.221.8	Equal pay for equal work		
331.222	Index-linked pay. Pay related to cost-of-living index etc.		
331.224	Specific pay supplements ⇒ 331.34		
331.224.2	Foreign service allowances		
331.224.4	Cost-of-living allowances		

331.312.64	Work with changing timetable. Shift work ◇ 331.312.64"345" Night shift	331.344.4	Catering ⇒ 331.424.4
331.312.8	Fixed number of hours and fixed schedule, but varying length	331.344.42	Subsidized meals. Meal vouchers. □ Luncheon vouchers
331.312.82	Part-time work. Reduced working time. Short-time working ⇒ 331.573	331.344.44 331.344.5 331.344.6	Free meals Special gifts. □ Christmas parcels Employee discount schemes. Allowances in kind ⇒ 331.211.7
331.312.84	Working time according to age	331.346	Expense allowances. Subsistence allowances
331.312.86	Reduced working time imposed by management. Lay offs	331.346.2	Travel allowances ⇒ 331.343.5
331.312.9	Unsocial hours	331.346.3	Day expenses
331.313	Paid working time	331.346.4	Separation allowances ⇒ 331.342.8
331.313.2	Actual, net amount of hours worked	331.346.5	Subsidies for use of private work facilities. □ Tools. Buildings. Private transport
331.313.4	Rest periods. Breaks	331.346.7	Allowances for use of personal clothes
331.313.6	Additional time worked. Paid overtime. □ Compulsory overtime. Voluntary overtime	331.348	Recreational and educational facilities. □ Study grants. Study trips ⇒ 331.426 ⇒ 35.088.6
331.314	Unpaid work	331.36	Career development. In-service training ⇒ 37.018.44 ⇒ 37.018.46 ⇒ 377
331.314.2	Unpaid overtime	331.361	Basic training ⇒ 331.108.38
331.315	Travelling time	331.361.2 331.361.3	Apprenticeships Instruction and supervision of trainees. □ Mentoring (guidance of a trainee by an experienced colleague)
331.316	Lost time. Idle time. Unproductive time	331.362	Training schools. Training workshops
331.316.2	Hours not worked due to circumstances beyond a person's control	331.363	Further education and retraining. □ Intensive courses. Refresher courses
331.32	Leave. Holidays. Other authorized time off ⇒ 35.083.53	331.364 331.366 331.368	Promotion of general education Admission of outsiders for training and information Training of supervisors. Training of foremen
331.322	Leave. Holidays	331.4	Working environment. Workplace design. Occupational safety. Hygiene at work. Accidents at work <i>Class here socioeconomic information only. For technical methods of accident prevention use 614.8.... For matters relating to work planning use 331.103...</i> ⇒ 35.088.2
331.322.2	Length of leave	331.41	Demands concerning the working environment
331.322.4	Time of leave. Staggering of holidays. □ Compulsory leave. Works vacations	331.42	Work premises. Working areas. Workplaces. Staff rooms. Rest rooms ⇒ 72.05 ⇒ 725.1
331.322.5	Special leave. □ Study leave. Sabbatical year. Leave for special activities. Maternity leave. Compassionate leave	331.421	Design of the work environment and workplace ⇒ 005.936.2
331.322.6	Paid leave	331.422	Indoor workplaces
331.322.7	Partially paid leave	331.422.2	Operational areas 331.422.2 = 331.432/438
331.322.8	Unpaid leave	331.422.4	Offices ⇒ 725.1
331.324	Holiday funds	331.423	Outdoor workplaces
331.326	Statutory holidays. Public holidays	331.424	Communal areas and hygiene facilities
331.326.2	Labour festivals and holidays. □ May Day. Bank holidays	331.424.2 331.424.4	Rest rooms. Staff rooms Canteens ⇒ 331.344.4 ⇒ 640.2
331.34	Fringe benefits. Supplementary earnings. Perquisites ⇒ 331.224 ⇒ 35.088.3	331.424.6	Washing facilities. Lavatories ⇒ 645.68 ⇒ 696.1/.4
331.341	Employer's contributions towards funds at work	331.426	Company sports facilities
331.342	Housing and accommodation facilities		
331.342.2	Housing of employees. □ Official quarters. Tied accommodation		
331.342.3	Low-interest loans for home purchases		
331.342.4	Assistance in search for accommodation		
331.342.5	Holiday houses for employees. Holiday houses for children of employees		
331.342.6	Removal assistance. Transfer allowances		
331.342.7	Residential allowance. Rent allowance		
331.342.8	Subsidy towards a family home located elsewhere. Separate home allowance ⇒ 331.346.4		
331.343	Transport facilities and concessions		
331.343.2	Company transport. □ Works buses		
331.343.3	Company cars		
331.343.4	Low-interest loans for the purchase of a car or other means of transport		
331.343.5	Allowances for use of private transport. Mileage allowances ⇒ 331.346.5		
331.343.6	Subsidies towards use of public transport. Assistance with fares		
331.343.7	Parking facilities		
331.344	Clothing. Food. Price discounts. Gifts		
331.344.2	Working clothes. □ Uniforms. Overalls ⇒ 614.895 ⇒ 687.157		

331.43	Exposing employees to mechanical and chemical hazards. Work under special conditions	331.467	Examination and determination of cause of accident
331.432	Work involving machinery	331.468	Responsibility for accidents
331.432.2	Work in various atmospheric pressures	331.468.2	Individual fault
331.432.22	Exposure to increased pressure. □ Caisson work	331.468.3	Fault of another person. Fault of a third party
331.432.24	Exposure to reduced pressure	331.468.4	Fault of management
331.432.4	Exposure to vibrations	331.47	Industrial health. Occupational health
331.432.6	Exposure to noise	⇒ 613.6	
331.433	Special climatic conditions ⇒ 551.58	331.471	Situation and conditions of hygiene at work
331.434	Lighting conditions ⇒ 628.9	331.472	Occupational diseases. Industrial diseases ⇒ 613.62 ⇒ 616-057
331.435	Exposure to radiation	331.472.2	According to type of work and occupation <i>Specify by :331.43...</i>
331.436	Work with noxious and dangerous materials	331.472.4	According to type of illness <i>Specify by :616...</i>
331.437	Work with dangerous tools	◇ 331.472.4:616.28-008.14	Occupational deafness
331.438	Work under other difficult circumstances	331.473	Criteria for recognition as occupational disease ⇒ 364.32:613.62
331.438.2	Work at great heights	331.48	Services at work for improvement of staff working conditions
331.438.3	Work at great depths	331.482	Industrial safety and health supervision
331.44	Influence of physical and psychological factors (work-related and non-work-related) on the employee ⇒ 159.944	331.483	Industrial, occupational health service
331.441	Strenuous, physically demanding work. □ Handling of heavy goods	331.483.1	First aid
331.442	Work with mentally adverse effects ⇒ 159.944.4	331.483.2	Medical examinations by company doctor
331.442.2	Work under time pressure	331.483.4	Inspections by company doctor ⇒ 614.4
331.442.4	Monotonous work	331.483.6	Responsibilities of the industrial health service towards specific groups of employees <i>Denote particular groups by -05... (Table 1k)</i>
331.443	Work-related factors. □ Workload. Job security	331.483.7	Special duties of the industrial health service. □ Duties concerning frequent absence, rehabilitation
331.445	Non-work-related factors <i>Denote personal characteristics by -05...</i> ◇ 331.445:159.9 Psychological influences ◇ 331.445:159.923 Influences of character ◇ 331.445:159.952 Effect of attention, concentration, interest ◇ 331.445:613 Influence of personal health (and habits) ◇ 331.445:663 Influence of alcoholic beverages, stimulants, narcotics	331.483.8	Shared industrial health service
331.45	Protection at work. Safety at work	331.484	Company social welfare service
331.451	Hygiene and medical care at work	331.485	Company psychology service
331.452	Accident prevention	331.5	Labour market. Employment
331.453	Safety measures for various types of job 331.453 ≡ 331.43	331.5.024.5	Employment policy. Policy towards labour market
331.454	Safety measures against various accident eventualities 331.454 ≡ 614.8	331.5.024.52	Manpower forecasting
331.456	Safety regulations outside the workplace. □ Legislation ⇒ 351.83	331.5.024.54	Manpower planning
331.458	Safety regulations at the workplace	331.52	Structure of the labour market
331.46	Accidents at work ⇒ 614.8	331.522	Supply and demand on the labour market
331.461	Hazards. Risks. □ Hazardous workplaces	331.522.2	Workplace potential
331.462	Accident-proneness	331.522.4	Workforce potential. Labour force. Manpower
331.463	Frequency of accidents. Distribution of accidents	331.523	Distribution of workplaces
331.464	Causes of accidents	331.524	Distribution of the labour force
331.464.1	Acts of God	331.526	Levels of employment. Employment situation, conditions
331.464.2	Accidents caused by technical failure	331.526.2	Full employment
331.464.3	Human error as cause of accidents	331.526.22	Overemployment. Labour shortage
331.464.34	Psychological factors as causes of accidents	331.526.3	Dual jobholding
331.464.36	Accidents due to negligence of safety regulations	331.526.4	Underemployment ⇒ 331.56
331.465	Accidents in connection with type of work <i>Specify by :331.43...</i>	331.526.6	Part-time employment ⇒ 331.312.82
331.466	Accidents according to place. □ Accidents on way to or from work	331.526.8	Moonlighting
		331.53	Employment mediation. Placement in work
		331.532	◇ 331.53.07 Employment offices. Employment agencies. Employment exchanges
		331.533	Direct employment services
		331.534	Indirect employment services. □ Temporary staff agencies
		331.535	Employment services stipulations. □ Work permits. Permission to dismiss
		331.536	Job search
			Vacancies

331.538	Employment service according to type of employees. □ Placement of low-ability, unsuitable, workshy persons	331.58	Encouragement of acceptance of available work. Filling of vacancies
331.54	Occupation. Vocation. Choice of occupation	331.582	Securing optimum utilization of existing workforce. □ Vocational rehabilitation (including disabled persons). Full employment of women, of older workers. Retraining assistance
331.542	Classification of occupations. Occupational groups	331.584	Encouraging applications for less attractive occupations
331.543	Analysis of vocation. Vocational profile	331.586	Encouraging vocational training and retraining
331.544	Vocational training opportunities	331.91	International regulation and organization of labour
331.545	Job prospects. □ Dying and nascent occupations. Jobs with a future	⇒ 351.83	
331.546	Job aptitude	332	REGIONAL ECONOMICS. TERRITORIAL ECONOMICS. LAND ECONOMICS. HOUSING ECONOMICS
331.547	Aptitude tests ⇒ 159.9.07	332.01	Theory. Doctrine. Fundamentals
331.548	Vocational guidance, advice, counselling. Career guidance	332.012	Kinds. Forms. Structures. Divisions. Branches. Systems and arrangements
331.55	Mobility of workforce	332.012.2	According to the socioeconomic system
331.552	Intraoccupational mobility. Change of job	332.012.23	In a market economy
331.553	Interoccupational mobility	332.012.24	In a planned economy
331.554	Mobility between various branches of the economy	332.012.3	According to types and forms of the property and sources of financing
331.556	Geographical mobility. Migration ⇒ 314.15	332.012.32	Private
331.556.2	Mobility within a state. Domestic migration	332.012.322	Owned by individuals, families, partnerships
331.556.22	Commuting ⇒ -057.62	332.012.324	Owned by corporations, organizations, institutions. □ Limited companies
331.556.4	International mobility. International migration of labour	332.012.327	Sole ownership by a legal or natural person
331.556.42	Mobility based on contracts between states	332.012.328	Mortmain
331.556.43	Free mobility	332.012.33	Publicly owned
331.556.44	Immigrant labour. □ Guest workers ('Gastarbeiter')	332.012.332	State ownership
331.556.46	Emigration of labour. □ Working abroad	332.012.334	Municipal. Provincial. Local authority
331.556.48	Frontier workers <i>Workers crossing a frontier to reach their workplace</i>	332.012.34	Cooperative, collective ownership
331.56	Unemployment	332.012.35	Socialized (publicly funded)
331.562	Structural unemployment. □ Technological unemployment	332.012.36	Mixed types
331.563	Cyclical unemployment ⇒ 330.33	332.02	Policy. Planning. Control. Measures. Instruments
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332.871	Administration of dwellings		<i>Specify agent of policy by :...</i>
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332.871.2	By representative of owner		$\diamond$ 334.021:351 Governments
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339.52	Foreign, international trade organizations		Re-exporting after processing into finished goods.
	◇ 339.52:334.788 Chambers of foreign commerce		Storage and bonded warehousing as trade
			⇒ 339.168.6

339.7	International finance		
	⇒ 336	339.743.053.2	Overvaluation
		339.743.053.3	Undervaluation
339.7.01	Theory	339.744	Convertibility
339.7.012	Systems, arrangements of international finance	339.745	Policy towards foreign exchange market
	339.7.012 ≡ 339.9.012	339.746	Exchange control. □ Foreign currency permits.
			Surrendering of currency holdings
339.72	International payments. International money and capital market	339.747	International currency crises
		339.748	Currency war
339.72.053	Balance of payments		
339.72.053.1	Foreign debt. Balance-of-payments deficit	339.9	International economy generally. International economic relations. Global economy
339.721	International liquidity	339.9.01	Theory of the international economy
339.722	Dealings in gold. Foreign exchange dealings. International money market		<i>The .01... auxiliaries which follow are also applicable at 339.5 and 339.7</i>
	⇒ 336.745	339.9.012	Systems, arrangements, of international economic relations
339.723	Payments from visible trade, from trade in goods		
	⇒ 339.5.053	339.9.012.23	International economic systems in a market economy
339.724	Payments from invisible trade, from services	339.9.012.24	International economic systems in a planned economy
339.725	Return on capital		
	⇒ 330.143	339.9.012.42	Free trade
339.726	Donations. Gratuitous transfers	339.9.012.421	Liberalization of trade
339.726.2	Nonmilitary aid. Foreign aid	339.9.012.435	Protectionism
	⇒ 339.96	339.9.012.436	Autarky. Economic self-sufficiency
339.726.3	Military aid	339.9.012.438	Colonial and neocolonial systems
339.726.5	Political debts. Indemnities. Reparations	339.9.012.438.2	Colonialist system
339.726.7	Contributions towards international organizations	339.9.012.438.4	Neocolonialist system
339.727	Dealings in capital. Provision of credit. Clearing. International capital market	339.91	International division of labour
	⇒ 336.778	339.92	International economic cooperation. International economic integration
339.727.2	Capital transactions. International loans. Capital movements	339.922	Integration as form of international and supranational economic cooperation
339.727.22	Import of capital. Capital inflow. Foreign investments	339.923	Economic unions. Economic communities. Economic blocs
339.727.24	Export of capital. Capital outflow. Investments abroad. □ Flight of capital. Hot money		◇ 339.923:061.1EU Economic provisions of the European Union (formerly the European Community)
339.727.3	Granting of international credit	339.924	Integration policies. Promotion of integration
339.727.4	Switch transactions		◇ 339.924:331.556.4 Promoting international migration of labour
339.727.6	Clearing transactions. Clearing. Settlement of international payments	339.926	Settling differences of a regional or international nature
	⇒ 336.768		
	⇒ 336.778		
	⇒ 339.742	339.94	Cooperation. Collaboration. Coproduction. Supply. Specialization. Internationalization of production
339.73	Organizations for international payments and clearance		
339.732	International banks	339.942	Intergovernmental level
339.732.2	International investment banks. International institutions for development	339.944	Company level. □ Joint ventures
339.732.4	International loan institutions. □ International Monetary Fund (IMF)	339.96	Development aid as form of international economic cooperation
339.732.44	International drawing rights. □ SDR (special drawing right)		⇒ 341.232
339.732.6	International clearing institutions. □ Bank for International Settlements (BIS)	339.97	International, supranational, multinational economic policy, planning, plans. International economic management
339.736	Ad-hoc associations. □ Paris Club		
339.737	Currency areas. □ (73) Dollar area	339.972	Adjustment, harmonization, coordination of state economic policies
339.738	Currency unions. Monetary unions	339.976	Coordination of national economic plans. International coordination of plans. Community planning and plans
339.74	Foreign exchange policy. Instruments of foreign exchange policy	339.977	International economic forecasting. International economic forecasts
339.742	Treaties and agreements as instruments of foreign exchange policy. □ Currency agreements. Payment treaties. Treaties and agreements on balance of payments, settlement, clearance	339.98	Economic antagonisms. Economic nationalism. Economic war
339.743	Exchange rates. □ Fixed exchange rate. Fluctuating exchange rate. Floating	339.982	Economic antagonisms
	⇒ 336.748	339.984	Economic nationalism
		339.986	Economic war
339.743.053	Equilibrium of exchange rates		⇒ 339.548

34 LAW. JURISPRUDENCE

The method of classification of section 34 is independent of the legislative systems of particular countries with different philosophies of law; it provides instead an arrangement and hierarchy based on generalized legal principles. Only in this way can a classification be offered that is equally applicable to any country

Legal systems valid in only part of a state may be denoted by place auxiliaries (3/9) and the form auxiliaries for legal sources (094)

◇ 34-055.2 Women and the law

◇ 347.736(494)(094.73) Bankruptcy in Swiss cantonal law

34.0...	SPECIAL AUXILIARY SUBDIVISION FOR CLASS 34 LAW. JURISPRUDENCE	340.111.5	Legal relationships
	<i>The auxiliaries listed under 34.0... are not applicable at 341.12 and 348 and their subdivisions. At some classes (e.g. 347.951) they have been adapted or extended to suit the special subject matter</i>	340.112	Ontology and structure of the law
		340.113	Legal terminology and linguistic usage
		340.114	Fundamental ideas, sources and objectives of the law. Justice. Law and power
		340.115	Methodology of law. Juridical methods ⇒ 34.06
34.01	Nature of law. Scope of law	340.116	Subdivision and classification of law.
34.02	Conditions of existence of law		Disciplines, faculties, branches of law. Legal specialities
34.021	Precedence, priority of laws	340.12	Trends in theory of law. Schools, doctrines of law. Philosophy of law. Theory of state and law
34.023	Conditions relating to subject of law		⇒ 340.15
34.025	Conditions relating to object of law		Theocratic and theological doctrines and schools
34.028	Conditions relating to form of law. □ Officially prescribed formalities. Openness of legal procedures and legal acts to the public	340.121	Doctrine of natural law. Doctrine of law of reason
		340.122	Historical schools of law
34.03	Operation, workings of law. Effects of law	340.123	Positivist and normativist schools. Pure theory of law
		340.124	Sociological and biological schools. Legal pragmatism
34.037	Legal, judicial measures to protect workings of law	340.125	Totalitarian and fascist doctrines and schools
34.038	Delivering judgement. Territorial and facultative legal competence	340.126	Marxist-Leninist doctrine of state and law
		340.128	
34.04	Reformation of law. Alterations, modifications, changes to law	340.13	Positive law. Legal norm. Enacted law. Statute law. <i>Ius scriptum</i>. Operation of legal norm. Rules of its application
34.043	Transmission of law	340.130	Sources of positive law
34.045	Annulment of law. Extinction of law ⇒ 340.133	340.131	Authority, validity of law. Coercive force of law.
		340.132	□ Rule of law
34.046	Inception and duration of law		Operation, interpretation, application of legal norm
34.047	Reformation of law. Reorganization	340.132.1	With respect to persons. Knowledge of the law
34.048	Nullity. Invalidity	340.132.2	With respect to place. Territorial effect.
34.05	Comparison <i>Use to denote difference from or similarity to other legal institutions or procedures</i> ⇒ 340.5		□ Application of laws in several countries.
			Reception of laws into new countries
34.06	Questions of legal method and technique ⇒ 340.115	340.132.3	◇ 340.132.2(44:493) Reception of French law in Belgium
		340.132.4	⇒ 347.113
34.07	External organization of law. Administrative organs. Authorities <i>Auxiliaries 3.07... (as at 35.07...)</i>	340.132.6	With respect to time. □ Retroactive effect of laws
		340.133	Transitional provisions for the introduction of new laws
34.08	Legal personnel. Functionaries. Officials. Staff <i>Auxiliaries 3.08... (as at 35.08...)</i> ⇒ 347.96	340.134	Interpretation and application of law
			Explicit and tacit abrogation of laws. Repeal. Modification. Occasional rescission, waiving. Moratoriums. Dead letter
34.096	Individual cases, issues, points of law	340.135	Legislation. Policy and technique of legislation, codification, preparation, drafting, editing of laws
		340.136	⇒ 341.018
340	LAW IN GENERAL. LEGAL METHODS AND AUXILIARY SCIENCES	340.137	⇒ 342.52
			Emergency legislation. Suspension of valid laws
340.1	Types and forms of law	340.14	Compilation, editing of collections of laws
340.11	Nature, methodology, elements of law	340.141	Harmonization of laws. Unification of laws
340.111	Concept, characteristics, and definition of the law		Unwritten law. Non-statute law. <i>Ius non scriptum</i>. Other sources of law
			Common law. Custom. Usage
			⇒ 347.997
			⇒ 39

340.142	Judgement, judicial decision, as source of law. Judicial precedent	341.1/8	Public international law. Law of nations
340.143	Jurisprudence as source of law. Role of jurisprudence, lawyers, jurists, in establishing legal norm	341.1	Law of international organizations
340.147	International treaties as a source of law	341.11	World parliament. World government with supranational authority
340.149	Other sources ◇ 340.149:82-84 Legal maxims. Obiter dicta	341.12	International governmental organizations without supranational authority <i>The auxiliaries listed under 34.0... are not applicable at 341.12 and its subdivisions 341.12.01/.05 ≡ 06.01/.05</i>
340.15	Historic types of law and legal systems. □ Primitive laws <i>Denote form of government by :321...</i>	341.121 341.123	League of Nations United Nations Organization (UNO. UN) ◇ 341.123.02 Members of the UN ◇ 341.123.04 Organs, committees of the UN ◇ 341.123.042 General Assembly of the UN ◇ 341.123.043 Committees with executive and consultative power, e.g. UN Security Council ◇ 341.123.045.9 Committees of the General Assembly with executive and consultative power, e.g. UN Economic and Social Council
340.5	Comparative law ⇒ 34.05		
340.6	Auxiliary legal sciences. Forensic psychology. Forensic chemistry. Forensic medicine		
340.61	Morals. Marriage. Sexual instinct. Procreation ⇒ 343.54 ⇒ 347.6	341.16	Specialized agencies of world government
340.62	Offences against the health, life, integrity of the person. Homicide, suicide, injury ⇒ 343.6	341.17 341.171 341.172	Regional associations or unions of states Supranational regional associations of states Non-supranational regional associations of states Regional associations involving partial surrender of sovereignty by member states Associations with specialized functions ◇ 341.176(4) Council of Europe
340.63	Forensic psychiatry. Mental diseases, madness, insanity ⇒ 159.97 ⇒ 343.95 ⇒ 616.89	341.174 341.176	
340.64	Questions relating to living persons. □ Age. Sex. Identity. Distinctive features, peculiarities <i>For identification in criminal cases see 343.982.323</i> ◇ 340.64:577.212 DNA-matching (as evidence of identity). 'Genetic fingerprinting' ◇ 340.64:577.212:81'342.1 Phonological identification. Voice prints (as evidence of identity)	341.18 341.181 341.182 341.184 341.184.81	Legal bases of international conferences, congresses ⇒ 005.745 Conferences of heads of state or heads of government. Summit conferences Conferences of foreign ministers or their deputies. Conferences of ambassadors Conferences of departmental ministers <i>341.184 ≡ 354.</i> Conferences of ministers of economic affairs ◇ 341.184.81:061.1 EU Conferences of economics ministers of the European Union Legal bases of interparliamentary conferences
340.65	Pretence of illness. Deliberately induced diseases. Sickness used as excuse. Concealed, dissimulated illness. Illnesses attributed to others. Shamming. Simulation	341.185	
340.66	Forensic medical evidence, testimony, of specialists	341.2 341.21 341.211	Persons and things in international law Subjects of international law Sovereign states. State sovereignty ⇒ 342.3 Equal rights. Equality of states Surrender of sovereignty Semisovereign states. Protectorates. Mandated territories Unitary states. Federal states. Confederations ⇒ 342.24 Neutral states. Neutralised states, rendered neutral by treaty. Permanent neutrality ⇒ 341.35 Other subjects of international law Churches ◇ 341.215.1:272 The Holy See International unions. Nongovernmental international organizations Natural persons, individuals in international law. World citizenship ◇ 341.215.4-054.72 Immigrants. Emigrants Intergovernmental organizations ◇ 341.217(7/8) Pan American Union ⇒ 061.1
340.67	Forensic chemistry. Forensic toxicology ⇒ 343.347 ⇒ 614.3 ⇒ 615.9	341.211.3 341.211.7 341.212 341.213	
340.68	Scientific and technical questions of legal character	341.214	
340.69	Subject specialists, expert witnesses: relations with justice, authorities ⇒ 343.148 ⇒ 347.948	341.215 341.215.1 341.215.2	
341	INTERNATIONAL LAW	341.215.4	
341.01	Theory of international law, law of nations		
341.018	Codification, implementation of international law. Promulgation. Coercion. Sanctions ⇒ 340.134	341.217	

341.218	Formation, recognition, transformation, end of states ⇒ 321.013 ⇒ 342.28	341.231.1	Declaration of the rights and duties of states. Offences against the rights and duties of states
341.218.1	Formation of states	341.231.12	Legal bases of peaceful coexistence
341.218.2	Recognition of new states. Recognition of governments	341.231.14	Human rights ⇒ 342.7 ⇒ 343.4
341.218.3	State succession ⇒ 341.223.2	341.231.3	Servitudes
341.218.4	Transformation of states. Changes in national territory. Partition of states	341.231.4	Rights and restrictions in constitutional matters
341.218.6	Ending, dissolution of states	341.231.5	Rights and restrictions in freedom of association and free cooperation with other states. Right of withdrawal
341.22	Things and property in international law	341.232	International aid, assistance, support. Cooperation. Mutual aid. Development aid ⇒ 339.96
341.221	Territories of a state	341.232.1	Defence aid. Help against attack ⇒ 355.357
341.221.1	River territory. Rivers as territory	341.232.2	Financial aid and cooperation
341.221.2	Inland waters, lakes as territory. Sea, maritime territory. Territorial waters ⇒ 341.225 ⇒ 551.468 ⇒ 556	341.232.3	Economic aid and cooperation
341.221.23	Coastal waters	341.232.4	Administrative aid
341.221.25	Foreshore	341.232.5	Technical aid
341.221.26	Bays. Gulfs. Inlets	341.232.7	Intellectual, cultural, social cooperation
341.221.27	Contiguous zones	341.233	Right of intervention
341.221.28	Continental shelf	341.233.1	Right of collective intervention. Financial control. Trusteeship
341.221.3	Colonial territory. Dependent, protected, mandated territory. Hinterland, outlying regions	341.233.2	Right of supervision and advisorship towards underdeveloped countries. International trusteeship. Trust territories
341.221.4	Internationalization. Extraterritorialization. Neutralization	341.233.5	Rejection of foreign intervention. National competence. Self-sufficiency
341.222	National frontiers, boundaries. □ Natural frontiers. Artificial frontiers. Historic frontiers. Strategic frontiers	341.234	Rights of minorities
341.223	Acquisition, disposal, loss of territory	341.236	Responsibility of states for the actions of their officials and nationals
341.223.1	Seizure, occupation. Discovery	341.237	Assistance given by the state in cases of individual claims by its nationals against a foreign state or the citizens of that state.
341.223.12	Accession		Acquired rights
341.223.14	Temporary occupation	341.238	Relations between states. International relations ⇒ 327
341.223.2	Annexation		
341.223.3	Voluntary disposal of territory. □ Sale. Cession. Exchange. Donation. Leasing. Concession	341.24	International legal acts. International agreements. International treaties
341.223.4	Plebiscite. Territorial referendum	341.241	Treaties ⇒ 341.85
341.223.7	Peaceful penetration	341.241.2	Treaties of alliance. Alliances
341.224	Waterways. International waterways. □ Canals. Rivers. Lakes	341.241.3	Mutual aid treaties
341.225	The high seas. Open seas ⇒ 341.221.2 ⇒ 347.79	341.241.4	Nonaggression treaties, pacts
341.225.1	Regulation and jurisdiction on the high seas	341.241.5	Treaties of guarantee (of territorial status quo)
341.225.3	Nationality of ships	341.241.6	Agreement for mutual consultation
341.225.5	International law of the sea	341.241.7	Cultural agreements
341.225.8	Law of fishing at sea	341.241.8	Commercial, trading, economic treaties ⇒ 339.542
341.226	Air. Air space. Ether, airwaves. International law of the air	341.242	Origin, conclusion and ratification of treaties
341.226.2	Air space. Air corridors ⇒ 347.82 ⇒ 351.814.2	341.242.1	Treaty-making power
341.226.5	Airwaves. Broadcasting ⇒ 347.83	341.242.2	Treaty-making procedures
341.227	Earth's interior. Tunnels. Subterranean tunnels. Submarine tunnels	341.242.21	Preparation of treaties
341.228	Polar regions	341.242.22	Conclusion of treaties
341.229	Space. Cosmic space. Outer space ⇒ 347.85	341.242.24	Ratification of treaties
341.23	Rights and duties of states. Legal bases of international coexistence, cooperation	341.242.25	Registration of treaties
341.231	Autonomy. Independence. Right of self-determination. Right to regulate own constitution. Right of legislation. Right of self-administration. Right of diplomatic representation. Petition for autonomy ⇒ 341.43	341.242.27	Publication, proclamation of treaties
		341.242.28	Maintaining secrecy. Secret treaties
		341.242.5	Reservations. Restrictions. Saving clauses. Escape clauses
		341.243	Guarantee of treaties
		341.244	Effects of treaties. Execution of treaty obligations
		341.245	Alteration of treaties. Modification, renegotiation. Annulment, dissolution of treaties. Expiry of treaties
		341.29	Questions of international law in particular cases

341.3	Law of war. International legal relations in war ◇ 341.3.018 Codification of law of war	341.355.2	Acts by belligerents against neutral trade. □ Right of search. Embargo
341.31	Aggression. State of war, belligerence. Declaration of war	341.355.4	Intervention by neutral states. □ Convoys
341.311	Aggression. Definition of aggression	341.356	War loans by neutral states
341.312	Causes of war ⇒ 355.013	341.357	Nationals of neutral states in the territories of belligerent states. □ Property of neutrals
341.312.5	Legal justification, legitimacy of war. Just war (bellum iustum)	341.358	Violation of neutrality
341.314	Ultimatum	341.36	Laws of various kinds of warfare ⇒ 355
341.315	Declaration of war	341.361	Laws of land warfare 341.361 ≡ 341.32
341.317	War zone. Theatre of war	341.362	Laws of sea warfare, marine warfare
341.32	Conduct and direction of the war ⇒ 355	341.362.1	Privateering. Prize. Booty. Piracy
341.321	Persons and property involved in war	341.362.2	Undersea warfare. □ Submarine warfare.
341.322	Licit and illicit conduct of war. Means of hostilities	341.362.3	Underwater mines
341.322.5	War crimes	341.363	Naval bombardment. Naval blockade
341.322.6	Reprisals	341.364	Laws of aerial warfare ⇒ 347.82
341.323	Siege. Bombardment. Blockade	341.37	Laws of colonial warfare
341.324	Occupation of territories. Administration and exploitation of resources of occupied territories	341.373	Agreements during the war
341.324.5	Requisition of property	341.375	Ceasefire. Truce. Armistice
341.324.6	Requisition of labour. Directed labour	341.38	Capitulation. Surrender
341.325	Envoys. Negotiators. Parlementaires (bearers of truce flag)	341.382	End and consequences of war. Postwar period. Return to state of peace ⇒ 341.223
341.326	Treason. Espionage. Subterfuges of war ⇒ 343.321 ⇒ 343.322 ⇒ 355.40	341.383	Peace treaty
341.328	Effect of war on international treaties	341.384	Right of postliminy. Restoration of former legal status of persons and property in liberated territories
341.33	Treatment of wounded and sick. Geneva Convention. International Red Cross. Military hospitals, casualty stations. Hospital vehicles, ships, etc. ⇒ 356.33 ⇒ 614.885	341.385	Occupation costs. Exaction of costs.
341.34	Capture and treatment of prisoners of war, hostages. Internment ⇒ 355.257.7	341.386	Reparations. Restitution. Indemnity. War debt
341.341	Capture and treatment of prisoners of war in general ◇ 341.341-052 Prisoners of war	341.388	Sequestration of enemy property
341.344	Internment of military personnel in enemy countries	341.39	Liquidation of enemy monetary assets
341.345	Internment of military personnel in neutral countries	341.4	Special jurisdictions, e.g. for sequestration, liquidation. Various tribunals, courts of arbitration ⇒ 341.636
341.346	Internment of civilians. □ Internment of persons whose beliefs or acts are prejudicial to the occupying power. Political imprisonment (as war policy) ◇ 341.346-052 Political prisoners, internees in wartime ⇒ 343.261	341.39	Civil war
341.348	Taking and treatment of hostages	341.4	International criminal law
341.35	Neutral states during war ⇒ 341.214 ⇒ 341.233	341.41/46	Criminal offences against general, shared law
341.351	Duties of neutral states	341.41	Criminal offences committed abroad
341.352	Rights of neutral states	341.411	Territoriality principle. Scene of the offence (locus delicti). Territorial and extraterritorial operation of criminal law
341.353	Permanent treaty-based voluntary neutrality	341.412	Personality principle. Laws of offender's place of origin
341.354	Military vehicles on neutral territory, in neutral air space etc. ⇒ 341.221 ⇒ 341.224/226	341.413	Universality principle. Jurisdiction of courts over acts irrespective of territoriality and personality principles
341.355	Trade, commerce of neutral states. □ Contraband of war	341.42	Effect and execution of foreign judgements, verdicts, sentences in criminal law
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		341.44	Extradition of suspects or convicted persons
		341.45	Other interstate assistance in lawsuits. □ Tribunals of enquiry. Provisional detention. Rogatory letters. Execution of foreign judgements
		341.456	International police ⇒ 351.74 ⇒ 355.357

341.46	Subjects of international law. Culpable persons. Criminals	341.68	Armed peace
341.461	Legal persons. Bodies corporate. Collective entities in public law. States. Governments ⇒ 347.193	341.7	Diplomatic law. Laws of diplomacy
341.462	Natural persons. Individuals	341.71	Heads of state and representatives of state in foreign affairs. Diplomats generally. □ Ambassadors. Chargés d'affaires
341.462.1	State functionaries. □ Diplomats. Agents, representatives of the public. Government officials. Military personnel	341.72	Duties of diplomats. Prerogatives, competence, powers of diplomats
341.462.2	Private persons. □ Writers. Journalists. Learned persons, scientists	341.73	Rights and immunities of diplomats and their staff. □ Extraterritoriality. Diplomatic immunity
341.48	Offences, acts punishable under international law. □ Conditions for existence of offence	341.74	Beginning, suspension, end of diplomatic missions
341.485	Genocide. Murder of national, ethnic, racial groups	341.75	Diplomatic ceremonial, protocol, etiquette. Rank. Precedence
341.49	Prosecution of offences. □ Punishment. Penalties. Jurisdiction (national or international)	341.76	Diplomatic business. Negotiations. Diplomatic instruments, documents. Protocols. Diplomatic language ⇒ 341.18
341.6	International arbitration. International adjudication, jurisdiction	341.78	Representatives analogous to diplomats. Specialized representatives
341.61	International disputes in general	341.781	Diplomatic missions. Attachés. □ Trade, military, press attachés
341.62	Peaceful settlement of disputes	341.782	Officials of missions
341.623	Good offices. Advice. Counsel. Mediation	341.783	Officials of international organizations. □ UN officials
341.624	Commissions of inquiry. International observers	341.785	Armed forces stationed abroad in peacetime
341.63	International arbitration	341.79	Members of foreign armed forces serving outside their own country in peacetime
341.631	Arbitration treaties	341.792	Members of security forces belonging to member states of a supranational organization
341.632	Treaties, on any subject, with arbitration clauses	341.794	Members of the armed forces delegated to serve a supranational organization
341.633	Obligatory, invariable arbitration. Occasional, facultative arbitration	341.8	Consular law ⇒ 339.522
341.634	Reservation clauses in arbitration. Provisos	341.81	Consuls generally. □ Consuls-general. Professional consuls. Vice-consuls. Honorary consuls. Relations with home country and host country
341.635	Compromise	341.82	Legal status of consuls. □ Duties, prerogatives, competence, powers of consuls
341.636	Courts of arbitration ⇒ 341.388	341.83	Rights and immunities of consuls. Speciality and scope of duties of consuls
341.637	Arbitration procedures	341.85	Consular treaties and conventions ⇒ 339.542 ⇒ 341.24
341.638	Execution of sentences. Sanctions	341.9	International private law. Conflict of laws ⇒ 341.4
341.64	Administration of international justice	341.91	Schools, doctrines of international private law
341.645	International courts	341.92	Kinds of conflict
341.645.2	International Court of Justice	341.922	Conflicts between different systems of private law
341.645.5	Courts of international organizations	341.924	Conflicts in time. Intertemporal law. □ Conflicts mobiles (conflicts in space and time). Acquired rights
341.649	International procedures other than court organization. Judicial personnel in international justice 341.649 ≡ 347.91/98	341.93	Rules of conflict
341.65	Peaceful means of compulsion	341.932	Establishment of rules of conflict
341.651	Retortion in general. Retaliatory measures	341.94	Application of rules of conflict in international private law
341.652	Reprisal	341.942	Interpretation and classification
341.653	Embargo	341.944	Renvoi
341.654	Peaceful blockade	341.945	Public policy
341.655	Economic sanctions. □ Boycott	341.947	Abuse of law
341.66	Measures for executing sentence. □ Compensation. Reparation		
341.67	Disarmament. Permanent, durable peace. Limitation of 'A', 'B', 'C' weapons: atomic (nuclear); biological (bacterial); chemical. Nonproliferation. Limitation of strategic arms ⇒ 355.019		
341.675	Limitation and abolition of methods of mass destruction		
341.678	International control of manufacture and sale of arms, munitions		

341.948	Application of foreign law	342.37	Constitutional monarchy. Parliamentary government
341.95	Status of aliens, foreigners, in private law		
341.96	Particular legal relationships in international private law	342.38	Republic. Republican form of state
◇ 341.96:347.51	Legal liability in international private law	342.382	Historic republics
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		342.384.2	Republics with parliamentary democracy
		342.384.4	Soviet republics. People's democratic republics
		342.384.9	Other forms of republic
341.98	Competence. Procedure. Judgement	342.392	Authoritarian states
341.981	Competence, jurisdiction. Conflicts of jurisdiction	342.393	Corporative states
341.983	Procedure. Proof	342.395	Transitional forms of government
341.985	Judgements. Sentences. Appeals	342.395.2	Provisional governments
341.987	Foreign instruments. □ Extent of validity. Where writ runs. Legalization	342.395.3	Governments in exile
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342.725	Linguistic rights. □ National language. Mother tongue. Official, co-official languages. Administrative languages	342.811 342.812 342.813 342.814 342.815	Voting age Property Education. Capacity. Competence Nationality. Race. Religion Disfranchisement. Exclusion from franchise. Suspension of franchise. Restriction of franchise
342.726	Rights of particular groups of persons <i>Specify by -05... (Table 1k)</i>		Residential qualification. Residence period
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342.736	Right to petition. Right to express grievances		
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342.744	Educational obligation. Compulsory schooling	342.843.6	Second ballot. Tie-breaking, 'run-off' election.
342.745	Duty to protect social order, public property. National discipline. Civil discipline		Failure to gain absolute majority. Hung vote (ballotage)
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343.123.11	Statements (reports, minutes)	343.157.5	Nullity appeal to uphold lawfulness. Appeal in the interests of justice. Application for annulment. Protest by the public prosecutor to uphold legality
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343.196	Appeal courts for criminal cases	343.236.1	Accomplishment, commission of offence
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343.2	Criminal law proper	343.24	Punishments. Educational, instructional measures ⇒ 343.211
343.2.01	Theory, concept, basis of criminal law. □ Justification of penal treatment. Deterrent, reformative, retributive functions. The right to punish	343.241	Punishment. Penalties ⇒ 343.8
343.21	Criminal law in general. Application and efficacy	343.242	Apportionment of punishments and educational measures
343.211	Basic tenets, axioms of criminal law	343.244	Educational, instructional measures. □ Corrective training. Constructive sentence. Community service
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343.211.3	Protection of human rights ⇒ 343.41	343.25	Physical punishments: corporal punishment, capital punishment (death penalty)
343.211.4	Equality before the law. Guarantee of lawfulness and justice	343.253	Carrying out of the death penalty. □ Stoning. Beheading. Crucifixion. Hanging. Strangulation. Shooting. Electrocution
343.211.5	Right of civilians to participate in creating criminal justice	343.254	Punishment not imposed by a court. □ Lynching. Blood feud. Talion
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347.78.04	Performing rights	Particular average	
347.781	Translation	General average	
347.781.5	Exhibition. Showing	Damage from collisions, running foul	
347.781.6	Amendments to rights. Sale. Reservation of rights	Maritime credit. Bottomry	
347.781.8	Literature. The written and spoken word	Maritime mortgage, hypothecation, pledging	
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347.785	Spoken works. □ Lectures. Courses.	Navigation. Shipping lanes. Maritime traffic rules. Lights, signals	
347.786	Speeches	Shipwreck. Loss at sea. Rescue. Lifeboat service. Ownership of wreck and cargo remnants. Stranded goods, flotsam, jetsam	
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347.788	Graphic arts. □ Drawings. Engravings.	Shipping according to purpose. □ Merchant shipping. Fishery. Auxiliary shipping. Pleasure shipping: sport, holiday cruising etc.	
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347.821.5	Air transport between any two states		Urgent, emergency, procedure. □ Putting under seal. Inventory
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347.823.35	Flying permission	347.921.3	Appellants, claimants, plaintiffs. Appellees, defendants, respondents
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347.998.74	Arbitration commissions (for industrial accidents etc.) ⇒ 331.109.64	349.41	Law of land, property ⇒ 332.2 ⇒ 347.23/.26
347.998.8	Special law courts and tribunals. Special or extraordinary courts	349.412	Ownership of land ⇒ 332.2 ⇒ 347.235
347.998.82	Travelling courts	349.412.2	Acquisition, transfer, disposal and expropriation of land
347.998.85	Administrative tribunals	349.412.22	Acquisition of land ⇒ 332.22
347.999	Commercial courts. Consular tribunals	349.412.24	Transfer of land ⇒ 347.232.8
348	ECCLESIASTICAL LAW. CANON LAW. RELIGIOUS LAW (OPTION) <i>The main place for religious and canon law is in Class 2; class general works about religious laws and canon law at 2-428 and 2-74 respectively; class works about a specific faith with the faith, adding the appropriate hyphen auxiliary. If preferred, works on religious and canon law may be classed here, with details denoted by colon combination. Follow one or other option consistently</i>	349.412.26	Loss of land property
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349	SPECIAL BRANCHES OF LAW. MISCELLANEOUS LEGAL MATTERS	349.414	Rights and obligations of landowners Land use and public planning of land use <i>Details by :332.3...</i> ⇒ 347.233
349.2	Labour law ⇒ 331	349.415	Land protection
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	◇ 349.22:331.108.6 Disciplinary matters ⇒ 347.454	349.42	Agrarian law ⇒ 338.43 ⇒ 347.243 ⇒ 347.453.1 ⇒ 63
349.23	Working conditions	349.422	Organizations for agricultural production and distribution
349.232	Wages <i>Details by :331.2...</i>	349.422.2	Agricultural cooperatives
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349.24	Circumstances of work	349.442	Building law
349.243	Health and hygiene arrangements <i>Details by :331.4...</i> ⇒ 331.45 ⇒ 331.483	349.444	Law of occupancy, dwellings ⇒ 347.254 ⇒ 347.453 ⇒ 365.6
349.244	Working restrictions. □ Categories of people not allowed to work <i>Details by -05...</i>	349.6	Environmental protection law
		349.7	Law of atomic, nuclear power

35 PUBLIC ADMINISTRATION. GOVERNMENT. MILITARY AFFAIRS

The .07/.08 special auxiliaries listed here are also applicable in the other divisions of class 3, as indicated at 3.07 and 3.08

35.07	ADMINISTRATIVE ARRANGEMENTS. PUBLIC AUTHORITIES	35.071.2	Growth. Development
		35.071.3	Rights. Privileges. Immunities. Constitution. Charter. Statutes
35.071	Foundation. Development	35.071.5	Territory. Area of jurisdiction
35.071.1	Origin. Setting up. Establishment. Foundation. Inauguration	35.071.51	Extent. Boundaries 35.071.55 Subdivision. Division into districts, regions etc.

35.071.6	Centralization. Decentralization	35.075.2	Executive committees, bureaux, offices
35.072	Relations with other organizations	35.075.5	Departments. Divisions. Sections
35.072.1	Independence. Autonomy. Centralization. Decentralization	35.075.51	Secretariats. Chancelleries
35.072.2	Intervention of superordinate authorities. Administrative supervision, tutelage. Collaboration. State control	35.075.52	Financial branch
35.072.21	Measures concerned with administrative action. □ Authorization. Suspension. Repeal	35.075.6	Regulatory bodies, authorities, inspectorates. 'Watchdogs'
35.072.22	Overstepping of powers. Exceeding of authority. Infringements	35.075.7	Joint, shared, cooperative organs (of several authorities). Consultative committees
35.072.23	Appeal against administrative measures	35.076	Competence, sphere of authority. Duties, powers of authorities
35.072.3	Cooperation between authorities. Official assistance	35.077	Functions, official actions of authorities
35.072.4	Dissolution. Suppression by superordinate authorities	35.077.1	Office management. Conduct of official business
35.072.5	Replacement of one authority by another	35.077.2	Administrative actions. Disputes
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35.073.1	Consequences in civil law of public decisions and policies	35.077.91	Destruction and preservation of documents
35.073.5	Wealth. Property. Economic management. Income from rents, interest	35.077.92	Use of languages. Official language(s)
35.073.51	Property not designated for public service	35.078	Kinds of administrative activity
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35.073.513	Hiring. Renting	35.078.16	Restrictions
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35.073.515.3	Guarding and protection of property	35.078.4	Direct intervention
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35.073.527	Debts	35.078.8	Withdrawal, abolition of administrative control. Derestriction. Decontrol
35.073.53	Needs, requirements. Resources. Supplies. Procurement, purchasing	35.08	PERSONNEL OF AUTHORITIES. OFFICIALS. STAFF. PUBLIC OFFICE: THEORY, ORGANIZATION. OFFICIALDOM AS A WHOLE ⇒ -05 ⇒ 331 ⇒ 352/354 ⇒ 658.3
35.073.531	Equipment. Furniture	35.081	Distinctive marks and characteristics of public office
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35.073.534	Consultations. Advice	35.081.71	Legal position of staff
35.073.537	Transport	35.081.72	Discussions with staff. Joint consultations (with equal representation)
35.073.54	Public buildings and land. Space requirements	35.081.73	Staff participation in management, in decision-making
35.073.541	Government offices	35.081.74	Permanent staff representation. Consultative councils. Staff councils
35.073.542	Buildings for civil servants	35.082	Admission to public office
35.073.543	Technical and industrial establishments	35.082.1	Aptitude requirements. Examinations. Examining boards. Probationary period
35.073.544	Land		
35.073.55	Regulations. Orders. Complaints. Criticism. Incidents. Accidents		
35.073.6	Fees. Payment for services. Taxes. Levies. Duties		
35.074	Formation, composition, dissolution of authorities		
35.074.1	Election and nomination of members		
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35.075.1	Organs of the highest level. Representative bodies. Deliberative assemblies, councils, colleges		

35.082.2	Personal characteristics as conditions of admission. ☐ Age. Sex. Health. Nationality. Beliefs	35.088.6	Training of officials ⇒ 331.348 ⇒ 331.36
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35.082.6	Oath of office. Swearing in		◇ 351:69 Administration of building, construction
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35.083	Duties and obligations of officials. Regulations of office	351.711	Public properties. Crown lands. State property. Municipal property. Management of public properties (administrative questions only)
35.083.1	Authority of administration over its staff. ☐ Discipline	351.712	Public works. Contracts, contractors ◇ 351.712:625.7 Public road works and contracts
35.083.3	Place of service. Residence. Domicile	351.712.1	Authorization and ordering of public works
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35.083.5	Duty to exercise one's office, to perform the work		
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35.083.7	Duty of obedience, loyalty	351.712.2.031	Modifications. Supplementary work.
35.083.8	Professional confidentiality. Official secrecy ⇒ 343.452	351.712.2.032	Acceptance. Official inspection and approval
35.083.91	Obligations regarding titles, decoration, affiliations. ☐ Permission or prohibition from accepting foreign honours	351.712.2.033	Obligations of authorities. Rights of contractor
35.083.92	Secondary employment, outside official duties	351.712.2.043	Obligations of contractor. Rights of authorities. ☐ Delivery terms. Penalty clauses
35.083.93	Marriage: permission or prohibition	351.712.2.045	Cession, transference of contracts.
35.083.94	Prohibition from accepting gifts, gratuities		Subcontracting
35.083.95	Duties in wartime	351.712.2.045	Recission, termination, cancellation of contracts
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35.084.8	Composition of the body of officials	351.712.5	Expropriation in the public interest. Compulsory purchase ⇒ 332.26 ⇒ 347.234 ⇒ 349.412.28
35.085	Competence, powers of officials. Sphere of authority. Responsibilities. Official actions	351.712.6	Removal of hindrances in public works
35.086	Responsibility, answerability of officials	351.712.7	Associations of landowners for joint execution of public works. ☐ Dyke-building associations, polder boards
35.087	Rights of officials. Status. Position. Entitlement to privileges	351.713	Administration of matters concerning fiscal law, taxes, levies ⇒ 336.2
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35.087.42	Supplements to salary. Allowances	351.741	Organization of police. Police service
35.087.43	Retirement pensions. Disablement pensions. Other assistance		◇ 351.741:351.941 Civil supervision of police forces. Including: Police complaints authorities
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35.087.5	Rights of dependants	351.743	Civic guard, militia. National guard
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35.088.2	Working conditions. Workplace. Offices ⇒ 331.4		
35.088.3	Economic and social provision for officials ⇒ 331.34		

351.746	Other kinds of police	351.755.3	Population registers. Records of domicile, residence
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358.118	Tractor-drawn artillery. Tracked, caterpillar-tracked, artillery	359.21	Combat personnel. □ Naval gunners. Rifle marines. Torpedo-men
358.119	Armoured units. Tank corps	359.22	Seamen. □ Deck hands. Helmsmen. Topmen. Sailmakers. Pilots
358.13	Technical services for artillery. □ Armourers. Chemical warfare, gas warfare, personnel	359.23	Engine-room personnel. □ Naval engineers. Stokers
358.2	Engineers' corps. Special units ⇒ 623		
358.23	Various units of engineers		
358.231	Pioneers		
358.232	Sappers. Miners, minelaying corps		
358.233	Pontoneers. Bridging troops		
358.234	Transport corps: rail, road, water ⇒ 355.69		

359.3	Troops and other personnel on naval service, aboard ship or on coastal defence ◇ 359.3:355.353.5 Coastal defence ◇ 359.3:355.424.3-051 Signallers ◇ 359.3:356 Marine infantry ◇ 359.3:358 Marine artillery ◇ 359.3:629.5.02-051 Naval constructors. Naval architects	359.5	Naval administrative and auxiliary services: personnel, organization, administration and supervisory services ◇ 359.5:27-725-051 Naval chaplains ◇ 359.5:342.56 Naval justice and judicial service ◇ 359.5:61-051 Doctors. Medical personnel ◇ 359.5:614.88-051 Ambulancemen ◇ 359.5:615-051 Pharmacists ◇ 359.5:785.13-051 Bandsmen ◇ 359.5-057.15 Recruitment personnel ◇ 359.5-057.17 Administrative personnel ◇ 359.5-057.212 Superintendents. Supervisors
359.38	Naval aviator corps. Fleet air arm ⇒ 623.7		
359.4	Naval technical services		
359.41	Construction of hull and machinery		
359.42	Hydraulic engineering		
359.43	Naval artillery	359.51	Inspection of administration

36 SAFEGUARDING THE MENTAL AND MATERIAL NECESSITIES OF LIFE.

□ Social work. Social aid. Housing. Insurance

⇒ 351.84

364

SOCIAL WELFARE

A large table of special auxiliaries has been provided to allow for complex numbers to be built up. Some of the examples provided are very detailed and, consequently, lengthy. This is to indicate how complex concepts can be identified, if required.

It is recommended that literature on people suffering from social welfare problems be filed under the problems. Use of the common auxiliaries Table 1k-05 is largely restricted to indicating age groups, whilst colon combination should be used for types of condition or disability, or particular causes of social problems.

On occasions it is necessary to change the standard citation order. In particular, the age group of people suffering from specified problems should be cited before the type of welfare activity applied is indicated.

Examples of this are:

364.63-053.2-73:351.755 Registration of abused children.

364.696-053.9-783.442:648.5 Home help cleaning services for the elderly.

For details of administrative arrangements in general in social welfare the special auxiliaries at 3.07/.086 and 35.07/.088.7 should be used, pending completion of the new schedule for management

- ◇ 364-12:316.323.6 Welfare capitalism
- ◇ 364-12:328.182 Interest group theory
- ◇ 364-17:141.82 Marxist critiques of welfare
- ◇ 364-17:329.11 Conservative critiques of welfare
- ◇ 364-3:061.1 State welfare agencies
- ◇ 364-3:061.1(100) International welfare agencies. Inter-gouvernemental welfare agencies
- ◇ 364-3:2 Religious welfare agencies
- ◇ 364-3-027.563 Commercial welfare agencies
- ◇ 364-3-027.564 Non-commercial welfare agencies. Charitable welfare agencies
- ◇ 364-322(100)OXF Oxfam
- ◇ 364-43-021.311 Generic social workers
- ◇ 364-43-021.321 Specialist social workers
- ◇ 364-47-058.833 Married partners as carers

- ◇ 364-47-058.837 Partners as carers. Co-habitees as carers
 - ◇ 364-624:336.23 Financing of welfare services from fees and contributions
 - ◇ 364-624:339.726.2 Financing of social welfare by overseas aid
 - ◇ 364-624:347.472 Financing of social welfare from gifts, donations
 - ◇ 364-646.6:336.72 Cooperative savings schemes
 - ◇ 364-646.6:336.773 Credit unions. Cooperative credit schemes
 - ◇ 364-72-27 Targeting of welfare provision
 - ◇ 364-781:316.472.4 Relationship between clients and workers
 - ◇ 364-781.3:303.621.32 Assessment of welfare client by interview
 - ◇ 364-781.3:303.63 Assessment of welfare client by observation
 - ◇ 364-781.3:303.643.2 Assessment of welfare client by analysis of written material
 - ◇ 364-781.8:616-036.83 Closure of case on recovery of client
 - ◇ 364-781.8:616-036.88 Closure of case on death of client
 - ◇ 364-782-022.326.5 Emergency intervention. Crisis intervention
 - ◇ 364-783.2"405" Long-term residential care
 - ◇ 364-787.4:373.22 Provision of crèches
 - ◇ 364-787.6:621.395.6 Provision of telephones
 - ◇ 364-787.6:654.924 Provision of alarms
- ⇒ 376

364-1

Theories of social welfare

364-11

Theories of causation

364-112

Social disorganization theory

364-12

Schools of thought

364-122

Diffusion theory

364-124

Welfare industrialization theory

364-126

Welfare convergence hypothesis. Welfare consensus

364-14

Representational conceptions. Typologies of welfare

Use alphabetical extension for the authors of typologies not listed

364-142	Residual conceptions	364-48	Advocates
364-144	Mixed conceptions. Welfare pluralism	364-482	Guardians ad litem
364-146	Institutional conceptions	364-5	Social welfare facilities
364-146.2	Welfare state	364-52	Assessment centres
364-15	Normative theories	364-53	Day centres. Drop-in centres
364-152	Individualist approach. Enterprise approach	364-54	Homes
364-154	Communitarian approach. Reformist approach	364-55	Sheltered accommodation
364-156	Collectivist approach. Structural approach	364-56	Refuges
364-17	Critical theories	364-57	Rehabilitation centres
364-172	Inverse care law	364-58	Hospices
364-174	Dependency culture	364-6	Contributions and payments
364-176	Poverty trap	364-62	Amounts payable. Premiums. Contributions
364-18	International welfare theories	364-622	Premium levels payable
364-182	Dependency theory	364-624	By whom payable. Social insurance liability.
364-184	World systems theory		Financing of social benefits. Financing of social welfare
364-186	Reciprocity	364-626	Redemption of premiums
364-2	Principles of assistance	364-64	Amounts receivable. Benefits receivable. Assistance receivable
364-21	Aims of assistance	364-642	Levels receivable
364-212	Preventive	364-644	By whom receivable
364-214	Remedial	364-644.2	Benefits to insured individual
364-216	Promotional	364-644.4	Benefits to family of insured
364-22	Persons as recipients of assistance.	364-644.6	Benefits to surviving dependants of insured
	Insured persons. Service users. Welfare clients	364-646	Types of benefits. Types of grants
364-23	Rights to assistance	364-646.2	Financial benefits. Financial aid. Financial grants. Pensions
364-232	Preconditions for assistance		⇒ 316.43
364-232.2	Regular payment of contributions as precondition for assistance	364-646.3	Benefits in kind
364-234	Restriction(s) on assistance	364-646.4	Reimbursement of costs
364-234.2	Repayable assistance. Returnable assistance.	364-646.6	Cooperative financial welfare
	Loans	364-646.8	Fiscal welfare. Tax system welfare
364-236	Requirements for assistance	364-7	Social welfare operations. Welfare effort. Social services. Service provision
364-25	Exclusions from assistance, contributions. Refusal of assistance. Exemption	364-72	Policy-setting activity
364-26	Influences on levels of assistance or contributions	364-72.2	Policy transfers
364-262	Ensuring of minimum subsistence level	364-73	Executive activity. Social administration
364-262.2	Definition of minimum subsistence level.	364-734	Review
	Poverty line	364-78	Social work. Work applied to social problems
364-264	Index linking	364-781	Case work. Clinical social work
364-266	Influence of further income, pensions, benefits on levels of assistance	364-781.1	Client referral
364-27	Targeting of assistance	364-781.12	Self-referral
364-28	Consequences of assistance	364-781.14	Third party referral
364-3	Social welfare agencies. Social welfare organizations	364-781.2	Assessment of risks
	<i>Use common auxiliaries Table 1e to denote the geographical scope of the organization's activities. To denote particular welfare agencies use alphabetic extension</i>	364-781.22	Risk to self
364-32	Voluntary agencies	364-781.24	Risk to others. Dangerousness
364-322	Volunteer bureaux	364-781.3	Assessment of needs
364-34	Co-operative welfare agencies	364-781.4	Statement of needs
364-36	Self-help welfare agencies	364-781.5	Allocation of care manager
364-362	Support groups	364-781.6	Package of services. Care plan
364-4	People as providers of social welfare assistance	364-781.7	Review of needs and services
364-42	Individuals themselves. Self help	364-781.72	Case conferences
364-43	Social workers. Social welfare workers	364-781.8	Closure of case. Disengagement
364-45	Care workers	364-781.82	Client transfer. Onwards referral
364-46	Visitors	364-781.9	Follow-up work
364-47	Carers	364-782	Intervention
		364-782.2	Removal of clients from normal environment
		364-782.4	Placement of clients
		364-782.42	Fostering
		364-782.44	Adoption
		364-783	Care work
		364-783.2	Residential care. Institutional care
		364-783.4	Non-residential care. Community care

364-783.42	Day care	364.3	Social benefits. Social allowances. Social insurance. Social assistance
364-783.44	Home care		<i>Social insurance systems vary from one country to another, especially relating to social assistance. The classification cannot reflect every variant. For private insurance see 368.9</i>
364-783.442	Home help. Personal social services		◇ 364.3:343.72 Abuse of social insurance. Social insurance fraud
364-784	Advisory work. Welfare information services		◇ 364.3-021.146.4 Lack of social insurance. Uninsured people
364-785	Therapeutic work. Therapy. Counselling ⇒ 615.85		◇ 364.3-23-021.311 Universal right to assistance
364-785.1	Aims of therapy		◇ 364.3-23-021.321 Special assistance. Discretionary assistance
364-785.12	Conciliation		◇ 364.3-232:336 Financial preconditions for assistance. Means testing
364-785.14	Social adjustment		◇ 364.3-232:347.171 Residence preconditions for assistance
364-785.2	According to number of people		◇ 364.3-232"4" Waiting period for assistance
364-785.22	Individual therapy. Individual counselling		◇ 364.3-234"4" Restrictions on duration of assistance
364-785.24	Group therapy		◇ 364.3-234-053 Age restrictions
364-785.4	Types of therapy		◇ 364.3-236:351.755 Requirement to register for assistance
364-785.6	Methods of therapy		◇ 364.3-28:330.526.5 Redistribution of wealth as consequence of social insurance
364-785.62	Behaviour therapy		◇ 364.3-622:331.215 Contributions based on levels of pay
364-785.64	Experience therapy		◇ 364.3-622:347.171 Influence of residence on contributions. Including: Living abroad
364-786	Rehabilitation work. Integration work		◇ 364.3-622-022.231 Minimum premiums payable
364-786.2	De-institutionalization		◇ 364.3-622-022.235 Maximum premiums payable
364-786.4	Occupational rehabilitation. Employment rehabilitation		◇ 364.3-622-264 Index-linked contributions
364-787	Support work		◇ 364.3-624:336.22 Financing of social insurance from general taxation. Public financing
364-787.2	Social support		◇ 364.3-624:339.13.027 Financing of social insurance from subsidies
364-787.22	Support for individuals		◇ 364.3-624-057.13 Amounts payable by self-employed persons
364-787.222	Visiting services		◇ 364.3-624-057.15 Amounts payable by employers
364-787.224	Sitting services. Minding		◇ 364.3-624-057.16 Amounts payable by employees
364-787.226	Outings. Trips		◇ 364.3-642-262 Amounts needed for subsistence
364-787.228	Respite		◇ 364.3-642-264 Index-linked benefits. Benefits based on cost of living
364-787.24	Support for families		◇ 364.3-642-62"4" Amounts receivable based on contribution period
364-787.242	Arrangements for family meetings or visits		
364-787.26	Support for communities. Community work. Neighbourhood work		
364-787.3	Emotional support. Emotional welfare		
364-787.32	Protection		
364-787.34	Victim support		
364-787.4	Employment support		
364-787.42	Sheltered work schemes		
364-787.422	Sheltered workshops		
364-787.424	Sheltered placements		
364-787.5	Mobility support. Help in meeting mobility needs		
364-787.52	Disability access provision		
364-787.522	Adaptations		
364-787.54	Mobility schemes		
364-787.6	Technical support		
364-787.7	Educational support. Educational welfare services		
364-787.8	Activity to promote opportunity. Equal opportunities activity		
364-787.82	Advocacy		
364-787.84	Positive action. Positive discrimination		
364-787.842	Quota systems		
364-787.86	Social and political mobilization		
364-787.862	Conscientization		
364-787.9	Health support. Health welfare services		
364-788	Provision of emergency shelter <i>Class here situations where emergency shelter is required. For homelessness see 364.682.42</i>		
364.2	Basic human needs. Requirements for well-being <i>For specific problems (previously at this number) see 364.6</i>	364.32	Insurance against incapacity to work.
◇ 364.2:159.9	Psychological needs		□ Health insurance
◇ 364.2:159.942	Emotional needs		◇ 364.32-22:613 State of health of insured persons
◇ 364.2:336	Financial requirements		◇ 364.32-232:61 Medical preconditions for assistance. Medical examination
◇ 364.2:37	Educational needs		Accident insurance
◇ 364.2:61	Health needs. Medical needs		Invalidity insurance
◇ 364.2:613.2	Nutritional requirements. Need for food		◇ 364.324-642:364.694-022.1 Levels of invalidity allowance receivable based on degree of disability
◇ 364.2:613.31	Need for drinking water		◇ 364.324-642-26:336.72 Influence of savings on levels of invalidity allowance
		364.326	Parental insurance. Maternity, paternity insurance
		364.334	Unemployment insurance
			◇ 364.334-642-26:347.627.4 Influence of remarriage on levels of unemployment allowance

- 364.35 Old age insurance. Pensions insurance**
 ◇ 364.35-642-053 Amounts receivable linked to age
 ◇ 364.35-642-26:347.627.1 Influence of death of partner on pension level
 ◇ 364.35-646.2:662.6/.9 Grants for pensioners to meet heating costs. Grants for pensioners to meet fuel costs
 ◇ 364.35-646.3:615 Free prescribed medication for pensioners
- 364.37 Surviving dependants' insurance**
- 364.38 Child benefit insurance**
 ◇ 364.38-646.3:637.14 Free milk for children
- 364.39 Minimum income insurance**
 ◇ 364.39-64-236:331 Requirement to work. Requirement to provide service in return for minimum income allowance
- 364.4 Fields of practice in social work**
For kinds of relief service (previously at this number) see 364-78
For general works only. For particular actions applied to problems use special auxiliaries under 364-7 with problems under 364.6. Use common auxiliaries Table 1k-05 to denote work with specific age groups and/or genders or colon combination to denote work with people with particular problems, or community group work
 ◇ 364.4:159.9 Work with people with mental health problems
 ◇ 364.4:316.334.55/.56 Community welfare work
 ◇ 364.4:316.334.55 Work with rural communities
 ◇ 364.4:316.334.56 Work with urban communities
 ◇ 364.4:331 Labour welfare work
 ◇ 364.4:343.265.2 Probation services
 ◇ 364.4:61 Medical social work
 ◇ 364.4:614.8.067.3 Work with people disabled as a result of accidents
 ◇ 364.4:616.899 Work with people with learning difficulties
 ◇ 364.4-022.326.5 Emergency relief services. Disaster relief services
 ◇ 364.4-05.5/.7 Children and family services
 ◇ 364.4-053.2 Child welfare. Work with children
 ◇ 364.4-053.6 Youth welfare
 ◇ 364.4-053.9 Elderly welfare services. Work with the elderly
 ◇ 364.4-616-036.86 Work with people disabled as a result of illness
- 364.6 Social welfare issues.** □ Social problems
For persons involved in social welfare (previously at this number) see 364-4
Social issues are denoted, under each heading, in a range of identifiable concepts indicating the absence or presence of ideal conditions. To indicate degrees, if these are not represented here, use the common auxiliaries Table 1k-02. Particularly relevant are:..
 -021.142 Existence. Non-existence.
 -021.146 Presence (or not).
 -021.146.2 Present. At hand.
- 021.146.4 Absent. Lacking.
 -021.161 Available. Unavailable. Extant.
 -021.331 Whole. Complete. Full. Entire. Comprehensive.
 -021.332 Partial. Incomplete.
 -021.333 Ideal. Perfect.
 -021.351 Limited. Limits. Unlimited.
 -022.12 Light.
 -022.15 Moderate.
 -022.18 Excessive. Severe.
 -022.251 Rare. Rarity.
 -022.252 Dearth. Shortage. Insufficient. Uncommon.
 -022.253 Adequate. Sufficient.
 -022.255 Abundant. Plenteous. Common.
 -022.257 Excess. Extreme. Excessive.
 -022.258 Acute. Chronic.
In order to prevent separation of related subjects, use colon combination to indicate people suffering from particular problems as well as the issues themselves, but use Table 1k-05 to denote the age groups and/or gender. For problems applying to particular ethnic groups use common auxiliaries Table 1f. To keep international problems separate from problems at individual/community level, intercalate place auxiliaries (Table 1e) after 364. For causation use -11 (theories of causation) from the special auxiliaries listed at 364
 ◇ 364(1-773).664 Debt in developing countries
 ◇ 364.664:331.56 Debt problems of unemployed people
 ◇ 364.682.42-11-053.6 Causes of teenage homelessness
- 364.61 Opportunity**
 Social injustice. Social deprivation
 ◇ 364.612(1-21)-78-22 Preventive social work applied to urban deprivation, inner-city deprivation
 ◇ 364.612(1-21)-787.6:061.2 Provision of social clubs in inner-city areas
 ◇ 364.612(1-21)-787.6:791.7 Provision of recreation grounds, playgrounds in inner-city areas
 ◇ 364.612(1-22)-78-24 Remedial social work applied to rural deprivation
 ◇ 364.612(1-22)-787.6:725.835 Provision of community centres, village halls in rural areas
 Disadvantage. Social disadvantage. Underprivilege
 ◇ 364.614:323.1 Social disadvantage due to racism
 ◇ 364.614:81'243 Social disadvantage due to language barriers
 ◇ 364.614:323.12 Acculturation work
 ◇ 364.614'234 Promoting opportunity through social pedagogy
 Special needs
 ⇒ 37.614.2
 Equal opportunity. Social justice
- 364.62 Psychological welfare**
For helpers, welfare workers (previously at this number) see 364-4
 ◇ 364.62-47-053.2 Psychological welfare of children as carers
 ⇒ 159.9

364.622	Psychological problems. Mental health problems ◇ 364.622:331.56 Psychological problems of the unemployed ◇ 364.622-53 Day centres for people with mental health problems ◇ 364.622-73-53:005.585 Evaluation of day centres for people with mental health problems ◇ 364.622-73-54:005.584.1 Inspection of mental health residential homes ◇ 364.622-73-54:005.584.1-022.345.2 Regular inspection, announced inspection of mental health residential homes ◇ 364.622-73-54:005.584.1-022.345.4 Irregular inspection, unannounced inspection of mental health residential homes ◇ 364.622-78-73:005.334.3 Mental health social services complaints handling ◇ 364.622-78-734-022.326.5 Mental health critical incident review ◇ 364.622-782.2 Removal of the mentally ill into care. Sectioning of the mentally ill ◇ 364.622-783.4-73:001.891 Monitoring of community care services for people with mental health problems ◇ 364.622-785.4:634 Horticulture therapy for people with mental health problems ◇ 364.622-785.4:792 Drama therapy for people with mental health problems	364.636	Bullying ◇ 364.636-784:654.15 Telephone advice lines, help lines for victims of bullying
		364.64	Interpersonal relationships. Family relationships
		364.642	Relationship problems. Problems within families ◇ 364.642:316.811.1 Marital discord ◇ 364.642:316.813.5 Welfare problems associated with divorce
		364.642.2	Co-dependency
		364.642.4	Abandonment ⇒ 343.623
		364.642.5	Running away
		364.642.6	Missing persons
		364.642.8	Bereavement
		364.65	Social conditions. Social issues <i>For recipients, persons in need of help (previously at this number) see 364-22</i>
		364.652	Social problems ◇ 364.652:314.117.3 Problem families ◇ 364.652:331.56 Unemployment as a welfare issue. Problems of unemployed people ◇ 364.652:34 Problems in relation to the law
		364.652.2	Social breakdown. Social insecurity ◇ 364.652.2:338.124.4 Social breakdown associated with economic crises, slumps ◇ 364.652.2:355.01 Social breakdown associated with war
364.624	Emotional problems ⇒ 159.942	364.652.4	Problems associated with social change ◇ 364.652.4:330.341.424 Problems associated with industrialization ◇ 364.652.4:330.341.426 Problems associated with de-industrialization ◇ 364.652.4:911.375.1 Problems associated with urbanization
364.624.2	Role adjustment ◇ 364.624.2:331.106.47 Emotional problems associated with retirement ◇ 364.624.2:618.4 Emotional problems associated with childbirth	364.654	Issues associated with social structure, social make up
364.624.4	Social isolation. Loneliness ◇ 364.624.4:314.151.3 Social isolation of migrants ◇ 364.624.4:316.624.2 Isolation due to asocial behaviour ◇ 364.624.4-787.222:343.26 Prison visiting services	364.658	Social well-being. Secure social conditions
		364.66	Financial resource issues ⇒ 336
364.624.6	Fear. Anxiety ◇ 364.624.6:364.63 Fear of abuse. Fear of ill-treatment ◇ 364.624.6:612.67 Fear of death. Distress of the dying ◇ 364.624.6:612.884 Fear of pain	364.662	Poverty ◇ 364.662:331.56 Poverty due to unemployment ◇ 364.662-54 Almshouses. Workhouses ◇ 364.662-646.2:339.726.2 Foreign financial aid to alleviate poverty ◇ 364.662-78:316.43 Social development welfare. Applied social development
364.628	Psychological well-being. Emotional security	364.664	Debt
364.63	Abuse. Ill-treatment. Risk of abuse ◇ 364.63:316.347 Racial abuse ◇ 364.63-027.553 Domestic abuse ◇ 364.63-028.16 Verbal abuse ◇ 364.63-053.2-73:351.755 Registration of abused children ◇ 364.63-053.2-782.2 Removal of abused children into care	364.668	Wealth. Financial well-being
		364.67	Educational welfare ◇ 364.67:373-021.161:336 Educational opportunity subject to financial resources ◇ 364.67:373-021.161:347.156 Educational opportunity subject to gender ◇ 364.67:373-021.161-053 Educational opportunity subject to age ⇒ 37
364.632	Physical abuse. Violence ◇ 364.632:616.89-008.441.45 Self harm, self mutilation as a welfare issue ⇒ 343.615	364.672	Educational welfare problems
364.633	Sexual abuse ⇒ 343.541	364.672.2	Illiteracy
364.634	Harassment. Emotional abuse	364.68	Environmental welfare
364.635	Neglect ⇒ 343.623	364.682	Environmental welfare problems ◇ 364.682:551.577.38 Welfare problems resulting from drought
364.635.2	Self neglect		

	◇ 364.682-787.9:641.16:551.577.38 Provision of drinking water to drought victims	365	ACCOMMODATION DESIRES AND THEIR FULFILMENT. SAFEGUARDING ACCOMMODATION
	◇ 364.682-788:685.531:556.166 Provision of tents as emergency accommodation for flood victims		⇒ 159.922.26
364.682.4	Housing problems		⇒ 331.342
	⇒ 365		⇒ 351.778.5
364.682.42	Homelessness. Destitution		⇒ 364.682.44
364.682.44	Unsatisfactory accommodation. □ Squalid accommodation. Slums		⇒ 643/645
	◇ 364.682.44:621.311-021.146.4 Accommodation lacking power		⇒ 69
	◇ 364.682.44:644.61-021.146.4 Accommodation lacking drinking water	365.2	⇒ 711
	◇ 364.682.44-787.9:644.61 Provision of drinking water supply	365.22	⇒ 728
364.682.442	Overcrowded accommodation		
364.682.444	Poorly constructed accommodation. □ Shanty towns	365.24	Human need for housing
364.69	Health welfare		Social requirements made of housing. Minimum social requirements for the suitability of a residence
	◇ 364.69:616-08-021.161 Availability or unavailability of medical treatment		Possibility of acquiring housing in view of the state of the housing market
	◇ 364.69-787.9:613:37 Provision of health education services	365.242	⇒ 332.85
	⇒ 61	365.244	⇒ 339.132/133
364.692	Health welfare problems	365.246	⇒ 351.778.5
	<i>Use common auxiliaries Table 1k-02 for severity of condition, e.g. -022.15 moderate, -022.18 excessive, severe</i>		Housing surplus
	◇ 364.692:615.2:615.015.6 Drug addiction as a welfare problem	365.26	Housing requirement met
	◇ 364.692:616-006 Welfare problems of people suffering from cancer	365.262	Housing deficiency. Housing shortage
	◇ 364.692:663.51:615.015.6 Alcoholism as a welfare problem	365.262.1	
	◇ 364.692:665.931:615.015.6 Substance abuse as a welfare problem	365.262.2	Housing acquisition
	◇ 364.692-787.9:641:613.391 Provision of food to the starving. Nutritional support	365.262.3	By own initiative
364.694	Physical disability	365.262.4	By purchasing
	◇ 364.694:614.8 Welfare problems of people disabled as a result of accidents...	365.262.6	By renting
	◇ 364.694:616.899 Welfare problems of people with learning difficulties	365.264	⇒ 347.453.3
	◇ 364.694-783.442:646.7 Personal care services for disabled people. Including: Washing, dressing, undressing		By building one's own home
	◇ 364.694-787.4:377 Training for disabled people	365.27	By offering services
	◇ 364.694-787.522:692.65 Provision of access ramps	365.28	By illegal occupation. By squatting
	◇ 364.694-787.54:636.044:617.751.98 Provision of guide dogs for the blind	365.4	Through placement by social services and social organizations. □ On socio-medical grounds. On merit, e.g. war veterans, valuable work
	◇ 364.694-787.54:656.015 Special parking schemes for disabled drivers	365.42	Housing adaptation. □ Because of change in marital status, social status
	◇ 364.694-787.54:656.132 Special bus services for the disabled	365.44	Enabling acquisition of suitable housing.
	◇ 364.694-787.54:656.132-027.565 Free bus travel for disabled people	365.45	□ By guaranteed mortgages and guarantees. By subsidized rents. By rent allowances
364.696	Physical frailty. Mobility difficulties	365.46	Housing desires
	◇ 364.696-053.9-783.442:635 Gardening services for the elderly	365.47	Relative to the location of the house within the estate structure and the site
	◇ 364.696-053.9-783.442:648.5 Cleaning services for the elderly	365.48	Relative to the location within the building
	◇ 364.696-053.9-783.442:698.1 Decorating services for the elderly	365.6	Relative to size, layout and type of construction
364.696.2	Bedridden condition	365.62	Relative to fixtures and fittings, communal provisions, services
364.696.4	Housebound condition	365.622	Relative to possible use and period of use
	◇ 364.696.4-783.44:641.5 Food preparation services for the housebound. Cooking services	365.623	Relative to the quality of the housing
		365.624	Security of the accommodation
		365.626	⇒ 349.444
			Threats to occupancy
			Through notice to quit
			Through rent disputes
			Through becoming uninhabitable. □ Neglect and disrepair. Through catastrophes
			Through expropriation. □ As a result of changed plans for redevelopment or traffic routes

365.64	Protection of the occupier	366.534	Consumer organization with government support
365.642	Through rent control. Sitting tenancy.	366.536	Consumer organization with business support
	□ Through subtenant protection. Through involvement of the rent tribunal committees	366.538	Consumer organization with mixed support
	⇒ 347.453		
	⇒ 351.778.542	366.54	Government consumer policy. Consumer protection by government
365.644	Through restoration or improvement of habitability. □ Renovation of old building	366.542	Consumer protection by legislative means
365.646	Through appeals against public measures	366.543	Special means of consumer redress provided by law
365.648	Through allocation of another residence	366.543.2	Small claims court
366	CONSUMERISM	366.543.4	Consumer 'ombudsman'
	<i>Denote consumers according to personal characteristics by -05... (Table 1k)</i>	366.544	Quality control as a tool of consumer policy
	◇ 366-053.2 Children as consumers	366.546	Liability as a tool of consumer policy
	◇ 366-058 Consumers according to social class and family circumstances	366.546.2	Product liability. Manufacturer's liability.
		366.546.4	Warranties. Guarantees
			Seller's liability
366.01	Consumer theory	366.56	Voluntary redress of consumer complaints and self-regulation by business
366.02	Role of the consumer in economy and society, and its history	366.6	Consumer information: information on behalf and in defence of consumer interests
366.1	Consumer behaviour		
366.12	Existing consumer behaviour. Buying and shopping behaviour	366.61	Consumer counselling
			◇ 366.61.07 Consumer advice centres
366.14	Changing the consumer behaviour. Consumer education	366.612	By public authorities
		366.613	By semipublic authorities
366.2	Consumer finance. Consumer expenditure	366.614	By organizations under private law
		366.615	By producers
366.4	Consumer hazards	366.62	Collection of information about goods and services
366.42	Consumer hazards connected with economic systems	366.622	Collecting the opinions of consumers
		366.624	Testing of goods and services
366.43	Consumer hazards from the production of products	366.624.2	Testing the individual product
366.432	Consumer hazards from lack of product variation or from product fixing	366.624.4	Comparative testing
366.433	Consumer hazards from unnecessary variation, unnecessary differentiation	366.626	Consumerist investigations of industries and service sectors
366.435	Consumer hazards from planned obsolescence		<i>Specify by :...</i>
366.436	Consumer hazards from inferior-quality products		◇ 366.626:661.12 Investigation of the pharmaceutical industry
366.438	Consumer hazards from dangerous products, lack of product safety	366.63	Communication levels for consumer information
		366.632	Personal communication. Interpersonal communication
366.44	Consumer hazards from the pricing of products	366.634	Group communication
	<i>Denote macroeconomic and mesoeconomic pricing concepts, such as monopoly pricing, by :338.5...</i>	366.636	Mass communication. Use of the mass media
		366.636.2	Consumerist comparative advertising
		366.636.4	Corrective advertising
366.46	Consumer hazards from sales or payment conditions, high-pressure salesmanship, sales rackets	366.64	Informative labelling
	<i>Specify by :658.8</i>	366.641	Quantity labelling
		366.641.2	Size labelling
366.48	Consumer hazards from misleading packaging or labelling	366.641.4	Quantity content labelling
	⇒ 343.532	366.641.6	Weight labelling
		366.641.62	Net weight labelling
366.482	Misleading size or shape of package	366.641.64	Drained weight labelling
366.484	Misleading information on packages, wrappers, labels	366.642	Composition labelling. Ingredients lists
		366.643	Nutrition labelling
366.484.2	Illustration of product	366.644	Performance labelling
366.484.4	Description of product	366.645	Price labelling. Unit price labelling
366.484.5	Claims for product	366.646	Date marking
		366.646.2	Production date
		366.646.4	'Sell by' date
		366.646.6	'Use by' date
366.5	Consumer protection. Types of consumer organization	366.647	Safety labelling. Warning labels
		366.648	User instructions
366.52	Self-protection by individual consumers	366.648.2	Indication on the product
		366.648.4	Information accompanying the product
366.53	Consumer organization		
	◇ 366.53(100) International consumer organization	366.65	Quality of consumer information
		366.652	Testable. Verifiable
366.532	Independent consumer organization	366.654	Not testable. Unverifiable

366.66	Effect of consumer information	368.025.12	Special kinds of premium. ☐ Investment premium. Risk premium. Natural premium
366.7	Consumer action in defence of consumer interests	368.025.13	Calculation of premium
366.72	Consumer complaints	368.025.14	Alteration of premium during term of insurance
366.74	Consumer lobbying and representation ◇ 366.74:35 with government	368.025.15	Refund of premium. Cancellation rebate. Indivisibility
366.76	Direct action	368.025.16	Payment of premiums. ☐ In advance, in arrears. Days of grace
366.762	Whistleblowing	368.025.17	Duration and frequency of premium payments
	<i>Drawing attention to abuses</i>	368.025.18	Making of payment. ☐ Onus. Place. Method: cash, cheque, credit transfer etc.
366.764	Picketing	368.025.2	Object of insurance: persons or things at risk of damage or injury
366.766	Boycotts	368.025.3	Object of insurance: property or item of property at risk of loss
368	INSURANCE. COMMUNAL PROVISION THROUGH SHARING OF RISK	368.025.4	Insured value. Valuation
368.01	Theory. Scientific principles of insurance. Actuarial theory	368.025.5	Sum assured
368.013	Kinds of insurance in general. Comparative studies. ☐ Insurance of certain risks. Assurance of certain sums	368.025.51	Full assurance. Partial insurance
368.02	Technique and methods of insurance	368.025.52	Insurance of the first, second, third etc. loss
368.021	Insurance management in general	368.025.53	Insurance of current value. Adjustable insurance
368.021.1	Recruiting of policyholders. Canvassing. Publicity	368.025.54	Overinsurance. Underinsurance. Insurance of true value
368.021.2	Arrangement of contract	368.025.55	Supplementary insurance
368.021.21	Proposals. Forms. Question sheets	368.025.56	Unlimited liability
368.021.22	Declaration. Proposer's duty of disclosure	368.025.57	Fixed-value insurances
368.021.24	Preliminary investigation, examination, at charge of insurer	368.025.6	Risk insured. Nature and classification of risks
368.021.25	Preliminary valuation	368.025.61	Determinate risk in general, simple (elementary), integral
368.021.26	Fixing of sum insured, amount of insurance, premium etc.	368.025.62	Alteration of risk during term of insurance
368.021.28	Cover note	368.025.63	Risks in time of strikes, civil disturbance, war
368.021.3	Notice of acceptance	368.025.65	Excluded risks
368.021.4	Checking the insurance contract	368.025.66	All-risk insurance. Comprehensive insurance
368.022	Insurance contracts in general	368.025.7	Damage. Loss. Abandonment
368.022.1	Forms of contract: oral, written. Wording ⇒ 347.135	368.025.71	Generalities. Causes
368.022.2	Delivery of the policy	368.025.72	Evidence, proof of loss or damage
368.022.4	Taking effect. Effective date	368.025.73	Prevention of loss. Limitation of damage
368.022.5	Duration. Time restrictions. Renewal	368.025.74	Notification of loss or damage
368.022.6	Restrictions of place. Extramural insurance. Insurance of property on journeys	368.025.75	Verification of loss or damage
368.022.8	Alterations and additions to contract	368.025.76	Extent of loss or damage. ☐ Total, partial loss. Write-off
368.022.9	Extinction of contract	368.025.8	Compensation for loss or damage. Indemnity
368.022.91	Extinction in general. Expiry	368.025.82	Calculation of indemnity
368.022.92	Termination of contract. Notice of cancellation. Waiver. Nullification. Annulment	368.025.84	Amount of indemnity
368.022.93	Redemption. Conversion. Transfer	368.025.85	Settlement of claims
368.022.96	Suspension	368.025.86	Agreement of parties to the settlement
368.023	Parties to the insurance contract	368.025.87	Payment of indemnity
368.023.1	Insurer. Underwriter	368.025.88	Redemption of claim. Surrender of policy. Lump-sum settlement
368.023.2	Insurant. Policyholder	368.025.89	Subrogation to the insured party, of beneficiary's claims on liable third parties
368.023.3	The insured	368.026	Kinds of policy
368.023.32	Personal, individual insurance	368.026.1	Policies covering own loss or damage
368.023.33	Collective, group insurance	368.026.3	Standard policies. 'All-in' policies
368.023.4	Beneficiaries	368.026.4	Policies from point of view of insured. ☐ Bearer policies. Policies for benefit of named persons
368.023.5	Intermediaries. ☐ Insurance salesmen. Agents. Brokers	368.026.5	Open policies
368.023.6	Assignees	368.026.6	Miscellaneous kinds of policy. ☐ Coupon policies. Profit-sharing policies. Pseudo-policies
368.023.7	Indemnitees	368.027	Reserves. Covering of liabilities. Covering funds
368.023.8	Allocatees	368.027.2	Covering from share capital
368.024	Insurance on one's own account, on others' account. Self-insurance	368.027.3	Reserves. Assets
368.024.1	For one's own account	368.027.4	Deposit of security bonds
368.024.2	For the account of a third party	368.027.7	Relation between covering funds and premium
368.024.4	Self-insurance	368.027.8	Use of profits
368.025	Details of insurance contracts	368.028	Commission
368.025.1	Premiums. Rates. Tariffs. Quotation. Payment	368.029	Reinsurance. Risk transference
368.025.11	Kinds of premium generally. ☐ Gross, net premiums. Fixed, variable premiums	368.03	Kinds of insurance firm, business, undertaking
		368.031	Insurance under public law. Official, state, national insurance
		368.032	Private insurance

368.032.1	Commercial insurance companies. Joint-stock companies	368.183	Insurance against robbery. Fidelity insurance.
368.032.2	Mutual insurance. Friendly societies, syndicates	368.185	Insurance against fraud, forgery, malversation
368.032.3	Mixed companies	368.186	Insurance of specific chattels against damage
368.032.5	Liability joint and several. Mutual liability		Insurance against breakage of objects.
368.032.7	Specific kinds of underwriting. □ Lloyd's of London	368.187	□ Insurance of glass
368.034	Mixed-economy organizations, part state-owned		Insurance against breakage and failure of machinery, equipment. Plant insurance
368.036	Public intervention in insurance business. Supervision. Approval	368.188	◇ 368.187:621.039 Insurance against faults in nuclear plant
368.04	Voluntary and compulsory insurance. Lack of insurance	368.19	Insurance against violent attack, bombings, explosions
368.041	Voluntary insurance		Insurance of other material things
368.042	Compulsory, obligatory insurance		◇ 368.19:646 Insurance of clothing. Wardrobe insurance
368.044	Lack of insurance. State of being uninsured	368.2	Transport insurance
368.07	Administrative arrangements. Organs of management		<i>Details by :... if not listed below</i>
	<i>Auxiliaries 3.07... (as at 35.07...)</i>		◇ 368.2:656.073 Dispatch insurance.
368.08	Administrative personnel	368.21	Insurance of goods forwarding services
	<i>Auxiliaries 3.08... (as at 35.08...)</i>	368.212	Insurance of land transport
368.1	Insurance of material things in general. Insurance of objects. Property insurance. Goods insurance. Insurance against loss or damage	368.212.1	Motor vehicle insurance. Passenger car insurance
368.11	Fire insurance. □ Property. Contents ⇒ 614.84	368.212.2	◇ 368.212.032.5 Liability
368.12	Various forms of insurance relating to buildings. Construction insurance	368.212.3	Comprehensive insurance
368.14	Insurance restricted to particular causes and consequences of fires. □ Lightning. Explosion. Arson. Damage from extinguishing fires	368.212.4	'All risks' insurance
368.15	Mortgage insurance. Insurance of mortgage debt against fire-damage of mortgaged property	368.213	Freight insurance
368.17	Insurance against loss or damage from natural causes, or similar ⇒ 368.11	368.214	Passenger insurance
368.171	Insurance against loss or damage due to the elements in general	368.216	Motorcycle insurance
368.172	Insurance against loss or damage due to bad weather ⇒ 368.51		368.213 ≡ 368.212
368.172.2	Insurance against storm damage		Goods vehicle insurance
368.172.3	Insurance against hail damage		368.214 ≡ 368.212
368.172.4	Insurance against frost damage		Railway insurance
368.172.5	Insurance against flood damage		368.216 ≡ 368.212
368.172.6	Insurance against earth tremors, land movements. □ Insurance against earthquakes, volcanic eruptions, landslides, avalanches, subsidence	368.22	Insurance of inland waterway transport
368.178	Insurance against biological damage, biodeterioration		<i>Class information about both sea and inland waterway transport at 368.23</i>
368.178.2	Insurance against dry rot	368.221	Hull insurance. Insurance of ship's fabric
368.178.4	Insurance against damage by animals	368.222	Insurance of merchandise
368.18	Insurance against loss or damage by human action. Insurance of particular objects	368.223	Cargo insurance. Freight insurance
368.182	Insurance against theft, burglary	368.224	Defrayment of towage costs for wrecks etc.
368.182.2	For private individuals	368.23	Insurance of sea transport. Marine insurance
368.182.3	For specific firms <i>Details by :...</i>	368.231	Hull insurance. Insurance of ship's fabric
368.182.4	For particular objects ◇ 368.182.4:629.322 Bicycle insurance	368.232	Insurance of merchandise
368.182.5	For securities	368.234	Cargo insurance. Freight insurance
368.182.6	For dispatch of valuables ⇒ 368.222	368.235	Insurance of valuables aboard ship
		368.236	Insurance against average, shipping losses ⇒ 347.796
		368.237	Insurance for safe arrival
		368.24	Insurance of air transport ⇒ 347.824.4
		368.26	Insurance of postal packets, letters, parcels etc.
		368.5	Agricultural insurance
		368.51	Insurance against weather risks, meteorological damage. Insurance against plant diseases, crop diseases, phytopathogens ◇ 368.51:632.116.3 Insurance against hail damage
		368.52	Insurance of animals. Livestock insurance ◇ 368.52:636.7 Insurance of dogs ◇ 368.52:638.1 Insurance of honeybees
		368.54	Harvest insurance. Insurance of growing crops

368.55	Forestry insurance. Timber insurance	368.912.14	Insurance in favour of the children. ☐ Dowry insurance. Study costs insurance. Capital payment at a certain date, e.g. on coming of age
368.8	Insurance of wealth, assets. Insurance of imponderables. Insurance against financial losses. Insurance of excluded risks. Various other kinds of insurance	368.912.2 368.912.4	Payment in the case of death Mixed life assurance. Payment both in the event of death and on reaching a certain date
368.81	Insurance against particular financial losses. ☐ Depreciation of investments	368.914	Pension insurance
368.811	Insurance of credit. Solvency insurance.	368.914.2	Life annuity. Old-age pension
368.811.3	Insurance of debt recovery	368.914.4	Life annuities with payment in favour of a second party falling due on the death of the insured. ☐ Widow's annuity pensions. Widower's annuity pensions. Orphan's pensions. Family member pensions. Partner pensions. Servant pensions
368.812	Insurance of guaranteed loans. Insurance of sureties. ☐ Bond, pledge, mortgage, bailment insurance	368.916	Insurance for services in the event of death. ☐ Insurance for funeral costs. Insurance for provision of gravestones
368.813	Insurance against losses in currency exchange		
368.814	Insurance against rent losses. Insurance against default of tenant		
368.814	Patent insurance		
368.815	Insurance of option moneys		
368.86	Insurance of civil liabilities. Third-party liability insurance 368.86 = 347.5	368.94	Private insurance against incapacity to work
368.87	Insurance against indirect damage and losses. ☐ Insurance against trade losses, industrial stoppages, interruption of business, shutdowns. Loss-of-profits insurance	368.941 368.941.2	Reason for incapacity to work Through illness 368.941.2 = 364.32
368.89	Other insurances Specify by :...	368.941.4 368.941.6 368.942	Through accident Through invalidity Private health insurance 368.942 = 364.32
368.891	Insurance against legal expenses. Legal protection insurance	368.943	Private accident insurance
368.892	Insurance against fiscal levies. Tax insurance Specify by :336.2...	368.944	Degree of incapacity to work resulting from illness or accident Details by .062
368.897	Insurance of profit share. Insurance of commission	368.945	Private invalidity insurance
368.9	Private insurance for individuals	368.95	Private insurance against unemployment
368.9.06	Medical aspects of insurance	368.96	Insuring parts of the body or qualities essential for carrying out a profession. ☐ A pianist's fingers. A dancer's legs
368.91	Life assurance	368.97	Travel insurance Details by (1/9) (Table 1e)
368.911	General life assurance matters		◇ 368.97(203) Air travel insurance
368.911.1	Financial aspects of life assurance		◇ 368.97(204) Sea travel insurance
368.911.11	Sum insured		◇ 368.97(21) Land travel insurance
368.911.12	Type of premium payment. ☐ Single payment. Continuous payment	368.971	Insurance against cancellation
368.911.14	Life assurance with surrender possibility	368.972	Insurance against cost of illness, accident or hospitalization. ☐ Transport home. Permanent invalidity. Death. Family visit
368.911.2	Life assurance according to the individual insured. ☐ Insurance for individuals who have been refused. Insurance for teetotallers	368.974	Insurance against loss of luggage, jewellery and money
368.911.4	Life assurance with profits	368.976	Insurance against bad weather. Reimbursement of money for the lack of sun, lack of snow etc.
368.911.5	Life assurance relating to medical examination		
368.911.6	Mortgage guarantee assurance		
368.911.7	Tontine assurance		
368.912	Capital-sum life assurance	368.98	Insurance against disruption of private life
368.912.1	Payment on reaching a certain age.	368.982	Insurance against divorce proceedings
368.912.12	Endowment insurance	368.984	Insurance against kidnapping, against unlawful detention
368.912.13	Payment to the person insured		
368.912.13	Payment to the spouse or partner		

37 EDUCATION

⇒ 364.67

37.0

FUNDAMENTAL TYPES AND PRINCIPLES OF EDUCATION

Throughout 37.0... the word 'school' can be taken to mean any educational institution

- ◇ 373.23.014.521 Denominational nursery schools
- ◇ 378.4.014.521 Denominational universities

37.01

Fundamentals of education. Theory. Policy etc.

37.011

Basic concepts

37.011.3

Educational situation

- ◇ 37.011.3-051 Educators. Teachers
- ◇ 37.011.3-052 Pupils. Students
- ⇒ -057.87
- ⇒ 37.016

37.011.33	Educational content (what is taught) ⇒ 37.016		
37.012	Methods of study of education. Pedagogic observation methods		
37.013	General educational and teaching theory. Principles of pedagogic activity. Practical pedagogics	37.014.543.3	◇ 37.014.543.1 :354 Central government support ◇ 37.014.543.1 -027.551 Privately supported education ◇ 37.014.543.1 -027.551:2 Education supported by religious organizations
37.013.2	Principles of educational (pedagogic) activity	37.014.543.5	Source of finance, support of students. □ Scholarships. Bursaries. Grants. Free places
37.013.3	Theory of the content of education	37.014.544	School fees, charges. Tuition fees
37.013.31	Formal education	37.014.55	Rationalization of education system
37.013.32	Material education	37.014.552	Authority of state, government, over education
37.013.41	Individual pedagogics		Centralization, decentralization of the educational system
37.013.42	Social pedagogics	37.014.553	Autonomy of educational establishments. □ Academic freedom
37.013.43	Cultural pedagogics	37.014.6	Quality control in educational establishments ◇ 37.014.6:005.6 Quality management. Total quality management (TQM)
37.013.44	Metaphysical pedagogics		◇ 37.014.6:005.642.1 Benchmarking
37.013.46	Theistic pedagogics	37.014.61	Methods for ensuring quality. □ Inspection ⇒ 37.014.611
37.013.73	Philosophic pedagogics	37.014.611	State inspection ◇ 37.014.611:35.075.6(1-2) Provincial, county and local inspection
37.013.74	Comparative pedagogics		◇ 37.014.611:35.075.6(1-3) Local, municipal inspection
37.013.75	Experimental pedagogics		◇ 37.014.611(410) In the UK: OFSTED. QAA
37.013.76	Descriptive and phenomenological pedagogics	37.014.614	Inspection by other interested parties <i>Denote specific religion by :2.</i>
37.013.77	Psychological pedagogics		◇ 37.014.614:2 Inspection by religious bodies
37.013.78	Sociological pedagogics	37.014.62	Governing bodies ⇒ 37.014.611
37.013.8	Particular branches of educational theory	37.014.621	Managing bodies. University Councils, Convocation, Senate
37.013.82	Orthopedagogics. Therapeutic pedagogics	37.014.623	Parents' consultative bodies. School guardianships, trusts
37.013.83	Andragogics. Theory of adult education	37.014.67	Students' councils, committees
37.014	Educational policy. Educational work and public life	37.014.68	Parent-teachers associations (PTAs)
37.014.1	Opportunity and obligation ⇒ 342.733	37.015	Disciplines of educational theory. Special pedagogic sciences
37.014.12	Right to be educated. Freedom to learn ⇒ 342.733		◇ 37.015:551.58 Educational climatology
37.014.13	Freedom to teach		◇ 37.015:57 Educational anthropology, biology, genetics
37.014.2	Significance of school for education of the people	37.015.3	Educational psychology. Educability
37.014.22	Combating of illiteracy. Literacy campaigns		◇ 37.015.3:005.32 Motivation of students
37.014.24	Schools and international relations		◇ 37.015.3:159.922 Differential psychology. Mental development of students
37.014.242	Exchange of teachers and pupils		◇ 37.015.3:159.922.7 Psychological development of adolescent students
37.014.25	International educational organization		◇ 37.015.3:159.937 Perception of students
37.014.3	School reform		◇ 37.015.3:159.95 Mental processes of students
37.014.4	Policy errors. □ Overexpansion of education		◇ 37.015.3:159.953 Memory and learning of students
37.014.5	Educational policy. Planning of education system		◇ 37.015.3:159.954 Creativity and imagination of students
	◇ 37.014.5:141.45 Atheistic policy, bias		◇ 37.015.3:159.955.6 Reasoning of students ⇒ 159.922.7 ⇒ 159.922.8 ⇒ 159.923.2 ⇒ 159.942
	◇ 37.014.5:2 Religious bias	37.015.31	Methods for development of mental, intellectual, imaginative capacity
	◇ 37.014.5:323.118 Racial segregation		◇ 37.015.31:17.022.1 Moral education
37.014.511	School segregation		◇ 37.015.31:316.42 Social education
	◇ 37.014.511:2 Religious segregation		◇ 37.015.31:379.83 Adventure training
	◇ 37.014.511:323.118 Racial segregation		◇ 37.015.31:57.081.1 Education through nature study and observation
37.014.511.3	Single sex schools		◇ 37.015.31:7 Artistic training
37.014.52	Religious aspects of educational policy	37.015.311	Intelligence training. Personality formation
37.014.523	Religious instruction in schools	37.015.312	Character training
37.014.524	School worship. Attendance at religious functions. School prayers		
37.014.525	Relations between the school and religious communities, church etc. □ School concordats. Inspection by religious authorities		
37.014.53	Social aspects of educational policy. □ Democratization. Equality of opportunity		
37.014.531	Compulsory education		
37.014.533	School leaving age		
37.014.54	Educational economics. Economics of education. Economic aspects of educational policy		
37.014.542	Planning of education system		
37.014.543	Financing of education system		
37.014.543.1	Source of finance, support of schools, colleges, universities ◇ 37.014.543.1 :352 Local government support		

- 37.016 Curriculum.** □ Subjects (at any level)
For curricula as form of document see (073), Table 1d
 ◇ 37.016:003-028.31 Rudiments of reading and writing
 ◇ 37.016:159.931/936 Development of the faculties, senses
 ◇ 37.016:172 Citizenship
 ◇ 37.016:316.774 Media studies
 ◇ 37.016:32 Politics
 ◇ 37.016:502/504 Study of the environment
 ◇ 37.016:511-028.31 Rudiments of number, arithmetic
 ◇ 37.016:53 Physics
 ◇ 37.016:74 Drawing. Design
 ◇ 37.016:808.1 Composition
 ◇ 37.016:808.5 Oral expression
 ◇ 37.016:81-028.31 Rudiments of language and speech
 ◇ 37.016:908 Study of one's locality. Area studies
 ⇒ 37.011.33
- 37.017 Aims and ideals of education**
 ◇ 37.017:17 Moral, ethical ideals
 ◇ 37.017:5/6 Scientific/technical awareness
- 37.017.4 Education for citizenship**
- 37.018 Educational organization. Basic forms, patterns, of education**
- 37.018.1 Education at home. Private tutoring**
- 37.018.3 Education in residential institutions, homes, boarding schools**
- 37.018.32 Boarding school system (in-boarders). □ Living in the institution. Educational children's villages (e.g. Pestalozzi)**
 ◇ 37.018.32*53 Weekly boarding
- 37.018.38 Boarding out system (out-boarders). □ Living with a family. Living in lodgings**
- 37.018.4 Organizational form of instruction**
- 37.018.42 Evening study. Evening school**
- 37.018.43 Remote education. □ Broadcast courses**
 ◇ 37.018.43:004 e-learning
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398.22	Sagas. □ Mythological tales. Christian legends. Historical tales. Hero tales	398.44	Dragons
398.23	Anecdotes. Popular humour ⇒ 82-36	398.46	Giants. Ogres. Dwarfs
398.25	Jests. Buffoonery	398.47	Witches. Witchcraft. Sorcery ⇒ 133.4 ⇒ 2-545
398.3	Popular beliefs and customs. Superstitions. Old wives' tales	398.5	Folk literature. Traditional drama
398.31	Beliefs and customs concerning fire. □ Beltane fires. Solstitial fires. Pyromancy	398.51	Folk books. Chap books
398.32	Beliefs and customs concerning particular places. □ Churchyards. Ruins. Springs. Forests	398.54	Folk plays. □ Mumming. Punch and Judy ⇒ 82-292
398.33	Beliefs and customs concerning particular times and festivals ⇒ 27-564	398.6	Riddles. Conundrums. Rebuses. Traditional jokes. Mottoes, devices, popular heraldry ⇒ 82-193
398.332	High days and holidays	398.7	Dream books. Popular interpretation of dreams ⇒ 2-544.8
398.332.1	Spring festivals		
398.332.12	Easter. Easter customs. □ Easter eggs	398.8	Folksongs. Traditional songs. Shanties ⇒ 784.4 ⇒ 82-144
398.332.16	Whitsun	398.81	Patriotic songs
398.332.19	Festivals and customs at seed times. □ May customs. Maypole. Walpurgis night	398.82	Songs of nature, the seasons, the passing year
398.332.2	Summer festivals. □ Summer solstice. Midsummer day	398.83	Songs of home, family, friendship, love
398.332.3	Autumn	398.831	Children's songs. Lullabies ⇒ 784.67
398.332.33	Harvest festivals and harvest customs. Harvest Thanksgiving	398.836	Friendship songs
398.332.34	Vintage. Vintage festivals	398.837	Farewell songs
398.332.36	All Saints' Day (All Hallows, Hallowmas). Halloween. All Souls' Day	398.838	Love songs
398.332.39	Late autumn. St Martin's Day. Miscellaneous late autumn festivals	398.84	Songs of community, dance, rambling, sport. Team songs ⇒ 784.66
398.332.4	Winter festivals	398.85	Songs of status, role, occupation. □ Work songs
398.332.41	Midwinter festivals. Winter solstice and associated customs	398.86	Comic songs. Street ballads ⇒ 784.72
398.332.412	St Nicholas	398.87	Songs with narrative or historical content. Folk ballads
398.332.414	Advent	398.88	Religious folksongs. Spirituals
398.332.416	Christmas and yuletide customs. □ Christmas trees	398.9	Proverbs. Popular maxims. Folk sayings. Taunts. Gibes
398.332.42	New Year's Eve.(Feast of St Sylvester). New Year's Day	398.91	Proverbs
398.332.44	Epiphany	398.94	Witticisms. Quips
398.332.46	St Valentine's Day	398.95	Popular names for people, plants, animals, places, streets etc. ⇒ 81'373.2
398.332.47	Festivals before and after Lent. □ Carnival. Shrove Tuesday (Mardi Gras) ⇒ 394.25		
398.34	Beliefs and customs in the home, household		
398.341	The house. □ Foundation stone. Topping out. Housewarming		
398.344	The household		
398.345	Garden		
398.4	The supernatural ⇒ 133		

5 MATHEMATICS AND NATURAL SCIENCES

Class here generalities about the pure sciences and the exact or mathematical sciences (including astronomy, mechanics and mathematical physics)

◇ 5(091) History of science

◇ 5-051 Practitioners of science. Scientists

502/504	ENVIRONMENTAL SCIENCE. CONSERVATION OF NATURAL RESOURCES. THREATS TO THE ENVIRONMENT AND PROTECTION AGAINST THEM	502.2	The environment as a whole <i>For nature study, observation of nature, natural history (previously at this number) see 57.081.1</i>
502	THE ENVIRONMENT AND ITS PROTECTION	502.21	Natural resources and energy ◇ 502.21:523.9 Solar energy ◇ 502.21:621.039 Nuclear energy
502.1	The environment and society. Conservation and protection in general	502.211	Protection of the living world, the biosphere ◇ 502.211:061 Organizations. Wildlife protection associations (e.g. World Wildlife Fund, WWF) ◇ 502.211:579 Microbial kingdom ◇ 502.211:582 Vegetation. Plant kingdom ◇ 502.211:592/599(204) Aquatic wildlife
502.11	Interaction, interdependence of environment and society. Mutual benefits, etc.		
502.12	Environmental awareness. 'Green' outlook. Green issues. 'Greenness'	502.22	The man-made environment ⇒ 71/72
502.13	Conservation measures and management		
502.131	Development	502.3/.7	Parts of the environment
502.131.1	Sustainable development ⇒ 332.146.2	502.3	Atmospheric environment <i>For nature and society, nature conservation and protection in general (previously at this number) see 502.1</i> ◇ 502.3:504.5:621.43.064 Atmospheric pollution by exhaust fumes from traffic ◇ 502.3:504.5:662.6/.7 Atmospheric pollution by fossil fuels ◇ 502.3:504.7 Global warming of atmosphere ⇒ 551.5 ◇ 502.3:551.510.534 Ozone layer ◇ 502.3:551.510.534:504.12 Ozone layer depletion ◇ 502.3:551.515 Atmospheric layers ◇ 502.3:613.15 Air quality. Including: Clean air ⇒ 546.217
502.132	Improvement		
502.14	Social, administrative, legislative measures on environmental conservation ◇ 502.14:061 Organizations to promote conservation ◇ 502.14:329.63 'Green' parties		
502.15	Environment in relation to planning and development ⇒ 71	502.5	Earth's surface. Landscape. Scenery <i>For nature complex as a whole, balance, danger, damage, threat of destruction (previously at this number) see 502.2 and 504</i>
502.17	Protection of the environment in general ⇒ 620.9	502.51	Hydrospheric environment ◇ 502.51:504.5 Hydrospheric pollution. Water pollution ◇ 502.51(26) Ocean and sea environment. Including: Sea floor. Sea water ◇ 502.51(26):504.5:665.6 Oil spillage at sea ◇ 502.51(28) Freshwater environment. Inland surface ◇ 502.51(282) Fluvial environment. Including: Rivers. Streams ◇ 502.51(282.02) Groundwater environment. Including: Springs. Aquifers ◇ 502.51(282.05) Fluvio-marine environment. Including: River mouths. Estuaries ◇ 502.51(285) Lakes. Reservoirs. Ponds ◇ 502.51(285.3) Swamps. Marshes ⇒ 551.46 ⇒ 574.6
502.171	Protection, rational use and renewal of natural resources ◇ 502.171:531.62 Energy conservation ◇ 502.171:546.212 Water conservation ◇ 502.171:662.997 Conservation of geothermal energy		
502.172	Protection of endangered living elements ◇ 502.172:502.211 Protection of endangered species. 'Red data'		
502.173	Control of usage, avoidance of waste, rationing of resources		
502.174	Restoration, salvage, reclamation, rescue measures. Wasteless and low-waste technology ◇ 502.174:627.533 Land reclamation ⇒ 620.97		
502.174.1	Recycling ⇒ 628.39		
502.174.2	Use of biodegradable materials		
502.174.3	Use of renewable energy resources ◇ 502.174.3:627.8.09 Use of hydroelectric power ⇒ 502.21		
502.175	Control of environmental quality. Control of pollution ⇒ 504.5 ⇒ 628.19 ⇒ 628.39 ⇒ 628.5		

- 502.52 Lithospheric environment**
For balance, self-correcting imbalance (previously at this number) see 502.2
 ◇ 502.52:551.21:502.13 Conservation of volcanic features
 ◇ 502.52:551.312.2:502.13 Conservation of marshes, moors
 ◇ 502.52:551.435.11:502.13 Conservation of gorges, canyons
 ◇ 502.52(210):502.13 Conservation of land formations. Coasts, etc.
 ◇ 502.52(23):502.13 Conservation of mountains, hills, slopes
- 502.521 Soil
 ⇒ 631.4
- 502.6 Glacial environment.** □ Ice. Snow
For protection of inanimate nature, landscape (previously at this number) see 502.17
 ⇒ 551.32
- 502.7 Earth's interior**
For protection of animate nature, wildlife conservation and protection (previously at this number) see 502.172
 ◇ 502.7:662.765.4 Conservation of oil deposits
- 504 THREATS TO THE ENVIRONMENT**
For environment science, environmentology (previously at this number) see 502
- 504.1 Direct damage. Depredation. Threat of depredation**
- 504.11 Mass depredation, despoliation. Destruction**
- 504.12 Depletion or exhaustion by over-exploitation or misuse**
 ◇ 504.12:502.7 Depletion of deposits
 ◇ 504.12:662.66 Depletion of coal deposits
 ◇ 504.12:502.7:662.765.4 Depletion of oil deposits
 ◇ 504.12:502.7:662.767 Depletion of natural gas reserves
 ◇ 504.12:639.2.058 Overfishing
- 504.121 Erosion
 ◇ 504.121:551.311.21 Erosion by flowing water
- 504.122 Deforestation
- 504.123 Desertification
- 504.2 Accidental damage by a natural or unknown agent**
 ◇ 504.2:614.84 Damage by fire (especially open-air fire)
- 504.3 Damage by a known natural or biological agent**
For atmospheric environment (previously at this number) see 502.3
 ◇ 504.3:632.5 Plant depredation. Including: Weeds. Epiphytes
 ◇ 504.3:632.6/.7 Animal depredation. Including: Vermin attack. Stampede
 ⇒ 582
 ⇒ 59
- 504.35 Genetic contamination**
 ◇ 504.35:599.742.73 Genetic contamination of Felidae spp., e.g. of wild cat *Felis silvestris* by hybridization with feral domestic cat *Felis catus*
 ⇒ 575.222.7
- 504.4 Damage from natural causes. Natural disasters. Natural hazards**
For hydrospheric environment (previously at this number) see 502.51 and 556. For water pollution see 502.51:504.5
 ◇ 504.4:551.21 Volcanic damage
 ◇ 504.4:551.515.3 Hurricane damage
 ◇ 504.4:556.166 Flood damage
 ◇ 504.4:556.167 Drought damage
- 504.5 Damage from harmful materials. Pollution.** □ Persistent organic pollutants (POPs). Persistent, toxic, bioaccumulative substances (PTBs)
For lithospheric environment (previously at this number) see 502.52, 551.4
 ◇ 504.5:551.577.13 Pollution by acid rain
 ◇ 504.5:628.3 Pollution by sewage
 ◇ 504.5:628.4.047 Pollution by radioactive waste
 ◇ 504.5:661.16 Pollution by pesticides
- 504.6 Damage from physical factors.**
 □ Effects of vibration, noise, heat
 ◇ 504.6:534.322.3 Damage caused by noise
- 504.61 Damage by man to the environment**
 ◇ 504.61:355.01 Damage caused by war
 ◇ 504.61:62 Damage caused by engineering works
 ◇ 504.61:69 Damage caused by building works
- 504.7 Global warming. 'Greenhouse effect'.**
 □ Effects of 'greenhouse gases'
For biospheric environment (previously at this number) see 502.211
 ◇ 504.7:582.26/.27 Algal blooming
- 504.8 Nuclear winter**
- 504.9 Wilful, wanton human damage to the environment. Vandalism**

51 MATHEMATICS

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| <p>51-3 COMPUTATION TECHNIQUES.
 □ Calculations. Programs. Devices</p> <p>51-32 Calculation without aids</p> <p>51-33 Use of tables
 ⇒ 519.66</p> | <p>51-35 Use of mechanical, graphical and other devices. □ Slide rules. Nomograms
 ⇒ 519.67</p> <p>51-37 Use of calculators, computers in mathematics
 ⇒ 004</p> |
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51-7	MATHEMATICAL STUDIES AND METHODS IN OTHER FIELDS OF KNOWLEDGE. □ Scientific mathematics. Actuarial mathematics Denote the field by :... ⇒ 519.862	510.65	Logico-mathematical theories. □ Formal arithmetic. Formal number theory
51-8	MATHEMATICAL GAMES AND RECREATIONS	510.66	Inference search and decision problems in logical and logico-mathematical calculi ⇒ 519.226
510	FUNDAMENTAL AND GENERAL CONSIDERATIONS OF MATHEMATICS	510.67	Theory of models ⇒ 519.86
510.2	Foundations of mathematics	510.8	General mathematical systems
510.21	General philosophical considerations. Critical aspects. Logical antinomies ⇒ 1	511	NUMBER THEORY
510.22	Set theory. Set theoretic approach	511.1	Arithmetic. Elementary number theory
510.222	Naive set theory	511.11	Construction of number systems (natural numbers, integers, real numbers, complex numbers). Number notations
510.223	Axiomatic set theory. Axiomatization of analysis	511.12	Integers and their manipulation. □ Addition. Subtraction. Multiplication. Division
510.225	Descriptive set theory ⇒ 515.128 ⇒ 517.517	511.13	Factors. Fractions. Ratios. Proportions. Percentages ⇒ 511.172
510.227	Theory of order types and of ordinal and cardinal numbers	511.14	Roots and their extraction. Irrationals and other incommensurables (elementary treatment). Surds. Rationalization. Imaginaries. Complex numbers ⇒ 511.2
510.23	Formalism (mathematical aspects)	511.17	Elementary number theory
510.24	Intuitionism (mathematical aspects)	511.172	Multiplicative structure of the ring $\mathbb{Z}$ of integers (including divisibility, least common multiple, most common divider, congruences, power residue, quadratic residue) ⇒ 511.13
510.25	Constructive approach. Constructive mathematics	511.174	Arithmetic functions (elementary theory)
510.26	Logical positivism ⇒ 165.731	511.176	Number sequences (including Faray sequences). Special numbers and polynomials (including Bernoulli numbers and polynomials)
510.27	Logical semantics. Semantic antinomies ⇒ 519.767	511.178	Combinatorial number theory (including elementary methods of partitions) ⇒ 511.344 ⇒ 519.1
510.5	Algorithms. Computable functions	511.2	Algebraic number theory ⇒ 511.14 ⇒ 511.331 ⇒ 511.38 ⇒ 511.522 ⇒ 511.525 ⇒ 511.8 ⇒ 512.62
510.51	Theory of algorithms: general considerations ⇒ 519.712	511.22	Local fields
510.52	Algorithmic complexity	511.221	Multiplicative group and ring of integers of local field as Galois modules
510.53	Algorithmic problems. Theories of hierarchies ⇒ 512.54.05	511.222	Filtrations and ramification
510.54	Algorithmic set theory	511.223	Local class field theory. Reciprocity laws
510.56	General theory of calculi	511.224	Galois representations for local fields ⇒ 512.547
510.57	Computable, recursive functions	511.225	Non-Archimedean analysis ⇒ 512.7 ⇒ 517
510.6	Mathematical logic ⇒ 16	511.225.1	p-adic analytic functions ⇒ 517.53
510.62	Logical languages. Logico-object languages	511.225.2	p-adic Banach algebras ⇒ 517.986.2
510.63	Logical theories. Logico-object theories. Classical logical systems	511.225.3	Special p-adic functions ⇒ 517.58
510.633	Propositional logic (and calculus) ⇒ 512.563	511.23	Algebraic number fields
510.635	Predicate logic (and calculus). □ Theories of types	511.231	Isolated questions of algebraic number fields theory
510.64	Nonclassical, formal systems of logic	511.232	Asymptotic distribution of fields
510.643	Modal logics	511.233	Discriminants of algebraic number fields
510.644	Multiple-value logics		
510.645	Formalization of traditional logics. □ Syllogistics		
510.646	Inductive logic ⇒ 162.3		
510.647	Probabilistic logic ⇒ 519.21		

- 511.234 Divisor class group. Divisor class number
 511.235 Special fields. $\square$ Low-degree fields. Parametric theory of fields
 511.236 Class field theory. $\square$ Abelian and cyclotomic fields. Abelian extensions of quadratic fields. Fields with complex multiplication. Gamma (G) extensions of fields. Brauer group of algebraic number fields
 511.237 Non-Abelian generalizations of class fields theory
 511.238 Galois theory of algebraic number fields
 $\Rightarrow$ 512.623.2
 511.24 Algebraic function fields of one variable over finite field
 $\Rightarrow$ 512.772
 511.26 Special algebraic numbers (including PV-numbers)
 511.3 Analytic number theory
 $\Rightarrow$ 511.524
 511.31 Elementary methods in analytic number theory
 $\Rightarrow$ 511.337
 511.313 Distribution of primes (elementary methods)
 511.315 Sequences. Densities of sequences. Shnirelman method
 511.317 Estimates on arithmetic functions. Extremal values of arithmetic functions
 511.32 Estimates on trigonometric sums and character sums. $\square$ Gauss sums. Klosterman sums. Dedekind sums. Weyl sums
 511.33 Multiplicative number theory
 $\Rightarrow$ 511.38
 511.331 Arithmetic Dirichlet series. $\square$ Zeta function. Dirichlet L-function. Dedekind L-function. Artin L-function
 $\Rightarrow$ 511.2
 $\Rightarrow$ 512.75
 $\Rightarrow$ 517.537.6
 511.333 Distribution of primes. $\square$ Distribution of primes in algebraic number fields
 511.333.2 Distribution of primes in commutative semigroups
 511.335 Dirichlet series and asymptotics of number functions
 511.337 Sieve methods. Large sieve method
 511.338 Lattice points in regions
 511.34 Additive number theory. Analytic methods
 $\Rightarrow$ 511.524
 511.344 Partitions
 $\Rightarrow$ 511.178
 $\Rightarrow$ 519.116
 511.346 Waring problem. Additive equations and systems
 511.348 Goldbach problem. Additive equations and systems on primes
 511.35 Asymptotics and estimates of number functions
 511.352 Multiplicative and additive functions
 511.38 Automorphic forms
 $\Rightarrow$ 511.2
 $\Rightarrow$ 511.33
 $\Rightarrow$ 512.7
 511.381 Automorphic forms of one variable. Modular forms of one variable
 511.382 Hilbert modular forms
 511.383 Siegel modular forms
 511.384 General approach to automorphic forms.
 $\square$ Representation-theoretic methods. Trace formula
 $\Rightarrow$ 512.817
 $\Rightarrow$ 515.178
 $\Rightarrow$ 517.986
p-adic automorphic forms
 511.386
 511.4 Diophantine approximations and inequalities. Theory of transcendental numbers. Geometry of numbers
 511.41 Continued fractions. Generalizations of continued fraction algorithms
 $\Rightarrow$ 511.72
 511.42 Diophantine approximations: general theory
 $\Rightarrow$ 511.46
 511.43 Analytic theory of Diophantine inequalities (circular method). Fractional parts. Uniform distributions. Applications to numerical analysis
 511.44 Approximation by rational and algebraic numbers to algebraic numbers (including Thue inequality and its generalizations)
 $\Rightarrow$ 511.523
 511.46 Theory of transcendental numbers
 $\Rightarrow$ 511.42
 511.462 Irrationality and linear independence of numbers, estimates of their measures
 511.464 Transcendence and algebraic independence of numbers, estimates of their measures
 511.48 Geometry of numbers
 511.5 Diophantine equations. Theory of forms
 511.52 Algebraic Diophantine equations
 $\Rightarrow$ 512.75
 $\Rightarrow$ 512.774
 511.522 Algebraic number theory methods in Diophantine equations. Norm equation.
 $\square$ Cyclotomic fields and last Fermat theorem
 $\Rightarrow$ 511.2
 511.523 Diophantine approximation methods in Diophantine equations (including Thue method and its generalizations)
 $\Rightarrow$ 511.44
 511.524 Analytic theory of Diophantine equations
 $\Rightarrow$ 511.3
 511.525 Equations over local number fields and rings.
 $\square$ Approximation of their solutions.
Local-global results
 $\Rightarrow$ 511.2
 $\Rightarrow$ 512.74
 511.526 Mathematical logic methods in Diophantine equations (including decidability, denumerability, enumeration of solutions)
 $\Rightarrow$ 510.65
Special Diophantine equations
 511.528
 511.53 Nonalgebraic Diophantine equations
 511.55 Theory of quadratic forms
 $\Rightarrow$ 512.647
 511.57 Theory of higher-degree forms and systems of forms
 $\Rightarrow$ 512.647
 $\Rightarrow$ 512.75
 $\Rightarrow$ 512.772
 $\Rightarrow$ 512.774

511.7	Metric number theory. Probabilistic number theory	512.536.7	Topological semigroups. $\square$ Compact and connected semigroups
511.72	Metric number theory $\Rightarrow$ 511.41	512.538	Various generalizations of associativity
511.75	Probabilistic number theory	512.54	Groups. Group theory
511.8	Noncommutative arithmetic $\Rightarrow$ 511.2 $\Rightarrow$ 512.55 $\Rightarrow$ 512.743	512.54.01	Axiomatizable classes of groups
511.82	Arithmetic of matrices	512.54.03	Elementary theory of various classes of groups
511.84	Arithmetic of quaternions	512.54.05	Algorithmic problems in group theory. The word problem
511.86	Arithmetic of central simple algebras	512.541	Abelian groups
511.88	Arithmetic of general algebras and rings	512.541.1	Purity and its generalizations. $\square$ Neatness. Algebraically compact groups
512	ALGEBRA	512.541.2	High subgroups. $\square$ E-groups
512.1	Elementary algebra: operations, formulae	512.541.3	Direct and subdirect sums. $\square$ Restricted groups
512.12	Addition. Subtraction. Multiplication. Division. Progressions	512.541.4	Extensions of Abelian groups. $\square$ Splitting of mixed Abelian groups
512.13	Identities and inequalities	512.541.5	Mappings of a group into itself and into other groups. $\square$ Automorphisms. Endomorphisms. Characteristic subgroups. Homomorphic images of a group. Equivalent Abelian groups. Quasi-isomorphisms
512.14	Binomial formula. Sum of powers of first n numbers etc. Powers of polynomials	512.541.6	Primary Abelian groups
512.5	General algebra	512.541.7	Torsion-free Abelian groups
512.53	Semigroups	512.541.8	Systems of generators. Factorization
512.531	Semigroups with finiteness conditions	512.542	Finite groups
512.532	Generating sets, relations and identities of semigroups	512.542.1	Generators and defining relations
512.532.2	Free semigroups, defining relations	512.542.2	Automorphisms of finite groups
512.532.3	Commutative semigroups	512.542.3	Finite p-groups
512.532.5	Idempotent semigroups	512.542.4	Finite solvable groups
512.533	Equivalences and complexes in semigroups. Homomorphisms	512.542.5	Finite simple groups. $\square$ Subgroups of nonsolvable groups
512.533.1	One-side regular equivalences	512.542.6	Arithmetic structure and normal structure of finite groups. $\square$ Extensions of finite groups. Normal series in finite groups. Sylow theorems. Factorizations. Normal complements
512.533.2	Homomorphisms of semigroups. $\square$ Congruences. Endomorphisms. Automorphisms. Correspondences. Products. Bands	512.542.7	Permutation groups $\Rightarrow$ 519.115.1
512.533.4	Special elements and complexes in semigroups	512.543	Relations between elements of groups
512.533.5	Ideals of semigroups. $\square$ Ideal equivalences. Homogroups	512.543.1	Systems of generators
512.533.7	Subsemigroups	512.543.2	Varieties of groups. $\square$ Identity relations
512.533.8	Lattices of subsemigroups, ideals and congruences of semigroups	512.543.5	Operations on groups. $\square$ Free, direct, regular, wreath products
512.534	Semigroups of transformations	512.543.7	Equations in groups. Embedding theorems in group theory. $\square$ Adjunction problem. Complete groups and completions
512.534.1	Semigroups of multivalued transformations (binary relations)	512.544	Relations between subgroups. Generalized solvable groups and finiteness conditions
512.534.2	Semigroups of single-value transformations	512.544.2	Lattices of subgroups. $\square$ Splitting of groups
512.534.3	Representation of semigroups by transformations	512.544.3	Normal series and systems. $\square$ Radicals in groups
512.534.4	Restrictive semigroups of mappings	512.544.4	Characteristic subgroups, automorphisms, endomorphisms
512.534.5	Semigroups of transformations of sets equipped with structures	512.544.5	Locally finite groups
512.534.7	Matrix semigroups. Linear semigroups	512.544.7	Approximations of groups
512.535	Inverse semigroups (generalized groups). $\square$ Complexes. Homomorphisms. Regular semigroups. Semiheaps and generalized heaps	512.545	Ordered groups
512.536	Semigroups with additional structures	512.546	Topological groups
512.536.1	Semigroups with operators. Omega semigroups	512.546.1	General theory of topological groups
512.536.2	Connection with theory of rings. Multiplicative semigroups of rings	512.546.2	Abelian topological groups
512.536.4	Semigroups with involution	512.546.3	Locally compact groups
512.536.5	Quasi-ordered and ordered semigroups. $\square$ Residual semigroups	512.546.4	Representations of topological groups
		512.546.6	Relations between subgroups in topological groups. Finiteness and related conditions
		512.546.8	Generalizations of topological groups
		512.547	Linear representations of abstract groups. Group characters
		512.547.2	Representations of finite groups
		512.547.4	Representations of infinite groups
		512.548	Generalizations of groups. Groupoids. $\square$ Quasigroups

- 512.55 Rings and modules**
 512.552 Associative rings and algebras
 512.552.1 Structures of rings
 512.552.2 Rings with chain conditions. $\square$ Artinian rings. Noetherian rings. Quasi-Frobenius rings
 512.552.3 Rings with conditions on ideals or subrings.
 $\square$ Skew fields
 512.552.4 Defining and identity relations in rings. Operations on rings
 512.552.5 Embeddings of rings. $\square$ Quotient rings
 512.552.6 Extensions of rings
 512.552.7 Semigroup and group rings
 512.552.8 Representations of rings
 512.553 Modules. $\square$ Homological classification of rings
 512.554 Nonassociative rings and algebras
 512.554.3 Lie rings. Lie algebras
 $\Rightarrow$ 512.81
 512.554.5 Alternative rings and closely related rings
 512.554.7 Jordan rings and algebras
 512.555 Ordered rings and modules
 512.556 General theory of topological rings, modules, fields
 $\Rightarrow$ 517.986.2
 512.558 Generalizations of rings

512.56 Lattices, including Boolean rings and algebras
 512.562 Partially ordered sets
 512.563 Boolean rings and algebras
 $\Rightarrow$ 517.987.3
 512.565 Types of lattice
 512.565.2 Distributive lattices
 512.565.3 Dedekind lattices and similar lattices
 512.565.6 Complemented lattices
 512.565.8 Complete lattices
 512.567 Representations of lattices
 512.568 Generalizations of lattices
 512.568.2 Semilattices
 512.568.3 Noncommutative lattices
 512.568.5 Multilattices

512.57 Universal algebras. Free algebras
512.58 Categories. Category theory
 $\Rightarrow$ 517.983.8
 512.581 General considerations of category theory.
 $\square$ Brandt groupoids. Ringoids. Inductive categories
 512.582 Functors
 512.583 Representations of categories
 512.585 Abelian categories
 512.587 General theory of species of structures (according to Ehresmann)

512.6 Special branches of algebra
512.62 Fields. Polynomials
 512.622 Polynomials, including binomials and prime factors
 512.623 General field theory
 512.623.2 Extensions of fields. $\square$ Derivations in fields. Galois extensions
 512.623.3 General Galois theory
 512.623.4 Valuations of fields
 512.623.5 Ordered fields
 512.624 Finite fields
 512.624.2 Polynomials over finite fields
 512.624.3 Equations and systems of equations over a finite field
 512.624.5 Linear sequences over a finite field
 512.624.7 Perfect difference sets
 512.628 Differential algebra. Difference algebra

512.64 Linear and multilinear algebra
 512.642 Vector spaces. Vector space theory
 512.643 Matrices and linear mappings. Matrix theory
 512.643.2 Determinants and their generalizations
 512.643.4 Matrix equations
 512.643.5 Eigenvalues of matrices
 512.644 Systems of linear equations and inequalities
 512.647 Multilinear algebra. Forms
 $\Rightarrow$ 511.55
 $\Rightarrow$ 511.57

512.66 Homological algebra
 512.662 Chain complexes. $\square$ Filtrations. Exact couples. Spectral sequences
 512.664 Derived functors. $\square$ Deformations of algebraic structures
 512.667 Algebraic analogues of various constructions from topology and algebraic geometry

512.7 Algebraic geometry
512.71 Commutative rings and algebras. Local theory. Foundations of algebraic geometry
 512.711 General commutative ring theory
 $\Rightarrow$ 512.55
 $\Rightarrow$ 512.62
 512.712 Valuations on commutative rings and theory of divisibility
 512.713 Arithmetic rings. Dedekind and Prüfer rings
 512.714 Polynomial rings
 512.715 Modules over commutative rings
 $\Rightarrow$ 512.553
 512.717 Local algebra. Local theory

512.72 Structure and variations of algebraic varieties. Fibre bundles. Vector bundles. Fibrations
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- 512.812 General Lie group theory. □ Properties, structure, generalizations of Lie groups and Lie algebras
- 512.813 Special classes of Lie groups. □ Compact, semisimple, solvable, nilpotent Lie groups and Lie algebras
- 512.814 Continuous subgroups of Lie groups. □ Cartan subgroups
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- 512.816 Lie transformation groups. □ Homogeneous spaces
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- 512.818 Theory of continuous pseudogroups (infinite Lie groups). □ Cartan pseudogroups. Cartan subgroups in infinite Lie groups
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- 514.112.4 Geometry of polygons, rectangles etc.
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□ Analytic geometry of straight lines, of curves
- 514.123 Analytic geometry in three-dimensional Euclidean space. □ Analytic geometry of planes, of surfaces of second order
- 514.124 Analytic geometry in multidimensional Euclidean space. □ Analytic geometry of linear varieties, of algebraic varieties
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517.982.22	Normed spaces. Banach spaces. Hilbert spaces	519.101	General theory of combinatorial analysis
517.982.23	Countably normed spaces. Fréchet spaces	519.11	Enumeration theory
517.982.24	Montel and kernel spaces	519.111	General methods of enumeration. □ Redfield-Polya-de Bruin theory
517.982.25	Geometry of topological linear spaces	519.111.1	Recurrences
517.982.26	Absolute potential theory ⇒ 517.956.224 ⇒ 519.217.5	519.111.2	Incidence algebras. Inversion formulae. The method of inclusion and exclusion
517.982.3	Linear functionals and conjugate spaces. □ Theory of moments	519.111.3	Combinatorics of formal power series. Generating functions
517.982.4	Theory of generalized functions (distributions)	519.114	Combinatorial sequences of numbers and polynomials
517.983	Linear operators. Linear operator equations	519.115	Enumeration problems in combinatorial analysis
517.983.2	Linear operators in infinite-dimensional linear spaces: general properties	519.115.1	Combinatorial problems on substitutions and permutations ⇒ 512.542.7
517.983.22	Linear operators in locally convex spaces	519.115.2	Enumeration of mappings and transformations
517.983.23	Linear operators in Banach spaces	519.115.3	Enumeration of integral matrices and their equivalence classes
517.983.24	Linear operators in Hilbert spaces	519.115.4	Enumeration of Boolean functions and disjunctive normal forms
517.983.26	Completely continuous and kernel operators	519.115.5	Problems on set partitions and set coverings. Finite topologies
517.983.27	Linear operators in semiordered spaces	519.115.6	Problems on paths in integral lattices
517.983.28	Perturbation theory of linear operators	519.115.8	Combinatorics of sequences
517.983.5	Linear equations in infinite-dimensional linear spaces	519.116	Combinatorial theory of number partitions
517.983.6	Vector functions. Operator functions	519.117	Combinatorial identities
517.983.8	Families of linear spaces. Categories of linear operators ⇒ 512.58	519.118	Combinatorial inequalities
517.984	Spectral theory of linear operators	519.119	Combinatorial problems in probability theory ⇒ 519.21
517.984.2	Spectral theory in general topological linear spaces and spaces with quasitopology	519.14	Combinatorial configurations and arrays
517.984.3	Spectral theory in Banach spaces	519.142	Matrix combinatorial problems. □ Hadamard matrices
517.984.4	Spectral theory in Hilbert spaces	519.143	Latin squares and their generalizations. Magic squares and cubes
517.984.5	Study of the spectrum of concrete operators, including differential and integral operators		
517.984.7	Operator extensions		

- 519.144 Block designs. □ Steiner systems
 519.145 Finite affine and projective geometries as block designs
 ⇒ 514.146
 519.145.4 Generalized polygons
 519.146 Combinatorics of arrangements of geometric objects
 ⇒ 514.174
 519.147 Packing and covering
 519.148 Problems on plane mosaics and tessellations
- 519.15 Combinatorial theory of choice**
 519.151 Matroids
 519.152 Transversals
 519.154 Ramsey theory
 519.156 Combinatorics of ordered sets
 ⇒ 512.562
 519.157 Extremal combinatorial problems
- 519.16 Algorithmic problems of combinatorial analysis**
 ⇒ 519.178
 519.161 General questions of combinatorial algorithms. Polynomial and NP-complete problems
 519.163 Constructions of efficient combinatorial algorithms. Search trees
 519.165 Combinatorial algorithms in algebraic problems
 519.168 Problems of optimum choice. Algorithms for extremal problems
- 519.17 Graph theory**
 519.171 General graph theory and representations of graphs
 519.172 Particular classes of graphs. Trees
 519.172.1 Trees
 519.172.2 Planar graphs
 519.172.3 Digraphs. Tournaments
 519.172.4 Regular graphs
 519.172.5 Bipartite graphs
 519.173 Topological and metric problems of graph theory. □ Connectivity. Planarity
 ⇒ 515.14
 519.174 Structural properties of graphs. Colourings
 519.174.1 Graphical partitions
 519.174.2 Walks in graphs
 519.174.3 Coverings and packings in graphs
 519.174.5 Factorization of graphs
 519.174.7 Colourings of graphs
 519.175 Classification and enumeration of graphs and digraphs
 519.176 Extremal problems of graph theory
 519.177 Matrix methods in graph theory. Spectral graph theory
 519.178 Algorithmic questions of graph theory
 ⇒ 519.16
 519.179 Generalizations of graphs
 519.179.1 Hypergraphs
 519.179.2 Nets
- 519.2 Probability. Mathematical statistics**
519.21 Probability theory. Stochastic processes
 519.211 Foundations and axioms of probability theory
 519.212 Abstract probability theory
 519.212.2 Combinatorial probabilities
 519.212.3 Geometrical probabilities
 519.213 Probability distributions and densities.
 □ Normal distribution. Characteristic functions. Measures of dependence. Infinitely divisible laws. Stable laws
 519.213.1 Normal distribution
 519.213.2 Characteristic functions, moments, semi-invariants and other characteristics
 ⇒ 517.5
 519.213.3 Measures of dependence
 519.213.5 Infinitely divisible laws
 519.213.7 Stable laws
 519.214 Limit theorems
 519.216 Stochastic processes in general
 519.216.3 Prediction theory
 519.216.5 Stopping times
 519.216.8 Martingales
 519.217 Markov processes
 519.217.1 General theory of Markov processes.
 □ Properties of sample functions
 Markov chains. Processes with finite or countable set of states
 Processes with independent increments
 Diffusion processes and their applications
 Additive functionals. Probabilistic potential theory
 ⇒ 517.982.26
 Transformation of Markov processes
 Boundary theory for Markov processes
 Controlled Markov processes
 Special stochastic processes
 Branching and epidemic processes
 Renewal processes
 Point random processes
 Gaussian processes and measures
 Stationary and harmonizable sequences and processes
 Extrapolation. Interpolation. Filtration
 Ergodic theorems
 Random functions of several variables
- 519.22 Statistical theory. Statistical models. Mathematical statistics in general**
 ⇒ 311
 Foundations of statistical theory
 Statistical dispersion and relationships.
 Statistical mean, deviations etc.
 Sufficiency. Sufficient statistics
 Distribution theory. □ Asymptotic theory.
 Characterization and structure theory
 Inference and decision theory. □ Likelihood.
 Bayesian theory. Fiducial probability
 ⇒ 510.66
- 519.23 Statistical analysis. Inference methods**
 Parametric methods
 Estimation of parameters and functionals
 Point estimation
 Confidence regions, tolerance, limits
 Hypothesis testing. □ Criteria. Discrimination
 Variance analysis. Covariance analysis
 Correlation analysis. Regression analysis
 Ranking and selection
 Paired and multiple comparisons
 Nonparametric methods
 519.234 ≡ 519.233.2/6
 Order statistics
 Paired comparison methods
 Statistics of dependent variables. Contingency tables
 Multivariate statistical methods
 519.237 ≡ 519.233.2/6
 Factor analysis
 Cluster analysis. Classification
- 519.24 Special statistical applications and models**
 Experimental design
 Optimal designs
 Block designs
 ⇒ 519.144
 Factorial designs
 Sampling. Sampling theory

519.244	Sequential methods. □ Optimal stopping. Cusum technique (cumulative sum technique)	519.66.099	Nine-figure tables. Tables to more than nine figures
519.245	Stochastic approximation. Monte Carlo methods	519.661	General tables. Ready reckoners etc.
519.246	Statistics of stochastic processes. □ Estimation of stochastic processes. Hypothesis testing. Statistics of point processes	519.662	Logarithm tables. Tables of elementary functions
519.246.8	Time series analysis. □ Autocorrelation. Regression	519.665	Tables of higher transcendental functions
519.248	Engineering statistics. Operational research statistics. □ Queuing theory. Quality control. Reliability	519.67	Machine, graphical and other methods of computational mathematics ⇒ 51-3
519.25	Statistical data handling	519.672	Mechanical methods of calculation. □ Slide rules. Desk calculators
519.252	Collection of data and planning of surveys	519.673	Solution of mathematical problems by use of modelling systems
519.253	Sample surveys ⇒ 519.243	519.674	Graphical methods of computation
519.254	Computation of data. Algorithms	519.675	Nomography. Nomographic devices
519.255	Presentation of data. □ Format of data	519.676	Probabilistic methods of computation
519.256	Preservation of original data. Data banks	519.677	Solutions of problems of mathematical analysis and applied problems
519.257	Use of statistical data	519.7	Mathematical cybernetics ⇒ 007 ⇒ 621.391
519.258	Types of statistical data	519.71	Control systems theory: mathematical aspects ⇒ 303.725.36 ⇒ 330.46
519.6	Computational mathematics. Numerical analysis ⇒ 004.3 ⇒ 51-3	519.711	General questions of control theory. □ Models. Modelling. Coding. Theory of nets
519.61	Numerical methods of algebra	519.712	Cybernetic problems of theory of algorithms. □ Universal machines ⇒ 510.51
519.612	Numerical methods for the solution of systems of linear alphabetic equations. □ Square matrix. General matrix	519.713	Automata
519.613	Numerical methods for matrix inversion	519.714	Synthesis questions in control theory. □ Synthesis problems for concrete classes of schemes. Methods of synthesis for different classes of function. Minimization of Boolean functions
519.614	Numerical methods for computing eigenvalues and eigenvectors of matrices	519.715	Analysis problems in control theory
519.615	Numerical methods for the solution of transcendental equations and systems of equations	519.716	Functional systems. □ Completeness problems. Lattice of closed classes. Metric problems
519.62	Numerical methods for solution of ordinary differential equations	519.717	Identity transformations. Transformation rules
519.622	Cauchy problems for ordinary differential equations	519.718	Stability. Reliability. Checking. □ Synthesis. Error-correcting systems. Tests
519.624	Boundary value problems for ordinary differential equations	519.72	Information theory: mathematical aspects ⇒ 303.725.37 ⇒ 330.47
519.63	Numerical methods for solution of partial differential equations	519.722	Entropy. Amount of information and its properties
519.632	Second-order equations of elliptic type	519.723	Asymptotic theorems of optimal coding. Shannon theory. Channels
519.633	Second-order equations of parabolic and hyperbolic type	519.724	Computation of information-theory characteristics for specific channels and messages. □ Channel capacity computation. Error-probability computation
519.635	Partial differential equations of higher order (and systems of such)	519.725	Algebraic theory of error-correcting codes
519.64	Numerical methods for solution of integral equations. Quadrature formulae	519.726	Nonhomogeneous codes for messages
519.642	Numerical methods for solution of integral equations	519.727	Methods of sequential coding
519.644	Quadrature formulae ⇒ 517.518.87	519.728	Methods of coding in continuous channels
519.65	Approximation. Interpolation ⇒ 511.4	519.728.2	Quantification of communications
519.651	Approximation of functions	519.728.4	Gaussian channels
519.652	Interpolation. Extrapolation	519.728.6	Channels with fading
519.653	Numerical differentiation	519.76	Semiotics. Mathematical theory of systems of symbols. Mathematical linguistics ⇒ 003 ⇒ 510.62 ⇒ 81'32
519.654	Least squares method	519.762	Syntactic study of systems of symbols
519.657	Empirical formulae	519.763	Meaningful interpretation of sets of symbols
519.66	Mathematical tables and their compilation ⇒ 51-3	519.764	Deciphering of systems of symbols
519.66.09	Tables according to degree of accuracy		
519.66.094	Four-figure tables		
519.66.095	Five-figure tables		

519.765	Mathematical linguistics in general. General mathematical investigations of languages	519.855	Parametric programming
519.766	Models for languages and language structures	519.856	Stochastic programming
519.766.2	Algebraological models. Set-theory models	519.857	Dynamic programming. □ Multistep games
519.766.4	Probabilistic-statistical models	519.86	Theory of economic-mathematical models
519.767	Semantics of a language: mathematical aspects ⇒ 510.27		⇒ 330.4
519.768	Mathematics of machine translation	519.862	⇒ 510.67
519.8	Operational research ⇒ 005.31 ⇒ 303.725.35 ⇒ 330.45	519.863	Descriptive models. □ Macroeconomic models.
519.81	Theory of utility and acceptability of solutions	519.865	Econometrics (mathematical) ⇒ 51-7
519.812	General theory of utility	519.866	Optimization models
519.813	Games of chance. Experimental games	519.868	Theory of economic behaviour. □ Exchange models. Equilibrium models. Theory of the firm. Portfolio selection
519.814	Theory of statistical solutions		Modelling of specific aspects of the economy.
519.816	Theory of decision-making	519.87	□ Price formation models. Cost-effectiveness models
519.83	Theory of games		Branching models
519.832	Antagonistic games. Matrix games	519.872	Mathematical models for operational research
519.833	Noncoalition games. □ Bimatrix games		Queuing theory. □ Service systems. Numerical simulation
519.834	Coalition games	519.873	Theory of reliability and reservation. Quality control
519.837	Dynamic games. □ Positional games	519.874	Theory of supplies and stock control. Inventory theory. □ Models of storage. Models of replacement
519.85	Mathematical programming ⇒ 004.42 ⇒ 519.7	519.876	Theory of large systems
519.852	Linear programming. □ Simplex method	519.876.2	Modelling of control processes
519.853	Nonlinear programming. □ Relaxation methods	519.876.3	Network planning
519.854	Discrete programming	519.876.5	Numerical imitation of systems. Simulation
		519.878	Search theory

52 ASTRONOMY. ASTROPHYSICS. SPACE RESEARCH. GEODESY

52-1/-8	PROPERTIES, PROCESSES, PARTS ETC. <i>These auxiliaries are applicable only at 52/524</i>	52-327	Rotation
52-1	Mode of treatment	52-33	Physical properties (except radiation)
52-12	Initial investigation	52-332	Dimensions. Shape parameters
52-121	Unidentified objects	52-332.2	Apparent angular dimensions
52-123	Initial hypotheses	52-332.4	Linear dimensions
52-125	Predictions	52-332.5	Oblateness parameters
52-126	Searches	52-332.6	Irregularities in shape
52-128	Discoveries	52-333	Mass
52-13	Observations and measurements. Reduced data	52-334	Physical bulk properties
52-14	Results of analysis. Evaluation of observations and measurements	52-334.2	Mechanical properties
52-16	Theoretical treatment and development	52-334.4	Viscosity
52-17	Numerical treatment. Simulation. Mathematical techniques	52-334.5	Opacity. Refractive index
52-3	Properties and phenomena, especially geometrical	52-334.6	Thermal conductivity
52-32	Position and motion	52-334.7	Electrical and magnetic properties
52-323	Position	52-335	Physical state variables
52-323.2	Angular coordinates	52-335.3	Density
52-323.3	Distance	52-335.5	Pressure
52-323.33	Parallax	52-335.7	Temperature
52-323.35	Light-time	52-336	Gravity field
52-323.4	Doppler shift. Radial velocity	52-337	Magnetic field
52-323.6	Position in space	52-338	Age
52-323.7	Distribution in space	52-34	Distribution of bodies in a system or systems
52-323.8	Orbital motion. Elements of orbit	52-35	Radiation from a body or system
52-325	Motion		⇒ 52-6
52-325.2	Proper motion		⇒ 52-7
52-325.4	Radial velocities	52-352	Apparent intensity (intensity as observed).
		52-353	Apparent magnitude. Flux density
			Absolute intensity (intensity as emitted).
			Absolute magnitude
		52-355	Spectrum. Spectral type
		52-357	Polarization
		52-36	Chemical composition. Abundances
		52-37	Biological properties
		52-38	Geometrical phenomena

- 52-383 Seasonal phenomena
52-384 Diurnal phenomena
- 52-4 Processes of bodies and systems**
52-42 Interactions between bodies within systems
52-423 Gravitational interactions
52-423.3 Tidal interactions
⇒ 521.16
52-423.4 Mass-flow. Exchange of mass
52-424 Collisions
52-425 Electromagnetic forces
52-43 Condensation. Accretion
52-44 Disintegration. Separation. Mass-loss
52-46 Bulk-matter processes
52-462 Conduction
52-464 Convection
52-466 Turbulence
52-468 Shock processes
52-47 Atomic processes
52-472 Excitation
52-473 Ionization
52-48 High-energy processes. □ Nuclear reactions
- 52-5 Stages in development of bodies and systems**
52-52 Origin. Formation. Cosmogony
⇒ 524.85
⇒ 550.2
⇒ 551.12
52-54 Evolution. Change of state or structure
52-55 Stability. Equilibrium
52-56 Variability
52-563 Regular variability. Pulsation
52-565 Irregular variability. Explosion
52-567 Isolated change. □ Quakes. Tremors. Seisms
52-59 Catastrophic change of state or structure. Final stages
- 52-6 Processes of radiation**
52-62 Emission
52-622 Thermal emission
52-624 Synchrotron radiation
52-626 Stimulated emission. Laser and maser emission
52-628 Luminescence
52-64 Radiative transfer
52-65 Propagation
52-652 Reflection
52-653 Refraction
52-655 Polarization. □ Faraday effect (Faraday rotation)
52-656 Interference. Diffraction. Coupling
52-657 Absorption. Reddening
52-658 Scintillation
52-67 Frequency and wavelength shifts. Doppler effect. Aberration
52-68 Velocity. Speed
52-683 Acceleration
52-687 Retardation. Deceleration
- 52-7 Character of radiation**
52-72 Corpuscular radiation
52-724 Protons and atomic nuclei
52-726 Plasma
52-728 Neutrinos
52-73 Very short-wave photons
52-732 Gamma-rays
52-735 X-rays (λ less than 1 nm)
52-739 Extreme ultraviolet (λ 1 nm to 50 nm)
52-74 Ultraviolet radiation (λ 50 nm to 300 nm)
52-75 Visible radiation (λ 0.3 μ m to 0.7 μ m)
52-76 Infrared radiation (λ 0.7 μ m to 1 mm)
52-77 Radio radiation (λ more than 1 mm)
52-78 Waves in plasmas
- 52-782 Shock waves
- 52-8 Parts and features of individual systems**
52-82 Interior. Central region. Head (of comet)
52-823 Core. Nucleus (of comet)
52-825 Mantle
52-827 Crust. Subsurface region
52-83 Surface. Surface features
52-834 Solid surface. Lithosphere. □ Mountains. Craters. Rills
52-836 Oceans. Hydrosphere
52-837 Photosphere
52-84 Regional features
52-842 Polar features
52-843 Mid-latitude features
52-844 Equatorial features
52-846 Apparent disc
52-847 Apparent edge
52-848 Terminator
52-85 Exterior regions
52-852 Atmosphere
52-853 Ionosphere
52-854 Magnetosphere
52-856 Spiral arms
52-857 Halo. Extended atmosphere
52-86 Exterior features. Surroundings
52-862 Rings and disc features
52-864 Tails
52-866 Filaments. Jets. Links
52-87 Satellites. Companions
52-88 Peculiar features
- 520 ASTRONOMICAL INSTRUMENTS AND TECHNIQUES**
⇒ 681.7
- 520.1 Observatories**
⇒ 727
- 520.2 Astronomical telescopes**
⇒ 681.751.2
- 520.2.01 Theory and general properties
520.2.01 $\equiv$ 681.7.01
520.2.02 Construction
520.2.03 Adjustment and testing. Instrumental errors
520.2.06 Optical components
520.2.06 $\equiv$ 681.7.06
520.2.07 Mechanical components
520.2.07 $\equiv$ 681.7.07
- 520.22 General-purpose optical telescopes**
520.222 Refractors. Long-focus telescopes
520.224 Reflectors. Reflecting telescopes in general. Foci
◇ 520.224.062.42 Conic section mirrors. E.g. parabolic mirrors
- 520.24 Solar instruments**
520.242 Coelostats. Heliostats. Siderostats
520.243 Solar towers
520.244 Spectroheliographs
520.245 Monochromators
520.246 Coronagraphs
520.248 Solar magnetographs
- 520.25 Astrometric instruments**
⇒ 521
520.252 Transit instruments
520.253 Meridian circles
520.254 Vertical circles
520.255 Zenith tubes (PZT, VZT, LZT etc.)
520.256 Modern astrolabes. Impersonal astrolabes
520.257 Astrographs. Astrographic cameras
520.259 Other astrometric instruments. □ Quadrants

520.27	Radio telescopes	520.86	Observation techniques for eclipses, transits, occultations
520.272	Simple telescopes		
520.272.2	Antennas. □ Reflectors. Feed. Control systems	520.87	Special techniques
520.272.4	Transmission systems	520.872	Interferometry
520.272.5	Receivers	520.874	Ranging techniques
520.274	Interferometer-type radio telescopes	◇ 520.874:621.375.82	Ranging by laser/maser
520.3	Auxiliary instruments. Recording instruments	◇ 520.874:621.396.96	Ranging by radar
	⇒ 621.397	520.876	Techniques used for non-ground-based observations
	⇒ 681.782.4		
520.32	Visual equipment	520.9	Other instruments or techniques
520.322	Oculars. Eyepieces	520.98	Planetariums
520.328	Micrometers		⇒ 681.782.7
520.33	Photographic instruments and equipment		⇒ 727
	<i>Specify by :77...</i>	520.99	Other simulators
◇ 520.33:771.319.5	Astronomical cameras	521	THEORETICAL ASTRONOMY. CELESTIAL MECHANICS
520.34	Photoelectric instruments and equipment		
	<i>Specify by :621.383...</i>	521.1	Celestial mechanics. General principles of dynamical astronomy
520.35	Spectroscopic instruments		
520.36	Interferometers	521.13	General problems
520.37	Recording and timing instruments used when making observations	521.131	General n-body problem
520.373	Analogue recorders	521.135	Restricted 3-body problem
520.374	Digital recorders	521.14	Theories of figures, and gravity fields and rotation of extended bodies
520.378	Timing equipment		
520.4	Measuring instruments, normally separate from telescopes	521.16	Theories of tidal interaction and other dissipative forces
520.42	Coordinate measuring instruments	521.17	Special problems
520.43	Densitometers. Microphotometers	521.172	Planetary theory (mutual perturbations important)
520.44	Spectrophotometers	521.174	Lunar theory (perturbations by distant third body important)
520.47	Blink comparators	521.176	Close-satellite theory (oblateness of primary important)
520.48	Automatic measuring instruments		
520.6	Instruments for specific environments	521.18	Special techniques
520.6.02	Aeroplane-borne instruments		⇒ 521.9
520.6.03	Balloon-borne instruments	521.181	Coordinate reference systems. □ Ecliptic. Equatorial. Galactic
520.6.04	Rocket-borne instruments	521.182	Numerical integration of orbits
520.6.05	Satellite-borne instruments	521.184	Integration of variations of orbit elements or parameters
520.6.07	Space-probe instruments	521.186	Averaging methods for long-term orbits
520.6.08	Instruments for observations from the Moon and other celestial bodies	521.19	Special techniques for development of analytical theories
520.62	Electromagnetic wave detectors.		<i>Specify by :51...</i>
	□ Spectrometers. Polarimeters		◇ 521.19:517.952
520.64	Plasma analysers (in situ). □ Ion analysers.		Using first-order partial differential equations
	Magnetometers		
520.66	Suprathermal particle detectors	521.3	Orbit determination
520.68	Dust detectors	521.31	Specification of orbits. Orbital elements and parameters
520.8	Techniques of observation, measurement, analysis	521.32	Initial orbit determination
520.8.07	Calibration	521.35	Orbit improvement
520.82	Photometry	521.8	Astrometric aspects of eclipses, transits, occultations
	⇒ 535.24		
	⇒ 681.782.43	521.81	Eclipses of the Sun and Moon
520.84	Spectrometry		
	⇒ 543.42	521.83	Satellite and ring phenomena
	⇒ 681.782.46	521.832	Eclipses of a satellite by its planet
520.849	Classification systems. □ Spectral classification. Luminosity classification		◇ 521.832:523.45-87*1
	◇ 520.849.07	521.834	Eclipses of Io by Jupiter
	Standard stars		Transits and occultations of satellites over and by planets
520.85	Polarimetry	521.835	Mutual phenomena of satellites
	⇒ 543.454	521.837	Ring phenomena
	⇒ 681.785.3		

521.84	Transits of planets across the Sun	523.41	Mercury
521.85	Occultations by the Moon	523.42	Venus
521.86	Occultations by planets and their satellites, and by minor planets	523.43	Mars
521.87	Occultations by comets	523.44	Minor planets (Planetoids. Asteroids)
521.89	Computation of rising, setting and local transit phenomena	523.45/48	Outer planets
521.9	Astrometry. Spherical astronomy	523.45	Jupiter
521.91	Coordinate and time systems in general	523.46	Saturn
521.92	Theory, determination and constants of precession and nutation	523.47	Uranus
521.93	Theory and determination of Earth rotation and polar motion	523.481	Neptune
521.932	Determination of time by astronomical observations	523.482	Pluto
	◇ 521.932:521.84 Determination by transit	523.489	Hypothetical planets
	◇ 521.932:528.37 Determination by altitudes	523.6	Interplanetary medium. Comets. Meteors. Meteorites
	◇ 521.932:681.786.3 Determination by sextant	523.62	Interplanetary medium
521.933	Irregularity of the Earth's rotation		◇ 523.62-65 Zodiacal light
521.934	Timescales. Systems of time reckoning	523.64	◇ 523.62-726 Interplanetary plasma. Solar wind
521.936	Latitude determination from astronomical observations	523.68	Comets
521.937	Movement of terrestrial poles. Variations of latitude and longitude	523.681	Meteors. Meteoroids. Meteorites
521.94	System of astronomical constants	523.682	Meteorites. Meteor remnants on Earth's surface
521.95	Reduction of astrometric observations	523.683	⇒ 552.6
521.96	Fundamental position and proper-motion systems	523.684	Meteors. Meteoroids. Shooting stars
521.97	Observational (instrumental) corrections	523.9	Meteor showers
	⇒ 520.25		Tektites. Australites. Moldavites
	⇒ 520.4		⇒ 549.691.1
521.98	Calculation of ephemerides		⇒ 552.64
523	THE SOLAR SYSTEM	523.92	The Sun. Solar physics
	◇ 523-52 Origin, formation of the Solar System	523.94	◇ 523.9:520.876 Observations of the Sun from space
523.2	General properties of the Solar System	523.942	Solar interior
523.21	Laws of planetary distribution. □ Bode's law	523.942.3	Quiet Sun
523.23	Conjunctions and oppositions	523.945	Photosphere
523.24	Motion of the Solar System in space	523.947	Granulation
523.3	Earth-Moon system		Chromosphere
523.31	The Earth as an astronomical body		Corona
	◇ 523.31-121 Unidentified objects in the vicinity of the Earth. Near-Earth objects (NEOs)	523.98	◇ 523.947:520.86 Observations of the corona during eclipses
523.34	Moon. Selenology	523.982	◇ 523.947-335.7 Temperature of the corona
523.4	Planets and their satellites. Planetology	523.983	◇ 523.947-355 Spectrum of the corona
	◇ 523.4-87 Satellites (of the Solar System planets)	523.985	Active Sun. Solar activity
	◇ 523.45-87*3 Jupiter satellite III (Ganymede)	523.987	Sunspots
523.41/43	Terrestrial, interior planets	523.988	Faculae. Plages
		524	Flares. Bursts
			Prominences
			Streamers
			STARS. STELLAR SYSTEMS. THE UNIVERSE
			◇ 524-86 Circumstellar features. Including: planets (other than those of the Solar System). Extrasolar planets
		524.1	Cosmic rays. Primary cosmic rays
		524.3	Stars
		524.31	Simple stars. Stars according to luminosity and spectral type
			◇ 524.31*1/*5 Stars according to luminosity class
			◇ 524.31*1 Supergiant stars. Class I stars
			◇ 524.31*2/*3 Giant stars. Class II and III stars
			◇ 524.31*4 Subgiant stars. Class IV stars
			◇ 524.31*5 Dwarf and subdwarf stars. Class V stars
			Stars according to spectral type
			Early-type stars. □ O stars. B stars

524.314	A stars. F stars	524.63	Local region of the Galaxy. Solar neighbourhood
524.316	Late-type stars. □ G stars. K stars. M stars	524.64	Galactic centre. Nucleus of the Galaxy
524.318	Other (non-main-sequence) classes of star. □ S stars. R stars. N stars	524.66	Spiral arms of the Galaxy
524.33	Variable stars	524.68	Halo of the Galaxy
524.331	Pulsating variables in general	524.7	Extragalactic systems ◇ 524.7-323.3 Distance of system ◇ 524.7-327 Rotation of system ◇ 524.7-52 Origin, formation of systems
524.332	RR Lyrae stars	524.72	Local Group of galaxies
524.333	Cepheids	524.722	Magellanic Clouds
524.334	Semiregular variables	524.724	Irregular galaxies in Local Group
524.335	Miscellaneous variables	524.726	Spiral galaxies in Local Group
524.336	Long-period variables. □ Mira stars. Me stars	524.728	Elliptical galaxies in Local Group
524.337	Novae. Explosive and eruptive variables	524.74	Types of galaxy
524.338	Special variable stars	524.77	Clusters and systems of galaxies
524.338.2	Shell stars. Envelope stars ⇒ 524.37	524.78	Intergalactic matter 524.78 ≡ 524.5
524.338.3	Symbiotic variables. Symbiotic stars	524.8	The Universe. Metagalaxy. Cosmology
524.338.5	Nebular variables. T Tau stars	524.82	Observational cosmology
524.338.6	Flare stars	524.822	Red-shift relation. Hubble constant
524.338.7	R Coronae Borealis stars	524.827	Sky-background radiation
524.35	Supernovae and related objects. Peculiar stars ◇ 524.35-76-121 Unidentified infrared objects believed to lie in the Galaxy	524.83	Fundamental theories of cosmology
524.352	Supernovae	524.85	Origin and formation of the Universe ⇒ 52-52
524.354	Objects believed to be supernovae remnants ◇ 524.354-423 Gravitational collapse of supernovae remnants	524.852	Big-bang theories
524.354.2	Crab nebula ◇ 524.354.2-62 Emission mechanism of Crab nebula ◇ 524.354.2-735 X-ray radiation from Crab nebula	524.854	Steady-state theories
524.354.4	Pulsars ◇ 524.354.4-327 Rotation of pulsars ◇ 524.354.4-563 Pulsation processes of pulsars	524.86	Origin and formation of the elements
524.354.6	Neutron stars ◇ 524.354.6-423 Gravitational collapse of neutron stars	524.88	Special hypotheses
524.354.7	White dwarf stars (as supernovae remnants)	524.882	Black holes
524.355	Other objects believed to be disintegrating or collapsing	524.883	Quasi-stellar objects (QSOs). Quasars
524.358	Wolf-Rayet stars	524.884	Antimatter
524.37	Planetary nebulae	528	GEODESY. SURVEYING. PHOTOGRAMMETRY. REMOTE SENSING. CARTOGRAPHY <i>The auxiliaries listed under 52-1/-8 are not applicable at 528 and its subdivisions</i>
524.38	Double and multiple stars. Binary systems	528-1/-7	Details of instruments, devices etc. 528-1/-7 ≡ 62-1/-7
524.382	Double stars (optical or visual doubles)	528.01	Preparatory work. □ Marking of stations. Signal construction
524.383	Visual binary systems	528.02	Geodetic measuring methods ⇒ 53.08
524.384	Spectroscopic binary systems	528.021	Methods of distance measurement
524.386	Eclipsing binary systems	528.021.1	Mechanical methods of distance measurement
524.387	Close binary systems. Very close binary systems	528.021.4	Optical and trigonometric methods of distance measurement
524.388	Multiple stars. Multiple systems	528.021.6/7	Methods of distance measurement by measuring the transit time of electromagnetic waves
524.4	Star clusters. Associations of stars	528.021.6	Electric methods of distance measurement. □ With tellurometer ⇒ 621.396.96
524.42	Associations of stars	528.021.7	Electro-optical methods of distance measurement. □ With geodimeter. With quantum amplifiers (laser etc.)
524.45	Open clusters. Galactic clusters	528.022	Methods of angular and direction measurement
524.47	Globular clusters	528.022.6	Measurement of oriented directions
524.5	Interstellar medium. Galactic nebulae	528.023	Methods for plotting position
524.52	Interstellar gas	528.024	Levelling methods
524.522	H I regions	528.026	Gravity measurement
524.523	H II regions	528.029.4/6	Frequency ranges and wavelengths of electromagnetic oscillations 528.029.4/66 ≡ 621.3.029.4/66
524.527	Interstellar molecules		
524.54	Emission nebulae		
524.57	Interstellar dust. □ Dark nebulae		
524.6	The Galaxy (Milky Way)		
524.62	Structural parameters of the Galaxy		

528.029.67/69	Light rays. Gamma-rays. Cosmic rays	528.234	Level spheroid
528.029.67	Light rays	528.235	Representations of approximate surfaces.
528.029.68	Gamma-rays		Geodetic projections. Map projections.
528.029.69	Cosmic rays		Graticule design. Grids
528.03	Auxiliary measurement	528.236	Coordinate systems. Conversion of coordinates
528.031	Determination of points	528.24	Relations between the shape of the Earth and its approximate surfaces
528.033	Simple auxiliary figures. □ Broken (traverse) line	528.241	Deflections of the vertical
528.034	Plumbing down	528.242	Geoid undulations
528.037	Centring measurements	528.247	Gravitational anomalies
528.06	Evaluation of measuring results	528.26	Arc measurement
528.061	Correction and reduction of measuring results	528.27	Gravity measurement
528.063	Geodetic computations		⇒ 531.51
528.065	Textual representation of survey results	528.28	Astro-geodetic determination of position. Geographical coordinates
528.066	Representation of survey results in numerical tables. Tables of coordinates. Trig lists. Station summaries		◇ 528.28.087.2 Astronomic-optical observation
528.067	Graphical representation of survey results. □ Field diagrams. Station diagrams		◇ 528.28.087.5 Astronomic-photographic position recording
528.08	General principles and theories of measurement and of construction of measuring instruments	528.281	Determination of latitude. Observation of polar altitudes
	528.08 ≡ 53.08	528.282	Determination of longitude
528.1	Theory of errors and adjustment in geodesy and photogrammetry	528.283	Determination of azimuth
	⇒ 519.233	528.288	Astronomical orientation. Astronomical connection
528.11	Theory of errors in observation	528.3	Geodetic surveying
528.13	Preparation of observation programmes respecting theory of errors	528.31/35	Planimetric determination of stations
528.14	Adjustment by least squares method	528.31	Outline and structure of the geodetic survey. □ Orientation. Point of origin. Laplace stations. Datum plane
528.16	Other methods of adjustment	528.32	Base-line measurement
528.162	Adjustment by successive approximation	528.33	Trigonometric networks
528.165	Graphical adjustment	528.331	Primary triangulation. First-order triangulation.
528.181	Treatment of adjustment problems by means of matrix and tensor calculus	528.335	First-order astro-geodetic networks
528.2	Figure of the Earth. Earth measurement. Mathematical geodesy. Physical geodesy. Astronomical geodesy	528.338	Filling-in of network. Triangulation of second and lower orders. □ Framework traverses
528.21	Fundamentals derived from potential theory. Level surfaces. Geoids	528.34	Planimetric stations. Trigonometric stations. Traverse stations
528.22	Methods of determining the figure of the Earth	528.341	Special methods of trigonometric determination of stations
528.221	Geometric (static) methods. Use of longitudinal and latitudinal measurements	528.342	Flare triangulation
528.223	Physical methods. Dynamic methods. □ Use of gravity measurements	528.343	Solar eclipse method
528.225	Astronomical methods. □ Use of precession, nutation. Use of irregularities in orbit of Moon, of satellites	528.344	Star occultation method
528.23	Mathematical surfaces approaching the shape of the Earth. Geodetic projections. Coordinate systems	528.35	Stellar triangulation
528.231	Spheres	528.37/38	Trilateration
528.232	Ellipsoids of revolution	528.37	Vertical determination of stations. Levelling
528.232.1	Reference ellipsoids and their basic values	528.371	Kinds of elevation and principles of levelling
528.232.2	Computations on the ellipsoid		Datum planes. Reference points. Gauges
528.232.22	Normal sections. Geodetic lines. □ Differential equations. Expansion in series. Staking-out curve. Loxodromes (rhumb lines). Great circles	528.38	Figuration and observation of level network
528.232.23	The geodetic triangle. Excess (spherical excess). Area. Transformation into the plane	528.381	Primary level network. First-order level network
528.232.24	The main problem of geodesy. Computation of geographical coordinates when azimuth and side are given. Inversion of the problem	528.385	Filling-in of level networks. Level networks of second and lower orders
528.233	Triaxial ellipsoids	528.388	Bench marks. Levelling stations
	528.233 ≡ 528.232	528.389	Special applications of levelling and depth measurement
		528.4	Field surveying. Land surveying. Cadastral survey. Topography. Engineering survey. Special fields of surveying
		528.41	Local geodetic networks
		528.411	Local triangulation networks. Detail triangulation
		528.412	Intercalation of single points. Determination of points by intersection

528.414	Traversing. Supplementary points ⇒ 528.335	528.53	Instruments and equipment for simultaneous measurement of angles and distances
528.42	Topographic surveying	528.531	Tacheometer (tachymeter) theodolites and accessories
528.421	Planimetric surveying	528.533	Plane tables
528.422	Levelling ⇒ 625.725	528.534	Alidades
528.425	Tacheometry. Topography. □ Study of landforms. Orientation on the ground	528.535	Topometers
528.44	Cadastral survey	528.54	Instruments and equipment for measurement of elevations, depths, inclinations
528.45	Urban surveying	528.541	Instruments and equipment for geometric levelling and inclination measurement. □ Levels. Spirit levels etc. Levelling rods
528.46	Surveying for soil improvement ⇒ 626.8 ⇒ 711	528.541.2	Automatic levels
528.47	Hydrographic surveying. Coast surveying	528.541.4	Simple inclinometers, clinometers, gradient meters
528.48	Engineering survey. Special fields of surveying ◇ 528.48:624.19 Surveying for tunnelling	528.543	Automatic height indicators
528.481	Observations of local ground displacements. □ Subsidence ⇒ 551.24 ⇒ 624.131	528.546	Instruments and devices for hydrostatic levelling
528.482	Observations of structures. Geodetic operations during erection and operation of structures. Observations of settling. Deformation measurement ⇒ 622.83 ⇒ 624.131.5	528.548	Direction-finding devices. Plumbing devices
528.484	Connection in underground survey. □ Galleries. Shafts. Plumbing-down. Depth measurements	528.56	Instruments and equipment for gravity measurement
528.486	Staking-out operations. Transfer of construction plan into terrain. Tracing	528.561	Pendulum instruments
528.5	Geodetic instruments and equipment <i>Details by special auxiliaries -1/-7 mentioned at 528-1/-7</i> ◇ 528.5-187 Accuracy ◇ 528.5-217 Ground plates. Base plates ◇ 528.5-539 Precision regulation ◇ 528.5-752.3 Stabilizing devices ⇒ 520.2/6 ⇒ 531.7 ⇒ 532.29 ⇒ 681.2 ⇒ 681.783	528.565	Gravimeters
528.51	Instruments and equipment for distance measurement ⇒ 531.71	528.58	Devices for marking-out. Signalling equipment. Devices for finding of marked points
528.511	Base apparatus. □ Invar wires. Invar tapes	528.59	Plotting instruments and equipment ⇒ 744.3
528.512	Geodetic equipment for length measurement, profiles etc. □ Subtense bars. Measuring tapes	528.7	Photogrammetry: aerial, terrestrial ⇒ 355.404.4 ⇒ 550.81 ⇒ 681.783.3 ⇒ 778.35
528.514	Optical instruments for distance measurement	528.71	Production of photograms
528.516	Electric instruments for distance measurement. □ Tellurometers	528.711	Methods of production of photograms
528.517	Electro-optical instruments for distance measurement. □ Geodimeters	528.711.1	Photographic production of photograms in central projection <i>Specify technical aspects of photography by :77...</i> ⇒ 528.835 ⇒ 771.319.5
528.519	Accessories for distance measurement. □ Stadia rods. Tacheometer (tachymeter) rods	528.714	Exterior influences on image formation. □ Illumination. Object brightness. Haze brightness. Image movement. Instrument vibration. Unsharpness
528.52	Instruments and equipment for angular measurement, direction measurement, staking-out	528.715	Photogrammetric survey flight ◇ 528.715:629.735 Photogrammetric survey aircraft ⇒ 528.837
528.521	Theodolites and accessories	528.716	Determination of camera orientation elements at the exposure station. □ Methods and instruments for measurement, control
528.526	Instruments for measurement of oriented directions	528.716.1	Planimetry. Determination of position
528.526.1	Magnetic instruments. □ Magnetic compasses	528.716.2	Altitude and altitude differences. Altimetry. □ Statoscope
528.526.6	Gyroscopic instruments. Surveying gyroscopes	528.716.3	Deviation of camera axis from the vertical (verticality). Tilt. □ Horizon camera. Solar periscope. Controlled camera mounting
528.528	Devices for staking-out angles and directions. □ Diopter, mirror, prism instruments. Protractors	528.716.4	Swing. Azimuth curvature of the flight line
		528.718	Practical procedures in production of terrestrial photograms. □ Relations between object distance, focal length, distance between exposures. Image scale. Base ratio

528.72	Restitution of photograms ◇ 528.72:528.236 Photogrammetric coordinate systems ⇒ 514.14	528.813	Interaction mechanisms. □ Absorption. Reflection ⇒ 537.87
528.721	Orientation and restitution of photograms in photographic central projection	528.83	Data acquisition
528.721.1	Orientation and restitution of single photograms	528.831	Resolution
528.721.15	Compilation of mosaics	528.831.1	Spatial resolution
528.721.16	Compilation of maps by tracing of photograms	528.831.2	Spectral resolution
528.721.2	Orientation and restitution of pairs of photograms	528.831.3	Intensity resolution
528.721.28	Stereophotograms and stereomosaics. □ Anaglyphs	528.831.4	Temporal resolution
528.722	Instruments for restitution of photograms in photographic central projection	528.831.5	Temperature resolution
528.722.2	Transformation printers	528.835	Remote sensors and their use <i>Denote particulars about the kind of remote sensor by .04... (listed under 528.8.04) and about general principles by .08... (see at 528.08)</i>
528.722.3	Restitution instruments for single photograms. □ Rectifiers. Photoangulators		◇ 528.835.042.3.089.6 Calibration of multispectral scanners ⇒ 528.711.1
528.722.4	Orthophoto instruments. □ Orthophotoscope. Orthoprojector	528.837	⇒ 778.35
528.722.6	Stereorestitution instruments. □ Photogoniometer-type instruments. Radial line instruments		Remote sensing platforms ◇ 528.837:629.73 Aircraft used for remote sensing ◇ 528.837:629.783 Artificial satellites used for remote sensing ⇒ 528.715
528.722.8	Instruments for measurement of plate coordinates and angles. □ Comparators. Radial triangulators	528.85	Data processing
528.73	Photogrammetric aerial triangulation. Determination of control points and pass points	528.851	On-board processing. Data transmission
528.731	Radial triangulation	528.852	Image processing. □ Conversion. Data compression. Correction of geometric distortions. Radiometric enhancement. Image enhancement. Stereo creation
528.732	Other methods of planimetric control from photograms	528.854	Image analysis. Pattern recognition. □ Segmentation. Feature extraction. Cluster analysis ⇒ 519.237.8
528.735	Spatial aerial triangulation	528.855	Ground investigations. Test sites
528.737	Control points. Transfer points	528.856	Geometric transformations. Mapping
528.738	Terrestrial methods of ground control determination	528.87	Interpretation. □ Visual requirements for image interpretation
528.74	Applications of photogrammetry ◇ 528.74:52 Applications in astronomy ◇ 528.74:528.93 Application to topographic cartography	528.871	Elements of image interpretation. □ Resolution. Size. Shape. Shadow. Colour. Pattern
528.77	Interpretation of photograms ⇒ 528.873	528.873	Interpretation of imagery <i>Class interpretation of particular applications under 528.88</i>
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528.8.041	Photographic central projection techniques <i>Specify technical aspects of photography by :77...</i> ⇒ 528.711.1		◇ 528.88:551.4 Application of remote sensing to geomorphology ◇ 528.88.041.3:632.2 Detection of plant injuries by means of infrared photography
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528.8.041.5	Multispectral photography	528.91	Theoretical cartography. Methodology of cartography
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528.8.042	Scanning techniques. □ Photographic, multispectral, thermographic, microwave scanning techniques	528.913	Principles of map design. Map editing ⇒ 528.235
528.8.044	Active techniques	528.914	Generalization
528.8.044.1	Side looking real aperture radar (SLAR)	528.915	Scales. Series of scales. Choice of scale
528.8.044.2	Side looking synthetic aperture radar (SAR)	528.916	Neat-line systems. Designation of map sheets. Sheet size
528.8.044.4	Scatterometer techniques		528.918
528.8.044.6	Light detection and ranging (LIDAR)	528.92	Practical cartography: processes, operations
528.8.044.8	Airborne profile recording (APR)	528.921	Execution of map design
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528.811	Nature of electromagnetic radiation ⇒ 535 ⇒ 537.86		

528.923	Cartographic drawing techniques. □ Drawing bases, supports. Cartographic drawing equipment ⇒ 744.43	528.932	Representation of elevation. Relief mapping. □ Hachures. Contour lines. Hill-shading. Hypsometric tints
528.924	Cartographic engraving with and without coating. □ Coating and coating base. Devices for cartographic scribing ⇒ 681.94	528.933	Colouring
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528.926	Cartographic colour separation methods	528.936	Map correction, revision, updating ◇ 528.936:528.4 Topographic report service
528.927	Cartographic reproduction ⇒ 655.3.026	528.94	Thematic cartography. Topical cartography ◇ 528.94:551.4 Geomorphological cartography
528.93	Topographic cartography. Chorographic cartography (landscape cartography). Representation of place and space ⇒ 003.62	528.951	Preparation of globes
528.931	Representation of planimetric detail. □ Settlement. Water features. Vegetation	528.952	Preparation of reliefs (three-dimensional)
		528.97	Style-sheets. Mapping specifications

53 PHYSICS

53.01	Theory and nature of phenomena	53.082.25	Use of hydrodynamic phenomena
53.02	General laws of phenomena	53.082.3	Use of aerostatic and aerodynamic phenomena
53.023	Empirical formulae	53.082.32	Use of aerostatic phenomena
53.03	Production and causes of phenomena	53.082.36	Use of aerodynamic phenomena
53.04	Effects of phenomena	53.082.4	Use of acoustic phenomena. □ Use of sound waves
53.043	Physical effects	53.082.5	Use of optical phenomena
53.044	Chemical effects	53.082.52	Use of photoelectric phenomena. Optoelectronic measurements
53.047	Biological effects	53.082.53	Use of reflection, refraction, absorption
53.05	Observation and recording of phenomena. Visual indication of phenomena	53.082.54	Use of interference
53.06	Application, utilization of phenomena	53.082.55	Use of polarization, double refraction (birefringence), dispersion in anisotropic substances
53.07	Apparatus for production and study of phenomena	53.082.56	Use of colour
53.072	Models	53.082.6	Use of thermal phenomena. □ Use of hot wires
53.08	General principles and theory of measurement and the design of measuring apparatus. Measurement methods ⇒ 531.7 ⇒ 681.2	53.082.7	Use of electrical, magnetic, nuclear phenomena
53.081	Units. Constants	53.082.72	Use of electrostatic phenomena, electrostatic force, capacitance
53.081.3	Absolute units	53.082.73	Use of electrokinetic phenomena. □ Use of piezoelectric phenomena
53.081.4	Relative and practical units	53.082.74	Use of electromagnetic, electrodynamic phenomena
53.081.5	Dimensions of units	53.082.75	Use of electrolytic, electrochemical phenomena
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53.081.7	Constants for specific substances	53.082.77	Use of discharge phenomena
53.082	Principles of measurement and of construction of instruments	53.082.78	Use of magnetic phenomena
53.082.1	Use of mechanical phenomena	53.082.79	Use of nuclear phenomena
53.082.12	Use of weight, static forces	53.082.8	Use of chemical phenomena
53.082.13	Use of dynamic phenomena	53.082.9	Use of biological phenomena. □ Biosensors
53.082.14	Use of pendulums, balance wheels	53.083	Methods of measurement. Instruments according to measurement method
53.082.15	Use of centrifugal force	53.083.1	Direct measurements
53.082.16	Use of gyroscopic devices	53.083.2	Resonance methods
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53.082.18	Use of friction	53.083.4	Null methods. Bridge methods
53.082.2	Use of hydrostatic and hydrodynamic phenomena	53.083.5	Compensation methods
53.082.22	Use of hydrostatic phenomena	53.083.6	Differential methods
		53.083.62	Using devices for measuring and indicating small differences
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		53.083.7	Remote measurement. Telemetry ⇒ 621.398 ⇒ 654.94
		53.083.8	Summation methods. Integral measurement
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53.084	Construction of instruments. Primary components <i>The -1/-9 auxiliaries from 62-1/-8 and 66-9 are applicable at .084 and its subdivisions to denote instrument details</i>	53.087.9	Devices for converting, calculating, counting, printing out
53.084.2	Receivers. Parts in contact with phenomena measured	53.088	Errors. Error correction. Evaluation, interpretation of measurements
53.084.4	Taring. Tare pans	53.088.2	Errors
53.084.6	Devices for magnification, amplification, of measured quantity	53.088.21	Errors due to observer
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53.084.82	Liquids: supply, reservoirs, conveyance, discharge	53.088.24	Errors due to transmission medium
53.084.83	Gases: supply, reservoirs, conveyance, discharge	53.088.3	Mean errors. Accuracy of measurement. Sensitivity. Correction of nonlinear inaccuracies by calibration curves <i>53.088.3 $\approx$ 53.088.2</i>
53.084.84	Sound waves: sources, distribution, reflection, damping	53.088.4	Limits of use of measuring instruments
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53.085	Indicating apparatus	53.089.6	Calibration and testing of instruments
53.085.1	Braking. Damping	53.089.62	Calibration equipment, accessories. ☐ Calibration rigs, tables
53.085.2	Zero point. Zero adjustment. Adjusting devices	53.089.68	Standard test objects, masters, gauges for testing and calibration of instruments
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53.085.31	Mechanical indicating devices. ☐ Needles. Pointers	53.091	Mechanical effects (except pressure)
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53.085.39	Accessories. ☐ Prisms. Mirrors	53.098	Magnetic effects
53.085.4	Scales	530.1	Basic principles of physics $\diamond$ 530.1:140.8 Philosophical outlook of physics. Weltanschauung
53.085.41	Rectilinear scales	530.11	The absolute in individual phenomena. ☐ The ether (aether)
53.085.42	Circular scales. Curved scales. Cylindrical scales	530.12	Relativity principle
53.085.49	Illumination of scales	530.13	Principles of constancy and variability
53.085.5	Springs $\Rightarrow$ 62-272	530.131	Principle of constancy
53.085.6	Casings. Housings $\Rightarrow$ 62-213	530.132	Principle of variability
53.086	Microscopic investigation $\Rightarrow$ 549.086 $\Rightarrow$ 57.086	530.14	Synthesis of principles of constancy and variability. Atomism in general
53.087	Observation. Recording	530.145	Quantum theory $\Rightarrow$ 535.14
53.087.2	Subjective observation	530.145.3	Quantum rules
53.087.21	Observation at normal distance	530.145.5	Determination of h
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53.087.251	Cinematographic observation	530.145.7	Perturbation calculation
53.087.252	Stroboscopic observation	530.15	Principle of action and reaction
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53.087.4	Objective observation. Automatic recording	530.152	Reaction
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53.087.5	Photographic recording		
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530.16	Causality and probability	531.19	Statistical mechanics
530.161	Principle of order in sequence of phenomena. Irreversibility. Heat death ⇒ -026.193	531.2	Statics. Forces. Equilibrium. Attraction ⇒ 531.35 ⇒ 531.42
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539.12-13	Decay. Energy change (Q value)	539.144.6	Half-life
539.12-14	Spin and moments	539.144.7	Isomers ⇒ 544.12
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539.12-164	Energy spectra	539.163	Radioactive series
539.12-17	Interactions with other particles and with matter. □ Scattering. Diffusion. Reflection. Absorption. Secondary radiations	539.164	Radionuclides and their characteristics. Decay schemata. □ Half-life. Energy spectrum
539.12-18	Beams. □ Energy spectrum. Polarization	539.165	Alpha-radioactivity and decay
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539.125.4	Protons	539.172	Individual nuclear reactions
539.125.46	Antiprotons	539.173	Fission
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539.125.52	Transport of neutrons. Neutron flux distribution	539.173.8	Products of fission <i>Specify by :546...</i>
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539.128	Simple particles (atomic number or charge less than 3) ⇒ 539.14	539.183	Properties of atoms in general
539.128.2	Deuterons (^2H)	539.183.2	Atomic mass. Isotopes ⇒ 539.143.2 ⇒ 54.027 ⇒ 621.039.3 ⇒ 621.384.8
539.128.3	Tritons (^3H)	539.183.3	Electronic configuration (shells etc.). Relationship to periodic table
539.128.4	Alpha-particles (^4He)	539.183.4	Spin and moments
539.128.53	Helium-3 (^3He)	539.183.5	Size. Dimensions. Radii
539.128.55	Helium-5 (^5He)	539.184	Internal mechanics of atoms
539.14	Nuclei	539.184.2	Energy levels and their structure
539.141	Nuclear forces	539.184.3	Ionization energies
539.142	Nuclear models	539.184.5	Excited states
539.142.2	Individual particle models. □ Shell models. Statistical models	539.186	Excitation and interaction of atoms. □ Interatomic forces
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539.143	Properties of nuclei	539.189	Analogous systems
539.143.2	Nuclear mass ⇒ 539.183.2	539.189.1	Mesic atoms
		539.189.2	Positronium
		539.19	Physics of single molecules
		539.192	General theory. Wave functions and methods of solution

539.193	General properties. □ Configuration and geometry of molecules. Bond lengths and angles in molecules. Moments	539.31	Generalities. Elastic forces. Elastic potential. Elastic region. Elastic limit
539.194	Internal mechanics of molecules. □ Energy relationships. Vibration and rotation in molecules. Electronic levels. Bonds. Resonance energies, potentials. Nuclear coupling. Polarizability	539.311	Elastic equilibrium conditions (elastostatics)
539.196	Interaction of molecules: with atoms, with other molecules. Excitation and dissociation of molecules. Intermolecular forces ⇒ 539.6	539.312	Conditions of elastic motion (elastokinetics)
539.198	Molecular beams and rays	539.313	Relationships between elastic forces and deformation
539.199	Macromolecules	539.319	Internal stresses in a body due to processing, mode of use etc. □ Stress distribution. Shrinkage stresses ⇒ 539.219.2
539.2	Properties and structure of molecular systems ⇒ 548.1	539.32	Elasticity moduli. □ Variation of modulus with conditions, e.g. temperature. Young's modulus
539.21	Properties of molecular systems	539.37	Deformation in general. □ Plane deformation. Three-dimensional deformation. Deformability
539.211	Surfaces of molecular systems. External structure. □ Fracture surfaces. Plane fracture. Conchoidal fracture. Laminar fracture. Fibrous fracture	539.371	Temporary deformation. Elastic deformation (spontaneously reversible deformation)
539.213	Amorphism. Amorphous state. Vitreous state. □ Theory, properties of the vitreous state	539.372	Nonspontaneously reversible deformation
539.214	Plasticity. Plastometry	539.373	Permanent, irreversible deformation.
539.215	Pulverulence. Powderiness. Specific surface. Friability. Colloidal distribution. Regularity, homogeneity. Granular fineness	539.374	Deformation of hardenable materials
539.216	Cellular structure. Fibrous structure. Laminar structure. Films. Thin films. Radiating structures	539.375	Plastic deformation. Viscous deformation.
539.217	Porosity. Permeability. Imperviousness	539.375.5	Properties of solids not cold-workable. Yielding
539.218	Compactness, closeness. Freedom from pores	539.375.6	Limit of deformation
539.219	Other properties and phenomena	539.376	Deformation at breaking
539.219.1	Irregularities. Impurities. Inclusions. Heterogeneity. □ Streaks. Gas emission	539.377	Wear in general ⇒ 539.43 ⇒ 539.538 ⇒ 620.178.16
539.219.2	Relationships between internal stresses and irregularities. □ Fissures. Cracks. Tears. Splits. Grooves. Notches. Flaking. Peeling. Exfoliation. Discontinuity. Noncohesion ⇒ 539.319	539.378	Deformation under constant load. Creep ⇒ 539.434
539.219.3	Diffusion. Autodiffusion	539.378.2	Influence of temperature on deformation.
539.22	Isotropy and anisotropy in molecular structure. Order-disorder. Texture of molecular systems and materials (especially synthetics) in general ⇒ 548.735	539.378.3	□ Hardening. Annealing. Quenching ⇒ 539.516
539.23	Production of thin films	539.378.6	Other types of deformation
539.231	Mechanical production. □ By spraying, sputtering	539.379	Welding by compression
539.232	Chemical production. □ By oxidation	539.38	Diffusion of bodies by compression
539.233	By adsorption	539.381	Disintegration. Dispersion
539.234	By evaporation and condensation	539.382	Effects of deformation. □ Volume: growth, shrinkage, constancy. Slip bands. Slip planes
539.24	Microstructure revealed by photomicrography. Structure of very small particles	539.383	Various causes of deformation. Graphic representation of deformation
539.25	Properties and structures revealed by ultramicroscopy	539.384	Generalities. Homogeneous and heterogeneous deformation
539.26	Properties and structures revealed by X-rays. Revelation of fine structure ⇒ 621.386	539.385	Deformation by extension. □ Stress-strain curves. Stress modulus. Influence of shape of specimen
539.27	Structure of molecular systems revealed by electron diffraction	539.386	Deformation by compression, crushing
539.3	Elasticity. Deformation. Mechanics of elastic solids ⇒ 620.1	539.387	Deformation by flexion
		539.388	Deformation by torsion. □ Warping. Torsion modulus
		539.388.1	Deformation by shearing. Shear
		539.388.2	Deformation by folding, winding, curving
		539.388.8	Other causes of deformation
		539.389	Deformation by fatigue
		539.389.1	Deformation by particular working methods.
		539.389.2	□ Deformation by drawing, rolling, stamping, pressing, punching, deep-drawing, drilling, forging
		539.389.3	Shrinkage and swelling according to moisture content
		539.389.4	Permanent deformation phenomena in solid bodies
		539.4	Yield point
			Hardening. □ Bauschinger effect
			After-effects. Ageing. Recovery.
			Accommodation
			Hysteresis
			Strength. Resistance to stress ⇒ 539.55 ⇒ 620.1

539.4.01	Theory. Various influences on strength	539.433	Resistance to vibration, oscillation
539.4.011	Theory of stress resistance	539.434	Creep resistance. Long-time creep strength.
539.4.011.1	Cohesion		Heat resistance. Resistance to quenching
539.4.011.2	Extension. Stretching	539.435	Natural stress limit. Endurance strength
539.4.011.22	Elastic limit		
539.4.011.25	Breaking point. Breaking stress	539.5	Properties of materials affecting deformability
539.4.012	Design theory. Permissible stresses		
539.4.013	Influence of shape of a body on its strength.	539.51	General properties of materials
	□ Notches. Corners. Waists. Constrictions	539.511	Malleability
539.4.014	Mechanical influences on strength. □ Tension.		⇒ 620.163.2
	Initial stress. Inherent stress	539.512	Weldability
539.4.015	Influence of structure on stress		⇒ 620.163.3
539.4.016	Influence of treatment, working		⇒ 620.179.2
	Details by :621.7... and :621.9...		⇒ 621.791.011
	◇ 539.4.016:621.785 Influence of heat treatment	539.514	Grindability. Crushability
539.4.019.1	Influence of time. Influence of rate of deformation	539.516	⇒ 620.163.4
539.4.019.2	Influence of dimensions. Influence of size		Hardenability
539.4.019.3	Physical influences		⇒ 620.163.6
			⇒ 620.179.3
539.41	Resistance to elastic deformation. Cohesive strength	539.52	Elasticity
539.411	Resistance to compression. Crush resistance.	539.53	Hardness. Softness. □ Hardness scales
	Buckling resistance	539.533	Resistance to penetration. Resistance to deformation.
539.412	Resistance to tension. Tear resistance	539.536	Hardness distribution
539.413	Resistance to flexion, bending	539.537	Surface hardness. Scratch hardness.
539.414	Resistance to torsion, twisting	539.538	Resistance to scraping
539.415	Resistance to shear. Resistance to punching	539.55	Impact strength
539.416	Resistance to creasing, folding, bending	539.551	Resistance to wear
539.417	Resistance to drawing. Resistance to rolling	539.554	Toughness and flexibility
539.419	Resistance to complex stresses. □ Resistance to dropping. Resistance to tumbling	539.557	Toughness. Ductility
			Cleavability
539.42	Fracture. Fracture mechanics	539.56	Flexibility. Pliability
539.421	Crack initiation. Crack propagation. Crack arrest. □ Stress intensity factor. Crack opening displacement. J integral	539.57	Brittleness. Friability. Fragility
		539.58	Consistency. Flow characteristics
539.422	Mode of fracture. □ Microstructure of fracture	539.6	Compressibility of solids
539.422.2	External appearance of the fracture		Intermolecular forces
539.422.22	Brittle fracture	539.61	⇒ 539.196
539.422.23	Ductile fracture. Sliding fracture. Shear fracture	539.62	Cohesion. Adhesion
539.422.24	Fatigue fracture, failure		Friction
539.422.25	Cleavage fracture. Cleavage separation fracture	539.621	⇒ 531.43
539.422.3	Structural fracture. □ Grain structure of fracture	539.622	Sliding friction
539.422.5	Fracture in composite materials	539.63	Rolling friction
539.424	Ultimate tensile strength. Fracture strength	539.67	Impact effects
539.43	Resistance to repeated stresses	539.8	Internal friction. Attenuability
539.431	Resistance to alternating stresses. Fatigue resistance	539.87	Other physico-mechanical effects
539.432	Resistance to repeated complex stresses	539.89	Mechanical effects of physical phenomena
	◇ 539.432:62-233.1 Stresses in transmission shafts	539.893	Effects of mechanical processes. Effects of pressure
			Production of very high pressures.
			Hypercompression

54 CHEMISTRY. CRYSTALLOGRAPHY. MINERALOGY

⇒ 340.67
 ⇒ 550.4
 ⇒ 66

54-1	State of substance	54-121	Elementary
	⇒ 62-403/-405	54-122	Molecular
54-11	State as regards occurrence and treatment	54-123	Atomic
54-112	Natural. Native. Raw. Untreated	54-124	Monomeric
54-114	Man-made. Synthetic. Artificial. Prepared	54-125	Isomeric
54-116	Processed. Treated. Converted. Modified	54-126	Polymeric. Oligomeric
54-12	State as regards atomic or molecular structure	54-127	Polycondensed
		54-128	Ionic. Ionized
		54-128.2	Anions

54-128.4	Cations	54-31	Oxides
54-13	Gaseous state	54-32	Acids
54-132	Rarefied	54-323	Meta-acid type
54-133	Compressed	54-324	Pyro-acid type
54-134	Vapour state	54-325	Ortho-acid type
54-135	Homogeneous gaseous phase containing more than one component 54-135.1/.5 $\equiv$ 54-145.1/.5	54-328	Poly acids. Heteropoly acids
54-138	Colloidal state with gaseous continuous phase. □ Mists. Smokes. Aerosols $\Rightarrow$ 544.772	54-36	Bases. Hydroxides
54-139	Supercritical fluid state	54-38	Salts. Analogous compounds
54-14	Liquid state	54-381	Basic salts
54-143	Molten	54-383	Neutral salts
54-144	Mesomorphic, paracrystalline state	54-384	Acid salts
54-145	Solutions (homogeneous liquid phase)	54-384.1	Primary acid salts
54-145.1	Degree of concentration, acidity etc.	54-384.2	Secondary acid salts
54-145.11	Dilute	54-384.3	Tertiary acid salts
54-145.12	Concentrated	54-386	Double salts. Complex compounds
54-145.13	Saturated	54-39	Peroxy compounds
54-145.14	Supersaturated	54-4	Chemicals. Reagents
54-145.15	Acid solution	54-41	Reagents in general
54-145.16	Basic solution	54-412	Precipitating agents
54-145.17	Neutral solution	54-414	Sorbents <i>Further developed for chromatography at 543.544-414</i>
54-145.2	Aqueous solution. Pure water as solvent	54-415	Mobile phase <i>Further developed for chromatography at 543.544-415</i>
54-145.3	Inorganic liquid (other than water) as solvent	54-42	Standard solutions
54-145.5	Means of expressing concentration	54-43	Indicators
54-145.51	Molar concentration. Molarity	54-44	Catalysts
54-145.52	Molal concentration. Molality	54-45	Buffers. Buffer solutions
54-145.53	Normal concentration. Normality	54-48	Degree of purity of chemicals
54-145.54	Mole fraction	54-481	Analytical grade
54-145.55	Activity. Coefficient of activity	54-482	Very pure
54-145.56	Volatility. Fugitivity. Partial pressures	54-483	Pure
54-145.561	Volatility. Fugitivity	54-484	Purified. Technical
54-145.562	Partial pressures	54-485	Commercial. Crude
54-145.8	Solvent. Solute	54-7	Types of radiation (wave and corpuscular)
54-145.82	Solvent	54-71	Ions
54-145.83	Solute	54-72	Elementary particles
54-148	Colloidal state with liquid continuous phase. □ Suspensions. Emulsions. Sols. Gels. Foams $\Rightarrow$ 544.773	54-722	Nuclei. Alpha-particles
54-149	Coarse suspensions	54-723	Nucleons
54-16	Solid phase	54-723.4	Protons
54-161	Amorphous state	54-723.5	Neutrons
54-162	Crystalline state	54-724	Electrons
54-165	Solid solutions. Homogeneous solid phase	54-73	Radio waves
54-165.2	Substitutional (subtractional) solid solutions	54-732	Microwaves
54-165.3	Interstitial solid solutions	54-74	Infrared rays
54-168	Colloidal state in solid medium $\Rightarrow$ 544.774	54-75	Visible light (visible radiation)
54-169	Loose solids	54-76	Ultraviolet rays
54-17	Allotropic states. Polymorphic states	54-77	X-rays
54-171	Stable	54-78	Gamma-rays (gamma-radiation)
54-172	Metastable	54-79	Other types of radiation
54-173	Enantiotropic modifications	54.01	Chemical substances and systems. Origin. Occurrence. Phases
54-174	Monotropic modifications	54.02	Composition. Structure. Isotopes
54-18	Special states as regards properties	54.021	Composition and formulae in general.
54-182	Colloidal state in general $\Rightarrow$ 544.77	54.022	Molecular formulae. Symbols etc.
54-183	Adsorbed $\Rightarrow$ 544.72	54.024	Structure and arrangement of atoms in the molecule. Structural formulae.
54-185	Mixtures. Heterogeneous systems in more than one phase		Radicals
54-188	Activated		
54-19	Intermetallic compounds. Intermetallides. Metallides. Berthollides. Daltonides		
54-3	Particular kinds of compound		

54.027	Isotopes (in general) <i>Specific isotopes are designated by adding .027 to the 546 number for the element, followed by an asterisk with the mass number</i> ◇ 546.26.027*14 Carbon 14 ◇ 546.791.027*235 Uranium 235	542.22	Preparation of chemicals. □ Mechanical separation. Comminution. Pulverization. Elutriation. Spatulas. Spoons. Mortars, pestles. Mills
54.03	Physical properties and constants. Mechanical and physical effects	542.23 542.231	Laboratory ware in general Glassware. □ Bottles, flasks, retorts, cups, funnels, tubes ⇒ 666.172 Porcelain vessels. Refractory vessels. Stoneware Metal vessels
54.04	Chemical properties and constants	542.232 542.233	Stands. Tripods. Tongs ⇒ 621.881
54.05	Production. Preparation. Isolation. Purification etc.	542.24	Glass, cork and rubber stoppers
54.051	General principles of procedure for production, sampling etc. <i>Not applicable at 544.77</i>	542.25 542.26	Flexible tubing. Caulking, packing materials
54.052	Mode of procedure, choice of method etc. <i>Not applicable at 544.77</i>	542.29 542.297.3	Various accessories Closing devices
54.053	Design of method or procedure. Technique used	542.3	Measurement of weight, mass. Measurement of volume. □ Apparatus: Balances. Weights. Hydrometers. Gauges. Graduated vessels. Pipettes. Measuring cylinders. Burettes ⇒ 531.73/75 ⇒ 681
54.055	Preparation by dry means. □ Grinding. Fusing	542.4	Heat and cold application
54.056	Preparation by wet means. □ Dissolving. Extracting	542.41	Laboratory ovens, furnaces 542.41 ≡ 66.041
54.057	Synthesis	542.42	Methods of heating Gas heating. Gas furnaces. Gas burners. Bunsen burners Heating by alcohol, kerosine etc. Special burners
54.058	Purification	542.423 542.424 542.425	Electric heating. Electric furnaces Indirect heating. □ Water baths. Steam baths. Sand baths
54.06	Analysis, investigation and handling in general 54.061/.066 ≡ 543.061/.066	542.45	Heat exchangers 542.45 ≡ 66.045
54.061	Qualitative investigation. Evidence. Identification experiments	542.455	Refrigeration and cooling. Refrigerators. Coolers
54.062	Quantitative investigation. Determination of quantity	542.46	Thermal treatment. □ Processes. Equipment 542.46 ≡ 66.046
54.063	Microanalysis. Semimicroanalysis	542.464	Roasting. Calcining
54.064	Trace analysis. Detection and determination of impurities	542.465 542.466 542.467 542.468	Fusion. Melting etc. Heating of liquids Boiling Pressure heating. Autoclave treatment
54.066	Investigation by means of specific reactions	542.47	Drying 542.47 ≡ 66.047
54.07	Apparatus and equipment for preparation, investigation and analysis	542.48	Distillation. Fractionation 542.48 ≡ 66.048
54.08	Measurement principles, methods, techniques. Instrumentation 54.08 ≡ 53.08	542.49	Other thermal operations. Thermal separation 542.49 ≡ 66.049
54.081	Units. Constants	542.5	Use of flames. Blowlamps ⇒ 536.46 ⇒ 543.41 ⇒ 662.62
542	PRACTICAL LABORATORY CHEMISTRY. PREPARATIVE AND EXPERIMENTAL CHEMISTRY	542.51	Nature and composition of flames. Oxidizing flames. Reducing flames
542.05	Preparative chemistry	542.53	Blowpipes. Blowlamps. Blow-torches
542.057	Synthesis	542.6	Working with liquids 542.6 ≡ 66.06
542.06	Experimental and investigative chemistry. Special techniques 542.06 ≡ 54.06		
542.07	Apparatus and equipment		
542.08	Principles and methods of measurement. Measuring instruments		
542.1	Chemical laboratories. □ Layout. Installations. Fittings		
542.2	Laboratory apparatus and instrumentation in general. Laboratory methods in general		
542.21	Arrangement and assembly of apparatus and accessories		

542.61	Solution. Solvents. Extraction. Leaching. Infusion
542.63	Mixing. Stirring. Maceration. Softening. Homogenization. Dilution. Concentrating
542.65	Solidification. Thickening
542.66	Separation. Clarification. Decantation. Settling
542.67	Purification. Filtration
542.68	Pressing
542.691	Washing
542.7	Working with gases
542.71	Production of gases
542.73	Collection of gases. Transfer of gases
542.74	Purification and solution of gases. □ Wash bottles. Woulfe-bottles 542.74 ≡ 66.074
542.743.1	Drying of gases
542.745.1	Washing of gases. Absorption gas purification
542.76	Gas storage. Gasometers 542.76 ≡ 66.076
542.762.3	Mercury gasometers
542.77	Rarefaction of gases
542.78	Compression, liquefaction of gases 542.78 ≡ 66.078
542.8	Physical, physicochemical and electrical operations <i>Other than those included at 542.2/7</i> 542.8 ≡ 66.08
542.816	Membrane methods. Dialysis. Ultrafiltration
542.85	Irradiation. Photochemical operations
542.86	Operations involving magnetic, electromagnetic and electric effects
542.87	Electrochemical methods
542.9	Chemical reactions. Special chemical processes 542.9 ≡ 66.09 542.9-9 ≡ 66-9 542.9'1/7 ≡ 542.91/97 <i>The 54-13/16 auxiliaries are not used in</i> 542.9
542.913	Synthesis ◇ 542.913-977 Pyrosynthesis
542.92	Decomposition. Cracking ◇ 542.92'43 Oxidative decomposition ◇ 542.92'7 Catalytic decomposition ◇ 542.92'73 Catalysts of decomposition. NOTE: Further specified by 66-92...
542.93	Reactions with water
542.934	Hydration
542.934.8	Dehydration ◇ 542.934.8'7 Catalytic dehydration
542.943	Oxidation ◇ 542.943-92 By reagents. Oxidants ◇ 542.943'7 Catalytic oxidation ◇ 542.943'78 Antioxidation. Inhibition of oxidation
542.943.2	Hydroxylation
542.952.16	Tautomerization
542.952.2	Rearrangement
542.97	Catalytic processes
542.973	Catalysts
542.978	Negative catalysis. Inhibition. Release

543	ANALYTICAL CHEMISTRY <i>The determination of constituents in the products to be analysed may be denoted by :546... or :547... with auxiliary .06 from 54</i> ◇ 543:553 Analysis of economic ores and minerals ◇ 543:615.2/3 Analysis of drugs ◇ 543:662.6 Analysis of fuels ◇ 543:669.14:546.26.06 Determination of carbon in steel
543-41	Reagents (materials) for chromatography
543-414	Sorbents
543-415	Mobile phase
543-43	Indicators
543.05	Sampling and preparation of samples for analysis. Separation and concentration of the analysis constituents
543.051	General principles and theory of sampling. □ Type of sample
543.052	Sampling procedure. □ Selection of sampling points. Sampling errors
543.053	Technique and apparatus for sample collection. □ Sampling tubes. Sampling vessels, bottles
543.054	Chemical preparation of the sample. Methods of isolation and concentration of the analysis constituents. Transfer of the analysis constituents into the shape convenient for determination 543.054.2/9 ≡ 66.02/09 ◇ 543.544.054.9 Reaction chromatography
543.054.9	Reaction chromatography
543.054.92	Pyrolytic chromatography
543.055	Dry treatment of samples, as a whole
543.056	Wet treatment of samples, as a whole
543.06	Analytical procedure. Special types and techniques
543.061	Qualitative analysis. Detection. Identification
543.062	Quantitative analysis. Estimation
543.063	Microchemical procedures
543.063.1	Semimicroprocedures
543.063.2	Microprocedures
543.063.21	Sub-microprocedure
543.063.22	Ultra-microprocedure
543.064	Trace analysis. Detection and estimation of impurities
543.066	Analysis by specific reactions
543.067	Special types of chemical analysis
543.067.2	Local analysis
543.067.22	Probe
543.067.3	Nondestructive analysis
543.067.4	Remote (control) analysis
543.067.5	Field (extralaboratory) analysis
543.068	Analysis techniques. Analysis control
543.068.2	Automatic analysis. Automatic analysis control ◇ 543.068.2:004.384 with a computer
543.068.3	Flow-injection analysis
543.068.5	Combined and hybrid methods
543.068.52	Combined
543.068.53	Hybrid
543.068.7	High performance methods ◇ 543.544.5.068.7 High performance liquid chromatography (HPLC)
543.068.8	Express methods
543.07	Apparatus and equipment for analysis
543.08	Measurement principles, methods and apparatus. Recording and processing of results 543.08 ≡ 53.08
543.084.852	Light sources

543.2	Chemical methods of analysis	543.243.32	Mercurimetry
543.21	Separation and determination by wet methods. Gravimetry. Systematic analysis ⇒ 543.557	543.243.4	With salts of barium
543.215	Cation separation and determination	543.243.5	With potassium cyanide
543.215.1	Group 1: alkali metals. Magnesium, ammonium. Cations devoid of common (group) reagent	543.243.6	With salts of lead
543.215.2	Group 2: alkaline earth metals precipitated by ammonium carbonate	543.243.9	With other compounds
543.215.3	Group 3: metals precipitated by ammonium sulphide	543.245	Complexometric titration (chelometric titration)
543.215.4	Group 4: metals precipitated by hydrogen sulphide in hydrochloric medium	543.245.2	With aminopolycarbonic acids
543.215.5	Group 5: metals precipitated by hydrochloric acid	543.245.3	With organophosphorus complexones (derivatives of amino alkylphosphonic acids)
543.215.6	Nonhydrogen sulphide analysis methods, e.g. ammonium-phosphate, acid-alkali, acetamide	543.245.9	With other complexones, e.g. phenylimine diacetic acid (PIDA)
543.216	Anion separation and determination 543.216 ≡ 546	543.247	Phase titration
543.216.131	Hydrochloric acid group	543.248	Weight titration
543.216.185	Phosphoric acid group	543.26	Air analysis. Detection of impurities in air 543.26 ≡ 543.27
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544.022.344.2	Two-dimensional defects. □ Dislocation networks. Defects of stacking. Grain boundaries	544.022.83	Agitation. Setting destruction
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544.144.5	Bond localization and delocalization. Conjugation		◇ 544.164.032.4 Thermochromism
544.144.52	Localized bonds. Two-centre bonds		◇ 544.164.032.52 Electrochromism
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544.146	Empirical characteristics of atoms and ions at bond formation		Electron energy loss spectra
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544.146.3	Coordination number	544.171.22	Compton profiles
544.146.5	Oxidation number. Oxidation state	544.171.23	Electron attachment spectra
544.146.6	Electronegativity. Electronegativity scales	544.171.24	Electron ejection (electron detachment) spectra
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	◇ 544.15:544.174.4 Rotational energies (levels). Rotational transitions ⇒ 544.17	544.173.4	Molecular ionization effectivity spectra
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 544.234.4 Liquation phenomena
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Denote physical properties of liquid crystals by .03
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 ◇ 544.3-971.2 Exothermic changes
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 ◇ 544.3.01 Phases. Phase states. NOTE: The subdivisions of .01 may be further specified as is shown at 544.344
 ◇ 544.3.03 Physical and physicochemical properties. Conditions and states. NOTE: Denote individual physical properties by :53...
 ◇ 544.3.031 Standard conditions and states
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 ◇ 544.3.032.1 Stability. Steadiness
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544.31 Thermodynamic functions of chemical elements and of their compounds
Denote thermodynamic functions of individual compounds, processes or bonds by the UDC numbers of these compounds etc. in colon combination with 544.3...
 ◇ 544.31:546.264-31 Thermodynamic functions of carbon dioxide
 ◇ 544.31.031 Standard thermodynamic functions
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- 544.312 Formation entropy. Absolute (third-law) entropy
 ◇ 544.312.031 Standard formation entropy (S°_{298})
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- 544.313.2 Isobaric-isothermic potential. Free energy. Gibbs' energy (G , ΔG)
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- 544.313.3 Isochoric-isothermic potential. Helmholtz's energy (F , ΔF)
- 544.313.4 Chemical potential. Partial molar isobaric potential (μ)
- 544.319 Other concepts related to thermodynamics
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- 544.32 Thermodynamic functions of chemical processes**
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- 544.322 Entropy of chemical processes. Entropy of reaction. Entropy change in chemical reaction
 ◇ 544.322.031 Standard entropy of reaction
 ◇ 544.322.032.3 Pressure (volume) effects on entropy of reaction
 ◇ 544.322.032.4 Temperature effects on entropy of reaction
- 544.323.2 Free energy of chemical processes
- 544.33 Thermochemistry**
- 544.332 Heat (enthalpy) of chemical processes. Enthalpy change in chemical reaction, ΔH . Additivity of reaction heats. Hess' law
Denote thermal effects of individual processes by the UDC numbers of the processes in colon combination with 544.332
If it is necessary to class these together under Thermochemistry use 544.332 in colon combination with the UDC numbers of the processes.
Details by special auxiliaries
 ◇ 544.332-971.2 Exothermic reactions
 ◇ 544.332-971.4 Endothermic reaction
 ◇ 544.332.031 Standard enthalpy of chemical process. Standard heat of reaction, ΔH°_{298}
 ◇ 544.332.032.4 Temperature effects on enthalpy. Kirchhoff's equation
- 544.332.2 Formation heats of chemical compounds
 ◇ 544.332.2-123 Atomic heat of formation
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 ◇ 544.332.2-971.4 Endothermic compounds
 ◇ 544.332.2.031 Standard heats of formation. Standard enthalpy of formation ($\Delta_f H^\circ_{298}$)
 $\Rightarrow$ 544.31
- 544.332.3 Heat of combustion
 $\Rightarrow$ 662.614.2
- 544.333 Specific heat. Atomic specific heat. Molar specific heat. Dulong and Petit's law
- 544.336 Low temperature chemistry
- 544.337 High temperature chemistry
- 544.338 Chemical means of producing thermal energy and methods of its accumulation
- 544.34 Chemical equilibrium**
Denote equilibria of individual systems by the UDC numbers of the systems in colon combination with 544.34
 ◇ 544.354:544.34 Ionic equilibrium
 ◇ 544.36:544.34 Acid-base equilibrium
 $\Rightarrow$ 544.4
- 544.341 Theory of chemical equilibrium
- 544.341.1 Forward and reverse reactions. Degree of reaction
- 544.341.2 Law of mass action (Guldberg and Waage's law). Equilibrium constant
 ◇ 544.341.2.032.3 Pressure effects on equilibrium constant
 ◇ 544.341.2.032.4 Temperature effects on equilibrium constant. Reaction isobar equation. Reaction isochore equation
 $\Rightarrow$ 544.354.081.7
 $\Rightarrow$ 544.421.081.7
- 544.341.3 Reaction extent. Degree of initial products change. Reaction quotient
- 544.341.5 Shift of equilibrium. Le Chatelier's principle
- 544.342 Equilibrium of homogeneous systems. Homogeneous equilibrium
Specify by -1
 ◇ 544.342-13 Equilibrium of gaseous systems. Gaseous mixture
 ◇ 544.342-14 Homogeneous equilibria of liquid systems
 $\Rightarrow$ 544.35
 $\Rightarrow$ 544.35
- 544.344 Heterogeneous equilibria. Phase equilibria
- 544.344.01 Phases. Phase composition. Phase states
Specify individual phases by adding -13/-16 to the system auxiliary, in decreasing numerical order
- 544.344.011 Single-phase systems
 ◇ 544.344.011-13 Gaseous phase
 ◇ 544.344.011-14 Liquid phase
 ◇ 544.344.011-16 Solid phase
- 544.344.012 Two-phase systems
 ◇ 544.344.012-14 Liquids
 ◇ 544.344.012-14-13 Liquid and gas
 ◇ 544.344.012-16-14 Solid and liquid
- 544.344.013 Three-phase systems
 ◇ 544.344.013-14 Liquids
 ◇ 544.344.013-14-13 Liquids and gases
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 ◇ 544.344.013-16-13 Solids and gases
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 ◇ 544.344.013-16-14-13 Solid, liquid and gaseous phases
- 544.344.014 Four- and multiphase systems
- 544.344.015 Phase state
 ◇ 544.344.015.032.1 Phase stability
 ◇ 544.344.015.032.1.032.4 Temperature effects. Thermal stability
- 544.344.015.2 Phase formation. Nucleation
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- 544.344.015.22 Phase partition. Phase surfaces (interfaces)
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- 544.344.015.3 State diagrams. Phase diagrams
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- 544.344.015.32 Phase boundaries. Liquidus. Solidus. Triple points. Critical points
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- 544.344.015.33 Singularities on phase equilibrium curves. Eutectics. Dystectics. Peritectics. Singularity points
- 544.344.015.34 Azeotropy
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544.362.4	Acid-base hardness and softness		Mass transfer. Transport (delivery, removal) of reactants and products species. Chemical transport. Diffusion stage (range) of reaction
544.362.42	Hard acid-bases	544.431.11	Kinetic stages of reaction. Kinetic range of reaction
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544.363	Polybasicity of acids and bases. Polyprotic acids and bases	544.431.12/.13	Intramolecular transfer. Intermolecular transfer
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544.4	Chemical kinetics. Catalysis	544.431.121.3	◇ 544.431.122:544.15 Activation energy
	◇ 544.4.032.7 Chemical action on the progress of reaction	544.431.122	◇ 544.431.122.032.1 Stability. Half-life
544.41	Chemical reaction classification		Transition state
	<i>Denote various aspects by colon combination or by special auxiliaries</i>	544.431.122.2	Hot state. Hot ground state
	◇ 544.41:544.342 Homogeneous reactions	544.431.122.3	Relaxation. Deactivation (quenching)
	◇ 544.41:544.342-13 In gaseous phase	544.431.123	Initiation of reaction
	◇ 544.41:544.342-145 In solutions	544.431.124	Charge transfer. Charge exchange
	◇ 544.41:544.344 Heterogeneous reactions. Reactions at interfaces	544.431.13	Electron transfer. Solvated electron
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544.412	Chemical reactions according to the nature of change	544.431.14	Atomization
544.412.1	Addition. Attachment	544.431.142	Ionization
544.412.2	Elimination. Detachment	544.431.143	Formation of chemically active species (free radicals)
544.412.3	Substitution. Exchange	544.431.15	Recombination of chemically active species
544.412.4	Rearrangement. Disproportionation	544.431.16	Multistage reactions. Complex reactions
544.42	Formal kinetics	544.431.2	Limiting stage. Rate determining stage. Slow stage. Reaction bottle-neck
	◇ 544.42.032 Reaction conditions. Reaction limits (ranges). Temperature and concentration reaction limits	544.431.22	Formation of intermediates (transient species of reaction)
544.421	Reaction rate. Absolute reaction rate. Relative reaction rate	544.431.24	Parallel reactions. Competing reactions.
	◇ 544.421.032.4 Dependence of reaction rate on temperature	544.431.5	Reaction channels
	◇ 544.421.032.76 Dependence of reaction rate on chemical structure of reactants. Substitution effects. Correlation relations. Inductive and mesomeric effects. Isotopic effect	544.431.52	Main reaction
	◇ 544.421.032.76:544.323.2 (Linear) relations (equations) of free energy. Hammett, Taft, Brown etc equations	544.431.53	Side (accompanying) reactions
	◇ 544.421.081.7 Rate constant of reaction. Kinetic resistance of process	544.431.6	Conjugated reactions. Induced reactions.
	⇒ 544.435.3	544.431.7	Chemical induction
544.421.2	Reaction half-life	544.431.8	Chain (self-propagating) reactions. Chain initiation, branching and inhibition (termination)
544.421.3	Inductive period. Retardation of reaction	544.432	Periodic (oscillatory) reactions.
		544.432.1	Belousov-Zschabotinskij reaction. Chaos, bifurcations and self-organization in chemical systems
		544.432.2	Reaction molecularity
		544.432.3	⇒ 544.422
		544.433	Monomolecularity
			Bimolecularity
			Termolecularity
			Heterolytic and homolytic reactions

544.433.2	Heterolytic (dipolar, ion, ion-molecular, ion-radical) reactions			<i>Denote chemical composition by special auxiliaries of 54 or by common auxiliary of materials -03 of Table 1k</i>
544.433.21	Nucleophilic			
544.433.22	Electrophilic			◇ 544.473:544.77 Colloidal (micellar) microheterogeneous catalysis
544.433.3	Homolytic (radical) reactions			◇ 544.473:577.15 Biocatalysis. Enzymatic catalysis
544.435	Dynamics of elementary reactions ⇒ 544.431			◇ 544.473-039.61 Catalysis by nonmetals
544.435.2	Molecular collisions. Molecular scattering. Collision theory. Chemical (reaction) collision (scattering)			◇ 544.473-039.63 Catalysis by metals
544.435.3	Reaction cross-section. Reaction probability			◇ 544.473-039.63-386 Metallocomplex catalysis
544.435.4	Reaction coordinate. Trajectory of reaction. Potential energy surface of reaction. Reaction profile. Reactants and product valleys. Barrier of reaction	544.474		Autocatalysis
		544.475		Negative catalysis (retardative action). Inhibition. Inhibitors. Retardants
544.436	Symmetry control in reaction kinetics	544.476		Mechanism of catalytic action
544.436.2	Orbital symmetry of reaction. Woodward-Hoffmann rule	544.476.2		Catalytic site. Active site. Turnover number
		544.476.4		Spillover
544.436.4	Concerted and nonconcerted reactions	544.478		Catalysts
544.437	Specificity of reaction. Selectivity of reaction			◇ 544.478-03 Catalysts by composition
544.437.2	Stereospecificity. Stereoselectivity ◇ 544.437.2:544.121.4 Asymmetric synthesis			◇ 544.478.01 Phase composition of catalyst
544.437.3	Regiospecificity. Regioselectivity			◇ 544.478.02 Catalyst structure and texture
				◇ 544.478.023.5 Catalyst dispersity
				◇ 544.478.023.57 Catalyst specific surface
544.45	Combustion. Explosion. Kinetics of combustion and of explosion ⇒ 542.943 ⇒ 544.431.7	544.478.1		Catalyst activity. Activation of catalyst. Promotion
		544.478.12		Activation of catalyst. Activators
544.452	Combustion ◇ 544.452:544.342 Homogeneous combustion ◇ 544.452:544.344 Heterogeneous combustion. Combustion at interfaces ◇ 544.452:544.47 Catalytic (surface-catalytic) combustion. Flameless combustion	544.478.13		Promotion of catalyst. Promoters
		544.478.3		Immobilization of catalyst. Fixation of catalyst
544.452.1	Flaming. Flash	544.478.32		Adsorptive or supported catalysts
544.452.12	Spontaneous flaming			◇ 544.478.32-039.662 Activated alumina
544.452.14	Ignition. Inflammation ⇒ 544.431.124	544.478.34		Catalyst supports
544.452.2	Flame propagation ⇒ 544.427	544.478.4		Catalyst selectivity
544.452.4	Flame stabilization	544.478.41		Selective catalysts
544.452.42	Flow combustion. Jet combustion. Torch process	544.478.42		Multifunctional catalysts
544.452.44	Flame extinction. Flameout	544.478.6		Catalyst poisoning. Catalyst poisoners. Contact poisoners
544.452.5	Perfect combustion. Burnout. Combustion completeness. Deflagration			Regeneration of catalytic activity. Catalyst regeneration
544.454	Explosion ◇ 544.454.032 Explosivity limits (range). Critical explosion parameters	544.478.7		
544.454.2	Thermal explosion			544.5
544.454.3	Detonation			Chemistry of high energy processes
544.454.4	Starvation			544.51
				Chemistry of excited states 544.51 ≡ 544.52 ⇒ 544.4
544.46	Topochemistry. Topochemical reactions. Mechanochemical reactions	544.52		Photochemistry ⇒ 577.34
544.461	Topochemical reactions. Topotactic reactions	544.522		Mechanisms of photochemical reactions ⇒ 544.43
544.463	Mechanochemical reactions	544.522.1		Primary photochemical act. Formation of primary photoproducts 544.522.1 ≡ 544.431.1
544.47	Catalysis. Catalytic reactions ◇ 544.47:544.342 Homogeneous catalysis ◇ 544.47:544.344 Heterogeneous catalysis ⇒ 544.723	544.522.12		Transfer, conversion and relaxation of photoexcitation energy
544.472	Catalysis according to the nature of the intermediate chemical interaction	544.522.121.2		Intramolecular transfer of photoexcitation energy. Radiationless internal (singlet-singlet) and intersystem (singlet-triplet) conversion. Formation of diradicaloid states
544.472.2	Acid-base (heterolytic) catalysis	544.522.121.3		Intermolecular transfer of photoexcitation energy. Resonance phototransfer. Excitones
544.472.3	Redox (homolytic) catalysis	544.522.122		Photoexcitation (one-photon and multiphoton, steady-state and flash or pulse). Photoactivation. Photogeneration of excited states (photo-pumping)
544.473	Catalysis according to the nature or composition of the catalysts <i>Denote individual catalytic substances of unknown composition by colon combination.</i>	544.522.122.2		Transition state. Transient species
		544.522.122.3		Hot state. Hot ground state
		544.522.123		Quenching (deactivation) of photoexcited states
		544.522.14		Photodissociation
		544.522.142		Photoatomization
		544.522.143		Photoionization
		544.522.15		Photogeneration of free radicals

544.522.16	Recombination of chemically active species	544.53	Chemistry of laser action. Laser induced chemical reactions
544.522.2	Secondary photochemical processes. Perrin's theory. Russell effect	544.531	Chemistry of laser action. Chemical reactions for laser action. Chemical lasers (chemically pumped lasers)
544.523	Quantitative characteristics of photochemical processes		◇ 544.531:667.28 Dye lasers
544.523.1	Minimum photochemical effect. Threshold energy (limit) of photochemical effect	544.532	Mechanism of laser chemical action
544.523.2	Quantum yield of photochemical processes		◇ 544.532:533.9.03 Producing laser plasma
544.523.7	Measurement of photochemical energy. Chemical photoactinometry		544.532/536 ≡ 544.522/526
544.525	Photosensitivity. Photosensitization	544.532.122	Multiphoton (multiquantum) laser excitation. Laser photon impact (nano-, picosecond)
544.525.2	Optical sensitization	544.536	Laser induced chemical reactions
544.525.4	Chemical sensitization	544.536.1	Laser induced synthesis
544.525.5	Desensitization	544.536.2	Laser induced photolysis
544.525.7	Sensitometry	544.537	Laser induced processes leading to changes in the properties of a substance
544.526	Individual photochemical processes. Photochemical reactions	544.537.224	Laser induced hole burning
	<i>Denote reactions and processes not mentioned in 544.526.1/8 by :542 and :544</i>	544.538	Physicochemical aspects of laser applications ⇒ 544.582.2
	◇ 544.526:542.917 Photoaddition	544.54	Radiation chemistry. Chemical action of ionizing radiation
	◇ 544.526:542.942/943 Photoredox processes		⇒ 539.1.044
	◇ 544.526:542.942 Photoreduction	544.541	Radiation stability. Radiation stabilization
	◇ 544.526:542.943 Photooxidation	544.542	Mechanisms of radiation induced reactions
	◇ 544.526:542.949.41 Photohydrolysis		544.542/546 ≡ 544.522/526
	◇ 544.526:542.952.1 Photoisomerization. Including: Phototautomerization. Phototropism	544.542.1	Primary radiation induced processes. Radiolysis. Radiolysis products. Generation, interactions and decay of chemically active species (of primary radiolysis products)
	◇ 544.526:542.952.2 Photorearrangement		<i>Specify the ionizing radiations by :539.1</i>
	◇ 544.526:542.952.6 Photopolymerization		◇ 544.542.1:539.122 Gamma-radiolysis. X-ray radiolysis
	◇ 544.526:542.952.614.2 Photodimerization. Excimers		◇ 544.542.1:539.124 Electron impact radiolysis
	◇ 544.526:544.412.2 Photoelimination. Photofragmentation. Photodetachment		◇ 544.542.1:539.125 Heavy particle bombardment radiolysis
	◇ 544.526:544.431.124 Photoinitiation		◇ 544.542.1:539.164 Radiolysis by heavy particles, formed at radioactive decay
	◇ 544.526:544.431.13 Charge phototransfer	544.542.122	Radiolysis by regime of irradiation (pulsed, steady-state)
	◇ 544.526:544.431.6 Photoinduced reactions	544.542.122.3	Hot atom implantation. Recoil atom chemical reactions
	◇ 544.526:544.723.2 Photoadsorption	544.542.16	Recombination of chemically active species. Reactions in spurs
	◇ 544.526:544.723.3 Photodesorption	544.542.2	Secondary radiation induced chemical processes
544.526.1	Photochemical synthesis ⇒ 577.355 ⇒ 581.132	544.543	Quantitative characteristics of radiation induced chemical processes
544.526.2	Photolysis (photodegradation, photodecomposition, photocleavage). Flash photolysis	544.543.1	Minimum radiation induced chemical effect. Threshold energy
544.526.5	Photocatalysis	544.543.2	Quantum yield of radiation induced chemical process (radiation yield)
544.527	Photochemical processes leading to change in the properties of a substance	544.543.7	Radiation actinometry
544.527.2	Photochromism. Photofading. Photobleaching	544.546	Radiation induced chemical reactions
544.527.22	Photochromism ⇒ 544.164		◇ 544.546:544.35 Radiation chemistry of aqueous solution
544.527.222	Transient photoabsorption from excited states. Triplet-triplet absorption		◇ 544.546:544.431.7 Radiation induced chemical chain reactions
544.527.224	Photochemical hole burning		◇ 544.546:544.433.2 Ion, ion-molecular, ion-radical radiation induced reactions
544.527.23	Photofading. Photochemical ageing	544.546.5	Radiation catalysis
544.527.24	Photobleaching	544.55	Plasma chemistry
544.527.25	Photostabilization		544.552/556 ≡ 544.522/526
544.527.5	Various photoeffects. Photoelectric effect		⇒ 533.9
544.528	Physicochemical aspects of photochemical applications	544.552	Mechanisms of plasma chemical reactions
544.528.2	Physical chemistry of photographic processes	544.552.1	Primary plasma chemical act. Primary products of plasma chemical reactions
	◇ 544.528.2:772.1 Silver process	544.552.11	Mass transfer. Transport (delivery, removal) of reactant and product species
	◇ 544.528.2:772.2 Non-silver and little-silver processes		Transfer of plasma excitation energy
	◇ 544.528.2:773 Photonics. Photographic process with organic dyes ⇒ 544.525	544.552.123	Quenching of plasma chemical reactions
544.528.22	Latent image formation		
544.528.23	Developing		
544.528.24	Fixation		

- 544.556 Individual plasma chemical reactions
 ◇ 544.556:542.943 Plasma oxidation
 ◇ 544.556:542.952.6 Plasma polymerization
- 544.556.1 Plasma chemical synthesis
- 544.557 Gas electrochemistry. High temperature gas chemistry
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- 544.558 Physicochemical aspects of plasma application
 ◇ 544.558:621.793.1 Plasma condensation
 ◇ 544.558:621.794.4 Reaction ion etching
- 544.56 **Interaction of magnetic fields with chemical processes**
- 544.561 Chemical radiophysics. Chemical pulsars.
 Radiodetection of chemical reactions
- 544.562 Mechanisms of interactions
 544.562/.565 ≡ 544.522/.525
- 544.562.121.2 Reactions via magnetically split multiplet (e.g. triplet) states
 Chemically induced dynamic polarization
- 544.566 Of electrons (CIDEP)
- 544.566.2 Of nuclei (CIDNP)
- 544.566.4 Action of magnetic field on chemical reactions.
- 544.567 Magnetochemical reactions
- 544.57 **Sonochemistry. Ultrasound chemical action**
 544.572/.576 ≡ 544.522/.526
- 544.576 Sonochemical reactions
- 544.58 **Radiochemistry. Nuclear chemistry. Chemistry of radioactive nuclides**
- 544.582 Chemistry of isotopes
- 544.582.2 Separation of isotopes
 ◇ 544.582.2:544.538 Laser separation of isotopes
- 544.582.22 Extraction and purification of radioisotopes
- 544.582.3 Production of tracer compounds and radioactive compounds
- 544.582.4 Isotope enrichment
- 544.582.6 Isotope effects
- 544.584 Discovery of new, superheavy elements and isotopes (nuclides)
 Specify by :546
 ◇ 544.584:546.798.32 Element 105. Nielsbohrium (hahnium)
- 544.584.2 Fast radiochemical methods of detection
- 544.584.4 Chemistry of radioactive substances at microconcentrations
- 544.584.6 Identification of new nuclides
- 544.586 Chemistry of new (nonclassical, hydrogen-like) atoms
- 544.586.2 Positronium chemistry
- 544.586.3 Muonium and pionium chemistry
- 544.586.4 Mesochemistry. Mesic chemistry
- 544.586.6 Quarkochemistry. Chemistry of hadrons
- 544.6 **Electrochemistry**
 ◇ 544.6:544.52 Photoelectrochemistry
 ◇ 544.6:57 Bioelectrochemistry
- 544.6.018 Electrochemical systems. Electrolytes
- 544.6.018.2 Electrochemical systems
- 544.6.018.21 Physical systems. □ Gravitational, allotropic systems
- 544.6.018.22 Concentration systems. □ Cationic, anionic systems. Systems with liquid interface
- 544.6.018.23 Chemical systems. □ Simple, complex, double systems
- 544.6.018.4 Electrolytes
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- 544.6.018.42 Electrolytes according to aggregative (physical) state
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- 544.6.018.43 Electrolytes according to nature of dissociation
- 544.6.018.432 Symmetric. Binary electrolytes
- 544.6.018.433 Asymmetric
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- 544.6.018.44 Electrolytes according to degree of dissociation
- 544.6.018.45 Electrolytes according to reaction of medium (acidic, alkaline etc.)
- 544.6.018.46 Electrolytes according to their conductance
- 544.6.018.462 Electrolytes with ionic conductance
- 544.6.018.462.2 Electrolytes with anionic conductance
- 544.6.018.462.4 Electrolytes with cationic conductance
- 544.6.018.462.42 Electrolytes with protonic conductance
- 544.6.018.463 Electrolytes with electronic conductance
- 544.6.018.464 Electrolytes with mixed (electron-ionic) conductance
- 544.6.018.469 Electrolytes with other kinds of conductance
- 544.6.018.47 Electrolytes according to composition
 ◇ 544.6.018.47-036.5 Polymeric. Polyelectrolytes
 ◇ 544.6.018.47-039.6 Inorganic electrolytes
 ◇ 544.6.018.47-039.7 Organic electrolytes
- 544.6.018.48 Electrolytes according to applications
- 544.6.018.482 Indifferent (extraneous) electrolytes
- 544.6.018.483 Matrix electrolytes
- 544.6.018.484 Electrolytes for salt bridge
- 544.6.018.49 Other types of electrolyte
- 544.6.076 Electrochemical apparatus and equipment
- 544.6.076.2 Electrochemical cells
 544.6.076.2 ≡ 544.6.076.3
- 544.6.076.224.4 Cells according to cathode
 ◇ 544.6.076.224.4-14 Cells with liquid cathode
- 544.6.076.228.2 Bipolar cells
- 544.6.076.242 Membrane cells
- 544.6.076.243 Diaphragm cells
- 544.6.076.3 Construction elements of electrochemical cells
- 544.6.076.32 Electrodes
 ◇ 544.6.076.32-033.5 Glass electrodes
 ◇ 544.6.076.32-034.791 Mercury electrodes
 ◇ 544.6.076.32-034.791.5 Amalgam electrodes
 ◇ 544.6.076.32-039.61 Nonmetallic electrodes
 ◇ 544.6.076.32-13 Gaseous electrodes in general
 ◇ 544.6.076.32-43 Reference electrodes. Indicator electrodes
- 544.6.076.321 Electrodes of first kind in general
- 544.6.076.322 Electrodes of second kind in general
- 544.6.076.323 Electrodes of third kind in general
- 544.6.076.324 Anodes. Cathodes
- 544.6.076.324.2 Anodes
- 544.6.076.324.4 Cathodes
- 544.6.076.326 Redox electrodes
- 544.6.076.327 Ion-selective electrodes
- 544.6.076.328 Construction characteristics of electrodes
- 544.6.076.328.1 Monopolar electrodes
- 544.6.076.328.2 Bipolar electrodes
- 544.6.076.328.3 Film electrodes
- 544.6.076.328.4 Dropping electrodes
- 544.6.076.328.5 Streaming electrodes
- 544.6.076.328.6 Fluidized(-bed) electrodes
- 544.6.076.328.7 Slot electrodes
- 544.6.076.328.8 Volume, spherical electrodes
- 544.6.076.328.92 Mobile, rocking, rotating electrodes
- 544.6.076.328.93 Scanning, scavenger electrodes
- 544.6.076.328.99 Other constructions of electrodes
- 544.6.076.34 Membranes and diaphragms in electrochemical cells
- 544.6.076.342 Membranes in electrochemical cells
- 544.6.076.343 Diaphragms in electrochemical cells

- 544.62 Phenomena and processes in bulk electrolyte in general. Transport phenomena. Transference numbers**
 ◇ 544.62:532.72 Diffusion in electrolytes. Coefficient of diffusion. Diffusion potential
- 544.622 Ionic mobility. Velocities of ions
- 544.623 Electric conductance of electrolyte. Ionic conductivity
Specify effects by .032
 ◇ 544.623.032.52 Dependence on field strength. Wien effect. Dependence on frequency. Dispersion of conductance. Debye-Falkenhagen effect
 ◇ 544.623.032.73 Influence of solvent and solutes. Effect of ionic atmosphere. Electrophoretic effect. Relaxation time. Debye-Hückel theory. Onsager equation
 ⇒ 537.311.3
- 544.63 Phenomena and processes in electrolyte-electrode system**
 ◇ 544.63:544.34 Electrode equilibrium
- 544.632 Electrochemical potentials
- 544.632.2 Real potentials. Work function of electron
- 544.632.3 Surface potential
- 544.632.4 Inner electrochemical potential. Galvani-potential. Potential differences at interfaces. Nernst potential
 ⇒ 544.634
- 544.632.5 External potential. Volta-potential
- 544.634 Electrode potentials in general. Electrode voltage
 ◇ 544.634.031 Standard electrode potentials
 ◇ 544.634.076.32 Electrode potentials according to composition and properties of electrode
 ◇ 544.634.076.32-034 Potentials of ion-metal electrode systems
 ◇ 544.634.076.32-039.6 Inorganic electrode systems
 ◇ 544.634.076.32-039.7 Organic electrode systems
 ◇ 544.634.076.32-386 Electrode systems of complex compounds
 ◇ 544.634.076.34 Membrane potential. Donnan potential
 ◇ 544.634.081 Potential scales. Hydrogen, absolute or Ostwald, correlative or null point scale
 ⇒ 544.632.4
- 544.634.2 Electrochemical series. Electrolytic pressure
- 544.635 Electromotive forces (emf). Electromotive force of cell. Potential difference (voltage)
- 544.636/638 Phenomena on electrode-electrolyte interface
- 544.636 Double electric layer at electrode-electrolyte interface
- 544.636.2 Structure of electric double layer: theory
- 544.636.21 Theory of condensed double layer (Helmholtz)
- 544.636.22 Theory of diffuse layer (Gouy-Chapman)
- 544.636.23 Theory of adsorption double layer (Stern). Electrode adsorption. Specific electrode adsorption
- 544.637 Electrocapillary phenomena
- 544.637.1 Relation between surface tension and potential difference at interface. Electrocapillary curve. Lippmann-potential. Billiter-potential
- 544.637.2 Zero-charge potential. Electrocapillary maximum potential. Chargeless surface potential. Null points
- 544.638 Electrokinetic phenomena. Electrokinetic (zeta) potential
- 544.638.2 Electro-osmosis. Streaming potential
- 544.638.3 Electrophoresis. Sedimentation (precipitation) potential
- 544.64 Nonequilibrium electrochemical processes in general**
 Faraday's laws. Electrochemical equivalent. Yield of substance by current. Current efficiency
 ⇒ 543.55
 Chemical effect of electric current. Electrolysis. Decomposition potential (voltage)
 ◇ 544.642.076.2 Electroplating baths (electrolyzers)
 Current-producing chemical reactions. Theory of galvanic cells (primary and secondary)
Denote individual current-producing reactions by :542.9....
Details by .0...
 ◇ 544.643.076.2 Galvanic cells. Accumulators
- 544.65 Transformations at electrolyte-electrode interface. Electrode processes**
 ◇ 544.65:544.4 Electrochemical reaction kinetics
- 544.651 Stages of electrode processes
- 544.651.1 Elementary stages
 544.651.1 ≡ 544.431.1
- 544.651.11 Transport (delivery, removal) of reactant species to or from electrode-electrolyte interface
- 544.651.13 Charge transfer. Interface transfer of charged species. Electron transfer. Discharge (ionization). Exchange current
 Accompanying reaction stages
 Chemical reactions preceding or following discharge act
 Phase transformations. □ Formation or destruction of crystalline lattice, gas evolution at electrode, gas diffusion into electrode
 Polarization. Emf of polarization. Overpotential. Underpotential
 544.652.1/2 ≡ 544.651.1/2
- 544.651.2 Hydrogen overpotential
- 544.651.22 Anodic overpotential
- 544.651.23 Cathodic overpotential
- 544.652.11 Transport overpotential. Diffusion overpotential. Concentration polarization
- 544.652.13 Electron transfer overpotential. Retarded discharge
- 544.652.23 Phase overpotential. Crystallization overpotential
- 544.652.6 Polarization emf
- 544.652.7 Passivation
- 544.652.8 Depolarization
- 544.653 Chemistry of electrode processes. Electrochemical reactions as electrode reactions. Electrochemical behaviour of substances. Electrochemical properties
Denote individual electrochemical reactions not mentioned in 544.653.1/3 by :542.9...
 ◇ 544.653:542.944.13 Electrochemical chlorination
 ◇ 544.653:542.944.14 Electrochemical bromination
 ◇ 544.653:542.952.614.2 Electrochemical dimerization
- 544.653.1 Electrosynthesis
Denote electrosynthesis of individual products by :546/547
 ◇ 544.653.1:546 Electrosynthesis of inorganic compounds
 ◇ 544.653.1:547 Electrosynthesis of organic compounds

- 544.653.2/3 Electrochemical oxidation-reduction. Redox process
- 544.653.2 Electrochemical oxidation. Anodic process. Anodic behaviour. Anodic oxidation
- 544.653.22 Electrochemical dissolution. Anodic dissolution. Anodic treatment. Electrochemical machining
- 544.653.23 Electrochemical oxidation. Anodizing. Film electroformation in general
- 544.653.3 Electrochemical reduction. Cathodic process. Cathodic behaviour. Cathodic reduction
- 544.654 Electrochemical evolution
 ◇ 544.654.076.324.2 Anodic evolution
 ◇ 544.654.076.324.2:546.13 Of chlorine
 ◇ 544.654.076.324.2:546.21 Of oxygen
 ◇ 544.654.076.324.4 Cathodic evolution
 ◇ 544.654.076.324.4:546.11 Of hydrogen
 ◇ 544.654.076.324.4:547 Of organic substances
 ⇒ 544.653.23
- 544.654.2 Electrodeposition of metals and alloys. Electrocrystallization
Denote electrodeposition of individual metals and alloys by :669...
- 544.7 Chemistry of surface phenomena and colloids**
- 544.72 Surface phenomena in contact systems (at interfaces)**
 ◇ 544.72.02 Structure. Composition
 ◇ 544.72.023 Structure of boundary layers and films. Degree of dispersity
 ◇ 544.72.023.2 Interfacial layers and films. Surface (backed) films. Unbacked films.
 NOTE: Denote aggregative (physical) state by -13/-16
 ◇ 544.72.023.22 Mono-, bi- and polymolecular (multimolecular) layers and films
 ◇ 544.72.023.221 Monomolecular layers and films
 ◇ 544.72.023.222 Bimolecular layers and films
 ◇ 544.72.023.223 Polymolecular (multimolecular) layers and films
 ◇ 544.72.023.23 Continuous films. Condensed and expanded films
 ◇ 544.72.023.232 Condensed films
 ◇ 544.72.023.233 Expanded films
 ◇ 544.72.023.25 Oriented films
 ◇ 544.72.023.26 Multicomponent films
 ◇ 544.72.023.27 Unbacked films
 ◇ 544.72.05 Formation of surfaces and interfaces. Film formation. Juvenile surfaces
 ⇒ 544.344.015.22
- 544.722 Chemical origin of contact (interfacial, surface) interactions (forces)
- 544.722.1 Nature of interfacial interactions
- 544.722.12 Lyophily
- 544.722.122 Hydrophily
- 544.722.123 Lipophily (oleophily)
- 544.722.13 Lyophobicity
- 544.722.132 Hydrophobicity
- 544.722.133 Lipophobicity (oleophobicity)
- 544.722.14 Amphiphily
- 544.722.15 Diphily
- 544.722.2 Surface activity. Chemical nature of surface tension
 ⇒ 532.61
- 544.722.22 Interfacial concentration. Abundant surface concentration
- 544.722.23 Redistribution on surface. Segregation
- 544.722.3 Wettability. Chemical nature of wetting. Spreading. Spreading ability
 ◇ 544.722.3.023.2 Wetting films
 ⇒ 532.64
- 544.722.4 Disjoining (wedging) pressure
- 544.722.5 Adherence of contacting surfaces
 ⇒ 539.61
- 544.722.52 Cohesiveness. Autocohesiveness. Juvenile contact
- 544.722.54 Adhesiveness. Tackiness. Sticking. Gluing
- 544.722.7 Lubricant action. Lubricity
 ◇ 544.722.7.023.2 Lubricant films
- 544.722.8 Washing action. Detergent (cleansing) action
 ⇒ 532.696.2
- 544.723 Adsorption. Chemical sorption (chemisorption)
 ⇒ 542.8
- 544.723.2 Adsorption. Physical adsorption
 ◇ 544.723.2.023.2 Adsorption films. Adsorption layers
 ◇ 544.723.2.023.221 Monomolecular (monolayer) adsorption
 ◇ 544.723.2.023.223 Polymolecular (multilayer) adsorption
- 544.723.21 Adsorbents. Adsorbers
- 544.723.212 Adsorptivity. Adsorption activity
- 544.723.213 Adsorption sites. Localized adsorption
- 544.723.214 Coverage. Adsorption stage
- 544.723.22 Adsorbates (substrates). Adsorbability. Adsorbate-adsorbate interactions
- 544.723.222 Coadsorption
- 544.723.223 Exchange adsorption
- 544.723.224 Displacement adsorption
- 544.723.23 Adsorbate-adsorbent interactions
- 544.723.232 Retention. Retention time. Relative retention. Preferential (selective) adsorption
- 544.723.233 Contact accommodation. Accommodation energy. Accommodation coefficients. Sticking probabilities (numbers)
- 544.723.237 Adsorptive destrengthening. Rehbinder effect
- 544.723.3 Desorption
- 544.723.5 Chemisorption
- 544.723.52 Activated adsorption
- 544.723.54 Formation of surface compounds
- 544.723.55 Dissociative adsorption
- 544.725 Adsorption and other surface phenomena in membranes (semipermeable, permselective films) and in diaphragms (capillary-porous bodies)
 ⇒ 544.774
- 544.725.2 Membrane transport phenomena. Membrane permeability. Membrane equilibrium. Donnan equilibrium
- 544.725.5 Capillary condensation
- 544.725.7 Capillary osmosis. Reverse (back) osmosis. Dialysis. Ultrafiltration. Hyperfiltration
- 544.726 Ion exchange in contact systems. Ion-exchange adsorption. Ion-exchange concentration. Ionites. Ion-exchange membranes
- 544.726.2 Ion-exchange capacity
- 544.726.3 Ion-exchange selectivity
- 544.77 Disperse and colloidal systems. Colloid chemistry**
Physicochemical properties are denoted by colon combination with 53 (See examples below).
Denote specific properties of colloids by .03... (See examples below)
 ◇ 544.77:535 Optical properties of colloids
 ◇ 544.77:537 Electrical properties of colloids
 ◇ 544.77.022 Structure of disperse systems

- ◇ 544.77.022.52 Globular, fibrillar structures
 ◇ 544.77.022.522 Globular structure
 ◇ 544.77.022.524 Fibrillar structure
 ◇ 544.77.022.532 Micellar structure. Micelles
 ◇ 544.77.022.533 Vesicular structure. Vesicles
 ◇ 544.77.022.535 Cellular structure. Honeycomb structure
 ◇ 544.77.022.537 Capillary-porous structure. Spongy structure
 ◇ 544.77.022.54 Periodic colloidal structures
 ◇ 544.77.022.542 Oriented structures. Schiller layers. Tactoids
 ◇ 544.77.022.544 Liesegang rings
 ◇ 544.77.022.8 Binding. Setting. Cross-linking
 ◇ 544.77.022.82 Setting. Structurization. Bound disperse systems
 ◇ 544.77.022.822 Gelation. Jelling (jellification). Transition to liquid-like state (to structured liquid). Sol-gel transformation. Coagulation structure. Coagulation contacts. Thixotropy. Rheopexy
 ◇ 544.77.022.823 Colloidal setting (dispersion binding). Transition to solid-like state. Condensation-crystallization structure. Phase contacts. Dilatancy
 ◇ 544.77.022.83 Agitation. Setting destruction
 ◇ 544.77.022.84 Swelling. Swelling ability
 ◇ 544.77.022.85 Syneresis
 ◇ 544.77.023.5 Dispersity and shape of particles, pores and capillaries
 ◇ 544.77.023.51 Monodispersity
 ◇ 544.77.023.52 Polydispersity
 ◇ 544.77.023.522 Size distribution of particles and pores
 ◇ 544.77.023.523 Particle size analysis. Dispersion analysis. Grading (grain-size, sieve) analysis. Sedimentation analysis
 ◇ 544.77.023.55 Shape of particles, pores and capillaries
 ◇ 544.77.03 Physical properties of colloids. Physical and mechanical effects of colloids
 ◇ 544.77.032.1 Colloid stability and stabilization
 ◇ 544.77.032.12 Aggregative stability
 ◇ 544.77.032.13 Sedimentation (dynamic) stability
 ◇ 544.77.032.14 Phase stability
 ◇ 544.77.032.16 Colloid protection. Emulsifying action
 ◇ 544.77.032.18 Protective colloids. Gold number
 ⇒ 544.35
- 544.77.051 Formation of colloids**
.051 from 54.051 is not applicable here
 544.77.051.1 Dispersing. Dispersion. Dispersive ability
 544.77.051.12 Emulsification. Emulsifying ability
 544.77.051.13 Foaming. Frothing. Foaming (frothing) ability. Foam multiplicity
 544.77.051.15 Pulverization. Spraying
 544.77.051.2 Deaggregation
 544.77.051.21 Peptization
 544.77.051.22 Deflocculation
 544.77.051.3 Decoagulation
 544.77.051.5 Formation of new phase. □ Condensational
 544.77.051.6 Formation of associated colloids
 ⇒ 544.777
 544.77.051.62 Micelle formation. Critical micelle concentration
 ⇒ 544.77.022.532
- 544.77.051.64 Vesicle formation
 ⇒ 544.77.022.533
 544.77.051.7 Solubilization (colloid dissolution)
 544.77.052 Destruction of colloids
.052 from 54.052 is not applicable here
 544.77.052.1 Coalescence (fusion of droplets or gas bubbles)
 544.77.052.12 Deemulgation. Emulsion breaking (stratification)
 544.77.052.13 Foam (froth) dying. Foam (froth) extinguishing
 544.77.052.2 Aggregation. Adhering of particles
 544.77.052.21 Coagulation. Curdling
 544.77.052.22 Flocculation. Flaking
 544.77.052.3 Coacervation
 544.77.052.5 Sedimentation. Settling. Flotation
 544.77.054.6 Purification of colloids
 544.772 Colloids with gaseous dispersion phase (medium). Aerosols
 ⇒ 54-138
 544.772.3 Liquid disperse phase. Mists. Fogs
 544.772.4 Solid disperse phase
 544.772.42 Fumes
 544.772.43 Dusts
 544.773 Colloids with liquid dispersion phase (medium). Dispersions
 ⇒ 54-148
 544.773.2 Gaseous disperse phase. Gaseous emulsions. Foams
 ◇ 544.773.2.023.2 Foamy films
 Liquid disperse phase. Emulsions
 ◇ 544.773.3.023.2 Emulsion films
 ⇒ 544.778.3
 544.773.32 Direct (straight) emulsions ('oil-in-water' type)
 544.773.33 Inverted emulsions ('water-in-oil' type). Emulsion inversion
 Critical emulsions
 Solid disperse phase
 544.773.4 Lyosols. Colloid solutions. Sols in general
 ⇒ 544.772
 544.773.422 Hydrosols
 544.773.423 Organosols. Liposols
 544.773.43 Gels. Coagels. Lyogels
 544.773.432 Hydrogels
 544.773.433 Organogels. Lipogels
 544.774 Colloids with solid dispersion phase (medium). Capillary-porous bodies. Solid colloids
 ⇒ 54-168
 ⇒ 544.725
 544.774.2 Gaseous disperse phase. Solid foams. Aerogels. Xerogels
 544.774.3 Liquid disperse phase. Solidified emulsions
 544.774.4 Solid disperse phase. Solid sols. Solid colloid solutions. Crystalline sols. Colloid glasses
 544.776 Semicolloids
 544.777 Molecular colloids. Polymeric (macromolecular) solutions. Jellies. Latexes
 544.778 Coarse disperse systems
 544.778.3 Suspensions. Slurries. Pastes
 544.778.4 Powders
- 546 INORGANIC CHEMISTRY**
Classification of compounds.
Most chemical compounds are to be considered as the result of an actual or hypothetical reaction of two oxides with each other, or of an oxide with a hydride. In this way, a chemical compound may be designated by the oxide or hydride radicals from which it is derived.
The chemical compound is denoted by means of the apostrophe. A binary compound is denoted by citing the number

546.133.1	Oxide of monovalent chlorine and its derivatives. Hypochlorites $\diamond$ 546.133.1-32 Hypochlorous acid HOCl
546.133.2	Hypothetical oxide of divalent chlorine
546.133.3	Oxide of hypothetical trivalent chlorine. Derivatives of chlorous acid HClO ₂ . Chlorites $\diamond$ 546.133.3-32 Chlorous acid
546.134	Chlorine dioxide ClO ₂ and its derivatives
546.135	Derivatives of pentavalent chlorine. Chlorates $\diamond$ 546.135-32 Chloric acid HClO ₃
546.136	Hypothetical derivatives of hexavalent chlorine
546.137	Derivatives of heptavalent chlorine. Perchlorates $\diamond$ 546.137-32 Perchloric acid HClO ₄
546.14	Bromine Br 546.14 $\equiv$ 546.13
546.15	Iodine I 546.15 $\equiv$ 546.13
546.16	Fluorine F 546.16 $\equiv$ 546.13 <i>except:</i>
546.169	Astatine At 546.169 $\equiv$ 546.13
546.17/.19	Nonmetals of the fifth group. Nitrogen group
546.17	Nitrogen N
546.171	Hydrogen compounds of nitrogen
546.171.1	Ammonia NH ₃ and its metal derivatives. Nitrides $\Rightarrow$ 546.39
546.171.5	Hydrazine N ₂ H ₄ and its derivatives
546.171.6	Azo derivatives
546.171.8	Hydrazoic acid HN ₃ and its derivatives. Azides
546.172	Hydroxylamine NH ₂ OH and its derivatives
546.172.5	Nitrous oxide N ₂ O and its derivatives
546.172.6	Nitric oxide NO and its derivatives
546.173	Derivatives of trivalent nitrogen. Derivatives of nitrogen trioxide. Nitrites $\diamond$ 546.173-31 Nitrogen trioxide N ₂ O ₃ $\diamond$ 546.173-325 Nitrous acid HNO ₂ $\diamond$ 546.173'131 Nitrogen trichloride NCl ₃
546.174	Derivatives of tetravalent nitrogen. Nitrogen dioxide NO ₂ . Dinitrogen tetroxide N ₂ O ₄
546.175	Derivatives of pentavalent nitrogen. Derivatives of nitrogen pentoxide. Nitrates $\diamond$ 546.175-31 Nitrogen pentoxide N ₂ O ₅ $\diamond$ 546.175-323 Nitric acid HNO ₃
546.176	Aqua regia. Mixture of hydrochloric acid and nitric acid
546.177	Compounds of nitrogen. Pernitric acid HNO ₄
546.18	Phosphorus P 546.18 $\equiv$ 546.17
546.19	Arsenic As 546.19 $\equiv$ 546.17
546.2	Nonmetals of the third, fourth, sixth group and zerovalent elements
546.21	Oxygen O
546.212	Water H ₂ O $\diamond$ 546.212*2 Heavy water D ₂ O
546.214	Ozone O ₃
546.215	Hydrogen peroxide H ₂ O ₂
546.216	Oxygen with positive valence. Tetravalent and hexavalent oxygen
546.217	Air $\Rightarrow$ 551.510.41

546.22/.24	Sulphur group. Chalkogens. Chalkogenides in general	546.272	Lower boroxides. Unsaturated boroxides. □ Boroxide B_2O_2 . Boron monoxide BO
546.22	Sulphur S	546.273	Trivalent boron and its derivatives. Borates ◇ 546.273-31 Boron trioxide B_2O_3 ◇ 546.273-324 Tetraboric acid $H_2B_4O_7$ ◇ 546.273-325 Boric acid H_3BO_3 . Boracic ◇ 546.273'131 Boron trichloride BCl_3
546.221	Derivatives of negative bivalent sulphur		
546.221.1	Hydrogen sulphide H_2S and its derivatives. Sulphides		
546.222	Hydrogen persulphides. Polysulphides		
546.223.1	Derivatives of hypothetical monovalent sulphur oxide. Sulphur suboxide S_2O	546.28	Silicon Si
546.223.2	Derivatives of hypothetical divalent sulphur oxide. Sulphur monoxide SO. Thiosulphates ◇ 546.223.2-325 Hyposulphurous acid H_2SO_2	546.281	Silicon hydride and its derivatives. Silanes. Silicides ◇ 546.281'261 Silicon carbide SiC
546.223.3	Derivatives of trivalent sulphur oxide ◇ 546.223.3-31 Sulphur sesquioxide S_2O_3	546.282	Lower silicon oxides. Unsaturated silicon oxides
546.224	Derivatives of tetravalent sulphur. Sulphites ◇ 546.224-31 Sulphur dioxide SO_2 ◇ 546.224-325 Sulphurous acid H_2SO_3	546.284	Silicic anhydride and its derivatives. Silicates ◇ 546.284-31 Silicon dioxide SiO_2 ◇ 546.284-325 Silicic acid H_2SiO_3 ◇ 546.284'161 Silicon tetrafluoride SiF_4 ◇ 546.284'161-32 Fluorosilicic acid H_2SiF_6
546.226	Sulphur trioxide and its derivatives. Sulphates ◇ 546.226-31 Sulphur trioxide SO_3 ◇ 546.226-324 Pyrosulphuric acid $H_2S_2O_7$ ◇ 546.226-325 Sulphuric acid H_2SO_4 ◇ 546.226'221-32 Thiosulphuric acid $H_2S_2O_3$	546.286	Derivatives of hexavalent silicon
546.227	Persulphuric acid $H_2S_2O_8$. Peroxosulphuric acid	546.287	Organosilicon compounds. Silico-oxalic acid HOOSi-SiOOH ⇒ 547.1'128
546.228	Polythionic acids in general. Dithionic acids	546.289	Germanium Ge 546.289 = 546.28
546.23	Selenium Se 546.23 = 546.22		
546.24	Tellurium Te 546.241/.248 = 546.221/.228	546.29	Zerovalent elements
546.249	Polonium Po 546.249 = 546.22	546.291	Helium He
		546.292	Neon Ne
546.25	Nonmetals of the third and fourth groups in general	546.293	Argon Ar
		546.294	Krypton Kr
546.26	Carbon C <i>For organic derivatives of carbon and its compounds and for compounds of carbon and halogens use 547</i> ◇ 546.26-162 Graphite. Diamond ⇒ 549.21	546.295	Xenon Xe
546.261	Carbides	546.296	Radon Rn
546.262	Lower and unsaturated oxides of carbon and their derivatives		
546.262.3	Divalent carbon and its derivatives ◇ 546.262.3-31 Carbon monoxide CO		
546.262.5	Carbonyl compounds in general ⇒ 546.725 ⇒ 546.735 ⇒ 546.745 ⇒ 546.765	546.3	Metals in general
546.263	Hypothetical trivalent carbon		
546.264	Tetravalent carbon. Carbonates ◇ 546.264-31 Carbon dioxide CO_2	546.30	Transition metals. Metalloids
546.265	Compounds of carbon with sulphur, selenium, tellurium and their derivatives, including those containing oxygen and halogens	546.302	Transition metals <i>Class here compounds of specified transition metals. For details about electron shells of transition metals see under 544 Physical Chemistry. Specific transition metals are classed within the appropriate metal groups</i> ◇ 546.302'221.1 Sulphide of transition metals
546.266	Cyanogen $N \equiv C - C \equiv N$ ⇒ 547.491	546.303	Metalloids <i>Class here compounds of specified metalloids. For properties and other details of metalloids see 544</i>
546.267	Hydrocyanic acid HCN and its derivatives. Cyanides		
546.268.1	Cyanic acid $N \equiv C - OH$ and its derivatives. Cyanates	546.31	Elements of the first group in general.
546.268.2	Isocyanic acid $O = C = NH$ and its derivatives. Isocyanates		Metals of the first group ⇒ 546.55
546.268.3	Fulminic acid $C \equiv N - OH$	546.311	Generalities on the alkali metals
546.268.5	Thiocyanic acid $N \equiv C - SH$ and its derivatives		
546.268.6	Isothiocyanic acid $S = C = NH$ and its derivatives	546.32	Potassium K ◇ 546.32-31 Potassium oxide K_2O ◇ 546.32-36 Potassium hydroxide KOH ◇ 546.32-39 Potassium peroxide K_2O_2 ◇ 546.32'131 Potassium chloride KCl ◇ 546.32'135 Potassium chlorate $KClO_3$ ◇ 546.32'173 Potassium nitrite KNO_2 ◇ 546.32'175 Potassium nitrate KNO_3 ◇ 546.32'185 Potassium phosphates ◇ 546.32'226 Potassium sulphate K_2SO_4 ◇ 546.32'226-384.1 Potassium hydrogen sulphate $KHSO_4$ ◇ 546.32'264 Potassium carbonate K_2CO_3 ◇ 546.32'264-384.1 Potassium hydrogen carbonate $KHCO_3$ ◇ 546.32'267 Potassium cyanide KCN
546.27	Boron B		
546.271	Boron hydrides and their derivatives. Boranes. Borides ◇ 546.271-386 Boron hydride. Boranates		

	◇ 546.32'273 Potassium borate $K_2B_4O_7$	546.655.3	Trivalent cerium
	◇ 546.32'273-39 Potassium perborate $KBO_2 \cdot H_2O_2 \cdot 3H_2O$	546.655.4	Tetravalent cerium
	◇ 546.32'284 Potassium silicates $K_2O \cdot xSiO_2$	546.656	Praseodymium Pr
	◇ 546.32'284'161 Potassium fluorosilicate K_2SiF_6	546.657	Neodymium Nd
		546.658	Promethium Pm
		546.659	Samarium Sm
546.33	Sodium Na	546.66	Yttrium group in general
	◇ 546.33'131 Sodium chloride NaCl. Table salt		546.66 $\approx$ 546.655
	$\Rightarrow$ 549.451.1, 553.631, 661.833		$\Rightarrow$ 546.650
	◇ 546.33'273 Sodium tetraborate $Na_2B_4O_7$. Borax	546.661	Europium Eu
546.34	Lithium Li	546.662	Gadolinium Gd
546.35	Rubidium Rb	546.663	Terbium Tb
546.36	Caesium Cs	546.664	Dysprosium Dy
546.37	Francium Fr	546.665	Holmium Ho
546.39	Ammonium radical	546.666	Erbium Er
546.4	Elements of the second group in general. Metals of the second group	546.667	Thulium Tm
546.40	Alkaline earth metals in general	546.668	Ytterbium Yb
546.41	Calcium Ca	546.669	Lutetium Lu
546.42	Strontium Sr	546.68	Gallium group in general
546.43	Barium Ba	546.681	Gallium Ga
546.44	Radium Ra		546.681 $\approx$ 546.683
546.45	Beryllium Be	546.682	Indium In
546.46	Magnesium Mg		546.682 $\approx$ 546.683
546.47/49	Zinc, cadmium and mercury group in general	546.683	Thallium Tl
546.47	Zinc Zn	546.683.1	Monovalent thallium
546.48	Cadmium Cd	546.683.3	Trivalent thallium
546.49	Mercury Hg	546.7	Metals of the sixth and seventh group. Base metals of the eighth group. Actinides
546.491	Monovalent mercury. Mercurous compounds	546.71	Elements of the seventh group in general. Metals of the seventh group. Manganese group
546.492	Divalent mercury. Mercuric compounds	546.711	Manganese Mn
546.55	Copper, silver and gold group in general	546.712	Divalent manganese
546.56	Copper Cu	546.713	Trivalent manganese
546.561	Monovalent copper. Cuprous compounds	546.714	Tetravalent manganese. Manganite
546.562	Divalent copper. Cupric compounds	546.716	Hexavalent manganese. Manganates
546.57	Silver Ag	546.717	Heptavalent manganese. Permanganates
546.571	Monovalent silver		◇ 546.717'32 Potassium permanganate $KMnO_4$
546.572	Divalent silver	546.718	Technetium Tc
546.59	Gold Au		546.718.2/.7 $\approx$ 546.712/.717
546.591	Monovalent gold	546.719	Rhenium Re
546.593	Trivalent gold		546.719.2/.7 $\approx$ 546.712/.717
546.6	Elements of the third group in general. Metals of the third group	546.72/.74	Base metals of the eighth group. Iron group in general
	<i>Numbers may be completed by adding a number 1,2,3 etc. indicating the valency of an element</i>		<i>For noble metals of the eighth group see 546.9</i>
	◇ 546.632 Bivalent scandium	546.72	Iron Fe
	◇ 546.633 Trivalent scandium	546.722	Divalent iron. Ferrous compounds
	$\Rightarrow$ 546.790	546.723	Trivalent iron. Ferric compounds
546.62	Aluminium Al	546.725	Carbonyl compounds of iron
546.63	Scandium Sc	546.726	Hexavalent iron. Ferrates
546.64	Yttrium Y	546.73	Cobalt Co
546.65	Rare earths in general	546.732	Divalent cobalt. Cobaltous compounds
	546.65 $\approx$ 546.655	546.733	Trivalent cobalt. Cobaltic compounds
546.650	Lanthanide group in general	546.735	Carbonyl compounds of cobalt
546.651	Cerium group in general	546.74	Nickel Ni
546.654	Lanthanum La	546.742	Divalent nickel
546.655	Cerium Ce	546.743	Trivalent nickel
		546.745	Carbonyl compounds of nickel
		546.75	Elements of the sixth group in general. Metals of the sixth group. Chromium group in general
		546.76	Chromium Cr
		546.762	Divalent chromium. Chromous compounds
		546.763	Trivalent chromium. Chromic compounds

546.765	Carbonyl compounds of chromium	546.81	Tin. Lead
546.766	Hexavalent chromium. Chromates	546.811	Tin Sn, in general
546.77	Molybdenum Mo	546.812	Divalent tin. Stannous compounds
546.772	Divalent molybdenum	546.814	Tetravalent tin. Stannic compounds
546.773	Trivalent molybdenum	546.815	Lead Pb, in general
546.774	Tetravalent molybdenum	546.816	Monovalent lead. Plumbous compounds
546.775	Pentavalent molybdenum	546.817	Divalent lead. Plumbic compounds
546.776	Hexavalent molybdenum	546.818	Trivalent lead
546.78	Tungsten W	546.819	Tetravalent lead
	546.78 $\equiv$ 546.77		$\diamond$ 546.819'817 Lead orthoplumbate Pb_3O_4
546.79	Radioactive elements. Radioactive isotopes.	546.82/.83	Titanium group in general
	Actinides in general	546.82	Titanium Ti
	<i>For notation of isotopes see note at 546</i>	546.822	Divalent titanium
	$\Rightarrow$ 539.16	546.823	Trivalent titanium
	$\Rightarrow$ 544.58	546.824	Tetravalent titanium. Titanates
546.790	Actinides in general (atomic number 89-103)	546.83	Zirconium. Hafnium
546.791	Uranium U	546.831	Zirconium Zr
	546.791.2/.6 $\equiv$ 546.772/.776		546.831 $\equiv$ 546.82
	$\diamond$ 546.791.027*234 Uranium 234 ^{234}U	546.832	Hafnium Hf
	$\diamond$ 546.791.027*235 Uranium 235 ^{235}U		546.832 $\equiv$ 546.82
	$\diamond$ 546.791.027*238 Uranium 238 ^{238}U	546.85	Elements of the fifth group in general.
	$\diamond$ 546.791.027 Uranium isotopes generally		Metals of the fifth group
546.795	Thorium Th		<i>For non metals of the fifth group in general see 546.17/.19</i>
	<i>Numbers 546.795/.797 may be completed by adding numbers from 1 to 8 indicating the valency of an element</i>	546.86/.87	Antimony group in general. $\square$ Antimony. Bismuth
	$\diamond$ 546.795.4 Tetravalent thorium	546.86	Antimony Sb
	$\diamond$ 546.795.4-36 Thorium hydroxide		546.86 $\equiv$ 546.17
	$\diamond$ 546.795.4'131 Thorium chloride ThCl_4	546.861	Antimony hydrides
	$\diamond$ 546.796.5 Pentavalent protactinium	546.863	Trivalent antimony and its derivatives
546.796	Protactinium Pa	546.865	Pentavalent antimony and its derivatives
546.797	Actinium Ac	546.87	Bismuth Bi
546.798	Transuranic elements		546.87 $\equiv$ 546.17
	<i>Elements with atomic number from 93 upwards</i>	546.88	Vanadium, Niobium and Tantalum group
	<i>Numbers 546.798.21/.28 and 546.798.291/.293 may be completed by adding a number from 1 to 8 indicating the valency of the element</i>	546.881	Vanadium V
546.798.2	Transuranic actinides	546.881.2	Divalent vanadium
546.798.21	Neptunium Np	546.881.3	Trivalent vanadium
546.798.22	Plutonium Pu	546.881.4	Tetravalent vanadium
546.798.23	Americium Am	546.881.5	Pentavalent vanadium
546.798.24	Curium Cm	546.882	Niobium Nb
546.798.25	Berkelium Bk		546.882 $\equiv$ 546.881
546.798.26	Californium Cf	546.883	Tantalum Ta
546.798.27	Einsteinium Es (atomic number 99)		546.883 $\equiv$ 546.881
546.798.28	Fermium Fm (atomic number 100)	546.9	Elements of the eighth group in general. Noble metals of the eighth group
546.798.29	Other transuranic actinides	546.91	Light platinum metals (platinum triad)
546.798.291	Mendelevium Md (atomic number 101)	546.92	Platinum Pt
546.798.292	Nobelium No (atomic number 102)	546.921	Monovalent platinum
546.798.293	Lawrencium Lr (atomic number 103)	546.922	Divalent platinum
546.798.3	Elements with atomic number 104 and higher. Superheavy elements	546.923	Trivalent platinum
546.798.31	Kurchatovium (atomic number 104)	546.924	Tetravalent platinum
	= Rutherfordium Rf	546.926	Hexavalent platinum
546.798.32	Dubnium Db/Hahnium Ha (atomic number 105)	546.93	Iridium Ir
	= Bohrium Bo, Nielsbohrium Ns, Pentium Unp		546.93 $\equiv$ 546.92
546.798.33	Seaborgium Sb/Eka-Wolfram (atomic number 106) (= Hexium Unh)	546.94	Osmium Os
	<i>Element with atomic number 106 and higher. Eka-Rhenium (atomic number 107)</i>	546.95	546.94 $\equiv$ 546.92
546.798.39	New discovered elements. Doubtful elements. Indefinable elements	546.96	Heavy platinum metals (ruthenium triad)
546.8	Elements of the fourth group in general. Metals of the fourth group and the fifth group	546.97	Ruthenium Ru
	$\Rightarrow$ 546.795	546.98	546.96 $\equiv$ 546.92
	$\Rightarrow$ 546.796		Rhodium Rh
			546.97 $\equiv$ 546.92
			Palladium
			546.98 $\equiv$ 546.92

547

ORGANIC CHEMISTRY

547'11/19 $\equiv$ 546.1/9547'2/8 $\equiv$ 547.2/8

Further details may be denoted:

(a) by apostrophe.

(b) by hyphen auxiliaries -1/-4 listed under 54.

(c) by point-nought auxiliaries .01/.08 listed under 54.

(d) by the condensed formula added directly to 547, e.g. 547CH₃O Methanol (methyl alcohol). Cf. 547.261

547-3

Types of organic compound

The -3 auxiliaries may be used with subdivisions 547.1/9, but not with 547 itself.

For organic compounds as a whole use 547.1-3... (see note at 547.1-3).

The -3 auxiliaries at 54 are applicable with addition of numbers indicated below

547-30

Oxygen-free compounds

547-302

Halogen derivatives

547-304

Nitrogenous compounds (C-N)

◇ 547.1-32-304.2 Amino acids

⇒ 547-327

547-304.1

Nitro derivatives. Nitroso derivatives

547-304.2

Amines. Imines. Ammonium bases. Ammonium salts

547-304.3

Hydrazines. Hydrazo derivatives. Diazonium salts

547-304.4

Azo derivatives. Diazo derivatives. Diazonium salts

547-304.5

Hydroxylamines

547-304.6

Oximes. Hydrazones. Semicarbazones. Azines (aldazines, ketazines)

547-304.7

Nitriles. Isonitriles

547-304.9

Other compounds. □ Triazines. Triazenes. Tetrazenes

547-305

Compounds containing sulphur directly linked to carbon

547-31/-39

Compounds containing oxygen (C-O)

547-311

Oxides. Epoxy compounds

547-312

Anhydrides

547-313

Lactides

547-314

Lactones

547-315

Esters of inorganic acids in general
⇒ 547-326

547-316

Aldehydes. Ketones. Acetals. Ketals

547-317

Diketopiperazines

547-318

Lactams

547-326

Esters in general
⇒ 547.315

547-327

Amides. Imides. Amidines. Hydroxamic acids.
Other substitutions by nitrogen in the carboxyl group

547'0

Details of the structure of organic compounds

These special auxiliaries are developed at various places in 547, e.g. at 547.52

547.022

Structure of organic compounds

547.022.1

Structure of organic compounds from the point of view of linking of the carbon atoms

547.022.11

Primary carbon atoms

547.022.12

Secondary carbon atoms

547.022.13

Tertiary carbon atoms

547.022.14

Quaternary carbon atoms

547.1

Classification of organic compounds.
Element-organic compounds

547.1-3

Types of organic compounds

Specify type using -3... auxiliaries at 54 and additions at 547. Use the highest number if more than one subdivision is applicable; the second group may be indicated by adding another hyphen auxiliary. Derivatives are indicated in the same way as the applicable functional groups themselves

◇ 547.1-316-315 Oxyaldehydes. NOTE: from -316 Aldehydes and -315 Esters of organic acids

◇ 547.1-32-304.2 Aminoacids. NOTE: from -32 Acids and -304.2 Amines

◇ 547.1-32 Acids

◇ 547.1-326 Esters

◇ 547.1-315 Esters of inorganic acids

547.1'1

Element organic compounds

547.1'1 $\equiv$ 546

Organic compounds of carbon with elements other than the halogens, nitrogen, oxygen or sulphur (which are classed under 547.2/9)

Classification of element organic compounds is based on the notation of the element in 546. The organic part is specified by the addition of '2/8

◇ 547.1'147'212 Diethyl zinc

547.1'11

Organic compounds of nonmetals in general

547.1'123

Organic compounds of selenium

547.1'124

Organic compounds of tellurium

547.1'127

Organic compounds of boron

547.1'128

Organic compounds of silicon

547.1'13

Organometallic compounds

547.1'147

Organic compounds of zinc

547.14

Comparison of cyclic and acyclic compounds

547.15

Organic compounds of the bivalent carbon atoms. Carbenes

547.16

Organic compounds not containing carbon.
□ Boroxynes, Borazines, Cyclosiloxanes. Phosphazenes

547.2

Acyclic compounds in general

547.21

Saturated hydrocarbons. Alkanes (paraffins) C_nH_{2n+2}

Details by special auxiliaries at 54

◇ 547.21.024 Alkyl groups

547.211

Methane CH₄

547.212

Ethane C₂H₆

547.213

Propane C₃H₈

547.214

Butane C₄H₁₀

547.215

Pentane C₅H₁₂

547.216

Hexane C₆H₁₄

547.217.1

Heptane C₇H₁₆

547.217.2

Octane C₈H₁₈

547.217.3

Nonane C₉H₂₀

547.217.4

Decane C₁₀H₂₂

547.217.5

Undecane C₁₁H₂₄

547.217.6

Dodecane C₁₂H₂₆

547.217.7

Tridecane C₁₃H₂₈

547.217.8

Tetradecane C₁₄H₃₀

547.217.9

Pentadecane C₁₅H₃₂

547.218.1

Hexadecane C₁₆H₃₄

547.218.2

Heptadecane C₁₇H₃₆

547.218.3

Octadecane C₁₈H₃₈

547.218.4

Nonadecane C₁₉H₄₀

547.218.5

Eicosane C₂₀H₄₂

547.218.6

Heneicosane C₂₁H₄₄

547.218.7

Docosane C₂₂H₄₆

547.218.8

Tricosane C₂₃H₄₈

547.218.9	Tetracosane $C_{24}H_{50}$	547.268.31	Hexacosanols $C_{26}H_{54}OH$. □ Ceryl alcohol.
547.219	Saturated hydrocarbons with 25 or more carbon atoms		Cerotin
547.22	Monohalogeno derivatives of saturated hydrocarbons	547.269	Monosubstituted derivatives of saturated hydrocarbons, containing sulphur bound directly to the carbon
547.221	Fluoro derivatives	547.269.1	Mercaptans $C_nH_{2n+1}SH$
	547.221 $\equiv$ 547.21		547.269.1 $\equiv$ 547.26
547.222	Chloro derivatives	547.269.3	Sulphonic acids, Sulphonic acids substituted in the $-SO_3H$ group
	547.222 $\equiv$ 547.21		
547.223	Bromo derivatives	547.269.4	Thiosulphonic acids
	547.223 $\equiv$ 547.21	547.269.6	Sulphinic acids. Sulphonic acids
547.224	Alkyl iodides	547.27	Ethers of the saturated hydrocarbons
	547.224 $\equiv$ 547.21		547.271/.278 $\equiv$ 547.261/.268
547.225	Iodoso compounds	547.271	Methyl ethers
547.226	Iodo compounds	547.272	Ethyl ethers
547.227	Iodonium compounds	547.279	Sulphur-containing analogues of ethers
547.23	Monosubstituted derivatives of saturated hydrocarbons containing nitrogen directly linked to carbon		<i>In mixed analogues of ethers denote component by apostrophe combination with subdivisions of 547.26</i>
547.231	Nitroso derivatives	547.279.1	◇ 547.279.12'261 Ethyl methyl sulphide
	547.231 $\equiv$ 547.21		Alkyl sulphides (alkyl thioethers)
547.232	Nitro derivatives	547.279.2	547.279.11/.18 $\equiv$ 547.261/.268
	547.232 $\equiv$ 547.21	547.279.3	Sulphonium compounds
547.233	Amines		Disulphides
547.233.1	Primary amines	547.279.4	547.279.31/.38 $\equiv$ 547.261/.268
	547.233.1 $\equiv$ 547.21	547.279.5	Polysulphides
547.233.2	Secondary amines		Sulphoxides. Sulphones
	547.233.2 $\equiv$ 547.21	547.28	Saturated monoaldehydes and monoketones. Analogues containing sulphur
547.233.3	Tertiary amines		Saturated monoaldehydes $C_nH_{2n}O$
	547.233.3 $\equiv$ 547.21	547.281	547.281 $\equiv$ 547.21
547.233.4	Quaternary ammonium compounds		Formaldehyde $HCHO$
	547.233.4 $\equiv$ 547.21	547.281.1	Acetaldehyde
547.234	Hydrazines	547.281.2	Monomer acetaldehyde
547.235	Other nitrogen derivatives containing two nitrogen atoms	547.281.21	Paraldehyde
547.235.2	Azo compounds	547.281.22	Metalddehyde
	<i>Further particulars may be denoted by special auxiliaries</i>	547.281.23	Thioaldehydes
	◇ 547.235.2'211 Azomethane	547.283.2	Saturated monoketones
547.235.3	Hydroxyhydrazines	547.284	547.284 $\equiv$ 547.21
547.235.4	Diazo- and isodiazocompounds	547.284.3	Propanone. Acetone CH_3COCH_3
547.235.5	Azoxy compounds containing the group $-N_2OH$	547.284.4	Butanones. Ethyl methyl ketone
547.235.6	Isonitramines with the group $-N_2O_2H$	547.286.2	Thioketones
547.236	Nitrogen derivatives containing three nitrogen atoms. Triazenes. Triazenes		547.286.2 $\equiv$ 547.21
547.237	Derivatives with nitrogen groups containing four or more nitrogen atoms. Tetrazanes. Tetrazenes	547.288	Condensation products of aldehydes and ketones with ammonia, hydrazine etc.
547.238	Hydroxylamines	547.288.1	Ammonium compounds
547.239	Nitriles (cyanides). Isonitriles (isocyanides)	547.288.15	Hexamine (hexamethylenetetramine)
547.26	Saturated monohydric alcohols $C_nH_{2n+1}OH$	547.288.2	Amino compounds
	◇ 547.26'11 Acid derivatives in general.	547.288.3	Hydrazine compounds. Hydrazones. Aldazines.
	Esters	547.288.4	Ketazines. Semicarbazones
	◇ 547.26'127 Esters of acids containing boron. NOTE: ('127 from 546.27)	547.29	Hydroxylamine compounds. Oximes
547.260.2	-OH groups in general		Monobasic saturated carboxylic acids
547.261	Methanol (methyl alcohol) CH_3OH		$C_nH_{2n}O_2$
547.262	Ethanol (ethyl alcohol) C_2H_5OH	547.29'05	<i>Denote esters by '26... (subdivisions of 547.26).</i>
547.263	Propanols (propyl alcohols) C_3H_7OH	547.29'051	<i>Denote anhydrides and peroxides by -312 and -39</i>
547.264	Butanols (butyl alcohols) C_4H_9OH	547.29'052	Derivatives of carboxylic acids
547.265	Pentanol (pentyl alcohols, amyl alcohols) $C_5H_{11}OH$	547.29'053	Isonitriles
547.266	Hexanols (hexyl alcohols) $C_6H_{13}OH$	547.29'054	Nitriles
547.267	Heptanols (heptyl alcohols) $C_7H_{15}OH$		Acyl halides
547.268	Octanols (octyl alcohols) and higher alcohols	547.291	Carboxylic acids which are substituted in the carboxyl group by ammonia, hydrazine etc.
547.268.11	Octanols (octyl alcohols) $C_8H_{17}OH$. □ Caprylic alcohol	547.292	Formic acid $HCOOH$
547.268.19	Hexadecanols (hexadecyl alcohols) $C_{16}H_{33}OH$. □ Cetyl (palmityl) alcohol	547.293	Acetic acid CH_3COOH
		547.294	Propionic acid CH_3CH_2COOH
		547.295	Butyric acids $CH_3(CH_2)_2COOH$
			Acids with 5 to 20 carbon atoms

547.295.1	Pentonic acids (valeric acids) $\text{CH}_3(\text{CH}_2)_3\text{COOH}$	547.335	Other nitrogen derivatives with two nitrogen atoms
547.295.2	Hexoic acids. $\square$ Caproic acid $\text{CH}_3(\text{CH}_2)_4\text{COOH}$	547.336	Nitrogen derivatives with three nitrogen atoms
547.295.4	Octoic acids. $\square$ Caprylic acid $\text{CH}_3(\text{CH}_2)_6\text{COOH}$	547.337	Nitrogen derivatives with four or more nitrogen atoms
547.295.5	Nonoic acids. $\square$ Pelargonic acid $\text{CH}_3(\text{CH}_2)_7\text{COOH}$	547.338	Hydroxylamines
547.295.6	Decoic acids. $\square$ Capric acid $\text{CH}_3(\text{CH}_2)_8\text{COOH}$	547.339	Nitriles (cyanides). Isonitriles (isocyanides)
547.295.71	Undecoic acids $\text{CH}_3(\text{CH}_2)_9\text{COOH}$	547.36	Unsaturated monohydric alcohols
547.295.72	Dodecoic acids. $\square$ Lauric acid $\text{CH}_3(\text{CH}_2)_{10}\text{COOH}$	547.361	Derivatives of the ethylene series
547.295.8	Tetradecoic acids. $\square$ Myristic acid $\text{CH}_3(\text{CH}_2)_{12}$	547.361.2	547.361 $\equiv$ 547.21
547.295.92	Hexadecoic acids. $\square$ Palmitic acid $\text{CH}_3(\text{CH}_2)_{14}\text{COOH}$	547.361.3	Vinyl alcohol $\text{C}_2\text{H}_3\text{OH}$
547.295.93	Heptadecoic acids. $\square$ Margaric acid	547.361.85	Allyl alcohol $\text{C}_3\text{H}_5\text{OH}$
547.295.94	Octadecoic acids	547.362	Phytol $\text{C}_{20}\text{H}_{39}\text{OH}$
547.295.96	Eicosoic acids. $\square$ Arachidic acid $\text{CH}_3(\text{CH}_2)_{18}\text{COOH}$	547.363	Derivatives of the acetylene series
547.296	Acids with more than 20 carbon atoms	547.364	547.362 $\equiv$ 547.21
547.296.12	Decosoic acids. $\square$ Behenic acid $\text{CH}_3(\text{CH}_2)_{20}\text{COOH}$	547.365	Derivatives of the series $\text{C}_n\text{H}_{2n-2}\text{O}$ containing no triple bonds
547.296.14	Tetracosic acids. $\square$ Lignoceric acid $\text{CH}_3(\text{CH}_2)_{22}\text{COOH}$	547.366	Derivatives of the series $\text{C}_n\text{H}_{2n-4}\text{O}$
547.296.16	Hexacosic acids. $\square$ Cerotic acid $\text{CH}_3(\text{CH}_2)_{24}\text{COOH}$	547.367	Derivatives of the series $\text{C}_n\text{H}_{2n-6}\text{O}$
547.299	Sulphur-containing analogues of the carboxylic acids	547.368	Less saturated monohydric alcohols
547.3	Acyclic unsaturated hydrocarbons and their monosubstituted derivatives		Sulphur-containing analogues of the unsaturated alcohols. Unsaturated mercaptans (thiols)
547.31	Unsaturated hydrocarbons		Other sulphur-containing unsaturated compounds. Unsaturated sulphonic and sulphinic acids
547.313	Hydrocarbons of the series C_nH_{2n} . Olefins (alkenes, alkenes)	547.37	Unsaturated ethers
	547.313 $\equiv$ 547.21	547.371	Ethers with one double bond
547.313.2	Ethylene C_2H_4	547.372	Ethers with one triple bond
547.313.3	Propene (propylene) C_3H_6	547.373	Ethers with two double bonds
547.313.4	Butenes (butylenes) C_4H_8	547.374	Less saturated ethers
547.313.5	Pentenes (amylenes) C_5H_{10}	547.379	Sulphur analogues of unsaturated ethers
547.314	Hydrocarbons of the series $\text{C}_n\text{H}_{2n-2}$ with a triple bond	547.38	Unsaturated monoaldehydes and monoketones. Ketenes
	547.314 $\equiv$ 547.21	547.381	Aldehydes of the ethylene series. Acrolein. Crotonaldehyde. Citronellal
547.314.2	Acetylene C_2H_2	547.382	547.381 $\equiv$ 547.21
547.315	Hydrocarbons of the series $\text{C}_n\text{H}_{2n-2}$ with two double bonds	547.383	Less saturated aldehydes. Citral
547.315.1	Hydrocarbons with cumulative double bonds. Allene	547.384	Unsaturated thioaldehydes, selenoaldehydes, telluroaldehydes
	547.315.1 $\equiv$ 547.21	547.385	Unsaturated ketones of the ethylene series.
547.315.2	Hydrocarbons with conjugated double bonds. $\square$ Divinyl. Piperylene. Isoprene	547.386	$\square$ Vinylalkyl ketones. Allyl acetone
	547.315.2 $\equiv$ 547.21	547.387	Less saturated ketones. $\square$ Phorone
547.315.3	Hydrocarbons with isolated double bonds. $\square$ Diallyl	547.388	Unsaturated thioketones
	547.315.3 $\equiv$ 547.21		Ketenes. $\square$ Dimethylketene
547.316	Hydrocarbons of the series $\text{C}_n\text{H}_{2n-4}$		Condensation products of unsaturated aldehydes and ketones with ammonia, hydrazine etc. $\square$ Hydrazones. Aldazines. Ketazines. Semicarbazones
547.317	Hydrocarbons of the series $\text{C}_n\text{H}_{2n-6}$	547.39	Monobasic unsaturated carboxylic acids
547.318	Less saturated hydrocarbons	547.39'05	Derivatives of acids
547.32	Monohalogeno derivatives of unsaturated hydrocarbons	547.39'052	547.39'051/'054 $\equiv$ 547.46'051/'054
	547.32 $\equiv$ 547.31	547.39'053	Nitriles
547.321	Fluoro derivatives	547.39'054.4	Acyl halides
547.322	Chloro derivatives		Amides
547.323	Bromo derivatives	547.391	Acids of the series $\text{C}_n\text{H}_{2n-2}\text{O}_2$ in general
547.324	Iodo derivatives	547.391.1	Acrylic acid $\text{C}_3\text{H}_4\text{O}_2$
547.33	Monosubstituted derivatives of unsaturated hydrocarbons containing nitrogen directly linked to carbon	547.391.2	Vinylacetic acid $\text{C}_4\text{H}_6\text{O}_2$
547.331	Nitroso derivatives	547.391.3	Crotonic acid. Methacrylic acid $\text{C}_4\text{H}_6\text{O}_2$
	547.331 $\equiv$ 547.31	547.391.4	$\text{C}_5\text{H}_8\text{O}_2$ acids. $\square$ Angelic acid. Tiglic acid
547.332	Nitro derivatives	547.391.5	$\text{C}_6\text{H}_{10}\text{BO}_2$ acids
	547.332 $\equiv$ 547.31	547.391.7	$\text{C}_7\text{H}_{12}\text{BO}_2$ acids
547.333	Amines	547.391.8	$\text{C}_8\text{H}_{14}\text{O}_2$ to $\text{C}_{10}\text{H}_{18}\text{O}_2$ acids
547.334	Hydrazines	547.391.9	$\text{C}_{11}\text{H}_{20}\text{O}_2$ to $\text{C}_{14}\text{H}_{26}\text{O}_2$ acids
		547.391.92	Undecylenic acids $\text{C}_{11}\text{H}_{20}\text{O}_2$
		547.392	$\text{C}_{15}\text{H}_{28}\text{O}_2$ and higher acids
		547.392.1	C_{15} acids. $\square$ Cimicinic acid
		547.392.2	C_{16} acids
		547.392.3	C_{17} acids

547.392.4	C ₁₈ acids. □ Oleic acid. Elaidinic acid C ₁₈ H ₃₄ O ₂	547.414	Polynitro compounds and nitrosonitro compounds
547.392.5	C ₁₉ acids. C ₂₀ acids	547.415	Polyamines
547.392.6	Acids with more than twenty carbon atoms 547.392.6 ≡ 547.296	547.415.1	Diamines 547.415.1 ≡ 547.21
547.392.611	C ₂₁ acids	547.415.3	Imines
547.392.612	C ₂₂ acids. □ Brucillic acid, brassidinic acid, cetoleinic acids	547.415.5	Triamines and polyamines
547.393	Acids of the propiolic acid series C _n H _{2n-4} O ₂ 547.393 ≡ 547.29	547.416	Halogeno derivatives and nitro derivatives of amines
547.393.4	Tetrollic acid C ₄ H ₄ O ₂	547.417	Polysubstituted hydrazino, diazo, azido compounds. Bistriazoethane
547.393.52	Sorbic acid C ₆ H ₈ O ₂	547.418	Polysubstituted derivatives of hydroxylamine. Alkylamine diisonitramines. Nitro-nitroso-hydroxylamino alkanes
547.394/.395	Unsaturated acids with two double bonds 547.394/.395 ≡ 547.391/.392		
547.394	Unsaturated acids with two double bonds, containing up to fourteen carbon atoms 547.394 ≡ 547.391	547.42	Polyhydric alcohols and mercaptans and their derivatives
547.395	Unsaturated acids with two double bonds, containing more than fourteen carbon atoms 547.395 ≡ 547.392	547.421	Hypothetical glycols (dihydric alcohols) and their derivatives, in general
547.395.4	Linoleic acid C ₁₈ H ₃₂ O ₂	547.422	Ethylene glycol (ethanediol) and its homologues
547.396	C _n H _{2n-6} O ₂ acids. Linolenic acid C ₁₈ H ₃₀ O ₂	547.422.2	Ethanediol and its homologues 547.422.2 ≡ 547.21
547.399	Sulphur analogues	547.422.22	Ethylene glycol (ethanediol) C ₂ H ₆ O ₂ ◇ 547.422.22-31 Ethylene oxide
547.4	Polysubstituted acyclic compounds. Acyclic compounds with mixed functions	547.422.23	Propylene glycol C ₃ H ₈ O ₂
547.41	Compounds not containing oxygen directly linked to carbon	547.423	Unsaturated α-glycols
547.411	Polyhalogeno compounds in general	547.424	Glycols in which the -OH groups are attached to two separated carbon atoms
547.412	Polyhalogeno derivatives of saturated hydrocarbons	547.425	Bivalent sulphur compounds
547.412.1	Polyhalogeno derivatives of methane	547.426	Trihydric alcohols and their derivatives
547.412.11	Dihalogeno derivatives CH ₂ HIg ₂	547.426.1	Glycerol (glycerine) C ₃ H ₈ O ₃
547.412.12	Trihalogeno derivatives CHHIg ₃	547.426.2	Ethers and esters of glycerol
547.412.122	Trifluoromethane CHF ₃	547.426.24	Esters of inorganic acids ◇ 547.426.24'171 Nitroglycerine C ₃ H ₅ (ONO ₂) ₃
547.412.123	Chloroform CHCl ₃	547.427	Higher polyhydric alcohols. Their relationships to the sugars. Higher sulphur compounds
547.412.124	Bromoform CHBr ₃	547.427.1	Tetrahydric alcohols. Erythritol C ₄ H ₁₀ O ₄ 547.427.1 ≡ 547.21
547.412.125	Iodoform CHI ₃	547.427.2	Pentahydric alcohols. Adonitol C ₆ H ₁₂ O ₅
547.412.13	Tetrahalogeno derivatives CHIg ₄	547.427.3	Hexahydric alcohols. Mannitol C ₆ H ₁₄ O ₆
547.412.132	Carbon tetrafluoride CF ₄	547.427.6	Higher sulphur compounds. □ Mercaptans. Mercaptals. Mercaptols. Sulphonic acids. Sulphinic acids. Sulphenic acids
547.412.133	Carbon tetrachloride CCl ₄		
547.412.134	Carbon tetrabromide CBr ₄		
547.412.135	Carbon tetraiodide CI ₄		
547.412.2	Dihalogeno derivatives of ethane and its homologues in general	547.43	Alcohols containing elements other than oxygen, sulphur etc., directly linked to carbon
547.412.3	Compounds of the type of ethylidene dichloride and isopropylidene dichloride	547.431	Compounds containing both halogens and hydroxyl groups, and their derivatives
547.412.4	Compounds of the type of 1,2-dichloroethane. Ethylene chloride C ₂ H ₂ Cl ₂	547.433	Nitrosoalcohols. Alkylene nitrosites. Alkylene nitrosates. Derivatives of trimethylethylene. Derivatives of trimethylbromoethylene. Derivatives of tetramethylethylene
547.412.5	Compounds of the type of 1,3-dichloropropane and 1,4-dichlorobutane	547.434	Nitroalcohols
547.412.6	Trihalogeno derivatives of ethane and its homologues	547.435	Amino alcohols
547.412.7	Polyhalogeno derivatives of ethane and its homologues	547.436	Amino derivatives of mercaptans, of sulphonic acids and other sulphur compounds. □ Taurine C ₂ H ₅ SO ₃
547.412.9	Completely halogenated derivatives of hydrocarbons	547.437	Hydroxyamino, hydrazino, diazo and azido compounds of alcohols. Diazomethane-disulphonic acid
547.413	Polyhalogeno derivatives of unsaturated hydrocarbons		
547.413.12	Dihalogeno derivatives of ethylene and its homologues in general 547.413.12 ≡ 547.412.2	547.44	Polyaldehydes. Polyketones. Polycarbonyl compounds
547.413.2	Dihalogeno derivatives of acetylene and its homologues and of the less saturated hydrocarbons	547.441	Dialdehydes. □ Glyoxal C ₂ H ₂ O ₂
547.413.4	Trihalogeno derivatives of unsaturated hydrocarbons. Trihalogeno derivatives of ethylene. □ Trichloroethylene	547.442	Diketones. □ Diacetyl C ₄ H ₆ O ₂ . Acetylacetone C ₅ H ₈ O ₂
547.413.5	Perhalogeno derivatives of unsaturated hydrocarbons. Perhalogeno derivatives of ethylene	547.443	Polyketones
		547.444	Ketoaldehydes. □ Methylglyoxal C ₃ H ₄ O ₂
		547.445	Diketenes. Ketoketenes. Acetylketenes

547.446	Halogeno derivatives of aldehydes and ketones. □ Chloral CCl_3CHO . Chloral hydrate $\text{CCl}_3\text{CH}(\text{OH})_2$	547.458.85	Tunicin. Mannosocellulose
547.447	Carbonyl compounds with nitroso, nitro, amino or hydrazino groups, linked to carbon via nitrogen or some other element. □ Amino derivatives. Muscarine $\text{C}_8\text{H}_{19}\text{NO}_3$	547.458.87	Hemicelluloses. Lichenin
547.448	Aldehyde and ketone compounds containing phosphorus, arsenic etc. <i>Specify compounds by use of apostrophe</i>	547.458.88	Pectin
547.449	Compounds containing a metal directly linked to carbon <i>Specify compounds by use of apostrophe</i>	547.459	Hydroxyketones and hydroxyaldehydes with nitroso, nitro, amino, hydrazino and other groups linked to carbon via nitrogen or some other element $547.459 \equiv 547.41$
547.45	Hydroxyaldehydes. Hydroxyketones	547.459.5	Glucosamines
547.451	Hydroxyaldehydes and hydroxyketones other than true sugars. □ Glycolaldehyde $\text{C}_2\text{H}_4\text{O}_2$. Aldols $\text{C}_4\text{H}_8\text{O}_2$	547.46	Polycarboxylic acids
547.452	Aldehydes and ketones, oxaldehydes and oxyketones containing sulphur groups	547.46'05	Polycarboxylic acids in which one or more carboxyl groups are replaced by nitrile groups or by acyl halide groups, or which are substituted in one or more carboxyl groups by ammonia, hydrazine etc.
547.452.2	Mercaptoaldehydes. Mercaptoketones.	547.46'051	IsoNitriles of polycarboxylic acids
547.452.3	Sulphides of aldehydes and ketones	547.46'052	Nitriles of polycarboxylic acids in general
547.452.4	Sulphonics, sulphinic and sulphenic acids of aldehydes and ketones. □ Acetaldehydisulphonic acid	547.46'053	Acyl halides of polycarboxylic acids in general
547.452.7	Aldehydes and ketones containing different sulphur groups. Sulphur containing oxaldehydes and oxyketones. Sulphonic acids of oxaldehydes and oxyketones	547.46'054	Polycarboxylic acids which are substituted in one or more carboxyl groups by ammonia, hydrazine etc.
547.454	Carbohydrates in general	547.46'054.4	Amides
547.455	Simple sugars (monosaccharides) <i>Denote sugar derivatives by apostrophe combination</i>	547.46'054.5	Imides. Imidic acids. Imidoethers. Imidoaldehydes
547.455.3	Trioses. Glyceraldehyde $\text{C}_3\text{H}_6\text{O}_3$	547.46'054.6	Amidines
547.455.4	Tetroses $\text{C}_4\text{H}_8\text{O}_4$. □ Erythrose. Threose	547.46'054.7	Hydroxylamine derivatives. □ Hydroxamic acids. Hydroxamic acids. Amidoximes
547.455.5	Pentoses $\text{C}_5\text{H}_{10}\text{O}_5$	547.46'054.8	Hydrazine derivatives. □ Hydrazidamines
547.455.522	Ribose	547.46'054.9	Derivatives of hydrazoic acids and other derivatives which are substituted by nitrogen in the carboxyl group
547.455.524	Arabinose	547.46'055	Derivatives of polycarboxylic acids containing two or more of the various groups given in subdivisions 547.46'052/054
547.455.526	Xylose	547.461	Saturated dicarboxylic acids in general
547.455.528	Lyxose	547.461.2	Oxalic acid. Oxalates. Oxamide
547.455.6	Hexoses $\text{C}_6\text{H}_{12}\text{O}_6$	547.461.3	1,3-Dicarboxylic acids. Malonic acid and its salts. Malonic diethyl ester
547.455.62	Aldohexoses with six carbon atoms	547.461.4	1,4-Dicarboxylic acids. Succinic acid and its derivatives $\text{C}_4\text{H}_4\text{O}_2$
547.455.623	Glucose	547.461.5	Glutaric acid and its homologues
547.455.624	Mannose	547.461.6	Adipic acid and its homologues
547.455.63	Ketohexoses with six carbon atoms	547.461.7	Pimelic acid and its homologues
547.455.633	Fructose	547.461.8	Dicarboxylic acids with carboxyl groups separated by more than five carbon atoms. □ Suberic acid. Azelaic acid. Sebatic acid
547.455.635	Sorbose	547.462	Unsaturated dicarboxylic acids
547.455.65	Inulin	547.462.3	Fumaric acid. Itaconic acid
547.455.7	Heptoses	547.463	Polycarboxylic acids
547.455.8	Octoses	547.463.1	Saturated tricarboxylic acids and derivatives
547.455.9	Higher monosaccharides	547.463.2	Unsaturated tricarboxylic acids
547.458	Polysaccharides	547.463.3	Saturated tetracarboxylic acids in general
547.458.1	Polysaccharide sugars in general	547.463.7	Unsaturated tetracarboxylic acids
547.458.2	Disaccharides. □ Sucrose (Saccharose. Cane sugar). Maltose. Trehalose	547.463.8	Pentacarboxylic acids. □ Of propane, propene, butane, butene
547.458.3	Trisaccharides. □ Dextrose	547.463.9	Hexacarboxylic acids. □ Of ethane, propane, pentane, hexane
547.458.41	Tetrasaccharides	547.464	Halogeno derivatives of carboxylic acids. □ Chloroformic acid. Chloroacetic acid
547.458.5	Non-sugar polysaccharides	547.465	Nitro and nitroso derivatives of carboxylic acids
547.458.6	Starch and other reserve substances	547.466	Amino derivatives of carboxylic acids. Amino acids
547.458.61	Starch	547.466.1	Amino acid condensations and linkages in general. Betaines. Lactams. Peptide links
547.458.63	Glycogen	547.466.2	α-Monoamino derivatives of saturated monocarboxylic acids. α-Amino fatty acids
547.458.68	Dextrin	547.466.22	Glycocoli. Glycine and its derivatives
547.458.7	Natural gums and mucilages	547.466.23	Alanine
547.458.8	Cellulose. Other carbohydrates of cell membranes	547.466.24	α-Aminobutyric acid
547.458.81	Cellulose. Hydrated celluloses	547.466.25	α-Aminovaleric acid. Valine
547.458.82	Cellulose esters. Nitrocelluloses. Acetylcelluloses. Viscose. Viscoid	547.466.26	α-Aminocaproic acid. Leucine. Isoleucine
547.458.83	Amyloids		
547.458.84	Lignocellulose. Other crude natural celluloses		

547.466.27	Higher α -amino acids	547.491.4	Mustard oil. Sinapic acid. Allyl thiocyanate
547.466.3	α -, γ -, δ -Monoamino fatty acids. Other monoamino fatty acids. α -Aminopropionic acid. γ -Aminobutyric acid. δ -Aminovaleric acid	547.491.6	Cyanamide
547.466.4	Diamino derivatives of fatty acids. Ornithine	547.491.7	Cyanazide
547.466.5	Amino derivatives of unsaturated monobasic acids. β -aminocrotonic acid	547.492	Carbonic acid derivatives in general
547.466.6	Amino derivatives of saturated dicarboxylic acids	547.493	Carbonic acid esters
547.466.63	Aspartic acids. Asparagine	547.494	Sulphur derivatives of carbonic acid
547.466.64	Glutamic acid. Glutamine	547.494.3	Trithiocarbonic acid
547.466.7	Amino derivatives of unsaturated dicarboxylic acids. Aminomaleic acid. Aminofumaric acid	547.495	Ammonia derivatives of carbonic acid
547.467	Hydrazino, azo, diazo, azido and hydroxyamino derivatives of carboxylic acids	547.495.1	Carbamic acid and its derivatives. Aminoformic acid. Urethanes
547.47	Hydroxy acids. Sulphonic, sulphinic and sulphenic acids of the carboxylic acids	547.495.2	Urea and its derivatives. Carbamide
547.471	Hydroxy derivatives of monobasic acids, in general	547.495.5	Biuret
547.472	α -Hydroxy fatty acids	547.495.6	Ureides
547.472.2	Glycolic acid	547.495.9	Guanidine and its derivatives. Nitroguanidine. Creatine. Creatinine. Arginine
547.472.3	Lactic acid. Lactates	547.496	Ammonia derivatives of thiocarbonic acids
547.473	Other monohydroxy acids. Lactones	547.496.3	Thiourea and its derivatives. Thiocarbamide
547.474	Polyhydroxy derivatives of fatty acids. Hydroxylactones	547.497	Hydrazine derivatives and di-imido derivatives of carbonic acid and thiocarbonic acids.
547.474.3	Glyceric acid. Glycidic acid	547.497.6	Semicarbazide
547.474.6	Gluconic acid. Mannonic acid	547.498	Hydroxylamine derivatives.
547.475	Hydroxy derivatives of unsaturated monobasic acids		□ Carbohydroxamic acid. Carbohydroximic acid
547.475.1	Hydroxy acids with one double bond		Carbonic acid derivatives of hydrazoic acid, of triazene, of triazane, of tetrazene.
547.475.124	Ricinoleic acid		□ Methylcarbonic azide. Carbamic azide.
547.475.2	Ascorbic acid		Carbamide imide azide. Carbazide. Aliphatic diazoamino compounds. Triazenedicarboxylic acid amidinic nitrile. Triazenedicarboxylic acid amidine-amide.
547.475.4	Hydroxy acids with one triple bond. α -Hydroxytetrollic acid. Ricinostearolic acid	547.5	Guanylnitrosoaminoguanilyltetrazene
547.475.5	Hydroxy acids with two or more double bonds	547.51	Isocyclic compounds in general.
547.476	Hydroxy derivatives of dicarboxylic acids	547.512	Mononuclear compounds in particular
547.476.2	Monohydroxysuccinic acids. Malic acids. Malates	547.513	Allicyclic compounds in general
547.476.3	Tartaric acids. Racemic acid	547.514	Cyclopropane group
547.476.6	Saccharic acids	547.517	Cyclobutane group
547.476.7	Derivatives of pimelic acid, suberic acid, sebacic acid	547.518	Cyclopentane group
547.477	Hydroxy derivatives of polycarboxylic acids. Citric acid		Heptamethylene or cycloheptane group.
547.478	Hydroxycarboxylic acids containing halogens, nitrogen or other nonmetals directly linked to carbon		□ Suberone. Unsaturated derivatives
547.478.9	Carboxylic acids containing sulphur, but not in the carboxyl group	547.52/59	Isocyclic compounds with one ring containing more than seven carbon atoms
547.48	Aldehydoacids. Ketoacids	547.52	Mononuclear compounds in general
547.481	Aldehydoacids in general. Glyoxylic acid	547.52'02/04	Aromatic compounds in general. Benzene nucleus in general
547.482	Aldehydodicarboxylic acids	547.52'021	Relative positions of the substituents
547.483	Aldehydohydroxy acids. Glucuronic acid	547.52'022	<i>Applicable throughout 547.52/59</i>
547.484	Ketoacids	547.52'023	Ortho. 1,2-Dialkylarenes
547.484.2	Saturated α -ketoacids. Pyruvic acid	547.52'031	Meta. 1,3-Dialkylarenes
547.484.3	Saturated β -ketoacids. Acetoacetic acid	547.52'032	Para. 1,4-Dialkylarenes
547.484.4	Saturated α -ketoacids. Laevulinic acid	547.52'033	Vicinal. 1,2,3-Trialkylarenes
547.485	Dicarboxylic ketoacids	547.52'041	Asymmetric. 1,2,4-Trialkylarenes
547.485.2	Oxalacetic acid	547.52'042	Symmetric. 1,3,5-Trialkylarenes
547.485.6	Ketoglutaric acids	547.52'043	1,2,3,4-Tetra-alkylarenes
547.486	Polycarboxylic ketoacids	547.521	1,2,3,5-Tetra-alkylarenes
547.487	Ketenic acids	547.523	1,2,4,5-Tetra-alkylarenes
547.488	Derivatives of aldehydoacids and ketoacids containing halogens, nitrogen or sulphur directly linked to carbon	547.524	Aromatic character. Constitution of benzene
547.49	Derivatives of cyanogen, of carbonic acid	547.525	Isomerism of the substitution products of benzene
547.491	Cyanogen compounds	547.526	Determination of position
	$\Rightarrow$ 546.26	547.53	Entry of substituents
		547.533	Genetic relationships between the benzene derivatives and compounds of other classes
			Benzene and its homologues.
			Hydrocarbons containing a benzene nucleus, in general
			<i>Denote aryl groups by .024</i>
			Benzene C_6H_6
			Toluene C_7H_8

547.534	C ₈ H ₁₀ hydrocarbons	547.55	Non-oxygenated derivatives of benzene hydrocarbons
547.534.1	Ethylbenzene	547.551	Aniline and its derivatives
547.534.2	Dimethylbenzenes. Xylenes	547.551.1	Aniline
547.535	C ₉ H ₁₂ hydrocarbons	547.551.2	Alkylanilines. Diphenylamine. Triphenylamine
547.535.1	Propylbenzene	547.551.3	Condensation products of aniline with aldehydes
547.535.2	Ethylmethylbenzenes	547.551.4	Acid derivatives of aniline
547.535.3	Trimethylbenzenes. Mesitylene	547.551.5	Substitution products of aniline obtained by the introduction of a halogen, nitroso, nitro or sulphonie group into the benzene nucleus, and their alkyl and acid derivatives
547.536	C ₁₀ H ₁₄ hydrocarbons	547.551.52	Aniline sulphonie acids. Sulphanilamides. Sulphonamides
547.536.1	Butylbenzene	547.551.55	Sulphinic acids. Sulphur oxides. Sulphones. Thiosulphonie acids
547.536.21	Methylpropylbenzenes. Cymenes	⇒ 547.542	
547.536.22	Diethylbenzenes	⇒ 547.543	
547.536.3	Ethylidimethylbenzenes	547.552	Homologues of aniline
547.536.4	Tetramethylbenzenes	547.552.1	Toluidine
	<i>Further particulars may be denoted by special auxiliaries</i>	547.552.2	Xylidine
	◇ 547.536.4'043 Durene	547.552.3	Pseudocumidine
547.537	Saturated hydrocarbons with more than ten carbon atoms. Benzene derivatives	547.553	Polyamines with amino groups attached to the nucleus
547.538	Unsaturated benzene hydrocarbons.	547.554	Aromatic amines with amino groups in aliphatic linkage
	□ Styrene C ₈ H ₈ . Phenylacetylene. Phenylallene	547.555	547.554.3/8 ≡ 547.533/538
547.539	Halogeno derivatives of benzene hydrocarbons	547.556	Hydroxylamine derivatives of aromatic hydrocarbons. Phenylhydroxylamine
547.539.1	Fluoro derivatives	547.556.3	Azoxy, azo and hydrazo compounds in general
547.539.2	Chloro derivatives	547.556.31	Azo compounds. Azoxy compounds
547.539.211.2	Chlorobenzene	547.556.33	Azobenzene
547.539.211.3	Benzyl chloride	547.556.35	Aminoazo compounds. Azo dyes. Aniline yellow. Helianthin. Chrysoidines
547.539.222.3	Benzylidene chloride	⇒ 667.28	
547.539.232.3	Benzotrichloride	547.556.35	Azoxy compounds. □ Azoxybenzene. Azoxytoluenes
547.539.251.2	Pentachlorobenzene	547.556.4	Hydrazo compounds
547.539.261	Hexachlorobenzene	547.556.5	Diazo compounds, hydrazines and their derivatives in general
547.539.3	Bromo derivatives. Bromobenzene	547.556.7	Diazo compounds
547.539.4	Iodo derivatives. Iodobenzene	547.556.8	Hydrazines. □ Phenylhydrazine
		547.556.9	Aliphatic-aromatic azo and hydrazo compounds. Hydrazone and formacyl compounds. Osazones
547.54	Sulphuric acid derivatives. Sulphurous acid derivatives of benzene hydrocarbons. Thiosulphuric acid derivatives of benzene hydrocarbons. Derivatives of nitric acid, of nitrous acid	547.557	Compounds whose molecule contains more than two nitrogen atoms directly linked
547.541	Benzenesulphonic acid. Benzyisulphonic acid and its homologues. Arysulphonic acid	547.557.1	Diazoamino compounds. Diazoaminobenzene
547.541.1	Monosulphonic acids. Benzenesulphonic acid. Benzyisulphonic acid	547.56	Phenols. Aromatic alcohols
547.541.2	Disulphonic acids	547.561	Generalities on monohydric phenols
547.541.3	Arysulphonic acids with more than two -SO ₃ H groups	547.562	Phenol (carbolic acid). Phenol ethers. Phenol esters. Phenetole
547.541.5	Arysulphonic acids substituted in all -SO ₃ H groups, except metal sulphonates and sulphonic acid esters	547.562.1	Phenol
547.541.51	Sulphohalogenides	547.562.3	Halogeno derivatives
547.541.52	Sulphonamides and their derivatives substituted in the -NH ₂ group	547.562.4	Functional derivatives of phenol
547.541.521	Sulphonamides SO ₂ (NH ₂) ₂		<i>Further particulars may be denoted by apostrophe combination</i>
547.541.522	Sulphohalogenamides	◇ 547.562.4'133	Sodium phenoxide
547.541.6	Arysulphonic acids containing nonsubstituted -SO ₃ H groups adjacent to substituted -SO ₃ H groups, except metal sulphonates and sulphonic acid esters	◇ 547.562.4'261	Anisole
547.542	Sulphinic acids. Sulphenic acids	◇ 547.562.4'562.1	Diphenyl ether
547.543	Thiosulphonic acids	547.563	Cresols. Other homologues of phenol with saturated sidechains. Thymol
547.544	Sulphoxides. Sulphones	547.564	Other substitution products of monohydric phenols
547.545	Nitro derivatives of benzene hydrocarbons. Nitroso compounds	547.564.1	Hydroxybenzenesulphonic acids.
547.546	Nitro compounds with -NO ₂ groups in the nucleus. □ Nitrobenzene. Dinitrobenzene. Nitrotoluenes	547.564.2	Phenolsulphonic acid
547.547	Nitro groups with -NO ₂ groups in the sidechain. □ Phenylnitromethane. Phenylnitroethane	547.564.3	Nitrosophenols
547.548	Halogenated nitro compounds. □ Chloronitrobenzenes	547.564.4	Nitrophenols. Picric acid. Dinitrocresols
547.549	Nitrosulphonic acids. Halogenosulphonic acids		Aminophenols. Anisidine. Phenetidine. Phenacetine
547.549.1	Nitrosulphonic acids		
547.549.2	Halogenosulphonic acids		

547.564.6	Oxyazo compounds. Sudan dyes. Azo derivatives of phenol. Aliphatic azo compounds	547.585	Acids with three or more carboxyl groups in the benzene nucleus. □ Benzenetricarboxylic acids. Benzenetetracarboxylic acids. Benzenepentacarboxylic acids. Mellitic acid
547.564.7	Hydrazo derivatives of phenol. Aliphatic hydrazo compounds	547.586.1	Acids with one or more of the carboxyl groups in saturated sidechains, in general
547.564.8	Diazo derivatives of phenol	547.586.2	Monobasic phenylaliphatic acids. Phenylacetic acid. Phenylalanine
547.564.9	Phenylhydrazines	547.586.3	Polybasic phenylaliphatic acids
547.565	Polyhydric phenols	547.586.4	Acids with one or more of the carboxyl groups in unsaturated sidechains, in general
547.565.2	Dihydric phenols. □ Pyrocatechol (o-dihydroxybenzene). Guaiacol. Veratrol. Cresol. Resorcinol (1,3-dihydroxybenzene). Styphnic acid. Orcinol. Hydroquinone (1,4-dihydroxybenzene)	547.586.5	Monobasic acids with one double bond. Atropic acid. Cinnamic acid
547.565.3	Trihydric phenols. Pyrogallol. Phloroglucinol	547.586.6	Monobasic acids with triple bonds
547.565.4	Tetrahydric phenols	547.586.7	Dibasic acids
547.565.6	Hexahydric phenols. □ Hexahydroxybenzene	547.587	Phenol-carboxylic acids
547.566	Phenols with unsaturated sidechains	547.587.1	Monohydroxy derivatives of monobasic nuclear carboxylic acids. □ Salicylic acid. Salol. Acetylsalicylic acid
547.566.1	Anethole	547.587.2	Polyhydroxy derivatives of monobasic nuclear carboxylic acids. □ Vanillic acid. Gyrophoric acid. Gallic acid. Tannin
547.566.2	Eugenol. Isoeugenol. Safrole. Isosafrole	547.587.3	Hydroxy derivatives of polybasic nuclear carboxylic acids
547.566.3	Myristicin	547.587.4	Hydroxy derivatives of carboxylic acids with carboxyl groups in saturated sidechains. □ Hydrocoumaric acids. Melilotic acid. Tyrosine
547.567	Quinones. □ Quinhydrone. Thymoquinone	547.587.5	Hydroxy derivatives of carboxylic acids with carboxyl groups in unsaturated sidechains
547.568	Aromatic alcohols. Phenol alcohols	547.587.51	Coumarin. Coumarinic acid. Coumaric acid
547.568.1	Aromatic alcohols. □ Benzyl alcohol. Cinnamyl alcohol. Methylphenyl carbinol. Phenylethylene glycol	547.587.52	Dihydroxycinnamic acids. Caffeic acid. Umbelliferone
547.568.5	Phenol alcohols. □ Oxybenzylalcohol	547.587.53	Trihydroxycinnamic acids. Sinapinic acid. Daphnetin. Aesculetin. Chrysotropic acid
547.569	Aromatic derivatives of hydrogen sulphide. □ Monothiophenols. Mercaptans	547.587.54	Piperic acids. Hydropiperic acids
547.57	Aromatic carbonyl compounds and their derivatives	547.587.6	Sulphur containing compounds of phenol-carboxylic acids
547.571	Aromatic aldehydes. □ Benzaldehyde. Cuminaldehyde. Terephthalaldehyde. Cinnamaldehyde	547.588	Alcohol acids. Phenol-alcohol acids
547.572	Aromatic ketones. □ Acetophenone	547.588.1	Alcohol acids with hydroxyl and carboxyl groups in the same sidechain. □ Mandelic acid. Tropic acid
547.573	Aromatic thioaldehydes. □ Aliphatic-aromatic thioketones. Thiobenzaldehyde. Thioacetophenone	547.588.2	Alcohol acids with hydroxyl and carboxyl groups in different sidechains
547.574	Condensation products of aromatic aldehydes. □ Oximes. Hydrazones	547.588.3	Phenol-alcohol acids. Meconine
547.575	Compounds containing both carbonyl and hydroxyl groups, in general	547.588.6	Sulphur containing compounds of alcohol acids and alcohol phenol acids
547.576	Hydroxyaldehydes. □ Salicylaldehyde. Anisaldehyde. Vanillin. Piperonal	547.589	Aldehydoacids. Ketoacids
547.577	Hydroxyketones. □ Hydroxyacetophenone. Dihydroxyacetophenone. Acetopyrocatechol	547.59	Hydrogenated mononuclear benzene hydrocarbons and their derivatives. Mononuclear hydro-aromatic compounds
547.578	Aldehydoalcohols. Ketoalcohols. □ Phenyl-lactic aldehyde. Phenyl tetrose	547.591	Generalities. Relationship to aromatic compounds. Partially hydrogenated aromatic compounds
547.579	Sulphur containing analogues	547.592/595	Derivatives of hydrogenated benzene hydrocarbons other than hydrogenated cymenes
547.58	Mononuclear aromatic acids. Aromatic acids in general	547.592	Hydrocarbons and their halogens, nitro and amino derivatives. Cyclohexane
547.581	Monobasic acids with carboxyl group directly attached to the ring	547.592.1	Cyclohexane, its halogeno derivatives, sulphur and acid derivatives and nitrogen derivatives etc.
547.581.2	Benzoic acid	547.592.12	Cyclohexane
547.582	Acid derivatives with substituents in the carboxyl group. □ Benzoyl chloride. Anhydrides. Esters	547.592.13	Halogeno derivatives
547.583	Derivatives formed by substitution in the benzene nucleus	547.592.14	Sulphurous acid derivatives. Analogous selenium or tellurium derivatives
547.583.1	Halogeno derivatives. □ Chlorbenzoic acids	547.592.15	Nitro derivatives. Amino derivatives
547.583.2	Sulphobenzoic acid. □ Saccharin	547.593	Hydrogenated benzene derivatives containing hydroxyl groups. Cyclohexanol. Naphthenol
547.583.4	Nitroso derivatives. Nitro derivatives	547.593.2	Hydroxy derivatives of cyclohexane. Homologues. Functional derivatives. Halogeno derivatives. □ Cyclohexanol. Naphthenol
547.583.5	Amino derivatives. □ Anthranilic acid		
547.583.6	Azoxy azo and hydrazo derivatives. □ Diazobenzoic acids, hydrazobenzoic acid, diazoaminobenzoic acid		
547.583.7	Azido derivatives		
547.584	Dicarboxylic acids with carboxyl groups linked to the nucleus. □ Phthalic acid. Phthalic anhydride. Isophthalic acid. Terephthalic acid. Phthalimide		

547.593.3	Hydroxy derivatives of cyclohexene. Cyclohexenol. Homologues, functional derivatives and halogeno derivatives		
547.593.4	Hydroxy derivatives of cyclohexadienes. Homologues, functional derivatives and halogeno derivatives		
547.594	Aldehydes. Ketones. Cyclohexanone	547.596.7	menthenols (isopulegol, terpineol, pinol hydrates, pinol, diosphenol), aminomenthenes (dihydrocarvylamine, vestrylamine). Unsaturated hydrocarbons (menthene). Carvomenthene. Ascaridole
547.594.1	Aldehydes. □ Cyclocitral, trimethyltetrahydrobenzaldehyde, dihydrobenzaldehyde		Doubly unsaturated derivatives of methane. Doubly unsaturated hydrocarbons. □ Menthadiene, menthadienone, carvone, carvone oxime, hydroxyaminocarvoxime, menthadienols, carveol, aminomenthadiene, carvylamine. Dihydrocymene, terpinenes, dipentene, limonene, phellandrene, carvestrene, sylvestrene
547.594.3	Saturated ketones. □ Cyclohexanone and its homologues, methylhexahydroacetophenone, hexahydropropionophenone, cyclohexanedione, methylcyclohexanedione, acetylmethylcyclohexanone, cyclohexanetrione, tetrachlorotetraketohexamethylene	547.597	Polycyclic terpenic compounds in general. Carane group
547.594.4	Unsaturated ketones. □ Cyclohexenone, isoacetophorone, acetylmethyltetrahydrobenzene, irone, ionone, heptachlorocyclohexenone, octachlorocyclohexenone, cyclohexenedione	547.598	Pinane group
547.594.5	Quinols (unsaturated ketoalcohols). □ Dimethylquinol, tribromomethylquinol, pseudophenols	547.599	Camphane group
547.595	Hydro-aromatic carboxylic acids	547.599.1	Saturated hydrocarbons and transformation products. Camphane. Isohydrocamphene
547.595'05	Derivatives of hydro-aromatic carboxylic acids according to carboxyl group 547.595'05 ≡ 547.46'05	547.599.2	Unsaturated hydrocarbons. □ Camphene, bornylene, fenchene and their transposition products. Camphenylene. Apocamphoric acid. Hydroxyfenchylic acid
547.595.2	Monocarboxylic acids of cyclohexane. Cyclohexanecarboxylic acid (hexahydrobenzoic acid). Halogen and amino derivatives. Hexahydrosalicylic acid. Hydroshikimic acid. Quinic acid. Dihydroxyhydroshikimic acid. Oxoheptanecarboxylic acid. Dioxocyclohexanecarboxylic acid	547.599.3	Amines. □ Bornylamine, neobornylamine, camphylamine, campholamine, fenchylamine, fencholenamine
547.595.3	Monocarboxylic acids of cyclohexene. Cyclohexenecarboxylic acids (tetrahydrobenzoic acids). Cyclogeranic acid. Shikimic acid. Sedanolic acid. Sedanonic acid. Oxocyclohexanecarboxylic acid	547.599.4	Alcohols. □ Borneol, isoborneol, fenchyl alcohol, isofenchyl alcohol, camphene glycol
547.595.4	Monocarboxylic acids of cyclohexadiene. Cyclohexadienecarboxylic acid (dihydrobenzoic acid)	547.599.6	Saturated ketones. Camphor and its derivatives. □ Camphoroxime
547.595.6	Dicarboxylic acids of cyclohexane. □ Hexahydrophthalic acid, hydroxyhexahydroisophthalic acid, dihydroxyhexahydroisophthalic acid, oxocyclohexanedicarboxylic acid, 2,3-succinoylsuccinic acid	547.599.7	Unsaturated ketones. □ Camphenone
547.595.7	Dicarboxylic acids of cyclohexene	547.599.8	Derived acids
547.595.8	Dicarboxylic acids of cyclohexadiene. □ Dihydroterephthalic acid	547.6	Polynuclear benzene derivatives and their hydrogenated derivatives. Polycyclic compounds in general
547.595.9	Hydro-aromatic polycarboxylic acids. □ Cyclohexanetricarboxylic acid, methylcyclohexadienetricarboxylic acid, dioxocyclohexanetetracarboxylic acid, isohydromellitic acid	547.61	Generalities on the isocyclic compounds with several nuclei
547.596	Derivatives of hydrogenated cymenes. Chemistry of the monocyclic terpenes	547.62	Polynuclear systems with rings directly linked
547.596.2	Saturated derivatives of menthane. □ Menthane, menthone, carvomenthone, menthadiene, menthanols (menthol, carvomenthol), terpin, terpin hydrate, cineol, menthanetriols, limonetrine, aminomenthanes, menthylamine, carvomenthylamine, pulegonamine, menthylhydrazine. Saturated hydrocarbons (menthanes)	547.621	Biphenyl series in general
547.596.4	Unsaturated simple derivatives of methane. □ Menthene, pulegone, isopulegone, dihydrocarvone, carvenone, carvotanacetone,	547.622	Biphenyl and its homologues
		547.623	Phenols and quinones of the biphenyl series
		547.624	Acids of the biphenyl series. □ Phenylbenzoic acid. Biphenylcarboxylic acid
		547.625	Amino and azo compounds of the biphenyl series. □ Congo dyes. Benzidine. Tolidine. Chrysamine
		547.626	Hydrogenated biphenyl derivatives. □ Tetrahydrobiphenyl
		547.628	Terphenyl. Triphenylbenzene. Tetraphenylbenzene
		547.629	Phenylcyclopropane compounds. Phenylcyclobutane compounds. Phenylcyclopentane compounds
		547.63	Polynuclear indirectly linked systems
		⇒ 547.551.2	
		547.631	Diphenylmethane group. Phenylfulvene group
		547.631.1	Diphenylmethane group in general Further particulars may be denoted by special auxiliaries
		547.631.2	◇ 547.631.1.05 Synthesis. Zincke's method. Friedel-Crafts reaction. Baeyer synthesis
			Compound of two benzene rings bound by a methylene group or substituted methylene group. □ Diphenylmethane. Diphenyl-2,2-propane

547.631.3	Nitro and amino derivatives of diphenylmethane and its homologues	547.638.2	Triphenylpropane group. □ 1,1,3- or 1,2,3-triphenylpropane, benzylidenesoxymethine, symmetrical triphenylglutaric acid
547.631.4	Phenols and alcohols	547.638.3	Tetraphenylpropane group.
547.631.6	Diphenylacetaldehyde. Benzophenone and its derivatives. Dyestuffs of the diphenylmethane series. Auramine. Pyronine. Maclurin	547.638.4	Hexaphenylpropane group
547.631.7	Carboxylic and hydroxycarboxylic acids. □ Diphenylacetic acid, benzhydrylcarboxylic acids, phenylphthalide	547.638.5	1,4-Diphenylbutane group. □ 1,2-Dibenzylethane, diphenyldiacetylene, diphenacyl-2,3-dibenzoylsuccinic acid, pulvic acid, vulpic acid
547.631.8	Hydrogenated derivatives of diphenylmethane. □ Hexahydrobenzophenone, methylenedihydroresorcinol	547.639	Triphenylbutane group. Tetraphenylbutane group. □ Dibenzoylstyrene, dibenzoylstilbene, desoxybenzoinpinacone, benzoinpinacone, dibenzylacetophenone, tribenzoylmethane
547.631.9	Phenylfulvene group		Compounds with benzene nuclei linked by aliphatic chains or by cyclic systems <i>For systems with several nuclei and directly linked rings use 547.62.</i> <i>For those that are linked by aliphatic chains use 547.631/.638</i>
547.632	Triphenylmethane group	547.64	Condensed cyclic systems in general.
547.632.2	Triphenylmethane		Spiranes
547.632.3	Homologues of triphenylmethane and of triphenylmethanol, and their halogeno, nitro and sulphonic acid derivatives ⇒ 547.633.4	547.642	Spiranes
547.632.4	Generalities on the amino derivatives of the triphenylmethane series	547.65	Binuclear systems in general
547.632.5	Malachite green group	547.651	Generalities on naphthalene and its derivatives. Isomerism of the compounds of naphthalene. Isomerism of the substitution products of naphthalene. Genetic relationships between naphthalene derivatives and compounds of other classes
547.632.6	Rosaniline group. Methyl violet. Fuchsine. Aniline blue	547.652	Naphthalene, its homologues and halogeno derivatives
547.633	Oxygenated and hydrogenated derivatives of triphenylmethane	547.653	Sulphonic and nitro derivatives etc. □ Naphthalenesulphonic acids
547.633.4	Hydroxy derivatives. Phenols and alcohols of the triphenylmethane series. □ Triphenylmethanol. Hydroxytetramethyldiaminotriphenylmethane, dihydroxytriphenylmethanol (benzaurine), dihydrotriphenylmethane, tetramethylrosamine, leucarine, aurine, leucorosolic acid, rosolic acid, eupittonic acid	547.654	Non-oxygenated naphthalene derivatives. □ Naphthylamine
547.633.5	Aldehydes and ketones of the triphenylmethane series	547.655	Hydroxy derivatives of naphthalene. □ Naphthaquinones. Naphthazarin
547.633.6	Acids of the triphenylmethane series. □ Triphenylmethanecarboxylic acid, triphenylmethanolcarboxylic acids, diphenylphthalide, phthalophenone, dimethylanilinophthalide, phenolphthalein, fluorane, fluorescein, eosin, gallein, rhodamine	547.656	Aldehydes and ketones of the naphthalene series
547.633.8	Hydrogenated derivatives of triphenylmethane	547.657	Acids of the naphthalene series. □ Naphthoic acid. Naphthalenecarboxylic acids
547.634	Tetraphenylmethane	547.659	Reduction products of naphthalene. □ Hydronaphthoic acids
547.636	Symmetrical diphenylethane group	547.66	Other systems with two condensed nuclei
547.636.2	1,2-Diphenylethane	547.661	Compounds with either one naphthalene nucleus and benzene nuclei, or several naphthalene nuclei
547.636.3	Hydrocarbons of the stilbene and tolane groups and their addition products	547.663	Systems with two nuclei consisting of a nucleus with six carbon atoms condensed with a nucleus having a different number of carbon atoms, in general
547.636.4	Amines of the diphenylethane series	547.664	Phenyltrimethylene and phenyltetramethylene group
547.636.5	Alcohols	547.665	Indene group. □ Indene. Indane (hydrindene)
547.636.6	Aldehydes. Ketones	547.667	Benzocycloheptane group
547.636.7	Acids. □ Phenylcinnamic acid, phenylhydrocinnamic acid, diphenylsuccinic acid, benzylidenephthalide, isobenzylidenephthalide, diphtalyl acid, diphtalylic acid, desylacetic acid	547.669	Other systems with two condensed nuclei
547.636.8	Hydrogenated derivatives of diphenylethane	547.67	Trinuclear systems
547.637	Triphenylethane group. Tetraphenylethane group. Pentaphenylethane group.	547.672	Anthracene and its derivatives. □ Anthrahydroquinone. Chrysarobin
547.638	Hexaphenylethane group	547.673	Anthraquinone and its derivatives. □ Alizarin. Purpurin
547.638.1	Compounds with two or more benzene nuclei linked by a chain of three or more carbon atoms or by other benzene nuclei	547.674	Compounds with benzene nuclei in addition to the anthracene nucleus. □ Phenylanthranol. Phenylanthracene. Phenolphthalidine
	1,3-Diphenylpropane group. □ 1,3-Dibenzoylacetone, dibenzoylmethane, debenzyl ketone, benzylideneacetophenone, benzylacetophenone, hydroxybenzylideneacetophenone, 1,3-dibenzoyl-2-hydroxypropylene, diphenylpropane-1,2,3-trione, dibenzylacetic acid, α,α' -diphenylglutaric acid, dibenzylglucolic acid	547.675	Phenylene group
		547.677	Phenanthrene and pyrenone group

547.678	Condensed trinuclear systems with benzene nuclei linked to nuclei with a different number of carbon atoms. Acenaphthene group. Fluorene group. Naphthindene group. Pentanthrene group. Indacene group	547.738	Five-membered rings containing several sulphur hetero-atoms or sulphur and another element except nitrogen
547.678.1	Diphenylene. Dibenzcyclobutadiene	547.739	Selenophen, tellurophen and their derivatives
547.678.2	Acenaphthene group. □ Acenaphthene, acenaphthylene, acenaphthenequinone	547.741.75	Pyrrole group
547.678.3	Fluorene group. □ Fluorenedibiphenylene-ethane, fluorenedibiphenylethylene, fluorenol, fluorenone, retene, acids of the fluorene group	547.74	Pyrroles in general
547.678.4	Naphthindene group	547.741	Pyrrole. Addition products of pyrrole after nitrogen
547.678.5	Pentanthrene group	547.742	Homologues of pyrrole. □ Methylated homologues, haemopyrrole
547.678.6	Indacene group	547.743	Non-oxygenated derivatives
547.678.7	Benzidine group	547.744	Hydroxy derivatives of pyrrole
547.68	Systems with more than three condensed nuclei	547.745	Carbonyl derivatives of pyrrole. Aldehydes. Oxycyclic ketones. Exocyclic ketones. Oxoheterocyclic derivatives. Pyrrolidone etc.
547.681	Pyrene group	547.746	Non-hydrogenated carboxylic acids of pyrrole. □ Pyrrolecarboxylic acids, pyrroledicarboxylic acids and their homologues, hydroxycarboxylic acids, ketocarboxylic acids
547.682	Naphthacene group. Naphthanthracene group	547.747	Hydrogenated carboxylic acids of pyrrole. □ Pyrrolinecarboxylic acids. Proline. Tropinic acid
547.683	Chrysene group	547.748	Multiring noncondensed systems in which the rings are directly linked to one another. □ Diperyl derivatives
547.684	Dinaphthanthracene group	547.749	Multiring noncondensed systems in which the rings are indirectly linked to one another
547.685	Picene group	547.75	Condensed systems with one pyrrole ring
547.686	Fluoranthene group	547.752	Indole and its nitrogen-containing derivatives
547.687	Di-indanoyl. Di-indanyl	547.753	Homologues of indole. □ 2-Methylindole, 3-methylindole (skatole), alkylindolines, alkylindolenines
547.688	Tris(trimethylene)benzene	547.754	Non-oxygenated derivatives. □ Hydrogen addition products of indole. Indoline. Halogenated derivatives of indole
547.689	Other groups	547.755	C-hydroxy derivatives of indole and indoline. □ Indoxyl, indolyethyl alcohol (tryptophol), trimethylindolinol
547.7	Heterocyclic compounds in general. Compounds with three-membered to five-membered rings in particular	547.756	Carbonyl derivatives of indole and of indoline. □ Indoxyl. Indolethylalcohol (tryptophol). Trimethylindolinol
547.71	Three-membered rings. Four-membered rings	547.757	Carboxylic acids of the indoles. □ Indolecarboxylic acid, indolylacetic acid, indolylpropionic acid, tryptophane, indoxyl acid
547.72	Five-membered rings	547.758	Indigo blue and associated derivatives of indole. □ Indigo white, indigo carmine
547.722	Furan, its homologues and non-oxygenated substitution products	547.759	Other condensed derivatives of pyrrole
547.723	Hydroxy derivatives of the furan group	547.76	Heterocyclic five-membered systems with one hetero-atom other than oxygen, sulphur or nitrogen. Compounds containing as hetero-atom iodine, nickel, boron etc.
547.724	Carbonyl derivatives of the furan group	547.77	Five-membered heterocyclic rings with two hetero-atoms, in general, except those with only oxygen or sulphur hetero-atoms.
547.725	Carboxylic acids of the furan group. □ Pyromucic acid, dehydromucic acid, pyrotritaric acid, carbopyrotritaric acid, methronic acid, furfurylacrylic acid	547.78	□ Pyrazoles. Pyrazoline. Pyrazolone. Antipyrine
547.727	Noncondensed systems of the furan type with several rings	547.781	Imidazoles. Isoxazoles. Oxazoles.
547.728	Condensed systems of the furan type with several rings	547.782	Isothiazoles. Thiazoles
547.729	Five-membered rings containing several oxygen hetero-atoms or oxygen and another element except nitrogen and sulphur	547.783	Imidazole (glyoxaline) group in general. □ Histamine
547.73	Five-membered rings containing sulphur as hetero-atom. Thiophene group	547.784	Hydroxy derivatives of imidazole
547.732	Compounds containing one thiophene ring. Thiophene group in general. □ Thiophene, its homologues, halogeno derivatives, sulphonic acid derivatives, nitro derivatives, amino derivatives and organometallic compounds, tetrahydrothiophene	547.785	Carbonyl derivatives of imidazole. Exocyclic derivatives. Glyoxalane. Ethylene-urea (glyoxalidone). Hydantoin. Allantoin.
547.733	Hydroxy derivatives. Keto derivatives. Carboxylic derivatives	547.786	Methyleneoxamide. Parabanic acid. Imides of keto-heterocyclic derivatives
547.734	Noncondensed systems of the thiophene type with several rings	547.787	Carboxylic acids of imidazoles. □ Histidine
547.735	Condensed systems of the thiophene type in general. □ Thionaphthens	547.788	Systems with several imidazole rings
547.736	Other condensed systems containing thiophene and benzene rings. □ Thiophthalan. Naphthiophen. Diphenylene sulphide	547.789	Isoxazole compounds
547.737	Condensed systems containing several thiophene rings. □ Thiophthen. Sulphetone		

547.787	Oxazole compounds	547.84	Heterocyclic compounds with six atoms in the ring, of which two are hetero-atoms other than nitrogen
547.788	Isothiazoles. Benzisothiazole		
547.789	Thiazole group. Other five-membered rings with two hetero-atoms	547.841	Dioxan and its derivatives 547.841.5/66 $\equiv$ 547.85/866
547.79	Five-membered heterocyclic systems with three or four hetero-atoms, except those with only oxygen or sulphur hetero-atoms. Triazoles. Methylene red. Tetrazoles	547.842	Dithian and its derivatives. Divinylene disulphide. Thianthrene 547.842.5/66 $\equiv$ 547.85/866
547.8	Heterocyclic compounds with six or more atoms in the ring	547.843	Thioxan (oxthian) compounds. 1,4-Thioxan. Phenoxthin. Naphthoxthin
547.81	Pyran and penthiophen group. Its analogues	547.85	Azines in general. Pyridazine and pyrimidine group
547.812	Pyrans	547.852	Pyridazines (1,2-diazines)
547.813	Systems with several noncondensed pyran rings	547.853	Pyrimidine compounds
547.814	Benzopyran compounds	547.854	Mononuclear pyrimidine derivatives containing oxygen. $\square$ Uracil. Thymine. Barbituric acid. Veronal. Thiouracil
547.815	Xanthen compounds		Uncondensed polynuclear pyrimidine systems
547.816	Other condensed systems of pyran. $\square$ Naphthopyrane. Dinaphthopyrane. Hydroxybinaphthylene. Hydroxyceroxene	547.855	Quinazoline and other condensed systems of pyrimidine rings and carbon rings
547.818	Compounds of penthiophen (thiopyran). $\square$ Thiocoumarine	547.856	Purine group
547.819	Heterocyclic compounds with a ring of six atoms including only one hetero-atom, the hetero-atom being other than oxygen, sulphur, selenium, tellurium or nitrogen	547.857	Purine and its methyl derivatives
547.82	Pyridine compounds in general. Uncondensed systems in particular	547.857.1	Halogeno derivatives. Other non-oxygenated derivatives
547.821	Pyridine. Pyridine compounds. $\square$ Picolines. Lutidines	547.857.2	Compounds with one oxygen atom linked to an esocyclic carbon atom. $\square$ Hypoxanthine
547.822	Non-oxygenated pyridine derivatives. $\square$ Piperidine	547.857.3	Compounds with two oxygen atoms linked to an esocyclic carbon atom. $\square$ Theophylline. Theobromine. Caffeine
547.823	Hydroxy derivatives of pyridine and of piperidine	547.857.4	Compounds with three oxygen atoms linked to an esocyclic carbon atom. $\square$ Uric acid
547.824	Carbonyl derivatives of pyridine and of hydrogenated pyridine. N-Alkyl pyridones. Pyridon. Hydroxypyromecazonic acid. Pyridoquinone. Pyromecazone. Piperidone. Triacetoneamine etc. Pyridine aldehyde. Pyridylalkyl ketones. Hydrogenated aldehydes and ketones of the pyridine series	547.857.5	Compounds with more than three oxygen atoms linked to esocyclic carbon atoms. $\square$ Alkylhydroxyisouric acid. Uric acid glycols
547.825	Thiolic and thionic derivatives of pyridine and of hydrogenated pyridines	547.857.6	Compounds which appear as imides of ketopurine compounds. $\square$ Adenine. Guanine
547.826	Non-hydrogenated carboxylic acids of the pyridine series	547.857.7	Carboxylic acids
547.827	Hydrogenated carboxylic acids of the pyridine series. Dihydrocarboxylic acids, tetrahydrocarboxylic acids, hexahydrocarboxylic acids. $\square$ Loiponic acid, scopolic acid, Eucaïne A, cincholoiponic acid, cincholoipine, meroquinene, granatic acid	547.857.8	C-aryl purine compounds
547.828	Systems in which pyridine rings are directly linked to other rings. $\square$ Bipyridyl. Bipiperidyl. Phenyl compounds of pyridine. Pyridyl pyrrole	547.86	Pyrazine, oxazine and thiazine compounds
547.829	Systems in which pyridine rings are indirectly linked to other rings. $\square$ Pyridoin. Pyridostilbene. Papaveric acid. Stilbazole	547.861	Pyrazine group in general. $\square$ Pyrazine
547.83	Condensed systems of the pyridine type	547.862	Systems with several uncondensed pyrazine nuclei. C-aryl pyrazine compounds. Nuclei of pyrazine and of benzene linked by intermediary groups. Nuclei of pyrazine and of unidazole linked by intermediary groups
547.831	Quinoline group. $\square$ Quinoline. Lepidine. Hydroxyquinolines	547.863	Condensed systems of pyrazine in general. Quinoxaline and other systems obtained from pyrazine by unilateral orthocondensation with carbon rings
547.832	Polynuclear quinoline derivatives. $\square$ Diquinolyl. Atophan	547.864	Phenazine group. $\square$ Induline. Mauveine. Safranil
547.833	isoQuinoline group	547.865	Other systems obtained from pyrazone by bilateral condensation with carbon rings
547.834	Other systems with two condensed pyridine nuclei. Tropine	547.866	Pyrazine in direct condensation with hetero rings
547.835	Acridine group	547.867	Oxazine compounds
547.836	Other condensed trinuclear systems	547.869	Thiazine, selenazine and analogous compounds. Thiodiphenylamine. Methylene blue
547.837	Systems with four condensed pyridine nuclei. $\square$ Anthraquinoline (alizarin blue). Quindoline. Apomorphine	547.87	Heterocyclic systems with six atoms in the ring, of which three are hetero-atoms
547.838	Systems with five or more condensed pyridine nuclei. $\square$ Indanthrene red. Flavanthrene	547.88	Heterocyclic systems with six atoms in the ring, having more than three hetero-atoms
		547.89	Heterocyclic systems with more than six atoms in the ring
		547.9	Natural substances. Substances of unknown composition $\Rightarrow$ 577.1 $\Rightarrow$ 615.2

547.91	Hydrocarbons. Oils. Fats. Waxes. Balsams. Gums. Carbohydrates. Glycosides ⇒ 665 ⇒ 678.4	547.962.6 547.962.7 547.962.9	Glutenins. Oryzenins Gliadins (prolamines). Zein Scleroproteins. Collagen. Keratin. Fibroin. Sericin. Spongin Compound proteins. Proteids Glycoproteins. Mucin. Mucoids Phosphoproteins. Casein. Vitellin. Ichthulin Nucleoproteins. Nucleic acids Nucleic acids Chromoproteins. Haemoglobin Protein derivatives and decomposition products Metaproteins Proteoses Peptones Polypeptides
547.912	Hydrocarbons of the carotene group. Petroleum hydrocarbons ⇒ 665.6	547.963 547.963.1 547.963.2 547.963.3 547.963.32 547.963.4 547.964 547.964.1 547.964.2 547.964.3 547.964.4	
547.913	Essential oils ⇒ 665.5		
547.913.2	Zingiberine. Selinene. Cadinene. Cedrene		
547.913.3	Farnesol. Nerolidol. Eudesmol		
547.913.4	Santanol. Eskantal and noreskantol group. Cedrenol		
547.913.5	Azulene		
547.913.6	Diterpenes		
547.913.7	Jasmone. Muscone. Civetone		
547.914	Plant juices. Balsams. Resins ⇒ 665.94	547.97 547.972	Pigments ⇒ 633.86 ⇒ 667 Natural xanthone, flavone and flavonol pigments ⇒ 547.81 Xanthone pigments. Gentisin Flavone pigments Flavonol pigments. Quercitrin. Quercetin. Xanthorhamnin. Rhamnetin. Rhamnazin Anthocyanins. Delphinidin Brazilin. Haematoxylin Cochineal ⇒ 638.3 Natural anthracene pigments. Purpurin Algal pigments. Litmus Berberine. Natural indigoid pigments. Indican. Isatan. Tyrian purple Other pigments Saffron Curcumin (turmeric) Chlorophylls Xanthophyll. Carotene (carotin)
547.915	Fats. Oils. □ Lipids		
547.916	Waxes		
547.917	Carbohydrates		
547.918	Glycosides. Amygdalin. Hesperidin. Saponin. Digitalin. Coniferin	547.972.1 547.972.2 547.972.3 547.973 547.974 547.975 547.976 547.977 547.978 547.979 547.979.3 547.979.4 547.979.7 547.979.8	
547.92	Steroids		
547.922	Cholesterol. Lanolin		
547.923	Unoxidized cholestanes. Cholestene. Neocholestene. Cholesterilene		
547.924	Oxidized cholestanes. Cholesterol. Cholestanone. Cholestenone. Cholestanecarboxylic acid		
547.925	Other zoosteroids		
547.926	Phytosteroids		
547.93	Bile substances. Bile acids. Bile pigments		
547.931	Glycocholic acid. Taurocholic acid		
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547.933	Cholanic acid		
547.937	Bile pigments. Bilirubin. Biliverdin. Bilifuchsin		
547.94	Natural alkaloids. Alkaloids of unknown composition ⇒ 547.8	547.98 547.99 547.991 547.992 547.993 547.994 547.995 547.996 548/549	Tannins Other substances Bitters. Aloin. Santonin. Picrotoxin Other vegetable substances. □ Humic acids. Lignin Animal toxins. Zootoxins ⇒ 591.145 Urine products. Urobilin ⇒ 591.149 ⇒ 612.461 Other nitrogenous substances of animal origin. □ Chitin. Heparin Other non-nitrogenous substances of animal origin
547.942	Cinchona bases		
547.943	Papaveraceae bases. Chelidonium		
547.944/945	Other bases of vegetable origin		
547.944.1	Strychnine bases		
547.944.2	Curare (urari). Curine		
547.944.3	Solanaceae bases		
547.944.5	Coca bases		
547.944.6	Liliaceae bases. Colchicine. Veratrine		
547.944.7	Aconite bases. Aconine. Aconitine		
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547.945.1	Ergot bases. Ergotamine. Ergotine		
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- 548.1.022/.024 Whorled crystals. Uniaxial crystals. Crystals with one optic axis
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- 548.1.022 Tetragonal system
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550.34.013	Theoretical investigation. □ Analysis. Models	550.348.098.2	Depth of focus
550.34.016	Experimental investigation. Measurements. Testing. Field experiments	550.348.098.3	Magnitude of source (energy)
550.34.03	Seismological programmes, systems, equipment	550.348.098.32	Macroseismic disturbances
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550.34.037	System auxiliary equipment. □ Telemetering equipment. Timing devices	550.348.098.5	Epicentral area or zone. Epicentre
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550.34.04	Network design and standards. □ Scales (Richter scale etc.)	550.348.098.62	Swarms. □ Earthquake clusters
550.34.042	Standards, scales and factors	550.348.098.64	Seismicity. Earthquake frequency. Seismic belts. Aseismic and seismic regions
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550.343	Prediction of natural phenomena	550.348.42	Man-made causes. □ Explosions
550.343.4	Seismic mapping. □ Isoseists (isoseisms). Homoseists (homoseisms, coseists)	550.348.43	Natural causes
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550.344.094.3	Particle and ground motion: surface, subsurface	550.348.436	Earthquakes. Seaquakes
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550.344.094.5	Seismic velocity	550.35	Terrestrial physics of rays and radiations ⇒ 551.521
550.344.094.6	Arrival and period	550.36	Terrestrial energetics and thermodynamics ⇒ 523.31 ⇒ 551.511
550.344.094.7	Wave processes. □ Acceleration. First motion. Emergence angle	550.360	Observation and measurement methods for the energy and thermodynamics of the Earth
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551.581.2	Climatic zones ⇒ (211/213)	551.589.6	Special phenomena. □ Ice Saints' Days
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551.581.23	Subtropical climate	551.590.21	Solar influences (except heat)
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◇ 551.582(083.4)	Statistical climatological tables	551.590.25	Phenomena attending meteors
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552.322	Hypabyssal calc-alkaline rocks. □ Granitic porphyries. Aplites. Pegmatites. Syenitic porphyries. Porphyritic gabbros	552.552	Silicified rocks
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552.47	Serpentine schists	552.62	Stony-iron meteorites. Lithosiderites. Pallasites
552.48	Eclogites. Amphibolites	552.63	Stone meteorites. Aerolites. Chondrites
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		553	ECONOMIC GEOLOGY. MINERAL DEPOSITS
			<i>Complex deposits may be denoted by adding '3'/9 at the number for the main component, with optional added entry 553'3'/9 ≡ 553.3/.9</i>
			◇ 553.411'435 Auriferous copper pyrites
			◇ 553.435'411 Auriferous copper pyrites [optional added entry]
			⇒ 549
			⇒ 622.3
		553.04	Mineral resources and reserves
		553.041	Potential resources. Promising areas in respect of various raw materials

553.042	Categories of reserve <i>Specify by adding appropriate designation, e.g. A, B, C1, C2, D1, D2</i>	553.25/26	Description and classification of mineral deposits according to various nongenetic considerations
553.043	Requirements	553.25	Primary and secondary minerals. Hypogene and supergene minerals
553.044	Probable reserves		
553.045	Commercial reserves	553.26	Distribution of ore minerals in deposits. Primary and secondary zoning. Erosion. Redeposition. Migration. Accumulation
553.046	Increase of reserves. Added reserves		
553.048	Computation, calculation of reserves		
553.06	Genetic classification of deposits	553.3/9	Ore and other mineral deposits. Natural resources
553.061	General problems		
553.061.2	Relation of mineralization to magmatic processes	553.3/4	Ore deposits. Metalliferous mineral deposits
553.061.3	Oil origin theory		
553.061.4	Enclosing rocks (country rocks). Wall rock	553.3	Ore deposits (metalliferous mineral deposits) in general. Iron and manganese ores
553.061.5	Syngenetic deposits: general description <i>For special cases see 553.062 and 553.068</i>		◇ 553.3.071 Native metal deposits
553.061.6	Epigenetic deposits: general description <i>For special cases see 553.064/.068</i>		◇ 553.3.072 Metal-bearing deposits, conglomerates, in general
553.061.7	Cap rocks (screen rocks)		◇ 553.3'3/9 Complex, polymetallic deposits.
553.062/.067	Endogenetic deposits: magmatic and postmagmatic		NOTE: See note and examples at 553 ⇒ 622.34
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553.063	Liquation-pneumatolytic deposits. Pegmatites	553.311	Magnetite-haematite deposits
553.064/.065	Postmagmatic deposits	553.311.2	Magnetite deposits
553.064	Pneumatolytic deposits	553.312	Haematite-limonite deposits
553.065	Hydrothermal deposits. □ Hypothermal, mesothermal, epithermal deposits	553.313	Chamosite deposits
553.067	Emanation deposits. Igneous rock deposits	553.314	Brown iron-ore deposits
553.068	Exogenetic deposits	553.315	Siderite deposits
553.068.1	Infiltration deposits	553.32	Manganese deposits
553.068.2	Sedimentary deposits	553.323	Manganese oxide, hydroxide, carbonite deposits. □ Pyrolusite. Psilomelane
553.068.3	Erosion deposits		
553.068.34	Residual deposits	553.4	Ore deposits other than iron and manganese
553.068.36	Weathering-crust deposits		
553.068.5	Placer and detrital deposits	553.41	Gold deposits. Silver deposits
553.068.7	Metamorphism and metamorphic deposits	553.411	Gold deposits. Auriferous conglomerates and complex deposits
553.068.9	Metamorphosed deposits		◇ 553.411.068.5 Gold placer deposits.
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553.07	Morphology and structure of ore bodies, deposits. Metallogeny	553.412	Silver deposits. Argentiferous deposits
553.071	Natural element deposits	553.43	Copper deposits. Cupriferous deposits
553.072	Element-bearing deposits or conglomerates in general	553.432	Quartz-sulphide deposits
553.073	Massive ore deposits	553.434	Cupriferous sandstones and slates
553.074	Impregnated ore deposits	553.435	Chalcocopyrite deposits
553.075	Stringer zones. Stockworks	553.44	Lead deposits. Zinc deposits
553.076	Bedded sedimentary deposits	553.444	Lead-zinc deposits of pyrite type
553.078	Metallogeny. Regularity of distribution of deposits	553.445	Lead deposits
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553.085	Macroscopic examination and appearance	553.45	Tin deposits. Stanniferous deposits
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553.2	Ore formation. Mineral formation ⇒ 622.03	553.461	Molybdenum deposits
553.21/.24	Physical and chemical processes causing mineral deposits	553.462	Tungsten deposits
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		553.491	Platinum-group deposits
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- 553.64** **Phosphate deposits**
553.641 Inorganic phosphate deposits. □ Apatite deposits
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- 553.65** **Deposits of emery and corundum (as abrasives)**
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- 553.67** **Deposits of silicates of magnesium, aluminium, iron etc.**
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553.676 Asbestos deposits. Chrysotile deposits. Deposits of other fibrous calcium and magnesium silicates
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- 553.73** **Ferruginous springs. Iron-bearing waters. Chalybeate springs. Arsenical springs**
- 553.74** **Sulphurous springs. Bitter mineral waters**
- 553.75** **Calcareous springs. Chalky waters**
- 553.76** **Radioactive springs**
- 553.77** **Salt springs. Brine springs. Waters with sodium chloride**
- 553.78** **Pure hot springs. Hot springs with salt content of up to 1 gram per litre**
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⇒ 551.23
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- 553.8** **Deposits of precious and semiprecious stones. Gemstone deposits**
⇒ 622.37
- 553.81** **Diamond deposits**
- 553.82** **Corundum deposits**
⇒ 553.65
553.823 Sapphire deposits
553.824 Ruby deposits
- 553.83** **Beryl deposits**
553.832 Emerald deposits
553.833 Aquamarine deposits
- 553.84** **Topaz deposits**
- 553.85** **Garnet deposits**
- 553.86** **Zircon deposits**
- 553.87** **Quartz deposits: amethyst, opal etc.**
⇒ 553.576
Amethyst deposits
Deposits of rock crystal. □ Smoky, yellow, brown rock crystal. Smoky topaz
Deposits of opal, chalcedony, jasper
- 553.872**
- 553.875**
- 553.878**
- 553.89** **Deposits of other precious and semiprecious stones**
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- 553.9** **Deposits of carbonaceous rocks. Hydrocarbon deposits**
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⇒ 622.323/.324
⇒ 622.33
- 553.91** **Graphite deposits. Graphite-bearing deposits**
- 553.93** **Anthracite deposits**
- 553.94** **Coal deposits**
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- 553.96** **Lignite deposits. Brown coal deposits**
- 553.97** **Peat deposits**
553.973 Sapropel deposits
- 553.98** **Hydrocarbon deposits. Presence of oil and gas**
◇ 553.98(26.03) Submarine oil and gas deposits
◇ 553.98(261.26.03) Oil and gas deposits in the North sea
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Petroleum deposits. Mineral oil deposits
Types of petroleum deposit. □ Traps. Pools
Anticlinal deposits (sheet-arch deposits)
Massive deposits. □ In buried crystalline ridges. In calcareous sandstone
Monoclinial deposits
Salt domes
- 553.981.23**
- 553.981.4**
- 553.982**
- 553.982.23**
- 553.982.231**
- 553.982.232**
- 553.982.233**
- 553.982.234**

- 553.983 Deposits of bituminous shales. Oil-shale deposits
553.984 Oil-sand deposits. Other oil-bearing rock deposits
553.985 Asphalt deposits. Hard bitumen deposits.
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553.986 Ozokerite (mineral wax) deposits. Hatchettine deposits
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553.99 **Deposits of fossil resins, copals. Amber deposits**
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⇒ 551.4
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556.011 Theoretical principles
556.012 Research. Methodology. Requirements etc.
556.013 Theoretical investigation. Use of models
556.02 **Practical work. Organization. Programmes. Projects etc.**
556.023 Laboratories. Laboratory work
556.024 Stations. Field work generally
556.025 Services and networks. Network design
556.028 Representative, experimental, model basins
556.04 **Observations. Data. Records**
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556.043 Data handling. □ Collection. Processing. Dissemination
556.044 Observational data on specific hydrological phenomena
556.045 Records
556.047 Hydrological analysis
556.048 Hydrological computation. □ Coefficients
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◇ 556.06(1/9) Regional forecasts
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556.114.7 Organic chemical content of water
Details by :547...
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◇ 556.162(1/9) Spatial distribution
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Normal runoff
Maximum runoff. Floods. Flood runoff
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556.32 **Subterranean, underground water. Vertical distribution**
556.322 Zone of aeration. Suspended water. Vadose water
556.324 Zone of saturation. Groundwater (in the strict sense)
556.33 **Aquifers. Water-bearing strata**
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556.336	Perched aquifers	556.536	Hydrodynamics of rivers. Fluvial hydraulics. □ Currents. Waves ⇒ 627.13 ⇒ 627.223.3
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556.343	Flow to wells. Drawdown	556.54	River mouths. Estuaries. Deltas. Fluvio-marine water
556.36	Springs ⇒ 553.7 ⇒ 627.12 ⇒ 628.112	556.541	⇒ 551.468.6 ⇒ 627.16
556.362	Depression springs	556.542	Properties of water in river mouths and estuaries. Fluvio-marine water 556.541 ≅ 556.11
556.363	Contact springs	556.545	Water balance in river mouths and estuaries
556.364	Artesian springs ⇒ 628.112.23	556.546	Estuarine regimes. □ Estuarine pollution. Exchange of fresh and saline water. Formation of brackish water ⇒ 502.51(282.05)
556.366	Fracture springs	556.546	Estuarine hydraulics and hydrodynamics ⇒ 532.5
556.367	Intermittent springs	556.55	Limnology. Lakes. Reservoirs. Ponds
556.368	Volcanic springs ⇒ 551.23	556.551	⇒ 627.17 ⇒ 627.81
556.38	Groundwater basins. Groundwater management ⇒ 556.18	556.552	Properties of water in lakes, reservoirs, ponds. Limnic water
556.382	Safe yield	556.555	Water balance in lakes etc.
556.383	Overdraft	556.556	Lake regimes. □ Lake level. Thermal regime. Ice regime
556.388	Pollution of groundwater. Protective measures ⇒ 628.19 ⇒ 628.394	556.557	Hydrodynamics of lakes. Lacustrine hydraulics. □ Currents. Waves
556.5	Surface-water hydrology. Land hydrology ⇒ 502.51(28)	556.56	Swamps. Marshes. Fens
556.51	Drainage basins. Catchment areas. River basins. Watersheds ⇒ 627.11	556.561	Properties of swamp water. Paludal water 556.561 ≅ 556.11
556.512	Water balance of drainage basins	556.562	Water balance of swamps
556.513	Area of drainage basins	556.565	Swamp regimes. Pollution of swamps ⇒ 502.51(285.3)
556.514	Shape of drainage basins	556.566	Hydraulics and hydrodynamics of swamps. Paludal hydraulics ⇒ 532.5
556.515	Slope of drainage basins		
556.516	Stream density		
556.52	Potamology. River systems ⇒ 627.15		
556.522	Classifications. Order of stream		
556.523	Large streams		
556.524	Tributaries		

56 PALAEOLOGY

- ◇ 56:57 Palaeobiology
- ◇ 56:581 General palaeobotany
- ◇ 56:591 General palaeozoology
- ◇ 56(29) Geographic palaeontology. Palaeontology by physiographic regions
- ◇ 56"61/62" Zone fossil study. Stratigraphic classification of fossils
⇒ 551.7, 57.072.2

56.01/.08	Theoretical aspects, techniques etc. 56.01/.08 ≅ 57.01/.08	561.233	Archaea (Archaeobacteria). □ Sulfolobus. Thermoplasma. Methanobacteria. Halobacteria
561	SYSTEMATIC PALAEOBOTANY 561 ≅ 582	561.24	Protista (Protoctista). Chromista. Protozoa
561.23	Bacteria	561.241	Myxomycota (Mycetozoa). Acellular slime moulds. Plasmodial slime moulds
561.231	Schizomycota (Bacteriophyta). Bacteria as plants If in doubt prefer 579.8	561.243	Hyphochytriomycota
		561.244	Oomycota (Oomycetes)
		561.245	Plasmodiophoromycota
561.232	Cyanophyta (Cyanophyceae, Cyanobacteria). Blue-green algae	561.26/.27	Algae. Phycology. □ Seaweeds. Pond scums. Freshwater phytoplankton. Marine phytoplankton

561.261	Chrysophyta. □ Golden-brown algae. Diatoms	561.47	Pinopsida (Coniferae, Coniferales).
561.263	Chlorophyta: Grass-green algae. Desmids.		Conifers. □ Taxaceae. Podocarpaceae.
	Charophytes		Araucariaceae. Pinaceae. Taxodiaceae.
561.268	Xanthophyta (Yellow-green algae)		Cupressaceae
561.272	Phaeophyta (Brown algae)	561.49	Gnetopsida. Gnetales
561.273	Rhodophyta: Red algae		Angiospermae (Magnoliophyta).
561.276	Pyrrophyta. Dinoflagellates. Cryptophytes.	561.5/.9	Angiosperms. Flowering plants.
	Cryptomonas. Peridinales. Peridinae.		Covered-seed plants. Later seed plants
	Ciliolagellatae		Monocotyledonae (Liliopsida)
	⇒ 593.162	561.52	Arecidae
561.277	Euglenophyta (Euglenoids)	561.521	Alismatidae
	⇒ 593.161.3	561.536	Commelinidae (Glumiflorae). Grasses.
561.279	Other algae or flagellates	561.542	Bamboos
			Zingiberidae
561.28	Fungi (Eumycota). True fungi. Moulds.	561.548	Liliidae
	Mycology	561.57	Anomaliae. Monocotyledons not provided for
561.281	Zygomycota (Zygomycetes). Mould fungi	561.599	elsewhere
561.282	Ascomycota		Dicotyledonae (Magnoliopsida)
561.283	Chytridiomycota (Chytridiomycetes,	561.6/.9	Dicotyledons in general
	Rumpomycetes)	561.61	Hamamelididae
561.284	Basidiomycota (Basidiomycotina)	561.62	Caryophyllidae
561.288	Mitosporic fungi. Deuteromycotina	561.66	Magnoliidae
	(Deuteromycetes, Fungi imperfecti). Asexual	561.67	Dilleniidae
	fungi. □ Hyphales. Hyphomycetes	561.68	Rosidae
561.29	Lichens	561.7	Asteridae
	<i>Class here treatments of lichens as a</i>	561.9	Other dicotyledons. Anomaliae
	<i>biological or nutritional group. Otherwise</i>	561.999	
	<i>class lichens with relevant fungal or algal</i>	562/569	SYSTEMATIC PALAEOZOOLOGY
	<i>partners</i>		562/569 ≡ 592/599
561.32	Bryophyta (Musci). Bryology. □ Mosses.		<i>Detailed subdivision is often made in only</i>
	Liverworts. Hornworts		<i>one place, according to whether the extinct</i>
561.35/.99	Tracheophyta. Vascular plants		<i>or living form is more important. If required,</i>
561.361	Rhyniopsida. Primitive vascular plants		<i>either section may be further subdivided like</i>
561.363	Zosterophylloids. Ancestors of microphyllous		<i>the other</i>
	plants		INVERTEBRATA IN GENERAL
561.364	Trimerophytoids. Ancestors of megaphyllous		Protozoa. Unicellular organisms.
	plants		□ Rhizopoda. Amoebozoa. Foraminifera.
561.37/.39	Pteridophyta. Ferns. Fern allies. Fernlike	562	Radiolaria. Flagellata
	spore plants	563.1	Coelenterata
561.374	Equisetopsida (Sphenopsida). □ Calamitales		Spongiaria (Porifera). Sponges
561.38	Lycopsidea. Club mosses and their relatives	563.3	Cnidaria
561.381	Lycopodiales. Club mosses. Lycopodiaceae	563.4	Anthozoa. Corals
561.382	Selaginellales. Selaginaceae. Fossil	563.5	Hydrozoa. □ Polyps. Medusas
	Selaginellites	563.6	Hydroidea
561.383	Isoetales. Isoetaceae. Quillwort	563.7	Graptolithida. Graptolites
561.391	Psilotopsida: Whisk ferns. □ Psilotaceae		Scyphozoa. Jellyfish
561.394	Filicopsida (Filicales). Ferns and their relatives		Ctenophora. Comb-jellies
561.4	Seed plants in general. Spermatophyta.	563.71	Echinoderma. □ Pelmatozoa. Crinoidea.
	Phanerogams. Siphonogamous	563.719	Featherstars. Sea-lilies
	embryophyta	563.73	Asteroidea. Starfish
561.41	Progymnospermopsida. Ancestors of	563.8	Ophiuroidea. Brittlestars
	gymnosperms	563.9	Echinoidea. Sea-urchins
561.42/.47	Gymnospermopsida. Plants with naked		Holothurioidea. Sea-cucumbers
	seeds	563.93	MOLLUSCA. SHELLFISH ETC.
561.42	Gymnospermae in general. Archaeosperma	563.94	Lamellibranchiata (Pelecypoda).
	(Archispermae). Naked-seed plants. The	563.95	Bivalves. □ Scallops. Oysters. Mussels. Clams
	first seed plants	563.96	Heterodonta
561.43	Cycadopsida (Cycladatae, Pteridospermae).	564	Nayadidae. Unionidae. Freshwater mussels
	Fern-like seed plants	564.1	Cardiidae. Cockles
561.44	Cycadaceae. Palm-ferns		Tridacnidae. Giant clams
561.45	Bennettitales (Cycadeoidales)	564.14/.17	
		564.141	
561.46	Ginkgoopsida. Ginkgoaceae. Ginkgo.	564.151	
	Maidenhair trees. Cordaitales	564.152	

564.18/19	Desmodonta	565.7	Insecta (Hexapoda). Insects
564.185	Myidae. Gapers (soft clams)	565.76	Coleoptera. Beetles
564.188	Pholadidae. □ Teredo. Ship-worm	565.77	Diptera. Flies
564.2	Solenaconchae. Scaphoda (tusk-shells)	565.78	Lepidoptera. □ Butterflies. Moths
564.3	Gastropoda. Univalves. □ Snails	565.79	Hymenoptera. □ Bees. Wasps. Ants
564.32	Streptoneura. Prosobranchia. Operculates	566/569	CHORDATA
564.323	Cyclobranchia. Docoglossa. Limpets	566.2	Tunicata
564.324	Aspidobranchia. Rhipidoglossa	566.3	Acrania (Leptocardia). Branchiostomidae
564.324.3	Bellerophonitidae. Bellerophons	566.4	Cephalochordata: Lancelets
564.325.3	Trochidae. Top shells	566.6	Myxini
564.329	Ctenobranchia. Pectinobranchia	566.61	Myxiniformes
564.329.1	Ptenoglossa. □ Scalaridae. Wentletraps	566.611	Myxinidae: Hagfishes
564.329.2	Taenioglossa. □ Hydrobiidae. Spire shells.	567/569	VERTEBRATA
564.329.39	Littorinidae. Winkles (periwinkles)	567	PISCES: FISHES
564.329.6	Cypraeidae. Cowries	567.1	Pteraspodomorphi
564.329.8	Rhachiglossa. □ Fasciolaridae.	567.2	Cephalaspodomorphi
564.38	Pyramidellidae. Fusidae. Buccinidae.	567.21	Petromyzontiformes
564.38	Columbellidae. Muricidae. Purpuridae. Mitridae.	567.211	Petromyzontidae: Lampreys
564.38	Volutidae. Olovidae. Harpidae	567.23	Anaspidiformes
564.38	Toxoglossa. □ Conidae. Cone shells	567.25	Galeaspidiformes
564.5	Pulmonata. Lung-bearing gastropods.	567.27	Cephalaspidiformes
564.51	□ Land snails. Garden snails. Slugs. Water snails (freshwater snails)	567.3	Chondrichthyes: Cartilaginous fishes
564.52	Cephalopoda	567.31	Elasmobranchii (Plagiostomata)
564.51	Tetrabranchiata	567.4/.5	Osteichthyes: Bony fishes
564.52	Nautiloidea. Nautiloids	567.411	Dipnoi: Lungfish
564.53	Ammonoidea. Ammonoids (ammonites)	567.413	Coelacanthimorpha
564.55	Dibranchiata	567.413.1	Coelacanthiformes. Coelacanth
564.56	Octopoda. Octopuses	567.5	Teleostei: Teleosts
564.58	Decapoda	567.511	Leptolepidimorpha
564.581	Belemnoidea. □ Belemnitidae. Belemnites	567.6	Amphibia (Batrachia): Amphibians in general
564.582	Sepioidea. □ Squids. Cuttlefish	567.8	Anura. Tailless amphibians. □ Toads. Frogs
564.6	Molluscoidea (Tentaculata)	567.9	Urodela. Tailed amphibians. □ Newts. Axolotls. Salamanders
564.7	Bryozoa. Ectoprocta. Polyzoa. □ Moss animalcules. Sea-mat	567.99	Stegocephali. □ Temnospondyli: Sclerocephalus, Ichthyostega, Eryops, Branchiosaurus
564.8	Brachiopoda. Lamp-shells	568	SAUROPSIDA
565	ARTICULATA	568.1	Reptilia. Reptiles
565.1	Vermes. Worms. Helminths	568.112	Lacertilia
565.14	Annelida. Segmented worms. □ Bristle-worms	568.112.11	Gekkonidae
565.2	Arthropoda in general	568.112.13	Agamidae
565.3	Branchiata. Crustacea. Crustaceans. □ Entomostraca. Phyllopora. Cirripedia. Barnacles	568.112.14	Iguanidae
565.33	Ostracoda	568.112.16	Anguidae
565.36	Malacostraca in general. □ Arthrostraca	568.112.17	Helodermatidae
565.38	Thoracostraca. □ Decapoda. Brachyura. Crabs. Leptostraca. Phyllocarida	568.112.18	Varanidae
565.39	Palaeostraca	568.112.19	Teiidae
565.391	Gigantosthraca. Eurypteridae. □ Sea-scorpions	568.112.21	Chamaeleontidae
565.392	Xiphosura. King-crabs (horseshoe crabs)	568.112.23	Lacertidae
565.393	Trilobita. Trilobites	568.112.27	Scincidae
565.4/7	Tracheata	568.112.31	Dolichosauridae
565.4	Arachnoidea. Arachnids. □ Mites. Spiders. Scorpions. Chelifers. Ricinulei	568.112.32	Aigialosauridae
565.6	Myriapoda. □ Millipedes. Centipedes	568.112.33	Mosasauroidea
		568.112.34	Paterosauridae
		568.114	Amphisbaenidae
		568.115	Ophidia: Snakes

568.12	Rhyncocephalia. □ Rhyncosauridae. Sphenodontidae. Thalattosauridae	569.312	Edentata. □ Extinct anteaters, armadillos. Glyptodon
568.13	Testudinata. Chelonia. □ Tortoises. Turtles	569.32	Rodentia and Lagomorpha. □ Extinct rodents. Extinct relatives of rats, mice, beaver. Extinct rabbits, hares
568.14	Crocodylia. Parasuchia		
568.15	Ichthyosauria. Ichthyosaurs (fish-lizards)		
568.16	Sauropterygia	569.35/38	Insectivora. □ Extinct insectivores
568.163	Plesiosauria. □ Plesiosaurs. Pliosaurs	569.5	Cetacea and Sirenia. □ Extinct marine mammals
568.164	Placodontia. Placodonts		
568.165	Mesosauria. Mesosaurs	569.61	Proboscidea. □ Extinct relatives of elephants. Gomphotheriidae (mastodons). Dinotherium. Mammoths
568.17	Theromorpha. □ Cotylosauria: Bradysaurus. Pareiasaurus etc. Pelycosauria: Edaphosaurus etc. Cynodonita. Dimetrodon	569.62	Hyracoidea. □ Arsinoitherium
568.18	Pterosauria. □ Pterodactyl. Pteranodon	569.63	Typotheria
568.19	Dinosauria. Dinosaurs. □ Theropoda: Tyrannosaurus, Plateosaurus, Allosaurus etc. Sauropoda: Brontosaurus, Diplodocus etc. Praedentata: Iguanodon, Stegosaurus, Triceratops, Psittacosaurus etc.	569.64	Toxodontia
568.2	Aves. Birds in general	569.65	Litopterna
568.211	Archaeornithes. □ Archaeopteryx	569.66	Amblypoda. □ Uintatherium
568.215	Hesperornithiformes. Hesperornis	569.67	Astrapotheroidea
568.221	Struthioniformes: Flightless birds		
568.221.1	Struthionidae: Ostrich	569.71	Condylarthra
568.221.2	Rheidae: Rheas		
568.221.7	Dinornithidae: Moas	569.72	Perissodactyla. □ Chalicotheriidae. Palaetheriidae. Brontotherium
568.23	Sphenisciformes. Gaviliformes. Podicipediformes. Procellariiformes. Pelecaniformes	569.722	Rhinocerotidae. □ Woolly rhinoceros. Hyracodon
568.24	Gruiformes. Charadriiformes. Ciconiiformes	569.723	Equidae. □ Extinct equines
568.25	Anseriformes	569.73	Artiodactyla
568.26	Galliformes. Columbiformes	569.731	Nonruminantia. □ Extinct relatives of pigs, hippopotamuses
568.27/278	Psittaciformes. Piciformes. Apodiformes. Caprimulgiformes. Coraciiformes	569.732	Bunoselenodontia. □ Anthracotherium
568.279.2	Accipitriformes	569.733	Tylopoda. □ Camelidae. Extinct relatives of camels
568.279.25	Strigiformes: Owls	569.735	Ruminantia. □ Extinct elks. Extinct bovines. Aurochs
568.28/29	Passeriformes: Passerines		
569	MAMMALIA. MAMMALS	569.74	Carnivora. □ Canidae. Ursidae. Mustelidae. Hyaenidae. Felidae. Extinct felines: Machairodus (Smilodon), sabre-toothed tiger; cave lion
569.1	Prototheria (Eplacentalia)		
569.2	Metatheria (Didelphia)	569.8	Primates. □ Extinct prosimians, simians
569.22/23	Marsupialia. Extinct marsupials	569.89	Hominidae. Hominids. □ Early humans and related forms. Homo sapiens (humans, man)
569.22	Diprotodontia. □ Vombatidae. Macropodidae. Potoroidae		
569.23	Polyprotodontia		
569.3/8	Eutheria (Placentalia, Monodelphia). □ Extinct placental mammals		

57 BIOLOGICAL SCIENCES IN GENERAL

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⇒ 59
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57.01	General laws. Theoretical aspects. Characteristics. Properties	57.014	Characteristics dependent on chemical composition and properties
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57.012.2	Macrostructure	57.016.2	Structural function. Supportive function
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57.016.6	Protective function	57.047	Biological factors. Biotic factors
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57.017.2	Origin. Formation. Localization. Arrangement	◇ 57.047:592/599	Effects of animals
57.017.22	Origin. Formation. Precursors	◇ 57.047:592/595	Effects of lower animals, invertebrates
57.017.3	Adaptability. Accommodation. Acclimatization. Ability to recover. Regeneration	⇒ 57.017	
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57.018.6	Colour. Pigmentation. □ Variations in pigmentation. Variegation	57.063	Taxonomic groups. Taxa
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57.018.7	Biogeometry. Topological relations <i>Denote details by Table 1k-02</i>	57.063.2	Classes and related groupings. □ Superclasses. Classes. Subclasses. Cohorts
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57.03	Pattern of property variations. □ Fluctuation. Cyclic variation. Equilibrium	57.063.4	Families and related groupings. □ Superfamilies. Families. Subfamilies. Tribes
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57.032	Variations arising under the influence of internal factors. □ Growth. Development. Differentiation ⇒ 57.041	57.063.6	Species and related groupings. □ Superspecies. Species
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57.045	Meteorological, climatic and cosmic factors <i>Denote details by :551.5...</i>	57.071	General theory
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		57.071.21	Fossilization and preservation of plants. □ Retention of shape. Silicification
		57.071.22	Fossilization and preservation of animals. □ Retention of colour
		57.071.23	Impressions. Imprints
		57.071.6	Problematic remains. Unidentified remains
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		57.071.64	Trace fossils. □ Animal tracks in rocks. Spoor. Footprints
		57.071.7	Influences of geological periods
		57.071.72	Succession of floras and faunas

- 57.071.74 Causes of extinction of organisms in various eras
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- 57.072 **Types of fossil**
- 57.072.2 Zone fossils
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- 57.072.4 Facies fossils
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- 57.08 **Biological techniques, experimental methods and equipment in general**
 ◇ 57.08:004 Use of computers for biological experiments
- 57.081 **Biological techniques in the field**
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- 57.081.1 Fieldwork. Observations and recordings in the wild
- 57.081.11 Field observation. Watching
 ◇ 57.081.11:598.2 Bird watching
- 57.081.12 Identification of specimens (in the field)
- 57.081.14 Recording of sights and sounds of nature
 ◇ 57.081.14:681.84 Sound recording
 ◇ 57.081.14:77.03 Photographic recording
- 57.081.15 Field experiments. □ Marking. Ringing, banding
 ◇ 57.081.15:654.937 Electronic tagging
- 57.081.17 Field measurement, recording
- 57.082 **Laboratory and museum techniques and equipment for plant and animal specimens and collections**
- 57.082.1 Specimen acquisition, identification, reconstruction
- 57.082.11 Methods of acquisition. □ Purchase. Exchange. Collecting. Capture
- 57.082.13 Identification and marking of specimens
- 57.082.14 Reconstruction, remodelling of plants or animals. □ Reconstructions of palaeobiological environment. Taxidermy. Skeletal reconstruction
- 57.082.2 Live specimens: care, nutrition, breeding.
 □ Housing of specimens. Vivariums
- 57.082.5 Preparation and preservation of specimens for collections
- 57.083 **Microbiological, virological, immunological methods and techniques**
- 57.083.1 Microbiological methods and techniques.
 □ Cultures of microorganisms
- 57.083.2 Virological methods and techniques.
 □ Cultures of viruses
- 57.083.24 Isolation of viruses
- 57.083.3 Immunological methods and techniques
- 57.083.32 Allergic reactions
- 57.083.33 Serological reactions
- 57.083.34 Fluorescent antibody tests
- 57.083.36 Cytotoxic tests
- 57.084 **Experiments and observations on whole living organisms, plants or animals**
- 57.085 **Experiments and observations on parts of organisms.** □ Experiments on organs, tissues: in vivo, in vitro
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- 57.086.13 Freezing. Cryotechniques. Cryoturbation.
 □ Freeze-drying
- 57.086.14 Dehydration. Embedding media
- 57.086.15 Cutting. Sectioning. Microtechnique. Microtomy
- 57.086.16 Staining. Contrasting. Impregnation
- 57.086.18 Mounting. Mounting agents
- 57.086.2 Optical microscopy techniques
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- 57.086.3 Electron microscopy techniques
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- 57.086.8 Methods and equipment for studying living cells and tissues
- 57.086.82 Preparation of living cells
- 57.086.83 Methods and equipment for the cultivation of cells and tissues. □ Tissue cultures
- 57.086.86 Methods and equipment for micromanipulations and microsurgery
- 57.086.88 Reconstruction of objects, organs and tissues from series of sections. □ Stereomicroscopy. Photomicrography
- 57.087 **Methods and techniques for parameter estimation. Recording of biological data**
- 57.087.1 Biometry. Statistical study and treatment of biological data
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- 57.087.2 Graphic recording techniques. Charting and mapping techniques (in biology)
- 57.087.3 Photographic methods and techniques. Cinematography. Recording and interpretation of biological images
- 57.088 **Special methods and techniques for studying biological molecules.**
 □ Separation. Centrifuging. X-ray study. Radioisotope methods
- 57.088.5 Methods and techniques for studying structures and properties. □ X-ray, resonance, hydrodynamic and sedimentation techniques
- 57.088.6 Methods and techniques for studying biochemistry. □ Allozyme electrophoresis. Restriction fragment length polymorphisms (RFLP)
- 57.088.7 Methods and techniques for studying genetics.
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- 57.089 **Biomedical methods.** □ Physiological, surgical, histopathological and related techniques
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- 57.089.2 Basic methods and techniques. □ Injection. Anaesthesia. Perfusion
- 57.089.3 Embryological techniques. □ Artificial fertilization. Transplantation of ova
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- 57.089.7 Necropsy. Dissection techniques
- 572 **ANTHROPOLOGY**
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- 572.021 Climate
- 572.021.1 Acclimatization

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572.027	Race and state. Political anthropology		◇ 572.9(=012) Mongoloid race
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572.1/.4	Anthropogeny. Human development in general. Origin of the human species ⇒ 573.5 ⇒ 575.8		◇ 573.01 General theories. E.g. dualism, vitalism
572.1	Unity of the human species. Monophyletic or polyphyletic origin. Monogenism. Polygenism	573.2	Theoretical biology
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572.512	Measurements and proportions of the body. □ Dimensions of the body	574	GENERAL ECOLOGY AND BIODIVERSITY. □ Biocoenology.
572.52	Pigmentation in humans. Integument ⇒ 57.018.6		Hydrobiology. Biogeography <i>Class here general and comparative studies of populations, biocoenoses and ecosystems</i> ⇒ 502.211 ⇒ 579.26 ⇒ 581.5 ⇒ 591.5 ⇒ 631.95
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572.523	Correlations of pigmentation of hair, skin, eyes. Pigmentation types	574.3	Populations and environment. □ Population dynamics
572.524	Organs of integument: skin, hair, nails	574.4	Terrestrial biocoenoses and ecosystems. Biogeocoenoses. □ Biogeochemical cycles. Food chains
572.525	Eyes. Colour of eyes	574.5	Hydrobiology. Aquatic biocoenoses and ecosystems. □ Food chains
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572.72	General proportions of the cranium. □ Comparisons with animal crania	574.63	Sanitary hydrobiology. □ Waste treatment. Sewage processing ⇒ 628.3
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572.783	Circulatory system		
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574.65	Technical and navigational hydrobiology	575.8	Evolution. Origin of species. Phylogeny
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574.9	Biogeography in general. Geographical distribution of organisms	575.822	Heredity and variation
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575.113.2	Allele. Allelism (allelomorphism).	575.856	Organism
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575.117	Expressivity and penetrance of genes	575.87	Origin of organs and functions
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575.118.2	Dominance	576	CELLULAR AND SUBCELLULAR BIOLOGY. CYTOLOGY
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575.13	Cytoplasmic inheritance. Extranuclear inheritance		Chemical composition. Physicochemical and molecular properties
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575.153	Genetic nuclear-cytoplasmic interaction	576.311	Cytoplasm
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575.17	Population genetics. Genetic processes in populations		Cytoplasmic matrix
575.18	Sex genetics. Genetic sex determination	576.311.33	Vacuolar systems of cytoplasm and its components. Endoplasmic reticulum (ergastoplasm). □ Golgi zone. Dictyosomes.
575.2	Variation		Ribosomes
575.21	Phenotypic variation. Phenotype	576.311.34	Various organelles of cytoplasm. □ Plastids.
575.22	Genotypic variation. Genotype		Leucoplasts. Chloroplasts. Chromatophores (chromatoplasts). Lyosomes. Peroxisomes
575.222	Combinative variation	576.311.346	Centrosome. Cell centre
575.222.2	Mating (crossing)	576.311.347	Mitochondria
575.222.4	Outbreeding	576.311.348	Organelles of special sense
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575.222.7	Hybridization. Hybrids	576.311.348.3	Myofibres (striated muscle)
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575.26	The rule of homologous series	576.32	Mechanical activity of the cell
575.27	Bacterial sex recombination and mutation. Bacterial transformation and transduction (only general topics)	576.321	Intracellular motility. Protoplasmic circulation.
575.28	Viral recombination and mutations (only general topics)		Plasma streaming. Cyclosis

576.322	Cell motility. Amoeboid movement. Ciliary and flagellary movements	577.1	Chemical bases of life. Biochemistry and bio-organic chemistry generally
576.323	Contraction. Contractile vacuole	577.11	Static biochemistry. Biopolymers and other biologically interesting substances
576.324	Deformations of cell body due to passive movements	577.112	Proteins
576.33	Irritability and sensitivity of the cell	⇒ 547.96	
576.333	Taxes. Tropisms (cytological basis)	577.112.34	Chemical reactions of the amino acids.
576.34	Cell nutrition. Cell metabolism	577.112.36	Chemical modification of the amino acids
576.341	Types of cell nutrition	577.112.37	Three-dimensional structure of the amino acids
576.342	Transport systems of the cell	577.112.38	Amino-acid complexes
576.343	Metabolic pathways of the cell	⇒ 577.121	Classification of the biologically interesting amino acids
576.344	Morphological foundations of cell metabolism	577.112.382	Aliphatic amino acids. □ Glycine. Alanine.
576.35	Cell reproduction. Cell division	577.112.383	Leucine. Isoleucine. Valine
576.351	Amitosis. Direct cell division	577.112.384	Oxyaminocarboxylic acids. □ Serine.
576.352	Endomitosis. Endoreproduction. Duplication of chromosomes without cell or nuclear division	577.112.385	Threonine
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576.354	Special kinds of cell multiplication or cell division. □ Meiosis	577.112.387	□ Aspartic acid (asparagine). Glutamic acid (glutamine)
576.356	Anomalies of cell division	577.112.388	Dibasic amino acids. □ Arginine. Lysine.
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576.371	Morphology of fertility. Germ cells	577.112.5	Cysteine. Methionine
576.372	Physiology and biochemistry of fertility	577.112.6	Aromatic amino acids. □ Phenylalanine.
576.38	Other types of cell function. Other cell activities	577.112.7	Tyrosine. Tryptophane
576.382	Secretory function of the cell. Cellular basis of secretion	577.112.8	Imino acids. □ Proline. Hydroxyproline
576.385	Pathological changes in the cell. Cell pathology. Cytopathology. Pathocytology	577.112.82	Other biologically interesting amino acids
576.385.2	Oedema. Hydropic degeneration of the cell	577.112.85	Chemical reactions of the proteins.
576.385.3	Deposition of substances. Infiltration	577.113	Determination of terminal and functional groups. Chemical modification of proteins
576.385.4	Pathology of cellular growth. □ Hypertrophy.	577.113.3	Polypeptide chain. Types of linkage in the proteins. Sequence of amino acids
576.385.5	Atrophy	577.113.4	Synthetic peptides and polypeptides
	Pathology of cell differentiation.	577.113.5	Protein complexes
	Carcinogenesis. Peculiarities of tumour growth	577.113.6	Protein classification
576.5	Cell interactions. Intercellular junctions. Cell populations. Cell behaviour in culture	577.113.7	Simple proteins. □ Albumins. Globulins.
576.52	Phenomena at cell surfaces. □ Cell adhesion. Cell contacts	577.113.8	Glutelins. Histones
576.53	Cell populations	577.113.9	Conjugated proteins. □ Phosphoproteins.
576.54	Cell interaction	577.114	Mucoproteins (glycoproteins). Chromoproteins (metalloproteins). Nucleoproteins. Lipoproteins
576.6	Cytoecology. Symbiosis of cells and organelles	577.114.3	Nucleic acids
576.7	Tissues and organs as a cellular set	577.114.4	Nucleosides, nucleotides and nitrogen bases.
576.8	Parasitology	577.114.5	Nucleotide composition of nucleic acids.
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576.88	Plant parasitology. Parasitic plants	577.115.4	Synthetic polynucleotides
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576.89	Animal parasitology. Parasitic animals.	577.115.7	Classification of the nucleic acids
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577'3	Cytological, cell and subcell details		Polysaccharide chain. Primary structure of the polysaccharides (for heteropolysaccharides)
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			Classification and nomenclature for polysaccharides and carbohydrates
			Lipids. Lipoids. Fats. Fat-like substances
			Biologically interesting fatty acids. Fatty acid composition of lipids
			Chemical reactions of lipids
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577.12	Dynamic biochemistry. Metabolism of biopolymers and other biologically active substances	577.16	Vitamins. Vitamin-like substances. Vitaminology ⇒ 591.133.1 ⇒ 612.015.6 ⇒ 616.39
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577.124.5	Biosynthesis of carbohydrates and polysaccharides		General studies of hormones Chemical structure of hormones Biogenesis, formation, metabolism of hormones Biological function of hormones Hormonal complexes. Hormonal receptors Synthetic hormones and their analogues Classification of hormones Plant hormones (phytohormones). □ Auxins. Gibberellins. Cytokinins. Dormins Animal hormones Hormones of invertebrates. Ergones Hormones of vertebrates Pituitary hormones (hypophyseal hormones) Thyroid hormones. Parathyroid hormones. □ Thyroxine. Calcitonin Adrenal hormones. □ Adrenalin. Noradrenalin. Aldosterone. Cortisone Sex hormones. □ Androgens, e.g. testosterone. Oestrogens. Progesterone Pancreatic, gastrointestinal and other hormones Pancreatic hormones. □ Insulin. Glucagon Gastrointestinal hormones. □ Gastrin. Secretin Thymus hormone Hormone-like substances (hormonoids). □ Neurotransmitters, e.g. histamine
577.125	Lipid metabolism. Lipoid metabolism. Fatty-acid metabolism	577.171 577.171.3 577.171.4	
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 Budding and/or appendaged bacteria
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 Spirochaetales
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 Spirillaceae
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 Gram-negative aerobic rods and cocci.
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 Pseudomonadaceae
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 Genera of uncertain affiliation
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 Genera of uncertain affiliation
 Gram-negative anaerobic cocci
 Veillonellaceae
 Gram-negative chemolithotrophic bacteria
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 Bacteria metabolizing sulphur
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- Methane-producing bacteria. Endospore-forming rods and cocci**
 Methane-producing bacteria
 Endospore-forming rods and cocci
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 Actinomycetaceae
 Mycobacteriaceae
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 Actinoplanaceae
 Dermatophilaceae
 Nocardiaceae
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 Micromonosporaceae
- Rickettsias. Mycoplasmas**
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 ⇒ 576.3
 ⇒ 581.17
- 581.81 General characteristics of plant tissues**
- 581.82 Various kinds of tissue**
 581.821 Tegumentary tissues. Epidermis. Outer skin of plant
 581.822 Secretory tissues
 581.823 Ground tissues. □ Parenchyma. Sclerenchyma. Collenchyma
 581.824 Vascular tissues. □ Xylem. Phloem
 581.825 Aerial tissues. □ Lenticels. Aerenchyma
 581.829 Other tissues
- 581.84 Structure of organs from a histological point of view. Cosmic factors in the structure of organs**
 581.84 ≡ 581.4
- 581.9 Geographic botany. Plant geography (phytogeography). Floras. Geographic distribution of plants**
 ⇒ 57.063.2
 ⇒ 581.527
- 581.91 Identification of floras**
- 581.92 Lists of species of flora**
- 581.93 Structure of floras (structural analysis)**
- 581.95 New localizations**
- 581.96 New species**
 581.961 Revisions of taxonomy

582

SYSTEMATIC BOTANY

For systematic palaeobotany see 561 which is parallel with 582

582-1

Concepts of general botany related to particular plants

582-1 ≡ 581

582.09

Classification according to size and form

Use either in the form listed here (as in the first example) or combined with 582.23/9 (as in the second example)

◇ 582.091(41) Trees of the British isles

◇ 582.628.091 Juglandales

Trees (large woody plants with trunk)

Shrubs (smaller woody plants without single trunk)

Undergrowth. Undershubs (e.g. brambles - Rosoideae)

Woody climbers (e.g. ivy - Araliaceae)

Herbaceous or non-woody plants (often called 'plants' or 'flowers')

582.23

Bacteria

582.231

Schizomycota (Bacteriophyta). Bacteria (as plants)

If in doubt prefer 579.8

582.232

Cyanophyta (Cyanophyceae, Cyanobacteria). Blue-green algae

582.233

Archaea (Archaeobacteria). □ Sulfolobus. Thermoplasma. Methanobacteria. Halobacteria

582.24

Protista (Protoctista). Chromista. Protozoa

582.241

Myxomycota (Mycetozoa). Acellular slime moulds. Plasmodial slime moulds

582.243

Hyphochytriomycota

582.244

Oomycota (Oomycetes)

582.244.1

Saprolegniales

582.244.2

Peronosporales

582.245

Plasmodiophoromycota

582.245.1

Plasmodiophorales

582.261/27

Algae. Phycology. □ Seaweeds. Pond

scums. Freshwater phytoplankton. Marine phytoplankton

Chrysophyta

582.261

Bacillariophyceae. Diatoms

582.261.1

Chrysophyceae: Golden-brown algae

582.261.2

Chlorophyta: Grass-green algae. Desmids

582.263

Chlorophyceae. Green algae. Ulva

582.263.1

Zygnemataceae: Conjugating green algae.

582.263.2

Volvocaceae. □ Spirogyra. Zygnema

582.263.3

Charophyceae. Charophytes. □ Stonewort

582.268

(Brittlewort). Chara

582.272

Xanthophyta: Yellow-green algae

582.276

Phaeophyta: Brown algae. □ Laminaria. Kelps.

582.273

Bladderwrack. Fucus

582.276

Rhodophyta: Red algae. □ Bangiophytes.

582.277

Florideophytes. Nemalion. Ceramium.

582.279

Gigartina. Gelidium

582.281

Pyrrophyta. Dinoflagellates. Cryptophytes.

582.281.21

Cryptomonas. Peridinales. Peridinae.

582.282

Ciliolagellatae

582.282.11

⇒ 593.162

582.277

Euglenophyta: Euglenoids

582.279

⇒ 593.161.3

582.28

Other algae or flagellates

582.281

Fungi (Eumycota, Eumycetes). True fungi.

582.281.21

Moulds. Mycology

582.282

Zygomycota (Zygomycetes)

582.282.11

Mucorales: Mould fungi

582.282.11

Ascomycota

582.282.11

Pneumocystidales. Protomycetales.

582.282.11

Schizosaccharomycetales. Taphrinales

- 582.282.12 Eurotiales
 582.282.123 Trichocomaceae (Aspergillaceae)
 582.282.123.2 Penicillium
 582.282.123.4 Aspergillus
 582.282.13 Lecanorales
 582.282.14 Leotiales
 582.282.16 Pezizales
 582.282.161 Helvellaceae
 582.282.162 Morchellaceae. Morels
 582.282.163 Tuberaceae. Truffles
 582.282.23 Saccharomycetales. □ Saccharomycetaceae.
 Saccharomyces cerevisiae: Brewer's yeast
 582.282.31 Other Ascomycota. □ Arthoniales.
 Dothideales. Ophiostomatales. Sordariales
 582.283 Chytridiomycota (Chytridiomycetes,
 Rumpomycetes)
 582.284 Basidiomycota (Basidiomycotina)
 582.284.1 Ustomycetes
 582.284.11 Ustilaginales. Smut fungi
 582.284.2 Teliomycetes
 582.284.21 Uredinales: Rust fungi
 582.284.3 Basidiomycetes
 582.284.4 Phragmobasidiomycetes
 (Heterobasidiomycetes). □ Auriculariales.
 Tremellales
 582.284.5 Holobasidiomycetes (Homobasidiomycetes)
 582.284.51 Agaricales. □ Agaricaceae. Mushrooms.
 Toadstools
 582.284.52 Boletales (Boletes). Poriales
 582.284.53 Cantharellales. □ Clavariaceae. Club fungi
 582.284.54 Nidulariales. Bird's nest fungi
 582.284.55 Lycoperdales. Puff Balls. Earth Stars
 582.284.56 Phallales. Stink-horns
 582.284.99 Other Basidiomycota
 582.288 Mitosporic fungi (Deuteromycotina,
 Deuteromycetes, Fungi imperfecti). Asexual
 fungi. □ Hyphomycetes (Hyphales)
- 582.29 Lichens**
*Class here treatments of lichens as a
 biological or nutritional group. Otherwise
 class lichens with relevant fungal or algal
 partners. For example Cetraria,
 Parmeliaceae (Iceland moss) should be
 classed at 582.282.13 Lecanorales*
- 582.3/99 Plantae**
- 582.32 Bryophyta (Musci). Bryology.** □ Mosses
 582.321 Hepatopsida (Hepaticae, Takakiopsida,
 Marchantiomorpha). Liverworts
 582.321.1 Jungermanniales
 582.321.2 Marchantiales
 582.322 Anthocerotophyta. Hornworts
 582.323 Sphagnopsida. Peat mosses. Bog mosses.
 Sphagnum
 582.324 Andreaeopsida. Granite mosses. Black mosses
 582.325 Bryopsida
 582.325.1 Polytrichales
 582.325.11 Eubryidae
 582.325.2 Bryales
 582.325.21 Hookeriales (Isobryales)
 582.325.23 Hypnobryales
 582.325.25 Buxbaumiales
 582.325.27 Schistostegales
 582.325.3 Other bryophytes
- 582.35/99 Tracheophyta. Vascular plants**
 582.361 Rhyniopsida. Primitive vascular plants
 582.363 Zosterophyllopsida. Ancestors of microphyllous
 plants
 582.364 Trimerophytids. Ancestors of megaphyllous
 plants
- 582.37/39 Pteridophyta. Ferns. Fern allies. Fernlike
 spore plants**
 582.374 Equisetopsida (Sphenopsida)
 582.374.2 Equisetales. □ Equisetaceae. Equisetum.
 Horsetails
 582.374.5 Calamitales. Extinct arborescent sphenopsids
- 582.38 Lycopsidea. Club mosses and their relatives**
 582.381 Lycopodiales. Club mosses. □ Lycopodiaceae
 582.382 Selaginellales. □ Selaginaceae. Fossil
 Selaginellites
 582.383 Isoetales. □ Isoetaceae. Quillwort
 582.383.1 Lepidodendraceae
 582.383.2 Sigillariaceae
 582.391 Psilotopsida: Whisk ferns. □ Psilotaceae
 582.394 Filicopsida (Filicales). Ferns and their relatives
 582.394.4 Ophioglossales
 582.394.41 Ophioglossaceae. □ Ophioglossum: Adder's
 tongue
 582.394.5 Mariattiales
 582.394.51 Marattiaceae
 582.394.7 Dennstaedtiaceae: Bracken
 582.394.722 Osmundaceae. Leptopteris. Osmunda
 582.394.726 Matoniaceae
 582.394.736 Cyathaceae. Cnemidaria. Cyanthea. Tree
 Ferns
 582.394.737 Hymenophyllaceae. Filmy ferns
 582.394.738 Gleicheniaceae
 582.394.739 Adantiaceae (Pteridaceae)
 582.394.742 Polypodiaceae. Colysis. Polypodium
 582.394.744 Schizaeaceae
 582.394.75 Marsiliaceae. □ Pilulariaceae: Pillwort
 582.394.77 Salviniales
 582.394.771 Salviniaceae. Free floating aquatics
 582.394.772 Azollaceae
- 582.4 Seed plants in general. Spermatophyta.
 Phanerogams. Siphonogamous
 embryophyta.**
- 582.41 Progymnospermopsida. Ancestors of
 gymnosperms**
- 582.42/47 Gymnospermopsida. Plants with naked
 seeds**
- 582.42 Gymnospermae in general. Archaeosperma
 (Archispermae). Naked-seed plants. The
 first seed plants**
- 582.43 Cycadopsida (Cycadatae, Pteridospermae).
 Fern-like seed plants**
- 582.44 Cycadaceae. Palm-ferns**
- 582.45 Bennettitales (Cycadeoidales)**
- 582.46 Ginkgoopsida. Ginkgoaceae. Ginkgo.**
 □ Maidenhair trees
 Cordaitales
- 582.47 Pinopsida (Coniferae, Coniferales). Conifers**
 582.471 Taxaceae. Yews. □ Taxus
 582.472 Podocarpaceae. Yellow woods. New Zealand
 white pine
 582.473 Araucariaceae. Monkey-puzzles. Norfolk Island
 pines. Araucaria
 582.475 Pinaceae. □ Pinus: Pines. Pitch pine. Picea:
 Spruce. Larix: Larch. Cedrus: Cedar. Firs.
 Pseudotsuga: Douglas fir
 582.476 Taxodiaceae. □ Swamp cypresses. Sequoias
 (Redwoods). Sequoiadendron (Big tree,
 Wellingtonia)
 582.477 Cupressaceae. □ Cupressus: Cypress. Thuja
 (Arbor.vitae, White cedar). Thujopsis.
 Juniperus: Juniper. Savin
 582.478 Other coniferae

582.49	Gnetopsida (Gnetales)	582.573.11	Agavaceae. □ Hostaceae
582.491	Ephedraceae. Ephedra	582.573.16	Alliaceae. □ Allium. Garlic. Onion
582.492	Welwitschiaceae. Welwitschia	582.573.21	Amaryllidaceae. □ Snowdrops. Narcissi.
582.493	Gnetaceae. Gnetum. Vinkella		Daffodils
582.5/.9	Angiospermae (Magnoliophyta).	582.573.26	Anthericaceae
	Angiosperms. Flowering plants.	582.573.31	Aphyllanthaceae
	Covered-seed plants. Later seed plants	582.573.36	Asparagaceae. □ Herreriaceae. Ruscaceae
582.52	Monocotyledonae (Liliopsida)	582.573.41	Asphodelaceae. □ Aloaceae: Aloes
582.521	Arecidae	582.573.46	Asteliaceae
582.521.1	Arecales	582.573.51	Blandfordiaceae
582.521.11	Palmae (Principes, Arecaceae). Palms	582.573.56	Convallariaceae
582.521.2	Cyclanthales	582.573.61	Cyanastraceae
582.521.21	Cyclanthaceae (Synanthae)	582.573.66	Doryanthaceae
582.521.3	Pandanales	582.573.68	Dracaenaceae. Dragon tree
582.521.31	Pandanaceae	582.573.71	Eriosphermaceae
582.521.4	Arales	582.573.76	Hemerocallidaceae
582.521.41	Araceae. Arums. □ Cuckoo-pint	582.573.81	Hyacinthaceae. □ Scilloideae. Bluebells.
582.521.42	Acoraceae		Hyacinths. Squill. Sea-onion
582.521.43	Lemnaceae. Duckweed	582.573.86	Hypoxidaceae. Curculigo
582.536	Alismatidae	582.573.88	Philesiaceae
582.536.1	Alismatales. □ Hydrocharitales, Najadales	582.573.90	Luzuriagaceae
582.536.11	Butomaceae	582.573.91	Phormiaceae. New Zealand flax
582.536.15	Alismataceae. □ Limnocharitaceae. Water plantain	582.573.96	Tecophilaeaceae. □ Ixioliriaceae
		582.573.98	Trilliaceae
582.536.21	Hydrocharitaceae. □ Frog-bit. Water thyme	582.574	Haemodorales
582.536.31	Potamogetonaceae	582.574.1	Haemodoraceae
582.536.32	Najadaceae. Waterweeds. Pondweeds	582.574.2	Philydraceae
582.536.33	Juncaginaceae	582.574.3	Pentastemonaceae
582.536.34	Aponogetonaceae	582.574.4	Pontederiaceae
582.536.4	Triuridales	582.574.5	Stemonaceae
582.536.41	Triuridaceae	582.574.6	Taccaceae
582.542	Commelinidae (Glumiflorae). Grasses.	582.575	Dioscoreales
	Bamboos	582.575.1	Dioscoreaceae. Yams
582.542.1	Cyperales	582.575.2	Smilacaceae
582.542.11	Graminae (Poaceae)	582.576	Velloziales
582.542.12	Cyperaceae. Sedges	582.576.1	Velloziaceae
582.542.2	Commelinales	582.59	Orchidales (Microspermae)
582.542.21	Rapateaceae	582.591	Corsiaceae
582.542.22	Commelinaceae. □ Tradescantia: Spider plant	582.594	Orchidaceae: Orchids. □ Paphiopedalum:
582.542.3	Eriocaulales		Lady's slipper. Vanilla. Orchis. Cattleya.
582.542.31	Eriocaulaceae. □ Pipewort		Cymbidium. Phalaenopsis. Dendrobium
582.542.4	Restionales. □ Restionaceae. Flagellariaceae	582.599	Anomaliae. Monocotyledons not provided for elsewhere
582.542.5	Juncales		
582.542.51	Juncaceae. Rushes	582.6/.9	Dicotyledonae (Magnoliopsida)
582.542.52	Thurniaceae	582.61	Dicotyledons in general
582.542.6	Hydatellales. Hydatellaceae	582.62	Hamamelididae
582.542.7	Typhales. □ Sparganiaceae. Typhaceae	582.621	Casuarinales. Casuarinaceae
582.548	Zingiberidae	582.628	Juglandales. Myricales. □ Rhoipteleaceae.
582.548.1	Bromeliales		Juglandaceae: Walnuts. Myricaceae. Sweet gale. Bog myrtle
582.548.11	Bromeliaceae. Pineapples	582.632	Fagales
582.548.2	Zingiberales	582.632.1	Betulaceae. □ Birches. Alders. Hornbeam.
582.548.21	Musaceae. □ Strelitziaceae. Heliconiaceae.		Hazelnut
	Musa: Banana. Strelitzia	582.632.2	Fagaceae. □ Beeches. Copper beech. Sweet chestnut. Oaks
582.548.23	Lowiaceae		Ticodendraceae
582.548.25	Cannaceae. □ Canna: Indian shot	582.632.3	Urticales
582.548.27	Zingiberaceae. Ginger	582.635	Ulmaceae. □ Elms. Wych-elms (witch-elms)
582.548.29	Marantaceae. Arrowroot	582.635.1	Moraceae. Mulberries. □ Ficus: Figs. Rubber plant. Artocarpus: Breadfruit
582.57	Liliidae	582.635.3	Cannabaceae (Cannabidaceae). □ Cannabis (Hemp). Hops
582.572	Liliales (Liliiflorae)		Urticaceae. Nettles
582.572.1	Alstroemeriaceae: Peruvian lily	582.635.38	Cecropiaceae
582.572.2	Burmanniaceae		Barbeyaceae
582.572.3	Colchicaceae	582.635.5	Eucommiales. Eucommiaceae
582.572.4	Calochortaceae	582.635.6	Hamamelidales
582.572.5	Campynemataceae	582.635.7	Myrothamnaceae. Myrothamnus
582.572.6	Melanthiaceae. Meadow saffron (Autumn crocus)	582.636	Hamamelidaceae. Wych-hazel (Witch-hazel)
		582.638	
582.572.7	Iridaceae. □ Geosiridaceae. Irises: Flags.	582.638.16	
	Crocuses. Gladioli	582.638.21	
582.572.8	Liliaceae. □ Lilies. Tulips		
582.573	Asparagales		

582.638.26	Platanaceae. Plane trees. □ American sycamore	582.681.26	Violaceae. Pansy (Heart's-ease)
582.643	Trochodendrales	582.681.31	Flacourtiaceae
582.643.1	Trochodendraceae. □ Tetracentraceae	582.681.36	Stachyuraceae
582.643.2	Cercidiphyllaceae	582.681.41	Passifloraceae. Passion-flower
582.643.3	Eupteleaceae	582.681.46	Caricaceae. Papaw
582.66	Caryophyllidae	582.681.51	Loasaceae
582.661	Caryophyllales (Centrospermae)	582.681.56	Datisceae
582.661.11	Didiereaceae	582.681.61	Begoniaceae
582.661.15	Chenopodiaceae. □ Halophytaceae. Goosefoot. Fat hen. Orache. Spinach. Sugar beet. Dysphania	582.681.66	Ancistrocladaceae
582.661.21	Amaranthaceae: Pigweed family. □ Foxtail	582.681.71	Cucurbitaceae. □ Gourds. Cucumber. Gherkin. Melons. Watermelon. Pumpkin. White bryony
582.661.26	Phytolaccaceae. □ Agdestidaceae. Barbeuiaceae. Gisekiaceae. Stegnospermataceae. Pokeweed. Inkberry	582.681.76	Elaeocarpaceae
582.661.31	Nyctaginaceae. □ Marvel of Peru	582.681.81	Salicaceae. □ Willows. Osiers. Poplars. Aspens
582.661.36	Aizoaceae. Hottentot fig	582.682	Euphorbiales
582.661.41	Portulacaceae. □ Hectorellaceae. Montia. Water blinks (Blinking chickweed)	582.682.1	Euphorbiaceae. Spurges. Dog's mercury
582.661.46	Basellaceae	582.682.2	Buxaceae. Box
582.661.51	Caryophyllaceae. □ Illecebraceae. Dianthus. Carnations. Pinks. Campions. Ragged robin. Sea purslane. Spurrey. Catchfly. Mouse-ear. Chickweed. Pearlwort. Sandwort. Soapwort. Stitchwort	582.682.3	Pandaceae
582.661.56	Cactaceae. Cactuses. □ Epiphyllum. Echinocactus. Giant cactus. Queen of the night. Opuntia. Prickly pear. Indian fig	582.682.4	Thymelaeaceae. □ Daphne. Spurge laurel
582.665	Polygonales	582.683	Capparidales. □ Batales (Batidales)
582.665.11	Polygonaceae. □ Dock. Sorrel. Bistort. Rhubarb. Knot-grass. Buckwheat	582.683.1	Capparidaceae (Capparaceae). Capers
582.668	Plumbaginales. □ Plumbaginaceae. Leadwort. Sea lavenders. Thrift. Sea pink	582.683.2	Cruciferae (Brassicaceae). □ Stocks. Cresses. Broccoli. Cauliflower. Water cress. Mustards. Radishes. Rape. Turnip. Wild cabbage. Rockets. Candytuft. Shepherd's purse. Woad
582.67	Magnoliidae	582.683.3	Resedaceae. □ Mignonette. Dyer's weed (Weld)
582.671	Nymphaeales. □ Nymphaeaceae. Ceratophyllaceae. Water lilies	582.683.4	Moringaceae. Moringa (Ben nut)
582.672	Piperales. □ Saururaceae. Piperaceae. Pepper	582.683.5	Bataceae
582.674	Aristolochiales. □ Aristolochiaceae: Birthworts. Hydnoraceae. Rafflesiaceae	582.683.6	Bretschneideraceae
582.675	Ranunculales. □ Papaverales	582.683.7	Akaniaceae
582.675.1	Ranunculaceae. □ Glaucidiaceae. Buttercups. Aconites. Monkshood. Crowfoot. Lesser celandine. Anemones. Spearwort. Marsh marigolds (Kingcups). Hellebores. Clematis. Old man's beard	582.683.8	Limnanthaceae
582.675.2	Coriariaceae. Tanner's tree (Ink tree)	582.684	Guttiferales
582.675.3	Berberidaceae. Barberry	582.684.1	Guttiferae (Clusiaceae). Garcinia family (St John's-wort family). □ Hypericaceae
582.675.4	Menispermaceae. Moonseed	582.684.2	Elatinaceae. Waterworts
582.675.5	Papaveraceae. □ Fumariaceae. Poppies. Chelidonium majus (Greater celandine)	582.684.3	Bonnetiaceae
582.677	Magnoliales. □ Laurales	582.685	Malvales
582.677.1	Magnoliaceae. Magnolias. Tulip tree	582.685.2	Malvaceae. Mallows. Cotton plant
582.677.2	Lauraceae. □ Cinnamon tree. Camphor tree. Bay laurel	582.685.3	Bombacaceae. Cotton-tree
582.677.3	Sabiaceae. □ Meliosmaceae	582.685.4	Tiliaceae. Lime tree (Linden)
582.677.4	Canellaceae (Winteranaceae). Canella. Warburgia	582.685.5	Sterculiaceae. □ Cacao: Cocoa beans. Cola tree (kola)
582.677.5	Annonaceae	582.685.6	Bixaceae. □ Cochlospermaceae. Annatto
582.677.7	Myristicaceae. □ Nutmeg. Mace	582.686	Nepenthales
582.677.8	Chloranthaceae	582.686.2	Nepenthaceae. Pitcher plants
582.678	Illiciales	582.686.3	Droseraceae. Sundew. Venus flytrap
582.678.1	Illiciaceae. Illicium. Japanese anise	582.687	Theales (including Diapensiales, Ebenales, Lecythidales)
582.678.2	Schisandraceae	582.687.11	Scytopetalaceae
582.678.3	Winteraceae. Drimys	582.687.16	Caryocaraceae
582.68	Dilleniidae	582.687.21	Theaceae. □ Asteropeiaceae. Camelliaceae. Tea plant
582.681	Violales. □ Salicales	582.687.26	Aquifoliaceae. Ilex (Holly)
582.681.11	Tamaricaceae. Tamarisk	582.687.31	Sarraceniaceae
582.681.16	Cistaceae. Rock-roses	582.687.36	Diapensiaceae
		582.687.41	Sapotaceae
		582.687.46	Ebenaceae. Ebony. Persimmon
		582.687.51	Styracaceae
		582.687.56	Lissocarpaceae
		582.687.61	Symplocaceae
		582.687.66	Lecythidaceae. □ Bertholletia: Brazil-nut
		582.688	Ericales
		582.688.1	Empetraceae. Crowberry
		582.688.2	Epacridaceae
		582.688.3	Ericaceae. □ Pyrolaceae. Monotropaceae. Azaleas. Rhododendrons. Gaultheria. Vaccinium. Bilberry (Whortleberry). Blueberry. Cranberry (Fenberry). Cavendishia. Erica. Heaths. Calluna. Heathers. Ling. Australian heath

582.688.4	Actinidiaceae	582.746.51	Aceraceae. Maple. Sycamore
582.689	Primulales	582.746.56	Hippocastanaceae. Horse chestnut
582.689.1	Myrsinaceae	582.746.61	Melanthaceae. Honeybush
582.689.2	Primulaceae. □ Primroses. Cowslips. Pimpernels. Yellow loosestrife	582.746.66	Anacardiaceae. □ Julianiaceae. Sumac. Poison ivy
582.689.3	Theophrastaceae. Jacquinia	582.751	Geraniales
582.7	Rosidae	582.751.1	Oxalidaceae. Wood sorrel. □ Lepidobotryaceae
582.711	Rosales	582.751.2	Geraniaceae. Geranium. Herb Robert. Cranesbills. Pelargonium. Storksbills
582.711.11	Saxifragaceae. □ Eremosynaceae. Parnassiaceae. Penthoraceae. Vahliaceae. Saxifrages. London pride. Anise	582.751.3	Tropaeolaceae
582.711.16	Crassulaceae. □ Stonecrops. Umbilicus. Pennywort	582.751.5	Balsaminaceae. Balsams. □ Impatiens
582.711.21	Cephalotaceae	582.755	Polygalales
582.711.26	Hydrangeaceae. □ Philadelphus: Mock orange	582.755.1	Malpighiaceae
582.711.31	Grossulariaceae. □ Escalloniaceae. Montiniaceae. Pterostemonaceae. Ribes. Currants. Gooseberries	582.755.2	Vochysiaceae
582.711.36	Pittosporaceae	582.755.3	Trigoniaceae
582.711.41	Byblidaceae	582.755.4	Tremandraceae
582.711.46	Brunelliaceae	582.755.5	Polygalaceae. Milkwort
582.711.51	Cunoniaceae	582.755.6	Xanthophyllaceae
582.711.56	Bruniaceae	582.755.7	Krameriaceae
582.711.61	Crossosomataceae	582.766	Celastrales
582.711.66	Columelliaceae	582.766.2	Icacinaceae
582.711.71	Rosaceae	582.766.3	Dichapetalaceae
582.711.711	Spiraeoideae. □ Spiraea. Meadowsweet	582.766.4	Corynocarpaceae
582.711.712	Rosoideae. □ Roses. Briars. Raspberry. Blackberry. Brambles. Strawberry. Cinquefoil. Geum (Avens)	582.766.5	Celestraceae. □ Hippocrateaceae. Spindle tree
582.711.713	Prunoideae. □ Almond. Apricot. Cherry. Peach. Plum. Damson. Blackthorn (Sloe)	582.766.6	Geissolomataceae
582.711.714	Malloideae. Malus. □ Apples. Pears. Crabtree (Crab. Crab-apple). Quince. Chaenomeles (Japonica). Cotoneaster. Rowan (Mountain ash). Hawthorn. Medlar	582.766.7	Stackhousiaceae
582.711.76	Connaraceae	582.766.8	Aextoxicaceae
582.721	Podostemales. Podostemaceae	582.766.9	Cardiopteridaceae
582.724	Protiales	582.776	Myrtales. □ Penaeaceae. Crypteroniaceae. Rhynchocalycaceae. Oliniaceae
582.724.1	Elaeagnaceae. □ Oleaster. Sea buckthorn	582.776.1	Lythraceae. Purple loosestrife.
582.724.2	Proteaceae	582.776.2	Sonneratiaceae. Punicaceae: Pomegranate
582.728	Santalales	582.776.4	Myrtaceae. □ Myrtles. Cloves. Pimento.
582.728.1	Olcaceae	582.776.5	Allspice. Eucalyptus. Gum trees
582.728.2	Santalaceae. Sandalwood	582.776.6	Melastomataceae. Astronia
582.728.3	Misodendraceae	582.776.7	Trapaceae. Horn nut
582.728.4	Loranthaceae. Mistletoe	582.782	Onagraceae. □ Evening primrose. Willowherb.
582.728.5	Balanophoraceae. □ Cynomoriaceae	582.782.1	Fuchsia
582.736	Leguminosae (Fabales). Legumes. Pulses	582.782.2	Combretaceae. Terminalia
582.736.1	Mimosoideae. □ Mimosa. Acacia (Wattles). Colophospermum (Balsam tree)	582.786	Rhamnales
582.736.2	Caesalpinioideae. □ Caesalpinia. Cassia. Senna	582.788	Rhamnaceae. □ Rhamnus: Buckthorn
582.736.3	Papilionoideae (Fabaceae). □ Lupins. Broom. Gorse (Furze). Medicks. Clover. Vetch. Lentils. Peas. Beans. Soya. Fenugreek. Castanospermum: Moreton Bay chestnut (Australian chestnut)	582.788.1	Vitaceae. Vines. Grapevines
582.741	Linales. □ Linaceae. Hugoniaceae. Flax (Linseed). Erythroxylaceae. Coca	582.788.2	Rhizophorales. Rhizophoraceae
582.746	Sapindales. □ Leitneriales	582.792	Cornales
582.746.11	Zygophyllaceae	582.792.1	Cornaceae. Dogwoods. □ Araliaceae.
582.746.16	Cneoraceae	582.792.2	Aucubaceae. Griselinaceae. Helwingiaceae.
582.746.21	Rutaceae. □ Rue. Citruses. Oranges. Lemons	582.794	Melanophyllaceae. Torricelliaceae
582.746.26	Simaroubaceae. □ Leitneriaceae. Quassia (Bitters)	582.794.1	Garryaceae
582.746.31	Irvingiaceae	582.794.2	Haloragidales
582.746.36	Burseraceae. □ Frankincense. Myrrh	582.913	Haloragidaceae (Haloragaceae). □ Myriophyllum. Water milfoils
582.746.41	Meliaceae. Cedrela	582.913.1	Gunneraceae. Gunnera
582.746.46	Sapindaceae. □ Ptaeroxylaceae. Soap-berry	582.913.2	Apiales
		582.913.3	Umbelliferae (Apiaceae). Carrot family (Parsley family). □ Sanicula. Chervil. Coriander.
		582.916	Hemlock. Caraway. Cumin. Celery. Parsley.
		582.916.11	Goutweed (Ground elder). Fennel. Dill. Carrot.
		582.916.16	Parsnip. Hogweed. Pimpinella (Anise)
			Araliaceae. Ivy
		582.9	Asteridae
		582.913	Callitrichales
		582.913.1	Hippuridaceae. Mare's tail
		582.913.2	Callitrichaceae. Mud water starwort
		582.913.3	Hydrostachyaceae. Hydrostachys
		582.916	Scrophulariales
		582.916.11	Buddlejaceae. Buddleia
		582.916.16	Oleaceae. □ Forsythia. Olea (Olive), Syringa (Lilac), Jasminum. Ash

582.916.21	Scrophulariaceae. □ Antirrhinum (Snapdragon). Veronica (Speedwell). Digitalis (Foxglove)			Sage. Clary. Melissa (Balm). Hyssop. Marjoram. Thyme. Mints (Spearment, Peppermint etc.). Ocimum. Basil
582.916.26	Orobanchaceae. Broomrape	582.933		Plantaginales. □ Plantaginaceae: Plantains (Ribwort)
582.916.31	Bignoniaceae. Calabash tree			Rubiales
582.916.36	Pedaliaceae. Sesame	582.936		Rubiaceae (Theligonaceae). □ Coffee.
582.916.41	Globulariaceae	582.936.1		Woodruff. Madder. Bedstraw. Lady's bedstraw.
582.916.46	Lentibulariaceae. □ Butterwort. Bladderwort			Cinchona (quinine)
582.916.51	Acanthaceae. Acanthus	582.936.2		Desfontainiaceae
582.916.56	Myoporaceae	582.936.3		Gelsemiaceae
582.916.61	Gesneriaceae. Saintpaulia (African violet)	582.971		Dipsacales
582.923	Gentianales	582.971.1		Caprifoliaceae. □ Carlemanniaceae.
582.923.1	Gentianaceae. Gentian			Honeysuckle (Woodbine). Elder. Viburnum.
582.923.2	Strychnaceae. Nux vomica			Guelder rose. Snowberry
582.923.3	Loganiaceae			Adoxaceae
582.923.4	Geniostomaceae	582.971.2		Valerianaceae. □ Triplotegiaceae. Valerians.
582.923.5	Apocynaceae. □ Dogbane. Vinca. Catharanthus. Periwinkle. Nerium (Oleander)	582.971.3		Corn-salad. Lamb's lettuce
582.923.6	Asclepiadaceae. Silkweed (Milkweed)	582.971.4		Dipsacaceae. □ Teasels. Scabious
582.926	Solanales	582.998		Asterales. □ Calycerales. Campanulales
582.926.1	Nolanaceae. Chilean bell-flower	582.998.1		Compositae (Asteraceae)
582.926.2	Solanaceae. □ Nightshades. Deadly nightshade (Belladonna). Henbane. Paprika. Capsicum. Hot peppers. Chillies. Tomato. Potato. Thorn-apple. Nicotiana (Tobacco). Petunia	582.998.12		Barnadesoideae
		582.998.14		Cichorioideae (Lactucoeideae). □ Chicory.
582.926.3	Convolvulaceae. □ Convolvulus (Bindweed). Batata (Sweet potato)	582.998.16		Endive. Hawkweeds. Dandelion. Lettuce
582.926.4	Polemoniaceae. □ Cobaeaceae. Jacob's ladder. Phlox			Asteroidae. □ Golden rod. Daisy. Aster. Edelweiss. Sunflower. Dahlia. Camomile (Chamomile). Tansy. Corn marigold. Yarrow (Milfoil). Chrysanthemum. Artemisias.
582.926.5	Hydrophyllaceae			Wormwood (Mugwort, Absinth). Tarragon.
582.929	Lamiales	582.998.2		Southernwood (Lad's love). Arnica. Groundsel.
582.929.1	Lennoaceae	582.998.3		Burdock. Thistles. Artichoke. Cornflower
582.929.2	Boraginaceae. □ Borage. Lungwort. Myosotis (Forget-me-not). Bugloss. Comfrey			Calyceraceae
582.929.3	Verbenaceae. □ Avicenniaceae. Cylocheilaceae. Phrymaceae. Stilbaceae. Verbena. Vervain	582.998.4		Campanulaceae. □ Pentaphragmataceae.
		582.998.5		Sphenocleaceae. Bell-flowers. Harebell.
582.929.4	Labiatae (Lamiaceae). □ Tetrachondraceae. Ajugoides. Rosmarinus (Rosemary). Ground ivy. Germander. Lavandula. Deadnettle. Salvia.	582.998.6		Lobelia. Rampion
				Goodeniaceae. □ Brunoniaceae
				Menyanthaceae. Buckbean
				Stylidiaceae. □ Donatiaceae. Stylidium.
				Forstera
		582.999		Anomaliae. Dicotyledons not provided for elsewhere

59 ZOOLOGY

◇ 59:069.029 Zoological gardens

59.01/08	Theoretical aspects, techniques etc. 59.01/08 ≡ 57.01/08	591.047	Transport of material in the body ⇒ 591.41
591	GENERAL ZOOLOGY	591.05	Physiological chemistry and metabolism in general. Composition of the body ⇒ 591.133.2 ⇒ 591.149
591-2/-9	Concepts of general zoology related to particular animals and classes of animal 591-2/-9 ≡ 592/599	591.1	Animal physiology ⇒ 612
591.04	Physics and chemistry of cells and organisms in general. Absorption	591.11	Blood. Body fluids in general. Circulation
591.041	Influence of nucleus and protoplasm. Irritability in general ⇒ 591.181	591.111	General chemical and physical properties of blood
591.044	Effect of physical and chemical factors of the environment on cells and organisms. □ Osmosis. Tropisms in general. Heliotropism. Phototropism. Effect of poisons. Effect of atmospheric pressure ⇒ 591.185 ⇒ 591.5	591.111.1	Composition of the blood. □ Erythrocytes (red corpuscles). Leucocytes (white corpuscles). Thrombocytes (blood platelets). Plasma (blood fluid). Serum. Fibrin
		591.111.2	Blood pigments. □ Haemoglobin. Haemocyanin. Echinochrome. Chlorocruorin. Haemerythrin. Achroglobin
		591.111.3	Congeaing, clotting, coagulation of blood. Haemolysis
591.046	Combined effects of cells and organs. Synergism. Division of labour	591.111.4	Physical properties of blood. □ Temperature. Viscosity. Osmotic pressure

- 591.111.7 Phagocytosis. Ingestion of foreign substances by leucocytes
- 591.111.8 Reticulo-endothelial defence systems. Antibodies. □ Precipitin reaction
- 591.112 Circulation of blood
- 591.112.1 Physiology of cardiac muscles. □ Stimulus formation. Sensitivity. Frequency of heartbeat
- 591.112.2 Activity of the heart. Valves. Cardiac sounds. Cardiac beat. □ Electrocardiogram
- 591.112.3 Blood pressure. Blood flow
- 591.112.4 Vascular nerves. Vasomotor fibres. Pressoreceptor nerves
- 591.112.5 Blood volume. Circulation time
- 591.112.6 Fluctuations without circulation
- 591.113 Haemolymph. Lymph of invertebrates
- 591.115 Fluid pressure inside a body. Turgor
- 591.12 Respiration. Breathing**
- 591.121 General physiology of respiration
- 591.122 Respiration over whole body surface. Cutaneous respiration
- 591.123 Respiration by aquiferous organs. Respiration by swim bladders
- 591.124 Intramolecular respiration
- 591.125 Respiration by branchiae (gills)
- 591.126 Respiration by tracheae (windpipes)
- 591.127 Respiration by lungs
- 591.128 Animal heat. □ Body temperature. Poikilothermal, 'cold-blooded' animals. Homeothermal, 'warm-blooded' animals. Temperature regulation. Insulation
- 591.13 Nutrition. Inanition**
⇒ 591.53
- 591.131 Seizure and comminution of food
- 591.131.1 Act of feeding in general. Feeding behaviour. Ways of ingesting food
⇒ 591.53
- 591.131.2 Killing of prey
- 591.131.3 Mastication. Saliva. Salivary acid
- 591.131.4 Digestion outside the mouth. Secretions for dissolution of nutrients outside the body
- 591.132 Digestion
⇒ 577.15
- 591.132.1 Mechanics of the stomach
- 591.132.2 Physiology of the stomach. □ Gastric juices
- 591.132.3 Rumination
- 591.132.4 Vomiting
- 591.132.5 Physiology of the midgut. □ Intestinal juice. Pancreatic juice. Bile (gall)
- 591.132.8 Regurgitation. Expulsion of indigestible elements in food through the mouth. Castings. Pellets
- 591.133 Physiological chemistry of foods and nutrition
- 591.133.1 Foods. Nutrients
- 591.133.11 Proteins
- 591.133.12 Fats
- 591.133.13 Carbohydrates
- 591.133.14 Water
- 591.133.15 Minerals
- 591.133.16 Vitamins
- 591.133.2 Synthetic metabolism. Assimilation of foods and nutrients. Enzymes. Ferments.
□ Anabolism. Catabolism
- 591.133.3 Food reserves and food storage in the animal body
- 591.133.31 Storage of fat
- 591.133.32 Storage of glycogen
- 591.133.33 Storage of protein
- 591.134 Growth
- 591.134.2 Rate of growth
- 591.134.3 Periodicity of growth
- 591.134.4 Course and duration of individual growth phase
- 591.134.5 Weight and bulk of the body
- 591.134.6 Ratio between body mass and body size
- 591.135 Physiology of the hindgut. □ Defecation. Evacuation of intestine
- 591.137 Uptake of nutrients and digestion in Protozoa
- 591.139 Senescence. Phenomena of ageing. Length of life. Rejuvenation. Death. Causes of death
- 591.14 Secretions. Excretions. Lymph. Glands in general**
- 591.141 Cutaneous secretions. □ Skin glands (function). Sweat glands. Lacrimal glands
- 591.142 Secretions in Protozoa
- 591.143 Serum. Serous fluids. Serous cavities
- 591.144 Lymphatic system of vertebrates (function). Spleen
- 591.145 Poisons. Poison glands. Toxins. Antitoxins
⇒ 591.431.6
⇒ 591.477.4
- 591.146 Secretions of the mammary glands. Milk
- 591.147 Internal secretions. Endocrine secretions. Hormones. □ Adrenalin. Insulin. Sex hormones. Androgens. Oestrogens
⇒ 577.17
- 591.148 Luminescence (bioluminescence)
⇒ 577.336
- 591.149 Excretions. Catabolism. □ Urine formation. Excreta (faeces, guano). Gaseous excretions. Sepia
- 591.15 Variation. Heredity**
⇒ 575.1
⇒ 575.2
⇒ 636.082
- 591.151 General polymorphism. Dimorphism
- 591.152 Geographic variation. Acclimatization
- 591.153 Heterophagous variation
- 591.155 Mimicry. Imitative (mimetic) variation
- 591.156 Sexual polymorphism
- 591.157 Variation of colour. Coloration (pigmentation).
□ Albinism. Melanism
⇒ 57.018.634
⇒ 572.522
- 591.158 Cross-bred (bastard) forms. □ Mongrels. Hybrids
- 591.159 Monstrous forms. Terata (monsters). Anomalies. Abnormalities
- 591.16 Generation. Procreation. Reproduction. Breeding. Biogenesis**
⇒ 57.017.5
⇒ 575
⇒ 591.56
⇒ 636.082
- 591.161 Abiogenesis (spontaneous generation)
⇒ 573.5
- 591.162 Parthenogenesis
- 591.163 Paedogenesis. Reproduction in immature stages. □ Larval reproduction. Pupal reproduction
- 591.164 Division (fission) in Protozoa
- 591.165 Vegetative propagation. Asexual reproduction.
□ Gemmation. Fragmentation
- 591.166 Gonochorism. Differentiation of sexes. Sexual reproduction in general
- 591.166.1 Mating. Copulation (including that of Protozoa)
- 591.166.5 Alternation of generations. □ Metagenesis. Alloëogenesis
- 591.167 Hermaphroditism. □ Autogamy. Autogamy
- 591.169 Regeneration. □ Sloughing. Moulting. Replacement

- 591.169.1 Physiological regeneration. Replacement of hair and teeth. Casting and renewal of antlers. Sloughing (casting of skin). Moulting. Replacement of epidermis, nails, claws
- 591.169.2 Occasional (reparative) regeneration. Repair of injuries
- 591.169.8 Autotomy. Self-amputation
- 591.17 Movement. Locomotion. Physiology of the skeleton and organs of locomotion**
- 591.171 Locomotion by legs. □ Walking. Running
- 591.172 Reptation. Crawling. Slithering
- 591.173 Swimming. Floating
- 591.174 Flying. Gliding
- 591.175 Physiology of the muscles
- 591.176 Skin. Pigment. Colouring matter
- 591.177 Mechanics of the skeleton and joints
- 591.178 Protoplasmic movement. Ciliary movement
- 591.18 Functions of the nervous system**
- 591.181 Excitability and physiology of the nerves in general. Electrophysiology
- 591.182 Nerves and muscles. □ Tonus. Reflex movements
- 591.185 Senses (reception). Tropisms. Taxes
⇒ 576.333
⇒ 591.044
- 591.185.1 Faculty of orientation
- 591.185.2 Mechanical senses. □ Touch (tactile sense). Sense of gravity (georeception). Geotaxis. Sense of balance (equilibrium). Sense of temperature (thermoreception). Thermotaxis. Sense of fluid flow (rheoreception). Rheotaxis. Chemical sensitivity (chemoreception). Chemotaxis
- 591.185.3
- 591.185.31 Sense of taste (gustation)
- 591.185.34 Sense of smell (olfaction)
- 591.185.5 Sense of hearing (audition)
- 591.185.6 Sense of light (photoreception). Sight. Vision. □ Phototaxis. Phototropism
- 591.185.62 Optical perception of direction
- 591.185.63 Optical perception of movement
- 591.185.67 Sense of colour
- 591.186 Hypnosis
- 591.187 Responses to stimuli in Protozoa
- 591.2 Diseases of (undomesticated) animals. Animal pathology**
Denote species by :592/599, diseases by :616...
591.2 ≡ 591.4
⇒ 636.09
- 591.3 Animal embryology. Animal ontogeny. Development of the individual organism**
- 591.31 Primary development of the egg.**
□ Fertilization. Segmentation
- 591.32 Formation of the embryonic rudiments**
- 591.33 Embryonic development without larval stage.** □ Ectoderm. Endoderm. Mesoderm. Mesenchyme
- 591.34 Metamorphosis. Larval stage**
- 591.341 Swarm cells (zoospores)
- 591.342 Larval forms of arthropods. □ Caterpillars. Maggots. Pupae. Chrysalises
- 591.343 Larval forms of Tunicata and vertebrates
- 591.35 Young animals**
- 591.35.02 Twinning research. □ Multiple births. Twins
- 591.351 Duration of development stage, to sexual maturity
- 591.36 Sexually mature animals**
- 591.366 Length of interval between breeding seasons (heats, ruts). Gestation period. Suckling period
⇒ 636.082.455
- 591.37 Hybridization. Crossing, cross-breeding. Interbreeding**
⇒ 591.158
⇒ 636.082.26
- 591.38 Inbreeding. Consanguineous breeding. Breeding in-and-in**
⇒ 636.082.25
⇒ 636.082.41
- 591.39 Experimental embryology. Mechanics of development**
Experiments on ova and spermatozoa
Experimental alteration of embryonic development. Determination of power of development
Experiments on larvae. Experimental alteration of metamorphosis
Experiments on surviving embryonic parts
- 591.391
- 591.392
- 591.393
- 591.398
- 591.4 Study of animal organs. Animal organology. Animal anatomy. Zootomy**
⇒ 611
- 591.4.068.1 Change of function
- 591.4.068.2 Atrophy
- 591.4.068.3 Transitory organs
- 591.4.068.4 Rudimentary organs
- 591.4.068.5 Homologous organs
- 591.4.068.6 Analogous organs
- 591.41 Circulatory organs. Circulatory system. Angiology**
- 591.411 Pericardium
- 591.412 Heart
- 591.413 Vessels. Arteries. Aorta
- 591.414 Veins. □ Portal vein
- 591.415 Capillaries
- 591.416 Vessels in the lower animals. Lacunae. Gastrovascular system
- 591.417 Echinoderm ambulacral system. Madreporites
- 591.42 Respiratory organs**
- 591.421 Nose
- 591.422 Larynx. Glottis. Vocal cords. Tympanum. Syrinx
- 591.423 Trachea (in vertebrates). Windpipe. Bronchi
- 591.424 Lungs
- 591.425 Pleura
- 591.426 Diaphragm
- 591.428 Branchiae. Branchial arches. Gills
- 591.429 Other respiratory organs. □ Tracheae (in invertebrates). Swim bladders. Siphons (of molluscs)
- 591.43 Organs of nutrition**
- 591.430 Organs of nutrition in Protozoa. □ Cytostome (cell mouth). Cytopyge (cytoproct, potential anus). Food vacuoles. Contractile vacuoles
- 591.431 Mouth
⇒ 591.145
⇒ 591.477.4
- 591.431.1 Proboscis for seizing food. Chitinous mandibles. Aristotle's lantern (in echinoderms)
- 591.431.2 Mouth appendages of arthropods
- 591.431.3 Tongue. Radula (in molluscs)
- 591.431.4 Teeth
- 591.431.5 Keratinous formations. □ Beaks. Horny plates. Baleen
⇒ 591.478

- 591.431.6 Oral glands. Salivary glands. Buccal glands.
Silk glands in Lepidoptera. Oral poison glands
⇒ 591.145
- 591.431.7 Cheek pouches (buccal pouches)
- 591.432 Pharynx. Oesophagus. □ Gullet. Crop.
Honey-bags
- 591.433 Stomach. Ventriculus. □ Gizzard
- 591.434 Gut. Intestine. □ Caecum. Rectum
- 591.435 Anus. Vent. Cloaca. □ Ink sac (in
cephalopods)
- 591.436 Hepar. Liver. Midgut glands of invertebrates.
□ Gall bladder
- 591.437 Pancreas. Hepatopancreas
- 591.438 Coeloms. Body cavities. Peritoneum
- 591.439 Corpora adiposa (fatty bodies)
- 591.44 Lymphatic system of vertebrates**
⇒ 591.144
- 591.441 Spleen
- 591.442 Lymphatic vessels. Lymph ducts. Lymph nodes
- 591.443 Thymus
- 591.444 Thyroid gland. Parathyroid gland
- 591.445 Suprarenal glands. Suprarenal medulla
- 591.446 Lymph glands
- 591.46 Urogenital system. Excretory organs.**
Sexual organs. Gonads. Sex glands
- 591.461 Excretory organs. □ Kidneys. Nephros
- 591.462 Urinary organs. □ Ureter. Urinary bladder
- 591.463 Male genital organs
- 591.463.1 Sperm. Spermatogenesis
- 591.463.2 Testes
- 591.463.3 Scrotum
- 591.463.4 Prostate
- 591.463.5 Spermatophores
- 591.463.6 Vas deferens
- 591.464 Male copulatory organs
- 591.464.1 Penis
- 591.464.3 Bulbus. Hectocotylus. Pterygopodia. Spadix.
Spicula
- 591.464.6 Secondary sexual characteristics. □ Scent
glands. Vocal organs
- 591.465 Female genital organs
- 591.465.1 Oögenetic processes
- 591.465.11 Ova. Eggs
- 591.465.2 Oviduct
- 591.465.3 Gonads. Ovaries
- 591.466 Uterus. Uterine glands. Shell glands (in
Platyhelminthes)
- 591.467 Female copulatory organs
- 591.467.1 Vagina
- 591.467.2 Vulva. Hymen
- 591.467.3 Ovipositor. Ovipositor spike
- 591.467.4 Receptaculum seminis
- 591.469 Mammae. Milk glands. Teats. Udders
- 591.47 Organs of motion, locomotion**
- 591.471 Skeleton
- 591.471.1 Skeleton formations in Protozoa.
□ Integuments. Scales. Shells. Endoskeleton
- 591.471.2 Skeletons of invertebrates
- 591.471.21 Skeletons of Spongiaria (Porifera). □ Sponges.
Spicules
- 591.471.22 Skeletons of Cnidaria. □ Nematophores.
Supportive lamellae. Calcareous structures of
corals
- 591.471.23 Skeletons of Echinoderma. □ Calcareous
plates. Pedicellariae. Spines
- 591.471.24 Skeletons of Mollusca. □ Operculum. Maxilla.
Shells. Cuttle-bone
- 591.471.25 Skeletons of Brachiopoda. □ Shells.
Calcareous spicules
- 591.471.26 Skeletons of Annelida. □ Integument.
Parapodia
- 591.471.27 Skeletons of Arthropoda. □ Chitinous
exoskeleton. Larval integument
- 591.471.28 Skeletons of Tunicata. □ Chorda dorsalis,
notochord. Mantle
- 591.471.29 Skeletons of other invertebrates
- 591.471.3 Osteology. Skeleton of vertebrates
- 591.471.31 Exoskeleton. □ Ganoid scales. Placoid scales.
Cutaneous teeth. Bony plates
- 591.471.32 Vertebral column and spinal cord
- 591.471.33 Ribs and breastbone
- 591.471.34 Scapular arch. Shoulder girdle
- 591.471.35 Pelvic girdle
- 591.471.36/37 Limbs. Extremities
- 591.471.36 Unpaired limbs
- 591.471.37 Paired limbs. □ Fins. Wings. Arms. Hands.
Legs. Feet
- 591.471.4 Skull. Cranium
- 591.471.41 Cartilaginous skull in elasmobranch fishes
(gristly fishes)
- 591.471.42 Skull bones in general
- 591.471.43 Visceral cranium. Facial skeleton
- 591.471.44 Cerebral cranium
- 591.472 Articulations, joints. Ligaments. Articular
membranes (in Arthropoda). Hinges (in
bivalves)
- 591.473 Muscle. Myology
- 591.473.1 Fibrils of unicellular animals. Myoneme
- 591.473.2 Muscular system of invertebrates.
□ Myoseptum. Ambulacral feet. Epithelial
muscle cells (in Cnidaria). Foot, siphon (in
Mollusca)
- 591.473.3 Muscular system of vertebrates. □ Skeletal
musculature. Voluntary, striated muscle.
Visceral musculature. Involuntary, smooth
muscle
- 591.473.9 Electric organs
- 591.474 Tendons
- 591.475 Mucus sac
- 591.476 Connective tissue. Parenchyma. Mesoderm.
Mesenchyme
- 591.477 Skin. Suckers. Mantle (in Mollusca etc.)
- 591.477.1 Ectoderm. □ Suckers. Chaetae. Mantle
- 591.477.2 Entoderm
- 591.477.3 Epidermal glands. Skin glands. □ Sweat
glands. Sebaceous glands. Mucus glands.
Scent glands. Lacrimal glands. Mammary
glands. Silk glands. Sting glands
- 591.477.4 Poisonous (venomous) organs. Stings
⇒ 591.145
⇒ 591.431.6
- 591.477.9 Luminous organs
- 591.478 Keratinous formations. □ Hair. Fur. Bristles.
Prickles. Claws. Talons. Hooves. Feathers.
Scales. Horns
For fur as a product see 675.6
⇒ 591.431.5
- 591.479 Pseudopodia. Cilia. Flagella. Cirri. Trichocysts
- 591.48 Nervous system**
- 591.481 Central nervous system
- 591.481.1 Brain of vertebrates. Encephalon.
□ Cerebellum
- 591.481.2 Hypophysis (pituitary body)
- 591.481.3 Pineal apparatus. □ Epiphysis (pineal body,
pineal gland). Parietal organ (pineal eye)
- 591.481.4 Ganglia in general
- 591.481.8 Neurons (neurones)
- 591.482 Spinal cord
- 591.483 Peripheral nervous system. Sympathetic
nervous system. Nerves

- 591.484 Organs of sight
 591.484.6 Eyes of vertebrates
 591.484.7 Parietal eye
 591.485 Organs of hearing. Ears
 591.486 Organs of smell. Nose
 591.487 Organs of taste. Chemoreceptors in general
 591.488 Mechanical sense organs. ☐ Thermoreceptors (organs of heat sense). Organs of touch (tactile sense). Organs of equilibrium (balance). Labyrinthine organs. Lateral line
- 591.49 Somatology. Shape of the animal body**
 591.491 Personae corporum. Zooids. Proglottides
 591.491.1 Giants, giant forms
 591.491.2 Dwarfs, dwarf forms
 591.492 Antimeres (actinomerer). Parameres. Metamerer (meromerer). Bodily symmetry
 591.493 Head
 591.494 Cephalothorax. Neck
 591.495 Thorax. Chest
 591.496 Abdomen. Hindquarters
 591.497 Tail
 591.498 Extremities. ☐ Limbs. Antennae, feelers. Fins. Flippers. Wings. Beaks, bills
 591.499 Body appendages. ☐ Elytra (wing-cases). Insect wings. Tentacles
- 591.5 Animal habits. Animal behaviour. Ecology. Ethology. Animal and environment. Bionomy**
 ◇ 591.5.081.15:654.937 Monitoring of animal behaviour by remote indication, e.g. electronic tagging
 ⇒ 502.211
 ⇒ 591.131.1
 ⇒ 631.95
- 591.51 Psychology of animals. Animal intelligence. Instinct**
 591.511 Theory and philosophy of animal psychology. Animal souls
 591.512 Instinct. Instinctive behaviour. ☐ Brood care. Courtship
 591.513 Learning. Learning mechanisms. Learned behaviour. ☐ Trial and error. Conditioning
- 591.52 The animal in relation to the area**
 ⇒ 574.91
 591.521 Style of habitation. ☐ Lairs. Dens. Underground dwellings. Burrows. Earths. Nests as dwellings
 591.522 Habitat
 591.523 Single migrations
 591.524 Special adaptations of animal species to territory (organizational adaptations)
 591.524.1 Special adaptations of aquatic animals
 591.524.2 Special adaptations of land-dwelling (terrestrial) animals
 591.525 Special adaptations of individual animals to changes in conditions of life (regulatory adaptation)
 591.526 Population density of an animal species. ☐ Reproduction rate. Birth rate. Destruction rate. Death rate
- 591.53 Nutrition. Animal nutrients, foodstuffs**
 ⇒ 591.131.1
 | 591.53.063 Diet. Dietary changes
 591.531 Classification of animals according to dietary preference
 ⇒ 591.13
 ⇒ 591.557
 591.531.1 Plant eaters in general. Herbivores. Phytophages
- 591.531.2 Animal eaters in general. Flesh eaters. Carnivores. Sarcophages. Predators
 591.531.22/.29 Carnivores according to prey
 591.531.22/.29 ≡ 592/599
 591.531.257 Insect-eaters (insectivores)
 591.531.3 Animal-and-plant eaters. Omnivores. Pantophages
 591.531.4 Animals with peculiar diets. ☐ Mud eaters. Sand eaters. Dust eaters. Carrion eaters. Scavengers. Dung eaters. Coprophages
 591.532 Classification of animals according to diversity of diet. ☐ Polyphages (diverse feeders). Oligophages (selective feeders). Monophages (specialized feeders)
 591.536 Assembly of food stores, larders. Food hoarders
 591.538 Nutritional requirements according to mode of life
- 591.54 Climatic conditions. Influence of seasons**
 591.541 Water
 591.542 Light. ☐ Cave animals (cavernicoles). Demersal (deep-sea) animals. Diurnal and nocturnal animals. Sleep
 591.543 Climate
 591.543.1 Temperature. Cold-weather (cryophilic) animals. Warm-weather (thermophilic) animals
 591.543.4 Alternation of climatic conditions. Seasons
 591.543.41 Dormant germs. Dormant eggs. Encysting
 591.543.42 Aestivation (estivation). Hibernation. Suspended animation
 591.543.43 Seasonal migration. ☐ Bird migration. Migration to spawning grounds
 591.544 Cosmic influences. ☐ Influence of Sun. Influence of Moon. Influence of cosmic radiation
 591.545 Seasons and reproduction. ☐ Spawning season. Rutting season. Mating season
 ⇒ 636.082.45
- 591.55 Animal societies. Animal sociology. Communal life**
 591.551 Relations between the sexes. ☐ Pairing (cohabitation) of sexes. Devices for location and recognition of the sexes by each other. Rut. Fighting among males, sparring. Courtship rituals
 ⇒ 591.464.6
 ⇒ 591.58
 591.552 Relations between members of a species. Gregarious formations. ☐ Herds. Flocks. Colonies. Associations
 591.553 Animal communities, biocoenoses
 ⇒ 574
 591.554 Animal fighting beyond the degree necessary for provision of food
 591.555 Sources and motives of social phenomena in animals
 591.555.1 Socially significant characteristics of individuals. ☐ Maternal instinct. Instinct of self-preservation. Volitional and emotional drives
 591.555.2 Group consciousness. Collective awareness
 591.555.3 Socially significant processes and norms. ☐ Reactions. Defences. Methods of influencing
 591.555.4 Aims and goals which have social significance
 591.556 Social relationships within the species
 591.556.1 The socially significant environment. Social context
 591.556.2 Relationships according to their direction. ☐ Approach, integration; estrangement, avoidance. Dependence; dominance. Tolerance; rivalry

- 591.556.3 Relationships according to the number of individuals involved
- 591.556.4 Relationships according to immediacy of action. □ Direct relationships. Indirect relationships
- 591.556.5 Relationships between groups and individuals
- 591.557 Relations between individuals of different species
⇒ 576.8
⇒ 581.557
⇒ 591.69
- 591.557.6 Symbiosis (living together with mutual advantage)
- 591.557.7 Commensalism (living together with one-sided advantage but without damage to the host).
□ Synoeketes
- 591.557.8 Parasitism (obtaining food at expense of host)
- 591.56 Provision for posterity. Care of young.**
Brood care
- 591.561 Passive brood care
- 591.562 Active brood care in general
- 591.563 Nest building. Egg cocoons. Insect structures
- 591.564 Conveying of eggs and embryos (larvae) to feeding grounds
- 591.565 Lodging of eggs and embryos in or on the body of the parent animal
- 591.566 Skin folds. Pouches. Marsupium
- 591.567 Viviparity
- 591.568 Care and feeding of the young
- 591.57 Defence. Attack**
- 591.571 Behaviour when in danger. □ Alarm response. Flight
- 591.572 Protective adaptations of the body
- 591.572.1 Shells and armour of body materials. Spines. Quills
- 591.572.4 Mucus and wax secretion
- 591.572.5 Stinging capsules. Poisonous spines. Irritant hairs. Poisonous juice. Venom. Repellent fluids. Sepia
- 591.572.6 Production of alarm calls. Warning signals
- 591.572.7 Warning and deterrent colours and markings
- 591.572.8 Protective imitation. Mimicry. Adaptive coloration
- 591.573 Autotomy as defence mechanism
- 591.574 Cleanliness
- 591.576 Protective colouring. Camouflage
- 591.577 Weapons for attack (offence)
- 591.58 Communication and expression by sound or ultrasonic vibrations, mime or other signals, smell, touch or chemical sense (chemoreception)**
- 591.582 Acoustic communication. □ Communication by sounds. Song. Croaks. Animal speech. Calls. Stridulation. Purring. Communication by ultrasonic vibrations
- 591.584 Communication through visual signals and movements. □ Mime. Body language. Displaying. Posture. Fleering. Grimacing
- 591.586 Communication through contact, touch. □ Use of feelers, antennae
- 591.6 Economic zoology. Applied zoology**
⇒ 636
⇒ 638
⇒ 639
- 591.61 Utility of animals**
- 591.611 Hunting and capture of animals
- 591.612 Training in captivity. Breaking in. Domestication
- 591.613 Breeding. Rearing
- 591.615 Protection and care of animals. □ Winter feeding
- 591.617 Deliberate use of animals by humans to combat other animals
⇒ 632.937
- 591.618 Deliberate use of animals by humans to control plants
⇒ 632.937
- 591.619 Diseases and pests of useful and harmful animals
- 591.65 Harmful animals**
⇒ 632.6/.7
- 591.67 Disease-spreading animals**
- 591.68 Animals that damage inanimate things**
- 591.69 Parasites**
⇒ 576.8
⇒ 581.557
⇒ 591.557
- 591.8 Animal histology**
Denote microstructure of a particular organ by :591.4...
- 591.81 Histology of the cell**
- 591.82 Histology of the connective tissue**
- 591.83 Histology of the cartilage (chondrus)**
- 591.84 Histology of osseous tissue (bone)**
- 591.85 Histology of blood and lymph.**
□ Chromatophores
- 591.86 Histology of muscle tissue (myoplasm)**
- 591.87 Histology of the epithelium**
- 591.88 Histology of the nerve tissue.** □ Nerve cells. Nerve fibres
- 591.9 Geographic zoology. Zoogeography. Fauna. Geographic distribution of animals**
Details by (1/9) (Table 1e)
◇ 591.9(23.03) High-mountain fauna
◇ 591.9(26) Marine fauna
⇒ 574.91
⇒ 591.523
- 591.91 Means of distribution**
- 591.92 Obstacles to distribution**
- 591.93 Significance of antecedents in distribution**
- 591.94 Faunal relicts**
- 591.95 Human interference with fauna**
- 592/599 SYSTEMATIC ZOOLOGY**
Detailed subdivision is often made in only one place, according to whether the extinct or living form is more important. If required, either section may be further subdivided like the other
⇒ 562/569
⇒ 57.06
- 592/599-1 Aspects of general zoology applied to particular animals**
592/599-1 ≡ 591
- 592 INVERTEBRATA**
- 593.1 Protozoa**
- 593.11 Rhizopoda in general**
- 593.12 Amoebozoa. Foraminifera**
- 593.121 Amoeba**

593.13	Heliozoa	593.9	Echinoderma
593.14	Radiolaria. Marine plankton	593.91	Pelmatozoa. Crinoidea. Featherstars. Sea-lillies
593.16	Flagellata (Mastigophora)	593.92	Asterozoa
593.161	Euflagellata. Trypanosomatidae	593.93	Asteroidea. Starfish
593.161.3	Euglenidae ⇒ 582.277	593.94	Ophiuroidea. Brittlestars
593.161.4	Chromomonadina	593.95	Echinoidea. Sea-urchins. Cake-urchins. Heart-urchins
593.161.52	Volvocidae	593.96	Holothurioldea. Sea-cucumbers
593.162	Dinoflagellata. Peridiniidae ⇒ 582.276	593.99	Hemichordata. Enteropneusta. Acorn worms
593.163	Cystoflagellata. Noctiluca	594	MOLLUSCA. SHELLFISH ETC.
593.17	Ciliata (Infusoria)	594.1	Lamellibranchiata (Pelecypoda). Bivalves. Mussels etc.
593.171	Holotricha. Paramaeciidae	594.11	Anisomyaria. Clams
593.172	Heterotricha. Stentoridae	594.121	Ostreidae. □ Oysters. Scallops
593.173	Oligotricha	594.124	Mytilidae. □ Sea mussels (common mussels)
593.174	Hypotricha	594.13	Homomyaria
593.175	Peritricha	594.14/.17	Heterodonta
593.176	Suctorina	594.141	Nayadidae. Unionidae. Freshwater mussels
593.19	Sporozoa	594.151	Cardiidae. Cockles
593.191	Gregarinida	594.18/.19	Desmodonta
593.192	Coccidiomorpha	594.185	Myidae. Gapers (soft clams)
593.193	Sarcosporidia	594.188	Pholadidae. Teredo. Ship-worm
593.194	Myxosporidia	594.2	Solenocoencha. Scaphopoda
593.195	Microsporidia	594.3	Gastropoda. Univalves. Snails etc.
593.3	Coelenterata	594.31	Amphineura. Chitons. Coat-of-mail shells
593.4	Spongiaria (Porifera). Sponges	594.32	Streptoneura. Prosobranchia. Operculates. □ Limpets. Top shells. Wentletraps. Cowries. Winkles (periwinkles). Whelks. Spire shells. Cone shells 594.32 ≡ 564.32
593.41	Myxospongia. Slime sponges	594.323	Cyclobranchia. Docoglossa. Limpets
593.42	Ceratospongia. Horny sponges	594.34	Heteropoda
593.43	Monactinellida (Halichondriidae). Breadcrumb sponge. Freshwater sponges	594.35	Opisthobranchia. □ Sea slugs
593.44	Tetractinellida	594.36	Nudibranchiata. □ Common grey sea slug. Sea lemon
593.45	Hexactinellida. Glass sponges	594.37	Tectibranchiata. Sea-hares
593.46	Calcispongiae. Purse sponges	594.38	Pulmonata. Lung-bearing gastropods. □ Land snails. Garden snails. Slugs. Water snails (freshwater snails)
593.5	Cnidaria	594.5	Cephalopoda
593.6	Anthozoa. Corals	594.51	Tetrabranchiata
593.61	Rugosa. Tetracoralla	594.52	Nautiloidea
593.62	Alcyonaria. Octocoralla. Precious corals. Fan-coral. Organ-pipe coral	594.53	Ammonoidea. Ammonites
593.63	Zoanthinaria. Hexacoralla	594.55	Dibranchiata
593.65	Actinaria. Sea anemones	594.56	Octopoda. Octopuses
593.66	Madreporaria	594.58	Decapoda
593.67	Tabulata	594.581	Belemnoidea. Belemnites
593.7	Hydrozoa. Polyps. Medusas	594.582	Sepioidea. □ Squids. Cuttlefish
593.71	Hydroidea	594.6	Molluscoidea (Tentaculata)
593.711	Hydrariae. Hydridae	594.7	Bryozoa. Ectoprocta. Polyzoa. □ Moss animalcules. Sea-mat
593.712	Tubulariae (Anthomedusae)	594.8	Brachiopoda. Lamp-shells
593.714	Campanulariae (Leptomedusae)		
593.715	Trachymedusae		
593.719	Graptolithes. Graptolites		
593.72	Siphonophora. Portuguese men-of-war		
593.73	Scyphozoa. Jellyfish. Scyphomedusae. Acalephae		
593.74	Stauromedusae. Calycosoa. Stalked jellyfish		
593.75	Rhizostomeae		
593.76	Lobomedusae		
593.77	Discomedusae		
593.8	Ctenophora. Comb-jellies. Sea-gooseberries. Sea-walnuts		

595	ARTICULATA	595.4/.7	Tracheata
595.1	Vermes. Worms. Helminths	595.4	Arachnoidea. Arachnids in general
595.11	Scolecida	595.41	Linguatulidae. □ Tardigrades (water bears)
595.12	Platyhelminthes. Flatworms	595.42	Acarina. □ Mites. Soil-mites. Itch-mites. Ticks. Sheep-tick. Fowl-tick
595.121	Cestodes. Tapeworms	595.43	Opiliones. □ Harvestmen (daddy-longlegs)
595.122	Trematodes. Flukes. □ Liver fluke	595.44	Araneida. Spiders. □ House-spider. Garden-spider. Money-spider. Bird-spiders. Tarantulas
595.123	Turbellaria. □ Triclad. Planarians. Polyclads	595.45	Pedipalpi. □ Whip-scorpions. Scorpion-spiders
595.124	Nemertini. Ribbon-worms	595.46	Scorpionidea. Scorpions
595.13	Nemathelminthes	595.47	Pseudoscorpionidae. False scorpions. Chelifers
595.131	Nematomorpha. □ Horsehair-worm (threadworm, hairworm)	595.48	Solifugae
595.132	Nematodes. □ Roundworms. Threadworm. Eelworm. Vinegar eel. Gapeworm. Hookworm	595.5	Protracheata (Onychophora). Peripatidae
595.133	Acanthocephali. Echinorhynchidae. Spiny-headed worms	595.6	Myriapoda
595.135	Chaetognatha. Arrow-worms. Sagitta	595.61	Diplopoda. Millipedes. □ Pselaphognatha. Chilognatha
595.14	Annelida. Segmented worms in general	595.62	Chilopoda. Centipedes
595.142	Chaetopoda. □ Polychaeta (bristle worms). Lugworms. Sea-mouse. Myzostomaria. Oligochaeta. Freshwater worms. Earthworms	595.63	Symphyla. Scolopendrellidae
595.143	Hirudinea. Leeches. □ Horse-leech. Duck-leech	595.64	Pauropoda
595.144	Echiuroidea. Spoonworms	595.7	Insecta (Hexapoda). Insects. Entomology
595.145	Sipunculoidea	595.71	Apterygogenea. □ Collembola (spring-tails, snow fleas). Thysanura (bristle-tails, silverfish)
595.18	Rotatoria. Rotifers (wheel animalcules)	595.72	Orthoptera
595.2	Arthropoda in general	595.721	Dermaptera. Forficulidae. Earwigs
595.3	Branchiata. Crustacea	595.722	Blattodea. Blattidae. Cockroaches. Blackbeetles
595.31	Entomostraca in general	595.724	Phasmodea. Phasmatidae. Stick-insects
595.315	Pantopoda. □ Pycnogonids (sea-spiders)	595.725	Mantodea. Praying mantis
595.32	Phyllopoda	595.727	Acridiidae. Short-horned grasshoppers. □ Common grasshoppers
595.323	Euphyllopoda. Fairy shrimps	595.728	Locustidae. Long-horned grasshoppers. □ Locusts
595.324	Cladocera. Daphnia (water fleas)	595.729	Gryllidae. □ Crickets. House-cricket. Field-cricket. Mole-cricket. Bush-crickets (great green grasshopper etc.)
595.33	Ostracoda	595.73	Archiptera
595.34	Copepoda. Cyclopes. Fish-lice	595.731	Thysanoptera (Physopoda). Thrips
595.35	Cirrepedia. Barnacles. □ Acorn-barnacle. Ship's barnacle	595.732	Corrodentia. □ Termites. Woodlice. Booklouse
595.36	Malacostraca in general	595.733	Odonata. □ Damsel-flies. Dragonflies
595.37	Arthrostraca	595.734	Ephemeroidea. Mayflies
595.371	Amphipoda. □ Freshwater shrimps. Sandhoppers. Sand-shrimps	595.74	Neuroptera. □ Lacewings. Sponge-flies. Ant-lions. Caddis-flies
595.373	Isopoda. □ Woodlice (slaters). Water-lice. Gribbles. Sea-slaters	595.75	Hemiptera (Rhynchota)
595.38	Thoracostraca	595.751	Aptera. □ Lice. Nits. Crab-lice ('crabs')
595.381	Cumacea. Cumaceans	595.752/.753	Homoptera
595.382	Stomatopoda. Squillidae	595.752	Phytophthires. □ Plant-lice. Aphids. Greenfly. Blackfly. Coccids. Scale-insects. Mealybugs. Lac-insects. Cochineal insects
595.383	Schizopoda. □ Mysids (opossum shrimps)	595.753	Auchenorhyncha. □ Cicadas. Fulgorids. Leaf-hoppers
595.384	Decapoda	595.754	Heteroptera. Hemiptera. □ Bugs. Plant-bugs. Barkbugs. Bedbugs (chinchies). Water-bugs. Water-boatmen. Pond-skaters
595.384.1	Macrura. □ Shrimps. Prawns. Lobsters. Crawfish (spiny lobster). Crayfish	595.76	Coleoptera. Beetles
595.384.2	Brachyura. Crabs in general		<i>The subdivision is based on that of Winkler</i>
595.384.4	Oxystomata		
595.384.5	Oxyrhyncha. □ Maiidae. Spider crabs		
595.384.6	Cyclometopa. □ Edible crabs. Swimming crabs (fiddler crabs)		
595.384.7	Catametopa. □ Pea crabs		
595.384.8	Anomura. □ Hermit crabs. Soldier crab		
595.386	Leptostraca. Phyllocarida		
595.39	Palaeostraca		
595.392	595.39 ≡ 565.39 Xiphosura. King-crabs		

- 595.762 Adephaga. ☐ Carabids (ground-beetles).
Water beetles. Diving beetles. Whirligigs.
Tiger-beetles
- 595.763/768 Polyphaga
595.763 Palpicornia. Staphylinoidea. ☐ Scavenger
beetles. Burying beetles. Carrion beetles. Rove
beetles. Skipjacks (click beetles, wireworms).
Ladybirds (ladybugs)
- 595.764 Lamellicornia. ☐ Scarabeidae. Dung-beetles.
Chafers. Cockchafer (maybug, rookworm).
Summer chafer (June bug). Garden chafer
(bracken clock). Lucanidae. Stag beetles.
Rhinoceros beetles
- 595.765 Sternoxia. ☐ Jewel beetle
595.766 Malacodermata. Fossipedes. Terebrilia.
☐ Glow-worms. Fireflies
- 595.767 Heteromera. ☐ Spanish fly (blister beetle).
Cardinal beetle. Hardback beetles
(mealworms)
- 595.768 Phytophaga. Rhynchophora
595.768.1 Phytophaga. Longhorn beetles. Timberman.
Musk beetle. Leaf beetles. Colorado beetle
Rhynchophora. Weevils. Bark beetle
- 595.768.2
- 595.77 Diptera. Flies**
595.771 Nematocera (thread-horns). ☐ Crane flies.
Gnats. Midges. Mosquitoes. Black flies
(Simuliidae)
- 595.772 Brachycera (short-horns). ☐ Tabanids.
Horsefly. Gadfly. Bottle. Soldier flies. Snipe
flies. Window flies
- 595.773 Cyclorapha
595.773.1 Aschiza. ☐ Syrphids. Drone-flies. Hover-flies
595.773.4 Schizophora. ☐ Bluebottles (blowflies).
Warble-flies. House-flies. Flesh-flies. Fruit-flies.
Dung-flies. Gall-flies. Leaf miners
- 595.774 Pupipara
595.775 Aphaniptera (Siphonaptera). ☐ Fleas. Chigoes
(chiggers, jiggers)
- 595.78 Lepidoptera. Butterflies. Moths**
595.782 Microlepidoptera. Smaller moths. ☐ Clothes
moth
- 595.783 Macrolepidoptera. Larger moths and butterflies
in general
- 595.785 Geometridae. Geometers (loopers)
595.786 Noctuidae (Caradrinidae). ☐ Underwings.
Cabbage moth. Turnip moth
- 595.787 Bombyces. ☐ Silkworm. Goat-moth. Nun-moth.
Emperor-moth. Lackey-moth. Eggar-moth.
Lappet-moth
- 595.788 Sphingidae. ☐ Hawk-moths. Death's head
hawk-moth. Eyed hawk-moth
- 595.789 Rhopalocera (Papilionina). Butterflies.
☐ Hairstreaks. Vanessa. Red admiral.
Painted lady. Camberwell beauty. Peacock
butterfly. Tortoiseshells. Swallowtail. Apollo.
White butterflies. Cabbage white. Brimstone
butterfly
- 595.79 Hymenoptera**
*The subdivision is based on that of
Schmiedeknecht*
- 595.792 Entomophaga. ☐ Ichneumon flies. Gall wasps.
Chalcid wasps
- 595.793 Symphyta. ☐ Sawflies. Borers. Wood-wasps
(horn-tails)
- 595.794 Aculeata. Stinging insects in general
595.795 Chrysididae. Ruby-tail wasps
595.796 Formicidae. Ants
595.797 Sphecidae. Sand-wasps. Digger wasps
595.798 Vespidae. Social wasps. ☐ Wasp (common
wasp). Hornet
- 595.799
- 596/599**
596.2
- 596.4**
596.6
596.61
596.611
- 597/599**
597.2/5
597.2
- 597.21**
597.211
- 597.3**
597.31
597.311
597.311.1
597.311.2
- 597.311.3
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597.523
- Apidae. Bees. ☐ Honeybee. Bumblebees
(humblebees). Leaf-cutting bees. Mining bees.
Carpenter bees
- CHORDATA**
Urochordata (Tunicata). ☐ Ascidiacea:
Sea squirts. Thaliasca: Salps. Larvacea:
Larvaceans
- Cephalochordata: Lancelets**
Myxini
Myxiniformes
Myxinidae: Hagfishes
- VERTEBRATA**
Pisces: Fishes. Ichthyology
Cephalaspidomorphi
597.2 $\approx$ 567.2
- Petromyzontiformes**
Petromyzonidae: Lampreys
- Chondrichthyes. Cartilaginous fishes**
Elasmobranchii
Galeomorphi
Heterodontiformes. Hornsharks
Lamniformes. ☐ Megamouth shark. Thresher
sharks. Basking sharks. Mako. Great white
shark
Orectolobiformes. ☐ Nurse sharks. Whale
sharks
Carcharhiniformes. ☐ Cat sharks.
Houndsharks
Squatinomorphi
Squatiniformes. Angel sharks
Squalomorphi
Hexanchiformes. ☐ Frilled sharks. Cow sharks
Squaliformes. ☐ Dogfish. Bramble Sharks
Pristiophoriformes: Saw sharks
Batoidea: Skates and Rays
Rajiformes: Skates
Pristiiformes: Sawfishes
Torpediniformes: Electric rays
Myliobatiformes. ☐ Sting rays. Eagle rays.
Manta rays
- Holocephali**
Chimaeriformes: Chimaeras (Ratfish)
- Osteichthyes (Bony fishes)**
Sarcopterygii
Dipnoi: Lungfishes
Ceratodontiformes. Australian lungfishes
Lepidosireniformes. South American and
African lungfishes
Coelacanthimorpha
Coelacanthiformes: Coelacanth
- Actinopterygii: Ray-finned fish**
Chondrostei
Polypteriformes: Bichirs
Acipenseriformes. ☐ Sturgeons. Paddle fishes
- Neopterygii**
Ginglymodi
Lepisosteiformes: Garfishes
- Halecomorphi**
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- Teleostei. Teleosts**
Osteoglossomorpha
Osteoglossiformes. ☐ Bonytongues. Butterfly
fishes. Arapaima

- 597.53 **Elopomorpha**
 597.531 Elopiformes. □ Tarpons
 597.533 Albuliformes. Including: Bonefishes. Spiny eels
 597.535 Anguilliformes: Eels. □ Conger. Moray
 597.537 Saccopharyngiformes: Deep sea eels
- 597.54 **Clupeomorpha**
 597.541 Clupeiformes. □ Anchovies. Herrings. Shads. Sardines
- 597.55 **Euteleostei**
 597.551 Ostariophysi
 597.551.1 Gonorhynchiformes. □ Milkfish. Beaked sandfishes
 597.551.2 Cypriniformes. □ Minnows and carps. Loaches. Chub
 597.551.21 Goldfish
 597.551.3 Characiformes: Characins. □ Piranhas
 597.551.4 Siluriformes: Catfishes
 597.551.5 Gymnotiformes. □ Electric eel
 597.552 Protacanthopterygii
 597.552.1 Esociformes. □ Pikes
 597.552.3 Osmeriformes. □ Smelts
 597.552.5 Salmoniformes
 597.552.51 Salmonidae
 597.552.511 Salmon
 Class here works dealing with Salmo salar (Atlantic salmon) and Oncorhynchus (Pacific salmon)
 597.552.512 Trout
 597.553 Stenopterygii
 597.553.1 Stomiiformes. □ Bristlemouths. Dragonfishes
 597.553.5 Ateleopdiformes: Jellynose fishes
 597.554 Scopelomorpha
 597.554.1 Aulopiformes. □ Spiderfishes. Lizardfishes
 597.554.5 Myctophiformes: Lanternfishes
 597.555 Paracanthopterygii
 597.555.1 Percopsiformes. □ Troutperches. Cavefishes
 597.555.3 Ophidiiformes. □ Caropids. Brotulas
 597.555.5 Gadiformes: □ Cods. Hakes
 597.555.51 Cod (Gadus sp.)
 597.555.7 Batrachoidiformes: Toadfishes
 597.555.9 Lophiiformes. □ Frogfishes. Batfishes. Deep sea angler fishes
 597.556 Acanthopterygii. Spiny finned fishes
 597.556.11 Atheriniformes. □ Rainbow fishes. Silversides
 597.556.13 Beloniformes. □ Flying fishes
 597.556.15 Cyprinodontiformes. □ Killifishes
 597.556.17 Lampridiformes. □ Opah. Ribbonfishes. Oarfishes
 597.556.19 Stephanoberyciformes. □ Gibberfishes. Whalefishes
 597.556.21 Beryciformes. □ Spiny fins. Squirrelfishes. Beardfishes
 597.556.23 Zeiformes. □ Dories
 597.556.25 Gasterosteiformes. □ Sand eels
 597.556.253 Sticklebacks
 597.556.255 Pipefishes and Seahorses
 597.556.27 Synbranchiformes. □ Swamp eels. Spiny eels
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 597.556.31 Scorpaeniformes. □ Gurnards. Rockfishes. Flatheads. Bullheads
 597.556.33 Perciformes
 597.556.331.1 Percoidei. □ Snooks. Perches. Sea basses. Sunfish. Jacks. Remoras. Snappers. Breams. Angelfishes
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 597.556.331.5 Labroides. □ Sclerids. Wrasses. Parrotfishes
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- 597.556.332.7 Gobiesocoides: Clingfish
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- 597.6 **Amphibia. Amphibians in general**
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- 598.115.32 Elapidae: Front-fanged snakes. □ Cobras. Mambas. Coral snakes. Sea snakes
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598.243.3 Scolopacidae: Sandpipers. □ Dunlin. Stints. Phalaropes. Curlews. Whimbrels. Godwits. Woodcock. Snipe
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598.243.8 Laridae: Gulls. □ Kittiwake. Terns
598.244 Ciconiiformes
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- 598.25 **Anseriformes**
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598.252.2 Geese. □ Brent goose. Barnacle goose. Canada goose. Greylag. Pink-footed goose
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- 598.26 **Galliformes. Columbiformes**
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598.261.6 Tetraonidae: Grouse. □ Prairie chicken. Ptarmigan
598.261.7 Phasianidae: □ Domestic fowl (chickens). Partridges. Pheasants. Peacock. Quails
598.261.8 Numididae: Guineafowl
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598.271.2 Loriidae: Lories
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598.272 Piciformes
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598.272.6 Picidae: □ Woodpeckers. Wrynecks. Sap-suckers
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598.274.12 Cuculidae: □ Cuckoos. Roadrunners. Coucals.
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- 598.276 Apodiformes
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 □ Condors
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 598.294.2 Passeridae: Old World Sparrows. □ Tree
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 598.295 Vireonidae: Vireos
 598.296 Fringillidae. Parulidae. Thraupidae.
 Emberizidae. Icteridae
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 Thraupidae: Tanagers
 598.296.2 Emberizidae: Buntings and New World
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 Emberizidae. Icteridae
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 Parulidae: New World warblers
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 Yellowhammer. Ortolan. Savannah sparrow
 Icteridae: New World blackbirds
MAMMALIA. MAMMALS
Prototheria (Eplacentalia). Egg-laying mammals
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 Ornithorhynchidae: Duck-billed platypus
Metatheria. Metatherian mammals
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Diprotodontia in general. □ Burramyidae.
 Pygmy possums. Petauridae. Gliding and
 Ring-tailed possums. Tarsipedidae. Honey
 possums
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 Vombatidae: Wombats
 Phalangeridae: Phalangiers
 Macropodidae: Kangaroo family. □ Kangaroos.
 Wallaroos. Wallabies. Quokkas
Polyprotodontia. □ Myrmecobiidae: Numbat.
 Peramelidae: Bandicoots. Notoryctidae:
 Marsupial mole. Dasyuridae: Australian
 marsupial cats. Thylacinae: Marsupial wolf.
 Caenolestidae: Shrew or Rat opossums
 Didelphidae: American opossums
Eutheria. Placental mammals
Pholidota. Xenarthra (Edentata)
 Pholidota. Manidae. Pangolins (scaly
 anteaters)
 Xenarthra (Edentata)
 Bradypodidae: Sloths
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Rodentia. Lagomorpha
 Rodentia. Rodents
 Sciuromorpha. Squirrel-like rodents
 Aplodontiidae: Mountain beaver (Sewellel)
 Scuridae. □ Squirrels. Chipmunks. Marmots.
 Flying squirrels. Souseliks. Prairie dogs
 Castoridae. Beavers
 Geomyidae: Pocket gophers
 Anomaluridae: Scaly-tailed squirrels
 Pedetidae: Spring hare (Springhaas)
 Heteromyidae. □ Kangaroo rats. Pocket mice
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 Gliridae. Dormice
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 Hesperominae: New World mice and rats.
 □ Field mice. Deer mice
 Cricetinae: Hamsters

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599.324.2	Thryonomyidae: Cane rats. Petromuridae: Rock rat (Dassie rat). Ctenodactylidae: Gundis	599.73	Artiodactyla. Even-toed ungulates
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599.324.3	Hystriidae: Old World porcupines	599.731.1	Suidae. □ Swine (pigs, hogs). Wild boar.
599.324.4	Erethizontidae: New World porcupines		Barbirusa (Pig-deer). Warthog
599.324.5	Capromyidae: Hutias and Coypus.	599.731.4	Hippopotamidae. Hippopotamuses
	Hydrochaeridae: Cabybara (Water-hog).	599.731.6	Tayassuidae: Peccaries
	Dinomyidae: Pacarana. Dasyproctidae: Pacas	599.733	Tylopoda
	Agoutis	599.733.1	Camelidae. Camels. □ Dromedary. Guanaco.
599.324.6	Chinchillidae: Chinchillas and Viscachas		Llama. Alpaca. Vicuna
599.324.7	Caviidae: Cavies. □ Guinea pigs. Maras	599.735	Ruminantia. Ruminants
599.324.8	Octodontidae: Degus	599.735.2	Tragulidae. Chevrotains (Mouse deer)
599.325	Lagomorpha	599.735.3	Cervidae. Antlered mammals
599.325.1	Leporidae: Hares and Rabbits	599.735.31	Cervinae. Old World and other deer. □ Red deer. Fallow deer. Chital. Sika. Père David's deer. Elk. Reindeer (Caribou). Moose
599.325.2	Ochotonidae: Pikas		Odocoileinae. New World and other deer.
599.35/38	Insectivora: Insectivorous mammals	599.735.34	□ Pampas deer. Mule deer. Pudu. Roe deer
599.362	Talpidae. Moles and Desmans		Muntiacinae. Muntjacs
599.363	Soricidae. Shrews. □ Alpine shrew. Pygmy shrew. Water shrew. Common shrew	599.735.36	Moschinae. Musk deer
599.365	Erinaceidae. Hedgehogs and Moon rats	599.735.38	Giraffidae. Giraffe and Okapi
599.38	Macroscelididae. Elephant shrews	599.735.4	Bovidae. Horned mammals (hollow-horned)
599.39	Dermoptera. Cynocephalidae. Colugos (Flying lemurs)	599.735.5	Bovinae. Cattle and Spiral horned antelopes.
		599.735.51	□ Oxen. Domestic ox (cattle). Yak. Aurochs (extinct). Bison. Buffalo. Eland. Kudu
599.4	Chiroptera. Bats		Caprinae. □ Sheep. Mouflon. Goats. Ibex.
599.41	Megachiroptera. Pteropodidae. Old world fruit bats. □ Flying foxes. Tomb bats	599.735.52	Musk-ox. Chamois
		599.735.53	Antilopinae: Small antelopes. □ Gazelles.
599.42/44	Microchiroptera	599.735.54	Dikdiks. Black buck. Springbok. Saiga
599.423	Rhinolophidae: Horseshoe bats		Hippotraginae. Large antelopes. □ Hartebeest.
599.424	Phyllostomidae: New World leaf-nosed bats.	599.735.6	Wildebeest (Gnus). Impala. Oryx. Addax
	□ Vampire bats		Antilocapridae. Pronghorns
599.426	Vespertilionidae: Vespertilionid bats.	599.74	Carnivora. Carnivorous mammals
	□ Pipistrelles	599.742	Fissipedia. Terrestrial carnivores
599.5	Cetacea and Sirenia	599.742.1	Canidae. Dogs. □ Jackals. Coyotes. Dingo
599.51/53	Cetacea. Whales	599.742.11	Wolves
599.51	Mysticeti. Baleen whales (Whalebone whales)	599.742.13	Domestic dogs
599.511	Balaenopteridae. Rorquals. □ Fin whales. Blue whale. Minke whale. Sei whale. Humpback whale	599.742.17	Foxes
		599.742.2	Ursidae. Bears. □ Black bear. Spectacled bear. Sun bear
599.512	Balaenidae. Right whales. □ Greenland whale (Bowhead)	599.742.21	Brown bear. □ Grizzly bear
599.515	Eschrichtidae: Grey whale	599.742.22	Polar bear
		599.742.28	Giant panda
599.53	Odontoceti. Toothed whales	599.742.3	Procyonidae. Raccoons and relatives.
599.532	Physeteridae: Sperm whales		□ Coatis. Lesser panda (Red panda). Kinkajou
599.533	Platanistidae: River dolphins	599.742.5	Mustelidae. □ Stoats. Weasels. Minks.
599.535	Ziphiidae: Beaked whales		Polecats. Ferrets. Martens. Sable. Wolverine (Glutton). Badgers. Skunks. Otters
599.536	Phocoenidae: Porpoises	599.742.6	Viverridae. □ Civets. Genets. Mongooses.
599.537	Delphinidae: Dolphins. □ Killer whale (Orca). Pilot whale	599.742.7	Meerkat. Linsangs. Binturong. Fossa
		599.742.71	Hyaenidae. Hyenas and Aardwolf
599.539	Monodontidae: Narwhal and Beluga (White whale)	599.742.711	Felidae. Cats
		599.742.712	Large felines ('Big cats')
		599.742.713	Lion
599.55	Sirenia. Sea-cows. □ Manatees. Dugongs. Steller's sea cow (extinct)	599.742.714	Tiger
		599.742.72	Leopard (Panther)
599.6/73	Ungulata: Hoofed mammals		Jaguar
599.61	Proboscidea. Elephantidae. Elephants. Mammoths (extinct)	599.742.73	Medium-sized felines. □ Puma (cougar, mountain-lion). Snow leopard
		599.742.75	Small felines. □ Ocelot. Wild cat. Domestic cat. Marbled cat. Serval. Margay
599.62	Hyracoidea. Procaviidae. Hyraxes (Dassies)	599.742.77	Lynx. Bobcat
		599.745	Cheetah
599.68	Tubulidentata. Orycteropodidae. Aardvark	599.745.1	Pinnipedia. Aquatic, marine carnivores
			Otariidae. Eared seals. □ Sea-lions. Fur seals

- 599.745.2 Odobenidae (Trichechidae). Walrus
 599.745.3 Phocidae. Earless seals. □ Common seal.
 Grey seal. Monk seals. Hooded seal. Elephant
 seals
- 599.78 **Scandentia, Tupaiidae. Tree shrews.**
 □ Tupai (Banxring)
- 599.8 **Primates**
- 599.81 **Prosimii. Prosimians**
- 599.811 Tarsidae. Tarsiers. □ Spectral tarsier
 599.813 Lorisidae. □ Lorises. Angwantibo. Potto.
 Galago (Bush-babies)
 599.814 Cheirogaleidae: Mouse lemurs and Dwarf
 lemurs
 599.815 Lemuridae: Large Lemurs. □ Ring-tailed lemur
 599.816 Indridae: Leaping lemurs. □ Indri. Sifakas
 599.817 Daubentoniidae: Aye-ayes
- 599.82 **Anthropoidi. Apes and monkeys**
- 599.821 Callitrichidae. Marmosets (Tamarins)
 599.822 Cebidae. New World monkeys. □ Capuchins.
 Spider monkeys. Titis. Sakis. Howlers. Squirrel
 monkeys. Uakaris
- 599.824 Cercopithecidae. Old World monkeys. □ Patas
 (Red monkey). Baboons. Gelada. Drills.
 Macaques. Barbary ape. Rhesus monkeys.
 Guenons. Mangabeys. Langurs. Colobus
 monkey. Proboscis monkey (Kahau)
- 599.88/89 **Hominoidea: Anthropoid apes and humans**
- 599.88 **Hylobatidae and Pongidae. Anthropoid apes**
- 599.882 Hylobatidae. □ Gibbons. Siamang
 599.883 Pongidae
 599.883.1 Orang utan
 599.883.4 Chimpanzees
 599.883.6 Gorilla
- 599.89 **Hominidae. Hominids.** □ Homo sapiens.
 Humans. Man
*Class here the position of humanity in
 nature and comparisons between humans
 and animals in general.*
*For general anthropology see 572; for
 human anatomy see 611; for physiology
 see 612; for pathology see 616*

6 APPLIED SCIENCES. MEDICINE. TECHNOLOGY

Class here generalities about the applied sciences and technology

60 BIOTECHNOLOGY

For general questions of the applied sciences (previously at this number) see 6

60-7	BIOTECH PLANT AND EQUIPMENT				<i>Note. The literature often deals with both the processes and the resultant organisms. Where the purpose is clearly experimental (e.g. transgenic mice), class the information here. Where the object is to produce useful modifications (e.g. GM crops), prefer 604.6. If in doubt, class all information about specific organisms at 604.6 and use 602.6 only for laboratory techniques</i>
601	FUNDAMENTAL CONCEPTS				
601.2	Biological fundamentals				
	◊ 601.2:575.853 Cell structure				
	◊ 601.2:576.31 Morphological behaviour				
	◊ 601.2:576.34 Metabolism. Metabolic design				
	◊ 601.2:577.121 Metabolites				
601.4	Genetic Fundamentals				
	◊ 601.4:575.116.4 Gene mapping				◊ 602.6:578/579 Modification of microorganisms generally
	◊ 601.4:575.224.4 Mutagenesis				◊ 602.6:579.8 Modification of bacteria
	◊ 601.4:575 Classical genetics				◊ 602.6:582.28 Modification of fungi
	◊ 601.4:577.212.3 DNA sequencing				◊ 602.6:582.5/.9 Modification of plants (flowering plants)
	◊ 601.4:577.213.3/.7 Synthesis of DNA				◊ 602.6:592/599 Modification of animals generally
	◊ 601.4:577.21 Molecular genetics				◊ 602.6:599.323.4MUS Modification of mice (Mus musculus). Creation of transgenic mice
602	PROCESSES AND TECHNIQUES				
602.1	Process control				
	◊ 602.1:004.891 Use of expert systems				
	◊ 602.1:519.673 Use of mathematical modelling	602.62			Physical methods
	◊ 602.1:53.082.9 Use of biosensors	602.621			Electroporation
	◊ 602.1:681.5 Use of automation technology	602.623			Microprecipitates
		602.625			Microinjection
602.3	Industrially important organisms	602.627			Biolithics (microprojectile bombardment)
	◊ 602.3:578.347 Bacteriophages	602.628			Fusion (liposomes as carriers)
	◊ 602.3:579.841.1 Pseudomonads				
	◊ 602.3:579.841.4 Methylophs	602.64			Biological methods. Use of vectors.
	◊ 602.3:579.852.11 Bacillus spp.				Transfection. Transduction
	◊ 602.3:579.852.13 Clostridia	602.641			Use of viral vectors
	◊ 602.3:579.85 Methanogens				◊ 602.641:578.822.9 Adeno-associated virus (AAV) as vector
	◊ 602.3:579.864 Lactic acid bacteria				◊ 602.641:578.826 Adenovirus as vector
	◊ 602.3:579.871 Corynebacteria				◊ 602.641:578.828 Retroviruses (including lentivirus) as vectors
	◊ 602.3:579.873.7 Streptomyces				Use of non-viral vectors
	◊ 602.3:58.086.83 Plant cell cultures				◊ 602.643:54-126 Polymers as vectors
	◊ 602.3:582.281.21 Mould fungi (filamentous fungi)	602.643			◊ 602.643:543.645.6 Peptides as vectors
	◊ 602.3:582.282.23 Yeasts				
602.4	Biotechnological processing. Bioprocessing	602.66			Custom gene construction
602.42	Upstream processing	602.68			Antibody engineering
	◊ 602.42:577.152.3 Fermentation (hydrolytic, proteolytic)	602.69			Other matters relating to genetic engineering. ◻ Deliberate production of sterile seed or organisms. Genetic Use Restriction Technology (GURT) or 'terminator technology'
	◊ 602.42-7 Equipment. Including: Bioreactors. Growth chambers. Fermentation vessels				
602.44	Downstream processing. Biotransformations. ◻ Separation, purification, refinement. Bioseparation	602.7			Cloning
	◊ 602.44:543.544.17:577.112 Use of affinity chromatography for separation and purification of proteins				<i>Class here the processes involved. For cloned organisms see 604.7</i>
	◊ 602.44-7 Separation and purification plant	602.72			Embryonic cloning
602.6	Genetic engineering. Insertion of modified genes. Production of transgenic organisms	602.74			◊ 602.72:612.646:602.9 Cloning of human embryos for stem cell research
	<i>Class here the processes involved. For genetically modified organisms see 604.6.</i>	602.8			Somatic cloning. Somatic cell nuclear transfer (SCNT)
					RNA interference. Gene silencing (disabling of specific genes)

- 602.9 Extraction and manipulation of pluripotent cells.** □ Stem cell research
 ◇ 602.9:611.018 Stem cell research on human cells
 ◇ 602.9:591.81 Stem cell research on animal cells
- 604 PRODUCTS**
- 604.2 Products of primary metabolism. Microbial products**
 ◇ 604.2:547.458 Polysaccharides
 ◇ 604.2:577.113.3 Nucleotides and related compounds
 ◇ 604.2:661.188.1 Glycerol
 ◇ 604.2:661.185 Biosurfactants
 ◇ 604.2:661.722 Ethanol
 ◇ 604.2:661.725.3 Isopropanol
 ◇ 604.2:661.725.4 Butanols
 ◇ 604.2:661.725.81 Polyols (other than glycerols)
 ◇ 604.2:661.727.4 Acetone
 ◇ 604.2:661.74 Organic acids, e.g. lactic acid, citric acid, gluconic acid, acetic acid
 ◇ 604.2:661.745 Amino acids
- 604.4 Products of secondary metabolism.** □ Fermentation and cell culture products
 ◇ 604.4:577.15 Enzymes
 ◇ 604.4:577.16 Vitamins
 ◇ 604.4:615.277.3 Anti-tumour agents. Carcinostatics
 ◇ 604.4:615.281.9 Antibacterials
 ◇ 604.4:615.331 Bacteriocins. Antimicrobial peptides. Lantibiotics
 ◇ 604.4:615.33 Antibiotics. Antimicrobials
 ◇ 604.4:615.371 Vaccines
 ◇ 604.4:636.085 Animal feedstuffs
 ◇ 604.4:662.6:579 Microbial biomass (as fuel)
 ◇ 604.4:663.051 Flavours and aromatic additives
 ◇ 604.4:664 Foods (for humans)
- 604.6 Genetically modified organisms (GMOs)**
Denote genetically modified (GM) versions of wild plants and animals by :58/59... and cultivated or domesticated versions by :63....
For genetic-engineering processes, see 602.6
 ◇ 604.6:582.282.23 GM Saccharomycetales (yeasts)
 ◇ 604.6:633.15 GM maize
 ◇ 604.6:633/635 GM crops
 ◇ 604.6:635.64 GM tomatoes
 ◇ 604.6:636.4 GM pigs, swine
 ◇ 604.6:636.52/58 GM chickens, domestic fowl
 ◇ 604.6:636 GM animals, livestock
- 604.7 Cloned organisms**
For cloning techniques, see 602.7
Denote cloned instances of wild plants and animals by :58/59... and cultivated or domesticated versions by :63...
 ◇ 604.7:636.32/38 Cloned sheep
 ◇ 604.7:636.4 Cloned pigs, swine
- 606 APPLICATIONS OF BIOTECHNOLOGY**
 ◇ 606:61 Medical applications
 ◇ 606:616-056.7 Biotech treatment of inherited disease. Gene therapy
 ◇ 606:616-056.7:602.8 By RNA interference (gene silencing), disabling of defective gene
 ◇ 606:616-056.7:602.8:616-002.17 Disabling of cystic fibrosis gene
 ◇ 606:616-056.7:602.8:616-006.6 Disabling of oncogene (cancer-causing gene)
 ◇ 606:62 Industrial applications
 ◇ 606:620.951 Energy from microbial processes
 ◇ 606:620.951:621.352.6 Bioelectrochemical fuel cells
 ◇ 606:623.458 Use in biological warfare
 ◇ 606:628 Application to public health engineering (also called environmental biotechnology)
 ⇒ 502/504
 ◇ 606:628.1 Drinking water management; preparation of potable water
 ◇ 606:628.3/4 Application to waste management
 ◇ 606:628.3 Sewage treatment and disposal
 ◇ 606:628.4 Refuse and solid waste treatment
 ◇ 606:63 Agricultural applications
 ⇒ 604.6, 604.7
 ◇ 606:631.528 Crop improvement (by applied genetics)
 ◇ 606:632 Biotech methods of plant protection
 ⇒ 581.2
 ⇒ 340.64:577.212
- 608 ISSUES.** □ Legal, economic, moral implications
- 608.1 Ethical aspects. Bioethics**
 ◇ 608.1:575:612.6.05 Bioethics and human genetics. Including: genetically engineered offspring ('designer babies')
- 608.3 Safety aspects. Biosafety**
 ◇ 608.3:001.893 Control. Supervision
 ◇ 608.3:35.075.6 Regulatory bodies
- 608.32 Dangers and precautions**
 ◇ 608.32:001.891.55 Field trials. Testing
 ◇ 608.32:005.52:005.334 Risk assessment in biotechnology
 ◇ 608.32:502/504 Environmental effects
 ◇ 608.32:502.131.1 Biotechnology and sustainability
 ◇ 608.32:504.35 Genetic contamination of biotech origin
 ◇ 608.32:504.35:582:633/635 Genetic contamination of (wild) vegetation by cross-pollination with GM crops
 ◇ 608.32:574.1 Impact on biodiversity
- 608.34 Consumer aspects**
 ◇ 608.34:663/664 Genetically modified drink and foodstuffs (GM food)
 ◇ 608.34:664:366.64 Labelling of GM foods
- 608.5 Biotechnology and intellectual property.** □ The concept of genetic inventions
 ◇ 608.5:347.77 Patenting of bioproducts
 ◇ 608.5:347.78 Copyrighting of bioinformation
- 608.7 Informative aspects**
 ◇ 608.7:316.774:070 Press coverage of biotechnology
 ◇ 608.7:316.774 Media coverage of biotechnology
 ◇ 608.7:659.3 Informing and educating the public

61 MEDICAL SCIENCES

For details of persons and personal characteristics in medicine add -05...

611/612	HUMAN BIOLOGY	611.018.16	Centrosome. Central body
611	ANATOMY. HUMAN AND COMPARATIVE ANATOMY ⇒ 343.93 ⇒ 572 ⇒ 591.3 ⇒ 591.4 ⇒ 616-091 ⇒ 743	611.018.18	Flagellae. Membrana cellulae. Cell membrane. Exoplasm. Cuticula. Intercellular substance
611.01	General anatomy. Anatomia generalis	611.018.2	Tela conjunctiva. Connective tissue ⇒ 611.74
611.012	Teratology. Malformations. Monsters. Anomalies. Teratogeny. □ Siamese twins. Foetus in foetu ⇒ 591.159 ⇒ 612.64 ⇒ 616-007	611.018.3	Cartilage. Fibrocartilage
611.013	Embryology ⇒ 591.3 ⇒ 612.646 ⇒ 618.32	611.018.4	Bone tissue. Osteogenesis ⇒ 611.71
611.013.1	Germ cells. □ Spermatozoa. Ovum ⇒ 611.63 ⇒ 611.65	611.018.44	Periosteum
611.013.2	Copulation. Fertilization ⇒ 576.37 ⇒ 612.6.06 ⇒ 612.63	611.018.46	Bone marrow
611.013.3	Germinal layers	611.018.5	Blood
611.013.31	Segmentation. Morula. Blastula. Germinal disc ⇒ 611.018.15	611.018.51	Erythrocytes. Red blood corpuscles
611.013.32	Gastrula. Development of the intestine	611.018.52	Erythroblasts. Thrombocytes. Blood platelets
611.013.37	Entoderm. Inner germinal layer	611.018.53	Leucocytes. White blood corpuscles. Phagocytes
611.013.38	Ectoderm. Outer germinal layer	611.018.54	Blood plasma. Blood serum
611.013.39	Mesoderm. Middle germinal layer	611.018.6	Muscle fibre ⇒ 611.73
611.013.395	Mesenchyma. Muscle fibre cells	611.018.61	Smooth musculature
611.013.4	Organa entodermalia. Organs of the inner germinal layer	611.018.62	Striated musculature
611.013.5	Organa ectodermalia. Organs of the outer germinal layer	611.018.63	Heart musculature
611.013.6	Organa mesodermalia. Organs of the middle germinal layer	611.018.7	Epithelium. Epithelial tissue
611.013.7	Forma externa embryonis. External form of the embryo	611.018.72	Glands
611.013.8	Adnexa embryonis. Appendicular organs of the embryo. □ Funiculus umbilicalis (umbilical cord). Sacchus vitellinus (vitelline sac). Allantois. Amnion. Chorion. Placenta	611.018.73	Mucosae. Mucous membrane
611.013.9	Embryologia experimentalis. Experimental embryology. Mechanisms of development. Causal morphology. Experimentally caused phenomena ⇒ 612.622 ⇒ 612.646	611.018.74	Endothelium
611.018	Histology. Histogenesis. Tissue structure, formation and differentiation ⇒ 591.8 ⇒ 612.014 ⇒ 616-091.8	611.018.8	Nerve tissue ⇒ 611.8
611.018.1	Cytology. Cell. Syncytium ⇒ 576.3 ⇒ 591.81	611.018.81	Structure of the nerve tissue. Connections of the nerve cells
611.018.11	Cytoplasm. Protoplasm. Trophospongium.	611.018.82	Nerve cells
611.018.13	Vacuolae. Intercellular bridges	611.018.83	Nerve fibres. Dendrites. Neurons
611.018.15	Nucleus. Nucleolus ⇒ 611.74	611.018.86	Nerve endings
611.018.15	Karyokinesis. Cell segmentation ⇒ 611.013.31	611.019	Comparative anatomy Specify by :59 ⇒ 591.4
		611.06	Characteristics of specific organs
		611.061	External characteristics
		611.068	Organs by functional and developmental characteristics 611.068 ≅ 591.4.068.1/6
		611.08	Anatomical techniques, experiments and observations 611.08 ≅ 57.08
		611.1/8	Systematic anatomy. Organs
		611.1	Angiology. Cardiovascular system. Blood vessels 611.1 ≅ 616.11/14
		611.11	Pericardium. Epicardium
		611.12	Heart. Endocardium. Myocardium
		611.13	Arteries. Aorta
		611.14	Veins
		611.16	Vasa capillaria. Capillary vessels
		611.2	Apparatus respiratorius. Systema respiratorium. Respiratory system 611.2 ≅ 616.21/26
		611.21	Nasus. Nose. □ Sinuses ⇒ 611.86

611.22	Larynx. Voice-box	611.638	Scrotum
611.23	Trachea, Windpipe. Bronchi	611.639	Spermatic cord
611.24	Pulmo. Lungs	611.64	Penis. □ Corpora cavernosa. Glans penis
611.25	Pleurae	611.65	Female genital organs
611.26	Diaphragm	⇒ 618	
611.28	Branchiae	611.651	Ovarium. Ovary. □ Ovarian follicles. Graafian follicles
611.29	Other organs. □ Swimming bladder	611.652	Corpora lutea
611.3	Apparatus digestorius. Systema digestorium. Digestive system.	611.653	Corpus atreticum. Degenerating follicle
	Alimentary canal	611.654	Corpus albicans
	611.3 ≡ 616.3	611.656	Oviducts (Fallopian tubes)
611.31	Cavum oris. Oral cavity. Mouth	611.66	Uterus. Womb
	⇒ 612.87	611.663	Cervix uteri
611.311	Rima oris. Mouth	611.664	Endometrium (uterine mucous membrane).
611.313	Lingua. Tongue	611.667	Myometrium (uterine muscular coat)
611.314	Dentes. Teeth. Dentition		Mesometrium. Rudimentary organs.
611.315	Velum palatinum. Palate	611.67	Degenerating parts
611.316	Glandulae oris. Salivary glands	611.671	Vagina and external female genitals
611.317	Labia oris. Lips	611.672	Vagina
611.318	Buccae (malae). Cheeks. Corpus adiposum buccae. Buccal pad of fat	611.673	Hymen
			External female genitals. □ Vulva. Mons pubis (mons veneris). Labia
611.32	Pharynx. Throat. Oesophagus. Gullet.	611.674	Female urethra
	Tonsillae palatinae. Tonsils		⇒ 611.62
611.33	Ventriculus. Stomach. □ Pylorus	611.675	Clitoris
611.34	Intestines. Guts. Bowels. □ Duodenum. Jejunum. Ileum. Caecum. Colon	611.69	Mammary glands. Breasts
611.35	Rectum. Anus. Perineum	611.7	Skeletal, locomotor and integumentary systems
611.36	Hepar. Liver. □ Ductus hepaticus. Hepatic duct. Vesica fellea. Gall bladder. Ductus choledochus. Bile duct	611.71	Osteology. Bones. Skeleton
			⇒ 611.018.4
611.37	Pancreas	611.711	Vertebral column. Spine (backbone)
611.38	Peritoneum. Omentum. Mesentery	611.712	Thorax. Chest. Rib-cage
611.4	Lymphatic system. Haemopoietic organs (haematopoietic organs).	611.713	Sternum. Breastbone
	Endocrine organs	611.714	Cranium. Skull. Bones of the head
611.41	Lien. Spleen	611.715	Cerebral bones of the skull. □ Occipital.
611.42	Systema lymphaticum. Lymphatic system.		Sphenoid. Temporals. Parietals. Frontal.
	□ Lymphatic vessels. Lymph glands	611.716	Ethmoid
611.428	Lymph nodes or glands. Closed follicles		Facial bones of the skull. □ Cheekbones (zygomatic bones). Maxilla (jawbones, mandible). Lacrimal bones. Nasal bones.
611.43	Endocrine organs (ductless glands).		Vomer
	□ Thymus	611.717	Bones of the upper extremities. □ Clavicle (collarbone). Bones of shoulders, arms, hands, fingers
611.44	Glandula thyroidea. Thyroid gland.	611.718	Bones of the lower extremities. □ Pelvic girdle.
	□ Glandulae parathyroidea. Parathyroid glands	611.72	Bones of thighs, legs, feet, toes
611.45	Glandulae suprarenales. Adrenal glands (suprarenal glands). Other ductless glands	611.72	Syndesmology. Articulations (joints).
611.47	Glomus caroticum. Carotid body. Paraganglia	611.73	Sutures
			611.72 ≡ 616.72
611.6	Urogenital system (genito-urinary system). Urinary and sexual organs		⇒ 611.018.3
611.61	Kidneys. Ureter	611.74	Myology. Muscular system. Skeletal musculature
611.62	Urinary bladder. Urethra		611.73 ≡ 616.74
	⇒ 611.674		⇒ 611.018.6
611.63	Male genital organs	611.75	Tenology. Ligaments. Tendons. Sinews.
611.631	Testes. Testicles	611.77	Fasciae. Aponeuroses
611.632	Epididymis		611.74 ≡ 616.74
611.633	Appendix testis. Paradidymis		⇒ 611.018.2
611.634	Vas deferens		Synovial bursae. Synovial sheaths
611.635	Ejaculatory tract	611.771	611.75 ≡ 616.76
611.636	Glandula vesiculosa	611.773	Dermatology. Common integument. Skin.
611.637	Prostate	611.774	Cutaneous glands
			⇒ 611.88
			⇒ 616.5
			Epidermis
			Sebaceous glands
			Sudoriferous glands (sweat glands)

611.778	Corium (dermis, cutis)	611.846	Accessory organs of the eye. □ Eye muscles.
611.779	Subcutaneous tissue		Lacrimal apparatus (tear glands)
611.78	Hair. Nails	611.847	Eyelids. Cilia. Eyelashes. Palpebral part of conjunctiva
⇒ 611.77		611.848	Supercilia. Eyebrows
611.781	Trichology. Hair. □ Pilar system. Lanugo.		
	Bristles. Beard	611.85	Ears. Organs of equilibrium and hearing
611.786	Nails	611.851	Inner ear in general
611.8	Nervous system. Sensory organs	611.852	Membranous labyrinth. □ Cochlea.
⇒ 611.018.8			Semicircular ducts
611.81	Central nervous system. Cerebrum, encephalon. Brain	611.853	Capsule of the bony labyrinth
611.812	Centres. Localizations	611.854	Tympanum. Ear-drum. Tympanic cavity
611.813	Prosencephalon. Forebrain. Telencephalon.	611.855	Tympanic membrane
	□ Cerebral cortex	611.856	Pharyngotympanic tube (Eustachian tube)
611.814	Diencephalon (thalamencephalon).	611.857	Auditory ossicles (bones of the ear). □ Malleus (hammer). Incus (anvil). Stapes (stirrup)
	□ Hypothalamus. Pituitary gland		External ear. □ Auricle (pinna). Ear lobe
611.815	Mesencephalon. Midbrain. □ Corpora quadrigemina. Quadrigeminal bodies. Cerebral peduncles	611.86	Olfactory organ (organ of smell). Nose.
611.816	Isthmus rhombencephali	□ Nostrils. Nasal mucosa	⇒ 611.21
611.817	Metencephalon. Hindbrain.	⇒ 612.86	
	□ Rhombencephalon. Cerebellum. Pons Varolii	611.87	Gustatory organs (organs of taste). Taste buds
611.818	Myelencephalon. Medulla oblongata.	⇒ 611.31	⇒ 612.87
	Cerebrospinal bulb		
611.819	Cerebral meninges	611.88	Tactile organs (organs of touch, feeling)
⇒ 611.829		⇒ 611.77	⇒ 612.88
611.82	Spinal cord. Spinal medulla. □ Grey matter.		
	White matter. Spinal meninges	611.89	Nerve ganglia
611.821	General form. Enlargements	⇒ 611.839	⇒ 616.834
611.822	Grey matter		Ganglia of the cranial nerves
611.824	White matter	611.891	Ganglia of the spinal nerves. Spinal ganglia
611.825	Anterior bundle. Lateral bundle	611.892	⇒ 611.832
611.826	Posterior bundle		Ganglia of the sympathetic nervous system
611.827	Roots of the spinal nerves	611.899	⇒ 611.4
⇒ 611.832			
611.828	Central canal of spinal cord	611.9	Anatomical topography. □ Somatology.
611.829	Spinal meninges		Parts, regions of the body. Regional anatomy
⇒ 611.819		⇒ 572.5	
611.83	Peripheral nervous system. Sensory nerves	611.91	Head. Cranial region
⇒ 611.018.8		611.92	Face. Facial region
611.831	Cranial nerves	611.93	Neck. Cervical region
⇒ 611.815		611.94	Chest. Thoracic region
⇒ 611.817		611.941	Breast. Pectoral region
⇒ 611.818		611.946	Back. Dorsal region
⇒ 611.891			
611.832	Spinal nerves	611.95	Abdominal region (ventral region). Belly
⇒ 611.827		611.951	Upper abdominal region
⇒ 611.892		611.955	Middle abdominal region. □ Umbilicus (navel)
611.833	Cervical nerves	⇒ 618.47	
611.834	Thoracic nerves	611.957	Inguinal region (lower abdominal region). Groin
611.835	Lumbar and sacral nerves	611.959	Lumbar region. Loins
611.836	Coccygeal nerve		
611.839	Sympathetic and parasympathetic nervous systems. Autonomic system	611.96	Pelvic region. Hips
611.839	611.839 ≡ 616.839	611.97	Upper extremities. □ Shoulders and arms (axillary, brachial, carpal regions etc.)
⇒ 611.89		611.976	Hands
611.84/88	Sensory organs	611.977	Fingers
⇒ 159.93			
611.84	Eyes. Organs of sight, vision	611.98	Lower extremities. Legs (femoral, crural, malleolar regions etc.)
⇒ 617.7		611.986	Feet
611.841	Outer coat (tunic) of the eye. □ Ocular part of conjunctiva. Cornea. Sclera	611.99	Coccygeal region
611.842	Uvea. Vascular coat of the eye. □ Iris. Pupil		
611.843	Retina. Optic bundle		
⇒ 611.83			
611.844	Eyeball. □ Lens. Vitreous body		

612	PHYSIOLOGY. HUMAN AND COMPARATIVE PHYSIOLOGY ⇒ 159.91 ⇒ 577 ⇒ 581.1 ⇒ 591.1 ⇒ 616-092	612.063 612.064 612.067	Stimulation. Positive influence Antagonistic, negative influence Effects of the system, organ or function ◇ 612.4.067:612.8.062 Effects of glands on the nervous system
612.01	General physiological problems	612.08	Physiological techniques, experiments and observations 612.08 ≡ 57.08
612.013	Life. Death. Vitalism ⇒ 128 ⇒ 573.3 ⇒ 612.67	612.1/.8	Systematic physiology
612.014	General physiology of the cell and the organism ⇒ 576.3 ⇒ 611.018 ⇒ 612.822	612.1	Blood. Cardiovascular, circulatory system ⇒ 616.15
612.014.1	Physiological chemistry of the cell. □ Aerobic and anaerobic cells	612.11	Properties of blood in general. Composition of blood
612.014.2	Histo-morphological properties of cells. □ Functions of cell components	612.111	Erythrocytes. Red blood corpuscles. □ Haemoglobin. Chemicals in corpuscles. Lecithin, urea, cholesterol. Red cell count, number, shapes
612.014.3	General physiology of the cell. □ Irritability. Fatigue	612.112	Leucocytes. White blood corpuscles. Amoeboid cells. □ Phagocytes. White cell count, number, shapes. Lymphocytes. Monocytes
612.014.4	Influence of environment on the cell. Tropisms ⇒ 615.83	612.113	Arterial blood
612.014.41	Effects of atmospheric pressure	612.114	Venous blood
612.014.42	Effects of electricity. Electrophysiology	612.115	Blood coagulation, clotting. □ Fibrin. Thrombin. Heparin
612.014.43	Effects of heat and cold. Temperature influences	612.116	Total blood volume
612.014.44	Effects of light. Phototropism	612.117	Physical properties of blood. □ Specific gravity, viscosity, colour of blood
612.014.45	Effects of vibrations, sound, noise. Acoustic influences	612.118	Biological properties of blood. □ Blood groups. Haemagglutination. Haemolysis
612.014.46	Effects of chemicals, of poisons. Toxic influences	612.119	Blood formation (haemopoiesis, haematopoiesis) ⇒ 612.356 ⇒ 612.41
612.014.461	Effects of water. □ Hydration	612.12	Chemical properties of blood
612.014.462	Physicochemical forces. □ Osmosis. Diffusion. Colloids. Alkalinity. Acidity. Viscosity. Surface tension. Adsorption	612.121	Reaction of blood. □ Acidity, alkalinity of blood
612.014.47	Effects of mechanical forces. □ Geotropism. Thigmotaxis	612.122	Blood carbohydrates and their derivatives. □ Blood glucose, glycogen
612.014.48	Other effects. □ Effect of ionizing radiations. Effect of X-rays (Roentgen rays)	612.123	Fats, lipoids, soaps, glycerine in blood
612.014.5	Constitutional effects	612.124	Blood albuminoids. Colloidal nitrogen compounds in blood
612.015	Physiological chemistry ⇒ 577.1 ⇒ 612.39	612.125	Non-colloidal nitrogen compounds. □ Serum pigments
612.015.1	Enzymes. Ferments ⇒ 577.325 ⇒ 663.1	612.126	Mineral substances in blood. □ Potassium
612.015.2	Normal composition of the body. □ Organ extracts. Organotherapy	612.127	Blood gases
612.015.3	Metabolism. □ Autolysis ⇒ 612.392	612.128	Enzymes ⇒ 577.15
612.015.4	Pigments	612.129	Other chemical substances
612.015.6	Effect and action of vitamins. Vitamin content of tissues and body fluids 612.015.6 ≡ 577.16	612.13	Circulation of the blood. Hydrodynamics of circulation
612.017	Immunity. Adaptation. Toxins. Antitoxins	612.133	Arterial circulation
612.018	Hormones. Internal secretions ⇒ 612.43 ⇒ 612.616.3 ⇒ 612.621.3	612.134	Venous circulation
612.019	Comparative physiology Specify by :59 ⇒ 591.1	612.135	Capillary circulation
612.06	Actions, effects, influences on or of the system, organ or function concerned. Interactions. Reciprocal effects	612.14	Blood pressure. Sphygmomanometry
612.062	Effects on the system, organ or function ◇ 612.4.062:612.8.067 Glands as affected by the nervous system	612.146	Effect of various substances on blood pressure and circulation
		612.15	Circulation rate in arteries, veins, capillaries. Measurement
		612.16	Pulse. Pulse rate. Sphygmographs
		612.17	Heart. Cardiology
		612.171	Mechanics of cardiac contraction. Cardiography. □ Heartbeat. Heart sounds
		612.172	Heart as muscle. Irritability, contractility, physiological morphology of heart. □ Cardiac rhythm
		612.173	Work of the heart: chemical, dynamic, thermal processes

612.174	Toxic effects on the heart	612.3	Alimentation. Eating. Digestion. Nutrition
612.176	Behaviour of the heart under stress ⇒ 612.232	612.31	Oesophagus. Oral cavity. Teeth. Salivary glands
612.178	Innervation of the heart. □ Syncopes. Cardiac reflexes	612.311	Mastication (chewing). Ingestion
612.18	Vasomotor nerves. Changes in volume of organs	612.312	Deglutition (swallowing) ⇒ 612.785
612.183	Vasoconstrictors	612.313	Salivary glands. □ Salivation. Saliva. Ptyalin
612.184	Vasodilators	612.314	Salivary venoms. Animal venoms in general <i>The main location for animal venoms is 591.145</i>
612.187	Vasomotor nerves in organs <i>Specify organ by the subdivisions of 612</i> ◇ 612.187.35 Influence of vasomotor nerves on the liver	612.315	Oesophagus
612.19	Action of various organs on blood circulation	612.32	Stomach. Gastric juices
612.2	Respiration. Breathing. Respiratory system ⇒ 612.793	612.321	Normal composition of gastric juice. □ Pepsin. Proteolytic enzymes
612.21	Respiratory movements. Mechanics of respiration. Pneumography	612.322	Gastric action on food. Digestion in the stomach
612.213	Influence of respiration on blood circulation ⇒ 612.146 ⇒ 612.19	612.323	Gastric secretion
612.215	Special physiology of the respiratory apparatus	612.323.4	Self-digestion of the stomach
612.216	Respiratory rate, frequency, rhythm, sounds	612.323.5	Effects of gastrectomy
612.216.3	Artificial respiration. Artificial changes of respiration ⇒ 612.232 ⇒ 615.816	612.325	Physiological morphology, histology, of the stomach
612.217	Physiology of the respiratory muscles, diaphragm. Mechanical processes of respiration	612.326	Pathological physiology of the stomach ⇒ 616.33
612.219	Sneezing. Laughing. Yawning	612.327	Motility of the stomach. □ Irritability of stomach as muscle. Evacuation of stomach. Aerophagy. Eructation (belching). Emesis (vomiting)
612.22	Chemistry of respiration. Gaseous exchange. Respiratory metabolism. □ Effect of oxygen on the organism	612.328	Innervation of the stomach. Action of nervous system on the stomach. Gastric sensitivity
612.223	Effects of atmospheric composition and pressure ⇒ 612.014.46	612.33	Intestines (guts). Midgut. Intestinal glands
612.23	Gaseous exchange in the blood	612.331	Normal intestinal juices. Intestinal gases
612.231	Expired air	612.332	Action of intestinal juice on food. Intestinal digestion
612.232	Asphyxia in physiology and forensic medicine. □ Artificial respiration ⇒ 612.176 ⇒ 614.824 ⇒ 615.816	612.332.7	Absorption and conversion of food in the intestines
612.233	Respiration in confined air	612.333	Intestinal secretion
612.234	Toxic effect of carbon monoxide	612.334	Action of chemicals and poisons on intestinal secretions and their removal. □ Laxatives ⇒ 615.246
612.24	Total lung capacity. Residual air	612.335	Physiological morphology, histology, of the intestines. □ Brunner's glands. Intestinal villi
612.25	Exhalation of water from the lungs	612.336	Pathological disturbances of the intestine. □ Parasites. Microorganisms ⇒ 576.88 ⇒ 576.89 ⇒ 616.34
612.26	Tissue respiration. Gaseous exchange between blood and tissues	612.337	Peristalsis. Intestinal movements
612.27	Effects of atmospheric pressure on living organisms ⇒ 613.643 ⇒ 616-001.11/.12	612.338	Innervation of the intestines. Action of nervous system on the intestines
612.274	Effect of high pressures	612.339	Peritoneum. Omentum. Abdominal cavity. Intra-abdominal pressure
612.275	Effect of low pressures	612.34	Pancreas
612.276	Effect of pressure on fermentation processes	612.341	Normal composition of pancreatic juice
612.28	Influence of the nervous system on respiration	612.342	Action of pancreatic juice on food
612.288	Respiratory reflexes ⇒ 612.833	612.343	Pancreatic secretion
612.291	Effects of blood pressure, circulation etc. via the carotid sinus	612.344	Action of chemicals on pancreatic secretion
		612.345	Physiological morphology of the pancreas
		612.346	Pathological physiology of the pancreas
		612.348	Innervation of the pancreas
		612.349	Pancreas as endocrine gland
		612.349.7	Islets of Langerhans
		612.349.8	Action of insulin
		612.35	Liver
		612.351	Chemical composition of liver. Hepatic circulation
		612.352	Action of liver on food and reserve materials. □ Blood sugar. Glycogen
		612.353	Chemical functions of liver. □ Urea formation

- 612.354 Action of poisons on liver. Toxic hepatic steatosis. Anti-toxic action of the liver
- 612.355 Temperature of the liver
⇒ 612.56
- 612.356 Haemopoietic (haematopoietic) function of the liver
⇒ 612.119
⇒ 612.41
- 612.357 Bile (gall). Bile secretion
- 612.357.1 Chemical composition of bile. □ Bile pigments: bilirubin, biliverdin, urobilin
- 612.357.2 Action of bile on intestines and contents
- 612.357.3 Bile secretion
- 612.357.4 Action of chemicals and poisons on bile secretion
- 612.357.5 Elimination of poisons by bile
- 612.357.6 Pathological physiology of liver. □ Biliary calculi (gallstones)
- 612.357.7 Excretion of bile. Physiology of bile ducts, gallbladder
- 612.357.8 Action of the nervous system on the bile ducts
- 612.358 Innervation of the liver. Action of nervous system on the liver
- 612.359 Ablation, regeneration, cicatrization and other morphological and histological phenomena in the liver
- 612.36 Large intestine. Defecation**
- 612.363 Appendix vermiformis
- 612.366 Rectum
- 612.38 Absorption.** □ Through skin, lungs, mucous surfaces etc. Imbibition. Osmosis. Dialysis. Diffusion
⇒ 612.791
- 612.39 Nutrition**
⇒ 612.015
⇒ 613.2
⇒ 615.24
- 612.391 Inanition. Hunger. Thirst
- 612.392 Foodstuffs in general
◇ 612.392.015.6 Physiological effect and action of vitamins
⇒ 612.015.3
- 612.392.1 Assimilation of carbon
- 612.392.2 Assimilation of nitrogen
- 612.392.3 Assimilation of water
- 612.392.4 Assimilation and conversion of sulphur, phosphorus, iron
- 612.392.5 Caloric value of foods
- 612.392.6 Mineral salts
- 612.392.7 Vegetable foods
- 612.392.8 Animal foods
- 612.392.9 Other foods. Artificial, synthetic foods
- 612.393 Action of stimulants, irritants, intoxicants, condiments. □ Alcohol. Coffee. Tea. Spices
- 612.394 Nutritional requirements and growth
- 612.395 Normal and minimum adult diet (nutritional requirement). □ Relation of diet to climate
- 612.395.1 Normal working requirement
- 612.395.2 Minimum requirement (subsistence diet)
- 612.395.5 In relation to climate and other external influences
- 612.395.6 Requirement in sickness, disease
- 612.396 Carbohydrates. □ Starches. Sugars. Glycolysis
- 612.397 Fats. Lipids. □ Lipoids. Cholesterol. Lipolysis
- 612.398 Proteins. Nitrogenous foodstuffs. □ Albumin. Albuminoids. Proteolysis
⇒ 641.12
- 612.398.19 Substances arising from the dissociation of albuminoids
- 612.398.192 Amino-acids
⇒ 547.49
- 612.398.193 Urea
⇒ 612.461.2
- 612.398.194 Ammonia and its derivatives
⇒ 612.461.26
- 612.398.195 Bases. Purine. Purine bodies
⇒ 612.461.25
⇒ 612.461.26
- 612.4 Glandular system. Secretion. Excretion**
Specify actions, effects and influences on or of the glands by the .06 auxiliaries of 612
◇ 612.4.062:612.93.067 Glands as affected by nutrition
- 612.4.05 Physics and chemistry of secretions
⇒ 577.17
⇒ 612.018
- 612.4.09 Physiological morphology of glandular activity
- 612.41 Haemopoietic (haematopoietic) system**
⇒ 612.119
⇒ 612.356
- 612.411 Spleen. Blood storage and filtering functions of spleen
- 612.419 Medulla. Marrow
- 612.42 Lymphatic system**
- 612.421 Chemical composition of lymph
- 612.423 Lymphatic circulation
- 612.428 Lymph glands
- 612.429 Reticulo-endothelial system
- 612.43 Endocrine physiology. Ductless glands. Endocrinology**
For experimental surgical techniques use the .089.6 special auxiliary of 612
◇ 612.432.089.64 Hypophysectomy
⇒ 612.018
⇒ 612.616.3
⇒ 612.621.3
- 612.432 Function of pituitary gland (hypophysis) in general
- 612.433 Anterior lobe. □ Antuitrin
- 612.434 Posterior lobe (neurohypophysis). □ Pituitrin
- 612.434.9 Intermedia. Chromatophorotrophic hormone
- 612.438 Thymus: chemistry, action etc.
- 612.44 Thyroid gland**
- 612.444 Effects of thyroid extracts, transplants, hormones
Specify affected system, organ by :612
◇ 612.444:612.8 Effects of thyroid extracts on the nervous system
- 612.447 Parathyroid glands
- 612.45 Adrenal glands (suprarenal glands). Other endocrine glands**
- 612.451 Corpora adrenalia (suprarenalia). Adrenal or suprarenal glands
- 612.452 Function of adrenal medulla. Adrenalin
- 612.453 Function of adrenal cortex. Cortisone
- 612.454 Effect of adrenal extracts, transplants, etc.
⇒ 612.444
- 612.455 Resection. Destruction
- 612.459 Other endocrine glands
- 612.46 Ren. Kidneys. Urine**
- 612.461 Chemical composition of urine
- 612.461.1 Reaction and physical properties. Urine analysis
- 612.461.2 Urea: Nitrogen compounds
- 612.461.21 Determination of urea: Ureometer
- 612.461.22 Total nitrogen determination
- 612.461.23 Nitrogen excretion. Nitrogen metabolism

- 612.461.25 Uric acid
612.461.26 Other nitrogenous substances in urine.
□ Allantoin, peptones, ammonia, enzymes in urine. Hippuric acid
612.461.27 Urinary pigments. □ Urochrome. Urobilin. Porphyrin
612.461.6 Inorganic substances, salts, gases in urine
612.461.8 Non-nitrogenous organic substances in urine
612.461.82 Sugar. Glycosuria. Lactosuria
612.462 Toxicity of urine
612.463 Secretion and volume of urine
612.464 Influence of poisons, chemicals, on secretion and excretion of urine
⇒ 615.254
612.465 Morphology and physiological histology of kidneys
612.466 Pathological physiology of renal function
⇒ 616.63
612.466.1 Urinary calculi. □ Stones in bladder
612.466.2 Secretion of urine in disease
612.466.6 Abnormal substances in urine
⇒ 616.63
612.466.7 Decomposition of urine
612.467 Excretion of urine. □ Bladder. Ureters. Urethra. Urination. Micturition

612.49 Other glands and secretory processes.
□ Carotid gland

612.5 Animal heat. Thermal processes
612.51 Origin of animal heat (thermogenesis). Calorimetry

612.52 Paths of heat loss
612.521 Heat loss by radiation through skin (cutaneous radiation)
612.523 Heat loss by evaporation in lungs (pulmonary evaporation)
612.524 Heat loss by evaporation through skin (cutaneous evaporation)

612.53 Regulation of body temperature. Thermotaxis
⇒ 612.792

612.54 Other influences on thermogenesis.
□ Thermal effects of baths, hunger, thirst, poisons

612.55 Variations in heat production and regulation
612.56 Body temperature. □ Thermal topography
612.57 Fever (pyrexia). Hyperthermia
⇒ 616.9

612.59 Effect of heat and cold on the organism. Adaptation to temperature
Specify by :612
◇ 612.59:612.82 Effect of heat and cold on the central nervous system
612.591 Effects of heat. □ Death from heat. Thermal rigor (heat cramp)
612.592 Effects of cold. □ Death from cold. Hypothermia. Cold cramp, rigor

612.6 Reproduction. Growth. Development
⇒ 611.6
612.6.05 Heredity, variation, breeding and related features
⇒ 575
612.6.051 Heredity. Atavism. Constancy of acquired characteristics
⇒ 575.1
612.6.052 Natural and artificial variation. Sexual variation
⇒ 575.2
612.6.052.2 Combinative variation
612.6.052.25 Inbreeding. Blood relationship
612.6.052.27 Hybrids and hybridization
612.6.052.4 Mutation. Mutagenesis
612.6.055 Natural and artificial selection
⇒ 575.827
612.6.057 Sexuality. Sex differentiation. Origin of the sexes
⇒ 57.017.5
612.6.058 Hermaphroditism (intersexuality)
612.6.06 Physiology of sex acts. Sexual intercourse, copulation, coitus
612.6.062 Sexual arousal, excitation. □ Engorgement
612.6.064 Sexual climax. Orgasm

612.61 Physiology of the male genital organs
◇ 612.61.064 Male orgasm
612.614 Physiology of the penis
◇ 612.614.062 Excitation of the penis.
Including: Engorgement of the penis. Penile erection
612.616 Testes. Testicles and their sheaths. Sperm. Seminal fluid
612.616.1 Castration in males and females
612.616.2 Physiology of spermatozoa
612.616.3 Testes as endocrine glands
⇒ 612.018
⇒ 612.43
612.616.31 Male sex hormones. □ Testosterone
612.616.38 Effects of testicular transplantations, extracts and synthetic androgens
Specify by :612
◇ 612.616.38:612.784 Effect on vocal register
612.617 Organs appendant to the testes.
□ Seminiferous tubules. Epididymis. Paradiidymis. Vas deferens. Seminal vesicles. Spermatic cord. Prostate gland. Scrotum

612.62 Physiology of the female genital organs
◇ 612.62.064 Female orgasm
612.621 Ovaries
612.621.1 Experimental and pathological lesions and removal of ovaries
612.621.3 Interstitial tissue of ovaries
⇒ 612.018
⇒ 612.43
612.621.31 Ovarian hormones. □ Oestrogens
612.621.38 Effects of ovarian extracts, transplants and synthetic oestrogens: experimental studies
Specify by :612
◇ 612.621.38:612.627 Effects on the uterus
612.621.5 Oestrus. Oestrous cycles
612.621.8 Graafian follicles
612.621.9 Ovulation
612.622 Ovum. □ Genesis, composition
⇒ 611.013.9
612.627 Oviducts (Fallopian tubes). Uterus. Vagina. Vulva
612.627.1 Vagina
612.627.5 Clitoris
◇ 612.627.5.062 Excitation and engorgement of the clitoris

612.63 Internal fertilization (impregnation). Conception. Pregnancy. Gestation. Parturition (childbirth)
⇒ 611.013.2
⇒ 612.63
⇒ 618
612.63.02 Duration and stages of pregnancy. Gestation
612.63.021 Gestation period. Duration of pregnancy
612.63.025 Stages of pregnancy
612.63.025.2 Early pregnancy
612.63.025.5 Mid-pregnancy

612.63.025.7	Late pregnancy	612.746	Pathological changes in muscular function. □ Contractures. Muscle tremors ⇒ 616.74
612.63.028	Labour	612.748	Innervation of muscles
612.63.03	Ovaries during pregnancy. □ Luteal cells. Progesterone. Pregnandiol	612.748.5	Nutrition of muscle. □ Muscular atrophy, degeneration, regeneration
612.63.04	Abnormal internal impregnation	612.75	Bones. Articulations (joints). □ Cartilage. Connective tissue. Fasciae. Synovial bursae. Tendon sheaths
612.631	Nidation of the ovum	612.751	Chemical composition
612.632	Single or multiple pregnancy	612.752	Nutrition of bones
612.64	Physiology of embryo, foetus and newborn. Developmental and teratological physiology Specify by :612	612.753	Growth, cicatrization, regeneration of bones
	◇ 612.64:612.17 Heart during development	612.754	Periosteum. Perichondrium
612.646	Embryo	612.755	Tendons
612.647	Foetus	612.76	Locomotion. Principles of animal mechanics
612.648	The newborn (neonate). □ Viability	612.761	Speed
612.649	Foetal appendages. □ Amnion. Umbilical chord. Urachus	612.763	Coordinated movements other than locomotion
612.65	Growth after birth Specify by :612	612.766	Human locomotion
	◇ 612.65:612.7 Growth of muscular system	612.766.1	Physiology of exercise, of work (ergology). Study of fatigue (ponology)
612.66	Influence of age on various physiological processes ⇒ 613.9	612.766.2	Physiology of rest
		612.769	Biological adaptation of movement
612.661	Puberty	612.78	Voice. Speech. Phonation
612.662	Menstruation	612.781	Sensitivity of the larynx
612.662.1	Hormonal control of menstruation	612.782	Motor function and innervation of the larynx. □ Movements of glottis, epiglottis
612.662.9	Cessation of menstruation. Menopause	612.783	Artificial larynx
612.663	Fertility. Infertility (sterility)	612.784	Vocal register. Vocal timbre. □ Range, pitch. Tone, resonance
612.664	Milk. Lactation. Mammary glands	612.785	Function of larynx in swallowing. Other laryngeal functions ⇒ 612.312
612.664.1	Chemical composition and physical properties of milk	612.789	Word formation, articulation. Speech. □ Ventriloquy ⇒ 81
612.664.3	Secretion of milk. □ Colostrum	612.79	Common integument. Skin
612.664.4	Effect of poisons, chemicals, on milk secretion. Their elimination	612.791	◇ 612.79.015 Physiological chemistry of skin
612.664.5	Physiological morphology of milk secretion	612.792	Absorption capacity of skin ⇒ 612.38
612.664.6	Pathological physiology of milk secretion	612.793	Cutaneous glands and secretions. Sweat ⇒ 612.524 ⇒ 612.53
612.664.7	Digestion of milk Class here only special works on the investigation of the properties of milk	612.794	Cutaneous respiration. □ Effect of cosmetics on skin ⇒ 612.2
612.664.8	Innervation of the mammary glands	612.795	Sensitivity of skin
612.67	Climacteric. Ageing. Senescence. Senility. Death ⇒ 57.017.6 ⇒ 612.013 ⇒ 613.98 ⇒ 616-053.9	612.796	Reaction of the skin to electricity and other stimuli
612.68	Longevity. Regeneration	612.798	Chromatophores. Skin pigments
612.7	Motor functions. Organs of locomotion. Voice. Integument	612.799	Trophic nerves of skin
612.72	Cilia	612.8	Growth and physiology of nails, hair, beard
612.73	Smooth musculature (involuntary muscle) 612.73 ≡ 612.74	612.8.01	Nervous system. Sensory organs
612.74	Striated musculature (voluntary, skeletal muscle)	612.8.02	Theories. Innervation
612.741	Muscular contraction. □ Myography. Muscle volume in contraction. Irritability of muscle	612.8.04	Inhibition (nervous)
612.742	Muscles after death. Rigor mortis. Autolysis	612.8.05	Effect of the nervous system on chemical and vegetative phenomena
612.743	Electrical phenomena in muscle		Effect of the nervous system on morphogenesis and evolution ⇒ 612.6.05
612.744	Chemistry of muscle. □ Composition of muscle. Myosin. Chemical effects of muscular contraction. Muscular fatigue	612.81	Peripheral nervous system
612.745	Dynamic and thermal effects of muscular contraction on the organism	612.811.1	Points of difference between sensory and motor nerves
612.745.1	Dynamic effects: Muscular energetics	612.811.3	Ganglia of the sensory nerves
612.745.3	Thermal effects	612.812	Recurrent sensitivity
612.745.5	Relations between work and heat	612.813	Electrical phenomena in nerves and muscles. □ Electrotonus
612.745.6	Occupational energetics	612.814	Chemical and thermal phenomena in nerves
		612.815	Physiological morphology, histology, of nerves

- 612.815.1 Sensory nerve endings
 612.815.2 Motor nerve endings
 612.816 Excitability of nerves and muscles. Neuromuscular physiology. □ Fatigue of the nerves
 612.817 Action of nerves on muscles and glands. □ Tonus
 612.818 Trophic nerves. Degeneration, regeneration, cicatrization
 612.819 Special physiology of nerves
- 612.82 Brain. Nerve centres in general**
 612.821 Physiological psychology
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 612.821.1 Reaction time. Psychometry
 612.821.2 Attention. Association. Memory. Imagination
 612.821.3 Instinct. Intelligence. Sensitivity in psychological sense. Feelings. Emotions
 612.821.4 Effect of poisons on intelligence. □ Effect of alcohol, anaesthetics
 612.821.5 Effect of disease on intelligence
 612.821.6 Psychic reflexes
 612.821.7 Sleep. Unconsciousness. The subconscious. □ Dreams. Hypnotism
 612.821.8 The senses in general. Theories of perception
 612.821.89 Sensory illusions. Optical illusions
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 612.822 Neurons (nerve cells) in general. Nerve centres
 ⇒ 611.018.8
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 612.823 Weight and morphology of the brain and nerve centres
 612.824 Circulation in the nerve centres. □ Cerebrospinal fluid
 612.825 Cerebral convolutions. Cerebral cortex
 612.825.1 Excitability of the cortex
 612.825.2 Psychomotor centres. Localization. □ Speech centre
 612.825.3 Cortical epilepsy
 ⇒ 616.853
 612.825.4 Action of the brain on vegetative functions
 612.825.5 Cortex as centre of sensation and perception. The senses (reception)
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 612.825.55 Hearing
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 612.825.56 Smell
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 612.825.57 Taste
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 612.825.59 Tactile sense
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 612.825.8 Intellectual work
 612.826 Basal ganglia. □ Commissures
 612.827 Cerebellum
 612.828 Medulla oblongata. Cerebrospinal bulb
 612.829 Physiology of the nervous system of invertebrates
- 612.83 Spinal cord**
 612.833 Reflex activity
 612.833.1/7 ≡ 612.1/7
 612.833.8 Action of higher nerve centres on reflexes. □ Conditioned reflexes
- 612.84 Physiological optics. Sight (vision)**
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 ⇒ 681.784
 612.841 External (fibrous) coat of eye. □ Cornea. Conjunctiva. Sclerotic (white of eye)
- 612.842 Middle (vascular) coat of eye. □ Iris. Pupil. Choroid. Ocular pigments. Circulation in the eye. Intra-ocular pressure
 612.843 Optic bundle. Retina
 612.843.1 Physiological morphology. Visual substances
 612.843.11 Individual structural elements
 612.843.111 Blind spot (porus opticus, optic disc)
 612.843.112 Macula lutea. Yellow spot
 612.843.116 Rod and cone layer
 612.843.116.1 Rods
 612.843.116.2 Cones
 612.843.14 Chemistry of light perception. Visual substances. □ Retinal red pigment (visual purple)
 612.843.2 Stimulation, irradiation, of the retina
 612.843.3 Light sense (photoreception). Colour vision
 612.843.31 Colour vision (chromatopsia)
 ⇒ 159.937.51
 ⇒ 535.64
 612.843.32 Colour discrimination. Colour difference threshold. □ Colour stimulation threshold. Chromatic threshold
 612.843.35 Colour contrast. Luminance contrast (brightness contrast). □ After-images
 612.843.36 Luminance perception. Adaptation
 612.843.363 Daytime vision. Bright adaptation. Photopic vision
 612.843.364 Night vision. Dark adaptation. Scotopic vision
 612.843.367 Dazzle
 612.843.4 Entoptic phenomena. □ Opacities. Streaks. Muscae volitantes. Retinal blood vessels. Phosphenes. Diffraction and polarization phenomena. Photisms
 612.843.5 Time effects in visual phenomena. □ Persistence of vision. Perception of movements. Flicker. Scintillation (flashes)
 612.843.6 Visual field (of the fixed eye). Visual acuity. □ Power of resolution. Recognition of detail. Sense of form, shape
 612.843.7 Visual perception. □ Spatial vision. Binocular vision. Stereoscopy
 612.844 Dioptrics. □ Astigmatism. Aberration
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 612.846.2 Laws of motion of the eyes
 612.846.3 Role of the oculomotor nerve
 612.846.4 Role of the trochlear nerve
 612.846.6 Role of the abducent nerve
 612.846.7 Visual field (of the moving eye)
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 612.847 Palpebral and lacrimal apparatus. □ Palpebrae. Eyelids. Cilia. Eyelashes. Tear ducts. Periorbita
- 612.85 Hearing (audition)**
 ◇ 612.85.01 Physiological theories, principles of hearing
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 612.851 Functions of external ear. Sebaceous glands
 612.854 Middle ear
 612.855 Tympanum. Ear-drum. Tympanic membrane
 612.856 Pharyngotympanic tube (Eustachian tube)
 612.857 Auditory ossicles (bones of the ear)
 612.858 Inner ear. Labyrinth. Auditory meatus
 612.858.1 Sound conduction in inner ear
 612.858.2 Sacculle. Utricle
 612.858.3 Semicircular ducts

- 612.858.4 Cochlea. Organ of Corti
 612.858.5 Auditory nerves
 612.858.6 Endolymph. Perilymph
 612.858.7 Conduction of acoustic stimuli in the brain.
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 Deafness
 612.858.8 Motor auditory reflexes
- 612.86 Smell (olfaction)**
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 ⇒ 611.86
- 612.87 Taste (gustation)**
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- 612.88 Sense of touch, feeling. Sense of equilibrium**
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 ◇ 612.88:612.75 Sensitivity of bones
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- 612.881 Spatial perception
 612.882 Sense of temperature. Sense of heat, cold
 612.883 Sense of pressure
 612.884 Sense of pain
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 612.886 Sense of equilibrium (balance)
 612.886.3 Vertigo
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 612.887 Anaesthesia. Hyperaesthesia. Synaesthesia
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- 612.89 Sympathetic nervous system**
 612.89.03 Trophic functions of the sympathetic nervous system
 612.89.05 Motor reflexes
 612.89.08 Regulation of tone
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- 612.891 Cervical ganglia. Cervical plexus
 612.892 Thoracic ganglia. Thoracic plexus
 612.893 Abdominal ganglia. Abdominal plexus
 612.896 Action of the sympathetic nervous system on the iris
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 612.897 Action of the sympathetic nervous system on the heart
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 612.898 Action of the sympathetic nervous system on the digestive tract
 612.899 Action of the sympathetic nervous system on the vascular system
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- 613 HYGIENE GENERALLY. PERSONAL HEALTH AND HYGIENE**
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- 613.1 Climatic factors**
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- 613.11 Adaptation to climate. Acclimatization**
- 613.12 Healthy climate. Unhealthy climate. Climates of various places**
 ◇ 613.12(23) Mountain climate
 ◇ 613.12(285.3) Swamp or marsh climate
- 613.13 Seasons. Times of day**
- 613.14 Need for air. Amount of air required in confined spaces**
- 613.15 Composition of atmosphere**
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- 613.16 Physical factors**
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 613.161 Atmospheric humidity, moisture. Atmospheric aridity, dryness
 613.162 Atmospheric pressure
 613.163 Sounds. Noises
 613.164 Light. Sunlight. Ultraviolet and infrared radiation
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 613.167 Magnetism. Electromagnetism
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- 613.2 Dietetics. Nutrition principles applied to feeding and foods**
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- 613.2.03 Dieting and feeding: procedure and precautions
 613.2.032.33 Feeding by stomach tube
 613.2.035 Diet indications and contraindications
 613.2.038 Trial dieting and feeding
 613.2.099 Food poisoning
- 613.21 Meals: number, times, intervals etc.**
- 613.22 Nutrition of children**
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- 613.221 Hyponutrition. Fasting. Dieting. Slimming**
 ⇒ 641.561
- 613.24 Hypernutrition. Overeating. Fattening. Putting on weight**
- 613.25 Vegetable foodstuffs**
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 613.263 Vegetable oils and fats
 613.268 Vegetable conserves. Fruit conserves
 613.269 ⇒ 664.8
- 613.27 Mineral foodstuffs**
 ⇒ 664.4
- 613.28 Foodstuffs of animal origin**
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 613.281 Eggs
 613.286 Milk. Milk regime. Lacto-vegetarian diet.
 613.287 Lacto-ovo-vegetarian diet
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- 613.287.1 Mother's milk
 613.287.2 Wet-nurse's milk
 613.287.3 Prepared or preserved human milk
 613.287.5 Cow's milk. □ Dried milk. Condensed milk. Skimmed milk. Buttermilk
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 613.287.8 Duration of breast-feeding
 613.287.9 Weaning
 613.288 Fats. Fatty foods. □ Butter. Cheese. Margarine
- 613.29 Other solid foods. Artificial foods**
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- 613.292 Restoratives. Tonics. Stimulants. Enriched food
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- 613.295 Effect of cooking processes
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- 613.298 Cooking utensils. Tableware
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- 613.3 Drinks. Curative drinks. Medicinal waters. Liquid diet**
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- 613.31 Water. Water cures. Dehydration. Ice**
- 613.32 Impurities in water and ice**
- 613.34 Effect of water and ice purification**
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- 613.38 Natural and artificial mineral waters**
- 613.4 Personal hygiene. Clothing**
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- 613.41 Baths in general**
- 613.42 Bathing in bathtubs**
- 613.43 Footbaths. Hipbaths (sitzbaths) etc.**
- 613.44 Showers. Shower-baths**
- 613.45 Sea-bathing. Saline baths**
- 613.46 Hot-air baths. Steam baths.** □ Russian baths. Turkish baths
- 613.47 Swimming baths**
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- 613.481 Type. Composition. Permeability. Absorbency. Thermal conductivity. Effect on skin
- 613.482 Quantity of clothing. Wearer's resistance to cold
- 613.483 Cleanliness of clothing
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- 613.485 Dyeing. Pigments in clothing
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- 613.489 Individual garments
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- 613.49 Toiletries. Cosmetics. Make-up. Lotions. Colorants**
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- 613.6 Occupational health hazards. Occupational health and hygiene**
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- 613.6.01 Publicity, propaganda and warnings against health hazards
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- 613.6.015 Psychological methods and media
- 613.6.02 Occupational health risks and causes of sickness or illness
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- 613.6.027 Risks and causes at work
- 613.6.06 Results, consequences of occupational sickness or illness
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- 613.6.069 Fatality. Fatal occupational sickness
- 613.62 Occupational diseases in general**
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- 613.632 Contact with toxic substances
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- 613.633 Contact with dust
- 613.634 Contact with other noxious substances.
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- 613.643 Pressure: mechanical, atmospheric
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- 613.644 Noise. Vibration
- 613.645 Light
- 613.646 Temperature. Humidity. Climate
- 613.647 Electricity
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- 613.648 Radiations
- 613.648.2 Non-ionizing radiations (other than light)
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- 613.65 Working with excessive strain on particular organs. Injurious posture. Lack or excess of exercise.** □ Repetitive strain injury (RSI). Lack or excess of exercise
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- 613.67 Military health and hygiene**
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- 613.692 Health and hygiene of travellers, explorers
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- 613.7** Health and hygiene of leisure, recreation, sleep
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- 613.71** Gymnastics. Physical culture
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- 613.72** Physical exercises and games
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- 613.76** Holidays. Leave
- 613.77** Festival days (religious holidays)
- 613.79** Rest. Sleep. Sleeplessness, insomnia
- 613.8** Health and hygiene of the nervous system. Health and ethics
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- 613.81** Alcoholic drinks. □ Abstinence. Temperance. Anti-alcohol movement. Teetotalism
- 613.83** Narcotics
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- 613.84** Tobacco. Smoking. □ Anti-smoking movement
- 613.86** Mental and moral health and hygiene.
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- 613.87** Social customs and usages from the point of view of hygiene. □ Hand-shaking. Kissing
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- 613.88** Sexual hygiene. Sexual education. Sex life
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- 613.881** Sexual excess. Debauchery, dissoluteness, promiscuity
- 613.882** Prostitution
- 613.884** Sexual moderation. Sexual abstinence. Continence. Chastity
- 613.885** Sexual orientation
- 613.885.1** Sexual ambivalence. □ Bisexuality
- 613.886** Auto-eroticism. Masturbation
- 613.888** Sexual hygiene
- 613.888.1** Protective measures. Precautionary measures
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- 613.888.15** Contraception. Birth control
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- 613.888.151.1** Traditional methods of contraception. □ Coitus interruptus
- 613.888.151.2** Birth control by artificial interruption of pregnancy (abortus provocatus)
- 613.888.151.3** Birth control by continuous abstinence (continence). Birth control by rhythm method
- 613.888.151.5** Birth control using contraceptive devices
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◇ 613.888.151.5:615.477.87 Use of intrauterine devices (IUDs)
- 613.888.151.7** Birth control by medical means. □ Oral contraceptives ('the pill')
- 613.888.151.9** Birth control by surgical means. □ Sterilization. Vasectomy
- 613.89** Celibacy. Marriage. Monogamy. Polygamy
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- 613.9** Health and hygiene in relation to race, age, sex
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- 613.94** Racial health. Eugenics
- 613.95** Child health and hygiene
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- 613.952** Health and hygiene of the newborn (neonate)
- 613.953** Health and hygiene in early infancy (up to 2 years)
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- 613.953.1** Milk feeding
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- 613.955** Health and hygiene of school-age children
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- 613.96** Health and hygiene of adolescents
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- 614** PUBLIC HEALTH AND HYGIENE. ACCIDENT PREVENTION
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- 614.1** Population. Depopulation
- 614.2** Public and professional organization of health
- 614.21** Hospitals. Infirmaries
- 614.212** Health clinics. Surgeries. Dispensaries
- 614.213** Nursing homes
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- 614.214** Convalescent homes
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- 614.216** Homes for chronic invalids. □ Homes for alcoholics, drug addicts, incurables
- 614.217** Isolation hospitals (for infectious, communicable diseases)
- 614.22** Regulations concerning dissection, vivisection
⇒ 57.089.6
- 614.23** Medical requirements and examination for suitability for a post. Issuing of official medical certificates (other than in forensic medicine)
- 614.25** Concerns, rights, duties of doctors
- 614.251** Rights. □ Freedom to practice. Privileges
- 614.252** Admission to the medical profession.
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- 614.253** Relations of doctors with colleagues, with patients. Medical ethics
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- 614.253.1** Doctors in public hospitals, universities, public laboratories and institutes
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614.253.4	Students	614.48	Sterilization. Disinfection
614.253.5	Attendants. Orderlies. Nurses. Midwives	614.6	Cemetery hygiene. Disposal of the dead ⇒ 351.776 ⇒ 718
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614.253.8	Patients	614.62	Cremation. Burning, incineration of corpses
614.253.83	Rights and obligations of patients. Doctor's responsibility for treatment. Neglect of patient. Withholding of treatment. Rejection of treatment	614.63	Transport of corpses
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614.253.89	Environment, family, particulars of patients	614.65	Effect of cemeteries on public health. □ Microbial contamination. Pollution of air, water
614.254	Professional life and activity of doctors. □ Surgery. Consultations. Visits. Holidays. Locums (locum tenentes)	614.66	Disinterment. Exhumation
614.255	Relations of doctors to the community, to organizations. Doctors in private and public undertakings	614.67	Postmortems. Autopsies ⇒ 616-091.5
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614.257	Doctors' fees	614.71	Hygiene of the air. Air pollution ⇒ 551.51 ⇒ 613.5
614.258	Medical organizations, professional societies	614.712	Purification of air. Effect of wind
614.26	Illegal practice of medicine. □ Quacks. Charlatans	614.715	Pollution by airborne dust, smoke
614.27	Sales of medicines, drugs ⇒ 615.12	614.718	Smells. Effluvia. Noxious emanations. Miasmas ⇒ 613.157
614.28	Sale of poisons, narcotics	614.72	Pollution by noxious gases
614.29	Official medical councils, commissions	614.73	Radioactive fallout ⇒ 614.876
614.3	Sanitary inspection and control ⇒ 343.347 ⇒ 351.773	614.75	Precautions with, and handling of, noxious and dangerous substances ⇒ 351.753 ⇒ 614.835 ⇒ 620.26
614.31	Inspection of food. □ Inspection of shops, markets ⇒ 612.39 ⇒ 613.2	614.76	Pollution of both soil and air. □ Decomposing animal and vegetable matter. Animal dung. Mucking out. Animal corpses
614.35	Inspection of medicines ⇒ 613.83	614.77	Soil. Effect of soil and its effluvia on health. Topography and health ⇒ 631.4
614.37	Inspection of colouring matter. □ Used in wallpapers, textiles, clothes, toys, kitchen utensils, furniture	614.777	Water. Hygiene of water ⇒ 628.1/3
614.38	Health control of public gatherings, meetings, places of assembly	614.778	Plants and human health ⇒ 614.448
614.39	National health services	614.779	Animals and human health ⇒ 614.449
614.4	Prevention and control of communicable (infectious, contagious) diseases. Prevention of epidemics ⇒ 351.77	614.78	Urban hygiene. Public health in towns ⇒ 351.777.8 ⇒ 628 ⇒ 696.1
614.44	Preventive and restrictive measures	614.79	Rural hygiene. Public health in the countryside
614.441	Obligation to notify diseases		
614.442	Measures affecting the air		
614.443	Measures affecting the soil. □ Drainage. Reclamation of marshes		
614.444	Measures against dust		
614.445	Measures affecting water. □ Dredging. Filling ditches		
614.446	Measures affecting humans. □ Control of disease carriers. Suspected cases		
614.447	Measures affecting public places, markets, transport		
614.448	Measures affecting plants, vegetation. □ Cultivation. Destruction		
614.449	Measures affecting animals. □ Destruction of vermin, parasites (rats, insects etc.). Rodent control. Protection of useful animals 614.449 ≡ 59		
614.45	Isolation of persons with communicable diseases		
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- 614.8 Accidents. Risks. Hazards. Accident prevention. Personal protection. Safety**
Denote persons involved in accidents by -05
- ◇ 614.8:59 Hazards etc. from animals
 - ◇ 614.8:631.3 Hazards etc. from agricultural machinery
 - ◇ 614.8:656.1 Road traffic accidents etc.
 - ⇒ 351.78
 - ⇒ 613.6
 - ⇒ 62-78
 - ⇒ 622.8
 - ⇒ 628.5
- 614.8.01 Publicity and propaganda media for accident prevention
- 614.8.013 Visual methods and media
Details by :654.91 and :659.13.
For signs, symbols, diagrams use colon combination with 003.6
- ◇ 003.628:614.876.013 Warning symbol for ionizing radiation, radioactivity
- 614.8.014 Acoustic, aural and oral methods and media
Further details may be denoted by colon combination with the subdivisions of 654.92 and 659.14
- 614.8.015 Psychological methods. Sustained campaigns
- 614.8.02 Occurrence and scale of accidents
- 614.8.026 Accidents in daily life and in the home
- 614.8.026.1 Risks
- 614.8.026.2 Causes
- 614.8.027 Accidents at work. Occupational accidents
- ⇒ 331.46
- 614.8.027.1 Risks
- 614.8.027.2 Causes
- 614.8.028 Scale, extent of accidents
- 614.8.028.2 Small-scale accidents
- 614.8.028.3 Moderate, medium-scale accidents
- 614.8.028.4 Large-scale, serious accidents
- 614.8.06 Results, consequences of accidents
- 614.8.067 Results, consequences of occupational accidents
- 614.8.067.1 Lost working days
- 614.8.067.3 Degree of disablement
- 614.8.069 Fatal accidents
- 614.8.084 Accident prevention, protection and safety
- ⇒ 614.88
- 614.8.086 Protection of the body or its parts against specific influences
- ⇒ 331.453
- 614.8.086.2 Against mechanical effects
- 614.8.086.3 Against dust
- 614.8.086.4 Against chemicals and gases
- 614.8.086.5 Against radiation
- 614.8.086.52 Against both visible and invisible radiation
- 614.8.086.53 Against visible radiation, glare
- 614.8.086.54 Against invisible radiation. □ Against infrared and ultraviolet radiation
- 614.81 Hazards from water.** □ Drowning. First aid in cases of drowning
- 614.818 Rescue apparatus. □ Lifeboats. Rescue rafts. Life-jackets. Life-belts
- 614.82 Hazards from various mechanical and physical causes**
- 614.821 Hazard of falling. □ Accidents due to falls, slipping etc. Safety nets, harness etc.
- 614.822 Falling, dropping, overbalancing of objects on to persons. Crushing
- 614.823 Subsidence, collapse, landslip. Burial, smothering
- 614.824 Suffocation. Asphyxiation. Gas poisoning
- ⇒ 612.232
- 614.825 Electrical hazards. Accidents due to electricity. Electric shock
- ⇒ 613.647
- 614.826 Mechanical shocks, impacts, collisions
- 614.83 Explosion hazards**
- 614.832 Hazards from excessive pressure, oversteering. Bursts. Blow-outs
- 614.833 Chemical explosions
- 614.834 Detection of explosive gas mixtures, dispersions
- 614.835 Hazards due to storage of explosive or flammable materials
- ⇒ 614.75
- 614.837 Addition of substances to inhibit explosions
- 614.838 Various other preventive measures against explosions. □ Safety valves
- 614.839 Restriction, limitation, localization of damage due to explosions
- 614.84 Fire hazards. Fires. Protection against fire**
- ⇒ 614.89
- 614.841 Fires: nature, causes, prevention, course, types
- 614.841.1 Nature of burning process, combustion, extinction
- 614.841.2 Causes of fire. Fire risks
- 614.841.22 Causes arising from nature of material.
- Spontaneous combustion
- 614.841.23 Natural external causes of fires. □ Lightning. Volcanic eruption
- 614.841.24 Technical defects as causes of fires. □ Fires from explosions. Electrical fires
- 614.841.25 Faulty construction, building, as cause of fires
- 614.841.26 Fires caused by humans. □ Fires caused by negligence. Deliberately caused fires
- ⇒ 343.76
- 614.841.27 Fires caused by acts of war, civil disturbance, riots
- 614.841.3 Fire prevention. Fire regulations. Limitation of fire damage
- 614.841.31 Instruction, education in fire precautions. Fire prevention by inspection
- 614.841.33 Fire prevention by construction, building design. □ Fire-resistant materials. Emergency exits
- ⇒ 621.316.98
 - ⇒ 699.81
- 614.841.34 Fire prevention at work. □ In offices, factories
- 614.841.4 Course, progress of fire. □ Type of fire. Particular cases of fire. Fire prevention and control in particular cases
- 614.841.41 Combustion of different materials.
- Flammability, ignitability, self-ignition.
- Course, progress of fire. Fire prevention and control
- 614.841.42 Fires and fire precautions in nature
- ◇ 614.841.42:630 Forest fires
 - ◇ 614.841.42(251.3) Prairie fires
 - ⇒ 632.187
- 614.841.43 Effect of gaseous, flammable material
- 614.841.44 Large, widespread fires. □ Conflagrations of groups of buildings, of large areas
- 614.841.45 Fires and fire protection in single buildings
- Denote particular industries by :66/68, otherwise specify by :69 and :72*
 - ◇ 614.841.45:664.7 Fires in flour mills
 - ◇ 614.841.45:692.4 Roof fires
 - ◇ 614.841.45:728.5 Hotel fires
 - ⇒ 699.81
- 614.841.46 Fires in rooms. Smaller fires

614.841.47	Fires outside buildings. □ Fires in storage areas, timberyards, dumps. Fires in excavations ⇒ 622.82	614.85	Policing of accidents. □ Technical organization. Patrol cars. Accident squads. Road accident units. Radio cars
614.841.48	Fires and fire protection in transport systems. □ Traffic, vehicle fires ◇ 614.841.48:629.7 Fires in aircraft ◇ 614.841.48:656.71 Fires at airports ⇒ 656.085.5	614.86	Traffic accidents. □ Accidents to persons caused by motor and other vehicles and craft. Road accidents. Rescue of travellers. Air search and rescue
614.841.492	Fires on humans and their clothing ⇒ 616-001.16/.17	614.862	Passenger protection and safety
614.842	Notification of fires. Fire extinguishing technique and tactics. Organization of firefighting	614.864	Protection and safety of pedestrians, bystanders etc.
614.842.4	Notification of fires. □ Fire alarms. Fire detection and alarm systems. Observation posts ⇒ 654.147 ⇒ 654.9	614.87	Exposure to ambient influences. Environmental hazards. Personal protection and safety ⇒ 613.6 ⇒ 616-001.1
614.842.6	Fire extinguishing technique and tactics. □ Extinguishing materials	614.872	Exposure to mechanical influences. □ Pressure. Sound. Vibrations
614.842.8	Organization of firefighting	614.873	Exposure to temperature variations, cold, heat
614.842.81	Preliminary help at fires. Further action, e.g. by firm's fire officers	614.875	Exposure to non-ionizing radiations. □ Light. Infrared rays. Ultraviolet rays
614.842.83	Fire brigades. Fire service	614.876	Exposure to ionizing radiations (radioactivity) ⇒ 614.73
614.842.84	Fire stations	614.878	Exposure to irritant substances. □ Chemicals ⇒ 614.8.084
614.842.86	Structure and personnel of fire brigade. Firemen's equipment, clothing etc.	614.88	First aid. Casualty and ambulance services
614.842.9	Emergency use of fire brigades, other than for fires	614.881	First aid posts. Fixed casualty stations
614.843	Firefighting equipment in general	614.882	First aid vehicles. Mobile casualty services
614.843.1	Fire hydrants ⇒ 621.643.51	614.883	Transport of the injured. Ambulances
614.843.2	Fire hoses	614.885	Volunteer assistance. Red Cross ⇒ 341.33
614.843.3	Connections. Stopcocks ⇒ 621.643.4	614.888	Equipment for accident services. □ Stretchers. Medicine chests. Resuscitation apparatus
614.843.4	Nozzles. Jets	614.89	Hazards to particular parts of the body. Personal protective equipment (PPE). Protective clothing etc. ⇒ 613.6 ⇒ 616-001 ⇒ 62-78 ⇒ 687.17 ⇒ 687.434
614.843.5	Small firefighting devices. □ Fire buckets. Stirrup pumps	614.891	Protection of the head and face. □ Protective hats, helmets, masks, veils
614.843.6	Hose reels. Other hose accessories	614.892	Protection of the ears. □ Ear-muffs. Ear-plugs
614.843.8	Foam devices. □ Foam generators, sprays	614.893	Protection of the eyes
614.844	Fire extinguishing installations. Combined fire alarm and extinguishing systems. □ Sprinkler systems	614.893.2	Protective glasses, spectacles, goggles ⇒ 681.738
614.845	Portable fire extinguishers. Other devices	614.893.6	Protective shields, screens, visors. □ Arc-welders' shields
614.845.1	Dry extinguishers. Powder extinguishers	614.894	Protection of respiratory system. □ Breathing apparatus
614.845.2	Wet extinguishers. Liquid extinguishers	614.894.2	Gas masks. Filter masks
614.845.4	Gaseous extinguishers	614.894.29	Filters for gas masks
614.845.5	Foam extinguishers	614.894.3	Masks for protection against dust ⇒ 628.511
614.846	Fire-extinguishing jets. Fire engines ⇒ 621.65/.69	614.894.4	Breathing apparatus with chemical and mechanical filters (where sufficient oxygen exists). Respirators without oxygen reservoir
614.846.1	Hand pumps	614.894.43	Filter. Filter pattern
614.846.2	Steam-powered pumps	614.894.7	Breathing apparatus for use in noxious gases, vapours, dusts, with oxygen shortage.
614.846.3	Motor-powered pumps (internal combustion engine)	614.895	Respirators with oxygen reservoir
614.846.4	Electric pumps		Protection of the body. Whole-body protection. □ Diver's suits. Protective clothes, aprons, overalls ⇒ 614.842.86 ⇒ 615.479 ⇒ 626.025 ⇒ 646 ⇒ 687.17
614.846.5	Other pumps		
614.846.6	Firefighting vehicles. Fire engines ⇒ 629.361 ⇒ 629.561.4		
614.846.9	Firefighting craft. Fire-floats ⇒ 627.357 ⇒ 629.561.4		
614.847	Rescue, escape and auxiliary equipment in firefighting		
614.847.1	Fire brigade ladders		
614.847.2	Mechanical firefighting equipment		
614.847.7	Rescue equipment. □ Escape chutes. Rescue ropes. Jumping sheets		
614.847.8	Special vehicles, e.g. for equipment		
614.847.9	Auxiliary equipment. □ Signalling apparatus. Floodlights. Tools		

- 614.896 Protection of the arms and hands. □ Protective sleevelets, gauntlets, gloves, mittens. Fingerstalls. Thimbles
⇒ 685.4
- 614.897 Protection of the legs and feet. □ Protective leggings, pads, shoes, boots
- 614.898 Protection of the skin generally. □ Protective lotions. Barrier creams
- 614.9 Animal health. Veterinary hygiene**
⇒ 591.2
⇒ 636.09
- 614.91 Preventive measures against communicable diseases in animals**
- 614.94 Accommodation, housing of animals**
- 614.95 Animal nutrition, feeding**
- 614.96 Transport of animals**
- 614.97 Hygiene of slaughterhouses, knackers' yards etc. Slaughtering methods**
- 614.99 Removal and disposal of animal bodies, carcasses**
- 615 PHARMACOLOGY. THERAPEUTICS. TOXICOLOGY**
⇒ 616-085
⇒ 661.12
- 615.01 Pharmacological details**
- 615.011 Physical, physicochemical and chemical properties of medicaments
- 615.012 Production of medicaments (drugs, medicines, remedies). Preparation and purification
- 615.014 Practical pharmacy. Pharmaceutical practice
- 615.014.2 Compounding and dispensing of medicaments
- 615.014.4 Storage. Preservation. Stabilization. Activation. Sterilization. Alteration
- 615.014.6 Coating. Encapsulation
- 615.014.8 Packaging and containers. Sealing and seals. Closing and closures. Labelling
- 615.015 Pharmacodynamics. General principles of pharmacology
- 615.015.1 Influence of chemical and physical factors
- 615.015.11 Relation between chemical composition and action
- 615.015.12 Relation between concentration and action
- 615.015.13 Relation between dose and response (action)
⇒ 615.015.3
- 615.015.14 Influence (or effect) of solvent used. □ Of constituents, vehicles, excipients
- 615.015.15 Time factors in pharmacological action
- 615.015.152 Immediate, instantaneous, rapid action
- 615.015.153 Slow, delayed action
- 615.015.154 Depot action (gradual release). □ By means of patches or subcutaneous implants
- 615.015.16 Physical factors. □ Temperature, pressure
- 615.015.2 Combined effects. Action of combination of drugs
- 615.015.3 Dose and action (response). Dosage. Dosimetry
⇒ 615.015.13
- 615.015.32 Homoeopathic dose. Homeopathy
- 615.015.33 Usual doses. Normal doses
- 615.015.34 Overdoses. Shock doses
- 615.015.35 Maximal doses. Toxic doses
- 615.015.36 Lethal (fatal) doses
- 615.015.37 Abortive doses. Divided doses (dosis refracta)
- 615.015.38 Multiple doses
- 615.015.4 Various modes of pharmacological action. Chemical and biochemical changes of substance in organisms. Biotransformation
- 615.015.5 Tolerance
- 615.015.6 Habituation. Dependence. Addiction
- 615.015.7 Tachyphylaxis
- 615.015.8 Resistance of microorganisms
- 615.017 General and nonspecific effect of individual medicaments
- 615.03 Clinical pharmacology.** □ Application, administration of medicaments. Medication
- 615.031 Localization
- 615.032 Methods and routes of administration
615.032 ≡ 611
- 615.033 Absorption and distribution. Diffusion and concentration. Blood level of medicaments
615.033 ≡ 611
- 615.034 Route of excretion
615.034 ≡ 611
- 615.035 Indication
- 615.035.1 Favourable or effective indication
- 615.035.2 Contraindication. □ Dangers in application
- 615.035.3 Abuse. Excessive use
- 615.035.4 Preventive administration. Prophylactic indication
- 615.035.7 Misuse. Erroneous administration
- 615.035.9 Urgent indication. Use in emergency
- 615.036 Clinical results. Statistics etc.
⇒ 615.06
- 615.038 Trial administration
- 615.039 Placebos. Dummy, pseudo-medication
- 615.06 Complication during treatment**
- 615.061 Previous treatment
- 615.065 Side-effects. □ Iatrogenic illness
- 615.07 Analysis. Standardization. Identification**
- 615.071 Visual inspection. Identification. Differentiation. Appearance
- 615.073 Physical analysis and tests
Further particulars may be denoted by colon combination with the subdivisions of 53
- 615.074 Chemical analysis and tests
Further particulars may be denoted by colon combination with the subdivisions of 54
- 615.076 Biological and microbiological analysis and standardization
- 615.077 Tests of preserving methods for medicaments. □ Aftertesting, checking. Packing, tinning.
- 615.099 Storage
Poisoning (intoxication). Addiction. Special toxicology
For general toxicology see 615.9
Use .099 with subdivisions of 615.2/3
⇒ 340.67
⇒ 616-099
- 615.1 General and professional pharmacy**
⇒ 614.2
- 615.11 Pharmacopoeias. Codices. Formularies. Prescriptions**
- 615.12 Pharmaceutical laboratories and pharmacies**
- 615.15 Pharmacists. Pharmaceutical chemists (dispensing chemists)**
- 615.19 Other people engaged in pharmaceutical profession.** □ Druggists (unqualified). Herbalists
- 615.2 Medicaments according to their principal action**
Chemical structure of substance may be denoted by the special auxiliaries '1, subdivided as 546, and '2/9 subdivided as 547.2/9.
Denote medicaments by name by adding A/Z

615.21/26	Pharmacodynamic actions	615.241	Agents affecting appetite, digestion. □ Digestants. Orexigens. Anorexiantes. Roborants. Tonics
615.21	Agents predominantly affecting the nervous system	615.242	Remedies for diseases of oral cavity and teeth
615.211	General anaesthetics	615.243	Remedies for diseases of stomach and duodenum, including ulcers. □ Antacids
615.212	Analgesics and antipyretics		Acids in general. Acida
615.212.3	Antipyretic analgesics	615.243.3	Antacids. Alkaline agents
615.212.4	Antipyretics. □ Antifebrilants, febrifuges	615.243.4	Emetics
615.212.7	Addictive analgesics. Narcotics	615.243.5	Anti-emetics
615.213	Antiepileptics. Anticonvulsants	615.243.6	⇒ 615.236
615.214	Psychopharmacological agents. Psychotropic agents ⇒ 613.83	615.244	Agents acting on liver and gall bladder. □ Choleretics. Cholagogues
615.214.2	Neuroleptics. Psycholeptics	615.245	Agents predominantly affecting the pancreas
615.214.21	Neuroplegics	615.246	Remedies for diseases of the intestine ⇒ 612.334
615.214.22	Tranquillizers. Ataractics		Adsorbents
615.214.23	Psychorelaxants	615.246.2	Laxatives (aperients). Purgatives. Cathartics
615.214.24	Sedatives. Hypnotics, soporifics	615.246.4	Physically acting evacuants. 'Lubricants'
615.214.3	Psychoanalitics	615.246.6	Carminatives. Remedies against flatulence, tympanites.
615.216	Agents predominantly affecting the somatic nerves (sensory and motor nerves), the striated musculature (voluntary, skeletal muscle)	615.246.7	Antidiarrhoeics, costive (binding) agents
615.216.2	Local anaesthetics	615.246.8	Detoxicants, antidotes
615.216.5	Myorelaxants. Skeletal-muscle relaxants	615.246.9	⇒ 615.279
615.216.6	Peripheral-nervous-system stimulants		
615.216.8	Agents predominantly affecting the sensory functions ⇒ 612.84/88	615.25	Agents predominantly affecting the endocrine glands, urogenital system, reproductive functions
615.217	Agents affecting the autonomic nervous system and smooth musculature (involuntary muscle). □ Antispasmodics. Ganglioplegics	615.252	Agents acting on endocrine glands 615.252 ≡ 612
615.217.2	Affecting the sympathetic nervous system. □ Sympathomimetics. Sympathicotonic.	615.252.349.7	Acting on the islets of Langerhans. □ Antidiabetica
615.217.3	Adrenergics. Sympatholytics. Adrenolytics Affecting the parasympathetic nervous system. □ Parasympathomimetics. Cholinergic agents. Parasympathicotonic. Vagotonic. Parasympatholytics. Anticholinergic agents. Neurotropics. Antispasmodics	615.254	Agents acting on the urogenital system ⇒ 612.464
615.217.4	Affecting the autonomic ganglion. □ Ganglion blockers and stimulants. Ganglioplegics	615.254.1	Diuretics. Saluretics
615.217.5	Affecting the smooth muscles. □ Myotropic antispasmodics	615.254.2	Antidiuretics
615.218	Histaminics. Antihistaminics. Anti-allergic agents	615.254.4	Renal function inhibitors
		615.254.7	Remedies against nephrolithiasis (kidney stones), urolithiasis (stones in urinary tract). □ Lithogogues. Litholytics. Lithotripts
615.22	Agents predominantly affecting the heart and circulation. Cardiovascular agents	615.256	Agents predominantly affecting the genital organs
615.221	Analeptics. Myocardiocics. Cardiocics	615.256.2	Agents affecting the reproductive functions of both sexes. □ Aphrodisiacs
615.222	Anti-arrhythmic agents. Cardiac depressants	615.256.3	Contraceptive substances. Spermaticides (spermicides) ⇒ 615.477.86/87
615.224	Agents affecting the coronary vessels. Coronary vasodilators	615.256.4	Agents affecting the male reproductive functions ⇒ 615.357
615.225	Agents predominantly affecting the blood vessels. □ Vasoconstrictors. Vasodilators	615.256.5	Agents affecting the female reproductive functions. □ Oestrogens. Gestagens. Abortifacients (ambolics). Remedies for menstrual disorders. Lactogenics ⇒ 615.357
615.23	Agents predominantly affecting the respiration and respiratory organs	615.26	Agents predominantly affecting the common integument
615.232	Respiratory stimulants	615.262	Agents affecting the skin
615.233	Cough remedies. □ Linctuses. Sternutatories. Errhines. Ptarmics	615.262.1	Astringents. Antiphlogistics (anti-inflammatory agents) ⇒ 615.276
615.234	Agents against asthma. □ Antiasthmatics. Antispasmodics. Antitussives	615.262.2	Emollients. Demulcents. Protectives. Skin softeners. Mucilaginous substances
615.235	Expectorants ⇒ 615.243.5	615.262.3	Epispastics. Rubefacients. Vesicants. Irritants
615.236	Anexpectorants ⇒ 615.243.6	615.262.5	Caustics. Corrosives. Keratolytics ⇒ 615.832.7 ⇒ 616-001.37
615.238	Remedies against common cold	615.263	Agents predominantly affecting cutaneous glands, hair, nails
615.24	Agents affecting the alimentary system, including oral cavity. Gastrointestinal agents ⇒ 612.39 ⇒ 613.2	615.263.2	Anhydrotics. Antiperspirants
		615.263.3	Diaphoretics. Sudorifics
		615.263.5	Agents affecting the sebaceous glands

615.263.6	Agents affecting the hair ⇒ 665.585		<i>Further particulars may be denoted by colon combination with the subdivisions of 577.15 and by alphabetic extension</i>
615.263.62	Hair tonics, restoratives. Agents promoting growth of hair	615.356	Vitamins
615.263.63	Remedies for hair diseases		<i>Further particulars may be denoted by colon combination with the subdivisions of 577.16 and by alphabetic extension</i>
615.263.64	Depilatories		
615.263.7	Agents affecting the nails ⇒ 665.584.7	615.357	Hormones
615.27	Agents affecting metabolism and cell functions		<i>Particulars may be denoted by colon combination with the subdivisions of 577.17 ⇒ 615.361</i>
615.272	Agents affecting metabolism. Biochemical agents	615.36	Therapy by means of body substances (except hormones)
615.273	Agents affecting blood and body fluids. □ Haemolytics. Coagulants, styptics. Anticoagulants	615.361 615.362	Organotherapy. Opothrapy Normal body fluids (except blood and serum). External secretion products in general
615.275	Agents affecting cell functions. □ Antirheumatics	615.363	Agents antagonistic to the function of an organ, a body fluid, a body ferment, or a secretion product
615.276	Anti-inflammatory agents generally. Antiphlogistics ⇒ 615.262.1	615.365 615.366	Anticytolsines. Anticytotoxins etc. Pathological products in general. □ Diseased organs. Extracts from diseased organs. Pathological exudates
615.277	Agents affecting abnormal cells. □ Carcinogens. Carcinostatics	615.368	Gynaecological substances
615.277.3	Anti-tumour agents. Carcinostatics. Antineoplastics	615.37	Immunotherapy ⇒ 614.47
615.277.4	Carcinogens		Pathogenic microorganisms. Parasites or their poisons. Vaccines
615.279	Antitoxins. Detoxicants. Antidotes	615.371 615.372	Cultures of microorganisms. Toxins. Toxoids. Pasteur's method
615.28	Agents against pathogenic microorganisms, parasites. Chemotherapeutics. Chemicals for hygienic purposes. □ Disinfectants. Antiseptics	615.373 615.375	Immune serum. Serotherapy Other immunizing agents
615.281	Antibacterials. Antivirals 615.281.8 ≡ 578.8 615.281.9 ≡ 579.8	615.38	Haemotherapy. □ Blood transfusion. Blood injection. Serum injection
615.281.8	Antivirals	615.381 615.382	Blood injection. Whole-blood transfusion Blood plasma. Plasma transfusion. Plasma expanders
615.281.9	Antibacterials	615.383 615.384 615.387	Blood serum. Serum transfusion Blood substitutes. Serum substitutes Preserved blood
615.281.934	Antispirochaetals		
615.281.981.11	Antirickettsials	615.4	Pharmaceutical preparations. Medical material. Equipment
615.282	Antimycotics. Fungicides 615.282 ≡ 582.28	615.45	Pharmaceutical preparations
615.282.84	Agents against Hyphomycetes. □ Candida albicans	615.451 615.451.1	Liquid preparations Galenicals. □ Extracts. Solutions. Infusions. Tinctures
615.283	Antiprotozoal agents 615.283 ≡ 593.1	615.451.2 615.451.3	Liquid preparations for internal use. □ Elixirs. Syrups. Emulsions Liquids for local, topical use. □ Collodions. Glycerines. Liniments. Embrocations. Skin lotions. Gargles. Mouthwashes. Aerosols. Sprays. Preparations for internal douches, enemas
615.283.21	Amoebicides	615.453	Solid preparations
615.284	Anthelminthics. Vermicides 615.284 ≡ 595.1	615.453.1 615.453.2 615.453.3 615.453.4 615.453.5 615.453.6 615.453.7 615.453.8	Dragees (sugar-coated pills) Powders. □ Insufflations. Dusting powders Granules. Grains Capsules. Cachets. Wafers Pills Tablets Pastilles. Troches. Lozenges Other solid preparations. □ Medicated sticks, pencils. Medicated tobaccos. Medicated teas
615.284.21	Anticestodals. Cestocides	615.454	Semisolid preparations
615.285	Agents against parasitic Arthropoda, Insecta 615.285 ≡ 595	615.454.1 615.454.2 615.454.3	Ointments. Pastes. Creams. Gels. Jellies Suppositories Cataplasms. Poultices
615.285.7	Agents against Insecta. Insecticides	615.456 615.457	Preparations for parenteral administration Ophthalmic preparations. Eye lotions. Collyria
615.3	Medicaments according to their origin		
615.31	Chemical substances in general		
615.32	Substances of natural origin and their mixtures. Drugs in the narrower sense		
615.322	Vegetable substances		
615.324	Animal substances		
615.326	Natural mineral substances		
615.327	Mineral water		
615.33	Antibiotics. Microbiological substances		
615.331	Bacterial antibiotics. □ Peptide antibiotics (bacteriocins), antimicrobial peptides. BLIS (bacteriocin-like inhibitory substances). Antibiotics (lanthionine-containing antibiotics)		
615.332	Produced by actinomycetes, streptomycetes		
615.334	Produced by fungi		
615.34	Catalytic antibodies (abzymes)		
615.35	Enzymes. Vitamins. Hormones		
615.355	Enzymes and their inhibitors		

615.458	Gaseous preparations (for inhalation, insufflation etc.)	615.477.5	Orthopaedic girdles, hosiery, belts, straps.
615.46	Medical materials. Surgical dressings	615.477.7	□ Support hose for varicose veins
	⇒ 616-7	615.477.8	Sanitary towels, napkins, tampons
615.461	Prosthetic material generally		Rubber and plastic articles for nursing, medicine, surgery
615.462	Plastics	615.477.82	Bulbs. Globes. Tubes
615.463	Gypsum, plaster and cement materials	615.477.83	Fingerstalls
615.464	Ceramic materials. Porcelain	615.477.84	Artificial teats for bottles. Dummies
615.465	Metallic materials. □ Pins. Nails		⇒ 615.473.84
615.466	Other surgical material. □ Bone. Ivory	615.477.85	Flexible sounds. Stomach tubes etc.
615.468	Dressings. Bandages	615.477.86	Male contraceptive devices. □ Condoms (sheaths)
	⇒ 616-089.4	615.477.87	Female contraceptive devices. □ Pessaries.
615.468.2	Gauze. Bandages. □ Sticking plasters.		Intra-uterine devices (IUDs)
	Gutta-percha tissue	615.478	Medical and surgical equipment for hospitals
615.468.4	Shredded wool and other absorbent materials.	615.478.1	Complete furnishings for hospitals and clinics
	□ Wadding. Cotton wool. Tow. Linen fluff	615.478.2	Furniture. Linen. □ Beds. Bedclothes. Sheets.
615.468.6	Thread for ligatures and sutures. □ Catgut		Pillows. Mattresses. Air beds. Water beds.
615.468.7	Dressings ready for use. □ Plaster-of-Paris dressings		Bedpans. Urinals (urine bottles)
			⇒ 645
615.47	Instruments. Apparatus. Equipment and furnishing		⇒ 684.4
	⇒ 616-7	615.478.3	Wheelchairs. Bath chairs. Trolleys. Other means of transport in hospitals
615.471	Medical and surgical instruments and equipment in general		⇒ 621.86/.87
	<i>Particulars may be denoted by colon combination with the subdivisions of 616 with addition of the -7 special auxiliary subdivisions listed under 616</i>	615.478.5	Incubators
	◇ 615.471:616.12-7 Cardiological instruments and equipment	615.478.6	Equipment for clinical examinations. Surgical, operation, theatre furniture
615.472	Surgical and medical instruments for therapeutic purposes	615.478.62	Examination and operation tables
615.472.1	Sets of instruments. Instrument cases	615.478.65	Operation chairs
615.472.2	Surgical needles. Clamps etc.	615.478.7	Various furniture and equipment
615.472.3	Knives. Scalpels. Lancets. Drills. Trephines.	615.478.72	Glass cupboards, wall cabinets, stands etc.
	Other cutting and sawing instruments	615.478.73	Sterilization apparatus. □ Autoclaves
615.472.4	Scissors. Forceps. Tweezers etc.	615.478.74	Disinfection apparatus. Incinerators, bins, containers for soiled dressings
615.472.5	Catheters. Dilators. Bougies		Spittoons. Suction apparatus for saliva removal
615.472.6	Instruments for tying off, obturation, tamponing, plugging	615.478.76	Basins, vessels for the operating theatre
	⇒ 616-089.45	615.478.77	Adjustable, portable and articulated stands, brackets
615.472.7	Instruments for cauterization	615.478.78	Various aseptic utensils for the operating theatre
615.472.9	Protective and wrapping material for surgical instruments	615.479	Clothing for doctors, nursing staff, patients
615.473	Appliances for introduction of medicaments. Apparatus for introduction and drainage of fluids, lavage		⇒ 685
615.473.1	For cleaning and treatment of the skin		⇒ 687
615.473.2	Needles for injection. Hypodermic needles. Cannulae	615.479.4	Clothing for doctors and nursing staff.
615.473.3	Hypodermic syringes		□ Operation clothes. Face masks. Aprons.
615.473.4	Ordinary syringes	615.479.5	Surgical gloves
615.473.5	Irrigators		Clothing for patients
615.473.8	Devices for administering, feeding. □ Medicine spoons. Droppers. Graduated glasses	615.8	Physiotherapy. Physical therapy. Radiotherapy. Other nonmedicinal therapeutic treatment
615.473.84	Infants' feeding bottles		Instrumentation and equipment
	⇒ 615.477.84	615.8-7	<i>Used for specific purposes with the main divisions of 615.8, but not as the main place which is at 615.4</i>
615.473.92	Various other appliances for introduction of medicaments		615.8-7 = 616-7
615.477	Orthopaedic material. Trusses. Rubber articles etc.	615.81	Mechanical methods and remedies
615.477.1	Orthopaedic footwear	615.811	Extractive treatment. Derivation. □ Bleeding.
	⇒ 685.348.4		Phlebotomy (venesection). Leeches. Cupping.
615.477.2	Artificial limbs. Prostheses. Crutches.	615.814	Induced abscess. Pyotherapy
	Walking-sticks	615.814.1	⇒ 615.324
615.477.21	Artificial arms, hooks etc.	615.814.5	Puncture and stinging treatment
615.477.22	Artificial legs		Needle puncture. Acupuncture
615.477.3	Orthopaedic corsets and correctives. Splints	615.815	Bee-stinging
615.477.4	Trusses. Suspensories. Other hernia supports, pads etc.	615.816	⇒ 638.178.8
			Passive hyperaemia. Bier's treatment
			Artificial respiration. □ Rhythmic movement of breast. Tickling of uvula. Rhythmic movement of tongue
			⇒ 612.216.3
			⇒ 612.232

- 615.816.2 Apparatus for continuous, prolonged artificial respiration in cases of paralysis of respiratory muscles. □ Iron lungs
⇒ 612.232
- 615.817 Cardiac massage. Pacemakers
- 615.82 Massage. Mechanotherapy**
⇒ 613.71
- 615.821 Rhythmic kneading or compression. Methodical percussion. Massage of reflex zones
- 615.823 Vibration therapy. Vibromassage
- 615.824 Friction. Rubbing. Beating with twigs etc.
- 615.825 Active and passive movement therapy (kinesiatrics). □ Curative gymnastics. Exercise therapy. Mechanotherapy. Thermomassage. Electromassage
- 615.827 Chiropractic
- 615.828 Osteopathy
- 615.83 Physical agents. Nature cures. Naturopathy**
⇒ 612.014.4
⇒ 613.16
- 615.831 Phototherapy. □ Treatment with light, lamps, sunlight. Heliotherapy (solar therapy). Sunbathing
⇒ 616.5-001.15
- 615.832 Thermotherapy. Heat treatment. Cold treatment
- 615.832.1 Thermaerotherapy. Aerothermotherapy. □ Hot air bath. Turkish baths. Roman baths. Dry sweating baths. Saunas
- 615.832.3 Radiant heat. Electrothermotherapy. Dowsing's bath. Photothermotherapy
- 615.832.4 Diathermy
- 615.832.5 Vapour baths (steam baths). Russian baths. Humid sweating baths
- 615.832.6 Hot-water bottles. Warm sandbags. Thermophores
- 615.832.7 Caulterization
⇒ 615.262.5
- 615.832.9 Cold treatment. Psychrotherapy. Cryotherapy. □ Internal and external treatment with ice. Ice bags. Cold water irrigation. Dry ice. Liquefied air
- 615.833 Hygrotherapy. Humidity, moisture treatment
- 615.834 Climatotherapy. □ Change of air, at sea shore, deserts, forests
⇒ 613.1
- 615.835 Aerotherapy. Air treatment. Pneumatotherapy. □ Altitude therapy. Oxygenation
- 615.835.7 Kneipp's cure. Kneippism
- 615.837 Therapy using sound. Ultrasonic therapy
- 615.838 Hydrotherapy. Balneotherapy. Water treatment. Water baths. Drink cures. Remedial spas
- 615.838.1 Partial, local baths. Hipbaths (sitzbaths) etc.
- 615.838.2 Hot baths. Tepid baths. Temperate baths. Alternating, hot/cold baths
- 615.838.3 Cold baths. Cold plunge. □ Indoor bathing. Sea, surf bathing. River bathing
- 615.838.4 Wet dressings. Wet packs. Wet friction, rub-down
- 615.838.5 Gas baths
- 615.838.7 Warm mud and sand baths. Peat baths
- 615.838.8 Prolonged bathing
- 615.838.9 Running-water baths. Showers. Douches. Irrigation. Drink cures
- 615.839 Thalassotherapy. Sea cure
- 615.839.1 Local treatment with seawater
- 615.839.2 Seawater baths. Warm seawater baths
- 615.839.4 Curative sea-voyages, cruises
- 615.84 Electrotherapy. Radiotherapy. Electrotherapeutics**
- 615.841 Electrotherapeutic apparatus and techniques in general
- 615.842 Static electricity. Franklinization
- 615.843 Direct, continuous current. Galvanization
- 615.844 Induced, faradic, alternating, undulating current
- 615.846 High-frequency current. D'Arsonvalisation
- 615.849 Radiotherapy. Radiology. Use of microwaves, Roentgen rays (X-rays), ionizing rays, corpuscular radiation
⇒ 621.386
- 615.849.1 Therapeutic application and action of radiation
For properties of radiation see 537.531 and 539.122/126.
For techniques see 539.164/166, 621.384.6 and 621.386
- 615.849.11 Electromagnetic waves. □ Microwaves. Ionizing rays. Röntgen rays (X-rays). Gamma rays
- 615.849.12 Corpuscular radiation. □ Using alpha particles, electrons, protons, neutrons, mesons
- 615.849.19 Other kinds of radiation. □ Laser beams (coherent light)
- 615.849.2 Radioactive substances. □ Radium and its isotopes
- 615.849.5 Technique of irradiation. □ Dosage. Indication
- 615.85 Various therapies**
- 615.851 Psychotherapy
⇒ 616.89
- 615.851.1 Psychic re-education. □ Suggestion. Autosuggestion. Psychoanalysis as therapy
⇒ 159.964.2
- 615.851.2 Hypnosis
- 615.851.3 Treatment through work. Occupational therapy
⇒ 615.825
- 615.851.4 Health guidance. □ Medical pedagogy. Education of the blind, deaf mutes, the retarded
⇒ 376
- 615.851.5 Change of milieu, scene, locality. □ Distractions. Travels
- 615.851.6 Group therapy
- 615.851.8 Mental discipline. Methodical exercises
- 615.852 Faith cures. Prayer, miracle cures
- 615.855 Thomsonianism
- 615.856 Perkinism. Metallotherapy
- 615.859 Clinotherapy. □ Treatment in rest clinics. Rest cures. Sleep cures
- 615.86 Treatment of psychoses. Treatment of the insane, mentally deranged**
- 615.862 Social therapies
- 615.863 Isolation. Seclusion
- 615.865 Continued supervision without restriction of freedom. Open-door methods
- 615.866 Restraining devices. □ Straitjackets. Manacles. Shackles
- 615.869 Special psychiatric treatment. □ Shock treatment (insulin) etc. Electroshock (electroconvulsive therapy/ECT)
- 615.874 Diet control. Abstinence as therapy. Dietotherapy
- 615.874.1 Abstinence. □ Temperance in indulgences, luxuries
- 615.874.2 Dietotherapy
⇒ 613.2
- 615.874.24 Restricted diet. □ Hunger cure (starvation diet). Raw diet. Low-mineral (mineral-reduced) diet.
- 615.874.25 Enriched diet. □ High-fat, high-mineral diets
- 615.89 Folk medicine. Popular, household remedies**

- 615.9 General toxicology. General studies of poisons and poisoning (intoxication)**
 ⇒ 340.67
 ⇒ 615.099
 ⇒ 615.246.9
 ⇒ 615.279
- 615.91 Poisons in general**
615.916 Inorganic poisons
Denote particulars by colon combination with subdivisions of 546 or by '1, subdivided as 546
- 615.917 Organic poisons**
Denote particulars by colon combination with subdivisions of 547.2/.9 or by '2/.9, subdivided as 547.2/.9
- 615.918 Vegetable poisons (phytotoxins)**
Denote particulars by colon combination with subdivisions of 58
- 615.919 Animal poisons**
Denote particulars by colon combination with subdivisions of 59
- 616 PATHOLOGY. CLINICAL MEDICINE**
The -00... and -0... auxiliaries are applicable throughout 616, 617 and 618
- 616-00 Morbid processes**
- 616-001 Traumatata. Injuries. Wounds**
616-001.1 Traumatata due to various physical agents
 ⇒ 614.87
- 616-001.11 Due to increased atmospheric pressure, compressed air.** □ Caisson disease (the bends)
- 616-001.12 Reduced atmospheric pressure.** □ Mountain sickness. Aviation sickness
- 616-001.14 General effect of light**
- 616-001.15 Local effect of light**
- 616-001.16 General effect of heat.** □ Sunstroke. Heatstroke
- 616-001.17 Local effect of heat.** □ Burns
- 616-001.18 General effect of cold.** □ Whole-body chilling, freezing. Hypothermia
- 616-001.19 Local effect of cold.** □ Frostbite
- 616-001.2 Electrical and radiation injuries**
- 616-001.21 Electrical injuries.** □ Lightning-strikes. Electrocution. Electric shock
- 616-001.26 Roentgen (X-ray) injuries**
 ⇒ 621.386
- 616-001.28 Effects of corpuscular radiation, of radioactive substances.** □ Radium. Mesothorium
- 616-001.3 Various mechanical and chemical injuries**
- 616-001.31 Impact injuries. Contusion. Bruising**
- 616-001.32 Crushing. Blast injuries. Crush syndrome**
- 616-001.33 Stretching. Distension**
- 616-001.34 Commotio. Concussion.** □ Jarring. Shaking
- 616-001.35 Compression. Constriction. Strangulation**
- 616-001.36 Traumatic shock.** □ Prolapse. Protrusion
- 616-001.37 Effect of chemicals.** □ Effect of caustic, corrosive agents
- 616-001.4 Open wounds. Abrasion. Laceration**
- 616-001.41 Wounds from sharp edges.** □ Cuts. Incisions
- 616-001.42 Contused wounds**
- 616-001.43 Lacerated wounds. Wounds from stretching, ripping.** □ Bites
- 616-001.44 Wounds caused by piercing (and cutting) instruments.** □ Stab wounds
- 616-001.45 Gunshot wounds. Bullet wounds**
- 616-001.46 Wounds with foreign bodies**
 ⇒ 616-003.6
- 616-001.47 Infected wounds**
Subdivide by -02:615.2
- 616-001.48 Avulsion. Detachment. Tearing away**
- 616-001.49 Venomous bites, stings. Poisoning from animal bites**
 ⇒ 616-022.913
- 616-001.5 Fractures. Breakage of bones.** □ Perforating fractures. Splinter fractures. Gunshot fractures
- 616-001.514 Compound fracture**
- 616-001.52 Callus. Ossification.** □ Pseudoarthrosis
- 616-001.6 Dislocation. Luxation. Traumatic displacement**
- 616-001.7 Sprains. Twisting. Wrenching. Diastasis. Separation of epiphysis**
- 616-001.8 Asphyxia. Suffocation.** □ Throttling. Strangling. Drowning
- 616-002 Inflammation. Irritation. Engorgement (hyperaemia). Mucous congestion**
- 616-002.1 Acute inflammation**
- 616-002.16 Parenchymatous, degenerative, alterative inflammation**
- 616-002.17 Interstitial, sclerosing, fibrous, retractive, atrophic inflammation.** □ Cystic fibrosis
- 616-002.18 Proliferating, hypertrophic, deforming, callous, productive, plastic, obliterative, adhesive, hyperplastic inflammation**
- 616-002.2 Chronic inflammation**
 616-002.2 ≡ 616-002.1
- 616-002.3 Purulent inflammation. Festering. Purulent effusion. Pus. Abscess. Pyosis. Suppuration**
- 616-002.31 Acute suppuration. Hot abscess**
- 616-002.32 Chronic suppuration. Chronic abscess**
- 616-002.33 Pustules. Pustular, variolar, acneform, impetiginous inflammation**
- 616-002.34 Boils. Furuncles**
- 616-002.35 Carbuncles. Anthrax carbuncles**
- 616-002.36 Phlegmon.** □ Gas gangrene. Gangrenous emphysema. Erysipelas
- 616-002.37 Exudative and gaseous suppuration**
- 616-002.4 Gangrene. Thanatosis. Decay, decomposition. Necrosis. Caries. Bone decay. Cold gangrene. Colliquation**
- 616-002.41 Caseation**
- 616-002.42 Gangrene. Dry gangrene. Mummification. Necrosis**
- 616-002.43 Fatty necrosis**
- 616-002.44 Ulcers. Chaps. Scabs**
- 616-002.46 Decubitus. Bed sores**
- 616-002.5 Tuberculous lesions. Tuberculosis**
 ⇒ 616.24-002.5
- 616-002.51 Acute tuberculosis**
- 616-002.52 Chronic tuberculosis.** □ Lupus
- 616-002.53 Cold tubercular abscess. Congestive abscess. Tuberculous suppuration**
- 616-002.54 Tuberculous ulceration, caries, necrosis**
- 616-002.56/.58 Parenchymatous, interstitial, proliferating tuberculous lesions**
 616-002.56/.58 ≡ 616-002.16/.18
- 616-002.6 Syphilis. Syphilides. Syphilitic rash**
 ⇒ 616.972
- 616-002.621 Primary syphilis. Chancre. Primary sore**
- 616-002.622 Secondary syphilis**
- 616-002.623 Tertiary syphilis**
- 616-002.66/.68 Parenchymatous, interstitial, proliferating syphilitic lesions**
 616-002.66/.68 ≡ 616-002.16/.18
- 616-002.7 Granulomatous and nodular inflammation**
- 616-002.71 Pseudotuberculosis**
- 616-002.73 Leprosy. Leproma**
- 616-002.77 Rheumatism**
 ⇒ 616.72-002.77
- 616-002.771 Acute rheumatism**
- 616-002.772 Chronic rheumatism**
- 616-002.773 Simple, chronic rheumatism. Fibrous chronic rheumatism. Rheumatic nodules**

- 616-002.775 Chronic, nodose, deforming rheumatism
616-002.78 Gout
⇒ 616.72-002.78
616-002.8 Inflammations due to plant parasites
616-002.8 ≡ 582
⇒ 616-008.98
616-002.828 Fungal, mycotic inflammation
◇ 616.5-002.828 Dermatomycoses (fungal infections of skin), e.g. athlete's foot
616-002.9 Inflammations due to animal parasites.
Parasitic cysts
616-002.93/.89 ≡ 593.1/599.89
⇒ 616-008.99
- 616-003 Regressive and reparative processes**
616-003.2 Discharge. Weeping. Accumulation. Transudate. Exudation. Infiltration. Efflux. Spontaneous evacuation. Elimination. Expulsion. Issue
616-003.4 Simple retention cysts
616-003.6 Complaints due to foreign bodies (grit, dust etc.). □ Exogenous fibrosis and pigmentation. Silicosis. Pneumoconiosis. Asbestosis
⇒ 616.24-003.6
616-003.7 Concretions. Calculi (stones). Lithiasis. Free-moving bodies
616-003.8 Anatomical degeneration. Tissue degeneration
616-003.829 Pigmentary degeneration. Autogenous pigmentation. □ Chromatosis. Melanosis
616-003.84 Deposits, infiltration, degeneration caused by calcareous and other inorganic substances. Calcification. Cretification
616-003.85 Decalcification
616-003.87 Horny degeneration. Keratosis
616-003.9 Cicatrization. Organization. Reparative processes
616-003.92 Scars (cicatrices). □ Keloids (cheloids)
616-003.93 Regeneration. Tissue renewal
616-003.94 Encapsulation. Encystment
- 616-004 Sclerosis. Hardening (induration). Cirrhosis**
⇒ 616.13-004.6
⇒ 616.36-004
616-004.6 Arteriosclerosis, local lesions
⇒ 616.61-004.6
- 616-005 Local circulatory disorders**
616-005.1 Haemorrhage
616-005.2 Congestion. Active, arterial hyperaemia. Acceleration of the circulation
616-005.3 Stasis. Passive, venous hyperaemia. Deceleration of the circulation
616-005.4 Local anaemia. Ischaemia. Arrested circulation (cessation of blood supply)
⇒ 616.155.194
616-005.5 Collateral circulation
616-005.6 Thrombosis. Blood clots
616-005.7 Embolisms
616-005.8 Infarction. Softening
616-005.9 Disturbances of the lymphatic system
616-005.91 Lymphorrhagia. Lymphorrhea
616-005.93 Lymphostasis
616-005.96 Elephantiasis
616-005.98 Oedema. Hydrops. Dropsy. Anasarca
- 616-006 Tumours. Neoplasms. Blastomata. Choristomata. Hamartomata. Oncology**
Special .03/.04 auxiliaries are applicable in the subdivisions of 616-006
616-006.03 Benign tumours in generalB
616-006.04 Malignant tumours in general
◇ 616-006.3.04 Malignant tumours from non-haemopoietic (non-haematopoietic) mesenchymal tissue. Sarcomata in generalB
616-006.2 Tumours of developmental or embryonic origin.
□ Teratomata. Dermoid cysts
- 616-006.3 Tumours derived from non-haemopoietic (non-haematopoietic) mesenchymal tissue
616-006.31 Angiomata. Angioblastomata. Angioblastic meningiomata. Fibroangiomata. Angiofibromata. Telangiectatic fibromata. Fibromata cavernosa. Myxofibroangiomata. Haemangiomata. Haemolymphangiomata. Naevi. Cavernous angiomata. Tumours. Angiosarcomata
616-006.32 Endotheliomata. Mesotheliomata
616-006.325 Myxomata. Mucous tumours. Collenomata. Gelatinous polyps. Gelatinous tumours. Colloid tumours. Myxoblastomata. Haemangioelastomyxomata. Myxosarcomata, myxomata sarcomatosum
616-006.326 Lipomata. Adipomata. Lipoblastomata. Lipomatosis. Lipofibromata. Lipomata fibrosa. Lipomyxomata. Lipochondromata. Angiolipomata. Fibrolipomata. Myxolipomata. Chondrolipomata. Lipomyomata. Steatomata. Collenomata myxomata. Liposarcomata
616-006.327 Fibromata. Fibroblastomata. Fibromatosis. Corpus fibrosum, connective tissue. Fibroosteomata. Myxofibromata, fibromyxomata. Mucous connective tissue tumours. Fibromata dura, dermatomata lenticulare. Dermatofibromata. Mollusci. Soft fibromata. Fibrosarcomata
616-006.328 Meningiomata. □ Angioblastic meningiomata
616-006.33 Chondromata. Chondroblastomata. Enchondromata. Ecchondromata. Chondrofibromata, fibrochondromata. Osteochondromata. Myxochondromata chondromyxomata. Chondrosarcomata
616-006.34 Osteomata. Osteonci. Osteogenic osteoblastomata. Osteosarcomata. Exostosis. Hyperostosis. Enostosis. Entostosis. Osteophytes. Osteophymata
616-006.35 Synoviomata
616-006.36 Myomata. Muscular tumours. Myoblastomata. Sarcoblastomata. Sarcogenic cell. Rhabdomyomata. Leiomyomata. Angiomyomata. Myosarcomata
616-006.38 Endoneurial and perineurial tumours.
□ Neurofibromata. Von Recklinghausen's disease. Neurofibrosarcomata
616-006.4 Haemopoietic (haematopoietic) tumours. Tumours of the nerve tissue
616-006.43 Tumours of erythropoietic tissue. Erythroblastomata
616-006.44 Tumours of lymphoid and reticular tissue.
□ Lymphomata. Adenia. Granulomata. Myelomata. Chloromata
616-006.48 Tumours of the nerve tissue.
□ Ependymomata. Neuroblastomata
616-006.5 Benign epithelial tumours. □ Papillomata. Adenomata. Polyps
616-006.52 Papillomata. Dendritic cancer
616-006.55 Adenomata
616-006.6 Malignant epithelial tumours. Carcinomata. Cancer
616-006.8 Tumours of special types
616-006.81 Melanomata. □ Pigmented moles or naevi
616-006.82 Chordomata
616-006.83 Ewing's tumour
616-006.85 Wilms's tumours. Nephroblastomata
- 616-007 Defective structural development. Physiological deformities. Malformations. Hemiterata. Oversized, undersized organs. Teratology. Monsters. Abnormalities**
⇒ 611.012

616-007.1	Developmental disorders and anomalies. Deviations in organic development, growth. □ Premature development. Precocity. Senilism. Retarded development. Infantilism	616-007.8	Compound autositic monsters: double, multiple etc.
616-007.2	Defective structural development. Deformities	616-007.9	Parasitic monsters. Unitary autositic monsters
616-007.21	Agensis. Stunted growth. Aplasia. Defective development. Hypoplasia. Dwarfism. Nanism. Microsomia. Arrested development	616-008	Functional and metabolic disorders <i>For use of auxiliary numbers see also under the individual organs</i>
616-007.22	Abnormal persistence of organs or parts	616-008.1	Specific functions of individual organs 616-008.1 ≡ 616-008.6
616-007.23	Atrophy. Diminution. Wasting. Resorption	616-008.2	Pulse. Retention. Abnormal sounds
616-007.24	Deformation. Deformities. Amorphism. Malformation	616-008.21	Local pulse. Local pulsations
616-007.251	Non-traumatic indentations. Perforation. Non-traumatic fissures. Non-traumatic ruptures. Dehiscence. Separation	616-008.22	Sphincteral disorders. □ Incontinence. Retention
616-007.253	Segmentation. Absence of opening. Fistula	616-008.28	Abnormal sounds peculiar to an organ ◇ 616.34-008.28 Intestinal sounds. Borborygmus (bowel noise, rumble)
616-007.254	Incomplete join. Fissure, cleft	◇ 616.72-008.28 Cracking of joints	
616-007.256	Duplication	616-008.3/.5	Functions of specific organs 616-008.3, 616-008.4 and 616-008.5 and their subdivisions are developed differently under individual organs
616-007.26	Insufficiency of an orifice. Dilatation. Distension. Malocclusion	616-008.6	Disorders of secretion, excretion, absorption, permeability. Syndrome peculiar to an organ. Auto-intoxication. Functional reactions. Dysfunction. Functional disorders
616-007.271	Stenosis. Stricture. Narrowing. Contraction. Pinching. Bending. Constriction. Atresia - closure of normal orifice. Obliteration. Imperforation. Lack of normal orifice	616-008.61	Increased activity. Hyperactivity. Overfunction. Excessive response
616-007.272	Obstruction. Occlusion. Closure	616-008.62	Abnormal persistence of a function or activity
616-007.274	Adherence. Coalescence. Symphysis. Ankyolosis. Fusion. Abnormal suture. Pathological union	616-008.63	Irregularity. Instability. Abnormal interruption. Dissociation. Perversion. Degeneration of function
616-007.281	Atony. Relaxation. Loosening. Effacement. Décollement. Detachment. Denudation. Tractability	616-008.64	Diminished activity. Loss of efficiency. Insufficiency. Hypexcitability. Inexcitability. Decompensation. Debility of an organ
616-007.285	Disappearance. Decay. Expulsion. Elimination. Evacuation. Issue. Spontaneous amputation. Loss. Spontaneous mutilation ⇒ 616-003.2	616-008.65	Inactivity. Loss of function
616-007.288	Atelectasis. Apneumatosi. Collapse of the lung	616-008.66	Compensatory, supplementary function
616-007.29	Abortion <i>Use only in conjunction with 618.31 to denote the abortion of an ectopic pregnancy</i> ⇒ 618.39	616-008.7	Body gases 616-008.7 ≡ 616-008.8
616-007.4	Developmental abnormalities of position	616-008.8	Body fluids, humours. Products of secretion or excretion
616-007.41	Displacement. □ Migration. Passage. Penetration. Ectopia. Dystopia. Abnormal abouchement. Abnormal implantation. Aberrant organs	616-008.81	Variations of physical properties of body fluids
616-007.42	Ptois. Blepharoptosis. Collapse	616-008.811	Quantity. Volume. □ Increase. Retention. Irregularity. Decrease
616-007.43	Hernia. Rupture. Prolapse. Descensus. Eversion. Inversion	616-008.811.1	Increase. Retention
616-007.44	Invagination. Intussusception. Proptosis	616-008.811.2	Continuity, steadiness
616-007.45	Heterotaxia. Visceral inversion. Situs inversus viscerum	616-008.811.3	Irregularity. Instability. Abnormal interruption. Perversion. Degeneration
616-007.46	Ecstrophy. Exstrophy. Evagination	616-008.811.4	Decrease. Suspension
616-007.47	Embedding. Inclusion. Impaction. Engagement. Interposition	616-008.811.6	Stasis. Passive hyperaemia. □ Stagnation. Congestion. Arrest. Slowing. Retention (in excretory outlet, lacuna)
616-007.5	Structural deviation, malorientation of organs. □ Club hand. Club foot	616-008.811.7	Transformation. Conversion
616-007.6	Developmental abnormalities of excess. Abnormal enlargement	616-008.811.8	Dissolution. Lysis
616-007.61	Hypertrophy. Hyperplasia. Macrosomia. Gigantism	616-008.811.9	Destruction
616-007.63	Dilatation. Widening. Distension	616-008.812	Odour. Smell
616-007.64	Ectasia. Stretching. Elongation. Diverticulum	616-008.813	Colour
616-007.7	General physical appearance. Morphological types. Figure. Size	616-008.814	Fluidity. Viscosity. Viscidity
616-007.71	Physical beauty	616-008.815	Coagulation. Coagulability
616-007.74	Physical disproportion. Ugliness	616-008.816	Sedimentation. Deposition
616-007.75	Hair colouring: blondes, brunettes etc. ⇒ 57.018.6 ⇒ 572.523	616-008.817	Density. Concentration
		616-008.818	Tension. Pressure
		616-008.819	Electrical and other properties
		616-008.82	Variations of inorganic constituents in body fluids ⇒ 546
		616-008.83	Variations of organic constituents in body fluids ⇒ 547
		616-008.831	Enzymes in body fluids
		616-008.84	Presence of various organic substances
		616-008.841.5	Presence of blood, of blood plasma
		616-008.841.7	Presence of serum

616-008.842/.848	Secretions of the respiratory, digestive, lymphatic, endocrine, urogenital, skeletal, locomotor, integumentary and nervous systems 616-008.842/.848 $\equiv$ 611.2/.8 (which in turn is roughly parallel with 616.2/.8)	616-022.38	Transmitted by plants, vegetables
616-008.842.4	Sputum. Expectoration	616-022.39	Transmitted by animals
616-008.843.1	Saliva	616-022.6	Diseases caused by unidentified viruses, germs, miasmas
616-008.846.1	Urine	616-022.7	Diseases caused by microorganisms, bacteria $\Rightarrow$ 616.98
616-008.847.9	Fat. ☐ Cellulite	616-022.8	Allergic symptoms caused by plants
616-008.85	Presence of other organic substances	616-022.82	Affections caused by particular plants 616-022.82 $\equiv$ 582
616-008.851	Erythrocytes. Red blood corpuscles	616-022.85	Affections caused by particular parts of plants
616-008.852	Haematoblasts. Blood platelets	616-022.852	Mechanical contact with plant hairs
616-008.853	Leucocytes. White blood corpuscles. Phagocytes	616-022.853	Contact with irritant hairs of plants. ☐ Nettle stings
616-008.854	Fragments of tissue. Desquamated cells	616-022.854	Contact with pollen
616-008.855	Coagulated fibrin	616-022.854.2	Hay fever and asthmatic symptoms caused by inhalation of pollen. Pollinosis
616-008.857	Concretions. Amyloid bodies	616-022.855	Contact with plant juices. ☐ Sap, latex, gum, resin
616-008.87	Presence of microorganisms, bacteria	616-022.856	Contact with plant thorns, spines, spikes
616-008.88	Presence of plant parasites	616-022.857	Contact with milled, ground parts of plants
616-008.89	Presence of animal parasites	616-022.9	Diseases and injuries caused by animals
616-008.9	Organs and tissues. Nutritional balance. Morbid metabolism. Disturbances of metabolism, of chemical composition of organs	616-022.912	Purely mechanical injuries. ☐ Bites. Tooth wounds. Pecks. Kicks. Hoof blows. Clawing. Crushing. Trampling
616-008.97	Carriers of microorganisms, bacteria	616-022.913	Injuries caused by toxic or irritant substances. ☐ Venomous bites. Insect bites, stings $\Rightarrow$ 616-001.49
616-008.98	Carriers of plant parasites $\Rightarrow$ 616-002.8	616-022.915	Injuries from electric phenomena on contact with animals. ☐ Shocks from electric eels, rays
616-008.99	Carriers of animal parasites $\Rightarrow$ 616-002.9		
616-009	Neural (nervous) disorders 616-009 $\equiv$ 616.8-009 Use only as auxiliary with subdivisions of 616, not at 616 itself. For neural disorders in general see 616.8-009	616-03	Various forms of disease, of treatment, of administration of medicaments The subdivisions of Table 1k -03 are not applicable under 616
616-011-09	General aspects of pathology	616-031	Localization. Distribution. Confinement to particular site. Absorption by, affinity for, arrest in, particular organs
616-01	Various aspects of disease, patients and medical intervention The following UDC numbers can be used both as main UDC numbers and as special auxiliaries	616-031.1	Distribution
616-018	Histological subdivisions 616-018 $\equiv$ 611.018 $\Rightarrow$ 616-091.8	616-031.2	Spatial localization. Appearance in specific parts of the body
616-02	Aetiology. Science of the causes of disease	616-031.3	Regional, or segmental localization
616-021	Various diseases not caused by living organisms. ☐ Constitutional organic diseases. Diatheses. Spontaneous, cryptogenic, idiopathic affections	616-031.4	Unilateral localization. Confined to one side
616-021.1	Constitutional organic diseases. General phenomena of a disease	616-031.5	Bilateral localization. Affecting whole body
616-021.2	Physiological, orthostatic functional disorders	616-031.6	Localization in relation to affected organ or part
616-021.3	Primary, protopathic, spontaneous, cryptogenic, idiopathic, affections. Diseases of unknown or uncertain origin	616-031.61	Central localization. Deepseated. Sub-... Intra-...
616-021.4	Aseptic affections	616-031.62	Peripheral. Superficial
616-021.5	Compensatory affections. Reactive affections	616-031.63	Surrounding. Around. Peri-...
616-021.6	Artificially induced affections. Experimental affections	616-031.64	Adjacent. Beside. Outside of. Para-... Extra-... Juxta-...
616-022	Diseases caused by living organisms. Bacterial, septic, parasitic affections	616-031.65	Pedicle, hilus, infundibulum of an organ $\diamond$ 616.61-031.65-006 Tumours of the renal hilus
616-022.1	Inoculation. Infection. Contamination For preventive inoculation see 614.47 and (for substances used) 615.37	616-031.66	Migratory erratic localization
616-022.3	Modes of contagion, infection, transmission	616-031.67	Ascending or antegrade localization
616-022.32	Air-borne	616-031.68	Descending or retrograde localization
616-022.33	Soil-borne	616-031.69	Interstitial localization $\Rightarrow$ 616-002.17
616-022.34	Dust-borne	616-031.7	Localization in relation to adjacent organs
616-022.35	Water-borne	616-031.8	Dimensions, extent of disease
616-022.36	Transmitted by humans. ☐ Germ carriers. In schools, public places, hospitals	616-031.81	Lesions over the whole body. Generalized, diffuse, total or complete lesions
		616-031.84	Localized, restricted, partial, circumscribed, incomplete lesions
		616-032	Route of entry, penetration, introduction $\diamond$ 616-032:611.3 Entry through the digestive system
		616-033	Route of dissemination, diffusion, dispersal. Passage $\diamond$ 616-033:611.1 Dispersal through the blood, cardiovascular system

616-033.1	Dissemination. Diffusion. Influence on adjacent organs	616-039.36	Gradual, progressive affections
616-033.2	Metastasis	616-039.37	Labile affections
616-033.3	Grafts	616-039.38	Mutation. Transformation of one disease into another
616-034	Path of elimination	616-039.4	Frequency
616-035	Indications. Choice of treatment. Suitability, appropriateness, advisability	616-039.5	Manifestation, appearance of disease
616-035.1	Advantages. Favourable conditions. Favourable indications. Lack of danger	616-039.71	Preventive measures
616-035.2	Drawbacks. Adverse conditions. Unfavourable indications (contraindications). Dangers. Difficulties. Harmfulness. Value. Importance	616-039.73	Conservative methods
616-035.3	Abuses	616-039.74	Provisional, temporary, emergency methods
616-035.4	Precautions	616-039.76	Radical, curative, permanent methods
616-035.7	Errors. Mistakes. Prejudices. □ Faults of diagnosis, technique	616-05	Persons and personal characteristics in pathology. Characteristics of the patient 616-05 ≡ -05
616-035.9	Suddenly occurring indications	616-051	Persons in active role. Medical staff
616-036	Evolution, course, prognosis of disease, treatment etc.	616-052	Persons in passive role. Patients
616-036.1	Character of affection. Course. Duration. □ Acute attacks. Galloping affections. Chronic affections. Latent affections	616-053	Diseases and treatment according to age
616-036.2	Occurrence, outbreak and spread of disease	616-053.2	Paediatrics. Diseases of infancy and childhood generally ⇒ 616-036.886-053.3
616-036.21	Pandemics. National, endemic, regional diseases	616-053.9	Geriatrics. Diseases of old age, presenility, senility
616-036.22	Epidemics. Epidemiology	616-055	Sexual incidence of disease. Diseases according to sex
616-036.23	Sporadic, isolated outbreaks	616-055.1	Diseases in males ⇒ 616.64/.69
616-036.25	Exotic, foreign diseases	616-055.2	Diseases in females ⇒ 618
616-036.3	Incubation period. Imminence of disease. Pre-...	616-056	Constitutional factors in disease. Predisposition. Constitution. Resistance
616-036.4	Initial stage of disease. Onset. Invasion	616-056.1	Disease according to general constitution, disposition, character
616-036.5	Asepsis. Antisepsis <i>Use always in conjunction with 616-089</i>	616-056.2	Disease according to physical condition, state of health
616-036.6	Period of morbid condition	616-056.25	Disease according to nutritional state
616-036.7	Maturation	616-056.3	Disease according to mental condition, psychological state
616-036.8	Final stage. Termination. Consequences. Results. Successes. Setbacks. Failures. Efficacy. Inefficacy. Survival. Statistics. Mortality in a specific disease	616-056.4	Disease according to temperament, complexion
616-036.81	Critical period. □ Crisis. Lysis (abatement)	616-056.7	Hereditary diseases, disposition. Genetically transmitted, congenital diseases
616-036.82	Convalescence. Recuperation	616-056.8	Diseases caused by habits, inclinations
616-036.83	Complete recovery, cure	616-057	Disease according to profession, occupation, job. Occupational diseases. Occupational medicine ⇒ 613.62
616-036.84	Spontaneous recovery	616-058	Social incidence of disease. Disease according to class. Social medicine
616-036.85	Delayed recovery. Retarded cure	616-06	Complications. Consequences. Repercussions. Concurrence, simultaneous state. Correlation. Previous illnesses. Previous state of health. Antagonism of two diseases
616-036.86	Incomplete, partial recovery. □ Residual weakness, debility. Disabilities. Invalidities. Sequelae. Crippled condition	616-07	Semeiology. Symptomatology. Signs and symptoms. Examination. Diagnosis. Propaedeutics
616-036.87	Relapse. Recurrence	616-071	Clinical investigation. Examination. Diagnosis
616-036.88	Death	616-071.1	Anamnesis. History
616-036.882	Death struggle. Death agony	616-071.2	Appearance. □ Facial expression. Inspection. Type. Physiognomy. Habitus. Bearing. Posture, decubitus ◇ 616-056.4-071.2 Facies scrofulosa ◇ 616.12-071.2 Cardiac facies
616-036.884	Apparent death	616-071.3	Clinical topography. □ Anthropometric indices. Mensuration. Goniometry. Anthropometry ◇ 616-053.5-071.3 Anthropometry of schoolchildren
616-036.885	Rapid death	616-071.4	Palpation. Pain as a disease sign. □ Succussion. Touch. Pressure
616-036.886	Sudden death. Unexpected death ◇ 616-036.886-053.3 Sudden infant death syndrome. Cot death ⇒ 616-053.2	616-071.5	Percussion. Pleximetry. □ Use of percussion rods. Mechanical reactions
616-036.887	Slow death. Lingered death		
616-037	Prognostic methods, signs, procedure 616-037 ≡ 616-07		
616-039	Special, abnormal, atypical, aberrant, hybrid, paradoxical forms. Pseudo-... In the form of... Peculiarities. Variations		
616-039.1	Occurrence of disease. Periodicity		
616-039.3	Continuity		
616-039.31	Paroxysmal attacks. □ Crises. Attacks. Violent colic		
616-039.32	Cyclic affections		
616-039.33	Continuous diseases		
616-039.34	Intermittent attacks. Discontinuous affections		
616-039.35	Repeated, serial, reiterative, recurrent affections		

- 616-071.6 Auscultation. □ Use of a stethoscope.
Phonendoscopy. Auscultatory percussion
- 616-071.7 Grattage. □ Scraping. Scratching
- 616-071.8 Examination under narcosis
- 616-072 Instrumental diagnosis
⇒ 616-7
⇒ 681.784
- 616-072.1 Endoscopy
- 616-072.2 Use of sound, catheter, bougie for diagnosis.
Dilatation. Lavage
- 616-072.3 Insufflation of air or gas-producing powder
- 616-072.4 Divulsion. Separation
- 616-072.5 Exploratory puncture, paracentesis. Sample taking. □ Blood sample
- 616-072.6 Instillations for diagnostic purposes. Vesicatory tests
- 616-072.7 Tests of function. □ Sensory acuity tests.
Artificially induced elimination. Permeability tests
- 616-072.8 Psychophysical and psychophysiological tests
- 616-073 Diagnosis by methods adopted from of physics
616-073 ≡ 53
⇒ 621.386
- 616-073.65 Thermometry. Temperature measurement
- 616-073.66 Calorimetry
- 616-073.7 X-ray diagnosis. Other electrodiagnostic methods
- 616-074 Chemical analysis and diagnosis
- 616-076 Microscopic and microchemical analysis and diagnosis. Biopsy
- 616-076.1 Staining
- 616-076.3 Counts. Counting methods
- 616-076.4 Ultramicroscopy
- 616-076.5 Cytodiagnosis. Cytoscopy
- 616-077.62 Supernumerary organs, parts, openings
- 616-078 Bacteriological, parasitological and humoral diagnosis
- 616-079 Special diagnostic techniques
- 616-079.2 Topographic diagnosis. Localization of affected part
- 616-079.4 Differential diagnosis. Distinguishing similar symptoms of different diseases
- 616-079.7 Retrospective diagnosis
- 616-08 Treatment**
- 616-082 Medical attendance. Administration of treatment
- 616-082.4 Admission, discharge, transfer of patient
- 616-082.8 The patient after discharge. Aftercare.
Follow-up
- 616-083 Hypurgy. Care of patient. Hygiene of the sick.
Nursing
616-083.1/.7 ≡ 613.1/.7
- 616-083.98 First aid. Emergency treatment
- 616-084 Prophylaxis. Public health measures.
Preventive treatment
616-084 ≡ 614
- 616-085 Therapy. Therapeutics. Pharmacology
616-085 ≡ 615
- 616-089 Operative treatment. Operative technique.
Surgery
For operative medicine generally see 617-089
Use -089 and its subdivisions only as auxiliaries with the subdivisions of 616/618
- 616-089.1 Generalities
- 616-089.11 Anatomical, physiological concepts in surgery.
Surgical topography
- 616-089.12 Operative routes and approaches
- 616-089.15 Indications. Choice, suitability, appropriateness, advisability of operations.
Errors. Mistakes
- 616-089.16 Evolution, course, general conditions in which operation is performed (in town, country, at home etc.)
- 616-089.163 Preoperative procedure. Preparation of the patient
- 616-089.165 Asepsis. Antisepsis
- 616-089.166 The operation
- 616-089.168 Completion. Consequences. Results. Success. Failure. Efficacy. Inefficacy. Statistics.
Post-operative care
- 616-089.17 Prognosis: methods, signs, procedures
- 616-089.2 Non-incisional treatment and apparatus
- 616-089.21 Extension, stretching. Counter-extension.
Reduction. Coaptation
- 616-089.22 Support. Immobilization. Restraint. Traction.
□ Splints. Plaster casts. Casings. Suspension apparatus
- 616-089.23 Orthopaedics. Orthodontics
⇒ 617.3
- 616-089.4 Dressings. Bandages. Tamponning
⇒ 615.468
- 616-089.41 Compresses. Poultices
- 616-089.42 Wadding, cotton-wool dressings
- 616-089.43 Binding. Bandaging. Bandages
- 616-089.44 Kinds of dressing. □ Dry dressings. Wet dressings. Antiseptic dressings
- 616-089.45 Tamponning. Surgical tampons
⇒ 615.472.6
- 616-089.48 Drainage
- 616-089.49 Renewal of dressings. Frequency of replacement
- 616-089.5 Surgical anaesthesia
Details by adding -03... (from 616-03)
⇒ 615.21
- 616-089.8 Operative treatment in the strict sense. Surgery
Instruments for particular operations are denoted by adding -7 (from 616-7) to the subdivision for the operation
- 616-089.81 Minor surgery
- 616-089.811 Preventive haemostasis. Ligature of extremities. Esmarch's tourniquet. Induced ischaemia
- 616-089.812 Clamping. Forcippresure. Vasotripsy. Angiotripsy
- 616-089.813 Compression. Expression. Torsion. Forced flexion
- 616-089.814 Ligation. Ligature. Filopressure
- 616-089.815 Curative tattooing
⇒ 391.91
⇒ 616.5-003.6
- 616-089.816 Aspiration. Suction
- 616-089.818.1 Filing. Scraping. Curettage. Surgical abrasion.
Scaling. Rasping. Grinding etc.
- 616-089.818.3 Stretching, dilatation
- 616-089.818.6 Esquilectomy. Sequestrotomy
⇒ 615.811
⇒ 615.814.1
⇒ 615.832
- 616-089.819 Minor surgery of body cavities and canals.
Minor intracanalicular, intracelial surgery
- 616-089.819.1 Catheterization
- 616-089.819.2 Indwelling catheter
- 616-089.819.3 Intubation
- 616-089.819.4 Cannulization
- 616-089.819.5 Dilatation. Divulsion. Bouginage. Introduction of sounds and other solid bodies (not powders) for therapeutic purposes
- 616-089.819.6 Aspiration, suction
- 616-089.819.8 Internal swabbing. Internal curettage
- 616-089.82 Evacuatory puncture, paracentesis. Lancing

616-089.84	Suture. Closure. Sewing up. Surgical synthesis. Padding	616-089.888.71	Symphysiotomy
616-089.843	Grafting, implantation, inclusion, reimplantation, transplantation, interposition, anastomosis <i>Denote the site of the graft by -031:...; denote by -032:... the implanted organ or the region from which it was taken</i>	616-089.888.73	Pubiotomy. Ischiopubiotomy. Hebotomy. Hebosteotomy
616-089.844	Autoplasty. Plastic surgery. Reconstruction. Organoplasty	616-089.888.81	Perforation. Extraction. Detachment. Expression of the placenta. Surgical delivery
616-089.85	Section. Incision. Incisional reduction. Organotomy	616-091	Pathological anatomy. Morbid anatomy ⇒ 611
616-089.851	Covering an organ with tissue e.g. in cases of fissure	616-091.1	The corpse (cadaver) and its alterations, changes. □ Putrefaction
616-089.852	Plicotomy	616-091.5	Autopsy. Necropsy. Postmortem examination ⇒ 614.67
616-089.853	Discission. Dissection. Dilaceration. Adhesiectomy. Exposure. Separation. Removal of duplication. Isolation. Synechotomy. Separation of coalescence (... lysis). Dissociation of organs (... rhexis)	616-091.6	Exhumation. Disinterment
616-089.855	Transfixion. Perforation	616-091.7	Injection of cadaver. Embalming. Preservation of corpse. Plastic regeneration of corpse
616-089.856	Operative suppression of function. Sterilization by an operation on the genital organs	616-091.8	Pathological histology. Histopathology. Cell pathology. Cytopathology. Morbid histology ⇒ 611.018 ⇒ 616-018
616-089.857	Massage of an exposed organ	616-092	Pathological physiology. Processes, mode of action of diseases. Pathogenesis. Study of origin of diseases <i>Use with subdivisions of 616/618; main place is 612</i>
616-089.86	Fistulization. Anastomosis. Artificial openings. Organostomy	616-092.11	Health and disease
616-089.87	Ablation. Resection. Extraction. Exeresis. Organectomy	616-092.12	Intermediate states between health and disease. □ Malaise, indisposition
616-089.872	Disarticulation	616-092.18	General pathology of the cell. Physiopathology of the cell ◇ 616-008.9-092.18 Pathological histochemistry
616-089.873	Partial resection or ablation. Shortening. Amputation	616-092.19	General defensive mechanisms of the organism. Role of individual organs in disease resistance
616-089.874	Trepanation. Trephinement ◇ 616.714.1-089.874 Trepanning (trephining) of cranium	616-092.4	Laboratory experiments. Experiments in vitro
616-089.875	Evisceration. Evidement. Exenteration	616-092.6	Experiments on human beings
616-089.876	Morcellation. Ecrasement	616-092.7	Experiments with microorganisms
616-089.877	Decortication. Denudation. Decapsulation. Removal of the periost. Removal of a saccus or enveloping membrane	616-092.8	Experiments with plants
616-089.878	Removal, extraction, enucleation of foreign bodies, calculi	616-092.9	Experiments on animals
616-089.879	Tripsis. trituration. Ecrasement, morcellation of foreign bodies, calculi	616-093/-098	Bacteriological and parasitological studies
616-089.881	Fixation. □ Internal fixation. ...pexy, ...desis	616-093	Microbial techniques
616-089.882	Replacement. Detorsion. Correction	616-093.75	Palliative methods. Mitigating methods
616-089.883	Exopexy. Exterioration. Evisceration	616-094	Microbial morphology ⇒ 579.23
616-089.884	Couching. Eversion. Reclination. Retraction. Suspension. Depression. Diastasis. Detachment. Disinsertion. Advancement. Reattachment	616-095	Microbial physiology ⇒ 579.22
616-089.886	Compression. Propulsion	616-097	Immunogenicity
616-089.888	Obstetric operations <i>Subdivisions .15/.81 may also be combined with 618.177</i>	616-097.1	Antigens
616-089.888.11	Artificial insemination	616-097.3	Antibodies
616-089.888.12	Reposition of the pregnant uterus	616-098	Microbial physics, chemistry, metabolism, catabolism
616-089.888.13	Perforation of the ovum	616-099	Poisoning. Intoxication <i>Use with subdivisions of 616/618 to denote local effects of poisoning; for toxicology see 615.099 and 615.9</i>
616-089.888.14	Surgical induction of labour. Abortion ◇ 618.39-089.888.14 Induction of abortion	616-7	Medical and surgical instrumentation and equipment <i>Applicable under 615.8 as well as 616/618</i> ⇒ 615.47 ⇒ 616-072
616-089.888.15	Acceleration of labour an abortion	616-71	Medical instruments and equipment in general. □ Diagnostic, measuring, testing and recording instruments and devices ◇ 616-072.1-71 Endoscopes ◇ 616.831-073-71 Electroencephalography
616-089.888.16	Retardation of labour and abortion	616-72	Surgical and therapeutic instruments ◇ 618.4-72 Obstetric operative instruments
616-089.888.17	Dilatation of the cervix	616-74	Reinforcement and reparative materials and equipment ◇ 616.314-74 Dental fillings
616-089.888.18	Incision of the externum		
616-089.888.19	Rupture of the membranes		
616-089.888.2	Manual extraction of the foetus. Accouchement forcé		
616-089.888.3	Use of forceps		
616-089.888.4	Version		
616-089.888.5	Embryotomy. Craniotomy. Decapitation. Basiotripsy		
616-089.888.61	Caesarean section. Sanger's operation		

- 616-76** **Corrective and protective appliances, aids etc.**
 ◇ 616.28-76 Hearing aids
 ◇ 616.314-76 Dental bridges
 ◇ 617.7-76 Ocular aids (spectacles, contact lenses)
- 616-77** **Prosthetic materials and parts. Artificial organs etc.**
 ◇ 616.314-77 Artificial teeth. Dentures
 ⇒ 615.461
- 616-78** **Machines and mechanical apparatus**
 ◇ 615.816-78 Artificial respiration apparatus (iron lungs)
- 616.1/.9** **Special pathology**
- 616.1** **Pathology of the circulatory system, blood vessels. Cardiovascular complaints**
- 616.11** **Pericardium. Epicardium**
- 616.11-002** Pericarditis. Inflammation of the pericardium
- 616.12** **Heart. Cardiac complaints.** □ Complaints of the atria, auricles, chambers, ventricles, valves
 ◇ 616.12-039 Heart abnormalities
 ◇ 616.12-07 Semeiology, symptomatology of the heart. Diagnosis of cardiac conditions
 ◇ 616.12-071.4 Palpation, touch in heart diagnosis
 ◇ 616.12-071.5 Percussion in heart diagnosis
 ◇ 616.12-071.6 Auscultation, stethoscopy in heart diagnosis
 ◇ 616.12-073.7 Electrocardiography. Electrocardiogram (ECG)
 ◇ 616.12-089 Heart surgery
- 616.12-007.4** Cardiopthesis. Cor pendulum. Dropheart. Cor mobile. Wandering heart
- 616.12-007.61** Hypertrophy of the heart. Cor bovinum
- 616.12-008** Functional disorders of the heart. Circulatory disorders
- 616.12-008.1** Disorders of cardiac activity
- 616.12-008.3** Changes in heartbeat, rhythm, pulse
- 616.12-008.31** Pulse rate
- 616.12-008.311** Increased pulse rate. Tachycardia
- 616.12-008.313** Atrioventricular dissociation. Fibrillation. Heart block
- 616.12-008.314** Decreased pulse rate. Bradycardia
- 616.12-008.315** Cardiac arrest. Heart failure. Stoppage of pulse
- 616.12-008.318** Changes in heart rhythm. □ Arrhythmia. Irregular pulse. Dropped beats. Pulsus paradoxicus. Pulsus intermittens
- 616.12-008.33** Arterial pulse. Arterial pressure (blood pressure). Amplitude, quality, tension of pulse
- 616.12-008.331.1** Hypertension. High blood pressure
- 616.12-008.331.2** Corrigan's pulse. Pulsus celer
- 616.12-008.331.4** Hypotension. Low blood pressure. Microsphygmia
- 616.12-008.331.5** Slow pulse. Unilateral pulse. Pulsus tardus
- 616.12-008.331.6** Dicrotic, catacrotic, polycrotic, catapolyrotic, monocrotic pulse
- 616.12-008.334** Elasticity of the arteries
- 616.12-008.34** Venous pulse. Jugular pulse. Venous blood pressure
- 616.12-008.35** Capillary pulse
- 616.12-008.37** Epigastric pulse
- 616.12-008.4** Mechanical disorders of the circulation
- 616.12-008.44** Hyposphyxia
- 616.12-008.45** Compensation
- 616.12-008.46** Cardiac insufficiency. Heart strain. □ Hyposystole. Asystole
- 616.12-009** Cardiac neurosis
- 616.12-009.11** Cardiac paralysis of the heart
- 616.12-009.17** Cardiac adynamia, asthenia. Cardiac weakness
- 616.12-009.2** Erethism. Cardiac irritability
- 616.12-009.3** Palpitations
- 616.12-009.7** Cardialgia. Cardiac pain. Precordial pain
- 616.12-009.72** Angina pectoris. Stenocardia. Sternalgia. Cardiac neuralgia
- 616.12-009.81** Cardiac reflex
- 616.12-009.86** Vasomotor disorders
- 616.121** Septa of the heart
- 616.122** Left heart
- 616.123** Right heart
- 616.124** Ventricles of the heart
- 616.125** Atria (auricles) of the heart
- 616.126** Endocardium. Lining membrane of the heart. Heart valves
- 616.126.3** Cardiac orifices and valves
- 616.126.32** Valvular incompetence. Valvular stenosis, narrowing
- 616.126.4** Atrioventricular orifices and valves
- 616.126.42** Mitral orifice and valve. □ Mitral incompetence. Mitral stenosis
- 616.126.5** Arterial valves of the heart
- 616.127** Myocardium. Heart muscle. Atrioventricular septum
- 616.127-005.8** Cardiac infarction. Myomalacia cordis. Softening of the heart
- 616.13** **Arteries**
- 616.13-002** Inflammation of the arteries. Arteritis. Endarteritis. Mesarteritis
- 616.13-004.6** Arteriosclerosis. Atherosclerosis. Hardening of the arteries. Calcification of the arteries
- 616.13-007.64** Aneurysm. Abnormal dilatation of arteries
- 616.131** Pulmonary artery
- 616.132** Aorta
- 616.132.1** Aortic arch. Bronchial arteries and veins
- 616.132.2** Coronary arteries
- 616.133** Carotid arteries
- 616.134** Subclavian artery
- 616.135** Thoracic aorta
- 616.136** Abdominal aorta
- 616.137** Iliac arteries
- 616.14** **Veins**
- ◇ 616.14-005.6 Thrombosis in veins. Including: Deep vein thrombosis (DVT)
- 616.141** Pulmonary veins
- 616.142** Cardiac veins
- 616.143** Subintestinal veins. Mesenteric veins
- 616.144** Cardinal veins. Main veins of the embryo
- 616.145** Superior vena cava. Brachiocephalic veins. □ Jugular, subclavian, axillary, thoracic veins
- 616.145.42** Brachial veins
- 616.146** Inferior vena cava
- 616.147** Iliac veins
- 616.147.1** Internal iliac vein
- 616.147.17** Haemorrhoidal veins. Haemorrhoidal plexus
- 616.147.17-007.64** Haemorrhoids (piles)
- 616.147.2** External iliac vein
- 616.147.3** Femoral, popliteal, saphenous, tibial veins. Veins of the legs and feet
- 616.147.3-007.64** Varices of the lower extremities. Varicose veins in legs
- 616.149** Portal vein
- 616.15** **Blood. Plasma. Serum. Sanguineous complaints. Clinical haematology**
- 616.15-002** Phlebitis. Venous inflammation
- 616.15-007.64** Plebectasia. Varicose veins. Varices

- 616.151 Variations in general and physical properties of blood
- 616.151.1 Variations of blood volume, total circulating blood or plasma. □ Blood dehydration. Plethora
- 616.151.4 Variations in the viscosity of the blood
- 616.151.5 Disorders of coagulation, clotting. Haemophilia
- 616.151.6 Abnormal sedimentation
- 616.152 Pathological variations of inorganic blood constituents
⇒ 616-008.82
- 616.152.11 Acidosis. Acidaemia. Variations of pH value of blood
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- 616.212 External nose. Nasal region. □ Nostrils (anterior nares). Nasal septum
- 616.213 Nasal conchae (turbinates)
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- 616.214.7 Respiratory region. Mucus-secreting region
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- 616.214.9 Nasal glands
- 616.215 Choanae. Posterior nares
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- 616.217 Region of the cribriform plate
- 616.218 Vomeronasal organ (organ of Jacobson)
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- 616.22-002 Laryngitis. Inflammation of the larynx
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- 616.22-009.1 Laryngeal paralysis. Laryngeal and glottal spasms, cramps. □ Vocal spasm. Singer's spasm
- 616.221 Laryngeal cartilages. □ Thyroid cartilage (Adam's apple). Epiglottis
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- 616.24-002.5 Pulmonary tuberculosis. Consumption of the lungs
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 Panting. Breathlessness. Suffocation.
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 ◇ 616.31-083 Care of the mouth and teeth.
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 ◇ 616.314-089.5 Dental surgical anaesthesia ⇒ 615.216
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 616.316.1 Submaxillary glands
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 616.321 Pharynx. Throat
 616.321-002 Pharyngitis. Pharyngeal inflammation. Sore throat
 616.322 Tonsils (palatine tonsils/faucial tonsils)
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 616.323 Pharyngeal tonsil. Adenoids
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 616.329 Oesophagus. Gullet
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616.33 Stomach. Gastric complaints
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 616.33-008.1 Disorders of gastric digestion
 616.33-008.28 Gastric borborygmus (stomach noises). Eructation (belching)
 ⇒ 612.327
 616.33-008.3 Emesis (vomiting). Gastric mucus. Upset stomach
 ⇒ 612.327
 616.33-008.4 Disorders of the appetite. □ Excessive appetite (hyperorexia). Loss of appetite (anorexia)
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 616.33-008.5 Disorders of sensation of thirst. □ Excessive thirst (polydipsia). Diminished thirst (hypodipsia)
 616.33-008.6 Disorders of gastric secretion and resorption.
 □ Auto-intoxication through the stomach
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 616.33-008.8 Chyme. Stomach contents. Gastric juice.
 □ Disorders of chemical processes in the stomach. Acidity, hyperacidity, hypoacidity
 616.33-009 Nervous gastric disorders. Nervous dyspepsia
 616.33-009.1 Gastric peristalsis. Stomach paralysis, spasms, cramps
 616.33-009.7 Gastralgia. Stomach ache
 616.333 Cardia (cardiac orifice/oesophageal orifice)
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 ◇ 616.34-085.451.3 Intestinal irrigation, lavage
 616.34-002 Enteritis. Gastroenteritis. Inflammation of the intestines and of gastrointestinal complex. Intestinal ulcers
 616.34-005.1 Enterorrhagia. Intestinal haemorrhage
 616.34-006 Intestinal tumours. Cancer of the bowel
 616.34-007.272 Intestinal blockage, obstruction, occlusion. Strangulated intestine. Ileus. Miserere (iliac passion)
 616.34-007.43 Intestinal hernia. Enterocoele
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 616.34-008.1 Disorders of intestinal digestion. Intestinal irregularity. Regurgitation
 616.34-008.3 Defecation. Evacuation of the bowels
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 616.34-008.89 Animal parasites in the bowels. Helminthiasis. Intestinal worms
 616.34-009 Nervous bowel disorders. Nervous intestinal catarrh
 616.34-009.1 Intestinal peristalsis. Intestinal paralysis, spasms, cramps
 616.34-009.74 Intestinal colic
 616.341 Small intestine
 616.342 Duodenum
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 616.344 Ileum
 616.345 Large intestine
 616.346 Caecum (and vermiform appendix)
 616.346.2 Appendix vermiformis (vermiform appendix)
 616.346.2-002 Appendicitis. Inflammation of the appendix vermiformis
 616.348 Colon
 616.349 Sigmoid colon
616.35 Rectum. Anus. Perineum
 616.351 Rectum
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 616.352 Anus. Anal sphincter
 616.352-008.22 Disorders of the anal sphincter. □ Faecal incontinence, retention
 616.352-009.613.7 Anal pruritus; itch
 616.353 Perineum (male)
 ⇒ 618.18

616.36 Liver. Hepatic complaints

- 616.36-002 Hepatitis. Inflammation of the liver
- 616.36-004 Cirrhosis of the liver
- 616.36-008.5 Icterus (jaundice)
- 616.36-008.6 Disorders of liver function. □ Of bile secretion
- 616.36-008.8 Bile (gall)

- 616.361 Bile-ducts (biliary ducts, tracts)
- 616.362 Hepatic ducts. Biliary glands. Vasa aberrantia hepatis
- 616.363 Junction of the bile ducts
- 616.364 Cystic ducts
- 616.366 Gallbladder

- 616.366-002 Cholecystitis. Inflammation of the gallbladder
- 616.366-003.7 Cholelithiasis. Biliary calculi (gallstones)
 - ◇ 616.366-003.7-039.31 Hepatic colic.
 - Gallstone colic

- 616.367 Common bile-duct (ductus communis choledochus). Duodenal papilla
- 616.368 Papilla or ampulla of Vater
- 616.369 Hepatobiliary capsule

616.37 Pancreas

- 616.379 Islets of Langerhans
- 616.379-008.64 Hypoinsulinism. □ Diabetes (diabetes mellitus)
 - ⇒ 615.252

616.38 Peritoneum. Omentum. Mesentery

- 616.381 Peritoneum. Abdominal lining
- 616.381-002 Peritonitis. Inflammation of the abdominal lining
- 616.381-003.217 Hydroperitoneum. Fluid in abdominal cavity.
- Ascites
- 616.381-007.274 Peritoneal adhesions

- 616.382 Omentum. Epiploon. Omental bursa
- 616.383 Mesentery
- 616.384 Chyle vessels
- 616.386 Subphrenic, subdiaphragmatic region

616.39 Dietetic diseases

- 616.391 Diseases due to lack or excess of inorganic substances in the diet. □ Avitaminoses and hypervitaminoses
- 616.392 Scorbutus. Scurvy (vitamin C deficiency).
 - Barlow's disease (infantile scurvy)
- 616.393 Inanition. Privation
- 616.395 Lack of water
- 616.396 Neglect
- 616.397 Beriberi (deficiency of vitamin B1, thiamine)
- 616.398 Pellagra (deficiency of nicotinic acid, niacin)
- 616.399 Corpora adiposa. Adipose bodies

616.4 Pathology of the lymphatic system, haemopoietic (haematopoietic) organs, endocrines**616.41 Haemopoietic (haematopoietic) system**

- 616.411 Spleen (lien). Splenic complaints
- 616.411-006 Tumours of the spleen
- 616.411-006.32 Endothelioma. Gaucher's disease
- 616.418 Haemolymph glands
- 616.419 Medulla (marrow). Myeloid tissue
 - ⇒ 616.71-018.46

616.42 Lymphatic system. Reticulo-endothelial system

- 616.423 Lymphatic vessels and lymph circulation generally
- 616.423-002 Lymphangitis. Inflammation of the lymph vessels
- 616.424 Thoracic duct. Receptaculum chyli

616.425**Large lymphatic trunks**

The remaining lymphatic vessels are classified with the appropriate organ by using -018.98

- ◇ 616.33-018.98 Lymphatic vessels of the stomach
- ⇒ 616.384

616.428

Lymph glands (lymph nodes). Lymphatic follicles in general

616.428-002

Lymphadenitis. Inflammation of the lymph glands. Bubo

616.43**Endocrine glands (ductless glands) in general****616.43-008.6**

Disorders of endocrine secretion

616.432

Pituitary gland

616.433

Anterior lobe (Pars anterior, Pars distalis)

616.434

Posterior lobe (Pars nervosa, neurohypophysis)

616.438

Thymus

616.44**Thyroid gland. Parathyroid glands****616.441**

Thyroid gland

616.441-006.5

Struma (single goitre)

616.441-008.61

Hyperthyroidism (thyrotoxicosis).
□ Exophthalmic goitre. Graves's disease
Hypothyroidism. □ Myxoedema.
Pachydermatous cachexia

616.441-008.64**616.447**

Parathyroid glands

616.448

Thyroglossal tract. Median thyroid.
Zuckerkandl's glands. Bochdalek's duct
Suprathyroid gland

616.449**616.45****Adrenal glands (suprarenal glands)****616.452**

Adrenal medulla

616.453

Adrenal cortex

616.47

Chromaffin system. □ Carotid body. Glomus coccygeum. Paraganglia. Zuckerkandl's bodies. Bursa of Fabricius

616.5**Skin. Common integument. Clinical dermatology. Cutaneous complaints**
⇒ 611.77**616.5-001.1**

Skin traumata due to external physical agents.
Exogenous dermatitis

616.5-001.15

Effect of light, of solar radiation on skin.

616.5-001.17

□ Actinodermatitis. Sunburn. Erythema
Local effects of heat on skin. □ Burns. Scalds.
Heat blisters

616.5-001.19

Local effect of cold on skin. □ Ice burns.
Frostbite

616.5-001.29

Local effect of corpuscular radiation and radioactive substances on skin. □ Radium burns

616.5-001.37

Effect of chemicals, caustics, corrosives on skin. □ Acid burns. Alkali burns

616.5-002

Organic dermatitis. Inflammation of the skin

616.5-002.1

Acute dermatitis. □ Blisters (vesicles, bullae).
Blains. Scales

616.5-002.3

Pyoderma. Skin abscesses. □ Pustules. Boils.
Furuncles. Carbuncles. Whitlows (felons)

616.5-002.4

Cutaneous gangrene. Gangrenous dermatitis.
□ Skin ulcers. Mal perforant. Decubitus. Bed sores

616.5-002.5

Cutaneous tuberculosis

616.5-002.52

Chronic cutaneous tuberculosis. Lupus

616.5-002.6

Cutaneous syphilis

616.5-002.621

⇒ 616.972

616.5-002.621

Chancres

616.5-002.9

Dermatozoonoses. Parasitic skin infections

616.5-003.42

Haematoma. Blood blisters

- 616.5-003.6 Cutaneous complaints due to foreign bodies.
Exogenous skin pigmentation. □ Tattooing
- 616.5-003.829 Autogenous skin pigmentation. □ Freckles (lentigo). Depigmentation. Albinism
- 616.5-003.875 Hard skin. □ Corns. Callosities
- 616.5-003.92 Scars (cicatrices). Weals. Keloids (cheloids)
- 616.5-004 Cutaneous sclerosis (dermatosclerosis).
□ Morbid hardening, induration, of the skin.
Scleroderma
- 616.5-005 Circulatory skin disorders. □ Angiodermatitis
- 616.5-005.96 Elephantiasis
- 616.5-005.98 Oedema. Anasarca (cutaneous dropsy)
- 616.5-006 Skin tumours. □ Moles. Warts. Verrucas.
Naevi. Birthmarks. Beauty spots
- 616.5-006.38 Von Recklinghausen's disease (neurofibromatosis). Subcutaneous nodules. Pachydermatocele
- 616.51 Erythematous, squamous skin diseases**
- 616.511 Erythroderma. Erythema. □ Reddening of skin. Exanthema. Rashes. Heat rash. Efflorescences. Flat skin eruptions
- 616.513 Dermatitis exfoliativa. Erythroderma exfoliativa
- 616.514 Urticaria. Nettle rash (hives)
- 616.515 Prurigo
- 616.516 Lichen. Neurodermatitis (nervous eczema)
- 616.517 Psoriasis
- 616.519 Angiofibroma contagiosum circumscriptum of the skin
- 616.52 Catarrhal, vesicular, papular skin complaints**
- 616.521 Eczema
- 616.522 Impetigo. Dermatitis vacciniformis
- 616.523 Herpes. Miliaria (prickly heat)
- 616.527 Pemphigus
- 616.529.1 Hereditary traumatic bullous dermatitis. Congenital pemphigus. Epidermolysis bullosa hereditaria. Hereditary traumatic pemphigus. Pemphigus with epidermal cysts
- 616.53 Sebaceous glands**
- 616.53-002 Sebaceous inflammation. □ Acne. Pustules. Spots
- 616.53-008.8 Disorders of sebum secretion. □ Seborrhoea (greasy skin). Comedones (blackheads)
- 616.56 Sudoriferous glands (sweat glands)**
- 616.56-008.8 Disorders of sweat secretion. □ Hyperhidrosis (excessive perspiration). Anhidrosis (lack of perspiration)
- 616.58 Other general skin complaints. □ Pernio (chilblains). Wrinkles**
- 616.59 Localized skin complaints. Parts, areas of the skin**
- 616.591 Epidermis. Skin surface
- 616.592 Corium (dermis, cutis). True skin
- 616.594 Pilar system. Hair
- 616.594.1 Head hair. Scalp. Hair follicles. Erector pili muscles. □ Hair loss. Baldness. Alopecia. Scurf. Dandruff (seborrhoea capitis)
- 616.594.7 Facial hair. Beard
- 616.594.8 Supercilia (eyebrows)
- 616.594.9 Other hair. Body hair
- 616.595 Squamae. Scales
- 616.596 Nails. Fingernails. Toenails. Ungual complaints. □ Ingrowing toenails
- 616.599 Stratum subcutaneum. Subcutaneous tissue
- 616.6 Pathology of the urogenital system. Urinary and sexual (genital) complaints**
- 616.61 Kidneys. Nephropathy. Renal complaints. Metanephrose**
- ◇ 616.61-78 Artificial kidney (blood-purifying, dialysis) machines
- 616.61-002 Nephritis. Inflammation of the kidneys
- 616.61-004 Renal sclerosis (nephrosclerosis). Hardening of the kidneys. Granular atrophy of the kidneys
- 616.61-005 Circulatory disorders of the kidneys. Renal haemorrhage, stasis etc.
- 616.61-008.6 Disorders of renal secretion, excretion.
□ Hyperactivity of kidneys. Kidney failure
- 616.611 Glomeruli
- 616.612 Tubules
- 616.612-008.316 Relationship of body temperature to pulse. Dissociated pulse
- 616.613 Renal calyces. Renal pelvis. Pyelopathy
- 616.614 Fibro-adipose cortex. Renal capsule. Perirenal region
- 616.615 Pararenal region
- 616.617 Ureters
- 616.62 Urinary bladder. Urethra. Vesical complaints**
- 616.62-002 Cystitis. Inflammation of the bladder
- 616.62-003.7 Vesical calculi. Cystolithiasis. Bladder stones, gravel
- 616.62-008.22 Disorders of micturition. □ Incontinence, retention of urine. Painful urination
- 616.63 Semeiology, symptomatology of urinary diseases. Disorders of the urinary apparatus. Diuresis**
- ⇒ 612.466
- 616.631 General
- 616.632 Pathological changes in inorganic constituents of urine
616.632 ≅ 546
- 616.633 Pathological changes in the organic constituents of urine
616.633.2/9 ≅ 547.2/9
- 616.634/635 Presence of various substances
- 616.634 Presence of various body fluids, secretions, excretions in urine
- 616.634.15 Haematuria
- 616.634.93 Pyuria
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- 616.89-008.447 Disorders of behaviour, conduct. □ Coquetry. Negligence, slovenliness. Dirtiness. Improperity. Indecency
- 616.89-008.45 Disorders of ideation (thought formation), abstraction, judgement. Disturbance of reasoning, imagination, ordering and association of ideas
- 616.89-008.451 Flight of ideas. Blankness of mind

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⇒ 616.89-008.441
- 616.891.7 Obsessive-compulsive psychoneurosis. Obsessional and compulsive neuroses (psychasthenia)
- 616.892 Organic psychoses. Psychoses of organic, nervous origin
- 616.892.2 General paralysis of the insane. Dementia paralytica
- 616.892.3 Senile psychosis. Senile dementia. Presbyophrenia
- 616.893 Acute mental confusion. Acute delirium (toxic, febrile, infectious exhaustion)
- 616.894 Chronic organic reactions. Chronic states of confusion. Organic dementia (organic mental deterioration)
- 616.895 Functional, emotional, affective psychoses. Psychogenic mental disorders. Deranged states
- 616.895.1 Manic-depressive psychosis
- 616.895.2 Mania (elation, euphoria, communicative excitement). Manic state
- 616.895.3 Cyclothymia. Variability of mood. Alternating, periodic, cyclic states
- 616.895.4 Melancholia. Hyposthenia. Depressive psychosis. Hypochondria (morbid low spirits). Pathological grief. Depressive state
For hypochondria as dread of disease see 616.89-008.441.1
⇒ 159.974
- 616.895.5 Manic or depressive stupor. Stuporous states
- 616.895.6 Mixed states. □ Agitated depression
- 616.895.7 Paranoia. Paranoid reactions with systematized delusions
- 616.895.8 Schizophrenia. Dementia praecox. □ Paranoid schizophrenia
- 616.896 Autism
- 616.899 Mental deficiency (amentia). Retarded mental development
- 616.899.2 Dull-wittedness and mental retardation
- 616.899.3 Feeble-mindedness. Moronism (mental age of 8 to 12 years)
- 616.899.5 Imbecility (mental age of 2 to 7 years)
- 616.899.6 Idiocy. □ Mongoloid idiocy
- 616.899.7 Cretinism. Cretinoid idiocy
- 616.9 Communicable diseases. Infectious and contagious diseases, fevers**
- 616.91 Acute infectious febrile diseases**
- 616.912 Variola. Alastrim. Smallpox. Cowpox
- 616.914 Varicella. Chicken pox
- 616.915 Morbilli. Measles
- 616.916.1 Rubella. German measles
- 616.917 Scarlatina. Scarlet fever
- 616.918 Miliaria. Miliary fever
- 616.92/.93 Various types of fever**
- 616.921.5 Influenza
- 616.921.8 Pertussis. Whooping cough
- 616.927 Typhoid fever. Enteric fever
- 616.927.7 Paratyphoid fever
- 616.928.5 Oriental sore (Delhi boil. Aleppo boil or ulcer)
- 616.928.8 Various tropical and exotic types of fever.
□ Five-day fever
- 616.931 Diphtheria
- 616.932 Cholera
- 616.934 Psilosis. Sprue. Intestinal moniliasis. Tropical stomatitis. Hill diarrhoea
- 616.935 Dysentery. Dysenteriform tropical diarrhoea
- 616.936 Malaria. Marsh fever. Paludism
- 616.937 Trypanosomiasis. □ Trypanosome fever. Sleeping sickness
- 616.94 Septic diseases. Septicaemia. □ Pyaemia. Sepsaemia. Phagedaenia. Hospital gangrene**
- 616.97 Venereal diseases. Venereology**
⇒ 616.6
- 616.971 Chancre. Soft sore
- 616.972 Syphilis
⇒ 616-002.6
⇒ 616.5-002.6
- 616.973 Gonorrhoea
- 616.974 Fourth venereal disease. Vincent's infection of the genital organs. Specific gangrenous and ulcerative balanoposthitis
- 616.975 Fifth venereal disease. Lymphogranuloma venereum. Frei's disease. Nicolas-Favre's disease. Lymphopathia venereum. Durand-Nicolas-Favre's disease
- 616.98 Specific infections. □ Infections due to particular organisms. Bacterial and viral infections**
Class here infections not provided for in 616.91/.97 and which cannot be classed with a particular part of the body
Specify by colon combination to 578/579
- ◇ 616.98:578.828 HIV Infection with Human immunodeficiency virus (HIV). Including: Acquired immune deficiency syndrome (AIDS)
- ◇ 616.98:579.852.11B. anth Infection with *Bacillus anthracis*. Anthrax

616.99	Parasitic infections of plant or animal origin <i>Class here infections not provided for in 616.91/97 and which cannot be classed with a particular part of the body</i> ⇒ 616-002.8/.9	617.577	The fingers
616.992	Infections with plant parasites. □ Phytoses. Mycoses 616.992 ≡ 582	617.578	Phalanges
616.993	Infections with animal parasites. Zoonoses	617.58	Lower limbs
616.993.1	Protozoal infections	617.581	Hips. Hip region
616.993.12	Amoebiasis ⇒ 616.34-008.89	617.582	Thigh
616.993.161	Leishmaniasis. Kala-azar	617.583	Knee
616.993.19	Sporozoa infections. □ Coccidiosis. Anaplasmosis. Piroplasmosis	617.584	Leg. Calf. Calf region
616.995.1	Worm infestations. Helminthiasis	617.585	Ankle. Tarsus
616.995.121	Tapeworm infestation	617.586	Foot
616.995.122	Liver fluke infestation. Lung trematode infestation. Blood trematode infestation. Bilharziasis. Schistosomiasis	617.587	Toes
616.995.132	Nematode infestations. □ Threadworm. Trichuriasis. Filariasis. Ascariasis. Hookworm. Ankylostomiasis. Oxyuriasis. Pinworm infestation	617.588	Phalanges
616.995.42	Acariasis. Mite infections. □ Tick fever. Itch-mite. Scabies. Mange	617.59	Tail
616.995.7	Insect infestations. □ Myiasis (fly-disease)	617.7	Ophthalmology. Eye disorders and treatment ⇒ 611.84 ⇒ 681.784
617	SURGERY. ORTHOPAEDICS. OPHTHALMOLOGY	617.7-001.15	Effect of light on the eye
617-7	Instrumentation and equipment 617-7 ≡ 616-7 <i>Used for specific purposes with the main divisions of 617, but not as the main place for these matters, which is at 615.4</i>	617.7-001.17	Burns of the eye
617.3	Orthopaedics in general ⇒ 616-089.23	617.7-001.31	Ocular contusion. Bruising of the eye. Black eye
617.5	Regional affections ◇ 617.5-089.844 Regional autoplasty. Plastic surgery of particular parts of the body. Reconstruction of particular organs 617.5 ≡ 611.9	617.7-001.32	Crushing of the eye
617.51	Head. Skull	617.7-001.37	Effect of caustic, corrosive agents on the eye
617.52	Face	617.7-001.4	Wounds of the eye
617.53	Neck	617.7-001.5	Rupture. Perforation of the eye
617.54	Trunk, thorax and thoracic wall	617.7-001.6	Dislocations of the eye
617.541	Chest	617.7-002	Ocular inflammation generally. Ophthalmia. Ophthalmitis. Bloodshot eye
617.542	Thoracic cavity	617.7-003.6	Foreign bodies (grit, dust etc.) in the eye
617.546	The back	617.7-006	Tumours of the eye
617.547	Spinal region	617.7-007	Malformations of the eye
617.55	Abdomen. Abdominal wall	617.7-007.681	Glaucoma ⇒ 617.749
617.551	Epigastric region. Epigastric fossa. □ Superior, lateral and posterior abdominal region. Hypochondrium. Traube's space	617.7-009.73	Photophobia. Ocular intolerance of light
617.552	Subperitoneal tissue	617.7-072.1	Ophthalmoscopy ◇ 617.7-072.1-7 Ophthalmoscopes
617.553	Retroperitoneal and peri-aortic regions	617.7-089	Treatment of the eye. Eye surgery
617.555	Umbilical region. Lateral abdominal region. Mesogastric region	617.7-089.24	Spectacles
617.556	Iliac fossa	617.7-089.243	Contact lenses
617.557	Inguinal region	617.7-089.25	Protective glasses. Anti-radiation glasses
617.559	Lumbar, sacral and coccygeal regions	617.7-089.26	Eye-patches
617.57	Upper limbs	617.7-089.28	Ophthalmic prostheses. □ Artificial eyes. Glass eyes etc.
617.571	Shoulder. Shoulder region. Deltoid region	617.7-089.87	Enucleation from the eye
617.572	Arms	617.71	Fibrous coats, tunics of the eye
617.573	Elbow	617.711	Conjunctiva ◇ 617.711-089.843 Transplantation of the conjunctiva
617.574	Forearm	617.711-002	Conjunctivitis. Inflammation of the conjunctiva
617.575	Carpus. Wrist joint. Anterior part of hand	617.713-002	Keratitis. Inflammation of the cornea
617.576	Hand. Metacarpus	617.715	Sclera. Sclerotic coat. White of eye
		617.72	Anterior segment of eye. Uvea
		617.721	Iris. Pigmented layers of eye. Pupil
		617.721.5	Pupil. Pupillary membrane
		617.721.6	Uvea. Corioides
		617.723	Choroid
		617.725	Ciliary body. Ciliary process
		617.726	Accommodation
		617.73	Retina. Optic nerve
		617.731	Optic nerve
		617.732	Optic disc. Papilla nervi optici
		617.735	Retina ◇ 617.735-072.1 Ophthalmoscopy of the retina
		617.735-002	Retinitis. Inflammation of the retina
		617.735-007.281	Detachment of the retina
		617.736	Macula lutea. Fovea centralis

- 617.74 **Dioptric apparatus of the eye**
 617.741 Lens
 | 617.741-004.1 Cataract. Opacification of the lens
 617.742 Zinn's zone. Petit's canal
 617.747 Vitreous humour. Vitreous body
 617.749 Aqueous humour. Anterior and posterior chambers
- 617.75 **Dysopia. Disturbances of vision. Eyesight defects**
 617.751 Visual acuity (VA). Degree of sharpness of sight
 ◇ 617.751-072.7 Determination of visual acuity. Sight tests. Test charts. Optotypes. Optometry
 617.751.6 Amblyopia. Dimness of vision in normal light (visual acuity between 6/7.5 and 3/60)
 617.751.9 Visual acuity less than 3/60. Virtual and total loss of acuity
 617.751.98 Blindness. Loss of sight
 617.753 Errors of refraction
 617.753.1 Hyperopia (hypermetropia) (mean spherical error positive). Far-sightedness. Long-sightedness
 617.753.2 Myopia (mean spherical error negative). Near-sightedness. Short-sightedness
 617.753.3 Astigmatism
 617.753.4 Presbyopia
 617.753.5 Anisometropia
Difference of sphere or cylinder exceeding 2.0 D
 617.753.6 Aniseikonia
 617.754 Visual field
 617.754.7 Scotoma. Dark spot (morbid blind spot)
 617.755 Disturbances of light sense (photoreception), of light sensitivity, of adaptation
 617.755.7 Day blindness (hemeralopia)
 617.755.8 Night blindness (nyctalopia)
 617.758.1 Concomitant squint (strabismus concomitans)
 ⇒ 612.846.8
 617.758.2 Heterophoria. □ Esophoria. Exophoria. Hyperphoria
 617.758.3 Polyopia. Multiple vision
 617.758.5 Micropsia. Macropsia. Megalopsia. Metamorphopsia
 617.758.6 Suppression. Inhibition
 617.758.7 Eccentric fixation
 617.758.8 Abnormal correspondence of the retina
- 617.76 **Accessory organs of the eye. Ocular adnexa. Orbital region**
 617.761 Extrinsic eye muscles and tendons
 ⇒ 612.846.8
 | 617.761-009.11 Squint (strabismus) in general
 | 617.761-009.24 Nystagmus
 | 617.761-009.27 Asynergy. Uncoordinated eye movements
 617.764 Lacrimal apparatus
 617.764.1 Lacrimal glands (tear glands)
 | 617.764.1-008.8 Tears. Shedding of tears. Weeping. Crying. Epiphora. Watering of eyes
 617.764.6 Lacrimal sac
- 617.77 **Eyelids. Palpebral region**
 617.774 Eyelashes
 617.775 Sebaceous glands. Glands of Zeis
 | 617.775-002.1 Hordeolum externum. External styte
 617.776 Meibom's glands
 | 617.776-002.1 Hordeolum internum. Internal styte
 | 617.776-002.2 Chalazion
- 617.777 Moll's glands
 617.78 **Bony orbit**
 618 **GYNAECOLOGY. OBSTETRICS**
 ⇒ 611.65
 ⇒ 612.6
 ⇒ 613.99
 | 618-089.888 Obstetric operations
 ⇒ 618.177-089.888.11
 ⇒ 618.39-089.888.14
 ⇒ 618.5-089.888...
- 618-7 **Instrumentation and equipment**
618-7 ≡ 616-7
Used for specific purposes with the main divisions of 618, but not as the main place for these matters, which is at 615.4
- 618.1 **Gynaecology. Pathology of the female**
For pathology of the male see 616.64
 ◇ 618.1-089 Gynaecological surgery. Gynaecological operations
- 618.11 **Ovaries**
 ◇ 618.11-089.87 Ovariectomy. Oophorectomy. Surgical removal of the ovaries. Ovarian sterilization
 | 618.11-006 Ovarian tumours. Ovarian cysts
 618.111 Follicles
 618.112 Ova. Ovules
 618.112.2 Corpus luteum. Corpus fibrosum. Corpus albicans. Atresia of the follicles
- 618.12 **Oviducts (Fallopian tubes)**
 ◇ 618.12-089.87 Salpingectomy. Surgical removal of Fallopian tubes. Tubal sterilization
- 618.13 **Periuterine organs. Female pelvis and peritoneum.** □ Uterine ligaments. Recto-uterine pouch
 Uterine ligaments
 Parametrium
 Perimetrium
 Pelvis. Cellular tissue and peritoneum
 Recto-uterine pouch. Pouch of Douglas
- 618.14 **Body of the uterus (womb). Uterine cavity. Uterine glands.** □ Endometrium (uterine mucous membrane). Myometrium (uterine muscular coat)
 ◇ 618.14-072.1 Endoscopy of the uterus. Intra-uterine examination
 ◇ 618.14-085.472.5 Use of dilators (laminaria) in the uterus
 ◇ 618.14-085.473 Irrigation, douching, lavage of the uterus
 ◇ 618.14-089 Treatment and surgery of the uterus
 ◇ 618.14-089.22 Uterine supports. Pessaries
 ◇ 618.14-089.819.8 Abrasion, curettage of the uterus
 ◇ 618.14-089.85 Hysterotomy. Incision of the uterus
- 618.14-002 Metritis. Endometritis. Inflammation of the uterus, of the endometrium
 618.14-005 Circulatory disorders of the uterus. Uterine bleeding, haemorrhage. Metrorrhagia. Metrostaxis. Uterine congestion. Uterine stasis. Uterine oedema
 618.14-006 Uterine tumours, cysts, myomata. Hysteromyoma. Cancer of the uterus
 618.14-007.44 Prolapse (proptosis) of the uterus. Hysteroptosis (metroptosis). Fallen womb

- 618.14-007.5 Structural deviation, malorientation of the uterus. Elongation, anteversion, retroversion, lateroversion etc. of the uterus
- 618.141 Myometrium. Uterine muscle
- 618.143 Fundus
- 618.144 Uterine horn. Cornua
- 618.145 Uterine glands
- 618.146 Cervix uteri. Isthmus of the uterus. Cervical canal
- 618.147 Cervical mucosa
- 618.148 Cervical glands
- 618.15 Vagina**
- 618.15-008.8 Vaginal secretions. Vaginal discharge. Leucorrhoea (the whites)
- 618.152 Hymen
- 618.16 External female genitals. Vulva**
- 618.161 Mons pubis
- 618.162 Labia majora
- 618.163 Labia minora
- 618.164 Female urethra
- 618.165 Clitoris
- 618.166 Skene's ducts
- 618.167 Bartholin's glands
- 618.17 Sexual life of the female. Functional disorders of the female genital organs**
- ◇ 618.17-083 Sexual hygiene in the female. Hygiene of the female sexual organs
- 618.17-008.11 Defloration. Loss of virginity. Rupture of hymen
- 618.17-008.14 Frigidity. Sexual unresponsiveness in the female
- 618.17-008.8 Menstruation. Menstrual periods
- 618.172 Menarche. Onset of menstruation
- 618.173 Menopause. Cessation of menstruation. Menopausal disorders
- 618.174 Menorrhagia. Excessively profuse menstruation. Metrorrhagia. Uterine bleeding other than normal menstruation
- 618.175 Dysmenorrhoea. Painful menstruation. Period pains
- 618.176 Amenorrhoea. Absence of menstruation
- 618.177 Sterility, infertility in the female
- ◇ 618.177-089.888.11 Artificial fertilization
- 618.179 Fertility in the female
- 618.18 Perineum in the female**
- 618.19 Mammary glands. Breasts**
- ◇ 618.19-089.87 Mastectomy. Amputation of breast
- 618.19-002 Mastitis. Mammary inflammation
- 618.19-006 Tumours of the breast. Breast cancer
- 618.191 Mamillae. Nipples. Areolae
- 618.191-002 Thelitis. Maxillary inflammation
- 618.2 Obstetrics. Midwifery. Tocology. Pregnancy (gravidity). Physiology of pregnancy. Cyesiology**
- ◇ 618.2-07 Diagnosis of pregnancy. Symptoms of pregnancy
- ◇ 618.2-082 Medical attendance on pregnant women
- ◇ 618.2-083 Care and hygiene of pregnant women
- 618.21 Genital organs in pregnancy**
- 618.231 Duration of pregnancy
- 618.232 Position of the foetus. Presentation of the foetus at birth
- 618.25 Multiple pregnancies. Twins etc.**
- 618.29 Characteristics of foetus at various stages of pregnancy, and of the infant at birth.**
- Size. Weight. Sex
- 618.3 Pathology of pregnancy**
- ◇ 618.3-06 Complications of pregnancy, e.g. vomiting (morning sickness)
- 618.3-008.6 Secretory disorders in pregnancy. Toxaemia of pregnancy. Hepatotoxaemia
- 618.31 Extra-uterine pregnancy. Ectopic pregnancy.**
- Tubal pregnancy
- 618.32 Pathology of the ovum and embryo**
- 618.32 ≡ 611.013
- 618.33 Pathology of the foetus**
- 618.331 Abnormal position of the entire foetus
- ⇒ 618.532
- 618.332 Abnormal position of a single part of the foetus
- ⇒ 618.532
- 618.333 Death and retention of the foetus
- 618.34 Pathology of the foetal appendages.**
- Vitelline sac. Allantois. Chorion. Amnion
- 618.36 Placenta**
- 618.38 Umbilical cord**
- 618.39 Premature and postmature birth. Miscarriage. Abortion**
- ◇ 618.39-021.3 Spontaneous abortion
- ◇ 618.39-085 Medically induced abortion. Termination of pregnancy by drugs
- ◇ 618.39-089.888.14 Surgically induced abortion. Operative termination of pregnancy
- ⇒ 343.62
- 618.4 Childbirth. Delivery. Parturition. Physiology of labour. Eutocia. Natural birth**
- ◇ 618.4-089 Treatment during labour. Preparation. Asepsis. Antisepsis. Position for delivery
- ◇ 618.4-089.5 Anaesthesia. Analgesia
- 618.41 Genital organs during confinement**
- 618.414.1 Uterine contractions
- 618.414.4 Labour pains
- 618.414.7 Taking up of the cervix. Effacement. Dilatation of the cervix. Discharge of mucus
- 618.414.8 Spontaneous rupture of the membranes. Discharge of amniotic fluid
- 618.43 Infant during confinement**
- 618.431 Intra-uterine foetal behaviour. □ Foetal sounds. Crying, breathing, hiccough (hiccup). Defecation
- 618.432 Position of the foetus. Presentation of the foetus at birth
- 618.436 Reduction in size, position and moulding of the head
- 618.437 Descent. Intrapelvic rotation
- 618.438 Expulsion. Delivery
- 618.439 Viability. Stillbirths
- 618.44 Foetal appendages during confinement and birth**
- 618.46 Placenta. Separation, expulsion of placenta. Afterbirth**
- 618.47 Umbilicus**
- 618.48 Umbilical cord**
- 618.492 Delivery of twins
- 618.495 Delivery of several infants. Triplets

618.498	Delivery of monsters. Deformed infants	618.536	Deviations from normal in the reduction in size, presentation or moulding of the head.
618.499	Postmortem delivery		Spontaneous version. Change in position
618.5	Pathology of parturition. Pathology of labour. Dystocia. Difficult birth	618.537	Abnormalities in emergence. Turning within the pelvis
	◇ 618.5-085 Medically induced labour.		
	Induction by drugs	618.54	Adnexal dystocia. Abnormalities of the foetal appendages
	◇ 618.5-085.82 Manipulatively induced labour. Induction by massage		
	◇ 618.5-089.888.14 Surgical induction of labour	618.56	Placental dystocia. Abnormalities of afterbirth, placenta, separation, expulsion
	◇ 618.5-089.888.2 Expression of the foetus. Manual extraction of the foetus		
	◇ 618.5-089.888.3 Use of obstetric forceps, lever	618.58	Separation and delivery of the umbilical cord
	◇ 618.5-089.888.4 Version of the foetus		
	◇ 618.5-089.888.5 Dismemberment of the foetus. Embryotomy	618.6	Normal course of puerperium: physiology, hygiene. Post-partum period. □ Involution of the genital organs. Lactation
	◇ 618.5-089.888.61 Caesarean section		
618.51	Uterine dystocia. Abnormalities of the female genital organs. □ Uterine inertia. Uterine and vaginal spasms	618.61	Involution of the sexual organs
		618.63	Lactation ⇒ 613.953.11
618.52	Pelvic dystocia. Abnormalities of the female pelvis ⇒ 616.718.19	618.7	Pathology of the puerperium, post-partum period
			Puerperal fever (childbed fever)
618.53	Foetal dystocia. Abnormalities of the foetus	618.7-002	
618.531	Abnormal size or shape. Hydrocephalus	618.71	Pathological involution of the sexual organs
618.532	Abnormal presentation or position ⇒ 618.331/.332	618.73	Pathology of lactation

62 ENGINEERING. TECHNOLOGY IN GENERAL

62-1/-9	CHARACTERISTICS AND DETAILS OF MACHINES, EQUIPMENT, PLANT, PROCESSES AND PRODUCTS <i>The term 'machine' is used in 62-1/-9 for brevity to include any tools, equipment and installations, and their components</i> <i>These special auxiliaries are applicable throughout 62/69 except where otherwise indicated.</i> <i>The -9... auxiliaries are listed at 66-9...</i>	62-112.81	Welded, soldered, brazed joints 62-112.81 ≡ 621.791
		62-112.813.6	Brazed joints
		62-112.82	Bonded ⇒ 621.792
		62-112.83	Screwed. Bolted ⇒ 621.882
		62-112.84	Riveted ⇒ 621.884
		62-112.85	Compression joints
		62-112.852	Longitudinal compression joints
		62-112.854	Transverse compression joints
		62-112.854.1	Shrunk-on joints. Shrink-fit connections
		62-112.86	Tenoned. Dovetailed
		62-112.87	Dowelled. Pinned
		62-112.88	Interlocking insertion connections
		62-112.9	Machines according to other kinds of construction
62-11	General design and construction characteristics. Configuration, layout, arrangement of main components	62-113	Single-operation, single-movement machines. □ Single-workpiece (e.g. of machine tools) ⇒ 62-589
62-111	Machines according to working axis, alignment	62-114	Machines with multiple action, operation, movement, function. □ Multi-workpiece (e.g. of machine tools)
62-111.1	Vertical		
62-111.2	Horizontal	62-115	Machines with variable function. Dual-purpose machines. Multipurpose machines. All-purpose, universal machines
62-111.3	Inclined. Oblique		
62-112	Machines according to type of construction	62-118	Crank-driven machines
62-112.3	Self-supporting construction	62-119	Other machines. □ Machines with oscillating drive. Machines with flexible drive
62-112.4	Simple construction. □ One-piece (monobloc) construction		
62-112.5	Sectional construction. Built-up, multi-unit, composite, compound structures	62-12	Reciprocating machines (piston engines) in general. Configuration, layout, arrangement of main components of reciprocating machines
62-112.6	Prefabricated construction. Assembly of unit parts, modules. Unit construction system		
62-112.8	Characterization according to the type of connection or joint		

62-121	Machines with one cylinder. Single-cylinder machines	62-14	Reciprocating machines (piston engines) according to mode of action of working fluids
62-122	Machines with parallel, in-line cylinders	62-141	Single-acting machines
62-123	Machines with obliquely angled cylinders. V-arrangement	62-142	Double-acting machines
62-124	Machines with axially aligned cylinders. □ End-to-end cylinder engines. Tandem engines	62-143	Uniflow machines
62-125	Machines with compound cylinders	62-144	Machines driven by explosion, combustion, spontaneous high pressure
62-127	Machines with parallel cylinders circularly arranged about a shaft, with oblique circular plate. Swashplate machines	62-144.3	Four-stroke machines
62-128	Machines with rotating cylinders	62-144.4	Two-stroke machines
62-129.1	Machines with oscillating cylinders	62-146	Compound machines. Machines with progressive effect. Action in series
62-129.2	Machines with coupled cylinders and two pistons	62-146.1	Single stage
62-129.3	Machines with one cylinder and two pistons	62-146.2	Two or more stages
62-129.5	Machines with auxiliary pistons	62-147	Machines with parallel effect. Action in parallel
62-129.6	Machines with stepped pistons (differential pistons)	62-15	Turbomachines according to mode of action of working fluids
62-129.8	Machines according to the valve arrangement	62-151	Direct action (impulsion)
62-129.81	Machines with L-head arrangement. Side valves on one side of the cylinder.	62-152	Reaction
62-129.82	Machines with T-head arrangement. Side valves on both sides of cylinder	62-153	Combined impulsion and reaction
62-129.83	Machines with I-head arrangement. Overhead valves	62-154	Axial admission
62-129.84	Machines with F-head arrangement. One side valve, one overhead valve.	62-155	Radial admission
62-129.87	Machines with cylinders without valves	62-156	Combined axial and radial admission
62-13	Rotary machines in general. Configuration, layout, arrangement of main components of rotary machines	62-157	Tangential admission
62-133	Machines with rotary, semirotary or pendulum motion, or with rotation-like action	62-158	Full or partial admission
62-133.2	Machines with simple rotor. Rotary piston machines with uniform rotation	62-158.1	Full admission
62-133.22	With sealing vanes in the housing wall	62-158.2	Partial admission
62-133.24	With sealing vanes in the rotor drum	62-159.3	Use of Magnus effect. Flettner rotors
62-133.26	With rotor rotating eccentrically	62-16/-17	Machines according to use of working fluid, vapour, steam
62-133.3	Rotary piston machines with variable rotation. Intermittent and variable-speed rotary piston machines	62-161	Machines with single-stage expansion
62-133.4	Machines with two or more rotors meshing like cogwheels	62-162	Machines with multistage expansion in one cylinder
62-133.5	Machines with helical rotor, with helicoidal rotors	62-164	Machines with multistage expansion in more than one cylinder
62-133.6	Machines with semirotary or oscillating rotor or vane	62-164.2	Machines with two-stage expansion
62-135	Turbine motors and driven parts in general. Turbomachines according to number of blade-rings <i>Denote number of blade-rings by .1/.4</i> ◇ 62-135.2 With two blade-rings ◇ 62-135.4 With four or more blade-rings	62-164.3	Machines with three-stage expansion
62-136	Turbomachines according to the type of casing	62-164.4	Machines with four-stage expansion
62-136.1	With single casing	62-168	Back pressure. Measurement, control, variation of clearance space
62-136.4	With more than one casing. With multiple casings	62-169	Machines with condensation (condensing engines)
62-137	Centrifugal motors and driven parts in general	62-171	Machines with free exhaust or mechanical extraction
62-14/-15	Characterization of machines according to the mode of action of the working medium (fluid)	62-172	Machines with intermediate extraction, bleeding, of working fluid. Exhaust recuperation
		62-173	Machines with heated, reheated or superheated fluid
		62-174	Machines according to steam pressure
		62-174.1	Machines with high-pressure steam
		62-174.2	Machines with low-pressure steam
		62-175	Machines with mixed working fluids. □ Steam and water mixed
		62-176	Machines with vapour other than steam. □ With mercury vapour
		62-18	Machines according to dimensions, mobility, weight, speed, precision

62-181	According to dimensions, size	62-192	Machines according to their reliability
62-181.1	According to linear dimensions	62-2	FIXED AND MOVABLE PARTS, COMPONENTS OF MACHINES
62-181.12	Length. Breadth (width)		<i>The 62-2... auxiliaries are not applicable at 633/635, 679.8 and their subdivisions</i>
62-181.13	Height	62-21	Fixed parts in general. Enclosures. Housings. Casings. Frames. Mountings. Supports
62-181.2	Large. Giant		⇒ 62-758
62-181.4	Small. Dwarf. Miniature	62-213	Machine enclosure or housing generally
62-181.48	Microminiature		⇒ 62-762
62-181.6	According to dimension ratios	62-213.1	Open
62-182	According to mobility, interchangeability, convertibility	62-213.2	Partially enclosed. □ Drip-proof. Splash-proof. Spray-proof. Water-jet-proof. Weatherproof
62-182.1	Integral part of structure	62-213.3	Closed. Wholly, totally enclosed
62-182.2	Fixed, stationary, immobile (without being integral)	62-213.32	Watertight. Immersible. Submersible
62-182.22	Built-in. Fitted	62-213.34	Gastight. Airtight. Flameproof. Explosion-proof
62-182.3	Mobile. Movable. Transportable	62-213.6	Machines in casings resistant to internal and external pressure
62-182.31	On air cushion. Hovering (ground effect)	62-214	Housing, casing or frame of machine components
62-182.32	On skids. On runners. Sliding		⇒ 62-76
62-182.34	On rolling elements. □ On balls. On rollers	62-214.2	Monobloc (one-piece)
62-182.35	On wheels (but not on rails)	62-214.3	Sectional. Composite
62-182.36	On flanged wheels (on rails)	62-214.6	Filled with oil or other liquid. □ Engine sumps
62-182.38	On caterpillar tracks	62-216	Elements of housings, frames, mountings
62-182.4	Portable. Hand-carried	62-216.5	Beds
62-182.5	Swivelling	62-216.6	Posts. Columns. Pillars
62-182.6	Foldable. Collapsible. Dismantlable. Stackable	62-216.62	Main supports
	⇒ 62-771	62-216.63	Guide supports
62-182.63	Foldable. Collapsible	62-216.64	Steady rests. Back stays
62-182.66	Dismantlable	62-216.7	Cross-members. Traverses
62-182.68	Stackable	62-216.72	Arms. Jibs. Outriggers. Extensions
62-182.7	Degree of alterability	62-216.74	Brackets
62-182.72	Adjustable (alterable as to dimensions, degree of precision etc, but performing the same function)		⇒ 62-219.4
62-182.73	With extensible parts	62-216.749	Counterweights
62-182.76	With detachable parts	62-216.8	Tables. Plates. Trays etc. □ Troughs. Tool receptacles
62-182.77	Made of built-up parts. □ Quick-release assemblies, e.g. for emergency	62-217	Bedplates. Baseplates. Foundation plates. Soleplates
62-182.78	Convertible. Adaptable (alterable to perform different functions)	62-218	Fixed foundations. Mountings
62-182.8	Removable. Detachable. Replaceable. Interchangeable	62-218.2	Masonry foundations, of brick, concrete etc., for mounting of machines
62-182.9	Degree of usability	62-218.3	Suspension from vertical surface. □ Wall mountings
62-182.92	Disposable. For single use	62-218.4	Suspension from horizontal surface. □ Ceiling mountings. Overhead mountings
62-182.94	Reusable. For repeated use	62-218.8	Fasteners. Attachments
62-183	According to weight		62-218.8 ≡ 621.88
62-183.2	Heavy	62-219	Adjustable or movable support or bearing arrangements for adjustable parts
62-183.4	Light	62-219.1	Platforms. Bearing-plates. Restplates
62-185	According to speed. Speed of rotation in general. Speed of processes	62-219.2	Slides. Slideways
	⇒ 66-94	62-219.3	Support columns, pillars, posts. Props. □ Tripods
62-185.3	Slow. Low-speed	62-219.4	Movable brackets. Cantilevers. Balance arms. Counterpoises
62-185.4	Fast. Rapid. High-speed		⇒ 62-216.74
62-185.6	With a wide speed range. □ Insensitive to racing. Insensitive to the dangers of critical speeds of rotation	62-219.5	Suspensions
62-185.7	Critical speed. Critical speed of rotation	62-219.51	Rigid suspensions
62-186	According to capacity, capability	62-219.52	Nonrigid, flexible suspensions
62-187	According to degree of precision, accuracy	62-219.521	On cords, wires, ropes etc.
	⇒ 621.713.1	62-219.523	On chains
	⇒ 621.88.07	62-219.527	Electromagnetic suspension
62-187.2	Low precision	62-22	Fixed parts associated with moving elements
62-187.3	Medium precision		
62-187.4	High precision		
62-188	According to dimensional variability		
62-188.2	Invariable. Unchangeable. Stable		
62-188.4	Variable. Changeable		
62-189.2	Dosing. Metering. Metered quantities		

62-222	Cylinders	62-229.4	Supports and mountings for particular objects. Stands. Frames. Horses. Trestles
62-222.1	Cylinder bodies	62-229.6	Feed mechanisms. □ Automatic magazines
62-222.2	Cylinder jackets	62-229.7	Devices for removing and transporting objects between successive operations. Transfer and handling devices within the machine.
62-222.4	Cylinder covers		□ Turning devices. Folding devices
62-224	Cylinder heads. Combustion and precombustion chambers. Manifolds	62-229.8	Discharging, unloading, emptying devices. Waste removal devices
62-224.2	Cylinder heads	62-23	Gear mechanisms. Mechanical power transmission elements. Conveying devices as parts of machines. Fastening, fixing devices
62-224.3	Auxiliary chambers. Precombustion chambers		⇒ 62-88
62-224.7	Headers. Manifolds. Inlet and exhaust chambers. Receivers. Inlet and exhaust arrangements in general		⇒ 621.8
62-225	Ducts. Nozzles. Jets	62-231	Construction of transmission mechanisms
62-225.1	Inlet and outlet ducts, ports, nozzles	62-231.2	Linear systems. Kinematic pairs
62-225.2	Throttling nozzles. Chokes	62-231.21	Systems without degrees of freedom.
62-225.3	Adjustable nozzles		□ Automatic coupling. Automatic centring
62-225.4	Flow-reversing nozzles	62-231.22	Systems with one degree of freedom.
62-225.5	Injection nozzles		□ Journal and bearing. Slide bar. Screw pair (bolt and nut)
	<i>Atomizing of flammable fuels through injection valves</i>	62-231.23	Systems with two degrees of freedom
62-225.6	Idling, slow-running nozzles	62-231.24	Systems with three degrees of freedom.
62-225.8	Nozzle design, construction		□ Ball-and-socket joints. Knuckle joints
	⇒ 621.643	62-231.25	Systems with four degrees of freedom
62-225.86	Nozzles for subsonic and supersonic flow	62-231.3	Linked mechanisms. Motive trains
62-225.862	Nozzles for subsonic flow	62-231.31	Four-bar chains (quadric cycle). □ Slider-crank mechanisms. Double crank mechanisms
	⇒ 677.051.125.262	62-231.32	Wedge chains. □ Wedge slide mechanisms. Couplings
62-225.864	Nozzles for supersonic flow	62-231.33	Multilink chains
	⇒ 621.165	62-232	Elements of reciprocating transmission mechanisms
62-226	Guide rings, blades, vanes of turbomachines and similar machines	62-232.1	Connecting rods. Coupling rods
62-226.1	Guide rings. Stators	62-232.2	Cranks. Crank discs. Return cranks
62-226.3	Guide blades. Guide vanes	62-232.3	Parallel (pantograph) mechanisms
	⇒ 62-253.5	62-233	Elements of rotary transmission mechanisms
62-226.4	Reversing blades	62-233.1	Axles. Shafts. Pivots. Knife edges
62-229	Tool and workpiece carriers, holders. Feed mechanisms. Emptying devices		⇒ 621.824
62-229.2	Tool holders, supports. Rests. Tool chucks etc.	62-233.11	Axles
62-229.22	Simple supports. Tool shanks. □ Morse taper shanks	62-233.12	Plain shafts
62-229.24	Multiple tool holders. Capstan heads. Turret heads	62-233.13	Eccentric shafts. Shafts with drive elements
62-229.3	Workpiece holders, supports. Clamps. Jigs etc.	62-233.132	Crankshafts
62-229.31	Clamping, gripping devices. Positioning, centring, locating devices. Worktables	62-233.133	Camshafts
62-229.311	Worktables with T-slots, fixing holes, built-in clamps	62-233.14	Shaped, profiled shafts
62-229.312	Various clamping devices. □ Fixing screws, bolts. Clamp plates. Vices		62-233.14 ≡ 621.824.4
62-229.313	Devices for distributing and equalizing the clamping forces	62-233.147	Power take-off shafts
62-229.314	Support plates. Wedges. Step blocks		<i>With profiled stub</i>
62-229.315	Magnetic clamping devices (permanent magnet or electromagnet). Vacuum or suction clamping devices	62-233.16	Pivots. Journals
62-229.316	Devices for adjustment, alignment. Positioning, centring, locating devices	62-233.18	Knife edges
62-229.32	Workpiece mountings and clamping devices primarily for rotating workpieces	62-233.2	Bearings. Bushings
62-229.321	Faceplates and their claw boxes		⇒ 621.822
62-229.322	Collets (conical grips) for gripping the outside of the workpiece	62-233.21	Bearings and bushings for axles and shafts
62-229.323	Workpiece chucks	62-233.22	Bearings for pivots and journals
62-229.324	Mandrels	62-233.23	Thrust bearings
62-229.33	Headstocks. Tailstocks. Centres	62-233.24	Bearings for knife edges
62-229.34	Dogs. Driving plates. Catches. Ratchet dogs	62-233.27	Ball bearings
62-229.35	Steady rests, back stays for workpieces	62-233.28	Roller bearings
62-229.38	Actuating devices and drives for clamping, other than those integral with the clamping device. □ Clamping spanners, keys, ratchets, lever assemblies. Electric, hydraulic, pneumatic pressure clamps	62-233.3/9	Gearing. Gears. Friction drives
			62-233.3/9 ≡ 621.833/839
		62-233.3	Toothed wheels. Toothed gears. Tooth profiles
		62-233.52	Cams
		62-233.6	Slide-bar transmissions. Link transmissions
		62-233.7	Devices with eccentrics, inclined planes, wedges, cranks
		62-233.8	Ratchet-and-pawl mechanisms
		62-233.9	Friction drives

62-234	Components for converting rectilinear to rotary motion. □ Z-axes. Swashplates <i>For crankshafts see 62-233.132</i>	62-312	Slide valves with invariable expansion
62-235	Flexible drives and transmissions <i>62-235 ≡ 621.85</i>	62-313	Distributing slide valves with expansion slide valves
62-236	Conveying devices as parts of machines <i>62-236 ≡ 621.86</i>	62-314	Reciprocating cylindrical valves (sleeve valves). Piston valves
62-236.58	Manipulators as parts of machines	62-315	Annular slide valves, for turbines
62-237	Cranes and hoists as machine components <i>62-237 ≡ 621.87</i>	62-32	Fluid control parts with rotary motion. Rotary valves. □ Regulating valves. Corliss valves
62-238	Fastening, fixing, securing devices. Devices for limiting movement <i>62-238 ≡ 621.88</i>	62-33	Fluid control parts with lifting motion. Lift valves. Drop valves etc.
62-24	Elements with reciprocating motion generally	62-331	Inlet valves
62-242	Pistons	62-332	Exhaust valves
62-242.2	Piston body	62-333	Valves according to kind of seat
62-242.3	Piston rings ⇒ 62-762.6	62-333.1	Valves with flat seat. Disc valves
62-242.4	Piston rods	62-333.2	Valves with conical seat. Poppet valves
62-25	Rotary elements generally	62-334	Balanced valves (pressure-compensated valves). □ Ring valves
62-251	Rotary pistons	62-335	Clack valves. Flap valves. Hinged valves (with or without spring)
62-252	Drums. Hubs. □ Cylindrical drums. Conical drums. Rotor hubs	62-336	Valve seats. Valve boxes
62-253	Rotors. Runners. Discs	62-34/-38	Regulating mechanisms for fluid control parts. Valve gears and actuators
62-253.5	Impeller blades. Vanes. Paddles. Buckets ⇒ 62-226.3	62-34	Valve gears and actuators generally. Motions generally, irrespective of particular type of valve
62-253.6	Screws. Propellers. Propeller blades	62-341	Motions with cranks, eccentrics, drag links
62-254	Wheels	62-342	Motions with gearwheels, cogwheels
62-254.2	Spoked wheels (wire wheels)	62-343	Motions with cams
62-254.3	Disc wheels	62-344	Motions with variable stroke, with feed or flow regulation
62-254.6	Components of wheels	62-347	Electrically actuated motions
62-254.61	Rims	62-35	Actuators for slide valve gear
62-254.62	Spokes	62-37	Actuators for rotary valve gear
62-254.63	Discs	62-38	Actuators for lift valve gear. Flap control gear
62-254.64	Hubs	62-39	Devices for positive closure of pipelines. Sprinkling devices. Flushing devices
62-26	Rigid oscillating elements	62-396	Devices for positive closure of pipelines <i>62-396 ≡ 621.646</i>
62-27	Elastic oscillating elements. Flexible moving parts	62-397	Sprinkling devices. Flushing devices <i>62-397 ≡ 621.647</i>
62-272	Springs	62-398	Various draining and overflow devices <i>62-398 ≡ 621.648</i>
62-272.2	Coil springs. □ Helical, spiral, cylindrical, conical springs	62-4	STATE, CONDITION, FORM OF MATERIALS, SURFACES, OBJECTS, PRODUCTS <i>The 62-4... auxiliaries are not applicable at 621.4 and subdivisions</i>
62-272.3	Leaf springs. Plate springs	62-403/-405	State of aggregation of materials <i>For process and operating characteristics and conditions according to state of aggregation use -911... (from 66-911...) ⇒ 54-1</i>
62-272.4	Annular springs		
62-272.41	Disc springs		
62-272.43	Ring springs		
62-272.5	Bar springs. Torsion springs		
62-272.6	Rubber springs		
62-272.8	Pneumatic and hydraulic springs		
62-272.82	Pneumatic springs. □ Air springs. Gas springs		
62-272.84	Hydraulic springs		
62-278	Diaphragms. Elastic membranes		
62-29	Components acting as tools		
62-3	FLUID CONTROL PARTS AND DRIVES. VALVES, CLOSURES ETC. <i>The 62-3... auxiliaries are not applicable at 626, 661.185, 662.2, 681.6 and their subdivisions</i>		
62-31	Fluid control parts with reciprocating motion. Slide valves		

62-403	Gaseous state (including solids and liquids in gaseous state) 62-403 $\equiv$ 54-13	62-422	Solid sections
62-403.2	Rarefied	62-422.1	Solid sections with curved boundaries. Rounds
62-403.3	Compressed	62-422.11	Circular. Round
62-404	Liquid state. Liquefied state 62-404 $\equiv$ 54-14	62-422.12	Curved polygonal. ☐ Fluted sections
62-404.1	Mobile (low viscosity)	62-422.13	Elliptical
62-404.2	Viscous	62-422.14	Lenticular
62-404.5	In solution	62-422.15	Drop-shaped. Streamlined
62-404.8	Colloidal state with liquid continuous phase. ☐ Suspensions. Emulsions. Sols. Gels. Foams	62-422.16	Oval
62-404.9	Semicolloidal state (coarse suspensions). ☐ Pastes. Mashs. Doughs. Sticky ...	62-422.2	Solid sections with straight boundaries in general. ☐ Beams. Posts. Rods
62-405	Solid state. Solidified state <i>Use only if needed to emphasize contrast with gaseous and liquid states</i>	62-422.3	Triangular
62-405.6	Loose solids. Pourable. Tippable. Particulate. Bulk material $\Rightarrow$ 62-492...	62-422.4	Quadrangular. Tetragonal
62-405.8	Porous. Cellular. Sponge-like	62-422.41	Square
62-408	Surface condition of objects $\Rightarrow$ 621.7.015 $\Rightarrow$ 621.9.01	62-422.42	Rectangular
62-408.2	Covered. Coated	62-422.44	Rhomboidal
62-408.3	Plated. Lined. Clad	62-422.45	Parallelogrammatic
62-408.4	Wrapped. Wound	62-422.46	Trapeziform (with two sides parallel)
62-408.6	Surface shape, form	62-422.47	Trapezoidal <i>With no sides parallel</i>
62-408.62	Plane. Flat	62-422.5	Pentagonal
62-408.64	Curved	62-422.6	Hexagonal
62-408.66	Irregular	62-422.7	With more than six sides. Straight polygonal
62-408.7	Smooth	62-422.9	Solid sections with both straight and curved sides. ☐ Quarter-rounds. Half-rounds. Round flats (half-lenticular)
62-408.8	Rough. Grained. Textured. Ribbed. Knurled. Grooved. Scored. Corrugated	62-423	Sections with shaped profiles. Angle sections
62-41	Flat objects. ☐ Blocks. Slabs. Plate. Sheet. Strip	62-423.1	I-sections. H-sections
62-412	Crude forms, for further processing. ☐ In metalworking: ingots, slabs, billets, bars, blanks. In timber, woodworking: logs	62-423.2	Angles. L-sections. V-sections. Quarter-circle channels
62-413	Thick plate and sheet (4.75 mm and above). ☐ Heavy plate. Armour plate. Thick board. Planks	62-423.3	S-sections
62-414	Medium-thick plate and sheet (3 mm to <4.75 mm)	62-423.4	T-sections
62-415	Thin plate and sheet (0.5 mm to <3 mm). ☐ Cardboard. Woven sheets. Webs	62-423.5	U-sections. C-sections. Semicircular channels $\Rightarrow$ 62-461
62-416	Very thin flat objects (less than 0.5 mm). ☐ Very thin sheet. Paper. Foil. Veneers	62-423.7	Z-sections
62-417	Uneven flat objects	62-423.9	Other shaped profile sections. ☐ A-sections
62-417.2	Corrugated plate and sheet. ☐ Corrugated iron	62-424	Sections with rail-type profile $\Rightarrow$ 625.143
62-417.4	Expanded plate and sheet	62-426	Single wires, threads, filaments. Unifilar $\Rightarrow$ 677-486
62-417.5	Curved or dished plate and sheet	62-427	Objects compounded of wires, threads, filaments. Multifilar
62-418	Strip. Band. Ribbon. Tape	62-427.4	Ropes. Cables. Cords. Braids $\Rightarrow$ 677-48 $\Rightarrow$ 679.7
62-419	Multilayer plate or strip. Compound, reinforced, layered sheet. Laminates	62-427.5	Nets. Mesh sieves. Similar objects $\Rightarrow$ 62-417.4 $\Rightarrow$ 62-472.2
62-419.3	Of more than two layers, joined with adhesive. ☐ Plywood	62-428	Helical, spiral, twisted objects. Screw-shaped objects
62-419.4	Double, duplex, two-layer (without adhesive). ☐ Bimetallic strip	62-43/-49	Objects not listed under 62-41/-42
62-419.5	Of more than two layers, joined without adhesive. ☐ Triplex sheet. Multiplex sheet. Sandwich construction	62-43	Solid objects of various shapes
62-419.7	Of two layers, joined with adhesive. ☐ Compound metal	62-431	Short prismatic objects. ☐ Building blocks. Bricks. Briquettes. Tiles
62-419.8	Reinforced with textiles, fabrics, fibres etc.	62-433	Edge shapes
62-42	Profiled objects. Sectional material	62-433.4	Plane. ☐ Chamfers. Bevels
		62-433.5	Curved. ☐ Mouldings. Roundings $\Rightarrow$ 62-44
		62-433.6	Combination of plane and curved
		62-434	Short cylindrical and conical objects
		62-434.1	Cylindrical objects
		62-434.5	Conical objects
		62-436	Spherical and spheroidal objects
		62-436.1	Spheres. Balls. Beads
		62-436.2	Spheroids (ellipsoids of rotation)
		62-436.4	Ellipsoids (three different axes)
		62-436.5	Ovoids

62-436.7	Barrel-shaped solid objects	62-514	Hand or foot controls
62-437	Streamlined objects	62-514.1	Hand controls (manual control) in general.
62-44	Shaped mouldings. Mitres. Dovetails.		Control handles
	□ Serrated mouldings	62-514.2	Control levers
	⇒ 62-433.5	62-514.3	Foot controls. Pedals etc.
62-45	Slotted, grooved, rabbeted (rebated), splined objects	62-514.4	Crankhandles
62-46	Hollow objects. Material of hollow section	62-514.5	Steering controls. □ Handlebars. Steering bars
62-461	Open hollow sections. Concave sections.	62-514.54	Main steering columns. Rigid steering bars
	□ Open channels, gutters	62-514.55	Steering linkages. Steering rods
	⇒ 62-423.5	62-514.58	Hand grips for steering controls
62-462	Closed hollow sections. □ Pipes. Tubes.	62-514.59	Handwheels. Steering wheels
	Hoses	62-515	Various control devices, particularly for vehicles
62-462.2	Seamed piping etc.	62-515.5	Actuated by movement of seats or footboards
62-462.3	Seamless piping etc.	62-515.7	Actuated independently of driver by a passenger, guard, instructor etc.
62-463	Closed hollow objects of irregular cross-section	62-519	Remote control. Remote steering
62-464	Spherical hollow objects. □ Globes. Balloons	62-52	Automatically operated or controlled machines and processes
	62-464 ≡ 62-436		⇒ 681.52/.53
62-465	Hemispherical, cylindrical and similar open hollow objects. □ Crucibles. Beakers. Pots. Bottles. Barrels	62-521	Mechanically controlled machines etc.
			⇒ 62-86
62-47	Perforated and annular objects	62-522	Fluidically, hydraulically, pneumatically controlled machines etc.
62-472	Objects with holes (punched, drilled, bored). Perforated objects	62-522.2	Hydraulically controlled (except electrohydraulically)
62-472.1	With blind holes		⇒ 62-82
62-472.2	With through holes. □ Sieves of perforated plate	62-522.7	Pneumatically controlled
	⇒ 62-427.5		⇒ 62-85
62-472.3	With depressions	62-523	Electrically controlled machines etc.
62-477	Annular objects. Rings		⇒ 62-83
62-478	Objects with cavities for weight reduction, air circulation etc. □ Hollow bricks	62-523.1	Electrostatically controlled
62-48	Forms and make-up of fibrous materials and products	62-523.2	Electromagnetically controlled.
	62-48 ≡ 677-48		Electrodynamically controlled
62-492	Pulverous and granular materials	62-523.3	Electrohydraulically controlled
62-492.2	Pulverous material. Powders. Dusts	62-523.6	Electrothermally controlled
62-492.3	Granular material. Grains. Granules. Shot. Grit	62-523.8	Electronically controlled
	⇒ 62-405.6	62-524	Thermally controlled machines etc. (except electrothermally)
62-493	Small objects of irregular form. □ Splinters. Shavings. Chips	62-526	Servo-controlled machines etc.
62-493.5	Flakes	62-527	Machines etc. acting at periodic intervals. Time controls
62-494	Objects made of irregular fibres. □ Felt products		⇒ 62-558
62-5	OPERATION AND CONTROL OF MACHINES AND PROCESSES	62-529	Automatic sequential control
	⇒ 621.3.07	62-53	Devices for positional, dimensional regulation (setting) etc.
	⇒ 621.316.7		
	⇒ 681.5	62-531	Control of mechanical quantities
62-50	Different types of control and regulation	62-531.1	Control of lengths
62-503.5	Determinate input control	62-531.2	Control of surfaces and cross-sections
62-503.51	Stabilizing control	62-531.3	Control of volumes. Control of capacities
62-503.54	Servo control	62-531.4	Control of angles. Control of position.
62-503.55	Program control and regulation		□ Centring devices
62-503.56	Optimal control and regulation	62-531.5	Control of masses
62-503.57	Adaptive control and regulation	62-531.6	Control of velocity. Control of acceleration
		62-531.7	Control of oscillation
		62-531.8	Control of pressures, forces
			⇒ 66.083
		62-531.9	Control of output. Control of energy
		62-533	Control of physical quantities
		62-533.4	Sound control. Tone control. Noise control
		62-533.5	Control of optical quantities
		62-533.52	Control of light intensity
		62-533.56	Control of colouring
		62-533.6	Control of thermal quantities. □ Control of temperature, heat
		62-533.7	Control of electrical quantities
62-51	General operation and control of machines and processes	62-533.8	Control of magnetic quantities

62-539	Precision regulation, adjustment (mechanical)	62-555.5	Controllers actuated by rays or beams
62-54	Methods of control. Elements connecting controller to point of control in general	62-555.6	Controllers actuated by the action of heat
62-541	Control by mechanical action in general	62-556	Controllers actuated by chemical phenomena
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62-753	Devices for compensating for expansion, tension, flexure, bending, torsional stress. □ Devices compensating for deformation due to wind, temperature change, frost, snow	62-762	Fluid-tight, pressure-tight joints, connections. Protective jointings. Packings. Glands. Seals <i>Denote packing materials by -03... (Table 1k)</i>
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62-757	Protection from damage by foreign bodies. Devices for deflecting or expelling foreign bodies	62-762.86	Scraper rings
62-757.2	Stone deflectors. Stone shields. □ Rail guards	62-763	Dirt traps. Mudguards, hubcaps etc. Dust extractors
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62-758.2	Protection against mechanical action	62-774.5	Standing places, surfaces. □ Bridges. Platforms. Rostrums ⇒ 621.869.3 ⇒ 69.057.6
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62-758.31	Dustproof containers	62-776	Cleaning devices
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620.172.253	Tensile testing by centrifugal force	620.178.151.4	Measurement of the diameter and depth of the impression (indentation). ☐ Accuracy of measurement. Shape of impression
620.172.254	High-speed tensile tests. Impact tensile tests ⇒ 620.178.7	◇ 620.178.151.4.08	Principles of measurement. Design of instruments for measuring indentation
620.173	Testing of compressive strength, with static or gradually increasing load. Compression tests. Crushing tests	620.178.151.6	Calculation of hardness values (maximum, minimum hardness). ☐ Hardness number
620.173.2	Determination of test values and material characteristics 620.173.21/.25 ≡ 620.172.21/.25	620.178.152	Impression methods. Indentation testing
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620.173.23	Determination of bulging	620.178.152.3	Cone and pyramid indentation tests. ☐ Vickers method
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620.174.22	Determination of deflection, elastic limit, modulus of elasticity	620.178.156	Various influences on hardness testing. ☐ Influence of temperature. Influence of apparatus or tool
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620.175.6	Determination of number of twists	620.178.32	Kinds of repeated dynamic stress test. Alternating stress tests. ☐ Tension-compression (push-pull) tests
620.176	Shear strength testing, with static or gradually increasing load. Shear tests. Punching tests	620.178.35	Repeated impact tests
620.176.2	Single and double shear tests. Shear resistance tests	620.178.37	Fatigue tests at low temperatures
620.176.21	Single shear tests	620.178.38	Fatigue tests at various temperatures
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620.193.23	Effect of air and water. Effect of humidity	⇒ 658.62	
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620.193.29	Permeability or imperviousness to air. Gas-tightness		
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620.194.23	Corrosion fatigue	620.3	Nanotechnology
620.194.3	Fretting corrosion. False brinelling		<i>Class here only general works. Specific technologies should be classed with the appropriate technology</i>
620.195	Influence of form and structure of the surface	620.9	Economics of energy in general
620.196	620.195 ≡ 620.193	⇒ 621.48	
620.196	Influence of foreign bodies and impurities on corrosion. Special forms of corrosion		
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	<i>Class information about particular kinds of machinery at the number for the subject</i>	621.039.4	Reactor engineering. Generalities on fission and fusion reactors
	<i>In 621.0, .0... is not used to denote special auxiliary subdivisions. The .0 subdivisions are not transferable to other 621... numbers</i>	621.039.5	Nuclear fission reactors
	⇒ 622.23.05 Mining machinery and equipment	621.039.5.0	Details of nuclear fission reactors
	⇒ 677.05 Textile machinery and equipment		621.039.5.01/.08 ≡ 621.039.51/.58
621.01	Mechanical engineering theory and principles. Mechanics as the basis of mechanical engineering	621.039.51	Reactor core theory and experiments. Reactor physics
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621.039	Applied nuclear science. Atomic energy and atomic industry. Nuclear engineering in general	621.039.512.2	Neutron multiplication
	⇒ 539.1	621.039.512.4	Neutron diffusion and moderation
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	⇒ 553.495	621.039.514.2	Without effect of delayed neutrons. Kinetics of fast reactors
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621.039.32	Isotope separation using chemical methods		Heat development in nuclear fission reactors. Fuel rating. □ Energy output. Temperature distribution. Thermal potential
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621.3.018.42	Frequency bands	621.3.029.422	0 to < 1 Hz; 8 to $> 3 \times 10^6$ m
621.3.018.422	Narrow bands	621.3.029.424	1 to < 10 Hz; 3×10^6 to $> 3 \times 10^7$ m
621.3.018.424	Wide bands	621.3.029.426	10 to < 100 Hz; 3×10^7 to $> 3 \times 10^8$ m. Power frequencies
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621.3.018.5	Synchronism. Asynchronism	621.3.029.45	100 to $< 30\,000$ Hz; 3×10^6 to $> 1 \times 10^4$ m
621.3.018.6	Mechanical oscillations. Hunting	621.3.029.5	Frequencies from 30 to $< 30\,000$ kHz. Wavelengths $10\,000$ m to > 10 m. High frequencies
621.3.018.7	Waveform. Distortion	621.3.029.51	30 to < 300 kHz; 1×10^4 to $> 1 \times 10^3$ m. Kilometric waves. Long waves
621.3.018.72	Sinusoidal waves	621.3.029.53	300 to < 3000 kHz; 1×10^3 to $> 1 \times 10^2$ m. Hectometric waves. Medium waves
621.3.018.73	Flat waves	621.3.029.55	3000 to $< 30\,000$ kHz; 1×10^2 to $> 1 \times 10^1$ m. Decametric waves. Short waves
621.3.018.74	Peaky waves	621.3.029.6	Frequencies from 30 MHz to < 3 THz. Wavelengths from 1×10^1 to $> 1 \times 10^{-4}$ m. Ultrashort waves
621.3.018.75	Waves of special form. □ Sawtooth waves. Rectangular waves. Step waves	621.3.029.62	30 to < 300 MHz; 1×10^1 to > 1 m. Very high frequency (VHF). Metric waves
621.3.018.756	Pulse waveforms	621.3.029.63	300 to < 3000 MHz; 1 to $> 1 \times 10^{-1}$ m. Ultrahigh frequency (UHF). Decimetric waves
621.3.018.78	Distortion	621.3.029.64	3 to < 30 GHz; 1×10^{-1} to $> 1 \times 10^{-2}$ m. Centimetric waves
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621.3.029.66	300 to < 3000 GHz; 1×10^{-3} to $> 1 \times 10^{-4}$ m. Decimillimetric waves	621.3.035.28	Deterioration of electrodes. Buckling. Formation of air bubbles
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		621.313.391	Homopolar, unipolar AC machines
		621.313.392	AC machines without commutator. Hysteresis machines
		621.313.5	Generators and motors based on the influence of electric or magnetic induction effects on fluid flow
		621.313.52	Generators
		621.313.522	Magnetohydrodynamic generators
		621.313.522.2	Employing the Faraday effect
		621.313.522.21	Open cycle
		621.313.522.22	Closed cycle
		621.313.522.3	Employing the Hall effect
		621.313.522.31	Open cycle

621.313.522.32	Closed cycle	621.314.5	Conversion of alternating current into direct-current and vice versa. Convertors. Invertors
621.313.523	Magnetoplasdynamic generators 621.313.523 $\equiv$ 621.313.522	$\Rightarrow$ 621.314.6	
621.313.524	Magnetogasdynamic generators 621.313.524 $\equiv$ 621.313.522	621.314.52	Motor generators
621.313.53	Flow motors. Induction pumps. Electrodynamic jet propulsion systems	621.314.53	Rotary convertors
621.313.532	Flow motors using flowing liquids	621.314.54	Cascade convertors
621.313.533	Flow motors using flowing gases	621.314.55	Convertors with rotary brushes. Rotating commutators
621.313.534	Flow motors using flowing plasmas.	621.314.57	Static convertors with discharge apparatus
	Electrogasdynamic jet propulsion systems	621.314.571	Choppers
621.313.8	Magnetoelectric machines. Permanent magnet machines	621.314.572	Invertors
621.313.82	Design. $\square$ Armature. Magnet. Flywheel	621.314.58	Reversible current convertors
621.313.84	Special-purpose magnetoelectric machines	621.314.6	Conversion of alternating current into direct current. Rectification. Rectifiers and rectifier equipment
621.314	Transformation. Transformers. Convertors. Rectifiers $\Rightarrow$ 621.313		<i>Class here information about complete apparatus or plant only. For substations with rectifiers use 621.311.44. Class active parts with the subject, e.g. for semiconductor devices use 621.382; for pool-cathode discharge tubes use 621.387.2</i>
621.314.1	Direct-current transformation (DC/DC transformation). Direct-current transformers. $\square$ Motor generators. Induction coils	621.314.62	Mechanical rectifiers. $\square$ Contact rectifiers
621.314.2	Alternating-current transformation (AC/AC transformation). Alternating-current transformers, convertors etc.	621.314.63	Dry rectifiers. Barrier-layer rectifiers. Solid-state, semiconductor rectifiers. $\square$ Selenium rectifiers
621.314.21	Alternating-current transformers	621.314.64	Electrolytic rectifiers
621.314.211	With air insulation and cooling	621.314.65	Low-pressure metal-arc rectifiers. Mercury-arc rectifiers $\Rightarrow$ 621.314.27 $\Rightarrow$ 621.387.2
621.314.212	With oil insulation and cooling	621.314.66	Atmospheric arc rectifiers (Marx rectifiers). High-pressure arc rectifiers
621.314.213	With insulation and cooling by means other than air and oil	621.314.67	Thermionic rectifiers. Glow-discharge rectifiers
621.314.214	Adjustable transformers. Variable-ratio transformers in general $\Rightarrow$ 621.316.722	621.314.671	Hot-cathode rectifiers. Thyatron rectifiers etc. $\Rightarrow$ 621.314.27 $\Rightarrow$ 621.387.13
621.314.214.2	Induction regulators	621.314.672	Glow-discharge rectifiers. Cold-cathode rectifiers
621.314.214.3	Transformers with variable turns ratio	621.314.69	Other rectifiers. $\square$ Point-plate rectifiers
621.314.214.5	Transformers with variable magnetic circuit	621.315	Transmission of electric energy. Power distribution and telecommunication lines. Conductors. Insulating materials. Accessories. Design, construction of lines
621.314.22	Transformers with ordinary closed ferromagnetic circuit $\diamond$ 621.314.22.08 Measuring transformers (instrument transformers)	621.315.1	Bare lines. Open-air lines. Overhead lines $\Rightarrow$ 621.315.24
621.314.222	Voltage transformers 621.314.222.1/.45 $\equiv$ 621.314.211/.214.5	621.315.14	Bare lines in general. $\square$ Stranded lines. Hollow, tubular lines
621.314.222.6	Power voltage transformers	621.315.17	Design, erection, maintenance of overhead lines $\Rightarrow$ 621.316.9
621.314.222.7	Boosting transformers	621.315.2	Insulated conductors in general. Cables (land, sea cables) $\diamond$ 621.315.2-777.6 Colour-coding of cables $\Rightarrow$ 621.315.3 $\Rightarrow$ 679.74
621.314.222.8	Instrument-type voltage transformers	621.315.21	Cables according to insulation, structure
621.314.223	Autotransformers 621.314.223 $\equiv$ 621.314.21	621.315.211	Cables according to insulation. $\square$ Solid, liquid (oil-filled), gas-pressure insulation
621.314.224	Current transformers 621.314.224.1/.45 $\equiv$ 621.314.211/.214.5	621.315.212	Concentric cable. Coaxial cable
621.314.224.8	Instrument-type current transformers	621.315.213	Nonconcentric cable. $\square$ Stranded, twisted cable. Segmental cable
621.314.225	Ring transformers 621.314.225 $\equiv$ 621.314.21	621.315.216	Insulated tubular conductors $\Rightarrow$ 679.743.4
621.314.228	Special-purpose transformers. $\square$ Transformers as balancers, compensators, equalizers. Interphase transformers (interphase reactors). Negative booster transformers	621.315.22	Armoured, sheathed cable. Armouring. Sheathing
621.314.23	Transformers with special closed ferromagnetic circuit, open ferromagnetic circuit, or none. Transformers with air gap, with air core. $\square$ Swinburne, Hedgehog, Tesla transformers (Tesla coils)	621.315.221	Sheathing
621.314.24	Transformation by means of rotating machinery	621.315.222	Mechanical protection. Armouring
621.314.241	Motor generators. Rotary convertors	621.315.223	Combined sheathing and armouring. Bedding (layer between sheathing and armouring)
621.314.25	Phase convertors. Phase transformers		
621.314.26	Frequency convertors. Frequency changers $\Rightarrow$ 621.372.63		
621.314.27	Simultaneous phase and frequency changing using grid-controlled current convertors. $\square$ Mercury-arc rectifiers. Thyratrons $\Rightarrow$ 621.314.65 $\Rightarrow$ 621.387.1		

621.315.223.5	Combined lead sheathing and steel tape armouring	621.315.616	Macromolecular insulants. □ Rubbers. Plastics. Synthetic resins
621.315.227	Serving (outermost protective layer), e.g. of jute	621.315.617	Dried liquid insulants. □ Paints. Varnishes. Drying oils. Silicones ⇒ 667.637.4
621.315.23	Underground cable: arrangements and laying	621.315.618	Gaseous insulants
621.315.24	Aerial cable: arrangements, erecting ⇒ 621.315.1	621.315.619	Insulators according to their structure. □ Rolled insulators. Laminated insulators
621.315.27	Floating cable	621.315.62	Insulators. Bushings
621.315.28	Underwater cable (submarine and river cables): arrangements, laying etc. □ Laying from cable-laying ships. Underwater cable recovery	621.315.622	Insulators according to shape. □ Cup (single-cup, double-cup etc.) insulators. Diabolo insulators
621.315.29	Accessories for laying and repair	621.315.623	Post, pin, pedestal insulators
621.315.3	Insulated wires. Flexible cords. Domestic flex. Interior wiring ◇ 621.315.3-777.6 Colour-coding of insulated wires ⇒ 621.315.2 ⇒ 679.74	621.315.624	Suspension insulators. Insulator strings
621.315.32	Insulated wires according to application ⇒ 621.3.056	621.315.625	Shackle insulators. Strain insulators
621.315.322	Heat-resistant wires. □ Ignition wires. Furnace leads. Soil-heating wires ⇒ 621.365	621.315.626	Bushings
621.315.323	Wires resistant to tropical climates	621.315.627	Spool insulators
621.315.324	Wires resistant to rough treatment. Heavy-duty flex	621.315.65	Insulator supports. Insulator mounting, fixing. □ Wall fixings. Mast fixings
621.315.325	Wires resistant to chemical corrosion	621.315.66	Poles, masts, pylons, towers for support of overhead lines ⇒ 624.97
621.315.33	Insulated wires according to covering material <i>Denote the insulating material by .621.315.61...</i>	621.315.67	Protective enclosures, conduits, boxes
621.315.34	Insulated stranded conductors. □ Flexible cord (flex). Litz wires ⇒ 621.315.14	621.315.671	Conduits. Pipes. Ducts
621.315.35	Armoured wires. □ Lead cable ⇒ 621.315.222	621.315.673	Distribution boxes. Ceiling roses ⇒ 62-758 ⇒ 621.316.36
621.315.36	Screened wires	621.315.674	Fixing of conduits, pipes, ducts. Conduit fittings
621.315.37	Laying, fixing of insulated wires ⇒ 621.3.049.73 ⇒ 621.316.3	621.315.68	Joining, splicing, connecting of wires. Cable junctions
621.315.5	Conductors and semiconductors according to material	621.315.682	Permanent junctions. Fixed cable connectors
621.315.55	Metal or alloy conductors	621.315.683	Rigid and articulated connections
621.315.551	With high melting points. Resistance alloys. □ Heating elements	621.315.684	Nonpermanent junctions. □ Removable connectors. Clamps. Screw holders. Terminals. Tag strips
621.315.552	With low melting points. Fusible alloys	621.315.685	Terminal boxes
621.315.553	With constant resistance	621.315.687	Cable junction details. Cable accessories. □ Junction boxes. Cable clips, connectors, shoes, lugs
621.315.554	Bimetallic conductors. □ In heat regulators. In relays	621.316	Distribution and regulation of electric energy. Switching. Commutation. Circuit-breaking. Protection
621.315.56	Carbon, e.g. graphite conductors	621.316.1	Distribution networks (in buildings etc.) <i>The .051... and .052... auxiliaries may be used under 621.316.1, substituting 'distribution' for 'transmission' in the .051/.052 texts.</i> <i>Denote type of current by .02...</i>
621.315.57	Other solid nonmetallic conductors	◇ 621.316.1.024	Direct-current networks
621.315.58	Liquid conductors	◇ 621.316.1.025	Alternating-current networks ⇒ 621.311.1 ⇒ 696.6
621.315.59	Conductors with very high resistance. Semiconductors	621.316.11	Methods of network calculation <i>For calculating boards (network analysers) see 621.316.313</i>
621.315.592	Electronic semiconductors	621.316.14	Mixed networks
621.315.6	Insulating materials. Insulators. Installation devices, supports etc. Distribution and transmission line accessories	621.316.15	Area covered by networks. Power consumption in relation to the size of the network
621.315.61	Insulating materials (insulants). Dielectric materials	621.316.17	Electric installations in buildings and on sites. Electric installations in vehicles and for traffic and conveying routes
621.315.611	Solid insulating materials in general	621.316.3	Switching. Commutation. Switchgear ⇒ 621.3.049.73 ⇒ 621.3.061/.062 ⇒ 621.315.37
621.315.612	Ceramic and similar materials. □ Porcelain. Glass. Vitreous enamel ⇒ 666.185 ⇒ 666.593 ⇒ 666.638 ⇒ 666.651	621.316.31	Switch and wiring diagrams. Layout of complete installations
621.315.613	Natural mineral insulants. □ Mica. Asbestos. Quartz	621.316.311	Indoor installations
621.315.614	Fibrous materials. □ Paper. Board. Textiles	621.316.312	Outdoor installations
621.315.615	Liquid insulants. □ Transformer oils. Circuit-breaker oils	621.316.313	Special installations for research and experiment. □ Network analysers

621.316.318	Illuminated switching-diagrams etc.	621.316.583	Holders for fluorescent tubes
621.316.34	Switchboards. □ Switchboard panels. Control consoles, desks. Switch cabinets. Distributing switchboards	621.316.584	Holders for special lamps. □ For bipost lamps
621.316.35	Busbars. Busbar systems	621.316.585	Electron tube holders
621.316.36	Protective arrangements for electric equipment. □ Boxes. Cases. Hoods. Covers ⇒ 62-758 ⇒ 621.315.673 ⇒ 621.315.687	621.316.7	Regulation of electric apparatus in general. Regulators <i>Denote methods of regulation by .07...</i> ⇒ 62-55
621.316.37	Central switch plant. General control, supervision posts. □ Emergency switchgear	621.316.71	Electric regulation and regulators of motion. Electric starting and braking ⇒ 62-573 ⇒ 621.337
621.316.5	Opening and closing of circuits. Circuit-breakers. Holders ⇒ 621.43.048	621.316.717	Electric starting in general. Starters
621.316.52	Devices for intermittent opening and closing of circuits. Interrupters (tremblers). □ Hammer breaks (Wagner's interrupters), electrolytic interrupters (Wehnelt-type) ⇒ 621.43.048	621.316.718	Control of running, motion, speed
621.316.53	Contactors (magnetic switches)	621.316.719	Electric braking in general
621.316.54	Switching devices <i>Details by .06...</i>	621.316.72	Regulation of electrical quantities in general
621.316.541	Plug-and-socket devices ◇ 621.316.541-78 With safety contacts	621.316.721	Current regulation. Current regulators
621.316.541.11	Sockets	621.316.722	Voltage regulation. Voltage regulators
621.316.541.12	Plugs	621.316.722.1	Devices for maintaining constant voltage. Voltage stabilizers
621.316.541.2	Plug contacts	621.316.722.2	Regulators for increasing voltage. Boosters
621.316.541.3	Plug switches	621.316.722.3	Regulators for decreasing voltage. Negative boosters
621.316.541.4	Plug changeover switches	621.316.722.4	Devices for dividing voltage. Voltage dividers
621.316.542	Switches	621.316.723	Impedance regulation. □ Regulation of resistance, inductance, capacitance
621.316.542.1	Lever switches	621.316.726	Frequency regulation. Frequency regulators
621.316.542.2	Rotary switches. Drum switches	621.316.727	Regulation of phase difference. Power-factor correction
621.316.542.3	Push-button switches	621.316.728	Electric power regulation. Power regulators. Load regulators
621.316.542.4	Pull switches. Cord switches	621.316.729	Synchronization. Synchronizers
621.316.542.6	Slider switches	621.316.73	Regulation of magnetic quantities in general. □ Induction regulators
621.316.542.7	Roller switches	621.316.76	Regulators according to method of action
621.316.542.9	Other forms	621.316.761.2	Regulation by compensation. Compensators ⇒ 621.313.226
621.316.543	Changeover switches	621.316.765	Regulation involving distinctive connections. □ Star-delta controllers. Series-parallel controllers
621.316.544	Combination switches. Selectors	621.316.771	Regulation by means of resistors ⇒ 621.316.8
621.316.545	Section switches. Isolators ⇒ 621.316.923	621.316.79	Electric regulation of nonelectrical quantities except motion ⇒ 621.316.71 ⇒ 621.316.73 ⇒ 662.927
621.316.546	Mercury switches. Mercury contact tubes	621.316.8	Resistors. Rheostats <i>Denote conductor and insulating materials by :621.315.5/.6</i>
621.316.56	Cell switches. Battery switches. Accumulator switches	621.316.82	Variable resistors. Rheostats ⇒ 621.316.544 ⇒ 621.316.71 ⇒ 621.317.727
621.316.57	Power circuit breakers (power switches), automatic switches. Safety switches <i>Details by .06...</i>	621.316.824	Adjusting devices
621.316.572	Switches mainly for protection of persons ⇒ 621.316.933.8	621.316.825	Variable resistors dependent on temperature. Thermistors. □ Barreters
621.316.572.2	Error voltage (EV) safety switches	621.316.84	Metallic resistors in general
621.316.572.4	Fault current safety switches	621.316.86	Nonmetallic solid resistors. □ Carbon
621.316.573	Overload releases (excess current switches). □ Motor protection switches. Line safety switches	621.316.87	Liquid resistors. □ Aqueous solution
621.316.574	Circuit-breakers with undercurrent releases	621.316.88	Gas resistors
621.316.577	Combined switches	621.316.89	Mixed and other types of resistor
621.316.578	Circuit-breakers actuated by some external means, independent of the closed circuit	621.316.9	Protection. Protective devices. Fuses. Earthing ⇒ 614.825 ⇒ 62-758.3 ⇒ 62-78
621.316.578.1	Time switches	621.316.91	Protection against overcurrents and overvoltages in general
621.316.58	Holders ◇ 621.316.58-213.2 Drip-proof, splash-proof holders ◇ 621.316.58-213.34 Gastight, explosion-proof holders ◇ 621.316.58-783.3 Holders with protection against accidental contact ⇒ 621.3.032.8 ⇒ 621.3.048	621.316.92	Protection against overcurrents and overvoltages of internal origin
621.316.581	Screw holders	621.316.923	Fuses
621.316.582	Pin holders. Bayonet holders	621.316.923.1	Open fuses

621.316.923.2	Cartridge fuses	621.317.2	Equipment for electrical laboratories. Laboratories for testing, calibration etc.
621.316.923.3	Plug fuses		◇ 621.317.2:621.315.2 Cable-testing laboratories
621.316.923.4	Horn fuses	621.317.3	Measurement of electrical quantities
621.316.923.5	Section fuses. Isolating fuses	621.317.31	Current measurement. Measurement of charge, quantity of electricity
621.316.923.6	Multiple fuses	621.317.32	Measurement of electromotive force, potential, field strength, voltage
621.316.925	Protection by relays. Selective protection ⇒ 621.318.5	621.317.33	Measurement of resistance, impedance, reactance, conductance, inductance, capacitance, permittivity
621.316.925.2	Protection by the Merz-Price system. Differential protection	621.317.333	Insulation testing
621.316.925.4	Protection by relays acting on the circuit-breakers ⇒ 621.316.57	621.317.337	Measurement of circuit quality. Q-factor measurement
621.316.925.42	Overvoltage relays	621.317.34	Measurement of transmission characteristics of networks and lines
621.316.925.43	Overcurrent relays	621.317.35	Waveform analysis, recording. Oscillographic methods. Measurement of harmonics and modulation
621.316.925.44	Thermal relays	621.317.36	Measurement of frequency, wavelength
621.316.925.45	Distance relays. Distance protective systems	621.317.37	Measurement of power factor and phase difference. Determination of phase sequence
621.316.926	Protective devices based on the heating of insulating or other material. □ Based on the heating of oil in transformers	621.317.38	Measurement of power and energy. □ Active power. Energy losses. Energy consumption
621.316.93	Protection against overcurrents and overvoltages of external origin. Surge protection. Lightning protection ⇒ 621.316.98/99	621.317.39	Electrical measurement of nonelectrical quantities ◇ 621.317.39:531.77 Electrical measurement of angular velocity
621.316.932	Protection against flashover. Flashover arc extinction	621.317.4	Measurement of magnetic quantities. Magnetic testing ⇒ 537.67
621.316.933	Surge protection. Surge arresters. Voltage arresters	621.317.41	Measurement of permeability, susceptibility
621.316.933.1	Spark gaps. Sphere gaps	621.317.42	Measurement of magnetic fields, magnetizing force, magnetic induction
621.316.933.2	Horn-type lightning arresters ⇒ 621.316.98	621.317.43	Measurement of magnetic losses, iron losses. □ Hysteresis losses. Eddy-current losses
621.316.933.3	Roller gaps. Cylinder gaps. Disc gaps. Needle gaps	621.317.44	Apparatus for magnetic measurement. □ Exploring coils (search coils). Magnetic balances. Magnetic needles. Magnetometers. Fluxmeters
621.316.933.4	Arresters using dielectrics	621.317.45	Direction indicators for lines of force. Polarity indicators
621.316.933.5	Liquid arresters ⇒ 621.316.938	621.317.49	Magnetic measurement of nonmagnetic quantities ⇒ 620.179.14
621.316.933.6	Discharge tubes. Gas-filled protectors. Ionic and cathode-drop arresters	621.317.6	Determination of relation between measured quantities. Establishment of characteristics, diagrams
621.316.933.8	Contact protectors	621.317.61	Measurement of relation between electrical quantities. □ Measurement of turns ratio (transformation ratio). Short-circuit characteristic
621.316.935	Protective devices based on inductance. Current-limiting reactors. Line choking coils. Earthing reactors. Arc-suppression coils. □ Petersen coils ⇒ 621.314.228 ⇒ 621.318.4	621.317.62	Measurement of relation between magnetic quantities. Plotting of hysteresis curves
621.316.936	Protective capacitors ⇒ 621.319.4	621.317.63	Measurement of relation between magnetic and electrical quantities
621.316.937	Combined protective devices. □ Combined capacitors and coils	621.317.66	Measurement of efficiency
621.316.938	Protective devices based on electrolytic action of current. Electrolytic lightning arresters	621.317.7	Electrical measuring apparatus and instruments. Electric indicators, meters, recorders ◇ 621.317.7:621.314.6 Rectifier meters ◇ 621.317.7:621.385 Electron-tube meters
621.316.94	Protection against transients of internal origin (harmonics, steep wave-fronts)	621.317.71	Apparatus for measurement of current intensity and quantity of electricity
621.316.95	Protection against combined effect of several electrical influences	621.317.711	Current-direction indicators
621.316.96	Protection against mechanical accidents. □ Against breakage of electric lines	621.317.712	Electrodynamometers
621.316.97	Protection against magnetic influences, mutual inductance, earth currents	621.317.713	Voltmeters, Coulometers
621.316.98	Lightning conductors ⇒ 551.594.2 ⇒ 621.316.933.2 ⇒ 621.316.938	621.317.714	Ammeters
621.316.99	Earthing ⇒ 621.316.935	621.317.715	Galvanometers
621.316.998	Counterpoises For aerial counterpoises use 621.396.679		
621.317	Electrical measurement (methods and apparatus) ⇒ 621.3.08		
621.317.1	Procedure of carrying out a given measurement		

621.317.716	Ammeter shunts	621.318.24	Magnetization. Devices for enhancing magnetization
621.317.72	Apparatus for measuring electromotive force, voltage, voltage gradient	621.318.25	Demagnetization
621.317.722	Electroscopes. Potential indicators. Voltage detectors. Polarity indicators	621.318.26	Devices for avoiding loss of magnetic properties. □ Keepers
621.317.723	Electrometers	621.318.3	Electromagnets
621.317.725	Voltmeters	621.318.34	Magnetization methods
621.317.726	Peak voltmeters (crest voltmeters)	621.318.35	Heating. Devices for avoiding eddy currents etc.
621.317.727	Voltage-matching apparatus	621.318.37	Electromagnets according to arrangements of poles. □ Solenoids
621.317.727.1	Voltage dividers	621.318.38	Electromagnetic devices
621.317.727.2	Potentiometers		<i>Denote purpose by :...</i>
621.317.727.3	Shunt resistors	◇ 621.318.38:61	Electromagnetic devices for medical use
621.317.728	Spark gap voltmeters	⇒ 621.86.062	
621.317.73	Apparatus for measuring resistance, impedance, inductance, capacitance, permittivity	621.318.381	Metal detectors
621.317.732	Resistance boxes	621.318.4	Inductor coils
621.317.733	Bridge instruments. □ Wheatstone bridges. Kelvin bridges	⇒ 621.315.5	⇒ 621.315.61
621.317.734	Ohmmeters	⇒ 621.316.935	
621.317.735	Apparatus for testing insulation	621.318.43	Inductors. Choking coils. Reactors.
621.317.736	Leakage indicators. Short-circuit indicators	□ Transducers	⇒ 621.375.3
621.317.737	Circuit-quality meters (Q-meters)	621.318.44	Coil-winding machines
621.317.738	Apparatus other than bridge instruments for measuring inductance, capacitance, permittivity	621.318.5	Relays
621.317.74	Apparatus for measurement of transmission quantities. □ Attenuation meters. Amplification meters. Standing-wave indicators. Level indicators. Coupling-factor indicators	621.318.51	⇒ 621.316.53
621.317.75	Apparatus for examining waveforms. Oscillographs	621.318.53	⇒ 621.316.925
621.317.752	Contact apparatus	621.318.56	⇒ 681.584.7
621.317.753	Electrostatic and electromagnetic oscillographs	621.318.562	Electrostatic relays
621.317.755	Cathode-ray oscillographs	621.318.563	Electrolytic relays
621.317.756	Apparatus for measuring damping decrements	621.318.57	Electromagnetic relays
621.317.757	Harmonic analysers. Distortion factor measurement		Direct-current relays
621.317.76	Frequency meters. Oscillation indicators. Wavemeters	621.318.58	Alternating-current relays
621.317.77	Phase-difference indicators. Power-factor meters. Phase-sequence indicators. Synchroscopes	621.319	Electronic relays. Electronic switches
621.317.78	Apparatus for measuring power, energy consumption etc.	621.319.1	⇒ 621.385
621.317.782	Load indicators	621.319.2	Photoelectric relays (light relays)
621.317.783	Volt-ampere meters	621.319.3	Applied electrostatics
621.317.784	Wattmeters		⇒ 537.2
621.317.785	Watt-hour meters. Electricity supply meters	621.319.32	Applied ferroelectricity
621.317.786	Electrolytic meters	621.319.33	Electrets
621.317.787	Ampere-hour meters (other than electrolytic meters). Time meters		Electrostatic machines. Electrostatic generators
621.317.787.1	Ampere-hour meters	621.319.34	Frictional electrostatic machines
621.317.787.2	Time meters	621.319.4	Influence machines (electrostatic induction machines). □ Wimshurst machines. Van de Graaff generators
621.317.79	Other measuring apparatus	621.319.43	Electrostatic machines with external feed (electric discharges)
621.317.791	Universal apparatus. Apparatus for measuring several quantities	621.319.44	Capacitors
621.317.792	Apparatus for investigations into atmospheric electricity	621.319.45	⇒ 621.316.936
621.317.794	Apparatus for measuring radiant energy. Bolometers	621.319.46	Continuously variable capacitors
621.318	Applied magnetism. Magnetic materials, apparatus. Coils. Relays	621.319.47	Capacitor banks. Capacitors with stepwise regulation
	⇒ 537.6/8	621.319.48	Electrolytic capacitors
	⇒ 669.018.58	621.319.5	Compressed-gas capacitors (pressure-type capacitors)
621.318.1	Magnetic materials in general	621.319.52	Vacuum capacitors
	⇒ 537.622	621.319.53	Nonvariable liquid capacitors
621.318.12	Hard magnetic materials	621.319.54	Other electrostatic apparatus. Surge generators
621.318.13	Soft magnetic materials	621.319.55	Direct-voltage circuits. □ Greinacher circuits.
621.318.14	Paramagnetic materials	621.319.7	Cockcroft circuits
621.318.2	Permanent magnets	621.319.71	Impulse circuits. Surge generators. Marx generators
		621.319.72	Alternating-voltage circuits for damped processes. Tesla circuits
			Relaxation circuits
			Electrostatic fields. Field generation. Field distribution
			Generation
			Regulation

621.319.74	Devices for avoiding unwanted fields and charges. Electrostatic screening, shielding ⇒ 621.316.99	621.332.33 621.332.34 621.332.35	Line suspension Suspension wires and cables Connection of contact wires with suspension wires and cables
621.32	Electric light sources <i>Denote individual parts of electric lamps by .032....</i> <i>Denote the purpose for which lamps are used by :...</i> ◇ 621.32:543.42 Lamps for spectrum analysis ◇ 621.32:615.831 Phototherapy lamps. Sunray lamps etc. ⇒ 628.9	621.332.4 621.332.41 621.332.42 621.332.43 621.332.5 621.332.52 621.332.53 621.332.6 621.332.8 621.333 621.333.3 621.333.4 621.333.41 621.333.42 621.336 621.336.2 621.336.3 621.336.4 621.336.5 621.336.6 621.336.7 621.336.8 621.337 621.337.1 621.337.2 621.337.3 621.337.4 621.337.5 621.337.6	Current supply methods other than overhead lines Underground current collection. Conduit system: side slot, central slot Conductor-rail system Lines for transmission by electromagnetic induction Line crossings. Points. Signal devices for crossings and points Line crossings without points Points Sectionalizing of lines. Length of sections. Switching of sections in and out of circuit Special devices on lines, suspensions etc. at bridges, crossings, viaducts, tunnels Electric traction motors ⇒ 621.313 Special magnetic circuits. Connection of the motor circuits Regeneration (recuperation) of electric energy ⇒ 62-593 ⇒ 620.97 ⇒ 621.337.5 By braking By use of down-gradients Current collection. Current-collecting devices and accessories ⇒ 62-831 Current collection. □ Contact pressure. Contact-breaking Current collectors for overhead lines. □ Bow, pantograph, trolley collectors. Collector shoes, rollers Adjustability, movement, play in overhead current collectors. □ Flexible overhead collectors Bases of overhead current collectors Derailment of overhead current collectors. □ Prevention of derailment. Restoration of contact Current collectors for conductor rails. Collector shoes Current collectors for underground, conduit system. Plough-type collectors Controls for electric locomotives and vehicles Control systems. □ Single-point control. Multipoint control Reversing switches. Controllers ⇒ 621.316.544 Electrical connection between locomotives and carriages Devices for control of speed and direction. Reversing devices Braking, deceleration devices. □ Electric brakes. Electromagnetic brakes ⇒ 62-592.3 ⇒ 621.333.4 Devices for control of rail points, line points, signals, by the vehicle and by hand from the vehicle ⇒ 625.151.6 ⇒ 656.259.1
621.321	Generalities. Comparison between various kinds of electric lamp		
621.325	Carbon arc lamps. Arc lamps with one carbon and one metal electrode ⇒ 537.523.5		
621.326	Incandescent lamps. Filament lamps		
621.326.1	Carbon filament lamps		
621.326.5	Lamps with nonmetallic incandescent elements (except carbon). □ With zirconia rods (Nernst lamps). With tantalum carbide elements		
621.326.7	Lamps with metallic filaments		
621.326.71	Vacuum general-purpose lamps (electric light bulbs)		
621.326.72	Gas-filled general-purpose lamps (electric light bulbs)		
621.326.73	Lamps with concentrated filaments. Spotlight lamps		
621.326.74	Lamps with more than one cap. □ Show window lamps		
621.326.75	Small, miniature lamps		
621.326.76	Lamps with variable illuminating power		
621.326.77	Economy lamps. Low consumption lamps		
621.326.78	Incandescent lamps with halogen-containing medium (tungsten halogen)		
621.327	Gas discharge light sources		
621.327.2	Electroluminescence sources		
621.327.4	Glow discharge lamps		
621.327.42	With inert gas filling. □ Neon discharge lamps		
621.327.43	With gas and metal vapour filling		
621.327.5	Arc discharge lamps		
621.327.52	With inert gas filling. □ Xenon discharge lamps		
621.327.53	With gas and metal vapour filling		
621.327.532	Sodium lamps		
621.327.534	Mercury lamps		
621.327.534.1	Low pressure mercury lamps		
621.327.534.15	Low pressure, mercury fluorescent lamps		
621.327.534.2	High pressure and extra-high pressure mercury lamps		
621.327.54	Halide lamps		
621.327.6	Discharge lamps with luminous electrodes (no positive column)		
621.327.7	Flash tubes (USA)		
621.327.9	Other gas-discharge light sources		
621.328	Compound lamps (blended-light lamps)		
621.33	Electric traction, locomotion, propulsion		
621.331	Electrification for traction. □ Railway electrification ◇ 621.331:625.31(44) Electrification of narrow-gauge railways in France		
621.331.3	Choice of type of current		
621.331.4	Choice of voltage		
621.332	Supply lines. Distribution of energy		
621.332.2	Supply and return lines		
621.332.21	Feeders		
621.332.22	Special return lines		
621.332.23	Use of track as return line (track-return lines)		
621.332.3	Overhead lines. Line suspension		
621.332.31	Contact wires. Trolley wires		
621.332.32	Structure and form of contact wires		
		621.35	Electrochemical technology <i>Denote individual parts for electrochemical equipment by .035...</i> ⇒ 544.6 ⇒ 66.087 ⇒ 661.13

621.351	Generation and storage of electric energy by electrochemical methods, in general	621.359.4	Electric precipitation (electroprecipitation) from gases. □ Cottrell process. Wet process ⇒ 537.2 ⇒ 669.162.252
621.352	Generation of electric energy by electrochemical methods. Primary voltaic cells (galvanic cells) and batteries	621.359.7	Colloidal dispersion by electric processes. Electrodialysis. Atomization ⇒ 66.044
621.352.1	Primary cells in general. Primary batteries	621.36	Thermoelectricity. Electric heating
621.352.2	Cells without oxidizing depolarizers. Daniell cells	621.362	Generation of electricity by heat.
621.352.3	Solid oxidizer cells (depolarizer cells). Leclanché cells	621.362.1	Thermoelectric generators
621.352.4	Liquid oxidizer cells (depolarizer cells)	621.362.2	Thermocouples
621.352.5	Cells with predominantly gaseous oxidizers (depolarizers)	621.363	Thermopiles
621.352.6	Cells with combustible materials. Fuel cells	621.365	Thermoelectric motors. Thermomagnetic (magneto-caloric) motors
621.354	Methods and equipment for commissioning and servicing of cells and batteries for generation or storage of electricity		Generation of heat by electricity. Electric heating <i>Denote individual parts of electric heating apparatus by .036...</i> ⇒ 666.1.031.5 ⇒ 669.187.2 ⇒ 669.712.3 ⇒ 683.9
621.354.2	Filling equipment. Draining equipment ⇒ 621.3.035.8	621.365.2	Generation of heat by electric arcs. Arc furnaces. Other arc heating apparatus ⇒ 621.791.75
621.354.3	Methods and conditions of service	621.365.3	Direct resistance heating (current flowing through the material). Electrode heating apparatus. Electrode furnaces etc.
621.354.32	Charging. Regeneration. Conservation. Buffering	621.365.31	Direct resistance heating of solid materials ◇ 621.365.31:621.884 Electric rivet heaters
621.354.33	Activation with electrolyte or other fluid	621.365.32	Direct resistance heating of noncoherent particles, granular and pulverous materials, powders. □ Graphitizing furnaces
621.354.34	Conditions and equipment for the improvement of charging and discharging (increasing efficiency, service life)	621.365.33	Direct resistance heating of liquids. □ Glass-melting baths. Electrode boilers
621.354.35	Methods for hermetically sealing galvanic cells	621.365.38	Direct resistance heating in small electrothermal apparatus
621.354.7	Use of batteries with regulator cells or counter-emf cells	621.365.4	Indirect resistance heating (current flowing through a thermal transmitter or heating element). Resistance furnaces. Electrothermal apparatus with heating elements
621.355	Storage of electricity by electrochemical methods. Accumulators. Secondary cells. Storage batteries (accumulator batteries)	621.365.41	Indirect resistance heating via solid metallic resistors or elements ◇ 621.365.41:669.28 Molybdenum heating elements
621.355.1	Installations. Maintenance in general	621.365.42	Indirect resistance heating via solid nonmetallic resistors or elements
621.355.2	Lead (lead-acid) accumulators in general	621.365.43	Indirect resistance heating via noncoherent particles, granular and pulverous heating elements. □ Slack coal furnaces
621.355.5	Acid accumulators with electrodes of lead and another metal	621.365.44	Indirect resistance heating via liquid heating elements. □ Salt bath electrode furnaces
621.355.7	Acid accumulators with electrodes of metals other than lead	621.365.46	Indirect resistance heating by directed infrared radiation ◇ 621.365.46:667.661 Infrared heating devices for drying coatings, varnishes ⇒ 621.384.3
621.355.8	Alkaline accumulators. □ Nickel-iron (Edison accumulators)	621.365.48	Indirect resistance heating in small electrothermal apparatus ◇ 621.365.48:644.19 Small domestic heating appliances with indirect resistance heating. Electric blankets etc.
621.357	Industrial electrochemistry. Electrolysis. Electrodeposition ⇒ 66.087	621.365.5	Inductive and dielectric (capacitive) heating in an alternating field. Induction furnaces and apparatus. Dielectric heating apparatus
621.357.1	Industrial electrolysis. □ Fusion electrolysis	621.365.51	Production of magnetic field in closed iron core
621.357.2	Industrial electrochemistry of inorganic compounds in general	621.365.52	Production of magnetic field without iron core. □ High-frequency induction furnaces <i>Denote frequency by .029...</i>
621.357.3	Industrial electrochemistry of organic compounds in general	621.365.53	Production of magnetic field with partial use of iron core
621.357.5	Pretreatment for electroforming and electroplating	621.365.55	Heating using electrostatic field. Dielectric heating
621.357.6	Electroforming (galvanoplasty). Forming by electrodeposition ⇒ 673.4		
621.357.7	Electroplating. Electroforming. Plating by electrodeposition ⇒ 621.793.3		
621.357.8	Electrobrightening. Pickling. Colouring. Anodizing ⇒ 667.761		
621.359	Industrial applications of electrocapillarity. Electroprecipitation ⇒ 532.63 ⇒ 537.361 ⇒ 66.087		
621.359.2	Electroendosmosis (electro-osmosis) ⇒ 537.362		
621.359.3	Electric precipitation (electroprecipitation) from liquids. Processes involving electrophoresis (cataphoresis, anaphoresis) ⇒ 537.242 ⇒ 537.363 ⇒ 543.54		

621.365.58	Inductive and dielectric heating in small electrothermal apparatus	621.372.45	Active dipoles
621.365.6	Mixed electric heating. Electric combined with other heating	621.372.5	Four-terminal networks (quadrupoles)
621.365.9	Other methods of electric heating. □ By cathode-ray furnaces	621.372.51	Transducing, matching quadrupoles. Coupled circuits
621.37	Technique of electric waves, electromagnetic waves, oscillations, pulses	621.372.52	Quadrupoles in feedback circuits ⇒ 621.373.121 ⇒ 621.375.13
	<i>Denote special features of electrical oscillation technique by .037...</i>	621.372.53	Quadrupoles in push-pull circuits
	◇ 621.37:615.849 Applications of electric waves etc. to radiotherapy	621.372.54	Quadrupole frequency filters
621.371	Unguided propagation of electromagnetic waves in free space, in interplanetary space or through various media ⇒ 537.876.2 ⇒ 621.391.812 ⇒ 621.396.94	621.372.542	Filters with one cut-off frequency. □ Low-pass filters. High-pass filters. Separating filters
621.371.3	Propagation between specified locations. Propagation paths	621.372.543	Filters with two cut-off frequencies. □ Band-pass filters. Band-stop filters
621.371.31	Free space paths. Mixed propagation paths. □ Signals originating below an ionized layer and subsequently passing through free space ⇒ 621.396.933	621.372.544	Filters with more than two cut-off frequencies. Filters with several pass-bands or band-stops
621.371.32	Propagation by direct radiation. Propagation by surface wave (Zenneck wave)	621.372.55	Distortion-correcting quadrupoles. □ Attenuation equalizers. Delay equalizers.
621.371.33	Propagation by indirect radiation with interposed reflection, refraction or diffraction. Propagation beyond the horizon	621.372.56	Phase compensators
621.371.332	Propagation by indirect radiation with reflection. □ Reflection due to atmospheric layers, meteorites, cosmic dust. Reflection from objects, e.g. radar echoes. Scattering	621.372.57	Quadrupole attenuators
621.371.333	Propagation by indirect radiation with refraction	621.372.58	Active quadrupoles ⇒ 621.375
621.371.334	Propagation by indirect radiation with diffraction	621.372.6	Gyrators
621.371.336	Skip distances (zones of silence). Hop distances	621.372.62	Multipoles
621.371.34	Propagation paths between two points on the earth and/or in the troposphere	621.372.63	Passive multipoles. □ Mixers
621.371.35	Propagation paths between two points, one or both situated outside the troposphere		Active multipoles. □ Frequency changers ⇒ 621.314.26 ⇒ 621.374.4
621.371.36	Propagation via intermediate links. □ Via satellites ⇒ 621.396.946	621.372.8	Waveguides
621.371.37	Usability of frequencies	621.372.81	◇ 621.372.8:621.315.61 Dielectric waveguides
621.371.372	Critical frequencies. Penetration frequencies	621.372.82	Generalities. □ Reflection in waveguides.
621.371.373	Maximum usable frequency (MUF)		Standing wave ratio. Matching of waveguides
621.371.374	Optimum working frequency (OWF). Optimum traffic frequency (FOT)	621.372.83	Kinds of waveguide and wavemode. □ Parallel-plate (stripline) waveguides.
621.371.375	Lowest usable frequency (LUF)	621.372.831	Rectangular, elliptical, circular waveguides.
621.371.376	Absorption limiting frequency (ALF)	621.372.832	Coaxial waveguides. Corrugated, ridge waveguides. Surface waveguides
621.371.38	Prediction of usable frequencies for a radiocommunication link	621.372.833	Coupling of waveguides
621.371.39	Propagation through water, through the earth, through other media	621.372.834	Coupling of two waveguides
621.372	Guided propagation of electric oscillations in general ⇒ 537.876.4 ⇒ 621.396.44 ⇒ 621.396.65	621.372.837	Multiterminal couplings
621.372.2	Theory of transmission lines ⇒ 621.3.09	621.372.85	Coupling between waveguides and other transmission systems
621.372.21	Homogeneous lines. □ Continuously loaded circuits (Krupp circuits)	621.372.851	Coupling between cavities and waveguides
621.372.22	Nonhomogeneous lines. □ Coil-loaded circuits (Pupin circuits)	621.372.852	Waveguide switches
621.372.4	Two-terminal networks (dipoles)	621.372.852.1	Obstacles. Loaded waveguides
621.372.41	Passive dipoles with both capacitance and inductance, with or without resistance. Oscillation circuits	621.372.852.2	Infinitely thin obstacles (obstacles of zero thickness). Irises
621.372.43	Passive dipoles for impedance setting or matching. Matching stubs	621.372.852.3	Obstacles of finite thickness. Waveguide filters
621.372.44	Nonlinear passive dipoles	621.372.852.4	Filters
		621.372.852.5	Phase changers (phase shifters). □ Squeeze sections
		621.372.852.6	Attenuators
		621.372.853	Polarizers
		621.372.855	Mode transformers. Mode filters
		621.372.86	Impedance transformers. Tuning posts, slug tuners etc.
		621.372.862	Composite structures. Loaded waveguides
		621.372.863	Waveguide terminations
		621.372.865	Input sections. Output sections. Measuring sections. Delay sections
		621.372.867	Input sections
		621.372.88	Output sections
		621.373	Measuring sections
			Delay sections
			Composite arrangement
			Generators of electric oscillations and pulses.
			Oscillators. Pulse generators
		621.373.1	Oscillators generally (irrespective of active element)
		621.373.12	Sinusoidal wave oscillators ⇒ 621.372.41
		621.373.121	Feedback oscillators

621.373.122	Negative-resistance oscillators	621.375.4 $\equiv$ 621.375.1
621.373.123	Transit time oscillators	$\Rightarrow$ 621.382.3
621.373.124	Heterodyne oscillators. $\square$ Tone oscillators	Dielectric amplifiers
621.373.13	Nonsinusoidal wave oscillators. Relaxation oscillators. Time bases	621.375.5 $\equiv$ 621.375.1
621.373.131	Multivibrators	Electromechanical amplifiers
621.373.132	Relaxation oscillators. Blocking oscillators. Squeggers	621.375.6 $\equiv$ 621.375.1
621.373.133	Feedback gating oscillators	Parametric amplifiers
621.373.134	Sawtooth wave generators	621.375.7 $\equiv$ 621.375.1
621.373.14	Triggered pulse generators	Quantum-mechanical amplifiers
621.373.3	Nonelectronic oscillators. $\square$ Electromechanical oscillators. Spark oscillators. Arc oscillators	Molecular amplifiers. Masers
621.373.4	Thermionic valve oscillators. Gas discharge tube oscillators	Masers for amplifying electrical oscillations
	621.373.4 $\equiv$ 621.373.1	Optical masers. Lasers
	Denote the active element by :...	Other types
	$\diamond$ 621.373.4:621.385.44 Tetrode oscillators	Modulation and demodulation. Processes and apparatus
621.373.422	Negative resistance valve oscillators	Class here particular modulators and demodulators not listed below, e.g. ring modulators
621.373.423	Transit-time valve oscillators	Amplitude modulation and demodulation
621.373.5	Oscillators using semiconductor elements	Amplitude modulators (double-sideband)
	621.373.5 $\equiv$ 621.373.1	Amplitude demodulators (double-sideband).
	Denote the active element by :...	AM detectors
	$\diamond$ 621.373.5:621.382.2 Oscillators using semiconductor diodes	Single-sideband systems
	$\diamond$ 621.373.5:621.382.3 Oscillators using transistors	Mixed, vestigial-sideband systems
621.373.52	Sinusoidal-wave oscillators using semiconductor elements	Frequency modulation and demodulation
621.373.54	Pulse oscillators using semiconductor elements	$\diamond$ 621.376.3.037.372.2 Modems
621.373.7	Parametric oscillators	Frequency modulators
621.373.8	Quantum-mechanical oscillators	Frequency demodulators. FM detectors
	621.373.8 $\equiv$ 621.375.8	Phase modulation and demodulation
621.373.9	Other oscillators	Pulse modulation and demodulation
621.374	Pulse processes. Frequency multipliers and dividers	Combined modulation systems
621.374.3	Pulse-operated circuits	Other types of modulation system
621.374.4	Frequency multiplying and dividing circuits	Delay devices. Electric information storage devices (stores)
	$\Rightarrow$ 621.314.26/27	$\diamond$ 621.377.037.3 Type and representation of information in storage devices
	$\Rightarrow$ 621.372.63	$\diamond$ 621.377.037.7 Details of electric pulse storage devices
621.374.5	Pulse-delay circuits	Delay devices
621.375	Amplifiers	Analog stores
	$\diamond$ 621.375:621.317 Amplifiers for electrical measurement	Digital stores, registers, storage devices
	$\diamond$ 621.375:621.397 Television amplifiers	Digital stores for variable data
	$\diamond$ 621.375.018.756 Pulse amplifiers	With separate information storage for separate information elements (one element per bit). $\square$ With bistable devices. With multistable devices
	$\diamond$ 621.375.024 Direct current amplifiers	With common storage medium for a number of information elements spatially confined to a discrete surface of the medium. $\square$ Magnetic tapes, wires, drums, discs, cards. Cathode-ray tube stores
	$\diamond$ 621.375.026 Power amplifiers (designation in watts)	With storage medium in which the information elements are propagated. $\square$ Using mechanical, acoustic, electromagnetic waves. Magnetic spin-echo stores
	$\diamond$ 621.375.029 Amplifier frequency ranges	Chemical information stores
	$\Rightarrow$ 681.583.7	Read-only stores (dead storage). $\square$ With card storage. With tape storage
	$\Rightarrow$ 681.84.086	Read-only stores with common storage medium for a number of information elements, each spatially confined to a discrete surface element of the common carrier
621.375.1	Amplifiers in general. Theory and properties	
621.375.12	Amplifiers according to type of coupling	621.38 Electronics. Electronic devices. Electron tubes. Photocells. Particle accelerators. X-ray tubes
621.375.121	Broadband coupling	Denote individual parts by .032....
621.375.122	Aperiodic (nonresonant) and resonant circuit coupling	Denote features and parts of electronic devices for the production of active rays and corpuscular beams by .038.6...
621.375.127	Push-pull amplifiers	$\Rightarrow$ 621.3.0385
621.375.13	Feedback amplifiers	$\Rightarrow$ 621.3.049.77
	$\Rightarrow$ 621.372.52	
621.375.132	Negative feedback	
621.375.133	Positive feedback	
621.375.134	Combinations of negative and positive feedback	
621.375.2/9	Various types of amplifier	
621.375.2	Thermionic valve amplifiers	
	621.375.2 $\equiv$ 621.375.1	
621.375.3	Magnetic amplifiers (transductor amplifiers)	
	621.375.3 $\equiv$ 621.375.1	
621.375.4	Amplifiers with electronic solid-state components. Semiconductor amplifiers. Diode amplifiers. Transistor amplifiers	

621.382	Electronic devices using solid-state effects. Semiconductor devices ⇒ 537.311.322 ⇒ 621.315.59	621.383.814	Image converter tubes with more than one conversion stage
621.382.2/3	Electrical semiconductor devices	621.384	Production and application of active rays and corpuscular beams ⇒ 621.039
621.382.2	Crystal diodes	621.384.2	Technical application of radioactive materials ⇒ 621.039.8
621.382.22	Point-contact diodes	621.384.3	Technical application of infrared rays
621.382.23	Junction diodes	621.384.4	Technical application of ultraviolet rays
621.382.232	With one p-n junction	621.384.5	Technical application of glow discharges
621.382.233	With more than one p-n junction. □ Four-layer diodes	621.384.6	Particle accelerators <i>The .038.6... auxiliaries are particularly applicable at 621.384.6 and its subdivisions</i> ⇒ 539.1.076 ⇒ 621.386.2
621.382.3	Transistors	621.384.63	Circular accelerators
621.382.32	Unipolar transistors (utilizing charge carriers of only one polarity)	621.384.633	Fixed-field accelerators. Cyclotrons in general
621.382.322	Analog transistors (control by pinch effect)	621.384.633.3	Weak-focusing constant-frequency cyclotrons
621.382.323	Field-effect transistors (control by external fields)	621.384.633.4	Weak-focusing frequency-modulated cyclotrons
621.382.33	Bipolar transistors (using both types of charge carrier with carrier injection)	621.384.633.5/6	Strong-focusing fixed-field accelerators. Fixed-field alternating-gradient (FFAG) accelerators. Azimuthally-varying-field (AFV) accelerators (including intersecting-beam accelerators, beam-stacking techniques)
621.382.332	Transistors with barrier layers (without intrinsic p-n junctions)	621.384.633.5	FFAG cyclotrons
621.382.332.2	Point-contact transistors	621.384.633.6	FFAG synchrotrons (magnetic field constant in time) <i>For conventional varying-field synchrotrons use 621.384.634</i>
621.382.332.3	Surface-barrier transistors	621.384.633.8	Microtrons
621.382.332.9	Other transistors with barrier layers (e.g. filamentary transistors)	621.384.633.9	Other fixed-field accelerators and cyclotrons
621.382.333	Junction transistors (with intrinsic p-n junctions)	621.384.634	Varying-field accelerators. Synchrotrons in general
621.382.333.3	Transistor triodes	621.384.634.3	Betatrions <i>For plasma betatrions use 621.384.637</i>
621.382.333.32	With one p-n junction (e.g. double-base triodes)	621.384.634.4	Constant-gradient or weak-focusing synchrotrons
621.382.333.33	With two p-n junctions (e.g. drift transistors)	621.384.634.5	Alternating-gradient or strong-focusing synchrotrons
621.382.333.34	With more than two p-n junctions	621.384.634.9	Other types of synchrotron (e.g. zero-gradient synchrotrons)
621.382.333.4	Transistor tetrodes	621.384.637	Circular accelerators using plasma as focusing or accelerating medium
621.382.334	Transistors with various kinds of contact. □ Transistors having simultaneously junction and point contacts. Spacistors	621.384.64	Linear accelerators (using high-frequency alternating accelerating fields)
621.383	Photoelectric tubes and cells ⇒ 535.215 ⇒ 535.247	621.384.643	Linear accelerators using discontinuous acceleration (drift tubes)
621.383.2	Photoelectric tubes. □ Photomultiplier tubes. Photoemissive cells	621.384.644	Linear accelerators using continuous acceleration. Waveguides. □ Helical accelerating structures
621.383.29	Photoelectric tubes combined with electron tube elements	621.384.647	Linear accelerators using plasma as focusing or accelerating medium
621.383.292	Photoelectric multiplier tubes. Photomultiplier tubes	621.384.65	Accelerators using steady or low-frequency accelerating fields
621.383.4	Photoconductive detectors <i>Denote the material by -03... (Table 1k)</i> ◇ 621.383.4-039.648'221 Cadmium sulphide ◇ 621.383.4-039.681.7'231 Lead selenide	621.384.653	Accelerators using electrostatic machines. □ Using Van de Graaff generators
621.383.45	Intrinsic photoconductive detectors	621.384.654	Cascade-rectified accelerators. □ Cockcroft-Walton accelerators
621.383.46	Extrinsic photoconductive detectors	621.384.658	High-voltage transformer accelerators. □ Resonant transformer types
621.383.47	Photon drag detectors	621.384.66	Auxiliary equipment and accessories for particle accelerators
621.383.48	Photoelectromagnetic detectors	621.384.663	Beam-transport systems. Beam-analysing systems <i>Class here devices used to transport the beam from accelerator to target area, or to split it for directing towards several target areas</i> ◇ 621.384.663:621.318 Analysing magnets
621.383.49	Other photoconductive detectors. E.g. detectors with μ -photoconductivity	621.384.664	Targets
621.383.5	Depletion layer photodetectors	621.384.665	Storage rings. Colliding-beam devices
621.383.51	Photovoltaic elements, solar cells and solar cell arrays (batteries, panels)		
621.383.52	Photodiodes		
621.383.53	Phototransistors		
621.383.6	Karolus cells. Kerr cells		
621.383.7	Photosensitive charge transfer devices (CTD)		
621.383.72	Charge coupled devices (CCD)		
621.383.74	Charge injection devices (CID)		
621.383.8	Image converters employing the photoelectric effect		
621.383.81	Image converter tubes employing the photoemissive effect		
621.383.811	Image converter tubes with electrostatic focusing		
621.383.812	Image converter tubes with magnetic focusing		
621.383.813	Image dissector tubes		

621.384.668	Beam-monitoring devices outside the accelerator ⇒ 621.3.038.616	621.386	X-ray apparatus. X-ray tubes <i>Denote individual parts by .032...</i> ⇒ 535.34 ⇒ 539.26 ⇒ 548.73 ⇒ 615.849 ⇒ 616-001.26 ⇒ 616-073 ⇒ 620.179.152.1 ⇒ 66.085.3 ⇒ 778.33
621.384.8	Particle separators. Mass spectrographs. Mass spectrometers	621.386.1	X-ray apparatus in general
621.385	Apparatus employing electron discharges. Electron tubes. Thermionic vacuum tubes (thermionic valves) <i>Denote individual parts of electron tubes by .032...</i> ⇒ 621.314.67 ⇒ 621.318.57 ⇒ 621.327.4	621.386.12	Universal apparatus
621.385.1	General principles and characteristics of electron tubes	621.386.14	Portable apparatus
621.385.2	Diodes (two-electrode tubes)	621.386.16	Apparatus for stereoradioscopy, stereoradiography, stereoradiogrammetry
621.385.29	Double diodes	621.386.17	Apparatus for cineradiography
621.385.3	Triodes (three-electrode tubes)	621.386.2	X-ray tubes ◇ 621.386.2:620.179.152.1 X-ray tubes for industrial radiography
621.385.39	Double triodes	621.386.7	Handling and use of X-ray tubes. Accessories
621.385.44	Tetrodes (four-electrode tubes)	621.386.8	X-ray apparatus other than tubes ◇ 621.386.8-784.7 Protective devices against radiation from X-ray apparatus ⇒ 614.89
621.385.45	Beam tetrodes	621.386.82	Radiochronometers. Dosimeters
621.385.5	Electron tubes with more than four electrodes	621.386.83	Accessories for radioscopic purposes. □ Beclere frames. Fluorescent screens
621.385.55	Pentodes (five-electrode tubes)	621.386.84	Accessories for radiography. Devices for locating foreign bodies by stereoscopic means. Cineradiography etc. □ Intensifying screens
621.385.56	Hexodes (six-electrode tubes)	621.386.85	Accessories for treatment with X-rays
621.385.57	Heptodes (seven-electrode tubes)	621.387	Gas-filled and vapour-filled discharge tubes (except for lighting). Ionization chambers. Ion counters <i>Denote individual parts by .032...</i> ⇒ 621.327 ⇒ 621.383
621.385.58	Octodes (eight-electrode tubes)	621.387.1	Discharge tubes with hot cathode (thermionic cathode)
621.385.59	Nonodes (nine-electrode tubes). Electron tubes with more than nine electrodes	621.387.12	Without means of controlling current flow
621.385.6	Velocity-modulation tubes. Klystrons. Travelling-wave tubes. Magnetrons and other microwave tubes which depend on transit time	621.387.122	Low-pressure gas or vapour types. □ Phanotrons. Hot-cathode rectifiers
621.385.62	Tubes controlled by a stationary field. □ Retarding-field tubes. Resonator tubes	621.387.123	High-pressure gas types. □ Tungar rectifiers. Rectigons
621.385.63	Tubes controlled by a travelling field. □ Forward-wave tubes. Electron-wave tubes. Backward-wave tubes. Carcinotrons	621.387.13	With one or more electrodes to provide electrostatic control. □ Gas-filled triodes, e.g. hot-cathode thyratrons, tacitrons
621.385.64	Magnetrons	621.387.14	With other means of controlling current flow. □ With magnetic control (permatrons). Control by current (plasmotrons)
621.385.65	Tubes with static crossed field. 'Magnetless' magnetrons (Versnel-type magnetrons)	621.387.2	Discharge tubes with pool-type cathode (liquid or solid)
621.385.8	Other types of hard-vacuum electron tube ⇒ 621.386.2	621.387.22	Without means of controlling current flow. □ Arc-discharge pool tubes, e.g. mercury-pool type
621.385.82	Tubes for production of positive rays (anode rays, canal rays)	621.387.23	With grid control. Grid-pool tubes. □ Mercury-pool type. Excitrons
621.385.83	Tubes with controlled electron beam	621.387.24	With igniter electrode to control the starting of the unidirectional current flow in each operative cycle. Ignitrons
621.385.831	Electron multipliers	621.387.25	With magnetic control. Hodelectrons
621.385.832	Tubes with directional and/or intensity beam control. Cathode-ray tubes	621.387.3	Discharge tubes with cold cathode (except radiation counters). Electrodeless discharge tubes ⇒ 621.387.4
621.385.832.2	Electromagnetic type of deflection	621.387.32	Without means of controlling current flow
621.385.832.4	Electrostatic type of deflection	621.387.322	Gas-filled glow tubes
621.385.832.5	Image tubes employing photoelectric effect. Camera tubes ⇒ 621.383.8	621.387.322.2	Voltage-stabilizing and control tubes. Neon stabilizers
621.385.832.52	With photoemissive effect. □ Dissector tubes. Iconoscopes. Orthicons		
621.385.832.54	With photovoltaic effect		
621.385.832.56	With photoconductive effect		
621.385.832.8	Special kinds of cathode-ray tube		
621.385.832.82	Storage tubes. □ Graphecons		
621.385.832.84	Beam-switch tubes. Frequency-multiplying tubes ⇒ 621.387.1		
621.385.832.86	Cathode-ray tubes with travelling-wave deflecting systems		
621.385.832.87	Cathode-ray counter tubes. □ Alfvén counters. Trochotrons		
621.385.833	Electron microscopes. Diffraction cameras ⇒ 537.533.35 ⇒ 778.317		
621.385.833.2	Electron microscopes		
621.385.833.4	Diffraction cameras		
621.385.834	Tuning-indicator tubes (magic eyes) ⇒ 621.387.322.3		

621.387.322.3	Indicator tubes. □ Two-electrode types: magic eyes. Multi-electrode types: inditrons, corona-discharge tubes ⇒ 621.385.834	621.391.822.4	Intermodulation noise
621.387.322.4	Rare-gas cartridges	621.391.823	Interference due to electric power systems and apparatus. Industrial interference. Man-made interference (accidental)
621.387.33	With control electrode	621.391.825	Deliberate interference. Jamming
621.387.332	Grid-glow tubes. Trigger tubes. Cold-cathode thyratrons. Neotrons. Pulsatrons	621.391.826	Echoes and deviations
621.387.333	Cold-cathode arc-discharge tubes. Strobotrons. Trigatrons	621.391.827	Crosstalk interference. Intermodulation hum
621.387.34	Multi-electrode types. □ Stepping tubes (scaling tubes). Dekatrons	621.391.83	Fidelity. Distortion
621.387.35	With external magnetic field. Penning gauges (cold-cathode ionization gauges)	621.391.832	Distortion of continuously varying signals. □ Linear distortion. Nonlinear distortion
621.387.36	Electrodeless discharge tubes. □ Radio-frequency indicators. Transmit-receive switches (TR switches, switching tubes)	621.391.833	Distortion of discontinuously varying signals, e.g. pulses. □ Characteristic distortion. Bias distortion. Fortuitous distortion
621.387.4	Radiation counters (ionization counters). Counter tubes	621.391.837	Picture quality as affected by transmitting or receiving transducers ⇒ 621.397.3
621.387.42	Gas-filled counter tubes	621.391.837.1	Resolution. □ Ghosting
621.387.422	Ionization chambers	621.391.837.2	Geometrical distortion
621.387.424	Geiger-Müller counters (Geiger counters)	621.391.837.3	Brightness. □ Contrast
621.387.426	Neutron counters	621.391.837.4	Dynamic effects. Spurious patterns. □ Interference patterns. Jitter. Bouncing. Jumping
621.387.44	Liquid counter tubes	621.391.84	Selectivity
621.387.46	Solid counter tubes. □ Crystal counters. Scintillation counters	621.391.88	Readability. Mutilation. □ Margin, e.g. in telegraphy
621.389	Other electronic apparatus and instruments	621.391.881	Margin, e.g. telegraphy
621.39	Telecommunication. Telegraphy. Telephony. Radiocommunication. Video technology and equipment. Telecontrol ⇒ 621.3.09 ⇒ 621.315 ⇒ 654	621.391.883	Quality figures. □ Signal-to-noise ratio
621.391	General questions of electrical communication engineering. Cybernetics. Information theory. Signal theory <i>Class here only if in relation to electrical communication</i> ⇒ 007 ⇒ 519.7	621.391.883.1	Signal-to-atmospherics ratio
621.391.1	Comparison of various systems ⇒ 621.394.13 ⇒ 621.396.1	621.391.883.2	Signal-to-noise ratio
621.391.3	Transmission of electric signals by conduction ⇒ 621.315 ⇒ 621.372	621.391.883.3	Signal-to-man-made-interference ratio
621.391.5	Transmission of electric signals by induction	621.391.883.6	Signal-to-echo ratio
621.391.6	Transmission of electric signals by various waves. □ By ultraviolet waves. By visible light. By infrared waves	621.391.883.7	Signal-to-crosstalk ratio
621.391.7	Secret systems of electrical communication	621.394	Telegraphy ⇒ 654.14
621.391.8	Quality of the received signal. Signal strength. Interference	621.394.1	Generalities
621.391.81	Propagation conditions. Signal strength	621.394.12	Telegraph systems with indicating receivers
621.391.812	Variations of signal strength due to free-space phenomena. □ Due to fluctuations (fading). Due to periodic, cosmic, atmospheric influences ⇒ 621.396.666	621.394.13	Comparison of various telegraph systems and apparatus ⇒ 621.391.1
621.391.814	Variations of signal strength due to guided transmission. Changes in transmission characteristics	621.394.14	Telegraph codes. □ Binary and pulse codes
621.391.816	Variations of signal strength due to terminal and/or intermediate equipment	621.394.18	Measures against mutilation of signals
621.391.82	Interference. Static	621.394.3	Telegraph switching systems
621.391.821	Interference due to atmospherics. Natural electric interference. □ From lightning	621.394.34	Subscriber telegraph switching systems
621.391.822	Interference due to noise	621.394.4	Telegraph systems. Multiplex telegraphy ⇒ 621.395.49
621.391.822.2	Thermal noise	621.394.41	Simplex telegraphy
621.391.822.3	Shot noise. □ Partition noise. Flicker effect	621.394.42	Time-division multiplex
		621.394.43	Systems with division by direction and amplitude. □ Duplex, diplex, quadruplex telegraphy
		621.394.44	Frequency-division multiplex (carrier telegraphy)
		621.394.45	Simultaneous use of lines with other parties
		621.394.5	Telegraph transmission
		621.394.52	Direct-current telegraphy
		621.394.54	Alternating-current telegraphy
		621.394.6	Telegraph apparatus
		621.394.61	Telegraph transmitters (senders)
		621.394.614	Transmitters with typewriter keyboard.
		621.394.615	Teleprinters as transmitters
		621.394.618	With dial
		621.394.62	Automatic transmitters and associated auxiliary apparatus
		621.394.623	Telegraph receivers
		621.394.624	Acoustic receivers. □ Sounders. Headphones
		621.394.625	Visual receivers (non-recording). □ Needle type. Mirror galvanometers. ABC dials
		621.394.625.1	Recording receivers
		621.394.625.2	Recording in code (except perforating). □ Ink, photographic, electric, magnetic recording
			Recording by perforating. Punched-tape receivers

621.394.625.3	Recording by direct printing. Teleprinters as recorders. Ticker-tape machines etc.	621.395.612	Microphones with induced current. Magnetostatic microphones ⇒ 621.395.62
621.394.625.4	Receiving mechanisms. Orientation devices	621.395.613	Microphones with variable impedance. □ Liquid-resistance microphones.
621.394.627	Alarm devices responding to specific codes or code sequences. □ SOS receivers		Contact-resistance microphones. Carbon microphones. Microphones with metallic contact resistance. Push-pull microphones
621.394.63	Telegraphic signalling and calling apparatus	621.395.614	Piezoelectric microphones. Crystal microphones
621.394.64	Telegraph repeaters	621.395.616	Electrostatic microphones. Capacitor microphones (condenser microphones)
621.394.642	Telegraph repeaters with solid moving parts. □ Contact type. Thermal type	621.395.617	Semiconductor microphones
621.394.643	Liquid-jet telegraph repeaters	621.395.618	Electronic microphones
621.394.646	Regenerative telegraph repeaters	621.395.619	Other types of microphone. □ Cathodophones.
621.394.65	Telegraph distributing apparatus. Telegraph switches		Microphones with liquid jets
621.394.66	Telegraph regulating apparatus. Synchronization apparatus. □ Phonic wheel	621.395.62	Telephone receivers <i>Class information about both transmitters and receivers at 621.395.61</i>
621.394.67	Telegraph code-changing apparatus. □ Code convertors		Acoustic receivers
621.394.68	Telegraph testing apparatus	621.395.623	Generalities. Vibrating diaphragms
621.394.7	Telegraph installations and networks	621.395.623.3	Receivers according to principle of action. □ Electromagnetic receivers. Electrodynamic receivers. Electrostatic receivers. Singing arc.
621.394.72	Telegraph stations. Telegraph offices	621.395.623.4	Piezoelectric receivers. Ionic receivers
621.394.73	Telegraph lines ⇒ 621.315		Horns. Earpieces
621.394.74	Telegraph networks	621.395.623.5	Receivers according to form of support. □ Hand receivers. Headphones. Earphones
621.394.75	Telegraph repeater stations	621.395.623.6	Loudspeakers
621.394.76	Telegraph exchanges. Connection installations. Telegraph switchboards		Recording systems in or for telephone receivers. □ Answering machines. Voicemail systems
621.394.9	Applications of telegraphy <i>Details by :...</i> ⇒ 656.254	621.395.623.7	Telephone calling apparatus
621.395	Telephony ⇒ 654.15	621.395.625	Calling apparatus at the sending end. □ Call-buttons. Push-buttons
621.395.12	Telephone circuit functions. □ Hunting. Engaged testing. Switching. Metering	621.395.63	Calling apparatus at the receiving end. □ Telephone bells, buzzers, warblers. Visual telephone signals, indicator lights
621.395.2	Private telephone systems, with or without connection to the public exchange	621.395.631	Accessories for calling apparatus. □ Indicators of termination of a call. Tone tests
621.395.3	Public telephone systems	621.395.632	Devices for calling intermediate stations
621.395.31	Telephone traffic studies. Provision of circuits and switches to carry traffic. Trunking and grading ⇒ 654.153	621.395.633	Devices for selective calling. □ Long-distance selection by the subscriber
621.395.32	Systems without exchange	621.395.634	Calling apparatus for automatic systems. □ Call selecting arrangements: Dials
621.395.33	Systems with manual exchange	621.395.635	Telephone repeaters ⇒ 621.395.72
621.395.34	Systems with automatic exchange	621.395.636	Telephone repeaters with solid moving parts. □ Repeater transmission. Transmission by rotary mechanism
621.395.345	Electronic switching. Electronic systems ⇒ 621.318.57	621.395.643	Liquid-jet telephone repeaters
621.395.347	Connections between automatic exchanges	621.395.65	Telephone switching apparatus. Telephone switchboards
621.395.348	Special connections between exchanges and subscribers	621.395.652	Press-button switches
621.395.348.4	Conference facilities	621.395.653	Switches with sliding contact. Knife switches
621.395.348.5	Connections for switching during subscriber's absence	621.395.654	Plug-and-bar switches
621.395.35	Semiautomatic telephony	621.395.655	Jack-and-plug switches. Key switches
621.395.36	Metering methods ⇒ 621.395.663	621.395.658	Automatic switches
621.395.37	Trunk systems. Toll systems	621.395.66	Apparatus for control, regulation, protection ⇒ 621.3.07
621.395.38	Signalling and pulsing	621.395.661	Capacitors. Coils. Transformers. Artificial lines (simulated lines)
621.395.4	Multiple telephony	621.395.663	Call counters (call meters). Prepayment apparatus ⇒ 621.395.36 ⇒ 681.118.1
621.395.42	Duplex telephony		Time-zone meters
621.395.43	Multiplex telephony. Phantom circuits	621.395.663.4	Prepayment apparatus (on payphones). □ Coin-slot mechanisms (slot machines) on telephones. Coin-operated telephones. Card payment telephones
621.395.44	Carrier telephony	621.395.663.6	Supervisory equipment. Regulating apparatus
621.395.49	Simultaneous telephony and telegraphy ⇒ 621.394.4		Voice-operated
621.395.5	Long-distance telephony by cable		
621.395.6	Telephone apparatus		
621.395.61	Telephone transmitters. Microphones <i>Class here information about both transmitters and receivers</i>		
621.395.611	Microphones according to form of support or specialized application. Microphones according to directional properties	621.395.664	
		621.395.664.1	

621.395.664.12	Echo suppressors	621.396.61	Radio transmitters
621.395.664.13	Reaction suppressors		<i>Class here information about both transmitters and receivers</i>
621.395.664.2	Band-width regulators	621.396.62	Radio receivers
621.395.664.3	Amplitude limiters. Amplitude filters		<i>Class information about both transmitters and receivers at 621.396.61</i>
621.395.665	Telephone volume control. ☐ Contrast expanders or compressors (companders)	621.396.621	Receiver circuits in general. ☐ Connections
621.395.666	Time switches to limit call duration. Prevention of re-operation. Delayed release	621.396.623	Radio receivers with acoustic output
621.395.667	Telephone tone control		621.396.623 $\equiv$ 621.395.623
621.395.669	Protective devices. ☐ Coherers. Telephone interference suppressors	621.396.624	Radio receivers with optical output
	$\Rightarrow$ 621.316.9	621.396.625	Recording radio receivers
621.395.7	Telephone installations	621.396.63	Radio calling apparatus
621.395.72	Telephone stations		621.396.63 $\equiv$ 621.395.63
621.395.721	Simple telephone stations	621.396.65	Apparatus for connecting radio systems with wired communication systems
621.395.721.1	Subscribers' installations in general		$\Rightarrow$ 621.372
621.395.721.2	Call boxes		$\Rightarrow$ 621.396.44
621.395.721.3	Wall telephones	621.396.66	Apparatus for control, regulation, protection
621.395.721.4	Table telephones		$\Rightarrow$ 621.3.07
621.395.721.5	Portable telephones. Mobile phones.	621.396.662	Radio tuning and regulating apparatus
	☐ Cellular phones	621.396.663	Radiogoniometers. Radiocompasses
	$\diamond$ 621.395.721.5:004.77 WAP (Wireless Application Protocol) phones	621.396.664	Supervisory, monitoring apparatus
621.395.721.7	Auxiliary equipment. Provision for writing. Directories. Arm supports	621.396.665	Radio volume control. Automatic gain control
621.395.722	Exchanges	621.396.666	Antifading devices. Regulation of amplification related to carrier wave
621.395.724	Repeater stations		$\Rightarrow$ 621.391.812
	$\Rightarrow$ 621.395.64	621.396.667	Radio tone control
621.395.73	Telephone lines	621.396.669	Protective devices and circuits. Radio interference suppressors
	$\Rightarrow$ 621.315		$\Rightarrow$ 621.316.9
	$\Rightarrow$ 621.395.5	621.396.67	Aerials. Radio antennas. Earth systems
621.395.74	Telephone networks. Grouping of lines and stations		$\Rightarrow$ 621.315.24
621.395.741	Long-distance cable networks		$\Rightarrow$ 624.97
621.395.742	Long-distance overhead lines	621.396.673	Unbalanced aerials. Aerials with earth or counterpoise
621.395.743	Local networks	621.396.674	Balanced aerials. Dipoles and frame aerials (loop aerials) without intentional directivity
621.395.9	Applications of telephony		Frame aerials. Loop aerials
	$\Rightarrow$ 654.1	621.396.674.1	$\Rightarrow$ 621.396.677.5
	$\Rightarrow$ 656.254	621.396.674.3	Dipoles
	$\Rightarrow$ 658.842	621.396.676	Special aerials for land vehicles, watercraft, aircraft
621.395.91	Automatic information services by telephone.		Directive aerials. Aerials with intentional directivity
	☐ Time check (speaking clock). Weather reports	621.396.677	Beam aerials. Aerial arrays
621.395.92	Hearing aids	621.396.677.3	Long-wire aerials (harmonic aerials). Wave aerials. ☐ Beverage aerials. Helical aerials
621.395.97	Broadcasting by telephone (programme transmission over lines)	621.396.677.4	Frame aerials (loop aerials) with intentional directivity. Direction-finding loops
621.396	Radiocommunication apparatus and methods (radio)		$\Rightarrow$ 621.396.674.1
	$\diamond$ 621.396.029 Frequency and wavelength ranges for radiocommunication	621.396.677.6	Adcock aerials
	$\Rightarrow$ 621.37	621.396.677.7	Waveguide and cavity radiators
	$\Rightarrow$ 654.177.19	621.396.677.8	Parasitic aerials. Secondary elements.
621.396.1	Generalities. Early experiments, e.g. by H R Hertz. Comparison of various radio systems		☐ Reflectors and directors. Mirrors. Lenses.
621.396.2	Transmission systems according to type of wave	621.396.678	Radomes (aerial housings)
621.396.21	Systems using damped waves or modulated waves. Telephonic radio systems	621.396.679	Improvised, makeshift aerials
621.396.22	Systems using continuous waves	621.396.69	Earth plates. Counterpoises
621.396.24	Systems using short waves		Various apparatus for radiocommunication.
621.396.25	Systems using long waves		Accessory apparatus in general
621.396.4	Multiple radio transmission. Directional radio transmission		$\Rightarrow$ 621.3.032.8
621.396.41	Duplex, diplex, multiplex radio transmission		$\Rightarrow$ 621.314.2
621.396.42	Transmission by switching the frequency		$\Rightarrow$ 621.316.58
621.396.43	Transmission by directed waves		$\Rightarrow$ 621.316.8
621.396.44	Transmission by radio-frequency waves using existing conductor (provided for other purposes). Wired radio	621.396.7	$\Rightarrow$ 621.318.4
	$\Rightarrow$ 621.3.052.7		$\Rightarrow$ 621.319.4
	$\Rightarrow$ 621.372		$\Rightarrow$ 621.385
	$\Rightarrow$ 621.396.65		Radio stations. Transmitting and receiving stations
621.396.6	Radiocommunication apparatus and circuits		$\diamond$ 621.396.7-022.51 Small radio stations
			$\diamond$ 621.396.7-022.56 Large stations.
			Broadcasting stations
			$\diamond$ 621.396.7-027.562 Amateur radio stations
			$\Rightarrow$ 654.191

621.396.73	Portable radio equipment	621.397.132.121	NTSC system (US National Television Systems Committee)
621.396.74	Radio broadcasting networks	621.397.132.122	Frequency multiplex system (FAM)
621.396.75	Radiogoniometric stations. Direction-finding radio stations	621.397.132.125	PAL (Phase Alternating Line) system
621.396.9	Special radio apparatus and methods. Radar	621.397.132.127	With line-sequential transmission. SECAM (Sequential And Memory) system
621.396.91	Radio information services. □ Time signal. Weather reports	621.397.132.129	Other simultaneous systems with single transmission channel. □ Valensi. Philips. Gargini
621.396.92	Bearing signals for explorers	621.397.132.2	Sequential colour systems. □ Field-sequential, line-sequential, dot-sequential
621.396.93	Radiocommunication with land vehicles, watercraft, aircraft. □ Bearing signals. Radio beacons. Ground control	621.397.132.5	System conversion
621.396.931	Radiocommunication with land vehicles	621.397.2	Transmission types
621.396.932	Radiocommunication with watercraft	621.397.22	Modulation methods
621.396.933	Radiocommunication with aircraft	621.397.23	Transmission methods
621.396.934	Radiocommunication with projectiles	621.397.27	Conversion to other television standards ⇒ 621.397.63
621.396.94	Radiocommunication through free space, through interplanetary space, through various media	621.397.3	Analysis and reproduction of image. Scanning
621.396.944	Submarine radiocommunication	621.397.31	Simultaneous analysis of all picture elements
621.396.945	Subterranean radiocommunication	621.397.33	Successive analysis of picture elements
621.396.946	Radiocommunication through free space. Radiocommunication through interplanetary space ⇒ 621.371.36	621.397.331	Image conversion and reconstitution. Picture signal generation ⇒ 621.391.837
621.396.949	Radiocommunication through other media <i>Specify medium by :...</i>	621.397.331.1	With mechanical control of light beam. Optomechanical methods. □ Nipkow discs. Mirror drums. Lensed discs. Prisms. Drum and traversing spots
621.396.96	Radar (radiolocation)	621.397.331.2	With electric control of light beam. Optoelectronic methods. Photoelectric image analysis
621.396.962	Radar systems according to type of wave	621.397.331.21	Using camera tubes with high velocity scanning beam. □ Iconoscopes
621.396.962.2	Continuous-wave radar	621.397.331.22	Using camera tubes with low-velocity scanning beam. □ Orthicons.
621.396.962.3	Radar by pulse-modulated wave. □ Pulse-Doppler. Transponders	621.397.331.23	Cathode-potential-stabilized emitrons
621.396.963	Methods of radar information display. Automatic response	621.397.331.24	Using image dissectors. □ Farnsworth cameras Using cathode-ray tubes ⇒ 621.385.832
621.396.963.3	Optical display. Cathode-ray tube display	621.397.331.26	Projection systems with electrical control of the light beam
621.396.963.4	Meter display	621.397.331.3	Intermediate film methods
621.396.963.5	Acoustic, audio indication	621.397.331.5	Nonoptical methods. □ Electromechanical, electrochemical, electrothermal methods
621.396.963.8	Automatic response systems ⇒ 627.712.63	621.397.332	Monochrome scanning techniques
621.396.965	Aerial scanning pattern	621.397.332.1	Linear scanning techniques: sequential (line-by-line), interlaced
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621.396.969	Applications of radar	621.397.4	Video equipment <i>Details by :621.397.131/.132</i>
621.396.969.1	Ascertaining range, azimuth (bearing), elevation (altitude), velocity	621.397.42	Video scanning and picture signal generation equipment. Video cameras ⇒ 681.772.7
621.396.969.3	Detection and location of objects (vehicles, craft, projectiles, obstacles etc.)	621.397.422	Stand still cameras
621.396.97	Sound broadcasting (technology) <i>For nontechnical aspects use 654.195</i> ⇒ 621.396.7	621.397.424	Motion picture cameras. Television cameras
621.397	Video technology. Television engineering. Video recording, transmission and reproduction. Video equipment and networks	621.397.43	Video recording and reproduction equipment <i>Class here registration equipment for recording and building up video programmes, e.g. Ampex</i>
621.397.12	Facsimile transmission. Phototelegraphy. Radiophoto	621.397.44	Video broadcasting equipment
621.397.121	Monochrome facsimile (black-and-white facsimile). Transmission of single monochrome pictures	621.397.442	Video transmitters
621.397.122	Colour facsimile. Transmission of single polychrome pictures 621.397.122 ≡ 621.397.132	621.397.444	Ancillary video transmission equipment. □ Transmission antennas. Transmission satellites <i>Class here equipment for wireless and cable transmission</i>
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621.397.131	Monochrome television (black-and-white television)		
621.397.132	Colour television		
621.397.132.1	Simultaneous colour systems		
621.397.132.11	With separate transmission channels for the primary colour signals		
621.397.132.12	With common transmission channel for the primary colour signals		

621.397.446	Video receivers. □ Television receivers	621.43.013.42	Pressure atomization by internal pressure. Jet atomization
621.397.45	Video replay equipment	621.43.013.43	Pressure atomization by external pressure. Injection
621.397.452	Video recorders (tape and disc)	621.43.013.6	Swirl. Turbulence
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621.397.46	Video display equipment	621.43.016	Matters relating to heat theory, thermodynamics. □ Temperatures. Heat transfer. Heat loss
621.397.48	Video border equipment. Peritelevision equipment. □ Television games. Viewdata (teletext) apparatus	621.43.018	Efficiency. Power
621.397.6	Video conversion equipment	621.43.018.2	Mechanical efficiency
621.397.63	Equipment for conversion to other television standards or other video recording systems ⇒ 621.397.27	621.43.018.3	Thermal efficiency from point of view of fuel consumption
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621.397.7	Video (facsimile and television) installations and networks ◇ 621.397.7-181.2 Large installations ◇ 621.397.7-182.3 Mobile installations ◇ 621.397.7-182.4 Portable installations	621.43.018.6	Attainable speed
621.397.74	Video networks	621.43.018.7	Power output. □ Units. Brake horsepower. Altitude performance. Indicated horsepower
621.397.742	Facsimile networks. □ Faxpost	621.43.018.8	Devices for measuring efficiency, power output. □ Brake dynamometers. Indicators ⇒ 531.78
621.397.743	Television networks. □ Cable television	621.43.019	Chemical and physicochemical principles. Combustion
621.398	Telecontrol (remote control). Telemetering. Remote indication ◇ 621.398:621.316 Remote control of power networks and switching ◇ 621.398:623.36 Remote mine-seeking devices ◇ 621.398:654.924 Remote alarm systems ◇ 621.398:654.93 Devices for teletransmission of orders, signals ◇ 621.398:656.1.05 Devices for remote road-traffic regulation. Remote control of traffic signals ◇ 621.398:656.257 Remote control of rail signals and points ◇ 621.398:681.116 Remote control of clock systems ⇒ 654.9	621.43.019.2	The explosive mixture. □ Combustibility. Air requirement. Limits of reaction. Coefficients
		621.43.019.4	Explosion (detonation) process
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621.565.93	Open heat exchangers. □ Trickling coolers. Spray coolers. Mixing coolers	621.6.022	Moist air
621.565.94	Closed heat exchangers. □ Moving coolers. Air coolers	621.6.023	Steam. Water vapour
621.565.95	Heat exchangers according to type of pipes and radiation source	621.6.027	Corrosive gases
621.565.952	Pipe heat exchangers. □ Spiral exchangers. Helical exchangers	621.6.028	Flammable, combustible gases
621.565.954	Plate heat exchangers. □ Spiral exchangers. Helical exchangers	621.6.03	Liquids
621.565.956	Block heat exchangers	621.6.031	Water
621.565.958	Exchangers with storage medium	621.6.032	Gassy, effervescent liquids (beer, mineral waters etc.)
621.57	Refrigerating machinery. Heat pumps <i>Details by .0... (from 621.5.0...) and -1/-9 (from 62-1/-8 and 66-9)</i> ◇ 621.574.013-932.2 Continuous-running absorption refrigerating machines ◇ 621.577.012-837 Electric-motor compression heat pumps	621.6.033	Slightly viscous liquids. Oils
621.572	Historically important refrigerating plant. □ Combination of regulator and compressor	621.6.034	Very viscous, syrupy liquids (tar, molasses etc.)
621.573	Refrigerating machines without liquid cooling medium. □ Air refrigerating machines	621.6.035	Liquids containing solids in suspension. Slurries
621.574	Refrigerating machines with liquid cooling medium. Refrigerating machines using vapour	621.6.036	Liquefied gases (liquefied air etc.)
621.576	Special designs of refrigerating plant. □ Rotary refrigerating machines (Audiffren system). Compression-absorption machines. Vacuum machines	621.6.037	Corrosive liquids
621.577	Heat pumps ⇒ 662.99	621.6.038	Flammable liquids
621.577.2	Using geothermal energy	621.6.04	Loose solids. □ Particles. Dusts. Powders. Grains. Granules. Bulk material
621.577.4	Using hydrothermal energy	621.6.05	Methods, installations and equipment for fluids handling
621.577.6	Using aerothermal energy	621.6.052	Pumping installations. Methods of pumping ◇ 621.6.052-932.2 Continuous pumping ◇ 621.6.052-932.4 Intermittent pumping ⇒ 621.65/.69
621.578	Additional plant. Auxiliary devices. Devices for simultaneous generation of cold and power	621.6.053	Main storage and distribution installations
621.58	Ice production. Use of low temperatures for solidifying liquids ⇒ 663.67	621.6.057	Filling, emptying, relief and overflow devices. Inlets. Outlets. Ventilation and evacuation devices ⇒ 621.646 ⇒ 621.648
621.59	Apparatus for generating extremely low temperatures. Cryogenic equipment. Liquefaction of gases (and related processes) ⇒ 661.91	621.6.057.2	Filling devices. Inlet orifices
621.592	Liquefaction of gases and gas mixtures that are not readily condensable	621.6.057.4	Emptying devices. Outlet orifices
621.593	Separation and fractionation of liquefied gas mixtures	621.6.057.6	Relief and overflow devices
621.594	Use of liquefied and solidified gases. □ Use of dry ice ⇒ 661.97	621.6.057.8	Ventilation devices. Evacuation devices
621.6	Fluids handling, storage and distribution plant and techniques <i>Details of machines, tools, equipment, processes etc. by -1/-9 (from 62-1/-8 and 66-9)</i> ⇒ 621.867.7/.8 ⇒ 622.69	621.6.058	Heating
		621.6.06	Fittings: types and forms ⇒ 621.643.4
		621.6.061	Straight connectors
		621.6.062	Curved connectors. Bends. Elbow pieces
		621.6.063	Junction pieces. □ T-pieces. Cross-pieces
		621.6.064	Bifurcated connectors. Y-pieces
		621.6.065	Adaptors. Reducers
		621.6.066	Pipe clips (saddle clips)
		621.6.067	Terminal plates. X-pieces (blank flanges)
		621.6.07	Design, construction, testing of fluids handling installations <i>Specify details by colon combination</i> ◇ 621.6.07:620.165.2 Tightness testing ◇ 621.6.07:620.197.6 Protective coating
		621.61	Blowers and blowing machines in general. Simple, reciprocating, rotary and jet blowers ⇒ 621.51 ⇒ 669.162.223
		621.611	Simple blowers
		621.611.1	Moist-air blowers. □ Bell blowers. Drum blowers. Water-barrel blowers
		621.611.2	Dry-air blowers. Bellows
		621.612	Reciprocating piston blowers in general
		621.613	Special designs of reciprocating blower. Compound blowers
		621.614	Blowers with rotary impellers, vanes
		621.615	Centrifugal blowers
		621.617	Blowers using jets or entrainment. Jet blowers. □ Liquid-jet blowers

621.63	Rotary air-movers. Fans. Air extractors	621.643.41	Tubular connections. Flexible connections
621.631	Rotary air-movers in general. Simple fans. □ Fans with wire cage	621.643.411	Permanent connections. □ Riveted, soldered, welded connections
621.634	Helical rotary fans. Axial-flow fans	621.643.412/.418	Disconnectable pipe joints
621.635	Centrifugal fans. Turbofans	621.643.412	Disconnectable flanged connections
621.636	Cross-flow fans. Tangential fans	621.643.412.8	Loose flanges
621.64	Fluids storage and distribution installations.	621.643.413	Socket connections. Spigot-and-socket joints
	□ Containers. Conduits. Pipes	621.643.414	Threaded connections. Screwed pipe joints
621.641	Fluid collection and storage in general ⇒ 628.13 ⇒ 66.061	621.643.415	Rapid, quick action connectors. □ Catch, hook, claw devices. Bayonet couplings ⇒ 621.889.8
621.642	Fluids storage vessels. □ Containers. Receptacles. Tanks. Reservoirs	621.643.416	Automatic couplings
621.642.1	Portable and movable vessels ⇒ 621.798.1 ⇒ 641.542	621.643.418	Articulated connections
621.642.11	Cans. Pots	621.643.43	Compensation devices for heat, thermal expansion. Pipe expansion joints
621.642.12	Jugs. Pitchers. Jars	621.643.5	Tapping points. Draw-off devices
621.642.14	Pails. Buckets	621.643.51	Standpipes. Hydrants ⇒ 614.843.1
621.642.15	Tubs	621.643.52	Fire hydrants ⇒ 614.843.1
621.642.16	Casks. Kegs. Barrels	621.643.53	Drinking fountains, troughs etc.
621.642.17	Bottles. Flasks. Cylinders ⇒ 666.171	621.643.54	Water cranes (e.g. by railways) ⇒ 629.4.082.4 ⇒ 656.065.5
621.642.2	Vessels and containers mounted on vehicles and craft ⇒ 629.356 ⇒ 629.463.3	621.643.8	Losses, leakages etc. in pipes, and their prevention
621.642.3	Stationary, fixed vessels and containers ⇒ 624.953 ⇒ 628.13	621.644	Pipelines and their construction ⇒ 628.147 ⇒ 656.56
621.642.31	Reservoirs open to the air	621.645	Fluids handling and distribution terminals ◇ 621.645.058 Heating at terminals
621.642.32	Tubs. Vats	621.646	Devices for opening and closing pipes. Valves. Flow regulating devices. Safety devices ⇒ 673.6
621.642.33	Casks. Barrels	621.646.2	Valves
621.642.34	Cisterns. Tanks	621.646.22	Screw valves
621.642.37	Subterranean reservoirs	621.646.23	Multiway valves
621.642.39	High-rise reservoirs. Raised storage tanks (for liquids other than water) <i>For water towers see 624.953, 628.134</i>	621.646.24	Automatic, self-acting valves. □ Check valves. Nonreturn valves
621.642.8	Storage losses and their prevention. □ Seepage. Leakage. Evaporation	621.646.25	Disc and flap valves. Choke valves (throttle valves). Butterfly valves
621.643	Fluids channelling and distribution devices. Conduits. Pipes <i>Specify conduits according to material by -03...</i>	621.646.28	Safety valves in general ⇒ 621.183.3 ⇒ 621.646.4
	◇ 621.643.1-033.3 Concrete open conduits. Concrete channels, gutters	621.646.3	Flow regulators. Consumption regulators
	◇ 621.643.2-033.3 Concrete closed conduits. Concrete pipes	621.646.38	Diaphragm flow regulators
	◇ 621.643.2-033.5 Glass pipes. Vitreous pipes	621.646.4	Pressure regulators. Pressure-reducing valves. High-pressure safety valves
	◇ 621.643.2-033.6 Ceramic pipes. Porcelain, earthenware pipes etc.	621.646.42	Diaphragm pressure regulators
	◇ 621.643.2-034.13 Cast iron pipes	621.646.43	Bell pressure regulators
	◇ 621.643.2-034.14 Steel pipes	621.646.44	Float pressure regulators
	◇ 621.643.2-034.2/.8 Non-ferrous metal pipes	621.646.49	Safety valves for very high pressures
	◇ 621.643.2-034.3 Copper pipes. Pipes of copper alloys, brass, bronze	621.646.5	Cut-off (shut-off) valves in general. □ Slide valves. Gate valves. Sluice valves
	◇ 621.643.3-036.4 Rubber hose	621.646.6	Taps. Cocks. Faucets
	◇ 621.643.3-036.5 Plastics hose ⇒ 621.774	621.646.7	Flow mixing devices
621.643.1	Open conduits. Channels. Gutters	621.646.72	Mixer taps. □ For blending hot and cold water
621.643.2	Closed conduits. Pipes. Pipelines. Tubes ⇒ 666.173	621.646.73	Mixer valves
621.643.3	Flexible pipes and tubes. Hose	621.646.74	Mixer slides (slide valves)
621.643.4	Pipe and hose connections. Compensation devices ◇ 621.643.4-762 Pressure-tight joints and jointing. Packings, sealants for pipe connections ⇒ 621.6.066 ⇒ 621.885	621.646.76	Mixers with flow meters
		621.646.8	Automatic, self-acting cut-off devices. Isolating valves
		621.646.9	Accessories for pipes, valves, cocks etc.
		621.646.92	Housings
		621.646.93	Seatings. Packings
		621.646.94	Devices for attenuating energy. Shock-absorbing, shock-reducing devices. Suppressors for water hammer, valve chatter
		621.646.95	Indicators and alarms for bursts, fractures, pressure irregularities. Leak detectors
		621.646.96	Water discharge devices. Air discharge devices

621.646.97	Devices for protection against freezing and blocking of pipes ⇒ 624.143	621.68	Various other pumps. □ Elastic-walled pumps. Peristaltic pumps. Special-purpose pumps. Measuring pumps (metering pumps)
621.646.98	Various other shut-off devices. □ Automatic closures permitting removal of tapping device. Key, bolt, locking closures	621.69	Devices for transporting, drawing and flow-reversing of liquids by direct action or by entrainment in a fluid
621.647	Spraying, sprinkling, flushing devices ⇒ 66.069.83	621.691	Devices for raising liquids by means of gas or compressed air in general. □ Monte-jus
621.647.2	Droplet-forming parts. Atomizers	621.692	Devices using pressure of gas other than air. □ Devices using compressed carbon dioxide
621.647.3	Spray and jet-forming devices. Nozzles ⇒ 614.843.4	621.693	Pulsometers
621.647.4	Rotary sprinklers ⇒ 614.844	621.694	Devices using entrainment in a fluid jet. Jet pumps ⇒ 621.527
621.647.7	Flushing devices (except containers and pipelines). Devices for intermittent swilling	621.694.2	Steam jet pumps. Air jet pumps. Gas jet pumps. Injectors. Ejectors. Jet condensers
621.647.9	Other devices. □ Floats	621.694.3	Liquid jet pumps. □ Water jet pumps ⇒ 621.647.3
621.648	Various draining and overflow devices	621.695	Raising of liquids by mixing with compressed air. Air-lift pumps
621.648.2	Draining devices for various vessels. □ Wastes for sinks, basins, baths	621.7	Mechanical technology in general: processes, tools, machines, equipment <i>Details of machines, tools, equipment, processes etc. by -1/-9 from 62-1/-8 and 66-9</i> ⇒ 621.9 ⇒ 658.2 ⇒ 658.5 ⇒ 669.056
621.648.3	Mechanisms for operating bottom drain valves in vessels of all kinds	621.7.01	Principles and theory of working and treatment of workpieces. Workability. Factors influencing working. Conditions. Faulty working .01 and its subdivisions are also applicable in 621.9
621.648.4	Overflow devices. □ For sinks, basins, baths	621.7.011	Workability. Formability. Stresses, flow etc. <i>Details by :539...</i>
621.648.5	Inserts. Filters	621.7.012	Working movement, stroke, travel. Effect on working etc. .012 is not applicable in 621.745 and 621.78
621.649	Devices for transporting, drawing and flow-reversing of liquids (except pumps and entrainment devices). Water-raising mechanisms	621.7.012.2	Height of drop. □ Of forge hammers
621.649.1	Scoops. Balers. Pivoted and counterpoised buckets (shadufs). Valved troughs etc.	621.7.012.3	Length of stroke, travel or traverse
621.649.2	Draw wells (bucket-and-rope devices). Swivel pails	621.7.012.4	Width, breadth of stroke
621.649.4	Norias. Bucket chains. Endless-chain elevators ⇒ 621.867.3	621.7.012.5	Depth of stroke. □ Shallow working. Deep working
621.649.5	Screw mechanisms. Archimedean screws	621.7.013	Angle, rake of tool. Direction of working (with or across the grain) .013 and its subdivision are not applicable in 621.78
621.649.6	Wheel elevators. □ Bucket wheels. Scoop wheels (with fixed or swinging scoops). Elevators with paddle-wheel drive	621.7.013.2	Angular set, inclination of tool
621.649.7	Siphon devices. Siphon pumps	621.7.013.3	Rake of tool: top-rake, side-rake
621.65	Pumps and pumping in general ◇ 621.65.05 Pumping methods, installations and equipment ◇ 621.65.058 Pumping with heating	621.7.013.4	Clearance angle. □ Positive, negative stroke or cut
621.651	Force pumps in general. Pumps with plungers or reciprocating pistons	621.7.013.5	Working along the grain
621.652	Pumps with disc pistons	621.7.013.6	Working across the grain
621.653	Pumps with plungers	621.7.014	Efficiency of tool or machine. Influencing factors, power required .014 and its subdivisions are not applicable in 621.78
621.654	Piston pumps with oscillating cylinder	621.7.014.2	Resistance of material. □ Frictional resistance
621.655	Pumps according to fluid distribution system. Valved and valveless pumps	621.7.014.3	Pressure applied to tool
621.656	Piston pumps without flywheels. Simplex, duplex and multiplex pumps	621.7.014.4	Penetration of tool
621.658	Diaphragm pumps	621.7.014.5	Rate of working. Speed of tool or machine
621.659	Combined reciprocating and rotary piston pumps	621.7.014.8	Formation of chips, flakes, flour, dust, other waste products ⇒ 621.775.8
621.66	Rotary and semirotary impeller machines and driven parts	621.7.015	Surface finish. Resultant condition of workpiece ⇒ 62-408 ⇒ 621.7.08 ⇒ 621.78.015
621.662	Rotary machines with simple, uniform impeller rotation. Vane pumps		
621.663	Rotary impeller machines with variable speed or intermittent rotation		
621.664	Machines with double or multiple intermeshing rotary impellers. Gear pumps		
621.665	Other rotary impeller machines. Screw pumps		
621.666	Machines with semirotary or oscillating impellers		
621.668	Rotary pumps with drums or scoop wheels		
621.67	Centrifugal pumps. Turbopumps		
621.671	Centrifugal pumps. Radial-flow pumps		
621.674	Helical rotary pumps (screw-propeller pumps). Axial flow pumps		
621.675	Turbopumps		

621.7.016	Working conditions. Ambient influences (atmosphere etc.)	621.7.043	Die forming
621.7.016.2	Hot working Specify temperature range by -97... from 66-97	621.7.044	High-energy-rate forming (HERF). Explosion forming etc.
621.7.016.3	Cold working Specify temperature range by -97... from 66-97	621.7.044.2	High-explosive forming and working
621.7.016.4	Dry working	621.7.044.3	Expanding-gas forming (pneumatic forming). Gas as forming medium
621.7.016.5	Wet working	621.7.044.4	Hydro-spark forming (electrohydraulic forming). Water or other liquid as forming medium
621.7.016.6	Working in protective or reactive ambient media. Working under controlled conditions 621.7.016.6 $\equiv$ 621.78.06	621.7.044.5	High-pressure forming (with plunger or other mechanical former)
621.7.018	Working allowances. Machining allowances	621.7.044.7	Electromagnetic forming and working
621.7.019	Faulty working. Working flaws, defects etc. and their testing Specify flaw details by :620.19... ◇ 621.7.019:620.191 Surface defects from faulty working. Roughness, pitting, discoloration etc. ◇ 621.746.019 Faulty casting and castings	621.7.06	Machinery: assemblies, main components, auxiliary units .06 is not applicable in 621.78, but is also applicable in 621.9
621.7.019.1	Cracks ⇒ 620.191.33	621.7.067	Transfer machines (units for handling stock between treatments) ⇒ 621.86/.87
621.7.019.2	Cavities ⇒ 620.192.3 ⇒ 620.192.4	621.7.068	Holding or disposing machines and units (between treatments). □ Coilers
621.7.019.3	Segregations. Inclusions ⇒ 620.192.4	621.7.07	Forming appliances, tools and accessories .07 and its subdivisions are also applicable in 621.9 ⇒ 621.715
621.7.019.4	Lack of fusion	621.7.071	Marking of work-lines and points
621.7.019.5	Imperfect shape. Distortion ⇒ 620.191.31 ⇒ 620.191.38	621.7.072	Master copies, models. Templates. Patterns and pattern-making ◇ 621.744.072 Foundry patterns (moulding patterns). Strickles, strickling gear ◇ 621.941.072 Copies and templates for lathework, turning
621.7.019.7	Surface defects	621.7.073	Form-reproducing, pattern-reproducing appliances. □ Dies. Permanent moulds ◇ 621.746.073 Permanent moulds for die-casting ◇ 621.983.073 Dies for cold sheet-working and deep drawing
621.7.019.73	Decarbonization	621.7.074	Ancillary forming appliances. □ Support and reinforcement formers etc. ◇ 621.98.074 Sheet-working formers and blocks
621.7.019.74	Exfoliation. Scaling	621.7.075	Form tools for plastic working and finishing of workpieces ◇ 621.981.075 Sheet-bending tools. Including: Machine tools
621.7.02	Preparation, cleaning and coating of workpiece for treatment. Pretreatment ⇒ 621.7.09 ⇒ 621.795	621.7.077	Manipulating, gripping, securing and other accessories. □ Tongs. Clamps ⇒ 621.881
621.7.022	Dry cleaning. Descaling by mechanical or physical means	621.7.079	Accessory materials for working and treatment. □ Lubricants. Pastes. Protective substances ⇒ 621.7.029
621.7.022.2	Brush cleaning. Wiping. Rubbing down	621.7.08	Measurement and testing generally .08 is not applicable in 621.78, but is also applicable in 621.9 ⇒ 53.08 ⇒ 531.7 ⇒ 621.713/.715
621.7.022.4	Flame priming. Burning off	621.7.09	Aftertreatment. Finishing etc. ⇒ 621.717
621.7.022.6	Ultrasonic cleaning	621.7.092	Aftertreatment 621.7.092 $\equiv$ 621.7.02 ⇒ 621.747 ⇒ 621.9.047/.048
621.7.023	Erosion cleaning. Blast cleaning. □ Sand-blasting. Shot-blasting ⇒ 621.547	621.71	Tolerances. Tool making. Assembling
621.7.024	Wet cleaning. Washing. Wet degreasing	621.713	Tolerances. Fits. Gauges
621.7.024.2	Spraying. Sluicing. Jetting with water etc.	621.713.1	Tolerances. Working allowances
621.7.024.4	Treatment with detergents, soap	621.713.2	Limits and fits
621.7.024.6	Treatment with solvents	621.713.3	Gauges. Gauge making in general ⇒ 531.718 ⇒ 681.2
621.7.025	Pickling. Chemical descaling. Chemical degreasing ⇒ 621.794 ⇒ 621.9.047		
621.7.025.2	In alkaline bath		
621.7.025.3	In acid bath		
621.7.025.4	In neutral or other kinds of bath		
621.7.025.6	Neutralizing		
621.7.025.7	Rinsing		
621.7.027	Electrolytic cleaning, derusting etc. ⇒ 621.357		
621.7.029	Coating and other preparation for working or treatment ⇒ 621.357.7 ⇒ 621.7.079 ⇒ 621.793/.795		
621.7.04	Treatment, working and forming techniques .04 and its subdivisions are not applicable in 621.78 and 621.791		

621.715	Tool making. Jig making ⇒ 62-229 ⇒ 621.7.07 ⇒ 621.9.02	621.742.45	Fillers and special additions (except binders and cements). □ Blacking (coal dust). Straw. Hay. Dung
621.715.2	Tools. Tool making in general	621.742.48	Binders. Cements ⇒ 666.9
621.715.4	Jigs. Jig making in general	621.742.5	Treatment of moulding sands, loams etc. □ Drying. Milling. Screening. Mixing ⇒ 621.926/928 ⇒ 66.047
621.717	Assembling, mounting in general	621.742.52	Drying (by natural and artificial means) <i>Details by :66.047...</i>
621.73/77	Chipless working ⇒ 621.9	621.742.53	Milling. Crushing. Grinding <i>Details by :621.926...</i>
621.73	Forge work. Forge layout and equipment. Open and die forging ⇒ 682 ⇒ 739	621.742.55	Treatment of old sands. □ Magnetic iron separation ⇒ 621.928.8
621.73.04	Forging methods and techniques	621.742.56	Screening. Sizing. Sieving <i>Details by :621.928...</i>
621.73.041	Hand forging	621.742.57	Mixing. Blending. Kneading <i>Details by :621.929...</i>
621.73.042	Open forging (machine forging), with hammer, press or roll	621.743	Cores. Core-making ◇ 621.743.073 Core boxes ◇ 621.743.074 Core reinforcements: core irons, core bars, core barrels
621.73.043	Die forging (closed forging)	621.743.4	Cores and (built-up) core materials ⇒ 621.742.4
621.73.045	Hammer forging	621.743.45	Reinforced cores ⇒ 621.743.074
621.73.046	Press forging	621.743.5	Core operations. Striking-up, sweeping of cores
621.73.047	Roll forging	621.743.52	In a core box
621.73.06	Machines and ancillary plant	621.743.53	With strickle board or loam board
621.73.07	Appliances and ancillary equipment for forging	621.743.54	Using core bars, barrels
621.73.073	Dies	621.744	Moulding. Patterns, moulds and mould-making ◇ 621.744.072 Foundry patterns (moulding patterns). Strickles, strickling gear
621.73.074	Anvils. Mandrels. Formers. Swage blocks etc.	621.744.3	Moulds (built-up moulds). Enclosure, maintenance, renewal
621.73.075	Hammers and special forging tools	621.744.33	Flasks or moulding boxes (to enclose or hold the mould) ⇒ 621.744.54
621.73.077	Accessory appliances. □ Tongs. Clamps. Manipulators	621.744.4	Moulding machines and their operation. Machine moulding generally ⇒ 621.979.3
621.733	Forge equipment, hearths and furnaces in general ◇ 621.733:621.365 Electric forge furnaces ◇ 621.733.074 Forging anvils ⇒ 62-6 ⇒ 66.04 ⇒ 662.6/8	621.744.42	Machines requiring hand lifting or hand drawing of patterns
621.735	Forging operations and techniques	621.744.43	True moulding machines. Machines with mechanical sand packing and mechanical lift or draw
621.735.2	Bending, straightening etc.	621.744.44	Machines with packing plate (sand-packing by pressure). Squeezers
621.735.3	Operations altering the cross-section. □ Upsetting. Drawing. Swaging. Flattening	621.744.45	Jolt-squeeze machines
621.735.4	Forming shoulders	621.744.46	Machines with double-ramming action (staple casting)
621.735.5	Forming elbows, cranks etc.	621.744.47	Jolt-ram machines. Vibration machines
621.735.6	Twisting	621.744.48	Sand-slingers. Other centrifugal machines
621.735.7	Dishing. Flanging ⇒ 621.984	621.744.49	Pressure-moulding machines
621.735.8	Forge welding ⇒ 621.791.42	621.744.5	Moulding operations and techniques, especially hand moulding
621.74	Foundry work. Foundry layout and equipment. Moulding and casting ◇ 621.74.02 Preparation of stock for casting ⇒ 669 ⇒ 672.1	621.744.52	General operation. □ Packing of drag and cope. Rapping and drawing the pattern. Core and chill insertion. Cutting the gates and runners. Closing the mould or flask
621.74.04	Techniques, special methods of foundry working and casting	621.744.53	Open sand moulding with pattern. □ Core moulding ⇒ 621.743
621.74.041	Sand casting in general	621.744.54	Flask and box moulding with pattern ⇒ 621.744.3
621.74.042	Centrifugal casting	621.744.55	Bedding-in, with strickle board or loam board.
621.74.043	Die casting		Template moulding ⇒ 621.744.072
621.74.045	Investment moulding and casting. Lost wax process (precision casting)		
621.74.046	Compound casting (in one or more than one metal)		
621.74.047	Continuous casting ⇒ 621.746.27		
621.742	Moulding sands, loams, mixes and their preparation		
621.742.4	Materials		
621.742.42	Moulding sands. Composition, properties, suitability etc.		
621.742.43	Loams. Composition, properties, suitability etc.		

621.744.56	Shell moulding	621.746.556	Staple casting
621.744.58	Drying and hardening the mould	621.746.558	Pouring with stirring
621.745	Melting (smelting). Furnaces, materials, processes ⇒ 662.92	621.746.56	Removal of slag and oxides. □ Skimming. Filtering ⇒ 66.067/.068
621.745.3	Furnaces. Main types, operation and control, maintenance etc. ⇒ 66.04 ⇒ 662.6/.9	621.746.58	Methods of obtaining sounder castings, even density etc. □ Use of pressure or vacuum. Chilling. Heating. Electrical means. Shaking. Jolting
621.745.31	Preheating installations and equipment	621.746.582	Direct pressure <i>Denote type of press by :621.979...</i>
621.745.32	Crucible furnaces. Pot furnaces. □ Rotary and rocking furnaces ⇒ 621.365	621.746.583	Using compressed gas or partial vacuum
621.745.33	Reverberatory furnaces (flame furnaces). Open-hearth furnaces	621.746.585	Electrical methods
621.745.34	Shaft furnaces. Cupola furnaces	621.746.586	Heating the riser
621.745.36	Small converters. Other types of foundry melting furnace	621.746.587	Heating the mould
621.745.37	Installations for duplex processes in foundry work	621.746.588	Mechanical means. □ Shaking. Jolting
621.745.4	Materials and additions used in foundry melting. □ Metals. Fluxes. Clinker. Slag	621.746.6	Solidifying. Standing time. Cooling. Cooling pits
621.745.5	Melting (smelting) procedure, processes, products ◇ 621.745.5:669.71 Aluminium smelting	621.747	Aftertreatment of castings. Finishing
621.745.55	Melting process. Effect of slags, scoria, sulphur in fuel etc. ⇒ 66.046.5	621.747.5	Aftertreatment operations
621.745.56	Molten metal or melt. □ Tapping, melting temperatures (fusion temperatures), yield ⇒ 621.746.51	621.747.51	Stripping of castings. Separation from moulds
621.745.57	Furnace (flue or stack) gases and their composition	621.747.52	Dressing-off. Trimming. Fettling. Removal of mould marks; runners, risers, cores
621.745.58	Slags. □ Chemical composition. Proportion of total yield	621.747.53	Sand removal. □ Preliminary cleaning-off. Brushing. Wiping
621.746	Casting. Transferring and pouring the molten metal ◇ 621.746.019 Casting defects	621.747.54	Cleaning, polishing. Grinding-off projections. Peeling. Tumbling etc.
621.746.2	General arrangements for casting, transfer of molten metal etc.	621.747.55	Blast cleaning. □ Sandblasting ⇒ 621.547
621.746.21	Casting pits	621.747.56	Chemical cleaning. Pickling etc. ⇒ 621.7.025
621.746.22	Conveyance of molten metal to the moulds.	621.747.57	Burning off. Blazing off. Decarbonizing. Desurfacing ⇒ 621.7.022.4 ⇒ 621.791.94
621.746.24	Batch casting	621.747.58	Repairing of damaged castings. □ Bonding.
621.746.27	Bringing the moulds to the ladles. Series casting	621.762	Welding. Reinforcing
621.746.27	Arrangements for continuous production of castings ⇒ 621.74.047	621.762.2	Powder metallurgy ⇒ 661.6 ⇒ 661.8 ⇒ 666.3/.7 ⇒ 669-138 ⇒ 669.018.9
621.746.3	Equipment for casting and pouring	621.762.3	Production, manufacture of powder for working and treatment
621.746.31	Casting tables	621.762.3	Treatment of powder before pressing and sintering ◇ 621.762.3:66.047 Drying of powder
621.746.32	Ladles	621.762.4	Pressing of powder and of pre-pressed compacts. Powder moulding etc.
621.746.33	Slag ladles. Slag trolleys	621.762.5	Sintering of compacts ⇒ 536.421.5 ⇒ 622.785 ⇒ 66.046.44
621.746.34	Pouring basin	621.762.8	Aftertreatment of sintered compacts. □ Working, fabricating, finishing
621.746.393	Ingot moulds	621.762.82	Sizing. Coining
621.746.395	Group plates. Funnel tubes. Channel bricks etc.	621.762.86	Impregnation of porous sintered compacts
621.746.4	Runners and gates. Risers. Shrinkage heads (dead heads)	621.763	Processes akin to powder metallurgy. □ Fibre metallurgy. Working from pellets, fabrics etc. ⇒ 621.775
621.746.42	Runners and gates	621.767	Spray forming. Dispersion metallurgy etc. ⇒ 621.793.7 ⇒ 678.027.7
621.746.43	Gate closures, stoppers. □ Plugs. Pins	621.77	Rolling, extruding, drawing and other plastic forming except forging and sheet-metal working ⇒ 621.73 ⇒ 621.98 ⇒ 621.992.7
621.746.46	Risers. Shrinkage-heads or dead-heads	621.771	Rolling. Rolling mills
621.746.5	Pouring operations		
621.746.51	General considerations. □ Temperatures. Methods of introducing molten metal. Rate of pouring. Pouring time		
621.746.55	Methods of pouring		
621.746.552	Top pouring (direct, from above)		
621.746.553	Side pouring (through several superimposed passages)		
621.746.554	Bottom pouring (from below)		
621.746.555	Arcuate, horn, tangential pouring		

621.771.06	Mill design and construction. Stand and roll arrangement	621.774.3	Seamless tube manufacture. □ Forged, rolled, drawn tube
621.771.061	Single-stand mills in general. Single-pass or single-roll mills	621.774.31	Pierced, trepanned tube. Tube drilled from billets
621.771.062	Double-pass mills	621.774.32	Punched tube. Tube sheared out, e.g. with hydraulic press
621.771.063	Multi-roll or cluster mills (with 4, 6, 12, 24 rolls)	621.774.33	Tube forged on a mandrel. Swaged tube
621.771.064	Planetary mills	621.774.34	Deep-drawn tube
621.771.065	Multi-stand mills with series or sets of roll pairs (continuous mills)	621.774.35	Rolled tube ⇒ 621.771.28
621.771.067	Transfer units (for handling stock between passes or operations)	621.774.36	Pilger-rolled or step-rolled tube
621.771.068	Coilers etc. (for holding strip or rod between operations) ⇒ 621.771.6	621.774.37	Drawn tube. □ Drawbench process. Pushbench process. Cast and subsequently drawn tube
621.771.07	Design and construction of rolls	621.774.38	Extruded tube
621.771.073	Smooth rolls (sheet or strip rolls)	621.774.4	Electrolytic manufacture of seamless tube ⇒ 621.357
621.771.073.8	Calibrating the rolls	621.774.5	Compound tube and pipe manufacture, using two or more metals. □ In layers. Lined tube manufacture
621.771.074	Patterned, grooved rolls (section rolls)	621.774.6	Bending, coiling, straightening of tube or pipe
621.771.2	Rolling and rolling mills according to product formed	621.774.7	Drawing down, tapering, expanding of existing tube or pipe. Narrowing, flaring, belling the end etc.
621.771.22	Mills for slabs, billets and other semis. Blooming mills. Cogging mills	621.774.8	Surface forming of tube or pipe. □ Tube patterning, ribbing, fluting
621.771.23	Mills for smooth (unpatterned) plate, sheet or strip	621.775	Ball, shot, pellet and metal wool manufacture
621.771.24	Mills for patterned, corrugated, ribbed plate, sheet or strip	621.775.2	Ball and round shot manufacture
621.771.25	Mills for bar, rod, wire rod ⇒ 621.778.1	621.775.6	Small shot, pellet and slug manufacture ⇒ 621.763
621.771.26	Mills for large sections and profiles. □ Beams. Girders. Rails	621.775.8	Chippings, fibres and metal wool manufacture ⇒ 62-48/-494 ⇒ 621.7.014.8
621.771.28	Mills for tube, pipe ⇒ 621.774.35	621.777	Extruding (strip and sectional material) <i>Denote shapes by -41/-42 from 62-4</i> ⇒ 621.979
621.771.29	Mills for tyres, wheel rims, wheel centres and other special products. Mills for tapered, conical products (spiral mills). Spoon mills. Pointing mills. Chain mills	621.777.2	Steady-pressure extrusion
621.771.6	Mills for special roll-forming and working processes	621.777.4	Impact extrusion
621.771.63	Mills for further working of strip or band into special sections or profiles. □ Channels. Ducts. Angle sections	621.777.8	Sheathing by extrusion ⇒ 621.771.8 ⇒ 621.772.8
621.771.65	Mills for further working of rolled bar, rod or other sectional material (except strip or band), e.g. into balls, shot ⇒ 621.771.25	621.778	Drawing and working of rod, wire and similar sectional material ⇒ 621.774.37 ⇒ 672.85 ⇒ 677.53 ⇒ 679.73
621.771.67	Mills for die-forming, tool-forming. □ Roto-flow. Die-rolling. Knurling ⇒ 621.992.7	621.778.021	Pointing (to allow the end to enter the die). Positioning etc.
621.771.8	Rolling on. Cladding, plating by rolling ◇ 621.771.8:669.718 Aluminium plating by rolling	621.778.05	Joints and cuts <i>Used especially at 621.778.4, 621.778.6</i>
621.772	Boilermaking. Making of pressure vessels, reaction vessels etc. ⇒ 621.18 ⇒ 621.73 ⇒ 621.98	621.778.1	Drawing: specific operations, plant and equipment ◇ 621.778.1.06 Machinery for drawing. Drawbenches and ancillary equipment ◇ 621.778.1.073 Forming appliances for drawing. Drawing dies
621.772.1	Heavy boilermaker's work. Iron and steel boilermaking	621.778.13	Bright drawing
621.772.2	Light boilermaker's work. Coppersmith's work	621.778.14	Drawing and polishing
621.772.4	Construction and design of boiler drum, pressure vessel. □ Seamless, welded, riveted, wound vessels	621.778.16	Matt drawing
621.772.8	Sheathing, stiffening, reinforcing with sheet metal	621.778.2	Bending, twisting, coil-making, straightening of wire etc. ⇒ 621.774.6 ⇒ 621.981/982
621.774	Tube and pipe manufacture generally ⇒ 62-462 ⇒ 621.643	621.778.23	Bending
621.774.1	Cast pipe and tube manufacture generally	621.778.24	Twisting of rod or single-strand wire
621.774.2	Seamed tube manufacture. □ Welded tube. Die-drawn tube	621.778.27	Coil-making. Winding, reeling. Coiling into springs ⇒ 62-272
		621.778.28	Straightening of rod, wire etc.

621.778.4	Joining, strengthening, reinforcing of and with wire. □ Wire-stitching. Binding with wire	621.78.082	Moderately rapid cooling. □ Cooling in still air
621.778.4.052	Joints: various types 621.778.4.052 ≡ 621.791.052	621.78.083	Rapid cooling. □ Cooling in turbulent air
621.778.5	Working of wire on continuous form. □ Working into barbed wire ⇒ 677.53	621.78.084	Sudden cooling in liquids. Quenching
621.778.6	Machining, cutting, splitting of rod and wire.	621.782	General layout and arrangement of plant and equipment for heating and heat treatment
621.778.8	Special working, forging of rod and wire Working of (thin) rod and wire into articles for general use, ornaments etc. ◇ 621.778.8:621.88 Into heavy fasteners, nails, screws etc. ◇ 621.778.8:686.86 Into office fasteners, clips, staples etc.	621.783	Furnaces, baths and other devices for heating and soaking ⇒ 621.784
621.78	Heating for working. Heat treatment. Hardening <i>The 621.7.012/014, 621.7.04/06 and 621.7.08 auxiliaries are not applicable here</i> ⇒ 66-97 ⇒ 66-98 ⇒ 66.04 ⇒ 669-15	621.783.2	Furnaces
621.78.012	Methods of heating according to heat source	621.783.22	Fixed-hearth furnaces (open fixed hearth). □ Batch furnaces. Continuous furnaces.
621.78.012.2	Open fire	621.783.23	Roll-over, roll-down furnaces. Soaking pits
621.78.012.3	Indirect heating (in electric resistance furnace, solid-fuel, liquid-fuel, gaseous-fuel furnace)	621.783.24	Movable-hearth furnaces. □ Walking-beam furnaces. Car-type hearth furnaces. Rotary hearth furnaces. Drum furnaces. Tunnel furnaces
621.78.012.5	Direct heat generation in the stock. □ By electric induction. By high-frequency	621.783.4	Closed-container furnaces. □ Pot furnaces. Crucible furnaces. Muffle furnaces. Bell furnaces. Chamber furnaces
621.78.012.8	Recuperated heat (from another source)	621.784	Heating and soaking baths. □ For tempering, liquid carburizing and solution treatment
621.78.013	Methods of heating to required stock or workpiece temperature	621.784.6	Baths and other devices for cooling. Baths in general ⇒ 621.783
621.78.013.6	Heating up	621.784.8	Quenching baths. Other quenching devices
621.78.013.7	Preheating	621.785	Devices and arrangements for slow and moderately rapid cooling. □ Cooling furnaces. Cooling pits. Cooling beds
621.78.013.8	Soaking. Compensating. Maintaining constant temperature	621.785.1	Heating and heat treatment processes and procedure ⇒ 669-15
621.78.014	Rate of heating. Heating time	621.785.3	Heating and soaking for subsequent hot working
621.78.014.5	Rapid heating	621.785.33	Annealing and related heat treatment processes
621.78.014.7	Slow heating	621.785.34	Black annealing
621.78.015	Surface condition after heating	621.785.36	Bright annealing
621.78.019	Heat-treatment defects	621.785.37	Processes involving phase transformation. □ Full annealing. Isothermal annealing. Normalizing
621.78.019.2	Combustion. Burning	621.785.4	Processes not involving phase transformation. □ Solution treatment (alone). Intermediate annealing. Stress relieving. Soft annealing. Graphitizing. Malleabilizing
621.78.019.3	Overheating. Coarse grain	621.785.5	Heat treatment methods aimed at obtaining a bainitic structure. □ Austempering (austenitizing). Isothermal transformation. Patenting of wire
621.78.019.4	Decarburization	621.785.51	Surface hardening. Case-hardening. Case-carburizing ⇒ 621.793.8
621.78.019.5	Insufficient heating	621.785.52	Case-carburizing (without hardening)
621.78.019.6	Weak spots. Weak skin	621.785.53	Cementation. Carbon case-hardening. Pack-hardening
621.78.019.7	Distortion	621.785.538	Surface hardening with other nonmetals. □ With nitrogen, e.g. nitriding. With sulphur
621.78.019.8	Rupturing. Cracks. Splits. Exfoliation. Corrosion	621.785.54	Denitriding
621.78.019.82	Exfoliation	621.785.56	Surface hardening by localized heating. □ Flame hardening. Electrothermal hardening ⇒ 669-155.4
621.78.019.84	Corrosion	621.785.57	Immersion hardening (dip hardening)
621.78.06	Ambient media for heating, soaking and cooling. Furnace atmospheres etc. <i>The 621.7.06 auxiliaries are not applicable here</i> ⇒ 66-97 ⇒ 66-98	621.785.6	Hardening
621.78.061	Vacuum (in vacuo) heating, soaking or cooling	621.785.616	Quench-hardening of steel. □ Martensitic hardening. Martempering
621.78.062	Gaseous media. Air. Protective and reactive atmospheres	621.785.68	Hardening according to degree of hardness attained
621.78.063	Water. Aqueous solutions or mixtures	621.785.681	Soft
621.78.064	Liquid air. Liquefied gases	621.785.683	Moderately hard
621.78.065	Oils and emulsions	621.785.685	Hard
621.78.066	Molten metals or salts	621.785.687	Very hard. Glass hard
621.78.067	Solid media		
621.78.08	Cooling, quenching. Cooling after various treatments <i>The 621.7.08... auxiliaries are not applicable at 621.78 and its subdivisions</i>		
621.78.081	Slow cooling. □ Cooling in the furnace		

621.785.7	Tempering and ageing ⇒ 667.76	621.791.3	Soldering. Brazing. Bronze-welding (braze-welding)
621.785.72	Tempering	621.791.31	Process according to heat carrier
621.785.78	Ageing. Precipitation hardening	621.791.311	Solid body. □ Bit, block, roller soldering
621.785.79	Complex, combined treatments	621.791.312	Liquid. □ Dip, salt-bath, strike-and-touch, ultrasonic, shaft soldering
621.785.796	Quenching and tempering (of steel)		Gas. □ Flame, hot gas, gas furnace soldering.
621.785.797	Ageing or temper-hardening combined with work-hardening	621.791.313	Electric current. □ Induction, arc (high frequency), resistance, oven soldering
621.785.9	Low-temperature treatment. Other (heat) treatments	621.791.317	Beam. □ Electron beam, laser beam, laser soldering
621.787	Work-hardening. Hardening by mechanical means. □ Drawing. Compression. Rolling. Impact (peening)	621.791.318	Process according to minimum surface soldering temperature (liquidus temperature)
621.787.6	Hardening by impact. Shot peening	621.791.35	Soft soldering (fusible solder below 450°) <i>621.791.35 ≡ 621.791.31</i> ⇒ 669.65°4
621.789	Other structure modification. Hardening, toughening etc. by other means. □ By cooling in a magnetic field. By ultrasonic vibrations	621.791.36	Hard soldering <i>Fusible solder from 450 up to 900°</i> <i>621.791.36 ≡ 621.791.31</i> ⇒ 669.35
621.79	Joining of materials. Surface treatment and finishing. Keeping and storing. Packing and packaging ◇ 621.79.02 Preparation, cleaning and coating for treatment	621.791.37	High temperature soldering (fusible solder over 900°C) <i>621.791.37 ≡ 621.791.31</i>
621.791/.792	Joining of materials	621.791.4	Welding by application of both heat and mechanical energy
621.791	Welding and allied techniques ◇ 621.791.011 Weldability ◇ 621.791.012.5 Deep welding ◇ 621.791.013 Angle, direction of torch ⇒ 621.735.8 ⇒ 621.771.8 ⇒ 621.78	621.791.42	Forge welding (hammer welding) <i>Alternatively, class under forge work at 621.735.8</i>
621.791.021	Seam preparation. Forming the edges	621.791.44	Heated-tool welding of metals
621.791.03	Welding plant and equipment	621.791.46/48	Welding of plastics
621.791.033	Soldering irons, copper bits etc.	621.791.46	Direct heated-tool welding
621.791.034	Blowpipes. Blowlamps. Torches, lances etc.	621.791.47	Indirect heated tool welding. □ Heat contact, heat impulse welding
621.791.035	Gas welding and cutting appliances	621.791.48	Hot gas welding. Hot air welding. □ String-bead welding. Extrusion welding
621.791.036	Thermit welding and cutting appliances	621.791.5	Autogenous welding. Gas welding
621.791.037	Electric welding and cutting appliances. □ Permanent nonconsumable electrodes. Point electrodes (tip electrodes). Rotatable electrodes (wheel electrodes) ⇒ 621.791.042	621.791.55	Gas fusion welding. □ Oxyacetylene fusion welding
621.791.039	Accessory equipment. □ Benches. Jigs. Fixtures. Electrode holders	621.791.56	Gas pressure welding. □ Oxyacetylene pressure welding
621.791.04	Filler materials and fluxes	621.791.6	Mould welding. Thermit welding (non-pressure Goldschmidt aluminothermic process)
621.791.042	Welding fillers and soldering fillers in rod or wire form. Rod electrodes. Wire electrodes	621.791.7	Beam and electric welding
621.791.044	Welding fillers and soldering fillers in sheet or plate form. Sheet electrodes. Plate electrodes	621.791.72	Beam welding. Particle beam or laser beam welding
621.791.046	Welding fillers and soldering fillers in powder or bonded form. □ Welding thermit	621.791.722	Electron-beam welding
621.791.048	Fluxes. □ Fluids. Powders. Pastes.	621.791.724	Laser beam welding (using incoherent light)
621.791.05	Joints. Welds. Cuts	621.791.725	Laser welding <i>Using coherent light</i>
621.791.051	Zones. □ Melting zone. Bonding zone	621.791.75	Arc welding (arc-fusion welding)
621.791.052	Joints	621.791.752	Open arc welding
621.791.052.2	Lap joints	621.791.752.2	Carbon-arc welding (Bernados process)
621.791.052.4	Butt joints. □ With square joint, single V joint	621.791.752.3	Metal-arc welding (Slavianoff system)
621.791.052.6	Strap joints	621.791.753	Submerged-arc welding. □ Elin-Hafegut process (EHV). Electro-Linde-Rapid process (ELLIRA)
621.791.053	Welds	621.791.754	Arc welding with protective gas <i>Specify the atmosphere by '1/9 parallel to 546.1/9</i>
621.791.053.2	Filler welds		◇ 621.791.754'264 Carbon dioxide
621.791.053.3	Plug welds		◇ 621.791.754'293 Argon
621.791.053.4	Slot welds		◇ 621.791.754'293'264 Mixed argon carbon dioxide
621.791.053.5	Edge welds	621.791.755	Plasma welding (tungsten plasma welding)
621.791.053.6	Butt welds with filler material	621.791.76/79	Resistance welding
621.791.053.7	Narrow slot welds	621.791.76	Conductive resistance stud welding
621.791.053.9	Welds without filler material	621.791.762	Resistance butt welding. □ Flash welding
621.791.054	Cuts	621.791.763	Spot and seam welding. □ Projection welding. Loop contact welding. Capacitor discharge welding
621.791.1	Welding by application of purely mechanical energy. □ Cold press welding. Friction welding. Ultrasonic welding		

621.791.77	Induction resistance stud welding (induction welding)	621.794.47	Aftertreatment of the material
621.791.79	Resistance fusion welding. Electrolytic welding	621.794.48	Recuperation and re-use of residues, acids
621.791.793	Electro-slag welding	621.794.5	Decorative chemical treatment. □ Chemical colouring, dyeing, bleaching
621.791.795	Chamber welding (channel welding). Enclosed resistance welding	◇ 621.794.5:666.29	Decoration by enamels and glazes
621.791.797	Electrolytic welding	⇒ 621.795.3	
621.791.9	Techniques related to welding	621.794.6	Protective chemical treatment
621.791.92	Overlaying. Building-up. Hard-facing	⇒ 620.197.2	
	621.791.92 ≡ 621.791	621.794.61	Protective oxidation. □ Blueing. Browning
	<i>Denote overlaying metal by :669...</i>	◇ 621.794.61:621.357.8	Protective anodizing. Anodic oxidation
	◇ 621.791.92:669.018.25 Overlaying etc. with exceptionally hard or wear-resisting alloys	621.794.62	Protective phosphating. Phosphatizing
	⇒ 621.771.8	621.795	Final surface treatment. Improving of surface.
	⇒ 621.785.54	□ Decoration. Colouring. Dusting. Marking	
621.791.923.6	Overlaying etc. by brazing	621.795.2	Surface brightening and decorating by grinding, polishing, turning etc.
621.791.94	Thermal cutting	◇ 621.795.2:621.923	Brightening by grinding and polishing
	621.791.94 ≡ 621.791	621.795.3	Surface colouring. □ Painting. Varnishing. Lacquering
621.791.945	Gas cutting. Oxygen cutting	⇒ 621.794.5	
621.791.947	Electric cutting	⇒ 667.6	
621.791.948	Gas-arc cutting	621.795.4	Dusting. Sprinkling with powder
621.792	Joining by adhesion. □ Bonding, gluing	⇒ 621.793.8	
	<i>Denote materials by colon combination, adding firstly the number for the base material, followed by the number(s) for the bonded layer(s)</i>	621.795.8	Marking. Imprinting. Stamping of marks
621.792.052	Joints	⇒ 621.798.7	
	621.792.052 ≡ 621.791.052	621.796	Keeping. Storing. Warehousing
621.792.053	Adhesive used	⇒ 62-74	
	◇ 621.792.053:678.061 Macromolecular adhesives. Rubber and plastics solutions	⇒ 621.86	
621.792.3	Metal-to-metal joining	⇒ 629.4.084	
621.792.4	Metal-to-nonmetal joining	⇒ 629.7.084	
621.792.6	Nonmetal-to-nonmetal joining	⇒ 658.78	
621.792.8	Pressure jointing. Caulking	⇒ 664.8.03	
	⇒ 62-762	⇒ 725.1	
621.793/795	Surface treatment	621.796.2	Keeping in the open. Heaps, dumps etc.
621.793	Metal coating and plating. Production of conducting, semiconducting, resistive, dielectric, magnetic coats and films. Metallization	621.796.3	Keeping in pits, caves, tunnels etc. (natural or man-made). Keeping in open or closed containers, tanks, bunkers etc.
621.793.1	Plating and film deposition by vacuum evaporation and condensation	621.796.4	Keeping in open sheds, barns etc. (open at sides)
621.793.3	Plating by chemical deposition	621.796.5	Keeping in tiered (multistorey) stacks, on racks or shelves (indoors)
	⇒ 621.357.7		⇒ 621.868.3
621.793.5	Coating in molten metal. Dip coating	621.796.6	Keeping in silos, hoppers, bulk-goods containers
621.793.6	Diffusion coating	621.796.8	Protective coverings, sheets, tarpaulins etc.
621.793.7	Coating by spraying. □ Flame spraying. Arc metallizing. Plasma metallizing	621.798	Packing and packaging equipment. Packs, containers, materials, machines etc.
621.793.71	Flame spraying	◇ 621.798-181.2	Large, giant-size packs
621.793.72	Arc metallizing	◇ 621.798-181.4	Small, miniature packs
621.793.74	Plasma metallizing	◇ 621.798-189.2	Packaging in portions, rations, doses
621.793.79	Other techniques of spray coating	◇ 621.798-974	Low temperature packaging. Frozen packs
621.793.8	Adhesion coating with metal (in foil or powder form)	◇ 621.798-982	Vacuum packs
	⇒ 621.792	◇ 621.798-986	Under moderate pressure. Pressure packs
	⇒ 621.795	⇒ 620.165.7	
	⇒ 667.649.2	⇒ 621.86	
621.794	Chemical finishing and surface treatments	⇒ 656.073.2	
	⇒ 620.197.2	⇒ 658.788.4	
	⇒ 669.056.9	⇒ 658.827	
621.794.4	Pickling. Etching. Mordanting	⇒ 659.154	
	⇒ 621.7.025	⇒ 675.059	
621.794.41	Plant and equipment: layout, installation etc.	⇒ 676.028.6	
	□ Etching machines. Vats. Tubs	⇒ 676.244	
621.794.42	Pickling agents. □ Pickling and mordanting baths. Etches. Lyes	⇒ 677.057.78	
621.794.44	Pickling and mordanting techniques and processes. Etching	621.798.1	Containers. Wrappers. Kinds of pack, package
	⇒ 621.357.8	⇒ 62-46	
621.794.46	Neutralizing. Rinsing	⇒ 621.642	
		⇒ 621.869.8	
		⇒ 685.51	

621.798.12	Heavy outer containers for solid goods or objects. □ Boxes. Cases. Crates. Chests. Packing cases ⇒ 674.6	621.798.5	Closing, fastening, securing apparatus and machines. □ Binding, tying machines. Box strapping machines. Stitching, sewing machines. Sealing machines
621.798.13	Containers mainly for transport of fluids and bulk material. □ Barrels. Casks. Drums ⇒ 674.4	621.798.6	Labelling. Labelling apparatus and machines
621.798.14	Lightweight containers, rigid or semirigid (outer or inner packaging)	621.798.64	Security devices for prevention of unauthorized opening. □ Labels as seals
621.798.141	Wicker containers. □ Baskets. Hampers	621.798.67	Label holders. □ Slip frames
621.798.142	Rigid, set-up boxes (e.g. paper or board boxes)	621.798.68	Gumming, moistening devices
621.798.144	Cans. Canisters. Tins. Small drums. Cylindrical boxes (e.g. paper or board tubes)	621.798.7	Marking, branding (except labelling) ⇒ 621.795.8
621.798.146	Tubes primarily for pastes	621.798.74	Seals and sealing devices for packages (e.g. sealing in wax). Seals (impresses) as marks
621.798.147	Small containers for liquids. □ Pots. Jars. Bottles. Flasks. Phials (vials) ⇒ 666.171 ⇒ 666.176	621.798.77	Rubber stamps. Branding tools
621.798.15	Flexible containers and wrappers. □ Sacks. Sachets. Bags. Envelopes. Wrappers. Liners ⇒ 676.24 ⇒ 676.821	621.798.8	Other packaging matters. Cleaning of packages. Cleaning equipment
621.798.16	Dispensing containers. Pressure-discharge, squeeze-discharge tubes, sachets etc. □ Tubes for pastes (e.g. toothpaste tubes)	621.8	Machine elements. Motive power engineering. Materials handling. Fixings. Lubrication <i>Class here information about power transmission, machine elements etc. treated generally. If they are treated as parts of specific kinds of machinery, class with the subject, adding the -1/-8 auxiliaries from 62-1/-8</i>
621.798.18	Coatings and other special packagings	621.8.02	Modes of transformation of motion <i>The kinds of conversion of motion listed below may also be understood to imply the reverse direction</i>
621.798.185	Dip coating as first, inner packaging	621.8.022	Transformation of continuous rectilinear motion
621.798.187	Secondary (outer) packaging, including dip coating	621.8.023	Transformation of reciprocating rectilinear motion
621.798.188	Special packaging	621.8.024	Transformation of continuous rotary motion
621.798.2	Ancillary packaging materials	621.8.025	Transformation of reciprocating rotary motion
621.798.22	Cushioning. Padding. Wadding. Shock-absorbing materials <i>Denote details by Table 1k-03...</i> ◇ 621.798.22-036.746.2 Polystyrene cushioning	621.8.026	Transformation of continuous motion in a defined track
621.798.23	Specially prepared or formed packings. □ Straw bonnets for bottles	621.8.027	Transformation of reciprocating motion in a defined track
621.798.24	Strings. Twines. Cords. Bands. Tapes. Strapping. Tying and securing in general	621.8.028	Transformation of constant-speed motion into variable-speed motion
621.798.26	Closures. Fastening, sealing, securing (except tying and strapping). □ Rivet, nail, staple, screw closures. Adhesion closures, sealing tapes. Soldered, welded closures. Press-and-twist closures, bayonet caps ⇒ 683.53	621.8.03	Forms of energy for generation, conduction (transfer), storage, transformation, control of mechanical forces and resistances <i>The .03 auxiliaries are not applicable in 621.86/.87</i>
621.798.27	Opening devices incorporated in the packaging. □ Ring-pulls, tear-off tabs, built-in keys	621.8.031	Static and dynamic mass forces and resistance. □ Gravity. Inertia. Friction
621.798.28	Dispensing devices incorporated in the packaging. □ Spouts. Pourers. Nozzles	621.8.032	Static-dynamic fluid forces. Hydraulic energy
621.798.3	Filling of containers: methods, machines. □ Combined filling and closing machines	621.8.033	Pneumatic forces. Air resistance
621.798.31	Filling by hand. Manual packing in general	621.8.034	Static-dynamic vibrational forces. Vibrational energy. □ Oscillatory energy. Pendulum energy. Acoustic vibrations
621.798.33	Cask-filling, barrel-filling machines	621.8.035	Radiant energy. Heat and light radiation
621.798.34	Sacking and bagging machines	621.8.036	Thermal energy. Forces of thermal expansion and contraction. Generation of pressure by heat
621.798.36	Centrifugal fillers	621.8.037	Electric energy. Electrostatic attraction
621.798.37	Special fillers for viscous liquids	621.8.038	Electromagnetic and magnetic forces
621.798.4	Packing and wrapping of specific objects or materials ◇ 621.798.4:662.81 Packing of fuel briquettes	621.8.039	Molecular forces. □ Resilience, spring power. Elastic, flexional, torsional energy
621.798.41	Packing and wrapping of sheet material (sheet stacking, positioning, covering)	621.81	Machine elements in general <i>The subdivisions 621.812/.818 are used only if it is necessary to denote these groupings of machine elements; otherwise, prefer the main numbers given in the references</i> ⇒ 62-2
621.798.42	Packing and wrapping of sectional or profile material	621.812	Permanent fixings ⇒ 621.791 ⇒ 621.792 ⇒ 621.88.084 ⇒ 621.884
621.798.422	Bundling and covering of bars and rods		
621.798.426	Reeling. Spooling. Winding on bobbins ⇒ 621.7.068 ⇒ 621.798.4		
621.798.44	Compression packing. Baling. Trussing		

621.813	Detachable fixings ⇒ 621.882 ⇒ 621.886 ⇒ 621.887 ⇒ 621.888.1/2	621.822.187	Bearings with elastic pad. Tilting pad bearings. Sliding pad bearings
621.814	Elastic fixings ⇒ 62-272 ⇒ 621.888.4 ⇒ 621.888.6	621.822.188	Bearings with self-adjusting surfaces
621.815	Shaft-hub connections ⇒ 621.824.4 ⇒ 621.88.084 ⇒ 621.886.1 ⇒ 621.886.6 ⇒ 621.887	621.822.2	Plain thrust bearings (longitudinal bearings)
621.816	Rotational elements ⇒ 62-592 ⇒ 621.821 ⇒ 621.822 ⇒ 621.824/.825	621.822.22	Collar thrust bearings. Collar step bearings
621.817	Gears ⇒ 62-585.2 ⇒ 621.83 ⇒ 621.85	621.822.24	Thrust bearings for collar journals. Collar bearings
621.818	Pipes and pipe fittings ⇒ 62-762 ⇒ 621.643 ⇒ 621.646	621.822.27	Thrust bearings according to lubrication 621.822.27 ≡ 621.822.17
621.82/.85	Drive systems. Drive elements ⇒ 62-8 ⇒ 629.03	621.822.28	Thrust bearings according to sliding surface 621.822.28 ≡ 621.822.18
621.82	Journals. Bearings. Shafts. Couplings. Connecting rods. Links ⇒ 621.816	621.822.4	Foot step bearings. □ Jewel bearings for fine mechanics
621.821	Journals. Pivots. Trunnions	621.822.5	Plain journal bearings (guide bearings)
621.822	Bearings. Bushes	621.822.52/.56	Journal bearings according to method of attachment
621.822.1	Plain bearings in general	621.822.52	Pedestal bearings. Plummer bearings. Pillow block bearings
621.822.11	Plain bearings according to load	621.822.53	Hanging bearings. Hanging bracket bearings
621.822.114	Plain bearings for mainly radial loads ⇒ 621.822.5	621.822.54	Eye-type bearings. Flange bearings
621.822.115	Plain bearings for mainly axial loads ⇒ 621.822.2	621.822.55	Wall bearings. Wall bracket bearings
621.822.116	Plain bearings for radial and axial loads. Combined radial/axial plain bearings (thrust bearings) ⇒ 621.822.4	621.822.56	Bracket bearings. Pillar bracket bearings
621.822.17	Plain bearings according to type of lubrication ⇒ 62-722	621.822.57	Journal bearings according to type of lubricant 621.822.57 ≡ 621.822.17
621.822.172	Hydrostatic plain bearings (pressure chamber bearings)	621.822.58	Journal bearings according to sliding surface 621.822.58 ≡ 621.822.18
621.822.173	Hydrodynamic plain bearings	621.822.59	Journal bearings according to bearing material
621.822.174	Aerostatic (gas-static) plain bearings	621.822.591	Solid bearings made out of a single material. One-metal bearings
621.822.175	Aerodynamic (gas-dynamic) plain bearings	621.822.592	Two-metal bearings. Two-layer bearings
621.822.176	Hybrid bearings (hydrostatic and hydrodynamic lubrication)	621.822.593	Three-metal bearings. Three-layer bearings
621.822.177	Plain bearings with solid lubricants. □ With graphite	621.822.6	Rolling contact bearings in general
621.822.178	Self-lubricating plain bearings. □ By means of the bearing material e.g. sintered bushing materials	621.822.61	Rolling contact bearings according to type of load
621.822.18	Plain bearings according to sliding surface	621.822.614	Rolling contact bearings for mainly radial loads ⇒ 621.822.74 ⇒ 621.822.84
621.822.181	Plain bearings with solid bushes. Single-lobe bearings	621.822.615	Rolling contact bearings for mainly axial loads ⇒ 621.822.75 ⇒ 621.822.85
621.822.182	Plain bearings with two wedge slits. Two-lobe bearings (bearings with oval clearings)	621.822.616	Rolling contact bearings for separate absorption of radial and axial loads ⇒ 621.822.76 ⇒ 621.822.86
621.822.183	Plain bearings with three wedge gaps. Three-lobe bearings	621.822.7	Ball bearings
621.822.184	Sectionalized-surface bearings with four or more wedge gaps (multi-wedge bearings)	621.822.71	Balls
621.822.186	Bearings with rigid pads. Pad-type bearings	621.822.72	Individual parts other than balls
		621.822.73	Ball bearings in general
		621.822.74	Radial-load ball bearings
		621.822.75	Axial-load ball bearings
		621.822.76	Combined, radial-axial ball bearings
		621.822.8	Roller bearings. Needle bearings
		621.822.81	Rollers. □ Cylinder rollers. Needle rollers. Tapered rollers. Spheroidal rollers. Collared rollers. Spring rollers (coiled)
		621.822.82	Parts other than rollers
		621.822.822	Cages
		621.822.823	Inner races
		621.822.824	Outer races
		621.822.828	Movable shoulders
		621.822.83	Roller bearings in general
		621.822.831	Cylinder rolling bearings in general
		621.822.832	Needle bearings in general
		621.822.833	Taper (conical) roller bearings in general
		621.822.835	Self-aligning roller bearings in general
		621.822.84	Radial-load roller bearings
		621.822.85	Axial-load roller bearings
		621.822.86	Combined, radial-axial roller bearings
		621.822.88	Combined ball and roller bearings. Ball-roller bearings
		621.822.9	Linear contact bearings. Longitudinal guideways. Straight guideways

621.824	Shafts. Axles	621.83.061.4	Transmissions for reversal of direction of motion. Reverse gears
621.824.2	Straight shafts. Smooth shafts. Plain shafts	621.83.062	Speed changes. □ Continuously variable speed. Stepwise speed change. Freewheel mechanisms (neutral gears). Overdrive transmissions
621.824.3	Eccentric shafts. Shafts with control elements	621.83.062.1	Continuously variable change of speed
621.824.32	Crankshafts ⇒ 621.827.2 ⇒ 621.837.7	621.83.062.2	Stepwise change of speed
621.824.33	Camshafts	621.83.062.6	Transmissions with freewheel mechanism
621.824.4	Shaped, profiled shafts ⇒ 621.815	621.83.062.7	Overdrive transmissions
621.824.42	Polygonal shafts. □ Shaft of uniform width	621.83.063	Speed ranges
621.824.43	Shafts with splines (with straight or involute flanks)	621.83.064	Output ranges
621.824.44	Key shafts (shafts with keyways)	621.83.065	Transmissions in relation to their axial position
621.824.47	Power take-off shafts (with profiles stub)	621.83.065.1	Coaxial transmissions
621.824.5	Flexible shafts	621.83.069.1	Transmissions with constant mesh (synchromesh gears)
621.824.6	Hollow shafts	621.83.069.2	Transmissions with preselectors. Automatic drives
621.824.8	Taper shaft elements. Shaft collars		
621.825	Couplings ⇒ 62-578 ⇒ 621.838	621.833	Toothed wheels (gearwheels). Toothed gears and gearing. Tooth profiles
621.825.2	Fixed (rigid) couplings	621.833.01	Running, working. Efficiency. Friction
621.825.5/7	Differential couplings	621.833.1	Cylindrical gears with parallel axes. Spur gears. Helical gears. Double-helical gears
621.825.5	Longitudinally elastic, torsion-resistant couplings. Extensible couplings	621.833.12	Gear racks and pinions. Toothed segments
621.825.52	Claw-type couplings	621.833.15	Cylindrical gears with external teeth. External gears
621.825.56	Couplings with external and internal tooth gearing. □ Toothed couplings. Curved-tooth couplings	621.833.16	Cylindrical gears with internal teeth. Internal gears (Annular gears)
621.825.6	Jointed couplings	621.833.2	Bevel gears. Other gears on shafts with intersecting axes
621.825.63	Universal couplings. Cardan joints	621.833.22	Bevel gears with straight and helical teeth
621.825.65	Ball-and-socket joints	621.833.24	Bevel gears with curved teeth
621.825.67	Disc joints. Hardy joints	621.833.28	Cylindrical gears on shafts with intersecting axes
621.825.7	Torsionally flexible (torsionally elastic) couplings	621.833.3	Gears on shafts with nonparallel, nonintersecting axes
621.825.72	Spring couplings 621.825.72 ≡ 62-272	621.833.35	Crossed helical gear pairs
621.825.74	Band couplings	621.833.36	Hypoid gears
621.825.76	Bellows couplings. □ Metal bellows couplings	621.833.38	Worm gears
621.825.78	Membrane couplings	621.833.4	Lantern wheel gears (lantern pinion gears)
621.826.22	Coil springs. Helical springs etc.	621.833.5	Noncircular gears
621.827	Connecting rods. Cranks. Eccentrics	621.833.51	Elliptical gears
621.827.1	Connecting rods	621.833.52	Hyperbolic gears
621.827.2	Cranks. Disc cranks. Countercranks ⇒ 621.824.32 ⇒ 621.837.7	621.833.6	Epicyclic gears. Hypocyclic gears. Planetary gears. Other gears and gear trains
621.827.5	Eccentrics	621.833.61	Planetary gears with three coaxial shafts. Differential gears
621.828	Articulated links, joints. Knuckles. Rockers. Toggle joints. Bolts and links	621.833.64	Planetary gears with more than three coaxial shafts
621.83	Mechanical transmissions. Gears, gearing. Cams. Sliding tracks. Ratchets. Clutches ⇒ 62-233.3/9 ⇒ 621.817	621.833.65	Planetary switch gear trains
621.83.05	Elements. Individual components	621.833.68	Branched variable gear trains. Variable shunt gear trains
621.83.052	Teeth	621.835	Cams. Cams operating on bars and levers
621.83.053	Gearing according to type of tooth flank. Flank profile	621.835.2	Cams ⇒ 62-233.52
621.83.054	Gearing according to direction of tooth flank	621.835.8	Swashplates ⇒ 62-234
621.83.058	Components for locking mechanisms	621.836	Guides. Guideways. Slide-bar transmissions. Slideways. Link transmissions
621.83.058.1	Ratchet wheels. Ratchet gears	621.836.3	Push-pull sheathed devices. □ Bowden cables
621.83.058.2	Locking springs. Return springs 621.83.058.2 ≡ 62-272	621.836.7	Maltese crosses
621.83.058.3	Pawls ⇒ 62-783.62	621.837	Drives with eccentrics, inclined planes, wedges, cranks
621.83.059	Components for friction gear mechanisms	621.837.2	Eccentric drives
621.83.059.1	Friction wheels. Friction discs	621.837.3	Inclined planes
621.83.059.2	Pressure mechanisms	621.837.4	Wedge transmissions
621.83.059.4	Intermediary elements 621.83.059.4 ≡ 62-4	621.837.7	Crank mechanisms ⇒ 621.824.32 ⇒ 621.827.2
621.83.06	Types of transmission. Speed. Output		
621.83.061	Speed reduction. Constant ratio. Speed increase		

621.838	Locking mechanisms. Engaging devices <i>Denote the type of control by -52...</i>	621.85.055.7	Chain connectors. □ Shackles. Clevises ⇒ 621.88
621.838.1	Locking mechanisms <i>Denote components by .058...</i> ⇒ 62-231.32 ⇒ 62-585.14	621.85.057	Controlling and guiding devices. □ Belt guides
621.838.14	Ratchet gears. Ratchet mechanisms <i>Movement in the direction of the locked force</i>	621.85.058	Tensioning and positioning devices for belts, chains etc.
621.838.15	Stop mechanisms <i>Movement contrary to the locked force</i>	621.85.058.2	Tensioning devices. □ Screw tensioners
621.838.16	Storage gears. □ Clamping mechanisms (free release). Spring mechanisms (forced release)	621.85.058.6	Positioning and displacing devices. □ Belt forks. Strikers
621.838.2	Mechanical engaging devices	621.85.06	Flexible drives according to performance, speed, output 621.85.06 ≅ 621.83.06
621.838.21	Devices with reciprocal engagement. □ Claw, dog and jaw clutches	621.852	Belt drives
621.838.22	Devices engaging by friction. Friction or slip clutches	621.852.4	Belt drives according to belt form
621.838.3	Electric and electromagnetic engaging devices. □ Bovet brakes	621.852.6	Belt drives according to design. □ Open belt drives. Crossed belt drives. Angle belt drives
621.838.4	Hydraulic engaging devices. Fluid couplings	621.854	Cable or rope drives ⇒ 621.85.054
621.838.5	Pneumatic engaging devices	621.855	Chain drives 621.855 ≅ 621.85.055 ⇒ 621.85.055
621.839	Friction drives. Frictionally engaged rolling contact gears <i>Denote components by .059...</i> ⇒ 531.43 ⇒ 62-585.13 ⇒ 621.838.22	621.86/87	Materials handling technology. Conveying equipment ⇒ 658.286.2
621.839.3	Friction wheel drives with constant transmission	621.86	Materials handling (mechanical handling) in general <i>The following auxiliaries are also applicable in 621.87. The .03 auxiliaries listed under 621.8.03 are not applicable here</i> ⇒ 62-237 ⇒ 621.6 ⇒ 625.11/5 ⇒ 629
621.839.5	Intermittent friction wheel gears	621.86.01	Theory. Loading
621.839.7	Adjustable friction wheel gears	621.86.016	Load. □ Deadweight. Limit load (maximum load). Carrying, lifting capacity. Overload
621.85	Power transmission flexible devices. Flexible transmissions	621.86.017	Machines according to mode of action, lifting or scooping stroke etc.
621.85.01	Theory. Principles ◇ 621.85.01:531.44 Theory, principles of sliding friction	621.86.017.2	High-stroke machines (acting mainly above the level on which the machine stands)
621.85.05	Flexible drive elements and devices	621.86.017.3	Low-stroke machines (acting mainly below the level on which the machine stands)
621.85.051	Pulleys and chain wheels	621.86.017.4	Machines with combined, compound action (at both high and low level)
621.85.051.22	Integral pulleys or wheels	621.86.03	Machines according to drive or propulsion system 621.86.03 ≅ 629.03
621.85.051.23	Parting pulleys, wheels	621.86.032	Wheeled machines
621.85.051.26	Cone pulleys. Stepped pulleys	621.86.033	Tracked machines. Caterpillar, crawler machines
621.85.051.32	Driving pulleys or wheels	621.86.05	Flexible transmission elements. □ Pulleys. Belts. Chains 621.86.05 ≅ 621.85.05
621.85.051.34	Driven pulleys or wheels	621.86.06	Load-bearing, carrying gear. Filling and emptying, loading and unloading devices ⇒ 621.861
621.85.051.42	Fast pulleys (pulleys fixed to shaft)	621.86.061	Lifting hooks. Lifting loops, eyes. Tongs. Traverses
621.85.051.44	Loose pulleys (idle or dead pulleys)	621.86.062	Adhesion devices. Lifting magnets. Suction, pneumatic, vacuum clamps for lifting ⇒ 621.318.38
621.85.051.5	Chain wheels. Sprockets	621.86.063	Grabs. Buckets (on cables etc.)
621.85.052	Belts (and belt appurtenances)	621.86.064	Scoops. Shovels. Dippers. Cutter heads. Suction heads. Skimmers. Graders. Scraper blades
621.85.052.4	Belts according to form, cross-section. □ Flat belts. Round belts. V-belts	621.86.065	Suspension lines. Sling ropes. Sling chains. Sling belts ⇒ 672.6 ⇒ 679.7
621.85.052.41	Flat belts	621.86.067	Bulk loading devices (including closures, feeders). □ Chutes. Funnels. Hoppers
621.85.052.42	Conical belts. V-belts		
621.85.052.43	Round belts		
621.85.052.44	Toothed belts		
621.85.052.7	Belt connectors, fasteners, clasps etc.		
621.85.052.8	Anti-slip and efficiency devices for belts		
621.85.054	Cables and ropes (and their appurtenances) 621.85.054 ≅ 621.85.052		
621.85.055	Chains (and chain appurtenances) ⇒ 672.6		
621.85.055.2	Chains according to design		
621.85.055.22	Integral chains		
621.85.055.24	Disconnected chains		
621.85.055.25	Roller chains		
621.85.055.26	Bushing chains		
621.85.055.27	Bushing-roller chains		
621.85.055.4	Chains according to link form		
621.85.055.41	Flat-link chains. Block-link chains		
621.85.055.44	Gear-link chains		
621.85.055.46	Jack-link chains. Hook-link chains		

621.86.068	Discharging devices. □ Stripper belts. Dump cars	621.867.26	Moving walkways. Travelators ⇒ 621.876.3 ⇒ 692.67
621.86.07	Mechanisms of handling equipment. Components of lifts etc.	621.867.3	Bucket conveyors. Bucket elevators ⇒ 621.649.4 ⇒ 621.879
621.86.071	Curves, bends in transfer lines or routes	621.867.31	With chain and fixed buckets
621.86.072	Crossover points in transfer lines	621.867.32	Elevators with chain and hinged buckets (tipping pails)
621.86.075	Handling equipment of variable length. □ Extensible devices. Telescopic devices	621.867.33	Band elevators with fixed buckets
621.86.078	Lift shaft equipment. Guide rails. Lift cages	621.867.36	Drag bucket conveyors with fixed drag buckets
621.86.078.1	Guide rails. Shafts. Basements	621.867.4/.6	Conveyors without traction mechanism
621.86.078.2	Lift cages	621.867.4	Worm, auger or screw conveyors. Worm elevators
621.86.078.3	Lift doors, gates. Gate and shaft locking devices	621.867.5	Chutes. Shaking conveyors
621.86.078.5	Signalling and control devices on lifts. Level or floor indicators	621.867.51	Chutes. Slides. Spiral slides (helter-skelters)
621.86.078.6	King bolts. Support, holding devices. Safety brakes etc.	621.867.52	Shaking, oscillating, vibrating conveyors. □ Shaker chutes. Rocking troughs
621.86.078.7	Various safety mechanisms. □ Devices to prevent overloading, overtravel or overshooting	621.867.53	Slide rod conveyors
621.86.078.8	Framework, armature of platform or cage	621.867.56	Tube chutes. Gravity tubes
621.86.08	Weighing and apportioning equipment	621.867.6	Roller conveyors. Rollerways ⇒ 629.314
621.861	Pulley hoists. Lifting tackle. Block and tackle ⇒ 621.86.05 ⇒ 621.86.06	621.867.61	Rollerways with idle rollers (nonpowered)
621.861.1	Simple pulley suspensions	621.867.63	Rollerways with driven, powered rollers. Roll tables
621.861.2	Rope tackle (normal block and tackle)	621.867.64	Disc-roller conveyors (wheel-roller conveyors)
621.861.3	Spur gear blocks (block and tackle with spur gears)	621.867.65	Ball-roller conveyors. Ball tables (castor tables)
621.861.4	Differential blocks	621.867.7/.8	Flow conveyors. Hydraulic conveyors ⇒ 621.643
621.861.5	Worm blocks (worm block and tackle)	621.867.7	Hydraulic conveyors. Hydraulic elevators. Trough or tube liquid-flow conveyors ⇒ 621.695
621.863	Winding gear, hoists	621.867.8	Pneumatic conveyors. Pneumatic elevators
621.863.2	Drum winding gear	621.867.81	Suction conveyors. Vacuum tubes ◇ 621.867.81.068 Discharging devices. Suction nozzles. Collectors. Discharging drums
621.863.6	Pulley, sheave or wheel winding gear	621.867.82	Pressure, air-pressure conveyors. Compressed-air tubes
621.864	Winches. Windlasses. Capstans	621.867.84	Conveyors with static compressed air ◇ 621.867.84:628.336.8 For sludge conveyance ⇒ 621.695
621.865	Accessories for manual load handling	621.867.85	Pneumatic troughs
621.865.1	Tools for clearing, sweeping, digging, scooping. □ Brooms. Pushers. Spades. Rakes. Shovels. Forks. Funnels	621.867.87	Pneumatic unit-load conveyors. □ Pneumatic post dispatch units. Message tubes ⇒ 656.86
621.865.2	Levering implements. □ Crowbars. Pinchbars	621.867.9	Centrifugal conveyors. □ Centrifugal drum conveyors. Centrifugal plate conveyors
621.865.4	Crossbars. Scissor levers	621.868	Floor conveyors. Materials handling vehicles
621.865.6	Ladders and outriggers with lifting devices. Devices for lifting sacks and bales	621.868.2	Free ranging floor and yard vehicles ⇒ 629.3
621.865.8	Handlers. Industrial robots	621.868.23	Vehicles without hoisting equipment. Castor trucks. Rolling frames, shelves etc.
621.866	Short-lift raising devices. Jacks ⇒ 621.869.3	621.868.232	With castors (no axle). Castor trucks with superstructure. Rolling frames, shelves
621.866.1	Lever, rack and slide-lift jacks	621.868.234	With one axle. Single-axle and single-wheel trucks. □ Wheelbarrows. Sack barrows
621.866.2	Screw jacks (gins)	621.868.238	With more than one axle. □ Hand carts, trolleys. Trailers. Forkless trucks
621.866.3	Air-cushion bellows. Lifting cushions	621.868.25	Vehicles with hoisting or lifting equipment
621.867	Continuous transfer equipment. Conveyors. Elevators. Chutes	621.868.254	With one axle. □ Sack barrows with hoists. Tank or barrel carts. Mechanical draymen's ladders
621.867.1/.3	Continuous conveyors with traction mechanism	621.868.258	With more than one axle, with a low lift. □ Platform trucks, trolleys. Low-lift stacking trucks (low fork-lift and portal-lift trucks)
621.867.1	Chain conveyors. □ With single or multiple lines	621.868.27	Vehicles with high-lift equipment ⇒ 629.364
621.867.13	Conveyors for sliding or dragging material. □ Scraper conveyors. Double chain conveyors. Drag chain conveyors	621.868.273	Crane trucks
621.867.15	Chain conveyors for carried materials. □ Stack conveyors. Circular conveyors. Table conveyors. Rocker, swing-tray conveyors ⇒ 621.876.3	621.868.274	High-lift trucks. High-lift fork trucks. Extending stackers
621.867.17	Tow-chain or pusher-chain conveyors. □ For chain-pulled trucks, trolleys	621.868.275	Fork-lift stackers. Side fork-lifts. Reach fork-lifts
621.867.2	Belt conveyors (conveyor belts) ⇒ 622.647.2		
621.867.21	Web conveyors (webbing belt conveyors)		
621.867.22	Steel band conveyors		
621.867.23	Cable or rope belt conveyors		
621.867.24	Mesh or netting belt conveyors. Wire mesh conveyors		
621.867.25	Apron conveyors		

621.868.277	Fork-lift extension devices	621.875	Transporter cranes. Portal cranes
621.868.3	High-bay stacking and retrieving equipment. Warehouse conveyors ⇒ 621.796.5		.04 from 621.874.04 is also applicable here ⇒ 622.69
621.868.6	Rail-type floor and yard vehicles. Rail trucks, wagons ⇒ 625.11/5	621.875.1	Transporter cranes
621.868.62	Special rail trucks for internal transport	621.875.4	Sliding or travelling platforms. Traverses (traversers) ⇒ 625.155
621.868.64	Hand-propelled rail truck installations	621.875.5	Portal cranes
621.868.66	Stationary-drive rail-truck haulage installations. □ Chain, cable, rope traction systems	621.876	Lifts. Escalators. Hoists ⇒ 622.67 ⇒ 626.5 ⇒ 692.66/67
621.868.68	Self-propelled rail truck installations	621.876.1	Platform lifts for shuttle operations. □ Goods lifts, service lifts. Passenger lifts, personnel elevators. Inclined hoists ⇒ 621.86.078 ⇒ 621.87.078 ⇒ 621.863 ⇒ 621.867
621.869	Auxiliary equipment for making-up unit loads. Loading and unloading devices, including those on vehicles		Vertical lifts
621.869.2	Unloading, discharging, emptying devices for rail vehicles. □ Tipplers. Dumpers (tipping wagons)	621.876.11	Goods lifts
621.869.3	Loading, filling equipment. □ Loading ramps. Loading platforms. Loading bridges. Lifting stages. Working platforms	621.876.113	Passenger lifts
621.869.8	Accessories for making-up unit loads ⇒ 621.798	621.876.114	Skip hoists. Bucket hoists ⇒ 621.863
621.869.81	Simple means of carrying. □ Baskets. Panniers	621.876.2	Continuously operating lifting installations. □ Continuous lifts (paternosters). Escalators (moving stairs) ⇒ 621.867
621.869.82	Pallets. Accessories for pallets	621.876.3	Cable cranes. Cable-excavators. Cable dredgers .04 from 621.874.04 is also applicable here ⇒ 625.577
621.869.83	Loading platforms. Stillages		Cable cranes
621.869.85	Roller stillages (trolley stillages)	621.877.3	Portal cable cranes. Bridge cable cranes
621.869.87	Loading frames, cradles	621.877.5	Cable excavators. Cable dredgers
621.869.88	Transfer containers. Freight containers. Transport containers Denote type of freight by colon combination ◇ 621.869.88:633.491 Containers for potatoes ⇒ 621.642 ⇒ 621.798.1 ⇒ 656.073.235	621.877.7	Excavation equipment for bulk materials. Earthmoving equipment
621.87	Cranes. Transporters. Lifts. Escalators. Levellers. Excavators. Dredgers 621.87.01/08 ≡ 621.86.01/08 The .03 auxiliaries listed under 621.8.03 are not applicable here ◇ 621.87:69.057.7 Cranes, transporters etc. for building sites ⇒ 62-237 ⇒ 624.96	621.878/879	Levelling equipment. Loaders. Scrapers Levelling and grading equipment Bulldozers Graders Dragline scrapers Loaders ⇒ 621.86.067 ⇒ 622.619
621.873	Cranes in general	621.878	Noncontinuous loaders. Batch, unit loaders. □ Scraper loaders. Drawhook (wide-grab) loaders. Shovel loaders, scoop loaders
621.873.1	Stationary, fixed cranes	621.878.2	Continuous loaders. □ Shaking, oscillating, vibrating loaders. Bucket elevator loaders
621.873.11	Stationary cantilever or jib cranes	621.878.23	Scrapers. □ Power shovels
621.873.12	Nonslewing cranes. □ Three-legged trestle cranes. Fixed-jib cranes	621.878.25	Digging machinery. Excavating machinery. Dredging machinery ⇒ 621.86.064 ⇒ 622.232 ⇒ 622.271.5 ⇒ 627.748
621.873.13	Part-slewing, limited-slew cranes. □ Wall-mounted cranes. Derrick cranes	621.878.27	Intermittent digging machines, with single bucket or shovel. Bucket dredgers
621.873.15	Fully slewing cranes. □ Luffing and slewing cranes	621.878.4	Universal bucket machines, with exchangeable attachments. Multipurpose digging machines
621.873.21/3	Mobile, travelling cranes		Single-purpose, specialized digging machines, with only one or two kinds of attachment
621.873.2	Cranes with limited mobility 621.873.2 ≡ 621.873.1	621.879	High-scoop and low-scoop rigid-arm digging machines ◇ 621.879.322.017.2 High-scoop. Face-shovel, crowd-shovel, luffing-shovel machines. Dipper-bucket machines etc. ◇ 621.879.322.017.3 Low-scoop. Back-hoe, back-acter machines. Drag-shovel machines etc.
621.873.3	Cranes with unlimited mobility on land 621.873.3 ≡ 621.873.1 ◇ 621.873.3:629.364.1 Cranes on road vehicles	621.879.3	
621.873.7	Floating cranes For floating cranes as vessels see 629.565.1 621.873.7 ≡ 621.873.1	621.879.31	
621.874	Bridge cranes ⇒ 624.8	621.879.32	
621.874.04	Crane trolleys, crabs (for bridge cranes) Also applicable in 621.875 and 621.877	621.879.322	
621.874.2	Overhead travelling cranes		
621.874.4	Goliath cranes		
621.874.7	Bridge cranes with special equipment ◇ 621.874.7:621.74 Foundry cranes		

621.879.323	Dragline excavators	621.88.084	Transverse press connections ⇒ 62-112.85 ⇒ 621.713.2 ⇒ 621.815 ⇒ 621.887
621.879.324	Grab machines. Mechanical grabs. Grapple-dredgers	621.88.086	Fastenings for rods, sections
621.879.325	Crane machines. Machines with gripping tongs	621.88.087	Tools for inserting or attaching fasteners
621.879.326	Loading shovel machines	621.88.088	Tools for loosening, extracting, removing fasteners ◇ 621.886.6.088 Tools for extracting keys and cotters
621.879.327	Skimmer machines	621.88.091	Forms of head
621.879.328	Digging machines with other kinds of attachment. □ Machines with dozers, rippers, tree-stump lifters, rams	621.88.091.1	Flat cylindrical or cheese head
621.879.4	Continuous excavators ⇒ 621.867	621.88.091.2	Raised (convex or oval) cylindrical, cheese or fillister head
621.879.41	Helical digging machines. Worm, auger or screw excavators	621.88.091.3	Domed, button, cup or snap head
621.879.44	Chain-elevator digging machines. Bucket-chain excavators. Ladder dredgers. Cutter-chain dredgers	621.88.091.4	Flat or raised (convex or oval) pan head
621.879.45	Suction excavators. Hydraulic, suction dredgers. □ Sand-pump dredgers. Sludgers	621.88.091.5	Raised (convex or oval) countersunk head
621.879.46	Delivery pumps (intermediate booster units)	621.88.091.6	Flat countersunk head
621.879.47	Shore pipes (pipeline dredgers)	621.88.092	Types of screw end <i>Form elements at the end of bolts and pins</i>
621.879.48	Rotary, bucket-wheel excavators	621.88.092.2	Flat, unpointed ends
621.88	Fastening, fixing devices. Fasteners ⇒ 621.778.8 ⇒ 624.078.4 ⇒ 672.88	621.88.092.3	Rounded, domed, tapered ends
621.88.07	Quality <i>Dimensional accuracy and surface finish</i> ⇒ 621.713.1 ⇒ 621.9.015	621.88.092.4	End with bolt, split pin or cotter holes
621.88.072	Coarse, category C product <i>Previously 'rough'</i>	621.88.092.5	End with point (cone point)
621.88.073	Medium coarse, category B product <i>Previously 'semibright'</i>	621.88.092.6	Recessed, cupped end
621.88.074	Medium, category A product <i>Previously 'bright' or 'fully machined'</i>	621.88.092.7	End with shaving slot
621.88.076	Fine, category F product	621.88.092.8	End with oval or coned half dog
621.88.082	Thread systems. Thread profiles ⇒ 621.99	621.88.092.9	End with drill point
621.88.082.1/3	Threads with triangular profiles (angular threads)	621.88.096	Design of components used for direct transmission of power to fixing devices ◇ 621.883.1.096.5 Multiple open-jaw spanners (bicycle spanner)
621.88.082.1	Metric threads	621.88.096.1	Single
621.88.082.13	Metric adjusting threads	621.88.096.2	Double
621.88.082.14	Metric MJ profile (modified ISO profile)	621.88.096.4	Cruciform. Four-way
621.88.082.15	Metric fine pitch	621.88.096.5	Multiple
621.88.082.17	Metric cutting threads	621.88.096.7	Cranked. Offset. Elbowed
621.88.082.2	Inch threads	621.88.096.8	90° offset
621.88.082.23	Whitworth threads	621.881	Clamping, fastening, gripping tools
621.88.082.25	Whitworth pipe threads	621.881.2	Vices. Vice clamps
621.88.082.3	Triangular threads other than metric and Whitworth threads. □ Steel conduit threads (tooth angle 80°)	621.881.3	Hand vices. Clamps. Cramps
621.88.082.4/6	Lead threads ⇒ 621.888.1	621.881.4	Tongs. Nippers. Pincers. Pliers
621.88.082.4	Flat threads. Trapezoidal threads	621.882	Threaded fasteners. Screws. Nuts and bolts. Washers ⇒ 621.99
621.88.082.5	Saw-tooth threads	621.882.1	Screws and bolts in general. □ Dimensions. Through bore
621.88.082.6	Rounded threads. □ Electrical threads ◇ 621.88.082.6-033.5 Glass threads	621.882.2	Screws, bolts according to head form. Screws and bolts for various materials
621.88.082.7/8	Screw threads that form the nut thread without cutting	621.882.21	Screws and bolts according to head form
621.88.082.7	Threads for screws to be screwed into nonmetallic workpieces. □ Wood screw threads. Chipboard screw threads. Threads for screws for plastics	621.882.211	Hexagon-head screws and bolts
621.88.082.8	Sheet-metal screw threads. Threads for self-tapping screws to be screwed into metallic workpieces	621.882.212	Hexagon screws with additional shapes. □ Flank screws. Collar screws. Cap screws
621.88.083	Shank-side or head-side thread end <i>Also applicable to internal thread</i> ⇒ 621.88.092	621.882.213	Square-head screws and bolts. □ Hammer-head screws. T-head screws
		621.882.214	Other polygonal-headed screws and bolts
		621.882.214.2	Screws and bolts with knurled or milled head. Thumb screws
		621.882.214.3	Triangle bolts
		621.882.215	Circular, round-head screws and bolts. Recessed, cup-head screws. Socket-head screws
		621.882.216	Screws and bolts with perforated head, with peg holes
		621.882.217	Eye-bolts and grooved, slot-headed screws
		621.882.218	Wing-headed screws. Fly or butterfly bolts
		621.882.219	Headless screws
		621.882.219.4	Headless screws. Studs, stud bolts. Grub screws
		621.882.219.6	Set screws with unthreaded portion

621.882.24/29	Screws and bolts for materials other than metals	621.888.2	Hinges. Pivots. Pintles
621.882.24	For wood		⇒ 683.36
	621.882.24 ≡ 621.882.21	621.888.4	Springs as fasteners
621.882.25	For leather		⇒ 62-272
	621.882.25 ≡ 621.882.21	621.888.6	Elastic supports. Rubber mountings, connectors etc.
621.882.27	For stone (brick)		⇒ 62-752
	621.882.27 ≡ 621.882.21	621.888.8	Tension jacks. Turnbuckles
621.882.3	Nuts	621.889	Interlocking insertion connections
	621.882.3 ≡ 621.882.21		⇒ 62-112.88
621.882.4	Washers		⇒ 621.813
621.882.449	Spring washers	621.889.4	Spring-loaded connections
621.882.5	Locking of screws and bolts. Securing devices for screws. □ Split pins. Lock-nuts. Nut retainers. Circlips	621.889.8	Insert-and-twist connections (bayonet connections)
621.882.6	Bolts (particularly anchor bolts) <i>The term 'bolts' in English includes many objects which in UDC are classed with screws, for which see 621.882.21 and subdivisions</i> ⇒ 62-229.312 ⇒ 621.828	621.89	Lubrication ⇒ 531.43 ⇒ 62-72
621.882.63	Stay bolts. Clamp or stirrup bolts. U-bolts	621.89.012	Properties <i>Denote properties other than those listed below by colon combination; especially :53... and :54...</i>
621.882.64	Fishbolts. Rag bolts. Foundation bolts. Expansion bolts	621.89.012.2	Lubricity. Capacity for film formation
621.883	Spanners. Wrenches. Screwdrivers	621.89.012.3	Oiliness
621.883.1	Open-jaw spanners. Open-ended wrenches	621.89.012.7	Properties of the film. □ Thickness
621.883.2	Ring spanners. Socket spanners	621.89.017	Testing <i>Details by :620.1...</i>
621.883.24	Ring spanners 621.883.241/.244 ≡ 621.882.211/.214	621.89.09	For various applications. Industrial applications in general. For mechanisms in general
621.883.243	Closed-end square wrenches	621.89.091	For precision mechanisms
621.883.26	Socket spanners (socket wrenches). Socket tee-wrenches 621.883.261/.264 ≡ 621.882.211/.214	621.89.092	For compressors, refrigerators
621.883.261	Hexagon socket wrenches	621.89.093	For light machinery and transmissions
621.883.27	Wrench keys (insertion keys, Allen keys) 621.883.271/.274 ≡ 621.882.211/.214	621.89.094	For heavy machinery and transmissions
621.883.3	Keys for slotted nuts, castle nuts. Insertion spanners. Tommies	621.89.095	For different temperatures and climatic conditions
621.883.4	Adjustable spanners. Monkey wrenches	621.89.095.1	For summer
621.883.43	Locked socket wrench. Ratchet wrench ⇒ 621.838.14	621.89.095.2	For winter
621.883.6	Pipe wrenches (cylinder wrenches)	621.89.095.4	For high temperatures
621.883.7	Screwdrivers (turnscrews)	621.89.095.5	For low temperatures
621.883.8	Tools for extracting broken screws	621.89.096	For cylinders
621.884	Rivets and riveting. Riveting tools ⇒ 621.976	621.89.097	For automobile, aircraft engines
621.884.2	Solid shank rivets	621.89.098	For steam turbines
621.884.4	Belt rivets	621.89.099	For other applications. Additives
621.884.6	Hollow rivets	621.891	Generalities. Friction in machines. Properties of lubricants ⇒ 532
621.884.7	Blind rivets <i>Head side not accessible</i>	621.892	Lubricants. □ Oils. Greases ◇ 621.892.012:536.468 Combustibility of lubricants ⇒ 665.637.6 ⇒ 665.765
621.884.8	Explosive-type rivets. Thermo-type rivets with bursting charge	621.892.1	Oils according to process of manufacture. □ Manufactured by polymerization.
621.885	Hook, eye, clip and ring fastenings and attachments. Clasps. Eyelets etc. ⇒ 621.643.066 ⇒ 621.643.4	621.892.2	Manufactured by hydrogenation Mineral hydrocarbon lubricants. Mineral oils and greases
621.886	Pins. Nails. Split pins. Keys and cotters ⇒ 621.813	621.892.21	Oils obtained from petroleum
621.886.1	Dowels, pins, studs (for existing holes)	621.892.22	Lignite oils. Shale oils
621.886.2	Nails. Brads. Tacks. Panel pins	621.892.25	Mineral greases
621.886.3	Cramps, staples etc.	621.892.28	Synthetic oils
621.886.5	Split pins. Cotter pins	621.892.3	Organic oils. Vegetable oils. Animal oils
621.886.6	Keys and cotters ⇒ 621.815	621.892.4	Mixtures of mineral and organic oils
621.887	Rings. Bushes, bushings. Collars, collets. Sleeves. □ Shrink-rings. Locating, centring rings. Adjusting rings. Spacing bushes. Split or spring bushes	621.892.5	Consistent, solidified oils and greases
621.888	Fixings with some movement	621.892.6	Soluble and emulsifiable oils. Oils for metalworking. Cutting oils etc.
621.888.1	Screws with some movement. Spindles	621.892.7	Graphite as lubricant. Graphite emulsions
		621.892.8	Oils and greases with special additives. □ Rubberized oils. High-vacuum greases. Oils and greases with high electrical conductivity. Lubricants with additives to prevent corrosion, lower solidification point etc.

621.892.9	Lubricants other than oils, greases and graphite. □ Metal film lubrication	621.9.044	High-energy-rate forming (HERF). Explosion forming ⇒ 621.7.044 ⇒ 621.98.044
621.893	Materials for decreasing friction. Antifrictional materials	621.9.047	Chemical and electrochemical working and machining. □ Etching. Electrolytic machining ⇒ 621.7.025
621.894	Anti-lubricant substances. Methods and materials for increasing the friction coefficient	621.9.048	Erosion machining, blasting. Spark erosion. Ultrasonic, vibration machining. Machining with particle beams. Electron-beam machining etc. ⇒ 621.547
621.895	Cutting oils 621.895 ≡ 621.892 <i>Class the use of cutting oils during the operation of specific machines at the corresponding subdivisions of 621.7 and 621.9 with the special auxiliary .079</i>	621.9.048.7	Electron-beam machining. Machining using other particle beams
621.896	Lubricating apparatus, containers, devices. □ Oil-cans. Grease-guns. Pumps <i>Class here independent lubricating devices. For lubricating devices as components of machines use the 62-72 auxiliary</i>	621.9.06	Machinery: design, layout, assemblies, main components, auxiliary units 621.9.06 ≡ 621.7.06
621.899	Reclamation, recuperation of used lubricants	621.9.07	Forming appliances. Copying appliances. Dies. Templates. Formers etc. 621.9.07 ≡ 621.7.07 ⇒ 621.715
621.9	Working or machining with chip formation. Abrasive working. Hammers and presses. □ Cutting. Grinding. Sheet working. Thread-forming <i>Details of machines, tools, equipment, processes etc. by -1/-9 (from 62-1/-8 and 66-9)</i> ⇒ 621.7	621.9.08	Measurement. Testing .08 is not applicable in 621.926/927 ⇒ 53.08 ⇒ 531.7 ⇒ 681
621.9.01	Principles and theory of tool-working and machining. Machinability. Workability etc. 621.9.01 ≡ 621.7.01	621.91	Cutting or machining with chip formation in general ⇒ 621.96
621.9.011	Machinability. Workability. Stresses and strains. Flow of material	621.911	Planing, shaping (technique, tools, equipment). Hand planes and chisels ⇒ 674.056
621.9.012	Working (machining) stroke, pass, travel, traverse	621.911.1	Hand planing, shaping: principles and technique
621.9.013	Angle, rake of tool. Direction of working	621.911.2	Hand planes. □ Fillisters. Ploughs. Spokeshaves. Adzes
621.9.014	Efficiency of tool, machine. □ Influencing factors. Power required	621.911.3	Cold chisels. Chipping chisels
621.9.015	Surface condition ⇒ 62-408 ⇒ 621.88.07	621.911.4	Scrapers
621.9.016	Working conditions (of material and surface)	621.911.5	Strippers. Drawshaves. Draw blades
621.9.019	Faulty working. Machining defects and their testing <i>Details by :620.19...</i>	621.911.7	Woodworking chisels. Gouges ⇒ 674.054
621.9.02	Cutting tools. Chip-forming tools ⇒ 621.713/.715	621.912	Planing machines. Shaping machines ◇ 621.912.025 Blades of planing and shaping machines
621.9.021	Hand tools for cutting etc. ⇒ 621.9.041	621.912.1	Planing machines with moving workpiece. □ Pedestal planers. Portal planers
621.9.022	Power tools for cutting etc. Machine tools ⇒ 621.9.042	621.912.2	Planing machines with traversing tool. □ Trying or truing machines. Thicknessing machines
621.9.025	Cutting-iron (tool proper). Types. Steel-tipped, hard-tipped tools etc.	621.912.3	Shaping machines (with horizontal quick-return ram)
621.9.025.1	Types of cutting tool. □ Profiled cutters (chasers). Pointed tools, for piercing and grooving. Edge tools. Face tools	621.913	Slotting and mortising machines
621.9.025.5	Cutting tools from tool steel ⇒ 669.018.25	621.914	Milling and hobbing
621.9.025.6	Steel-tipped tools. Tool-steel tips (brazed or welded)	621.914.1	Milling technique
621.9.025.7	Hard-tipped tools. Hardmetal, cermet and diamond tips ◇ 621.9.025.7:669.018.25 Hardmetal tips ◇ 621.9.025.7:669.018.9 Cermet tips	621.914.2	Milling cutters (mills), hand or machine. □ End-mills. Side-mills
621.9.029	Used, worn tools and their treatment ◇ 621.9.029:621.923.6 Tool grinding	621.914.3	Milling machines
621.9.04	Technique and special methods of tool-working and machining	621.914.4	Combined boring and milling machines
621.9.041	Hand tool-working ⇒ 621.9.021	621.914.5	Hobbing technique
621.9.042	Power tool-working ⇒ 621.9.022	621.914.6	Hobbing cutters. Hobs ◇ 621.914.6:621.833 Gear hobs
		621.914.7	Special profiling machines ◇ 621.914.7:621.882.1 Screw-cutting millers
		621.918	Filing and rasping. Technique, tools, machines and equipment ⇒ 674.055
		621.918.2	Files (especially hand files)
		621.918.4	Rasps. Grating tools
		621.918.8	Filing and rasping machines
		621.919	Broaching. Broaches ⇒ 621.951.7
		621.919.2	Broaches (especially hand broaches)

621.919.3	Broaching machines	621.924.3	Face-grinding machines and equipment with rotary disc. □ Rotary sanders, finishers 621.924.3 ≡ 621.924.5
621.92	Abrasive materials and tools. Grinding and allied techniques. Crushing, separating, mixing of abrasives	621.924.5	Cylindrical grinding and honing machines
621.921	Abrasives. Grinding materials. Polishing materials ⇒ 666.792	621.924.56	External grinding machines
621.921.1	Metallic abrasives. Metal powders, pastes etc.	621.924.57	Internal grinding machines
621.921.2	Abrasive grits and sands. Carborundum and other silicon abrasives and carbides. □ Emery. Sand. Glass. Basalt	621.924.6	Special profile grinding machines ◇ 621.924.6:621.833 Gear grinders
621.921.3	Other inorganic abrasives. □ Polishing rouge (hydrated iron oxide). Tripoli powder (tripolite, diatomite, infusorial earth). Corundum. Pumice. Chalk, whiting. Red lead (minium)	621.924.7	Cleaning and polishing in revolving drums. Barrel finishers. Tumblers
621.921.34	Diamond	621.924.8	Brush cleaning. Scratch and scaling brushes and brushing machines. Metal brushes etc.
621.921.4	Vegetable polishing materials. □ Cork. Wood particles	621.924.9	Erosion forming, blasting equipment. Sandblasting. Shot-blasting etc. Blast machines and equipment ⇒ 621.547
621.921.5	Leather, buff, chamois etc.	621.924.9.023	Abrasive particles. □ Sand. Shot
621.921.6	Papers and cloths coated with abrasive. □ Emery paper. Glasspaper. Sandpaper ⇒ 676.265	621.924.9.024	Blasting nozzle, jet
621.921.7	Textile fabrics for polishing. Polishing cloths, felts etc.	621.924.91	Blast machines, blowers: main types
621.921.8	Rubbers and plastics for polishing	621.924.911	Open blast machines. Hose or free-jet sandblasters.
621.922	Abrasive tools. □ Grinding and polishing slips, wheels, discs, stones <i>For power tools see 621.924</i>	621.924.912	Chest or box blast machines
621.922.02	Grinding, polishing tools or stones. □ Types. Manufacture. Treatment	621.924.913	Drum blast machines
621.922.023	Abrasive sticks, steels etc. Whetstones	621.924.914	Rotary table, turntable blast machines
621.922.024	Abrasive boards, flat slabs. Grinding slips. Oilstones	621.924.915	Sprocket-table blast machines (suction-type or pressure-type) with revolving or swinging nozzles
621.922.025	Abrasive wheels, discs, drums etc. Buffing discs. Sanding discs. Grinding wheels. Grindstones	621.924.916	Rolling-table blast machines
621.922.079	Accessory materials. □ Pastes. Oils	621.924.918	Centrifugal blast machines
621.922.1/.8	Abrasive tools according to material 621.922.1/.8 ≡ 621.921.1/.8	621.924.93	Water-jet blasting (wet blasting) ⇒ 621.7.024.2
621.923	Grinding, polishing and allied processes (and associated equipment)	621.924.95	Enclosures for blasting. □ Housings. Casings. Cabinets
621.923.04	Special grinding and polishing techniques 621.923.04 ≡ 621.9.04	621.926	Comminution (especially of hard materials). □ Crushing. Grinding. Milling. Pulverizing <i>Class here comminution in general, as well as comminution of hard materials</i> ⇒ 621.927 ⇒ 622.73 ⇒ 662.923 ⇒ 664.73
621.923.045	Grinding between centres	621.926.08	Degree of comminution (size reduction) 621.9.08 is not applicable at 621.926 and its subdivisions
621.923.046	Centreless grinding	621.926.082	Preliminary reduction to lumps, chunks. Coarse reduction, breaking, crushing
621.923.1	Grinding of flat and curved surfaces in general. Surface-grinding, with wheel (as substitute for milling, filing etc.)	621.926.083	Intermediate, secondary reduction. Breaking to smaller pieces. Medium fine crushing
621.923.12	Centreless grinding of cylindrical objects	621.926.084	Reduction to granular form. Fine crushing
621.923.2	Rough grinding. Preliminary trimming, burring etc.	621.926.085	Reduction to particle size. Fine grinding
621.923.3	Intermediate, semi-fine grinding. Facing	621.926.086	Reduction to fine powder. Superfine grinding. Pulverization
621.923.4	Fine grinding. Finishing. Smoothing	621.926.087	Mechanochemical (tribochemical) activating size reduction ⇒ 539.215
621.923.5	Honing (and hones). Superfinishing	621.926.1	Breaking and crushing by hand. □ With hammers, pounders, pestles and mortars ⇒ 621.927.1
621.923.6	Edge and tip grinding. Tool grinding, sharpening	621.926.2	Mechanical breakers and crushers in general. □ Jawbreakers. Breakers with shearing anvil
621.923.7	Polishing. Lapping. Buffing. Burnishing (extra-bright, mirror-finish). Roll-polishing	621.926.3	Crushers and mills with rollers, cones, balls etc. □ Roller mills (gyratory breakers, rolling crushers). Ring mills. Centrifugal ball mills. Mortar mills. Pug mills. Edge runners
621.923.9	Other special grinding or polishing techniques	621.926.4	Impact crushing mills. □ Stamp mills. Stamp batteries. Star crushers. Hammer mills. Disintegrators
621.924	Grinding and polishing machines (for cleaning, sharpening, providing surface key etc.). □ Bench grinders. Brush cleaning. Sandblasting	621.926.5	Ball mills with screens. Ball drums. Barrel or tube mills. Vibratory ball mills. Aerofall mills. Compound mills
621.924.1	Grinding machines, bench grinders, machine grinders in general		
621.924.2	Grinders with flat or plane surface and linear motion. □ Reciprocating, oscillating grinders, sanders etc. Continuous grinders, with endless band or belt		

621.926.7	Mills with shear or rubbing action. Friction mills. □ Disc mills. Cone mills. Bell mills. Drag mills	621.928.3	Centrifuges. Centrifugal separators. Cyclone separators and classifiers. □ Hydrocyclones. Gas cyclones. Whizzers. Hydroextractors ⇒ 621.928.93 ⇒ 648.32 ⇒ 66.022.63 ⇒ 66.067.5
621.926.76	Worm, screw or auger mills ⇒ 621.927.76	621.928.4	Separation by means of water, washing. Water separators, graders, classifiers. □ Elutriators ⇒ 622.75/77
621.926.8	Superfine reduction mills with ancillary separating devices	621.928.44	Concentrators. Thickeners ⇒ 66.082.2
621.926.88	Air-swept pulverizers (pulverizers with air current for particle removal). Eddy mills. Vortex mills	621.928.45	Classifying or sizing by combined water current and sedimentation
621.927	Comminution of soft materials. □ Crushing. Grinding. Milling. Pulverizing ⇒ 631.333.92 ⇒ 631.361.8 ⇒ 631.363.2 ⇒ 664.7	621.928.46	Settling. □ Settling cones. Settling ponds
621.927.08	Degree of comminution or size reduction 621.927.08 ≡ 621.926.08 621.9.08... is not applicable at 621.927 and its subdivisions	621.928.47	Washing in water current, running water. Strainers
621.927.1	Crushing and grinding of soft materials by hand ⇒ 621.926.1	621.928.48	Trough washing. Table washers. Launder washers ⇒ 622.763
621.927.3	Grinding plant and equipment with millstones	621.928.5	Flotation separation and separators. Separation using interfacial tension ⇒ 622.765
621.927.4	Milling, crushing and grinding plant with rollers	621.928.6	Separation and classification by air current. Vortex separators. Air-current separators
621.927.6	Cleaning devices and equipment	621.928.7	Electrostatic separation and separators ⇒ 621.359
621.927.7	Small special-purpose mills. □ Coffee-grinders. Pepper mills 621.927.7 ≡ 621.926.7 ⇒ 641.512	621.928.8	Magnetic separation and separators. Magnets as separators
621.927.76	Worm or screw mills. □ Meat mincers	621.928.9	Dust separation, extraction and collection. Dust extractors. Dust collectors ⇒ 628.511
621.928	Separating, sorting, sifting, screening equipment. □ Sieves, screens, meshes. Separators. Extractors ⇒ 622.72 ⇒ 66.06 ⇒ 66.07	621.928.92	Dust separation without filter. Gravity-type extractors and collectors
621.928.028	Separating parts or elements. Screen decks, accessories	621.928.93	Centrifugal dust separation. Cyclone dust extractors and collectors
621.928.028.1	Bar or grating screens. Grizzlies	621.928.94	Dust separation with cloth filter (dust bags, tubes etc.)
621.928.028.2	Perforated plates. Riddle-plates	621.928.95	Dust separation with dry filter (except cloth)
621.928.028.3	Wire mesh or gauze screens	621.928.96	Dust separation using resonance effect, baffles etc.
621.928.028.4	Screening cloths or filters	621.928.97	Dust separation using liquids. □ Using wet filters
621.928.028.7	Control screens (oversize, guard, check screens)	621.929	Mixing and blending of solids or semisolids. Amalgamation of solids with liquids ⇒ 66.063
621.928.1	Separation technique in general. Sifting methods based on various properties	621.929.2	Blade, knife, paddle mixers. □ Blungers
621.928.11	Separation, sorting, picking by hand. □ Stone-picking	621.929.3	Worm or screw mixing machines
621.928.12	Density difference methods. Gravity sorting	621.929.6	Special mixers for dry materials
621.928.13	Size difference methods. Sizing, size grading. Dimensional sorting	621.929.7	Vibrating, rocker, shaker mixers
621.928.14	Methods based on difference of resistance to fluid flow. Separation by air or liquid current	621.93	Saws, sawing and similar cutting techniques ⇒ 621.96/98 ⇒ 631.342 ⇒ 674.053
621.928.15	Methods based on elasticity differences	621.93.02	Design, construction of saw tools, blades, holders etc.
621.928.16	Methods based on hardness, brittleness, friability, cleavage	621.93.021	Hand saws and slicers in general, including handles and frames
621.928.17	Separation methods based on adhesion	621.93.022	Machine saws and slicers in general, including holders and frames
621.928.18	Precipitation methods	621.93.023	Blades in general: shape and parts (except teeth)
621.928.2	Sieves. Screens. Riddles. Sieving and screening techniques and equipment	621.93.023.1	General outline or contour
621.928.21	Stationary sieves and screens	621.93.023.2	Stiffening, tensioning of blade ⇒ 621.93.066
621.928.22	Continuous sieves and screens. □ Endless mesh belts	621.93.023.3	Cross-section of blade
621.928.23	Reciprocating sieves and screens. Jigging, shaking, vibrating screens. Rocker screens etc.	621.93.023.4	Body of blade
621.928.24	Cylindrical and prismatic sieves and screens. Rotary screens. Drum or trommel screens. Conical sieves etc.	621.93.023.5	End of blade
621.928.26	Roll screens (roll-bar screens). □ Live-roll grizzlies		

621.93.024	Saw teeth. Blades according to teeth. Toothed blades (except heat saws). □ Setting. Truing. Sharpening	621.937	Slicers. Slicing ⇒ 641.512
621.93.025	Blades, teeth of special material and construction	621.937.2	Hand tools and equipment for slicing and cutting ⇒ 672.7
621.93.025.7	Hard-edged blades. Hard-tipped teeth. □ Hardmetal, cermet, diamond tips	621.937.3	Reciprocating machine slicers and cutters
621.93.026	Sawing using abrasion or heat. □ Wire rope saws. Abrasive cutter discs (cutting sheets). Heat saws (friction or fusion saws). Electric arc saws	621.937.4	Circular, rotary slicers and cutters
621.93.026.2	Abrasion saws, sawing	621.937.5	Band slicers and cutters
621.93.026.3	Friction saws, sawing: blades not toothed	621.937.6	Cutting wires. Tools with cutting wires. □ Cheese cutters
621.93.026.4	Friction saws, sawing: blades with blunt teeth. Fusion cutting saws	621.941	Lathes. Lathework. Turning, facing, boring etc. ◇ 621.941.025 Turning tools, cutting-irons etc. ⇒ 674.057
621.93.026.6	Electric arc saws, sawing (with toothed blades)	621.941.21	Facing lathes
621.93.06	Design, lay out, main components of machinery. Ancillary units	621.941.22	Centre lathes. Engine lathes
621.93.066	Tensioning devices. Blade tensioners	621.941.23	Semiautomatic and automatic lathes. Capstan lathes. Turret lathes
621.931	Saws, sawing in general	621.941.24	Universal lathes. Contouring, form-turning lathes. Copying, repetition lathes
621.932	Hand saws and sawing	621.941.25	Screw-cutting lathes. Brass-finishers' lathes
621.932.2	Hand stiff saws. One-man cross-cut saws. Woodworkers' saws	621.941.26	Precision lathes. Watchmakers' lathes
621.932.3	Pull saws (slack saws). Two-man cross-cut saws. Rip saws	621.941.27	Special-purpose lathes. □ Wheel-turning lathes. Lathes for turning cranks, multiaxial components. Turning and boring lathes. Horizontal boring machines
621.932.4	Smaller span saws, frame saws, bow saws. □ Metal-cutting saws. Hacksaws. Locksmiths' frame saws. Gardeners' frame saws. Pruning and lopping saws. Woodworking frame saws. Hand fret-saws. Coping saws	621.941.28	Vertical lathes. Vertical turning and boring mills
621.932.5	Larger span saws, frame saws, bow saws. □ Foresters' saws. Felling and logging saws. Builders', carpenters', joiners' frame saws. Hand turning saws, sweep saws. Mitre saws. Bevel saws	621.95	Drills. Drilling, boring, reaming etc. ⇒ 622.24 ⇒ 674.055 ⇒ 679.8.055
621.933	Reciprocating machine saws (stroke-cut machine saws). Power-driven stroke saws	621.951	Mechanical drilling in general. Hand boring tools. Hand drills, reamers etc.
621.933.2	Rigid, stiff-blade reciprocating saws. □ Nibbling saws. Scroll saws	621.951.2	Awls. Bradawls. Spikes. Other plain boring tools
621.933.3	Pull, draw-stroke or slack-blade reciprocating saws. Machine saws with spring counterpull	621.951.3	Gimlets. Augers. Other twist-edged boring tools. Drill bits for wood, earth etc. Centre bits. Spoon bits. Dowel bits
621.933.4	Bow saw machines. Hacksaw machines. Power-drive bow saws and hacksaws	621.951.4	Drills and bits chiefly for metalwork. □ Flat-end drills. Pointed-tip drills. Centre drills. Tap borers. Twist drills
621.933.5	Machine turning saws, sweep saws. Scroll saws with turning stroke. Power fret-saws. Jig saws	621.951.41	Flat-end drills, single-cutting or double-cutting
621.933.6	Gang saws. Saw gangs	621.951.42	Pointed-tip drills, common bits
621.933.7	Nibbling saws with gang feed	621.951.43	Centre drills
621.934	Circular saws. Rotary saws	621.951.44	Pin drills. Tap borers
621.934.2	Circular saws according to the design of the blade. □ Plate saws. Worm saws. Saw with internal and external cutting edge	621.951.45	Twist drills
621.934.3	Circular saws with work fed to the saw blade (non-stroke circular saws)	621.951.47	Countersinks. Rose bits
621.934.4	Circular saws with saw blade brought to the workpiece (circular stroke saws). □ Traversing circular saws. Hand circular saws	621.951.5	Hand and breast drills and braces. Hand drills. Breast drills. Bit braces (carpenters' braces)
621.934.5	Circular saws with both work feed and saw traverse	621.951.6	Engineers' ratchet braces (and drilling pillars)
621.934.8	Oscillating circular saws (wobble saws, drunken saws)	621.951.7	Reamers. Reaming ⇒ 621.919
621.934.9	Dished and domed saws. □ Concave saws. Barrel saws. Cylinder saws	621.952	Drilling machines
621.935	Band saws	621.952.2	Radial drilling machines
621.935.1	Vertical band saws	621.952.3	Vertical drilling machines (column and pillar drilling machines)
621.935.2	Horizontal band saws	621.952.5	Reamer drills (drilling-reaming machines)
621.935.3	Veneer band saws	621.952.6	Tapping drills (drilling tapping machines)
621.936	Rotary flexible saws. Cable saws. Chain saws	621.952.7	Capstan, turret drills (drilling machines)
621.936.4	Cable saws. Wire rope saws	621.952.8	Deep-drilling machines
621.936.6	Chain saws	621.953	Portable drilling machines (hand-held power drills)
		621.957	Centring devices. Centre punches
		621.957.2	Centre-drilling devices ⇒ 621.951.43
		621.957.3	Centre-punching devices ⇒ 621.961.3
		621.957.6	Centring machines
		621.96	Cutting without chip formation. □ Punching, blanking, shearing ⇒ 621.91

621.961	Punches, punching and blanking in general ◇ 621.961.025.7 Hard-tipped punches	621.979	Presses: types, design, construction ◇ 621.979:621.73.06 Forging presses ◇ 621.979:621.983.5 Embossing presses ◇ 621.979.07 Press-tools. Dies, punch-tools and other formers
621.961.2	Cutting, perforating and blanking punches, punching	621.979.1	Direct-action presses without stock displacement during press stroke. □ Lever presses. Toggle presses. Eccentric, cam and crank presses. Rack presses. Screw, fly and friction presses. Impact presses, stamps. Ram, piston or plunger presses
621.961.3	Noncutting punches, punching. Pin punches ⇒ 621.957.3	621.979.2	Direct-action presses with stock displacement during press stroke. □ Rotary, drum or roller presses. Calendaring presses. Continuous, belt or band presses. Extruding presses. Batch-work ram presses
621.961.4	Drifts. □ For aligning of rivet holes	621.979.21	Rotary drum (roller) presses. Calendaring presses
621.962	Hand implements for perforating. □ Hand punches. Hole clippers	621.979.23	Continuous (belt, band) presses
621.963	Power punching and blanking machines. □ Machine punches, perforators ⇒ 621.979	621.979.25	Extruding presses. Extruding processes ⇒ 621.777
621.965	Shears and shearing in general. □ For parting, paring, trimming	621.979.26	Ram presses with displacement of stock (batch work)
621.966	Hand and bench shears. □ Guillotine trimmers. Snips. Secateurs 621.966 ≡ 621.967	621.979.27	Screw presses, fly presses with continuous displacement of stock
621.967	Power shears. Machine shears. □ Power guillotines, trimmers, knibblers	621.979.3	Indirect-action presses. □ Jolt-ram and jolt-squeeze presses. Centrifugal presses. Injection-moulding presses ⇒ 621.744.4 ⇒ 678.057
621.967.1	Parallel-bladed shears. Bloom and billet shears. Curve-cutting shears	621.979.6	Presses for separation and expression of liquids. □ Filter presses. Drying presses ⇒ 66.067.4 ⇒ 66.068
621.967.2	Oblique-bladed shears, scissors. □ Lever shears. Jaw shears	621.979.6.068	Separation devices. □ Filter vessels, containers. Draining tubs. Strainers
621.967.3	Rotary disc shears. Circular-bladed shears. Strip shears. Flying shears	621.979.62	Closed presses. □ Frame presses. Screen presses. Trough presses
621.967.8	Combination punching shears (blanking shears)	621.979.63	Open presses
621.968	Cleaving, slitting and splitting implements and machines. □ Axes. Hatchets. Choppers. Adzes ⇒ 631.342 ⇒ 672.719 ⇒ 674.053	621.979.64	Semi-open presses. □ American box presses. Step or stage presses
621.97	Hammers and presses. Design and construction of tools and machines (for cold working, hot working, forming, pressing, stamping etc.)	621.98	Sheet working, bending, stamping etc. (chiefly of metals). Sheet-metal working tools, machines and processes generally ⇒ 621.771.23 ⇒ 621.96 ⇒ 621.97
621.972	Hand hammers. Mallets. Mauls ◇ 621.972.075 Hammer heads (weight, form, general balance etc.) ⇒ 674.058	621.98.04	Techniques of sheet working 621.98.04 ≡ 621.7.04
621.972.3	Single-head hammers. □ Pein, claw hammers etc. Club hammers. Scaling and chipping hammers	621.98.043	Die forming. Die pressing. Die stamping
621.972.4	Two-hand hammers (metal and stone-head hammers). Sledge-hammers	621.98.044	High-energy-rate forming (HERF). □ Explosion forming. Electromagnetic forming
621.972.7	Mallets. Mauls (malls)	621.981	Bending. Curving. Corrugating. Folding. Flanging. Beading. Seaming
621.974	Mechanical hammers. Machine, power hammers	621.981.2	Bending in roll-bending machines. Round-bending, curving machines
621.974.062	Lift and drop mechanism of mechanical hammers. □ Friction-operated mechanism	621.981.3	Corrugating, crimping machines and processes. Production of corrugated sheet
621.974.063	Hammer weight, ram or tup	621.981.6	Folding, flanging, beading machines and processes. Pleating and creasing presses and tools. Edge curling. Seaming
621.974.1	Lever hammers (e.g. for open forging, sheet-working). □ Front or face hammers. Side hammers. Tail hammers	621.982	Straightening. Flattening. Planishing. □ Stretch-flattening. Beating out
621.974.2	Drop hammers (falling tups) and stamps (e.g. for die forging, pile driving)	621.983	Cold hammer and press forming of sheet, strip and plate. □ Die forming. Stamping. Beating. Deep drawing. Embossing. Stretch forming ⇒ 621.97
621.974.3	Portable, hand-held power hammers	621.983.1	Cold hammer working and beating (with or without formers)
621.974.4	Steam, air and gas hammers. Hydraulic hammers		
621.974.5	Spring-operated hammers		
621.974.8	Forging machines for complete forging operations. □ Reciprocating impact hammers. Pressure-forging, squeeze-forging machines. Roll forging presses		
621.976	Riveting equipment ⇒ 621.884		
621.976.2	Riveter-hammers. Rivet dies. Rivet dollies		
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622.276.42	Gas drive. Air drive	622.32	Liquid and gaseous minerals
622.276.43	Water flooding <i>Introduction of water into the oil-bearing strata by means of pressure tubes with the object of forcing the oil in the direction of the hoisting tubes</i>	622.321	Liquid minerals in general
622.276.5	Lifting methods. □ Spontaneous oil production. Gas or air lift. Lifting by pumps, scoops etc.		
622.276.6	Stimulation methods. Influencing the area near the well. Borehole treatment		
622.276.63	Treatment with acids		
622.276.64	Pressure injection of surface-active agents into oil-bearing strata		

622.322	Mineral springs 622.322 $\equiv$ 553.7	622.412	Means of detecting firedamp $\Rightarrow$ 622.817
622.323	Petroleum (with or without natural gas)	622.413	Physical properties of mine atmosphere
622.324	Gases	622.414	Nature of air flow. Air pressure. Measurement, instruments (e.g. manometers, anemometers)
622.324.5	Natural gas. Methane	622.416	Calculation of air quantities in mines (in relation to staff, output etc.)
622.324.6	Carbon dioxide	622.417	Survey and measuring stations
622.324.7	Helium and other inert (rare) gases	622.418	Air conditioning $\Rightarrow$ 622.481
622.33	Solid minerals of organic origin. Carboniferous minerals. Coals. Carbonaceous rocks $\Rightarrow$ 662.62	622.42	Natural ventilation
622.331	Peat	622.43	Primitive ventilation, by heat and pumps
622.332	Lignite. Brown coal	622.44	Ventilation by fans, blowers, compressors $\Rightarrow$ 621.51/.54 $\Rightarrow$ 621.61/.63
622.333	Coals. Black coal generally	622.45	Introduction and distribution of air through shafts, galleries and workings. Airways
622.335	Anthracite. Graphitized coal	622.451	Mine ventilation networks
622.336	Graphite. Black lead. Plumbago	622.452	Introduction of air through shafts. Ventilation shafts
622.337	Bitumen. Asphalt. Bituminous shales. Oil-shale. Ozokerite (mineral wax). Resins	622.453	Air distribution in the mine. Airways. Ducts
622.339	Other organic minerals. $\square$ Amber	622.454	Methods, schemes of ventilation
622.34	Metals and metalloids. Ores (metalliferous minerals) <i>In the subdivisions derived from 553... denote complex polymetallic deposits by adding '3/9, as described at 553</i>	622.455	Airtight barriers, dams. $\square$ Air-doors. Airlocks. Brattices. Duct and shaft closures and seals
622.341	Iron ores. Manganese ores 622.341.1/.2 $\equiv$ 553.31/.32	622.456	Air leakages and their control
622.342	Gold ores. Auriferous conglomerates. Silver ores. Argentiferous conglomerates 622.342.1/.2 $\equiv$ 553.411/.412	622.457	Ventilation of underground workings. $\square$ Ventube ventilation. Auxiliary (local) ventilation
622.343/.349	Other ores 622.343/.349 $\equiv$ 553.43/.499	622.458	Ventilation of opencast workings
622.345	Tin ores	622.46	Other ventilation installations. $\square$ Ozone apparatus. Air filters
622.348.1	Nickel ores	622.47	Lighting, illumination of mines. Miners' lamps $\Rightarrow$ 628.9 $\Rightarrow$ 683.8
622.349.2	Aluminium ores	622.471	Lamps with exposed flames. $\square$ Lanterns. Oil lamps. Acetylene lamps
622.349.5	Radioactive minerals. $\square$ Uranium	622.473	Safety lamps. $\square$ Davy lamps. Petrol or gasoline safety lamps. Acetylene safety lamps
622.35	Natural stones. Building stones. Monumental stones. Freestones 622.35 $\equiv$ 553.5 $\Rightarrow$ 679.85	622.474	Portable electric lamps. $\square$ Battery lamps. Cap or helmet lamps. Hand lamps (electric torches)
622.36	Various inorganic economic minerals and earths (mainly nonmetalliferous) 622.36 $\equiv$ 553.6	622.475	Lamp rooms. Lamp cleaning, maintenance, trimming, filling etc.
622.363.1	Rock salt (halite) and other sodium salts	622.476	Permanent illumination underground, especially by acetylene and other nonelectric methods
622.364	Phosphates. Phosphate rock	622.477	Permanent electric lighting <i>Details by :621.32...; :628.9...</i>
622.37	Precious and semiprecious stones. Gemstones 622.37 $\equiv$ 553.8 $\Rightarrow$ 679.87	622.478	Insulation and similar lighting problems in mines <i>Details by :621.315.6... or :621.316.93...</i>
622.4	Ventilation, air conditioning, heating and illumination of mines $\Rightarrow$ 622.8	622.48	Heating, water supply, signalling and other accessory services and installations
622.41	Ventilation and air conditioning of mines in general	622.481	Heating of mineshafts
622.411	Principles. Theoretical data. Types of damp and dust encountered. Quantities of air required. Air distribution	622.485	Water supply
622.411.3	Mine gases, damp	622.489	Signalling, communications equipment $\Rightarrow$ 621.39 $\Rightarrow$ 628.975 $\Rightarrow$ 654 $\Rightarrow$ 658.284
622.411.4	Composition of firedamp	622.5	Mine drainage. Unwatering (dewatering) $\Rightarrow$ 622.84
622.411.5	Mine dusts. Sources of dust formation $\Rightarrow$ 616.24-003.6 $\Rightarrow$ 622.807 $\Rightarrow$ 628.511	622.51	Sources, composition and flow of mine water
622.411.51	Air sampling. Dust determination and estimation. Composition of atmosphere. Instruments. $\square$ Konimeters (conimeters), dust counters; filter methods	622.512	Determination of water flow into mines
622.411.52	Carbonaceous dusts. Coal dust	622.513	Surface measures
622.411.53	Stone dusts	622.514	Underground measures to limit or stop water flow into mines

622.515	Dams. Gate doors. Watertight barriers	622.64	Continuous and semicontinuous conveying
622.516	Sumps. Lodges	622.646	Ore chutes, drawing holes and openings.
622.517	Removal of sludge and slime		Loading out of ore
622.53	Mine water drainage	622.647	Conveyors etc.
622.531	Surface installations. □ Cornish pumps. Beam engines		⇒ 621.867
622.532	Underground pumps	622.647.1	Drag or scraper conveyors
622.534	Pipelines for drainage	622.647.2	Belt conveyors
	<i>Details by :621.64...</i>	622.647.3	Bucket conveyors
622.536	Unwatering during shaft sinking	622.647.4	Worm, auger or screw conveyors
622.537	Draining of flooded workings	622.647.5	Trough conveyors
622.538	Removal of acid waters	622.647.6	Roller conveyors
622.539	Other equipment for water raising	622.647.7	Shaking, oscillating, vibrating conveyors
	<i>Details by :621.649...</i>	622.647.8	Bunker conveyors. Conveyor trains
	◇ 622.539:621.649.6 Bucket wheels	622.647.9	Ejection conveyors. Centrifugal conveyors
		622.648	Pipeline conveying with and without carrying medium
622.58	Drainage in opencast mining		⇒ 621.643/644
622.583	Methods of drainage. □ Filters. Grout curtains	622.648.2	⇒ 621.867.7/8
622.584	Surface drainage		Hydraulic conveying. Hydraulic winding. Coal pumps
622.585	Primary drainage in opencast mines	622.648.3	Pipeline conveying without carrying medium
622.586	Continuous drainage	622.648.6	Pneumatic transport arrangements
622.587	Decreasing the pressure of subdeposit waters and observation of groundwater		⇒ 621.867.8
622.6	Haulage and handling. Hoisting. Winding. Surface transport	622.67	Hoisting. Winding. Vertical transport and haulage
	<i>The auxiliaries listed under 622.06... are not applicable at 622.6 and its subdivisions. Denote haulage, transport etc. according to motive power or propulsive force by -8... (from 62-8...)</i>		⇒ 621.876
622.61	Underground transport, haulage and handling in general	622.67.05	Elements and devices for flexible transmission
	◇ 622.61:622.341 Underground transport of iron ores		622.67.05 ≡ 621.85.05
622.611	Haulage and handling at the working face	622.67.06	Parts of conveying devices
622.612	Loading. Filling of cars, tubs etc.		622.67.06 ≡ 621.86.06
622.619	Loading machines, equipment. □ Loaders. Duckbills. Scrapers. Transloaders	622.67.07	Parts of lifts, hoists
	⇒ 621.878.4		622.67.07 ≡ 621.86.07
622.62	Haulage on roadways and tracks. Trams and tubs. Locomotives. Cable railways	622.672	Ladders, stairs and other fixed devices
622.621	Devices for signalling. Central control and blocking systems	622.673	Winding gear
622.622	Primitive haulage. □ In baskets or sacks. By humans or pack animals	622.673.1	Winding engines. Winders
622.624	Gravity inclines (brake inclines)		⇒ 621.863
	⇒ 622.625.5	622.673.2	Headframes. Winding towers
	⇒ 622.646/647	622.673.3	Head sheaves. Pulleys
622.625	Underground railways and tramways. Cable, rope and chain railways	622.673.4	Overhead gears
622.625.1	Tracks, rails. Track construction	622.673.6	Ropes. Cables
	622.625.1 ≡ 625.1	622.673.7	Tensioning devices
622.625.2	Rolling stock. Locomotives etc.	622.673.8	Winding drums. □ Koepe hoists
622.625.24	Rolling stock. Mine cars, trucks, wagons	622.674	Cage guides. Guide beams (spears). Bunttons
622.625.28	Mine locomotives	622.675	Loading stations. Shaft stations. Landings
622.625.5	Cable, rope and chain railways. Cableways	622.675.2	Pit bottoms (shaft bottoms)
622.625.5.05	Haulage ropes, cables	622.675.3	Pit-banks
622.625.5.06	Haulage chains	622.675.4	Decking and undecking of the winding cage.
622.625.51	Haulage engines. Incline engines		□ Scotchies. Car rams
622.625.52	Direct (inclined) haulage	622.675.5	Turning at loading and banking stations.
622.625.53	Main and tail haulage		□ Tipplers
622.625.54	Endless-rope haulage		⇒ 621.869.2
622.625.57	Suspended cableways	622.675.6	Car cleaning
622.627	Underground trackless transport. Underground roads and vehicles. Shuttle cars	622.675.7	Shaft or pit doors
	<i>Details by :625.7/8, :629.3...</i>	622.675.8	Kep gears. □ Keps (catches). Tilting decks
	⇒ 622.26	622.676	Various hoisting and lifting devices
			622.676 ≡ 621.876
		622.678	Use of hoists. Raising and lowering
		622.678.2	Raising and lowering of personnel
		622.678.5	Skip winding. Cage winding
		622.678.52	Cage winding. □ Cage catchers
		622.678.53	Skip winding
		622.678.54	Skip-and-cage winding
		622.678.6	Disposal of ore, coal etc.
		622.678.7	Lowering of supplies, e.g. timber, machines
		622.68	Opencast transport
		622.682	Conveyor transport
		622.683	Rail transport
		622.684	Motor transport
		622.685	Hydraulic transport
		622.686	Pneumatic transport

622.69	Surface storage, handling and transport of minerals (at or from the mining plant, pithead, well, quarry or mill, including pipeline transport) <i>Denote the state of the mineral to be stored or transported by '1... from 622'1</i> ⇒ 622.063.8 ⇒ 621.875	622.7.092	Control of products <i>Details by :543...</i> ◇ 622.7.092:543.632.22 By combustion and ignition in air. Ash determination
622.691	Storage, transport etc. of natural gases. □ Gasholders. Gas pipelines 622.691 ≡ 622.692	622.7.094 622.7.097	Control of quantities produced (output, yield) Control of tailings, waste products
622.691.2	Storage of natural gases ◇ 622.691.2-404 Storage of natural gases in liquid, liquefied state ◇ 622.691.2-405 Storage of natural gases in solid, solidified state	622.72	Sorting. Sifting. Picking ⇒ 621.928 Hand picking Mechanical sorting Electronic sorting ◇ 622.725:535.3 Sorting by reflection, refraction etc.
622.692	Storage, transport etc. of liquid minerals ◇ 622.692:622.323 Storage, transport etc. of petroleum ⇒ 628.12/14 ⇒ 665.6	622.721 622.723 622.725	
622.692.1	Collection in the field, at the mine-head, pit-head or well-head	622.73	Comminution. Crushing. Grinding. Milling. Pulverizing ◇ 622.73:621.926.3 Comminution in crushers and mills with rollers, cones, balls etc. ⇒ 621.926
622.692.2	Storage of liquid minerals. □ Containers: tanks, reservoirs. Storage losses <i>Details by :621.642...</i>	622.74	Classification and sizing by means of sieves and screens. Sieves and screens for separating, grading, washing ⇒ 621.928.2
622.692.4	Pipeline transport of liquid minerals. Liquid pipelines ⇒ 621.644	622.75/77	Processing by means of liquids and gases. Wet processing. Separation by settling of solids under gravity ⇒ 621.928.3/8 Classification by settling, in general Sedimentation in water. Methods and equipment using weak water current Elutriation. Methods and equipment using a vertical or almost vertical water current ⇒ 622.762 Wet centrifugal separation. Hydrocyclone separation ⇒ 621.928.3
622.692.5	Liquids terminals and their operation ⇒ 621.645	622.751 622.752	Jigging. Jigging machines Trough, table or launder washers (mainly horizontal water current) ⇒ 621.928.48
622.692.6	Other means of surface transport of liquid minerals, oils etc. <i>Details by :629... and :656...</i>	622.753	Concentrating tables. Tabling Fixed tables. □ Cornish buddles. Slime tables Band or belt tables. Vanners Shaker tables. Rockers Circular tables. □ Funnel tables Strakes. Tyes Flow tables Tabling in connection with flotation ⇒ 622.765.6
622.693	Storage, transport etc. of solid minerals, ores	622.755	Flotation. Concentration methods based on surface tension ⇒ 621.928.5
622.693.2	Storage of solid minerals, ores. Containers. □ Bins. Sacks. Silos. Bunkers. Hoppers	622.762 622.763	
622.693.4	Pipeline transport of fluosolids, suspensions, slurries, muds etc.	622.764 622.764.2 622.764.3 622.764.4 622.764.5 622.764.6 622.764.7 622.764.8	
622.693.6	Other means of surface transport of solid minerals and ores <i>Details by :621.86/87, :629... and :656...</i>	622.765	
622.7	Treatment, processing of minerals. Mineral dressing. Ore preparation <i>Details of processing variables and characteristics by -9... (from 66-9). Denote mineral by :622.3...</i> ◇ 622.7:622.337 Treatment of bitumens, e.g. oil-shale ⇒ 622.027 ⇒ 669.052	622.765.061 622.765.063 622.765.066 622.765.2 622.765.3 622.765.4 622.765.5 622.765.6 622.765.8	Flotation media, reagents etc. □ Frothers. Depressants. Activators. Regulators. Collectors Kinds of flotation. □ Bulk, selective, direct, backward, primary, secondary, cleaner flotation Defrothing. Removal of flotation agents Oil flotation. Processes using large quantities of oil with preliminary mixing of ore, pulp, oil and water Skin flotation (flotation using surface tension alone) Froth flotation Other flotation processes. □ Jigging, ionic, sinter and salt flotation. Processes combining oil flotation and magnetic separation (e.g. Murex process) Flotation in connection with tabling ⇒ 622.764.8 Adhesion processes
622.7.01	General problems <i>The auxiliaries listed under 622.01... are not applicable at 622.7 and its subdivisions</i>		
622.7.012	Consumption. Output. Efficiency 622.7.012 ≡ 66.012		
622.7.016	Chemical and textural composition of minerals. Methods of analysis, assay		
622.7.017	Technological properties of minerals. □ Washability		
622.7.05	Preliminary treatment. Transfer of mineral to concentration or separation plant		
622.7.06	Separation media. Concentration reagents. □ Preparation, introduction and recovery <i>The auxiliaries listed under 622.06... are not applicable at 622.7 and its subdivisions</i>		
622.7.09	Supervision and control by means of sampling ◇ 622.7.09:620.113 Sampling processes, basis and errors		

- 622.766 Separation using heavy media. Sink-float processes
 622.766.1 Pure liquids, e.g. carbon tetrachloride
 622.766.2 Solutions, e.g. of calcium chloride
 622.766.4 Suspensions. Emulsions. □ Barytes suspension. Sand suspension. Sludge. Mud
 622.766.7 Separation using dense media cyclones
 622.767 Pneumatic separation. Air-current separators
 ⇒ 621.928.6
 622.767.52 Processes using weak air currents or none. Sedimentation
 622.767.53 Processes using essentially vertical air currents
 622.767.55 Processes using rotary, cyclonic air currents. Centrifugal air separation. □ Trickling, centrifugal and vibrating screens. Screens with air circulation
 ⇒ 621.928.3
 622.767.62 Pneumatic jigs
 622.767.63 Processes using essentially horizontal air currents
 622.767.64 Pneumatic tables
 622.767.66 Processes using dispersion in gases. Air-buoyancy processes. □ Air-sand process
 622.771 Mechanical separation. □ Dry separation, based on shape or surface form. Methods using elasticity. Methods based on cleavability, hardness, fragility. Centrifugal separators. Spiral or screw separators
 ⇒ 621.928.1/3
 ⇒ 622.755
 622.772 Combined separation processes.
 □ Leach-precipitation-flotation (Mostowitsch process)
 622.775 Separation using solvents. □ Cyanidation
 622.776 Amalgamation
 622.777 Electrostatic separation
 ⇒ 621.359
 ⇒ 621.928.7
 622.778 Magnetic separation
 ⇒ 621.928.8
622.78 Heat treatment of ores and minerals
 ⇒ 669.162.1
 622.781 Preparation of ores and additions for roasting, sintering, shaping etc. Mixing with fuel, binding agents etc.
 622.782 Roasting. Calcining
 622.784 Suspension, flash or fluosolid roasting
 622.785 Sintering
 622.788 Manufacture of shaped agglomerates (briquettes, pellets, nodules). Briquetting, pelletizing etc. □ Sintering of green shapes
Details by :662.8.05...
622.79 Further preparation of ores and minerals. Aftertreatment. Storage. Disposal
 ⇒ 669.16
 622.791 Chemical treatment. Weathering. Corrosion
 622.793 Treatment of water
 622.794 Elimination of excess water and dust
 622.794.2 Filtration of slimes or overflow. Dewatering apparatus. □ Suction, centrifuge, filter apparatus
 ⇒ 66.067.3
 622.794.3 Removal of slimes, slurries, mud etc.
 622.794.4 Drying
 ⇒ 66.047
 622.794.7 Dust separation (dedusting) by gravity, centrifugal force, filters etc.
For dedusting as a health and safety measure in mines see 622.807.1
 ⇒ 621.928.9
 622.795 Mixing and blending processes. Mixing of ores, concentrates
 ⇒ 621.929
622.8 Mining hazards, accidents and damage. Health protection and safety in mines
 ⇒ 331.4
 ⇒ 351.823.3
 ⇒ 614.8
 622.807 Underground dust control in general. Dust precautions
 ⇒ 622.411.5
 ⇒ 628.511
 622.807.1 Dry dust collection. Dedusting
 ⇒ 622.794.7
 622.807.2 Wet dust collection and suppression. Spraying
 622.807.4 Water infusion. Infusion with other liquids
 622.807.5 Steam infusion. Infusion with other vapours
 622.807.7 Use of wetting agents
 622.807.8 Dust control by improving working processes
622.81 Gas and dust explosions underground. Mine explosions and safety measures
 ⇒ 614.83
 ⇒ 622.412
 622.812 Firedamp explosions
 622.814 Coal dust explosions. Sulphide dust explosions
 622.816 Outbursts of gases and liquids from boreholes
 622.817 Prevention and control of explosions
 622.817.3 Dusting with stone or rock dust
 622.817.4 Gas emission control
 622.817.5 Grading of mines according to dust and gas content. Mine categories
 622.818 Measures for containing, localizing damage. Automatic barriers. Sealing-off
622.82 Underground fires. Fires in mines. Mine firefighting
 ⇒ 614.84
 622.822 Mine fires and fire hazards in general
 622.822.2 Spontaneous fires (fires due to spontaneous combustion). Exogenetic fires (externally caused fires). Fire detection and control
 622.822.3 Fire prevention
 622.824 Fires and fire hazards in mine supports, accessory installations and equipment underground
 622.826 Fires and fire hazards at surface storage, handling and transport installations. Fires in spoil-banks, slagheaps, stockpiles, tanks etc.
 622.827 Fires and fire hazards in gas, oil and equipment of boreholes. Borehole fires. Well fires
622.83 Ground and rock movements. Rock pressure. Subsidence. Geostatic and geodynamic phenomena
 ⇒ 550.3
 ⇒ 551.24
 ⇒ 622.02/.03
 622.831 Pressure and displacement effects. Fractures, faults etc.
 622.831.2 Compression effects. □ Abutment pressure. Stress caused by cavities, shafts
 622.831.3 Strains and stresses. Rock pressure. Heave. Rockbursts
 622.831.6 Caves breaking surface. Forming of fissures, cavities etc.
 622.833 Ground movements associated with shaft sinking, drilling and driving operation. □ Creep. Stress distribution
 ◇ 622.833:622.25 Caused by shaft sinking

622.834	Ground movements caused by exploitation and working. □ Collapses. Cave-ins. Subsidence. Upheavals ◇ 622.834:622.273.21(091) Historical accounts of ground movements in relation to stowing and filling	622.88	Control, regulation, reconstruction of mining damage. □ Damage assessment, evaluation. Compensation. Restoration of mining
622.837	Damage due to mining operations ◇ 622.837:624.21 Damage to bridges	623	MILITARY ENGINEERING <i>Details of military machines and equipment by -1/-9 (from 62-1/-8 and 66-9)</i> ⇒ 355/359
622.838	Special safety measures against subsidence, rock movement etc. ◇ 622.838:69.059.22 Measures against damage to buildings	623.09	Special defence and assault equipment (of fortifications, vehicles, ships etc.)
622.838.5	Safety pillars	623.093	Armouring. Armoured equipment, vehicles etc.
622.838.6	Safety measures according to method of working. □ Harmonic extraction. Retreating methods. Continuous methods	623.094	Armament. Arming (with weapons). Armed vehicles etc.
622.839	Damage falsely attributed to mining. □ Damage due to weight of buildings, subsoil movement, vibration	623.1/7	Land-based military engineering. Artillery
622.84	Flooding and hydrologic regime in relation to rock movements ⇒ 622.5	623.1/3	Field engineering. Fortifications. Defence works
622.841	Influence of mining on water sources. □ Groundwater variations, exhaustion	623.1	Fortresses. Permanent fortifications. Fortified lines, areas ⇒ 711.559.6 ⇒ 725.1 ⇒ 728.81
622.843	Cave-ins and subsidence in relation to drainage	623.11	Principles. Theory. □ Sections. Profiles. Contours. Plans. Layout. Flanking defences (outworks) in general. Defilading
622.844	Cave-ins and subsidence in relation to dissolution	623.12	Details of fortresses and permanent fortifications. □ Masonry, walling. Earthworks. Cover from fire. Blindage. Armouring
622.847	Water penetration. Flooding (inundation)	623.126	Various defence constructions. Gun emplacements. □ Casemates. Bastions. Turrets. Gun mountings
622.85	Damage to vegetation and animals due to mining ⇒ 502.172	623.127	Approach obstacles. □ Palisades. Tank traps. Ditches
622.86	Mining accidents. Safety precautions. Relief and rescue work ⇒ 614.82 ⇒ 614.89	623.13	Fortified lines, strong points and communications
622.861	Accidents involving the miner's own movement or action. □ Falls. Injuries from tools	623.15	Temporary fortifications and fieldworks in general ⇒ 355.424.1
622.862	Accidents due to external causes. □ Injuries from explosions, fires (burns), rock falls etc. Hazards from water, gas. Drowning. Suffocation. Injuries from vehicles, moving machinery. Electric shocks	623.2	Assault and defence works. Siege works. Field fortifications. Trench works
622.864	Precautions, safety measures against mining accidents in general	623.21	Investment. Preparations. Establishing camp
622.865	Emergency exits, escapeways etc. (provision, inspection, maintenance)	623.22	Circumvallation. Countervallation
622.867	Relief organization. Rescue work (methods, equipment etc.). □ First aid posts. Personnel equipment, e.g. breathing apparatus, gas masks	623.23	Parallels. Batteries
622.868	Recovery work after fires etc. Unsealing	623.24	Trenches. Zigzags, saps etc. □ Communication trenches. Simple excavations. Foxholes, dugouts ⇒ 624.134
622.87	Occupational health hazards in mining. Preventive and palliative measures, e.g. modified working methods, control of ambient conditions ◇ 622.87:613.633 Pathogenic effects of dusts in mines ◇ 622.87:613.633:616.24-003.6 Dusts in mines as cause of pneumoconiosis ◇ 622.87:613.644 Injurious effects of noise and vibrations in mining. Including: Effects of percussion tools ◇ 622.87:613.65 Injurious posture or working position in mining. Including: Prevention of muscular strain, cramp, etc. ⇒ 613.6 ⇒ 616-057	623.25	Breaching batteries. Breaches. Assault works
		623.3	Land minefields. Field obstacles. Flooding
		623.36	Land minefields ⇒ 623.454.36
		623.363	Minelaying. □ Tapes and markers. Lanes and paths
		623.365	Mine breaking. Mine clearing
		623.368	Mine detection
		623.38	Field obstacles. Traps. □ Wire entanglements. Pitfalls
		623.39	Flooding as means of defence

623.4	Armaments. Ordnance. Arms. Weapons. Guns. Artillery. Small arms. Side arms. Armoured and other service vehicles. Ammunition and war gases. Guided missiles and rockets		
623.4.01	Design fundamentals. Characteristics and tests of weapons and military equipment	623.4.068	Devices for reducing noise, flash etc. Noise suppressors, dampers (silencers). Flash eliminators
623.4.011	Military characteristics	623.4.069	Extractors. Ejectors
623.4.012	Geometric parameters. Choice of size ⇒ 62-181	623.4.07	Various questions concerning firearms. Accessories. □ Tamping tools, ramrods, bolt levers. Pouches, ammunition holders. Holsters, pistol cases
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623.466	Flight curves. Stability. Control	623.55.024	Observation. Ranging. Fuse (fuze) adjustment
623.466.3	Flight curves. □ Ballistic curves. Pursuit curves (hound-and-hare, fixed-lead pursuit). Proportional navigation. Constant bearing (collision course)	623.55.025	Fire control devices
		623.55.027	Guiding of missiles
623.466.33	Ballistic curves	623.55.028	Target making
623.466.34	Pursuit curves. □ Pure (hare and hound) pursuit curves. Fixed lead pursuit curves		
623.466.36	Proportional navigation	623.551	Fire power. Fire control
623.466.38	Constant bearing (collision course)	623.552	Land firing (firing from ground)
623.466.5	Stability and control	623.553	Underwater firing
623.467	Launching of missiles and rockets ◇ 623.467.018 Prelaunch testing	623.554	Firing at sea
		623.555	Infantry firing
623.467.2	Tower and silo launching	623.556	Firing from aircraft
623.467.22	Tower launching	623.557	Bombing (aerial bombing) ⇒ 623.428.82 ⇒ 623.451.74
623.467.24	Silo launching		
623.467.3	Ramp, rail and table launching		
623.467.5	Gun or tube launching. Sled launching. Multiple launching		
623.467.7	Vehicle, ship and raft launching ⇒ 623.437.448		
623.467.8	Underwater launching ⇒ 623.827.2		

623.56	Effects of and on projectiles. Experimental studies	623.62	Radio and radar countermeasures
623.561	Effects on animate objects, living targets	623.623	Weapon systems or complexes as countermeasures against enemy radio and radar systems
623.562	Effects on inanimate objects, nonliving targets		
623.562.1	On fixed targets, earthworks etc.	623.624	Other radio and radar countermeasures. □ Neutralization. Jamming ⇒ 623.77
623.562.2	On masonry		
623.562.3	On armour-plating	623.64	Topography. Cartography. Surveying ⇒ 355.47 ⇒ 528 ⇒ 623.4.022 ⇒ 91
623.562.4	On concrete		
623.562.5	On movable targets (except armour), e.g. vehicles	623.642	Terrain and its tactical properties. Topographic ground reconnaissance. □ Visual surveying
623.562.6	In water, against underwater objects	623.643	Topographic maps. Map work. Orientation. Target referencing
623.562.7	Against explosives, propellants	623.644	Special maps and topographic documents. □ Maps for troops and headquarters. Route maps. Intelligence maps. Town plans (street maps). Aerial photographs, photoplans, photomaps
623.565	Effects on projectiles. Fragmentation. Blast		Devices and instruments for orientation of troops in the field. Surveying instruments
623.565.1	In confined spaces	623.646	Methods of determining initial geodetic data to provide support for combat actions
623.565.2	In the air	623.647	
623.565.3	At the target. Damage to projectile	623.67	Clearing and making of obstructions. Ammunition and bomb disposal
623.565.5	In earth		
623.565.6	In water	623.68	Protection, demolition and repair of buildings, construction works etc.
623.566	Underwater ballistics ⇒ 623.94/.95	623.683	Protection. □ With sandbags. With concrete blocks
623.566.1	Torpedoes	623.685	Demolition or destruction as means of defence ◇ 623.685:624.21 Bridge demolition ⇒ 623.454.37
623.566.2	Underwater bombs and shells	623.687	Repair and restoration of essential structures
623.566.3	Depth charges. Charges on sea bed	623.7	Military and naval aviation. Camouflage etc. ⇒ 358.4 ⇒ 359.38
623.566.4	Missiles and rockets, in water		
623.566.5	Underwater explosions. □ Effects on sea bed, on surface (water projected above surface)	623.74	Military and naval aircraft <i>Details by :629.7....</i> <i>Denote armouring and armament by .09...</i>
623.567	Controlled projectile trajectories ⇒ 531.55	623.741	Lighter-than-air aircraft (aerostats) in general
623.567.2	Initial stage	623.742	Balloons. □ Moored (e.g. barrage balloons). Towed, free balloons ⇒ 623.748
623.567.5	Mid-course	623.743	Airships (dirigibles)
623.567.8	Terminal stage	623.746	Heavier-than-air aircraft (aerodynes) ◇ 623.746-519 Heavier-than-air aircraft with remote control. Pilotless, unmanned aircraft
623.57	Behaviour of mountings during firing. □ Recoil ⇒ 623.4.067 ⇒ 623.435	623.746.1	Heavier-than-air aircraft according to type
623.59	Ranges. Firing practice equipment	623.746.12	Landplanes
623.591	Ranges in general. □ Firing ranges (shooting ranges, butts). Bombing ranges. Proving grounds (test ranges). Underwater test ranges	623.746.13	Seaplanes
623.592	Simulators and trainers	623.746.14	Amphibious aircraft
623.593	Targets	623.746.16	Gliders. Sailplanes
623.593.2	Fixed targets	623.746.17	Gyroplanes. Autogiros. Helicopters
623.593.5	Moving targets	623.746.2	Reconnaissance aircraft. Observation aircraft
623.593.51	On land	623.746.3	Fighter aircraft (fighters)
623.593.52	On water	623.746.34	Fighter-bombers
623.593.53	In the air	623.746.35	Fighter-interceptors
623.594	Markers' shelters and covers	623.746.4	Bomber aircraft (bombers)
623.6	Field engineering equipment and operations. Lines of communication. Radar countermeasures. Military topography	623.746.43	Dive-bombers
623.61	Military communications. Troop control	623.746.47	Suicide aircraft
623.611	Apparatus. Equipment <i>Details by :621.39...</i>	623.746.5	Transport aircraft. Freight, supply, troop, personnel carrier aircraft. Hospital aircraft
623.612	Technique. Organization <i>Details by :654...</i>	623.746.6	Anti-submarine aircraft
623.613	Field cable (wire) communication lines	623.746.7	Trainer aircraft
623.615	Communication centres: apparatus, crosses etc.	623.746.8	Tanker aircraft
623.616	Document communication stations. Military post stations		
623.618	Automatic troop control systems. Military information systems. □ Command posts. Control centres. Information networks		
623.618.7	Devices for interconnection of military computers (including mobile computers) with communication channels		

623.748	Devices for obstructing enemy aircraft. □ Barrages. Cables. Nets ⇒ 623.742	623.827	Underwater vessels. Submersibles. Submarines ⇒ 629.58
623.76	Air defence technique ⇒ 355.469	623.827.2	Missile-launching submarines ⇒ 623.467.8
623.762	Anti-aircraft defence systems	623.827.3	Attack submarines
623.764	Antimissile, anti-rocket defence systems	623.827.5	Minelaying submarines
623.765	Air defence control systems ⇒ 623.618	623.827.6	Midget submarines
623.77	Camouflage ⇒ 355.585 ⇒ 358.36 ⇒ 677.027.468	623.827.7	Landing submarines
623.772	Effect of weather conditions	623.828	Transport, depot and auxiliary vessels
623.773	Passive camouflage. Deceptive coloration, designs etc.	623.828.2	Landing craft. Amphibious transport craft
623.775	Artificial fog. Smoke screens ⇒ 623.454.5 ⇒ 662.17	623.828.5	Transport ships. □ Freight, supply, troop, personnel carrier vessels. Cargo ships. Troopships
623.776	Blackout ⇒ 355.585.6 ⇒ 628.978.7 ⇒ 645.398	623.828.6	Hospital ships
623.777	Active camouflage. Decoy construction. Dummy structures ⇒ 623.829.5	623.828.7	Training ships
623.8/.9	Naval engineering ⇒ 355.46	623.828.8	Maintenance and repair ships. Power station ships
623.8	Naval construction. Naval bases. Warships and other vessels ⇒ 629.5	623.828.9	Other auxiliary craft and vessels. □ For harbour entrance control. For barracks
623.8.01/.08	Design details, parts etc. 623.8.01/.08 ≡ 629.5.01/.08	623.829	Net and boom layers. Minelayers. Anti-mine defence ships. Other types of warship
623.81	Naval dockyards and workshop establishments. Coastal and floating installations and forts ⇒ 627.096 ⇒ 627.3 ⇒ 629.5.08	623.829.1	Net-laying and boom-laying vessels ⇒ 623.93 ⇒ 623.95
623.816	Temporary installations. □ Floating harbours. Mulberry harbours	623.829.2	Minelaying vessels
623.817	Coastal, foreshore and offshore strongpoints. □ Marine forts, redoubts. Sea walls	623.829.3	Minesweepers. Mine-hunting vessels
623.818	Floating bases. Floating batteries	623.829.5	Decoy craft. Q-ships (mystery ships)
623.82	Combat craft (fighting ships) and auxiliary vessels. Warships	623.829.8	Lake, river craft (shallow draught)
623.821	Combat force command ships. Battleships	623.9	Material and equipment of naval forces. Protection and armament of vessels. Submarine defences <i>For particular types of armoured or armed vessel use 623.8... with .09...</i>
623.821.1	Combat force command ships	623.91	Armouring, armour-plating of vessels. Armouring of all kinds, including conning towers, turrets, decks
623.821.2	Battleships	623.92	Partitioning of hull and superstructure. □ Cellular construction. Compartments. Bulkheads
623.821.3	Battle-cruisers	623.93	Outer defence of vessels and ports. Defence against submarines in general ⇒ 355.465 ⇒ 62-757.5 ⇒ 623.829.1
623.821.4	Pocket-battleships	623.931	Booms. Outrigger protection
623.822	Pursuit vessels. Cruisers. Aircraft-carriers	623.932	Anti-torpedo nets. Anti-submarine nets
623.822.3	Cruisers	623.933	Other anti-torpedo and anti-mine devices ⇒ 623.959
623.822.7	Aircraft-carriers	623.935	Methods and devices for concealment of vessels ⇒ 623.454.5 ⇒ 623.77
623.823	Destroyers. Corvettes. Frigates. Light anti-submarine vessels. □ Convoy escort vessels	623.936	Defence systems against mass-destruction weapons ⇒ 623.454.86
623.823.1	Destroyers	623.94	Naval artillery and armaments ⇒ 623.4 ⇒ 623.554
623.823.2	Corvettes	623.94.01/.08	Armament details 623.94.01/.08 ≡ 623.4.01/.08
623.823.3	Submarine-chasers	623.941	Artillery installations on board
623.823.6	Frigates	623.941.1	Munitions magazines
623.824	Armed coastal defence vessels. Gunboats. Sloops	623.941.2	Munitions hoists
623.825	Torpedo-carrying surface vessels. Torpedo boats. Coastal motorboats. Motor launches. E-boats	623.941.3	Gun positions on board. □ Conning towers. Fixed gun turrets, half-turrets. Sponsons ⇒ 623.433 ⇒ 623.434.5
623.826	Small patrol and reconnaissance vessels. Scout vessels. Fishery escort, patrol and protection vessels		

- 623.941.4 Mobile turrets
⇒ 623.432
- 623.942 Shipboard training and practice firing. Training equipment
- 623.946 Naval torpedoes, tubes and accessories
◇ 623.946-5 Control mechanisms. Steering, discharge, depth-control gear
◇ 623.946-8 Type of drive, propulsion
◇ 623.946.082 Warheads
⇒ 355.465
⇒ 623.451.72
⇒ 623.566.1
- 623.946.2 Spar-type, boom-type, towed and other early forms of torpedo
- 623.946.3 Self-propelled torpedoes (surface or submarine)
- 623.946.5 Special torpedoes. □ Net-cutting torpedoes
- 623.946.6 Torpedo tubes (surface or submarine)
- 623.95 Mines and depth charges. Minelaying. Minesweeping**
⇒ 355.465
- 623.952 Submarine mines in general. Mines moored or at rest on sea bed. Ground mines
- 623.953 Free-floating mines. Contact mines
- 623.955 Depth charges. □ Control of explosion depth. Depth charge release gear
⇒ 623.566.3
- 623.956 Detonation. Detonating mechanisms. Setting for time etc.
- 623.957 Minelaying, mine-sowing techniques
⇒ 623.829.2
- 623.958 Minesweeping, mine-searching and clearing techniques
⇒ 623.829.3
- 623.958.2 Mine searching. Mine detection
⇒ 623.983
⇒ 681.883
- 623.958.3 Mine sweeps
- 623.959 Methods of countering, deactivating mines.
□ Mine disposal. Degaussing (demagnetizing of ships)
⇒ 623.933
- 623.97 Naval attack and defence methods not included elsewhere**
- 623.971 Fire-ships and similar destructive devices. Infernal machines
- 623.974 Frogmen
- 623.98 Various applications of science. Special instruments, equipment**
Details by :...
- 623.983 Detectors of underwater objects (e.g. submarine detectors). □ Asdic. Sonar
◇ 623.983:621.396 With radiocommunication. Radio beacons, buoys etc.
◇ 623.983:681.883.43 With hydrophones
⇒ 681.883
- 624 CIVIL AND STRUCTURAL ENGINEERING IN GENERAL.**
□ Substructures. Earthworks. Foundations. Tunnelling. Bridge construction. Superstructures
Under 624 is classed information about the practice of civil and structural engineering, whereas 69 is for building materials, trades and construction, and 72 is for architectural design and kinds of building according to purpose

- 624-1 Civil engineering details**
-11... and -18... from 62-11 and 62-18 are applicable at 624 and subdivisions
624-1 ≡ 624.1 except for -11 and -18
- 624-15 Foundations**
- 624-2/-9 Details of construction machinery and equipment**
Details by -2/-9 (from 62-2/-8 and 66-9)
- 624.01/07 Special auxiliary subdivision for class 624 Civil and structural engineering**
The 624.0... special auxiliaries (except .02) have similar meanings under 69; preferably, use .01, .04 and .07 (which relate to all structures generally) mainly under 624, but .03 and .05 (which relate primarily to buildings) mainly under 69
- 624.01 Structures and structural parts according to material and method of construction**
⇒ 624.9
⇒ 691
⇒ 693
⇒ 694
- 624.011 Structures in materials of organic origin
- 624.011.1 Wooden structures
⇒ 674
- 624.011.2 Wood structures with substructures of other material
- 624.011.7 Structures in organic materials other than wood
- 624.011.72 Structures in straw, reed, bamboo etc.
- 624.011.73 Structures in cork and similar materials
- 624.011.74 Structures in paper, pasteboard etc.
□ Structures in tarred paper, roofing felt
- 624.011.75 Structures in other fibrous materials.
□ Structures in canvas (sailcloth), felt (e.g. iron felt), woven material
- 624.011.77 Structures in natural rubber
- 624.011.78 Structures in plastics
- 624.011.8 Structures in materials of animal origin.
□ Structures of skins (hide, leather), dung, shells, bones
- 624.011.9 Structures in organic mineral materials.
□ Structures in bitumen, asphalt
- 624.012 Masonry structures in natural stone, cast stone, artificial masses, ceramic materials
- 624.012.1 Structures in natural stone. Natural stone masonry
624.012.11/12 ≡ 553.5/6
- 624.012.2 Structures in artificial stone. Cast stone masonry
- 624.012.25 Structures in reinforced artificial stone
- 624.012.26 Structures in prestressed artificial stone
- 624.012.3 Structures in precast concrete
Denote lightweight by -183.4
⇒ 666.97
⇒ 691.32
- 624.012.35 Structures in precast reinforced concrete
- 624.012.36 Structures in prestressed precast concrete
- 624.012.4 Structures in concrete, in-situ concrete (ordinary concrete), mass concrete
Denote lightweight by -183.4
⇒ 691.32
- 624.012.44 Structures in mass concrete
- 624.012.45 Structures in reinforced concrete
⇒ 666.982
- 624.012.46 Structures in prestressed concrete
- 624.012.6 Glass structures
624.012.6 ≡ 666
- 624.012.8 Structures in earth, mud, clay, brick

624.012.81	Structures in dried earth, adobe, mud. Pisé-de-terre	624.042.63	Stresses due to construction. Enforced stresses
624.012.82	Structures in baked clay. Structures in ordinary and refractory bricks	624.042.68	Effects of notches, holes or openings on the conditions of stress
624.014	Metal structures ⇒ 693.8	624.042.7	Stress due to ground movements, earth tremors (earthquakes), seismic effects
624.014.1	Cast iron structures	624.042.8	Dynamic loads
624.014.2	Steel structures ◇ 624.014.2.078.4 Jointed steel structures (riveted, welded, screwed)	624.044	Calculation of deformation, strain (elastic and plastic deformations)
624.014.27	Tubular-steel structures	624.046	Structural strength. Load-bearing capacity
624.014.7	Structures in light metal (and light metal alloys)	624.046.2	Carrying capacity
624.014.9	Structures in other metal	624.046.3	Stability
624.016	Structures in composite materials. Composite structures	624.046.4	Rigidity
624.02	Structural parts and elements <i>For structural parts and elements of buildings prefer 692...</i> 624.02 ≡ 692	624.046.5	Safety factor. Load factor
624.024	Roofs. Kinds of roof	624.047	Movable and stationary structures
624.03	Size, permanence, location and shape of structures <i>See note at 624.01/07</i> 624.03 ≡ 69.03	624.047.1	Movable structures ⇒ 69.033
624.034.5	Underwater structures	624.047.2	Stationary structures
624.035.2	Mountain, hillside structures	624.048	Model statics. Use of models in structural design
624.035.4	Underground (subterranean) structures	624.05	Site, organization and plant. Construction methods 624.05 ≡ 69.05 <i>See note at 624.01/07</i>
624.04	Structural design. Graphical and analytical statics for investigation and calculation of structures ⇒ 531.2	624.07	Structural elements. Load-bearing members
624.041	Establishing the general design. Systems and grids. Statics of structures in general	624.071	Elements for taking only a single axial stress
624.041.1	Internally and externally statically determinate structures	624.071.2	Flexible elements. Elements for suspended structures. Flexible carrying members
624.041.2	Statically indeterminate structures in general	624.071.22	Ropes. Suspension ropes
624.041.3	Internally statically determinate, externally statically indeterminate structures	624.071.23	Cables. Suspension cables
624.041.4	Externally statically determinate, internally statically indeterminate structures	624.071.232	Carrying cables
624.041.5	Internally and externally statically indeterminate structures	624.071.234	Back-stay cables
624.041.6	Support reactions. Shear forces. Moments. Influence lines. Deflection curves. Diagram of resistance moments	624.071.25	Chains. Suspension chains
624.041.61	Support reactions. Lateral, shear forces. Shear lines. Intersection forces	624.071.3	Rigid elements
624.041.62	Moments. Moment lines	624.071.32	Elements under tensile stress
624.041.63	Influence lines	624.071.322	Carrying members
624.041.65	Elastic, deflection curves. Deflection lines	624.071.324	Suspension rods. Overhead rails
624.041.66	Comparability conditions. Lines of pressure. Thrust lines	624.071.34	Elements for taking compressive stress (without buckling stress). □ Compression bars
624.041.68	Stress distribution diagrams. Resistance moment diagrams	624.072	Elements for taking biaxial stresses
624.042	Calculation of loads and stresses	624.072.1	Plates and discs in general ⇒ 624.073
624.042.1	Loads and stresses in general. Internal stresses. Foundation pressure. Uplift	624.072.2	Elements subject to bending stress. Beams. Girders. Joists etc.
624.042.12	Internal stresses	624.072.21	Overhanging beams, one end free. Booms. Brackets. Cantilevers
624.042.2	Static loading. Dead load or weight (structure plus floor load plus internal fittings)	624.072.22	Fixed beams, supported at two or more points. Truss and trestle beams. Lintels
624.042.3	Incidental loads. Live, superimposed or moving loads. □ From vibrations, impact, traffic	624.072.221	Simply-supported beams
624.042.4	Loads from wind, snow, ice, water	624.072.222	Beams fixed at one or both ends
624.042.41	Wind loads. □ Wind pressure. Wind suction	624.072.223	Elastically fixed beams
624.042.42	Snow loads	624.072.226	Hanging trusses
624.042.43	Ice loads	624.072.227	Composite trusses
624.042.44	Water loads. □ Rainwater loads	624.072.23	Continuous beams on three or more supports with constant or variable moments of inertia
624.042.5	Temperature effects. Thermal stresses etc.	624.072.232	Beams on fixed supports allowing free rotation of beam
624.042.6	Various other static effects	624.072.233	Beams on supports subject to elastic settlement and rotation
624.042.62	Stresses due to deformations	624.072.233.5	Beams on continuous elastic foundation or bedding. □ Beams on railway track
		624.072.238	Continuous beams hinged together to be statically determinate. □ Gerber girders
		624.072.24	Plate girders
		624.072.3	Elements subject to bending or flexural stress and axial stress (without buckling stress)
		624.072.31	Short columns. Piers. Jambes. Thick walls
		624.072.32	Arches. Arched beams and girders
		624.072.322	Fixed arches etc.
		624.072.323	Hinged arches etc.
		624.072.323.6	Hinged arches with tension tie

624.072.326	Continuous arches. Continuous arched beams and girders	624.074.4	Continuous membrane structures. Stressed skin structures. Shells
624.072.327	Bowstring arches (tied arches) etc.	624.074.42	Stressed skin structures. Folded plate structures (prismatic structures)
624.072.328	Arches etc. partially stressed during construction. Prestressed arches	624.074.43	Shells
624.072.33	Frames. Plane frameworks. Parts of frames and frameworks <i>Denote statical aspects by :624.04..., structural aspects by :624.07...</i>	624.074.432	Axially symmetrical shells. Cupola shells
624.072.331	Hinged frames	624.074.433	Cylindrical shells. Vault shells
624.072.332	Simply supported frames. Rampant frames	624.074.434	Translation shells. Paraboloid shells
624.072.335	Continuous frames	624.074.435	Conoid shells
624.072.336	Portal structures. Frameworks. □ Portal frames. Trusses. Vierendeel girders	624.074.437	Composite shells
624.072.338	Parts, components of frames and frameworks. □ Tie-beams. Posts. Chords. Nodes	624.074.438	Parts of shells. □ Shell surfaces. Edges
624.072.4	Elements subject to combined stresses	624.074.5	Three-dimensional frameworks (space frames). Lattice or grid structures. □ Geodesic domes. Lamella roofs
624.072.44	Bent or angled members. Corner pieces	624.075	Elements whose state of stress must be checked for stability
624.072.45	Fixed or restrained beams subject to bending or flexural stress and tensile stress	624.075.2	Elements subject to buckling stress
624.072.46	Curved beams and girders (e.g. for bridges curved in plan)	624.075.22	Compression members (braces, struts etc.) in frameworks
624.072.7	Girders of closed hollow section. Hollow girders. □ Box girders. Tubular girders (girders of round section)	624.075.23	Long columns. Pillars, posts, stanchions
624.072.9	Bracings. □ Longitudinal bracings. Transverse bracings. Cross bracings. Diagonal bracings ◇ 624.072.9:624.042.41 Windbraces	624.075.24	Articulated, jointed columns (composed of several members joined)
624.073	Plane load-bearing structures. Plates. Slabs. Discs ⇒ 624.072.1	624.075.25	Compression chords of frameworks
624.073.11	Freely supported plates	624.075.3	Elements subject to instability, tilting, overturning
624.073.12	Fixed or clamped plates	624.075.4	Elements subject to both bulging and buckling
624.073.13	Plates on point supports (plates on pillars, columns etc.)	624.078	Structural connections, joints, links and their calculation
624.073.131	Plates on one support	624.078.3	Masonry joints (and their calculation)
624.073.132	Plates on more than one support	624.078.32	Joints allowing play, movement. □ Expansion joints. Running joints. Settlement joints
624.073.135	Plates on mushroom-headed supports (e.g. mushroom ceilings)	624.078.34	Construction joints. □ Horizontal, course joints. Vertical, butt joints
624.073.136	Plates on supports without mushroom heads (e.g. slab ceilings)	624.078.36	Rigid joints (with no play)
624.073.2	Plates on elastic supports	624.078.37	Dummy joints
624.073.4	Continuous plates	624.078.38	Joint filling (preformed, sealing fillers)
624.073.5	Plate-like frame or grid structures. □ Beam grillages (e.g. timber gratings). Diagonal grids. Ribbed slabs, floors	624.078.4	Connections of structural members (and their calculation). Timber, steel joints etc. Junction points
624.073.6	Plates with interruptions. □ Plates with holes, openings	624.078.41	Form of interface between members. □ Butt, lap, dovetail joints
624.073.8	Thin plates, layers, panes, panels. Thin-plate structures. Thin walls	624.078.42	Connections with pins, brads, dowels, nails, stud-bolts
624.074	Three-dimensional load-bearing structures (space structures)	624.078.43	Glued joints
624.074.1	Bridge-shaped structures	624.078.44	Riveted joints
624.074.2	Domes. Cupolas	624.078.45	Welded joints
624.074.22	Flat cupolas. Semi-cupolas	624.078.46	Screwed, bolted joints (nut-and-bolt joints)
624.074.23	Caps	624.078.48	Compound and multiple joints (using mixed methods). □ Glued and pinned joints
624.074.24	Surmounted cupolas	624.078.49	Joints and connections by mechanical action
624.074.25	Pointed cupolas	624.078.492	Prestressing connections
624.074.27	Ribbed cupolas. Umbrella cupolas	624.078.494	Shrinkage connections ⇒ 621.88.084
624.074.28	Parts of cupolas. □ Soffits. Squinches. Pendentives. Lanterns	624.078.5	Structural bearings (and their calculation)
624.074.3	Vaults. Vaulting	624.078.52	Fixed bearings. □ Rocker, pivot, pin bearings
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624.078.82	Stiffening beams. Stiffeners	624.131.524	Bearing capacity. Permissible earth deformation
624.078.84	Stiffening rings	624.131.526	Settlement analysis
624.1	Substructures. Earthworks. Foundations. Tunnelling	624.131.53	Earth pressures, active and passive. Pressure distribution. Soil stability. Planes of rupture. Landslips
624.12	Rock engineering and mechanics	624.131.532	Active earth pressure, thrust
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624.121	Rock mechanics	624.131.535	Effect of vibrations on earth pressures
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624.121.532	Mountain pressure	624.131.537	Soil stability. Planes of rupture (slip surfaces). Landslips
624.121.537	Stability. Planes of rupture. Slip surfaces	$\Rightarrow$ 624.139.3	
624.121.54	Mountain movements. Deformations. Rheology	624.131.54	Soil rheology. Ground movements and deformations
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624.127	Securing of rock slopes. Protective rock engineering in mountainous terrain	624.131.55	Soil vibrations, perturbations. Earth tremors
624.13	Geotechnics. Soil mechanics. Earthworks. Frozen ground engineering	$\Rightarrow$ 550.34	
624.131	Soil mechanics. Structural soil technology. Technical and economic studies	624.131.551	Causes of earth tremors
624.131.1	Origin of soil. Engineering geology. Mineralogical, geological and biological factors in assessing soils	624.131.552	Effect of earth tremors
624.131.2	Classification of soils for engineering	624.131.6	Water in the soil (groundwater).
	$\Rightarrow$ 552		☐ Groundwater movement: flow, percolation, seepage etc. Effects: erosion, capillary effects, pressures
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624.131.21	Cohesionless soils (frictionless soils). ☐ Stones. Pebbles. Shingle. Gravel. Sand. Rock dust. Rock flour	624.131.63	Flow to wells, sumps and drains
624.131.22	Cohesive soils. ☐ Clays. Mud. Silt. Loam. Marl (soil)	624.131.64	Seepage forces
624.131.23	Loess. Loess clay	624.131.65	Uplift pressures
624.131.24	Naturally cemented soils. ☐ Conglomerates. Bog iron ore	624.131.66	Piping and internal erosion
624.131.25	Rock-type soils. ☐ Sandstone. Shale. Slate. Schist. Limestone. Marl(stone). Chalk. Granite	624.131.67	Filters. Filtering basins
624.131.26	Topsoil. Humus	624.131.7	Applications of soil mechanics. Earth or soil as building or structural material. Earth as aquifer in construction work
624.131.27	Other soils with organic admixtures. ☐ Slime. Ooze. Bog soils. Sapropel. Peat. Coal	624.131.8	Technical and economic preparatory work
624.131.3	Soil survey, examination. Soil sampling. Soil testing	$\Rightarrow$ 528	
624.131.31	Survey of terrain. Site investigation	624.132	Removal and transport of soil. Earthmoving operations
624.131.32	Geophysical exploration techniques		<i>For earthmoving equipment see</i>
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624.131.33	Scrapes. Trenches	624.132.3	Loosening and removal of soil by hand
624.131.34	Trial bores. Boreholes		Mechanical loosening and removal of soil
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624.131.35	Pile-driving tests	624.133/138	Transport of soil
624.131.36	Soil sampling	624.133	Formation and securing of earthworks
624.131.37	Soil analysis. Laboratory tests. Model tests	624.134	Lowering of ground level. Excavation. Setting out, clearing, stripping and auxiliary works
624.131.38	Field tests and investigations		Cutting of channels, trenches, ditches.
624.131.381	Penetration tests (deep sounding)		Trenching
624.131.382	Foundation soil tests	$\Rightarrow$ 623.24	
624.131.383	Trial loading of bearing surfaces (vertical and horizontal)	$\Rightarrow$ 626.131	
624.131.384	Trial loading and test driving of piles	624.134.1	Earth cutting methods. ☐ Layer, terrace, face and side excavation
624.131.385	Dynamic investigations	624.134.2	Cuttings without timbering
624.131.386	Earth pressure measurements	624.134.4	Cuttings with timbering
624.131.387	Pore water pressure measurements	624.135	Raising of ground level. Mounds. Filling and levelling of large areas. Fills
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624.131.5	Soil statics. Soil dynamics		$\Rightarrow$ 627.5
624.131.52	Foundation pressure. Soil loading. Pressure or load distribution. Hydrodynamic forces. Bearing capacity	624.136.1	Methods of tipping
624.131.522	Pressure or load distribution. Foundation pressure	624.136.2	Embankments of earth, clay and similar material
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624.137	Banks etc. Retaining structures. Transition works between different ground levels.	624.143.3	Measures against ice formation (icing). Anti-icing methods (mechanical, thermal, chemical)
624.137.2	Securing of earth masses	624.143.5	Removal of ice. De-icing methods
624.137.4	Banks. Batters. Slopes. Banquettes (berms) ⇒ 625.736	624.143.8	Use of chemical antifreeze agents and freezing-point depressants ⇒ 62-715
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624.137.6	Retaining walls. Revetments ⇒ 624.164		⇒ 551.578.46
624.137.7	Anchorage. Rock anchors ⇒ 624.078.7		⇒ 614.8
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624.138.1	Improvement of the bearing capacity of soils (ground improvement). Soil compaction, consolidation, stabilization		Damage caused by loading with hoar-frost, snow etc. Countermeasures
624.138.2	Theory		⇒ 62-753
624.138.22	Mechanical methods	624.144.4	⇒ 62-757.3
624.138.23	Infilling. Ramming. Tamping. Punning. Rolling. Vibrating. Vibroflotation		Control of snow drifting. Snow barriers. □ Fences. Hedges. Galleries
624.138.24	Mixing soil with various materials. Soil-cement stabilization		⇒ 625.164
624.138.26	Injection (pressure grouting)	624.144.5	Removal of snow and ice. Snow melting
624.138.29	Insertion of piles		⇒ 625.174
624.138.3	Explosive compaction (deep blasting)	624.144.53	Methods of snow and ice removal (mechanical, thermal, chemical)
624.138.34	Physical methods	624.144.55	Equipment for snow and ice removal. □ Snow ploughs. Snow blowers. Snow melters
624.138.4	Dewatering, drainage of ground		⇒ 629.362.5
624.138.5	Chemical methods. □ Stabilization with resins, soaps, salts etc.	624.144.8	Measures to prevent or minimize slipping or skidding on ice and snow
624.139	Electrochemical methods. Electro-osmosis		⇒ 625.089.1
624.139.2	Frozen ground engineering		⇒ 625.095
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624.139.24	⇒ 622.271.7		⇒ 629.3.027.543
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624.139.3	Effects of frozen ground on engineering works and structures		⇒ 551.326
624.139.34	Frost heave. Differential swelling of soil	624.146	⇒ 626.144
624.139.36	Subsidence, settling, caving, collapse caused by thawing. Frost boil		⇒ 627.224
624.139.38	Effects of freeze and thaw on the bearing capacity of the ground	624.147	⇒ 627.73
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624.139.6	Control of frozen ground	624.148.3	Ice and snow as foundation and structural or engineering materials. Artificial compaction of snow
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624.139.64	Control of ice formation (icing) in the ground. Artificial frost zones	624.148.7	Special materials for cold climates. □ Special properties
624.139.65	Control of ice formation (icing) in streams and rivers		Special adaptations and designs for cold climates
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624.139.7	Frozen soil as structural or engineering material	624.15	Foundation engineering. Substructure work
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624.143.2	Measures against frost (antifrost, defrosting methods)	624.151.5	⇒ 626.86
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		624.152.2	Excavating and embanking operations for foundations. Foundation trenches
		624.152.5	Excavation of foundation. □ Stripping of turf, vegetation. Clearing topsoil
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624.153.5	Foundations in the dry, without soil stabilization	624.155.152	Impact pile drivers. Pile hammers
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624.153.524	Individual column foundations (pad foundations). Tower foundations. Foundations for high-rise buildings	624.156	Grillages, footings, platforms and other special foundations in the dry ⇒ 624.153.52
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624.154.53	Concrete bored piles (with recovery of casing or drilling tube). □ Compacting of the concrete by vibration, compressed air	624.157.2	Well foundations in open water. Caisson foundations without compressed air. Open caissons. Box caissons ⇒ 624.156.8
624.154.54	Special designs of bored pile. □ Bored piles with permanent (non-recovered) casing. Vibrated concrete bored piles. Precast bored piles <i>Denote named systems by alphabetic extension</i>	624.157.3	Caisson foundations with compressed air. Pneumatic caissons
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624.154.64	Insertion or placing of piles	624.157.33	Airlocks and airlock operation. □ Supply of air, compressed air etc.
624.154.68	Extraction or withdrawal of casing tubes	624.157.34	Working chambers. Drainage and pumping of chambers
624.154.8	Structural details of bored piles. □ Foot, head, skin, shaft, shoe, point etc. of pile	624.157.35	Cutting curbs, edges, equipment of caissons
624.155	Prefabricated, driven piles (impact-driven piles)	624.157.36	Various other equipment. □ Equipment for soil removal
624.155.1	Driven piles in general	624.157.37	Recompression (decanting) airlocks
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624.155.112	Timber piles (treated or untreated)	624.157.39	Foundation-laying with diving bells
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624.155.114	Steel piles	624.157.5	Pneumatic and hydraulic foundation-laying without caissons
624.155.116	Various materials. Composite piles	624.157.51	Compressed air foundations without caissons
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624.155.12	Pile-driving plan. □ Depth of drive. Pile spacing. Pile rake	624.159.1	Special foundations
624.155.15	Use of equipment for driving, loosening and extraction of piles, including sheet piles. Pile driving and extraction	624.159.11	Foundations for impact or dynamic loads
		624.159.14	Foundations resistant to earthquakes, landslips or mining subsidence
		624.159.2	Damage, deterioration, failure of foundations. □ Baring, erosion, scour, undermining, settling of foundations
		624.159.3	Protection against scour, underwashing etc. □ Stone-filled fascines, mattresses. Chained blocks
		624.159.4	Strengthening, reinforcement of foundations. Underpinning of structures

624.16	Abutments. Piers		
624.164	Abutments. □ Wing walls. End supports ⇒ 624.137.5	624.191.86	Tunnel chambers, engine rooms, workshops
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624.164.3	Dead, hidden abutments. Buried, blind abutments	624.191.94	Climate engineering in tunnels. Ventilation, air conditioning etc. Lighting 624.191.94 ≅ 622.4
624.164.4	Stepped abutments. Hollow abutments. Closed abutments. Recessed abutments. Projecting abutments. Reduced abutments. Arch abutments. Rectangular abutments. Trapezoidal abutments	624.191.947	Tunnel lighting
624.164.5	Recesses in abutments	624.191.95	Tunnel drainage, dewatering 624.191.95 ≅ 622.5
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624.164.7	Anchorage abutments	624.196	Tunnels etc. for conveying fluids, e.g. water, oil ⇒ 621.643.2 ⇒ 621.644 ⇒ 625.745.2 ⇒ 626.132.4 ⇒ 628.2
624.166	Piers (structural supports). Pylons For piers as separate structures see 627.332		
624.166.2	Piers of equal resistance in general		
624.166.3	River piers. Land piers		
624.166.32	River piers. □ Pier noses. Cutwaters. Breakwaters. Ice guards	624.2/8	Bridge construction The auxiliaries listed under 624.21.0... are applicable throughout 624.21/8 ⇒ 725.95
624.166.33	Land piers. □ End piers. Shore piers		
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624.166.5	Recesses in piers	624.21	Bridges in general. Structure, parts etc. ⇒ 625.712.36 ⇒ 625.745.1 ⇒ 626.132/133
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624.166.7	Multi-column piers (clustered piers)		
624.166.8	Pylons (structural supports). Hinged piers (rocker piers) For pylons as separate structures see 624.971	624.21.03	Bridges according to certain characteristics: size, permanence, location, shape etc. Generally parallel to 69.03
624.166.82	Pylons	624.21.033	Dismantlable, transportable and temporary bridges. □ Bailey bridges ⇒ 624.87
624.166.84	Rocker piers	624.21.036	Bridges according to layout, basic form, shape
624.19	Tunnels. Tunnelling. Galleries. Gallery construction	624.21.036.1	Bridges straight in plan
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	◇ 624.19.035.2 Mountain tunnels	624.21.036.3	Bridges curved in plan
	◇ 624.19.035.4 Underground, subterranean tunnels ⇒ 625.712.35	624.21.036.4	One-span, single-span bridges
	◇ 624.19.074.353 Capping vaults ⇒ 622.25/28	624.21.036.5	Bridges with several independent superstructures
624.191	Tunnelling. Gallery excavation and construction	624.21.036.6	Bridges with continuous superstructures
624.191.1	Technical and economic preparatory work. □ Geological and topographic survey. Selection of tunnel direction, gradients. Sections. Statics. Designing. Scheduling. Costing	624.21.036.7	Bridges with load-bearing grillage
624.191.2	Execution of tunnel works. Tunnelling methods	624.21.036.8	Multistorey bridges
624.191.21	Open-pit (cut-and-cover) tunnelling	624.21.037	Viaducts
624.191.22	Gallery construction in general. □ Tunnel driving. Timbering	624.21.09	Parts of bridges
624.191.24	Pilot headings (pilot tunnels)	624.21.095	Bridge carriageways
624.191.242	Driving of pilot headings	624.21.095.2	Carriageway position in relation to girders or supporting structure
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624.191.3	Breaking ground. □ Cutting. Blasting 624.191.3 ≅ 622.23	624.21.095.22	In middle position. Deck level between upper and lower booms or chords
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624.191.7	Spoil removal, haulage, transport 624.191.7 ≅ 622.6	624.21.095.3	Technical design
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624.191.81	Facing, lining of tunnels. Tunnel supports, rings, tubing	624.21.095.323	Orthotropic carriageway surface (deck-plate, longitudinal ribs and cross-girders)
624.191.82	Structural elements of tunnels. □ Tunnel walls, partitions, niches. Tunnel floors, deckings. Shafts. Tunnel entrances, portals	624.21.095.325	Track for railways, tramways ⇒ 625.14
		624.21.095.33	Bridge pavement
		624.21.095.332	Drainage
		624.21.095.334	Heating
		624.21.096	Bridge railings, handrails, parapets, balustrades
		624.21.097	Bridge accessories. □ Traffic control devices on bridges: signals, barriers, tollgates etc. Lighting equipment on bridges

624.26	Flat or slab bridges	624.91	Superstructures according to roof design <i>Denote roof details by .024..., kind of building by :72...</i>
624.27	Girder bridges	624.92	Superstructures according to system of construction. ☐ Solid, rigid, flexible, shell construction
624.271	Simple beam bridges		Structures with one bay or span
624.272	Bridges with stiffened main girders	624.921	Structures with several bays or spans
624.273	Bridges with composite girders (indented or doweled)	624.922	
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624.28	Truss-framed bridges. Strut-frame bridges	624.943	Superstructures in constructional framework
624.281	Trussed beam bridges with truss above deck	624.943.2	Constructional composite framework superstructures
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624.283	Trussed and strutted bridges (with deck between trusses)		
624.3	Truss bridges (open-web, lattice girder bridges)	624.95	Storage structures <i>Denote coating or sealing by -761...</i> ⇒ 621.642.3 ⇒ 631.24
624.31	Bridges with parallel booms or chords. Lattice bridges	624.952	Storage structures for gaseous materials. Gasholders
624.32	Bridges with curved booms or chords. Bowstring girder bridges	624.953	Storage structures for liquids. ☐ Tanks. Water towers. Overground reservoirs ⇒ 627.81 ⇒ 628.13
624.34	Bridges with pin-jointed members. Hinged girder bridges	624.954	Storage structures for bulk solids. ☐ Bunkers. Silos
624.35	Cantilever bridges	624.96	Supporting structures for materials handling, conveying, transporting. ☐ Support gantries. Hoist towers <i>Details by :621.8...</i>
624.5	Cable bridges. Suspension bridges <i>Denote flexible and rigid elements for suspension parts by .071.2... and .071.3...</i> ◇ 624.5.071.232 Carrying cables. Main suspension cables ◇ 624.5.071.234 Back-stay cables ◇ 624.5.071.324 Suspension rods. Suspension hangers. Suspension links	624.97	Tall slender structures. ☐ Towers. Chimneys. Masts ⇒ 625.745.7
624.52	Suspension bridges without stiffening or incompletely stiffened	624.971	Freestanding, self-supporting towers. Pylons (cable towers). Freestanding masts. Chimney stacks etc. ⇒ 624.166.8 ⇒ 625.745.9
624.53	Rigid suspension bridges. Tied cantilever bridges. Stayed girder bridges	624.972	Structures with additional support. Guyed, stayed, strutted masts etc.
624.6	Arch bridges ◇ 624.6.072.322 Fixed-arch bridges ◇ 624.6.072.323 Hinged-arch bridges	624.974	Structures suspended from a central tower or core. Umbrella structures ⇒ 692.231.3
624.8	Movable bridges ⇒ 621.874	625	CIVIL ENGINEERING OF LAND TRANSPORT. RAILWAY ENGINEERING. HIGHWAY ENGINEERING <i>Details of transport engineering works by -1/-9 (from 62-1/-8 and 66-9)</i>
624.82	Drawbridges. Bascule bridges	625.02	Theory of road pavement design ⇒ 625.75 ⇒ 625.8
624.822	Drawbridges	625.021	Flexible pavements. ☐ Macadam. Tarmacadam. Gravel
624.824/828	Bascule bridges	625.022	Rigid pavements. ☐ Concrete
624.824	Bascule bridges with fixed axis	625.023	Stabilized soil pavements. ☐ Mixed-in-place. Lean concrete
624.825	Roll bascule bridges		
624.826	Folding bridges	625.03	Interaction between vehicle and track or road: research, experiments etc. ⇒ 629.3.015 ⇒ 629.4.015
624.828	Counterweight or counterpoise chamber	625.031	Geometry of contact between vehicle and track. ☐ Rail and tyre sections
624.83	Swing bridges (pivot bridges, turn bridges)	625.032	Chemical and physical (especially mechanical) effects between vehicle and track
624.84	Lift bridges	625.032.3	Effects of contact between track and wheel considered in isolation ⇒ 531.4
624.85	Traversing bridges (retractable roller bridges)		
624.86	Transporter bridges. ☐ Ferry bridges. Wharf bridges. Loading and unloading bridges		
624.87	Floating bridges ⇒ 624.21.033		
624.872	Boat bridges		
624.873	Pontoon bridges (punts, lighters, cylindrical floats)		
624.876	Raft bridges		
624.9	Superstructures (except bridges). Superstructures complete in themselves <i>Denote details at 624.91/.97 by .01/.07 listed under 624.01/.07</i>		

625.032.32	Surface properties of wheel and track. □ Roughness	625.08	Road construction plant and machinery. Finishing. Maintenance. Repairs. Minor improvements
625.032.37	Wheel slip	625.083	Filling of joints, voids
625.032.39	Corrugation of track	625.084	Rolling flat. Road rollers
625.032.4	Mechanical effects between track and vehicle in a straight path. □ Effects of uneven track, of wheels out of round, of unbalanced masses. Hunting, lurching etc. Vehicle vibration	625.085	Other compacting equipment. □ Punnens. Power rammers. Frog rammers
625.032.42	With stationary vehicle	625.086	Planing of surface. Planers
625.032.43	With moving vehicle. □ Vehicle lurching, hunting, vibration	625.087	Scarifying of surface. Scarifiers
625.032.44	Effects of track elasticity	625.088	Auxiliary machines and equipment for preparing surfacing materials
625.032.5	Mechanical effects between track and vehicle in a curved path	625.089	Finishing. Surfacing. Repairs. Minor improvements
625.032.6	Mechanical effects special to particular places. □ Effects of crossing bridges, level crossings	625.089.1	Surfacing. Surface dressing. □ Sealing. Spreading of chippings. Gritting (as part of road construction) ⇒ 624.144.8 ⇒ 625.75
625.032.7	Risk of overturning, derailment or skidding of vehicle	625.089.11	Wearing surface in general. □ Sealing
625.032.8	Effects of, and reactions to, vehicle movement. Resistance to motion. Tractive resistance	625.089.12	Surface dressing. □ Addition, spreading of chippings. Gritting
625.032.82	Resistance to motion and reactions caused by braking forces	625.089.2	Repairs to road surface
625.032.821	Aerodynamic and rolling resistances ⇒ 533.68	625.089.4	Minor improvements
625.032.823	Effects of acceleration and deceleration	625.089.42	Raising of level
625.032.824	Gradient resistance	625.089.43	Lowering of level
625.032.825	Curve resistance	625.089.45	Widening
625.032.826	Braking resistance	625.089.46	Narrowing
625.032.86	Equations of motion. Requisite motive, propulsive or tractive forces. Tractive effort (output performance). Energy dissipation	625.089.47	Realignment
625.033	Stresses in the vehicle and the track	625.089.48	Alteration of type of road pavement
625.033.3	Track resistance	625.094	Roads and parts of roads in special colours
625.033.37	Track deformation. Displacement of the track. Creep and misalignment of the rails ⇒ 625.143.58	625.095	Skidding. Antiskid properties ⇒ 624.144.8 ⇒ 625.089.1
625.033.38	Bearing capacity of substructure. Track stability	625.096	Roadworks in relation to traffic disruption. Confining works to one side of road at a time etc.
625.033.4	Stresses in the vehicle. Effects of traction, propulsive and other forces	625.097	Dust formation ⇒ 628.511
625.04	Influences on tracks and roads. Properties of tracks and roads in relation to these influences	625.098	Traffic noise (from interaction with surface). Surface noise ⇒ 628.517.2 ⇒ 656.053.7
625.041	Influence of construction method	625.11/5	Railways. Tramways. Cableways
625.042	Atmospheric and other physical influences (and track or road properties in relation to these)	625.1	Railways in general. Permanent way. Track construction ⇒ 621.33 ⇒ 624.19 ⇒ 624.2/8 ⇒ 627.35 ⇒ 629.4 ⇒ 656.2 ⇒ 711.75 ⇒ 725.1
625.042.1	Influence of water	625.11	Theoretical investigations. Route planning and specification ⇒ 528.4
625.042.2	Influence of frost ⇒ 624.142	625.111	General studies of line layout. Projects and plans for lines etc.
625.042.35	Influence of condensation	625.112	Track gauge. Junction of lines of differing gauge ⇒ 629.488.4
625.042.4	Influence of air	625.113	Studies of longitudinal section. □ Gradients. Curves. Transition curves in the horizontal and vertical planes
625.042.5	Influence of light on the surface. □ Diffusion (scattering) of light. Absorption of light. Reflection of light	625.114	Studies of track cross-section. □ Calculation of earthworks. Clearance. Train gauge
625.06	Binders, binding materials and their use ⇒ 625.75 ⇒ 625.8	625.115	Theoretical studies of superelevation and gauge widening ⇒ 625.144.2 ⇒ 625.176
625.061	Binders in molten form. Hot binders		
625.062	Binders in solution. □ Road oils. Cutbacks		
625.063	Binders in emulsion form		
625.065	Binders in pulverous, powder form		
625.066	Coating. Impregnation, pretreatment of materials		
625.067	Grouting method. Pouring of binder over aggregate		
625.068	Mixing methods. □ Plant mix. On-site, in-situ mixing		
625.07	Mineral aggregates		
625.072	Coarse aggregates. □ Gravel. Shot. Grit		
625.073	Fine aggregates. □ Sand		
625.074	Fillers. Powders. Flours		

625.12	Substructure. Foundations and subgrade (formation)	625.146	Workshops and depots for track components
625.122	Construction and consolidation of earthworks. Methods of protection. □ Cuttings. Embankments. Fills. Slips (landslides). Wash-outs	625.15	Track junctions and accessories
625.123	Drainage installations. Flood precautions. Lowering of groundwater, water table. Ditching. Frost precautions etc.	625.151	Branchings. Turnouts. Junctions. Points and switch crossings
625.14	Superstructure, track. Permanent way in general	625.151.2	Layout and structure of junctions. Number of turnouts. Trailing of switches. Design of tongue rail or blade etc.
625.141	Ballast. Ballasting materials. □ Depth and profile of the ballast	625.151.3	Control of points. Switch actuation: manual, automatic etc. ⇒ 656.25
625.142	Rail supports. Transverse and longitudinal sleepers. Ties	625.151.34	Remote operation. Signal-box mechanisms
625.143	Rails. Rail fastenings	625.151.5	Various locking and interlocking devices.
625.143.1	Shapes, profiles, types, sections and weight of rails. Length of rails. Rail classification	625.151.6	Securing of points and their transmissions
625.143.2	Quality of rail metal. Manufacture, testing, heat treatment of rails. Quality standardization ⇒ 006 ⇒ 620.1	625.151.8	Points controlled from the vehicle ⇒ 621.337.6 ⇒ 656.259.1
625.143.3	Rail fractures and fissures ⇒ 620.19	625.152	Switch crossings. Other parts of branchings. □ Crossing vees, frogs. Wing rails. Guide rails.
625.143.4	Rail joints and fishplates. Fishbolts	625.153	Temporary, flying track connections
625.143.46	Insulation of rail joints	625.153	Line, rail, track crossings 625.152 ≅ 625.151
625.143.48	Continuous welded rail. Jointless rail	625.154	Slip switches 625.153 ≅ 625.151
625.143.5	Fastening of rails to sleepers. Rail creep	625.155	Turntables. Turn bridges
625.143.51	Elastic fastenings for rails. Tension fastenings	625.156	Transfer tables. Traversers ⇒ 621.875.4
625.143.52	Baseplates or soleplates on transverse sleepers. Fastening of rails to baseplates	625.156.2	Buffer stops and similar devices ⇒ 629.4.028.2 ⇒ 656.259.1
625.143.525	Elastic, resilient baseplates	625.156.3	Buffer stops
625.143.53	Rail chairs or seatings on transverse sleepers. Fastening of rails into chairs	625.156.4	Wheel stops. Scotch blocks. Skid shoes
625.143.54	Fastening of rails, baseplates and chairs to transverse sleepers, with clips, spikes, screws, track bolts etc.	625.156.6	Sand drag
625.143.56	Fastening of rails to longitudinal sleepers and to isolated supports. Transverse connections for these structures, to maintain track gauge. Rail braces. Spacing ties	625.156.7	Stop or brake blocks
625.143.57	Fastenings and connections for sleeper rails and for compound, multipiece rails	625.156.8	Retarder or check points
625.143.58	Anticreep measures and devices ⇒ 625.033.37	625.157	Rail brakes
625.144	Track laying	625.16	Various accessories. □ Distance indicators
625.144.1	Track design and equipment. Spacing of sleepers. Length and location of joints. Rail length	625.161	Secondary installations. Access roads. Level crossings
625.144.2	Gauge-clearance. Cant or superelevation. Track laying in curves. Measuring devices for curves ⇒ 625.115	625.162	Houses. Lodges. Huts. □ Crossing-keepers' houses. Plate-layers' huts
625.144.3	Construction of junction tracks	625.162.1	Level crossings. Track enclosure. Demarcation ⇒ 656.216.2
625.144.4	Track-laying operations. □ Pegging-out, staking-out. Ballasting. Rail bending. Packing and tamping of sleepers	625.162.2	Level-crossing barriers, gates: design, installation
625.144.5	Machines for track laying and installation. □ Machines for transport and installation of prefabricated track sections (rail and sleeper assemblies). Rail-lifting, ballasting, tamping machines	625.162.4	Operation of the gates
625.144.6	Machines and equipment for operations on individual track components. □ Drills, tongs, nips, saws etc. Appliances for bending, straightening, hoisting, planing, shaping. Spiking, screwing and bolting tools	625.162.5	Road traffic signs at level crossings. Warning and safety devices. □ Signals. Acoustic alarms (audible warning systems) ⇒ 656.055 ⇒ 656.25
625.144.7	Hand tools for work on track. □ For packing of sleepers	625.162.6	Devices for checking correct operation of level crossings
625.145	Appliances for dismantling track, removing and transporting rails	625.162.7	Roadway, carriageway at level crossings. Intersections with tramway lines etc.
		625.163	Visibility of level crossings. Accident prevention at level crossings. Elimination of level crossings
		625.164	Cultivation and planting of railway land
		625.165	Snow-shields and other protective structures. Protection against rock-falls. Protective galleries etc. ⇒ 62-757 ⇒ 624.12 ⇒ 624.144.4
		625.17	Access roads. Station approaches. Yards, courtyards. Gardens etc.
		625.171	Track maintenance, repair, replacements
			Supervision and inspection of the track ⇒ 629.464.47

625.172	Running maintenance. Condition of line.	625.528	Installation of cables. Coupling dogs, claws
625.173	Maintenance of gauge	625.54	Overhead suspension railways (with rigid tracks). Telfer lines. Suspension monorails etc.
	Repairs and replacements. □ Correction, alignment of rails, of track, of curves. Correction of cant, superelevation. Renewal of materials, ballast		⇒ 625.34
	⇒ 625.144.5/.7		⇒ 629.435
625.173.2	Correction of rails and tracks. □ Surveying. Alignment	625.543	Supports and substructure
625.173.6	Renewal of materials for the permanent way. Reclamation and reconditioning of old materials. Building-up and repair of the track by welding	625.548	Cars and clamping devices
625.173.7	Renewal of the ballast. Cleaning of the ballast	625.57	Aerial ropeways and cableways. Cable telfers
625.174	Clearing the track of snowfall	625.571	Preliminary work. Planning. Routes etc.
	⇒ 624.144.5	625.572	Cableways with movable supports. Travelling cableways
	⇒ 629.464.2	625.573	Cableway masts, supports. Anchorages. Counterpoises (balance weights). Tensioning devices
625.175	Weed-killing	625.574	Driving gear. Machinery
625.176	Alteration of gauge		<i>Details by -8... or :621.86...</i>
	⇒ 625.112	625.575	Loading and unloading arrangements
	⇒ 625.115	625.576	Aerial ropeways and cableways with load moving in a horizontal or inclined plane
625.3	Railways of special construction	625.576.2	Simple ropeways. □ Tow-lifts
	⇒ 625.5	625.576.3	Monocable system (single suspension and traction cable)
	⇒ 629.43	625.576.4	Bicable system (separate suspension and traction cables)
	⇒ 656.33	625.577	Aerial ropeways and cableways with vertical movement of load within a span. Blondins
625.31	Narrow-gauge railways		⇒ 621.877
625.311	Between 1000 mm and 1435 mm gauge	625.578	Cars and clamping devices
625.312	Up to 1000 mm gauge	625.578.2	Cars. Gondolas. Seat frames (on ski-lifts etc.)
625.32	Adhesion railways for steep gradients	625.578.4	Freight carriers, containers, skips
625.33	Rack railways	625.578.7	Engaging of cable. Shoes, clamps etc.
	⇒ 629.4.033	625.71.8	Highway engineering
	⇒ 629.421.6		⇒ 351.712
625.34	Nonsuspended monorail systems, with gyroscope stabilizer		⇒ 351.811
	⇒ 625.54		⇒ 711.73
625.35	Broad-gauge railways	625.71.8].05	Site organization and plant. Construction methods
625.39	Other railways of special construction.		625.71.8].05 ≡ 69.05
	□ Atmospheric railways. Compressed-air railways. Railways with skids, sledge runners. Railways with endless screw (helical drive)	625.7	Highways in general. Roads. Carriageways. Streets. Rural and urban road systems
	⇒ 629.4.034	625.71	Kinds of ordinary road according to importance and purpose
625.4	Elevated railways. Underground railways. Suburban railways. Town railways. Tramways	625.711	Roads in open country
	⇒ 629.43	625.711.1	National roads, routes. Major roads. Trunk roads. Arterial roads
	⇒ 656.34	625.711.2	By-roads. Minor roads. Access roads for farms, mine workings etc. Country lanes, tracks etc.
625.41	Overhead railways on viaducts. Elevated railways		⇒ 625.814
625.42	Underground railways	625.711.3	Highways made especially for motor vehicles. Motorways
625.45	Suburban railways. Town railways	625.711.4	Cycle tracks
625.46	Tramways	625.711.5	Tracks for pedestrians and pack animals (together). Pack roads. □ Mule tracks
625.5	Cable railways. Funicular railways. Suspension railways. Aerial ropeways and cableways. Telfers	625.711.6	⇒ 625.814
	⇒ 621.87		Footpaths. Pedestrian paths, walks
	⇒ 656.35		⇒ 625.734.2
625.52	Cable-traction railways (with traction by cable, rope or chain). Gravity planes (self-acting inclines, jinny-roads). Funicular railways	625.711.7	Tracks for horse riding. Rides. Bridle paths
	⇒ 621.087.3	625.711.8	Other special types
	⇒ 622.016.4	625.711.812	Roads on high ground (along or near watersheds or divides). Hill roads. Mountain roads
	⇒ 629.4.034	625.711.813	Roads in valleys. Roads through flat country
625.527	Cables. Ropes. Chains	625.711.814	Steep roads, gradients, ascents (transition from valley to high ground, or connecting two valley roads). Serpentine roads (with sharp curves, hairpin bends). Roads through mountain passes
	⇒ 672.6		
	⇒ 679.7		

625.711.816	Bypasses. Ring roads. Orbital routes	625.714	Roads along watersides (embankments). Causeways. Towpaths ⇒ 626.32 ⇒ 627.18 ⇒ 627.35
625.711.82	Desert roads. Roads over sand dunes	625.714.7	Filling, covering or narrowing of watercourses in the interests of road traffic
625.711.83	Swamp, marsh roads. Meadow roads	625.717	Pavements for airports and airfields. Airstrips ⇒ 711.553.9 ⇒ 711.77
625.711.84	Forest roads and trails	625.717.2	Take-off and landing pads or strips
625.712	Urban roads	625.717.3	Runways. Taxiways
625.712.1	Major roads in towns. Main roads. Urban arterial roads. Principal streets (avenues, boulevards etc.)	625.717.5	Hard standings. Aprons. Loading and traffic areas
625.712.12	Communication routes between localities. Urban cross routes, through-roads	625.717.7	Aircraft parking and servicing areas
625.712.14	Major traffic roads, distributor roads	625.72	Planning and routeing of roads ◇ 625.72:528.486 Transfer of road plan to terrain. Staking-out and tracing ⇒ 711.73
625.712.15	Ceremonial, triumphal routes. Roads for processions, parades	625.721	Layout on map. Use of contour lines. Design of road network (radial, concentric, triangular, rectangular etc.)
625.712.16	Streets for markets and fairs	625.723	Longitudinal profiles (lengthwise sections) ⇒ 625.732
625.712.18	Shopping streets	625.724	Changes of direction. Transition curves
625.712.2	Minor roads in towns. Secondary roads. Side streets. Back streets. Urban lanes, alleys	625.725	Levelling. Gradients. Ascents. Descents. Embankments. Cuttings ⇒ 528.422
625.712.25	Service roads (to industrial and commercial premises). Private access roads, drives (to courtyards, garages etc.) ⇒ 625.712.7 ⇒ 625.746.3	625.726	Mass-haul diagrams. Quantities of spoil and fill
625.712.27	Dead ends. Cul-de-sacs. No-through-roads	625.73	Parts of roads. Design. Dimensions
625.712.28	Residential streets	625.731	Subgrade. Foundation. Earthworks. Cuttings. Transport of fill. Formation
625.712.3	Other types of road, street	625.731.1	Subgrade
625.712.31	Gateways. Archways ⇒ 725.941 ⇒ 725.961	625.731.2	Earthworks. Cuttings. Ground improvement (soil compaction, consolidation, stabilization) ⇒ 624.138
625.712.32	Galleries. Colonnades ⇒ 72.012.75	625.731.3	Drainage of subgrade, subsoil and foundation <i>Details by :626.86...</i>
625.712.33	Covered ways. Arcades ⇒ 725.1	625.731.4	Filling. Fill
625.712.34	Pedestrian underpasses and overpasses. Subways. Walkways	625.731.6	Formation. Preparation of road bed. Trenching for bed. Removal of earth
625.712.35	Underground roads. Traffic underpasses. Roads in tunnels ⇒ 624.195 ⇒ 725.95	625.731.7	Rigid foundation or base. □ Concrete and reinforced concrete foundations
625.712.36	Elevated roads. Traffic overpasses ⇒ 725.95	625.731.8	Flexible foundation or base
625.712.37	Tiered, multilevel roads	625.731.81	Rolled stone foundations
625.712.4	Open areas. Squares. Esplanades. City walls, ramparts ⇒ 711.61 ⇒ 712.25	625.731.82	Stone pavement foundations. □ Bottom layer of sand, ash, gravel. Bottom layer of rough cast
625.712.43	Open areas surrounding buildings	625.731.85	Foundations with bituminous binders (black base foundations)
625.712.44	Open areas with road traffic. Squares as traffic routes. Traffic roundabouts	625.731.86	Use of existing roads as foundation for new surface 625.731.86 ≡ 625.8
625.712.45	Ceremonial squares. Open areas for festivities, parades etc.	625.731.87	Use of fibrous materials, bitumen paper or other insulating layers
625.712.46	Open areas for markets and fairs. Market squares etc.	625.731.9	Intermediate layer. Binding layer
625.712.47	Open areas for games, recreation. Playgrounds	625.732	Transverse profiles (cross-sections). □ Camber. Superelevation. Stepping (multilevel roads) ⇒ 625.723
625.712.48	Residential squares etc.	625.732.2	Concave cross-section
625.712.5	Public promenades, gardens, parks ⇒ 712.2	625.732.3	Convex cross-section. Cambered
625.712.6	Sites reserved for special purposes	625.732.4	Inclined cross-section. Superelevation
625.712.61	Storage depots ⇒ 725.1	625.732.5	Roads with carriageways at different levels. Stepped roads
625.712.62	Stopping places for animal-drawn vehicles	625.732.8	Partially paved roads
625.712.63	Parking areas for motor vehicles. Car parks etc.	625.733	Flanks, shoulders or haunches ⇒ 625.889
625.712.64	Parking places for cycles. Cycle stands. Cycle racks	625.734	Footways. Median strips. Refuges
625.712.68	Military parade grounds	625.734.2	Footways (sidewalks, pavements) ⇒ 625.711.6 ⇒ 625.88
625.712.7	Inclines. Ramps ⇒ 625.712.25 ⇒ 625.746.3		
625.712.8	Flights of steps in streets. Passenger lifts (elevators) in streets ⇒ 621.876 ⇒ 625.5		

625.734.3	Median strips (central reserves)	625.746.52	Control towers
625.734.4	Refuges (islands)	625.746.53	Traffic control devices
625.735	Ditches. Drainage channels ⇒ 624.134		◇ 625.746.53:656.053.2 For restricting speed.
625.736	Embankments. Slopes, rakes or batters ⇒ 624.137.2		Traffic-calming measures. Including: Road humps ⇒ 625.745.691
625.737	Retaining walls ⇒ 624.137.5	625.748	Street furniture. Ancillary works not directly connected with road traffic ⇒ 725.94
625.738	Parapets. Barriers. Guard rails	625.748.2	Small permanent structures. □ Shelters. Kiosks. Public lavatories, urinals ⇒ 628.425 ⇒ 725.1
625.739	Road junctions. □ Forks, bifurcations. Intersections. Crossroads	625.748.28	Underground garages and filling stations ⇒ 625.748.54 ⇒ 725.1
625.739.3	Slip roads		Small temporary structures
625.739.4	Points of merging	625.748.3	Temporary fences, screens (hoardings)
625.74	Methods of road construction. Ancillary works	625.748.32	Temporary offices and huts for roadworks
625.745	Ancillary works as parts of roads	625.748.34	Other street furniture
625.745.1	Bridges ⇒ 624.2/8	625.748.4	Seats. Benches
625.745.11	Footbridges. Pedestrian bridges	625.748.42	Receptacles for litter. Litter bins ⇒ 628.463
625.745.12	Bridges for pedestrian and vehicular traffic. Road and tramway bridges. Viaducts ⇒ 624.21.095.32	625.748.44	Post boxes. □ Pillar boxes ⇒ 656.843
625.745.2	Water conduits, open or closed. Channels. Culverts ⇒ 624.196	625.748.46	Other installations and street furniture
625.745.3	Road gutters. Drainage grips ⇒ 628.251	625.748.5	Public clocks
625.745.4	Drains. Gratings. Inspection holes (manholes). Traps. Shafts. Vents 625.745.4 ≡ 628.2	625.748.51	Public fountains
625.745.5	Protective boundaries and barriers. Safety markings. Edge markers (edge stones, spur posts). Guardstones, fenders. Reflectors etc. ⇒ 625.888	625.748.52	Filling stations. Petrol pumps ⇒ 625.748.28 ⇒ 725.1
625.745.55	Crash barriers	625.748.55	Vending machines
625.745.6	Signs. Signposts ⇒ 625.746.5	625.748.56	Fire alarms
625.745.62	Distance indicators. □ Milestones	625.748.57	Taxicab call boxes
625.745.63	Direction signs	625.748.6	Advertising devices. Advertisement pillars etc.
625.745.64	Frontier and border signs. Boundary signs	625.75	Road surface treatment ⇒ 625.06 ⇒ 625.089.1 ⇒ 625.8
625.745.65	Place-name signs	625.752	Treatment with bitumen, asphalt
625.745.66	Street-name signs	625.753	Treatment with oil
625.745.67	House-number plates	625.754	Treatment with tar
625.745.68	Display hoardings. Billboards	625.755	Treatment with bitumen-tar mixtures
625.745.691	Regulatory signs ⇒ 625.746.53	625.756	Treatment with bitumen-rubber mixtures
625.745.693	Warning, danger signs	625.76	Maintenance and repair of highways. Road mending. Street cleansing ◇ 625.76.08 Machinery. Equipment
625.745.7	Various other masts, poles, columns ⇒ 624.97	625.761	Maintenance of stone-paved roads, block or sett paving
625.745.8	Street lighting supports. □ Lampposts. Lamp standards. Branched lighting columns ⇒ 628.971 ⇒ 725.94	625.762	Maintenance of macadamized or metalled roads
625.745.9	Supports and suspensions for wires, cables etc. □ Telegraph poles ⇒ 624.971	625.763	Maintenance of wood block roads
625.746	Other ancillary works connected with use of the road by traffic	625.764	Maintenance of monolithic cement or concrete roads
625.746.1	Junction and levelling of road pavement with permanent way of tramways, railways etc.	625.765	Maintenance of bituminous, asphalt roads
625.746.2	Street exits and entrances of underground railways etc.	625.766	Maintenance of footways, sidewalks
625.746.3	Lowering of footway to permit vehicular access ⇒ 625.712.25 ⇒ 625.712.7	625.767	Maintenance of cycle tracks
625.746.5	Ancillary works for traffic regulation. □ Traffic bollards. Signalling devices, road signs etc. Road and lane markings (white lines). Road studs ⇒ 625.745.6 ⇒ 654.05 ⇒ 656.05	625.768	Continuous minor maintenance
		625.768.1	Road cleansing. Road sweeping. □ Brooms. Mechanical sweepers ⇒ 351.777.614 ⇒ 628.462
		625.768.4	Removal of fallen leaves, litter
		625.768.5	Removal of snow, ice ⇒ 629.362.5
		625.768.6	Gritting of roads. Precautions against ice ⇒ 624.144.8
		625.768.7	Watering of roads. Sprinkler vehicles
		625.768.8	Maintenance of street lighting

625.77	Planting of highways and open areas.	625.859.4	Pavements of fine mixtures, with granular material, sand etc.
625.771	Roadside planting	625.86	Pavements of broken stones, gravel, grit, earth
625.772	Conditions and planting methods. □ Single, double, multiple rows. Spacing of trees	625.861	Materials used. Kinds of stone, debris etc.
625.774	Irregular planting (not in rows)	625.862	Preparation and storage of materials.
	Maintenance and protection of planted areas from traffic, road works etc. □ Drainage. Irrigation. Tree guards. Pruning		Stone-breaking (by hand, with hammer; mechanically) ⇒ 621.926 ⇒ 622.73
625.78	Underground services. □ Pipelines. Cables. Conduits. Distribution networks	625.863	Laying of broken stone pavements etc.
625.782	Cross-sections of underground installations. Complexes of several supply and disposal channels	625.87	Various other kinds of pavement
625.783	Services following direction of road	625.871	Metal paving (e.g. iron, steel)
625.784	Services not following direction of road	625.872	Pavements of vitrified brick
625.79	Other matters relating to highways. □ Projecting structures. Overhangs. Eaves	625.873	Mosaic paving
625.8	Hard surfacing of highways. Pavement, carpet or wearing course. Preparation and use of materials ⇒ 625.02 ⇒ 625.75	625.874	Pavements of reinforced slabs
		625.877	Pavements of fibrous materials
625.81	Primitive roads. Earth roads etc.	625.878	Rubber pavements <i>Denote binders by .06...</i>
625.814	Roads of earth, gravel, sand, clay, trass etc. ⇒ 625.711.2 ⇒ 625.711.5	625.878.2	Pavements of vulcanized rubber sheet
625.815	Roads with slab pavements	625.878.4	Pavements of vulcanized rubber slabs
625.815.1	Natural stone slabs. Flagstones	625.878.5	Pavements of vulcanized rubber blocks
625.815.2	Burnt clay (hard slabs)	625.878.7	Pavements of rubber and asphalt
625.815.5	Cement	625.88	Construction of footways, sidewalks <i>625.881/.887 = 625.81/.87</i> ⇒ 625.734.2
625.815.9	Other artificial stone	625.888	Kerbs. Kerbstones. Edge stones
625.82	Stone pavements. Brick pavements	625.889	Verges other than footways ⇒ 625.733
625.821	Kinds of stone used	626/627	HYDRAULIC ENGINEERING AND CONSTRUCTION. WATER (AQUATIC) STRUCTURES ⇒ 351.79 ⇒ 551.46 ⇒ 556 ⇒ 711.76
625.821.1	Natural stones. Setts. Pavement of large and small stones		
625.821.2	Clinker (klinker) bricks	626	HYDRAULIC ENGINEERING IN GENERAL <i>Details of hydraulic engineering works by -1/-2 and -4/-9 (from 62-1/-2, 62-4/-8, and 66-9). The 62-3... auxiliaries are not applicable at 626 and subdivisions</i>
625.821.3	Various ceramic and clay products. Ceramic tiles. Clay tiles. Bricks	626-3	Cross-sectional parts of hydraulic constructions
625.821.4	Slag	626-31	Upstream approaches and parts <i>626-31 = 626-33</i>
625.821.5	Concrete	626-33	Downstream approaches and parts
625.821.9	Other artificial stone	626-331	Spayed sidewalls
625.822	Manufacture of paving stones and setts ⇒ 679.85	626-332	Permeable floor
625.823	Laying of stone paving	626-333	Impermeable floor
625.83	Wood block pavements. Cork pavements	626-335	Stream control installations. □ Sheet piling.
625.831	Kinds of wood used	626-337	Sluice gates. Wells
625.833	Laying of wood paving	626-338	Energy dissipators. □ Baffle walls. Blocks
625.836	Cork pavings. Artificial wood pavings	626.0	Scouring of foundations. Settlement of floor and apron
625.84	Concrete pavements	626.01	Special auxiliary subdivision for class 626 Hydraulic engineering in general <i>In 626.0, .0 is not used to denote special auxiliary subdivisions. The .0 subdivisions are not transferable to other 626... numbers</i>
625.841	Unreinforced concrete	626.02	Scientific principles of hydraulic engineering
625.843	Reinforced concrete		Underwater work in general ⇒ 624.157 ⇒ 627.02 ⇒ 627.24 ⇒ 627.25
625.844	Cement-bound macadam. Trass-macadam		
625.848	Concrete road with joints		
625.85	Asphalt pavements. Bituminous pavements <i>Denote binders by .06...</i>		
625.851	Pavements of mastic or melted asphalt		
625.852	Pavements of compressed asphalt		
625.853	Pavements of asphalt slabs or tiles		
625.855	Pavements of asphalt with mineral aggregates (gravel, sand etc.). □ Bituminous soil stabilization		
625.856	Bitumen-macadam pavements		
625.857	Pavements of asphalt-tar mixtures		
625.858	Pavements of stones and tar. Tarmacadam pavements		
625.859	Pavements of tar with other minerals		
625.859.3	Pavements of coarse mixtures, with broken stones, rubble, chippings etc.		

626.022	Equipment for underwater work. □ Diving bells. Divers' shafts. Diving ships (diver-support vessels) ⇒ 629.58	626.24	Drawing, raising water by mechanical means. Water-raising mechanisms ⇒ 621.649 ⇒ 621.65/.69
626.024	Divers' work	626.25	Feeder channels, ditches. Feeder sluices, weirs
626.025	Diving suits. Breathing equipment. Ancillary equipment	626.26	Discharge, drainage installations
626.027	Working methods. □ Use of tools under water	626.28	Particular structures on or in canals
626.1	Inland navigable waterways. Canals. Canalized rivers ⇒ 626.9 ⇒ 627.4	626.281	Lock and sluice approaches and basins. Passing places. Turning basins. Lead-in piers, jetties
626.11	Generalities. Types of canal	626.283	Safety gates
626.112	Canals without locks. Canals in level terrain	626.3	Shape and transverse dimensions of canals. Cross-sections
626.113	Canals with locks. □ Summit canals. Junction canals	626.31	Water-bearing section. □ Wetted perimeter. Wetted cross-section. Canal width and depth. Mean water depth. Headroom under bridges. Angle of side slope
626.114	Lateral canals. Canals parallel with rivers	626.32	Towpaths ⇒ 625.714
626.12	Planning and routeing of canals. □ Longitudinal profiles (lengthwise sections). Bends, curves. Widening at bends. Crossings, intersections of canals	626.33	Spoil-banks
626.13	Canal construction	626.34	Counter drains
626.131	Excavation of canal bed. Embankments and cuttings for canals. □ Excavators for long trenches. Dredging ⇒ 621.879 ⇒ 624.134 ⇒ 631.312.63	626.35	Canal banks
626.132	Underground canals (tunnel canals). Aqueducts	626.4	Locks. □ Layout. Location. Dimensions ⇒ 627.45
626.132.4	Tunnel canals (underground canals) ⇒ 624.196	626.41/.46	Chamber locks
626.132.5	Canal aqueducts (canal bridges) ⇒ 624.2/.8	626.41	Chamber locks in general. Various designs
626.133	Access, approaches, to canals. Crossings of roads and waterways. Emptying, draining of canals ⇒ 621.649 ⇒ 621.65/.69 ⇒ 625.714 ⇒ 626.32	626.413	Single locks
626.134	Lining, sealing of canals. Making watertight. Local seepage. General, invisible seepage. Lining materials	626.414	Double or twin locks. Foursquare locks (for two or more vessels abreast)
626.14	Maintenance of canals	626.415	Locks for trains of barges
626.142	Maintenance dredging	626.416	Locks for negotiating large differences of level. □ Staircase locks (flights of locks). Shaft locks (high-lift locks)
626.143	Cleaning out. □ Clearing of weeds	626.419	Other kinds of lock. □ Switch locks. Dead-end locks. Turn-around locks. Safety locks or guard gates
626.144	Removal of ice ⇒ 624.145	626.42	Structural parts of chamber locks. Timber locks. Masonry locks
626.15	Planting of canal slopes and embankments	626.421	Fixed parts. □ Chamber walls, sills, groundsills. Head bays
626.17	Reinforcement of canal slopes and embankments ⇒ 624.137	626.422	Moving parts
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626.2	Water supply (feed water) for summit canals ⇒ 626.81/.84	626.422.21	Mitre gates
626.21	Quantity of water necessary. Water consumption. Water losses	626.422.22	Flap gates
626.22	Supply from natural watercourses and lakes	626.422.23	Radial or segmental gates
626.23	Supply from artificial water storage. Canal reservoirs	626.422.24	Vertical-lift gates
		626.422.25	Roller gates
		626.422.3	Ship caissons (sliding caissons)
		626.422.4	Accessories of lock gates. □ Bearings. Anchorages. Operating gear. Sluices and valves for filling and emptying of locks
		626.423	Buildings for lock machinery. Engine houses
		626.43	Construction of chamber locks
		626.44	Equipment. □ Towing installations. Safety ladders. Mooring rings. Bollards. Fenders. Arrester chains
		626.45	Operation and maintenance of locks. Lock operation time. Water consumption. Water losses, leakage, seepage
		626.46	Arrangements for saving water in locks. □ Floats. Water-saving chambers. Lock reservoirs
		626.49	Locks without chambers

626.5	Lifting gear and inclined planes for vessels	626.812	Supply from natural watercourses. □ Spring water. River water
626.51	Vertical lifting gear for vessels. Barge lifts. Boat hoists. □ Hydraulic lifts. Float operated lifts. Counterweight lifts ⇒ 621.876	626.813	Supply from effluent or drainage channels
626.52/.54	Inclined planes, ramps, slopes for vessels	626.814	Supply from lakes, reservoirs, navigation canals
626.52	Inclined planes with haulage cradles	626.816	Water extraction using wells, dams, drainage structures
626.53/.54	Wet haulage with hoisting troughs (moving locks)	626.82	Irrigation canals, channels etc. ⇒ 621.643
626.53	Haulage with troughs longitudinal to the slope	626.821	Routeing of channels. Dimensions of cross-sections. Types of channel
626.54	Haulage with troughs transverse to the slope	626.822	Construction of channels
626.7	Traction and propulsion on rivers and canals <i>Class here hydraulic engineering aspects, including installations. Class vessels at 629.5...</i>	626.823	Ancillary works
626.71	Conditions affecting vessel movement. □ Tractive or propulsive effort. Water resistance. Realized speed or headway	626.823.2	Inverted siphons. Culverts
626.72	Propulsion by human power	626.823.3	Superpassages
626.73	Towage. Haulage	626.823.4	Devices for excluding and ejecting silt from irrigation channels. □ Silt excluders. Silt ejectors. Sedimentation tanks. Filtering devices
626.731	Towing by human or animal power (manhauling etc.)	626.823.5	Sluices (floodgates). □ Inlet sluices. Outlet sluices. Distributing sluices
626.732	Mechanical hauling with endless chain or cable	626.823.6	Measuring weirs, flumes, glaciis
626.74	Propulsion by sail and by water current. Sailing. Drifting, floating (e.g. of rafts, log trains). Downstream navigation ⇒ 629.512 ⇒ 629.525	626.823.7	Details of conduits and pipes. □ Couplings. Valves ⇒ 621.643.4 ⇒ 621.646
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		626.824	Measurement, distribution, rating of water. Rotation (alternating use of water sources). Recuperation, re-use of water
		626.826	Water losses in irrigation channels
		626.83	Waterworks for irrigation. Pumping stations. □ Water-raising machinery
		626.84	Methods of irrigation
		626.841	Irrigation by flooding. □ Irrigation of fields with sewage, effluent, waste water. Check system ⇒ 628.381 ⇒ 631.863 ⇒ 631.879.2
		626.842	Irrigation by overflow or spillover. □ Furrow and groove irrigation. Bed or strip irrigation
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626.88	Hydraulic works for fish-breeding ⇒ 639.2/3	627.094	Fishing ports and facilities
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626.886	Fish ponds (for particular species) ⇒ 628.357.3 ⇒ 639.311	627.1	Natural inland waters ⇒ 351.79 ⇒ 556
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626.93	Impounded canals at sea level (maritime canals with entrance or exit locks)	627.133	Metering, gauging of watercourses (quantity and velocity measurements) ⇒ 681.121
626.95	Maritime canals with several levels or reaches. Maritime summit canals	627.14	Control and regulation of torrents and turbulent rivers or streams. □ Obstructions. Reinforcement. Damming
627	NATURAL WATERWAY, PORT, HARBOUR AND SHORE ENGINEERING. NAVIGATIONAL, DREDGING, SALVAGE AND RESCUE FACILITIES. DAMS AND HYDRAULIC POWER PLANT Details of natural waterway and harbour works, hydraulic power plant etc. by -1/-9 (from 62-1/-8 and 66-9) ◇ 627(26) Marine, sea, ocean works and facilities ◇ 627(28) Inland water, river and lake works and facilities	627.141	Control of torrents. □ Causes. Countermeasures
627.01	Fundamental principles ⇒ 532 ⇒ 556	627.142	Control of turbulent, fast-flowing rivers and streams. □ Rapids. Narrows. Freshets

627.15	Generalities on rivers and streams ⇒ 551.435.1 ⇒ 556.52/.53 ⇒ 627.222.4 ⇒ 627.4	627.222.1	Sea bed. Submarine hydrography ⇒ 551.35 ⇒ 551.462
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627.417 Revetment with stone paving, rubble, concrete, asphalt surfacing etc.
627.418 Stabilization of steep or vertical banks.
□ Sheet piling. Campshedding. Retaining walls. Embankment walls
- 627.42 Training works. Improvement of rivers by modification of bed or channel**
627.421 Narrowing and restriction of watercourse.
□ River groynes. Spurs. Ground sills. Training walls. Secondary dykes (wattle dykes). Mattresses
⇒ 624.136
⇒ 627.514
627.422 Straightening of course. Cutting, excavation works
⇒ 624.134
⇒ 627.47
627.423 Channel clearing, removal of obstacles. Dredging
⇒ 627.513
⇒ 627.74

- 627.424 Closing off tributaries, affluents, former channel
627.425 Mixed systems of groynes and low dykes
- 627.43 Improvement of rivers by damming works. Weirs. Gated dams**
 ◇ 627.43:626.882 Weirs with fish passes
 ⇒ 351.793
 ⇒ 627.82
- 627.431 Fixed weirs
627.432 Movable weirs
627.432.3 Bar or beam weirs. □ Stop-log weirs. Needle, pin, post weirs. Blind weirs
627.432.4 Drum and shutter weirs
627.432.5 Sluice weirs. Bridge weirs
627.432.6 Cylindrical weirs. □ Segment weirs. Sector weirs. Rolling-gate weirs
627.432.7 Siphon spillways
- 627.44 Narrows. Sluiceways**
- 627.45 Sluices, floodgates in rivers**
 ⇒ 626.4
- 627.47 Diversion of watercourses**
 ⇒ 627.422
- 627.5 Other waterway protection and improvement works. Dykes. Coast protection. Land drainage and reclamation**
 ◇ 627.5:624.145 Coast protection from damage caused by floating ice
- 627.51 Flood defences**
 ⇒ 351.792
- 627.511 Causes of flooding. Regime of rainfall and watercourses. Nature of terrain
627.512 Spates, floodwaters in watercourses. Prediction of spates
627.513 Preventive works. □ Channel clearing. Dredging
 ⇒ 627.423
 ⇒ 627.74
- 627.514 Longitudinal dykes, embankments. Levées.
 □ Plan, layout. Design. Construction
 ⇒ 624.136
- 627.514.2 River dykes. □ Summer and winter river dykes. Lateral dykes. Spur dykes (wing dams)
627.514.3 Sea dykes. □ Summer and winter sea dykes. Ring dykes. Cluster dykes
627.514.6 Dyke construction works
627.514.61 Dyke ramps
627.514.62 Passings. Dyke openings. Dyke blows. Dyke breaches
627.514.63 Dyke sluices. Overflows. Spillways
627.515 Maintenance and protection of dykes. Reinforcement of dykes
627.516 Flood control. Floodwater information services. Flood warnings, alarms. Opening of sluiceways, floodgates
627.517 Dyke breaches, bursts. Repair and reconstruction of dykes
 ⇒ 351.793.18
- 627.52 Shore protection. Coast protection. Defence of coast from effects of wind and waves**
 ⇒ 627.222
 ⇒ 627.235
- 627.521 Natural changes in shores. □ Erosion. Undermining. Drifting. Deposition
627.522 Protective structures against effects of waves. Revetment. □ Fascining. Paving
 ⇒ 624.137.4/5
- 627.523 Protective structures against accumulation of sand and pebbles. Structures for stabilizing the shore
 ⇒ 627.223.7
 ⇒ 627.746
- 627.523.1 Deflection, diversion of littoral currents
627.523.2 Coastal groynes. Shore groynes
 ⇒ 627.235
 ⇒ 627.421
- 627.523.3 Periodic removal of pebbles etc.
627.524 Protective structures against effects of wind. Construction of dunes
 ⇒ 551.311.3
 ⇒ 551.435.74
- 627.524.1 Effects of the formation of dunes. Migratory dunes. Artificial control of sand deposit. Sand-catching fences
627.524.2 Dune protection. Dune cultivation. □ Horizontal and vertical coverings. Planting
- 627.53 Flood-prone lowlands and their protection**
627.532 Drainage. Drying out
 ⇒ 626.86
- 627.533 Land reclamation. Aggradation. Polders
 ⇒ 626.87
 ⇒ 627.514
- 627.533.1 Aggradation. Land warping. Soil deposition
627.533.13 Reclamation of land from river silt
627.533.14 Reclamation of land (silt deposit) on the shore. Marsh formation
627.533.2 Polders (reclaimed land) in general.
 □ High-risk polders
- 627.533.3 River polders
627.533.4 Sea polders
627.533.5 Deep polders. Other kinds of polder
627.533.6 Canals within polders. Drainage channels in polders
 ⇒ 626.861
- 627.533.7 Canals outside polders. Surrounding channels
627.534 Pumping stations. Water-raising machinery. Pumping operations
627.536 Depoldering. Conversion or reversion of polders to unprotected land
- 627.7 Navigational facilities. Channel maintenance. Marine salvage and rescue facilities**
627.7.06 Special features: lighting etc.
627.7.062 Lighted. Lit. With lights
627.7.063 Unlit. Without lights
- 627.71 Navigational facilities. Navigation aids and equipment. Lane marking for shipping**
 ⇒ 654.9
 ⇒ 656.05
- 627.711 General siting and arrangement of facilities and equipment
627.712 Fixed markers, indicators and guides. Radio and radar beacons (fixed)
627.712.2 Channel width and depth markers. Stakes etc.
627.712.3 Fairway condition, direction and distance markers
627.712.33 Channel diversion signs
627.712.35 Direction signs, indicators, pointers
627.712.36 Identification signs. Landmarks
627.712.38 Distance markers
627.712.4 Warning markers and signs
627.712.42 Obstruction warning (surface or underwater)
627.712.44 Sound signal devices. Audible warning devices
627.712.6 Radio, radar beacon guides (fixed)
627.712.62 Radio beacon guides
 ⇒ 621.396.93

627.712.63	Automatic response beacons. Radar beacons ⇒ 621.396.963.8	627.722.4	Fog or poor visibility signals (sound) ⇒ 627.713.44
627.712.7	Guide, aligning, sighting markers	627.722.6	Radio (fog, storm) signals
627.712.8	Electric (h.f.) guide cables	627.722.8	Underwater sound signals ⇒ 681.883
627.713	Floating markers, indicators and guides. Buoys 627.713 = 627.712 ⇒ 629.516	627.723	Weather signals, bulletins and forecasts for shipping ⇒ 551.50
627.713.2	Marker buoys	627.724	Flag signals ⇒ 654.912
627.713.42	Obstruction warning buoys	627.724.2	Signal flags (hoistable)
627.713.44	Sound buoys. Bell buoys ⇒ 627.722.4	627.724.5	Stern flags. Pennants. Burgees etc. (for identification)
627.713.6	Radio, radar buoys	627.725	Shore communication equipment
627.714	Anchoring or mooring devices for floating markers and vessels in channels. □ Buoy moorings. Special anchors ⇒ 627.338	627.725.2	Semaphores
627.715	Lighthouses and lightships (positioning, equipment etc.) ◇ 627.715(210) On mainland (promontory, coast) ◇ 627.715(26) At sea. In the open sea ◇ 627.715(28) On inland waters (rivers, canals, lakes) ◇ 627.715.032 Fixed structures. Lighthouses ◇ 627.715.034 Floating structures. Lightships ⇒ 725.1	627.725.3	Heliographs
627.715.2	Positioning. Systems of arrangement	627.725.5	Light (lamp) signalling, signal masts etc.
627.715.22	Lateral system	627.725.6	Shore, coastal telegraphs
627.715.23	Cardinal system	627.725.7	Traffic control lights. □ For locks, ferries
627.715.24	Axial system	627.726	Long-distance communication and signalling between shore and ships ⇒ 621.396.6 ⇒ 621.396.96
627.715.3	Light intensity, visibility, colour	627.727	Distress signals. Distress rockets. Other SOS facilities on board ⇒ 627.77
627.715.31	Visibility	627.728	Remote control (telecontrol) of ships ⇒ 621.398 ⇒ 654.93
627.715.32	Colour	627.73	Ice-breaking service, facilities and equipment ⇒ 627.224.7 ⇒ 629.561.5
627.715.4	Spread or sweep of light. □ All-round light. Sector light. Beam. Revolving light	627.74	Dredging, dragging and sluicing. Related channel and harbour maintenance work ◇ 627.74.082 In tidal waterways, harbours ◇ 627.74.084 In nontidal waterways, harbours ⇒ 627.423 ⇒ 627.513
627.715.5	Duration of signal. □ Continuous signal. Intermittent signal (flash). Wax-and-wane signal	627.741	Dredging of rocky beds
627.715.8	Lantern. Lighthouse chamber. Optical and mechanical equipment ⇒ 628.9	627.742	Dredging of hard, resistant beds
627.715.82	Special optical equipment for light intensification	627.743	Dredging of sandy beds
627.715.85	Mechanical equipment. □ Rotators	627.744	Dredging of muddy, silty beds
627.715.88	Igniting and extinguishing equipment ◇ 627.715.88-519 Remote control devices ◇ 627.715.88-523 Automatic electrical devices	627.745	Sluicing (for channel maintenance) ⇒ 627.322
627.717	Pilotage. Pilot stations and equipment. Manoeuvring techniques and regulations ⇒ 656.071.7	627.746	Shore clearance. Bottom clearance ⇒ 626.143 ⇒ 627.356 ⇒ 627.523
627.717.2	Pilot stations and equipment ⇒ 629.561.7	627.747	Rock removal. □ Blasting ⇒ 622.235
627.717.3	Channel pilotage. □ Port entry and exit. River or canal pilotage	627.748	Floating equipment for channel and harbour maintenance. □ Dredging vessels. Sludge vessels ⇒ 621.879
627.717.5	Manoeuvring techniques and regulations	627.76	Marine salvage. Refloating, rescue of ships ◇ 627.76.02 Work underwater, by divers ◇ 627.76.082 In tidal waters ◇ 627.76.084 In nontidal waters ⇒ 629.566
627.718	Towing and tug facilities in navigable channels ⇒ 626.75 ⇒ 627.768 ⇒ 629.561.1 ⇒ 656.076.7	627.763	Search and marking gear, anchors etc. for sunken vessels ⇒ 627.713.42
627.72	Nautical signalling. Weather signals. Shore communication ⇒ 654.9 ⇒ 656.05	627.764	Salvage of cargo and property from total wrecks in situ
627.721	Signals indicating tides, water levels	627.765	Repair and salvage of damaged and foundered (but refloatable) vessels ⇒ 627.36 ⇒ 629.5.083.5
627.722	Signals for emergencies, poor visibility, bad weather. Fog, storm and wind signals. □ Signal rockets, cannon etc. Audible signals (gongs, bells, sirens, horns, foghorns). Radio signals	627.767	Righting, refloating of capsized vessels
627.722.2	Emergency signals. □ Rockets. Cannons		

627.768	Towing-off, refloating of (relatively undamaged) grounded vessels ◇ 627.768.082 On the tide. Using the tide ◇ 627.768.084 Without the aid of tide ⇒ 627.718	627.8.066 627.8.068	Crest of dam. Freeboard Inspection galleries in dams etc. ⇒ 624.19
627.77	Marine rescue, life saving. Safety and survival of seafarers ⇒ 614.81 ⇒ 627.727 ⇒ 629.5.067 ⇒ 656.6.08	627.8.07 627.8.09	Structural elements of dams etc. 627.8.07 ≡ 624.07 Utilization of dams for water power (hydraulic energy) ◇ 627.8.09:621.311.214 For pumped-storage hydroelectric power stations ⇒ 627.85
627.772	Lifeboat service, facilities, stations and equipment (land based) ⇒ 629.527	627.81	Reservoirs. Storage ponds ⇒ 556.155 ⇒ 556.55 ⇒ 624.953
627.773	Rocket apparatus, stations and equipment	627.815	Storage capacity
627.775	Air-sea rescue facilities. □ Use of helicopters	627.816	Equalizing reservoirs
627.777	Shipboard or shore rescue devices. Life-saving equipment. Life-preservers in general ⇒ 614.818 ⇒ 626.823.93	627.82	Dams. Barrages <i>Class here information about dams for increasing the water level; for damming as river improvement see 627.43</i> ⇒ 627.132 ⇒ 627.51
627.777.1	Escape devices in the hull. □ Airlocks	627.824	Earth dams. Rockfill dams. Gravity dams
627.777.2	Lifeboats, liferafts (on board)	627.824.2	Earth dams without core (including earth dams with surface reinforcement)
627.777.3	Lifebuoys. Other buoyant furniture or fittings	627.824.3	Earth dams with core. Rockfill dams
627.777.4	Life-belts. Life-jackets (lifevests)	627.824.7	Gravity dams
627.778	Underwater escape systems. □ Airlocks, escape hatches in submarines	627.825	Arch dams
627.8	Dams. Hydraulic power plant works ◇ 627.8:624.1 Substructures, earthworks, foundations etc. for dams ⇒ 351.793 ⇒ 621.311.21 ⇒ 627.51 ⇒ 628.1	627.826/.828	Dams with distributed support
627.8.01	Dams etc. according to material and method of construction 627.8.01 ≡ 624.01	627.826	Multiple arch dams
627.8.03	Dams etc. in particular conditions. Location and shape of dams	627.827	Multiple dome dams
627.8.034	Water in relation to dams etc.	627.828	Cellular dams. Buttress dams. Dams of special construction
627.8.034.43	Water level at dams	627.83	Operating installations of dams. Draw-off and overflow structures. Spillways Overflow spillways
627.8.034.6	Properties and composition of water in relation to structures in water	627.831	Bottom-discharge conduits. Low-level outlets
627.8.034.7	Detritus. Transported material. Precipitation. Deposits. Sediment. Silt	627.832	Water outlets. □ Gates. Sluices
627.8.034.8	Cleaning affected parts of structure	627.833	Shaft spillways. Morning glory spillways
627.8.034.9	Protection from effects of water and air. □ Coatings, linings, facings etc. for dams. Sealing, draining of dams etc.	627.834	Siphon spillways. Spillways with movable flaps
627.8.035.4	Underground structures for dams etc.	627.835	Energy dissipators. Stilling pools
627.8.036	Dams etc. according to layout, ground plan, shape	627.838	
627.8.036.1	Straight	627.84	Intake structures and conduits. □ Intake canals. Head race canals. Forebays. Pressure pipelines. Penstocks. Closing devices
627.8.036.2	Oblique. Skew	627.841	Intake canals. Head race canals. Forebays
627.8.036.3	Curved	627.842	Tunnels. Pressure tunnels
627.8.04	Structural design. Graphical statics. Analytical statics 627.8.04 ≡ 624.04	627.844	Pressure pipelines. Penstocks
627.8.05	Site organization for dam construction: methods and equipment 627.8.05 ≡ 69.05 ⇒ 624.13 ⇒ 627.02 ⇒ 693 ⇒ 694	627.845	Closing devices
627.8.06	Individual parts or structural areas of dams and reservoirs	627.846	Surge tanks
627.8.064.2	Downstream face	627.85	Power houses ⇒ 621.311.21 ⇒ 627.8.09
627.8.064.3	Upstream face	627.86	Tail race canals. Outlet or discharge canals
627.8.064.4	Reservoir bottom	627.88	Ancillary structures. □ Gratings. Rakes. Log chutes. Sand traps ◇ 627.88:626.882 Fishways
627.8.065	Core walls	628	PUBLIC HEALTH ENGINEERING. WATER. SANITATION. ILLUMINATING ENGINEERING <i>Details of machinery, equipment and processes by -1/-9 (from 62-1/-8 and 66-9)</i> ⇒ 351.77 ⇒ 502 ⇒ 614.1/.7

628.1	Water supply. Water treatment. Water consumption	628.112.1	Low-pressure intake. Horizontal intake. Shallow wells
◇ 628.1(1-2) Local systems		628.112.2	High-pressure and relatively high-pressure intake. Vertical intake. Deep wells
◇ 628.1(1-21) Urban systems		628.112.22	Shaft wells
◇ 628.1(1-22) Rural systems		628.112.23	Artesian wells
◇ 628.1(1-28) Intercommunal systems		⇒ 556.364	
◇ 628.1(1-3) Regional systems		628.112.24	Tubular (bore) wells. Bore holes
⇒ 556.18		628.112.3	Underground river-bed water intake. Infiltration intake. Infiltration into the ground. Ray intakes
⇒ 614.777		628.112.4	Intake of spring water. □ Intake facilities: stone reservoirs, water-receiving chambers
⇒ 626.8		628.112.5	Extraction of water from dunes
⇒ 627.8		628.113	Water from surface sources. □ Intake from surface sources. Intake facilities. Types of water intake
⇒ 628.3		◇ 628.113(26) From seas and oceans. Sea water receiving devices	
⇒ 631.67		◇ 628.113(282) From rivers. River intakes in general	
⇒ 663.6		◇ 628.113(285.2) From lakes, water storage reservoirs	
⇒ 696.1			
⇒ 725.1			
628.1.03	Types of water		
	<i>Use to class systems according to the type of water supplied</i>		
	<i>Denote physical, chemical and biological properties by :53..., :54..., :57/59</i>		
628.1.032	Natural water. Raw water	628.113.3	Coastal-type intakes
628.1.033	Main water in general. Potable water. Domestic potable water supply	628.113.4	Riverbed-type intakes
628.1.034	Nonpotable water	628.113.6	Floating and portable intake facilities
628.1.034.2	For industry. Industrial water supply	628.113.7	Bucket intakes
628.1.034.3	For agriculture. Agricultural water supply	628.116	Obtaining of water from the atmosphere
628.1.034.4	For public service needs	628.116.2	Collection of rainwater
	◇ 628.1.034.4:614.84 For fire-fighting	628.116.3	Melting of snow and ice
	◇ 628.1.034.4:625.768.7 For street-watering		
	◇ 628.1.034.4:712.5 For fountains and other water features	628.12	Water lifting. Water delivery pumping stations and installations
			<i>Denote types of pump by :621.65/.69</i>
628.1.036	Spring water and groundwater with special properties	628.13	Storage of water. Water reservoirs
	⇒ 553.7		<i>The '18 auxiliaries at 628.1'18... are particularly applicable here</i>
	⇒ 556.3		⇒ 624.953
	⇒ 613.38	628.132	Reservoirs in general. Pressureless reservoirs. Pure water reservoirs
	⇒ 615.327	628.134	Water towers. Water-pressure reservoirs. Water-pressure columns
	⇒ 615.838	628.135	Pneumatic water-pressure installations
628.1.036.4	Natural mineral waters		
628.1.036.5	Natural medicinal waters	628.14	Water conveyance from intake to distribution network. Water conduits and water-supply networks
628.1.037	Rainwater	628.143	Water conduits. Water-supply pipelines. Aqueducts. Pipe joints
628.1.038	Water of improved quality, of higher purity. Distilled water		<i>Details by :621.643...</i>
628.1'1	Special characteristics of waterworks systems, e.g. of water conduits and water piping networks	628.144	Water-supply networks for delivering water to consumers. Piping of water-supply network
628.1'11	By mode of water feed		<i>The apostrophe auxiliaries from 628.1, especially '13 and '14, are particularly applicable here</i>
628.1'112	Forced-feed		◇ 628.144'132 Branched networks
628.1'113	Gravity-feed		◇ 628.144'14 Zonal networks
628.1'12	By hydraulic regime	628.144.2	Outdoor water-supply networks, e.g. street, courtyard
628.1'122	Pressure	628.144.22	Water-main networks. Main pipelines
628.1'123	Pressureless	628.144.23	Distributive networks (water supply to individual house branches)
628.1'13	By shape	628.144.3	Indoor water-supply networks
628.1'132	Branched (dead-end)	628.145	Crossings and interconnections of water conduits. □ Aqueduct, bridge and level crossings. Siphons. Tunnels. Culverts
628.1'133	Annular (closed)		628.145.2/.5 ≡ 626.823.82/.85
628.1'14	By zone. Zonal waterworks systems		⇒ 628.26
628.1'15	Uniflow systems. Single-pass systems	628.145.8	Interconnections (to and from water conduits)
628.1'16	Repeated water utilization systems	628.146	Network facilities
628.1'17	Return systems		<i>Details by :621.646..., excepting 621.646.97</i>
628.1'18	By reservoir function		
	<i>Used mainly at 628.13</i>		
628.1'181	Regulating		
628.1'182	Stand-by (for storage)		
628.1'183	Stand-by regulating		
628.11	Water-supply sources. Intake, collection, catchment facilities		
	⇒ 556		
628.112	Groundwater. Spring-water. Wells		

628.147	Construction and maintenance of water conduit systems		◇ 628.164-926.33'264 Treatment by calcined soda. Elimination of noncarbonate hardness
628.147.1	Construction		◇ 628.164-926.41 Treatment by slaked lime. Lime-soda method. Elimination of carbonate hardness
628.147.2	Technical maintenance. Maintenance in good repair. Elimination of defects and restoration of the capacity of installations and their equipment		◇ 628.164.046 Thermal methods of hardness elimination
628.147.22	Protection against corrosion and rust		◇ 628.164.081.312.32 Cation method
628.147.23	Treatment of deposits. Cleaning, washing and disinfection of individual facilities and pipe networks	628.165	Water desalination and distillation <i>Denote reagents by -92... and methods by .0...</i>
628.147.24	Thawing of frozen conduits and pipes		◇ 628.165.048 Distillation
628.147.25	Breakdowns and damages: their detection and repair ⇒ 628.179.3	628.166	◇ 628.165.081.312.32 Cation method ⇒ 628.161.2 Disinfection of water <i>Denote reagents by -92... and methods by .0...</i>
628.16	Water treatment: processes, installations and equipment. Water purifying facilities in general <i>Denote reagents used, according to their chemical composition, by -926/-927 from 66</i> ⇒ 502.51 ⇒ 543.3 ⇒ 614.777 ⇒ 725.1	628.167	◇ 628.166-926.57 By silver ◇ 628.166.085 By irradiation ◇ 628.166.094.3-926.214 By ozone. By ozonation Removal of odours and after-tastes from water <i>Denote reagents by -92... and methods by .0...</i>
628.16.0	Individual treatment processes and their equipment 628.16.04/.09 ≡ 66.04/.09	628.168	◇ 628.167.069.84 Degassing
628.16.04	Heat treatment operations	628.168.2	Other types of water treatment
628.16.046	Thermal processes and plant	628.168.3	Deactivation
628.16.048	Distillation		Water stabilization to prevent corrosion and deposition (of salts)
628.16.06	Mechanical methods of purification, and equipment	628.168.4	Addition of various chemicals to alter water properties, e.g. to improve potable water qualities <i>Denote reagents by -92... and methods by .0...</i>
628.16.065.2	Precipitation by reagents. Coagulation ◇ 628.16.065.2-926.62'226 By aluminium sulphate	628.169	◇ 628.168.4.094.412 Addition of fluoride. Fluoridation
628.16.067.1	Filtration	628.17	Treatment and utilization of deposit
628.16.08	Physical and physicochemical methods, and equipment	628.171	Water requirement and consumption. Water losses and their prevention. Water economy
628.16.084	Vibration processes		Water requirement and consumption. Water supply and discharge balance
628.16.087	Electrochemical processes		◇ 628.171'313" Prospective water consumption (requirement)
628.16.094	Oxidative-reductive processes. Regulation of pH etc.		◇ 628.171.033 Household drinking water consumption (requirement)
628.16.094.3	Oxidation ◇ 628.16.094.3-926.214 Ozonation	628.179	◇ 628.171.034.2 Industrial water consumption (requirement)
628.16.098.4	Anaerobic processes		Water economy. Water losses and their prevention
628.16.099.72	Decolorizing	628.179.2	628.179.0 ≡ 66.0
628.161	Purification of raw water in general		Multiple use of water. Return and repeated water supply. Use of sewage in return and repeated water supply. Sewageless production
628.161.1	Requirements as to the quality and properties of water	628.179.3	<i>The '16 and '17 auxiliaries are particularly applicable here</i>
628.161.2	Water purification by removal of specific substances <i>Specify by :546... and :547...</i> ◇ 628.161.2:546.21 Removal of oxygen ◇ 628.161.2:546.72 Removal of iron ⇒ 628.165	628.19	Water losses. Water loss control ⇒ 628.147.25
628.161.3	Water purification by specific reagents <i>Use to denote general points on the use of individual reagents for water purification. Class the use of reagents for specific operations or methods of purification at the operation or method</i> ◇ 628.16.065.2-926.62'226 Coagulation by aluminium sulphate ◇ 628.164-926.33'264 Softening by soda	628.191	Pollution of water sources and supply (causes, prevention and removal) ⇒ 556.388
628.164	Water softening. Reduction of hardness. Prevention of carbonate deposits ◇ 628.164-92 Reagent method of water softening	628.193	Causes of water pollution <i>Specify causes by :...</i> ◇ 628.191:628.39 Water pollution caused by sewage disposal
		628.194	Water pollution with specific materials ◇ 628.193:546.815 Water pollution with lead and its compounds ◇ 628.193:665.61 Water pollution with oil
			Pollution at a certain stage of water supply, distribution or treatment ◇ 628.194:628.11 Pollution of water catchment areas

- 628.196 Prevention and removal of water pollution
Denote methods by :...
 ◇ 628.196:654.9 Breakdown alarms
- 628.2 Urban water removal. Town drainage. Sewerage**
- 628.21 Sewerage systems**
 628.211 Combined sewerage systems
 628.212 Separate sewerage systems. □ Problem of entrance of extraneous rainwater (infiltration)
 628.213 Pumping systems
 628.214 Ejector systems
 628.217 Overflow methods
- 628.22 Calculation of shape and size of sewers**
 ⇒ 628.311
 628.221 Rainfall factors. □ Drainage area. Rainfall calculation, distribution, intensity
 ⇒ 551.577/.579
 628.222 Surface water runoff factors. □ Amount, rate and duration of runoff. Time of concentration. Quantity and rate of discharge. Dry-weather flow
- 628.23 Ventilation of sewers**
 ⇒ 628.8
 628.231 Ventilating shafts in streets
 628.232 Lamp holes. Inspection shafts. Ventilating columns
 628.234 Ventilation pipes outside buildings
 628.235 Ventilation pipes inside buildings
 628.237 Sewer gas
 628.238 Mechanical aids to ventilation
- 628.24 Design and construction of sewers**
 628.241 Main sewers. Intercepting sewers. Branch mains
 628.241.2 Outfall sewers. Ocean outfalls
 628.242 Site drains. House drains (collection lines)
 ⇒ 696.13
 628.245 Drainpipes. Waste pipes
 628.246 Pipe connections, joints, junctions
 628.247 Construction of pipe sewers. Laying and replacement of pipes
 628.248 Construction of monolithic in-situ sewers
- 628.25 Ancillary works to sewers and drains**
 628.251 Small branches. Street connections and traps. Road gully traps
 ⇒ 625.745.3
 628.252 House connections. Private, domestic branches
 628.253 Inspection chambers. Manholes. Lamp holes. Covers. Ladders. Step-irons
 628.257 Valves and penstocks for sewers
 628.258 Overflows and regulators. □ Storm overflows (separation weirs)
 628.259.2 Devices for the separation of liquid from solid matter
- 628.26 Crossings of watercourses. Siphons. Sag pipes**
 ⇒ 626.823.8
 ⇒ 628.145
 628.262 Inverted siphons
 628.264 Bridge siphons
- 628.27 Blockage and fracture of sewers**
- 628.28 Clearing, cleaning and flushing of sewers**
 628.282 Flushing installations. Reservoirs for flushing
 628.284 Gully emptying. Gully sucking
 628.285 Use of floating cleaners. □ Cleaning barges
 628.287 Various cleaning devices. □ Brooms. Scrapers. Sludge lifters
 628.288 Cleaning with pressurized water jets
- 628.29 Raising and discharging of sewage.**
 □ Water-raising devices. Pumping stations. Ejectors
 ⇒ 621.65/.69
 628.292 Elevators. Pumping stations and ejectors
- 628.3 Sewage. Treatment, disposal, utilization of sewage**
 628.3.0 ≡ 66.0
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 628.3.03 Kinds of sewage by origin
 628.3.03 ≡ 628.1.03
 628.3.033 Domestic sewage
 628.3.034.2 Industrial sewage
- 628.31 Sewage properties, treatment and purification in general**
 628.311 Quantity, volume of sewage
 ⇒ 628.22
 628.312 Composition and properties of sewage. Constituents
 628.312.1 Sewage solids
 628.312.2 Substances in solution
 628.312.3 Degree of pollution, toxicity. Biochemical oxygen demand
 628.312.4 Chemical oxygen demand
 628.312.5 Sewage analysis. Determination of sewage composition
Specify by :546/547
 628.313 Sampling of sewage
 628.315 Sewage treatment systems and stages
 628.315.1 Comprehensive treatment
 628.315.2 Primary and secondary treatment
 628.315.21 Primary (including preliminary) treatment
 628.315.23 Secondary (biological) treatment
 628.315.3 Tertiary treatment
 628.316 Treatment by removal of pollutants
 628.316 ≡ 628.16
The kinds of treatment in the subdivisions may be further specified by method, using the .0 auxiliaries. Individual processes are classed at 628.33/.35
 ◇ 628.316.6.094.3 Sewage disinfection by oxidation
 ◇ 628.349.094.3 Sewage oxidation
 Sewage purification by removal of specific substances
 ◇ 628.316.12:546.49 Removal of mercury
 ◇ 628.316.12:547.562.1 Removal of phenol. Dephenolizing of sewage
 628.316.6 Disinfection of sewage
 ◇ 628.316.6.094.3 Sewage disinfection by oxidation
 628.316.7 Removal of odours and after-tastes from sewage
- 628.32 Sewage treatment plants. Sewage works**
 ⇒ 696.135/.138
- 628.33 Physical and mechanical sewage treatment**
 628.334 Screening. Grease and grit removal
 ⇒ 621.928
 628.334.1 Screening
 628.334.11 Bar screens with manual cleaning
 628.334.12 Bar screens with mechanical cleaning
 628.334.13 Fine screens
 628.334.14 Disintegration of solids
 628.334.2 Grit removal. □ Sand traps
 628.334.21 Sand traps
 628.334.26 Grit washing. Grit handling. Grit disposal
 628.334.3 Grease removal. Separation by gravity, air current, vacuum
 628.334.4 Foam, scum removal

628.334.5	Sedimentation process. Sedimentation or settling tanks	628.352	Septic tanks in general
628.334.6	Stormwater tanks	628.353	Biofiltration. Biofilters
628.335	Coagulation, flocculation (without adding chemicals) ⇒ 628.345	628.353.1	Methods of purification through trickling biofilters. Trickling biofilters <i>Details by -9... from 66</i>
628.336	Sludge. Handling and disposal of sludge	628.353.12	Biofilter load ◇ 628.353.12-189.2 Dosage ◇ 628.353.12-531.3 Control, regulation of feed
628.336.1	Raw sludge: composition, properties <i>Details by :53..., :54..., :57...</i>	628.353.13	Application of sewage to filter bed
628.336.3	Sludge processing in general. Sludge preparation. Sludge conditioning <i>Denote methods of processing not given in 628.336.3/5 by the .0 auxiliaries</i>	628.353.14	Ventilation of biological filters
628.336.31	Sludge concentration	628.353.15	Filter media
628.336.32	Condition of sludge for filtration or drying. □ Elutriation	628.353.153	Bacterial flora on the filter medium. Biofilm ⇒ 628.355
628.336.4	Dewatering and filtration of sludge. □ Drying beds ◇ 628.336.4.047 Heat drying ◇ 628.336.4.067.1 Filtration. Filters ◇ 628.336.4.067.5 Centrifuging	628.353.16	Treatment of filter effluent <i>Details by :628.33...</i>
628.336.5	Sludge digestion. Methane tanks. Digesters. Septic tanks ◇ 628.336.5.098.2 Aerobic digestion ◇ 628.336.5.098.4 Anaerobic digestion	628.353.2	Contact beds
628.336.51	Mesophilic digestion <i>Details by -95... and -965.8... from 66</i>	628.353.3	Aerated filters
628.336.52	Thermophilic digestion	628.354	Sand filtration. Sand filters
628.336.53	Sludge heating	628.354.1	Intermittent sand filters
628.336.54	Sludge circulation. Sludge discharge	628.354.2	Continuous sand filters. Magnetite filters
628.336.55	Supernatant liquor: withdrawal, disposal	628.355	Activated sludge in general. Bacterial flora of biological sewage treatment in general. Chemistry and physiology of the microorganisms ⇒ 628.353.153
628.336.56	Addition of other substances to digestion tanks. □ Addition of refuse, screenings, grease skimmings, nightsoil	628.356	Methods of sewage purification by activated sludge. Aeration in general
628.336.57	Digested sludge: composition, properties	628.356.1	Diffused-air aeration
628.336.58	Control chambers for digestion. Pipe galleries	628.356.2	Mechanical aeration
628.336.6	Sludge gas (methane/carbon dioxide mixture): quantity, composition, properties ⇒ 628.385	628.356.3	Aeration tanks
628.336.7	Sludge disposal or destruction ⇒ 628.381	628.356.4	Re-aeration
628.336.71	Incineration of sludge. Gasification of sludge	628.356.5	Secondary settlers
628.336.72	Barging to sea	628.357	Biological ponds. Sewage ponds. Self-purification
628.336.73	Land fills. Lagoons	628.357.1	Lagoons for complete treatment of sewage
628.336.8	Conveyance and transporting of deposit formed during and after (as a result of) processing of sewage, sludge and digested sludge ◇ 628.336.8:621.643.2 Through pipes	628.357.2	Lagoons for effluent treatment
628.337	Electrical treatment and purification of sewage ⇒ 628.349.087	628.357.3	Fish ponds ⇒ 626.886 ⇒ 639.311
628.34	Chemical sewage treatment	628.357.4	Ponds for treatment by aquatic plants
628.342	Mixing with chemicals. Addition of chemicals. Stirrers, mixers etc.	628.358	Final sewage purification process. Deep sewage purification. Fine purification
628.345	Precipitation. Chemical coagulation ⇒ 628.335	628.38	Utilization of sewage and of sewage constituents. Utilizable sewage constituents ⇒ 628.477 ⇒ 696.139
628.349	Other chemical methods ◇ 628.349.081.312.3 Purification by ion exchange resins ◇ 628.349.087 Electrochemical purification ◇ 628.349.087.2 By electroreduction ◇ 628.349.087.3 By electro-oxidation ◇ 628.349.087.4 By electroflotation ◇ 628.349.087.5 By electrocoagulation ◇ 628.349.087.7 By electrolysis ◇ 628.349.094.3 Sewage oxidation	628.381	Utilization of sewage in agriculture ⇒ 626.841 ⇒ 631.863
628.35	Biological and biochemical treatment of sewage. Equipment for biological and biochemical treatment	628.383	Fats and oils from sewage
628.351	Preliminary treatment of sewage before biological purification. Securing of microflora activity ⇒ 628.312.3	628.385	Gas, methane from sewage ⇒ 628.336.6
		628.386	Extraction of other valuable substances from sewage
		628.387	Utilization of the final effluent
		628.388	Use of radioactive wastes ⇒ 621.039.77 ⇒ 628.477.047
		628.39	Disposal of untreated, unusable waste into natural environments. Pollution of water, soil and air. Preventive measures ⇒ 351.777.6 ⇒ 543.3 ⇒ 574.63 ⇒ 614.7 ⇒ 628.19 ⇒ 628.47

628.393	Sewage collectors. Sludge stores	628.4.046	Disease-carrying refuse. Infectious wastes. □ Wastes from hospitals, clinics, operating theatres, surgeries. Wastes from pathological and bacteriological establishments. Corpses as waste. Refuse from medical research and nursing establishments ⇒ 620.266.2 ⇒ 628.513
628.394	Disposal of sewage into natural waters (seas, rivers, lakes) <i>Details by (26/28) from Table 1e</i> ⇒ 556.388 ⇒ 628.472.7 ⇒ 628.515 ⇒ 628.541	628.4.047	Radioactive waste ⇒ 620.267 ⇒ 621.039.7 ⇒ 628.388
628.395	Disposal of sewage into the atmosphere ⇒ 551.510.4 ⇒ 628.511 ⇒ 628.512 ⇒ 628.52	628.4.06	Analytical procedure <i>Details by :543...</i>
628.396	Disposal of sewage into the soil, geological strata, caverns. Underground disposal ⇒ 628.472.4 ⇒ 628.516 ⇒ 628.542	628.4.061	Qualitative analysis
628.398	Disposal and burial of especially harmful materials, e.g. radioactive materials	628.4.062	Quantitative analysis
		628.4.08	Measurement, determination, evaluation of waste. Calculation of amount of refuse
628.4	Urban hygiene. Wastes. Refuse. Rubbish. Garbage. Collection and disposal of town wastes	628.42	Latrines. Cesspools. Collection and disposal of excreta
628.4.02	Design, organization, management etc.	628.422	Bucket latrines
628.4.03	Origin and quantity of refuse	628.423	Pit latrines
628.4.032	Domestic refuse. □ Solid waste from households, hotels, restaurants etc. Kitchen waste, food scraps etc. Waste paper. Fuel waste, ashes, cinders. Worn-out and broken objects	628.424	Other arrangements. Soil-spreading systems (earth closets) etc.
628.4.033	Highway refuse. □ Refuse from street cleaning. Litter. Rubber detritus. Street dirt. Dust. Cigarette ends. Fallen leaves. Market refuse. Detritus from sanding, salting etc. in winter	628.425	Public lavatories in general ⇒ 625.748.2 ⇒ 696.14 ⇒ 725.1
628.4.034	Business, industrial and trade refuse (waste not deriving directly from production processes). □ Packing materials. Fuel waste. Office refuse. Cotton waste. Rags	628.427	Collection, conveying and disposal of excreta
628.4.035	Bulky refuse (Waste too bulky for ordinary containers). □ Discarded furniture. Large boxes	628.46	Urban cleansing. Collection and removal of refuse <i>Denote equipment and vehicles by :621.6... and :629...</i>
628.4.036	Builders' waste. Construction debris. Demolition debris ⇒ 69.059.6	628.462	Urban cleansing. Street cleansing. □ Road sweeping. Watering of roads. Sprinkling ⇒ 351.777.614 ⇒ 625.768.1
628.4.037	Derelict cars, wrecks. Tyres. Scrap metal	628.463	Collection of refuse. Refuse collection systems. □ Collection (storage) of refuse in containers. Refuse bins. Dustbins (garbage cans). Bulk rubbish containers. Rubbish skips ◇ 628.463:692.758 Use of refuse chutes ⇒ 625.748.44
628.4.038	Industrial waste. Manufacturing, factory waste (Waste deriving from production processes). □ Spoilt raw products. Faulty batches. Rejects. Casting sand. Fuel waste ⇒ 658.567	628.464	Collection of refuse in vehicles. □ Refuse carts, vans, lorries, trucks. Rubbish boats, barges
628.4.04	Kinds of refuse <i>Details by -03... from Table 1k, -403/-405 from 62-403/-405, etc.</i>	628.465	Removal, transport, conveying of refuse
628.4.042	Readily decomposed refuse. Biodegradable wastes. □ Food scraps. Organic waste. Excreta. Garden and park refuse. Straw. Textiles. Leather	628.468	Depots for refuse collection
628.4.043	Refuse not readily decomposed. Barely degradable and nondegradable wastes. □ Wood. Pulp products. Slag. Clinker. Builders' waste. Rubble. Glass. Metal. Plastics. Rubber	628.47	Disposal and utilization of urban refuse ⇒ 351.777.61 ⇒ 614.78 ⇒ 628.38/39
628.4.045	Chemical and explosive refuse. □ Poisonous, toxic wastes. Substances containing mercury, cyanide, chromium etc. Concentrated acids and alkalis. Noxious decomposition products, e.g. fluorine, chlorine, nitrous fumes ⇒ 620.26	628.472	Disposal of refuse by dumping, tipping
		628.472.2	Uncontrolled public tipping
		628.472.3	Controlled tipping. Refuse tips, dumps
		628.472.4	Infilling of ground with refuse and sludge ⇒ 628.396
		628.472.7	Dumping of refuse in sea, lakes or watercourses ⇒ 628.394 ⇒ 628.515 ⇒ 628.541
		628.473	Disposal of refuse by composting
		628.473.2	Decomposition of organic matter. Study of rotting
		628.473.3	Composting methods. □ Surface composting. Composting in silos. Composting in fermentation cells

628.473.4	Design and structure of compost works. Processes for separate or combined decomposition of solid and liquid refuse	⇒ 622.411.5 ⇒ 622.8/.81 ⇒ 625.097 ⇒ 662.613.13 ⇒ 662.96 ⇒ 699.87
628.473.6	Biology and hygiene of composting	
628.474	Disposal of refuse by burning, incineration ⇒ 662.65 ⇒ 662.734	
628.474.3	Burning in incineration plant	628.511.1 Dust in general
628.474.32	Processes involved in combustion	628.511.12 Dust according to origin
628.474.33	Methods of combustion of refuse	628.511.122 Dust as intended product of a process (dust as end-product)
628.474.35	Combined combustion of refuse and sludge	628.511.123 Dust as unwanted by-product
628.474.36	Combined combustion of refuse and solid fuel	628.511.124 Dust of geological origin
628.474.37	Combustion of heavy industrial waste	628.511.125 Dust of cosmic origin
628.474.38	Heat recovery from refuse incineration	628.511.13 Dust according to nature and content
628.474.5	Kinds and sizes of incineration plant. Arrangement of furnaces	628.511.132 Organic dusts. □ Sugar dust. Cereal dust. Pollen
628.474.7	Operation and details of incineration plant	628.511.133 Inorganic dusts. □ Metallic dust. Stone dust. Mineral dust
628.474.72	Bunkers. Conveying equipment. Dust separation	628.511.134 Street dust
628.474.74	Extraction of valuable constituents. □ Metal separators	628.511.135 Textile dust
628.474.75	Fire grate. Fire gas. Conveying of flue gas	628.511.137 Composite dusts
628.474.76	Purification of exhaust gas <i>Details by :66.074...</i>	628.511.2 Prevention of dust formation
628.474.77	Treatment of slag. □ Slag disposal equipment	628.511.4 Precipitation and filtering of dust
628.475	Disposal of refuse by gasification (pyrolysis)	628.512 Measures against toxic gases and vapours. □ Exhaust ducts ⇒ 62-784.2 ⇒ 623.459.8
628.475.3	Processes involved in gasification	
628.475.4	Methods of gasification of refuse	
628.475.5	Gas yield	628.513 Measures against infections, disease-carrying exhalations. Measures against bacteria ⇒ 620.266.2 ⇒ 628.4.046
628.475.7	Gasification plant 628.475.7 ≡ 628.474.7	
628.477	Utilization of treated and untreated waste, refuse, residues ⇒ 628.38	628.514 Measures against nuisance from steam, water vapour ⇒ 628.84 ⇒ 697.9
628.477.1	Direct use of untreated waste	
628.477.2	Production of industrially useful substances from waste. □ Production of protein, of alcohol	628.515 Measures against pollution of water ⇒ 614.777 ⇒ 628.394 ⇒ 628.472.7 ⇒ 628.541
628.477.3	Use of compost in agriculture ⇒ 631.879	
628.477.4	Quality requirements for compost. Methods for evaluating compost	
628.477.6	Recovery, recycling and re-use of waste and scrap. Salvage of scrap ⇒ 658.567	628.516 Measures against pollution of the soil ⇒ 351.777.6 ⇒ 614.76/.77 ⇒ 628.396
628.477.7	Utilization of ash, slag, residual coke and tars from gasification ⇒ 662.613	628.517 Measures against acoustic nuisances
628.477.8	Utilization of waste heat in incineration	628.517.2 Measures against noise. Noise control ⇒ 351.759.4 ⇒ 534.83 ⇒ 656.053.7 ⇒ 699.844
628.48	Other matters	
628.482	Questions of materials ⇒ 620.2	628.517.4 Measures against vibrations ⇒ 699.84
628.483	Questions of corrosion ⇒ 620.193 ⇒ 620.197	628.518 Measures against optical, electrical and other nuisances. Measures against heat, high temperatures ◇ 628.518:535.21 Against light, visible radiation ◇ 628.518:539.16 Against radioactivity, ionizing radiation
628.5	Measures against industrial and other nuisances ⇒ 613.6 ⇒ 614.7	
628.51	Measures against industrial nuisances of various kinds	628.52 Measures against industrial fumes, effluvia, smells, odours ⇒ 628.512/.514 ⇒ 66.099.73 ⇒ 699.87
628.511	Measures against dust ⇒ 543.275 ⇒ 551.510.4 ⇒ 613.633 ⇒ 614.715 ⇒ 614.893/.894 ⇒ 62-758.31 ⇒ 62-784.4 ⇒ 621.7.014.8 ⇒ 621.928.9 ⇒ 622.23.057	628.521 Siting of factories ⇒ 658.21 628.522 Use of tall chimneys ⇒ 624.971 ⇒ 697.8 628.523 Condensation of fumes in water 628.524 Combustion of fumes in furnaces

628.53	Measures against smoke and soot. Smoke abatement	628.87	Comfort requirements. Indoor climate and body heat regulation ⇒ 612.53
628.532	Against smoke from domestic fires	628.88	Indoor climates associated with different types of heating, cooling and ventilating system. Measurements of installations
628.538	Against smoke from steam-powered generators, power houses	628.882	With independent room heaters ⇒ 697.2
628.539	Against smoke from particular industries	628.883	With central heating systems using hot air or gas as conductor (not emitted into rooms) ⇒ 697.385
628.54	Treatment of industrial solid wastes	628.884	With central heating systems using hot water ⇒ 697.445
628.541	Discharge of wastes into watercourses ⇒ 628.394 ⇒ 628.472.7 ⇒ 628.515	628.885	With central heating systems using steam ⇒ 697.515
628.542	Burial	628.889	Effect of indoor climate measurements on ventilation and air-conditioning installations ⇒ 697.9
628.545	Incineration	628.89	Influence of design and structure of building on indoor climate
628.55	Artificial conditioning of the natural atmosphere. □ Fog and mist control. Fog dispersal. Cloud dispersal ⇒ 551.509.6	628.9	Lighting. Illumination engineering ⇒ 535 ⇒ 621.32 ⇒ 622.47 ⇒ 683.8
628.58	Measures against other nuisances 628.58 ≡ 699.8	628.9.02	Light from celestial bodies and the sky
628.6	Drainage and sanitary installations of buildings (particularly public health requirements) ⇒ 351.778.5 ⇒ 613.5 ⇒ 644.68 ⇒ 645.68 ⇒ 696.14	628.9.021	Sunlight. Daylight
628.8	Indoor climate. Air conditioning. Heating. Ventilating <i>Mainly public health aspects, and particularly in large spaces, e.g. tunnels, arcades</i> ⇒ 551.584.6 ⇒ 62-784.2 ⇒ 622.4 ⇒ 628.23 ⇒ 644.1 ⇒ 697	628.9.025	Moonlight
628.8.02	The human factor. Influences of occupancy	628.9.026	Starlight
628.81	Heating generally, especially of large spaces	628.9.03	Luminescent light sources. □ Chemiluminescence. Phosphorescence. Fluorescence. Electroluminescence ⇒ 535.37 ⇒ 661.143 ⇒ 667.526 ⇒ 667.637.4
628.82	Cooling generally, especially of large spaces	628.9.04	Light from artificial light sources
628.83	Ventilation, especially of large spaces. Air distribution and movement	628.9.041	Light from electric light sources <i>Specify by :621.32...</i>
628.84	Air conditioning: technique and plant	628.9.041.8	Combined light sources
628.85	Physical criteria determining indoor climates: definition, nature, measurement	628.9.045	Flaming torches (links). Flambeaux. Candles ⇒ 662.126 ⇒ 665.14 ⇒ 665.194
628.852	Temperatures. □ Air temperature. Wall temperature. Mean radiant temperature	628.9.047	Light from lamps using combustion or other chemical reactions (lamps with solid, liquid or gaseous combustibles) ⇒ 683.8
628.853	Humidity	628.9.048	Light from atomic light sources
628.853.2	Absolute humidity	628.9.06	Systems and methods of lighting
628.853.3	Relative humidity	628.9.061	Direct lighting
628.854	Air movement. Air circulation. Ventilation	628.9.062	Indirect lighting (Concealed lighting. Reflected lighting)
628.854.2	Air velocity	628.9.063	Semi-indirect lighting
628.854.3	Ventilation rate	628.9.064	Directed or projected lighting. □ Spotlighting. Floodlighting
628.86	Composite criteria determining indoor climates: definition, nature, measurement	628.9.066	Transillumination
628.862	Directly measurable values. □ Temperatures, thermometer readings. Kata value (cooling rate)	628.91	Technique of light production
628.863	Derivable values (not directly measurable). □ Comfort value	628.92/97	Illumination engineering. Technique of application of light
		628.92	Natural lighting. Daylighting of interiors
		628.921	Generalities. Design and calculation. Daylight factor (sky factor)
		628.922	Side lighting in general
		628.924	Fenestration. Location, number, dimensions and distribution of windows
		628.925	Influence of light transmission properties of materials on natural lighting. Type of glazing
		628.925.2	Transparent glazing

628.925.3	Translucent glazing	628.971.8	Lighting of transport and communication spaces (except fixed street lighting). □ Lighting of parking areas, airstrips, terminuses, depots
628.925.5	Prismatic glazing (ribbed glass)		
628.925.7	Effect of dirt on natural light transmission	628.971.9	Lighting for outdoor work
628.926	Influence of reflection properties of materials on natural lighting. □ Effect of dirt on reflection of natural light	628.972	Interior lighting in general
		628.973	Interior lighting predominantly for atmosphere. Mood lighting. Lighting for psychological and aesthetic effect
628.927	Influence of diffusion properties of materials (and of vapour, fog, smoke etc.) on natural lighting		
628.928	Top lighting (overhead lighting). □ Roof lighting. Lighting by skylights	628.973.1	Domestic, room lighting. Lighting in the home, dwellings, restaurants etc.
628.93	Artificial lighting	628.973.2	Lighting of shops, shop windows, showrooms
628.932	Use of point sources, line sources, plane or surface sources or diffusers	628.973.3	Lighting of theatres, cinemas etc.
628.932.1	Use of line sources. □ Strip lighting	628.973.4	Lighting of museums, galleries etc.
628.932.2	Use of plane or surface sources or diffusers. □ Panel lighting	628.973.5	Lighting of churches
628.932.3	Use of point sources. □ Light bulbs	628.974	Festive lighting and illuminations. Lighting for publicity, advertising. Architectural use of lighting
628.933	Use of three-dimensional diffusers. □ Globes		
628.935	Influence of light transmission properties of materials on artificial lighting. □ Effect of dirt on artificial light transmission	628.974.6	Festive lighting and illuminations (e.g. of parks, fairgrounds, pleasure gardens). Lighting to enhance public buildings, fountains etc.
		628.974.7	Lighting for publicity, advertising. Luminous signs (neon signs etc.) ⇒ 659.135
628.936	Influence of reflection properties of materials on artificial lighting		
628.937	Influence of diffusion properties of materials (and of vapour, fog, smoke etc.) on artificial lighting	628.974.8	Architectural use of lighting
		628.975	Lighting for signals, warning and direction signs ⇒ 627.71 ⇒ 627.72 ⇒ 654.91 ⇒ 656.054/055 ⇒ 656.25
628.94	Luminaires (lamps, lighting fittings). Light projectors. Beam-directing devices		
	<i>Class complete fittings here</i>	628.976	Lighting of work areas, other than general room lighting. Localized lighting (for concentrated work, study). □ Work lamps. Reading lamps. Desk lamps. Lamps on machines ⇒ 53.085.49
	<i>Denote sources and systems of lighting by .02/06</i>		
628.941	Fixed lamps	628.977	Interior lighting predominantly for work
628.941.3	Ceiling lamps	628.977.1	Lighting of factories, workshops, workrooms
628.941.4	Wall lamps	628.977.2	Lighting of offices, drawing offices, art rooms, schoolrooms, libraries, ticket offices etc.
628.941.6	Standing lamps, fixed		
628.941.8	Hanging lamps	628.977.4	Lighting for health care and medical treatment. □ Hospital lighting
628.943	Movable lamps connected to mains supply (electric or gas). □ Mains table lamps. Standard lamps	628.977.7	Lighting of large halls, concourses etc.
		628.977.8	Interior lighting of vehicles. Lighting of traffic installations, garages, signal boxes etc.
628.944	Portable lamps (battery-powered or with portable fuel supply). □ Lanterns. Flashlights. Electric torches		
		628.978	Special-purpose lighting
628.946	Light projectors	628.978.3	Fluorescent lighting
628.946.1	Mirror projectors, reflecting	628.978.6	Emergency lighting. Safety lighting with reserve power
628.946.2	Lens projectors		
628.946.3	Combined mirror-lens projectors	628.978.7	Air-raid precautionary lighting. Obscured lighting for black-outs ⇒ 355.585
628.946.8	Accessories for reflectors, projectors and refractors		
628.947	Reflectors without own light source		
628.948	Building arrangements. Structural fittings. □ Recesses. Screens. Glass ceilings	628.979	Specific effects of lighting. Effects on humans, on other organisms ◇ 628.979:159.93 Lighting and the psychology of sensory perception. Psychological effect of lighting on humans ◇ 628.979:612.014.44 Physiological effect of lighting on humans
628.95	Accessories for lighting fittings. Parts, materials etc.		
628.952	Shades. Diffusers. Reflectors. Decorative lamp hangings ⇒ 535.345.6		
628.953	Light guides as part of illuminating devices	628.98	Evaluation of light (physical, technical, economic, physiological and psychological aspects)
628.955	Materials for lighting fittings. Optical characteristics of materials ⇒ 666.24/.26		
628.97	Lighting according to the object illuminated. Special-purpose lighting	628.981	Evaluation of light sources ◇ 628.981:621.326 Evaluation of light from electric incandescent lamps, filament lamps
628.971	Exterior lighting ⇒ 625.745.8 ⇒ 725.94	628.984	Evaluation of luminaires (lamps, lighting fittings)
		628.987	Evaluation of lighting installations, complete systems ◇ 628.987:628.977.2:744.2 Evaluation of lighting installations in drawing offices
628.971.6	Street lighting, fixed		
628.971.7	Lighting of squares for public events. Lighting of sports grounds, stadiums etc.		

629	TRANSPORT VEHICLE ENGINEERING <i>The .0 special auxiliaries listed at 629.0... are applicable throughout 629. They provide a common framework for more specific .0 auxiliaries which are listed at each main division of 629 and which differ in detail to suit the main categories of vehicle (land, rail, water, air). Two or more .0 auxiliaries may be directly combined under a main number. Details of machine characteristics, components, operation and control, power sources etc. by -1/-9 (from 62-1/-8 and 66-9), when these are not enumerated in the .0 auxiliaries</i> ⇒ 625 ⇒ 656	629.031 Propulsion systems using muscle and natural forces 629.032 Wheel drive or propulsion systems 629.033 Propulsion systems for particular kinds of vehicle, e.g. track and rack-and-pinion propulsion 629.034 Propulsion systems for particular kinds of vehicle, e.g. rope, cable, chain and worm-gear propulsion 629.035 Propeller or screw propulsion systems 629.036 Rocket or jet propulsion systems 629.038 Combined propulsion systems
629.01	Vehicle design principles: characteristics, capabilities, tests 629.012 Dimensions. Dimensional design 629.013 Weights. Tonnage. Capacity. Payload 629.014 Functional capabilities. Mode of use <i>Use the 629.014... auxiliaries only for occasional or special capabilities or uses not provided for in the direct subdivisions of 629.3/9</i>	629.04 Interior layout: accommodation, installation, equipment for persons and freight ⇒ 629.06 ⇒ 674.25 629.042 Interior layout and finish. Compartments. Cabins etc. 629.043 Crew arrangements, stations and quarters. Control compartment 629.044 Entry and exit arrangements. □ Gangways. Ladders 629.045 Freight (cargo): loading, unloading, stowage and tie-down arrangements 629.046 Passenger comfort, service and hygiene arrangements <i>Further details by :64... and :684...</i>
629.014.1	General functional capabilities.	
	□ Multifunctional. Multipurpose	
	⇒ 629.9	
629.014.17	Amphibious	
	⇒ 629.94	
629.014.2	Active vehicles. □ Tractor (towing), pusher, carrier, launcher vehicles	
629.014.3	Passive vehicles. □ Towed, pushed, carried, launched vehicles	
629.014.4	Trains (of active and passive vehicles)	
629.014.5	Articulated vehicles	
629.014.6	Passenger vehicles in general	
629.014.7	Freight (cargo) vehicles in general	
629.014.8	Vehicles designed for special purpose	
629.015	Mechanics and dynamics in relation to medium, environment	
629.016	Performance characteristics, capabilities. Assessment	
629.017	Operational reliability, fitness for use.	
	□ Stability. Controllability. Manoeuvrability.	
	Braking capacity	
629.018	Vehicle test procedures and equipment	
	⇒ 629.072	
	⇒ 629.081	
	⇒ 629.086	
629.02	Vehicle structure, construction principles, general layout and parts ⇒ 674.25	
629.021	Construction principles and properties (of whole or parts)	
629.022	General layout, configuration, relation of parts to whole vehicle	
629.023	Vehicle shell, framework. Structural members	
629.024	Bodywork	
629.025	Special vehicle structures	
629.026	Layout of propulsion unit structures	
	⇒ 629.03	
629.027	Support, suspension, shock-absorbing structures. Wheels etc.	
629.028	Attachments. Couplings. Mountings and other auxiliary structures	
629.03	Propulsion systems <i>Further details by -8... from 62-8 and/or :621...</i> ⇒ 629.026	
		629.05 Guidance, control-initiation and navigation systems and instruments (vehicle-borne) ⇒ 629.062 ⇒ 629.066 629.051 Principles and purpose of guidance and navigation systems 629.052 Means of operation of guidance systems 629.052.2 Acoustic systems. □ Sonar. Echo systems ⇒ 681.883 629.052.3 Radio navigation systems, including television and radar. Laser systems ◇ 629.052.3:621.396.75 Use of radiogoniometric stations in navigation. Direction-finding radio stations ◇ 629.052.3:621.396.93 Use of bearing signals. Radio beacons. Ground control ◇ 629.052.3:621.396.96 Radar navigation systems ◇ 629.052.3:621.396.962.3 Navigational use of transponders 629.052.4 Electromagnetic systems 629.052.5 Infrared systems 629.052.6 Optical systems 629.052.7 Inertial systems 629.052.9 Other systems or means of operation 629.053 Special systems of control for particular types of vehicle 629.054 Vehicle-borne instruments ⇒ 53.08 ⇒ 621.317 ⇒ 681.2 629.056 Means of operation of navigation systems

629.056.2	Astronomical (celestial) navigation systems ⇒ (083.5) ⇒ 681.786.3	629.072.182	Great circle sailing Use with 629.5
629.056.6	Inertial navigation systems	629.072.184	Rhumb-line sailing Use with 629.5
629.056.8	Satellite navigation systems	629.072.19	Homing
629.056.82	Doppler systems (e.g. TRANSIT)	629.073	Operational techniques for special conditions, e.g. weather, state of the route or terrain, engine failure 629.073 ≡ 656.052.5
629.056.84	Time-difference systems	629.075	Departure techniques. Operations for starting to move, getting under way, take-off, launch etc. ⇒ 629.085
629.058	Operational characteristics (of systems and instruments)	629.076	Operations en route, during the journey
629.06	Vehicle auxiliary systems and devices ⇒ 629.04	629.077	Arrival techniques. Operations for slowing and stopping, berthing, landing etc. ⇒ 629.087
629.062	Control-linking and reinforcing systems. Servo controls etc. ⇒ 629.05	629.078	Operations for on-board maintenance of the vehicle (by the crew)
629.063	Auxiliary supply and feed systems Details by -6/-72 from 62, and by :621.6...	629.08	Base equipment, installations and corresponding technical procedures ⇒ 656.06 ⇒ 725.1
629.063.2	Gas supply and feed systems	629.081	Workshop, vehicle shed or depot installations and equipment ⇒ 629.018
629.063.4	Water supply and feed systems	629.082	Ancillary service or supply installations, depots and stores (for fuel, water etc.)
629.063.6	Fuel supply and feed systems	629.083	Maintenance, handling, servicing and repair of vehicles
629.063.7	Lubricant supply and feed systems	629.084	Spare parts and replacements. □ Issue. Stores. Stock control ⇒ 658.78
629.064	Auxiliary power systems in general	629.085	Base equipment and technical procedure for starting, take-off, launching ⇒ 629.075
629.064.2	Pneumatic systems	629.086	Base equipment and techniques for en-route control and tests: tracking etc. ⇒ 629.018 ⇒ 629.076
629.064.3	Hydraulic systems	629.087	Base equipment and technical procedure for stopping, slowing, landing ⇒ 629.077
629.064.5	Electric systems Details by :621.3...	629.3	Land vehicle engineering (excluding rail vehicles) ⇒ 621.86/.87 ⇒ 623.43
629.064.7	Auxiliary engines	629.3.01	Vehicle design: principles, characteristics, capabilities, tests
629.065	Materials-handling equipment. □ Refuelling systems and devices. Revictualling systems and devices Details by :621.86/.87 ⇒ 629.045	629.3.012	Dimensions. Dimensional design
629.066	Observation signalling, communication and identification devices ⇒ 629.05	629.3.012.2	Length
629.067	Vehicle safety and protection systems Further details by :614.8... and :656.08... ⇒ 629.047	629.3.012.21	Wheelbase
629.069	Other auxiliary systems forming part of the vehicle. □ Armament systems	629.3.012.3	Breadth
629.07	Technical aspects of vehicular operations (operational techniques), including performance For nontechnical aspects use 656.052 Use .07... to denote equipment, devices, technical modifications etc. for the listed operations ⇒ 629.01	629.3.012.31	Track width
629.071	Preliminary operations and preparations. Operations after coming to rest, after arrival	629.3.012.4	Height
629.072	Driving, navigating, piloting operations in general. □ Test runs (trial runs)	629.3.012.5	Measures of volume
629.072.1	Navigation. Route planning	629.3.013	Weights. Capacity. Tonnage. Payload
629.072.11	Principles of navigation	629.3.013.1	Vehicle weight (weight of construction)
629.072.112	Course plotting	629.3.013.4	Total weight
629.072.114	Dead reckoning	629.3.013.5	Carrying capacity. Payload. Tonnage
629.072.116	Position lines (Sumner lines). Fixes	629.3.013.521	Gross tonnage
629.072.12	Determination of various features	629.3.013.522	Net tonnage
629.072.122	Determination of geographical position	629.3.013.55	Load capacity
629.072.124	Determination of height above sea level	629.3.013.56	Useful load
629.072.126	Determination of azimuth	629.3.014	Functional capabilities and mode of use of vehicles Use the auxiliaries listed at 629.014... only for occasional or special capabilities or uses not provided for in the direct subdivisions of 629.3
629.072.128	Determination of elevation. □ Elevation from a fixed point		
629.072.15	Determination of relative position		
629.072.152	Heading. Bearing. Distance		
629.072.154	Height above ground Use with 629.7		
629.072.17	Determination of speed and velocity		
629.072.172	Ground speed. Air speed. Sea speed		
629.072.174	Angle and rate of climb or descent Use with 629.7		
629.072.18	Route planning		

629.3.015	Mechanics and dynamics in relation to the medium <i>For mechanics as such use 531/534</i> ⇒ 625.03	629.3.023.24	Floor. Floor hatches
629.3.015.3	Aerodynamics	629.3.023.25	Doors. Hatches. Shutters. Locks. Steps. Footboards
629.3.015.4	Strength requirements	629.3.023.26	Windows. Windscreens. Window panes. Window accessories ◇ 629.3.023.26-787.3 Windscreen wipers
629.3.015.5	Vibrations. Prevention and control of vibrations ⇒ 629.3.027.3	629.3.023.27	Floor connection pieces. Platforms. Transition equipment for articulated vehicles
629.3.015.6	Noise. Prevention and control of noise	629.3.023.29	Other parts of the body
629.3.016	Performance	629.3.023.9	Surface coatings
629.3.016.3	Range	629.3.024	Body shapes (construction)
629.3.016.5	Speed	629.3.024.1	Body forms
629.3.016.8	Characteristics and capabilities under special conditions	629.3.024.11	Single-decker
629.3.017	Operational reliability. Roadworthiness	629.3.024.12	One-and-a-half deck
629.3.017.2	Stability	629.3.024.13	Double-decker
629.3.017.3	Manoeuvrability	629.3.024.18	Multidecker
629.3.017.5	Braking capacity	629.3.024.3	Front
629.3.018	Vehicle test procedures and equipment	629.3.024.4	Top
629.3.018.2	Laboratory and special equipment. Testbench ⇒ 629.3.081.3	629.3.024.5	Side
629.3.018.4	Strength tests: fatigue, vibration, impact etc.	629.3.024.6	Sections
629.3.018.7	Running and operating tests	629.3.024.61	Longitudinal sections
629.3.02	Vehicle structure <i>Engines and motors are under 621</i> <i>Parts not listed under 629.3.0 may be denoted by the -1/-9 auxiliaries (from 62-1/-8 and 66-9) in combination with .02 or its subdivisions</i> ◇ 629.3.02-59 Braking devices ⇒ 674.25	629.3.024.62	Cross sections
629.3.021	Construction principles and properties (of whole or parts)	629.3.024.7	Rear
629.3.021.2	Vehicle construction types	629.3.024.8	Bottom
629.3.021.21	Vehicles with separate body and underframe	629.3.025	Special structures for land vehicles. Air deflector plates
629.3.021.24	Monocoque or tubular construction	629.3.025.2	Stabilizers
629.3.021.242	Monocoque construction	629.3.025.3	Spoilers
629.3.021.243	Tubular construction	629.3.026	Layout, positioning of power units
629.3.022	General layout and configuration. Relation of parts to whole. Layout	629.3.026.1	Power units at the front of the vehicle
629.3.022.4	Vehicles according to the number of wheels, tracks, runners etc. <i>Use especially when the number differs from the usual for the type of vehicle</i> ◇ 629.322.022.44 Four-wheeled pedal cycles ◇ 629.331.022.43 Three-wheeled cars	629.3.026.11	Parallel to driving direction
629.3.022.41	One-wheeled, one-tracked etc.	629.3.026.12	Transfer units
629.3.022.42	Two-wheeled, two-tracked etc.	629.3.026.13	Inclined to driving direction
629.3.022.43	Three-wheeled, three-tracked etc.	629.3.026.2	Power units in the middle of the vehicle
629.3.022.44	Four-wheeled, four-tracked etc.	629.3.026.3	Power units at the back of the vehicle
629.3.022.45	Five-wheeled, five-tracked etc.	629.3.026.4	Under-floor power units
629.3.022.46	Six-wheeled, six-tracked etc.	629.3.027	Support, suspension and shock-absorbing structures. Wheels etc.
629.3.022.48	Multiwheeled, multitruacked etc.	629.3.027.1	Axles, with parts (supporting axles, rotating parts)
629.3.023	Shell, framework and structural members	629.3.027.11	Front axle
629.3.023.1	Chassis	629.3.027.12	Rear axle
629.3.023.11	Chassis frame (complete)	629.3.027.14	Bogies
629.3.023.12	Longitudinal runner	629.3.027.18	Forks for cycles or motorcycles
629.3.023.13	Cross member	629.3.027.2	Steering mechanism 629.3.027.2 ≈ 621.83
629.3.023.15	Framework attachments	629.3.027.233.8	Worm-gear steering
629.3.023.151	Attachments for engine suspension	629.3.027.3	Springs. Suspension. Shock absorbers <i>Specify by -272... or -567...</i>
629.3.023.153	Attachments for wheel suspension	629.3.027.4	Wheels <i>Denote materials by -03... (Table 1k)</i>
629.3.023.2	Body, and constructional components of the body ◇ 629.3.023.2-036.5/8 Body of synthetic material	629.3.027.41	Wheels according to function
629.3.023.21	Body framework, as a whole	629.3.027.412	Driving wheels
629.3.023.211	Arches and struts	629.3.027.414	Trailing wheels
629.3.023.212	Connecting members (cross connection pieces)	629.3.027.415	Steered, directional wheels
629.3.023.22	Roofs. Walls. Partitions. Fillings	629.3.027.418	Spare wheels
629.3.023.23	Tarpaulins. Folding roofs. Collapsible hoods	629.3.027.42	Wheels according to position
		629.3.027.421	Position in relation to the vehicle
		629.3.027.421.1	Front wheels
		629.3.027.421.3	Centre wheels
		629.3.027.421.5	Rear wheels
		629.3.027.422	Number of wheels in combination
		629.3.027.422.1	Simple, single wheels
		629.3.027.422.3	Twin, double wheels. Wheels in pair
		629.3.027.422.5	Multiple wheels
		629.3.027.44	Wheels according to kind
		629.3.027.441	Fixed wheels
		629.3.027.443	Removable, detachable wheels
		629.3.027.445	Sprung, elastically mounted wheels
		629.3.027.45	Wheels according to shape
		629.3.027.451	Straight wheels

629.3.027.453	Dished wheels	629.3.027.74	Tracks
629.3.027.454	Disc wheels	629.3.027.75	Striding mechanisms
629.3.027.455	Spoked wheels	629.3.027.8	Caps and covers as protection against dirt (for wheels, running gear) <i>Details by -76...</i>
629.3.027.48	Components of wheels		◇ 629.3.027.8-763 Mudguards
629.3.027.481	Rims	629.3.028	Couplings, mountings and other auxiliary constructions
629.3.027.481.1	Fixed	629.3.028.3	Couplings. Coupling between trailer and truck (articulated vehicles excepted)
629.3.027.481.2	Detachable	629.3.028.32	Towing couplings
629.3.027.481.3	Split	629.3.028.34	Couplings for airpipes, gaspipes, fuelpipes etc.
629.3.027.482	Components of rims	629.3.028.4	Mountings
629.3.027.483	Spokes	629.3.03	Propulsion systems <i>Denote further particulars of engines by 62-..., especially 62-8..., with details by :621...</i>
629.3.027.484	Discs	629.3.031	Propulsion systems using muscle power and natural forces
629.3.027.485	Hubs	629.3.031.2	Muscle-powered propulsion
629.3.027.487	Wheel bolts and nuts	629.3.031.3	Wind-driven propulsion
629.3.027.5	Tyres	629.3.032	Wheel-drive propulsion
629.3.027.51	Tyres according to type	629.3.032.2	Configuration of drive wheels
629.3.027.511	Rigid tyres	629.3.032.22	Rear-axle drive. Rear-wheel drive
629.3.027.512	Elastic tyres	629.3.032.24	Front-axle drive. Front-wheel drive
629.3.027.513	Solid tyres	629.3.032.26	Drive through more than one axle. Four-wheel drive. Six-wheel drive
629.3.027.514	Pneumatic tyres	629.3.033	Propulsion systems through surface contact other than wheels, e.g. tracked
629.3.027.514.1	With inner tube	629.3.035	Propeller or screw propulsion <i>629.3.035 ≡ 629.7.035</i>
629.3.027.514.2	Tubeless	629.3.036	Reaction or jet propulsion <i>629.3.036 ≡ 629.7.036</i>
629.3.027.514.3	Diagonal	629.3.038	Combined propulsion systems
629.3.027.514.4	Radial	629.3.04	Interior layout. Accommodation. Installations and equipment (for persons and freight) ⇒ 674.25
629.3.027.52	Tyre components	629.3.042	Interior layout and finish
629.3.027.521	Outer cover, casing	629.3.042.1	Compartments
629.3.027.521.1	Of textile	629.3.042.2	Cabins
629.3.027.521.2	Of rubber on textile inlay	629.3.043	Crew arrangements. Control compartment. Driving cabin
629.3.027.521.3	With wire, metal cord inlay	629.3.044	Entry/exit arrangements. Mounting/dismounting arrangements
629.3.027.521.4	With puncture-resistant protective inlay	629.3.045	Arrangements for loading/unloading, stowage and tie-down of freight (cargo)
629.3.027.522	Flange. Rim	629.3.046	Passenger comfort, service and hygienic arrangements <i>Details by :64... or :684...</i>
629.3.027.523	Profile. Tread		◇ 629.3.046:684.434.7 Seats ⇒ 656.075
629.3.027.523.1	All-purpose	629.3.047	Personal protection and safety installations and equipment ⇒ 614.86 ⇒ 629.3.067
629.3.027.523.2	Snow profile	629.3.047.1	Body restraints
629.3.027.523.3	Terrain profile	629.3.047.12	Air bags
629.3.027.523.4	Slicks	629.3.047.14	Safety belts
629.3.027.524	Inner tubes	629.3.047.16	Child restraints
629.3.027.525	Valves	629.3.047.2	Emergency escape devices
629.3.027.526	Repair devices. Puncture repair outfits	629.3.048	Conditioning systems. Lighting ⇒ 628.8/9
629.3.027.53	General layout	629.3.048.3	Air conditioning
629.3.027.531	Tyres according to construction	629.3.048.4	Ventilation
629.3.027.531.1	Tyres cast in one piece	629.3.048.7	Heating
629.3.027.531.2	Tyres with ratio between height and width	629.3.048.8	Lighting ⇒ 629.3.066.2
629.3.027.531.3	Tyres with interchangeable tread	629.3.05	Guidance, control-initiation and navigation systems and instruments (vehicle-borne)
629.3.027.532	Tyres according to pressure	629.3.051	Principles and purpose of guidance and navigation systems
629.3.027.532.1	High-pressure tyres		
629.3.027.532.2	Medium-pressure tyres		
629.3.027.532.3	Low-pressure tyres		
629.3.027.533	Tyres with special emergency properties		
629.3.027.533.1	Tyres filled with rubber balls, springs etc.		
629.3.027.533.2	Tyres filled with foam		
629.3.027.533.3	Tyres filled with gas		
629.3.027.533.4	Tyres filled with liquid		
629.3.027.533.5	Multichamber tyres		
629.3.027.533.6	Protection against punctures, tyre damage. Nail catchers		
629.3.027.534	Self-sealing tyres		
629.3.027.535	Tyres with reinforcements (internal or external)		
629.3.027.54	Various devices for tyres		
629.3.027.542	Indicating devices		
629.3.027.542.2	Devices to indicate wear		
629.3.027.542.3	Devices to indicate pressure loss		
629.3.027.542.4	Devices to indicate temperature		
629.3.027.543	Devices to prevent skidding		
629.3.027.543.2	Spikes. Studs		
629.3.027.543.3	Chains		
629.3.027.6	Supports. □ Stands. Stabilizing legs. Balancing devices		
629.3.027.7	Parts taking the place of wheels		
629.3.027.71	Skids. Runners		
629.3.027.72	Cylinders		
629.3.027.73	Balls		

629.3.052	Means of operation <i>For subdivisions see 629.052...</i>	629.3.065	Materials-handling equipment. ☐ Refuelling systems and devices. Revictualling systems and devices <i>Details by :621.86/.87</i> ⇒ 629.3.045
629.3.053	Special systems of control for particular types of vehicle	629.3.065.2	Cargo-handling gear
629.3.054	Vehicle-borne instruments ⇒ 53.08 ⇒ 621.317	629.3.065.22	Derricks
629.3.054.2	Instruments for monitoring drive mechanism and fuel	629.3.065.23	Cranes
629.3.054.22	Tachometer ⇒ 531.775	629.3.065.25	Hoisting equipment
629.3.054.23	Pressure gauges ⇒ 531.787	629.3.065.5	Special kinds of equipment
629.3.054.232	For admission valves of the engine	629.3.066	Observation, signalling, communication and identification equipment
629.3.054.233	For vacuum in the engine	629.3.066.1	Observation equipment
629.3.054.234	For various gaseous propellants	629.3.066.2	External lighting ⇒ 656.057.8
629.3.054.235	For various fluid propellants	629.3.066.22	Navigation lights
629.3.054.236	For fuel	629.3.066.28	Search lights
629.3.054.237	For lubricants	629.3.066.3	Signalling and communication equipment
629.3.054.239	Other pressure gauges	629.3.066.32	Radio equipment
629.3.054.24	Thermometers ⇒ 536.5	629.3.066.34	Visual equipment
629.3.054.241	Free-air thermometers	629.3.066.342	Flags
629.3.054.242	Inlet-air thermometers	629.3.066.345	Signalling lamps. Signalling lights
629.3.054.243	Cylinder-head thermometers	629.3.066.347	Equipment for signal rockets
629.3.054.244/.249	Other thermometers <i>629.3.054.244/.249 ≡ 629.3.054.234/.239</i>	629.3.066.36	Acoustic equipment
629.3.054.25	Instruments for measuring various physical or chemical quantities. ☐ Instruments for measuring concentration or chemical composition	629.3.066.362	Whistles
629.3.054.26	Instruments for measuring mechanical or thermal energy ⇒ 531.78	629.3.066.363	Sirens
629.3.054.27	Instruments for measuring electrical quantities ⇒ 621.317.7	629.3.066.367	Megaphones
629.3.054.28	Consumption meters <i>629.3.054.284/.289 ≡ 629.3.054.234/.239</i>	629.3.066.5	Intercommunication systems
629.3.054.29	Quantity regulation <i>629.3.054.294/.299 ≡ 629.3.054.234/.239</i> ⇒ 681.12	629.3.066.51	One-way systems. ☐ Address systems
629.3.054.3	Instruments for monitoring other systems and mechanisms <i>Specify by .0...</i> ◇ 629.3.054.3.027.2 for the steering mechanism ◇ 629.3.054.3.048.7 for the heating system ◇ 629.3.054.3.066.2 for external lighting	629.3.066.52	Two-way systems. ☐ Telephones
629.3.054.4	Instruments for measuring relationship of vehicle to the external environment <i>629.3.054.4 ≡ 629.7.058.4</i>	629.3.066.8	Identification markings. ☐ Name plates. Number plates. Nationality plates
629.3.054.45	Speedometers	629.3.067	Vehicle safety and protection systems ⇒ 629.3.047 ⇒ 656.08
629.3.054.5	Instruments for monitoring internal spaces of vehicle. ☐ Fire detectors	629.3.067.2	Bumpers
629.3.054.6	Instruments for detecting particular external conditions, e.g. freezing, icing	629.3.067.3	Roll bars. Door bars
629.3.058	Operational characteristics of systems and instruments <i>629.3.058 ≡ 629.7.058</i>	629.3.067.4	Crush zone
629.3.058.85	Panels. Dashboards	629.3.069	Other auxiliary systems forming part of the vehicle. ☐ Armament systems
629.3.06	Vehicle auxiliary systems and devices	629.3.07	Technical aspects of vehicular operations (operational techniques), including performance. <i>Use the subdivisions of .07 to denote equipment, devices, technical modifications etc. for the listed operations.</i> <i>For nontechnical aspects use 656.1 with .052</i>
629.3.062	Control linking/reinforcing systems. Servo control(s) etc.	629.3.071	Preliminary operations and preparations.
629.3.062.2	Ballasting systems	629.3.072	Operations after coming to rest, after arrival
629.3.062.22	Replacement systems	629.3.072.1	Driving, navigating operations in general. ☐ Test runs (trial runs). Training
629.3.062.24	Trimming systems	629.3.072.2	Navigational computations
629.3.062.3	Power-plant control systems	629.3.072.4	Driving technique
629.3.063	Auxiliary supply and feed systems <i>For subdivisions see 629.063...</i>	629.3.072.8	Manoeuvring techniques
629.3.064	Auxiliary power systems (in general): pneumatic, hydraulic, electrical etc. <i>For subdivisions see 629.064...</i>	629.3.078	Training of drivers
		629.3.078	Operations for on-board maintenance of the vehicle (by the crew)
		629.3.08	Plant, installations, equipment and related techniques
		629.3.081	Vehicle factories and workshops
		629.3.081.1	Vehicle factories. Plant and installations for vehicle construction and assembly ⇒ 658.52
		629.3.081.2	Workshops. Garages. Repair stations
		629.3.081.3	Equipment <i>Specify by :621...</i> ⇒ 629.3.018.2
		629.3.081.4	Vehicle manufacturing and repairing techniques
		629.3.081.7	Finishing operations

629.3.082	Ancillary service or supply installations. Depots, stores (for fuel, water etc.)	629.34	Vehicles for the transport of passengers
629.3.082.1	Ancillary industry	629.341	Omnibuses (buses)
629.3.083	Maintenance, repair, modification of vehicles	629.342	Coaches (long distance)
629.3.083.4	Maintenance. Servicing	629.343	Minibuses
629.3.083.5	Repairs	629.344	Taxis
629.3.083.6	Conversion	629.345	Hire cars
629.3.083.7	Modernization	629.346	Ambulances
629.3.083.8	Vehicle recovery	629.349	Other
629.31	Simple transport	629.35	Freight vehicles. Goods vehicles
629.311	Carried conveyances (by humans or animals). □ Packs. Litters. Stretchers. Sedan-chairs	629.351	<i>Specify kinds of freight or goods by :...</i> Trucks (lorries)
629.312	Wheeled conveyances propelled, pushed, or drawn by humans. □ Hand-carts. Trolleys. Barrows. Baby-carriages ('prams'). Wheelchairs. Rickshaws. Scooters	629.351.1	Open trucks
629.313	Wheeled vehicles drawn by animals. □ Carts. Wagons. Gigs. Carriages <i>Denote special purpose vehicles by .014.8 and :629.34/.37.</i> <i>Denote the number of wheels by .022.4...</i>	629.351.2	Closed trucks ◇ 629.351.2:636 Cattle trucks
629.314	Conveyances with load borne directly by the axles. Roller conveyances	629.352	Vans. Delivery vans
629.316	Runner mounted (sliding) conveyances. □ Sleds. Sleighs <i>For motor vehicles on runners use the subdivisions of 629.33/.37 with .027.71</i>	629.353	Pickups. Tip- or dump-trucks
629.32	Cycles. Saddled vehicles. Pedal vehicles	629.354	Container trucks. Vehicles for containerized goods
629.322	Pedal cycles in general. Bicycles <i>Denote cycles with other than two wheels by .022.4...</i> ◇ 629.322.022.41 Monocycles ◇ 629.322.022.43 Tricycles ⇒ 629.371.1 ⇒ 629.373.1	629.355	Vehicles with cooling devices for perishable goods. Refrigerator vans
629.322.2	Touring and sports cycles	629.356	Tankers and vehicles for transporting bulk or granular goods
629.322.4	Cycles for more than one rider	629.357	Transporters for very heavy or outsize loads. E.g. vehicle and boat transporters, massive timber and tree transporters, heavy structure transporters
629.322.42	Cycles for two riders. Tandems	629.359	Other special vehicles for transporting freight
629.322.48	Cycles for more than two riders	629.36	Special purpose vehicles
629.322.7	Cycles with detachable sidecars or trailers. Sidecars and trailers for cycles	629.361	Vehicles for welfare and amenity in general ◇ 629.361:027.3 Mobile libraries ◇ 629.361:393.92 Funeral cars. Hearses ◇ 629.361:614.846.6 Fire-fighting vehicles ◇ 629.361:628.4 Refuse collection vehicles ◇ 629.361:791.096 Mobile cinemas
629.322.8	Cycle-barrow and cycle-rickshaw combinations	629.362	Vehicles for control or safety of traffic
629.323	Pedal vehicles with seats rather than saddles	629.362.1	Vehicles for driving instruction
629.326	Motorcycles in general ⇒ 629.371.1	629.362.2	Patrol and police vehicles
629.326.2	Light motorcycles. Mopeds	629.362.5	Snow-clearing vehicles
629.326.3	Motor scooters	629.363	Workshop, vehicle-recovery and repair vehicles
629.326.5	Touring and sports motorcycles	629.363.1	Breakdown vehicles
629.327	Motorcycle combinations	629.363.2	Vehicles for measurement
629.327.2	Motorcycle-sidecar combinations. Sidecars for motorcycles	629.364	Vehicles with special structures
629.327.4	Motorcycle-trailer combinations. Trailers for motorcycles	629.364.1	Cranes
629.33	Motor vehicles. Automotive engineering (in general)	629.364.2	Lifting vehicles
629.331	Motorcars. Automobiles	629.365	Earthmoving vehicles and machines
629.331.1	Closed cars: saloons, sedans, hatchbacks	629.366	Agricultural vehicles ⇒ 631.3
629.331.2	Estate cars. Station wagons	629.367	Track-laying (tracked, half-track) vehicles
629.331.3	Convertibles. Open cars	629.369	Other special purpose vehicles
629.331.4	Old-timers	629.37	Special performance vehicles with two or more wheels. Special terrain vehicles with two or more wheels
629.331.5	Experimental cars	629.371	Vehicles for racing or sporting purposes
629.332	Trailers, semitrailers (small, medium, large) and articulated vehicles in general <i>For special-purpose trailers or articulated vehicles add .014.3 or .014.5 to the number for the vehicle</i>	629.371.1	Racing or sporting vehicles with two or three wheels
629.334	Mobile homes. Caravans. Campers. Motorhomes. Motor caravans ◇ 629.334.014.3 Towed caravans. Trailer homes ⇒ 728.76	629.371.11	Racing bicycles, cycles
		629.371.12	Cross-country bicycles
		629.371.13	Race-mopeds
		629.371.14	Cross-country mopeds
		629.371.15	Racing motorcycles
		629.371.16	Cross-country motorcycles
		629.371.2	Racing, sporting vehicles with more than three wheels
		629.371.21	Racing vehicles
		629.371.22	Rally vehicles
		629.371.23	Sporting vehicles
		629.371.26	Go-karts
		629.373	Vehicles for special off-road use. Special terrain vehicles

- 629.373.1 Pedal cycles for special terrain. □ Mountain bikes
- 629.373.2 Tricycles
- 629.373.3 Off-road vehicles
- 629.4 Rail vehicle engineering. Locomotives. Rolling stock. Railway yards. Installations and services concerned with rolling stock**
⇒ 625.1/5
- 629.4.01 Design, characteristics and capabilities of rail vehicles. Tests
- 629.4.012 Dimensions. Gauges
- 629.4.012.2 Arrangements for conforming to track gauges
- 629.4.012.4 Arrangements for conforming to loading gauges
- 629.4.012.5 Dimensions of bodies, chassis etc.
- 629.4.013 Weights. Tonnage. Capacity. Payloads
- 629.4.013.2 Weight loading. Size loading
- 629.4.013.22 Fuel weights. Bunkering capacity
- 629.4.013.24 Freight loads. Payloads
- 629.4.014 Functional capabilities and mode of use of vehicles
Use the auxiliaries listed at 629.014... only for occasional or special capabilities or uses not provided for in the direct subdivisions of 629.4
- 629.4.015 Mechanics and dynamics of rolling stock.
□ Interaction of rolling stock with track (with reference to rolling stock)
For interaction between track and rolling stock with reference to track use 625.03
- 629.4.016 Performance characteristics and capabilities of rail vehicles. Assessment of power, speed etc. Theoretical studies of speed and efficiency
- 629.4.016.1 Design formulas
- 629.4.016.12 Tests of consumption. Choice of fuel or power source
- 629.4.016.2 Useful energy. Installed power. Efficiency and fuel consumption
- 629.4.016.3 Comparisons of technical values and efficiency according to method of traction adopted. Range
- 629.4.016.5 Speed
- 629.4.016.52 Low speeds. Up to 50 km/h (31.1 mile/h)
- 629.4.016.54 Medium and high speeds. 51 - 200 km/h (31.2 - 124 mile/h)
- 629.4.016.56 Very high speeds. Over 200 km/h (124 mile/h)
- 629.4.017 Operational reliability. Stability. Handling. Braking capacity
⇒ 62-592
⇒ 629.4.077
- 629.4.018 Tests: methods and equipment
- 629.4.02 Structure of rail vehicles. Ideas and principles of construction. General layout and components
- 629.4.021 Principles of construction
- 629.4.021.2 Construction techniques
- 629.4.021.21 Vehicles with separate body and underframe
- 629.4.021.24 Monocoque construction. Tubular construction
- 629.4.021.26 Riveted construction. Welded construction
- 629.4.022 Layout according to gauge
- 629.4.022.2 For broad gauge railways
- 629.4.022.4 For standard gauge railways
- 629.4.022.6 For narrow gauge railways
- 629.4.023 Construction elements. Shell. Framework
- 629.4.023.1 Underframe. Bodywork. Superstructure. Intercommunication
- 629.4.023.11 Underframe
- 629.4.023.14 Body. Bodywork
- 629.4.023.15 Roof. Superstructure
- 629.4.023.16 Communicating parts
- 629.4.023.163 Gangway floorplates
- 629.4.023.165 Gangways
- 629.4.023.17 Construction elements for entry and closure.
□ Doors
- 629.4.023.174 Automatic closing devices
- 629.4.023.176 Steps. Folding steps
- 629.4.023.18 Sides of vehicles. Windows. Shutters etc.
- 629.4.023.2 Elements forming the base and framework of vehicles
- 629.4.024 External structure. Body shell, coachwork etc.
⇒ 674.25
- 629.4.025 Structures peculiar to railway rolling stock
- 629.4.026 Arrangement of propulsion or power units
- 629.4.027 Support, suspension and shock-absorbing structures. Wheels
- 629.4.027.1 Support in general
- 629.4.027.11 Axles. □ Type of axle. Function and form of axles. Parts of axles. Axle boxes
- 629.4.027.2 Bogies
- 629.4.027.23 Two-axle bogies
- 629.4.027.25 Bogies with more than two axles
- 629.4.027.26 Counterbalanced bogies. Compensating balances
- 629.4.027.27 Linking between bogies. Guides for bogies
- 629.4.027.3 Suspension. Springs. Vertical shock absorbers. Dampers
- 629.4.027.31 Types of spring
◇ 629.4.027.31-272.2 Coil springs
◇ 629.4.027.31-272.3 Laminated springs
◇ 629.4.027.31-272.82 Pneumatic (air) springs
- 629.4.027.32 Types of suspension
- 629.4.027.35 Shock absorbers. Dampers
- 629.4.027.4 Wheels. Tyres
- 629.4.027.41 Types of wheel
- 629.4.027.412 Driving wheels
- 629.4.027.414 Load-bearing wheels
- 629.4.027.416 Guiding wheels
- 629.4.027.43 Tyres in general. Types of tyre
- 629.4.027.432 Resilient, elastic wheels. Pneumatic wheels
- 629.4.027.434 Metal tyres. Flanges
- 629.4.027.44 Positioning and pressing-on of the wheel to the axle
- 629.4.027.45 Locking and securing devices for fixing wheels to axles
- 629.4.027.5 Wheelsets. □ Sets of traction wheels. Sets of carrying wheels. Adjustable gauge wheelsets
- 629.4.028 Coupling devices, buffers, connectors etc. Traction equipment. Drawgear
- 629.4.028.1 Drawbars
- 629.4.028.11 Couplings. □ Screw couplings. Impact couplings
- 629.4.028.12 Buffing gear
⇒ 625.156
- 629.4.028.21 Buffers (horizontal shock absorbers)
- 629.4.028.22 Coupling by striking the buffers
- 629.4.028.3 Combined buffing and drawgear. Couplings with buffers
- 629.4.028.31 Automatic couplings with central buffer
- 629.4.028.4 Other methods of coupling
- 629.4.028.41 Coupling between engine and wagons, between locomotive and tender etc.
- 629.4.028.42 Connectors for services (air and steam pipes, electric cables etc.)
- 629.4.028.5 Transitional couplings. Mixed couplings
- 629.4.028.8 Components of couplings
- 629.4.028.81 Coupling head
- 629.4.028.82 Fastenings
- 629.4.028.83 Uncoupling and disconnecting devices. Locating devices for couplings
- 629.4.028.84 Coupling arms

629.4.03	Propulsion systems. Power units	629.4.052	Means of operation
629.4.031	Propulsion systems using natural forces, muscular strength, wind etc.		<i>For subdivisions see 629.052...</i>
629.4.032	Propulsion through wheels. Wheel drive	629.4.053	Special systems of control for particular kinds of rail vehicle
629.4.033	Tracked propulsion. Toothed-wheel propulsion. Rack-and-pinion propulsion ⇒ 625.33 ⇒ 629.421.6	629.4.053.3	Automatic train control, with repeater signals shown on the engine
629.4.034	Propulsion by cables, chains (cable-traction railways). Propulsion by endless screw (helical drive) ⇒ 625.39 ⇒ 625.52	629.4.054	Vehicle-borne instruments 629.4.054 ≡ 629.3.054
629.4.035	Propulsion by propeller or airscrew 629.4.035 ≡ 629.7.035	629.4.054.2	Instruments for monitoring drive mechanisms and fuel
629.4.036	Propulsion by reaction pressure. Jet propulsion 629.4.036 ≡ 629.7.036	629.4.054.23	Pressure gauges
629.4.038	Combined propulsion systems	629.4.054.3	Instruments for monitoring other systems and mechanisms. □ Specify by .0...
629.4.04	Interior layout of rail vehicles. Accommodation, installations and equipment (for persons and freight) ⇒ 674.25	629.4.054.45	Speedometers
629.4.042	Interior layout and finish generally. Compartments. Berths. Saloons	629.4.058	Operational characteristics of systems and instruments
629.4.042.2	Interior partitions	629.4.06	Auxiliary systems and equipment mounted on the vehicle or engine
629.4.042.4	Compartments. Berths. Corridors	629.4.062	Control linking and reinforcing systems. Power assistance (e.g. servomotors)
629.4.042.42	Non-corridor stock	629.4.063	Auxiliary supply and feed systems <i>For subdivisions .063.2/.7 see 629.063...</i>
629.4.042.44	Compartments with central corridor	629.4.063.8	Equipment for sanding, greasing, wiping the rails
629.4.042.46	Compartments with side corridor	629.4.064	Auxiliary power systems <i>For subdivisions see 629.064...</i>
629.4.042.5	Soundproofing. Thermal insulation. Fireproofing	629.4.065	Materials-handling systems. □ Refuelling systems and devices <i>Further details by :621.86/.87</i> ⇒ 629.4.045
629.4.042.6	Floors. Surfacing. □ Linoleum. Carpets	629.4.066	Observation, signalling, communication and identification devices ⇒ 654.9
629.4.042.7	Luggage racks	629.4.067	Safety and protection systems on rail vehicles
629.4.043	Accommodation for train crew and staff. Driving and control compartments. Driver's or engineer's cabin. Footplate	629.4.067.2	Equipment for emergency lighting on engines, carriages, trains
629.4.043.2	Layout of control positions for various kinds of traction	629.4.067.3	Automatic monitoring, safety devices ◇ 629.4.067.3-783.52 Automatic emergency braking devices. Dead man's handle
629.4.043.4	Layout of compartments, positions for train staff	629.4.067.4	Antiskid devices acting on the wheels
629.4.044	Arrangements for boarding and alighting, entry and exit ⇒ 629.4.023.16 ⇒ 629.4.023.17	629.4.067.5	Devices for clearing obstacles. □ Guard-irons (rail guards, cowcatchers). Snowploughs ⇒ 629.464.22
629.4.045	Freight handling and storage. Arrangements for loading, unloading, stowage, dunnage. Securing and loss prevention for loose materials	629.4.067.6	Warning devices. □ Sirens. Klaxons
629.4.046	Arrangements for passenger comfort and service <i>Details by :64... and :684...</i>	629.4.069	Other auxiliary systems forming part of the vehicle. □ Armament systems
629.4.046.3	Seats	629.4.07	Technical aspects of operations (operational techniques), including performance <i>For nontechnical aspects use 656.2 with .052</i> <i>Use the subdivisions of .07 to denote equipment, devices, technical modifications etc. for the listed operations</i>
629.4.046.4	Accessories. □ Window winders and adjusters. Blinds. Fixed and folding surfaces, tables. Ashtrays. Mirrors	629.4.071	Preliminary operations and preparations. Operations after coming to rest, after arrival
629.4.046.7	Sanitary installations. Lavatories. Washing facilities. Other hygienic arrangements	629.4.071.2	Adaptation of locomotives for work required: backing, hauling (single and/or multiple)
629.4.047	Installations and equipment for personal protection and safety	629.4.071.5	Duty cycle for engines
629.4.047.2	Alarm signals. □ Communication cords	629.4.072	Driving, running operations in general. □ Running trials and training
629.4.047.4	Fire extinguishers	629.4.073	Operational techniques for special conditions, e.g. weather, climate, terrain, engine or power failure 629.4.073 ≡ 656.052.5
629.4.048	Climatic control. Air conditioning. Heating. Ventilation. Lighting ⇒ 628.8/.9	629.4.075	Departure techniques. Operations for starting-up, pulling away ⇒ 629.4.085
629.4.048.3	Air conditioning	629.4.076	Operations en route, while running
629.4.048.4	Ventilation	629.4.077	Arrival techniques. Operations for slowing and stopping ⇒ 629.4.087
629.4.048.7	Heating		
629.4.048.8	Lighting		
629.4.05	Guidance and control systems on rail vehicles		
629.4.051	Functional characteristics and principles. Purpose		

629.4.078	Operations for on-board maintenance (by the crew)	629.423.3	Components of electric locomotives ⇒ 621.33
629.4.08	Installations at base or depot. Equipment and techniques for servicing rail vehicles <i>The .08 subdivisions provide general aspects which have equivalents in other forms of transport. For aspects peculiar to rail see 629.47/48</i> ⇒ 656.06	629.423.31	Electric motors ◇ 629.423.31:621.313.2 Direct current motors
629.4.082	Ancillary service and supply installations.	629.423.32	Power transmission. Transmission elements. Power conversion systems <i>Denote controls by :621.337...</i>
629.4.082.2	Supply depots and stores	629.423.33	Transmission of electric energy. Conductors. Current collectors <i>Details by :621.336...</i>
629.4.082.3	Supply of fuels (solid, liquid and gaseous fuels)	629.424	Rail traction by internal combustion engine. Diesel rail traction ◇ 629.424-82 Diesel-hydraulic traction engines
629.4.082.4	Supply of electric energy	629.424.1	Various kinds of diesel locomotive 629.424.1 ≡ 629.421
629.4.082.5	Supply of water ⇒ 621.643.54	629.424.2	Self-propelled diesel carriages, railcars. Multiple-unit diesel trains. Diesel motor-carriage units
629.4.082.6	Supply of lubricants. Greasing bays	629.424.3	Components of diesel locomotives ⇒ 621.436
629.4.083	Supply of sand. Sandboxes	629.424.4	Electric equipment in diesel locomotives <i>Specify by :621.3...</i>
629.4.083	Maintenance. Handling, servicing and repair of rail vehicles ⇒ 629.48	629.425	Locomotives with internal combustion engines other than diesel <i>Details by :621.43...</i> 629.425 ≡ 629.421
629.4.084	Spare parts and replacements. □ Stock control and methods of issue. Spares depots and stores ⇒ 621.796	629.426	Locomotives with gas turbine engines 629.426 ≡ 629.421 ⇒ 621.438
629.4.085	Base equipment and technical procedure for starting ⇒ 629.4.075	629.427	Motorized trucks and trolleys. Track motor cars. Gang cars ⇒ 621.43
629.4.086	Base equipment and technical procedure for control and tests en route (e.g. for running trials) ⇒ 629.4.072	629.428	Other kinds of locomotive. Engines with special motor systems. □ Reaction (rocket) locomotives ⇒ 621.45
629.4.087	Base equipment and technical procedure for stopping, braking ⇒ 629.4.077		
629.41	Motive power department	629.43	Rail transport using special engines. Engines and vehicles for town and suburban transport. Rail vehicles for short-distance transport generally ⇒ 625.3/5
629.42	Kinds of traction. Motive power units		
629.421	Motive power units according to purpose and construction	629.431	Engines and vehicles for elevated railways
629.421.1	Locomotives for passenger trains	629.432	Engines and vehicles for underground railways
629.421.2	Locomotives for goods trains	629.433	Engines and vehicles for tramways. Trams, tramcars
629.421.3	General-purpose locomotives. Locomotives for mixed trains	629.434	Engines and vehicles for town and suburban (surface) railways
629.421.4	Shunting locomotives	629.435	Engines and vehicles for monorails
629.421.5	Works locomotives. Industrial locomotives	629.435.2	For tracked monorails
629.421.6	Mountain-railway locomotives. Rack-and-pinion engines ⇒ 625.33 ⇒ 629.4.033	629.435.4	For suspended monorails
629.421.7	Articulated locomotives	629.436	Engines and vehicles for funiculars (cable railways)
629.421.8	Locomotives without fireboxes. Locomotives using superheated water. Locomotives with steam reservoirs	629.437	Air-cushion rail vehicles. Hovertrains ⇒ 629.576
629.422	Steam engines and tenders ⇒ 621.13	629.438	Engines for special railway systems
629.422.1	Various kinds of steam locomotive 629.422.1 ≡ 629.421 <i>Details by -1/-9 (from 62-1/-8 and 66-9)</i>	629.438.2	Vehicles for very wide-gauge railways (rail vehicles for carrying road vehicles and other rail vehicles)
629.422.2	Self-propelled steam carriages, railcars	629.438.4	Tubular railways (powered by compressed air or vacuum). Atmospheric railways
629.422.3	Components of steam engines <i>Specify by :621.134.1/8 and :621.18..., with further detail by -1/-9 (from 62-1/-8 and 66-9)</i>	629.439	Engines and vehicles for other special rail systems. Engines with propeller, flywheel, gyroscope, compressed-air drive, nuclear power etc. ⇒ 629.4.035
629.422.7	Tenders		
629.423	Electric locomotives <i>Details by -83... or :621.3...</i>	629.44	Passive stock. Hauled rail vehicles in general
629.423.1	Various kinds of electric locomotive 629.423.1 ≡ 629.421	629.45	Passenger stock. Passenger carriages, coaches
629.423.2	Self-propelled electric carriages, railcars. Multiple-unit electric trains. Electric motor-carriage units	629.451	Passenger coaches for regional (short haul) trains ⇒ 629.457

629.452	Coaches articulated at a common axle or bogie. Articulated trains. Talgo system	629.464	Railway maintenance vehicles
629.454	Coaches for international trains. Coaches for long hauls, with special accommodation etc.	629.464.2	Snow-clearing rail vehicles ⇒ 625.174
629.454.2	Coaches with special accommodation (for long hauls)	629.464.22	Railway snow ploughs
629.454.22	Sleeping cars	629.464.25	Snow blowers
629.454.25	Coaches with reclining berths (couchettes)	629.464.26	Rotary snow ploughs
629.454.3	Restaurant cars. Dining cars. Refreshment cars	629.464.3	Crane wagons ⇒ 621.873.2
629.454.34	Restaurant cars with automatic vending machines, with self-service	629.464.4	Track maintenance wagons
629.454.36	Buffet coaches. Bar coaches. Cafeteria coaches	629.464.42	Track-laying wagons
629.454.4	Saloon coaches (Pullman cars)	629.464.43	Ballast wagons ⇒ 625.173.7
629.454.5	Platform coaches. Observation cars. Canopy coaches	629.464.44	Plate-layers' trolleys ⇒ 625.173.2
629.454.52	Platform coaches (double-deckers, split-level cars) ⇒ 629.457.4	629.464.45	Maintenance trains. Repair and rescue trains. Workshop wagons
629.454.54	Observation cars	629.464.46	Wagons for calibration of weighbridges. Weighbridge test wagons (track scale test cars) ⇒ 681.264.3
629.454.56	Canopy coaches	629.464.47	Rail testing and measuring vehicles ⇒ 625.171
629.456	Coaches with special equipment	629.464.48	Wagons for transporting rail vehicles. Dollies. Track charge vehicles ⇒ 629.4.027.52
629.456.2	Medically equipped coaches: ambulance coaches, hospital coaches. Hospital trains	629.466	Special-purpose vehicles and trains <i>Specify by :...</i> ◇ 629.466:659.134.3 Trains for publicity, advertising
629.456.3	Coaches for entertainment. □ Theatre coaches. Cinema coaches. Video coaches. Television coaches	629.467	Multipurpose rail vehicles. Vehicles for railway staff, personnel
629.456.5	Exhibition coaches. Library coaches		
629.456.6	Instruction coaches. Coaches with teaching equipment		
629.457	Coaches for high density of passengers		
629.457.2	Suburban stock. Commuter coaches	629.47	Rail depots. Engine sheds. Sidings and their equipment
629.457.4	Platform coaches (double-deckers, split-level cars)		
629.46	Freight stock. Miscellaneous rail vehicles	629.48	Large railway workshops and factories. Construction, assembly and repair shops for rail vehicles. Central workshops ⇒ 629.4.083
629.462	Vans		
629.462.1	Luggage vans	629.482	For construction and repair of engines, locomotives
629.462.2	Goods vans	629.483	For construction and repair of passenger carriages, coaches
629.462.3	Postal vans, for parcel post (postal packages)	629.484	For construction and repair of goods wagons, freight cars
629.463	Goods wagons or trucks (freight cars)	629.486	For repair of bogies and axles
629.463.1	Closed and covered goods wagons	629.487	Specialized technical equipment for workshops
629.463.12	Insulated, heated and refrigerated wagons	629.488	Maintenance, modification and repair of rolling stock (techniques and technology). Frequency and extent of maintenance operations
629.463.122	Insulated (isothermic) wagons		Maintenance of rolling stock generally
629.463.123	Heated wagons	629.488.2	Work bays
629.463.124	Refrigerator wagons	629.488.22	Inspection and repair pits
629.463.125	Cold-store (mechanically refrigerated) wagons	629.488.23	Installations for removal and replacement of axles
629.463.2	Specialized covered goods wagons (except hopper wagons). □ Wagons with sliding partitions. Wagons with openable end walls. Fruit and vegetable wagons. Stock cars, cattle trucks. Stable-wagons (rail horse-boxes)	629.488.24	Washing and cleaning installations
629.463.3	Tank and container wagons. Rail tankers etc.	629.488.25	Installations for disinfection, fumigation, deinfestation (insect destruction), decontamination
629.463.32	Tank-wagons for liquids	629.488.26	Regulating, running-in operations
629.463.33	Tank-wagons for compressed or liquified gas	629.488.27	Upkeep, preservation, dismantling, demolition of rolling stock
629.463.34	Wagons for the transport of molten metals	629.488.3	Upkeep. Long-term maintenance. Preservation
629.463.36	Bulk-material container wagons with pressure discharge. □ Dust container wagons	629.488.32	Dismantling. Demolition
629.463.37	Bulk-material container wagons with openable roof and self-discharge	629.488.36	Equipment and installations for conversion of gauge ⇒ 625.112 ⇒ 629.4.012
629.463.4	Wagons for transport of automobiles. Car transporters	629.488.4	Installations for changing axles and bogies
629.463.5	Wagons with opening roofs	629.488.42	Devices for adjusting gauges
629.463.6	Open goods wagons. Other kinds of wagon	629.488.45	Mounting on rails. Rerailers
629.463.62	Ordinary flat wagons (flat trucks or open trucks)	629.488.5	Removal from rails. Derailers
629.463.63	Wagons for transport of lorries, trailers and containers. Container wagons	629.488.6	
629.463.64	Special wagons for transport of heavy loads. Wagons for exceptionally large objects		
629.463.65	Tipper wagons		
629.463.66	Hopper wagons		

629.5 Watercraft engineering. Marine engineering. Boats. Ships. Boatbuilding and shipbuilding
 ⇒ 623.8

629.5.01	Naval architecture. Ship design: principles, capabilities, tests	629.5.015.23	Profile resistance. Eddy
629.5.011	Rules and regulations for design and classification of ships. Tonnage measurement ◇ 629.5.011:005.71 Classification societies ⇒ 351.813.22	629.5.015.24	Wave resistance
629.5.012	Dimensions. Dimensional design	629.5.015.25	Induced resistance
629.5.012.2	Length	629.5.015.26	Shallow water resistance
629.5.012.22	Overall length	629.5.015.3	Aerodynamics affecting design ⇒ 533.6
629.5.012.23	Length between perpendiculars	629.5.015.4	Strength requirements. Hydrodynamic and structural design requirements ⇒ 624.04 ⇒ 629.5.018.4 ⇒ 629.5.023.4
629.5.012.24	Waterline length	629.5.015.5	Ship vibrations. Prevention and control of vibrations. Protection from vibrations
629.5.012.3	Breadth. Beam	629.5.015.6	Ship noise. Prevention and control of noise. Protection from noise
629.5.012.32	Overall breadth	629.5.016	Ship performance, capabilities and propulsion characteristics
629.5.012.33	Moulded breadth	629.5.016.2	Diving depth
629.5.012.4	Height dimensions	629.5.016.3	Range of sailing. Radius of sailing
629.5.012.42	Overall height	629.5.016.5	Ship speed
629.5.012.44	Depth	629.5.016.51	Design speed specifications. □ Service speed. Trial speed
629.5.012.442	Draught	629.5.016.7	Propulsion assessment
629.5.012.444	Freeboard	629.5.016.8	Characteristics and capabilities under special conditions
629.5.012.6	Calculated measures	629.5.017	Operational reliability. Seaworthiness. Stability
629.5.012.62	Centre of buoyancy	629.5.017.1	Reliability and stability requirements
629.5.012.63	Metacentric radius	629.5.017.12	Wind forces
629.5.012.64	Centre of gravity	629.5.017.13	Grounding stability
629.5.012.65	Metacentric height	629.5.017.14	Docking stability
629.5.012.7	Ship coefficients	629.5.017.15	Loss of stability due to bulk cargo
629.5.012.71	Block coefficient	629.5.017.16	Loss of stability due to water on deck
629.5.012.72	Prismatic coefficient	629.5.017.161	Water in deck cargo
629.5.012.73	Mainframe coefficient	629.5.017.162	Free water surfaces
629.5.012.74	Waterplane coefficient	629.5.017.17	Full icing allowance
629.5.013	Weights. Displacement. Carrying capacity	629.5.017.2	Ship seakeeping. Stability and controllability
629.5.013.1	Weight of construction	629.5.017.21	Pitching. Slamming
629.5.013.2	Light displacement. Cubic displacement	629.5.017.22	Rolling
629.5.013.3	Weight of fuel and supplies	629.5.017.23	Yawing
629.5.013.4	Deadweight	629.5.017.24	Drift
629.5.013.5	Carrying capacity	629.5.017.25	Stability and controllability at engine failure
629.5.013.52	Register tonnage	629.5.017.26	Damping forces and moments. Heave motion
629.5.013.521	Gross tonnage	629.5.017.28	Control forces
629.5.013.522	Net tonnage	629.5.017.3	Manoeuvrability
629.5.013.55	Container capacity	629.5.017.5	Braking capacity
629.5.013.57	Load waterline. Loadline mark	629.5.018	Testing: procedures, instruments, equipment
629.5.014	Functional capabilities and mode of use of vehicles <i>Use the auxiliaries listed at 629.014... only for occasional or special capabilities or uses not provided for in the direct subdivisions of 629.5</i>	629.5.018.1	Model basins and their equipment, including wave producers and similar equipment
629.5.015	Mechanics in relation to the medium <i>For mechanics as such use 531/534</i> ⇒ 629.5.017	629.5.018.12	Towing tanks. □ Wave tanks. Depressurized towing tanks
629.5.015.1	Hydrostatics affecting design. Stability ⇒ 532.3	629.5.018.13	Basins of special purpose
629.5.015.11	Buoyancy	629.5.018.15	Cavitation tunnels
629.5.015.12	Ship static stability	629.5.018.2	Laboratory equipment and instruments for conducting experiments and tests
629.5.015.13	Ship dynamic stability	629.5.018.22	Ship models with or without propelling agents
629.5.015.14	Intact stability	629.5.018.23	Towing devices
629.5.015.141	Transverse stability	629.5.018.26	Special measuring and monitoring equipment and instruments for conducting experiments and tests
629.5.015.142	Longitudinal stability	629.5.018.4	Strength tests: fatigue, vibration, impact etc.
629.5.015.143	Stability at large angles of heel	629.5.018.7	Tests
629.5.015.144	Capsizing. Loss of stability	629.5.018.71	Model tests
629.5.015.15	Damage stability and its calculations	629.5.018.72	Full-scale testing
629.5.015.152	Lost buoyancy	629.5.018.75	Ship trials
629.5.015.154	Constant displacement	629.5.018.8	Scale effects. Influence of model basin walls
629.5.015.17	Heel	629.5.02	Structure of craft or vessel. Layout and parts
629.5.015.2	Hydrodynamics affecting design. Resistance ⇒ 532.5	629.5.021	Construction principles and properties <i>Denote ships according to the main material of construction by .021-03... using Table 1k</i>
629.5.015.21	Friction resistance	629.5.021.1	Definition of ship's architectural type and design
629.5.015.22	Pressure resistance		

629.5.021.12	Full scantling ships	629.5.023.8	Floors
629.5.021.122	Smooth deck ships	629.5.023.82	Continuous floors
629.5.021.123	Three island ships	629.5.023.84	Impenetrable floors
629.5.021.125	Well-deck ships	629.5.023.86	Bracket floors
629.5.021.127	Raised deck or raised quarter-deck ships	629.5.023.9	Surface coatings
629.5.021.13	Ships with excessive freeboard. Shelter-deck ships	629.5.024	Hull and superstructure
629.5.021.15	One-deck ships	629.5.024.1	Hull shapes
629.5.021.16	Ships with two or more decks	629.5.024.11	V-form
629.5.021.17	Lightened deck ships	629.5.024.12	U-form
629.5.021.18	Ships constructed for ice navigation (with ice strengthening of hull), other than icebreakers ⇒ 629.561.5	629.5.024.3	Forebodies
629.5.021.5	Inflatable craft	629.5.024.31	Bow
629.5.022	General layout and configuration	629.5.024.32	Bulbous bow
629.5.022.1	Single-hull craft	629.5.024.5	Midsection hull form
629.5.022.2	Twin-hull craft	629.5.024.7	Aft end structure
629.5.022.22	Catamarans	629.5.024.71	Screw arrangements
629.5.022.25	Small waterplane twin hull (SWATH) ships	629.5.024.711	Single screw
629.5.022.3	Three-hull and multihulled craft. Trimarans	629.5.024.712	Twin screw
629.5.022.5	Hinged ships. E.g. split hoppers	629.5.024.715	Special screw arrangements
629.5.023	Shell, framework and structural members	629.5.024.72	Propeller hull clearance
629.5.023.1	Framing systems	629.5.024.75	Special stern forms
629.5.023.12	Longitudinal framing system	629.5.024.752	Sternbulb
629.5.023.121	Girders	629.5.024.755	Twisted tail configuration
629.5.023.121.1	Centre girders	629.5.025	Special structures. Stabilizers, hydrofoils etc. ⇒ 629.5.062.2
629.5.023.121.2	Side girders	629.5.025.1	Hydrofoils generally
629.5.023.122	Bottom longitudinals	629.5.025.2	Stabilizers
629.5.023.123	Side longitudinals	629.5.025.22	Roll stabilizers
629.5.023.124	Panting stringers	629.5.025.24	Gyroscopic stabilizers
629.5.023.125	Deck longitudinals	629.5.025.26	Rolling tanks
629.5.023.13	Transverse framing system	629.5.025.4	Design of hydrofoils or stabilizers ◇ 629.5.025.4-023.221 Arched ◇ 629.5.025.4-023.3V V-form ◇ 629.5.025.4-023.44 Rectangular ◇ 629.5.025.4-023.446 Trapeziform ◇ 629.5.025.4-023.711 Flat
629.5.023.131	Double bottom bracket floors	629.5.025.5	Layout of the hydrofoil or stabilizer in relation to the hull
629.5.023.17	Composite framing system	629.5.025.6	Angles of the hydrofoil or stabilizers
629.5.023.2	Shell envelope plating	629.5.025.7	Surface characteristics
629.5.023.21	Keel	629.5.025.72	Area
629.5.023.22	Bottom shell	629.5.025.73	Profile. Camber
629.5.023.23	Bilge shell	629.5.025.74	Twist
629.5.023.24	Side shell	629.5.025.76	Aspect ratio
629.5.023.241	Sheerstrakes	629.5.025.8	Elements of hydrofoil
629.5.023.242	Ice belt	629.5.026	Control-unit and propulsion-unit structures
629.5.023.243	Bulwark	629.5.026.1	Number of propulsion units (engines, motors)
629.5.023.27	Shell openings	629.5.026.11	Single engine
629.5.023.3	Kinds of deck	629.5.026.12	Double engine
629.5.023.311	Upper tween deck	629.5.026.13	Three engines
629.5.023.312	Lower tween deck	629.5.026.14	Four or more engines
629.5.023.313	Raised quarter-deck	629.5.026.2	Engine room. Boiler room. Engine-room workshops and stores
629.5.023.38	Deck covering	629.5.026.22	Aft
629.5.023.4	Primary structure and members	629.5.026.23	Midships
629.5.023.44	Longitudinal members: stringers	629.5.026.24	Forward
629.5.023.441	Side stringers	629.5.026.3	Engine control stations
629.5.023.46	Transverse members: spars, struts, bracing	629.5.026.4	Tunnels. Ducts
629.5.023.461	Side transverses	629.5.026.41	Propeller-shaft tunnel
629.5.023.462	Web frames supporting side stringers	629.5.026.42	Pipe tunnels. Tubing tunnels
629.5.023.463	Bulkheads	629.5.026.47	Funnels. Exhaust outlets
629.5.023.463.1	Collision bulkhead	629.5.026.5	Hovering installations
629.5.023.463.2	After peak bulkhead	629.5.026.53	Skirts
629.5.023.463.7	Bulkhead stiffeners	629.5.026.57	Fan devices
629.5.023.463.8	Bulkhead doors	629.5.026.8	Masts and rigging
629.5.023.5	Deck openings	629.5.028	Attachments, couplings, mountings and other auxiliary structures
629.5.023.53	Hold hatches. □ Hatch coamings	629.5.028.2	Pushing devices
629.5.023.54	Skylights	629.5.028.3	Towing devices
629.5.023.7	Hull fittings	629.5.028.5	Ice-breaking attachments ⇒ 629.561.5
629.5.023.71	Port lights. Windows	629.5.028.7	Anchors and mooring gear
629.5.023.72	Manholes. Inspection holes		
629.5.023.73	Doors. Ports		
629.5.023.731	Watertight doors		
629.5.023.734	Cargo ports		
629.5.023.735	Cargo doors. Light doors		
629.5.023.736	Ramps		
629.5.023.737	Hatch covers. Manhole covers		

629.5.028.71	Anchor gear	629.5.043.122	Lookout posts
629.5.028.711	Anchors	629.5.043.13	Chart room
629.5.028.712	Anchor chains and accessories	629.5.043.14	Radio room. Telecommunications room
629.5.028.714	Anchor lifting equipment. Capstans.	629.5.043.15	Control station ⇒ 629.5.026.3
	Windlasses		
	⇒ 621.864	629.5.043.16	Command centre
629.5.028.715	Hawse pipe	629.5.043.17	Steering tiller rooms
629.5.028.717	Stoppers	629.5.043.2	Rooms for pilots and port authorities. Office rooms
629.5.028.72	Mooring equipment		
629.5.028.721	Bits. Bollards	629.5.043.3	Crew's working spaces
629.5.028.722	Mooring lines	629.5.043.32	Workrooms
629.5.028.724	Buffer devices. Fenders	629.5.043.34	Service spaces. Store rooms
629.5.03	Ship propulsion systems	629.5.043.36	Lockers. E.g. for lamps, paint, flags
	<i>Denote further particulars of engines by 62-..., especially 62-8..., with details by :621...</i>	629.5.043.4	Detention spaces. Cells
		629.5.044	Entry/exit, embarkation/disembarkation arrangements: ladders, gangways etc.
629.5.031	Propulsion systems using muscle power and natural forces	629.5.045	Arrangements for loading/unloading, stowage and tie-down of freight (cargo)
629.5.031.2	Muscle-powered propulsion	629.5.045.2	Cargo compartments. Holds
629.5.031.21	Oars	629.5.045.22	Dry cargo holds
629.5.031.22	Paddles	629.5.045.23	Refrigerating holds
629.5.031.3	Wind-driven propulsion	629.5.045.24	Luggage compartments
629.5.031.32	Sails	629.5.045.25	Mail compartments
629.5.031.34	Flettner rotors	629.5.045.26	Compartments for transportation of valuable goods
629.5.032	Wheel-drive propulsion	629.5.045.27	Compartments for transportation of explosion-hazardous and fire-hazardous goods
629.5.032.1	Single paddlewheel	629.5.045.3	Holds for transportation of animals
629.5.032.2	Twin paddlewheel	629.5.045.4	Cargo tanks. Removable tanks
629.5.034	Chain or cable propulsion	629.5.045.6	Compartments for ship's stores
629.5.035	Propeller or screw propulsion	629.5.045.61	Forepeak
629.5.035.5	Propeller kinds and design	629.5.045.62	Afterpeak
629.5.035.55	Adjustable pitch. Controllable pitch	629.5.045.63	Double-bottom tanks
629.5.035.58	Contrarotating propellers	629.5.045.64	Fresh-water tanks
629.5.035.6	Voith-Schneider propulsion	629.5.045.65	Fuel compartments
629.5.035.8	Propeller shafts	629.5.045.651	Coal bunkers
629.5.036	Reaction or water-jet propulsion	629.5.045.652	Fuel tanks
629.5.037	Special types of propulsion. E.g. airscrew propulsion, propulsion units of air-cushion vehicles	629.5.046.6	Health and medical service facilities. □ Sickbays
629.5.038	Combined propulsion systems		
629.5.04	Interior layout. Accommodation. Installations and equipment	629.5.046.7	Hygienic facilities ◇ 629.5.046.7:643.521 Bathrooms ◇ 629.5.046.7:643.523 Washrooms ◇ 629.5.046.7:643.524 Lavatories
629.5.042	Interior layout. Accommodation. Installations and equipment		
629.5.042.2	Cabins ◇ 629.5.042.2-057.1 Cabins for staff ◇ 629.5.042.2-057.68 Cabins for passengers	629.5.047	Personal protection and safety installations and equipment ⇒ 629.5.067
629.5.042.3	Saloons. Messrooms ◇ 629.5.042.3-057.1 Saloons for staff ◇ 629.5.042.3-057.68 Saloons for passengers	629.5.047.2	Emergency escape devices
629.5.042.5	Service compartments ◇ 629.5.042.5:643.3 Gallies ◇ 629.5.042.5:643.37 Pantries ◇ 629.5.042.5:643.42 Bars ◇ 629.5.042.5:643.47 Storerooms for provisions ◇ 629.5.042.5:643.53 Lounges ◇ 629.5.042.5:643.554 Laundries ◇ 629.5.042.5:643.9 Storage spaces ◇ 629.5.042.5:687.53 Hairdressing saloons. Barbers' shops	629.5.047.5	Provisions for life-jackets, inflatable dinghies etc.
629.5.042.8	Circulation areas ◇ 629.5.042.8:692.622 Stairs ◇ 629.5.042.8:692.81 Doorways ◇ 629.5.042.8:72.052.7-025.72 Corridors	629.5.048	Conditioning systems. Lighting ⇒ 628.8/9 ⇒ 697
629.5.042.82	Doorways	629.5.048.3	Air conditioning
629.5.042.84	Corridors	629.5.048.4	Ventilation
629.5.042.86	Stairs	629.5.048.7	Heating and cooling systems for ship's spacing
629.5.043	Crew arrangements, stations and quarters. Control compartments	629.5.048.72	Steam heating
629.5.043.1	Navigational spaces	629.5.048.73	Water heating
629.5.043.12	Bridge. Wheelhouse	629.5.048.75	Electric heating
		629.5.048.78	Cooling
		629.5.048.8	Lighting ⇒ 628.9 ⇒ 629.5.066.2
		629.5.05	Guidance, control-initiation and navigation systems and instruments (vehicle-borne)
		629.5.051	Principles and purpose of navigation systems etc.
		629.5.051.5	En-route control. □ Automatic pilot
		629.5.051.53	Collision avoidance
		629.5.051.8	Final control and guidance (coastal/harbour approach)
		629.5.052	Means of operation <i>For subdivisions see 629.052...</i>

629.5.053	Special systems of control for particular types of ship	629.5.066.36	Acoustic equipment
629.5.054	Instruments on board	629.5.066.362	Whistles
◇ 629.5.054.03	Instruments for propulsion systems (power plant)	629.5.066.363	Sirens
629.5.058	Operational characteristics of systems and instruments	629.5.066.364	Fog horns
629.5.058 ≡ 629.7.058		629.5.066.366	Typhons
629.5.058.42	Depth measurement. Sounding	629.5.066.367	Megaphones
629.5.06	Ship auxiliary systems and devices	629.5.066.5	Intercommunication systems
629.5.061	Directional control systems and devices	629.5.066.51	One-way systems. E.g. address systems
629.5.061.1	Steering gear	629.5.066.52	Two-way systems. E.g. telephones, speaking tubes
629.5.061.11	Rudders	629.5.066.54	Engine-room telegraphs
629.5.061.112	Ordinary-type rudders	629.5.066.8	Identification markings
629.5.061.113	Semibalanced rudders	629.5.067	Ship safety and protection systems
629.5.061.114	Balanced rudders	⇒ 627.77	
629.5.061.115	Kitchen rudders	⇒ 629.5.047	
629.5.061.116	Flettner rudders	⇒ 656.08	
629.5.061.117	Hydrofoil rudders	629.5.067.2	Lifeboats (as part of equipment)
629.5.061.117.2	Counter rudders	629.5.067.3	Boat-handling gear. □ Davits
629.5.061.12	Rudder elements	629.5.067.7	Railings and awnings
629.5.061.13	Rudder design types	629.5.067.8	Systems for fire protection and fire extinguishing. Firefighting equipment
629.5.061.131	Hollow (two-layer rudders)	Specify by :614.84...	
629.5.061.132	Solid (one-layer rudders)	629.5.068	Special auxiliary systems forming part of the ship
629.5.061.14	Steering controls. □ Tillers. Steering wheels	629.5.068.2	Systems for provision of water
629.5.061.15	Action rudders	629.5.068.3	Sewage and drainage systems
629.5.061.16	Steering engines	629.5.068.4	Refuse disposal systems
629.5.061.17	Bow thrusters	629.5.069	Other auxiliary systems forming part of the craft
629.5.061.2	Elevators (to control diving of a submarine)	629.5.07	Technical aspects of ship operations (operational techniques), including performance.
629.5.062	Control linking/reinforcing systems. Servo control systems	Use the subdivisions of .07 to denote equipment, devices, technical modifications etc. for the listed operations.	
629.5.062.2	Ballasting systems	For nontechnical aspects use 656.6 with .052	
629.5.062.22	Replacement systems. Submerge and surfacing systems	629.5.071	Preliminary operations and preparations for sailing. Operations after berthing, docking
629.5.062.23	Heeling systems	629.5.072	Navigating, piloting and steering operations in general. □ Ship trials
629.5.062.24	Trimming systems	629.5.072.1	Navigational computations. Course plotting
629.5.062.26	Double-bottom tank ballasting system	629.5.072.182	Great circle sailing
629.5.062.3	Power-plant control systems. □ Reversers	629.5.072.184	Rhumb-line sailing
629.5.062.7	Special control systems for docking and mooring (e.g. for ferries)	629.5.072.4	Manoeuvring technique. Manoeuvres
629.5.063	Auxiliary supply and feed systems	629.5.072.8	Instruction. Training of navigational officers
For subdivisions see 629.063...		629.5.073	Operational techniques for special conditions
629.5.064	Auxiliary power systems (in general): pneumatic, hydraulic, electrical etc.	629.5.073 ≡ 656.052.5	
For subdivisions see 629.064...		629.5.073.2	Blind manoeuvring
629.5.065	Materials-handling equipment. □ Refuelling systems and devices. Revictualing systems and devices	629.5.073.4	Sailing in ice conditions
Further details by :621.86/.87		629.5.073.5	Sailing in rough weather
⇒ 629.5.045		629.5.073.7	Sailing with sudden changes in conditions, e.g. engine failure
629.5.065.2	Cargo-handling gear	629.5.075	Departure techniques. Operations for starting to move, getting under way
629.5.065.22	Cargo derricks and their rigging	⇒ 629.5.085	
629.5.065.23	Deck cranes	629.5.075.1	Leaving harbour. Casting-off
629.5.065.25	Hoisting lines	629.5.075.2	Clearing a dock or a lock
629.5.065.26	Cargo blocks	629.5.075.3	Weighing anchor
629.5.065.5	Special kinds of equipment	629.5.076	Operational techniques en route
629.5.066	Observation, signalling, communication and identification equipment	629.5.077	Arrival techniques. Operations for mooring, docking and anchoring
629.5.066.1	Observation devices. □ Periscopes	⇒ 629.5.087	
629.5.066.2	External lighting	629.5.077.1	Coming from the sea. Pilotage in restricted waterways. Mooring (berthing)
629.5.066.22	Navigation lights	629.5.077.2	Docking. Raising on the slip
629.5.066.28	Search lights	629.5.077.3	Anchoring
629.5.066.3	Signalling and communication equipment	629.5.078	Operations for shipboard maintenance of the vessel (by the crew)
629.5.066.32	Radio equipment	629.5.078.4	In port or harbour
629.5.066.34	Visual equipment	629.5.078.6	En route, at sea etc.
◇ 629.5.066.34:683.8	Signalling lamps.	629.5.08	Shore, port and harbour equipment, installations and related techniques
◇ 629.5.066.34:929.9	Flags	⇒ 627.3	
◇ 629.5.066.34:929.929:003.292.6	SEM		
Semaphore			
629.5.066.347	Equipment for signal rockets and flares		

629.5.081	Shipbuilding yards and plants: their equipment and production techniques	629.524.2	Ordinary type
629.5.081.2	Shipbuilding and repairing facilities	629.524.4	Racing or sport type
629.5.081.22	Stocks	629.525	Sailboats, sailing dinghies and small sailing yachts
629.5.081.24	Slipways. Launching facilities	629.525.2	Open sailing boats. Dinghies
629.5.081.3	Docking facilities	629.525.3	Small sailing cruisers
629.5.081.32	Drydocks	629.525.4	Racing sailboats (single hull, catamaran, trimaran)
629.5.081.322	Graving docks. Building docks		
629.5.081.323	Marine railways	629.526	Motorboats and small motor launches (with inboard or outboard motor)
629.5.081.324	Mechanical lift docks		Motor cabin cruisers
629.5.081.326	Floating dry docks	629.526.2	Racing or sporting motorboats. Speedboats
	◇ 629.5.081.326-214.3 Sectional floating dry docks	629.526.4	Lifeboats
629.5.081.328	Drydock equipment and machinery	629.527	Ordinary lifeboats
	◇ 629.5.081.328:621.65 Pumps for drydocks	629.527.2	Unsinkable boats
	◇ 629.5.081.328:626.422 Sluices, floodgates for drydocks	629.527.3	Lifeboat accessories
		629.527.4	Multipassenger motor launches (mainly for waterways). Water-buses
		629.528	
629.5.081.34	Fitting-out basins. Wetdocks		
629.5.081.4	Shipbuilding and repairing techniques	629.53	Sailing vessels. Larger sailing ships
629.5.081.5	Launching	629.532	Ancient sailing vessels (with or without oars).
629.5.081.52	Preparation of ship for launching		□ Galleons. Caravels
629.5.081.53	Slipway launching		Smaller seagoing sailing vessels. □ Cutters.
629.5.081.532	Fore-and-aft launching	629.533	Sloops. Ketches. Dhows
629.5.081.534	Broadside launching		Larger sailing ships
629.5.081.54	Floating in a building dock	629.534	Full square-rigged ships. □ Brigs. Frigates
629.5.081.7	Fitting out and finishing operations	629.534.2	Mixed-rigged ships. □ Barques (barks).
629.5.082	Supply installations on shore, at base: storage and delivery of fuel, lubricant, water etc.	629.534.3	Brigantines. Schooner brigs
629.5.083	Survey and maintenance. Repair. Modifications		Fore-and-aft rigged ships. Schooners
	<i>For techniques see 629.5.081...</i>	629.534.4	Sailing ships with auxiliary engines
629.5.083.2	Classification survey	629.535	Sailing steamers
629.5.083.4	Maintenance	629.535.2	Sailing motor-ships
629.5.083.5	Ship repairs	629.535.4	
629.5.083.6	Conversion	629.54	Motor-ships in general. Seagoing motor vessels. Motor vessels for large inland waters. Coasters
629.5.083.65	Wartime adaptation of merchant ships (WAMS)		Craft carrying passengers and freight
629.5.083.7	Modernization	629.541	Ferries
629.5.084	Spare parts and replacements. □ Issue.	629.541.2	◇ 629.541.2:629.331 Car ferries
	Stores. Stock control		◇ 629.541.2:629.4 Train ferries
629.5.085	Shore, quay or wharf equipment and techniques for clearing port, unmooring, casting off. Final maintenance before sailing	629.541.4	Passenger ships
		629.541.41	Line ships
629.5.086	Shore or base equipment and techniques for control of sailings and trial runs en route	629.541.42	Cruise liners
		629.541.6	Hospital ships
629.5.087	Shore or base equipment and techniques for ships entering, mooring, berthing, anchoring, docking etc. Booms etc.	629.542	Freight ships
		629.542.2	General cargo vessels
		629.542.4	Refrigerator ships
		629.542.6	Ships for transportation of wood (timber)
629.51/58	Types of watercraft or vessel and their construction. Boats and ships	629.543	Tankers
		629.544	Container ships
629.51	Floating structures with no (specific) means of propulsion	629.545	Ships with special means of cargo handling
629.512	Simple moving structures. □ Rafts. Log rafts	629.545.2	Roll-on/roll-off ships
629.514	Stationary structures. □ Houseboats. Floating hotels. Platforms	629.545.4	Barge- or LASH (Lighter Aboard SHip) carriers
	⇒ 629.563.2	629.545.6	Dock ships
	⇒ 728.77	629.546	Ships for the transportation of mass goods
629.516	Buoys	629.546.2	Bulk carriers
	⇒ 627.713	629.546.4	Ore carriers
		629.546.6	Ore-bulk-liquid carriers
629.516.2	Mooring buoys	629.55	Waterway (river and canal) craft
629.516.4	Buoys for storage of liquids	629.552	River and canal ferries
		629.553	River and canal passenger (cruising, touring) ships
629.52	Small craft and boats		Motor barges. Waterway (river and canal) motor freighters
629.521	Primitive and simple small boats. □ Piroques. Dugouts. Coracles	629.555	Waterway (river and canal) tankers
629.522	Boats propelled by pole or stern-oar		
629.522.2	Punts	629.556	
629.522.4	Gondolas	629.56	Special purpose vessels and craft
629.523	Boats propelled by paddle	629.561	Service craft
629.523.2	Kayaks	629.561.1	Tugs
629.523.4	Canoes	629.561.11	Harbour tugs
629.524	Rowboats, rowing dinghies and racing rowing boats (sculls and oars)	629.561.12	Seagoing tugs
		629.561.2	Pushers

629.561.4	Firefighting ships ⇒ 614.846.9	629.7.01	Air/space vehicle design principles, characteristics and tests ⇒ 533.6 ⇒ 629.7.021
629.561.5	Icebreakers ⇒ 629.5.028.5	629.7.012	Dimensions. Dimensional design
629.561.7	Pilot craft. Harbour administration service craft	629.7.012.2	Overall length
629.562	Fishery vessels	629.7.012.3	Breadth or span
629.562.2	Trawlers	629.7.012.5	Depth or diameter
629.562.3	Seiners	629.7.013	Weights (overall or gross weights, payloads etc.)
629.562.4	Luggers	629.7.013.1	Construction weight
629.562.5	Mackerel fishing boats	629.7.013.2	Empty weight. Tare weight
629.562.6	Drifters	629.7.013.3	Fuel weight
629.562.7	Whalers	629.7.013.5	Payload weight
629.562.8	Fishing factory ships	629.7.014	Functional capabilities and mode of use <i>Use the auxiliaries listed at 629.014... only for occasional or special capabilities or uses not provided for in the direct subdivisions of 629.7.</i> <i>Additional subdivisions of .014.1 for use under 629.7 are listed here</i>
629.563	Ships for offshore purposes	629.7.014.16	VTOL (Vertical Take-Off and Landing). STOL (Short Take-Off and Landing)
629.563.2	Drilling platforms	629.7.014.18	Manned
629.563.21	Semisubmersibles	629.7.015	Mechanics and dynamics in relation to the medium. □ Aerodynamic design problems. Strength requirements
629.563.22	Jack-up platforms	629.7.015.3	Aerodynamic problems affecting design <i>Specify problems by :533.6...</i>
629.563.23	Gravity platforms	629.7.015.4	Strength requirements. Aerodynamic and structural design requirements <i>Details by :539.4...</i> ⇒ 629.7.018.4
629.563.24	Tension leg platforms	629.7.015.7	Aeromechanical (flight mechanics) problems affecting design
629.563.25	Guyed towers	629.7.016	Flight characteristics and capabilities (design considerations). Performance. Assessments
629.563.3	Drilling ships	629.7.016.2	Altitude
629.563.4	Ships for the maintenance and marking of shipping lanes	629.7.016.3	Range
629.563.42	Dredgers. Bed-clearing vessels	629.7.016.5	Speed
629.563.422	Bucket dredgers	629.7.016.52	Subsonic (up to Mach 1)
629.563.424	Suction dredgers	629.7.016.53	Transonic (Mach 0.8 to Mach 1.3)
629.563.44	Buoy-laying vessels	629.7.016.54	Supersonic (Mach 1 to 4.99)
629.563.45	Lightships	629.7.016.55	Hypersonic (Mach 5 upwards)
629.563.5	Crane ships	629.7.016.7	Propulsion assessment
629.563.6	Cable-laying ships. Pipe-laying ships	629.7.016.8	Characteristics and capabilities under special conditions
629.563.7	Oil recovery vessels. Oil skimmers	629.7.016.82	At large angles of attack
629.563.8	Auxiliary vessels for assistance and supply	629.7.016.84	Hovering characteristics
629.563.82	Supply vessels	629.7.016.86	Transition characteristics
629.563.84	Diving ships	629.7.017	Operational reliability. Airworthiness. Spaceworthiness ⇒ 533.6.013 ⇒ 656.085.1
629.564	Weather, research, exploration and expeditionary ships	629.7.017.1	Reliability, effectiveness. Vulnerability. Accuracy. Serviceability. Effects of use etc.
629.564.2	Space- or cosmic-research ships	629.7.017.2	Stability. Controllability (attitude) ⇒ 533.6.013.4 ⇒ 629.7.062.2
629.564.3	Oceanographic research ships	629.7.017.21	Pitch. Longitudinal stability
629.564.4	Hydrographic research vessels	629.7.017.22	Roll. Lateral stability
629.564.5	Marine-biological research vessels	629.7.017.23	Yaw
629.564.7	Exploration and expeditionary ships. E.g. polar research vessels	629.7.017.25	Stability and controllability at engine failure
629.565	Special craft for in-port facilities, cargo handling, ship repairs	629.7.017.26	Damping forces and moments
629.565.1	Floating cranes	629.7.017.27	Stability derivatives
629.565.2	Floating docks	629.7.017.28	Control forces
629.565.3	Pontoons and lighters	629.7.017.3	Manoeuvrability
629.566	Marine salvage and recovery vessels ⇒ 627.76	629.7.017.5	Braking capacity ⇒ 629.7.062.5
629.57	Surface-skimming craft <i>Denote special purpose craft by :629.54/.56</i>	629.7.018	Vehicle test procedures and equipment. Aeronautical laboratory equipment. Aircraft models
629.572	Hydrofoil craft. Hydroplanes		
629.574	Surface effect ships (SES)		
629.576	Air cushion vehicles (ACV). Hovercraft ⇒ 629.437		
629.58	Underwater craft. Submersibles.		
	Submarines ⇒ 623.827		
629.582	Diving bells		
629.584	Submersible devices. □ Bathyspheres.		
	Bathyscaphes		
629.585	Submarines		
629.7	Air and space transport engineering. Aeronautics and aircraft. Rockets and missiles. Astronautics and spacecraft ⇒ 347.823.34 ⇒ 351.814 ⇒ 623.7 ⇒ 656.7		

629.7.018.1	Wind tunnels. Shock tubes ⇒ 533.6.071	629.7.025.35	Flaps. Airbrakes. Steel parachutes
629.7.018.2	Laboratory equipment for moving models and free-flight tests	629.7.025.36	Elevators. Elevons
629.7.018.3	Aeroballistic ranges. Rocket sleds. Other flight test facilities	629.7.025.37	Rudders
629.7.018.4	Structural strength tests. □ Fatigue, vibration, impact tests <i>Details by :620.17...</i> ⇒ 629.7.015.4	629.7.025.38	Auxiliary control surfaces. □ Tabs
629.7.018.7	Flight tests	629.7.025.4	Planform of wings and stabilizers. Sweepback and sweepfore. Shapes: tapers, triangles, deltas etc.
629.7.018.72	Tethered tests	629.7.025.5	Arrangement of wings, stabilizers etc. in relation to bodies <i>Denote vertical, horizontal, inclined by -111...</i>
629.7.018.74	Flying-model tests	629.7.025.52	Above and separate from body
629.7.018.77	Flying laboratories and research vehicles	629.7.025.522	Rigid struts
629.7.02	Air/space vehicle structure. Construction principles. General layout and parts	629.7.025.524	Rogallo wings
629.7.021	Construction principles and properties (of whole or parts)	629.7.025.54	High-wing position on body
629.7.021.1	Monocoque	629.7.025.55	Mid-wing position, normal to surface
629.7.021.2	Semi-monocoque (stressed skin)	629.7.025.56	Low-wing position
629.7.021.3	Space-frames	629.7.025.58	Encircling body and separate from it: annular, semiannular
629.7.021.4	Solid	629.7.025.6	Angles of wings, stabilizers etc.
629.7.021.5	Inflatable	629.7.025.7	Surface characteristics of wings, stabilizers etc. □ Area, profile, camber, twist. Taper ratio. Aspect ratio (span-chord ratio)
629.7.021.6	Retractable	629.7.025.8	Elements of wings, stabilizers etc.
629.7.022	General layout and configuration. Relation of parts to whole	629.7.025.82	Leading edges
629.7.023	Shell. Framework. Structural members	629.7.025.84	Tips. Wingtips
629.7.023.2	Skin. Coatings, panels, openings etc. Outer doors. Windows. Transparent hoods etc.	629.7.025.86	Root
629.7.023.22	Coatings. Surfaces	629.7.025.88	Trailing edges
629.7.023.24	Fixed coverings, fairings, fillets. Nacelles	629.7.026	Dynamic lifting and control structures. Propulsion unit structures
629.7.023.25	Removable panels, covers. Hinged panels. External doors	629.7.026.1	Number of units, engines or motors
629.7.023.26	Transparent panels, windows, hoods	629.7.026.11	Single-unit. Single-engine
629.7.023.27	Openings in the skin. E.g. for cooling ducts, fuel ducts, antennas	629.7.026.12	Double-unit. Twin-engine
629.7.023.4	Structural members. Frames. Ribs, struts, bracing etc.	629.7.026.13	Triple-unit. Three-engine
629.7.023.42	Ribs. Formers	629.7.026.14	Four-unit. Four-engine
629.7.023.44	Longitudinal members. □ Longerons. Stringers	629.7.026.5	Rotary lifting and control devices ⇒ 629.7.035
629.7.023.46	Transverse members. □ Spars. Struts. Bracing. Strong points	629.7.026.55	Propeller devices
629.7.023.8	Attachments between structural parts	629.7.026.56	Rotor devices
629.7.024	Bodies	629.7.026.57	Fan devices
629.7.024.1	Body shapes. Types of fuselage or hull	629.7.026.58	Rotary stabilizers
629.7.024.11	Single fuselage. Single hull	629.7.026.6	Propulsion-unit structures according to position
629.7.024.12	Twin fuselage. Twin hull	629.7.026.62	Above and separate from wing
629.7.024.15	Capsules	629.7.026.63	Attached directly to wing
629.7.024.18	Complex shapes. □ Clusters	629.7.026.64	Within wing
629.7.024.2	Merged wing-bodies. □ Re-entry structures designed for lifting and re-entry	629.7.026.65	Attached directly to body
629.7.024.3	Forebodies. □ Noses. Cones	629.7.026.66	Within body
629.7.024.5	Main bodies. Centre sections	629.7.026.67	Separate from body
629.7.024.7	Afterbodies. Tail booms	629.7.026.68	On rotor tips
629.7.024.8	Bases	629.7.027	Undercarriages. Take-off, landing and mooring gear on vehicles ⇒ 629.7.085/087
629.7.025	Lifting and control surfaces. Aerofoils. Wings. Stabilizers etc.	629.7.027.2	Main undercarriage assembly, wheel and track
629.7.025.1	Wings in general	629.7.027.23	Wheels. Castoring units. Bogies etc.
629.7.025.11	Single: monoplane	629.7.027.24	Special undercarriages. □ Special accessories, shock absorbers for undercarriages ◇ 629.7.027.24-182.8 Detachable undercarriages ◇ 629.7.027.24.021.6 Retractable undercarriages
629.7.025.12	Double, multiple: biplane, triplane etc.	629.7.027.3	Skids, skis and attachments, on aircraft etc.
629.7.025.14	Mainplanes: nose-to-tail, for tailless configurations	629.7.027.4	Floats and attachments on aircraft etc. Hydrofoils on aircraft etc. ⇒ 629.572
629.7.025.15	Mainplanes for other configurations	629.7.027.5	Protective devices for body, tail and wingtips on take-off and landing
629.7.025.16	Foreplanes	629.7.027.6	Arrester hooks on aircraft etc. Similar short-landing devices ⇒ 629.7.087.22
629.7.025.17	Tailplanes		
629.7.025.2	Stabilizers in general		
629.7.025.3	Control surfaces in general		
629.7.025.32	All-moving controls		
629.7.025.33	Slats. Spoilers		
629.7.025.34	Ailerons		

629.7.027.7	Mooring devices and gear on aircraft etc. ⇒ 629.7.087.27	629.7.046	Passenger comfort and service arrangements <i>Details by :64... and :684...</i>
629.7.028	Attachments. Couplings. Auxiliary structures	◇ 629.7.046:643.3	Galleys
629.7.028.2	Attachments (external, to skin). Separating mechanisms	◇ 629.7.046:643.521	Bathrooms
629.7.028.3	Towing devices on aircraft etc.	◇ 629.7.046:643.523	Washrooms
629.7.028.4	Mountings and accessory booms or arms. □ Mountings etc. for solar-cell paddles on spacecraft	◇ 629.7.046:643.524	Lavatories
629.7.028.6	Auxiliary structures. Turrets, radomes etc.	629.7.047	Personal protection and safety installations and equipment
629.7.03	Propulsion systems in air/space vehicles	629.7.047.2	Emergency escape devices. □ Ejector seats. Capsules
629.7.031	Aircraft propulsion using natural forces, muscular strength etc.	629.7.047.4	Safety belts
629.7.031.2	Muscle-power propulsion	629.7.047.5	Life-belts. Life-jackets. Life-vests. Inflatable dinghies etc. (in case of ditching)
629.7.031.7	Radiation pressure	629.7.047.8	Protective clothing. □ Pressure suits. Anti-g (anti-gravity) suits. Space suits
629.7.031.8	Electric and magnetic effects. Electrogravitics	629.7.048	Life-support systems. Climate control (vehicle-borne systems). Air conditioning. Cabin pressurization. Oxygen supply. Heating. Cooling. Lighting
629.7.035	Airscrew propulsion: propeller, rotor, fan ⇒ 62-8 ⇒ 621.4 ⇒ 629.7.026.5	629.7.048.3	Cabin air-conditioning, pressurization systems
629.7.035.2	Piston-engine airscrew propulsion 629.7.035.25/.27 ≡ 629.7.035.5/.7	629.7.048.4	Oxygen supply, air-cleaning systems
629.7.035.25	Piston-propeller propulsion	629.7.048.7	Heating, cooling, temperature-control systems
629.7.035.26	Piston-rotor propulsion	629.7.048.8	Lighting ⇒ 628.9
629.7.035.3	Turbine-engine airscrew propulsion 629.7.035.35/.37 ≡ 629.7.035.5/.7	629.7.05	Guidance, control-initiation and navigation systems and instruments (vehicle-borne)
629.7.035.35	Turboprop propulsion	629.7.051	Principles. Purpose
629.7.035.37	Turbofan propulsion	629.7.051.2	Initial control and guidance systems (for take-off and ascent)
629.7.035.5	Propellers: kinds and design	629.7.051.5	Mid-course or en-route control and guidance systems. □ Automatic pilot
629.7.035.52	Tractor propellers	629.7.051.53	Collision avoidance. Bad-weather avoidance
629.7.035.53	Pusher propellers	629.7.051.56	Terrain avoidance
629.7.035.55	Variable-pitch propellers. Propellers with adjustable or controllable pitch. Feathering etc.	629.7.051.8	Final or homing control and guidance systems. Approach, landing, rendezvous
629.7.035.56	Reversing propellers. Braking propellers	629.7.052	Means of operation <i>For subdivisions see 629.052...</i>
629.7.035.58	Contrarotating propellers	629.7.054	Instruments on board 629.7.054 ≡ 629.3.054
629.7.035.6	Rotors	629.7.054.2	Instruments for monitoring propulsion mechanism and fuel
629.7.035.62	Main rotors	629.7.054.23	Pressure gauges
629.7.035.67	Tail rotors	629.7.054.3	Instruments for monitoring other systems and mechanisms <i>Specify by .0...</i>
629.7.035.7	Fans	629.7.054.4	Instruments for measuring the relationship of the vehicle to the external environment 629.7.054.4 ≡ 629.7.058.4
629.7.036	Propulsion by reaction pressure. Jet propulsion. Rocket propulsion etc. ⇒ 621.45 ⇒ 629.76	629.7.054.42	Altimeters
629.7.036.2	Propulsive duct (athodyd)	629.7.054.45	Air-speed indicators
629.7.036.22	Ramjet	629.7.054.47	Artificial horizons
629.7.036.24	Pulse-jet	629.7.058	Basic characteristics of operation of observation and indicating devices
629.7.036.3	Turbojet	629.7.058.4	Functional capabilities of systems and instruments
629.7.036.5	Rocket propulsion	629.7.058.42	Height or altitude measurement ⇒ 531.717.3 ⇒ 531.719.4
629.7.036.7	Electric and electromagnetic propulsion	629.7.058.43	Distance or range measurement
629.7.036.72	Electrothermal propulsion. Arc jet (plasma arc)	629.7.058.44	Indication of direction
629.7.036.73	Electrostatic, ion propulsion. Photon propulsion etc.	629.7.058.45	Speed measurement
629.7.036.74	Electrodynamic propulsion. Plasma jet (plasma pinch using magnetohydrodynamic forces)	629.7.058.47	Indication of attitude (pitch, roll, yaw)
629.7.036.8	Nuclear propulsion ⇒ 621.039.578	629.7.058.48	Drift
629.7.038	Combined propulsion systems	629.7.058.5	Operating functions
629.7.04	Interior layout of air/space vehicles. Accommodation. Installations. Equipment	629.7.058.52	Neither transmitting nor receiving (independent)
629.7.042	Interior layout and finish. Compartments, cabins etc.	629.7.058.53	Receiving only. □ Passive homing
629.7.042.2	Internal skin. □ Coatings, linings. Bulkheads. Inner doors	629.7.058.54	Transmitting and receiving. □ Radar
629.7.042.3	Main (saloon) arrangements	629.7.058.55	Transmitting only. □ With radionumber-beacon systems locating vehicle for tracking etc.
629.7.043	Crew accommodation. Captain's, pilot's compartments. Flight deck. Cockpit etc.	629.7.058.6	Methods of comparing measured parameters or received data
629.7.044	Arrangements for boarding and alighting, entry and exit. Steps. Gangways etc. (vehicle-borne)		
629.7.045	Freight handling and storage. Arrangements for loading, unloading, stowage of cargo. Securing, tying down etc.		

629.7.058.62	Amplitude comparison, including directional systems	629.7.072.172	Air speed of aircraft
629.7.058.64	Frequency comparison	629.7.072.174	Angle and rate of climb or descent
629.7.058.66	Phase comparison	629.7.073	Operational techniques for special conditions 629.7.073 $\equiv$ 656.052.5
629.7.058.68	Travel-time comparison	629.7.073.2	Blind flying
629.7.058.7	Output. Display	629.7.073.6	Flying in bad terrain, mountainous country etc.
629.7.058.72	Aural systems. ☐ 'Talk-down' landing systems	629.7.073.7	Flying with sudden changes in conditions, e.g. engine failure
629.7.058.74	Visual display. ☐ Pictorial map display. Simulated display	629.7.075	Take-off or launch techniques $\Rightarrow$ 629.7.085
629.7.058.75	Direct-to-control systems	629.7.075.4	Boost separation
629.7.058.76	Via-telemetry systems	629.7.075.6	Stage separation
629.7.058.77	Recorded output	629.7.076	Operations in flight. Flight performance
629.7.058.8	Elements. Mountings. Panels	629.7.076.2	Climb phase. Climb manoeuvres
629.7.058.82	Elements. Gyro inertial units. Mountings	629.7.076.4	Cruise phase. Course-changing manoeuvres
629.7.058.85	Panels. Dashboards etc.	629.7.076.46	Collision course. Interception course
629.7.06	Auxiliary systems and devices on air/space vehicles	629.7.076.6	Orbit phase. Orbiting manoeuvres $\Rightarrow$ 521.3
629.7.062	Control linking and reinforcing systems. Servo control etc.	629.7.076.66	Rendezvous
629.7.062.2	Attitude control systems 629.7.062.21/.23 $\equiv$ 629.7.017.21/.23	629.7.076.8	Re-entry phase. Re-entry calculations, manoeuvres
629.7.062.22	Roll control systems	629.7.077	Descent and landing techniques $\Rightarrow$ 629.7.087
629.7.062.3	Power-plant control systems. ☐ Thrust. Reverser	629.7.078	Operations for on-board maintenance (by the crew)
629.7.062.4	Air-flow control systems (other than airbrakes). Laminar flow	629.7.08	Ground or base equipment, installations and techniques $\Rightarrow$ 625.717 $\Rightarrow$ 656.01/.09 $\Rightarrow$ 656.71 $\Rightarrow$ 725.1
629.7.062.5	Airbrake control systems $\Rightarrow$ 629.7.017.5	629.7.081	Hangar and workshop installations and equipment generally
629.7.062.7	Undercarriage, landing-gear control systems	629.7.082	Ancillary service and supply installations. Depots and stores for fuel, water, inflation gas etc.
629.7.063	Auxiliary supply and feed systems For subdivisions see 629.063...	629.7.083	Maintenance. Handling, service and repair of air/space vehicles
629.7.064	Auxiliary power systems in general For subdivisions see 629.064...	629.7.084	Spare parts and replacements. Stock control, issue etc. Spares depots and stores $\Rightarrow$ 621.796
629.7.065	Materials-handling equipment. ☐ Refuelling systems and devices. Revictualing systems and devices Further details by :621.86/.87 $\Rightarrow$ 629.7.045	629.7.085	Base equipment and technical procedure for take-off, launching
629.7.066	Observation, signalling, communication and identification equipment and devices $\Rightarrow$ 654.9	629.7.085.2	For take-off from ground surface or ship. ☐ Rails, cables, catapults. Take-off trolleys. Launching pads. Silos
629.7.066.1	Observation devices. ☐ Periscopes. Astrodomes	629.7.085.22	Rails
629.7.066.3	Signalling, indicator devices	629.7.085.23	Cables
629.7.066.5	Intercommunication systems	629.7.085.24	Catapults
629.7.066.8	Identification markings. Instruction markings	629.7.085.25	Take-off trolleys
629.7.067	Safety and protection systems on air/space vehicles. ☐ Icing prevention. Devices for removing ice and rain. Fire-extinguishing systems $\Rightarrow$ 614.8 $\Rightarrow$ 62-776 $\Rightarrow$ 62-787	629.7.085.26	Launching pads
629.7.067.5	Against ice and rain. ☐ Removal systems	629.7.085.27	Silos
629.7.067.8	Against accidents. ☐ Explosion-suppression systems. Fire-extinguishing systems Details by :614.8...	629.7.085.4	For take-off from sea or other water surface
629.7.069	Other auxiliary systems forming part of the vehicle. ☐ Armament systems	629.7.085.5	For take-off from underwater
629.7.07	Technical aspects of operations (operational techniques), including performance Use the subdivisions of .07 to denote equipment, devices, technical modifications etc. for the listed operations. For nontechnical aspects use 656.7 with .052	629.7.085.6	For take-off from aircraft
629.7.071	Preliminary operations and preparations.	629.7.085.7	For take-off from balloons
629.7.072	Operations after landing and coming to rest Piloting, navigating operations in general. ☐ Test flights. Training 629.7.072 $\equiv$ 656.052 except 656.052.5	629.7.085.8	For take-off from extraterrestrial base. ☐ From artificial satellites, space stations, lunar surface, other celestial bodies
629.7.072.116	Position-fixing in aircraft	629.7.086	Base equipment and technical procedure for trial and test flights. Tracking equipment
629.7.072.154	Height above ground, of aircraft	629.7.087	Base equipment and technical procedure for landing, arresting, mooring
		629.7.087.2	For landing within atmosphere, on ground surface or ship
		629.7.087.22	Arrester gear. ☐ Arrester nets. Arrester barriers. Arrester cables $\Rightarrow$ 629.7.027.6
		629.7.087.27	Mooring devices. ☐ Mooring masts. Mooring pylons. Mooring towers $\Rightarrow$ 629.7.027.7

629.7.087.4	For landing on sea or other water surface	629.762.2	Ballistic missiles
629.7.087.6	For recovery by airborne catchment devices	629.762.5	Nonballistic missiles
629.7.087.8	For landing on extraterrestrial sites. □ On artificial satellites, space stations, lunar surface, other celestial bodies	629.764	Space vehicle boosters
		629.765	High-altitude sounding rockets and near-space probes
629.73	Aeronautical engineering. Aircraft	629.78	Spaceflight engineering. Astronautics. Spacecraft
	⇒ 533.61/.66		
629.733	Lighter-than-air aircraft (aerostats)	629.782	Fringe-flight winged spaceplanes (designed for controlled flight through atmosphere, orbiting manoeuvres and return to Earth)
629.733.3	Balloons		Artificial satellites. □ Navigational satellites. Communications or relay satellites
629.733.5	Airships (dirigibles)	629.783	Ferry and rendezvous space vehicles (for transporting personnel and equipment to orbital and interplanetary stations etc.)
629.734/.735	Heavier-than-air aircraft (aerodynes)		Space probes (for moon, circumlunar, planetary, circumplanetary, sun and circumsolar flight)
629.734	Nonpowered aerodynes	629.784	Space stations (orbital, for research; not for landing or return)
629.734.3	Fixed-wing (including flexible, variable-shape etc.)		Orbital (near-Earth) space stations
629.734.33	Gliders. Sailplanes	629.785	Interplanetary and lunar vehicles for landing, without return
629.734.333	Gliders		Interplanetary and lunar vehicles for landing and return
629.734.334	Sailplanes		
629.734.34	Kites		
629.734.4	Moving-wing. □ Rotorchutes	629.786	
629.734.5	Hybrid (combination of fixed and moving wings)	629.786.2	
629.734.7	Wingless. □ Parachutes	629.787	
629.735	Powered aerodynes. Aeroplanes. Rotorcraft etc.	629.788	
629.735.3	Fixed-wing aeroplanes (including tilt-wing, variable shape etc.)		
629.735.33	Landplanes (only as distinct from seaplanes)	629.9	Convertible vehicles, designed for normal use in or on more than one medium
629.735.35	Seaplanes (Float-seaplanes. Flying boats)		<i>Vehicles designed primarily for use in or on a single medium are classed in 629.3/.7</i>
629.735.4	Moving-wing, dynamic devices. Rotorcraft		
629.735.43	Autogiros. Gyroplanes	629.92	Vehicles for rail or road travel (rail-road vehicles)
629.735.45	Helicopters		
629.735.47	Ornithopters	629.94	Vehicles for land or water travel. Amphibious vehicles
629.735.5	Hybrid (combination of fixed and moving wings)		
629.735.7	Wingless powered aircraft except rockets and missiles. □ 'Flying bedstead'. Flying platform	629.95	Vehicles for land or air travel
		629.97	Vehicles for water or air travel
			⇒ 629.735.35
629.76	Rocket and missile engineering. Rockets. Missiles		
	◇ 629.76.038 Multistage (propulsion) rockets and missiles		
	⇒ 621.454		
629.762	Missiles (ballistic and nonballistic)		
	⇒ 623.4.08		
	⇒ 623.46		

63 AGRICULTURE AND RELATED SCIENCES AND TECHNIQUES. FORESTRY. FARMING. WILDLIFE EXPLOITATION

The -1/-9 auxiliaries from 62-1/-8 and 66-9 are also applicable throughout 63, with the exception of -1... and -2... in 633/635

630	FORESTRY	630*12	Forest biology
	<i>The subdivisions of 630 are a selection from the Forest Decimal Classification (IUFRO, Vienna).</i>	⇒ 57	
	<i>For fruit-tree growing see 634.1/.6</i>	630*13	Forest zoology in general
		⇒ 59	
630*1	Environmental factors in forestry. Forest biology	630*14	Systematical zoology of forests
		⇒ 592/599	
630*11	Site factors in forestry	630*15	Animal ecology of forests. □ Game and fish management. Hunting. Shooting. Fishing
	⇒ 631.96	⇒ 591.5	
630*111	Forest climate and microclimate. □ The atmosphere. Meteorology	⇒ 639	
	⇒ 551.5	630*16	Forest botany in general
	⇒ 630*42	⇒ 581	
630*113	Situation. Forest topography. □ Gradient and aspect	630*17	Systematic botany of forests
	⇒ 551.4	⇒ 582	
630*114	Forest soil	630*18	Plant ecology of forests
	⇒ 631.4	⇒ 581.5	
630*116	Forest hydrology. Water conservation, soil conservation and erosion	630*2	Silviculture
	⇒ 556	⇒ 631.962	
		630*22	Silvicultural systems
		630*221	High forest systems. □ Clear-felling system. Successive generation fellings. Continuous forest

630*222	Coppice, pollarding and lopping systems	630*43	Forest fires
630*226	Changes of silviculture system. Conversion (by system or species)		⇒ 614.841.42
630*228	Constitution and composition of stands. Forms of stand	630*432	⇒ 631.588
630*23	Regeneration and formation of stands	630*433	Prevention and control of forest fires
630*231	Natural regeneration	630*434	Salvage and disposal of fire-damaged timber
630*232	Artificial regeneration. □ Tree nursery practice. Sowing, planting and propagation of trees. Cuttings of trees	630*436	Revegetation etc. of burnt areas
630*233	Afforestation of particular types of land	630*44	Beneficial effects of fire. Uses of fire in silviculture and other forms of husbandry
	⇒ 627.414	630*45	Damage by harmful plants. Virus diseases
	⇒ 627.524	630*46	Damage by animals
	⇒ 630*9	630*5	Damage to tree growth by man
630*234	Formation of forest by natural succession		Measurements of trees. Tree growth, increment. Development and structure of stands
630*235	Underplanting. Advance planting etc. Nurses. Formation of mixtures	630*51	⇒ 528.48
630*236	Care of plantations in initial stages	630*52	Systems and units of measurement (including conversion tables)
630*237	Improvement, amelioration of forest sites	630*53	Measurements of stem dimensions and volume (of trees, stands, forests and timber)
	⇒ 631.6	630*54	Special measurements of trees and stands.
630*24	Tending of stands and trees	630*55	□ Crown dimensions. Stand density
630*241	Cleanings		Assessment of forest site quality
630*242	Thinnings	630*56	Age determination of trees
630*245	Tending of trees. □ Pruning		⇒ 902.67
	⇒ 631.542		Increment. Development and structure of stands
630*25	Treatment of defective, derelict or very open stands	630*6	Forest management. Business economics of forestry. Administration and organization of forest enterprises
630*26	Combinations of forestry with agriculture and pastoral husbandry. Irrigated and floodland forests. Shelterbelts etc. and their treatment		⇒ 631.16
630*27	Ornamental arboriculture. Parks, arboreta. Hedges and hedge plants	630*62	Methods of management. Working plans.
	⇒ 635.055	630*64	Annual or periodic yield. Yield regulation
	⇒ 712	630*65	Forestry as business
630*28	Forest husbandry. Growing and collection of forest products except wood. □ Bark, fruit and seed crops. Saps. Leaf crops. Osier, bamboo, cane growing	630*66	Special business problems of timber growing
		630*67	Costing of logging operations
630*3	Work science and work studies in forestry. Harvesting of wood		Assessment of financial results. Bookkeeping and accountancy. Business statistics.
630*31	Logging and transport: general	630*68	Short-term planning and financing
630*32	Felling and related operations	630*681	Administration and organization of forest enterprises
	⇒ 621.93	630*682	National (federal, state) forest services and administration
630*33	Degree of utilization and waste. Clearing the felling site		Private forest administration. Staff questions in private forestry
630*34	Storage of wood in the forest. Storage at log dumps or landings	630*7	Marketing of forest products. Economics of forest transport and the wood industries
630*35	Performance measurements (for determining piece rates in felling and related operations)	630*8	Forest products and their utilization
630*36	Tools, machines and equipment for felling and related operations		⇒ 674
630*37	Transport	630*81	Wood and bark. Their structure and properties
	<i>For materials handling equipment see 621.86/87</i>	630*82	Conversion, shaping, assembly and finishing of wood
630*38	Forest engineering. □ Buildings, roads and bridges in forests. Protective works. Forest drainage, irrigation, waterways and stream training	630*83	Timber industries and products. Uses of wood.
	⇒ 626.8	630*84	Manufacture of wooden objects
	⇒ 627.42		Preservation and other treatments to improve the properties of wood. Damage by biological agencies and its control. Seasoning.
	⇒ 627.53	630*85	Timberyard practice (handling and storage)
	⇒ 630*26	630*86	Grading of wood and wood products. Timber quality
630*4	Forest injuries and protection		Pulp industries. Composite materials made wholly or partly of woody matter. Chemical utilization of wood
	⇒ 368.55	630*88	Economy in the use of wood. Replacement of wood by competitive materials
	⇒ 502.172	630*89	Other (minor) forest products
	⇒ 632	630*9	Forests and forestry from the national point of view. Forestry policy. Afforestation policy.
630*41	General technique of forest protection. Types of injury. Methods of damage and pest control		Social economics of forestry
630*42	Injuries from inorganic agencies (excluding fire)	630*92	⇒ 332.3
			⇒ 630*233
			Forest ownership and ownership policy

630*93	Public supervision and regulation of forestry, including legislation for these purposes	631.117.4	Research farms. Experimental farms.
630*94	Other methods to implement forest policy. □ Subsidies. Credits. Forest insurance. Advisory services. Publicity. Education. Research. Associations	631.117.6	Agricultural research stations Demonstration farms. Pilot farms
630*95	Forest taxation	631.12	Farm equipment and materials generally (selection, acquisition, maintenance etc.) ⇒ 631.2 ⇒ 631.3 ⇒ 631.8
630*96	Forest labour questions. Labour questions in wood transport and the wood industries		
630*97	International forest policy. Other international collaboration	631.14	Farming and farms according to products and production
631/635	FARM MANAGEMENT. AGRONOMY. HORTICULTURE	631.143	Farming without livestock
631	AGRICULTURE IN GENERAL	631.145	Factory and farm enterprises combined. Industrialized farming generally ⇒ 631.17
631.1	Farm management and organization ⇒ 332	631.147	Biotechnic, ecological farming (for alternative, intermediate and soft technology). Organic farming and farms
631.1.016	Establishment, growth, decline of farms or estates. Amalgamation, conversion etc.	631.15	Farm production. Farm management. Farm administration
631.1.017	Size and extent of farm or estate		Systems of farm production and management
631.1.017.1	Large farms, plantations, ranches etc.	631.151	Extensive farming system
631.1.017.3	Small farms. Smallholdings. Farmsteads ◇ 631.1.017.3(410.518) Smallholdings and their organization in the Scottish highlands. Crofts and crofting	631.151.1	Intensive farming system
631.1.02	Special conditions, agreements etc.	631.151.2	Specialized farm production system
631.1.027	Marketing and delivery conditions	631.151.3	Diversified farm production system
631.11	Farms, farm properties and farming systems ◇ 631.11(1-773) Farms in underdeveloped areas	631.151.4	Integrated farm production system
631.111	Farm location, site and land features (terrain) ◇ 631.111(23.01) Hill farms	631.151.6	Vertically integrated farm production system
631.111.1	Farm location theory	631.151.62	Horizontally integrated farm production system
631.111.2	Layout and suitability of farm property (for specific crops etc.) ◇ 631.111.2:633.63 Suitability for sugar beet production	631.152	Principles of farm production management. □ Expansion. Modernization. Technical innovations. Regrouping. Administrative modifications <i>Details by :005...</i>
631.111.3	Farmland quality	631.153	Planning of farm production ⇒ 631.58
631.111.4	Farm location in specialized areas ◇ 631.111.4(1-751.2) Farms in nature reserves	631.153.3	Choice of cultivation and husbandry system ◇ 631.153.3:631.582 Crop rotation ◇ 631.153.3:631.584 Intercropping
631.112	Farm acquisition. Farm purchase, sale <i>Details by :658.6/8</i>	631.153.4	Choice of work method. Division of labour. □ Use of manual labour, animals or machines
631.115	Farms according to tenure, ownership and supervision. Systems of management	631.153.5	Supervision of production plans
631.115.1	Private farms	631.153.7	Changes in cultivation and production. Conversion to other systems
631.115.11	Family-run farms	631.155	Farm production control
631.115.13	Partnership farms	631.155.1	Farm production goals and quotas
631.115.17	Leasehold farms. Tenant farms. Sharecrop farms ⇒ 347.453.1	631.155.12	Farm production goals
631.115.2	Manager farms. Corporation farms	631.155.13	Farm production quotas
631.115.6	Collective farms	631.155.2	Produce supplies
631.115.7	Public enterprise farms. State, communal and municipal farms	631.155.3	Produce losses
631.115.71	State farms	631.155.4	Produce surpluses
631.115.73	Communal farms. Municipal farms	631.155.6	Subsidies ◇ 631.155.6:633.1 Grain subsidy
631.115.75	Mixed enterprise farms	631.157	Farm business and office administration
631.115.8	Agricultural cooperatives ⇒ 334.732 ⇒ 631.173	631.158	Administration of farm work and personnel <i>Details by :005...</i>
631.115.9	Farms according to other management or operating systems. □ Contract farms. Prison farms	631.16	Farm finances and economic assessment
631.117	Instructional, research and demonstration farms	631.162	Farm financial administration. Accounting, bookkeeping etc. <i>Details by :657...</i>
631.117.2	Instructional farms. School farms. Training farms ⇒ 373.6:63 ⇒ 378.663	631.164	Agricultural estimation and valuation
		631.164.2	Principles of estimation
		631.164.23	Investment of working capital
		631.164.24	Valuation of crops (including land value)
		631.164.27	Valuation of by-products and wastes
		631.164.28	Estimation of damage
		631.164.6	Estimation and valuation of whole agricultural enterprises
		631.165	Estimation of harvests and harvest failures

- 631.17 **Agricultural and farm technology**
 ◇ 631.17:661.15 Use of agricultural chemicals. Chemization
 ⇒ 631.145
 ⇒ 631.348
 ⇒ 631.53.027.2
 ⇒ 632.934
- 631.171 Mechanization and automation of farming and agriculture
- 631.172 Use of powered machinery
 ⇒ 631.37
- 631.173 Organization of farm equipment and special services
- 631.173.2 Machine-tractor stations (MTS)
- 631.173.4 Repair and technical stations (RTS)
- 631.173.6 Services for joint use of machines
- 631.2 **Farm buildings, structures and installations. Buildings for livestock, produce, machinery and equipment**
 ⇒ 624.9
 ⇒ 628
 ⇒ 69
 ⇒ 72.059
- 631.21 **Farmsteads.** □ Groups of farm buildings. Farmyards. Outfarms. Multipurpose farm buildings
 ◇ 631.21:628.4 Sanitary installations in farm buildings
- 631.22 **Livestock buildings and structures. Rooms and fittings**
 ⇒ 636.083.1
 ⇒ 728.96
- 631.222 Stables. Similar buildings for hoofed animals
- 631.223 Sheds. Similar buildings for cloven-footed animals. □ Cowsheds. Bull pens. Goat and sheep pens. Pig sties
- 631.224 Kennels. Other buildings for beasts of prey
- 631.225 Houses and structures for rabbits and rodents
- 631.227 Houses and structures for poultry and other birds. Aviary structures
- 631.229 Houses and structures for other livestock
- 631.23 **Plant breeding and nursery structures**
 ⇒ 631.344
 ⇒ 631.544
 ⇒ 635.03
 ⇒ 728.98
- 631.234 Greenhouses (glasshouses)
- 631.236 Hotbed structures. Forcing houses
- 631.24 **Buildings for storage, handling and processing of produce.** □ Stores. Storage sheds. Granaries. Barns. Haybarns. Silos
 ⇒ 621.796.4
 ⇒ 631.56
 ⇒ 725.1
- 631.248 Pits and other stores for manure and waste.
 □ Manure pits. Manure tanks. Compost pits. Compost heaps. Waste pits
 ⇒ 631.333.92
- 631.25 **Sheds and other buildings for farm machinery and equipment.** □ Tractor sheds etc. Farm workshops. Forges. Toolsheds
- 631.27 **Farmland boundary divisions. Enclosures**
- 631.271 Hedgerows. Hedges
 ⇒ 712.41
- 631.272 Walls
 ⇒ 692.212
- 631.273 Fencing. Railings. Gates. Stiles. □ Electrified fences
 ⇒ 692.88
- 631.274 Trenches. Ditches. Dikes. Crossings over them. □ Cattle grids
- 631.3 **Agricultural machines, implements and equipment**
Details by -1/-9 (from 62-1/-8 and 66-9)
- 631.3.02 Components of farm machines and implements. Tools. Machine parts. Attachments
Further subdivided at 631.312.02..., 631.33.02..., 631.352.02..., 631.354.02..., 631.361.02...
- 631.3.05 Machines and implements with devices for measuring, gauging, sampling etc.
- 631.3.06 Combinations of machines and implements. Sets of machines etc. for use in sequence
 ◇ 631.356.2.06 Combinations of root harvesters with other machines (e.g. root crop topper + root crop lifter + root crop collector + loader and tipper)
- 631.3.07 Farm machines and implements according to the type of drive and power transmission
 ⇒ 62-8
- 631.3.071 Animal-drawn (e.g. horse-drawn, ox-drawn) implements. Hand implements
- 631.3.072 Towed implements for three-wheel, four-wheel and tracked or caterpillar tractors
- 631.3.074 Towed implements for two-wheel tractors
- 631.3.075 Equipment mounted on vehicles or craft
Denote types of vehicle by :629...
- 631.3.076 Implements with their own motive power. Self-propelled machines and implements, except tractors. Motorized devices
- 631.3.077 Cable-traction implements (rope or chain drive, winch-operated etc.)
- 631.31 **Soil cultivation machines and implements. Auxiliary machines and implements**
 ⇒ 631.51
- 631.311 General machines and implements
- 631.311.3 Post-hole diggers. Plant-hole borers
- 631.311.5 Trenchers and ditchers. Trench-digging equipment. Machines for simultaneous trenching and drain laying. Other machines for land improvement
- 631.311.7 Uprooting machines and implements (for removing roots, stumps, bushes etc.). Rooting, grubbing, stumping devices. Stone pickers
- 631.311.8 Soil-shifting, soil-mixing, soil-regenerating and soil-sterilizing appliances
- 631.312 Ploughs
- 631.312.02 Components of ploughs
- 631.312.021 Plough bodies. □ Ploughshares. Tines. Mouldboards. Furrow clearers. Plough discs
- 631.312.024 Ancillary parts of ploughs. □ Knife coulters. Disc coulters. Skim coulters. Manure-burying coulters
- 631.312.07 Ploughs according to motive power
 ◇ 631.312.076 Powered, motorized ploughs
 ◇ 631.312.077 Cable-traction ploughs
- 631.312.2 Animal-drawn ploughs with shares
- 631.312.3 Ploughing machines. Ploughs with movable tools. □ Rotary ploughs. Gyrotillers. Spading machines
- 631.312.4 Tractor ploughs with shares
- 631.312.5 Subsoiling ploughs. Deep-digging ploughs
- 631.312.6 Special ploughs
- 631.312.62 Ploughs modified for ridging, potato lifting, root-crop lifting, drawing furrows, hoeing
- 631.312.63 Ploughs with shares, for forming irrigation and drainage channels. Implements for cutting channels

631.312.68	Grubbing ploughs. Ploughs for land clearance	631.333.5	Machines for spreading pulverous (powder-form) and granular fertilizers
631.312.69	Paring or stubbing ploughs. Skim ploughs.	631.333.6	Farmyard manure spreaders. Muckspreaders. Machines for spreading animal dung, compost and peat
631.312.8	Weeding ploughs	631.333.7	Seed and fertilizer distributors in combination with supplementary feed units. Planting machines with supplementary feed units.
631.313	Disc ploughs	631.333.92	Equipment for supplying nutrients to plant roots
631.313	Harrow		Equipment for comminuting farmyard manure.
631.314	Rollers for soil treatment. □ Clod crushers.		Manure storage pits. Treatment plant for farmyard and liquid manure. Devices for preparation of compost
631.315	Scrubbers		⇒ 621.927
631.315	Hand tools for soil treatment. Weeding tools.		⇒ 628.473
631.315.1	Garden tools		⇒ 631.248
631.315.1	Digging hoes or Dutch hoes. Spud hoes	631.334	Sowing and planting machines combined with fertilizer distributors. Combine drills
631.315.2	Draw-hoes (pulling hoes). Hand cultivators.	631.335	Equipment for propagating plants by layering, grafting etc.
631.315.2	Rotary hand-harrows. Ridging tools. Digging hooks	631.336	Equipment for transplanting. □ For transplanting trees
631.315.4	Weeding tools	631.338	Equipment for plant culture other than in the ground. □ Equipment for plant culture without soil, e.g. hydroponics
631.315.6	Soil rakes. Garden rakes		
631.315.7	Spades. Shovels. Garden trowels. Digging forks	631.34	Machines and equipment for care and protection of plants
631.316	Cultivators. Machine hoes	631.342	Cutting tools. □ Shears. Clippers. Secateurs. Pruning knives. Billhooks and sickles for plant care. Axes. Hatchets. Saws
631.316.2	Cultivators (e.g. rotary cultivators)		⇒ 621.93
631.316.4	Machine hoes (for soil working in and between the rows). Machines and implements for thinning and singling plants		⇒ 621.96
631.316.5	Pasture aerators. Turf cutters		⇒ 672.71
631.316.6	Machines for uprooting or cutting weeds.	631.343	Soil treatment equipment for the care and protection of plants. □ Soil heating devices
631.317	Weeding machines		⇒ 624.143
631.317	Milling cultivators		⇒ 631.344.8
631.319	Machines and implements for soil cultivation for planting and tending plants	631.344	Devices for covering and protecting plants
631.319.2	Machines and implements for making channels, furrows and ridges (for planting, sowing, manuring). Implements for drawing furrows		⇒ 631.23
631.319.3	Machines and implements for making holes for planting, and covering them. Dibbling implements (for planting and care of row crops)		⇒ 631.544
631.319.4	Mulching machines	631.344.1	Cloches
	⇒ 631.544.7	631.344.2	Cold frames
631.33	Machines and implements for sowing and planting. Fertilizer distributors	631.344.3	Hotbeds. Heated frames. Forcing beds
631.33.02	Components. Details	631.344.4	Cold greenhouses (glasshouses)
631.33.021	Containers. □ Seed boxes for drills. Hoppers for fertilizer distributors	631.344.5	Heated greenhouses (glasshouses).
631.33.022	Feed and distributor mechanisms		Hothouses. Forcing houses
631.33.024	Devices for introducing the seed into the soil. □ Coulters. Seed tubes. Press wheels	631.344.7	Shading devices. Mats
631.331	Machines for sowing	631.344.8	Heating, ventilating and watering equipment for forcing houses, glasshouses etc.
631.331.1	Seed broadcasters		Details by :621.647... and :628.8...
631.331.4	Ring rollers for making furrows	631.345	Supports for plants. □ Sticks, canes, stakes.
631.331.5	Seed drills. □ Band sowing (strip sowing) machines. Machines for drilling in rows. Cluster and group sowing machines	631.346	Wires and cords. Training and tying devices
631.331.92	Equipment for seed preparation	631.347	Pots, boxes etc.
631.331.93	Machines for producing seed tapes	631.347.1	Watering appliances. Irrigation equipment
631.332	Planting machines. Machines and equipment for planting-out. Devices for preparing small plants and seedlings. Machines for making soil blocks	631.347.2	Pipelines. Hoses. Vehicles for transporting hoses. Hose reels
631.332.5	Machines for pricking-out seedlings, planting pot plants etc.	631.347.3	Watering appliances. Watering cans
631.332.7	Potato planters	631.347.4	Spraying equipment. Sprinklers. Sprays, spray guns, jets
631.332.8	Special planting machines and equipment	631.347.8	Rotary sprinklers
631.332.81	Machines for planting in furrows	631.348	Devices for watering roots or supplying nutrients. Root-feeding devices
631.332.87	Planting sticks		Machines and equipment for plant protection.
631.333	Fertilizer distributors. Manure spreaders. Machines and equipment for fertilizer preparation and distribution		□ Pest deterrents, scarecrows, bird-scarers etc. Traps, snares. Chemical pest and disease control. Fumigators. Smoke generators.
	⇒ 631.816.3		Herbicide and pesticide sprays. Weed-killing equipment. Flame guns
631.333.4	Equipment for spreading liquid manure. Manure tanks, pumps, sprays. Slurry tankers	631.35	Specify by :632.93...
			Harvesting equipment. Harvesters
			⇒ 631.55

631.351	Hand tools for harvesting. Simple harvesting tools. □ Scythes and sickles for harvesting. Pitchforks. Hay and straw forks	631.361.3	Winnowers
631.352	Mowing machines in general ⇒ 631.354	631.361.4	Clover hullers. Pod removers, shellers
631.352.02	Mower components	631.361.5	Machines for seed or grain polishing etc.
631.352.022	Cutting mechanisms	631.361.7	Machines and equipment for preparing fruits
631.352.027	Swathing attachments and collecting devices	631.361.72	Machines for splitting and separating flesh from kernels. For seed removal. For stoning fruit. For husk removal. For shelling nuts etc.
631.352.92	Mowing machines for plants growing in water. Reed cutters. Reed mowers. Rush cutters	631.361.8	Machines and equipment for cutting vegetables and fruit. □ Chopping machines. Dicing machines. Comminuting machines. Grinders, crushers, squeezers. Juice extractors ⇒ 621.927 ⇒ 664.7
631.352.93	Grass mowers. Lawnmowers. Lawn edgers	631.361.9	Machines for removing the coverings from vegetables and fruit. □ Peeling machines. Machines for removing stalks
631.352.94	Mowing machines for operating in ditches and on banks	631.362	Machines and equipment for cleaning and sorting produce. Containers, sieves etc. ⇒ 621.928 ⇒ 66.047 ⇒ 66.067
631.352.95	Hedge-trimming machines	631.362.1	Simple devices. Shovels. Sieves. Tanks
631.353	Haymaking machines (except mowers)	631.362.3	Cleaning and sorting machines. □ Blowers, centrifugal separators etc. Sorters and graders
631.353.2	Hay rakes. Hay turners. Hay sweeps. Hand rakes. Horse rakes. Buckrakes. Reel-type rakes. Reel and fork-type tedders. Combined rakes and turners. Side rakes and swath turners	631.362.5	Machines for removing the covering of seed crops (hull, husk, chaff removal etc.). Machines for picking and processing hops
631.353.3	Hay loaders. Hay sweeps with tying device. Pick-up balers ⇒ 631.364	631.362.6	Machines for washing produce. Centrifugal machines. Machines for removing chemical residues (e.g. pesticides from fruits) ⇒ 632.95.028
631.354	Machines for cutting cereal crops. Binders. Combine harvesters	631.363	Machines and equipment for preparing feeding stuffs or fodder
631.354.02	Machine components	631.363.1	Ensilage, fermentation equipment. Silos (for fodder)
631.354.022	Cutting mechanisms. Fingers. Finger bars. Knives. Knife sections	631.363.2	Comminuting machines. Grinders, crushers, squeezers (for fodder). Fodder chopping machines. Machines for pressing, compressing, cubing etc. Food mashers ⇒ 621.927 ⇒ 664.7
631.354.024	Binding devices. Packers. Needles. Knotters	631.363.5	Equipment for heat treatment of fodder. □ Steamers for fodder
631.354.025	Binder conveyors. Canvas conveyors. Chain conveyors. Loaders. Grain collectors, bins	631.364	Equipment for bundling, trussing, tying etc. Bagging machines. Packing machines
631.354.026	Threshing mechanisms	631.365	Equipment for drying, airing, ventilation of produce. Driers. Crop ventilators ⇒ 664.723
631.354.028	Crop lifters (fingers). Crop dividers. Sheaf counters. Devices for animal protection	631.367	Equipment and gear for covering clamps, containers etc. for the storage of agricultural products
631.354.1	Binders	631.37	Machines and equipment for providing power and traction. Agricultural transport equipment ⇒ 631.565
631.354.2	Combine harvesters	631.371	Machines and equipment for providing motive power, drive ⇒ 62-8
631.354.3	Pickers. Headers. Machines for collecting the heads of clover crops etc.	631.372	Powered vehicles. □ Tractors <i>Details by :629...</i> ⇒ 629.366
631.354.4	Sheaf binders	631.373	Nonpowered vehicles. Farm carts. Trailers <i>Details by :629...</i>
631.355	Maize harvesters. □ Pickers. Shellers. Huskers	631.374	Conveying equipment. Loading and unloading gear (other than structural fixtures) ⇒ 621.86/.87
631.356	Machines and implements for gathering row crops	631.4	Soil science. Pedology. Soil research ⇒ 622.011.4 ⇒ 624.131.4 ⇒ 626.8
631.356.2	Root harvesters. □ Lifting devices. Topping devices (e.g. beet toppers) ⇒ 631.3.06		
631.356.4	Potato harvesters		
631.358	Harvesters for other crops <i>Denote particular crops by :633/635</i>		
631.359	Equipment and machines for measuring crop yield		
631.36	Machines and equipment for the preparation and storage of crops or produce. Machines and equipment for the preparation of feeding stuffs ⇒ 631.56 ⇒ 664.7		
631.361	Machines and equipment for separating or removing parts of crops		
631.361.02	Components		
631.361.022	Threshing devices, drums, sieves etc.		
631.361.025	Cleaning devices, shakers etc.		
631.361.026	Grain binders. Reaper-binders. Baling presses		
631.361.027	Straw conveyors		
631.361.1	Hand tools for binding sheaves, cutting chaff, threshing. □ Flails		
631.361.2	Threshing machines (threshers)		

631.41	Chemical and physicochemical soil properties ◇ 631.41:631.811 Soil properties as regards plant nutrition. Nutrient content of soil	631.431.7	Compaction
631.412	Physicochemical properties of soil in general	631.432	Groundwater. Agricultural hydrology ⇒ 556.3 ⇒ 626.8 ⇒ 628.1
631.413	Electrochemical properties of soil. Ion exchange	631.432.1	Water table. Phreatic surface
631.413.2	Fixation	631.432.2	Soil moisture ⇒ 556.142 ⇒ 631.423.2
631.413.3	Salt accumulation. Salt content	631.432.3	Permeability. Lessivage. Leaching. Mobility of soil constituents
631.413.4	Cation exchange	631.433	Soil atmosphere
631.414	Colloidal properties of soil	631.433.1	Soil aeration
631.414.2	Colloidal state. Hardening. Loosening. Impenetrable layers (hardpan)	631.433.2	Conditions of anaerobiosis. Soil reduction
631.414.3	Absorption. Adsorption	631.433.3	Soil respiration
631.415	Soil reaction. Soil acidity and alkalinity. Hydrogen ion concentration in soil (pH value) ⇒ 543.241 ⇒ 543.554.2	631.433.5	Composition of soil atmosphere
631.415.2	Acidity. Acid soils	631.434	Soil structure. □ Soil porosity
631.415.3	Alkalinity. Alkaline soils	631.435	Granulometric composition of soil. Soil texture
631.415.7	Wild plants as indicators of soil reaction	631.436	Thermal properties of soil
631.416	Inorganic chemistry of soil. Nature of various soil components	631.436.1	Transfer of heat
631.416.1	Nitrogen	631.436.2	Specific heat
631.416.2	Phosphoric acid	631.436.5	Effect of heat
631.416.3	Other nonmetals and their compounds 631.416.3 ≡ 546	631.436.6	Effect of cold, frost ⇒ 551.34
631.416.4	Potassium	631.437	Electrical and magnetic properties of soil <i>Details by :537...</i>
631.416.5	Sodium	631.438	Radioactivity of soil
631.416.7	Calcium	631.44	Classification of soils <i>Physiographical and geological details by :55...</i>
631.416.8	Other metals and their compounds 631.416.8 ≡ 546		◇ 631.44:551.432 Mountain soils ◇ 631.44:552.524 Loess soils
631.417	Organic chemistry of soil	631.44.06	Catenas. Soil associations
631.417.1	Carbon content	631.442	Traditional classification of soils (mainly according to texture and chemical composition) <i>Mixed soils may be denoted by combining the following subdivisions by means of the apostrophe</i>
631.417.2	Humus content		Sandy soil (arenaceous soil) ◇ 631.442.1'2 Silico-calcareous soil. Calcareous sands
631.417.4	Carbon:nitrogen ratio		Lime-bearing soil (calcareous soil)
631.417.7	Other organic compounds 631.417.7 ≡ 547		Silty soil. Loam
631.417.8	Organometallic compounds. Chelates		Clay soil (argillaceous soil) ◇ 631.442.4'1 Clay-sand soil (argillaceous sandy soil). Allitic soil. Sandy clay ◇ 631.442.4'2 Clay-lime soil (calcareous clay). Marl
631.418	Soil solution	631.442.1	Organic soils
631.42	Soil analysis and experiments	631.442.2	Stony soils
631.421	Field and pot experiments. Statistical methods. Lysimetry	631.442.3	Genetic classification of soils
631.422	Qualitative soil analysis 631.422 ≡ 631.423	631.442.4	Hydromorphic soils. □ Tundra soils. Moorland soils. Bog soils. Peats. Gleys. Vlei soils
631.423	Quantitative soil analysis ⇒ 54.062 ⇒ 543.06		Podzolic soils. Podzols
631.423.2	Water content. Moisture content ⇒ 631.432.2		Brown soils. Brown earths
631.423.5	Dissolved salts		Chernozemic soils. Chernozem (black earth).
631.423.6	Composition of soil air		Arid soils. Desert soils
631.423.7	Determination of exchangeable bases		Mediterranean soils. □ Krasnozems (red soils). Terra rossa
631.425	Physical soil analysis 631.425 ≡ 631.43		Tropical and subtropical soils. □ Tropical red loams. Subtropical black soils (tirs, regur, grumusol)
631.427	Biological soil analysis ⇒ 631.46		Rendzinas. Calcareous brown soils
631.427.1	Macrobiological analysis. □ Indicator plants		Other soil types. □ Marine soils. Skeletal soils (lithosol, regosol, neosol, presol)
631.427.2	Microbiological analysis. Examination of microflora and microfauna		
631.427.3	Analysis methods using living plants. □ Neubauer technique	631.445	Soil fertility and toxicity ⇒ 632.12
631.427.4	Analysis by means of microbes. Bioassay	631.452	Soil fertility
631.43	Physical and mechanical properties of soil	631.453	Soil toxicity
631.431	Mechanical properties of soil		
631.431.1	Density		
631.431.2	Deformation. Shear		
631.431.3	Plough draught. Resistance to ploughing		
631.431.4	Cohesion. Compressibility		
631.431.5	Plasticity. Viscosity. Sticky point		
631.431.6	Swelling and shrinking. Cracking		

631.454	Manurial requirement of soil ⇒ 631.8	631.52	Improvement of plant strains
631.458	Impoverishment of soil. Soil exhaustion ⇒ 631.584.9	631.522	Biology of plant reproduction 631.522 ≡ 581.16
631.459	Erosion ⇒ 551.3.053 ⇒ 632.125	631.524	Variation of characteristics ⇒ 581.15
631.46	Soil bacteriology. Soil biology ⇒ 631.427	631.526	Groups and types of cultivated plant
631.461	Zymotic microbes of the soil. Soil bacteria. Soil microbiology ⇒ 579.64 ⇒ 579.8 ⇒ 582.231	631.526.1	Genus
631.461.1	Ammonifying bacteria. Ureolytic bacteria. Proteolysis. Ammonification	631.526.2	Species and subspecies
631.461.2	Nitrite-forming bacteria	631.526.3	Variety
631.461.3	Nitrate-forming bacteria. Nitrification	631.526.4	Genetic strain
631.461.4	Denitrification	631.527	Improvement by selection (natural and artificial selection). Selective plant breeding
631.461.5	Nitrogen-fixing bacteria. Nitrogen fixation	631.527.2	Mass selection. Rapid methods of selection
631.461.6	Decomposition of non-nitrogenous compounds. □ Decomposition of cellulose, other carbohydrates	631.527.3	Pedigree selection
631.461.7	Conversion of inorganic substances, minerals. □ Sulphur bacteria. Iron bacteria. Phosphate bacteria	631.527.5	Hybrids, inbreeding and heterosis
631.462	Soil sterilization, disinfection	631.527.6	Clonal selection
631.463	Antibiotics. Microbial antagonism	631.527.7	Selection of polyploid species
631.464	Vitamins in soil	631.527.8	Selection methods
631.465	Enzymes in soil	631.527.81	Emasculation
631.466	Other microflora. □ Fungi. Mycorrhiza. Yeasts. Rod-shaped bacteria (Actinomycetales). Algae	631.527.82	Pollination
631.467	Microfauna. □ Protozoa. Nematodes	631.528	Improvement by modification and mutation (natural and induced). Applied genetics ⇒ 575
631.468	Macrofauna and mesofauna. Animals in the soil ⇒ 59	631.528.1	Mutation of genes and chromosomes. Mutagenesis
631.47	Surveying, classification and utilization of land	631.528.2	Polyploidization
631.472	Soil profiles. Soil profile development	631.528.6	Production of mutations. Genetic modification ◇ 631.528.6:633 Genetic modification of field crops ◇ 631.528.6:633.34 Genetic modification of soya
631.472.5	Litter. Organic horizons	631.529	Acclimatization
631.472.6	Mineral soil in general. A-C horizons	631.53	Plant propagation. Vegetative propagation ⇒ 581.165
631.472.7	Soil layers. □ Topsoil. Subsoil	631.53.01	Seed
631.472.8	Micropedology	631.53.02	Seed production. Collection, storage and treatment of seeds
631.473	Soil surveys	631.53.027	Seed treatment
631.48	Pedogenesis. Soil formation ⇒ 551.435.1 ⇒ 551.435.2	631.53.027.2	Chemical methods
631.482	Soil formation by physical and mechanical processes. □ Alluvial soils. Diluvial soils ⇒ 551.311.234	631.53.027.3	Physical methods
631.483	Soil formation by chemical means	631.53.027.32	Using temperature. □ Vernalization
631.484	Soil formation by biological means. □ Plant, bacterial, animal action	631.53.027.33	Using electricity
631.485	Transported and displaced soil ⇒ 551.311.24	631.53.027.34	Using radiation
631.5	Agricultural operations	631.53.03	Cultivation of young plants, seedlings. □ Pricking out. Nurseries ⇒ 635.03
631.51	Soil preparation and ploughing ⇒ 631.31	631.53.04	Sowing. Planting. Transplanting
631.51.02	Time of working on the soil	631.53.041	Direct sowing, planting
631.51.021	Basic tillage	631.53.042	Broadcasting
631.51.022	Work before sowing or planting	631.53.048	Sowing rate. Quantity of seed
631.51.023	Work during growth	631.531	Seeds and spores ⇒ 631.576.3
631.511	Soil preparation without mechanical devices. □ With spade, hoe, rake	631.532/535	Vegetative propagation
631.512	Soil preparation with mechanical tools. □ With animal or motor power	631.532	Propagation by separation of part of a growing plant ⇒ 581.165.7
631.513	Soil preparation with explosives	631.532.1	Using bulbs
		631.532.2	Using tubers. Division of tubers into parts with one eye each
		631.532.3	Division or separation of clumps
		631.533	Detachment of separate plants
		631.533.1	Suckers from the collar or stump
		631.533.2	Stolons
		631.533.3	Terminal and proliferous buds. □ Plantlets. Bulbils
		631.534	Propagation by layering
		631.535	Propagation by cutting. Cuttings
		631.538	Renovation of orchards and vineyards etc. Rejuvenation

631.54	Tending and care of plants	631.55.03	Special harvesting methods
⇒ 635.04		631.55.032	Harvesting before maturity. Harvesting unripe
631.541	Grafting	631.55.034	Multiple harvesting. First, second crop etc.
631.541.1	Choice of stock (recipient plant) and scion or graft. Compatibility	631.55.035	Gleaning
631.541.3	Grafting methods. □ Cleft, splice, saddle and side grafting. Inlaid crown grafting. Herbaceous grafting	631.552	Cutting. Mowing. Scything
631.541.4	Grafting according to the position of the graft on the stock. □ Grafting on the root, crown, stem, apex	631.553	Haymaking. Haycock and rick building
631.541.5	Shield grafting (with shield-buds)	631.554	Harvesting, reaping of grain crops
631.541.6	Grafting by approach (inarching)	631.555	Maize harvesting
631.541.7	Tying-in and protection of the graft	631.558	Harvesting of other crops
631.541.8	Internal grafting		Specify by :633/635
631.542	Cutting back. Pruning	631.559	Harvest yield
⇒ 630*245		631.56	Preparation of produce. Treatment after harvesting
631.542.1	Winter pruning	631.561	Threshing. Separation of fruit from husk.
631.542.2	Summer pruning		Hulling, shelling, paring etc.
631.542.3	Cutting, trimming of plants for particular purposes. □ Cutting for growth. Removal of diseased parts	631.562	631.561 ≡ 631.361
631.542.32	Cutting for shaping		Cleaning and sorting of produce. Winnowing.
◇ 631.542.32:712.41	Shaping of trees, bushes, shrubs. Including: topiary	631.562	Sieving. Peeling. Washing etc.
631.543	Planting	631.563	631.562 ≡ 631.362
⇒ 631.53.04			Preparation and storage of produce
631.543.1	General questions. □ Positioning, situation of plant. Effect of light and shade	631.563.2	⇒ 631.24
631.543.2	Density of planting. Spacing	631.563.3	Drying
631.543.3	Planting methods		⇒ 66.04
631.543.5	Removal of surplus plants. Clearing. Thinning	631.563.4	Reducing in size. Comminuting
631.543.8	Kinds of planting. Site in general. Spacing (of rows; of plants). Planting for shade, as shelter. Care and protection of plantations	631.563.6	⇒ 621.926/927
⇒ 630*26		631.563.8	Mixing
⇒ 630*8			⇒ 621.929
⇒ 631.544.7		631.563.9	Fermenting
631.544	Protection and shelter of plants. Cultivation under cover		⇒ 577.151
⇒ 631.23		631.564	Preparing and preserving for storage
⇒ 631.344		631.565	⇒ 664
631.544.1	Cultivation under cloches		Storage. □ In silos
631.544.2	Hotbed cultivation. Cultivation in hot and cold frames		⇒ 621.796
631.544.4	Greenhouse (glasshouse) cultivation		Packing
◇ 631.544.4-977	Hothouse cultivation		⇒ 621.798
631.544.7	Cultivation under various coverings. Protection and covering of soil. □ Transparent sheeting. Mulching and mulches. Shelter plants. Windbreaks		⇒ 631.364
⇒ 626.846			Transport
⇒ 631.319.4			⇒ 629
⇒ 631.543.8			⇒ 631.37
631.546	Natural and artificial forms, shapes. □ Staked, vertical, erect forms. Espalier forms. Trellised forms. Wall training. Arbours, bowers. Pergolas. Horizontal cordon training	631.57	⇒ 656
631.547	Phases and control of plant growth		Products, by-products and semifinished products of agriculture
631.547.1	Germination. Seedlings	631.571	⇒ 630*8
631.547.2	Growth	631.572	⇒ 635.07
631.547.3	Full development	631.573	Wood. Bark
631.547.4	Flowering. Blossoming. □ Artificial stimulation of flowering	631.574	Stems. Fibres. Straw
⇒ 631.811.98		631.575	Roots. Tubers. Bulbs
631.547.5	Fruiting. Fructification	631.576	Leaves. Foliage
631.547.6	Ripening	631.576.2	Flowers. Blossoms
631.547.66	Maturing	631.576.3	Fruits. Parts of fruit
631.547.67	Artificial ripening		Whole fruits and nuts
631.55	Gathering. Harvesting. Crop		Seeds as products. Cereals. Nuts without shells
⇒ 631.35		631.576.4	⇒ 631.531
		631.577	Fruit skins, peel, rind. Husk. Fibres. Bran
		631.58	Plant juices, saps, gums, essences etc.
			Particular cultivation systems
		631.581	⇒ 631.153
		631.582	Fallowing
		631.582.1	Crop rotation
		631.582.2	Monoculture
		631.582.3	Shifting cultivation. Land rotation
		631.582.9	Three-course rotations
		631.583	Other systems. □ Rough grazing
		631.584	Intensive cultivation
			Intercropping. Catch cropping. Exhaustive cropping
		631.584.2	Catch crop system
		631.584.4	Cover crop system
		631.584.5	Mixed cropping. Intercropping

631.584.9	Exhaustive cropping	631.81.036	Biodynamic processes
631.585	Pastoral system. Transhumance	631.81.095.337	Requirement for certain elements. Essential micronutrients. Trace elements
631.586	Dry farming. Use of deeper soil levels	631.81.095.338	Lack of certain elements. Mineral deficiency
631.587	Cultivation on irrigated land	631.811	Plant nutrition
631.588	Cultivation using special phenomena. □ Use of electricity, radio and acoustic waves, artificial light, heat (soil warming), fire ⇒ 630*43	631.811.98	Plant stimulants (for artificial promotion of growth without manuring). Plant growth regulators, hormones ⇒ 581.192 ⇒ 631.547.4
631.589	Cultivation in artificial media	631.812	Production, preparation, storage etc. of fertilizers. Constitution and form of fertilizers <i>Denote state of aggregation by -403/-405 and -492... from 62</i>
631.589.2	Hydroponics	631.813	Composition, reaction (pH value), toxicity and reciprocal reactions of fertilizers ⇒ 543
631.589.3	Sand cultures. Other inert media	631.814	Loss of fertilizer principles. Changes in manurial value
631.589.5	Plastics media	631.815	Economic aspects
631.6	Rural engineering ⇒ 556.18 ⇒ 626.8 ⇒ 627.5 ⇒ 628.1 ⇒ 630*237 ⇒ 630*38	631.816	Application and distribution of fertilizers
631.6.02	Soil and water conservation. Erosion control	631.816.1	Doses. Rates of application
631.6.03	Water: composition and properties	631.816.2	Time of application (e.g. before, at, after sowing) <i>Denote specific seasons or months by "32..." from Table 1g</i>
631.61	Clearing and exploitation of particular terrains ⇒ 626.87	631.816.3	Method of application. □ Broadcasting. Placement. Injection. Spraying (in aqueous solution). Dusting (aerial application) ⇒ 631.333
631.611	Reclamation of marginal land in general, grassland, heaths etc.	631.82	Mineral amendments. Artificial fertilizers in general
631.612	Reclamation of stony and rocky land. Reclamation of dunes	631.821	Lime. Chalk. Gypsum ⇒ 666.92
631.613	Reclamation of sloping land. □ Terracing	631.822	Marl
631.614	Reclamation of virgin forest and bush	631.823	Polyelectrolytes. Soil conditioners
631.615	Reclamation of swamps and moorland	631.824	Magnesium
631.616	Reclamation of land below sea level. Empoldering ⇒ 627.53	631.825	Iron sulphate
631.617	Reclamation of deserts and arid lands	631.826	Silt from inundations. Alluvium ⇒ 627.533.1
631.618	Reclamation of industrial and mined land	631.828	Other amendments. □ Salt. Iodine. Copper
631.62	Drainage ⇒ 626.86 ⇒ 627.532 ⇒ 628.2 ⇒ 631.432 ⇒ 632.123	631.829	Combustion gases. Carbon dioxide
631.67	Irrigation ⇒ 626.81/.84 ⇒ 628.1	631.83	Potassium fertilizers
631.671	Water requirement	631.831	Wood ash. Plant ash. Peat ash ⇒ 674.88
631.672	Water supply - 631.672 ≡ 628.1	631.832	Chlorides
631.674	Irrigation methods	631.833	Sulphates
631.674.1	Surface irrigation	631.834	Crude potash salts
631.674.2	Flood irrigation	631.835	Sylvinite
631.674.3	Stream diversion	631.836	Silicates
631.674.4	Subsurface irrigation	631.837	Carbonates
631.674.5	Sprinkler irrigation	631.84	Nitrogen fertilizers
631.674.6	Trickle irrigation. Drip irrigation	631.841	Ammonium fertilizers. Amides
631.675	Rate and time of irrigation. Irrigation parameters	631.842	Nitrate fertilizers. Saltpetres (potassium and sodium nitrates) etc.
631.8	Fertilizers, manures and soil enrichment. Promotion of growth. Plant stimulants ⇒ 631.454 ⇒ 661.152	631.847	Vegetable nitrogen fertilizers. Bacterial and mycorrhizal inoculation of soil
631.8.022.3	Additives 631.8.022.3 ≡ 66.022.3	631.85	Phosphate fertilizers
631.81	Generalities. Plant nutrients. Production and use of fertilizers, manures	631.851	Mineral phosphates. Raw phosphates
631.81.031	Absorption, uptake of nutrients	631.852	Bone meal and similar products
631.81.033	Conversion and distribution in the plant	631.853	Basic slag. Thomas slag ⇒ 669.184.28
		631.854	Guano (bird and bat droppings). Fish guano (fish meal) ⇒ 664.957
		631.855	Superphosphate ⇒ 631.89

631.857	Electrophosphates. Condensed phosphates	631.963.3	Artificial stands in general
631.858	Silicophosphates. Rhenania phosphates		
631.859	Other phosphate fertilizers. □ Sodium, potassium, ammonium, calcium, magnesium phosphates	632	PLANT DAMAGE, INJURIES. PLANT DISEASES. PESTS, ORGANISMS INJURIOUS TO PLANTS. PLANT PROTECTION
			⇒ 368.51
631.86	Organic manures in general. Fertilizers of biological origin		⇒ 576.8
631.861	Manures of animal origin in general		⇒ 630*4
631.862	Farmyard manure. Animal dung. Stable muck (including sullage, liquid manure)	632.03	Determination and assessment of damage
631.863	Human excrement, sewage	632.038	Extent of damage
	⇒ 626.841		⇒ 368.025.76
	⇒ 628.381		
	⇒ 631.879.2	632.08	Biological techniques. Experimental methods etc.
631.867	Other fertilizers of animal origin. □ Fish waste. Blood. Slaughterhouse offal		632.08 ≡ 57.08
631.87	Other organic manures	632.1	Nonparasitic complaints. Plant diseases of complex origin
631.871	Manures of plant origin in general		
631.872	Vegetable detritus. □ Leaves. Forest litter. Leafmould. Humus. Straw	632.11	Unfavourable meteorological or climatic conditions. Damage by weather
631.873	Vegetable material of aquatic origin		⇒ 551.5
631.874	Green manuring. Soil-improving crops		⇒ 632.931.2
631.875	Manuring with cut plants	632.111	Extremes of temperature. Cold. Heat
631.877	Manuring with oil-seed cakes. □ With groundnut cake	632.111.5	Frost
631.878	Peat, lignite, coal as fertilizers	632.111.6	Cold. Low temperatures
	⇒ 674.88	632.111.7	Freezing of soil
631.879	Other wastes as fertilizers	632.111.8	Heat. High temperatures
	⇒ 628.477.3	632.112	Drought
631.879.1	Urban refuse, garbage as fertilizer		⇒ 57.045:55...
631.879.2	Sewage as fertilizer		⇒ 631.671
	⇒ 626.841	632.115	Light
	⇒ 628.381	632.115.1	Excessive light
	⇒ 631.863	632.115.2	Lack of light. Shade
631.879.3	Industrial wastes as fertilizers	632.115.3	Sun scorch. Sun scald
	◇ 631.879.3:663.48 Brewery wastes as fertilizers	632.116	Precipitation and condensation
	⇒ 628.54	632.116.1	Rain
631.879.4	Compost	632.116.2	Snow. Avalanches
	⇒ 628.473	632.116.3	Hail
631.89	Compound fertilizers. Mixed manures	632.116.4	Fog. Mist
631.893	Purely inorganic fertilizing mixtures	632.116.5	Dew
631.894	Purely organic fertilizing mixtures (mixed manures)	632.116.6	Rime. Hoar-frost. Glazed frost
631.895	Mixtures of organic and inorganic fertilizers	632.117	Wind
631.92	Geophysical effects of cultivation. Influence of agriculture on climate, soil, watercourses, erosion, landslips etc.	632.117.1	Storms. Cyclones
		632.117.2	Constant wind
631.95	Agrarian ecology	632.118	Electrical and radioactive phenomena
	<i>Class here only ecological studies from an agricultural point of view</i>	632.118.1	Lightning
	⇒ 581.5	632.118.2	Electricity. Earth currents. Electric fields
	⇒ 591.5	632.118.3	⇒ 550.37
631.96	Types of plantation. Appraisal of site		Radioactivity
	⇒ 630*11	632.12	Unfavourable soil conditions
	⇒ 634.8.047		⇒ 631.45
	⇒ 635.04	632.121	Nutrient deficiency. Chlorosis
631.961	Plantations in relation to site		632.121 ≡ 546
631.962	Kinds of stand and of woods	632.122	Injurious or toxic constituents
	⇒ 630*22	632.122.1	Inorganic injurious substances. Toxic minerals
631.962.2	Density		632.122.1 ≡ 546
631.962.3	Pure stands	632.122.2	Organic injurious substances
631.962.4	Mixed stands		632.122.2 ≡ 547
631.962.5	Even-aged stands	632.123	Humidity. □ Soil moisture. Floods. Waterlogging
631.962.6	Uneven-aged stands		⇒ 631.432
631.963	Origin and development of stands etc. Plant migration. Invasion and naturalization of plant species	632.124	⇒ 631.62
		632.125	Unfavourable physical soil conditions
631.963.2	Natural stands in general		⇒ 631.43
		632.126	Soil degradation. □ Erosion. Landslides
			⇒ 551.3.053
			⇒ 631.459
			Seismic disturbances. Earthquakes. Earth tremors
			⇒ 550.34

- 632.15 Atmospheric impurities. Chemical or industrial pollution**
⇒ 628.395
⇒ 628.5
- 632.151 Smoke. Smog. Gases
- 632.152 Airborne solid particles, dusts. □ Volcanic ash
- 632.153 Waste waters. Seawater. Irrigation water etc.
- 632.154 Injuries due to plant protection preparations, pesticides etc.
⇒ 632.95.024
- 632.155 Injuries due to fertilizers
- 632.16 Various physiological disorders or complex diseases**
- 632.161 Degeneration
⇒ 581.148
- 632.163 Dying back. Withering
- 632.164 Abortion or failure of crop. □ Washing away of pollen by rain
- 632.165 Lodging or laying of crop. □ Beating down by wind
- 632.166 Premature fall of leaves, flowers, fruits
- 632.167 Physiological diseases during storage
- 632.168 Teratology. Malformations
⇒ 581.2
- 632.18 Other nonparasitic complaints**
- 632.184 Mechanical injuries. Wounds
- 632.187 Fire damage. Damage by bombs
- 632.2 Symptomatology of plants**
Only general matters are classed here. If the causal agent is known, class injuries under 632.1 and 632.3/.8
- 632.21 Galls. Cecidia
- 632.23 Tumours. Canker
- 632.24 Wilts. Blights
- 632.25 Rots. Moulds. Mildews. Damping-off
- 632.26 Leaf spots. Discolorations. Necrosis
- 632.27 Witches' broom (tufted growths)
- 632.3 Bacterial and virus diseases of plants**
632.3.01/.08 ≡ 57.01/.08
⇒ 578
⇒ 579
- 632.35 Bacterial diseases of plants**
Subdivide alphabetically by name of genus and species of bacterium. See Table 1h
- 632.38 Virus diseases of plants**
Subdivide alphabetically by name of host plant and/or name of virus. See Table 1h
- 632.4 Fungal, algal diseases of plants. Plant mycoses etc.**
632.4.01/.08 ≡ 57.01/.08
632.43/.49 ≡ 582.23/.29 and/or subdivide alphabetically by name of genus and species of fungus etc. See Table 1h
- 632.5 Injurious plants (i.e. injurious to other plants)**
632.5.01/.08 ≡ 57.01/.08
Subdivide 632.51/.53 alphabetically by name of genus and species of plant (See Table 1h)
- 632.51 Weeds**
- 632.52 Other nonparasitic plants.** □ Scrub. Brush. Intermediate hosts of parasites (plants harbouring pests). Poisonous plants
⇒ 615.918
- 632.53 Parasitic plants. Epiphytes. Climbing plants**
- 632.6 Animals injurious to plants (except Insecta)**
632.6.04/.05 ≡ 591.04/.05
632.6.08 ≡ 57.08
632.62/.69 ≡ 592/599 and/or subdivide alphabetically by name of genus or species of animal. See Table 1h
◇ 632.69 *Microtus agrestis* The field vole
Microtus agrestis as injurious to plants, or
◇ 632.693.234.3 *Muridae* (mice, rats, voles etc.) as injurious to plants
⇒ 591.65/.69
- 632.7 Insects injurious to plants**
632.7.04/.05 ≡ 591.04/.05
632.7.08 ≡ 57.08
632.71/.79 ≡ 595.71/.79 and/or subdivide alphabetically by name of genus or species of insect. See Table 1h
⇒ 591.65/.69
- 632.727 Common grasshoppers as injurious to plants
- 632.752 Plant-lice, aphids, greenfly etc. as injurious to plants
- 632.76 Beetles**
◇ 632.76 *LEP* The Colorado beetle
Leptinotarsa decemlineata as injurious to plants, or
- 632.8 Other injuries and diseases of plants. Complaints due to unknown causes**
- 632.9 Control of plant diseases and pests**
- 632.91 Generalities**
Experimental methods
Plant protection services
Plant protection measures. □ Legislation
Import. Export. Plant quarantine
Inspection methods
Forecasting. Prognosis
- 632.92 Invasion and spread of plant diseases and pests**
- 632.93 Methods of control of plant diseases, pests, damage**
◇ 632.93:631.462 Soil disinfection for disease and pest control
◇ 632.93:631.53.02 Seed treatment for disease and pest control
- 632.931 Methods of cultivation. Modification of climatological and meteorological factors. Destruction of infected plants and vectors
Cultivation methods for disease and pest control etc.
Details by :631.5/.9
◇ 632.931.1:631.51 Soil preparation
Action on meteorological or climatic conditions. Modification of weather
632.931.2 ≡ 632.11
- 632.931.21 Modification of temperature (heat, cold)
- 632.931.4 Destruction of infected plants and of vectors, carriers
- 632.934 Chemical methods of disease and pest control
⇒ 631.348
⇒ 632.982
- 632.934.1 Spraying, dusting, atomizing of pesticides
- 632.934.2 Fumigation
- 632.934.3 Systemic treatment. Endotherapy
- 632.934.4 Poison baits
- 632.935 Physical and mechanical methods of disease and pest control etc.

632.935.1	Control of temperature (raising, lowering) for disease and pest control <i>For control of temperature as injurious factor in itself, use 632.931.21</i>	632.954 632.955	Herbicides. Weedkillers Preparations for destruction of worms. Vermicides. Nematicides
632.935.2	Use of water for disease and pest control. □ Hydration. Dehydration. Dipping	632.958	Preparations for destruction of vertebrate pests. □ Reptile, bird, fish, mammal poisons. Rodenticides etc.
632.935.3	Use of pressure and vacuum for disease and pest control	632.98	Implementation of treatment
632.935.4	Use of radiations, acoustic waves etc. for disease and pest control	632.981	Kinds of pest-control preparation ⇒ 62-403/-405 ⇒ 62-492
632.935.7	Mechanical methods of disease and pest control. □ Physical collection and destruction of pests. Protection of crops with barriers, guards, screens. Packing. Wrapping. Hermetic storage	632.981.1 632.981.2 632.981.3	Sprays. Suspensions. Solutions. Emulsions Dusts. Powders. Granular preparations Adjuvants. Auxiliary ingredients. □ Wetting agents. Adhesives. Emulsifiers. Diluents. Bulking agents
632.936	Psychological and sensory pest control ⇒ 631.348	632.981.4 632.981.5 632.982	Synergists Compatibility Methods of application. Apparatus ⇒ 631.347 ⇒ 631.348 ⇒ 632.934
632.936.1	Light traps	632.982.1 632.982.2	Sprinkling. Spraying Mist application. Nebulization. Atomization. Aerosols
632.936.2	Attractants. Baits	632.982.3	Dry spreading. Broadcasting. Dusting, e.g. sulphuring
632.936.3	Repellents	632.982.4	Aerial application ⇒ 656.7
632.936.4	Scarecrows	632.982.5 632.982.6 632.982.7	Injection. □ Soil injection. Plant injection Smoke-generating. Fumigation Hot-water treatment. Scalding, sterilization
632.936.5	Acoustic methods	632.983 632.983.1	Time of application Treatment at vegetative stage. Summer spraying. Leaf spraying etc.
632.937	Natural limitation of plant diseases and pests. Biological control ⇒ 591.617/.619 ⇒ 632.938	632.983.2 632.983.3	Treatment at dormant stage. Winter spraying Treatment at time of sowing or planting
632.937.1	Natural enemies of plant diseases and pests. Organisms inimical to pests. Meteorological and climatic conditions unfavourable to pests	633/635	SPECIFIC CROPS. HORTICULTURE <i>The special auxiliaries -1/-2 from 62-1/-2 are not applicable in 633/635, where different -1/-2 auxiliaries are used</i>
632.937.2	Natural balance of populations. □ Fluctuations. Disturbances of natural balance. Protection of natural allies	633	FIELD CROPS AND THEIR PRODUCTION <i>Denote seasonal varieties by "32..."</i> ◇ 633"321" Spring varieties ◇ 633"321:324" Alternative varieties ◇ 633"324" Winter varieties ⇒ 582 ⇒ 634 ⇒ 635
632.937.3	Use of natural enemies of pests. □ Importation. Breeding. Use of natural occurrence	633-1/-2	Equipment, operations etc. for field crops. Plant damage and diseases <i>633-1/-2 ≡ 631/632</i> <i>The 62-1... and 62-2... auxiliaries are not applicable at 633 and its subdivisions</i>
632.938	Immunology. □ Plants resistant to diseases and pests. Immunization of plants	633.1	Cereals. Grain crops
632.939.1	Sterilization of insects ⇒ 632.935.4	633.11 633.111	Wheats. Triticum Hexaploid species. Triticum aestivum. □ Common wheat Tetraploid species Diploid species
632.95	Plant protection preparations. Pesticides	633.112 633.113	Buckwheat. Fagopyrum
632.95.02	Effect of pesticides	633.12	Oats. Avena sativa
632.95.021	According to stage of development of pest	633.13	Ryes. Secale cereale
632.95.021.1	Eggs. Ovicides	633.14	Maizes. Indian corns. Zea mays. Sweet corn ⇒ 635.67
632.95.021.2	Larvae. Larvicides		
632.95.021.3	Adults		
632.95.024	Physiological effect of pesticides. Toxicology of pesticides. Toxicity to pests, other animals, humans. Effect on plants. Phytotoxicity ⇒ 632.154		
632.95.025	Mode of action of pesticides		
632.95.025.1	By contact. Contact poisons		
632.95.025.2	By ingestion. Stomach poisons		
632.95.025.3	By inhalation. Respiratory poisons		
632.95.025.4	By antagonism		
632.95.025.5	By synergism		
632.95.025.8	Resistance to pesticides		
632.95.026	Preventive and curative treatment		
632.95.026.1	Preventive treatment		
632.95.026.2	Eradicant treatment		
632.95.026.3	Curative treatment		
632.95.027	Pesticide dosage		
632.95.028	Persistence of pesticides and their effects. Pesticide residues ⇒ 631.362.6		
632.951	Preparations for destruction of arthropods and molluscs. Insecticides. Acaricides. Molluscicides		
632.952	Antifungal preparations. Fungicides		
632.953	Antibacterial and antiviral preparations. Bactericides. Viricides		

633.16	Barleys. <i>Hordeum sativum</i>	633.42	Brassica and celeriac roots. □ Kohlrabi root, swedes, turnips. <i>Brassica. Celeriac. Apium graveolens rapaceum</i>
633.17	Millet. Sorghums		
633.171	Proso or common millet. <i>Panicum miliaceum</i>		
633.172	Digitaria. □ Crab grass. <i>Fonio. Iburu</i>	633.43	Carrot. <i>Daucus carota</i>. Turnip-rooted chervil. <i>Chaerophyllum bulbosum</i>
633.173	Pearl millet. <i>Pennisetum typhoideum</i>		
633.174	Sorghum. <i>Sorghum vulgare</i>	633.44	Parsnip. <i>Pastinaca sativa</i>
633.175	Foxtail millet. <i>Setaria italica</i>		
633.177	Teff. <i>Eragrostis tef</i>	633.49	Potatoes. Sweet potatoes. Jerusalem artichokes
633.179	Other millets. □ African millet. <i>Eleusine coracana. Manna grass. Glyceria fluitans. Wild rice. Zizania aquatica</i>		<i>Only field aspects are classed here. The main numbers are 635.21/.24</i>
633.18	Rices. <i>Oryza</i>		⇒ 633.68
633.18.03	Water rice. Paddy rice	633.491	Potato. <i>Solanum tuberosum</i>
633.18.05	Upland rice	633.492	Sweet potato. <i>Ipomoea batatas</i>
633.19	Other cereal and corn crops	633.493	Cassava. <i>Manioc. Manihot utilissima</i>
633.2	Forage grasses. Meadow and pasture grasses	633.494	Girasol (Jerusalem artichoke). <i>Helianthus tuberosus</i>
633.2.03	Kinds of meadow. Grasslands	633.495	Taro. <i>Colocasia esculenta</i>
	⇒ 636.083.314	633.496	Yautia, tannia or coco-yam. <i>Xanthosoma</i>
	⇒ 636.084.22	633.498	<i>Coleus rotundifolius</i>
633.2.031	Meadows of hay	633.5	Textile and fibre plants
633.2.032	Natural meadow	633.51	Cotton and similar fibres
633.2.033	Pastures		633.51 ≡ 677.2
633.21	Meadow grasses. <i>Poa</i>	633.511	Cotton. <i>Gossypium</i>
633.22	Cocksfoot. <i>Dactylis</i>	633.52	Flax and similar fibres
633.23	Bent. <i>Agrostis</i>		633.52 ≡ 677.1
633.24	Timothy grass. <i>Phleum</i>	633.521	Flax. Line plant. <i>Linum usitatissimum</i>
633.25	Cereals as green forage crops	633.522	Hemp. <i>Cannabis sativa</i>
	633.25 ≡ 633.1	633.525.1	Ramie. China grass. <i>Boehmeria nivea</i>
	<i>Applies only to cereals cultivated for green forage or fodder</i>	633.526.2	Agave fibres. □ Sisal. <i>Agave sisalana</i>
		633.528	Fibres from monocotyledon fruits. □ Coconut fibre. <i>Cocos nucifera. Loofah. Vegetable sponge. Luffa</i>
633.3	Forage plants except grasses	633.529.3	Fibres from water plants. Marine fibres. □ Sea wool. <i>Posidonia australis. Sea grass. Zostera marina</i>
633.31/.37	Leguminosae. Forage legumes		⇒ 639.64
633.31	Lucernes. <i>Medicago</i>	633.58	Plants yielding fibres for basketwork, wickerwork, coarse weaving
633.32	Clovers. <i>Trifolium</i>		633.58 ≡ 677.5
633.33	Cow peas. <i>Vigna</i>	633.584.3	Willows. Osiers. <i>Salix</i>
633.34	Soya (soy bean). <i>Glycine</i>	633.584.5	Bamboo. <i>Bambusa. Rattan. Calamus</i>
633.35	Pulses. <i>Vicia</i>. □ Lentils. Beans. Peas	633.6	Sugar and starch plants
	⇒ 635.65		⇒ 664.1/.2
633.36/.37	Other leguminosae	633.61	Sugar cane. <i>Saccharum officinarum</i>
633.361	Sainfoin. <i>Onobrychis viciaefolia</i>	633.63	Sugar beet. <i>Beta vulgaris</i>
633.365	French honeysuckle. <i>Hedysarum coronarium</i>	633.64	Sugar maple. <i>Acer saccharum</i>
633.367	Lupins. <i>Lupinus</i>	633.65	Sugar palm. <i>Arenga saccharifera</i>
633.368	Groundnut (peanut). <i>Arachis hypogaea</i> (as forage plant)	633.68	Starch plants
633.37	Other forage legumes		⇒ 633.49
633.39	Other forage plants. Nonleguminous forage plants	633.681	⇒ 664.27
633.4	Edible roots and tubers. Root crops	633.683	Arrowroot. <i>Maranta arundinacea</i>
	⇒ 635.1	633.685	Sago palm. <i>Metroxylon sagu</i>
633.41/.44	Beets. Carrots. Parsnips	633.686	Yam. <i>Dioscorea</i> and <i>Rajania</i>
	<i>Only field cultivation aspects are classed here. The main numbers are 635.11/.14</i>	633.687	<i>Canna</i>
633.41	Beets. Red and yellow beets. <i>Beta vulgaris</i>		<i>Amorphophallus</i>
	⇒ 633.63	633.7	Plants yielding stimulants. Beverage plants
633.412	Beetroot. <i>Beta vulgaris conditiva</i>		⇒ 663.9
633.415	Yellow salad beet. <i>Beta vulgaris alba</i>		
633.416	Fodder beet (mangelwurz)l)		

633.71	Tobaccos. <i>Nicotiana</i>	633.861	Plants yielding yellow dye. □ Saffron. <i>Crocus sativus</i> . Turmeric root. <i>Curcuma longa</i> . Sumach. <i>Rhus</i> . Fustic. <i>Chlorophora tinctoria</i> . Dyer's broom. <i>Genista tinctoria</i>
633.72	Teas. <i>Camellia sinensis</i>	633.862	Plants yielding blue dye. □ Dyer's woad. <i>Isatis tinctoria</i> . Litmus plant. <i>Chrozophora tinctoria</i> . Indigo plant. <i>Indigofera</i>
633.73	Coffees. □ <i>Coffea arabica</i> . Other <i>Coffea</i> species	633.863	Plants yielding red dye. □ Madder. <i>Rubia tinctorum</i> . Safflower. <i>Carthamus tinctorius</i> . Henna. <i>Lawsonia inermis</i>
633.74	Cacao (cocoa or chocolate tree). <i>Theobroma cacao</i>	633.864	Plants yielding green dye. □ Buckthorn. <i>Rhamnus</i>
633.75	Opium poppy. <i>Papaver somniferum</i>	633.865	Plants yielding brown dye
633.76	Kola. <i>Cola acuminata</i>	633.87	Tanning plants. Plants yielding tannin ⇒ 675.041
633.77	Maté. <i>Ilex paraguayensis</i>	633.871	Anacardiaceae. □ Sumach. <i>Rhus</i> (as tannin source). Tanner's myrtle. <i>Coriaria</i>
633.78	Chicory. <i>Cichorium intybus</i>	633.872	Fagaceae. □ Oak. Chestnut. Beech
633.791	Hop. <i>Humulus lupulus</i>	633.873	Plants bearing gall apples. □ Oak apples
633.792	Betel. <i>Piper betle</i>	633.874	Gambier shrub. <i>Uncaria gambier</i>
633.8	Aromatic plants. Condiment plants. Oleaginous plants. Dye plants. Tanning plants. Medicinal plants	633.875	Leguminosae. □ Acacia species. Cassia species
633.81	Aromatic plants. Plants yielding perfume ⇒ 665.52	633.876	Mangroves. □ Rhizophoraceae (true mangroves). Other mangroves
633.811	Perfumed flowers. □ Roses. Jasmine. Violets	633.877	Conifers. □ Fir bark. Hemlock bark. Larches. Pines. Phylloclades (e.g. emblic)
633.812	Perfumed herbaceous plants, shrubs. □ Lavender. Rosemary. Geranium (pelargonium). Patchouli	633.878	Other barks yielding tannin. □ Casuarinaceae. Myricaceae. Salicaceae. Betulaceae. Ulmaceae. Moraceae. Proteaceae
633.814	Perfumed woods. □ Sandalwood. Cedar wood	633.879	Plants which yield tannin from parts other than the bark
633.815	Perfumed roots. □ Orris. Vetiver	633.879.1	Plants with leaves yielding tannin
633.816	Balsam plants. Incense plants. □ Frankincense	633.879.4	Plants with fruits yielding tannin
633.819	Other perfume plants. □ Abel musk. Musk mallow. <i>Hibiscus abelmoschus</i>	633.879.6	Tannin woods. Plants yielding wood extract
633.82	Aromatic herbs. Condiment plants ⇒ 635.7 ⇒ 664.5	633.879.9	Plants with roots yielding tannin
633.821	Vanilla. <i>Vanilla planifolia</i>	633.88	Medicinal plants. Medicinal herbs ⇒ 615.2
633.822	Mints. <i>Mentha</i>	633.881	Plants yielding drugs affecting the heart and circulation. □ Digitalis (foxglove)
633.823	Wintergreen. <i>Gaultheria</i>	633.882	Plants yielding drugs affecting the respiratory system. □ Tussilago (coltsfoot)
633.824	Bay leaf. <i>Laurus nobilis</i>	633.883	Plants yielding drugs affecting the digestive system. □ Emetics, e.g. ipecacuanha. Purgatives, e.g. jalap. Tonics, e.g. quassia
633.825	Ginger. <i>Zinziber officinale</i>	633.884	Plants yielding substances acting on the glands. □ Cholagogues. Diaphoretics
633.826	Turmeric plant. <i>Curcuma longa</i>	633.885	Plants yielding antipyretics. □ Cinchona (as source of quinine)
633.827	Star anise. <i>Illicium verum</i>	633.886	Plants yielding drugs affecting the urogenital system
633.83	Spice plants	633.887	Plants yielding substances acting on the skin. Emollients, astringents etc.
633.831	Allspice (pimento). <i>Pimenta dioica</i>	633.888	Plants yielding drugs affecting the nervous system. □ Valerian. <i>Papaver somniferum</i> (opium poppy). <i>Cannabis sativa</i> (Indian hemp, hashish). <i>Belladonna</i> (deadly nightshade). <i>Nux vomica</i> (as source of strychnine)
633.832	Clove. <i>Eugenia aromatica</i> or <i>caryophyllus</i>	633.9	Other plants with industrial uses
633.833	Cinnamon. <i>Cinnamomum zeylanicum</i>	633.91	Rubber plants ⇒ 678.03
633.834	Nutmeg. <i>Myristica fragrans</i>	633.912	Euphorbiaceae
633.84	Pungent condiment (hot spice) plants	633.912.1	Hevea (<i>Euhevea</i>)
633.841	Pepper. <i>Piper nigrum</i>	633.912.11	Para rubber tree. <i>Hevea brasiliensis</i>
633.842	Spanish pepper. Paprika. Pepper from pimiento. <i>Capsicum annuum</i> ⇒ 635.649	633.912.7	Ceara and Manicoba rubber. Manihot species
633.843	Cayenne pepper. Chilli pepper. <i>Capsicum frutescens</i>	633.913.1	Rubber-yielding Apocynaceae. Rubber-yielding vines
633.844	Mustard. Brassica	633.913.2	Rubber-yielding Moraceae. □ Ficus species (India-rubber plant etc). Species of <i>Castilloa</i> , <i>Morus</i>
633.845	Caper. <i>Capparis spinosa</i>	633.913.3	Rubber-yielding Compositae. □ Guayule
633.846	Horseradish. <i>Cochlearia armoracia</i>	633.913.4	Rubber-yielding Asclepiadaceae
633.85	Oleaginous plants. Oil-yielding plants 633.85 ≡ 665.3		
633.852.52	Groundnut (peanut). <i>Arachis hypogaea</i> (as oil source)		
633.852.73	Olive. <i>Olea europaea</i> (as oil source)		
633.853.74	Sesame. <i>Sesamum indicum</i>		
633.855.34	Oil palms (various <i>Palmae</i> species)		
633.86	Dye plants. Plants yielding pigments, dyestuffs ⇒ 547.97 ⇒ 667.272/276		

633.93	Gum and resin plants	634.25	Peaches. <i>Prunus persica</i>
633.931	Gum arabic. Arabin-rich saps. Gum acacia.	634.26	Nectarine. <i>Prunus persica nectarina</i>
633.932	Acacia arabica. Acacia senegal	634.3	Rutaceous and moraceous fruits.
633.933	Cerasin-rich saps. Cherry gum. <i>Prunus</i> species		Citrous fruits in general
633.934	Bassorin-rich saps. Tragacanth. <i>Astragalus gummifer</i>	634.31	Oranges. □ <i>Citrus aurantium</i> . Sour orange.
633.936	Saps rich in both cerasin and bassorin		Seville orange. Bergamot. Sweet orange
633.937	Gum of firs and spruces. <i>Abies</i> and <i>Picea</i>		(<i>Citrus sinensis</i>)
633.938	species	634.32	Various citrous fruits
633.939	Balata. <i>Mimusops balata</i>	634.322	Mandarins. Tangerines. <i>Citrus reticulata</i> .
633.94	Plants yielding turpentine and rosin.	634.323	Clementine. <i>Satsuma</i> etc.
	□ Conifer resins. Turpentine tree (terebinth).	634.323.5	Shaddock (pompelmoose). <i>Citrus grandis</i> or
	<i>Pistacia terebinthus</i> . Pines, larches and firs.	634.324	maxima
	<i>Pinus</i> , <i>Larix</i> and <i>Abies</i> species		Grapefruit. <i>Citrus paradisi</i>
	633.94 ≡ 665.94		Kumquat. <i>Fortunella</i>
633.944.6	Plants yielding copal. <i>Agathis</i> and <i>Copaifera</i>	634.33	Lemon. Citron. Lime
	species etc.	634.331	Citron. <i>Citrus medica</i>
633.95	Other sap-yielding plants	634.334	Lemon. <i>Citrus limon</i>
633.954	Catechu plants. □ <i>Acacia catechu</i>	634.337	Lime. <i>Citrus aurantifolia</i>
633.956	Camphor tree. <i>Cinnamomum camphora</i>	634.36	Moraceae
634	HORTICULTURE GENERALLY	634.37	Fig. <i>Ficus carica</i>
	<i>The .0 special auxiliaries listed under</i>	634.38	Mulberries. <i>Morus</i> species
	<i>634.8.0... and 635.0... are also applicable</i>	634.39	Artocarpus
	<i>throughout 634</i>	634.392	Breadfruit. <i>Artocarpus communis</i>
	⇒ 635	634.393	Jackfruit. <i>Artocarpus heterophylla</i>
634-1/-2	Horticultural equipment, operations	634.4	Other pulp fruits
	etc. Plant damage and diseases	634.41	Annonaceae. □ <i>Annona</i> species. Soursop.
	<i>The 62-1... and 62-2... auxiliaries are not</i>		Sugar apple. Custard apple etc. North
	<i>applicable at 634 and its subdivisions</i>		American papaw or pawpaw. <i>Asimina triloba</i>
	<i>634-1/-2 ≡ 631/632</i>	634.42	Myrtaceae. □ <i>Guava</i> etc. <i>Psidium</i> species.
634.1	Fruit growing generally		Rose apple etc. <i>Syzygium</i> or <i>Eugenia</i> species
	<i>For garden plants and horticulture generally</i>	634.43	Sapotaceae fruits. □ <i>Sapodilla</i> . Sapote
	<i>see 635</i>	634.44	Anacardiaceae fruits
	◇ 634.1.047 Orchards: siting, layout etc.	634.441	Mango etc. <i>Mangifera</i>
	◇ 634.1.054 Fruiting shrubs or bushes	634.442	Ciruela. Mombin (hog plum etc.). <i>Spondias</i>
	◇ 634.1.055 Fruit trees	634.45	Ebenaceae. □ Persimmon. <i>Diospyros</i>
	◇ 634.1.076 Fruit (yields)	634.46	Leguminosae fruits. □ Tamarind. Carob (St
634.10	Pome fruits in general		John's bread). <i>Cassia</i>
634.11	Cultivated apples. <i>Malus sylvestris</i>,	634.47	Guttiferae
	subspecies mitis	634.471	Mangosteen etc. <i>Garcinia</i>
634.12	Wild apples. □ Crab apple. Paradise apple	634.472	Mammee-apple. <i>Mammea americana</i>
634.13	Pears. <i>Pyrus communis</i>	634.48	Rubiaceae. Voavanga. <i>Vanguaria</i>
634.14	Quinces. <i>Cydonia oblongata</i>. Japanese		madagascariensis
	quince. <i>Chaenomeles</i>	634.5	Nuts
634.15	Medlars. <i>Mespilus germanica</i>	634.51	Walnuts. <i>Juglans</i>
634.16	Loquat or Japanese medlar. <i>Eriobotrya</i>	634.52	Hickory. <i>Carya</i>. □ Pecan
	<i>Japonica</i>	634.53	Chestnuts. <i>Castanea</i>
634.17	Hawthorn. <i>Crataegus</i>	634.54	Hazelnut. <i>Corylus avellana</i>
634.18	Service apple. <i>Sorbus domestica</i>	634.55	Almonds. □ True almond. <i>Prunus</i>
634.19	Other pome fruits		<i>amygdalus</i> . Indian almond (myrobalan).
634.2	Stone fruits. Drupes in general		<i>Terminalia catappa</i>
634.21	Apricots. <i>Prunus armeniaca</i>	634.57	Other nut-like fruits
634.22	Plums. □ <i>Prunus domestica</i> and others.	634.572	Ginkgo. <i>Ginkgo biloba</i>
	Domestic plum and prune. Damson.	634.573	Cashew nut. <i>Anacardium occidentale</i>
	Greengage. Cherry plum. Sloe	634.574	Pistachio. <i>Pistacia vera</i>
634.23	Cherries	634.575	Brazil nut. <i>Bertholletia excelsa</i>
634.232	Sweet cherry (common cherry). <i>Prunus avium</i>		
	(<i>Cerasus avium</i>)		
634.233	Sour cherry (Morello cherry). <i>Prunus cerasus</i>		
	(<i>Cerasus vulgaris</i>)		

- 634.58 Groundnut (peanut). *Arachis hypogaea* (as fruit)
- 634.6 Various tropical and subtropical fruits**
- 634.61 **Palmaceae. Palm fruits**
- 634.613 Palm fruits generally. □ Date. Sugar palm.
Nipa palm. Betel nut. Rattan palm. Palmyra palm
- 634.614 Oil palm. *Elaeis guineensis*
- 634.616 Coconut. *Cocos nucifera*
- 634.63 **Olive. *Olea europaea***
- 634.64 **Pomegranate. *Punica granatum***
- 634.65/66 Fruits of Caricaceae, Basellaceae, Lauraceae, Rhamnaceae, Myricaceae, Polygonaceae, Euphorbiaceae and Ericaceae**
- 634.651 Papaya (true papaw or pawpaw). *Carica papaya*
- 634.652 Sapindaceae
- 634.652.2 Litchi (lychee). *Litchi sinensis*
- 634.652.3 Rambutan. *Nephelium lappaceum*
- 634.653 Avocado (alligator pear). *Persea americana*
- 634.662 Jujube. *Ziziphus jujuba*
- 634.663 Bog myrtle. *Myrica gale*
- 634.67 Fruits of Bombacaceae, Mellaceae, Geraniaceae, Solanaceae**
- 634.672 Durian. *Durio zibethinus*
- 634.673 Mellaceae fruits. □ Langsats. *Lansium*
- 634.674 Fruits of Geraniaceae, Oxalidaceae and Malpighiaceae. □ Bilimbi. *Carambola*
- 634.675 Fruits of Solanaceae. □ Tree tomato. *Cyphomandra betacea*. *Physalis*. *Solanum*
⇒ 633.491
⇒ 635.21
- 634.7 Small fruits of shrubs and herbaceous plants. Berries**
- 634.71 Bramble fruits. *Rubus***
- 634.711 Raspberry. *Rubus idaeus*
- 634.712 Oriental raspberry. *Rubus phoenicolasius*
- 634.713 Blackberry. *Rubus fruticosus* or *nigrobaccus*
- 634.714 Loganberry. *Rubus loganobaccus*
- 634.715 English blackberry
- 634.717 Dewberry. *Rubus caesius*
- 634.718 East Asian *Rubus* species
- 634.72 Currants. *Ribes***
⇒ 634.8.078
- 634.721 Currants (*Ribes*) in general
- 634.722 Red currant. *Ribes rubrum*
- 634.723 Black currant. *Ribes nigrum*. Other black currants
- 634.724 *Ribes aureum*
- 634.726 Gooseberry. *Ribes grossularia*
- 634.727 *Ribes oxycanthoides*
- 634.73 Fruits of the heath family. Vaccinaceae.**
□ Blueberries. Cranberries
- 634.735 Canadian blueberry. *Vaccinium canadense*
- 634.738 Cowberry. Red whortleberry. *Vaccinium vitis-idaea*
- 634.74 Other shrubs bearing berries**
- 634.741 Amelanchier
- 634.742 Buffalo berry. *Elaeagnus argentea*
- 634.743 Oleaster. *Elaeagnus multiflora*
- 634.744 Caranda. *Carissa carandas*
- 634.745 Guelder rose. *Viburnum opulus*
- 634.746 Barberry. *Berberis vulgaris*
- 634.747 Elderberry. *Sambucus nigra*
- 634.75 Strawberries. *Fragaria***
- 634.753 Wild strawberry. *Fragaria vesca*
- 634.77 **Fruits of herbaceous species**
- 634.771 *Musa* species in general
- 634.772 Plantains or cooking bananas
- 634.773 Common banana. *Musa paradisiaca*
- 634.774 Pineapple. *Ananas comosus* or *Ananas sativa*
- 634.775 Fruits of Cactaceae. Cactus fruits
- 634.776 Other fruits of herbaceous species
- 634.776.1 Tree tomato. *Cyphomandra betacea*
- 634.776.3 Passion fruit. *Passiflora edulis*
- 634.776.4 Water chestnut. *Trapa*
- 634.776.5 Fruit of Zingiberaceae
- 634.8 Viticulture. Grape vines. Vineyards**
◇ 634.8-051 Vine growers. Wine growers. Grape growers. Vine dressers
⇒ 663.2
- 634.8.03 Young vines (cultivation and propagation)
- 634.8.032 Vine-growing in the open
- 634.8.034 Vine-growing under glass (in hothouse)
- 634.8.037 Vine nurseries and their management
- 634.8.04 Mature vines (planting etc.)
- 634.8.042 Open-air cultivation
- 634.8.044 Cultivation under glass. Hothouse cultivation
- 634.8.047 Vineyards (site, layout etc.)
- 634.8.05 Vines according to ecological features.
Hardiness, resistance etc. of vines
Vine yields. Grape harvest
Grapes: appearance, size of bunches etc.
Grape juices. Juice yields
Dried grapes. □ Currants. Raisins. Sultanas
⇒ 634.72
⇒ 634.864
- 634.8.09 Crosses, hybrids of vines. Bastard forms of vine
- 634.8.092 Natural hybrids. Spontaneous hybrids
- 634.8.093 Artificial hybrids
- 634.8.094 Direct producers
- 634.84 Vitis species for viticulture. *Vitis vinifera* and others. Botanic classification of vines. Ampelography. Wine and grape regions**
- 634.841 Muscadinia
- 634.842 Evvites
- 634.842.1 Labruscae
- 634.842.2 Ripariae
- 634.842.3 Rupestrae
- 634.842.4 Cinerascensae
- 634.842.5 Aestivalae
- 634.842.6 Labruscoideae
- 634.842.7 Viniferae
- 634.842.8 Labruscoideae asiaticae
- 634.85 Vintage or wine-press vines. Grapes for winemaking**
- 634.852 Vines yielding light-skinned grapes. Green or white grapes for wine
- 634.853 Vines yielding dark-skinned grapes. Red, purple or black grapes for wine
- 634.86 Vines yielding grapes for eating (fresh, prepared or preserved) and soft drinks**
- 634.862 Vines yielding light-skinned grapes. Green or white grapes for eating
- 634.863 Vines yielding dark-skinned grapes. Red, purple or black grapes for eating
- 634.864 Vines yielding grapes for use dried (as currants or raisins). Seedless grapes (for sultanas)
⇒ 634.8.078
- 634.865 Vines yielding grapes for juice extraction (for soft drinks, grape juice)
- 635 GARDEN PLANTS. GARDENING**
⇒ 630*27
⇒ 634
⇒ 712

635-1/-2	Gardening equipment, operations etc.	635.115	Yellow salad beet. <i>Beta vulgaris alba</i>
	Plant damage and diseases	635.116	Fodder beet. Sugar beet as fodder
	<i>The 62-1... and 62-2... auxiliaries are not applicable at 635 and its subdivisions</i>	635.12	Brassica and celeriac roots
	635-1/-2 $\equiv$ 631/632	635.125	Kohlrabi root. <i>Brassica oleracea</i> $\Rightarrow$ 635.348
635-167	Garden irrigation, watering	635.126	<i>Apium graveolens dulce</i> (turnip). Swede.
635.01	General principles and practice of horticulture	635.127	Rutabaga. <i>Brassica campestris napobrassica</i>
	$\Rightarrow$ 631.1	635.128	<i>Apium graveolens secalinum</i> (turnip). <i>Brassica campestris esculenta</i>
635.012	Experimental and research horticulture	635.13	Carrot. <i>Daucus carota</i>. Turnip-rooted chervil. <i>Chaerophyllum bulbosum</i>
635.015	Practical gardening (generally). □ Technique. Equipment. Tools	635.14	Parsnip. <i>Pastinaca sativa</i>. Turnip-rooted parsley. <i>Petroselinum crispum</i>
635.017	Professional and commercial garden management. Park, estate and market gardening	635.15	Radish. <i>Raphanus sativus</i>
635.018	Amateur, hobby, home, allotment gardening $\Rightarrow$ 712.28	635.16	Various roots
635.03	Young plants (seedlings, saplings etc.) and their care. Nurseries	635.162	Horseradish. <i>Cochlearia hispanica</i>
	635.032/.034 $\equiv$ 635.042/.044	635.163	Rampion. <i>Campanula rapunculus</i>
	$\Rightarrow$ 630*232	635.164	Skirret (water parsnip). <i>Sium sisarum</i>
	$\Rightarrow$ 631.23	635.165	Wayaka plant (yam bean). <i>Pachyrrhizus erosus</i>
	$\Rightarrow$ 631.53.03	635.166	Black salsify (viper's grass). <i>Scorzonera hispanica</i>
635.037	Nurseries	635.167	Golden thistle. <i>Scolymus hispanicus</i>
635.04	Mature plants and their care. Gardens: site, layout etc.	635.168	Salsify (oyster plant). <i>Tragopogon porrifolius</i>
	$\Rightarrow$ 631.54	635.17	Rat-tailed radish. <i>Raphanus caudatus</i>
	$\Rightarrow$ 631.96	635.18	Chinese artichoke. <i>Stachys sieboldii</i>
635.042	Plants growing in the open, in beds etc.	635.2	Edible tubers and bulbs
635.043	Plants growing in pots, boxes or tubs	635.21/.24	Potatoes. Sweet potatoes. Jerusalem artichokes
635.044	Plants growing under glass or other cover (protected or forced)		$\Rightarrow$ 633.49
635.047	Gardens: site, layout etc. $\Rightarrow$ 712.27	635.21	Potato. <i>Solanum tuberosum</i>
635.05	Types and forms of garden plant	635.22	Sweet potato. <i>Ipomoeas batata</i>
635.051	Flowering plants (phanerogams) in general	635.23	Cassava. Manioc. <i>Manihot utilissima</i>
635.051.2	Tender plants	635.24	Girasol (Jerusalem artichoke). <i>Helianthus tuberosus</i>
635.051.3	Half-hardy plants	635.25/.26	Bulb crops
635.051.4	Hardy plants	635.25	Onion. <i>Allium cepa</i>
635.051.7	Deciduous plants	635.26	Other bulbs and tubers
635.051.8	Evergreen plants	635.261	Leek. <i>Allium ampeloprasum porrum</i>
635.052	Herbaceous plants (non-woody plants) in general	635.262	Garlic. <i>Allium sativum</i>
	635.052 $\equiv$ 635.051	635.263	Shallot. <i>Allium ascalonicum</i>
635.053	Climbing and trailing plants. Creepers	635.264	Welsh onion. <i>Allium fistulosum</i>
	635.053 $\equiv$ 635.051	635.265	Chives. <i>Allium schoenoprasum</i>
635.054	Woody plants (shrubs and trees) in general	635.3	Plants with edible stalks, leaves or flowers
	635.054 $\equiv$ 635.051		$\Rightarrow$ 635.48
635.055	Trees (woody plants with trunks)	635.31	Asparagus. <i>Asparagus officinalis</i>
	635.055 $\equiv$ 635.051	635.32	Flowers and buds as vegetables.
635.058	Nonflowering plants (cryptogams). Mosses, ferns etc.		□ Artichoke (globe artichoke). <i>Cynara scolymus</i>
635.07	Plant yields (whole plants, parts etc.)	635.33	Brassicas in general
	$\Rightarrow$ 631.57	635.34	Cabbage and related plants
635.071	Whole plants. Whole-plant yields	635.341	Red cabbage. <i>Brassica capitata rubra</i>
635.072	Stems. Branches. Twigs. Bark etc.	635.342	White cabbage. <i>Brassica capitata alba</i>
635.073	Roots. Tubers. Bulbs	635.343	Savoy cabbage. <i>Brassica bullata</i>
635.074	Leaves. Foliage. Greenery	635.344	Sea kale. <i>Crambe maritima</i>
635.075	Flowers. Buds. Blossoms	635.347	Other kales
635.076	Fruits. Fruit yield	635.348	Kohlrabi. <i>Brassica oleracea canlo-rapa</i> (aerial parts) $\Rightarrow$ 635.125
635.077	Saps. Juices. Other live yields		
635.078	Dried products		
635.1/.8	Vegetables. Kitchen gardening	635.35	Cauliflower. <i>Brassica oleracea botrytis</i>
	$\Rightarrow$ 712.27	635.356	Broccoli. <i>Brassica botrytis</i>
635.1	Edible roots. Root vegetables		
	$\Rightarrow$ 633.41/.44		
635.11	Beets. Red and yellow beets. <i>Beta vulgaris</i>		
635.112	Beetroot. <i>Beta vulgaris rubra</i>		

635.36	Brussels sprouts. <i>Brassica oleracea gemmifera</i>	635.67	Sweet corn (Maize. Indian corn). <i>Zea mays</i> (especially <i>Zea mays saccharata</i>). Maize for popcorn ⇒ 633.15
635.4	Other green vegetables. Leaf vegetables	635.7	Aromatic herbs. Plants for seasoning ⇒ 633.82 ⇒ 664.5
635.41	Spinach. <i>Spinacia oleracea</i>	635.71	Aromatic flowers and leaves. □ Thyme. <i>Thymus vulgaris</i> . Rosemary. <i>Rosmarinus officinalis</i> . Basil. <i>Ocimum basilicum</i> . Marjoram. Oregano. <i>Origanum vulgare</i> . Sweet marjoram. <i>Origanum majorana</i> . Saffron. <i>Crocus sativus</i> . Artemisia. Wormwood (absinth). <i>Artemisia absinthium</i> . Southernwood (Old man). <i>Artemisia abrotanum</i> ⇒ 635.74
635.42	Chard. Spinach beet. <i>Beta vulgaris cicla</i>	635.72	Mints. <i>Mentha</i>
635.43	Orach (orache). <i>Atriplex hortensis</i>	635.74	Other seasoning plants. □ Tarragon. <i>Artemisia dracunculus</i> . Sage. <i>Salvia officinalis</i> . Savory. Hyssop. Woodruff
635.44	Chinese mustard. <i>Brassica rugosa</i>	635.75	Aromatic seeds. □ Coriander seed. <i>Coriandrum sativum</i> . Cumin. <i>Cuminum cyminum</i> . Caraway seed. <i>Carum carvi</i> . Aniseed. <i>Pimpinella anisum</i> . Fennel seed. <i>Foeniculum vulgare</i> . Dill seed. <i>Anethum graveolens</i>
635.45	Sorrel. <i>Rumex</i>	635.78	Garnishing herbs
635.46	Purslane. <i>Portulaca oleracea</i>	635.781	Parsley. <i>Petroselinum crispum</i>
635.47	Talinum	635.782	Chervil. <i>Anthriscus cerefolium</i>
635.48	Rhubarb. Rheum	635.8	Edible fungi
635.49	Other edible leaves. □ Dandelion. <i>Taraxacum officinale</i> . Fennel. <i>Foeniculum vulgare</i> . Borage. <i>Borago officinalis</i>	635.82	Mushrooms. <i>Agaricus campestris</i> . Gill fungi in general
635.5	Salad plants. Salad vegetables	635.83	Edible boleti. <i>Boletus edulis</i> and similar species. Pore fungi in general
635.52	Lettuce. <i>Lactuca sativa</i> . □ Butterhead, cos and crisp lettuce	635.84	Morels. <i>Morchella</i>
635.53	Celery. <i>Apium graveolens dulce</i> and <i>secalinum</i>	635.85	Truffles. Tuber
635.54	Chicory. <i>Cichorium intybus</i>	635.89	Other fungi
635.55	Endive. <i>Cichorium endlva</i>	635.898	Poisonous fungi
635.56	Piquant salads. Cresses. □ Nasturtium. <i>Cardamine</i> . <i>Tropaeolum</i> genera	635.9	Ornamentals. Decorative plants. Ornamental gardening
635.57	Lamb's lettuce (corn-salad). <i>Valerianella oltioria</i>	635.91	Ornamentals for indoor use ⇒ 745.9
635.6	Edible fruits and seeds. Pulses <i>Especially gourd fruits, 635.61/.63. For most fruits see 634</i>	635.912	Cut or picked flowers, stems, greenery etc.
635.61	Melons and similar plants	635.913	Treated or dried ornamentals. Immortelles
635.611	Melon. Cantaloup. <i>Cucumis melo</i>	635.914	Pot plants. Box plants. Tub plants. Other growing plants (for indoor use)
635.615	Watermelons. <i>Citrullus vulgaris</i>	635.915	Flower-windows. Indoor window-boxes etc.
635.62	Cucurbita and similar plants	635.917	Lower plants. Mosses. Ferns
635.621	Cucurbita pepo. □ Pumpkin. Squash. Courgette	635.918	Conservatory plants. Glasshouse plants
635.622	Winter squash	635.92	Ornamentals grown and used outdoors
635.623	Vegetable marrow	635.921	Pot plants, box plants, tub plants grown outdoors
635.624	Cucurbita maxima	635.922	Garden display ornamentals (bed and border)
635.626	Cucurbita ficifolia	635.923	Rockeries. Rock-garden plants
635.627	Bottle gourd etc. <i>Lagenaria</i>	635.924	◇ 635.923(23) Alpine flowers
635.628	Loofah. Luffa	635.925	Wild and woodland garden ornamentals
635.63	Cucumber. <i>Cucumis sativus</i> . Gherkin	635.926	Park and public garden ornamentals
635.64	Tomatoes and similar plants. □ Tomato. <i>Lycopersicum esculentum</i> ⇒ 633.842	635.927	Pond and lake plants. Water plants
635.646	Aubergine (egg plant). <i>Solanum melongena</i>	635.928	Background and enclosure plants. Hedges. Ground-cover. Shrubberies
635.648	Okra. <i>Hibiscus esculentus</i>		Lawn and turf features. □ Grass paths, patches
635.649	Pimiento (sweet pepper). <i>Capsicum annuum</i> ⇒ 633.842		
635.65	Pulses. <i>Vicia</i> and others ⇒ 633.33/.35		
635.651	Broad bean. <i>Vicia faba</i>		
635.652	French bean. Green beans. Haricot beans. <i>Phaseolus vulgaris</i>		
635.653	Lima bean. <i>Phaseolus lunatus</i>		
635.654	Dolichos and other pulses. □ Other <i>Phaseolus</i> species. Runner bean (scarlet runner). Cow peas (<i>Vigna</i>). Pigeon pea		
635.655	Soya bean (soy bean). <i>Glycine</i>		
635.656	Peas in general. <i>Pisum sativum</i>		
635.657	Chick-pea (gram). <i>Cicer arietinum</i>		
635.658	Lentils. <i>Lens esculenta</i>		

636	ANIMAL HUSBANDRY AND BREEDING IN GENERAL. LIVESTOCK REARING. BREEDING OF DOMESTIC ANIMALS	636.064	Biological properties, characteristics 636.064 $\equiv$ 57.017
636.01	Origin, evolution and domestication of animals kept by humans. Taming of animals. Biological bases of rearing and training	636.066	Temperament of animals
636.02	Animals kept for breeding, showing and scientific uses	636.068	Internal (organic) characteristics, modifications etc. 636.068.1/.6 $\equiv$ 591.4.068.1/.6
636.022	Pedigree breeding stock. Show stock. Evaluation, judging of pure-bred animals	636.08	General questions of breeding and rearing
636.025	Animals bred or kept for permanent exhibition or display (e.g. in zoos, aquariums, terrariums) $\Rightarrow$ 791.82	636.081	Various procedures. Acquisition and marking of animals
636.028	Animals bred or kept for scientific research, observation, experiments $\Rightarrow$ 57.082	636.081.1	Acquisition of animals. ☐ Selection. Purchase. Capture $\Rightarrow$ 639.1.081
636.03	Animals bred or kept for food or other produce (productivity, yield). Farm and ranch animals in general	636.081.2	Marking of animals. ☐ Marks of ownership. Tagging. Branding
636.033	Animals kept for slaughtering, for meat and fat. Meat animals. Fatstock. Meat and fat yields	636.082	Breeding of animals. Practical breeding. Genetic aspects $\Rightarrow$ 575 $\Rightarrow$ 591.15 $\Rightarrow$ 591.16
636.034	Animals kept for live yield of food produce. Dairy and egg-laying stock. Milk and egg yields	636.082.1	General questions. Role of external conditions, environment, in breed development
636.035	Animals kept for live yield of wool, textile hairs etc. Wool or hair yield, composition and quality	636.082.2	Choice of stock for selective breeding $\Rightarrow$ 575.827
636.036	Animals kept for other products (e.g. industrial raw materials)	636.082.21	Genealogical records, stud books etc. Stock registers. Pedigrees
636.04	Animals bred or kept for work and service or pleasure of humans	636.082.22	Improvement of quality of livestock
636.042	Work animals in general $\Rightarrow$ 636.13/.15	636.082.23	Selection: mass, individual, lineal
636.042.2	Draught and yoke animals	636.082.24	Choice of mating pairs
636.042.4	Pack animals	636.082.25	Rearing of pure breeds. ☐ Line breeding. Inbreeding $\Rightarrow$ 591.38 $\Rightarrow$ 636.051
636.042.6	Passenger-carrying animals	636.082.26	Cross-breeding. Outbreeding. Hybridization $\Rightarrow$ 591.37 $\Rightarrow$ 636.052
636.043	Service animals for patrol, liaison and control duty	636.082.3	Structure of herds and flocks. Rotation, replacement pattern $\Rightarrow$ 591.55 $\Rightarrow$ 636.05
636.043.2	Patrol and reconnaissance animals	636.082.31	Composed uniquely of males
636.043.3	Tracker animals	636.082.32	Composed uniquely of females
636.043.5	Communication and message-carrying animals	636.082.34	Composed uniquely of castrated animals. Gelded animals. Spayed animals
636.043.7	Herd-control and round-up animals	636.082.35	Composed uniquely of young animals
636.044	Rescue, guide and guard animals for protection of persons and property (e.g. guide dogs; guard dogs or watchdogs)	636.082.4	Breeding methods and techniques. Reproduction
636.045	Pets generally. Animals kept for companionship or decoration. Household animals, mascots etc. (e.g. peafowl; pet dogs, lapdogs; pet cats)	636.082.41	Consanguineous breeding
636.046	Hunting and sporting animals <i>For transport animals see 636.042</i> $\Rightarrow$ 798	636.082.45	Oestrus. Season, heat, rut $\Rightarrow$ 591.545
636.046.2	Riding animals	636.082.453	Mating. Coupling. Service
636.046.3	Hunting animals. Hunters	636.082.453.5	Artificial insemination. ☐ Instruments. Apparatus
636.046.5	Racing animals	636.082.454	Fertilization. Impregnation. Fertility. Sterility
636.046.8	Fighting animals	636.082.455	Gestation $\Rightarrow$ 591.366
636.047	Performing animals. Circus and stage animals	636.082.456	Birth
636.049	Animals bred or kept for ceremonial and other uses (e.g. ritual, sacred animals)	636.082.46	Egg-laying. Laying period
636.05	Animals according to breeding	636.082.47	Incubation, brooding, sitting. Hatching of chicks
636.051	Pure-bred animals. Thoroughbreds in general $\Rightarrow$ 636.082.25 $\Rightarrow$ 636.12	636.082.471	Natural hatching. Brooding birds
636.052	Cross-bred animals. Cross-breeds. Mongrels. Hybrids $\Rightarrow$ 636.082.26	636.082.474	Artificial incubation. Incubators. Artificial hatcheries
636.06	Characteristics, constitution and special biological features of domestic animals	636.083	Care of animals in general
636.061	External characteristics and traits 636.061 $\equiv$ 57.018	636.083.1	Management and maintenance of buildings for rearing livestock $\Rightarrow$ 631.22
		636.083.14	Litter
		636.083.18	Comfort and freedom of movement of enclosed animals
		636.083.3	Particular methods and equipment for care of animals

636.083.31	Care and rearing methods in general	636.085.4	Value, inventory and storage of feedstuffs.
636.083.312	Rearing in permanent buildings. □ In stalls, stables, byres		Determination of volume and weight of feedstuffs
636.083.312.5	Rearing in batteries. Rearing in cages	636.085.5	Individual kinds of feedstuff
636.083.313	Rearing in temporary structures, enclosures. □ In pens, folds, camps	636.085.51	Green fodder in general ⇒ 636.086
636.083.314	Rearing in the open. Pasturing, putting to grass ⇒ 636.084.22	636.085.52	Silage
636.083.32	Devices for tethering animals. □ Collars. Halters. Chains. Clasps	636.085.53	Roughage in general
636.083.35	External body care. Grooming. □ Curry-combs. Brushes. Driers. Tail bandages	636.085.54	Concentrated feedstuffs in general ⇒ 636.086/.087
636.083.36	Washing of animals	636.085.55	Compound feedstuffs
636.083.38	Exercising, walking etc. of animals	636.085.57	Artificial, synthetic feedstuffs
636.083.4	Care of hooves, horns, coats	636.085.6	Preparation of fodder for consumption. Cooking arrangements, fodder kitchens ⇒ 643.396
636.083.42	Care of hooves. Shoeing ⇒ 682.1	636.085.7	Preservation and storage of feedstuffs ⇒ 631.24 ⇒ 631.563
636.083.43	Care of horns. Dehorning	636.085.8	Experimental feeding
636.083.45	Care of coats. □ Clipping. Shearing	636.086	Specific vegetable feedstuffs 636.086.1/.4 = 633.1/.4 ⇒ 636.085.51 ⇒ 664.78
636.083.5	Guarding and supervision of animals on pastures. Shepherd etc.	636.086.5	Sprouted grain
636.083.6	Protection of animals from the weather (heat, cold, wind, rain etc.) and ground conditions	636.086.7	Other vegetable feedstuffs
636.083.7	Harness ⇒ 685.1	636.086.74	Fruit-bearing plants 636.086.741/.747 = 634.1/.7
636.083.8	Muzzles. Collars. Reins. Leashes	636.086.75	Market garden plants 636.086.751/.756 = 635.1/.6
636.084	Feeding of animals	636.086.8	Chemically treated cellulose feedstuffs
636.084.1	Feeding of young animals. □ Suckling. Weaning	636.087	Industrial by-products as feedstuffs. Feedstuffs of animal origin. Additives
636.084.13	Artificial milk-feeding (bottle-feeding etc.)	636.087.2	Industrial by-products as feedstuffs. □ From starch-works, breweries, distilleries. Draff, malt, yeast etc. ⇒ 631.879.3 ⇒ 663.14
636.084.2	Feeding according to locality	636.087.3	Feedstuffs high in particular constituents
636.084.21	Feeding in stables, stalls etc.	636.087.6	Feedstuffs of animal origin. □ Milk, meat, fish products. Bone meal
636.084.22	Feeding at pasture. Grazing. Pasturing ⇒ 636.083.314	636.087.7	Additives. Stimulants. Feeding supplements (salt, lime, vitamins, proteins etc.)
636.084.3	Provision of water. Watering of animals ⇒ 626.81	636.087.8	Antibiotic feeds. Biological preparations. Hormones
636.084.4	Various questions of feeding and nutrition generally	636.088	Training, domestication (taming) and adaptation of animals for particular purposes
636.084.41	Quantity of food. Rations	636.09	Veterinary science. □ Diseases, enemies and pests of domestic and useful animals: control and treatment. Zootechnical, protective measures <i>For diseases of other animals see 591.2</i> ◇ 636.09-051 Veterinary practitioners ◇ 636.1.09 Diseases of horses (and other equines), and their treatment ◇ 636.1.09:616.24 Pulmonary diseases of horses etc., and their treatment ⇒ 614.2/.6
636.084.42	Feeding methods. Choice of feedstuff		
636.084.421	Feeding with dry feedstuffs	636.092	Rapacious enemies of domestic animals
636.084.422	Feeding with moist feedstuffs	636.092.1	Humans as injurious to domestic animals. Cruelty to animals. Theft of animals. Poaching etc. ⇒ 179.3 ⇒ 343.58 ⇒ 343.77
636.084.423	Rationed feeding. Hand-feeding	636.092.2	Stray or prowling domestic animals as injurious to other domestic animals (e.g. cats as enemies of birds, poultry)
636.084.424	Ad-lib feeding. Self-feeding	636.092.3	Wild animals, carnivores, predators as injurious to domestic animals
636.084.429	Feeding times. Periodicity of feeding <i>Specify times or periods by "3..." and "5..." (Table 1g)</i>	636.093	Insects as injurious to domestic animals (prevention and control)
636.084.5	Special feeding		
636.084.51	Feeding in mating season, at gestation, lactation, egg-laying		
636.084.52	Feeding for production. Fattening. Feeding for yield of milk, eggs, wool etc.		
636.084.55	Feeding for maintenance		
636.084.7	Feeding and watering devices		
636.084.74	Devices for distributing feed. □ Mangers. Feeding troughs, hoppers		
636.084.741	Troughs		
636.084.742	Racks		
636.084.743	Hoppers etc. for feeding measured quantities		
636.084.744	Self-feeders		
636.084.75	Watering devices. □ Drinking troughs. (Self-regulating) cisterns		
636.084.76	Combined feeding and watering devices		
636.085	Feedstuffs, fodder in general		
636.085.1	Composition of feedstuffs		
636.085.2	Nutritive value of feedstuffs. □ Assimilability. Taste, flavour. Calorific value. Digestibility. Starch equivalent		
636.085.3	Analysis of feedstuffs (general questions)		

- 636.1 Domestic equines. Horses**
- 636.11 Oriental horses.** □ Arab, Turkmen, Persian, Tartar, Berber horses
- 636.12 Trotting and racing horses**
⇒ 636.051
- 636.122 Thoroughbreds
- 636.127.1 Trotters
- 636.128 Standardbreds (American trotters). □ Morgan horse
- 636.13 Saddle horses.** □ Riding, hunting, cavalry horses. Hunters
◇ 636.13.046:798.252 Polo ponies (i.e. polo horses)
- 636.14 Carriage horses. Coach horses**
- 636.15 Draught horses. Cart horses. Dray horses**
- 636.152 British draught horses. □ Clydesdales, Shires, Suffolk punches
- 636.153 German and Dutch draught horses
- 636.154 French and Belgian draught horses
- 636.158 American draught horses
- 636.16 Ponies**
- 636.162 British ponies. □ Shetland and Galloway ponies. Exmoor, Dartmoor and New Forest ponies. Welsh ponies
- 636.168 American ponies
- 636.18 Other equines**
- 636.182 Asses. Donkeys
- 636.183 Mules. Hinnies
- 636.184 Zebras
- 636.2 Large ruminants. Cattle, oxen**
◇ 636.2.053 Young oxen. Calves
- 636.22 British cattle breeds**
- 636.222 English beef breeds. □ Beef shorthorns. Durham, Teeswater, Lincoln cattle. Red Shorthorns. Hereford, Polled Hereford and Sussex cattle
- 636.223 Scottish, Welsh and Irish beef breeds.
□ Aberdeen Angus, Galloway and Highland cattle. Welsh Black cattle
- 636.224 English dairy breeds. □ Channel Islands cattle. Jersey, Guernsey, Alderney cattle
- 636.225 Scottish and Irish dairy breeds. □ Ayrshire, Dunlop, Cunningham and Kerry cattle
- 636.226 English dual-purpose breeds. □ Devon, Redpoll, Norfolk cattle
- 636.23 German, Dutch, Danish and Swiss cattle breeds**
- 636.234 German dairy breeds. □ Holstein, Friesian, Oldenburg cattle
- 636.235 Dutch and Danish cattle breeds
- 636.237 Swiss cattle breeds
- 636.24 French and Belgian cattle breeds**
- 636.242 French beef breeds
- 636.244 French dairy breeds
- 636.245 Belgian and Flemish cattle breeds
- 636.27 Other European cattle breeds**
- 636.271 Russian cattle breeds
- 636.272 Spanish cattle breeds
- 636.273 Hungarian cattle breeds
- 636.275 Finnish cattle breed
- 636.276 Swedish cattle breeds
- 636.28 Non-European cattle breeds**
- 636.282 American beef breeds. □ Texas Longhorns
- 636.283 American dairy breeds. □ American Holderness cattle. French Canadian cattle
- 636.29 Other large ruminants**
- 636.291 Zebus
- 636.292 Bison family. □ American buffalo (Bison bison). Cattalo (hybrid of bison bull and ox cow). European bison (wisent)
Species of wild cattle
Buffalo
Yak
Gayal
Musk ox
Antelope
Deer. Reindeer
Camels. Dromedaries
Other Camelidae. Llama
- 636.3 Small ruminants. Sheep. Goats**
- 636.32/38 Sheep**
- 636.32 British sheep breeds**
- 636.321 British longwool sheep
- 636.323 British shortwool sheep
- 636.325 British mountain sheep
- 636.33 Dutch and German sheep breeds**
- 636.331 Dutch sheep breeds
- 636.333 German sheep breeds
- 636.34 French and Flemish sheep breeds**
- 636.346 Pyrenean sheep breeds
- 636.35 Italian sheep breeds**
- 636.36 Spanish sheep breeds. Merinos etc. Other European merinos**
- 636.37 Other European sheep breeds**
◇ 636.37(48) Scandinavian sheep breeds
Tsigaya sheep
Ratzka sheep
- 636.38 Non-European sheep breeds**
◇ 636.38(5) Asiatic sheep breeds
- 636.39 Goats**
- 636.4 Pigs. Swine**
- 636.42 English pig breeds**
- 636.43 German, Dutch and Hungarian pig breeds**
- 636.44 French and Swiss pig breeds**
- 636.45 Italian pig breeds.** □ Neapolitan and Tuscan pigs
- 636.46 Spanish and Portuguese pig breeds**
- 636.47 Other European pig breeds**
- 636.475 Russian pig breeds
- 636.48 Non-European pig breeds. American pig breeds. Others**
- 636.5 Poultry**
- 636.52/58 Domestic fowl. Chickens**
- 636.52 British chicken breeds**
- 636.53 German, Austrian and Dutch chicken breeds**
- 636.54 French and Belgian chicken breeds**
- 636.55 Italian chicken breeds. Mediterranean chicken breeds**
- 636.56 Spanish chicken breeds**
- 636.57 Other European chicken breeds**
- 636.58 Non-European chicken breeds**
- 636.581 American chicken breeds
- 636.584 Rhode Island Red chickens
- 636.585 Asiatic chicken breeds
- 636.587 Special chicken breeds. □ Bantams
- 636.588 Other chicken breeds. □ Fighting cocks. Fancy breeds

- 636.59 Other types of poultry**
 636.592 Turkeys
 636.593 Guinea fowl
 636.594 Pheasants
 636.595 Peafowl
 636.596 Pigeons and doves
 636.597 Ducks
 636.597.2 English duck breeds. □ Aylesbury duck
 636.597.4 French duck breeds
 636.597.8 Other duck breeds. □ Muscovy ducks. Pekin ducks
 636.598 Geese
 636.598.3 German goose breeds. □ Emden geese
 636.598.4 French goose breeds. □ Toulouse geese
 636.598.7 Sebastopol geese
 636.598.8 Non-European goose breeds
- 636.6 Birds (except poultry and game) bred or kept by humans**
 ⇒ 598.2
- 636.61 Birds bred or kept for plumage**
- 636.68 Ornamental and song birds. Cage and aviary birds in general**
- 636.7 Dogs**
- 636.72 Working dogs**
 Or use 636.7.042
- 636.74 Guard dogs. Watchdogs. Sheepdogs. Police dogs**
 Or use 636.7.044
- 636.75 Hunting dogs. Hounds**
 Or use 636.7.046
- 636.76 Pet dogs. Lapdogs**
 Or use 636.7.045
- 636.8 Cats**
- 636.9 Other animals kept by humans**
- 636.91 Guinea-pigs (cavies)**
- 636.92 Domestic rabbits**
 ⇒ 639.112
- 636.93 Fur animals**
 ⇒ 636.035
 ⇒ 637.62
 ⇒ 639.11
- 636.931 Marsupials kept for fur
 636.931.1 American opossum (possum)
 636.932 Rodents kept for fur
 636.932.2 Beaver
 636.932.3 Coypu. Nutria
 636.932.4 Chinchillidae. Chinchillas. Vizcachas (viscachas), woolly mice
 636.933 Ungulates kept for fur. □ Karakul sheep
 636.934 Predators, carnivores, kept for fur
 636.934.2 Foxes. Vulpines
 636.934.3 Raccoon dog
 636.934.4 Raccoon
 636.934.5 Mustelidae. □ Martens. Sable. Polecat (fitchew). Mink. Skunk
 ⇒ 636.96
- 636.95 Various land and amphibious animals kept mainly for food**
 Specify by :592/599
 ◇ 636.95:594.3 Edible snails
 ◇ 636.95:597.82 Edible frogs
- 636.96 Various small land mammals. □ Hedgehog. Ferret. Mongoose**
 Except those at 636.91/.93, and primates
 Specify by :599...
- 636.97 Various larger land mammals. □ Elephant. Primates**
 Except those at 636.1/.8
 ◇ 636.97:599.8 Primates kept by humans. Domesticated apes and monkeys
- 636.98 Reptiles and other vertebrates. □ Reptilia, e.g. lizards, snakes, tortoises. Amphibia. Fish, e.g. goldfish**
 Except those at 636.1/.97
 Details by :597/599
 ⇒ 636.95
 ⇒ 639.34
- 636.99 Other animals kept by humans.**
 □ Invertebrates
 Except those in 636.1/.98
 ◇ 636.99:595.142 Chaetopoda. Various worms etc.
 ◇ 636.99:595.143 Hirudinea. Leeches
 ⇒ 639.54
- 637 PRODUCE OF DOMESTIC (FARMYARD) ANIMALS AND GAME**
 ⇒ 664.9
 ⇒ 665.2
- 637.02 Processing machines and equipment for animal produce**
 Machines and equipment for preliminary treatment. □ Strainers. Separators. Cleaners
 637.022 Equipment for temperature treatment and for preservation. □ Pasteurizers. Sterilizers
 637.023 Equipment for altering the physical condition of produce. □ Homogenizers. Churns
 637.024 Machines and equipment for further working before concentrating, drying or shaping
 637.025 Devices and equipment for concentrating or drying of produce
 637.026 Devices for shaping, cutting or grading (by size). □ Moulds. Cutters. Graders
 637.027 Equipment for aftertreatment. □ For cultures, additives, dressing, wrapping
- 637.03 Methods and procedures for treatment of animal produce**
 Further subdivided under 637.5
- 637.04 Chemical composition and ingredients of animal produce**
 637.041 Ratio of food-value ingredients to whole product
 637.041.3 High food value content
 637.041.5 Moderate, average food value content
 637.041.7 Low food value content
 637.043 Fat content
 637.044 Carbohydrate content. □ Lactose content
 637.045 Protein content
 637.046 Content in mineral salts (e.g. calcium)
 637.047 Content in biochemical substances
 ◇ 637.047:577.15 Enzyme content
 ◇ 637.047:577.16 Vitamin content
- 637.05 Qualities and properties of animal produce (except composition)**
 637.052 General appearance. Surface texture. Colour
 637.053 Physical state. □ Body texture. Feel. Density. Viscosity. Greasiness
 Details by :53...
 637.054 Chemical properties (other than composition)
 Details by :54/543
 ⇒ 637.04
- 637.055 Biological/bacteriological properties**
 ◇ 637.055:579.8... Content of specific microorganisms

637.056	Keeping qualities ⇒ 637.066	637.133.2	Preheating and heating of milk
637.057	Flavour. Taste. Aroma	637.133.3	Pasteurization
637.058	Properties induced by preservatives or other artificial means	637.133.4	Sterilization of milk
		637.133.7	Chemical preservation of milk. Other preservation methods
637.06	Faults, defects, contamination, adulteration of animal produce	637.134	Treatment of milk to alter physical condition. Homogenization
637.061	General deterioration. □ Staleness. Unfreshness	637.136	Bacteriological treatment of milk ⇒ 637.146
637.062	Appearance and colour defects. □ Discoloration. Streakiness. Ropiness	637.136.3	Culturing
637.063	Physical, body texture and consistency defects. □ Excessive greasiness. Dryness	637.136.5	Fermentation
637.064	Chemical or physicochemical defects. □ Sediment. Gassiness. Overacidity	637.137	Condensing and drying of milk
637.065	Bacterial defects. □ Mouldiness. Contamination	637.138	Treatment of processed milks with additives, flavourings etc.
637.066	Decomposition. Putrefaction. Rottenness	637.14	Processed milks
637.067	Flavour, taste and aroma defects. □ Rancidity. Sourness. Bitterness	637.141	Processed whole milk
637.068	Adulterations. Impurities	637.141.3	Pasteurized milk. Sterilized milk
637.07	Examination, testing, analysis of animal produce	637.141.4	Homogenized milk
	<i>Class here methods of examination etc.</i>	637.141.8	Milk with added flavourings. □ Chocolate milk drinks. Fruit milk drinks
	<i>For testing of particular properties see 637.04/.06</i>	637.142	Concentrated milks
		637.142.2	Condensed milk, cream, whey and other milk products ⇒ 637.148 ⇒ 637.344
		637.142.3	Evaporated milk
637'6	Produce of livestock, bred or kept animals	637.142.5	Coagulated concentrated milk
	637'6 ≡ 636	637.143	Dried milk. Milk powder. Reconstituted milk
	◇ 637.12'639 Goat's milk	637.143.2	Dried milk, cream powder and the like
	◇ 637.5'62 Beef (i.e. ox meat) and beef products	637.143.6	Reconstituted milk (from milk powder)
		637.144	Dietary milks. Special milks for infants and children
		637.146	Cultured, fermented and sour milk products ⇒ 663.1
637'7	Produce of game and hunted animals	637.146.1	Bacterial and yeast treatment of milk in general
	637'7 ≡ 639.1	637.146.2	Milk products from combined acid and alcohol fermentation. □ Gioddu. Araka. Huslanka
	◇ 637.54'724 Meat from partridge, grouse etc.		Kefir
	◇ 637.55'712 Meat of small furred animals. Rabbit and hare meat	637.146.21	Mazun. Koumiss
	◇ 637.61'711.7 Hides of large predators	637.146.23	◇ 637.146.23'61 Koumiss from mare's milk
637'8	Produce of fish and other aquatic animals	637.146.3	Milk products from acid fermentation. □ Leben. Skyr
	637'8 ≡ 639.2	637.146.33	Butter cultures, starters. Cheese cultures
	◇ 637.5'845.1/.3 Whalemeat	637.146.34	Yoghurt. Acidophilus milk
	◇ 637.56'87 Flesh of aquatic molluscs, oysters	637.147	Skim milk ⇒ 637.144 ⇒ 637.247
	⇒ 639.38		Casein from skimmed milk. □ Lactic acid and rennet casein. Sodium lactate
637.1	Dairying and dairy produce in general	637.147.2	Coprecipitates of skimmed milk
637.11	Milking. Milking equipment	637.148	Cream. Cream products ⇒ 637.187 ⇒ 663.674
637.112	Factors affecting milk yield and quality. □ Times and frequency of milking		Whipping cream
	◇ 637.112"32" Seasonal variations ⇒ 636.084.52	637.148.2	Coffee cream
637.114	Milking by hand	637.148.4	Sterilized cream
637.115	Milking by machine. Mechanical milk-handling systems	637.148.6	
637.12	Milk (fresh, unprocessed milk)	637.18	Milk and cream substitutes
637.123	Colostrum	637.181	Vegetable milk
637.13	Milk processing	637.182	Milk from animal fats
637.131	Initial treatment and improvement of milk for processing	637.187	Cream from skim milk with added fat. Synthetic cream
637.131.2	Purification. Clarification. Straining. Filtering	637.2	Butter and buttermaking
637.131.3	Deodorizing of milk		◇ 637.2.024 Butter churns
637.131.5	Skimming. Separating. Creaming of milk		◇ 637.2.067 Rancidity of butter
637.131.6	Standardizing the composition	637.23	Buttermaking processes
637.131.8	Treatment with additives. □ Vitaminizing	637.233	Cream or milk treatment for buttermaking
637.133	Temperature treatment and preservation of milk	637.233.1	Initial treatment. Standardizing and separating of cream or milk ⇒ 637.131
637.133.1	Cooling and freezing of milk		

- 637.233.6 Temperature treatment and preservation of cream or milk. □ Pasteurization. Sterilization. Cooling
⇒ 637.133
- 637.234 Butter formation. Butter working, shaping etc.
- 637.234.2 Butter churning
- 637.234.3 Butter washing
- 637.234.5 Working and salting of butter
- 637.234.7 Shaping, moulding and cutting of butter
- 637.236 Ripening of butter. Butter cultures
- 637.238 Colouring and treatment of butter with additives
- 637.238.2 Colouring of butter
⇒ 667.777
- 637.238.4 Additives for aroma, flavour etc.
- 637.24 By-products of buttermaking**
- 637.247 Buttermilk
- 637.25 Types and forms of butter**
- 637.251 Fresh butter (not stored or processed)
- 637.252 Cold-stored, freezer-stored butter
- 637.253 Stable forms of butter, e.g. for tropical climates
- 637.255 Semifluid butters. □ Ghee
- 637.27 Reprocessing of butter and butter products. Processed or reworked butter**
- 637.28 Butter substitutes. Butter oil**
⇒ 664.3
- 637.3 Cheese and cheesemaking**
◇ 637.3.022 Cheese curd-forming and curd-treating equipment. Curd-breakers
◇ 637.3.027 Cheese moulds
- 637.33 Cheesemaking processes**
- 637.333 Cream or milk treatment for cheesemaking.
□ Initial treatment. Temperature treatment and preservation. Skimming. Pasteurization
- 637.334 Cheese formation and shaping
- 637.334.2 Renneting, coagulation and setting
- 637.334.3 Curd and its treatment. □ Cutting, scalding, cooking of curd. Pitching and piling of curd. Cheddaring
- 637.334.5 Cheese milling, grinding, sorting
- 637.334.7 Cheese moulding, hooping, pressing. Bandaging of cheese
- 637.336 Cheese ripening, maturing
- 637.337 Dressing. Oiling, waxing and washing cheese
- 637.338 Colouring and treatment of cheese with additives
- 637.338.2 Colouring of cheese
⇒ 667.777
- 637.338.4 Additives for aroma, flavouring etc.
- 637.34 By-products of cheesemaking**
- 637.344 Whey, as cheese by-product
⇒ 637.142.2
- 637.345 Milk sugar (lactose)
- 637.347 Whey butter. Whey cream
- 637.35 Types and forms of cheese**
- 637.352 Fresh soft cheeses. Cream cheese
- 637.353 Blue and white mould cheeses (soft and semi-soft)
- 637.353.5 Blue mould cheeses. □ Gorgonzola. Stilton. Roquefort
- 637.353.7 White mould cheeses. □ Brie. Camembert
- 637.354 Hard cheeses
- 637.354.6 Scalded cheeses
- 637.354.62 Slightly acidified cheeses. Cheeses ripened with rod bacteria. □ Emmental. Gruyere
- 637.354.64 Strongly acidified cheeses. □ Parmesan
- 637.354.8 Cheeses from low-temperature cooking. Salted cheeses
- 637.354.82 Open-texture cheeses. □ Gouda. Edam
- 637.354.84 Close-texture cheeses. □ Cheddar
- 637.356 Acid-casein cheeses
- 637.356.2 Acid-casein cheeses eaten fresh. □ Cottage cheese. Gervais. Quark
- 637.356.4 Ripened acid-casein cheeses. □ Harzer. Mainzer
- 637.357 Acid-albumin cheeses. □ Schabziger.
- 637.358 Gammelost
- 637.358 Processed cheese
- 637.38 Cheese substitutes**
- 637.4 Eggs. Egg products**
Denote eggs of specific birds etc. (especially other than domestic fowl or chicken) by '6... or '7...
◇ 637.4'659.7 Domestic duck eggs
◇ 637.4'727 Wild waterfowl eggs
- 637.41 Structure and parts of eggs**
- 637.411 Whole eggs
- 637.412 Egg shell
- 637.413 Egg white. Albumen
- 637.414 Egg yolk
- 637.417 Wind-eggs
Eggs laid with soft shell or without shell
- 637.43 Treatment of eggs**
- 637.433 Physical methods of preserving eggs in the shell. □ Cooling, heating, pasteurizing, sterilizing of eggs
- 637.434 Chemical methods of preserving eggs in the shell. □ Preserving in liquids, e.g. water glass, isinglass. Preserving in gases
- 637.435 Techniques for packing eggs (in their shells)
⇒ 621.798
- 637.437 Shelling of eggs. Removal of egg shells
⇒ 641.514.3
- 637.438 Processing of egg constituents (egg pulp)
⇒ 637.413/.414
- 637.48 Processed and synthetic egg products. Egg substitutes**
- 637.481 Dried egg. Egg powder
- 637.482 Chemical or therapeutic substances containing egg. □ Iodized eggs
- 637.483 Synthetic egg white
- 637.484 Synthetic yolk
- 637.5 Meat. Flesh products for food**
Denote meat and meat products from specific animals by '6/'8 as instructed at 637'6, 637'7 and 637'8
◇ 637.5'62 Beef (i.e. ox meat) and beef products. Veal and veal products
◇ 637.5'64 Pork (i.e. pig meat) and pork products
◇ 637.5'848 Marine turtle meat and its products
- 637.5.03 Methods of treatment of meat and flesh
- 637.5.031 Preparatory treatment. Cleaning etc. of meat
- 637.5.032 Ageing of meat. □ By enzymes
- 637.5.033 Preservation and storage of meat in general
- 637.5.034 Smoke curing. Smoking and drying of meat
- 637.5.035 Chemical curing. Salting, pickling etc. of meat
- 637.5.037 Refrigeration, freezing of meat
⇒ 637.513.8
- 637.5.039 Special treatment for consumption. Tenderizing etc.
- 637.51 Fresh meat. Meat substitutes**
- 637.512 Selection of animals and preparation for slaughter
- 637.513 Slaughtering and meat packaging techniques and equipment

637.513.1	Slaughtering and meat packaging plants and procedure ⇒ 637.53	637.53	Disposal and utilization of carcasses. Knackery ⇒ 637.513.1
637.513.11	Home slaughtering	637.54	Meat of poultry and game birds 637.54 ≡ 637.51 ◇ 637.54'65 Meat of domestic birds, poultry ◇ 637.54'72 Meat of game birds ◇ 637.54'724 Meat from partridge, grouse etc.
637.513.12	Slaughterhouses. Abattoirs	637.55	Venison. Game meat (except birds) 637.55 ≡ 637.51 ◇ 637.55'71 Meat of specific game animals ◇ 637.55'711.4 Meat of wild reindeer etc. ◇ 637.55'712 Meat of small furred animals. Rabbit and hare meat
637.513.14	Meat packaging plants	637.56	Flesh of fish, crustaceans (shellfish) etc. 637.56 ≡ 637.51 ◇ 637.56'81/83 Flesh of specific fishes ◇ 637.56'87/88 Flesh of aquatic molluscs and crustaceans ⇒ 639.38
637.513.18	Trimming of carcasses	637.6	Animal products other than food ◇ 637.6'6/8 Products of specific animals. NOTE: if not implicit in direct divisions 637.61/69 ⇒ 665.93 ⇒ 675
637.513.2	Slaughterhouse equipment, appliances	637.61	Skins. Pelts. Hides. Shells ◇ 637.61'698 Skins etc. of reptiles and some other vertebrates ◇ 637.61'848 'Tortoiseshell' (i.e. shell of marine turtles) ⇒ 675.86, 679.92 ⇒ 639.386
637.513.22	Killing and stunning devices. □ Humane bolt killer. Slaughter pistols, cartridges. Killing masks. Electric stunning and killing	637.62	Hair. Bristles. Wool ⇒ 636.93 ⇒ 675.82/83 ⇒ 677.021.153 ⇒ 677.3
637.513.27	Slaughterhouse tools. □ Knives. Saws. Splitting and skinning devices	637.63	Feathers. Down. Bird skins ⇒ 675.84 ⇒ 677.39
637.513.3	Devices for disembowelling, eviscerating. Scraping and shaving tools. Scalding devices. Boilers, autoclaves etc.	637.64	Horns. Claws. Talons. Beaks ⇒ 675.86
637.513.4	Cutting and chopping devices. □ Meat-cutting and slicing machines. Bacon slicers. Chopping and mincing mills ⇒ 621.927.76 ⇒ 637.523.4	637.65	Bones. Teeth. Tusks. Ivory
637.513.5	Weighing machines for meat	637.66	Blood. Gut
637.513.66	Containers for fresh meat	637.661	Animal blood ⇒ 631.867
637.513.68	Devices for hanging meat	637.663	Animal intestine. Gut ⇒ 675.85 ⇒ 677.39
637.513.8	Refrigerating installations. Freezers. Cold stores ⇒ 637.5.037	637.68	Civet. Musk. Castor (castoreum) ⇒ 665.529.4
637.514	Types of fresh meat. Joints and cuts of meat	637.691	Edible nests of birds
637.514.5	Minced meat ⇒ 637.521.47	637.692	Raw materials for the preparation of endocrine hormones ⇒ 663.15
637.514.7	Bones for food purposes	638	KEEPING, BREEDING AND MANAGEMENT OF INSECTS AND OTHER ARTHROPODS
637.514.8	Marrow	638.08	General questions of breeding and rearing 638.08 ≡ 636.08
637.514.9	Offal. Byproducts of carcass trimming and dressing		
637.518	Meat substitutes		
637.52	Processed, cured, prepared and cooked meats. Their production ⇒ 664.91/93		
637.521	Ready-to-cook meat products and their production		
637.521.4	Types of ready-to-cook meat product		
637.521.42	Natural ready-to-cook meat products		
637.521.44	Breaded meat products. □ Hamburger. Schnitzel		
637.521.47	Minced meat products ⇒ 637.514.5		
637.522	Selection and preparation of raw meat for processing		
637.523	Sausage production ⇒ 637.524		
637.523.2	Meat processing techniques for sausage production		
637.523.3	Sausage-making machinery and equipment		
637.523.31	Devices for mixing and kneading meats and pastes		
637.523.35	Salting tubs		
637.523.37	Boilers. Steamers. Cooking ovens		
637.523.38	Drying chambers. Smoking chambers		
637.523.4	Special machines and appliances for cutting, dicing and mincing sausage-meat and its products ⇒ 637.513.4		
637.523.7	Sausage-filling machines. Linkers		
637.524	Kinds of sausage products		
637.525	Cured, dried and smoked meats (except sausages)		
637.525.2	Bacon		
637.525.3	Ham		
637.528	Prepared and cooked meats other than those mentioned above. □ Brawn. Pickled meats. Jellied meats		

638.1	Bee-keeping. Apiculture	638.14.06	Honey production ⇒ 638.16
638.11	Establishment of apiary. Preliminary considerations. □ Site. Acquisition of bees, hives, equipment. Apiary buildings. Business management systems	638.14.07	Apiary and hive records and results
		638.141	Apiary management in general
		638.141.3	Apiary tools and equipment ⇒ 638.11 ⇒ 638.163
638.12	The honeybee (<i>Apis mellifera</i>). Classification, anatomy, physiology, psychology, behaviour and scientific study in general <i>For other Apis species see 595.799</i> <i>Denote details not listed below by :591...</i>	638.141.4	Kinds of apiary and their management
		638.141.42	Home apiaries
		638.141.43	Out-apiaries
		638.141.44	Migratory apiaries and beehouses. Migratory beekeeping in general
638.121.1	Queen bees	638.141.7	Colony management. □ Systems employing two or more queens per hive. Compound hives
638.121.2	Worker bees		
638.121.3	Drones	638.142	Hives and their parts
638.123	Subspecies, varieties, races of <i>Apis mellifera</i>	638.142.2	Fixed-comb hives and skeps
638.124	The bee colony as a whole, as an organism. □ Social behaviour of honeybees. The hive-mind, corporate personality. Spirit of the beehive ⇒ 591.552 ⇒ 591.56	638.142.3	Movable-comb hives
		638.142.31	Hives according to the method of adding combs
		638.142.38	Hive parts. □ Hive stands, frames, bodies, combs
		638.142.8	Ventilation devices. Hive entrances. Take-off platforms for bees ⇒ 638.147.7
638.124.2	The seasonal cycle. Natural growth, development, multiplication, decline and death of bee colonies	638.143	Cleaning and preserving of hives and their parts. □ Sterilizing. Painting
638.124.22	The seasonal cycle ⇒ 638.14.01	638.144	Feeding of bees. Foods
638.124.25	Multiplication of colonies. Swarming	638.145	Rearing and breeding of bees, including queens. Introduction of queen bees <i>Class human actions here. For actions of bees see 638.124</i>
638.124.3	Communication between bees by dances and other means. Social behaviour in foraging		
638.124.4	Reception, treatment, storage and use of food and other materials by bees. Reserves. Economy of the hive. □ Reception, treatment, storage and use of nectar, honeydew, saccharine foods, honey, water, pollen, propolis etc. Wax secretion and comb-building	638.145.3	Selection of bee colonies. Building up and improving strains of bees
		638.145.4	Rearing of bees. Stimulation and building up of colonies. Rearing of workers and drones
		638.145.5	Rearing of queen bees. □ Mating arrangements. Artificial insemination of queen bees
638.124.5	The home or hive. □ Selection, cleaning, modification etc. by bees. Temperature and humidity control, ventilation (fanning) by bees. Guarding, defence by bees ⇒ 591.57 ⇒ 638.142.8 ⇒ 638.147.7	638.145.6	Queen transport and introduction. Queen candy
		638.145.7	Collection, packaging and dispatch of bees
		638.147	Other manipulations
		638.147.1	Control of robbing and drifting by bees ⇒ 638.124.6
638.124.6	Relations between bee colonies. Colony consciousness. Robbing, fighting by bees ⇒ 638.147.1	638.147.2	Basic manipulations
		638.147.21	Opening the hive and subduing the bees.
638.124.7	Division of labour in the bee colony	638.147.22	Examination, handling and transfer of combs
638.124.8	Properties, qualities of colonies. □ Colony balance. Queenrightness. Queenlessness. Degree of docility. Honey-getting capacity. Morale	638.147.23	Shaking, brushing, driving bees from combs
			Uniting colonies with each other and with combless bees
		638.147.24	Adding, taking away bees (not comb) to/from a stock
		638.147.25	Getting brood combs or super-combs drawn
638.13	Foraging by bees. Bee-forage. Bee-pasture ⇒ 638.121.2	638.147.26	Taking and hiving of swarms. Hiving of combless bees
638.132	Nectar. Nectar-yielding plants ◇ 638.132(213.5) Tropical nectar plants ◇ 638.132(497.2) Bulgarian nectar plants ⇒ 581.135.4 ⇒ 581.135.5	638.147.27	Finding, marking, removal and destruction of queen bees. Finding, selection, destruction or transference of queen cells
		638.147.4	Swarm prevention and control (manipulations not involving increase)
638.135	Propolis. Propolis-yielding plants	638.147.5	Artificially increasing the colony
638.138	Pollen. Pollen-yielding plants	638.147.7	Artificial modification of ventilation, temperature and humidity in the hive ⇒ 638.124.5 ⇒ 638.142.8
638.14	Apiary and colony management. Hive-work and manipulation. Hives and their parts. Breeding and rearing of bees	638.147.8	Preparations and manipulations for the honey-flow
638.14.01	Seasons. The beekeeper's year ⇒ 638.124.22		
638.14.03	Flora. Bee-pasture ⇒ 638.13	638.15	Diseases of bees. Pests, enemies of bees. Injuries to bees
638.14.05	Wax production ⇒ 638.171	638.15-02/-099	General pathology of bees <i>638.15-02/-099 ≡ 616-02/-099</i>

638.153	Diseases affecting adult bees	638.222	Cultivated varieties of silkworm moth (Bombycidae subfamily). □ Bombyx mori
638.154	Diseases affecting the brood, bees in immature phases	638.224	Wild, semicultivated varieties of silkworm moth (mainly Saturniinae). □ Silkworms producing tasar (tusser), muga and shantung silks.
638.155	Harmful plants Specify by :582...		Antheraea species. Attacus species
638.157	Harmful insects. Other animal enemies Specify by :592/599	638.226	Processional varieties (mainly Thaumetopoeinae subfamily). □ Anaphe species. Eucheira species
638.158	Injuries and poisoning of bees (other than by harmful animals and plants). □ Injuries by flood, fire, extreme temperatures, damp, starvation, asphyxiation	638.228	Other families and subfamilies of silk-spinning moths of commercial importance. □ Yponomeuta subfamily. Pieridinae subfamily
638.158.2	Poisoning by sprays, dusts, fumes etc.		
638.16	Honey ⇒ 638.14.06 ⇒ 663.393 ⇒ 664.164	638.23	Nutrition of silkworms. Food plants ⇒ 638.244
638.162	Composition, properties and quality of honey ◇ 638.162:532.13 Viscosity of honey ◇ 638.162:543.92 Organoleptic testing of honey. Aroma and flavour of honey ◇ 638.162:577.1 Biochemistry of honey ◇ 638.162:615.28 Antiseptic properties of honey	638.232	Mulberry leaves
638.163	Extraction and treatment of honey. □ Clarification. Liquefaction. Preservation. Bottling. Packing. Transport	638.234	Plant leaves other than mulberry Specify by :582... or :634/635 ◇ 638.234:634.37 Fig leaves
638.165	Forms and varieties of honey	638.236	Effect of plant diseases, pests etc. on silkworms ⇒ 632
638.165.22	Comb honey. Section honey. Chunk honey	638.238	Physiological and biochemical aspects of silkworm nutrition. □ Essential amino acids. Vitamins ⇒ 591.13
638.165.23	Extracted, strained or pressed honey	638.24	Sericulture proper. Silk farms
638.165.8	Honey according to forage plant furnishing most of the nectar 638.165.8 ≡ 582	638.241	Nursery management in general. □ Rearing rooms. Equipment
638.165.847.52	Honey from nectar of Abietoideae, fir trees	638.242	Incubation. Hatching. Separation of newly hatched larvae
638.165.891.248	Honey from nectar of Ericoideae, heaths and heathers	638.243	Hygiene, disinfection, cleaning in silk farms
638.166	Alterations and defects in honey. Altered or deficient honey	638.244	Feeding of silkworms. □ Feeding programmes. Maintaining freshness ⇒ 638.23
638.167	Use of honey in foods, beverages, medicaments, other products ◇ 638.167:615.324 Use of honey in pharmacy ◇ 638.167:664.14 Use of honey in sugar confectionery, sweets	638.245	Care of the silkworms
638.17	Beeswax and other honeybee products	638.247	Rising of silkworms. Spinning process
638.171	Beeswax. □ Composition. Properties. Processing. Uses ⇒ 638.14.05 ⇒ 665.231	638.25	Diseases of silkworms. Pests, enemies of silkworms. Injuries to silkworms. Infections. Harmful plants. Harmful insects and other animals. Poisoning etc.
638.178.2	Pollen. Pollen pellets. Pollen-trapping ⇒ 581.33	638.252	Diseases, especially infectious diseases due to viruses and microorganisms
638.178.8	Bee venom ⇒ 591.145 ⇒ 615.814.5	638.255	Harmful plants Specify by :582...
638.19	Honeybees as pollinating agents. Use of bees by humans for pollination ⇒ 581.162.3	638.257	Harmful insects. Other animal enemies Specify by :592/599
638.2	Silkworm cultivation. Sericulture. Silk farming ⇒ 677.37	638.258	Injuries and poisoning by other agents. □ Heat or cold, damp, asphyxiation, sprays, dusts as harmful agents
638.21	Generalities. Design and equipment of sericultural establishments. Economics of sericulture	638.26	Silkworm seed (egg) collecting
638.22	Silkworms. Varieties of silkworm moth	638.262	Silkworm seed (egg)
638.221	Life cycle of the silkworm ⇒ 591	638.264	Cocoons for seed collecting
638.221.2	Egg stage	638.27	Cocoons, for silk industry ⇒ 677.021.153
638.221.4	Larval stage (the silkworm)	638.271	Cocoons
638.221.6	Pupa. Chrysalis	638.272	Bave and brin (single thread of raw silk). Physical and chemical properties of raw silk. Constituents (fibroin, sericin etc.)
638.221.8	Adult stage. Moth	638.272.6	Defects of bave. Evenness
		638.273	Collection, preservation and storage of silk
		638.28	Various products of the silk farm. Waste silk. □ Silkworm gut. Other waste products
		638.3	Cochineal insects (Dactylopius coccus) ⇒ 547.975 ⇒ 667.271
		638.4	Rearing of other insects. □ Ants. Glow-worms. Performing fleas. Crickets. Insects injurious to pests

638.5	Rearing of arachnids	639.1.081.24	Stationary hunting with lures, decoys, stool-pigeons etc. ⇒ 639.1.081.98
638.52	Cheese mites (<i>Tyroglyphus siro</i>)	639.1.081.25	Hunting with lights, mirrors etc.
638.54	Spiders (<i>Araneida</i>) ⇒ 677.375	639.1.081.3	Hunting with auxiliary animals. Hunting with hounds etc.
638.8	Keeping insects for ornamental purposes	639.1.081.31	Hunting with weapons and animals
639	HUNTING. FISHING. FISH BREEDING ⇒ 343.77 ⇒ 347.246 ⇒ 347.26 ⇒ 351.765 ⇒ 799	639.1.081.32	Hunting without weapons. Coursing. Hawking (falconry) etc.
639.1	Hunting	639.1.081.5	Hunting in open country
639.1.02	Various kinds of game	639.1.081.6	Hunting in forests
639.1.021	Furred game	639.1.081.7	Hunting in marshes and on water
639.1.022	Feathered game. Game birds	639.1.081.8	Hunting on horseback. Riding to hounds
639.1.03	Formation of game stock. Game breeds and breeding	639.1.081.9	Other methods of hunting. Hunting equipment ⇒ 623.442.6 ⇒ 623.444.6
639.1.031	Breeds. Inbreeding	639.1.081.98	Means of luring game. □ False bird calls, whistles. Decoy animals. Bait ⇒ 639.1.081.19 ⇒ 639.1.081.24
639.1.032	Crossing	639.1.09	Diseases of game. Pests, enemies of game
639.1.033	Hybridizing	639.1.091	Diseases and their treatment. Epizootics ⇒ 591.2
639.1.034	Egg laying. Nesting	639.1.092	Pests, enemies of game. □ Humans as injurious to game. Theft of game. Poaching etc. Domestic animals gone wild. Feral cats, dogs etc. Wild animals, carnivores, predators as injurious to game
639.1.04	Rearing and care of game 639.1.04 ≡ 636.083	639.1.093	Insects as injurious to game
639.1.05	General questions of hunting	639.11/.16	Animals hunted
639.1.052	Hunting grounds	639.11	Mammals as game Class see mammals at 639.24... ⇒ 639.13
639.1.053	Nature of hunting grounds. Appraisal and management of hunting grounds 639.1.053 ≡ 639.2.053	639.111	Large wild animals. Big game generally ⇒ 636.29
639.1.055	Restrictions on hunting	639.111.1	Deer. Wild pig (wild boar). Elk (moose)
639.1.055.2	In relation to time. □ Seasonal restrictions. Open and close seasons	639.111.2	Chamois. Izard. Ibex. Mouflon (wild sheep)
639.1.055.3	In relation to place. □ Conservation areas. Wildlife sanctuaries	639.111.4	Reindeer. Eland. Wapiti
639.1.055.4	In relation to means of pursuit and capture. Prohibited hunting methods	639.111.5	Antelope. Gazelle
639.1.055.5	Riding animals and vehicles for hunts	639.111.6	Wild and domesticated bovines. □ Cattle. Oxen. Bison. Buffalo. Yak
639.1.055.8	Sick and wounded game animals. Tracking of wounded animals	639.111.7	Large carnivores, predators. □ Large felines (Lion. Tiger. Leopard etc.). Hyena. Jackal. Coyote. Wolf. Bear
639.1.057	Persons involved in hunting. □ Owners of chases, hunting reserves. Masters of hunts. Hunt supervisors. Auxiliary personnel. Beaters	639.111.8	Elephant. Hippopotamus. Rhinoceros. Giraffe. Okapi. Wild equines. Zebra
639.1.058	Damage associated with hunting, and compensation. Damage caused by hunts. Damage caused by game	639.112	Small furred mammals. Small game generally. □ Rabbit. Hare. Beaver ⇒ 636.92/.93
639.1.06	Devices for care of game stock	639.113	Small carnivores, predators. □ Fox. Badger. Otter. Small felines (Wild cat. Lynx etc.). Mustelidae. Martens. Sable. Polecat (fitchew). Weasel. Stoat (ermine) ⇒ 636.934
639.1.07	Fencing. Supervision	639.12	Game birds. Wildfowl ⇒ 636.59
639.1.081	Hunting methods and procedures. Capture and destruction of animals	639.121	Snipe. Woodcock
639.1.081.1	Trapping. Traps and snares	639.123	Pheasant
639.1.081.11	Nets, particularly fishing nets	639.124	Gallinaceous game. □ Partridge. Grouse. Wild turkey
639.1.081.12	Dry-land nets. Drop nets. Bird nets	639.125	Capercaillie. Black grouse. Hazel hen
639.1.081.13	Snares. Springes. Nooses. Lariats, lassoes. Bolas	639.126	Small wild birds
639.1.081.15	Lime-twigs	639.127	Aquatic birds. Waterfowl
639.1.081.16	Cages	639.127.1	Shore birds. Waders. Gressorial birds
639.1.081.17	Spring traps. Gins. Fall traps. Pitfalls	639.127.2	Swimming birds of marsh, river and coast. □ Wild ducks. Geese. Swans
639.1.081.18	Spring guns. Explosives	639.127.3	Sea birds Except those at 639.127.9
639.1.081.19	Other methods of capture or destruction. □ Electric traps. Laying poison. Baits. Lures ⇒ 639.1.081.98	639.127.9	Cormorant. Pelican
639.1.081.2	Hunting operations. Shooting etc.	639.128	Birds of prey. Predatory birds
639.1.081.21	Hunting in battue. Driving, beating, rounding up. Hunting with shewels		
639.1.081.22	Still-hunting. Stalking		
639.1.081.23	Hunting from ambush, from hides. Hunting in migration season		

- 639.13** **Seals. Walrus. Sea-lions**
 ⇒ 639.24
- 639.14** **Crocodiles. Alligators**
- 639.15** **Other reptiles.** □ Tortoises
 ⇒ 639.248
- 639.16** **Other animals**
- 639.18** **Products of game animals, other than food**
 639.18 ≡ 637.6
- 639.181 Skins. Pelts. Hides. Shells
 ⇒ 675
- 639.2** **Fishing. Fisheries**
 ⇒ 799.1
- 639.2.03 Methods of increasing fish stocks
 639.2.03 ≡ 639.3.03
- 639.2.04 Methods of fish rearing
 639.2.04 ≡ 639.3.04
- 639.2.05 General questions of fishing
- 639.2.052 Fishing grounds, sites
- 639.2.052.2 Inland waters. Freshwater fishing
 ⇒ 639.21
- 639.2.052.3 Seas. Oceans
- 639.2.052.32 Inshore sea fishing. Coastal fishing
 ⇒ 639.22
- 639.2.052.34 Fishing in the high seas. Ocean fishing
- 639.2.053 Condition of fishing grounds. Evaluation and
 surveying of fishing grounds. Fish stock
 assessment. Forecasting, prediction of fishing
 yields
- 639.2.053.1 Oceanographic and meteorological factors
 affecting fishing
 ⇒ 551.46
 ⇒ 551.5
- 639.2.053.2 Fish stocks. Composition and density
- 639.2.053.3 Biological factors affecting fishing. Fish
 migration. Tagging, marking
- 639.2.053.4 Artificial measures to enrich fish stocks.
 □ Releasing young fish. Transplanting schools
 of fish
- 639.2.055 Restrictions on fishing
 639.2.055 ≡ 639.1.055
- 639.2.057 Persons involved in fishing
- 639.2.058 Damage to fisheries by fishing activities.
 Overfishing
- 639.2.06 Fishing vessels: their equipment and use
 For the construction of vessels see 629.562
- 639.2.064 Well vessels
- 639.2.065 Fishing trawlers
- 639.2.067 Handling and treatment of catch on board.
 □ Icing of catch
 ⇒ 664.951.1
- 639.2.068 Processing and preservation of catch on board
 (except icing). On-board processing plant,
 equipment. Factory ships
 ⇒ 664.95
 ⇒ 665.213/.215
- 639.2.07 Fish wharfs. Equipment for unloading catches
 ⇒ 627.33
 ⇒ 627.35
- 639.2.081 Fishing methods and equipment
- 639.2.081.1 Catching fish with nets, baskets, explosives
 etc.
- 639.2.081.11 Fishing nets
- 639.2.081.117 Filtering gear. Towed gear. Surrounding gear.
 Beach seines
- 639.2.081.16 Baskets. Fyke nets. Trap nets. Fish weirs etc.
- 639.2.081.18 Explosives
- 639.2.081.19 Electric fish-catching. Fish-catching using
 poisons
- 639.2.081.2 Catching fish with weapons, e.g. firearms
 ⇒ 639.2.081.5
- 639.2.081.3 Catching fish with auxiliary animals
- 639.2.081.4 Angling. Line fishing. Fishing rods and lines
- 639.2.081.42 Fishing rods. Holders and containers for fishing
 rods
- 639.2.081.43 Fishing lines. Traces
- 639.2.081.44 Hooks. Hook holders. Hook pouches
- 639.2.081.45 Corks. Floats. Sinkers
- 639.2.081.46 Bait. □ Fishing flies
- 639.2.081.5 Harpoon fishing. Harpoons and harpoon guns.
 Spear-fishing
- 639.2.081.7 Fish detection apparatus. Methods and
 equipment for locating fish (e.g. echolocation,
 asdic, sonar)
 ◇ 639.2.081.7:681.883.42 Fish location with
 echo-sounding apparatus
- 639.2.081.8 Attracting fish with light
- 639.2.09 Diseases of fish. Pests, enemies of fish
 639.2.09 ≡ 639.1.09
- 639.21** **Fishing for freshwater species. Fishing
 inland waters**
 ◇ 639.21:597.211 Lampreys
 ◇ 639.21:597.423 Sturgeons. Paddle fishes
 ◇ 639.21:597.535 Eels
 ◇ 639.21:597.551.2 Minnows and carps.
 Loaches. Chub
 ◇ 639.21:597.551.4 Catfishes
 ◇ 639.21:597.552.1 Pikes
 ◇ 639.21:597.552.3 Smelts
 ◇ 639.21:597.552.511 Salmon
 ◇ 639.21:597.552.512 Trout
 ◇ 639.21:597.556.331.1 Snooks. Perches.
 Sea basses. Sunfish. Jacks. Remoras.
 Snappers. Breams. Angelfishes
- 639.22** **Fishing for deep-sea and inshore species**
 ◇ 639.22:597.315.3 Dogfish. Bramble sharks
 ◇ 639.22:597.317 Skates and rays
 ◇ 639.22:597.541 Anchovies. Herrings.
 Shads. Sardines
 ◇ 639.22:597.551.3 Piranhas
 ◇ 639.22:597.551.4 Siluriformes: Catfishes
 ◇ 639.22:597.555.5 Cods. Hakes
 ◇ 639.22:597.555.51 Cod (*Gadus* sp.)
 ◇ 639.22:597.555.9 Frogfishes. Batfishes.
 Deep sea angler fishes
 ◇ 639.22:597.556.31 Gurnards. Rockfishes.
 Flatheads. Bullheads
 ◇ 639.22:597.556.331.1 Snooks. Perches.
 Sea basses. Sunfish. Jacks. Remoras.
 Snappers. Breams. Angelfishes
 ◇ 639.22:597.556.332.3 Blennies
 ◇ 639.22:597.556.333.7 Mulletts
 ◇ 639.22:597.556.334.1 Barracudas.
 Mackerels and tunas. Swordfish. Marlin
 ◇ 639.22:597.556.35 Flounders. Dabs.
 Halibut. Plaice. Soles. Turbot
 639.239 Other marine fishes. Angler fishes (*Lophiidae*)
- 639.24** **Hunting and capture of marine mammals
 and reptiles**
 639.245/.247.3 ≡ 599.5/.73
 639.245.1/.3 Hunting of whales. Whaling
 639.247.453 Hunting of seals in the water
 ⇒ 639.13
 639.248 Marine turtles
 ⇒ 675.86
- 639.25** **Inshore and coastal fishing**
 Only general questions are classed here

- 639.27** **Collection of aquatic molluscs.** □ Oysters
639.27 ≡ 639.4
Class studies of both capture and breeding in 639.4
- 639.28** **Capture of aquatic crustaceans etc.**
639.28 ≡ 639.5
Class studies of both capture and breeding in 639.5
- 639.29** **Harvesting or fishing of other marine products.** □ Sponges. Corals. Vegetable marine products: seaweed etc.
Class studies of both capture and breeding in 639.6
- 639.3** **Fish breeding. Pisciculture**
⇒ 626.88
- 639.3.03 Reproductive processes of fish. Propagation of fish
- 639.3.032 Hybridization
- 639.3.034 Fertilization. Laying of eggs. Fish eggs. Fish larvae
- 639.3.034.1 Natural fertilization
- 639.3.034.2 Artificial fertilization
- 639.3.04 Methods of rearing fish
- 639.3.041 Hatching out
- 639.3.041.1 Natural hatching
- 639.3.041.2 Artificial hatching. □ Incubation apparatus
⇒ 626.884
- 639.3.043 Feeding, nutrition of fish. Fish foods
- 639.3.043.13 Controlled feeding, rationing
- 639.3.043.14 Fattening
- 639.3.043.2 Foods. Diets
- 639.3.045 Acclimatization, adaptation of fish (from fresh to seawater and vice versa)
- 639.3.05 General questions of fish rearing
639.3.05 ≡ 639.1.05
- 639.3.06 Installations and equipment for fish breeding
- 639.3.07 Setting-up and running fish breeding stations
- 639.3.09 Diseases of fish. Pests, enemies of fish
639.3.09 ≡ 639.1.09
- 639.31** **Fish breeding in fresh water**
- 639.311 Fish culture in ponds
⇒ 626.886
⇒ 628.357.3
- 639.312 Fish culture in lakes
- 639.313 Fish culture in flowing water. □ Stocking of rivers
- 639.32** **Fish breeding in seawater and brackish water. Stocking of sea areas**
- 639.33** **Storage and transport of live fish in general. Fish tanks, aquariums**
- 639.34** **Breeding of fish for ornament. Ornamental fish.** □ Goldfish. Tropical fish
⇒ 636.045
⇒ 636.98
- 639.37** **Culture of individual types of fish. Products of fish culture in general**
- 639.371/.374 Individual types of fish and sea mammal
639.371/.374 ≡ 639.21/.24
- 639.38** **Various products of fish and other marine animals**
⇒ 637'8
⇒ 637.5'848
⇒ 637.56
- 639.381 Flesh and fat in general. □ Blubber. Oil, e.g. fish liver oil
⇒ 665.213/.215
- 639.382 Fish eggs. Fish roes. Caviar
- 639.383 Whalebone
- 639.385 Fish scales
- 639.386 Skins of fish and other marine animals. Hides of whale, seal etc. Swim bladders. Tails and fins
⇒ 637.61
- 639.387 Fish bones. Fishbone
- 639.388 Ambergris
⇒ 665.529.2
- 639.389 Other products. □ Teeth of fish and sea mammals
- 639.4** **Breeding of aquatic molluscs (shellfish)**
For shellfish which are Crustacea see 639.5
- 639.4.03 Reproductive processes. Propagation
639.4.03 ≡ 639.3.03
- 639.4.04 Rearing methods
639.4.04 ≡ 639.3.04
- 639.4.05 General questions of shellfish rearing
639.4.05 ≡ 639.1.05
- 639.4.06 Sites (culture beds) and installations. Containers for rearing
- 639.4.061 Natural beds
- 639.4.062 Beds for spawning and propagation. Collecting places
- 639.4.063 Beds for rearing
- 639.4.064 Beds for fattening and colouring
- 639.4.09 Diseases of shellfish. Pests, enemies of shellfish
639.4.09 ≡ 639.1.09
- 639.41** **Culture of oysters (and other pearl sources)**
- 639.411 Oysters
- 639.412 Pearl oysters. Pearl mussels
- 639.42** **Mussel culture. Edible mussels**
- 639.44** **Culture of other shellfish (edible bivalves).**
□ Clams. Cockles. Scallops
- 639.45** **Culture of marine gastropods.** □ Whelks. Winkles
- 639.5** **Breeding of aquatic crustaceans, sea urchins, leeches etc.**
639.5.03/.04 ≡ 639.3.03/.04
639.5.05 ≡ 639.1.05
639.5.06 ≡ 639.4.06
639.5.09 ≡ 639.1.09
- 639.51** **Breeding of aquatic crustaceans**
- 639.512 Shrimps and prawns
- 639.515 Spiny lobster, crawfish (Loricata)
- 639.516 Lobster
- 639.517 Crayfish (Astacura)
- 639.518 Edible crabs
- 639.519 Other crabs. □ Hermit crab. Cocoa palm crab. Sand crabs
- 639.53** **Breeding of sea urchins**
- 639.54** **Breeding of leeches**
⇒ 636.99:595.143
- 639.55** **Breeding of sea cucumbers**
- 639.56** **Marine worms of the Eunicidae and Nereidae families. Palolo worms (East Asia and Japan)**
- 639.6** **Other marine products**
- 639.61 Sponges
- 639.62 Coral
⇒ 679.93
- 639.63 Other animal products
- 639.64 Marine plants. Seaweeds. Algae
⇒ 633.529.3

64 HOME ECONOMICS. DOMESTIC SCIENCE. HOUSEKEEPING

Under 64 is classed information about the household, including household articles from the point of view of their use; also housekeeping operations and domestic work on a commercial scale (e.g. hotel-keeping, catering, cleaning, laundering)

The -1/-9 auxiliaries from 62-1/-8 and 66-9 are applicable throughout 64

- ◇ 64-182.1 Integral parts of building or structure
- ◇ 64-182.22 Fitted. Built-in
- ◇ 64-182.3 Mobile. Movable. Transportable
- ◇ 64-182.7 With adjustable, movable parts. Convertible. Adaptable
- ◇ 64-52 Automatic control

64.01	General principles of home economics. Theory and practice of organizing households. Theory of housekeeping and household administration	64.012.4	Household management <i>Details by :005...</i>
64.011	General principles of housekeeping	64.012.5	Execution. Implementation
64.011.1	Purpose and targets of housekeeping. Policy. Education of management. Relations between targets of housekeeping and targets of individuals and groups ⇒ 640.52	64.012.6	Collaboration. Cooperation. Sharing of experience. Problem solving
64.011.2	Evaluation. Establishment of indicators, standard values, target values <i>Specify methodological questions by :303.2...</i>	64.012.7	Inspection. Supervision. Control ⇒ 658.562
64.011.22	Indicators. Actual values. □ Standard of living. Level of supply. Level of mechanization	64.012.8	Maintaining secrecy. Protective measures. Security
64.011.23	Target values	64.013	Sociological and psychological aspects. □ Preference. Aspiration. Satisfaction
64.011.3	Resources. Means ⇒ 658.2	64.016	Development. Evolution ⇒ 640.16
64.011.32	Material resources	64.017	Size. Extent. Number of persons belonging to the household
64.011.34	Human resources ⇒ 331 ⇒ 64.04 ⇒ 658.3	64.018	Quality. Grade
64.011.4	Productivity. Effectiveness. Rationalization. Cost-consciousness	64.03	Money management in the household. Buying. Consumer interests, policy and research ⇒ 330.567.2
64.011.5	Manual work. Mechanization and automation in the household	64.031	Planning expenditure. Household budgeting <i>Specify by :657...</i> ⇒ 64.012
64.011.54	Mechanization	64.031.2	Household budget
64.011.6	State, circumstances of the household	64.031.3	Financial education. Budgeting advice
64.011.61	Stability	64.031.4	Available means, resources
64.011.62	Continuity	64.031.42	Income
64.011.63	Discontinuity	64.031.44	Consumer credit --- ◇ 64.031.44:332 Land loans. Property loans. Mortgages ⇒ 336.77
64.011.64	Change	64.031.6	Household expenditure. Financial management. Cost estimates. Closing the accounts. Balancing the books
64.011.65	Trouble. Breakdown	64.033	Shopping. Consumption. Consumer behaviour
64.011.7	Errors. Failures. Faults. Deficiencies. Misuse	64.033.2	Analysis. Research
64.011.8	Improvements. Reforms. Reorganization	64.033.3	Decision making
64.012	Methods in housekeeping. Planning. Organization ⇒ 001.8 ⇒ 303	64.033.4	Comparison of costs
64.012.1	Observations. Investigations. Tests	64.033.5	Buying behaviour
64.012.2	Plans. Programmes. Aids to planning	64.033.52	Routine buying
64.012.28	Decision-making	64.033.54	Impulse buying
64.012.282	Decision-making processes	64.033.56	Buying as result of a genuine decision, not as a matter of routine
64.012.282.2	Routine decisions	64.033.6	Behaviour during use or consumption
64.012.282.4	Impulsive decisions	64.034	Ordering, buying by ordering in writing or by telephone, or from travelling salesmen or commercial representatives
64.012.284	Decision-making techniques	64.035	Kinds of account. Payment methods ⇒ 336.71
64.012.284.2	Budget planning <i>Planning in relation to time and money</i>	64.036	Income and consumption studies
64.012.284.4	Planning to a target-date, deadline	64.038	Means and methods
64.012.3	Forms of organization. Organizational structure	64.038.1	Measures under public law
64.012.32	Forms of organization	64.038.2	Measures under private law
64.012.34	Organizational structure	64.038.4	Consumer representation
64.012.341	Demarcation of functions and tasks. □ Specialization. Differentiation of functions.		
64.012.343	Extension of functions Centralization. Coordination		

64.04	Housekeeping services, jobs, tasks ⇒ 64.011.34 ⇒ 647	64.066.24	Consumption of water
		64.066.26	Consumption of materials
		64.066.4	Consequences of usage and consumption. □ Wear and tear of things treated. Health and environmental consequences
64.041	Services by persons who belong to the household	64.066.8	Safety ⇒ 614.8
64.041.1	Services by members of the family or relations <i>Details by -055.5/7</i>	64.067	Appliances according to utilization of additives
64.041.3	Services by fellow occupants	64.068	Ventilation and suction devices ◇ 641.53.068 On cookers ◇ 648.525.068 On vacuum cleaners
64.041.4	Services by residential staff	64.069	Purchase. Installation. Maintenance. Removal. Costs
64.042/.045	Services, housekeeping jobs by those not belonging to the main household ⇒ 364.4	64.069.2	Selection of makes and models
64.042	Partial services by those not belonging to the main household	64.069.3	Installation. Arrangement and adaptation of the room, of space. Conditions for installation and connection
64.042.1	By family members	64.069.4	Maintenance
64.042.2	By neighbours, neighbourhood	64.069.5	Repairs
64.042.3	By strangers. By acquaintances	64.069.6	Guarantee. Customer service
64.042.4	By domestic help, external helpers	64.069.7	Disposing
64.042.5	By collective arrangement. □ Customer services in blocks of flats	64.069.72	Sale. Re-sale. Exchange
64.042.6	By maintenance and repair companies. □ Window cleaners. Laundries. Dry cleaners. Clothing repairers	64.069.74	Giving away
64.043	Complete outside service. A-Z service	64.069.76	Scrap
64.044	Special services by people or organizations outside the household	64.069.8	Costs. Profits
64.044.3	Service centres	64.08	Moving house. Removals
64.045	Regular (e.g. daily) delivery of foodstuffs, newspapers, hot meals etc.	640	TYPES OF HOUSEHOLD AND HOUSEHOLD MANAGEMENT
64.047	Household tasks. Housework. Domestic work. □ Allocation of tasks. Division of labour. Introduction of role specialization. Household management. Care for feeding, clothing, housing, family, plants, pets, garden. Requirements and upkeep of the household <i>Specify by colon combination</i>	640.1	Private households
		640.12	Household management method. □ Functions of the private household
64.048	Study of work and working methods in the household. □ Analysis of work. Design, installation and equipment of the workplace in the household. Fitting, storage and arrangement of furniture, equipment, appliances, provisions and aids for various household tasks ⇒ 612.76	640.16	Households according to family stage
		640.162	Nuclear family households according to family stage
64.05	Home economics. Housekeeping, furnishing and equipping a home etc. for particular kinds of person 64.05 ≡ -05 ◇ 641.053.2 Food for children	640.162.2	Preliminary phase
		640.162.4	Living together, married or unmarried, without children
64.06	Household appliances and machines. Labour-saving devices in the home	640.162.6	Living together, married or unmarried, with children living at home <i>Specify by -053...</i>
64.061	Purpose and possible applications	640.162.7	Living together, married or unmarried, children having left home
64.061.2	Labour saving	640.162.8	Incomplete nuclear family
64.061.4	Time saving	640.163	Households with families spanning more than one generation
64.061.6	Programmed applications	640.164	Households having more than one family
64.062	Kinds of control	640.165	Households with alternative types of family
64.063	Kinds of loading, setting up 64.063 ≡ 648.23.063	640.166	Households without a nuclear family
64.064	Kinds of unloading 64.064 ≡ 648.23.064	640.166.2	Households of independent persons. Single person households
64.065	Inserts	640.166.4	Households of more than one person
64.066	Consumption. Consequences of usage. Safety measures	640.2	Housekeeping in companies ⇒ 331.424.4 ⇒ 640.6
64.066.2	Consumption	640.23	In commercial companies
64.066.22	Consumption of energy	640.25	In public companies
		640.4	Hotel and catering industry. Hospitality management
		640.41	Tourist accommodation. Guest accommodation ⇒ 728.5
		640.412	Hotels
		640.414	Motels
		640.415	Botels (providing mooring and overnight facilities)

- 640.416 Inns. □ Coaching inns. Caravanserais (khans)
640.418 Hostels. □ Youth hostels
- 640.42 Boarding houses. Guest houses. Furnished houses, flats and rooms for rent**
- 640.43 Restaurants. Inns**
⇒ 642.5
- 640.432 Restaurants
640.433 Fast food restaurants. Self-service restaurants
640.434 Slot-machine restaurants
640.435 Snack bars. Refreshment rooms. Cafeterias. Cake and confectionery shops
640.436 Shops with facilities for the immediate consumption of food
640.437 Sales points for particular foodstuffs
640.438 Other businesses with catering facilities
- 640.44 Tea rooms. Coffee bars. Wine bars. Bars**
640.441 Tea rooms. Coffee bars
640.442 Coffee houses. Cafés
640.443 Bars. Wine bars. Public houses. Pubs
640.444 Milk bars
640.445 Shops with facilities for the immediate consumption of drinks
640.446 Stalls, open-air stands for drinks
640.447 Ice cream sales
- 640.45 Catering service. Provision of ready meals**
⇒ 641.557
640.452 To private households
640.454 To large households
640.456 To commercial households. □ Airports. Film sets
- 640.46 Clubs. Societies. Casinos**
- 640.5 Large households (institutional households)**
- 640.52 Large households according to primary function**
640.522 Institutional households. □ Hospitals
Function is not primarily housekeeping
640.522.2 Health care institutions
640.522.4 Educational establishments
640.522.5 Lodgings
640.523 Club households
Function is not primarily housekeeping and includes only parts of it
640.525 Collective, cooperative households
Function is always the solving of domestic problems
- 640.53 Large households according to the function fulfilled by the household with regard to the primary group**
- 640.54 Large households according to range of services**
640.542 Complete households
640.544 Incomplete households
- 640.55 Large households according to the authority and the power that takes economic decisions**
- 640.57 Large households according to economic decisions**
640.572 Profitability principle
640.574 Consumption principle
640.576 Principle of preserving capital
- 640.6 Centralized households. □ In service flats**
⇒ 640.2
- 640.7 Households in country houses, weekend houses and in weekend communities**
- 640.8 Temporary households. □ Of tourist parties**
- 640.85 On walking tours and journeys**
- 640.87 On walking tours, journeys and in camps with a large number of participants**
- 641/642 FOOD. COOKING. DISHES. MEALS**
⇒ 338.439
⇒ 612.39
⇒ 613.2
⇒ 614.31
⇒ 63
⇒ 663
⇒ 664
- 641 FOOD. COOKING. DISHES**
◇ 641.033 Purchasing of food
◇ 641.053.2 Food for children
- 641.1 Foodstuffs from the point of view of properties. Nutritional value**
- 641.12 Proteins. Nitrogenous foodstuffs. □ Meat. Eggs. Nuts. Vegetables**
⇒ 612.398
⇒ 613.262
- 641.13 Carbohydrates. □ Cereals. Fruits. Sugars**
⇒ 612.396
⇒ 613.263
- 641.14 Fats**
⇒ 612.397
⇒ 613.268
⇒ 613.288
- 641.16 Water**
⇒ 612.392.3
⇒ 613.31
- 641.17 Salts. Mineral constituents. Ashes**
⇒ 612.392.6
⇒ 613.27
- 641.18 Vitamins. Growth-promoting substances. Spices. Condiments. Stimulants**
⇒ 612.015.6
⇒ 612.393
⇒ 613.291
⇒ 613.292
- 641.3 Foodstuffs according to source or season**
◇ 641.3:613.26 Vegetable foodstuffs
◇ 641.3*324" Winter foodstuffs
- 641.4 Preservation of foodstuffs in the home**
- 641.42/.44 Preparation and preservation processes**
641.42/.44 ≡ 664.8.02/.04
641.437 Preservation by refrigeration
641.447 Preservation by drying, dessication
- 641.45 Improvement, preservation and aftertreatment of beverages**
641.45 ≡ 663.05
- 641.48 Home-made preserves, conserves**
Details by :664.8/.9
- 641.5 Preparation of foodstuffs and meals. Cookery**
◇ 641.5:642.14 Cooking of midday meal
◇ 641.5.06 Cooking appliances. Cooking utensils

641.51/.54	Cooking of foodstuffs ⇒ 641.55/.56	641.53.09	Cooking or kitchen heating equipment according to kind of heat transfer
641.51	Preparatory treatment of foodstuffs. Dressing foodstuffs ◇ 641.51.06 Food-preparing equipment, machines ⇒ 641.544	641.53.092	Mainly by radiation
641.512	Cutting. Slicing. Chopping. Mincing. Shredding. Comminution ⇒ 621.927.7 ⇒ 621.937 ⇒ 621.968	641.53.093	Mainly by conduction
641.513	Washing. Scrubbing ⇒ 621.928.4	641.53.094	Mainly by convection
641.514	Scraping. Peeling. Coring. Straining. Mixing. Beating etc. ⇒ 631.361.7/.8	641.53.095	By combined radiation and convection
641.514.2	Scraping. Scouring	641.532	Cooking with equipment using solid fuel
641.514.3	Shelling. Peeling	641.533	Cooking with equipment using liquid fuel. □ Cooking with oil stoves, spirit stoves etc.
641.514.4	Stoning. Coring. Destalking	641.534	Cooking with gas-fired equipment
641.514.6	Straining. Sieving. Colandering ⇒ 621.928.2	641.535	Cooking with electric equipment ⇒ 621.365
641.514.7	Squeezing. Crushing. Centrifuging	641.536	Cooking with equipment using mixed, combined heat sources (e.g. gas-electric cookers)
641.514.8	Mixing. Blending. Kneading. Beating. Whipping ⇒ 621.929	641.538	Cooking with steam and hot-water equipment. □ Steam generators
641.516	Opening of containers. Removal of closures, lids, corks etc. Can opening. Bottle opening ⇒ 672.466 ⇒ 683.564	641.539	Cooking with other heating equipment. □ Cooking with solar energy. Infrared cooking. Microwave cooking
641.517	Measurement and weighing of foodstuffs	641.54	Kitchen utensils. Cooking vessels. Kitchen accessories <i>Denote power source by -8... (from 62)</i>
641.518	Preparatory treatment with various accessories. □ Boning. Plucking. Shaping. Pricking	◇ 641.542.4-83	Steamers with electric heating element
641.52	Cooking processes using heat or cold	641.542	Cooking vessels and similar equipment ⇒ 621.642.1
641.521	Heating of food in general ⇒ 641.53/.54	641.542.1	Water boilers. Kettles
641.522	Heating in fat	641.542.2	Pots. Pans. Saucepans. Cooking dishes
641.522.2	Frying in fat	641.542.22	Pans with inserts
641.522.3	Stewing in fat	641.542.23	Casseroles
641.522.4	Baking in fat. Roasting	641.542.26	Coffee heaters. Tea heaters. Coffee machines. Percolators. Coffee, tea brewing devices. Infusers. Samovars etc. ◇ 641.542.26.065
641.523	Moist heating	641.542.3	Inserts. Filters etc.
641.523.3	Stewing without fat	641.542.4	Pressure cookers. Autoclaves
641.523.6	Braising	641.542.5	Steamers. □ Tiered steamers. Potato and vegetable steamers
641.524	Wet heating. Heating in liquids	641.542.6	Water-bath cooking vessels (double boilers, bains-marie). Sand-bath cooking vessels
641.524.2	Boiling in water. Poaching	641.542.7	Hot-air cabinets
641.524.3	Boiling in other liquids. □ Boiling in milk	641.542.72	Pans for frying, baking
641.524.4	Distilling	641.542.73	Frying pans
641.524.5	Scalding	641.542.74	Baking dishes. Cake dishes
641.524.6	Infusion	641.542.75	Oil pans. Chip pans
641.525	Steaming. Cooking in steam ◇ 641.525-987 Steaming under high pressure	641.542.75	Baking tables. Frying tables. Deep fryers. Automatic baking devices
641.526	Dry heating	641.542.8	Thawing apparatus. Warming apparatus ⇒ 641.546.2
641.526.2	Heating in hot air. Baking	641.544	Accessories for preparing foodstuffs ⇒ 641.51
641.526.3	Grilling on a gridiron	641.544.2	Small accessories
641.526.4	Roasting on a spit	641.544.22	Kitchen cutlery. Kitchen spoons etc.
641.526.5	Toasting. Grilling	641.544.24	Sieves. Strainers. Colanders
641.526.6	Gratinating. Putting a crust on	641.544.26	Cutting, chopping boards for bread, meat, etc.
641.526.7	Burning (roasting). □ Roasting coffee beans, chestnuts	641.544.3	Baking, frying and roasting accessories
641.527	Warming	641.544.5	Devices for handling, cleaning etc. of fish, meat, poultry
641.527.2	Direct warming, e.g. of drinks and other prepared foods	641.544.6	Shaping devices ◇ 641.544.6:637.2
641.527.4	Indirect warming. □ Warming in the can. Warming in a water bath (bain-marie)	641.544.7	Butter formers
641.528	Cooling	641.544.72	Supports, stands in general. Mats. Covers. Handling devices for vessels
641.528.2	Leaving to stand	641.544.74	Stands for containers. Asbestos mats. Heat distributors. Warm jackets for cooking vessels
641.528.6	Freezing <i>Only of items eaten frozen</i>	641.544.75	Cake stands
641.53	Cooking with various heating appliances <i>Denote kitchen firing equipment (kitchen ranges, stoves, cookers etc.) by .06</i> ⇒ 683.9	641.544.77	Lids for containers. Lid fasteners. Lid holders
			Devices for handling, holding securely, hanging-up. □ Grips. Handles. Pot-lifting devices. Pot holders. Tongs for pots. Oven cloths

641.544.8	Equipment for cleaning foodstuffs. Buckets, bowls, basins. Containers for use in sinks	641.565	Dishes conforming to religious and other prescriptions. Dishes for particular faiths, cults etc.
641.546	Equipment for keeping food hot, cold or fresh		◇ 641.565:28 Islamic dishes
641.546.2	Equipment for keeping food hot	641.566	Dishes for particular festivals, feast days
641.546.22	Heated compartments. Hotplates etc.	641.568	National dishes, specialities etc.
641.546.26	Insulated containers. Insulated flasks. Vacuum flasks		<i>Details by (4/9)</i>
641.546.28	Devices for warming by chemical means or heat elements	641.8	Preparation of main and side dishes. Dressing of prepared dishes. Condiments
641.546.3	Equipment for keeping food cold. □ Porous containers, covers etc. (e.g. butter coolers, milk coolers). Cooling with ice, chemicals, other means	641.81	Main dishes
641.546.4	Cooling and freezing equipment ⇒ 621.56	641.82	Appetizers. Side dishes. Soups
641.546.42	Equipment and machines for preparation of ice cubes, crushed ice	641.822	Appetizers. Entrées. Hors d'oeuvres
641.546.43	Ice compartments. Ice boxes	641.824	Side dishes
641.546.44	Refrigerators	641.827	Soups
641.546.45	Deep-freeze cabinets, chests	641.83	Salads. Compotes. Foods eaten raw, uncooked
641.546.46	Combined refrigerators and deep-freezers. Fridge-freezers	641.84	Bread with dressing or filling. Sandwiches. Bread and butter, spread etc.
641.546.5	Devices, containers, racks etc. for keeping food fresh (over a long period). □ Covers for sliced ends. Bread bins. Storage jars and boxes. Egg racks. Guards against insects	641.85	Desserts. □ Sweets. Puddings. Fruit. Cheese ⇒ 664.686
641.548	Transport and distribution of food. Conveying and serving arrangements. Food conveyors, stands, trolleys (including heated trolleys). Food vehicles. Food hoists ⇒ 621.86/.87 ⇒ 643.64	641.87	Beverages. Drinks. □ Table water. Beer. Wine. Coffee. Tea <i>Specify by :663...</i> ⇒ 613.3
641.55/.56	Preparation of meals, courses, dishes ⇒ 613.21 ⇒ 641.51/.54	641.88	Garnishings. Condiments ⇒ 613.291 ⇒ 664.5
641.55	Preparation of meals in general ◇ 641.55(083.12) Recipes. Cookery books	641.881	Salt
641.552	Preparation of meals with regard to simplicity, preparation time	641.882	Acids. Vinegar
641.553	Preparation of meals with regard to cost. Economical meals	641.883	Sweeteners. Sugar etc.
641.554	Preparation of ordinary meals for daily needs	641.885	Spices. Condiments 641.885 = 664.5 <i>Specify by :664.5...</i> ⇒ 633.82 ⇒ 635.7
641.555	Preparation of ordinary meals for special needs. □ Meals for workers engaged in heavy work	641.887	Sauces. Gravies
641.557	Preparation of ready-cooked meals for delivery or taking away ⇒ 640.45 ⇒ 642.3	642	MEALS AND MEALTIMES. TABLEWARE ⇒ 392.8
641.558	Preparation of meals in cases of need. □ Meals for soup-kitchens	642.09	Menus. Wine lists etc.
641.56	Preparation of individual dishes. Special dishes. Dietary foods ◇ 641.56:27-565.3 Christmas food ◇ 641.56:26 Jewish diets ◇ 641.56(44) French cooking ◇ 641.56-053.3 Diets for infants, babies ◇ 641.56-055.26 Diets for expectant mothers ◇ 641.56-056.24 Diets for invalids ◇ 641.56-056.253 Weight-increasing diets ◇ 641.56-056.255 Weight-decreasing diets ⇒ 641.55	642.1	Main meals. Normal mealtimes
641.561	Diets for healthy persons. Weight-reducing, slimming diets ⇒ 613.24	642.12	Morning meal. Breakfast
641.562	Children's, infants', babies' diets and dishes ⇒ 613.22	642.14	Midday meal. Lunch or dinner
641.563	Diets and dishes for the sick, invalids	642.16	Evening meal. Dinner
641.564	Vegetarian dishes ⇒ 613.261	642.17	Late evening or bedtime meal. Supper
		642.2	Light meals. Snacks <i>Denote time of day or night by "344..." and "345..."</i> ◇ 642.2"344.7" Afternoon light meal. Afternoon tea
		642.3	Meals on excursions or journeys. □ Packed lunches. Picnics
		642.4	Meals on festive occasions. Feasts. Banquets ⇒ 394.1

642.42	Festive meals for small groups. Small lunches and dinners	642.739.8	Food carriers on tracks. Mobile, rotary food carriers. Turntables
642.43	Festive meals for large groups. □ Meals for receptions, balls	642.74	Table linen. □ Table cloths. Serviettes, napkins
642.46	Cocktails. Cocktail parties	642.78	Table decoration. □ Table ornaments, flowers, vases. Table lighting. Centre-pieces
642.48	Table talk. Toasts. Speeches	643/645	THE HOME. HOUSEHOLD FITTINGS AND FURNISHINGS
642.5	Meal arrangements for larger numbers of persons, for parties of tourists and other groups. Self-service meals etc. ⇒ 640.43	643	THE HOME. THE DWELLING ◇ 643.012 Planning, organization in the home ⇒ 332.8 ⇒ 365 ⇒ 351.778.5 ⇒ 728
642.51	Fixed price meal. Table d'hôte	643.1	Site. □ Location in relation to traffic. Proximity of churches, schools, shopping areas. Proximity to services, water, gas, electricity
642.52	American plan. Full day's meals at fixed price. Full board (pension)	643.2	Ground plans. Size, position and connection of rooms ⇒ 72.012 ⇒ 72.05
642.53	European plan. Menu with fixed prices for various dishes. A la carte	643.3	Kitchens
642.54	Club system. Choice of combinations of courses for fixed price	643.31	Position of kitchen in the dwelling. □ Lighting factors
642.57	Meals where guests serve themselves with various dishes. □ Buffet, cafeteria, self-service systems	643.35	Kitchens in private households
642.59	Other matters. □ Tips. Cover charges	643.36	Large kitchens, e.g. in restaurants, hospitals
642.6	Table-laying. Waiting. Serving	643.37	Rooms ancillary to kitchens. □ Larders. Pantries ⇒ 643.47
642.61	Setting or laying of table	643.392	Combined cooking and living areas. Kitchen-parlours
642.63	Serving, waiting at table. Carving at table	643.394	Sculleries. Washing-up rooms
642.7	Tableware. Table cover, accessories. Table decoration ⇒ 642.74 ⇒ 642.739.2	643.396	Fodder kitchens. Kitchens for animal food ⇒ 636.085.6
642.72	Tableware. Vessels for food. Crockery. China, glassware etc. ⇒ 666.11/.28 ⇒ 666.5	643.398	Kitchenettes. Kitchen alcoves. Recesses for cooking
642.721	Sets, services of tableware	643.4	Dining rooms and ancillary areas
642.727	Individual pieces of tableware	643.41	Eating areas. □ Dining rooms. Breakfast rooms. Banqueting halls
642.727.2	Plates. Bowls. □ Soup plates. Pudding plates. Side plates. Slop bowls. Finger bowls	643.42	Drinking areas. □ Bars. Drinking saloons
642.727.3	Cups. Saucers	643.45	Butlers' pantries
642.727.4	Glasses	643.47	Food storage rooms ⇒ 643.37
642.727.5	Tankards. Mugs	643.48	Food preparation rooms
642.727.7	Serving plates. Serving dishes. □ Tureens. Sauce boats	643.5	Sleeping, washing, living areas ◇ 643.5-055.1 Gentlemen's rooms ◇ 643.5-055.2 Ladies' rooms
642.727.8	Decanters. Carafes. Punch bowls. Jugs. □ Inserts. Stoppers etc. Drip catchers etc.	643.51	Sleeping areas
642.73	Cutlery. Table accessories. Serving implements	643.512	Simple bedrooms
642.732	Cutlery. Table silver ⇒ 672.713 ⇒ 672.76	643.513	Bed-sitting rooms. □ Sitting rooms with alcove containing bed
642.732.2	Eating implements in general	643.515	Bedrooms and bed-sitting rooms with alcoves for bath, shower and/or lavatory
642.732.22	Knives	643.52	Washing areas
642.732.23	Forks	643.521	Bathrooms
642.732.24	Spoons	643.522	Shower baths. Shower rooms. Saunas
642.732.5	Ladles. Serving spoons. Slices. □ Soup ladles. Sauce ladles. Fish slices. Cake slices	643.523	Washrooms
642.732.7	Knives and other cutting implements for serving. □ Cake knives. Poultry shears	643.524	Lavatories
642.734	Condiment containers. Cruets. Casters	643.527	Dressing rooms. Changing rooms
642.738	Containers for other foodstuffs <i>Specify type of food by :...</i> ◇ 642.738:637.2 Butter dishes	643.528	Clothes closets. Built-in wardrobes. Wash closets. Bed closets
642.739.2	Protective tableware. □ Table mats. Stands for decanters, glasses. Beer mats. Coasters		
642.739.3	Plate stands. Plate mats. Plate racks		
642.739.4	Plate warmers		
642.739.7	Trays. Salvers		

- 643.53** Living and reception areas. □ Living rooms. Drawing rooms. Sitting rooms. Parlours
- 643.54** Children's, guests' and staff areas
 643.542 Children's rooms. □ Nurseries. Schoolrooms
 643.544 Guest rooms. Visitors' rooms
 643.548 Servants' rooms. Maids' rooms
- 643.55** Areas for particular activities
 643.552 Studies. Writing rooms
 643.553 Reading rooms. Libraries (library rooms)
 643.554 Housework areas. □ Sewing, needlework rooms. Laundries, washing rooms. Ironing rooms
 643.555 Work and hobby areas. Workshops. Woodwork, carpentry rooms etc.
 643.556 Games rooms, e.g. for billiards, table tennis
 643.557 Areas for artistic activities. Artists' studios
 643.558 Large areas for sport, dancing etc.
 643.558.2 Rooms for sport. Gymnasiums
 643.558.8 Ballrooms
- 643.56** Entrance, access and connecting areas in dwellings. □ Halls. Passages. Stairwells
- 643.57** Open spaces in dwellings. Ancillary structures
 643.572 Open spaces. Sheltered sitting areas. Patios etc.
 643.573 Terraces. Roof terraces
 643.573.2 Terraces
 643.573.4 Roof terraces
 643.575 Verandahs. Balconies (open)
 643.577 Loggias. Wintergardens. Sun lounges (closed)
- 643.58** Spaces in annexes, outbuildings. □ Areas in sheds, garages etc. Separate washhouses, laundries etc.
- 643.6** Permanent, built-in communication and conveying systems
- 643.62** Communication systems. □ Telephones. Speaking tubes. Door knockers, bells etc.
 ⇒ 654.9
 ⇒ 692.9
- 643.64** Conveying systems. □ Lifts. Hoists. Serving hatches. Chutes
 ⇒ 621.86/.87
 ⇒ 641.548
 ⇒ 692.6
- 643.8** Underground areas. Basements. Cellars
 ⇒ 624.1
- 643.82** Cellars for heating installations. Cellars for fuel. Coal cellars. Cellars with built-in oil tanks
 ⇒ 697
- 643.85** Storage cellars. □ Wine cellars
- 643.88** Protection areas in cellars. Shelters in cellars
 ⇒ 355.58
 ⇒ 699.852
- 643.9** Storage spaces. Attics. Lofts.
 □ Box-rooms. Drying lofts
- 644** INSTALLATIONS FOR HEALTH AND COMFORT IN DWELLINGS
- 644.1** Indoor climate control. Heating. Ventilation. Air conditioning
 ⇒ 628.8
 ⇒ 697
- 644.11** Type of heating
Details by :697...
- 644.12** Equipment for fires in rooms, kitchens.
 □ Pokers. Tongs. Coal scuttles. Ash bins. Fenders
 ⇒ 645.494
 ⇒ 683.9
- 644.19** Small domestic heating appliances.
 □ Bed-warming appliances. Footwarmers. Heated fabrics, rugs, blankets. Linen warmers, airers
- 644.3** Domestic lighting
 ⇒ 628.9
- 644.34** Natural lighting
- 644.36** Artificial lighting
Details by :628.93...
 ◇ 644.36:628.93.062 Indirect artificial lighting
 ⇒ 621.32
 ⇒ 683.8
- 644.6** Domestic water supply and sanitary installations
 ⇒ 628.1
- 644.61** Drinking water supply
 ⇒ 696.11
- 644.62** Domestic hot water supply
 ⇒ 683.97
 ⇒ 696.4
- 644.65** Drainage
 ⇒ 696.13
- 644.68** Hygienic and sanitary installations and fittings
Details by :696.14...
 ⇒ 628.6
 ⇒ 645.68
- 645** FURNITURE AND HOUSEHOLD FITTINGS
- 645.1** Floor coverings, fixings etc.
Denote material by -03... (Table 1k)
 ⇒ 692.53
 ⇒ 693.7
- 645.13** Floor coverings. □ Carpets
 ◇ 645.13-037.87 Linoleum floor coverings
 Floor coverings, carpets laid in one piece
 Large pieces. Floor coverings of more than one width
 Runners
 Mats. Rugs. Doormats
 Small pieces. □ Carpet tiles
- 645.132**
- 645.133**
- 645.135**
- 645.136**
- 645.138**
- 645.15** Underlays. Underfelts
- 645.16** Fixings and fastenings for floor coverings.
 □ Stair rods. Carpet stretchers
- 645.2** Wall coverings and their fittings
 ⇒ 684.75
- 645.23** Wall coverings. Wall hangings. □ Wall drapes, arrases, tapestries, carpets
 645.23 ≡ 645.13
 Wall rugs
- 645.236**
- 645.26** Fastenings and fittings for wall coverings
- 645.3** Door and window furnishings and fittings
 ⇒ 683.3

645.31	Furnishings for doors and windows.	645.425	Special types of bed (according to construction). □ Chair beds. Camp beds. Hammocks
645.312	Curtains and blinds		⇒ 685.533/.534
645.312.2	Curtains. Window drapes	645.427	Accessories for beds. □ Insect nets. Canopies.
645.312.4	Heavy curtains		Bed curtains. Valances
645.313	Net curtains. Mesh curtains. Lace curtains	645.428	Sheet holders. Bed holders. Bed airers etc.
645.315	Roller blinds		
645.316	Venetian blinds	645.43	Tables in general. □ Dining tables
	Awnings	645.431	Ordinary tables. Large tables
645.33	Curtain rails, rods and fittings. □ Brackets.	645.432	Pull-out tables. □ Sliding tables. Leaf tables
	Curtain rings. Curtain hooks	645.433	Portable, dismantlable or collapsible tables.
645.34	Fastening and guiding of curtain cords and roller-blind belts. □ Clips and tensioners.		Folding tables. □ Gateleg tables
	Adjusting, braking and locking devices	645.434	<i>Details by -182...</i>
645.35	Drawing devices for curtains, hangings, blinds. Pulls, winders etc. for opening, closing, raising, lowering	645.436	Small tables. □ Side tables. Occasional tables.
		645.438	Whatnots. Bedside tables
645.36	Safety devices. Window bars, grilles etc. to prevent falling out	645.439	Counters. Sales tables etc.
			One-legged tables. Pedestal tables
645.37	Devices for opening and closing of doors and windows. Door openers and closers. Door lifters	645.44	Transportable tables. Tea trolleys. Serving trolleys etc.
		645.442	
645.38	Protective devices against damage, soiling etc. □ Door stops, buffers. Kick plates. Finger plates	645.443	Working tables. Desks
		645.444	Writing tables: Writing desks. Typing tables
645.39	Other door and window fittings.		Reading desks. Lecterns. Choir stalls
	□ Accessories for flowers etc. as window furniture	645.447	Combinations of tables with seating arrangements. □ School benches and similar furniture
645.392	Devices for protection from condensation, rain, draught. □ Draught excluders. Weather strips		Working tables etc. for special purposes.
645.393	Insulation equipment	645.45	□ Workbenches. Laboratory benches. Kitchen tables
645.393.4	Insulating windows. Double windows. Double glazing	645.45.09	<i>Specify by :...</i>
645.395	Protection against insects. □ Insect screens. Insect nets	645.45.092	Storage furniture. □ Cupboards. Cabinets
645.396	Devices for preventing breaking-in, burglary	645.45.093	Cupboards according to arrangement, accessories
645.398	Equipment for use in wartime. □ Protective plating. Antigas seals. Blackout equipment	645.45.094	Cupboards with shelves and/or drawers
	⇒ 355.585	645.45.097	Cupboards with suspension arrangements only
			Cupboards with storage and suspension devices
645.4	Furniture and accessories	645.451	Cupboards with mirrors
	◇ 645.4-182.22 Built-in, fitted furniture		Clothes cupboards. Wardrobes. Linen cupboards. Cupboards for footwear, hats, umbrellas etc.
	◇ 645.4-182.4 Portable furniture	645.452	◇ 645.451.097 Wardrobes with mirrors
	◇ 645.4-182.63 Foldable, collapsible furniture		Shelves. Racks. Boards. Stands. Supports
	◇ 645.4-182.66 Dismantlable furniture	645.453	645.452 ≡ 684.46
	◇ 645.4-182.68 Stackable furniture		Bookcases. Filing cabinets. Cabinets with desk flaps
	◇ 645.4-182.7 Convertible furniture		⇒ 022.4
	◇ 645.4.053.2 Children's furniture	645.454	⇒ 684.457
	⇒ 672.94		Parts of cupboards. Book supports etc.
	⇒ 674.23	645.455	□ Doors. Shelves. Rails. Racks
	⇒ 684		Chests, chests of drawers, trunks for general storage
645.41	Seating. Chairs	645.456	Cupboards, chests etc. for special purposes
645.411	Simple chairs. Simple stools	645.456.1	Dressing tables
645.412	Armchairs. Easy chairs. Sofas. Settees.	645.456.3	Display cabinets. Show cases. Display tables
	Rocking chairs	645.456.5	Sideboards
645.413	Chairs, seats for special purposes	645.459	Other types of cupboard and similar furniture.
	<i>Specify by :...</i>		□ Kitchen cabinets. Bedside cabinets
	◇ 645.413:629.7.046 Aircraft seats	645.46	Accessories for furniture. □ Rolling and sliding devices. Castors. Closures. Protective and ornamental plates
	◇ 645.413:687.53 Barbers' chairs		
645.414	Multiple seating. Bench seats	645.47	Supports for body and head
	⇒ 684.436	645.471	Bolsters. Pillows. Cushions
645.416	Long-legged chairs, stools. □ Bar stools	645.475	Mattresses. □ Futons. Palliasses
	◇ 645.416.053.2 Children's high chairs	645.477	Special equipment for beds. □ Headrests.
645.417	Pouffes and similar seats. Footstools etc.		Backrests
645.42	Furniture for lying down, reclining	645.48	Bedclothes. Table linen. Towels etc.
645.421	Beds. Bedsteads		⇒ 687.268
645.422	Cradles. Cots. Children's beds	645.481	Bedclothes, bedding in general
645.423	Divans. Divan beds		

645.482	Bed covers. Blankets. Quilts. Counterpanes. Bedspreads, eiderdowns etc. Duvets. Sleeping bags ⇒ 685.533.8	646.24	Repair. Patching. Darning
645.483	Bed linen. □ Sheets. Quilt covers. Pillowcases (pillowslips)	646.26	Knitting
645.486	Bathroom and kitchen textiles. □ Towels. Dishcloths. Drying cloths (tea-towels)	646.27	Crocheting
645.49	Other household articles	646.4	Clothing. Garments ⇒ 685.2/.4 ⇒ 687/687.3
645.492	Mirrors ⇒ 686.7	646.42	Underclothing. Underwear
645.493	Accessories for pictures. □ Picture or photograph frames. Picture hooks. Picture stands, easels ⇒ 645.5	646.43	Outer clothing. □ Suits. Jackets
645.494	Screens and guards. Fireguards	646.45	Outdoor clothing. Overcoats
645.496	Ash trays. Spittoons	646.46	Footwear
645.497	Ladders. Platform steps. Stepladders	646.47	Clothes for particular occasions. □ Ceremonial wear. Morning dress. Sports clothes. Working clothes
645.497.1	Standing ladders ◇ 645.497.1-182.73 Extending standing ladders	646.48	Accessories for clothing. □ Scarves. Belts. Aprons. Handkerchiefs. Gloves. Muffs
645.497.2	Free-standing ladders. □ Platform steps. Stepladders	646.5	Headgear. Hats etc. ⇒ 687.4
645.497.5	Suspended ladders. □ Rope ladders	646.6	Care and protection of clothing ⇒ 648
645.497.6	Fixed ladders	646.61	Cleaning materials. Spot or stain removers
645.498	Letter boxes. Provisions boxes. Security boxes	646.62	Cleaning without washing. □ Beating. Brushing. Vacuum cleaning
645.5	Decorative articles. Ornaments ◇ 645.5:681.113.1 House clocks ◇ 645.5:75.051 Pictures	646.63	Chemical cleaning (dry cleaning) ⇒ 648.28
645.6	Complete room furnishings. Suites of furniture ◇ 645.6:643.51 Bedroom furniture. Bedroom suites	646.65	Cleaning of footwear. Shoe cleaning. Shoe polishing and polishes ⇒ 665.921
645.63	Kitchen furniture and equipment	646.67	Protection, preservation of clothing, especially furs. □ Dust and moth excluders (dust covers, bags etc.). Chemical protection (moth repellents etc.)
645.68	Furniture and equipment of bathrooms, washrooms, lavatories ⇒ 628.6	646.68	Devices for hanging, stretching and pressing of clothing. □ Clothes hangers, stretchers, presses. Shoe trees
645.681	Fixed furniture. □ Wash stands (washbasins) ⇒ 645.4	646.7	Body care. Beauty care. Accessories for dressing and undressing ⇒ 687.5
645.682	Utensils for bathrooms, washrooms	646.71	Body care and cleanliness in general. Bathing, showering etc. ⇒ 613.4 ⇒ 644.68 ⇒ 645.68
645.682.2	Soap dishes, containers. Dispensers of liquid soap	646.72	Care of the hair and beard ⇒ 672.714/.718
645.682.3	Towel holders. Towel dispensers. Towel-dispensing machines	646.73	Care of the teeth
645.687	Commodities for lavatories. □ Cleaning equipment. Toilet paper. Toilet-paper dispensers. Chamber pots. Commodes	646.74	Care of the nails
646/649	PERSONAL REQUISITES. HOUSEHOLD MANAGEMENT ◇ 646/649-055.2 Women and the home	646.75	Care of the skin
646	CLOTHING. BODY CARE ⇒ 391 ⇒ 613.4 ⇒ 687	646.78	Accessories for dressing and undressing. □ Bootjacks. Shoehorns. Buttonhooks. Garment holders and removers
646.1	Clothing materials (selection, purchase etc.)	647	DOMESTIC STAFF
646.2	Making and mending of clothes at home. Domestic dressmaking	647.2	Staff working in the house. Household servants and supervisors
646.21	Needlework. Sewing <i>Details by :687...</i> ◇ 646.21:687.021 Measuring ◇ 646.21:687.051.5 Patterns ◇ 646.21:687.053 Sewing machines	647.21	Supervisory staff. □ Housekeepers. Managers. Receptionists
		647.25	Domestic service staff Butlers. Major-domos. Maitres d'hotel Kitchen staff. □ Cooks. Chefs Au pairs
		647.251	
		647.252	
		647.255	

- 647.258 Staff for the care of children or invalids.
□ Nannies. Wet-nurses. Nurses
- 647.29 **Other domestic staff. Casual staff**
Class here specialist staff, especially those in large households
◇ 647.29:687.53 Hairdressers
◇ 647.29:78 Musicians
- 647.3 **Staff working outside the house.**
□ Gatekeepers. Chauffeurs. Gardeners
- 648 **WASHING. LAUNDRY. CLEANING**
Class under 648 both household processes and industrial processes, and their related equipment
◇ 648.012 Planning and organization of laundry, cleaning etc.
◇ 648.04 Kinds of service
◇ 648.06 Washing, cleaning appliances and machines
⇒ 613.4/.5
- 648.1 **Laundry and laundries in general**
⇒ 643.554
⇒ 643.58
⇒ 725.1
- 648.11 **Laundries: organization, services**
648.111 Laundries and launderettes
648.111.2 Commercial laundries
648.111.3 Coin-operated laundries. Launderettes
648.111.4 Self-service laundries. Public washhouses
648.112 Types of laundry service. □ Complete service. Repairs
648.114 Collection and delivery of washing
- 648.12 **Identification of laundry items.** □ Laundry marks. Laundry marking devices
- 648.13 **Sorting and bundling of laundry**
- 648.17 **Storage and retrieval of soiled laundry**
- 648.18 **Washing agents. Washing solutions. Substances for other treatments (bleaching, starching etc.)**
⇒ 661.185/.187
⇒ 661.43
⇒ 664.2
- 648.19 **Pretreatment of laundry.** □ Disinfection
- 648.2 **Soaking. Washing. Chemical cleaning**
- 648.21 **Soaking. Soaking vats, tubs**
- 648.22 **Washing by hand**
648.223 Washing appliances. □ Washtubs. Washboards
648.224 Wash boilers. Coppers
- 648.23 **Mechanical washing. Washing machines. Washing plant**
648.23.063 Kinds of loading
648.23.063.2 Top-loading
648.23.063.3 Front-loading
648.23.063.4 Side-loading
648.23.064 Kinds of unloading
648.23.064.1 Self-emptying
648.23.064.2 Tipping-out
648.234 Simple washing machines with fixed tub and agitator
648.235 Drum-type, rotary washing machines
648.237 Washing plant, installations
648.238 Multipurpose washing machines (e.g. washing and spin-drying)
- 648.24 **Rinsing**
648.241 Rinsing agents
- 648.242 Rinsing by hand
648.243 Machine rinsing
◇ 648.243-932.2 Continuous machine rinsing
- 648.25 **Starching**
648.251 Starches
Details by :677.027.6...
648.252 Starching by hand
648.253 Machine starching. Starching machines
- 648.26 **Bleaching**
- 648.27 **Blueing. Optical brightening**
648.271 Blueing agents
648.274 Agents for improving colour
648.276 Optical brighteners
- 648.28 **Chemical cleaning (dry cleaning)**
⇒ 646.63
- 648.3 **Water extraction. Drying**
- 648.32 **Water extraction and extractors**
648.322 Wringers (mangles)
648.326 Spin driers (centrifugal driers)
⇒ 648.235
- 648.33 **Drying (by evaporation). Drying devices**
648.333 Washing lines and accessories. □ Tensioning devices. Winders. Hooks. Poles
648.334 Clothes pegs. Peg containers
648.335 Drying frames for washing. Drying stands. Clothes horses
648.336 Drying cupboards, cabinets. Drying machines
648.336.2 Drying cupboards. Drying cabinets
648.336.6 Drying machines. □ Tumbler driers
- 648.37 **Shaking machines**
- 648.4 **Ironing. Collection and packing of washing**
- 648.41 **Ironing, smoothing, pressing in general. Theory of ironing**
- 648.42 **Hand pressing. Irons (smoothing irons). Ironing boards, tables**
- 648.43 **Machine pressing. Ironing machines. Pressing mangles**
⇒ 677.057.73
⇒ 687.054
648.431 Calender-feed machines
648.432 Ironing machines for flatwork
648.433 Roller ironing machines
648.436 Pressing and shaping machines for shaped articles
648.437 Gloss ironing machines
- 648.46 **Folding machines**
648.462 Folding machines for flatwork
648.466 Folding machines for shaped articles
- 648.47 **Collection and storage of finished laundry**
- 648.48 **Packing and wrapping of laundry parcels**
Details by :621.798...
- 648.5 **Cleaning and care of houses and dwellings**
Especially interiors and house contents
◇ 648.5.04 Cleaning services
◇ 648.5.06 Cleaning machines and equipment
⇒ 613.5
- 648.52 **Cleaning of walls, floors, furniture etc.**
648.523 Polishing equipment and machines
648.523.1 Floor polishing equipment. □ Polishing brushes. Scrubbers. Nylon discs. Cleaning, polishing cloths. Parquet floor cleaning felts
648.523.5 Floor polishing machines

- 648.524 Sweeping and brushing equipment. □ Brooms. Dustpans and brushes. Mops and buckets
⇒ 687.9
- 648.525 Vacuum cleaners
- 648.526 Dusting equipment. □ Dusters. Feather dusters. Dust mops
- 648.527 Equipment for cleaning and care of carpets and other coverings
- 648.527.5 Carpet sweepers. Carpet sweeping machines. Combined sweepers and vacuum cleaners
- 648.527.7 Machines for steaming of carpets. Machines for wet cleaning of carpets
- 648.527.8 Machines for dry cleaning of carpets. Carpet shampooing machines
- 648.528 Beating equipment. □ Carpet beaters and beating machines. Combined beaters and vacuum cleaners
- 648.529 Auxiliary equipment for cleaning and care. □ Kneeling pads. Protective devices
- 648.53 Window cleaning. Cleaning and care of glass panes, mirrors, doors etc. and their frames**
- 648.532 Window demisters, deicers
⇒ 629.067
- 648.533 Equipment for window cleaning by hand. □ Window squeegees
- 648.536 Window-cleaning machines and devices
- 648.538 Safety devices, supports, harnesses etc. to prevent falling
⇒ 614.821
- 648.54 Cleaning and care of crockery, pots and pans. Washing-up materials**
- 648.544 Washing-up, rinsing and drying equipment. □ Basins. Dish mops. Pot scourers. Draining racks
- 648.545 Washing-up machines (dishwashers)
- 648.55 Cleaning and care of metal objects, cutlery**
- 648.56 Cleaning equipment for glass objects and components**
- 648.58 Cleaning agents and other materials for home care. □ Cleaners. Rinses. Polishes**
- 648.6 Disinfection. Disinfectants**
◇ 648.6:645.475 Disinfection of mattresses
◇ 648.6:696.13 Disinfection of house drains
⇒ 615.28
- 648.7 Pest extermination. Vermin control.**
□ Traps (e.g. mousetraps). Poisons. Pesticides. Insecticides
⇒ 631.348
⇒ 632.95
⇒ 639.1.081.1
⇒ 661.16
- 649 DOMESTIC CARE OF CHILDREN, INVALIDS, GUESTS**
⇒ 173.5
- 649.1 Domestic child care**
⇒ 37.018.1
⇒ 613.95
- 649.16 Baby care. Care of infants in the first year of life**
⇒ 613.221
⇒ 641.562
- 649.8 Domestic care of the sick, invalids**
- 649.9 Care of guests and visitors. Hospitality**
⇒ 392.7

654**TELECOMMUNICATION AND TELECONTROL (ORGANIZATION, SERVICES)**

Details of telecommunications machinery by -1/-9 (from 62-1/-8 and 66-9)

- ⇒ 338.47
⇒ 347.83
⇒ 351.817
⇒ 355.343.1
⇒ 358.236
⇒ 621.39
⇒ 622.489
⇒ 623.61
⇒ 656

654.01**General public services****654.02****Telecommunications routes, lines**

654.021

Number and frequency of communications

654.022

Spatial distribution of communications. Routes

654.024

Communications by particular persons or services

◇ 654.024:35 Government communications

654.025

Communications to a large number of persons
⇒ 654.19

654.026

Communications for particular purposes

654.027

Communications according to speed, rapidity

654.027.3

Rapid transmission of messages

654.027.5

Delayed transmission of messages

654.028

Communications according to form (language, code, secrecy)

654.029.3

Indication of recipient, addressee (e.g. telephone number, telegraphic address)

654.03**Tariffs. Charges. Rates. Payment**

654.031

General principles of tariff. Alterations, increases, reductions in charges

654.032

Tariffs. Scales of charges

654.032.1

Tariffs according to number, frequency and length of communications

654.032.2

Tariffs according to place of origin and distance of communications

654.032.3

Tariffs according to time

◇ 654.032.3"345" Night rates

◇ 654.032.3"38" Holiday rates

654.032.4

Tariffs according to category of person. Rates for officials, for the press etc.

654.032.6

Tariffs for particular purposes. Tariffs according to object of communication

654.032.7

Tariffs according to speed, rapidity of communication (e.g. express rate)

654.032.8

Tariffs according to form of communication

654.032.9

Additional charges. □ Charges for insurance, for registration, for prepaid reply

654.033

Recording and control of tariff estimates

654.034

Methods of payment. □ Prepayment meters (slot machines, coin boxes). Payment by card

654.036

Places for settlement. Accounting centres

654.038

Refunds

654.039

Documents. Forms

654.04**Faults. Interruptions, delays. Complaints. Repairs****654.05****Signals. Call signs. Warning tones etc.**

⇒ 654.9

654.06**Auxiliary services using telecommunications. □ Time check**

⇒ 621.395.9

⇒ 621.396.9

654.07**Administration. Management**

⇒ 005.1

654.071

General administrative service. Staff.

□ Engineers. Operators

654.072	Administration of public use of communications. □ Booking. Connection. Transmission of messages. Supervision, control of use	654.195.6	Radio sound broadcasting
654.073	Delivery of messages to recipient, addressee	654.197	Television broadcasting
654.077	Administrative arrangements. □ Material and financial resources. Offices, accommodation. Furniture. Equipment 654.077 ≡ 35.07	654.9	Signalling. Various telecommunication services. Remote control (telecontrol) ⇒ 62-519 ⇒ 621.398 ⇒ 627.71/.72 ⇒ 629.066 ⇒ 654.05 ⇒ 656.05 ⇒ 656.25
654.1	Telegraphy. Telephony. Radio. Television		
654.14	Line telegraphy (cable, wire services) ⇒ 621.394	654.91	Optical, visual signalling ⇒ 628.975
654.141	Telegraph offices	654.912	Methods of producing optical effects. □ Signalling with colours, shapes, flags, smoke, light sources, rockets ⇒ 627.724 ⇒ 662.12
654.142	Telegraphy by private line		Signalling using written or printed characters
654.143	Telegraphy by public line 654.143 ≡ 654.153	654.913	Optical indicating and alarm devices. □ Indicator boards
654.144	Manual and automatic telegraph traffic 654.144 ≡ 654.154	654.915	Remote writing equipment ⇒ 654.145 ⇒ 681.613.5
654.145	Communication by teleprinter. Telex service ⇒ 654.916 ⇒ 681.613.4	654.916	
654.147	Special-purpose telegraphy for alarm and warning systems. □ Telegraphy for police, for fire services ⇒ 614.842.4 ⇒ 654.9	654.92	Acoustic signalling. Bell systems. Alarm systems
654.15	Line telephony ⇒ 621.395	654.922	Methods of sound production. □ Signal whistles, sirens, horns, hooters. Signal bells. Explosions, detonations. Speech sounds. Talking machines, indicators
654.151	Telephone exchanges. Telephone stations	654.923	Bell systems. Chimes etc. ◇ 654.923-83 Electrically operated bell systems ◇ 654.923-86 Mechanically operated bell systems
654.151.2	Telephone exchanges		Alarm systems and installations
654.151.4	Telephone stations. □ Call boxes	654.924	Automatic intruder and burglar alarm systems ⇒ 62-759.8
654.152	Private telephony. Internal telephone services (within a building or group of buildings)	654.924.3	Automatic fire alarm systems
654.153	Public telephony	654.924.5	Awakening devices. Alarms set for a predetermined time. Time signals (e.g. in schools, factories, offices) ⇒ 681.11.033.2
654.153.2	Local telephone traffic	654.925	Other acoustic signalling devices
654.153.3	Interlocal telephone traffic (calls between different exchanges). Trunk calls	654.926	Transmission of acoustic signals through various media
654.153.4	Long-distance telephone traffic. International calls	654.927	Transmission through air Transmission through water Transmission through the earth Reception of acoustic signals
654.154	Manual and automatic telephone traffic	654.927.1	
654.154.2	Manual telephone traffic	654.927.2	
654.154.3	Semiautomatic telephone traffic	654.927.3	
654.154.4	Automatic telephone traffic	654.928	
654.157	Emergency call services	654.93	Remote transmission of commands, messages, signals. Remote indication
654.16	Radiocommunication ⇒ 621.396 ⇒ 654.19	654.931	Mechanical apparatus for transmission of messages and signals
654.164	Radiotelegraphy (wireless telegraphy) 654.164 ≡ 654.14	654.935	Transmission of messages and signals using lamps, drop indicators or bell systems
654.165	Radiotelephony (wireless telephony) 654.165 ≡ 654.15	654.937	Transmission of messages and signals by electrical means
654.17	Facsimile and television transmission (point to point) ⇒ 621.397 ⇒ 654.197	654.938	Selection and control devices for indicator installations
654.171	Facsimile transmission	654.94	Remote indication of particular phenomena, e.g. temperature ⇒ 681.12
654.172	Television transmission		
654.173	Combined television and telephony	655	GRAPHIC INDUSTRIES. PRINTING. PUBLISHING. BOOK TRADE ⇒ 0 ⇒ 681.6 ⇒ 76 ⇒ 774/777
654.19	Broadcasting ⇒ 621.396/.397		
654.191	Broadcasting stations, studios ⇒ 621.396.7 ⇒ 621.397.7		
654.195	Sound broadcasting ⇒ 621.396.44 ⇒ 621.396.97		
654.195.2	Broadcasting over lines. Cable sound broadcasting		

655.02	Working processes <i>Further subdivided at some subdivisions of 655, e.g. 655.28.02..., 655.3.02...</i>	655.225.3	Quality and properties of relief plates etc. □ Depth of relief. Shape and size of image elements (e.g. dots). Tonal rendering
655.021	Preparatory operations	655.225.8	Production of relief surfaces according to method of representation. □ By line. By half-tone dot, autotype
655.023	Defects. Faults. Difficulties		
655.024	Processes concerning colour production	655.226	Production of flat, planographic printing surfaces. Lithography
655.025	Processes concerning the use of materials other than paper	655.226.1	Earlier planographic processes. Manual lithography. Transfer processes
655.026	Processes concerning the manufacture of specific printing products	655.226.2	Mechanical, photographic, electric, electronic and other processes for transferring the image on to planographic plate or cylinder.
655.027	Processes concerning the reproduction of illustrations		Photolithography
655.028	Finishing processes	655.226.5	Quality and properties of planographic plates etc. □ Shape of image elements. Tonal rendering
655.06	Products <i>Subdivided at some subdivisions of 655, e.g. 655.28.06..., 655.3.06...</i>		Production of diagraphic printing surfaces
655.1	Printing industry in general	655.227	Screen production
655.11	History of printing. Discovery of book-printing techniques	655.227.3	Stencil production
		655.227.7	Production of collographic and similar printing surfaces. □ Collotype. Aquatone
655.15	Printing houses, presses and related businesses ◇ 655.15:005.1 Organization of printing houses	655.228	Production of other kinds of printing surface
		655.229	
655.2	Graphic reproduction processes. Production of printing surfaces. Typesetting, composing	655.24/28	Production of type matter, letterpress
655.22	Production of printing surfaces (except composing) ⇒ 655.24/28 ⇒ 774 ⇒ 778.18	655.24	Type styles. Characters ⇒ 003 ⇒ 004.223 ⇒ 681.615.2
655.222	Reproduction, duplication of printing surfaces. Duplicating of plates, blocks, type. Conversion of printing surfaces of one process into those of another	655.241	Type body size
655.222.2	Stereotyping in metal	655.242	Type weight. □ Light, medium, bold face
655.222.22	Forming of the printing surfaces ⇒ 681.617.21	655.243	Typographic measuring systems. □ Point system
655.222.23	Moulding, casting processes. □ Centrifugal casting. Injection moulding <i>Details by :621.74...</i>	655.244	Typefaces. □ Antiqua (Old Style). Sanserif. Blackletter
655.222.25	Hardening of stereotypes by surface treatment	655.245	Special characters. □ Logotypes. Ligatures
655.222.3	Nonmetal stereotyping	655.246	Type sorts other than letter forms. □ Signs, rules, braces, dashes etc. Fleurons and other ornaments
655.222.32	Stereotyping in rubber	655.247	Type specimen books and sheets
655.222.34	Stereotyping in plastics	655.249	Lists of founts, typefaces. Sets of characters
655.222.4	Electrotyping ⇒ 621.357	655.25	Production of information carriers for typesetting. Manuscripts, originals, copy for printing
655.222.6	Production of relief printing plates by methods other than casting or moulding. □ Use of photosensitive materials	655.251	Rules for the production and treatment of copy etc. Copy preparation. □ Correction marks ◇ 655.251(035) Style manuals
655.222.8	Conversion of printing surfaces of one process into those of another. Conversion methods	655.253	Creation of information medium for typesetting. Physical production of copy (e.g. handwritten, typed, by information storage devices)
655.224	Production of recessed, intaglio printing surfaces. Gravure	655.254	Processing and conversion of copy
655.224.1	Earlier intaglio processes. □ Manual engraving. Etching. Heliogravure	655.254.2	Processing of copy
655.224.2	Mechanical, photographic, electric, electronic and other processes for transferring the image on to intaglio plate or cylinder. □ Photogravure	655.254.22	Editing ⇒ 070.4 ⇒ 808.2
655.224.3	Quality and properties of intaglio plates etc. □ Depth of recessing (cell depth). Shape and size of cells or pits. Tonal rendering	655.254.24	Technical preparation for composition. □ Marking up. Layout. Paste-up
655.225	Production of original relief printing surfaces. Process engraving	655.254.4	Conversion of copy. Production of machine-readable record. Text-processing (word-processing) operations
655.225.1	Manual production of relief printing surfaces. □ Line engraving. Cutting or incising	655.26	Typography. Graphic design. Kinds of text (material to be set)
655.225.2	Mechanical, photographic, electric, electronic and other processes for transferring the image on to relief plate or cylinder. Photoengraving	655.262	Typographic design of the text. □ Aesthetic questions. Text division and layout.
		655.267	Arrangement of illustrations Kinds of text. □ Straight or mixed material. Foreign language or script. Tabular material. Formulae
		655.28	Typesetting. Composing ⇒ 681.616

655.28.02	Operations. Processes	655.3.022.6	Printing according to mechanical principle. □ Flat-bed, cylinder, rotary. Direct and indirect or offset
655.28.021	Preparatory operations	655.3.022.7	Form of paper or other printable surface. □ Sheet. Web or roll. Containers. Other shaped objects
655.28.022	Individual procedures in copy preparation and setting	655.3.023	Printing problems. □ Difficulties with forme, ink, machine. Mackle, blurring, slurring. Friars, monks
655.28.022.1	Scientific studies	655.3.024	Colour printing <i>By number of colours</i>
655.28.022.2	Basic principles and techniques of composing. □ Line formation, lineation. Justification. Word division. Column formation. Make-up of pages and sections (gatherings or signatures). Imposition	◇ 655.3.024.2	Two-colour printing. Duplex printing
655.28.022.3	Inception of setting operation. Data input	◇ 655.3.024.3/5	Multicolour printing. Process printing
655.28.022.31	Input by manual keyboarding	◇ 655.3.024.3	Three-colour printing
655.28.022.32	Input by data carrier	◇ 655.3.024.4	Four-colour printing
655.28.022.36	Use of electronic data processing. Input from computer	655.3.025	Printing on surfaces other than paper ◇ 655.3.025:678.5 Printing on plastics in general ⇒ 666.1.058.4 ⇒ 677.027.5
655.28.022.37	Remotely controlled input. Teletype setting	655.3.026	Presswork for particular kinds of product <i>Specify by .066...</i>
655.28.022.5	Typesetting proper. Assembly and positioning of characters, matrices, type	655.3.027	Printing of pictorial matter, illustrations. Line and half-tone work
655.28.022.7	Processes for the fixing and output of the finished setting (e.g. casting of type, developing and fixing of film). Output of data carrier (cast type, photocomposed matter etc.)	655.3.028	Finishing work. □ Dusting. Drying
655.28.022.8	Correction and revision. □ Proofs. Proofreading. Proof correction	655.3.06	Products of presswork. Printed matter
655.28.023	Difficulties	655.3.062	Quality and characteristics of printed matter
655.28.026	Typesetting for particular kinds of product <i>Specify by .066...</i>	655.3.062.2	Pictorial quality. □ Fidelity of reproduction. Contrast. Sharpness
655.28.027	Insertion of pictures into text. Setting, arrangement, integration of illustrations ⇒ 655.533	655.3.062.3	Defects of printed matter. Blemishes. □ Distortion. Slurring. Showthrough. Poor registration. Interference patterns, e.g. moiré
655.28.028	Finishing work. □ Clearing and distributing of type. Tying up of set type. Storage of live matter, data carriers etc. Standing type	655.3.062.9	Spoiled sheets. Discards
655.28.06	Products of composing. Set type. Live matter	655.3.066	Products of printing
655.28.062	Quality and characteristics of matter. □ Errors. Defects	655.3.066.11	Books. Pamphlets
655.28.066	Typeset products 655.28.066 ≡ 655.3.066	655.3.066.12	Newspapers
655.281/.282	Mechanical typesetting and casting. Slug-casting	655.3.066.13	Periodicals
655.281	Line setting and casting (slug casting) processes. □ Linotype. Typograph. Ludlow	655.3.066.14	Notices. Announcements
655.282	Character-setting and casting processes. □ Monotype	655.3.066.22	Artistic reproductions
655.285	Composing by hand. □ Manual typesetting with metal type. Composing with dry transfer lettering	655.3.066.23	Jobbing work. Advertising matter
655.286	Typewriter composition ⇒ 681.612.84	655.3.066.24	Posters. Signs
655.287	Photocomposition (phototypesetting, filmsetting) ⇒ 681.616.7	655.3.066.25	Packaging
655.3	Printing operations and techniques. Presswork ⇒ 667.52 ⇒ 676 ⇒ 681.62	655.3.066.252	Labels. Tags
655.3.02	Operations. Processes	655.3.066.26	Calendars
655.3.021	Preparatory operations. □ Make-ready. Preparation of machine, forme, ink, paper	655.3.066.27	Transfers
655.3.022	Printing procedure	655.3.066.28	Transparent items
655.3.022.1	Scientific studies	655.3.066.32	Postcards. Greetings cards
655.3.022.16	Trial printing	655.3.066.34	Tickets
655.3.022.2	Stages of production. □ Proofing. Perfecting (backing up)	655.3.066.35	Forms
655.3.022.3	Inking the forme	655.3.066.356	Pressure-transfer forms. Carbonized forms
655.3.022.5	Physics of the impression area. □ Imprinting forces. Rolling pressure. Deformation phenomena	655.3.066.36	Currency. Banknotes. Securities. Postage stamps etc.
		655.3.066.43	Maps. Globes
		655.3.066.51	Three-dimensional representations
		655.3.066.52	Microimages
		655.3.066.53	Scented items
		655.3.066.9	Other products <i>Specify by :...</i>
		655.32	Relief printing, presswork
		655.322	Letterpress work
		655.326	Other kinds of relief printing
		655.326.1	Flexography (rubber-plate printing)
		655.326.3	Vacuum printing
		655.326.6	Linocut printing
		655.33	Diagraphic printing. Screen and stencil work
		655.332	Screen printing. Silk-screen printing (serigraphy) ⇒ 666.1.058.4
		655.335	Stencil printing

655.34	Planographic printing	655.52	Relations between publisher and author. Size of edition or print run. Royalties. Revised editions. Free copies etc.
655.341	Collotype printing		
655.342	Lithographic printing (from stone plates)		
655.343	Direct planographic printing from metal plates. □ Zincography	655.521	Contract between author and publisher ⇒ 347.788.1
655.344	Indirect planographic printing. Offset lithography	655.522	Royalties. □ Author's royalties. Publisher's royalties
655.35	Intaglio printing. Printing from recessed surfaces	655.523	Collaboration between author and publisher during printing and publication. Corrections. Illustrations
655.352	Intaglio printing from engraved plates		
655.353	Intaglio printing from etched plates. Gravure, photogravure printing	655.524	Contracts and agreements with authors' associations
655.36	Other printing techniques	655.525	Intermediaries between author and publisher. □ Literary agents. Arrangements relating to translation and adaptation rights
655.366	Ancillary operations. □ Ruling. Pagination. Folding. Cutting. Gluing. Coating. Stamping	655.527	Printing arrangements. □ Limited print run. Numbered copies. Author's free copies. Surplus or extra copies (overs) in a print run
655.39	Other printing and reproduction processes		
655.392	Electrostatic printing and similar processes ⇒ 778.1	655.53	Details of books, reports, periodicals, other publications
655.395	Reproduction by means of colour reaction		
655.4	Publishing and bookselling in general ⇒ 686.1	655.531	Format and quality of paper
655.41	Publishing in general. Publishing houses. Publishers	655.532	Type style and printing details. □ Typeface. Size of print area. Column width
655.411	Scientific publishing	655.533	Illustrations. Pictorial matter ⇒ 655.28.027
655.413	Specialized publishing. Publication of works on special branches of knowledge or for particular reader groups, professions etc. ◇ 655.413:050 Periodical publishing ◇ 655.413:62 Technical publishing ◇ 655.413:78 Music publishing ⇒ 087	655.534	Binding, casing, cover. Book spine ⇒ 686.1
655.418	Various publishing matters	655.535	Parts of books, reports, periodicals and other publications
655.418.1	Publishing on commission, via agent	655.535.2	Bibliographic information in publications. □ Title page. Contents lists. Bibliographic strip (in periodicals). Erratum sheets (corrigenda)
655.418.3	Publishing by the author. Self-publishing	655.535.28	Colophon. Printer's and publisher's imprints. Identification number. □ International Standard Book Number (ISBN)
655.42	Bookselling in general. Selling of publications. General, antiquarian and wholesale bookselling	655.535.4	Foreword. Introduction. Preamble. Preface. Editorial remarks
655.421	Retail bookselling. General bookshops	655.535.5	Text or body of publication. □ Textual divisions: chapters, sections, articles, paragraphs etc. Footnotes. References (in text)
655.422	Bookselling by commercial travellers, representatives	655.535.6	Advertisements. Advertisement section ⇒ 659.131
655.423	Export bookselling. Mail order bookselling	655.535.7	Indexes. Lists of figures, illustrations. Lists of references (separate from text)
655.424	Bookstalls. Sale of newspapers and periodicals. Newsagents. Newsvendors. Street news vending	655.54	Selling price of publications ⇒ 655.5.03
655.425	Antiquarian bookselling. Second-hand bookselling ⇒ 090.1	655.55	Launching of publications. Promotion, publicity. Advertising of books etc. Reviews, press notices ⇒ 659.1
655.426	Specialized bookselling ◇ 655.426:61 Medical bookselling	655.56	Sales organization ⇒ 070.33 ⇒ 656.886
655.427	Loaning, lending of books		
655.428	Bookselling by intermediaries	655.561	Organization of distribution. □ Distribution via wholesalers, factors etc. Direct mailing, dispatch by post (e.g. of periodicals)
655.428.1	Bookselling by commission houses	655.563	Terms of supply and delivery. □ Standing order. Sale or return
655.428.2	Bookselling by wholesale suppliers	655.567	Stocks of publications
655.428.3	Discount bookselling	655.57	Methods of publication
655.5	Various other questions of publishing and bookselling ⇒ 001.81 ⇒ 179.1 ⇒ 808.2	655.571	Publication on subscription
655.5.03	Prices. Tariffs 655.5.03 ≡ 658.8.03	655.572	Publication in parts or instalments
655.51	Selection of manuscripts. Offers to publish. Submission, soliciting, acceptance of manuscripts	655.578	Simultaneous publication in periodicals and as separates. Serialization in newspapers ⇒ 050 ⇒ 070.448
655.512	Appraisal of manuscripts. Publishers' readers. Commercial prospects, saleability. Target readership		

655.58	Publishing and selling by societies, guilds, book clubs etc.	656.022.9	Area of operation. Places served or visited. Stops, stations, ports of call
655.59	Publishing by organs of state, public authorities, utilities etc. Government publishing and presses	656.025	Transport of persons and goods
656	TRANSPORT AND POSTAL SERVICES. TRAFFIC ORGANIZATION AND CONTROL <i>Details of characteristics of transport and postal equipment by -1/-9 (from 62-1/-8 and 66-9)</i> ⇒ 338.47 ⇒ 338.48 ⇒ 343.346 ⇒ 347.763 ⇒ 347.79 ⇒ 347.8 ⇒ 351.81 ⇒ 354.4 ⇒ 355.69 ⇒ 629 ⇒ 658.286	656.025.2	Transport of persons. Passenger transport ⇒ 656.072
656.01	General organization	656.025.4	Transport of goods. Freight transport ⇒ 656.073
656.013	Stations. Terminals. Motorway complexes. □ Hotel, restaurant facilities	656.025.6	Transport of persons and goods on a large scale
656.015	Stopping places. Resting places. Parking places	656.027	Transport according to speed, rapidity. Slow and express transport
656.017	Technical aspects of vehicular operations (operational techniques), including performance	656.029.4	Transport with shared traffic. □ Transport on common routes
656.017.1	Preliminary operations and preparations. Operations after coming to rest, after arrival	656.03	Tariffs. Transactions. Rates. Charges. Tolls. Transport costs and fares ⇒ 658.8.03
656.017.2	Driving, navigating operations in general ◇ 656.017.2:005.963.1 Training of drivers	656.031	Principles, structure of tariff
656.017.21	Navigational computations	656.031.2	Gross tariff
656.017.22	Driving technique	656.031.3	Net tariff
656.017.24	Manoeuvring techniques	656.031.4	Cost price tariff
656.017.26	Operations for on-board maintenance of the vehicle (by the crew)	656.031.5	General principles of tariffing 656.031.5 ≡ 658.8.035
656.02	Transport routes, services and special characteristics	656.031.8	Alteration of tariff. □ Increase, raising of rates. Reduction, lowering of rates
656.021	Traffic frequency. Traffic density. Traffic observation	656.032	Tariffs for persons and their luggage
656.021.2	Frequency of transport. Traffic density	656.032.1	Tariffs according to number and frequency of journeys
656.021.22	Low density	656.032.13	Season ticket rates (for single route or for network). □ Zone tickets
656.021.24	High density. □ Traffic congestion	656.032.14	Tariffs for given number of journeys (e.g. for six-trip or ten-trip tickets)
656.021.5	Route capacity	656.032.2	Tariffs according to locality and distance. □ Transit or through-rate. Return fare. Rate for round trips, tours
656.022	Routes. Lines. Services	656.032.3	Tariffs according to time of journey ◇ 656.032.3:345" Night rates
656.022.1	Regular routes. Scheduled services	656.032.34	Holiday rates. □ Rates for public holidays, rest days, other holidays. Holiday excursion rates ◇ 656.032.34:792 Rates for theatre visits ◇ 656.032.34:796 Rates for visits to sporting events
656.022.2	Temporary routes or services	656.032.4	Tariffs according to category of person. Concessionary fares ◇ 656.032.4-053.4 Child's fare ◇ 656.032.4-053.9 Old person's fare ◇ 656.032.4-059.2 Group fare. Party fare
656.022.3	Irregular services. Occasional, special services	656.032.5	Tariffs for groups of persons, for parties
656.022.31	Taxi service	656.032.6	Tariffs for luggage
656.022.32	Charter service	656.032.7	Tariffs according to speed, rapidity of transport ◇ 656.032.7-026.512.1 Express rates
656.022.33	Excursions. Holiday travel	656.032.8	Tariffs according to form of travel accommodation. Rates according to class of travel ◇ 656.032.8-021.477 Second class. Club class. Coach class ◇ 656.032.8-021.478 Third class. Economy class. Steerage ◇ 656.032.8-021.479 First class. Luxury class
656.022.4	Opening, alteration and discontinuation of routes	656.032.9	Special tariffs ◇ 656.032.9:645.41 Charges for seat reservation ◇ 656.032.9:645.421 Charges for sleeping accommodation ◇ 656.032.9-027.565 Free travel ◇ 656.032.9-027.566 Excess charges
656.022.5	Schedules, timetables. Arrival and departure	656.033	Tariffs for freight. Goods rates
656.022.6	Unforeseen or accidental modifications to journey. □ Cancellation, interruption, postponement of journey. Extension, rerouting, diversions of journey	656.033.1	Tariffs according to number and frequency of journeys
656.022.8	Routes according to importance and mode of transport	656.033.2	Tariffs according to locality and distances
656.022.81	Main routes	656.033.3	Tariffs according to time of journey. □ Night rates. Rates for public holidays
656.022.82	Secondary routes. Feeder routes		
656.022.83	National routes (internal routes) and services		
656.022.86	Cabotage. Coastal navigation		
656.022.87	Combined air and sea transport		
656.022.88	Combined air and rail transport. □ For freight: air-rail through-transit freight service		

656.033.4	Tariffs according to consignors and consignees of the goods. □ Special rates for public authorities	656.052.2	Operations and procedures before departure or after arrival
656.033.6	Tariffs for specific purposes. □ Tariffs according to nature of the goods. Rates by volume, by properties, for produce, for bulk goods. Charges for particular objects, e.g. bicycles ⇒ 620.2	656.052.3	Moving off. Starting. Taking off <i>Use with direct divisions of 656, adding (2...) auxiliaries (Table 1e) if required. See examples at 656.7</i>
656.033.7	Tariffs according to speed, rapidity of transport. Normal and express goods rates	656.052.4	Control during journey. En-route navigation. En-route driving. En-route manoeuvres
656.033.8	Special tariffs. □ Rates for collecting, bulking. Rates for carriage with limited liability. Dead freight	656.052.42	Journey in straight line. Rectilinear travel
656.033.9	Extra charges. □ Charges for insurance, registration etc. Rates for reply-paid and cash-on-delivery (COD) service etc. Charges for loading, transshipment, warehousing etc. Port or harbour dues, wharfage	656.052.43	Speed. Duration. Distance
656.034	Tariffs for transport of animals 656.034 = 656.032	656.052.44	Speed changes. □ Acceleration. Deceleration, braking
656.035	Collection of money, fares	656.052.45	Direction changes in general
656.035.2	Methods of fare collection	656.052.46	Direction changes in the horizontal plane. □ Turning. Reversing
656.035.21	Ticket offices. Sale of tickets by operating staff	656.052.47	Direction changes in the vertical plane. Changes of altitude, height, depth. □ Climbing, ascent. Diving, descent
656.035.22	Self-service ticket machines ⇒ 681.173	656.052.48	Travel with regard to obstacles. □ Hazard avoidance. Obstacle avoidance. Manoeuvring with regard to other vehicles
656.035.3	Ticket control. Ticket inspection	656.052.5	Travel in particular conditions, especially meteorological. Travel in various weathers
656.035.4	Measures against irregularities. Penalties, fines for travelling without valid ticket etc.	656.052.52	Travelling 'blind'
656.035.6	Payment for reserved seats, berths, cabins etc. Reservation dockets	656.052.53	Travel through fog, mist, rain
656.035.7	Time of payment	656.052.54	Travel in the cold, on ice, on snow
656.035.71	Prepayment. Payment in advance	656.052.55	Travel during storms, gales
656.035.77	Subsequent payment. Payment in arrears	656.052.56	Travel on bad surfaces
656.036	Places for settlement	656.052.57	Travel under sudden changes in conditions. □ In eddies, whirlwinds. Loss of anchor
656.037	Customs matters <i>Details by :339.543...</i>	656.052.59	Acrobatic travel. Stunt riding, driving etc. Aerobatics
656.038	Delivery of goods	656.052.7	Approach navigation. Completion of journey
656.039	Travel documents. Tickets. Vouchers	656.052.72	Approach navigation ◇ 656.61.052.72 Harbour approach ◇ 656.7.052.72 Airfield approach
656.039.1	Tickets etc. for passengers and luggage in general	656.052.74	Completion of journey. Stopping. □ Docking ◇ 656.61.052.74 Stopping of ship. Landing. Docking. Mooring. Anchoring ◇ 656.7.052.74 Stopping of aircraft or spacecraft. Descent. Landing. Mooring
656.039.4	Documents for goods transport. Freight documents. □ Bills of lading. Cargo manifests. Way-bills. Consignment notes ⇒ 347.795	656.052.8	Instruction, practice, training in vehicle control
656.039.6	Documents for transport of animals	656.052.9	Journey reports. Logs
656.039.7	Customs and excise documents. □ Documents relating to trade quotas	656.053	Traffic restrictions and control. Regulation of traffic
656.04	Damage. Delays. Complaints. Liability ⇒ 347.796 ⇒ 347.824	656.053.1	Measures to reduce volume of traffic
656.043	Damage. Delays in transit. Complaints	656.053.2	Restrictions on speed. Speed limits
656.045	Liability	656.053.3	Restrictions on direction. □ One-way systems. Roundabouts
656.045.2	General questions. Kinds of liability. □ Absolute liability (liability without fault). Liability for negligence, delay, stopovers	656.053.4	Restrictions on vehicle weight or dimensions. Prohibition of specific kinds of vehicle
656.045.3	Limitation of liability	656.053.41	Restrictions on length, width, height of vehicles
656.045.4	Exclusion of liability. Nonliability. Saving clauses ('exemption from liability' clauses)	656.053.42	Restrictions on vehicle weight
656.045.5	Liability of fixed-location organizations. Liability of stations, ports etc.	656.053.43	Restrictions on type of load
656.045.6	Liability of the carrier: driver, captain, pilot etc.	656.053.431	With persons as load. Limit to number of passengers
656.045.7	Liability of owner or keeper	656.053.432	With goods, freight. Load dimension and weight limits
656.045.8	Liability of unlawful driver or pilot	656.053.5	Waiting restrictions. Prohibition of stopping
656.05	Traffic. Driving. Traffic signs and signals ⇒ 351.81	656.053.6	Restrictions on staying (longer-term). Parking restrictions. Prohibition of parking, mooring, landing
656.051	General principles of traffic movement	656.053.63	Regulation of parking, mooring, landing. □ Parking meters. Mooring fees
656.052	Driving of vehicles. Navigation. Manoeuvres <i>For technical aspects of navigation use 629.072.1</i> ⇒ 629.05	656.053.7	Restrictions on traffic noise. □ Restriction of audible warning systems (e.g. of use of horn on motor vehicles) ⇒ 351.759.4 ⇒ 534.836 ⇒ 625.098 ⇒ 628.517.2
656.052.1	Navigation. Route planning		

656.053.8	Obstruction of traffic routes. Closing of thoroughfares to traffic. Keeping access to buildings clear	656.065.8	Revictualling. Supply of food and drink
656.054	Traffic signals ⇒ 654.9	656.07	Administration, management of transport, traffic etc. <i>The 656.07... auxiliaries are not applicable at 656.82</i>
656.054.1	Signals given by traffic officers	656.071	General administration. Grades and functions
656.054.2	Signals given to traffic officers	656.071.1	Command. Senior technical staff. □ Captain, pilot. Engineers. Mechanics. Radio, telegraph operators
656.054.3	Signals made by route users to each other. □ Indication of speed changes, braking, halting, direction changes ⇒ 656.057.88	656.071.2	Lower-grade technical staff. □ Assistants, mates. Able seamen. Firemen, stokers
656.054.8	Information signs	656.071.3	Administrative services and staff on the vehicle. □ Accountancy section, pursers etc. Domestic, catering section. Handling of baggage and goods. Inspection, security service
656.055	Fixed, permanent traffic signs 656.055 ≡ 656.053 ⇒ 625.745.6 ⇒ 625.746.5	656.071.4	Station and port services. □ Directorate (stationmaster, harbourmaster, airport controller). Accounts section. Traffic control. Domestic, catering section. Baggage, luggage handling. Porters. Inspection, security section
656.055.9	Warning signs. Prohibitory and compulsory signs. □ Danger, hazard signs. Signs for bends, junctions, level crossings etc. Traffic priority signs	656.071.6	Hygienic and medical services
656.055.96	Other signs. □ Signs for traffic islands (refuges), direction of traffic flow, skid hazards etc. Route signs. Direction signs. Distance signs. Place-name signs	656.071.7	Pilotage. □ Map reading ◇ 656.071.7-051 Guides. Pilots
656.056	Movable, temporary signs. Halt signs ⇒ 625.746.5	656.071.8	Maintenance and repair service. Cleaning service
656.056.4	Traffic lights	656.072	Passenger services and formalities ⇒ 656.025.2
656.057	Lights and beacons. Light signals (except traffic lights) ⇒ 628.975	656.072.1	Bookings, ticket sales. Registration of passengers
656.057.1	Lights for marking out path (flare-paths etc.)	656.072.14	Bookings via booking agencies ⇒ 656.079
656.057.3	Lights for various purposes. Illuminated signs 656.057.3 ≡ 656.053	656.072.2	Boarding (entraining, embarkation etc.). Changing, transfer between routes or vehicles. Alighting, disembarkation
656.057.4	Guiding lights, beacons. □ Lights at harbour entrance ⇒ 627.71	656.072.27	Stopovers. Enforced stay. Awaiting connections
656.057.7	Small illuminated signalling devices (fixed or independent). Intermittent lights (flashing beacons), e.g. for fire service, for police	656.072.3	Distribution of passengers on the vehicle
656.057.8	Vehicle lighting and lamps ⇒ 628.94	656.072.4	Classification of passengers
656.057.83	Lights according to colour <i>Denote specific colours by -026.613... (Table 1k)</i> ◇ 656.057.83-026.613.1 Red lights	656.072.41	Passengers according to class of travel
656.057.87	Lamps on vehicles, for illuminating the road. Headlamps	656.072.42	Passengers according to number
656.057.88	Exterior lights on vehicles. Position lights (side-lights, rear-lights). Illuminated direction indicators ⇒ 656.054.3	656.072.44	Passengers according to kind of journey. □ Tourists
656.06	Ancillary services	656.072.45	Passengers on official business ◇ 656.072.45:355.08 Military personnel as passengers
656.062	Service stations in general 656.062.6/.8 ≡ 662.6/.8 ⇒ 656.1.016	656.072.46	Passengers according to destination
656.064	Provision of fuel and supplies to service and filling stations	656.072.5	Supervision, inspection of passenger transport. □ Passport control. Quarantine control. Inspection by the operator ⇒ 656.073.5
656.064.6/.8	Supply of various fuels (to filling stations) 656.064.6/.8 ≡ 662.6/.8	656.072.6	Care of passengers. □ Accommodation. Catering. Entertainment
656.065	Provision of fuel and supplies to vehicles ⇒ 629.082 ⇒ 629.084	656.072.7	Relations with passengers. Supervisory measures. Conduct, behaviour of passengers
656.065.2	Supply of spare parts	656.073	Goods transport services and formalities ⇒ 656.025.4
656.065.3	Supply of fuel. Refuelling. Bunkering. Tanking up. Filling ⇒ 725.1	656.073.1	Booking and registration of freight. Acceptance of goods for transit
656.065.4	Supply of lubricants	656.073.2	Freight handling ⇒ 621.798
656.065.5	Supply of coolants ⇒ 621.643.54	656.073.21	Loading
656.065.6	Filling with buoyant gas	656.073.23	Transfer of load. Transshipment
656.065.7	Vehicle wash facilities	656.073.235	Container transport
		656.073.24	Change of destination. Change of means of transport
		656.073.25	Unloading
		656.073.27	Temporary storage. Warehousing
		656.073.28	Transport within terminal or port
		656.073.3	Distribution of goods in the vehicle
		656.073.37	Empty running of conveyance

656.073.4	Classification of goods or freight <i>Denote shape or form of goods by -4... (from 62-4...)</i>	656.076.5	Lending, loan of vehicles. Other arrangements with a third party
656.073.41	Particular objects as freight ◇ 656.073.41:629.322 Bicycles as freight	656.076.6	Laying-up of vehicles. Interruption of service
656.073.42	Freight according to ownership. □ Passenger luggage. Property of third parties. Property of the crew, carrier etc.	656.076.7	Assistance, auxiliary services to vehicles (not emergency). □ Towage, tug service ⇒ 626.7 ⇒ 627.718 ⇒ 656.087
656.073.43	Freight according to various characteristics. □ According to value (Valuables, money etc. Goods of medium or little value). Postal items. Packages. Heavy or bulky goods	656.076.8	Change of vehicle use. Use in different operation
656.073.435	Fragile goods	656.078	Measures for promotion and development of transport. Competition and coordination in transport
656.073.436	Dangerous goods or freight ⇒ 614.75 ⇒ 620.26 ⇒ 656.833.3	656.078.1	Cooperation between transport undertakings. □ Sharing, pooling of traffic. Common operation of routes, lines. Joint organization, equipping etc.
656.073.438	Corpses or carcasses as freight ⇒ 614.63	656.078.2	Agreements with consignors or forwarding agents (giving preference to carrier)
656.073.44	Freight according to kind of vehicle required <i>Specify by :629...</i>	656.078.3	Facilities for dispatch and payment
656.073.45	Freight according to kind of stowage required	656.078.4	Rebates. Discounts. Price reductions, concessions
656.073.451	Stowage in holds	656.078.8	Competition. □ Rival forms of transport.
656.073.452	Stowage in bunkers	656.079	Competition in the same form of transport
656.073.453	Stowage in specially protected compartments, in strongrooms		Travel agencies, bureaux ⇒ 656.072.14
656.073.454	Stowage in compartments for explosive materials or ammunition	656.08	Accidents. Assistance in accidents <i>The 656.08... auxiliaries are not applicable at 656.82</i> ⇒ 614.85/.86 ⇒ 656.04
656.073.455	Stowage in refrigerated compartments. Cold storage	656.081	General matters. Investigation
656.073.456	Stowage in tanks	656.082	Derailments. Overturning of vehicles. Falling of vehicles (e.g. aircraft crashes)
656.073.458	Stowage on deck	656.083	Breaks, ruptures of couplings
656.073.46	Freight according to destination and length of journey	656.084	Collisions, crashes. □ Collisions with other vehicles or craft. Collisions with obstacles. Running over
656.073.48	Freight according to management and disposal. □ Undelivered goods. Goods refused by carrier. Lost and found (lost property). Losses and thefts in transit. Unclaimed goods ⇒ 347.7 ⇒ 347.232.1 ⇒ 351.759.5 ⇒ 656.878	656.085	Various other accidents. □ Sinking. Foundering
656.073.5	Supervision and inspection of goods traffic. □ Official inspection, customs. Inspection by the operator. Inspection by other agencies	656.085.1	Accidents due to intrinsic defects of vehicle. Unfitness for use (Unroadworthiness, Unseaworthiness, Unairworthiness). Vehicle wear and tear ⇒ 656.091.2
656.073.7	Management of goods	656.085.2	Accidents due to natural forces generally. □ Due to earthquake, flood, bad weather, ice, fog
656.073.8	Damage to goods or freight	656.085.3	Shipwreck
656.073.85	Intrinsic defects. Damage caused by natural forces	656.085.4	Beaching. Stranding. Running aground
656.073.851	Intrinsic defects. □ Faults of construction. Wear and tear (scuffing, cracking etc.). Leakage. Short weight. Melting. Ullage	656.085.5	Fires. Conflagrations
656.073.852/.856	Damage caused by natural forces <i>656.073.852/.856 ≡ 656.085.2/.6</i>	656.085.6	Explosions
656.073.86	Damage caused by humans <i>656.073.86 ≡ 656.086</i>	656.086	Damage caused by humans
656.074	Services and formalities for transport of animals <i>656.074 ≡ 656.072</i>	656.086.1	Damage caused by crew, vehicle personnel
656.075	Domestic services on transport. □ Catering services. Stewards ⇒ 656.071.8 ⇒ 656.072.6	656.086.13	Through carelessness
656.076	Kind of use of vehicles	656.086.14	Through mischief. Barratry
656.076.1	Private use of vehicles. Pleasure vehicles. Use of vehicles on private business, in commerce	656.086.15	Through excessive fatigue or drunkenness
656.076.2	Use of vehicles for public transport. Use by common carriers	656.086.16	Through navigational error
656.076.3	Charter of vehicles. Hiring of vehicles (with driver)	656.086.2	Damage caused by hostilities, enemy action etc. (e.g. seizure, blockade, armed force. Piracy, theft)
656.076.4	Leasing, hiring, rental of vehicles (without driver)	656.086.3	Damage caused by shore or ground personnel (servicing, maintenance staff etc.)
		656.087	Assistance to vehicles in emergency. Towing ⇒ 656.076.7
		656.088	Vehicle recovery, refloating etc. ⇒ 627.76
		656.089.2	Exercises, practice in safety measures. □ Lifeboat drill
		656.091	Documents, certificates, licences relating to vehicles and their operation

656.091.1	Licences and permits authorizing the operation of vehicles. □ Driving licences. Navigating, pilots' licences	656.14	Pedestrian traffic. Carriage by persons, animals
656.091.2	Documents relating to the vehicle. Vehicle registration documents, log books. Certificates of fitness for service (certificates of roadworthiness, seaworthiness, airworthiness) ⇒ 347.823.34 ⇒ 629.017 ⇒ 656.085.1	656.142 656.145	Pedestrian traffic Transport on animals. Riding. Pack-animals, sumpters
656.091.3	Permits relating to the engine and equipment of the vehicle	656.18	Transport by pedal cycle, by motorcycle. Cycle traffic
656.091.4	Documents relating to use of the vehicle. □ Accommodation permits (e.g. berthing papers)	656.183 656.183.5 656.186 656.187	Transport by pedal cycle Transport by motorized pedal cycle, moped Transport by motorcycle Transport of goods by motorcycle with sidecar, motorized tricycle
656.091.5	Documents indicating ownership and nationality	656.19	Other kinds of road transport. Wheelless traffic. □ Sledge traffic
656.091.6	Excise documents. Customs clearance papers. Customs permits (triptyques) etc.	656.2	Rail transport. Rail traffic ⇒ 351.812 ⇒ 625.11.5
656.091.7	Safe-conduct documents ⇒ 351.755.61 ⇒ 351.756	656.21	Operation of railway tracks, buildings, stations ⇒ 725.1
656.11/5	Land transport	656.211	Siting and layout of passenger stations, halts, stops
656.1	Road transport	656.211.3 656.211.4 656.211.5	Junction stations Terminus stations Details of passenger stations. □ Halls. Waiting rooms. Platforms. Booking offices. Barriers. Facilities for passengers and trains
656.11	Highways, roads, bridges and their use. Organization of road traffic ◇ 656.11:625.711.1 Organization of traffic on trunk roads. Arterial traffic ◇ 656.11.021.24 Road traffic congestion ◇ 656.11.021.24:159.942.6 Strong emotions caused by road traffic congestion. Including: 'Road rage' ⇒ 351.811 ⇒ 625.71.8	656.211.6 656.211.7	Station transport. Vehicles for connections between stations. □ Station buses Connections with shipping services. Railway ferries
656.11.053.1	Measures to reduce volume of road traffic. □ Financial measures. Tolls, road pricing, congestion charging	656.212 656.212.4 656.212.5 656.212.6	Siting and layout of goods stations Shunting tracks Marshalling yards. Train formation Handling, loading tracks. Goods lifting and handling equipment
656.117	Temporary use of roads for non-transport purposes. □ Use of roads for meetings, ceremonies, fairs, street markets	656.212.7	Details of goods stations. □ Loading platforms. Halls. Covered loading areas. Business offices
656.12	Wheeled transport. Haulage, cartage, carriage. Conveyances (including animal-drawn, horse-drawn traffic)	656.212.8 656.213	Weighing installations. Load gauges Special stations. □ Coaling stations. Stations for livestock, stockyards. Dock stations. Private stations, sidings
656.121	Public conveyances in towns. □ Vehicles for hire	656.216 656.216.2	Individual installations on the track Level crossings. Crossing gates ⇒ 625.162 ⇒ 656.265
656.125	Goods transport. Haulage, cartage. Parcel carriage and delivery. Carts and vans ⇒ 656.025.4 ⇒ 656.073	656.216.3	Stopping places
656.128	Private conveyances. Private carriages (especially horse carriages)	656.22	Commercial organization of railways. Train services
656.13	Motor transport. Motor-vehicle traffic ◇ 656.13.052.8 Instruction in use of motor vehicles. Driving instruction. Driving schools	656.222	Running of trains ⇒ 656.022
656.131	Public motor vehicles. Motor vehicles for hire. Taxis ⇒ 656.076.31.4	656.222.1 656.222.2 656.222.3	Itinerary. Route. Places and areas served Speed limits. Loads Make-up of trains. Type of rolling stock. Class of carriage
656.132	Motoromnibuses (omnibuses). Bus services	656.222.4 656.222.5	Track diagrams. Capacity of lines Passenger services. Indicators of route. Schedules, timetables ⇒ 656.022.5
656.135	Haulage by motor vehicles. Goods transport by lorries, vans ⇒ 656.025.4 ⇒ 656.073	656.223 656.223.1	Use and distribution of rolling stock Passenger stock. Passenger coaches and trains
656.136	Motor-transport trains. Vehicles with trailers. Road trains	656.223.2	Freight stock. Goods wagons and trains. □ Exchange between networks, and return of empty wagons ⇒ 656.078.1
656.137	Agricultural tractors. Other special-purpose motor vehicles ⇒ 631.371		
656.138	Private motor transport. Private cars, automobiles		

656.224	Passenger train services. □ Departure. Arrival. Control ⇒ 656.025.2 ⇒ 656.072	656.27	Operation of branch lines with light traffic (but belonging to main networks). Operation of light railways. Simplification of services and procedures
656.225	Freight or goods train services. □ Bulking, make-up of loads ⇒ 656.025.4 ⇒ 656.073	656.3	Transport by special railways, industrial railways, works railways
656.226	Baggage, luggage service. Parcel service ⇒ 656.073.42/.43	656.33	By railways of special construction 656.33 ≡ 625.3
656.228	Publicity. Service to customers ⇒ 658.818 ⇒ 659.11	656.34	By elevated and underground railways, tramways etc. 656.34 ≡ 625.4 Alternatively, class tramway services at 656.4
656.23	Financial and commercial matters of railways ◇ 656.23.03 Railway tariffs, rates, fares	656.35	By aerial ropeways and cableways. By telfers etc. 656.35 ≡ 625.5
656.236	Charges for transport other than the rail journey itself. □ Charges for collection and delivery. Charges for connections with land or water transport	656.4	Transport by light railways 656.4 ≡ 656.2
656.25	Safety measures. Signals ⇒ 629.4.066 ⇒ 629.4.067	656.5	Other forms of land transport ⇒ 621.86/.87 ⇒ 622.6
656.251	Signals in general. Signal shapes, colours	656.52	Transport and traffic inside large buildings ⇒ 658.286.2
656.252	Hand signals and manually controlled signals. Signals on trains. Surveillance of trains in general. Checking of trains in stations. □ Signals along the train. Head and tail signals. On-board station indicators	656.56	Conveying of fluids (liquids, gases) by pipeline ⇒ 621.643/.644 ⇒ 622.691/.692
656.253	Fixed signals on tracks and at stations. Station indicators (destination boards etc.) at stations. Signals at tunnels and crossings	656.6	Transport by water ⇒ 347.79 ⇒ 351.813 ⇒ 626/627
656.254	Long-distance communications systems. Telecommunications in rail transport ⇒ 629.4.05	656.61	Transport by sea. Maritime navigation. Shipping ◇ 656.61.052 Control of vessels. Navigation. Manoeuvres ◇ 656.61.052:551.326 Navigation etc. through floating ice ◇ 656.61.052.4 En-route navigation at sea
656.254.1	Telegraphy. Telephony. Radio. Television Details by :654.1...	656.611	Premises. Buildings
656.254.2	Audible warnings. Special warning devices	656.611.2	Shipping offices
656.254.4	Communication between stations and trains during the journey	656.612	Vessels. Ships. Merchant fleet. □ Composition of fleet. Shipping losses, wrecks
656.254.5	Dispatching system	656.613	Secondary transport for shipping. Connecting, feeder services (by land, air or water)
656.254.6	Protection of trains in emergency	656.614	Types of transport
656.255	Shuttle operation on single-track lines	656.614.2	Passenger transport by water. Passenger shipping ⇒ 656.025.2 ⇒ 656.072
656.256	Block systems	656.614.3	Goods transport by water. Freight, cargo shipping ⇒ 656.025.4 ⇒ 656.073
656.257	Central control of signals and points. Loose and rigid connections. □ Signal boxes ⇒ 625.151.3	656.615	Operation of ports, harbours ⇒ 627.2 ⇒ 656.071.4
656.258	Locking, engaging. Release, disengaging	656.615.8	Tug service in harbours ⇒ 656.076.7
656.259	Other safety devices	656.616	Operation of docks ⇒ 627.3
656.259.1	Safety devices actuated by the train. □ Retarders. Rail brakes ⇒ 621.337.6 ⇒ 625.151.6 ⇒ 625.156	656.618	Coastal, inshore shipping. Coasting. □ Tugboat navigation. Barging and towing ⇒ 656.022.86 ⇒ 656.076.7
656.259.2	Release, disengagement of signals and brakes on the train by means of the track	656.62	Transport by inland waterways. River and canal traffic 656.621/.626 ≡ 656.611/.616
656.259.3	Safety devices for electric trains, acting by current reduction or cut-out		
656.259.4	Block signals actuated by the train		
656.259.5	Explosive signals for railways. Rail detonators		
656.259.6	Locking of signals and points. Free-fall signals		
656.26	Ancillary operations, services		
656.261	Collection and delivery of goods. Delivery service		
656.262	Railway buffets, restaurants, refreshment rooms. Railway hotels		
656.265	Gatekeeper, crossing-keeper duties ⇒ 656.216.2		

656.627	Bridge and lock services. Other ancillary works ⇒ 627.7	656.816.31	Postal sorting installations. Sorting machines
656.627.2	Bridge services. Operation, maintenance etc. of bridges ⇒ 624.2/8	656.82	Management and general operation of postal service <i>The 656.07... and 656.08... auxiliaries listed under 656 are not applicable here</i>
656.627.3	Lock services. Operation, maintenance etc. of locks ⇒ 626.4	656.83	General conditions of acceptance of mail
656.627.4	Boat hoist services. Operation, maintenance etc. of lifting gear for vessels ⇒ 626.5	656.831	Preparation of mail. Making-up of packets etc.
656.628	Traction on rivers and canals. Inland tugboat navigation. Barging and towing on inland waterways	656.831.2	Addresses. Name and address of recipient. Additional information (e.g. name and address of sender)
656.629	Local, short-distance water transport. □ Water taxis. Water buses. River boats. Passenger steamers	656.831.3	Use of tie-on or adhesive labels, transparent address panels (e.g. in window envelopes) etc.
656.65	Transport on lakes. Lake navigation and traffic	656.831.4	Items without addresses or with simplified addresses
656.66	Transport by ferryboat. Ferry services. Ferry traffic	656.831.5	Packaging. Characteristics of packages. □ Weight. Size, dimensions. Sealing
656.7	Transport by air. Air traffic ◇ 656.7:632.982.4 Flights for crop spraying ◇ 656.7:656.86 Transport of mail by air. Airmail service ◇ 656.7:659.138.6 Flights for advertising, publicity ◇ 656.7.052.3 Take-off. Ascent ⇒ 533.6.015.1 ◇ 656.7.052.3(21) Taking off from land ◇ 656.7.052.3(26/28) Taking off from water ◇ 656.7.052.4 In-flight operations ◇ 656.7.052.74 Descent. Landing. Mooring ⇒ 533.6.015.2 ⇒ 347.82 ⇒ 351.814 ⇒ 656.834.3	656.831.7	Combined, bundled items. Items fastened together. Attachments. Enclosures
656.71	Landing places. Airfields. Aerodromes. Airports and ancillary services 656.71 = 656.61 ⇒ 625.717 ⇒ 725.1	656.832	Acceptance of mail
656.713	Feeder services	656.832.1	Unconditional acceptance of mail
656.73	Commercial air traffic	656.832.2	Conditional acceptance of mail
656.74	Noncommercial air traffic	656.832.4	Refusal of mail. Exclusion from postal service
656.8	Postal services and administration ◇ 656.8:061.1(100) International postal conventions (e.g. Universal Postal Union, UPU) ⇒ 351.816	656.832.6	Restrictions on import, export and transit
656.80	Generalities on the postal service	656.833	Restrictions on acceptance of mail
656.801	Organization of the postal service	656.833.1	Maximum or minimum size or weight
656.803	Charges. Rates. Fees. Expenses	656.833.2	Maximum or minimum sum
656.81	Branches, departments of the postal service ◇ 656.81:005.912 Office management aspects ⇒ 656.07	656.833.3	Restrictions on account of content of items. □ Fragile or perishable contents. Liquids. Live animals. Dangerous packages. Packages with prohibited contents ⇒ 614.75 ⇒ 620.26 ⇒ 656.073.436
656.811	Personnel of the postal service	656.833.6	Offensive, indecent, immoral items. Obscene mail. Items prejudicial to law and order ⇒ 343.542
656.815	Administrative and operational services. Technical services	656.833.8	Items unsuitable for postal conveyance because of shape or any other reason
656.815.1	Administrative departments of the postal service. Central and regional administration	656.834	Mail liable to surcharge. Mail subject to special treatment
656.815.3	Postal service points. Post offices. Sub-post-offices etc.	656.834.1	Registered items. Registered mail
656.815.6	Technical departments of the postal service	656.834.3	Urgent or express items. Air mail
656.816	Operational plant, equipment	656.834.4	Mail with advice of shipment, delivery, payment. Mail with transit note etc.
656.816.3	Technical installations and equipment of the postal service	656.834.8	Recorded-delivery items ⇒ 656.883.1
		656.835	Bulky, cumbersome mail. Fragile items
		656.835.1	Prepayment of postage. Franking. Postage stamps. Philately
		656.835.11	Prepayment of postage with printed validation.
		656.835.111	Postage stamps. Prepaid postal stationery
		656.835.112	Postage stamps
		656.835.113	Postage stamps with general validity. □ Special issues: commemorative stamps, charity stamps etc.
		656.835.114	Postage stamps for particular kinds of mail (e.g. parcel post stamps) or transit (e.g. air mail stamps)
		656.835.115	Postage stamps for official business etc.
		656.835.116	Postage stamps for government use, for armed forces, for various organizations
		656.835.117	Stamps without prepayment value. □ Postage-due stamps. Fiscal stamps.
		656.835.12	Revenue stamps
			Stamps for private postal services. □ Hotel postage stamps
			Form of issue of postage stamps. Blocks, rolls, sheets etc. of stamps
			Prepaid postal stationery (with embossed, impressed or printed-on postage stamps).
			Prepaid postcards, letter cards, envelopes etc.

656.835.13	Reply coupons	656.883	Postal money collection and payment services
656.835.16	Sale of postage stamps. □ Sale at post offices, by agents etc. Stamp vending machines. Day of issue. Period of sale ⇒ 681.13	656.883.1	Postal money collection service ⇒ 656.834.4
656.835.168	Sale of used postage stamps	656.883.11	Cash-on-delivery service
656.835.17	Validity of postage stamps. Misuse, forgery, falsification, substitution etc. of stamps	656.883.12	Collection of bills
656.835.18	Questions relating to design and production of postage stamps. □ Designs, art work, lettering, colours of stamps. Printing and misprints. Gumming. Perforation	656.883.15	Special collection services
656.835.3	Prepayment without use of postage stamps	656.883.3	Postal remittance and payment service
656.835.31	Lump-sum prepayment. Postage subscription	656.883.31	Postal orders <i>Class here telegraphic postal orders</i>
656.835.33	Prepayment using franking machines ⇒ 681.173.7	656.883.32	Post-office orders. Money orders. Postal notes
656.835.34	Prepayment in cash	656.883.35	Special postal payment services
656.835.91	Philately. Stamp collections <i>For stamp collecting as hobby use</i> 379.824:656.835.91 ⇒ 686.82	656.884	Postal cheque clearance and credit transfer service. Post-office giro ⇒ 336.717.1
656.84	Posting of mail	656.885	Post-office savings bank service ⇒ 336.722
656.841	Posting in post offices, over the counter	656.886	Post-office subscription service for serials, newspapers, periodicals ⇒ 050
656.842	Posting at postal vehicles and craft (handing in to vans, boats etc.)	656.887	⇒ 070.33 ⇒ 655.56/571
656.843	Posting in post boxes. Kinds of post box, pillar box etc. Collection, clearing of mail from post boxes ⇒ 625.748.46	656.889	Post-office passenger transport service.
656.845	Other modes of posting	656.96	Post-office conveyance of persons
656.847	Hours of opening of post offices. Hours of counter business, of posting		Non-postal services
656.848	Handling of mail after posting	657	General forwarding agencies.
656.848.3	Date stamping. Cancelling, postmarking		Intermediaries, brokers, commission agents for carrier services 656.96 ≡ 656
656.85	Dispatch of mail. Exchange of mail between countries	657.05	ACCOUNTANCY ⇒ 005.521
656.851	Preparation of mail for dispatch. Sorting and routing of mail. Coding of address, destination (post codes) ⇒ 681.187	657.1	⇒ 336.717.16
656.852	Dispatch of mail	657.11	⇒ 336.722.6
656.853	Arrival and receipt of mail (at office of destination)	657.12	⇒ 347.719
656.86	Transport, conveyance of mail ⇒ 621.367.87 ⇒ 656.11.7	657.122	⇒ 347.728.1
656.87	Distribution, delivery of mail	657.123	
656.871	General conditions of delivery	657.124	
656.872	Ordinary delivery to addressee. House delivery of mail	657.126	
656.873	Collection of mail by addressee. Post restante. Private boxes, PO box etc.	657.15	Management accounting
656.874	Delivery by courier (special messenger). □ Express delivery	657.16	
656.878	Undeliverable items. Returned mail. (Redirection. Return to sender etc.)	657.2	Bookkeeping methods, systems, techniques
656.88	Individual postal services	657.21	Single-entry bookkeeping
656.881	Letter-post service		Double-entry bookkeeping
656.881.1	Letter-post items with personal communications. Letters, postcards etc.	657.212	Transfer bookkeeping
656.881.3	Letter-post items without personal communications	657.213	Tabular bookkeeping
656.881.33	Printed matter	657.22	American system of bookkeeping
656.881.34	Commercial papers		Duplicate, simultaneous bookkeeping.
656.881.35	Samples of merchandise	657.221	□ Kalamazoo system. Loose-leaf bookkeeping
656.881.4	Insured mail. Letters and packets with declared value ⇒ 368.26	657.222	
656.882	Parcel post service	657.223	
		657.224	Financial, budgetary bookkeeping
		657.225	Public sector bookkeeping in combination with double-entry bookkeeping
			Account keeping. Recording, entering of transactions. Booking of entries ◇ 657.2:005.41 Recording, entering of transactions during development
			Accounts: definitions, functions, purpose. Sets of accounts. Debit, credit. Incomings, outgoings Debit Credit
			Opening, keeping, closing and reopening of account books Opening and organization of account books Keeping, running of account books Reorganization of account books during trading (within business or financial year) Closure of account books (e.g. following liquidation) Reopening of account books

657.23	Cross-entries. Adjustments. Cancellations. Correction of errors	657.372	Balancing of accounts. Methods, techniques for preparation of balance sheet ◇ 657.372:005.1 Theory of balances. Static and dynamic balances
657.24	Evidence of transactions. Bookkeeping documents	657.372.1	Drawing-up of balance sheets in general
657.242	Documents of cash transactions. □ Receipts. Notes of quittance. Cancellation of bills, invoices	657.372.12	Profit-and-loss analysis ⇒ 657.44
657.243	Documents referring to goods movements. Dockets for goods inwards, goods outwards	657.372.14	Closing of books and carrying forward of balances
657.243.1	Bills. Invoices	657.372.2	Valuation ⇒ 332.64 ⇒ 657.92
657.243.2	Consignment notes. Dispatch advice notes. Bills of lading, shipping etc.	657.372.3	Adjustments for decrease of value. Depreciation, amortization. Writing off
657.244	Credit documents (as bookkeeping evidence) ⇒ 336.777 ⇒ 347.457 ⇒ 347.74	657.372.4	Adjustments for increase of value
657.244.1	Bills of exchange. Money orders, drafts. Promissory notes. Letters of credit	657.372.5	Goodwill ⇒ 347.715 ⇒ 657.421.3
657.244.2	Cheques. Money transfer documents	657.375	Kinds of statement or balance sheet
657.244.5	Warrants (for goods or money)	657.375.1	Annual statements. Trading balances
657.244.6	Commission notes	657.375.2	Fiscal balances
657.244.7	Lists of bills for discount	657.375.3	Special, extraordinary statements. □ Balances at formation, amalgamation, reorganization, takeover, liquidation, bankruptcy, insolvency
657.246	Bookkeeping documents of internal transactions. Internal vouchers etc.	657.375.6	Consolidated balance sheets
657.25	Physical presentation of books of account. Layout, arrangement of accounts. □ Bookkeeping scripts	657.375.7	Condensed balance sheets
657.28	Methods of payment	657.4	Accounts. Accounting systems. Cost accounting ◇ 657.4:005.51 Accounting plan. Compulsory systems of accounting
657.282	Cash payments	657.41/.45	Financial accounting. Accounts
657.286	Remittances	657.41	Capital accounts ⇒ 330.14 ⇒ 347.72.033 ⇒ 658.14
657.3	Budgets. Estimates. Closure of accounts. Business records. Balances	657.411	Own capital (of enterprise) ⇒ 658.147
657.31	Budgets. Estimates. Financial plans ◇ 657.31:005.584.1 Budgetary control ⇒ 336.14	657.411.1	Capital accounts. Private capital
657.312	Kinds of budget (general, current, variable)	657.411.3	Share capital 657.411.3 ≡ 336.763.2
657.312.2	Financial budgets	657.411.6	Reserve capital
657.312.3	Business, trading budgets	657.411.8	Profit-and-loss accounts (from point of view of own capital) ⇒ 657.44
657.312.5	Fixed and variable budgets	657.412	Outside capital (borrowed, loan capital) ⇒ 658.148
657.313	Calculations before and after trading. Comparison of estimated and actual expenditure. Variations from estimates	657.412.1	Non-negotiable bonds. Debentures. Mortgages. Pledges ⇒ 336.763.3 ⇒ 347.27 ⇒ 347.728.3 ⇒ 347.798
657.32	Books of account in general. Account books	657.412.2	Negotiable bonds
657.322	Cash books	657.412.4	Long-term and short-term bonds 657.412.4 ≡ 657.424
657.33	Memorandum books. Day books. Journals	657.412.6	Insurance reserves. □ Life-insurance reserves. Fire-insurance reserves
657.34	Ledgers	657.412.7	Provident funds. □ Welfare, pension, sickness-benefit funds
657.342	Posting journal entries to the ledger	657.42	Accounts of working, floating funds
657.35	Subsidiary books of account	657.421	Fixed assets. Invested capital ⇒ 658.152
657.352	Order books. Books of purchases (bought books). Sales books ⇒ 657.422.8 ⇒ 658.813	657.421.1	Tangible assets, their conveyancing and amortization (writing off). Real estate and movables
657.353	Secondary, auxiliary cash books	657.421.3	Intangible assets and their amortization (writing off). □ Rights, concessions, patents etc.
657.354	Bill books. Bill diaries (due-date registers). Bank books (passbooks). Bank transfer books		Goodwill. Trade secrets ⇒ 347.715 ⇒ 657.372.5
657.356	Stock or warehouse books ⇒ 657.371 ⇒ 658.787		
657.36	Trial balances. Rough balance. Debit-and-credit balance. Interim balance		
657.37	Annual closure of accounts		
657.371	Stocktaking. Inventories ⇒ 657.356 ⇒ 658.787		

657.422	Floating, circulating assets. Liquid capital ⇒ 658.153.2	657.471	Costs. Kinds of cost. Cost centres. Cost units
657.422.1	Money. Cash. Funds in bank. Open credit. Call money	657.471.1	Kinds of cost. Production costs ◇ 657.471.1:311.14 Overhead expenses. Indirect costs. Administrative costs. Operation costs. Average costs
657.422.2	Bills, receivable and payable		◇ 657.471.1:311.15 Marginal cost. Marginal utility
657.422.3	Credits for redemption		◇ 657.471.1:658.8 Marketing costs
657.422.4	Various securities		◇ 657.471.1:659.1 Publicity costs
657.422.5	Coupons for encashment		◇ 657.471.1-022.214 Single cost
657.422.6	Stocks of raw materials. Intermediate products (half-finished goods) in store ⇒ 658.272/.273	657.471.11	Material costs. Materials pricing. Stock valuation. Current market price. Average price. FIFO (first in first out). LIFO (last in first out). Other systems
657.422.7	Stocks of finished products (goods for sale) ⇒ 658.62 ⇒ 658.784	657.471.12	Labour costs. Wage and salary costs. Social security costs
657.422.8	Purchase accounts. Sales accounts. Turnover or trading accounts ⇒ 657.352 ⇒ 657.444	657.471.122	Social security costs. Social benefits costs
657.423	Trading or operating assets. Working capital	657.471.13	Site costs. Accommodation costs
657.423.2	Forward transactions, futures (in securities or goods) ⇒ 336.764.2	657.471.14	Plant costs. Costs of machinery, equipment
657.423.3	Work in progress (orders in hand). Construction, manufacture etc.	657.471.16	Notional costs (expenses assumed for costing purposes) Offsetting of internal services
657.423.5	Agencies. Offices. Branches	657.471.6	Cost centres
657.424	Investments ⇒ 336.767	657.471.61	General cost centres
657.424.2	Short-term investments (against security, mortgage etc.)	657.471.62	Production cost centres. □ Machinery as cost centre
657.424.4	Long-term investments	657.471.63	Production services cost centres
657.425	Redemption or amortization accounts	657.471.64	Materials cost centres
657.429	Various assets	657.471.65	Administration or management cost centres
657.43	Personal accounts	657.471.66	Marketing or sales cost centres
657.431	Kinds of personal account. □ Current accounts. Interest accounts ⇒ 336.717.4	657.471.7	Cost unit accounting
657.432	Debitor accounts. □ Payments to suppliers	657.471.72	Total cost
657.433	Creditor accounts. □ Liabilities, commitments. Payments from customers	657.471.74	Orders cost. Job cost
657.435	Partnership accounts	657.471.76	Single output cost. Unit cost
657.437	Accounts for agents, representatives etc. for purchasing and selling	657.474	Processing of costs
657.439	Other personal accounts	657.474.2	Relations between commercial accounting and cost accounting
657.44	Profit-and-loss accounts ⇒ 347.72.035 ⇒ 657.372.12 ⇒ 657.411.8 ⇒ 658.155	657.474.22	Cost distribution sheet. Cost ledger
657.442	Results of various business operations. Incidental profits and losses. Unforeseen expenses	657.474.23	Operational (factory) bookkeeping
657.443	Various commissions and interest	657.474.5	Calculation of costs. Costing. Calculation methods
657.444	Sales accounts ⇒ 657.422.8	657.474.51	Actual cost calculation
657.445	Variations of stock	657.474.52	Standard costing
657.446	Nonoperational expenses and revenues. □ Taxes. Insurance	657.474.55	Direct and indirect costing. Marginal costing
657.447	Turnover. Calculation of business	657.474.57	Calculation methods ◇ 657.474.57:338.46 Service costing
657.45	Other kinds of account	657.474.571	Calculation by division. Equivalence methods of calculation
657.451	Distinction between general and special accounts or group and subsidiary accounts	657.474.573	Additions and surcharge calculation
657.454	Adjustment accounts. Clearing accounts. Controlling accounts. Contra accounts	657.478	Evaluation, appraisal of costing
657.47	Costing. Cost accounting ◇ 657.47:005.591.46 Centralized cost accounting ⇒ (085.6) ⇒ (085.7) ⇒ 338.5 ⇒ 338.58 ⇒ 658.7 ⇒ 658.8	657.478.2	Estimates (preliminary, interim and final)
		657.478.5	Uniform methods of cost accounting
		657.478.7	Short-range profit estimating
		657.478.8	Cost comparisons. Comparison over a period of time. Projected and actual costs. Method comparison
		657.479	Special problems of cost accounting
		657.479.2	Effect of capacity usage ratio and output level on costs
		657.479.3	Cost reduction or lowering, e.g. through mass production
		657.479.4	Accounting for residues (wastes), by-products, linked products
		657.479.5	Discretionary factors
		657.6	Official examination and verification of accounts. Auditing ◇ 657.6-051 Auditors ◇ 657.6-051-057.3 Internal auditors (staff auditors) ◇ 657.6-051-057.54 External auditors ⇒ 005.942 ⇒ 336.146

657.62	Analysis and verification of balance sheets	658.14/17	Financial matters in general ◇ 658.14/17(047.32) Financial reports
657.63	Examination, verification, audit of account books	658.14	Financing. Funding of enterprises. Raising of capital ◇ 658.14:005.21 Financial policy ⇒ 330.14 ⇒ 657.41
657.631.6	Relations with the client. Reports on state of finances	658.141	Level of capital. Amount of capital
657.631.8	Compulsory auditing. Statutory obligation of auditing	658.144	Direct issue of securities, shares etc.
657.632	Frauds and errors ⇒ 343.537 ⇒ 657.23	658.147	Own capital. Share capital 658.147 ≡ 657.411
657.633	Technical details of auditing. □ Detailed audit. Internal checks	658.148	Outside capital. □ Borrowed, loan capital. Capital from third parties. Short-term capital (negotiable bonds). Long-term capital (debentures, mortgages) 658.148 ≡ 657.412
657.635	Auditors' certificates, reports, statements	658.149.3	Conversions. Transfers
657.92	Appraisal. Assessment. Technique of estimating and valuation ◇ 657.92:336.2 Valuation assessment for taxation ◇ 657.92:657.47 Valuation for costing ⇒ 657.372.2	658.149.5	Repayment. Redemption
657.922	Valuation technique, particularly for real estate (property appraisal) ⇒ 332.64	658.15	Private financial management. Financial administration of enterprises
658	BUSINESS MANAGEMENT, ADMINISTRATION. COMMERCIAL ORGANIZATION ◇ 658:005.51 Corporate planning ⇒ 005.71 ⇒ 33 ⇒ 347.7	658.152	Capital investment. Fixed assets. Use of funds 658.152 ≡ 657.42
658.1	Forms of enterprise. Finances	658.153	Circulating, floating assets. Business capital. Liquid assets
658.11	Kinds and forms of enterprise ⇒ 334	658.153.2	Kinds of business capital 658.153.2 ≡ 657.422
658.112	Site, location, place of business	658.153.8	Rate of circulation of business capital
658.112.2	Principal site (head office, main works etc.)	658.155	Profit. Loss. Earning power. Trends in profitability. Yield. Trading profit, return ⇒ 657.44
658.112.3	Subsidiary sites (branch offices, works etc.)	658.155.2	Distribution of profits. Dividends
658.114	Forms of enterprise in general. Private businesses ⇒ 334.722 ⇒ 347.72	658.155.3	Losses
658.114.1	Sole proprietorship. Ownership by single individual. Individual traders. One-man businesses	658.155.4	Reserves. Reserve funds
658.114.2	Partnership	658.16	Financial reorganization
658.114.25	Trading partnerships (Sociétés en nom collectif)	658.167	Realization of capital. Liquidation of assets
658.114.3	Companies with share capital in general	658.168	Capital increase and reduction. Financial reconstruction. Fusion
658.114.32	Limited partnerships (Sociétés en commandite. Kommanditgesellschaften)	658.168.1	Increase of share capital
658.114.4	Companies with limited liability. Joint stock companies	658.168.2	Reduction of share capital
658.114.5	Alliances in general. □ Syndicates. Holdings. Concerns. Consortiums. Joint ventures	658.168.3	Financial reconstruction
658.114.6	Private monopolies of production. Cartels. Trusts ⇒ 334.75 ⇒ 334.73	658.168.5	Fusion. Amalgamation. Merger ⇒ 005.591.45 ⇒ 330.341.44 ⇒ 334.758 ⇒ 347.72.04
658.114.7	Cooperatives	658.168.52	Merger by transfer of shares
658.114.8	Associations, unions (including trade unions), institutions, foundations etc. active as enterprises	658.168.526	Purchase of shares of another company
658.115	Public, state and nationalized enterprises	658.168.55	Merger by transfer of goods, merchandise
658.115.31	Monopolies in accordance with public law. Public utilities	658.17	Extraneous expenses. □ Expenses for cultural, social, political purposes
658.115.33	Public companies with share investment. Mixed-ownership enterprises	658.18	Saving measures in general
		658.2	Installations and plant. Premises. Buildings, works, factories. Materials ◇ 658.2:368.1 Insurance of premises, factories etc. ◇ 658.2:628.517.2 Noise control in relation to premises, factories etc. ⇒ 69 ⇒ 725
		658.21	Siting. Situation. Location. Relocation. Removal ◇ 658.21:658.272 Siting in relation to procurement of raw materials ⇒ 332 ⇒ 628.521 ⇒ 711.454

658.218	Site. □ Space available. Light, air. Soil properties. slope of ground	658.5	Production engineering and planning. Design. Production management and control
658.219.4	Relocation, removal of premises. Resiting of factories		◇ 658.5:005.336.3-021.471 Exemplary production. Optimal production
658.23	Plan and use of site. Layout, disposition of buildings		◇ 658.5:005.336.3-021.473 Imperfect production. Low standard production
658.231	Future expansion. Provision for enlargement		◇ 658.5:005.51 Production programmes
658.232	Adaptation to requirements of production, operation		◇ 658.5:005.584.1 Production control
658.233	Internal disposition and interrelationships of the building		⇒ 338.3
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658.234	Area of surface occupied	658.51	Organization of production
658.26	Energy sources. Power supply. Provision of heat and water	658.511	Production analyses
658.261	Generation of own power or motive force	658.511.2	Product analysis
	⇒ 620.9	658.511.3	Process analysis
	⇒ 621.1/5	658.511.4	Form, design analysis
658.262	Distribution of power or motive force on site (including that obtained externally)	658.511.5	Analysis of means of production (machines, tools). Plant layout (manufacturing layout).
658.264	Provision of heat		Machine plotting
	◇ 658.264:621.365 Use of electric heating	658.511.8	Job instructions. Operational write-up
658.265	Water supply	658.512	Production plan, schedule. Determination of sequence of operations (sequencing). Routeing
	<i>Details by :628.1...</i>	658.512.2	Designing. Shape, design, construction of products. Industrial design
	◇ 658.265:628.1.033 Supply of main-water, potable water		⇒ 7.05
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658.27	Primary means of production. Manufacturing plant, materials etc.	658.512.4	Subcontracting
658.272	Raw and auxiliary materials in general	658.512.6	Preparation of production work
658.273	Initial, intermediate, semifinished products etc.		Distribution of work. Allocation of workload.
658.274	Machinery and equipment in general		Work schedules and deadlines. Machine loading
	<i>For specific machinery and equipment use the relevant section, e.g. 621.7, 621.9, 66.013, 67, 68</i>	658.512.62	⇒ 331.103
658.28	Ancillary, secondary facilities for production. Associated services	658.513	Deployment of labour force. □ Individual working. Group working. Working at more than one workplace or station
658.284	Communications equipment. Installations for control, signalling, information	658.513.2	Supervision of production work. □ Follow-up.
	⇒ 654	658.513.3	Progressing. Expediting. Scheduling
658.286	Materials handling, conveying, transport	658.513.4	Materials schedule
	⇒ 621.798	658.513.5	Operation schedule
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658.286.2	Internal transport and conveying. Flow of materials. □ Supply and conveying of workpieces, tools etc.	658.514.4	Execution of production work. Dispatching
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658.286.4	External transport and conveying		Distribution of work according to time allowed
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658.29	Other matters relating to management and organization of installation, plant, premises	658.516.3	Standardization of products
658.3	Human relations in the enterprise	658.52	Standardization of equipment, tooling
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658.310.42	Staff relations. Interpersonal relations	658.522	Kinds of production
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658.310.8	Personnel management and administration	658.524	◇ 658.52:338.364 Automated production
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	◇ 658.310.8-051 Personnel managers	658.527	Single-item production. One-off production
			Special-order production. Jobbing. Making to order
			Batch or lot production. Series production
			Mass production
			Continuous production. Line production. Flow production
		658.53	Performance and time data
			⇒ 005.642.5
		658.531	Determination of production time.
		658.532	Measurement of working time
			Records of work completed

658.56	Quality control. Control of tool issue, materials, products etc.	658.64	Services
658.562	Control of operation. Production supervision. Quality control and inspection ⇒ 005.584.1	658.69	Exchanges in kind (barter) ⇒ 339.185 ⇒ 347.452
658.562.2	Control and inspection of materials	658.7	Buying. Purchasing. Procurement. Management of stock ◇ 658.7:339.182 Terms and conditions of delivery ⇒ 64.03
658.562.3	Control and inspection of work, operations, processes	658.7.03	Buying prices and tariffs 658.7.03 ≡ 658.8.03 substituting 'buying' for 'selling'
658.562.4	Quality control and inspection of materials and products ⇒ 658.772	658.71	Purchasing policy and plans
658.562.42	Inwards inspection. Inspection on receipt	658.711	Preparations for buying. Sources of supply. Analysis of requirements. Availability. Details of suppliers, products, services
658.562.44	Intermediate inspection. Control of work in progress. Process control	658.712	Purchasing and procurement policy
658.562.47	Final inspection. Inspection of end-product. Acceptance sampling	658.713	Methods of procurement. □ Purchase. Leasing. Hiring. Exchange
658.562.5	Control and inspection of equipment, machinery	658.715	Inviting of quotations, tenders. Inquiries for price. Price analysis, comparison
658.563	Control of tool issue	658.715.2	Submission of tenders
658.566	Control of materials issue	658.716	Centralized purchasing systems (in large firms, public administration etc.). Buying departments
658.567	Treatment of wastes, residues, rejects, scrap ⇒ 628.4.038 ⇒ 628.477 ⇒ 628.54	658.72	Purchasing methods
658.567.1	Utilization of waste and scrap	658.721	Contract purchasing
658.567.3	Regeneration and recycling of waste or scrap ⇒ 620.28	658.722	Direct purchasing, from manufacturer or producer
658.567.5	Rendering waste harmless. Neutralizing waste. Destroying, disposal of waste (on the premises of the enterprise). □ Decontaminating, disinfecting, draining waste	658.724	Indirect purchasing, through agent, factor, dealer
658.58	Servicing and maintenance work ⇒ 62-7 ⇒ 658.818.3	658.727	Purchasing at auctions
658.581	Maintenance and care of machinery	658.74	Ordering ◇ 658.74:005.336.3-021.471 Optimum quantities
658.588	Damage. Faults. Defects ⇒ 368.1 ⇒ 614.8 ⇒ 620.19	658.742	Release of orders
658.588.1	Preventive measures	658.75	Follow-up of orders (chasing)
658.588.2	Damage assessment. Troubleshooting. Fault tracing	658.77	Receipt of goods. Acceptance of delivery
658.588.4	Damage to materials and machinery. Faulty materials. Failure of machinery	658.771	Control of quantity of goods received
658.588.5	Damage through interruption of production	658.772	Quality control in goods received
658.588.5	Damage in relation to staff. Replacement of staff	658.773	Invoice checking and control
658.588.8	Repairs	658.78	Stock and stores management. Warehousing. Dispatch ⇒ 621.796
658.589	Innovations. Replacements	658.782	Sorting and marking of stock
658.6	Commercial organization and practice. Goods. Services ⇒ 330.12	658.783	Stock on hand. Stock levels
658.62	Goods. Merchandise. Wares. Products ◇ 658.62:005.336.3-021.471 Quality products. Goods of high quality ◇ 658.62:005.336.3-021.473 Bad, unserviceable products. Goods of poor quality ◇ 658.62(1-82) Goods according to source of supply ◇ 658.62(520-82) Goods from Japan. Japanese products ⇒ 620.2	658.784	Stores of semifinished and finished products (sales stock) ⇒ 657.422.61.7
658.624	New products	658.785	Stores of materials, supplies
658.625	Old, used second-hand goods	658.786	Handling, treatment of goods in store
658.626	Branded, proprietary goods	658.787	Stock control. Stocktaking. Inventories. Stock records ⇒ 657.356 ⇒ 657.371
658.627	Dead stock. Substandard goods. Job lots ⇒ 658.849.4	658.788	Dispatch. Consignment. Forwarding of goods ⇒ 347.463 ⇒ 656
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658.8	Marketing. Sales. Selling. Distribution	658.818.6	Guarantee of taking back goods. Guarantee of refund
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◇ 658.8:005.32	Sales psychology	⇒ 659.1	
◇ 658.8:005.51	Sales planning	658.821	Promotional methods. □ Creation of demand.
◇ 658.8:005.52	Market research, surveys etc.		Combating of competition
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◇ 658.8:339.18	Sales agreements and conditions	658.827	Presentation and packaging of goods. Display wrapping
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658.8.03	Selling prices and tariffs		⇒ 681.138
658.8.031	Calculation of selling prices	658.842	Sale by telephone
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658.8.031.4	Factory prices. Manufacturer's prices	⇒ 658.882	
658.8.031.5	Wholesale prices	658.843.2	Part payment. Payment on account. Payment by instalments
658.8.031.6	Middleman's prices	658.843.4	Hire purchase
658.8.031.7	Retail prices	⇒ 658.91	
658.8.033	Rebates. Discounts	658.843.5	Credit sale
⇒ 658.881.3		658.845	Sale by means of samples, pattern books. Sale using illustrated literature, catalogues etc.
658.8.035	Sales tariffs. Scales of charges		Sales at exhibitions. Sending on approval
⇒ 656.031.5		658.846	Selling by sales representatives, brokers, factors, agents, commercial travellers
658.8.035.2	Fixed or flat-rate tariffs. Payment of fixed unit price regardless of quantity consumed	658.849	Other methods of selling
658.8.035.3	Tariffs with progressive reduction of unit price	658.849.2	Direct selling by manufacturer
658.8.035.4	Proportional tariffs	⇒ 339.35	
658.8.035.5	Tariffs with progressive increase of unit price	658.849.4	Occasional sales. Selling of job lots, bargain goods
658.8.035.7	Tariffs with standing charge for minimum consumption and fixed price per unit of extra consumption	⇒ 339.375	
658.8.036	Tariffs dependent on factors other than quantity consumed	⇒ 658.627	
658.8.036.2	Tariffs according to locality and distance	658.849.5	Second-hand trade. Junk business. Sale of used goods
658.8.036.3	Tariffs according to time. □ Peak rates. Off-peak rates	⇒ 339.375	
658.8.036.4	Tariffs according to category of consumer	⇒ 655.425	
658.8.036.5	Tariffs for groups of consumers. □ Purchasing associations, collectives	658.849.7	Sale by auction. Sale by court order
658.8.036.6	Tariffs for special purposes. Tariffs according to nature of merchandise. □ Charges for dangerous, perishable goods	658.85	Technique of selling. Salesmanship. Sales talk etc.
658.8.036.7	Tariffs according to speed or promptness of delivery	⇒ 659.144	
658.8.036.8	Tariffs according to form or class of delivery. □ De luxe rate	658.86/87	Distribution channels (wholesale, retail)
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658.8.038	Free supply (gratis, free of charge)	⇒ 339.33	
658.8.07	Sales departments	⇒ 339.34	
658.81	Sales organization	658.87	Retail selling. Retailing
658.811	Sales management	⇒ 339.37	
658.813	Processing of orders. □ Order books. Delivery notes. Invoices and invoicing	658.871	Retail outlets. Shops. Stores
⇒ 657.352		658.872	Mail order business
658.815	Supervision of order handling. Filling, clearing of orders. Progressing of orders	658.879	Other retailing methods. □ Street vending.
658.817	Evaluation of sales results. Sales reports. □ Reporting of lost orders		Door-to-door selling. Car boot sales
658.818	Service to customer. After-sales service	658.88	Payments. Collection of money due. Credit management
658.818.2	Customer advice. Consultancy (on choice of goods, services etc.)	658.881	Cash payment
⇒ 659.2		658.881.3	Discount for cash payment. □ Discount vouchers. Trading stamps
658.818.3	Maintenance and repair service. Servicing	⇒ 658.8.033	
⇒ 658.58		658.882	Sale on account. Credit sales. Conditions for credit
658.818.4	Settlement of claims and complaints	⇒ 336.77	
658.818.5	Exchange of goods	⇒ 351.825	
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658.884	Debt collection. Issue of demands, due notices. Coercion	659.113.5 659.113.6	Forecasting, projections of results Execution of campaign. □ Estimates. Designs. Production of publicity material
658.89	Purchasers as objects of sales effort. Prospective customers, clients	659.113.7	Results and effectiveness of publicity. Effectiveness of medium and distribution. Checking, monitoring of results
658.891	Customers according to profession, occupation ◇ 658.891:614.23 Doctors as prospective customers	659.113.8	Management of advertising media. Supervision of various modes of publicity
658.893	Customers according to channel and type of demand ◇ 658.893:339.37 Retailers as prospective customers	659.117 659.117.1	Target audience of publicity. Advertising targets Targets according to profession, occupation ◇ 659.117.1:614.23 Doctors as advertising targets
658.9	Other commercial activities ◇ 658.9:347.779.3 Commercial patent agencies	659.117.3	Targets according to channel and type of demand ◇ 659.117.3:339.37 Retailers as advertising targets
658.91	Letting. Rental. Hire. Leasing ◇ 658.91:338.5 Rates of hire. Rental and leasing costs, tariffs ◇ 658.91:339.18 Rental and leasing agreements. Terms of rental or lease ⇒ 339.187.6	659.118	Advertising tactics. □ Catching the attention. Informative advertising. Persuasion. Enticement, allurement. Authoritarian form. Obligation
658.97	Subscription arrangements. Continuous sale	659.12 659.121 659.122	Concept and design in advertising Establishing of objectives Formation and development of concepts, ideas. Conceptualization. □ Image. Theme. Motif
659	PUBLICITY. INFORMATION WORK. PUBLIC RELATIONS	659.123	Textual design and compilation. Production of copy (copywriting). □ Advertising jingles, slogans, catchphrases ◇ 659.123:791.641.3 Writing for audiovisual advertisements (film, television), commercials ⇒ 659.131.2 ◇ 659.123-051 Advertising copywriters
659.1	Publicity. Advertising ◇ 659.1:005.1 Publicity methods and methodology. Theory and practice of advertising ◇ 659.1:005.21 Advertising policy ◇ 659.1:005.32 Psychology of advertising ◇ 659.1:005.336.1 Economic efficiency (cost-effectiveness) etc. of advertising ◇ 659.1:005.743 Publicity departments. Advertising departments ◇ 659.1:338.5 Advertising rates, prices ◇ 659.1-051 Practitioners of publicity, advertising ⇒ 655.55 ⇒ 658.82	659.125 659.126 659.127 659.127.3 659.127.4 659.127.6 659.127.8	Pictorial design (except trademarks). □ Use of graphic art. Use of photography ⇒ 659.131.3 Trade marks. Logos. House colours etc. ⇒ 347.772 Forms of advertising, publicity Creation of trust, confidence in the company or advertiser Creation of demand. □ Creation of trust in the product or service. Promotional offers Maintaining of confidence. Retention of customers, of brand loyalty Maintaining the visibility of the product. □ Product placement (arranging for the product to be featured in entertainment media) General make-up of advertisements. □ Eye-catching. Copy layout <i>Class here general problems concerning advertising media only. For specific matters see 659.13/.16</i> ◇ 659.128:655.26 Typography
659.11	Publicity campaigns, their execution. Advertising tactics. Target audience of publicity ⇒ 659.441	659.128	
659.111	Coverage and distribution of publicity		
659.111.2	Methods of distribution. □ Hand-outs. Postal distribution. Help-yourself displays		
659.111.3	Scope and coverage of publicity. Degree of repetition. Duration of effect, impact		
659.111.31	Advertising drives. Advertising campaigns for specific occasions and events		
659.111.32	Special days for advertising a single commodity, merchandise or service ◇ 659.111.32:655.4 Book days	659.13/.16	Publicity and advertising media
659.113	Planning and management of publicity	659.13	Visual advertising media
659.113.2	Market research, surveys	659.131	Advertising in publications ⇒ 050 ⇒ 070.36 ⇒ 070.485 ⇒ 655.535.6
659.113.21	Sources of data. Statistics and their evaluation		
659.113.22	Research by interviewing		
659.113.23	Questions of competition		
659.113.25	Research into consumer habits, demand, market trends etc.	659.131.1	Published advertisements in general. Published announcements
659.113.26	Presentation and evaluation of market analyses	659.131.2	Textual matter in advertisements. □ Content and style. Prose. Verse. Headlines. Slogans ⇒ 659.123
659.113.3	Publicity planning, organizational aspects. □ Coordination of publicity, production and sales	659.131.3	Pictorial matter in advertisements. Illustrations ⇒ 659.125
659.113.4	Determination of campaign territory		

659.131.4	Make-up and technical details of advertisements. □ Size. Format. Layout	659.138.6	Advertising by aeronautical means. Use of aircraft for publicity. □ Towing banners.
659.131.7	Placing of advertisements ◇ 659.131.7:004.738.5 Internet advertising ◇ 659.131.7:070 Newspaper advertising		Dropping leaflets. Skywriting (using vapour trail). Other aerial stunts ⇒ 656.052.59 ⇒ 656.7 ⇒ 659.134.7
659.131.8	Publicity in the general and editorial sections of publications. □ Articles, discussions, reports, reviews etc. Advertising features	659.138.7	Monuments, statues for advertising
659.132	Publicity by means of one's own publications	659.138.8	Scenes, tableaux, puppets etc. for advertising
659.132.1	Publicity periodicals published by the firm. House journals. Customer magazines or bulletins	659.14	Auditory, acoustic advertising media
659.132.2	Prospectuses and other ephemeral publications	659.144	Direct oral, spoken publicity. The human voice as means of advertising, announcing etc. □ Sales talk. Barkers. Town criers ⇒ 659.151 ⇒ 658.85
659.132.21	Explanatory and informative printed matter. □ Instruction booklets. Directions for use. Catalogues. Brochures	659.144.1	Individual advertising. Propaganda efforts. Sales talks by salesmen or representatives
659.132.22	Printed advertising matter. □ Prospectuses. Leaflets. Handbills	659.144.2	Advertising in conferences
659.132.27	Inserts and enclosures (in other publications)	659.145	Indirect oral publicity. Broadcast and recorded advertisements, announcements etc. □ Talks. Radio drama. Newscasts. Advertisers' announcements (commercials). Music, song
659.132.4	Direct mail (mail shots). □ Circulars. Letters. Trade cards		Advertising and publicity using projected and electronic images (whether or not combined with sound). □ Audiovisual publicity ◇ 659.148:778.25 Advertising using slides, transparencies ◇ 659.148:778.53.099 Advertising using silent film ◇ 659.148:778.534.4 Advertising using sound film
659.133	Advertising by means of posters, placards, signs	659.148	
659.133.1	Bill-posting on fixed sites. □ Posters on pillars, hoardings, walls		Advertising and publicity by demonstration and exhibition
659.134	Mobile advertising media. Advertising using vehicles		Demonstrations of the use of a product ⇒ 659.144
659.134.1	Advertising in public transport	659.151	Trade exhibitions. Exhibitions for publicity purposes ⇒ 069.9
659.134.2	Advertising on itinerant carriers. Sandwich-men. Special clothing for advertising	659.152	Personal presentations. □ Guided works tours ◇ 659.153:687 Presentations of garments. Fashion shows (with models, mannequins) ⇒ 005.584.1 ⇒ 005.963.3
659.134.3	Advertising on land vehicles. Land vehicles as advertising devices	659.153	Wrapping, packaging, decoration of goods for advertising purposes. Marking, identification ⇒ 621.798 ⇒ 659.126
659.134.6	Advertising on ships, boats. Floating objects as advertising devices	659.154	Devices and systems for display of goods. □ Shop fittings
659.134.7	Advertising using balloons (captive or dirigible), aircraft, fireworks ⇒ 659.138.6	659.157	Glass cases. Showcases. Display cabinets etc. Display frames, trestles, stands, racks etc. (e.g. for shoes, hats)
659.134.8	Flags, pennants for advertising. Devices on vehicles for distributing, scattering publicity material	659.157.2	Adjustable supports and shelves. Wall brackets
659.135	Publicity and advertising using light (and luminous effects) ⇒ 628.974.7	659.157.5	Various shaped display devices. □ Clothes models. Busts. Articulated dummies. Fabric stands
659.135.1	Continuously illuminated signs	659.157.6	Shop-window display
659.135.2	Intermittently illuminated signs. Flashing signs	659.157.8	Eye-catching devices
659.135.3	Advertisements projected on to screens, ground, clouds etc. ⇒ 659.148	659.158.2	Window dressing. Window display equipment
659.135.4	Movable shades and screens for advertising	659.158.3	Decorative materials (e.g. papers, fabrics, grass matting, cork chippings)
659.135.9	Self-luminous or luminescent materials in advertising	659.158.4	Advertising matter on the window pane. □ Transfers, stickers
659.136	Various mechanisms for visual advertising	659.158.6	
659.136.1	Advertising on particular objects ◇ 659.136.1:662.536.2 Advertising on matchboxes	659.16	Promotion methods with free gifts, premiums, discounts etc.
659.136.2	Advertising on calendars ⇒ 659.162	659.162	Free gifts (not the product) for publicity purposes ◇ 659.162:686.853 Advertising calendars ⇒ 659.136.2
659.136.3	Advertising on menu holders and similar frames and stands	659.163	Free samples of the product
659.136.4	Advertising on revolving displays, discs, leaves etc.		
659.136.7	Advertising on other mechanisms. □ Advertising on prisms, panels, slats or louvres (showing one face at a time). Advertising toys, novelties. Viewers		
659.138	Special visual devices for attracting attention		
659.138.4	Devices for printing advertisements on paved surfaces		
659.138.5	Automatic sign-writing and message-relaying devices. □ Electric bulletin boards		

659.164	Promotional bonuses, premiums	659.24	Technical information and consultancy ◇ 659.24:621.3 Information and consultancy in electrical engineering
659.166	Promotional discounts, rebates, price reductions		
659.167	Sales, especially clearance sales, as promotional method	659.25	Information and consultancy in sciences and pure studies ◇ 659.25:39 Information and consultancy in ethnography
659.168	Competitions, contests as promotional method		
659.18	Advertising and publicity according to kind of originator	659.27	Information about persons in general
659.181	Collective, group advertising. Communal publicity	659.28	Inquiries for information. Requests for consultation 659.28 ≡ 659.2
659.182	Advertising and publicity by authors and inventors		
659.183	Advertising and publicity by manufacturers, producers	659.3	Mass communication. Informing, enlightening of the public at large ⇒ 001.92 ⇒ 374.7
659.184	Advertising and publicity by traders, dealers		
659.185	Advertising and publicity by craftsmen, artisans	659.4	Public relations (PR) ⇒ 32.019.5 ⇒ 354.36
659.186	Advertising and publicity by consumers, customers		
659.187	Advertising and publicity by professional promoters (advertising agents, publicists)	659.42	Public relations according to source or initiating agency
659.19	Other methods of publicity and advertising	659.43	Public relations according to purpose or target audience
659.2	Information work. Advisory and consultancy services ◇ 659.2:061.1 Official, government advisory services ◇ 659.2:061.2 Private advisory services ⇒ 002 ⇒ 005.572 ⇒ 005.942 ⇒ 364-784 ⇒ 366.61 ⇒ 658.818.2	659.44	Public relations strategy, tactics and execution. Ideas, concepts and planning. Public relations media ◇ 659.44(1-194) External public relations ◇ 659.44(1-4) National public relations ◇ 659.44(100) International public relations Strategy and tactics in public relations ⇒ 659.11
659.21	Technique of information work	659.441	Approach, scope and duration in public relations
659.22	Technique of consultancy and advising	659.441.1	Planning, sources, effectiveness, impact of public relations
659.23	Business and commercial intelligence, information. Business consultancy	659.441.3	Tactics in public relations
659.231	Economic intelligence and advice ⇒ 658.8.012.12	659.441.8	Concept and design in public relations.
659.232	Services for ascertaining and confirming addresses	659.442	Execution ⇒ 659.12
659.233	Information concerning solvency of persons and firms (credit-worthiness)	659.443/.446	Public relations media
659.235	Management consultancy and business information. Advice on business management and organization ⇒ 658	659.443	Visual public relations media
		659.444	Auditory, acoustic public relations media
		659.445	Public relations by demonstration and exhibition
		659.446	Public relations methods with free gifts etc.

66 CHEMICAL TECHNOLOGY. CHEMICAL AND RELATED INDUSTRIES

Under 66 is classed the industrial production of chemical and metallurgical products, as well as the processing of foodstuffs, other agricultural products and suchlike. Also under 66 are classed the products themselves: e.g. under 661 the products of the chemical industry in the strict sense, and under 664.8/9 the products of the food preservation industry.

Pure and theoretical chemistry, and the related preparation of small quantities of chemical substances, are classed at 54

The industrial production of substances which are not individually provided for under 66 may be specified at the most appropriate 66 subdivision by :546... or :547....

The -2/-8 auxiliaries from 62-2/-8 are applicable throughout 66, except for -3 under 661.185 and 662.2

66-9

VARIABLES, CONDITIONS AND CHARACTERISTICS OF PRODUCTION PROCESSES, PLANT AND EQUIPMENT

The -9 subdivisions are used to specify the characteristics or variables of processes, not processes themselves, e.g.

a) -98 Pressure characteristics, variables.

*.083 Operations involving pressure.
b) .047-912 Drying in the fluidized state.
.096.5-932.2 Continuous fluidization
The -9... auxiliaries are applicable throughout 62/69*

66-91

State of material: characteristics and variables

66-911	State of aggregation <i>Use with processes. For description of products use -4... from 62-4...</i> 66-911.3/6 $\equiv$ 54-13/-16	66-933.4	Mechanized, by machine (power-aided, but without automatic or computer control)
66-911.3	Gaseous state	66-933.6	Automated (using automatic or computer-controlled machines)
66-911.33	Compressed, liquefied gas	66-935	Related or interdependent processes
66-911.34	Vaporous state	66-935.2	Related or interdependent processes with the same primary or raw material
66-911.38	Colloid with gaseous dispersion medium. Aerosols. Smokes	66-935.3	Related or interdependent processes with the same end-products
66-911.4	Liquid	66-935.4	Processes with reclamation, recovery, regeneration
66-911.48	Colloid with liquid dispersion medium. Emulsions. Gels. Foams	66-935.5	Processes with enrichment, concentration, refining
66-911.6	Solid	66-936	Processes according to phase state
66-911.69	Particulate	66-936.3	Homogeneous processes
66-912	Processes in the fluidized state $\Rightarrow$ 66.096.5	66-936.33	Reactions between gases or vapours
66-913	State of humidity	66-936.35	Reactions between liquids
66-913.1	Dry	66-936.4	Heterogeneous processes
66-913.2	Damp. Humid. Moist	66-936.42	Reactions between gases and liquids
66-913.3	Wet	66-936.43	Reactions between gases and solids
66-914	Concentration characteristics $\Rightarrow$ 66.082	66-936.46	Reactions between liquids and solids
66-914.3	Low concentration. Dilute	66-936.48	Reactions between solids
66-914.5	Medium concentration	66-94	Direction, velocity, rate, duration of processes etc.
66-914.7	High concentration. Concentrated	66-942	Parallel processes. Uniflow processes
66-915	pH value $\Rightarrow$ 544.354.3	66-943	Cross current. Criss-cross processes
66-915.1	Acid	66-944	Opposed, reversed processes. Counterflow processes
66-915.2	Neutral	66-945	Open- or closed-circuit processes
66-915.3	Alkaline	66-945.2	Open-circuit, open-chain processes
66-916	rH value	66-945.3	Closed-circuit processes. Circular, circulating, cyclic processes
66-916.1	Oxidizing	66-946	Static or moving processes
66-916.2	Reducing	66-946.1	Static
66-92	Reactions in terms of reagents. Reagents	66-946.3	Moving. Agitated $\Rightarrow$ 66-912
66-921	Processes involving molecules of the same type. Auto-, self- processes. Homo-processes	66-947	Rapid. Accelerated. Transitory
66-922	Processes involving molecules of different types	66-947.1	Very rapid
66-922.2	Block processes	66-947.2	Rapid
66-922.3	Inoculation processes. Graft processes	66-947.3	Accelerated
66-926	Inorganic reagents 66-926 $\equiv$ 546 with the use of the apostrophe for compounds as at 546	66-947.5	Short (duration). Transitory
66-927	Organic reagents 66-927 $\equiv$ 547 with the use of the apostrophe for compounds as at 547	66-948	Slow. Prolonged. Delayed
66-93	Processes, machines and equipment according to mode of working, continuity, associations, media	66-948.1	Slow
66-931	Processes, plant and equipment according to directness	66-948.2	Prolonged. Extended
66-931.2	Direct	66-948.3	Retarded. Delayed
66-931.4	Indirect	66-948.5	Lengthy. Long-lasting
66-932	Processes, plant and equipment according to continuity	66-95	Simple and multiple processes. Single-stage and multistage processes
66-932.1	Single processes	66-951	Simple, simplified processes
66-932.2	Continuous	66-952	Multiple, complex, repeated processes
66-932.3	Semicontinuous	66-953	Single-stage processes
66-932.4	Discontinuous	66-954	Multistage processes. Step, cascade processes
66-932.42	Impulsive. Intermittent	66-956	Single-bath processes
66-933	Processes according to degree of mechanization	66-957	Multi-bath processes
66-933.2	By human effort (without power-aided tools or machines). $\square$ Manual (by hand)	66-958	Combined processes
		66-96	Processes in terms of altering product properties. Modification in general
		66-962	Processes without purification or refining
		66-963	Processes with purification or refining
		66-965	Particulars of processes according to the form or condition of the material to be treated
		66-965.1	Loose. Flocculent. In the form of flakes

66-965.2	Coiled. In a coil	66.012.2	Consumption of raw and auxiliary materials.
66-965.3	In a hank or skein		Material balance
66-965.4	As ribbon or tape	66.012.3	Energy consumption. Energy balance
66-965.6	According to condition of stress or tension	66.012.4	Capacity. Output. Production cycle
66-965.61	Stressed. Under tension	66.012.42	Capacity
66-965.62	Not stressed. Not under tension. Slack. Limp	66.012.43	Output
66-965.81	Open. Uncovered	66.012.44	Intensity
66-965.82	Closed. Covered	66.012.45/.47	Production cycle
66-966	Processes according to durability (permanence, fastness)	66.012.45	Preparatory work
66-966.1	Durable. □ Wash-resistant	66.012.46	Throughput. □ Rate. Duration
66-966.2	Not durable	66.012.47	Duration of discharging operations. Duration of finishing
66-966.5	Fast (colourfast)		Yield. Transformation degree. Selectivity
66-967	Processes according to liquor ratio	66.012.5	Process efficiency. Productivity. Chemical engineering economics
66-967.1	Low liquor ratio		
66-967.2	Medium liquor ratio	66.013	Chemical works generally. Chemical factories. Plant. Equipment. Auxiliary services
66-967.3	High liquor ratio		⇒ 658.2
66-97	Thermal characteristics. Temperature.		⇒ 725.1
	Temperature range	66.013.2	Degree of plant integration
	<i>Denote specific scale and value by abbreviation and figure, following an asterisk (see Table 1h)</i>	66.013.22	Integrated works
	◇ 66-97°C0 0°C (freezing point of water)	66.013.25	Nonintegrated works
	◇ 66-97°F32 32°F (freezing point of water)	66.013.5	Plant design and construction, start-up and adjustment
	◇ 66-97°K273 273 K (freezing point of water)	66.013.6	Power supply
	◇ 66-97°C100 100°C (boiling point of water)		◇ 66.013.6:621.22 Water power supply
	◇ 66-97°F212 212°F (boiling point of water)		◇ 66.013.6:621.311 Electricity power supply
	◇ 66-97°K373 373 K (boiling point of water)		⇒ 620.4
66-971	Thermodynamic characteristics		⇒ 658.26
66-971.2	Exothermic	66.013.7	Chemical processing services: water, steam and gas for use in processing
66-971.4	Endothermic		⇒ 621.1
66-973	Very low temperature		⇒ 628.1
66-974	Low temperature		⇒ 661.9
66-975	Room temperature		⇒ 662.76
66-976	Medium, moderate temperature	66.013.8	Industrial hygiene. Sanitation. Other accessory services
66-977	High temperature		⇒ 613.6
66-978	Very high, extremely high temperature		⇒ 614.8
66-98	Pressure. Pressure range	66.014	Chemical composition
	⇒ 66.083	66.017	Materials and their properties
66-982	Vacuum. In vacuo		⇒ 620.1/.2
66-983	Very low pressure	66.018	Resistance of materials to external effects and actions
66-984	Low pressure		66.018 ≡ 669.018
66-985	Atmospheric pressure. Normal pressure	66.018.4	Resistance to thermal effects
66-986	Medium, moderate pressure	66.018.86	Resistance to radiation
66-987	High pressure. Hyperbaric	66.019	Defects of materials
66-988	Very high, extremely high pressure		<i>Specify by :620.19...</i>
66.0	CHEMICAL ENGINEERING IN GENERAL	66.02	Chemical processing operations and equipment in general
66.01	Fundamentals of chemical engineering		<i>Each of the subdivisions of 66.02/.09 represents both a process and the related equipment. The exceptions are 66.023 denoting reactors whereas reactions are denoted by 66.09, and 66.041 denoting furnaces whereas roasting and calcining are denoted by 66.046.4</i>
66.011	Process design. Calculation of processes. Modelling. Optimization. Mathematical methods in chemical engineering. New features in processes, installations etc.		
66.012	Production and process control, regulation and supervision. Efficiency. Performance. Chemical engineering economics	66.021	Standard processes
	<i>Further details by :658.5... or -5...</i>		<i>If preferred, specific processes may be grouped under each subdivision of 66.021 by adding .022/.099</i>
66.012.1	Measurement. Control. Instrumentation		◇ 66.021.2.063.8 Mixing of liquids
	⇒ 53		◇ 66.021.2.071.4 Mixing of gases
	⇒ 681.2		◇ 66.021.3.048 Distillation. Evaporation
			◇ 66.021.3.065.5 Crystallization
			◇ 66.021.3.081.3 Adsorption

66.021.1	Hydromechanical processes in general.	66.026.4	Head-creating equipment (pumps etc.) ⇒ 621.65
66.021.2	Hydromechanical apparatus in general		
66.021.3	Momentum transfer. Fluid flow and mixing	66.028	Introduction of reagents. Charge. Filling. Feeding devices. Charging devices. Filling devices. Discharge. Discharging devices <i>Specify operation and control by -5...</i>
66.021.4	Thermokinetic processes in general. Heat interchange. Heat transfer ⇒ 66.045		◇ 66.028-531.3 Consumption control. Correlation of reagents. Correlation control ⇒ 681.533.38 ⇒ 681.542.2
66.022	Individual methods of treatment. Preparatory operations. Additives <i>The 66.022... auxiliaries are not applicable at 666.29 and its subdivisions</i>	66.028.2	Dosage. Dosing devices. Dosing tanks
66.022.1	Pretreatment in general	66.03	Materials, characteristics, processes and equipment <i>The .03 auxiliaries are developed at various subdivisions of 66 for specific industries</i>
66.022.3	Additives. Treatment with additives		
66.022.31	Additives according to solubility	66.04	Heat treatment operations and equipment <i>See the note under 66.02</i>
66.022.314	Soluble additives		
66.022.316	Insoluble additives	66.040	Heat treatment according to physical state. Phase changes ⇒ 536.42
66.022.32	Bulking materials. □ Aggregates. Extenders. Fillers	66.040.2	Heat treatment in the solid phase ⇒ 66.046.4
66.022.33	Additives affecting colour. Colour stabilizers ⇒ 667.622	66.040.26	Phase change solid to liquid
66.022.34	Additives affecting hygroscopicity. □ Humectants. Desiccants. Hydrophobylizing and hydrophobic additives. Waterproofing additives ⇒ 66.047.8	66.040.27	Phase change solid to gas ⇒ 66.049.6
66.022.35	Additives affecting solubility	66.040.3	Heat treatment in the liquid phase ⇒ 66.046.6
66.022.36	Additives conducive to aggregate state preservation. Aggregate state stabilizers (of suspensions, emulsions, foams etc.) ⇒ 66.097.8	66.040.36	Phase change liquid to solid
66.022.362	Stability	66.040.37	Phase change liquid to gas
66.022.38	Additives affecting other physical properties	66.040.4	Heat treatment in the gaseous phase ⇒ 66.046.9
66.022.384	Additives affecting caking capacity. Anticaking agents	66.040.46	Phase change gas to solid
66.022.385	Dust inhibitors	66.040.47	Phase change gas to liquid
66.022.387	Additives affecting combustibility. Combustion inhibitors		
66.022.389	Other additives affecting physical properties <i>Specify by :53...</i>	66.041	Furnaces. Ovens. Kilns ◇ 66.041-6 Furnaces according to fuels, heat sources ◇ 66.041-69 Heating mechanisms and controls ◇ 66.041-982 Vacuum furnaces ◇ 66.041-987 High-pressure furnaces ⇒ 621.365 ⇒ 621.733 ⇒ 621.745.3 ⇒ 621.783 ⇒ 683.9
66.022.39	Additives affecting biological properties	66.041.1	Processes preliminary to furnace treatment
66.022.392	Additives affecting organoleptic reactions (smell, taste, texture) 66.022.392 = 543.92	66.041.13	Preparation of materials for furnace treatment
66.022.393	Denaturation	66.041.132	Preheating of materials
66.022.4	Impregnation ⇒ 620.197.7	66.041.15	Preparation of furnace. Annealing of furnace <i>Specify the cooling process by -71...</i>
66.022.5	Mechanical treatments <i>Details by :621.9...</i>	66.041.2	Furnace atmospheres
66.022.6	Individual methods of treatment for substances in different aggregate states. □ For phase separation	66.041.3	Methods of heating. Types of firing <i>Specify by -6...</i> ◇ 66.041.3-65 Electric furnaces ⇒ 621.365
66.022.62	Methods using gravity	66.041.34	Externally heated furnaces
66.022.63	Methods using centrifugal forces	66.041.35	Internally heated furnaces. Flues etc. for internal heating
66.023	Chemical equipment in general	66.041.36	Furnaces with heating gas passing through the charge
66.023.2	Apparatus for chemical transformations. Reaction apparatus. □ Vats. Carboys	66.041.37	Furnaces with direct contact between charge and heat-exchange liquid
66.023.23	Reaction columns, towers	66.041.4/5	Furnaces according to design
66.025	Fittings for reservoirs and reaction vessels. □ Valves. Gates. Closures. Manholes. Inspection panels. Connections for measuring instruments. Adjoining of pipes. Inlets. Outlets. Packings	66.041.41	Furnaces with open vessels for the charge (pots, cauldrons, crucibles)
66.026	Conveyance, transportation and corresponding facilities <i>Specify aggregate state of transported material by -91...</i> ◇ 66.026-911.6 Conveyance of solids		
66.026.2	Pipes. Pipelines ⇒ 621.643		

66.041.43	Reverberatory furnaces. Flame furnaces.	66.043	Constructional materials for furnaces etc.
66.041.44	Open-hearth furnaces	66.043.1	Refractories for furnace construction and lining
66.041.454	Shaft furnaces		<i>Specify resistance to external actions by .018...</i>
66.041.455	Tubular furnaces	◇ 66.043.1.018.841	Acid resistant refractories
66.041.456	Muffle furnaces	◇ 66.043.1.018.842	Alkali resistant refractories
	Crucible furnaces	◇ 66.043.1.018.85	Gas resistant refractories
	◇ 66.041.456-956 With a simple crucible	66.043.5	Furnace materials according to chemical properties. □ Basic, alkaline or acidic materials
	◇ 66.041.456-957 With a multi-unit crucible	66.044	Repair of furnaces etc. Patching, relining
	◇ 66.041.456-958 Combined	66.044.3	Rendering of furnaces
66.041.46	Retorts	66.045	Heat exchangers. Heat insulation. Cooling and refrigerating devices
66.041.47	Heating chambers		⇒ 66.021.4
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66.048.4	Complete distillation followed by fractional condensation	66.061.16	Solution of solids
		66.061.18	Solvents. Extractants ⇒ 66.061.38 ⇒ 66.063.93 ⇒ 667.629.2
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		66.061.34	Extraction from solids. Leaching
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		66.063	Mixing of liquids. Dilution. Concentration. Homogenization
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66.063.5	Wetting. Moistening. □ Wetting agents ⇒ 661.185.72 ⇒ 667.629.5 ⇒ 678.049.7	66.067.124	Porous filter media. □ Ceramic, metalloceramic (cermet) filter materials ⇒ 66.081.6
66.063.6	Formation of liquid dispersion systems <i>For stabilization of dispersions add .022.36</i> ⇒ 66.069.85	66.067.16	Filtration according to consistency (viscosity) of liquid <i>Specify by -911.4...</i> ◇ 66.067.16-911.43 Filtration of melts ◇ 66.067.16-911.48 Filtration of colloidal suspensions ◇ 66.067.16-911.49 Filtration of coarse suspensions, slimes
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66.063.7	Mixing and kneading of plastic substances ⇒ 621.929 ⇒ 678.023.3	66.067.3	Filters according to design features and details ◇ 66.067.3-229.6 Charging devices of filters ◇ 66.067.3-957 Battery filters ◇ 66.067.3.022.63 Centrifugal filters
66.063.72	Softeners, plasticizers and their use <i>Specify by :547...</i> ◇ 66.063.72:547.26 Alcohols as softeners ⇒ 667.621.7 ⇒ 678.049.1	66.067.31	Filter funnels
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66.063.93	Dilution. Thinning. Diluents ⇒ 66.082.4 ⇒ 667.629.2	66.067.34	Drum filters
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66.065.32	Thermal treatment. Cooling. Heating ◇ 66.065.32-977 Heat treatment	66.067.92	Removal of slime. Desliming ◇ 66.067.92.094.7 Desliming with simultaneous neutralization by alkali
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66.067.12	Filtering media <i>Denote specific materials by -03....</i> <i>Denote cleaning devices by -776</i>	66.069.856	Foam prevention. Foam prevention devices <i>Denote foam prevention agents, foam quenchers, antifoam additives by .022.3...</i>
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66.067.123.3	Filter papers		
66.067.123.8	Gauzes. Sieves. Perforated sheet materials		

66.071	Gas mixing. Gas separation. Dispersion gas system formation	66.074.41	Introduction of gases into saturation vessels etc.
66.071.4	Gas mixing. Gas mixers	66.074.43	Solvent liquids for gases: properties, condition
66.071.5	Gas humidification ⇒ 66.074.31	66.074.44	Circulation of solvent liquid
66.071.6	Separation of gases and gas mixtures. Gas separators in general ◇ 66.071.6.081 Gas separation by sorption ◇ 66.071.6.081.6 Gas separation by partition or membrane, by diffusion. Membrane gas separation. Gas diffusion units ⇒ 66.074.1/2	66.074.46	Separation of products from the gas
66.071.62	Separation of gas mixtures by deep cooling ◇ 66.071.62.048.4 Separation of gas mixtures by fractional condensation	66.074.48	Treatment of waste or effluent gases
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66.074.2	Purification of gases by removal of solid particles. □ By removal of dust, soot, fumes ⇒ 621.928.9	66.075	Carburation. Carburation media. Various additions to gas 66.075 ≡ 662.76.075
66.074.3	Purification of gases by removal of vaporous and gaseous admixtures. Removal of various substances <i>Specify substances not in 66.074.3... by :...</i> ◇ 66.074.3:546.264'131-31 Purification by removal of phosgene	66.076	Collection, accumulation and storage of gases. □ Reservoirs for gas storage. Gasholders Wet gasholders. Bell-type gasholders. Telescopic gasholders Dry gasholders Cylinders, bottles and similar containers for gas ◇ 66.076.5-464 Spherical containers ◇ 66.076.5-465 Cylindrical containers
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66.074.35	Separation, removal of oxygen (from gases)	66.078	Raising the pressure of gases. Compression and liquefaction of gases
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66.074.38	Removal of organic admixtures. Purification by removal of volatile and strong odour substances <i>Specify removal of organic compounds by :...</i> ◇ 66.074.38:547.563 Cresol ◇ 66.074.38:662.715.1 Wood tar ◇ 66.074.38:678 Macromolecular materials	66.078.5	Compressed-gas generators. Regulating devices
66.074.391	Separation, removal of acidic gases in general (from gases)	66.08	Physical and physicochemical operations and equipment. Physical and physicochemical effects on chemical processes <i>See the note under 66.02</i>
66.074.395	Separation, removal of cyanogen and hydrocyanic acid (from gases)	66.081	Sorption processes in general. Membrane processes
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66.074.4	Solution (dissolution) of gases in saturation vessels. □ Extraction of solid products ⇒ 66.061.12 ⇒ 66.069.82	66.081.3	Adsorption. Separative chromatography 66.081.3 ≡ 661.183 ⇒ 543.544
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		66.081.5	Desorption
		66.081.6	Membrane processes <i>Denote semipermeable membranes by -278</i>
		66.081.62	Dialysis. Dialysers ⇒ 66.087.97

66.081.63	Reversed osmosis. Ultrafiltration. Hyperfiltration ⇒ 66.087.92
66.082	Processes affecting concentration level ⇒ 66-914
66.082.2	Concentration ⇒ 621.928.44 ⇒ 66.065.8
66.082.3	Maintaining uniform concentration
66.082.4	Dilution ⇒ 66.063.93
66.083	Processes involving pressure ⇒ 66-98
66.083.2	Pressure-increasing processes. Pressurizing. Compression ⇒ 66.078
66.083.3	Maintaining uniform pressure
66.083.4	Pressure-reducing processes. Decompression. Rarefaction ⇒ 66.077
66.083.8	Degassing in vacuo ⇒ 66.069.84
66.084	Operations involving vibrations. Vibration-based processes
66.084.2	Processes using vibration or shock power. □ Shock wave processes
66.084.6	Processes using very low frequency vibration. Infrasonic processes
66.084.7	Processes using sound vibrations
66.084.8	Processes using very high frequency vibrations. Ultrasonic processes
66.085	Processes involving irradiation. Photochemical processes ⇒ 664.8.039.5
66.085.1	Long-wave radiation. □ Infrared radiation
66.085.2	Visible radiation. Light
66.085.3	Short-wave radiation. □ Ultraviolet radiation. X-radiation. Gamma-radiation ⇒ 621.386
66.085.5	Corpuscular radiation. □ Alpha-radiation. Beta-radiation. Neutron radiation
66.086	Operations involving magnetic, electromagnetic and electric effects
66.086.2	Operations involving electric effects. □ Electrostatic, high frequency processes
66.086.4	Operations involving electromagnetic, magnetic effects. □ Magnetochemical processes. Magnetic induction
66.087	Electrochemical processes
66.087.2	Electroreduction
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66.087.5	Electrocoagulation
66.087.7	Electrolysis and galvanization 66.087.7 ≡ 621.357
66.087.9	Application of electrocapillarity. Electrosedimentation 66.087.9 ≡ 621.359
66.087.92	Electroendosmosis. Electro-osmosis
66.087.97	Electrodialysis
66.088	Processes using electric discharges. Processes using lasers. Processes using plasma. Plasmatic and plasmochemical processes

66.09	Chemical technical reactions. Special chemical and biochemical processes <i>The great variety of chemical reactions makes it impossible to denote each of them by a direct division of 66.09. The following methodological instructions explain how to denote reactions.</i> 1) Many of the texts in 66.09 comprise both the general names of reactions and the specific names of commonly observed reactions. If need be, the latter can be denoted by methodological means, e.g. .095.118 Deacylation. .095.118:547.291 Deformylation. .095.118:547.292 Deacetylation. 2) Reactions with elimination of a particular group are denoted by .091.8. See examples at 66.091.8. 3) To designate a reagent, the appropriate -92 auxiliary is used, e.g. .094.3-926.217 Oxidation by air. .095.253-927.5 Alkylolation. Benzylolation. Phenylethylolation. -92... should be used only when the resources of 66.09 have been exhausted
66.091	Principal types of chemical transformation. Synthesis. Reactions of inorganic and organic compounds in general Reactions of inorganic compounds in general Reactions of organic compounds in general Synthesis in general ◇ 66.091.3-977 Pyrosynthesis Substitution, exchange in general Rearrangement in general Addition in general Elimination in general <i>Denote elimination of specified groups by combining auxiliaries</i> ◇ 66.094.25.091.8 Dehydrogenation ◇ 66.094.522.091.8 Desulphurization ◇ 66.095.13.091.8 Elimination of acyloxy group ◇ 66.095.81.091.8 Denitration ◇ 66.096.3.091.8 Dedeuteration. Detritiation
66.092	Decomposition processes. Cracking. Cleavage. Destruction ◇ 66.092-977 Thermal decomposition. Thermal cracking. Pyrolysis ◇ 66.092-977-922 Copyrolysis ◇ 66.092.085 Radiation decomposition. Radiolysis ◇ 66.092.088 Decomposition in electric discharges. Electrocracking. Plasmochemical decomposition. Plasmochemical pyrolysis ◇ 66.092.094.25 Hydrogenolysis. Destructive hydrogenation. Hydrocracking ◇ 66.092.094.3 Oxidative decomposition. Oxygenolysis ◇ 66.092.097.3 Decomposition catalysts ⇒ 66.094.94 ⇒ 665.64
66.092.1	Decomposition in general. Cracking without catalysts
66.092.2	Preliminary processes for decomposition. Preheating
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66.092.81	Processes for further refining of cracking products. Reforming ⇒ 665.64	66.094.48	Dehalogenation For Wurtz-Fittig reaction use 66.095.253.094.48 ◇ 66.094.48.094.25 Reductive dehalogenation (hydrodehalogenation)
66.092.89	Cracking to coke. Coking	66.094.5	Introduction or elimination of nonmetals
66.092.9	Other processes of decomposition. □ Distillation with zinc dust	66.094.52	Introduction or elimination of sulphur and selenium
66.093	Reactions and treatment with water. Addition and extraction of water	66.094.522	Introduction of sulphur. Sulphuration (sulphurization) ◇ 66.094.522.094.2 Reductive sulphuration
66.093.1	Solvation	66.094.523	Introduction of thiol-, thionic groups
66.093.2	Humidification. Moistening	66.094.524	Introduction of sulphuric acid. □ Sulphation. Sulphonation. Sulphoxidation. Sulphamation. Sulphonylation
66.093.4	Hydration	66.094.525	Introduction of sulphurous acid. □ Sulphination. Sulphenation
66.093.41	Physical-chemical hydration	66.094.529	Introduction of selenium. Selenation
66.093.43	Chemical hydration ◇ 66.093.43-92 Hydrating agents ◇ 66.093.43.097 Catalytic hydration	66.094.53	Introduction of phosphorus
66.093.48	Chemical dewatering. Chemical elimination of water. Dehydration For cyclodehydration use 66.095.252.093.48	66.094.533	Phosphorylation. □ Transphosphorylation ◇ 66.094.533.094.3 Oxidative phosphorylation
66.094	Treatment with various agents	66.094.55	Introduction of boron
66.094.1	Oxidation-reduction reactions. Redox reactions. Disproportionation (dismutation, rearrangement). Cannizzaro reaction	66.094.57	Introduction of silicon
66.094.2	Reduction in general. Reduction with various reducing agents Denote reducing agents by -92...	66.094.6	Acidification. Treatment with acids, in general
66.094.25	Hydrogenation	66.094.7	Alkalization. Basification. Treatment with bases, in general
66.094.29	Other reduction processes. Mixed reduction processes	66.094.72	Alkaline fusion
66.094.3	Oxidation. Oxygenation ◇ 66.094.3-921 Autoxidation. Self-oxidation ◇ 66.094.3-926.214 Oxidation by ozone. Ozonation. Ozonolysis ◇ 66.094.3-926.217 Oxidation by air ◇ 66.094.3-926.22 Oxidation by sulphur and sulphur compounds ◇ 66.094.3.094.72 Oxidizing alkaline fusion ◇ 66.094.3.097 Catalytic oxidation ◇ 66.094.3.097.8 Antioxidation. Inhibition of oxidation ◇ 66.094.3.098 Biochemical oxidation	66.094.8	Neutralization. Removal of acids or bases. Deacidification. Debasification
66.094.32	Hydroxylation. Glycolization	66.094.94	Exchange decomposition. Solvolysis ⇒ 66.091.4
66.094.39	Epoxidation	66.094.941	Hydrolysis ◇ 66.094.941-977 Pyrohydrolysis. Steam hydrolysis ◇ 66.094.941.094.7 Saponification. Alkaline hydrolysis
66.094.4	Introduction and elimination of halogens Halogens may be specified in the subdivisions of 66.094.4 by combining auxiliaries ◇ 66.094.42.094.414 Exchange of hydrogen for bromine	66.094.942	Alcoholysis. Reactions with spirits. □ Methanolysis. Ethanolysis. Phenolysis
66.094.41	Halogenation in general. □ Perhalogenation. Transhalogenation. Halogenolysis	66.094.943	Acidolysis. Reactions with acids. □ Acetolysis. Nitrolysis. Formolysis. Trifluoroacetolysis
66.094.412	Fluorination in general. □ Perfluorination	66.094.944	Ammonolysis. Reactions with ammonia ◇ 66.094.944.094.3 Ammoxidation (oxidative ammonolysis, oxidative amination, cyanodemethylation, aminooxidation, oxyammonolysis) ⇒ 66.095.8
66.094.413	Chlorination in general. □ Perchlorination. Chlorolysis (chlorinolysis) ◇ 66.094.413.094.3 Oxidizing chlorination. Oxychlorination ◇ 66.094.413.094.328 Chlorodehydroxylation	66.094.944.2	Aminolysis
66.094.414	Bromination in general. □ Perbromination. Bromolysis (brominolysis)	66.094.944.3	Amidolysis
66.094.415	Iodination in general. □ Periodination. Iodolysis	66.094.944.4	Hydrazinolysis (hydrazination)
66.094.42	Direct halogenation. Exchange of hydrogen for halogen	66.094.945	Thiolysis (mercaptolysis) ⇒ 66.094.522/525
66.094.43	Mutual substitution of different halogens	66.094.946	Cyanolysis
66.094.44	Replacement of oxygen-containing groups by halogens ⇒ 66.095.85	66.095	Various processes. Special organic chemical processes
66.094.45	Introduction of hydrohalogenic acid. Hydrohalogenation. Addition of CHHg, CHlg to multiple links	66.095.1	Acylation. Acetalization. Acyloxylation. Alkoxylation
66.094.46	Addition of halogens	66.095.11	Acylation in general. □ Transacylation. Introduction of acetyl, formyl and other groups. Formylation. Reimer-Tiemann reaction. Acetylation. Acetoacetylation. Malonylation. Oxalylolation. Succinylation. (succinoylation). Ethoxalylolation (ethoxaloylation). Acylation according to Friedel-Crafts ◇ 66.095.11-927.5 Benzoylation ◇ 66.095.11.094.2 Reductive acylation ◇ 66.095.11.094.25 Hydroacylation ◇ 66.095.11.094.522 Thioacylation. Thioformylation ◇ 66.095.11.095.252 Cycloacylation ◇ 66.095.11.095.832 Aminoacylation ◇ 66.095.11.096.3 Deuteroacylation. Deuteroformylation

66.095.118	Deacylation. □ Deacetylation. Deformylation	◇ 66.095.253.095.832-927.5 Aminobenzylation
66.095.12	Acetal formation. Acetalization (acetalation, ketalization). □ Acetonation (isopropylidenation). Glycosidation. Thioacetalization. Glucosidation (glucosylation)	◇ 66.095.253.095.833 Amidoalkylation (acylaminoalkylation). Amidomethylation (acylaminomethylation). Carbamoylethylation (carbamylethylation)
66.095.13	Formation of compound esters. □ Introduction of acyloxy-group. Acyloxylation (acoxylation). Acetoxylation	66.095.255 Introduction of alkenylo-radicals. Alkenylation. Olefination. □ Methylenation. Vinylation. Allylation. Prenylation. Wittig reaction
	◇ 66.095.13-927.5 Introduction of cyclic acyloxy-group. Aroyloxylation. Benzoyloxylation	66.095.256 Alkinylation. □ Ethynylation (ethynation). Propargylation
66.095.14	Formation of simple ethers. □ Alkoxylation. Methoxylation	66.095.257 Arylation. □ Phenylation. Tolylation. Hetarylation
	◇ 66.095.14-927.5 Aroxylation	66.095.26 Polymerization in general. Formation of macromolecular (high molecular) compounds
	◇ 66.095.14.094.41 Halogenalkoxylation	◇ 66.095.26-921 Homopolymerization
	⇒ 66.095.253	◇ 66.095.26-922 Copolymerization
66.095.17	Salt formation	◇ 66.095.26-922.2 Block copolymerization
66.095.2	Various kinds of molecular transformation	◇ 66.095.26-922.3 Inoculated copolymerization. Graft copolymerization
66.095.21	Isomerization. □ Beckmann transformation. Racemization	◇ 66.095.26-936.3 Homogeneous polymerization
	◇ 66.095.21.094.25 Hydroisomerization	◇ 66.095.26-936.4 Heterogeneous polymerization
	◇ 66.095.21.095.252 Cycloisomerization	◇ 66.095.26-977 High temperature polymerization
	◇ 66.095.21.097 Catalytic isomerization	◇ 66.095.26.085 Irradiation polymerization. Photochemical polymerization
66.095.216	Tautomerization	◇ 66.095.26.097 Catalytic polymerization
66.095.22	Rearrangement	66.095.261 Polymerization methods. Chain length
66.095.25	Cyclization. Alkylation	<i>In the subdivisions, denote copolymerization (co-...) by -922...</i>
66.095.251	Homologization	66.095.261.1 Addition polymerization
66.095.252	Cyclization (ring closing, ring formation)	66.095.261.2 Polycondensation (condensation polymerization)
	◇ 66.095.252.091.7 Cycloaddition. Diene synthesis. Diels-Alder addition	66.095.261.3 Telomerization
	◇ 66.095.252.093.48 Cyclodehydration	66.095.261.4 Oligomerization
	◇ 66.095.252.094.25 Hydrocyclization (reductive cyclization)	◇ 66.095.261.4.095.252 Cyclo-oligomerization
	◇ 66.095.252.094.258 Dehydrocyclization (cyclodehydrogenation)	66.095.261.42 Dimerization
	◇ 66.095.252.094.3 Oxidative cyclization	66.095.261.43 Trimerization
66.095.252.4	Aromatization	66.095.262 Polymerization in relation to the medium
66.095.252.5	Anellation (inner)	<i>Denote phase by -911...</i>
66.095.253	Alkylation. □ Methylation. Ethylation. Propylation. Butylation. Dialkylation. Polyalkylation. Transmethylation. Alkylation after Friedel-Crafts	◇ 66.095.262-911.3 Polymerization from gaseous monomers
	◇ 66.095.253-927.5 Aralkylation (aryalkylation). Benzylation. Phenylethylation. Tritylation (triphenylmethylation)	◇ 66.095.262-911.48 Suspension polymerization. Emulsion polymerization
	◇ 66.095.253-927.7 Pyridylethylation. Pyranylation	66.095.262.1 Polymerization without solvents and diluent. Block polymerization in mass
	◇ 66.095.253.094.25 Hydroalkylation (reductive alkylation)	66.095.262.2 Polymerization with solvents
	◇ 66.095.253.094.32 Hydroxyalkylation. Hydroxymethylation. Hydroxyethylation. Hydroxybutylation	66.095.268 Depolymerization
	◇ 66.095.253.094.41 Halogenalkylation (alkylhalogenation). Halomethylation	66.095.3 Condensation
	◇ 66.095.253.094.48 Wurtz-Fittig reaction	◇ 66.095.3-921 Autocondensation (self-condensation)
	◇ 66.095.253.095.13 Carbalkoxyalkylation (alkoxycarbonylalkylation). Carbethoxymethylation (ethoxycarbonylmethylation). Acetoxymethylation. Acetoethylation	◇ 66.095.3.094.25 Hydrocondensation (reductive condensation)
	◇ 66.095.253.095.14 Alkoxyalkylation. Alkoxy-methylation	◇ 66.095.3.094.258 Dehydrocondensation (oxidative condensation)
	◇ 66.095.253.095.14-927.5 Phenoxyalkylation	◇ 66.095.3.095.252 Cyclocondensation (cyclic condensation). Tishler reaction
	◇ 66.095.253.095.252 Cycloalkylation. Cyclopentylation. Cyclohexylation	66.095.31 Condensation without elimination. □ Aldol condensation. Acyloin condensation
	◇ 66.095.253.095.64 Carboxyalkylation. Carboxymethylation. Carboxyethylation	66.095.32 Condensation with water elimination. □ Crotonization. Perkin reaction
	◇ 66.095.253.095.832 Aminoalkylation. Aminomethylation. Aminoethylation. Transaminomethylation	66.095.34 Condensation of aldehydes and ketones with derivatives of ammonia. Preparation of Schiff's bases. □ Oxymation. Osazonation
		66.095.35 Condensation with hydrohalogenic acid elimination
		66.095.36 Condensation with hydrogen sulphide elimination
		66.095.37 Condensation with alcohol elimination. □ Ester condensation

66.095.5	Various processes of addition, especially by multiple links ⇒ 66.091.7	66.095.85	Sandmeyer and Gattermann reactions. Diazo-group exchange
66.095.53	Addition of HOHg. Hypohalogenation. Halogenhydrination ⇒ 66.094.41	66.095.86	Coupling reaction. Azo coupling (diazo coupling)
66.095.54	Addition of NOHg. Nitrosohalogenation. Addition of NO ₂ Hg. Nitrohalogenation 66.095.542/.545 ≡ 66.094.412/.415 to identify the halogen	66.095.91	Introduction of solubilizing groups in general ⇒ 66.094.524 ⇒ 66.095.253 ⇒ 66.095.3
66.095.55	Addition of HCN. Hydrocyanation (cyanhydration) ⇒ 66.095.842	66.095.92	Named organic reactions not classifiable elsewhere in 66.095 Specify by alphabetic extension (Table 1h)
66.095.6	Carbonylation. Carboxylation	66.096	Other reactions and processes
66.095.62	Introduction of carbonyl group (except acylation). Carbonylation. Ketonization ◇ 66.095.62.094.25 Hydroformylation. Oxosynthesis. Roelen reaction ◇ 66.095.62.094.32 Hydroxycarbonylation. Ketolization. Hydroxyketonation ◇ 66.095.62.095.252 Cycloketonization ⇒ 66.095.11	66.096.1	Treatment with inert gases (rare gases)
66.095.64	Obtaining of carboxyl compounds by introduction of carbon dioxide. Carboxylation ◇ 66.095.64.094.25 Hydrocarboxylation (hydrocarbonylation)	66.096.2	Treatment with carbonic acid or with compounds forming carbonic acid
66.095.7	Formation and reactions of organometallic compounds	66.096.3	Isotope exchange. □ Hydrogen exchange. Deuteration. Deuterium exchange. Deuterolysis. Tritiation (tritilation)
66.095.71	Formation of organometallic compounds. Metalation	66.096.4	Complex formation. Introduction of complex (coordination) bonded atoms. Formation of 'sandwiched structure'. Formation of molecular compounds
66.095.712	Formation of organic compounds of metals Denote elimination by .091.8 66.095.712 ≡ 546 for specific metals	66.096.5	Fluidization in general
66.095.712.46	Formation of magnesium organic compounds. Grignard reagents For Grignard reaction see 66.095.72	66.097	Catalytic processes. Catalysts. Catalyst carriers
66.095.712.49	Formation of mercury organic compounds. Mercuration	66.097.3	Catalysts Denote fixed and mobile by -946.1 and -946.3. Details by :546/547 and/or -03... ◇ 66.097.3-039.672 Catalysts on iron basis ◇ 66.097.3-039.7 Organic catalysts
66.095.713	Formation of organic compounds of semimetals 66.095.713 ≡ 546 for specific semimetals	66.097.36	Poisoning
66.095.713.289	Formation of germanium organic compounds. Germanylation. Germylation	66.097.38	Regeneration
66.095.718	Demetalation	66.097.4	Initiation agents
66.095.72	Reactions of organometallic compounds. □ Grignard reaction, Nesmejanov reaction	66.097.5	Media. Carriers
66.095.8	Introduction of nitrogen containing groups. Azotization. Cyanation	66.097.6	Activators. Promoters
66.095.81	Nitration ◇ 66.095.81.094.3 Oxynitration (oxidative nitration) ⇒ 66.094.943	66.097.8	Negative catalysts. Inhibition. Release. Stabilization of reaction. Stabilizers. Inhibitors. Passivators. Releasing agents
66.095.82	Nitrosation	66.098	Biochemical processes
66.095.83	Amination. Amidation. Diazotization	66.098.2	Aerobic processes
66.095.832	Amination. □ Transamination ◇ 66.095.832.094.25 Reductive amination ◇ 66.095.832.095.257 Aramination (arylamination) ◇ 66.095.832.095.712.49 Aminomercuration	66.098.4	Anaerobic processes
66.095.832.2	Imination	66.099	Other processes (operations) not mentioned in 66.02/.098
66.095.833	Amidation ◇ 66.095.833.095.11 Acetamidation	66.099.2	Granulation. Flaking ⇒ 62-492.3 ⇒ 621.775
66.095.833.2	Obtaining of amides of acids. Imidation	66.099.4	Preservation. Conservation
66.095.834	Diazotization	66.099.5	Treatment with powders. □ Dusting
66.095.84	Cyanation. Thiocyanation	66.099.6	Colouring of products. □ Dyeing. Staining
66.095.842	Cyanation (nitrylation). Substitution of CN for halogen, sulphide or carboxyl group ⇒ 66.094.944.3 ⇒ 66.095.55	66.099.7	Decolorization. Deodorization. Disinfection
66.095.843	Introduction of cyanic N≡C-O-, thiocyanogen N≡C-S-, isocyanic O=C=N- etc. groups. Thiocyanation (rhodanation)	66.099.72	Decolorizing. □ Blanching. Bleaching
		66.099.73	Deodorizing ⇒ 628.52
		66.099.74	Disinfection
		661	CHEMICALS Scope: the production of chemicals; also the chemicals themselves (chemical products), especially from the point of view of their further processing and utilization Denote individual inorganic and organic constituents by :546... and :547...
		661.1	Special chemical industries
		661.11	Fine chemicals in general Chemicals produced in a relatively pure state, usually in small amounts
		661.12	Pharmaceutical products ⇒ 615

661.13	Chemicals for the electrotechnical industry ⇒ 621.35	661.16.032.2	Contact
661.14	Photochemical materials (other than plates, films, papers) ⇒ 771.7	661.16.032.3	Fumigant
661.143	Luminescent (photoluminescent) and phosphorescent materials. Luminophors ⇒ 628.9.03 ⇒ 667.637.4	661.16.032.5	Systemic
661.15	Production of agricultural chemicals ⇒ 661.16	661.16.034	Chemicals according to stage of development or mode of action
661.15'1/9	Specific nutrients and microadditives	661.16.034.1	Nonselective chemicals
661.15'1	Nitrogen	661.16.034.3	Selective chemicals
661.15'11	Nitrate	661.16.034.5	Chemicals which are both ovicides and adult-insecticides
661.15'12	Ammonia	661.16.034.6	Ovicides
661.15'13	Amide	661.16.034.7	Larvicides
661.15'14	Nitrate and ammonia	661.16.034.8	Chemicals for killing adults
661.15'15	Ammonia and amide	661.162	Chemicals for controlling plants
661.15'16	Nitrate and amide	661.162.2	Weedkillers. Herbicides
661.15'2	Phosphorus	661.162.6	Plant growth and development regulators. Phytohormones. □ Absciscins. Gibberellins. Cytokinins
661.15'3	Potassium	661.162.62	Defoliant
661.15'4	Calcium	661.162.63	Chemicals for drying plants. Desiccants
661.15'5	Magnesium	661.162.65	Chemicals against plant lodging. Retardants
661.15'6	Sulphur	661.162.66	Growth and blossoming stimulants
661.15'7	Organic substances not containing nitrogen 661.15'7 ≅ 547	661.163	Antifungal, antibacterial, antiviral chemicals. Chemicals against plant diseases
661.15'9	Microadditives 661.15'9 ≅ 546	661.163.2	Antifungal chemicals. Fungicides
661.152	Fertilizers. Inorganic fertilizers in general ⇒ 631.8	661.163.3	Antiseptics
661.152.2	Simple fertilizers in general Class specific simple fertilizers under 661.5/.8	661.163.4	Bactericides. Viricides
661.152.3	Complex fertilizers Fertilizer content may be denoted by '1/9 ◇ 661.152.3'1'2 Nitrogen-phosphate (NP) fertilizers ◇ 661.152.3'1'2'3 Nitrogen-phosphate-potassium (NPK) fertilizers ◇ 661.152.3'11'2 Nitrophos ◇ 661.152.3'13'7 Urea-formaldehyde fertilizers	661.163.6	Chemicals controlling the development of microorganisms. □ Phytoalexins
661.152.32	Compound fertilizers	661.164	Chemicals for controlling arthropods ⇒ 661.168
661.152.33	Mixed and mixed-compound fertilizers	661.164.2	Chemicals for killing arthropods. Insecticides and acaricides
661.152.4	Organic-inorganic fertilizers	661.164.6	Growth regulators for arthropods
661.152.5	Micronutrients. Fertilizers containing microadditives Denote specific fertilizers by adding '9... ◇ 661.152.5'927 Boric fertilizers ◇ 661.152.5'971.1 Manganese fertilizers	661.165	Chemicals for control of molluscs
661.155	Chemicals for animal feeding. Artificial feedstuffs and feed additives	661.165.2	Molluscicides. Lymacides
661.155.2	Feedstuffs based on chemicals. Concentrates	661.165.6	Growth regulators for molluscs
661.155.3	Feed additives	661.166	Chemicals for controlling nemathelminthes
661.155.4	Animal growth stimulants. Fattening stimulants	661.166.2	Nematicides
661.155.5	Industrial wastes used for feedstuffs and feed additives	661.166.6	Growth regulators for nemathelminthes
661.155.6	Feedstuffs from natural substances mixed with chemical feed additives	661.167	Chemicals for controlling vertebrates
661.155.8	Feedstuff conservants. Feedstuff stabilizers	661.167.2	Chemicals for controlling rodents. Rodenticides
661.158	Veterinary products	661.167.5	Chemicals for controlling birds
661.16	Pest-control chemicals. Pesticides. Disinfectants Denote additives by .022.3... ◇ 661.16.022.39 Synergists	661.167.7	Chemicals for controlling fish. Ichthyocides
661.16.03	Special characteristics and effects	661.168	Repellents. Attractants. Antifeedants
661.16.032	Chemicals according to penetration into pests	661.168.2	Repellents
661.16.032.1	Stomach	661.168.4	Attractants
		661.168.5	Antifeedants
		661.169.2	Complex-effect chemicals
		661.169.23	Chemicals for seed-treatment
		661.169.25	Soil sterilizers
		661.169.5	Chemicals for protecting against unfavourable climatic or meteorological conditions, e.g. sun scorch
		661.17	Chemical compositions and mixtures for technical purposes
		661.172	Agents for preventing and removing scale and deposit
		661.173	Antistatic agents
		661.174	Agents against fire and combustion
		661.175	Anti-icing and antifreezing agents
		661.177	Hydraulic fluids. □ Brake fluids. Shock-absorbing fluids
		661.18	Colloids. Adsorbents. Surface-active agents. Detergents. Wetting, foaming, emulsifying and flotation agents. Soaps. Glycerol
		661.182	Colloids 661.182 ≅ 544.77
		661.183	Adsorbents

661.183.1	Adsorbents with special properties. Ion-exchangers. Cationic and anionic exchangers	661.187.4	Production of soft soaps, potassium soaps
661.183.12	Ion-exchanging adsorbents. Ionites (ion-exchangers)	661.187.5	Production of hard soaps, sodium soaps
661.183.122	Derived from carbon	661.187.6	Production of Marseilles soaps
661.183.123	Derived from synthetic resins and other synthetic materials. Polymeric ion-exchangers	661.187.7	Soaps according to composition
661.183.123.2	Cationic exchangers	661.187.71	Tallow soaps. Lime soaps
661.183.123.3	Anionic exchangers	661.187.72	Rosin soaps
661.183.123.4	Amphoteric ion-exchangers. Polymeric ampholyte	661.187.73	Oleic-acid soaps and similar soaps
661.183.123.6	Chelating ion-exchangers. Complex ion-exchangers	661.187.74	Semi-cold soaps. Cold-stirred soaps
661.183.123.7	Electronic ion-exchangers. Redox-ionites	661.187.76	Soaps containing synthetic detergents or water softeners. Hard-water soaps
661.183.124	Derived from natural or synthetic silicates	661.187.8	Soaps according to their use
661.183.126	Derived from cellulose and its derivatives	661.187.83	Household soaps
661.183.2	Activated carbon	661.187.84	Soaps for use on the skin ⇒ 665.58
661.183.3	Other organic adsorbents	661.187.842	Toilet soaps
661.183.4	Natural silicates. Natural earths	661.187.843	Shaving soaps
661.183.5	Other natural minerals. Natural oxides	661.187.845	Medicated soaps
661.183.6	Synthetic silicates. Synthetic zeolites	661.187.85	Abrasive soaps. □ Pumice soaps
661.183.7	Silica gel	661.187.86	Laundry soaps
661.183.8	Synthetic inorganic oxides. Activated alumina. □ Aluminium hydroxide gel	661.187.88	Soaps for other uses ◇ 661.187.88:621.778.1 Soaps for wire drawing ⇒ 621.892
661.184	Semicolloids	661.188	Glycerol industry
661.185	Surface-active agents (surfactants). □ Wetting, foaming and emulsifying agents. Detergents. Flotation agents ⇒ 648.18 ⇒ 66.063.5 ⇒ 661.187	661.188.1	Glycerol (glycerine)
661.185-3	Mixtures	661.188.2	Recovery of glycerol from waste liquors in stearate production
661.185.2	Organic ionic agents	661.188.3	Recovery of glycerol from soap liquors (spent lyes), from soap production in open vessels
661.185.22	Anionic surfactants	661.188.4	Recovery of glycerol from soap liquors (spent lyes), from soap production in closed vessels
661.185.23	Cationic surfactants	661.188.5	Production of colourless technical glycerol
661.185.3	Amphotytic (amphoteric) agents	661.188.6	Production of chemically pure glycerol. Pharmaceutical glycerol
661.185.4	Organic non-ionic surfactants	661.188.7	Other processes for glycerol manufacture. □ Synthetic glycerol production. Glycerol from fermentation processes
661.185.5	Inorganic surfactants	661.188.8	Glycerol derivatives
661.185.6	Detergents and cleansers (powder, liquid or paste). Scouring powders	661.2	Production of sulphur and its derivatives
661.185.7	Wetting, foaming and emulsifying agents	661.21	Sulphur
661.185.72	Wetting agents ⇒ 66.063.5	661.211	Production of sulphur from native sulphur ores. □ Flowers of sulphur. Rod sulphur
661.185.74	Foaming agents ⇒ 66.069.852	661.214	Production of sulphur from raw materials with chemically bonded sulphur. Recovered sulphur. Gaseous (by-product) sulphur
661.185.76	Emulsifying agents ⇒ 66.063.61	661.214.1	Sulphur from pyrites and other sulphides and concentrates
661.185.8	Flotation agents ⇒ 622.765.061	661.214.2	Sulphur from gases including waste gases from various production processes. Sulphur from oil, coal, shales etc.
661.187	Soap industry <i>Denote state and shape of soap by -4... (from 62-4...)</i> ◇ 661.187-404 Liquid soap ◇ 661.187-404.9 Soft soap ◇ 661.187-412 Bar soap. Cake soap ◇ 661.187-492 Soap powder ◇ 661.187-493.5 Soap flakes	661.214.3	Sulphur from gypsum, anhydrite and other sulphates
661.187.032	Preparation of the lye. Evaporation	661.214.9	Other methods of sulphur production. □ From etching solutions, acid sludge and the wastes of processes using sulphuric acid
661.187.033	Saponification. Boiling ⇒ 66.094.941.094.7	661.22	Sulphur compounds with oxidation number -2
661.187.034	Liming. Salting out. Grinding. Adjusting	661.221	Carbon sulphide (carbon disulphide)
661.187.035	Artificial figging. Dyeing. Perfuming. Filling	661.222	Carbonyl sulphide COS
661.187.036	Draining. Running off. Pumping out	661.224	Hydrogen sulphide
661.187.037	Shaping. Cooling. Forming of cakes of soap etc.	661.225	Sulphides. Disulphides. Polysulphides
661.187.047	Drying --- (from 66.047)	661.23	Halogen compounds of sulphur. □ Sulphur chlorides, bromides, fluorides ⇒ 661.4
661.187.2	Raw materials for soap. □ Fatty acids. Rosin. Alkalies	661.24	Sulphurous acid. Sulphites. Sulphurous anhydride. Sulphur compounds with oxidation number +4
		661.241	Sulphurous acid

661.243	Sulphites. Hydrosulphites (bisulphites)	661.44	Chlorates in general. Chloric acid
661.245	Sulphur oxide dichloride. Thionyl chloride. Thionyl fluoride. Thionyl bromide	661.45	Chlorine compounds not given elsewhere
661.248	Sulphur dioxide (sulphurous acid anhydride)	661.452	Chlorine oxides
661.25	Sulphuric acid ◇ 661.25-404.512 Concentrated sulphuric acid	661.453	Chlorites in general. Chlorous acid
661.251	Oleum (fuming sulphuric acid). Sulphuric acid monohydrate	661.454	Perchlorates in general. Perchloric acid
661.253	Production of sulphuric acid by means of nitrous sulphuric acid	661.46	Bromide and bromide compounds
661.257	Production of sulphuric acid by contact process ⇒ 661.248	661.461	Bromine
661.27	Compounds of sulphur trioxide with halogens and nitrogen	661.462	Bromides in general
661.28	Thiosulphates. Dithionites. Dithionates. Polythionates	661.463	Hypobromites in general
661.29	Other compounds of sulphur Specify by :546... ⇒ 661.494 ⇒ 661.639.6	661.464	Bromates in general. Bromic acid
661.3	Production of soda and potash, alkalis	661.47	Iodine and iodine compounds
661.33	Production of soda and potash. Soda and potash industry	661.471	Iodine extraction
661.332	Potassium carbonate. Potash ⇒ 661.832	661.472	Iodides in general
661.333	Sodium carbonate. Calcined soda. Soda in general ⇒ 661.833	661.473	Hypoiodites in general
661.334	Dense soda	661.474	Iodates in general. Iodic acid
661.335	Soda crystals	661.48	Fluorine and fluorine compounds
661.336	Sodium hydrocarbonate (bicarbonate)	661.481	⇒ 661.8'036
661.34	Production of alkalis. Alkalis in general. Potassium hydroxide. Sodium hydroxide	661.482	Fluorine
661.342	Potassium hydroxide (caustic potash)	661.487	Fluorides in general
661.343	Sodium hydroxide (caustic soda)	661.488	Hydrogen fluoride. Hydrofluoric acid
661.4	Production of halogens and halogen compounds, inorganic peroxo-compounds ⇒ 661.23 ⇒ 661.723 ⇒ 661.8'04	661.49	Inorganic per-compounds. Peroxo-salts in general ⇒ 661.8'024
661.41	Production of chlorine	661.491	Hydrogen peroxide
661.413	Production of chlorine by oxidation of hydrochloric acid or chlorides by means of various oxidants. Chemical methods of chlorine production Specify oxidants by -92... ◇ 661.413-926.21 Production of chlorine by oxidation of hydrogen chloride gas. Deacon process	661.493	Peroxocarbonates (percarbonates). Peroxocarbonic acid (percarbonic acid)
661.417/.418	Electrochemical production of chlorine	661.494	Peroxosulphates (persulphates). Peroxosulphuric acid (persulphuric acid). Caro's acid
661.417	Electrolysis of fused chlorides	661.495	Peroxoborates (perborates). Peroxoboric acids
661.418	Production of chlorine and alkalis by electrolysis of fused solutions	661.496	Peroxophosphates (perphosphates). Peroxophosphoric acids (perphosphoric acids)
661.419	Production of hydrogen chloride and hydrochloric acid	661.5	Production of nitrogen compounds. Fixation of nitrogen. Nitrogen industry in general ⇒ 631.461 ⇒ 661.938 ⇒ 661.98
661.419.1	Hydrogen chloride	661.52	Production of ammonium salts. Production of nitrogen fertilizers in general ⇒ 661.635.13
661.419.4	Hydrochloric acid ◇ 661.419.4.081.2 Absorption of hydrogen chloride. Absorbers	661.521	Ammonium chloride (sal ammoniac)
661.42	Chlorides. Natural salts	661.522	Ammonium sulphate
661.43	Hypochlorites in general. Hypochlorous acid. □ Stain removers and bleaches based on hypochlorites ⇒ 648.18	661.523	Ammonium carbonates. Mixtures of carbonates. □ Ammonium bicarbonate. Ammonium carbamate
		661.525	Ammonium nitrate. Ammonia saltpetre
		661.526	Mixtures (fertilizers) on the basis of ammonium nitrate. □ Ammonium sulphate-nitrate
		661.53	Ammonia
		661.56	Nitric acid. Nitrides in general
		661.561	Production of nitric acid from sodium nitrate and sulphuric acid
		661.565	Synthesis from oxygen and nitrogen
		661.566	Production of diluted nitric acid by oxidation of ammonia. Conversion of ammonia ◇ 661.566.081.2 Absorption of nitrogen dioxide by water
		661.567	Direct synthesis of concentrated nitric acid from nitrogen oxides
		661.57	Nitrogen compounds not mentioned above
		661.571	Nitrides. Azides. Metal amides
		661.572	Hydrazine and its derivatives
		661.573	Hydroxylamine and its derivatives

661.574	Nitrous acid. Nitrites in general ⇒ 661.8'042	661.66	Production of carbon and its inorganic compounds ⇒ 661.33 ⇒ 661.8'062 ⇒ 661.97
661.575	Chloramines	661.664	Hydrocyanic acid (prussic acid). Cyanides. Cyanates ⇒ 661.8'037 ⇒ 661.8'038
661.6	Production of various nonmetals and semimetals (metalloids), and their compounds	661.665	Production of carbides in general
661.63	Production of phosphorus and its compounds. Phosphorus industry in general	661.665.1	Silicon carbide (carborundum)
661.631	Phosphorus	661.665.2	Heavy metal carbides ⇒ 669.018.25
661.632	Natural phosphates. Production of phosphoric fertilizers in general	661.665.3	Boron carbide
661.632.1	Chemical processing by means of mineral acids. Wet method	661.666	Pure carbon. Graphite. Carbon blacks
661.632.2	Production of superphosphates	661.666.2	Graphite
661.632.7	Thermal processing of natural phosphates. Thermal phosphates	661.666.3	Mineral black
661.632.9	Phosphates of animal origin ⇒ 664.932.7 ⇒ 664.957 ⇒ 665.937	661.666.4	Technical carbon. Carbon black. □ Flameblack. Lampblack
661.634	Production of phosphorus-containing acids. □ Orthophosphoric acid. Polyphosphoric acids	661.666.5	Vegetable blacks. □ Wood charcoal
661.635	Phosphorus-containing salts. Orthophosphates in general ⇒ 661.8'045.5	661.666.6	Animal charcoal. Bone charcoal. Blood charcoal ⇒ 665.938
661.635.1	Phosphates of alkali metals	661.668	Metal carbonyls in general
661.635.11	Potassium phosphates	661.669	Other carbon compounds. □ Insertion compounds
661.635.12	Sodium phosphates	661.67	Ultramarine and derived pigments. □ Ultramarine (lapis lazuli). Lazurite ⇒ 667
661.635.13	Ammonium phosphates	661.68	Production of silicon and its compounds ⇒ 661.183.6 ⇒ 661.665.1 ⇒ 661.8'065
661.635.3	Mixtures	661.683	Water glass. Sodium and potassium silicates
661.635.4	Alkaline earth phosphates	661.685	Silicohydrides. Silicides
661.635.41	Calcium phosphates	661.686	Silicon compounds containing halogens. □ Silicon chloride. Silicon bromide. Silicon fluoride ⇒ 661.488
661.635.42	Magnesium phosphates	661.687	Silicon compounds from nitrogen. □ Silicon nitride
661.635.6	Condensed (dehydrated) phosphates in general	661.691	Selenium. Selenium compounds ⇒ 661.8'055
661.635.64	Metaphosphates	661.692	Tellurium. Tellurium compounds ⇒ 661.8'056
661.635.66	Pyrophosphates	661.693	Germanium and its compounds ⇒ 661.8'067
661.635.68	Polyphosphates	661.699	Other nonmetals or semimetals and their compounds
661.636	Phosphorus hydride. Phosphides	661.7	Production of organic substances. Organic chemicals <i>Denote organic products which are not given in 661.7... by :547...</i> ⇒ 662.749.3 ⇒ 662.75 ⇒ 665.6 ⇒ 665.9
661.638	Phosphorus compounds containing halogens	661.71	Simple organic compounds ⇒ 665.71
661.639.2	Phosphites. Hypophosphates. Hypophosphites	661.715	Hydrocarbons
661.639.3	Phosphorus oxides	661.715.1	Aliphatic hydrocarbons in general
661.639.5	Phosphorus nitrides	661.715.2	Saturated aliphatic hydrocarbons 661.715.2 ≡ 547.21
661.639.6	Phosphorus sulphides. Phosphorus sesquisulphide. Phosphorus pentasulphide	661.715.3	Unsaturated aliphatic hydrocarbons 661.715.3 ≡ 547.31
661.64	Production of arsenic, arsenites, arsenic oxides and salts ⇒ 661.8'046	661.715.4	Cyclic hydrocarbons in general
661.641	Arsenic	661.715.5	Monocyclic hydrocarbons except benzene and its homologues. Polymethylenes. □ Cyclopentane. Cyclopentadiene. Cyclohexene
661.642	Arsenic compounds with oxidation number +3 in general. Arsenious acid		
661.642.1	Diarsenic trioxide (white arsenic, commercial arsenic)		
661.642.5	Arsenites		
661.643	Arsenic pentoxide compounds in general. □ Arsenic pentoxide. Arsenic acid. Arsenates and analogous compounds		
661.644	Arsenic hydrides. Arsenic halogenides		
661.645	Arsenic compounds with sulphur. Arsenic sulphides		
661.65	Production of boron and boron compounds ⇒ 661.8'061		
661.651	Processing of ores containing borates. □ Processing of datolite, boracite, ulexite (boronatrocaltite), tincal, kernite		
661.652	Disodium tetraborate (sodium borate, borax)		
661.654	Boric acid		
661.657	Boron compounds other than borates and boric acid		

661.715.6	Polycyclic hydrocarbons other than those with exclusively aromatic nuclei ⇒ 662.749.3	661.725.7	Monohydric unsaturated aliphatic alcohols 661.725.7 ≡ 547.36
661.715.7	Aromatic hydrocarbons	661.725.8	Polyhydric alcohols. Cyclic alcohols
661.716	Mixtures of aliphatic hydrocarbons	661.725.81	Polyhydric alcohols (polyols) <i>For the production of glycerol see 661.188</i> 661.725.81 ≡ 547.42 other than compounds listed in 661.725.82... and glycerol
661.716.1	Mixtures of saturated hydrocarbons		
661.716.2	Mixtures of unsaturated hydrocarbons	661.725.82	Dihydric alcohols (glycols, diols) in general.
661.716.3	Mixtures of saturated hydrocarbons with unsaturated hydrocarbons		Dihydric alcohol ethers
661.717	Organic nitrogen compounds	661.725.822	Ethylene glycol
661.717.2	Nitro compounds of hydrocarbons	661.725.823	Propylene glycol
661.717.3	Amines	661.725.84	Alcohols containing other elements
661.717.4	Nitriles	661.725.841	Alcohols containing halogen
661.717.5	Amides	661.725.844	Amino alcohols (oxyamines)
661.717.52	Urea	661.725.85	Cyclic alcohols
661.717.53	Acrylamide	661.725.852	Phenol
661.717.55	Lactams. □ Caprolactam	661.725.9	Mixtures of alcohols
661.717.6	Organic isocyanates	661.726	Ethers other than diethyl ether. Esters of alcohols (except methanol and ethanol) with oxygen-containing inorganic acids 661.726 ≡ 661.725
661.718	Organic compounds of various nonmetals		
661.718.1	Phosphorus compounds	661.727	Aldehydes. Ketones
661.718.2	Arsenic compounds	661.727.1	Formaldehyde
661.718.32	Antimony compounds	661.727.2	Acetaldehyde
661.718.33	Bismuth compounds	661.727.3	Butyl aldehyde
661.718.4	Boron compounds	661.727.4	Acetone
661.718.5	Silicon compounds	661.727.6	Methyl ethyl ketone
661.718.6	Germanium compounds	661.727.7	Acrylic aldehyde
661.719	Organic compounds of sulphur, selenium, tellurium	661.727.8	Cyclic aldehydes and ketones. □ Furfural
661.719.2	Sulphur compounds	661.728	Cellulose and its derivatives <i>For processes in the production of cellulose use 676.02...</i> ⇒ 547.458.8 ⇒ 676.16 ⇒ 677.46 ⇒ 678.54
661.719.3	Selenium compounds		
661.719.4	Tellurium compounds		
661.72	Alcohols and their derivatives ⇒ 663.5		
661.721	Methanol (methyl alcohol)	661.728.7	Cellulose
661.722	Ethanol (ethyl alcohol). □ Denatured alcohol	661.728.8	Cellulose derivatives. Cellulose esters in general
661.723	Halogen derivatives of simple hydrocarbons. Organohalogens. □ Organochlorines. Chlorofluorocarbons (CFCs, Freons). Polychlorinated biphenyls (PCBs). Dioxins ⇒ 661.4	661.728.81	Cellulose formate
661.723.2	Halogen derivatives with one carbon atom	661.728.82	Cellulose acetate
661.723.3	Chloral hydrate. Chloral	661.728.84	Cellulose xanthate. Viscose
661.723.6	Halogen derivatives with two carbon atoms	661.728.85	Other organic cellulose esters
661.723.61	Monohalogen derivatives	661.728.86	Cellulose nitrate
661.723.62	Dihalogen derivatives. Dichloroethylene	661.728.88	Cuprammonium derivatives
661.723.63	Trihalogen derivatives. Trichloroethylene	661.728.89	Other cellulose derivatives. □ Cellulose ethers
661.723.64	Tetrahalogen derivatives. Tetrachloroethane	661.729	Organic peroxides
661.723.65	Pentahalogen derivatives. Pentachloroethane	661.74	Organic acids, their salts, esters etc. <i>Denote salts and esters of particular acids (...ates) by adding the digit 4 to the number for the acid</i> ◇ 661.742.14 Salts and esters of acrylic acid. Acrylates ⇒ 665.12
661.723.66	Hexahalogen derivatives. Hexachloroethane	661.741	Monobasic saturated carboxylic acids
661.723.7	Saturated derivatives with more than two carbon atoms 661.723.7 ≡ 547.21	661.741.1	Acetic (ethanoic) acid ⇒ 661.8'071.2
661.723.8	Unsaturated derivatives with more than two carbon atoms 661.723.8 ≡ 547.31	661.741.2	Formic acid ⇒ 661.8'071.1
661.724	Diethyl ether (ethyl ether, sulphuric ether)	661.741.3	Propionic (propanoic) acid ⇒ 661.8'071.3
661.725	Alcohols other than methanol and ethanol. Polyhydric alcohols	661.741.4	Butyric acid ⇒ 661.8'071.4
661.725-11/-12	Esters of alcohols with oxygen-containing inorganic acids and their salts 661.725-11/-12 ≡ 546.1/-2	661.741.5	Valeric acid
661.725.2	Propyl alcohol	661.742	Monobasic unsaturated carboxylic acids
661.725.3	Isopropyl alcohol (isopropanol)	661.742.1	Acrylic (propenoic) acid
661.725.4	Butanols (butyl alcohols)	661.742.2	Methacrylic acid
661.725.5	Amyl alcohols ⇒ 663.551.7	661.743	Dicarboxylic acid, saturated and unsaturated
661.725.6	Higher monohydric saturated aliphatic alcohols (with more than five carbon atoms). □ Hexanols. Heptanols. Octanols	661.743.1	Oxalic acid ⇒ 661.8'074.2
		661.743.2	Succinic acid

661.743.3	Adipic acid	661.8'027.3/9	Salts of specific metals 661.8'027.3/9 $\equiv$ 546.3/9
661.743.4	Sebacic acid	661.8'03	Halogen compounds. Cyanide compounds. Hydride $\Rightarrow$ 661.4
661.743.7	Maleic acid	661.8'032	Chlorine compounds
661.743.73	Maleic anhydride	661.8'032.1	Chloride
661.743.78	Fumaric acid	661.8'032.2	Oxychlorine compounds
661.744	Aromatic acids.	661.8'032.22	Hypochlorite
661.744.1	Benzoic acid $\Rightarrow$ 661.8'078.2	661.8'032.23	Chlorite
661.744.2	Phthalic acids	661.8'032.25	Chlorate
661.744.22	Phthalic acid (orthophthalic acid) $\Rightarrow$ 661.8'078.4	661.8'032.27	Perchlorate
661.744.23	Isophthalic acid	661.8'034	Bromine compounds 661.8'034 $\equiv$ 661.8'032
661.744.24	Terephthalic acid	661.8'035	Iodine compounds 661.8'035 $\equiv$ 661.8'032
661.744.5	Salicylic acid $\Rightarrow$ 661.8'078.3	661.8'036	Fluorine compounds 661.8'036 $\equiv$ 661.8'032
661.745	Amino acids	661.8'037	Cyanide compounds
661.746	Hydroxy acids. Polyhydroxycarboxylic acids	661.8'037.1	Cyanide
661.746.1	Glycolic (hydroxyacetic) acid	661.8'037.2	Isocyanate
661.746.2	Lactic acid $\Rightarrow$ 661.8'074.3	661.8'037.3	Cyanate
661.746.3	Tartaric (dihydroxysuccinic) acid	661.8'037.4	Fulminate
661.746.4	Gluconic acid	661.8'037.5	Cyanamide
661.746.5	Citric acid $\Rightarrow$ 661.8'074.5	661.8'038	Thiocyanate
661.747	Oxo acids	661.8'039	Hydride
661.747.1	Acetoacetic acid	661.8'04	Pentavalent element compounds
661.77	Heterocyclic compounds	661.8'041	Nitride $\Rightarrow$ 661.5
661.78	Organometallic and organo-semimetallic compounds	661.8'042	Nitrite
661.782	Organo-semimetallic compounds	661.8'043	Nitrate
661.782.2	Of germanium	661.8'044	Other nitrogen compounds
661.782.3	Of arsenic	661.8'044.3	Azide
661.782.4	Of antimony	661.8'044.5	Amide
661.782.5	Of bismuth	661.8'045	Phosphorus compounds $\Rightarrow$ 661.63
661.782.6	Of tellurium	661.8'045.3	Phosphite
661.782.7	Of polonium	661.8'045.4	Metaphosphate
661.783/789	Various organometallic compounds 661.783/789 $\equiv$ 546.3/9	661.8'045.5	Orthophosphate
661.784.7	Zinc ethyl	661.8'046	Arsenic compounds $\Rightarrow$ 661.64
661.8	Metallic compounds in general. Salts. Mineral compounds <i>To denote specific compounds (oxides, bases, salts) not allowed for elsewhere, add the '0 auxiliaries listed below to the direct division denoting the metal of the anion. Thereafter, basic compounds (salts) and double or complex compounds (salts) may be distinguished by adding '08 and '09</i> $\diamond$ 661.862'027 Aluminates (salts corresponding to the oxide Al_2O_3)	661.8'046.1	Arsenide
661.8-11/-18	Valency (of the metallic element in the compound) $\diamond$ 661.8-11 Monovalent $\diamond$ 661.8-12 Divalent	661.8'046.3	Arsenite
661.8'01	Compounds of special importance	661.8'046.5	Arsenate
661.8'02	Oxides	661.8'047	Antimony compounds
661.8'022	Oxide	661.8'05	Sulphur, selenium and tellurium compounds. Chalcogenides
661.8'023	Hydroxide	661.8'051	Sulphur compounds $\Rightarrow$ 661.2
661.8'024	Peroxide	661.8'051.1	Sulphide
661.8'027	Salts with corresponding oxide as an acid residue	661.8'051.6	Polysulphide
661.8'027.1	Acids. Heteropoly acids 661.8'027.1 $\equiv$ 546	661.8'052	Sulphite. Hydrosulphite
	$\diamond$ 661.878'027.118.5 Phosphotungstic acid	661.8'053	Sulphate. Hydrogen sulphate (bisulphate)
	$\diamond$ 661.878'027.1 Tungstic acid	661.8'053.2	Sulphate
661.8'027.2	Metallic compounds with more electropositive metals 661.8'027.2 $\equiv$ 546	661.8'053.7	Hydrogen sulphate
	$\diamond$ 661.887'027.2 Bismuthates	661.8'054	Other sulphur compounds
		661.8'054.1	Pyrosulphite. Pyrosulphate
		661.8'054.2	Thiosulphate
		661.8'054.3	Dithionite
		661.8'054.5	Peroxosulphate (persulphate)
		661.8'055	Selenium compounds
		661.8'056	Tellurium compounds
		661.8'061	Boron compounds $\Rightarrow$ 661.65
		661.8'062	Carbon compounds (inorganic) $\Rightarrow$ 661.66
		661.8'065	Silicon compounds
		661.8'067	Germanium compounds
		661.8'07	Organic salts, esters etc. $\Rightarrow$ 661.74
		661.8'071	Salts of saturated fatty acids with fewer than ten carbon atoms

661.8'071.1	Formate	661.85	Lead, copper, silver and gold compounds
661.8'071.2	Acetate	661.852	Lead compounds
661.8'071.3	Propionate		◇ 661.852'022 Lead oxides. NOTE: (derived from 661.8'022, further subdivided here)
661.8'071.4	Butyrate		◇ 661.852'022.1 Massicot. Yellow lead. Amorphous or molten lead oxide
661.8'074	Various salts of aliphatic acids		◇ 661.852'022.2 Litharge. Crystalline lead oxide
661.8'074.2	Oxalate		◇ 661.852'022.3 Litharge of silver
661.8'074.3	Lactate		◇ 661.852'022.4 Red lead (minium). Pure lead oxide
661.8'074.5	Citrate		◇ 661.852'062 Carbon compounds of lead. NOTE: (derived from 661.8'062, further subdivided here)
661.8'075	Salts of aliphatic acids with longer chains		◇ 661.852'062.28 Basic lead carbonate. Lead white. Cerussite
661.8'075.2	Palmitate	661.856	Copper compounds
661.8'075.3	Stearate	661.857	Silver compounds
661.8'075.6	Oleate	661.859	Gold compounds
661.8'075.7	Linoleate		
661.8'075.9	Resinate		
661.8'078	Salts of aromatic acids		
661.8'078.2	Benzoate		
661.8'078.3	Salicylate		
661.8'078.4	Phthalate		
661.8'08	Basic salts		
	◇ 661.856'08 Basic copper salts	661.86	Aluminium compounds. Compounds of rare earths
661.8'09	Double and complex salts	661.862	Aluminium compounds
	<i>When denoting double salts, the higher main subdivision from 661.8 should be used, with further subdivision as 546. Divide 661.8'09 like 546, e.g. 661.842'045.5'091.6 Calcium fluoride phosphate</i>		◇ 661.862'027 Aluminates ⇒ 669.712.3
661.8'099.9	Mixtures	661.863/868	Rare earth compounds. Compounds of scandium, yttrium, gallium, indium etc.
		661.863.1	Scandium compounds
		661.864.1	Yttrium compounds
661.83	Alkali metal compounds	661.865	Rare earth compounds in general
	⇒ 661.3	661.865.4	Lanthanum compounds
661.832	Potassium compounds in general. Potassium industry	661.865.5	Cerium compounds
	◇ 661.832'043 Potassium nitrate (nitre, saltpetre)	661.865.6	Praseodymium compounds
	⇒ 661.332	661.865.7	Neodymium compounds
	⇒ 661.342	661.865.8	Promethium compounds
661.833	Sodium compounds	661.865.9	Samarium compounds
	◇ 661.833'032.1 Sodium chloride. Rock salt (halite)	661.866.1	Europium compounds
	⇒ 661.333	661.866.2	Gadolinium compounds
	⇒ 661.336	661.866.3	Terbium compounds
	⇒ 661.343	661.866.4	Dysprosium compounds
	⇒ 661.652	661.866.5	Holmium compounds
	⇒ 661.683	661.866.6	Erbium compounds
	⇒ 664.41	661.866.7	Thulium compounds
661.834	Lithium compounds	661.866.8	Ytterbium compounds
661.835	Rubidium compounds	661.866.9	Lutetium compounds
661.836	Caesium compounds	661.868.1	Gallium compounds
		661.868.2	Indium compounds
		661.868.3	Thallium compounds
661.84	Compounds of group II metals	661.87	Manganese group compounds, iron group compounds and group VI metal compounds
661.841	Generalities on alkaline earth metals	661.871	Manganese compounds
661.842	Calcium compounds	661.872	Iron compounds
661.843	Strontium compounds		◇ 661.872'02 Iron oxides in general
661.844	Barium compounds		◇ 661.872'022 Iron oxide. NOTE: (derived from 661.8'022, further subdivided here)
661.845	Beryllium compounds		◇ 661.872'022-12 Iron (II) oxide (ferrous oxide)
661.846	Magnesium compounds		◇ 661.872'022-13 Iron (III) oxide (ferric oxide)
661.847	Zinc compounds		◇ 661.872'022.21 Natural red iron oxide. Haematite. Red ochre. Terra di Siena (red chalk, ruddle)
	◇ 661.847.2 Zinc oxides		◇ 661.872'022.23 Synthetic red iron oxides. English red. Venetian red. Pompeian red. Naples red. Indian red. Japanese red. Chinese red
	◇ 661.847.22 Zinc white		◇ 661.872'022.3 Yellow iron oxide. Yellow ochre
	◇ 661.847.22-027.332 Grey zinc oxide (as waste from zinc white). Zinc grey. Pearl grey. Stone grey		◇ 661.872'022.4 Black iron oxide. Iron black
	◇ 661.847.27 Zincates		◇ 661.872'022.5 Other iron oxides. Umber
	◇ 661.847.277.3 Cobalt green		◇ 661.872'023 Iron hydroxides
	◇ 661.847.511 Zinc sulphide. Lithopone. Sulphopone		◇ 661.872'037.1 Iron cyanide. NOTE: (derived from 661.8'037.1, further subdivided here)
	◇ 661.847.532 Zinc sulphate. White vitriol		
661.848	Cadmium compounds		
661.849	Mercury compounds		
	◇ 661.849'051.1511 Mercuric sulphide. Cinnabar		

	◇ 661.872'037.19 Complex cyanides	661.94	Ozone
	◇ 661.872'037.194.1 Calcium ferrocyanide	661.96	Hydrogen
	◇ 661.872'053.2 Iron sulphates. NOTE: (derived from 661.8'053.2, further subdivided here)	661.968	Hydrides in general ⇒ 661.8'039
	◇ 661.872'053.2-12 Ferrous sulphate. Green vitriol. Copperas	661.97	Production of carbon oxides, carbonic acid, phosgene
	◇ 661.872'053.29 Complex sulphates. Iron alum	661.974	Carbonic acid
661.873	Cobalt compounds	661.975	Carbonic dioxide(II) (carbon monoxide)
661.874	Nickel compounds	661.977	Phosgene
661.875	Chromium compounds	661.98	Oxides of nitrogen
	◇ 661.875'02 Chromium oxides	661.981	Nitric hemioxide (nitrous oxide, laughing gas)
	◇ 661.875'027.1 Chromic acid	661.982	Nitric oxide
661.877	Molybdenum compounds	661.984	Nitrogen dioxide (nitrogen tetroxide)
661.878	Tungsten compounds	661.986	Nitrosyl chloride
661.879	Compounds of radioactive elements	661.99	Production of other gases
661.879.1	Uranium compounds	662	EXPLOSIVES. FUELS
661.879.2	Radium compounds	662.1/4	Explosives ⇒ 622.235 ⇒ 623.45
661.879.93	Neptunium compounds		
661.879.94	Plutonium compounds	662.1	Fireworks. Pyrotechnic and related devices ⇒ 623.4.083.3
661.88	Group IV and Group V metal compounds	662.11	Display fireworks
661.881	Tin compounds	662.111	Pyrotechnic components. □ Fuses, touchpapers, primers for fireworks. Charges (producing light, colour, bangs, whistles)
661.882	Titanium compounds	662.112	Stationary flares. Fire sprays, fountains etc. Bengal lights
	◇ 661.882'022-13 Titanium sesquioxide	662.113	Fireworks producing noise and light. □ Bangers. Squibs. Crackers, firecrackers.
	◇ 661.882'022-14 Titanium dioxide. Titanium white	662.114	Roman candles. Rains of fire. Sparklers. Indoor fireworks
661.883	Zirconium and hafnium compounds	662.115	Flying fireworks. □ Rockets (skyrockets). Flying dragons
661.883.1	Zirconium compounds	662.116	Whirling, spinning fireworks. □ Catherine wheels
661.883.2	Hafnium compounds	662.12	Mounted firework displays. Pyrotechnic mosaics, pictures, set-pieces
661.884	Thorium compounds	662.122	Pyrotechnic signalling devices. Signal flares. Flare apparatus ⇒ 654.912
661.884.1	Thorium, recovered (isolated) from the compounds of radioactive elements. Mesothorium	662.123	Star signals. Very lights
661.886	Antimony compounds	662.124	Signal rockets
661.887	Bismuth compounds	662.126	Flares for photographic flashlight ⇒ 771.448.2
661.888	Vanadium, niobium and tantalum compounds	662.126	Flaming torches as signals ⇒ 628.9.045 ⇒ 665.194
661.888.1	Vanadium compounds	662.13	Pyrotechnic devices for rescue purposes. □ Pyrotechnic devices for lifebuoys
661.888.2	Niobium compounds	662.14	Incendiary devices and substances. □ Firebrands, Greek fire ⇒ 623.454.72
661.888.3	Tantalum compounds	662.15	Blasting, demolition devices. □ Petards. Explosive cartridges
661.89	Compounds of metals of the platinum and ruthenium groups	662.16	Gas-generating fireworks. War gases
661.892	Platinum compounds	662.161	Gas-generating fireworks and equipment, e.g. for police use
661.893	Iridium compounds	662.164	Production of war gases ⇒ 623.459
661.894	Osmium compounds	662.164.1	Inorganic war gases Specify by :546...
661.896	Ruthenium compounds	662.164.2	Organic war gases Specify by :547...
661.897	Rhodium compounds		
661.898	Palladium compounds		
661.9	Production of gases ⇒ 66.07 ⇒ 662.76		
661.91	Compressed, liquefied and solidified gases in general		
	◇ 661.91-403.3 Compressed gases		
	◇ 661.91-404 Liquefied gases		
	◇ 661.91-405 Solidified gases ⇒ 621.51 ⇒ 621.59		
661.92	Atmospheric air		
661.93	Production of oxygen, nitrogen and the inert (rare) gases		
661.937	Oxygen		
661.937.1	Oxygen-rich mixtures. Oxygen-rich air		
661.937.2	Pure oxygen		
661.937.6	Production of oxygen from chemical compounds		
661.938	Nitrogen		
661.939	Inert gases (rare gases) 661.939 = 546.29		

662.17	Smoke-generating devices (producing smokes, mists, fogs) ⇒ 623.454.5 ⇒ 623.775 ⇒ 631.348	662.221.1	Black powder. Ordinary blasting powder <i>For gunpowder as propellant see 662.33</i> ⇒ 662.24
662.18	Special fireworks for indoor use. □ Stage, theatre fires and flares ⇒ 662.113	662.221.2	Nitrate powders in which one or more constituents of ordinary black powder have been replaced by other substances
662.19	Other types of firework	662.221.21	With saltpetre (potassium nitrate) replaced. □ Replacement by sodium nitrate, ammonium nitrate
662.2	High explosives ⇒ 622.235 ⇒ 623.45	662.221.22	With sulphur replaced. □ Potassium xanthate powders
662.2.3	Mixtures of high explosives <i>The 62-3... auxiliaries are not applicable at 662.2 and its subdivisions</i>	662.221.23	With charcoal replaced. □ Replaced by chaff, sugar
662.2-31/-38	Explosive mixtures with organic substances 662.2-31/-38 ≡ 662.231/.238 ◇ 662.237.3-381 Mixture of TNT and hexogen	662.221.3	Black powder with additives. □ With added hydrocarbons or carbohydrates
662.2-39	Explosive mixtures with inorganic and various inactive substances	662.221.4	Ammonium nitrate explosives ⇒ 661.525
662.2-391	Mixtures with nitrates	662.222	Powders based on chlorates or perchlorates
662.2-392	Mixtures with chlorates	662.222.1	Explosive powders containing chlorates
662.2-393	Mixtures with perchlorates	662.222.2	Explosive powders containing perchlorates
662.2-395	Explosive mixtures with azides (e.g. lead azide)	662.23	High explosives derived from organic substances
662.2-397	Explosive mixtures with active inorganic substances. □ Mixtures with metal powder, e.g. aluminium	662.231	Nitrated carbohydrates
662.2-398	Explosive mixtures with inactive substances. □ Kieselguhr	662.231.21	Nitrocellulose. Guncotton (pyroxylin)
662.2.03	Production processes	662.231.23	Explosive mixtures with nitrocellulose as main constituent. □ Nitrocellulose mixtures with black powder
662.2.032	Nitration	662.231.3	Nitrocellulose from various raw materials. □ From wood cellulose. From cotton
662.2.033	Washing. Drying. Separation, removal of acids	662.232	Nitrated aliphatic alcohols and ethers
662.2.034	Recovery and treatment of spent acids and by-products	662.232.1	Nitroglycerine in general
662.2.035	Pulverizing. Milling, grinding	662.232.2	Dynamite. □ Ordinary dynamite, with kieselguhr. Dynamite with active additives
662.2.036	Mixing. Kneading. Rolling. Pressing. Moulding. Casting. Stabilization	662.232.4	Nitrated glycols
662.2.037	Cutting. Slicing	662.232.5	Pentaerythritol tetranitrate (PETN)
662.2.038	Surface treatment. Impregnation. Paraffination (waxing)	662.232.6	Nitroisobutylglyceryl trinitrate
662.2.099	Finishing. □ Granulation. Polishing. Graphitizing. Drying. Dust removal	662.232.7	Nitrated chlorohydrins
662.21	Mode of action and properties of high explosives	662.233	Explosive gelatins. Gelignite
662.215	Physical properties and their testing	662.233.1	Nitroglycerine and nitrocellulose mixtures (without other active substances)
662.215.1	Combustion of explosives. □ Deflagration. Detonation	662.233.2	Nitroglycerine and nitrocellulose mixtures with other active substances (e.g. with added ammonium nitrate)
662.215.2	Performance. Efficiency. Explosive yield. □ Lead block test. Lead plate test. Explosive pressure. Brisance. Shattering or fragmentation effect	662.234	Other aliphatic nitro derivatives. □ Nitromethane. Nitrated aliphatic aldehydes and ketones.
662.215.3	Density of explosive charge	662.235	Nitrated aliphatic acids. Nitrated aliphatic amines
662.215.4	Initiation of detonation	662.235.1	Nitrated phenols and aromatic alcohols. Picric acid and its derivatives. □ Picrates
662.215.5	Sensitivity to percussion, impact. Sensitivity to friction	662.235.2	Picric acid
662.216	Chemical properties and their testing	662.235.3	Mixtures of picric acid with other substances
662.216.1	Chemical stability of explosives	662.235.5	Picrates
662.216.2	Hygroscopicity of explosives	662.235.8	Nitrated phenols (nitrophenols)
662.216.3	Combustion products of explosives. □ Explosion gas: quantity and composition. Combustion residues	662.236	Nitrated aromatic alcohols
662.216.4	Heat of combustion of explosives	662.236.5	Various nitrated aromatic compounds
662.216.5	Ignition temperature of explosives	662.236.7	Nitrated sulphides. □ Hexanitrodiphenyl sulphide
662.216.6	Afterburning of explosives	662.237	Nitrated aromatic amines. □ Hexanitrodiphenylamine. Tetranitromethylaniline (tetryl)
662.22	High explosives derived from inorganic substances. Mixtures of substances inert at normal temperatures	662.237.2	Nitrated aromatic hydrocarbons. □ Nitrated coal-tar derivatives
662.221	Explosive powders containing nitrates	662.237.3	Nitronaphthalene
		662.237.5	Trinitrotoluene (TNT)
		662.238	Nitrobenzene and its derivatives
		662.238.1	Nitrated heterocyclic compounds
		662.238.3	Cyclotrimethylene trinitramine (RDX, hexogen, cyclonite)
		662.24	Nitrated heterocyclic alcohols
			Permitted explosives (safe explosives) for use in mines ⇒ 622.235 ⇒ 662.221.1

662.241	Safe explosives resembling black powder	662.41	Initiators: substances ignited by flame, friction or percussion for initiating an explosion
662.242	Safe explosives based on ammonium nitrate		
662.243	Safe explosives with low gelatin content		
662.244	Safe explosives with high gelatin content	662.411	Special properties of initiators. ☐ Sensitivity to percussion, impact. Sensitivity to friction. Sensitivity to heat
662.245	Sheathed safe explosives		
662.25	Explosives formed at time of use. Special explosives with liquid or gaseous constituents	662.412	Fulminates. ☐ Silver fulminate. Mercury fulminate
662.251	Sprengel explosives	662.413	Azides. ☐ Silver azide. Lead azide
662.251.1	Sprengel explosives with solid constituents	662.414	Various detonating compounds. ☐ Lead styphnate. Tetrazene
662.251.2	Sprengel explosives with liquid constituents. ☐ With nitric acid, picric acid, carbon disulphide, benzene, nitrobenzene	662.415	Initiator compositions. Priming mixtures. ☐ Compositions based on fulminates, azides etc.
662.252	Safe explosives with gaseous constituents. ☐ With nitrogen dioxide, liquid oxygen (LOX), liquid air	662.42	Priming devices. ☐ Percussion caps. Igniters ⇒ 623.454.2
662.3	Propellant powders. Gunpowders	662.421	Mechanically operated priming devices
662.31	Special properties of gunpowder	662.421.1	Friction primers. ☐ Friction caps, tubes, igniters
662.311	Physical properties and their testing ⇒ 531.5 ⇒ 623.5	662.421.2	Percussion primers. Percussion caps
662.311.1	Combustion of propellants. ☐ Disruptive or brisant powders. Progressive powders	662.421.21	Paper caps (e.g. for toy guns) ⇒ 623.447
662.311.2	Performance. Efficiency	662.421.22	Metal caps
662.311.3	Density of propellant charge	662.422	Chemical priming devices. Chemical igniters
662.311.4	Flammability, ignitability of propellants	662.423	Electric priming devices. ☐ Spark, filament and arc igniters ⇒ 621.3.032.4
662.311.5	Sensitivity to percussion, impact. Sensitivity to friction	662.43	Detonators for high explosives
662.312	Chemical properties and their testing	662.44	Firing tubes for guns
662.312.1	Chemical stability of propellants	662.45	Matches (match-cords). String-type fuses (fuzes) ⇒ 623.454.1
662.312.2	Hygroscopicity of propellants		
662.312.3	Combustion products of propellants. ☐ Explosion gas: quantity and composition. Combustion residues	662.451	Quick-match fuses
662.312.4	Heat of combustion of propellants	662.452	Slow-burning fuses (slow-match fuses). Safety fuses
662.312.5	Ignition temperature of propellants	662.46	Detonating fuses
662.312.6	Afterburning of propellants. Muzzle flash	662.47	Projectile fuses. Fuses for shells etc.
662.313	Mechanical properties and their testing. ☐ Size and shape of particles	662.471	Wooden time fuses
		662.472	Metal time fuses. Perforated fuses. Insertable fuses. Conduits for detonation cord
662.32	Early types of gunpowder	662.473	Percussion fuses. Impact fuses
662.33	Black powder. Ordinary gunpowder. Cannon and musket powders ⇒ 662.221.1	662.474	Combined fuses. Impact firing caps
662.34	Various gunpowders resembling black powder, with inorganic constituents	662.48	Firing devices. Firing mechanisms
662.341	Gunpowders containing nitrates. ☐ Ammonal ⇒ 661.525	662.481	Firing devices with chemical reaction
662.342	Gunpowders containing chlorates	662.482	Mechanical firing devices. Firing mechanisms
662.343	Gunpowders containing perchlorates	662.5	Igniters. Kindlers. Matches. Lighters
662.35	Smokeless propellant powders	662.51	Primitive methods of producing fire, flame. ☐ Wood friction kindlers. Flint-striking devices. Tinder-boxes. Touchwood. Amadou
662.351	Smokeless powders based on nitrocellulose. ☐ Guncotton (pyroxylin)		
662.351.6	Nitrocellulose with additives to reduce muzzle flash	662.52	Various kindlers. ☐ Kindlers with air (fire syringes), with hydrogen, with fulminating primers
662.352	Nitroglycerine smokeless powders. Smokeless powders of nitrocellulose and nitroglycerine. Cordite	662.53	Matches (stick matches). Vestas ⇒ 661.63
662.352.6	Nitroglycerine powder with additives to reduce muzzle flash	662.53.03	Raw materials
662.353	Smokeless powders of nitrocellulose with other explosives. ☐ With nitrobenzene. With nitronaphthalene	662.53.031	Wood and other materials for matchsticks
662.4	Initiators, primers for explosives. Detonators	662.53.035	Flammable substances for match-heads
		662.53.036	Impregnating substances for matchsticks
		662.53.05	Processes and equipment for match manufacture
		662.53.052	Machines for cutting of matchsticks and insertion into holders on machines
		662.53.053	Methods and equipment for insertion of stems into conveyor or frame
		662.53.054	Impregnation of matches. ☐ Colouring. Dipping. Paraffining. Drying
		662.53.055	Conveyor bands, trays etc. for matches

662.53.056	Separation and distribution of matches	662.612.5	Combustion temperature. □ Temperature distribution in the flame. Temperature distribution in solid fuels
662.53.057	Devices for filling of matchboxes		
662.53.058	Devices for printing and stamping of matches	662.613	Combustion products. Combustion residues ⇒ 628.477.7
662.531	Sulphur matches. Pyrophoric pastes	662.613.1	Ash. Clinker. Flue dust
662.532	Phosphorus matches. □ Phosphorus-headed wooden matches. Safety matches. Wax matches (wax vestas). Wax tapers. Fuse matches. Paper matches	662.613.11	Ash. □ Content, composition. Fusion point of ash
662.533	Non-phosphorus matches	662.613.12	Clinker. Slag. Combustion residues containing unburnt fuel. □ Tapping of liquid slag. Recovery, utilization, gasification of residues ⇒ 66.046.58
662.534	Headless matches	662.613.13	Solid matter in smoke and combustion gases: soot, flue dust, fly ash, fly coke. □ Separation. Removal. Use ⇒ 628.511
662.536	Friction strips and matchboxes and books (with friction strips)	662.613.4	Liquid combustion products: tar. □ Tar recovery
662.536.1	Friction strips for matches	662.613.5	Combustion gases and suspensions: smoke, flue gases. □ Composition. Dew point. Enthalpy ⇒ 662.96
662.536.2	Matchboxes. Match books ⇒ 688.94	662.614	Heat balance ⇒ 662.998
662.58	Devices for aiding ignition, promoting combustion. Firelighters	662.614.2	Useful heat. Heat utilization. Combustion efficiency. Calorific value ⇒ 536.66 ⇒ 62-66
662.581	Solid firelighters	662.614.4	Heat losses
662.581.05	Processes and equipment for firelighter manufacture 662.581.05 ≡ 662.8.05	662.614.42	Heat losses through combustion products
662.582	Liquid firelighters	662.614.422	Heat losses through gaseous combustion products
662.584	Firelighters with pyrophoric metals or alloys. □ Friction spark-producers	662.614.423	Heat losses through incomplete combustion (unburnt fuel etc.)
662.59	Other devices for producing fire. □ Pocket lighters ⇒ 683.88 ⇒ 683.98	662.614.44	Heat losses through conduction
662.591	Lighters using solar heat. Magnifying lenses (burning glasses) ⇒ 681.722	662.614.45	Heat losses through radiation
662.592	Other nonelectric lighters. □ Cigar and cigarette lighters. Flint lighters (with or without fuel supply)	662.614.49	Other heat losses. □ Heat losses in unsteady-state operation. Heating-up losses
662.593	Electric lighters. □ With electric spark, filament or arc ignition	662.614.5	Remainder of heat balance
662.6/9	Heat economy. Fuel economy. Fuels. Heating ⇒ 536 ⇒ 62-6 ⇒ 621.4 ⇒ 66.04	662.62	Solid fuels in general ⇒ 662.71/74
662.6	Heat or fuel economy in general. Combustion. Natural fuels	662.63	Wood and other vegetable matter as fuels
662.61	Combustion ⇒ 662.9	662.631	Softwoods as fuels
662.611	Theory of combustion ⇒ 536.46	662.632	Hardwoods as fuels
662.611.2	Stoichiometric relations in combustion. Measurement and control of combustion	662.636	Grasses and their residues as fuels. □ Bagasse, reeds, straw as fuels ⇒ 664.113
662.611.22	Complete combustion	662.637	Rinds and barks as fuels
662.611.23	Incomplete combustion	662.638	Wood waste, sawdust as fuels
662.611.3	Oxygen and air requirements. □ Excess air	662.64	Peat and lignite as fuels
662.611.4	Smokeless combustion	662.641	Peat as fuel ⇒ 553.97 ⇒ 622.331
662.612	Dynamics of combustion	662.641.03	Preparation. □ Peat cutting, comminution
662.612.1	Ignition process	662.641.1	Low-moor peat
662.612.12	Ignition. Ignition temperature. Ignition properties	662.641.2	High-moor peat
662.612.13	Slow oxidation. Self-ignition	662.642	Lignite as fuel ⇒ 553.96 ⇒ 622.332
662.612.16	External ignition. Spark ignition	662.642.1	Fossilized wood. Fibrous lignite. Xylite
662.612.2	Chemistry of combustion process. Combustion reactions	662.642.2	Brown coal. Brown lignite. Earthy lignite
662.612.3	Physics of combustion process	662.642.3	Black lignite
662.612.31	Homogeneous reactions. Gas combustion	662.65	Various low-grade fuels. □ Refuse as fuel.
662.612.32	Heterogeneous reactions. □ Reaction time (combustion time)		Coal wastes, slurry, middlings, tailings. Animal dung as fuel ⇒ 628.474/477

662.66	Hard, bituminous coals and anthracite as fuels	662.75	Liquid fuels
662.66.016	Caking capacity of coal	662.753	Mineral oils as fuels ⇒ 665.6
662.661	Long-flame lean coal. Gas flame coal	662.753.1	Light oils
662.662	Long-flame fat coal. Gas coal	662.753.2	Middle oils
662.663	Fat coal proper. Forge coal	662.753.3	Heavy oils
662.664	Short-flame fat coal. Coking coal	662.754	Organic chemicals as liquid fuels. □ Alcohol, methanol as fuels
662.665	Lean coal. Short-flame anthracite coal	662.756	Animal and vegetable oils as fuels
662.667	Anthracite as fuel	662.756.2	Animal oils as fuels
662.67	Shale as fuel	662.756.3	Vegetable oils as fuels. □ Resin oils
662.7	Fuel technology. Processed fuels	662.757	Colloidal, gelatinized fuels
⇒ 665.6		662.758	Composite liquid fuels. Blended, mixed liquid fuels
662.71/74	Solid fuels	662.758.2	Mixtures of liquid fuels only
⇒ 662.62		662.758.3	Solutions of gases as fuels
		662.758.4	Solutions of solids as fuels
662.71	Fuel technology of wood	662.76	Gaseous fuels. Gas producers
662.711	Charcoal production in stacks, piles	⇒ 66.071	⇒ 661.9
662.712	Charcoal production in ovens. Wood distillation. Wood tar production	⇒ 662.749	
662.715	Wood tar and its processing	662.76.032	Gas producers in general
662.715.1	Wood tar	662.76.034	Processes and equipment for gas production with by-product recovery. □ Separation of lean and rich gases
662.715.7	Pyroligneous acid	662.76.035	Processes and equipment for simple gas producers
662.715.8	Wood-tar pitch	662.76.036	Processes and equipment for complex gas producers. Gas-generating plant with secondary producers
662.73	Fuel technology of peat, lignite, refuse etc.	662.76.037	Processes and equipment for using incandescent incombustible materials
662.731	Peat distillation and carbonization	662.76.075	Carburation of fuel gases. Carburation media. Various additions to gas
662.732	Lignite distillation and carbonization	662.76.075.2	Carburation by passing the gas through the carburation medium
662.734	Carbonization of refuse, wastes ⇒ 628.474	662.76.075.3	Carburation by atomization of the carburation medium 662.76.075.3 ≡ 621.43.03
662.735	Peat tar and its processing. □ Peat-tar pitch	662.76.075.4	Carburation by surface action
662.736	Sapropelite tar and its processing	662.76.075.5	Other carburation processes and equipment. □ Carburation using solid agents
662.737	Brown-coal tar and its processing. □ Brown-coal pitch	662.76.075.8	Materials for carburation and odorization. Carburants. Various additions to gas (e.g. odorants for warning of gas leaks)
662.739	Technology of mixed fuels	662.761	Producer gas (air gas)
662.74	Fuel technology of hard, bituminous coal. Coke production	662.762	Semi-water gas. Mond gas
662.741	High-temperature carbonization of hard coal (above 1000°C). High-temperature coking	662.763	Water gas
662.741.1	Coking in heaps	662.763.1	Blue water gas (BWG)
662.741.2	Coking in retorts or similar ovens	662.763.2	Carburetted water gas (CWG)
662.741.3	Coking in special ovens at high temperature. Metallurgical coke (blast furnace coke). Foundry coke	662.764	Gas from coal distillation. Coal gas. Illuminating gas. Town gas (blend of coal gas and water gas)
662.741.31	Coking in ovens without heat recovery and without by-product recovery	662.764.2	Crude gas
662.741.32	Coking in ovens without heat recovery but with by-product recovery (recuperative ovens). Condensers for tar and ammonia recovery. Waste-heat ovens	662.764.3	Purified gas
662.741.33	Coking in ovens with heat recovery but without by-product recovery (regenerative ovens)	662.764.7	Precipitation in gas mains, gaspipes (gums, resins etc.). □ Preventive measures
662.741.34	Coking in ovens with recovery of heat and by-products (regenerative and recuperative ovens)	662.765	Gas from thermal processing of other solid or liquid fuels
662.741.35	Quenching and cooling of coke (wet or dry methods). Heat recovery	662.765.1	Wood gas
662.742	Medium-temperature and low-temperature carbonization of hard coal ◇ 662.742.048.3 Fractional distillation	662.765.2	Peat gas. Brown-coal gas
662.742.1	Medium-temperature carbonization of hard coal (about 700°C)	662.765.3	Resin gas
662.742.2	Low-temperature carbonization of hard coal (about 450°C)	662.765.4	Oil gas
662.749	Coke. Coal tar ⇒ 662.76	662.765.5	Air-gas mixtures. □ Carburized gas. Aerogene gas
662.749.2	Coke	662.766	Acetylene
662.749.3	Coal tar and its processing. Gas tar	662.766.3	Acetylene production by action of water on calcium carbide
662.749.38	Coal-tar pitch	662.766.31	Water-feed generators in general
662.749.39	Pitch coke	662.766.32	Generators without automatic control of water flow

662.766.33	Generators with automatic control of water flow	662.9	Furnace and combustion engineering ⇒ 662.61
662.766.34	Contact-type generators with bell	662.91	Domestic and personal heating appliances in general ⇒ 644.1 ⇒ 683.9 ⇒ 697
662.766.35	Contact-type generators with bell and closure between carbide container and water	662.917	Personal warming devices. Heating the human body or its parts. □ Hot-water bottles, jars etc.
662.766.37	Special contact-type generators with bell. □ Kipp's apparatus. Immersion-drop-type acetylene generators	662.919	Heated objects, e.g. stones Heating in the open air. □ Open-air stoves, braziers
662.766.38	Carbide-immersion-type acetylene generators	662.92	Industrial firing in general ⇒ 621.18 ⇒ 621.745 ⇒ 66.041.3 ⇒ 66.042.3
662.766.5	Processes following acetylene generation. □ Aeration. De-aeration. Removal and use of acetylene lime sludge	662.921	Heat sources in general. □ Fireboxes (fire chambers). Combustion chambers. Heating surfaces. Grate surfaces
662.766.6	Dissolved acetylene. □ Solution of acetylene, e.g. in acetone. Filling material for dissolved acetylene	662.922	Furnace draught in general. □ Natural draught. Chimneys: shape, structure, cleaning
662.766.7	Mixtures of acetylene with gases other than hydrocarbons. □ Oxyacetylene mixtures	662.923	Artificial, forced draught (and its regulation)
662.767	Various gaseous hydrocarbons as fuels. □ Natural gases as fuels ⇒ 665.612 ⇒ 665.72	662.923.1	Draught regulators, throttles etc.
662.767.1	Methane as fuel	662.923.3	Bellows. Blowers
662.767.2	Biogas as a fuel	662.924	Devices for introducing air and steam. Humidification, moistening
662.767.3	Propane as fuel	662.925	Preheating of combustion air. Heat storage ⇒ 662.995
662.767.4	Butane as fuel	662.926	Cooling of combustion chamber walls. Insulation of combustion chamber ⇒ 662.998
662.767.7	Mixtures of lower hydrocarbons as fuels	662.927	Temperature control in general. Heat regulators ⇒ 536.58 ⇒ 621.316.79 ⇒ 681.536.5
662.769	Other gases and gas mixtures as fuels	662.928	Fire doors, fire bridges and related devices
662.769.1	Blast-furnace gas as fuel	662.929	Firing control. Stocking, charging. Ash and clinker removal. Slagging. Furnaceman's, stoker's equipment
662.769.2	Hydrogen as fuel. Detonating gas and acetaldehyde as fuels	662.93	Solid-fuel furnaces
662.8	Mechanical treatment for solidification of fuels. Briquetting ⇒ 622.788	662.932	Grates. Charging devices for furnaces ⇒ 66.042.83
662.8.05	Processes and equipment for fuel solidification	662.932.1	Components of grates. □ Grate bars, segments etc. Devices for ash and clinker removal. Slagging devices, scavengers ⇒ 66.042.86
662.8.051	Pretreatment of briquetting fuel. □ Mixing of different grades of fuel	662.932.2	Simple horizontal grates
662.8.052	Briquetting without binders	662.932.3	Grates with air injection, bellows, blowers. □ Undergrate blast systems
662.8.053	Briquetting with binders	662.932.4	Inclined grates. Stepped grates
662.8.053.2	Briquetting with inorganic binders. □ With clay, water glass, gypsum	662.932.5	Other kinds of fixed grate. □ Underfeed grates.
662.8.053.3	Briquetting with organic binders. □ With emulsions, asphalt, pitch, resins, latex	662.932.6	Zoned grates. Double grates. Basket grates.
662.8.053.4	Briquetting with mixtures of inorganic and organic binders. □ With cellulose waste liquor	662.932.61/.65	Hollow grates. Grates with external cooling
662.8.054	Mixing, blending, kneading processes	662.932.66	Automatically movable grates
662.8.055	Pressing, compaction of fuel blend. Forming, shaping of briquettes	662.932.67	Components of movable grates. Types of movable grate 662.932.61/.65 = 662.932.1/.5
662.8.056	Coating and impregnation of briquettes. Additives, e.g. for smoke abatement ⇒ 662.892	662.932.68	Tipping grates. Rocking grates
662.8.057	Final treatment of briquettes. □ Carbonization of briquette binders. Drying. Cooling	662.932.69	Rotary, revolving grates
662.81	Briquettes	662.932.7	Travelling grates. Chain grates
662.812	Peat briquettes	662.932.8	Other movable grates. Removable, transportable grates
662.813	Lignite briquettes. Brown-coal briquettes	662.932.8	Accessories for removable grates. □ Levellers, skimmers. Slag dams
662.814	Coal briquettes		Fuel feed. Charging devices for furnaces. □ Mechanical stokers. Scoop-type, overfeed, underfeed stokers
662.815	Coke briquettes		
662.816	Charcoal briquettes		
662.818	Briquettes from waste		
662.84	Shaped coal. Paris coal. Cylindrical, spherical, hollow briquettes etc.		
662.87	Pulverous and pulverized fuels. □ Coal dust as fuel		
662.88	Impregnation. Solidification of liquid fuels. Briquetted chemicals as fuels. □ Solid alcohol. Solid petroleum. Briquetted metaldehyde		
662.892	Smoke dispellents. Substances added to fuels at time of combustion ⇒ 662.8.056 ⇒ 662.966		

662.933	Pulverized-fuel furnaces ⇒ 662.87	662.963	Purification of emissions by washing, scrubbing ⇒ 66.074.5
662.933.1	Comminution, pulverizing devices combined with the furnace. Burners for pulverized fuel. Combustion chambers for pulverized fuel. Ash and clinker removal. Slagging ⇒ 621.926	662.964	Devices for separating solid and incandescent particles in smoke
662.933.2	Pulverized-fuel furnaces with centrifugal fuel dispersion	662.965	Precipitation from smoke
662.933.4	Pulverized-fuel furnaces with fuel dispersion by injection	662.966	Methods of smoke prevention (smoke abatement). Smoke combustion. Smoke consumption (in the strict sense). Smoke dispellents ⇒ 662.892
662.933.8	Combinations of pulverized-fuel furnaces with other types (e.g. with grate firing). Supplementary firing	662.98	Heating of apparatus and utensils. Various heating methods ⇒ 542.4 ⇒ 621.365 ⇒ 621.745 ⇒ 644.1 ⇒ 683.9
662.936	Furnaces for chunk-form solid fuel (not pulverized) <i>Details by -66...</i>	662.981	Solid-fuel heating appliances. □ Solid-fuel ovens, stoves, hotplates
662.939	Other kinds of solid-fuel furnace. □ Ten-Brink furnaces	662.982	Liquid-fuel heating appliances. □ Liquid-fuel burners, stoves etc. Spirit burners
662.94	Liquid-fuel furnaces ⇒ 683.944	662.983	Gas-fired heating appliances. □ Gas burners, rings. Bunsen burners
662.941	Construction details	662.986	Heating by direct contact. □ Heating by injection of hot gas, air, steam. Heating by contact with hot liquids or solids
662.941.2	Liquid-fuel burners in general	662.987	Indirect heating. Heat transfer media. □ Baths of hot water, molten salts, molten metals ⇒ 536.24
662.941.5	Control devices, regulators for liquid-fuel furnaces	662.989	Heating by means of chemical reactions (heat of reaction)
662.941.8	Special devices for viscous and colloidal fuels	662.99	Heat recovery. Use of natural heat. Prevention of heat loss. Insulation ⇒ 62-68 ⇒ 620.97 ⇒ 621.186.8 ⇒ 621.577 ⇒ 662.957
662.942	Liquid-fuel furnaces with centrifugal fuel dispersion (and other mechanical dispersion except injection)	662.992.8	Combustion control and testing
662.944	Liquid-fuel furnaces with fuel dispersion by injection of air, steam, gas	662.992.82	Control and testing in the combustion chamber
662.946	Wick burners	662.992.84	Control and testing of combustion gas, flue gas
662.947	Liquid-fuel furnaces with large absorbent surface	662.993	Utilization of heat from incandescent residues
662.948	Liquid-fuel furnaces with fuel vaporization (before combustion) ⇒ 662.951.2	662.994	Utilization of waste gas
662.948.3	Vaporizing devices for liquid fuels ⇒ 621.43.03	662.995	Heat storage in general ⇒ 662.925
662.949	Furnaces for simultaneous combustion of both liquid and solid fuels	662.997	Utilization of heat from natural phenomena. □ Use of geothermal energy. Volcanic heat. Hot springs, geysers. Solar heat ⇒ 620.91 ⇒ 621.472 ⇒ 697.329
662.95	Gaseous-fuel furnaces. Gas-fired furnaces ⇒ 683.945	662.998	Industrial heat insulation and insulators. Heat loss ⇒ 66.045.3 ⇒ 662.614 ⇒ 662.926
662.951.2	Gas burners in general ⇒ 683.8771.878	663	INDUSTRIAL MICROBIOLOGY. INDUSTRIAL MYCOLOGY. ZYMURGY, FERMENTATION INDUSTRY. BEVERAGE INDUSTRY. STIMULANT INDUSTRY ⇒ 178 ⇒ 351.761 ⇒ 579 ⇒ 613.3 ⇒ 613.8 ⇒ 641.87
662.951.22	Low-pressure gas burners without air supply	663.03	Special processes <i>The 663.03... auxiliaries are applicable only in 663.1/5</i>
662.951.23	Low-pressure gas burners with air injection		
662.951.25	High-pressure gas burners with air injection		
662.951.26	High-pressure gas burners with gas injection		
662.953	Furnaces for rich gases		
662.954	Furnaces for lean gases		
662.955	Gaseous-fuel furnaces with combustion in stages. Semi-gas furnaces		
662.957	Gaseous-fuel furnaces with heat recovery (regenerative and recuperative furnaces) ⇒ 662.99		
662.959.2	Furnaces for combustion of both gaseous and liquid fuels		
662.959.3	Furnaces for combustion of both gaseous and solid fuels		
662.959.4	Special devices for burners in steam generators with internal combustion ⇒ 621.181.2		
662.959.5	Gaseous-fuel furnaces with flue gas recirculation. Drum gas heaters		
662.959.6	Gaseous-fuel furnaces for flameless surface combustion		
662.96	Smoke consumption. Smoke abatement. Treatment of combustion products ⇒ 628.511 ⇒ 662.613.5 ⇒ 662.8.056		

663.031	Raw materials, secondary materials and their preparation. □ Milling. Grinding. Crushing. Water treatment. Steam treatment	663.14.038	Processes and equipment for treatment of yeast
663.031.1	Ferments 663.031.1 ≡ 663.1	663.14.038.3	Aftertreatment to increase fermentative capacity and other properties
663.031.2/.4	Raw materials 663.031.2/.4 ≡ 663.14.031.2/.4	663.14.038.5	Defects of yeast, and means of improvement. □ Flocculation
663.031.8	Liquors forming bases of blended drinks. □ Sugar syrups ⇒ 663.812 ⇒ 664.14	663.14.038.7	Conversion of brewer's and distiller's yeast into baker's yeast
663.032	Equipment for preparation of mash and wort (for yeast processes and other processes)	663.14.039.3	Factors affecting fermentation. □ Concentration. Acidity. Pressure. Temperature
663.033	Fermentation vessels and accessories	663.14.039.4	Foam, froth formation
663.035	Extraction and pressing of yeast	663.142	Fermentation processes without aeration. □ Viennese method
663.036	Preservation and storage of yeast	663.143	Fermentation processes with aeration
663.05	Additions. Preservation. Aftertreatment <i>Class here the use of additives in the beverage industry generally</i> <i>The 663.05... auxiliaries are not applicable at 663.9 and its subdivisions</i> ⇒ 664.8.022	663.15	Industrial fermentation processes. Production of enzymes (other than by yeast) ◇ 663.15:661.715 Production of hydrocarbons by fermentation 663.15 ≡ 577.15
663.051	Additions for improvement of flavour. Aromatic supplements	663.16	Vitamin production by fermentation 663.16 ≡ 577.16
663.052	Additions for improvement of appearance or physical properties. □ Additions to improve viscosity, foaming capacity ⇒ 667.777	663.18	Industrial bacteriology. Various processes using microorganisms (microbes) ⇒ 579
663.053	Preservation methods for beverages 663.053 ≡ 664.8.03	663.2	Wine. Winemaking. Oenology ⇒ 634.8
663.054	Diluents for beverages	663.21	Wines in general. Wines according to origin ◇ 663.21(44-82) French wines
663.057	Methods and equipment for infusing beverages with carbon dioxide. Carbonation, aeration of beverages	663.22	Special wines. Imitation wines
663.058	Various aftertreatments	663.221	White wines
663.058.2	Natural maturing, ageing of beverages. □ Cellaring. Means of preservation in cellars	663.222	Rosé wines
663.058.3	Artificial maturing, ageing of beverages	663.223	Sparkling wines. □ Champagnes
663.058.4	Mixing of beverages	663.224	Muscatel wines
663.059	Bottling and final treatment before consumption ⇒ 683.5	663.226	Raisin wines
		663.229	Imitation wines
663.1	Microbiological industries. Science and technique of applied microbiology. Applied mycology ⇒ 577.15 ⇒ 579.6 ⇒ 582.23 ⇒ 582.28 ⇒ 637.146 ⇒ 66.098	663.23	Wine substitutes ⇒ 663.3
		663.233	Artificial wines produced by fermentation
		663.236	Nonalcoholic wine substitutes. Grape juice, must
663.12	Yeasts ⇒ 582.282.23	663.24	Wine products
663.123	Brewing yeast. □ Ascending and descending yeasts	663.241	Brandy. □ Cognac ⇒ 663.55 ⇒ 663.83
663.125	Pure yeasts. Wine yeasts	663.242	Wine vinegar ⇒ 661.741.1
663.126	Baker's yeast. Pressed yeast	663.25	Winemaking
663.127	Nutrient yeasts	663.251	Characteristics and properties of wine
663.128	Harmful yeasts. □ Apiculatus yeast. Mould yeast	663.252	Winemaking proper
		663.252.1	Grape harvesting. Grape picking. Grape preparation, storage, drying, stripping
663.13	Methods and equipment for production of pure yeast cultures	663.252.2	Crushing and pressing of grapes. Extraction of must
		663.252.3	Operations before fermentation. □ Heating and cooling of must. Sugaring, chaptalization. Addition of alcohol. Removal of acid
663.14	Fermentation of yeast	663.252.4	Fermentation of wine. Alcoholic ferments ⇒ 663.12
663.14.031	Raw materials (and their treatment)	663.252.5	Racking, tunnning of wine
663.14.031.2	Carbon compounds	663.252.6	Pressurage (second pressing). □ Decanting.
663.14.031.3	Nitrogen compounds		Dregs, lees of wine ⇒ 663.26
663.14.031.4	Nutrient salts (except nitrogen salts) and acids (either as additives or produced by fermentation)	663.253	Constituents of wine. □ Alcohol. Sugar. Acids. Esters
663.14.036	Preservation of yeast and fermentation liquors		

663.255	Winemaking equipment and materials.	663.4	Beers. Brewing. Malting
663.255.1	Cellarage equipment	663.41	Types of beer. □ Hopped beer. Unhopped beer. Spruce beer. Ginger beer
663.255.2	Grape staking and stripping devices (e.g. combs). Grape crushing and squeezing devices. Grape mills	663.42	Raw materials for beer
663.255.3	Fermentation vats. □ Perret vats	663.421	Cereals for brewing. Brewing barley. Germinative capacity ⇒ 633.1
663.255.4	Apparatus for treatment of must. □ Apparatus for sulphite treatment, sulphite removal, concentration	663.422	Barley substitutes. □ Rice. Maize. Starch flour. Sugar. Wheat. Oats
663.255.5	Wine presses	663.423	Hops. Lupulin ⇒ 633.791
663.255.6	Wine pumps	663.424	Hop substitutes. □ Bitter barks, woods, leaves. Pungent buds and berries. Quassia. Wormwood. Aloes
663.255.7	Wine filters	663.43	Malt preparation. Malting ⇒ 664.786
663.256	Pasteurization equipment. □ Wine boilers, sterilizers	663.432	Washing and steeping of barley for malting. Treatment of steep water ◇ 663.432.033 Steeps. Steeping tanks. Aerating, washing devices
663.256.1	Preservation and storage of wine	663.433	Germination, sprouting of barley. Production of green malt
663.256.2	Wine preservation processes. □ Pasteurization of wine. Chemical preservation of wine	663.433.1	Malting processes and equipment
663.256.27	Storage of wine. Wine vessels. □ Wine vats, casks, bottles	663.434	Drying and kilning of green malt
663.257	Preparation and maintenance of wine vessels. □ Cleaning. Washing. Sulphurization	663.436	Malt cleaning devices. Dressing, screening and polishing devices for malt. Sleeve filters. Dust filters. Husk removal
663.257.1	Maintenance and improvement of wine	663.439	Products of malting. Malts. □ Crystal malt. Coloured malt. Black malt
663.257.2	Drawing off of wine	663.44	Preparation of wort and beer
663.257.3	Filtering of wine	663.442	Grinding or crushing of malt. □ Grist cases. Malt mills. Crushed-malt hoppers
663.257.4	Clarifying, fining of wine	663.443	Mashing and brewing
663.258	Plastering of wine. Muting, sulphite treatment of wine	663.443.2	Mashing equipment. □ Coppers. Fining vats. Lauter tubs
663.258.2	Defects, deterioration, adulteration of wine	663.443.3	Brewing by infusion. Successive mashes. Small beer
663.258.2	Natural defects of wine	663.443.4	Brewing by decoction. □ Bavarian method (Munich method). Swabian method (Augsburg-Nuremburg method)
663.258.3	Accidental adulteration of wine. □ Flavour defects. Harsh taste. Contamination by vessel. Woody taste, corkiness	663.444	Boiling and hopping of wort. Wort boilers
663.258.4	Wine diseases. □ Mould in wine	663.445	Cooling of wort
663.258.8	Adulteration of wine (deliberate) and its control ⇒ 343.347 ⇒ 614.3	663.45	Fermentation of wort. Storage, lagering
663.26	Use of winemaking residues. □ Use of rapes, grape refuse, marc, lees. Products, e.g. marc brandy, marc vinegar ⇒ 663.252.6	663.451	Spontaneous fermentation
663.28	Establishments, premises connected with wine production	663.452	Fermentation by addition. Bottom fermentation. Production of ordinary beers, lagers, export beers
663.283	Oenological establishments	663.46	Preservation and other treatments
663.284	Vintners	663.461	Preservation of beer. □ Pasteurization of beer. Chemical preservation of beer
663.285	Individual premises. □ Fermentation rooms. Wine cellars, stores, shops	663.465	Preparation for retail sale. Transfer to bottles, containers ⇒ 663.059 ⇒ 683.5
663.3	Cider. Perry. Other fruit wines, sap wines ⇒ 634.1 ⇒ 663.81	663.47	Production of particular kinds of beer
663.31	Apple wine. Cider	663.471	English beers. □ Stout. Porter. Brown ale. Light ale. Pale ale. Bitter
663.316	Nonalcoholic cider. Apple juice	663.472	German beers. Austrian beers. □ German March beer, Weissbier. Munich, Pilsen, Dortmund beers
663.317	Alcoholic cider	663.473	Belgian beers. □ Lambic. Gueuse-Lambic
663.34	Pear wine. Perry	663.474	French beers
663.346	Nonalcoholic perry. Pear juice	663.475	Rice beers. Japanese sake
663.347	Alcoholic perry	663.476	Sorghum beers. Millet beers. African pombe. Romanian braga
663.35	Berry wines. □ Redcurrant, gooseberry, strawberry wines <i>Specify by :634.7...</i>	663.477	American maize beers
663.36	Other fruit wines. □ Cherry, rose-hip, rhubarb wines. Mixed fruit wines	663.478	Malt wines and extracts
663.37	Sap wines	663.478.1	Barley wine
663.373	Wood-sap wines (e.g. birch-sap wine)	663.478.2	Malt extract
663.374	Wines from sap of flowering shoots. □ Palm wines. Agave wine (Mexican pulque)		
663.393	Honey wine. Mead ⇒ 638.16		

663.478.3	Malt vinegar ⇒ 661.741.1		<i>Denote process techniques by the 66.0... auxiliaries.</i>
663.479.1	Kvass (Russian rye beer)		<i>Denote the use by :...</i>
663.48	Brewery wastes, residues. Brewery by-products. □ Draff. Malt combings ⇒ 631.879.3	663.634	◇ 663.63:663.4 Brewery water Distilled water as commercial product ⇒ 628.16.048
663.5	Potable alcohol. Spirituous liquors (ardent spirits). Alcohol distillation, refining ⇒ 661.72 ⇒ 663.241	663.635	Brine, seawater as commercial product ◇ 663.635.048 Distillation of seawater ⇒ 551.464.09 ⇒ 628.165 ⇒ 663.648
663.51	Alcohol: nature, formation, types of alcohol. Spirits in general	663.64	Mineral and medicinal waters as beverages (effervescent and still) ◇ 663.64.05 Additions, preservation and aftertreatment of mineral and medicinal waters ◇ 663.64.057 Carbonation. Aeration ◇ 663.64.059 Bottling and final treatment before consumption
663.52	Alcohol production processes generally. Alcohol distilleries, refineries	663.642	Artificial mineral water. Carbonated table water (soda water) ⇒ 613.38
663.53	Conversion of amylaceous (starch-containing) and cellulosic substances into sugar. Saccharification of malt, wood, peat	663.646	Natural mineral water as commercial product
663.531	Processing of grain. Production of sacchariferous mash. □ Maceration. Boiling ⇒ 664.7	663.648	Artificial brine, seawater as commercial product ⇒ 663.635
663.532	Processing of roots, tubers, starch		
663.533	Mould treatment of amyliferous substances	663.67	Edible ice. Ice cream
663.534	Saccharification of cellulosic substances. □ Saccharification of wood	663.672	Natural ice. Ice collection, transport and storage ⇒ 551.32
663.54	Production of alcoholic liquors from sacchariferous substances	663.673	Manufactured ice. Artificially produced ice ⇒ 621.56/.58
663.541	Processing of sugar beet	663.674	Flavoured ices. Ice cream
663.542	Processing of girasols (Jerusalem artichokes)		
663.543	Processing of residues from sugar refineries. □ Production of sugar spirits, rum ⇒ 664.153	663.8	Main ingredients for the blending of drinks. Fruit and vegetable juices. Syrups. Liqueurs. Soft drinks <i>The auxiliaries listed under 663.03... are not applicable at 663.8 and its subdivisions</i> ◇ 663.8.051 Additions for improvement of flavour. Aromatic supplements (for blended drinks) ◇ 663.8.054 Diluents (for blended drinks)
663.55	Distillation. Rectification. Treatment of liquors containing formed alcohol ⇒ 66.048		
663.551.1	Simple distillation	663.81	Fruit juices. Vegetable juices. Syrups for soft drinks ⇒ 663.236 ⇒ 663.3 ⇒ 664.857
663.551.2	Distillation with rectification	663.812	Sugar syrups as bases for blended drinks ⇒ 663.031.8 ⇒ 664.14
663.551.4	Rectification of crude alcohol	663.813	Natural fruit and vegetable juices ◇ 663.813:634.711 Raspberry juice
663.551.5	Production of particular spirituous liquors <i>Denote particular kinds of spirit by :63... or :663...</i> ◇ 663.551.5:633.1 Grain spirits. Gin, vodka etc. ◇ 663.551.5:634.21 Apricot brandy ◇ 663.551.5:663.43 Malt spirit. Whisky ⇒ 663.241 ⇒ 663.543 ⇒ 663.83	663.814	Berry syrups ◇ 663.814:634.711 Raspberry syrup
663.551.6	Distillation residues. □ Residues from distillation of amylaceous substances	663.815	Syrups from large fruits ◇ 663.815:634.31 Orange syrup
663.551.7	Fusel. Fusel oil ⇒ 661.725.5	663.816	Syrups from flowers, blossoms ◇ 663.816:634.31.075 Syrup from orange blossom
663.6	Water for beverages and other industrial uses. Mineral waters. Medicinal waters. Edible ice <i>The auxiliaries listed under 663.03... are not applicable at 663.6 and its subdivisions</i> ⇒ 543.3 ⇒ 553.7 ⇒ 556.3 ⇒ 613.3 ⇒ 628.1 ⇒ 641.87	663.817	Almond syrup. Almond milk. Orgeat syrup
663.63	Water and its treatment for special use <i>Class here only works from the point of view of chemical technology. The main number for water treatment is 628.16</i>	663.82	Drinks containing solid fruit (or pieces of fruit)
		663.83	Liqueurs <i>Obtained by infusion in alcohol of fruits or plants</i> ⇒ 663.241 ⇒ 663.551.5
		663.831	Anisette
		663.832	Chartreuse and similar liqueurs
		663.833	Liqueurs from fruit stones, nuts

663.834	Cream liqueurs	663.92	Cocoa substitutes. □ Cola (kola) ⇒ 633.76
663.837	Aperitifs		
663.837.1	Bitters	663.93	Coffee. Caffeine ⇒ 633.73 ⇒ 641.87
663.837.3	Quinines		
663.837.5	Vermouths	663.931	Commercial varieties of coffee. Green coffee
663.85	Synthetic soft drinks	663.932	Drying of coffee beans. Storage and transport
663.86	Sparkling, fizzy soft drinks ⇒ 663.64	663.933	Coffee processing
663.865	Solid preparations for the production of fizzy soft drinks	663.933.1	Pretreatment of coffee beans
663.87	Mixed drinks. □ Grogs. Cocktails	663.933.2	Dehusking, hulling, stoning of coffee beans. Sifting
663.88	Health drinks. □ Liquorice water. Tonics	663.933.3	Blending of coffees
663.9	Chocolate. Cocoa. Coffee. Tea. Tobacco <i>The auxiliaries listed under 663.03... and 663.05... are not applicable at 663.9 and its subdivisions</i> ⇒ 633.7 ⇒ 641.87	663.933.4	Coffee roasting (and roasters)
		663.933.6	Glazing, polishing of coffee beans. Coating, impregnating of coffee beans
663.91	Cocoa. Chocolate ⇒ 633.74 ⇒ 641.87	663.933.7	Grinding of coffee beans ⇒ 621.927.7
663.91.05	Machinery, apparatus and equipment for production of cocoa products	663.933.8	Decaffeination
663.911	Raw materials. Auxiliary materials	663.935	Deterioration of coffee (e.g. mildewing). Adulteration of coffee
663.911.1	Raw materials of vegetable origin. □ Cocoa. Pure carbohydrates. Alcohols	663.937	Production of caffeine
663.911.3	Raw materials of animal origin. □ Milk	663.938	Coffee products
663.911.5	Auxiliary materials	663.938.3	Ordinary roasted coffee
663.913	Preparation of raw materials	663.938.4	Ground coffee
663.913.1	Preparation of raw cocoa. □ Disinfection, cleaning. Drying, roasting, cooling of cocoa beans. Breaking, shelling, reduction of cocoa ⇒ 621.926	663.938.5	Decaffeinated coffee
663.913.5	Preparation of other raw materials. □ Preparation of sugar, nuts, milk, fruit	663.938.7	Coffee essence
663.913.7	Mixing and kneading of ingredients	663.938.8	Coffee extract. Other solid coffee preparations
663.914	Production of cocoa powder, cocoa butter and cocoa-containing preparations	663.94	Coffee substitutes. Coffee supplements
663.914.1	Defatting the cocoa mass	663.941	Chicory ⇒ 633.78
663.914.2	Treatment of defatted cocoa	663.942	Cereal coffee. Acorn coffee
663.914.3	Solubilization of cocoa	663.95	Tea. Theine ⇒ 633.72 ⇒ 641.87
663.914.4	Preparation of cocoa butter		
663.914.5	Production of cocoa-containing preparations. □ Cocoa powders. Cocoa pastes	663.951	Commercial varieties of tea. □ Black tea. Green tea
663.915	Production of chocolate	663.952	Various kinds of tea and their manufacture
663.915.1	Refining of crude chocolate	663.952.031	Tea manufacturing processes
663.915.2	Refining of chocolate mass	663.952.031.1	Withering
663.915.3	Moulding of chocolate	663.952.031.2	Steaming. Panning
663.916	Production of chocolate confectionery, sweets (chocolates) ⇒ 664.144/149 ⇒ 664.68	663.952.031.3	Rolling
663.916.1	Production of basic compositions. □ Production of fondant, caramel, sugary compositions	663.952.031.4	Fermentation
663.916.2	Moulding of chocolate confectionery	663.952.031.5	Firing. □ Polishing
663.916.3	Enrobing, coating of chocolate confectionery	663.952.031.6	Sorting. Cutting
663.916.5	Decoration of chocolate confectionery	663.952.031.7	Sifting. Grading. Picking over by hand
663.917	Extraction of theobromine	663.952.031.8	Bulking
663.918	Products of the cocoa and chocolate industry	663.952.2	Fully fermented tea
663.918.1	Intermediate cocoa products. □ Cocoa mass. Cocoa press cakes. Cocoa butter	663.952.5	Unfermented tea
663.918.2	Cocoa products. □ Cocoa powder. Cocoa concentrates, extracts	663.952.6	Partly fermented tea
663.918.3	Cocoa by-products. □ Shells	663.952.7	Kinds of tea according to special processes
663.918.4	Kinds of chocolate	663.952.71	Brick tea. Paled tea. Faggot tea. Cake tea. Ball tea
663.918.5	Chocolate confectionery, sweets (chocolates)	663.952.72	Powdered tea. □ Yenchá. Hikacha (ceremonial teas)
		663.952.73	Scented tea
		663.952.74	Broken tea. □ Tea fluff. Warehouse dust
		663.954	Blending of commercial teas ◇ 663.954:543.927 Tea tasting
		663.954.2	Blending methods. Cutting and mixing of teas
		663.954.3	Tea blends
		663.954.5	Packing methods
		663.954.56	Tea bags
		663.955.1	Deterioration of tea
		663.955.2	Adulteration of tea. □ Artificial colouring.
		663.958	Blending with other leaves
		663.958.7	Various tea products
		663.958.8	Tea essences
			Solid extracts and other tea preparations
		663.96	Tea substitutes. Tea-like products

663.97	Tobacco industry ⇒ 178.7 ⇒ 351.761 ⇒ 613.84 ⇒ 633.71 ⇒ 688.93	664.1	Sugar. Molasses. Glucose etc.
663.97.05	Tobacco processes and equipment	664.1.03	Extraction and purification of juice ⇒ 633.6
663.97.051	Preparation of raw tobacco	664.1.031	Preliminary treatment of sugar plants. □ Washing. Heat treatment (of whole plants) ⇒ 664.1.04
663.97.051.1	Pretreatment. Drying of tobacco. Making into hands. Fermentation	664.1.032	Comminution, cutting, grating of sugar plants
663.97.051.6	Sorting, screening, cleaning of tobacco. Blending of tobaccos	664.1.033	Juice extraction from sugar plants. Extracting equipment
663.97.051.7	Roasting and cooling of tobacco	664.1.034	Extraction by pressing, by milling. Rolling and crushing mills etc.
663.97.051.8	Chemical treatment of tobacco and tobacco products. □ Nicotine extraction. Tobacco preservation. Improvement of tobacco colour, aroma	664.1.035	Other extraction methods. □ Extraction by diffusion, centrifuging, maceration. Use of solvents
663.97.052	Treatment of tobacco for snuff, chewing, smoking	664.1.037	Purification of juice. □ Mechanical purification. Filtering
663.97.052.3	Comminution, shredding of tobacco	664.1.038	Treatment of juice with chemical agents. □ Clarification, defecation of sugar juices. Treatment with lime, carbon dioxide, sulphur dioxide ⇒ 66.09
663.97.052.4	Tobacco stripping, deribbing. Flattening the ribs	664.1.038.8	Treatment and extraction of clarification residues. Treatment of waste-waters (sweet waters)
663.97.052.5	Machines for cutting tobacco	664.1.039	Treatment of juice by physical or physicochemical methods
663.97.052.6	Machines for spinning, dressing tobacco	664.1.04	Heat treatment (of juice). Thermal processes <i>Subdivisions as at 66.04...</i> ◇ 664.1.048 Distillation. Evaporation
663.97.053	Machines and equipment for cigar production	664.1.05	Filtration of syrup. Boiling, refining, crystallization
663.97.054	Machines and equipment for cigarette production	664.1.052	Filtration of syrup
663.97.054.2	Cigarette-filling machines with continuous extruder or pressure chamber	664.1.053	Boiling of filtered syrup. Boiling methods
663.97.054.4	Cigarette machines with continuous rod of formed tobacco	664.1.054	Crystallization of syrup
663.97.056	Cutting equipment for cigars and cigarettes	664.1.055	Separation of sugar from syrup. □ By centrifuging ⇒ 66.022.63
663.97.057	Production of cigarette wrappers, papers. Production of filters, tips, mouthpieces	664.1.056	Removal of harmful substances
663.971	Snuff	664.1.057	Sugar refining. Refined, white sugar
663.972	Chewing tobacco	664.1.058	Candy sugar
663.973	Cut tobacco in general. Smoking tobacco	664.11	Cane sugar
663.973.2	Tobacco according to width and length of strands. Cut of tobacco (fine, coarse etc.)	664.111	Sugar cane as raw material ⇒ 633.61
663.973.4	Tobacco mixtures, blends	664.113	Treatment of bagasse and cane juice ⇒ 662.636
663.973.5	Pressed tobacco (plug)	664.117	Sugar-cane products. Juice. Crude sugar. Brown sugars. Various kinds of cane sugar <i>For molasses see 664.15</i>
663.973.7	Nicotine-free tobacco (denicotinized tobacco)	664.12	Beet sugar
663.974	Cigarettes	664.121	Sugar beet as raw material ⇒ 633.63
663.975	Cigars. Cigarillos. Cheroots	664.123	Treatment of slices (cosettes). □ Extraction. Drying
663.976	Filters and filter-tips for cigars and cigarettes	664.126	Intermediate sugar-beet products. □ Juice. Boiling products. Run-off syrups
663.977	Chemical treatment of tobacco smoke. Smoke dispellers	664.127	Sugar-beet products. □ Crude sugar. Various kinds of beet sugar
663.98	Tobacco substitutes	664.13	Various other sugars ⇒ 633.64/65 ⇒ 664.16
663.99	Industries based on various stimulant plant products. Narcotics ⇒ 178.8 ⇒ 351.761 ⇒ 613.83 ⇒ 615.212 ⇒ 633.888	664.132	Maple sugar
663.991	Opium ⇒ 633.75	664.133	Palm sugar
663.992	Hashish. Marijuana ⇒ 633.522	664.134	Sorghum sugar. Durra sugar
664	PRODUCTION AND PRESERVATION OF SOLID FOODSTUFFS <i>Denote the state, condition or form of foodstuffs by -4... (from 62-4...)</i> ◇ 664-404 Liquid state ◇ 664-41 Slabs. Bars. Strips ◇ 664-492.2 Powdered ◇ 664-492.3 Granulated ◇ 664-493 Chipped. Shredded ◇ 664-493.5 Flakes ⇒ 667.777	664.135	Milk sugar. □ Sugar from buttermilk
		664.136	Malt sugar (maltose)
		664.138	Raffinose. Melitose
		664.14	Processing of sugar. Sugar products. Sugar confectionery (sweets) ⇒ 663.812

664.141	Casting, moulding, cutting of sugar	664.162.037	Filtration of hydrolysed starch (after neutralization) 664.162.037 $\equiv$ 66.067
664.141.6	Manufacture of sugarloaves. Loaf sugar	664.162.038	Colour development during preparation. ☐ Precautions against colour development
664.141.7	Production of sugar in lumps, cubes, blocks (except sugarloaves)	664.162.048	Distillation. Evaporation. Concentration of glucose Subdivisions as at 66.048...
664.141.8	Use of low-grade products of refining	664.162.052	Filtration of glucose (after concentration) 664.162.052 $\equiv$ 66.067
664.141.81	Production of low-grade loaf sugar (chunks, bastard sugar)	664.162.065	Solidification, crystallization of glucose Subdivisions as at 66.065...
664.141.82	Production of low-grade powdered sugar. Ground sugar	664.162.1/6	Methods of glucose production according to the kind of raw material
664.143	Sugar confectionery (sweets) in general	664.162.1	Glucose production from potato starch
664.143.1	Boiling of sugar. Evaporating pans. Equipment for cold production of syrup. Sugar syrups	664.162.2	Glucose production from maize starch
664.143.2	Machines and equipment for drawing sugar	664.162.4	Glucose production from rice, tapioca, other starches 664.162.4 $\equiv$ 664.2
664.143.3	Kneaders and mixers for confectionery. Machines for preparing, treating raw materials	664.162.6	Glucose production from wood, from cellulose
664.143.4	Steaming equipment. Crystallizing, candying equipment. Lozenge pans	664.162.7	Glucose, fructose and maltose syrups
664.143.5	Moulds for confectionery	664.162.71	Glucose-fructose syrups
664.143.6	Moulding, forming machines for confectionery. ☐ Combined moulding and wrapping machines	664.162.72	High-fructose syrups
664.143.7	Machines for filled sweets (sugar-coated sweets): pralines, dragées, bonbons	664.162.75	Glucose-galactose syrups
664.144/149	Various sugar confectionery products $\Rightarrow$ 663.916 $\Rightarrow$ 664.8.035.5	664.162.76	High-maltose syrups
664.144	Filled sweets. Bonbons. Gums (chewing gum etc.). Fondants. Dragées. Pastilles. Lozenges. Sweets with soft or nut centres etc.	664.162.8	Sweeteners (substitutes for saccharose)
664.145	Marzipan, nougat etc.	664.162.81	Natural sweeteners
664.146	Caramelized confectionery. Toffees	664.162.82	Artificial sweeteners
664.147	Barley sugar	664.164	Mixtures containing glucose. ☐ Invert sugar. Artificial honey $\Rightarrow$ 638.16
664.148	Sweetened liquorice. Liquorice lozenges. Cough sweets. Throat lozenges etc.	664.165	Various hexoses. ☐ Galactose. Fructose
664.149	Other sugar confectionery	664.2	Starch. Amylaceous materials $\Rightarrow$ 633.68 $\Rightarrow$ 648.18
664.15	Molasses. Treacle. Syrup	664.2.03	Treatment of starch sources before desiccation $\Rightarrow$ 664.7
664.151	Kinds of molasses	664.2.031	Preliminary treatment (before crushing and grinding). ☐ Washing, cleaning, maceration of starch sources
664.151.1	Cane sugar molasses	664.2.032	Comminution, crushing, grinding of starch sources. Comminution with addition of chemicals
664.151.2	Beet sugar molasses	664.2.033	Processes of opening-up of cells by fermentation
664.151.3	Molasses from other sugars 664.151.3 $\equiv$ 664.13	664.2.034	Methods of increasing starch yield. Methods of eliminating other substances
664.152	Extraction of sugar from molasses	664.2.035	Washing of starch pulp. ☐ Cylindrical sieves. Sets of sieves
664.153	Processing of molasses to obtain alcohol, rum etc. $\Rightarrow$ 663.543	664.2.036	Bleaching processes for starch pulp. Other intermediate treatment
664.154	Production of foodstuffs using molasses	664.2.037	Separation of crude starch
664.157	Molasses syrups. Treacle	664.2.038	Purification of crude starch. ☐ Scrubbers for starch
664.16	Dextrins. Monosaccharides (monoses). Glucose. Various other products of hydrolysis of starch. Sweeteners (substitutes for saccharose)	664.2.047	Drying of starch Subdivisions as at 66.047...
664.161	Dextrins	664.2.05	Operations after drying of starch
664.161.1	Dextrin production by roasting, torrefaction	664.2.051	Deodorization of starch
664.161.2	Roasting with use of chemicals	664.2.052	Bleaching of dry starch
664.161.3	Dextrin production using dilute chemicals (without roasting)	664.2.053	Blueing of starch
664.161.4	Dextrin production using diastases. Biological methods	664.2.054	Screening, bolting, grinding of starch
664.161.7	Kinds of dextrin according to physical properties. ☐ Coloured dextrin. White dextrin (soluble, insoluble)	664.2.055	Forming, milling, pearling of starch. Products of various forms. ☐ Powdered, crystal, lump starch. Pearl starch. Starch flakes
664.161.8	Kinds of dextrin according to source, origin 664.161.8 $\equiv$ 664.2	664.2.057	Liquid and semi-liquid starch. ☐ Suspensions of starch in water (starch milk). Starch paste
664.162	Glucose. Various other products of hydrolysis of starch. Sweeteners (substitutes for saccharose)	664.2.058	Preparation of thin-boiling starch (soluble starch)
664.162.031	Preliminary treatment of glucose sources. ☐ Preparation of raw materials. Preparation of starch milk	664.2.058.2	Preparation by physical and mechanical means
664.162.036	Hydrolysis of starch milk. Neutralization. Saccharification. Conversion equipment	664.2.058.4	Preparation by chemical methods

664.2.059	Preparation of starch that swells in cold water (pre-gelatinized starch)	664.44	Substitutes for sodium chloride. Salt mixtures (e.g. with phosphates)
664.22	Potato starch (farina) ⇒ 633.491	664.48	Other edible minerals (including those with organic additives). □ Calcium salts
664.23	Cereal starches in general ⇒ 633.1	664.5	Spices. Condiments ⇒ 613.291 ⇒ 633.82/.84 ⇒ 635.7 ⇒ 641.88
664.231	Wheat starch	664.51	Pepper (Piper species)
664.232	Rye starch	664.515	Pepper substitutes. Adulterations of pepper
664.233	Buckwheat starch. Oat starch. Barley starch	664.52	Hot and aromatic peppers (capsicum, pimenta species)
664.235	Other cereal starches 664.235 ≡ 633.17	664.521	Hot peppers. □ Chillies. Paprika. Cayenne pepper
664.236	Gluten	664.522	Aromatic peppers. □ Allspice (pimento). Tabasco pepper
664.24	Rice starch	664.53	Mustard
664.25	Maize starch (corn starch)	664.54	Nutmeg
664.26	Starches from various fruits. □ Nut starches. Soya starch. Bean starch. Lentil starch	664.55	Ginger
664.27	Other edible starches. □ Arrowroot starch. Cassava starch (Manioc starch. Tapioca starch). Sago starch. Banana starch. Yam starch. Canna starch. Starches from various stems, stalks, tubers, bulbs ⇒ 633.68	664.56	Cinnamon. Cassia
664.28	Starch derivatives	664.563	Cinnamon substitutes. Adulterations of cinnamon
664.29	Various products associated with starch. □ Pectin	664.57	Vanilla
664.3	Edible oils and fats. Margarine. Protein foodstuffs	664.572	Vanilla substitutes. Adulterations of vanilla
664.3.032	Plant and equipment for oil and fat processing. □ Emulsifying and homogenizing equipment. Churns. Washing equipment. Moulds and cutters	664.58	Seasoning plants preserved in vinegar. □ Gherkins. Capers. Pickles. Preserved olives. Other preserved vegetables ⇒ 664.8
664.3.033	Processes. □ Emulsifying. Churning. Washing, working, mixing of fat products. Salting. Pasteurization, sterilization. Moulding and cutting of fat products. Colouring. Flavouring	664.59	Other spices and condiments
664.31	Edible oils and fats in general	664.6/.7	Technology and processing of cereal grains
664.314	Edible oils in general	664.6	Baking. Bread. Flour confectionery
664.315	Edible fats in general	664.61	Bakeries. □ Kinds of bakery. Bakery equipment
664.32	Edible animal oils and fats	664.64	Raw and intermediate materials for baked products
664.324	Edible animal oils. □ Animal-oil emulsions	664.64.016	Nonchemical properties of bakery materials. Quality: its determination and evaluation
664.325	Edible animal fats. □ Margarine of animal origin	664.64.016.3	Physical properties. □ Structure, particle size of bakery materials. Colour. Content of other flours
664.33	Edible vegetable fats	664.64.016.7	Biological properties. □ Taste, aroma, nutritive value of bakery materials
664.336	Margarine of vegetable origin	664.64.016.8	Technological properties. Baking quality
664.34	Edible vegetable oils. □ Vegetable-oil emulsions. Mayonnaise	664.64.019	Spoiling of bakery materials. Deterioration. □ Discoloration
664.35	Mineral oils and fats used in the food industry	664.641	Meal and flour. □ Composition. Selection ⇒ 664.761
664.36	Mixtures of edible fats. □ Mixtures of butter and margarine	664.641.1	Cereal flours ⇒ 633.1
664.38	Protein foodstuffs. Edible proteins. □ Gelatin	664.641.12	Wheat flour
664.4	Edible minerals ⇒ 613.27	664.641.14	Rye flour
664.41	Edible salt (sodium chloride). □ Cooking salt. Table salt ⇒ 549.451.1 ⇒ 553.631 ⇒ 622.363.1 ⇒ 661.42 ⇒ 661.833.321	664.641.15	Maize flour
		664.641.18	Rice flour
		664.641.19	Other cereal flours
		664.641.2	Other flours. □ Potato flour. Soya flour. Buckwheat flour ⇒ 633.34 ⇒ 633.491
		664.641.4	Mixtures of cereal flour with other flours
		664.642	Baker's yeast. Leaven ⇒ 663.126
		664.643	Artificial leavening agents. Baking powders ⇒ 664.654.2

664.644	Other raw materials for baked products. Flour improvers ⇒ 664.746	664.66.016/.019	Properties and spoiling of bakery products 664.66.016/.019 ≡ 664.64.016/.019
664.644.1	Salt ⇒ 664.41	664.66.03	Storage and preservation of bakery products
664.644.2	Yeast food. Fermentation stimulants. Mineral products	664.661	Bread, rolls and buns exclusively or mainly from whole-grain wheat, wholemeal wheat or wheat flour. □ Wholemeal bread. White bread. Bread rolls. Currant buns
664.644.3	Fatty materials. Shortening	664.661.2	Plain bread without special additives other than improving agents
664.644.4	Protein-rich materials. □ Milk. Dried milk (milk powder). Eggs. Egg white (albumen)	664.661.21	Loaf bread
664.644.5	Enzyme-rich materials. □ Amylase preparations. Malt extracts. Fungal amylases. Proteinase preparations	664.661.26	Rolls. Muffins etc.
664.644.6	Carbohydrate-rich materials. □ Sugars. Saccharified flours	664.662	Bread exclusively or mainly from whole-grain rye, rye wholemeal or rye flour. Rye bread. Pumpernickel
664.644.7	Water-absorbent materials. Dextrinized flours. Gums. Alginates. Carob seed flour etc.	664.663	Bread from other meals or flours. □ Barley bread. Millet bread. Maize bread
664.644.8	Softeners and emulsifiers. □ Polyethylene stearate. Lecithin	664.664	Bread-like products requiring special process or recipe ⇒ 664.68
664.646	Dusting powders for baking pans. Pan grease. Pan coatings	664.664.2	Unleavened bread
664.647	Various bakery materials. □ Emulsion-type improvers for white bread. Semifinished products as fillings. Pastry fillers	664.664.3	Dry baked products
664.647.6	Cake mixes (ready-made)	664.664.32	Toast
		664.664.33	Rusks
		664.664.34	Crispbread. Krustenbrot. Waffelbrot
		664.664.4	Fat-rich products. □ Scones. Brioches. Butterstollen
664.65	Baking: preparation, procedure and plant. Bread. Flour confectionery	664.665	Bread-like products with special properties or for dietetic purposes. Dietetic bread. □ Children's rusks ⇒ 613.2 ⇒ 615.874 ⇒ 664.681
664.65.05	Bakery machinery	664.667	Spiced breads. □ Gingerbread
664.651	Preparation of raw materials and ingredients	664.668	Bread-like feeding stuffs for animals. □ Dog biscuits ⇒ 636.084/.087
664.652	Addition of leavening agents		
664.653	Mixing, kneading and handling of dough		
664.653.1	Mixing and kneading of dough		
664.653.3	Weighing and dividing of dough		
664.653.4	Moulding and treatment of dough (especially for breadmaking) <i>Details by :664.66... and :664.68...</i> ◇ 664.653.4:664.683.6 For cake production	664.68	Manufacture and treatment of biscuits, wafers, pastries, cakes, puddings
664.653.5	Direct conversion of grain into bread	664.681	Biscuits. Wafers. Waffles. Crackers
664.654	Natural and artificial leavening of dough. Fermentation, proving of dough ⇒ 663.1	664.682	Dry flour confectionery products. Pastries. □ Sweet pastries. Salted pastries. Cheese pastries
664.654.1	Fermentation of dough with biological agents. □ Fermentation with yeast, with leaven	664.683	Fancy flour confectionery. Cakes. Fancy pastries
664.654.2	Leavening of dough by chemical and physical means. □ Leavening with baking powder, with solid carbon dioxide ⇒ 664.643	664.683.6	Oven-baked products from batter (not dry). Cakes
664.654.3	Accelerated fermentation of dough. Addition of fermentation stimulants (yeast foods)	664.683.7	Fried products from dough or batter. □ Fritters. Pancakes. Doughnuts
664.654.6	Fermentation rooms and cabinets for dough	664.684	Filled bakery products. Pies. Tarts. Pasties. Turnovers. Pastry rolls with fillings (e.g. sausage rolls)
664.655	The baking process. Other finishing of bakery products. Influence of these processes on the products ◇ 664.655.041 Bakery ovens	664.685	Special baked products without flour ⇒ 664.14
664.655.1	Baking in ovens	664.685.4	Products from almond paste, coconut paste etc.
664.655.3	Frying in fat	664.685.6	Meringues
664.655.5	Surface treatment of products after baking. □ Washing	664.686	Puddings and their ingredients ⇒ 641.85
664.655.6	Utensils and equipment for baking. □ Baking pans. Baking boards	664.69	Cereal paste products. Pasta. Processed cereals
664.656	Treatment of baked products. □ Cooling, freezing. Filling. Decorating. Cutting, wrapping	664.691	Raw materials for pasta
664.657	Prevention of staleness in baked products. Treatment of stale products ⇒ 664.66.019	664.692	Manufacturing processes for pasta
		664.694	Pasta products. □ Macaroni. Vermicelli. Spaghetti. Noodles
		664.696	Processed cereals. Ready-to-serve cereals. Breakfast cereals ⇒ 664.78
664.66	Bakery products in general. Bread and allied products ⇒ 664.68	664.696.1	Flaked cereals
		664.696.2	Puffed cereals. □ Popcorn
		664.696.3	Shredded cereals

- 664.696.4 Granular ready-to-serve cereals
- 664.7 Cereal technology. Flour and corn milling. Grain processing**
 ⇒ 621.927
 ⇒ 663.531
 ⇒ 664.2.03
- 664.71 Milling of wheat and rye. Flour mills**
- 664.71-11 Wheat
 664.71-12 Rye
 664.71.05 Milling machinery
- 664.715 Properties and structure of grain in relation to milling
- 664.717 General principles of milling. Comparison of milling processes
- 664.72 Storage, cleaning and conditioning of grain**
 664.723 Cleaning of grain before storage. Aspiration and drying of grain
 ⇒ 631.365
 664.724 Storage of grain
 ⇒ 631.24
 ⇒ 631.563
 ⇒ 725.1
 664.726 Final cleaning of grain (just before milling).
 □ Aspiration, separation, washing of grain
 664.727 Special chemical and physical treatment of grain. □ Bleaching of whole grain. Sterilization of grain. Entoleters
- 664.73 Grinding of grain. Corn milling**
 ⇒ 621.926/927
- 664.731 Primitive grinding methods. □ Pestle and mortar. Handmills with discs or rolls
- 664.732 Grinding with stones and discs. Millstones
- 664.733 Grinding with rollers. Roller mills. Cylindrical crushers, converters, reducers
- 664.734 Comminution by pounding, stamping etc. Hammer mills
- 664.738 Deflaking devices (detachers)
- 664.74 Sifting, grading, purification of milling products. Other flour milling operations**
- 664.741 Sifting, sieving, bolting of flour etc.
- 664.742 Grading of flour with metal sieves, cloth screens
- 664.743 Grading and purification of semolina
- 664.744 Ancillary equipment and machinery
- 664.746 Treatment of flour to improve baking quality, colour
 ⇒ 664.644
- 664.748 Other treatment of milling products.
 □ Sterilization. Denaturation. Enrichment
- 664.76 Various products and by-products from milling and cleaning**
 ⇒ 663.942
- 664.761 Flour (as milling product)
 ⇒ 664.641
- 664.762 Semolina
 ⇒ 664.743
- 664.764 Bran
- 664.765 Germ (cereal germ)
 ⇒ 665.33
- 664.768 Offals. Waste products of milling
- 664.78 Technology of cereals other than wheat and rye**
 ⇒ 636.086
 ⇒ 664.696
- 664.78.01 General principles of cereal processing
- 664.78.03 Storage and preservation of cereals
 ⇒ 664.724
- 664.782 Processing of rice. Rice milling
- 664.782.2 Primitive methods of rice processing. Manual devices. □ Scalding of rice
- 664.782.4 Modern rice-milling processes. □ Cleaning, drying, husking of rice. Separation. Paddy machines. Polishing of rice
- 664.782.7 Other treatments of rice. □ Parboiling, conversion, precooking of rice. Malek process
- 664.782.8 Rice products. □ Rice flour. Milled rice. Parboiled rice. Precooked rice. Rice by-products, e.g. animal feed
- 664.784 Processing of maize. Maize milling (corn milling)
- 664.784.8 Maize products. Maize by-products, e.g. animal feed
- 664.785 Processing of oats. Oat milling
- 664.785.3 Production of oat flakes. □ Cleaning, grading, drying of oats. Hulling, separation of hulls and groats. Rolling of groats into flakes
- 664.785.8 Oat products. □ Rolled oats. Oatmeal. Oat by-products, e.g. animal feed
- 664.786 Processing of barley. Barley milling
 ⇒ 663.43
- 664.786.3 Production of pot barley and pearl barley.
 □ Cleaning, bleaching, separating, hulling of barley. Pearling of barley. Aftertreatment
- 664.786.8 Barley products. □ Pot barley. Pearl barley. Barley by-products, e.g. animal feed
- 664.788 Processing of other cereals and similar grains.
 □ Processing of sorghum (milo), buckwheat, millet
- 664.8/9 Food preservation technique**
- 664.8 Preservation of organic edible substances (in general). Preservation of plant, vegetable products**
- 664.8.022 Treatment and preparation of raw materials for food preservation
Class here the use of additives in foodstuffs generally. For specific foods with additives, see the various subdivisions of 664
Subdivisions as at 66.022...
 ◇ 664.8.022.1 Pretreatment of raw foodstuffs.
 NOTE: Details by :641.51...
 ◇ 664.8.022.3 Pretreatment with preservatives and other additives
 ⇒ 663.05
- 664.8.022.7 Culinary treatment, cooking of prepared foodstuffs for preservation
 664.8.022.7 ≡ 641.52
- 664.8.03 Food storage and preservation
 ⇒ 621.796
- 664.8.031 Food storage in pits, silos, open containers, vessels
- 664.8.032 Food storage in closed containers, vessels. Enclosed food stores, lofts
- 664.8.033 Food preservation in a vacuum
- 664.8.034 Food preservation by smoking
- 664.8.035 Food preservation by chemicals
- 664.8.035.1 Preservation by gases
- 664.8.035.2 Preservation by salting
- 664.8.035.3 Preservation by salting and smoking
- 664.8.035.4 Preservation in acids (e.g. vinegar)
- 664.8.035.5 Preservation in sugar syrups. □ Candying. Crystallizing
 ⇒ 664.144/149
- 664.8.035.6 Preservation in alcoholic liquids
- 664.8.035.7 Preservation with antiseptics. □ Preservation with formaldehyde, benzoates, salicylates, phenols, borates, sulphites, milk of lime
- 664.8.036 Food preservation by heating

664.8.036.1	Preservation by cooking in general	664.844	Dried vegetables. Desiccated vegetables
664.8.036.2	Superheating. Sterilization. Autoclaving.		<i>Denote the kind of vegetable by :635...</i>
	Aseptic preservation	664.848	Preserved edible fungi (including dried fungi).
	⇒ 66.046.8		□ Preserved mushrooms and truffles
664.8.036.3	Preservation by pasteurization in general	664.849	Vegetable pastes, purées
664.8.036.5	Heating of food in cans. Preserving cans, tins.		<i>Denote the kind of vegetable by :635...</i>
	Canned food	664.85	Preservation of fruit (including nuts)
	◇ 664.8.036.5:543 Chemical examination of	664.853	Fruits preserved by particular methods
	canned foods. Including: Analysis of air in		<i>Denote the kind of fruit by :634...</i>
	cans		◇ 664.853.035:634.53 Chemically preserved
	⇒ 672.46		chestnuts
664.8.036.52	Food canning in general	664.854	Fruit preserved by drying. Dried and desiccated
	⇒ 664.9		fruit
664.8.036.522	Testing of keeping qualities		<i>Denote the kind of fruit by :634...</i>
664.8.036.523	Spoilage of canned foods. Bulging of can	664.856	Conserves produced from fruit juice. Fruit
	(swells). Bacteriological examination of canned		jellies
	food		<i>Denote the kind of fruit by :634...</i>
	<i>Details by :579.8...</i>	664.857	Fruit syrups. Juice concentrates
	⇒ 614.31		<i>Denote the kind of fruit by :634...</i>
	⇒ 579.67		⇒ 663.81
664.8.036.53	Apparatus for preservation. Can filling and		⇒ 665.56
	closing machines. Cooking and cooling	664.858	Jams. Marmalades. Fruit preserves, conserves
	equipment for canned food		<i>Denote the kind of fruit by :634...</i>
	⇒ 672.462	664.86	Preservation of other plant products.
664.8.036.58	Canned foods in general. □ Keeping qualities.		□ Preservation of aromatic plants, juice drinks,
	Testing		oils
664.8.036.6	Heating of food in bottles, jars. □ Hermetic	664.87	Concentrated foods in general. Vegetable
	sealing		concentrates in particular
	⇒ 666.17		⇒ 664.932
664.8.036.7	Heating of food in containers other than cans	664.871	Soup products in general. □ Soup powders.
	and bottles		Soups. Sauces. Gravies
664.8.037	Food preservation by refrigeration. Storage and	664.872	Yeast preparations and products
	transport of refrigerated food		⇒ 663.12
	⇒ 621.56/.58	664.874	Malt preparations and products
	⇒ 656.073.44		⇒ 663.43
	⇒ 66.045.5	664.9	Preservation of animal products
	⇒ 725.1		⇒ 637
664.8.037.1	Chilling of food. Cool storage (cold storage	664.9.022	Pretreatment and preparation of raw materials
	above freezing). Treatment with ice		for food preservation
664.8.037.5	Freezing of food (deep freezing). Freezer		<i>Subdivisions as at 66.022.... See the</i>
	storage (cold storage below freezing)		<i>examples at 664.8.022</i>
	⇒ 621.565	664.9.022.7	Culinary treatment, cooking of prepared
664.8.037.59	Use of frozen foods. □ Thawing (defrosting)		foodstuffs for preservation
664.8.038	Food preservation by coating		664.9.022.7 ≡ 641.52
664.8.039	Other methods of food preservation	664.9.03	Storage and preservation
	⇒ 66.08		664.9.03 ≡ 664.8.03
664.8.039.3	Preservation by pressure	664.9.047	Preservation by drying, desiccation
664.8.039.4	Preservation by vibration. Preservation by		<i>Subdivisions as at 66.047...</i>
	ultrasound	664.91	Preservation of meat in general
664.8.039.5	Preservation by radiation. □ Preservation by		⇒ 637.52
	ultraviolet radiation	664.92	Dried, smoked and salted meats
664.8.039.7	Preservation by biological processes.	664.921	Dried meat. □ Biltong. Pemmican
	□ Preservation by fermentation	664.922	Salted meat
664.8.047	Preservation by drying. Desiccation of food.	664.923	Smoked meat
	□ Freeze-drying	664.93	Canned meat. Preserved meat products
	<i>Subdivisions as at 66.047...</i>	664.931	Meat preserved in cans. Canned meat.
664.83	Preservation of root vegetables, potatoes		□ Pressed meat (e.g. corned beef)
	etc.	664.932	Concentrated meat products
664.833	Root vegetables, potatoes etc. preserved by	664.932.1	Soup products from meat in general
	storage or by chemicals		⇒ 664.871
	664.833 ≡ 664.8.03	664.932.2	Meat extracts, stocks. □ Liebig's extract. Beef
664.834	Root vegetables, potatoes etc. preserved by		teas, broths. Consommés
	drying. Desiccated root vegetables, potato etc.	664.932.7	Powdered meat. Meat meal
	□ Potato powder	664.933	Canned meat products. □ Canned sausage,
664.84	Preservation of other vegetables, legumes		ham, jellied meat
	◇ 664.84:635.65 Preservation of pulses	664.934	Potted meats. Meat pastes
664.843	Vegetables preserved by particular methods	664.935	Meat preserved in vinegar. Pickled meat
	<i>Denote the kind of vegetable by :635...</i>		
	◇ 664.843.035:635.64 Chemically preserved		
	tomatoes, aubergines, pimientos etc.		

664.95	Preservation of fish. Preserved fish products ⇒ 639.2.068	665.13	Generalities on waxes ⇒ 665.92
664.951	Preservation of fish by salting, curing, kippering etc. <i>Details by :639.2...</i>	665.135	Synthetic waxes and wax substitutes
664.951.1	Preliminary storage of fish in ice ⇒ 639.2.067	665.14	Manufacture of candles in general ⇒ 628.9.045 ⇒ 665.587
664.951.2	Salted fish ◇ 664.951.2:597.541HER Salted herrings	665.14.059.2	Preparation of candle wicks
664.951.3	Smoked, kippered fish	665.14.059.4	Drawing of candles
664.951.31	Hot-smoked fish ◇ 664.951.31:597.541HER Hot-smoked herrings (kippers)	665.14.059.5	Casting of candles. Other forming, shaping of candles
664.951.32	Cold-smoked fish ◇ 664.951.32:597.522.511 Cold-smoked salmon	665.15	Tallow candles
664.951.4	Fish preserved in oil	665.16	Stearine candles (stearic acid candles)
664.951.5	Marinated fish. Pickled fish	665.17	Paraffin candles. Spermaceti candles
664.951.51	Cold-marinated fish ◇ 664.951.51:597.541HER Cold-marinated herrings. Pickled herrings	665.18	Wax candles. Beeswax candles
664.951.53	Boiled marinated fish ◇ 664.951.53:597.535 Jellied eels	665.19	Other solid fats for use in illumination ⇒ 662.88
664.951.6	Sterilized fish products. Fishcakes etc. ◇ 664.951.6:597.541SAR Pilchards (sardines) in oil	665.194	Flaming torches (links) ⇒ 628.9.045 ⇒ 662.126
664.951.7	Shellfish and crustacean preserves	665.2	Oils, fats and waxes of animal origin ⇒ 637.043 ⇒ 665.529
664.952	Fish sausage. □ Cod liver sausage. Canned crab sausage	665.21	Oils of animal origin
664.953	Pastes, spreads of fish and crustaceans. Fish paste	665.213	Fish oils ⇒ 639.381
664.955	Fish roes (hard and soft roes). □ Caviar	665.214	Fish liver oils
664.956	Drying of fish. Dried fish. □ Stockfish	665.215	Marine mammalian oils. Train oil. □ Whale oil
664.957	Fish meal ⇒ 631.854	665.216	Amphibian oils. Alligator oil. Crocodile oil
664.958.2	Fish salads	665.218	Horn oils. Hoof oils
664.958.8	Crustacean salads	665.219	Other oils of animal origin
664.959	Other fish products ⇒ 665.213/.214	665.22	Fats of animal origin. Tallow
664.97	Preservation of other animal products 664.97 ⇒ 637	665.221	Beef tallow. Mutton tallow. Greaves
665	OILS. FATS. WAXES. ADHESIVES. GUMS. RESINS ⇒ 547.91	665.222	Pig fat. Lard
665.1	Generalities on oils, fats, waxes and products derived from them <i>Denote chemical processes in treatment of oils, fats, waxes etc. by .09... (from 66.09...)</i>	665.223	Fats of various mammals. □ Horse fat. Deer and reindeer tallow. Camel fat. Bear fat
665.11	Generalities on products derived from oils, fats and waxes	665.224	Fats of various birds. □ Chicken fat. Duck fat. Goose fat
665.112	Characteristic properties. □ Vitamin content. Nutritional value. Rancidity	665.225	Fats of other creatures. □ Reptile fats (e.g. tortoise and turtle fat). Fish fats. Insect fats
665.114	Unsaponifiable matter	665.228	Carcass fats. Bone fats. Skin fats. Degras ⇒ 637.53 ⇒ 675.044
665.117	Oil cake. Extracted meal. Cake from low-pressure expellers	665.23	Waxes of animal origin
665.12	Fatty acids. Generalities on stearic acid production	665.231	Beeswax ⇒ 638.171
665.122	Production of fatty acids. Separation from glycerol. Saponification	665.237	Waxes from marine mammals. Spermaceti
665.123	Synthesis of fatty acids	665.238	Cholesterol. Lanolin. Wool fat (wool grease)
665.124	Distillation of fatty acids	665.3	Vegetable oils, fats, waxes, phosphatides ⇒ 633.85 ⇒ 665.52
665.126	Separation of solid fatty acids from liquid fatty acids	665.32	Nondrying vegetable oils
665.127	Various fatty-acid products. □ Saturated and unsaturated fatty acids. Stearine pitch	665.321	Nondrying oils of cryptogams. □ Mushroom oils. Ergot oil. Lactarius oil
665.128	Production of solid fatty acids from liquid fatty acids. Conversion of oleic acid to isooleic acid	665.322	Nondrying oils of gymnosperms and conifers. □ Juniper berry oil
		665.323	Nondrying oils of monocotyledons. □ Chufa tuber oil. Banana oil. Sabadilla oil
		665.324	Nondrying oils of certain dicotyledons. □ Hazelnut oil. Acorn oil. Urticales oils.
		665.325	Ranales oils
		665.325.1	Nondrying oils of Rosales, Pandales, Geraniales, Sapindales, Rhamnales and Malvales
			Rosales oils, Rosaceae oils in general. □ Almond oil

665.325.2	Groundnut oil (peanut oil, arachis oil)	665.347.8	Compositae oils. Sunflower oil
665.325.3	Leguminosae oils other than groundnut oil	665.347.9	Other Compositae oils
665.325.5	Geraniales oils	665.348	Other drying vegetable oils 665.348 $\equiv$ 582
665.325.6	Sapindales oils	665.349	Undetermined vegetable drying oils. □ Mohamba oil
665.325.7	Sapindineae oils. Balsamineae oils		
665.325.9	Malvales oils		
665.326	Nondrying oils of Parietales, Opuntiales, Myrtiflorae and Umbelliflorae	665.35	Vegetable fats and oils with high melting point
665.327	Nondrying oils of Metachlamydeae and Sympetalae	665.351	Fats of cryptogams
665.327.3	Olive oil	665.353	Fats from monocotyledons
665.328	Other nondrying vegetable oils	665.353.3	Palm fats
665.329	Undetermined nondrying vegetable oils. □ Chinese butter oil. Ojoc oil. Ceje oil	665.353.4	Palm oil
665.33	Semi-drying vegetable oils ⇒ 664.765	665.353.6	Coconut oil. Coconut butter. □ Copra
665.331	Semi-drying oils of cryptogams	665.354	Fats from dicotyledons
665.332	Semi-drying oils of gymnosperms and conifers	665.355	Fats of Rosales, Pandales, Geraniales, Sapindales, Rhamnales and Malvales
665.333	Semi-drying oils of monocotyledons. □ Cereal oils	665.356	Fats of Parietales, Opuntiales, Myrtiflorae and Umbelliflorae
665.333.4	Maize oil (corn oil)	665.357	Fats of Metachlamydeae and Sympetalae
665.334	Semi-drying oils of certain dicotyledons	665.358	Other vegetable fats 665.358 $\equiv$ 582
665.334.9	Rape oil (rapeseed oil, colza oil)	665.359	Undetermined vegetable fats
665.335	Semi-drying oils of Rosales, Pandales, Geraniales, Sapindales, Rhamnales and Malvales	665.36	Vegetable waxes. □ Carnauba wax (Brazilian wax)
665.335.2	Soyabean oil	665.37	Vegetable phosphates. □ Lecithin
665.335.5	Castor oil	665.38	Special products
665.336	Semi-drying oils of Parietales, Opuntiales, Myrtiflorae and Umbelliflorae	665.381	Synthetic oils
665.337	Semi-drying oils of Metachlamydeae and Sympetalae	665.382	Sulphonated oils, fats, fatty acids and waxes
665.338	Other semi-drying vegetable oils 665.338 $\equiv$ 582	665.383	Modified oils. □ Boiled oils. Polymerized oils. Oxidized oils
665.339	Undetermined semi-drying vegetable oils	665.5	Essential oils. Perfumery. Cosmetics
665.34	Drying vegetable oils	665.52/54	Essential oils
665.341	Drying oils of cryptogams	665.52	Natural essential oils ⇒ 665.3
665.342	Drying oils of gymnosperms and conifers	665.522	Essential oils of cryptogams
665.343	Drying oils of monocotyledons. □ Asparagus (seed) oil	665.523	Essential oils of gymnosperms and conifers
665.344	Drying oils of certain dicotyledons	665.524	Essential oils of monocotyledons
665.344.2	Juglandaceae oils	665.525	Essential oils of dicotyledons
665.344.3	Moraceae oils	665.526	Essential oils of Rosales, Geraniales, Sapindales, Malvales, Parietales, Myrtiflorae and Umbelliflorae
665.344.7	Ranunculaceae oils	665.527	Essential oils of Metachlamydeae
665.344.8	Rhoeadales oils	665.528	Essential oils of other plants 665.528 $\equiv$ 58
665.344.9	Papaveraceae oils. Sarraceniales oils. □ Poppy seed oil	665.528.1	Essential oils of undetermined plants
665.345	Drying oils of Rosales, Pandales, Geraniales, Sapindales, Rhamnales and Malvales	665.529	Scents of animal origin. Aromatic animal oils ⇒ 665.2
665.345.1	Rosales oils. Rosaceae oils. □ Rose-hip oil. Currant (Ribes) oil. Raspberry (seed) oil. Blackberry (seed) oil. Mountain ash berry (seed) oil. Strawberry (seed) oil. Gooseberry (seed) oil	665.529.2	Ambergris ⇒ 639.388
665.345.2	Papilionatae oils. □ Acacia oil	665.529.4	Musk. Civet ⇒ 637.68
665.345.4	Linseed oil. Flax seed oil	665.53	Principal constituents of essential oils <i>Details by :547..., especially :547.913...</i>
665.345.5	Euphorbiaceae oils in general	665.54	Synthetic essential oils. □ Artificial musks
665.345.6	Wood oil. China wood oil. Tung oil	665.56	Fruit essences ⇒ 664.857
665.345.8	Geraniales oils. □ Ngart oil	665.562	Natural fruit essences
665.345.9	Sapindales oils. Rhamnales oils. Malvales oils	665.564	Synthetic fruit essences
665.346	Drying oils of Parietales, Opuntiales, Myrtiflorae and Umbelliflorae	665.57	Perfumery. Perfumes. Perfuming
665.347	Drying oils of Metachlamydeae and Sympetalae	665.572	Natural perfumes <i>Details by :665.52...</i>
665.347.1	Diapensiales oils. Ericales oils. Primulales oils. Plumbaginales oils. Arbutus oil. Bilberry or whortleberry (seed) oil	665.574	Synthetic perfumes <i>Details by :547...</i>
665.347.3	Contortae oils	665.577	Perfuming (of various substances)
665.347.4	Tubiflorae oils in general. Labiatae oils		
665.347.5	Solanaceae oils. Scrophulariaceae oils		
665.347.7	Campanulatae oils. Lobelia oil		

665.58	Cosmetics. Toiletries ⇒ 613.49 ⇒ 646.7 ⇒ 687.5	665.612	Gaseous constituents of crude oil (including natural gas) ⇒ 662.767 ⇒ 665.72
665.582	Toilet waters. Bath additives. Other perfumed substances	665.613	Liquid constituents of crude oil
665.582.2	Toilet waters. Scented waters	665.614	Colloidal mixtures, emulsions in crude oil
665.582.3	Deodorants. Antiperspirants	665.615	Other liquid constituents. □ Brine in crude oil
665.582.4	Bath additives. □ Bath salts. Bath essences. Bath oils	665.617	Solid constituents of crude oil
665.582.7	Smelling salts. Smelling sachets. Scent bags	665.62	Treatment of crude oil and natural gas for working-up, conversion and refining
665.582.8	Perfumed substances. Perfume pencils, sticks. Cologne, menthol sticks etc.	665.622	Separation of crude oil from gas, solids and brine
665.583	Products for care of mouth and teeth	665.622.2	Separation of natural gas from crude oil containing solids and brine
665.583.2	Mouthwashes. Breath sweeteners. □ Mouthwash tablets. Mouth sprays	665.622.3	Separation of solids from crude oil containing brine
665.583.4	Products for the care of the teeth. □ Toothpastes. Toothpowders	665.622.4	Separation of oil from brine
665.584	Skin creams. Make-up. Powders. Nail cosmetics	665.622.7	Working-up of crude products in the field
665.584.2	Skin creams. Face lotions. Make-up	665.625	Stabilization (fractionation under increased pressure)
665.584.22	Solid emulsion-based creams. □ Solid creams with additives: vitamin creams, hormone creams etc.	665.63	Working-up processes for crude oil and its fractions, mainly by distillation <i>Details of distillation processes and apparatus by .048... (from 66.048...)</i>
665.584.223	Shaving creams and creams for the care of the skin after shaving ⇒ 665.585.9	665.632	Obtaining and working-up of gases. Cracking and reforming of gases
665.584.24	Liquid emulsion-based creams. □ Liquid creams with additives: vitamin creams, hormone creams etc.	665.633	Obtaining and working-up of gasolines (motor spirit, petrol)
665.584.25	Skin lotions. Face lotions	665.633.2	Light gasoline (boiling point 100°C)
665.584.26	Skin oils, greases, salves	665.633.7	Heavy gasoline (naphtha) (boiling range 100-200°C)
665.584.28	Make-up. Face cosmetics ⇒ 665.584.4	665.633.8	Preparation of gasolines with various boiling points. □ By redistillation
665.584.288	Make-up pencils. Lipsticks	665.634	Obtaining and working-up of kerosine (boiling range 150-250°C)
665.584.4	Powders. □ Talcum powders. Baby powders. Face powders	665.635	Obtaining and working-up of gas oil (boiling range 200-350°C)
665.584.7	Products for nail care, manicure. Nail cosmetics. □ Nail varnish. Nail varnish removers	665.637	Obtaining and working-up of long residues or topped crudes
665.584.8	Eyelash and eyebrow make-up. □ Mascara. Eyebrow pencils	665.637.2	Paraffinic products <i>For kerosine (commonly called paraffin) see 665.634, 665.74</i>
665.585	Products for hair care	665.637.3	Asphaltic products
665.585.2	Protectives against hair and scalp complaints. Dandruff remedies. Remedies for hair loss, baldness. Hair conditioners	665.637.4	Mixed-base products
665.585.3	Hair colorants and decolorants. Hair dyes, tints etc. Hair bleaches	665.637.5	Fractions of long residue
665.585.4	Hair dressings. Pomades. Brilliantines. Hair fixatives. Hair oils, creams etc.	665.637.6	Production of lubricating oil ⇒ 621.892 ⇒ 665.765
665.585.5	Hair cleansers. Hair shampoos	665.637.7	Production of paraffin wax
665.585.6	Hair lotions	665.637.8	Production of asphalt bitumen
665.585.7	Permanent-wave preparations	665.64	Cracking and reforming ⇒ 66.092 ⇒ 66.092.81
665.585.8	Hair removers. Depilatories	665.642	Thermal cracking and reforming processes
665.585.9	Products for skin care after shaving (after-shaves). Styptics ⇒ 665.584.223	665.644	Catalytic cracking and reforming processes ◇ 665.644.097.3 Catalysts for cracking and reforming ◇ 665.644.097.38 Catalyst regeneration
665.587	Aromatic fumigants. Incenses. Fumigant powders, papers. Perfumed candles, charcoal etc. ⇒ 665.14	665.644.2	Catalytic cracking processes
665.6/7	Processes and products of the petroleum and allied industries ⇒ 621.89 ⇒ 662.7	665.644.4	Catalytic reforming processes. □ Platforming. Cat-forming. Houdriforming. Hydroforming. Thermoform process
665.6	Mineral oil technology. Technology of petroleum and allied products ⇒ 662.753	665.65	Conversion processes other than cracking and reforming. Synthesis. Removal of unwanted constituents by high-temperature processes <i>Denote processes not listed under 665.65... by .09... (from 66.09...)</i> ◇ 665.65.095.81 Nitration
665.61	Crude oil ⇒ 622.276/279		

665.652	Processes yielding larger molecules (if specific to mineral oil)	665.7.033.2	Hydrocarbon compounds
665.652.2	Polymerization	665.7.033.22	Paraffins. Paraffinic products
665.652.4	Alkylation	665.7.033.23	Naphthenes. Naphthenic products
665.652.5	Condensation	665.7.033.25	Aromatic products
665.652.7	Synthesis from nonmineral oil sources. □ Fischer-Tropsch synthesis. Hydrogenation of carbon monoxide. Hydrogenation of solids	665.7.033.28	Asphaltic products
665.652.8	Reaction with various chemicals. □ Halogenation. Oxidation. Sulphurization	665.7.033.5	Products according to specific elements contained. □ Sulphur, nitrogen content
665.654	Degradation processes	665.7.033.8	Products according to heaviness or lightness. □ Molecular weight. Volatility. Melting range. Density
665.654.2	Hydrogenation splitting. Hydrocracking	665.7.035	Application properties
665.656	Processes which change the carbon skeleton structure without generally affecting molecular weight	665.7.035.2	Compatibility (e.g. solvent power, cloud point). Capillary and colloidal properties (e.g. surface tension)
665.656.2	Isomerization to branched chains (aliphatics)	665.7.035.3	Burning and ignition properties. □ Knocking. Soot formation. Flash point. Illuminating power. Calorific value
665.656.4	Cyclization	665.7.035.5	Chemical and physical stability. □ Oxidation stability. Storage stability. Heat stability. Colour stability
665.656.6	Aromatization	665.7.035.6	Rheological properties. □ Oiliness, flow, viscosity. Viscosity index. Penetration properties. Ring-and-ball
665.658	Processes mainly not affecting the carbon skeleton nor its dimensions	665.7.035.7	Optical, organoleptic and other sensory properties. □ Odour. Taste. Colour. Fluorescence. Toxicity
665.658.2	Hydrogenation. Hydrofining	665.7.035.8	Electrical properties. □ Dielectric, insulating, antistatic properties
665.658.4	Dehydrogenation	665.7.038	Auxiliary products. Additives which influence properties
665.658.6	Catalytic processes	665.7.038.1	Additives according to composition. □ Ash-free additives
665.66	Refining. Removal of unwanted constituents by low-temperature processes	665.7.038.2	Surface-active additives (surfactants). Dispersants. □ Detergents. Anti-ring-sticking additives. Emulsifiers. Foam or froth inhibitors
665.662	Physical refining processes	665.7.038.3	Knock-preventing additives. Octane-number improvers. Cetane-number improvers
665.662.2	Adsorption. □ With fuller's earth, bauxite, silica gel, bleaching earth	665.7.038.5	Antioxidants. Corrosion inhibitors. Wear-preventing additives. Metal passivators and deactivators
665.662.3	Selective extraction processes	665.7.038.62	Extreme pressure additives. Oiliness additives
665.662.5	Cooling, crystallization and subsequent working-up of the products ⇒ 665.637.7	665.7.038.64	Viscosity-index improvers (VI improvers). Pour-point depressants
665.663	Physicochemical refining processes	665.71	Petrochemicals (as a group) ⇒ 546/547
665.664	Chemical refining processes	665.72	Gases. □ Liquefied gases ⇒ 662.767 ⇒ 665.612
665.664.2	Refining using inorganic chemicals. □ Alkali washing. Sulphuric-acid refining. Refining with salts. Sweetening processes	665.73	Gasolines. Other products with boiling point in gasoline range. □ White spirit
665.664.3	Refining using organic chemicals. □ With amines. With phenoxides	665.733	Power gasolines
665.666	Refining processes according to the removed constituents	665.734	Gasolines with specific boiling points. Test gasolines
665.666.2	Removal of specific hydrocarbons	665.74	Kerosine. Cut-back material for bitumen
665.666.4	Removal of substances containing sulphur. □ Removal of thioalcohols, mercaptans	665.75	Gas oils. Diesel fuel. Fuel oils
665.666.5	Removal of other substances containing oxygen. □ Removal of naphthenic acids, naphthenates, phenols, cresols	665.752	Gas oils
665.666.6	Removal of substances containing nitrogen	665.753	Jet fuels. Diesel fuels
665.668	Refining according to the required effect. □ For bleaching, deodorizing, corrosion prevention	665.754	Furnace oils. Fuel oils
665.7	Products of the mineral oil industry. Mineral oils and similar products	665.76	Lubricating oils. Greases. Special fluids. Fractions other than hydrocarbons
665.7.03	Products according to origin and properties. Additives influencing properties	665.761	Oils, greases etc. according to manufacture
665.7.032	Products according to origin	665.761.2	Distillate oils
665.7.032.2	Unmodified natural products	665.761.4	Residual oils
665.7.032.5	Products derived by modification from natural products (excluding products from mineral oil) ⇒ 662.71/.74	665.761.6	Highly refined oils. □ White oil. Liquid paraffin
665.7.032.52	From wood ⇒ 662.71	665.765	Lubricating oils. Greases ⇒ 621.892 ⇒ 665.637.6
665.7.032.53	From peat ⇒ 662.73		
665.7.032.54	From lignite, brown coal		
665.7.032.56	From coal (hard coal) ⇒ 662.74		
665.7.032.57	From bituminous sand (athabasca) or shale ⇒ 665.775		
665.7.033	Products according to composition or specific constituents		

665.766	Nonhydrocarbon fractions. □ Phenolic products: cresols, creosote <i>Details by :547...</i>	665.944.2	Ester resins. Benzo-resins
665.767	Special-purpose fluids ◇ 665.767:620.197 Corrosion-inhibiting fluids	665.944.22	Benzoin
665.77	Solid and semisolid products	665.944.23	Balsam resins
665.772	Waxes. Paraffin wax ◇ 665.772.032.2 Earth wax. Ozokerite. Cerisine wax	665.944.24	Storax (styrax)
665.772.3	Microcrystalline waxes. Petrolatum. Petroleum jelly	665.944.26	Aloes. Acaroid resin
665.772.5	Macrocrystalline waxes	665.944.27	Dragon's blood. Other benzo-resins
665.775	Asphaltic bitumen	665.944.28	Gum resins from Persian-Afghan Umbelliferae. □ Asafoetida. Ammoniacum
665.775.4	Asphalt mixtures	665.944.3	Resene-containing resins. Burseraceous resins
665.775.5	Emulsions	665.944.32	Burseraceous gum resins. □ Myrrh
665.777	Carbonaceous products	665.944.33	True resins. True burseraceous resins in general. Elemi
665.777.2	Soot. Carbon black	665.944.34	Anacardiaceous resins
665.777.4	Petroleum coke. □ Electrode coke	665.944.35	Dipterocarpaceous resins
665.9	Miscellaneous organic chemical industries	665.944.36	Dammar resins
665.92	Wax polishes and cleaning preparations	665.944.4	Acid resins in general
665.921	Wax polishes. Similar solid or paste-like materials. □ Leather blacking ⇒ 646.65	665.944.5	Coniferous resins
665.922	Liquid wax polishes	665.944.52	Physiological coniferous resins. Primary exudation resins. □ Sandarac
665.926	Preparations for both cleansing and polishing	665.944.53	Pathological coniferous resins. Secondary exudation resins in general
665.93	Adhesives. Glues. Gelatin. Proteins ⇒ 637.6	665.944.54	Pine resins. Turpentine
665.931	Glues in general ⇒ 665.943	665.944.55	⇒ 665.948
665.931.7	Gelatin	665.944.57	Spruce resin
665.932	Hide glue. Skin glue	665.944.58	Recent fossil coniferous resins
665.933	Bone glue	665.944.6	Fossil coniferous resins. □ Amber
665.934	Extraction of glues and greases from animal wastes	665.944.61	Resins of Caesalpinioideae. Copals
665.935	Fish glue. Isinglass	665.944.62	True copals in general
665.936	Liquid glue	665.944.63	East African copals. Trachylobium copals
665.937	Bone processing in general	665.944.64	West African copals. Copaiba copals
665.938	Ivory black. Animal charcoal (bone black, bone charcoal) ⇒ 661.183.2 ⇒ 661.666.6	665.944.68	American copals
665.939	Other adhesives ⇒ 666.968	665.944.7	Copaiba balsams
665.939.1	Albumins. □ Ovalbumin (egg white, albumen)	665.944.72	Various resins of specific composition
665.939.2	Proteins. □ Casein	665.944.73	Resinol resins. □ Guaiac resin
665.939.3	Vegetable adhesives in general	665.944.74	Aliphatic resins. Lacs. □ Shellac. Thapsia resin
665.939.35	Algal products	665.944.75	Coloured resins. □ Gamboge
665.939.4	Various protein derivatives. Amino acids	665.944.76	Enzyme resins. □ Japan lac. China lac
665.939.5	Adhesives from synthetic resins and other plastics	665.944.77	Convolvulaceous resins
665.939.56	Adhesives made from synthetic polycondensation products 665.939.56 ≅ 678.6	665.944.9	Latex resins
665.939.57	Adhesives made from synthetic polymerization products 665.939.57 ≅ 678.7	665.947	Resins of uncertain constitution
665.939.6	Adhesives made from starch	665.947.1	Resin pitch. Turpentine products
665.94	Gums. Resins. Mucilages. Resinous plant products ⇒ 633.93/.94	665.947.2	Resin pitch
665.941	Gums in general	665.947.3	Colophony
665.941.1	Water-soluble gums. □ Gum arabic (gum acacia)	665.947.4	Resins from cellulose pulps. Liquid rosin. Tall oil ⇒ 676.085
665.941.2	Water-insoluble gums. □ Tragacanth. Bassora gum	665.947.6	Lignin
665.941.3	Partially water-soluble gums. □ Cherry gum	665.947.8	Pine tar
665.943	Birdlime. Fly-glue	665.948	Chemically modified natural resins
665.944	Resins. Gum resins. Oleoresins	665.948.1	Essential oils occurring in coniferous oleoresins ⇒ 665.944.54
		665.948.6	Oil of turpentine
		665.948.7	Wood turpentine oil
		666	Pine oil
		666.016	GLASS INDUSTRY. CERAMICS. CEMENT AND CONCRETE
		666.016.1	Compositions, formulae of mixtures
		666.016.2	Batch composition. Mixing unfired materials, raw materials. Compositions. Mixtures
		666.1	Composition of products. □ Cement proportions. Formulae for glass
			Glass industry. Glass technology. Glass manufacture in general. Glass in the comprehensive sense
			<i>The .03... and .05... auxiliaries are listed here under 666.1 but are applicable throughout 666.1/.28.</i>
			<i>The other .0... auxiliaries are listed at 66.0...</i>
			◇ 666.1.01 Glass engineering science

666.1.03	Processes and plant for melting and working glass	666.1.036.3	Glass forming by shaking
666.1.031	Glass melting	666.1.036.4	Drawing of flat, sheet glass
666.1.031.1	Glass melting in general. Processes during glass melting	◇ 666.1.036.4-71	Cooling equipment
666.1.031.11	Sintering of the batch	⇒ 666.1.038.3	
666.1.031.12	Charging. Charging installations and machines	⇒ 666.1.036.8	
666.1.031.13	Melting. Fritting	⇒ 666.15	
666.1.031.14	Colouring. Decolorizing	666.1.036.5	Drawing of glass as cylinders, tubes, bars or rods (from the bath or with blowing pipe). Nozzle drawing for production of glass silk. Production of hollow glass fibres
	⇒ 666.1.058		⇒ 666.173
	⇒ 666.124		⇒ 666.188
	⇒ 666.24		⇒ 666.189.21
666.1.031.15	Refining. Homogenization. Mixing, stirring, blocking	666.1.036.6	Glass forming by inflation, foaming (by physical or chemical means)
666.1.031.16	Thermal homogenization of the melt before forming	⇒ 666.189.3	
666.1.031.2/6	Glass-melting furnaces in general	666.1.036.7	Float processes for glass forming.
666.1.031.2	Tank furnaces for glass melting	□ Flotation on, or immersion in, a bath of molten material	
666.1.031.3	Pot and crucible furnaces for glass melting	⇒ 666.163	
666.1.031.4	Special glass-melting furnaces. □ Bushings for glass fibre manufacture	666.1.036.8	Machines and devices for drawing and forming of flat, sheet glass
666.1.031.5	Electric glass-melting furnaces	⇒ 666.1.036.4	
	<i>Details by :621.365...</i>	666.1.036.9	Equipment for forming glass whether or not in contact with a molten material
666.1.031.6	Other kinds of furnace and process.	666.1.037	Subsequent shaping of glass
	□ Laboratory furnaces. Vertical furnaces.	666.1.037.1	Rod-drawing process for glass fibre production
	Round furnaces	⇒ 666.189.2	
666.1.031.7	Ancillary equipment for feeding molten glass to forming machines. □ Libbey-Owens suction rotating bowl	666.1.037.2	Transformation of glass fibres to obtain stratified material
666.1.031.8	Ladling and measuring-out of molten glass.	666.1.037.3	Various methods of treatment. □ Sintering of glass powder. Leaching-out of components
666.1.032	Feeders for molten glass. Blow cases	666.1.037.4	Cracking off. Fusing. Melting off (e.g. of capillary tubes)
	Glass forming in general. Glass blowing in general. Glass blowing without compressed air in particular. Hot decorative shaping of hollow glass	⇒ 666.1.053.2	
666.1.032.3	Mouthblowing of glass (without compressed air)	666.1.037.5	Fusing of glass to metal
666.1.032.5	Generalities on moulds for injecting, pressing, casting or blowing glass, with or without use of compressed air or suction. Glass mould technology	666.1.037.6	666.1.037.5 ≡ 669
666.1.032.8	Surface decoration. Sticking. Introduction of bubbles. Sealing	666.1.037.7	Fusing of glass to nonmetals (e.g. ceramics)
666.1.032.82	Surface decoration of glass during forming		Mechanical production of glass ampoules, phials etc. □ Mechanical production of glass tablet bottles, test tubes, miniature light bulbs
666.1.032.83	Sticking of glass masses		⇒ 666.176
	⇒ 666.1.054.1	666.1.037.8	Lamp working of glass
666.1.032.84	Intentional introduction of bubbles into glass	⇒ 542.53	
666.1.032.85	Sealing, uniting of materials with glass during forming. □ Production of millefiore and glass with inclusions.	⇒ 666.174	
666.1.033	Glass blowing with compressed air	666.1.037.822	Lamp working using enamels
	<i>Denote characteristics of glass-blowing machinery by -2/-8 (from 62-2/-8)</i>	666.1.037.823	Production of glass figurines
	⇒ 666.17	⇒ 666.189.1	
666.1.034	Casting of glass	666.1.037.9	Other techniques for final shaping of glass.
	⇒ 666.164	□ Flaring. Bending	
666.1.035	Glass forming by injection, pressing, rolling, stamping (processes and plant)	666.1.038	Thermal treatment of glass. Annealing of glass. Strengthening by heat treatment or by crystallization. Reheating
666.1.035.1	Injection of glass	⇒ 66.04	
666.1.035.2	Pressing of glass	666.1.038.2	Lehrs. Annealing kilns
	⇒ 666.185	666.1.038.3	Strengthening of glass articles by rapid cooling.
666.1.035.3	Rolling of glass, in general		Thermal toughening of glass. Strengthening by controlled crystallization
	⇒ 621.771	666.1.038.4	Reheating furnaces. Toughening furnaces
	⇒ 666.162	666.1.038.5	Toughening of glass fibres during drawing
666.1.035.4	Continuous rolling of glass	666.1.038.6	Thermal coloration of glass. Striking of coloured glass. Development of colour by reheating (e.g. in ruby and yellow glasses by silver staining)
666.1.035.5	Rolling of ornamental, patterned glass		
666.1.035.7	Casting rollers	666.1.038.8	Devitrification processes
666.1.035.8	Forming of solid glass objects	666.1.039	Strengthening of glass products by modification of the structure or of the composition of the surface layers
666.1.036	Glass forming by drawing, centrifuging, pulverizing, foaming etc. (processes and plant)		⇒ 666.1.053.65
666.1.036.2	Glass drawing by centrifugal processes.		⇒ 666.112.8
	Drawing by gaseous entrainment		
	⇒ 666.189.2		

666.1.05	Subsequent working of glass or glass articles. Glass finishing in general	666.1.058.4	Transfers on glass. Printing and spraying of glass. Screen-printing of glass (e.g. silk-screen process) ⇒ 655.3.025 ⇒ 655.332 ⇒ 681.625.7
666.1.053	Mechanical or hand working of glass ⇒ 621.9	666.1.058.5	Firing on
666.1.053.2	Cutting off. Separating ⇒ 666.1.037.4	666.1.058.6	Mordant colours and lustres on glass. Yellow staining. Iridizing
666.1.053.3	Glass drilling, planing, turning, milling. Screw cutting of glass	666.1.058.7	Colouring of glass by filling connecting pores with pigment
666.1.053.4	Embossing and grooving of glass	666.1.058.8	Glass painting (with conventional colours). □ Eglomisé work (back-painting of glass). Cold painting and lacquering of glass
666.1.053.5	Grinding and polishing of glass (in general) ⇒ 621.92 ⇒ 666.224 ⇒ 666.285	666.11/.28	Kinds of glass. Glass articles. Glassware <i>The 666.1.0... auxiliaries are applicable throughout 666.11/.28</i> ⇒ 642.72
666.1.053.6	Engraving. Etching. Matt finishing	666.11	Glass. Glass articles in general ◇ 666.11.016.2 Composition of glasses ◇ 666.11.017 Properties of glasses
666.1.053.61	Mechanical engraving of glass. □ Diamond-point engraving. Scribing ⇒ 666.281	666.112	Glasses according to composition
666.1.053.62	Mechanical matt finishing of glass. Sandblasting of glass ⇒ 621.547 ⇒ 666.284	666.112.2	Alkali-silica glasses. Water glass ⇒ 661.683
666.1.053.63	Chemical finishing of glass. □ Matt etching of glass. Badging of glass by etching	666.112.3	Alkali-lime-silica glasses
666.1.053.65	Modification of the chemical structure or the composition of the surface layers. Production of antireflection coatings (nonglare glass). Improving the surface of glass ⇒ 666.1.039 ⇒ 666.1.056 ⇒ 666.266.4	666.112.4	Lead glasses ⇒ 666.21
666.1.053.8	Fragmentation of glass	666.112.5	Glasses containing iron and alumina
666.1.054	Joining of glasses or glass articles ⇒ 666.1.037.4 ⇒ 666.155 ⇒ 666.157 ⇒ 681.7.024	666.112.6	Barium glasses
666.1.054.1	Joining in the cold state. □ Cementing. Sticking ⇒ 665.93 ⇒ 666.1.032.83 ⇒ 666.968	666.112.7	Borosilicate glasses. □ Pyrex glass ⇒ 666.117.3/.4
666.1.054.2	Joining in the hot state. □ Welding. Soldering ⇒ 621.791 ⇒ 666.1.037.4	666.112.8	Glasses of almost pure silica. □ Vycor glass ⇒ 666.1.039 ⇒ 666.192
666.1.055	Coating of articles with glass layers. □ Spraying with molten glass. Dipping in molten glass. Vitrification of a coating ⇒ 666.29	666.112.93	Alkali-free glasses
666.1.056	Coating of glass with other substances. Metallizing of glass. Production of protective films on glass ◇ 666.1.056:546.284-31 Coating of glass with silica (silicon dioxide). Siliconizing of glass ◇ 666.1.056:669.228 Silvering of glass (in the strict sense) ⇒ 666.1.053.65 ⇒ 686.4 ⇒ 686.7	666.113	Glasses according to oxide components <i>Denote a specific oxide by colon combination with the number for the element from 546... and further oxides by replacing the digits 546 with an apostrophe; the numbers are cited in descending order</i> ◇ 666.113:546.62'47'41'33'32'28'27 Glass containing oxides of aluminium (Al_2O_3), calcium (CaO), sodium (Na_2O), potassium (K_2O), silicon (SiO_2), boron (B_2O_3) and zinc (ZnO)
666.1.056.2	Metallizing of glass by chemical precipitation	666.117	Glasses according to technical properties
666.1.056.3	Electrochemical metallizing of glass	666.117.1	Glasses with characteristic working properties. □ Short glasses. Long glasses
666.1.056.4	Physicochemical metallizing of glass	666.117.2	Glasses with particularly high or low melting temperature. Soft glasses
666.1.056.6	Metallizing of glass by physical methods	666.117.3	Glasses with particularly high resistance to thermal shock. Heat-resistant glasses. Oven glass ⇒ 666.112.7
666.1.056.8	Powder coating of glass	666.117.4	Glasses with particularly high chemical durability. Chemical-resistant glasses ⇒ 666.112.7
666.1.058	Staining and painting of glass. Other techniques of applying colour to glass ⇒ 666.1.031.14	666.119	Glass defects ⇒ 620.19
666.1.058.1	Pigments for glass staining. Glass colours. Black stain ⇒ 666.24 ⇒ 667.783 ⇒ 748	666.119.2	Defects in molten glass. Defects of body, of material. Inhomogeneities, colour defects etc.
		666.119.3	Defects in hollow and pressed glass
		666.119.4	Defects in flat glass
		666.119.6	Defects in optical glass. □ Short finish. Sleek Staining
		666.12	Raw materials for glass making and glass working (in general)
		666.122	Raw materials for the introduction of acid glass components

666.122.2	Silica-containing raw materials. □ Quartz. Sand	666.171	Glass bottles, carboys etc.
666.122.3	Alumina-containing raw materials. □ Clay. Feldspar		⇒ 621.642.17
666.122.4	Raw materials containing boron oxides. Borax		⇒ 621.798.147
666.122.5	Raw materials containing phosphorus oxides	666.171.9	⇒ 683.5
666.123	Raw materials for the introduction of basic glass components	666.171.92	Parts of glass bottles
666.123.2	Alkali-containing raw materials. □ Soda. Potash		Bottle necks
666.123.3	Alkaline-earth-containing raw materials. Natural rocks as raw materials, in general		◇ 666.171.92:621.88.082.6 Screw-threaded bottle necks (rounded thread)
666.123.4	Raw materials containing heavy metals (e.g. lead, zinc)	666.172	⇒ 666.285
666.124	Raw materials containing colouring, decolorizing, opalizing or refining components	666.173	White or coloured hollow glassware
	⇒ 666.1.031.14		Glass tubes
	⇒ 666.24		⇒ 666.1.036.5
666.125	Batch	666.174	⇒ 621.643.2
	◇ 666.125.016.1 Batch composition		Apparatus glass for lamp working. Glass apparatus (chemical, physical etc.) and other glass articles produced by lamp working
666.127	Glass cullet. Glass in blocks. Scrap. Cullet. Broken glass. Glass chips		⇒ 542.1/.2
666.128	Glass bars. Glass rods. Flashing knobs for casing or filigree work. Rods and beads for further working	666.174.2	⇒ 542.53
	⇒ 666.188		⇒ 666.1.037.8
	⇒ 666.25		Articles for chemical, physical and other purposes
	⇒ 666.27		◇ 666.174.2:536.51 Glass thermometers
		666.174.5	◇ 666.174.2:542.48 Glass equipment for distillation
666.15	Glass sheets and plates in general. Sheet glass. Flat glass in general		Sealing glasses (e.g. for joining glass to metal). Intermediate glasses for fusing different kinds of glass. Solder glass
	⇒ 666.1.036.4	666.175	Glass envelopes etc. for lighting fittings.
	⇒ 686.6		□ Glass globes and lampshades. Glass bulbs, e.g. for incandescent lamps (electric light bulbs). Glass tubes for fluorescent and discharge tubes
	⇒ 691.615		⇒ 621.3.032.7
666.151	Glass in sheets. Sheet glass in general		⇒ 628.95
666.152	Glass Cylinders	666.176	Glass ampoules, phials etc. □ Glass tablet bottles. Test tubes
666.153	Sheet glass with non-flat surfaces. Bent and convex glasses. Ribbed glass		⇒ 615.014.8
666.155	Laminated glass. Safety glass. Bulletproof laminated glass		⇒ 621.798.147
	⇒ 666.1.054		⇒ 666.1.037.7
	⇒ 666.166		
	⇒ 666.181	666.18	Other simple glass products
	⇒ 666.25	666.181	Toughened glass
666.157	Double and multiple glass sheets (with air or other layers). Double glazing. Multiple glazing. Multilayer insulating glasses	666.185	⇒ 666.155
	⇒ 666.1.054		Pressed glass. □ Pressed household glassware. Building, architectural glass. Glass bricks, tiles etc. Glass insulators
666.16	Rolled flat glass. Plate glass. Ornamental glass. Cast glass. Wired glass		⇒ 666.1.035.2
	⇒ 686.6		⇒ 666.17
666.161	Rough-cast glass	666.188	⇒ 691.618.5
666.162	Plate glass		Glass bars. Glass rods
	⇒ 666.1.035.3		⇒ 666.1.036.5
666.163	Float glass	666.189.1	⇒ 666.128
	⇒ 666.1.036.7		Shaped glass objects. □ Glass figurines. Christmas-tree decorations
666.164	Ornamental glass (cast fancy glass). Cast glass in general. □ Pressed and cut glass. Hammered glass. Fluted, ribbed and other patterned glasses	666.189.2	<i>For glass eyes see 617.7-089.28</i>
	⇒ 666.1.034		⇒ 666.1.037.823
	⇒ 666.153		Glass drawn into fibres and articles made from them
	⇒ 666.264		⇒ 666.1.036.2
666.166	Armoured glass. Reinforced glass. □ Wired glass. Fused glass with metallic inlay. Chauvel glass	666.189.21	⇒ 666.1.037.1
	⇒ 666.155		⇒ 677.52
	⇒ 666.266.3		Glass fibres and their products. Textile glass
666.17	Hollow glass in general (irrespective of production process). Blown hollow glass in particular	666.189.211	◇ 666.189.21:666.192 Quartz fibres
	⇒ 666.1.033	666.189.212	◇ 666.189.21:666.22 Optical fibres
	⇒ 666.185	666.189.213	⇒ 681.7.068
		666.189.23	⇒ 666.1.036.5
			Discontinuous glass fibres. Spun yarn. Glass wool
			Continuous glass fibres. Textile glass. Fibre glass yarns and woven fabrics
			Hollow glass fibres
			Glass cord, braid

666.189.24	Porous glass with interconnected pores	666.25	Flashed glasses. Cased glasses. Glasses coloured on the surface. Frosted, satin-finish, pearl, iridescent glasses. Hollow glass with interlayer, e.g. of plastics. Hollow laminated glass ⇒ 628.955 ⇒ 666.128 ⇒ 666.155
666.189.3	Porous glass with separated pores. Foam glass. Inflated glass ⇒ 666.1.036.6		
666.189.4	Powdered, pulverous glass. Glass dust. Glass granules. Spangle glass ⇒ 621.921 ⇒ 676.265		
666.19	Vitreous silica (fused silica). Fused basalt. Slags etc.	666.26	Light-scattering glasses in general. Various other glasses ⇒ 666.284 ⇒ 666.291
666.192	Vitreous silica (fused silica). Silica glass. Quartz. Rock crystal ⇒ 666.112.8 ⇒ 666.189.21	666.263	Opaque glasses. □ Opal glass. Opaline. Opalescent glass. Translucent glass. Milk glass. Marble glass ◇ 666.263:666.25 Flashed opal glass ⇒ 628.955 ⇒ 666.291.7
666.193	Fused basalt. Basalt wool		Coloured opal glass
666.195	Fused zirconia	666.263.4	Ornamented glass (reworked glass). Glasses decorated by fusing on of different kinds of glass ⇒ 666.164 ⇒ 748
666.198	Mineral fibres. Mineral wool	666.264	Luminescent glass. Fluorescent glass
666.21	Lead glass (lead crystal). Crystal glass. Light crystal. Heavy crystal ⇒ 666.112.4		Various glasses with particular characteristics
666.22	Optical glasses ⇒ 535.399 ⇒ 681.7.02/03	666.265	Antique glass
666.221.3	Optical crown glass	666.266	Electrically conductive glass (with conductive ingredient or with inlays). Glass that can be electrically heated ⇒ 666.166
666.221.4	Optical flint glass	666.266.1	Glass with metallic coating for other purposes. □ For reflectance. For nonreflectance ⇒ 666.1.053.65 ⇒ 686.7
666.221.6	Special glasses	666.266.3	Photosensitive and photoplastic glass. Glass with internal images formed by radiation and/or chemical or physical treatment
666.223	Optical glass blanks. □ Pressed lenses		Pieces of glass joined by lead or other suitable material. Glass mosaics. Glasses for mosaic production ⇒ 693.74 ⇒ 72.04 ⇒ 738.5 ⇒ 748
666.224	Ground optical glasses ⇒ 666.1.053.5		
666.227	Various optical glass products	666.266.4	
666.227.3	Spectacle glasses		
666.227.5	Contact lenses	666.266.5	
666.227.7	Optical glasses as parts of measurement and control equipment		
666.227.8	Optical glasses as part of image transmission equipment		
666.23	Strass (paste). Paste stones. Imitation gems of glass ⇒ 671.16 ⇒ 679.82	666.268	
666.231	Strass (paste). Imitation gems. Simulated precious stones		
666.233	Imitation diamonds of glass		
666.24	Transparent coloured glasses. Glasses coloured in the mass (coloured throughout). Stained glass. Glasses with selective absorption ⇒ 628.955 ⇒ 666.1.031.14 ⇒ 666.1.058.1 ⇒ 666.124 ⇒ 667.783 ⇒ 748	666.27	Glass beads. Small glass articles. Glass ornaments ⇒ 666.128 ⇒ 666.189.1
666.241	Red glasses. Ruby glasses. Orange glasses	666.28	Engraved glasses. Cut glasses. Matt glasses. Decoratively finished glasses in general
666.242	Yellow glasses. Blue glasses. Green glasses. Aventurines	666.281	Engraved glasses ⇒ 666.1.053.61
666.245	Violet glasses. Grey glasses. Black glasses	666.284	Matt glasses ⇒ 666.1.053.62 ⇒ 666.26
666.246	Glasses transmitting particular radiations	666.285	Ground or cut hollow glassware ⇒ 666.1.053.5
666.246.6	Glasses with variable transmission		
666.247	Glasses not transmitting particular radiations. Glass opaque to certain wavelengths ⇒ 614.893.2 ⇒ 681.7.066.6 ⇒ 681.738	666.29	Enamels. Glazes. Silicate colours. Engobes ⇒ 666.1.055 ⇒ 669.056.93
666.248	Glasses transmitting certain radiation in only one direction. One-way glass	666.29.022	Preparation of raw materials for enamels etc. <i>The 66.022... auxiliaries are not applicable at 666.29 and its subdivisions</i>
666.249	Special coloured glasses. □ Glasses with narrow absorption bands (e.g. didymium and neodymium glasses)	666.29.022.1	Pretreatment. Weighing
		666.29.022.2	Comminution of raw materials ⇒ 621.926
		666.29.022.4	Mixing of raw materials

666.29.022.5	Separating, screening, sieving of raw materials ⇒ 621.928	666.3-1	Ceramic materials and objects according to various characteristics <i>The 62-1... auxiliaries are not applicable at 666.3/.7</i>
666.29.022.7	Production and ageing of slips	666.3-12	Structure
666.29.022.71	Fineness of grinding. Regulation of slips	666.3-121	Very fine-grained ceramic materials
666.29.022.73	Ageing of slips	666.3-122	Fine-grained ceramic materials
666.29.031	Melting of enamels etc. Cooling the melts	666.3-123	Coarse-grained ceramic materials
666.29.041	Melting furnaces for enamels, glazes etc. <i>Subdivisions as at 66.041...</i>	666.3-127	Porous ceramic materials
666.29.042	Ancillary equipment for melting furnaces <i>Subdivisions as at 66.042...</i>	666.3-128	Dense ceramic materials. Ceramics of compact, close-grained structure
666.29.047	Drying. Drying ovens <i>Subdivisions as at 66.047...</i>	666.3-13	Resistance to physical, chemical and other influences
666.29.05	Processes and equipment for application of enamels, glazes etc.	666.3-131.9	Impervious ceramics
666.29.051	Processes and equipment for wet application. □ Dipping, pouring on, brushing, spraying of enamels etc. Edge-lining of pottery	666.3-134.1	Acid-resistant ceramics
666.29.052	Processes and equipment for application by powdering. □ Sifting. Sieves	666.3-134.2	Alkali-resistant ceramics
666.29.053	Processes and equipment for stoving and cooling ⇒ 666.29.041/.042	666.3-135	Heat-resistant ceramics. Ceramics resistant to high temperatures
666.29.054	Cleaning off of enamelled or glazed ware	666.3-16	High-strength ceramics
666.29.056	Decoration of enamelled or glazed ware. □ Painting, metallizing, gilding, printing of glazed ware	666.3-18	According to method of production and treatment
666.291	Enamel fluxes. Glazing fluxes. Colouring agents. Opacifiers	666.3-182	Hand-moulded ceramics
666.291.1	Raw materials <i>666.291.1 ≡ 666.12</i>	666.3-183	Cast ceramics
666.291.3	Enamel colours. Colouring agents, stains ⇒ 667.622	666.3-183.2	Slip-cast ceramics
666.291.5	Ceramic colours	666.3-183.4	Fusion-formed ceramics
666.291.7	Opacifiers and their action ⇒ 666.263	666.3-184	Pressed, moulded ceramics
666.291.9	Special glazing fluxes. Cold glazes	666.3-184.2	Extruded ceramics
666.293	Enamelling of metals (especially ferrous metals)	666.3-184.4	Compression-moulded ceramics
666.293.2	Effect of shape or form on enamelling	666.3-186	Dried, unfired ceramics
666.293.3	Cleaning of metal surfaces (sheet metal and castings)	666.3-187	Fired, sintered ceramics
666.293.35	Chemical pickling	666.3-188	Glazed ceramics
666.293.52	Enamels for sheet steel	666.3-4	State, condition, form of ceramic materials and objects <i>Subdivisions as at 62-4...</i>
666.293.521	Ground-coat enamels	666.3.015	Reactions. Behaviour of ceramic materials during working and treatment. □ Shrinkage
666.293.522	Cover-coat enamels	666.3.016	Composition of ceramic mixes, batches <i>Subdivisions as at 666.016...</i>
666.293.53	Enamels for cast iron <i>666.293.53 ≡ 666.293.52</i>	666.3.017	Technical properties of ceramics (physical and mechanical properties) and their testing
666.293.55	Matt enamels. Granite enamels. Other kinds of enamel	666.3.019	Material defects in ceramics <i>Specify by :620.19...</i>
666.293.57	Methods of preventing the enamelling of certain parts of the metal	666.3.022	Preparation of raw materials. Production of paste for ceramics <i>The auxiliaries listed under 66.022... are not applicable at 666.3/.7</i>
666.295	Ceramic glazes	666.3.022.1	Pretreatment of raw materials
666.295.2	Colourless glazes. Salt glazes	666.3.022.2	Comminution of raw materials. Pulverizing elements ⇒ 621.926
666.295.3	White glazes	666.3.022.4	Preparation of the clay
666.295.4	Coloured glazes	666.3.022.5	Separating, screening, sieving of raw materials ⇒ 621.928
666.295.5	Crystal glazes	666.3.022.6	Preparation and treatment of the paste and slip
666.295.6	Crackled glazes	666.3.03	Shaping of ceramics
666.295.7	Matt glazes	666.3.032	Shaping processes
666.295.8	Vapour glazes	666.3.032.2	Shaping by hand. Hand-moulding
666.297	Engobes ⇒ 666.3.052	666.3.032.4	Slip-casting
666.3/.7	Ceramics <i>The -1 auxiliaries listed under 666.3, and -2/-9 (from 62-1/-8 and 66-9) are applicable throughout 666.3/.7. The .0 auxiliaries listed under 66 are also applicable, with the exceptions and additions given under 666.3.0...</i>	666.3.032.5	Fusion-forming
666.3	Ceramics in general. Ceramic raw materials	666.3.032.6	Pressing. Moulding. □ Extrusion. Compression-moulding
		666.3.032.8	Ramming
		666.3.034	Finishing work before firing <i>Details by :621.9...</i>
		666.3.04	Heat-treatment operations. Thermal processes <i>Subdivisions as at 66.04...</i>
			◇ 666.3.041 Furnaces, kilns for ceramics
			◇ 666.3.042 Ancillary equipment for kilns
			◇ 666.3.046.4 Firing of ceramics
			◇ 666.3.047 Drying of ceramics. Drying kilns
		666.3.05	Slip coating. Glazing. Decoration

666.3.052	Coating. Engobing ⇒ 666.297	666.64	Earthenware (porous ceramic ware). Pottery. Clayware
666.3.053	Glazing of ceramics <i>Details by :666.295...</i>	666.641	Unglazed porous ware. Pottery. Clayware. Terracotta (unglazed)
666.3.056	Decoration of ceramics. □ Painting. Underglaze. On-glaze decoration ⇒ 686.4	666.642/647	Glazed porous ware
666.3.058	Printing, lithography on ceramics	666.642	Products with coloured body and opaque glaze
666.32/36	Ceramic raw materials	666.642.1	Faience
666.32	Clays for ceramics	666.642.3	Majolica
666.321	China clay. Kaolin	666.642.4	Delftware
666.322	Plastic clays. □ Bentonite	666.642.6	Glazed terra cotta
666.323	Clay shales and schist	666.643	Products with white or cream body and translucent glaze. Fine, white earthenware. □ Semiporcelain
666.324	Refractory clays	666.645	Sanitary earthenware ⇒ 666.596
666.325	Non-plastic clays (lean, short clays). □ Sandy clays. High-lime clays. Feldspathic clays	666.646	⇒ 666.632
666.327	Argillaceous earths. Marls. Clay-like earths. Fuller's earth	666.647	Earthen wall tiles
666.36	Non-plastic raw materials ⇒ 666.647	666.65	Special ceramic bodies for electrical use
666.362	Chamotte. Grog. Crushed bricks, broken saggars and other ceramic scrap	666.651	Insulators. Dielectrics ⇒ 621.315.612
666.363	Silica	666.652	Semiconducting ceramic bodies
666.364	Raw materials that lower the melting point. Fluxing agents <i>Details by :553...</i>	666.653	Ferromagnetic ceramic bodies
	◇ 666.364:553.55 Limestone. Chalk	666.654	Thermoelectric ceramic bodies
	◇ 666.364:553.613 Feldspar	666.655	Piezoelectric ceramic bodies
666.368	Accessory materials in the ceramics industry <i>Materials that do not form part of the body but are used as accessories in production</i>	666.658	Synthetic micas and micaceous minerals
666.5	Porcelain. China <i>Scope: ceramic ware with partly vitrified body, translucent in thin layers, white or almost white after firing</i> <i>The special auxiliaries listed under 666.3 are also applicable here</i>	666.7	Coarse ceramics. Heavy clayware. Refractories. Hard ceramic materials. Mixtures of ceramics and other materials <i>The special auxiliaries listed under 666.3 are also applicable here</i> ⇒ 691.4
	◇ 666.5(510-82) Chinese porcelain. China ware in the strict sense ⇒ 642.72 ⇒ 645.5 ⇒ 738	666.71	Solid bricks and blocks
666.51	Hard porcelain. □ Hard porcelain ware for household, hotel use. Ovenware	666.712	Building bricks
666.52	Soft-paste porcelains. Soft porcelain. □ Bone china	666.714	Facing bricks
666.59	Technical porcelains	666.715	Facing tiles
666.591	Laboratory porcelain	666.716	Radial bricks. Chimney bricks
666.592	Refractory porcelain	666.72	Hollow bricks and blocks
666.593	Electrical porcelain. □ Porcelain for insulators ⇒ 621.315.612	666.73	Drainage and sewer pipes
666.596	Porcelain for sanitary ware. Sanitary porcelain ⇒ 666.645	666.74	Roofing tiles
666.597	Porcelain for dental use. Dental porcelain	666.75	Quarry tiles. Flooring and paving tiles
666.6	Stoneware. Earthenware (porous ceramic ware) <i>The special auxiliaries listed under 666.3 are also applicable here</i>	666.76	Refractory bodies and products ⇒ 66.042.21 ⇒ 66.043.1 ⇒ 666.946.6
666.61	Stoneware	666.762	Refractory bodies and products according to composition
666.63	Technical stoneware	666.762.1	Refractories of alumina, aluminium silicate. Fireclay refractories
666.631	Laboratory stoneware. Stoneware for chemical use	666.762.11	Alumina. Corundum
666.632	Stoneware for sanitary use. Sanitary stoneware ⇒ 666.645	666.762.12	Bauxite. (High-alumina) diaspor. Gibbsite. Boehmite
666.635	Stoneware floor tiles	666.762.14	Mullite
666.638	Stoneware for electrical use ⇒ 621.315.612	666.762.15	Sillimanite. Kyanite. Andalusite
		666.762.16	Fireclay refractories
		666.762.2	Refractories of silica, quartzite, quartzite-and-clay
		666.762.3	Refractories of magnesite, forsterite, spinel, dolomite. Chrome-magnesite refractories
		666.762.4	Refractories of chromic oxide. Chromite refractories
		666.762.5	Refractories of zirconium dioxide. Zircon refractories
		666.762.6	Refractories of oxides of alkaline-earth metals. Refractories of magnesium, beryllium, thorium etc.

666.762.8	Refractories of carbon, graphite, carbides	666.9.015.82	Volume stability after hardening. □ Cracking.
666.765	Natural refractory building materials in general		Swelling, expansion
	<i>Specify by :553...</i>	666.9.015.83	Incompatibility of materials
	◇ 666.765:553.546 Quartzite	666.9.017	Technical properties of cements etc. (physical and mechanical properties) and their testing
	◇ 666.765:553.571 Sandstone		◇ 666.9.017:539.4 Strength. Resistance to deformation
666.767	Refractory mortars in general		◇ 666.9.017:620.173 Testing of compressive strength of cements etc.
666.77	Chemically resistant heavy clay bodies and products. Corrosion-resistant clayware	666.9.019	Material defects. Resistance to attack
666.772	Alkali-resistant clayware		<i>Specify by :620.19...</i>
666.774	Acid-resistant clayware	666.9.02	Processing in general. Raw materials and their preparation. Apparatus and equipment
666.79	Abrasives. Ceramic fuels. Mixtures of ceramics and other materials	666.9.022	Raw materials. □ Pretreatment, handling and preparation generally. Crushing and grinding. Compounding and mixing. Grading, cleaning etc.
666.792	Abrasives. □ Abrasive oxides, e.g. corundum. Abrasive carbides, e.g. silicon carbide (carborundum). Abrasive borides, nitrides		<i>The 66.022... auxiliaries are not applicable at 666.9 and its subdivisions</i>
	⇒ 621.921	666.9.022.1	Natural raw materials
	⇒ 676.265		<i>Details by :553.5... and :553.6...</i>
666.794	Ceramic fuels	666.9.022.2	Artificial raw materials. □ Slags
	⇒ 621.039.54	666.9.022.3	Preparation, crushing and grinding
666.798	Mixtures of ceramics and other materials		<i>Details by :621.926...</i>
666.798.2	Ceramic-metallic mixtures. Cermets		⇒ 666.94.052
	⇒ 669.018.9	666.9.022.4	Mixing and blending raw materials for burning in kilns
666.9	Gypsum, lime and cement industries. Hard-setting materials. Plasters and compositions. Mortar and concrete		<i>Details by :621.929...</i>
	⇒ 691.3	666.9.022.41	Dry processes
	⇒ 691.5	666.9.022.42	Wet processes
	⇒ 693	666.9.022.5	Grading, sizing and cleaning of aggregates, fillers etc. Sieves, separators etc.
666.9-1	Materials according to nature, state, properties. Bulk material (as distinct from shaped products)		<i>Details by :621.928...</i>
	<i>For shaped products use -4... (from 62-4...)</i>	666.9.022.6	Admixtures, additives for aggregates.
	⇒ 62-405.6		□ Pigments. Special cements
666.9-12	Materials according to structural characteristics, texture etc.	666.9.022.7	Water as raw material for preparation of mixes
666.9-121	Fine texture. Superfine. Smooth		⇒ 666.972.17
666.9-122	Medium texture. Sandy. Gritty	666.9.023/.028	Apparatus and equipment generally
666.9-123	Coarse texture. Rough		<i>Subdivisions as at 66.023/.028</i>
666.9-13	Impervious materials: resistant to physical, chemical or other influences		◇ 666.9.023.2 Reaction vessels
	666.9-13 ≡ 620.193		◇ 666.9.026.2 Piping
666.9-16	High-strength materials	666.9.03	Compounding of prepared raw materials. Processing and special treatment
666.9.01	Chemical engineering principles and practice. Plant and services in cement industry etc. Properties, defects and testing of materials in cement industry etc.		<i>The subdivisions are further developed at 666.97.03...</i>
	666.9.01 ≡ 66.01 with addition of .015...	666.9.031	Preparation of mix
666.9.015	Reactions. Behaviour of materials undergoing change	666.9.033	Processing of mix. □ Compaction. Moulding. Protection
	⇒ 666.9.03/.05	666.9.035	Treatment, additions, measures relating to setting and hardening. □ Addition of hydraulic constituents
666.9.015.2	Heating, burning, calcination reactions	666.9.038	Protective and other special measures during and after hardening
666.9.015.3	Behaviour of materials in application, working (laying, floating, rendering). Workability. Floatability etc.	666.9.04	Heat treatment. □ Burning. Calcination
			<i>Subdivisions as at 66.04...</i>
666.9.015.4	Setting reactions and changes generally	666.9.05	Handling and treatment of processed material
666.9.015.42	Reactions in mortar. □ Hydration. Hydrated compounds	666.9.051	Initial treatment of processed material. Cooling and storage of caustic lime, cement clinker etc.
666.9.015.43	Reciprocal effects between reactions and setting process	666.9.052	Further treatment of bulk material. □ Slaking of lime. Grinding of cement, clinker. Batch handling and delivery of ready-mixed concrete
666.9.015.5	Setting time. Setting rate. Setting medium. Defective setting	666.9.053	Colouring of bulk material
666.9.015.6	Factors influencing setting		<i>For colouring of formed products see 666.9.058</i>
666.9.015.64	Effects of chemical composition	666.9.055	Forming, fabricating, bonding etc. of hard-set products
666.9.015.66	Effects of additions, admixtures		<i>Details by :621.7...</i>
666.9.015.67	Effects of temperature. □ Effects of heat, cold, frost	666.9.056	Machining. Cutting. Tooling. Bush-hammering etc.
			<i>Details by :621.9...</i>
666.9.015.7	Behaviour of materials during hardening	666.9.058	Coating, dyeing, impregnating etc. Painting and other surface finishing
666.9.015.8	Behaviour of materials after setting and hardening		

666.9.059	Storage, handling and transport of finished products ⇒ 621.86/.87 ⇒ 658.78	666.946.1	Rapid-hardening cements. Quick-setting cements
666.91	Gypsum and anhydrite industry. Gypsum plasters and plaster products ⇒ 666.941 ⇒ 691.31 ⇒ 691.553 ⇒ 693.6	666.946.2	Expanding cements. Nonshrinking cements
666.913	Gypsum and anhydrite plasters (chemical type)	666.946.3	Cements resistant to various influences
666.914	Plaster products for building	666.946.4	Sand cement. Other lean cements
666.914.4	Constructional plaster products. □ Plasterboard. Plaster blocks	666.946.5	Masonry cement
666.914.5	Compositions for renderings and facings. □ Stucco materials ⇒ 666.96	666.946.6	Refractory cement ⇒ 666.76 ⇒ 666.948.4
666.92	Lime industry. Lime and lime products ⇒ 631.821 ⇒ 691.51 ⇒ 693.6	666.946.7	Injection cement. Grouting cement ⇒ 666.97.033.12/.14
666.923	Limes according to source	666.947	Fused cements ⇒ 666.948.3
666.924	Lime products (forms for building or other use). Caustic lime. □ Quicklime. Air-slaked lime. Hydrated lime. Slaked lime. Lime putty. Limewash (whitewash)	666.948	High-alumina cements
666.925	Hydraulic lime	666.948.2	Sintered high-alumina cement
666.94	Cement industry ⇒ 666.97 ⇒ 691.32 ⇒ 691.54	666.948.3	Fused high-alumina cement
666.94.052	Clinker, its crushing and grinding <i>Details by :621.926...</i> ⇒ 666.9.022.3	666.948.4	Refractory high-alumina cement
666.941	Weakly calcined cement (without sintering in kiln). □ Natural Roman cement (Parker's cement). Magnesia cement (Sorel cement). Dolomite cement. Gypsum cement	666.948.5	Sulphoaluminous cement
666.942	Portland cement	666.95	Pozzolanas. Substances with similar effect ⇒ 691.54
666.942.2	Ordinary Portland cement (natural)	666.951	Natural pozzolanas. Trass. Diatomaceous earths (Diatomite. Kieselguhr) ⇒ 553.53
666.942.3	Portland cements with special properties (artificial). □ Rapid-hardening, high-early-strength, low-heat, air-entraining and sulphate-resisting Portland cements	666.952	Artificial substances. Artificial pozzolanas. □ Blast-furnace slag. Furnace ash. Fly ash. Converted gaize. Brick dust
666.942.4	Iron-oxide-rich Portland cements.	666.953	Limed pozzolanas
666.942.5	Manganese-oxide-rich Portland cements	666.96	Hard-setting cement compositions, agglomerates and imitation stone products (for rendering, facing, bedding, filling, bonding etc.) <i>Denote aggregates by .022.32 (listed under 66) and the common auxiliaries of materials (Table 1k-03)</i> ◇ 666.96.022.32 Aggregates for hard-setting compositions ◇ 666.96.022.32-032 Mineral aggregates ◇ 666.96.022.32-032.7 Bituminous aggregates ◇ 666.96.022.32-035 Aggregates of organic origin (vegetable, animal) ◇ 666.96.022.32-035.3 Aggregates of wood, cork, sawdust etc. ⇒ 666.914.5 ⇒ 691.3/.5 ⇒ 692.53 ⇒ 693.7
666.942.6	Clay-rich, alumina-rich Portland cement ⇒ 666.948	666.961	Asbestos cement. Similar fibrous or felted cement compositions and products
666.942.7	Silica-rich Portland cement	666.962	Magnesium-based and similar compositions (for jointless floorings, etc.)
666.942.8	Chalk-rich Portland cement. Other limestone-rich Portland cements.	666.963	Terrazzo. Similar agglomerates with inorganic cement base (for floorings etc.)
666.943	Magnesia-rich Portland cement	666.964	Mastics. Agglomerates containing coarse or fine aggregates with bituminous binders
666.943.1	White and coloured Portland cements	666.965	Sand-lime products (calcium silicate products). Imitation sandstone. Other imitation stone products. □ Lightweight lime concrete.
666.943.2	Slag-containing cements. Blast-furnace and sulphated cements	666.966	Imitation stones of slag, clinker etc.
666.943.3	Portland blast-furnace cement (up to 20% blast-furnace slag)	666.968	Graphite-based compositions and pastes.
666.943.4	Blast-furnace cement (20-80% blast-furnace slag)	666.97	Plumbago compositions
666.944	Clinker slag cement (over 85% blast-furnace slag)		Bedding, bonding and jointing compounds or cements (for specific materials)
666.944.2	Sulphated cement. Supersulphated cement		Mortar and concrete industry. Concrete elements. Factory production of concrete ⇒ 624.012.3/.4 ⇒ 666.94 ⇒ 691.3 ⇒ 693.5
666.945	Pozzolanic cements. Trass cements ⇒ 691.54		
666.946	Artificial pozzolanic cements		
	Cements manufactured simultaneously with other materials. □ Basset cement.		
	Anhydrite-based cement		
	Cements with special properties		

666.97.03	Preparation of mix. Processing and treatment in works or factory	666.972.123	Coarse aggregate
666.97.031	Preparation of mix (cement, mortar, concrete mix). Batching and mixing in factory or central plant	666.972.124	Fine aggregate
666.97.031.2	Manual, hand mixing. Hand mixing equipment	666.972.125	Lightweight aggregate ⇒ 666.973
666.97.031.3	Machine mixing. Cement and concrete mixers (for batch mixing). Batch mixers	666.972.126	Special graded aggregates. □ Gap-graded aggregate
666.97.031.4	Continuous mixing. Continuous cement and concrete mixers	666.972.127	Aggregates for construction of nuclear reactors
666.97.031.5	Colloidal mixing. Colloidal mixers. Other grout mixers	666.972.16	Concrete admixtures. Various agents
666.97.031.6	Retempering, remixing after start of hardening	666.972.162	Water-repellent agents
666.97.031.9	Consistency, stiffness of concrete etc.	666.972.163	Surface-hardening agents
666.97.033	Processing of mix. □ Compaction. Moulding	666.972.167	Frost-protection agents
666.97.033.1	Compaction of concrete mix	666.972.168	Air-entraining agents
666.97.033.11	Ramming, tamping, rodding. Rammers. Tampers. Punners. Rammed concrete	666.972.17	Mixing water for concrete. Ratio of water to cement ⇒ 666.9.022.7
666.97.033.12	Pouring. Floating. Poured, floated concrete	666.972.2	No-slump concrete (moist-earth consistency)
666.97.033.13	Pumping. Pumped concrete	666.972.3	Low-slump concrete (plastic consistency)
666.97.033.14	Injection. Spraying. Grouting. Injected, sprayed or grouted concrete ⇒ 666.946.7	666.972.4	High-slump concrete (pourable, fluid consistency)
666.97.033.16	Vibration. Vibrators. Vibrated concrete	666.972.5	Concrete with special properties
666.97.033.17	Centrifuging, spinning. Spun concrete	666.972.52	Water-resistant concrete. □ Waterproof concrete
666.97.033.3	Special processing, compaction etc. for underwater use, hydraulic works. Underwater concrete	666.972.53	Frost-resistant concrete. □ Air-entrained concrete
666.97.033.4	Placing and supporting the reinforcement for concrete <i>For use with 666.98</i>	666.972.54	Low-heat concrete
666.97.035	Measures during setting and hardening of concrete	666.972.55	Wear-resistant concrete. Hard concrete
666.97.035.4	Vacuum treatment for water extraction. Vacuum concrete	666.972.7	Heavy concrete (concrete with heavy aggregate)
666.97.035.5	Curing of concrete. Artificial hardening of concrete	666.973	Lightweight concrete. Concrete with low unit weight
666.97.037	Striking of formwork. Removal of concrete objects from moulds. Time of removal	666.973.2	Lightweight concrete with special mineral and/or nonmineral aggregate. □ Sawdust concrete. Cork concrete. Nailable concrete. Breeze concrete
666.97.038	Protective treatment of concrete after hardening	666.973.3	Lightweight concrete with only coarse aggregate. No-fines concrete
666.97.05	Bulk handling and transport. Fabrication and finishing of formed products	666.973.4	Concrete with single-size, gap-graded lightweight aggregate
666.97.052	Batch handling. Delivery of ready-mixed concrete (at the site)	666.973.6	Aerated, airblown concrete. Cellular concrete. Insulating concrete
666.97.052.2	Plant-mixed concrete. Delivery in special containers or trucks without mixers	666.974	Special concretes
666.97.052.3	Truck-mixed concrete. Mixing in transit. Truck mixers	666.974.2	Refractory concrete. Fire-resistant concrete
666.97.053	Continuous delivery of ready-mixed concrete (at the site). □ Delivery of concrete through pipelines, by pump, with compressed air. Pneumatic concrete placers	666.974.4	Coloured concrete
666.97.055	Fabrication, bonding etc. of concrete	666.974.6	Concretes containing special cements. □ Concrete containing high-alumina cement
666.97.056	Machining. Cutting. Tooling. Bush-hammering etc.	666.98	Reinforced cement and concrete. Prestressed concrete <i>Denote the placing and supporting of the reinforcement by .033.4 (from 666.97.033.4)</i>
666.97.058	Coating, dyeing, impregnating etc. Painting and other surface finishing of concrete	666.981	Reinforced cement products
666.971	Mortars	666.982	Reinforced concrete. □ Prestressed concrete products, beams etc. ⇒ 624.012.45/.46 ⇒ 691.328 ⇒ 693.55/.56
666.971.3	Lime mortar (nonhydraulic mortar)	666.982.2	Reinforced concrete
666.971.4	Cement mortar (hydraulic mortar)	666.982.24	Concrete reinforcement. □ Steel rods, mesh
666.972	Concrete. Cement concrete	666.982.3	Pretensioned prestressed concrete
666.972.1	Concrete according to aggregate and admixture composition	666.982.4	Prestressed concrete
666.972.11	Aggregate/cement ratio	666.982.7	Wood concrete, bamboo concrete etc.
666.972.112	Lean concrete (high aggregate, low cement)	667	COLOUR INDUSTRIES. □ Manufacture and use of dyes, inks, paints etc. ⇒ 535.6 ⇒ 547.97 ⇒ 621.795.3 ⇒ 676.024.72 ⇒ 676.026.72 ⇒ 677.027 ⇒ 678.029.8
666.972.114	Fat concrete (low aggregate, high cement)		
666.972.12	Aggregates for concrete. □ Form (size, shape), weight of aggregate		
666.972.122	All-in aggregate (ballast, crusher-run aggregate). Ungraded aggregate		

667-12	Colours. Colour tones <i>Denote specific colours by -026.613...</i> <i>(Table 1k)</i> ◇ 667-12-026.613.1 Red colour group	667.287.6 667.289	Nitro and nitroso dyestuffs Other synthetic organic dyestuffs <i>Details by :547...</i>
667.2	Dyestuffs industry. Manufacture and use of dyestuffs ⇒ 677.016.42 ⇒ 677.027.4 ⇒ 677.047.4	667.4/.5 667.4	Inks. Writing materials Writing inks ⇒ 003.57
667.21	Auxiliary materials	667.41	Ordinary writing inks
667.214	Mordants	667.42	Black copying inks
667.215	Developing agents	667.43	Copying inks. Aniline inks
667.216	Brightening agents	667.44	Coloured inks
667.217	Fixing agents	667.45	Permanent inks (indelible inks)
667.22	Pretreatment of material to be dyed	667.46	Sympathetic inks (invisible inks)
667.24	Dyeing processes	667.47 667.488	Metallic inks. □ Gold ink. Silver ink Inks for ball-point pens
667.26	Inorganic dyestuffs	667.49	Other kinds of ink. □ Powdered inks. Ink tablets
667.27	Natural organic dyestuffs	667.5	Printing inks (black or coloured). Lithographic inks. Writing and drawing pencils. Coloured crayons
667.271	Dyestuffs of animal origin ⇒ 547.975 ⇒ 638.3	667.5.03 667.5.032	Colouring matter. Binders. Additives Colouring matter (dyes, pigments) for printing inks. Coloured printing inks according to pigment
667.272/.276	Dyestuffs of plant origin. Vegetable dyestuffs ⇒ 633.86	667.5.033	Binding agents for printing inks. Printing inks according to binders and vehicles
667.272	Dyestuffs from whole plants (e.g. litmus)	667.5.034 667.5.036	Additives for printing inks Moistening and washing agents for graphic use
667.273	Dyestuffs from the wood. □ Fustic. Catechu. Logwood	667.52	Printing inks (black or coloured) ◇ 667.52:681.624.4 Inks for rotary letterpress printing machines ⇒ 655.3
667.274	Dyestuffs from bark or roots (e.g. madder, turmeric)	667.521 667.521.1	Black printing inks Ordinary black printing inks. Carbon black (lampblack) ⇒ 661.666.4
667.275	Dyestuffs from leaves, flowers, fruit (e.g. indigo, woad) ⇒ 667.286.2	667.521.2 667.521.4	Fine black printing inks Carbon inks
667.276	Dyestuffs from twigs (e.g. gambier or catechu)	667.522 667.523 667.524	Coloured printing inks Aniline printing inks Planographic printing inks. □ Lithographic inks. Offset printing inks ⇒ 667.55
667.28	Synthetic organic dyestuffs. □ Aniline dyes	667.525	Intaglio printing inks. □ Inks for etching, engraving, gravure
667.281	Azo, azoxy and hydrazo dyestuffs	667.526	Printing inks for other purposes. Special printing inks. □ Stencil inks. Screen printing inks. Textile printing inks. Security printing inks (for printing of share certificates etc.) Luminescent printing inks ⇒ 628.9.03 ⇒ 677.027.5 ⇒ 677.047.5
667.281.1	Monoazo dyestuffs	667.528	Varnishes and lacquers for printed matter
667.281.2	Diazo dyestuffs	667.53	Inks for ink pads. Stamping inks (e.g. for rubber-stamping). Duplicating inks (e.g. hectographic inks). Ribbon inks (e.g. pigments for typewriter ribbons)
667.281.5	Polyazo dyestuffs	667.55	Lithographic inks ⇒ 667.524
667.281.6	Polyazo dyestuffs from diazotized amines by autocoupling. □ Bismarck brown. Chrysoid dyestuffs	667.56	Pencils. Crayons ⇒ 686.862.2 ⇒ 75.023
667.281.74	Stilbene dyestuffs	667.562 667.564 667.565	Pencils. Lead pencils (graphite pencils) Ink pencils. Aniline-dye chalks and pencils Coloured pencils. Crayons
667.281.8	Hydrazo dyestuffs		
667.282	Anthracene and anthraquinone dyestuffs		
667.282.1	Simple anthracene dyestuffs		
667.282.4	Anthracene dyestuffs with carbocyclic ring systems		
667.282.5	Anthracene dyestuffs with condensed heterocyclic ring systems		
667.282.7	Anthracene sulphur-containing dyestuffs		
667.283	Acridines. Diarylmethane dyestuffs. Triarylmethane dyestuffs. Phthaleins. Pyronines. Hydroxyketone dyestuffs		
667.283.1	Acridine dyestuffs		
667.283.3	Auramines. Diarylmethane dyestuffs		
667.283.4	Triarylmethane dyestuffs		
667.283.6	Hydroxyketone dyestuffs		
667.283.7	Phthaleins		
667.283.8	Pyronines. Xanthenes		
667.284	Azines. Oxazines. Thiazines. Indulines. Indophenols. Indamines		
667.284.6	Indophenols		
667.285	Sulphur-containing dyestuffs ⇒ 667.282.7		
667.286	Vat dyestuffs		
667.286.2	Indigo dyestuffs ⇒ 667.275		
667.287	Quinoline, cyanine, phthalocyanine, nitro and nitroso dyestuffs		
667.287.4	Quinoline dyestuffs. Cyanines		

667.6	Coatings and coating techniques. Paints. Varnishes. Lacquers <i>Subdivisions of 667.6 may be combined using the apostrophe to replace the digits 667.6 in the second and subsequent elements in the compound; the elements are cited in descending numeric order</i> ◇ 667.629.6'215.3 Anti-skinning agents for stand oil ◇ 667.661.23'443'332.4 Spray guns for spray application of oil paints	667.621.7	Other essential constituents of the binder system. □ Plasticizers ⇒ 66.063.72 ⇒ 667.629
		667.622	Colorants, pigments, dyes in coatings <i>Denote colours by -12...</i> ⇒ 66.022.33 ⇒ 666.291.3
		667.622.1	Inorganic pigments in coatings (except metals)
		667.622.11	Mineral pigments <i>Details by :661.8...</i> ◇ 667.622.11:661.847 Pigments of zinc compounds (e.g. zinc white) ◇ 667.622.11:661.852 Pigments of lead compounds (e.g. red lead, minium) ◇ 667.622.11:661.872 Pigments of iron compounds (e.g. iron oxides)
667.61	General matters. □ Properties. Tests. Defects		
667.612	Properties and testing of coating materials before or during application		
667.612.2	Particle size in coatings. Coarseness of grain ⇒ 539.215		
667.612.3	Behaviour of coating materials in storage. □ Separating and sedimentation. Thickening. Gelation. Redispersibility (response to agitation)	667.622.2	Metallic pigments <i>Details by :669...</i> ◇ 667.622.2:669.35 Copper alloys as pigments
667.612.6	Behaviour of coating materials during application. Film-forming properties. □ Workability. Brushability. Spreadability. Leafing	667.622.3	Organic pigments
667.612.8	Behaviour of coating materials while drying. Drying time	667.622.31	Natural organic pigments
667.613	Properties and testing of coatings after application and drying. Properties of applied coatings (including primers, undercoats) ◇ 667.613:535.685 Water resistance of coatings	667.622.32	Synthetic organic pigments
		667.622.33	Dyes for lake formation
667.613.2	Weathering of coatings. Behaviour under various weathering conditions ◇ 667.613.2:620.199 Accelerated, artificial weathering of coatings. Short-term testing of coatings ⇒ 535.686	667.622.4	Dyes soluble in the binder 667.622.4 ≡ 667.622.3
667.613.3	Behaviour of coatings towards liquids ◇ 667.613.3:547.294 Behaviour towards butyric acids	667.622.5	Fillers. Extenders. Dyestuff bases
667.613.4	Behaviour of coatings towards gases and vapours ◇ 667.613.4:546.171.1 Behaviour towards ammonia	667.629	Auxiliaries, additives to coatings ⇒ 667.621.7
667.613.5	Biological properties of coatings (and their investigation) ◇ 667.613.5:620.193.8 Behaviour towards biological action. Resistance to biological attack	667.629.2	Solvents. Thinners. Diluents ◇ 667.629.2.061.18 Solvents ◇ 667.629.2.063.93 Thinners. Diluents
667.613.7	Defects in coatings	667.629.3	Driers. Siccatives ⇒ 66.047.8
667.62	Constituents of coatings	667.629.38	Drying retarders
667.621	Vehicles and binders. Binders as coatings, without pigment	667.629.4	Stabilizers (e.g. agents to prevent settling) ⇒ 66.022.36
667.621.2	Aqueous, water-based binders	667.629.5	Emulsifiers. Wetting agents ⇒ 66.022.34 ⇒ 66.063.5 ⇒ 66.063.61 ⇒ 661.185
667.621.22	Inorganic aqueous binders. □ Lime. Water glass	667.629.6	Antiskinning agents
667.621.26	Organic aqueous binders. □ Proteins, e.g. albumin. Carbohydrates. Gums	667.629.8	Agents affecting surface finish. Agents producing special surface effects, textures
667.621.3	Asphalt. Bitumen. Tar	667.629.82	Agents for special gloss effects. □ For high gloss, satin finish, matt finish
667.621.4	Fatty oils and waxes as binders	667.629.83	Agents for producing uneven, irregular surfaces. □ For wrinkled effect. For frosted effect
667.621.42	Animal and vegetable oils as binders. Special products from oils	667.629.84	Agents for producing broken, cracked surfaces (crazed effect, craquelure)
667.621.44	Waxes as binders	667.629.88	Agents for reducing chalking
667.621.5	Fatty oils with modified composition, as binders. □ Boiled oils. Oxidized oils. Lightly sulphurized oils. Factice	667.629.9	Other auxiliary substances. □ Flow improvers. Poisonous, toxic agents (bactericides, fungicides etc.) ⇒ 667.637.4
667.621.53	Polymerized oils. □ Stand oil	667.63	Coatings: production and use
667.621.6	Macromolecular binders. □ Natural and synthetic resins as binders	667.632	Manufacture of coatings. □ Processes and plant <i>Details by :02/09 (from 66.02/09)</i> ◇ 667.632.022 Preparation of raw materials for coatings ◇ 667.632.022.5 Mechanical treatment. Comminution etc. ◇ 667.632.046 Thermal treatment. Heating, boiling. Varnish cooking
		667.633	Coatings according to composition <i>Denote the colour by -12.....</i>
		667.633.2	Coatings according to kind of binder 667.633.2 ≡ 667.621

- 667.633.22 Water-based coatings. □ Water paints. Whitewashes
- 667.633.24 Oil-based and wax-based coatings. Oil paints
- 667.633.26 Resin-based coatings. □ Lacquers. Varnishes
- 667.633.4 Coatings according to colorant, pigment, dye
- 667.633.41 Coatings with inorganic pigments (except metals)
Details by :661.8...
- 667.633.42 Coatings with metallic pigments
Details by :669...
- 667.633.6 Coatings according to auxiliaries, additives
Details by :667.629...
- 667.634 Coatings according to method of application
- 667.634.2 Coatings for brushing. Brushing paints and varnishes
- 667.634.3 Coatings for spraying. Spray paints and varnishes
- 667.634.4 Coatings for dip-coating. Dipping paints and varnishes
- 667.634.5 Coatings for dabbing, stippling etc. Stippling paints
- 667.634.6 Coatings for roller application. Roller paints and varnishes
- 667.634.7 Coatings for spatula application. □ Filler. Glazing putty. Sanding surfacer
- 667.634.8 Coatings for application by tumbling
- 667.635 Coatings according to method of drying
- 667.635.3 Coatings for drying at normal temperature. Quick-drying (rapid-drying) coatings
- 667.635.5 Coatings for drying at elevated temperature
- 667.636 Coatings according to substrate (material to be painted)
- 667.636.2 Coatings for inorganic materials. □ For metals and other minerals
⇒ 669.056.9
- 667.636.4 Coatings for organic materials. □ For wood and other fibrous materials. For rubbers and plastics
- 667.636.8 Coatings for covering existing coatings
- 667.637 Coatings according to properties and use
- 667.637.2 Coatings resistant to particular influences. Protective coatings
- 667.637.22 Coatings resistant to mechanical influences. □ Wear-resistant coatings
- 667.637.23 Coatings resistant to other physical and chemical influences
- 667.637.232 Coatings resistant to heat and cold
◇ 667.637.232:621.565 Coatings for refrigerating plant
- 667.637.233 Coatings resistant to water, moisture
- 667.637.234 Coatings resistant to smoke, flue gases
- 667.637.236 Coatings resistant to substances with dissolving and/or swelling action. □ Coatings resistant to hydrocarbons, alcohols, oils
- 667.637.237 Coatings resistant to acids or alkalis
- 667.637.25 Coatings resistant to atmospheric influences
- 667.637.27 Coatings resistant to rust, corrosion
- 667.637.4 Coatings for special purposes
◇ 667.637.4:536.522.3 Temperature-indicating paints
◇ 667.637.4:620.193.8 Coatings with biological action. Bactericidal, fungicidal paints
⇒ 667.629.9
◇ 667.637.4:621.315.617 Insulating lacquers
◇ 667.637.4:628.9.03 Luminescent paints (luminous paints)
◇ 667.637.4:699.812 Fire-resistant coatings
- 667.637.8 Coatings for production of particular surface effects
667.637.8 ≡ 667.629.8
- 667.638 Coatings according to function in the coating system
- 667.638.2 Primers. Priming paints
- 667.638.4 Undercoat paints. Intermediate coats. Fillers. Putties
- 667.638.6 Final coats (top coats). Finishing paints
- 667.64 **Coating technique, processes**
⇒ 698.12
- 667.644 Coating application technique
667.644 ≡ 667.634
- 667.644.3 Application by spraying. □ Spray-painting (paint spraying)
- 667.645 Drying of the coating
667.645 ≡ 667.635
- 667.645.3 Drying at normal temperature. Quick-drying (rapid-drying) methods
- 667.646 Coating technique according to substrate (material to be painted)
667.646 ≡ 667.636
- 667.647 Coating technique according to properties of coating
- 667.647.2 Coating according to binder
667.647.2 ≡ 667.621
- 667.647.22 Coating with water-based paints etc.
- 667.647.4 Coating according to colorant, pigment, dye
Details by :661.8...
- 667.648 Production of the individual coats
- 667.648.1 Preliminary work. □ Removal of old coats (paint stripping). Smoothing. Filling
- 667.648.2 Rust-proofing. Application of antirust undercoat
- 667.648.3 Priming
- 667.648.4 Application of undercoats, intermediate coats
- 667.648.6 Application of final coat (top coat)
- 667.648.8 Finishing work. □ Sanding. Polishing. Stoving
- 667.649.2 Application of foil (leaf). Application of transfers (decalcomania)
⇒ 669.058
⇒ 669.216
⇒ 676.263.4
- 667.649.4 Decorative, ornamental painting. Production of surface patterns, designs. □ By stencilling. By rolling-on
⇒ 744.34
- 667.649.6 Lettering in paint. Signwriting. Sign painting
⇒ 744.9
- 667.653 The finished coat. Coating layer
- 667.653.2 Finished coats according to binder
667.653.2 ≡ 667.621
- 667.653.24 Finished oil-based coat
- 667.653.4 Finished coats according to the colouring constituents
Details by :661.8...
- 667.657 Function of the coat
- 667.657.1 Colouring effect
- 667.657.2 Protective effect
- 667.657.3 Imitation of other materials
Specify by :...
◇ 667.657.3:674.031.929.3 Imitation of teak
- 667.657.4 Coats for special purposes
Denote the purpose by :...
- 667.657.8 Coats with special surface effect
667.657.8 ≡ 667.629.8
- 667.658 Individual coats, layers
667.658 ≡ 667.638
- 667.658.4 Undercoats. Intermediate layers
- 667.661 Tools, equipment for coating operations
- 667.661.1 Equipment for preparation and storage of coating materials
- 667.661.13 Grinding, milling devices. Paint mills. Mixing devices for paints etc.

667.661.17	Containers for coating materials. Paint and varnish containers etc. ◇ 667.661.17:621.642.17 Bottles for coating materials
667.661.2	Equipment for applying the coating. □ Paint brushes. Sprays 667.661.2 ≡ 667.634 ⇒ 687.978 ⇒ 75.022.2
667.661.23	Spray guns. Paint sprays
667.661.3	Devices for producing special surface effects, textures 667.661.3 ≡ 667.629.8
667.661.34	Devices for producing crazed effects, craquelure etc.
667.661.4	Equipment for producing patterns, designs, lettering. □ Stencils. Rollers ⇒ 667.649.4/.6
667.661.5	Equipment for application of foil (leaf) and transfers (decalcomania) ⇒ 669.058 ⇒ 669.216 ⇒ 676.263.4
667.661.6	Special equipment. □ Spraying booths. Turntables
667.691	Agents and methods for removal of coatings. Stripping agents. □ Paint removers. Paint strippers
667.7	Colouring, staining etc., of various materials (except printing and painting) ⇒ 677.027.4
667.74	Colouring of wood. Wood staining
667.741	By colour etching
667.742	Chemical wood etching. Strengthening of the grain. Grain colour development. □ Ageing with acids. Ageing with ammonia fumes (smoking)
667.743	Colouring by percolation (infiltration)
667.75	Colouring, staining of natural and artificial stone, cement etc.
667.76	Colouring, staining of metal ⇒ 621.785.7 ⇒ 669.056.9
667.761	By electrical means ⇒ 621.357.8
667.762	By chemical means ⇒ 621.794.5
667.763	By thermal means ⇒ 535.662.1
667.767	Eloxidation colouring (of aluminium)
667.77	Colouring of other materials
667.772	Colouring, staining of leather. Leather dyeing ⇒ 675.026.1 ⇒ 675.046.1
667.773	Colouring, staining of rubber and plastics
667.777	Colouring, dyeing of foodstuffs ⇒ 637.238.2 ⇒ 637.338.2 ⇒ 663.052
667.78	Colouring, staining of transparent materials
667.781	Colouring of liquids
667.783	Production of light filters. □ Use of gelatin films, foils, plates ⇒ 666.1.058.1 ⇒ 666.24

669	METALLURGY ⇒ 539.4 ⇒ 539.5 ⇒ 546.3/.9 ⇒ 620.18 ⇒ 621.7/.9 ⇒ 622 ⇒ 661.8
669-1	State of the metal or alloy, according to treatment <i>The 62-1... auxiliaries are not applicable at 669 and its subdivisions.</i> <i>For the state, condition or form of metallurgical products (and semifinished materials) use -4... (from 62-4...)</i>
669-11	Metals in natural state
669-122	Rolled metals ⇒ 669-177
669-124	Drawn metals
669-126	Extruded metals
669-13	Worked or wrought metals. Hot-formed metals
669-131	Worked or wrought metals generally
669-132	Beaten metals
669-133	Hammered metals
669-134	Forged metals
669-135	Pressed metals ⇒ 669-126
669-136	Stamped metals
669-138	Metals prepared by powder metallurgy. □ Green compacts. Sintered compacts ⇒ 621.762
669-14	Cast metals ⇒ 621.74
669-141	Sand-cast metals
669-142	Centrifugally cast metals
669-143	Die-cast, injection-moulded metals
669-144	Chill-cast metals
669-145	Precision-cast metals. □ Investment-moulded metals
669-146	Compound-cast metals
669-147	Continuously cast metals
669-15	Heat-treated metals <i>Details by :621.785...</i> ⇒ 669-97
669-151	Heated metals generally
669-151.6	Rate of heating
669-151.7	Soaked metals
669-151.8	Defectively heated
669-151.82	Burnt
669-151.83	Overheated
669-151.84	Decarburized
669-151.86	Modified in surface composition. □ Dezincked
669-153	Annealed metals
669-153.3	Black annealed
669-153.4	Bright annealed
669-153.6	Annealed by processes involving phase transformation
669-153.7	Annealed without phase transformation
669-154	Melted, molten metals
669-155	Case-carburized metals. Surface-hardened metals
669-155.1	Case-carburized without hardening
669-155.2	Case-hardened. Pack-hardened
669-155.3	Nitrided. Carbo-nitrided. Cyanided metals
669-155.4	Surface-hardened by localized heating. Flame-hardened ⇒ 621.785.54
669-156	Hardened, quenched metals

669-156.2	Hardened in air jet	669.014	Reactions involving metals
669-156.3	Hardened in water or aqueous solutions	669.014.6	Behaviour of the individual elements
669-156.4	Hardened in liquid air or similar medium	669.014.7	Reactions according to where they take place
669-156.5	Hardened in oil	669.014.72	Reactions in the combustion zone
669-156.6	Hardened in molten metals or salts	669.014.74	Reactions between the evolved gases and the ores and fluxes
669-156.7	Hardened in solid media. □ In sand	669.014.76	Reactions in the zone of formation of the metal
669-157	Tempered and aged metals	669.014.8	Irregularities in the reactions. □ Irregular working of furnace: too hot, too cold. Excess or shortage of ores. Explosions. Eruptions
669-157.2	Tempered metals	669.015	Kinds of product
669-157.8	Aged metals. Age-hardened metals. Precipitation-hardened metals		<i>Class individual products under 669.1/8</i>
669-157.9	Metals treated by combined processes	669.015.2	Intermediate products
669-158	Cooled metals	669.015.3	Crude metal products
669-158.8	Rate of cooling	669.015.4	Pure metal products
669-159.2	Low-temperature treated metals	669.015.5	Super-pure metal products
669-17	Metals structurally modified (chiefly crystal structure) by means other than heat treatment alone	669.015.7	Gases. Blast-furnace gas
669-172	Single crystal, monocrystal metals	669.015.8	Solid by-products ⇒ 66.046.58/59
669-174	Recrystallized metals	669.015.9	Losses
669-175	Metals with abnormal grain size	669.017	Physical metallurgy of individual metals
669-176	Metals with preferred orientation	669.017.1	Alloy systems ⇒ 669.055
669-177	Work-hardened metals. Strain-hardened metals ⇒ 669-122	669.017.11	Phase diagrams <i>For the iron-carbon phase diagram use 669.112.2.</i> <i>Use 669.055.017.11 for phase diagrams of other systems</i>
669-178	Metals electromagnetically modified, hardened etc.	669.017.12	Binary alloys
669-19	Metals characterized by other special features <i>Developed at 669.15-19... and 669.35-19...</i>	669.017.13	Ternary alloys
669-2/-9	Details of machinery, plant and products <i>The -2/-9 auxiliaries (from 62-1/-8 and 66-9) are applicable throughout 669</i>	669.017.14	Quaternary alloys
	◇ 669-2 Components of machines, apparatus, equipment	669.017.15	Alloys of higher complexity
	◇ 669-4 State, condition, form of metallurgical products (and semifinished materials) ⇒ 669-1	669.017.16	Structures of alloys ⇒ 539.2 ⇒ 548.7
	◇ 669-403 Gaseous state	669.017.3	Phase transformations
	◇ 669-404 Liquid state. Molten state	669.018	Metals and alloys according to their general properties ⇒ 669.055
	◇ 669-405 Solid state. Solidified state	669.018.2	Metals and alloys with special mechanical properties
	◇ 669-408 Surface condition of metal objects	669.018.23	Easily machinable. Free-machining metals and alloys ⇒ 669.018.26
	◇ 669-41 Flat metal objects (e.g. ingots, plate, sheet, strip)	669.018.24	Bearing metals and alloys
	◇ 669-42 Metal sections. Profiled metal objects	669.018.25	Exceptionally hard or wear-resistant metals and alloys (including hard metals) ⇒ 661.665.2 ⇒ 669.018.9
	◇ 669-462 Closed hollow sections of metal (e.g. metal pipes, tubes, hoses)	669.018.252	Metals and alloys for processing with chip formation ⇒ 621.9.025
	◇ 669-492 Metal powders, grains, shot	669.018.253	Metals and alloys for shearing. □ For blanking, punching, stamping out
	◇ 669-5 Operation and control of machines and processes	669.018.254	Metals and alloys for pressing, upsetting, cold-heading tools etc. ⇒ 621.7.073
	◇ 669-6 Fuel and other heat source characteristics of machinery and installations	669.018.255	Metals and alloys for drawing dies, cold rolls etc.
	◇ 669-8 Motive power or force characteristics of machines and apparatus	669.018.256	Wear-resisting metals and alloys. □ For rock-drilling, crushing, disintegrating tools
	◇ 669-97 Thermal characteristic data. Temperature. Temperature range ⇒ 669-15	669.018.258	Metals and alloys for hot working and die casting
	◇ 669-98 Pressure. Pressure range	669.018.26	Easily deformable metals and alloys. Deep drawing alloys. Soft alloys (e.g. soft steel) ⇒ 669.018.23 ⇒ 669.127.5
669.01/.09	Metallurgical properties, processes and equipment <i>669.01/.09 ≡ 66.01/.09</i> <i>.05 is further developed here</i>	669.018.27	Metals with high elasticity (e.g. metals for springs)
669.01	Metallurgy in general. □ Production and significance of metals ⇒ 620.18	669.018.28	Casting alloys (used principally in the cast state)
669.013	Metallurgical plant, works <i>Subdivisions as at 66.013...</i>		

669.018.29	Structural, constructional metals and alloys ⇒ 691.7	669.05	Metallurgical processes and resultant characteristics <i>Despite the notional appearance, the .051/.054 subdivisions are for use only as main numbers; they may not be detached and applied to the direct divisions of 669. The .05 auxiliaries are further developed in 669.2/.8 at the numbers for specific metals (especially .053 and .054.3)</i>
669.018.291	Ordinary (heavy) structural and constructional metals and alloys	669.051	Generalities on ores and metallurgy ⇒ 622.34
669.018.292	High-strength structural and constructional metals and alloys	669.052	Generalities on treatment of ore mixtures ⇒ 622.7
669.018.298	Metals and alloys for hardening by heat treatment	669.053	Extraction from ores. Extraction metallurgy <i>For subdivisions, see individual metals in 669.2/.8</i> ◇ 669.3.053 Extraction metallurgy of copper. Copper extraction
669.018.4	Metals and alloys with special thermal properties	669.053.2	Dry extraction of metals. Pyrometallurgy
669.018.41	Low-temperature metals and alloys	669.053.4	Wet extraction of metals. Hydrometallurgy
669.018.42	Metal and alloys with low melting point	669.054	Purification. Refining. Recovery ⇒ 669.04
669.018.44	Creep-resistant metals and alloys	669.054.1	Refining by chemical means
669.018.45	Metals and alloys with high melting point ⇒ 669.018.9	669.054.2	Refining by physical means
669.018.47	Metals and alloys with special expansion properties. □ Metals and alloys with low expansion, e.g. invar	669.054.3	Special refining processes for particular metals <i>669.054.3 ≡ 669.2/.894</i> <i>For subdivisions, see individual metals in 669.2/.8</i>
669.018.48	Non-strain-ageing metals and alloys	669.054.7	Electrolytic refining with the metal as anode
669.018.5	Metals and alloys with special electrical and magnetic properties	669.054.72	In molten salt baths
669.018.52	Metals and alloys with low electrical resistance ◇ 669.018.52:621.315.1 Metals and alloys for bare lines, overhead lines	669.054.76	In aqueous solutions
669.018.54	Metals and alloys with high electrical resistance	669.054.8	Recovery of metals from scrap, waste products or alloys
669.018.55	Metals and alloys with constant electrical resistance	669.055	Alloys <i>1. To denote alloys according to properties, add .018 and its subdivisions to the number for the alloy or a group of alloys, e.g. 669.14.018.841 Acid-resistant steels. [669.21/.23].055.018 Alloys of noble metals, according to properties. 669.3.055.018.24 Copper alloys as bearing metals. [669.85/.86].055.018 Alloys of rare metals, according to properties.</i>
669.018.57	Metals and alloys for thermocouple elements		<i>2. To denote alloys according to composition, class at the number for the main constituent, followed by .055; denote further constituents by colon combination with the subdivisions of 669.2/.8, in order of diminishing quantity (or in UDC order if preferred or if the ration of the constituents is unknown)</i>
669.018.58	Metals and alloys with special magnetic properties		◇ [669.21/.23].055 Alloys of noble metals
669.018.582	Metals and alloys for permanent magnets		◇ 669.21.055:669.22 Alloys of gold and silver
669.018.583	Metals and alloys for the magnetic circuit in electric machines		◇ 669.21.055 Gold alloys
669.018.584	Nonmagnetic metals and alloys		◇ 669.3.055:669.5:669.6 Alloys of copper, zinc and tin
669.018.6	Metals and alloys with other special physical properties. □ With special acoustic or optical properties		◇ 669.3.055 Copper alloys ⇒ 669.017.1 ⇒ 669.018
669.018.62	Weldable metals and alloys	669.056	Fabrication and finishing of metals and alloys <i>Denote details by :...</i>
669.018.672	Low-density metals and alloys		◇ 669.3.056:621.771.2 Rolling of copper
669.018.674	High-density metals and alloys		◇ 669.3.056 Finishing of copper
669.018.7	Metals and alloys with special chemical properties	669.056.9	Surface finish, coating, colouring etc. of specific metals and alloys ⇒ 667.6 ⇒ 667.636.2 ⇒ 667.76
669.018.72	Pyrophoric metals and alloys		669.056.91 Oxidized finish. Anodized finish
669.018.73	Metals and alloys with reducing properties (e.g. for production of hydrogen)		669.056.92 Dyed finish
669.018.74	Non-sparking metals and alloys		
669.018.8	Corrosion-resistant metals and alloys <i>669.018.8 ≡ 620.193</i>		
669.018.821	Metals and alloys resistant to climatic effects in general		
669.018.822	Metals and alloys resistant to effects of air, of oxygen		
669.018.841	Metals and alloys resistant to inorganic acids		
669.018.842	Metals and alloys resistant to inorganic alkalis		
669.018.85	Metals and alloys resistant to hot gases and vapours		
669.018.891	Metals and alloys resistant to ageing		
669.018.9	Pseudo-alloys and similar metallic combinations. □ Cermets ⇒ 666.798.2 ⇒ 669.018.25 ⇒ 669.018.45		
669.019	Material defects and their prevention <i>Specify by :620.19...</i>		
669.04	Heating, melting and refining processes and plant <i>Subdivisions as at 66.04...</i> ◇ 669.046 Thermal processes and plant in metallurgy. Heating and melting processes ◇ 669.049.4 Separation of solid and liquid metallic phases by partial fusion		

669.056.93	Ceramic finish. Vitreous enamelling ⇒ 666.29	669.112.21	Above liquidus
669.056.95	Rubber or plastics bonded finish ⇒ 678.066	669.112.22	Iron-cementite region (metastable system)
669.056.96	Paint finish. Lacquer finish	669.112.221	Beginning of solidification (liquidus)
669.056.99	Metal finishes (on metals and alloys). □ Cladding, plating with metal ⇒ 669.058	669.112.222	Delta range
669.057	Uses, applications of specific metals and alloys <i>Denote details by :...</i> ◇ 669.3.057:663.4 Applications of copper in brewing ◇ 669.3.057 Applications of copper	669.112.223	End of solidification (solidus)
669.058	Metals and/or alloys applied as finish (on metal or any other material). □ Metal coatings, platings, claddings <i>For metals as recipients of cladding or plating see 669.056.99</i> ⇒ 621.357 ⇒ 621.793 ⇒ 66.087 ⇒ 667.649.2 ⇒ 667.661.5 ⇒ 669.216 ⇒ 686.4	669.112.224	Eutectic (ledeburite)
669.058.4	Coatings of the metal and/or its alloys on nonmetallic materials ◇ 669.71.058.4:676 Aluminium coatings on paper and board ◇ 669.71.058.4 Aluminium coatings on nonmetals	669.112.227	Gamma range
669.058.5	Coatings of the metal and/or its alloys on the same metal or its alloys	669.112.227.1	Austenite
669.058.6	Coatings of the metal and/or its alloys on other metals or their alloys	669.112.227.3	Transformation of austenite
669.058.61	From vapours	669.112.227.31	Pre-eutectoid transformation
669.058.65	Immersion method. In a metal bath	669.112.227.32	Pearlite formation
669.058.66	With metal powder	669.112.227.33	Bainite formation
669.058.67	By spraying the metal	669.112.227.34	Martensite formation
669.058.68	Plating by cladding, sheathing, overlaying	669.112.228	Alpha range. □ Ferrite. Tertiary cementite
669.058.7	Electrolytic deposits of the metal and/or its alloys on other metals or their alloys	669.112.24	Iron-graphite region (stable system)
669.09	Other metallurgical processes <i>Subdivisions as at 66.09...</i> ◇ 669.094.1 Reduction of metals in general. Hydrogenation ◇ 669.094.22 Reduction using charcoal, coke and other substances with high carbon content ◇ 669.094.23 Reduction using metals, in general (e.g. with aluminium. Aluminothermy, thermite process)	669.112.241	Beginning of precipitation of graphite
669.1	Ferrous metallurgy. Iron and steel ◇ 669.1.017 Physical metallurgy of ferrous metals	669.112.243	Graphite-austenite eutectic
669.11	Iron and its alloys with carbon alone ⇒ 669.15	669.112.245	Precipitation of graphite from gamma solid solution
669.111	Condition of carbon in iron and steel ◇ 669.111:669.15 Effect of other alloying elements	669.112.247	Transformation from cementite to graphite
669.111.2	As elementary carbon (graphite)	669.12	Pure iron ⇒ 546.72 ⇒ 669.17
669.111.3	As carbide. Iron carbides. □ Cementite	669.127	Forms of pure iron
669.111.4	As dissolved carbon	669.127.3	Electrolytic iron
669.112	Form of iron and carbon in various types of iron and steel	669.127.4	Chemically produced iron. □ Iron carbonyl. Ferrum reductum
669.112.2	Iron-carbon system (iron-carbon phase diagram) ◇ 669.112.2:669.15 Effect of other alloying elements ⇒ 669.017.11	669.127.5	Soft iron. Other technically pure grades of iron ⇒ 669.018.26
		669.127.8	Special forms of pure iron. □ Pyrophoric iron. Colloidal iron. Iron in thin films
		669.13	Cast iron <i>Denote cast irons according to their properties by .018...</i> ⇒ 669.15-196 ⇒ 669.162.275.1
		669.131	Types of cast iron
		669.131.2	White cast iron
		669.131.4	Mottled cast iron
		669.131.6	Grey cast iron
		669.131.7	Spheroidal graphite cast iron
		669.131.8	Malleable cast iron
		669.132	Influence of nonmetals other than carbon
		669.136	Fabrication and finishing of cast iron
		669.136.9	Coatings of cast iron ⇒ 666.293
		669.138	Cast iron applied as coating
		669.14	Alloys of iron with carbon (other than cast iron). Steel in general <i>Denote steels according to their properties by .018...</i> ◇ 669.14.018.291 Structural steels. Concrete steels ◇ 669.14.018.298 Case-hardening steels, heat-treatable steels ◇ 669.14.018.41 Low-temperature steels ◇ 669.14.018.44 Creep-resistant steels ◇ 669.14.018.48 Non-strain-ageing steels ◇ 669.14.018.582 Permanent magnet steels ◇ 669.14.018.583 Dynamo steels. Transformer steels ◇ 669.14.018.584 Nonmagnetic steels ◇ 669.14.018.62 Weldable steels ◇ 669.14.018.8 Corrosion-resistant steels. Stainless steels ◇ 669.14.018.891 Non-ageing steels Steels according to method of production Steels produced by direct processes

669.141.2	Wrought iron made by the decarburization of pig iron. Steel made by the decarburization of pig iron or by remelting of scrap	669.162.212	Blast-furnace lining materials
669.141.22	Wrought iron	669.162.212.2	Stack lining
669.141.23	Damask steel	669.162.212.3	Dam stone. Tymp stone. Baffle stone
669.141.24	Mild steel	669.162.212.4	Bosh lining
669.141.243	Open-hearth steel (Siemens-Martin steel)	669.162.212.6	Hearth lining
669.141.243.3	Acid open hearth steel	669.162.213	Armouring of blast furnaces. Mantle ring. Supporting structure
669.141.243.4	Basic open hearth steel	669.162.214	Cooling of blast furnace
669.141.243.6	Tandem furnace steel	669.162.215	Furnace tops. Gas offtakes
669.141.244	Bessemer steel	669.162.215.2	Furnace tops
669.141.244.1	Acid Bessemer steel	669.162.215.4	Gas offtakes
669.141.244.2	Basic Bessemer steel	669.162.216	Tap-hole fittings. Mixtures for plugging tap-hole. Slag notch
669.141.245	Steel made by various pneumatic processes. □ Rotor process. Kaldo process	669.162.216.2	Tap-hole fittings
669.141.246	Crucible steel	669.162.216.4	Mixtures for plugging tap-hole
669.141.247	Electric steel (electric-furnace steel)	669.162.216.6	Slag notch
669.141.25	Steel for castings	669.162.218	Securing the wall and hearth against break-outs
	669.141.25 ≡ 669.141.24	669.162.22	Supply of blast to the furnace
669.141.3	Carburized steel. Cemented steel. Shear steel	669.162.221	Tuyeres. Tuyere connections
669.146	Fabrication and finishing of steel	669.162.222	Blast main. Blast-furnace gas main. Explosion valves
669.148	Steel (and alloy steel) applied as coating	669.162.223	Blowers
669.15	Alloys of iron with elements other than carbon. Alloy steel. Alloyed cast iron. Ferroalloys		⇒ 621.61
	<i>Denote alloying components by apostrophe combination, according to the instructions at 669.055</i>	669.162.224	Drying of blast. Oxygen enrichment of blast
	⇒ 669.11	669.162.225	Regulation of moisture in blast. Steam injection
669.15-192	Carbon-free ferrous alloys	669.162.227	Working at high or top pressure
	◇ 669.15'24'25-192 Carbon-free iron-nickel-cobalt alloys	669.162.228	Blast volume, pressure and consumption. Total cross-section at which blast is applied. Number of tuyeres and emergency tuyeres
669.15-194	Low-carbon ferrous alloys. Alloy steels	669.162.23	Blast heating. Hot-blast stoves
	◇ 669.15'24'26-194 Low-carbon alloys of steel, nickel and chromium.	669.162.231	Brick stoves with intermittent heating
	Chromium-nickel steel	669.162.232	Cast iron hot-blast stoves
	◇ 669.15'28-194 Molybdenum steel	669.162.233	Continuous-pipe stoves
669.15-194.2	Low-alloy steels	669.162.236	Equalizers
669.15-194.3	High-alloy steels	669.162.238	Operation of hot-blast stoves. Heat balance
669.15-194.5	Steels according to structure	669.162.24	Charging of blast furnaces. Furnace hoists
669.15-194.52	Hypoeutectoid. Hypoppearlitic	669.162.242	Charging by means of inclined hoists, skip bridges, bucket elevators
669.15-194.53	Eutectoid. Pearlitic	669.162.244	Furnace hoists
669.15-194.54	Hypereutectoid. Hyperpearlitic	669.162.25	Cleaning and utilization of blast-furnace gas
669.15-194.55	Martensitic		⇒ 669.162.275.3
669.15-194.56	Austenitic	669.162.251	Utilization of blast-furnace gas in recuperators
669.15-194.57	Ferritic	669.162.252	Gas cleaning. Gas cleaners
669.15-194.58	Ledeburitic		⇒ 621.928.9
669.15-194.591	Bainitic		⇒ 66.074
669.15-196	High-carbon ferrous alloys. Alloyed cast iron	669.162.26	Operation of blast furnaces
	◇ 669.15'24'25-196 Nickel-cobalt cast iron	669.162.261	Putting into operation
	⇒ 669.13	669.162.261.2	Drying of lining
669.15-198	Ferroalloys	669.162.261.3	Charging. Lighting. Blowing-in
	◇ 669.15'74-198 Ferromanganese	669.162.262	Calculation of charge and burden
669.16	Production of pig iron and ferroalloys	669.162.262.3	Burdening
	◇ 669.16.013 Pig iron plant	669.162.262.5	Calculation of the burden. □ Slag rates
	⇒ 622.79	669.162.262.6	Calculation of additions
669.162	Production of pig iron. First smelting	669.162.263	Reactions and processes in the blast furnace
669.162.1	Raw materials and additions. Their preparation for use in blast furnace. Burden	669.162.263.2	Course of the reactions
	<i>Class only general matters here</i>	669.162.263.4	Reactions in the blast furnace
	⇒ 622.78		⇒ 669.014
669.162.12	Iron ores	669.162.264	Irregularities in blast-furnace-operation and their prevention
669.162.14	Additions	669.162.265	Acid smelting in the blast furnace
669.162.16	Coal and coke for pig iron production	669.162.266	Tapping of the blast furnace
669.162.2	Blast furnaces. Operation of blast furnaces	669.162.266.2	Tapping of pig iron
	⇒ 66.041.44	669.162.266.4	Tapping of slag. Transport of slag
669.162.21	Generalities on blast furnaces	669.162.267.2	Changeover to other grades of pig iron
669.162.211	Dimensions of blast furnaces. Profiles	669.162.267.3	Reduction of output. Damping down. Blowing out
		669.162.267.4	Operation with various fuels. Injection of solid, liquid or gaseous fuels
		669.162.267.6	Desulphurization of pig iron
		669.162.267.7	Desiliconization of pig iron

669.162.267.8	Elimination or recovery of elements other than sulphur or silicon from pig iron <i>Specify elements by :669...</i> ◇ 669.162.267.8:669.292 Recovery of vanadium	669.182.71	Refining
669.162.27	Economy and output of blast furnaces <i>Details by .012...</i>	669.182.73	Puddling
669.162.275	Composition of blast-furnace products	669.183	Steel produced in reverberatory furnaces.
669.162.275.1	Pig iron. Iron cast in the first melt ⇒ 669.13		Open-hearth steel (Siemens-Martin steel)
669.162.275.2	Blast-furnace slags	669.183.2	Open-hearth processes
669.162.275.3	Blast-furnace gas ⇒ 669.162.25	669.183.21	Open-hearth furnaces
669.162.275.4	Flue dust	669.183.211	Generalities. Structural details <i>Denote fuels, heat sources by -6... (from 62-6...)</i>
669.162.275.8	Blast-furnace by-products. □ Recovery of zinc	669.183.211.1	Open-hearth furnaces. Modified open-hearth furnaces
669.162.28	Studies of the blast-furnace process	669.183.211.2	Hearths
669.162.281	Theories of the blast furnace	669.183.211.3	Furnace walls. Crown. Roof
669.162.282	Reduction experiments	669.183.211.4	Charging holes. Furnace doors
669.162.283	Study of blast furnace reactions	669.183.211.5	Tap-hole. Tapping spout. Pouring hole
669.162.3	Charcoal blast furnaces 669.162.3 ≡ 669.162.2	669.183.212	Blocks. Burners. Gas and air uptakes
669.162.4	Electric blast furnaces 669.162.4 ≡ 669.162.2	669.183.212.2	Blocks for gas combustion
669.162.6	Production of synthetic pig iron	669.183.212.3	Burners for fuel oil
669.162.8	Other methods of production of pig iron	669.183.212.4	Gas uptakes
669.163	Production and utilization of iron cast in the first melt. □ Blast-furnace castings. Pigs. Ingots. Tubes ⇒ 621.74	669.183.212.5	Air uptakes
669.168	Production of ferroalloys. □ Production in blast furnace, in electric furnace, by aluminothermy ⇒ 669.719	669.183.212.6	Roof lances for oxygen injection, for decarburization etc.
669.17	Production of pure iron ⇒ 546.72 ⇒ 669.12	669.183.213	Regenerators
669.173	Production of pure iron by chemical means. □ By reduction of iron oxide or iron carbonyl	669.183.213.1	Design, arrangement of regenerators
669.177	Electrolytic production of pure iron	669.183.213.2	Checkers. Checker grids. Checker bricks
669.177.035	Components of electrolytic production plant 669.177.035 ≡ 621.3.035	669.183.213.3	Sizes of gas chambers and air chambers. Ratio of sizes
669.177.035.4	Electrolytes, baths for electrolytic production	669.183.213.4	Slagging of checkers. Slag chambers. Slag pockets
669.177.035.45	Melts	669.183.213.8	Reaction in regenerators. Cleaning of regenerators
669.179	Other methods of production of high-purity iron	669.183.214	Gas ports. Air ports. Valves
669.18	Production of steel ◇ 669.18.013 Steel production plant. Steelworks	669.183.214.2	Gas ports
669.181	Direct processes for production of wrought iron, sponge iron and steel	669.183.214.3	Air ports
669.181.2	Direct processes for production of wrought iron and steel. □ Catalan process	669.183.214.6	Valves (reversing valves)
669.181.22	Direct processes in pits	669.183.214.62	Gas valves
669.181.23	Direct processes in hearths	669.183.214.64	Air valves
669.181.24	Direct processes in shaft furnaces	669.183.215	Furnace anchoring. Chamber anchoring
669.181.25	Direct processes in rotary furnaces, crucibles, muffles, retorts	669.183.217	Regenerative firing, recuperative firing in open hearth furnaces
669.181.26	Direct methods in reverberatory furnaces	669.183.217.2	Regenerative firing
669.181.27	Direct methods in electric furnaces	669.183.217.4	Recuperative firing
669.181.28	Slags from direct processes	669.183.218	Operation of open-hearth furnaces. Melting process
669.181.4	Production of sponge iron	669.183.218.5	Melting process (behaviour of elements)
669.181.42	Treatment of ores with reducing agents but without flux	669.183.22	Refining with ore
669.181.44	Treatment of ores with reducing agents and fluxes	669.183.23	Scrap process
669.181.46	Treatment of ores with salts and fluxes	669.183.24	Combined processes
669.182	Production of steel by cementation. Indirect processes for production of wrought iron	669.183.28	Open-hearth slags
669.182.3	Production of cemented steel, shear steel, faggotted steel, damask steel	669.183.3	Acid open-hearth process 669.183.3 ≡ 669.183.2
669.182.7	Indirect production of wrought iron	669.183.4	Basic open-hearth process 669.183.4 ≡ 669.183.2
		669.183.5	Multiplex processes. Duplex processes
		669.183.52	Converting as primary process
		669.183.54	Open-hearth as primary process
		669.183.57	Other duplex processes
		669.183.58	Triplex processes.
		669.184	Converter/open-hearth/electric furnace
			Converter process. Principles of converting ⇒ 66.041.498
		669.184.1	Acid Bessemer processes
		669.184.12	Acid converters
		669.184.121	Dimensions. Volume. Shape
		669.184.122	Parts of converters. □ Casing, nose, body, bottom. Converter trunnion rings. Blast boxes.
			Tuyere blocks
		669.184.123	Suspension of converters. Tipping arrangements
		669.184.124	Converter bottoms. Materials of bottom
		669.184.125	Lining of converter walls

669.184.126	Arrangement of converter plants. Converter chimneys	669.21	Gold
669.184.13	Method of operation. Reactions in acid converter	669.213	Extraction of gold
669.184.132	Method of operation	669.213.1	Extraction of auriferous sands and ores by comminution and washing
669.184.132.1	Normal, too hot and overblown melt	669.213.2	Extraction of gold by mercury.
669.184.132.3	Cooling by addition of scrap or ore	669.213.3	□ Amalgamation. Distillation
669.184.135	Reactions in acid converter	669.213.4	Wet extraction of auriferous sulphides.
669.184.136	Control of process by study of flames	669.213.5	Roasting smelting. Second roasting. Melting in the presence of lead. Cupellation
669.184.14	Blast supply to converters	669.213.6	Extraction of gold by chlorination.
669.184.141	Blast consumption. Blast pressure	669.213.7	□ Chlorination plant. Screens. Filters.
669.184.142	Blast cross-section. Nozzle diameter of tuyere. Number of tuyeres	669.214	Chlorination towers
669.184.143	Drying of blast. Heating of blast	669.215	Extraction of gold by bromine
669.184.144	Composition of blast (air blast, other gases, etc.). Direction of blast	669.216	Extraction of gold by leaching with cyanide
669.184.15	Waste gases and fumes from converter processes (and similar processes)	669.217	Electrolysis of cyanide solution
669.184.16	Losses in converter process. □ Oxidation loss. Ejection. Converter dust (flue dust, roof dust)	669.218	Separation of gold from accompanying metals. Gold refining
669.184.18	Acid Bessemer slags	669.219	Gold alloys
669.184.2	Basic Bessemer process (Thomas process) 669.184.2/26 ≡ 669.184.1/16	669.220	◇ 669.215'3 Alloys of gold and copper ⇒ 669.791.5
669.184.28	Basic Bessemer slags (Thomas slags) ⇒ 631.853	669.221	Fabrication and finishing of gold. □ Production of gold leaf, foil. Production of gold sheet ⇒ 667.649.2 ⇒ 667.661.5
669.184.288	Recovery of various components ◇ 669.184.288:669.292.054.8 Recovery of vanadium	669.222	Applications of gold ⇒ 671.11 ⇒ 737.1
669.184.4	Mixers. Pig iron mixers	669.223	Gold applied as coating or plating. Gilding.
669.186	Production of crucible steel (cast steel)	669.223.1	Gold-plating
669.186.2	Crucibles. Crucible materials ◇ 669.186.2:666.762 Crucible manufacture (according to composition) ◇ 669.186.2:666.762.1 Crucibles of alumina or fireclay	669.223.2	Silver
669.186.3	Crucible furnaces	669.223.3	Extraction of silver
669.186.4	Charges for production of crucible steel	669.223.4	Extraction of silver by lead ⇒ 669.443.4
669.186.5	Working methods. Operation. Reaction	669.224	Extraction of silver with mercury.
669.186.52	Furnace equipment. Charging	669.225	□ Amalgamation. Distillation ⇒ 669.791.5
669.186.55	Reactions in the crucible ⇒ 66.046.5	669.226	Extraction of silver by solution and precipitation
669.186.58	Crucible slags	669.227	Refining of silver
669.187	Production of electric steel (electric-furnace steel)	669.228	Silver alloys
669.187.2	Electric steel furnaces. Electric process ⇒ 621.365	669.229	Fabrication of silver ⇒ 739.1
669.187.2.036	Individual parts of electric steel furnaces 669.187.2.036 ≡ 621.3.036	669.230	Applications of silver ⇒ 671.11 ⇒ 737.1
669.187.23	Furnace bottoms and lining	669.231	Silver applied as coating or plating. Silvering.
669.187.24	Furnace roofs and linings. Lances for oxygen lancing	669.231.6	Silver-plating ⇒ 666.1.056
669.187.25	Melting processes ⇒ 669.046	669.231.7	Metals of the platinum group
669.187.28	Electric steel slags	669.232	Platinum
669.187.3	Acid electric steel furnaces. Acid electric process 669.187.3 ≡ 669.187.2	669.233	Fabrication and finishing of platinum.
669.187.4	Basic electric steel furnaces. Basic electric process 669.187.4 ≡ 669.187.2	669.234	□ Platinum sponge. Platinum black
669.187.5	Electric remelting processes and furnaces	669.235	Applications of platinum
669.187.52	Vacuum remelting processes	669.236	Iridium
669.187.56	Electro-slag refining	669.237	Osmium
669.187.58	Plasma remelting	669.238	Palladium
669.2/8	Nonferrous metallurgy. Nonferrous metals	669.239	Rhodium
669.2	Nonferrous metals in general	669.240	Ruthenium
669.21/23	Noble metals ⇒ 671 ⇒ 739.1	669.241/29	Various metals of the 4th, 5th, 6th and 8th groups of the periodic table
		669.242	Nickel
		669.243	Extraction of nickel
		669.243.1	Enrichment smelting of nickel in general
		669.243.2	Treatment of sulphide matte
		669.243.3	Treatment of Swedish nickel and cobalt ores ⇒ 669.253.2
		669.243.4	Treatment of arsenical ores
		669.243.5	Treatment of New Caledonian ores. Garnier process
		669.243.6	Treatment of Canadian ores, Sudbury ores, others

669.243.7	Extraction of nickel as nickel carbonyl. □ Langer process. Mond process	669.347	Electrolytic refining
669.243.82	Wet methods in general. □ Leaching. Precipitation	669.35	Copper alloys
669.243.87	Electrolysis of crude nickel	◇ 669.35'24	Alloys of copper and nickel. Cupro-nickel
669.243.88	Electrothermal production of nickel. Ferronickel production	◇ 669.35'5	Alloys of copper and zinc. Brass. Tombac
669.244	Refining of nickel. Production of high-purity nickel	◇ 669.35'5'6	Alloys of copper, zinc and tin. Gunmetal
669.245	Nickel alloys	◇ 669.35'6	Alloys of copper and tin. Bronzes
669.248	Nickel applied as coating etc. Nickel plating	669.35-191	Alpha structure ◇ 669.35'6-191 Alpha bronze
669.25	Cobalt	669.35-192	Beta structure ◇ 669.35'6-192 Beta bronze
669.253	Extraction of cobalt	669.35-193	Gamma structure ◇ 669.35'5-193 Gamma brass
669.253.1	Direct production		
669.253.2	Extraction of cobalt during nickel production ⇒ 669.243.3	669.369	Surface finish, coating, colouring of copper
669.255	Cobalt alloys	669.37	Applications of copper ◇ 669.37:621.315.55 Copper conductors
669.26	Chromium	669.38	Copper and/or its alloys applied as coating etc. Copper plating
669.264	Refining of chromium. Production of high-purity chromium	669.384	Copper coating of nonmetallic materials
669.265	Chromium alloys	669.386	Nonelectrolytic copper coating and plating (of metals other than copper itself)
669.268	Chromium applied as coating etc. Chromium plating	669.387	Electrolytic copper coating and plating ⇒ 621.357 ⇒ 621.793
669.27	Tungsten	669.4	Lead
669.274	Refining of tungsten. Production of high-purity tungsten	669.43	Extraction of lead
669.275	Tungsten alloys ⇒ 661.665.2	669.431	Dry extraction of lead. Dry processing of galena (lead sulphide)
669.28	Molybdenum	669.431.1	Preliminary treatment of galena
669.284	Refining of molybdenum. Production of high-purity molybdenum	669.431.2	Reduction of oxygen compounds
669.29	Other metals of the 4th, 5th and 6th groups of the periodic table	669.431.3	Smelting with iron
669.292	Vanadium	669.431.4	Other direct methods. □ Roasting and reaction
669.292.5	Vanadium alloys	669.431.6	Treatment of residues, dross, dust
669.293	Niobium	669.432	Processing of other lead ores
669.294	Tantalum	669.436	Wet processing of lead ores (except electrolysis)
669.295	Titanium	669.44	Refining of lead
669.296	Zirconium	669.443.1	Removal of copper from crude lead
669.297	Hafnium	669.443.2	Removal of tin, arsenic, antimony from lead. Softening, improving of lead
669.298	Thorium	669.443.3	Removal of zinc and cadmium from lead
669.3	Copper	669.443.4	Removal of gold and silver from lead ⇒ 669.223.1
669.33	Extraction of copper	669.443.6	Removal of bismuth from lead. Debismuthing
669.331	Dry extraction of copper. Roasting. Pretreatment	669.447	Electrolytic refining of lead
669.332	Production of copper matte	669.48	Lead and/or its alloys applied as coating etc.
669.332.1	In coke blast furnace	669.5	Zinc
669.332.2	In reverberatory furnaces	669.53	Extraction of zinc
669.332.3	Pyritic smelting	669.531	Dry extraction of zinc
669.332.4	Semi-pyritic smelting. Partial pyritic smelting	669.531.2	Muffle processes in general. Horizontal retort processes
669.332.5	Orkla process	669.531.4	Shaft-furnace processes. Vertical retort processes
669.332.8	Smelting in electric furnace	669.531.5	Processes with rotary tube furnaces. Rotary kiln process (Waelz process)
669.333	Production of crude copper	669.531.6	Extraction of zinc oxide
669.333.1	Reduction of oxide in shaft furnace	669.531.7	Treatment of zinc ore in the blast furnace
669.333.2	Reduction of oxide in reverberatory furnace	669.531.8	Electrothermal methods. □ Snyder and De Laval process
669.333.3	Smelting in converters	669.536	Wet extraction of zinc
669.333.4	Treatment of crude copper matte	669.537	Electrolytic extraction of zinc
669.333.5	Refining of black copper. Rosette copper	669.537.2	Electrolysis of molten zinc chloride
669.333.6	Fire refining of blister copper. Production of malleable copper	669.537.3	Electrolysis with soluble anodes
669.333.7	Treatment of ores containing metallic copper		
669.334	Wet extraction of copper. □ Chloridizing. Sulphatization. Cementation. Refining		
669.337	Electrolytic processes		
669.34	Refining of copper. Production of high-purity copper		
669.343	Special processes. □ Fire refining. Refining melt		

669.537.4	Electrolysis of zinc sulphate solution, of zinc oxalate solution	669.714.7	Electrolytic refining (metal to be refined serving as anode) 669.714.7 $\equiv$ 669.713.7
669.537.5	Electrolysis of zinc chloride solution	669.715	Aluminium alloys ◇ 669.715.018.8 Corrosion-resistant aluminium alloys
669.537.6	Electrolysis of ammoniacal or alkaline zinc solutions	669.716	Fabrication and finishing of aluminium
669.54	Refining of zinc. Production of pure zinc	669.716.9	Surface finishing, coating, colouring of aluminium
669.55	Zinc alloys	669.717	Applications of aluminium and its alloys ◇ 669.717:629.7 Aluminium for air and space vehicles ⇒ 624.014.7 ⇒ 691.77
669.58	Zinc applied as coating or plating. Galvanizing	669.718	Aluminium applied as coating, cladding, plating
669.586	Galvanizing (nonelectrolytically)	669.718.7	Electrolytic aluminium coating or plating on other metals
669.586.5	Hot-dip galvanizing	669.719	Aluminothermy. Production of metals with high melting points. Production of high temperatures by oxidation of aluminium ⇒ 621.791.6 ⇒ 669.17
669.587	Electrogalvanizing	669.721	Magnesium
669.6	Tin	669.721.3	Extraction of magnesium
669.63	Extraction of tin	669.721.31	By chemical means
669.631	Extraction of crude tin	669.721.36	By electrothermal means
669.64	Refining of tin. Production of pure tin	669.721.37	By electrolytic means
669.647	Electrolytic refining	669.721.4	Refining of magnesium
669.648	Recovery of tin, especially from tin plate	669.721.41	Refining by chemical means
669.65	Tin alloys	669.721.42	Refining by physical means
◇ 669.65.018.24	White bearing metal (Babbitt's metal)	669.721.47	Electrolytic refining
◇ 669.65'4	Alloys of tin and lead. Soft solders (from a metallurgical point of view)	669.721.5	Magnesium alloys
⇒ 621.791.3		669.721.9	Production of metals with high melting points. Production of high temperatures by oxidation of magnesium
669.68	Tin applied as coating etc. Tin plating	669.725	Beryllium
669.684	Tin coating or plating of nonmetallic materials	669.73	Cadmium
669.686	Nonelectrolytic tinning, tin plating	669.733	Extraction of cadmium from zinc ores
669.686.5	Immersion method of tin plating. Dipping.	669.733.1	By dry processes. Distillation of residues or brown powder with carbon
669.687	Production of white metal	669.733.2	By wet processes
669.687	Electrolytic tin plating	669.74	Manganese
669.7	Light metals in general	669.743	Extraction of manganese
◇ 669.7.018	Light metals according to their general properties. Alloys of light metals in general	669.743.1	By smelting
669.71	Aluminium	669.743.2	Production of pure carbon-free manganese
669.711	Generalities. Surveys of the aluminium industry	669.75	Antimony
669.712	Preparation of intermediate products from raw materials. Production of alumina	669.753	Extraction of antimony
669.712.1	Alkaline treatment. Processes for aluminates (except aluminate production in electric furnace)	669.753.1	From sulphide ores
669.712.2	Acid treatment. Production of alumina from aluminium salts	669.753.3	By smelting in blast furnace. Hungarian process
669.712.3	Electrothermal processes. Processes involving compounds other than aluminates or aluminium salts as intermediate products (e.g. nitrides, carbides, sulphides) ⇒ 621.365 ⇒ 661.862	669.753.7	By electrolytic processes
669.712.4	Other processes. □ Decomposition of aluminium silicate by superheated steam	669.754	Refining of antimony
669.712.5	After treatment of alumina and aluminium hydroxide. Purification of alumina	669.76	Bismuth
669.713	Extraction of aluminium and aluminium alloys from aluminium compounds	669.763	Extraction of bismuth
669.713.1	Production by purely chemical, nonelectrolytic methods	669.763.1	Dry processes
669.713.6	Electrothermal production	669.763.2	Wet processes
669.713.7	Electrolytic production	669.763.3	Extraction of bismuth from mixed ores, intermediate products and residues
669.713.72	Fused-salt-bath electrolysis	669.771.78	Various nonmetal constituents of alloys. Metalloids
669.713.76	Electrolysis of aqueous solutions		<i>Only from the point of view of their influence on metals and alloys. Class production of metalloids as such under 661</i>
669.714	Refining of aluminium and aluminium alloys	669.774	Polonium
669.714.1	Refining by chemical means. □ Use of fluxes, gases	669.775	Sulphur
669.714.2	Refining by physical means <i>Details by .04...</i> ◇ 669.714.2.048 By distillation	669.776	Selenium
		669.777	Tellurium
		669.778	Arsenic
		669.779	Phosphorus

669.781	Boron	669.849	Rhenium
669.782	Silicon. Silicothemy	669.85/.86	Rare earth metals
669.783	Germanium		669.85/.86 $\equiv$ 546.65/.66
669.784	Carbon	669.871	Gallium
669.785	Gases in metals in general	669.872	Indium
669.786	Nitrogen	669.873	Thallium
669.787	Oxygen	669.88	Alkali metals
669.788	Hydrogen		$\Rightarrow$ 661.3
669.791	Mercury	669.882	Potassium
669.791.5	Mercury alloys. Amalgams	669.883	Sodium
	$\Rightarrow$ 669.215	669.884	Lithium
	$\Rightarrow$ 669.223.3	669.885	Rubidium
669.793	Scandium	669.886	Caesium
669.794	Yttrium	669.89	Alkaline earth metals
669.82	Uranium series	669.891	Calcium
669.822	Uranium	669.892	Strontium
669.823	Neptunium	669.893	Barium
669.824	Plutonium	669.894	Radium
669.848	Technetium		

67 VARIOUS INDUSTRIES, TRADES AND CRAFTS

The -1/-9 auxiliaries from 62-1/-8 and 66-9 are applicable throughout 67 except for -1... under 678, and -2... under 679.8

67.01	Generalities on the individual industry	671.121.9	Other decorative objects. ☐ Articles of guilloche, lattice metalwork. Pendants. Locketts
67.014	Chemical constitution and properties of materials	671.123	Jewellery for the head. ☐ Tiaras. Diadems
67.017	Physical and mechanical properties, workability and serviceability of materials	671.125	Various regalia, insignia. ☐ Sceptres. Crowns. Sword or dagger hilts, sheaths. Batons, staffs of office
	$\Rightarrow$ 620.17		Badges and similar decorations
67.019	Faults, defects, damage of materials	671.128	$\Rightarrow$ 355.134.2
	$\Rightarrow$ 620.19		$\Rightarrow$ 671.42
67.02	Manufacturing processes		$\Rightarrow$ 929.71
67.03	Raw materials	671.13	Other articles of precious metals. ☐ Chains, cases, boxes (e.g. snuffboxes). Pens and nibs of precious metals
67.04	Auxiliary materials		$\Rightarrow$ 672.86
67.05	Machinery. Plant. Tools	671.14	Articles coated with gold or silver.
	$\Rightarrow$ 621.7		☐ Articles of gold plate or silver plate. Articles of rolled gold or rolled silver
	$\Rightarrow$ 621.9		
67.06	Products	671.15	Working of precious stones, gems, pearls
67.08	By-products. Wastes		$\Rightarrow$ 679.8.02
671	ARTICLES OF PRECIOUS METALS, GEMS, PRECIOUS STONES	671.152	$\Rightarrow$ 679.87
	$\Rightarrow$ 669.21/.23	671.153	Diamond cutting
	$\Rightarrow$ 679.87	671.157	Cutting other precious stones
671.1	Gold and silver articles. Articles of other precious metals. Jewellery	671.16	Working of pearls
	$\Rightarrow$ 391.7		Articles of nonprecious stones. Articles of paste, imitation gems
	$\Rightarrow$ 681.11		$\Rightarrow$ 666.23
	$\Rightarrow$ 739.1		$\Rightarrow$ 679.86
	$\Rightarrow$ 739.2	671.18	Jewel settings
671.11	Gold and silver articles. Engraved and embossed articles	671.2	Small articles of various metals and alloys
671.12	Jewellery	671.21	Fancy jewellery. ☐ Imitation gold and silver. Polished steel. Jet
671.121	Brooches. Bracelets. Rings etc.	671.22	Various metal objects other than jewellery
671.121.2	Brooches. Decorative pins, clasps	671.4	Manufacture of coins and medals. Minting
671.121.3	Bracelets. Closures, buckles etc. for bracelets. ☐ Wristwatch bands		$\Rightarrow$ 737
671.121.4	Ear jewellery. Ear-placers. ☐ Earrings. Ear-clips	671.4.02	Manufacturing processes for coins and medals
671.121.5	Rings (finger rings). ☐ Wedding rings. Signet rings	671.4.024	Production of blanks
671.121.6	Key-rings. Security devices for watches, bracelets etc.	671.4.025	Testing. Adjusting to weight. Automatic weighing
		671.4.026	Milling the rim

671.4.027	Soft annealing	672.61	Chains in general. Welded chains in particular. □ Chain components: links, joints
671.4.028	Striking. Edge stamping. Correcting the impression	672.62	Unwelded chains
671.41	Coins	672.63	Form and shape of chain links
	⇒ 336.746	672.64	Special types of chain
671.411/.418	Metal coins	672.641	Vaucanson chains. Ladder chains
	671.411/.418 ≡ 669.1/.894	672.642	Galle chains. Roller chains
671.412.1	Gold coins	672.643	Dismantlable chains
671.419	Nonmetallic coins	672.65	Chains for special purposes. □ Towing chains. Transmission chains. Bicycle chains
671.42	Medals. Medallions. Plaques. Plaquettes	672.66	Anchoring devices. Anchors. □ Anchor shanks, arms, flukes, stocks
	671.42 ≡ 671.41	672.67	Drags and drums for securing of chains
	⇒ 355.134.2	672.7	Cutlery. Cutting and piercing tools and weapons
	⇒ 671.128	672.71	Cutting and piercing tools and weapons.
672	ARTICLES OF IRON AND STEEL IN GENERAL		□ Knives in general
672.1	Articles of cast iron and steel		⇒ 621.93
	⇒ 621.74		⇒ 621.96
672.11	Building components and fittings of cast iron and steel		⇒ 631.342
	⇒ 682.6		⇒ 635-1/-2
	⇒ 691.7		⇒ 682.3
672.13	Artistic and ornamental cast iron work	672.711	Cutting, piercing, stabbing weapons.
	⇒ 673.3		□ Daggers. Combat knives
	⇒ 682.64		672.711 ≡ 623.444
	⇒ 739.4	672.712	Pocket knives. □ Penknives. Clasp-knives.
672.3	Wrought or forged iron and steel ware		Jack-knives
	⇒ 682	672.713	Table knives. Kitchen knives. Erasing knives
	⇒ 683		⇒ 642.732
672.31	Welded or riveted articles of steel.	672.714	Scissors. Shears. Secateurs
	□ Containers. Vats. Tanks	672.715	Razors
672.32	Drawn steel articles		⇒ 646.72
	⇒ 621.983/.984		⇒ 687.53.052
672.33	Zinc-coated (galvanized) steel articles.	672.715.1	Simple razors. □ Cut-throat razors. Hair trimmers with inset blade
	Tin-coated steel articles	672.715.3	Safety razors. Razor blades
	⇒ 621.357	672.716	Surgical instruments
	⇒ 621.793		⇒ 615.472.3/.4
672.4	Tinware. Articles of tinplate and sheet metal	672.718	Shearing machines, clippers. Hair clippers
			⇒ 687.53.052
672.41	Articles of plates sheet metal, e.g. tinplate (tin-coated steel sheet) or zinc-coated steel sheet (galvanized steel). □ Buckets. Tanks	672.719	Other cutting tools. □ Hatchets. Choppers.
			Axes
			⇒ 621.968
			⇒ 631.342
			⇒ 674.053
672.42	Articles of uncoated steel sheet	672.73	Grinding and sharpening devices. □ Knife sharpeners. Strops
672.43	Articles of enamelled steel sheet	672.76	Forks. Spoons
	⇒ 666.293		⇒ 642.732
672.46	Preserving cans, tins. Food cans	672.8	Small metal articles
	⇒ 664.8/.9	672.81	Pins. □ Steel pins (for dressmaking etc.). Hatpins. Scarf and tie pins. Safety pins
	⇒ 664.8.036.5		⇒ 671.121.2
672.462	Closing of cans. Methods of closure		⇒ 687.53.053.5
672.464	Opening of cans. □ Attached devices.	672.82	Needles. □ Sewing needles. Darning needles.
	Removable strips. Ring-pull tops		Packing needles. Saddler's needles. Bodkins.
672.466	Can openers (tin openers). Hand devices and machines for opening cans		Awls. Knitting needles. Crochet hooks
	⇒ 641.516		⇒ 687.053.2
	⇒ 683.564	672.83	Fasteners
672.47	Various small articles. □ Sleeves, tubes, cases, capsules of sheet metal		⇒ 687.078
672.49	Meshes and sieves made from sheet metal	672.831	Simple hooks
	⇒ 672.85	672.834	Carabiners. Spring rings
672.6	Chains. Anchors	672.835	Eyes. Rings
	⇒ 62-235	672.836	Buckles
	⇒ 621.85.055	672.837	Fasteners other than buckles, hooks and eyes.
	⇒ 627.371		□ Zip fasteners (zippers)
	⇒ 629.5.028.7		

- 672.85** Articles of metal wire and strip. □ Wire mesh, sieves. Wire netting. Wire springs. Corset rods
⇒ 621.778
⇒ 672.49
- 672.86** Steel pen nibs
⇒ 671.13
⇒ 686.862.1
- 672.87** Metal buttons
⇒ 688.2
- 672.88** Metal fixing devices. □ Screws, bolts etc. Nails. Tacks
672.88 ≡ 621.88
- 672.9** Various metal articles
- 672.92** Umbrella frames
- 672.93** Metal boxes, chests, caskets
⇒ 683.34
- 672.94** Metal furniture
⇒ 645.4
⇒ 682.5
⇒ 684.4
- 673** **ARTICLES OF NONFERROUS METALS (EXCEPT PRECIOUS METALS)**
- 673.1** Articles of copper and copper alloys in general
- 673.2** Articles of wrought copper and brass
- 673.3** Artistic castings of nonferrous metals.
□ Artistic bronze work
⇒ 669.35'6
⇒ 672.13
⇒ 739.4
- 673.4** Articles produced by electroforming (galvanoplasty)
⇒ 621.357.6
- 673.5** Bells. Bellfounding. □ Carillons
⇒ 780.633.1
- 673.6** Fittings for pipes and conduits. Taps. Valves
673.6 ≡ 621.646
- 673.7** Other parts of pipelines, systems
673.7 ≡ 621.647
- 673.8** Articles of tin, lead, zinc, light metals etc. □ Collapsible tubes
- 674** **TIMBER AND WOODWORKING INDUSTRY**
Denote state and form of wood products by -4... from 62-4...
◇ 674-412 Round timber. Stems, logs etc.
◇ 674-413 Thick sheets. Planks, boards etc.
◇ 674-414 Medium-thick sheets. Thin planks, boards etc.
◇ 674-415 Thin sheets, e.g. fretworking sheets
◇ 674-416 Very thin sheets. Veneers
◇ 674-419 Composite boards and sheets. Laminated board
◇ 674-419.3 Plywood
◇ 674-422 Convex profiles. Solid sections. Rounds. Dowel rod. Strips. Mouldings. Beadings etc.
◇ 674-423 Concave profiles. Hollow mouldings. Angle sections etc.
⇒ 630'83
⇒ 684
⇒ 694
- 674.02** **Woodworking processes**
- 674.021 Marking. Measuring
674.022 Fixing. Clamping
674.023 Trimming. Sawing. Cutting. Barking
674.024 Chiselling
674.025 Drilling. Boring
674.026 Planing
674.028 Assembling. Bending. Wood joints
674.028.1 Joining wooden parts without additional means.
□ Scarfed joints. Mortise-and-tenon joints. Dovetailing. Mitring
674.028.2 Dowelled joints. Pegged joints
674.028.3 Joints with ring connectors or frame angles
674.028.4 Screwed joints
674.028.5 Nailed joints
674.028.6 Cold bending of wood
674.028.7 Hot bending of wood
674.028.9 Other wood joints. □ Glued joints
- 674.03** **Species of wood**
⇒ 582
⇒ 630
- 674.031 Hardwoods. Deciduous, broadleaf timbers
674.031.1/3 According to hardness
674.031.1 Hard woods
674.031.19 Very hard woods
674.031.2 Semihard woods
674.031.3 White woods. Soft broadleaf woods
Not softwoods, for which see 674.032
674.031.5/9 Hardwoods according to botanical families
674.031.52/9 ≡ 582.52/9
Additional detail is given here
674.031.628.2 Juglandaceae. □ Carya (Hickory)
674.031.632.1 Betulaceae. □ Betula (Birch)
674.031.632.2 Fagaceae. □ Fagus (Beech). Castanea (Chestnut). Quercus (Oak)
674.031.635.1 Ulmaceae. □ Ulmus (Elm). Celtis (Hackberry)
674.031.638.26 Platanaceae. □ Platanus (Plane)
674.031.681.81 Salix (Willow)
674.031.682.2 Buxaceae. □ Buxus (Box, Boxwood)
674.031.685.3 Bombaceae. □ Ochroma (Balsa)
674.031.685.4 Tiliaceae. □ Tilia (Lime, Linden)
674.031.687.46 Ebenaceae. □ Diospyros (Ebony)
674.031.728.2 Santalaceae. □ Santalum (Sandalwood)
674.031.746.11 Zygophyllaceae. □ Guaiacum. Lignum vitae
674.031.746.41 Meliaceae. □ Chloroxylon. Satinwood
674.031.746.413 Swietenioideae. □ Swietenia (Mahogany)
674.031.746.51 Aceraceae. □ Acer (Maple)
674.031.751 Geraniaceae
674.031.776.2 Myrtaceae. □ Eucalyptus (Gum-tree wood)
674.031.916.165 Fraxinus (Ash)
674.031.929.3 Verbenaceae. □ Tectona (Teak)
674.032 Softwoods. Conifer timbers
⇒ 674.031.3
- 674.032.3/4 Softwoods according to botanical families
674.032.3/4 ≡ 582.31/4
Additional detail is given here
674.032.471 Taxaceae. □ Taxus (Yew)
674.032.475.2 Abies (Fir (deal))
674.032.475.3 Larix (Larch)
674.032.475.4 Pinus. (Pine (deal). Pitch pine)
674.032.475.5 Picea (Spruce)
674.032.475.6 Tsuga (Hemlock wood)
674.032.475.7 Pseudotsuga (Douglas fir)
674.032.475.8 Cedrus (Cedar)
674.032.476 Taxodiaceae. □ Sequoia (Redwood)
674.037 Artificially coloured woods. Dyed, stained woods. □ Imitation mahogany, rosewood, ebony. Wood with enhanced grain
674.038 Woods from the standpoint of their intended use

674.038.1	Woods with various characteristics, according to condition of the various parts of the tree	674.056	Tools, machines for planing. □ Planes. Scrapers ⇒ 621.911/.912
674.038.15	Characteristics and defects connected with growth. □ Compression wood. Wound wood. Spiral-grained wood	674.057	Tools, machines for turning and thread-cutting. □ Lathes. Thread-cutters ⇒ 621.941 ⇒ 621.99
674.038.16	Wood from various parts of the tree. □ Rootwood	674.058	Tools, machines for striking, nailing, bending, joining 674.058 ≡ 674.028
674.038.17	Texture of the wood. Composition as regards heartwood formation. □ Heartwood. Mature wood. Close-textured timber. Alburnum (sapwood)	674.058.5	Nailing devices. Hammers. Nailing machines ⇒ 621.972 ⇒ 621.974
674.038.18	Organically changed wood. □ Rotten wood. Decayed wood	674.06	Products of the timber industry. Timber
674.038.19	Characteristics in relation to the ecology of the tree	674.061	Specimens of timber. □ Small, clear specimen (free of knots, defects). Large structural size
674.038.3	Wood according to kind of preparation. □ Cleft wood; split logs. Sawn wood, planks	674.07	Surface treatment of wood 674.076/.077 ≡ 667.6/.7 674.078.2 ≡ 665.92 ⇒ 667.74
674.038.4	Floated timber. Dry timber. Impregnated timber	674.09	Processing roundwood into converted timber
674.038.5	Woods for various purposes. □ Structural timber. Joinery timber Denote purpose by :... ◇ 674.038.5:780.6 Wood for musical instruments	674.092	Processing of roundwood by hand, especially with rip-saws
674.038.6	Woods according to processing, working. □ Sawn, planed, turned, bent wood	674.093	Sawmills. Sawmilling industry
674.038.7	Woods according to texture	674.093.2	Classification of sawmills according to the special purpose. Sawmills' ancillary businesses
674.038.71	Closeness of rings in timber. □ Fine-ringed, close-ringed timber. Broad-ringed timber	674.093.24	Planing shop
674.038.74	Wood from various growth periods. □ Early wood. Late wood	674.093.26	Veneer shop
674.038.75	Fineness of structure. Fineness of grain	674.093.3	Log yards
674.038.77	Wood with low shrinkage	674.093.4	Yards for converted timber. Stacks. Layers
674.038.78	Textures giving effect of aesthetic, decorative value. □ Moiré effect. Floral effect. Burrs. Knots. Bird's-eye pattern	674.093.6	Sawing in the sawmill. Cutting. Sawing-shop
674.04	Seasoning, impregnation and other treatment of wood	674.1	Carpentry
674.045	Moistening	674.11	Structural, constructional carpentry. □ Roof trusses. Floors. Door frames ⇒ 624.011.1 ⇒ 694
674.046	Scalding. Steam treatment against biological attack	674.12	Ship's carpentry
674.047	Seasoning of timber	674.2	Joinery ◇ 674.2-419.3 Joinery with plywood
674.047.1	Air drying. Natural seasoning	674.21	Constructional joinery. □ Joinery for fixtures of buildings, partitions, doors, windows ⇒ 694.6
674.047.3	Artificial drying. Kiln drying	674.23	Furniture-making ⇒ 645.4 ⇒ 684.4
674.048	Impregnation of wood	674.24	Marquetry. Inlay work. □ Preparation of the veneers ⇒ 684.6
674.048.2	Devices and machines for impregnating	674.25	Vehicle joinery. Coach-building joinery ⇒ 629.02
674.048.3	Impregnation against vegetable attack (injurious plants)	674.26	Lattice work. Fences. Palings. Trellis. Ceiling and close lathing
674.048.4	Impregnation against animal attack	674.3	Cartwrighting. Wheelwrighting
674.048.5	Sealing processes	674.31	Wooden wheels. □ Hubs, naves. Spokes. Fellies (felloes), rims. Tenons ⇒ 629.3.027.4
674.049.2	Hardening. Hardening equipment	674.32	Bodywork. Vehicle frame ⇒ 629.3.023.2
674.049.3	Fireproofing. Reduction of flammability, combustibility. Flame-retardant treatment	674.4	Cooperage. Cask, barrel making ⇒ 621.798.13
674.05	Woodworking tools. Woodworking machines Denote further details at the subdivisions by :621.9...		
674.053	Tools, machines for trimming, cutting and barking. □ Saws. Axes ⇒ 621.93 ⇒ 621.968 ⇒ 631.342 ⇒ 672.719		
674.054	Tools, machines for mortising, cutting out. □ Chisels. Gouges ⇒ 621.911.7		
674.055	Tools, machines for rasping, filing, boring. □ Files. Rasps. Drills. Braces ⇒ 621.918 ⇒ 621.95		

674.41	Parts of casks, vats and other wooden vessels. □ Staves. Heads. Hoops	674.88	Processing of peat (except as fuel and for textile industry). □ Peat mould, mull (horticultural peat). Peat litter. Peat bales <i>For peat as fuel and for spinning use 662.64 and 677.194</i> ⇒ 631.831 ⇒ 631.878
674.42	Kinds of wooden vessel. □ Casks, barrels. Wooden vats, tubs. Wooden buckets, pails	675	LEATHER INDUSTRY (INCLUDING FUR AND IMITATION LEATHER) <i>The auxiliaries listed under 675.0... are not applicable at 675.92 and its subdivisions</i> ⇒ 637.61 ⇒ 685 ⇒ 686.1 ⇒ 687 ⇒ 745.53
674.5	Wooden articles and their manufacture. □ Wooden household articles. Wooden toys ⇒ 688.7	675.01	Theory of leather and fur production. Properties. Defects. Tests
674.6	Box-making. Wooden packaging ⇒ 621.798.12	675.011/013	Process design. Process control. Plant. Services <i>675.011/013 ≅ 66.011/013</i>
674.61	Wooden boxes, crates, packing cases. Wooden chests (as packaging)	675.014	Chemical composition, properties and analysis (of hides, skins, leather, fur)
674.62	Wooden trunks. Wooden chests (strongboxes) ⇒ 685.511	675.014.2	Hide substances ◇ 675.014.2:547.96 Determination of proteins (collagen etc.)
674.63	Wattle or wickerwork containers. Wicker furniture. □ Wicker boxes, hampers, baskets	675.014.4	Tanning substances. Tannin
674.64	Basketwork covers, casings (e.g. for bottles) ⇒ 677.54	675.014.6	Fibre basis, properties and determination
674.7	Manufacture of round wooden articles. Turnery	675.015	Structure of hides, skins, leathers, fur and their constituent collagen (protein) fibres
674.71	Masts. Poles. Round posts. □ Cable poles. Telegraph poles	675.015.2	Macrostructure of hides, skins, leather, fur
674.72	Staves. Rods. Canes. □ Fishing rods	675.015.22	Epidermis
674.73	Walking sticks and canes. Stocks, handles ⇒ 685.557	675.015.23	Derma
674.75	Tool handles. Helves	675.015.24	Flesh
674.76	Bentwood. □ For chair and furniture parts. For pulleys	675.015.26	Hair
674.77	Turned wooden articles. Turnery proper. □ Turned tool handles, shafts, furniture parts	675.015.4	Chemical structure, properties, composition of collagen (protein) fibres
674.8	Production and processing of wood chips, shavings, waste. Wood compositions. Processing of cork and peat	675.015.6	Molecular and microstructural properties of collagen (protein) fibres
674.81	Consolidated wood. Compressed wood waste	675.016	Properties and investigation of finishes and auxiliary materials in leathers and fur
674.812	Compressed wood. Densified wood ◇ 674.812-419.3 Compressed plywood	675.017	Physical and mechanical properties of leather, their determination and testing
674.815	Compressed wood chip materials in the form of boards, battens, planks, mouldings etc.	675.017.1	Topography of leather
674.816	Compositions of wood waste, sawdust. □ Xylolith (plastic of sawdust and Sorel cement). Plastic wood ⇒ 691.334	675.017.12	Side
674.817	Products of compressed wood pulp. Wood fibre materials ◇ 674.817-415 Fibreboard sheet	675.017.13	Butt. Bend. Rump. Back
674.82	Wood wool. Wood shavings. Sawdust ⇒ 621.798.22 ⇒ 677.195	675.017.14	Belly. Shoulder
674.83	Processing of cork. □ Cork stoppers (bottle corks) ◇ 674.83-41 Cork in sheets, blocks etc. ◇ 674.83-493 Cork crumbs	675.017.15	Horse front. Horse butt. Shell
674.84	Synthetic cork. Cork substitute	675.017.16	Head. Cheek. Shank
674.87	Processing of conifer needles (e.g. into needle wool) ⇒ 677.195	675.017.17	Offal hide (low quality)
		675.017.2	Dimensional properties and classification not determined by production. Size. Mass
		675.017.3	Parameters determined during the production process. Sampling. Test methods
		675.017.4	Strength of leather. Resistance to mechanical influences <i>Details by :620.17...</i>
		675.017.5	Optical, thermal, electrical and other properties
		675.017.55	Optical properties <i>Details by :535...</i>
		675.017.56	Thermal properties <i>Details by :536...</i>
		675.017.57	◇ 675.017.56:536.495 Heat resistance Electrical and magnetic properties <i>Details by :537...</i>
		675.017.6	Properties with respect to fluids (liquids, vapours, gases). □ Resistance to fluids. Permeability

675.017.62	Properties (including impermeability) with respect to gases, air, steam and other vapours ⇒ 620.193.29	675.026.22	Filling
675.017.63	Properties (including impermeability) with respect to water and other liquids ⇒ 620.193.19	675.026.23	Currying. Stuffing. Oiling. Grossing
675.017.8	Serviceability. Properties affecting use, suitability	675.026.24	Laying-away
675.017.82	Hide, skin thickness	675.026.26	Drying. Airing off
675.017.83	Hair covering. Fur (pile) height	675.026.27	Conditioning
675.017.84	Elasticity. Springiness. 'Give'	675.026.3	Coating. Waxing. Grain finishing. Colour finishing. Final finishing
675.017.85	Form stability. Creasing, crumpling properties	675.026.4	Cutting out
675.017.86	Textural properties. Felting etc.	675.026.6	Stacking
675.017.87	Surface, finish properties. Soilure	675.028	Special processes or treatments (other than those above). □ Lustring. Finishing of hair pile. Finishing of linings
675.017.88	Wear resistance in general	675.03	Raw skins, hides and pelts according to origin
675.019	Defects, faults, damage (in raw and finished leather and fur), their detection and remedy	⇒ 636/639	
675.019.1	Visible defects (surface defects) 675.019.1 ≡ 620.191	⇒ 675.06	
675.019.2	Internal defects. □ Consistency. Shrinkage 675.019.2 ≡ 620.192	675.031	Skins, hides of domestic animals (and allied wild species)
675.019.3	Defects, faults, damage due to ambient influences 675.019.3 ≡ 620.193	675.031.1	Hides of oxen. Cattle hide
675.019.5	Detection of defects, faults, damage	675.031.11	Cowhide
675.019.7	Removal of defects etc.	675.031.113	Calfskin
675.02	Working and production methods, processes and treatment of leather and fur	675.031.2	Hides of sheep. Sheepskin
675.023	Treatment of raw skins and fur (before tanning)	675.031.3	Hides of goats. Goatskin
675.023.1	Preliminary treatment. □ Flaying. Curing. Sterilization. Sorting	675.031.4	Hides of swine. Pigskin
675.023.2	Washing. Soaking. Degreasing. Oil removal	675.031.5	Hides of equines. □ Horse hide. Donkey or ass hide
675.023.3	Liming, unhairing, fleshing, scudding	675.031.6	Hides of camels
675.023.32	Liming	675.031.8	Skins of small domestic animals (and allied wild species). □ Hare skins. Rabbit skins. Cat skins. Skins of dog, dingo etc.
675.023.33	Hair removal (depilation). Unhairing. Chemical loosening of hair. □ Sweating. Scalding	675.032	Skins of large wild land-mammals. □ Tiger skins. Bear skins. Deer skins. Elephant hides
675.023.34	Fleshing	675.033	Skins of small wild land-mammals. □ Skins of mink. Skins of fox. Skins of stoat or ermine
675.023.36	Scudding	675.034	Reptile skins. □ Crocodile skins. Snakeskin
675.023.37	Plucking the bristles	675.035	Fish skins. Skins of marine mammals. □ Sharkskin. Skins of whales, dolphins etc.
675.023.4	Cutting or dressing hair on raw skin, raw fur	675.036	Sealskin
675.023.5	Deliming ◇ 675.023.5:54-32 Deliming with acids ◇ 675.023.5:54-38 Deliming with salts	675.039	Bird skins. Skins of fowl
675.023.6	Bating	675.04	Tanning and auxiliary materials used in leather and fur production
675.023.7	Pickling. Salting. Drenching	675.041	Natural vegetable tanning agents. □ Leaves. Bark. Wood. Roots
675.024	Tanning and tawing processes	⇒ 633.87	
675.024.3	Vegetable tanning	⇒ 675.014.4	
675.024.4	Mineral tanning. Tanning with organic and synthetic substances. Oil tanning	675.042	Mineral and synthetic tanning agents <i>Denote mineral (inorganic) agents by :546... and synthetic organic agents by :547...</i>
675.024.43	Chrome tanning	675.043	Chemical auxiliaries
675.024.44	Iron tanning	675.043.2	Water
675.024.46	Tanning with alums, with silicon, with zirconium, with titanium compounds	675.043.3	Acids. Oxides. Bases. Salts
675.024.47	Tanning with organic and synthetic substances. □ Formaldehyde tanning	675.043.4	Surface-active agents. Plasticizers
675.024.48	Oil tanning	675.043.6	Protein substances ⇒ 547.96
675.024.49	Tanning with mixtures, e.g. tannin and chemical admixtures	675.043.7	Carbohydrates ⇒ 547.455 ⇒ 547.458
675.024.5	Electrical tanning	675.043.8	Macromolecular materials (polymers). Other chemicals (irrespective of function)
675.024.6	Partial tanning. Tanning of different sides by different processes	675.044	Processed animal and vegetable fats. □ Oxidized fats. Degras. Sulphonated, saponified and polymerized fats and oils. Fatty acids ⇒ 665.228 ⇒ 665.382/383
675.024.7	Mixed, combined tanning methods	675.045	Oils. Fats. Greases. Leather oils (mixtures of fats and oils)
675.024.9	Retanning	675.045.2	Mineral oils ⇒ 665.7
675.025	Mechanical treatment and processing of skins, leather and fur (especially after tanning) 675.025 ≡ 675.055		
675.026	Finishing of skins, leather, fur		
675.026.1	Dyeing. Staining. □ Preparation for dyeing ⇒ 667.772		
675.026.2	Impregnation. Filling. Currying. Stuffing. Oiling. Drying. Conditioning		

675.045.3	Animal and vegetable fats and oils ⇒ 665.2/3	675.06	Finished leathers and furs (according to origin) and their applications 675.06 ≡ 675.03
675.045.4	Synthetic oils and greases ⇒ 665.381	675.07	Skins, leathers and furs with special characteristics
675.046	Finishing materials	675.072	Soft, pliable leather
675.046.1	Colorants. Pigments. Dyes. Bleaching agents ⇒ 667.772	675.073	Hard, rigid leather
675.046.3	Waxes	675.074	Strong, hard-wearing leather
675.046.8	Enamel and other coatings. Varnish, lacquer etc.	675.075	Leathers with special physical properties. □ Optical, thermal, electrical properties 675.075 ≡ 675.017.5
675.05	Equipment, machines for hide and skin processing, for leather and fur production	675.076	Leathers with special protective properties against fluids 675.076 ≡ 675.017.6
675.052	Equipment, machines for production of tanning agents	675.076.3	Waterproof leather
675.053	Equipment, machines for treatment of raw skins and fur (before tanning)	675.08	Wastes from leather and fur production (including slaughter and tannery wastes) ⇒ 675.8
675.053.1	Equipment, machines for preliminary treatment and processing. □ Flaying equipment. Curing equipment. Sterilization equipment. Sorting equipment	675.081	Wool. Hair. Flocks. Bristle. Flesh
675.053.2	Washing equipment. Soaking equipment. Steeping and drumming equipment. □ Vats. Paddles. Drums	675.082	Fats. Degras. Flesh tallow. Wool fat
675.053.3	Equipment, machines for hair removal (depilation). Unhairing machines. Fleshing equipment. Scudding equipment	675.083	Hide and skin wastes
675.053.4	Equipment, machines for cutting hair on hides, skins, raw fur	675.084	Horns. Hooves. Bones etc.
675.054	Equipment, machines for tanning and tawing	675.085	Entrails. Guts. Intestines
675.054.3	Tanning pits. Layers. Handlers	675.086	Shavings. Cuttings. Scraps. Shreds. Leather dust
675.054.5	Electrical tanning equipment	675.087	Bird feathers
675.055	Equipment, machines for mechanical treatment of skins, leather and fur (especially after tanning)	675.088	Tanning wastes. Liming wastes. Other processing wastes. □ Bark. Extracts. Spent tan liquor. Lime slab. Spent salt
675.055.1	Equipment, machines for general and preliminary treatment. □ Samming machines. Shaving machines	675.1	Leather and leatherware generally. Special-purpose leathers
675.055.2	Equipment, machines for improving the body texture	675.11	Technical and engineering leathers generally. □ Belting leather. Packing leather. Leather for machine components
675.055.22	Setting-out machines	675.12	Saddle leather. Harness leather. Leather for human accoutrements. □ Belt and strap leather. Orthopaedic leather. Leather for protective wear ⇒ 614.89 ⇒ 615.477.1 ⇒ 685.1/2 ⇒ 685.348.4 ⇒ 687.1/4
675.055.24	Staking machines. Stakers	675.14	Footwear leather. Boot and shoe leathers ⇒ 685.3
675.055.26	Rollers	675.15	Garment leathers. Gloving leathers. Headwear, millinery leathers. Lining leathers. Smallwear leathers ⇒ 675.18 ⇒ 685.4 ⇒ 687.1/4
675.055.27	Kicking equipment. Beating machines	675.16	Leather for travel and sports equipment. Leather for floor and wall coverings. Leather for upholstery and household use ⇒ 685.5/8
675.055.3	Equipment, machines for improving the surface	675.162	Bag hide. Case hide. Trunk leather Details by :685.51...
675.055.34	Smoothing machines for furs. Ironing equipment. Embossing equipment	675.164	Leather for sports and gymnasium equipment Details by :685.6...
675.055.35	Buffing machines	675.166	Leather for floor and wall coverings Details by :645.1/2 ⇒ 691.18
675.055.36	Glazing machines. Brushing machines	675.167	Upholstery leather. Furnishing leather Details by :684.7...
675.055.5	Equipment, machines for splitting. Equipment, machines for measuring and marking	675.168	Household leathers. □ Leathers for cleaning, polishing
675.055.6	Equipment, machines for cutting, clicking. Perforating machines. Punches		
675.056	Equipment, machines for finishing of skins, leather, fur		
675.056.1	Dyeing equipment. □ Vats. Paddles. Drums. Brushes. Stencils		
675.056.2	Impregnation equipment. Filling equipment. Currying equipment. Oiling equipment. Drying equipment. Conditioning equipment 675.056.2 ≡ 675.026.2		
675.056.3	Coating equipment. Waxing equipment. Final finishing equipment		
675.056.4	Cutting-out tools. Pliers. Shears		
675.056.5	Pilers		
675.058	Equipment, machines for special processes or treatments (other than those given above). □ Lustring equipment. Finishing equipment for hair pile, for linings		
675.059	Packaging equipment. Machines for packing leather in kips ⇒ 621.798		

675.18	Fancy, ornamental, decorative leathers. Artistic smallware leathers ⇒ 675.15	675.83	Wool waste processing industry ⇒ 637.62 ⇒ 677.3
675.2	Leather according to method of production, processing or special finish	675.84	Feather waste processing industry ⇒ 637.63 ⇒ 688.46
675.22	Split leather	675.85	Industries based on the processing of soft-tissue wastes (offal, entrails). □ Processing of blood, gut, fat ⇒ 637.661/.663
675.23	Rawhide. Raw leather	675.86	Industries based on the processing of hard-tissue wastes (skeletal wastes) ⇒ 637.61'848 ⇒ 637.64/.65
675.24	Leathers with special tanning. □ Oil-tanned leathers (e.g. chamois). Mineral-tanned leathers: chrome-tanned, alum-tanned, tawed leather	675.861	Industries processing bones
675.25	Mechanically treated leathers. □ Embossed leather. Buff. Perforated, punched leather	675.862	Industries processing teeth, tusks etc. Ivory industry ⇒ 688.1
675.26	Leather with special finish	675.863	Industries processing horns
675.261	Dyed, stained leather	675.864	Industries processing whalebone (baleen)
675.262	Morocco (saffian leather)	675.866	Industries processing oyster shells, tortoiseshell ⇒ 679.92
675.263	Natural grain leather	675.868	Industries processing fish scales. □ Manufacture of imitation pearls
675.264	Corrected grain leather. Artificially grained leather	675.87	Other industries based on the processing of leather and fur production wastes. □ Skin and fur pieces. Tanning wastes. Spent lime. Lime slab. Waste waters. Spent liquors
675.265	Suede. Nubuk (unglazed)	675.92	Imitation leather industry <i>The auxiliaries listed under 675.0... are not applicable at 675.92 and its subdivisions</i>
675.266	Parchment	675.92.01	General matters of process planning and control. Properties, faults etc.
675.267	Glazed leather. Polished leather	675.92.02	Manufacturing processes and operations
675.268	Coated leather. Enamelled hide. Patent leather	675.92.022	Material or stock preparation. Comminution, shredding, milling. Screening. Batching or metering. Plasticization
675.27	Russia leather. Sandal leather. Similar leathers	675.92.023	Mixing and blending. Preparation of impregnating, cementing and surface-strengthening compounds (grounds) etc.
675.28	Bonded, laminated leather. Multi-ply leather	675.92.023.2	Mixing and blending
675.6	Fur production and products <i>Denote details of fur production by the .0... auxiliaries at 675.0...</i> ⇒ 591.478	675.92.023.4	Preparation of impregnating and cementing compounds. Preparation of grounds (surface-strengtheners) etc.
675.61	Floor pelage. Wall pelage. Carpet and rug furs. Hanging furs. Quilting furs. Upholstery furs ⇒ 645.1/.2 ⇒ 684.73	675.92.023.5	Base preparation
675.62	Garment furs. Clothing fur. Lining and trimming fur ⇒ 685.3/.4 ⇒ 687	675.92.024	Heating. Drying. Wetting
675.68	Fur for decorative and other minor articles ◇ 675.68:688.42 Fur for artificial flowers ◇ 675.68:688.72 Fur for toys, dolls ◇ 675.68:745.5 Fur for handicrafts	675.92.024.2	Heating. Drying
675.8	Industries based on the processing of leather and fur production wastes. Industries based on the processing of by-products of slaughtering (carcass by-products). □ Waste collection, reprocessing, utilization <i>Use the .0... auxiliaries from 675.0... where appropriate</i> ◇ 675.862.026.1 Dyeing, bleaching ivory ◇ 675.862.08 Wastes of the ivory industry ⇒ 637.6 ⇒ 675.08	675.92.024.3	Wetting
675.81	Leather waste processing industry. □ Flesh wastes, e.g. glue. Leather shavings, cuttings, dust	675.92.026	Impregnating. Cementing. Surface-strengthening. Coating and deposition of films etc. based on macromolecular compounds ⇒ 678.026
675.82	Industries processing hair and bristle wastes ⇒ 687.9 ⇒ 688.47	675.92.026.2	Impregnating
		675.92.026.3	Cementing
		675.92.026.4	Mixing, dilution and cementing
		675.92.026.5	Grounding (surface strengthening)
		675.92.026.7	Coating. Deposition of films
		675.92.026.8	Gelation
		675.92.027	Moulding and forming processes
		675.92.027.2	Calendering
		675.92.027.3	Sinking. Extruding
		675.92.027.4	Stretching. Spinning
		675.92.027.6	Forming by casting and pressing
		675.92.027.7	Cutting out and stamping of blanks (blanking)
		675.92.027.8	Stock concentration, adjustment, dewatering. Collection of fibrous and cementing material

675.92.028	Curing, vulcanizing, reclaiming. Other chemical processes	676.014	Chemical constitution, composition (of pulps, paper etc.)
675.92.028.2	Vulcanization	676.014.2	Inorganic constituents (not additives) <i>Specify by :543... and :546...</i>
675.92.028.3	Rate and state of vulcanization (heat relations, criteria etc.)	◇ 676.014.2:543.613.2	Moisture or water content
675.92.028.4	Faults and difficulties in vulcanization or curing. □ Gas development. Shrinkage spots. Blooming. Scorching. Undervulcanization. Overvulcanization	◇ 676.014.2:543.632.22	Ash. Ash analysis
675.92.028.5	Devulcanization	◇ 676.014.2:546.17	Nitrogen content
675.92.028.6	Reclaiming processes	◇ 676.014.2:546.22	Sulphur content
675.92.028.7	Drying and vulcanization	676.014.3	Organic constituents
675.92.029	Finishing processes <i>675.92.029 ≡ 678.029</i>	676.014.32	Functional groups. □ Carboxyl content. Carbonyl content
675.92.03	Raw materials and semifinished products (initial and intermediate products)	676.014.33	Monosaccharides. Polysaccharides <i>Details by :547.45...</i>
675.92.031	Crude latex	676.014.34	Other organic constituents. □ Encrusting substances (not additives). Low-molecular accessory wood constituents <i>Details by :547...</i>
675.92.032	Coagulates. □ Raw natural rubber. Block rubber. Crêpe. Sheet rubber <i>675.92.032 ≡ 678.032</i>	◇ 676.014.34:547.914	Balsams. Resins
675.92.033	Rubber stocks. Moulding compounds	Reaction. Solubility. Chemical constants.	
675.92.035	Textiles. Fibres. Nonwoven fabrics. Paper. Cellulose. Wastes. □ Wastes from the leather industry. Shoemaking wastes	Odour. Taste ⇒ 543.92 ⇒ 66-915	
675.92.04	Additives. Accessory materials <i>675.92.04 ≡ 678.04</i>	676.014.361	Bleachability. Degree of delignification. □ Roe chlorine number. Permanganate number. Kappa number
675.92.049.9	Other minor additives. □ Reinforcing agents.	676.014.362	Reaction. Acidity. Alkalinity
675.92.05	Ageing inhibitors. Blowing agents	676.014.363	Solubility
675.92.052	Machines and equipment for imitation leather production	676.014.364	Odour. Taste
675.92.052	Equipment for preparation of materials. Comminuting equipment. Screening and batching equipment etc.	676.014.365	Copper number etc.
675.92.053	Equipment for preparing and impregnating stocks, grounds etc. □ Mixers. Blenders. Cement agitators, kneaders	676.014.4	Additives etc. □ Loadings. Pigments. Coatings. Size. Softening agents. Impregnating agents
675.92.054	Heating and drying plant. Wetting devices	676.014.8	Molecular, structural and morphological properties
675.92.056	Equipment for application of impregnating and cementing compounds, of grounds (surface-strengtheners) etc. □ Dipping tanks (jiggers). Spreaders. Mixing tanks	676.014.82	Molecular weight. Chain length. Degree of polymerization. Viscosity
675.92.057	Moulding and extruding equipment, presses etc.	676.014.84	Molecular structure. Molecular forces. Micellar and intermicellar structure. Crystallinity. Lattice structure
675.92.058	Curing and vulcanizing plant. Reclaiming, reactivation plant	676.017	Physical and mechanical properties and their testing. Processing properties. Serviceability etc.
675.92.06	Finished products of imitation leather <i>675.92.06 ≡ 678.06</i>	676.017.2	General physical and mechanical properties and constants
675.922	Rubber-type imitation leather	676.017.22	Dimensions. □ Surface area. Volume
675.923	Cardboard-type imitation leather	676.017.25	Weight. Substance (grammage). Specific gravity. Bulk
675.924	Plastic leather	676.017.27	Structure of sheet. □ Fibre orientation. Capillary structure
675.925	Imitation leather with nonwoven backing	676.017.28	Surface properties. □ Smoothness. Two-sidedness
675.926	Imitation leather with woven backing	676.017.3	Drainage properties. Fibre fractionation. Shive content. Zero-fibre content. Schopper-Riegler number. Freeness value
675.927	Imitation leather with paper backing	676.017.4	Strength properties
675.928	Plastics-type imitation leather (used for shoes, clothing, haberdashery etc.)	676.017.42	Strength properties of the material as a whole. □ Resistance to tension, flexion, shear, folding, winding. Breaking strength <i>Specify by :539...</i>
676	PULP, PAPER AND BOARD INDUSTRY <i>The auxiliaries listed under 676.0... are not applicable at 676.8 and its subdivisions</i>	676.017.43	Strength properties of specific regions
676.01	Generalities on pulp and paper production. Properties and testing	676.017.432	Bonding strength
676.011	Process design. Development of processes. □ Flow charts. Theory	676.017.435	Resistance to picking and peel-off
676.012	Process supervision. Consumption. Throughput. Yield	676.017.44	Strength properties under various conditions. □ Wet strength
676.012.2	Consumption of raw and auxiliary materials	676.017.5	Acoustic, optical, thermal, electrical and magnetic properties
676.012.3	Heat consumption. Energy consumption	676.017.54	Acoustic properties
676.012.4	Capacity. Throughput	676.017.55	Optical properties <i>Details by -026.613... and/or :535...</i>
676.012.5	Yield	◇ 676.017.55:535.31	Reflectance
676.013	Mill, plant, installations (generally). □ Design, construction, power source of mills	◇ 676.017.55:535.34	Transparency. Opacity
		◇ 676.017.55:535.68	Colour fastness, e.g. yellowing

676.017.56	Thermal properties <i>Details by :536...</i> ◇ 676.017.56:536.468 Combustibility	676.022.62	Digesting and cooking. Factors influencing digestion. □ Circulation of liquor, of steam. Volume of liquor. Ratio of chips to liquor. Temperature. Time
676.017.57	Electrical properties <i>Details by :537...</i> ◇ 676.017.57:537.226 Dielectric properties	676.023.1	Bleaching. Bleaching process. □ Bleaching temperature, time
676.017.58	Magnetic properties	676.024	Mechanical aftertreatment. Handling of pulp slurries. Paper and board stock preparation
676.017.6	Physical behaviour towards gases and liquids	676.024.1	Handling of slurry stock. Flow systems
676.017.62	Porosity. Permeability to air, gas, vapour	676.024.13	Pumping and conveying
676.017.63	Sizing properties generally. Permeability to water. Water resistance	676.024.14	Thickening. Dewatering. Diluting
676.017.64	Permeability to inks	676.024.15	Proportioning. Metering
676.017.65	Permeability to fats, oil, turpentine, kerosine	676.024.16	In-process storage
676.017.66	Water absorption. Adsorption. Swelling in water. Cobb test, Bendtsen test etc.	676.024.17	Mixing. Agitation
676.017.67	Absorption of other liquids. Swelling in liquids other than water	676.024.2	Washing. Washing in digester
676.017.68	Dimensional stability. □ Shrinking. Expansion	676.024.3	Fibre dispersion (before washing, screening etc.). Dispersion of waste paper
676.017.7	Physical and mechanical processing properties. Physical serviceability	676.024.4	Separation of dirt. Screening and cleaning. Despecking. Fractionation
676.017.72	Printability	676.024.5	Disintegration of pulp sheets, waste paper, broke and trim etc. □ Soaking. Defibration
676.017.73	Runnability	676.024.6	Beating and refining
676.017.74	Foldability. Wrappability. Mouldability	676.024.7	Chemical treatment of stock. Addition of nonfibrous additives and constituents
676.017.75	Erasability. Sealability	676.024.72	Colouring. Dyeing (in stock). Optical bleaching ⇒ 667.2
676.017.76	Varnishability	676.024.73	Loading. Filling
676.017.77	Washability. Filterability	676.024.74	Sizing (in stock and wet end)
676.017.8	Chemical processing properties. Chemical serviceability. □ Acetylation ability. Nitration ability. Chemical aggressivity. Toxicity	676.024.76	Addition of additives for improving paper properties. □ For improving absorbency, softness, opacity, whiteness, preservation (keeping) properties
676.019	Faults. Defects. Degradation	676.024.78	Addition of additives for combating manufacturing problems. □ For improving dewatering, drainage. For prevention of slime, foam (froth)
676.019.1	Surface defects. Visible faults, flaws 676.019.1 ≡ 620.191	676.024.82	De-aeration
676.019.2	Internal defects. Internal faults, flaws	676.026	Production of pulp, paper and board
676.019.3	Damage caused by various agents. Resistance to attack 676.019.3 ≡ 620.193	676.026.1	Flow and flow systems in approach flow and wet end. □ Flow spreading, distribution, concentration. Flow in stuff box, machine chest
676.02	Manufacturing processes	676.026.2	Formation and water removal on the wire
676.021	Handling, pretreatment, conservation of fibrous raw materials <i>Applicable only at 676.1 and its subdivisions</i>	676.026.21	Formation theory. Influencing factors. □ Drainage on the wire. Fibre orientation. Influence of shake. Fibre bonding. Influence of suction
676.021.2	Receipt at mill. Handling and storage in mill area	676.026.22	Papermaking by hand. □ Watermarking ⇒ 676.026.4 ⇒ 676.026.74
676.021.3	Preparatory treatments	676.026.23	Papermaking on fourdrinier machines
676.021.32	Reduction to processing size. □ Splitting, sawing of wood	676.026.24	Papermaking on cylinder machines (vat machines)
676.021.33	Preliminary moistening, soaking, steaming	676.026.25	Papermaking on combined machines. □ Yankee MG combined. Cylinder combined
676.021.34	Sorting. Cleaning. Removal of unwanted parts. Barking	676.026.26	Papermaking on other machines for wet processes. □ On machines for nonwovens by wet process
676.021.36	Comminution. Chipping. Shredding. Crushing	676.026.27	Papermaking by dry processes (air felting)
676.021.38	Finishing of chips, shreds etc. □ Cleaning. Drying	676.026.3	Sheet transfer and draw
676.021.8	Handling of materials. Conveyance to place of preprocessing, storage, use. Storage	676.026.4	Pressing and water removal in the presses. Molette watermarking ⇒ 676.026.22 ⇒ 676.026.74
676.022	Pulping processes <i>Applicable only at 676.1 and its subdivisions.</i> <i>Details of processes by -9... (from 66-9...)</i>	676.026.5	Drying. Removal of vapour and condensates. Cooling
676.022.2	Preliminary treatment (before grinder, digester etc.)	676.026.52	Transfer of water from sheet to air
676.022.3	Charging of pulping equipment. □ Feeding. Packing	676.026.53	Removal of moisture from the felts. Bottom-felt drying systems
676.022.4	Mechanical pulping (in grinders, defibrators etc.)	676.026.54	Removal of condensate from the cylinders. Removal of moist air from machine room. Heating and ventilating. Cooling
676.022.43	Grinding of logs		
676.022.44	Defibration of chips		
676.022.48	Various mechanical pulping processes. □ Pulping by explosion, by rotating brushes, by ultrasonics. Hammermilling		
676.022.6	Chemical pulping: Other pulping methods		
676.022.61	Pretreatment in digester. Prehydrolysis. Impregnation. Diffusion. Evacuation		

676.026.7	Sheet treatment, on and off machine	676.041	Accessory materials and additives for pretreatment of fibrous raw material
676.026.71	Smoothing. Calendering	676.042	Pulping chemicals and additives. □ Sodium hydroxide. Sodium sulphite. Calcium bisulphite. Limestone
676.026.72	Surface improvement and other protective treatments ⇒ 667.6/.7	676.043	Bleaching agents. □ Chlorine compounds. Calcium hypochlorite
676.026.721	Surface colouring	676.044	Additives used in mechanical aftertreatment, in stock preparation etc. Agents for washing, drying, filling, sizing (e.g. kaolin for filling). Agents for improving paper properties. For prevention of foam (froth)
676.026.722	Sizing	676.046	Additives to promote dewatering, drainage on the wire. Agents for improving paper surface. For surface sizing. For coating, waxing, varnishing, lacquering. For impregnation. For lamination (e.g. starch for board lamination)
676.026.723	Coating. Surface protection. □ Dip coating. Brush coating. Blade coating. Spray coating. Roll coating. Electrostatic coating. Bituminizing. Waxing. Solvent coating	676.05	Equipment, machines for production of pulp, paper and board <i>676.05 is roughly parallel with 676.02 from which additional subdivisions may be derived</i>
676.026.724	Internal protection. □ Impregnation. Saturation. Parchmentation. Vulcanizing	676.051	Equipment, machines for handling, pretreatment and storage of raw materials (pulpwood, vegetable fibres etc.)
676.026.73	Laminating of paper and board layers. Laminating of paper and board with films, foils. □ Adhesive methods. Hot-melt methods. Heat-and-pressure bonding	676.051.2	Equipment, machines for receipt at mill, for handling and storage in mill area
676.026.74	Patterning. □ Embossing. Imitation watermarks. Graining. Marbling. Application of abrasive surface. Application of flock, velour ⇒ 676.026.22 ⇒ 676.026.4 ⇒ 777.6	676.051.3	Equipment, machines for preparatory treatment
676.026.75	Elasticizing of paper. Making extensible, stretchable paper. □ Crêping	676.051.32	Equipment, machines for reduction to processing size. □ Splitters. Saws
676.026.76	Compressing. Consolidating (e.g. consolidation of fibreboard)	676.051.33	Equipment, machines for moistening, soaking, steaming
676.026.77	Conditioning and damping. Maturing	676.051.34	Equipment, machines for sorting, cleaning, barking. □ Washing drums for wood. Rag thrashers, willows
676.028	Finishing. Aftertreatment	676.051.36	Equipment, machines for comminution. □ Chippers. Shredders. Crushers
676.028.2	Unwinding and rewinding. Splicing and web joining	676.051.38	Equipment, machines for screening, cleaning, drying of chips, rags, cut straw etc. □ Chip screens. Cleaners. Dusters. Driers. Presses
676.028.3	Slitting the roll (rewinding)	676.051.8	Equipment, machines for handling of materials, for conveyance to place of storage, use etc. □ Conveyors. Elevators. Storage bins. Silos
676.028.4	Handling of sheets. □ Cutting. Trimming. Counting. Sorting. Folding	676.052	Equipment, machines for pulping processes
676.028.6	Wrapping and packing of rolls, sheets etc. ⇒ 621.798	676.052.1	Equipment, machines for preparation of liquor, cooking etc. ◇ 676.052.1:66.023.23 Absorption towers ◇ 676.052.1:66.045.1 Heat exchangers ◇ 676.052.1:66.067.3 Filters
676.028.8	Stacking, storage and handling prior to dispatch	676.052.2	Equipment, machines for preliminary treatment (before grinding, digesting etc.). □ Steaming vessels. Pressure chambers
676.03	Fibrous raw materials	676.052.3	Equipment, machines for charging of grinders, digesters etc. □ Chip packers. Chip distributors
676.031	Hardwood. Deciduous, broadleaf wood 676.031 ≡ 674.031	676.052.4	Equipment, machines for mechanical pulping. □ Grinders. Defibrators. Hammer mills
676.032	Softwood. Conifer wood 676.032 ≡ 674.032	676.052.6	Digesters
676.034	Vegetable fibres other than wood	676.053.1	Equipment, machines for bleaching of pulp. □ Bleach hollanders. Bleaching towers
676.034.2	Grass, straw etc.	676.054.1	Equipment, machines for mechanical aftertreatment. Equipment, machines for handling of pulp slurries, stock suspensions. □ Thickeners
676.034.22	Cereal straw	676.054.2	Equipment, machines for washing of pulp. □ Washing drums. Wash filters. Diffusers. Vacuum filters
676.034.23	Esparto. Alfa	676.054.3	Equipment, machines for fibre dispersion (before washing, screening etc.)
676.034.24	Flax. Hemp. Jute. Ramie	676.054.4	Equipment, machines for separation of dirt, for screening and cleaning, for despecking, for fractionation. □ Knotters. Sandtraps
676.034.25	Sugar cane (bagasse)		
676.034.26	Bamboo		
676.034.27	Reeds		
676.034.4	Fibres from bast and bark. □ Kozo (mulberry tree). Mitsumata. Gampi		
676.034.5	Leaf fibres. □ Sisal. Manila hemp. New Zealand flax. Tobacco		
676.034.6	Fibres from seeds. □ Linters		
676.034.8	Other vegetable fibres. □ Agricultural waste other than straw. Seaweed. Peat. Half-fossil fibres. Fossil fibres		
676.035	Organic synthetic and regenerated fibres		
676.036	Inorganic fibres		
676.038	Wastes and secondary fibres		
676.038.2	Waste paper. Waste board. Broke and trim		
676.038.3	Rags and textile waste (including wool and silk)		
676.038.4	Wood waste. □ Sawdust. Slabbing chips. Barks		
676.04	Nonfibrous raw materials and additives. Processing agents <i>Details of materials by :54..., :55..., :66..., :67...</i> ◇ 676.04:663.63 Process water		

676.054.5	Disintegrators and breakers for pulp sheets, waste paper, broke and trim etc. □ Pulpers. Hydrapulpers. Kollergangs. Defibrators	676.058.3	Cutters. Trimmers. Counting devices. Sorters. Folding devices
676.054.6	Beaters. Refiners	676.058.6	Equipment, machines for wrapping and packing of rolls, sheets etc. □ Balers
676.054.65	Beaters. □ Hollanders. Rod mills. Jordan beaters	676.058.8	Equipment, machines for stacking, storing etc.
676.054.66	Refiners. Curlators	676.06	Products according to specific characteristics, properties
676.054.7	Equipment, machines for chemical treatment of stock, for addition of fillers and additives, for dyeing, filling, sizing	676.062	Products according to overall permanence and durability. Permanent/durable paper etc. ◇ 676.062.014.362 pH of permanent/durable products ◇ 676.062.017.42:539.416 Folding strengths of permanent/durable products
676.054.72	Equipment, machines for preparation and addition of dyeing solutions	676.064	Products according to principal physical and chemical characteristics. Products according to composition <i>676.064.1 and 676.064.2 should be used only when the relevant main number does not distinguish between paper and board</i> ◇ 676.237.064.2 Drawing board
676.054.73	Equipment, machines for preparation and addition of loadings, fillers	676.064.1	Paper
676.054.74	Equipment, machines for preparation and addition of size (in stock)	676.064.2	Board
676.054.82	De-aerators	676.064.4	Wood-free products
676.056	Equipment, machines for sheet-making. Paper machines. Board machines	676.064.5	Wood-containing products
676.056.1	Equipment, machines for the approach flow. □ Stuff boxes. Machine chests	676.064.7	Single-ply, single-layer products
676.056.22	Equipment for papermaking by hand. □ Moulds. Deckles	676.064.8	Two-ply, double-layer products. Multi-ply, multilayer products ⇒ 676.264
676.056.23	Fourdrinier machines	676.065	Products according to bleach and colour properties
676.056.24	Cylinder machines (vat machines)	676.065.1	Unbleached products
676.056.25	Combined machines	676.065.2	Bleached products
676.056.26	Other machines for wet processes	676.065.5	Uncoloured products (not dyed or surface-coloured)
676.056.27	Machines for dry processes, for air felting etc.	676.065.6	Coloured products (dyed etc.)
676.056.3	Equipment, machines for forming of sheet, for couch, for sheet transfer and draw	676.067	Products according to sheet-treatment characteristics
676.056.31	Forming board. Suction boxes. Shaking equipment, for fibre orientation	676.067.1	According to finish
676.056.32	Couch rolls. Suction couch rolls. Dandy rolls. Watermark rolls. Transfer rolls	676.067.11	Uncalendered products
676.056.34	Pick-ups. Lick-ups	676.067.13	Machine-finished products (M.F.)
676.056.38	Equipment, machines for lamination of sheets, plies	676.067.14	Machine-glazed products (M.G.)
676.056.4	Equipment, machines for pressing and water removal in the presses, for Molette watermarking	676.067.15	Supercalendered products (S.C.)
676.056.42	Extractor rolls. Couch presses. Place presses. Suction presses. Smoothing presses. Baby rolls. Manchones. Reverse presses	676.067.16	Gloss-finished products (without coating)
676.056.45	Wires for paper machines and board machines. Equipment for wires	676.067.2	According to surface improvement, protection
676.056.46	Felts for paper machines and board machines. Equipment for felts	676.067.21	Unsize products
676.056.47	Equipment for leading, guiding the paper web. □ Dancing rolls	676.067.22	Sized products
676.056.48	Auxiliary equipment. □ Deckle straps. Deckle boards. Deflectors. Doctors. Rope carriers. Break indicators. Alarm devices. Apparatus for preventing static electricity	676.067.23	Uncoated products
676.056.49	Machinery for circulation of backwater, white water. Save-alls etc.	676.067.24	Coated products (without gloss calendering) ⇒ 676.226 ⇒ 676.26
676.056.5	Equipment, machines for drying. Driers	676.067.25	Coated products with supercalendering or high-gloss finishing
676.056.7	Equipment, machines for sheet treatment	676.067.4	According to ruling (e.g. ruled, graph-marked). According to watermarking
676.056.71	Equipment, machines for smoothing. Calenders, rolls, presses for finishing	676.08	Recovery of waste liquor. Regeneration of chemicals. Disposal of wastes.
676.056.72	Equipment, machines for surface improvement and protective treatment with chemicals. □ For surface colouring, sizing, coating	676.082	By-products. Treatment of effluent Waste liquors and their use (untreated). Recovery of pulping and bleaching chemicals. Regeneration. Composition. □ Green liquor. White liquor
676.056.73	Laminators. Hot presses etc.	676.083	By-products of pulping, obtained by fermentation processes. □ Alcohols. Organic acids
676.056.74	Equipment, machines for patterning. □ Embossing, marbling devices	676.084	Lignosulphonic acids. Lignosulphonates.
676.056.752	Crêping machines etc.	676.085	Degradation products of lignosulphonic acids
676.056.77	Conditioners for pulp, paper and board sheets		Tall oil. Tall oil skimmings. Products from digester gas relief ⇒ 665.947.3
676.058	Equipment, machines for roll finishing, for cutting, trimming, wrapping, packing etc.	676.087	Recovery of heat and vapour
676.058.1	Unwinders. Rewinders. Splicers. Roll changers		
676.058.2	Slitters		

676.088	Effluent treatment and disposal. Recovery of fibres, fillers, size etc. Treatment and disposal of waste gases <i>Details by :628.5...</i>	676.252.2	Towelling paper. Toilet tissue (lavatory paper). Napkin paper. Face tissues. Cellulose wadding ⇒ 676.267.6
676.1	Pulps	676.252.3	Blotting paper and board. Filter paper
676.15	Mechanical pulps. Woodpulp	676.252.32	Blotting paper and board ⇒ 686.851
676.153	Groundwood pulps	676.252.33	Filter paper and board. □ Paper for chromatography. Paper for electrophoresis. Paper for air and gas filtration
676.154	Mechanical pulp from chips	676.254	Paper for electrical purposes
676.157	Mechanical pulp from other processes	676.254.2	Paper with electrical insulation properties
676.16	Chemical pulps. Semichemical pulps. Other not purely mechanical pulps ⇒ 661.728	676.254.22	Electrical insulating paper
676.163	Sulphite pulps. □ Bisulphite pulp. Acid pulp	676.254.24	Condenser paper. Capacitor paper
676.164	Pulps prepared by alkaline processes. □ Sulphate pulps. Kraft pulp. Soda, lime and other alkaline pulps	676.254.26	Cable paper
676.166	Other chemical pulp. □ Pulp prepared by other acid methods, by use of chlorine, by use of organic solvents, by hydrotropic pulping	676.254.4	Electrical conductive paper
676.168	Other kinds of pulp	676.255	Cigarette paper. Other paper for the tobacco industry
676.2	Paper and board ◇ 676.2-027.33 Recycled paper and board	676.256	Base paper for various purposes. □ For light-sensitive, photosensitive paper. For carbon paper ◇ 676.256:691.145 Base paper for wallpaper
676.22	Printing paper and board ◇ 676.22:655.32 Paper for letterpress printing ◇ 676.22:655.344 Paper for offset litho printing ◇ 676.22:655.35 Paper for gravure printing ⇒ 686.8	676.258	Papers for other industrial uses. □ Indicator papers for chemical reactions (e.g. litmus paper). Textile substitutes. Spinning papers. Nonwovens
676.222	Newsprint	676.26	Coated paper. Laminated paper. Wet-strength paper. Impregnated paper. Decorative paper ⇒ 676.226
676.223	Book and magazine papers	676.262	Coated paper and board (generally). □ Chromo paper. Art paper. Photographic paper (light-sensitive, photosensitive paper). Heliographic paper
676.224	Special printing papers	676.263	Coated copy and transfer paper. Adhesive, adhesion-reducing and waxed paper and board
676.224.1	Featherweight paper. India paper. Bible paper	676.263.2	Coated copy paper. Computer paper
676.224.2	Cheque paper (safety paper). Banknote paper. Postage-stamp paper	676.263.3	Carbon and other copying papers. □ Carbon paper (with pigment coating). Carbon-impregnated paper ('carbonless' copying paper)
676.224.3	Map paper	676.263.4	Transfer paper. □ Decalcomania paper ⇒ 667.649.2 ⇒ 667.661.5
676.224.4	Music-printing paper	676.263.5	Adhesive and other sticking papers. □ Label and poster papers. Gummed paper.
676.224.5	Label-printing paper	676.263.6	Pressure-sensitive adhesive paper
676.224.9	Other special printing papers. □ Luxury printing paper	676.263.8	Adhesion-reducing paper. Release paper
676.226	Coated printing papers ⇒ 676.26	676.264	Waxed and paraffin paper and board
676.227	Printing board. □ Ivory board. Bristol board. Carte-de-visite board (visiting-card board) ⇒ 676.27	676.265	Laminated paper and board. Papers with two or more plies, layers. Lacquered paper and board ◇ 676.264:669 Metal-foil paper ⇒ 676.064.8 ⇒ 686.126
676.23	Writing paper. Drawing paper and board etc. ⇒ 686.8		Papers coated with powder, flock etc. □ Abrasive papers (Carborundum paper. Sandpaper. Emery paper. Glasspaper). Flock paper. Velour paper ⇒ 621.921.6 ⇒ 666.189.4 ⇒ 666.792
676.232	Writing paper. Draft paper. Copy paper. Notepaper. Bond paper etc.	676.266	Wet-strength paper. Impregnated paper etc.
676.233	Ledger paper. Account-book paper etc.	676.266.2	Vegetable parchment
676.234	Typing paper. Air-mail paper etc.	676.266.3	Wet strength paper
676.236	Stamped paper	676.266.4	Oiled paper etc.
676.237	Drawing paper and board. Watercolour paper. Crayon paper ⇒ 744.36	676.266.7	Asphalt paper. Bitumen paper etc.
676.24	Greaseproof, packaging and wrapping paper ⇒ 621.798.15	676.267	Decorative papers (treated by mechanical means)
676.242	Glassine. Greaseproof paper ⇒ 676.266.2	676.267.2	Patterned paper. □ Marble paper
676.242.2	Glassine. Transparent paper. Tracing paper	676.267.6	Crêpe paper. Clupak ⇒ 676.252.2
676.242.3	Greaseproof paper. Pergamyn. Butter paper	676.267.7	Embossed paper
676.244	Sack paper. Bag paper. Wrapping paper. Liners		
676.248	Special kinds of packaging paper and wrapping paper		
676.25	Sanitary paper. Papers for various industrial and other uses		
676.252	Sanitary paper. Absorbent paper		

676.27	Board ⇒ 676.227 ⇒ 676.262 ⇒ 676.264	676.816.8	Paper streamers ⇒ 688.756
676.272	Boxboard. Folding boxboard	676.817	Lace paper. Imitation lace. Perforated paper
676.273	Solid fibre container board. Corrugated board	676.818	Paper and board for forms, tickets, postage stamps, receipts etc.
676.274	Corrugating medium. Fluting. Liner board. Kraft liner board	676.82	Objects of paper or thin board, folded and glued
676.275	Millboard. Chipboard	676.82.05	Equipment and machines
676.277	Pressboard. Polished board. Presspahn. Moulded products	676.821	Paper bags. Paper sacks ⇒ 621.798.15
676.278	Other boards	676.822	Envelopes. Letter cards etc.
676.28	Fibre building board. Hardboard. Wallboard. Roofing and flooring felt ⇒ 691.14	676.823	Document cases
676.282	Hard building board (pressed)	676.824	Containers of folded or pressed paper. Folders
676.283	Soft building board (not pressed)	676.825	Lampshades of folded paper
676.284	Insulating board	676.826	Wrappers
676.285	Acoustic board	676.828	Folded paper cups
676.287	Roofing felt. Flooring felt	676.83	Objects of paper or thin board, rolled and glued. □ Tubes. Spools. Reels. Paper cones. Conical lampshades
676.8	Paper ware. Cardboard industry. Objects made from paper or cardboard <i>The auxiliaries listed under 676.0... are not applicable at 676.8 and its subdivisions</i>	676.83.05	Machines
676.81	Objects made from cut paper or thin board	676.83.052	Tube-winding machines. Cutting devices
676.81.02	Methods of production 676.81.02 ≡ 676.81.05	676.84	Assembled objects of paper or board in general. □ Cardboard boxes
676.81.05	Equipment and machines	676.84.02	Methods of production 676.84.02 ≡ 676.84.05
676.81.052	Equipment, machines for stamping. Stamping dies. Longitudinal and transverse cutters	676.84.05	Machinery and equipment
676.81.053	Equipment, machines for perforating. □ Punches. Drilling devices	676.84.051	Equipment, machines for preparatory processes
676.81.054	Equipment, machines for bending, folding, crêping. Devices for joining by crimping, serration, perforation	676.84.051.1	Equipment, machines for scribing, for grooving without bending, for indenting, for slotting and notching (without waste), for shearing, for dividing (without waste), for edge cutting, for ruling, for arching, for bending, for profiling. Combined devices
676.81.055	Goffering machines. Pleating machines (e.g. for production of lampshades and paper garlands)	676.84.051.2	Equipment, machines for bending of board for producing grooves and slots. Equipment, machines for pressing, for crimping
676.81.056	Equipment, machines for forming, pressing, embossing	676.84.051.3	Equipment, machines for stamping, for punching, for perforating. Equipment, machines for cutting out, for blanking, for slotting (with waste) etc.
676.81.057	Equipment, machines for cutting of fringes and trimming of edges and borders	676.84.051.5	Equipment, machines for producing complete cut-outs for boxes. Automatic pressing and stamping machines
676.81.058	Equipment, machines for application of adhesive. Gumming and gluing machines	676.84.052	Equipment, machines for assembling of board into boxes etc.
676.811	Adhesive labels. Tie-on labels	676.84.052.1	Cardboard-box machines in general. Folding, forming machines. Devices for gumming, gluing (with assembling)
676.812	Paper shreds. Paper flock. Confetti ⇒ 688.756	676.84.052.2	Equipment, machines for producing boxes with drawers, trays. □ Matchbox machines
676.813	Common types of cut card, visiting cards etc. Postcards. Picture postcards	676.84.052.3	Embossing and stamping presses for boxes
676.814	Playing cards	676.84.053	Equipment, machines for assembling with metal fasteners (with rivets, staples etc.)
676.815	Cards for punching. Punched cards	676.84.054	Equipment, machines for applying glue. □ For coverings, linings, edge and corner pieces
676.815.2	Punched cards for manual information retrieval. Edge-punched, slotted cards etc. Optical coincidence cards ⇒ 686.846	676.84.059	Equipment, machines for finishing
676.815.4	Punched cards for the control of machines (e.g. accounting machines, textile machines) ⇒ 004.081.1 ⇒ 677.054.87	676.85	Objects made from paper and board in sheets, stamped and moulded. □ Stamped bottle caps, covers. Stamped boxes, cases
676.816	Paper tapes. Paper rolls	676.86	Objects moulded from paper pulp
676.816.2	Adhesive paper tapes. Sealing tape	676.861	Objects of lacquered board. □ Boxes, caskets, furniture of lacquered board
676.816.3	Paper rolls for telegraphy (teleprinter tape, ticker tape)	676.863	Figures, stucco, ornaments of paper pulp
676.816.5	Paper rolls for punching. Rolls for punched-tape machines (e.g. tape-typewriters, accounting machines) ⇒ 004.081.2	676.865	Solid objects of paper pulp
676.816.6	Paper rolls for spinning of paper thread etc. ⇒ 677.57	676.866	Flat objects of paper pulp
		676.867	Dished, concave objects of paper pulp. □ Plates. Dishes

676.868	Various hollow objects of paper pulp. □ Containers, beakers, bottles. Buckets, pails. Tubular objects of paper pulp: spools, cigar holders etc.	677.014.8	Molecular and structural properties of fibrous materials and their testing
676.87	Various chemically treated or impregnated objects	677.016	Examination of finished textiles with regard to results of treatment. Examination of dyestuffs and textile auxiliaries
676.872	Articles of vulcanized fibre	677.016.1	Examination of textile finishing processes in general
676.874	Waterproof, greaseproof, boilable paper containers	677.016.2	Examination of preparatory treatments (their effects). □ Effects of cleaning, of bleaching
677	TEXTILE INDUSTRY <i>Details of textile machinery and processes by -1/-9 (from 62-1/-8 and 66-9). The subdivisions of -48 are listed here</i> ⇒ 687	677.016.4	Examination of dyeing and printing processes (their effects)
677-48	Forms and shapes of textile materials	677.016.41	Colour determination and measurement
677-485	Spinning solution	677.016.42	Determination of type of dyestuff ⇒ 667.2
677-486	Forms and shapes of fibres and filaments	677.016.43	Determination of quantity of dyestuff applied
677-486.1	Short natural fibres or cut staple (also staple fibre)	677.016.47	Colour-fastness and its determination
677-486.2	Continuous filaments	677.016.6	Examination of finishing effects
677-486.3	Profiled and hollow fibrous materials	677.016.62	Qualitative analysis of the type of dressing agents applied
677-486.31	Profiled	677.016.64	Quantitative analysis of applied dressing agents
677-486.32	Hollow	677.016.65	Determination of the distribution of dressing agents
677-486.4	Forms of fibrous material	677.016.67	Determination of finishing effects
677-486.411	Straight. Stretched	677.016.8	Examination of dyestuffs and textile auxiliaries ⇒ 677.027.42 ⇒ 677.04
677-486.412	Crimped. Curled	677.017	Physical, chemical and mechanical processing properties. Serviceability of textile materials. Testing of these properties
677-486.5	Fibrous materials in flock form	677.017.2	Dimensions. Weight. □ Relationships between dimensions and weight
677-486.6	Fibrous materials in fleece form. Web	677.017.3	Arrangement and number of fibres or single filaments in rovings, slivers, yarns etc. Arrangement and number of fibres or yarns in sheet textiles. □ Fibre distribution.
677-486.7	Fibrous materials in sliver form	677.017.4	Smoothness. Twist. Fabric structure. Warp and weft. Structure of knitted goods, nets, felts etc. Behaviour towards mechanical stresses. Strength. Elongation. Elasticity <i>Specify properties by :539.3/6, and testing by :620.17...</i>
677-486.8	Slubbings. Rovings	677.017.5	Acoustic, optical, thermal, electrical properties
677-487	Forms and shapes of yarns	677.017.54	Acoustic properties. □ Sound insulation
677-487.1	Monofilament	677.017.55	Optical properties <i>Details by :535...</i> ⇒ 677.019.1
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677-487.5	Forms of yarn package	677.017.57	Electrical properties <i>Details by :537...</i>
677-487.51	Loose	677.017.6	Physical behaviour towards gases, liquids, solids. Behaviour towards radiation other than light
677-487.511	Rolls	677.017.62	Behaviour towards gases. □ Permeability to air, steam, other gases
677-487.512	Hanks	677.017.63	Behaviour towards liquids
677-487.513	Cakes	677.017.632	Wettability. Sorption ⇒ 532.63
677-487.52	Packages formed on tubes, beams etc.	677.017.633	Repellent properties
677-488	Forms of fabric, including film	677.017.635	Dimensional changes (due to liquids). Dimensional stability. Shrinkage
677-488.2	Plain fabrics	677.017.636	Permeability to liquids
677-488.3	Pile fabrics	677.017.65	Behaviour towards solids. □ Dust permeability
677-489	Blends of fibres	677.017.67	Behaviour towards radiation
677.01	Generalities on the textile industry. Properties, faults and testing of textile materials. Examination of dyestuffs, textile auxiliaries and finishing effects	677.017.7	Processing properties and their testing. Processability
677.011	Operating and process planning and other general questions. □ Statistical methods in testing. Specifications. Standards	677.017.8	Serviceability and its testing
677.014	Chemical composition of fibrous materials. Chemical properties and their testing (including fibre analysis). Molecular and structural properties and their testing	677.017.82	Resistance to abrasion. □ Pilling
677.014.2/3	Chemical composition and properties determined by chemical test methods.	677.017.83	Crease recovery. Crease retention. Pleat retention
677.014.2	□ Molecular weight. Ash content. Inorganic materials content. Resistance to chemicals		
677.014.3	Chemical composition and properties determined by physical test methods. □ Liquid, moisture content		
677.014.5	Identification and differentiation of fibrous materials. Qualitative fibre analysis		
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677.017.84	Resistance to light and atmospheric influences. Resistance to storage, to tropical climates ⇒ 677.016.47	677.021.125.5	Aftertreatment
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677.027.47	Aftertreatment of dyeings	677.027.562.4	Conversion printing
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677.027.51	Preparatory processes	677.027.562.8	Special processes. □ Lithography. Painting of fabrics
677.027.511	Designing	677.027.566	Embossed printing. □ Colour embossing. Colouring over embossments. Imitation embroidery ⇒ 677.027.615.17 ⇒ 677.027.623.56
677.027.512	Apparatus. Production of printing blocks	677.027.57	Aftertreatments for printing. □ Treatment with metal salts and synthetic resins
677.027.512.1	Production of printing blocks	677.027.571	Developing. Fixing (where not included in printing process)
677.027.512.2	Engraving and photochemical treatment of plates	677.027.572	Washing-out of surplus dyes, chemicals, thickeners
677.027.512.3	Production of screens	677.027.573	Soaping
677.027.512.5	Production of rollers	677.027.575	Aftertreatment of prints to improve colour fastness <i>Details by :535.68...</i>
677.027.512.6	Preparation of the doctor blade	677.027.58	Printing on nontextile materials
677.027.513	Preparation of dyes and auxiliaries	677.027.6	Finishing
677.027.513.1	Preparation of printing pastes	677.027.61	Mechanical finishing
677.027.513.2	Preparation of thickeners	677.027.611	Milling. Fulling
677.027.514	Preparation of the textile materials	677.027.612	Crabbing. Blowing. Decatizing
677.027.514.1	Priming with mordants, naphthols, oils, leuco-esters	677.027.613	Raising (nap raising etc.). Shearing. Sueding. Brushing. Beating. Breaking
677.027.514.2	Protective measures against doctor streaks etc.	677.027.615	Calendering. Pressing. Forming ◇ 677.027.615-966.1 Permanent finish
677.027.514.4	Fastening the cloth to the printing table (by pinning, by gluing)	677.027.615.1	Calendering
677.027.514.6	Chlorination of wool before printing ⇒ 677.027.264	677.027.615.11	Roller calendering. □ Chasing
677.027.52	Printing processes according to the dye used	677.027.615.12	Friction calendering. Chintzing
677.027.523	Direct printing. Discharge printing. Resist printing	677.027.615.13	Beetling
677.027.523.1	With directly absorbed dyes	677.027.615.15	Calendering with fluted rollers. Rippling. □ Schreinerer

677.027.615.16	Moiré finishing	677.027.625.16	Protection against high temperatures and fire. □ Improvement of resistance to ironing.
677.027.615.17	Embossing. Goffering ⇒ 677.027.566 ⇒ 677.027.623.56		Reduction of flammability. Fireproofing
677.027.615.2	Pressing. Pleating. Boarding. Setting	677.027.625.17	Protection against electrical effects. □ Prevention of static charges (antistatic proofing). Alteration of electrical conductivity
677.027.615.21	Pressing		Alteration of resistance to chemical influences
677.027.615.22	Pleating	677.027.625.2	Towards gases
677.027.615.23	Boarding. Setting. Thermosetting	677.027.625.21	Towards acids, bases, salts
677.027.618	Mechanical shrinking and stretching	677.027.625.22	Towards oxidizing, reducing agents (bleaches)
677.027.62	Chemical finishing. □ Impregnating. Chemical modification of textile materials	677.027.625.23	Towards solvents, swelling agents
677.027.621	Preparatory processes	677.027.625.24	Alteration of resistance to biological influences
677.027.622	Alteration of chemical properties (including dyeing properties) and structural properties of fibres	677.027.625.3	<i>Details by :58... and :59...</i>
677.027.622.1	Etherification. Esterification. Saponification	677.027.625.5	Alteration of resistance to combined influences
677.027.622.2	Treatment with aldehydes, amides, amines	677.027.625.51	Reduction of felting of wool and other animal hairs
677.027.622.3	Immunization. Animalization. Philanization	677.027.625.52	Protection against climatic influences
677.027.622.4	Chlorination	677.027.625.53	Increasing resistance to wear
677.027.622.5	Modification by copolymers (graft polymerization)	677.027.625.54	Increasing resistance to ageing
677.027.622.6	Alteration of swelling capacity of fibres (reducing, increasing)	677.027.625.55	Increasing resistance to soiling
677.027.623	Alteration of external properties of textiles	677.027.65	Coating (by means of steaming, spraying, spreading, doctoring, pouring, sprinkling) ⇒ 621.793
677.027.623.1	Improvement of handle, feel	677.027.651	Coating with substances of low molecular weight. □ Coating with metals, drying oils, paraffins, waxes, tar, bitumen
677.027.623.2	Weighting. Loading. Filling	677.027.652	Coating with substances of high molecular weight <i>Specify by :678.5/7</i>
677.027.623.3	Alteration of volume. □ Bulking	677.027.653	Coating with fibres (flock). Flock coating
677.027.623.4	Stiffening	677.027.66	Laminating. Backing
677.027.623.5	Alteration of appearance	677.027.8	Final processes. □ Grading. Pressing. Measuring. Packaging. Stitching. Labelling
677.027.623.51	Delustring. Lustring. Imparting translucency	677.03	Raw materials of the textile industry <i>Specify by :633.5..., :636..., :638.2..., :661.728..., :676..., :678... etc.</i>
677.027.623.52	Mercerizing	677.04	Textile auxiliary materials
677.027.623.53	Other caustic treatments. □ Lixivation	677.042	Generally applied textile auxiliaries
677.027.623.54	Alteration in transparency and opalescence. Linerizing. □ Organdie finish	677.042.1	Water-softening agents ⇒ 543.32 ⇒ 628.164 ⇒ 661.187.76
677.027.623.56	Crêping. Embossing ⇒ 677.027.566 ⇒ 677.027.615.17	677.042.2	Surface-active agents (surfactants), e.g. wetting agents, detergents ⇒ 661.185
677.027.623.58	Application of colour and fluorescent brighteners during chemical finishing	677.042.4	Protective colloids. Fibre-protecting agents (e.g. ultraviolet absorbers). Swelling agents. Conditioning agents
677.027.623.7	Alteration of odour, smell	677.042.5	Solvents
677.027.625	Alteration of resistance to external influences	677.042.6	Buffers. Catalysts. Stabilizers
677.027.625.1	To physical influences	677.042.7	Oxidizing agents. Reducing agents
677.027.625.11	To purely mechanical forces	677.044	Special textile auxiliaries
677.027.625.111	To tension, pressure, shear. □ Increasing of dry and wet strength. Increasing of sewing strength	677.044.1	Auxiliaries for the manufacture of man-made filaments and fibres
677.027.625.112	To bending and folding (crease resistance). □ Creaseproofing. Improving of crease retention	677.044.12	Stabilizing agents. □ Ultraviolet stabilizers. Heat stabilizers
677.027.625.113	To rubbing, chafing. □ Increasing of abrasion resistance	677.044.13	Softening agents. Hardening agents
677.027.625.12	Alteration of surface tension between liquids and textiles	677.044.132	Softening agents
677.027.625.121	Permeable water-repellent, oil-repellent and dirt-repellent finishes	677.044.134	Hardening agents
677.027.625.122	Increasing of absorbency and wetting power	677.044.14	Agents for delustring during spinning. Mass-colouring agents (spin-dyeing agents). Optical brightening agents
677.027.625.13	Proofing of fabrics ⇒ 677.027.65 ⇒ 677.027.66	677.044.15	Crimping agents. Agents for forming hollow fibres
677.027.625.131	Proofing against gases. □ Making impervious to air	677.044.16	Agents for precipitation or coagulation of the thread
677.027.625.132	Proofing against liquids. □ Waterproofing	677.044.17	Modifiers. Doping agents
677.027.625.133	Proofing against solids. □ Down-proofing. Dust-proofing	677.044.2	Auxiliaries for fibre and yarn processing
677.027.625.135	Proofing against radiation of all types	677.044.22	Lubricants. Softening agents
677.027.625.14	Protection of the fibrous material itself against radiation of all types	677.044.23	Sizing agents
677.027.625.15	Improvement of heat-retaining and cold-retaining properties (alteration of thermal conductivity). Modification of infrared reflectivity	677.044.25	Smoothing agents. Waxing agents

677.047	Textile dyeing and finishing auxiliaries 677.047 $\equiv$ 677.027	677.051.151.26	Equipment, machines for processing tow. Shaking and refining apparatus
677.047.1	Auxiliaries for all or several stages of dyeing and finishing Other than those at 677.042... and 677.044...	677.051.151.27	Equipment, machines for recovery of fibres from waste. Shive decorticators
677.047.2	Auxiliaries for preparation and pretreatment (for dyeing and finishing)	677.051.151.5	Equipment, machines for production of fibres from leaves
677.047.4	Auxiliaries for dyeing $\Rightarrow$ 667.2	677.051.151.7	Equipment, machines for production of fibres from fruits
677.047.5	Auxiliaries for textile printing $\Rightarrow$ 667.526	677.051.152	Equipment, machines for production of fibres from seeds. Gins (cotton gins)
677.047.6	Auxiliaries for finishing	677.051.153	Equipment, machines for production of wools, hairs and silks
677.05	Textile machinery and equipment Details by -1/-9 (from 62-1/-8 and 66-9)	677.051.154	Equipment, machines for production and extraction of mineral fibres
677.051.1	Equipment, machines for manufacturing man-made filaments and fibres. Machines and accessories for preparing natural fibres for spinning	677.051.155	Sorting machines and equipment
677.051.12	Equipment, machines for manufacturing man-made filaments and fibres	677.051.156	Equipment, machines for scouring, washing, chemical cleaning of fibres (with aqueous solutions or organic solvents)
677.051.122	Equipment, machines for treatment of raw materials prior to dissolution or melting 677.051.122 $\equiv$ 677.021.122	677.051.157	Equipment, machines for carbonizing of fibres
677.051.123	Equipment, machines for forming of spinning solution or melt, for treatment prior to extrusion 677.051.123 $\equiv$ 677.021.123	677.051.158	Equipment, machines for oiling, batching, brightening of fibres. Equipment, machines for other dressings
677.051.125	Equipment, machines for extrusion, for subsequent treatment of extruded fibres	677.051.16	Equipment, machines for cutting, breaking, opening, cleaning, blending
677.051.125.1	Apparatus for feeding and final handling of the spinning solution	677.051.162	Equipment, machines for cutting
677.051.125.2	Spinning machines	677.051.163	Equipment, machines for breaking or tearing (including rag pulling, fibre recovery). ☐ Tow-to-top machines
677.051.125.22	Bobbin and rotary spinning machines	677.051.164	Equipment, machines for opening, for cleaning. ☐ Bale breakers. Willows. Scutchers. Devils
677.051.125.23	Centrifugal spinning machines. Pot spinning machines	677.051.166	Equipment, machines for blending. Fibre blenders
677.051.125.24	Spinning machines with continuous aftertreatment	677.051.17	Equipment, machines for carding and web dividing
677.051.125.26	Equipment, machines for solution spinning	677.051.173	Roller cards
677.051.125.262	Spinnerets	677.051.174	Flat cards
677.051.125.264	Spinning bath. Steam chamber. Thread guides. Godets. Drawing rollers	677.051.178	Parts of cards. ☐ Carding parts: cylinders, workers, strippers, fancies, flats, doffers. Card clothing. Web dividers and condensers
677.051.125.27	Equipment, machines for melt spinning	677.051.18	Equipment, machines for hackling, combing, backwashing, drawing
677.051.125.272	Spinning heads	677.051.182	Spreaders
677.051.125.274	Blowing box. Thread guides. Drawing rollers	677.051.183	Hackling machines
677.051.125.277	Apparatus for various melt spinning methods	677.051.185	Combing machines
677.051.125.28	Apparatus for reaction spinning	677.051.187	Backwashing machines
677.051.125.3	Drawing machines. Draw twisters	677.051.188	Machines for drawing slivers. Drawing frames
677.051.125.4	Uptwisters	677.052	Equipment, machines for spinning, twisting, cabling. Equipment, machines for manufacturing textured yarns
677.051.125.5	Equipment, machines for aftertreatment	677.052.2	Roving frames. Flyer frames
677.051.125.56	Equipment, machines for surface treatment $\Rightarrow$ 677.052.78	677.052.3/5	Spinning machines
677.051.125.7	Driers	677.052.3	Ring spinning frames
677.051.125.8	Cutters for staple fibres	677.052.31	Ring spinning frames according to formation of cop. ☐ With upright spindles. With suspended spindles
677.051.127	Machines for film. ☐ For split film	677.052.32	Ring spinning frames for woollen and condenser yarns
677.051.128	Collecting apparatus	677.052.33	Ring spinning frames for cotton and worsted yarns
677.051.15/18	Equipment, machines for producing and extracting natural fibres and for preparing fibres for spinning	677.052.35	Ring spinning frames for bast fibre yarns
677.051.15	Equipment, machines for production and extraction of natural fibres	677.052.36	Ring spinning frames for direct spinning of man-made fibres (tow-to-yarn)
677.051.151	Equipment, machines for production of fibres from stems, leaves, fruits. Equipment, machines for decorticating and retting	677.052.4	Various spinning machines
677.051.151.2	Equipment, machines for production of fibres from stems	677.052.43	Can spinning machines. Pot spinning machines. Centrifugal spinning machines
677.051.151.22	Rippling machines. Deseeding machines. Bundling apparatus	677.052.45	Flyer spinning machines. Cap spinning machines
677.051.151.23	Equipment, machines for retting	677.052.47	Gill spinning machines
677.051.151.25	Equipment, machines for production of long vegetable fibres. ☐ Decorticating machines. Breaking machines. Scutching machines	677.052.48	Open-end spinning machines
		677.052.481	Machines for preparation of feed material 677.052.481.5/8 $\equiv$ 677.051.15/18

677.052.482	Machines for fibre separation and cleaning. Trash extraction machines	677.053.277	Machines for laying yarns in cans
677.052.483	Devices for fibre transfer and conveying	677.053.28	Equipment, machines for making-up of yarns. □ Machines for moistening, for brightening yarns
677.052.484	Devices for yarn formation	677.053.29	Parts. Accessories
677.052.485	Yarn take-up devices. Doffing devices	677.053.292	Parts. □ Thread guides. Tensioners. Yarn cleaners. Creeling devices
677.052.49	Other spinning machines	677.053.296	Accessories. □ Tubes and other supports. Knotting apparatus
677.052.5	Mule spinning machines (self-acting mules)	677.053.7	Equipment, machines for warping, sizing, beaming etc. of yarns (and folded yarns)
677.052.6	Doublers and twisters ⇒ 679.7.05	677.053.72	Warping creels. Warp beam stands
677.052.62	Flyer doublers and twisters	677.053.73	Sectional warpers
677.052.63	Ring doublers and twisters	677.053.74	Beam warping machines
677.052.64	Uptwisters (twisting frames with several decks)	677.053.75	Sizing machines
677.052.65	Two-for-one twisters	677.053.76	Beaming machines
677.052.66	Other twisting machines ◇ 677.052.66-137 Centrifugal twisting machines	677.053.77	Equipment, machines for preparing the warp for further processing. □ Leasing machines. Reaching-in machines. Drawing-in machines. Reeding or denting machines. Warp tying machines. Drop-wire pinning machines
677.052.68	Stranding machines	677.054	Looms. Weaving machines and auxiliary devices ⇒ 681.124
677.052.78	Equipment, machines for manufacture of textured yarns. □ Gear crimpers. Extrusion texturing machines	677.054.3	Looms. Weaving machines
677.052.9	Parts of spinning, twisting, cabling, texturing machines	677.054.32	Wide looms 677.054.32 ≅ 677.054.8
677.052.93	Spindles. Flyers. Ring rails. Rings and travellers	677.054.33	Narrow-fabric looms. Smallware looms 677.054.33 ≅ 677.054.8
677.052.94	Drafting devices	677.054.35	Circular looms 677.054.35 ≅ 677.054.8
677.052.942	Roller drafting devices	677.054.39	Other types of loom
677.052.943	Slip-draft devices	677.054.5	Weaving machines according to fabrics manufactured
677.052.944	Twist tube drafting devices	677.054.52	Machines for weaving plain fabrics
677.052.945	Gill drafting devices	677.054.53/54	Machines for weaving pile fabrics
677.052.946	Porcupine roller drafting devices	677.054.53	Machines for weaving weft-pile fabrics (e.g. chenille)
677.052.947	Other drafting devices	677.054.54	Machines for weaving warp-pile fabrics
677.052.948	Parts of drafting devices. □ Rollers. Roller coverings. Weighting devices	677.054.57	Machines for weaving special types of carpet. □ Machines for weaving Axminster carpets
677.052.95	Various motions and devices ◇ 677.052.95-59 Yarn stop motions	677.054.58	Machines for knotting carpets
677.052.952	Thread guides	677.054.8	Weaving mechanisms and motions
677.052.953	Thread cleaners	677.054.82	Warp let-off and fabric take-up motions. Temples
677.052.954	Yarn brakes. Tension regulators	677.054.823	Warp let-off motions. □ Warp beam brakes. Devices for regulating warp tension
677.052.955	Antibalooning devices, separators	677.054.825	Fabric take-up motions. □ Cloth rollers
677.052.956	Thread-feeding devices. Rails. Automatic donning and doffing devices	677.054.827	Temples
677.052.96	Tubes. Bobbins. Pirns etc.	677.054.83	Shedding mechanisms, motions
677.052.97	Other parts	677.054.832	Tappet motions
677.052.972	Clearing rollers. Clearing plates. Pneumatic clearers	677.054.833	Dobbies
677.052.974	Apparatus for wetting, wet spinning or wet twisting. □ Troughs	677.054.834	Jacquard machines
677.052.976	Devices for aftertreatment	677.054.835	Shedding motions for multiphase shed
677.052.98	Special parts of particular types of machine	677.054.836	Combined shedding motions (including selvedge-forming unit)
677.052.982	Special parts of roving frames	677.054.838	Parts of shedding motions. Shedding accessories
677.052.983	Special parts of ring spinning frames	677.054.838.2	Harness. Healds etc.
677.052.984	Special parts of special spinning frames	677.054.838.3	Jacquard attachment. □ Jacquard harness. Comber boards
677.052.985	Special parts of mule spinning machines	677.054.838.4	Devices for control of shedding. □ Cards. Card cylinder. Needles, hooks, looping wheels. Pegged cylinder. Knives
677.052.985.2	Headstock	677.054.838.5	Tappets. Shaft-lifting levers
677.052.985.4	Quadrant. Winding block	677.054.838.6	Reverse motions. Spring reverse motions
677.052.985.6	Carriage	677.054.838.7	Devices for shed clearing
677.052.985.8	Shaper rail. Cam plate. Shaper plates. Winding motions (winding faller, counter faller)	677.054.84	Picking mechanisms, motions. Weft-changing motions
677.052.988	Special parts of machines for manufacturing textured yarns	677.054.842	Devices for insertion of weft
677.053	Equipment, machines for winding, reeling, balling, warping, sizing, beaming etc. of yarns	677.054.842.2	Shuttles. Pirns
677.053.2	Equipment, machines for winding, reeling, balling etc. of yarns and folded yarns		
677.053.23	Winding machines		
677.053.27	Equipment, machines for reeling, balling, assembly winding etc.		
677.053.273	Hank-winding machines		
677.053.275	Wrapping machines. Machines for winding thread on cards		
677.053.276	Assembly-winding devices		

677.054.842.3	Grippers	677.055.568	Parts peculiar to large circular knitting machines
677.054.842.4	Needles for weft insertion	677.055.6	Warp knitting machines
677.054.842.5	Jets for weft insertion	677.055.62	Flat warp knitting machines. □ Tricot knitting machines. Milanese machines. Raschel machines (double-rib looms). Galon (galloon) machines
677.054.843	Shuttle boxes and their components		
677.054.844	Motions for insertion of weft	677.055.64	Circular warp knitting machines. □ Maratte machines
677.054.844.2	Picking motions. □ Pick-and-pick		
677.054.844.6	Non-picking motions. □ Pneumatic motions. Hydraulic motions. Electromagnetic motions	677.055.68	Parts peculiar to warp knitting machines. □ Warp beams. Knitting actions (e.g. needle bars, sinker bars). Pattern devices (fall plates etc.)
677.054.844.8	Parts of picking motions. □ Picking shaft (picking stick). Lug strap (picker)		
677.054.845	Sleys (slays). □ Reeds. Shuttle guides	677.055.7	Other types of knitting machine. □ Waltex
677.054.846	Weft-changing motions. □ Shuttle-changing mechanisms	677.055.8	Loop-forming elements
677.054.847	Motions for weft or shuttle replenishment. □ Magazine, battery. Weft-cutting motions. Cutters	677.055.82	Needles
677.054.85	Safety devices. Stop motions	677.055.84	Sinkers
677.054.87	Equipment, machines for producing cards. □ Card-punching machines ⇒ 676.815.4	677.055.86	Sliders
677.054.89	Other auxiliary devices, parts and accessories for weaving machines	677.056	Other fabric-forming machines
677.055	Knitting machines and equipment	677.056.1	Tufting machines
677.055.3	Straight-bar knitting machines	677.056.2	Machines for the manufacture of waddings and fulled felts
677.055.32	Cotton-type knitting machines	677.056.22	Machines for production of waddings
677.055.34	Rib knitting machines	677.056.24	Machines for production of fulled felts
677.055.38	Parts peculiar to straight-bar machines	677.056.3	Machines for production of needle felts
677.055.382	Needle-and-sinker devices. □ Needle bar. Sinker and jack. Catch bar	677.056.4	Machines for production of nonwovens 677.056.4 = 677.026.4
677.055.383	Fashioning devices	677.056.5	Machines for production of lace ⇒ 677.055.62
677.055.384	Welt turning and rib transfer devices	677.056.6	Embroidery machines
677.055.385	Yarn carriers (selection). Friction boxes etc.	677.056.7	Machines for producing knotted fabrics, braidings and trimmings
677.055.386	Regulating devices	677.056.73	Machines for production of knotted fabrics, fringes
677.055.387	Draw-off mechanisms	677.056.75	Machines for production of braidings. Braiding machines
677.055.388	Pattern devices. □ Lace-decking mechanisms. Plush devices. Striping devices	677.056.77	Machines for production of trimmings
677.055.4	Flat-bar knitting machines	677.057	Equipment, machines for textile dyeing, printing and finishing
677.055.42	With one needle bed	677.057.1	Equipment, machines for processes for all stages of dyeing and finishing
677.055.44	With two needle beds	677.057.11	Equipment, machines for control and preliminary fault removal, for inspection of goods
677.055.45	V-bed knitting machines. Lamb-type knitting machines	677.057.12	Equipment, machines for removal of water and air
677.055.46	Purl knitting machines	677.057.13	Equipment, machines for tensioning and drying. Driers
677.055.47	Home knitting machines	677.057.14	Equipment, machines for humidifying, damping, spraying
677.055.48	Parts peculiar to flat-bar machines. □ Needle-bed assembly. Carriage and cam systems. Pattern devices (e.g. striping devices)	677.057.15	Steaming apparatus. Steamers
677.055.5	Circular knitting machines	677.057.17	Rope pilers, openers and spreaders. Cutting, piling and rolling equipment
677.055.52	Sinker-wheel knitting machines. Loop-wheel machines	677.057.2	Equipment, machines for preparatory treatment (for textile finishing)
677.055.522	With horizontal needles	677.057.21	Equipment, machines for preparation of goods. □ Sewing machines ⇒ 687.053
677.055.524	With vertical needles	677.057.22	Singeing equipment
677.055.528	Parts peculiar to sinker-wheel and loop-wheel machines. □ Sinker-wheel assembly. Pattern devices (e.g. striping devices, plating attachments)	677.057.25	Equipment, machines for cleaning. □ Washing equipment and machines
677.055.54/56	Circular knitting machines (excluding sinker-wheel machines and loop-wheel machines)	677.057.26	Equipment, machines for caustic bathing ⇒ 677.057.65
677.055.54	Small circular knitting machines (diameter up to 165 mm)	677.057.27	Equipment, machines for bleaching
677.055.542	Single cylinder machines without dial	677.057.4	Equipment, machines for dyeing ⇒ 667.2
677.055.546	Double cylinder machines	677.057.41	Equipment for preparation of dye liquor
677.055.548	Parts peculiar to small circular knitting machines. □ Cylinder assembly. Dial assembly. Cam systems. Pattern devices (e.g. striping devices)	677.057.42	Dyeing vats, vessels
677.055.56	Large circular knitting machines (diameter greater than 165 mm). □ Interlock machines	677.057.43	Dyeing apparatus. □ Pin frames. Pressure jig equipment
677.055.562	Single-cylinder machines without dial	677.057.44	Dyeing machines. □ Winch-and-vat. Jigs (jiggers)
677.055.564	Single-cylinder machines with dial		
677.055.566	Double-cylinder machines		

677.057.46	Equipment for special dyeing processes. □ Dyeing and steaming chambers	677.071.5	Products of extraction and processing of natural fibres. □ Retted, decorticated, ginned, reeled fibres
677.057.47	Apparatus and equipment for continuous dyeing	677.071.6	Products of cutting, breaking, opening, cleaning, blending. □ Cut natural fibres.
677.057.5	Equipment, machines for printing on textiles ⇒ 655.3.025 ⇒ 677.057.77	677.071.7	Broken natural fibres Products of carding. □ Web. Batt. Card sliver.
677.057.51	Equipment, machines for production of printing pastes	677.071.8	Slubbings Products of hackling, combing, drawing and backwashing. Tops. Rovings. Slivers
677.057.52	Equipment for printing by hand	677.072	Yarns. Plied yarns. Cords. Ropes
677.057.53	Printing machines. □ Block printing machines. Screen printing machines. Roller printing machines ⇒ 681.625.7	677.072.2	Rovings manufactured in flyer frames
677.057.54	Equipment for photomechanical printing	677.072.3	Ring-spun yarns ◇ 677.072.3:677.022.6 Yarns for making twisted threads
677.057.6	Equipment, machines for finishing	677.072.32	Yarns from woollen, condenser spinning. Woollen yarns. Condenser yarns
677.057.61	Equipment, machines for mechanical finishing	677.072.33	Yarns from cotton and worsted spinning
677.057.611	Equipment, machines for milling, decatizing, damping	677.072.34	Yarns from bast fibre spinning processes
677.057.612	Equipment, machines for raising, cropping, polishing	677.072.35	Yarns from schappe spinning processes
677.057.613	Equipment, machines for brushing, beating, breaking	677.072.36	Man-made fibre yarns from direct spinning processes
677.057.615	Equipment, machines for calendering, pressing, shaping	677.072.39	Fancy yarns. □ Voile yarns. Crêpe yarns. Jaspé
677.057.617	Equipment, machines for mechanical shrinkage	677.072.4	Yarns from special spinning processes. □ Open-end yarns. Self-twist yarns
677.057.62	Equipment, machines for padding. □ Foulards (padding mangles)	677.072.5	Yarns from spinning mules
677.057.63	Equipment, machines for coating. □ Equipment for evaporation of metals and other substances. Equipment for spraying, spreading, doctoring, pouring, sprinkling. Roller applicators	677.072.6	Twisted and plied yarns. □ Threads. Cords. Ropes ◇ 677.072.6:677.024 Yarns for weaving ◇ 677.072.6:687.023 Sewing yarns
677.057.64	Equipment, machines for polymerization and condensation	677.072.61	Twisted and plied yarns etc. according to degree of twisting. □ Spun core yarns. Gimp yarns
677.057.65	Mercerizing machines ⇒ 677.057.26	677.072.62	Plain twisted and plied yarns
677.057.7	Equipment, machines for final processes	677.072.63	Twisted fancy yarns. □ Ondé yarns. Twisted crêpe yarns
677.057.71	Equipment for checking, control ⇒ 658.562	677.072.68	Cords. Ropes
677.057.72	Measuring equipment and machines ⇒ 681.2	677.072.7	Textured yarns. □ Falsetwist yarns.
677.057.73	Laying machines. Doubling machines. Ironing machines ⇒ 648.43 ⇒ 687.054	677.073	Gear-crimped yarns. Jet-bulked yarns
677.057.74	Lapping machines. Wrapping machines	677.073.2	Products of winding, reeling, balling, warping, sizing, beaming etc. of single and plied yarns
677.057.75	Combined machines (e.g. measuring and doubling machines)	677.073.7	Wound yarns. Reeled yarns. Balled yarns
677.057.77	Edge-printing machines ⇒ 677.057.5	677.074	Warped yarns. Sized yarns. Beamed yarns
677.057.78	Pressing machines. Packing machines ⇒ 621.798		Woven fabrics, textiles ◇ 677.074:664.742 Screen cloth for grading flour
677.07	Products of the textile industry <i>Details of form and shape by -48....</i> <i>Denote applications by :...</i> ◇ 677.07:645 Fabrics for furnishing ◇ 677.07:66.067.123.2 Fabrics for filters	677.074.1	Design and construction aspects
677.071	Products of spinning of man-made fibres. Products of preparation of natural and man-made fibres (before spinning)	677.074.13	Type of yarn <i>Specify by :677.072...</i>
677.071.2	Products obtained from spinning man-made fibres	677.074.14	Setting. Set of warp. Set of weft
677.071.25	Filaments. Staple fibres	677.074.15	Systems of warp and weft
677.071.252	Continuous filaments	677.074.16	Design of weave. Types of weave
677.071.258	Staple fibres	677.074.162	Basic weaves. Weaves derived from basic weaves
677.071.27	Man-made staple fibres manufactured from sheets. Split fibres	677.074.162.3	Plain weaves. Tabby weave. Calico weave. Repp weave. Panama weave
677.071.28	Man-made tow	677.074.162.5	Twill weave. Honeycomb weave
677.071.51.8	Products obtained from preparation of natural fibres before spinning	677.074.162.7	Sateen weave. Satin weave
		677.074.164	Combination weaves
		677.074.164.2	Striped weaves
		677.074.164.3	Check weaves
		677.074.164.4	Shading weaves
		677.074.164.5	Crêpe weaves
		677.074.164.7	Damask weaves
		677.074.166	Reinforced weaves. Double weaves. Multiple weaves
		677.074.167	Reform weaves. Tertiary weaves
		677.074.168	Other weaves
		677.074.168.2	Pleated weaves. Quilted weaves
		677.074.168.4	Brocade weaves

677.074.168.5	Pile weaves. Velvet weaves. Plush weaves.	677.077.5	Printed fabrics
677.074.168.6	Terry weaves. Looped weaves	677.077.6	Finished fabrics
677.074.17	Leno weaves. Net weaves	677.077.611	Milled and fulled fabrics. □ Flannel
	Colour composition. □ Single shade	677.077.613	Raised, sheared fabrics etc. □ Velour. Velvet
	(self-coloured). Patterned	677.077.623.52	Mercerized fabrics
677.074.3/5	Fabrics according to dimensions, form and surface structure	677.077.625.13	Proofed fabrics. □ Waterproof fabrics
677.074.3	Wide fabrics. Narrow fabrics. Smallwares. Tubular fabrics	677.08	Wastes and by-products of the textile industry
677.074.32	Wide fabrics		
677.074.33	Narrow fabrics. Smallwares. □ Ribbons. Tapes. Straps	677.1/5	Textile fibres
677.074.35	Fabrics from circular looms	677.1/3	Natural fibres
677.074.5	Plain fabrics. Pile fabrics ⇒ 677-488		⇒ 677.51
677.074.52	Plain fabrics	677.1/2	Vegetable fibres
677.074.53	Weft pile fabrics. □ Fabrics with chenille weft		⇒ 633.5
677.074.54	Warp pile fabrics. □ Double-pile carpets. Terry fabrics. Towelling	677.1	Bast fibres (fibres from dicotyledon stems). Hard fibres (fibres from monocotyledon leaves and fruit)
677.074.57	Special types of carpet. □ Axminster carpets		⇒ 633.52
677.074.58	Knotted carpets. Machine-knotted carpets. □ Ghiordes carpets. Smyrna carpets. Persian carpets. Senné carpets ◇ 677.074.58(55-86) Persian carpets		⇒ 677.2
677.074.6	Bobbinets. Tulle	677.11/15	Fibres from dicotyledon stems. Bast fibres
677.075	Knitted fabrics ⇒ 687.3	677.11	Flax. Linen fibres (fibres from line plant)
677.075.3	Flat weft knitted fabrics	677.12	Hemp
677.075.4	Flat knitted fabrics	677.13	Jute
677.075.42	Plain flat knitted fabrics		⇒ 677.143
677.075.44	Rib flat knitted fabrics	677.14	Various lignified stem fibres from dicotyledons
677.075.46	Purl flat knitted fabrics	677.141	Sunn hemp
677.075.5	Circular-weft knitted fabrics. □ Terry knitted fabrics. Interlock fabrics	677.142	Kenaf. Bimli. Gambo. Deccan. Ambari
677.075.6	Warp-knitted fabrics. □ Tricot knitted fabrics. Milanese knitted fabrics	677.143	Other malvaceous fibres
677.076	Other products of the textile industry	677.146/149	Other lignified stem fibres from dicotyledons 677.146/149 ≅ 582.61/9
677.076.1	Tufted fabrics	677.15	Various nonlignified stem fibres from dicotyledons
677.076.12	Tufted fabrics produced with only one needle	677.151	Ramie fibres. China grass
677.076.14	Tufted fabrics produced with more than one needle	677.152	Orties. Nettle fibres
677.076.2	Waddings. Filled felts	677.153	Yercum. Mudar
677.076.22	Waddings	677.154	Bauhinia fibres
677.076.24	Filled felts. Wool felts. Hair felts	677.156/159	Other nonlignified stem fibres from dicotyledons 677.156/159 ≅ 582.6/9
677.076.3	Needle felts		
677.076.4	Nonwovens	677.16	Leaf and leaf-sheath fibres of monocotyledons
677.076.42	Fibre-bonded fabrics. Fibre fleeces	677.161	Musaceae fibres. □ Manila hemp (abaca). Banana fibres
677.076.44	Strengthened, reinforced nonwovens. □ Needle fleecès	677.162	Agave fibres. □ Sisal. Henequen
677.076.45	Sew-knit webs. Stitch-bonded fabrics	677.163	Fourcroya fibres (Furcraea)
677.076.49	Other nonwovens	677.164	Liliaceae fibres. □ Phormium fibre (New Zealand flax). Yucca fibre
677.076.5	Lace	677.165	Bromeliaceae fibres. □ Pineapple fibres. Caroa
677.076.52	Handmade lace ⇒ 746.21	677.166	Fibres of Pandanaceae (screw pines)
677.076.56	Machine-made lace	677.167	Stipa fibres. □ Esparto grass (alfa) ⇒ 677.542
677.076.6	Embroidery	677.168	Leaf fibres of Palmaceae. Palm-leaf fibres. Raffia
677.076.62	Handmade embroidery ⇒ 746.3		
677.076.64	Spatula embroidery	677.17	Stem fibres of monocotyledons (from stalks, petioles, trunks)
677.076.66	Machine-made embroidery	677.172	Gramineae fibres. Grasses. Straw
677.076.7	Knotted fabrics. Braided fabrics. Braidings. Trimmings	677.173	Cyperaceae fibres. Reeds
677.076.73	Knotted fabrics. Nets ⇒ 746.25	677.174	Palmaceae fibres. Piassava fibres
677.076.75	Braided products. □ Braided cords. Facings. Shoelaces. Fringes	677.175	Juncacea fibres. Canes. Rushes
677.076.77	Trimnings	677.177	Tillandsia (limes) fibres. Crin vegetal
677.076.9	Other textile fabrics. □ Alcantara	677.178	Fibres from stalks, stems, trunks of other monocotyledons 677.178 ≅ 582.52
677.077	Products of textile dyeing, printing and finishing 677.077 ≅ 677.027		
677.077.4	Dyed fabrics		

677.18	Fibres from monocotyledon fruits. □ Coir (coconut fibre). Luffa (loofah) fibres	677.4	Man-made fibres ⇒ 677.5 ⇒ 678.5/7
677.19	Other vegetable fibres	677.46/47	Man-made fibres from natural polymers
677.191	Fibres from tree bark	677.46	Man-made fibres from cellulose or other polysaccharides ⇒ 661.728 ⇒ 676.1
677.192	Fibres from mosses and ferns	677.461	Cellulose nitrate rayon
677.193	Fibres from water plants. Marine fibres. □ Seaweed. Sea grass	677.462	Cuprammonium rayon
677.194	Peat fibres ⇒ 674.88	677.463	Viscose rayon
677.195	Fibres obtained in the form of chips or shavings. Fibres from conifer needles ⇒ 674.82 ⇒ 674.87	677.464	Man-made fibres from cellulose acetate
677.2	Vegetable hairs	677.465	Man-made fibres from other cellulosic esters
677.21	Cotton ◇ 677.21.08 Wastes and by-products of the cotton industry. Cotton wadding	677.466	Man-made fibres from cellulose ethers
677.22	Other seed hairs. □ Akund floss	677.467	Other cellulosic fibres
677.23	Fruit hairs. Vegetable down. □ Kapok	677.469	Man-made fibres from polysaccharides other than cellulose and its compounds (e.g. polyuronides). □ Alginate fibres
677.3	Animal fibres ⇒ 637.62	677.47	Man-made fibres from proteins and other natural nitrogenous materials
677.31	Wool from sheep ◇ 677.31:636.36 Merino wool etc.	677.472	Fibres from proteins. □ Casein fibres. Soya fibres
677.312	Fleece wool in general. Untreated wool	677.473	Fibres from other nitrogenous materials. Chitin and its derivatives
677.313	Long wool (for combing, on the worsted system)	677.494	Synthetic fibres 677.494 ≡ 678
677.314	Short wool (for carding, on the woollen system)	677.494.6	Fibres from polycondensates
677.316	Skin wool. Pulled wool. □ Slipe wool. Sweated wool	677.494.674	Polyester fibres
677.318	Recovered wool. Shoddy. Wool wastes	677.494.675	Polyamide fibres. Nylon
677.32	Wool from Auchenidae. □ Alpaca. Vicuna. Llama	677.494.7	Fibres from polymerizates
677.33	Goat hair. □ Cashmere. Mohair	677.494.742	Polyolefin fibres. □ Polyethylene fibres.
677.34	Camel hair	677.494.745.3	Polypropylene fibres
677.35	Other hairs and bristles	677.499	Polyacrylonitrile fibres (acrylic fibres).
677.351	Cattle hair	677.5	Polyvinylidene dinitrile fibres
677.352	Hog bristle. Pig bristle	677.51	Other synthetic fibres
677.353	Dog hair. Cat hair	677.51	Mineral and metal fibres. Braiding materials. Rubber threads. Paper yarns
677.354	Hare fur. Rabbit fur	677.51	Natural mineral fibres ⇒ 677.1/3
677.354.5	Agora rabbit hair	677.511	Asbestos fibre
677.356	Beaver hair	677.512	Pele's hair (natural lava fibres)
677.358	Horse hair (mane and tail)	677.52	Artificial mineral fibres
677.37	Silk ⇒ 638.2	677.521	Glass fibres ⇒ 666.189.21
677.371	Cultivated silk. True silk (from silkworm of Bombyx mori)	677.522	Siag, rock and artificial lava fibres Specify by :552...
677.371.1	Raw silk (filament, nett). Grège	677.523	Ceramic fibres
677.371.2	Thrown silk. □ Tram. Organzine	677.53	Metal fibres ◇ 677.53:669.21 Of gold and its alloys. Gold thread ⇒ 621.778
677.371.3	Half-boiled-off silk. Souple silk	677.531	Fabrics of precious metals
677.371.4	Boiled-off silk. Scoured silk. Degummed silk	677.532	Metallic gauze. Wire gauze. Gauze for screens. Gauze of copper, brass, steel etc.
677.371.5	Weighted silk	677.533	Wire netting. Mesh of brass, steel, galvanized steel etc.
677.371.7	Schappe	677.534	Screens of perforated metal (punched or expanded metal)
677.371.8	Silk noils. Bourette	677.536	Wire mesh with links sealed with cement or terracotta
677.372/373	Wild silks	677.54	Vegetable braiding and plaiting materials ⇒ 633.58 ⇒ 677.1
677.372	Wild silk from Antheraea pernyi. Tasar (tusser) silk	677.541	Straw and similar products. □ Straw matting. Plaited straw
677.373	Wild silk from other silkworm species	677.542	Esparto grass (for matting) ⇒ 677.167
677.373.1	Eri silk. Ailanthus silk (from Samia species)	677.543	Willow shoots. Osiers (for basket-making)
677.373.2	Fagara silk (form Attacus atlas)		
677.373.5	Anaphe silk. Nest silk (from Anaphe species)		
677.375	Other natural silks. □ Spider silk. Silk from molluscs. Sea-silk ⇒ 638.54		
677.39	Other fibres of animal origin ◇ 677.39:637.63 From feathers etc. ◇ 677.39:637.663 From intestine (gut)		

677.544	Rushes and reeds (of Juncaceae and Cyperaceae)	678.021	Processes prior to compounding and/or forming
677.545	Bamboo. Rattan	678.021.1	Precoagulation processes. Treatment of latex
677.546	Shavings. □ Rice straw	678.021.12	Preservation of latex
677.547	Leaves. Leaf parts	678.021.13	Handling. Purifying. □ Removal of monomers. Fractionation. Screening. Skimming
677.548	Bass. Linden bass	678.021.14	Dilution with water
677.55	Threads and fibres of rubber and similar elastic materials	678.021.15	Concentration
	677.55 ≡ 678	678.021.16	Homogenization. Blending
677.554	Threads of natural rubber	678.021.17	Compounding
677.557	Threads from synthetic rubber	678.021.172	Thickening
677.57	Paper threads and yarns	678.021.174	Sensitizing
678	INDUSTRIES BASED ON MACROMOLECULAR MATERIALS. RUBBER INDUSTRY. PLASTICS INDUSTRY	678.021.18	Artificial dispersion. Redispersion
	⇒ 547	678.021.2	Coagulation. Spontaneous coagulation
	⇒ 66.0	678.021.3	Coagulation by mechanical processes
678-1	Properties and composition of specific materials	678.021.5	Coagulation by smoking
	<i>Details by :53... and :54...</i>	678.021.6	Coagulation by addition of chemical products (or of natural materials containing such products)
	<i>The 62-1... auxiliaries are not applicable at 678 and its subdivisions</i>	678.021.7	Coagulation by drying
678-13	Copolymers	678.021.8	Coagulation by biological methods
	⇒ 678.7	678.021.9	Coagulation by other processes. □ Electrical, ultrasonic, heat processes of coagulation. Coagulation by freezing out. Agglomeration. Aggregation
678-19	Mixtures of macromolecular materials	678.022	Treatment of coagulates
678-2/-9	Details of machinery, plant and products	678.022.1	Treatment of coherent coagulates in general
	<i>The special auxiliaries -2/-9 (from 62-2/-8 and 66-9) are applicable throughout 678</i>	678.022.2	Treatment before rolling out or sheeting
	◇ 678-2 Machine components	678.022.3	Crêpe rolling process
	◇ 678-4 Shape and form of products	678.022.4	Preparation of double, four-ply and sole crêpes
	◇ 678-5/-8 Operation, control, servicing, accessories and power sources of plant and machinery	678.022.5	Drying of crêpes
	⇒ 678.05	678.022.7	Sheet rolling process
678.0	General information. Properties. Analysis and testing. Manufacturing processes, plant and equipment. Products and their applications	678.022.8	Smoking of the sheets
678.01	Properties. Defects. Constitution. Serviceability. Sampling. Analysis. Testing	678.022.9	Treatment of incoherent coagulates, pearls etc.
	◇ 678.01:53 Physical properties, defects, testing etc.	678.023	Size reduction. Comminution. Compounding. Kneading. Separation. Screening
	◇ 678.01:54 Chemical properties, defects, testing etc.	678.023.1	Kneading. Masticating
678.011	Manufacturing variables. Other factors affecting properties	678.023.2	Pulverizing. Grinding. Separation. Screening
678.012	Chemical analysis. Determination of constituents, molecular complexity etc.		⇒ 621.926/928
	<i>Denote chemical constituents by :546... and :547...</i>	678.023.3	Mixing. Blending
678.016	Serviceability tests (under practical conditions)		⇒ 66.063.7
	<i>Details by :620.16...</i>	678.024	Conversion into preforms, blanks etc. for moulding and forming
678.017	Testing of mechanical properties	678.025	Intermediate operations. □ Preheating. Cooling. Purification. Filtration
	<i>Details by :620.17...</i>	678.026	Impregnation, coating and cladding with macromolecular compound. Reinforcing. Proofing. Bonding. Cementing. Decorating etc.
678.019	Material defects. Resistance to attack		⇒ 678.029.6
	678.019 ≡ 620.19	678.026.2	Impregnation (with macromolecular compound)
	⇒ 678.028.4	678.026.3	Coating and cladding (with macromolecular compound). □ Dipping. Spraying. Flame spraying. Spreading, spread coating. Electrodeposition
678.02	Manufacturing processes and operations	678.027	Forming operations
	◇ 678.02:66.095.26 Polymerization processes (in rubber and plastics industries)	678.027.2	Calendering
	◇ 678.02:66.095.261.1 Addition polymerization	678.027.3	Extrusion
	◇ 678.02:66.095.261.2 Polycondensation	678.027.4	Spinning
	⇒ 66.09		⇒ 677.021.125.2
		678.027.5	Film casting
		678.027.6	Dipping
		678.027.7	Moulding. Press moulding. Casting (in moulds)
		678.027.72	Compression moulding. Premoulding. Breathing
		678.027.73	Transfer moulding
		678.027.74	Injection moulding. Jet moulding. Offset injection moulding
		678.027.75	Cold moulding
		678.027.76	Blow moulding. Fluid-pressure forming
		678.027.77	Casting (in moulds)
			⇒ 678.027.5
		678.027.78	Explosion moulding

678.027.9	Special non-plastic forming operations	678.03	Raw materials (primaries) and semiprocessed materials (intermediates) for the rubber and plastics industries
678.027.92	Deep drawing		⇒ 633.91
678.027.93	Low-pressure moulding		⇒ 66
678.027.94	Lay-up methods. □ Laminating. Pressing-in	678.031	Latex (crude latex)
678.027.95	Powder sintering moulding. Engels method		678.031 ≡ 678.021.1
678.027.96	Moulding of foam articles	678.031.3	Purified latex
678.027.97	Skiving	678.031.5	Concentrated latex
678.027.98	Orientation	678.032	Coagulates. □ Raw, natural rubber (untreated, uncured). Block, crêpe, sheet rubber
678.028	Curing. Vulcanizing. Reclaiming. Other chemical operations during or after forming		⇒ 678.062
678.028.2	Processes	678.032.1	Coagulum. Cup lumps. Untreated rubber
678.028.22	Hot curing in sulphur bath		678.032.12/19 ≡ 678.021.2/9
678.028.24	Vulcanization in sulphur monochloride solution. Cold curing	678.032.15	Coagulum obtained by smoking
678.028.25	Steam vulcanization	678.032.2	Block rubber. Slabs. Worms
678.028.26	Vulcanization with colloidal sulphur. Peachey process	678.032.3	Crêpes
678.028.27	Dry-heat vulcanization. □ Vacuum vulcanization. High-frequency vulcanization. Infrared vulcanization	678.032.4	Sole crêpes
678.028.28	Self-cures. Vulcanization at room temperature	678.032.5	Dried crêpes
678.028.292	Vulcanization and curing in solution	678.032.6	Lower-grade crêpes. Scrap rubber. Earth rubber. Compo. Skim rubber
678.028.295	Solarization. Vulcanization and curing by means of light	678.032.7	Sheet rubber (unsmoked)
678.028.296	Radiation vulcanization. Radiation curing. □ Gamma-vulcanization. Electron vulcanization. Thermoradiation vulcanization	678.032.8	Smoked sheets
678.028.297	High-speed vulcanization	678.033	Compounded materials. Formulae
678.028.298	Post-vulcanization	678.034	Preforms of various shapes. □ Pellets. Tablets. Chips. Chunks
678.028.3	Rate and state of vulcanization and curing	678.04	Auxiliary materials. Additives
678.028.4	Faults and difficulties in vulcanization and curing		⇒ 66.022.3
	⇒ 678.019	678.041	Additives for latex
678.028.5	Devulcanization	678.042	Solvents
678.028.6	Recycling process. Reclaiming process		<i>Details by :546... and :547...</i>
	⇒ 678.046.1	678.043	Vulcanizing agents. Cross-linking agents
	⇒ 678.46		⇒ 661.2
678.029	Machining. Fabricating. Finishing		⇒ 661.49
678.029.3	Machining. Fashioning		⇒ 661.691
	⇒ 621.9		⇒ 661.692
678.029.4	Joining, reinforcing, strengthening of surfaces		⇒ 661.729
678.029.42	Cementing		⇒ 661.8'02
	⇒ 621.792	678.044	Accelerators. Catalysts. Initiators. Activators
678.029.43	Welding		⇒ 66.097
	⇒ 621.791	678.045	Vulcanization retarders. Polymerization inhibitors. Chain regulators
678.029.44	Sewing. Stitching	678.046	Fillers. Extenders. Compounding ingredients exerting more than one action
678.029.45	Mechanical attachment. □ Attachment with screws, staples, rivets	678.046.1	Rubber-containing materials. □ Regenerated rubber
	⇒ 621.88		⇒ 678.028.6
678.029.46	Strengthening. Mechanical reinforcing		⇒ 678.46
678.029.5	Bonding. Uniting with other materials	678.046.2	Carbon black (lampblack). Graphite
	◇ 678.029.5:669 Bonding to metals	678.046.3	Mineral fillers. Inorganic fillers. □ Metals. Kaolin
	◇ 678.029.5:677 Bonding to textiles	678.046.5	Vegetable fillers (except rubber). □ Wood waste. Cotton waste
678.029.6	Impregnation, coating and cladding with protective materials		678.046.6
	⇒ 678.026		Animal fillers. □ Proteins. Leather waste, trimmings. Hide waste. Wool
678.029.7	Finishing. Polishing. Surface hardening etc.	678.046.7	Synthetic fillers
678.029.71	Baking. Drying	678.047	Colouring materials. Dyes. Pigments
678.029.72	Seasoning		⇒ 667
678.029.73	Annealing		⇒ 678.029.8
678.029.74	Antistatic treatment	678.048	Antioxidants. Preservatives. Stabilizers
678.029.76	Polishing. □ Press polishing. Mechanical polishing		⇒ 66.094.3.097.8
678.029.77	Chemical polishing with solvents or vapours		⇒ 66.097.8
678.029.78	Surface hardening		⇒ 678.041
678.029.8	Decoration. Bleaching. Dyeing. Colouring	678.049	Softeners. Plasticizers. Wetting agents. Lubricants. Other minor additives
	⇒ 667	678.049.1	Softeners in general
	⇒ 678.047		<i>For specific substances see 661.7...</i>
678.029.93	Dusting with powders		⇒ 66.063.72
678.029.94	Deodorization	678.049.2	Asphalts. Bitumens
678.029.98	Final operations		⇒ 665.775

678.049.3	Other mineral oil products. □ Liquid paraffin. Paraffins. Petroleum jelly	678.066	Impregnated goods. Proofed goods. □ Impregnated, proofed fabrics
678.049.4	Vegetable oils. Fatty acids and their derivatives. Soaps ⇒ 661.187 ⇒ 665.3	678.067	Formed objects. Mouldings. Castings
678.049.5	Animal oils ⇒ 665.21	678.069	Decorated products. Painted products. Surface-finished products
678.049.6	Resins. Resin oils. Abietic acid	678.07	Classification according to special characteristics
678.049.7	Wetting agents. Emulsifiers. Peptizers ⇒ 66.063.5/6	678.072	Thermosetting materials in general ⇒ 678.6
678.049.8	Odorants ⇒ 66.022.392 ⇒ 665.577	678.073	Thermoplastic materials in general ⇒ 678.5 ⇒ 678.7
678.049.9	Other minor additives	678.074	Elastomers. Rubbery materials in general ⇒ 678.4 ⇒ 678.7
678.049.91	Fireproofing agents	678.075	Plastomers ⇒ 678.5
678.049.92	Antifreezing agents	678.076	Soft materials
678.049.93	Lubricants. Mould lubricants	678.077	Hard materials. Rigid materials
678.05	Plant. Machinery. Equipment	678.078	Stereo structure. □ Stereospecific polymers. Atactic polymers
678.051	Plant etc. for processes prior to compounding	678.09	Materials resulting from particular processes 678.09 ≡ 66.09
678.052	Plant etc. for treatment of coagulates, crude rubber etc.	678.094.2	Reduction products
678.053	Grinding plant. Mixing plant. Classifying plant 678.053 ≡ 678.023	678.094.25	Hydrogenated derivatives
678.054	Preforming plant and machinery. □ Tableting machines. Briquetting machines	678.094.3	Oxidation products. Oxygenated derivatives
678.055	Plant etc. for intermediate operations. Heating and drying plant	678.094.413	Chlorinated derivatives
678.056	Equipment for coating. □ Dipping plant. Spraying plant 678.056 ≡ 678.026	678.095.21	Isomerization products
678.057	Forming plant 678.057 ≡ 678.027 ⇒ 621.979.3	678.095.268	Depolymerization products
678.057.2	Calenders. Doubling calenders. Doubling rollers	678.4	Natural rubber. Other natural macromolecules containing only carbon and hydrogen
678.057.3	Extruding machines. Extruders	678.43	Derivatives with low sulphur content. Soft vulcanized rubber
678.057.7	Moulding presses. Moulding machines	678.44	Derivatives with high sulphur content. Hard vulcanized rubber. □ Ebonite. Vulcanite
678.058	Curing plant. Vulcanization plant	678.46	Regenerated sulphur derivatives. Reclaimed rubber ⇒ 678.028.6 ⇒ 678.046.1
678.058.1	Vulcanization pans	678.47	Derivatives of natural rubber (other than vulcanized)
678.058.2	Vulcanization autoclaves	678.471	Containing no additional elements ◇ 678.471.095.252 Cyclo-rubbers
678.058.3	Vulcanization presses	678.472	Containing additional hydrogen only. Hydrogenated derivatives
678.058.4	Hot-air stoves for vulcanization	678.473	Containing halogens. Halogen derivatives
678.058.5	Vulcanizers for repairs. Vulcanizers for used, worn objects	678.474	Hydrohalogenated derivatives
678.058.6	Recycling plant. Reclaiming plant	678.475	Containing elements of the 6th and/or 5th group
678.059	Machining plant. Fabricating plant. Finishing plant 678.059 ≡ 678.029 ⇒ 621.9	678.476	Alkylated derivatives
678.06	Applications of macromolecular materials. Finished products	678.477	Diels-Alder addition products
◇ 678.06:615.4	Macromolecules for medical material and equipment ⇒ 615.462	678.479	Metal-containing derivatives
◇ 678.06:621.798	Macromolecules for packaging	678.48	Gutta-percha. Balata. Chicle. Guayule etc.
◇ 678.06:687	Use of macromolecules in clothing industry and beauty culture ⇒ 677.46	678.481	Gutta-percha
678.061	Latex products. Compositions and mixings. □ Lubricants. Greases. Paints. Lacquers. Adhesives ⇒ 678.031	678.482	Balata
678.062	Uncured products. Unvulcanized products. Crude rubber products ⇒ 678.032	678.483	Chicle
678.063	Cured products. Vulcanized products	678.484	Guayule
678.065	Tyres, tracks and similar traction devices ⇒ 629.3.027.5 ⇒ 629.3.027.7	678.5	Plastics Especially semisynthetic plastics based on cellulose, proteins and other natural macromolecules containing carbon, hydrogen and oxygen or other elements ⇒ 678.7

678.54	Plastics based on cellulose or its derivatives ⇒ 661.728 ⇒ 677.46			<i>Here are listed the functional groups themselves (678.6 with ...2 to ...7 and ...82 to ...86) and selected combinations</i> ⇒ 665.939.56 ⇒ 677.494
678.542	Plastics based on cellulose or regenerated cellulose	678.62	Polycondensates synthesized from -CHO groups	
678.542.3	Regenerated cellulose. □ Viscose.	678.622	-CHO with -CHO. Aldehyde-aldehyde polycondensates	
678.542.7	Cuprammonium cellulose Vulcanized fibres ⇒ 676.872	678.63	Polycondensates synthesized from cyclic -OH groups	
678.544	Plastics based on cellulose esters. □ Cellulose nitrate, celluloid. Cellulose acetate	678.632	Aryl -OH with -CHO. Phenol-aldehyde polycondensates. Phenoplasts. Phenolic polycondensates	
678.546	Plastics based on single cellulose ethers	678.64	Polycondensates synthesized from acyclic -OH groups	
678.547	Plastics based on mixed cellulose ethers (e.g. with methyl and ethyl alcohol)	678.642	Acyclic -OH with -CHO. Alcohol-aldehyde polycondensates	
678.55	Plastics based on compounds (other than cellulose) containing only carbon, hydrogen and oxygen	678.643	Acyclic -OH with cyclic -OH. Alcohol-phenol polycondensates	
678.552	Starch-base plastics	678.644	Acyclic -OH with acyclic -OH. Alcohol-alcohol polycondensates, diol-diol polycondensates. □ Alkylene oxide polycondensates. Polyethylene oxide	
678.554	Alginate-base plastics	678.65	Polycondensates synthesized from -NH groups	
678.555	Agar-agar-base plastics	678.652	=NH with -CHO. Amine-aldehyde polycondensates. Aminoplasts. Aniline, urea and melamine polycondensates	
678.557	Lignin-base plastics	678.653	=NH with cyclic -OH. Amine-phenol polycondensates	
678.56	Plastics based on proteins or other natural nitrogenous compounds (with or without oxygen and/or sulphur)	678.654	=NH with acyclic -OH. Amine-alcohol polycondensates	
678.562	Plastics based on plant proteins. □ Plastics based on soya protein, groundnut (peanut) protein	678.66	Polycondensates synthesized from -N=C=O groups	
678.563	Plastics based on other nitrogenous substances of plant origin	678.662	-N=C=O with -CHO. Isocyanate-aldehyde polycondensates	
678.565	Casein plastics. □ Casein-formaldehyde	678.663	-N=C=O with cyclic -OH. Isocyanate-phenol polycondensates	
678.567	Plastics based on keratin, gelatin (collagen), other scleroproteins	678.664	-N=C=O with acyclic -OH. Isocyanate-alcohol, diisocyanate-diol polycondensates. Polyurethanes	
678.569	Plastics based on other nitrogenous substances of animal origin	678.665	-N=C=O with =NH. Isocyanate-amine, diisocyanate-diamine polycondensates	
678.58	Plastics based on pitches, resins and other natural materials not conventionally regarded as macromolecular	678.666	-N=C=O with -N=C=O. Isocyanate-isocyanate polycondensates	
678.582	Plastics based on oils, fats, waxes Specify by :665...	678.67	Polycondensates synthesized from -COOH groups	
678.584	Plastics based on bituminous substances, asphalts, tars ⇒ 665.775	678.672	-COOH with -CHO. Acid-aldehyde polycondensates	
678.586	Plastics based on natural resinous substances. □ Plastics based on copals, resin pitch Specify by :665.94...	678.673	-COOH with cyclic -OH. Acid-phenol polycondensates	
678.6	Synthetic polycondensation products. Polycondensates. Synthetic resins. Synthetic fibres etc. <i>Synthetic polycondensates are here regarded as formed by the interaction of at least two components, each comprising a functional group contained in one or more compounds.</i> <i>The functional groups are denoted by ...2 to ...7 and ...82 to ...86, added in descending order to 678.6 starting with the numerically highest one involved in the reaction.</i> <i>Thus, since 3 denotes cyclic -OH groups and 2 denotes -CHO groups, their polycondensation products (the phenol-aldehydes or phenoplasts) have the main number 678.632.</i> <i>Similarly 678.675 denotes the polyamide polycondensates such as nylons, with the further addition of 4 (for acyclic -OH) giving 678.675.4 as the number for polyester polyamides.</i>	678.674	-COOH with acyclic -OH. Acid-alcohol polycondensates. Polyester polycondensates. Alkyd polycondensates. Glyptals	
		678.675	-COOH with =NH. Acid-amine polycondensates. Polyamide polycondensates. Nylon	
		678.676	-COOH with -N=C=O. Acid-isocyanate polycondensates	
		678.677	-COOH with -COOH. Acid-acid polycondensates	
		678.68	Polycondensates synthesized from =CO, halogen-containing, sulphur-containing and other functional groups	
		678.682	Polycondensates synthesized from halogen-containing groups	
		678.683	Polycondensates synthesized from =CO groups. Ketone polycondensates	

678.684	Polycondensates synthesized from sulphur-containing groups or sulphides.	678.746.8	Containing two or more different hetero-elements. ☐ Polynitrostyrene.
678.684.82	Polysulphide polycondensates	678.747	Polyvinylphenyl phosphate
678.685	Polysulphide-halide polycondensates.		Monolenic aromatic polymerizates with a reactive =C in at least one ring. ☐ Coumarone polymers. Polycoumarone
678.685.82	☐ Thiokol		Subdivide broadly as 678.746...
678.686	Polycondensates synthesized from =CH groups	678.76	Multilenic polymerizates (two or more reactive =C or one or more =C)
678.686	Benzyl chloride polycondensates	678.761	Multilenic aliphatic polymerizates in general
678.686	Polycondensates synthesized from epoxy groups. Polyepoxides. Epoxy resins	678.762	Multilenic hydrocarbon polymerizates. Diene polymers. ☐ Polybutadiene
678.7	Synthetic polymerization products. Polymerizates. Synthetic rubbers	678.763	Multilenic halide polymerizates.
	◇ 678.7-13 Synthetic copolymers	678.764	Polyhaloprenes. ☐ Polychloroprene. Neoprene
	⇒ 665.939.57	678.765	Multilenic oxygen-containing polymerizates
678.71	Hydrocarbon polymerizates in general	678.766	Multilenic polymerizates containing nitrogen, sulphur or other hetero-elements
678.74	Monolenic polymerizates (one reactive =C)	678.767	Multilenic aromatic or cyclic polymerizates without reactive =C in the rings
678.741	Aliphatic, acyclic polymerizates in general	678.768	Multilenic aromatic or cyclic polymerizates with reactive =C in at least one ring
678.742	Monolenic hydrocarbon polymerizates	678.769	Multilenic polymerizates with three or more =C
678.742.2	Polyethylene		Multilenic polymerizates with =C. Acetylene polymers
678.742.3	Polypropene. Polypropylene	678.8	Various macromolecular materials not included in 678.4/.7
678.742.4	Polybutene. Polybutylene. Polyisobutylene	678.82	Synthetic macromolecular compounds obtained partly by saturation of carbon-carbon bonds. Mixed polymerizates-condensates.
678.742.5	Polypentenenes		☐ Sulphone-olefin polymers
678.742.6	Polyhexenes	678.83	Organic polymers not included elsewhere
678.743	Monolenic halide polymerizates	678.84	Silicon-containing polymers. ☐ Silicones.
678.743.2	Vinyl halide polymers. Polyvinyl halides.		Organosiloxane polymers. Siloxanes
	☐ Polyvinyl fluoride (PVF). Polyvinyl chloride (PVC)	678.85	Phosphorus-containing polymers. Phosphonate polymers
678.743.3	Vinylidene halide polymers	678.86	Polymers based on other inorganic elements
678.743.4	Tetrahalo-olefin polymers.	679	INDUSTRIES BASED ON VARIOUS PROCESSABLE MATERIALS
	Polytetrahaloethylenes.	679.7	Cable and cordage industries
	☐ Polytetrafluoroethylene (PTFE) and homologues		⇒ 62-427
678.743.5	Allyl halide polymers. Polyallyl halides	679.7.02	Methods of manufacturing, processing and treatment
678.744	Monolenic oxygen-containing polymerizates		⇒ 677.02
678.744.3	Unsaturated carboxylic acid polymers and their esters	679.7.022	Methods of manufacturing, processing and treatment of wire, yarns or tape (strapping)
678.744.32	Acrylic polymers and their esters. ☐ Polyacrylic acid		⇒ 621.778
678.744.33	Methacrylic polymers	679.7.022.1	Mechanical treatment
678.744.34	Dicarboxylic acid polymers and their esters		Details by :621.7... and :621.9...
678.744.4	Unsaturated alcohol ester polymers.		◇ 679.7.022.1:621.771.25 Rolling
	☐ Polyvinyl acetate (PVAC)		◇ 679.7.022.1:621.914 Milling
678.744.5	Ether polymerizates	679.7.022.3	Heat treatment
678.744.6	Ketone polymerizates. Aldehyde polymerizates		Details by :621.78...
678.744.7	Alcohol polymerizates. ☐ Polyvinyl alcohol (PVAL)	679.7.022.4	Mechanical and heat treatment. ☐ Sintering.
678.745	Monolenic polymerizates containing nitrogen, sulphur and other elements		Stretching (of plastics)
678.745.2	Amine polymerizates. Imine polymerizates.	679.7.022.5	Chemical treatment. Pickling
678.745.3	Polyamines. Polyimines. ☐ Polyvinyl amine		⇒ 621.794
678.745.3	Cyanide polymerizates. Nitrile polymerizates.	679.7.022.6	Cleaning (of drawn wire)
678.745.6	☐ Polyacrylonitrile. Polyvinyl cyanide	679.7.022.8	Coating
678.745.6	Sulphur-containing polymerizates. ☐ Thioether polymers. Thioalcohol polymers		◇ 679.7.022.8:621.793 Metallizing
678.745.7	Containing elements other than nitrogen and sulphur		◇ 679.7.022.8:621.795.3 Lacquering
678.745.8	Containing two or more different hetero-elements. ☐ Oxyhalo polymerizates.	679.7.023	Methods for the manufacture of stranded wires or twisted yarns
678.746	Polyacrylic amide. Polymethacrylic amide		⇒ 621.778.24
678.746	Monolenic aromatic polymerizates without reactive =C in the ring(s)		⇒ 677.022.6
678.746.2	Hydrocarbon polymerizates. ☐ Polystyrene		
678.746.3	Halide polymerizates		
678.746.4	Oxygen-containing polymerizates. ☐ Ether polymerizates. Ketone polymerizates. Phenol polymerizates. Alcohol polymerizates		
678.746.5	Nitrogen-containing polymerizates. ☐ Amine polymerizates. Imine polymerizates.		
	Polyamines. Polyimines		
678.746.6	Sulphur-containing polymerizates		
678.746.7	Containing other hetero-elements		

679.7.023.2	According to the number of wires (conductors), cores and yarns to be stranded (laid up)	679.727	Design, manufacture and mounting of strand and rope terminals, attachments, fittings and connections
679.7.023.3	Stranding in one operation	679.728	Testing of raw materials, semimanufactured products and end-products <i>Details by :620.1...</i>
679.7.023.4	Stranding in more than one operation		
679.7.024	Methods of covering		
679.7.024.2	Winding. Lapping		
679.7.024.7	Sheathing	679.73	Manufacture of wires, wire strands and wire ropes (especially of steel wires, steel wire strands and steel wire ropes) <i>Denote materials by -03... (Table 1k)</i>
679.7.025	Methods of conditioning		
679.7.025.2	Drying	679.732	Manufacture of wires and wire strands, according to function
679.7.025.3	Dipping	679.733	Covering of wires, strands and wire ropes (especially those made of steel), according to function. □ Armouring
679.7.025.4	Impregnating		
679.7.025.5	Applying bitumen. Greasing	679.734	Manufacture of cores in wire strands and wire ropes (especially those made of steel)
679.7.025.6	Liming. Chalking	679.735	Manufacture of wires, wire strands and wire ropes (especially those made of steel), according to construction
679.7.025.7	Conditioning by heat treatment	679.736	Manufacture of wires, wire strands and wire ropes (especially those made of steel), according to field of application
679.7.025.8	Other methods of conditioning	679.737	Design, manufacture and mounting of strand and rope terminals, fittings, connections and attachments
679.7.027	Methods of joining wires and sheaths/coverings ⇒ 621.79	679.738	Testing of raw materials, semimanufactured products and end-products <i>Details by :620.1...</i>
679.7.027.2	Riveting		
679.7.027.3	Soldering	679.74	Manufacture of wire and cable for conducting electricity (conductors for power and signals) <i>Denote materials by -03... (Table 1k)</i> ⇒ 621.315
679.7.027.4	Welding. Melting. Fusing		
679.7.027.5	Sticking. Gluing	679.742	Manufacture of conductors, according to function. □ As main, neutral, earthing, auxiliary conductor
679.7.027.6	Twisting together		
679.7.027.8	Wrapping with wire	679.743	Covering (including liquid and gaseous layers) of the conductors, cores and stranded cores, according to function
679.7.028	Special processes or treatment	679.743.1	Electrical insulation
679.7.028.2	Cable tray winding and unwinding	679.743.2	Thermal insulation
679.7.028.4	Coiling. Winding on drums	679.743.3	Preventing the ingress of moisture (lead/aluminium cover)
679.7.028.5	Stretching, smoothing and pressing of paper coils	679.743.4	Mechanical protection
679.7.028.7	Colouring ⇒ 621.794 ⇒ 621.795	679.743.5	Chemical protection
679.7.028.8	Marking. Inking. Printing	679.743.6	Electrical screening
679.7.029	Other methods	679.743.8	Filling
679.7.03	Raw materials <i>Denote the material by -03... (Table 1k)</i>	679.744	Manufacture of cores (not being conductors). □ Fillings
679.7.032	Raw materials for wire and yarns (not for sheaths or coverings)	679.745	Manufacture of wires and cables, according to construction
679.7.034	Raw materials for sheaths and coverings	679.746	Manufacture of wires and cables according to field of application
679.7.05	Machinery, equipment, installations, plant and tools ⇒ 677.052.6	679.746.2	Transmission and distribution cables for electrical energy
679.7.052	Equipment, machines for manufacturing separate wires, yarns or tape (such as strapping) 679.7.052 ≡ 679.7.022	679.746.3	Installation wire and cable
679.7.053	Equipment, machines for stranding. Stranders. □ Quadders	679.746.4	Wiring. Wiring systems
679.7.054	Equipment, machines for covering 679.7.054 ≡ 679.7.024	679.746.5	Cables for transmission of information
679.7.055	Equipment, machines for conditioning 679.7.055 ≡ 679.7.025	679.746.52	Cables for telecommunication
679.7.057	Equipment, machines for for joining wires and sheaths/coverings	679.746.522	Cables for telephony
679.7.058	Special installations or machines 679.7.058 ≡ 679.7.028	679.746.523	Cables for pulse code modulation
679.7.059	Other equipment and machines	679.746.524	Cables for central antenna television (CATV)
		679.746.528	Cables for special telecommunication purposes. □ Microphone cables. Aerial cables
679.72	Manufacture of fibre yarn, fibre strands, fibre cordage and similar products (such as strapping) <i>Denote materials by -03... (Table 1k)</i> ⇒ 677.021/.022	679.746.55	Submarine cables. River cables
679.722	Manufacture of fibre yarn and strands	679.747	Design, manufacture and mounting of parts necessary for the application of wires and cables
679.724	Manufacture of cores in strands		
679.725	Manufacture of fibre yarn, fibre strands, fibre cordage and similar products, according to construction. □ Ply. Lay	679.748	Testing of raw materials and end-products <i>Details by :620.1...</i>
679.726	Manufacturing of fibre yarn, fibre strands, fibre cordage and similar products, according to field of application ◇ 679.726:621.798 For the packaging industry		

679.76	Manufacture of optical fibre cables	679.829	Other artificial stones. □ Magnesia-based, terrazzo-based artificial stones. Artificial sandstone
679.8	Stone technology. Stone industry ◇ 679.8-026.613 Colours of synthetic stones ⇒ 622	679.85	Natural stone. Building stones. Monumental stones. Freestones 679.85 ≡ 553.5 ⇒ 622.35
679.8.02	Stoneworking processes and operations 679.8.021/.027 ≡ 679.8.051/.057 ⇒ 671.15	679.854.8	Marble products
679.8.022	Grinding and polishing of stones	679.86	Various nonprecious minerals and earths 679.86 ≡ 553.6
679.8.028	Surface treatment of stones	679.861	Clay. Kaolin. Feldspar. Others
679.8.028.2	Cleaning	679.867	Asbestos. Mica. Vermiculite. Others
679.8.028.3	Impregnation	679.87	Precious and semiprecious stones 679.87 ≡ 553.8 ⇒ 549.091 ⇒ 622.37 ⇒ 671
679.8.05	Tools and machines for stoneworking	679.89	Other stones. □ Very hard stones
679.8.051	Planing equipment. Milling equipment	679.9	Technology of other natural processable materials
679.8.052	Grinding equipment. Polishing equipment	679.91	Amber. Meerschaum (sepiolite) ⇒ 553.673 ⇒ 679.86
679.8.053	Cutting equipment	679.92	Shells (of animals). □ Tortoiseshell. Mother of pearl ⇒ 637.61 ⇒ 639.41 ⇒ 675.86
679.8.054	Turning equipment. Lathes	679.93	Corals ⇒ 639.62
679.8.055	Drilling equipment	679.931	Genuine corals
679.8.056	Cleaning equipment. Splitting equipment	679.932	Black corals. □ Gorgona. Antipatharia. Gerardia
679.8.057	Equipment for trimming, roughening, bush-hammering	679.933	Corals of vegetable origin
679.8.06	Stone products	679.937	Vegetable opal. Tabashir
679.82	Artificial stones made by chemical synthesis. Synthetic precious gems. Stones, manufactured in laboratories or factories, the chemical composition of which corresponds to that of natural stones ◇ 679.82.02:66.046.5 Fusion processes. E.g. Verneuil process (flame fusion). Flux fusion ⇒ 549.091 ⇒ 666.23 ⇒ 671 ⇒ 681.11.035.2	679.94	Corozo nut. Vegetable ivory
679.822	Synthetic corundums		
679.823	Synthetic oxides other than corundums. □ Synthetic quartz		
679.824	Synthetic spinels		
679.826	Synthetic diamonds		
679.827	Synthetic beryls. □ Synthetic emeralds		

68 INDUSTRIES, CRAFTS AND TRADES FOR FINISHED OR ASSEMBLED ARTICLES

The -1/-9 auxiliaries from 62-1/-8 and 66-9 are applicable throughout 68 except for -1 under 687, and -3 and -6 under 681.6

681	PRECISION MECHANISMS AND INSTRUMENTS	681.11.031.1	Energy sources and storage
681.1	Apparatus with wheel or motor mechanisms	681.11.031.11	Mechanical energy stores. □ Gravity systems, weights. Elastic systems, springs
681.11	Horology. Watch and clock making <i>Denote details of watch and clock mechanisms by -1/-9 (from 62-1/-8 and 66-9)</i> ◇ 681.11-187.4 High precision. Precision watches and clocks ◇ 681.11-218 Mountings. Suspension. Watch and clock fastenings ◇ 681.11-752 Devices for damping vibrations, shock ⇒ 006.924 ⇒ 531.761 ⇒ 623.542 ⇒ 671.1 ⇒ 739.3	681.11.031.12	Chemical energy sources and stores. □ Batteries. Power cells
681.11.03	Components. □ Watch and clock movements. Clockwork	681.11.031.13	Nuclear energy sources
681.11.031	Devices providing energy, motive power. Winding mechanisms	681.11.031.2	Devices for energy conversion. Winding mechanisms etc.
		681.11.031.21	Mechanical devices. □ Manual winders, keys. Automatic winders (self-winding devices). Oscillating weights
		681.11.031.22	Electric devices. □ Electromagnet. Electric motor
		681.11.031.23	Thermodynamic devices. □ Bimetallic devices
		681.11.031.24	Optical devices. □ Photoelectric cells. Rectifiers
		681.11.032	Regulators, governors (especially oscillators and escapements)
		681.11.032.1	Periodic regulators
		681.11.032.11	Harmonic oscillators. □ Pendulums. Elastic oscillators (balance wheel and spring etc.). Electromagnetic oscillators. Masers
		681.11.032.12	Relaxation oscillators. Time-base oscillators
		681.11.032.13	Rotary regulators. □ Conical pendulums. Centrifugal governors. Flywheels

681.11.032.2	Nonperiodic regulators. □ Water clocks (clepsydrae). Sand glasses	681.113.9	Special-purpose nonportable timekeepers
681.11.032.3	Energy distributors. □ Escapements	681.113.91	Astronomical clocks
681.11.032.4	Timing devices. Index regulators. Trimming condensers etc.	681.113.92	Chronometers for navigation (e.g. ships' chronometers)
681.11.033	Indicating devices	681.113.93	Quartz crystal clocks
681.11.033.1	Optical, visual indicating devices. □ Dials. Visual displays. Leaves. Tablets. Pointers. Clock hands	681.113.94	Atomic, molecular clocks
681.11.033.2	Acoustic, auditory indicating devices. □ Door bells. Door chimes. Alarm bells. Talking clock ⇒ 654.925 ⇒ 780.633.1	681.114	Portable timekeepers (worn on the person). Watches
681.11.033.3	Tactile indicating devices. □ Blind man's watches	681.114.1	Pocket watches. Fob watches
681.11.034	Transmission devices	681.114.8	Wristwatches
681.11.034.1	Clock and watch trains	681.114.9	Other kinds of watch. □ Ring watches. Brooch watches. Pendent watches
681.11.034.2	Wheel mechanisms. Wheels, arbors, pinions etc. □ Movements. Striking mechanisms	681.116	Clock systems
681.11.034.3	Motion works. □ Cannon pinions. Minute wheels. Ratchets and clicks	681.116.1	Master clocks
681.11.034.4	Hand setting mechanisms	681.116.2	Subordinate, slave clocks
681.11.034.5	Transmission devices for special purposes. □ For engaging and disengaging alarm. For calendar, date displays	681.116.3	Synchronous clocks
681.11.034.6	Devices for transmitting, amplifying, converting impulses in timers, chronographs	681.118	Chronographs. Time control and transmission devices ⇒ 681.174/.175
681.11.034.7	Devices for transmission of time signals over a distance	681.118.1	Timing devices. □ Chronoscopes (stopwatches etc.). Telephone timers. Parking meters ⇒ 621.395.663
681.11.035	Bodies. Frames. Frameworks	681.118.5	Time control mechanisms. Time switches. □ Program control clocks. Delayed exposure devices on cameras
681.11.035.1	Plates. Bridge and balance cocks	681.118.7	Time recording devices. Recording clocks. □ Time stamps. Telltales (watchmen's control clocks). Other visual-record timers
681.11.035.2	Bearings. □ Spring bearings. Magnetic bearings. Bouchons. Jewels. Ball bearings ◇ 681.11.035.2-752 Shock absorbers	681.118.8	Clock-like drive mechanisms. □ Mechanisms in recording instruments. Mechanisms in barographs (recording barometers)
681.11.035.3	Assembly devices. □ Studs. Screws. Snails	681.12	Reckoners of consumption. Industrial gauges. Flowmeters. Counters etc. ⇒ 681.138.8
681.11.036	Cases and accessories ⇒ 671.121	681.12.08	Measurement methods 681.12.08 ≡ 53.08
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681.11.036.2	Watch cases	681.121.1	Fluid measuring vessels
681.11.036.3	Clock glasses. Watch glasses	681.121.2	Reciprocating piston meters (positive meters)
681.11.036.4	Outside control devices. Buttons, push pieces etc.	681.121.3	Rotary, oscillating piston meters (semipositive meters)
681.11.05	Clockmakers' tools. Horological equipment and machinery	681.121.4	Rotary, turbine meters (inferential meters)
681.11.054	Machines, machine tools. □ Lathes. Jacot tools. Jewel setting tools	681.121.5	Tipping, tilting flowmeters
681.11.057	Measuring, checking, regulating, timing devices. □ Watch timers. Movement simulators	681.121.7	Flowmeters working on various principles
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681.111/.118	Kinds of timekeeper. Clocks, timepieces, chronometers etc.	681.121.74	Partial flow meters (indicating relative proportions of different fluids). Proportion meters
681.111	Timekeepers with no mechanical drive. Ancient kinds of timekeeper	681.121.8	Rate-of-flow meters. Current meters
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681.111.3	Clepsydrae. Water clocks	681.121.84	Differential-head meters (pressure-difference meters)
681.111.4	Sand clocks. Sand glasses. Hour glasses	681.121.85	Auxiliary and partial flow meters
681.111.6	Candle clocks. Lamp clocks	681.121.87	Rate-of-flow meters for open channels
681.111.8	Clocks with verge (crown wheel) escapement. Clocks with foliot escapement	681.121.88	Discharge meters
681.113	Nonportable timekeepers	681.121.89	Other kinds of rate-of-flow meter or method
681.113.1	House clocks. □ Wall, table clocks. Pendulum clocks. Longcase clocks (grandfather, grandmother clocks)	681.122	Meters for gases and vapours
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681.113.3	Exterior clocks. □ Tower clocks. Steeple clocks	681.122.3	Meters for dry gases
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681.124.076	Zero-setting devices	681.17	Registering, recording and control devices ⇒ 681.626
681.124.077	Automatic release of an operation. □ Shut-off of machines. Cut-out after reaching specified total	681.171	Mechanical tills with recording mechanisms. Cash registers
681.124.1	Counting devices according to kind of counting mechanism. □ Pointer mechanisms. Drum roller mechanisms. Disc mechanisms	681.171.07	Cash register calculating mechanisms
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681.125	Calorimeters	681.171.085	Indicating, signal mechanisms. Shutters. Display
681.128	Level or stage indicators for vessels, containers	681.171.087	Registering, recording, printing mechanisms. Mechanisms for issuing bills or receipts
681.128.2	Indicators with gauge glass	681.172	Money changing, sorting, distributing, counting machines
681.128.3	Indicators with float	681.172.2	Machines for issuing, distributing, paying out money. □ Automated teller machines (ATMs). Cashpoints. Change-giving mechanisms ⇒ 621.395.663.6 ⇒ 681.13
681.128.4	Indicators working on hydrostatic principles	681.172.3	Money sorting machines
681.128.5	Indicators working on manometric principles (with air pressure or suction)	681.172.4	Coin tubes, sleeves. Automatic cash containers
681.128.6	Indicators working on electrical principles. □ By measuring varying resistance, varying capacitance, varying inductance. Thermoelectric indicators. Photoelectric indicators	681.172.5	Money counting and wrapping devices
681.128.7	Indicators with immersed (stationary or moving) bodies. □ Graduated gauging rods. Vane wheels	681.172.6	Money testing, verifying machines
681.128.8	Indicators working by various specialized methods	681.173	Ticket machines. Cancelling machines. Franking machines ⇒ 656.035.22
681.128.82	Indicators working by vibration	681.173.2	Ticket printing and issuing apparatus ⇒ 681.625.834
681.128.83	Indicators working by weight. Fluid weighing devices	681.173.4	Devices for cancelling travel and other tickets. Ticket punches, perforating devices etc.
681.128.84	Indicators with magnetic indicating device	681.173.6	Fare receiving and checking devices (for public transport etc.)
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681.13	Slot machines. Vending machines. Automats ◇ 681.13:625.748.54 Money-operated petrol pumps ◇ 681.13:641.87 Coin-operated drinks dispensers ◇ 681.13:654.172 Television, video slot machines ◇ 681.13:654.195.6 Radio slot machines ◇ 681.13:658.713 Slot machines for automatic hiring ◇ 681.13:658.713:681.752 Slot machines for optical instruments, opera glasses etc. ◇ 681.13:681.12 Slot meters for gas, water, electricity (prepayment meters) ◇ 681.13:681.2 Slot machines for special-purpose measuring devices ◇ 681.13:681.2:612.14 Slot machines for measuring blood pressure ◇ 681.13:681.26 Slot machines for weighing. Coin-operated weighing machines ◇ 681.13:681.6-6 Coin-operated printing mechanisms ◇ 681.13:681.845 Gramophone, phonograph slot machines. Jukeboxes, nickelodeons ◇ 681.13:685.82 Slot machines for games of chance. Including: Fruit machines (one-armed bandits) ◇ 681.13:681.13:77 Slot machines for producing photographs ◇ 681.13:778.5 Slot machines for presenting pictures. Including: For moving pictures (kinescope, mutoscope etc.) ◇ 681.13:794.85 Slot machines for games of skill. Including: Pinball machines ⇒ 621.395.663.6 ⇒ 681.172.2	681.173.9	Other kinds of checking and control machines
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		681.175	Time control mechanisms <i>For horological aspects see 681.118.1/.7</i>
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		681.175.3	Automatic starters, starting gates, traps (for horse, dog races)
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		681.176	Machines for lotteries, games of chance, betting etc. Score, stake and ballot counting devices. □ Voting machines. Devices for drawing lottery, raffle tickets etc. Totalizators
		681.178	Control devices for activating machinery. Other control devices. □ Devices for checking identity, credit cards, tickets etc.
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		681.182	Sorting devices for grading according to weight
		681.183	Sorting devices for sorting according to various physical characteristics 681.183 ≡ 53
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		681.187	Mail-sorting machines ⇒ 656.851
		681.188	Cryptographic devices. Ciphering, encoding machines. Deciphering, decoding machines ⇒ 003.26

681.2	Instrument-making in general. Instrumentation. Measuring instruments and their manufacture. Balances. Weighing devices	681.269	Other kinds of weighing mechanism, according to use or purpose
	<i>Denote purpose of instrument by :.... Denote details of instrument by -1/-9 (from 62-1/-8 and 66-9)</i>	681.269.1	Letter balances
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	◇ 681.2-5 Operation, control, regulation of instruments	681.337	Ancillary analog equipment. Input and output sections
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681.2.081	Units. Constants		⇒ 007
681.2.082	Principles of measurement. Use of mechanical, optical, electrical phenomena etc.		⇒ 331.103.255
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681.264	Compound-lever platform scales (for large commercial loads). Weighbridges	681.5.015.33	Prony's method
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681.584.76	Relays based on ionic devices	681.587.72	Electromechanical actuators
681.584.77	Electrostatic relays	681.587.73	Electromachine actuators
681.584.78	Magnetic relays	681.587.75	Electrical actuators with clutches
681.584.781	Contact relays in general	681.587.78	Electromagnetic actuators
681.584.782	Electromagnetic relays. □ Reed relays		
681.584.783	Magnetolectric relays	681.6	Graphic reproduction machines and equipment
681.584.784	Electrodynamic relays		<i>The 62-3... and 62-6... auxiliaries are not applicable at 681.6 and its subdivisions</i>
681.584.785	Magnetostrictive relays		⇒ 655.1/3
681.584.787	Magnetic amplifier relays		⇒ 76
681.584.8	Radioactive relays		⇒ 774/777
681.586	Transducers. Sensors		⇒ 778.1
681.586.2	Mechanical transducers and sensors. Gyroscopic devices	681.6-3	Printing processes
681.586.3	Fluidic, hydraulic, pneumatic transducers	681.6-32	Relief, letterpress printing
681.586.34	Hydraulic transducers and sensors	681.6-33	Diagraphy. Screen and stencil printing
681.586.35	Pneumatic transducers and sensors	681.6-34	Planographic printing
681.586.37	Electric-to-fluidic transducers. Electric-to-pneumatic transducers	681.6-35	Intaglio printing
681.586.4	Acoustic transducers and sensors	681.6-6	Graphic reproduction machines according to various characteristics
681.586.48	Ultrasonic transducers	681.6-61	Equipment according to colour of ink
681.586.5	Optical transducers and sensors	681.6-612	Individual colours
681.586.54	Acousto-optical transducers		<i>Denote colours by -026.613... (Table 1k)</i>
681.586.57	Electro-optical transducers	681.6-62	Equipment according to number of colours
681.586.59	Magneto-optical transducers	◇ 681.6-62-623	Three-colour equipment
681.586.6	Thermal transducers and sensors	681.6-63	Machines according to the state of the ink etc. used
681.586.62	Thermomechanical transducers. □ Bimetallic transducers	681.6-631	Using mobile liquid inks
681.586.67	Thermoelectric transducers. □ Thermocouple transducers	681.6-632	Using viscous inks
681.586.69	Thermistors	681.6-633	Using solid materials. □ Using metal foil, metal powder
681.586.7	Electrical and magnetic transducers	681.6-64	Inkless printing (relief, embossed effect). Embossing
681.586.71	Potentiometric, resistance transducers		
681.586.72	Semiconductor transducers. □ Hall effect transducers	681.6.09	Combined machines (for more than one printing process)
681.586.73	Optoelectronic transducers. □ Photoelectric transducers		
681.586.74	Electrochemical and electrolytic transducers	681.61	Typewriters. Shorthand or stenographic machines. Composing machines (typesetting machines). Equipment for production of printing surfaces
681.586.76	Thermionic valve transducers		
681.586.77	Electrostatic transducers	681.61.06	Details of typewriters and composing machines
681.586.772	Capacitance transducers	681.61.061	Types and matrices
681.586.773	Piezoelectric transducers		
681.586.774	Ferroelectric transducers. □ Rochelle salt transducers		

681.61.062	Internal mechanisms. Movement and control devices	681.613.7	Typewriters for book or ledger entries. Flat-platen typewriters
681.61.062.1	Control of type bars	681.613.8	Tabulating typewriters. Typewriters for tables, tabular material
681.61.062.2	Shift mechanisms	681.615	Machines and equipment for production of type. Printing type
681.61.062.3	Carriage-operating mechanisms	681.615.2	Sorts of type. □ Characters. Symbols. Justifying material. Spacing. Rules
681.61.062.4	Line-spacing mechanisms	681.615.4	Machines and equipment for casting of type. Typefoundry equipment
681.61.062.5	Back-spacing mechanisms	681.616	Composing or typesetting machines and equipment
681.61.062.6	Carriage return mechanisms	◇ 681.616.061	Matrices
681.61.062.7	Roller-turning mechanisms	◇ 681.616.063	Medium for automatic control, e.g. punched tape
681.61.062.8	Mechanisms for holding and guiding the type. □ Type bars. Type wheels	◇ 681.616.065	Keyboards of composing machines
681.61.063	Carriage details of typewriters. Register bands (control devices) of composing machines	◇ 681.616.068	Justifying devices in composing machines
681.61.063.2	Margin controls	681.616.1	Composing tools and equipment. □ Chases. Quoins. Composing frames. Composing sticks. Galleys
681.61.063.3	Devices for indicating the position of the carriage	681.616.2	Early typesetting equipment, for composing by hand
681.61.063.4	Construction and guiding of the carriage	681.616.3	Mechanisms for composing and distributing ready-made type
681.61.063.5	Paper-guiding devices. □ Feed rollers. Guide plates	681.616.4	Machines for composing and casting of type, by matrix
681.61.063.6	Rollers. Bearings for the rollers	681.616.42	Line composing and casting machines (slug-casting machines). □ Linotype. Intertype
681.61.064	Inking mechanisms	681.616.422	Magazines. Matrix magazine mechanisms
681.61.064.3	According to form of inking device	681.616.423	Composing mechanisms
681.61.064.34	Mechanisms with inking ribbon. Ribbons (typewriter ribbons)	681.616.424	Feed mechanisms
681.61.064.35	Mechanisms with ink pads	681.616.425	Casting mechanisms
681.61.064.37	Mechanisms with ink rollers	681.616.426	Mechanisms for alignment and locking of matrices
681.61.065	Keys. Keyboards	681.616.427	Mechanisms for finishing the cast slugs (removal of burr, beard)
681.61.067	Devices relating to type and composed matter	681.616.429	Matrices and spacing for line-casting machines
681.61.067.1	Mechanisms for adjusting the visibility of the type. Blackness. Density. Type impact control	681.616.43	Single-letter composing and casting machines (character-casting machines). □ Monotype machines
681.61.067.2	Sound absorption fitments	681.616.432	Die cases
681.61.067.3	Underlining or line drawing mechanisms	681.616.435	Type casting mechanisms
681.61.067.4	Deletion and erasure mechanisms	681.616.436	Type line collecting mechanisms
681.61.067.6	Duplicating and multiple copying mechanisms	681.616.47	Machines for assembly of cast type, for office relief-printing machines. □ Multigraph machines
681.61.067.7	Counters for characters, words, lines and pages	681.616.48	Machines for distributing cast type, for office relief-printing machines
681.61.067.9	Other parts. □ Type cleaning devices. Copy holders	681.616.5	Machines for embossing using movable matrices or dies. Die-stamping machines
681.61.068	Alignment devices	681.616.6	Keyboard typesetting machines
681.612	Typewriters in general	681.616.7	Photographic typesetting machines. Phototypesetting, photocomposing or filmsetting machines and equipment
681.612.1	Type-bar machines with covered type basket	⇒ 655.287	Special devices for composing
681.612.2	Type-bar machines with open type bed	681.616.8	Machines and equipment for producing controlling medium (e.g. tape) for composing machines. □ Punches for paper tape
681.612.3	Typewriters with type wheel, cylinder, drum etc., with keys	681.616.82	Equipment for the programming of composing machines by means of data processing
681.612.4	Typewriters with type wheel, cylinder, drum etc., with single type control device (e.g. pointer, indicator)	681.617	Machines and equipment for production of relief printing surfaces
681.612.5	Other typewriters with mechanical action. □ With type-carrying plate. With type-carrying striking bars. With type on endless band or chain, or on rule or strip. With spherical type-carrier or ball	⇒ 655.222	
681.612.7	Electric typewriters	⇒ 655.225	
681.612.8	Accessories. Tabulators	681.617.2	Machines and equipment for production of stereotypes (including those of rubber and plastics)
681.612.84	Justifying mechanism. Typewriters with justification	681.617.21	Flong and other moulding materials
⇒ 655.286			
681.612.9	Other typewriters. □ Silent, noiseless typewriters. Automatic typewriters for copying etc.		
681.613	Specialized typewriters, according to purpose		
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681.613.1	Braille typewriters		
681.613.2	Typewriters producing special form of record or symbols. Perforating typewriters (producing punched record, e.g. punched tape)		
681.613.3	Shorthand or stenographic machines		
681.613.4	Teletype machines. Teleprinters		
⇒ 621.394.625.3			
⇒ 654.145			
681.613.5	Typewriters for musical notation		

681.617.22	Machines and equipment for production of stereo moulds	681.62.067.3	Various mechanisms. Clamping grippers. Suction grippers. Point devices
681.617.23	Machines and equipment for casting and processing of stereos	681.62.067.4	Devices to prevent staining, soiling of paper
681.617.24	Machines and equipment for processing of relief printing surfaces of all kinds	681.62.067.5	Delivering, stacking mechanisms for printed sheets
681.617.26	Equipment for coating or plating of stereos (and other relief surfaces). □ Equipment for chromium-plating	681.62.067.6	Collating, sorting mechanisms for printed sheets. □ Delivery trays. Sorting racks
681.617.3	Machines and equipment for galvanoplastic duplication of blocks (electrotyping)	681.62.067.7	Mechanisms for folding and cutting the printed sheets ⇒ 621.967.8
681.617.4	Machines and equipment for production of original relief blocks (etching, process engraving, photoelectric engraving)	681.62.067.8	Heaters. Dryers. Antistatic devices
681.617.6	Machines and equipment for production of curved relief surfaces	681.621	Printing machines for office, administrative and laboratory use. Duplicating machines
681.617.8	Equipment for reduction or enlarging of images. □ Pantographs	681.621.1	Duplicating machines. □ Copying presses. Document copiers. Hectographs. Spirit duplicators
681.618	Machines and equipment for production of non-relief printing surfaces ⇒ 655.224 ⇒ 655.226/227	681.621.3	Stencil duplicating machines
681.618.3	Machines and equipment for production of planographic printing surfaces	681.621.4	Duplicating machines with printing type
681.618.4	Machines and equipment for production of intaglio printing surfaces	681.621.5	Planographic duplicators, direct and offset. Rotary direct lithographic printers. Offset duplicators
681.618.5	Machines and equipment for production of diagraphic printing surfaces (stencils and screens)	681.621.6	Addressing machines
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681.62.063	Printing mechanisms. Fastening of printing surfaces	681.621.65	Addressing machines using spirit process
681.62.063.3	Platens	681.621.7	Laboratory printing machines (for duplicating write-ups of experiments)
681.62.063.4	Tables	681.621.8	Machines for special duplicating processes (electrical, chemical)
681.62.063.5	Impression cylinders, rollers	681.621.83	For electrical duplicating processes
681.62.063.6	Fastening of printing surfaces (to the table or cylinder)	681.621.84	For chemical duplicating processes
681.62.064	Mechanisms for inking and de-inking the printing surface	681.622	Hand printing presses. Platen presses
681.62.064.1	Inking mechanisms in general	681.622.1	Early, wooden presses
681.62.064.2	Ink reservoirs	681.622.2	Single-colour platen presses. Platen presses for monochrome printing
681.62.064.3	Actual inking devices. □ Brushes. Capillary devices. Pads. Rollers	681.622.3	Multicolour platen presses. Platen presses for colour printing
681.62.064.4	Devices for pulverized and solid inks	681.622.4	Platen presses for proofing
681.62.064.5	Damping mechanisms	681.622.6	Platen presses for embossing
681.62.064.6	Holding devices for the inker	681.624	Relief printing presses. Letterpress printing machines <i>For relief platen presses see 681.622</i>
681.62.064.7	Replacement of inkers	681.624.1	Flat-bed cylinder presses
681.62.064.9	De-inking devices. Wipers	681.624.4	Rotary letterpress machines
681.62.066	Mechanisms for feeding the material to be printed (paper feed etc.)	681.624.41	Sheet-feed rotary letterpress machines
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681.62.066.2	Feed of individual sheets. Sheet feeders in general	681.624.5	Web-feed rotary letterpress machines for colour printing (multicolour printing)
681.62.066.3	Sheet feeders with friction rollers, friction bands etc.	681.624.8	Flexographic printing machines
681.62.066.5	Sheet feeders with needles, metal brushes etc.	681.625	Non-relief printing presses. Special-purpose printing machines
681.62.066.6	Devices for lifting individual sheets by suction	681.625.1	Intaglio printing presses
681.62.066.7	Devices for raising, lowering etc. of the stack	681.625.11	Copper-plate printing presses. Gravure presses
681.62.066.8	Automatic feed mechanisms for cards and envelopes	681.625.12	Flat-bed intaglio presses
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		681.625.21	Planographic hand presses
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681.625.78	On rubber or plastics		
681.625.8	Printing presses for particular products ⇒ 655.28.026	681.65	Accessories for printing works ⇒ 667.52 ⇒ 676.22
681.625.811	Book presses	681.651	Various masses and materials
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681.625.823	Jobbing presses	681.652.3	Metal printing plates
681.625.825	Printing presses for packaging, labels	681.652.4	Printing plates of other materials (e.g. rubber, plastics) <i>Denote material by -03...</i>
681.625.827	Transfer printing machines	681.652.5	Flexible printing plates
681.625.834	Printing machines for travel tickets ⇒ 681.173.2	681.652.6	Rigid curved printing surfaces ⇒ 681.62.061
681.625.835	Form printing machines	681.654	Rollers for inking, damping, cleaning etc.
681.625.836	Banknote printing machines	681.654.3	Wiping rollers. Cleaning rollers
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681.625.9	Other special purpose printing machines. □ Computer printers. Machines for desktop publishing ⇒ 004.356.2	681.7	Optical apparatus and instruments <i>Details of optical instruments by -1/-9 (from 62-1/-8 and 66-9)</i> ⇒ 520 ⇒ 77
681.625.91	Impact printers	681.7.01	Theory and general characteristics of optical instruments
681.625.912	Dot-matrix impact printers	681.7.012	Aberrations
681.625.914	Line printers	681.7.013	Characteristics and performance of optical instruments ⇒ 77.022.8
681.625.918	Plotters	681.7.013.1	Ray path
681.625.92	Non-impact printers <i>Denote particulars of processes not listed here by :77...</i> ◇ 681.625.92:771.429 Scanning printers, electronic printers etc. ◇ 681.625.92:772.938 Electrographic printers	681.7.013.2	Light-gathering power
681.625.922	Thermal printers. □ Thermal dye transfer printers	681.7.013.22	Aperture
681.625.923	Ink-jet printers	681.7.013.24	Luminous intensity
681.625.924	Laser printers, light-emitting-diode (LED) printers, and similar page printers ◇ 681.625.924:772.16 Laser printers using dry silver processes	681.7.013.25	Light transmittance
681.625.97	Ruling machines. Machines for printing grids, graph paper	681.7.013.3	Field of vision. Vignetting
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681.626.11	Hand stamps. Hand printing devices	681.7.013.7	Depth of focus
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681.628.4	Bronzing machines and devices	681.7.022	Chipless working. □ Casting. Moulding. Rolling
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681.628.7	Machines for pretreatment and aftertreatment of material to be printed	681.7.023.4	Milling
681.628.8	Machines for washing the rollers	681.7.023.5	Edge shaping
681.628.9	Other accessories	681.7.023.7	Grinding and polishing
681.628.93	Conveying mechanisms. □ Conveying tables. Transport rollers	681.7.024	Assembly of optical glasses. Bonding. □ Fine cementing. Optical contacting. Wringing. Fusion. Soldering. Welding
681.628.94	Measuring apparatus. □ Block-height measuring gauges. Etch-depth gauges. Magnifiers	681.7.025	Mounting and retention for working. □ Coarse cementing. Mechanical and pneumatic clamping
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681.7.03	Materials for the manufacture of optical apparatus. Optical glasses and other materials ⇒ 666.22	681.7.062.2	Plane mirrors
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681.7.042	Abrasives. Grinding materials	681.7.064	Apertures. Plates. Filters. Divided circles. Divisions
681.7.043	Polishing agents. □ Rouge. Chromium oxide. Rare-earth oxides	681.7.064.2	Stops (diaphragms). Apertures
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681.7.046	Solvents. □ Spirit. Acetone. Benzene. Ether	681.7.064.42	Absorption filters
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681.7.048	Coolants. Lubricants (oils etc.)	681.7.064.44	Polarization interference filters. Lyot filters
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681.7.054.2	Tools for working the optical surfaces. □ Carriers, supports (e.g. lens holders). Grinding and polishing tools. Pressure tools. Setting, adjusting devices and tools	681.7.066.2	Spherical lenses
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681.7.067.252.5	Special lenses	681.7.072.2	Mountings for optical elements (except spectacle lenses), according to design. □ Simple mountings. Mountings with provision for centring. Mountings with means of focusing. Mountings with adhesives, sealing media
681.7.067.252.6	Zoom lenses	681.7.072.4	Frames, rims and fittings for spectacle lenses ◇ 681.7.072.4-034 Metal frames
681.7.067.253	Photographic mirror objectives (catoptric objectives)	681.7.072.42	Frames for spectacle lenses
681.7.067.254	Anamorphic lenses (anamorphotes)	681.7.072.43	Individual elements of spectacle frames. □ Rims. Bridges. Side-arms. Earpieces
681.7.067.26	Optical projection objectives 681.7.067.26 ≡ 681.7.067.25	681.7.072.44	Fittings, e.g. for rimless spectacles
681.7.067.27	Cinematographic objectives 681.7.067.27 ≡ 681.7.067.25	681.7.072.6	Optical benches and fittings
681.7.067.28	Microscope objectives	681.7.073	Casings. Tubes
681.7.067.281	Dry objectives	681.7.074	Tube heads
681.7.067.282	Immersion objectives	681.7.075	Adjusting, focusing devices. Drives. □ Coarse drives. Racks. Fine drives. Micrometer screws, gear drives
681.7.067.283	Microscope mirror objectives (catoptric objectives)	681.7.076	Supports. Backings (e.g. of glass, of metal)
681.7.067.285	Catadioptric objectives	681.7.077	Stages. □ Sliding stages. Cross stages. Rotating stages
681.7.067.288	Special types of objective. □ Contrast objectives. Interference microscope objectives. Photomicrographic objectives	681.7.078	Stands for optical instruments
681.7.067.3	Oculars (eyepieces)	681.7.08	General principles and theory of optical measurement and the design of optical measuring apparatus 681.7.08 ≡ 53.08
681.7.067.32	Oculars according to number of components ◇ 681.7.067.322 Oculars with two components	681.72	Magnifiers. Microscopes
681.7.067.33	Huygens oculars	681.722	Magnifiers ⇒ 662.591
681.7.067.34	Ramsden oculars	681.723	Microscopes
681.7.067.35	Kellner oculars	681.723.2	Microscopes according to their mode of action
681.7.067.36	Orthoscopic oculars. Compensating oculars. Periplanatic oculars	681.723.21	Transmitted-light microscopes
681.7.067.37	Projection oculars. Projectives	681.723.22	Incident-light microscopes
681.7.067.38	Quartz oculars. Measuring oculars. Special oculars. □ Microscope micrometers. Autocollimation oculars	681.723.23	Contrast microscopes
681.7.067.5	Inverting systems, particularly for telescopes	681.723.24	Projection microscopes. Solar microscopes
681.7.067.7	Condensers	681.723.25	Ultramicroscopes
681.7.067.72	Bright-field condensers	681.723.26	Interference microscopes. □ Michelson micro-interferometer. Interference magnifier.
681.7.067.73	Dark-field condensers	681.723.27	Crystal optic eyepiece
681.7.067.74	Mixed-light condensers (with light of various colours)	681.723.28	Fluorescence microscopes
681.7.067.75	Change-over condensers	681.723.4	Polarization microscopes. □ Orthoscopes. Conoscopes
681.7.067.76	Vertical illumination condensers (top illumination condensers)	681.723.7	Microscopes according to design. □ Monocular microscopes. Binocular microscopes. Pancratic microscopes (with zoom magnification). Autocollimation microscopes
681.7.067.77	Special condensers	681.723.78	Microscopes according to their application. □ Universal microscopes. Dissection microscopes
681.7.067.772	Contrast condensers. Phase contrast condensers	681.73	Pocket microscopes. Travelling microscopes
681.7.067.773	Interference condensers	681.731	Spectacles. Monocles. Contact lenses
681.7.067.775	Pancratic (zoom) condensers	681.732	Monocles
681.7.067.778	Ultraviolet condensers	681.733	Spectacles (with optical effect) in general. With monofocal, bifocal, multifocal lenses
681.7.068	Fibre-optic devices ⇒ 666.189.21	681.735	Pince-nez. Lorgnettes
681.7.068.1	Fibre-optic devices according to flexibility (rigid; flexible)		Contact lenses (scleral and corneal lenses)
681.7.068.2	Single fibres		

681.738	Spectacles without optical effect. □ Safety glasses. Protective glasses. Goggles. Sunglasses ⇒ 614.893.2 ⇒ 666.247 ⇒ 681.7.066.6	681.783	Geodetic and photogrammetric measuring instruments ⇒ 528.5
681.75	Telescopes. Binoculars	681.783.2	Optical geodetic measuring instruments. □ Levels. Theodolites. Tacheometers (tachymeters). Laser rangefinders. Setting-out prisms
681.751	Monocular telescopes	681.783.3	Photogrammetric instruments ⇒ 528.7
681.751.2	Astronomical telescopes ⇒ 520.2 ⇒ 681.782	681.783.32	Aerial photogrammetric equipment
681.751.3	Terrestrial telescopes. □ Observation telescopes. Sighting telescopes (telescopic sights). Periscopic telescopes	681.783.322	Surveying instruments. □ Serial air-survey cameras (automatic cameras). Navigation instruments. Laboratory instruments
681.752	Binocular telescopes. Binoculars. □ Prism binoculars. Field glasses. Galilean binoculars. Opera glasses	681.783.323	Transformation instruments
681.756	Telescopes with concave mirror objectives. Reflecting telescopes	681.783.325	Evaluation instruments. □ Topographic mapping instruments. Stereocomparators. Radial triangulators. Stereoscopes
681.758	Radio telescopes	681.783.33	Terrestrial instruments
681.77	Cameras. Projectors. Collimators	681.783.332	Surveying instruments. □ Ballistic measuring cameras
681.772	Cameras	681.783.335	Evaluation instruments (stereo mapping instruments)
681.772.2	Photographic cameras. Still cameras ⇒ 771.31	681.784	Testing and measuring instruments for medical and ophthalmic optics ⇒ 616-072 ⇒ 616-7 ⇒ 617.7
681.772.3	Cinematographic cameras (cine cameras) ⇒ 778.53	681.784.2	Apparatus for determining the properties of spectacle lenses (objective methods). □ Ophthalmometers. Keratoscopes. Skiascopes
681.772.7	Television cameras. Video cameras ⇒ 621.397.42	681.784.3	Apparatus for determining visual acuity (VA) and the properties of spectacle lenses (subjective methods). □ Refraction-measuring spectacle lenses (trial lenses). Trial cases. Phoropters. Sight-test equipment (e.g. test charts, test objects). Sight-testing instruments. Cross-cylinders. Cylinder compensators. Instruments for testing eye coordination. Optometers
681.775	Projectors ⇒ 628.946	681.784.4	Other instruments for measuring and observing the eye
681.775.2	Still projectors ⇒ 778.27	681.784.41	Ophthalmoscopes
681.775.22	Diascopes	681.784.43	Instruments for measuring the field of vision
681.775.23	Episcopes	681.784.44	Instruments for testing colour vision
681.775.24	Epidiascopes	681.784.45	Instruments for measuring the internal pressure of the eye. Tonometers
681.775.5	Cinematographic projectors ⇒ 688.746 ⇒ 778.55	681.784.46	Adaptometers
681.775.7	Television projectors	681.784.47	Accommodometers
681.777	Collimators. □ Autocollimators	681.784.48	Exophthalmometers. Pupillometers
681.78	Optical and optoelectric measuring and observation instruments	681.784.5	Instruments for matching lenses to the eye
681.782	Optical astronomical instruments ⇒ 520 ⇒ 681.751.2	681.784.6	Instruments for measurements concerned with spectacles and spectacle lenses. □ Vertex refractometers. Spherometers
681.782.2	Astronomical observation and measuring instruments	681.784.7	Optical therapy devices. □ Ray lamps. Light condensers. Orthoptic and pleoptic instruments
681.782.22	Astrographs	681.784.8	Aids to optical observation and diagnosis. □ Diagnostic magnifiers. Gonioscopes. Surgical microscopes. Endoscopes. Special colorimeters. Devices for counting blood corpuscles. Fundus cameras
681.782.23	Astrolabes	681.785	Instruments for measuring optical properties. Optical instruments for determining nonoptical properties
681.782.24	Transit instruments	681.785.2	Refractometers
681.782.25	Meridian circles	681.785.22	Limiting-angle refractometers. □ Abbé refractometers. Immersion refractometers. Sphere refractometers. Jewel refractometers
681.782.26	Zenith telescopes	681.785.23	Deflection refractometers. □ Difference refractometers. Hilger refractometers. Chance refractometers. Microscope refractometers. Spectrometers
681.782.27	Vertical circles		
681.782.4	Auxiliary equipment ⇒ 520.3		
681.782.42	Instruments for solar observation. □ Coelostats. Spectroheliographs. Coronagraphs. Helioscopes, heliometers. Heliostats. Heliotropes, heliographs. Solar magnetographs ⇒ 520.24		
681.782.43	Photometers		
681.782.44	Interferometers		
681.782.45	Micrometers		
681.782.46	Spectrographs		
681.782.47	Guidance and control equipment		
681.782.5	Evaluation equipment. Comparators		
681.782.7	Planetariums. Domes ⇒ 520.98 ⇒ 727		
681.782.8	Simulators		

681.785.24	Reflection refractometers	681.84.081	Devices for recording and reproduction of sound waves
681.785.3	Polarimeters. Polariscopes. Ellipsometers	681.84.081.3	Styluses. Gramophone needles. Recording styluses. Transducing styluses. Stylus-changing devices
681.785.4	Photometers ⇒ 771.376	681.84.081.4	Sound boxes. Pickups
681.785.42	Photometers according to their function and wavelength range	681.84.081.42	Direct mechanical sound boxes (acoustic sound boxes)
681.785.422	Filter photometers	681.84.081.45	Pneumatic, compressed-air sound boxes
681.785.423	Spectrophotometers	681.84.081.47	Electromagnetic pickups
681.785.43	Photometers for measuring absorption	681.84.081.48	Condenser-microphone sound boxes. Electrodynamical, electrostatic, piezoelectric and other electric pickups
681.785.44	Photometers for measuring emission. □ Arc photometers. Spark photometers	681.84.082	Control of the stylus. Other devices related to the pickup
681.785.45	Photometers for luminescence. Fluorescence photometers. Phosphorescence photometers	681.84.082.2	Control, guidance of needle or stylus
681.785.46	Photometers for Raman radiation	681.84.082.3	Needle holders. Stylus holders, cartridges
681.785.47	Photometers for measurement of scattered light. Scattered-light photometers. □ Turbidity photometers	681.84.082.4	Devices for automatic control of pickup
681.785.5	Spectroscopic apparatus ⇒ 543.42	681.84.082.5	Stopping and starting mechanisms
681.785.52	Filter equipment	681.84.082.9	Other devices. □ Cueing devices, for indicating particular passages in a recording. Marking and signalling devices
681.785.53	Prism spectroscopic apparatus. □ Dispersion prisms (e.g. Abbé, Amici, Rutherford prisms). Prism spectroscopes. Prism spectrometers. Prism spectrographs. Prism monochromators. Prism spectrophotometers. Instruments for producing optimal colours using prisms. Instruments for mixing spectrally dispersed light using prisms	681.84.083	Various physical processes of sound recording and reproduction (except direct mechanical or acoustic)
681.785.55	Spectroscopic apparatus with grating. □ Diffraction gratings. Grating spectroscopes. Grating spectrometers. Grating spectrographs. Grating monochromators. Grating spectrophotometers. Instruments for producing optimal colours using gratings. Instruments for mixing spectrally dispersed light using gratings	681.84.083.5	Optical sound recording ⇒ 778.534.4
681.785.57	Interference spectroscopic apparatus. □ Interference spectroscopes. Interference spectrometers. Fourier spectrometers	681.84.083.7	Sound recording by electrical resistance
681.785.58	Evaluation apparatus for spectrograms. Auxiliary apparatus for calibration. □ Spectral-line photometers	681.84.083.8	Magnetic sound recording
681.785.6	Other instruments for measuring optical properties. □ Densitometers. Goniophotometers. Instruments for determining focal lengths, apertures. Instruments for determining image properties, aberrations. Instruments for determining optical transmission functions. Twin-telescope instruments	681.84.084	Devices for acoustic transmission of sound waves. □ Tone arms. Horns, trumpets. Speaking tubes. Sound reflectors
681.786	Instruments for measuring geometrical properties. □ Length-measuring, distance-measuring instruments. Angle-measuring, dividing instruments. Goniometers. Coordinate-measuring instruments. Area-measuring instruments	681.84.085	Design and construction of recording studios and equipment
681.786.3	Angular measurement and dividing instruments. □ Goniometers. Sextants ⇒ 656.61.052	681.84.086	Sound amplification and amplifiers. □ Megaphones. Resonators. Reflectors. Modulators ⇒ 621.375
681.787	Interferometric instruments	681.84.087	Various devices for modifying the sound
681.787.2	Interference refractometers	681.84.087.2	Regulation
681.787.5	Interference comparators. □ Väisälä interferometers. Dowell interferometers	681.84.087.3	Mixing. Superimposition
681.787.6	Multiple-beam interferometers	681.84.087.4	Distortion and its correction
681.787.7	Special interferometers. □ Corner-cube interferometers. Angled-mirror interferometers	681.84.087.7	Stereo acoustics in sound recording and reproduction. Achievement of depth and directional effect
681.787.8	Polarization interferometers	681.84.089	Other devices
681.8	Technical acoustics	681.84.089.1	Record holders or supports (on players). □ Turntables. Spindles
681.84	Sound recording and reproduction ⇒ 621.395.625	681.84.089.2	Casings, containers for record players. Cabinets, cases etc. Portable, carry-case machines
		681.84.089.3	Record storage, containers. □ Protective sleeves, drums. Record cases, racks
		681.84.089.5	Speed indicators
		681.84.089.7	Record cleaning devices
		681.84.089.8	Reworking of recordings. □ Splicing, joining, editing of tapes
		681.841	General questions of sound recording. □ Analysis of sound. Measurement of pitch and volume. Test records
		681.842	Sound reproducing machines, audio equipment in general ◇ 681.842:004.357 Players for digital sound files, e.g. MP3 players ◇ 681.842-022.52 Miniature audio devices, e.g. personal stereo
		681.843	Cylinder-playing machines. Cylinder phonographs
		681.844	Disc-playing machines. Disc phonographs. Gramophones. Record players

681.845	Multiple-record machines. Reproducers for more than one cylinder or disc. ☐ Automatic record changers. Jukeboxes ⇒ 681.135.5	681.883.065	Buoy-borne systems
681.846	Sound recording and reproducing machines according to physical process employed	681.883.066	Towed systems
681.846.3	Direct mechanical recording and reproduction (acoustic recording and reproduction)	681.883.067	Free-floating systems
681.846.5	Optical recording and reproduction	681.883.068	Airborne systems
681.846.7	Electrical or magnetic recording and reproduction. Magnetic recorders	681.883.07	Sonar systems and equipment according to purpose
681.846.73	Winding and unwinding of the recording medium (tape, wire etc.)	681.883.072	Sonar systems for research
681.846.8	With electrical-optical sound recording and reproduction. Optical recorders. ☐ Compact-disc recorders	681.883.074	Sonar systems for navigation
681.847	Sound recording and reproducing machines according to application	681.883.078	Special-purpose sonar systems. ☐ Military sonar. Naval sonar
681.847.2	Dictation machines	681.883.2	Sonar systems according to type of radiation
681.847.4	Keyboard-operated sound reproducers. ☐ Voders	681.883.22	Pulsed sonar systems
681.848	Sound reproducing machines combined with other equipment (e.g. with cinematograph equipment, clocks, furniture) ⇒ 681.11.033.2	681.883.24	Continuous-radiation sonar systems
681.85	Production of recording media. Records <i>Denote materials by -03... (Table 1k)</i> ◇ 681.854-036.7 Synthetic polymerizate gramophone discs ◇ 681.855-034.71 Aluminium compact discs ⇒ (086.7)	681.883.26	Bearing by means of nondirectional sonar systems
681.85.025	Optical and optical-electric transcription	681.883.27	Bearing by means of directional sonar systems
681.85.027	Electrical methods of duplication. Rerecording. Remastering	681.883.28	Binaural bearing
681.853	Manufacture of cylindrical records	681.883.4	Sonar equipment (including countermeasures and navigation systems)
681.854	Manufacture of disc records (gramophone records)	681.883.41	Sonars
681.855	Manufacture of compact discs for sound recording	681.883.42	Echo-sounding apparatus
681.857	Manufacture of magnetic tapes for sound recording	681.883.43	Sound locators. Hydrophones
681.858	Manufacture of optical tapes for sound recording	681.883.44	Sonobuoys
681.88	Acoustic detection, location, direction-finding and ranging devices. Sound locators. Hydrophones. Sonar	681.883.45	Underwater communication. Supersonic telegraphy. Acoustic telemetering systems
681.881	In the atmosphere: by the binaural process, with flat sensors	681.883.47	Countermeasures against sonar equipment. ☐ Noise generators. False targets
681.882	In the atmosphere: according to the group principle	681.883.48	Sonar navigation systems
681.883	Sonar methods and equipment. Underwater object location. Echo ranging and sounding ⇒ 629.052.2	681.883.6	Sonar units
681.883.02	Methods of target search	681.883.7	Display and recording devices (indicators and recorders)
681.883.022	Active sonar methods	681.883.74	Auditory devices
681.883.024	Passive sonar methods	681.883.75	Visual observation devices
681.883.03	Resolution	681.883.77	Target classification equipment
681.883.04	Range	681.884	Fixing and location by measuring on large bases
681.883.05	Means of sweeping or scanning	681.89	Other acoustic devices and instruments
681.883.052	Step sweep (step-by-step sweep)	681.892	Devices for detecting mechanical vibrations (e.g. in geological research, in machine testing)
681.883.053	Circular scanning	681.9	Engraving and sculpting machines. Reproduction of relief forms ⇒ 73 ⇒ 76
681.883.054	Lateral scanning	681.92	Equipment for pyrography, chasing, engraving etc.
681.883.06	Sonar systems and equipment according to location	681.922	Equipment for engraving or sculpting by heat. ☐ For pyrography, pyrogravure, poker-work, branding
681.883.062	Shipborne systems	681.923	Embossing equipment. Equipment for chasing metal surfaces
681.883.062.2	Surface-ship systems	681.924	Notching, grooving equipment
681.883.062.4	Submarine-borne systems	681.93	Equipment for reproducing relief forms
681.883.063	Sea-bed systems	681.932	Equipment for reproducing sculptures (facsimile, reduction, enlargement)
681.883.064	Dipping systems	681.933	Devices for taking the measurements of a sculpture
		681.934	Equipment for producing maquettes, models. Equipment for copying the contours of mouldings. Equipment for taking mouldings of parts of the human body and other plastic objects
		681.94	Graving, engraving equipment
		681.944	Equipment for milling or otherwise engraving cylindrical surfaces, rollers etc. Cog cutting machines
		681.945	Machines for cutting milled or knurled gripping surfaces on switching wheels, knobs etc.

681.946	Machines for cutting rectangular or parallel lines on plane surfaces. Hachure, cross-hatching machines	683.334	Padlocks
681.947	Engine-turning, guilloche equipment	683.335	Safety locks (with various kinds of key). □ Deadlocks
682	SMITHERY. BLACKSMITHERY. FARRIERY. HAND-FORGED IRONWORK ⇒ 621.73 ⇒ 683 ⇒ 739	683.336	Locks with various safety devices
		683.336.2	Combination locks
		683.336.3	Check locks. Locks which reveal unauthorized opening
		683.336.4	Time locks
		683.336.5	Locks with alarm device
		683.336.6	Locks operated by electric, electronic or magnetic devices
682.1	Farriery. Shoe-smithing ⇒ 636.083.42	683.336.9	Types of lock according to the auxiliary components
682.1.03	Shoeing nails	683.336.96	Mortise locks and other locks with retractable element
682.1.05	Farriers' tools. □ Tongs. Hammers. Rasps	683.336.98	Locks opened by pressure or impact perpendicular to the lock cover (by pushing or pulling)
682.12	One-piece horseshoes		Locks according to application
682.13	Multiple horseshoes	683.338	Door locks
682.14	Soft shoeing. □ Hoof protectors. Soft inserts	683.338.2	Locks for furniture, boxes, suitcases
682.15	Shoes for animals other than equines. □ Cloven-hoof shoes	683.338.3	Attachment locks
		683.338.4	Locking devices for prisoners. □ Handcuffs.
682.17	Calks, calkins. Claws	683.338.48	Manacles. Shackles
682.18	Methods and devices for care of the hooves		
682.3	Toolsmithing. Edge-tool forging ⇒ 621.73 ⇒ 672.3 ⇒ 672.71	683.34	Containers and areas for the safekeeping of valuables ⇒ 672.93
		683.341	Fortified boxes. Strongboxes. □ Cash boxes.
682.5	Furnishing ironwork ⇒ 672.94 ⇒ 683.3 ⇒ 739.4	683.342	Deed boxes
		683.343	Safes
			Safes as part of the building. □ Wall safes.
682.6	Building ironwork ⇒ 672.11 ⇒ 691.71.8	683.35	Strongrooms. Safe deposits
		683.352	Holding devices in general. Catches
682.62	Railings. Balconies. Balustrades	683.352.2	Door holders
682.64	Ornamental ironwork. Signs, plaques etc.	683.352.3	Door holders with play, for retaining shut or ajar
683	IRONMONGERY. HARDWARE. LOCKSMITHING. BOTTLING. LAMPS. HEATING APPLIANCES ⇒ 672.3 ⇒ 691.8	683.352.4	Door holders without play. □ Ball catches
			Door holders with alarm devices. □ Shop door holders with bells
683.1	Ironmongery. Hardware <i>Here are classed general questions and information about the making of various kinds of hardware. Individual items of hardware are classed at the specific number for the subject, e.g. 621.88</i> <i>Fastening and fixing devices (clamps, screw fixings etc.)</i>	683.352.6	Safety chains
		683.353	Window securing devices (window catches, locks etc.)
		683.36	Hinges. Pintles. Door and window furniture. □ Butt hinges. Rollers for sliding doors ⇒ 621.888.2
683.3	Locksmithing. Door furniture ⇒ 684.419 ⇒ 739.4	683.37	Closers, checks, retainers etc. for doors and windows
		683.372	Door closers
683.31	Bolts. Latches	683.373	Automatic closing devices for windows, doors, shutters (activated by weather, timers etc.)
683.311	Simple bolts		Door and window checks (slamming preventers), alone or combined with closers
683.312	Multiple bolts	683.374	Arresters or retainers for windows, doors, shutters in the open position
683.313	Safety bolts	683.376	Devices for skylights, grilles, sliding windows, roller blinds etc.
683.33	Locks. Keys	683.378	
683.33.07	Components of locks	683.5	Bottling and dispensing accessories ⇒ 621.642.17 ⇒ 621.798.147 ⇒ 663.059 ⇒ 666.171
683.33.071	Lock grips. Lock handles. Lock releases	683.51	Serving and drinking vessels and accessories. □ Bottles. Flasks. Jugs. Jars
683.33.072	Lock components. □ Bolts. Keepers. Tumblers. Keyholes	683.53	Closures. Stoppers, corks, caps lids etc.
683.33.073	Devices to prevent removal of key	683.531	Stoppers for closure of storage vessels (particularly liquor storage vessels)
683.33.074	Lock safeguards. Keyhole closures	683.532	Stoppers with devices other than for sealing. □ Stoppers with pourers, sprays, applicators
683.33.078	Keys	683.533	Cap closures
683.331	Simple locks. Locks with day bolt	683.534	Ball closures

683.535	Stirrup clips	683.878	Blue-flame gas burners ⇒ 621.791
683.54	Devices to prevent refilling of bottles. Nonreturn valves	683.88	Ignition and extinguishing devices
683.55	Seals and other safeguards for closures	683.882	Ignition devices for lamps with liquid or solid fuels
683.56	Washing, filling and sealing of containers. Opening and dispensing devices	683.883	Extinguishing devices for lamps with liquid or solid fuels. □ Extinguishers. Blowers. Snuffers
683.561	Cleaning of bottles, glasses, beer piping	683.884	Ignition and extinguishing devices for gas lamps
683.561.2	Cleaning of bottles	683.887	Catalytic gas self-igniters
683.561.4	Cleaning of drinking vessels	683.888	Pilot lights and their design. □ Climbing flames. Flashback lines
683.561.5	Cleaning of beer pipelines		
683.562	Filling apparatus. Bottle and cask filling	683.89	Other parts and attachments. □ Lamp cylinders. Windshields. Holders
683.562.2	Bottling plant for liquor etc. Pressure filling. Siphoning		
683.562.3	Barrel, cask filling plant	683.9	Heating appliances. Stoves ⇒ 621.365 ⇒ 641.53 ⇒ 644.12 ⇒ 66.041 ⇒ 662.9 ⇒ 697
683.563	Closing, wiring, capping of bottles		
683.564	Devices for opening the closures of bottles and other containers. Cap and lid openers ⇒ 641.516 ⇒ 672.466	683.92	Fireplaces, stoves and kitchen ranges in general
683.564.2	Cork extractors. Corkscrews	683.922	Fuelling devices
683.564.3	Devices for opening cap closures. Bottle openers	683.923	Air supply. Draught regulation
683.564.4	Devices for opening tins and jars. Can openers (tin openers). Preserve jar openers	683.927	Stoves and kitchen ranges with boiler for hot water supply, with dish warmer etc.
683.564.5	Devices for extracting bungs from barrels. Bung-drawers	683.929	Ash removal. Cleaning
683.564.8	Devices for opening boxes and cases. Case openers	683.93	Open fireplaces for space heating. Hearths. Grates ⇒ 697.243.3
683.565	Dispensing equipment. Devices for tapping, drawing, siphoning etc.		
683.565.1	Dispensing devices. □ Beer engines. Beer pumps	683.94	Closed fires for space heating. Stoves
683.565.2	Air compressors for beverage dispensing. Beer pressure apparatus	683.943	Solid-fuel stoves
683.565.3	Devices for carbon dioxide and other gases	683.944	Liquid-fuel stoves ⇒ 662.94
683.565.4	Devices for dispensing directly from bottles. □ Siphon bottles	683.945	Gas-fired stoves, heaters. Gas fires ⇒ 662.95
683.565.6	Pressure regulators. Splash protection	683.946	Stoves for solid, liquid or gaseous fuels
683.566	Barrel closures. Bungs etc. Bung securers	683.95	Kitchen ranges. Cooking stoves
683.567	Dispensing taps. Cocks, taps, faucets ⇒ 621.646.6	683.953	Solid-fuel cooking stoves
683.568	Devices for measuring and indicating amounts of liquid	683.954	Liquid-fuel cooking stoves
683.58	Beer mats. Beer reckoners, reference marks etc.	683.955	Gas-fired cooking stoves. Gas cookers
		683.956	Cooking stoves for several fuels (e.g. coal-or-gas-fired)
683.8	Fuel or flame lamps ⇒ 628.9	683.957	Electric cooking stoves
683.86	Details of fuel lamps ⇒ 628.95	683.96	Small portable heating appliances
683.862	Lamp reservoir	683.97	Water heaters. Hot water tanks. □ Continuous-flow water heaters (geysers) ⇒ 644.62 ⇒ 696.4
683.864	Fuel supply (fuel feed)		
683.865	Attachment of reflectors, refractors, shades etc.	683.972	Water circulation systems. Hot water tanks
683.867	Wicks and accessories. □ Wick trimmers	683.973	Direct contact of water container with flames. Kitchen boilers, coppers
683.868	Incandescent bodies, mantles etc. Mantle holders	683.974	Indirect heating through combustion gases
683.87	Burners	683.975	Heating by means of immersion of element. Immersion heaters
683.871	Bright wick burners in general		
683.872	Blue-flame wick burners	683.98	Accessories for heating appliances. □ Igniters (e.g. gas poker). Fire irons ⇒ 662.58/.59
683.873	Arrangement of the wick in the burner. □ Wick sleeves. Wick adjusters. Wick fitting devices		
683.874	Slow burners (in which the gas produced in the wick is not immediately combusted)	684	FURNITURE AND ALLIED INDUSTRIES. FURNITURE MANUFACTURE. UPHOLSTERY <i>Denote details, components etc. of furniture by -1/-9 (from 62-1/-8 and 66-9)</i> ⇒ 645 ⇒ 674
683.875	Bowl burners. Flat flame burners		
683.876	Vapour burners (for volatilized liquid fuel). Hydrocarbon vapour burners. □ Volatilizers. Evaporators		
683.877	Incandescent gas burners. Acetylene burners ⇒ 662.951.2		

684.4	Furniture. Furniture design and manufacture	684.434.8	Thrones. Chairs of state. Similar massive chairs
◇ 684.4:022.9	Design and manufacture of library furniture	684.435	Stools
◇ 684.4:615.478.2	Design and manufacture of hospital furniture	684.436	Multiple seating. □ Benches. Settees. Pews
◇ 684.4:645.63	Design and manufacture of kitchen furniture	684.44	Tables. Desks. Workbenches
◇ 684.4:005.936.33	Design and manufacture of office furniture	684.442	Tables
⇒ 005.936.33:645.4		684.444	Desks. Writing tables. Bureaux. Lecterns
⇒ 672.94		684.446	Counters. Shop counters. Other service counters
⇒ 674.23		684.447	Workbenches. Worktables. Laboratory benches. Butcher's tables etc.
684.4.03	Reproduction furniture. Furniture in imitation historic styles	684.45	Cabinets. Chests. Cupboards. Room dividers
684.4.034	Reproduction Renaissance furniture	684.452	Chests. Coffers
684.4.04	Furniture according to material used in construction	684.453	◇ 684.452.07 Ottomans (upholstered chests)
	<i>Details by -03...</i>	684.453.2	Cupboards
684.4.05	Furniture-making: methods and equipment	684.453.5	Storage cupboards
684.4.051	Preliminary treatment of material	684.453.6	Wardrobes. Linen cupboards
684.4.053	Shaping. Cutting. Drilling. Bending		Lockers (for cloakrooms, factories, schools etc.)
684.4.057	Preparatory surface treatment. □ Planing. Grounding. Filing	684.453.7	Bedside lockers. Bedside cabinets
684.4.058	Assembling. Joining	684.453.8	Special-purpose cupboards. □ Cocktail cabinets
684.4.059	Final surface treatment. Finishing. □ Painting. Staining. Polishing	684.454	Kitchen cabinets. Dressers. Sideboards
	<i>Details by :621.794/.795, :665.92 or :667.6/.7</i>	684.455	Drawer cabinets
684.4.07	Upholstered furniture	684.455.2	Chests of drawers
	⇒ 684.73	684.455.3	Dressing tables
684.41	Furniture components. Furniture accessories	684.455.5	Filing cabinets
684.412	Bases. Support components. Pedestals. Legs. Feet. Struts etc.	684.455.6	Plan cabinets
684.414	Horizontal surfaces	684.456	Display cabinets. Showcases. Special-purpose cabinets (e.g. radio and gramophone cabinets)
684.415	End-pieces. Back-pieces. Side-pieces	684.457	Bookcases
684.415.2	End-pieces	684.458	Room dividers. Combined shelf and cabinet units
684.415.6	Back-pieces	684.46	Shelving. Racking. Stands. Supports
684.415.8	Side-pieces. □ Arms	684.462	Shelving. Racking
684.416	Closures. Covers. □ Doors. Lids. Flaps	684.466	Hall stands. Coat stands. Hat stands. Umbrella stands
684.417	Drawers, trays etc.	684.468	Special-purpose stands
684.419	Furniture accessories. Cabinet hardware. □ Handles. Knobs		<i>Denote purpose by :...</i>
	⇒ 683.3	684.49	Minor articles of furniture
684.42	Beds. Cots. Divans. Couches	684.494	Screens. Fireguards. Fenders
684.422	Beds	684.496	Baskets. □ Linen baskets. Waste paper baskets
	◇ 684.422-182.6 Folding beds	684.498	Vertical display boards. □ Blackboards. Notice boards
	⇒ 685.533	684.6	Veneering. Marquetry. Inlay etc.
684.426	Cots. Cradles		⇒ 674.24
684.428	Divans. Day beds. Studio couches	684.61	⇒ 745.51
	⇒ 684.433.3	684.62	Veneering. Affixing other materials.
684.43	Seating. Chairs. Seats. Benches. Stools	684.63	□ Affixing plastics, metal
684.432	◇ 684.43.07 Upholstered seating	684.64	Inlay
	Plain chairs (not upholstered). Straight-back chairs	684.65	Marquetry
684.433	Easy chairs. Armchairs. Settees. Sofas. Suites	684.7	Relief work. □ Beadings. Trimmings
684.433.2	Armchairs. Club chairs. Fireside chairs		Painted artistic finishes and similar
684.433.3	Settees. Sofas		Upholstery. Bedding. Covers. Hangings
	⇒ 684.428	684.72	⇒ 698.7
684.433.4	Television chairs	684.721	⇒ 745
684.433.7	Rocking chairs	684.722	Bedding. Mattresses. Quilts
684.434	Special types of chair	684.723	Mattresses without springs. Bolsters. Pillows
684.434.2	High-legged chairs. □ Children's high chairs. Workbench chairs	684.73	Spring mattresses
684.434.5	Pedestal chairs. Swivel chairs		Eiderdowns. Bed covers
684.434.7	Vehicle seats		Upholstery
			⇒ 675.16
			⇒ 675.61
			⇒ 684.4.07

684.74	Hangings for furniture. Loose covers for furniture	685.34.013	Research methods. Range of length and width of shoes
684.75	Hangings for rooms. □ Window hangings. Wall hangings. Arrases ⇒ 645.2/.3 ⇒ 691.15 ⇒ 745.52	685.34.016	Shoe design. □ Modelling. Construction
684.76	Hangings for special purposes. □ Funeral drapes. Hangings for indoor and outdoor festivities ⇒ 688.76	685.34.017	Physical, mechanical and processing properties of shoes. Testing methods ⇒ 531.7 ⇒ 620.113
685	SADDLERY. FOOTWEAR. GLOVING. TRAVEL, SPORTS, GAMES AND OTHER EQUIPMENT	685.34.019	Defects and damage. Control and correction <i>Specify defects in main materials by .03, in auxiliary materials by .04; production defects by .02</i>
685.1	Saddlery. Harness making ⇒ 629.313 ⇒ 636.083.7 ⇒ 675.12	685.34.02	Production processes ◇ 685.34.02-932.2 Continuous processes
685.1.05	Saddler's tools	685.34.021	Preparation processes ⇒ 621.923.6 ⇒ 665.9
685.11	Vehicle covers	685.34.021.2	Production, measurement and cutting of patterns
685.12	Saddlery in the strict sense. Saddles and harness for riding and coach horses	685.34.021.3	Lasts. Measurement of lasts
685.121	Saddles	685.34.022	Processes which are repeated at separate stages of shoe production
685.122	Saddle accessories. □ Saddle cloths. Girth belts	685.34.022.2	Splitting (equalizing). Skiving of parts
685.123	Head harness. Bridles, reins and accessories. Blinkers	685.34.022.3	Roughing. Glazing. Milling
685.126	Coach harness	685.34.022.4	Gluing
685.14	Harness making. Saddles and harness for working horses and other equines. □ Pack saddles. Panniers. Draft harness. Sledge harness	685.34.022.5	Evaporation
685.15	Spurriery. Spur and stirrup making. Other accessories. □ Bits. Buckles. Horse brasses	685.34.022.6	Stamping. Marking
685.16	Special harness making. Equipment for animals other than equines (e.g. oxen, camels)	685.34.023	Special methods for manufacturing shoes or shoe parts of polymer materials ◇ 685.34.023-036.664 Manufacture from polyurethane
685.17	Whips. Straps. Thongs. Traces etc.	685.34.024	Punching, cutting and working of shoe components
685.2	Accoutrements. Beltmaking. Leather and other equipment ⇒ 646.4 ⇒ 675.12 ⇒ 687.38	685.34.024.3	Punching, cutting of materials for shoe components
685.22	Equipment worn round the waist. Belts. Belt buckles and attachments	685.34.024.4	Upper devices and equipment
685.23	Equipment worn over the shoulder. □ Sashes. Baldrics. Rifle slings. Braces (trouser suspenders)	685.34.024.5	Working of sole parts
685.24	Equipment for protecting the arms and hands. □ Oversleeves. Arm protectors. Muffs ⇒ 687.39	685.34.025	Manufacture of uppers. Shoe assembly
685.25	Equipment for protecting the body. □ Leather aprons. Plastrons. Life-jackets ⇒ 614.818 ⇒ 614.895 ⇒ 687.17	685.34.025.2	Manufacture of uppers
685.3	Footwear. Other aids to locomotion ⇒ 646.4 ⇒ 675.14 ⇒ 675.62	685.34.025.4	Shoe assembly. □ Pinching and shaping of uppers over last. Attachment of sole and heel
685.34	Footwear industry. Shoemaking. Bootmaking	685.34.027	Finishing of shoes and components
685.34.01	Shoe fashion, design. Properties and defects. Testing methods	685.34.027.2	Shoe sole finishing
685.34.012	Fashion	685.34.027.3	Out-lasting. Attachment of heel from inside
		685.34.027.5	Upper finish
		685.34.027.6	Finishing of shoes
		685.34.027.7	Decorative finish
		685.34.03	Main materials used in the manufacture of shoes 685.34.03 ≡ -03 <i>For properties of shoe materials see corresponding fields of application. E.g. For physical and mechanical properties of leather and their testing see 675.017</i>
		685.34.04	Auxiliary materials used in the manufacture of shoes
		685.34.042	Auxiliary materials for assembly of shoes. □ Sewing threads. Shoe nails. Adhesives. McKay pitch
		685.34.043	Finishing agents. □ Paint. Lacquer. Wax ⇒ 665 ⇒ 667.2 ⇒ 667.63
		685.34.045	Accessories. Decorations. □ Bands. Straps. Buckles
		685.34.046	Antislip, antislid devices. □ Studs. Spikes
		685.34.05	Shoemaking machinery, tools and equipment ◇ 685.34.05-2 Components of shoe machines ◇ 685.34.05-5 Operation and control of shoe machines
		685.34.051	Machines and appliances for preparatory processes 685.34.051 ≡ 685.34.021 ⇒ 621.923.6

685.34.052	Machines, tools and instruments used in various stages of shoe manufacture 685.34.052 $\equiv$ 685.34.022	685.343/.348	Footwear for particular purposes <i>For the use of the apostrophe '... see the note at 685.341</i>
685.34.052.2	Splitting machines for upper leather	685.343	Shoes for daily use
685.34.053	Machines for production of shoes or shoe parts from polymer materials	685.343.2	Outdoor shoes
685.34.054	Machines for punching, cutting and working of shoe parts 685.34.054 $\equiv$ 685.34.024	685.343.4	House shoes
685.34.055	Machines for the manufacture of uppers and for assembly of shoes 685.34.055 $\equiv$ 685.34.025	685.343.8	Overshoes. Galoshes
685.34.057	Machines for finishing of shoes and shoe parts 685.34.057 $\equiv$ 685.34.027	685.344	Footwear for festive occasions. ☐ Fancy shoes. Party shoes ◇ 685.344:793.3 Dancing shoes
685.34.07	Shoe parts. Intermediate products <i>Specify materials by -03...</i>	685.345	Professional footwear ⇒ 614.897 ⇒ 685.341.4
685.34.072	Uppers. Parts of uppers	685.345.2	Footwear for industrial occupations 685.345.2 $\equiv$ 6
685.34.073	Soles. Parts of soles. ☐ Soles. Heels	685.345.222	Footwear for miners
685.34.074	Protective devices	685.345.26	Footwear for workers in the chemical industry
685.34.08	Wastes from shoe production ◇ 685.34.08-035.51 Leather waste	685.345.4	Footwear for performers. ☐ Stage shoes, e.g. buskins. Ballet shoes. Footwear for circus performers
685.341	Footwear according to various properties <i>Denote combinations of subdivisions .1/.8 with each other or with 685.343/.348 and their subdivisions by apostrophe (preferably cited in descending numeric order)</i> ◇ 685.341.83'72'42 Waterproof women's boots ◇ 685.346'852'371.1 Boys' canvas sport shoes	685.345.5	Military footwear
685.341.1	Shoes according to manufacturing or production methods	685.346	Footwear for sporting activity
685.341.12	Mass-produced shoes	685.346.4	Bathing shoes. Beach shoes
685.341.14	Handmade shoes	685.346.6/.9	Shoes for particular sports and games 685.346.6/.9 $\equiv$ 796/799
685.341.2	Shoes according to manufacturing process	685.346.633.2	Football boots
685.341.22	Sole-sprung bottoms	685.346.652	Climbing boots and shoes
685.341.223	Screwed soles	685.348	Hygienic and orthopaedic footwear
685.341.224	Nailed soles	685.348.2	Prophylactic footwear (to prevent foot ailments). ☐ Footwear with orthopaedic or hygienic devices
685.341.23	Sewing processes	685.348.4	Orthopaedic footwear <i>Specify the complaint by :616...</i> ⇒ 615.477.1
685.341.25	Chemical processes	685.348.42	For deformed feet
685.341.252	Glued soles	685.348.43	For shortened legs
685.341.253	Vulcanized soles	685.36	Specialized footwear and accessories for travel on ice, snow, water ⇒ 685.65
685.341.254	Injection-moulded soles	685.361	Ice skates and accessories
685.341.3	Footwear according to the main material 685.341.3 $\equiv$ -03	685.362	Roller skates and related devices. ☐ Skateboards
685.341.353	Wooden shoes. Clogs	685.363	Snowshoes. Skis. Accessories (e.g. ski sticks)
685.341.355.1	Leather footwear	685.363.1	Snowshoes
685.341.355.3	Footwear of imitation leather	685.363.2	Skis
685.341.364	Rubber footwear	685.363.3	Accessories for maintenance and repair. ☐ Ski waxes
685.341.371.1	Canvas footwear	685.363.4	Skins and other antislip devices
685.341.4	Footwear according to protection properties ⇒ 685.345	685.363.7	Sticks, poles for skiing and snowshoeing. Ski sticks
685.341.42	Waterproof	685.364	Waterskis. Accessories
685.341.43	With protection against impact	685.37	Equipment for walking and moving in special terrains. Devices for amplifying jumping force. ☐ Stilts. Pogo sticks
685.341.47	Resistant to slipping, sliding	685.38	Walking and locomotion aids for the infirm and injured
685.341.6	Footwear according to components ⇒ 685.34.07	685.381	Crutches ⇒ 616-089.22
685.341.7	Footwear according to shape	685.382	Wooden legs. Wooden feet ⇒ 615.477.22
685.341.72	Boots	685.383	Wheelchairs
685.341.73	Shoes	685.4	Gloving. Glovemaking ⇒ 646.4 ⇒ 675.15 ⇒ 675.62
685.341.74	Half-shoes. ☐ Randed shoes. Moccasins	685.41	Gloves of common leathers. ☐ Kid gloves
685.341.75	Slippers	685.42	Long leather gloves, gauntlets.
685.341.76	Sandals		Elbow-length, evening gloves. Ladies' fancy gloves
685.341.79	Other shapes		
685.341.8	Footwear for wearers of a particular sex or age		
685.341.82	Men's footwear		
685.341.83	Women's footwear		
685.341.85	Children's footwear		
685.341.852	Boys' footwear		
685.341.853	Girls' footwear		
685.341.86	Footwear for toddlers and babies		

685.43	Furred and lined leather gloves. Fur gloves. Mittens. Muffs. Padded gloves. Boxing gloves. Others	685.558	Sticks and umbrellas convertible into seats, tables etc. □ Shooting sticks
685.44	Gloves of textiles, cut and sewn	685.59	Other travel goods. □ Label holders or protectors. Insect guards, mosquito nets. Field dispensaries. First aid kits
685.45	Rubber gloves		
685.46	Entirely woven gloves. Knitted gloves. Lace gloves. Others	685.6	Sports and games equipment. Equipment and installations for athletics, gymnastics, physical training ⇒ 796.022
685.5	Travel goods. Camping equipment ⇒ 675.16	685.61	General questions
685.51	Luggage. □ Trunks. Suitcases. Bags ⇒ 688.33	685.618	Score-keeping and indicating devices. Scoreboards
685.511	Trunks and chests in general ⇒ 674.62	685.619	Equipment for laying out sports fields, grounds. Equipment for marking out playing areas ⇒ 712.257
685.512	Suitcases. Valises. Portmanteaus		
685.513	Overnight bags. Holdalls. Grips. Carpet bags. Kitbags	685.62	Equipment and installations for games of motion and skill. □ Skipping ropes. Hoops and sticks. Spinning tops and whips. Marbles. Quoits. Yoyos
685.514	Saddlebags. Knapsacks. Rucksacks. Haversacks. Satchels	685.626	Badminton equipment
685.516	Cases for utensils. Canteens	685.628	Equipment for bowls, skittles, ninepins etc. □ Bowling alley equipment
685.517	Picnicking equipment ⇒ 641 ⇒ 642		
685.53	Camping equipment	685.63	Equipment and installations for ball and throwing games
685.531	Tents	685.632	Equipment for high ball games played with a large ball (basketball, volleyball etc.). □ Balls. Nets. Protective equipment
685.531.05	Components. Equipment	685.633	Equipment for football etc. (association football, rugby football, American football, pushball etc.). □ Balls. Posts. Nets
685.531.051	Tent frames	685.634	Equipment for ball and racket games (tennis, rounders etc.). □ Balls. Nets. Rackets
685.531.053	Tent skins and edge connections	685.635	Equipment for bat, mallet, club ball games (golf, polo, hockey, baseball, cricket etc.). □ Balls. Pucks. Bats. Clubs. Sticks
685.531.054	Tent posts, poles	685.637	Equipment for throwing (without target). □ Discuses. Weights, shots. Hammers. Spears. Javelins
685.531.055	Tent pins, pegs (for anchoring to the earth)	685.638	Equipment for aimed throwing. □ Boomerangs. Wooden missiles, clubs
685.531.056	Links between tent and tent pins. □ Guy ropes. Toggles	685.639	Other equipment for ball and throwing games. □ Marks. Butts. Targets
685.531.057	Groundsheets	685.64	Equipment and installations for gymnastics, athletics, acrobatics
685.531.2	Ridge tents. Hike tents	685.641	Gymnastics equipment. □ Dumbbells. Indian clubs. Chest expanders. Spring grips
685.531.3	Large tents. Wall tents. Marquees. Frame tents	685.643	Equipment for jumping, balancing and similar exercises
685.531.4	Bell tents. Conical tents	685.643.2	Equipment for jumping (high jumping, long jumping, pole vaulting). □ Jumping frames, bars, ropes. Vaulting poles
685.531.5	Lean-to tents. Tents as extensions. □ Tents extending from vehicles	685.643.3	Equipment for vaulting. Vaulting horses and boxes
685.532	Temporary shelters. □ Bivouacs	685.643.4	Hurdles
685.533	Portable bedding and sleeping equipment ⇒ 645.425 ⇒ 684.422-182.6	685.643.5	Spring boards. Bars and blocks. Starting blocks
685.533.2	Camp beds	685.643.6	Mats, gym mats. Mattresses
685.533.5	Air beds. Air mattresses (inflatable)	685.643.8	Equipment for balancing exercises. □ Tightropes. Balancing poles. Beams
685.533.8	Sleeping bags ⇒ 645.482	685.644	Horizontal and parallel bars
685.534	Hammocks	685.645	Cross-beams, 'gallows'
685.535	Portable, folding seats. Camp stools. Collapsible chairs	685.646	Trapezes. Rings. Ropes. Wall bars. Roundabouts, 'giant strides'. Other equipment
685.538	Travel cushions. Headrests. Travel seats	685.647	Equipment for acrobatic and contortionist exercises. □ Trampolines
685.54	Load-carrying devices. Equipment for lifting or carrying heavy weights. □ Hooks, grapples, straps. Container belts (with pouches etc.). Bandoliers ⇒ 685.22 ⇒ 685.23	685.648	Equipment for training, practice 685.648 ≡ 79
685.55	Umbrellas etc. Walking sticks	685.648.683	Equipment for boxing training. □ Punchbags. Sparring dummies
685.55.07	Components	685.648.712.3	Equipment for rowing practice
685.55.072	Frames. Handles. Grips. Tips. Spikes		
685.55.076	Opening and closing mechanisms		
685.551	Umbrellas (for protection from rain)		
685.552	Parasols. Portable sunshades		
685.554	Special designs of umbrella and shade. Garden and market shades. Awnings. Canopies		
685.555	Collapsible, telescoping umbrellas and shades. Pocket umbrellas etc.		
685.557	Sticks. Canes. Walking sticks ⇒ 674.73		

685.65	Equipment and installations for ice skating, skiing, tobogganing etc. ⇒ 685.36	685.87	Equipment and accessories for ball-and-cue games. □ Billiard tables, balls, cues. Equipment for variants of billiards
685.659	Winter sports installations	685.88	Table-tennis equipment. Ping-pong tables, nets, balls, bats
685.659.1	Ice skating installations. Ice rinks etc.	686	BOOKBINDING. METALLIZING. MIRROR-MAKING. STATIONERY
685.659.3	Skiing installations	686.1	Bookbinding ⇒ 655.4 ⇒ 655.534
685.659.32	Ski jumps. Artificial jumping slopes for skiing	686.1.01	Theory. Limitations. Defects 686.1.01 ≡ 621.7.01
685.659.33	Hoists, lifts, elevators for ski slopes. □ Ski-lifts, ski-hoists, ski-tows. Chair-lifts, chair-hoists ⇒ 625.57	686.1.019	Defects and difficulties in binding
685.659.39	Other equipment. □ Equipment for dry skiing. Dry ski runs	686.1.02	Binding processes 686.1.02 ≡ 686.1.05
685.659.4	Sledge, sleigh, toboggan runs	686.1.03	Materials and accessories of bookbinding
685.659.5	Bobsleigh runs	686.1.031	Materials for binding together the sheets and sections. □ Thread. Wire. Rings. Spirals. Combs. Gauze
685.659.6	Equipment for ice games (for curling, ice hockey, ice baseball etc.)	686.1.032	Materials for the binding, case or cover ◇ 686.1.032:665.93 Adhesives ◇ 686.1.032:676.267.2 Patterned paper. Marbled paper
685.68	Boxing and wrestling equipment. Boxing and wrestling rings ⇒ 685.43	686.1.033	Materials for the tooling, ornamentation. Colours and metal foils ◇ 686.1.033:667.5 Binder's inks
685.7	Mountaineering, climbing equipment. Swimming equipment ⇒ 796	686.1.039	Tools and other accessories of bookbinding
685.72	Mountaineering, climbing equipment. □ Grapnels, grappling irons. Climbing irons, pitons etc. Ropes and roping equipment. Ice axes, picks. Crampons	686.1.05	Binding machines
685.73	Swimming equipment and installations. □ Frogman flippers. Buoyancy devices (rings, water wings etc.). Water games equipment, goals etc.	686.1.051	Machines for handling the sheets and sections
685.8	Equipment for board and table games ⇒ 688.72 ⇒ 794.022	686.1.051.2	Jogging machines. Joggers
685.81	Card game equipment and accessories	686.1.051.3	Machines for applying adhesive. Gumming, gluing, pasting machines. Machines for end-papering, tipping-in and edge-gluing
685.811	Playing cards (manufacture, appearance and types)	686.1.051.5	Machines and equipment for gathering the sheets, sections or webs
685.811.2	Appearance (patterns, designs, suits etc.)	686.1.051.54	Gathering machines ◇ 686.1.051.54-79 Collating mechanism of gathering machines
685.811.3	Traditional playing cards. Standard cards. Cards to be matched. Combination cards	686.1.051.58	Inserting machines
685.811.4	Fancy playing cards with standard suits (illustrated, commemorative, instructional playing cards)	686.1.052	Binders' presses. □ Bundling presses. Baling presses
685.811.5	Fancy playing cards without reference to standard suit systems. □ Quartets, seven families game. Mexican lottery	686.1.053	Folding machines
685.811.6	Tarot cards (traditional, illustrated, simplified)	686.1.054	Machines and equipment for binding together the sheets and sections
685.811.7	Fortune-telling cards (with commentaries, signs of the zodiac, other figures and illustrations)	686.1.054.2	Machines and equipment for sewing with thread. □ Sewing frames. Stitching machines
685.811.8	Magic cards (conjurers', illusionists' cards).	686.1.054.3	Machines and equipment for fastening with wire. □ Stapling machines. Machines for spiral binding, ring binding
685.82	Equipment and accessories for games of chance. □ Dice. Dice beakers etc. Roulette equipment. Equipment for lotto, tombola, housey-housey, bingo etc.	686.1.054.7	Machines for unsewn binding (adhesive binding)
685.84	Board game accessories	686.1.055	Machines and equipment for processing the sewn book (uncased)
685.841	Chess sets, boards and pieces	686.1.055.2	Machines for rounding the spine
685.842	Boards and pieces for minor board games not dependent on chance. □ For draughts, halma	686.1.055.3	Machines for consolidating, compacting, pressing
685.843	Boards and pieces for backgammon, dominoes and other table games	686.1.055.7	Gluing machines. Casing machines
685.85	Equipment and accessories for pattern games or puzzles requiring patience or dexterity ⇒ 688.727.9	686.1.055.8	Edge-colouring and gilding machines
		686.1.056	Machines and equipment for cutting and guillotining paper and board. □ Paper saws, cutters, guillotines, trimmers
		686.1.057	Machines and equipment for production of book covers, cases. □ Gold blocking presses. Embossing presses
		686.1.058	Machines and equipment for fixing the case or cover to the sewn book. □ Casing machines. Soft-binding, brochure machines. Book presses (for pressing after casing-in). Machines for applying dust jackets, wrappers

686.1.059	Special purpose binding machines and mechanisms working in combination to produce seminished or finished products in industrial or mass-production book binding. In-line series of machines	686.77	Spherical mirrors. Silvered globes. Multifaceted reflectors
686.11	Binding in the strict sense. Traditional, hand bookbinding. Cases for books in roll form. Volumn cases	686.8	Office equipment. Stationery. Writing, drawing and artists' accessories ⇒ 005.936.33 ⇒ 005.936.41 ⇒ 676.22/.23
686.12	Mechanized, mass-production bookbinding	686.81	Writing blocks. Ledgers. Exercise books
686.121	Kinds of brochure, pamphlet etc.	686.811	Writing blocks. Carbon pads, blocks
686.121.1	Consisting of loose sheets	686.812	Ledgers. Carbon-interleaved books
686.121.11	With spiral binder	686.814	Paper rolls as stationery. Continuous stationery
686.121.12	With toothed (plastics) binder	686.816	Exercise books
686.121.13	With threads or laces	686.82	Albums. □ Adhesive mounts, corners ⇒ (0.029.1)
686.121.14	With ring binder	686.83	Folders. Files. Portfolios. □ Document folders, wrappers. Correspondence files
686.121.2	Consisting of single sections or signatures	686.84	Card index equipment ⇒ 686.84 ⇒ 686.874
686.121.3	Consisting of more than one section	686.842	Equipment for standing or vertical indexes (e.g. in cabinet form)
686.122	Kinds of book binding, book-type binder etc.	686.843	Equipment for visible indexes
686.122.3	Hand-finished bindings	686.845	Equipment for book-form indexes
686.122.6	Mass-produced bindings	686.846	Manual punched-card index equipment. □ Sorting equipment for manual punched-card indexes ⇒ 676.815.2
686.122.61	Board bindings. Bindings with special paper coatings	686.851	Blotters. Blotting pads ⇒ 676.252.32
686.122.62	Fabric and artificial leather bindings	686.853	Calendars. □ Tear-off calendars. Perpetual calendars
686.122.64	Leather bindings	686.854	Newspaper stands. Reading stands
686.122.66	Plastics bindings	686.855	Journal cases. Protective covers for newspapers
686.122.8	Appurtenances of books. □ Book catches, fasteners. Protective covers for books. Dust jackets. Slip cases. Bookmarks	686.86	Office equipment
686.122.9	Other book-type binders or bindings. □ Screw binders. Clamp or springback binders	686.861	Inkwells. Ink bottles ⇒ 667.4
686.125	Bindery techniques for other products. □ Mounting of maps and pictures	686.862	Writing implements ⇒ 003.58
686.126	Gluing of paper and other materials to paper, board etc. Laminating (with clear film; with metal foil) etc. ⇒ 676.264	686.862.1	Pens. Nibs. Penholders
686.4	Metallizing. Metallic varnishing <i>Metallizing of nonmetals only is classed here</i> ⇒ 621.357 ⇒ 621.793 ⇒ 666.1.056 ⇒ 667.649.2 ⇒ 669.058	686.862.2	Pencils in general ⇒ 667.56
686.41	Gilding. Application of gold	686.862.21	Graphite pencils ('lead' pencils)
686.43	Silvering. Application of silver	686.862.22	Crayons. Coloured pencils
686.45	Bronzing. Application of bronze	686.862.23	Copying pencils. Ink pencils. Inerasable pencils
686.49	Application of other metals to nonmetals 686.49 ≡ 669	686.862.3	Propelling pencils. Refillable pencils
686.493	Coppering. Application of copper	686.862.4	Fountain pens
686.5	Framing. Production of picture frames	686.862.5	Ballpoint pens
686.6	Glazing. Glass-cutting ⇒ 666.15/.16 ⇒ 698.3	686.862.6	Felt-tip pens. Fibre-tip pens. □ Felt-tip pens with fluorescent ink. Marking pens. Text markers
686.7	Mirrors. Mirror making ⇒ 645.492 ⇒ 666.1.056 ⇒ 666.266.4	686.862.8	Other writing instruments
686.71	Hand mirrors. Pocket mirrors. Shaving mirrors	686.862.81	Charcoal
686.72	Wall mirrors (framed and unframed). Mirrors as parts of furniture (e.g. of wardrobes)	686.862.82	Chalk
686.74	Mirrors with curved surfaces. Concave and convex mirrors. Mirrors of cylindrical or conical section	686.864	Auxiliary equipment for drawing and writing. □ Rulers. Set squares. Protractors. French curves. Drawing pins (thumbtacks)
		686.865	Various items of office equipment
		686.865.2	Machines for cutting, comminuting and perforating paper
		686.865.22	Guillotines
		686.865.24	Shredders
		686.865.26	Punches (perforators)
		686.865.4	Paperweights
		686.865.5	Fastening and stapling equipment. □ Paper clips and clamps. Staplers
		686.865.7	Page holders. Page turners. Line and column indicators

686.865.9	Writing aids for the blind and other handicapped persons. □ Braille equipment	687.051.2	Measuring devices
686.866	Stamping and marking equipment. Moistening, gumming and stamp-affixing equipment ⇒ 681.173.7	687.051.3	Patterns for cutting out and measuring
686.866.4	Dispensers for sheets of paper and cutters for rolls of paper or stamps	687.051.4	Tailors' dummies. Dressmakers' dummies
686.866.5	Moistening equipment	687.051.5	Production of patterns (e.g. paper patterns). □ Templates for drawing patterns. Production of patterns by moulding
686.866.6	Gumming and affixing devices. □ Stamp affixers	687.052	Cutting-out devices
686.867	Equipment for opening and sealing letters	687.053	Sewing machines. Sewing equipment ⇒ 677.057.21
686.867.1	Letter openers	687.053.1	Devices relating to the thread. □ Thread guides. Tensioners. Thread feed
686.867.5	Devices for inserting contents into envelopes	687.053.2	Needles. □ Sewing-machine needles ⇒ 672.82
686.867.6	Sealing equipment. □ Sealing wax. Seals. Signets ⇒ 736.3	687.053.3	Shuttles. Loopers. Bobbins
686.868.7	Pencil sharpeners. Pencil-sharpening machines	687.053.4	Devices relating to the fabric to be sewn. □ Fabric feed. Hemmers. Pleating, gathering devices
686.868.8	Materials for correcting and erasing	687.053.5	Sewing-machine propulsion ◇ 687.053.5-83 Electric propulsion ◇ 687.053.5-872.7 Propulsion by hand or foot
686.868.81	Rubbers	687.053.6	Sewing machines according to construction
686.868.83	Erasing knives	687.053.62	Tacking machines. Long-stitch machines
686.868.85	Ink remover	687.053.63	Chain-stitch machines
686.868.87	Correction fluid	687.053.64	Double lock-stitch machines
686.87	Office furniture and fittings ⇒ 005.936.33 ⇒ 645.4	687.053.65	Machines for chain stitch and lock stitch
686.872	Containers and furniture for papers, forms etc. Stationery cabinets	687.053.66	Machines for specialized stitches
686.873	Book supports, troughs, shelves. Portable, movable and rotary bookcases	687.053.661	Machines for decorative stitches
686.874	Card index furniture. Card cabinets ⇒ 686.84	687.053.663	Hem-stitch machines
686.878	Waste-paper bins, baskets	687.053.664	Machines for scalloping, edging
686.88	Drawing and artists' accessories ⇒ 741.02 ⇒ 75.022	687.053.665	Overcast-stitch machines
686.881	Paintboxes for watercolours, for oil colours. Paint dishes. Palettes	687.053.666	Looping, binding-off machines
686.884	Modelling materials. Modelling kits, tools etc.	687.053.67	Swing-needle machines
686.886	Sets of drawing or mathematical instruments	687.053.68	Embroidery machines
686.887	Pen or pencil boxes, trays and stands	687.053.7	Sewing machines according to use
686.888	Slates or tablets for erasable writing. □ Blackboards. Wallboards. Blackboard wipers	687.053.72	Sewing machines for hard or tough fabrics. Sewing machines for leather, especially footwear ⇒ 685.34.055
687	CLOTHING INDUSTRY. GARMENT MANUFACTURE. BEAUTY CULTURE <i>The 687.02/078 auxiliaries are applicable only at 687.1/3 and subdivisions</i> ⇒ 646.4 ⇒ 675.15 ⇒ 675.62 ⇒ 677 ⇒ 746.4	687.053.73	Buttonhole-sewing machines
687-1	Kinds of textile material <i>687-11/-15 ≡ 677.1/5</i> <i>The 62-1... auxiliaries are not applicable at 687 and its subdivisions</i>	687.053.74	Machines for sewing various items on to fabrics. □ Machines for sewing straw, bands on hats, beads, buttons
687.01	Principles, aesthetics, design and other fundamentals	687.053.75	Machines for sewing straw items. □ Machines for sewing straw mats, straw bottle-covers
687.016	Workmanship. Cut. Style	687.053.76	Machines for sewing quilted covers
687.02	Working procedures	687.053.77	Machines for repair of fabrics. Darning equipment etc.
687.021	Measurement. Production of patterns	687.053.78	Machines for sewing carpets etc.
687.022	Cutting out the fabric	687.053.9	Auxiliary processes and devices relating to sewing. □ Thimbles. Needle cases. Pincushions ⇒ 614.896
687.023	Sewing	687.054	Pressing and damping equipment. Smoothing irons. Ironing boards ⇒ 648.42/43 ⇒ 677.057.73
687.03	Raw materials for the clothing industry. Fabrics. Trimmings	687.056	Auxiliary equipment. □ Marking equipment. Pinning-up equipment
687.05	Tools and machinery for the clothing industry	687.072	Made-to-measure clothes
687.051	Tools and machinery for preparatory work	687.073	Ready-made clothes
		687.076	Stiffening in articles of clothing. Backing. Interlining. Padding
		687.078	Fastening of clothing ⇒ 672.83
		687.078.2	Hooks and eyes
		687.078.3	Press fasteners in general
		687.078.4	Button holes
		687.078.6	Cord or string fasteners. Hook fasteners. Clamp or grip fasteners. Laces and eyelets
		687.078.7	Zip fasteners

687.079	Other matters. Size adjustment of articles of clothing	687.242	Underpants ⇒ 687.254.5
687.1	Outer clothing. Tailoring	687.244	Baby linen. Layette. □ Napkins (nappies). Bibs
687.11	Men's outer clothing	687.245	Handkerchiefs
687.112	Suits	687.246	Nightclothes
687.112.2	Jackets	687.25	Clothing accessories
687.112.3	Waistcoats	687.251	Shirt fronts. Dickies
687.112.4	Trousers. □ Shorts. Knee breeches	687.252	Detachable collars. Collar stiffeners
687.119	Other articles of men's outer clothing. □ Smocks. Pullovers. Sweaters	687.253	Cuffs. Cuff protectors
687.12	Women's outer clothing	687.254	Underwear. Hosiery
687.122	Dresses. Frocks. Suits	687.254.5	Underwear in general. □ Vests. Slips.
687.123	Skirts. Trousers. □ Slacks. Shorts. Culottes	687.254.8	Petticoats. Knickers. Briefs. Combinations
687.124	Blouses. Pullovers. Sweaters	687.254.81	Hose and accessories
687.129	Other articles of women's outer clothing	687.254.85	Hose. □ Stockings. Tights. Socks. Footless hose
687.13	Children's outer clothing	687.255	Suspenders. Garters etc.
687.131	Boys' clothing 687.131 ≡ 687.11	687.256	Bustles. Crinolines. Fathinales
687.132	Girls' clothing 687.132 ≡ 687.12	687.26	Brassieres. Corsets. Bodices
687.14	Sports clothes Denote the sport by :796/799 ◇ 687.14:797.2 Swimwear ⇒ 796.022.7	687.268	Household linen Bed linen. Table linen. Bathroom textiles. Kitchen textiles 687.268 ≡ 645.48
687.15	Formal dress. Professional clothes. Working clothes. House clothes ⇒ 7.022.7	687.3	Knitwear. Ties. Devices for securing and protecting clothing ◇ 687.3-037.21 Cotton knitwear ◇ 687.3-037.21-037.31 Wool-and-cotton knitwear ◇ 687.3-037.31 Woollen knitwear ◇ 687.3-037.37 Silk knitwear ◇ 687.3-037.47 Synthetic knitwear ⇒ 677-489 ⇒ 677.075
687.151	Formal dress	687.37	Neckwear and accessories. □ Neckties (ties). Cravats. Rings, pins, clips, for neckties
687.151.1	Men's formal dress. □ Evening dress. Dress suits. Tail coats. Frock coats. Dinner jackets (tuxedos). Morning coats	687.38	Devices for holding, securing and protecting garments. □ Braces (trouser suspenders). Belts. Sleeve holders. Cycle clips ⇒ 685.22 ⇒ 685.23
687.151.2	Women's formal dress. □ Evening dresses, gowns. Ball gowns	687.39	Other protective garments ⇒ 685.24
687.152	Uniforms ⇒ 355.14 ⇒ 355.665	687.392	Neck and throat protectors. □ Scarves.
687.153	Official dress. Gowns. Robes ⇒ 27-525.4 ⇒ 371.896	687.393	Mufflers. Neckerchiefs
687.157	Professional clothes. Working clothes ⇒ 615.479.4	687.394	Head protectors. Ear protectors
687.158	House clothes. □ Housecoats. Dressing gowns. Bathrobes. House jackets	687.4	Arm protectors. Leg protectors. □ Wristlets. Knee-pads. Gaiters. Leggings. Legwarmers
687.16	Theatrical and motion picture costumes. Clothes for stage, film, television ⇒ 792.024:687.16	687.4.02	Headgear. Hatmaking. Millinery The 687.02/078 auxiliaries listed under 687 are not applicable at 687.4 and its subdivisions
687.17	Workwear (garments for general protection). Overwear ⇒ 614.89 ⇒ 685.25	687.4.022	Hatmaking processes
687.172	Coats. Overcoats. Topcoats	687.4.023	Hatmaking by hand
687.172.1	Men's coats	687.4.05	Machine manufacture of hats
687.172.2	Women's coats	687.42	Hatmaking machinery, tools and equipment
687.173	Overwear for protection against the weather. □ Raincoats. Windcheaters. Anoraks. Parkas	687.43	Men's hats (felt, fur, straw hats etc.)
687.174	Workwear. Overwear for work and sport	687.431	Men's caps
687.175	Garments for protection against mechanical effects. □ Against bites, stings, impact, missiles	687.432	Caps without peaks. □ Berets. Fezzes
687.179	Aprons	687.433	Peaked caps
687.18	Parts of garments	687.434	Caps for protection from the sun. Tropical hats. □ Pith helmets (sola topis)
687.182	Linings. Interlinings ⇒ 687.076	687.44	Protective headgear for special purposes. □ Firemen's helmets. Diving helmets
687.183	Pockets	687.45	Military headgear. □ Helmets. Shakos. Bearskins ⇒ 623.445.2
687.2	Linen wear. Lingerie		Women's headgear. Women's hats, caps, bonnets. □ Bathing caps. Hat veils
687.21	Linen wear in general		
687.24	Washable linen wear		
687.241	Shirts		

687.46	Children's headgear	687.53.058.8	Protective gowns, capes for haircutting. Neck rests. Other equipment
687.48	Hat trimmings. Hat decorations		
687.5	Beauty culture industries <i>The 687.02/078 auxiliaries listed under 687 are not applicable at 687.5 and its subdivisions</i> ⇒ 391.6 ⇒ 613.49 ⇒ 646.7 ⇒ 661.187.84 ⇒ 665.58	687.54	Nail care. Manicure. Pedicure
687.5.01	Aesthetics. Fashion. Style	687.54.05	Nail care equipment
687.5.02	Working processes	687.54.052	Devices for cutting and shortening nails. □ Nail scissors. Nail clippers. Nail files
687.5.04	Auxiliary materials and products	687.54.053	Devices for polishing and cleaning nails
687.5.05	Machinery, tools and equipment	687.54.053.2	Nail polishing equipment
687.51	Manufacture of false hair. Wigmaking (boardwork) ⇒ 688.75	687.54.053.6	Nail cleaners
687.512	Wigs	687.54.058	Nail care sets. Combined nail care equipment. Manicure sets ⇒ 688.33
687.513	Chignons. Hair knots. Hair plaits	687.55	Cosmetology. Beauty care
687.516	Toupees. Hairpieces	687.552	Medicinal beauty care
687.517	False beards and moustaches	687.552.2	Skin care
687.518	False eyelashes	687.552.3	Hair care
687.53	Hair care. Beard care. Hairdressing profession. Barbering	687.553	Aesthetic care
687.53.02	Working processes 687.53.02 ≅ 687.53.05	687.553.2	Making-up. Use of cosmetics
687.53.03	Use of hair care products Specify by :665.585...	687.56	Other auxiliary equipment for beauty culture
687.53.04	Auxiliary materials. □ Alum stone (shaving blocks)	687.561	Containers and stands for beauty care
687.53.05	Hair care machinery, tools and equipment	687.561.5	Cases for face powder. Powder compacts
687.53.051	Lathering equipment ⇒ 687.974	687.561.6	Containers for pomades, creams, oils, salves
687.53.052	Haircutting equipment. Shaving equipment. □ Hair scissors. Haircutting machines, clippers. Razors ⇒ 672.715 ⇒ 672.718	687.563	Equipment for toilet preparations in powder form
687.53.053	Equipment for dressing hair and beard	687.563.2	Powder puffs. Powder papers
687.53.053.2	Curling irons or tongs	687.563.3	Powder sprinklers, puffers etc.
687.53.053.4	Equipment for waving and curling hair (except curling irons)	687.563.6	Cases for make-up sticks, pencils
687.53.053.41	Simple equipment. Hair rollers ◇ 687.53.053.41-65 Electrically heated hair rollers	687.566	Scent bags, pads. Scent or perfume bottles
687.53.053.42	Permanent waving equipment, electric or chemical. Hair curlers	687.8	Artificial furriery ⇒ 675.6/7.8
687.53.053.43	Heaters for waving equipment	687.84	Artificial furriery according to raw material. □ Artificial furs of feathers, animal hair, wool, synthetics
687.53.053.45	Other hair curling equipment. □ Pinching clamps. Grips. Cooling equipment	687.9	Brush industry. Brooms etc. ◇ 687.9-1 Brushes according to kind of fibre ⇒ 687-1 ⇒ 648.524 ⇒ 675.82
687.53.053.47	Auxiliary devices. □ Curl wrappers, papers etc.	687.9.05	Brush industry tools and equipment
687.53.053.48	Caps or hoods for waving hair. Hair nets	687.92	Brushes according to construction
687.53.053.5	Hair combs (back combs). Hairpins. Hair grips. Hair slides. Decorative objects for supporting the finished hairstyle ⇒ 672.81	687.921	Brushes without handles in general
687.53.054	Equipment for washing and drying hair. Devices for hair colouring. Combs	687.922	Brushes with handles in general
687.53.054.2	Equipment for washing hair. Shampooing equipment	687.924	Rotary brushes
687.53.054.3	Equipment for drying hair. Hair-driers (including hand-held equipment)	687.925	Brushes with a reservoir for supplying liquid or cleaning materials
687.53.054.4	Devices for hair colouring	687.93	Brooms
687.53.054.5	Combs. Rake combs. Hair brushes	687.95	Cleaning brushes. □ Brushes for cleaning clothes, furniture, carpets
687.53.054.6	Devices for cleaning combs	687.96	Scouring brushes. □ Brushes for bottles, glasses, barrels. Scrubbing brushes
687.53.058	Other equipment for the hairdressing profession	687.97	Special-purpose brushes
687.53.058.3	Shaving mugs, bowls	687.971	Hairbrushes
		687.972	Toothbrushes. Nailbrushes
		687.974	Shaving brushes ⇒ 687.53.051
		687.977	Polishing brushes. □ Shoe brushes
		687.978	Paint brushes. Artists' brushes
		688	FANCY GOODS. TOYS. DECORATIVE ARTICLES ⇒ 745
		688.1	Ivory work ⇒ 675.862
		688.2	Buttons. Button manufacture ⇒ 672.87

688.21	Buttons in general	688.721.25	Special types of doll. □ Glove puppets. Dolls for Punch and Judy shows. Dolls which drink, swim etc.
688.211	Buttons of plastics, horn, glass etc.	688.721.27	Accessories for dolls. □ Dolls' clothes, furniture, prams. Toy equipment, e.g. toy telephones
688.212	Cloth buttons. Washable buttons. Other buttons for clothes	688.721.3	Paper dolls. Dressing, cut-out and folding figures
688.27	Collar fastenings. Cuff fastenings	688.722	Dolls' houses and accessories. Dolls' house rooms and equipment
688.272	Collar fastenings and safety devices. □ Collar studs	688.723	Jumping jacks. Puppets on strings. Marionettes
688.273	Cuff fastenings and safety devices. □ Cuff links	688.724	Moving figures. Mechanical toys
688.3	Fancy goods. Souvenirs	688.724.2	Self-propelled figures. Climbing figures
688.3.072	Fittings for purses, cases etc.	688.724.3	Figures moved by hand
688.3.073	Fasteners for bags, cases etc.	688.724.4	Toys with mechanically moving jointed figures
688.3.077	Decoration for handbags, purses etc.	688.725	Tops (spinning tops)
688.31	Cases. Caskets	688.726	Nonmoving toy figures. □ Toy soldiers. Toy animals
688.312	Spectacle cases	688.727	Moving toys (e.g. toy vehicles)
688.313	Cases for field glasses, binoculars etc.	688.727.1	◇ 688.727-8 According to motive power
688.314	Protective covers for watches	688.727.1	Land vehicle toys (other than rail vehicles)
688.316	Caskets for jewellery		Specify by :629.3...
688.317	Key containers	688.727.2	Watercraft toys
688.33	Tollet bags. Beauty cases		Specify by :629.5...
	⇒ 685.51	688.727.31	Toy aircraft, aeroplanes etc.
	⇒ 687.54.058		Specify by :629.7...
688.34	Tool containers. Tool bags	688.727.35	Inflatable toys
688.35	Bags. Portfolios	688.727.38	Bubble-making toys. Smoke-blowing toys
688.351	Document cases. Briefcases. Portfolios	688.727.4	Projectiles. Catapults
688.353	Shopping bags. String bags	688.727.5	Optical toys. Colour toys
688.359	Ladies' handbags	688.727.7	Magnetic and electric toys
688.36	Purses. Wallets	688.727.8	Toy railways. Train sets
688.362	Purses	688.727.81	Tracks. Curves. Points. Accessories
688.4	Artificial flowers, plants and feathers. Goods made of hair		688.727.81 ≡ 625.1
688.42	Artificial flowers	688.727.82	Toy rolling stock. Toy locomotives
688.429	Artificial fruits		Specify by :629.4...
688.43	Artificial garlands	688.727.9	Constructional games. Jigsaw puzzles
688.44	Artificial wreaths, crosses etc.		⇒ 685.85
688.46	Articles made of feathers. Artificial feathers	688.728	Toy armour. Toy helmets. Red Indian outfits.
	⇒ 675.84		Harnesses for toy horses
688.47	Goods made of hair. □ Flowers, lockets, medallions etc. of hair	688.729.1	Hoops
	⇒ 675.82	688.729.2	Toys for sand games. □ Sand moulds
688.5	Fans	688.73	Musical toys. Children's musical instruments
688.6	Light shades. Lampshades. Reflectors. Screens. Covers for flower pots		⇒ 780.6
688.61	Light shades and reflectors. Lampshades	688.731	Individual musical toys
	⇒ 628.952		688.731 ≡ 681.8
688.62	Screens	688.731.181	Toy trumpets
688.621	Screens against the light	688.732	Musical instruments as parts of toys.
688.622	Fire screens		Noise-making toys. □ Rattles. Cap pistols
688.623	Draught screens		⇒ 623.447
688.63	Covers for flower pots	688.74	Theatre equipment. Panorama equipment. Circus equipment. Music hall equipment. Recreation park equipment
688.7	Toys. Articles for amusement, tricks. Decorations		⇒ 792.023
	⇒ 674.5	688.741	Theatre equipment. □ Stage sets, curtains, scenery. Equipment for moving them
688.72	Toys	688.742	Theatre accessories. Equipment for stage effects
	◇ 688.72:621.715.2 Toy tool sets	688.742.2	Stage platforms. On-stage effects. □ Artificial grass, clouds, rain
	◇ 688.72:648.524 Toy sweeping equipment, brooms etc.	688.742.3	Equipment for better audibility of words and visibility of acting. Acoustic aids
	⇒ 623.447	688.742.4	Equipment for the production of special noise effects. □ Thunder machines. Wind machines
	⇒ 685.61.8	688.742.5	Stage lighting. Projected images etc.
688.721	Dolls and accessories for dolls	688.742.6	Equipment for the production of special light effects. □ Fire effects. Lightning effects
	⇒ 688.726	688.742.8	Equipment for lifting and lowering people.
688.721.2	Dolls in general. Doll-like toy animals. □ Teddy bears		Trap-doors. Theatre machinery

688.743	Equipment for panoramas, dioramas, stereoramas etc.	688.783	Relief decorations
688.744	Equipment for circuses etc.	688.788	Special ornamental objects. Wall decorations. □ Special picture frames, mountings.
688.745	Equipment for marionette theatres, shadow theatres, Punch-and-Judy shows. Equipment for sideshows ⇒ 792.9		Ornamental tiles. Ornamental letters and numbers
688.746	Equipment for cinemas. Accessories for presenting films ⇒ 778.55 ⇒ 791	688.79	Other decorative effects. Imitations
688.747	Equipment for peepshows, kaleidoscopes, other shows with interplay of colours (e.g. colour music)	688.792	Lighting effects. □ Patterns produced by projected light (reflected). Transmitted light effects (e.g. imitation stained glass)
688.749.8	Equipment for mazes	688.793	Colour effects. □ Changing pictures. Trick pictures
688.75	Costumes and disguises. Party accessories	688.794	Line effects. Linear patterns
688.751	Masks. □ Velvet masks. Half-masks ⇒ 391.8	688.795	Decorative surface effects. □ Marbling. Batik pattern
688.752	Fancy dress. Masquerade costumes. Dominoes	688.796	Imitations of various objects, three-dimensional effects etc.
688.753	Grotesque and comic figures. Animal costumes	688.796.2	Imitations of natural substances and patterns. □ Imitation wood grain
688.754	Disguises ⇒ 687.51	688.796.4	Imitations of artistic works, artificial patterns. □ Textured reproductions of paintings
688.755	Party novelties. Party favours	688.797	Varying, changing patterns. Patterns on changing surfaces
688.756	Accessories for parties, public festivities etc. □ Confetti. Paper streamers. Crackers ⇒ 676.812 ⇒ 676.816.8	688.9	Miscellaneous small articles
688.757	Surprises. Illusions. Jokes	688.92	Holders, clasps, hooks etc. □ Flower holders, pins, etc. Umbrella holders, stands. Carrying handles, rings etc.
688.76	Decorations for ceremonies, celebrations, public festivities. □ Bunting. Festive emblems ⇒ 666.189.1	688.93	Smokers' requisites
688.77	Equipment for amusement areas ⇒ 791.7	688.932	Pipes (tobacco pipes). Hookahs. Narghiles
688.771	Roundabouts	688.933	Accessories and components of tobacco pipes. □ Cleaning devices for pipes. Pipe cleaners
688.773	Swings. See-saws	688.934	Accessories for cigars and cigarettes. □ Cigar holders. Cigarette holders. Cigar cutters
688.774	Rocking horses. Other toy animals for riding. Hobby horses	688.935	Containers for tobacco. Cigar and cigarette cases
688.775	Slides. Switchbacks. Roller coasters etc. Water chutes. Chair-o-planes. Big wheels. Moving floors. Other equipment	688.936	Ashtrays. Smokers' stands
688.78	Articles for decorative effects. Wall decorations	688.94	Match containers. Match dispensers ⇒ 662.536.2
688.782	Decoration of surfaces. □ By burning: poker-work	688.96	Money boxes. Cash boxes
		689	AMATEUR HANDICRAFTS. TECHNICAL HOBBIES ◇ 689:621.396 Amateur radio ⇒ 379.826 ⇒ 7.073 ⇒ 7.077

69 BUILDING (CONSTRUCTION) TRADE. BUILDING MATERIALS. BUILDING PRACTICE AND PROCEDURE

Under 69 is classed information about the building industry generally, including building materials, components, crafts and execution.

Manufacture of building materials and components is classed in 66/67, civil engineering questions in 624/628 and architectural questions in 72

The -1/-9 auxiliaries from 62-1/-8 and 66-9 are applicable throughout 69

◇ 69:614 Public health aspects of buildings. Including: 'Sick building syndrome'

◇ 69:614:725.1:005.912 'Sick' office blocks

⇒ 349.44, 351:69, 351.785

◇ 69-115 Multi-purpose buildings

◇ 69-5 Operation and control of machines and processes in buildings. Including: Intelligent buildings

69.0

SPECIAL AUXILIARY SUBDIVISION FOR 69 BUILDING, CONSTRUCTION TRADE. BUILDING MATERIALS. BUILDING PRACTICE AND PROCEDURE

The .0 special auxiliaries have similar meanings under 624 and 69; preferably use .03 and .05 (which relate primarily to buildings) mainly under 69, but .01, .04 and .07 (which relate to all structures generally) mainly under 624

69.01	Buildings and parts of buildings according to material 69.01 $\equiv$ 624.01 <i>See note at 69.0.</i> <i>The .01 auxiliaries are not applicable at 691 and 693 and their subdivisions</i>	69.05	Site, organization and plant. Industrialized building and erection. Site tests. Building failures. Building durability and maintenance
69.03	Size, permanence, location and shape of buildings	69.051	Selection and investigation of site. State of site at start of work $\Rightarrow$ 624.131.31
69.032	Size of buildings in general (large, medium, small) $\Rightarrow$ 69.053	69.052	Access roads to site. Completed roads. Temporary roads
69.032.2	Multistorey buildings generally	69.053	Measurements, dimensions of buildings. Building height, width (frontage) and depth $\Rightarrow$ 69.032
69.032.21	Buildings according to number of floors	69.054	Marking out and setting out of planned structures and traffic areas
69.032.22	High buildings. Skyscrapers	69.055	Arrangement and organization of site
69.032.4	Very small buildings	69.055.2	Storage of materials and units. Protection against wet and heat. Equipment for preheating or cooling before placing
69.032.6	Very light buildings	69.055.6	Temporary buildings on the site. ☐ Site offices. Workers' accommodation. Huts for equipment and tools
69.033	Dismantlable, transportable and temporary buildings. Inflatable, deflatable structures (pneumatic, pressurized or air structures) $\Rightarrow$ 351.778.534	69.055.7	Services $\diamond$ 69.055.7:696.1 Water supply $\diamond$ 69.055.7:696.6 Electricity supply
69.033.1	Dismantlable (and deflatable) buildings and structures	69.056	Preparation of parts for industrialized construction. Industrialized building systems
69.033.15	Inflated, deflatable (rubber or plastics) structures	69.056.3	Industrialized building systems according to interchangeability. ☐ Open (interchangeable) systems. Closed (noninterchangeable) systems
69.033.2	Transportable buildings	69.056.5	Industrial building systems according to type of unit
69.033.6	Temporary buildings. Emergency, makeshift buildings, huts	69.056.52	Precast concrete plate structures (wall-like plates)
69.033.8	Permanent (as opposed to temporary) buildings	69.056.53	Slab structures (room-high slabs)
69.034	Buildings related to water	69.056.55	Skeleton structures. ☐ Stanchions. Frames $\Rightarrow$ 624.94
69.034.2	Floating buildings	69.056.56	Cell structures. Box structures (room-size cells)
69.034.3	Buildings in water (not floating)	69.057	Industrialization of construction. Procedures for erection of prefabricated parts and whole structures
69.034.4	Buildings partly in water	69.057.1	Prefabrication of building parts on or near site. Field fabrication
69.034.5	Underwater buildings	69.057.2	Assembly on site
69.034.6	Properties and composition of water as affecting structures in water. ☐ Evaporation	69.057.4	Erection, lifting, sinking, floating of single parts or complete structures
69.034.7	Deposits. Rubble. Boulders. Silt. Flotsam	69.057.43	Fixing and arranging of parts
69.034.8	Cleaning of the affected parts	69.057.44	Erection of masts, chimneys, tanks and stationary structures for the support of moving loads
69.034.9	Protection against effects of water and air	69.057.45	Erection by lifting methods. ☐ Erection by hoists, jacks, platforms etc. Lift-slab method
69.034.92	Coatings. Linings. Facings	69.057.46	Lowering into position of units or complete structures on the ground
69.034.93	Packing. Stuffing. Sealing	69.057.47	Floating into position and sinking of large units or complete structures. ☐ Positioning of bridge parts, underwater tunnels, pipelines
69.034.96	Drainage	69.057.5	Shuttering (falsework, formwork or moulds). Centering
69.035	Buildings in mountainous country. Underground buildings $\Rightarrow$ 624.127 $\Rightarrow$ 624.144	69.057.51	Types of shuttering
69.035.2	Buildings on slopes. Mountain and hillside buildings	69.057.52	Forms. Shuttering for walls, partitions and generally upright structural parts. Vertical and horizontal movable shuttering. Sliding forms. Climbing framework
69.035.4	Underground buildings. Subterranean buildings	69.057.53	Shuttering for floors. Shuttering for horizontal parts of structures generally
69.036	Buildings according to layout, ground plan	69.057.55	Inflatable, deflatable shuttering. Pneumatic formwork
69.036.1	Straight structures		
69.036.2	Skew structures		
69.036.3	Curved structures		
69.036.4	Structures with a single opening		
69.036.5	Structures with several independent superstructures		
69.036.6	Structures with connected or continuous superstructures		
69.036.7	Structures with load-bearing grillage		
69.036.8	Multistorey load-bearing structures		
69.04	Graphical and analytical statics for investigation and calculation of buildings. Strength, stability etc. 69.04 $\equiv$ 624.04 <i>See note at 69.0</i>		

69.057.58	Centering and forms for arches. Shuttering for piers, columns, chimneys, conduits, ducts etc.	691.1	Materials of organic origin
69.057.59	Permanent shuttering. Self-supporting shuttering	691.11	Wood. Timber. Wood products
69.057.6	Construction scaffolding. Working platforms and staging. □ Inflatable, deflatable scaffolding ⇒ 62-774.5	691.112	Hardwoods
69.057.7	Plant and equipment for handling and transport on site	691.113	Softwoods
	◇ 69.057.7:621.867 Conveyors	691.115	Wood waste. Sawdust. Wood wool
	◇ 69.057.7:621.873 Cranes (in general)	691.116	Plywood. Wood veneers. Laminated wood. Laminated board
	◇ 69.057.7:621.876 Lifts. Hoists	691.12	Plant stems. Straw (thatch). Reeds. Bamboo
	◇ 69.057.7:625.31 Narrow-gauge railways	691.13	Cork. Cork substitutes
	◇ 69.057.7:629.353 Dump trucks	691.14	Paper. Pulp products
69.058	Site tests. Investigation of complete structures	691.145	⇒ 676.28
	⇒ 624.04		Wallpapers
69.058.2	Measurements of deformation	691.15	⇒ 698.16
69.058.3	Measurements of the movement of complete structures	691.15	Other fibrous vegetable substances.
69.058.4	Measurements of settlement	691.16	□ Textiles. Fabrics
69.058.5	Measurements of stress	691.17	Bituminous materials. □ Asphalt. Tar. Pitch. Bituminous felt, board
69.058.6	Measurements of uplift	691.173	Macromolecular materials. Rubbers, plastics etc.
69.058.7	Measurements of temperature	691.175	Rubber
69.058.8	Measurements of the dynamic effects of loads	691.175	Plastics
69.059	Maintenance of buildings	691.178	691.175.5/.8 ⇒ 678.5/.8
69.059.1	Maintenance in general. □ Inspection. Cleaning	691.18	Gums. Resins and similar adhesives
69.059.2	Damage and repair	691.18	Materials of animal origin
69.059.22	Damage. Dilapidation		⇒ 675.166
69.059.25	Restoration. Repair	691.2	Natural stones. Other mineral materials
69.059.28	Collapse. Total loss	691.21	Natural stones (according to origin and composition)
69.059.3	Strengthening. Reinforcement. Alteration. Conversion	691.212	691.21 ⇒ 553.5
69.059.32	Strengthening of foundations and/or structure (by structural means, injection etc.). □ Shoring. Underpinning	691.213	Plutonic stones. □ Granite. Syenite
69.059.35	Conversion. Structural alteration	691.213.5	Volcanic stones and lavas
69.059.38	Extensions to existing buildings or storeys. (Connected) annexes. Addition of storeys	691.214	Pumice. Pozzolana. Perlite
69.059.4	Life of buildings	691.214.2	Metamorphic stones. Schists etc.
69.059.5	Displacement, moving, raising of complete buildings	691.214.8	Slates (e.g. roof slates)
69.059.6	Demolition. Rubble	691.215	Marble
	⇒ 628.4.036	691.216	Sedimentary and carbonate stones.
69.059.62	Demolition. □ Dismantling for reassembly elsewhere	691.217	□ Limestone. Sandstone
	<i>Denote procedures by :...</i>	691.22	Calcareous sandstones
	◇ 69.059.62:621.791.034 Using torches, lances etc.	691.22	Siliceous stones
	◇ 69.059.62:621.9.048.7 Particle-beam machining	691.26	Sand, gravel and similar aggregates
	⇒ 711.164	691.261	691.223/.224 ⇒ 553.623/.624
69.059.63	Clearance of rubble or debris	691.263/.266	Various inorganic economic minerals and earths
69.059.64	Utilization of rubble or debris	691.263.5	Clays. Feldspars. Similar aluminium silicate deposits
69.059.7	Reconstruction. Rebuilding (using some elements of the original building)	691.27	691.261 ⇒ 553.61
69.07	Structural elements. Load-bearing members	691.27	Various salts and other compounds
	69.07 ⇒ 624.07	691.3	691.263/.266 ⇒ 553.63/.66
	See note at 69.0		Gypsum. Alabaster
691	BUILDING MATERIALS. BUILDING COMPONENTS	691.31	Fibrous and foliate silicates
	◇ 691:620.1 Testing of building materials		691.27 ⇒ 553.67
	◇ 691:699.8 Building materials with protective and special properties	691.311	Artificial stone. Concrete. Various agglomerates
	◇ 691:699.844 Soundproofing materials. Acoustic insulating materials		⇒ 666.96/.97
	◇ 691-4 State, condition, form of materials	691.316	⇒ 691.54
	⇒ 624.01		Artificial stone. Reconstituted stone. Gypsum building products. Calcium silicate stone
			⇒ 666.91
			Gypsum and anhydrite building materials
			◇ 691.311-419 Gypsum plasterboard
			◇ 691.311-431 Gypsum blocks
			Artificial sand-lime bricks. Calcium silicate stone. Lightweight lime-concrete bricks

691.32	Concrete. Concrete building elements ⇒ 666.94 ⇒ 666.97	691.421.77	Chemically resistant bricks. Alkali-resistant bricks. Acid-resistant bricks ⇒ 666.77
691.322	Aggregates (including lightweight aggregates). Breeze. Clinker. Slag. Pumice. Concrete aggregates in general	691.424	Roofing tiles
691.327	Unreinforced concrete <i>Denote concrete building elements of particular form by -4... from 62-4...</i>	691.43	Glazed, porous or semiporous ware
691.327.3	Unreinforced lightweight concrete	691.432	Salt-glazed ware in general
691.327.32	Unreinforced lightweight aggregate concrete <i>Specify the aggregate by :691...</i> ◇ 691.327.32:691.213.5 Pumice concrete	691.433	Stoneware ◇ 691.433-431 Stoneware bricks. Stoneware tiles (paving tiles) ◇ 691.433-462 Stoneware pipes
691.327.33	Aerated concrete. Foamed concrete	691.434	Glazed earthenware. Faience. Terracotta. Whiteware
691.327.332	Aerated concrete	691.434.2	Faience, terracotta for facings etc.
691.327.333	Foamed concrete	691.434.3	Whiteware. □ Sanitary ware. Wall tiles
691.328	Reinforced concrete. Prestressed concrete. Reinforced concrete elements ⇒ 666.982	691.49	Other ceramic products. □ Porcelain products
691.328.1	Reinforced concrete (concrete reinforced with steel). Reinforced concrete elements	691.5	Binders. Bedding materials
691.328.2	Prestressed concrete. Prestressed concrete elements	691.51	Limes ⇒ 666.92
691.328.3	Reinforced lightweight concrete	691.53	Mortars in general
691.328.4	Concrete with reinforcement other than steel	691.54	Cements. Pozzolanic materials ⇒ 666.944 ⇒ 666.95 ⇒ 691.3
691.328.42	Concrete reinforced with nonferrous metals	691.541	Roman cement
691.328.43	Concrete reinforced with inorganic materials (other than metals)	691.542	Portland cement
691.328.44	Concrete reinforced with organic materials ◇ 691.328.44:691.12 Concrete reinforced with bamboo	691.544	Other cements. □ High-alumina cement etc.
691.328.5	Reinforced cement products. Asbestos cement products etc.	691.545	Pozzolanic materials
691.33	Concretes and artificial agglomerates with inorganic binders other than cement	691.55	Plasters
691.332	Concretes and agglomerates with pozzolanic binders and lime or gypsum additives. Silicate concrete. Lime concrete ⇒ 691.316	691.552	Mud. Earth. Dung. Blood. Other primitive mixtures
691.332.2	Reinforced silicate concrete	691.553	Gypsum plasters. Anhydrite plasters ◇ 691.553:693.63 Gypsum plaster for stucco ⇒ 666.91
691.332.3	Prestressed silicate concrete	691.554	Lime plasters
691.332.5	Aerated silicate concrete. Foamed silicate concrete. Lightweight lime concrete	691.555	Cement plasters
691.333	Concretes and agglomerates with pozzolanic binders and alkalis or hydraulic activators. Soil-silicate concrete	691.557	Synthetic plasters
691.334	Concretes and agglomerates with magnesium-based binders. Xyloolith ⇒ 674.816	691.57	Paints. Distempers. Varnishes. Stains. Preservatives ⇒ 667.6
691.34	Concretes and artificial agglomerates with organic binders. □ Synthetic resin concretes. Plastic concretes	691.58	Bedding, filling and jointing materials. Levelling compounds. Mastics. Putties. Adhesives
691.4	Earth. Cob. Clayware. Stoneware. Ceramic building materials ⇒ 666.7	691.6	Glass. Window glass. Plate glass. Enamel etc. 691.61/.62 ≡ 666.1/.297
691.41	Dried earth. Adobe. Mud. Pisé-de-terre. Cob (unburnt, puddled clay)	691.615	Sheet glass. Flat glass in general
691.42	Heavy ceramics or clayware. Burnt clay	691.616	Rolled flat glass
691.421	Bricks (ceramic) ◇ 691.421-478 Perforated bricks	691.618.5	Pressed glass. □ Building, architectural glass. Glass bricks, tiles
691.421.2	Ordinary bricks (clay bricks)	691.624	Transparent coloured glass. □ Sun glass.
691.421.24	Long-fired bricks. Clinker bricks	691.629	Heat-absorbing glass
691.421.4	Facing bricks. □ Glazed bricks ⇒ 692.232.4	691.7	Enamels. Glazes
691.421.7	Refractory bricks. Chemically resistant bricks	691.713	Metallic building materials 691.71/.78 ≡ 669.1/.894 with .018... as at 669.018... ⇒ 672.11 ⇒ 682.6
691.421.76	Refractory bricks (fire-resisting bricks). Firebricks ⇒ 666.76	691.714	Cast iron
			Steel ◇ 691.714.018.8 Stainless steel. Noncorrosive steel

691.72	Nonferrous metals in general	692.233.3	Pediments. Gables
691.73	Copper. Copper alloys. □ Bronzes. Brass. Gunmetal	692.237	Details of floor and wall junctions.
691.74	Lead	692.238	Spandrel-beams. Window-spandrels
691.75	Zinc	692.239	Balconies. Canopies. Other cantilevered projections
691.77	Light metals	692.24	Details at ground level. □ Entrances
691.8	Simple units. Builders' hardware ⇒ 682.6 ⇒ 683	692.241.2	Internal walls. Cross-walls
691.81	Prefabricated simple units	692.241.3	Load-bearing cross-walls
691.87	Building materials for strengthening and reinforcing in general ⇒ 693.554 ⇒ 693.564	692.25	Non-load-bearing cross-walls
691.88	Fixing accessories. □ Fasteners. Dowels. Nails. Bolts. Screws. Clips. Brackets ⇒ 621.88	692.251	Partitions
692	STRUCTURAL PARTS AND ELEMENTS OF BUILDINGS	692.253	⇒ 694.5
692.1	Foundations of buildings ⇒ 69.055	692.256	Light slab partitions
692.113	Earthworks in general ⇒ 624.13	692.253:691.311-419	Framed partitions
692.115	Foundations. Substructure in general ⇒ 624.15	692.253:691.311-419	Denote covering materials by :691..., and framing by :693... or :694...
692.2	Walls. Partitions ◇ 692.2:693.2 Brick or block walls ◇ 692.2:693.3 Cob walls ◇ 692.2:694.5 Light timber walls	692.253:694.5	Gypsum plasterboard partitions
692.21	Boundary walls and party walls in general	692.256	Timber-framed partitions.
692.212	Boundary walls enclosing open spaces, gardens etc. (including gateways) ⇒ 712.35	692.29	Stud partitions
692.214	Party walls ◇ 692.214:699.81 Fire-resistant party walls ◇ 692.214:699.844 Soundproof party walls ⇒ 347.238.2	692.292	Temporary partitions. Structural plywood partitions. Glazed partition screens etc.
692.22	Walls below damp-proof course level. Parts below ground generally ◇ 692.22:692.115 Foundation walls ◇ 692.22:699.82 Damp-proof courses	692.294	Wall details in general
692.23	Exterior, external walls. Enclosing walls above ground generally. Walls in elevation. Façades ⇒ 692.23	692.295	⇒ 692.233
692.231.2	Load-bearing exterior walls	692.297	Wall projections. Corbels. Offsets. Cornices.
692.231.3	Non-load-bearing exterior walls	692.298	String-courses. Plinths. Dadoes. Wainscoting and skirtings (interior)
692.232	Exterior walls and facings. Weather-resisting parts	692.299	Wall junctions. □ Corners. Angles. Quoins. Splays
692.232.1	Solid walls	692.4	Walls curved in plan
692.232.2	Cavity walls. Double-shell walls	692.41	Wall piers. Buttresses. Pilasters. Isolated pillars and columns
692.232.3	Walls with impervious skins	692.412	Wall recesses. Chases
692.232.4	Weather-resisting parts. Claddings. Facings	692.415	Wall openings and their surroundings in general. □ Voids. Doorheads. Lintels. Reveals.
692.232.42	With small units. □ With tiles. With slates. Weather boarding	692.42/47	Sills. Thresholds
692.232.44	With large units. □ With sheets	692.42	⇒ 692.8
692.232.45	Curtain walling. Curtain walls	692.421	Roofs and roofing. Roof accessories
692.232.7	Infilling walls to framed structures Denote materials by :691... ◇ 692.232.7:691.327.3 Infilling walls of unreinforced lightweight concrete	692.421	Denote purpose by :72...
692.233	Details of exterior walls. Various related parts according to location. Projections etc. ⇒ 692.29	692.422	◇ 692.44:726 Cupolas for churches
692.233.2	Details of wall-heads and roof junctions. Eaves. Cornices. Parapets	692.423	⇒ 624.042
		692.425	⇒ 624.072
		692.426	The roof as protective covering.
		692.43	□ Weather-resisting features of roofs
		692.432	Relation to climate. Precautions against tropical rain and other elements. Snow protection etc.
		692.433	◇ 692.412:551.5 Climate factors
		692.435	◇ 692.412:696.121 Roof drainage
			Roof slope. Run and rise. Angle of inclination
			Roof coverings. Roof materials
			Shapes of roofs. Roof construction
			Plane framework systems. High-pitched and low-pitched roofs
			Rafter-pitched roofs. Lean-to roofs.
			□ Collar-beam roofs. Ridged or saddle roofs.
			Span roofs with purlins, with trusses etc.
			Hipped roofs. □ Curbed roofs with upper part hipped.
			Hipped roofs with upper part gabled
			Curb roofs. Mansard roofs. Gambrel roofs.
			Demi-curb roofs
			Shed roofs. North-light roofs. Sawtooth roofs
			Large-span roofs
			Flat roofs. Roof terraces (terrace roofs). Sealed flat roofs
			Flat roofs
			Roof terraces (terrace roofs)
			Sealed flat roofs

692.44/47	Three-dimensional framework systems	692.6	Stairs. Ramps. Lifts. Escalators
692.44	Cupolas. Domes. Vaulted roofs	692.61	Generalities. Dimensions. Pitch. Number of steps per flight. Headroom
	⇒ 624.074.2		
	⇒ 624.074.3	692.62	Types of stairs. Fixed ladders
692.442	Schwedler domes	692.622	Stairs, stairways, staircases according to location. □ Exterior steps. Terrace steps. Interior staircases
692.443	Zimmerman domes		Fixed ladders. □ Step-irons. Rung-ladders. Ladder-stairs
692.444	Domes with ribs	692.623	Staircases and stairways according to design or shape. □ Straight flights. Winding stairs (spiral staircases etc.)
692.445	Latticework domes. Network domes		
692.446	Lamella roofs	692.625	
692.447	Pavilion roofs. Steeples		
692.45	Shell roofs	692.63	Parts of stairs and staircases
	⇒ 624.074.4	692.631	Newels of helical stairs
692.46	Folded-plate roofs	692.632	Newel posts
692.47	Suspended roofs	692.633	Handrails. Balusters and balustrades. Banisters. Landing railings
692.48	Roof trusses. Plate-roof girders. Frame trusses. Structural parts of trusses	692.635	Solid steps. Treads. Risers
	⇒ 624.078.4	692.637	Strings and carriages
692.484	Parts of trusses with mixed members	692.64	Landings. Platforms
692.49	Roof parts. Roof details	692.65	Inclined planes. Ramps
692.492	Roof openings. □ Skylights. Dormers. Lucarnes. Lanterns. Roof lights	692.66	Lifts
692.493	Ridge turrets		⇒ 621.876
692.497	Patent glazing	692.67	⇒ 696.6
692.5	Floors. Intermediate floors. Floorings. Ceilings		Moving stairs. Escalators
	⇒ 624.074	692.7	⇒ 621.876.3
692.51	Floors resting on the ground or on sleeper walls	692.71	Accessory parts of constructions
	◇ 692.51:699.82 Damp-proofing membranes	692.711	Chimneys. Chimney stacks. Flues
692.52	Suspended and elevated floors. Subfloors. Deckings. Indoor balconies and galleries	692.713	⇒ 697.8
692.522	Concrete floors and similar fire-resisting floors. Flat slab floors. Ribbed slab floors. Precast beam floors. Hollow tile floors	692.74	Fireplaces (hearths)
	⇒ 624.073.136	692.75	Smoke flues
	⇒ 624.073.5	692.756	Ventilation shafts
692.526	Timber floors	692.758	Various chutes and access facilities
	◇ 692.526:694.4 Solid timber floors		Fuel chutes. Coal chutes
	◇ 692.526:694.5 Wooden joist floors	692.77	Refuse chutes
692.528	Deckings. □ Ribbed panels of sheet steel		⇒ 628.463
692.53	Floorings. Floor finishes (wearing surface) including skirtings and coves of same material	692.78	⇒ 628.47
	⇒ 645.1	692.782	Service ducts for pipes etc. Cable ducts. Heating ducts. Walkable ducts
	⇒ 692.292	692.783	Means of access to services. □ Horizontal covers
	⇒ 693.7		Covers or gratings for trenches and ducts (continuous access)
692.533	Floorings of metal or other minerals	692.784	Manhole frames. Manhole access covers
	⇒ 692.522		◇ 692.783:621.315.23 For electric cable conduits
692.533.1	Floorings of composition, concrete etc.		◇ 692.783:628.253 For sewers or drains generally
692.533.2	Terrazzo floorings		◇ 692.783:696.134.2 For house drains
	⇒ 666.963	692.79	Small access boxes. Clap boxes for water mains, gas mains
692.533.3	Mosaic and similar floorings	692.8	⇒ 621.646.92
692.533.4	Tiled floorings. Small slab floorings. Block floorings		Trapdoors. Cellar flaps
692.533.5	Flagstone floorings. Similar large slab floorings	692.81	Doors. Gates. Windows. Service openings
692.535	Floorings of nonmineral materials	692.811	⇒ 683.3
692.535.1	Timber floorings. □ Parquet floorings		⇒ 692.299
692.535.5	Cork floorings. Linoleum floorings. Fibreboard floorings	692.812	Doors. Gates
692.535.6	Rubber floorings. Plastics floorings		Hinged or pivoted gates and doors. □ Swing doors. Revolving doors
692.535.7	Floor grilles (of metal or wood)	692.813	Sliding gates and doors. □ Lift gates. Portcullises
692.54	Ceilings. Suspended ceilings	692.814	Grille-type gates and doors. □ Rolling grille. Articulated scissor-type grille
	⇒ 692.52		Folding or telescopic gates and doors

692.815	Combined types of gate or door	693.1	Masonry in natural stone. Rubble masonry, slightly or not dressed. Rough rubbles, erratics
692.817	Doors according to purpose		
692.817.1	Entrance doors (external doors)		
692.817.3	Room doors (internal doors)	693.12	Stone cutting and dressing. Dimensions of stones. Face-work
692.817.7	Jib doors. Concealed doors (secret doors)		
692.82	Windows	693.15	Stone walling. □ Random, random-coursed and regular work. Stones set dry (dry stone walling) or in mortar. Bond. Shape of stones. Bedding. Joints
	⇒ 692.299		
	⇒ 692.492		
	⇒ 698.3		
692.821	Casement windows. Top-hung casements. Bottom-hung casements. Pivoted windows	693.2	Masonry of treated material, stones, blocks etc.
692.822	Sliding windows. Sash windows		
692.823	Combined types of window. □ Sliding-folding windowsm	693.21	Masonry of dressed natural stones. Cut stone masonry. Ashlar masonry
692.824	Display windows. Shop windows		
692.826	Composite windows projecting in plan. Bay windows. Bow windows. Corner windows	693.22	Bricklaying. Brick bonds. Mortar joints. Pointing
692.827	Special details. □ Safety frames. Screens (flyscreens). Gauze, protectors of various types	693.24	Masonry of loam bricks (air-dried), loam blocks, loam ashlar
692.828	Arrangements to protect against dirt and condensation on windows	693.25	Masonry of brickwork (burnt bricks)
692.83	Shutters. Blinds	693.27	Masonry of blocks, purpose-made bricks etc. Masonry of prefabricated units in concrete (or similar artificial masses)
692.831	Rigid shutters. Lattice shutters. Venetian shutters		
692.833	Blinds. Draw blinds. Venetian blinds	693.28	Block-laying (as opposed to bricklaying)
692.834	Roller shutters (for windows)		
692.836	Awnings. Sun canopies. Weather-protection canopies	693.29	Composite masonry. Masonry of various kinds of natural and/or artificial stones, blocks, units etc.
692.84	Ventilating and sealing devices for doors and windows	693.3	Masonry of stabilized earth. Pisé-de-terre (rammed-earth construction). Cob
692.842	Ventilation devices and fittings		
	⇒ 697.921.2		
692.843	Sealing devices. Draught excluders etc.	693.5	Concrete construction work. Concreting. In-situ concrete
692.848	Airtight, gastight sealing devices		⇒ 666.94
	<i>Details by :699.87</i>		⇒ 666.97
692.86	Fanlights	693.54	Concreting technique in general. Plain concrete (mass concrete)
692.87	Service and observation openings. Closures. Grilles	693.542	Batching. Mixing. Proportioning of mix. Control of consistency
692.871	Service hatches, grilles		Batching
	⇒ 692.827	693.542.3	Additions, admixtures. Air entrainment.
692.876	Observation and inspection openings and closures. Panels, peepholes etc.	693.542.4	Waterproofers. Accelerators. Plasticizers
692.88	Fencing. Railings. Palings. Turnstiles. Admission gates	693.542.5	Mixing. Use of concrete mixers. Control of consistency
	⇒ 712.35	693.542.51	Mixing by hand
692.9	Ancillary parts, fixtures and fittings	693.542.52	Machine-mixing. Use of concrete mixers
	◇ 692.9:628.94 Lamp fittings	693.542.53	Ready-mixed concrete
	◇ 692.9:631.345 Supports for plants	693.546	Placing and compacting the concrete
	◇ 692.9:673.5 Fittings for bells, carillons, chimes	693.546.1	Discontinuous or batch placing (e.g. by barrows or trucks)
	⇒ 691.88	693.546.2	Placing by chute, conveyor or similar devices
692.92	Exterior fixtures and fittings. □ Posts, masts, and their sockets	693.546.3	Placing by pumping, spraying, injecting or similar methods
	◇ 692.92:621.316.98 Fixtures for lightning conductors	693.546.4	Compaction by vibration. Vibrated concrete. □ With external vibrators on formwork.
692.97	Interior fixtures and fittings. □ Hooks. Pegs. Brackets	693.546.5	Pervibration. Transvibration
	⇒ 645	693.546.7	Ramming. Rodding. Other compaction methods
	⇒ 646	693.547	Finishing, screeding, floating (of concrete)
	⇒ 684		Curing the concrete (including steam curing), heating, cooling and other measures during setting and hardening
693	MASONRY AND RELATED BUILDING CRAFTS	693.547.1	Curing methods
	⇒ 666.9	693.547.12	Water or wetted coverings
		693.547.13	Membranes or special solutions
		693.547.14	Steam curing
		693.547.2	Vacuum treatment. Vacuum concrete

693.547.3	Cold-weather concreting. Concrete construction in frost ⇒ 624.139	693.62	Plasters according to use and purpose ◇ 693.62:692.2 Wall plasters ◇ 693.62:692.54 Ceiling plasters ◇ 693.62:699.82 Waterproof renderings ◇ 693.62:699.844 Soundproof plastering
693.547.4	Cooling of mass concrete	693.621	Interior plastering
693.547.6	Hot-weather concreting. □ Wetting of surfaces during curing. Wet coverings	693.625	Exterior rendering. Mortar coats. Textured surfaces. Parging. Pargeting. Rough-cast finish. Spatterdash. Pebbledash. Scraped finish
693.547.7	Protection from vibration or other mechanical influences	693.63	Stucco
693.547.8	Removal or striking of formwork. Striking time. Cleaning of forms	693.64	Treatment of plastered surfaces (technical and artistic). □ Scagliola (imitation marble). Sgraffito (graffito, scratch work)
693.548	Surface treatment of the concrete. Exposed concrete. □ Surface roughening, bush-hammering etc. Washing-out (exposed aggregate finish)	693.65	Graining
693.55	Reinforced concrete construction ⇒ 666.982	693.66	Rejointing. Repointing
693.552	Batching, mixing of reinforced concrete 693.552 ≡ 693.542	693.69	Various special finishes and facings
693.554	Reinforcement (steel)	693.7	Floor tiling. Wall tiling. Paving. Paviour's trade. Asphalt work ⇒ 666.96
693.554.1	Preparation and assembly of reinforcement ◇ 693.554.1:621.791 Welding of reinforcement ◇ 693.554.1:621.791.052.4 Butting of reinforcement ◇ 693.554.1:621.882 Bolting, screwing of reinforcement ◇ 693.554.1:621.981 Bending of reinforcement	693.73	Laying of jointless floorings. □ Laying of terrazzo, macadam floorings
693.554.2	Prefabricated reinforcement. □ Reinforcing cages. Reinforcing framework	693.74	Laying of tiles, slabs, prefabricated mosaics etc. on ceilings, walls, sanitary installations, floors ⇒ 666.268
693.554.3	Position of reinforcement in concrete	693.741	Laying of mosaics
693.554.6	Placing and supporting the reinforcement. Anchorages. Hooks and eyes. Loops. Bar spacers <i>Details by :621.88...</i>	693.745	Laying of quarry tiles
693.556	Placing and compacting the reinforced concrete 693.556 ≡ 693.546	693.75	Paving. Paviour's trade
693.557	Curing the reinforced concrete (including steam curing). Heating, cooling and other measures during setting and hardening 693.557 ≡ 693.547	693.78	Asphalting. Asphalter's trade
693.558	Surface treatment of the reinforced concrete	693.8	Structural work in steel and other metals
693.56	Prestressed concrete construction	693.81	Iron and steel work
693.562	Batching, mixing of prestressed concrete 693.562 ≡ 693.542	693.813	Structural ironwork
693.564	Prestressing reinforcement. Prestressed steel. Prestressing	693.814	Structural steelwork
693.565	Tensioning. Pretensioning (methods and devices) ◇ 693.565:536.41 Pretensioning by thermal expansion ◇ 693.565:621.866 Pretensioning with jacks	693.814.1	Various structural steel members. □ Prestressed steel members ◇ 693.814.1:072.2 Steel girders
693.566	Placing and compacting the concrete 693.566 ≡ 693.546	693.814.2	Connections in steelwork. Steel joints 693.814.2 ≡ 624.078.4
693.567	Curing the concrete (including steam curing), heating, cooling and other measures during setting and hardening 693.567 ≡ 693.547	693.814.3	Tubular steel work
693.568	Surface treatment of the concrete	693.82	Structural work in nonferrous metals in general 693.82 ≡ 693.814
693.6	Plasterer's trade. Finishing work ⇒ 666.91/.92	693.827	Structural work in light metals (e.g. aluminium)
693.61	Plastering	693.9	Skeleton structures with panel infillings. Constructions combining various materials. Composite construction
693.611	Plasters according to their constituents	693.94	Timber-framed skeleton structures
693.612	Plasters with admixtures	693.95	Concrete-framed skeleton structures
693.613	Preparation of surfaces to receive plaster. Plaster base, lathing	693.955	Skeleton structures with frame of reinforced concrete
693.614	Application of plasters. Methods of application	693.956	Skeleton structures with frame of prestressed concrete
		693.97	Skeleton structures of metal
		693.972	Steel-framed skeleton structures
		693.977	Skeleton structures with frame of light metal
		693.98	Constructions combining various materials. Composite construction
		694	TIMBER CONSTRUCTION. CARPENTRY. JOINERY ⇒ 674

694.1	Timber construction in general. Carpentry. Carpenter's work	696.135	Sewage disposal for isolated buildings. Sewage purification. Septic tanks ⇒ 628.32
694.12	Fixing materials. Fasteners	696.136	Soakaways (for surface water)
694.14	Wood joints 694.14 ⇒ 624.078.4	696.138	Conservancy systems. Cesspools ⇒ 628.42
694.3	Temporary timber construction. □ Sizing, hewing, assembling of parts	696.139	Solid refuse disposal. □ Incinerators for local use ⇒ 628.542
694.4	Heavy timber construction. □ Sizing, hewing, assembling of parts for heavy load-bearing members, frameworks, girders, beams, trusses	696.14	Sanitary installations and fittings in buildings. Lavatory fittings ⇒ 628.6 ⇒ 644.68
694.5	Light timber construction. □ Sizing, hewing, assembly of parts for light frameworks, walls, roofs, floors	696.141	Water closets. Urinals. Bidets ⇒ 628.425
694.6	Joinery. □ Wood finishings, grounds, backings. Battens, fixings. Panelled work. Doors. Windows. Trim. Staircases. Boarded floors ⇒ 674.21	696.141.1	Water closets
694.7	Cabinet work. Ornamental fixed joinery ⇒ 684.4 ⇒ 684.6 ⇒ 749.1	696.141.2	Urinals
696	EQUIPMENT, SERVICES, INSTALLATIONS IN BUILDINGS (SANITARY, GAS, STEAM, ELECTRICAL). PIPE FITTER. PLUMBER. METAL WORKER. DRAINLAYER. ELECTRICIAN. OTHER TRADES	696.141.3	Slop-sinks
696.1	Pipe fitter's trade. Other metalwork trades ⇒ 628.1	696.141.5	Bidets
696.11	Water supply installations. □ Installations for drinking water ⇒ 621.642/.646 ⇒ 644.61	696.142	Sinks and accessories
696.113	Storage cisterns in buildings	696.143	Washbasins
696.115	Water supply layout to fittings. Pipes and pipe sizes	696.144	Other ablutionary installations
696.117	Taps. Stop valves. Fittings	696.144.1	Footbaths
696.12	Drainage above ground	696.144.2	Shower-baths. Shower cabinets
696.121	Roof drainage. Rainwater gutters, pipes and special fittings. Rainwater heads etc.	696.144.4	Sitzbaths
696.122	Waste pipes. Soil pipes. Ventilating pipes ⇒ 628.2	696.144.5	Full-length baths (including bath panels)
696.13	Drainage below ground. Site drains. House drains. Local refuse disposal ⇒ 628.242	696.144.7	Jacuzzis
696.132	Layout of drainage systems. □ Depth. Alignment. Gradient. Size	696.146	Special sanitary fittings for hospitals, dental surgeries etc. ⇒ 615.478
696.133	Drainpipes and fittings	696.2	Gas installations. Gas fitter's trade. Gas service pipes, fittings. Gas metering ⇒ 621.6.02 ⇒ 621.643 ⇒ 681.122 ⇒ 683.8 ⇒ 697.245
696.133.1	Drainpipes. □ Bends, junctions, branches, connections of drainpipes	696.3	Steam installations ⇒ 621.18 ⇒ 697.5
696.133.2	Ventilation of site drains. □ Fresh-air-inlet and vent pipe (induct and educt vents). Gas checks. Mica flaps	696.4	Hot water supply ⇒ 644.62 ⇒ 683.97 ⇒ 697.3/.4
696.133.3	Traps (water seals, gas checks). Gulleys. Interceptors	696.42	Boiler plant and equipment
696.134	Drain laying. Construction of manholes (inspection chambers). Drainlayer's trade	696.43	Hot water piping and connections. □ Sizes. Layout. Insulation
696.134.1	Construction of drains	696.45	Heat-emitting installations. □ Towel rails
696.134.2	Manholes (inspection chambers). Rodding (access) eyes ⇒ 692.783	696.46	Heat storage installations. □ Hot water cylinders
696.134.5	Testing for watertightness. Standard of workmanship	696.48	Systems and appliances for heating water
		696.5	Pneumatic, compressed air and vacuum installations ⇒ 621.5
		696.6	Electrical installations. Electric wiring systems. Electrician's trade ⇒ 621.316.1
		697	HEATING, VENTILATION AND AIR CONDITIONING OF BUILDINGS ⇒ 628.8 ⇒ 644.1 ⇒ 662.6/.9 ⇒ 683.9

697.1	Heating of buildings generally. Principles, requirements, calculations. External and internal influences ⇒ 551.5 ⇒ 612.5 ⇒ 699.86	697.31	General principles. Economy. Layout and design
697.11	General principles. Properties of buildings from heating point of view (measurements, results). Temperature levels. Fuel consumption	697.32	Boilerhouse or furnace-room plant. Fuel storage space
697.12	Temperature and air-change requirements with special reference to heat-loss calculation ⇒ 628.87	697.325	Stoves
697.13	External influences on heat loss from buildings	697.326	Boilers (including combined hot water and central heating types) ⇒ 621.18
697.132	Exterior temperature, climate, season etc. (with reference to heat-loss calculation)	697.328	Heat storage apparatus. Calorifiers, steam or hot water
697.133	Heat transmission losses. □ Influence of humidity and other factors	697.329	Solar heating devices. Other heat transfer and storage apparatus ⇒ 662.997 ⇒ 697.7
697.134	Exposure to wind (especially as factor in heat-loss calculation)	697.33	Distribution of heating medium. Pipework. □ Sizes. Layout. Insulation
697.135	Air infiltration (through windows, ventilation openings, doors etc.). Heat loss from draughts	697.34	District heating (zonal heating). Heat distribution over a distance to several buildings or premises ⇒ 697.384 ⇒ 697.434 ⇒ 697.444 ⇒ 697.514
697.137	Moisture, humidity factors in heat loss	697.347	Measurement, metering of heat consumption (by the various users). Control of heat consumption by use of calorifiers ⇒ 681.125
697.14	Internal influences on heat loss from buildings. □ Effects of layout, occupancy etc. on heat loss	697.35	Heat-emitting apparatus. □ Radiators. Convector heaters. Unit heaters
697.2	Space heating by individual heat-generating appliances in rooms. □ By fires, stoves, panel-heaters ⇒ 644.11 ⇒ 662.9 ⇒ 683.9	697.352	Types of radiator. Design, location and other factors influencing heating capacity
697.24	Space heating by fuel-burning heaters	697.353	Radiant and panel heaters. Design, location and other factors influencing heating capacity
697.243	Solid fuel heaters	697.355	Convectors. □ Grilles. Casings
697.243.3	Primarily radiant solid-fuel heaters. □ Open fires. Fireplaces, hearths, grates ⇒ 683.93	697.356	Unit heaters
697.243.5	Primarily convecting solid-fuel heaters. Solid-fuel stoves	697.357	Heating coils. Gilled pipes
697.244	Liquid-fuel heaters. □ Kerosine-fired (paraffin-fired) heaters	697.358	Heating capacity. Measuring methods. Differential radiation. Convection coefficients
697.244.3	Primarily radiant liquid-fuel heaters	697.38	Heating by circulation of hot air or gases (not in direct contact with the air in the space to be heated). Primitive heating systems. □ Hypocausts. Flue gases
697.244.5	Primarily convecting liquid-fuel heaters	697.381	General principles. Economy. Design and layout
697.244.7	Combined radiant-convactor liquid-fuel heaters	697.382	Stoves and furnaces for warm-air heating (nonventilating). □ Fan furnaces
697.245	Gas-fired heaters. Gas fires	697.383	Distribution, circulation of hot air or gases. Sizes, layout, insulation etc. of ducts Details by :621.64... and :662.998
697.245.3	Primarily radiant gas fires (normally requiring flues)	697.384	District heating (zonal heating), by hot air or gases only ⇒ 697.34
697.245.5	Primarily convecting gas-fired heaters	697.385	Heat-emitting apparatus designed for indirect hot air or gas circulation systems 697.385 ≡ 697.35
697.245.7	Combined radiant-convactor gas-fired heaters (flueless). Gas panel heaters. Gas-fired radiators	697.39	Adventitious heating. Total efficiency of heating installation. □ Heating by chimney flues
697.246	Combined heating methods (e.g. coal-and-gas-fired heaters)	697.4	Hot-water central heating. Central heating with other liquids, e.g. oil ⇒ 696.4
697.27	Electric space heaters ⇒ 621.365	697.43	Low-pressure hot-water heating systems
697.273	Primarily radiant electric heaters. Electric fires	697.431	General principles. Calculations. Economy. Design
697.275	Primarily convecting electric heaters. Electric radiators (filled)	697.432	Boiler plant and equipment 697.432 ≡ 697.32
697.278	Electric storage heaters	697.433	Piping and connections (sizes; layout, insulation etc.)
697.3	Central heating in general (including combined hot water supply and space-heating systems) ⇒ 696.4		

697.433.2	Gravity circulation	697.575	Heat-emitting apparatus specially designed for exhaust-steam central heating systems
697.433.4	Accelerated circulation		
697.434	District heating (zonal heating), by low-pressure hot water only ⇒ 697.34	697.7	Other methods of heating. □ Solar heating ⇒ 697.329
697.435	Heat-emitting apparatus designed for low-pressure hot-water central heating 697.435 ≅ 697.35	697.8	Chimneys. Flues ⇒ 624.971 ⇒ 628.522 ⇒ 662.922/923 ⇒ 692.71
697.44	High-pressure hot-water heating systems		
697.441	General principles. Calculations. Economy. Design	697.81	Flues (internal). □ Flue dimensions. Masonry flues. Pipe flues
697.442	Boiler plant and equipment 697.442 ≅ 697.32	697.83	Interior chimneys
697.443	Piping and connections (sizes, layout, insulation etc.)	697.84	Exterior chimneys. □ Chimney stacks. Chimney pots
697.444	District heating (zonal heating), by high-pressure hot water only ⇒ 697.34	697.85	Freestanding chimney stacks. □ Factory chimney stacks ⇒ 624.042
697.445	Heat-emitting apparatus designed for high-pressure hot-water central heating 697.445 ≅ 697.35		
697.47	Methods of accelerating the hot water circulation or head on system	697.87	Draught regulation in flues
697.49	Use of other circulating liquids. □ Oil as circulating liquid	697.88	Soot removal. Ash removal. Chimney sweeping
697.5	Steam central heating (by live and exhaust steam) ⇒ 621.1	697.9	Ventilation. Air conditioning ⇒ 628.8
697.51	Live-steam central heating generally. □ Principles. Economy. Design	697.91	General principles. Plant, equipment etc.
697.511	General principles. Calculations. Economy. Design	697.911	Principles. Calculations. General design
697.512	Boiler plant and equipment 697.512 ≅ 697.32	697.912	Plant and machinery in general
697.513	Piping and connections (sizes, layout, insulation etc.)	697.92	Ventilation proper
697.514	District heating (zonal heating), by live steam only ⇒ 697.34 ⇒ 697.574	697.921	Introduction and extraction of air. Air inlets. Air outlets
697.515	Heat-emitting apparatus designed for live-steam central heating systems	697.921.2	Natural ventilation. □ Via windows, louvres, airbricks
697.52	Vacuum steam-heating systems 697.52 ≅ 697.51	697.921.4	Mechanical ventilation. □ Exhaust system. Plenum system
697.53	Low-pressure steam central heating systems 697.53 ≅ 697.51	697.922	Distribution of fresh air. Air ducts and equipment (design and construction)
697.54	High-pressure steam central heating systems 697.54 ≅ 697.51	697.93	Humidity control
697.55	Atmospheric or vapour central heating systems 697.55 ≅ 697.51	697.931	Humidity (influence, measurement, control)
697.56	Subatmospheric or differential central heating systems 697.56 ≅ 697.51	697.932	Humidification. Humidifiers. Direct humidifying systems
697.57	Heating by exhaust steam ⇒ 621.186.8	697.933	Reduction of moisture content. Dehumidifiers
697.571	Principles. Calculation. Economy. Design	697.934	Drying of air
697.572	Special devices for removing air and condensation water	697.94	Air-conditioning units in general. Cleansing, washing and filtration of air ⇒ 628.511
697.573	Piping and connections (sizes, layout, insulation etc.)	697.941	Air cleaners generally
697.574	District or zonal heating by exhaust steam. □ Using steam from power stations ⇒ 697.34 ⇒ 697.514	697.942	Air filters (for removal of airborne particles). Dust filters etc.
		697.943	Odour removal. Deodorization of air. □ Deodorization by absorption on carbon
		697.945	Centrifugal collectors. Dynamic collectors
		697.946	Electric precipitation. Electrostatic air cleaners ⇒ 621.359.4
		697.947	Ozone generators. Ozonizers
		697.948	Ultraviolet-ray equipment
		697.95	Circulation and movement of air
		697.952	Natural air circulation. □ By differential air temperature. By wind
		697.953	Air circulation by mechanical means. Fans ◇ 697.953:621.634 Helical fans. Axial-flow fans ⇒ 621.5
		697.956	Air circulation by heat sources in the room, e.g. fires, stoves
		697.957	Movement of air flow ⇒ 628.854

697.97	Temperature conditioning. Cooling and heating of air	699.826	Waterproof structures. □ Waterproof doors.
697.971	Cooling generally	699.828	Waterproof windows
697.972	Cooling by ice		Drying out of buildings
697.973	Cooling by liquid evaporation	699.83	Protection against weathering, wind, storms. Weatherproof construction
697.974	Cooling by use of well water		
697.975	Cooling by refrigeration	699.84	Protection against vibration and noise
	Details by :621.57...	699.841	Protection against seismic effects, earth tremors. Earthquake-resistant construction
697.978	Heating of introduced air		Insulation against relatively slight vibrations
	⇒ 697.38	699.842	Protection against acoustic effects, noise.
		699.844	Soundproofing. Sound insulation and absorption
			⇒ 628.517.2
697.98	Dust and fume removal (building aspects, installations)	699.85	Civil defence construction
	⇒ 628.511		⇒ 355.58
697.982	Exhaust systems		Underground defence shelters
	⇒ 621.61/.63	699.852	⇒ 643.88
697.983	Collectors		Defence shelters above ground
697.984	Hoods	699.853	
697.986	Duct and piping design, layout and construction	699.86	Thermal insulation of buildings
698	FINISHING AND DECORATING TRADES	699.865	Insulation against cold. Cold-resistant construction
	⇒ 747	699.866	Insulation against heat. Heat-resistant construction
698.1	Painting and decorating trades		⇒ 699.812
698.12	Painting of buildings. Decorating painting. Painting and distempering. Housepainter's trade	699.868	Prevention of damage from thermal expansion.
	⇒ 667.64		Expansion joints
	⇒ 691.57	699.87	Protection against various toxic, noxious and biological effects
698.16	Paperhanger's trade		Gasproof, fume-proof, smokeproof construction
	⇒ 684.75	699.871	◇ 699.871:546.296 Construction with safeguards against radon gas
	⇒ 691.145		⇒ 62-784.2
698.3	Glazing. Glazier's trade		⇒ 623.459.72
	⇒ 666.15/.16		⇒ 628.52
	⇒ 686.6		⇒ 628.53
	⇒ 748.5	699.872	Dustproof construction. Measures against harbouring dirt and dust
698.7	Ancillary decorating trades. Soft furnishing and upholstery trades		⇒ 62-784.4
	◇ 698.7:645.1 Fitting of floor coverings.		⇒ 628.511
	Carpet laying. Linoleum laying	699.874	Rot-proof construction. Measures against microorganisms. □ Measures against fungi, moulds, dry rot
	◇ 698.7:645.3 Hanging of curtains, blinds etc.		Protection against vegetation, moss and roots
	⇒ 684.7		Pest-proof, vermin-proof construction
699.8	Protection of and in buildings. Emergency measures. Precautions	699.876	
699.81	Fire protection. Fireproofing of buildings. Firefighting installations and equipment	699.878	
	⇒ 614.84	699.88	Protection against various physical influences
699.812	Fireproof construction of buildings. Use of fire-resistant materials. Fire-coats, protective paints	699.885	Protection against light
	⇒ 667.637.4	699.887	Protection against electrical and radioactive effects
699.814	Design of buildings for fire prevention. □ Fire doors. Baffles. Smoke outlets	699.887.2	Protection against lightning
699.816	Firefighting installations in buildings. □ Built-in extinguishing systems	699.887.3	Protection against ionizing radiation. Radiation shielding (e.g. in atomic plant)
	Details by :614.843/.847		⇒ 667.661.2
699.82	Protection against moisture. Damp-proof and waterproof construction	699.887.5	⇒ 75.022
			Protection against electrical radiation.
699.822	Damp-proof courses (against capillary moisture, rising damp)	699.888	□ Protection against X-rays
			Antimagnetic construction. Magnetism-proofing and protection

7 THE ARTS. RECREATION. ENTERTAINMENT. SPORT

The auxiliaries listed under 7.0... are not applicable at 77 and its subdivisions, except for 7.04...

Denote exhibition catalogues of art objects by (083.824) from Table 1d

Denote the work of particular artists or practitioners by alphabetic extension (Table 1h)

◇ 730Rodin The sculptures of Rodin

◇ 75.037.2(083.824) Exhibition catalogues of Cubist paintings

7.01	THEORY AND PHILOSOPHY OF ART. PRINCIPLES OF DESIGN, PROPORTION, OPTICAL EFFECT ◇ 7.01:111.852 Aesthetics of art. Taste in art	7.021.1 7.021.2	Preparation of ground Laying out. Rough drawing. Sketching. Preliminary presentation
7.011	Taste and social significance of art	7.021.22 7.021.23 7.021.3	Two-dimensional. □ Drawings. Sketches Three-dimensional. □ Models. Designs Technique, according to medium <i>The following numbers should be used for the techniques themselves, not for the materials used to effect those techniques, which are listed below at 7.023. They apply in particular to Class 75 for special techniques in painting, and will normally be appended only to 75</i>
7.011.2	Esoteric versus popular art. Social significance of art		
7.011.22	Esoteric, highbrow art. Art of and for the initiated few. Art for art's sake. 'Ivory tower' art		
7.011.26	Popular art. Art for the people, for the many		
7.011.28	Kitsch. The cult of bad taste		
7.011.3	Public, controlled, official art versus free, individual creation. Commissioned art. Art to order	7.021.32	Painting in the strict sense (with liquid medium) <i>Expand, as required, using Table 1k-03 - Common auxiliaries of materials</i> ◇ 7.021.32-035.676.332.066.3 In watercolour. Aquarelle ◇ 7.021.32-035.676.332.4 In oils
7.011.4	Utility aspect of art. Pure, academic versus applied art		
7.012	Design. Composition ◇ 7.012-025.14 Mixed use ◇ 7.012-027.37 Mass-produced	7.021.33	Special painting techniques <i>Expand, as required, using Table 1k-03 - Common auxiliaries of materials</i> ◇ 7.021.33-035.676.332.6 Painting in lacquer ◇ 7.021.33-035.8 Painting in wax
7.012.23	Approaches		
7.013	Harmony in general. Proportion. □ Principles of unity, order, symmetry, variety, monotony. Harmonic systems. Laws of geometric proportion. Dimensional coordination. Modular coordination. Form. Rhythm and line	7.021.333	Fresco <i>Painting on wet plaster</i>
7.014	The Orders. □ The Orders of classical architecture	7.021.34	Painting, pigmenting, with solid (stick) medium. □ Drawing. Crayoning <i>Expand, as required, using Table 1k-03 - Common auxiliaries of materials</i> ◇ 7.021.34-035.675.2 With lead pencil. Silver point ◇ 7.021.34-035.683 With charcoal ⇒ 741
7.014.1	Modules. Intercolumniation	7.021.346	Sgraffito
7.014.11	Modules	7.021.4	Execution. Application of pigment
7.014.12	Intercolumniation	7.021.42	Craftsmanship
7.014.3	Tuscan Order	7.021.45	Machine manufacture
7.014.4	Doric Order	7.021.5	Modelling
7.014.5	Ionic Order	7.021.7	Woodcarving
7.014.6	Corinthian Order	7.021.8	Finishing processes. □ Fixing. Drying. Varnishing
7.016	Profiles. Mouldings. Tracery effects	7.022	Tools. Equipment. □ Chisels etc. Machines and accessories for shaping
7.016.2	Terminal, skyline features or embellishments. □ Finials. Pinnacles. Pendants. Bosses	7.022.1	Tools, equipment for preparation of the support (base). □ Stretchers. Easels. Drawing boards. Drawing tables ⇒ 7.023.11
7.016.4	Motifs. Enrichment by motifs. □ Motifs from nature, e.g. foliage motifs	7.022.2	Tools, equipment for application of the pigment. □ Brushes. Spatulas. Palette knives. Spraying devices
7.017	Optical effects	7.022.3	Palettes. Paintboxes
7.017.2	Chiaroscuro. Light and shade	7.022.7	Special clothing ⇒ 614.89 ⇒ 687.15
7.017.4	Colour. Texture. □ Harmony of colour		
7.017.6	Surface effect	7.022.8	Accommodation. Work space. Studios
7.017.9	Other optical effects. □ Perspective effect. Illusions. Corrections. Refinements	7.022.81	Open-air work
7.02	TECHNIQUE. CRAFTSMANSHIP. □ Working methods. Drawings, models, accessories. Materials available. Preservation, repair. Conversion		
7.021	Working methods. Processes. Execution		

7.022.82	Workshops. Studios	7.025.8	Change of location. □ Exchange. Transfer. Loan for temporary exhibition
7.023	Materials. Choice of materials <i>For building materials, at 72, denote kinds of material by :691...</i>	7.026	Reproduction. Copying. Making of facsimiles, replicas, casts, miniatures. Printing of graphics. Taking prints, pulls ⇒ 738 ⇒ 748.5
7.023.1	Materials <i>For materials in all other contexts, use -03... (Table 1k)</i>	7.026.2	Models of existing works ⇒ 7.021.23 ⇒ 7.061
	◇ 7.023.1-032.5 Rock. Stone ◇ 7.023.1-032.548 Marble ◇ 7.023.1-033.21 Plaster ◇ 7.023.1-034 Metals ◇ 7.023.1-034.21 Gold ◇ 7.023.1-034.22 Silver ◇ 7.023.1-034.35'6 Bronze ◇ 7.023.1-035.3 Wood. Including: Wooden surfaces. Wood panels ◇ 7.023.1-035.437 Drawing paper and board. Watercolour paper. Crayon paper ◇ 7.023.1-035.5 Materials of animal origin ◇ 7.023.1-035.51 Leather (natural leather) ◇ 7.023.1-035.561 Bone ◇ 7.023.1-035.562 Ivory ◇ 7.023.1-035.563 Horn ◇ 7.023.1-035.566 Oyster shell. Tortoiseshell ◇ 7.023.1-035.813 Wax in general ◇ 7.023.1-036.5 Plastics ◇ 7.023.1-037 Textiles ◇ 7.023.1-037.1 Canvas. Linen ◇ 7.023.1-037.21 Cotton	7.027	Kinds of plastic art form. Methods of sculptural representation In the round. □ Statues. Busts
		7.027.1	
		7.03	ARTISTIC PERIODS AND PHASES. SCHOOLS, STYLES, INFLUENCES
		7.03'0	Phases of development <i>The '0 auxiliaries, which are parallel with =... '0 (Table 1c) and 81'0, are for use mainly in 7.032, where they may be needed in creating a useful date sequenceC</i>
		7.03'01	Old period. Archaic periodC
		7.03'02	Classical periodC
		7.03'04	Middle periodC
		7.03'06	Late periodC
		7.031	Prehistoric art. Primitive art (ancient and modern) ⇒ 903 ⇒ 904
7.023.11	Support (base, underlying surface)	7.031.1	Prehistoric art <i>Specify period by "... (Table 1g)</i>
7.023.2	Pigments applied in liquid form <i>Expand, using Table 1k-03</i>		◇ 7.031.1"632/633" Palaeolithic and Mesolithic art (Old and Middle Stone Age) ◇ 7.031.1"634" Neolithic art (New Stone Age) ◇ 7.031.1"636" Copper Age art ◇ 7.031.1"637" Bronze Age art ◇ 7.031.1"638" Iron Age art
	◇ 7.023.2-035.675 Inks. Washes ◇ 7.023.2-035.676.332.2 Water-based colours. Including: Gouache (body colour, poster colors) ◇ 7.023.2-035.676.332.2.066.3 Emulsified colours. Including: Tempera (egg tempera) ◇ 7.023.2-035.676.332.4 Oil colours ◇ 7.023.2-035.676.332.6 Lacquer. Varnish	7.031.2	Primitive art (ancient and modern) <i>Details by (=...) (Table 1f) or by (1/9) (Table 1e) and "... (Table 1g) if preferred</i>
7.023.3	Pigments applied in solid form <i>Expand, as required, using Table 1k-03 - Common auxiliaries of materials</i>		◇ 7.031.2(=414/45) African primitive art ◇ 7.031.2(=821.173) Mayan art ◇ 7.031.2(=822.21) Aztec art ◇ 7.031.2(=87) Pre-Columbian art (pre-Spanish art) of ancient Central and South American civilizations ◇ 7.031.2(=873) Ancient Peruvian art. Including: Inca. Tiahuanaco. Nazca
	◇ 7.023.3-035.675.5 With chalk ◇ 7.023.3-035.675.62 With lead pencil. Silverpoint ◇ 7.023.3-035.676.293 Siccatives ◇ 7.023.3-035.676.332.6 Varnishes ◇ 7.023.3-035.683 With charcoals	7.031.3	Indigenous, aboriginal art <i>Details by (=...) (Table 1f), or by (1/9) (Table 1e) and "... (Table 1g), if preferred</i>
7.024	Accessories	7.031.4	Folk or peasant art <i>Details by (=...) (Table 1f), or by (1/9) (Table 1e) and "... (Table 1g) if preferred</i>
7.024.4	Titles. Legends. Captions. Inscriptions		◇ 7.031.4(=214.58) Roman. Gypsy art ◇ 7.031.4(=81/87) American Indian (Amerindian) art
7.024.5	Mountings. Base pedestals. Frames	7.032	Ancient cultures of the Old World. Styles of Antiquity <i>Details by (1/9) (Table 1e) and "... (Table 1g). In order to maintain the correct order when number-building, e.g. for Ancient Greek art, the '0 auxiliaries listed under 7.03 may be used. This will overcome the problems of the automatic filing of dates in a straight numerical sequence, which</i>
7.025	Damage, deterioration, removal. Preservation, protection		
7.025.1	Damage. Destruction		
7.025.2	Deterioration. Dilapidation. Decay. Neglect		
7.025.21	Remains. Ruins. Fragments		
7.025.22	Neglect. Disuse. Misuse		
7.025.23	Deformation. Distortion		
7.025.3	Care. Upkeep. Preservation. Conservation		
7.025.4	Repair. Restoration. Renovation. Renewal. Cleaning. Other technical measures		
7.025.5	Change of use. Conversion		
7.025.7	Loss or unauthorized removal. □ Loss in transit. Loss by theft		

creates difficulties when dates from both the pre-Christian and Christian eras are needed.

If the citation order 'date - place' is preferred, so as to place Roman art after Greek, for example, this may be achieved by citing the date before the area number. In this case, the '0 auxiliaries must precede the date, as in an automated system the numerals file in ascending order and this will create difficulties with dates that cover the pre-Christian and Christian eras.

If preferred, the individual auxiliaries for Cyprus (393.7), Phoenicia (394.5), Arabia (394.7) and Carthage (397.3) may be used

- ◇ 7.032(31) Oriental. Asiatic
- ◇ 7.032(315) Chinese (including Tibetan, Central Asian, Korean). Mongolian. Manchurian
- ◇ 7.032(32) Ancient Egyptian
- ◇ 7.032(34) Indian and south-east Asian. Sinhalese, art of Sri Lanka (Ceylon)
- ◇ 7.032(352) Assyrian
- ◇ 7.032(354) Babylonian (Chaldaean). Including: Sumerian. Akkadian
- ◇ 7.032(355) Iranian (Persian). Including: Elamite. Median. Achaemenid. Seleucid (Parthian). Sassanian. NOTE: If preferred, the individual auxiliaries for Media (353), Parthia (356), Sassania (357) and Elam (359) may be used
- ◇ 7.032(358) West Asiatic (Mesopotamian. Iranian)
- ◇ 7.032(37) Ancient Italic and Roman. NOTE: If preferred, the individual auxiliaries for Etruria (375) and Rome (376) may be used
- ◇ 7.032(37)'01 Early Roman
- ◇ 7.032(37)'02 Classical Roman
- ◇ 7.032(37)'04 Late Roman Empire
- ◇ 7.032(37)'06 Survival after fall of Rome in the West
- ◇ 7.032(38) Grecian. Ancient Greek
- ◇ 7.032(38)'01"637" Archaic Greek. Including: Mycenaean
- ◇ 7.032(38)'02"-05/-04" Classical Greek
- ◇ 7.032(38)'04"-03/-01" Hellenistic period. Pre-Roman and Roman Hellenistic
- ◇ 7.032(391)"637" Aegean: Bronze Age. Including: Minoan, Cretan. Helladic. Cycladic
- ◇ 7.032(392/393) Of Asia Minor
- ◇ 7.032(392) Trojan
- ◇ 7.032(394) South Arabian (pre-Islamic). Phoenician (Tyre, Sidon). Carthaginian. Cyprian. Syrian. Palestinian
- ◇ 7.032'01"637"(38) Archaic Greek. Including: Mycenaean
- ◇ 7.032'01"637"(391) Aegean: Bronze Age. Including: Minoan, Cretan. Helladic. Cycladic
- ◇ 7.032'02"-05/-04"(38) Classical Greek
- ◇ 7.032'04"-03/-01"(38) Hellenistic period. Pre-Roman and Roman Hellenistic
- ◇ 7.032'02"-0246/-0027"(37) Republican Rome
- ◇ 7.032'02"-0027/+0069"(37) Early Roman Empire

7.033/038

Styles: mediaeval to modern

These subdivisions listed below provide a primarily chronological sequence of styles. If a geographical grouping is required, the

common auxiliaries of place (Table 1e) may be intercalated after the number for the style in general

- ◇ 7.033.4(410.1) Romanesque - England (in general)
- ◇ 7.033.4(410.1)1 Romanesque - England - early (Anglo-Saxon) [instead of: 7.033.41(410.1)]
- ◇ 7.033.4(430) Romanesque - Germany (in general)
- ◇ 7.033.4(430)1 Romanesque - Germany - early (Merovingian, Frankish) [in stead of: 7.033.41(430)]

7.033

- 7.033.1
- 7.033.2
- 7.033.3
- 7.033.32
- 7.033.33
- 7.033.34
- 7.033.35
- 7.033.36
- 7.033.37
- 7.033.38
- 7.033.39
- 7.033.4

Mediaeval (Christian, Islamic)

- Early Christian
- Byzantine
- Islamic
- Ummayyad. Caliphate
- Abbasid. Samanid. Buwayid (Persian), Tulunid (Syro-Egyptian). Aghlabid (West Islamic)
- Fatimid: Syro-Egyptian, North African, Sicilian
- Seljuk. Almoravid (West Islamic)
- Almohad (West Islamic)
- Mongol (Iraq-Iran). Ayyubid (Syro-Egyptian). Hafsids, Naerid (West Islamic). Ayyulid (Yemen). Mogul (India)
- Timurid, Safavid (Iraq-Iran). Mamluk (Syro-Egyptian). Rasulid (Yemen)
- Ottoman
- Romanesque
 - ◇ 7.033.4(410.1) English Romanesque
 - ◇ 7.033.4(415) Irish, Celtic Romanesque
 - ◇ 7.033.4(430) German Romanesque. Rhineland Romanesque (Rhenish)
 - ◇ 7.033.4(44) French Romanesque. Including: Norman. Burgundian. Cluniac
 - ◇ 7.033.4(450) Romanesque in Italy and Sicily
 - ◇ 7.033.4(46) Romanesque of the Iberian peninsula
- Early, First Romanesque
 - ◇ 7.033.41(410.1) In England: Anglo-Saxon
 - ◇ 7.033.41(430) In Germany: Merovingian (Frankish)
 - ◇ 7.033.41(450) In Italy: Ostrogothic
 - ◇ 7.033.41(46) In the Iberian peninsula: Visigothic
- Middle Romanesque
 - ◇ 7.033.42(410.1) In England: Viking. Norse
 - ◇ 7.033.42(450) In Italy: Lombard (Langobardic)
 - ◇ 7.033.42(46) In the Iberian peninsula: Asturian. Mozarabic
- Carolingian
- Ottonian
- Late Romanesque
 - ◇ 7.033.44(410.1) Norman art in England. Anglo-Norman
- Late Mediaeval, especially in western Europe. Gothic
 - ◇ 7.033.5(410.1) English Gothic
 - ◇ 7.033.5(430) German Gothic
 - ◇ 7.033.5(44) French Gothic
 - ◇ 7.033.5(46) Gothic in the Iberian peninsula
- Early, Transitional, Primary Gothic
 - ◇ 7.033.51(410.1) Early English Gothic
 - ◇ 7.033.51(44) In France: Transitional. Primary. Lancet
 - ◇ 7.033.51(46) In the Iberian peninsula: Transitional, Cistercian. First, French

7.033.52	Middle, High, Secondary Gothic	◇ 7.033.52(410.1) In England: Decorated	◇ 7.035(410) Post-Renaissance styles in Britain
		◇ 7.033.52(44) In France: Secondary. Rayonnant	◇ 7.035(410) "17" Earlier period (18th century)
		◇ 7.033.52(46) In the Iberian peninsula: Secondary. Radiant (Rayonnant)	◇ 7.035(410) "18" Later period (19th century). Victorian
7.033.53	Late, Tertiary Gothic	◇ 7.033.53(410.1) In England: Perpendicular	◇ 7.035(430) Post-Renaissance styles in Germany
		◇ 7.033.53(44) In France: Tertiary. Flamboyant	◇ 7.035(44) Post-Renaissance styles in France
		◇ 7.033.53(46) In the Iberian peninsula: Tertiary. Flamboyant	◇ 7.035(460) Post-Renaissance styles in Spain
7.033.54	Later Gothic (transitional to Renaissance)	◇ 7.033.54(410.1) In England: Early Tudor. See also 7.034.4(410.1)	◇ 7.035(73) Post-Renaissance styles in the USA
		◇ 7.033.54(430) In Germany: 16th century Gothic. Sondergotik	7.035.1 Existing national and transitional styles
			◇ 7.035.1(460) In Spain: Neoclassical
			◇ 7.035.1(73) In USA: First National, Federal
			7.035.2 Classicism. Classic and archaic revival
			◇ 7.035.2(410) Classic revival in Britain
			◇ 7.035.2(460) In Spain: Empire
			◇ 7.035.2(73) In USA: Second National. Classic revival
7.034	Renaissance. Baroque. Rococo. Colonial		7.035.23 Graeco-Roman influence
	◇ 7.034(410) Renaissance and Colonial in Britain (and British colonies). NOTE: Alternatively, use the place auxiliaries (Table 1e) for the individual colonies		◇ 7.035.23(410) In Britain: Adam style
	◇ 7.034(410.1) English Renaissance	7.035.24 Roman influence and transition to Empire style	◇ 7.035.24(44) In France: Directoire
	◇ 7.034(44) French Renaissance	7.035.25 Egyptian and Oriental influences	◇ 7.035.25(410) In Britain: Regency
	◇ 7.034(450) Italian Renaissance		◇ 7.035.25(44) In France: Empire
	◇ 7.034(7) Renaissance styles in North America. Colonial styles	7.035.26 Neo-Greek (Néo-Grec)	
	◇ 7.034(71) In Canada: French Colonial	7.035.27 Italianate	
	◇ 7.034(73) In USA: English Colonial	7.035.3 Mediaevalism. Gothic revival. Other mediaeval influences	
7.034.3	Earliest phase of Renaissance		◇ 7.035.3(410) Mediaevalism in Britain. Gothic revival. Pre-Raphaelitism
7.034.4	Early Renaissance		◇ 7.035.3(460) In Spain: Elizabethan
	◇ 7.034.4(410.1) In England: Transitional. Tudor	7.035.36/37 Mainly German mediaevalist movements	
	◇ 7.034.4(450) In Italy: Quattrocento (15th century) styles. Including: Florentine, Lombardic schools	7.035.36 Gründerzeit. Nazarenes (Nazarener)	
		7.035.37 Biedermeier	
7.034.5	High Renaissance	7.035.4 Renaissance, Baroque and Rococo revivals	
	◇ 7.034.5(410.1) In England: Early Renaissance. Including: Elizabethan. Jacobean	7.035.5 Eclecticism	
	◇ 7.034.5(44) In France: 16th century; Italian, Roman influence	7.035.7 Symbolistic trends	
	◇ 7.034.5(450) In Italy: Cinquecento (16th century) styles. Including: Roman school	7.035.9 Return to traditional (national, domestic) forms. Traditionalism with contemporary design	
7.034.6	Late Renaissance	7.035.92 'Arts and Crafts' influences	
	◇ 7.034.6(410.1) In England: Stuart	◇ 7.035.92(410) Arts and Crafts movement in Britain	
	◇ 7.034.6(44) In France: 17th century; Classical, Bourbon	7.035.93 Art Nouveau. Jugendstil	
	◇ 7.034.6(450) In Italy: Mannerist. Including: Venetian school		
7.034.7	Baroque		
	◇ 7.034.7(410.1) In England: Baroque influence. Queen Anne		
	◇ 7.034.7(44) In France: Baroque influence. The Grand Manner: Louis XIV, Régence		
7.034.8	Rococo. Rocaille		
	◇ 7.034.8(410.1) In England: Georgian		
	◇ 7.034.8(44) In France: Perruque, Rocaille, Rococo, Louis XV		
7.034.9	Subsequent phases		
	◇ 7.034.9(44) In France: Louis XVI. First Empire		
7.035	Post-Renaissance. □ Revivals (classical, mediaeval etc.), chiefly 19th century. Neoclassicism. Revivalism. Classicism and the Romantic movement. Historicism. Eclecticism		
			7.036 Modern art movements (from late 19th century). Realism, impressionism and subsequent 20th century trends
			<i>If a geographical or chronological grouping is required, the common auxiliaries of place and time may be intercalated after .036 in the subdivisions below</i>
			◇ 7.036(430).7 German Expressionism. Including: Wopswede. Der Blaue Reiter. Der Sturm. Die Brücke
			◇ 7.036(44).45 In France: Revue Blanche
			◇ 7.036(492).7 Dutch Expressionism. Including: De Ploeg. De Bergense School
			◇ 7.036(492).81 Experimentelen
			◇ 7.036(493).7 Belgian Expressionism. Including: Sint Maertens Laethem
			◇ 7.036 "189" Modern art movements of the 1890s, fin de siècle
		7.036.1 Realistic trends. Objective realism. Naturalism. Accurate, detached representation	
		7.036.2 Impressionist movements. Impressionism. Luminism. Pleinairism. Intimism	

7.036.3	Functionalism. 'Fitness for purpose' etc.	7.038.53	Intermedia
7.036.4	Post-Impressionism. Neo-Impressionism	◇ 7.038.53:77	Photography as an artistic medium
7.036.43	Divisionism. Pointillism. Luminism	◇ 7.038.53:78	Image and sound/music
7.036.45	Symbolism. Nabis	◇ 7.038.53:791	Film, video as an artistic medium
	◇ 7.036.45(44) In France: Revue Blanche		
7.036.5	Expressive monumentalism	7.038.531	Happenings. Fluxus. Environment. Body art. Performance art
7.036.7	Expressionist movement. Expressionism	7.038.532	Concrete poetry. Mail art. Book art. Rubber stamp art
	◇ 7.036.7(430) German Expressionism. Including: Wopswede. Der Blaue Reiter. Der Sturm. Die Brücke	7.038.54	Arte povera. Concept art
	◇ 7.036.7(492) Dutch Expressionism. Including: De Ploeg. De Bergense School	◇ 7.038.54(410.1)	In England: Art and language
	◇ 7.036.7(493) Belgian Expressionism. Including: Sint Maertens Laethem	7.038.541	Earth art. Land art. Narrative art. Spurensicherung
7.036.8	New Expressionism	7.038.542	Structuralism
7.036.81	Cobra	7.038.543	Multiple art
	◇ 7.036.81(492) Experimentelen	7.038.544	Space art
7.036.82	Internationale Situationniste	7.038.55	Installation art. Installations
		7.038.6	Postmodernism
7.037	Transition between Expressionism and abstract art	⇒ 141.78	
	<i>If a grouping by country is required, the common auxiliaries of place (Table 1e) may be intercalated after .037</i>	7.04	SUBJECTS FOR ARTISTIC REPRESENTATION. ICONOGRAPHY. ICONOLOGY. DETAILS AND FINISHES. DECORATION. ORNAMENT
	◇ 7.037(410.1).3 In England: Vorticism		<i>Divide by direct addition of other .0 auxiliaries, by -03... (Table 1k), and by :6... and :7...</i>
	◇ 7.037(430).7 In Germany: Neue Sachlichkeit		◇ 7.04-035.3 Timber details
	◇ 7.037(436).5 In Austria: Wiener Schule		◇ 7.04.017 Optical effects of details and decoration (light and shade, colour, texture etc.)
	◇ 7.037(450).5 In Italy: Pittura metafisica		
	◇ 7.037(73).7 In USA: Precisionism. The Immaculates	7.041	Human form (face and body). Life studies
7.037.1	Fauvism	⇒ 743	
	⇒ 7.036.7	7.041.2	Studies of the human form. □ Head. Bust
7.037.2	Cubism. □ Orphism. Purism. Simultaneism. Synchronism. Musicalism	7.041.3	The nude
7.037.3	Futurism	7.041.5	Portraits
	◇ 7.037.3(410.1) In England: Vorticism	7.041.53	Self-portraits
7.037.4	Dada (Dadaism, artistic Nihilism)	7.041.6	Arrangement of clothing. Drapery
7.037.5	Surrealism. Fantastic art	7.041.7	Scenes from life (court life, rustic life). Genre. Narrative art
	◇ 7.037.5(436) In Austria: Wiener Schule	7.041.8	Occupations. Trades. Professions
	◇ 7.037.5(450) In Italy: Pittura metafisica	◇ 7.041.8:34	Representations of the legal profession
7.037.7	Neorealism. Magical realism		
	◇ 7.037.7(430) In Germany: Neue Sachlichkeit	7.042	Animal life. Fauna
	◇ 7.037.7(73) In USA: Precisionism. The Immaculates	◇ 7.042:599.723.2	Representations of horses
		◇ 7.042:598.2	Representations of birds
7.038	Abstract art. Non-figurative art	7.043	Plant life. Flora. □ Trees. Flowers. Fruit
	<i>If a grouping by country is required, the place auxiliaries (Table 1e) may be intercalated after .038</i>	◇ 7.043:582.632.2	Representations of oak trees
	◇ 7.038(410.1).54 In England: Art and language	◇ 7.043:582.711.712	Representations of roses
	⇒ 7.036.7	⇒ 7.049.6	
7.038.1	Geometric abstraction	7.044	Historical representations. Reportage.
7.038.11	Constructivism	□ Scenes of royalty (coronations etc.)	
7.038.12	Rayonnism	State functions. Public ceremonies.	
7.038.14	Suprematism	Pageants etc. Military, martial subjects.	
7.038.15	Non-objectivity	War scenes. Battles	
7.038.16	Bauhaus. Functionalism	7.045	Symbols. Allegories. Emblems.
7.038.17	Art deco	Devices, heraldic designs. Monograms	
7.038.18	Concrete art	Dances of Death (Danses macabres)	
7.038.2	Expressionist abstraction	7.046	Epic, mythological, legendary and religious representations
7.038.21	Tachism. Nuclearism. Informal art	Mythology	
7.038.3	Kinetic art. Op art. Technological art. Light art		
7.038.4	New abstraction		
7.038.41	Nul. Zero. The New Style		
7.038.42	Minimal art. Hard edge		
7.038.5	Realism. New realism		
7.038.51	Pop art. New Realism, hyperrealism. Photographic realism		

7.046.2	Legends. Fantasies. Heroes	7.072.2	Study of art. □ Art historians, scholars, theorists
7.046.3	Religious subjects	7.072.3	Art criticism. □ Art writers, critics, reviewers ⇒ 82-95
	◇ 7.046.3:27-312.3 Nativity scenes	7.072.4	Art judges and judging. Art juries ⇒ 7.092
	◇ 7.046.3:27-312.8 Calvary scenes, Pieta ⇒ 2-276.63	7.072.5	Expert opinions, reports (technical etc.). Scientific experts
7.047	Nature representations. Scenery. Views. Panoramas. Landscapes. Seascapes	7.073	Appreciation. □ Arts lovers. Sports lovers. Enthusiasts, fans. Audiences. Spectators
7.048	Pattern and line ornament. Systems of ornamentation (coloured or not) ⇒ 7.038	7.074	Collecting and collectors ⇒ 371.387 ⇒ 379.824
7.048.1	Arabesques. Grotesques	7.075	Commercial management. □ Agents. Dealers. Managers. Producers. Impresarios. Promoters
7.048.3	Nonfigurative art. Nonrepresentational art	7.077	Amateur participation and participants (creative, performing etc.)
7.049	Other subjects	7.078	Patronage. Sponsorship. Encouragement ⇒ 35.078 ⇒ 351.854
7.049.1	Ideas. Emotions	7.079	Arts festivals <i>Class here general information about arts festivals. For festivals of particular art forms, add the auxiliary to the number for the art form</i> ◇ 791.65.079 Film festivals ◇ 82:7.079 Literature festival
	◇ 7.049.1:177.1 Representations of chivalry	7.08	CHARACTERISTIC FEATURES, FORMS, COMBINATIONS ETC. (IN ART, ENTERTAINMENT AND SPORT) <i>See specific uses at 78.08, 792.08, 794.08 and 796.08</i>
	◇ 7.049.1:177.6 Representations of friendship, love		
7.049.2	Satirical subjects. Caricatures. Grotesque figures <i>For grotesques in the strict sense use 7.048.1</i> ⇒ 741.5		
7.049.6	Still life (nature morte) ⇒ 7.043		
7.05	APPLICATIONS OF ART (IN INDUSTRY, TRADE, THE HOME, EVERYDAY LIFE). INDUSTRIAL, COMMERCIAL DESIGN ETC. <i>.05 is modified and further subdivided under various divisions of class 7. Specific applications of art and design may be brought together under 7.05 itself by :...</i> ◇ 7.05:62 Art and design in industry ◇ 7.05:64 Art and design in the house, home ⇒ 658.512.2 ⇒ 745/749		
7.06	VARIOUS QUESTIONS CONCERNING ART	7.091	Performance, presentation (in original medium)
7.061	Forgery. Counterfeiting. Fakes. Plagiarism ⇒ 343.533 ⇒ 347.788.4 ⇒ 7.026	7.091.1	Auditions. Hearing, testing of candidates or applicants
7.063	Lost art. Missing art	7.091.2	Rehearsals. □ Reading rehearsals. Movement rehearsals. Dress rehearsals
7.07	OCCUPATIONS AND ACTIVITIES ASSOCIATED WITH THE ARTS AND ENTERTAINMENT	7.091.3	Studio performances
7.071	Creative and interpretative occupations. Professional artists, performers, players ⇒ 7.077	7.091.4	Festival performances
7.071.1	Creative artists. Professional artists (e.g. painters, sculptors, composers)	7.091.5	Introduction of new works. First performances. First nights. Premières
7.071.2	Performing, interpretative artists. Performers, players. □ Directors. Conductors. Instrumentalists. Singers. Dancers. Actors	7.091.6	Repertory performances. Repeat performances
7.071.3	Copyists. Restorers. Arrangers, adapters, transcribers. Translators	7.091.8	Touring exhibitions, performances
7.071.4	Teachers. Instructors. Lecturers	7.092	Competitions. Contests. Records. Championships
7.071.5	Training and education (including visits, excursions, study tours) ⇒ 371.233	7.093	Prediction of outcome. □ Gambling, betting (recreational aspects) ◇ 794.4.093 Gambling at card games ◇ 798.4.093 Gambling on horse races ⇒ 174.6
7.072	Expertise	7.094	Screen, film, motion picture presentation (e.g. of ballets, circuses, operas, theatre)
		7.096	Sound radio presentation (e.g. of concerts, films, drama, opera)
		7.097	Television presentation (e.g. of films, ballet, theatre)

71 PHYSICAL PLANNING. REGIONAL, TOWN AND COUNTRY PLANNING. LANDSCAPES, PARKS, GARDENS

⇒ 911.37

711	PRINCIPLES AND PRACTICE OF PHYSICAL PLANNING. REGIONAL, TOWN AND COUNTRY PLANNING	711.42	Kinds of town, locality, settlement ⇒ 911.375.3
711-1	General principles and practice 711-1 ≡ 711.1 ◇ 711.453.4-167 Extension planning of harbour towns	711.423	Settlements of primitive type ⇒ 903.3 ⇒ 903.4
711.1	General principles and practice. Land development. Plans	711.426	Towns, localities, settlements according to building materials <i>Details by :693/694</i> ◇ 711.426:694.1 Constructed in wood, timber
711.11	Preliminary phases	711.427	Imaginary towns, localities, settlements. Ideal cities
711.12	Kinds of development	711.43	Localities according to size and status
711.121	Unplanned development	711.432	Capital cities. Seats of government.
711.122	Planned development (including plans)		Metropolises ⇒ 711.451
711.13	Population: movement, distribution, densities <i>Class here population as a factor in physical planning</i> <i>Details by :314...</i>	711.433	Cities. Major towns (population over 100 000)
		711.434	Large towns
		711.435	Small towns
		711.436	Civil parishes
		711.437	Villages
		711.438	Hamlets
711.14	Land use ⇒ 332.3	711.45	Towns, localities, settlements according to function, chief industry etc. Centres of various industries ⇒ 911.372.4
711.142	Land classification. Evaluation of soil quality etc.		Administrative centres. Regional capitals (county towns etc.) ⇒ 711.432
711.143	Land management. Economic use of land		Trading, commercial towns. Business centres
711.144	Acquisition of land ⇒ 332.22	711.451	Traffic, transport, communications centres
711.16	Planning programme and progress	711.452	Centres of land transport. □ Railway centres, junctions
711.163	Replanning. Improvement	711.453	Ports. Harbour (dock) towns
711.164	Clearance. Demolition. Thinning out ⇒ 69.059.62 ⇒ 7.025.1	711.453.1	Airport towns
711.167	Extension. Enlargement	711.453.4	Industrial towns. Producing, manufacturing centres ⇒ 628.521 ⇒ 658.21
711.168	Rebuilding. Reconstruction. Redevelopment. □ Rebuilding war-damaged, fire-damaged, flood-damaged areas	711.453.9	Health, holiday, pleasure resorts ◇ 711.455(210.5) Coastal, seaside resorts ◇ 711.455"324" Winter resorts
		711.454	Holiday camps
711.2	Regional planning		Fortified towns and localities. Defence bases
711.21	Planning on a supranational scale	711.455	Religious centres. Cathedral towns
711.22	Planning on a federal, union scale		Centres of learning. University towns
711.24	Planning on a national, state scale	711.455.8	Residential towns. Dormitory towns
711.25	Planning on a provincial, county scale	711.459.6	Units, areas within the town or settlement. □ Zones. Districts. Quarters. Localities
711.27	Planning on a district, local scale	711.46	Zoning in general (especially legal aspects and ordinances)
711.28	Planning for townships, parishes	711.47	Concentration and differentiation. Kinds of area ◇ 711.52:902 Areas of archaeological interest
711.3	Country planning. Rural development <i>Denote the kind of rural land by (25...)</i> ⇒ 712.2	711.48	Civic areas, wards
711.4	Town planning. Urban development. Built-up areas. Conurbations. Settlements	711.5	Central areas. Town centres, nuclei
711.41	Forms, shapes of built-up areas or urban units	711.51	Units according to function ⇒ 911.375.4
711.412	Radiating. Concentric	711.52	Civic (administrative) areas
711.413	Gridiron plan. Grid development	711.522	Trading, commercial, business, distributing areas
711.416	Linear development. Ribbon development ◇ 711.416-121 Unplanned ribbon development	711.523	Shopping centres, areas ⇒ 725.1
711.417	Garden cities. New towns. Satellite towns	711.55	Offices. Business centres
711.417.2	Garden cities ⇒ 711.582.5	711.551	
711.417.4	New towns	711.552	
711.417.5	Satellite towns	711.552.1	
		711.552.3	

711.553	Traffic, transport, communications areas	711.7	Traffic routes (layout, planning)
711.553.1	Land transport areas	711.73	Highways. Roads, streets
711.553.12	Railway areas. □ Depots. Junctions		⇒ 625.7/8
711.553.17	Road transport areas. □ Bus depots	711.75	Railways
711.553.2	Parking spaces. Special parking areas		⇒ 625.1
	⇒ 72.051.6		
	⇒ 725.1	711.76	Canals. Waterways
711.553.4	Port, harbour districts. Quay, dockside areas		⇒ 626/627
711.553.9	Airport areas. □ Airfields. Runways. Landing strips	711.77	Airways
	⇒ 625.717		⇒ 625.717
	⇒ 629.7.08	711.8	Public utilities. □ Distribution lines. Supply lines
	⇒ 725.1		◇ 711.8(24) Subsurface utilities
711.554	Location of industry. Industrial districts	712	PLANNING OF LANDSCAPE (NATURAL AND DESIGNED). PARKS. GARDENS
711.555	Districts containing health services (hospitals, asylums etc.)		⇒ 630*27
711.556	Districts for segregation, isolation		⇒ 635
	◇ 711.556:725.1:343.811 Prison, penitentiary, reformatory areas	712.2	Planning of the landscape pattern in general
	◇ 711.556:725.1:614.21 Isolation hospital areas		◇ 712.2(1-751.2) Planning of national parks
711.557	Hotel, boarding districts		⇒ 625.712.5
711.558	Areas for pleasure, entertainment, amusement, recreation generally. □ Theatre districts (theatreland). Fairgrounds	712.24	Agricultural land (as part of landscape pattern). □ Hedged land. Strip cultivation. Terraced development
711.558.5	Lidos		
711.559.6	Military, defence localities	712.25	Open spaces. Unbuilt areas (in or near towns or villages)
	⇒ 623.1		⇒ 625.712.4
711.56	Localities with religious purpose. Precincts centred on church or cathedral		⇒ 711.61
711.57	Intellectual, educational localities. Centres of learning. Localities centred on libraries, museums, galleries, schools, colleges, universities. Exhibition, exposition districts	712.252	Commons and greens
	⇒ 371.61	712.253	Planning of parks, public gardens (with or without play and sports facilities)
711.58	Community, neighbourhood units.		◇ 712.253:58:069.029 Planning of botanic gardens
	Residential areas		◇ 712.253:59:069.029 Planning of zoological gardens
711.581	Neighbourhood units		⇒ 727
711.582	Residential units. □ Housing estates (housing schemes)	712.254	Squares. □ Open squares. Garden squares
711.582.5	Garden suburbs (residential)		<i>For enclosed, urban squares see 711.61</i>
	⇒ 711.417.2	712.256	Playgrounds. Recreation grounds
711.582.7	Temporary settlements. □ Camps. Settlements of caravans, houseboats		⇒ 791.7
711.585	Undesirable, unhealthy areas. □ Criminal districts. Slums	712.257	Playing fields. Sports grounds
	◇ 711.585-164 Slum clearance		<i>Denote particular sports by :796...</i>
711.586	Uninhabitable, condemned areas	712.26	Private grounds, parks. Enclosed gardens. Estates
711.6	Arrangements of buildings in towns. Site planning. Grouping	712.27	Market gardens. Nursery gardens. Orchards. Fruit and vegetable gardens. Kitchen gardens
711.61	Squares. Other open spaces		⇒ 635.1/8
	⇒ 625.712.4	712.28	Smallholdings. Allotments (leisure gardens)
	⇒ 712.25	712.3	Landscape gardening in general (planning, layout, design, execution)
711.62	Arrangement of buildings	712.31	Minor modelling of ground surface.
711.621	Terraces, terrace grouping (with party walls)		□ Banks. Terraces. Placing of rock
	⇒ 728.31		<i>Details of earthworks by :624.13...</i>
711.622	Semidetached	712.35	Boundaries. □ Walls. Fences. Hedges.
	⇒ 728.34		Ditches. Ha-has
711.623	Detached		⇒ 692.212
	⇒ 728.37		⇒ 692.88
711.63	Apportionment. Site division. □ Blocks. Parcels	712.36	Approaches. □ Roads. Drives. Paths. Walks. Avenues
711.64	Alignment. Frontages. Building line. Spacing. Density		⇒ 625.7/8
711.65	Height of buildings. Skylines	712.4	Planting design. Vegetation, plant forms (arrangement, planting)
711.68	Small, incidental structures. Town furnishings. Street furniture		
	⇒ 625.748	712.41	Trees. Bushes. Shrubs
	⇒ 725.94		

- 712.42** Herbaceous beds and borders. Flower beds as part of the enclosed garden. Carpet bedding, ground cover. Turf, lawns
 ◇ 712.42:635.923 Rockeries. Rock, alpine gardens
- 712.5** Water features
 ◇ 712.5:725.948 Fountains
 ◇ 712.5(282.3) Waterfalls. Cascades
 ◇ 712.5(285.3) Ponds
- 712.6** Structural features in general
Denote particular features by :624..., :69..., :72...
- 712.7** Garden ornaments
 ◇ 712.7:681.111.1 Sundials as garden ornaments
- 718** CEMETERIES. GRAVEYARDS. CREMATORIALS. OTHER PLACES FOR DISPOSAL OF THE DEAD (PLANNING, DESIGN, UPKEEP ETC.)
- 719** PRESERVATION OF RURAL AND URBAN AMENITIES GENERALLY
 ◇ 719:061.2 Organizations for amenity protection
 ⇒ 502

72 ARCHITECTURE

⇒ 624/628

⇒ 69

- 72.01** Theory and philosophy of architecture. Principles of design, proportion, optical effect
Subdivisions as at 7.01 with additions listed below
 ◇ 72.01:111.852 Aesthetics of architecture. Architectural taste
- 72.012** Design. Composition. Disposition, relationships, dimensional requirements of buildings
 ⇒ 658.233
- 72.012.1** Project. □ Plan and accommodation requirements. Choice of site. Conditions imposed
- 72.012.18** According to manner of provision (affecting cost)
- 72.012.183** Special requirements
 ◇ 72.012.183-025.14 Buildings for mixed use
 ◇ 72.012.183-027.37 Mass-produced
 ◇ 72.012.183-027.39 Custom-built
- 72.012.4** Relationships of parts to whole
- 72.012.42** Main part (in relation to the whole)
- 72.012.43** Wings (in relation to the whole)
- 72.012.44** Outbuildings (in relation to the whole)
- 72.012.5** Relationship of exterior to interior. Interaxial divisions at points of support. Openings. Fenestration. Bay divisions
- 72.012.7** Semi-enclosed spaces. Open-air parts
- 72.012.72** Open above. Unroofed. □ Courtyards. Atria. Patios. Areas. Lightwells
- 72.012.75** Open at side. Covered spaces and ways. □ Verandahs. Loggias. Terrace walks. Arcades. Cloisters. Open porches. Patios
 ⇒ 625.712.32
- 72.012.8** Interiors. Enclosed spaces and parts. □ Halls. Galleries. Corridors
- 72.012.9** Space for equipment, furniture, storage units
- 72.02** Technique. □ Working methods. Drawings, models, accessories. Materials available. Preservation, repair. Conversion
Subdivisions as at 7.02
- 72.03** Periods and phases of architecture. Schools, styles, influences
Subdivisions as at 7.03...
- 72.04** Architectural details and finishes. Decoration. Ornament
Subdivisions as at 7.04
- 72.05** Parts, spaces and rooms planned for specific uses
Use subdivisions at 643.3/.5 colonized to 72.05 wherever possible
 ◇ 72.05:643.523 Washing facilities
 ◇ 72.05:643.524 Lavatories
 ◇ 72.05:643.527 Changing rooms
 ◇ 72.05:643.528 Cloakrooms
- 72.051** General space and access considerations. External communication and approach
- 72.051.6** Approaches to buildings. □ Space for setting down, loading, unloading, parking, garages
 ⇒ 711.553.2
- 72.051.8** Enclosure. Special measures for security and protection
 ⇒ 725.96
- 72.052** Entrance and circulation space. Internal access and communication
- 72.052.2** Entrances. Exits
- 72.052.4** Concourses. □ Entrance halls. Forecourts. Vestibules. Lobbies. Foyers. Promenades
- 72.052.5** Waiting space. Waiting rooms
- 72.052.7** Circulation areas
 ◇ 72.052.7-025.71 Vertical circulation. Including: Stairways. Escalators. Liftwells
 ◇ 72.052.7-025.72 Horizontal circulation. Including: Corridors. Passages. Ambulatories. Gangways
- 72.053** Assembly space generally. □ Halls, auditoriums (as parts of buildings)
- 72.054** Business, administrative, instructional spaces. Office space. Storage space
 ◇ 725.1.054 Council chambers
 ◇ 727.054:373 School classrooms
- 72.055** Layout of spaces for specialized technical activity. Rooms according to type of fixed equipment, fittings installed
Specify the activity by :...
 ◇ 725.1.055:614.21:617-089 Hospital operating theatres
 ◇ 725.1.055:621.9 Machine shop space and layout in factories
- 72.056** Parts, spaces specific to particular kinds of building. □ Spaces for isolation, seclusion, privacy
 ◇ 725.1.056:343.811 Prison cells

- 72.059** **Ancillary spaces, annexes etc.**
 □ Laundry-handling space. Fuel stores (e.g. coal bunkers). Spaces and rooms for animals
 ⇒ 631.2
- 721** **BUILDINGS GENERALLY.**
 □ Programming. Design. Parts for specific uses. Rooms
- 725** **PUBLIC, CIVIL, COMMERCIAL, INDUSTRIAL BUILDINGS. SECULAR ARCHITECTURE GENERALLY**
 ⇒ 658.2
- 725.1** **Civic buildings. Legislative buildings. Public service buildings. Public utility buildings**
- ◇ 725.1:005.912 Office buildings
 - ◇ 725.1:328.1 Parliament buildings
 - ◇ 725.1:336.24 Customs houses. Customs posts
 - ◇ 725.1:336.71 Banks
 - ◇ 725.1:336.761 Stock exchanges. Chambers of commerce
 - ◇ 725.1:339.175-026.912 Structures for open-air markets
 - ◇ 725.1:339.176.5 Showrooms (as buildings)
 - ◇ 725.1:339.3 Commercial, business buildings
 - ◇ 725.1:339.37 Shops. Stores
 - ◇ 725.1:339.337 Wholesale stores. 'Cash and carry' stores
 - ◇ 725.1:339.372 Retail shops
 - ◇ 725.1:339.372.2 Department stores
 - ◇ 725.1:339.372.5 Self-service shops. Supermarkets
 - ◇ 725.1:339.372.7:646.72 Hairdressing salons
 - ◇ 725.1:339.378.2 Shopping centres. Shopping malls, arcades
 - ◇ 725.1:34 Judicative buildings. Law courts
 - ◇ 725.1:341.71 Embassies
 - ◇ 725.1:341.78 Legations
 - ◇ 725.1:341.81 Consulates
 - ◇ 725.1:342.3-057.341 Official residences of heads of state. Royal and presidential palaces
 - ◇ 725.1:342.3-057.343 Official residences of ministers of state
 - ◇ 725.1:343.811 Prisons. Jails (gaols). Penitentiaries. Reformatories
 - ◇ 725.1:343.815 Correctional establishments for juveniles. Juvenile reformatories
 - ◇ 725.1:343.819.1 Open prisons
 - ◇ 725.1:351.74 Police stations
 - ◇ 725.1:351.814 Air traffic control structures
 - ◇ 725.1:352 Provincial and municipal administrative offices. Including: Town halls. Civic centres
 - ◇ 725.1:354 Government office buildings
 - ◇ 725.1:355 Military buildings
 - ◇ 725.1:364-5 Health and welfare buildings
 - ◇ 725.1:364-54 Homes for chronic invalids, incurables
 - ◇ 725.1:364-54-053.2 Children's homes
 - ◇ 725.1:364.694-57 Disablement rehabilitation centres
 - ◇ 725.1:613.41-027.555 Public bath buildings (except swimming baths)
 - ◇ 725.1:613.47 Swimming baths
 - ◇ 725.1:613.47-026.911 Indoor (covered) swimming baths
 - ◇ 725.1:613.47-026.912 Outdoor (open-air) swimming baths
 - ◇ 725.1:614.21 Hospitals. Health centres. Clinics. Nursing homes. Infirmaries. Surgeries. Dispensaries. Sanatoriums
 - ◇ 725.1:614.64 Mortuaries
 - ◇ 725.1:614.842.84 Fire stations
 - ◇ 725.1:615.838 Spa buildings. Kursaals
 - ◇ 725.1:62 Industrial buildings generally. Factories. Mills. Works. Yards
 - ◇ 725.1:621.796 Storage buildings. Warehouses. Depots
 - ◇ 725.1:621.796-026.911 Covered warehouses, depots
 - ◇ 725.1:621.796-026.912 Open-air depots
 - ◇ 725.1:623.1 Forts. Fortresses
 - ◇ 725.1:625.712.61 Storage buildings. Depositories. Depots
 - ◇ 725.1:625.748 Toll houses. Toll gates
 - ◇ 725.1:627.31 Dock buildings. Dock structures
 - ◇ 725.1:627.32 Marine, coastal, waterside buildings and structures. Port, harbour structures
 - ◇ 725.1:627.33 Mooring and berthing structures. Including: Wharfs. Quays. Piers. Jetties. Landing stages
 - ◇ 725.1:627.715 Lighthouses
 - ◇ 725.1:628.12 Waterworks
 - ◇ 725.1:628.425 Public lavatories
 - ◇ 725.1:629.7.081 Hangars. Aircraft repair sheds
 - ◇ 725.1:629.7.087.27 Mooring and supporting structures for airships, missiles
 - ◇ 725.1:631.24 Silos. Grain elevators
 - ◇ 725.1:640.43 Restaurants. Cafes
 - ◇ 725.1:640.435 Quick-service bars. Including: Snackbars. Buffets. Cafeterias. Coffee bars
 - ◇ 725.1:640.443 Licensed premises. Including: Public houses. Taverns. Brasseries
 - ◇ 725.1:640.46 Casinos
 - ◇ 725.1:640.5 Buildings for communal eating. Including: Dining halls. Refectories. Canteen buildings
 - ◇ 725.1:654.141 Telegraph offices
 - ◇ 725.1:654.191 Television studios
 - ◇ 725.1:656 Buildings for transport, traffic
 - ◇ 725.1:656.015 Garages. Buildings and facilities for parking vehicles (generally)
 - ◇ 725.1:656.062 Garages. Coach, bus stations. Including: Service stations, depots. Repair shops
 - ◇ 725.1:656.065.3 Filling stations
 - ◇ 725.1:656.065.7 Vehicle wash facilities
 - ◇ 725.1:656.132 Bus garages. Tram depots
 - ◇ 725.1:656.132.013 Coach and bus stations, terminals
 - ◇ 725.1:656.21 Railway stations (generally)
 - ◇ 725.1:656.21:625.42 Underground stations
 - ◇ 725.1:656.212 Freight, goods stations
 - ◇ 725.1:656.222.5 Passenger stations
 - ◇ 725.1:656.257 Other railway buildings. Including: Signal boxes. Railway repair shops, sheds
 - ◇ 725.1:656.66 Ferry-houses
 - ◇ 725.1:656.71 Airport buildings and structures
 - ◇ 725.1:656.71.013 Air passenger and freight buildings
 - ◇ 725.1:656.815.3 Post offices
 - ◇ 725.1:656.851 Sorting offices
 - ◇ 725.1:66.013 Chemical industry buildings
 - ◇ 725.1:663.4 Breweries
 - ◇ 725.1:671.4 Mints
 - ◇ 725.1:688.743 Buildings for panoramas, dioramas

- ◇ 725.1:692 Factories for the construction of building components
 ◇ 725.1:711.455 Health and holiday resort buildings. Including: Seaside, pier buildings
 ◇ 725.1:711.553.9 Landing-strip, runway structures
 ◇ 725.1:711.558.5 Lido buildings and structures
 ◇ 725.1:78/79 Public entertainment and recreation buildings
 ◇ 725.1:78.096 Concert buildings. Concert halls
 ◇ 725.1:782.096 Opera houses
 ◇ 725.1:791.096 Cinemas. Film theatres
 ◇ 725.1:792.096 Theatres. Playhouses
 ◇ 725.1:792.096-026.912 Open-air theatres
 ◇ 725.1:792.7.096 Cabarets. Dance halls. Music halls, etc.
 ◇ 725.1:792.73.096 Cabarets. Night clubs
 ◇ 725.1:793 Dance halls (palais de dance)
- 725.2 Ancillary buildings, structures etc.**
 ◇ 725.2:624 Bridges. Viaducts. Flyovers. Tunnels. Underpasses. Other structures associated with traffic
- 725.21 Pavilions**
 ◇ 725.21:069.9 Exhibition pavilions
 ◇ 725.21:78 Bandstands. Music pavilions
 ◇ 725.21:796.358 Cricket pavilions
- 725.22 Follies**
- 725.23 Winter gardens. Palm houses, etc.**
 725.826 Stadiums, amphitheatres, hippodromes and similar buildings
 ◇ 725.826:791.862 Bull rings
 ◇ 725.826:796.42 Athletics stadiums
- 725.83 Assembly buildings**
 725.832 Congress, conference, lecture buildings or halls
 725.835 Community centres. Village halls
- 725.84 Buildings for indoor games and amusements**
- 725.85 Buildings, halls for physical recreation and training.** □ Gymnasiums. Covered courts
- 725.86 Rinks.** □ Ice rinks. Roller-skating rinks
- 725.87 Water sport buildings. Water sport structures.** □ Yachthouses. Boathouses. Ramps. Slips
- 725.88 Riding school buildings and structures**
- 725.89 Ancillary buildings and structures for sport.**
 □ Pavilions. Shelters
 725.891 Ancillary buildings and structures for shooting and archery ranges. □ Shooting butts
 725.892 Ancillary buildings and structures for wheel racing and animal racing. Race track structures
 725.893 Ancillary buildings and structures for outdoor games (ball games) and athletics. Sports pavilions
 ◇ 725.893:796.358 Cricket pavilions
- 725.91 Exhibition buildings, pavilions, halls etc.**
- 725.94 Monuments. Memorials. Street furniture. Various freestanding structures (as architectural features)**
 ⇒ 625.748
 ⇒ 628.971
 ⇒ 711.68
 ⇒ 726.825
- 725.941**
- 725.942**
- 725.945**
- 725.948**
- 725.95**
- 725.96**
- 725.961**
- 725.962**
- 725.963**
- 726**
- Triumphal arches and arcades
 ⇒ 625.712.31
 Triumphal columns. Obelisks. Stelae
 Memorials. □ War memorials. Cenotaphs.
 Commemorative statues, plaques etc.
 ⇒ 726.8
 Fountains. Drinking fonts, troughs, pumps etc.
 ⇒ 621.643.53
- Bridges. Viaducts. Flyovers. Tunnels. Underpasses. Other structures associated with traffic**
 ⇒ 624.2/.8
 ⇒ 625.712.35/.37
- City, town gateways. City walls. Moats etc.**
 Gateways. Monumental gates
 ⇒ 625.712.31
 Monumental screen walls. Balustrades.
 Railings
 City walls. Ramparts
- RELIGIOUS, ECCLESIASTICAL ARCHITECTURE. SACRED AND FUNERARY BUILDINGS**
 ◇ 726:2-523.4 Temples. Pagodas. Sanctuaries. Shrines
 ◇ 726:2-523.6 Religious houses, monasteries, etc. NOTE: Denote religion by 21/29
 ◇ 726:2-586.5 Hermitages
 ◇ 726:26-523.4 Synagogues. Jewish temples
 ◇ 726:27-523.2 Parts of Christian churches. Including: Nave. Aisles. Crossings. Transepts
 ◇ 726:27-523.4 Christian church architecture (generally). NOTE: Denote denominations by 271/279
 ◇ 726:27-523.41 Cathedrals
 ◇ 726:27-523.42 Churches
 ◇ 726:27-523.43 Separate, free-standing chapels
 ◇ 726:27-523.46 Side-rooms. Including: Side-chapels. Chancies. Saints' chapels
 ◇ 726:27-523.47 Screens
 ◇ 726:27-523.48 Sacristies. Vestries. Crypts
 ◇ 726:27-523.5 Cloisters. Chapterhouses. Baptistries
 ◇ 726:27-523.6 Monastic buildings. Religious houses. Including: Monasteries, abbeys, convents
 ◇ 726:27-526.1 Pulpits
 ◇ 726:27-526.2 Altars
 ◇ 726:28-523.4 Mosques
- 726.8 Funerary architecture. Sepulchral monuments, memorials etc.**
 ⇒ 718
 ⇒ 903.5
- 726.81 Funerary, cemetery, mortuary chapels**
- 726.82 Sepulchral monuments**
 726.821 Tombs. Sepulchres. Crypts. Vaults
 726.822 Mausoleums
 726.823 Ossuaries. Charnel houses. Towers of silence
 726.825 Memorial features. □ Gravestones, crosses, slabs
 ⇒ 725.94
 Rock tombs
 Sarcophagi

726.83	Crematoriums ⇒ 718	728.2	Multi-family dwellings
726.832	Columbaria	728.22	Blocks of flats (tenements, service flats etc.). Apartment blocks
726.84	Mass burial features. □ Grave mounds, caves etc. Catacombs	728.222	Large apartment blocks. □ Blocks of low-rent flats
726.85	Pyramids and similar monuments	728.224	Smaller blocks (with less than ten flats)
726.9	Other religious buildings and structures	728.226	Service flats. Luxury flats
726.95	Pilgrimage shrines. Isolated crosses	728.24	Maisonettes (two-storey apartments in blocks or houses)
727	BUILDINGS FOR EDUCATIONAL, SCIENTIFIC, CULTURAL PURPOSES	728.27	Boarding houses. Apartment houses (with permanent or semipermanent residents)
	◇ 727:001.32 Buildings of scientific and learned societies	728.3	Single-family dwellings. Urban houses
	◇ 727:02 Documentation, library and archive buildings	728.31	Terraced houses (town houses)
	◇ 727:069 Museums	728.34	Semidetached houses
	◇ 727:069:7 Art galleries	728.37	Detached houses (fully detached)
	◇ 727:373 Schools	728.38	Bungalows. Semibungalows
	◇ 727:373.018.32 Boarding schools	728.4	Residential buildings of firms and other organizations. Halls of residence. Club buildings. Residential hostels
	◇ 727:373.23 Nursery schools	728.5	Buildings for tourist accommodation. Buildings for guest accommodation ⇒ 640.41
	◇ 727:373.3 Primary schools	728.6	Rural domestic architecture. Smaller rural dwellings
	◇ 727:373.5 Secondary schools		
	◇ 727:376 Special schools	728.61	Cottages and similar small dwellings
	◇ 727:378.4 Universities	728.67	Farmhouses
	◇ 727:378.6 Vocational and specialist training schools, colleges. Technical and professional academies	728.7	Occasional and mobile dwellings
	◇ 727:520.1 Astronomical observatories	728.76	Wheeled dwellings. Mobile homes. Caravans ⇒ 629.334
	◇ 727:520.98 Planetariums	728.77	Floating dwellings (barge, pontoon etc.). Houseboats ⇒ 629.514
	◇ 727:550.34.034 Seismological observatories. Seismometric stations	728.8	Isolated large domestic buildings
	◇ 727:550.8.024 Geophysical observatories or field stations	728.81	Castles, fortified or semifortified ⇒ 623.1
	◇ 727:551.501.9 Meteorological stations, observatories	728.82	Châteaux, unfortified castles. Country seats, mansions
	◇ 727:58:069.029 Botanic garden buildings. Including: Plant houses. Herbariums	728.83	Manor houses
	◇ 727:59:069.029 Zoological garden buildings. Including: Animal houses	728.84	Country houses, villas
	◇ 727:59:069.029:639.33 Aquariums	728.9	Outbuildings. Sheds. Other domestic dependencies
	◇ 727:621.396.7-022.56 Broadcasting stations	728.94	Outbuildings for keeping and storing tools, produce, vehicles. □ Toolsheds. Domestic garages
	◇ 727:7.022.82 Studios for artistic activities	728.96	Outbuildings for animals. □ Stables. Pigeonhouses. Kennels ⇒ 631.22
	◇ 727:791.022.8 Film studios	728.98	Glasshouses. Greenhouses. Hothouses. Conservatories. Verandahs ⇒ 631.23
728	DOMESTIC ARCHITECTURE. HOUSING. RESIDENTIAL BUILDINGS	728.991	Summerhouses. Gazebos. Garden shelters
728.1	Housing and dwellings generally (architectural aspects)		
	◇ 728.1(1-21) Urban housing. Dwellings in towns		
	◇ 728.1(1-22) Rural housing. Dwellings in the country ⇒ 332.8 ⇒ 347.238 ⇒ 347.254 ⇒ 351.778.5 ⇒ 643		
728.18	Temporary housing and dwellings. □ Huts. Shacks. Shanties. Log cabins		

73 PLASTIC ARTS

73.02	Technique of plastic arts. Craftsmanship <i>Subdivisions as at 7.02</i>	73.03	Periods and phases of plastic arts. Schools, styles, influences <i>Subdivisions as at 7.03...</i>
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73.04	Subjects for plastic or sculptural representation. Iconography. Iconology <i>Subdivisions as at 7.04...</i>	738.3	Stoneware
730	SCULPTURE IN GENERAL. STATUARY. □ Statues. Effigies ⇒ 2-526.64	738.4	Enamel (vitreous enamel). Cloisonné. Enamel ornaments
736	GLYPTICS. SIGILLOGRAPHY	738.5	Mosaics
736.2	Glyptography. Glyptics. Lapidary work. Engraving of gems and semiprecious stones. Cameos. Inscriptions ⇒ 739.2	738.6	Terracotta
736.3	Sigillography. Sphragistics. □ Seals. Signets. Stamps. Dies ⇒ 686.867.6 ⇒ 929.6	738.8	Various ceramic objects
737	NUMISMATICS ⇒ 671.4	738.81	Decorative tiles
737.1	Coins <i>Denote material by -03...</i> ⇒ 336.74	738.82	Wall tablets. Flagstones
737.14	Coins according to finish. □ Milled. Perforated	738.83	Ceramic lamps
737.2	Medals. Medallions	738.84	Ceramic statues, figurines, statuettes
737.22	Medals of honour. Decorations	739	METAL ARTS. □ Chased, cast, beaten, embossed, repoussé, wrought metal objects ⇒ 669.21/.23 ⇒ 682 ⇒ 671
737.23	Commemorative medals	739.1	Goldsmith's and silversmith's art
737.24	Competition, contest medals	739.2	Jewellery (metalworking aspects) ⇒ 736.2
737.27	Medals as charms. Talismans. Amulets	739.3	Artistic horology ⇒ 681.11
737.3	Tokens. Jettons. Counters	739.4	Artistic ironwork. Artistic locksmith's work. Decoratively worked ferrous metal objects: forged, beaten or cast ⇒ 672.13 ⇒ 682.5 ⇒ 683.3
738	CERAMIC ARTS. POTTERY ⇒ 666.3/.7	739.5	Artistic work in various other metals (bronze, copper, tin etc.)
738.1	Porcelain	739.7	Artistic armoury, gunsmithing, weaponsmithing. □ Chasing, engraving of weapons and armour ⇒ 623.44 ⇒ 681.94
738.2	Glazed earthenware	739.8	Other artistic metal objects. Curios. Bric-a-brac
738.22	Faience		
738.23	Majolica		

74 DRAWING. DESIGN. APPLIED ARTS AND CRAFTS

741/744	DRAWING	742.5	Shadow delineation. Perspective projection of shadows. Sciagraphy ⇒ 514.182.3
741	DRAWING IN GENERAL	743	ANATOMICAL DRAWING ⇒ 611 ⇒ 7.041
741.02	Drawing technique. Working methods. Drawing equipment. Drawing materials <i>Subdivisions as at 7.02...</i> ⇒ 686.86 ⇒ 686.88	744	LINEAR AND GEOMETRIC DRAWING. TECHNICAL DRAWING
741.5	Caricature. Cartoons. Satirical and humorous drawings ◇ 741.5:32 Political cartoons	744.1	Drawing office practice, procedure, organization
741.7	Silhouettes. Scissor-cuts	744.2	Drawing office layout. □ Division. Arrangement of drawing boards, tables. Lighting ⇒ 628.977.2
741.9	Collections of drawings. Sketchbooks	744.3	Drawing equipment
742	PERSPECTIVE IN DRAWING	744.32	Furniture. □ Drawing boards. Drawing tables. Frames. Stands. Drafting machines
742.1	Perspective in general. □ Plane of delineation. Reference points	744.34	Drawing instruments. □ Rules, straightedges. Set squares. T-squares. Compasses. Dividers ⇒ 667.649.6 ⇒ 686.8
742.3	Axonometric and isometric perspective		
742.4	Aerial perspective. Bird's-eye view		

744.36	Drawing materials ⇒ 667.4 ⇒ 676.237	746	FANCY WORK. ART NEEDLEWORK
744.4	Drawing methods, rules, conventions	746.1	Weaving ⇒ 677.024
744.42	Kinds of drawing. □ Technical drawings and diagrams. Charts. Graphs	746.2	Lacemaking. Lace. Tatting. Filet. Tulle
744.43	Execution, technique of drawing. □ Thickness of stroke or line. Use of different strokes. Use of colour. Dimensioning. Lettering. Titling	746.21	Lacemaking ⇒ 677.026.52 ⇒ 677.076.52
744.44	Diagrams, directions, instructions for assembling, mounting, operating	746.22	Point lace. Tenerife lace. Tatting
744.5	Reproduction, duplicating of drawings. □ Making blueprints etc. ⇒ 772.21 ⇒ 778.1	746.25	Braided, plaited, knotted work. Fringes (macramé), tassels, braids, pompons ⇒ 677.026.73 ⇒ 677.076.73
744.8	Storage, preservation and protection of drawings etc.	746.26	Filet. Netting. Mesh. Filet lace. Embroidered net
744.9	Ornamental, heraldic and display lettering. Signwriting (e.g. for posters, shop signs) ⇒ 667.649.6 ⇒ 766	746.27	Egyptian network
745/749	INDUSTRIAL AND DOMESTIC ARTS AND CRAFTS. APPLIED ARTS <i>Class under 745/749 information about the design, decoration and ornamentation of all kinds of object</i> <i>Denote particular objects by :...</i> ◇ 745.51:645.45 Cupboards, wardrobes, shelves as joiner's art ⇒ 684.45	746.28	Tulle
745	DECORATIVE HANDICRAFTS ⇒ 688	746.3	Embroidery ⇒ 677.026.62 ⇒ 677.076.62
745.5	Handicrafts according to material	746.31	Counted-thread methods
745.51	Artistic woodwork. □ Artistic joinery. Marquetry ⇒ 674 ⇒ 684.6 ⇒ 749.1	746.32	Drawn-thread work
745.52	Artistic textile work. □ Decorative fabrics. Rugs, tapestries etc. Printed and painted fabrics. Batik. Passementerie, decorative edging. Gimping, galloon etc. ⇒ 684.75	746.33	Whitework methods
745.53	Artistic leather work ⇒ 675	746.34	Coloured embroidery
745.54	Artistic paper work. □ Papercraft. Origami. Decorative papers (e.g. wallpaper). Papier maché work ⇒ 676	746.35	Embroidered pleating (smocking)
745.55	Artistic objects of ivory, horn, shell etc.	746.36	Quilting. Marseilles embroidery
745.56	Artistic objects of salt and flour. □ Figures made of dough	746.4	Tailoring. Dressmaking. Sewing. Knitting. Crochet ⇒ 687
745.9	Floral arts. Flower arrangement and compositions ⇒ 635.91	746.5	Beadwork (pearls etc.)
745.92	Flower arrangement for vase or bowl display	746.6	Coarse fibre work. □ Work with bast, raffia
745.93	Flower arrangement and floral decoration without containers. □ Bouquets. Garlands. Wreaths	746.7	Wickerwork. Basketwork. Cane weaving
745.94	Dressing and flat display of flowers, dried seeds etc. (e.g. in pictures and albums)	747	INTERIOR DECORATION ◇ 747:643 Decoration of particular rooms ◇ 747:643.53 Decoration of living and reception rooms ◇ 747:725/728 Interior decoration of particular kinds of building ◇ 747:725.1:640.43/.44 Interior decoration of restaurants, cafes, canteens etc. ⇒ 159.937.51 ⇒ 535.6 ⇒ 698
		748	VITREOUS ARTS. ARTISTIC GLASSWARE AND CRYSTAL WARE ⇒ 666.1/.2 ⇒ 75.023
		748.5	Artistic glazing. Decorative windows. □ Stained-glass work ⇒ 698.3
		748.6	Glass etching. Etched, engraved glass
		749	ARTISTIC FURNITURE, HEATING AND LIGHTING DEVICES
		749.1	Artistic furniture (antique and modern) ⇒ 684.4.03 ⇒ 745.51
		749.2	Artistic heating and lighting devices

75 PAINTING

Denote the work of particular artists by alphabetic extension (Table 1h)

◇ 75 Rubens Works of Rubens

⇒ 535.6

75.01	Theory and philosophy of painting <i>Subdivisions as at 7.01...</i> ◇ 75.01:111.852 Aesthetics of painting. Taste in painting	75.053	Panoramas. Dioramas. Vistas. Transparent, translucent pictures. Peepshows. Cosmoramas. Pleoramas
75.02	Technique. Craftsmanship <i>Subdivisions as at 7.02</i>	75.054	Theatrical painting. Scene-painting
75.03	Periods and phases of painting. Painting schools, styles, influences <i>Subdivisions as at 7.03...</i>	75.055	Sculpture painting. Vase painting
75.04	Subjects for painting <i>Subdivisions as at 7.04...</i>	75.056	Miniature painting. Book illustration (especially for printed books) ⇒ 766
75.05	Applications of painting	75.057	Manuscript decoration. □ Illumination, colouring of pictures, initial letters ⇒ 091.31 ⇒ 744.9
75.051	Pictures in general. Panel painting	75.058	Ornamental painting and decorative art. □ Monochrome pictures, grisaille. Red chalk drawings, sanguines. Depiction of relief objects (as optical illusion, trompe l'oeil) ⇒ 747
75.052	Murals. Frescoes. Paintings on walls, ceilings etc. ⇒ 72.04 ⇒ 7.021.333		

76 GRAPHIC ARTS. GRAPHICS

If required, apply .0 special auxiliaries listed at 7.0...

⇒ 655

⇒ 681.6

761	RELIEF TECHNIQUES	762.2	Etching
761.1	Woodcut	762.23	Aquatint
761.2	Metal cut	763	PLANOGRAPHIC TECHNIQUES. LITHOGRAPHY (AUTOLITHOGRAPHY)
762	INTAGLIO TECHNIQUES	766	APPLIED GRAPHIC ARTS. COMMERCIAL GRAPHICS ◇ 766:659.133.1 Poster design. Posters as graphic art ⇒ 744.9
762.1	Engraving		
762.11	Copper engraving. Steel engraving		
762.12	Dry point		
762.13	Mezzotint		

77 PHOTOGRAPHY AND SIMILAR PROCESSES

⇒ 528.7

⇒ 535.24

⇒ 535.6

⇒ 661.14

⇒ 681.7

⇒ 791.43/.45

77-1/-8	CHARACTERISTICS OF PHOTOGRAPHIC MACHINERY, EQUIPMENT 77-1/-5 ≡ 62-1/-5 77-7/-8 ≡ 62-7/-8	77.018	Permanence
		77.018.2	Resistance of photographic materials and photographs
		77.018.22	Resistance to humidity and water
		77.018.23	Resistance to heat and cold
		77.018.24	Resistance to abrasion
		77.018.25	Resistance to particulate matter. □ Resistance to sand, dust
		77.018.26	Resistance to light or other radiation <i>Specify by :535-1/-7</i> ◇ 77.018.26:535-2 Resistance to visible light ◇ 77.018.26:535-31 Resistance to ultraviolet radiation
77.0	PHOTOGRAPHIC THEORY, TECHNIQUES AND VARIOUS GENERAL CONSIDERATIONS <i>The auxiliaries listed under 7.0... are not applicable at 77 and its subdivisions, except for 7.04...</i>	77.018.27	Resistance to chemicals and chemical products in close proximity ◇ 77.018.27:667.6 Resistance to paints
77.01	Theory, principles and nature of photographic phenomena		

77.018.28	Resistance to fungus, insects and other forms of biological attack	77.021.55	Protective coatings. Anti-abrasion coatings. Antistatic coatings
77.018.4	Ageing phenomena <i>Forms of deterioration occurring at any time after creation of the original image</i>	77.021.56	Light-filter coatings ⇒ 771.534.26
77.018.41	Fading. Regression. □ Colour shift	77.021.57	Stripping coatings ⇒ 771.526
77.018.43	Staining of areas containing no silver	77.021.58	Backing layers
77.018.44	Discoloration of areas containing silver. □ Browning. Yellowing. Spotting. Silvering.	77.021.6	Coatings of nonsensitive materials (e.g. for image transfer)
77.018.46	Fog	77.021.9	Pretreatment of light-sensitive materials prior to exposure
77.018.47	Dimensional changes. □ Shrinkage. Stretching		
77.018.47	Breakage. Tearing. Curling. Separation of emulsion from support	77.022	Operations associated with exposure ⇒ 771.2/.44
77.018.6	Procedures to enhance permanence, performed during processing of the original image ⇒ 77.026.6	77.022.1	Choice and arrangement of model and subject. Photographic perspective. Pictorial appreciation
77.018.61	Removal of residual processing chemicals. □ Hypo removal ⇒ 77.023.8	77.022.4	Choice and arrangement of lighting. □ Lamps. Screens
77.018.62	pH control ⇒ 77.023.22	77.022.5	Determination of exposure. Theory of exposure measurement ⇒ 771.376
77.018.63	Protective coatings applied to the image surface. □ Protective varnishes, laminants ⇒ 77.024.3	77.022.6	Adjustment, setting of the shutter
77.018.65	Toning	77.022.7	Adjustment of the aperture
77.018.66	Hardening	77.022.8	Adjustment of distance setting. Depth of field (focus), theory and calculation ⇒ 681.7.013
77.018.67	Lubrication ⇒ 778.586		
77.018.68	Stabilization bath to prevent coupler printout	77.023	Processing work. Operations following exposure (with consideration of materials used) ⇒ 771.4 ⇒ 771.7
77.019	Deleterious influences and defects due to processing	77.023.1	Preparation of the darkroom
77.02	Photographic operations ⇒ 771	77.023.2	Preparation of baths and developing apparatus
77.021	Manufacture of light-sensitive, photosensitive materials. Treatment prior to exposure	77.023.22	pH control in processing solutions ⇒ 77.018.62
77.021.1	Manufacture of light-sensitive materials. Emulsions, additions etc.	77.023.3	Handling and treatment of exposed material before developing
77.021.11	Manufacture of emulsions and light-sensitive mixtures	77.023.38	Treatment before developing. □ Latensification. Desensitization.
77.021.12	Special emulsions and mixtures of emulsions. □ Emulsions without vehicle. Fine-grain emulsions. Emulsions for reversal processing ⇒ 771.531	77.023.4	Prehardening. Reduction, e.g. Sterry effect Development (developing) ⇒ 771.72
77.021.13	Special additions to emulsions and light-sensitive mixtures. □ Sensitizers. Hardening agents ⇒ 771.534.21 ⇒ 771.534.3 ⇒ 773.5	77.023.41	Chemical development
77.021.14	Melting before coating (second melting)	77.023.412	Development methods according to apparatus. □ Dish development. Tank development.
77.021.15	Application of the sensitive mixture to the support ⇒ 77.021.5 ⇒ 771.531 ⇒ 771.534.13	77.023.413	Machine development (including spray processing)
77.021.16	Drying, including paper	77.023.415	Two-bath development. Split development
77.021.17	Cutting. Perforating. Identification ◇ 77.021.17:621.961.2 Film perforating machines	77.023.415.2	Development influencing sensitometric properties
77.021.18	Packing and confection (into spools, film packs etc.)	77.023.415.3	Development influencing grain size, sharpness, resolution and contrast transmission function
77.021.3	Application by the user (with ready-made, manufactured mixtures). Coating by hand	77.023.415.5	Development influencing image colour ⇒ 778.6
77.021.5	Types of coating on sensitive materials	77.023.415.7	Development influencing mechanical properties
77.021.51	Substrate. Underlayers. □ Baryta layer. Plastic coatings	77.023.417	Development techniques for special conditions ◇ 77.023.417(211) Development in cold climates ◇ 77.023.417(213) Development in hot climates
77.021.53	Sensitive coatings	77.023.417.12	Development at low temperatures (below 18°C)
77.021.54	Intermediate coatings. Separating layers	77.023.417.14	Development at high temperatures (18°C and above)
		77.023.417.2	Development in strong light. Daylight development
		77.023.417.3	Rapid development

77.023.42	Physical development (silver deposition from the developer at the sites of silver nuclei)	77.026.345	Chemical analysis of dyes
77.023.421	Before fixing	77.026.4	Mechanical treatment and repair of photographs
77.023.422	After fixing	77.026.41	Dismounting
77.023.43	Development with gases, e.g. ammonia	77.026.42	Dry cleaning
77.023.44	Semi-dry development	77.026.43	Flattening. Uncurling
77.023.46	Development by physical agencies. □ Electrolytic development. Heat development	77.026.44	Laying down (reparation of tears)
77.023.47	Electrographic development ⇒ 772.93	77.026.45	Emulsion transfer
77.023.48	Activation processing	77.026.46	Retouching. Scratch removal
77.023.5	Toning before fixing	77.026.47	Facing
77.023.6	Transfer of electrographic images ⇒ 772.93	77.026.48	Ultrasonic cleaning
77.023.7	Work on the developed image	77.026.5	Chemical restoration of photographs
77.023.71	Hardening	77.026.51	Refixing and rewashing
77.023.72	Stopping	77.026.52	Intensification. Regeneration
77.023.73	Clearing and fixing	77.026.53	Reduction
77.023.74	Other work on the developed image. □ Intensification. Reduction. Bleaching. Toning. Subsequent reversal ⇒ 77.018.6 ⇒ 77.026.5	77.026.54	Bleaching and redevelopment
77.023.8	Removal of the fixing salts	77.026.55	Tone modification
77.023.81	Washing. Washing accelerators	77.026.56	Wet cleaning
77.023.83	Drying. Drying accelerators	77.026.57	Stain removal
77.023.9	Other work following exposure. □ Quality control	77.026.58	Fumigation
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77.053	Exposures with combined lighting. Night exposures	77.075	Photographs on nonmetallic mineral supports (except glass). □ On porcelain, ceramics, stone Specify by -03...
77.053.3	Daylight with continuous artificial light	77.076	Photographs on wood or woody materials. □ Photographs on cork
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77.067	Colour photographs ⇒ 778.6		
77.068	Sizes, formats of photographs (pictures and bases)		
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771.13	Photo-finishing laboratories. Developing rooms. Darkrooms. Printing rooms	771.319.2	Pinhole cameras ◇ 771.319.2:778.33 Pinhole cameras for radiography
771.15	Research establishments and laboratories ⇒ 001.89:005.71	771.319.3	Automatic photographic machines ⇒ 681.133
771.16	Factories. Workshops	771.319.4	Camera obscura
771.2	Studio and laboratory equipment, lighting, fittings etc. ⇒ 77.026.8 ⇒ 771.4	771.319.5	Astronomical and photogrammetric cameras ⇒ 520.33 ⇒ 528.711.1 ⇒ 778.32
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771.22	Lighting of studios. □ Windows. Skylights. Curtains, screens etc. ⇒ 628.9	771.319.7	Cameras incorporating processing, developing devices. Instant-print cameras (e.g. Polaroid)
771.23	Darkroom fittings. Fixed equipment for developing, fixing, washing etc. ⇒ 771.43	771.32	Plate or film holders, carriers. Dark slides. Adapters (e.g. for sheet film, to fit roll-film cameras)
771.24	Darkroom lighting and lights	771.33	Devices for daylight loading, changing and positioning
771.26	Copying-room equipment. Reproduction-room equipment	771.331	Changing devices (for plates or sheet film)
771.27	Projection-room equipment. Enlarging-room equipment	771.332	Spools for roll-film cameras
771.29	Motion-picture studio (film studio), laboratory and workshop equipment	771.333	Film cassettes. Film cartridges
771.3	Exposure apparatus. Cameras. Enlargers ⇒ 778.53	771.334	Film packs
771.31	Cameras and other exposure apparatus For cine cameras see 778.53 ⇒ 681.772.2	771.34	Stands. Tripods
771.311	Studio cameras	771.35	Optical parts of photographic cameras
771.312	Portable stand cameras	771.351	Photographic camera objectives ⇒ 535.31 ⇒ 681.7.067.25
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771.429	Other printers. □ Scanning printers. Electronic printers	771.531.2	Papers
771.43	Developing, fixing, washing and drying equipment	771.531.22	Printing-out papers. Daylight papers
771.431	Developing dishes	771.531.221	Salted papers
771.432	Developing troughs, tanks and machines	771.531.222	Albumen papers
771.435	Washing equipment	771.531.223	Collodio-chloride papers
771.436	Drying and glazing equipment	771.531.224	Silver gelatin papers. Gelatino-chloride papers. Chloro-citrate papers
771.44	Lighting and light sources for exposure and printing ◇ 771.44:628.92 Natural lighting. Daylight exposure etc. ◇ 771.44:628.93 Artificial lighting ⇒ 621.32 ⇒ 628.9	771.531.225	Self-toning papers
771.447	Continuous artificial lighting. Lamps or other light sources	771.531.226	Arrowroot papers
771.448	Flash lighting. Flashlight sources	771.531.23	Developing-out papers
771.448.2	Flash powders (magnesium, aluminium etc.) ⇒ 662.124	771.531.231	Silver gelatin papers
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771.5	Photographic materials, negative and positive. Emulsion vehicles. Bases or supports. Sensitized materials and their characteristics ⇒ 77.021	771.531.24	Resin-coated papers
771.51	Emulsion vehicles (nonsensitive materials used in production of light-sensitive layers)	771.531.25	Non-silver papers
771.52	Bases or supports for sensitive materials ⇒ 77.07	771.531.251	Carbon tissue
771.521	Glass bases	771.531.252	Papers with special sensitivity. □ Sensitivity to heat. Sensitivity to pressure
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771.524	Metal bases	771.531.28	Developer-incorporated papers
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		771.531.35	Cine film (motion-picture film)
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		771.534.13	Sensitization. Increasing of sensitivity by various means (especially digestion and chemical sensitization) ◇ 771.534.13:546.593 Gold sensitization
		771.534.14	Hypersensitization (of coated products)
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771.534.232	Orthochromatic sensitization (extending to yellow)		◇ 771.722:771.537.6 Developers for influencing resolution
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771.534.32	Desensitization by means of dyes		<i>Specify by :546...</i>
771.534.5	Sensitivity measurements. Sensitometry		◇ 771.74:546.33'223.2 Sodium thiosulphate ('hypo')
	⇒ 535.24		
771.534.51	Systems of sensitometry	771.76	Washing aids. □ Hypo clearing agents. Hypo eliminators
	<i>Denote named systems by alphabetic extension (Table 1h), e.g. ASA, DIN, Hurter-Driffield, Scheiner</i>	771.762	Salt solutions
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771.534.58	Various effects due to exposure to excessive radiations. □ Herschel, Becquerel, Villard effects	771.78	Chemicals and baths for printing and other finishing processes
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771.537.2	Optical properties of the image area. □ Colour		Processes using silver salts on plain or coated paper. □ Calotype
771.537.3	Structural properties of the image area.	772.12	
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771.537.6	Rendering of detail. Definition. □ Resolution. Sharpness. Halation. Development effects. Edge effects	772.15	Processes using silver salt emulsions
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771.537.8	Chemical properties of the developed image.	772.152	Processes using gelatin emulsions
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	□ Composition. Preparation. Water quality	772.2	Processes using metals other than silver group
771.72	Developers. Developing agents and additions	772.21	Simple ferroprussiate paper processes (blueprint, cyanotype processes)
	◇ 771.72.023.41.019 Defects in developers for chemical processes		
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772.22	Ferroprussiate processes using a colloidal substance. Cyanofor processes with gum containing an iron salt	773.712	Diazo compounds and other agents used in diazotype ⇒ 77.021.1 ⇒ 771.728
772.23	Ferrogallie processes	773.712.1	Light-sensitive diazo compounds
772.25	Manganese salt processes	773.712.2	Coupling components
772.27	Cobalt salt processes	773.712.5	Agents for modifying physical characteristics of coating or developing solutions (single components) ⇒ 77.021.13
772.29	Processes using metals other than the silver and iron groups		Humidifying agents. Wetting agents. Agents for increasing the rate of development
772.293	Processes using platinum group	773.712.6	Agents for modifying the physical or chemical nature of the developed image. □ Toners.
772.294	Processes using chromium group	773.712.7	Mordants
772.295	Processes using lead group metals		Compounds used in developing diazo prints.
772.296	Processes using tin group metals		Developers and additions
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772.95	Recording of images using electrochemical and electrothermal phenomena	773.714.1	Colloid hardening
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773	PHOTOGRAPHIC SYSTEMS OR PROCESSES USING ORGANIC COMPOUNDS ⇒ 774 ⇒ 778.6	773.714.3	End effects. □ Vesicular image or Kalvar process
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773.3	Differential tanning, dyeing, imbibition	773.74	Light-sensitive dye processes (bleach-out processes)
773.4	Oil printing	773.75	Silver-dye bleach processes
773.5	Dye processes	773.8	Inking, pigment application processes. □ Bromoil process. Bromoil transfer process
773.51	Processes using chromium salts	773.9	Light-sensitive colloid, resin and polymer processes generally
773.52	Processes using chromium and vanadium salts	773.91	Dichromated processes
773.6	Enamels. Photoceramics	774	PHOTOMECHANICAL PROCESSES GENERALLY ⇒ 655.22 ⇒ 773 ⇒ 778.18
773.7	Processes using dye-forming organic compounds and light-sensitive dyes. Dyeline processes	774.2	Preparation of supports generally
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773.711	General photochemistry and physics of diazo processes	774.4	Exposure to light. □ Stripping (washing out). Preparing plate for printing
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		774.7	Processes for rendering tonal gradation (halftone processes)
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774.9	Planographic processes using colloidal layers	777.2	Photographic processes for production of relief plates (line and halftone blocks)
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776.43	Preparing plate for printing. □ Desensitizing. Etching to affect moisture-retaining properties		Halftone screen processes. Screens and their manufacture. □ Line screens (e.g. crossline screens). Grained screens. Variable-density and contact screens
776.44	Tonal reversal of image (black to white). □ Reversal processes, e.g. Vandyke process. Offset deep-etch (intaglio litho) processes	777.33	⇒ 77.024.1
776.6	Transfer processes for photolithographic images		Production of discontinuous tone by photoelectric scanning
	⇒ 655.34	777.4	⇒ 621.383
776.7	Processes for rendering tonal gradation (halftone processes)		Duplication, reproduction of plates
	◇ 776.7:777.3 Screen processes (halftone screening)	777.5	⇒ 655.22
776.8	Processes for colour photolithography (i.e. multicolour photolithography)	777.6	⇒ 681.617
	⇒ 774.8		Moulding processes. □ Woodburytype
777	INTAGLIO AND RELIEF PROCESSES. PHOTOGRAVURE. LETTERPRESS OR RELIEF PLATES (LINE AND HALFTONE BLOCKS)	777.7	Photofiligranes (Woodbury). Imitation watermarks. Lithophany etc.
			⇒ 676.026.74
777.1	Photographic processes for production of intaglio plates		Photosculpture. Processes for making photographs in relief, models etc.
	⇒ 655.224	778	SPECIAL APPLICATIONS AND TECHNIQUES OF PHOTOGRAPHY
	⇒ 762		◇ 778:61 Medical photography
777.11	Intaglio processes according to sensitive material	778.1	Photocopying. Photographic reproduction and printing.
777.111	Photogravure using resists		Reprography
777.114	Production of halftone printing surfaces		⇒ 655.392
777.12	Preparation of support		⇒ 681.6
777.14	Etching	778.11	⇒ 77.026.1
777.15	Retouching of the etched surface		⇒ 77.039
777.17	Graining. Screen processes. □ Photoglyphy		⇒ 771.317
777.18	Processes for colour photogravure (i.e. multicolour photogravure) printing		⇒ 771.318
	⇒ 774.8	778.111	⇒ 778.1...
			Contact printing and copying
			◇ 778.11:772.21 Copying by ferroprussiate processes (blueprint, cyanotype)
			◇ 778.11:773.7 Copying by dyeline processes (e.g. diazotype)
			Transmission printing (with light passing through the original)

778.112	Reflex copying ⇒ 773.716	778.22	Optical parts of projection equipment. □ Lenses. Condensers
778.12	Projection printing and copying	778.23	Special light sources ◇ 778.23:621.326 Incandescent lamps for projection
778.122	Same-size copying		
778.123	Enlarging ⇒ 77.068.4 ⇒ 771.318.5	778.24	Projection screens
778.124	Reduction	778.25	Lantern slides. Positive transparencies. Film strips
778.14	Microcopying. Microphotographic technique and equipment ◇ 778.14:002 Microcopying in documentation ◇ 778.14.071 Micro-opaques (e.g. on paper or card). Microcards ◇ 778.14.072 Microtransparencies (e.g. on glass or film). Microfilm	778.251	Lantern slides on glass
778.142	Microcopies. Microphotographs (on roll, sheet, fiche etc.)	778.252	Film strips
778.142.5	Updatable microfilm	778.253	Lantern slides on film
778.143	Hard copy enlargements of microforms ⇒ 772.16	778.254	Accessories for making slides
778.148	Reading apparatus. Microfilm readers ⇒ 778.27	778.26	Various accessories. □ Framing devices for slides
778.148.2	Projection-type microform readers ◇ 778.148.2:778.274 Diascopic readers	778.27	Projectors: various types ⇒ 681.775.2 ⇒ 778.21
778.148.6	Direct-viewing readers	778.274	Diascopes (slide projectors), for transparencies
778.15	Special methods for modifying the image ⇒ 771.537	778.275	Episcopes (for opaques)
778.151	Geometrical modifications. □ Rectification. Distortion	778.276	Epidiascopes (for transparencies or opaques)
778.153	Texture effects	778.3	Techniques with mainly scientific applications
778.155	Modification of tonal range	778.31	Techniques in which the object is photographed on an enlarged scale ⇒ 537.533.35 ⇒ 57.086.2 ⇒ 57.086.3 ⇒ 681.72
778.156	Tone separation	778.312	Photomacrography (magnifications up to 10 times)
778.156.4	Tone separation by masking. □ Person process	778.315	Photomicrography (using optical microscope) ⇒ 77.035
778.156.6	Tone separation by special development	778.317	Electron micrography ⇒ 621.385.833
778.18	Reproduction or process photography ⇒ 655.22 ⇒ 774/777	778.32	Telephotography (including astronomical applications) ⇒ 520.33 ⇒ 771.319.5 ⇒ 771.351.4
778.182	Process photography according to kind of photographic image or picture	778.33	X-ray and gamma-ray photography. Radiography ⇒ 621.386
778.182.2	Halftone process photography		Radiographic apparatus
778.182.3	Line process photography	778.332	Intensifying screens <i>Class here photographic uses only. For fluorescent screens in general use</i> 535.371.07
778.182.4	Crossline process photography	778.333	
778.183	Process photography according to photographic material used	778.34	Photography by radiation other than light, X-rays and gamma-rays ⇒ 772.9
778.183.2	Process photography on dry plate, film, or negative paper	778.342	Ultraviolet-ray (UVR) photography
778.183.3	Process photography on wet plate	778.344	Infrared-ray (IRR) photography
778.183.4	Process photography on collodion-emulsion plate	778.347	Photography using particle beams, corpuscular radiation. □ Ionography. Electronography. Autoradiography ◇ 778.347:539.163 Autoradiography and autoradiographs (records of radionuclides in tracer work) ⇒ 77.026.13
778.186	Colour separations (for graphic arts or photomechanical processes) ⇒ 771.317 ⇒ 778.6	778.35	Aerial photography <i>For photogrammetry see 528.7</i>
778.19	Other copying, reproducing and printing techniques. □ Printing by scanning. Electronic printing	778.36	Panoramic photography ⇒ 771.316
778.2	Projection. Projectors and screens ⇒ 681.775 ⇒ 778.45 ⇒ 778.55 ⇒ 791.42		
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- 778.37 **High-speed and ultra-high-speed photography**
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- 778.38 **Holography (photographic aspects).**
 □ Special photographic materials. Techniques for preparing holograms
- 778.4 **Stereophotography. Flat photographic reproduction with effect of solidity or relief**
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 ⇒ 778.534.1
- 778.41 **Stereophotographic image separation and recombination**
- 778.42 **Stereoscopic cameras (stereocameras). Stereoscopic adapters, lens attachments**
 ⇒ 771.315
- 778.43 **Making, processing, finishing of stereoscopic photographs.** □ Stereoscopic prints. Stereo pairs
- 778.44 **Stereoscopic viewers.** □ Stereoscopes. Stereoscopic spectacles
- 778.45 **Stereoscopic projection**
- 778.5 **Cinematography. Motion-picture photography**
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- 778.51 **Early and primitive animated pictures (including drawings and other nonphotographic forms).** □ Flickers. Mutoscopes. Zoetropes
- 778.52 **Photographs at fixed intervals, allowing reconstitution of motion. Chronophotography**
 ⇒ 778.534.8
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- 778.53 **Cinematographic filming.** □ Exposure, registration (takes, shots)
 ⇒ 681.772.3
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- 778.53.099 Capturing of motion picture images without sound record (silent filming)
- 778.531 Cameras with intermittent film movement
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- 778.533 Camera parts and details
- 778.533.1 Film channel, guide. Film feed, transport. Feed rollers, sprocket wheels
- 778.533.2 Film gate
- 778.533.3 Shutter
- 778.533.4 Film magazines. Film reels
- 778.533.5 Film take-up and rewind devices
- 778.533.6 Operating mechanisms (mechanical, electric etc.)
- 778.534 Special kinds, systems and methods of cinematography
- 778.534.1 Stereoscopic cinematography. Other methods producing spatial effect. Wide-screen and panoramic systems
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- 778.534.2 Colour cinematography (multicolour cinematography)
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- 778.534.42 Sound record separate from film
- 778.534.45 Sound film (with sound track on the film). Optical and magnetic sound tracks
- 778.534.46 Stereophonic sound film. Stereophonic sound tracks (optical and magnetic)
- 778.534.48 Synchronization of picture and sound. Post-synchronization. Dubbing. Rerecording. Sound mixing
 ⇒ 778.588.8
- 778.534.6 Trick film work. Special effects. □ Animated cartoons
- 778.534.7 Production of composite pictures (e.g. using front or back projection)
- 778.534.8 Running speed. Taking frequency. Frame frequency
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- 778.535 Filming of titles and subtitles
- 778.54 **Direct viewing apparatus for pictures of moving objects**
- 778.55 **Cinematographic projection and projectors**
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- 778.551 Projectors with intermittent film movement
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- 778.553 Projector parts and details
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- 778.554 Special methods of projection
 778.554 ≡ 778.534
- 778.57 **Test films**
- 778.58 **Treatment of cinematograph film**
- 778.582 Splicing, cutting, joining etc.
- 778.583 Perforation of printed film
- 778.585 Alteration of the picture
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- 778.588 Copying, printing of cine film. Copying of internegatives and duplicate negatives.
 □ Equipment. Laboratories. Copies, prints
- 778.588.2 Contact printing. Contact copying machines and their parts
- 778.588.3 Optical printing. Reduction. Enlargement
- 778.588.5 Fades and wipes. Equipment for printing fades and wipes
- 778.588.7 Copying, printing of optical sound tracks
- 778.588.8 Subsequent overprinting of captions and sound track
- 778.6 **Colour photography (multicolour photography)**
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 ⇒ 77.023.415.5
 ⇒ 77.067
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 ⇒ 778.186
 ⇒ 778.534.2
- 778.62 **Monochrome colour processes**
- 778.621 Toning. Dyeing. Mordanting
- 778.622 Coloured residual images
- 778.623 Wash-off reliefs (tanned)
- 778.624 Dye-bleach processes
- 778.625 Chromogenic development
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- 778.63 **Direct colour photography processes**
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- 778.631 Processes using interference. □ Lippmann's process
- 778.632 Processes using prismatic dispersion
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778.66	Subtractive colour processes	778.682	Physiological and psychological factors in colour fidelity ⇒ 159.937.51 ⇒ 612.84
778.661	Production of coloured images on multilayer films by separate treatment of the individual layers		
778.662	Production of coloured images on multilayer films with simultaneous colour formation in all layers	778.8	Photography with multiple or continuous exposures. Trick photography. Illusions etc.
778.664	Colour pictures obtained by superposed printing of separation positives. Dye imbibition processes	778.81	Distorted photographs. Trick photography
778.665	Formation of coloured image by bleaching-out of mixture of dyes in an emulsion. Dye degradation processes. □ Dye-bleach processes	778.82	Multiple photographs of the same object (not intended for reconstitution of motion) ⇒ 778.52
778.666	Dye diffusion transfer processes		
778.668	Two-colour subtractive processes	778.85	Superimposed photographs of similar objects or persons for visualization of common characteristics
778.67	Copying of colour photographs ⇒ 77.026.14		
778.671	Copying screen plates and screen films		

78 MUSIC

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78.036.9	Jazz and related styles. □ Blues. Jazz. Dixieland. Boogie-woogie	78.081.1	First movements: allegro, presto, vivace etc.
78.04	Subjects for musical representation. Programme music. Illustrative, descriptive or representational music ◇ 785.11.04 Symphonic programme music. Symphonic poems (tone poems)	78.081.2	Second movements: andante, andantino, adagio, largo etc.
78.05	Applications of music in industry and trade, the home and everyday life	78.081.3	Third movements: minuet, scherzo, allegretto, rondo etc.
		78.081.4	Last movements. Finales
		78.082/.086	Forms of instrumental work
		78.082	Symphonies. Sonatas. Concertos
		78.082.1	Symphonies
		78.082.2	Sonatas
		78.082.3	Sonatinas

- 78.082.4 Concertos
Use .082.4 to class concertos according to the principal instrument. The main number for concertos is 785.6
- 78.083 Other, shorter forms**
- 78.083.1 Suites, divertimenti etc.
- 78.083.2 Variations. Preludes. Fugues
- 78.083.3 Fantasias. Arabesques. Moments musicaux. Toccatas. Humoresques. Capriccios etc.
- 78.083.4 Meditations. Nocturnes. Serenades. Songs without words. Berceuses
- 78.083.5 Rhapsodies. Ballades
- 78.083.6 Other short forms. □ Scherzos. Impromptus. Bagatelles
- 78.083.7 Concert-études
- 78.083.82 Transcriptions. Paraphrases. Arrangements
⇒ 78.088
- 78.083.83 Potpourris. Medleys
- 78.084 Marches. March-like forms**
- 78.085 Dance forms**
- 78.085.1 Forms derived from early dances. □ Minuets. Giges. Gavottes. Pavaues
- 78.085.2 Older ballroom dances. □ Waltzes. Polkas. Mazurkas. Quadrilles. Galops
- 78.085.3 Modern dances
- 78.085.321 Tangos
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- 78.085.324 Pasodobles
- 78.085.325 Rumbas
- 78.085.4 Children's dances
- 78.085.5 Ballets
- 78.085.7 National dances. Character dances
- 78.086 Separate parts of dramatic musical works (other than vocal music)**
- 78.086.1 Overtures. Preludes
- 78.086.2 Intermezzos. Entr'actes
- 78.086.4 Finales
- 78.087 Forms in relation to the instruments used**
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- 78.087.1 For one solo instrument
- 78.087.2 For solo instrument with accompaniment (other than orchestral accompaniment)
◇ 780.8.087.2:780.614.33:780.616.433 Works for violin or viola d'amore with piano accompaniment
- 78.087.3 For several solo instruments
78.087.3 ≡ 785.7
- 78.087.4 For one instrument as accompaniment
◇ 780.8.087.4:780.616.433 Piano part of a work with piano accompaniment
- 78.087.5 Works for learning and practice. □ Exercises. Scales. Studies
- 78.087.6 Vocal music
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- 78.087.612 For woman's voice
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- 78.087.612.3 Mezzosoprano
- 78.087.612.6 Contralto
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- 78.087.613 For man's voice
- 78.087.613.1 Tenor
- 78.087.613.2 Baritone
- 78.087.613.3 Bass
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78.087.62/.67 ≡ 785.72/.77
- 78.087.68 Choirs
- 78.087.681 Children's choirs. Boys' choirs
- 78.087.682 Women's choirs
- 78.087.683 Men's choirs
- 78.087.684 Mixed choirs
- 78.087.685 Solos with chorus
- 78.088 Transcriptions, arrangements for other instruments or voices (without other alteration)**
⇒ 78.083.82
⇒ 78.089.7
- 78.089 Format of published musical works**
- 78.089.1 Libretti. Legends. Commentaries
⇒ 82-293
- 78.089.6 Scores. Parts
- 78.089.7 Piano reductions
⇒ 78.088
- 78.089.8 Collections of works according to composer
- 78.09 Types of performance or presentation**
Subdivisions as at 7.09...
- ◇ 78.091 Live performances. Concerts
- ◇ 784.087.68.092 Competitions for choirs
- ◇ 785.11.091.2 Orchestral rehearsals
- 780.6 Musical instruments and accessories**
Specific musical instruments are classed together with music intended for them. This classification of musical instruments is based on the system published in 1914 by Erich von Hornbostel and Curt Sachs. Under this system - now accepted by musicologists all over the world - instruments are categorized according to the way in which the sound is produced
- 780.6.01 Musical instruments according to pitch
- 780.6.011 Higher than soprano. Sopranino
- 780.6.012 Soprano
- 780.6.013 Alto
- 780.6.014 Tenor
- 780.6.015 Baritone and bass
- 780.6.015.1 Baritone
- 780.6.015.2 Bass
- 780.6.016 Lower than bass. Contrabass, double-bass
- 780.6.02/.06 Parts of instruments
- 780.6.02 General construction
- 780.6.021 Resonance devices. □ Resonators
- 780.6.022 Sound boxes. Soundboard bridges. Soundboards
- 780.6.023 Sound-producing and damping devices
- 780.6.024 Mutes
- 780.6.03 Components of keyboard instruments
- 780.6.031.2 Frames. Wreath-planks. Wreath-blocks
- 780.6.031.3 Cases
- 780.6.031.6 Air cavities
- 780.6.031.7 Tension devices for soundboards
- 780.6.033 Stringing. Tuning devices
- 780.6.034 Keyboards, manuals. Fingerboards, frets etc. Keys
- 780.6.035 Transposition devices. Couplers. Registers
- 780.6.036 Action. □ Hammer heads. Jacks. Hammers. Quills
- 780.6.037 Pedals. Stops
- 780.6.039 Other devices and mechanical aids for playing keyboard instruments
- 780.6.04 Components of stringed instruments. Accessories
- 780.6.041 General construction. □ Ribs. Backs. Soundposts. Bass-bars
- 780.6.042 Bows
- 780.6.043 Bridges
- 780.6.05 Components of wind instruments
- 780.6.052 Reeds

780.6.056	Tone-altering devices for wind instruments	780.616.3	Plucked instruments plucked with the keyboard. □ Spinets. Virginals
780.6.056.2	Mouthpieces, beaks, ligatures etc.	780.616.31	Harpsichords (clavicytheriums)
780.6.056.5	Valves	780.616.4	Struck board instruments
780.6.057	Wind supply for organ pipes. □ Organ bellows.	780.616.41	Board instruments struck with the hand.
780.6.06	Organ wind-chests. Organ pipes, reeds		□ Dulcimers. Santirs. Santors
780.6.06	Components of mechanical musical instruments etc.	780.616.43	Instruments struck via the keyboard
780.6.061	Accessories for music-roll machines. Music rolls	780.616.431	Clavichords
780.6.062	Mechanical playing devices for keyboard instruments. □ Pinned cylinder mechanisms.	780.616.432	Fortepianos
	Perforated roll mechanisms	780.616.433	Pianofortes (upright pianos, grand pianos)
780.6.063	Changing devices for music discs, cylinders etc. (for musical boxes)	780.62	Membranophones
780.6.065	Plucking mechanisms. Sound-producing devices	780.621	Tubular drums
780.6.069	Other accessories. □ Music discs, cylinders etc. (for musical boxes)	780.621.1	Cylindrical drums
780.6.07	Accessories for percussion instruments.	780.621.2	Conical drums. □ Bata drums. Ntenga drums
	□ Drumsticks. Tympani sticks. Wire brushes	780.621.3	Barrel drums. □ Dhola drums
780.6.08	Accessories for electric and electronic instruments	780.621.32	Tablas
		780.621.4	Waisted drums. □ Chango drums. Dugdugi drums. Kalunga drums. Tsuzumis
780.6.081	Toggle switches. Microphones. Pick-ups	780.621.5	Goblet drums. □ Darbeks. Tarabukas
780.6.081.1	Electromagnetic pickups	780.621.6	Footed drums. □ Panhuéhuets.
780.6.081.2	Electrostatic pickups		Talpanhuéhuets
780.6.081.3	Photoelectric pickups	780.621.7	Long drums. □ Chocó drums. Handle drums
780.6.081.4	Piezoelectric pickups	780.622	Vessel drums. □ Naqaras. Talambas
780.6.081.5	Gizmotrons	780.622.1	Orchestral and band drums. □ Kettledrums
780.6.081.6	Vibrating devices	780.622.11	Timpani
780.6.081.7	Air microphones and contact pickups	780.622.12	Bass drums
780.6.082	Accessories for electromechanical instruments	780.622.13	Gong drums
780.6.082.1	Tone wheels. Perforated discs	780.622.14	Tenor drums
780.6.083	Signal processing devices and amplifiers.	780.622.15	Side drums
	□ Preamplifiers. Speakers	780.622.16	Cocktail drums
780.6.084	Loudspeakers. □ Telharmonium. Choralceco	780.622.17	Timbales
780.6.087	Wah-wah pedals. Fuzz pedals	780.623	Frame drums. □ Bodhrans. Gagaku drums.
780.6.088	Phasers. Fuzz phasers		Kelontongs. Tambourines. Terbangs. Uchiwa daikos
780.61/.66	Individual musical instruments	780.624	Friction drums. □ Ingungus. Moshupianes.
780.61	Chordophones	780.628	Rommelpots
780.611	Bows. □ Simple bows. Resonated bows.	780.63	Mirlitons. □ Eunuch flutes. Kazoos. Zobos
	Mouth bows. Ground bows. Multiple bows	780.631	Idiophones
780.612	Lyres. □ Box lyres. Bowl lyres		Stamping idiophones. □ Sticks. Tap shoes.
780.613	Harps		Tubes
780.613.1	Bow (arched) harps. Angle harps	780.632	Stamped idiophones. □ Boards. Stamped pits
780.613.3	Frame harps	780.633	Shaken idiophones. □ Jingles. Rattles
780.613.31	Orchestral harps	780.633.1	Bells
780.614	Lutes		⇒ 673.5
780.614.1	Plucked lutes		⇒ 681.11.033.2
780.614.11	Round-backed. □ Bouzouki. Cobza. Mandolin.	780.633.11	Pellet bells. □ Claw bells. Hawk bells
	Sitar. Tambura. Theorbos. Uti	780.633.13	Clapper bells
780.614.13	Flat-backed. □ Balalaikas. Banduras. Banjos.	780.633.131	Handbells
	Samisens. Ukeleles	780.633.133	Church bells
780.614.131	Guitars	780.633.14	Struck bells
780.614.3	Bowed lutes	780.634	Percussion instruments. □ Boards. Celestes.
780.614.31	Fretted bowed lutes. □ Viols		Glockenspiels. Gongs. Lithophones. Maracas.
780.614.33	Unfretted bowed lutes	780.634.3	Metallophones. Slit drums. Triangles. Tubular bells. Xylophones
780.614.331	Fiddles	780.635	Steel drums (pans)
780.614.332	Violins		Concussion instruments. □ Castanets.
780.614.333	Violas	780.636	Clappers. Claves. Cymbals. Wind chimes
780.614.334	Violoncellos (cellos)		Friction instruments. □ Musical glasses, glass harps. Musical saws
780.614.335	Double basses. Contrabass viols	780.637	Scraped instruments. □ Bone scrapers.
780.615	Zithers		Washboards
780.615.1	Simple zithers. Stick zithers. Raft zithers. Tube zithers. Trough zithers	780.638	Plucked idiophones (linguaphones). □ Jew's harps. Sansas
780.615.2	Long zithers. □ Ch'ins. Kotos	780.64	Aerophones (wind instruments)
780.616	Board instruments	780.641	Blow hole instruments
	<i>Class here keyboard instruments in general</i>	780.641.1	End-blown flutes
780.616.1/.3	Plucked board instruments	780.641.2	Side-blown flutes. □ Fifes
780.616.1	Plucked board instruments plucked with the hand. □ Aeolian harps. Psalteries	780.641.21	Orchestral flutes. Piccolos
		780.641.3	Nose flutes
		780.641.4	Vessel flutes. □ Ocarinas

- 780.641.5 Multiple flutes. □ Double flutes. Triple flutes
 780.641.6 Panpipes
 780.642 Whistle mouthpiece instruments. □ Flageolets.
 Whistles. Whistle flutes
 780.642.1 Recorders
 780.643 Single reed instruments. □ Bassett horns.
 Hornpipes
 780.643.1 Clarinets
 780.643.11 Multiple clarinets
 780.643.2 Saxophones
 780.644 Double reed instruments. □ Shawms
 780.644.1 Oboes. □ Oboe d'amore. Cor anglais
 780.644.2 Bassoons. □ Contrabassoons
 780.645 Bagpipes
 780.646 Cup mouthpiece instruments
 780.646.1 Trumpets. □ Alphorns
 780.646.2 Trombones. □ Sackbuts
 780.646.3 Horns. □ Hunting horns. Post horns. Valve
 horns (French horns)
 780.646.4 Tubas
 780.646.5 Cornetts. Serpents
 780.646.6 Bugles. Cornets. Flugelhorns. Key bugles.
 Saxhorns
 780.646.7 Other brass band instruments. □ Euphoniums.
 Sousaphones
 780.647 Free reed instruments
 780.647.1 Mouth organs. □ Harmonicas. Melodicas
 780.647.2 Accordions. Concertinas. □ Melodeons. Piano
 accordions
 780.647.3 Harmoniums
 780.648 Bull-roarers and buzzers
 780.649 Organs
 780.649.1 Portative organs. Regals
 780.65 **Mechanical, electrical and electronic
 instruments**
 780.651 Mechanical instruments
 ◇ 780.651:780.61 Mechanical stringed
 instruments (stroked, struck or plucked)
 ◇ 780.651:780.634 Carillons. Mechanical
 glockenspiels. Chime-producing
 movements. Instruments with several
 methods of sound excitation
 ◇ 780.651:780.64 Mechanical aerophones
 ◇ 780.651:780.647 Mechanical accordions.
 Melodeons
 ◇ 780.651:780.647.1 Mechanical harmonicas
 780.651.1 Hurdy Gurdies. Geigenwerks (mechanical
 harpsichords)
 780.651.2 Automatic instruments. □ Automatic harps.
 Automatic pianos (pianolas). Orchestrions.
 Self-playing xylophones
 780.651.3 Mechanical organs. Barrel organs
 780.651.7 Musical boxes
 780.653 Electronic instruments. Electrical
 (electroacoustic) instruments
 780.653.1 Electroacoustic instruments
 ◇ 780.653.1:780.613 Electric harps
 ◇ 780.653.1:780.614.13 Electric banjos
 ◇ 780.653.1:780.614.131 Electric guitars
 ◇ 780.653.1:780.614.131-023.843
 Hollow-bodied electric guitars
 ◇ 780.653.1:780.614.131-026.76 Solid-bodied
 electric guitars
 ◇ 780.653.1:780.614.33 Electric violins
 ◇ 780.653.1:780.614.334 Electric cellos
 ◇ 780.653.1:780.614.335 Electric double
 basses
 ◇ 780.653.1.081.1:780.616.43
 Electromagnetic instruments struck via the
 keyboard
 ◇ 780.653.1.081.1:780.616.433
 Electromagnetic pianos. Including:
 Neo-Bechstein Flügel. Radioklavier.
 Elektrophonisches Klavier. Variachord.
 Crea-Tone
 ◇ 780.653.1.081.2:780.616.431 Electrostatic
 clavichords. Including: Clavinets
 ◇ 780.653.1.081.2:780.616.433 Electrostatic
 pianos. Including: Elektrochord. Pianatron.
 Minipiano. Dynatone
 ◇ 780.653.1.081.6:780.634 Electric
 percussion instruments. Including: Electric
 Glockenspiels. Vibraphones
 780.653.2 Electromechanical instruments. □ Birotron.
 Mellotron
 ◇ 780.653.2.082.1:780.649 Electromechanical
 organs. Including: Celluphones. Organovas.
 Syntronic organs
 780.653.21 Photoelectric film instruments. □ Clavivox.
 Cross-Grainger free music machine. Variofon
 780.653.3 Electronic instruments
 780.653.31 Monophonic electronic instruments. □ Ondes
 martenots. Theramins
 780.653.32 Partially polyphonic instruments. □ Hellertions.
 Melochords
 780.653.33 Fully polyphonic instruments
 ◇ 780.653.33:780.616.433 Electronic pianos
 ◇ 780.653.33:780.647 Electronic accordions
 ◇ 780.653.33:780.649 Electronic organs
 Synthesizers
 ⇒ 004.4
 780.653.362 Analog synthesizers, e.g. Moog
 780.653.364 Digital synthesizers, e.g. Yamaha (r)
 780.66 **Other forms of musical instrument**
 ◇ 780.66:612.7 The human voice as a
 musical instrument
 780.661 Improvised musical instruments
 ◇ 780.661:642.732.24 Spoons as musical
 instruments
 780.7 **Music teaching aids. Other music
 accessories**
 780.71 **Music teaching and practice aids**
 780.712 Beat or tempo indicators
 780.712.1 Metronomes
 780.712.2 Batons
 780.73 **Pitch indicators.** □ Tuning forks. Pitch pipes
 780.74 **Note indicators for keyboards**
 780.76 **Devices for recording of played music.
 Music roll perforators**
 780.78 **Accessories for sheet music**
 780.782 Music holders. Music desks
 780.783 Music stands
 780.784 Page turners for music
 780.8 **Music for individual instruments**
Use colon combination with 780.6...
 781 **THEORIES OF MUSIC. GENERAL
 QUESTIONS**
 ⇒ 78.01
 781.1 **Scientific theories of music.
 Mathematical, physical, physiological,
 psychological bases. Musical
 acoustics**
 781.15 **Theory of rhythm**
 781.2 **General theory of music**
 781.21 **General principles. Musical grammar**

- 781.22 Sound from musical point of view. □ Tone. Timbre. Tonality. Keys. Chords. Intervals
- 781.24 Musical notation. Clefs etc.
- 781.4 Harmony. Counterpoint. Melody
- 781.41 Harmony. Thorough bass. Figured bass. Modulation
- 781.42 Counterpoint. Canon. Fugue
- 781.43 Melody. Melodic structure
- 781.5 Musical forms. Kinds of musical work
Details by .08...
- 781.6 General questions of composition and interpretation
- 781.62 Rhythm. Metre. Syncope. Dynamics. Intonation
- 781.63 Instrumentation. Orchestration
- 781.64 Transposition. Instrumental reduction
- 781.65 Improvisation
- 781.66 Accompaniment
- 781.68 Interpretation. □ Tempo. Expression. Phrasing. Nuance
- 781.7 Theories and forms of national music (of various countries or peoples)
- 781.8 Theories and forms of the music of antiquity
- 781.91 Topics concerning musical instruments.
□ Tuning. Tuning up
- 781.97 Indexes of musical works
For bibliographies of works about music, use 016:78 or 78(01)
- 781.972 Inventories of musical sources
Works showing the location of the manuscript or the earliest edition of musical works
- 781.973 Catalogues of musical works
- 782/785 KINDS OF MUSIC
- 782 DRAMATIC MUSIC. OPERA
⇒ 792.5
⇒ 82-293
- 782.1 Grand opera (with sung recitative)
- 782.6 Opéra comique (with spoken dialogue)
- 782.7 Comic opera. Opera buffa
- 782.8 Operettas. Stage plays with songs (musicals)
- 782.9 Other dramatic music
◇ 782.9:792.78 Pantomime music
◇ 782.9:792.82 Ballet music
◇ 782.9:793.386 Music for masques
- 783 CHURCH MUSIC. SACRED MUSIC. RELIGIOUS MUSIC
◇ 783:272 Roman Catholic church music
◇ 783:[274/278] Protestant church music
◇ 783:273.4(4/9) Anglican church music
⇒ 27-535
⇒ 27-535.7
⇒ 27-535.8
- 783.1 Sacred instrumental music (e.g. for organ, for orchestra)
- 783.2 Liturgical music. Liturgical song
- 783.21 Masses. Communion services
- 783.22 Common of the mass. □ Introit. Kyrie. Gloria in excelsis. Gradual. Credo. Sanctus. Benedictus. Agnus Dei. Ite missa est. Communion. Elevation
- 783.23 Liturgical music for special occasions.
□ Proper of the season. Offertories. Graduals. Sequences. Tropes
- 783.24 Choral services. □ Antiphons. Compline. Vespers
- 783.25 Congregational singing. Accompaniment of congregational singing
- 783.26 Litanies
- 783.27 Psalms. Psalters. Lamentations
- 783.28 Liturgical plays. Mysteries. Music for high feasts, Christmas, Easter etc.
- 783.29 Masses for the dead. Requiem. Te Deum. High Masses. Solemn Masses. Magnificats. Processions. Processional and recessional music
- 783.3 Sacred oratorios. Passion music. Sacred cantatas
- 783.4 Church anthems, motets and chants (not mentioned elsewhere). □ Anthems etc. in the manner of Palestrina
- 783.5 Gregorian and Anglican chant. Protestant chorales
- 783.51 Gregorian chant. Plainsong
- 783.52 Anglican chant
- 783.53 Protestant chorales
- 783.6 Sacred choral music, songs, hymns
- 783.65 Popular religious songs. □ Sacred folksongs. Carols
- 783.7 Evangelical songs
- 783.8 Choral music for cantor or precentor and choristers
- 783.9 Congregational songs, canticles, hymns, sequences
- 784 VOCAL MUSIC
- 784.1 A cappella music. Unaccompanied vocal music. Rounds
- 784.12 Simple songs
- 784.13 Songs with refrains
- 784.15 Madrigals
- 784.2 Vocal extracts from larger works.
□ Arias from operas, oratorios etc. Recitatives
- 784.3 Vocal chamber music. □ Songs. Ballades. Lieder
- 784.4 Popular, traditional, historic songs.
□ Folksongs. Drinking songs. Street cries, ballads
⇒ 398.8
⇒ 78.011.26

784.5	Secular oratorios and cantatas	785.1	Ensemble music for relatively large numbers of instruments
784.6	Community songs. Children's songs	785.11	Orchestral music. Music for symphony orchestra
784.66	Community songs. Songs of particular social groups, occupations etc. ◇ 784.66:378.18 Students' songs	785.12	Music for wind bands (e.g. brass bands)
784.67	Children's songs	785.13	Military music
784.68	Songs for particular seasons and festivals 784.68 ≡ 398.33	785.15	Hunting music
784.69	Other songs for various occasions 784.69 ≡ 39	785.16	Music for small orchestras
784.692.5	Wedding songs	785.161	Music for dance bands, jazz bands etc.
784.7	Other kinds of song	785.6	Concertos
784.71	Martial songs. National anthems. Patriotic and political songs ◇ 784.71(438) Polish patriotic songs, national anthem etc.	785.7	Chamber music. Music for several instruments concertante ⇒ 78.087
784.72	Humorous, comic songs. Satirical songs	785.72	Duos
784.75	Exotic songs. Negro songs. Black minstrel songs	785.73	Trios
784.9	Singing technique. Vocal exercises. Singing practice. □ Solfeggi, sol-fas	785.74	Quartets
785	INSTRUMENTAL MUSIC. SYMPHONIC MUSIC. GROUPING OF INSTRUMENTS. ENSEMBLE MUSIC	785.75	Quintets
		785.76	Sextets
		785.77	Septets
		785.78	Octets
		785.79	Nonets

79 RECREATION. ENTERTAINMENT. GAMES. SPORT

- ◇ 79.07 Occupations and activities associated with entertainment, sport etc. NOTE: Details from 7.07
 ◇ 79.091 Performance, presentation
 ◇ 79.092 Competitions. Contests. Records. Championships

791	CINEMA. FILMS (MOTION PICTURES) <i>For public entertainments, spectacles, displays (previously at this number) see 394.2</i> <i>Denote details by Table 1d - Common auxiliaries of form, and by the special auxiliaries listed at 7 (if not otherwise provided for)</i> ◇ 791(049.32) Critical reviews of film ◇ 791.077 Amateur film (motion pictures). Including: Home movies	791-56 791-561 791-57 791-58 791.1 791.11 791.12	Glamour girls Dumb blondes Animated, cartoon characters Talking animals Principal stages of cinema development Silent era <i>Specify period by Table 1g - Common auxiliaries of time</i> ◇ 791.11"189" Silent films of the 1890s Post-silent era <i>Specify place, period and national characteristics by common auxiliary tables 1e, 1f and 1g</i> ◇ 791.12(540) Films of the post-silent era in India ◇ 791.12(=21)(73) Indian films of the post-silent era shot in USA ◇ 791.12"194" Films of the 1940s Genre, subject, style of film Artistic styles <i>Denote details of style by .0... (from 7.0...)</i> ◇ 791.21.036.1"199" Realist films of the 1990s ◇ 791.21.037.5 Surrealistic style in films Genre (style defined by subject and background) <i>For issues and background elements other than genre see 791.23. Any combination of more than one of the following terms of</i>
791-2	Films according to characteristics of presentation		
791-21	Feature-length films. Main features		
791-22	Shorts. Second features, B-movies		
791-23	Prequels		
791-24	Sequels		
791-25	Remakes		
791-5	Character types	791.2	
791-51	Heroes	791.21	
791-511	Tough guys		
791-512	He-men		
791-52	Heroines		
791-53	Rebels		
791-54	Bad guys	791.22	
791-55	Bad girls		
791-551	Gangsters' molls		
791-552	Femmes fatales. Seductresses		

- genre may be made by adding a colon. Any genre term can be subdivided alphabetically according to a named place or person
- ◇ 791.221.5:791.239 Crime film with entertainment world as background
 - ◇ 791.227MAL Malcolm X as plot subject in film
 - ◇ 791.237.6NEW New York as background in film
- 791.221/.228 Fictional genres
- 791.221.1 Musicals
- 791.221.2 Comedy
- 791.221.21 Slapstick
- 791.221.22 Gag
- 791.221.23 Screwball
- 791.221.24 Spoof
- 791.221.25 Black comedy
- 791.221.26 Satire
- 791.221.27 Situation comedy
- 791.221.28 Tragi-comedy
- 791.221.3 Tragedy
- 791.221.4 Melodrama
- 791.221.41 Weepies
- 791.221.5 Thriller. Suspense. Crime
- ◇ 791.221.5:327.88 Suspense films about political conspiracies
 - ◇ 791.221.5:343.123.12 Detective films
 - ◇ 791.221.5:343.321 Spy films
 - ◇ 791.221.5:343.611 Murder films
 - ◇ 791.221.5:343.72-057.34 Films about fraud carried out by civil servants
 - ◇ 791.221.5:343.811 Prison films
- 791.221.51 Film noir
- 791.221.6 Violence
- 791.221.7 Action
- 791.221.71 Disaster films
- 791.221.72 Group jeopardy
- 791.221.8 Fantasy
- ◇ 791.221.8:82-311.9 Science fiction films
- 791.221.9 Horror
- ◇ 791.221.9:392.28 Vampire films
 - ◇ 791.221.9:398.42 Ghost films
- 791.222 War films
- ◇ 791.222:323.26 Resistance movements in war depicted in films
 - ◇ 791.222:355.46 Naval war films
 - ◇ 791.222:355.48(597.7:73) Films about the history of warfare between North Vietnam and USA
 - ◇ 791.222:623.459 Chemical warfare in films
- 791.223 Conventionalized geographical genres
- 791.223.1 Westerns
- ◇ 791.223.1:791.221.2 Comic westerns
 - ◇ 791.223.1(450-82) Westerns produced in Italy ('spaghetti westerns')
- 791.223.2 Easterns
- 791.224 Films representing historical events
- ◇ 791.224(450)*15" Films depicting Italy in the sixteenth century
- 791.224.1 Period and costume drama
- 791.224.2 Swashbucklers
- 791.225 Epics
- ◇ 791.225:22 Biblical epics
 - ⇒ 791.232
- 791.226 Pornographic films, erotica
- 791.227 Biographical themes treated as entertainment or fiction in film
- For documentary treatment of biography use 791.229:929. Subdivide alphabetically according to a named person*
- ◇ 791.227CAP Films about Al Capone
- 791.227.1 Films a clef. Films inspired by real persons
- Subdivide alphabetically according to title of film representing pseudonymous person*
- ◇ 791.227.1CIT Citizen Kane
- 791.228 Animation. Cartoons
- 791.229 Factual and anthologizing genres
- 791.229.1 Compilation films (anthologies of extracts)
- 791.229.2 Documentary film
- ◇ 791.229.2:597.3 Documentary films about sharks
- 791.229.4 Newsreels
- 791.23 Issues and background elements
- Class here issues raised in fictional films, and settings and milieux as background or plot motif. For documentary treatment of these subjects see 791.229. For genre see 791.22*
- 791.231 Politics as plot/background in film
- ◇ 791.231:327.323.3 Communism in film
- 791.232 Religion as plot/background in film
- ◇ 791.232:26 Judaism in film
- 791.233 Social issues in film
- Specify type of person by Table 1k-05 - Common auxiliaries of persons and personal characteristics*
- ◇ 791.233:141.72 Feminist films
 - ◇ 791.233:316.613.42 Depiction of friendship in films
 - ◇ 791.233:392.5-058.12 Films depicting weddings of upper-class persons
 - ◇ 791.233:613.885 Depiction of same-sex (homosexual) relationships in films. 'Coming out' films
- 791.234 Moral issues in film
- ◇ 791.234:176.5 Prostitution in films
 - ◇ 791.234:178.1 Depiction of alcoholism in films
- 791.235 Psychological and physical issues in film
- ◇ 791.235:159.972 Mental disorder in films
- 791.236 Nature as plot/background in film
- ◇ 791.236:591.611 Hunting and capture of animals depicted in films
- 791.237 Travel and locations as plot/background in film
- ◇ 791.237:910.4(729.726:729.828) Films depicting travel between Antigua and Grenada
- 791.237.2 Transport as plot/background in film
- ◇ 791.237.2:725.1:656.71 Airports in films
- 791.237.21 Road movies
- 791.237.4 Buildings as plot/background in film
- ◇ 791.237.4:725.1:614.21 Hospitals in films
- 791.237.6 Places as plot/background in film
- Specify place by Table 1e - Common auxiliaries of place*
- ◇ 791.237.6(421.4) North London as plot/background in films
- 791.238 Sport and games as plot/background in film
- ◇ 791.238:797.2 Swimming in films
- 791.239 Entertainment world as plot/background in film
- ◇ 791.239:791 Film world as background in films
- 791.24 Stylistic devices
- 791.241 Series of films
- ◇ 791.241BON James Bond films
- 791.242 Serial films. Films intended to be shown in separate episodes
- 791.243 Cliff-hanger
- 791.244 The chase
- 791.246 'Homage' (allusion to, or imitation of, scenes or images in other films)

791.3	Aesthetics and film	791.62	Production: technical aspects <i>For public ceremonies, processions, commemorations (previously at this number) see 394.4. For parades see 394.5. For military parades see 355.511.44</i>
791.31	Relationship of film and other arts ◇ 791.31:78 Relationship of film and music		◇ 791.62:005.71MGM Film studios, Metro-Goldwyn-Mayer (MGM)
791.32	Theory and criticism of film (as a subject) <i>Denote details of style by .0... (from 7.0...)</i> ◇ 791.32.072.3 Criticism of film ◇ 791.32.072.36.7 Existentialist criticism of film		◇ 791.62:330.322.12 Private investment in film production. NOTE: Specify place by Table 1e - Common auxiliaries of place
791.321	Adverse criticism of films. □ 'Turkeys', flops		◇ 791.62(44) Film production in France
791.4	Society and film <i>For visual displays and exhibitions (previously at this number) see 069.9</i>	791.621	Equipment
		791.621.1	◇ 791.621:628.977 Lighting in film production
		791.622	Locations, studios, sound stages
791.41	Relationship of film and other media ◇ 791.41:621.397 Relationship of film and television		Cinematography <i>Specify techniques of Cinematography by Class 77</i>
		791.623	Post-production
		791.623.1	Editing
791.42	Films appealing to specific social groups <i>For slide shows, 'magic lantern' shows (previously at this number) see 778.25:069.9, 069.12</i> ◇ 791.42:316.346.2-055.2 Women's pictures ◇ 791.42:316.347(=2/=8) Social differentiation according to race. Including: 'Blaxploitation'	791.63	Production: aesthetic aspects ⇒ 628.974.6 ⇒ 662.11 ⇒ 394.5
		791.631	Producing <i>Specify protagonists by -051 from Table 1k-05 - Common auxiliaries of persons</i>
			◇ 791.631-051 Film producers
791.43	Influence of film on its social audience <i>For cinema in general (previously at this number) see 791</i> <i>Specify type of person by Table 1k-05 - Common auxiliaries of persons</i> ◇ 791.43:32.019.52 Political propaganda by means of film ◇ 791.43:351.751.5 Censorship of film, ratings classification ◇ 791.43:351.751.5-053.6 Films for teenagers ◇ 791.43:659.1 Film and advertising ◇ 791.43:791.225-053.6 Influence of violent films on teenagers	791.632	Scripting ◇ 791.632:792.026 Film scripts ◇ 791.632-051 Scriptwriters
		791.633	Direction ◇ 791.633-051 Directors
		791.634	Art direction
		791.634.2	Titling. Title sequences (including credit titles): aesthetic aspects
		791.635	Film acting <i>Specify protagonists by -051 from Table 1k-05 - Common auxiliaries of persons</i> ◇ 791.635:792.028.57 Walk-on parts in film. NOTE: Specify roles of acting by :792... ◇ 791.635:929 Biographical treatment of stars ◇ 791.635-051 Actors ◇ 791.635-053.2 Child actors ◇ 791.635-055.2 Actresses ◇ 791.635.091.1 Auditions. NOTE: Details of performance by .0... (from 7.0) ⇒ 792.028
791.5	Education and film		Other artistic elements ◇ 791.636:78 Music in film
791.51	Film as part of educational curriculum <i>Specify level of education according to Class 373...</i> ◇ 791.51:373.5 Film as taught at secondary school level ⇒ 373.2.016 ⇒ 373.3.016	791.636	
791.52	Film as a medium for education ◇ 791.52:371.686 Film as audio-visual aid, teaching material	791.64	Distribution <i>Specify place by Table 1e - Common auxiliaries of place</i>
791.53	Corporate bodies, institutions, collections of film <i>Expand type of institution by Class 06 ...</i> ◇ 791.53:069 Museums of film		◇ 791.64:330.322.14 Public investment in distribution ◇ 791.64(1-21) Distribution of films in urban districts
		791.641	Promotion, marketing ◇ 791.641:658.62 Merchandising related to films ◇ 791.641:659.133 Film posters
791.6	Film as industry <i>For public festivals, pageantry etc. (previously at this number) see 394</i>	791.641.3	Advertising, commercial film
		791.641.5	Trailers
791.61	Film industry in general <i>For fairs, festivities etc. (previously at this number) see 394.46, 394.6</i> ◇ 791.61:330.322.14 Public investment in the film industry. NOTE: Specify management aspects of industry by Class 33 ◇ 791.61:331.45 Health and safety in the film industry	791.65	Exhibition. Transmission <i>For carnivals, masquerades (previously at this number) see 394.5</i> ◇ 791.65:339.15.054.22 Commercial success at the box office. NOTE: Specify aspects of performance by Class 7. ◇ 791.65.079 Film festivals, awards ◇ 791.65.091.5 Film premieres

- 791.7** **Recreation grounds, playgrounds etc. Funfairs.** . □ Swings. Slides. Helter-skelters. Roundabouts. Switchbacks. Scenic railways
⇒ 688.77
⇒ 712.256
- 791.75** **Theme parks**
◇ 791.75:94 Historical theme parks
- 791.8** **Menageries. Circuses. Exhibitions and performances involving trained or fighting animals**
⇒ 636.046/.047
- 791.82** **Menageries**
⇒ 636.025
- 791.83** **Circuses**
⇒ 792.73
- 791.86** **Fights involving animals**
791.862 Bull fighting
791.866/.869 Other kinds of animal fighting. □ Bear baiting. Cock fighting
791.866/.869 ≡ 636/639
- 791.9** **Other amusements**
- 792** **THEATRE. STAGECRAFT. DRAMATIC PERFORMANCES**
⇒ 782
⇒ 82-2
- 792.01** **Theory and philosophy of theatre and stage**
 Subdivisions as at 7.01...
◇ 792.011.2 Social significance of theatre
- 792.02** **Theatrical technique.** □ Production. Staging. Stage craft. Stage management
792.021 Stage design and decor in general
792.022 Mechanical equipment and services. □ Stage machinery. Lighting equipment. Theatre workshops
792.023 Stage equipment and properties (props). Scenery. Furniture etc.
⇒ 688.74
792.024 Actors' properties (props) and equipment
◇ 792.024:687.16 Theatrical costumes
792.024.3 Make-up
⇒ 665.584.28
⇒ 687.51
792.026 Scripts. Libretti. Musical works for stage use
792.026.1 Script reading. □ Analysis. Assessment. Adjustments
792.026.2 Individual parts. Roles
792.026.8 Casting
792.027 Artistic direction. Continuity techniques
792.027.2 Dramatic direction
⇒ 792.2
792.027.4 Musical direction
⇒ 792.5
792.027.5 Dance direction
⇒ 792.8
⇒ 793
792.028 Acting. Performing
792.028.3 Acting technique. □ Speech. Movement. Gesture
⇒ 808.5
792.028.4 Interpretation of roles. □ Playwright's or actor's own conception
792.028.5 Roles or parts according to relative importance
792.028.52 Leading roles
792.028.54 Supporting roles
792.028.57 Minor roles. Walk-on parts
792.028.6 Type roles or parts
792.028.61 Individual-personality or 'character' roles
792.028.62 Heroic roles
- 792.028.63 Romantic and sentimental roles
792.028.64 Tragic roles
792.028.65 Melodramatic roles. Tragi-comic roles
792.028.66 Comedy roles
792.028.67 Comic, burlesque, parody roles
792.028.7 Singing roles
⇒ 78.087.6
792.028.8 Dancing roles
⇒ 792.8
⇒ 793.3
- 792.03** **Theatre and dramatic art according to periods, styles**
 Subdivisions as at 7.03...
- 792.05** **Theatres and stages (types, layout)**
⇒ 725.1
792.051 Outdoor, open-air theatres. Mobile theatres
792.052 Theatre premises (indoor). □ Exits. Entrances. Box offices. Foyers
792.053 Large theatres. Auditorium and stage layout
792.054 Small studio or workshop theatres. Experimental theatres
792.055 Types of stage or arena
792.055.1 Miniature stages. Booths. Stalls
792.055.2 Small platform or rostrum stages
792.055.3 Early historical types of stage. □ Stages of Hans Sachs, of Shakespeare
792.055.4 Open (round) stage. □ Arena. Theatre-in-the-round
792.055.6 Conventional modern stage (with proscenium wings, curtain and backcloth)
792.055.7 Quick transformation stages. □ Revolving stages
- 792.06** **Various questions concerning theatre**
 Subdivisions as at 7.06...
◇ 792.061 Ethical and legal contraventions. Plagiarism
- 792.07** **Theatrical profession and associated occupations and activities**
 Subdivisions as at 7.07...
◇ 792.071.1 Scriptwriters. Librettists
◇ 792.071.2 Performing and interpretative occupations. Professional stage and theatre artists
◇ 792.071.2.027 Stage directors
◇ 792.071.2.028 Stage actors, players, performers
◇ 792.072.3 Theatre writers, critics
◇ 792.073 Theatregoers. Theatre audiences. Spectators etc.
◇ 792.075 Commercial management of theatre. Theatre organization and publicity. Managers. Producers. Impresarios. Promoters. Agents
◇ 792.077 Amateur theatricals and players. Amateur dramatics and performers
- 792.08** **Theatre pieces, plays, shows (according to participation, size or scale, length etc.)**
⇒ 82-2
792.081 Solo. Monologue. Pas seul etc.
792.082 Duo. Dialogue. Pas de deux etc.
792.085 Full-cast. Large-scale. □ 'Spectaculars'
792.086 Reduced or small-scale. Intimate pieces
792.088 Short pieces. One-act plays
- 792.09** **Performances. Rehearsals. Competitions. Presentation in various media**
 Subdivisions as at 7.09...
◇ 792.091.8 Theatre tours

792.2	Spoken drama. Plays	793.2	Particular social recreations, requiring preparations
792.21	Tragedies	793.22	Harvest festivals. Christmas festivities. New Year festivities. Birthday celebrations. Wedding festivities
792.22	Comedies. Farces etc. 792.22 $\equiv$ 82-22	793.24	Tableaux vivants. Charades etc.
792.23	Melodramas. Thrillers	793.3	Art of movement. Dance $\Rightarrow$ 394.3 $\Rightarrow$ 78.085 $\Rightarrow$ 792.8 $\Rightarrow$ 796.412
792.24	Representational or informative plays 792.24 $\equiv$ 7.04	793.31	Folk, national, regional dances. Round dances. Country dances
792.241	Character and genre plays	793.32	Art of movement
792.244	Historical plays	793.322	Old, archaic dances. Modern dance. ☐ Character dances. Expressive dance (Ausdruckstanz)
792.245	Allegorical, morality plays	793.324	Artistic dances
792.246	Religious plays. Nativity, miracle, mystery, passion plays	793.327	Eurhythmic $\Rightarrow$ 792.87
792.25	Sketches. Extempores. Improvised and occasional pieces 792.252/.254 $\equiv$ 82-252/-254	793.33	Social dances for individual couples. Ballroom dancing
792.28	Propagandist or didactic plays	793.34	Social dances for groups or teams. ☐ Formation dancing. Square dancing
792.5	Musical productions $\Rightarrow$ 782	793.35	Dances associated with popular music and club culture. ☐ Disco dancing
792.54	Opera. Music drama $\diamond$ 792.54:782.1 Grand opera $\diamond$ 792.54:782.7 Opera buffa. Comic operas $\diamond$ 792.54:782.8 Operettas. Stage plays with songs	793.38	Balls. Public dances
792.57	Musicals (with dance, spectacle etc.)	793.386	Masques. Masked balls. Costume, fancy-dress balls
792.7	Variety. Vaudeville. Music hall. Cabaret. Revues. Pantomimes	793.5/.7	Parlour games. Party games
792.73	Variety. Music hall, Cabaret $\Rightarrow$ 791.83	793.5	Games involving forfeits
792.75	Revue (satirical, topical etc.)	793.7	Games not characterized by action. ☐ Instructive games. Memory games. Games testing observation, presence of mind etc. Forbidden words (e.g. yes-no games). Puzzle games. Riddles. Conundrums
792.78	Theatrical entertainments combining drama, singing, dancing, variety etc. ☐ Pantomime ('Christmas' pantomime)	793.8	Scientific recreations. Conjuring. Sleight of hand. Illusionism. Magic tricks
792.8	Presentations of art of movement. Choreography $\Rightarrow$ 793.3	794	BOARD AND TABLE GAMES (OF THOUGHT, SKILL AND CHANCE)
792.82	Ballet productions	794.02	Practical play and equipment
792.83	Staged dance Details by :793.3...	794.021	Practical procedure. Method and rules of play
792.87	Eurhythmic displays	794.022	Equipment. ☐ Board. Table. Pieces. Cards $\Rightarrow$ 685.8
792.9	Special kinds of theatre $\Rightarrow$ 688.745	794.05	Strategy of play. Tactics. Analysis etc.
792.92	Wordless performances. Mime	794.08	Participation (number of players), combinations etc. ☐ Solo, duo, team playing 794.081/.085 $\equiv$ 796.081/.085
792.94	Total theatre. Multimedia, audience-involving productions	794.088	Special playing arrangements
792.95	Son et lumière	794.088.2	Remote play. ☐ By correspondence. By telephone
792.96	Shadow, silhouette theatre. Shadow plays	794.088.4	Blind play
792.97	Puppet and marionette theatre. ☐ Punch-and-Judy shows $\Rightarrow$ 688.723/.724	794.088.6	Quick games
793	SOCIAL ENTERTAINMENTS AND RECREATIONS. ART OF MOVEMENT. DANCE $\Rightarrow$ 394.3	794.088.8	Simultaneous games
		794.09	Competitions, contests, matches etc. (and their presentation) Subdivisions as at 7.09...
		794.1	Chess
		794.12	Chess games as a whole. Phases
		794.122	Opening
		794.125	Middle game
		794.127	End game

794.14	Chess compositions	796.012.13	Speed capacity
794.2	Minor board games not dependent on chance	796.012.2	Coordinative, psychophysical capacities
794.24	Games played on a checkered board	796.012.21	Proficiency. Agility. Dexterity
794.242	Draughts (checkers) <i>Played on board of 64 squares</i>	796.012.23	Mobility. Flexibility. Suppleness
794.244	Other games on a 64-square board	796.012.26	Perceptive capacities in movement
794.246	Games on a board with more than 64 squares. □ Halma	796.012.262	Capacity for orientation
794.25	Games with pieces and boards of special design	796.012.265	Feeling for movement. Capacity for rhythmic movement
794.3	Table games with pieces or counters and an element of chance. □ Backgammon. Ludo. Dominoes. Mah-jong. Snakes and ladders	796.012.266	Balance ⇒ 796.012.45
794.4	Card games	796.012.3	Characteristics of movement
794.41	Partner card games. □ Bridge. Whist	796.012.32	Precision, accuracy in movement
794.42	Card-combination games. □ Poker. Baccarat. Vingt-et-un	796.012.33	Economy of movement. Effectiveness of movement. Rationality of movement
794.43	Card-pattern, sequence and matching games. □ Patience	796.012.35	Rhythm of movement. Accompaniment of movement. Harmony of movement. Naturalness of movement
794.45	Card games with specially designed or illustrated cards	796.012.36	Characteristics of movement flow. □ Space. Time. Dynamics
794.5	Pattern games or puzzles requiring patience or dexterity. □ Solitaire. Abacus-type puzzles. Manipulative puzzles	796.012.37	Repetition of movement: cyclic and acyclic
794.54	Jigsaw puzzles and similar puzzles	796.012.38	Structure of movement
794.8	Minor aiming games. Children's miniature games. □ Shovelboard (shuffleboard). Quoits	796.012.382	Symmetrical movement
794.83	Table games imitating sports. □ Table football	796.012.384	Asymmetrical movement
794.85	Pin-table games. Pinball	796.012.4	Basic forms of movement. Process of movement. Kinds, types of movement
794.9	Games of pure chance	796.012.41	Basic horizontal and vertical movements
794.91	Dice	796.012.412	Basic horizontal movements
794.92	Roulette	796.012.412.3	Stepping. Striding
794.93	Lotto (housey-housey, bingo) and similar random choice games. Tomboia	796.012.412.4	Walking. □ Marching
796	SPORT. GAMES. PHYSICAL EXERCISES ⇒ 379.8 ⇒ 685.6 ⇒ 685.7	796.012.412.5	Running. □ Jogging
796.01	Philosophy. Theory. Purpose. Methodology. Ethics. Aesthetics	796.012.412.8	Slowing down. Stopping
796.011	Philosophical, scientific, theoretical principles of physical culture and sport	796.012.414	Basic vertical movements
796.011.1	Aims of sporting activity (occupation) in personality formation and development	796.012.414.2	Bouncing
796.011.3	Content of training. Training methods. Training institutions	796.012.414.3	Hopping
796.011.5	Ethics of sport. Fair play	796.012.414.6	Jumping. □ Jumping upwards, downwards, horizontally
796.011.7	Sports aesthetics. The aesthetic in sport	796.012.42	Basic movements on and with the extremities or other parts of the body
796.012	Theory of movement in sport. Motor functions in sport	796.012.422	Basic movements on the extremities. □ Clambering. Climbing. Hanging. Leaning ⇒ 796.012.46
796.012.1	Motor capacities. Dynamic characteristics. Bodily characteristics	796.012.424	Basic movements with the extremities. □ With arm, hand, leg, foot
796.012.11	Strength capacity. Power	796.012.426	Basic movements on and with the torso
796.012.12	Endurance capacity ◇ 796.012.12"403" Endurance over a short period ◇ 796.012.12"405" Endurance over a long period	796.012.428	Basic movements on and with the head and neck
		796.012.43	Rotating movements
		796.012.432	Rotation around a fixed axis. □ Circling. Upstart. Spinning
		796.012.434	Rotation around a free axis. □ Flip. Somersault. Salto. Roll
		796.012.44	Basic movements with or on apparatus, animals or other persons
		796.012.442	Throw. Toss
		796.012.443	Push, put. Kick
		796.012.444	Hit. Stroke. Thrust. Shove. Bowl
		796.012.45	Basic movements with balance. □ Balancing. Hanging, hovering suspended. Gliding ⇒ 796.012.266
		796.012.46	Basic forms of movement related to force. □ Holding. Gripping
		796.012.462	Lifting. Carrying
		796.012.464	Pressing. Stemming. Pushing
		796.012.47	Movement techniques
		796.012.471	Start. Departure
		796.012.475	Handover. Changeover
		796.012.476	Turning technique. Technique on bends
		796.012.478	Finish. Run in. Finishing sprint
		796.012.5	Technique of movement. Form of movement. Phases of movement
		796.012.51	Run-up. Take-off. Delivery. Kick-off

796.012.53	Swing. Swinging. Mount. Flight phase. Support phase	796.015.24	Operational plan
796.012.55	Dismount. Recovery. Landing. Falling	796.015.26	Training cycles
796.012.562	Technique while standing, upright	796.015.28	Training structure
796.012.564	Technique on the ground, floor, mat ◇ 796.853.23.012.564 Judo Ne-Waza	796.015.3	Training periods. Stages of training
796.012.57	Ball handling. □ Technique with ball, shot, bowl	796.015.31	Preparatory period
796.012.572	Styles and forms of ball throwing	796.015.32	Competitive period
796.012.572.2	Styles of ball throwing	796.015.33	Transition period
796.012.572.22	One-handed styles of throwing a ball	796.015.36	Stages of training
796.012.572.24	Two-handed styles of throwing a ball	796.015.361	Basic training
796.012.572.4	Forms of ball throwing. □ While standing, moving. At close or long range. Free throw. Throw in	796.015.363	Build-up training
796.012.573	Styles and forms of striking a ball	796.015.365	High-achievement training and training for promising young people
796.012.573.2	Styles of striking a ball. □ One-handed or two-handed. Forehand or backhand	796.015.367	Training for highest levels of performance
796.012.573.4	Forms of striking a ball. □ Smash. Lob	796.015.38	Training down
796.012.574	Styles and forms of kicking a ball	796.015.4	Training organization. Management of training
796.012.574.2	Styles of kicking a ball	796.015.5	Methods, forms of training
796.012.574.4	Forms of kicking (or heading) a ball	796.015.52	Strength training. Muscle power training. □ Strength development
796.012.574.42	Kicking a dead ball	796.015.53	Speed training
796.012.574.43	Kicking a rolling ball	796.015.54	Endurance training
796.012.574.44	Kicking a ball in the air	796.015.542	Interval training ◇ 796.015.542"403" Short interval method ◇ 796.015.542"405" Long interval method
796.012.574.45	Penalty kick. Shot at goal	796.015.544	Continuous method
796.012.574.47	Drop kick. Punt	796.015.546	Methods alternating the load. □ Alternating-speed running
796.012.574.48	Heading a ball	796.015.55	Supplementary training
796.012.575	Passing a ball	796.015.56	Compensatory training
796.012.576	Catching, receiving, taking over a ball. □ Tackling. Winning a ball	796.015.57	Aerobic and anaerobic training
796.012.577	Ball possession	796.015.572	Aerobic training
796.012.577.2	Ball control. Ball grip	796.015.574	Anaerobic training
796.012.577.4	Moving the ball. □ Dribbling. Directing a ball. Tapping a ball	796.015.576	Hypoxic training
796.012.577.6	Feinting, dummieing with a ball	796.015.58	Circuit training. Station training. Team training
796.012.577.8	Ball juggling	796.015.59	Other methods and forms of training. □ Imitative training. Performance training.
796.012.6	Movements for sports practice. Physical exercises	796.015.6	Mental training. Autogenous training
796.012.61	Exercises for general development	796.015.62	Load. Physical capacity
796.012.62	Special exercises	796.015.628	Apportionment of effort
796.012.63	Exercises in preparation for competition	796.015.64	Overload
796.012.65	Design and composition of sports exercises	796.015.68	Gradual approach to peak performance
796.012.652	Compulsory exercises	796.015.68	Condition as regards training. Supervision of degree of training
796.012.656	Choreography of sports exercises ⇒ 793.3	796.015.8	Performance in sports
796.012.66	Formations	796.015.83	Problems of finding and encouraging young participants ('new blood')
796.012.67	Exercise sequences	796.015.84	Performance conditions and ability
796.012.8	Body positions	796.015.85	Preparedness for performance. Development of performance
796.012.81	Standing. □ Handstand. Headstand	796.015.86	Factors influencing performance ◇ 796.015.86:616-001 Injuries
796.012.82	Squatting. On all fours	796.015.868	Analysis of movement. Analysis of exercise
796.012.83	Sitting	796.017	Dangerous sports. Extreme sports
796.012.85	Kneeling	796.02	Technical matters. Practice and exercise facilities. Sports gear, equipment. Materials. Protection against accidents, hazards
796.012.86	Lying	796.021	Technical procedures
796.012.87	Supported. Hanging	796.021.2	Preparations
796.012.88	Stretched. Bent	796.022	Equipment for games, gymnastics, sports. Sports gear and accessories for competitions, contests ⇒ 623.44 ⇒ 685.6/.7
796.015	Methodology of sport. Sports training. Performance in sports. Load	796.022.7	Sports clothing ⇒ 687.14
796.015.1	Theory and methodology of training	796.023	Choice of materials
796.015.11	Principles of training	796.023.1	Surface material on which the sport is performed. □ Ground surface. Floor surface. Playing surface. Track surface <i>Use especially for nontraditional or unusual surfaces</i> <i>Specify the surface material by -03...</i>
796.015.13	Types of training		
796.015.132	Fitness training		
796.015.134	Training in technique		
796.015.136	Training in tactics		
796.015.14	Training system		
796.015.15	Process of training. Coaching		
796.015.2	Training programme. Planning of training		
796.015.22	Training plan ◇ 796.015.22"403" Short term plan ◇ 796.015.22"405" Long term plan ◇ 796.015.22"450" Annual plan		

796.025	Damage or destruction, preservation and upkeep of sports materials, equipment, gear etc.	796.071.4	Professional sports cadres. ☐ Sports teachers. Practice leaders. Trainers. Guides
796.028	Maintenance of sports facilities, sports grounds	796.071.42	Professional sports leaders. Sports officials. Sports guides. Coaches. Attendants
796.03	Sports movement. Sports activity from various points of view	796.071.43	Professional sports teachers. Trainers
796.032	Olympic movement. Olympic concept, idea	796.072	People with sporting knowledge and expertise
796.032.2	Olympic Games	796.072.2	Sports scientists. Sports theorists
796.034	Sport according to participation	796.072.3	Sports reporters. Sports writers
◇ 796.034-056.26	The physically handicapped, disabled as participants in sport	796.072.4	Referees. Linesmen. Timekeepers. Judges
796.034.2	Mass sport. National sport. Popular sport	796.073	Sports connoisseurs. Sports enthusiasts. Spectators. Sports fans
796.034.6	Top-level sport. Competitive sport	796.075	Commercial organizers. Managers. Promoters
796.035	Recreational sport. Sport for health. Keep-fit	796.077	Amateurism. Amateur sportsmen, players. Amateur trainers etc.
796.038	Sports communication. Sports relations	796.077.2	Amateur sportsmen, players
796.05	Manner of contest. Combative style. Strategy. Tactics	796.077.4	Amateurs involved with sport. ☐ Amateur teachers, coaches, trainers, attendants
796.051	Sporting strategy and objectives	796.077.4 ≡ 796.071.4	
796.052	Sporting tactics. Perception, grasp of tactics	796.077.5	Trainee sportsmen. Sports pupils, students
796.052.1	Setting tactical objectives. ☐ Tactical plan. Tactical discussion	796.078	Patronage, sponsorship of sport. Encouragement of sport
796.052.2	Tactical actions and systems	796.08	Participation (number of players), combinations etc.
796.052.22	Formation tactics. Formation systems. ☐ Attacking formations. Defensive formations	796.081	Solo sport. Individual participants (with or without competitors)
796.052.24	Match tactics. Contest tactics	796.082	Two participants (opposed, or as duo)
796.052.242	Attacking tactics. Attacking systems. ☐ Free play. Rapid attack. Rotating play	796.082.1	One-against-one. Singles
796.052.244	Defensive tactics. Defensive systems. ☐ Covering. Man-to-man marking. Marking space	796.082.2	Two together. Pairs
796.054	Tactical measures and capabilities	◇ 796.342.082.1	Tennis singles
796.054.2	Tactical capabilities	796.083	Three participants
796.054.22	Feint. Dummy	796.084	Four individual participants
796.054.23	Positional play. Placing. Change of rhythm or tempo	796.084.2	Pairs (two-a-side). Doubles players
796.054.4	Manoeuvres	◇ 796.342.084.2	Tennis doubles
796.054.6	Tactical measures	796.084.4	Four-a-side
796.054.62	Substitution of players	796.085	Normal complement of teams, sections, groups. Team. Crew
796.054.63	Time-out	796.087	Groups, teams with unspecified number of members
796.054.64	Delaying tactics. Wasting time. ☐ Keeping possession of the ball	◇ 796.417.2.087	Acrobatics in groups
796.056	Roles and tasks of sportsmen, players	796.088	Participants in games and matches in which the playing areas or the apparatus do not conform to the usual regulations of the sport
796.056.1	Individual sportsman, player	◇ 796.323.088	Mini-basketball
796.056.2	Team player	796.09	Contests. Records
796.056.22	Defender	796.092	Records. Best, superlative performances
796.056.222	Goalkeeper. Goaltender	796.092.2	Records. ☐ World records (international records). National records. Event records. Track records
796.056.24	Attacker. ☐ Striker. Winger	796.092.27	Technical qualities (of vehicle or equipment)
796.056.7	Other roles, tasks of players. ☐ Captain. Coxswain	796.092.29	Scoring methods. Evaluation of results
796.056.8	Relationships of sportsmen, players. ☐ Team-mates. Partners	796.092.292	Placing by evaluation
796.058	Analysis of execution of strategy and tactics	796.092.292.2	Order of merit system
796.058.2	Analysis of training	796.092.292.3	Scoring tables
796.058.4	Analysis of competition	796.092.293	Point scoring. Point tables
796.06	Organization of sport, in general	796.092.298	Units of measurement in the scoring of sports results. ☐ Measurable and nonmeasurable units
796.062	Organization and management of sport	796.093	Contests. Matches. Competitions. Races
796.063	Rules and regulations. ☐ Rules of competition. Rules of games and sports	796.093.1	Contests. Championships
796.063.4	Competition arrangements. Formalities. Competition protocol. ☐ Regulations. Opening. Announcement of results	796.093.11	Conditions of participation
796.065	Functions in sport (and their execution). ☐ Refereeing	796.093.111	Conditions of participation according to age groups, sex and age
796.07	Kind of participation in sport	Specify by -053... and/or -055...	
<i>Details of personal characteristics by -05...</i>		796.093.112	Conditions of participation based on qualifications, selection, weight etc.
796.071	Professionalism. Professional sportsmen and cadres	796.093.113	Disqualification
796.071.2	Professional sportsmen, players	796.093.117	Conditions of participation according to classes or categories of vehicle

796.093.2	Duration of contests <i>Specify by "..."</i> ◇ 796.093.2"414.24" Lasting 24 hours ◇ 796.093.2"420.6" Lasting 6 days	796.093.645.1	Modern pentathlon <i>Denote each discipline by :...</i> ◇ 796.093.645.1:796.864 Fencing ◇ 796.093.645.1:799.311.2 Pistol shooting
796.093.3	Courses. Distances <i>Specify distances by adding (181...)</i> ◇ 796.421.093.3(181km20) 20 kilometre walk	796.094	Screen, film, motion-picture presentation of sports events
796.093.32	Circuits	796.096	Sound radio presentation of sports events. □ Radio commentaries
796.093.33	Straight course	796.097	Television presentation of sports events
796.093.34	Relay course <i>Denote the number of legs (stages) by adding the appropriate digit</i>	796.1	Outdoor games and recreation ⇒ 379.84
796.093.35	Hurdling course. Obstacle course. Slalom course	796.11	Children's games
796.093.352	Hurdling course	796.12	Countryside and beach games
796.093.354	Obstacle course	796.13	Singing games. Round games. Dancing games. Running games
796.093.356	Slalom courses in general	796.14	Running and agility games. □ Vaulting. Leap-frog. Hide-and-seek. Tag
796.093.356.1	Super-giant slalom	796.15	Games, recreation with model vehicles <i>Denote competitions by adding .093</i> ⇒ 629 ⇒ 681.5 ⇒ 689
796.093.356.2	Giant slalom	796.152	Games, recreation with model land vehicles
796.093.356.4	Slalom. Special slalom	796.154	Games, recreation with model water vehicles (model boats, ships)
796.093.356.6	Parallel slalom	796.156	Games, recreation with flying models (model aircraft etc.)
796.093.36	Course with signals, markers. □ With buoys	796.2	Games of motion and skill with special equipment
796.093.37	Courses at heights or depths <i>Specify the height or depth by (181...)</i> ◇ 797.215.093.37(181m8) Skin diving at 8 metres	796.21	Games with ropes (e.g. skipping ropes), hoops, swings
796.093.38	Long distance competitions ◇ 797.553.093.38 Long distance hang-gliding	796.22	Spinning-top games
796.093.4	Contests according to form and development	796.24	Games with quoits, corks, sticks and throwing rings
796.093.42	Kinds of match	796.25	Cup-and-ball games. Diabolo. Yoyos
796.093.422	Training game, match	796.26	Games involving throwing at targets
796.093.423	Friendly game, match	796.262	Darts
796.093.426	Championship game, match. Championships	796.27	Croquet. Pall-mall
796.093.427	Cup game, match. Cup competition	796.28	Games with bowls, skittles
796.093.43	Round-robin tournaments or elimination tournaments	796.282	Skittles
796.093.432	Round-robin competitions	796.283	Bowling. Ten-pin bowling
796.093.434	Elimination competitions. Knock-out matches. Qualifying matches. □ Heats. Finals. Time trials	796.285	Games involving bowling at a target. □ Bowls. Jeu de boules (pétanque). Boccia
796.093.45	Remote contests. Correspondence contests	796.3	Ball games ⇒ 797.25
796.093.46	Tournaments. Tourneys. Regattas	796.31/32	With ball thrown by hand
796.093.48	Open or closed competitions	796.31	High ball games with small ball. □ Fives. Field ball. Balle indienne. Jägerball. Balle cavalière
796.093.482	Open competitions	796.32	High ball games with large ball
796.093.484	Closed competitions. Invitation competitions	796.322	Handball
796.093.5	Contests according to manner and location	796.323	Basketball. Korfball. Netball
796.093.52	On closed tracks	796.323.2	Basketball
796.093.54	On roads, streets	796.323.4	Netball. Korfball. Korbball
796.093.55	In open country. Cross-country. Cross events	796.325	Volleyball
796.093.56	On hills	796.328	Other high ball games with large ball. □ Flyball. Fistball. Slingball
796.093.57	Classic races and contests ◇ 796.61.093.57(44) Tour de France	796.33	Ball games in which the ball is played with foot and hand
796.093.58	Contests held out-of-doors or indoors	796.332	Association football (soccer)
796.093.582	Out-of-doors	796.333	Rugby football and related games
796.093.584	Indoors	796.333.3	Rugby union football
796.093.6	Contests according to number of events, disciplines. Combined competitions	796.333.4	Rugby league football
796.093.61/62	Combined competitions in disciplines of one type of sport ◇ 796.42.093.61 Combined athletic events	796.333.5	Gaelic football
796.093.615	Five events ⇒ 796.093.645.1		
796.093.62	Ten events		
796.093.64	Combined competitions in disciplines of several sports		
796.093.642	In two types of sport ◇ 796.92.093.642 Biathlon		
796.093.643	In three different sports. □ Triathlon (running, swimming, cycling)		
796.093.645	In five types of sport		

796.333.6	Australian football	796.422.12	Short-distance running (less than 1000m). Sprints
796.333.7	American football. Canadian football	◇ 796.422.12.093.3(181m100) 100 metre sprint	
796.334	Pushball	◇ 796.422.12.093.344(181m100) 4x100 metres relay	
796.34	High ball games with striking implement. Racket games	◇ 796.422.12.093.352 Hurdling	
796.342	Tennis. Lawn tennis	Middle-distance running (1000m to less than 10 000m)	
796.344	Games played with shuttlecock. Badminton	◇ 796.422.14.093.354(181m2000) 2000 metre steeplechase	
796.345	Pelota, jai alai etc.	Long-distance running (10 000m and over)	
796.346	Squash. Rackets	◇ 796.422.16.093.3(181km42) 42 kilometre flat event. Marathon (42.195km)	
796.35	Low ball games with striking implement	796.43	Field events
796.352	Golf	796.431	Jumping
796.352.2	Conventional golf	796.431.1	High jump
796.352.4	Miniature golf. Mini-golf	796.431.2	Long jump
796.354	Hurling. Camogie. Shinty	796.431.25	Triple jump. Hop-step-and-jump
796.355	Hockey (field hockey)	796.431.3	Long high jump
	⇒ 796.966	796.431.4	Pole vault
796.356	Lacrosse	796.433	Throwing
796.357	Baseball and softball	796.433.1	Shot putting. Weight throwing
796.357.2	Baseball	796.433.2	Javelin throwing
796.357.4	Softball	796.433.3	Discus throwing
796.358	Cricket	796.433.4	Hammer throwing
796.38	Table ball games	796.433.7	Throwing of other objects (e.g. balls, poles)
796.382	Billiards and similar games. □ Snooker. Pool	796.5	Touring. Walking. Mountaineering. Orientation sports. Camping
796.386	Table tennis. Ping-pong	⇒ 338.48	
796.39	Other ball games	796.51	Walking. Rambling. Hiking. Marching
796.4	Gymnastics. Acrobatics. Athletics	⇒ 338.488.1:796.51	
	⇒ 371.7	796.52	Climbing
	⇒ 613.71	796.524	Hill climbing. Hill walking
	⇒ 615.825	796.525	Mountaineering
	⇒ 796.8	796.526	Rock climbing
796.41	Gymnastics	796.54	Camping. Tenting
796.411	Free exercises. Mass exercises	⇒ 355.716	
796.412	Rhythmic gymnastic exercises	⇒ 338.481.2:796.54	
	⇒ 793.3	796.55	Caving. Potholing
796.412.2	Competitive gymnastics. Artistic gymnastics	796.56	Orientation sports. Orienteering
796.412.22	Without apparatus	796.57	Touring with vehicles. □ Touring with bicycle, motorcycle, motorboat, motorcar, skis, rowing boat, scull, kayak, canoe
796.412.24	With apparatus. □ With hoops, ropes, sticks		⇒ 338.48
796.413	Balancing exercises (with or without apparatus)	796.6	Wheel sports. Cycling. Roller sports
796.413.12	On tightrope	⇒ 796.7	
796.414	Exercises on fixed equipment	796.61	Cycling
796.414.2	On the horizontal bar	◇ 796.61.093.52 Cycle racing on tracks	
796.414.4	On parallel bars	◇ 796.61.093.52.093.2"420.6" Six-day track race	
796.414.5	On asymmetric bars	◇ 796.61.093.54 Cycle racing on roads, streets. Circuit racing. Criterium racing	
796.414.6	On the horse, pommel horse	◇ 796.61.093.55 Cross-country cycle racing. Cyclo-cross	
796.416	Trapeze and ring exercises. Climbing and hanging exercises (on rope, ladder, wall-bars etc.)	796.62	Artistic cycling. Figure-cycling
796.417	Acrobatic exercises. □ Floor exercises	796.63	Cycle ball
796.417.2	Acrobatics	796.64	Cycle polo
796.417.3	Floor gymnastics	796.69	Roller sport
796.417.4	Agility exercises. Tumbling (on the ground or on a mat)	⇒ 685.362	
796.418	Jumping exercises	Roller skating	
796.418.1	Free jumping exercises	796.691	796.691.2/.4 ≡ 796.912/914
796.418.2	Skipping-rope exercises. Skipping	796.692	Roller hockey
796.418.3	Vaulting-horse exercises. Vaulting	796.694	Skateboarding
796.418.7	Exercises on the trampoline. Trampolining	796.696	Roller-skiing. Grass-skiing
796.42	Athletics in general	796.698	⇒ 796.92
	Denote distances by .093.3 with (181...)		Roller sport with sails. □ Windskiating
796.421	Walking		
	◇ 796.421.093.3(181km20) 20 kilometre walking event		
796.422	Running		
796.422.1	Kinds of race		
	Specify by .093.3..., denoting the distance by (181...)		

796.7	Motoring. Motorcycling	796.9	Winter sports. Ice games. Skiing. Sledging
796.71	Motoring. Motor sport	796.91	Ice skating
	◇ 796.71.093.52 Track racing		⇒ 685.36
	◇ 796.71.093.54 Road racing. Rallying		⇒ 725.86
	◇ 796.71.093.55 Autocross	796.912	Artistic skating. Figure skating
796.712	Stock-car racing	796.914	Speed skating
796.714	Go-karting		◇ 796.914.093.3(181m500) 500 m. speed skating
796.72	Motorcycling	796.92	Skiing. □ On-piste skiing. Cross-country skiing
	◇ 796.72.093.52 Speedway racing		◇ 796.92:796.56 Ski orienteering
	◇ 796.72.093.55 Motocross		◇ 796.92:796.57 Ski touring
796.74	Motorcycle polo		⇒ 796.696
796.8	Combat sports. Self-defence sports. Heavy athletics. Trials of strength	796.921	Snow-shoeing
	<i>Denote weight by 'kg...'</i>	796.922	Nordic skiing. Langlauf. Cross-country skiing
796.81	Freestyle wrestling (all-in wrestling)		◇ 796.922.093.3(181km50) 50 km. cross-country event
796.814	Sambo		◇ 796.922.093.34(181km10) 4x10 km. cross-country relay
796.815	Sumo		◇ 796.922.093.612 Nordic combination (15 km. cross-country and ski jumping)
	⇒ 796.853.2		◇ 796.922.093.642 Biathlon
796.817	National forms of wrestling. □ 'Schwingen'. Glima	796.925	Ski-jumping. Speciality jumping. Ski flying
796.819	Other types of wrestling. □ Pankration	796.926	Alpine skiing. Downhill skiing
796.82	Olympic wrestling. Graeco-Roman wrestling		◇ 796.926.093.33 Downhill race
796.83	Boxing		◇ 796.926.093.356 Slalom. NOTE: For subdivisions see 796.093.356...
796.835	Kick-boxing. All-style boxing	796.927	◇ 796.926.093.61 Combined event
796.836	Savate. French boxing		Ski acrobatics. Trick skiing. Freestyle skiing.
796.837	Chinese and Korean boxing styles		□ Ski ballet. Hot-dog skiing
	⇒ 796.855	796.928	Ski patrol events. Patrouillenlauf
	⇒ 796.856	796.929	Ski sailing
796.838	Thai boxing (Burmese boxing)	796.93	Ski-joring (ski-kjöring)
796.839	Other types of boxing	796.94	Skibobbing and alpine surfing
796.85	Oriental (Asiatic) combat and self-defence sports. Martial arts	796.942	Skibobbing
796.853	Budo sports. Japanese combat and self-defence styles	796.947	Monoskiing
	Budo without weapons	796.948	Alpine surfing. Snow surfing. Snowboarding
796.853.2	Judo	796.95	Sledging. Sleighing. Tobogganing
796.853.23	◇ 796.853.23.012.564 Judo Ne-Waza	796.951	Sledging. Tobogganing
796.853.24	Aikido	796.952	Bobsleighing
796.853.25	Jujitsu (jujutsu, jiu-jitsu)	796.958	Ice sailing. □ Ice yachting. Ice surfing. Windskiing
796.853.26	Karate	796.96	Ice games
796.853.4	Budo with weapons	796.962	Target games on ice. □ Curling. Ice skittles
796.853.42	Kendo	796.964	Bandy
796.853.45	Kyudo	796.966	Ice hockey
	⇒ 799.322	796.97	⇒ 796.355
796.855	Wushu (Kung-fu). Chinese combat and self-defence styles		Sports contests on ice and snow. □ Races on cycles, motorcycles, motorcars, bobcats, animals
	⇒ 796.837		⇒ 796.6/.7
796.856	Korean combat and self-defence styles. □ Taekwondo. Hapkido		⇒ 796.95
	⇒ 796.837		⇒ 798.2/.9
796.857	Indonesian combat and self-defence styles. □ Pentjak-silat	797	WATER SPORTS. AERIAL SPORTS
796.859	Other oriental (Asiatic) combat and self-defence styles. □ Bando. Arnis. Kuntao. Kalaripayat		<i>The 796.0... auxiliaries are also applicable at 797 and its subdivisions</i>
796.86	Fencing	797.1	Watercraft sports. Boating
796.862	With foil, electric foil		◇ 797.1.093.46 Regattas
796.864	With épée, electric épée	797.11	Pleasure boating. Pleasure cruising
796.865	With sabre	797.12	Boating without sails
796.88	Weightlifting	797.122	Kayaking. Canoeing. Paddling
796.89	Other types of strength and self-defence sports	797.122.2	Kayaking
	Power sports. □ Bodybuilding. Powerlifting	797.122.3	Canoeing. □ Canadian canoeing
796.894	Tug-of-war	797.122.5	Paddling
796.896		797.123	Rowing
		797.123.1	Sculling
		797.123.2	Rowing with oars

797.14	Sailing. Yachting	797.561.2	Parachute jumping. □ Target jumping. Style jumping. Free-fall jumping (skydiving)
797.142	In centre-board or drop-keel boats. □ Dinghy sailing. Jolly-boating	797.561.3	Flying with parachute. □ Paragliding
797.144	In keel-boats. Yachting	797.562	Bungee jumping
797.146	Sailing in multihulled boats	797.563	Sport with rocket-propelled devices
797.146.2	Catamaran sailing		
797.146.3	Trimaran sailing	798	RIDING AND DRIVING. HORSE AND OTHER ANIMAL SPORTS
797.148	Sailboarding (windsurfing)		<i>The 796.0... auxiliaries are also applicable at 798 and its subdivisions</i>
797.15	Motorboating		⇒ 636.046
797.151	With inboard motor		⇒ 636.088
797.152	With outboard motor		⇒ 685.1
797.17	Sports using other watercraft etc.		⇒ 725.1:78/79
797.175	Sports using water cycles, pedal boats, podoscaphes	798.2	Riding
797.176	Waterskiing	798.22	Dressage
	◇ 797.176:797.553 Parasailing on waterskis	798.23	Riding exercises. Schooling
	◇ 797.176.093.356 Slalom course	798.24	Riding in schooling lanes, manèges
797.176.2	Waterski acrobatics	798.25	Mounted games (equestrian games etc.)
	<i>Denote the movements by .012... (from 796.012...)</i>	798.252	Polo (equestrian polo)
	◇ 797.176.2.012.434 Salto	798.256	Riding tournaments. Jousting
	◇ 797.176.2.012.55 Landing		⇒ 394.7
797.178	Surfing (surf-riding)	798.26	Equestrian jumping. Showjumping
797.18	Team games with boats. □ Canoe polo. Water jousting	798.27	Acrobatic riding. Circus riding
797.2	Swimming. Diving. Aquatic games	798.28	Cross-country riding. Riding for hunting
797.21	Swimming (without artificial aids)		⇒ 338.488.1:656...
	<i>Denote distances by .093.3 with (181...)</i>	798.29	Three-day eventing (dressage, cross-country and show-jumping)
797.212	Swimming strokes. Swimming disciplines		
797.212.2	Breast stroke	798.4	Horse racing. Mounted racing
797.212.3	Back stroke	798.42	Flat racing
797.212.4	Freestyle swimming. Crawl		
797.212.6	Butterfly stroke. Dolphin stroke	798.44	Steeplechase
797.212.7	Medley swimming	798.6	Driving with horses. Coaching, carriage-driving. Trotting
797.212.8	Swimming in open water (river, lake, sea)		Driving teams of horses
	◇ 797.212.8.093.38 Long distance swimming	798.64	◇ 798.64.082.2 Teams of two horses
797.214	Rescue swimming, life saving		◇ 798.64.083 Teams of three horses. Troika
	⇒ 614.81	798.66	Driving contests. Trotting, harness racing
797.215	Underwater swimming. Skin diving	798.662	Driving contests
797.217	Artistic swimming. Water ballet. Synchronized swimming		◇ 798.662.093.354 Obstacle races
797.22	Swimming with artificial aids		◇ 798.662.093.55 Cross-country races
797.25	Aquatic games		Trotting races
797.253	Water polo	798.664	Gymkhanas
797.259	Other ball games played in water	798.668	
797.26	Diving	798.8	Canine sports. Dog racing etc.
797.262	Springboard diving	798.9	Other animal contests, e.g. with birds
797.264	High diving		◇ 798.9:636.596 Pigeon racing
797.5	Aerial sports	799	FISHING. HUNTING. SHOOTING AND TARGET SPORTS
797.54	Ballooning. Flying in dirigibles, airships		<i>The 796.0... auxiliaries are also applicable at 799 and its subdivisions</i>
797.55	Flying in aeroplanes	799.1	Fishing
797.551	In gliders. Gliding		<i>Details by :639.2...</i>
797.551.2	Stunt flying, aerobatics in gliders	799.11	Freshwater fishing
	◇ 797.551.2.012.434 Rotation around a free axis (looping the loop)	799.12	Fishing with rod and line. Angling
	◇ 797.551.2.012.66 Formation flying	799.13	Net fishing in inland waters
797.553	Hang-gliding. Flying by kite	799.15	Coastal fishing. □ Shrimping. Crab fishing
797.555	In powered aeroplanes (fixed-wing aircraft)	799.16	Sea fishing
797.555.2	Stunt flying, aerobatics in powered aircraft	799.2	Hunting
	<i>Details as at 797.551.2</i>		<i>Details by :639.1...</i>
797.556	In helicopters or other aircraft with movable wings	799.3	Shooting. Target sports
	<i>Specify by :629.735...</i>		⇒ 623.44
797.56	Other sports with flying and jumping equipment		
797.561	Parachuting (as sport)		

799.31	Shooting with firearms	799.313	With military firearms
799.311	With sports weapons	799.315	With air guns
799.311.2	With pistols and revolvers	799.32	Shooting with other weapons
799.311.4	With sport rifles (special sports weapons)	799.322	Archery. ☐ Crossbow shooting
799.312	With hunting weapons (shotguns etc.). ☐ Clay-pigeon shooting. Skeet shooting		⇒ 796.853.45
799.312.2	Shooting at a moving target	799.324	Throwing at a target
799.312.5	Clay pigeon shooting (with hunting weapons)	799.324.2	With throwing stick, boomerang
799.312.52	Trap shooting. Olympic shooting	799.324.4	With spear
799.312.54	Skeet shooting	799.324.6	With catapult
		799.326	Shooting with a blowpipe (blowgun)

8 LANGUAGE. LINGUISTICS. LITERATURE

80 GENERAL QUESTIONS RELATING TO BOTH LINGUISTICS AND LITERATURE. PHILOLOGY

Class here works about philology in the broadest sense, including study of literary texts as materials of general cultural history. For philology in the narrow sense (linguistics, especially historical) see 81

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|---------------|--|---------------|--|
| 801 | PROSODY. AUXILIARY SCIENCES AND SOURCES OF PHILOLOGY | 801.655.6 | Amorphous verse. Free rhythms |
| | | 801.655.7 | Free verse |
| 801.6 | Prosody: metre, rhythm, rhyme and verse pattern | 801.656 | Alternating, combined patterns |
| | <i>The prosody of particular languages may be classed by using 811...:801.6 for linguistic studies or 821...:801.6 for literary studies. Further particulars may be denoted by common auxiliaries of form (Table 1d)</i> | 801.66 | Rhyme |
| | ◇ 801.66(03) Rhyming dictionaries | ⇒ 801.632 | |
| | ◇ 811.111:801.66(03) English rhyming dictionaries | ⇒ 82-1 | |
| | ⇒ 81'342 | 801.661 | General concept. □ Purity. Consonance. Symbolism |
| 801.61 | Generalities | 801.662 | Degree, extent of rhyme (number of rhyming syllables). □ Monosyllabic, disyllabic, trisyllabic rhyme |
| | ⇒ 81'342.8 | 801.663 | Linguistic-grammatical classification of rhymes. □ True, broken, split rhyme |
| | ⇒ 81'342.9 | 801.664 | Position of rhyme in the line (end, middle, beginning) |
| 801.611 | Quantity | 801.665 | Rhyme plan or scheme. Rhyme pattern (of the stanza) |
| 801.612 | Stress, accent (word and phrase) | 801.67 | Stanzas, strophes, couplets, verses (of a poem) |
| 801.613 | Rhythm | 801.671 | Generalities on stanzas. Strophe, antistrophe etc. |
| | ⇒ 81'367.2 | 801.671.1 | Rhythmic grouping |
| 801.614 | Verse melody | 801.671.2 | Periods (sentences) |
| 801.615 | Verse harmony | 801.671.4 | Stanzas according to parity. □ Isometric. Metabolic |
| 801.63 | Metre. Metrical measures, feet | 801.671.5 | Interconnection, links between stanzas |
| 801.631 | Metrical measures in general | 801.671.51 | Refrains |
| 801.631.1 | Arsis and thesis. Rise and fall. □ Anacrusis | 801.671.6 | Correspondence between stanzas |
| 801.631.2 | Pause. Caesura. Central and terminal pause | 801.672 | Classical strophes. □ Sapphic. Asclepiadic. Alcaic |
| 801.631.3 | Enjambement | 801.673 | Germanic verse forms |
| 801.631.4 | Hiatus | 801.673.1 | Nordic verse forms. □ Edda stanzas |
| 801.631.5 | Prosodic licence | 801.673.2 | English verse forms |
| | ⇒ 81'342.62 | 801.673.3 | German verse forms. □ Nibelungen, Minnesänger stanzas |
| 801.631.51 | Contraction. □ Elision. Synaloepha. Crasis. Apocope. Syncope. Aphaeresis. Syneresis (synaeresis) | 801.675 | Romance verse forms |
| | Prolongation. □ Diaeresis | 801.675.1 | French verse forms. □ Quatrain. Rondeau. Rondel. Villanelle |
| 801.631.6 | Verse endings | 801.675.2 | Italian verse forms. □ Ottava rima. Terza rima. Madrigal. Canzone. Sonnet. Triolet |
| 801.631.61 | Catalexis | 801.675.3 | Spanish verse forms. □ Decima. Glosse. Cancion |
| 801.631.62 | Rhyme quantities. Grammatical endings (from a prosodic point of view) | 801.676 | Slavic verse forms. Other western forms |
| 801.632 | Alliteration. Assonance | 801.677 | Oriental verse forms (of Indo-European and Semitic literatures) |
| | ⇒ 801.66 | 801.677.1 | Indic verse forms |
| 801.633 | Disyllabic feet. □ Iambs. Trochees. Spondees. Pyrrhics | 801.677.3 | Persian verse forms |
| 801.634 | Trisyllabic feet. □ Anapaests. Dactyls. Tribachs. Molossi. Cretics. Bacchii | 801.677.6 | Arabic verse forms |
| 801.635 | Polysyllabic feet. □ Choriamb. Ionics. Dochmii | 801.678 | Other Asian verse forms. African verse forms |
| 801.65 | Verse line and syllable patterns (according to numerical characteristics) | 801.7 | Auxiliary sciences and studies of philology |
| 801.651 | Rhythmic measures or units. □ Monopody, dipody, tripody etc. | ◇ 801.7:81'22 | Semiotics as philological auxiliary science. Study of philology through scripts etc. |
| 801.653 | Stress or accentuation. Arsis. Accentual verse. □ Diarsis, triarsis etc. | 801.73 | Hermeneutics. Textual criticism, interpretation, explanation. Recension. Exegesis |
| 801.654 | Metrical quantity. Classical or quantitative verse. □ Pentameters, hexameters, iambic trimeters etc. | ⇒ 27-271.5 | |
| 801.655 | Lines according to number of syllables. Syllabic verse | ⇒ 27-273 | |
| 801.655.1 | Lines of fewer than ten syllables | ⇒ 27-275 | |
| 801.655.2 | Decasyllables | ⇒ 27-277.2 | |
| 801.655.3 | Dodecasyllables. Alexandrines | | |
| 801.655.5 | Blank verse | | |

801.8	Sources of linguistics and philology. Collections of texts	◇ 808.5:159.925 Mimicry. Expressiveness. Including: Gesture, body language
801.81	Oral sources, mainly popular ◇ 801.81:398.21 Folk tales etc. as philological source material	◇ 808.5:37.016:808.5 Training. Including: Voice development, phonetic training
801.82	Written sources. Literary texts ◇ 801.82:821.112.2-2Schiller Dramatic works of Schiller as philological source material	◇ 808.5:612.2 Breathing and breath control ◇ 808.5:612.761 Speed, rapidity of speech ◇ 808.5:612.78 Voice and diction. Including: Pitch, tone, timbre ◇ 808.5:612.789 Articulation etc.
801.83	Mixed sources	◇ 808.5:616.89-008.434 Speech disorders. Including: Stammering, lisping
808	RHETORIC. THE EFFECTIVE USE OF LANGUAGE	◇ 808.5:81'342.9 Modulation of speech. Accentuation, rhythm, intonation ⇒ 159.946.3 ⇒ 347.781.8 ⇒ 37.016:81-028.31 ⇒ 612.78 ⇒ 792.028.3 ⇒ 81'342.2 ⇒ 81'355
808.1	Authorship. Literary activity and technique. □ Creative writing. Writing in publishable form <i>Class here comprehensive works on preparation and submission of manuscripts</i> ⇒ 070.4 ⇒ 655.251(035) ⇒ 655.254.22 ⇒ 655.51 ⇒ 808.2	
808.2	Editing. Adapting and arranging copy for publication ⇒ 070.4 ⇒ 655.254.22	808.51 Public speaking. Oratory. □ After-dinner speaking. Platform speaking. Making speeches and toasts for special occasions ⇒ 27-475.2 ⇒ 808.53
808.25	Ghost-writing (ghosting) <i>Producing text to be published under another person's name (usually by editing or adapting oral material)</i>	808.53 Debating and public discussion
808.5	Rhetoric of speech. Art or technique of oral expression <i>Class here voice, expression, gesture. Denote details by colon combination if required</i>	808.54 Recitation. Art of the diseur <i>Class here oral interpretation. For choral speaking use 808.55</i> 808.543 Storytelling 808.545 Reading aloud 808.55 Theatre and stage oratory and effect. Choral speaking 808.56 Conversation

81 LINGUISTICS AND LANGUAGES

Class here works about philology in the sense of (especially historical) linguistics. Class historical descriptions of particular languages or language groups at the number for the language or group, with the addition of -112 (from 81-112)
The -11/-26 and '1/'443 special auxiliaries may be used with 81 itself to denote the subject matter of linguistics in general, and also in combination with the subdivisions of 811 for linguistic aspects of particular languages or language groups.

Denote various aspects, e.g. relationship to other disciplines by colon combination

- ◇ 81:001.82 Language classifications (as a subject), e.g. areal, typological, genetic, functional
- ◇ 81:1 Philosophy of linguistics
- ◇ 81:39 Ethnolinguistics
- ◇ 81:572 Languages and anthropology
- ◇ 811.11-112 Philology of Germanic languages

81-11	Schools and trends in linguistics <i>-11 and its subdivisions may be added to 81'1/'443</i> ⇒ 81'32 ⇒ 81'33	81-116.6 Generative linguistics (N.A. Chomsky) ⇒ 81'344.1 ⇒ 81'364 ⇒ 81'372
81-112	Diachronic linguistics (historical linguistics)	81-116.7 Tagmemics. Tagmemic grammar (K.L. Pike)
81-114	Synchronic linguistics (static linguistics)	81-117 London school (J.R. Firth, systematic grammar of M.A.K. Halliday)
81-115	Evolutionism. Comparative linguistics ⇒ 81'362	81-119 Other linguistic schools and trends
81-116	Structuralism. Structural linguistics	81-13 Methodology of linguistics. Methods and means <i>For methods identified with schools and trends use 81-11, e.g.</i> <i>Comparative (general) method 81-115.</i> <i>Diachronic method 81-112.</i> <i>Distributive method 81-116.</i>
81-116.2	Geneva school (F. de Saussure)	
81-116.3	Prague school	
81-116.4	Copenhagen school (L. Hjelmslev).	
81-116.5	Glossematics	
	American structuralism (Z.S. Harris)	

	Formal method 81-116.4. Functional method 81-116.3. Glossematic method 81-116.4. Historical method 81-112. Historical-comparative (causal) method 81-115. Method of confrontation 81-116. Method of immediate constituents 81-116.6. For psycholinguistic method see 81'23		
81-132	Method of string analysis	81'23	Psycholinguistics. Psychology of language ⇒ 159.946.3
81-133	Descriptive method	81'232	Language acquisition. Developmental linguistics
81-139	Other methods	81'233	Development of speech ability in the individual
81-2	CHARACTERISTIC FEATURES OF LANGUAGE	81'234	Pathology of speech. Neurolinguistics
81-21	Languages of so-called primitive peoples	81'234.2	Aphasia
81-22	Languages of so-called developed and civilized peoples	81'24	Practical knowledge of languages
81-23	Living languages	81'242	Native language. Native speakers
81-24	Dead, extinct languages	81'243	Foreign language. Acquired language. Second language. Non-native speakers
81-25	Spoken languages. Languages without extant literature	81'244	Official languages. State-recognized languages
81-26	Written languages. Literary languages	81'246	Mono- and multilingualism
81'0	ORIGINS AND PERIODS OF LANGUAGE. PHASES OF DEVELOPMENT <i>Use only in combination with subdivisions of 811. '03, '05, '07 and '09 are not generally applicable: they are reserved for specific applications (e.g. at Latin, Greek, Hebrew, Egyptian)</i>	81'246.1	Monolingualism
		81'246.2	Bilingualism
		81'246.3	Multilingualism (plurilingualism) in general. Trilingualism in particular
81'01	Old period. Archaic period	81'25	Theory of translation ⇒ =030 ⇒ 347.78.034 ⇒ 519.768 ⇒ 655.525 ⇒ 81'253 ⇒ 81'322 ⇒ 81'322.4
81'02	Classical period	81'253	Interpreting (oral translation). □ Simultaneous translation
81'04	Middle period	81'255	Written translation
81'06	Modern period	81'255.2	Literal translation. □ Keys. Cribes ◇ 81'255.2:6 Technical translation
81'08	Revived language	81'255.4	Stylistic, literary translation
81'1	GENERAL LINGUISTICS <i>Class here introductions, monographs etc. on (general) linguistics as well as documents comprising papers on various linguistics subjects</i>	81'26	Language planning. Control of language. Standardization of language ⇒ 81'272
81'16	Metatheory of linguistics	81'27	Sociolinguistics. Usage of language
81'2'44	SUBJECT FIELDS AND FACETS OF LINGUISTICS	81'271	Usage and purity of language
81'22	General theory of signs in relation to linguistics. Semiology. Semiotics	81'271.1	Correct usage and misuse of spoken and written forms
81'221	Nonverbal communication ⇒ 811.93	81'271.12	Correct usage
81'221.2	Mime, facial expression and gesture languages	81'271.14	Deviations (incorrectness). Solecisms. Slips (inadvertencies)
81'221.22	Natural mime, facial expression and gesture languages	81'271.16	Acceptability of spoken and written forms. Acceptable utterances (as judged by natives)
81'221.24	Conventionalized mime and gesture languages. □ Deaf, deaf-mute gesture languages	81'271.2	Purity of language
81'221.4	Visual code-sign languages	81'272	Applied sociolinguistics. Language policy ⇒ 81'26
81'221.5	Aural sign languages. Sound-code languages	81'276	Social dialects. Vocational slang. Special vernacular forms, idioms, jargons etc. associated with particular social strata or user groups <i>Denote further details at the subdivisions by :... and -05... Class social dialects, jargons etc. in particular languages by adding '276 or its subdivisions to the language numbers in 811...</i>
81'221.7	Tactile sign languages (touch or pressure) ⇒ 003.24		◇ 811.133.1'276.6:34 Language of French lawyers ◇ 811.162.1'276.3-053.5 Language of Polish schoolchildren

81'276.1	Language or idiom of particular social classes. Sociolects	81'324	Statistical linguistics
81'276.11	Language or idiom of the upper strata. Cultured, educated parlance	81'33	Applied linguistics
81'276.12	Language or idiom of the lower strata. Common, popular, familiar parlance	81'34	Phonetics. Phonology
81'276.16	Polite, refined language of caste, court or other hierarchy	81'342	Phonetics
81'276.2	Vulgar parlance. Slang in general ⇒ 81-25	⇒ 801.6	
81'276.3	Language or idiom of a particular age group or sex Specify by -053... and -055... ◇ 81'276.3-053.2 Infant or child talk. Mothers' or nurses' baby talk, prattle ◇ 81'276.3-053.5 Schoolchildren's talk ◇ 81'276.3-053.6 Teenage language, idiom or slang ◇ 81'276.3-055.1 Male parlance or idiom ◇ 81'276.3-055.2 Female parlance or idiom	⇒ 81'35	
81'276.4	Secret language, jargon or cant	81'342.1	Acoustic phonetics. Analysis and properties of (vocal) sound. □ Pitch. Intensity. Quantity. Timbre ⇒ 534
81'276.45	Thieves' cant or slang. 'Argot', 'Rotwelsch' etc.	81'342.2	Articulatory phonetics. Phonation, articulation, utterance of (vocal) sounds ⇒ 159.946.3 ⇒ 612.78
81'276.5	Occupational slangs ◇ 81'276.5:355.11 Soldiers' slang ◇ 81'276.5:378.18 Students' slang	81'342.21	Voice. Vocal organs or larynx
81'276.6	Professional and technical language or parlance. Technolects ◇ 81'276.6:34 Legal language ◇ 81'276.6:355 Military language ◇ 81'276.6:62 Technological and engineering language	81'342.22	Analysis of phonation: breathing, duration, articulation
81'28	Dialectology. Geographical linguistics. Areal linguistics	81'342.3	Auditive phonetics
81'282	Dialects. Local and regional language. Variants. Vernaculars Class dialects of particular languages by adding '282 (or its subdivisions) and appropriate (1/9) auxiliaries denoting localities or regions to the language numbers in 811...	81'342.4	Speech sounds: vowels, consonants ⇒ 003.28 ⇒ 81'344.2/4
81'282.2	Local (district) dialects ◇ 811.161.2'282.2(477.87) Subcarpathian Russian (dialect of Ukrainian)	81'342.41	Vowels (sonants)
81'282.3	Regional, provincial dialects ◇ 811.134.2'282.3(468.1) Andalusian dialect of Spanish	81'342.415	Diphthongs
81'282.4	National variants or dialects of a language outside the country of origin	81'342.42	Consonants
81'282.8	Contact languages. Pidgins and creoles Class here collective information about pidgins and creoles. For individual pidgins and creoles see the subdivisions of =1/=8 (Table 1c) or the subdivisions of 811, which may be derived from them by replacing the equals sign with 811 and a point ◇ 811.111.8 English-based pidgins and creoles. NOTE: derived from =111.8	81'342.5	Sounds according to articulation position (tongue, lips, etc.) Gutturals. Velars Velopalatals Palatals Dentals Labials Cerebrals Nasals
81'286	Geographical linguistics. Dialect geography. Distribution in specific language areas ◇ 81'286(084.42) Language and word atlases	81'342.51	Phonetic (sound) changes, mutations and laws
81'32	Mathematical linguistics ⇒ 519.76	81'342.52	Phonetic laws (changes due to modified articulation factors). □ Grimm's law. Verner's law
81'322	Computational linguistics	81'342.53	Phonetic changes without modified articulation factors ⇒ 801.631.5
81'322.2	Natural language processing	81'342.54	General changes. □ Analogy. Assimilation. Aphaeresis. Apocope. Syncope. Metathesis
81'322.3	Parsing	81'342.55	Vowel changes. □ Umlaut. Crasis. Ablaut. Monophthongization and diphthongization.
81'322.4	Machine translation	81'342.56	Vowel sandhi and elision. Vowel harmony
81'322.5	Speech analysis	81'342.57	Consonantal changes. □ Assimilation. Dissimilation. Doubling. Interchange. Consonantal sandhi and elision
81'322.6	Speech synthesis	81'342.6	Syllables and syllabification (syllabification)
		81'342.61	Word cohesion. Euphony. Word accent. Word tone. □ Enclisis. Proclisis ⇒ 801.61
		81'342.62	Phrase and sentence modulation. Rhythm and intonation of speech ⇒ 801.61 ⇒ 808.5 ⇒ 81'367.2
		81'342.621	Phonology. Phonemics For types of language according to phonological classification use 81'442
		81'342.622	Generative phonology ⇒ 81-116.6 ⇒ 81'364 ⇒ 81'372
		81'342.623	Phonemes ⇒ 81'342.4
		81'342.7	Archiphonemes
		81'342.8	Cenemes
		81'344.1	Phonological system
		81'344.2	Distinctive (phonological) opposition
		81'344.22	
		81'344.24	
		81'344.3	
		81'344.32	

- 81'344.34 Evolution of a phonological system.
Phonemicization
- 81'344.4 Phonological features
⇒ 81'342.4
- 81'344.6 Phonology of representational function
- 81'344.62 Phonemic functions. □ Delimitative function.
Culminative function
- 81'35 Graphemics. Orthography. Spelling**
⇒ 003.03
⇒ 003.08
⇒ 003.2/.3
⇒ 37.016:81...
⇒ 81'342
- 81'351 Alphabet (in general)**
⇒ 003.23
⇒ 003.33/.34
- 81'352 Vowel and diphthong representation**
- 81'353 Consonant representation**
- 81'354 Systems of orthography. Phonetic orthography. Spelling reform**
- 81'355 Pronunciation**
⇒ 808.5
- 81'36 Grammar**
◇ 81'36-112 Historical grammar
⇒ 37.016:81...
⇒ 81-116.7
⇒ 81-117
- 81'362 Comparative grammar**
- 81'364 Generative grammar. Transformational grammar.** □ Finite-state grammar.
Phase-structure grammar
⇒ 81-116.6
⇒ 81'344.1
⇒ 81'372
- 81'366 Morphology. Flexions (inflexions)**
For types of language according to morphological classification use 81'443
- 81'366.2 Declension. Types of declension. Declension systems
- 81'366.3 Conjugation. Types of conjugation.
□ Objective conjugation. Subjective conjugation
- 81'366.5 Grammatical categories
⇒ 81'367.7
- 81'366.52 Gender
- 81'366.521 Masculine
- 81'366.522 Feminine
- 81'366.523 Neuter
- 81'366.53 Number
- 81'366.531 Singular
- 81'366.532 Dual
- 81'366.533 Plural
- 81'366.54 Case
- 81'366.541 Nominative
- 81'366.542 Vocative
- 81'366.543 Genitive
- 81'366.544 Accusative
- 81'366.545 Dative
- 81'366.546 Ablative
- 81'366.547 Locative
- 81'366.548 Instrumental
- 81'366.549 Other cases (e.g. in Caucasian languages).
□ Aessive. Illative. Caritive
- 81'366.55 Comparison. □ Comparatives. Superlatives
- 81'366.56 Person. □ First, second, third person
- 81'366.57 Voice
- 81'366.571 Active voice
- 81'366.573 Passive voice
- 81'366.574 Reflexive voice. Middle voice
- 81'366.58 Tense and aspect
- 81'366.581 Present
- 81'366.582 Past
- 81'366.582.2 Imperfect
- 81'366.582.3 Past historic. Preterite. Passé défini
- 81'366.582.4 Aorist
- 81'366.582.5 Perfect. Present perfect. First perfect
- 81'366.582.6 Pluperfect. Past perfect
- 81'366.582.7 Past anterior. Second past perfect. Double pluperfect
- 81'366.584 Future
- 81'366.584.2 Present future. First future
- 81'366.584.3 Future perfect. Second future
- 81'366.587 Kinds of action. □ Perfective. Cursive, durative, imperfective. Iterative. Terminative
- 81'366.59 Mood
- 81'366.591 Indicative
- 81'366.592 Optative
- 81'366.593 Imperative
- 81'366.594 Conjunctive. Subjunctive
- 81'366.596 Conditional
- 81'367 Syntax**
⇒ 81'366.54
- 81'367.2 Sentences generally: rhythm, stress, intonation, modulation
⇒ 81'342.9
- 81'367.3 Types of clause or sentence
- 81'367.32 According to the attitude of the speaker to the expressed contents
- 81'367.321 Affirmative sentences
- 81'367.322 Interrogative sentences
- 81'367.323 Optative sentences
- 81'367.33 According to the degree of their complexity. Clause complements
- 81'367.332 Simple clauses and their parts
- 81'367.332.2 Noun clauses
- 81'367.332.3 Verb clauses
- 81'367.332.6 Subject
- 81'367.332.7 Predicate
- 81'367.332.8 Copula
- 81'367.333 Clause complements: object, attributes (noun, adjective)
- 81'367.334 Incomplete clauses. Clause equivalents
- 81'367.335 Compound clauses. Sentences (with coordinate and subordinate clauses)
- 81'367.335.1 Parataxis. Coordination
- 81'367.335.2 Hypotaxis. Subordination
- 81'367.335.3 Ellipsis
- 81'367.4 Phrases from the syntactical viewpoint.
□ Word groupings and groups. Types and elements of phrases. Noun phrase (NP). Verb phrase (VP). Syntagms
⇒ 81'373.74
- 81'367.5 Construction and phrasing of sentences and clauses
- 81'367.51 Clause arrangement. Positioning of main and subordinate clauses in compound sentences.
□ Logical order, inverted order of clauses
- 81'367.52 Word arrangement. Positioning of parts of speech within clause. □ Logical order, inverted order of words
- 81'367.6 Parts of speech
⇒ 81'373.614
- 81'367.62 Lexical words. Autosemantic words
- 81'367.622 Substantives. Nouns
- 81'367.622.11 Common nouns
- 81'367.622.12 Proper nouns
- 81'367.622.13 Concrete nouns
- 81'367.622.14 Abstract nouns

81'367.622.15	Nomina agentis. Agent nouns	81'373.22	Names in nature (other than geographical or topographical) <i>Specify by :...</i> ◇ 81'373.22:524.3 Names of stars
81'367.622.16	Nomina actionis. Nouns of action	81'373.23	Names of persons, groups of persons, peoples etc. Anthroponymy
81'367.622.17	Nomina instrumenti	81'373.231	Given (personal) names. Forenames, Christian names etc.
81'367.622.18	Nomina loci	81'373.232	Family names
81'367.622.19	Nomina temporis	81'373.232.1	Surnames. Individual surnames
81'367.622.21	Collective nouns	81'373.232.2	Patronymics. Matronymics
81'367.622.22	Diminutives. Hypocorisms	81'373.232.3	Names of house(hold)s: royal, noble etc.
81'367.622.23	Augmentatives	81'373.232.4	Names of kinship groups (derived from common ancestor) ◇ 81'373.232.4(411) Scottish clan names, or ◇ 811.152.2'373.232.4 Scottish clan names
81'367.623	Adjectives	81'373.233	Names of groups with a common social bond, status or rank. □ Caste names
81'367.623.1	Concrete adjectives	81'373.234	Names of peoples or races with distinctive ethnic and linguistic characteristics <i>Specify by (=...) (Table 1f)</i> ◇ 81'373.234(=81'=82) Names of North American Indian tribes
81'367.623.2	Abstract adjectives	81'373.235	Names of inhabitants of countries, regions, localities etc. <i>Specify by (1/9) (Table 1e)</i>
81'367.623.3	Denominative adjectives	81'373.237	Names of occupations, professions
81'367.623.4	Numerative adjectives	81'373.4	Various categories of words ⇒ 81'373.72
81'367.623.6	Possessive adjectives	81'373.42	Words related in form, meaning, stem and sound
81'367.623.7	Demonstrative adjectives	81'373.421	Synonyms
81'367.623.8	Interrogative adjectives	81'373.422	Antonyms
81'367.623.91	Indefinite adjectives	81'373.423	Homonyms. □ Homographs. Homophones. Full homonyms
81'367.623.92	Relative adjectives	81'373.424	Stem-related or root-related words. Etymologically related and cognate words ⇒ 81'373.6 ⇒ 81'374.2-29:81'373.6
81'367.624	Adverbs. □ Relative adverbs. Adverbial particles	81'373.43	Neologisms
81'367.625	Verbs and verb forms	81'373.44	Archaisms
81'367.625.1	Transitive verbs	81'373.45	Foreign words. Borrowed or loan words. Calques (loan-translations). Half-translations (loan-blends) ⇒ 81'373.613
81'367.625.2	Intransitive verbs	81'373.46	Special subject words. □ Technical words ⇒ 81'276.6
81'367.625.3	Finite verbs	81'373.47	Exclamatory words. Oaths. Swearwords ⇒ 179.5 ⇒ 81'367.628
81'367.625.4	Infinitive, participles, gerunds etc.	81'373.48	Fashionable (mode) words. Clichés. Buzzwords. Slogan words etc.
81'367.625.41	Infinitive	81'373.49	Euphemisms
81'367.625.42	Supine	81'373.6	Etymology ⇒ 81'374.2-029:81'373.6
81'367.625.43	Participles. □ Present, past, future participles	81'373.61	General etymological laws
81'367.625.44	Verbal adjectives. Gerundives	81'373.611	Word formation: stems or roots, affixes, reduplication. Derivation. Basic words and derivatives
81'367.625.45	Verbal nouns. Gerunds	81'373.612	Word meanings and their changes (semantic laws)
81'367.625.5	Auxiliary verbs	81'373.612.2	Metaphoric use of words. Metaphor (in the strict sense). Metonymy. Synecdoche
81'367.626	Pronouns	81'373.612.4	Pleonasm
81'367.626.1	Personal pronouns	81'373.613	Borrowing ⇒ 81'373.45
81'367.626.2	Possessive pronouns	81'373.614	Etymology of parts of speech <i>Specify by '367.6...</i>
81'367.626.3	Demonstrative pronouns	81'373.7	Phraseology
81'367.626.4	Relative pronouns	81'373.72	Idioms. Idiomatic expressions
81'367.626.5	Interrogative pronouns	81'373.74	Phrases from the lexical viewpoint ⇒ 81'367.4
81'367.626.6	Indefinite pronouns		
81'367.626.7	Reciprocal pronouns		
81'367.626.8	Reflexive pronouns		
81'367.627	Numerals. □ Cardinals. Ordinals		
81'367.628	Interjections ⇒ 81'373.47		
81'367.63	Grammatical words (function words, synsemantic words)		
81'367.632	Articles. □ Definite, indefinite articles		
81'367.633	Prepositions		
81'367.634	Conjunctions		
81'367.635	Particles		
81'367.7	Syntactic role and use of different word categories and grammatical categories ⇒ 81'366.5 ⇒ 81'373		
81'37	Semantics		
81'371	Kinds of meaning. □ Denotation (referential meaning). Connotation (emotive meaning). Contextual meaning		
81'372	Generative semantics. Deep and surface structure ⇒ 81-116.6 ⇒ 81'344.1 ⇒ 81'364		
81'373	Lexicology ⇒ 81'367.7		
81'373.2	Names according to types of object they denote. Onomastics		
81'373.21	Names of places. Geographical and topographical names. Toponymy		

81'374	Lexicography. Compilation and organization of dictionaries <i>Class dictionaries themselves with the subject, adding the form auxiliary (038); see the examples under (038) in Table 1d</i>	811.112.5	Dutch (Flemish in Belgium and north-eastern France)
81'374.2	Lexicography according to the subject coverage of the dictionary ◇ 81'374.2:81'373.421 Dictionaries of synonyms (as subject) ◇ 81'374.2:54 Chemical dictionaries (as subject) ◇ 81'374.2-029:81'373.6 Etymological dictionaries (as subject)	811.12 811.124	Italic languages Latin <i>Denote phases by '0...</i> ◇ 811.124'01 Old Latin ◇ 811.124'02 Classical Latin ◇ 811.124'03 Vulgar Latin ◇ 811.124'04 Mediaeval Latin ◇ 811.124'06 Modern Latin (e.g. for scientific nomenclature)
81'374.7	Order (arrangement) of dictionary entries	811.13	Romance languages
81'374.72	Formal order	811.131.1	Italian
81'374.721	Alphabetical order	811.133.1	French <i>Denote phases by '0...</i> ◇ 811.133.1'01 Old French ◇ 811.133.1'03 Anglo-Norman
81'374.722	Word-stem or word-root order		Spanish ◇ 811.134.2'282.3(468.1) Andalusian dialect of Spanish
81'374.73	Concept (semantic) order	811.134.2	Portuguese
81'374.74	Order according to the graphic structure of characters	811.134.3	
81'374.8	Lexicography according to the number of languages ◇ 81'374.8-022.214 Monolingual (one-language) lexicography ◇ 81'374.8-022.215 Bilingual lexicography ◇ 81'374.8-022.218 Multilingual lexicography	811.14	Greek (Hellenic) <i>Denote phases by '0...</i> ◇ 811.14'02 Classical Greek ◇ 811.14'03 Koine (New Testament Greek) ◇ 811.14'04 Byzantine Greek ◇ 811.14'06 Modern Greek
81'38	General stylistics		
81'42	Text linguistics. Discourse analysis		
81'44	Typological linguistics <i>For linguistic families and groups see 811...</i>	811.15	Celtic languages
81'442	Types of language according to phonological classification (typology)	811.16 811.161.1	Slavic languages Russian
81'442.2	Vowel languages (languages having less than 70 per cent consonant phonemes)	811.17	Baltic languages
81'442.3	Consonant languages (languages having more than 70 per cent consonant phonemes)	811.21/.22 811.214.21/.22 811.214.21 811.214.22	Indo-Iranian languages Hindustani (Hindi-Urdu complex) Hindi Urdu
81'443	Types of language according to morphological classification	811.29	Dead Indo-European languages (not listed elsewhere)
81'443.2	Isolating languages		
81'443.3	Agglutinative languages	811.34	Dead languages of unknown affiliation, spoken in the Mediterranean and Near East (except Semitic)
81'443.4	Inflected (flexional) languages		
81'443.5	Polysynthetic languages		
811	LANGUAGES 811.1/.9 ≡ =11/=9 <i>The subdivisions of 811 can be derived from =1/=9 (Table 1c Languages) by substituting a point for the equals sign. Only a few examples of languages and language groups are enumerated here.</i> <i>Further specification of individual languages or language groups by adding '0/'44 and/or -2/-26</i> ◇ 811.111-26 English as literary language ◇ 811.124'02'36 Grammar of classical Latin ◇ 811.134.2'24 Practical knowledge of Spanish ◇ 811.163.1'366.53 Number (singular, dual, plural) in Old Church Slavic	811.35 811.361 811.41 811.411.16 811.411.21 811.42 811.43 811.432.875 811.45 811.51 811.511.141 811.521 811.531 811.541	Caucasian languages Basque (Euskera, Euskara) Afro-Asiatic (Hamito-Semitic) languages Hebrew <i>Denote phases by '0...</i> ◇ 811.411.16'02 Biblical Hebrew ◇ 811.411.16'03 Mishnaic ◇ 811.411.16'04 Mediaeval Hebrew. Rabbinic ◇ 811.411.16'08 Modern Hebrew Arabic <i>Denote phases by '0...</i> ◇ 811.411.21'02 Classical Arabic ◇ 811.411.21'06 Modern ('Standard') Arabic Nilo-Saharan languages Congo-Kordofanian (Niger-Kordofanian) languages Swahili group Khoisan languages Ural-Altaic languages Hungarian Japanese Korean Ainu
811.1/.2	Indo-European languages		
811.11	Germanic languages		
811.111	English <i>Denote phases by '0...</i> ◇ 811.111'01 Anglo-Saxon (Old English) ◇ 811.111'04 Middle English		
811.112.2	German (High German, Standard written German) <i>Denote phases by '0...</i> ◇ 811.112.2'04 Middle High German (c. 10th-14th century)		

811.55	Palaeo-Siberian languages	811.81/.82	North-American Indian languages
811.56	Eskimo-Aleut languages	811.87	Central and South American Indian languages
811.57	Dravidian languages	811.9	Artificial languages
811.58	Sino-Tibetan languages	811.92	Artificial languages for use among human beings. International auxiliary languages (interlanguages)
811.581	Chinese languages		
811.61	Austro-Asiatic languages	811.93	Artificial languages used to instruct machines. Programming languages. Computer languages
811.621	Malayo-Polynesian languages		<i>Use only for the study of such languages from a linguistic point of view</i>
811.622	Oceanic languages		<i>Specify by alphabetic extension (Table 1h)</i>
811.71	Indo-Pacific (non-Austronesian) languages		⇒ 004.43
811.72	Australian languages		⇒ 004.655
811.8	American Indian (Amerindian) languages		

82 LITERATURE

Philological aspects of literature may be denoted by colon combination with 801 and its subdivisions 801.6/.83.

Linguistic aspects of literature may be denoted by colon combination with 81....

Further particulars may be denoted by common auxiliaries, colon combination etc.

- ◇ 82:001.8 Systematics of literature
- ◇ 82:1 Philosophy of literature
- ◇ 82:111.852 Aesthetics of literature. Literary taste
- ◇ 82:176.8 Literary ethics. Including: Plagiarism
- ◇ 82:7.079 Literature festivals
- ◇ 82(091) Literary history in general

82-1/-9	LITERARY FORMS. GENRES	82-193.1	Riddles
82-1	POETRY. POEMS. VERSE		⇒ 398.6
82-12	Dramatic verse. □ Dramatic poems.	82-193.2	Epigrams. Madrigals (in the literary sense)
	Verse dramas		⇒ 784.15
	⇒ 82-2		⇒ 801.675.2
82-13	Epic poems	82-193.3	Sonnets
			⇒ 801.675.2
82-132	Comic epics. Comic heroic poems	82-193.4	Rondels, roundels or roundelays (rondeaux).
82-133	Bucolic poems. Pastoral poems. Idylls.		Triolets
	Eclogues		⇒ 801.675.1
82-14	Lyric poems	82-193.5	Nuptial poems. □ Epithalamia. Prothalamia
82-141	Religious odes. Hymns. Songs of praise.	82-193.6	Epitaphs. Memorial inscriptions
	Ritual verse. Bacchic odes, dithyrambs.	82-194	Rhyming almanacs
	Metaphysical, mystical poems	82-2	DRAMA. PLAYS
82-142	Heroic odes. Eulogies		⇒ 792
82-143	Elegies. Laments. Monodies. Threnodies		⇒ 82-12
82-144	Ballads	82-21	Tragedies
	⇒ 398.8	82-22	Comedies
	⇒ 82-146.2	82-221	Comedy in the strict sense
82-145	Moral, philosophical poems		<i>Drama with a happy ending but not inconsistent with seriousness</i>
82-146	Sentimental, comic songs etc.	82-222	Comedies, i.e. amusing plays. Vaudevilles.
82-146.1	Drinking songs. Songs celebrating love		Lustspiele. □ High comedy. Aristocratic
82-146.2	Sentimental ballads. Love songs		comedy. Low, low-life comedy. Comedy of
82-146.3	Comic songs		intrigue
82-147	Cantatas. Poems to be set to music	82-223	Farce. Boisterous comedy. Situation
82-15	Didactic poems		comedy
82-16	Descriptive poems	82-224	Series of unconnected sketches. Revues
82-17	Satirical and humorous poems	82-225	Parody. Satire
82-191	Allegorical poems. Verse fables, parables etc.		⇒ 82-17
82-192	Melodic verse. Verse with music. □ Poems		⇒ 82-7
	with jazz	82-23	Melodramas. Thrillers. Mysteries.
	⇒ 82-147		Popular theatre
82-193	Shorter poetic forms	82-24	Historical plays
		82-25	Various kinds of play

- 82-251 Religious drama
⇒ 82-291.1
⇒ 82-97
- 82-252 Fairytale, fantasy plays
- 82-254 Plays for presentation on festive occasions. Occasional plays
- 82-27 Monologues. Recitations. Plays for a single performer (one-handers)
- 82-28 Tendentious, didactic, propagandist plays
- 82-291 Mediaeval and Renaissance drama
82-291.1 Liturgical drama. □ Religious, biblical plays. Mystery, miracle plays. Nativity, passion plays
⇒ 82-251
⇒ 82-97
82-291.2 Secular drama. □ Moralities. Moral plays with deterrent examples. Satirical farces. Play of fools
- 82-292 Theatre of the people. Folk dramas
⇒ 398.54
- 82-293 Musical drama. Libretti
⇒ 78.089.1
⇒ 782
82-293.1 Grand opera
82-293.2 Comic opera. Operettas
82-293.4 Music drama. Drama declaimed with musical accompaniment, Melodrama (in the strict sense)
⇒ 82-23
- 82-3 FICTION. PROSE NARRATIVE
- 82-31 Novels. Full-length stories
82-311.1 Psychological novels. Introspective, first-person novels
82-311.2 Representational novels of customs and characters. Depiction of everyday life. Realistic, slice-of-life novels
82-311.3 Novels dependent on plot. Adventure novels
82-311.4 Social novels. Picaresque novels. Simpliciads
82-311.5 Frivolous, comic novels
82-311.6 Historical novels. Political novels. War novels
82-311.7 Tendentious, didactic, propagandist novels
82-311.8 Travel novels. Exotic novels
82-311.9 Scientific novels. Science fiction
82-312.1 Philosophical novels. □ Existential novels
82-312.2 Religious novels. Novels with mystical or moral themes
82-312.3 Bucolic novels. Stories of country life
82-312.4 Crime novels. Detective novels. Novels of mystery, suspense. Thrillers
82-312.5 Popular, cheap, sensational fiction
82-312.6 Autobiographical novels. Fiction-style biography, semi-biography
82-312.7 Epistolary novels. Dialogue novels
82-312.8 Symbolic, cabalistic novels
82-312.9 Fantasy, fantastic novels
82-313.1 Satirical novels
82-313.2 Utopian novels
- 82-32 Short stories. Novellas
82-32 ≡ 82-31
- 82-34 Tales
⇒ 398.2
- 82-341 Comic stories
⇒ 82-311.5
- 82-342 Tales with a theoretical or practical moral. Fables
- 82-343 Myths. Legends. Fairytales
- 82-344 Tales of the fantastic, the supernatural, the macabre
- 82-36 Brief tales. Anecdotes
⇒ 398.23
- 82-39 Ancient (e.g. mediaeval) romances
- 82-4 ESSAYS
- 82-5 ORATORY. SPEECHES. □ Sermons. Addresses. Lectures. Toasts
- 82-6 LETTERS. ART OF LETTER-WRITING. CORRESPONDENCE. GENUINE LETTERS. OTHER WORKS IN EPISTOLARY FORM
- 82-7 PROSE SATIRE. HUMOUR, EPIGRAM, PARODY ETC.
- 82-82 Polygraphies. Selections, extracts. Anthologies. Curiosa
⇒ 08
- 82-83 Philosophical or discursive dialogues. Conversations on various subjects
- 82-84 Maxims. Gnostic utterances. Sententiae. Aphorisms. Pithy sayings. Adages. Proverbs. Thoughts. Table talk. Obiter dicta
- 82-845 Wit. Repartees. Word plays. Puns
- 82-9 VARIOUS OTHER GENRES
- 82-91 Popular literature. □ Popular editions. Simplified versions
⇒ 087.6
⇒ 398.5
- 82-92 Periodical literature. Writings in serials, journals, reviews. Journalism. Occasional literature. Polemic, political writings. Pamphleteering
⇒ 050
⇒ 070
- 82-93 Literature for children. Juvenile literature
⇒ 087.5
- 82-94 History as literary genre. Historical writing. Historiography. Chronicles. Annals. Memoirs. Journals. Diaries. Biography. Autobiography
⇒ 82(091)
⇒ 929
- 82-95 Criticism as a literary genre. Reviews. Literary histories
⇒ 82.09
- 82-96 Works of science and philosophy as literature
- 82-97 Religious literature in general
⇒ 27-29
⇒ 82-251
⇒ 82-291.1
- 82-98 Mottoes, slogans, devices. Personal maxims. Inscriptions. Rebuses. Puzzles. Anagrams
- 82-991 Dissertations

82-992	Descriptions of travels	82.0	THEORY, STUDY AND TECHNIQUE OF LITERATURE <i>These special auxiliaries may be used with the numbers for the literature of individual languages or for literary forms or both</i> ◇ 82-2.09 Criticism of dramatic works ◇ 821.112.2-2.09 Criticism of German dramatic works
82-993	Licentious, salacious, titillating literature. Erotica. Pornography		
82-994	Offbeat, eccentric literature. Inspirational, visionary, insane literature		
82-995	Literary fabrications, hoaxes. Pastiches. Imaginary, supposed works ⇒ 098.3		
82'0	ORIGINS AND PERIODS OF LANGUAGES. PHASES OF DEVELOPMENT 82'0 ≡ 81'0	82.02	Literary schools, trends and movements <i>For schools and trends in linguistics see 81-11...</i> <i>Specify by "... (Table 1g), (1/9) (Table 1e) or by alphabetic extension (Table 1h)</i>
82'01	Old or early periods. Earliest forms	82.09	Literary criticism. Literary studies
82'04	Middle period. Mediaeval	82.091	Comparative literary studies. Comparative literature
82'06	Modern period		
82'282	Literature in dialects, local and regional language, variants and vernaculars 82'282 ≡ 81'282	821	LITERATURES OF INDIVIDUAL LANGUAGES 821.1/9 ≡ 11/9 <i>The subdivisions of 821 can be derived from 11/9 (Table 1c Languages) by substituting a point for the equals sign. Only a few examples are enumerated here. Denote literature in dialect by '282 and subdivisions. Use (4/9) to class regional or national literatures, either at 821 itself or under the literature of a particular language</i> ◇ 821(494) Swiss literature in general ◇ 821.133.1(494) Swiss literature in the French language ◇ 821.133.1'282.4(494) Literature in the Swiss dialect of French
82.../Z	Works of specific authors <i>The number for literature in the relevant language is selected from 821... and specific authors denoted by name, abbreviated if desired</i> ◇ 821.111SHAK Works of Shakespeare ◇ 821.133.1-2MOL Dramatic works of Molière		
82.../Z1/7	Kinds of edition		
82.../Z1/7].01	In original languageD		
82.../Z1/7].02	In bilingual or annotated editionD		
82.../Z1/7].03	Translated text. Transliterated (e.g. romanized) textD		
82.../Z1/7].05	Abridgements. ParaphrasesD	821.111	English literature ◇ 821.111(73) American literature, i.e. literature of the USA in the English language ◇ 821.111(411) Scottish literature in the English language ◇ 821.111'282.3(414) Scottish literature in Lallans (Lowland Scots dialect of English)
82.../Z1/7].07	Interpretations. RecensionsD		
82.../Z1/7].08	Language and style (of author or work). □ Vocabulary. Grammar. Stylistic featuresD		
82.../Z1/7].091	Sources of the work. Author's source material (works imitated, adapted etc.)D	821.112.2	German literature ◇ 821.112.2(436) Austrian literature in German
82.../Z1/7].092	Continuations, sequels (with different title or by other writers)D	821.133.1	French literature ◇ 821.133.1(71) Canadian literature in French
82.../Z1/7].093	Vindications. ApologiasD	821.134.2	Spanish literature ◇ 821.134.2(7/8) Hispanic American literature ◇ 821.134.2(8) South American literature in Spanish ◇ 821.134.2(82) Argentine literature
82.../Z1	Complete works or sets		Portuguese literature ◇ 821.134.3(81) Brazilian literature
82.../Z2	Incomplete collections		
82.../Z3	Selections. Anthologies		
82.../Z4	Contributions to others' works. Joint works		
82.../Z411	Introductions. Forewords. Prefaces. Presentation inscriptions. Dedications		
82.../Z412	Joint, composite, collective works (edited by the writer or with contributions by the writer)		
82.../Z5	Anonymous works attributed to an author. Pseudonymous, cryptonymous works	821.134.3	
82.../Z6	Works in the author's name but attributed to others. Spurious works		
82.../Z7	Individual works <i>Add title, abbreviated if desired</i> ◇ 821.111SHAK7ROM Shakespeare's Romeo and Juliet ◇ 821.111SHAK7ROM.03=112.2 Romeo and Juliet, translated into German	821.14	Greek (Hellenic) literature ◇ 821.14'02 Classical Greek literature ◇ 821.14'06 Modern Greek literature
		821.161.1	Russian literature
		821.521	Japanese literature
		821.922	Literature in Esperanto

9 GEOGRAPHY. BIOGRAPHY. HISTORY

Table 1e - Common auxiliaries of place and Table 1g - Common auxiliaries of time will be needed continuously throughout the class. In addition 1k-05 may be needed to express personal characteristics, Table 1k-03 for the materials used in the making of artefacts in Archaeology and Table 1f - Ethnic grouping for topics in History. Table 1d - Common auxiliaries of form will be constantly needed and in Geography the subdivisions of (084.3) may be useful to express form and scale of maps
 ◇ 9-05 Persons and personal characteristics in geography, biography and history

902

ARCHAEOLOGY

Methods and techniques for the study of past civilizations and cultures based on discovery and interpretation of material remains or relics

902.01

Typology. Comparative study of form. Stratigraphy. Comparative study of location. Associations. Closed finds. Seriation

902.03

Archaeology according to conditions, location etc. of site and excavations

902.03 ≡ 69.03

These special auxiliary subdivisions may be used to extend the principal subdivisions below. They presuppose the citation order: technique - special branch of archaeology. Should the opposite order be required, the colon should be used

- ◇ 902.4.034 Recording techniques in underwater archaeology
- ◇ 902.034:902.4 Underwater archaeology - recording techniques

902.034

Underwater archaeology

- ◇ 902.4.034 Recording techniques in underwater archaeology

902.034.4

Archaeology of remains partially underwater

902.035

Deep underground, mountainside, cave and rockshelter archaeology

- ◇ 902.66.035 Geochronological dating techniques in cave archaeology

902.2

Archaeological fieldwork and sites generally

⇒ 351.853

⇒ 502

902.3

Field archaeology methods and techniques. □ Excavation, preservation techniques

902.4

Recording techniques in archaeology

902.6

Dating techniques in archaeology

902.62

Dating by stratigraphy

902.63

Dating by sequence. Dating by seriation

902.64

Dating by typology

902.65

Physicochemical dating techniques.

- Radiocarbon dating. Palaeomagnetic dating. Dating by chemical analysis. Dating by patination, obsidian hydration, thermoluminescence

902.66

Geochronological dating techniques.

- Glacial varve sequence, sedimentation rate

902.67

Palaeontological dating techniques (palaeobotanic and palaeozoological).

- Palynology (pollen analysis). Dendrochronology (tree-ring dating)

⇒ 630*55

903

PREHISTORY. PREHISTORIC REMAINS, ARTEFACTS, ANTIQUITIES

Interpretation and synthesis of the material relics of ancient humans, their culture forms and civilizations

- ◇ 903*63* Archaeological, prehistoric, protohistoric periods and ages
- ◇ 903-03 Materials of remains and artefacts. NOTE: See -03... (Table 1k) for enumeration of materials
- ◇ 903-032.42 Gold remains
- ◇ 903-033.64 Earthenware remains
- ◇ 903-034.4 Copper remains

Three different special auxiliary tables may be used with 903 and its subdivisions.

These concepts are introduced by -4 to express the shape and form of remains, .0 to express the materials from which objects are made and ' (apostrophe) to express the type of culture and level of civilization. All these may be used in combination or independently according to need. Prefer special subdivisions at .01/.08 where available, e.g. class flaked objects in 903.01, not 903-4.

The citation order (reverse of the filing order) is: 1. Type of culture and level of civilization. 2. Materials and techniques. 3. Shape and form of remains

- ◇ 903.21*12.05 Cast metal tools of hunting peoples
 - ◇ 903.23*15.02 Wheel-made pottery of nomadic peoples
 - ◇ 903.25-032.42 Gold ornaments
 - ◇ 903.26*14.08-035.3 Carved wooden objects of worship of pastoral peoples
 - ◇ 903.5*16 Burial remains of advanced farming cultures
- ⇒ 7.031

903-4

Shape and form of remains

903-4 ≡ 62-4

903-422.11

Round objects

903-462

Tubes

903.01

Stone and bone objects according to shaping method. □ Flaked, chipped, polished objects

903.02

Pottery according to method of manufacture. □ Moulded, ring-formed, wheel-made pottery

903.03

Glass according to method of manufacture

903.04

Textiles according to method of manufacture. □ Plaited, knitted, woven textiles

903.05

Metal artefacts according to method of manufacture. □ Cast and wrought metal

903.07

Artefacts according to quality of execution

903.08

Ornamentation. Decoration. □ Painted, carved, incised

903'1	Prehistoric culture forms	903.8	Intentional deposits (other than in graves). □ Hoards. Votive deposits. Bog finds
903'12	Hunting and fishing cultures		
903'13	Primitive farming cultures	904	CULTURAL REMAINS OF HISTORICAL TIMES
903'14	Pastoral cultures		<i>Antiquities, artefacts of ancient, mediaeval and modern times</i>
903'15	Nomadic cultures		<i>Denote subject or field of interest by colon combination with 0/9.</i>
903'16	Advanced farming cultures		<i>Specify location by (1/9), date by "...", materials by -03 of Table 1k</i>
903'18	Town, city cultures (civilizations)		◇ 904:6 Industrial antiquities
903.2	Artefacts		◇ 904:621 Antique machinery and tools
903.21	Tools. Implements		◇ 904:624.21 Historic bridges and their remains
903.22	Weapons		◇ 904:72 Historic architecture and remains
903.23	Vessels. Jars. Urns. Bowls		◇ 904:725.826 Remains of stadiums, amphitheatres etc.
903.24	Clothing		⇒ 7.031
903.25	Personal ornaments		
903.26	Objects of worship. Religious, cult objects	908	AREA STUDIES. STUDY OF A LOCALITY
903.27	Artistic remains. □ Cave paintings ⇒ 7.031.1		<i>Class here comprehensive or multidisciplinary studies of an area - its geography, history, customs, culture, economy etc.</i>
903.28	Food remains. Kitchen middens		<i>Comprehensive historical studies are classed at 94 'General history'</i>
903.3	Prehistoric dwellings		<i>Specify locality by (3/9) (Table 1e)</i>
903.4	Prehistoric settlements		◇ 908(411) Comprehensive studies of Scotland
903.5	Burial remains. □ Prehistoric graves, barrows (tumuli)		◇ 908(414.45) Comprehensive studies of Midlothian
903.6	Prehistoric stone monuments		
903.7	Places of worship. □ Temples, altars		

91 GEOGRAPHY. EXPLORATION OF THE EARTH AND OF INDIVIDUAL COUNTRIES. TRAVEL. REGIONAL GEOGRAPHY

Information about a specific subject in relation to place is classed with the subject, using (1/9)

Table 1e

- ◇ 331.2(450) Salaries and wages in Italy
- ⇒ 308
- ⇒ 528
- ⇒ 55

910	GENERAL QUESTIONS. GEOGRAPHY AS A SCIENCE. EXPLORATION. TRAVEL		◇ 910.4(091) Historical accounts of voyages etc.
910.1	Science of geography. Methodology (theory, systems, methods)		◇ 910.4(100) International and world travel
910.2	Kinds and techniques of geographical exploration. □ Field work. Exploration and travels	911	◇ 910.4(430:450) Travels between Germany and Italy
910.21	Preparation. Planning		GENERAL GEOGRAPHY. SCIENCE OF GEOGRAPHICAL FACTORS (SYSTEMATIC GEOGRAPHY). THEORETICAL GEOGRAPHY
910.22	Equipment. Instruments. Baggage. Clothing	911.2	Physical geography
910.26	Evaluation or Interpretation of literature, maps and other documents from a geographical point of view ◇ 910.26:528.77 Geographical interpretation of aerial photograms		◇ 911.2:556 Hydrogeography
910.27	Geographical mapping. Thematic mapping in the field		◇ 911.2:581.9 Plant geography (phytogeography) as branch of physical geography
910.3	Exploration of particular geographical features ◇ 910.3:551.324 Exploration of glaciers	911.3	⇒ 574.9
910.4	Voyages of discovery. □ Journeys. Travels. Expeditions ◇ 910.4:796.5 Journeys in the form of walking, mountaineering, camping	911.37	⇒ 581.9
			⇒ 591.9
			Human geography (cultural geography). Geography of cultural factors
			◇ 911.3:30 Social geography
			◇ 911.3:314 Demographic geography
			◇ 911.3:32 Political geography
			◇ 911.3:33 Economic geography
			Settlements. Settlement geography
			⇒ 658.21
			⇒ 71

911.372	Site, types, function, forms of settlement.	911.375.635	Surroundings, environs of town. Peripheral area
911.372.2	Deserted settlements	911.375.64	Districts, precincts, quarters of town <i>Specify function by colon combination and characterize inhabitants using -05... (Table 1k)</i>
	Settlements according to spatial distribution and situation		
	◇ 911.372.2(23.07-04) Altitude limits of settlements		◇ 911.375.64:725.1:354 Government, ministry districts
911.372.3	Types of settlement		◇ 911.375.64:725.1:656.21 Railway station districts
	<i>Specify type by colon combination</i>		◇ 911.375.64:725.1:78/79 Entertainment and amusement districts
	◇ 911.372.3:316.324.22 Settlements of hunters and gatherers		◇ 911.375.64:728 Residential districts
911.372.31	Single settlements		◇ 911.375.64:728-054 According to predominant ethnic characteristics, nationality, citizenship etc.
911.372.32	Grouped settlements		◇ 911.375.64:728-058 According to predominant social class, civil status of inhabitants
911.372.4	Settlements according to their function		◇ 911.375.64:728-058.34 Poor quarters. Districts of low-income groups ⇒ 711.5
	◇ 911.372.4:27-788 Monastic settlements	911.375.67	Administrative division of a town or city
	◇ 911.372.4:35 Administration centres as settlements		
	◇ 911.372.4:796.9 Winter sports resorts as settlements		
	⇒ 711.45		
	⇒ 711.451		
	⇒ 711.455		
	⇒ 711.55		
	⇒ 711.56		
	⇒ 911.375.4		
911.372.5	Pattern, plan (perspective), elevation of settlements	911.5/9	Theoretical geography
911.372.7	Utilities and services of settlements.	911.5	Typological geography <i>Study of geographical landscape types, and divisions based on them</i>
	Infrastructure of settlements		
911.372.9	Deserted settlements. Desertion, abandonment of settlements	911.52	Natural landscape
911.373	Rural settlements (their study and geography)	911.53	Cultural landscape
911.374	Special types of settlement	911.6	Individual spatial divisions and regional units
911.374.1	Small suburban settlements	911.7	Comparative geography. □ Comparative regional research
911.374.2	Country settlements functioning as centres		
911.374.3	Nonintegrated shopping centres	911.8	Normative geography <i>Norms and laws of form, ecology, development etc.</i>
	<i>Outside built-up areas</i>	911.9	Applied, practical geography ◇ 911.9:502 Application of geography to wildlife conservation and protection
911.374.4	Industrial and manufacturing centres		
	<i>Outside built-up areas</i>		
911.374.5	Traffic settlements. □ Railhead settlements		
911.374.6	Residential settlements of urban type (in open country)	912	NONLITERARY, NONTTEXTUAL REPRESENTATIONS OF A REGION. □ Pictures. Graphs. Diagrams. Profiles. Cartograms. Maps. Atlases. Globes (as expressions of geographical knowledge) 912 ≡ (08) ◇ 912(6)"17" Nontextual representations of Africa in the 18th century ⇒ 528 ⇒ 778.35
911.374.7	School and church settlements (in open country)		
911.374.8	Fortified settlements and nobles' estates (in open country)	912.43	Cartographic images, maps, etc. (as expressions of geographical knowledge)
911.375	Urban settlements (their study and geography).	912.643	Globes
	□ Towns. Cities	913	REGIONAL GEOGRAPHY
911.375.1	Urban growth. Urbanization		◇ 913(100) Regional geography of the world
911.375.12	Growth of new districts. Urban sprawl		◇ 913(21) Geography of land areas as a whole
911.375.3	Types of town or city		◇ 913(26) Geography of oceans and seas as a whole
	<i>Specify size by (188) from Table 1e and date by "..." from Table 1g</i>		◇ 913(3) Geography of the ancient world
	⇒ 711.417		◇ 913(315) Geography of ancient China
	⇒ 711.42		◇ 913(32/39) Geography of other individual countries of the ancient world
911.375.4	Towns or cities according to function		◇ 913(38) Geography of ancient Greece
	◇ 911.375.4:378.4 University or college towns		◇ 913(399.7) Geography of Pre-Columbian America
	⇒ 711.45		
	⇒ 711.47		
	⇒ 711.55		
	⇒ 711.57		
	⇒ 911.372.4		
911.375.5	Townscape. □ Plan, layout, elevation, skyline (of town or city)		
911.375.6	Internal division and uses of urban area.		
	□ Sectors, quarters, districts of town		
911.375.62	Functional divisions, uses of urban area		
	◇ 911.375.62:656 Area occupied by traffic and transport facilities		
911.375.63	Town divisions in relation to its centre		
911.375.631	Town centre. City part (inner city)		
911.375.632	Suburbs. Suburban areas		

- ◇ 913(4/9) Descriptions of individual regions and countries of the modern world (whether or not from the viewpoint of specialist, academic geography)
- ◇ 913(4) Geography of Europe
- ◇ 913(41) Geography of the British Isles
- ◇ 913(410) Geography of Great Britain and the United Kingdom
- ◇ 913(430) Geography of Germany
- ◇ 913(436) Geography of Austria
- ◇ 913(5) Geography of Asia
- ◇ 913(7) Geography of North and Central America
- ◇ 913(73) Geography of the United States of America
- ◇ 913(9) Geography of Oceania, the polar regions, Australasia etc.
- 929**
- BIOGRAPHICAL AND RELATED STUDIES**
- Denote subject or speciality by colon combination with 0/9. The citation order may be Biography - subject, or Subject - biography, exchanging the position of the numbers on either side of the colon, as preferred. To ensure consistency one of these alternatives should be selected and adhered to. Specify country by (1/9), period by "...", individuals by A/Z (Table 1h)*
- ◇ 929:27-36 Saints' lives. Hagiography.
NOTE: Or 27-36:929, if an arrangement of Subject - biography is preferred
- ◇ 929:272-732.2 Biographies of popes
- ◇ 929:75 Biographies of painters
- ◇ 929Grotius Biographies of Hugo Grotius (Huig de Groot)
- ◇ 929Charles(44)*5 Biographies of King Charles V of France
⇒ (092)
⇒ 012
⇒ 82-94
- 929.5**
- Genealogy**
- 929.51**
- Theoretical genealogy**
⇒ 575.1
- 929.511 Theory of descendancy, of descendancy tables
- 929.512 Theory of ancestry, of ancestry tables.
Ascendancy tables
- 929.513 Theory of collateral relationship, of consanguinity
- 929.52**
- Special genealogy. Family histories**
Details by common auxiliaries, here and at subdivisions
- ◇ 929.52:81*373.232 Family names and their origins
- ◇ 929.52SAC History of the Sackville family
- ◇ 929.522(410.3) Ascendancy tables of Welsh families
- ◇ 929.522"17" Ascendancy tables for the 18th century
- 929.521 Tables of descendants. Descendancy tables.
Tables of lineage. Family trees
- 929.522 Tables of ancestors. Ascendancy tables.
Pedigree. Ancestral lineage
- 929.523 Proof of blood relationship. Consanguinity tables
- 929.53**
- Sources of genealogy.** □ Registers of births, deaths and marriages. Baptismal registers. Burial registers. Records of legal status. Civil registers
⇒ 27-735
⇒ 27-774-735
⇒ 347.183
⇒ 351.755
- 929.532 Ecclesiastical registers. Church registers
◇ 929.532:27-725 Lists of clergy (as genealogical source)
- 929.535 Registers of secular, civil authorities
◇ 929.535:351.755 Civil registry (of police etc.) as genealogical source
- 929.55**
- Funerary inscriptions. Epitaphs**
⇒ 726.82
⇒ 82-193.6
- 929.6**
- Heraldry.** □ Blazonry. Armorial bearings. Coats of arms
⇒ 736.3
- Theory and principles**
The shield or scutcheon and its various shapes
Heraldic tincture. Armorial colours
The divisions of the field
Devices. Charges. Crests. Bearings
Ordinaries and common charges. □ Chevrons, lozenges
Natural charges. □ Human figures, animals, plants
Man-made objects, artefacts as charges.
□ Castles, weapons
Fantastic forms as charges. □ Dragons
House marks, letters
Accessories. □ Headgear (helms, crowns, coronets). Marks of rank. Badges, emblems
- Armorial according to content**
Nature of the armorials
Form of the armorials
Families
Families, civil and ecclesiastical bodies
Families and civil bodies
Families and ecclesiastical bodies
Civil and ecclesiastical bodies
Civil corporate bodies
Ecclesiastical corporate bodies
- Heraldic sources**
Official objects
Seals
Coins and medals
⇒ 737
Charters. Documents. Diplomas
Public and private buildings
Works of art
Specify by :73/76
- 929.64**
- 929.64'2
929.64'4
- 929.642
929.643
929.644
929.645
929.646
929.647
929.648
- 929.65**
- 929.651/653
929.651
929.652
- 929.653
929.655
929.657
- 929.7**
- 929.7.031
929.7.032
929.7.034
929.7.035
929.7.036
929.7.037
929.7.038
- 929.71**
- 929.73**
- 929.731
929.732
929.733
929.735
- Nobility. Titles. Peerage**
Nature, characteristics, establishment and structure of the nobility
Creation, investiture. Acquisition of nobility.
Half blood nobility
Proof, verification of nobility
Rights, prerogatives, privileges of the nobility
Duties, obligations of the nobility
Disclaimers. Dormancy and extinction of titles
Marks of nobility. Insignia. Dress
- Orders of chivalry. Decorations**
⇒ 671.128
- Degrees of nobility. Rank. Precedence**
Royalty. Monarchs. Kings, queens. Princes, princesses. Archdukes, archduchesses. Grand dukes, grand duchesses
Hereditary nobility. Dukes. Marquesses. Earls. Viscounts. Barons. Baronets. Counts. Margraves
Personal (nonhereditary) nobility. Life peers. Knights
Landed gentry. Squires

929.737	Civic nobility. Respected, revered families (not strictly nobility)	929.925	Merchant navy flags
		929.927	Flags of organizations, societies
929.9	Flags. Standards. Banners	929.929	Other flags. □ International flags (United Nations). Yachting flags. Signal flags. Code flags
929.921	National flags. National colours		
929.922	Local and regional flags. □ Town, municipal flags		
929.923	Official flags and standards of individuals. □ Of heads of state	929.93	Flags of the nobility and the orders of chivalry
929.924	Flags and standards of national armed forces. □ Army. Navy. Air force		929.93 ≡ 929.7

93/94 HISTORY

⇒ (091)

⇒ (093)

930	SCIENCE OF HISTORY. HISTORIOGRAPHY <i>Class here works about historiography and the state of historical studies in particular places and times</i> ◇ 930(485)"18" State of historical studies in Sweden in the 19th century		◇ 930.2:81'22 Semiotics (as ancillary historical science)
		930.23	Historical criticism
		930.24	Historical chronology ⇒ 006.92
		930.25	Archivistics. Archives (including public and other records) ⇒ 005.92
930.1	History as a science. □ Theory and philosophy of history. Structure and form of history	930.251	Archival techniques and organization. □ Establishment and administration of repositories
930.2	Methodology of history. Ancillary historical sciences. □ Studies based on written sources, records, inscriptions ◇ 930.2:003.071 Epigraphy (as ancillary historical science). Study of inscriptions ◇ 930.2:003.072 Palaeography (as ancillary historical science). Study of ancient writing ◇ 930.2:003.074 Diplomatics (as ancillary historical science). Critical study of the internal characteristics, style and form of documents	930.253	Archive repositories and their contents. □ History, scope, arrangement and description of contents ◇ 930.253:352 Local government archives. Municipal archives
		930.255	Archive lists and catalogues
		930.85	History of civilization. Cultural history ⇒ 008 ⇒ 130.2 ⇒ 17.023.36

94 GENERAL HISTORY

The history of individual countries and peoples is expressed through the use of Table 1e - Common auxiliaries of place, Table 1g - Common auxiliaries of time, and, where appropriate, Table 1f - Common auxiliaries of ethnic grouping and nationality, for peoples who possess a common history but cannot be geographically limited, e.g. the Hittites, the Jews, the Gipsies (see the last two examples below). Where a number is given in the Common auxiliaries of place, e.g. (367) Regions of the Slavs, that number should be preferred. For some countries alternative Special auxiliary subdivisions are available if a shorter notation is required, but these are not recommended

- ◇ 94(100) World history (chronological summation of facts). NOTE: Class here also popular histories of special classes of event, e.g. 'Great disasters of the world'
- ◇ 94(100)"/05" Ancient history in general. History of ancient peoples
- ◇ 94(100)"05/..." Mediaeval and modern history in general
- ◇ 94(100)"1914/1918" First World War, 1914-18. NOTE: If preferred, place under History of Europe at 94(4)"1914/1918"
- ◇ 94(100)"1939/1945" Second World War, 1939-45. NOTE: If preferred, place under History of Europe at 94(4)"1939/1945"
- ◇ 94(3) History of the ancient world. NOTE: Alternatively class areas of the ancient world with the history of their modern counterparts
- ◇ 94(315) History of ancient China (to ca. 600 AD)
- ◇ 94(32) History of ancient Egypt (to 640 AD)
- ◇ 94(34) History of ancient India (to 647 AD). Including: Ancient Indo-China
- ◇ 94(35) History of Ancient Medo-Persia. Including: Chaldaea, Mesopotamia, Media, Persia, Assyria, Babylonia
- ◇ 94(36) History of the north, west and east European peoples to 476 AD
- ◇ 94(363) History of the Germanic tribes. Including: Teutons, Saxons, Angles, Franks, Goths
- ◇ 94(364) History of the Celts
- ◇ 94(366) History of the Britons
- ◇ 94(367) History of the Slavs

- ◇ 94(37) History of ancient Rome (to 476 AD)
- ◇ 94(38) History of ancient Greece (to 323 AD)
- ◇ 94(397) History of ancient Africa. Including: Mauretania, Numidia, Carthage, Libya, Aethiopia
- ◇ 94(398) History of south-east Europe. Including: Dacia, Vindelicia, Rhaetia, Noricum, Pannonia, Illyria, Dalmatia
- ◇ 94(4+7) History of the West
- ◇ 94(4) History of Europe
- ◇ 94(4)"0375/1492" European Middle Ages (in the wider sense) ca. 375-1492
- ◇ 94(4)"1492/1914" Modern history of Europe, ca. 1492-1914
- ◇ 94(4)"192/199" History of Europe from 1919. Twentieth-century European history in general
- ◇ 94(4)"1945/..." History of Europe from 1945. Post-war Europe
- ◇ 94(=214.58) History of the Romany peoples, gipsies
- ◇ 94(=411.16) History of the Jewish people

94(=822.21) HISTORY OF THE AZTECS**94(=873.122) HISTORY OF THE INCAS****94(41/99) HISTORY OF INDIVIDUAL COUNTRIES**

If preferred, the ancient history of an individual country, e.g. China, Greece, may also be collocated here, rather than scattered through the use of (3) - Areas of the ancient world

94(416)

94(417)

94(420)

94(41) HISTORY OF THE BRITISH ISLES

Class here the history of the current administrative divisions of the United Kingdom

94(411)

◇ 94(410.111) History of Greater London
History of Scotland

◇ 94(411)".../1057" Gaelic period, to 1057

◇ 94(411)"1057/1314" Efforts at control by England, 1057-1314. Battle of Bannockburn, 1314

◇ 94(411)"1314/1424" Independent Kingdom, 1314-1424

◇ 94(411)"1424/1542" From James I to James V, 1424-1542

◇ 94(411)"1542/1603" Reformation, 1542-1603

◇ 94(411)"1603/1707" Personal union with England, 1603-1707

◇ 94(411)"1707/1837" Act of Union. House of Hanover, 1707-1837

◇ 94(411)"1837/2000" Victorian period and twentieth centuries, 1837-2000

◇ 94(411)"2001/..." Twenty-first century, 2001-...

94(415)

History of Ireland

Alternative use 94(416) and 94(417) to file information about the two areas separately

◇ 94(415)".../1086" Early history, to 1086

◇ 94(415)"1086/1171" Separate kingdoms, 1086-1171

◇ 94(415)"1171/1399" House of Plantagenet, 1171-1399

◇ 94(415)"1399/1485" Houses of Lancaster and York, 1399-1485

◇ 94(415)"1485/1603" House of Tudor, 1485-1603

◇ 94(415)"1603/1691" House of Stuart, up to the Treaty of Limerick, 1603-1691

◇ 94(415)"1691/1800" From the Treaty of Limerick to the Act of Union, 1691-1800. Including: The House of Hanover

◇ 94(415)"1801/1900" The nineteenth century, 1801-1900

◇ 94(415)"19" The twentieth century, 1901-

◇ 94(415)"1901/1920" Up to the Government of Ireland Act, 1920

- ◇ 94(415)"1920/..." From 1920. Northern Ireland and the Irish Free State (later Republic of Ireland). NOTE: Alternatively use 94(416) and 94(417) to file information about the two areas separately

History of Northern Ireland

History of the Irish Free State and Republic of Ireland

History of England and Great Britain

Class here the history of the United Kingdom as a whole. For the history of the countries of the UK individually, see

94(411), 94(415) and 94(429) as well as 94(420). See also 94(41) and the note there

◇ 94(420)".../1066" Prehistory, Roman and Anglo-Saxon period

◇ 94(420)"1066/1154" Norman kings, 1066-1154

◇ 94(420)"1154/1399" House of Plantagenet, 1154-1399

◇ 94(420)"1399/1485" Houses of Lancaster and York, 1399-1485. Wars of the Roses, 1455-1485

◇ 94(420)"1485/1603" House of Tudor, 1485-1603. Tudor and Elizabethan periods

◇ 94(420)"1603/1714" House of Stuart, 1603-1714. Interregnum, 1649-1660. The Commonwealth. Cromwell. The Restoration

◇ 94(420)"1714/1837" House of Hanover, 1714-1837. Georgian and Regency periods

◇ 94(420)"1837/..." Victoria and House of Windsor, 1837-

◇ 94(420)"1837/1901" Victoria, 1837-1901

◇ 94(420)"1901/1910" Edward VII, 1901-1910

◇ 94(420)"1910/1936" George V, 1910-1936

◇ 94(420)"1936" Edward VIII, 1936

◇ 94(420)"1936/1952" George VI, 1936-1952

◇ 94(420)"1952/..." Elizabeth II, 1952-

History of Wales

94(429)

94(430)**94(44)****94(5)**

94(54)

HISTORY OF GERMANY**HISTORY OF FRANCE****HISTORY OF THE ORIENT. HISTORY OF ASIA**

History of the Indian subcontinent

◇ 94(54)"0647/1774" Asian dynasties and European penetration, 647-1774. Muslim conquests. Mogul period

◇ 94(54)"1765/1947" British India, 1765-1947. Including: East India Company, Sepoy mutiny, control by British Crown

◇ 94(54)"1947/..." Independence and partition, 1947-. Including: States of India, Pakistan, Bangladesh. NOTE: Alternatively use 94(540), 94(549.1) and 94(549.3) to class information about these countries separately

- 94(540) History of India
⇒ 94(54)"1747/..."
- 94(549.1) History of Pakistan
⇒ 94(54)"1947/..."
- 94(549.3) History of Bangladesh
⇒ 94(54)"1947/..."
- 94(569.4) HISTORY OF PALESTINE AND ISRAEL**
 ◇ 94(569.4)".../0070" Early history, to 70 AD
 ◇ 94(569.4)"0070/0638" Pre-Islamic period, 70-638
 ◇ 94(569.4)"0638/1917" Mediaeval and modern Palestine, 638-1917
 ◇ 94(569.4)"1917/1948" Palestine under British control and mandate, 1917-1948
 ◇ 94(569.4)"1948/..." Independence, 1948-. State of Israel. NOTE: Alternatively use 94(569.4) to class the history of Israel
 ⇒ 94(=411.16)
- 94(591) History of Burma
 ◇ 94(591)".../1826" Early history, to 1826
 ◇ 94(591)"1826/1885" Period of British conquest, 1826-1885
 ◇ 94(591)"1885/1948" British rule, 1885-1948
 ◇ 94(591)"1948/..." Independence, 1948-
- 94(595) History of Malaysia, Malaya, Straits Settlements and Singapore
- 94(6) HISTORY OF AFRICA**
 94(62) History of Egypt and the Sudan
 94(620) History of Egypt
 ◇ 94(620)"0640/1517" Period of Arab rule, 640-1517
 ◇ 94(620)"1517/1882" Period of Ottoman rule, 1517-1882
 ◇ 94(620)"1882/1922" British occupation and protectorate 1882-1922
 ◇ 94(620)"1922/..." Independence, 1922-
- 94(624) History of Sudan
 ◇ 94(624)".../1820" From the Arab invasion, 7th century, to 1820
 ◇ 94(624)"1820/1956" Period of Egyptian and British rule, 1820-1956. Anglo-Egyptian Sudan
 ◇ 94(624)"1956/..." Independence, 1956-
- 94(664) History of Sierra Leone
 ◇ 94(664)".../1787" Early history, to 1787
 ◇ 94(664)"1787/1896" British colony, 1787-1896
 ◇ 94(664)"1896/1961" Colony and protectorate, 1896-1961
 ◇ 94(664)"1961/..." Independence, 1961-
- 94(665.1) History of the Gambia
 ◇ 94(665.1)".../1807" Early history, to 1807
 ◇ 94(665.1)"1807/1965" British colony, 1807-1965
 ◇ 94(665.1)"1965/..." Independence, 1965-
- 94(667) History of Ghana
 ◇ 94(667)".../1874" Early history, to 1874
 ◇ 94(667)"1874/1957" British colony, 1874-1957. Gold Coast
 ◇ 94(667)"1957/..." Independence, 1957-
- 94(668.1) History of Togo, Togoland
 ◇ 94(668.1)".../1894" Early history, to 1894
 ◇ 94(668.1)"1894/1914" German period, 1894-1914
 ◇ 94(668.1)"1914/1960" Anglo-French period, 1914-1960
 ◇ 94(668.1)"1960/..." Independence, 1960-
- 94(669.1) History of Nigeria
 ◇ 94(669.1)".../1886" Early history, to 1886
 ◇ 94(669.1)"1886/1960" British colony and protectorate, 1886-1960
 ◇ 94(669.1)"1960/..." Independence, 1960-
- 94(671.1) History of Cameroon
 ◇ 94(671.1)".../1884" Early history, to 1884
 ◇ 94(671.1)"1884/1916" German period, 1884-1916
 ◇ 94(671.1)"1916/1959" Anglo-French period, 1916-1959
 ◇ 94(671.1)"1960/..." Independence, 1960-
- 94(676) History of the east African states (and British East Africa)
- 94(676.1) History of Uganda
 ◇ 94(676.1)".../1894" Early history, to 1894
 ◇ 94(676.1)"1894/1962" British period, 1894-1962
 ◇ 94(676.1)"1962/..." Independence, 1962-
- 94(676.2) History of Kenya
 ◇ 94(676.2)".../1895" Early history, to 1895
 ◇ 94(676.2)"1895/1963" British period, 1895-1963
 ◇ 94(676.2)"1963/..." Independence, 1963-
- 94(678) History of Tanzania (Tanganyika and Zanzibar)
 ◇ 94(678)"1964/..." United republic, 1964-
- 94(678.1) History of Zanzibar
 ◇ 94(678.1)".../1700" Early history, to 1700
 ◇ 94(678.1)"1700/1890" Arab rule, 1700-1890
 ◇ 94(678.1)"1890/1963" British protectorate, 1890-1963
 ◇ 94(678.1)"1963/..." Independence, 1963-
- 94(678.2) History of Tanganyika
 ◇ 94(678.2)".../1884" Early history, to 1884
 ◇ 94(678.2)"1884/1916" German period, 1884-1916
 ◇ 94(678.2)"1916/1961" British period, 1916-1961
 ◇ 94(678.2)"1961/..." Independence, 1961-
- 94(680) History of South Africa
 ◇ 94(680)".../1488" Early history, to 1488
 ◇ 94(680)"1488/1814" Period of exploration and settlement, 1488-1814
 ◇ 94(680)"1814/1909" Nineteenth and early twentieth centuries, 1814-1909
 ◇ 94(680)"1909/1961" Union of South Africa, 1909-1961
 ◇ 94(680)"1961/..." Republic of South Africa, 1961-
- 94(688.1) History of Namibia
 94(688.3) History of Botswana
 94(688.5) History of Lesotho
 94(688.7) History of Swaziland
 94(689) History of Rhodesia and Nyasaland
 ◇ 94(689)".../1888" Early history, to 1888
 ◇ 94(689)"1888/1953" British control, 1888-1953
 ◇ 94(689)"1953/1963" Federation of Rhodesia and Nyasaland (Central African Federation), 1953-1963
 ◇ 94(689)"1964/..." Post-federation, 1964-
- 94(689.1) History of Zimbabwe
 94(689.4) History of Zambia
 94(689.7) History of Malawi
- 94(7) HISTORY OF NORTH AND CENTRAL AMERICA**
 94(71) History of Canada
 ◇ 94(71)".../1760" Early history and French rule, to 1760
 ◇ 94(71)"1760/1791" British rule, 1760-1791
 ◇ 94(71)"1791/1841" Constitutional or representative government, 1791-1841
 ◇ 94(71)"1841/1867" The Union, 1841-1867
 ◇ 94(71)"1867/1931" The Federation, 1867-1931
 ◇ 94(71)"1931/..." Self-governing Canada, 1931-. Dominion of Canada

- 94(72) History of Mexico
 94(728) History of Central America
 94(728.2) History of British Honduras (Belize)
 ◇ 94(728.2)"1502/1535" Discovery, exploration and conquest, 1502-1535
 ◇ 94(728.2)"1535/1821" Colonial period, 1535-1821
 ◇ 94(728.2)"1821/1900" Nineteenth century, 1821-1900
 ◇ 94(728.2)"19" Twentieth century, 1901-2000
 ◇ 94(728.2)"20" Twenty-first century, 2001-...
- 94(729) History of the Caribbean territories. West Indies
 ◇ 94(729)"1492/1608" Discovery, exploration, conquest, 1492-1608
 ◇ 94(729)"1608/1800" Seventeenth and eighteenth centuries, 1608-1800
 ◇ 94(729)"1801/1902" Nineteenth century, 1801-1902
 ◇ 94(729)"1902/..." Twentieth century, 1902-
- 94(73) History of the United States of America
 ◇ 94(73)".../1607" From discovery to 1607. Including: Pre-Columbian period, discovery (by Norsemen, Columbus etc.), explorations
 ◇ 94(73)"1607/1775" Colonial period, 1607-1775
 ◇ 94(73)"1775/1789" War of Independence. Founding of the Union. 1775-1789
 ◇ 94(73)"1789/1809" Constitutional period, 1789-1809. Including: Administrations of Washington, Adams, Jefferson
 ◇ 94(73)"1809/1845" Early nineteenth century, 1809-1845. Including: Administrations of Madison, Monroe, Adams, Jackson, van Buren, Harrison, Tyler
 ◇ 94(73)"1845/1861" Mid nineteenth century, 1845-1861. Including: Administrations of Polk, Taylor, Fillmore, Pierce, Buchanan
- 94(73)"1861/1865" Administration of Abraham Lincoln, 1861-1865. American Civil War
 ◇ 94(73)"1865/1901" Later nineteenth century, 1865-1901. Including: Administrations of Andrew Johnson, Grant, Hayes, Garfield, Cleveland, B. Harrison, McKinley
 ◇ 94(73)"1901/..." History of the United States from 1901
 ◇ 94(73)"1901/1953" Early twentieth century, 1901-1953. NOTE: To express the administration of individual presidents, use the dates that they held office
 ◇ 94(73)"1953/..." Later twentieth century, 1953-. NOTE: To express the administrations of individual presidents, use the dates that they held office
 History of individual states of the USA
- 94(734/739)
- 94(8)**
94(9)
 94(931)
- HISTORY OF SOUTH AMERICA**
HISTORY OF OCEANIA, THE POLAR REGIONS, AUSTRALASIA ETC.
 History of New Zealand
 ◇ 94(931)".../1840" Maori period. Early history, to 1840
 ◇ 94(931)"1840/1907" Colonial period, 1840-1907
 ◇ 94(931)"1907/..." Dominion status, 1907-
- 94(94)
- History of Australia
 ◇ 94(94)".../1788" Early history, to 1788
 ◇ 94(94)"1788/1851" Settlement and early growth, 1788-1851
 ◇ 94(94)"1851/1901" Colonial administration, 1851-1901
 ◇ 94(94)"1901/1945" Commonwealth (to the end of the Second World War), 1901-1945
 ◇ 94(94)"1945/..." Later twentieth century, 1945-